TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		25	476.8		
		30	403.4		
		35	240.0		
		40	145.7		
		45	56.68		
		50	23.50		
		60	2.050		
$M_n = 135 \times 10^3$	120 (5 min)	25	255.2	Optically active hot stage optical microscopy	301
		30	157.5		
		35	103.3		
		40	41.7		
		45	23.6		
		50	6.01		
		55	2.02		
		60	0.467		
Poly(oxypropylene) isotactic	120 (20 min)	45.3	162.9	Hot stage optical microscopy; nucleation rates listed in text	11
$M_v = 86.0 \times 10^3$		47.7	109.2		
(Dow Chem.)		50.2	63.2		
		52.6	38.3		
		55.0	20.0		
4.2. COMPLEXES					
Poly(oxyethylene)/resorcinol		25	3030	Hot stage optical microscopy; unbanded spherulites	303
stoichiometry, 2 : 1 complex		30	4440		
PEO; $M_w = 6000$		35	7110		
		40	10800		
		45	15550		
		50	21230		
		55	27500		
		60	33170		
		62	34240		
		65	32900		
		67	31100		
		69.5	22540		
		71	16060		
		72.3	11050		
		72.8	720		
		72.7	8047		
		25.1	1490	Banded spherulites	303
		27.3	1740		
		30.2	2430		
		32.5	2740		
		35	3320		
		37.5	4020		
		40	4950		
		42.4	6020		
		44.9	6530		
		47.5	7490		
		50	6950		
		52.5	6230		
		54.7	5200		
		57.5	3270		
PEO; $M_w = 35000$		26	15990	Unbanded spherulites	302
		30.3	27680		
		35	139700		
		42.5	391110		
		47.2	784000		
		52.	1459150		
		57.5	2711400		
		62.5	3478600		
		67.4	4145700		
		71.4	3961000		
		74.2	3269600		
		78	1928000		

TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		80	1141300	Unbanded spherulites	302
		82	481000		
		82.5	117200		
		84.0	43100		
		85	14650		
		25	2160	Unbanded spherulites	302
		28	2400		
		30	3000		
		32	3580		
		35.5	4270		
		37.7	4940		
		42.4	5670		
		45.7	6530		
		50	7240		
		54	8020		
		57.3	8580		
		60	8220		
		62.5	6310		
		64	4170		
		65	3460		
		66	2660		
		67	1460		
Poly(oxyethylene)/resorcinol stoichometry 2 : 1 complex		45	15.34×10^3	Optical hot stage microscopy; two spherulitic form of PEO: α and β; other growth features were established α-type	303
		47.5	17.96		
		50	20.10		
		52.5	24.0		
		55	27.1		
		57.5	30.7		
		60	32.8		
		62.5	34.2		
		65	35.5		
		45	4.35×10^3	β-type	303
		47.5	4.21		
		50	6.03		
		52.5	7.14		
		55	8.6		
		57.5	11.40		
		60	12.51		
		60	14.32		
4.3. BLENDS					
Poly(oxyethylene)/poly[1-(methoxycarbonyl)-1-methylethylene] $M_w = 20 \times 10^3$ PEO/PMMA					
100/0	80	49.0	3333	Hot stage microscopy	6
		50.0	1950		
		51.0	1183		
		52.0	600		
		53.0	333		
		54.0	167		
		55.0	33		
95/5	80	47.0	3300	Hot stage microscopy	6
		48.0	2300		
		49.0	1583		
		50.0	1067		
		51.0	683		
		52.0	433		
		53.0	250		
		54.0	150		
		55.0	33		
90/10	80	47.0	1883	Hot stage microscopy	6
		48.0	916		
		49.0	650		
		50.0	483		

TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		51.0	283		
		52.0	216		
		53.0	116		
		54.0	50		
		55.0	17		
80/20	80	47.0	800	Hot stage microscopy	6
		48.0	600		
		49.0	367		
		50.0	233		
		51.0	200		
		52.0	100		
		53.0	33		
		54.0	17		
70/30	80	49.0	167	Hot stage microscopy	6
$M_w = 116 \times 10^3$		50.0	177		
0/100		51.0	100		
		52.0	67		
		53.0	17		
Poly(oxyethylene)/Poly[1-(methoxycarbonyl)-1-methylene] PEO/PMMA					
100/0		47.6	10189	Hot stage microscopy;	278
$M_w = 2.0 \times 10^3$		49.0	6793	homopolymer and blends	
		48.5	6203		
		49.4	5333		
		49.9	4771		
		51.4	2386		
		52.5	1094		
		53.3	477.2		
		54.2	252.6		
90/10		46.3	3200	Hot stage microscopy;	278
		47.3	2189	homopolymer and blends	
		48.5	2049		
		49.3	1319		
		50.4	1263		
		51.4	701.8		
		52.4	421.1		
		53.4	252.6		
		54.3	140.4		
80/20		46.3	926.3	Hot stage microscopy;	278
		47.3	729.8	homopolymer and blends	
		48.4	561.4		
		49.3	477.2		
		50.5	449.1		
		51.4	252.6		
		52.5	252.6		
		53.4	84.2		
		54.3	56.1		
70/30		46.3	364.9	Hot stage microscopy;	278
		47.3	280.7	homopolymer and blends	
		48.2	280.7		
		49.3	168.4		
		50.3	140.4		
		51.4	0.0		
		53.4	0.0		
Poly(oxyethylene)/poly[1-methylcarbonyl)-1-methy(ethylene] $M_w = 10 \times 10^3$					
PEO/PMMA		48.7	2570	Hot stage microscopy;	278
100/0		49.7	1583	homopolymer and blends	
		50.3	1106		
		53.2	153		
		55.5	0.		
90/10		47.2	689	Hot stage microscopy;	278
		47.8	536.2	homopolymer and blends	
		48.8	374.5		
		50.0	255.3		
		50.8	187.2		

TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		51.8	76.6		
		53.0	42		
80/20		42.9	740.4	Hot stage microscopy;	278
		45.0	434.0	homopolymer and blends	
		45.9	331.9		
		46.8	280.8		
		47.8	221.3		
		48.8	119.1		
		50.4	59.6		
70/30		41.1	417.0	Hot stage microscopy;	278
		42.0	280.8	homopolymer and blends	
		43.0	246.8		
		44.1	195.7		
		45.0	153.2		
		45.8	136.2		
		46.8	110.6		
		47.7	68.1		
		48.9	51.1		
60/40		39.2	119.1	Hot stage microscopy;	278
		41.1	93.6	homopolymer and blends	
		42.9	76.6		
		44.0	68.1		
		45.0	42.6		
		45.8	25.5		
		46.7	17.0		
Poly(oxyethylene)/poly([1-(methoxycarbonyl)-1-methylethylene]		54		Hot stage microscopy;	278
M_w(PEO) = 100000; M_w(PMMA) = 110000		53	167	homopolymer and blends	
PEO/PMMA wt.% ratio: 100/0		52			
		51	483		
		50	1134		
		49	1600		
		48	2550		
		47			
	90/10	53	67	Hot stage microscopy;	278
		52	117	homopolymer and blends	
		51	217		
		50	275		
		49	357		
		48	547		
		47	667		
		46			
	80/20	50		Hot stage microscopy;	278
		49	133	homopolymer and blends	
		48	900		
		47	300		
		46	333		
		45	433		
		43	734		
		42			
	70/30	49	63	Hot stage microscopy;	278
		48	102	homopolymer and blends	
		47	102		
		46	140		
		45	163		
		44	190		
		43	250		
		42	267		
		41	305		
		40			
		39			
		47	37		
	60/40	46	43	Hot stage microscopy;	278
		45	57	homopolymer and blends	
		44	80		
		43	98		

TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		42			
		41	102		
		40			
		39	125		
		38			
Poly(oxyethylene)/poly[1-(methoxycarbonyl)-1-methylethylene]		54	257	DSC measurements; rate constants	
M_w(PEO)100000; M_w(PMMA)110000		53	1278	in text, Avrami exponents	305
PEO/PMMA		52	546	$2.3 \leq n \leq 3.2$ respectively	
100/0		51	348		
		50	186		
		49	108		
		48	120		
		47	48		
90/10		53	1998	DSC measurements; rate constants	305
		52	996	in text, Avrami exponents	
		51	744	$2.3 \leq n \leq 3.2$ respectively	
		50	324		
		49	258		
		48	186		
		47	120		
		46	84		
80/20		50	2356	DSC measurements; rate constants	305
		49	1986	in text, Avrami exponents	
		48	1752	$2.3 \leq n \leq 3.2$ respectively	
		47	1149		
		46	744		
		45	606		
		43	192		
		42	204		
70/30		49	4344	DSC measurements; rate constants	305
		48	2292	in text, Avrami exponents	
		47	1992	$2.3 \leq n \leq 3.2$ respectively	
		46	1470		
		45	1308		
		44	1008		
		43	1044		
		42	684		
		41	564		
		40	432		
		39	342		
60/40		47	4776	DSC measurements; rate constants	305
		46	3906	in text, Avrami exponents	
		45	3222	$2.3 \leq n \leq 3.2$ respectively	
		44	2562		
		43	2082		
		42	1932		
		41	1542		
		40	1368		
		39	1140		
		38	1020		
Poly(oxyethylene)/poly(1-acetoxyethylene)		48.6	6167		278
PEO; $M_w = 2.0 \times 10^3$		49.5	5325		
PEO/PVA		50.9	2690		
100/0		51.7	1976		
		52.7	1116		
		53.6	530		
		54.6	164		
		55.5	73		
90/10		46.0	4630		278
		46.9	3879		
		47.8	3550		
		48.6	2214		
		49.7	2086		
		50.6	1537		

TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		51.7	1262		
		52.7	549.0		
		53.5	347.7		
		54.5	91.5		
80/20		45.9	3824		278
		46.8	3001		
		48.6	1738		
		49.7	1281		
		50.6	878.4		
		51.5	622.2		
		52.5	237.9		
		53.6	183.0		
		54.6	0.0		
60/40		45.7	1756		278
		46.5	1354		
		47.6	969.9		
		48.5	768.6		
		49.5	512.4		
		50.5	274.5		
		51.5	201.3		
		52.6	73		
Poly(oxyethylene)/poly([1-(methoxycarbonyl)-1-methylethylene]					
PEO; $M_w = 990000$; PMMA, $M_w = 525000$					307
PEO/PMMA					
100/0		11	67.3		
		18	78		
		22.5	56.5		
		26	56.6		
		30.7	37		
		36	24		
		41.7	9		
		45	5.2		
		50	1.72		
		55	0.041		
		57	0.018		
92/8		15.5	450		307
		22.5	272		
		26	217		
		31	130		
		36	75		
		41	23		
		49	11		
		53	1.3		
		55	0.43		
		60	0.014		
85/15		50	22		307
		51	11.5		
		52.5	7.2		
		54	3.2		
		56	0.82		
		58	0.18		
		60	0.014		
70/30		35	550		307
		40	381		
		45	200		
		48	96		
		51	35		
		54	9.5		
		56	2.6		
		58	0.50		
PEO = 990000; PPMA = 125000		16	23		307
100/0		22	16.8		
		28	11.5		
		31	10		
		36	8.6		

TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		41	5.2		
		43.5	3.0		
		46.5	2.4		
		49.5	1.4		
		51.5	0.83		
		54.5	0.42		
		56.5	0.17		
		57.5	0.10		
		58.5	0.06		
92/8		23	108		307
		30	58		
		35	52		
		42	70		
		46	11		
		49	12		
		51	5.4		
		54	2.0		
		56	0.66		
		58	0.19		
		60	0.064		
85/15		48.5	7.7		307
		50	3.7		
		54	9.05		
		57.6	3.8		
		58.2	0.46		
		58.3	0.19		
		59.3	0.10		
		60	0.041		
70/30		35	611		307
		40	446		
		45	229		
		48	110		
		50	52		
		52	17		
		54	11		
		56	2.93		
		57.3	1.46		
		58	0.56		
		59.5	0.22		
Poly[oxy(1-oxo-2,2-dimethyltrimethylene)] poly(1,1-difluoroethylene); (poly(pivalolactone)/poly(vinylidene fluoride) PPVL/PVF_2; $M_v = 250 \times 10^3 / M_n = 45 \times 10^3$; $M_w = 140 \times 10^3$					
100/0, w/w	260 (3 min)	180	6.2×10^3	DSC isothermal rates, selected values; consult reference for more details on Regime growth	308
		184	3.17×10^3		
		186	2.40×10^3		
		189	1.22×10^3		
		192	0.60×10^3		
		197	0.157×10^3		
		200	0.066×10^3		
		205	2.0×10^3		
		207	1.35×10^3		
		210	7.45×10^3		
		215.5	2.2×10^3		
70/30, w/w		169	3.52×10^3	DSC isothermal rates, selected values; consult reference for more details on Regime growth	308
		174	2.08×10^3		
		177	1.49×10^3		
		183	6.86×10^2		
		190	2.65×10^2		
		195	8.90×10		
		200	3.00×10		
		205	8.5		
		210	3.25		
		215	1.55		
20/80, w/w	260 (3 min)	165	2.21×10^2	DSC isothermal rates, selected values; consult reference for more details on Regime growth	308
		170	1.47×10^2		
		174	1.24×10^2		

TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		178	8.84×10		
		182	5.87×10		
		186	3.60×10		
		189	2.32×10		
		192	1.54×10		
		196	8.08×10		
		199	4.22		
		202	2.81		
		204	1.57		
10/90, w/w		160	9.4×10	DSC isothermal rates, selected values; consult reference for more details on Regime growth	308
		166	7.01×10		
		170.1	5.99×10		
		174	4.38×10		
		179	3.22×10		
		184	1.81×10		
		190	9.52		
		196	4.13		
		200	2.39		
		205	1.11		
		206	0.72		
50/50, w/w	260 (3 min)	170	1.53×10^3	DSC isothermal rates, selected values; consult reference for more details on Regime growth	308
		174	1.11×10^3		
		178	6.35×10^2		
		182	4.19×10^2		
		185	2.69×10^2		
		190	1.27×10^2		
		195	4.00×10		
		200	1.10×10		
		205	4.73		
		210	2.15		
		215	0.75		
30/70, w/w		164	6.99×10^2	DSC isothermal rates, selected values; consult reference for more details on Regime growth	308
		168	5.21×10^2		
		172	3.93×10^2		
		182	1.25×10^2		
		188	5.80×10		
		192	2.62×10		
		197	1.00×10		
		200	6.6		
		205	3.3		
		210	1.0		
Poly (1,1-difluoroethylene)/poly(butylene succinate) PVF_2, $M_w = 140 \times 10^3$; PBSU, $M_n = 54 \times 10^3$					413
wt.% :PVF_2 / PBSU	190 (5 min)			Hot-stage microscopy DSC, T_m and T_m^0 and enthalpy values listed in Ref. 413. Spherulitic morphology in all blends	
100/0		150	197		
		155	61.1		
		160	10.8		
80/20		145	185		
		150	116		
		155	36		
60/40		140	222		
		145	110		
		150	44.3		
		155	13.0		
40/60		140	133		
		145	65		
		150	17.4		
Poly(butylene succinate)/poly (1,1-difluoroethylene) PBSU, $M_w = 54 \times 10^3$; PVF_2, $M_w = 140 \times 10^3$					413
wt.% PBSU / PVF_2	190 (5 min)			Hot-stage microscopy DSC, T_m and T_m^0 and enthalpy values listed in Ref. 413. Spherulitic morphology in all blends	
100/0		80	1943		
		85	980		
		90	456		
		100	73.2		

TABLE 4. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
80/20		80	1842		
		85	969		
		90	434		
		95	169		
60/40		80	1470		
		85	710		
		90	248		
		95	130		
Poly(oxyethylene); PEO/poly(octafluoropentoxytrifluoroethoxyphosphazene); PPZ/poly[oxy(1-chloromethyl)ethylene]; PECH					
PEO, $M_w = 5 \times 10^6$		53.8	609	Hot stage microscopy; bulk crystallization and morphology also recorded	309
PPZ, -unlisted		55.1	400		
PECH, $M_w = 7 \times 10^6$		56	286		
PEO/PPZ: 100/0		57.1	215		
		58	166		
		58.8	142		
75/25		52.3	1100	Hot stage microscopy; bulk crystallization and morphology also recorded	309
		54	563		
		55.1	429		
		56	286		
		57	239		
		57.9	193		
50/50		49.3	1402	Hot stage microscopy; bulk crystallization and morphology also recorded	309
		50.6	1045		
		57.9	770		
		53.2	536		
		54.2	426		
		55.2	302		
PEO/PECH: 75/25		49.2	905	Hot stage microscopy; bulk crystallization and morphology also recorded	309
		50.5	693		
		51.5	526		
		52.5	435		
	53.5	321			
		54.3	271		
PEO/PPZ/PECH: 66/17/17		49.7	633	Hot stage microscopy; bulk crystallization and morphology also recorded	309
		50.7	528		
		51.7	405		
		52.7	336		
		53.6	279		
		54.5	242		
PEO/PPZ/PECH: 50/25/25		48.9	600	Hot stage microscopy; bulk crystallization and morphology also recorded	
		50	477		
		51	422		
		51.8	370		
		52.7	313		
		53.6	266		

TABLE 5. POLY(CARBONATES)

Polymer	T_c (°C)	G (nm/s)	Remarks	Refs.
Poly(oxycarbonyloxy-1,4-phenylene-isopropylidene-1-4-phenylene)	190	0.0833	Hot stage optical microscopy	310
	195	0.042		
Poly(oxycarbonyloxy-1,4-phenylenemethylene-1,4-phenylene)	175	16.67	Hot stage optical microscopy	310
	180	10.00		
	185	5.00		
	190	3.33		
Poly(oxycarbonyloxy-1,4-phenylenethio-1,4-phenylene)	145	20.0	Hot stage optical microscopy	310
	150	15.00		
	160	8.33		
	170	5.85		
	180	4.17		

TABLE 6. POLY(ESTERS)

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
6.1. HOMOPOLYMER MELTS					
Poly(hydroxybutyrate)		23	21.2	Measurements by Atomic Force Microscopy in thin polymer films; lamellar growth noted from spherulitic–melt interfaces; polymer characterization given in Ref. 103(a)	103
		30	52.3		
		33	78.6		
		38	134		
		42	283		
		45.4	425		
		55	995		
		67.4	4117		
		80	3930		
		100	3239		
		107	1787		
		115	902		
		119.5	425		
		127	150		
		129.5	86		
Poly(oxydecamethyleneoxysebacoyl)					
$M_v = 10.3 \times 10^3$	100	67.1	87.35	Hot stage optical microscopy; nucleation densities given in text	105
		68.1	21.67		
		69.1	6.451		
		70.1	1.884		
		71.1	0.583		
		72.0	0.083		
Poly(oxyethyleneoxyadipoyl)					
$M_v = 10.3 \times 10^3$		20	66.346		311
		40	18.00		
		44	13.169		
		47	9.002		
Poly(oxyethyleneoxyterephthaloyl)		123	4.0	Hot stage optical microscopy; crystallization from the glassy state at T_g	185
		140	17.5		
		145	34.0		
		161	56.0		
		170	70.0		
		190	64.5		
		200	42.0		
		210	21.8		
		120	7	Linear growth rates by laser light scattering; nucleation densities are provided in the test	312
		130	18		
		140	37		
		150	61		
		160	105		
		170	140		
		180	151		
		190	100		
		200	140		
		210	151		
		220	6		
		140	13	0.5 kbar	312
		150	21		
		160	23		
		170	41		
		150	2.5	1.5 kbar	
		160	8		
		170	16		
		180	30		
		160	2	2 kbar	
		170	9.5		
		180	24		
		190	34		
Poly(oxyethyleneoxyterephthaloyl)		39.6	147.0	Hot stage optical microscopy	313
		40.9	140.0		
		42.3	158.5		
		50.3	166.0		

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TABLE 6. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		53.4	150.2		
		57.0	153.4		
		57.6	120.0		
		58.7	116.0		
		61.4	76.81		
		67.6	74.18		
		72.0	51.34		
		75.2	26.17		
		81.8	5.801		
		58.0	3.001		
$M_n = 16.8 \times 10^3$	294	220	41.08		314
		240	3.100		
$M_n = 19.0 \times 10^3$		120	6.84	Hot stage optical microscopy	315
		130	15.0		
		140	47.4		
		150	75.2		
		160	93.2		
		170	112.2		
		180	112.8		
		190	117.4		
		200	85.0		
$M_n = 27.4 \times 10^3$		120	3.28	Hot stage optical microscopy	315
		130	8.50		
		140	19.9		
		150	36.8		
		160	56.2		
		180	72.1		
		190	63.6		
		200	43.9		
		210	20.2		
$M_n = 39.1 \times 10^3$		120	1.54	Hot stage optical microscopy	315
		130	4.40		
		140	8.26		
		150	14.2		
		160	18.0		
		170	27.8		
		180	26.6		
		190	19.1		
		200	13.4		
Poly(oxymethyleneoxysuccinyl)		16.4	12.00	Hot stage optical microscopy	316
$M_w = 5.98 \times 10^3$		23.0	31.17		
		26.3	62.34		
		28.5	44.18		
		35.6	91.35		
		36.3	91.83		
Poly(oxytetramethylene oxysuccinyl); PTMS		$\bar{M}_w = 1.14 \times 10^5$ $\bar{M}_n/\bar{M}_n = 3.1$			
	200(10 min)	89	162.7	Rate of spherulite growth; hot stage	405
		90	146.4	microscopy; $T_m^0 = 132$°C;	
		91	119.5	$T_g = -38$°C; DSC, ^{1}HNMR	
		92	102.3		
		93	82.1		
		94	71.6		
Showa High Polymer Co., Japan		95	54.0	IR characterization of polymer;	
		97	35.7	Density-Temp. results listed	
		100	19.2		
		102	7.7		
		105	3.85		
Poly(oxy-1-oxo-2-methylethylene)/poly(L-lactic acid)					
$M_v = 150 \times 10^3$		115.2	62.2	Hot stage microscopy;	316
		120.2	71.1	Regimes I and II crystallization	
		126.0	83.2	ranges cited	
		130.3	82.8		
		134.4	63.8		
		139.0	57.4		
		143.8	49.3		

TABLE 6. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
$M_v = 260 \times 10^3$		119.5	49.9	Hot stage microscopy;	316
		128.9	61.8	Regimes I and II crystallization	
		134.7	55.2	ranges cited	
		138.3	48.3		
$M_v = 350 \times 10^3$		114.2	27.9	Hot stage microscopy;	316
		119.0	43.2	Regimes I and II crystallization	
		125.0	49.7	ranges cited	
		129.4	55.2		
		133.5	46.1		
		143.1	34.5		
		150.1	20.8		
$M_v = 690 \times 10^3$		113.5	11.9	Hot stage microscopy;	316
		119.0	27.7	Regimes I and II crystallization	
		128.4	41.4	ranges cited	
		134.4	38.8		
		138.1	30.9		
		143.1	19.0		
		147.7	15.6		
		149.9	15.1		
		154.9	6.1		
		156.8	4.8		
Poly(oxyterephthalaloyloxydecamethylene)	160 (5 min)	120.0	5.534	Hot stage optical microscopy;	317
$M_n = 10.0 \times 10^3$		121.0	3.451	nucleation densities listed	
		123.0	1.592		
		125.0	0.6217		
		126.0	0.2550		
6.2. BLENDS					
Poly(oxy-1-oxohexamethylene)/poly(1-chloroethylene) PCL/PVC					
100/0	100 (5 min)	25	696.6	Hot stage optical microscopy	318
		30	533.3		
		35	381.7		
		20	50		
75/25		25	39.5	Hot stage optical microscopy	318
		30	29.2		
		35	19.5		
70/30		20	23.7	Hot stage optical microscopy	318
		25	21.6		
		30	13.2		
		35	8.5		
65/35		20	8.3		
		25	7.4		
		30	5.4		
		35	3.3		
60/40		20	2.75	Hot stage optical microscopy	318
		25	2.2		
		30	1.7		
		35	1.2		
50/50		20	0.22	Hot stage optical microscopy	318
		25	0.24		
		30	0.20		
		35	0.14		
Poly(β-hydroxybutyrate)					
isotactic/atactic		35	38.2	Optical hot stage microscopy;	319
isotactic: $M_w = 445000$		50	405	DSC melting characterization	
$M_n = 137000$		60	907	also made	
atactic: $M_w = 21000$		65	1103		
$M_w = 11000$		70	1958		
isotactic/actactic 100/0		76	2582		
		84	3162		
		87.5	3435		
		90	3533		
		95	3322		
		100	2980		

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TABLE 6. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		110	1778		
		114	1348		
		120	763		
		124	454		
		130	199		
		134.5	86.9		
		140	18.9		
90/10		35	27.0	Optical hot stage microscopy;	319
		45	200	DSC melting characterization	
		55	469	also made	
		60	797		
		65	1096		
		70	1508		
		74	1814		
		77	2000		
		80	2196		
		83.5	2382		
		87	2480		
		90	2489		
		94	2425		
		97	2365		
		100	2155		
		108	1980		
		105.5	1747		
		107	1592		
		110	1371		
		112	1184		
		117	753		
		126	332		
		130	177		
		142	29		
75/25		34.4	16.4	Optical hot stage microscopy;	319
		48.3	154	DSC melting characterization	
		52	295	also made	
		58	468		
		70	936		
		74	1177		
		77.5	1526		
		84	1612		
		87.5	1699		
		90.5	1797		
		100	1595		
		105	1439		
		110	1206		
		115.5	875		
		120.5	521		
		130	155		
60/40		35.5	37.9	Optical hot stage microscopy;	319
		49.7	131.2	DSC melting characterization	
		55	272	also made	
		60	402		
		65	630		
		70	750		
		77	1044		
		80	1174		
		84	1250		
		90	1347		
		94.5	1410		
		97	1398		
		100	1276		
		104	1132		
		107	1032		
		110	910		
		115	644		
		120	642		

TABLE 6. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		130	143		
		137	309		
50/50		40.4	2.96	Optical hot stage microscopy;	
		50	12.00	DSC melting characterization	
		61	291.6	also made	
		65	433.0		
		70.4	617.6		
		75	769.4		
		80.5	921.3		
		84.5	985.7		
		90.0	1126		
		97	1126		
		100	1067		
		105.0	901.0		
		135	318.1		
6.3. COPOLYMERS					
Poly(oxyethyleneoxyterephthalate)/poly(1,4-cyclohexane dimethylene terephthalate) (random copolymers)					
100/0, mol%		160	50	Hot stage optical microscopy	320
		165	54	$T_m = 280°C$; $T_g = 76°C$	
		170	76		
		180	85		
		185	79.9		
		190	79		
		195	78.1		
		200	63		
		205	57		
		210	40.7		
		215	35		
		220	22		
		225	14		
		230	10		
95/5, mol/mol		160	38	Hot stage optical microscopy;	
		165	48	$T_m = 269°C$; $T_g = 75.5°C$	
		170	54.5		
		175	56.6		
		180	52		
		185	50		
		190	40.7		
		195	39		
		200	31.6		
		205	24.4		
		210	15.3		
		215	9.2		
		220	6		

TABLE 7. POLY(AMIDES)

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
Poly(iminopentamethylene iminoadipoyl) Nylon 5,6					
	282 (30 min)	220.5	1742	Hot stage optical microscopy;	321
		222	1385	morphology depends upon	
		224.5	1210	crystallization temperature and	
		226	1090	and is illustrated in text	
		230	476		
		235	161.7		
		236	160.0		
		239	121.7		
		240	96.7		
		242	55.0		

TABLE 7. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
Poly(iminomethyleneiminoadipoyl) Nylon 96					
positive spherulites	282 (30 min)	200	1684	Hot stage optical microscopy; spherulitic morphology depends upon crytallization temperature and is illustrated in text	321
		203	1334		
		205	1134		
		207.5	897		
		213	532		
		215	500		
		218	257		
		225	80.8		
		230	10.84		
		231	8.00		
		234	2.67		
Poly(iminohexamethylene) iminoadipoyl) Nylon 66					
positive spherulites	295 (30 min)	150	22000	Hot stage optical microscopy; spherulitic growth	322
$M_n = 12.7 \times 10^3$		175	18000		
		180	16500		
		190	13000		
		200	9500		
		210	6,000		
		220	3,200		
		230	1,500		
Poly(iminohexamethyleneiminoadipoyl) Nylon 66					
$M_n = 11.6 \times 10^3$	295	241	166.7	Hot stage optical microscopy; nucleation densities cited in text	323,325
	295	247	58.35		
	295	250	13.84		
	295	252	10.50		
	285	247	66.08		
	262 (10 min)	251	14.24	Negative spherulites	326,327
		256	83.35		
		257	13.34		
		259	9.168		
		261	6.668		
		263	4.167		
		265	2.500		
Poly(iminohexamethyleneiminoadipoyl) Nylon 66					
$M_n = 12.9 \times 10^3$	300 (30 min)	246	106.68		328
		248	56.34		
		253	10.84		
$M_n = 13.7 \times 10^3$	300 (30 s)	141	13503		329
		160	13669		
		180	12119		
		199	8901		
		215	5167		
		230	2117		
		234	1530		
		237	920		
		239	765		
		241	471		
		244	368		
$M_n = 14.6 \times 10^3$	280	241	283	Nucleation densities listed in text	330
		243	230		
		245	180		
		248	33		
		252			
Poly(iminohexamethyleneiminoadipoyl) Nylon 66					
	300	241.5	204		322
		243	175		
		245	128		
		248	58.34		
		252	5.50		
	315	241.5	280.0		322
		243	168.4		
		245	113.4		
		248	56.6		

TABLE 7. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		252	6.33		
		50	3650.7		
$M_w = 17.2 \times 10^3$		100	4706.6	Positive spherulites	331
		142	6751.3		
		160	6101.2		
		178	5201.0		
		198	3700.7		
		200	12900.6		
		228	466.7		
$M_n = 25.5 \times 10^3$	300 (30 s)	180.0	1435.6	Positive spherulites	329
		200.0	7951.5		
		211.0	5284.4		
		220.0	2733.8		
		230.0	1615.3		
		235.5	680.13		
		240	483.4		
Poly(iminohexamethyleneiminosebacoyl) Nylon 6,10					
$M_n = 11.6$ and 26.3×10^3		200.0	253.4	Hot stage optical microscopy	331
		205.0	120.0		
		212.0	36.67		
		217.0	13.34		
		205.0	186	Negative spherulites	
		212.0	130		
		217.0	7		
Poly(imino(1-oxohexamethylene)) Nylon 6					
$M_n = 27.4 \times 10^3$	300 (30 s)	102.3	900	Positive spherulites	329
		112.5	1478		
		122.0	2033		
		130.0	2767		
		140.5	3133		
		150.5	2933		
		159.0	2450		
		170.0	1431		
		180.0	710		
		182.0	610		
$M_v = 24.7 \times 10^3$		90	375	Positive spherulites	322
		95	610		
		101	640		
		107	1108		
		117	1571		
		124.5	1933		
		135	2438		
		141	2438		
		148	2105		
		157	1790		
		172	983		
		184	305		
$M_w = 49 \times 10^3$		40	80	Hot stage optical microscopy; positive spherulites	312
		41	60		
		42	50		
		43	38		
		44	27		
		45	20		
		46	15		
		47	10		
		48	5.8		
		49	4.2		
		50	2.8		
		51	1.8		
$M_w = 15 \times 10^3$		39	162	Hot stage optical microscopy; positive spherulites	312
		40	138		
		41	102		
		42	100		
		43	81		

TABLE 7. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		44	62		
		45	35		
		46	27		
		47	21		
		48	15		
		49	8		
		50	6		
		51	1.7		
$M_w = 7 \times 10^3$		40	197	Hot stage optical microscopy;	312
		41	15	positive spherulites	
		42	100		
		43	75		
		44	58		
		45	42		
		46	28		
		47	18		
		48	13		
		49	6.7		
		50	4.2		

TABLE 8. POLY(SILOXANES)

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
8.1. HOMOPOLYMER MELTS					
Poly(oxydimethylsilylene-1,4-phenylenedimethylsilylene)					
TMPS	200 (5 min)	23	316	Hot stage optical microscopy;	333
$M_v = 8.7 \times 10^3$		30	530	selected molecular weights;	
		40	900	see text	
		50	1450		
		60	1640		
		70	1740		
		80	1460		
		90	959		
		100	417		
		110	161		
		120	14.5		
		130	0.593		
$M_v = 372 \times 10^3$	200 (5 min)	24	80.0	Hot stage optical microscopy;	333
		30	113.0	selected molecular weights;	
		40	166	see text	
		60	328		
		70	342.0		
		80	246		
		90	146		
		100	65.0		
		110	20.0		
		130	0.124		
		134	0.0365		
$M_v = 1400 \times 10^3$	200 (5 min)	24	41.5	Hot stage optical microscopy;	333,388
		40	165	selected molecular weights;	
		51	280	see text	
		60	328		
		70	302		
		80	240		
		90	146		
		100	70.6		
		110	23.4		
8.2. COPOLYMERS					
Poly(oxydimethylsilylene-1,4-phenylenedimethylsilylene-block-poly(oxydimethylsilylene)					
TMPS/DMS		21.5	5.25	Hot stage optical microscopy;	182
100/0	200 (5 min)	22.5	5.92	crystallinites and densities	

TABLE 8. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
$M_w = 141 \times 10^3$		23.0	7.87	also listed	
		24.0	7.63		
		40.0	24.75		
		60.0	41.00		
		70.0	40.00		
		79.0	30.28		
		82.0	30.33		
		90.0	21.53		
		105.0	6.03		
		110.0	3.17		
		116.0	14.17		
		119.0	5.75		
		120.0	5.28		
		121.0	4.30		
		125.0	0.21		
		130.0	0.44		
90/10		− 20.0	2.58	Hot stage optical microscopy;	182
		− 10.0	17.67	crystallinites and densities	
		0.0	75.33	also listed	
		10.0	240.00		
		20.0	531.6		
		30.0	851.6		
		40.0	1420.0		
		45.0	1645.0		
		50.0	1850.0		
		55.0	1883.4		
		60.0	1700.0		
		70.0	1515.0		
		80.0	970.0		
		90.0	418.3		
		100.0	120.2		
		110.0	13.52		
		115.0	2.70		
80/20		− 20.0	33.5	Hot stage optical microscopy;	182
		− 10.0	142.2	crystallinites and densities	
		0.0	360.0	also listed	
		10.0	643.3		
		20.0	1041.7		
		30.0	1346.7		
		35.0	1498.4		
		40.0	1613.4		
		42.0	1630.0		
		45.0	1636.7		
		50.0	1625.0		
		55.0	1541.7		
		60.0	1396.7		
		65.0	1225.0		
		70.0	1018.4		
		75.0	798.3		
		80.0	585.0		
		90.0	241.7		
		95.0	129.7		
		100.0	58.8		
		105.0	21.3		
		110.0	5.62		
		115.0	0.925		
		120.0	0.177		
50/50		− 20.0	501.6	Hot stage optical microscopy;	182
		− 10.0	558.3	crystallinites and densities	
		0.0	550.0	also listed	
		10.0	351.6		
		20.0	260.0		
		25.0	215.0		
		30.0	173.3		
		38.0	111.6		

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TABLE 8. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		42.0	86.8		
		45.0	69.3		
		50.0	45.2		
		55.0	26.8		
		62.5	9.58		
		70.0	2.18		
40/60		−20.0	278.3	Hot stage optical microscopy;	182
		−15.0	283.3	crystallinites and desnsities	
		10.0	280.0	also listed	
		0.0	235.0		
		10.0	165.0		
		20.0	109.83		
		23.5	94.83		
		30.0	75.00		
		35.0	51.33		
		40.0	35.33		
		45.0	22.83		
		50.0	12.17		
		55.0	8.17		
		58.5	3.25		
		67.5	0.32		
30/70		−20.0	261.7	Hot stage optical microcopy;	182
		−10.0	188.3	crystallinites and densities	
		0.0	119.84	also listed	
		10.0	66.50		
		20.0	35.67		
		25.0	24.00		
		30.0	15.83		
		35.0	9.250		
		40.0	5.133		
		45.0	3.067		
		50.0	0.558		

TABLE 9. OTHERS

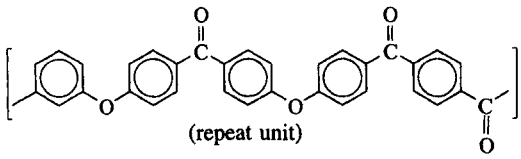
Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
9.1. HOMOPOLYMER MELTS					
Poly(aryl ether ketone ether ketone ketone)		176	1350	Stage I spherulite growth conventional	335
PEKEKK; 100/0, w/w		179	1627	microbeam SALS; synchrotron time	
		193	1840	resolved SAXS	
		210	1900		
		228	1850		
		247	1693		
		263	1540		
		285	1607		
(repeat unit)		296	1515		
		313	1055		
		327	773		
		338	718		
		335	189		
		183	1862	Stage II spherulite aggregates	
		188	2042	(higher rate)	
		204	2217		
		224	2050		
		242	2017		
		272.5	1833		
Poly(oxy-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene) PEEK					
$T_m = 340$°C		240	980	Hot stage microscopy; crystallinity,	402
$T_g = 144$°C		260	540	lamellar dimensions, $t_{1/2}$ values	
		277	264	also given in text	
		287	161		

TABLE 9. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
Poly(oxy-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene) PEEK ICI grade 150P					
	400 (15 min)	270	14.60×10^2	Hot stage microscopy with video accessories; $T_m^0 \sim 395$°C	267
		275	12.88×10^2		
		280	9.91×10^2		
		285	6.95×10^2		
		290	5.26×10^2		
		295	3.60×10^2		
		298	2.90×10^2		
		300	2.42×10^2		
		305	1.91×10^2		
		310	1.05×10^2		
		315	0.627×10^2		
		320	0.364×10^2		
		325	0.171×10^2		
	415 (15 min)	270	12.0×10^2	Note decrease for higher fusion condition	267
		275	10.4×10^2		
		280	7.94×10^2		
		285	5.37×10^2		
		290	3.97×10^2		
		295	2.97×10^2		
		298	2.50×10^2		
		300	1.97×10^2		
		305	1.54×10^2		
		310	0.845×10^2		
		315	0.384×10^2		
		320	0.275×10^2		
		325	0.119×10^2		
Poly(thio-1,4-phenylene) Ryton	330 (2 min)	220	74.5	Hot stage and video micrometer assembly with optical microscope; DSC measurements; computer modeling	227
		225	54.4		
		230	37.6		
		235	23.2		
		240	13.59		
Poly(thio-1,4-phenylene) Fortron	345 (2 min)	250	7.81	Bulk nucleation density; density measured Avrami exponent ~ 3	227
		255	4.92		
		260	2.48		
		265	0.95		
Poly(thio-1,4-phenylene)		84	1387	Hot stage microscopy; DSC rates listed in text Avrami $2 < n < 3$ values	227
$M_w = 24000$		79	1243		
$M_w/M_n = 1.4$		74	995		
		69	783		
		64	557		
		59	429		
		54	276		
		48.7	148		
$M_w = 49000$		88	490		227
		83	410		
		78	300		
		73	232		
		68	244		
		63	100		
		58	67		
		52.5	20.5		
$M_w = 63000$		92	414		227
$M_w/M_n = 1.5$		87	356		
		81.8	244		
		76.7	159		
		71.8	130		
		67	79		
		62	41		
		56.6	20		
		255.3			
Poly(thio-1,4-phenylene)					
$M_w = 56000$		225		Growth rates measured as function of polymer end group	337
		230			
		235			

TABLE 9. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		240			
		245			
		250			
Na^+ washed		255			337
		225			
		235			
		240			
		245			
		250			
Mn^{2+} washed		225			
		230			
		235			
		245			
		250			
		255			
		260			
Acid washed		225			
		230			
		240			
		250			
		255			
		260			
Poly(thio-1,4-phenylene) Phillips Pet.		100	0.87	Hot stage microscopy; Regimes I and	152
$M_w = 15 \times 10^3$		110	10.0	II crystallization noted	
		120	91.0		
		140	680.0		
		150	1200.0		
		160	1200.0		
		170	2300.0		
		180	2300.0		
		190	2200.0		
		200	1700.0		
		210	1200.0		
		220	710.0		
		230	390.0		
		240	160.0		
		250	53.0		
		260	12.0		
		270	1.30		
		280	0.094		
$M_w = 51 \times 10^3$		115	1.40	Hot stage microscopy; Regimes I and	152
Phillips Pet. Co.		120	5.2	II crystallization noted	
		125	10.0		
		130	25.0		
		140	73.0		
		150	160.0		
		160	250.0		
		170	300.0		
		180	280.0		
		190	260.0		
		200	240.0		
		210	180.0		
		220	98.0		
		230	48.0		
		240	21.00		
		250	7.90		
		260	1.80		
		270	0.13		
Poly(thio-1,4-phenylene)		102	8.5	Crystallization rates from DSC peak	338
		106	3.0	time (mins); measurements below	
		110	1.5	T_{max} were made by upquenching	
		114	0.9	from the amorphous state; $T_m^0 \sim 330$°C	
		118	0.71	using Hoffman–Weeks plot	
		120	0.54		
		235	0.36		

TABLE 9. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		240	0.53		
		245	0.82		
		250	1.30		
		255	2.40		
		260	5.20		
		265	8.00		
		270	17.5		
Random copolymer with 5% metamethylene linkages		110	6.71	$T_m^0 \sim 290$°C	338
		115	3.16		
		120	1.90		
		125	1.23		
		130	0.88		
		135	0.70		
		140	0.53		
		145	0.46		
		200	0.39		
		210	0.59		
		220	1.2		
		230	2.30		
		235	2.81		
		240	5.01		
		245	9.47		
Random copolymer with 8% metamethylene linkages		105	14.93	$T_m^0 \sim 283$°C	338
		110	5.80		
		120	1.97		
		130	1.09		
		140	0.67		
		150	0.47		
		190	1.75		
		200	2.52		
		205	4.19		
		210	5.26		
		215	7.40		
		220	9.62		
		225	23.02		
Random copolymer with 10% metamethylene linkages		105	13.4	$T_m^0 \sim 273$°C	338
		110	7.10		
		120	2.65		
		130	1.34		
		140	0.845		
		150	0.620		
		160	0.520		
		170	0.490		
		180	0.760		
		190	1.11		
		200	1.80		
		205	2.54		
		210	3.81		
		215	6.21		
		220	10.2		
Poly(thio-1,4-phenylene)		219.7	120	Optical hot stage microscopy	338
$M_w = 63000$		225	102		
TCB, 0 mol%		230	75		
		235	50		
		240	33		
		245	22		
		250	14		
$M_w = 75000$		220	104	Optical hot stage microscopy	338
TCB, 0.13 mol%		225	76		
		230	54		
		235	38		
		245	16		
		250	11.4		
		255	5.7		
$M_w = 65000$		220	89	Optical hot stage microscopy	338

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TABLE 9. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
TCB, 0.2 mol%		225	64		
		230	44		
		235	30		
		240	22		
		245	20		
		250	8.8		
		255	4.3		
catena-Poly(selenium)					
Selenium (Gallard Schlesinger Chemical Manufacturing Co.)	600 (2 h) and ice water quenched	80	0.171	Hot stage optical microscopy; purity = 99.9999%	54
		90	0.808		
		100	1.535		
		110	2.383		
		120	3.083		
		130	5.834		
		140	4.884		
		150	2.85		
Selenium (Yokozawa Co.) 99.999%	230 (1 hr) and quenched	80.0	0.27	Spherulitic structures; growth rates of chlorinated polymer listed in text	414
		87	0.50		
		90	0.65		
		100	1.70		
		105	2.72		
		110	4.10		
		120	7.68		
9.2. IONOMERS					
Poly(thio-1,4-phenylene) ionomers		225	175	PPS with end-group counterions; growth rates in order $PPSCa^{2+}$>$PPSH^{+}$> $PPSZn^{2+}$> $PPSNa^{+}$; Growth rates obtained by hot stage optical microscopy; rate constants documented	338
PPS/Na^{+}		230	129		
		235	92		
		240	62		
		245	43		
		250	52		
		255	14		
PPS/Ca^{2+}		225	142		338
		230	106		
		235	71		
		240	49		
		245	81		
		250	16		
		255	28		
PPS/Zn^{2+}		225	123		338
		230	93		
		235	63		
		240	42		
		245	28		
PPS/Zn^{2+}		250	15.6		338
		255	9.0		
PPS/H^{+}		225	107		338
		230	79		
		235	53		
		240	36		
		245	22		
		250	10		
		245	43		
		158.3	113.0	Regime II crystallization; Transition II to III at 160°C approx.	340
		159.0	72.95		
		161.3	32.13		
		165.2	15.70		
		166.2	3.22		
		170.0	0.74		
		175.0	0.07		
		175.1	0.14[4]		
Poly(oxyethylene) complex with *p*-nitrophenol ($M_w = 6000$)		28	6660	Hot stage optical microscopy	303
		30.5	8470		
		34	11310		

TABLE 9. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		36.8	11860		
		41	12760		
		45	10820		
		49	7940		
		53	3870		
		60	1174		
		63	237		
		65	76.6		
		67.2	65.0		
		69.2	30.4		
Poly(oxyethylene)/poly(oxypropylene)	Blend composition				341
$M_w = 5 \times 10^6/1.5 \times 10^6$	353 (10 min)			Rates measured on hot-stage microscope;	
100/0		53	450	blends were prepared by solution	
		54	217	casting from acetonitrile; two	
		55	122	crystallization regimes II and III	
		56	52	were reported along with interfaced	
		57	23	energy values	
		58	12		
		59	4.5		
30/20		52	833		341
		53	500		
		54	300		
		55	120		
		56	55		
		57	25		
		58	8.5		
60/40		51	1417		341
		52	850		
		53	517		
		54	250		
		55	117		
		56	80		
		57	33		
40/60		50	1967		341
		51	1317		
		52	967		
		53	517		
		54	300		
		55	115		
20/80		52	617		341
		53	383		
		54	300		
		56	57		
		57	17		
		58	12		

3. Rates of Bulk Crystallization (Avrami Constants)

TABLE 10. POLY(DIENES), POLY(ALKANES), POLY(VINYLS)

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
10.1. MELTS AND SOLUTIONS							
C198 alkane		72.6	18			Half-time, listed as $t_{0.1}$, *i.e*,	233
		73	25			10% transformation *not* 50%	
		74.2	62			as cited here in column	
		74.7	101				
		75	112				
		75.2	123				
		76.2	410				
		76.5	516				

TABLE 9. *cont'd*

Polymer	T_f (°C)	T_c (°C)	G (nm/s)	Remarks	Refs.
		36.8	11860		
		41	12760		
		45	10820		
		49	7940		
		53	3870		
		60	1174		
		63	237		
		65	76.6		
		67.2	65.0		
		69.2	30.4		
Poly(oxyethylene)/poly(oxypropylene)	Blend composition				341
$M_w = 5 \times 10^6/1.5 \times 10^6$	353 (10 min)			Rates measured on hot-stage microscope;	
100/0		53	450	blends were prepared by solution	
		54	217	casting from acetonitrile; two	
		55	122	crystallization regimes II and III	
		56	52	were reported along with interfaced	
		57	23	energy values	
		58	12		
		59	4.5		
30/20		52	833		341
		53	500		
		54	300		
		55	120		
		56	55		
		57	25		
		58	8.5		
60/40		51	1417		341
		52	850		
		53	517		
		54	250		
		55	117		
		56	80		
		57	33		
40/60		50	1967		341
		51	1317		
		52	967		
		53	517		
		54	300		
		55	115		
20/80		52	617		341
		53	383		
		54	300		
		56	57		
		57	17		
		58	12		

3. Rates of Bulk Crystallization (Avrami Constants)

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10.1. MELTS AND SOLUTIONS							
C198 alkane		72.6	18			Half-time, listed as $t_{0.1}$, *i.e*,	233
		73	25			10% transformation *not* 50%	
		74.2	62			as cited here in column	
		74.7	101				
		75	112				
		75.2	123				
		76.2	410				
		76.5	516				

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		76.7	605				
		78.3	973			Molecular chain extension starts here	
		79.2	2063				
		80.3	571				
		81.4	1538				
		82.3	539				
		83.4	3893				
Poly(1-butenylene)	0 (30 min)	−80		$2.8 \leq n \leq 5.4$	4.8×10^{-16}	Measured by ^{1}HFT-NMR;	342
cis-1,4-Poly(butadiene)		−70			3.4×10^{-14}	$t_{0.5}$ values not listed	
$M_w = 3.15 \times 10^5$		−60			2.2×10^{-12}		
$M_w/M_n = 3 \pm 0.4$		−50			3.25×10^{-9}		
cis content − 96.5%		−40			7.45×10^{-9}		
		−30			3.97×10^{-8}		
		−25			3.74×10^{-9}		
		−20			4.5×10^{-11}		
		−15			6.1×10^{-13}		
Poly(1-butene)		65	57.6				
$M_n = 73 \times 10^3$		70	60			Nucleation densities cited in text	251
$M_w = 750 \times 10^3$		80	96				
		85	222				
		90	456				
		92	708				
		95	1572				
		100					
Poly(1-butene); unfractionated		95	7500	3.05	1.59×10^{-6}	Dilatometry	251
		98	15000	3.10	2.98×10^{-7}		
		102	60000	3.43	2.51×10^{-9}		
		105	189600	3.80	2.00×10^{-13}		
$M_v = 409 \times 10^3$		80	43	1.9	5.10×10^{-4}	Calorimetry; DSC	121
		78	32	2.0	6.89×10^{-4}		
		74	25	1.9	16.3×10^{-4}		
		72	22	2.0	14.0×10^{-4}		
Poly(1-butene)	170 (30 min)	95	750	3.05		Dilatometric rate constants	306
isotactic; unfractionated		98	15000	3.10		expressed in text in min^{-n}	
		102	60000	3.43			
		105	189600	3.80			
Solution concn.							
0.5 g/dl		58	27000	~2.4		Crystallized from amyl acetate	
1.0 g/dl		58	12600	~2.4		solution; k values listed in min^{-n}	
2.0 g/dl		58	9600	~2.4			
Poly(1-butene)		90	55	1.9	3.20×10^{-4}		121
$M_v = 964 \times 10^3$		88	37	2.0	5.64×10^{-4}		
		80	15	2.0	18.0×10^{-4}		
		78	12	1.9	50.2×10^{-4}		
$M_v = 679 \times 10^3$		86	80	1.9	1.55×10^{-4}		121
		84	65	2.0	1.98×10^{-4}		
		82	44	1.9	4.37×10^{-4}		
		78	29	2.2	4.48×10^{-4}		
		76	22	2.1	9.84×10^{-4}		
		72	18	2.2	11.3×10^{-4}		
		70	14	2.0	33.3×10^{-4}		
$M_v = 349 \times 10^3$		84	99	2.0	0.708×10^{-4}		121
		82	73	2.0	1.62×10^{-4}		
		80	58	2.1	1.08×10^{-4}		
		78	45	2.1	2.04×10^{-4}		
		76	38	2.3	1.80×10^{-4}		
		70	24	2.1	9.15×10^{-4}		
		66	17	2.1	20.8×10^{-4}		
$M_v = 289 \times 10^3$		80	101	1.9	1.03×10^{-4}		121
		78	86	1.9	1.28×10^{-4}		
		76	62	1.9	2.19×10^{-4}		
		70	43	2.0	3.12×10^{-4}		
		68	38	2.0	4.17×10^{-4}		
		66	33	2.2	3.64×10^{-4}		
		62	26	2.0	11.1×10^{-4}		

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
$M_v = 96 \times 10^3$		68	136	1.9	0.54×10^{-4}		
		66	101	2.0	0.86×10^{-4}		
		62	52	2.0	2.43×10^{-4}		
		60	33	2.0	6.85×10^{-4}		
		58	25	1.8	19.4×10^{-4}		
Poly(1-butenylene); hydrogenaed Phillips Pet Co. $M_w = 5.10 \times 10^3$		97.0	40			DSC; for sample details consult text; crystallinity, T_m listed in tables	344
		99.0	44				
		100.0	66				
		101.0	78				
		102.0	98				
		103.0	126				
		104.0	186				
$M_w = 13.3 \times 10^3$		95.0	34			DSC; for sample details consult text; crystallinity, T_m listed in tables	344
		97.0	40				
		99.0	64				
		100.0	73				
		101.0	89				
		102.0	135				
$M_w = 60.7 \times 10^3$		89.0	59			DSC; for sample details consult text; crystallinity, T_m listed in tables	344
		91.0	92				
		92.0	125				
		93.0	173				
		94.0	262				
		95.0	427				
		96.0	876				
$M_w = 340.0 \times 10^3$		89.5	68			DSC; for sample details consult text; crystallinity, T_m listed in tables	344
		91.5	135				
		92.5	180				
		93.5	282				
		94.5	498				
		95.5	1056				
		96.5	2334				
Poly(1-butenylene) *trans*	157	120		1.26	0.064	DSC; melt to condis crystal; transformation I: two "Avrami" regions exist; nucleation time and crystallinities are listed	18
		119		1.13	0.77		
		117		0.62	27.4		
		115		0.47	119		
		113		0.41	3010		
		108		0.35	8050		
		120		0.32	28.7	Transformation region II	
		119		0.22	96.6		
		117		0.19	310		
		115		0.21	543		
		113		0.19	1090		
		108		0.14	1340		
		105		0.08			
		55		1.74	0.052	DSC, condis to rigid crystal transformation; region II	
		54		1.41	0.922		
		53		1.10	10.2		
		52		0.74	273		
		49		0.28	1490		
		45		0.15	930		
		55		0.12	68.4		
		54		0.08	360		
$M_n = 49 \times 10^3$ (GPC)	100 (15 min)	30.0	420	2.6		Dilatometry; values of k in original text	185
		35.0	1200	2.6			
		40.0	3300	2.6			
		45.0	1200	2.6			
$M_n = 59 \times 10^3$		30.0	180	4.4		Dilatometry; values of k in original text	185
		35.0	450	4.5			
		40.0	3360	3.2			
		45.0	16800	2.6			
$M_n = 70 \times 10^3$		30.0	174	5.0		Dilatometry; values of k in oritinal text	185
		35.0	456	4.0			
		40.0	3300	2.9			
		45.0	14400	2.8			

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
$M_n = 77 \times 10^3$		30.0	210	3.6		Dilatometry; values of k in original text	185
		35.0	246	3.1			
		40.0	3300	3.1			
		45.0	9600	2.0			
$M_n = 94 \times 10^3$		30.0	216			Dilatometry; values of k in original text	185
		35.0	660	3.2			
		40.0	3900	2.2			
		45.0	14400	2.1			
$M_n = 106 \times 10^3$		30.0	420	3.1		Dilatometry; values of k in original text	185
		35.0	1560	2.6			
		40.0	5160	2.4			
Poly(1-butenylene) *trans*		45.0	19200	2.6			
	100 (15 min)	45.0	12840	2.8		k_n values listed in text	343
		42.0	5100	2.7			
$M_v = 30 \times 10^3$		40.0	2280	2.8			
		37.5	1050	2.8			
		35.0	600	2.2			
		32.5	360	2.2			
		27.5	192				
Poly(1-butenylene) *cis* 1,4		213.0	13×10^2			^{1}HFT-NMR measurements from toluene solution @ $0.32 \leq v_2 \leq 1.0$ volume fractions; $t_{0.5}$ (s) is replaced by $t_{0.1}$ (s) in column 4	345
$M_n = 740000$		214.3	18×10^2				
$M_w/M_n = 1.25$;		214.8	22×10^2				
$v_2 = 1.0$							
$v_2 = 0.79$		204.7	5.5×10^2				
		205.7	9.9×10^2				
		207.3	14×10^2				
$v_2 = 0.59$		199.2	5.0×10^2				345
		201.7	14×10^2				
		203.0	23×10^2				
$v_2 = 0.42$		198.3	11×10^2				345
		200.5	30×10^2				
		201.7	44×10^2				
$v_2 = 0.32$		197.8	18×10^2				345
		198.5	22×10^2				
		200.0	49×10^2				
Poly(chlorotrifluoroethylene)		180	180			Dilatometry	346
		186	480	3			
		191	1500	3			
		94	2820	3			
		195	4200	3			
		196	5220	3			
		197	8580	3			
		197	11160	3			
Poly(ethylene) fraction		105.75	2.4	4		Crystallization measurements carried out in solution by dilatometry rate data from transformation plots; v_2 = polymer volume, fraction used in solution	167
n-Hexadecane		107.5	5.4				
$M_n = 3 \times 10^5$		108.5	19.2				
$v_2 = 0.540$		110.5	102.0				
Polyethylene fractions							
α-Chloronaphthalene		120	0.13	4			324
$M_n = 4000$		121	0.17				
$v_2 = 1$		122	0.21				
		123	0.52				
		124	1.56				
		125	9.60				
$v_2 = 0.70$		109	0.14				324
		110	0.16				
		111	0.18				
		112	0.25				
		113	0.72				
		114	2.22				
$v_2 = 0.46$		101	0.14				324
		102	0.17				
		103	0.24				
		104	0.30				

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		105	0.54				
$v_2 = 0.30$		98	0.20	4			324
		100	0.36				
		101	0.5				
		102	0.81				
		103	1.82				
$M_n = 10000$		124	0.13	3			324
$v_2 = 1$		125	0.16				
		126	0.34				
		127	0.72				
		128	3.30				
$M_n = 10000$		109	0.78	3			324
$v_2 = 0.70$		110	0.10^8				
		111	0.15				
		113	0.30				
		116	0.81				
$M_n = 10000$		106	0.20	3			324
$v_2 = 0.46$		108	0.36				
		109	0.54				
		111	1.80				
$M_n = 10000$		102	0.35	3			324
$v_2 = 0.30$		103	0.47				
		104	0.84				
		105	1.14				
$M_n = 100000$		122	0.09	3			324
$v_2 = 1$		123	0.10^4				
		124	0.13^8				
		125	0.16				
		126	0.27				
		127	0.54				
		128	1.62				
$M_n = 100000$		113	0.08^4	3			324
$v_2 = 0.70$		115	0.22				
		116	0 27				324
		117	0.36				
		118	0.72				
$M_n = 100000$		105	0.26	3			324
$v_2 = 0.46$		107	0.54				
		108	0.84				
		110	1.80				
$M_n = 100000$		102	0.33				324
$v_2 = 0.30$		103	0.42				
		104	0.57				
		105	0.72				
		106	2.16				
$M_n = 250000$		124	0.19^2	3			324
$v_2 = 1$		125	0.39				
		126	0.66				
		127	1.33				
		128	4.19				
$M_n = 250000$		113	0.16	3			324
$v_2 = 0.70$		114	0.25				
		115	0.36				
		116	0.57				
$M_n = 250000$		105	0.30	3			324
$v_2 = 0.46$		107	0.66				
		108	0.90				
		110	3.12				
$M_n = 250000$		102	0.18	3			324
$v_2 = 0.30$		103	0.21				
		104	0.27				
		105	0.38^4				
		106	0.54				
Poly(ethylene) fractions							
α-Chloronaphthalene		98.0	0.26	4		Dilatometry:	336

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TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
$M_n = 14000$		98.9	5.40	4		crystallinity = 85%	
$v_2 = 0.003$		100.1	14.70	4			
		100.9	48.60	4			
		101.9	156.00	4			
$M_n = 20000$		98.0	2.40	4		Crystallinity = 90%	336
$v_2 = 0.003$		99.0	6.48	4			
		99.9	16.20	4			
		100.9	42.00	4			
$M_n = 47000$		98.25	492	3			336
$v_2 = 0.003$		99.2	9.60	3			
		100.1	27.60	3			
		100.9	69.00	3			
		101.8	186.00	3			
$M_n = 190000$		98.1	9.60	3		Crystallinity = 95%	336
$v_2 = 0.003$		98.9	28.20	3			
		100.0	36.00	3			
		100.9	84.00	3			
		102.0	420.00	3			
$M_2 = 1200000$		96.4	2.55	3		Crystallinity = 75%	336
$v_2 = 0.003$		98.4	9.60	3			
		99.4	25.50	3			
		100.4	66.00	3			
Poly(ethylene) fractions							
Decalin		81.8	2.40	4		Dilatometry;	334
$M_n = 100000$		82.8	3.90	4		crystallinity = 95%	
$v_2 = 0.003$		83.7	6.30	4			
		85.2	19.80	4			
		86.4	60.30	4			
		87.2	126.00	4			
Xylene		87.2	3.00	4		Crystallinity = 90%	334
$M_n = 100000$		88.1	5.10	4			
$v_2 = 0.003$		89.0	12.00	4			
		90.0	24.00	4			
Tetralin		87.9	2.16	3		Crystallinity = 90%	334
$M_n = 100000$		89.4	3.72	3			
$v_2 = 0.003$		90.5	10.20	3			
		91.5	21.00	3			
		92.4	45.00	3			
		93.7	156.00	3			
α-Chloronaphthalene		98.9	90.00	3			334
$M_n = 100000$		100.0	270.00	3			
$v_2 = 0.003$		100.9	780.00				
		102.0	3720.0				
Poly(ethylene) Marlex 50		122	528			Dilatometry	347
		123	750				
		124	1080				
		125	1560				
		126	3420				
		127	11400				
		128	34800				
		129	198000				
		130	1140000				
Poly(ethylenes)							
$M_n = 1.69 \times 10^3$		151.7	328	1.3		Pulsed NMR;	348
$M_w = 120 \times 10^3$		151.0	364	1.3		pressure = 1 kbar	
$M_v = 90.5 \times 10^3$		153.0	571	1.5			
$M_z = 599 \times 10^3$		154.3	969	1.6			
		154.3	930	1.6			
		155.0	1380	1.6			
		155.3	1610	1.6			
		157.0	7210	1.7			
		173.0	295	1.2		2 kbar	348
		173.8	322	1.3			
		175.0	485	1.5			
		177.0	895	1.5			

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		177.3	989	1.5			
		179.0	3580	1.7			
		195.0	384	1.3		3 kbar	348
		197.0	521	1.4			
		197.3	594	1.4			
		199.0	785	1.5			
		199.3	950	1.6			
		201.0	1800	1.7		4 kbar	348
		217.0	260	1.6			
		219.0	367	1.7			
		221.0	670	1.7			
		221.3	806	1.7			
		223.0	2880	1.7			
		227	350	1.5		5 kbar	348
Poly(ethylene) HDPE	200 (2 min)	124	165	2.32		Bulk rates by DSC; crystallinity	349
		124	165	2.35		melting points and density	
		125	338	2.29		data are given inthe original	
Dow 08064N		126	750	2.05		references; t_i is proportional	
		127	1920	2.24		to t_{y2} (see original text)	
Dow 10062N		120	51	2.53			349
		121	81	2.44			
		122	162	2.60			
Phillips Marlex EHM 6006		120	42				349
		121	46				
		122	60				
		123	136	2.46			
		124					
		125	351	2.20			
Phillps Marlex EHM 6030		120	36	2.25			349
		121	56	2.72			
		122	108	2.76			
		123	222	2.56			
		124	308	2.48			
Phillips Marlex TR-855		122	66	3.18			349
		123	78	2.46			
		124	156	2.58			
DuPont Alathon 7480		122	165	1.89			349
		123	263	1.93			
		124	510	1.96			
		125	998	1.85			
Soltex T60-1000		120	54	2.36			349
		121	100	2.61			
		122	222	2.63			
		123	496	2.18			
		105		2.26			
Soltex T60-2000		123	240	2.32			349
		124	480	2.50			
		125	518	2.25			
		126	1500	2.47			
ARCO Super Dylan 5503A		120	45	1.96			349
		121	96	2.28			
		121.5	150	2.23			
		122	261	2.60			
ARCO Super Dylan 5503B		120	48	2.07			349
		121	81	1.98			
		122	252	2.62			
		125	714				
		123	370				
		120	250				
Sinclair 2907		117	62.5			Transmitted light intensity;	256
		115	31.3			(a) atmospheric pressure	
		113	19.2				
		112	7.14				
		143	1429			(b) 0.5 kbar	256
		141	476				

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		138	189				
		137	137				
		135	62.6				
		133	28.8				
		157	154			(c) 1 kbar	256
		156	769				
		153	250				
		151	115				
		149	75				
		147	42.9				
		145	19.2				
		167	2000			(d) 1.5 kbar	256
		165	769				
		163	303				
		161	149				
		159	107				
		157	57.8				
		156	44.1				
		155	33.3				
		177	1250			(e) 2.0 kbar	256
		176	769				
		174	500				
		172.5	189				
		170	88.5				
		168	50.0				
		167	34.1				
		166	25.6				
$M_v = 166 \times 10^3$	160 (15 min)	127.6	6780	1.6		DSC; rates are also given in	
		126.6	3000	1.6		text after quenching; some	343
		126.1	2190	1.6		dilatometry results also available;	
		125.6	1500	1.4		k_n values are also cited	
		125.1	1080	1.2			
		124.1	420				
	170 (30 min)	95	7500	3.05		Dilatometry from amyl acetate	306
		98	15000	3.10		solution; k_n values listed	
		102	60000	3.43		in text	
		105	189600	3.80			
Poly(ethylene)							
I: $M_n = 52 \times 10^{-3}$		10(II)	12000	Varies		Dilatometry; pressure	188
$M_w/M_n = 2.7$		12(II)	3600	between		1000 kg/cm^2; ΔT_c (°C) is	
II: $M_n = 54 \times 10^{-3}$		17(II)	90	1 and 3339		cited; not t_c; see text	
$M_w/M_n = 3.0$		6(II)	24000	with pressure		2000 kg/cm^2; pressures up	
						to 5000 kg/cm^2 reported	
Poly(ethylene), unfractionated		150.6	357.0	1.9			
$M_w = 64000$		152.1	1200.0	0.9		Pressure = 840 kg/cm^2	339
$M_n = 14000$		153.6	10242.0	20.0			
		171.6	76.3	1.7		1950 kg/cm^2	339
		175.6	586.0	2.1			
		177.5	2100.0	2.1			
		193.4	46.3	1.3		3400 kg/cm^2	339
		195.8	55.2	1.3			
		197.3	99.0	1.6			
		199.2	100.0	1.7			
	220.3	18.7		1.0		5100 kg/cm^2	339
	222.7	19.4		1.0			
	225.2	22.3		1.1			
	227.6	47.2		1.2			
	229.1	52.4		1.2			
	226.6	21.6		1.2		5300 kg/cm^2	
	228.6	25.4		1.2			
	230.5	47.5		1.3			
	232.4	67.8		1.3			
	196.6			2.8			
Poly(ethylene) fractions	198.8			2.7	2.8	Avrami exponent of extended	263
$M_w = 15500$	186.6			2.9		chain crystallization in	

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
$M_w/M_n = 1.2–1.3$	187.5			2.9		parentheses; calorimetry	
$M_w = 150000$	188.5			3.2		(DTA) @ 2.44 kg/cm^2	
$M_w/M_n = 1.2–1.3$	183.6			3.3			
	185.3			3.0			
	186.3			3.2			
	187.3			3.3			
					6000		
					1080		
					114		
					60		
Poly(ethylene), fractionated	125.4		906	3.1	1.45×10^{-4}	Dilatometry; col. #5 lists primary	304
$M_v = 18.5 \times 10^3$	125.8		1212	3.1	5.51×10^{-5}	process; 4 lists time (min^{-n})	304
	126.9		3216	3.1	3.27×10^{-6}		
	127.6		5826	3.0	8.68×10^{-7}		
	127.8		8220	3.0	3.30×10^{-7}		
	128.5		18180	2.8	1.04×10^{-7}		
	128.6		21300	2.7	9.01×10^{-8}		
	126.6			3.5	4.69×10^{-6}		
$M_v = 7.3 \times 10^3$	127.5			3.5	5.62×10^{-7}		304
	128.5			3.5	3.66×10^{-8}		
	129.5			3.5	1.18×10^{-9}		
	130.5			3.5	3.42×10^{-12}		
	131.0			3.5	5.62×10^{-14}		
$M_v = 1.8 \times 10^3$	128.6			3.5	7.48×10^{-6}		304
	128.7			3.5	1.86×10^{-6}		
	129.5			3.5	8.30×10^{-8}		
	130.0			3.9	5.65×10^{-10}		
	130.3			3.8	9.52×10^{-11}		
Poly(ethylene) fractions	124.5	96		2.6	2.40×10^{-11}	Calorimetry; fractions from	284
$M_n = 25.0$	125.5	174		2.3	6.00×10^{-2}	Marlex 60; column 5 listed	
$M_n = 41.0$	126.5	282		2.2	2.3×10^{-2}	as time (min^{-n})	
$M_w/M_n = 1.6$	127.5	636		2.6	1.5×10^{-3}		
		128.0	810	2.9	3.6×10^{-4}		
		129.0	2100	2.4	1.4×10^{-4}		
		130.5	4200	3	2.0×10^{-6}		
Poly(ethylene), fractions							
$M_n = 5.9 \times 10^3$		125.1	53	4		Dilatometry; column 4 gives	344
		126.1	280			values of $t_{0.1}$(min); note: n is	
		127.1	1500			independent of T_c for each	
		128.1	5000			fraction	
$M_n = 14 \times 10^3$		126.1	4.2	3			344
		127.1	20				
		128.1	130				
		129.1	1050				
		130.1	5600				
$M_n = 20 \times 10^3$		125	2.3	3			344
		126	4				
		127	6.8				
		128	18				
		129	76				
		130	220				
$M_n = 47 \times 10^3$		125	2.5	3			344
		126	4				
		127	6.5				
		128	12				
		129	35				
		130	190				
$M_n = 122 \times 10^3$		125	4.7	3			344
		127	9.5				
		128	17				
		129	36				
		130	130				
		131	670				
		132	10700				

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TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
$M_n = 200 \times 10^3$		125	3.4	3			344
		126	4.8				
		127	7.5				
		128	15				
		129	35				
		130	75				
$M_n = 284 \times 10^3$		125.1	6.5	3			344
		126.1	13				
		127.1	23				
		128.1	45				
		129.1	100				
		130.1	320				
		131.1	1600				
		132.1	17000				
$M_n = 600 \times 10^3$		125.1	9	3			344
		126.1	16				
		127.1	32				
		128.1	65				
		129.1	180				
		130.1	460				
		131.1	2000				
		132.1	21000				
$M_n = 1.2 \times 10^6$		125.1	17	3			344
		126.1	37				
		127.1	120				
		128.1	390				
		129.1	1400				
		130.1	7500				
$M_n = 8.0 \times 10^6$		125.1	20	3			344
		126.1	62				
		127.1	260				
		128.1	1020				
		129.1	4600				
		130.1	36000				
		116	0.053	2		Calorimetry (DSC); column 4 gives $t_{0.25}$ (min), not the half-time listed in column 4	332
		117	0.080				
		118	0.14				
		118	0.14				
		119	0.21				
		120	0.51				
		121	0.91				
		122	2.00				
		123	4.35				
		124	10.5				
		125	31.3				
		125.5					
		126	83.3				
$M_n = 5.0 \times 10^6$		116	0.048	2			
		117	0.091				332
		118	0.13				
		119	0.26				
		120	0.43				
		121	1.02				
		122	2.17				
		123	4.17				
		124	10.9				
		125	27.8				
		126	90.9				
$M_n = 3.0 \times 10^6$		116	0.05	2			332
		117	0.077				
		118	0.12				
		119	0.25				
		120	0.42				
		121	0.87				
		122	2.38				

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		123	4.76				
		124	10.0				
		125	29.4				
		126	76.9				
$M_n = 8.0 \times 10^5$		116	0.041	3			332
		117	0.08				
		118	0.14				
		119	0.26				
		120	0.49				
		121	0.99				
		122	1.85				
		123	4.0				
		124	8.3				
		125	21.7				
		126	55.5				
		127	133.3				
		127.5	227.3				
$M_n = 4.25 \times 10^5$		116	0.047	3			332
		117	0.07				
		118	0.13				
		119	0.26				
		120	0.40				
		121	0.63				
		122	1.49				
		123	2.38				
		124	5.0				
		125	10.3				
		126	29.3				
		126.5	38.5				
		127	66.7				
		127.5	102.0				
$M_n = 1.07 \times 10^5$		116	0.055	3			332
		117	0.083				
		118	0.16				
		119	0.28				
		120	0.44				
		121	0.76				
		122	1.28				
		123	2.50				
		124	4.2				
		125	8.3				
		126	18.2				
		127	33.3				
		127.5	66.7				
		128	156.3				
$M_n = 1.95 \times 10^4$		116	0.043	3			332
		117	0.063				
		118	0.13				
		119	0.19				
		120	0.30				
		121	0.47				
		122	0.71				
		123	1.20				
		124	1.92				
		125	3.3				
		125.5	5.9				
		126	15.6				
		127	71.4				
$M_n = 11.4 \times 10^3$		117	0.045	3			332
		118	0.067				
		119	0.11				
		120	0.20				
		121	0.33				
		121.5					
		122	0.53				

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		122.5					
		123	0.71				
		123.5					
		124	1.54				
		124.5					
		125	4.0				
		125.5	5.9				
		126	11.5				
		126.5	25.0				
		127	66.7				
$M_n = 8.0 \times 10^3$		118	0.043	4			332
		119	0.077				
		120	0.13				
		121	0.24				
		122	0.40				
		123	0.77				
		124	2.4				
		124.5	4.8				
		125	10.9				
		125.5	28.6				
		126	83.3				
$M_n = 5600$		113	0.063	4			332
		114	0.093				
		115	0.13				
		116	0.16				
		117	0.22				
		118	0.26				
		119	0.38				
		120	0.67				
		121	1.4				
		121.5	2.2				
		122	4.8				
		122.5	13.5				
		123	27.0				
		123.5	55.6				
		124	196.0				
$M_n = 3.9 \times 10^3$		111	0.034				332
		112	0.076				
		113	0.10				
		114	0.14				
		115	0.19				
		116	0.24				332
		117	0.38				
		118	0.71				
		119	1.79				
		120	6.67				
		121	52.6				
		122	357				
		123	2000				
$M_n = 2.9 \times 10^3$		111	0.054				332
		112	0.085				
		113	0.12				
		114	0.16				
		115	0.21				
		116	0.30				
		117	0.50				
		118	1.16				
		119	4.35				
		120	25.0				
Poly(ethylenes)							
$M_w = 51.5 \times 10^3$	155 (15 min)	123.4	28.5	Not listed	Expressed as	LDI techniques see Ref. 94; SAXS	350
$M_w/M_n = 1.5$		124.5	45.0		$t_{0.5}$ only	dimensions and σ_e values	
0 branches/1000 CH_2's		125.5	96.5			tabulated and regimes cited	
		126.5	188				
		127.5	323				

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		127.8	439				
		129.0	993				
		130.0	8130				
		131.2	4330				
$M_w = 89.9 \times 10^3$	155 (15 min)	119.6	76.5	Not listed	Expressed as	LDI techniques see Ref. 94; SAXS	350
$M_w/M_n = 1.53$		120.5	133		$t_{0.5}$ only	dimensions and σ_e values	
6.67 branches/1000 CH_2's		121.5	188			tabulated and regimes cited	
		122.7	358				
		123.8	636				
		125.0	1453				
$M_w = 41.8 \times 10^3$	155 (15 min)	116.5	60	Not listed	Expressed as	LDI techniques see Ref. 94; SAXS	350
$M_w/M_n = 1.17$		117.5	137		$t_{0.5}$ only	dimensions and σ_e values	
15.5 branches/1000 CH_2's		118.5	243			tabulated and regimes cited	
		120.0	389				
		120.8	680				
		121.8	1239				
		122.6	2778				
		123.7	3985				
		127.7	4167				
$M_w = 54.7 \times 10^3$	155 (15 min)	111.5	76.5	Not listed	Expressed as	LDI techniques see Ref. 94; SAXS	350
$M_w/M_n = 1.69$		112.5	98.0		$t_{0.5}$ only	dimensions and σ_e values	
13.64 branches/1000 CH_2's		113.5	133			tabulated and regimes cited	
		144.5	173				
		115.5	263				
		116.7	330				
		118.5	873				
		119.7	1686				
		121.0	6710				
$M_w = 65.5 \times 10^3$	115 (15 min)	97.5	58				350
$M_w/M_n = 2.1$		98.3	98				
32 branches/1000 CH_2's		99.3	182				
		100.3	304				
		101.2	573				
		102.5	840				
		103.3	1570				
		104.4	1908				
		105.3	3310				
		106.3	3345				
		107.3	3378				
10.2. BLENDS AND COPOLYMERS							
Poly(ethylene)/poly(1-phenylethylene)							
PE/PS	175	122	126	1.57		Listed in text; DSC, mech.	351
		123	144	1.86		properties text crystallinity	
		124	264	2.29		and morphology	
		125	540	2.44			
		126	1140	2.26			
Rigidex 50/Carinex SR61		122	96	1.88			351
50/50, w/w		123	125	1.94			
		124	228	2.32			
		125	444	2.59			
		126	9000	2.36			
Poly(ethylene-*co*-1-acetoxyethylene) Hostalen, GC 6400							
0 wt.% VA		126.9	1071			Calorimetry: data interpolated	356
MI = 4.5 g/10 min		125.9	704			from original plots	
		125.1	367				
		124.1	138				
		123.2	90.5				
5.5 wt.% VA		102.2	1628				356
MI = 1.6 g/10 min		101.2	878				
		100.4	589.5				
		99.4	318				
		98.4	158				
		97.5	92				
		96.5	60				

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TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
14.0 wt.% VA		87.6	1334				356
MI = 6.5 g/10 min		86.6	770				
		85.7	553.0				
		84.8	353				
		83.9	241				
17 wt.% VA		80.1	2444				356
MI = 108 g/10 min		89.1	1735				356
		78.3	1098				
		77.4	731				
		76.4	467.5				
		75.6	297				
		74.6	183				
		73.7	120				
		72.6	83.5				
		71.7	55.5				
		70.7	35.6				
19 wt.% VA		81.0	2200				356
MI = 98 g/10 min		80.2	1560				
		79.3	900				
		78.3	605				
		77.3	389.4				
		76.5	246.6				
		75.5	155				
		74.5	97.2				
		73.5	62.4				
19 wt.% VA		81.1	904				356
MI = 2.7 g/10 min		80.3	689				
		79.3	508				
		78.4	302				
		77.4	169.2				
24 wt.% VA		68.6	716				356
MI = 19 g/10 min		67.8	639				
		66.8	438.6				
		65.9	343.2				
Polypropylene/ethylene–propylene rubber							
	230 (2 min)	126	188	2.45	3.58×10^{-6}	Rates by DSC; spherulite growth rates also reported (see Section 2)	403
		128	268	2.55			
		130	455	2.72	8.80×10^{-8}		
		132	774	2.78	7.60×10^{-9}		
PP/EPR		134	1169	2.51			
100/0, w/w		136	1892	2.58	4.20×10^{-9}		
Melt flow index = 15; Montell Polyolefins G. Natta Res. Centre, Ferrara, Italy							
PP/EPR	230 (2 min)	120	188	2.48	1.95×10^{-6}		
		122	288	2.36			
		124	431	2.24	8.38×10^{-7}		
		126	678	2.30			
		128	1068	2.34	6.57×10^{-8}		
		130	1693	2.40			
		132	2700	2.38	5.83×10^{-9}		
PP/EPR	230 (2 min)	120	123	2.68	1.77×10^{-6}		
40/60, w/w		122	244	2.79			
		124	376	2.59	2.15×10^{-7}		
		126	596	2.52	1.04×10^{-7}		
		128	1045	2.57			
		130	1780	2.56	5.37×10^{-9}		
Poly(1-hexadecene), PHD isotactic		36		4.3	7.30×10^{-4}	Measurements by DSC; SAXS data included; units of k are (min^{-n}) *not* (s^{-n}) and independent of T_c for these comb-like polymers; $T_m = 58$°C	352
$M_v = 7800$		38			2.45×10^{-3}		
isotacticity = 43%		40			5.70×10^{-3}		
		41			1.32×10^{-2}		
		42			7.64×10^{-2}		
		43			4.10×10^{-1}		
$M_v = 2800$		36		3.7 ± 0.6	8.05×10^{0}		352
isotacticity = 62%		38			9.60×10^{-1}		
		40			1.49×10^{-1}		
		41			6.49×10^{-2}		
		42			2.65×10^{-2}		

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		43			9.95×10^{-3}		
		44			3.45×10^{-3}		
		45			8.86×10^{-4}		
$M_v = 1800$		36		3.3 ± 0.4	2.51×10^{1}		352
isotacticity = 91%		38			2.89×10^{0}		
		40			3.57×10^{1}		
		41			1.36×10^{-1}		
		42			5.46×10^{-2}		
		43			1.22×10^{-2}		
		44			2.96×10^{-3}		
		45			2.46×10^{-4}		
Poly(isoprene) *cis*		0	188160	3.6		Dynamometer for $t_{0.1}$(s);	166,353
room temperature; crosslinked		−10	30480	3.6		10% transformation	
at 0.5 MR; 20% gel		−15	14820	3.5		elongation ratio;	
		−20	10560	3.5		$\alpha = 1$	
		0		2.45		$\alpha = 2$	
		−10		2.90			
		−15		2.95			
		−20		2.85			
		0		2.10		$\alpha = 3$	
		−10		2.15			
		−15		2.15			
		−20		2.10			
		0		1.65		$\alpha = 4$	
		−10	9060	1.60			
		−15		1.57			
		−20	4668	1.55			
		0	21240	1.40		$\alpha = 5$	
		−10	4896	1.55			
		−15	3186	1.37			
		−20	1896	1.40			
		0		1.35		$\alpha = 6$	
		−10	2107	1.30			
		−15		1.35			
		−20	654	1.13			
Poly(isoprene) *cis*							
Natural Rubber		−5	55200				104,354,355
		−11	19800				
		−16	12000				
		−17	10200				
		−22	9000				
		−27	9000				
		−33	14400				
trans Gutta Percha		−38	23400				109,359,355
		35	768				
		40	1260				
		45	6780				
		47	11700				
		49	19800				
		51	31800				
		53	64500				
		55	129000				
		57	291000				
		59	720000				
$[\eta] = 2.8$ in toluene at 30°	100 (30 min)	20.5	475		6.25×10^{-9}	Dilatometry; other grades	109,359,355
		26.5	641		3.20×10^{-9}	of polymer are presented	
		30.5	919		9.44×10^{-11}		
		36.5	2160		7.50×10^{-11}		
		40.5	3995		1.22×10^{-11}		
		45.0	11417				
Poly(isoprene) *trans*-1,4	100 (30 min)	20.5			9.30×10^{-3}	Rate results from dilatometry	242
LMF Spalding Co. R-308		26.5			3.6×10^{-8}	measurements	
		30.5			1.84×10^{-8}		
		36.5			3.12×10^{-9}		
		40.5			6.40×10^{-11}		

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TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		45.0			2.21×10^{-11}		
HMF Spalding Co.		20.5			6.25×10^{-8}		242
		26.5			3.20×10^{-8}		
		30.5			9.44×10^{-10}		
		36.5			7.50×10^{-10}		
		40.5			1.22×10^{-10}		
		45.0			5.56×10^{-12}		
Balta		20.5			9.26×10^{-8}		242
		26.5			3.60×10^{-8}		
		30.5			1.84×10^{-8}		
		36.5			1.89×10^{-9}		
		40.5			2.12×10^{-10}		
		45.0			2.73×10^{-12}		
Poly(isoprene) *trans*-1,4	100 (5 min)	30.0			240×10^{-3}	Transformation rates by DSC;	357
$M_n = 49 \times 10^3$		32.5			108×10^{-3}	heat of fusion and melting	
		35.0			462×10^{-3}	points also provided	
		37.5			570×10^{-4}		
		40.0			102×10^{-5}		
		42.5			216×10^{-6}		
		45.0			474×10^{-7}		
$M_n = 59 \times 10^3$		30.0			360×10^{-3}		357
		32.5			78×10^{-3}		
		35.0			156×10^{-5}		
		37.5			60×10^{-5}		
		40.0			96×10^{-6}		
		42.5			498×10^{-7}		
		45.0			168×10^{-7}		
$M_n = 70 \times 10^3$		30.0			74.3×10^{-3}		357
		32.5			169×10^{-3}		
		35.0			210×10^{-4}		
		37.5			84×10^{-4}		
		40.0			330×10^{-6}		
		42.5			252×10^{-7}		
		45.0			126×10^{-7}		
$M_n = 77 \times 10^3$	100 (5 min)	30.0			198×10^{-2}		
		35.0			102×10^{-3}		
		37.5			360×10^{-5}		
		40.0			384×10^{-6}		
		42.5			474×10^{-7}		
		45.0			60×10^{-7}		
$M_n = 94 \times 10^3$	100 (5 min)	32.5			120×10^{-3}		357
		35.0			174×10^{-4}		
		37.5			558×10^{-5}		
		40.0			420×10^{-5}		
		42.5			114×10^{-5}		
		45.0			450×10^{-6}		
$M_n = 106 \times 10^3$	100 (5 min)	30.0			150×10^{-3}		357
		32.5			276×10^{-4}		
		35.0			120×10^{-5}		
		37.5			414×10^{-5}		
		40.0			78×10^{-5}		
		42.5			234×10^{-6}		
		45.0			144×10^{-7}		
		45.0	11417		5.56×10^{13}		
Vulcanized rubber							
Extension ratios: 1×1		−27		3.0	10^{-9}	Biaxial crystallization X-ray	358
1.5×1.5				2.4		kinetics; Avrami analysis	
2×2				1.7			
2.5×2.5				1.1	2.7×10^{-5}		
3×3				0.7	2.6×10^{-3}		
Poly (1-phenylethylene)		50			3.97×10^{-5}	Solvent, dimethyl phthalate, (DMP);	400
isotactic (90%)		55			3.63×10^{-5}	DMSO &TP solvents;	
$M_w = 4 \times 10^5$		60			5.75×10^{-5}	$T_{max} = \sim 70$°C in DMP;	
		65			5.74×10^{-5}	polymer–solvent interaction	
		70			6.67×10^{-5}	is DMP > DMSO > TP	

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		75			5.72×10^{-5}		
		80			4.67×10^{-5}		
		90			1.46×10^{-5}		
Poly(1-phenylethylene) isotactic	228.0 (1 min)	180	168.0			DSC measurements; see text	359
$M_n = 850 \times 10^3$	230.0 (1 min)		198.0				
	234.0 (1 min)		222.0				
	237.0 (1 min)		282.0				
	227.5 (1 min)		372				
	240.0 (1 min)		414				
	243.0 (1 min)		594				
	229.5 (1 min)		780				
$M_w = 850 \times 10^3$	233.0 (1 min)	170.0	210			DSC	359
		180.0	210				
		160.0	288				
		150.0	330				
		190.0	4200				
Poly(1-phenylethylene); syndiotactic							
$M_w = 803000$	300 (1 min)	110	15343	3.00		Rate values listed in text in s^{-1} vs. T_c; rate is MW dependent; $T_m \sim 543$K; $T_g \sim 373$ K	398
		115	4783				
		120	1972				
		125	988				
		135	326				
		145	144				
		155	78				
		170	43				
		180	32				
		190	25				
		207	22				
		220	23				
		225	25				
		230	29				
		235	37				
		240	49				
		246	79				
		250	164				
$M_w = 562000$	300 (1 min)	110	2440	2.88			398
		115	1005				
		120.5	504				
		130	187				
		150	56				
		160	36				
		175	26				
		188	21.3				
		202	21				
		215	24.5				
		220	27.5				
		230	40				
		235	56				
		240	87.8				
		250	485				
$M_w = 348000$	300 (1 min)	110	920				398
		115	436				
		120	237				
		125	144.5				
		134.5	70.7				
		146	42.0				
		156	29.6				
		166	22				
		175	18.8				
		182.5	17.9				
		190	16.6				
		200	18.2				
		212	19.5				
		230	35.7				
		235	54.5				

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		239	87.8				
		245.5	167.6				
		250	472				
Poly(1-phenylethylene), PS (syndiotactic)							
$M_w = 609000$; 1.5 mol%	330 (3 min)	149	19			DSC measurements for $t_{0.5}$	360
neutralized with various		155	11			values; T_g identical for all	
counterions cations Na^+ ions		160	6.8			materials; the counterion	360
		165	5.1			affects rate as $Na^+ < K^+$	
		170	4.1			$< Rb^+ < Cs^+$ for isothermal	
		175	4.0			changes	
		181	3.97				
		189	6.1				
		200	13.4				
		204	21.3				
		210	30.3				
K^+ ions		154.5	10.4				
		160	6.5				
		165	4.8				
		170	3.9				
		175	3.6				
		181	3.6				
		190	5.0				
		200	8.1				
		205	11.8				
		209.5	16.8				
		215	244.4				
		219.5	35.7				
Rb^+ ions		154.5	9.7				
		160	5.9				
		165	4.1				
		170	3.3				
		175	2.9				
		182	3.0[5]				360
		190	3.9				
		200	6.0				
		204	8.5				
		210	10.6				
		215	15.1				
		220	20.1				
		225	30.2				
Cs^+ ions		150	6.0				360
		165	4.5				
		170	3.6				
		175	3.1				
		182	2.9				
		185	2.9				
		200	3.3				
		209.5	5.6				
		214.5	7.0				
Cs^+ ions		220	9.9				360
		225	14.0				
		232	27.7				
H^+ ions		210	1.4				360
		214	1.8				360
		220	2.8				
		225	4.5				
		231.5	11.9				
Poly(1-phenylethylene)/poly[oxy(2,6-dimethyl-1,4-phenylene)]							
syndiotactic		245	1360			DSC measurements; hot stage	361
$M_w = 129000$		244	1210			microscopy spherulitic growth	
SPS/PPO; 100/0, w/w		243	925			rates listed	
		242	676				
		241	571				
		240	489				

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		239	437				
		237	257				
		236	219				
		235	137				
$M_n = 32000$		240	1289				361
$M_w = 244000$		239	1154				
SPS/PPO; 90/10		238	905				
		236	695				
		235	560				
		234	462				
SPS/PPO; 80/20		240	1923				361
		239	1450				
		238	1294				
		236	1042				
		235	824				
Poly(1-phenylethylene)/poly(1-methoxyethylene) PVME							
SPS/PVME; 90/10		237	1502				361
		236	1450				
		235	1065				
		233	719				
		232	645				
		231	486				
SPS/PVME; 80/20		237	2060				361
		236	1850				
		235	1602				
		234	1444				
		232	1165				
		231	1113				
Poly(1-phenylethylene)(S)/poly(*trans*-1-butylene)(B)/poly(imino-1-oxohexamethylene)(C)-*co*-poly(ε-caprolactam)							
Poly(ε-caprolactam) (100%)	250 (3 min)	37	64	2.63 ± 0.10		DSC measurement under nitrogen; rate constants given in min^{-n}; also spherulitic growth rates listed; nucleation dependent upon block size and type	362
		39	125				
		40	177				
		41	223				
		43	518				
		45	923				
		47	2374				
Wt.% composition ratios:		39	170	2.31 ± 0.11			362
35S/15B/50C		40	359				
		43	719				
		45	1564				
27S/15B/58C		37	86	2.54 ± 0.08			362
		38	115				
		39	151				
		40	200				
		41	269				
		43	549				
33S/06B/61C		37	76	2.34 ± 0.15			362
		39	151				
		40	169				
		41	213				
		43	420				
		45	858				
205/15B/65C		38	96	2.03 ± 0.03			362
		39	132				
		40	175				
		43	572				
		44	691				
9S/14B/77C		36	76	1.92 ± 0.06			362
		37	94				
		38	113				
		39	165				
		41	317				
		42	398				
		43	581				
Poly(propylene)							

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TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
$M_v = 50.5 \times 10^3$		135	474			Dilatometry	363
		140	1524				
		145	6870				
		150	26400				
		155	117000				
$M_v = 96.8 \times 10^3$		133	636				363
		134	894				
		135	1356				
		136	2124				
		137	3564				
		138	5256				
		139	8094				
		140	11154				
		141	16320				
$M_v = 307 \times 10^3$		135	642				363
		140	2280				
		145	11400				
		150	32400				
		155	152400				
$M_v = 447 \times 10^3$	190 (10 min)	137.1	6600	2.9		Hot stage optical microscopy	343
		134.1	2820	2.9			
		133.0	2220	2.9			
		130.2	780	3.1			
$M_v = 447 \times 10^3$	190 (10 min)	140.0	8280	2		DSC; rates are also given in test after quenching; results also available; k_n values are also cited	343
		137.5	5340	2			
		135.0	2400	2			
		132.5	1344	2			
		130.0	756	2			
		127.5	486	2			
		125.0	378	2			
		122.5	288	2			
		120.0	222	-			
		117.0	168	-			
$M_v = 447 \times 10^3$ (nucleated with 1 wt.% indigo)	190 (10 min)	145.0	990	2		DSC; Rates are also given in text after quenching, some dilatometry results also available; k_n, n values are also cited	343
		143.5	720	2			
		142.5	612	2			
		141.5	498	2			
		140.5	438	2			
Poly(propylene) Shell Carlona White							
$M_n = 196 \times 10^3$	270 (30 min)	75	3.0	4		DLI (depolarized light intensity) rate measurements	94(b)
		100	6.7	4			
		110	11.1	4			
		125	100.0	3			
Poly(propylene)							
$M_w = 300 \times 10^3$		47	2045			0.3 g/100 ml decalin	364
		50	3296	4			
		51	3952	4			
		52	4528	4			
		53	5494	4			
		54	6740	4			
		65	2068	4		0.3 g/100 ml tetralin	
		67	2892	4			
		69	3863	4			
		70	4685	4			
		71	5717	4			
		72	7339	4			
Poly(propylene) isotactic	200 (5 min)	120	91			DSC measurements to give rate data; interfacial energy $\sigma_e = 92$ erg/cm^2; average rate constant listed; $T_m^0 = 184.3$°C; $\sigma_e = 77$ erg/cm^2; average rate constant listed; $T_m = 178.7$°C	365
$M_w = 947000$; $M_w/M_n = 8.7$		123	161				
Tacticity = 96.5%; $[\eta] = 3.48$ dl/g		125	267				
		128	427				
		130	825				
		132	1508				
$M_w = 129000$; $M_w/M_n = 11$		120	56				

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
Tacticity = 97%; $[\eta] = 0.66$ dl/g		123	100				
		125	165				
		128	256				
		130	599				
		132	1271				
		135	1789				
Poly(propylene) isotactic	200 (5 min)	124	24			Hot stage microscopy;	365
Tacticity[m] = 99.8;		125	30			$T_m^0 = 186.4$°C;	
[mm] = 99.7; [mmm] = 99.4		126	52			$\sigma_e = 125$ erg/cm^2	
$M_v = 44350$; $[\eta] = 53$ dl/g		127	64				
		128	99				
		129	137				
		130	192				
		132	512				
		134	749				
		135	1621				
		136	2041				
Tacticity[m] = 99.77;		125	29			$T_m^0 = 177.0$°C;	365
[mm] = 99.7; [mmm] = 99.4		126	42			$\sigma_e = 69$ erg/cm^2	
$M_v = 16940$; $[\eta] = 26$ dl/g		127	50				
		128	67				365
		130	154				
		132	276				
		133	610				
		135	638				
		138	1729				
Poly(propylene), unfractionated		140	8280	2	3.50×10^{-5}	Calorimetry (DSC): values in	179
$M_v = 4.47 \times 10^5$		137.5	5340		9.52×10^{-5}	column 5 in min^{-n}	
		135.0	2400	2	4.02×10^{-4}		
		132.5	1344	2	1.35×10^{-3}		
		130.0	756	2	5.39×10^{-3}		
		127.5	486	2	1.05×10^{-2}		
		125	378	2	1.80×10^{-2}		
		122.5	288	2	3.16×10^{-2}		
		120	222				
		117	168				
Poly(propylene), unfractionated		95	38	2.96		Values for log k are reported;	200
		100	48	3.05		isotactic index 95–97%;	
		105	66	3.05		$[\eta] = 1.64$ in decalin at 135°C	
		110	78	3.05			
		115	115	3.03			
		117	130	2.96			
		120	181	3.08			
		122	242	3.12			
		125	407	3.03			
		127	561	2.97			
		130	917	3.05			
		132	1529	3.03			
		135	2073	2.98			
Poly(propylene), unfractionated		133	636	5.82×10^{-4}		Dilatometry: k for $n = 3$ (min)	
Profax, $P = 2300$		134	894	2.10×10^{-4}		in column 6	364
		135	1356	6.01×10^{-5}			
		136	2,124	1.56×10^{-5}			
		137	3564	3.31×10^{-6}			
		138	5256	1.03×10^{-6}			
		139	8094	2.82×10^{-7}			
		140	11148	1.08×10^{-7}			
		141	16320	3.45×10^{-8}			
$P = 1200$		135	474	1.43×10^{-3}			394
		140	1524	4.23×10^{-5}			
		145	6870	4.62×10^{-7}			
		150	26400	8.14×10^{-9}			
		155	117000	9.35×10^{-11}			
$P = 7300$		135	642	5.66×10^{-4}			394
		140	2280	1.26×10^{-5}			

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TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		145	11400		1.01×10^{-7}		
		150	32400		4.40×10^{-9}		
		155	152400		4.23×10^{-11}		
Poly(propylenes) isotactic		114	670			Isothermal crystallizations	366
metallocene type		116.5	870			measured by DSC; times @	
$M_w = 678-288$ K		118	1090			20% not 50% conversion are	
		120.5	1500			listed; average curve, digitized	
		123	2445			values	
		126	4315				
		128.8	8430				
		130.7	14520				
		132.5	27130				
Poly(propylene)/poly(ethylene)							
PE: $M_w = 307 \times 10^{-3}$; $M_n = 10.2 \times 10^3$							
$M_w = 307 \times 10^3$; $M_n = 15.6 \times 10^3$							
i-PP/HDPE: 100/0		122	152	3		DSC	6
		123	187	3			
		124	237	3			
		125	305	3			
		127	456	3			
		130	981	3			
		131	1320	3			
		132	1763	3			
90/10		123	138	5		DSC	6
		124	210	4			
		125	309	4			
		126	354	3			
		127	675	3			
		128	668	3			
		130	1140	3			
		132	1863	3			
70/30		123	124	4		DSC	6
		124	215	5			
		125	355	3			
		126	450	3			
		127	675	3			
		128	850	3			
		130	1210	3			
		131	2040	3			
60/40		125	420	3		DSC	6
		126	566	3			
		126	647	3			
		127	763	3			
50/50		122	119	3		DSC	6
		123	163	3			
		124	242	3			
		125	330	3			
40/60		125	558	3		DSC	6
		125.5	847	3			
		126	1285	3			
30/70		122	91	3		DSC	6
		123	110	3			
		124	230	3			
		125	523	3			
		125	614	3			
		126	940	3			
		126	1777	3			
10/90		122	110	3		DSC	6
		123	190	3			
		124	441	3			
		124.5	730	3			
		125	1260	3			
		125.5	1650	3			
		126	2880	3			

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
0/100		122	91	2		DSC	6
		123	120	2			
		124	223	2			
		125	354	2			
		122.5	548	2			
		126	900	2			
		126.5	1415	2			
		127	3750	2			
PP/LDPE: 100/0		127.8	573.1			DSC measurements	277
		125.8	349.0				
		122.8	189.6				
90/10		127.8	704.8			DSC measurements	277
		125.8	401.6				
		122.8	207				
80/20		127.8	848.2			DSC measurements	277
		125.8	439				
		122.8	231.6				
60/40		125.8	788.1			DSC measurements	277
		122.8	219.8				
		127.8	1130.0				
		122.8	355.1				
40/60		127.8	1440.6			DSC measurements	277
		125.8	805.6				
		122.8	340.2				
Poly(propylene)/poly(ethylene) low density; melt flow indices 2.9/6.7 g/min wt.% *i*-PP/LDPE							
100/0	200 (10 min)	125	137			Two roll mill-blended; DSC	367
		130	479			measurements radial spherulitic	
		132	856			growth rates also measured; Avrami	
		135	2001			exponent $1.09 \leq n \leq 3.86$; $\log k_n$	
95/5		125	133			tabulated for various n's	367
		130	415				
		132	594				
		135	1441				
90/10		125	217				367
		130	709				
		132	1123				
		135	2257				
85/15		125	193				367
		130	559				
		132	1254				
		135	1849				
80/20		125	241				367
		130	747				
		138	1212				
		135	2684				
Poly(propylene-*co*-ethylethylene)							
isotactic; 0 wt.% 1-butene		120	130	3.4	2.16	Calorimetry; values in column 6	298
		121	175	3.5	2.29	tabulated here as $-1/3 \log k$	
		122	230	3.3	2.41		
		123	295	3.0	2.52		
		124	380	2.9	2.63		
		125	460	3.0	2.72		
		126	550	3.0	2.79		
		127	750	2.9	2.93		
		128	1010	2.7	3.06		
		129	1300	2.7	3.17		
13.5 wt.% 1-butene		113	175	2.6	1.93		298
		114	190	2.4	1.95		
		115	250	2.4	2.05		
		116	295	2.5	2.11		
		117	430	2.4	2.25		
		118	503	2.3	2.30		
		119	642	2.4	2.39		
		120	750	2.4	2.45		
		121	1070	2.4	2.58		

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TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		122	1190	2.6	2.62		
29.8 wt.% 1-butene		100	355	2.2	2.02		298
		101	535	2.3	2.05		
		102	640	2.2	2.11		
		103	720	2.3	2.15		
		104	960	2.4	2.24		
		105	1050	2.3	2.27		
34.5 wt.% 1-butene		103	270	2.7	2.24		298
		104	330	2.7	2.32		
		105	360	2.7	2.35		
		106	420	2.7	2.41		
		107	500	2.7	2.48		
		108	660	2.7	2.59		
53.3 wt.% 1-butene		76	390	2.3	2.04		298
		77	420	2.3	2.06		
		78	560	2.3	2.16		
		79	670	2.3	2.22		
		80	810	2.3	2.28		
		81	1100	2.4	2.49		
Poly(propylene-*stat*-ethylene)							
ethylene content; 2.7 mol%		100	78			Crystallization rates measured by	368
		105	252			DSC; linear growth rate	
		110	654			measurements also listed	
		115	2460				
		125	7865				
5.7 mol%		100	64				368
		105	133				
		110	369				
		115	1203				
		120	2453				
6.9 mol%		100	49				368
		105	87				
		110	245				
		115	816				
		120	2453				
8.7 mol%		100	44				368
		105	72				
		110	153				
		120	958				
11.0 mol%		100	23				368
		105	35				
		110	69				
		115	216				
		120	750				
Poly(propylene)		120	161	3		Poly(propylene) crystallinity 61%;	368
		121	199	3		this polymer is basic to the	
		122	242	3		blend study immediately below it	
		123	299	3			
		124	366	3			
		125	443	3			
		126	554	3			
		127	688	3			
		128	845	3			
		129	992	3			
Poly(propylene) in poly(propylene) poly[imino(1-oxohexamethylene)] Nylon 6							
		124	60	3		Poly(propylene) crystallinity	368
		125	72	3		72.6%, polyamide 6, 23.1%	
		126	92	3			
		127	117	3			
		128	156	3			
		129	182	3			
Poly(propylene) in poly(propylene)/poly(imino(1-oxohexamethylene)]/nylon 6 poly(propylene) modified with 10% maleic acid							
		122	100	4		Poly(propylene) crystallinity	368
		123	125	4		82.2%	
		124	155	4			

TABLE 10. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		125	196	4		DSC studies	
		126	245	4			
		127	304	4			
		128	357	4			
		129	466	4			
		190	0.016				
Poly (propylene)/poly[imino(1-oxo-hexamethylene)]; PP in PP/PA							
70/30 wt.%; 0% anhydride		124	60	3		Compatabilizing agent: maleic anhydride used at 1–10 wt.% levels. Only DSC rates listed; PP, 0%, used as reference	250
		125	72	3			
		126	92	3			
		127	117	3			
		128	156	3			
		129	182	3			
70/30 wt.%; 10% PP modified with anhydride		122	100	4			250
		123	125	4			
		124	155	4			
		125	196	4			
		126	245	4			
		127	304	4			
		128	357	4			
		129	446	4			
Poly [imino (1-oxo-hexamethylene)]/poly (propylene) PA in PP/PA wt.%: 0% compatabilizer							
		203	170	3		Maleic acid compatabilizer in PP: 0–10 wt.%; reference PA rates in text DSC rate studies, thermal and mechanical properties in text	250
		204	203	3			
		205	276	3			
		206	329	3			
		207	419	3			
		208	636	3			
10% compatabilizer		203	213	3			250
		204	252	3			
		205	324	3			
		206	499	3			

TABLE 11. POLYOXIDES

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
11.1. HOMOPOLYMER MELTS							
Poly(oxyethylene)	100 (10 min)	54.7	432			Dilatometry/crystallinity values cited for $t_{0.1}$ transformation	369
$M_v = 23 \times 10^3$		55.6	780				
$M_w/M_n = 1.2$ (GPC)		57.0	2640				
		58.1	7200				
		59.4					
$M_v = 42 \times 10^3$	100 (10 min)	54.7					369
$M_w/M_n = 1.3$		55.6					
		57.0	960				
		58.1	2940				
		59.4	16200				
$M_v = 84 \times 10^3$							
$M_w/M_n = 1.21$	100 (10 min)	54.7	960				369
		55.6	1440				
		57.0	3600				
		58.1					
		59.4					
$M_v = 130 \times 10^3$	100 (10 min)	54.7					369
$M_w/M_n = 1.35$		55.6					
		57.0	462				
		58.1	1200			Dilatometry/crystallinity values cited for $t_{0.1}$ transformation	369
		59.4	4620				
$M_v = 240 \times 10^3$		54.7					369
$M_w/M_n = 1.22$		55.6					369

TABLE 11. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		57.0	780				
		58.1	1380				
		59.4					
$M_v = 540 \times 10^3$	100 (10 min)	54.7					
$M_w/M_n = 1.53$		55.6					
		57.0	522			Dilatometry/crystallinity	369
		58.1	660				
		59.4					
$M_v = 1.570 \times 10^3$	100 (10 min)	54.7					369
$M_w/M_n = 1.85$		55.6				Dilatometry/crystallinity values	369
		57.0	558			cited for $t_{0.1}$ transformation	
		58.1	960				
		59.4	558				
$M_w = 0.3 \times 10^3$	88 (15 min)	−13.0	1146	2.3	See text	DSC and dilatometry; polymer	370
		−14.0	888	2.5		sources listed in text	
		−15.0	600	1.7			
		−20.0	354				
$M_w = 0.4 \times 10^3$		+0.3	360	2.3			370
		−1.0	270	2.2			
		−1.9	1140	2.2			
		−3.9	720	2.2			
		−5.4	660	2.3			
		−6.4	600				
		−7.1	240				
$M_w = 0.6 \times 10^3$		16.9	3870	2.1			370
		14.9	600	1.2			
		14.1	564	1.2			
		12.9	378				
$M_w = 1.0 \times 10^3$		30.4	2460	1.9			370
		30.1	1170	1.7			
		26.8	600	1.8			
		25.0	444				
$M_w = 1.5 \times 10^3$		45.1	14400	2.1			370
		44.5	7380	2.1			
		43.5	3210	2.2			
		42.4	2280	2.0			
		39.8	1020	1.7			
		30.4	240				
$M_w = 2.0 \times 10^3$		49.3	12120	2.0			370
		48.2	6000	1.8			
		48.0	3540	2.3			
		46.5	2340	2.3			
		45.5	1506	1.7			
		43.1	1380	2.7			
$M_w = 3.0 \times 10^3$		49.4	12540	1.9			370
		48.5	3360	1.7			
		46.9	1560	1.7			
		45.5	990	1.7			
		44.5	660	1.5			
$M_w = 4.0 \times 10^3$		51.0	24480	1.8			370
		49.9	3060	1.7			
		48.5	1440	1.5			
		46.9	1044	1.4			
		44.5	456				
$M_w = 6.0 \times 10^3$		55.7	19860	1.8			370
		52.2	1344	1.7			
		49.9	642	1.5			
		48.5	480	1.4			
		46.9	426	-			
$M_w = 15 \times 10^3$		58.1	15120	2.1			370
		57.4	6720	2.2			
		55.7	2520	1.9			
		53.2	900	1.4			
		52.2	672	1.4			
$M_w = 20 \times 10^3$		59.3	13140	2.2			370

TABLE 11. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		59.0	6900	2.5			
		58.1	4884	2.1			
		57.1	1476	2.0			
		56.0	1200	1.9			
		55.0	612	1.8			
$M_w = 12.6 \times 10^3$ Carbowax 6000	$T_m + 25$ (60 min)					Shear dilatometer; couette, geometry; shear rate (s^{-1})	167
		55.6		2.9	10^{-6}	0	
		56.7		3.2	10^{-6}	0	
		56.7		3.7	10^{-9}	1.4	
		56.7		5.0	10^{-10}	5.4	
		56.8		5.1	10^{-10}	10.9	
		56.5		4.8	10^{-7}	28.0	
		56.3		6.1	10^{-8}	45.7	
		56.9		6.2	10^{-9}	56.1	
		56.6		6.4	10^{-9}	93.4	
		58.1		2.6	10^{-8}	0	
		58.0		3.7	10^{-9}	1.3	
		58.1		4.0	10^{-9}	5.3	
		58.1		5.0	10^{-10}	10.8	
		58.2		4.7	10^{-10}	11.0	
		58.2		5.0	10^{-8}	19.4	
		58.1		6.7	10^{-10}	32.5	
		58.1		6.6	10^{-10}	54.8	
		58.5		6.2	10^{-10}	84.1	
		59.0		2.2	10^{-7}	0	
		59.8		2.9	10^{-10}	0	
		59.5		4.4	10^{-13}	3.0	
		59.4		4.3	10^{-12}	5.3	
		59.0		5.2	10^{-11}	8.6	
		59.0		5.2	10^{-11}	9.3	
		59.4		5.6	10^{-14}	10.8	
		59.6		5.8	10^{-14}	11.0	
		59.0		6.0	10^{-12}	20.2	
		50.0		6.3	10^{-12}	35.7	
		59.1		6.1	10^{-12}	40.0	
		59.6		6.7	10^{-14}	44.6	
		60.0		6.6	10^{-14}	68.9	
$M_w = 72.2 \times 10^3$ WSR-N10 Union Carbide	$T_m + 25$ (60 min)					Shear rate (s^{-1})	167
		58.5		3.2	10^{-6}	0	
		58.0		3.0	10^{-5}	0	
		57.8		3.3	10^{-5}	0	
		58.6		3.9	10^{-8}	5.4	
		58.8		5.2	10^{-9}	7.6	
		59.1		5.3	10^{-8}	17.0	
		59.2		6.7	10^{-10}	25.2	
		57.8		5.5	10^{-7}	37.6	
		59.2		5.8	10^{-10}	36.5	
		59.6		3.2	10^{-7}	0	
		59.8		2.9	10^{-6}	0	
		59.5		3.0	10^{-7}	0	
		59.6		3.9	10^{-8}	1.4	
		59.6		3.3	10^{-7}	3.0	
		59.5		3.5	10^{-8}	5.4	
		59.2		4.5	10^{-9}	8.4	
		59.6		4.4	10^{-8}	9.6	
		59.6		3.5	10^{-8}	10.8	
		59.6		5.9	10^{-12}	17.1	
		59.9		5.0	10^{-8}	19.8	
		59.6		4.1	10^{-8}	27.1	
		59.8		3.9	10^{-7}	38.9	
		61.1		2.9	10^{-8}	0	
		61.0		3.1	10^{-8}	0	
		61.0		4.1	10^{-9}	1.4	
		60.9		4.8	10^{-10}	3.1	

TABLE 11. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		61.0		4.9	10^{-11}	5.4	
		61.0		5.0	10^{-10}	9.1	
		61.8		5.1	10^{-11}	9.2	
		61.3		5.1	10^{-9}	28.5	
$M_w = 3.5 \times 10^3$	69 (15 min)		39.3	2.7	0.11	Slightly degraded after the	60(a)
$M_w/M_n < 1.2$			40.4	2.9	4.5×10^{-1}	determinations; DSC	
			41.7	2.6	1.6×10^{-2}	measurements	
			43.5	2.5	3.3×10^{-3}		
			44.4	2.6	3.0×10^{-4}		
			45.3	2.6	8.0×10^{-5}		
			46.4	2.9	2.3×10^{-6}		
$M_w = 4.5 \times 10^{-3}$			35.0	3.0	1.99		60(a)
$M_w/M_n < 1.04$			37.0	3.1	0.93		
			39.0	3.1	0.34		
			40.4	3.1	0.19		
			42.7	3.0	2.6×10^2		
			43.4	2.8	4.2×10^3		
			45.0	2.9	2.5×10^4		
$M_w = 7.0 \times 10^3$			39.8	2.4	2.1		60(a)
$M_w/M_n < 1.2$			41.2	2.4	1.2		
			43.4	2.1	0.23		
			44.9	2.1	7.7×10^{-2}		
			47.0	2.4	8.2×10^{-3}		
			49.4	2.4	3.8×10^{-4}		
$M_w = 9.0 \times 10^3$			40.1	1.9	1.85	DSC	60(a)
$M_w/M_n = 1.07$			43.4	2.0	0.68		
			48.4	2.3	2.6×10^{-2}		
			49.6	2.1	8.3×10^{-4}		
$M_w = 20.0 \times 10^3$			44.4	2.5	0.49		60(a)
$M_w/M_n = 1.3$			46.3	2.7	0.13		
			48.4	2.4	2.2×10^{-2}		
			51.7	2.4	7.4×10^{-4}		
			53.2	2.1	3.1×10^{-6}		
$M_w = 100.0 \times 10^3$			39.8	2.2	6.7		60(a)
M_w/M_n is unknown			43.4	2.2	2.7		
			47.3	2.3	0.58		
			49.4	2.3	0.15		
			51.5	2.1	1.6×10^{-2}		
			53.9	2.1	1.0×10^3		
			39.5	2.47	8.23	DSC used for bulk rate constants;	
			40.0	2.50	2.96	two stage Avrami rates cited;	
			45.6	3.00	0.90	Stage I	
			46.8	2.68	0.50		
			47.9	2.87	0.20		
			48.9	3.08	3.65×10^{-2}		
			50.0	3.04	3.71×10^{-3}		
			51.3	2.85	1.95×10^{-4}		
			51.9	3.46	1.14×10^{-6}		
			54.4	3.33	4.38×10^{-9}		
			55.2	3.22	2.40×10^{-11}		
			45.6	0.95	1.16	Stage II	
			46.8	1.12	0.82		
			47.9	1.06	0.56		
			48.9	1.02	0.30		
			50.0	0.95	0.14		
			51.3	0.98	4.21×10^{-2}		
			51.9	1.02	1.04×10^{-2}		
			54.4	0.86	3.88×10^{-3}		
			55.2	0.79	1.06×10^{-3}		
$M_v = 23 \times 10^3$	100 (10 min)	54.7	432			Dilatometry; crystallinity values	60(a)
$M_w/M_n = 1.2$ (GPC)		5.6	780			cited as $t_{0.1}$(s), not $t_{0.5}$	
		57.0	2640				
		58.1	7200				
		59.4					
$M_v = 42 \times 10^3$	100 (10 min)	54.7				Dilatometry; crystallinity values	60(a)
$M_w/M_n = 1.3$		55.6				cited as $t_{0.1}$(s), not $t_{0.5}$	

TABLE 11. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		57.0	960				
		58.1	2940				
		59.4	16200				
$M_v = 84 \times 10^3$	100 (10 min)	54.7	960			Dilatometry; crystallinity values cited as $t_{0.1}$(s), not $t_{0.5}$	160(a)
$M_w/M_n = 1.21$		55.6	1440			Dilatometry; crystallinity values cited as $t_{0.1}$(s), not $t_{0.5}$	160(a)
		57.0	3600				
		58.1					
		59.4					
$M_v = 130 \times 10^3$	100 (10 min)	54.7				Dilatometry; crystallinity values cited as $t_{0.1}$(s), not $t_{0.5}$	160(a)
$M_w/M_n = 1.35$		55.6					
		57.0	462				
		58.1	1200				
		59.4	4620				
$M_v = 240 \times 10^3$		54.7				Dilatometry; crystallinity values cited as $t_{0.1}$(s), not $t_{0.5}$	160(a)
$M_w/M_n = 1.22$		55.6				Dilatometry; crystallinity values cited as $t_{0.1}$(s), not $t_{0.5}$	160(a)
		57.0	780				
		58.1	1380				
		59.4					
$M_v = 540 \times 10^3$	100 (10 min)	54.7				Dilatometry; crystallinity values cited as $t_{0.1}$(s), not $t_{0.5}$	160(a)
$M_w/M_n = 1.53$		55.6					
		57.0	522				
		58.1	660				
		59.4					
$M_v = 1570 \times 10^3$	100 (10 min)	54.7					
$M_w/M_n = 1.85$		55.6				Dilatometry; crystallinity values cited as $t_{0.1}$(s), not $t_{0.5}$	160(a)
		57.0	558				
		58.1	960				
		59.4	558				
$M_v = 20 \times 10^3$	88 (15 min)	59.3	12960	2.21		DSC; rates are also given in text after quenching; some dilatometry results also available; k_n values are also cited	343
		59.0	6900	2.5			
		58.1	4884	2.1			
		57.1	1476	2.0			
		56.0	1206	1.			
		55.0	612	1.8			
		54.0	540				
		52.0	492				
		48.2	300				
$M_n = 1.6 \times 10^3$	100 (15 min)	43.3	600			Dilatometry; selected values only listed	295
$M_w/M_n = 1.05$		44.8	3120				
		45.9	73200				
$M_n = 4.1 \times 10^3$		49.2	2760				295
$M_w/M_n = 1.05$		50.9	9600				
		53.4	277200				
$M_n = 20 \times 10^3$		54.8	1140				295
$M_w/M_n = 1.2$		56.3	2640				
$[\eta] = 0.374$		57.0	8100				
		57.5	17940				
		58.2	33360				
$M_n = 200 \times 10^3$		55.9	840				295
$M_w/M_n = 1.2$		57.8	3360				
$[\eta] = 1.92$		58.2	3492				
		58.9	12180				
		59.7	27840				
$M_n = 600 \times 10^3$		55.0	1140				295
$M_w/M_n = 1.2$		57.2	5760				
$[\eta] = 3.65$		58.2	6540				
		58.9	18900				
		59.4	25680				
		60.2	91200				
Poly(oxyethylene)							
0% silica	120 (45 min)	53.7	2334			Dilatometry; Me_3SiCl-treated silica	11
		54.7	3873				
		55.8	7381				

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TABLE 11. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		56.8	15071				
		57.8	32221				
2% silica		53.7	1941				11
		54.7	3533				
		55.8	6888				
		56.8	14065				
		57.8	31488				
5% silica		53.7	1770				11
		54.7	3297				
		55.8	6578				
		56.8	13432				
10% silica		53.6	1282				11
		54.7	2128				
		55.7	3873				
		56.8	8093				
		57.8	17707				
Poly(oxymethylene)							
$M_w = 66 \times 10^3$		140.0	36			DSC measurements of rate, average	340
$M_w = 41 \times 10^3$		142.5	48			$t_{0.5}$ values interpolated from	
$M_w = 35 \times 10^3$		145.0	72			mean line of data of 3 samples;	
		147.5	90			lamellar thickness and	
		150.0	185			spherulitic growth rates listed	
		152.5	420			in text	
		155.0	1800				
		157.5	6000				
		160.0	24900				
Poly(oxymethylene–oxyethylene)							
$M_v = 9.4 \times 10^3$	80 (4 h)	19.5	4.0	2.8		Rate constants also listed in text	297
		22	5.7	2.8			
		25	10.5	4.1			
		28	22	4.1			
Unfractioned		35	1400	3.0			
$M_v = 6.7 \times 10^3$		19.5	240	2.8		Dilatometry; two crystal	372
		22	342	2.8		modifications	
		25	630	4.1			
		28	13200	4.1			
		35	84000	3.0			
$M_v = 10 \times 10^3$		0	26	2.07	10	DSC	372
		2	28	2.93	8.8		
		4	25	2.19	11		
		6	28	2.65	8.8		
		8	26	2.34	10		
		10	34	2.40	6.0		
		12	75	2.00	1.2		
		14	126	2.00	0.43		
		15	135	1.90	0.38		
$M_v = 30 \times 10^3$		0	16	2.03	27	DSC	372
		2	16	2.05	27		
		4	14	1.90	35		
		8	31	2.26	7.2		
		10	25	1.90	11		
		14	39	1.90	4.5		
		16	34	2.12	6.0		
		19	101	1.95	0.67		
		21	290	2.00	0.085		
Poly(oxypropylene)	125 (10 min)	49.7	12600	3.1		Hot stage optical microscopy	343
$M_v = 300 \times 10^3$		48.5	10680	3.0			
		47.5	8580	3.0			
		46.7	6600	3.6			
		45.5	3540	3.3			
		42.8	2520	3.1			
		40.3	1650	3.0			
		38.7	1200	2.7			
$M_v = 300 \times 10^3$	125 (10 min)	50.0	15660	2.1		DSC; rates are also given in text	343
		47.5	8640	2.1		after quenching; some diletometry	
		45.0	4500	2.2		results also available	

TABLE 11. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		42.5	2580	2.0			
		40.0	1620	2.0			
		38.5	1200	2.1			
		35.0	2.0				
After quenching		−53.5	1920	2.2			
		−52.0	990	2.3			
		−50.0	480	2.1			
Poly(oxytetramethylene)							
$M_n = 13.4 \times 10^3$	260	208.8		2.6	8.8×10^{-8}	DSC; polymerization method;	373
		209.8			2.8×10^{-8}	catalyst and T_m(optical)	
		211.1			7.4×10^{-9}	given in text	
		212.0			3.1×10^{-9}		
		212.7			1.5×10^{-9}		
		212.9			9.2×10^{-10}		
		213.5			5.6×10^{-10}		
Poly(oxytrimethylene)							
$M_n = 79 \times 10^3$	80 (15 min)	−50.2	579.7			DSC	290
		−44.4	333.5				
		−39.3	242.4				
		−36.7	224.6				
		−33.8	200.6				
		−28.7	189.4				
		−26.5	201.8				
		−23.6	208.0				
		−18.9	276.1				
		−14.5	369.9				
		−10.2	658.6				
		−6.9	1195.2				
		−4.0	2371.5				
		−0.4	8000.0				
11.2. BLENDS							
Poly(oxyethylene)/Poly([1-(methoxycarbonyl)-1-methylethylene]						PEO10/PMMA (isotactic)	
100/0		46.6	49.1				219
		49.3	114.1				
		50.1	213.7				
		51.0	417.8				
		52.0	766.3				
		53.0	2631				
		53.9	3076				
90/10		46.4	85.9				219
		47.3	125.9				
		48.1	206.5				
		49.2	317.0				
		50.1	382.9				
		51.0	1671				
		52.0	3681				
		53.0	4615				
80/20		43.2	202.0				219
		44.2	251.8				
		45.2	798.9				
		46.2	967.7				
		47.1	2047				
		48.1	3681				
		49.1	4615				
		50.1	6185				
60/40		39.2	346.8				219
		40.1	417.8				
		41.1	540.5				
		42.1	656.5				
		43.1	1312				
		44.2	1312				
		45.2	2298				
		46.1	3076				
		47.2	9230				

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TABLE 11. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		48.1	3681				
		49.0	6185				
Poly(oxyethylene)/poly(octafluoropentoxytrifluoroethoxyphosphazene)/poly[oxy(1-(chloromethyl)ethylene]							
PEO, $M_w = 5 \times 10^6$; PPZ, Unlisted M_w; PECH, $M_w = 7 \times 10^5$							
PEO/PPZ: 100/0	100 (5 min)	52		2.1	2.51×10^{-2}	Measurements made by DSC and optical hot stage microscopy; T_m values and morphology documented units of k are unspecified in text-assumed to be in seconds	309
		53		2.0	5.25×10^{-2}		
		54		2.1	2.34×10^{-2}		
PEO/PPZ: 75/25		52		1.6	2.09×10^{-2}		
		53		1.7	0.93×10^{-2}		
		54		1.7	0.30×10^{-2}		
PEO/PPZ: 50/50		52		1.5	1.07×10^{-2}		309
		53		1.6	0.58×10^{-2}		
		54		1.7	0.27×10^{-2}		
PEO/PECH: 75/25		52		2.0	0.30×10^{-3}		309
		53		2.3	0.15×10^{-3}		
		54		2.1	0.85×10^{-3}		
PEO/PECH: 50/50		50		1.8	0.15×10^{-3}		309
		51		1.9	0.85×10^{-3}		
		52		2.1	0.43×10^{-3}		
PEO/PPZ/PECH: 66/17/17		52		2.0	1.66×10^{-2}		309
		53		2.1	0.50×10^{-2}		
		54		1.9	0.28×10^{-2}		
PEO/PPZ/PECH: 50/25/25		50		1.9	2.29×10^{-2}		309
		51		2.0	0.72×10^{-2}		
		52		2.1	0.17×10^{-2}		

TABLE 12. POLY(CARBONATES)

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)							
Bistan AF		35	598.5	3		DSC; solvent induced crystallization in CCl_4 and acetone; crystallinity cited; film thickness; 20000 & 10000 nm	374
$M_v = 62.5 \times 10^3$		40	268.9	3			
		25	2308.5	3			
		40	2964.1	3			
Poly(oxydecamethyleneoxysebacoyl)	135 (15 min)	73.7	35400	2.7		k_n values listed in text	343
$M_n = 14.4 \times 10^3$		73.2	16560	2.6			
$M_v = 28.6 \times 10^3$		72.8	13140	2.6			
		72.0	6780	2.6			
		71.8	6120	2.9			
		70.8	2100	2.2			
		69.2	900	1.6			
		68.2	678	1.6			
		67.2	444	1.5			
		66.6	396				
		65.8	324				
Poly(oxyethyleneoxyadipoyl)	125 (15 min)	39.0	28800	2.4		k_n values listed in text DSC; rates are also given in text after quenching some dilatometry; results also available	343
$M_w = 20 \times 10^3$		37.5	10800	2.5			
		36.8	8580	2.4			
		36.1	7800	2.4			
		35.6	6000	2.4			
		34.2	2880	2.7			
		33.2	1020	2.4			
		32.1	2052	2.7			
		29.9	1090	2.4			
		27.8	756	2.4			
		25.7	438	1.9			
		23.5	336				
		21.6	252				
		18.5	120				

Table 12. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
Poly(oxyethyleneoxysebacoyl)	135 (15 min)	71.4	10440	2.8		DSC rates are also given in text	343
$M_n = 14.4 \times 10^3$		70.9	7020	2.5		after quenching; some dilatometry	
		70.1	3960	2.6		resultsavailable; k_n values are	
		69.6	3540	2.6		also cited	
0.5% soln. in tricresol		68.6	2220	2.5			
of 0.74 dl/g		67.6	1440	2.5			
		65.7	648	2.2			
		64.7	510	2.0			
		63.5	408	2.0			
		62.5	294	1.9			

TABLE 13. POLY(ESTERS)

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
13.1. HOMOPOLYMER MELTS							
Poly (ethylene-2,6-naphthalate)	280 (10 min)	230	110			DSC measurements. Column 4	412
PEN (5); intrinsic viscosity		235	170			contains T_m (time to attain	
~0.6 dl/g		240	358			max.heat flow and crystallinity,	
		250	1104			X_C); proportional to $t_{0.5}$	
	320	230	107			Range of crystalline order	
		235	254			$0.4 < X_C < 0.45$. Non-isothermal	
		240	395			data are listed in text ; spherulitic	
		250	1042			morphology is discussed	
PEN (6); intrinsic viscosity	280	230	147				
~0.7 dl/g		235	223				
		240	374				
		250	799				
	320	230	165				
		235	338				
		240	527				
		250	1232				
Poly(oxyethyleneoxyterephthaloyl)							
$M = 6.65 \times 10^3$		227	72	2.6		DSC; k (min^{-n}) listed in text;	375
		230	111.6	3.1		crystallinities also cited	
		235	349.2	3.4			
$M = 11.75 \times 10^3$		227	73.2	2.6			375
		230	133.2	3.4			
		232	177.0	3.4			
		235	297.0	3.9			
$M = 8.10 \times 10^3$		227	46.2	2.4			375
		230	88.2	3.4			
		232	126.0	3.9			
		235	244.8	4.0			
$M = 7.10 \times 10^3$		227	58.2	2.6			
		230	112.2	3.1			
		232	163.8	3.8			
		235	328.2	4.0			
$M = 5.35 \times 10^3$		225	81.0	2.0			375
		227	138.6	3.3			
		232	441.0	3.5			
		235					
$M_w = 81.7 \times 10^3$	270 (30 min)	225	906			0.0288 mol% Sb_2O_3	376
		230	1812				
		235	4536				
		240	9816				
$M_w = 69.7 \times 10^3$	270 (30 min)	225	486			0.0288 mol% Sb_2O_3	376
		230	1392				
		235	3348				
		240	8418				

TABLE 13. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
$M_w = 49.0 \times 10^3$	270 (30 min)	235	1398			0.0288 mol% $Ca(OOCCH_3)_2$	376
		240	3498				
		242	4158				
$M_w = 30.6 \times 10^3$	270 (30 min)	235	978				376
		240	2340				
		242	3408				
oriented fibers	RT quench	95	17190			Fibers, $f_a = 0.032$[a]; crystallinity and density listed	377
		100	7020				
		105	4800				
		110	900				
		115	600				
		95	14400			$f_a = 0.044$; crystallinity and density listed	
		100	3540				
		105	840				
		110	660				
		115	300				
		95	1920			$f_a = 0.123$; crystallinity and density listed	
		100	480				
		105	360				
		110	180				
DuPont, Type A Mylar	277	95	1380			DSC; crosslinked with 1 mrad γ-radiation	378
$M_w = 35.5 \times 10^3$		100	720				
$[\eta] = 0.580$		105	240				
		110	150				
		230	360				
		235	780				
		240	210				
		95	2400			2 mrad γ-radiation	
		100	1200				
		105	480				
		110	240				
		230	600				
		235	1260				
		240	5400				
Intrinsic viscosity		95	1080			3 mrad γ-radiation	
$[\eta] = 0.573$		100	450				
		105	234				
		110	132				
		230	240				
		235	600				
		240	1620				
$M_n = 12.2 \times 10^3$	295.0	228.3		2.0	3.3×10^{-6}	DSC; polymerization method, catalyst and T_m (optical) given in text	373
		230.2			1.6×10^{-6}		
		232.6			5.1×10^{-7}		
		233.7			5.4×10^{-7}		
		234.6			3.3×10^{-7}		
		236.8			1.2×10^{-7}		
		237.2			1.5×10^{-7}		
		239.8			4.8×10^{-8}		
Poly(oxyethylene oxyterephthaloyl)		107	120			Crystallized from glassy state; average $t_{0.5}$ values, interpolated	379
(Various data compiled by author)		108	100				
		110	80				
		112	564				
		116	371				
		125	139				
		205	59				
		216	173			Nucleated with low mol. wt. poly(propylene)	
		227	393				
		232	549				
		236	734				
		237	937				
		113	1110			Nonnucleated polymer	
		114	862				
		117	655				
		124	342				

TABLE 13. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		133	175				
		202	72			Nucleated with poly(propylene)	
		215	213				
		226	444				
		231	607				
		233	810				
		235	1044				
		172	77			Nucleated with LLDPE	
		188	83				
		203	144				
		215	258				
		220	390				
		224	566				
		227	706				
		229	868				
		231	994				
		232	1134				
		215	362			Nonnucleated polymer; these	
		219	538			constitute $t_{0.5}$(s) averages	
		223	700			compiled from the literature	
		224	809				
		226	953				
		227	1080				
		228	1259				
Poly(oxyhexamethylene-terephthaloyl) PET	285	26.6	222		10	Avrami equation does not fit data;	
		26.9	243		50	Malkin equation was used (see Ref.);	
		27.6	439		100	crystallization temp. expressed as	
		25.5	444		125	$T_m - T_c$ (°C); Pressure MPa;	
		25.1	720		150	$T_m = 264.2 - 0.5058P$;	
						$dT_m/dP = 0.506$/MPa;	
						$T_c = 234.6 - 0.5330$P; @10MPa;	
						$dT/dt = 2.5$	
Poly(oxyhexamethyleneoxyterephthaloyl)							
$M_n = 11.8 \times 10^3$	186	131.5		2.9	1.2×10^{-8}		373
		132.7			6.1×10^{-9}		
		134.0			1.4×10^{-9}		
		135.2			5.3×10^{-10}		
		136.6			1.6×10^{-10}		
		138.0			3.1×10^{-11}		
Poly(oxy-1-oxo-2,2-dimethylethylene)							
$M_n = 71 \times 10^3$	250 (5 min)	186.0	84	2.7		Dilatometry, DSC thermograms	380
$M_w = 163 \times 10^3$		188.5	162	2.5		and $t_{0.1}$ values also given; higher	
		191.0	192	2.4		fractions up to $M_n = 241 \times 10^3$	
		194.0	492	2.3		are measured	
		197.0	582	2.3			
Poly(oxy-1-oxyethylene)							
$M_w = 20\text{–}145 \times 10^3$		201	27.8		4 (approx.)	DSC	371
		203	38.4				
		205	52.2				
		207	66.2				
Poly(oxy-1-oxohexamethylene)						Shear dilatometer, shear rate(s^{-1})	
$M_w = 9.4 \times 10^3$	$T_m + 24$ (60 min)	48.1		3.0	10^{-10}	0	167
		48.0		3.7	10^{-10}	1.4	
		48.0		3.9	10^{-9}	2.9	
		48.3		3.7	10^{-9}	5.2	
		48.1		3.7	10^{-7}	10.9	
		48.1		4.0	10^{-8}	14.6	
		48.6		5.0	10^{-10}	16.0	
		48.4		5.5	10^{-11}	26.9	
		49.8		2.9	10^{-9}	0	
		49.8		2.9	10^{-8}	1.3	
		49.8		3.1	10^{-8}	2.8	
		49.8		4.5	10^{-11}	5.1	
		49.8		4.2	10^{-10}	8.9	
		49.8		4.3	10^{-9}	10.8	

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TABLE 13. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		49.5		4.7	10^{-11}	11.2	
		49.8		4.9	10^{-11}	14.9	
		49.8		5.1	10^{-11}	26.6	
		49.9		5.0	10^{-11}	48.0	
		50.9		2.9	10^{-8}	3.0	
		50.9		3.2	10^{-9}	5.5	
		50.9		4.0	10^{-10}	11.0	
		50.8		4.1	10^{10}	14.3	
		50.8		4.8	10^{-12}	16.0	
		50.8		4.8	10^{-11}	24.9	
		50.9		5.0	10^{-11}	47.2	
		50.9		4.1	10^{-10}	15.3	
						Crystallinity with time of transformation cited	
Poly(oxypentamethyleneoxyterephthaloyl)							
$M_n = 13.4 \times 10^3$	260*	210.1		2.8	7.1×10^{-9}	Dilatometry	343
		211.7			2.1×10^{-9}		
		212.1			1.5×10^{-9}		
		213.6			3.7×10^{-10}		
		214.3			2.6×10^{-10}		
		214.7			1.5×10^{-10}		
		214.9			8.9×10^{-11}		
Poly(decamethylene-4,4′-diphenoxyterephthaloyl) PDDPT							
	265*	234	30	3		DSC studies; transformation after conditioning @265°C from the nematic state	247
		235	48	3			
		236	60	3			
		238	108	3			
		239	144	3			
	240	246	2				
		241	324	2			
	280*	237	72	3		Transformation from the nematic state after conditioning @280°C	
		238	90	3			
		239	156	3			
		240	264	2			
		241	486	2			
	300*	237	84	3		Transformation from the isotropic state after conditioning @300°C	
		238	132	2			
		239	258	2			
		240	522	2			
Poly(hexamethylene-4,4′-diphenoxyterephthaloyl) PHDPT							
	190*	171.5	57	3		DSC rate; transformation from nematic state after conditioning @190°C	248
		172	78	3			
		172.5	102	3			
		173	144	3			
	173.5						
	200*	171	54	3		Transformation from nematic state after conditioning @200°C	
		171.5	54	3			
		172.5	78	3			
		173	102	3			
		173.5	144	3			
		174	192	3			
		174.5	264	3			
	240*	170	42	2		Transformation from the isotopic state after conditioning @240°C	
		171	42	2			
		171.5	48	2			
		173	78	2			
		173.5	90	2			
		174	114	2			
		174.5	174	2			
		175	288	2			

* Denotes mesophase conditioning temperature; not T_f.

TABLE 13. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
Poly(hexamethylene-4,4′-diphenoxyterephthaloyl PHXDPT							
	280*	262	60	2		DSC studies transformation from the nematic state after conditioning @280°C	248
		262.5	60	2			
		263	78	2			
		263.5	90	2			
		264	102	2			
		264.5	120	2			
	285*	262	30	2		Transformation from nematic state after conditioning @285°C	
		262.5	36	2			
		263	48	2			
		263.5	66	2			
		264.5	90	2			
		265	96	2			
		265.5	138	2			
13.2. BLENDS AND COPOLYMERS							
Poly(oxy-1-oxohexamethylene)/poly(1-chloroethylene)							
PCL/PVC: 70/30	100 (5 min)	5	216	3	4.36×10^{-2}		318
		15	372		1.41×10^{-2}		
		25	1200		1.95×10^{-3}		
65/35		5	600	3	1.99×10^{-3}		
		15	900		7.60×10^{-4}		
		25	2220		7.57×10^{-5}		
60/40		5	2760	3	4.57×10^{-6}		
		15	2100		8.50×10^{-6}		
		25	3480		2.62×10^{-6}		
50/50		5	37800	3	2.45×10^{-9}		
		15	28800		3.98×10^{-3}		
		25	35700		3.16×10^{-9}		
Poly(oxyethyleneoxyterephthaloyl)/poly(ethylene-1,2-diphenoxyethane-1,2-dicarboxylate)							
PET/PC: 0/100, w/w		80	0.164			DSC used; rates expressed in s^{-1}; $T_m = 288$°C; $T_g = 72$°C	
		90	0.159				
		100	0.154				
		120	0.123				
		130	0.109				
		140	0.089				
		150	0.078				
		170	0.064				
33/67, w/w		80	0.105			$T_m = 284$°C; $T_g = 71$°C	
		90	0.109				
		100	0.108				
		110	0.095				
		120	0.083				
		130	0.064				
		150	0.045				
		170	0.004				
50/50, w/w		80	0.078			$T_m = 274$°C; $T_g = 71$°C	
		90	0.081				
		95	0.083				
		100	0.083				
		110	0.077				
		120	0.074				
		130	0.062				
		140	0.06				
		150	0.055				
		170	0.051				
67/33, w/w		80	0.07			$T_m = 276$°C; $T_g = 70$°C	
		90	0.072				
		100	0.077				
		105	0.076				
		110	0.072				
		120	0.061				

* Denotes mesophase conditioning temperature, not T_f.

TABLE 13. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
		130	0.049				
		140	0.045				
		150	0.04				
0/100, w/w		100	0.018			$T_m = 280$°C	
		110	0.02				
		120	0.027				
		130	0.031				
		140	0.03				
		160	0.02				
		170	0.018				
Poly(oxyethyleneoxyterephthaloyl)/poly(oxybutyleneoxyterephthaloyl)							
PET/PBT: 100/0, w/w	280 (5 min)		119	2920		DSC measurement; t_c values are listed in text; relation to k (s^{-n}) is not specified	382
		129	940				
		139	423				
		150	234				
		159	183				
		169	169				
		195	247				
		205	334				
		216	693				
		224	1256				
		229	2098				
90/10, w/w		100	2017				382
		105	615				
		110	500				
		195	133				
		221	115				
		120	157				
		125	131				
		130	107				
		135	83				
		210	282				
		216	369				
		223	1016				
		225	1290				
		230	3105				
75/25, w/w		75	2000				382
		80	494				
		85	485				
		90	327				
		94	176				
		100	161				
		105	110				
		110	107				
		115	88				
		120	54				
		125	51				
		185	119				
		206	232				
		210	337				
		215	515				
		217	587				
		218	602				
		222	805				
		227	1888				
Poly(oxyethyleneoxyterephthaloyl)/Poly(oxycarbonyloxy-1,4-phenylene-isopropylidene-1,4-phenylene)							
PET/PC	285 (5 min)	210			13.10×10^3	DSC measurements; Avrami n variable; optical micrographs included on text	395
$M_n = 16000$		215			9.76×10^3		
$M_w = 57000$		220			5.74×10^3		
$M_n = 24000$		225			3.42×10^3		
100/0, w/w		230			2.14×10^3		
Additive modified		220			22.26×10^3	With antioxidant Itganox, 2 ppm	
		225			11.82×10^3		
		230			6.68×10^3		
		235			3.42×10^3		

TABLE 13. *cont'd*

Polymer	T_f (°C)	T_c (°C)	$t_{0.5}$ (s)	n	k (s^{-n})	Remarks	Refs.
80/20, w/w		220			20.62×10^3		395
		225			10.92×10^3		
		230			6.25×10^3		
		235			2.90×10^3		
60/40, w/w		220			24.46×10^3		395
		225			13.77×10^3		
		230			8.13×10^3		
		235			3.50×10^3		
40/60, w/w		215			23.96×10^3		395
		220			14.90×10^3		
		225			8.83×10^3		
		230			4.88×10^3		
60/40, w/w	280 (1 min)	230			18.90×10^3	Effect of fusion on rate	395
	(5 min)				17.05×10^3		
	(10 min)				13.94×10^3		
	(15 min)				12.22×10^3		
	(20 min)				8.77×10^3		
	(30 min)				6.65×10^3		
	300 (1 min)	230			7.26×10^3	Effect of fusion on rate	395
	(5 min)				5.50×10^3		
	(10 min)				2.46×10^3		
	(20 min)				1.23×10^3		
	(30 min)				0.08×10^3		
Poly(butylene naphthalate terephthalate)-*co*-poly(hydroxybutylnaphthalate)							
BHBT/PHBN: 100/0, w/w	$T_m + 40$ (6 min)	161	65	2.5	2.91×10^{-5}	DSC measurements	
62.9/37.1, w/w		164	91	2.7	7.52×10^{-6}	DSC measurements	
		167	133	2.9	1.49×10^{-6}		
		170	204	3.0	3.12×10^{-8}		
100/0, w/w		161	85	2.5	9.30×10^{-4}	DSC measurements	
55.5/44.5, w/w		164	27	2.8	1.41×10^{-4}	DSC measurements	
		167	201	2.8	6.23×10^{-7}		
		170	336	2.9	1.08×10^{-7}		
100/0, w/w		161	97	2.7	6.19×10^{-6}	DSC measurements	
45.6/54.5, w/w		164	151	2.9	9.61×10^{-7}	DSC measurements	
		167	250	2.9	2.33×10^{-7}		
		170	471	3.0	2.10×10^{-8}		
Poly(butylene naphthalate terephthalate)-*co*-poly(hydroxybutylnaphthalate)							
BHBT/PHBN: 100/0, w/w	$T_m + 40$ (6 min)	161	65	2.5	2.91×10^{-5}	DSC measurements	404
62.9/37.1, w/w		164	91	2.7	7.52×10^{-6}	DSC measurements	404
		167	133	2.9	1.49×10^{-6}		
		170	204	3.0	3.12×10^{-8}		
100/0, w/w		161	85	2.5	9.30×10^{-4}	DSC measurements	404
55.5/44.5, w/w		164	27	2.8	1.41×10^{-4}	DSC measurements	404
		167	201	2.8	6.23×10^{-7}		
		170	336	2.9	1.08×10^{-7}		
100/0, w/w		161	97	2.7	6.19×10^{-6}	DSC measurements	404
45.6/54.5, w/w		164	151	2.9	9.61×10^{-7}	DSC measurements	404
		167	250	2.9	2.33×10^{-7}		
		170	471	3.0	2.10×10^{-8}		

TABLE 14. POLY(AMIDES)

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly(imino(1-oxo-hexamethylene)) Nylon 6		205	1080			Dilatometry	289
$M_n = 14.6 \times 10^3$		207	2040				
		209	3600				
		210	4980				
		212	6000				
		215	12600				
Poly[imino(1-oxohexamethylene)] Nylon 6		48	1848	~2		DSC rate measurements from the glassy state; rate constants listed for non-integer n values	401
		49	1608				
		49.5	1230				

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TABLE 14. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
		51.5	750				
		52.5	493				
		57.6	93				
		60.2	45				
		62.9	21.6				
Poly[imino(1-oxohexamethylene)]		241		3	1.48×10^{-9}	Rate constants in (min^{-n})	323
$M_n = 11.6 \times 10^3$		247		3	1.71×10^{-10}		
		250		3	1.39×10^{-11}		
		252		4	6.71×10^{-16}		
		243		3	3.06×10^{-9}		
		247		3	7.40×10^{-10}		
		250		3	7.40×10^{-11}		
		252		3	3.52×10^{-11}		
		253		4	5.71×10^{-16}		
		247		3	1.53×10^{-9}		
		250		3	6.50×10^{-10}		
		252		3	6.00×10^{-11}		
		253		3			
		254		4	5.48×10^{-16}		
		247		3	1.90×10^{-9}		
		250		3	5.10×10^{-10}		
		252		3	3.75×10^{-10}		
		255		3			
		257		4	2.70×10^{-14}		
$M_n = 17 \times 10^3$		178.0	98	2.3	Values cited in	DSC; induction times cited in text	383
		182.1	122	2.1	text in (min^{-n})		
		186.1	160	2.0			
		189.2	262	2.0			
		193.2	400	19			
		195.3	560	1.8			
		199.3	960	2.05			
		203.3	3000	1.9			
		210.4	13100	2.35			
		176.0	24	4.5		With 0.1% kaolin	
		178.0	30	4.7			
		182.1	42	4.7			
		184.1	53	4.6			
		186.1	60	4.8			
		189.2	120	4.4			
		191.2	186	3.6			
		193.2	304	3.3			
		195.3	325	3.4			
		197.3	600	3.0			
		199.3	920	3.1			
		203.3	2100	2.7			
		184.1	32	3.2		With 0.01–0.5% talc	
		186.1	36	3.25			
		191.2	76	4.15			
		195.3	120	4.4			
		199.3	192	4.6			
		201.3	340	5.4			
		203.3	475	5.4			
		205.3	760	5.6			
		207.3	1230	6.1			
Poly[imino(1-oxohexamethylene)] Nylon 6		204	107	3		This polyamide is used in the blend below; see text	250
		205	166	3			
		207	255	3			
		208	271	3			
		209	398	3			
Poly[imino(1-oxohexamethylene)] Nylon 6		203	170	3			250
		204	203	3			
		205	276	3			
		206	329	3			
		207	419	3			
		208	636	3			

TABLE 14. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly[imino(1-oxohexamethylene)] Nylon 6		29.5	77.2			DSC measurements of $t_{0.5}$ and k;	384
linear	240 (10 min)	33.4	665			morphological study reported	
$M_n = 16000$		35.6	560				
$M_w = 29000$		38.5	432				
DP = 270		41.2	605				
		45.0	280				
3-arm branched "dendrimers"	240 (10 min)	26.5	384				
		30.6	605				
$M_n = 19000$		32.3	493				384
$M_w = 36000$		35	385				
DP = 300		41	278				
6-arm branched "dendrimers"		23	868				
		28.3	646				
$M_n = 21000$		30.8	510				384
$M_w = 42000$		33.1	416				
		37	309				
Poly[imino(1-oxo-dodecamethylene)] Nylon 10							
$M_v = 15.7 \times 10^3$	220 (15 min)	163	168	4.3		DSC; k values listed in text	385
$[\eta] = 1.34$ dl/g		164	240	4.9			
(*m*-cresol at 20°C)		165	330	4.9			
		166	498	5.2			
		167	600	5.3			
		168	996	4.9			
		169	1200	4.7			
$M_v = 29.6 \times 10^3$		164	192	5.2			385
		165	240	5.3			
		166	348	5.0			
		167	546	5.3			
		168	750	5.5			
		169	996	5.6			
$M_v = 41.2 \times 10^3$		161	348	4.9			385
$[\eta] = 1.70$ dl/g		162	426	4.9			
		163	666	5.6			
		164	858	5.9			
		165	996	5.0			
$M_v = 9.5 \times 10^3$	220 (2 h)	162	360	4.4		Dilatometry; k values also cited	385
$[\eta] = 0.92$ dl/g		165	696	4.9		in (min^{-n})	
(*m*-cresol at 20°C)		167	1128	4.2			
		169	3000	5.7			
$M_v = 23.9 \times 10^3$		162	522	5.0			385
$[\eta] = 1.80$ dl/g		165	1110	5.6			
		167	2070	4.7			
		169	3750	4.5			
$M_v = 23.9 \times 10^3$		162	396	4.7		Contains 0.1% kaolin	385
$[\eta] = 1.80$ dl/g		165	720	4.4			
		167	1332	4.7			
		169	2400	4.6			
Poly(iminoadipoyl iminohexamethylene) Nylon 66							
$M_n = 12.7 \times 10^3$		162	1.4			Light depolarization technique;	94
		175	1.85			spherulite growth from the melt	
		187	3.03				
		200	4.0				
		212	9.1				
		224	16.7				
		238	40.0				
		20	10000			Growth from glassy state; spin line	94
		29	6600			quenched and subsequently	
		34	1000			crystallized; fiber birefringence	
		40	500			development using LDT on hot	
		44	380			stage assembly between	
		60	33			crossedpolars.Amorphous	
		70	6.3			condition verified X-ray diffraction	
		80	4.3				

TABLE 14. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
14.1. BLENDS OF POLY(AMIDES)							
Poly(iminohexamethyleneaminoadipoyl)/poly[2-2'-bis(3,4-dicarboxy-phenoxy)phenylpropane-2-phenylene bisimide]							
	300 (5min)	237	29	$2.4 \leq n \leq 3.0$	k values listed in text as min^{-n}	DSC studies under nitrogen; textures are spherulitic $T_m^0 = 295$ C	410
(Nylon 66/PEI) wt%							
100/0		238.5	34				
		240	44				
		242	79				
		244	105				
95/5		237	38			T_m^0 unlisted	
		238.5	50				
		240	55				
		242	94				
		244	158				
90/10		237	39			$T_m^0 = 287$ C	
		238.5	58				
		240	69				
		242	100				
		244	165				
		246	300				
85/15		236	32			T_m^0 unlisted	
		237	40				
		238.5	40				
		240	71				
		242	111				
		244	198				
		246	294				
75/25		237	29			$T_m^0 = 281$ C	
		238.5	40				
		240	52				
		242	69				
		244	100				
50/50		237	34			$T_m^0 = 273$ C	
		238.5	42				
		240	65				
		242	94				
		244	120				
Poly[imino(1-oxo-hexamethylene)]/poly(iminohexamethyleneimioadipoyl)/[imino(1-oxohexamethylene)]							
Nylon 6/Nylon 66/caprolactam (mol/mol/mol)							
6/6.6/4.4	$T_m^0 + 30$(10–20 min)	207	12.3			DSC rate measurements; Polymers characterized by pyrolysis, NMR; T_m^0 using Flory copolymer theory (see Ref. 215)	386
		209	22.7				
		210.5	42.1				
		212	76.3				
6/6.6/8.2	$T_m^0 + 30$ (10–20 min)	205	14.8				386
		206.5	26.4				
		208	52.7				
		210	84.7				
6/6.6/17.5	$T_m^0 + 30$ (10–20 min)	205	19.6				386
		206.5	34.6				
		208	52.3				
		194	93.4				
6/66-4,4	$T_m^0 + 30$ (10–20 min)	195	12.3				386
		198	22.7				
		201	40.8				
		204	74.4				
6/PXD.6-4.1	$T_m^0 + 30$ (10–20 min)	182	12.4				386
		186	23.9				
		189	46.5				
		192	71.0				
6/MXD.6-3.4	$T_m^0 + 30$ (10–20 min)	189.5	13.2				386
		193	27.0				
		199	53.8				
		202.5	128				

TABLE 14. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
6/I-3.8	T_m^0 + 30 (10–20 min)	182	27.0				386
		185	53.8				
		188	75.9				
		191	194				
		194	190.4				
6/MXD.I-3.9	T_m^0 + 30 (10–20 min)	175	30.8				386
		187	110.6				
6/6.T-4.5	T_m^0 + 30 (10–20 min)	168	53.5				386
		171	69.5				
		174	96.0				
		177	142.6				
		180	198.2				

TABLE 15. POLY(URETHANES)

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly(oxycarbonyliminotetramethylene iminocarbonyl oxyhexamethylene)							
$M_w = 10 \times 10^3$	195 (10 min)	169	3540	2.1		k_n values listed in text	343
		167	1650	2.1			
		165	1110	2.0			
		163	630	1.9			
		162	360	2.0			
Poly(oxycarbonyliminohexamethylene iminocarbonyl-bis(oxyethylene)							
	150 (10 min)	109	17280	3		k_n values listed in text	343
		107	8880	3			
$[\eta] = 0.81$ in		105	4860	3			
0.5% *m*-cresol		103.5	3060	3			
		100	960	3			
		95	630	3			
After quenching		14.2	5580				
		16.2	3960				
		17.8	3000				
		20.3	1950				
		22.3	1320				
		25.6	600				
		27.2	432				
		28.2	330				
Poly(oxycarbonyliminohexamethylene iminocarbonyl-tris(oxyethylene))							
	140 (10 min)	85	4260	2		k_n values listed in text	343
		81	2160	2			
		78	1350	2			
		75	930	2			
		72	630	2			
		69	420	2			
After quenching		14.7	3960				
		16.1	2940				
		19.0	1320				
		22.0	960				
		24.5	630				
		26.0	390				

15.1. BLENDS OF POLY(URETHANES)

Poly[imino(1-oxohexamethylene)] Nylon 6; 70/30 blend with 10% polypropylene with maleic anhydride							
		203	213	3			
		204	252	3			
	205	324	3				
		206	497	3			
		28.0	300				

TABLE 16. POLY(SILOXANES)

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly(oxydimethylsilylene) DMS	20	−55.6	900	2.2			343
		−6.7	660	2.1			
$M_v = 400 \times 10^3$		−58.0	480	2.0			
		−59.6	420	2.1			
		−60.5	318	2.1			
		−63.0	228	2.0			
		−65.0	174	2.0			
		−67.0	138				
		−115	1080	2.0			
After quenching		−109	108				
Poly(oxydimethylsilylene-1,4-phenylenedimethylsilylene) TMPS							
$M_w = 15.8 \times 10^3$	200 (5 min)	100	82			Other bulk transformation methods DSC and SAXS have been used on these same fractions	387,388
		110	2553				
		115	6383				
		120	24000				
		126.5	111540				
$M_w = 37.5 \times 10^3$		100	54				387,388
		110	1580				
		115	4110				
		120	10000				
		126.5	55050				
$M_w = 109 \times 10^3$		100	105				387,388
		110	2857				
		115	5405				
		120	15000				
		126.5	56075				
$M_w = 245 \times 10^3$		100	316				387,388
		110	8571				
		115	13636				
		120	30000				
		126.5	120000				
$M_w = 380 \times 10^3$		100	3191				387,388
		110	20000				
		115	35294				
		120	99174				
		126.5	272727				
Poly(oxydimethylsilylene)	20	−55.6	900	2.2			343
		−56.7	660	2.1			
$M_v = 400 \times 10^3$		−58.0	480	2.0			
		−59.6	420	2.1			
		−60.5	318	2.1			
		−63.0	228	2.0			
		−65.0	174	2.0			
		−67.0	138				
		−115	1080	2.0			
After quenching		−109	108				

TABLE 17. POLY(PHOSPHAZENES)

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly[bis(4-fluorophenoxy) phenylphosphazene]							
$M_n = 0.16 \times 10^6$	215 (5 min)	120	4	~2.0		DLI technique cross-polars with Chino controller and accessories; textures recorded	389
$M_w = 1.18 \times 10^6$		122.5	6.5				
		125	15				
		127.5	63				
		129.5	2200				
Poly[bis(trifluoroethoxy) phosphazene)]							
		229.8		~2.0 ± 0.1	1810	DSC isotropic thermotropic (2D) transformation rates: $10^7\ ks^{-2}$	406
		230.2			913		
		230.5			644		

TABLE 17. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
		231.0			477		
		231.5			186		
		228.6		$\sim 2.0 \pm 0.1$	8760	DLI isotropic → 2D transformation rates: $10^7\,ks^{-2}$	
		229.6			3410		
		230.6			1030		
		231.1			707		
		231.6			289		
		232.6			60.8		
		233.6			17.0		
		234.6			2.97		
		73.0		$\sim 2.0 \pm 0.1$	40600	DSC thermotropic (2D) → 3D crystalline transformation rates: $10^7\,ks^{-2}$	
		73.3			18000		
		73.6			6270		
		73.9			2210	Small birefringence changes negated the DLI measurement	
		74.2			927		
Poly[bis(phenylphenoxy) phosphazene]							
	290 (10 min)	237.5		$\sim 2.0 \pm 0.1$	49300	DLI method for isotropic → 2D thermotropic transition rate: $10^7\,ks^{-2}$	407
		237.9			16600		
		238.3			4080		
		238.7			1630		
		239.1		$\sim 2.0 \pm 0.1$	776		
		239.5			199		
		239.9			53.0		
		240.3			27.6		
		240.7			10.7		
		198.5		$\sim 2.0 \pm 0.1$	48000	DLI method for 2D thermotropic → 3D crystalline transformation rate: $10^7\,ks^{-2}$	
		198.9			29200		
		199.6			10700		
		200.0			5210		
		200.4			1750		
		200.9			450		
		201.4			71.1		
Poly[di(*p*-methylphenoxy) phosphazene]							
		101	7.2			*in situ* SAXS synchrotron radiation rate measurements; induction times by WAXS and SAXS are listed in text	17
		107	54				
		112	89				
		112.5	104				
		117.5	150				
		118.5	154				
		129	263				
		132.5	984				

TABLE 18. OTHERS

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly(oxy-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene)							
PEEK Poly(etheretherketone)		188.8[a]	5.1	1.8×10^{-7}	50[b]	DSC; Nonisothermal rates from glassy state; crystallinity, density listed	380
		182.3	5.7	4.7×10^{-10}	20		
		175.2	4.7	1.6×10^{-8}	10		
		171.3	6.9	6.6×10^{-16}	5		
		167.1	7.5	7.7×10^{-19}	2		
		164.4	7.8	2.9×10^{-4}	1		
Poly(oxy-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene)							
PEEK Poly(etheretherketone)		160	126	2.5–2.6		DSC; k values in text in min^{-n}	408
original polymer		164	54				
$M_n = 14000$		168	24				
		172	18				
Melt crystallized		290	21.6	2.3–2.4			408
		300	40.2				
		310	136.2				

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TABLE 18. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Solvent treated and desorbed		168	570	1.1–1.2			408
		172	522				
		176	450				
		180	330				
		184	276				
		190	222				
		194	180				
Poly[oxy-1-methyl-2-oxyethylene]; poly(L-lactic acid)							
	200 (20 min)	80		2.62	1.25×10^{-3}	DSC measurements; various analysis of kinetic data are presented	145
		90		2.73	1.19×10^{-3}		
		100		2.82	2.97×10^{-3}		
		110		2.66	2.81×10^{-3}		
		120		3.24	1.88×10^{-3}		
Poly[oxy-1-methyl-2-oxyethylene]; poly(L-lactic acid)							
$M_v = 700000$	220 (10 min)	70			5.79×10^{-6}	Light depolarization technique; morphology and X-ray diffraction reported; k expressed t^{-1}s by the authors; values are not listed	390
		75			3.96×10^{-5}		
		80			1.33×10^{-4}		
		85			4.44×10^{-4}		
		90			1.49×10^{-3}		
		95			3.35×10^{-3}		
		100			6.79×10^{-3}		
		109.5			4.48×10^{-3}		
		115			4.10×10^{-3}		
		120			2.81×10^{-3}		
		125			1.73×10^{-3}		
		130			6.61×10^{-4}		
		134.5			4.29×10^{-4}		
		140			2.37×10^{-4}		
		145			1.01×10^{-4}		
		149.5			5.30×10^{-5}		
		155			2.14×10^{-5}		
		159			1.85×10^{-5}		
		165			1.41×10^{-5}		
Poly(L-lactide-*co*-meso-lactide)		85	88	2.5		Kinetics made using DSC	391
$M_n = 10.1 \times 10^3$; wt. (0% meso)		90	420	2.5			
		95	270	2.7			
		100	228	2.4			
		105	174				
		110	114				
		115	210	2.7			
		120	240				
		125	306	3.4			
		130	522	3.2			
		135	374	2.4			
$M_n = 15.7 \times 10^3$; wt. (0% meso)		90	684				391
		100	288				
		110	240				
		120	342				
		130	804				
$M_n = 8.8 \times 10^3$; wt. (3% meso)		85	1434				391
		90	660				
		95	486				
		100	564				
		105	516				
		110	360				
		115	414				
		120	492				
		25	690				
Poly(L-lactide)							391
$M_n = 11.4 \times 10^3$; wt. (3% meso)		100	1668				
		105	1176				
		110	1182				
		115	1332				
$M_n = 5.8 \times 10^3$; wt. (6% meso)		100	684				391
		110	648				

TABLE 18. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
		120	696				
$M_n = 11.4 \times 10^3$; wt. (6% meso)		110	2640				391
Poly(L-lactide-*co*-meso-lactide)							
$M_n = 12.3 \times 10^3$;		90	<60			Kinetics measured by DSC	391
(wt. 0% meso);		100	<60				
wt. 6% talc additive		110	<60				
		120	<60				
$M_n = 10.5 \times 10^3$;							
(wt. 3% meso);		90	120				391
wt. 6% talc additive		100	84				
		110	<60				
		120	126				
$M_n = 12.2 \times 10^3$;		90	342				391
(wt. 6% meso);		100	180				
wt. 6% talc additive		110	234				
		120	882				
$M_w = 8.1 \times 10^3$		90	528	2.3			391
(wt. 9% meso)		100	360				
wt. 6% talc additive		110	600				
		120	≥900				
Poly(L-lactide-*co*-meso-lactide)							
(wt. 9% meso)						Kinetics measured by DSC	391
$M_n = 3.9 \times 10^3$;							
wt. 2% talc additive		90°C	1134	2.7			391
$M_n = 4.6 \times 10^3$;							
wt. 6% talc additive			762	2.3			391
$M_n = 5.5 \times 10^3$;							
wt. 11% talc additive			600	2.3			391
$M_n = 5.3 \times 19^3$;							
wt. 2% talc additive			444	2.2			391
wt. 2% talc additive			1176				391
wt. 6% talc			876				391
wt. 11% talc			732				391
wt. 21% talc			732				391
Poly(thio-1,4-phenylene)		208.8	215.5				
		218.3	163.6				
		217.3	144.8				
		227	92.9				
		235.8	43.4				
		245.1	19.5				
		254.1	7.41				
		263.3	1.68				
		272.7	0.125				
Poly(thio-1,4-phenylene)		250	170	2.6		DSC	392
Phillips Pet. Co. research grade R-4		245	100	2.9			
$M_w = 51 \times 10^3$		240	48	2.3			
		235	34	2.3			
		230	26	2.6			
		220	17	2.5			
		215	14	2.6			
Phillips Pet. Co. Resin grade V-1		250	240	2.0			392
$M_w = 15 \times 10^3$		245	140	2.4			
		240	60	2.2			
		235	40	2.1			
$M_w = 15 \times 10^3$		220	22	2.5			392
		215	17	2.2			
Poly(thio-1,4-phenylene) Ryton V-I (Phillips Pet. Co.)							
	320	240	97.3		64.8×10^{-6}	DSC measurement	392
		243	101		59.1×10^{-6}		
		245	142		29.8×10^{-6}		
		249	207		12.6×10^{-6}		
		255	601		1.1×10^{-6}		
δ-irradiated polymer	320	240	76.6		170×10^{-6}	DSC measurement	392
		243	120		98×10^{-6}		
		245	137		70×10^{-6}		

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TABLE 18. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
		251	329		5.7×10^{-6}		
		255	507		1.7×10^{-6}		
Poly (thio-1,4-phenylene) (PPS homopolymer) 0% metamethylene linkages		102	8.5			Crystallization rates listed as DSC peak times (mins); measurements below	411
		106	3.0				
		110	1.5				
		114	0.9				
		118	0.71			T_{max} were made by upquenching from the amorphous state ($T_m^0 \sim 330$°C using Hoffman-Weeks plot)	
		120	0.55				
		235	0.36				
		240	0.53				
		250	1.30				
		245	0.82				
		255	2.40				
		260	5.20				
		265	8.00				
		270	17.5				
Random copolymer with 5% metamethylene linkages		110	6.71			$T_m^0 \sim 290$° C	
		115	3.16				
		120	1.90				
		125	1.23				
		130	0.88				
		135	0.77				
		140	0.53				
		145	0.46				
		200	0.39				
		210	0.59				
		220	1.20				
		230	2.30				
		235	2.81				
		240	5.01				
		245	9.47				
		250	23.5				
Random copolymer with 8% metamethylene linkages		105	14.93			$T_m^0 \sim 283$° C	
		110	5.80				
		120	1.97				
		130	1.09				
		140	0.670				
		150	0.470				
		190	1.75				
		200	2.52				
		205	4.19				
		210	5.26				
		215	7.40				
		220	9.62				
		225	23.02				
Random copolymer with 10% metamethylene linkages		105	13.4			$T_m^0 \sim 273$° C	
		110	7.10				
		120	2.65				
		130	1.34				
		140	0.845				
		150	0.620				
		160	0.520				
		170	0.490				
		180	0.760				
		190	1.11				
		200	1.80				
		205	2.54				
		210	3.81				
		215	6.21				
		220	10.20				

TABLE 19. COMPOSITES

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly(thio-1,4-phenylene) unreinforced		220	6.89×10^7	$2.2 \leq n \leq 2.7$		Modeled from rate data; $t_{1/2}$ values are graphical; Avrami n variable	393
		225	4.19				
		230	4.11				
		235	1.35				
		240	1.10				
With commercial glass (70% wt.)		220	11.23×10^7	$2.1 \leq n \leq 2.5$			393
		225	7.24				
		230	5.78				
		235	5.72				
		240	3.14				
Unsized glass (35% wt.)		220	163×10^7	$2.3 \leq n \leq 2.5$			393
		225	7.14				
		230	5.46				
		235	3.17				
		240	0.81				
Sized glass		220	17.0×10^7	$2.2 \leq n \leq 2.5$			393
		225	12.1				
		230	9.11				
		235	4.9				
		240	1.84				
Commercial carbon (66% wt.)		220	38.4×10^7	$2.3 \leq n \leq 2.5$			393
		225	14.1				
		230	7.83				
		235	5.61				
		240	6.87				
Unsized AS4 (36% wt.)		220	2.54×10^7	$2.1 \leq n \leq 2.4$			393
		225	1.55				
		230	6.67				
		235	0.93				
		240	0.37				
Sized AS4 (39% wt.)		220	204.8×10^7	$2.0 \leq n \leq 2.4$			393
		225	165.3				
		230	75.3				
		235	28.3				
		240	15.8				
Graphitized Thornel (35% wt.)		220	158.3×10^5	$1.6 \leq n \leq 1.7$			393
		225	79.3				
		230	28.90				
		235	13.95				
		240	6.08				
Poly(thio-1,4-phenylene) Unreinforced		220	176				394
		225	258				
		230	438				
		235	800				
		240	1409				
Unsized AS4, 54 wt.%		220	209				394
		225	317				
		230	395				
		235	516				
		240	718				
Sized AS4, 64%		220	294				394
		225	311				
		230	540				
		235	457				
		240	700				
Graphitized Thornel, 65%		220	232				394
		225	195				
		230	183				
		235	198				
		240	222				
Phillips prepreg 66 wt.%		220	294				394
		225	413				
		230	540				
		235	457				
		240	700				

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TABLE 19. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly(oxyethyleneoxyterephthaloyl) modified		220			22.0×10^3	DSC measurements; fibers	395
		225			12.0×10^3	enhanced rate of crystallization	
		230			6.5×10^3		
		235			3.37×10^3		
Modified/unsized Kevlar		225			17.60×10^3		395
		230			9.14×10^3		
		235			4.23×10^3		
Modified/unsized glass		220			17.25×10^2		395
		225			10.45×10^2		
		230			5.53×10^2		
		235			2.34×10^2		
19.1. COMPOSITES OF BLENDS							
Poly(oxyethyleneoxyterephthoyl)/Poly(oxycarbonyloxy-1,4-phenylene-isopropylidene-1,4-phenylene) PET/PC							
Poly(carbonate) 80/20, w/w		220			20.4×10^3	DSC measurements	396
Unreinforced		220			20.4×10^3		
36% wt. Kevlar		220			13.1×10^3		
49% wt. Kevlar		220			11.4×10^3		
68% wt. Kevlar		220			5.96×10^3		
Unreinforced		230			6.09×10^3		
36% wt. Kevlar		230			3.82×10^3		
68% wt. Kevlar		230			0.77		
Poly(oxyethyleneoxyterephtholy)/Poly(oxycarbonyloxy-1,4-phenylene-isopropylidene-1,4-phenylene) PET/PC							
		220			24.4×10^3		397
		225			13.9×10^3		
60/40, w/w		230			8.3×10^3		
		235			3.4×10^3		
Unsized Kevlar		215			11.4×10^3		397
		220			6.8×10^3		
		225			4.5×10^3		
		230			2.3×10^3		
Unsized glass		215			10.9×10^3		397
		220			6.35×10^3		
		225			3.9×10^3		
		230			2.02×10^3		
Poly(oxyethyleneoxyterephthoyl)/Poly(oxycarbonyloxy-1,4-phenylene-isopropylidene-1,4-phenylene) terephthalate)							
		220			20.7×10^3		396
Poly(carbonate) 80/20, w/w		225			11.3×10^3		
		230			6.03		
		235			3.1		
80/20 unsized Kevlar		215			17.7×10^3		396
		220			11.5×10^3		
		225			6.8×10^3		
		230			4.0×10^3		
80/20 unsized glass		215			15.6×10^3		396
		220			9.5×10^3		
		225			5.5×10^3		
		230			3.12×10^3		
Poly(thio-1,4-phenylene) unreinforced		205			124	DSC measurements; optical	394
		210			215	microscopy of composites	
		215			356		
		220			513		
		225			914		
		230			1022		
		235			2350		
		240			2760		
Unsized AS4 36 and 38 wt.%		220			171		394
		225			246		
		230			435		
		235			788		
		240			1400		
Sized AS4 39 wt.%		220			122		394
		225			197		
		230			289		
		235			511		

TABLE 19. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly(thio-1,4-phenylene) unreinforced		220			176	DSC measurements Avrami variable; optical microscopy of composites	394
		225			258		
		230			438		
		235			799		
		240			1409		
Unsized Kevlar 58 wt.%		220			116		394
		225			172		
		230			232		
		235			335		
		240			570		
Phillips prepreg 55 wt.%		220			43		394
		225			60		
		230			86		
		235			163		
		240			327		
Sized Kevlar 53% wt.%		220			26		394
		225			52		
		230			69		
		235			126		
		240			253		
Poly(thio-1,4-phenylene) poly(phenylene sulfide) (PPS)							
100/0, w/w		250	58.6			DSC measurements	227
		257	106.3				
		263	215.3				
		268	497.1				
		269	558.4				
PPS/Fortron glass		251	58.5				227
		257	110.7				
		263	242.0				
		268	558.4				
Poly(thio-1,4-phenylene) (unreinforced)		220	167				397
100/0, w/w		225	261				
		230	440				
		235	796				
		240	1391				
With sized glass		220	167				397
35/65, w/w		230	440				
		240	988				

D. REFERENCES

1. J. D. Hoffman, G. T. Davies, J. I. Lauritzen Jr., in: N. B. Hannay (Ed.), "Treatise on Solid State Chemistry", vol. 3, Plenum, New York, 1976, p. 497.

1(a) K. Armistead, G. Goldbeck-Wood, Ad. Polym. Sci., **100**, 219 (1992).

2. L. Mandelkern, in: Chap. 4, "Physical Properties of Polymers", Am. Chem. Soc. Publication, 1984.
3. I. C. Sanchez, J. Macromol. Sci. C: Rev. Macromol. Chem., **10**, 113 (1974).
4. See, for example, Faraday Disc. Chem. Soc., "Organization of Macromolecules in the Condensed State", 68 (1979); Faraday Disc. Chem. Soc., 95 (1993).
5. J. H. Magill in: Part A, J. M. Schultz (Ed.) "Treatise on Materials Science and Technology", vol. 10, Academic Press, New York, 1977.
6. See, for example Martuscelli et al., E. Martuscelli, R. Palumbo, M. Kriszewski (Ed.), "Polymer Blends: Processing, Morphology and Properties", Plenum, New York, 1981.
7. X. Li, S.L. Hsu, J. Polym. Sci., Polym. Phys. Ed., **22**, 1331 (1984).
8. J. Roovers in "Methods of Experimental Physics", 16c, Chap. 16, Academic Press, New York, 1980, p. 275.
9. J. H. Magill, Makromol. Chem., **187**, 455 (1985).
10. H. Tadokoro, Polymer, **25**, 147 (1984).
11. J. H. Cole, L. E. St. Pierre, J. Polym. Sci., Polym. Symp., **63**, 205 (1978).
12. G. E. Molau in: S. L. Aggarwal (Ed)., "Block Copolymers", Plenum, New York, 1970, p. 102.
13. C. Robinson, Trans. Faraday Soc., **52**, 571 (1956).
14. P. J. Flory, Proc. Roy. Soc. A, **234**, 73 (1956).
15. M. Dosiere (Ed.), "Crystallization of Polymers", NATO ASI Series C, vol. 405, Klumer Academic, Dordrecht, 1993.

TABLE 19. *cont'd*

Polymer	T_f (°C)	T_c(°C)	$t_{0.5}$ (s)	n	k (s^n)	Remarks	Refs.
Poly(thio-1,4-phenylene) unreinforced		220			176	DSC measurements Avrami	394
		225			258	variable; optical microscopy	
		230			438	of composites	
		235			799		
		240			1409		
Unsized Kevlar 58 wt.%		220			116		394
		225			172		
		230			232		
		235			335		
		240			570		
Phillips prepreg 55 wt.%		220			43		394
		225			60		
		230			86		
		235			163		
		240			327		
Sized Kevlar 53% wt.%		220			26		394
		225			52		
		230			69		
		235			126		
		240			253		
Poly(thio-1,4-phenylene) poly(phenylene sulfide) (PPS)							
100/0, w/w		250	58.6			DSC measurements	227
		257	106.3				
		263	215.3				
		268	497.1				
		269	558.4				
PPS/Fortron glass		251	58.5				227
		257	110.7				
		263	242.0				
		268	558.4				
Poly(thio-1,4-phenylene) (unreinforced)		220	167				397
100/0, w/w		225	261				
		230	440				
		235	796				
		240	1391				
With sized glass		220	167				397
35/65, w/w		230	440				
		240	988				

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1. J. D. Hoffman, G. T. Davies, J. I. Lauritzen Jr., in: N. B. Hannay (Ed.), "Treatise on Solid State Chemistry", vol. 3, Plenum, New York, 1976, p. 497.

1(a) K. Armistead, G. Goldbeck-Wood, Ad. Polym. Sci., **100**, 219 (1992).

2. L. Mandelkern, in: Chap. 4, "Physical Properties of Polymers", Am. Chem. Soc. Publication, 1984.
3. I. C. Sanchez, J. Macromol. Sci. C: Rev. Macromol. Chem., **10**, 113 (1974).
4. See, for example, Faraday Disc. Chem. Soc., "Organization of Macromolecules in the Condensed State", 68 (1979); Faraday Disc. Chem. Soc., 95 (1993).
5. J. H. Magill in: Part A, J. M. Schultz (Ed.) "Treatise on Materials Science and Technology", vol. 10, Academic Press, New York, 1977.
6. See, for example Martuscelli et al., E. Martuscelli, R. Palumbo, M. Kriszewski (Ed.), "Polymer Blends: Processing, Morphology and Properties", Plenum, New York, 1981.
7. X. Li, S.L. Hsu, J. Polym. Sci., Polym. Phys. Ed., **22**, 1331 (1984).
8. J. Roovers in "Methods of Experimental Physics", 16c, Chap. 16, Academic Press, New York, 1980, p. 275.
9. J. H. Magill, Makromol. Chem., **187**, 455 (1985).
10. H. Tadokoro, Polymer, **25**, 147 (1984).
11. J. H. Cole, L. E. St. Pierre, J. Polym. Sci., Polym. Symp., **63**, 205 (1978).
12. G. E. Molau in: S. L. Aggarwal (Ed)., "Block Copolymers", Plenum, New York, 1970, p. 102.
13. C. Robinson, Trans. Faraday Soc., **52**, 571 (1956).
14. P. J. Flory, Proc. Roy. Soc. A, **234**, 73 (1956).
15. M. Dosiere (Ed.), "Crystallization of Polymers", NATO ASI Series C, vol. 405, Klumer Academic, Dordrecht, 1993.

16. See, for example, "Polymer Liquid Crystals", Faraday Disc. Chem. Soc., 79 (1984).
17. J. H. Magill, C. Riekel, Makromol. Chem. Rapid Comm., **7**, 186 (1986).
18. J. Grebowicz, S. Z. D. Cheng, B. Wunderlich, J. Polym. Sci., Part B (Polym. Phys.), **24**, 675 (1986).
19. See, for example, M. Blume and D. E. Moncton, Phys. Today, **38**, 69 (1985).
20. G. Elsner, C. Riekel, H. G. Zachmann, Adv. Polym. Sci., **67**, 3 (1985).
21. J. H. Magill, J. M. Schultz, J. S. Lin, Colloid Polym. Sci., **264**, 193 (1986).
22. R. Kimmich, R. Bachus, Colloid Polym. Sci., **260**, 911 (1982).
23. W. Prieske, C. Riekel, M. H. J. Koch, H. G. Zachmann, Nuclear Instr. Meth., **208**, 435 (1983).
24. H. W. Spiess, J. Adv. Polym. Sci., **66**, 23 (1985).
25. D. T. Grubb, J. J-H. Liu, M. Gaffrey, D. H. Bildrback, J. Polym. Sci., Polym. Phys. Ed., **22**, 367 (1984).
26. L. Mandelkern, "Crystallization of Polymers," McGraw Hill, New York, 1961.
27. P. J. Flory, J. Am. Chem. Soc., **84**, 2857 (1962).
28. A. Keller, Rep. Prog. Phys., **31**, 623 (1968).
29. J. J. Point, A. J. Kovacs, Macromolecules, **13**, 399 (1980).
30. G. M. Stack, L. Mandelkern, I. G. Voigt-Martin, Macromolecules, **17**, 321 (1984).
31. B. Wunderlich "Macromolecular Physics," vol. 2, Academic, New York, 1976 and other volumes in this series.
32. P. H. Geil, "Polymer Single Crystals", Wiley-Interscience, New York, 1962.
33. See, for example, Ref. 5.
34. B. Wunderlich, "Macromolecular Physics", vol. 1, Academic, New York, 1973.
35. See, for example, Ref. 5, Section F, p. 264, et. seq.
36. Z. Tadmor, C. G. Gogos, "Principles of Polymer Processing", Wiley, New York, 1979.
37. A. Ziabicki, "Fundamentals of Fiber Formation", Chap. 3, Wiley, New York, 1976, p. 149.
38. E. Ratajski and H. Janeschitz-Kriegl, Colloid Polym. Sci., **274**, 938 (1996).
39. R. Gachter, H. Muller (Ed.) "Plastic Additives Handbook-Stabilizers, Processing Aids, Fillers Reinforcements, Colorants for Thermoplastics", Carl Hanser Verlag, Munchen, 1985.
40. A. K. Fritzsche, F. P. Price in: S. L. Aggarwal (Ed.) "Block Copolymers", Plenum, New York, 1969, p. 252.
41. G. Eder, H. Janeschitz-Kriegel, Colloid Polym. Sci., **226**, 1087 (1988).
42. W. M. Leung, R. St. J. Manley, A. R. Panaras, Macromolecules, **18**, 760 (1984).
43. M. Cooper, R. St. John Manley, Polym. Lett., **11**, 363 (1973).
44. V. F. Holland, P. H. Lindenmeyer, J. Polym. Sci., **57**, 589 (1962).
45. J. G. Fatou, in Encyl. Polym. Sci. and Eng., Suppl. vol. (2nd Ed., J. Wiley and Sons, Inc. (1989).
45a. A. Keller, G. Goldbeck-Wood in "Comprehensive Polymer Sci., 2nd Suppl." Chap. 7, 241 (S. L. Aggarwal and S, Russo, Eds.) Elsevier, New York, 1996.
45b. D. H. Hoffman and R. L. Miller, Polymer, **38**, 3151 (1997).
46. A. S. Vaughan, D. C. Bassett in: C. Booth and C. Price, Eds., "Comprehensive Polym. Sci.," vol. 2, Chap. 12, 413, Pergamon Press, 1989.
47. I. G. Voigt Martin, E. W. Fischer, L. Mandelkern, J. Polym. Sci., Polym. Phys. Ed., **18**, 2347 (1980).
48. F. Khoury, E. Passaglia, in: N. B. Hannay (Ed.) "Treatise on Solid State Chemistry", Plenum, vol. 3, Chap. 6, New York, 1976.
49. W. Dietz, Colloid Polym. Sci., **259**, 413 (1981).
50. A. Keller, A. O'Connor, Disc. Faraday Soc., **25**, 114 (1958).
51. D. C. Bassett, A. M. Hodge, Proc. Roy. Soc. A, **377**, 61 (1981).
52. H. D. Keith, F. J. Padden, Jr., Polymer, **25**, 28 (1983); J. Appl. Phys., **30**, 1479 (1959).
53. C. Briske, N. H. Hawthorne, Trans. Faraday Soc., **63**, 1546 (1967).
54. R. G. Crystal, J. Polym. Sci. A-2, **8**, 1755, 2153 (1970).
55. I. Ye. Bolotov, S. B. Fischeleva, Phys. Metals Metallog. USSR (Eng. trans.), **20**, 147 (1965).
56. P. J. Barnham, E. D. T. Atkins, I. A. Nieduszynski, Polymer, **15**, 762 (1974).
57. I. Ezek, I. Korritta, K. Leble, Soc. Phys. Crystallogr., **2**, 653 (1957).
58. J. G. Morley, Glass Technol, **6**, 69, 77 (1965).
59. H. W. Morse, J. D. H. Donnay, Am. Mineral, **21**, 391 (1932).
60. J. H. Magill, D. J. Plazek, J. Chem. Phys., **46**, 3757 (1967); J. H. Magill, H.M. Li, J. Crystal Growth, **20**, 135 (1973).
61. D. M. Sadler, Polymer, **24**, 140 (1983); Polym. Comm., **27**, 140 (1986); Nature, **326**, 147 (1987).
62. M. Bark, H. G. Zachmann, R. Alamo, L. Mandelkern, Makromol, Chemie, **193**, 2363 (1992).
63. C. Santa Cruz, N. Stribeck, H. G. Zachmann, F. J. Balt-Calleja, Macromolecules, **24**, 5980 (1991).
64. E. Ergoz, J. G. Fatou, L. Mandelkern, Macromolecules, **5**, 147 (1972).
65. See, for example, Ref. 1, Fig. 17; N. Okui et al. (to be published).
66. A. H. Kovacs, A. Gonthier, Kolloid Z. U. Z. Polym., **250**, 530 (1972); A. J. Kovacs, A. Gonthier, C. Straupe, J. Polm. Sci., Symp., **50**, 283 (1875).
67. K. S. Lee, G. Wegner, Makromol. Chem. Rapid Comm., **6**, 203 (1985).
68. G. Ungar, J. Stejny, A. Keller, I. Bidd, M. C. Whiting, Science, **229**, 386 (1985).
68a. G. Ungar, X. Zeng, G. M. Brooke, S. Mohammed, Macromolecules, **31**, 1875 (1998).
69. L. Mandelkern in: C. Booth and C. Price (Eds.), "Comprehensive Polym. Sci.," vol. 2, chap. 11, Pergamon Press, 1989, p. 363.
70. A. N. Kolmogorov, Isz. Akad. Nauk SSSR, Ser. Math., **1**, Fig. 3, 355 (1937).
71. V. Göler, F. Sacks, G. Sacks, Z. Phys., **77**, 281 (1932).

72. U. R. Evans, Trans. Faraday Soc., **41**, 365 (1945).
73. M. Avrami, J. Chem. Phys., **7**, 1103 (1939); **8**, 212 (1940); **9**, 177 (1941).
74. L. Mandelkern, "Crystallization of Polymers", Chap. 8, McGraw Hill, New York, 1964.
75. M. G. Dobb, J. E. McIntyre, Ad. Polym. Sci., **60/61**, 61 (1984).
76. G. Broza, J. Petermann, Praktische Metallographic, **17**, 175 (1987).
77. F. L. Binsbergen, J. Crystal Growth, **16**, 249 (1972).
78. J. N. Hay, Z. J. Prezekop, J. Polym. Sci., Polym., Phys. Ed., **17**, 951 (1979).
79. A. Galeski, J. Polym., Sci., Polym. Phys. Ed., **19**, 721 (1981).
80. L. E. Levine, K. Lakshmi Narayan and K. F. Kelton, J. Mat. Sci., **12**, 124 (1997).
81. A. Galeski, E. Piorkowska, J. Polym. Sci., Polym. Phys. Ed., **21**, 1313 (1983).
82. M. Birnboim, J. H. Magill, G. C. Berry, in: R. Lenz, R. S. Stein (Ed.), "Electromagnetic Scattering" Gordon and Breach, New York, 1967, p. 413.
83. A. Misra, R. S. Stein, Polym. Lett. B, **10**, 473 (1972).
84. S. Go, D. Peiffer D., Mandelkern, R. S. Stein, J. Polym. Sci., Polym. Phys. Ed., **15**, 1189 (1977).
85. See, for example, Ref. 34, Chap. 3, p. 322.
86. D. C. Martin, E. L. Thomas, Polymer, **36**, 1743 (1995).
87. B. Lotz, J. C. Wittmann, A. J. Lovinger, Polymer, **37**, 4979 (1996).
88. Q. Zong, D. Innis, K. Kjoller, V. B. Elings, Surf. Sci. Lett., **290**, L688 (1993).
89. S. N. Magonov, V. Elings, V. S. Papkov, Polymer, **38**, 297 (1997).
89a. S. N. Magonov, Polymer Sci., Series B, **38**, 34 (1996).
90. F. H. Mueller, H. Martin, J. Polym. Sci. C, **6**, 83 (1964).
91. N. Bekkedahl, Rubber Chem. Technol., **40** (3), 35 (1967).
92. J. H. Magill, Res. Dev. (London), **11**, 30 (1962).
93. A. Keller, G. R. Lester, L. D. Morgan, Phil. Trans. Roy. Soc. London, **247**, 1 (1954).
94. J. H. Magill, Polymer, **2**, 221 (1961), See also ref. 5, p.322; (b) Polymer, **3**, 35 (1962).
95. Z. Ding, J. E. Spruiell, J. Polym. Sci., Part B: Polym. Phy., **34**, 2783 (1996); P. Supaphol, J. E. Spruiell, *ibid*, indem., **36**, 681 (1998).
96. V. G. Baranov, Disc. Faraday Soc., **49**, 137 (1970).
97. W. H. Cobb, R. L. Burton, J. Polym. Sci., **10**, 276 (1953).
98. K. P. Mamedov, A. D. Nurieve, Soviet Phys. Crystallog. (Eng. Trans.), **12**, 605 (1968).
99. A. Peterlin, E. Roechl, J. Appl. Phys., **34**, 102 (1963).
100. R. W. Warfield, Makromol. Chem., **89**, 269 (1965).
101. R. Vignaud, J. M. Schultz, Polymer, **27**, 651 (1986).
102. R. C. Hirst, H. Y. Chen, Rubber Chem. Technol., **46**, 22 (1973).
103. See, for example, P. J. Barham, in Ref. 15, p. 153.
103a. J. K. Hobbs, T. J. McMaster, M. J. Miles, P. J. Barham, Polymer, **39**, 2437 (1998).
104. L. A. Wood, N. Bekkdahl, J. Res. Natl. Bur. Std., **36**, 487 (1946).
105. P. J. Flory, A. D. McIntyre, J. Polym. Sci., **18**, 592 (1955).
106. J. H. Magill, S. V. Peddada, G. McManus, Polym. Sci. Eng., **21**, 1 (1981).
107. R. P. Sheldon, J. Polym. Sci. B, **1**, 655 (1963).
108. See, for example, L. E. Alexander, "X-ray Diffraction Methods in Polymer Science", Chap. 3, Wiley Interscience, NY, 1969.
109. R. L. Miller, Encycl. Polym. Sci. Technol., **4**, 459 (1966).
110. L. Alexander, S. Oldberg, G. R. Taylor, J. Appl. Phys., **26**, 1068 (1955).
111. G. Farrow, I. M. Ward, Polymer, **1**, 330 (1960).
112. F. L. Binsbergen, Doctoral Thesis, "Nucleation in the Crystallization of Polyolefins", Univ. Groningen, 1969.
113. F. P. Price, Encycl. Polym. Sci. Technol., **8**, 63 (1969).
114. A. J. Kovacs, Ric. Sci., **25**, 668 (1955).
115. R. L. Cormia, F. P. Price, D. Turnbull, J. Chem. Phys., **37**, 1333 (1962).
116. A. Sharples, Polymer, **3A**, 250 (1962).
117. J. M. Schultz, R. W. Hendricks, J. Appl. Cryst., **11**, 551 (1978).
118. H. G. Zachmann, G. Wutz, in Ref. 15, p. 403.
119. A. Ryan, J. L. Stanford, W. Bass, T. M. W. Nye, Polymer, **38**, 759 (1997).
120. H. G. Zachmann, R. Gehrke, W. Prieske, C. Riekel, Proc. Symp. Natl. Lab., DESY, Hamburg, October 28–31, 1984.
121. J. D. Hoffman, J. J. Weeks, J. Chem. Phys., **37**, 1723 (1962).
122. P. J. Barham, A. Keller, E. L. Otun, P. A. Holmes, J. Mat. Sci., **19**, 2781 (1985).
123. W. Fronk. B. Heise, H. R. Schubach, W. Wilke, Colloid Polym. Sci., **262**, 99 (1984).
124. L. Mandelkern, Polym. Sci. Eng., **7**, 232 (1967).
125. J. M. Schultz, E. W. Fischer, O. Schaumberg, H. A. Zachmann, J. Polym. Sci., Polym. Phys. Ed., **18**, 239 (1980).
126. P. J. Flory, D. Y. Yoon, K. A. Dill, Macromolecules, **17**, 862 (1984).
127. J. F. Nagle, P. D. Gujrati, M. Goldstein, J. Phys. Chem., **88**, 4599 (1984).
128. R. C. Domszy, M. Glotin, L. Mandelkern, J. Polym. Sci., Polym. Symp., **71**, 151 (1984).
129. C. G. Vonk, J. Polym. Sci., C. Polym. Lett., **24**, 305 (1986).
130. C. M. Guttman, E. A. DiMarzio, J. D. Hoffman, Polymer, **22**, 1466 (1981).
131. J. Gardella, Jr., J. S. Chen., J. H. Magill, D. M. Hercules, J. Am. Chem. Soc., **105**, 4536 (1983).
132. R. L. Schmitt, J. A. Gardella, J. H. Magill, R. L. Chin, Polymer, **28**, 1243 (1987).
133. T. W. Ebbesen, Physics Today, **49** (6), 26 (1996).
134. See, for example, review by F. P. Price in: A. C. Zettlemoyer (Ed.), "Nucleation", Chap. 8, Dekker, New York, 1969.
135. F. C. Frank, M. Tosi, Proc. Roy Soc. A (London), **263**, 323 (1960).
136. F. P. Price, J. Polym. Sci., **42**, 49 (1960).
137. J. J. Point, Macromolecules, **12**, 770 (1974); J. J. Point, M. C. Colet, Am. Chim, Fr., **15**, 221 (1990).
138. S. S. Pollack. J. H. Magill, J. Polym. Sci. A-2, **7**, 551 (1969).
139. R. L. Miller, Kolloid Z. Z. Polym., **225**, 62 (1968).

140. D. H. Jones, A. J. Latham, A. Keller, M. Girolamo, J. Polym. Sci., Polym. Phys. Ed., **11**, 1759 (1973).
141. J. J. Point, J. J. Janimak, J. Crystal Growth, 131, 50 (1993).
142. P. J. Phillips in Ref. 15, p. 201 and others.
143. C. Devoy, L. Mandelkern, L. Bouland, J. Polym. Sci. A-2, **7**, 869 (1970).
144. J. H. Magill, J. Appl. Phys., **35**, 3249 (1964); J. H. Magill in: L. A. Kleintjens, P. J. Lemstra (Ed.), "Intergration of Fundamental Polymer Science and Technology", Elsevier, New York, 1986, p. 480.
144a. N. Okamato, M. Orguni, Solid State Comm., **99**, 1 (1996).
145. E. Urbanovici, H. A. Schneider and H. J. Cantow, J. Polym. Sci. Part B: Polym. Phys. Ed., **35**, 359 (1997).
146. C. R. Snyder, H. Marand, M. L. Mansfield, Macromolecules, **29**, 7508 (1996).
147. J. D. Hoffman, Polymer, **24**, 3 (1983).
148. E. Passaglia, E. A. DiMarzio, Polymer, **27**, 510 (1986).
149. R. L. Miller, E. G. Seeley, J. Polym. Sci., Polym. Phys. Ed., **20**, 2297 (1982).
150. G. J. Rensch, P. J. Phillips, N. M. Vatansever, and V. A. Gonzalez, J. Polym. Sci., Polym. Phys. Ed., **24**, 1943 (1986).
151. D. Roitman, H. Marand, R. L. Miller, J. D. Hoffman, J. Phys. Chem., **93**, 6919 (1989).
152. A. J. Lovinger, D. D. Davies, F. J. Padden, Jr., Polymer, **26**, 1595 (1985).
152a. J. D. Hoffman, R. L. Miller, Macromolecules, **21**, 3038 (1988).
153. J. M. Schultz, W. H. Robinson, G. M. Pound, J. Polym. Sci., Part: A-2, **5**, 511 (1967).
154. L. Mandelkern, J. M. Price, M. Goplan, J. G. Fatou, J. Polym. Sci, A-2, **4**, 385 (1966).
155. J. H. Magill, Macromol. Rapid Comm., **19**, 65 (1998); (also Macromol. Chem. and Phys., **11**, 2365 (1998).
156. M. Hikosaka, Polymer, **31**, 458 (1990).
157. A. Keller, M. Hikosaka, S. Rastogi, A. Toda, P. J. Barham, G. Goldbeck-Wood, J. Mat. Sci., **29**, 2579 (1994).
158. D. C. Bassett, B. Turner, Phil. Mag., **29**, 285, 295 (1974).
159. J. H. Magill, J. Inorganometallic Polym., **2**, 213 (1992).
159a. J. J. Point, Macromolecules, **30**, 1375 (1997).
160. A. Keller, J. Polym. Sci., Symp., **51**, 7 (1975).
160a. S. Z. D. Cheng, B. Wunderlich, J. Polym. Sci. Part B: Polym. Phys. Ed., **24**, 595 (1986).
161. T. Kajiyama, I. Ohki, A. Takahara, Macromolecules, **28**, 4768 (1995).
162. T. Ozawa, Polymer, **12**, 150 (1971); Thermochim. Acta, **100**, 109 (1986).
163. K. Nakamura, K. Imada, M. Takanayangi, Int. J. Polym. Mat., **2**, 71 (1972).
164. K. Harnisch and H. Muschik, Colloid Polym. Sci., **261**, 208 (1983).
165. A. Ziabicki, Appl. Polym. Symp., **6**, 1 (1967).
166. M. R. Mackley, A. Keller, Phil. Trans. Roy. Soc., **278**, 29 (1975).
167. C. H. Sherwood, F. P. Price, R. S. Stein, J. Polym. Sci., Polym., **63**, 77 (1978).
168. H. H. George, A. Holt, A. Buckley, Polym. Engrg. Sci., **23**, 95 (1983).
169. H. G. Kim, L. Mandelkern, J. Polym. Sci. A-2, **6**, 181 (1968).
170. R. Ono, K. Miyasaka, K. Ishikawa, J. Polym. Sci., Polym. Phys. Ed., **11**, 1477 (1973).
171. Y. Long, R. A. Shanks, Z. H. Stachurski, Prog. Polym., **20**, 651 (1995).
172. See for example, P. J. Flory, "Principles of Polymer Chemistry," Chapter 12, Cornell Univ. Press, Ithaca, 1953, p. 495.
173. K. Binder, Advances in Polymer Sci., **112**, 181 (1994).
174. L. A. Utracki, "Polymer Alloys and Blends" (Thermodynamics and Rheology) Carl-Hauser Verlag, Munich, 1989.
175. L. A. Utracki, "Encyclopedic Dictionary of Commercial Polymer Blends," Chem-Tec Publishing, Toronto, 1996.
176. A. B. Desia, G. L. Wilkes, J. Polym. Sci., Symp., **46**, 291 (1974).
177. N. K. Dvoyashkin, A. L. Maklakov, V. S. Smirnov, Polym. Sci. (USSR), (Engl. Trans.), **21**, 2985 (1979).
178. R. Legras, J. M. Dekoninck, A. Vanzieleghem, J. P. Mercier, E. Nield, Polymer, **27**, 109 (1986).
179. F. L. Binsbergen, J. Polym. Sci. C, **59**, 11 (1977).
180. J. Rault, E. Robelin, J. Phys., **43**, 1437 (1982).
181. H. A. Battaerd, J. Polym. Sci., Symp., **49**, 149 (1975).
182. H. M. Li, J. H. Magill, J. Polym. Sci., Polym. Phys. Ed., **16**, 1059 (1978).
183. H. W. Starkweather, P. Zoller, J. Polym. Sci. Polym. Phys. Ed., **29**, 751 (1982); see also P. Zoller, VI/475, "Polymer Handbook", 3rd Edition, Wiley NY, 1989.
184. M. Hikosaka, Jap. J. Appl. Phys., **20**, 617, (1981).
185. L. H. Palys, P. J. Phillips, J. Polym. Sci., Polym. Phys. Ed., **18**, 829 (1980).
186. B. C. Edwards, P. J. Phillips, D. Sorenson, J. Polym. Sci., Polym. Phys. Ed., **18**, 1737 (1980).
187. G. K. L. Davies, M. C. M. Cucarella, J. Mat. Sci., **15**, 1557 (1980).
188. T. Hatakeyama, H. Kanetsuna, H. Kaneda, T. Hashimoto, J. Macromol. Sci.-Phys. B, **10**, 359 (1974).
189. M. R. Kamal, P. G. Lafleur, Polym. Eng. Sci., **24**, 692 (1984).
190. R. Thomann, J. Kressler, B. Rudolf, R., Mülhaupt, Polymer, **37**, 2635 (1996).
190a. J. He, P. Zoller, J. Polym. Sci., Polym. Phys. Ed., **32**, 1049 (1994).
190b. C. A. Hieber, Int. Polym. Processing, **12**, 249 (1997).
190c. H. Ito, Y. Tsutsumi, K. Minagawa, J. Takimoto, K. Koyama, Colloid Poly. Sci., **273**, 811 (1995).
191. R. Reinitzer, Monatsh Chem., **9**, 421 (1888).
192. P. J. Flory, G. Ronca, Mol. Cryst. Liq. Cryst., **54**, 311 (1979).
193. P. G. DeGennes, "The Physics of Liquid Crystals" Clarendon Press, Oxford, 1974.
194. D. Acierno, A. A. Collyer (Eds.) "Rheology and Processing of Liquid Crystalline Polymers," Chapman and Hall, London, 1996.
195. See for example: Prog. Polym. Sci. (Reviews) **21**, pps. 775–1211 (1996).
196. L. A. Utracki, Ad. Chem. Series, **239**, 77 (1994).
197. S. Hanna, T. Lemmon, R. J. Spontak, A. H. Windle, Polymer, **33**, 3 (1992).

198. H. C. Langelaan, A. Posthuma de Boer, Polymer, **37**, 5667 (1996).
199. M. R. Mackley, A. Keller, J. Polymer Sci., Part B, Polym. Phys., **30**, 335 (1992); J. Mat. Sci., **10**, 1501 (1975).
200. M. D. Wolkowicz, J. Polym. Sci: Polym. Symp., **63**, 365 (1978).
201. J. Rietveld, A. J. McHugh, J. Polym. Sci., Polym. Letters, **21**, 919 (1983).
202. S. Liedauer, G. Eder, H. Janeschitz-Kriegl, P. Jerschow, W. Geymayer, E. Ingolic, Int. Polym. Processing, **8**, 236 (1993).
203. G. Eder, H. Janeschitz-Kriegl, G. Krobath, Progr. Colloid and Polym. Sci., **80**, 1 (1989).
204. A. J. McHugh and W. S. Yung, J. Polym. Sci., Part B: Polym. Phys., **27**, 431 (1989).
205. A. C. Bushman and A. J. McHugh, J. Polym. Sci., Part B: Polym. Phys., **34**, 2393 (1996).
206. A. N. Beris, B. J. Edwards, "Thermodynamics of Flowing Systems," Oxford Press, NY, 1994.
207. J. D. Hoffman, Polymer, **20**, 1071 (1979).
208. A. J. Pennings, A. M. Kiel, Kolloid Z. u. Z. Polym., **205**, 160 (1965).
208a. See for example, Jan Smook, Ph D Thesis, "Preparation and Properties of Ultra-High Strength Polyethylene Fibers", University of Groningen, June 1984.
208b. M. Kunz, M. Dreshler, M. Möller, Polymer, **36**, 1331 (1995).
209. T. Takashi, M. Ishamura, I. Ishimoto, Polym. Lett., **8**, 651 (1970).
210. J. C. Whittmann, B. Lotz, J. Polym. Sci., Polym. Phys. Ed., **23**, 205 (1985).
211. S. Yan, J. Petermann and D. Yang, Polym. Bull., **38**, 87 (1997).
212. A. J. Greso, P. J. Phillips, Polymer, **35**, 3373 (1994).
213. T. W. Owen, D. Hull, Plast. Polym., **42**, 19 (1974).
214. J. K. Su, J. Mark, Macromolecules, **10**, 120 (1977).
215. P. J. Flory, J. Chem. Phys., **15**, 3976 (1947).
216. R. Gaylord, J. Polym. Sci., Polym. Phys. Ed., **14**, 1827 (1976).
217. R. G. Alamo, E. K. M. Chan, L. Mandelkern, I. G. Voigt-Martin, Macromolecules, **25**, 6381 (1992).
218. I. C. Sanchez, J. Polym. Sci., Part C, **59**, 109 (1977).
219. J. VanRuiten, F. vanDieren, V. B. F. Mathot, in Ref. 15, p. 481.
220. G. Goldbeck-Wood in Ref. 15, p. 249.
221. G. Goldbeck-Wood in: E. A. Colbourn (Ed.) "Computer Simulation of Polymers," Longman, London, 1994.
222. J. N. Hay, Z. J. Przekop, J. Polym. Sci., Polym. Phys. Ed., **17**, 951 (1979).
223. R. J. Tabar, A. Wasiak, S. D. Hong, T. Yuana, R. S. Stein, J. Polym. Sci., Polym. Phys. Ed., **19**, 49 (1981).
224. K. W. Mahin, K. Hanson, J. W. Morris, Acta. Metall., **28**, 443 (1980).
225. N. Billon, J. M. Escleine, J. M. Haudin, Colloid Polym. Sci., **267**, 668 (1989).
226. N. Billon, J. M. Haudin, Ann. Chim., **15**, 1 (1990).
227. N. A. Mehl, L. Rebenfeld, J. Polym. Sci., Part B: Polym. Phys., **31**, 1677, 1687 (1993); **57**, 187 (1995).
228. M. Negahban, ASME, AMD-Vol. 203: "Current Research in the Thermo-Mechanics of Polymers in the Rubbery-Glassy State", 1995, p. 45.
229. M. Negahban, J. Eng. Sci., **35** (3), 277 (1997).
230. M. Negahban, A. S. Wineman, R-J. Ma, Int. J. Engrng. Sci., **31**, 93 (1993).
231. A. Stevenson (1989) in: N. P. Cheremisinoff, (Ed.), "Crystallization of Elastomers" vol. 2, Marcel Dekker, NY, p. 61.
232. C. M. Hsuing, M. Cakmak, Polym. Eng. Sci., **31**, 1372 (1991).
233. S. J. Organ, A. Keller, G. Ungar, Ref. 15, p. 81.
234. S. J. Sutton, A. S. Vaughan, D. C. Bassett, Polym. Comm., **37**, 5735 (1996).
235. B. J. Barham, Amer. Chem. Soc., Polym. Preprints, April, **36**, 301 (1995).
236. A. Toda, H. Kiko, J. Polym. Sci., Part B: Polym. Phys., **27**, 53 (1989).
237. R. M. Gohil, K. C. Patel, R. D. Patel, Polymer, **15**, 402 (1974).
238. E. H. Andrews, P. J. Owens, A. Singh, Proc. Soc. A (Lond.), **234**, 79 (1971).
239. P. J. Phillips, N. Vatansever, Macromolecules, **20**, 2138 (1987).
240. G. J. Rensch, P. J. Phillips and N. Vatansever, J. Polym. Sci., Part B: Polym. Phys., **24**, 1943 (1986).
241. E. Fischer, J. F. Henderson, J. Polym. Sci. A-2, **5**, 377 (1967).
242. C. K. L. Davies, A. J. Carter, M. C. M. Cucarella, Ong. Eng. Long, K. M. Pickard and S. V. Wolfe, Progress in Rubber and Plastics Technology, **5** (2), 112 (1989).
243. Ni Ding, E. J. Amis, Macromolecules, **24**, 3906 (1991).
244. M. Johnsen, J. Lehmann, Kolloid Z. Z. Polym., **230**, 317, 325 (1969).
245. S. Z. D. Cheng, J. Chen, S. X. Wu, A. Zhang, M. A. Yandrasits, Q. Zhuo, R. P. Quirk, A. Habenschuss and P. R. Zschak, Ref. 15, p. 51.
246. X. Gao, E. J. Amis, Amer. Chem. Soc. Polymer Preprints, April, **36**, 329 (1995).
247. I. Campoy, C. Marco, M. A. Gomez, J. G. Fatou, Macromolecules, **25**, 4392 (1992).
248. I. Campoy, M. A. Gomez, C. Marco, J. G. Fatou, Eur. Polym. J., **29**, 1161 (1993).
249. J. Powers, J. D. Hoffman, J. J. Weeks, F. A. Quinn Jr., J. Res. Natl. Bur. Std., **69A**, 335 (1965).
250. I. Campoy, J. M. Arribas, M. A. Zapoita, C. Marco, M. A. Gomez, J. G. Fatou, Eur. Polym. J., **31**, 475 (1995).
251. R. D. Icenogle, J. Polym. Sci., Polym. Phys. Ed., **23**, 1369 (1985).
252. F. P. Price, J. Chem. Soc., **74**, 311 (1952).
253. A. J. Lovinger, J. Polym. Sci., Polym. Phys. Ed., **18**, 793 (1980).
254. H. T. Tseng, P. J. Phillips, Macromolecules, **18**, 1565 (1985).
255. J. D. Hoffman, L. T. Frolen, G. S. Ross, J. I. Lauritzen Jr., J. Res. Nat. Bur. Std., **79a**, 671 (1975).
256. F. P. Price, J. Phys. Chem., **64**, 169 (1960).
257. W. Banks, J. N. Hay, A, Sharples, G. Thompson, Polymer, **5**, 163 (1964).

258. T. Naono, J. Sci. A, (Hiroshima Univ.), **24**, 653 (1960).
259. J. D. Hoffman, G. S. Ross, L. J. Frolen, private communication.
260. S. Hoshino, F. Meineek, J. Powers, R. S. Stein, J. Polym. Sci. A, **3**, 3041 (1965).
261. B. von Falkai, H. A. Stuart, Kolloid-Z., **162**, 138 (1959).
262. H. D. Keith, F. J. Padden Jr., J. Appl. Phys., **35**, 1270, 1286 (1964).
263. L. Marker, P. M. Hays, G. P.Tilley, R. M. Early, O. J. Sweeting, J. Polym. Sci., **38**, 107 (1959).
264. S. Z. D. Cheng, B. Wunderlich, J. Polym. Sci. B, Polym. Phys, **24**, 577 (1986).
265. E. Ratajski, PhD Thesis, Linz University, Linz, Austria, 1993; H. Janeschitz-Kriegel, E. Fleischmann, W. Geymayer in: J. Karger-Kocsis (Ed.), "Polypropylene: Structure, Blends and Composites A", Chapman and Hall, London, 1995, p. 295.
266. See Ref. #6, p.49.
267. M. Chen, J-Y. Chen, J. Polym. Sci., Polymer Phys., **36**, 1335 (1998).
268. A. S. Kenyon, R. C. Gross, L. Wurstner, J. Polym. Sci., **40**, 159 (1959).
269. J. Boon, G. Challa, D. W. van Krevelen, J. Polym. Sci. A-2, **6**, 1835 (1968).
270. T. Suzuki, A. J. Kovacs, Polym. J., **1**, 82 (1970).
271. T. Suzuki, A. Kovacs, Polym. J., **1**, 82 (1970); see also R. L. Miller (Ed.), "Flow Induced Crystallization", Breach Science Pub. Ltd., London, 1979, p. 31.
272. B. C. Edwards, P. J. Phillips, Polymer, **15**, 351 (1974).
273. A. de Boer, G. O. R. Alberda van Ekenstein, G. Challa, Polymer, **16**, 930 (1975).
274. G. O. R. Alverda van Ekenstein, Y. Y. Tan, G. Challa, Polymer, **26**, 283 (1985).
275. Z. Bartczak, A. Geleski, M. Pracella, Polymer, **27**, 537 (1986).
276. E. Martuscelli, C. Sellitti, C. Silvestre, Makromol. Chem., Rapid Comm., **6**, 125 (1985).
277. E. Martuscelli, Polym. Eng. Sci., **24**, 563 (1984)
278. E.Martuscelli, Polymer, **25**, 1099 (1984).
279. G. S. Y. Yeh, S. L. Lambert, J. Polym. Sci. A-2, **10**, 1183 (1972).
280. Z. Bartczak, A. Galeski, Polymer, **31**, 2027 (1990).
281. J. Petermann, H. Gleiter, J. Polym. Sci. Phys. Ed., **11**, 359 (1973).
282. Y. Long, Z. H. Stachurski and R. A. Shanks, Polymer. Internat., **26**, 143 (1991).
283. P. J. Phillips, B. C. Edwards, J. Polym. Sci., Polym. Phys. Ed., **13**, 1819 (1975).
284. J. N. Hay, P. A. Fitzgerald, M. Wiles, Polymer, **17**, 1015 (1976).
285. B. C. Edwards, P. J. Phillips, Polymer, **15**, 491 (1974).
286. E. G. Lovering, J. Polym. Sci. C, **30**, 329 (1970).
287. E. G. Lovering, J. Polym. Sci. A-2, **8**, 747 (1970).
288. J. Huang, H. Marand, Macromolecules, **30**, 1069 (1997).
289. M. Inoue, J. Polym. Sci., **61**, 343 (1962).
290. E. Perez, A. Bello, J. G. Fatou, Makromol. Chem., **186**, 439 (1985).
291. W. J. Barnes, W. G. Luetzel, F. P. Price, J. Phys. Chem., **65**, 80 (1961).
292. S. F. Dzhaliov, J. Phys. Chem. (USSR), **39**, 10, 1376 (1965).
293. F. Rybnikar, Coll. Czech. Chem. Commun., **31**, 4080 (1966).
294. F. De Candia, V. Vittoria, Makromol. Chem., **155**, 17 (1972).
295. D. R. Beech, C. Booth, I. H. Hillier, C. J. Pickles, Eur. Polym. J., **8**, 799 (1972).
296. A. J. Kovacs, C. Straupe, A. Gonthier, J. Polym. Sci. (Polym. Symp.), **59**, 31 (1977).
297. R. Archambault, R. E. Prud'homme, J. Polym. Sci. Polym. Phys. Ed., **18**, 35 (1980); **27**, 537 (1986).
298. L. Crispino, E. Martuscelli, M. Pracella, Macromol. Chemie, **181**, 1747 (1980); See also J.G. Fatou in Ref. 332.
299. C. Booth, D. V. Dodgson, I. H. Hillier, Polymer, **11**, 11 (1970).
300. S. Aggarwal, L. Marker, W. L. Kollar, R. Geroch, J. Polym. Sci. A-2, **4**, 715 (1966).
301. J. H. Magill, Makromol. Chem., **86**, 283 (1965).
302. L. Paternostre, M. Dosiere, D. Villiers, P. Damman, Cl. Bourgaux in Ref. 15, p. 549.
303 P. Damman and J. J. Point, Ref 15, p. 539.
304. W. Banks, M. Gordon, A. Sharples, Polymer, **4**, 61 (1963).
305. E. Martuscelli, Polymer, **25**, 1099 (1984).
306. K. S. Sastry, R. D. Patel, Eur. Polym. J., **8**, 63 (1972).
307. G. C. Alfonso, T. P. Russell, Macromolecules, **19**, 1143 (1986).
308. J. Huang, A. Prasad, H. Marand, Polymer, **35**, 1896 (1994).
309. C. del Rio, J. L. Acosta, Polym., Bull., **38**, 63 (1997).
310. B. von Falkai, W. Rellensmann, Makromol. Chem., **75**, 112 (1964).
311. M. Takayanagi, Mem. Fac. Eng. Kyushu. Univ., **16** (3), 111 (1957).
312. P. J. Phillips, H. T. Tseng (private communication).
313. M. Takayanagi, N. Kusumoto, J. Chem. Soc. Japan (Ind. Chem. Sect.), **62**, 587 (1959).
314. F. D. Hartley, F. W. Lord, L. B. Morgan, Proc. Roy. Soc. A (London), **247**, 23 (1954).
315. F. van Antwerpen, Doctoral Thesis, "Kinetics of Crystallization Phenomena of Spherulites in Polyethyleneterephthalate", Delft Technical Univ. 1971.
316. R. Vasanthakumari, A. J. Pennings, Polymer, **24**, 175 (1983).
317. A. Sharples, F. L. Swinton, Polymer, **4**, 119 (1963).
318. C. J. Ong, F. P. Price, J. Polym. Sci., Polym. Symp., **63**, 59 (1978).
319. R. Pearce, G. R. Brown, R. H. Marchessault, Polymer, **35**, 3984 (1994).
320. N. Okui (private communication).
321. J. H. Magill, J. Polym. Sci., Part A, **3**, 1995 (1965).
322. J. H. Magill, Polymer, **6**, 367 (1965).
323. F. D. Hartley, F. W. Lord, L. B. Morgan, Ric. Sci. Suppl. A, **25**, 577 (1955).
324. J. G. Fatou, E. Riande, R. Garcia Valdecasas, J. Polym. Sci., **13**, 2103 (1975).
325. F. P. Price, "Fundamental Phenomena in the Materials Sciences", Surface Phenomena Chem. Biol., **3**, 85 (1966).

326. E. H. Boasson, J. M. Woestenenk, J. Polym. Sci., **24**, 57 (1957).
327. F. Khoury, J. Polym. Sci., **33**, 389 (1958).
328. J. V. McLaren, Polymer, **4**, 175 (1963).
329. B. B. Burnett, W. F. McDevit, J. Appl. Phys., **28**, 1101 (1957).
330. J. V. McLaren, Polymer, **4**, 175 (1963).
331. C. R. Lindegren, J. Polym. Sci., **50**, 181 (1961).
332. J. G. Fatou, C. Marco, L. Mandelkern, Polymer, **31**, 1685 (1976); See also J. G. Fatou, "Morphology and Crystallization of Polyolefins", Chap.8, in Handbook of Polyolefins, M. Dekker, New York, 1993, p. 155.
333. J. H. Magill, J. Appl. Phys., **35**, 3249 (1964).
334. E. Riande, J. G. Fatou, Polymer, **17**, 795 (1976).
335. W. Wang, J. M. Schultz and B. S. Hsiao, J. Polym. Sci., Part B: Polymer Phys., **34**, 3095 (1996).
336. E. Riande, J. G. Fatou, Polymer, **17**, 99 (1976).
337. B. G. Risch, S. Srivavas, G. L. Wilkes, J. F. Geibal, C. Ash, S. White, M. Hicks, Polymer, **37**, 3623 (1996).
338. L. C. Lopez, G. L. Wilkes, J. E. Giebel, Polymer, **30**, 147 (1989).
339. M. Kyotani, H. Kanestuna, J. Poly. Sci., Phys. Ed., **12**, 231 (1974).
340. C. J. Plummer, P. Menu, N. Cudre-Mauroux, H. H. Kausch, J. Appl. Polym. Sci., **55**, 489 (1995).
341. E. Morales, M. Salmeron, J. L. Costa, J. Polym. Sci., Part B: Polym. Phys., **34**, 2715 (1996).
342. G. Feio, J. P. Cohen-Addad, J. Polym. Sci., Part B: Polym. Phys., **26**, 389 (1988).
343. Yu K. Godovsky, G. L. Slonimsky, J. Polym. Sci., Polym. Phys. Ed., **12**, 1053 (1974).
344. J. C. Hser, S. H. Carr, Polym. Eng. Sci., **19**, 436 (1979).
345. G. Feio, G. Buntinx, J. P. Cohen-Addad, J. Polym. Sci., Part B: Polym. Phys., **27**, 1 (1989).
346. F. Rybnikar, Coll. Czech. Chem. Commun., **27**, 2307 (1962).
347. L. Mandelkern, in: R. H. Doremus, B. W. Roberts, D. Turnbull (Eds.), "Growth and Perfection of Crystals", Wiley, New York, 1958, p. 467.
348. D. R. Brown, J. Jones, J. Polym. Sci., Polym. Phys. Ed., **22**, 655 (1984).
349. D. M. Hoffman, B. M. McKinley, Polym. Eng. Sci., **25**, 567 (1985).
350. W. S. Lambert, P. J. Phillips, Polymer, **37**, 3585 (1996).
351. A. Aref-Azar, J. N. Hay, J. Marsden, N. Walker, J. Polym. Sci., Polym. Phys. Ed., **18**, 637 (1980).
352. B. Pena, J. A. Delgado, A. Bello, E. Perez, Polymer, **35**, 3039 (1994).
353. J. F. Oth, P. J. Flory, J. Am. Chem. Soc., **80**, 1927 (1958).
354. L. Mandelkern, F. A. Quinn, Jr., D. E. Roberts, J. Am. Chem. Soc., **78**, 926 (1956).
355. W. Cooper, G. Vaughan, Polymer, **4**, 329 (1963).
356. M. Johnsen, G. Nachtrieb, H. G. Zachmann, Kolloid Z. u Z. Polymere, **240**, 756 (1970).
357. E. G. Lovering J. Polym. Sci. A, **2**, 2197d (1970).
358. R. Ono, K. Miyasaka and K. Ishikawa, J. Polym. Sci., Polym. Phys. Ed., **11**, 1477 (1973).
359. N. Overbergh, D. Sadler, A. Keller, J. Polym. Sci., **15**, 1485 (1977).
360. E. B. Orler, B. H. Calhoun, R. B. Moore, Macromolecules, **29**, 5965 (1996).
361. S. Cimmino, E. DiPace, E. Martuscelli, C. Silvestre in Ref. 15, p. 527.
362. V. Balsamo, F. von Gyldenfeldt, R. Stadler, Macromol. Chem. Phys., **197**, 3317 (1996).
363. J. H. Griffith, B. G. Ranby, J. Polym. Sci., **38**, 107 (1959).
364. J. K. Shah, L. E. Lahti, Ind. Eng. Chem. Prod. Res. Dev., **12**, 304 (1973).
365. A. Celli, E. D. Zanotto, Thermochimica Acta, **260/270**, 191 (1995); Data courtesy of A. Celli (in 1997), "Montell G. Natta Research Center", 44100 Ferrara, Italy.
366. M. J. Galante, L. Mandelkern, R. G. Alamo, A. Lehtinen, R. Paukkeri, J. Thermal Analysis, **47**, 913 (1996).
367. F. Avalos, M. A. Lopez-Manchado, M. Arroyo, Polymer, **37**, 5681 (1996).
368. S. Laihonen, U. W. Gedde, P. E. Werner and J. Martinez-Salazar, Polymer, **38**, 361 (1997).
369. J. Q. G. MacLaine, C. Booth, Polymer, **16**, 680 (1975).
370. Yu K. Godovsky, G. L. Slonimsky, N. M. Garbar, J. Polym. Sci. C, **38**, 1 (1972).
371. C. C. Chu, Polymer, **21**, 1480 (1980).
372. M. Neron, A. Tardif, R. E. Prudhomme, Eur. Polym. J., **12**, 605 (1976).
373. M. Gilbert, F. J. Hybart, Polymer, **13**, 327 (1972).
374. E. Turska, H. Janeczek, Polymer, **20**, 855 (1979).
375. P. C. Vilanova, S. M. Rivas, G. M. Guzman, Polymer, **26**, 423 (1985).
376. B. Gunther, H. G. Zachmann, Polymer, **24**, 1008 (1983).
377. G. C. Alfonso, M. P. Verdona, A.Wasiak, Polymer, **19**, 711 (1978).
378. G. S. Fielding-Russell, P. S. Pillai, Kolloid Z., Z. Polymer, **250**, 2 (1972).
379. L. Bouland, Plast. Eng., **43**, 39 (1987).
380. C. Borri, S. Brucker, V. Creseenzi, G. Della Fortuna, A. Mariano, P. Scarazzato, Eur.Polym. J., **7**, 515 (1971).
381. R. C. Domszy, M. Glotin, L. Mandelkern, J. Polym. Sci., Polym. Symp., **71**, 151 (1984).
382. Y. Yu, K-J. Choi, Polym. Eng. Sci., **37**, 91 (1997).
383. G. Gurato, D. Gaidano, R. Zannetti, Makromol. Chem., **79**, 231 (1978).
384. B. G. Risch, G. L. Wilkes, J. M. Warakomski, Polymer, **34**, 2328 (1993).
385. F. Maneschalchi, R. Rossi, A. Mathussi, Eur. Polym. J., **9**, 601 (1973).
386. W. H. Jo, D. H. Baik, J. Polym. Sci., Part B: Polym. Phys., **27**, 673 (1989).
387. J. H. Magill, Polym. Lett., **6**, 853 (1968).
388. J. H. Magill, J. Polym. Sci. A-2, **5**, 89 (1967).
389. T. Masuko, R. Okuizumi, K. Yonetake, J. H. Magill, Macromolecules, **22**, 4636 (1989).
390. C. Marega, A. Marigo, V. D. Noto, R. Zannetti, Macromol. Chem., **193**, 1599 (1992).
391. J. J. Kolstad, J. Appl. Polym., Sci., **62**, 1079 (1996).

392. N. Bulakh, J. P. Jog, J. Macromol. Sci. B: Phys., **34**, 15 (1995).
393. G. D. Desio, L. Rebenfeld, J. Appl. Polym. Sci., **45**, 2005 (1992).
394. G. P. Desio, L. Rebenfeld, J. Appl. Polym. Sci., **44**, 1989 (1992).
395. V. E. Reinsch, L. Rebenfeld, J. Appl. Polym. Sci., **59**, 1913 (1996).
396. V. E. Reinsch, L. Rebenfeld, J. Appl. Polym. Sci., **59**, 1929 (1996).
397. V. E. Reinsch, L. Rebenfeld, J. Appl. Polym. Sci., **59**, 1939 (1966).
398. R. D. Wesson, Polym. Eng. Sci., **34**, 1157 (1994).
399. Y. F. Wang, D. R. Lloyd, Polymer, **34**, 2324 (1993).
400. T. Sasaki, M. Kurita, T. Yabu, T. Takahashi, Macromolecules, **28**, 8528 (1995).
401. W. Kozlowski, J. Polym. Sci., Part C, **#38**, 47 (1972).
402. D. J. Blundell, B. N. Orbon, Polymer, **24**, 953 (1983).
403. Y. Yokoyama, T. Ricco, J. Applied Polym. Sci., **66**, 1007 (1997).
404. C-S. Wang, C-S. Lin, - Lai, Polymer Bull., **39**, 185 (1997).
405. T. Miyata, T. Masuko, Polymer, **38**, 4003 (1997).
406. R. J. Ciora, Jr., J. H. Magill, Macromolecules, **23**, 2350 (1990).
407. R. J. Ciora. Jr., J. H. Magill, Macromolecules, **23**, 2359 (1990).
408. E. M. Woo, W. J. Chang, C-H. Hong, C-Y., Chen, Macromol. Chem. Phys., **197**, 2359 (1996).
409. E. Martuscelli, M. Pracella, W. P. Yue, Polymer, **25**, 1099 (1984).
410. J. H. Lee, S.-G. Lee, K.-Y. Choi, J. Liu, Polymer J. **30**, 531 (1998).
411. S. S. Wu, D. S. Kalika, R. R. Lamonte, S. Makhija, J. Macromol, Sci. B: Phys., **35**, 157 (1996).
412. S. W. Lee, M. Cakmak, J. Macromol. Sci. B, **37**, 501 (1998).
413. J.-C. Lee, H. Tazawa, T. Ikehara, T. Nishi, Polymer J., **30**, 327 (1998).
414. S. Hamada, T. Sato, T. Shirai, Bull. Chem. Soc. Japan, **40**, 864 (1967).

Isomorphous Polymer Pairs

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A. INTRODUCTION

This chapter is intended to provide a list of known pairs of isomorphous monomeric units in synthetic crystalline polymer systems; we shall regard as "isomorphism" the statistical copresence of different units within the crystallites. Additional information will be also given, namely:

(a) the approximate range of relative concentrations where isomorphism is observed;
(b) the chain conformation in the crystalline state, the type of isomorphism (see below);
(c) the physicochemical techniques used in the investigation (see Section B); and
(d) the references to the scientific contributions wherein the isomorphism is demonstrated, implied, or claimed.

Systems in which the molar fraction of either type of monomeric units does not exceed about 5% have been disregarded. Cocrystallization of hydrogenated and deuterated units has not been considered as relevant in this context either.

Although we are aware of the growing expansion and importance of copolymer systems producing mesophases (92), we have chosen to limit our list to isomorphism in the classical crystallographic sense. In particular, polymer systems were regarded as crystalline either (i) when off-meridional X-ray reflections appear on the nonequatorial layers, or (ii) if the sample is shown to undergo a transition to a mesophase system before melting, as may be revealed by DSC methods. Sometimes distinctions between real crystalline systems and mesophases may be difficult, such as in the case of several polyesters. In all instances in which some degree of uncertainty is implied, an asterisk has been inserted.

The isomorphous polymer pairs are subdivided according to the scheme already proposed in a review article on the subject (2). Two basic classes of macromolecular isomorphism were distinguished, namely isomorphism of whole homopolymer chains (Class 1) and isomorphism of monomeric units within copolymer chains (Class 2) (2,62,63). Class 1 may be further subdivided into three different subclasses: subclass 1.1, where the monomeric units differ in chemical constitution (copolymers in the usual sense) (Table 1.1); subclass 1.2, where the monomeric units have randomly opposite, nonsuperimposable configurations (Table 1.2); and subclass 1.3, where the monomeric units, although stereochemically identical, assume different conformations at random. In spite of the scientific relevance of the last subclass in the context of polymer structure, we have preferred to omit the corresponding table, in view of both, the relatively low relevance of the issue for many people interested in polymer isomorphism, and of the wide ensemble of polymers that are presumably affected by this crystallographic disorder under appropriate circumstances. Also, for the sake of greater generality, we have included in subclass 1.2 (Table 1.2) all the cases where the isomorphous units are related either by geometrical stereoisomerism or by random head-to-head, head-to-tail enchainment. Concerning Class 2, the subclass consisting of stereochemically identical chains with randomly opposite orientations of their axes within the crystal is not

reported for the same reasons given above for subclass 1.3. Consequently, only isomorphous systems consisting of stereochemically different homopolymer chains are reported. In each case, the two units of the isomorphous pair are separated by a slash.

From a crystallographic viewpoint the following two types of polymer isomorphism may be distinguished (the different monomeric units will be designated as A and B).

Type 1: Homopolymers A_N and B_M have a similar crystal structure, with sufficiently small differences in unit cell dimensions. Binary systems, whether copolymers or homopolymer mixtures, display the same structure at any intermediate composition, except for a continuous change in cell dimensions (isomorphism in a true sense). The homopolymer chains may possess helical conformations with slightly different ratios of monomeric units per helix pitch, in which case intermediate helix conformations should appear at intermediate compositions.

Type 2: The two homopolymers have different crystal structures and in each of them some amount of monomeric units of the other may be accommodated. Accordingly, the structure of the homopolymer corresponding to the monomeric units with prevailing amount is observed, and an abrupt change in the structure of the second homopolymer may take place at a suitable composition. The simultaneous presence of both structures may also be observed within some intermediate composition range, in analogy with low-molecular weight systems (isodimorphism).

These types correspond to the first two types of polymer isomorphism suggested by Natta (62,63), who also proposed a third type whenever one of the two homopolymers is unable to crystallize by itself. We have thought it expedient to include these caseswithin Type 2, particularly in view of the difficulty in ascertaining the complete inability of either homopolymer to crystallize. It should be added in this context that no assumption concerning thermodynamic stability of the following isomorphous systems is implicit here; on the other hand, the crystal packing and friction forces may well be strong enough in most cases as to prevent any appreciable solid–solid transformation.

Concerning the partition of the data within each separate heading (1.1, 1.2, etc.), we tried to conform to the standard criteria adopted in this book. Each monomeric unit is assigned to the class corresponding to the higher ranking monomer.

Whenever a crystallizable copolymer is constituted by a regularly repeating sequence of monomeric units, isomorphism is usually ruled out in view of the very presence of constitutionally repeating units, i.e., of an effective homopolymer.

Several vinyl copolymers show macromolecular isomorphism when the side groups are sterically ordered. If, as it always happens in the examples known hitherto, the order is of the isotactic type, we have called them "isotactic" (copolymers).

Several cases are known of polymers having long, regular side groups (comb-like copolymers or the like) that may promote crystallization in spite of lack of configurational order in the main chain. These systems are not considered as belonging to the present list if crystallization is basically due to regular packing of the side chains only.

Whenever the conformation of the chain skeleton may be described as a regular helix, we have used the symbolism recommended by the document "Terminology Relating to Crystalline Polymers" of IUPAC (1984) and adopted, e.g., by R. L. Miller in the chapter "Crystallographic Data for Various Polymers" of this book. As an example, the helical conformation of the crystalline isotactic copolymers of 3-methyl-1-butene and 4-methyl-1-pentene (see Ref. 14), corresponding to four monomeric units in one turn, each unit having two skeletal atoms, is designated as $2*4/1$. However, referring to the general notation p/q, we have not considered as helical those conformations where both p and q are equal to unity (i.e., one repetitive unit in one turn). In these instances, qualifications such as "extended", "planar zig-zag", etc. are used to characterize the conformation. Also, whenever the helix structure is not explicitly reported by the author, the general term "helical" is adopted.

It may happen that, for a given concentration of the two monomeric units, more than one crystalline phase is observed. Under these circumstances, the corresponding chain conformations are reported.

For a discussion of macromolecular isomorphism, see Refs. 2,90,91, and some of the papers quoted therein.

B. TECHNIQUES

XR	Qualitative evaluation of crystallinity by diffraction methods
XR1	Structural analysis by diffraction methods (including observation of continuous changes of lattice spacings)
MP	Optically determined melting point
TA	Thermal analysis (DSC, DTA, etc.)
DM	Dynamical–mechanical measurements (e.g., mechanical and dielectric relaxation experiments)
MA	Morphological analysis (electron microscopy, SAXS etc.)
D	Density determination by flotation techniques
IR	Infrared spectroscopy
ED	Electron diffraction
CA	Conformational analysis

C. TABLES OF ISOMORPHOUS PAIRS OF MONOMER UNITS

TABLE 1. ISOMORPHOUS UNITS WITHIN THE SAME MACROMOLECULES (COPOLYMERS)

Monomer giving isomorphous copolymers	Mol% of first component	Type of isomorphism	Chain conformation	Techniques	Refs.
1.1. ISOMORPHOUS UNITS WITH DIFFERENT CHEMICAL CONSTITUTION					
1.1.1. MAIN-CHAIN CARBON COPOLYMERS					
Copoly(alkenamers)					
trans-1-Pentenamer/*trans*-1-octenamer	100–0	1	Extended, planar zig-zag	XR1	20,57,60
trans-1-Pentenamer/*trans*-1-heptenamer	100–0	1	Extended, planar zig-zag	XR1	20,57,60
trans-1-Octenamer/*trans*-1-dodecenamer	100–0	1	Extended, planar zig-zag	XR1	20,57,60
Copoly(alkenes)					
Ethylene/propylene	100–70	2	Planar zig-zag	MP, XR1, XR, TA	5,6,71,77, 8,93, 77
Propylene/*n*-1-hexene	100–50	2	Isotactic (2*3/1)*	XR, TA, DM	67
Propylene/*n*-1-butene	89–3	2	Isotactic (2*3/1)	XR1, MP	84
	100–0	2	Isotactic (2*3/1)	XR1, MP, MA	41
n-1-Butene/3-methyl-1-butene	100–0	1	Isotactic (2*4/1) and (2*29/8)	XR1	83
n-1-Butene/*n*-1-pentene	94–16	1	Isotactic (2*3/1)	XR1, MP	84
n-1-Butene/*n*-1-hexene	100–30	2	Isotactic (2*3/1) and (2*29/8)	XR1, MP	84
n-1-Butene/4-methyl-1-pentene	100–75	2	Isotactic (2*3/1) and (2*29/8)	XR1, MP	84
	75–50	2	Isotactic (2*3/1) and (2*7/2)		
	50–0	2	Isotactic (2*7/2)		
n-1-Butene/*n*-1-octene	100–50	2	Isotactic (2*3/1) and (2*29/8)	XR1, MP	84
n-1-Butene/*n*-1-decene	100–90	2	Isotactic (2*3/1) and (2*29/8)	XR1, MP	84
n-1-Butene/*n*-1-dodecene	100–90	2	Isotactic (2*3/1) and (2*29/8)	XR1, MP	84
3-Methyl-1-butene/4-methyl-1-pentene	100–50	2	Isotactic (2*4/1)	XR1, MP	70
	50–0	2	Isotactic (2*4/1) and (2*7/2)		
4-Methyl-pentene/1-pentene	50–100	2	Isotactic (2*4/1)	XR1, MP,	85
			(2*4/1)	XR1	15
	0–50	2	Isotactic (2*7/2)	TA, DM	41
4-Methyl-1-pentene/*n*-1-hexene	100–20	2	Isotactic (2*7/2)	TA, DM	41
4-Methyl-1-pentene/4-methyl-1-hexene	100–0	1	Isotactic (2*7/2)	XR1, MP	3
4-Methyl-1-pentene/*n*-1-octene	100-70	2	Isotactic (2*7/2)	XR1, MP	85
4-Methyl-1-pentene/*n*-1-decene	100-85	2	Isotactic (2*7/2)	XR1, MP	85
Copoly(vinyls) (nonhalogenated)					
Ethylene/vinyl alcohol	100–0	2	Planar zig-zag	XR1	12
				XR1, DM	58
Ethylene/(ethylene + CO)	100–50	1	Planar zig-zag	XR1	11,17,18
Isopropyl vinyl ether/*sec*-butyl vinyl ether	100–0	1	Isotactic (2*17/5)	ED, MA	1
				XR, MP	3
Copoly(vinyls) (halogenated)					
Ethylene/tetrafluoroethylene	100–0	2	Planar zig-zag	XR1, TA, MA	34
				XR1	88,55
				CA	29
Vinyl fluoride/vinylidene fluoride	100–0	1	Planar zig-zag	XR1, MP	61
Vinyl fluoride/tetrafluoroethylene	100–20	2	Planar zig-zag	XR1, MP	61
Vinylidene fluoride/trifluoroethylene	100–90		Planar zig-zag	XR, TA, DM	94
	88–15	2	$(TGTG')_n$		
	15–0		Helical		
	52		Planar zig-zag, (2*3/1)	XR1	53
	55		Planar zig-zag, $(TGTG')_n$	XR1, IR	80
	100–0		Planar zig-zag, $(TGTG')_n$	XR1	52
Vinylidene fluoride/tetrafluoroethylene	100–0	2	Planar zig-zag	XR1, TA, MA	34
				XR1	55
Vinylidene fluoride/chlorotrifluoroethylene	100–0	2	Planar zig-zag, (2*17/1)	XR1	55
Vinylidene fluoride/bromotrifluoroethylene	100–10	2	Planar zig-zag	XR1	55
Trifluoroethylene/tetrafluoroethylene	100–0	2	Helical, planar zig-zag	XR1, TA, MA	34
Tetrafluoroethylene/chlorotrifluoroethylene	100–0	1	Helical	XR1, TA	56
Tetrafluoroethylene/bromotrifluoroethylene	100–55	2	Helical	XR1, TA	56
Tetrafluoroethylene/hexafluoropropylene	100–93	2	Helical	XR1, TA	87
Vinyl chloride/vinylidene chloride	100–85, 44–0	2	Planar zig-zag, $(TGTG')_n$	XR1, TA, MA	65

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TABLE 1. *cont'd*

Monomer giving isomorphous copolymers	Mol% of first component	Type of isomorphism	Chain conformation	Techniques	Refs.
Copoly(styrenes)					
Styrene/*p*-methylstyrene	100–50	2	Isotactic (2*3/1)	XR1, MP	62,63
Styrene/*o*-methylstyrene	100–80	2	Isotactic (2*3/1)	XR1, MP	62,63
Styrene/*o*-fluorostyrene	100–0	1	Isotactic (2*3/1)	XR1, MP	62,63
Styrene/*p*-fluorostyrene	100–50	2	Isotactic (2*3/1)	XR1, MP	62,63
	50–0		Isotactic (2*4/1)		
Styrene/*p*-ethylstyrene	100–85	2	Isotactic (2*3/1)	XR1, MP	62,63
Styrene/*p*-chlorostyrene	100–85	2	Isotactic (2*3/1)	XR1, MP	62,63
Styrene/*p*-bromostyrene	100–85	2	Isotactic	XR1, MP	62,63
1.1.2. MAIN-CHAIN HETEROATOM COPOLYMERS					
Copoly(ethers)					
Trioxane/1,3-dioxolane	100–84	2	(2*9/5)*	XR1, TA, D	21,22
Copoly(aldehydes)					
Acetaldehyde/propionaldehyde	100–0	1	Isotactic copolymer (2*4/1)	XR1	78
Acetaldehyde/*n*-butyraldehyde	100–0	1	Isotactic copolymer (2*4/1)	XR1	78
Acetaldehyde/isobutyraldehyde	100–0	1	Isotactic copolymer (2*4/1)	XR1	78
Propionaldehyde/*n*-butyraldehyde	100–0	1	Isotactic copolymer (2*4/1)	XR1	78
n-Butyraldehyde/isobutyraldehyde	100–0	1	Isotactic copolymer (2*4/1)	XR1	78
n-Butyraldehyde/*n*-heptanal	100–0	1	Isotactic copolymer (2*4/1)	XR1	78
Copoly(carbonates)					
Hydroquinone/chlorohydroquinone/$COCl_2$	100–0	1	Extended, planar zig-zag	XR, MP	73
Toluenehydroquinone/chlorohydroquinone/$COCl_2$	100–0	1	Extended, planar zig-zag	XR, MP	73
Copoly(esters)					
Ethylene glycol, sebacic acid/undecanedioic acid/dodecanedioic acid	Equimolar proportion of the three acids	2	Extended, planar zig-zag	XR	31
Ethylene glycol, terephthalic acid/*p*-hydroxybenzoic acid	100–90, 40–0	2	Extended, distorted planar	XR, TA, D	50 48
Ethylene glycol, terephthalic acid/isophthalic acid	100–70, 30–0	2	Extended, distorted planar	XR, TA, ED, MA	39
				D	48
Ethylene glycol, terephthalic acid/*p*-hydroxyoxypropoxybenzoic acid	100–0	2	Extended, distorted planar	XR	43
				XR, MP	44
Ethylene glycol, terephthalic acid/1,4-bis-(*p*-carboxyphenoxy)butane	only 50	2	Extended, distorted planar	XR	45
Ethylene glycol, 1,4-bis-(*p*-carboxyphenoxy) butane/*p*-γ-hydroxypropoxybenzoic acid	only 50	2	Extended, distorted planar	XR	46
Tetramethylene glycol/decamethylene glycol, adipic acid	30–0	2	Extended, planar zig-zag	MP	28
Pentamethylene glycol/decamethylene glycol, adipic acid	40–0	2	Extended, planar zig-zag	MP	28
Hexamethylene glycol, adipic acid/decamethylene glycol, sebacic acid	100–0	2	Extended, planar zig-zag	XR1, TA	42
Hexamethylene glycol, sebacic acid/decamethylene glycol, adipic acid	100–0	2	Extended, planar zig-zag	XR1, TA	42
Decamethylene glycol, adipic acid/sebacic acid	100–70	2	Extended, planar zig-zag	MP	28
Decamethylene glycol/hydroquinone, adipic acid	100–70	2	Extended, planar zig-zag	MP	28
Decamethylene glycol/diethylene glycol, adipic acid	100–70	2	Extended, planar zig-zag	MP	28
4,4′-Dihydroxybiphenyl, pimelic acid/azelaic acid	100–0	2	Extended, planar zig-zag	XR1, TA	86
4,4′-Dihydroxybiphenyl, azelaic acid/decanedioic acid	100–60, 25–0	2	Extended, planar zig-zag	XR1, TA	86
4,4′-Dihydroxybiphenyl, decanedioic acid/tetradecanedioic acid	100–0	2	Extended, planar zig-zag	XR1, TA	86
4,4′-Dihydroxybiphenyl, dodecanedioic acid/tetradecanedioic acid	100–0	2	Extended, planar zig-zag	XR1, TA	86
Lactic acid R/S	100–85	2*	Extended, planar zig-zag*	XR, MA, TA	30
Copoly(amides)					
Hexamethylenediamine, adipic acid/sebacic acid	100–60, 50–0	2*	Extended, planar zig-zag	XR	7
Hexamethylenediamine, adipic acid/cyclohexanedicarboxylic acid	100–70	2	Extended, planar zig-zag	MP	72

TABLE 1. *cont'd*

Monomer giving isomorphous copolymers	Mol% of first component	Type of isomorphism	Chain conformation	Techniques	Refs.
Hexamethylenediamine, adipic acid/terephthalic acid	100–0	2	Extended, planar zig-zag	MP XR TA	96,66,72 23 40
Hexamethylenediamine, adipic acid/ nonamethylene diamine, azelaic acid	100–60, 50–0	2	Extended, planar zig-zag	XR	7
Hexamethylenediamine/decamethylene-diamine, adipic acid	100–70, 60–0	2*	Extended, planar zig-zag	XR	7
Hexamethylenediamine, adipic acid/deca-methylene diamine, sebacic acid	100–65, 50–0	2*	Extended, planar zig-zag	XR	7
Hexamethylenediamine/*trans*-1,4-cyclohexanebis-methylamine, adipic acid	100–70*	2	Extended, planar zig-zag	MP	72
Hexamethylenediamine, suberic acid/4-benzene-diacetic acid	40–0*	2	Extended, planar zig-zag	MP	95
Hexamethylenediamine/decamethylene-diamine, adipic acid	100–0	2	Extended, planar zig-zag	XR	7
Hexamethylenediamine/decamethylenediamine, sebacic acid	100–0	2	Extended, planar zig-zag	XR MP	7 28
Hexamethylenediamine, sebacic acid/*p*-phenylene-dipropionic acid	100–0	2	Extended, planar zig-zag	XR, MP	81
Hexamethylenediamine, sebacic acid/ 3-(*p*-carboxy-methyleneoxy) phenylpropionic acid	100–0	2	Extended, planar zig-zag	XR, MP,	81
Hexamethylenediamine, sebacic acid/ 4-(*p*-carboxy-methylene)phenylbutyric acid	100–0	2	Extended, planar zig-zag	MP	81
Hexamethylenediamine/*N,N'*-diisobutyl-hexamethylene diamine, sebacic acid	100–0	2	Extended, planar zig-zag	XR, MP	89
Hexamethylenediamine, sebacic acid/1,4-(dicarboxy-methyleneoxy)phenylene	100–0	2	Extended, planar zig-zag	XR, MP	81
Hexamethylenediamine, undecanedioic acid/ 6-oxa-hendecanedioic acid	100–0	1	Extended, planar zig-zag	MP	74
Hexamethylenediamine, undecanedioic acid/ 6-thio-hendecanedioic acid	100–0	1	Extended, planar zig-zag	MP	74
Hexamethylenediamine/α,α'-*p*-xylenediamine, sebacic acid	50–0*	2	Extended, planar zig-zag	MP	95
Hexamethylenediamine, *p*-phenylenedipropionic acid/4-(*p*-carboxymethylene)phenylbutyric acid	100–0	2	Extended, planar zig-zag	XR, MP	81
Hexamethylenediamine, *p*-phenylene-dipropionic acid/3-(*p*-carboxymethyleneoxy)-phenylpropionic acid	100–0	2	Extended, planar zig-zag	XR, MP	81
Hexamethylenediamine, *p*-phenylene-dipropionic acid/*p*-bis-(carboxymethyleneoxy)-phenylene	100–0	2	Extended, planar zig-zag	XR, MP	81
Hexamethylenediamine, 3-(*p*-carboxymethyleneoxy) phenylpropionic acid/*p*-bis-(carboxymethyleneoxy) phenylene	100–0	2	Extended, planar zig-zag	MP	81
Heptamethylenediamine, adipic acid/terephthalic acid	100–0	2	Extended, planar zig-zag	MP XR, MP, DM	96 19
Heptamethylenediamine/4-oxa-heptamethylene-diamine, adipic acid	75, 50, 25	1	Extended, planar zig-zag	XR, MP, DM	19
Heptamethylenediamine, adipic acid/4-oxa-hepta-methylenediamine, terephthalic acid	75, 50, 25	2	Extended, planar zig-zag	XR, MP, DM	19
Heptamethylenediamine, terephthalic acid/4-oxa-heptamethylenediamine, adipic acid	75, 50, 25	2	Extended, planar zig-zag	XR, MP, DM	19
Heptamethylenediamine/4-oxa-heptamethylene-diamine terephthalic acid	75, 50, 25	2	Extended, planar zig-zag	XR, MP, DM	19
Octamethylenediamine, adipic acid/terephthalic acid	100–0	2	Extended, planar zig-zag	MP	96
Octamethylenediamine/2,2'-*p*-phenylene-bis-ethylamine, adipic acid	100–0*(100–60)	2	Extended, planar zig-zag	MP	95
Octamethylenediamine/2,2'-*p*-phenylene-bis-ethylamine, azelaic acid	100–0*(100–10)	2	Extended, planar zig-zag	MP	95
Octamethylenediamine, sebacic acid/ 4-benzenedipropionic acid	100–0	2	Extended, planar zig-zag	MP	95

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TABLE 1. *cont'd*

Monomer giving isomorphous copolymers	Mol% of first component	Type of isomorphism	Chain conformation	Techniques	Refs.
Octamethylenediamine/2,2′-*p*-phenylene-bis-ethylamine, sebacic acid	100–0*(100–10)	2	Extended, planar zig-zag	MP	95
Octamethylenediamine/2,2′-*p*-phenylene-bis-ethylamine, suberic acid	100–0*(100–30)	2	Extended, planar zig-zag	MP	95
Octamethylenediamine/bis(4-aminocyclohexyl) methane, dodecanedioic acid	100–0*(90, 50)	2	Extended, planar zig-zag	TA	16
Nonamethylenediamine, adipic acid/terephthalic acid	100–0	2	Extended, planar zig-zag	MP	96
Nonamethylenediamine/bis(4-aminocyclohexyl) methane, octanedioic acid	100–0	2	Extended, planar zig-zag	TA	16
Nonamethylenediamine/bis(4-aminocyclohexyl) methane, dodecanedioic acid	100–0	2	Extended, planar zig-zag	XR, TA	16
5-Methyl-nonamethylenediamine/bis-(4-amino-cyclohexyl)methane, dodecanedioic acid	100–0*(90, 50)	2	Extended, planar zig-zag	XR, TA	16
Decamethylenediamine, adipic acid/sebacic acid	100–0	2	Extended, planar zig-zag	XR	7
Decamethylenediamine, adipic acid/terephthalic acid	100–0	2	Extended, planar zig-zag	MP	96
Decamethylenediamine/bis(4-aminocyclohexyl) methane, dodecanedioic acid	100–0*(90, 50)	2	Extended, planar zig-zag	TA	16
Undecamethylenediamine, adipic acid/terephthalic acid	100–0	2	Extended, planar zig-zag	MP	96
Dodecamethylenediamine, adipic acid/terephthalic acid	100–0	2	Extended, planar zig-zag	MP	96
Dodecamethylenediamine/bis(4-aminocyclohexyl) methane, dodecanedioic acid	100–0*(90, 50)	2	Extended, planar zig-zag	XR, TA	16
1,4-Bis(aminomethyl)cyclohexane (*cis/trans*), adipic acid	100–0	2	Twist boat, planar zig-zag	XR1	69
1,4-Bis(aminomethyl)cyclohexane (*cis/trans*), sebacic acid	100–0	2	Twist boat, planar zig-zag	XR1	69
1,4-Bis(aminomethyl)cyclohexane (*cis/trans*), dodecanedioic acid	100–0	2	Twist boat, planar zig-zag	XR1	69
α,α′-*p*-Xylenediamine, nonanedioic acid/5-oxa-nonanedioic acid	100–0	1	Extended, planar zig-zag	MP	74
α,α′-*p*-Xylenediamine, nonanedioic acid/5-thio-nonanedioic acid	100–0	1	Extended, planar zig-zag	MP	74
α,α′-*p*-Xylenediamine, undecanedioic acid/6-oxa-undecanedioic acid/6-thio-undecanedioic acid	All proportions of the three acids	1	Extended, planar zig-zag	XR, MP, D	74
α,α′-*p*-Xylenediamine, triadecanedioic acid/7-oxa-triadecanedioic acid/6-oxa-triadecanedioic acid	All proportions of the three acids	1	Extended, planar zig-zag	MP	74
α,α′-*p*-Xylenediamine, triadecanedioic acid/7-thio-triadecanedioic acid	100–0	1	Extended, planar zig-zag	MP	74
α,α′-*p*-Xylenediamine, pentadecanedioic acid/6-oxa-pentadecanedioic acid	100–0	1	Extended, planar zig-zag	MP	74
α,α′-*p*-Xylenediamine, pentadecanedioic acid/8-tia-pentadecanedioic acid	100–0	1	Extended, planar zig-zag	MP	74
α,α′-*p*-Xylenediamine, 7-oxa-triadecanedioic acid/7-thio-triadecanedioic	100–0	1	Extended, planar zig-zag	MP	74
4-Oxa-heptamethylenediamine, adipic acid/terephthalic acid	75, 580, 25	2	Extended, planar zig-zag	XR, MP, DM	19
Caprolactam/4(-aminomethylcyclohexyl) carboxylic acid	100–70, 50–0	2	Extended, planar zig-zag	MP XR, MP	51 68
Caprolactam/4(-aminomethyl)benzoic acid	100–70	2	Extended, planar zig-zag	MP	51
Caprolactam/γ-4(-aminomethyl)propionic acid	100–80, 15–0	2	Extended, planar zig-zag	XR, TA, IR	75
Caprolactam-ω-laurolactam	100–70, 30–0	2	Extended, planar zig-zag	XR1, TA, D, IR TA DM	74 36 38
Copoly(peptides)					
α-Methyl-*S*-glutamate/γ-benzyl-*S*-glutamate	100–0	2	α-helix, (3*18/5)	XR1	35,47
α-Methyl-*S*-glutamate/γ-methyl-*R*-glutamate	100–0	2	α-helix, (3*18/5)	XR1, IR	59
α-Benzyl-*S*-glutamate/*S*-phenylalanine	80	2*	α-helix, (3*18/5)	MA, ED	79
Butyl-*S*-aspartate/benzyl-*S*-aspartate	87–0	1	α-helix, (3*4/1)	XR1, TA, IR, DM	82

TABLE 1. *cont'd*

Monomer giving isomorphous copolymers	Mol% of first component	Type of isomorphism	Chain conformation	Techniques	Refs.
Copoly(diacetylenes)					
(General formula $[(=\overset{R}{\overset{\vert}{C}}-C\equiv C-\overset{R}{\overset{\vert}{C}}=)_n\ (=\overset{R'}{\overset{\vert}{C}}-C\equiv C-\overset{R'}{\overset{\vert}{C}}=)_{1-x})]_n$					
R R′					
p-Tolylsulfonyloxymethyl/*p*-fluorosulfonyloxymethyl	100–0	1	*trans*-Planar	XR1	26
p-Tolylsulfonyloxymethyl/*p*-chlorosulfonyloxymethyl	100–80	1	*trans*-Planar	XR1	24,25
p-Tolylsulfonyloxymethyl/*p*-bromosulfonyloxymethyl	100–93	1	*trans*-Planar	XR1	25
p-Tolylsulfonyloxymethyl/*p*-naphthylsulfonyloxymethyl	100–90	1	*trans*-Planar	XR1	4
p-Tolylsulfonyloxymethyl/*p*-methoxysulfonyloxymethyl	100–90	1	*trans*-Planar	XR1	4
Methylcarbazolyl/methylcarbazolyl, methylanthryl	100–94	1	*trans*-Planar	XR1	27
Copoly(phosphazenes)					
(General formula $[(RO)_x(R'O)_{2-x}PN]_n$)					
R R′					
Phenyl/*p*-ethylphenylene	100–85/15–0	2	(*cis*–*trans*)$_2$	XR1, TA	33
4-Chlorophenylene/4-methylphenylene	50 only	1*	(*cis*–*trans*)$_2$	XR1, TA	9
4-Chlorophenylene/4-ethylphenylene	50 only	1*	(*cis*–*trans*)$_2$	TA	9
4-Chlorophenylene/phenyl	50 only	2	(*cis*–*trans*)$_2$	XR1, TA	9
4-Chlorophenylene/3-methylphenylene	50 only	2*	(*cis*–*trans*)$_2$	TA	9
4-Chlorophenylene/3-chlorophenylene	50 only	2	(*cis*–*trans*)$_2$	TA	9
4-Methylphenylene/3-methylphenylene	50 only	2	(*cis*–*trans*)$_2$	TA	9
3-Chlorophenylene/4-methylphenylene	50 only	2	(*cis*–*trans*)$_2$	TA	9
3-Chlorophenylene/3-methylphenylene	50 only	1*	(*cis*–*trans*)$_2$	TA	9
1.2. ISOMORPHOUS UNITS WITH DIFFERENT CONFIGURATIONS AND/OR WITH HEAD-TO-HEAD, OR HEAD-TO-TAIL CONSTITUTIONAL DISORDER					
Poly(vinyl alcohol)	Atactic polymer		Planar zig-zag	XR1	12,13,14
Poly(vinyl fluoride)	Atactic polymer with constitutional disorder		Planar zig-zag	XR1	64
Poly(vinylidene fluoride)	Constitutional disorder		Planar zig-zag	XR1, CA	29
Poly(chlorotrifluoroethylene)	Atactic polymer		(2*14/1)	XR1	13
			(2*16/1)	XR1	54
Poly(ethylene-*alt*-chlorotrifluoroethylene)	Atactic polymer with constitutional disorder		Extended, distorted planar	XR1	76
Poly(*p*-carboxyphenyl methacrylate)	Atactic polymer		Extended, distorted planar	XR1, ED, MA, TA	10
Poly(butadiene-*alt*-acrylonitrile)	Atactic polymer		Extended, distorted planar	XR, IR	32

TABLE 2. ISOMORPHISM OF MACROMOLECULES

Monomer giving isomorphous copolymers	Mol% of first component	Type of isomorphism	Chain conformation	Techniques	Refs.
Poly(propylene)/poly(1-butene)	100–80 and 20–0	2	Isotactic polymers (2*3/1)	DM, MA, TA, ED	37
Poly(4-methylpentene)/poly(4-methylhexene)	100–75 and 20–0	1	Isotactic polymers (2*7/2)	XR1, MP	3
Poly(1-propylvinyl ether)/poly(*sec*-butyl vinyl ether)	100–0	1	Isotactic polymers (2*17/5)	XR1, MP	3
Poly(styrene)/poly(styrene-*co*-*p*-methyl styrene) (70/30, mol/mol)	50	1	Isotactic polymers (2*3/1)	XR1, IR	63
Poly(oxy-4,4′-biphenylene oxy dodecanoyl)/poly(oxy-4,4′-biphenylene oxy-tetradecanedioyl)	100–0	2	Planar zig-zag	XR1, TA, MA	86

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D. REFERENCES

1. G. C. Alfonso, L. Fiorina, E. Martuscelli, E. Pedemonte, S. Russo, Polymer, **14**, 373 (1973).
2. G. Allegra, I. W. Bassi, Adv. Polym. Sci., **6**, 549 (1969).
3. G. Allegra, I. W. Bassi, C. Carlini, E. Chiellini, G. Montagnoli, Macromolecules, **2**, 311 (1969).
4. D. J. Ando, D. Bloor, Makromol. Chem. Rapid Commun., **1**, 385 (1980).
5. C. H. Baker, L. Mandelkern, Polymer, **7**, 7 (1966).
6. C. H. Baker, L. Mandelkern, Polymer, **7**, 71 (1966).
7. W. O. Baker, C. S. Fuller, J. Am. Chem. Soc., **64**, 2399 (1942).
8. I. W. Bassi, P. Corradini, G. Fagherazzi, A. Valvassori, Eur. Polym. J., **6**, 709 (1970).
9. J. J. Beres, N. S. Schneider, C. R. Desper, R. E. Singler, Macromolecules, **12**, 566 (1979).
10. A. Blumstein, S. B. Clough, L. Patel, R. B. Blumstein, E. C. Hsu, Macromolecules, **9**, 243 (1976).
11. M. M. Brubaker, D. D. Coffman, H. H. Hochn, J. Am. Chem. Soc., **74**, 1509 (1952).
12. C. W. Bunn, S. Peiser, Nature, **159**, 161 (1947).
13. C. W. Bunn, J. Appl. Phys., **25**, 820 (1954).
14. C. W. Bunn, Nature, **161**, 929 (1948).
15. T. W. Campbell, J. Appl. Polym. Sci., **5**, 184 (1961).
16. R. W. Campbell, H. W. Hill, Jr., Macromolecules, **8**, 238 (1975).
17. Y. Chatani, T. Takizawa, S. Murahashi, Y. Sakata, Y. Nishimura, J. Polym. Sci., **55**, 811 (1961).
18. Y. Chatani, T. Takizawa, S. Murahashi, J. Polym. Sci., **62**, 527 (1962).
19. F. B. Cramer, R. G. Beaman, J. Polym. Sci., **21**, 237 (1956).
20. G. Dall'Asta, G. Motroni, G. Carella, It. Pat., **773**, 657 (1966).
21. M. Droscher, Makromol. Chem., **178**, 1195 (1977).
22. M. Droscher, K. Hertwig, H. Reimann, G. Wegner, Makromol. Chem., **177**, 1695 (1976).
23. O. B. Edgar, R. H. Hill, J. Polym. Sci., **8**, 1 (1952).
24. V. Enkelmann, Makromol. Chem., **179**, 2811 (1978).
25. V. Enkelmann, J. Mat. Sci., **15**, 951 (1980).
26. V. Enkelmann, Makromol. Chem., **184**, 1945 (1983).
27. V. Enkelmann, G. Schleicr, H. Heichele, J. Mat. Sci., **17**, 533 (1982).
28. R. D. Evans, H. R. Mighton, P. J. Flory, J. Am. Chem. Soc., **72**, 2018 (1950).
29. B. L. Farmer, J. B. Lando, J. Macromol. Sci. Phys. B, **11**, 89 (1975).
30. E. W. Fischer, H. J. Sterzel, G. Wegner, Kolloid Z. Z. Polym., **251**, 980 (1973).
31. C. S. Fuller, J. Am. Chem. Soc., **70**, 421 (1948).
32. J. Furukawa, E. Kobayashi, K. Uratani, Y. Iseda, J. Umemura, T. Takenaka, Polym. J., **4**, 358 (1973).
33. S. Futamura, J. K. Valaitis, K. R. Lucas, J. W. Fieldhouse, T. C. Cheng, D. P. Tate, J. Polym. Sci., Polym. Phys. Ed., **18**, 767 (1980).
34. Y. L. Gal'perin, D. Y. Tsvankin, Vysokomol. Soedin. A, **18**, 2691 (1976).
35. R. Gehani, J. Watanabe, S. Kasuya, I. Uematsu, Polym. J., **12**, 871 (1980).
36. Yu. K. Godovskii, I. I. Dubovik, S. L. Ivanova, V. V. Kurashev, T. Frunze, G. L. Slonimskii, Vysokomol. Soedin. A, **19**, 392 (1977).
37. R. M. Gohil, J. Petermann, J. Makromol. Sci. B: Phys., **18**, 217 (1980).
38. J. Goodman, A. H. Kehayoglou, Eur. Polym. J., **19**, 321 (1983).
39. M. Hachiboshi, T. Fukuda, S. Kobayashi, J. Macromol. Sci. Phys., **3**, 525 (1969).
40. E. D. Harvey, F. J. Hybart, Polymer, **12**, 711 (1971).
41. W. A. Hewett, F. E. Weir, J. Polym. Sci. A, **1**, 1239 (1963).
42. G. J. Howard, S. Knutton, Polymer, **9**, 527 (1968).
43. M. Ishibashi, Polymer, **5**, 103 (1964).
44. M. Ishibashi, J. Polym. Sci. A, **2**, 4361 (1964).
45. M. Ishibashi, J. Polym. Sci. B, **2**, 791 (1964).
46. M. Ishibashi, J. Polym. Sci. B, **2**, 789 (1964).
47. T. Kanamori, K. Itoh, A. Nakajima, Polym. J., **1**, 524 (1970).
48. T. Kawaguchi, Y. Nukushina, Kobunshi Kagaku, **20**, 529 (1963).
49. A. H. Kehayoglou, Eur. Polym. J., **19**, 183 (1983).
50. R. W. Lenz, J. Jin, K. A. Feichtinger, Polymer, **24**, 327 (1983).
51. M. Levine, S. C. Temin, J. Polym. Sci., **49**, 241 (1961).
52. A. J. Lovinger, T. Furukawa, G. T. Davis, M. G. Broadhurst, Polymer, **24**, 1225 (1983).
53. A. J. Lovinger, G. T. Davis, T. Furukawa, M. G. Broadhurst, Macromolecules, **15**, 323 (1982).
54. Y. Miyamoto, C. Nakafuku, T. Takemura, Polym. J., **3**, 122 (1972).
55. G. Moggi, P. Bonardelli, J. C. J. Bart, J. Polym. Sci., Polym, Phys. Ed., **22**, 357 (1984).
56. G. Moggi, P. Bonardelli, C. Monti, J. C. J. Bart, J. Polym. Sci., Polym. Phys. Ed., **23**, 1099 (1985).
57. G. Motroni, G. Dall'Asta, I. W. Bassi, Eur. Polym. J., **9**, 257 (1973).
58. K. Nakame, M. Kameyama, T. Matsumoto, Polym. Eng. Sci., **19**, 579 (1979).
59. A. Nakajime, T. Hayashi, K. Itoh, T. Fujiwara, Polym. J., **4**, 10 (1979).
60. G. Natta, I. W. Bassi, G. Faghcrazzi, Eur. Polym. J., **5**, 239 (1969).
61. G. Natta, G. Allegra, I. W. Bassi, D. Sianesi, G. Caporiccio, E. Torti, J. Polym. Sci. A, **3**, 4263 (1965).
62. G. Natta, Makromol. Chem., **35**, 93 (1960).
63. G. Natta, P. Corradini, D. Sianesi, D. Morero, J. Polym. Sci., **51**, 527 (1961).
64. G. Natta, I. W. Bassi, G. Allegra, Acc. Naz. Lincei Rend., **31**, 350 (1961).
65. K. Okuda, J. Polym. Sci. A, **2**, 1749 (1964).
66. H. Plimmer, R. J. W. Reynolds, L. Wood, H. Hargreaves, I. C. I. Ltd. British Pat. Appl., 604/49.

67. A. Piloz, A. Douillard, J. Y. Decroix, J. P. May, G. Vallet, Polym. J., **6**, 506 (1974).
68. F. R. Prince, E. M. Pearce, R. J. Fredericks, J. Polym. Sci. A-1, **8**, 3533 (1970).
69. F. R. Prince, R. J. Fredericks, Macromolecules, **5**, 168 (1972).
70. F. P. Roding, E. R. Walter, J. Polym. Sci., **37**, 555 (1959).
71. M. J. Richardson, P. J. Flory, J. B. Jackson, Polymer, **4**, 221 (1963).
72. J. Ridgway, J. Polym. Sci. A-1, **8**, 3089 (1970).
73. J. D. Rubin, J. Polym. Sci. A, **1**, 1645 (1963).
74. K. Saotome, H. Komoto, J. Polym. Sci. A-1, **4**, 1475 (1966).
75. S. W. Shalaby, E. A. Turi, P. J. Harget, J. Polym. Sci., Polym. Chem. Ed., **14**, 2407 (1976).
76. J. P. Sibilia, L. G. Roldan, S. Chandrasekaran, J. Polym. Sci. A, **10**, 549 (1972).
77. H. W. Starkweather Jr., F. A. Van Catledge, R. N. MacDonald, Macromolecules, **15**, 1600 (1982).
78. A. Tanaka, Y. Ozumi, K. Hatada, S. Endo, R. Fujishije, Polymer Lett., **2**, 181 (1964).
79. Y. Tajima, M. Anderson, P. H. Geil, Int. J. Biol. Macromol., **1**, 98 (1979).
80. K. Tashiro, K. Takano, M. Kobayashi, Y. Chatani, H. Tadokoro, Polymer, **22**, 1312 (1981).
81. T. C. Tranter, J. Polym. Sci. A, **2**, 4289 (1964).
82. Y. Tsujita, M. Fukagawa, I. Uematsu, Polym. J., **14**, 773 (1982).
83. A. Turner Jones, Polym. Lett., **3**, 591 (1965).
84. A. Turner Jones, Polymer, **7**, 23 (1966).
85. A. Turner Jones, Polymer, **6**, 249 (1965).
86. J. Watanabe, W. R. Krigbaum, Macromolecules, **17**, 2288 (1984).
87. J. J. Woeks, R. K. Eby, E. S. Clark, Polymer, **22**, 1496 (1981).
88. F. C. Wilson, H. W. Starkweather, Jr., J. Polym. Sci., Polym. Phys. Ed., **11**, 919 (1973).
89. E. L. Wittbecker, R. C. Houtz, W. W. Watkins, Ind. Eng. Chem., **40**, 875 (1948).
90. B. Wunderlich, "Macromolecular Physics", Academic Press, New York, 1973, Ch. 2, p. 147.
91. B. Wunderlich, "Macromolecular Physics", Academic Press, New York, 1980, Ch. 10, p. 255.
92. B. Wunderlich, J. Grelowicz, Adv. Polym. Sci., "Liquid Crystal Polymers II/III", Springer-Verlag, Berlin, 1984, Ch. 1, p. 4.
93. B. Wunderlich, D. Poland, J. Polym. Sci. A, **1**, 357 (1963).
94. T. Yagi, M. Tatemoto, J. Sako, Polym. J., **12**, 209 (1980).
95. A. J. Yu, R. D. Evans, J. Am. Chem. Soc., **81**, 5361 (1959).
96. A. J. Yu, R. D. Evans, J. Polym. Sci., **42**, 249 (1960).

Miscible Polymers

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A. DEFINITION OF MISCIBILITY

A rigorous definition of miscibility is used in the following tables. In order to be classified as miscible polymers, there must be some evidence in the literature that the polymers are miscible on a submolecular (segmental) scale. If the polymer mixture contains several phases, the polymers will be listed as miscible even if segmental miscibility data exists in only *one* of these phases, but this is noted under Comments in the Tables. Tables 1–7 list miscibility in the *amorphous* state. A polymer pair will be listed as miscible even if one or both of the polymers forms a single-component crystalline phase in the mixture; a note about the existence of crystallinity, however, is made under Comments. Evidence for miscibility of a polymer with only one microphase in a microphase separated block copolymer is also noted. In other cases, literature data indicated miscibility with a known block copolymer, but no data were shown which indicate whether the mixture still contained the second microphase. In those cases, no comments are made in these Tables, and readers will have to make judgements of their own. Since miscibility on a submolecular or segmental scale includes cocrystallization, some mixtures that have proven cocrystallization behavior are shown in Table 8.

There is some controversy in the literature about the dimensions of the submolecular scale of segment mixing that indicates true miscibility in the amorphous state of polymers. Followers of one school of thought feel that every segment of polymer A must be surrounded by the numbers of segments of polymers A and B whose fraction is governed by the volume fraction of the two polymers present in that phase. This is the random mixing assumption implicit in the Flory–Huggins theory of polymer solutions and mixtures. These ideas are much too rigorous for real solutions, even solutions of small molecules. It is well known that clustering occurs in real solutions, so that volume elements containing an unexpectedly large number of one of the solution components (in solutions of small molecules) or segments of one of the polymers (in polymer mixtures) can and do occur. Thus, data generated by some solid state NMR techniques and some fluorescence techniques that can show the proximity of two polymers over a distance of 1–2 nm have often been misinterpreted to show immiscibility in a miscible polymer system. In fact, data showing that a large proportion of the nearest neighbor segments of a particular polymer belong to the same polymer could simply indicate clustering in a homogeneous phase.

B. DATA COLLECTION

The following tables include information from the third edition of "Polymer Handbook", additional information from an incomplete polymer–polymer miscibility database compiled by one of us (SK), and the data from papers (but not patents) published in the English language and listed in "CA Selects–Polymer Blends" and Science Citation Index through December 1995 as compiled by one of us (SHG). Note that Table 8 contains less complete information for the earlier years than the other tables because most of the information in this table comes from the incomplete database. The inclusion of patents and papers in languages other than English by SHG would have necessitated a great deal more time and additional data compilers.

C. ARRANGEMENT OF THE TABLES

The following eight tables have essentially the same format as those compiled for the third edition of "Polymer Handbook". Each table contains a particular mixture only once. The column labeled "Polymer I of" is an alphabetical listing. The column labeled "Polymer II of" is an alphabetical listing of the polymers miscible with the appropriate polymer from the first column. The common names of polymers were used whenever possible; sometimes the Chemical Abstracts or IUPAC names of polymers also appear. Unfortunately, we could not cross-list the many possible names of the polymers. Thus, it may be necessary to look for a particular polymer under several names in these tables.

The column labeled Methods very briefly indicates the method(s) used by the experimenters to determine miscibility in amorphous phases. A large number of such methods exist, but two of these are most commonly used. One of these involves the determination of glass transition temperatures (T_g's) by methods such as DSC, DTA, dynamic mechanical measurements, dilatometry, or other methods. The observation of a single T_g for the amorphous portion of a polymer mixture, especially if it is intermediate between those of the constituent polymers and varies monotonically between these values as the mixture composition changes, has generally been considered a reliable indication of a miscible polymer pair. This method is difficult to use when the T_g's of the constituent polymers are close together, perhaps within 10°C of each other. Furthermore, the glass transition region of a miscible polymer pair is often broader than that of the unmixed polymers. Some experimenters have interpreted this as an indication of two overlapping glass transition regions, but we have taken the existence of such a broad glass transition region as an indication of miscibility. We feel that a broad glass transition region probably indicates the existence of composition fluctuations rather than two overlapping T_g's.

The criterion of sample clarity has often been used as an indication of miscibility. This criterion will not work if the constituent polymers have the same refractive index or if they are so immiscible that they form phases so large that they do not scatter light appreciably. Furthermore, semicrystalline mixtures are always turbid, even if their amorphous phases are completely miscible. In addition, on at least one occasion, a set of polymers was turbid at low temperatures, clear at intermediate temperatures, and turbid again at high temperatures only because their refractive indices became almost identical at the intermediate temperatures.

All single methods, not all of them described here, of determining polymer–polymer miscibility sometimes give ambiguous results. Unless many experimenters have used a variety of methods to determine miscibility, the inferences drawn from the data may be wrong. This is one of the reasons for listing the methods used and the references that describe the experiments, results, and interpretation. Some of the mixtures listed in these tables, especially when only a single mixture has been reported in a single reference, may turn out to be less miscible than stated. This is especially possible because some of the methods that were used in the original investigations do not necessarily probe miscibility as such. For example, a negative volume change on mixing, that is, a shrinkage on mixing, could possibly be observed if each of the coexisting phases contains both polymers, though in different proportions, and if each of the coexisting phases has a negative volume change on mixing. Also, FTR methods generally involve the observation of absorption band shifts due to hydrogen bonding between polymers; however, this hydrogen bonding could conceivably occur on phase boundaries or within coexisting phases each of which contains both polymers, but in different proportions.

The Comments column in the Tables indicates compositions of copolymers, whether the samples were semicrystalline, and other relevant data.

The eight tables are:

1. Chemically dissimilar polymer pairs miscible in the amorphous state at room temperature. Some of these polymers may be copolymers, but the polymer pairs may not have any monomer units in common. One or both of the polymers may be semicrystalline.
2. Polymer pairs containing one monomer in common (at least one of these must be a copolymer), miscible in the amorphous state at room temperature. One or both of the polymers may be semicrystalline.
3 Chemically, dissimilar polymer triads and tetrads miscible in the amorphous state at room temperature. It seemed reasonable to place mixtures of three or four polymers that were miscible in a separate table.
4. Polymer pairs miscible in the amorphous state at room temperature. Molecular weight dependence investigated. These are usually polymer pairs that are miscible when the molecular weights are low and immiscible as when the molecular weights are increased.
5. Polymer pairs that appear to have high temperature miscibility although immiscible at or below room temperature (Upper Critical Solution Temperature [UCST] behavior).

6. Polymer pairs miscible at room temperature that appear to have a lower critical solution temperature (LCST) above room temperature. These polymer pairs are also listed in one of the earlier tables, usually Table 1 or 2.
7. Polymer pairs that appear to have both lower and upper critical solution temperatures.
8. Polymer pairs that cocrystallize and form mixed crystals. These blends are generally composed of polymers with similar subunits that can substitute for each other in the same unit cells; this is generally called "isomorphous replacement". This Table is probably incomplete even though cocrystallization is expected to be rare.

Abbreviations

DSC	Differential scanning calorimetry
FTIR	Fourier-transform infrared spectroscopy
IGC	Inverse gas chromatography
LS	Light scattering
MW	Molecular weight (unspecified)
M_n	Number-average molecular weight
M_w	Weight-average molecular weight
NMR	Nuclear magnetic resonance spectroscopy
NRET	Nonradiative energy transfer fluorospectroscopy
PALS	Positron annihilation lifetime spectroscopy
SAXS	Small-angle X-ray scattering
SALS	Small-angle light scattering
SANS	Small-angle neutron scattering
SEM	Scanning electron microscopy
T_g	Glass transition temperature
T_m	Melting temperature
WAXD	Wide-angle X-ray diffraction

D. TABLES

TABLE 1. CHEMICALLY DISSIMILAR POLYMER PAIRS MISCIBLE IN THE AMORPHOUS STATE AT ROOM TEMPERATURE

Polymer I of	Polymer II of	Method	Comments	Refs.
2-Acrylamido-2-methylpropanesulfonic acid	Vinylpyridine	Single T_g	II was 2- or 4-vinylpyridine (2 blend systems); formed complexes	354
Acrylic acid	Acrylamide	NMR	Formed complexes	1012
	Acrylamide-*co*-*N*,*N*-dimethylacrylamide	Fluorescence study	Formed complexes; II had 6.0–55 mol% acrylamide	859
	Caprolactam (nylon 6)	Single T_g; WAXD	Semicrystalline	1140
	N,*N*-dimethylacrylamide	Fluorescence study	Formed complexes	859
	Ethylene glycol	Single T_g; FTIR; NMR	Semicrystalline when > 50% II	374,712,953, 1115,1128
	Vinyl alcohol	FTIR; T_m-depression; NMR	Semicrystalline	197,1211,1255,1256,1257
	Vinyl methyl ether	Single T_g	Formed 1:1 complexes	160
	N-vinylpyrrolidone	NMR	Formed complexes	1128
Acrylic acid-*co*-ethylene	Caprolactam (nylon 6)	Single T_g	I had 19 wt.% acrylic acid	564
Acrylic acid-*co*-styrene	Caprolactone	Single T_g	I had 11.4–20.8 wt.% acrylic acid	383
	Ester	Single T_g	I had 20% acrylic acid; II was nylon 6, nylon 11 or nylon 12 (3 miscible blend systems)	471,884
	Ethyl methacrylate	Single T_g; IGC	I had 2.65–20 wt.% acrylic acid	72,1001
	Ethyl methacrylate-*co*-4-vinylpyridine	Single T_g; IGC	I had 20 mol% acrylic acid	1001
	Ethylene glycol	Single T_g	I had 21 wt.% acrylic acid	384
	Isobutyl methacrylate-*co*-4-vinylpyridine	Single T_g; IGC	I had 20 mol% acrylic acid	1001
	Methyl methacrylate	Single T_g	I had 8–20 wt.% acrylic acid	72,382,384
	n-Propyl methacrylate	Single T_g	I had 2.65–8.84 wt.% acrylic acid	70
	Vinyl methyl ether	Single T_g	I had 8–20 wt.% acrylic acid	576
Acrylonitrile-*co*-butadiene	Cellulose acetate-butyrate	Some compositions appeared homogeneous	–	30,249,452,651
	Chloroprene	Single T_g when I had 18% acrylonitrile; sorption of solvent vapors	Two phases in electron micrograph when I had 18% acrylonitrile may indicate a crystalline phase; two torsional transitions when I had ≥ 28% acrylonitrile	93,797
	Epichlorohydrin	Single T_g	II had 26–54 wt.% acrylonitrile	966
	Nitrocellulose	Transparent films when I had 28.6–44.4% acrylonitrile	Immiscible when I had 18.4% acrylonitrile	93,296,398,399 452
	Vinyl acetate-*co*-vinyl chloride	Single loss peak when ≤ 40% vinyl acetate in II; clear films for some mixtures of commercial polymers	Two loss peaks when ≥ 50% vinyl acetate in II	296,452 539,651

6. Polymer pairs miscible at room temperature that appear to have a lower critical solution temperature (LCST) above room temperature. These polymer pairs are also listed in one of the earlier tables, usually Table 1 or 2.
7. Polymer pairs that appear to have both lower and upper critical solution temperatures.
8. Polymer pairs that cocrystallize and form mixed crystals. These blends are generally composed of polymers with similar subunits that can substitute for each other in the same unit cells; this is generally called "isomorphous replacement". This Table is probably incomplete even though cocrystallization is expected to be rare.

Abbreviations

DSC	Differential scanning calorimetry
FTIR	Fourier-transform infrared spectroscopy
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MW	Molecular weight (unspecified)
M_n	Number-average molecular weight
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NRET	Nonradiative energy transfer fluorospectroscopy
PALS	Positron annihilation lifetime spectroscopy
SAXS	Small-angle X-ray scattering
SALS	Small-angle light scattering
SANS	Small-angle neutron scattering
SEM	Scanning electron microscopy
T_g	Glass transition temperature
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	Acrylamide-*co*-*N*,*N*-dimethylacrylamide	Fluorescence study	Formed complexes; II had 6.0–55 mol% acrylamide	859
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	N,*N*-dimethylacrylamide	Fluorescence study	Formed complexes	859
	Ethylene glycol	Single T_g; FTIR; NMR	Semicrystalline when >50% II	374,712,953, 1115,1128
	Vinyl alcohol	FTIR; T_m-depression; NMR	Semicrystalline	197,1211,1255,1256,1257
	Vinyl methyl ether	Single T_g	Formed 1:1 complexes	160
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Acrylic acid-*co*-styrene	Caprolactone	Single T_g	I had 11.4–20.8 wt.% acrylic acid	383
	Ester	Single T_g	I had 20% acrylic acid; II was nylon 6, nylon 11 or nylon 12 (3 miscible blend systems)	471,884
	Ethyl methacrylate	Single T_g; IGC	I had 2.65–20 wt.% acrylic acid	72,1001
	Ethyl methacrylate-*co*-4-vinylpyridine	Single T_g; IGC	I had 20 mol% acrylic acid	1001
	Ethylene glycol	Single T_g	I had 21 wt.% acrylic acid	384
	Isobutyl methacrylate-*co*-4-vinylpyridine	Single T_g; IGC	I had 20 mol% acrylic acid	1001
	Methyl methacrylate	Single T_g	I had 8–20 wt.% acrylic acid	72,382,384
	n-Propyl methacrylate	Single T_g	I had 2.65–8.84 wt.% acrylic acid	70
	Vinyl methyl ether	Single T_g	I had 8–20 wt.% acrylic acid	576
Acrylonitrile-*co*-butadiene	Cellulose acetate-butyrate	Some compositions appeared homogeneous	–	30,249,452,651
	Chloroprene	Single T_g when I had 18% acrylonitrile; sorption of solvent vapors	Two phases in electron micrograph when I had 18% acrylonitrile may indicate a crystalline phase; two torsional transitions when I had ≥ 28% acrylonitrile	93,797
	Epichlorohydrin	Single T_g	II had 26–54 wt.% acrylonitrile	966
	Nitrocellulose	Transparent films when I had 28.6–44.4% acrylonitrile	Immiscible when I had 18.4% acrylonitrile	93,296,398,399 452
	Vinyl acetate-*co*-vinyl chloride	Single loss peak when ≤40% vinyl acetate in II; clear films for some mixtures of commercial polymers	Two loss peaks when ≥50% vinyl acetate in II	296,452 539,651

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Vinyl chloride	Single T_g; IGC	I had 29–45 wt.% acrylonitrile	9,79,85,161,196, 452,484,520,561, 614,615,632,682,755 799,860,867,885,1186
	Vinyl chloride	Single T_g; microscopy	I was hydrogenated; I had 38 wt.% acrylonitrile	1191
	Vinyl chloride, head-to-head	Single T_g	I had 23.5–56.6 wt.% acrylonitrile	158
Acrylonitrile-*co*-butadiene-*co*-methyl acrylate	Ethylene-*alt*-maleic anhydride	Single T_g; microscopy	I was made from a copolymer of AN/MA (75/25) grafted onto BD/AN (70/30) copolymer	1156
Acrylonitrile-*co*-butadiene-*co*-styrene	Propylene-*co*-vinyl chloride	Single T_g	I was Blendex 701; II had 3.2 or 3.8% propylene; electron micrographs may indicate two phases	107
Acrylonitrile-*co*-methyl methacrylate	*N,N*-dimethylacrylamide	Single T_g; transparency	I had 70 wt.% acrylonitrile (AN)	648
	Maleic anhydride	Single T_g; transparency	I had 70 wt.% AN	648
	Vinyl chloride	Single T_g; transparency	II had 2–17 wt.% AN	1091
	N-vinylpyrrolidone	Single T_g; transparency	I had 70 wt.% AN	648
Acrylonitrile-*co*-methyl methacrylate-*co*-α-methylstyrene	Vinyl chloride	Single T_g	I had 8/58/34 or 32.3/8.1/59.6/ acrylonitrile/methyl methacrylate/ α-methylstyrene	93,1078
Acrylonitrile-*co*-α-methylstyrene	Alkyl methacrylate	Single T_g; transparency	I had 30 wt.% acrylonitrile (AN); II had an alkyl group of acetonyl, chloromethyl, 2-chloroethyl, 2-hydroxyethyl, 2-hydroxypropyl, methoxymethyl, methylthiomethyl or tetrahydrofurfuryl (8 miscible blend systems)	266,268,279
	n-Butyl methacrylate	Single T_g	I had 12 wt.% AN	152
	n-Butyl methacrylate-*co*-methyl methacrylate	Single T_g; transparency	I had 30 wt.% AN; II had 70 wt.% methyl methacrylate	289
	Ethyl methacrylate	Single T_g	I had 10–28 wt.% AN	152,277
	Ethyl methacrylate-*co*-methyl methacrylate	Single T_g; transparency	I had 30 wt.% AN; II had 30 or 60 wt.% methyl methacrylate	287
	Ethylene, chlorinated	Single T_g	Depended on AN and Cl contents	154
	Maleic anhydride-*co*-styrene	Single T_g	Region of miscibility was sensitive to MW	1011
	Methyl methacrylate	Single T_g	I had 10–37 wt.% AN; I had 30 wt.% AN and II was atactic or isotactic (Ref. 277); I had 6.5–30 wt.% AN (Ref. 1010)	152,277, 1010
	Methyl methacrylate-*co*-2,2,6,6-tetramethyl-piperidinyl methacrylate	Single T_g; transparency	I had 30 wt.% AN; II had $\geq$76.2 wt.% methyl methacrylate	272,274
	n-Propyl methacrylate	Single T_g	I had 17–20 wt.% AN	152
	Vinyl chloride	Single T_g	I had 30 wt.% AN; miscible when melt blended or cast from methyl ethyl ketone; immiscible when cast from tetrahydrofuran; I had 11.9–30 wt.% AN (Ref. 1010)	435,684, 1010
Acrylonitrile-*co*-*p*-methylstyrene	Acetonyl methacrylate	Single T_g; transparency	I had 43–61 wt.% AN	118
	2-Bromoethyl methacrylate	Single T_g; transparency	I had 14–61 wt.% AN	608
	n-Butyl methacrylate	Single T_g; transparency	I had 13.6 wt.% AN	285
	2-Chloroethyl methacrylate	Single T_g; transparency	I had 18.3–46 wt.% AN	613
	Chloromethyl methacrylate	Single T_g; transparency	I had 13–42 wt.% AN	263
	Cyclohexyl methacrylate	Single T_g; transparency	I had $\leq$22 wt.% AN	120
	2,6-Dimethyl-1,4-phenylene ether	Single T_g; transparency	I had $\leq$5.0 wt.% AN	275
	Ethyl methacrylate	Single T_g; transparency	I had 12–33 wt.% AN	285
	Methyl methacrylate	Single T_g; transparency	I had 12–34 wt.% AN	285
	Methylthiomethyl methacrylate	Single T_g; transparency	I had 19–34 wt.% AN	269
	n-Propyl methacrylate	Single T_g; transparency	I had 6–28 wt.% AN	285
	Tetrahydrofurfuryl methacrylate	Single T_g; transparency	I had 12–48 wt.% AN	286
	Tetrahydropyranyl-2-methyl methacrylate	Single T_g; transparency	I had 5–29 wt.% AN	119

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
Acrylonitrile-*co*-styrene	Acetonyl methacrylate	Single T_g; transparency	I had 33–58 wt.% AN	121
	Arylate	Single T_g	Miscibility the greatest when AN = 25%	7
	Arylate of tetramethylbisphenol-A	Single T_g	I had 4–13 wt.% AN	1048
	Benzyl methacrylate	Single T_g	I had 7.8–24.3 wt.% AN	1180
	2-Bromoethyl methacrylate	Single T_g; transparency	I had 10–58 wt.% AN	608
	Butylene adipate	Single T_g	I had 13–32.3 wt.% AN	228
	Butylene sebacate	Single T_g	I had 13–20 wt.% AN	228
	n-Butyl methacrylate-*co*-methyl methacrylate	Single T_g; transparency	I had 22 wt.% AN; II had 70 wt.% methyl methacrylate	289,405
	tert-Butyl methacrylate-*co*-methyl methacrylate	Single T_g	Miscibility depended on compositions of I and II	621
	Caprolactone	Single T_g; FTIR; microscopy; LS	I had 13–30 wt.% AN	228,295,458,565,729, 1021,1047,1097,1145,1185
	2-Chloroethyl methacrylate	Single T_g; transparency	I had 12.5–43 wt.% AN	613
	Chloromethyl methacrylate	Single T_g; transparency	I had 12–37 wt.% AN	262
	Copolyarylate of Tetramethylbisphenol-A and tetrabromobisphenol-A	Single T_g	Miscibility depended on compositions of I and II	
	Copolycarbonate of bisphenol-A and tetramethylbisphenol-A	Single T_g	I had $\leq$ 18 wt.% acrylonitrile; II had $\leq$ 22 wt.% bisphenol-A	431
	Cyclohexyl methacrylate	Single T_g; transparency	I had $\leq$ 20 wt.% AN	119
	Cyclohexyl methacrylate-*co*-methyl methacrylate	Single T_g	Miscibility depended on compositions of I and II	621
	2,6-Dimethyl-1,4-phenylene ether	Single T_g	I had $\leq$ 10.5 wt.% AN	457,1022
	2,6-Dimethyl-1,4-phenylene ether, benzoylated	Single T_g	I had 18 mol% AN; II had 35 or 86% benzoylated units	776
	Ethylene adipate	Single T_g	I had 25–28 wt.% AN	228
	Ethyl methacrylate	Single T_g; transparency	I had 5.5–28.0 wt.% AN (Ref. 229); I had 9–34 wt.% AN (Ref. 112)	112,229,489
	Ethyl methacrylate-*co*-methyl methacrylate	Single T_g; transparency	I had 22 wt.% AN; II had 30 or 50 wt.% methyl methacrylate	288
	Ethyl methacrylate-*co*-2,2,6,6-tetramethyl-piperidinyl methacrylate	Single T_g; transparency	I had 30 wt.% AN; II had $\leq$ 13.3 wt.% 2,2,6,6-tetramethyl-piperidinyl methacrylate	274
	Ethyloxazoline	Single T_g; transparency	I had 25–40 wt.% AN; two T_g's when I had 8, 23.5 or 70% AN	426,652
	Hexamethylene sebacate	Single T_g	I had 16.2 wt.% AN	228
	2-Hydroxyethyl methacrylate	Transparency	I had 22 wt.% AN	277
	2-Hydroxypropyl methacrylate	Single T_g; transparency	I had 22 wt.% AN	279
	Itaconic anhydride-*co*-methyl methacrylate	Single T_g	Miscibility depended on compositions of I and II	920
	Methoxymethyl methacrylate	Single T_g; transparency	I had 30 wt.% AN	269
	Methyl methacrylate	Transparency	I had 9.4–34.4 wt.% AN (Ref. 789); I had 9.5–28 wt.% AN (Ref. 229); I had 9–39 wt.% AN (Ref. 164)	56,164,229, 392,566, 598,651,670,741, 782,789
	Methyl methacrylate-*co*-*N*-phenylitaconimide	Single T_g	Miscibility depended on compositions of I and II	169
	Methyl methacrylate-*co*-phenyl methacrylate	Single T_g	Miscibility depended on compositions of I and II	621
	Methyl methacrylate-*co*-2,2,6,6-tetramethyl-piperidinyl methacrylate	Single T_g; transparency	I had 22 or 30 wt.% AN; II had $\leq$ 31.4 wt.% 2,2,6,6-tetramethyl-piperidinyl methacrylate	272,274
	Methylthiomethyl methacrylate	Single T_g; transparency	I had 9–36 wt.% AN	267
	Novolac	Single T_g	I had 25 wt.% AN; II was formaldehyde + 13/17/70 mol% *p-tert*-butylphenol/*m*-cresol/*o*-cresol	221
	Oxy-1,4-phenylene-sulfonyl-1,4-phenylene oxy-2,6-diisopropyl-1,4-phenylene isopropylidene 3,5-diisopropyl-1,4-phenylene	Modulus-temperature	I had 13–16% AN	685
	Phenyl acrylate	Single T_g	I had 14.6–34.1 wt.% AN (Ref. 717) or 11.5–32 wt.% AN (Ref. 1171)	717,1171, 1180

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	n-Propyl methacrylate	Single T_g	I had 5.7–19.5 wt.% AN (Ref. 229) or 9–24 wt.% AN (Ref. 1090)	229, 1090
	Tetrahydrofurfuryl methacrylate	Single T_g; transparency	I had 9–48 wt.% AN	281,286
	Tetrahydropyranyl-2-methyl methacrylate	Single T_g; transparency	I had ≤ 31 wt.% AN	116
	Vinyl chloride	Single T_g	I had 11.5–26 wt.% AN	437
Acrylonitrile-*co*-vinylidene chloride	Caprolactone	Single T_g	I was usually Saran F (80% vinylidene chloride); semicrystalline when ≤ 70 wt.% I	31,825,887
	Hexamethylene terephthalate	Single T_g	Probably semicrystalline; I was Saran F(80% vinylidene chloride)	36
Allyl alcohol-*co*-styrene	Ester	Single T_g	I had M_W = 1.4–2.1 kg/mol and had 1.3–7.7 wt.% OH groups; II was butylene adipate, butylene sebacate, caprolactone, 1,4-cyclohexane dimethylene succinate, decamethylene sebacate, 2,2-dimethyl-1,3-propylene adipate or hexamethylene sebacate (7 blend systems); high OH content of I favored miscibility	46,869
Amic acid	3,3′-Diamino-4,4′-benzidine isophthalamide	Transparency; FTIR	I was PARC-TPI	1071
	Phenol–formaldehyde	Clear before and after curing	I was made from 4,4′-methylene-dianiline and pyromellitic dianhydride	1
	Imide	Single T_g	II was Polyimide P84; miscible over certain composition range	538
Amideimide	Novolac	Single T_g	I was made from rosin	466
	Resol	Single T_g	I was made from rosin maleic anhydride Diels–Alder adduct with 4,4′-diamino-diphenylmethane	468
	Shellac	Single T_g	I was made from rosin-imidoamino acid with 4,4′-diaminodiphenylmethane	469
n-Amyl methacrylate	Vinyl chloride	Single T_g; transparency	–	319,649,812,850
	Vinyl chloride, chlorinated	Single T_g; transparency	II had 65 or 68 wt.% Cl	319,810
	Vinyl chloride-*co*-vinylidene chloride	Single T_g; transparency	II was Saran, $\geq$ 80% vinylidene chloride	407,812
Aramide	Amide	Single T_g	I was made from 2,2,4-trimethyl-1,6-hexanediamine and terephthalic acid (Trigamid T); semicrystalline; II was nylon 6, nylon 6,6, nylon 6,12, nylon 6,9 or nylon 6,10 (5 miscible blend systems)	211,212, 595
	Butylene terephthalate	Single T_g	I was Trogamid T; semicrystalline	596
	Caprolactam-*co*-laurolactam	Single T_g	I was Trogamid T; II had 43.8–55.6 mol% caprolactam; semicrystalline	213,214
	Caprolactam-*co*-2-pyrrolidinone	Single T_g	I was Trogamid T; II had 93.5 mol% caprolactam; semicrystalline	213,214
	Copolyester	Single T_g	I was Trogamid T; II was made from 1,4-cyclohexanedimethanol/ethylene glycol/terephthalic acid (PETG 6763)	596
	Cycloaliphatic amide	Single T_g	I was Trogamid T; II was made from 1,4- and 1,3-benzenedicarboxylic acids and a mixture of 1,6-hexanediamine and bis(4-aminocyclohexyl)methane (Bexloy AP C-803)	212
	Ethylene terephthalate	Single T_g	I was Trogamid T; semicrystalline	596
Aramide	Amide	Single T_g	I was made from 1,6-hexanediamine and a mixture of isophthalic and terephthalic acids (Zytel 330); semicrystalline; II was nylon 6, nylon 6,6, nylon 6,9 or nylon 4,6 (4 miscible blend systems)	212,213
	Cycloaliphatic amide	Single T_g	I was Zytel 330; II was Bexloy AP C-803	212
Aramide	Amide	Single T_g; T_m-depression	I was poly[2,2′-bis(trifluoromethyl)-4,4′-biphenylene dicarboxamide]; II was nylon 6, nylon 4,6, nylon 6,6, nylon	1013

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
			6,9, nylon 6,10 or nylon 11 (6 blend systems)	
Arylate	Aramide	Single T_g	I was made from bisphenol-A and a mixture of terephthalic acid/isophthalic acid/5-*tert*-butyl-isophthalic acid = 44/44/12 (Polyarylate C); II was Trogamid T	188
	Aramide-S	Single T_g	I was Polyarylate C; II was made from 5-*tert*-butylisophthalic acid/isophthalic acid 90/10 and 1,12-diaminododecane (Aramide-S)	189
	Aramide-M	Single T_g	I was made from bisphenol-A and a mixture of terephthalic acid/isophthalic acid/5-*tert*-butyl-isophthalic acid = 42.5/42.5/15 (Polyarylate D); II was made from adipic acid and *m*-xylene-diamine/hexamethylene diamine 80/20 (Aramide-M)	189
	Aramide	Single T_g	I was Polyarylate D; II was Trogamid T	189
	Aramide-S	Single T_g	I was Polyarylate D; II was Aramide-S	189
	Aramide-M	Single T_g	I was made from bisphenol-A and a mixture of terephthalic acid/isophthalic acid/5-*tert*-butyl-isophthalic acid = 37.5/37.5/25 (Polyarylate E); II was Aramide-M	189
	Aramide	Single T_g	I was Polyarylate E; II was Trogamid T	189
	Aramide-S	Single T_g	I was Polyarylate E; II was Aramide-S	189
	Aramide	Single T_g	I was made from bisphenol-A and a mixture of terephthalic acid/5-*tert*-butylisophthalic acid 50/50 (Polyarylate F); II was Trogamid T	189
	Aramide	Single T_g	I was Polyarylate F; II was Aramide-S	189
	Carbonate of bisphenol-A	Single T_g	I was made from bisphenol-A and a mixture of terephthalic acid/isophthalic acid 50/50 (Ardel D-100); miscibility arised from transesterification	687
	Copolyester	Single T_g	I was Ardel D-100; II was made from cyclohexane dimethanol and terephthalic acid/isophthalic acid (80–87% tere) (Kodar A-150) and another copolymer with tere/iso ratio of 70/30	687
	Copolyester	Single T_g	I was Ardel D-100; II was made from cyclohexane dimethanol, ethylene glycol and terephthalic acid (molar ratio 1:2:3, PETG-6763)	291,687
	Ethylene terephthalate	Single T_g	I was Ardel D-100; miscibility arised from transesterification	687
	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g	I was Ardel D-100; miscibility arised from transesterification	687
Arylate of tetramethyl-bisphenol-A	Butylene sebacate	Single T_g	Semicrystalline	907
	Hexamethylene sebacate	Single T_g	Semicrystalline	906
Aryl ether ketone	Etherimide	Single T_g	I was made from 4,4′-difluoro-benzophenone and hydroquinone; II was Ultem 1000	334
	Etherimide	Single T_g	I was made from *p*-phenoxybenzoyl chloride; II was Ultem 1000	334
Aryl sulfone	Imide	Single T_g	I was Radel A; II was XU-218	375
γ-Benzyl glutamate	γ-Butyl glutamate	Dynamic mechanical methods; DSC	Miscible as random coils but not as α-helics	813
Bis(methyoxyethoxyethoxy) phosphazene	Acrylic acid-*co*-methyl methacrylate	Single T_g; transparency	II had 10, 20 or 30 mol% acrylic acid	1087
	Methacrylic acid-*co*-methyl methacrylate	Single T_g; transparency	II had 20 mol% methacrylic acid	1087
	Styrene-*co*-styrene-4-carboxylic acid	Single T_g; transparency	II had 75 mol% styrene	1087

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Styrene-*co*-*p*-vinylphenol	Single T_g; transparency	II had 78 mol% styrene	1087
	p-Vinylphenol	Single T_g; transparency	–	1087
Bis(methoxyethoxyethoxy-ethoxy)phosphazene	*p*-Vinylphenol	Single T_g; transparency	–	1087
3-Bromo-2,6-dimethyl-1,4-phenylene ether	*p*-Bromostyrene	Single T_g; microscopy	–	402
	p-Bromostyrene-*co*-styrene	Single T_g; microscopy	II had $\leq$85 mol% styrene	402
3-Bromo-2,6-dimethyl-1,4-phenylene oxide-*co*-2,6-dimethyl-1,4-phenylene ether	*p*-Bromostyrene	Single T_g; microscopy	I had $\leq$22 mol% 2,6-dimethyl-1,4-phenylene ether	402
	Styrene	Single T_g; microscopy	Miscible up to full bromination of I when I had MW $\leq$40 kg/mol and II had MW $\leq$30 kg/mol	402,530
2-Bromoethyl methacrylate	Ester	Single T_g	Semicrystalline; II was butylene adipate, caprolactone, 2,2-dimethyl-1,3-propylene adipate, 2,2-dimethyl-1,3-propylene succinate, ethylene adipate, or hexamethylene sebacate (6 miscible blend systems)	1137
	Tetrahydrofurfuryl methacrylate	Single T_g; transparency	–	606
p-Bromostyrene-*co*-styrene	2,6-Dimethyl-1,4-phenylene ether	Single T_g; microscopy	I had $\geq$53 mol% styrene	402
Butadiene	Butadiene	Single T_g	I was 1,4-polybutadiene; II had 63% 1,2-	701
	Diene-*co*-ethylene-*co*-propylene	Single T_g	Commercial samples; many mixtures had two T_g's	92,440,451
	Isoprene	Single T_g; microscopy; light scattering; NMR	Depended on microstructure; immiscible when I and II were both highly cis; I was deuterated (Ref. 335)	47,80,149,237,249, 335,520,542,548,549, 757,770,771,787,854, 901,1032,1063
Butadiene-*co*-styrene	*tert*-Butylstyrene	Single dynamic mechanical loss peak	I had 25% styrene; II had $M_w = 0.835$ kg/mol	138
	Glycerol ester of hydrogenated rosin	Single dynamic mechanical loss peak	I had 25% styrene; II had $M_w = 0.5$ kg/mol	139
1-Butene	1-Pentene	Single T_g	–	190
	Propylene	Single T_g	Semicrystalline	190,502,657
Butyl acrylate	Ethylene, chlorinated	Single T_g; transparency	II had 49.8 or 62.1% Cl; immiscible when II had 48.0% Cl	100
	4-Hydroxystyrene-*co*-styrene	Single T_g	I was *tert*-butyl; II had 28-66 mol% 4-hydroxystyrene	897
	Vinyl chloride	Single T_g; transparency	Some papers showed two T_g's (Refs. 812, 866) or phase separation in solution (Ref. 425)	425,655,812,845, 846,850,851 866
	Vinyl chloride-*co*-vinylidene chloride	Single T_g; transparency	II was Saran ($\geq$80% vinylidene chloride)	812
Butyl acrylate-*co*-methyl methacrylate	Vinyl chloride	Single T_g; electron microscopy	I was Degalan V26(90% methyl methacrylate) (Ref. 886); I had 75% butyl acrylate) (Ref. 654)	654,886
Butyl methacrylate	*p*-(2-Hydroxy-hexafluoroisopropyl) styrene-*co*-styrene	Single T_g; transparency	II had 83.3–98.25 mol% styrene; immiscible when II had 99.0 mol% styrene	103,572,646, 807
	p-(2-Hydroxy-hexafluoroisopropyl)-α-methylstyrene-*co*-styrene	NMR	II had 1.5 mol% OH-containing monomer	89
	Vinyl chloride	Single T_g; transparency; miscible in solution	I was *n*-butyl, isobutyl or *sec*-butyl	319,425,649, 653,846,810,850
	Vinyl chloride-*co*-vinylidene chloride	Single T_g; transparency	II was Saran, $\geq$80% vinylidene; I was *n*-butyl, isobutyl or *sec*-butyl	409,812
n-Butyl methacrylate-*co*-methacrylic acid	Styrene-*co*-4-vinylpyridine	Single T_g; FTIR	I had 8 or 18 mol% methacrylic acid; II had 15, 22 or 28 mol% 4-vinylpyridine	985
n-Butyl methacrylate-*co*-4-vinylpyridine	Styrene-*co*-methacrylic acid	Single T_g; FTIR	I had 10.6, 18.9 or 26 mol% 4-vinylpyridine; II had 12, 24 or 29 mol% methacrylic acid	198,985
N-Butyl-3-hydroxymethyl carbazoyl methacrylate	ω-Hydroxybutyl-3,5-dinitrobenzoyl methacrylate	Single T_g	–	693
tert-Butylstyrene	Cyclohexyl acrylate	Single T_g; transparency	–	765
	Cyclohexyl methacrylate	Single T_g; transparency	–	765

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Isoprene	Single dynamic mechanical loss peak	I had $M_w = 0.835$ kJ/mol II was natural rubber	138
	3,3,5-Trimethyl-cyclohexyl methacrylate	Single T_g; transparency	-	765
Butylene adipate	Epichlorohydrin	Single T_g	Semicrystalline	223
	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g; melt transparent; IGC	Semicrystalline when ≤ 60 wt.% II	332,1205
	Vinyl acetate-*co*-vinylidene chloride	Single T_g	II was Saran C (≥ 80% vinylidene chloride); semicrystalline	31
	Vinyl bromide	Single T_g	-	150
	Vinyl chloride	Single T_g	Semicrystalline when ≤ 70 wt.% II	900
Butylene dodecamethylene dicarboxylate	Vinyl chloride	Single T_g	Semicrystalline	872
Butylene 2,6-naphthalene dicarboxylate	Ethylene 2,6-naphthalene dicarboxylate	Single T_g	Blend had ≥ 80 wt.% I	1238
Butylene sebacate	Epichlorohydrin	Single T_g	Semicrystalline	223
	Vinyl chloride	Single T_g	Semicrystalline	872
Butylene terephthalate	Arylate	Single T_g; X-ray scattering	Semicrystalline	1040
	Butylene oxide-γ-butylene terephthalate	Single T_g; dielectric loss measurements; microscopy	II had 84 wt.% butylene terephthalate	554,703
	Copolyester	Single T_g	Semicrystalline when ≥ 25 wt.% I; II was CP-350(bisphenol-A-neopentyl glycol terephthalate)	768
	Copolyester	Single T_g	Semicrystalline; II was made from bisphenol-A and 1/1 or 1/3 terephthaloyl chloride/isophthaloyl chloride	438,439
	Decamethylene-4,4′-terephthaloyl-dioxydibenzoate	Single T_g; microscopy; T_m-depression	0–50% II in blend	660,661,662
	Ethylene terephthalate	Single T_g	Semicrystalline	219,220,916
	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g; melt transparent	Semicrystalline	688,689,735
	Vinyl acetate-*co*-vinylidene chloride	Single T_g	II had ≥ 80% vinylidene chloride	36
	Vinyl chloride	Single T_g	Semicrystalline	36,686
	Vinyl chloride, chlorinated	Single T_g	Semicrystalline	36
Caprolactam	Acrylic acid	Single T_g	Semicrystalline	378
	Hexamethylene adipamide	Single T_g; enthalpy recovery kinetics	Semicrystalline	215,830
	Methacrylic acid	SAXS	Semicrystalline	371
	Styrene, sulfonated and Li-neutralized	Single T_g	II had 9.8 mol% sulfonated groups; semicrystalline	583,584
	Styrene, sulfonated and Mn-neutralized	Single T_g; T_m-depression	II had 10.2 mol% sulfonated groups; Semicrystalline	523
	Styrene, sulfonated and Zn-neutralized	Single T_g; T_m-depression	II had 10.1 mol% sulfonated groups; semicrystalline	524
Caprolactone	Acrylonitrile-*co*-*tert*-butylstyrene	Single T_g; SALS; microscopy	II had 19.8–39.6 wt.% acrylonitrile	1175
	3,3-Bis(chloromethyl)oxetane	Single T_g	Semicrystalline; II was Penton	301,445
	Chlorostyrene	Single T_g	II was *o*-chlorostyrene or mixed *o*- and *p*-isomers; semicrystalline when ≥ 21% I	18,32
	Epichlorohydrin	Single T_g	Semicrystalline	82,223
	Ester	DSC; SALS; X-ray diffraction; microscopy	II was made from terephthaloyl chloride and 4,4′-bis(6-hydroxy-hexyloxy) biphenyl	821
	Ethylene, chlorinated	Single T_g	Semicrystalline when ≤ 90% II; II had ≥ 30% Cl; two T_g's when II had 25% Cl	32,55,191 295
	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g; melt transparent; IGC	Semicrystalline when >50 wt.% I	82,193,332,982
	Nitrocellulose	Single T_g	≥ 50% I; semicrystalline when ≥ 50% I	82,352,353
	Propylene, chlorinated	Single T_g	I had 66 wt.% Cl; semicrystalline when ≥ 50 wt.% I	18,32
	Vinyl acetate-*co*-vinylidene chloride	Single T_g	II was Saran C (≥ 80 mol% vinylidene chloride); semicrystalline	31,887
	Vinyl bromide	Single T_g	Semicrystalline when >30% I	147

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Vinyl chloride	Single T_g; NMR	Semicrystalline when $\geq$30% I	82,352,353,446 628,635,872,900, 1082
	Vinyl chloride, chlorinated	Single T_g	II had 67.2 wt.% Cl; semicrystalline when $\geq$50 wt.% I	54
	Vinyl chloride-*co*-vinylidene chloride	Single T_g	Semicrystalline; II had 79% or 88% vinylidene chloride; some samples of II were Saran B	31,825,868, 873,887
	Vinylidene fluoride	T_m-depression	Semicrystalline	386
	Vinyl methyl ether	Single T_g		1145
Caprolactone-*co*-ethylene terephthalate	Ethylene, chlorinated	Single T_g	I had 33–89 wt.% ethylene terephthalate; II had 67.2 wt.% Cl	525,892
	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g; FTIR	II had 48–82 wt.% ethylene terephthalate	1243
	Vinyl chloride	Single T_g	I had 33 or 44 wt.% ethylene terephthalate	525
	Vinyl chloride, chlorinated	Single T_g	I had 33 wt.% ethylene terephthalate	525
Carbon monoxide-*co*-ethyl acrylate-*co*-ethylene	Vinyl chloride	Single T_g	Semicrystalline when I had >60% ethylene; I had 4.0–19.6% carbon monoxide, 5.5–35.4% ethyl acrylate, and 34.2–79.6% ethylene	692
Carbon monoxide-*co*-2-ethylhexyl acrylate-*co*-ethylene	Vinyl chloride	Single T_g	I had 17.8/34.2/47.9 or 18.6/39.6/41.8 = carbon monoxide/2-ethylhexyl acrylate/ethylene	692
Carbon monoxide-*co*-ethylene-*co*-vinyl acetate	Chlorostyrene	Single T_g	I was Elvaloy 741; II was a mixture of *o*- and *p*- isomers	18
	Propylene, chlorinated	Single T_g	I was Elvaloy 741	18
	Vinyl chloride	Single T_g	Semicrystalline when I had >60% ethylene; I was Elvaloy 741 or 10.5/66.7/22.8 or 10.2/58.2/31.5 = carbon monoxide/ethylene/vinyl acetate	21,40,92
Carbonate of bisphenol-A	Alkylene terephthalate	Single T_g	II had an alkylene group of 2-ethyl-2-methylpropylene, 1,7-heptylene, 1,9-nonylene or 1,5-pentylene (4 blend systems); miscibility arised from *trans*-esterification	1074
	Biphenyl-4,4′ylene sebacate	Single T_g	Up to 40 wt.% II in blend	387
	Butylene adipate	Single T_g	Semicrystalline	174,176
	Butylene terephthalate	Single T_g; FTIR	–	1132
	p-tert-Butylphenol formaldehyde	Single T_g	–	1057
	Caprolactone	Single T_g; SANS; SAXS; FTIR; NMR	Semicrystalline, especially when $\geq$ 25 wt.% II	174,176,645 948,949,950
	Caprolactone-*co*-ethylene terephthalate	Single T_g; FTIR	II had 48 or 58 wt.% ethylene terephthalate	1120,1242
	Carbonate	Single T_g; transparency	II was carbonate of bisphenol-E, bisphenol-F, bisphenol-Z, bisphenol chloral, dimethylbisphenol-A, dimethylbisphenol-Z, norborane, or tetramethylbisphenol-F (8 miscible blend systems)	427
	Copolyester	Single T_g; melt transparent	Semicrystalline when cast from solvent up to 80 wt.% II; II was prepared from bisphenol-A and 1/1 = terephthalic/isophthalic acid	585
	Copolyester	Single T_g	II was made from terephthalic acid and 1,4-cyclohexane-dimethanol and up to 60 mol% ethylene glycol	23,1093
	Copolyester	Single T_g; melt transparent	Semicrystalline; II was KODAR A150, made from 1,4-cyclohexanedimethanol and 1/4 = isophthalic/terephthalic acid	174,556 581,1251
	Copolyester	Single T_g	II was Hytrel 5556, a multiblock copolymer with poly(tetramethylene ether)glycol terephthalate soft segment	248

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
			and tetramethylene terephthalate hard segment	
	1,4-Cyclohexane-dimethylene succinate	Single T_g	Semicrystalline	174,175
	1,4-Cyclohexane-dimethylene terephthalate	Single T_g; melt transparent	Semicrystalline	174,486,581
	Cyclohexyl methacrylate-*co*-methyl methacrylate	Single T_g	II had 6.1–43.8 wt.% cyclohexyl methacrylate	574,622,1039
	Ethylene adipate	Single T_g	Semicrystalline	174,176
	Ethylene succinate	Single T_g	Semicrystalline	174,176
	Ethylene terephthalate	Single T_g	Semicrystalline; ≥ 50 or 75% II; two T_g's when $\leq 60\%$ II	337,593 603,1151
	p-(2-Hydroxy-hexafluoroisopropyl) styrene-*co*-styrene	Single T_g; transparency	II had 29–78 mol% styrene; immiscible when II had 83 mol% styrene	571,646, 807
	Maleic anhydride-*co*-styrene	Light scattering	II had 8% maleic anhydride	712
	Methacrylic acid-*co*-styrene	Single T_g	II had 2–23 wt.% methacrylic acid	10,11
	Methyl methacrylate	Single T_g	Blends prepared by precipitation of THF solutions in *n*-heptane or cast from hot THF solutions	29,109, 481,486,715
	Methyl methacrylate-*co*-*p*-methylstyrene	Single T_g	II had ≥ 70 wt.% methyl methacrylate	357
	Methyl methacrylate-*co*-*N*-phenylmaleimide	Single T_g	II had ≥ 70 wt.% methyl methacrylate	357
	Methyl methacrylate-*co*-phenyl methacrylate	Single T_g	II had 16–63 wt.% phenyl methacrylate	622
	Methyl methacrylate-*co*-styrene	Single T_g	II had ≤ 14 wt.% styrene	577,930
	Novolac	Single T_g	II had MW = 1 kg/mol and was made from formaldehyde and 13/17/70 mol% = *p*-*tert*-butylphenyl/*m*-cresol/*o*-cresol	221
	Phenyl methacrylate	Single T_g	–	480,622
	Pyrrole	Single T_g	Blend had 7% II	856
	Vinylidene chloride-*co*-vinyl chloride	Single T_g	II had 13.5 wt.% vinyl chloride; semicrystalline when ≤ 25 wt.% I	871
Carbonate of hexafluorobisphenol-A	Alkyl methacrylate	Single T_g; transparency	II had an alkyl group of methyl, ethyl or *n*-propyl (3 miscible blend systems)	225,430
	Isopropyl methacrylate	Single T_g; transparency	≤ 70 wt.% I in blend	430
	Methyl methacrylate-*co*-α-methylstyrene	Single T_g	II had 81 or 95 wt.% methyl methacrylate	930
Carbonate of tetramethyl-bisphenol-A	Acrylonitrile-*co*-α-methylstyrene	Single T_g	II had 4–16.5 wt.% acrylonitrile	1010
	Acrylonitrile-*co*-styrene	Single T_g; FTIR	II had < 13% acrylonitrile	8,225,1069
	Allyl alcohol-*co*-styrene	Single T_g	II had < 19.1 wt.% allyl alcohol	225
	Butylene sebacate	Single T_g	Semicrystalline	227
	Caprolactone	Single T_g	Semicrystalline	227
	Carbonate of bisphenol-A	Single T_g; transparency; dielectric spectroscopy	–	51,427,1124
	Carbonate	Single T_g; transparency	II was carbonate of bisphenol-AP, bisphenol-E, bisphenol-F, bisphenol-S, bisphenol-Z, bisphenol chloral, dimethylbisphenol-A, dimethylbisphenol-Z, norbonane, tetramethylbisphenol-F or tetramethylbisphenol-P (11 miscible blend systems)	427
	Ester	Single T_g	II was decamethylene decamethylene dicarboxylate, decamethylene sebacate, ethylene adipate, hexamethylene decamethylene dicarboxylate, or hexamethylene sebacate (5 miscible blend systems)	227
	Maleic anhydride-*co*-styrene	Single T_g	II had 8% maleic anhydride	225
	Methyl methacrylate-*co*-styrene	Single T_g	II had < 50 wt.% methyl methacrylate	575,884,1194
	α-Methylstyrene	Single T_g	–	227
	p-Methylstyrene-*co*-tetrahydrofurfuryl methacrylate	Single T_g	II had ≤ 50 mol% tetrahydrofurfuryl methacrylate	270

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Styrene-*co*-tetrahydrofurfuryl methacrylate	Single T_g	II had ≤ 53 mol% tetrahydrofurfuryl methacrylate	270
Cellulose	Acrylonitrile	NMR; single T_g; microscopy; WAXD	–	560,627,1208 556
	Caprolactone	NMR	–	625,1075
	Ethylene glycol	Single T_g; FTIR	–	
	Vinyl acetate	Single T_g	$\geq 70\%$ I in blend	626
	Vinyl alcohol	NMR; FTIR; T_m-depression	–	560,623,744,1075
	4-Vinylpyridine	Single T_g, NMR	–	559,1208
Cellulose acetate	2,6-Dimethyl-1,4-phenylene ether, phosphonate derivative	Single T_g	II had 9.5% P	87
	2,6-Dimethyl-3-bromo-1,4-phenylene ether, phosphonate derivative	Single T_g	II had 10.5% P	87
	Styrene, phosphonate derivative	Single T_g	II had 8.0% P	87
	Styrene, phosphorylated	Single T_g	II had more than 1 phosphoryl group per 4 styrene units	550
	Styrenephosphonate ester	Single T_g	Blend initially prepared was heterogeneous, became miscible when annealed above T_g	792
	styrene-*co*-vinylidene chloride phosphonylated, chloromethylated	Single T_g; electron microscopy	I was Eastman 394–45; II had 7% P, 14.5% Cl	88
	styrene-*co*-inylidene chloride phosphonylated	Single T_g	II had 8.6% P	87
	4-Vinylpyridine	Single T_g; cast films transparent	Semicrystalline; I had 39.4% acetyl content or had 10H/2 glucose	27,86
Cellulose acetate-butyrate	Ethylene-*co*-vinyl acetate	Single T_g; transparent films	I had 18.5/45 = acetic acid/butyric acid and II had 80% vinyl acetate; two T_g's when II had 70% vinyl acetate	93,452
	Hydroxybutyrate-*co*-valerate	Single T_g; WAXD; NMR	II had 10 mol% valerate; miscible when blend had 20–50% II, and partially miscible when blend had 60–80% II	84
Cellulose acetate propionate	Alkylene glutarate	Single T_g	The alkylene group was ethylene, diethylene, hexamethylene, pentamethylene, tetramethylene or trimethylene (6 miscible blend systems)	927
Cellulose triacetate	Highly subsitituted cyanoethyl cellulose	Single T_g; FTIR; microscopy	–	1026
Chitin	*N*-Vinylpyrrolidone	Single T_g; FTIR	–	1133
Chitin-*g*-2-methyl-2-oxazoline	Vinyl alcohol	Single T_g; FTIR	–	913
	Vinyl chloride	Sinlge T_g; FTIR	Blend had ≤ 10 wt.% II	914
Chitosan	Ethylene oxide	T_m depression	–	1245
Chitosan acetate	Ethylene oxide	FTIR; microscopy	–	1227
p-Chloro-(3-ethenoxypropyl) cinnamate	*p*-Methoxy-(6-ethenoxyhexyl) cinnamate	Single T_g	–	515
	p-Methoxy-(6-ethenoxypropyl) cinnamate	Single T_g	–	515
	p-Methyl-(3-ethenoxypropyl) cinnamate	Single T_g	–	515
1-Chloroethyl methacrylate	Alkyl methacrylate	Single T_g; transparency	The alkyl group of II was methyl, ethyl, *n*-propyl, isopropyl, *n*-butyl or tetrahydrofurfuryl (6 miscible blend systems)	1152
	Ester	Single T_g; transparent melt	II was butylene adipate, caprolactone, 2,2-dimethyl-1,3-propylene adipate, 2,2-dimethyl-1,3-propylene sebacate, 2,2-dimethyl-1,3-propylene succinate or hexamethylene sebacate (6 miscible blend systems)	1154
2-Chloroethyl methacrylate	Alkyl methacrylate	Single T_g; transparency	The alkyl group of II was methyl, ethyl or tetrahydrofurfuryl (3 miscible blend systems)	612
	Ester	Single T_g; transparent melt	II was butylene adipate, caprolactone, 2,2-dimethyl-1,3-propylene adipate, 2,2-dimethyl-1,3-propylene sebacate, ethylene adipate or hexamethylene sebacate (6 miscible blend systems)	609,611
	N-vinylpyrrolidone	Single T_g	–	607

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
Chloromethyl methacrylate	Alkyl methacrylate	Single T_g	II had an alkyl group of methyl, ethyl, *n*-propyl, isopropyl, acetonyl or tetrahydrofurfuryl (6 miscible blend systems)	123,273
	Carbonate of bisphenol-A	Single T_g	Miscibility depended on method of preparation	271
	Ester	Single T_g; transparent melt	II was butylene adipate, caprolactone, 2,2-dimethyl-1,3-propylene adipate, 2,2-dimethyl-1,3-propylene sebacate, 2,2-dimethyl-1,3-propylene succinate, ethylene adipate, ethylene succinate or hexamethylene sebacate (8 miscible blend systems)	609,611
	N-vinylpyrrolidone	Single T_g	–	607
Chloroprene	Ethylene-*co*-methyl acrylate	Single T_g; phase contrast microscopy	Semicrystalline	396
	Isoprene, epoxidized	Single T_g; FTIR		597
3-Chloropropyl methacrylate	Acrylonitrile-*co*-*p*-methylstyrene	Single T_g	II had 15–33 wt.% AN	1105
	Acrylonitrile-*co*-styrene	Single T_g	II had 12–35 wt.% AN	1105
	Ester	Single T_g	II was butylene adipate, caprolactone, 2,2-dimethyl-1,3-propylene adipate or hexamethylene sebacate (4 miscible blend systems)	1111
	Tetrahydrofurfuryl methacrylate	Single T_g	–	1104
o-Chlorostyrene	Styrene	Single T_g	–	396
o-Chlorostyrene-*co*-*p*-chlorostyrene	2,6-Dimethyl-1,4-phenylene ether	Single T_g; transparency	I had 23-64% *p*-chlorostyrene	16
	Styrene	Single T_g	After quenching from 145°C; I had 71–92 mol% *o*-chlorostyrene; two T_g's when I had $\leq$62 mol% *o*-chlorostyrene	131,803
o-Chlorostyrene-*co*-*o*-fluorostyrene	2,6-Dimethyl-1,4-phenylene ether	Single T_g; transparency; NMR	I had 14–40 mol% *o*-chlorostyrene; two T_g's when I had >36 mol% *o*-chlorostyrene	22,840, 843
p-Chlorostyrene-*co*-*o*-fluorostyrene	2,6-Dimethyl-1,4-phenylene ether	Single T_g; transparency; NMR	I had 13–74 mol% *p*-chlorostyrene; two T_g's when I had >74 mol% *p*-chlorostyrene	22,840, 843
p-Chlorostyrene-*co*-styrene	2,6-Dimethyl-1,4-phenylene ether	Single T_g	I had $\geq$34.7 mol% styrene; two T_g's when I had <34.7 mol% styrene	243,413, 749
Collagen hydrolyzate	Ethyl acrylate	Single T_g	–	578
Copolyester	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g	I was cyclohexylene dimethylene tere/isophthalate	689
	p-(2-Hydroxy-hexafluoroisopropyl) styrene-*co*-styrene	Single T_g; clear films	I was KODAR PETG (made from terephthalic and isophthalic acid and ethylene glycol and 1,4-dihydroxymethyl cyclohexane; II had 60 mol% styrene; immiscible when II had $\geq$83 mol% styrene	571,646
		Single T_g; clear films	I was VITEL PE-307; II had 83–90.3 mol% styrene	646
		Single T_g	I was VITEL VPE-5545; II had 83 mol% styrene; immiscible when II had 90.3 mol% styrene	646
	Vinyl chloride	Single T_g for II with soft segments of I	I was Hytrel, a multiblock copolymer with poly(tetramethylene ether) glycol terephthalate soft segments and tetramethylene terephthalate hard segments; hard segments crystalline; two T_g's after 130°C annealing; Refs. 343, 344 showed single T_g to 50 or 60 wt.% I	343,344, 617
	Vinyl chloride	Single T_g	I was made from adipic acid, hexane-1,6-diol and neopentanediol and had $M_n = 2$ kg/mol	331
Copolyester-carbonate	Ethylene terephthalate	Single T_g	I was from 1/2 = terephthalate/bisphenol-A with ester and carbonate linkages	3

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Vinyl chloride	Single T_g	I was made from adipic acid and hexane-1,6-diol and neopentane-diol and diphenyl carbonate	333
Copolyether-amide	*N*-vinylpyrrolidone	Single T_g; transparency	Studied up to 30% II; I was made from isophthaloyl chloride and *cis*-4,4′-diamino-dibenzo-18-crown-6	720
Copolymer of bromohydroquinone and 4,4′-(decamethylenedioxy) bibenzoic acid	Copolymer of *p*-hydroxybenzoic acid and 3-methyl-1,6-hexanediol-terephthalic acid	Single T_g; microscopy	Transition temperature from a mesophase to an isotopic phase changed continously and smoothly depending on the composition	380
	Copolymer of 4,4′-(decamethylenedioxy)-bibenzoic acid and 4,4′-(ethylenedioxy)diphenol	Single T_g; microscopy	Transition temperature from a mesophase to an isotopic phase changed continously and smoothly depending on the composition	380
2-Cyanoethyl methacrylate	Alkyl methacrylate	Single T_g	II had an alkyl group of methyl, acetonyl or tetrahydrofurfuryl (3 miscible blend system)	1234
	2-Ethyl-2-oxazoline	Single T_g	Blend had ≥ 80 wt.% I	1236
	2-Hydroxyethyl methacrylate	Single T_g; FTIR	–	1235
	2-Hydroxypropyl methacrylate	Single T_g; FTIR	Blend had ≤ 50 wt.% I	1235
	Tertiary amide	Single T_g	II was *N*,*N*-dimethylacrylamide, *N*-methyl-*N*-vinylacetamide or *N*-vinyl-2-pyrrolidone (3 blend systems); formed complexes	1236
Cyanomethyl methacrylate	Alkyl methacrylate	Single T_g; FTIR	II had an alkyl group of acetonyl, 2-hydroxyethyl or 2-hydroxypropyl (3 miscible blend systems)	1233, 1235
	2-Ethyl-2-oxazoline	Single T_g	Blend had ≥ 60 wt.% I	1236
	Tertiary amide	Single T_g	Complexation formation; II was *N*-vinylpyrrolidone, *N*,*N*-dimethylacrylamide or *N*-methyl-*N*-vinylacetamide (3 miscible blend system)	1236
1,4-Cyclohexanedimethylene succinate	Epichlorohydrin	Single T_g	Semicrystalline	223
	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g; transparent melt	Semicrystalline ≥ 50 wt.% I	332
	Vinyl chloride-*co*-vinylidene chloride	Transparent melt; large T_m-depression	II was Saran (86.5% vinylidene chloride)	868
	Vinyl chloride	Single T_g	Semicrystalline ≤ 30 wt.% II	898
1,4-Cyclohexanedimethylene 1,4-Cyclohexanedicarboxylate	Arylate	Single T_g; PALS	II was Ardel D100	1251
	Carbonate of bisphenol-A	Single T_g' PALS	II was Ardel D100	1251
Cyclohexyl methacrylate	Vinyl chloride	Single T_g; NMR	–	641
	Vinyl chloride-*co*-vinylidene chloride	Single T_g	Semicrystalline ≥ 75% II; II had 86.5 wt.% vinylidene chloride	409,870
	2-Vinylnaphthalene	Single T_g; excimer fluorescence	I had low MW; II had MW = 70 kg/mol	231
Decamethylene decamethylene dicarboxylate	Vinyl chloride	Single T_g	Semicrystalline	872
Decamethylene sebacate	Epichlorohydrin	Single T_g	Semicrystalline	223
Dianhydrosorbityl dicarboxylate	Dianhydrosorbityl dicarboxylate	Single T_g; microscopy	6 Miscible blend systems based on different polyesters; the dicarboxylate was adipate, oxalate, sebacate, suberate or succinate	925
2,3-Dichloro-1-propyl acrylate	Ethyl acrylate-*co*-glycidyl methacrylate	Single T_g	II had 65% ethyl acrylate	181
2,2-Dichloroethyl methacrylate	Alkyl methacrylate	Single T_g; transparency	The alkyl group of II was methyl, ethyl, *n*-propyl, isopropyl, *n*-butyl, isoamyl, cyclohexyl or tetrahydrofurfuryl (8 miscible blend systems)	1110
2,6-Diethyl-1,4-phenylene ether	Styrene	Single T_g	–	134
N,*N*-dimethylacrylamide	Allyl alcohol-*co*-styrene	Single T_g; FTIR	Formed complexes; II had 4.5 or 6.5 wt.% OH groups	974
	Alkyl methacrylate	Single T_g; FTIR	II had an alkyl of chloromethyl, 2-chloroethyl, 3-chloropropyl,	1108

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
			2-bromoethyl or 2-iodoethyl (5 miscible blend systems)	
	Vinylidene fluoride	Single T_g	Semicrystalline ≥ 60 wt.% II	241
N,N-Dimethylacrylamide-*co*-ethylene	Vinyl chloride	Single T_g	I had 71.4 or 73.1 wt.% ethylene; two T_g's when I had ≥ 75.1 wt.% ethylene	563
2,6-Dimethyl-1,4-phenylene ether	Acrylic acid-*co*-styrene	Single T_g	II had 4.7–8.5 wt.% acrylic acid	921,1022
	p-Bromostyrene-*co*-fluorostyrene	Single T_g; FTIR	II had 11–73 mol% *p*-bromostyrene	547,834
	Cyclohexylmaleimide-*co*-styrene	Light scattering	I had 83 wt.% styrene	712
	o-Fluorostyrene-*co*-*p*-fluorostyrene	Single T_g; transparency	II had 10–38% *para* isomer; two T_g's when II had 5 or ≥ 43% *para* isomer	412,835
	o-Fluorostyrene-*co*-styrene	Single T_g; transparency	II had ≥ 9 mol% styrene	836,839,842
	p-Fluorostyrene-*co*-styrene	Single T_g; transparency	II had ≥ 44 mol% styrene	531,532,836,839,842
	p-(2-Hydroxy-hexafluoroisopropyl) styrene-*co*-styrene	Single T_g	II had 77.9–90.3 mol% styrene	571,572,646,807
	Luoloylmeleimide-*co*-styrene	Light scattering	II had 81 wt.% styrene	712
	Maleic anhydride-*co*-styrene	Single T_g; transparency; microscopy	II had 5.9–12.5 wt.% maleic anhydride	232,233, 1022,1026
	Methacrylic acid-*co*-styrene	Single T_g	II had 5.9 or 7.8 mol% methacrylic acid	917,921
	2-Methyl-6-benzyl-1,4-phenylene ether	Single T_g; transparency	–	751
	Methyl methacrylate-*co*-styrene	Single T_g	II had 2–20 wt.% methyl methacrylate	258,461,1022
	2-Methyl-6-phenyl-1,4-phenylene ether	Single T_g	–	752,753
	α-Methylstyrene	Single T_g	–	134,459,691
	α-Methylstyrene-*co*-styrene	Single T_g	All compositions of II	459,754
	o-Methylstyrene	NMR	NMR showed heterogeneity on a 3-nm scale	195
	p-Methylstyrene	Single T_g	–	195,235,788
	p-Nitrostyrene-*co*-styrene	Single T_g	II had <22 mol% nitrostyrene	155,158
	N-phenylmaleimide-*co*-styrene	Single T_g; light scattering	II had 84 or 90 wt.% styrene	712,1102
	Styrene	Single T_g; transparency; NMR	–	39,201,526,589,663, 750,753,784,1207
	Styrene, head-to-head	Single T_g	–	463
	Styrene, iodinated	Single T_g	II had ≤ 47% iodination	261
	Styrene, sulfonated and metal-neutralized	Single T_g	II had up to 7.8 mol% sulfonation (Ref. 677); II had 2.8 mol% sulfonation and was neutralized by Na, Zn, Ni or Mn, and miscible when blend had ≤ 40 wt.% I (Ref. 1200)	677, 1200
	Styrene, syndiotactic	Single T_g	–	954
	Styrene-*b*-isoprene-*b*-styrene	Single T_g for mixed with styrene blocks of II	II had block MW 15-51-17 and 78-52–80 kg/mol	568
	Styrene-*co*-2,2,6,6-tetramethyl-piperidinyl methacrylate	Single T_g	II had ≤ 24 wt.% 2,2,6,6-tetramethylpiperidinyl methacrylate	257
	Urethane acrylate	Single T_g; FTIR	I and II formed interpenetrating network	567
2,6-Dimethyl-1,4-phenylene ether, carboxylated	Styrene	Single T_g	11.8 mol% of repeat units in I carboxylated; two T_g's when 35.0 mol% repeat units in I carboxylated	874
2,6-Dimethyl-1,4-phenylene ether, brominated phosphonylated	Cellulose acetate	Single T_g; electron microscopy	II was Eastman 394–45	88
2,6-Dimethyl-1,4-phenylene ether, modified	Epichlorohydrin, modified	Single T_g	Both I and II had at least 25% carbazolyl or 3,5-dinitrobenzoyl groups	664
2,6-Dimethyl-1,4-phenylene ether, sulfonylated	Acrylonitrile-*co*-styrene	Single T_g	I had 20–26 wt.% sulfonylation	410
	p-Bromostyrene	Single T_g	I had >12 mol% sulfonylation	1216,1220
	o-Bromostyrene-*co*-*p*-bromostyrene	Single T_g	High *p*-bromostyrene content and high degree of sulfonylation favored miscibility	1216
	p-Chlorostyrene	Single T_g	II had >9 mol% sulfonylation	1217,1220
	o-Chlorostyrene-*co*-*p*-chlorostyrene	Single T_g	Miscibility depended on compositions of I and II	1217
	o-Chlorostyrene-*co*-*o*-fluorostyrene	Single T_g	Miscibility depended on compositions of I and II	1218

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	p-Chlorostyrene-*co*-*o*-fluorostyrene	Single T_g	Miscibility depended on compositions of I and II	1218
	o-Fluorostyrene	Single T_g	I had 4–17 mol% sulfonylation	1220
	p-Fluorostyrene	Single T_g	I had 17–66 mol% sulfonylation	1220
	o-Fluorostyrene-*co*-*p*-fluorostyrene	Single T_g	Miscibility depended on compositions of I and II	1215
	p-Fluorostyrene-*co*-styrene	Single T_g	I had 20–26 wt.% sulfonylation	410
	Phenylvinyl *sec*-butyl ether-*alt*-*N*-phenylmaleimide	Single T_g	I had 68–87 mol% sulfonylation	1219
	Phenylvinyl ethyl ether-*alt*-*N*-phenylmaleimide	Single T_g	I had 61–91 mol% sulfonylation	1219
	Styrene	Single T_g	I had <31 wt.% sulfonylation	410
α,α-Dimethyl-β-propiolactone (pivalolactone)	Vinyl chloride	T_m-depression	Semicrystalline	32,33
Dimethylsiloxane	Ethylene-*co*-methyl acrylate	Single T_g; FTIR	I had a small amount of vinyl groups; II had 21 wt.% methyl acrylate	1182
Dimethylsiloxane, modified	Butyl methacrylate	Single T_g; FTIR	One of the methyl groups in the repeating unit of I was changed to 4-hydroxy-4,4-bis (trifluoromethyl) butyl group	127
	Ethylene glycol	Single T_g; FTIR	One of the methyl groups in the repeating unit of I was changed to 4-hydroxy-4,4-bis(trifluoromethyl) butyl group	127
2,6-Dipropyl-1,4-phenylene ether	Styrene	Single T_g	–	134
Dodecamethylene adipamide, *N*,*N*-dimethyl substituted	*p*-(2-Hydroxy-hexafluoroisopropyl) styrene-*co*-styrene	Single T_g	Semicrystalline when $\leq$ 50% substitution on I	454
Dodecamethylene adipate	Epichlorohydrin	Single T_g	Semicrystalline	223
	Vinyl chloride	Single T_g	Semicrystalline	872
Dodecamethylene decamethylene dicarboxylate	Vinyl chloride	Single T_g	Semicrystalline	872
Dodecamethylene dodeca-methylene dicarboxylate	Vinyl chloride	Single T_g	Semicrystalline	872
Enaminonitrile	Ethylene glycol	Single T_g	–	587
	Tertiary amide	Single T_g	II was *N*,*N*-dimethylacrylamide, 2-ethyl-2-oxazoline or *N*-vinyl-2-pyrrolidone (3 miscible blend systems)	587
	4-Vinylpyridine	Single T_g	–	587
Epichlorohydrin	Acrylonitrile-*co*-styrene	Single T_g	II had 18 or 25 wt.% acrylonitrile	305
	Alkyl acrylate	Single T_g	II had an alkyl group of ethyl or *n*-propyl (2 miscible blend systems)	224,226
	Alkyl methacrylate	Single T_g	II had an alkyl group of ethyl, *n*-propyl, *n*-butyl or cyclohexyl (4 miscible blend systems)	224
	Butyl acrylate rubber	Single T_g	II was Hycar 4043	226
	Ethyl acrylate rubber	Single T_g	II was Hycar 4051EP	226
	Ester	Single T_g	II was ethylene adipate, ethylene sebacate or 2,2-dimethyl-1,3-propylene adipate (3 miscible blend systems)	223,278
	Glycidyl methacrylate	Single T_g	–	864
	Glycidyl methacrylate-*co*-methyl methacrylate	Single T_g	II had 76% glycidyl methacrylate	864
	Hexamethylene sebacate	Single T_g	–	864
	Methacrylonitrile-*co*-methyl methacrylate	Single T_g	II had 30–40 mol% methacrylonitrile	864
	Methyl methacrylate	Single T_g	Broad T_g might due to separate phases with differing T_g's within the mixture	135,224
	Vinyl acetate	Single T_g	–	303
Epichlorohydrin-*co*-ethylene glycol	Neopentyl glycol adipate	Transparency	I was Hydrin 200; refractive indices not equal in the temperature range investigated	288
Ester	Ester	Single T_g; microscopy	13 Miscible blend systems based on different linear aliphatic polyesters	925
	Ester	Melting behavior	Both polymers I and II were liquid crystalline; II had one more ethylene unit in the main chain	979

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
Ester, hyperbranched and 3,5-dihydroxyphenyl-terminated	Aramide	Single T_g	II was Trogamid T or MXD-6	1125
	Amide	Single T_g	II was nylon-6	1125
	Butylene adipate	Single T_g	–	1125
	Copolyester	Single T_g	II was made from terephthalic acid and a mixture of ethylene glycol and 1,4-cyclohexane dimethanol	965
Ester, hyperbranched and 3,5-diacetoxyphenyl-terminated	Butylene terephthalate	Single T_g	–	1125
	Copolyester	Single T_g	II was made from terephthalic acid and a mixture of ethylene glycol and 1,4-cyclohexane dimethanol	1125
Ether ether ketone	Arylketone	Single T_g	II had 25, 50 or 57% ketone content	333
	Etherketone	Single T_g	I was oxy-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene; II was oxy-1,4-phenylenecarbonyl-1,4-phenylene; semicrystalline	739
	Etherimide	Single T_g; microscopy; WAXD; T_m-depression	I was Victrex 450G; II was Ultem 1000; semicrystalline	172,946,1037 1178
	Ether sulfone	Single T_g	–	883
Ether ether ketone, sulfamidated	Amideimide	Single T_g; FTIR	I had degree of sulfamidation of 1.33; II was Torlon 4000T	1061
Ether ether ketone, sulfonated	Amideimide	Single T_g; FTIR	I had degree of sulfonation of 0.42, 0.53 or 1.00; II was Torlon 4000T	414,1061
	Etherimide	Single T_g; FTIR	II was Ultem 1000	414,1061
Ether ether sulfone, phenolphthalein	Epoxy	Single T_g; microscopy	Miscible when II was cured by amine; immiscible when II was cured by anhydride	1023
Etherimide	Amideimide	Single T_g	I was Ultem 10000; II was Torlon 4203L	1147
	Ethylene terephthalate	Single T_g; microscopy; T_m-depression	I was Ultem 1000	942,1054,1127
	Imide	Single T_g	I was Ultem 1000; II was based on diphenylpyromellitic dianhydride (DPPMDA) (3 different imides)	240
Etherimide ether	Thioetherimide ether	NMR	I was made from 4,4′-oxydianiline (ODA) and 4,4′-(1,4-phenylene-dioxy)diphthalic acid; II was made from ODA and 4,4′-thio-diphthalic anhydride	1000
Ether ketone ketone	Etherimide	T_m-depression	II was Ultem 1000	946
Ether sulfone	Aramid	Single T_g	II was made from isophthaloyl chloride and bis(4-aminophenyl) ether	562,600
	Imide	Single T_g; NMR	II was Polyimide XU218	372
	Imide	Single T_g	II was Matrimide 5218	510,940
	N,N′-(oxydi-*p*-phenylene) isophthalamide	Single T_g	–	1031
	Phenylene ether ether ketone	Single T_g	–	1184
	Phenylene ether sulfone ketone	Single T_g	–	1184
	Phenylene sulfone ether sulfone ketone	Single T_g	–	1184
	Phenylene thioether ether ether ketone	Single T_g	–	1184
	m-Phenylene isophthalamide	Single T_g	Prepared by *in situ* blending	1030
Ether urethane	Vinyl chloride	Single T_g	20 or 40 wt.% II; I was Adiprene L-100 and 4,4′-methylene-bis-2-chloroaniline	344
3-Ethenoxypropyl cinnamate	*p*-Methyl-(3-ethenoxypropyl) cinnamate	Single T_g	–	515
Ethyl acrylate	Novolac	Single T_g	II had MW about 1 kg/mol; II was formaldehyde and 13/17/70 mol% = *p-tert*-butylphenol/*m*-cresol/*o*-cresol or formaldehyde and 15/17/68 mol% = 2-*tert*-butylphenol/*m*-cresol/*o*-cresol	221

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Hexafluoroacetone-*co*-vinylidene fluoride	Single T_g	II had 8 mol% vinylidene fluoride	411,1059
	Styrene, sulfonated	Single T_g; FTIR	II had 7.2 mol% sulfonated units	1198
	Styrene-*co*-*p*-vinylphenol	Single T_g; FTIR	II had >4.46 mol% vinylphenol	1198
	Vinyl chloride	Single T_g	*In situ* polymerized II; other mixtures formed two phases (Refs. 296, 426 for example)	846
	Vinyl chloride-*co*-vinylidene chloride	Single T_g; transparency	II was Saran ($\geq$80% vinylidene chloride); I had MW = 50 kg/mol; two T_g's when I had MW = 140 kg/mol and II was Saran (86.5 wt.% vinylidene chloride)	812,870
	Vinylidene fluoride	Single T_g; transparency	Semicrystalline $\geq$25 wt.% II	56,80,844
Ethyl acrylate-*co*-methacrylonitrile-*co*-α-methylstyrene	Vinyl chloride	Single T_g; transparency	I had 2/40/58 = ethyl acrylate/methacrylonitrile/α-methylstyrene or unknown composition	422,424
Ethyl acrylate-*co*-methyl methacrylate	Vinyl chloride	Single T_g; electron microscopy	I was Polaroid K120N (95% methyl methacrylate)	886
Ethyl acrylate-*co*-4-vinylpyridine	Styrene-*co*-styrene sulfonic acid	Single T_g; transparency	I and II both $\geq$5% of second monomer (data obtained $\leq$10% substitution)	210,773
	Vinyl chloride	Single T_g	I had 2–14 mol% 4-vinylpyridine	137
	Vinyl chloride, sulfonated	Single T_g	I had 6.2 mol% 4-vinylpyridine; II had 3.7 mol% sulfonic acid	137
(*N*-ethylcarbozol-3-yl)methyl acrylate	2-(3,5-Dinitrobenzoyl) oxyethyl methacrylate	Single T_g; NMR	–	760
(*N*-ethylcarbozol-3-yl)methyl methacrylate	2-(3,5-Dinitrobenzoyl) oxyethyl methacrylate	Single T_g; NMR	–	761
Ethyl methacrylate	Novolac	Single T_g; FTIR	II had MW about 1 kg/mol and was formaldehyde and 13/17/70 mol% = *p*-*tert*-butylphenol/*m*-cresol/*o*-cresol	221,434
	Ethyl acrylate	Single T_g	–	156
	p-(2-Hydroxy-hexafluoro-isopropyl)styrene-*co*-styrene	Single T_g; transparency	II had 90.3–98.9 mol% styrene	103,571,646
	Vinyl acetate	Transparency	Transparent film when I <25 wt.% or when content of I toward 100 wt.%	634
	Vinyl chloride	Single T_g; transparency	–	234,319,649, 812,846
	Vinyl chloride-*co*-vinylidene chloride	Single T_g; transparency	II was Saran ($\geq$80% vinylidene chloride) or 86.5 wt.% vinylidene chloride; semicrystalline $\leq$30% I	409,812, 870
	Vinylidene fluoride	Single T_g	Semicrystalline when $\geq$40% II; two T_g's when >60% II	244,361,476 630,695
Ethylene, chlorinated	Acrylonitrile-*co*-butadiene	Single T_g	Depended on Cl content and acrylonitrile content	96,161
	Hexamethylene terephthalate	Single T_g	I had 48% Cl; immiscible with I had 36% Cl	36
	Methyl methacrylate	Single T_g; transparency; electron microscopy	I had $\geq$50 wt.% Cl; II was atactic, isotactic or syndiotactic (Ref. 1204)	99,296,812 848,1204
	Neopentyl glycol adipate	Single T_g	I had 42 or 48 wt.% Cl; immiscible when I had $\leq$36 wt.% Cl	288
	Vinyl chloride, chlorinated	Single T_g	I and II with identical chlorine content were immiscible	818
Ethylene, chlorosulfonated	Vinyl chloride	Single T_g	I had 1% S as SO_2Cl and 42 wt.% Cl; stated immiscible when I had 30 wt.% Cl in Ref. 296	200
Ethylene adipate	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g; transparent melt	Semicrystalline when $\geq$50 wt.% I	332
Ethylene-*co*-methacrylic acid	Ethylene glycol	FTIR	I had 55 wt.% methacrylic acid	287
	Ethylene glycol-*co*-propylene glycol	Single T_g; FTIR	I had 18, 32, 44 or 55 wt.% methacrylic acid; II had 70 wt.% ethylene oxide	494
	2-Ethyl-2-oxazoline	FTIR	I had 44 wt.% methacrylic acid	512
	Styrene-*co*-2-vinylpyridine	FTIR	I had 32 wt.% methacrylic acid; II had 70 wt.% 2-vinylpyridine	495

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Tetramethylene glycol	FTIR	I had 55 wt.% methacrylic acid	496
	Vinyl ethyl ether	FTIR	I had 55 wt.% methacrylic acid	496
	Vinyl methyl ether	Single T_g; FTIR	I had 18, 32, 44 or 55 wt.% methacrylic acid	494
	2-Vinylpyridine	FTIR	Miscible when I had 18 wt.% methacrylic acid and formed complexes when I had 44 or 55 wt.% methacrylic acid; formed complexes when I had 32 wt.% methacrylic acid and was $\leq 35\%$ I in blend	494
Ethylene-*co*-methacrylic acid, zinc neutralized	Vinyl methyl ether	FTIR	I had 44 wt.% methacrylic acid	732
	4-Vinylpyridine	NMR	I had 15 wt.% methacrylic acid and 60% zinc-neutralized	52
Ethylene-*co*-methyl acrylate *co*-unspecified acid	Vinyl chloride	Single T_g	I was Vamac N-123	395
Ethylene orthophthalate	Vinyl acetate	Single T_g	–	829
Ethylene glycol	Copolyester-polyurethane	Single T_g	–	363
	Epoxy resin	Single T_g	Miscible when II was uncured; immiscible when II was cured (Ref. 313)	313, 1118,1119
	Ethersulfone	Single T_g; light scattering	–	317,534,743
	Ethyl methacrylate	Single T_g	–	130
	Ethylene-*co*-vinyl acetate	Single T_g	II had 87.1 wt.% vinyl acetate (Ref. 132); II had 73.4 wt.% vinyl acetate, miscible with I of $M_w = 20$ and 100 kg/mol (Ref. 328)	132, 328
	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g; NMR	Semicrystalline when >40% I	363,364,690
	p-(2-Hydroxy-hexafluoroisopropyl) styrene-*co*-styrene	Single T_g	II had 77.9 mol% styrene and I/II was 1/8-1/2; semicrystalline and no amorphous phase	806
	Methacrylic acid	Single T_g; NMR	Formed complexes in water and dimethylsulfoxide	374,1128
	Methacrylic acid-*co*-styrene, Na-neutralized	FTIR	I had 5.6–17.9 mol% acid	1098
	Methyl methacrylate	Single T_g; FTIR; NMR	I was miscible with atactic or syndiotactic II and immiscible with isotatic II; semicrystalline when >20 wt.% I	124,389,511 552,759, 1192
	Resorcinol	NMR	–	51
	Styrene, sulfonated and Na-neutralized	Single T_g	II had 2.9–8.8 mol% sulfonation; miscible when blend had low I content	498
	Styrene-*co*-*p*-vinylphenol	Single T_g	II had 13.1 mol% styrene and I/II was 1/8–1/1; semicrystalline at higher I content and no amorphous phase miscibility	806
	Styrene-*co*-vinylphenol trifluoromethyl carbinol	Single T_g	II had 84.4 mol% styrene and I/II was 1/8–1/4 or II had 54.5 mol% styrene and I/II was 1/8–1/2; semicrystalline at higher I content and no amorphous phase miscibility	806
	Sulfone	Transparency; light scattering	I had $M_n = 3.5$–4.0 kg/mol; immiscible when I had $M_n = 200$ kg/mol	796
	Vinyl acetate	Single T_g	Ref. 397 stated miscible when ≥ 40 wt.% II in blend; Ref. 311 stated completely miscible if I had $M_w = 20$ kg/mol, and miscible up to 50 wt.% II if I had $M_w = 100$ kg/mol	311,397, 553
	Vinyl chloride	Single T_g; T_m-depression; NMR; IGC	Semicrystalline	418,1126,994
	Vinylnapthalene	Single T_g	Semicrystalline when >50% I; two amorphous phases when 25–50% I	178,580,679
	p-Vinylphenol	FTIR; NMR; X-ray diffraction	Semicrystalline	668,888,1150, 1149
Ethylene-*co*-sulfur dioxide-*co*-vinyl acetate	Vinyl chloride	Single T_g; transparency	I had 72.7/8.8/18.5 = ethylene/sulfur dioxide/vinyl acetate; immiscible when	339

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
			I had 89.0/3.2/7.8 = ethylene/sulfur dioxide/vinyl acetate	
Ethylene terephthalate	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g	T_g's of I and II were close; semi-crystalline	689
	Liquid crystalline copolyester	Single T_g	II was copolymer of ethylene terephthalate (40%) and hydroxybenzoic acid (60%) (Ref. 898) or a segmental block copolyester (Ref. 1099)	898, 1099
	Ethylene-2,6-naphthalene dicarboxylate	Single T_g	Miscibility arised from *trans*-esterification	912
Ethylene-*co*-vinyl acetate	Acrylic acid-*co*-alkyl acrylate	Single T_g	II had an alkyl group of *n*-butyl, ethyl, ethylhexyl or methyl (4 blend systems); miscibility depended on compositions of I and II	924
	3-Alkyl thiophene	Microscopy	I had 20 wt.% of vinyl acetate; II had an alkyl group of dodecyl or octyl; blend had < 30 wt.% II	1033,1095
	Aniline, *N*-octadecylated	Microscopy	I had 20 wt.% vinyl acetate; II was 60.2 mol% alkylated	1246
	Chloroprene	Single T_g; microscopy	I had 28 wt.% vinyl acetate	1081
	Hexafluoroacetone-*co*-vinylidene fluoride	Single T_g	I had $\geq$ 70% vinyl acetate; II had 8.9 mol% hexafluoroacetone	336
	Novolac	Single T_g; FTIR	I had 14–50 wt.% vinyl acetate	1130
	Propylene, chlorinated	Single T_g	I had 40, 44.3 or 45 wt.% vinyl acetate; II had 40 or 50 wt.% Cl	499,500
	Vinyl chloride	Single T_g	I had 65–75 wt.% vinyl acetate; two T_g's when I had 40–45 wt.% vinyl acetate; ambiguous when I had 45–65 wt.% vinyl acetate	216,222, 325,330, 452,541, 672,756
	Vinyl chloride, chlorinated	Single T_g	Stated that there was a miscible region when I had 0.4–0.75 mol% ethylene and II had 0.4–0.8 mol% vinyl chloride	747
Ethylene-*co*-vinyl acetate-*co*-unnamed monomer	Vinyl chloride	Single T_g	Semicrystalline; I was du Pont de Nemours PB3041 (>60% ethylene)	41
Ethylene-*co*-vinyl alcohol	Alkyl methacrylate	Single T_g; FTIR	II had an alkyl group of methyl, *n*-octyl or *n*-decyl (3 miscible blend systems); I had 21 mol% vinyl alcohol	1054
	Amide	Single T_g	I had 33 mol% vinyl alcohol; II was nylon 12 or nylon 6,12 (2 miscible blend systems)	4
	Copolyamide	Single T_g	II had 60 mol% nylon 6 units and 40 mol% nylon 12 units; I had 59 mol% vinyl alcohol	909
	4-Vinylpyridine	Single T_g	I had 68 mol% vinyl alcohol	1045
2-Ethylhexyl methacrylate	Styrene-*co*-hexyl methacrylate	Single T_g	I had 3–5% acrylic acid; II had 30% styrene and 5–15% 3-(dimethyl-amino)-2,2-dimethylpropyl methacrylate	766
N-ethyl-3-hydroxymethyl-carbazoyl methacrylate	ω-Hydroxyalkyl-3,5-dinitrobenzoyl methacrylate	Single T_g	II had an alkyl group of butyl, hexyl, pentyl or propyl (4 blend systems)	694
2-Ethyl-2-oxazoline	Acrylic acid	Single T_g	Formed complexes in water, methanol or dioxane	517
	Acrylic acid-*co*-styrene	Single T_g	II had 74 mol% acrylic acid	517
	Allyl alcohol-*co*-styrene	Single T_g; FTIR	II had 4.5 or 6.5 wt.% of hydroxyl group	974
	Aramid	Optical microscopy; light scattering	–	533
	Ether sulfone	Light scattering, transparency	–	599
	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g; transparency	–	426
	Maleimide-*co*-styrene	Single T_g	II had 9.5–48.2 mol% maleimide	188
	Novolac	Single T_g	Formed complexes in methanol or acetone	517
	Sulfone, carboxylated	Single T_g	II had $\geq$ 0.93 degree of carboxylation	254

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	p-Vinylphenol	Single T_g	Formed complexes in dioxane	517,857
	Vinylidene fluoride	Single T_g	>50 wt.% II	243
2-Ethyl-6-propyl-1,4-phenylene ether	Styrene	Single T_g	–	134
2-Fluoroethyl methacrylate	Alkyl methacrylate	Single T_g	II had an alkyl group of methyl or ethyl (2 miscible blend systems)	1153
3-Fluoropropyl methacrylate	Alkyl methacrylate	Single T_g	II had an alkyl group of methyl or tetrahydrofurfuryl (2 miscible blend systems)	1155
	Ester	Single T_g	II was butylene adipate or 2,2-dimethyl-1,3-propylene adipate (2 miscible blend systems)	1155
Glycidyl methacrylate	Acrylonitrile-*co*-styrene	Single T_g	II had 32.3–58.8 wt.% AN	1007
	Methyl methacrylate	Single T_g	II had $M_w \leq 4.25$ kg/mol	1007
	α-Methylstyrene	Single T_g	II had $M_w = 3.5$ kg/mol	1009
Glycidyl methacrylate-*co*-methyl methacrylate	Acrylonitrile-*co*-styrene	Single T_g	Miscibility depended on compositions of I and II	1007
Glycidyl methacrylate-*co*-styrene	Carbonate of tetramethylbisphenol-A	Single T_g	I had 1.6–37.1 wt.% GMA	1007
	2,6-Dimethyl-1,4-phenylene ether	Single T_g	I had 1.6–18.6 wt.% GMA	1007
	Vinylidene fluoride	Single T_g	I had ≤35.7 wt.% GMA	1009
Hexadecamethylene dodecamethylene dicarboxylate	Vinyl chloride	Single T_g	Semicrystalline	872
Hexafluoroisopropyl methacrylate	Alkyl methacrylate	Single T_g	II had an alkyl group of methyl or ethyl (2 miscible blend systems)	1153
Hexamethylene adipamide (nylon 6,6)	Ethylene terephthalate	FTIR; NMR	Blends prepared by coprecipitation; blends prepared by melt blending showed no sign of hydrogen-bonding	656
Hexamethylene adipate	Ester	Single T_g; microscopy	II was butylene sebacate or butylene succinate (2 miscible blend systems)	925
Hexamethylene isophthalate	Amide	Single T_g	Polymer II was nylon 6,5, nylon 6,7, nylon 6,8 or nylon 6,9 (4 miscible blend systems)	962
Hexamethylene sebacate	Vinyl acetate-*co*-vinylidene chloride	Single T_g	Semicrystalline; II was Saran C (≥80% vinylidene chloride)	31
	Vinyl bromide	Single T_g	–	150
	Vinyl chloride	Single T_g	Semicrystalline when ≥50% I	872,900
Hexamethylene terephthalate	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g	Semicrystalline when ≥75% I	735
	Vinyl chloride	Single T_g	Semicrystalline; deposited from some solvents as two phases	36
1-Hexene	1-Pentene	Single T_g	–	190
Hexyl methacrylate	Vinyl chloride	Single T_g; transparency	Especially when II was polymerized *in situ* (Ref. 846); stated as immiscible in Ref. 812	812,846, 850
	Vinyl chloride-*co*-vinylidene chloride	Single T_g; transparency	II was Saran (≥80% vinylidene chloride)	409,812
3-Hydroxybutyrate	Cellulose acetate-butyrate	Single T_g	Semicrystalline	719,933,1183
	Cellulose acetate-proprionate	Single T_g; microscopy	Semicrystalline	719,1183
	Cellulose tributyrate	Single T_g; microscopy	Semicrystalline	1183
	Epichlorohydrin	Single T_g; microscopy; T_m depression	Semicrystalline	987,1183,1201
	Ethylene glycol	Single T_g	Semicrystalline	37,464,1237
	Ethylene glycol-*co*-lactide	Single T_g	II had 15 wt.% ethylene glycol	1241
	Lactide	Single T_g; microscopy	–	923,1079
	Methyl methacrylate	Single T_g	Blend had ≤20 wt.% I	1103,1183,1189
	Vinyl acetate	Single T_g	Semicrystalline	292,465
	Vinyl alcohol	NMR	Blend had a large amount of I	1239
	p-Vinylphenol	Single T_g; FTIR	–	1043
	Vinylidene fluoride	Single T_g; T_m-depression	Semicrystalline	407,414
3-Hydroxybutyrate-*co*-3-hydroxyvalerate	Acrylonitrile-*co*-butadiene-*co*-styrene	Single T_g	I had 16% hydroxyvalerate	180
	Vinyl chloride	Single T_g	I had 16 or 18% hydroxyvalerate	180,952
Hydroxyether of bisphenol-A (Phenoxy)	2,2-Dichloroethyl methacrylate	Single T_g; FTIR	–	1017
	2,2-Dichloroethyl methacrylate-*co*-methyl methacrylate	Single T_g; FTIR	II had ≤62 wt.% methyl methacrylate	1017

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Ether sulfone	Light scattering	–	714
	Ethylene oxide-*co*-propylene oxide	Single T_g	II had $\geq$22 mol% ethylene oxide	1025
	Methyl methacrylate	Single T_g; NMR	II was atactic, isotactic or syndiotactic	112,218, 775,910
	Vinyl methyl ether	Single T_g; IGC	–	690,819,820,996
	Vinylpyridine	Single T_g; FTIR	II was 2- or 4-vinylpyridine (2 miscible blend systems)	981
	N-vinylpyrrolidone	Single T_g; transparency: FTIR	–	1253
	Urethane	Single T_g	II was made from 1/1/2 = polycaprolactone diol (M_n = 2.1 kg/mol)/ 1,4-butanediol/4,4′-diphenylmethane diisocyanate	730
Hydroxyether of bisphenol-A, benzoylated	*N*-vinylpyrrolidone	Single T_g; FTIR	–	1002
Hydroxyether of bisphenol-A, modified	2-Vinylpyridine	Single T_g; FTIR	OH groups in I was converted to benzoate, acetate or methoxy group	989
Hydroxyether of phenolphthalein	Butylene terephthalate	Single T_g	–	310
	Carbonate of bisphenol-A	Single T_g	–	310
	Ether sulfone	Single T_g; FTIR	–	306
	Ethylene glycol	Single T_g	–	309
β-Hydroxyethyl-3,5-dinitrobenzoyl methacrylate	*N*-alkyl-3-hydroxymethylcarbazoyl methacrylate	Single T_g	II had an alkyl group of butyl, decyl, dodecyl, ethyl, hexadecyl, hexyl, methyl, octyl or tetradecyl (9 miscible blend systems)	693
	N-(2-hydroxyethyl)carbazoyl methacrylate	Single T_g	–	693
2-Hydroxyethyl methacrylate	Acrylamide-*co*-styrene	Single T_g; NRET	II had 40, 47 or 52 mol% acrylamide	955,1248
p-(2-Hydroxy-hexafluoroisopropyl) -α-methylstyrene-*co*-styrene	Butyl acrylate	Single T_g	I had $\geq$1.2 mol% OH-containing monomer	90
	Butyl methacrylate	Single T_g	I had >2.8 mol% OH-containing monomer	1166
	Methyl methacrylate	Single T_g	I had $\geq$1.9 mol% OH-containing monomer	90
	Methyl methacrylate-*co*-2-vinylpyridine	Single T_g	I had $\geq$6 mol% OH-containing monomer	377
p-(2-Hydroxy-hexafluoroisopropyl)styrene-*co*-styrene	Aramide	Single T_g; transparency	I had 60 mol% styrene; two T_g's when I had 83 mol% styrene; II was iminotrimethylhexamethylene-iminoterephthaloyl	571,646
	Butyl methacrylate	Single T_g; NRET	I had >1 mol% OH-containing monomer; formed complexes when I had 18 mol% OH	1166
	α,ω-Decanedicarboxylic-*co*-hexanediamine acid (nylon 6,12)	Single T_g	Semicrystalline	454
	Hydroxypropyl L-glutamate	Single T_g; light scattering; microscopy	Blend had <50 vol% II	971
	Methyl methacrylate	Single T_g; transparency; NMR; FTIR; WAXD; NRET	I had >8 mol% OH; formed complexes when I had 24 mol% OH; II was isotactic or syndiotactic (Ref. 1116,1117)	571,646, 1021,1116, 1117,1241
	Vinyl acetate	Single T_g; transparency	I had 90.3 mol% styrene	571,646
	Vinyl methyl ether	Single T_g; transparency	I had 83.3–99.9 mol% styrene	571,572,646
	Vinyl methyl ketone	Transparency	I had 90.3 mol% styrene	567
2-Hydroxypropyl methacrylate	Tertiary amide	Single T_g; FTIR	Formed complexes; II was dimethylacrylamide, *N*-methyl-*N*-vinylacetamide or ethyloxazoline (3 blend systems)	975
Imide	Ether ketone ketone	Single T_g	I was made from pyromellitic dianhydride (PMDA) and 4,4′-bis(3-aminophenoxy) biphenyl	1179
	Ether ketone	Single T_g	I was the same as the previous system; II was Victrex 220P; blend had $\leq$50 wt.% I	1179
	Imide	X-ray diffraction	I was made from PMDA and oxydianiline; II was made from	391

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
			PMDA and *p*-phenylenediamine; <90% II in blend	
	Imide	Single T_g	I was made from 2,2′-bis(3,4′-dicarboxyphenyl)hexafluoro-propane dianhydride (6FDA) and 2,2′-bis(3-aminophenyl)-hexafluoropropane (3,3′-6F-diamine); II was made from 6FDA and 2,2′-bis (4-aminophenyl)hexafluoropropane (4,4′-6F-diamine)	130
	Imide	Single T_g	I was made from 4,4′-biphthalic anhydride (BPDA) and 3,3′-6F-diamine; II was made from BPDA and 4,4′-6F-diamine	130
	Imide	Single T_g	I was made from 3,3′4,4′-benzophenone-tetracarboxylic dianhydride (BTDA) and 3,3′-6F-diamine; II was made from BTDA and 4,4′-6F-diamine	130
	Imide	Single T_g	I was made from 3,3′,4,4′-diphenyl-ethercarboxylic dianhydride (OPDA) and 3,3′-6F-diamine; II was made from OPDA and 4,4′-6F-diamine	130
	Imide	Single T_g	I was made from PMDA/BPDA/3,3′-6F-diamine = 25/25/50; II was made from PMDA/BPDA/4,4′-6F-diamine = 25/25/50	130
	Imide	Single T_g	I was made from PMDA/BTDA/3,3′-6F-diamine = 25/25/50; II was made from PMDA/BTDA/4,4′-6F-diamine = 25/25/50	130
	Imide	Single T_g	I was made from PMDA/OPDA/3,3′-6F-diamine = 25/25/50; II was made from PMDA/BTDA/4,4′-6F-diamine = 25/25/50	130
	Imide	Single T_g; WADX; SAXS	I was made from 3,3′,4,4′-diphenyltetra-carboxylic dianhydride (DBTA) and oxydianiline (DOA); II was made from 4,4′-thiodiphthalic anhydride and ODA	1244
	Imide	Single T_g	I was made from DBTA and 3,3′-hexa-fluoroisopropylidene dianiline (3,3′-HIPD); II was made from DPTA and 4,4′-hexafluoroisopropylidene dianiline (4,4′-HIPD)	957
	Imide	Single T_g	I was made from 5,5′-[2,2,2-trifluoro-1-(trifluoromethyl)ethylidene] bis-1,3-isobenzofuranedione (A) and HIPD; II was made from A and 4,4-HIPD	957
	Imide	Single T_g	I was made from BTDA and 3,3′-HIPD; II was made from BTDA and 4,4′-HIPD	957
2-Iodoethyl methacrylate	Acrylonitrile-*co*-*p*-methylstyrene	Single T_g	II had 21–42 wt.% AN	1106
	Acrylonitrile-*co*-styrene	Single T_g	II had 13–51 wt.% AN	1106
	Ester	Single T_g	II was butylene adipate, caprolactone, 2,2-dimethyl-1,3-propylene adipate or hexamethylene sebacate (4 miscible blend systems)	1111
	Tetrahydrofurfuryl methacrylate	Single T_g	–	1107
cis-Isoprene	Butadiene-*co*-1,2-vinylbutadiene	Single T_g	II had ≥ 32.3% vinylbutadiene	420
	cis-Pentenamer	Single T_g	–	77
	Vinylethylene	Single T_g; NMR	II had 86% 1,2-butadiene	570,697,700,810,811
Isoprene, chlorinated	Ethylene-*co*-vinyl acetate	Single T_g; transparency	Semicrystalline; commercial samples	503,665
Isoprene, modified by hexafluoroisopropanol group	Alkyl methacrylate	Single T_g; FTIR	II had an alkyl group of methyl or *n*-butyl (2 miscible blend systems)	959
	Ethylene-*co*-methyl acrylate	Single T_g; FTIR	II had 40 or 75 wt.% methyl acrylate	959

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Ethylene-*co*-methyl methacrylate	Single T_g; FTIR	II had 50 wt.% methyl methacrylate	959
	Ethylene-*co*-vinyl acetate	Single T_g; FTIR	II had 35 or 70 wt.% vinyl acetate	959
	Methyl acrylate	Single T_g; FTIR	–	959
	Methyl acrylate-*co*-styrene	Single T_g; FTIR	II had 28 wt.% methylacrylate	959
Isoprene-*stat*-*N*-(2-hydroxyethyl) carbazole methacrylate	Styrene-*stat*-*N*-itaconimidyl-3,5-dinitrobenzoate	Single T_g	Polymer had at least 20 mol% of electron-donor or electron-acceptor group	963
Isoprene-*b*-methyl methacrylate	Acrylonitrile-*co*-styrene	Single T_g	II had 22 wt.% acrylonitrile	1049
	p-(2-Hydroxy-hexafluoroisopropyl)-α-methylstyrene-*co*-styrene	microscopy	II had $\geq$ 1.6 mol% OH groups	1254
Isopropenyl methyl ketone	*p*-Vinylphenol	Single T_g; NMR; FTIR	–	1035
Isopropyl acrylate	Isopropyl methacrylate	Single T_g; transparency	–	453
Maleic acid-*co*-styrene	Vinyl butyral	Transparency	I was Daidou Kogyo Styrite HS-2	452
Maleic anhydride-*alt*-α-methylstyrene	2,6-Dimethyl-1,4-phenylene ether, sulfonylated	Single T_g	II had < 55 mol% sulfonylation	838
Maleic anhydride-*co*-styrene	Caprolactone	Single T_g	I had 14 or 25 wt.% maleic anhydride; semicrystalline	192
	Carbonate of tetramethylbisphenol-A	Single T_g	I had < 12% maleic anhydride	1008
	2,6-Dimethyl-1,4-phenylene ether	Single T_g	I had < 10% maleic anhydride	1008
	2,6-Dimethyl-1,4-phenylene ether, sulfonylated	Single T_g	I was alternating copolymer and had < 55 mol% sulfonylation	838
	Ethyl methacrylate	Single T_g	I had 0–33 wt.% maleic anhydride	73
	Methyl methacrylate	Single T_g; NMR; FTIR	I had 8–33 wt.% maleic anhydride	73,999
	n-Propyl methacrylate	Single T_g	I had 0–18.1 wt.% maleic anhydride	73
	Vinyl methyl ether	Single T_g	I had 4.7 wt.% maleic anhydride	862
Maleimide-*alt*-α-methylstyrene	2,6-Dimethyl-1,4-phenylene ether, sulfonylated	Single T_g	Depended on degree of sulfonylation of II	837
Methacrylate, liquid crystalline	Methacrylate, liquid crystalline	Single T_g; microscopy	I had electron-donating (carbazolyl methylene)aniline groups; II had electron-accepting groups	1077
Methacrylic acid-*co*-α-methylstyrene	Butyl methacrylate-*co*-4-vinylpyridine	Single T_g; FTIR	I had 50.6 mol% methacrylic acid (MA); II had 18 or 26 mol% vinylpyridine (VP)	1131
	Ethyl methacrylate-*co*-4-vinylpyridine	Single T_g; FTIR	I had 50.6 mol% MA; II had 23 mol% VP	1131
	Methyl methacrylate-*co*-4-vinylpyridine	Single T_g; FTIR	I had 50.6 mol% MA; II had 8 mol% VP	1131
Methacrylic acid-*co*-styrene	Ethyl methacrylate	Single T_g; fluorescence study	Miscible when cast from toluene or dioxane; immiscible when cast from chloroform; I had 93.4–96.7 mol% styrene	103
Methacryloyl glycine-*co*-styrene	Ethyl methacrylate	Fluorescence study	Miscible when cast from toluene; immiscible when cast from chloroform; I had 93.8–97.2 mol% styrene	103
Methacrylonitrile-*co*-α-methylstyrene	Vinyl chloride	Single T_g	I was azeotropic	423
Methacrylonitrile-*co*-*p*-methylstyrene	*n*-Butyl methacrylate	Single T_g	II had 12–28 wt.% of methacrylonitrile (MAN)	944
	Ethyl methacrylate	Single T_g	II had 12–55 wt.% MAN	944
	Methyl methacrylate	Single T_g	II had 19–70 wt.% MAN	944
	Isopropyl methacrylate	Single T_g	I had 19–34 wt.% MAN when II had $M_n = 161$ kg/mol; I had 16–34 wt.% MAN when II had $M_n = 41.6$ kg/mol	944
	n-Propyl methacrylate	Single T_g	I had 12–43 wt.% MAN	944
Methacrylonitrile-*co*-styrene	*t*-Butyl methacrylate-*co*-methyl methacrylate	Single T_g	Miscibility depended on composition of I and II	1139
	Cyclohexyl methacrylate-*co*-methyl methacrylate	Single T_g	Miscibility depended on composition of I and II	1139
	Ethyl methacrylate	Single T_g	I had 4.5–44 wt.% methacrylonitrile (MAN)	943
	Methyl methacrylate	Single T_g; transparency; fluorescence spectroscopy studies especially at the low MAN end	Miscibility range varied in various	970,986, 1018, 1139

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Isopropyl methacrylate	Single T_g; transparency	I had 12–38 wt.% MAN when II had $M_n = 161$ kg/mol; I had 8–43 wt.% MAN when II had $M_n = 41.6$ kg/mol	943
	n-Propyl methacrylate	Single T_g; transparency	I had 2–32 wt.% MAN	943
4-[[6-(Methacryloyloxy)hexyl]-oxy]-*N*-[(9-methyl-2-carbazolyl) methylene]aniline	4-[[6-(Methacryloyloxy)hexyl] oxy]-*N*-(4′-nitrobenzylidene) aniline	Single T_g	Polymer I and polymer II were liquid crystalline	1076
4-[[1-[4(-Methoxyphenyl)azo] phenyl]-4-oxy]-3-propyl] oxy]styrene	4-[[1-[(4-cyanobiphenyl)-4′)oxy]-3-propyl]oxy]styrene	Single T_g	Blend showed liquid crystallinity	1042
p-Methoxystyrene-*co*-*p*-vinyl-phenol	Alkyl acrylate	Single T_g; FTIR	II had an alkyl group of methyl or ethyl (2 miscible blend systems); I had 33–63 mol% vinylphenol	1177
	Vinyl acetate	Single T_g; FTIR	I had 33–63 mol% vinylphenol	1177
Methyl acrylate	Epichlorohydrin	Single T_g	–	226
	Epichlorohydrin-*co*-ethylene glycol	Single T_g	I had 50 mol% epichlorohydrin (Hydrin 200)	226
	Ethylene glycol	Single T_g	–	226
	Nitrocellulose	Single T_g	–	416
	Vinyl acetate	Single T_g; transparency; NMR	Probably formed two phases when deposited from some solvents (Refs. 601)	453,601,615 1196
	Vinyl chloride-*co*-vinylidene chloride	Single T_g; transparency	II was Saran ($\geq 80\%$ vinylidene chloride); I had MW = 1.0 kg/mol; two T_g's when I had MW = 576 kg/mol in Ref. 870	812,870
	Vinylidene fluoride	Single T_g; transparent melt	Semicrystalline	844,1122
Methyl acrylate-*co*-vinylcarbazole	Methyl methacrylate-*co*-2(3,5-dinitrobenzoyl)oxyethyl methacrylate	Single T_g	Both I and II had >27 mol% monomer with interactive groups	705
	Styrene-*co*-2(3,5-dinitrobenzoyl) oxyethyl methacrylate	Single T_g	Both I and II had >12 mol% monomer with interactive groups	173
Methyl 1-bicyclobutane-carboxylate	Acrylonitrile-*co*-styrene	Single T_g; transparency	II had 24–38 wt.% acrylonitrile	202
	Ethylene glycol	Single T_g; transparency	≤ 20 wt.% II	202
2-Methyl-6-ethyl-1,4-phenylene ether	Styrene	Single T_g	–	134
α-Methyl-α-ethyl-β-propiolactone	Vinyl chloride	Single T_g	Semicrystalline when blend had 75% I	33
Methyl methacrylate	Acrylonitrile-*co*-methyl methacrylate-*co*-styrene	Single T_g	II had 40/39.1/20.9 or 61.1/19.6/19.3 = styrene/methyl methacrylate/ acrylonitrile	620
	4′-[[2-(Acryloyloxy)ethyl]-ethylamino]-4-nitroazobenzene	Single T_g; NMR	–	1228
	4′-[[2-(Acryloyloxy)ethyl]-ethylamino]-4-chloro-4-nitro-azobenzene	Single T_g; NMR	–	1228
	p-tert-Butylphenol-*co*-formaldehyde	Single T_g; FTIR; WAXD	–	647,1056
	Carbon monoxide-*alt*-propylene	Single T_g; FTIR; NMR	–	1229
	Carbonate	Single T_g; transparency	II was carbonate of bisphenol-F or bisphenol-AF (2 miscible blend systems)	426
	Carbonate of bisphenol chloral	Single T_g; NMR	–	110.426,673,983
	Copolycarbonate of bisphenol-A and hexafluorobisphenol-A	Single T_g; transparency	II had 4.5–14.0 wt.% hexafluoro-bisphenol-A	430
	p-Chlorophenol-*co*-formaldehyde	Single T_g; FTIR	–	647
	Epichlorohydrin-*co*-ethylene glycol	Single T_g	–	224
	Ethyl acrylate	Single T_g	–	156
	Ethyl methacrylate	Single T_g	Made from slurries of polyethyl methacrylate and methyl methacrylate; inherently unstable, phase separation upon heating	763
	Formaldehyde-*co*-*p*-nitrophenol	Single T_g; FTIR	–	647
	Formaldehyde-*co*-phenol	Single T_g; FTIR	–	647
	Hexafluoroacetone-*co*-vinylidene fluoride	Single T_g	–	443

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	N-Maleimide-*co*-styrene	Single T_g; FTIR	II had 8, 14 or 21 wt.% *N*-maleimide	1213
	Methoxymethyl methacrylate	Single T_g	–	264
	Methyl acrylate	Single T_g	–	156
	α-Methylstyrene	Single T_g	–	167,765
	Nitrocellulose	No phase separation in solution or films	–	199,296,452,651
	Novolac	Single T_g; FTIR	II had MW about 1.0 kg/mol and was formaldehyde and 13/17/70 mol% = *p-tert*-butylphenol/*m*-cresol/*o*-cresol or formaldehyde and 15/17/68 mol% = 2-*t*-butylphenol/*m*-cresol/*o*-cresol; I was isotactic, atactic or syndiotactic	221,434,473
	Propylene glycol	Single T_g; SAXS	I had MW = 550 kg/mol; II had MW = 1.0–3.9 kg/mol	527
	Tetrafluoroethylene-*co*-vinylidene fluoride	Single T_g; T_m-depression	II had 80 mol% vinylidene fluoride	113,324
	Trifluoroethylene-*co*-vinylidene fluoride	Light scattering; FTIR; T_m-depression	II had 58 or 75 mol% vinylidene fluoride	713,1066
	Vinyl acetate	Single T_g; FTIR; LS; microscopy	Miscible when cast from chloroform or cyclohexanone; immiscible when cast from tetrahydrofuran or melt blended	302,602, 726,777, 778,928, 1092
	Vinyl acetate-*co*-vinyl chloride	Single T_g; transparency	II had 5.9–15.1 vol% vinyl acetate	236,296,452, 497,651
	Vinyl chloride	Single T_g; transparency	Especially when II was polymerized *in situ* (Ref. 846); I of all tacticities miscible with II (Ref. 338); increase in isotactic content and mol. wt. of I reduced miscibility with II (Ref. 893)	66,338,368, 641,642,653, 812,846,850, 892,893,895
	Vinyl chloride, chlorinated	Single T_g; FTIR	II had 63–68 wt.% Cl; I was atactic, isotactic of syndiotactic (1204)	506,833,1204
	Vinyl chloride-*co*-vinylidene chloride	Single T_g; transparency	II was Saran (>80% vinylidene chloride) or 86.5 wt.% vinylidene chloride; semicrystalline when ≥20% II and I was isotactic, and when ≥75% II and I was atactic for the second II	409,506, 812,870
	Vinyl chloroacetate	Single T_g	–	179
	Vinylidene fluoride	Single T_g; NMR	Semicrystalline when >35–65% II; I was isotactic, atactic or syndiotactic	124,293, 320,342,361, 569,618,630, 643,644,696
Methyl methacrylate, imidized	Acrylonitrile-*co*-styrene	Single T_g	I had 53–91 wt.% imidized units; II had 5.7–33 wt.% acrylonitrile; miscibility depended on extent of imidization of I and acrylonitrile content of II	230,1123
	Vinyl chloride	Single T_g	I had 53–85 wt.% imidized units	230
Methyl methacrylate-*co*-styrene	Copolycarbonate of bisphenol-A and tetramethylbisphenol-A	Single T_g; transparency	I had ≤33 wt.% methyl methacrylate; II had ≤15 wt.% bisphenol-A	431
	Copolycarbonate of hexafluoro-bisphenol-A and tetramethyl-bisphenol-A	Single T_g	I had 9, 20 or 33 wt.% methyl methacrylate; II had 20 wt.% hexafluorobisphenol-A	1194
Methyl methacrylate-*co*-vinylpyridine	Acrylic acid-*co*-butyl acrylate	Single T_g; FTIR	I had 13–20 mol% vinylpyridine; II had 13–20 mol% acrylic acid	385
2-Methyl-2-oxazoline	Allyl alcohol-*co*-styrene	Single T_g; FTIR	II had 6.5 wt.% OH group	978
	Hydroxyether of bisphenol-A	Single T_g; FTIR	–	978
	2-Hydroxypropyl methacrylate	Single T_g; FTIR	–	978
	Vinyl alcohol	Single T_g; FTIR	–	913
	Vinyl chloride	Single T_g	I <50 wt.% in blend	442
	Vinyl fluoride	Single T_g	I <50 wt.% in blend	442
	p-Vinylphenol	Single T_g; FTIR	–	978
2-Methyl-1-pentene sulfone	Novolac	Single T_g	II had MW = 1.0 kg/mol and was formaldehyde and 13/17/70 mol% = *p-tert*-butylphenol/*m*-cresol/*o*-cresol or formaldehyde and 15/17/68 mol% = 2-*tert*-butylphenol/*m*-cresol/*o*-cresol	221

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
2-Methyl-6-propyl-1, 4-phenylene ether	Styrene	–	–	134
α-Methyl-α-*n*-propyl-β-propiolactone	Vinyl chloride	Single T_g	Semicrystalline when > 50% I	32,33
α-Methylstyrene	Alkyl methacrylate	Single T_g; transparency	II had an alkyl group of cyclohexyl, ethyl or *n*-propyl (3 miscible blend systems)	116,765
	Cyclohexyl acrylate	Single T_g; transparency	–	765
p-Methylstyrene	Cyclohexyl acrylate	Single T_g; transparency	–	765
	Cyclohexyl methacrylate	Single T_g; transparency	–	116,765
	2,6-Dimethyl-1,4-phenylene ether, bromobenzylated	Single T_g; transparency	II had 9, 16 or 37% bromination	261
α-Methylstyrene-*co*-4-(2-hydroxyethyl)-α-methylstyrene	Alkyl acrylate	Single T_g	I had 6, 41, 72 or 89 mol% α-methylstyrene; II had an alkyl group of ethyl, *tert*-butyl or methyl (3 blend systems)	968
	Alkyl methacrylate	Single T_g	I had 6, 41, 72 or 89 mol% α- methylstyrene; II had an alkyl group of ethyl, *tert*-butyl or methyl (3 blend systems)	968
	Dialkyl itaconate	Single T_g	I had 6, 41, 72, or 89 mol% α-methylstyrene; II had a dialkyl group of dimethyl, diethyl, di-*tert*-butyl or di-*n*-propyl (4 blend systems)	968
	Vinyl acetate	Single T_g	I had 6, 41, 72 or 89 mol% α-methylstyrene	969
	4-Vinylpyridine	Single T_g	I had 6, 41, 72 or 89 mol% α-methylstyrene	969
	N-Vinyl pyrrolidone	Single T_g	I had 6, 41, 72 or 89 mol% α-methylstyrene	969
α-Methylstyrene-*co*-[4-(1,1,1-trifluoro-2-hydroxyethyl)-α-methylstyrene	Alkyl acrylate	Single T_g	I had 23, 47, 60, 84 or 93 mol% α-methylstyrene; II had an alkyl group of methyl, ethyl or *tert*-butyl (3 blend systems)	968
	Alkyl methacrylate	Single T_g	I had 23, 47, 60, 84 or 93 mol% α-methylstyrene; II had an alkyl group of methyl, ethyl or *tert*-butyl (3 blend systems)	968
	Dialkyl itaconate	Single T_g	I had 23, 47, 60, 84 or 93 mol% α-methylstyrene; II had a dialkyl group of dimethyl, diethyl, di-*n*-propyl or di-*tert*-butyl (4 blend systems)	968
	Vinyl acetate	Single T_g	I had 23, 47, 60, 84 or 93 mol% α-methylstyrene	969
	4-Vinylpyridine	Single T_g	I had 23, 47, 60, 84 or 93 mol% α-methylstyrene	969
	N-Vinyl pyrrolidone	Single T_g	I had 23, 47, 60, 84 or 93 mol% α-methylstyrene	969
N-Methyl-*N*-vinylacetamide	Allyl alcohol-*co*-styrene	Single T_g; FTIR	II had 4.5 or 6.5 wt.% OH group; formed complexes in 2-butanone	973
Monomethyl itaconate	Thiocarboate	Single T_g	II had methyl, ethyl or propyl side groups	669
Natural rubber, epoxidized	Acrylic acid-*co*-ethylene	Single T_g	I had 50 mol% epoxidation; II had 6 mol% acrylic acid; blend had > 50 wt.% II	1134
	Alkyl acrylate-*co*-vinylidene chloride	Single T_g	I had 50 mol% epoxidation; II had 10% alkyl acrylate	400
	Chloroprene	Single T_g	I had 50 mol% epoxidation	1065
	Ethylene, chlorinated	Single T_g	I had 50 mol% epoxidation; II had 25 or 48% Cl	543
	Ethylene, chlorosulfonated	Single T_g	I had 25 mol% epoxidation; II was Hypalon 40	590
	Novolac	Single T_g	I had 50 mol% epoxidation	401
	Propylene, chlorinated	Single T_g	I had 50 mol% epoxidation	400
	Vinyl chloride	Single T_g	I had 50 mol% epoxidation	507,827,828
	Vinyl chloride, plasticized	Single T_g	I had 50 mol% epoxidation; II had 40 phr dioctyl phthalate	826

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
Neopentyl glycol adipate (2,2-dimethyl-1,3-propylene adipate)	Hydroxyether of bisphenol-A (Phenoxy)	Single T_g; transparency	–	332
	Vinyl acetate-*co*-vinyl chloride	Single T_g; transparency	II had 2 or 17 wt.% vinyl acetate	276
	Vinyl chloride	Single T_g; transparency	–	277
	Vinyl chloride-*co*-vinylidene chloride	Single T_g	Semicrystalline; II was Saran (86.5 wt.% vinylidene chloride)	868
	Vinylidene fluoride	Clear melt although refractive indices very different; T_m-depression	Semicrystalline	287
Neopentyl glycol succinate (2,2-dimethyl-1,3-propylene succinate)	Epichlorohydrin	Single T_g; transparency	–	223
	Hydroxyether of bisphenol-A (phenoxy)	Single T_g; transparency	Semicrystalline $\leq$ 20 wt.% II	332
	Vinyl chloride	Single T_g; transparency	–	900
	Vinyl chloride-*co*-vinylidene chloride	Single T_g	$\leq$ 60 wt.% I; possible second T_g when >60 wt.% I; semicrystalline	868
Nitrocellulose	Ester	Single T_g; FTIR	I had 12.62 or 13.42% N; II was butylene adipate, caprolactone, ethylene adipate or valerolactone (4 miscible blend systems)	393
	Vinyl acetate	Single T_g; clear films	–	236,421,452
Novolac	*n*-Butyl methacrylate	Single T_g; FTIR	–	434
	Vinyl acetate	Single T_g	I had MW about 1.0 kg/mol and was formaldehyde and 15/17/68 mol% = 2-*tert*-butylphenol/*m*-cresol/*o*-cresol	221
	Vinyl methyl ether	Single T_g	I had MW about 1.0 kg/mol and was formaldehyde and 13/17/70 mol% = *p*-*tert*-butylphenol/*m*-cresol/*o*-cresol	221
Phenolphthalein polyether ether ketone	Epoxy	Single T_g	–	312
	Hydroxyether of bisphenol-A (phenoxy)	Single T_g	–	308,312
	Sulfone	Single T_g	–	308,311
	N-Vinylpyrrolidone	Single T_g	–	307
Phenolphthalein polyether ether sulfone	*N*-Vinylpyrrolidone	Single T_g	–	307
Phenol–formaldehyde	*N,N*-Dimethylacrylamide	Single T_g	Formed complexes from acetone, dioxane and ethyl acetate	881
	Methyl methacrylate-*co*-styrene	Single T_g	II had >20 wt.% methyl methacrylate	433
Phenyl acrylate	Vinyl benzoate	Single T_g; transparency	–	535
Phenylene, sulfonated	Ethyl acrylate-*co*-4-vinylpyridine	Single T_g	–	592
p-Phenylene benzobisoxazole	Amide	Single T_g; dielectric measurements; WAXS; LS	II was Zytel 330 (Du Pont)	945
	Amide	Single T_g; microscopy; dielectric measurements	II was nylon-6,6; semicrystalline	1084
2,2-(*m*-Phenylene)-5,5′-bibenzimidazole	Arylate	Single T_g	–	125
	Carbonate of bisphenol-A	FTIR	–	594
	Imide	Single T_g; FTIR	II was made from 3,3′,4,4′-benzophenonetetracarboxylic dianhydride and 3,3′-diaminobenzophenone (LARC TPI)	299
	Imide	Single T_g; FTIR	II was 2,2′-bis(3,4-dicarboxyphenoxy)phenylpropane-2-phenylene bisimide(Ultem 1000)	507,509,748, 823
	Imide	Single T_g; FTIR	II was made from 3,3′,4,4′-benzophenone tetracarboxylic dianhydride and 5,6-amino-1-(4′-aminophenyl)-1,3,3′-trimethylindam (XU 218)	115,507, 748
	Imide	Single T_g; FTIR	II was made from benzophenone-tetracarboxylic dianhydride and 4/1 mixture of 2,4-toluenediisocyanate	507,781

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
			and 4,4′-diphenylmethanediisocyanate(PI 2080)	
	Imide sulfone	Single T_g; FTIR	II was made from 3,3′,4,4′-benzophenone-tetracarboxylic dianhydride and 3,3′-diaminodiphenyl sulfone	105,369, 492
	Imide sulfone, fluorinated	Single T_g; FTIR	$\geq$ 80% I	369
	Siloxaneimide, segmental copolymer	Single loss tangent peak	II was 3,3′,4,4′-benzophenone-tetracarboxylic dianhydride and 3,3′-diaminodiphenylsulfone based polyimide and dimethylsiloxane segmental copolymer	105
m-Phenylene isophthalamide	Aramide	Single T_g	II was a copolyamide of 1,6-hexanediamine, caprolactam and terephthalic acid (BASF Ultramid T) or Selar 3426 (Du Pont) (2 blend systems)	993
	Hexamethylene isophthalamide	Single T_g	Semicrystalline	81,993
	Heptamethylene isophthalamide	Single T_g	Semicrystalline	81
	m-Xylene adipamide	Single T_g	Semicrystalline	993
Phenylene sulfide	Phenylene sulfide ether	Single T_g	Semicrystalline	926
p-Phenylene terephthalamide	Amide	Single T_g; FTIR	Semicrystalline; II was nylon 6 or nylon 6,6 (2 miscible blend systems)	477
	p-Phenylene 1,3,4-oxadizole	Miscroscopy; FTIR	–	1080
n-Propyl acrylate	Vinyl chloride	Single T_g; transparency	–	850
n-Propyl methacrylate	Vinyl chloride	Single T_g; transparency	–	319,649,812,850
	Vinyl chloride-*co*-vinylidene chloride	Single T_g; transparency	Semicrystalline when $\geq$ 65% II; II was Saran ($\geq$ 80% vinylidene chloride) or 86.6 wt.% vinylidene chloride	409,812, 870
Propyloxazoline	Styrene	Single T_g	<25 wt.% I in blend	442
Propylene	Hydrogenated oligocyclopentadiene	Melting point depression; WAXS; SANS	–	550
Propylene, chlorinated	2,2-Dimethyl-1,3-propylene adipate	Single T_g; transparency	I had 67 wt.% Cl	287
Rubber, chlorinated	Vinyl methyl ether	Phase diagram	–	250
Styrene	Carbonate of tetramethylbisphenol-A	Single T_g; light scattering; NMR; PALS	Semicrystalline after long annealing	316,359,428, 741,1100,1230,1231
	2-Chlorostyrene	Negative volume change of mixing	–	814
	Copolycarbonate of bisphenol-A and tetramethylbisphenol-A	Single T_g; transparency	II had $\geq$ 85 wt.% tetramethylbisphenol-A	431
	Cyclohexyl acrylate	Single T_g	–	765
	Cyclohexyl methacrylate	Single T_g; transparency	–	116,621,765
	Cyclohexyl methacrylate-*co*-methyl methacrylate	Single T_g	II had $\geq$ 21.6% cyclohexyl methacrylate	621
	2,6-Dimethyl-1,4-phenylene ether, bromobenzylated	Single T_g	II had $\leq$ 76% bromination	259
	2,6-Dimethyl-1,4-phenylene ether, sulfonated	Single T_g	II had < 1.9% sulfonation	345
	2,6-Dimethyl-1,4-phenylene ether-*co*-2,3,6-trimethyl-1,4-phenylene oxide	Single T_g	I was atactic or isotactic; II had < 20 mol% 2,3,6-trimethyl-1,4-phenylene oxide	638
	Ethyl methacrylate	Single T_g	Miscible when melt blended	72
	o-Fluorostyrene-*co*-*p*-fluorostyrene	Single T_g	II had 10 or 23 mol% *p*-fluorostyrene	1144
	α-Methylstyrene	Single T_g	–	710,722,822,865,875
	n-Propyl methacrylate	Single T_g	Miscible when melt blended	72
	Vinyl methyl ether	Single T_g; transparency; NMR; attenuated total reflectance IR	Two phases when deposited from some solvents	43,44,62,126,475, 565,616,964
	Vinyl methyl ether	Single T_g	I was syndiotactic; blend had < 20 wt.% I	954
	Vinyl methyl ether	SANS; NMR	I was atactic or isotactic	915,918,1199
Styrene, sulfonated	Alkylene glycol amine-terminated	FTIR	I had 4.5 mol% sulfonation; II had M_n = 0.9 kg/mol (Jeffamine ED-900)	246
	N,*N*-Dimethylacrylamide	Transparency; NMR	I had 12 mol% sulfonation	487,1064
	2,6-Dimethyl-1,4-phenylene ether	Single T_g	I had < 2.6% sulfonation	345
	10,12-Docosadiynylene diisonicotinate	Single T_g; FTIR	I had 9 mol% sulfonation	990,991
	Ethylene oxide	Single T_g	I had 1.35 or 4.51 mol% SO_3H; II end-capped with propylamine groups	861

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Bis[2-[2-(2-methoxyethoxy)ethoxy] ethyl itaconate	Single T_g; FTIR	I had 14 mol% sulfonation	1086
	Methyl methacrylate	Single T_g; transparency	I had 12 mol% sulfonation	487
	Methyl methacrylate-*co*-4-vinylpyridine	NMR	I had 9.9% sulfonation; II had 11.1% vinylpyridine	604
	Urethane	FTIR	I had 7.9 mol% sulfonation	801
Styrene, sulfonated and metal-neutralized	Amide	Single T_g; FTIR; SAXS	I was lightly sulfonated and Mg- or Mn-neutralized; II was nylon 6	1038
	Amide	Single T_g; FITR; SAXS	I had 9.7 mol% sulfonation and Li-neutralized; II was nylon 4	1138
	Amide	Single T_g	I had 9.8 mol% sulfonation and Li-neutralized; II was nylon 6,6 or nylon 6,10 (2 blend systems)	1135
	Amide	Single T_g; FTIR; SAXS	I had 5.4 or 9.8 mol% sulfonation and Li-neutralized; II was nylon 6	1168,1223
	Amide	Single T_g; NMR; FTIR; SAXS	I had 6.5–9.0 mol% sulfonation and Zn-neutralized; II was nylon 6	1083,1223
	Ethyl acrylate-*co*-4-vinylpyridine	FTIR	I had 2.1–7.3 mol% sulfonation and Zn-neutralized; II had 2.4–10.6 mol% vinylpyridine	1176
Styrene-*co*-styrene-4-carboxylic acid	*N,N*-Dimethylacrylamide	Single T_g; transparency	I had 25 or 54 mol% acid groups	487
	Methyl methacrylate	Single T_g; transparency	I had 25 mol% acid units	487
Styrene-*co*-4-vinylbenzene phosphonic acid	Styrene-*co*-4-vinylbenzene phosphonic acid diethyl ester	Single T_g	Both I and II had 7.5 or 13.3 mol% phosphorus moiety	1250
Styrene-*co*-4-vinylbenzoic acid, K-neutralized	5,7-Dodecadiyne-1,12-dicarboxylic acid, K-neutralized	Single T_g	I had 26 mol% acid group	990
Styrene-*co*-4-vinylbenzoic acid, triethylammonium ionomer	5,7-Dodecadiyne-1,12-dicarboxylic acid, triethylammonium ionomer	Single T_g; FTIR	I had 26 mol% acid group	990,991
Styrene-*co*-4-vinylphenyl-dimethylsilanol	*n*-Butyl methacrylate	Single T_g; FTIR	I had 9–34 mol% silanol group	1112,1114
	N-Vinyl-2-pyrrolidone	Single T_g; FTIR	I had $\geq$4 mol% silanol group; formed complex when II had $\geq$34 mol% silanol group	1113
Styrene-*co*-*p*-vinylphenol	Bis[2-[2-(2-methoxyethoxy) ethoxy]ethyl] itaconate	Single T_g; FTIR	II had 22 mol% vinylphenol	1086
	n-Butyl methacrylate	Transparency; FTIR; fluorescence study	I had 8, 25, 43 or 75 wt.% vinylphenol	103,876
	Caprolactone	Single T_g	I had 10–20.4 wt.% vinylphenol	905,1206
	N,N-Dimethylacrylamide	Transparency; FTIR; single T_g; NMR	I had 22, 42 or 82 mol% vinylphenol	487,1064
	Ethyl methacrylate	Transparency; fluorescence study	I had >0.93 mol% vinylphenol	103
	n-Hexyl methacrylate	FTIR; LS	I had 8, 25 or 43 wt.% vinylphenol	876,922
	Methyl methacrylate	Single T_g; transparency; fluorescence study; FTIR; NMR	I had >0.93 mol% vinylphenol; II was atactic, isotactic or syndiotactic	103,487,1055
	Vinyl butyl ether	FTIR	I had 48 wt.% vinylphenol	734
	4-Vinylpyridine	Single T_g	I had 50 mol% vinylphenol	832
Styrene-*co*-4-vinylpyridine	Isoprene, sulfonated	Single T_g; transparency	II had 5–10% substituted and partly cyclized; immiscible when II had 2% substituted	210,896
N,N'-Sulfo-*p*-phenylene terephthalamide	Vinyl alcohol	Single T_g	–	972
	4-Vinylpyridine	Single T_g	–	972
	N-Vinylpyrrolidone	NMR	–	972
Sulfone of hexafluorobisphenol-A	Methyl methacrylate	Single T_g	I had M_w = 27 kg/mol and II had M_w = 1.21 kg/mol; blends of polymers of higher MW showed LCST	931
Sulfone of tetramethylbisphenol-A	Acrylonitrile-*co*-α-methylstyrene	Single T_g	II had 9–19 wt.% AN when I had M_w = 4.8 kg/mol; II had 16–19 wt.% AN when I had M_w = 12.4 kg/mol	908,931
	Acrylonitrile-*co*-styrene	Single T_g	II had 11–25 wt.% AN when I had M_w = 4.8 kg/mol; II had 13–20 wt.%	931

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
			AN when I had $M_w = 12.4$ kg/mol	
	Maleic anhydride-*co*-styrene	Single T_g	II had 17 or 18 wt.% maleic anhydride when I had $M_w = 4.8$ kg/mol	931
Tetrafluoroethylene, sulfonated	Ethyl acrylate-*co*-4-vinylpyridine	Single T_g	I had equivalent weight of 1.155 kg/mol	591
Tetrafluoroethylene-*alt*-vinyl alcohol	Alkyl methacrylate	Single T_g; FTIR	II had an alkyl group of methyl or ethyl (2 miscible blend systems)	961
	Methyl acrylate	Single T_g; FTIR	–	961
	Vinyl acetate	Single T_g; FTIR	–	961
Tetrahydrofurfuryl methacrylate	Allyl alcohol-*co*-styrene	Transparency	II had 5.4–6.0% OH	253
	Epichlorohydrin	Single T_g; transparency	–	117
	Hydroxyether of bisphenol-A (phenoxy)	Single T_g; transparency	–	253
	Vinyl chloride	Single T_g; transparency	–	281
	Vinyl chloride-*co*-vinylidene chloride	Single T_g	II had 88% vinylidene chloride	117
Tetrahydropyranyl-2-methyl methacrylate	Allyl alcohol-*co*-styrene	Transparency	II had 5.4–6.0 wt.% OH	118
	Epichlorohydrin	Single T_g; transparency	–	117
	Hydroxyether of bisphenol-A (phenoxy)	Single T_g; transparency	–	118
	Vinyl chloride	Single T_g; transparency	–	117
	Vinyl chloride-*co*-vinylidene chloride	Single T_g	II had 88% vinylidene chloride	117
	p-Vinylphenol	Single T_g; transparency	–	118
2,2,2-Trichloroethyl methacrylate	Alkyl methacrylate	Single T_g; transparency	II had an alkyl group of methyl, ethyl, *n*-propyl, isopropyl, cyclohexyl or tetrahydrofurfuryl (6 miscible blend systems)	1110
3,3,5-Trimethyl cyclohexyl methacrylate	3,3,5-Trimethyl-cyclohexyl acrylate	Single T_g; transparency	–	765
Trifluoroethylene	Vinylidene fluoride	Single T_g	Semicrystalline	252
Trimethylene adipate	Vinyl chloride	Single T_g; transparency	When $\geq 75\%$ II	872
Urethane	Carbonate of bisphenol-A	Single T_g	I was based on polycarbonate diols and/or polypropylene glycol; miscible when blend had 10–20 wt.% I	997
	Ethylene glycol	FTIR	I was made from a mixture of 2,4- and 2,6-tolyldiisocyanate and 1,4-butanediol	141,347
	Ethylene-*co*-methyl acrylate	Single T_g; FTIR	I was Estane 58311; II had 21 wt.% methyl acrylate	1181
	Ethylene glycol-*co*-propylene glycol	Single T_g; FTIR	I was made from a mixture of 2,4- and 2,6-tolyldiisocyanate and 1,4-butanediol; II had 70 wt.% ethylene glycol	145
	Methacrylic acid-*co*-styrene	Single T_g	I had polyether segment with MW = 1.0 or 2.0 kg/mol; II had 10.5% methacrylic acid, Li-neutralized	707
	Urethane	Single T_g	I and II were separate polymers; I was 4,4′-diisocyanate diphenyl methane (MDI)/dipropylene glycol = 9/10 mol ratio; II was poly G55–56 polyoxyethylene-*co*-oxypropylene, 45% ethylene oxide, MW = 2.0 kg/mol/MDI/1/1	521
	Vinyl chloride	Single T_g	Blend had ≤ 30 wt.% I	1029
	N-vinylpyrrolidone	Single T_g for II with hard segments of I; soft segment T_gs also observed; clear films	I was segmented, made from methylene bis(4-phenyl isocyanate) and poly(ethylene glycol) and *cis*-2-butene-1,4-diol and butane-1,4-diol	206
Urethane, functionalized	Acrylic acid-*co*-styrene	Single loss peak; microscopy	I had tertiary amine group; II had 15 or 30 mol% acrylic acid; linear blend of I and II or interpenetrating polymer network of I and II	346
	2-Methyl-5-vinylpyridine-*co*-styrene	Single T_g	I carried pendant carboxylated groups neutralized by ammonium ion	706

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
Valerolactone	Vinyl acetate-*co*-vinylidene chloride	Single T_g	Semicrystalline; II was Saran C ($\geq$ 80% vinylidene chloride)	31
	Vinyl bromide	Single T_g	–	150
	Vinyl chloride	Single T_g	Semicrystalline when $\geq$ 50% I	32,34,35
Vinyl acetate	2-Bromoethyl methacrylate	Single T_g; transparency	Miscible when cast from methyl ethyl ketone; immiscible when cast from tetrahydrofuran	610
	Chloroalkyl methacrylate	Single T_g; transparency	II had chloromethyl or 2-chloroethyl group (2 blend systems)	610
	Hexafluoroacetone-*co*-vinylidene fluoride	Single T_g	II had 8.9 mol% hexafluoroacetone	748,1073
	Maleic anhydride-*co*-styrene hydrolyzed	Single T_g	1.8–43% hydrolyzed	67
	Vinyl nitrate	Single T_g; phase contrast microscopy	–	12
	Vinylidene fluoride	Single T_g	Semicrystalline	53,57
Vinyl acetate-*co*-vinyl chloride	*n*-Butyl methacrylate-*co*-isobutyl methacrylate	Single T_g	Miscibility range depended on compositions of I and II; I had $\geq$ 81 wt.% vinyl chloride	748
	Hexyl acrylate-*co*-methyl acrylate	Single T_g	I had 5–23.7 vol% vinyl acetate; II had 15–88 vol% hexyl acrylate; miscibility depended on compositions of I and II	379
	n-Butyl methacrylate	Single T_g	I had 5.9–15.1 wt.% vinyl acetate	497
Vinyl alcohol	Ethylene imine	Single T_g	Semicrystalline	1024
Vinyl alcohol	Methacrylic acid	NMR	–	1255,1256,1257
Vinyl alkanoate	Vinyl alkanoate	Single T_g; microscopy	The alkanoate was butyrate, caprate, caproate, caprylate or laurate (5 miscible blend systems)	925
Vinyl butyral	*N*-Vinylpyrrolidone	Single T_g; transparency	$\leq$ 50% II	209
	Urethane elastomer	Single T_g	II from methylene bisdiphenyl diurethane and polytetramethylene ether glycol; miscible with hard segment	762
Vinyl chloride	Acetonyl methacrylate	Single T_g; transparency	–	123
	Acrylic acid-*co*-carbitylacrylate-*co*-α-methylstyrene-*co*-styrene	Single T_g	II was an oligomer with $M_n = 1.07$ and $M_w = 3.32$ kg/mol	683
	Acrylonitrile-*co*-*n*-butyl acrylate	Optical microscopy	II had 53, 67 or 84 mol% butyl acrylate	655
	Alkyl acrylate	Single T_g	II had pentyl or propyl group (2 blend systems)	851
	Alkyl methacrylate	Single T_g	II had cyclohexyl or isopropyl group (2 blend systems)	319
	Butadiene, epoxidized	Single T_g	II had 50 mol% epoxidation	546
	Butadiene-*co*-styrene, epoxidized	Single T_g	II had 45 wt.% styrene; epoxidation > 40 mol%	544
	Butyl acrylate-*co*-ethyl acrylate	Single T_g	II had 22.9, 43.5 or 52.8% butyl acrylate	851
	Butyl acrylate-*co*-hexyl acrylate	Single T_g	II had 47.8% butyl acrylate	851
	Butyl acrylate-*co*-methyl acrylate	Single T_g	II had 44.6% butyl acrylate	851
	Butyl methacrylate-*co*-methyl methacrylate	Single T_g	II had 30 or 70% methyl methacrylate	880
	Caprolactone-*co*-L-lactide	Single T_g	II had $\leq$ 40 wt.% lactide	831
	Caprolactone-*co*-lignin	Single T_g; microscopy	II was a 7-armed starlike polymer containing 20 wt.% lignin derivative	984
	Caprolactone-*co*-methylcaprolactone	Single T_g	Whatever composition of II	831
	2,2-Dimethyl-1,3-propylene sebacate	Single T_g; transparency	–	611
	Ethyl acrylate-*co*-hexyl acrylate	Single T_g	II had 41.7 or 43.2% ethyl acrylate	851
	Ethyl acrylate-*co*-methyl methacrylate	Single T_g	II had 10% ethyl acrylate	875
	Ethylene-*co*-vinyl acetate	Permittivity measurements	II had 70% vinyl acetate; composition dependent; miscible when blend contained $\geq$ 25% II	678
	Methyl acrylate-*co*-hexyl acrylate	Single T_g	II had 9.7% methyl acrylate	851
	Methyl methacrylate-*b*-styrene	–	No gross phase separation when I had $M_n = 43$ kg/mol; PMMA and PS block had the same $M_n = 40$ kg/mol in II	376

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Nitrile rubber, carboxylated	Single T_g	II was Krynac-221	1170
	Pentyl acrylate-*co*-propyl acrylate	Single T_g	II had 48.1% propyl acrylate	851
	Vinyl acetate	Single T_g	Ref. 59 stated miscible when cast from methyl ethyl ketone and immiscible when cast from tetrahydrofuran; Ref. 183 stated blends cast from both THF and MEK were immiscible	59,183
Vinyl chloride, chlorinated	Alkyl methacrylate	Single T_g; transparency	I had 65 or 68 wt.% Cl; II had *n*-butyl, cyclohexyl, ethyl, *n*-hexyl, *n*-propyl or isopropyl group (6 blend systems)	319,812
	Butadiene, epoxidized	Single T_g	II had 20 or 43 mol% epoxidation	779
	Butadiene-*co*-acrylonitrile	Single T_g	Depended on Cl-content and acrylonitrile content	96,152,355,356
	Ethylene terephthalate	Single T_g	Probably semicrystalline	36
	Hexamethylene sebacate	Single T_g	I had 67.2 wt.% Cl; semicrystalline when $\geq$ 50 wt.% II	54
	Hexamethylene terephthalate	Single T_g	Semicrystalline	36
	Maleic anhydride-*ran*-styrene	Single T_g	I was Goodrich CPVC 3010; II had 86% styrene	68,69
	α-Methyl-α-*n*-propyl-β-propiolactone	Single T_g	I had 67.2 wt.% Cl	54
	Urethane	Single T_g	I had 67 wt.% Cl; II had a soft poly (tetramethylene adipate) segment capped with 2-hydroxyethyl acrylate	246
	Valerolactone	Single T_g	I had 67.2 wt.% Cl	54
Vinyl chloride-*co*-vinylidene chloride	Alkyl methacrylate	Single T_g	I had 12% vinyl chloride; II had phenyl or isopropyl group (2 blend systems)	409
	Butylene adipate	Single T_g	I had 13.5 wt.% vinyl chloride; semicrystalline	873
	Butylene sebacate	Single T_g	I had 13.5 wt.% vinyl chloride; semicrystalline	873
	Dodecamethylene adipate	Single T_g	I had 13.5 wt.% vinyl chloride; semicrystalline	873
	Dodecamethylene decamethylene dicarboxylate	Single T_g	I had 13.5 wt.% vinyl chloride; semicrystalline	873
	Dodecamethylene dodecamethylene dicarboxylate	Single T_g	I had 13.5 wt.% vinyl chloride; semicrystalline	873
	Hexadecamethylene dodecamethylene dicarboxylate	Single T_g	I had 13.5 wt.% vinyl chloride; semicrystalline	873
	Hexamethylene sebacate	Single T_g	I had 13.5 wt.% vinyl chloride; semicrystalline	873
Vinyl fluoride	Vinylidene fluoride	Single T_g	Semicrystalline; later work found the blends immiscible (Ref. 298)	252,298
Vinyl *p*-methoxycinnamate	*p*-Chloro-(3-ethenoxypropyl) cinnamate	Single T_g	–	515
	Vinyl *p*-chlorocinnamate	Single T_g	–	515
	Vinyl cinnamate	Single T_g	–	515
Vinyl *p*-methylcinnamate	*p*-Chloro-(3-ethenoxypropyl) cinnamate	Single T_g	–	515
	p-Cyano-(3-ethenoxypropyl) cinnamate	Single T_g	–	515
	3-Ethyoxypropyl cinnamate	Single T_g	–	515
	Vinyl *p*-chlorocinnamate	Single T_g	–	515
	Vinyl cinnamate	Single T_g	–	515
Vinyl methyl ether	Acrylonitrile-*co*-styrene	Single T_g	II had < 10-11% acrylonitrile	576
	Alkyl acrylate	Single T_g	II had an alkyl group of ethyl, *n*-propyl or *n*-butyl (3 miscible blend systems)	941
	2-Chlorostyrene	Single T_g; FTIR; SAXS	–	1203
	Epichlorohydrin	NMR, dielectric spectroscopy	–	911,1129
	Maleic anhydride-*co*-styrene	Single T_g	II had 15% maleic anhydride	576
	Methyl methacrylate-*co*-styrene	Single T_g	II had < 60 mol% methyl methacrylate	108,321
	α-Methylstyrene	Light scattering	–	318
	α-Methylstyrene-*co*-4-(2-hydroxyethyl)-α-methylstyrene	Single T_g	II had 6–89 mol% α-methylstyrene	170

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	α-Methylstyrene-*co*-4-(1,1,1-trifluoro-2-hydroxyethyl)-α-methylstyrene	Single T_g	II had 0–96 mol% α-methylstyrene	170
	α-Methylstyrene-*co*-styrene	Light scattering	II had 77.5 or 88 wt.% styrene	318
	m-Methylstyrene-*co*-styrene	FTIR; light scattering	II had 76.5 or 87.5 wt.% styrene	1028
	o-Methylstyrene-*co*-styrene	FTIR; light scattering	II had 76.2 or 86.8 wt.% styrene	1028
	p-Methylstyrene-*co*-styrene	Light scattering	II had $\geq$61 mol% styrene	318,1016
	Nitrostyrene-*co*-styrene	Single T_g	II had $\leq$25.6 mol% nitrostyrene	158,166,192
	Styrene-*co*-1-vinylnaphthalene	Light scattering	II had 17.4, 23, 50 or 80 wt.% styrene; depended on composition of blend	381,406
	Styrene-*co*-2-vinylnapthalene	Single T_g; FTIR; light scattering	II had 19.2, 51 or 84 wt.% styrene	318,1175
	Vinyl alkanoate	Single T_g; IGC	The alkanoate was butyrate or propionate (2 miscible blend systems)	988
	1-Vinylnaphthalene	Single T_g; FTIR	II had M_n = 5.4 kg/mol; blend contained 65–85 wt.% II	407
	Vinylidene fluoride	Single T_g; transparent melt	Semicrystalline	58
Vinyl methyl ketone	Vinylidene fluoride	Single T_g; transparent melt	Semicrystalline	58
p-Vinylphenol	Acetonyl methacrylate	Single T_g; transparency	–	118
	4-Acetoxystyrene	Single T_g; FTIR	–	340
	Amide	Single T_g	Polymer II was nylon 6, nylon 6,6 or nylon 11 (3 miscible systems); semicrystalline	1088
	Butyl acrylate-*co*-*t*-butyl acrylate	Single T_g	II had 64 mol% butyl acrylate	897
	Butylene adipate	Single T_g; FTIR; NMR	Semicrystalline	667,919,1088,1089
	Butylene terephthalate	Single T_g; FTIR	Semicrystalline	1088,1089
	Caprolactone	Single T_g; FTIR; IGC	Semicrystalline	143,667,1088,1096
	Cellulose tripropionate	Single T_g	–	1088
	Copolyester	Single T_g; FTIR	II was five different copolyesters	1089
	Dialkyl itaconate	Single T_g; FTIR	II had a dialkyl group or methyl, ethyl, bis[2-[2-(2-methoxyethoxy)ethoxy] ethyl or bis[2-(2-methoxyethoxy)ethyl (4 miscible blend systems)	1034,1086
	N,N-dimethylacrylamide	Single T_g	Formed complexes in dioxane or acetone	487,795,857, 1088,1089
	2,2-Dimethylpropylene terephthalate	Single T_g; FTIR	–	1088,1089
	Ethyl acrylate	FTIR	–	143
	γ-Ethyl L-glutamate	FTIR; microscopy	–	639,902
	Ethyl methacrylate	Single T_g; FTIR	–	146,282,733,1164
	Ethyl methacrylate-*co*-methyl methacrylate	Single T_g; FTIR	II had 30, 60 or 70 wt.% MMA	1162,1164
	Ethylene adipate	Single T_g; NMR	Semicrystalline	667,919,1089
	Ethylene 2,6-naphthalene dicarboxylate	Single T_g; FTIR	–	1088,1089
	Ethylene succinate	Single T_g	Semicrystalline	919
	Ethylene terephthalate	Single T_g; FTIR	Semicrystalline	1089
	Ethylene-*co*-vinyl acetate	Single T_g; FTIR	II had 70% vinyl acetate	143,144,588
	Hexamethylene sebacate	Single T_g	Semicrystalline	667
	n-Hexyl methacrylate-*co*-methyl methacrylate	Single T_g; FTIR	II had 20, 38 or 48 wt.% *n*-hexyl methacrylate	958
	Hexylene-*m*-xylene dicarboxamide	Single T_g	–	1088
	2-(2-Methoxyethoxy)ethyl methacrylate	Single T_g; FTIR	–	1086
	γ-Methyl L-glutamate	FTIR; microscopy	–	639,902
	Methyl acrylate	FTIR; NMR	–	143,889,894,1195,1224
	Methyl methacrylate	Single T_g; FTIR; NMR	–	146,282,487,733,1164,1224
	n-Propyl acrylate	FTIR	–	143,641
	n-Propyl methacrylate	FTIR; single T_g	–	146,282,733
	Isopropyl methacrylate	Single T_g	–	280
	Styrene-*co*-4-vinylbenzenephosphonic acid diethyl ester	Single T_g; FTIR; NMR	II had >7.5 mol% phosphorus groups	1249
	Styrene-*co*-4-vinylpyridine	Single T_g	II had 50 mol% 4-vinylpyridine	832
	Styrene-*co*-2-vinylpyridine	Single T_g; FTIR	Formed complexes in ethanol; II had 70 mol% 2-vinylpyridine	976
	Tetrahydrofuran	FTIR; transparency	–	734

TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Tetrahydrofurfuryl methacrylate	Single T_g	–	282
	2,2,5-Trimethylene terephthalamide	Single T_g	–	1088
	Vinyl acetate	Single T_g; FTIR	–	143,144,588
	Vinyl methyl ether	Single T_g; FTIR	–	1150
	Vinylpyridine	Single T_g; FTIR	Formed complexes; II was 2- or 4-vinylpyridine (2 miscible systems)	832,976
	N-vinylpyrrolidone	Single T_g	Formed complexes from methanol	857
Vinyl propionate	Ethyl acrylate	Single T_g; transparency; IGC	–	60,61
N-vinylpyrrolidone	Alkyl methacrylate	Single T_g	II had an alkyl group or methyl or ethyl; miscible when cast from chloroform (Ref. 1141); immiscible when cast from DMF (Ref. 607)	607,1141
	Allyl alcohol-*co*-styrene	Single T_g; transparency	II had 5.4–6.0 or 7.3–8.0 wt.% OH	284,977
	Aramide	Single T_g; transparency	Polymer II was five different types of aramides	947
	Benzimidazole	Single T_g; transparency	–	947
	n-Butyl methacrylate-*co*-2-hydroxyethyl methacrylate	Single T_g; transparency	II had $\geq$ 20.9 mol% 2-hydroxyethyl methacrylate	501
	Haloalkyl methacrylate	Single T_g; FTIR	II had a haloalkyl group of chloromethyl, 2-chloroethyl, 3-chloropropyl, 2-bromoethyl or 2-iodoethyl (5 miscible blend systems)	607,1109,1141
	Copolyamide	Single T_g	II had 1/1/1 = caprolactam/hexamethylene adipamide/hexamethylene sebacamide structural unit	315
	Dialkyl itaconate	Single T_g	II had an dialkyl group of dimethyl, di-2-chloroethyl or di-3-chloropropyl (3 miscible blend systems)	1141
	Epoxy resin	Single T_g; FTIR	–	370
	Ethersulfone	Single T_g	–	307
	Ethylene glycol	Single T_g	I had MW = 360 kg/mol; II had MW = 0.3 kg/mol	780
	Epichlorohydrin	Single T_g	–	300
	Ethyl methacrylate-*co*-2-hydroxyethyl methacrylate	Single T_g; transparency	II had $\geq$ 7.5 mol% 2-hydroxyethyl methacrylate	265
	2-Hydroxyethyl methacrylate	Single T_g; transparency	–	284
	2-Hydroxyethyl methacrylate-*co*-methyl methacrylate	Single T_g; transparency	II had $\geq$ 1.9 mol% 2-hydroxyethyl methacrylate	501
	2-Hydroxypropyl methacrylate	Single T_g; transparency	–	284,977
	Methacrylic acid	NMR	Formed complexes	1128
	Monoalkyl itaconate	FTIR	II had an alkyl group of benzyl or ethyl (2 blend systems); formed complexes	95,938
	Sulfone	Single T_g; transparency	–	255
	Sulfone, carboxylated	Single T_g; transparency	II had degree of carboxylation 0.43–1.93	255
	Vinyl alcohol	Single T_g; NMR	Semicrystalline	624,659,805,856,890,998
	Vinyl alcohol-*co*-vinyl acetate	Single T_g	II had $\geq$ 70% vinyl alcohol	207
	Vinyl chloride	Single T_g	–	300
	Vinyl formal	Single T_g: FTIR	–	349,1044
Vinylidene fluoride	Acetonyl methacrylate	Single T_g	Semicrystalline	122
	N,N-Dimethylacrylamide	Single T_g	Semicrystalline	242
	Ethyl methacrylate-*co*-methyl methacrylate	T_m-depression	II had 16–80 wt.% methyl methacrylate	283
	Methacrylic acid-*co*-methyl methacrylate	Single T_g; T_m-depression	Semicrystalline; II had 2.2–9.5 mol% methacrylic acid	1094
	Methyl methacrylate-*co*-styrene	Single T_g	II had < 13 vol% styrene	388
	N-Methyl-*N*-vinylacetamide	Single T_g	Semicrystalline	242
	Methoxymethyl methacrylate	Single T_g	Semicrystalline	267
	Pivalolactone	T_m-depression	Semicrystalline	1038
	N-Vinylpyrrolidone	Single T_g	Semicrystalline	17,94,241,408
Vinylpyridine	Hydroxyalkyl methacrylate	Single T_g; FTIR	I was 2- or 4-vinylpyridine; II had an alkyl group of 2-hydroxyethyl or 3-hydroxypropyl (4 miscible systems)	937

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Monoalkyl itaconate	Single T_g; FTIR	I was 2- or 4-vinylpyridine; II had an alkyl group or methyl or ethyl (4 miscible systems); formed complexes	939,1014, 1212
	Vinyl acetate-*co*-vinyl alcohol	Single T_g; FTIR	I was 2- or 4-vinylpyridine (2 miscible systems); II had 29–88 mol% vinyl alcohol	934,935,991
	Vinyl butyral	SALS	I was 2- or 4-vinylpyridine (2 miscible systems)	1003
2-Vinylpyridine	Aramide	Single T_g; FTIR	II was made from hexamethylene-diamine and a mixture of tereph-thaloyl and isophthaloyl dichlorides	767
4-Vinylpyridine	Methylocellulose	Single T_g; NMR	–	559
	Sulfone, carboxylated	Single T_g; transparency	II had degree of carboxylation 0.43–1.93	256
	p-Phenylene terephthalamide	Transparency; microscopy	–	329
2,3-Xylenyl methacrylate	2,6-Xylenyl methacrylate	Single T_g	–	1158

TABLE 2. POLYMER PAIRS CONTAINING ONE MONOMER IN COMMON, MISCIBLE IN THE AMORPHOUS STATE AT ROOM TEMPERATURE

Polymer I of	Polymer II of	Method	Comments	Refs.
Acrylic acid-*co*-styrene	Methyl methacrylate-*co*-styrene	Single T_g	Miscibility depended on compositions of I and II	72
Acrylonitrile-*co*-butadiene	Acrylonitrile-*co*-butadiene	Single T_g; clear films; electron microscopy	Difference in composition between I and II $\leq$22% acrylonitrile	19,47,101,106, 149,390,452
	Acrylonitrile-*co*-styrene	Single T_g	–	162
	Acrylonitrile-*co*-vinylidene chloride	Single T_g	I had 23.6–49.6 vol% AN; II had 19 wt.% AN	966
Styrene-*co*-maleic anhydride--*co*-acrylonitrile	Acrylonitrile-*co*-styrene	Single T_g	Miscibility depended on compositions of I and II	965
	Maleic anhydride-*co*-styrene	Single T_g	Miscibility depended on compositions of I and II	965
Acrylonitrile-*co*-methyl methacrylate	Acrylonitrile-*co*-*N*-phenylitaconimide	Single T_g	–	168
	Itaconic anhydride-*co*-methyl methacrylate	Single T_g	Miscibility depended on compositions of I and II	920
	Methyl methacrylate-*co*-*N*-phenylitaconimide	Single T_g	–	168
Acrylonitrile-*co*-methyl methacrylate-*co*-α-methylstyrene	Methyl methacrylate	Single T_g	I had 20/20/60 = acrylonitrile/methyl methacrylate/α-methylstyrene	633
Acrylonitrile-*co*-methyl methacrylate-*co*-styrene	Acrylonitrile-*co*-styrene	Light scattering	–	358
	Methyl methacrylate	Light scattering	–	358
Acrylonitrile-*co*-α-methylstyrene	Acrylonitrile-*co*-methyl methacrylate	Single T_g	–	152
	Acrylonitrile-*co*-styrene	Single T_g	I had 32.3 wt.% acrylonitrile; II had 27 wt.% acrylonitrile	155,684
Acrylonitrile-*co*-α-methylstyrene-*co*-styrene	Acrylonitrile-*co*-α-methyl styrene-*co*-styrene	Clear films	Difference in composition between I and II < 12% acrylonitrile	769
Acrylonitrile-*co*-styrene	Acrylic acid-*co*-styrene	Single T_g	–	380
	Acrylonitrile-*co*-benzyl methacrylate	Single T_g	–	1180
	Acrylonitrile-*co*-fumaronitrile-*co*-styrene	Single T_g	I had 32–40 wt.% AN; II had 11–17 wt.% fumaronitrile	1222
	Acrylonitrile-*co*-methyl methacrylate	Single T_g	–	164,620
	Acrylonitrile-*co*-*N*-phenylitaconimide	Single T_g	–	168
	Acrylonitrile-*co*-styrene	Single T_g; no visible phase separation	Difference in composition between I and II < 3.5% acrylonitrile	485,582
	p-(2-Hydroxy-hexafluoroisopropyl)-styrene-*co*-styrene	Single T_g; clear film	I had 70% styrene; II had 90.3% styrene	571,646
	Maleic anhydride-*co*-styrene	Single T_g	I had 44.4–94.6 wt.% styrene; II had 50–91.5 wt.% styrene	25,462,554, 745,1008

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TABLE 1. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	Monoalkyl itaconate	Single T_g; FTIR	I was 2- or 4-vinylpyridine; II had an alkyl group or methyl or ethyl (4 miscible systems); formed complexes	939,1014, 1212
	Vinyl acetate-*co*-vinyl alcohol	Single T_g; FTIR	I was 2- or 4-vinylpyridine (2 miscible systems); II had 29–88 mol% vinyl alcohol	934,935,991
	Vinyl butyral	SALS	I was 2- or 4-vinylpyridine (2 miscible systems)	1003
2-Vinylpyridine	Aramide	Single T_g; FTIR	II was made from hexamethylene-diamine and a mixture of tereph-thaloyl and isophthaloyl dichlorides	767
4-Vinylpyridine	Methylocellulose	Single T_g; NMR	–	559
	Sulfone, carboxylated	Single T_g; transparency	II had degree of carboxylation 0.43–1.93	256
	p-Phenylene terephthalamide	Transparency; microscopy	–	329
2,3-Xylenyl methacrylate	2,6-Xylenyl methacrylate	Single T_g	–	1158

TABLE 2. POLYMER PAIRS CONTAINING ONE MONOMER IN COMMON, MISCIBLE IN THE AMORPHOUS STATE AT ROOM TEMPERATURE

Polymer I of	Polymer II of	Method	Comments	Refs.
Acrylic acid-*co*-styrene	Methyl methacrylate-*co*-styrene	Single T_g	Miscibility depended on compositions of I and II	72
Acrylonitrile-*co*-butadiene	Acrylonitrile-*co*-butadiene	Single T_g; clear films; electron microscopy	Difference in composition between I and II $\leq$22% acrylonitrile	19,47,101,106, 149,390,452
	Acrylonitrile-*co*-styrene	Single T_g	–	162
	Acrylonitrile-*co*-vinylidene chloride	Single T_g	I had 23.6–49.6 vol% AN; II had 19 wt.% AN	966
Styrene-*co*-maleic anhydride--*co*-acrylonitrile	Acrylonitrile-*co*-styrene	Single T_g	Miscibility depended on compositions of I and II	965
	Maleic anhydride-*co*-styrene	Single T_g	Miscibility depended on compositions of I and II	965
Acrylonitrile-*co*-methyl methacrylate	Acrylonitrile-*co*-*N*-phenylitaconimide	Single T_g	–	168
	Itaconic anhydride-*co*-methyl methacrylate	Single T_g	Miscibility depended on compositions of I and II	920
	Methyl methacrylate-*co*-*N*-phenylitaconimide	Single T_g	–	168
Acrylonitrile-*co*-methyl methacrylate-*co*-α-methylstyrene	Methyl methacrylate	Single T_g	I had 20/20/60 = acrylonitrile/methyl methacrylate/α-methylstyrene	633
Acrylonitrile-*co*-methyl methacrylate-*co*-styrene	Acrylonitrile-*co*-styrene	Light scattering	–	358
	Methyl methacrylate	Light scattering	–	358
Acrylonitrile-*co*-α-methylstyrene	Acrylonitrile-*co*-methyl methacrylate	Single T_g	–	152
	Acrylonitrile-*co*-styrene	Single T_g	I had 32.3 wt.% acrylonitrile; II had 27 wt.% acrylonitrile	155,684
Acrylonitrile-*co*-α-methylstyrene-*co*-styrene	Acrylonitrile-*co*-α-methyl styrene-*co*-styrene	Clear films	Difference in composition between I and II < 12% acrylonitrile	769
Acrylonitrile-*co*-styrene	Acrylic acid-*co*-styrene	Single T_g	–	380
	Acrylonitrile-*co*-benzyl methacrylate	Single T_g	–	1180
	Acrylonitrile-*co*-fumaronitrile-*co*-styrene	Single T_g	I had 32–40 wt.% AN; II had 11–17 wt.% fumaronitrile	1222
	Acrylonitrile-*co*-methyl methacrylate	Single T_g	–	164,620
	Acrylonitrile-*co*-*N*-phenylitaconimide	Single T_g	–	168
	Acrylonitrile-*co*-styrene	Single T_g; no visible phase separation	Difference in composition between I and II < 3.5% acrylonitrile	485,582
	p-(2-Hydroxy-hexafluoroisopropyl)-styrene-*co*-styrene	Single T_g; clear film	I had 70% styrene; II had 90.3% styrene	571,646
	Maleic anhydride-*co*-styrene	Single T_g	I had 44.4–94.6 wt.% styrene; II had 50–91.5 wt.% styrene	25,462,554, 745,1008

TABLE 2. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	N-phenylitaconimide-*co*-styrene	Single T_g	–	164
	N-phenylmaleimide-*co*-styrene	Single T_g	–	25,26
3-Bromo-2,6-dimethyl-1,4-phenylene ether	3-Bromo-2,6-dimethyl-1,4-phenylene ether-*co*-2,6-dimethyl-1,4-phenylene ether	Single T_g; microscopy	II had $\geq$48 mol% 3-bromo-2,6-dimethyl-1,4-phenylene oxide	402
3-Bromo-2,6-dimethyl-1,4-phenylene ether-*co*-2,6-dimethyl-1,4-phenylene ether	2,6-dimethyl-1,4-phenylene ether	Single T_g; microscopy	I had $\geq$62 mol% 2,6-dimethyl-1,4-phenylene ether	402
p-Bromostyrene-*co*-styrene	Styrene	Single T_g	I had $\geq$90 mol% styrene	402,785
Butadiene	Butadiene	Single T_g	Semicrystalline; I was SKD (87% *cis*-1,4) and II was SKBM (Ref. 251); I had 50% vinyl-1,2, 35% *cis*-1,4 and 15% *trans*-1,4, and II had 98% *cis*-1,4 (Ref. 736); I had 50% vinyl-1,2, 35% *cis*-1,4 and 15% *trans*-1,4, and II had 55% *trans*-1,4, 35% *cis*-1,4 and 10% vinyl-1,2 (Ref. 736)	251,736
	Butadiene-*co*-styrene	Single T_g	II had >70% butadiene; immiscible when II had <60% butadiene	92,149,182, 237,238,520,548, 549,737,770,771, 787,882
Butadiene-*co*-styrene	Butadiene-*co*-styrene	Single T_g; microscopy	Difference in composition between I and II <20% styrene; two T_g's when composition difference >20%	182,519,520, 455,757,771
	Styrene	Single dynamic mechanical loss peak	I had 25% styrene; II had $M_w \leq 917$; $\leq$50% II; two loss peaks when 75% II	138,139
Butyl acrylate-*co*-butyl methacrylate	Butyl acrylate-*co*-butyl methacrylate	Clear films	Difference in composition between I and II 25% butyl acrylate; immiscible when composition difference 50% butyl acrylate	453
Butyl acrylate-*co*-*N*-hydroxyethylcarbozolyl acrylate	Butyl acrylate-*co*-β-hydroxyethyl-3,5-dinitrobenzoyl acrylate	Single T_g	I had 6 or 48 mol% electron donating carbozoyl groups; II had 10 or 63 mol% electron withdrawing dinitrobenzoyl groups	723
Butyl acrylate-*co*-ethyl acrylate	Butyl acrylate-co-ethyl acrylate	Clear films	Difference in composition between I and II 25% ethyl acrylate; immiscible when composition difference 50% ethyl acrylate	453
	Ethyl acrylate	Clear films	I had 75% ethyl acrylate; immiscible when I had $\leq$50% ethyl acrylate	453
Butyl acrylate-*co*-methyl methacrylate	Butyl acrylate-*co*-methyl methacrylate	Single T_g; clear films	Difference in composition between I and II $\leq$10% methyl methacrylate	447,448
n-Butyl methacrylate	*n*-Butyl methacrylate-*co*-styrene	Single T_g; clear films	II had $\geq$90 wt.% *n*-butyl methacrylate	238
n-Butyl methacrylate-*co*-*p*-chlorostyrene	*n*-Butyl methacrylate-*co*-*p*-chlorostyrene	Single T_g; microscopy	–	76
n-Butyl methacrylate-*co*-*N*-hydroxyethylcarbazoyl acrylate	*n*-Butyl methacrylate-*co*-β-hydroxyethyl-3,5-dinitrobenzoyl acrylate	Single T_g	I had 9 mol% carbazoyl groups; II had 9 mol% dinitrobenzoyl groups	723
Butyl methacrylate-*co*-methyl methacrylate	Butyl methacrylate-*co*-methyl methacrylate	Clear films	Difference in composition between I and II $\leq$20–30% butyl methacrylate, depending on compositions of I and II	76,447
n-Butyl methacrylate-*co*-styrene	*n*-Butyl methacrylate-*co*-styrene	Single T_g; microscopy		76
	Styrene	Single T_g; clear films	I had $\geq$90 wt.% styrene	239
Caprolactam-*co*-caprolactone	Caprolactone-*co*-laurolactam	Single T_g	I and II had various compositions	1126
Caprolactam-*co*-caprolactone-*co*-laurolactam	Caprolactam-*co*-laurolactam	Single T_g	I and II had various compositions	992
2,6-Dimethyl-1,4-phenylene ether, sulfonylated	2,6-Dimethyl-1,4-phenylene ether, sulfonylated	Single T_g	Difference in degree of sulfonylation between I and II in the range of 20–26 wt.%	410
Dimethylsiloxane, functionalized	Dimethylsiloxane, functionalized	Single T_g	I had 4–15 mol% electron-donating carbazoyl or *N*-methylaniline groups; II had 4–15 mol% electron-accepting	723

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TABLE 2. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
			dinitrobenzoyl groups; donor/acceptor = 1/1	
Epichlorohydrin	Epichlorohydrin-*co*-ethylene glycol	Single T_g	I was Hydrin 100; II was Hydrin 200	373
Ethyl acrylate-*co*-ethyl methacrylate	Ethyl acrylate-*co*-ethyl methacrylate	Clear films	I and II had 25, 50 or 75% ethyl acrylate	453
	Ethyl methacrylate	Clear films	I had 75 wt.% ethyl methacrylate; immiscible when I had $\leq$ 50 wt.% ethyl methacrylate	453
Ethyl acrylate-*co*-methyl methacrylate	Ethyl acrylate-*co*-methyl methacrylate	Clear films	Difference in composition between I and II < 20–27% ethyl acrylate depending on compositions of I and II	447
Ethylene, chlorinated	Ethylene, chlorinated	Single T_g; clear films	Difference in composition between I and II $\leq$ 1.2–13.3 wt.% Cl depending on compositions of I and II	97,98, 636,817
	Ethylene-*co*-methyl acrylate	Single T_g; phase contrast microscopy	Semicrystalline when I had 25 wt.% Cl; I had 25 or 48 wt.% Cl	396
	Ethylene-*co*-vinyl acetate	Single T_g; microscopy	I had 35.4–52.6 wt.% Cl; II had 50–60 wt.% ethylene	849
Ethylene-*co*-vinyl acetate	Ethylene, low density	Single T_g	I:II = 50 : 50	1172
	Ethylene-*co*-vinyl acetate	Single T_g; microscopy	I and II had 7–100 wt.% vinyl acetate	925
	Ethylene-*co*-vinyl alcohol	Single T_g; FTIR	I had 21 mol% vinyl acetate; II had 21 mol% vinyl alcohol	960
	Vinyl acetate-*co*-vinyl chloride	Single T_g	–	745,746
Ethylene-*co*-vinyl chloride	Ethylene-*co*-vinyl chloride	Single T_g	Difference in vinyl chloride or ethylene content of I and II < 15 mol%	70,71
	Vinyl chloride	Single T_g	I had > 80 mol% vinyl chloride	70
Ethylene naphthalene-2,6-dicarboxylate	Ethylene naphthalene-2,6-dicarboxylate-*co*-4-hydroxybenzoate	NMR	II had 20 mol% 4-hydroxybenzoate	1027
Ethyl methacrylate-*co*-methyl methacrylate	Ethyl methacrylate-*co*-methyl methacrylate	Single T_g	–	98
o-Fluorostyrene-*co*-*p*-fluorostyrene	*o*-Fluorostyrene-*co*-*p*-fluorostyrene	Single T_g	I and II had 10 and 23 mol% *p*-fluorostyrene, respectively	1144
o-Fluorostyrene-*co*-styrene	*o*-Fluorostyrene-*co*-styrene	Single T_g	I had 18 or 49% *o*-fluorostyrene; II had 40 or 77% *o*-fluorostyrene	716,1144
	Styrene	Single T_g	I had 18, 40 or 49% *o*-fluorostyrene and II had M_n = 15 kg/mol I had 18% *o*-fluorostyrene and II had M_n = 110 kg/mol	716,1144
p-Fluorostyrene-*co*-styrene	Styrene	Single T_g	I had 8 or 16 mol% *p*-fluorostyrene; II had M_n = 40 or 326 kg/mol	1144
Hydrocarbon polymer, chlorinated	Hydrocarbon polymer, chlorinated	Clear films	One series derived from reduction of polyvinyl chloride and one series derived from chlorination of polyethylene	75
p-Hydroxybenzoic acid-*co*-6-hydroxy-2-naphthoic acid-*co*-terephthalic acid-*co*-hydroquinone	*p*-Hydroxybenzoic acid-*co*-6-hydroxy-2-naphthoic acid	Single T_g; X-ray diffraction	X-ray diffraction showed no significant transesterification occurred in the mixture	92
Hydroxybutyrate	Hydroxybutyrate	Microscopy; SAXS	I was isotactic; II was atactic	903
	Hydroxybutyrate-*co*-hydroxyvalerate	Microscopy; melting behavior	II had 3.8–22.3 mol% hydroxyvalerate	1142,1143
Isoprene	Isoprene-styrene block copolymer	Electron microscopy	I had M_n = 6.5–53.9 kg/mol; II was diblock, triblock or four-arm star copolymer having total M_n = 26–622 kg/mol	378
Maleic anhydride-*co*-styrene	Acrylonitrile-*co*-styrene	Single T_g, transparency, microscopy	I had 15.3 wt.% maleic anhydride; II had 15.0, 19.5 or 25.0 wt.% acrylonitrile	436,555
	Maleic anhydride-*co*-styrene	Single T_g	Miscibility occurred when difference in MA contents between I and II was < 2.5 vol%	1159
	Methyl methacrylate-*co*-styrene	Single T_g	Miscibility depended on compositions of I and II	1008
Maleimide-*co*-styrene	Styrene-*co*-2-vinylpyridine	Single T_g; FTIR	I had 86 wt.% styrene; IH had 30 wt.% styrene	1214

TABLE 2. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
Methacrylic acid-*co*-methyl methacrylate	Methyl methacrylate-*co*-4-vinylpyridine	Phase diagram	I had 91.48 or 77.7 mol% methyl methacrylate; II had 80.2 mol% methyl methacrylate	198
Methacrylic acid-*co*-styrene	Butyl methacrylate-*co*-methacrylic acid	Single T_g	I and II had 13.7 and 20.2 mol% methacrylic acid, respectively; both were Na-neutralized	1051
	Styrene-*co*-4-vinylpyridine	Phase diagram	I had 89.5 mol% styrene; II had 83.9 mol% styrene	198
Methacrylonitrile-*co*-styrene	Methacrylonitrile-*co*-methyl methacrylate	Single T_g	Miscibility depended on compositions of I and II	970
	Methacrylonitrile-*co*-styrene	Single T_g	Miscibility depended on compositions of I and II	970
[4-(Methacryloyloxy)-butyl]-pentamethyldisiloxane	[4-(Methacryloyloxy)-butyl]-pentamethyldisiloxane-*co*-methyl methacrylate	Single T_g; clear films	II had 67% methyl methacrylate	64
[4-(Methacryloyloxy)-butyl]-pentamethyldisiloxane-*co*-methyl methacrylate	Methyl methacrylate	Single T_g; clear films	I had 67 or 75% methyl methacrylate	64
Methyl acrylate	Methyl acrylate-*co*-vinyl acetate	Clear films	II had 25–75% methyl acrylate	453
Methyl acrylate-*co*-methyl methacrylate	Methyl acrylate-*co*-methyl methacrylate	Clear films	Difference in composition between I and II 28–48% methyl acrylate when I and II had degree of polymerization = 400 and 20–35% methyl acrylate when I and II had degree of polymerization = 3000, depending on compositions of I and II	447
Methyl acrylate-*co*-vinyl acetate	Vinyl acetate	Clear films	I had 25–75% vinyl acetate	453
Methyl methacrylate	Ethyl methacrylate-*co*-methyl methacrylate	Single T_g	I was isotactic; II had $\leq$45% ethyl methacrylate	727
	Methyl methacrylate	Single T_g	I was isotactic; II was sydiotactic	727
	Methyl methacrylate-*alt*-styrene	Single T_g; microscopy	Deuterated blends were miscible; miscibility affected by deuteration of I	1006
Methyl methacrylate-*co*-(*N*-ethylcarbazol-3-yl)methyl methacrylate	Methyl methacrylate-*co*-2-[(3,5-dinitrobenzoyl)oxy]ethyl methacrylate	Single T_g	I and II had similar acceptor or donor content	1160
Methyl methacrylate-*co*-styrene	Acrylonitrile-*co*-styrene	Single T_g	–	164,404
	Maleic anhydride-*co*-styrene	Single T_g	–	404
	Methyl methacrylate-*co*-*N*-phenylitaconimide	Single T_g	–	168
	Methyl methacrylate-*co*-styrene	Single T_g; microscopy	–	76,444
	N-Phenylitaconimide-*co*-styrene	Single T_g	–	168
p-Nitrostyrene-*co*-styrene	*p*-Nitrostyrene-*co*-styrene	Single T_g	–	158
Novolac resin	Novolac resin	Single T_g	I was formaldehyde and 13/17/70 mol% = *p*-*tert*-butyl phenol/*m*-cresol/*o*-cresol; II was formaldehyde and 15/85 mol% = 2-*tert*-butyl phenol/*o*-cresol; I and II had MW 0.8–1.6 kg/mol	221
N-Phenylmaleimide-*co*-styrene	4-Cyanostyrene-*co*-styrene	Single T_g	I had 39 wt.% styrene; II had 58.5–83.1 mol% styrene	187
	2-Cyanostyrene-*co*-4-cyanostyrene-*co*-styrene	Single T_g	II had 23.5 or 22.7 mol% styrene; ratios of 2-cyanostyrene/4-cyano-styrene were 25:75 or 78:22	187
Styrene	Styrene	Single T_g	I had MW = 0.8 kg/mol; II had MW = 8420 kg/mol; I/II = 31.7/68.5	102
	Styrene, iodinated	Transparency	II had degree of iodination of 6 or 15%	261
Styrene-*co*-styrenesulfonic acid	Styrene-*co*-*p*-vinylphenol	Single T_g; FTIR	I had 86 mol% styrene; II had 22 mol% vinylphenol	1086
	Styrene-*co*-4-vinylphenol	FTIR	I had 1.8–9.9 mol% acid; II had 1.9–9.9 mol% pyridine	1190
Styrene-*co*-vinylphenol	Styrene-*co*-vinylphenol	FTIR	–	349
	Vinylphenol	Single T_g; FTIR	I had > 11 mol% OH groups	1247,1088
Sulfone	Sulfone, carboxylated	Single T_g	Degree of carboxylation of II $\leq$ 1.3	1258
Vinyl acetate-*co*-vinyl chloride	Vinyl chloride	Single T_g; microscopy of fibers	I had 78 or 86 wt.% vinyl chloride	449
	Vinyl chloride, chlorinated	Single T_g	–	745

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TABLE 2. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
Vinyl acetate-*co*-vinyl sterate	Vinyl acetate-*co*-vinyl sterate	Single T_g; microscopy	I and II had 21–98 wt.% vinyl acetate	925
Vinyl chloride	Propylene-*co*-vinyl chloride	Single T_g; microscopy of fibers	II had 89 or 91 wt.% vinyl chloride	449
Vinyl chloride, chlorinated	Vinyl chloride	Single T_g	Up to 61.3 or 65.2% Cl in I with dependence on mol% CCl_2 groups	91,504,505
	Vinyl chloride, chlorinated	Single T_g	Difference in composition between I and II $\leq$ 3–4% Cl depending on composition of I and II with dependence on number of CCl_2 groups	98,504

TABLE 3. CHEMICALLY DISSIMILAR POLYMER TRIADS (AND TETRADS) MISCIBLE IN THE AMORPHOUS STATE AT ROOM TEMPERATURE

Polymer I of	Polymer II of	Polymer III of (and Polymer IV of)	Method	Comments	Refs.
Acrylic acid-*co*-styrene	Ethylene glycol	Methyl methacrylate	Single T_g	I had 12.2, 20.8 or 33.1 mol% acrylic acid	384
Acrylonitrile-*co*-butadiene	Vinyl chloride	Vinyl chloride-*co*-vinylidene chloride	Single T_g	Semicrystalline; I had 30 or 40 wt.% acrylonitrile; III had 65 wt.% vinylidene chloride	860
Acrylonitrile-*co*-butadiene-*co*-styrene	Carbonate of bisphenol-A	Propylene-*co*-vinyl chloride	Single T_g	I was Blendex 701; II was Merlon M39, M-60; III had 3.2 or 3.8% propylene; I/II/III = 1/1/2; two T_g's when I/II/III = 3/7/10	107
Acrylonitrile-*co*-methyl methacrylate	Acrylonitrile-co-styrene	Maleic anhydride-*co*-styrene	Single T_g	I had 12% acrylonitrile; II had 25% acrylonitrile; III had 25% maleic anhydride	74
	Acrylonitrile-*co*-styrene	Methyl methacrylate-*co*-*N*-phenyl-itaconimide	Single T_g	I had 68 or 95 vol% MMA; II had 63 or 70 vol% styrene; III had 12 or 45 vol% MMA; three miscible binary pairs; there was an immiscibility loop	967
	Acrylonitrile-*co*-styrene	Methyl methacrylate-*co*-styrene	Single T_g	I had 2.7–7.4 vol% acrylo-nitrile; II had 14.5–49.5 vol% acrylonitrile; III had 11.2–16.0 vol% styrene	163,165
Acrylonitrile-*co*-α-methylstyrene	Acrylonitrile-*co*-styrene	Vinyl chloride	Single T_g	I had 32.3 wt.% acrylonitrile; II had 27 wt.% acrylonitrile; >25% I	684
	Ethyl methacrylate	Methyl methacrylate	Single T_g	I had 30 wt.% acrylonitrile	290
Acrylonitrile-*co*-*N*-phenylitaconimide	Acrylonitrile-*co*-styrene	*N*-phenylitaconimide-*co*-styrene	Single T_g	I had 6 or 9 vol% acrylonitrile; II had 22, 44 or 65 vol% acrylonitrile; III had 12 or 42 vol% styrene	157
Acrylonitrile-*co*-styrene	Butylene adipate	Carbonate of bisphenol-A	Single T_g	I had 12 or 25% acrylonitrile	738
	Caprolactone	Carbonate of bisphenol-A	Single T_g	–	738
	Caprolactone	Hydroxyether of bisphenol-A (phenoxy resin)	Single T_g	I had 5.7–30 wt.% AN	1209,1052
	Caprolactone	Maleic anhydride-*co*-styrene	Single T_g	I had 15 wt.% AN; III had 14 wt.% maleic anhydride	1210
	Caprolactone	Vinyl chloride	Single T_g	Miscible when I/II/III = 68/16/16	351
	Carbonate of bisphenol-A	Cyclohexane dimethylene succinate	Single T_g	–	738
	Ethyl methacrylate	Maleic anhydride-*co*-styrene	Single T_g	I had 25% acrylonitrile; III had 14–25% acrylonitrile	74
	Ethyl methacrylate	Methyl methacrylate	Single T_g	I had 30 wt.% acrylonitrile; some blends showed LCST	280
	Methyl methacrylate	Vinyl chloride	Single T_g	I had 23 wt.% acrylonitrile	349

TABLE 3. *cont'd*

Polymer I of	Polymer II of	Polymer III of (and Polymer IV of)	Method	Comments	Refs.
n-Amyl methacrylate	*n*-Butyl methacrylate	Vinyl chloride	Single T_g	Blend had >70 wt.% III	1157
	n-Propyl methacrylate	Vinyl chloride	Single T_g	Blend had >70 wt.% III	1157
Arylate	Butylene therphthalate	Hydroxyether of bisphenol-A (phenoxy)	Single T_g	I was Arilef U-100; III >30 wt.%	7,208,366
3-Bromo-2,6-dimethyl-1,4-phenylene ether-*co*-2,6-dimethyl-1,4-phenylene ether	*p*-Bromostyrene-*co*-styrene	2,6-dimethyl-1,4-phenylene ether	Single T_g; microscopy	≤37 mol% brominated monomer in I and in II	402
	p-Bromostyrene-*co*-styrene	Styrene	Single T_g; microscopy	≤16 mol% brominated monomer in I and in II	402
	2,6-Dimethyl-1,4-phenylene ether	Styrene	Single T_g; microscopy	≤45 mol% brominated monomer in I	402
	p-Bromostyrene-*co*-styrene	2,6-Dimethyl-1,4-phenylene ether (polymer IV of styrene)	Single T_g; microscopy	≤26 mol% brominated monomer in I and in II	402
p-Bromostyrene-*co*-styrene	2,6-dimethyl-1,4-phenylene ether	Styrene	Single T_g; microscopy	≤29 mol% *p*-bromostyrene in I	402
Butylene terephthalate	Carbonate of bisphenol-A	Hydroxyether of bisphenol-A (phenoxy)	Single T_g	–	608,681
	Hydroxyether of bisphenol-A (Phenoxy)	Methyl methacrylate	Single T_g, microscopy	Blend had >50 wt.% II	1053
Caprolactone	Carbonate of bisphenol-A	Carbonate of tetramethylbisphenol-A	Single T_g	Miscible at all compositions	1070
	Carbonate of bisphenol-A	Hydroxyether of bisphenol-A (phenoxy)	Single T_g	Generally miscible when >60 wt.% I in blend	125
	Hydroxyether of bisphenol-A (Phenoxy)	Vinyl methyl ether	Single T_g	Miscible at all compositions; all blends showed LCST	303
	Vinyl chloride	Vinyl chloride, chlorinated	Single T_g	II had 67.2 wt.% Cl; generally miscible when I >40% at high II/III ratio and miscible when I >26% at low II/III ratio	20
Carbonate of bisphenol-A	Carbonate of bisphenol chloral	Methyl methacrylate	Single T_g	–	427
	Carbonate of tetramethylbisphenol-A	Acrylonitrile-*co*-styrene	Single T_g	III had 2.7–19.7 wt.% AN	1068
	Carbonate of tetramethylbisphenol-A	Methyl methacrylate-*co*-styrene	Single T_g	III had 4.5–58.5 wt.% MMA	1068
	Carbonate of tetramethylbisphenol-A	Styrene	Single T_g	–	488,1068
Carbonate of tetramethylbisphenol-A	Methyl methacrylate	Methyl methacrylate-*co*-styrene	Single T_g	–	884
2-Chlorostyrene	Cyclohexyl acrylate	Styrene	Single T_g; light scattering	Three miscible binary pairs; there was an immiscibility loop in the phase diagram	1167
Copolyester	Epoxy	Hydroxyether of bisphenol-A (phenoxy)	Single T_g, microscopy	II/III = 1/1; I was made from 65/30/5 = terephthalate/isophthalate/sebacate and 70/20/10 = ethylene glycol/resorcinal di(β-hydroxy ethyl)ether/poly(tetramethylene ether)glycol; II was 4/1 by wt. = Araldite 6099/Araldite 6010	2
Epichlorohydrin	Ethylene glycol	Methyl methacrylate	Single T_g	Miscible at all compositions	573
	Methyl methacrylate	Vinyl acetate	Single T_g	Miscible at all compositions; all blends showed LCST	304
Etherimide	Amideimide	Ether ether ketone, sulfonated	Single T_g	I was Ultem 1000; II was Torlon 4000T; III had degree of sulfonation of 0.53 or 1.00	415
	Ether ether ketone	Liquid crystalline polymer	Single T_g	I was Ultem 1000; II was Victrex 450G; III was HX4000	78
Ether urethane	Oxytetramethylene-*co*-oxyditetramethylene-*co*-terephthalic acid	Vinyl chloride	Single T_g	I was Adiprene L-100 and 4,4′-methylene-bis-2-chloroaniline: I/II/III = 1/1/2; two T_g's when I/II/III = 1/2/1 and 1/1/1	344

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TABLE 3. *cont'd*

Polymer I of	Polymer II of	Polymer III of (and Polymer IV of)	Method	Comments	Refs.
Ethyl methacrylate	Methyl methacrylate	*p*-Vinylphenol	Single T_g; FTIR; microscopy	Blend had ≥ 60 wt.% III	1161,1163
	Methyl methacrylate	Vinylidene fluoride	Single T_g	Blend had 30–80 wt.% III	474
Ethylene, chlorinated	Natural rubber, epoxidized	Vinyl chloride	Single T_g	II was 50 mol% epoxidized	394
Ethylene, chlorosulfonated	Natural rubber, epoxidized	Nitrile rubber, carboxylated	Single T_g	I had 35% Cl and 1% S (Hypalon-40); II was 50 mol% epoxidized; III was Krynac-211	702
Ethylene glycol	Vinyl acetate	*p*-Vinyl phenol	Single T_g; FTIR	Three miscible binary pairs; there was an immiscibility loop in the phase diagram	491,1240
Hexafluoroacetone-*co*-vinylidene fluoride	Methyl methacrylate	Vinyl acetate	Single T_g	I had 9 mol% hexafluoro-acetone	1073
Methyl acrylate	Vinyl acetate	*p*-Vinylphenol	Single T_g	Miscible at all compositions	1240
Methyl acrylate-*co*-styrene	Vinyl acetate	*p*-Vinylphenol	Single T_g	I had 28 wt.% styrene	1240
Methyl methacrylate	Vinyl acetate	Vinylidene fluoride	Single T_g	–	956
Natural rubber, epoxidized	Nitrile rubber, carboxylated	Chloroprene	Single T_g	I was 50 mol% epoxidized; II was Krynac-211; I/II/III = 1/1/1	14
Neopentyl glycol adipate	Vinyl chloride	Vinylidene fluoride	Single T_g	III < 50 wt.%	490
Styrene-*co*-4-vinylpyridine	Vinyl acetate	*p*-Vinylphenol	Single T_g	I had 30 wt.% styrene	1240

TABLE 4. POLYMER PAIRS MISCIBLE IN THE AMORPHOUS STATE AT ROOM TEMPERATURE, MOLECULAR WEIGHT DEPENDENCE INVESTIGATED

Polymer I of	Polymer II of	Method	Comments	Refs.
Acrylonitrile-*co*-α-methylstyrene	Carbonate of bisphenol-A	Single T_g; transparency	I had 9–12 wt.% AN; miscible when II had $M_w = 3.8$ kg/mol; immiscible when II had $M_w = 9.9$ kg/mol	930
Acrylonitrile-*co*-styrene	Carbonate of bisphenol-A	Single T_g; transparency	I had 24.6 wt.% AN and II had $M_w = 3.8$ kg/mol; miscible when I had $M_w = 3.137$ kg/mol and immiscible with I had $M_w = 6.085$ kg/mol	932
3-Bromo-2,6-dimethyl-1,4-phenylene ether	Styrene	Single T_g; transparency	I had MW ≤ 40 kg/mol; II had MW ≤ 30 kg/mol	529
n-Butyl methacrylate	Styrene	Single T_g; transparency	I had $M_w = 320$ kg/mol: II had MW = 171 kg/mol; two T_g's when II had MW = 1100 kg/mol	193
Carbonate of bisphenol-A	Ester	Single T_g	II was made from 4,4′-(2-norbornyl-idene)di-phenol and a mixture of terephthalic acid and azelaic acid; miscible with I had $M_w = 28.1$ kg/mol; immiscible when I had $M_w = 73.2$ kg/mol	1232
2,6-Dimethyl-1,4-phenylene ether	Alkylstyrene	Transparency	I had $M_w = 5$–500 kg/mol and II was a ring-alkyl (with various carbon atoms) styrene. Raising MW of II eventually produced a transition from clear to hazy or cloudy films in each system except that containing poly-α-methylstyrene. With this resin, clear single-T_g films were obtained even with M_w of I = 500 kg/mol and M_w of II = 800 kg/mol	403
	p-Bromostyrene	Transparency	II had $M_n = 29.1$ kg/mol and $M_w = 32$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 6.2 ± 2.4 kg/mol. When II had $M_n = 6.4$ kg/mol and $M_w = 7.04$ kg/mol, M_w limit for miscibility of I was 14 ± 4 kg/mol	403
	p-tert-Butylstyrene	Transparency	II had $M_n = 113$ kg/mol and $M_w = 289$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 4.5 kg/mol ± 0.7 kg/mol	403

TABLE 4. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
	4-Ethylstyrene	Transparency	II had $M_n = 139$ kg/mol and $M_w = 158$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 22.4 kg/mol	403
	4-Methoxystyrene	Transparency	II had $M_n = 282$ kg/mol and $M_w = 457$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 3.6 kg/mol	403
	4-Methylstyrene	Transparency	II had $M_n = 118$ kg/mol and $M_w = 265$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 35.5 kg/mol	403
	4-*n*-Propylstyrene	Transparency	II had $M_n = 128$ kg/mol and $M_w = 267$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 14 ± 4 kg/mol	403
	Vinylbenzylchloride	Transparency	II had $M_n = 16.8$ kg/mol and $M_w = 29.3$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 14 ± 4 kg/mol	403
	4-Vinylbiphenyl	Transparency	II had $M_n = 79$ kg/mol and $M_w = 153$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 7 ± 1.6 kg/mol	403
	Vinylnaphthalene	Transparency	II had $M_n = 13$ kg/mol and $M_w = 32.6$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 10.2 kg/mol	403
	Vinyltoluene	Transparency	II had $M_n = 20.7$ kg/mol and $M_w = 40.4$ kg/mol; M_w limit for miscibility in 50/50 blend of I with II was 69 ± 25 kg/mol	403
Ethylene	Dimethylsiloxane	Cloud point curve	I was oligomeric polyethylene with $M_w = 0.254$ to 2.234 kg/mol and II was oligomeric polydimethylsiloxane with $M_w = 0.505$ to 4.5 kg/mol	584
Ethylene glycol	Propylene glycol	Miscibility stated	I and II both had MW < 1 kg/mol	65
	Vinyl acetate	Single T_g	I had $M_w = 20$ kg/mol miscible with II; I had $M_w = 100$ kg/mol miscible with II when blend had $\leq 50\%$ II	328
	Sulfone	Scattering turbidimetry	Blends with II > 40 wt.% were homogeneous when I had MW = 3.5–4.5 kg/mol and II had MW = 35.6 kg/mol. Blends were heterogeneous at all compositions when I had MW = 200 kg/mol	796
Hydroxybutyrate	Lactide	Microscopy	Miscible when II had $M_n = 1.759$ kg/mol; immiscible when II had $M_n = 159.4$ kg/mol	923
cis-Isoprene	Styrene	Single dynamic mechanical loss peak	I was natural rubber; II had $M_w \leq 0.375$ kg/mol; two peaks when II had $M_w \geq 0.6$ kg/mol	140
	Vinyl cyclohexane	Single dynamic mechanical loss peak	I was natural rubber; II had $M_w \leq 0.375$ kg/mol; two peaks when II had $M_w \geq 0.65$ kg/mol	140
Methyl methacrylate	Carbonate of bisphenol-A	Single T_g; transparency	Miscible when M_w of I/II were 2.4 kg/mol/ 38 kg/mol, 10.55 kg/mol, 38 kg/mol, 60 kg/mol/9.9 kg/mol, 105 kg/mol;/9.9 kg/mol; immiscible when M_w of I/II were 20.3 kg/mol;/38 kg/mol, 33.5 kg/mol;/25.9 kg/mol, 33.5 kg/mol;/38 kg/mol and 60 kg/mol;/38 kg/mol	930
	Carbonate of hexafluorobisphenol-A-*co*-carbonate of tetramethyl-bisphenol-A	Single T_g; transparency	II had 60 wt.% HFPC and $M_w = 33$ kg/mol, and miscible when I had M_w = 4–9.9 kg/mol; when II had M_w = 103 kg/mol, it was miscible with I having $M_w = 4$ kg/mol and 11 kg/mol only	1194
	Styrene	Single T_g, transparency	Miscible when I had $M_w = 2.4$ kg/mol and II had $M_w = 2.95$ kg/mol; blends showed UCST when I and II had higher M_w (Ref. 929)	640,804, 929
	α-Styrene	Single T_g; transparency	Immiscible when I had $M_w = 6.7$ kg/mol and II had $M_w = 105.5$ kg/mol; miscible when I and II had lower M_w	929
	Sulfone	Single T_g; transparency	Miscible when I had $M_w = 1.21$ kg/mol or and 2.4 kg/mol and II had $M_w = 12$ kg/mol; immiscible when I had $M_w = 4.25$ kg/mol and II had $M_w = 34$ or 44 kg/mol	931

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TABLE 4. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
Methyl methacrylate-*co*-styrene	Styrene	Single T_g; transparency	I had 26% styrene and II had MW = 0.6 kg/mol or I had 62.6% styrene and II had MW ≤ 2.1 kg/mol or I had 80% styrene and II had MW ≤ 4 kg/mol	557,804
α-Methylstyrene	Styrene	Single T_g; transparency; neutron scattering	Miscible when I had M_w = 55 kg/mol and II had M_w = 22 or 52 kg/mol; immiscible when II had M_w = 341 kg/mol (Ref.929)	38,42,63, 159,203, 691,710, 721,822, 865,929
	Styrene, deuterated	Single T_g	I had M_n = 25.8–78.3 kg/mol; II had M_n = 36.1–84.1 kg/mol	671
α-Methylstyrene-*co*-styrene	Styrene	Single T_g	I had M_n = 59–818 kg/mol; II had 3–79 mol% α-methylstyrene	159
Propylene	Ethylene-*co*-propylene	Single T_g	Miscible when I had M_w = 33 kg/mol and II had M_w = 44.5 kg/mol; immiscible when I and II had higher M_w	1101
Styrene	Carbonate of bisphenol-A	Single T_g; transparency	I had M_w = 0.58, 0.68, 0.95 or II had M_w = 3.8, 9, 25.9 or 38 kg/mol; miscibility depended on M_w and composition of blend	930
	Carbonate of dimethylbisphenol-A	Single T_g; transparency	II had M_w = 30 kg/mol; miscible when I had M_w = 0.58, 0.68 or 0.95 kg/mol; immiscible when I had M_w = 2.95 kg/mol	930
	Carbonate of tetramethylbisphenol	Single T_g; transparency	II had M_w = 31 kg/mol; miscible when I had M_w = 2.95 kg/mol; partially miscible when I had M_w = 9.2 or 17.5 kg/mol; immiscible when I had M_w = 341 kg/mol	930
	o-Chlorostyrene-*co*-*p*-chlorostyrene	Single T_g	I had M_n = 2.2–93 kg/mol; II had 7–100 mol% *o*-chlorostyrene	131
	Vinyl methyl ether	Single T_g	I had mol. wt. 0.8–233 kg/mol; II had mol. wt. 1.0–97.5 kg/mol	878,724
	2-Vinyl naphthalene	Single T_g	I had MW = 2.2 kg/mol; II had MW ≤ 70 kg/mol	731
p-Vinylphenol	Diethyl itaconate	Single T_g	II had M_w = 9.4 kg/mol and was miscible with I having M_w = 1.5–7 kg/mol, 9–11 kg/mol;, 22 or 30 kg/mol; II had M_w = 61 kg/mol and was miscible with I having M_w = 1.5–7 kg/mol and 9–11 kg/mol	1034

TABLE 5. POLYMER PAIRS THAT APPEAR TO HAVE HIGH TEMPERATURE MISCIBILITY, ALTHOUGH IMMISCIBLE AT OR BELOW ROOM TEMPERATURE (UCST BEHAVIOR)

Polymer I of	Polymer II of	Comments	References
Acrylonitrile-*co*-styrene	Sulfone of tetramethylbisphenol-A	I had 9 wt.% AN and II had M_w = 4.8 kg/mol; I had 11 wt.% AN and II had M_w = 12.4 kg/mol	931
Aramide	Imide	I was Aramid 34I; II was Polyimide UR	1136
Butadiene	Butadiene	I had 98% *cis*-1,4; II had 71% vinyl-1,2 and 19% *trans*-1,4	736
	Butadiene, perdeuterated	Both I and II were *cis*-1,4	49
	Butadiene-*co*-styrene	I was *cis*-1,4; II had 23–30 wt.% styrene	362,650,737, 797,1121
	Styrene	II was hydrogenated or partly deuterated, I and II had MW = 2.0 kg/mol (Ref. 516); I had M_w = 0.96 kg/mol and II had M_w = 9.0 kg/mol (Ref. 808)	516,808
	Terpene resin, hydrogenated	I had 10–70.4% 1,2-content	1062
Butadiene-*co*-styrene	Styrene	II was hydrogenated or partly deuterated; II was 68.4 wt.% styrene and had MW of 10 kg/mol and II had MW of 100 kg/mol	516

TABLE 5. *cont'd*

Polymer I of	Polymer II of	Comments	References
Caprolactone	Ester	II was made from adipoyl chloride and 4,4′-bis(6-hydro-hexyloxy)biphenyl	821
Carbonate of bisphenol-A	Methacrylic acid-*co*-styrene	II had 4–23 wt.% methacrylic acid	10
	Liquid crystalline polymer of biphenyl-4,4′-ylene sebacate	Blend had 50–80 wt.% II	387
	Segmented copolyetherester	I had a hard tetramethylene terephthalate segment and a soft polytetramethylene-ether-glycol terephthalate segment	1015
Chlorostyrene-*co*-styrene	2,6-dimethyl-1,4-phenylene oxide	I had 32.9 mol% styrene	234,863
Cyclohexyl methacrylate	Cyclohexyl acrylate	–	765
2,2-Dimethyl-1,3-propylene sebacate	Haloalkyl methacrylate	II had a haloalkyl group of 2-bromoethyl, 2-chloroethyl, 3-chloropropyl or 2-iodoethyl (4 blend systems)	611,1137,1111
Dimethylsiloxane	Methyl ethyl siloxane	–	341
	Phenyl methyl siloxane	–	469
Epoxy	Acrylonitrile-*co*-butadiene, carboxyl terminated	I was Epon 828, $M_n = 0.38$ kg/mol; II had 17 wt.% acrylonitrile, $M_n = 3.5$ kg/mol	492
	2,6-Dimethyl-1,4-phenylene oxide	II was diglycidyl ether of bisphenol-A	1148
Ether ether sulfone	Ether ether ketone-*co*-ether ether sulfone	II had 50% ether ether sulfone	740
Etherimide	Ethylene terephthalate-*co*-ethylene naphthalate	The higher the ethylene naphthalate content in II, the lower the UCST	1072
Ethyl methacrylate	Ethyl acrylate	–	766
Ethylene	Propylene	II was atactic; no UCST when II was isotactic; from high temperature phase diagrams	452
Ethylene, chlorinated	Ethylene, chlorinated	Cl contents = 40-50 wt.%	816
Hemicellulose	Lignin	Blend had 10, 80 or 90 vol% of II	1188
Isophthalamide-6I (iminoiso-phthaloyl-iminohexamethylene)	Hexamethylene adipamide (nylon 6,6)	Interchange reaction slow	899
Isoprene	Vinylethylene	II was syndiotactic and had 92% vinyl-1,2-	363
Maleic anhydride-*co*-styrene	Sulfone of tetramethylbisphenol-A	I had 14 wt.% maleic anhydride; II had $M_w = 4.8$ kg/mol	931
Methyl methacrylate	Carbonate of dimethylbisphenol-A	I had $M_w = 1.21$ or 2.4 kg/mol; II had $M_w = 33$ kg/mol	930
	Carbonate of tetrachlorobisphenol-A	I had $M_w = 1.21$ kg/mol; II had $M_w = 41$ kg/mol	930
	Carbonate of tetramethylbisphenol-P	I had $M_w = 1.21$ or 2.4 kg/mol; II had $M_w = 31$ kg/mol	930
	Hydroxybutyrate	–	1189
	Styrene	I had $M_w = 2.03$–6.0 kg/mol; II had $M_w =$ 1.95–9.2 kg/mol	558,929,1146
	Sulfone of tetramethylbisphenol-A	I had $M_w = 1.21$ or 2.4 kg/mol; II had $M_w = 4.8$ or 12.4 kg/mol	931
	Sulfone of tetramethylhexafluoro-bisphenol-A	I had $M_w = 1.21$ kg/mol; II had $M_w = 40$ kg/mol	931
	Sulfone of tetramethylbisphenol-P	I had $M_w = 1.21$ kg/mol; II had $M_w = 40$ kg/mol	931
	Vinyl chloride	–	674
1,3-Phenylene adipamide	Hexamethylene adipamide (nylon 6,6)	I was made from *m*-phenylenediamine and adipic acid; interchange reaction slow	899
Styrene	Caprolactone	I had $M_n = 0.95$ kg/mol	629
	Carbonate of bisphenol chloral	I had $M_w = 0.58$ or 0.68 kg/mol; II had $M_w = 30$ kg/mol	930
	α-Methylstyrene	UCST increased with molecular weight	514
	o-Methylstyrene	–	24
	p-Methylstyrene	–	24,786
	m-Methylstyrene	–	24
	Styrene, brominated	II had 27 or 29 mol% brominated units	784
	Styrene, deuterated	–	48,879
	Sulfone of tetramethylbisphenol-A	I had $M_w = 1.21$ or 2.4 kg/mol; II had $M_w = 4.8$ or 12.4 kg/mol	931
	Sulfone of tetramethylbisphenol-P	I had $M_w = 1.21$ kg/mol; II had $M_w = 40$ kg/mol	931
	Sulfone of tetramethylhexafluorobisphenol-A	I had $M_w = 1.21$ kg/mol; II had $M_w = 40$ kg/mol	931
	Vinyl methyl ether	By extrapolation of solution data	168
Vinyl chloride	Acrylonitrile-*co*-methyl methacrylate-*co*-α-methylstyrene	II had 32.3 wt.% AN, 8.1 wt.% MMA and 59.6 wt.% α-MS; blend had 50–80 wt.% II	1078
Vinylidene fluoride	Methyl methacrylate	–	809
Vinylidene fluoride-*co*-tetrafluoroethylene	Vinylidene fluoride-*co*-hexafluoroacetone	I had 80 mol% vinylidene fluoride; II had 92 mol% vinylidene fluoride	951

TABLE 6. POLYMER PAIRS MISCIBLE AT ROOM TEMPERATURE THAT APPEAR TO HAVE A LOWER CRITICAL SOLUTION TEMPERATURE (LCST) ABOVE ROOM TEMPERATURE

Polymer I of	Polymer II of	Comments	Refs.
Acrylic acid, Li-neutralized	Ethylene oxide	0.1–0.4 mole fraction of I was neutralized	1115
Acrylic acid-*co*-styrene	Ethyl methacrylate	I had 8.84 wt.% acrylic acid	72
	Methyl methacrylate	I had 2.65–20.0 wt.% acrylic acid	384,856
	n-Propyl methacrylate	I had 2.65–8.84 wt.% acrylic acid	856
Acrylonitrile-*co*-butadiene-*co*-methyl acrylate	Ethylene-*alt*-maleic anhydride	I was Barex 210	1156
Acrylonitrile-*co*-α-methyl styrene	Acetonyl methacrylate	I had 30 wt.% acrylonitrile (AN)	266
	n-Butyl methacrylate	I had 12 wt.% AN	152
	n-Butyl methacrylate-*co*-methyl methacrylate	I had 30 wt.% AN; II had 70 wt.% methyl methacrylate	289
	Carbonate of tetramethylbisphenol-A	I had 4–16.5 wt.% AN	1010
	Ethyl methacrylate	I had 30 wt.% AN (Ref. 277); II had 10–28 wt.% AN (Ref. 152)	152,277
	Ethyl methacrylate-*co*-methyl methacrylate	I had 30 wt.% AN; II had 60 wt.% methyl methacrylate	289
	Methoxymethyl methacrylate	I had 30 wt.% AN	266
	Methyl methacrylate	I had 30 wt.% AN and II was atactic or isotactic (Ref. 277); I had 31.3, 33.1, 34.8 or 41.3 wt.% AN (Ref. 790); I had 10–37 wt.% AN (Ref. 152)	152,277,790, 1010
	Methyl methacrylate-*co*-2,2,6,6-tetramethyl-piperidinyl methacrylate	I had 30 wt.% AN; II had $\geq$ 76.2 wt.% methyl methacrylate)	272,274
	Methylthiomethyl methacrylate	I had 30 wt.% AN	268
	n-Propyl methacrylate	I had 17–20 wt.% AN	152
	Tetrahydrofurfuryl methacrylate	I had 30 wt.% AN	281
	Vinyl chloride	I had 11.9–30 wt.% AN	1010
Acrylonitrile-*co*-*p*-methylstyrene	Acetonyl methacrylate	I had 43.0–61.0 wt.% AN	121
	n-Butyl methacrylate	I had 13.6 wt.% AN	285
	Cyclohexyl methacrylate	I had 15.8–21.3 wt.% AN	120
	Ethyl methacrylate	I had 26.5–32.3 wt.% AN	285
	n-Propyl methacrylate	I had 21.3–26.5 wt.% AN	285
	Tetrahydrofurfuryl methacrylate	I had 32.3–46.5 wt.% AN	285
	Tetrahydropyranyl-2-methyl methacrylate	I had 23.8–26.5 wt.% AN	119
Acrylonitrile-*co*-styrene	Acetonyl methacrylate	I had 33–58 wt.% AN	121
	Acrylonitrile-*co*-methyl methacrylate	Some blends showed LCST	620
	Butyl methacrylate-*co*-methyl methacrylate	Some blends showed LCST	405
	Caprolactone	I had 24–28 wt.% AN	565,1145
	Copolycarbonate of bisphenol-A and tetramethylbisphenol-A	Some blends showed LCST, depending on compositions of I and II	431
	Cyclohexyl methacrylate	I had 13.4–19.8 wt.% AN	120
	2,6-Dimethyl-1,4-phenylene ether, benzoylated	I and 18 mol% AN; II had 35% benzoylated unit	776
	Ethyl methacrylate	I had 5.5–28 wt.% AN	229,489
	Maleic anhydride-*co*-styrene	Some blends showed LCST	436,462,555
	Methoxymethyl methacrylate	I had 30 wt.% AN	258
	Methyl methacrylate	I had 9.4, 9.8, 27.0, 28.2, 30.5 or 34.4 wt.% AN (Ref. 506); I had 29.4–39.3 wt.% AN (Ref. 164)	56,164, 460,506,566
	Methyl methacrylate-*co*-2,2,6,6-tetramethyl-piperidinyl methacrylate	I had 22 or 30 wt.% AN; II had 69, 74 or 85.5 wt.% methyl methacrylate	272,274
	Methyl methacrylate-*co*-*N*-phenylmaleimide	I had 24.6 wt.% AN; II had 23.5 wt.% *N*-phenylmaleimide	185
	Methylthiomethyl methacrylate	I had 34.5 wt.% AN	269
	Phenyl acrylate	I had 14.6, 29.9 or 34.1 wt.% AN	717
	n-Propyl methacrylate	I had 5.7–19.5 wt.% AN	229,1090
	Tetrahydrofurfuryl methacrylate	I had 34.5–46.6 wt.% AN	286
	Tetrahydropyranyl-2-methyl methacrylate	I had 2.2 or 22–30.5 wt.% AN	119
	Vinyl chloride	I had 11.5–26 wt.% AN	437
Aniline, *N*-octadecylated	Ethylene-*co*-vinyl acetate	I was 60.2% alkylated; II had 20 wt.% vinyl acetate	1246
Aramide	Caprolactam-*co*-laurolactam	I was Trogmid T; II had 43.8 mol% caprolactam	213
Arylsulfone	Imide	I was Radel A; II was XU 218	375
2-Bromoethyl methacrylate	Vinyl acetate	Blends cast from methyl ethyl ketone	610
Butadiene	Butadiene	I had 50% vinyl-1,2, 35% *cis*-1,4 and 15% *trans*-1,4; II had 55% *trans*-1,4, 35% *cis*-1,4 and 10% vinyl-1,2	736
	Isoprene	I had $\leq$ 59% 1,4 content and II was *cis*-1,4 isomer (Ref. 810); I was deuterated and II was protonated (Ref. 335)	335,810
Butyl acrylate	Ethylene, chlorinated	–	100
	Vinyl chloride	–	655,845,850,851

TABLE 6. *cont'd*

Polymer I of	Polymer II of	Comments	Refs.
Butyl methacrylate	*p*-(2-Hydroxy-hexafluoroisopropyl)styrene-*co*-styrene	II had 90.3–90.8 mol% styrene	646
N-butyl-3-hydroxymethyl-carbazoyl methacrylate	ω-Hydroxybutyl-3,5-dinitrobenzoyl methacrylate	–	298
p-tert-Butylstyrene	Cyclohexyl acrylate	–	765
	Cyclohexyl methacrylate	–	765
Caprolactone	Ethylene, chlorinated	II had 30 or 42.1 wt.% Cl	53,191
	Vinylidene fluoride	Blends rich in I showed LCST	386
Carbon monoxide-*co*-ethyl acrylate-*co*-ethylene	Vinyl chloride	I had 13.8/7.41/78.8 = carbon monoxide/ethyl acrylate/ethylene	692
Carbonate of bisphenol-A	Caprolactone	–	56,177,645
	Cyclohexyl methacrylate-*co*-methyl methacrylate	II had $\leq$ 3.3 wt.% cyclohexyl methacrylate (Ref. 574); II had 14.7–34.3 wt.% cyclohexyl methacrylate (Ref. 81)	81,574
	Methyl methacrylate	Blends prepared by precipitation of tetrahydrofuran solutions by *n*-heptane; or by solution casting from tetrahydrofuran at high temperature	109,481,486
	Methyl methacrylate-*co*-*p*-methylstyrene	II had $\geq$ 70 wt.% methyl methacrylate	357
	Methyl methacrylate-*co*-*N*-phenylmaleimide	II had $\geq$ 70 wt.% methyl methacrylate	357
	Methyl methacrylate-*co*-phenyl methacrylate	II had 16–63 wt.% phenyl methacrylate	81
	Methyl methacrylate-*co*-styrene	II had $\leq$ 14 wt.% styrene	577
Carbonate of hexafluorobisphenol-A	Alkyl methacrylate	II had an alkyl group of ethyl, *n*-propyl or isopropyl (3 blend systems)	427,430
Carbonate of tetramethylbisphenol-A	Acrylonitrile-*co*-styrene	II had 2 or 11.5% acrylonitrile	225
	Methyl methacrylate-*co*-styrene	II had 4.5, 32.5, 33.5 or 35 wt.% methyl methacrylate (Ref. 575); II had 4.5–33.5 wt.% methyl methacrylate (Ref. 429)	429,575,884
	Styrene	I had M_w = 341 kg/mol; II had M_w = 33 kg/mol	359,428,741,930
Carbonate of tetramethylbisphenol-P	Styrene	I had M_w = 9.2 kg/mol; II had M_w = 31 kg/mol	930
Caprolactone	Styrene-*co*-*p*-vinylphenol	II had 10 wt.% vinylphenol	905
	Vinyl methyl ether	–	1145
Cellulose acetate	4-Vinylpyridine	I was 10H/2 glucose	86
1-Chloroethyl methacrylate	Alkyl methacrylate	II had an alkyl group of ethyl, *n*-propyl, isopropyl or *n*-butyl (4 blend systems)	1152
2-Chloroethyl methacrylate	Ethyl methacrylate	–	612
Chloromethyl methacrylate	Alkyl methacrylate	II had an alkyl group of ethyl, isopropyl or *n*-propyl (3 blend systems)	273
o-Chlorostyrene	Styrene	–	15
o-Chlorostyrene-*co*-*p*-chlorostyrene	Styrene	I had 71–92 mol% *o*-chlorostyrene	803
2,2-Dichloroethyl methacrylate	Alkyl methacrylate	II had an alkyl group of *n*-propyl, isopropyl, *n*-butyl or isoamyl (4 blend systems)	1110
2,6-Dimethyl-1,4-phenylene ether	Acrylic acid-*co*-styrene	II had 6.3 or 8.5 wt.% acrylic acid	1022
	Acrylonitrile-*co*-styrene	II had 6.3 wt.% acrylonitrile	1022
	o-Chlorostyrene-*co*-*p*-chlorostyrene	–	16
	o-Chlorostyrene-*co*-*o*-fluorostyrene	II had about 14–40 mol% *o*-chlorostyrene	840,841
	p-Chlorostyrene-*co*-*o*-fluorostyrene	II had 66–74 mol% *p*-chlorostyrene	840,841
	o-Fluorostyrene-*co*-*p*-fluorostyrene	II had 10–38% *p*-fluorostyrene	835
	o-Fluorostyrene-*co*-*p*-bromostyrene	II had 68 or 73 mol% bromostyrene	834
	o-Fluorostyrene-*co*-styrene	II had 9–20 mol% styrene	839,842
	p-Fluorostyrene-*co*-styrene	II had about 22–54 mol% styrene	531,839,842
	Maleic anhydride-*co*-styrene	II had 4.6–9.9 wt.% MA (Ref. 1226); II had 7.3–12.5 wt.% MA (Ref. 1022)	1022,1226
	Methyl methacrylate-*co*-styrene	II had 12.5–20 wt.% methyl methacrylate	1022
2,6-dimethyl-1,4-phenylene ether, sulfonylated	*o*-Fluorostyrene	I had 4–17 mol% sulfonylated units	1220
	p-Fluorostyrene	I had 17–66 mol% sulfonylated units	1220
	Phenylvinyl *sec*-butyl ether-*alt*-*N*-phenyl-maleimide	I had 87 mol% sulfonylated units	1219
	Phenylvinyl ethyl ether-*alt*-*N*-phenylmaleimide	I had 87 or 91 mol% sulfonylated units	1219
Dodecamethylene decamethylene dicarboxylate	Vinyl chloride	–	872
Dodecamethylene dodecamethylene dicarboxylate	Vinyl chloride	–	872
	Vinyl chloride-*co*-vinylidene chloride	II had 13.5 wt.% vinyl chloride	873

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TABLE 6. *cont'd*

Polymer I of	Polymer II of	Comments	Refs.
Epichlorohydrin	Alkyl acrylate	II had an alkyl group of ethyl or *n*-propyl (2 blend systems)	224
	Alkyl methacrylate	II had an alkyl group of methyl, ethyl, *n*-propyl, *n*-butyl or cyclohexyl (5 blend systems)	224
	Ethyl acrylate rubber	–	226
	Vinyl acetate	–	305
	N-vinylpyrrolidone	–	300
Ether sulfone	Imide	II was Matrimide 5218	510
Ethyl acrylate	Hexafluoroacetone-*co*-vinylidene fluoride	II had 8 mol% hexafluoroacetone	411,1059
	Vinylidene fluoride	–	56,844
2-Ethylhexyl methacrylate, modified	Hexyl methacrylate-*co*-styrene, modified	I had 3–5% acrylic acid; II had 30% styrene and 5–15% 3-(dimethylamino) 2,2-dimethylpropyl methacrylate	766
N-ethyl-3-hydroxymethylcarbazoyl methacrylate	ω-Hydroxyalkyl-3,5-dinitrobenzoyl methacrylate	II had an alkyl group of butyl, hexyl, pentyl or propyl (4 blend systems)	694
Ethyl methacrylate	*p*-(2-hydroxy-hexafluoroisopropyl)styrene-*co*-styrene	II had 90.3–98.8 mol% styrene	646
	Vinyl chloride-*co*-vinylidene chloride	II had 86.5 wt.% vinylidene fluoride	870
	Vinylidene fluoride	I was syndiotactic or atactic; LCST below T_m of II if I had high MW isotactic	56,695,802
Ethyloxazoline	Aramid	–	532
	Ether sulfone	–	599
Ethylene, chlorinated	Ethylene-*co*-vinyl acetate	I had 35.4–52.6 wt.% Cl; II had 40–45 wt.% vinyl acetate	849
	Methyl methacrylate	I had 50–52 wt.% Cl	99,848
Ethylene orthophthalate	Vinyl acetate	I had $M_n = 9.5$ kg/mol; no LCST when I had $M_n = 5.1$ kg/mol	829
Ethylene ether	Enaminonitrile	–	585
	Ether sulfone	–	317,534,743,853
	Ethylene-*co*-vinyl acetate	–	129
	Sulfone	M_n of I = 3.5–4.5 kg/mol; II was Udel P1700	796
Ethylene-*co*-vinyl acetate	Hexafluoroacetone-*co*-vinylidene fluoride	I had 70% vinyl acetate; II had 8.9 mol% hexafluoroacetone	336
	Vinyl chloride	I had 30 or 37 wt.% ethylene	631,678
2-Fluoroethyl methacrylate	Alkyl methacrylate	II had an alkyl group of methyl or ethyl (2 blend systems)	1153
3-Fluoropropyl methacrylate	Methyl methacrylate	–	1155
Hexadecamethylene dodecamethylene dicarboxylate	Vinyl chloride	–	872
n-Hexyl methacrylate	Styrene-*co*-*p*-vinylphenol	II had 7 or 9 wt.% vinylphenol	922
Hydroxyether of bisphenol-A (phenoxy)	Alkyl methacrylate	II had an alkyl group of methyl, tetrahydrofurfuryl or tetrahydropyranyl-2-methyl (3 blend systems)	111,118,253
	Ether sulfone	–	714,764
	Ethylene oxide-*co*-propylene oxide	II had 22, 50 or 66 mol% ethylene oxide	1025
	Vinyl methyl ether	–	818
Hydroxyether of phenolphthalein	Butylene terephthalate	–	310
	Ether sulfone	–	306
	Ethylene glycol	–	309
β-Hydroxyethyl-3,5-dinitrobenzoyl methacrylate	*N*-alkyl-3-hydroxymethylcarbazoyl methacrylate	II had an alkyl group of butyl, decyl, dodecyl, ethyl, hexadecyl, hexyl, octyl or tetradecyl	693
2-(Hydroxyhexafluoroisopropyl) styrene-*co*-styrene	Methyl methacrylate	I had 90.3–96.1 mol% styrene	646
	Vinyl methyl ether	I had 90.3–99.9 mol% styrene	646
Hydroxypropyl Glutamine	Styrene-*co*-*p*-(hexafluoro-2-hydroxy-2-propyl)-α-methylstyrene	II had 98.5 mol% styrene; blend had < 50 vol% I	971
2-Iodoethyl methacrylate	Tetrahydrofurfuryl methacrylate	–	1107
Isoprene	*cis*-Butadiene-*co*-1,2-vinylbutadiene	II had ≤ 24.3% vinylbutadiene	420
Maleic anhydride-*co*-styrene	Ethyl methacrylate	I had 0–33 wt.% maleic anhydride	73
	Methyl methacrylate	I had 8, 10, 13 or 33 wt.% maleic anhydride	73
	n-Propyl methacrylate	I had 0–18.1 wt.% maleic anhydride	73
Methacrylonitrile-*co*-*p*-methylstyrene	*n*-Butyl methacrylate	I had 15–25 wt.% MAN	944
	Ethyl methacrylate	I had 40, 46 or 53 wt.% MAN	944
	Methyl methacrylate	I had 34, 53, 56 or 68 wt.% MAN	944
	Isopropyl methacrylate	I had 22–31 wt.% MAN and I had $M_n = 161$ kg/mol; II had 17–31 wt.% MAN and I had $M_n = 41.6$ kg/mol	944
	n-Propyl methacrylate	I had 37 or 40 wt.% MAN	944
Methacrylonitrile-*co*-styrene	Ethyl methacrylate	I had 5.6, 35.4 or 40.1 wt.% MAN	943

TABLE 6. *cont'd*

Polymer I of	Polymer II of	Comments	Refs.
	Methyl methacrylate	I had 46.6–62.8 wt.% MAN	1018
	n-Propyl methacrylate	I had 2–32 wt.% MAN	943
Methyl acrylate	Vinylidene fluoride	–	56,844
Methyl methacrylate	Acrylonitrile-*co*-methyl methacrylate-*co*-styrene	II had 40.0/39.1/20.9 or 61.1/19.6/19.3 of S/MMA/AN	620
	Carbonate of bisphenol chloral	–	110,930
	Copolycarbonate of bisphenol-A and hexafluorobisphenol-A	II had 5.3–14 wt.% hexafluorobisphenol-A	430
	Epichlorohydrin-*co*-ethylene glycol	II had 50 mol% ethylene glycol	224
	Ethylene, chlorinated	II had 56–68 wt.% Cl; I was atactic, isotactic or syndiotactic	1204
	Ethylene glycol	–	224
	Hexafluoroacetone-*co*-vinylidene fluoride	–	443
	N-maleimide-*co*-styrene	II had 8, 14 or 21 wt.% *N*-maleimide	1213
	Styrene-*co*-*p*-vinylphenol	I was syndiotactic; II had 5 mol% vinylphenol	1055
	Sulfone of hexafluorobisphenol-A	I had $M_w = 2.4$ kg/mol;, 1.21 or 4.25 kg/mol; II had $M_w = 27$ kg/mol or 56 kg/mol	931
	Tetrafluoroethylene-*co*-vinylidene fluoride	II had 80 mol% vinylidene fluoride	324
	Vinyl acetate	Blends cast from chloroform or cyclohexanone	302,777,778,1092
	Vinyl chloride	–	368,506,653,742
	Vinyl chloride, chlorinated	II had 63, 65 or 68 wt.% Cl	506,833
	Vinyl chloride-*co*-vinylidene chloride	I was atactic or isotactic; II had 86.5 wt.% vinylidene chloride	506,870
	Vinylidene fluoride	LCST above decomposition temperature of I	55
Methyl methacrylate, imidized	Acrylonitrile-*co*-styrene	I had 53–91 wt.% imidized unit; II had 5.7–33 wt.% acrylonitrile; some blends showed LCST	230
α-Methylstyrene	Carbonate of tetramethylbisphenol-A	I had $M_w = 3.5$ or 6.7 kg/mol; II had $M_w = 33$ kg/mol	930
	Carbonate of tetramethylbisphenol-P	I had $M_w = 3.5$ or 6.7 kg/mol; II had $M_w = 31$ kg/mol	930
	Ethyl methacrylate	–	765
	Methyl methacrylate	–	167,765
	n-Propyl methacrylate	–	765
	Styrene	–	710,722,865,929
	Tetrahydropyranyl-2-methyl methacrylate	–	116
α-Methylstyrene-*ran*-4-(2-hydroxyethyl)-α-methylstyrene	Alkyl acrylate	II had an alkyl group of methyl, ethyl or *tert*-butyl (3 blend systems)	968
	Alkyl methacrylate	II had an alkyl group of methyl, ethyl or *tert*-butyl (3 blend systems)	968
	Dialkyl itaconate	II had an alkyl group of methyl, ethyl, *n*-propyl, *n*-butyl or *tert*-butyl (5 blend systems)	968
	Vinyl acetate	–	969
	Vinyl pyrrolidone	–	969
	4-Vinylpyridine	–	969
α-Methylstyrene-*stat*-[4-(1,1,1-trifluoro-2-hydroxyethyl)-α-methylstyrene	Alkyl acrylate	II had an alkyl group of methyl, ethyl or *t*-butyl (3 blend systems)	968
	Alkyl methacrylate	II had an alkyl group of methyl, ethyl or *t*-butyl (3 blend systems)	968
	Dialkyl itaconate	II had an alkyl group of methyl, ethyl, *n*-propyl, *n*-butyl or *t*-butyl (4 systems)	968
	Vinyl acetate	–	969
	Vinyl pyrrolidone	–	969
	4-Vinylpyridine	–	969
Neopentyl glycol adipate (2,2-dimethyl-1,3-propylene adipate)	Hydroxyether of bisphenol-A (phenoxy)	–	332
Neopentyl glycol succinate (2,2-dimethyl-1,3-propylene succinate)	Hydroxyether of bisphenol-A (phenoxy)	–	332
	Vinyl chloride-*co*-vinylidene chloride	II was Saran, 86.5 wt.% vinylidene chloride; LCST above T_m when ≤ 50 wt.% I	868
Oxymethylene	Novolac	–	858
Phenol-formaldehyde	Methyl methacrylate-*co*-styrene	II had 30.4–83.5% MMA	433
p-Phenylene benzobisoxazole	Nylon	II was Zytel 330 (Du Pont) or nylon 6,6 (2 blend systems)	945
2,2′-(*m*-Phenylene)-5,5′-bibenzimidazole	Imide	II was Ultem 1000 or XU218 (2 blend systems)	114

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TABLE 6. *cont'd*

Polymer I of	Polymer II of	Comments	Refs.
n-Propyl methacrylate	Vinyl chloride-*co*-vinylidene chloride	II had 86.5 wt.% vinylidene chloride	870
Styrene	Copolycarbonate of bisphenol-A and tetramethylbisphenol-A	II had $\geq$ 85 wt.% tetramethylbisphenol-A	431
	o-Fluorostyrene-*co*-*p*-fluorostyrene	I had $M_n = 7$ kg/mol; II had 10 or 33 mol% *p*-fluorostyrene	1144
Styrene, sulfonated and Zn-neutralized	Caprolactam (nylon 6)	I had 6.5, 8.1 or 9.0 mol% sulfonated units	1083
Styrene, sulfonated and neutralized	Caprolactam (nylon 6)	I was neutralized by Li, Mg or Zn	1223
Vinyl acetate	Hexafluoroacetone-*co*-vinylidene fluoride	–	1073
	Maleic anhydride-*alt*-styrene, hydrolyzed	II was 1.8–43% hydrolyzed; LCST increased with increasing degree of hydrolysis	67
Vinyl acetate-*co*-vinyl chloride	*n*-Butyl methacrylate	I had 5.9–15.1 vol% vinyl acetate	497
	Hexyl acrylate-*co*-methyl acrylate	I had 5–23.7 vol% vinyl acetate; II had 15–88 vol% hexyl acrylate; some blends showed LCST	379
	Methyl methacrylate	I had 5.9–15.1 vol% vinyl acetate	497
Vinyl butyral	Vinylpyridine	II was 2- or 4-vinylpyridine (2 blend systems)	1003
Vinyl chloride	Acrylonitrile-*co*-*n*-butyl acrylate	II had 53, 67 or 84 mol% butyl acrylate	655
	Alkyl acrylate	II had an alkyl group of *n*-propyl or *n*-amyl (2 blend systems)	851
	Alkyl methacrylate	II had an alkyl group of *n*-butyl or *n*-amyl (2 blend systems)	649,653
	Butyl acrylate-*co*-ethyl acrylate	II had 22.9, 43.5 or 52.8% butyl acrylate	851
	Butyl acrylate-*co*-hexyl acrylate	II had 47.8% butyl acrylate	851
	Butyl acrylate-*co*-methyl acrylate	II had 44.6% butyl acrylate	851
	Butyl acrylate-*co*-methyl methacrylate	II had 75% butyl acrylate	654
	Butyl methacrylate-*co*-methyl methacrylate	II had 30 or 70% methyl methacrylate	653
	Ethyl acrylate-*co*-hexyl acrylate	II had 41.7 or 43.2% ethyl acrylate	851
	Methyl acrylate-*co*-hexyl acrylate	II had 9.7% methyl acrylate	851
	Pentyl acrylate-*co*-propyl acrylate	II had 48.1% propyl acrylate	851
Vinyl methyl ether	Acrylonitrile-*co*-styrene	II had $\leq$ 10–11% acrylonitrile	575
	Alkyl acrylate	II had an alkyl group of ethyl or *n*-butyl (2 blend systems)	941
	2-Chlorostyrene	–	1203
	Hydrogeneous styrene-*b*-deuterated styrene	–	1019
	Maleic anhydride-*co*-styrene	II had < 15% maleic anhydride	575
	α-Methylstyrene	–	318
	α-Methylstyrene-*co*-4-(2-hydroxyethyl)-α-methylstyrene	II had 6–89 mol% α-methylstyrene	170
	α-Methylstyrene-*co*-4-(1,1,1-trifluoro-2-hydroxyethyl)-α-methylstyrene	II had 0–93 mol% α-methylstyrene	170
	α-Methylstyrene-*co*-styrene	II had 77.5 or 88 wt.% styrene	318
	m-Methylstyrene-*co*-styrene	II had 76.5 or 87.5 wt.% styrene	1028
	o-Methylstyrene-*co*-styrene	II had 76.2 or 86.8 wt.% styrene	1028
	p-Methylstyrene-*co*-styrene	II had 61–95 wt.% styrene	318,1016,1028
	Methyl methacrylate-*co*-styrene	II had < 60 mol% methyl methacrylate	108
	Nitrostyrene-*co*-styrene	II had $\leq$ 25.6 mol% Nitrostyrene	165,192
	Styrene-*co*-1-vinyl naphthalene	II had 17.4 wt.% styrene	318
	Styrene-*co*-2-vinyl naphthalene	II had 19.2 wt.% styrene	318
Vinyl methyl ketone	Styrene-*co*-4-vinylstilbene	Cloud point was sensitive to the *cis*- fraction of 4-stilbenzyl groups	365
p-Vinylphenol	4-Acetoxystyrene	–	340
2-Vinylpyridine	Aramide	II made from hexamethylenediamine and a mixture of isomers of phthalic acid	767
	2-Hydroxyethyl methacrylate	–	937
4-Vinylpyridine	Vinyl acetate-*co*-vinyl alcohol	II had 32–55 mol% vinyl alcohol	1046
Vinylidene fluoride	Methoxymethyl methacrylate	–	267

TABLE 7. POLYMER PAIRS THAT APPEAR TO HAVE BOTH A LOWER CRITICAL SOLUTION TEMPERATURE AND A UPPER CRITICAL SOLUTION TEMPERATURE

Polymer I of	*Polymer II of*	Comments	Refs.
Acrylonitrile-*co*-styrene	Acrylonitrile-*co*-butadiene	I had 25 wt.% acrylonitrile; II had 40 wt.% acrylonitrile	636
Butadiene	Butadiene-*co*-styrene	II had 45 wt.% styrene	797
Caprolactone	Acrylonitrile-*co*-styrene	II had 12.4–26.4 wt.% acrylonitrile	1193

Table 7. *cont'd*

Polymer I of	Polymer II of	Comments	Refs.
Carbonate of bisphenol-A	Methyl methacrylate	Blends showed a LCST-UCST loop	72,479,1085
Dimethylsiloxane	Methylphenylsiloxane	–	467
2-Hydroxyethyl Acrylate-*co*-liquid crystalline acrylate	2-Hydroxyethyl acrylate-*co*-liquid crystalline acrylate	Both I and II had 6% 2-hydroxyethyl acrylate; the liquid crystalline polyacrylates had different spacer groups	1252
o-Methylated phenol formaldehyde	Methyl methacrylate	I had 20 or 50% of OH group methylated	432
	Methyl methacrylate-*co*-styrene	I had 20 or 50% of OH group methylated; II had 50 or 60% methyl methacrylate	432
Styrene	2,6-Dimethyl-1,4-phenylene ether, carboxylated	II had 8.0–10.3 mol% carboxyl group	148

TABLE 8. POLYMER PAIRS THAT *CO*-CRYSTALLIZE

Polymer I of	Polymer II of	Method	Comments	Refs.
Butylene terephthalate	Butylene glycol-*b*-butylene terephthalate	T_g and T_m by DSC, dielectric loss measurements	Two T_g's when I was Valox 310 (General Electric) having $M_n = 20$ kg/mol and II contained 44–67 wt.% butylene terephthalate; single T_g and single T_m when II contained 85 wt.% butylene terephthalate; single T_m indicated that only one type of crystal was present	554
	Butylene glycol-*b*-butylene terephthalate	T_g and T_m by DSC, dielectric loss measurements, microscopy	Single T_g and single T_m when I was Valox 310 (General Electric) with $M_n = 33$ kg/mol, II contained 84 wt.% butylene terephthalate and I/II = 1/3 to 3/1; thermal analysis and morphological evidence pointed to extensive *co*-crystallization in this mixture; two T_g's when II contained 50 wt.% butylene terephthalate	703
	Ester–ether segmented polymer	Dielectric loss peak and T_m by DSC	I was Valox 310 (General Electric) with $M_n = 20$ kg/mol; II was comprised of a poly(butylene oxide) soft segment and a hard segment 4GT-butylene terephthalate, and had total MW = 25 to 30 kg/mol; single beta dielectric loss peak and single T_m were found when II was 44 or 67 wt.% 4GT; single T_m between the T_m's of the pure components indicated *co*-crystallization; two dielectric loss peaks and two T_m when II was 84 wt.% 4GT	704
Ether ether ketone	Ether ether ether ketone	T_g and T_m by DSC	Single T_g and single T_m when I contained 33% ketone linkages and had melt flow = 3.0 dg/min at 400°C and II had melt flow = 10 dg/min; I/II = 1/1	333
	Ether ether ketone ketone	T_g and T_m by DSC	Single T_g and single T_m when I contained 33% ketone linkages and had melt flow = 3.0 dg/min at 400°C and II contained 50% ketone linkages and had melt flow = 4.0 dg/min; I/II = 1/1	333
	Ether ketone	T_g and T_m by DSC	Single T_g and single T_m when I contained 33% ketone linkages and had melt flow = 3.0 dg/min and II contained 50% ketone linkages and had melt flow = 1.5 dg/min; I/II = 1/1	333
	Ether ketone	T_g and T_m by DSC; WAXD	Single composition-dependent melting endotherms obtained upon quenching when I had intrinsic viscosity = 1.124 dl/g in sulfuric acid at 30°C and II had intrinsic viscosity = 1.107 dl/g in sulfuric acid at 30°C, but it did not happen when other thermal histories were imposed	739
	Ether-*co*-ketone	T_g and T_m by DSC	Single T_g and single T_m when I contained 33% ketone linkages and had melt flow = 3.0	333

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TABLE 8. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
			dg/min and II contained 57% ketone linkages and had melt flow = 1.5 dg/min; I/II = 1/1	
	Ether ketone (polymer II), etherimide (polymer III)	T_g and T_m by DSC	I had reduced viscosity = 1.05 dl/g in 1 wt.% conc. sulfuric acid at 23°C; II had reduced viscosity = 1.9 dl/g in the same condition as for I; III was Ultem 1000 from GE with reduced viscosity = 0.5 dl/g in 0.2 wt.% chloroform at 25°C; single T_g and single T_m for I/II/III = 25/25/50	334
Ethylene	Butadiene, hydrogenated	DSC, SALS and SEM	I was narrow MW fraction with $M_w = 113$ kg/mol and $M_n = 105$ kg/mol; II contained 2.2–5.5 mol% ethyl branches and had $M_w = 103$–420 kg/mol and $M_n = 83$–156 kg/mol; stated that I and II *co*-crystallized upon rapid crystallization from the melt when II contained up to 2 mol% of ethyl branches	13
	Ethylene	DSC and SALS	I was polyethylene and II was UHMWPE with MW = 5,000 kg/mol; I was LLDPE (GRSN-7047 from Union Carbide with $M_w = 134$ kg/mol and $M_w/M_n = 4$), or I was HDPE (Chemplex 6186 with $M_w =$ 134.8 kg/mol and $M_w/M_n = 7.4$); only a single composition dependent T_m was observed; SALS showed the *co*-crystallization took place; but when I was LDPE ($M_w =$ 114 kg/mol and $M_w/M_n = 5.2$), separate crystals were formed	483
		DSC, SAXS and X-ray diffraction	I was linear LLDPE and II was HDPE; co-crystallization was evidenced by a single T_m by DSC; stated that X-ray diffraction peak width, SAXS and Raman spectra were all consistent with a single crystallizing species; but for HDPE/LDPE and LLDPE/ LDPE, separate crystals were seen with two T_m's being found by DSC	783
			I was HDPE with $M_w = 100$ kg/mol and $M_n = 11$ kg/mol and II was LLDPE with $M_w = 350$ kg/mol and $M_n = 16$ kg/mol; after the blend with I/II = 25/75 by weight was crosslinked either by irradiation or by peroxides in the liquid state above a critical degree of cross-linkage, the DSC curve of the blend showed only one single melting point on heating and only one single crystallization peak on cooling; Refs. 45, 348, 478, 579 and 676 also discussed the miscibility of PE/PE blends	360
	Ethylene, deuterated	T_m by DSC	Single T_m varying smoothly with composition and no gross segregation occurred	217
Ethylene	Ethylene-*co*-vinyl acetate	DSC, SALS and SEM	I was narrow MW fraction with $M_w = 113$ kg/mol and $M_n = 105$ kg/mol; II contained 1.12–6.6 mol% of acetate branches and had $M_w = 60.3$–105 kg/mol and $M_n =$ 23.5–43.6 kg/mol; stated that I and II co-crystallized upon rapid crystallization from the melt contained up to 2 mol% of acetate branches	13
Ethylene-*co*-vinyl acetate	Ethylene-*co*-vinyl acetate	DSC, SALS and SEM	I and II had $M_w = 60.3$–105 kg/mol, $M_n =$ 23.5–43.6 kg/mol, and they were different in the content of acetate branches; stated that these copolymers containing about 1–2 mol% of acetate branches were found to co-crystallize with one another	13

TABLE 8. *cont'd*

Polymer I of	Polymer II of	Method	Comments	Refs.
Glutamate	Glutamate	Dynamic mechanical measurements, X-ray diffraction	I or II was poly(L-glutamate) or poly(D-glutamate) having m number of methylene group in the ester group of a general formula as $-CH_2CH_2COO-(CH_2)_m-C_6H_5-$, m might be 1, 2 or 3 so the component polymer would be represented as Lm or Dm; the MW of both components were 135 to 380 kg/mol and $M_w/M_n = 1.1$ to 1.2; stated that binary blends of L2, D2, L3 and D3 were miscible and formed isomorphous mixed crystals at all compositions, whereas other pairs, with the exception of L1/D1, were immiscible	717
Isopropyl vinyl ether, isotactic	*sec*-Butyl vinyl ether, isotactic	Unit cell dimension from X-ray diffraction	Co-crystallization at all compositions	605
4-Methyl-1-hexene, isotactic	4-Methyl-1-pentene, isotactic	Unit cell dimensions from X-ray diffraction	Co-crystallization only below 25 wt.% I and above 75 wt.% I	605
Tetrafluoroethylene	Tetrafluoroethylene-*co*-perfluoroalkyl vinyl ether	Dynamic mechanical measurements	II had 98–99 mol% tetrafluoroethylene; single α-transition from DMA	1173
Trifluoroethylene-*co*-vinylidene fluoride	Trifluoroethylene-*co*-vinylidene fluoride	DSC, X-ray and phase-contrast microscopy	No phase separation by phase-contrast microscopy, a sharp single X-ray peak, intermediate in spacing between those of the individual copolymers, and clearly different from their superposition, and only one ferroelectric transition and only one melting point by DSC, indicated co-crystallization within the same lattice when I was 52/48 mol% vinylidene fluoride (VDF)/trifluoroethylene (TFE), II was 65/35 mol% VDF/TFE, or I was 65/35 mol% VDF/TFE, and II was 73/27 mol% VDF/TFE; but two X-ray peaks and two ferroelectric transitions and two melting points when I was 52/48 and II was 73/27 mol% VDF/TFE	800
Vinyl alcohol	Vinyl alcohol	DSC	I and II had a diad syndiotacticity of 54 and 58 mol%, respectively	1197
Vinylidene fluoride-*co*-tetrafluoroethylene	Vinylidene fluoride		I had 9.1, 14.3 or 17.6 mol% tetrafluoroethylene	980
	Hexafluoroacetone-*co*-vinylidene fluoride	DSC	I had 80 mol% vinylidene fluoride; II had 92 mol% vinylidene fluoride; single melting peak	951

E. REFERENCES

1. J. Adduci, D. K. Dandge, J. Kops, Am. Chem. Soc., Polym. Prepr., **22** (2), 109 (1981).
2. S. M. Aharoni, Macromolecules, **11**, 277 (1978).
3. S. M. Aharoni, J. Macromol. Sci. B: Phys., **22**, 813 (1983).
4. T. O. Ahn, H. M. Jeong, Pollimo, **11**, 112 (1987).
5. T. O. Ahn, S. Lee, H. M. Jeong, Eur. Polym. J., **25**, 95 (1989).
6. T. O. Ahn, S. Lee, H. M. Jeong, K. Cho, Angew. Makromol. Chem., **192**, 133 (1991).
7. T. O. Ahn, Y. J. Lee, S. M. Lee, H. M. Jeong, J. Macromol. Sci. B: Phys., **29**, 91 (1990).
8. T. O. Ahn, B. V. Nam, S. Lee, H. M. Jeong, Polym. Commun., **32**, 415 (1991).
9. A. B. Aivazov, K. G. Mindiyarov, Y. V. Zelenev, Y. G. Oganesov, V. G. Raevskii, Vyosokomol. Soedin. Ser. B., **12**, 10 (1970).
10. S. Akiyama, K. Ishikawa, H. Fujiishi, Polymer, **32**, 1673 (1991).
11. S. Akiyama, W. J. MacKnight, F. E. Karasz, J. Nambu, Polym. Commun., **28**, 236 (1987).
12. S. Akiyama, N. Inaba, R. Kaneko, Chem. High Polym. Japan, **26**, 529 (1969).
13. R. G. Alamo, R. H. Glaser, L. Mendelkern, J. Polym. Sci., Polym. Phys. Ed., **26**, 2169 (1988).
14. R. Alex, P. P. De, S. K. De, Polym. Commun., **31**, 366 (1990).
15. P. S. Alexandrovich, F. E. Karasz, W. J. MacKnight, J. Macromol. Sci. B: Phys., **17**, 501 (1980).
16. P. S. Alexandrovich, F. E. Karasz, W. J. MacKnight, Polymer, **18**, 1022 (1977).
17. G. C. Alfonso, A. Turturro, M. Pizzoli, M. Scandola, G. Ceccorulli, J. Polym. Sci., Polym. Phys. Ed., **27**, 1195 (1989).

TABLE 8. *cont'd*

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4-Methyl-1-hexene, isotactic	4-Methyl-1-pentene, isotactic	Unit cell dimensions from X-ray diffraction	Co-crystallization only below 25 wt.% I and above 75 wt.% I	605
Tetrafluoroethylene	Tetrafluoroethylene-*co*-perfluoroalkyl vinyl ether	Dynamic mechanical measurements	II had 98–99 mol% tetrafluoroethylene; single α-transition from DMA	1173
Trifluoroethylene-*co*-vinylidene fluoride	Trifluoroethylene-*co*-vinylidene fluoride	DSC, X-ray and phase-contrast microscopy	No phase separation by phase-contrast microscopy, a sharp single X-ray peak, intermediate in spacing between those of the individual copolymers, and clearly different from their superposition, and only one ferroelectric transition and only one melting point by DSC, indicated co-crystallization within the same lattice when I was 52/48 mol% vinylidene fluoride (VDF)/trifluoroethylene (TFE), II was 65/35 mol% VDF/TFE, or I was 65/35 mol% VDF/TFE, and II was 73/27 mol% VDF/TFE; but two X-ray peaks and two ferroelectric transitions and two melting points when I was 52/48 and II was 73/27 mol% VDF/TFE	800
Vinyl alcohol	Vinyl alcohol	DSC	I and II had a diad syndiotacticity of 54 and 58 mol%, respectively	1197
Vinylidene fluoride-*co*-tetrafluoroethylene	Vinylidene fluoride		I had 9.1, 14.3 or 17.6 mol% tetrafluoroethylene	980
	Hexafluoroacetone-*co*-vinylidene fluoride	DSC	I had 80 mol% vinylidene fluoride; II had 92 mol% vinylidene fluoride; single melting peak	951

E. REFERENCES

1. J. Adduci, D. K. Dandge, J. Kops, Am. Chem. Soc., Polym. Prepr., **22** (2), 109 (1981).
2. S. M. Aharoni, Macromolecules, **11**, 277 (1978).
3. S. M. Aharoni, J. Macromol. Sci. B: Phys., **22**, 813 (1983).
4. T. O. Ahn, H. M. Jeong, Pollimo, **11**, 112 (1987).
5. T. O. Ahn, S. Lee, H. M. Jeong, Eur. Polym. J., **25**, 95 (1989).
6. T. O. Ahn, S. Lee, H. M. Jeong, K. Cho, Angew. Makromol. Chem., **192**, 133 (1991).
7. T. O. Ahn, Y. J. Lee, S. M. Lee, H. M. Jeong, J. Macromol. Sci. B: Phys., **29**, 91 (1990).
8. T. O. Ahn, B. V. Nam, S. Lee, H. M. Jeong, Polym. Commun., **32**, 415 (1991).
9. A. B. Aivazov, K. G. Mindiyarov, Y. V. Zelenev, Y. G. Oganesov, V. G. Raevskii, Vyosokomol. Soedin. Ser. B., **12**, 10 (1970).
10. S. Akiyama, K. Ishikawa, H. Fujiishi, Polymer, **32**, 1673 (1991).
11. S. Akiyama, W. J. MacKnight, F. E. Karasz, J. Nambu, Polym. Commun., **28**, 236 (1987).
12. S. Akiyama, N. Inaba, R. Kaneko, Chem. High Polym. Japan, **26**, 529 (1969).
13. R. G. Alamo, R. H. Glaser, L. Mendelkern, J. Polym. Sci., Polym. Phys. Ed., **26**, 2169 (1988).
14. R. Alex, P. P. De, S. K. De, Polym. Commun., **31**, 366 (1990).
15. P. S. Alexandrovich, F. E. Karasz, W. J. MacKnight, J. Macromol. Sci. B: Phys., **17**, 501 (1980).
16. P. S. Alexandrovich, F. E. Karasz, W. J. MacKnight, Polymer, **18**, 1022 (1977).
17. G. C. Alfonso, A. Turturro, M. Pizzoli, M. Scandola, G. Ceccorulli, J. Polym. Sci., Polym. Phys. Ed., **27**, 1195 (1989).

18. D. Allard, R. E. Prud'homme, J. Appl. Polym. Sci., **27**, 559 (1982).
19. A. M. Ambler, J. Polym. Sci., Polym. Chem. Ed., **11**, 1505 (1973).
20. B. Ameduri, R. E. Prud'homme, Polymer, **29**, 1052 (1988).
21. E. W. Anderson, H. E. Bair, G. E. Johnson, T. K. Kwei, F. J. Padden Jr., D. Williams, in: S. L. Cooper and G. M. Estes, (Eds.), "Multiphase Polymers", Am. Chem. Soc., Adv. Chem. Ser., vol. 176, Washington, DC, 1979.
22. M. Andreis, Z. Veksli, R. Vukovic, New Polym. Mater., **3**, 197 (1992).
23. Anonymous, Res. Discl., **229**, 182 (1983).
24. M. Antonietti, S. Lang, H. Sillescu, Makromol. Chem., Rapid Commun., **7**, 415 (1986).
25. Y. Aoki, Macromolecules, **21**, 1277 (1988).
26. Y. Aoki, Macromolecules, **23**, 2309 (1990).
27. P. Aptel, I. Cabasso, J. Appl. Polym. Sci., **25**, 1969 (1980).
28. C. A. Arnold, D. H. Chen, Y. P. Chen, R. A. Waldbauer, M. E. Rogers, J. E. McGrath, High Performance Polym., **2**, 83 (1990).
29. A. Asano, K. Takegoshi, K. Hikichi, Polym. J., **24**, 555 (1992).
30. R. M. Asimova, P. V. Kozlov, V. A. Kargin, S. M. Vtorygin, Vysokomol. Soedin., **4**, 544 (1962).
31. M. Aubin, Y. Bedard, M. F. Morrissette, R. E. Prud'homme, J. Polym. Sci., Polym. Phys. Ed., **21**, 233 (1983).
32. M. Aubin, D. Bussieres, D. Duchesne, R. E. Prud'homme, in: K. Solc, (Ed.), "Polymer Compatibility and Incompatibility – Principles and Practices", MMI Press, Harwood 1982.
33. M. Aubin, R. E. Prud'homme, Macromolecules, **13**, 365 (1980).
34. M. Aubin, R. E. Prud'homme, J. Polym. Sci., Polym. Phys. Ed., **19**, 1245 (1981).
35. M. Aubin, R. E. Prud'homme, Polymer, **22**, 1223 (1981).
36. M. Aubin, R. E. Prud'homme, Polym. Eng. Sci., **24**, 350 (1984).
37. M. Avella, E. Martuscelli, Polymer, **29**, 1731 (1988).
38. M. Baer, J. Polym. Sci. A, **2**, 417 (1964).
39. H. E. Bair, Polym. Eng. Sci., **10**, 247 (1970).
40. H. E. Bair, E. W. Anderson, G. E. Johnson, T. K. Kwei, Am. Chem. Soc. Polym. Prepr., **19**, 143 (1978).
41. H. E. Bair, D. Williams, T. K. Kwei, F. J. Padden Jr. Org. Coatings Plastics Chem. Prepr., **37**, 240 (1977).
42. D. G. H. Ballard, M. G. Rayner, J. Schelten, Polymer, **17**, 640 (1976).
43. M. Bank, J. Leffingwell, C. Thies, Macromolecules, **4**, 43 (1971).
44. M. Bank, J. Leffingwell, C. Thies, J. Polym. Sci. A, **2**, 1097 (1972).
45. P. J. Barham, M. J. Hill, A. Keller, C. C. A. Rosney, J. Mater. Sci. Lett., **7**, 1271 (1988).
46. R. S. Barnum, S. H. Goh, D. R. Paul, J. W. Barlow, J. Appl. Polym. Sci., **26**, 3917 (1981).
47. G. M. Bartenev, G. S. Kongarov, Rubber Chem. Technol., **36**, 668 (1963).
48. F. S. Bates, G. D. Wignall, Macromolecules, **19**, 269 (1968).
49. F. S. Bates, G. D. Wignall, S. B. Dierker, Macromolecules, **19**, 1938 (1986).
50. G. Beaucage, R. S. Stein, T. Hashimoto, H. Hasegawa, Macromolecules, **24**, 3443 (1991).
51. C. Belaribi, G. Marin, P. Monge, Eur. Polym. J., **22**, 481 (1986).
52. L. A. Belfiore, Am. Chem. Soc., Polym. Prepr., **29** (1), 17 (1988).
53. R. E. Belke, I. Cabasso, Polymer, **29**, 1831 (1988).
54. G. Belorgey, M. Aubin, R. E. Prud'homme, Polymer, **23**, 1051 (1982).
55. G. Belorgey, R. E. Prud'homme, J. Polym. Sci., Polym. Phys. Ed., **20**, 191 (1982).
56. R. E. Bernstein, C. A. Cruz, D. R. Paul, J. W. Barlow, Macromolecules, **10**, 681 (1977).
57. R. E. Bernstein, D. R. Paul, J. W. Barlow, Polym. Eng. Sci., **18**, 683 (1978).
58. R. E. Bernstein, D. C. Wahrmund, J. W. Barlow, D. R. Paul, Polym. Eng. Sci., **18**, 1220 (1978).
59. D. E. Bhagwagar, C. J. Serman, P. C. Painter, M. M. Coleman, Macromolecules, **22**, 4654 (1989).
60. C. Bhattacharyya, S. N. Bhattacharyya, B. M. Mendal, J. Indian Chem. Soc., **63**, 157 (1986).
61. C. Bhattacharyya, M. Maiti, B. M. Mandal, S. N. Bhattacharyya, Macromolecules, **22**, 4062 (1989).
62. A. K. Bhowmick, C. C. Kuo, A. Manzur, A. MacArthur, D. McIntyre, J. Macromol. Sci., B: Phys., **25**, 283 (1986).
63. P. Black, D. J. Worsfold, J. Appl. Polym. Sci., **18**, 2307 (1974).
64. T. F. Blahovici, G. R. Brown, L. E. St. Pierre, Polym. Eng. Sci., **22**, 1123 (1982).
65. C. Booth, C. J. Pickles, J. Polym. Sci., Polym. Phys. Ed., **11**, 595 (1973).
66. M. Bosma, G. ten Brinke, T. S. Ellis, Macromolecules, **21**, 1465 (1988).
67. M. Bosma, E. J. Voremkamp, G. Ten Brinke, G. Challa, Polymer, **29**, 1694 (1988).
68. L. G. Bourland, J. Vinyl Technol., **10**, 183 (1988).
69. L. G. Bowlard, D. M. Braunstein, J. Appl. Polym. Sci., **32**, 6151 (1986).
70. T. N. Bowmer, A. E. Tonelli, Macromolecules, **19**, 498 (1986).
71. T. N. Bowmer, A. E. Tonelli, J. Vinyl Technol., **8**, 98 (1986).
72. G. R. Brannock, J. W. Barlow, D. R. Paul, J. Polym. Sci., Polym. Phys. Ed., **28**, 871 (1990).
73. G. R. Brannock, J. W. Barlow, D. R. Paul, J. Polym. Sci., Polym. Phys. Ed., **29**, 413 (1991).
74. G. R. Brannock, D. R. Paul, Macromolecules, **23**, 5240 (1990).
75. D. Braun, J. Herth, G. P. Hellmann, Angew. Makromol. Chem., **171**, 53 (1989).
76. D. Braun, D. Yu, P. R. Kohl, X. Gao, L. N. Andradi, E. Manager, G. P. Hellmann, J. Polym. Sci., Polym. Phys. Ed., **30**, 577 (1992).

77. H. G. Braun, G. Rehage, Angew. Makromol. Chem., **131**, 107 (1985).
78. R. E. S. Bretas, D. G. Baird, Polymer, **33**, 5233 (1992).
79. W. Breuers, W. Hild, H. Wolff, W. Burmeister, H. Hoyer, Plast. Katusch., **1**, 170 (1954).
80. R. M. Briber, F. Khoury, Polymer, **28**, 38 (1987).
81. J. Brisson, B. Breault, Macromolecules, **24**, 495 (1991).
82. G. L. Brode, J. V. Koleske, J. Macromol. Sci. A: Chem., **6**, 109 (1972).
83. N. Y. Buben, V. J. Goldanskii, L. Y. Zlatkevich, V. G. Nikolskii, V. G. Raevskii, Polym. Sci. USSR, **9**, 2575 (1967).
84. C. M. Buchannan, S. C. Gedon, A. W. White, M. D. Wood, Macromolecules, **25**, 7373 (1992).
85. R. Buchdahl, L. E. Nielson, J. Polym. Sci., **15**, 1 (1955).
86. I. Cabasso, Am. Chem. Soc., Org. Coatings Chem. Prepr., **45**, 359 (1981).
87. I. Cabasso, J. Jagur-Gordinski, D. Vofsi, in: K. C. (Eds.), "Polymer Alloys, Blends, Blocks, Grafts and IPN's, Klempner, D., and Frisch", Plenum Press, New York, 1977.
88. I. Cabasso, J. Jagur-Gordinski, D. Vofsi, J. Appl. Polym. Sci., **18**, 1969 (1974).
89. G. C. Campbell, D. L. Vanderhart, Y. Feng, C. C. Han, Macromolecules, **25**, 2107 (1992).
90. X. Cao, M. Jiang, T. Yu, Makromol. Chem., **190**, 117 (1989).
91. B. Carmoin, G. Villoutreix, R. Berlot, J. Macromol. Sci., B: Phys., **14**, 307 (1977).
92. R. Caspary, Kautschuk Gummi Kunst., **25**, 249 (1972).
93. R. Casper, L. Morbitzer, Angew. Makromol. Chem., **58/59**, 1 (1977).
94. G. Ceccorulli, M. Pizzoli, M. Scandola, G. C. Alfonso, A. Turturro, Polymer, **30**, 1251 (1989).
95. L. C. Cesteros, J. M. Rego, J. J. Vazquez, I. Katime, Polym. Commun., **31**, 152 (1990).
96. Z. Chai, F. E. Karasz, Macromolecules, **25**, 4716 (1992).
97. Z. Chai, R. Sun, Polymer, **24**, 1279 (1983).
98. Z. Chai, R. Sun, F. E. Karasz, Macromolecules, **25**, 6113 (1992).
99. Z. Chai, R. Sun, D. J. Walsh, J. S. Higgins, Polymer, **24**, 263 (1983).
100. Z. Chai, D. J. Walsh, Makromol. Chem., **184**, 1459 (1983).
101. L. A. Chandler, E. A. Collins, J. Appl. Polym. Sci., **13**, 1585 (1969).
102. S. S. Chang, Polym. Commun., **29**, 33 (1988).
103. C. T. Chen, H. Morawetz, Macromolecules, **22**, 159 (1989).
104. F. L. Chen, E. M. Pearce, T. K. Kwei, Polymer, **29**, 2285 (1988).
105. Y. P. Chen, D. H. Chen, C. A. Arnole, D. A. Lewis, J. F. Pollard, J. S. Graubeal, T. C. Ward, J. E. McGrath, Am. Chem. Soc., Polym. Prepr., **29** (2), 370 (1988).
106. F. S. Cheng, J. L. Kardos, Am. Chem. Soc., Polym. Prepr., **10**, 615 (1969).
107. J. T. Cheng, G. J. Mantell, J. Appl. Polym. Sci., **23**, 1733 (1979).
108. Y. Y. Chien, E. M. Pearce, T. K. Kwei, Macromolecules, **21**, 1616 (1988).
109. J. S. Chiou, J. W. Barlow, D. R. Paul, J. Polym. Sci., Polym. Phys. Ed., **25**, 1459 (1987).
110. J. S. Chiou, D. R. Paul, J. Appl. Polym. Sci., **33**, 2935 (1987).
111. J. S. Chiou, D. R. Paul, J. Appl. Polym. Sci., **42**, 279 (1991).
112. J. S. Chiou, D. R. Paul, J. W. Barlow, Polymer, **23**, 1543 (1982).
113. J. W. Cho, Polym. Int., **24**, 173 (1991).
114. S. Choe, W. J. MacKnight, F. E. Karasz, Am. Chem. Soc., Polym. Mater. Sci. Eng., **59**, 702 (1988).
115. S. Choe, D. J. Williams, W. J. MacKnight, F. E. Karasz, Polym. Mater. Sci. Eng., **56**, 827 (1987).
116. Y. F. Chong, S. H. Goh, Eur. Polym. J., **27**, 501 (1991).
117. Y. F. Chong, S. H. Goh, J. Appl. Polym. Sci., **43**, 437 (1991).
118. Y. F. Chong, S. H. Goh, J. Appl. Polym. Sci., **44**, 633 (1992).
119. Y. F. Chong, S. H. Goh, Polymer, **33**, 127 (1992).
120. Y. F. Chong, S. H. Goh, Polymer, **33**, 132 (1992).
121. Y. F. Chong, S. H. Goh, Polymer, **33**, 1289 (1992).
122. Y. F. Chong, S. Y. Lee, S. H. Goh, Eur. Polym. J., **26**, 1145 (1990).
123. Y. F. Chong, S. Y. Lee, S. H. Goh, Eur. Polym. J., **26**, 1207 (1990).
124. T. S. Chow, Macromolecules, **23**, 333 (1990).
125. W. H. Christiansen,D. R. Paul, J. W. Barlow, J. Appl. Polym. Sci., **34**, 537 (1987).
126. C. W. Chu, L. C. Dickinson, J. C. W. Chien, Polym. Bull., **19**, 265 (1988).
127. E. Y. Chu, E. M. Pearce, T. K. Kwei, T. F. Yeh, Y. Okamoto, Makromol. Chem., Rapid Commun., **12**, 1 (1991).
128. H. H. Chuah, T. Kyu, T. Helminiak, Polym. Mater. Sci. Eng., **56**, 58 (1987).
129. T. S. Chung, P. N. Chen, Sr., J. Appl. Polym. Sci., **40**, 1209 (1990).
130. T. S. Chung, R. H. Vora, M. Jaffe, J. Polym. Sci., Polym. Chem. Ed., **29**, 1207 (1991).
131. S. Cimmino, F. E. Karasz, W. J. MacKnight, J. Polym. Sci., Polym. Phys. Ed., **30**, 49 (1992).
132. S. Cimmino, E. Martuscelli, M. Saviano, C. Silvestre, Polymer, **32**, 1461 (1991).
133. S. Cimmino, E. Martuscelli, C. Silvestre, M. Canetti, C. Dedalla, A. Seves, J. Polym. Sci., Polym. Phys. Ed., **27**, 1781 (1989).
134. E. P. Cizek, US Pat., 3,383,435 (1968) to General Electric Co.
135. J. N. Clark, J. S. Higgins, C. K. Kim, D. R. Paul, Polymer, **33**, 3137 (1992).
136. M. B. Clark Jr., C. A. Burkhardt, J. A. Gardella, Jr., Macromolecules, **22**, 4495 (1989).
137. S. D. Clas, A. Eisenberg, J. Polym. Sci., Polym. Phys. Ed., **22**, 1529 (1984).

138. J. B. Class, S. G. Chu, J. Appl. Polym. Sci., **30**, 805 (1985).
139. J. B. Class, S. G. Chu, J. Appl. Polym. Sci., **30**, 825 (1985).
140. J. B. Class, S. G. Chu, J. Appl. Polym. Sci., **30**, 815 (1985).
141. M. M. Coleman, J. Hu, Y. Park, P. C. Painter, Macromolecules, **29**, 1659 (1988).
142. M. M. Coleman, J. Y. Lee, C. J. Serman, Z. Wang, P. C. Painter, Polymer, **30**, 1298 (1989).
143. M. M. Coleman, A. M. Lichkus, P. C. Painter, Macromolecules, **22**, 586 (1989).
144. M. M. Coleman, C. J. Serman, P. C. Painter, Macromolecules, **20**, 226 (1987).
145. M. M. Coleman, P. J. Shrovanek, P. C. Painter, Macromolecules, **21**, 59 (1988).
146. M. M. Coleman, Y. Xu, P. C. Painter, Am. Chem. Soc., Polym. Mater. Sci. Eng., **64**, 28 (1991).
147. M. M. Coleman, H. Zhang, Y. Xu, P. C. Painter, Am. Chem. Soc., Polym. Prepr., **32** (2), 44 (1991).
148. G. Cong, Y. Huang, W. J. MacKnight, F. E. Karasz, Macromolecules, **19**, 2675 (1986).
149. P. J. Corish, Rubber Chem. Technol., **40**, 324 (1967).
150. P. Cousin, R. E. Prud'homme, Eur. Polym. J., **18**, 957 (1982).
151. P. Cousin, R. E. Prud'homme, in: D. R. Paul, L. H. Sperling, (Eds.), "Multicomponent Polymer Materials", Am. Chem. Soc., Adv. Chem. Ser., vol. 211, Washington, DC, 1986.
152. J. M. G. Cowie, E. M. Elexpuru, Eur. Polym. J., **28**, 623 (1992).
153. J. M. G. Cowie, E. M. Elexpuru, J. H. Harris, I. J. McEwen, Makromol. Chem., Rapid Commun., **10**, 691 (1989).
154. J. M. G. Cowie, E. M. Elexpuru, I. J. McEwen, J. Polym. Sci., Polym. Phys. Ed., **29**, 407 (1991).
155. J. M. G. Cowie, E. M. Elexpuru, I. J. McEwen, Polymer, **33**, 1993 (1992).
156. J. M. G. Cowie, R. Ferguson, M. D. Fernandez, M. J. Fernandez, I. J. McEwen, Macromolecules, **25**, 3170 (1992).
157. J. M. G. Cowie, R. Ferguson, I. J. McEwen, Polym. Commun., **32**, 293 (1991).
158. J. M. G. Cowie, M. D. Fernandez, M. J. Fernandez, I. J. McEwen, Eur. Polym. J., **28**, 145 (1992).
159. J. M. G. Cowie, M. D. Fernandez, M. J. Fernandez, I. J. McEwen, Polymer, **33**, 2744 (1992).
160. J. M. G. Cowie, M. T. Garay, D. Lath, I. J. McEwen, Brit. Polym. J., **21**, 81 (1989).
161. J. M. G. Cowie, J. H. Harris, Polymer, **33**, 4592 (1992).
162. J. M. G. Cowie, J. H. Harris, I. J. McEwen, Macromolecules, **25**, 5287 (1992).
163. J. M. G. Cowie, D. Lath, Polym. Commun., **28**, 300 (1987).
164. J. M. G. Cowie, D. Lath, Makromol. Chem., Macromol. Symp., **16**, 103 (1988).
165. J. M. G. Cowie, G. Li, R. Ferguson, I. J. McEwen, J. Polym. Sci., Polym. Phys. Ed., **30**, 1351 (1992).
166. J. M. G. Cowie, I. J. McEwen, L. Nadvornik, Macromolecules, **23**, 5106 (1990).
167. J. M. G. Cowie, S. Miachon, Macromolecules, **25**, 3295 (1992).
168. J. M. G. Cowie, V. M. C. Reid, I. J. McEwen, Polymer, **31**, 905 (1990).
169. J. M. G. Cowie, V. M. C. Reid, I. J. McEwen, Polymer, **31**, 486 (1990).
170. J. M. G. Cowie, A. A. N. Reilly, Polymer, **33**, 4814 (1992).
171. J. M. G. Cowie, S. Saeki, Polym. Bull., **6**, 75 (1981).
172. G. Crevecoeur, G. Groeninckx, Macromolecules, **24**, 1190 (1991).
173. G. Crone, A. Natansohn, Polym. Bull., **27**, 73 (1991).
174. C. A. Cruz, J. W. Barlow, D. R. Paul, Macromolecules, **12**, 726 (1979).
175. C. A. Cruz, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **25**, 1549 (1980).
176. C. A. Cruz, D. R. Paul, J. W. Barlow, J. Appl. Polym. Sci., **24**, 2101 (1979).
177. C. A. Cruz, D. R. Paul, J. W. Barlow, J. Appl. Polym. Sci., **23**, 589 (1979).
178. E. Cuddkhy, J. Moacanin, A. Rembaum, J. Appl. Polym. Sci., **9**, 1385 (1965).
179. S. Das, F. Rodriguez, Am. Chem. Soc., Polym. Prepr., **28** (2), 140 (1987).
180. P. B. Dave, N. J. Ashar, R. A. Gross, S. P. McCarthy, Am. Chem. Soc., Polym. Prepr., **31** (1), 442 (1990).
181. D. D. Davis, G. H. Taylor, T. K. Kwei, J. Appl. Polym. Sci., **26**, 2001 (1981).
182. H. K. de Decker, D. J. Sabatine, Rubber Age, **99** (4), 73 (1967).
183. H. de Jager, G. ten Brinke, Macromolecules, **24**, 3454 (1991).
184. M. T. DeMeuse, E. C. Chenevey, Z. H. Ophir, J. J. Rafalko, M. I. Haider, Am. Chem. Soc., Polym. Mater. Sci. Eng., **59**, 1714 (1988).
185. M. T. DeMeuse, M. Jaffe, Am. Chem. Soc., Polym. Prepr., **30** (2), 540 (1989).
186. B. D. Dean, J. Appl. Polym. Sci., **30**, 4193 (1985).
187. B. D. Dean, J. Appl. Polym. Sci., **33**, 2259 (1987).
188. B. D. Dean, J. Appl. Polym. Sci., **34**, 887 (1987).
189. B. D. Dean, J. E. Harris, J. Appl. Polym. Sci., **46**, 745 (1992).
190. J. Y. Decroix, A. Piloz, Rev. Caoutsch. Plast., **54**, 91 (1977).
191. G. Defieuw, G. Groeninckx, H. Reynaers, Polymer, **30**, 595 (1989).
192. G. Defieuw, G. Groeninckx, H. Reynaers, Polymer, **30**, 2158 (1989).
193. G. Defieuw, G. Groeninckx, H. Reynaers, Polymer, **30**, 2164 (1989).
194. G. DiPaola-Baranyi, P. Degre, Macromolecules, **14**, 1456 (1981).
195. L. C. Dickinson, H. Yang, C. W. Chu, R. S. Stein, J. C. W. Chien, Macromolecules, **20**, 1757 (1987).
196. D. Dimitrova, A. Aivazov, Y. Zelenev, God. Vissh. Khim. Tekhnol. Inst., Burgas. Bulg., **8**, 57 (1971) [Chem. Abstr., **79**, 127188d (1973)].
197. L. Dinililuc, C. DeKesel, C. David, Eur. Polym. J., **28**, 1365 (1992).

198. S. Djadoun, Polym. Bull., **9**, 313 (1983).
199. A. Dobry, F. Boyer-Kawenoki, J. Polym. Sci., **2**, 90 (1974).
200. C. P. Doube, D. J. Walsh, Polymer, **20**, 1115 (1979).
201. E. Drioli, A. Apicella, H. B. Hopfenberg, E. Martuscelli, L. Nicolais, Chim. Ind. (Milan), **60**, 975 (1978).
202. X. Drujon, G. Reiss, H. K. Hall, Jr., New Polym. Mater., **3**, 283 (1992).
203. D. J. Dunn, S. Krause, J. Polym. Sci., Polym. Lett. Ed., **12**, 591 (1974).
204. M. Ebert, R. W. Garbella, J. H. Wendroff, Makromol. Chem., Rapid Commun., **7**, 65 (1986).
205. S. L. Edie, H. Marand, Am. Chem. Soc., Polym. Prepr., **32** (3), 329 (1991).
206. S. H. Egboh, Am. Chem. Soc., Polym. Mater. Sci. Eng., **49**, 45 (1983).
207. J. I. Eguiazabal, E. Calahorra, M. Cortazar, G. M. Guzman, Makromol. Chem., **187**, 2439 (1986).
208. J. I. Eguiazabal, J. J. Iruin, M. Cortazar, J. Appl. Polym. Sci., **32**, 5945 (1986).
209. J. I. Eguiazabal, J. J. Iruin, M. Cortazar, G. M. Guzman, Makromol. Chem., **185**, 1761 (1984).
210. A. Eisenberg, P. Smith, Z. L. Zhou, Polym. Eng. Sci., **22**, 1117 (1982).
211. T. S. Ellis, Polymer, **29**, 2015 (1988).
212. T. S. Ellis, Macromolecules, **22**, 742 (1989).
213. T. S. Ellis, Polymer, **31**, 1058 (1990).
214. T. S. Ellis, Polym. Eng. Sci., **30**, 998 (1990).
215. T. S. Ellis, Polymer, **33**, 1469 (1992).
216. C. Elmqvist, S. E. Svanson, Eur. Polym. J., **11**, 789 (1975).
217. A. D. English, P. Smith, D. E. Avelson, Polymer, **26**, 1523 (1985).
218. R. Erro, M. Gaztelumendi, J. Nazabal, New Polym. Mater., **3**, 87 (1992).
219. A. Escala, E. Balitzer, R. S. Stein, Am. Chem. Soc., Polym. Prepr., **19** (1), 152 (1978).
220. A. Escala, R. S. Stein, in: S. L. Cooper, G. M. Estes, (Eds.), "Multiphase Polymers", Am. Chem. Soc., Adv. Chem. Ser., vol. 176, Washington, DC, 1979.
221. S. R. Fahrenhlotz, T. K. Kwei, Macromolecules, **14**, 1076 (1981).
222. D. Feldman, M. Rusu, Eur. Polym. J., **10**, 41 (1974).
223. A. C. Fernandes, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **29**, 1971 (1984).
224. A. C. Fernandes, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **32**, 5481 (1986).
225. A. C. Fernandes, J. W. Barlow, D. R. Paul, Polymer, **27**, 1788 (1986).
226. A. C. Fernandes, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **32**, 6073 (1986).
227. A. C. Fernandes, J. W. Barlow, D. R. Paul, Polymer, **27**, 1799 (1986).
228. A. C. Fernandes, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **32**, 5357 (1986).
229. M. E. Fowler, J. W. Barlow, D. R. Paul, Polymer, **28**, 1177 (1987).
230. M. E. Fowler, D. R. Paul, L. A. Cohen, W. T. Freed, J. Appl. Polym. Sci., **37**, 513 (1989).
231. C. W. Frank, W. Zin, in: C. E. Hoyle, J. M. Torkelson, (Eds.), "Photophysics of Polymers", Am. Chem. Soc., Symp. Ser., vol. 358, Washington, DC, 1987.
232. J. R. Fried, G. A. Hanna, Polym. Eng. Sci., **22**, 705 (1982).
233. J. R. Fried, G. A. Hanna, H. Kalkanoglu, in: K. Solc, (Ed.), "Polymer Compatibility and Incompatibility – Principles and Practices", MMI Press, Harwood, 1982.
234. J. R. Fried, F. E. Karasz, W. J. MacKnight, Macromolecules, **11**, 150 (1978).
235. J. R. Fried, T. Lorenz, A. Ramdas, Polym. Eng. Sci., **25**, 104 (1985).
236. F. Friese, Plaste Kautsch., **15**, 646 (1968).
237. K. Fujimoto, N. Yoshimiya, Nippon Gomu Kyokaishi, **38**, 284 (1965); Rubber Chem. Technol., **41**, 669 (1968).
238. K. Fujimoto, N. Yoshimura, Nippon Gomu Kyokaishi, **39**, 919 (1966); Rubber Chem. Technol., **41**, 1109 (1968).
239. K. Fujioka, N. Noethiger, C. L. Beatty, Y. Baba, A. Kagemoto, in: C. D. Han (Ed.), "Polymer Blends and Composites in Multiphase Systems", Am. Chem. Soc., Adv. Chem. Ser., vol. 206, Washington, DC, 1984.
240. T. Fukai, J. C. Yang, T. Kyu, S. Z. D. Cheng, S. K. Lee, S. L. Hsu, F. W. Harris, Polymer, **33**, 3621 (1992).
241. M. Galin, Makromol. Chem., Rapid Commun., **5**, 119 (1984).
242. M. Galin, Makromol. Chem., **188**, 1391 (1987).
243. M. Galin, Makromol. Chem., Rapid Commun., **8**, 411 (1987).
244. M. Galin, L. Maslinko, Eur. Polym. J., **23**, 923 (1987).
245. M. E. Galvin, Macromolecules, **24**, 6354 (1991).
246. D. Gracia, J. Polym. Sci., Polym. Phys. Ed., **24**, 1577 (1986).
247. A. Garton, S. Wang, R. A. Weiss, J. Polym. Sci., Polym. Phys. Ed., **26**, 1545 (1988).
248. M. Gaztelumendi, I. Mondragon, J. Nazabal, Makromol. Chem., Macromol. Symp., **20/21**, 269 (1988).
249. B. D. Gesner, in: N. M. Bikales, (Ed.), "Encyclopedia of Polymer Science and Technology", Wiley-Interscience, New York, vol. 10, 694 (1969).
250. A. V. Gevorkyan, M. E. Ovsepyan, A. S. Safarov, Arm. Khim. Zh., **31**, 107 (1978).
251. Y. K. Godovskii, N. P. Bessonova, Vysokomol. Soedin. A, **21**, 2293 (1979); Polym. Sci. USSR, **21**, 2531 (1979).
252. Y. K. Godovskii, Z. F. Zharikova, Y. M. Malinskii, Vysokomol. Soedin. A, **23**, 133 (1981); Polym. Sci. USSR, **23**, 149 (1981).
253. S. H. Goh, Polym. Bull., **17**, 221 (1987).
254. S. H. Goh, W. W. Y. Lau, C. S. Lee, Polym. Commun., **32**, 202 (1991).
255. S. H. Goh, W. W. Y. Lau, C. S. Lee, Polym. Bull., **26**, 319 (1991).
256. S. H. Goh, W. W. Y. Lau, C. S. Lee, Polym. Bull., **29**, 524 (1992).
257. S. H. Goh, S. Y. Lee, Eur. Polym. J., **23**, 315 (1987).
258. S. H. Goh, S. Y. Lee, Thermochim. Acta, **123**, 3 (1988).
259. S. H. Goh, S. Y. Lee, Eur. Polym. J., **24**, 923 (1988).
260. S. H. Goh, S. Y. Lee, Eur. Polym. J., **25**, 571 (1989).

261. S. H. Goh, S. Y. Lee, Eur. Polym. J., **25**, 997 (1989).
262. S. H. Goh, S. Y. Lee, J. Appl. Polym. Sci., **41**, 1391 (1990).
263. S. H. Goh, S. Y. Lee, Eur. Polym. J., **26**, 715 (1990).
264. S. H. Goh, S. Y. Lee, Polym. Bull., **23**, 643 (1990).
265. S. H. Goh, S. Y. Lee, Makromol. Chem., **191**, 3081 (1990).
266. S. H. Goh, S. Y. Lee, Polym. Commun., **31**, 463 (1990).
267. S. H. Goh, S. Y. Lee, Y. F. Chong, Polym. Bull., **25**, 257 (1991).
268. S. H. Goh, S. Y. Lee, Y. F. Chong, M. K. Neo, C. L. Leong, Macromolecules, **24**, 806 (1991).
269. S. H. Goh, S. Y. Lee, C. L. Leong, J. Appl. Polym. Sci., **44**, 199 (1992).
270. S. H. Goh, S. Y. Lee, S. M. Low, Eur. Polym. J., **28**, 661 (1992).
271. S. H. Goh, S. Y. Lee, M. K. Neo, J. Appl. Polym. Sci., **40**, 2243 (1990).
272. S. H. Goh, S. Y. Lee, K. S. Siow, J. Appl. Polym. Sci., **31**, 2055 (1986).
273. S. H. Goh, S. Y. Lee, K. S. Siow, M. K. Neo, Polymer, **31**, 1065 (1990).
274. S. H. Goh, S. Y. Lee, K. S. Siow, C. L. Pua, J. Appl. Polym. Sci., **33**, 353 (1987).
275. S. H. Goh, L. S. Lim, Eur. Polym. J., **26**, 711 (1990).
276. S. H. Goh, T. T. Nguyen, K. S. Siow, J. Appl. Polym. Sci., **29**, 1463 (1984).
277. S. H. Goh, D. R. Paul, J. W. Barlow, Polym. Eng. Sci., **22**, 34 (1982).
278. S. H. Goh, D. R. Paul, J. W. Barlow, J. Appl. Polym. Sci., **27**, 1091 (1982).
279. S. H. Goh, K. S. Siow, J. Appl. Polym. Sci., **32**, 3407 (1986).
280. S. H. Goh, K. S. Siow, Thermochim. Acta, **105**, 191 (1986).
281. S. H. Goh, K. S. Siow, J. Appl. Polym. Sci., **33**, 1849 (1987).
282. S. H. Goh, K. S. Siow, Polym. Bull., **17**, 453 (1988).
283. S. H. Goh, K. S. Siow, Polym. Bull., **20**, 393 (1988).
284. S. H. Goh, K. S. Siow, Polym. Bull., **23**, 205 (1990).
285. S. H. Goh, K. S. Siow, S. Y. Lee, Eur. Polym. J., **27**, 921 (1991).
286. S. H. Goh, K. S. Siow, K. S., S. Y. Lee, Eur. Polym. J., **28**, 657 (1992).
287. S. H. Goh, K. S. Siow, Y. H. Leong, Eur. Polym. J., **21**, 915 (1985).
288. S. H. Goh, K. S. Siow, T. T. Nguyen, A. Nam, Eur. Polym. J., **20**, 65 (1984).
289. S. H. Goh, K. S. Siow, K. S. Yap, J. Appl. Polym. Sci., **29**, 99 (1984).
290. S. H. Goh, K. S. Siow, K. S. Yap, Thermochim. Acta, **102**, 281 (1986).
291. A. Golovoy, M. F. Cheung, Polym. Eng. Sci., **29**, 85 (1989).
292. P. Grece, E. Martuscelli, Polymer, **30**, 1475 (1989).
293. R. A. Grinsted, J. L. Koenig, J. Polym. Sci., Polym. Phys. Ed., **28**, 177 (1990).
294. J. Grobelny, D. M. Rice, F. E. Karasz, W. J. MacKnight, Macromolecules, **23**, 2139 (1990).
295. G. Groeninckx, M. Vandermaliene, G. Defieuw, H. Reynaers, H., Am. Chem. Soc., Polym. Prepr., **29** (1), 438 (1988).
296. G. Grose, K. Friese, quoted in K. Friese, Plaste Kaut., **12**, 90 (1965).
297. G. Guerra, S. Choe, D. J. Williams, F. E. Karasz, W. J. MacKnight, Macromolecules, **21**, 231 (1988).
298. G. Guerra, F. E. Karasz, W. J. MacKnight, Macromolecules, **19**, 1935 (1986).
299. G. Guerra, D. J. Williams, F. E. Karasz, W. J. MacKnight, J. Polym. Sci., Polym. Phys. Ed., **26**, 301 (1988).
300. Q. Guo, Makromol. Chem., Rapid Commun., **11**, 279 (1990).
301. Q. Guo, Makromol. Chem., **191**, 2639 (1990).
302. Q. Guo, Polym. Commun., **31**, 217 (1990).
303. Q. Guo, Eur. Polym. J., **26**, 1329 (1990).
304. Q. Guo, Eur. Polym. J., **26**, 1333 (1990).
305. Q. Guo, Polym. Commun., **32**, 62 (1991).
306. Q. Guo, Eur. Polym. J., **28**, 1395 (1992).
307. Q. Guo, Eur. Polym. J., **28**, 1049 (1992).
308. Q. Guo, J. Huang, T. Chen, Polym. Bull., **20**, 517 (1988).
309. Q. Guo, J. Huang, T. Chen, Polym. Commun., **31**, 115 (1990).
310. Q. Guo, J. Huang, T. Chen, Z. Feng, Polym. Commun., **31**, 240 (1990).
311. Q. Guo, J. Huang, T. Chen, H. Zhang, Y. Yang, C. Hou, Z. Feng, Polym. Eng. Sci., **30**, 44 (1990).
312. Q. Guo, J. Huang, B. Li, T. Chen, H. Zhang, Z. Feng, Polymer, **32**, 58 (1991).
313. Q. Guo, X. Peng, Z. Wang, Polymer, **32**, 53 (1991).
314. Q. Guo, H. Xu, D. Ma, S. Wang, Eur. Polym. J., **26**, 67 (1990).
315. Q. Guo, M. Yu, Z. Feng, Polymer, **33**, 893 (1992).
316. W. Guo, J. S. Higgins, Polymer, **31**, 699 (1990).
317. W. Guo, J. S. Higgins, Polymer, **32**, 2115 (1991).
318. C. S. Ha, W. J. Cho, R. J. Roe, Pollimo, **14**, 322 (1990).
319. C. S. Ha, R. E. Prud'homme, W. J. Cho, Pollimo, **14**, 506 (1990).
320. B. R. Hahn, J. H. Wendroff, Polymer, **26**, 1619 (1985).
321. J. L. Halary, M. H. Leviet, T. K. Kwei, E. M. Pearce, Macromolecules, **24**, 5939 (1991).
322. J. L. Halary, J. M. Ubrich, L. Monnerie, H. Yang, R. S. Stein, Polym. Commun., **26**, 73 (1985).
323. J. L. Halary, J. M. Ubrich, J. M. Nunzi, L. Monnerie, R. S. Stein, Polymer, **25**, 956 (1984).
324. N. Hamazaki, J. W. Cho, S. Miyata, Polymer J., **23**, 333 (1991).
325. C. F. Hammer, Macromolecules, **4**, 69 (1971).
326. C. C. Han, M. Okada, Y. Muroga, B. J. Bauer, Q. Tran-Cong, Polym. Eng. Sci., **26**, 1208 (1986).
327. C. C. Han, H. H. Yang, J. Appl. Polym. Sci., **33**, 1199 (1987).
328. C. D. Han, H. S. Chung, J. K. Kim, Polymer, **33**, 546 (1992).
329. M. Hara, G. J. Parker, Polymer, **33**, 4650 (1992).

330. D. Hardt, Brit. Polym. J., **1**, 225 (1969).
331. D. Hardt, C. Sueling, C. Linder, L. Morbitzer, Angew. Chem. Int. Ed., **21**, 174 (1982).
332. J. E. Harris, S. H. Goh, D. R. Paul, J. W. Barlow, J. Appl. Polym. Sci., **27**, 839 (1982).
333. J. E. Harris, L. M. Robeson, J. Polym. Sci., Polym. Phys. Ed., **25**, 311 (1987).
334. J. E. Harris, L. M. Robeson, J. Appl. Polym. Sci., **35**, 1877 (1988).
335. M. Hasegawa, S. Sakurai, M. Takenaka, T. Hashimoto, C. C. Han, Macromolecules, **24**, 1813 (1991).
336. M. Hasegawa, S. Akiyama, Polymer J., **20**, 471 (1988).
337. P. M. Hendrichs, J. Tribone, D. J. Massa, J. M. Hewitts, Macromolecules, **21**, 1282 (1988).
338. M. Hendrioulle-Granville, K. Kyuda, R. Jerome, Ph. Teyssie, F. C. de Schryver, Macromolecules, **23**, 1202 (1990).
339. J. J. Hickman, R. M. Ikeda, J. Polym. Sci., Polym. Phys. Ed., **11**, 1713 (1973).
340. S. Himuro, N. Sakamoto, S. Arichi, Polym. J., **24**, 1371 (1992).
341. H. Horiuchi, S. Irie, T. Nose, Polymer, **32**, 1970 (1991).
342. D. J. Hourston, I. D. Hughes, Polymer, **18**, 1175 (1977).
343. D. J. Hourston, I. D. Hughes, J. Appl. Polym. Sci., **21**, 3099 (1977).
344. D. J. Hourston, I. D. Hughes, J. Appl. Polym. Sci., **26**, 3467 (1981).
345. D. T. Hsieh, D. G. Peiffer, Polymer, **33**, 1210 (1992).
346. K. H. Hsieh, L. M. Chou, J. Appl. Polym. Sci., **38**, 645 (1989).
347. J. Hu, Y. Park, P. C. Painter, M. M. Coleman, Am. Chem. Soc., Polym. Prepr., **29** (1), 321 (1988).
348. S. R. Hu, T. Kyu, R. S. Stein, J. Polym. Sci., Polym. Phys. Ed., **25**, 71 (1987).
349. J. Huang, Q. Guo, Makromol. Chem., Rapid Commun., **11**, 613 (1990).
350. J. C. Huarng, K. Min, J. L. White, Polym. Eng. Sci., **28**, 1085 (1988).
351. J. C. Huarng, K. Min, J. L. White, Polym. Eng. Sci., **28**, 1590 (1988).
352. D. S. Hubbell, S. L. Copper, J. Appl. Polym. Sci., **21**, 303 (1977).
353. D. S. Hubbell, S. L. Copper, in: S. L. Cooper, G. M. Estes, (Eds.), "Multiphase Polymers", Am. Chem. Soc., Adv. Chem. Ser., vol. 176, Washington, DC, 1979.
354. M. B. Huglin, J. M. Rego, Polymer, **31**, 1269 (1990).
355. W. Huh, F. E. Karasz, Polym. Mater. Sci. Eng., **60**, 792 (1989).
356. W. Huh, F. E. Karasz, Macromolecules, **25**, 1057 (1992).
357. K. Ikawa, S. Hosoda, Polym. Networks Blends, **1**, 103 (1991).
358. K. Ikawa, S. Hosoda, Polym. J., **22**, 643 (1990).
359. K. H. Iller, W. Heckmann, J. Hambrecht, Colloid Polym. Sci., **262**, 557 (1984).
360. K. H. Illers, E. Koehnlein, Makromol. Chem., **187**, 2725 (1986).
361. R. L. Imken, D. R. Paul, J. W. Barlow, Polym. Eng. Sci., **16**, 593 (1976).
362. T. Inoue, F. Shomura, T. Ougizawa, K. Miyasaka, Rubber Chem. Technol., **58**, 873 (1985).
363. M. Iriarte, E. Espi, A. Etxeberria, M. Valero, M. J. Fernandez-Berridi, Macromolecules, **24**, 5546 (1991).
364. M. Iriarte, J. I. Iribarreu, A. Etxeberria, J. J. Iruin, Polymer, **30**, 1160 (1989).
365. M. Irie, R. Iga, Makromol. Chem., Rapid Commun., **7**, 751 (1986).
366. J. J. Iruin, J. I. Eguiazabal, G. M. Guzman, Eur. Polym. J., **25**, 1169 (1989).
367. K. Ishikawa, S. Kawahara, S. Akiyama, Polym. Networks Blends, **1**, 1 (1991).
368. H. Jager, E. J. Vorenkamp, G. Challa, Polym. Commun., **24**, 290 (1983).
369. V. Janarthanan, F. E. Karasz, W. J. MacKnight, Polymer, **33**, 3388 (1992).
370. V. Janarthanan, G. Thyagarajan, Polymer, **33**, 3593 (1992).
371. J. Janicke, A. Wlochowicz, Angew. Makromol. Chem., **168**, 9 (1989).
372. J. Grobelny, D. M. Rice, F. E. Karasz, W. J. MacKnight, Polym. Commun., **31**, 86 (1990).
373. D. Jaroszynska, R. Gaczynski, Plaste Kautsch., **23**, 581 (1976).
374. S. H. Jeon, T. Ree, J. Polym. Sci., Polym. Chem. Ed., **26**, 1419 (1988).
375. K. Jeremic, F. E. Karasz, W. J. MacKnight, New Polym. Mater., **3**, 163 (1992).
376. R. J. Jerome, Albert, Ph. Teyssie, Am. Chem. Soc., Polym. Prepr., **27** (1), 72 (1986).
377. M. Jiang, X. Cao, W. Chen, T. Yu, Polym. Bull., **21**, 599 (1989).
378. M. Jiang, X. Cao, T. Yu, Polymer, **27**, 1923 (1986).
379. B. S. Jin, S. C. Kim, D. S. Lee, Polym. J., **24**, 1189 (1992).
380. J. I. Jin, E. J. Choi, K. Y. Lee, Polym. J., **18**, 99 (1986).
381. Y. Jin, R. Y. M. Huang, J. Appl. Polym. Sci., **36**, 1799 (1988).
382. W. H. Jo, C. A. Cruz, D. R. Paul, J. Polym. Sci., Polym. Phys. Ed., **27**, 1057 (1989).
383. W. H. Jo, H. G. Kim, J. Polym. Sci., Polym. Phys. Ed., **29**, 1579 (1991).
384. W. H. Jo, Y. K. Kwon, I. H. Kwon, Macromolecules, **24**, 4708 (1991).
385. W. H. Jo, S. C. Lee, M. S. Lee, Polym. Bull., **21**, 183 (1989).
386. W. H. Jo, S. J. Park, I. H. Kwon, Polym. Int., **29**, 173 (1992).
387. W. H. Jo, H. Yim, I. H. Kwon, T. W. Son, Polym. J., **24**, 519 (1992).
388. W. H. Jo, J. T. Yoon, S. C. Lee, Polym. J., **23**, 1243 (1991).
389. E. John, T. Ree, J. Polym. Sci., Polym. Chem. Ed., **28**, 385 (1990).
390. A. H. Jorgenson, L. A. Chandler, E. A. Collins, Rubber Chem. Technol., **46**, 1087 (1973).
391. J. H. Jou, P. T. Huang, Polym. J., **22**, 909 (1990).

392. R. H. Jung, D. J. Stein, Prepr. IUPAC Symp. Aberdeen, 411 (1973).
393. J. J. Jutier, E. Lemieaux, R. E. Prud'homme, J. Polym. Sci., Polym. Phys. Ed., **26**, 1313 (1988).
394. S. N. Kaklas, D. D. Sotiropoulou, J. K. Kallitsis, N. K. Kalfoglou, Polymer, **32**, 66 (1991).
395. N. K. Kalfoglou, Angew. Makromol. Chem., **118**, 19 (1983).
396. N. K. Kalfoglou, A. G. Margaritis, Angew. Makromol. Chem., **125**, 135 (1984).
397. N. K. Kalfoglou, D. D. Sotiropoulu, A. G. Margartis, Eur. Polym. J., **24**, 389 (1988).
398. L. E. Kalinina, V. I. Alekseenko, S. S. Voyutskii, Kolloid. Zh., **18**, 691 (1956); Colloid J. USSR, **18**, 689 (1956).
399. L. E. Kalinina, V. I. Alekseenko, S. S. Voyutskii, Kolloid. Zh., **18**, 180 (1956); Colloid J. USSR, **18**, 171 (1956).
400. J. K. Kallistsis, N. K. Kalfoglou, Angew. Makromol. Chem., **148**, 103 (1987).
401. J. K. Kallistsis, N. K. Kalfoglou, J. Appl. Polym. Sci., **37**, 453 (1989).
402. R. P. Kambour, J. T. Bendler, R. C. Bopp, Macromolecules, **16**, 753 (1983).
403. R. P. Kambour, P. E. Gundlach, I. C. W. Wang, D. M. White, G. W. Yeager, Polym. Commun., **29**, 170 (1988).
404. H. W. Kammer, J. Kressler, B. Kressler, D. Scheller, H. Kroschwitz, G. Schmidt-Naake, Acta Polym., **40**, 75 (1989).
405. H. W. Kammer, J. Piglowski, Acta Polym., **40**, 363 (1989).
406. P. Kanakalatha, K. Vijayan, M. K. Sridhar, A. K. Singh, Polymer, **24**, 621 (1983).
407. D. P. Kang, W. J. Cho, Pollimo, **13**, 638 (1989).
408. D. P. Kang, C. S. Ha, W. J. Cho, J. Polym. Sci., Polym. Chem. Ed., **27**, 1401 (1989).
409. D. P. Kang, C. S. Ha, R. E. Prud'homme, W. J. Cho, Pollimo, **12**, 634 (1988).
410. H. S. Kang, W. J. MacKnight, F. E. Karasz, Am. Chem. Soc., Polym. Prepr., **28** (2), 134 (1987).
411. Y. Kano, S. Akiyama, Polym. Bull., **29**, 97 (1992).
412. F. E. Karasz, W. J. MacKnight, J. Appl. Polym. Sci., **32**, 4423 (1986).
413. F. E. Karasz, W. J. MacKnight, J. J. Tkacik, Am. Chem. Soc., Polym. Prepr., **15**, 415 (1974).
414. R. J. Karcha,R. S. Porter, J. Polym. Sci., Polym. Phys. Ed., **27**, 2153 (1989).
415. R. J. Karcha, R. S. Porter, Polymer, **33**, 4866 (1992).
416. V. A. Kargin, J. Polym. Sci., Part C., **4**, 1601 (1963).
417. I. A. Katime, M. S. Anasagasti, M. C. Peleteiro, R. Valenciano, Eur. Polym. J., **23**, 907 (1987).
418. J. D. Katsaros, M. F. Malone, H. H. Winter, Polym. Bull., **16**, 83 (1986).
419. S. Kawahara, S. Akiyama, Polym. J., **22**, 361 (1990).
420. S. Kawahara, S. Akiyama, K. Ueda, Polym. J., **21**, 221 (1989).
421. T. Kawai, Kogyo Kagaka Zasshi, **59**, 779 (1956).
422. J. F. Kenney, J. Polym. Sci., Chem. Ed., **14**, 123 (1976).
423. J. F. Kenney, in: L. H. Sperling (Ed.), "Recent Advances in Polymer Blends, Grafts, and Blocks", Plenum Press, New York, 1974, p. 117.
424. J. F. Kenny, Org. Coatings Plastics Chem. Prepr. Am. Chem. Soc., **37**, 615 (1977).
425. R. J. Kern, J. Polym. Sci., **33**, 524 (1958).
426. H. Keskkula, D. R. Paul, J. Appl. Polym. Sci., **31**, 1189 (1986).
427. C. K. Kim, D. R. Paul, Macromolecules, **25**, 3097 (1992).
428. C. K. Kim, D. R. Paul, Polymer, **33**, 1630 (1992).
429. C. K. Kim, D. R. Paul, Polymer, **33**, 2089 (1992).
430. C. K. Kim, D. R. Paul, Polymer, **33**, 4929 (1992).
431. C. K. Kim, D. R. Paul, Polymer, **33**, 4941 (1992).
432. H. I. Kim, E. M. Pearce, T. K. Kwei, Macromolecules, **22**, 3498 (1989).
433. H. I. Kim, E. M. Pearce, T. K. Kwei, Macromolecules, **22**, 3374 (1989).
434. H. I. Kim, J. R. Pennachia, T. K. Kwei, E. M. Pearce, E. M., Am. Chem. Soc., Polym. Prepr., **32** (3), 343 (1991).
435. J. H. Kim, J. W. Barlow, D. R. Paul, Polym. Eng. Sci., **29**, 581 (1989).
436. J. H. Kim, J. W. Barlow, D. R. Paul, J. Polym. Sci., Polym. Phys. Ed., **27**, 223 (1989).
437. J. H. Kim, J. W. Barlow, D. R. Paul, J. Polym. Sci., Polym. Phys. Ed., **27**, 2211 (1989).
438. M. Kimura, R. S. Porter, Org. Coatings, Plastics Chem. Prepr. Am. Chem. Soc., **45**, 84 (1981).
439. M. Kimura, R. S. Porter, G. Salee, J. Polym. Sci., Polym. Phys. Ed., **21**, 367 (1983).
440. L. A. Kleintjens, unpublished results, quoted in Ref. K58.
441. S. Klotz, R. H. Schuster, H. J. Cantow, Makromol. Chem., **187**, 1491 (1986).
442. S. Kobayashi, M. Kaku, T. Saegusa, Macromolecules, **21**, 334 (1988).
443. S. Kobayashi, S. Tasaka, S. Miyata, Kobunshi Ronbunshu, **44**, 695 (1987).
444. P. R. Kohl, A. M. Seifert, G. P. Hellmann, J. Polym. Sci., Polym. Phys. Ed., **28**, 1309 (1990).
445. J. V. Koleske, in: D. R. Paul, S. Newman (Eds.), "Polymer Blends; (Vol. 2)", Academic Press, New York, Chap. 22 (1978).
446. J. V. Koleske, R. D. Lundberg, J. Polym. Sci. A-2, **7**, 795 (1969).
447. F. Kollinsky, G. Markert, Makromol. Chem., **121**, 117 (1969).
448. F. Kollinsky, G. Markert, in: N. A. J. Platzer (Ed.), "Multicomponent Polymer Systems", Am. Chem. Soc., Adv. Chem. Ser., vol. 99, Washington, DC, 1971.
449. M. Kolowski, W. Laskawski, Angew. Makromol. Chem., **91**, 29 (1980).
450. R. Koningsveld, H. A. G. Chermin, M. Gordon, Proc. Royal Soc. A, **319**, 331 (1970).
451. R. Koningsveld, L. A. Kleintjens, H. M. Schoffeleers, Pure Appl. Chem., **39**, 1 (1974).
452. K. Kosai, T. Higashino, Nippon Setchaku Kyokai Shi, **11**, 2 (1975).

453. K. Kosai, T. Higashino, N. Nishioka, Shikizai Kyokaishi, **50**, 76 (1977).
454. D. L. Kotzev, E. M. Pearce, T. K. Kwei, J. Appl. Polym. Sci., **29**, 4443 (1984).
455. G. Kraus, K. W. Rollman, in: N. A. J. Platzer (Ed.), "Multicomponent Polymer Systems", Am. Chem. Soc., Adv. Chem. Ser., vol. 99, Washington, DC, 1971.
456. S. Krause, N. Roman, J. Polym. Sci., Part: A, **3**, 1631 (1965).
457. J. Kressler, H. W. Kammer, Acta Polym., **38**, 600 (1987).
458. J. Kressler, H. W. Kammer, Polym. Bull., **19**, 283 (1988).
459. J. Kressler, H. W. Kammer, Herzog, H. Heyde, Acta Polym., **41**, 1 (1990).
460. J. Kressler, H. W. Kammer, K. Klostermann, Polym. Bull., **15**, 113 (1986).
461. J. Kressler, H. W. Kammer, U. Morgenstern, B. Litauszki, W. Berger, Makromol. Chem., **191**, 243 (1990).
462. J. Kressler, H. W. Kammer, G. Schmidt-Naake, K. Herzog, Polymer, **29**, 686 (1988).
463. M. Kryszewski, J. Jachowciz, M. Malanga, O. Vogl, Polymer, **23**, 271 (1982).
464. Y. Kumagai, Y. Doi, Polym. Deg. Stab., **35**, 87 (1992).
465. Y. Kumagai, Y. Doi, Polym. Deg. Stab., **36**, 241 (1992).
466. A. K. Kundu, S. S. Ray, B. Adhikari, S. Maiti, Eur. Polym. J., **22**, 369 (1986).
467. A. K. Kundu, S. S. Ray, S. Maiti, S., Eur. Polym. J., **22**, 821 (1986).
468. A. K. Kundu, S. S. Ray, R. N. Mukherjea, S. Maiti, Angew. Makromol. Chem., **143**, 197 (1986).
469. C. M. Kuo, S. J. Clarson, Am. Chem. Soc., Polym. Prepr., **31**, 550 (1990).
470. C. M. Kuo, S. J. Clarson, Macromolecules, **25**, 2192 (1992).
471. J. A. Kuphal, L. H. Sperling, L. M. Robeson, J. Appl. Polym. Sci., **42**, 1525 (1991).
472. N. Kuwahara, M. Ishizawa, S. Saeki, M. Kaneko, Rept. Prog. Polym. Phys. Japan, **19**, 9 (1976).
473. T. K. Kwei, J. Polym. Sci., Polym. Lett. Ed., **22**, 307 (1984).
474. T. K. Kwei, H. L. Frisch, W. Radigan, S. Vogel, Macromolecules, **10**, 157 (1977).
475. T. K. Kwei, T. Nishi, R. F. Roberts, Macromolecules, **7**, 667 (1974).
476. T. K. Kwei, G. D. Patterson, T. T. Wang, Macromolecules, **9**. 780 (1976).
477. T. Kyu, T. J. Chen, H. S. Park, J. L. White, J. Appl. Polym. Sci., **37**, 201 (1989).
478. T. Kyu, S. R. Hu, R. S. Stein, J. Polym. Sci., Polym. Phys. Ed., **25**, 89 (1987).
479. T. Kyu, D. S. Lim, J. Polym. Sci., Polym. Lett. Ed., **27**, 421 (1989).
480. T. Kyu, D. Park, W. Cho, J. Appl. Polym. Sci., **44**, 2233 (1992).
481. T. Kyu, J. M. Saldanha, J. Polym. Sci., Polym. Lett. Ed., **26**, 33 (1988).
482. T. Kyu, J. M. Saldanha, J. Polym. Sci., Polym. Phys. Ed., **28**, 97 (1990).
483. T. Kyu, P. Vadhar, J. Appl. Polym. Sci., **32**, 5575 (1986).
484. V. Landi, Appl. Polym. Symp., **25**, 223 (1974).
485. V. Landi, Rubber Chem. Technol., **45**, 222 (1972).
486. C. J. T. Landry, P. M. Hendrichs, Macromolecules, **22**, 2157 (1989).
487. C. J. T. Landry, D. M. Teegarden, Macromolecules, **24**, 4310 (1991).
488. C. J. T. Landry, H. Yang, J. S. Machell, Polymer, **32**, 44 (1991).
489. D. Lath, J. M. G. Cowie, E. Lathova, Polym. Bull., **28**, 361 (1992).
490. W. W. Y. Lau, S. H. Teoh, S. H. Goh, Brit. Polym. J., **20**, 323 (1988).
491. C. Le Menestrel, D. E. Bhagwagar, P. C. Painter, M. M. Coleman, M. M., J. F. Graf, Macromolecules, **25**, 7107 (1992).
492. D. S. Lee, G. Quin, Polym. J., **21**, 751 (1989).
493. H. S. Lee, T. Kyu, Am. Chem. Soc., Polym. Prepr., **30** (1), 308 (1989).
494. J. Y. Lee, P. C. Painter, M. M. Coleman, Macromolecules, **21**, 346 (1988).
495. J. Y. Lee, P. C. Painter, M. M. Coleman, Macromolecules, **21**, 954 (1988).
496. J. Y. Lee, P. C. Painter, M. M. Coleman, Am. Chem. Soc., Polym. Prepr., **29** (1), 313 (1988).
497. M. S. Lee, S. C. Lee, S. H. Chae, W. H. Jo, Macromolecules, **25**, 4339 (1992).
498. S. C. Lee, M. S. Lee, W. H. Jo, J. Polym. Sci., Polym. Phys. Ed., **29**, 759 (1991).
499. S. S. Lee, K. H. Chung, K. S. Kim, J. C. Jung, Angew. Makromol. Chem., **164**, 171 (1988).
500. S. S. Lee, K. H. Chung, K. S. Kim, J. C. Jung, Angew. Makromol. Chem., **165**, 89 (1989).
501. S. Y. Lee, M. Y. Low, S. H. Goh, Eur. Polym. J., **27**, 1379 (1991).
502. T. H. Lee, H. Marand, Am. Chem. Soc., Polym. Prepr., **32** (3), 316 (1991).
503. J. Leffingwell, C. Thies, H. Gertzman, Am. Chem. Soc., Polym. Prepr., **14**, 596 (1973).
504. M. H. Lehr, Polym. Eng. Sci., **25**, 1056 (1985).
505. M. H. Lehr, Polym. Eng. Sci., **26**, 947 (1986).
506. E. Lemieux, R. E. Prud'homme, R. Forte, R. Jerome, Ph. Teyssie, Macromolecules, **21**, 2148 (1988).
507. L. Leung, D. J. Williams, F. E. Karasz, W. J. MacKnight, Polym. Bull., **16**, 457 (1986).
508. W. Li, L. Shi, D. Shen, Y. Wu, J. Zheng, Polymer, **30**, 604 (1989).
509. K. Liang, G. Banhegyi, F. E. Karasz, W. J. MacKnight, J. Polym. Sci., Polym. Phys. Ed., **29**, 649 (1991).
510. K. Liang, J. Grebowicz, F. E. Karasz, W. J. MacKnight, J. Polym. Sci., Polym. Phys. Ed., **30**, 465 (1992).
511. S. A. Liberman, A. de S. Gomes, E. M. Macchi, J. Polym. Sci., Polym. Chem. Ed., **22**, 2809 (1984).
512. A. M. Lichkus, P. C. Painter, M. M. Coleman, Macromolecules, **21**, 2636 (1988).

513. D. S. Lim, T. Kyu, J. Chem. Phys., **92**, 3944 (1990).
514. J. L. Lim, R. J. Roe, Polymer, **29**, 1227 (1988).
515. A. A. Lin, T. K. Kwei, A. Reiser, Macromolecules, **22**, 4112 (1989).
516. J. L. Lin, D. Rigby, R. J. Roe, Macromolecules, **18**, 1609 (1985).
517. P. Lin, C. Clash, E. M. Pearce, T. K. Kwei, M. A. Aponte, J. Polym. Sci., Polym. Phys. Ed., **26**, 603 (1988).
518. B. Litauszki, G. Schmidt-Naake, J. Kressler, H. W. Kammer, Polym. Commun., **30**, 359 (1989).
519. D. I. Livingston, J. E. Brown Jr., Proc. Int. Congr. Rheol. 1968, **4**, 25 (1970).
520. D. I. Livingston, R. L. Rongone, Proc. 5th Int. Rubber Technol. Conf. Brighton, 337 (1967); Chem. Abstr., **72**, 67965y (1970).
521. R. J. Lockwood, L. M. Alberino, in: K. N. Edwards, W. F. Gum (Eds.), "Urethane Chemistry and Applications", Am. Chem. Soc., Symp. Ser., vol. 172, Washington, DC, 1981.
522. D. J. Lohse, Polym. Eng. Sci., **26**, 1500 (1986).
523. X. Lu, R. A. Weiss, Macromolecules, **24**, 4381 (1991).
524. X. Lu, R. A. Weiss, Macromolecules, **25**, 6185 (1992).
525. D. Ma, R. E. Prud'homme, Polymer, **31**, 917 (1990).
526. W. J. MacKnight, J. Stoelting, F. E. Karasz, in: N. A. J. Platzer (Ed.), "Multicomponent Polymer Systems", Am. Chem. Soc., Adv. Chem. Ser., vol. 99, Washington, DC, 1971.
527. E. M. Macchi, S. A. Liberman, A. S. Gomes, Makromol. Chem., **187**, 573 (1986).
528. J. M. Machado, R. N. French, Polymer, **33**, 439 (1992).
529. J. M. Machoda, R. N. French, Polymer, **33**, 760 (1992).
530. A. Maconnachie, R. P. Kambour, R. C. Bopp, Polymer, **25**, 357 (1984).
531. Y. Maeda, F. E. Karasz, W. J. MacKnight, Netsu Sokutei, **16**, 65 (1989).
532. Y. Maeda, F. E. Karasz, W. J. MacKnight, R. Vukovic, J. Polym. Sci., Polym. Phys. Ed., **24**, 2345 (1986).
533. M. Maejima, H. Nakamura, T. Inoue, T. Sakai, Sen'i Gakkaishu, **45**, 516 (1989).
534. M. Maejima, Y. Takeda, T. Inoue, T. Sakai, Sen'i Gakkaishu, **44**, 444 (1988).
535. N. Maiti, S. Dutta, S. N. Bhattacharyya, B. M. Mandel, Polym. Commun., **29**, 363 (1988).
536. S. Makhija, E. M. Pearce, T. K. Kwei, Polym. Mater. Sci. Eng., **63**, 498 (1990).
537. S. Makhija, E. M. Pearce, T. K. Kwei, F. Liu, Polym. Eng. Sci., **30**, 798 (1990).
538. S. Makhija, E. M. Pearce, T. K. Kwei, J. Appl. Polym. Sci., **44**, 917 (1992).
539. S. Manabe, P. Murakami, M. Takayanagi, Mem. Faculty Eng. Kyushu Univ., **28**, 295 (1969).
540. H. Marand, M. Collins, Am. Chem. Soc., Polym. Prepr., **31** (1), 552 (1990).
541. K. Marcincin, A. Ramonov, V. Pollak, J. Appl. Polym. Sci., **16**, 2239 (1972).
542. A. I. Marei, E. A. Sidorovich, Polym. Mechanics, **1** (5), 55 (1965).
543. A. G. Margaritis, J. K. Kallitisis, N. K. Kalfoglou, Polymer, **28**, 2122 (1987).
544. A. G. Margaritis, J. K. Kallitisis, N. K. Kalfoglou, Polymer, **30**, 2253 (1989).
545. A. G. Margaritis, N. K. Kalfoglou, Polymer, **28**, 497 (1987).
546. A. G. Margaritis, N. K. Kalfoglou, Eur. Polym. J., **29**, 1043 (1988).
547. Y. H. Mariam, K. P. W. Pemawansa, K. B. Bota, Am. Chem. Soc., Polym. Prepr., **27** (1), 271 (1986).
548. P. A. Marsh, A. Voet, L. D. Price, Rubber Chem. Technol., **40**, 359 (1967).
549. P. A. Marsh, A. Voet, L. D. Price, T. J. Mullens, Rubber Chem. Technol., **41**, 344 (1968).
550. E. Martuscelli, M. Canetti, A. M. Bonfatti, A. Seves, Polymer, **32**, 641 (1991).
551. E. Martuscelli, G. B. Demma, in: E. Martuscelli, R. Palumbo, M. Kryszewski, (Eds.), "Polymer Blends: Processing, Morphology and Properties", Plenum, New York, 1980, p. 101.
552. E. Martuscelli, C. Silvestre, M. L. Addonizio, L. Amelino, Makromol. Chem., **187**, 1557 (1986).
553. E. Martuscelli, C. Silvestre, C. Gismondi, Makromol. Chem., **186**, 2161 (1985).
554. L. M. Martynowicz-Hans, J. Runt, J., Am. Chem. Soc., Polym. Prepr., **27** (1), 269 (1986).
555. J. Maruta, T. Ougizawa, T. Inoue, Polymer, **29**, 2056 (1988).
556. P. Masi, D. R. Paul, J. W. Barlow, Rheol. Proc., **3**, 315 (1980).
557. D. J. Massa, in: S. L. Cooper, G. M. Estes (Eds.), "Multiphase Polymers", Am. Chem. Soc., Adv. Chem. Ser., vol. 176, Washington, DC, 1979, p. 433.
558. D. J. Massa, Am. Chem. Soc., Polym. Prepr., **29** (1), 434 (1988).
559. J. F. Masson, R. St. J. Manley, Macromolecules, **24**, 5914 (1991).
560. J. F. Masson, R. St. J. Manley, Macromolecules, **25**, 589 (1992).
561. M. Matsuo, C. Nozaki, Y. Jyo, Polym. Eng. Sci., **9**, 197 (1969).
562. M. Matsuura, H. Saito, S. Nakata, Y. Imai, T. Inoue, Polymer, **33**, 3210 (1992).
563. M. Matzner, L. M. Robeson, E. W. Wise, J. E. McGrath, Makromol. Chem., **183**, 2871 (1982).
564. M. Matzner, D. L. Schoher, R. N. Johnson, L. M. Robeson, J. E. McGrath, in: H. B. Hopfenberg (Ed.)., "Permeability of Plastic Films and Coatings to Gases, Vapors, and Liquids", Plenum, New York, 1979, p. 125.
565. L. P. McMaster, Macromolecules, **6**, 760 (1973).
566. L. P. McMaster, in: N. A. J. Platzer, (Ed.), "Copolymers, Polyblends and Composites", Am. Chem. Soc., Adv. Chem. Ser., vol. 142, Washington, DC 1975, p. 43.
567. P. C. Mengujoh, H. L. Frisch, J. Polym. Sci., Polym. Chem. Ed., **27**, 3363 (1989).
568. G. C. Meyer, G. E. Tritscher, J. Appl. Polym. Sci., **22**, 719 (1978).

569. J. Mijovic, H. L. Luo, C. D. Han, Polym. Eng. Sci., **22**, 234 (1982).
570. J. B. Miller, K. J. McGrath, C. M. Roland, C. A. Trask, Macromolecules, **23**, 4543 (1990).
571. B. Y. Min, E. M. Pearce, Org. Coatings, Plastics Chem. Prepr., (Am. Chem. Soc.), **45**, 58 (1981).
572. B. Y. Min, E. M. Pearce, T. K. Kwei, Am. Chem. Soc., Polym. Prepr., **24** (2), 441 (1983).
573. K. E. Min, J. S. Chiou, J. W. Barlow, D. R. Paul, Polymer, **28**, 1721 (1987).
574. K. E. Min, D. H. Lee, H. M. Jung, Polym. Bull., **24**, 221 (1990).
575. K. E. Min, D. R. Paul, Macromolecules, **20**, 2828 (1987).
576. K. E. Min, D. R. Paul, J. Polym. Sci., Polym. Phys. Ed., **26**, 2257 (1988).
577. K. E. Min, D. R. Paul, J. Appl. Polym. Sci., **37**, 1153 (1989).
578. I. Mindru, M. Olteanu, L. Dehe, V. Trandafir, Rev. Roum. Chim., **28**, 683 (1983).
579. L. Minkova, M. Mikhailov, Colloid Polym. Sci., **265**, 1 (1987).
580. J. Moacanin, E. Cuddihy, A. Rembaum, in: N. A. J. Platzer (Ed.), "Plasticization and Plasticizer Processes", Am. Chem. Soc., Adv. Chem. Ser., Vol. 48, Washington, DC, 1965 p. 159.
581. R. N. Mohn, D. R. Paul, J. W. Barlow, C. A. Cruz, J. Appl. Polym. Sci., **23**, 575 (1979).
582. G. E. Molau, J. Polym. Sci., Part B, **3**, 1007 (1965).
583. A. Molnar, A. Eisenberg, Polym. Commun., **32**, 370 (1991).
584. A. Molnar, A. Eisenberg, Macromolecules, **25**, 5774 (1992).
585. I. Mondragon, M. Cortazar, G. M. Guzman, Makromol. Chem., **184**, 1741 (1983).
586. I. Mondragon, M. Gaztelumendi, J. Nazabal, Polym. Eng. Sci., **28**, 1126 (1988).
587. J. A. Moore, J. H. Kim, Macromolecules, **25**, 1427 (1992).
588. E. J. Moskala, J. P. Runt, M. M. Coleman, in: D. R. Paul, L. H. Sperling (Eds.), "Multicomponent Polymer Materials", Am. Chem. Soc., Adv. Chem. Ser., vol. 211, Washington, DC, 1986, p. 77.
589. M. Mucha, J. Jachowicz, M. Kryszewski, Acta Polym., **32**, 156 (1981).
590. S. Mukhopadhyay, S. K. De, J. Mater. Sci., **25**, 4027 (1990).
591. R. Murali, A. Eisenberg, J. Polym. Sci., Polym. Phys. Ed., **26**, 1385 (1988).
592. R. Murali, A. Eisenberg, R. K. Gupta, F. W. Harris, Am. Chem. Soc., Polym. Prepr., **27** (1), 343 (1986).
593. S. R. Murff, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **29**, 3231 (1984).
594. P. Musto, L. Wu, F. E. Karasz, W. J. MacKnight, Polymer, **32**, 3 (1991).
595. M. E. Myers, A. M. Wims, T. S. Ellis, J. Barnes, Macromolecules, **23**, 2807 (1990).
596. V. M. Nadkarni, V. L. Shingankuli, J. P. Jog, Polym. Eng. Sci., **28**, 1326 (1988).
597. J. B. Nagode, C. M. Roland, Polymer, **32**, 505 (1991).
598. K. Naito, G. E. Johnson, D. L. Allara, T. K. Kwei, Macromolecules, **11**, 1260 (1978).
599. H. Nakamura, J. Maruta, T. Ohnaga, T. Inoue, Polymer, **31**, 303 (1990).
600. S. Nakata, M. Kakimoto, Y. Imai, T. Inoue, T., Polym. J., **22**, 80 (1990).
601. A. K. Nandi, B. M. Mandal, S. N. Bhattacharyya, Macromolecules, **18**, 1454 (1985).
602. A. K. Nandi, B. M. Mandal, S. N. Bhattacharyya, S. K. Roy, Polym. Commun., **27**, 151 (1986).
603. T. R. Nassar, D. R. Paul, J. W. Barlow, J. Appl. Polym. Sci., **23**, 85 (1979).
604. A. Natansohn, A. Eisenberg, Am. Chem. Soc., Polym. Prepr., **27** (1), 349 (1986).
605. G. Natta, G. Allegra, I. W. Bassi, C. Carlini, E. Chiellini, Montagnoli, Macromolecules, **2**, 311 (1969).
606. M. K. Neo, S. H. Goh, Macromolecules, **24**, 2564 (1991).
607. M. K. Neo, S. H. Goh, Polym. Commun., **32**, 200 (1991).
608. M. K. Neo, S. H. Goh, Eur. Polym. J., **27**, 915 (1991).
609. M. K. Neo, S. H. Goh, Eur. Polym. J., **27**, 927 (1991).
610. M. K. Neo, S. H. Goh, Polymer, **33**, 2012 (1992).
611. M. K. Neo, S. H. Goh, Polymer, **33**, 3203 (1992).
612. M. K. Neo, S. Y. Lee, S. H. Goh, J. Appl. Polym. Sci., **43**, 1301 (1991).
613. M. K. Neo, S. Y. Lee, S. H. Goh, Eur. Polym. J., **27**, 831 (1991).
614. L. E. Nielson, J. Am. Chem. Soc., **75**, 1435 (1955).
615. L. E. Nielson, S. N. Chinai, unpublished work, quoted in: L. E. Nielson, "Mechanical Properties of Polymers", Reinhold, New York, 1962, p. 173.
616. T. Nishi, T. K. Kwei, Polymer, **16**, 285 (1975).
617. T. Nishi, T. K. Kwei, T. T. Wang, J. Appl. Phys., **46**, 4157 (1975).
618. T. Nishi, T. T. Wang, Macromolecules, **8**, 909 (1975).
619. T. Nishi, T. T. Wang, T. K. Kwei, Macromolecules, **8**, 227 (1975).
620. M. Nishimoto, H. Keskkula, D. R. Paul, Polymer, **30**, 1279 (1989).
621. M. Nishimoto, H. Keskkula, D. R. Paul, Macromolecules, **23**, 3633 (1990).
622. M. Nishimoto, H. Keskkula, D. R. Paul, Polymer, **32**, 1274 (1991).
623. Y. Nishio, T. Haratani, T. Takahashi, R. St. J. Manley, Macromolecules, **22**, 2547 (1989).
624. Y. Nishio, T. Haratari, T. Takahashi, J. Polym. Sci., Polym. Phys. Ed., **28**, 355 (1990).
625. Y. Nishio, N. Hirose, T. Takahashi, Polym. J., **21**, 347 (1989).
626. Y. Nishio, R. St. J. Manley, Macromolecules, **21**, 1270 (1988).
627. Y. Nishio, S. K. Roy, R. St. J. Manley, Polymer, **28**, 1385 (1987).
628. S. Nojima, H. Tsutsui, M. Urushihara, W. Kosaka, N. Kato, T. Ashida, Polym. J., **18**, 451 (1986).
629. S. Nojima, Y. Terashima, T. Ashida, Polymer, **27**, 1007 (1986).

630. J. S. Noland, N. N. Hsu, R. Saxon, J. M. Schmitt, in: N. A. J. Platzer (Ed.), "Multicomponent Polymer Systems", Am. Chem. Soc., Adv. Chem. Ser., vol. 99, Washington, DC, 1971, p. 15.
631. E. Nolley, D. R. Paul, J. W. Barlow, J. Appl. Polym. Sci., **23**, 623 (1979).
632. Y. G. Oganesov, V. S. Osipchuk, K. G. Mindiyarov, V. G. Rayevskii, S. S. Voyutskii, Vysokomol. Soedin. Ser. A, **11**, 896 (1969); Polym. Sci. USSR, **11**, 1012 (1969).
633. O. Olabisi, A. G. Farnham, in: S. L. Copper, G. M. Estes, (Eds.), "Multiphase Polymers", Am. Chem. Soc., Adv. Chem. Ser., vol. 176, Washington, DC, 1979, p. 559.
634. J. Y. Olayemi, M. K. Ibiyeye, J. Appl. Polym. Sci., **31**, 237 (1986).
635. C. J. Ong, F. P. Price, J. Polym. Sci., Symp., **63**, 45 (1978).
636. H. J. Oswald, E. T. Kubu, SPE Trans., **3**, 168 (1963).
637. T. Ougizawa, T. Inoue, Polym. J., **18**, 521 (1986).
638. P. Padunchewit, D. R. Paul, J. W. Barlow, in: L. A. Utracki, R. A. Weiss (Eds.), "Multiphase Polymers: Blends and Ionomers", Am. Chem. Soc., Symp. Ser., vol. 395, Washington, DC, 1989, p. 84.
639. P. C. Painter, W. L. Tang, J. F. Graf, B. Thomson, M. M. Coleman, Macromolecules, **24**, 3929 (1991).
640. R. R. Parent, E. V. Thompson, J. Polym. Sci., Polym. Phys. Ed., **16**, 1829 (1978).
641. J. F. Parmer, L. C. Dickinson, J. C. W. Chien, R. S. Porter, Am. Chem. Soc., Polym. Prepr., **29** (1), 476 (1988).
642. J. F. Parmer, L. C. Dickinson, R. S. Porter, Am. Chem. Soc., Polym. Prepr., **27**(1), 295 (1986).
643. G. D. Patterson, T. Nishi, T. T. Wang, Macromolecules, **9**, 603 (1976).
644. D. R. Paul, J. O. Altamirano, in: N. A. J. Platzer (Ed.), "Copolymers, Polyblends and Composites", Am. Chem. Soc., Adv. Chem. Ser., vol. 142, Washington, DC, 1975, p. 371.
645. D. R. Paul, J. W. Barlow, C. A. Cruz, R. N. Mohn, T. R. Nassar, D. C. Wahrmund, D. C., Am. Chem. Soc., Org. Coatings. Plastics Chem. Prepr., **37** (1), 130 (1977).
646. E. M. Pearce, T. K. Kwei, B. Y. Min, J. Macromol. Sci., Chem. A, **21**, 1181 (1984).
647. J. R. Pennacchia, E. M. Pearce, T. K. Kwei, B. J. Bulkin, J. P. Chen, Macromolecules, **19**, 973 (1986).
648. S. Percec, L. Melamud, J. Appl. Polym. Sci., **41**, 1853 (1990).
649. P. Perrin, R. E. Prud'homme, Polymer, **32**, 1468 (1991).
650. S. S. Pestov, V. N. Kuleznev, V. A. Shershnew, Kolloid. Zh., **40**, 705 (1978); Colloid J. USSR, **40**, 581 (1978).
651. R. J. Peterson, R. D. Corneliussen, L. T. Rozelle, Polym. Prepr., **10**, 385 (1969).
652. J. L. Pfennig, H. Keskkula, D. R. Paul, J. Appl. Polym. Sci., **32**, 3657 (1986).
653. J. Piglowski, Eur. Polym. J., **24**, 905 (1988).
654. J. Piglowski, Acta Polym., **41**, 518 (1990).
655. J. Piglowski, K. H. Kammer, Acta Polym., **41**, 431 (1990).
656. L. Z. Pillon, L. A. Utracki, D. W. Pillon, Polym. Eng. Sci., **27**, 562 (1987).
657. A. Piloz, J. Y. Decroix, J. F. May, Angew. Makromol. Chem., **54**, 77 (1976).
658. Z. H. Ping, Q. T. Nguyen, J. Neel, Makromol. Chem., **190**, 437 (1989).
659. Z. H. Ping, Q. T. Nguyen, J. Neel, Makromol. Chem., **191**, 185 (1990).
660. M. Pracella, E. Chielline, D. Dainelli, Makromol. Chem., **190**, 175 (1989).
661. M. Pracella, E. Chiellini, G. Gralli, D. Dainelli, Mol. Cryst. Liq. Cryst., **153**(A), 525 (1987).
662. M. Pracella, D. Dainelli, G. Gralli, E. Chielline, Makromol. Chem., **187**, 2387 (1986).
663. W. M. Prest, R. S. Porter, J. Polym. Sci., Part A-2, **10**, 1639 (1972).
664. C. Pugh, V. Percec, Macromolecules, **19**, 65 (1986).
665. A. Purcell, C. Thies, Am. Chem. Soc., Polym. Prepr., **9**, 115 (1968).
666. R. Qian, X. Gu, Angew. Makromol. Chem., **141**, 1 (1986).
667. C. Qin, L. A. Belfiore, Am. Chem. Soc., Polym. Prepr., **31** (1), 263 (1990).
668. C. Qin, C., A. T. N. Pires, L. A. Belfiore, Polym. Commun., **31**, 177 (1990).
669. D. Radic, L. Gargallo, Thermochim. Acta, **180**, 241 (1991).
670. V. Ramaswamy, H. P. Weber, Appl. Optics, **12**, 1581 (1973).
671. A. Rameau, Y. Gallot, P. Marie, B. Farnoux, Polymer, **30**, 386 (1989).
672. B. G. Ranby, J. Polym. Sci., Polym. Symp., **51**, 89 (1975).
673. P. C. Raymond, D. R. Paul, J. Polym. Sci., Polym. Phys. Ed., **28**, 2103 (1990).
674. I. N. Razinskaya, B. P. Shtarkman, A. E. Chalykh, I. N. Sapozhnikova, Vysokomol. Soedin, Ser. B, **27**, 871 (1985).
675. I. N. Razinskaya, L. I. Vidyakina, T. I. Radbil', B. P. Shtarkman, Vysokomol. Soedin. Ser. A, **14**, 968 (1974); Polym. Sci. USSR, **14**, 1079 (1972).
676. M. Ree, T. Kyu, R. S. Stein, J. Polym. Sci., Polym. Phys. Ed., **25**, 105 (1987).
677. R. A. Register, T. R. Bell, J. Polym. Sci., Polym. Phys. Ed., **30**, 569 (1992).
678. G. S. Rellick, J. P. Runt, J. Polym. Sci., Polym. Phys. Ed., **24**, 279 (1986).
679. A. Rembaum, J. Mocanin, E. Cuddihy, J. Polym. Sci., Part C, **4**, 529 (1963).
680. P. M. Remiro, J. Nazabal, J. Appl. Polym. Sci., **42**, 1639 (1991).
681. P. M. Remiro, J. Nazabal, J. Appl. Polym. Sci., **42**, 1475 (1991).
682. R. A. Reznikova, A. D. Zaionchkovskii, S. S. Voyutskii, Zh. Tekh. Fiz., **25**, 1045 (1955).
683. P. B. Rim, E. B. Orler, Macromolecules, **20**, 433 (1987).
684. M. Rink, L. Monfrini, P. Gasparini, A. Pavan, Rubber Process. Appl., **3**, 145 (1983).
685. L. M. Robeson, quoted in reference S32.
686. L. M. Robeson, J. Polym. Sci., Polym. Lett. Ed., **16**, 261 (1978).
687. L. M. Robeson, J. Appl. Polym. Sci., **30**, 4081 (1985).

688. L. M. Robeson, A. B. Furtek, Am. Chem. Soc., Org. Coatings Plastics Chem. Prepr., **37**, 136 (1977).
689. L. M. Robeson, A. B. Furtek, J. Appl. Polym. Sci., **23**, 645 (1979).
690. L. M. Robeson, W. F. Hale, C. N. Merriman, Macromolecules, **14**, 1644 (1981).
691. L. M. Robeson, M. Matzner, L. J. Fetters, J. E. McGrath, in: L. H. Sperling (Ed.), "Recent Advances in Blends, Grafts and Blocks", Plenum, New York, 1974, p. 281.
692. L. M. Robeson, J. E. McGrath, Polym. Eng. Sci., **17**, 300 (1977).
693. J. Rodriguez-Parada, V. Percec, Macromolecules, **19**, 55 (1986).
694. J. Rodriguez-Parada, V. Percec, J. Polym. Sci., Polym. Chem. Ed., **24**, 579 (1986).
695. E. Roerdink, G. Challa, Polymer, **21**, 1161 (1980).
696. E. Roerdink, G. Challa, Polymer, **19**, 173 (1978).
697. C. M. Roland, Macromolecules, **20**, 2557 (1987).
698. C. M. Roland, Rubber Chem. Technol., **61**, 866 (1988).
699. C. M. Roland, J. Polym. Sci., Polym. Phys. Ed., **26**, 839 (1988).
700. C. M. Roland, C. A. Trask, Rubber Chem. Technol., **62**, 896 (1989).
701. J. Roovers, P. M. Toporowski, Macromolecules, **25**, 1096 (1992).
702. A. Roychoudhury, P. P. De, A. K. Bhowmick, S. K. De, Polymer, **33**, 4737 (1992).
703. J. Runt, L. Du, L. M. Martynowicz, D. M. Brezny, M. Mayo, Macromolecules, **22**, 3908 (1989).
704. J. Runt, L. M. Martynowicz-Hans, L. Du, M. Mayo, Am. Chem. Soc., Polym. Prepr., **28** (2), 153 (1987).
705. E. Russell, G. Crone, A. Natansohn, Polym. Bull., **27**, 67 (1991).
706. M. Rutkowska, J. Appl. Polym. Sci., **42**, 1695 (1991).
707. M. Rutkowska, A. Eisenberg, Macromolecules, **17**, 821 (1984).
708. M. Rutkowska, A. Eisenberg, J. Appl. Polym. Sci., **33**, 2833 (1987).
709. J. H. Ryou, H. D. Kim, D. S. Lim, C. S. Ha, W. J. Cho, Pollimo, **16**, 225 (1992).
710. S. Saeki, J. M. G. Cowie, I. J. McEwen, Polymer, **24**, 60 (1983).
711. H. Saito, Y. Fujita, T. Inoue, Polym. J., **19**, 405 (1987).
712. H. Saito, T. Inoue, J. Polym. Sci., Polym. Phys. Ed., **25**, 1629 (1987).
713. H. Saito, M. Takahashi, T. Inoue, J. Polym. Sci., Polym. Phys. Ed., **26**, 1761 (1988).
714. H. Saito, D. Tsutsumi, T. Inoue, Polym. J., **22**, 128 (1990).
715. J. M. Saldanha, T. Kyu, Macromolecules, **20**, 2840 (1987).
716. W. Salomons, G. ten Brinke, F. E. Karasz, Polym. Commun., **32**, 185 (1991).
717. M. Sankarapandian, K. Kishore, Macromolecules, **24**, 3090 (1991).
718. S. Sasaki, M. Nagao, M. Gotoh, J. Polym. Sci., Polym. Phys. Ed., **26**, 637 (1988).
719. M. Scandola, G. Ceccorulli, M. Pizzoli, Macromolecules, **25**, 6441 (1992).
720. E. Schchori, J. Jagur-Grodzinski, J. Appl. Polym. Sci., **20**, 1665 (1976).
721. J. Schelten, W. Schamtz, D. G. H. Ballard, G. W. Longman, M. G. Rayner, G. D. Wignall, Prepr. Int. Microsymp. Crystal. Fusion Polym., Louvain-la-Neuve, 1 (1976).
722. H. A. Schneider, P. Dilger, Polym. Bull., **21**, 265 (1989).
723. H. A. Schneider, U. Epple, B. Leikauf, H. N. Neto, New Polym. Mater., **3**, 115 (1992).
724. H. A. Schneider, B. Leikauf, Thermochim. Acta, **114**, 165 (1987).
725. I. A. Schneider, E. M. Calugaru, Eur. Polym. J., **11**, 861 (1975).
726. W. Schnek, D. Reichert, H. Schneider, Polymer, **31**, 329 (1990).
727. J. A. Schroeder, F. E. Karasz, W. J. MacKnight, Polymer, **26**, 1795 (1985).
728. J. W. Schurer, A. DeBoer, G. Challa, Polymer, **16**, 201 (1975).
729. C. G. Seefried Jr., J. V. Koleske, J. Testing Eval., **4**, 220 (1976).
730. C. G. Seefried Jr., J. V. Koleske, F. E. Critchfield, Polym. Eng. Sci., **16**, 771 (1976).
731. S. N. Semerek, C. W. Frank, in: Han, C. D. (Ed.), "Polymer Blends and Composites in Multiphase Systems", Am. Chem. Soc., Adv. Chem. Ser., vol. 206, Washington, DC, 1984, p. 77.
732. C. J. Serman, P. C. Painter, M. M. Coleman, Am. Chem. Soc., Polym. Prepr., **29** (1), 315 (1988).
733. C. J. Serman, P. C. Painter, M. M. Coleman, Polymer, **32**, 1049 (1991).
734. C. J. Serman, X. Xu, P. C. Painter, M. M. Coleman, Polymer, **32**, 516 (1991).
735. R. W. Seymour, B. E. Zehner, J. Polym. Sci., Polym. Phys. Ed., **18**, 2299 (1980).
736. K. Shah, K. Min, J. L. White, Polym. Eng. Sci., **28**, 1277 (1988).
737. K. Shah, J. L. White, K. Min, Polym. Eng. Sci., **29**, 586 (1989).
738. V. S. Shah, J. D. Keitz, D. R. Paul, J. W. Barlow, J. Appl. Polym. Sci., **32**, 3863 (1986).
739. C. K. Sham, G. Guerra, F. E. Karasz, W. J. MacKnight, Polymer, **29**, 1016 (1988).
740. C. K. Sham, C. H. Lau, D. J. Williams, F. E. Karasz, W. J. MacKnight, Brit. Polym. J., **20**, 149 (1988).
741. M. T. Shaw, J. Appl. Polym. Sci., **18**, 449 (1974).
742. S. Shen, J. M. Torkelson, Macromolecules, **25**, 721 (1992).
743. Y. D. Shibanov, T. M. Radzhabov, Vysokomol. Soedin., Ser. B, **30**, 281 (1988).
744. M. Shibayama, T. Yamamoto, C. F. Xiao, S. Sakurai, A. Hayami, S. Nomura, Polymer, **32**, 1010 (1991).
745. T. Shiomi, F. E. Karasz, W. J. MacKnight, Macromolecules, **19**, 2274 (1986).
746. T. Shiomi, T. Eguchi, I. Ishimatsu, K. Imai, Macromolecules, **23**, 4978 (1990).
747. T. Shiomi, F. E. Karasz, W. J. MacKnight, Macromolecules, **19**, 2644 (1986).

748. T. Shiomi, M. Suzuki, M. Tohyama, K. Imai, Macromolecules, **22**, 3578 (1989).
749. A. R. Shultz, B. M. Beach, Macromolecules, **7**, 902 (1974).
750. A. R. Shultz, B. M. Gendron, J. Appl. Polym. Sci., **16**, 461 (1972).
751. A. R. Shultz, B. M. Gendron, J. Polym. Sci., Symp. Ed., **43**, 89 (1973).
752. A. R. Shultz, B. M. Gendron, in: J. Chiu, (Ed.), "Polymer Characterization by Thermal Methods of Analysis", Dekker, New York, 1974, p. 175.
753. A. R. Shultz, B. M. Gendron, J. Macromol. Sci., Chem. A, **8**, 175 (1974).
754. A. R. Shultz, A. L. Young, J. Appl. Polym. Sci., **28**, 1677 (1983).
755. Y. J. Shur, B. Ranby, J. Appl. Polym. Sci., **19**, 2143 (1975).
756. Y. J. Shur, B. Ranby, J. Appl. Polym. Sci., **19**, 1337 (1975).
757. Y. F. Shutilin, Vysokomol. Soedin., Ser. B, **23**, 708 (1991).
758. A. Siegmann, G. Cohen, Z. Baraam, J. Appl. Polym. Sci., **37**, 1481 (1989).
759. C. Silvestre, S. Cimmino, E. Martuscelli, F. E. Karasz, W. J. MacKnight, Polymer, **28**, 1190 (1987).
760. A. Simmons, A. Natansohn, Macromolecules, **25**, 1272 (1992).
761. A. Simmons, A. Natansohn, Macromolecules, **25**, 3881 (1992).
762. T. F. Sincock, D. J. David, Polymer, **33**, 4515 (1992).
763. D. P. Singh, S. Kalachandra, D. T. Turner, J. Polym. Sci., Polym. Phys. Ed., **26**, 1795 (1988).
764. V. B. Singh, D. J. Walsh, J. Macromol. Sci., Phys. B, **25**, 65 (1986).
765. W. Siol, Makromol. Chem., Macromol. Symp., **44**, 47 (1991).
766. W. Siol, U. Terbrack, Angew. Makromol. Chem., **185/186**, 259 (1991).
767. D. J. Skrovanek, M. M. Coleman, Polym. Eng. Sci., **27**, 857 (1987).
768. E. L. Slagowski, E. P. Chang, J. J. Tkacik, Polym. Eng. Sci., **21**, 513 (1981).
769. R. J. Slocombe, J. Polym. Sci., **26**, 9 (1957).
770. G. L. Slonimskii, J. Polym. Sci., **30**, 625 (1958).
771. G. L. Slonimskii, N. K. Komskaia, Zhur. Fiz. Khim., **30**, 1746 (1956).
772. K. L. Smith, A. E. Winslow, D. E. Peterson, Ind. Eng. Chem., **51**, 1361 (1959).
773. P. Smith, A. Eisenberg, J. Polym. Sci., Polym. Lett. Ed., **21**, 223 (1983).
774. S. Y. Soh, J. Appl. Polym. Sci., **44**, 371 (1992).
775. S. Y. Soh, J. Appl. Polym. Sci., **45**, 1831 (1992).
776. M. Song, Y. Huang, G. Cong, Polym. Commun., **32**, 155 (1991).
777. M. Song, H. Liang, B. Jiang, Polym. Bull., **23**, 615 (1990).
778. M. Song, F. Long, Eur. Polym. J., **27**, 983 (1991).
779. D. D. Sotiropoulou, K. G. Gravalos, N. K. Kalfoglou, J. Appl. Polym. Sci., **45**, 273 (1992).
780. R. Spindler, D. F. Shriver, Macromolecules, **19**, 347 (1986).
781. S. Stankovic, G. Guerra, D. J. Williams, F. E. Karasz, W. J. MacKnight, Polym. Commun., **29**, 14 (1988).
782. D. J. Stein, R. H. Jung, K. H. Iller, H. Hendus, Angew. Makromol. Chem., **36**, 89 (1974).
783. R. S. Stein, S. McGuire, M. Satkowski, D. Esnault, Am. Chem. Soc., Polym. Prepr., **30** (2), 270 (1989).
784. J. Stoelting, F. E. Karasz, W. J. MacKnight, Polym. Eng. Sci., **10**, 133 (1970).
785. G. R. Strobl, J. T. Bendler, R. P. Kambour, A. R. Shultz, Macromolecules, **19**, 2683 (1986).
786. A. Stroeks, R. Paquail, E. Nies, Polymer, **32**, 2653 (1991).
787. G. V. Struminskii, G. L. Slonimskii, Zhur. Fiz. Khim., **30**, 1941 (1956).
788. A. C. Su, J. R. Fried, in: D. R. Paul, L. H. Sperling, "*Multicomponent Polymer Materials*", *Am. Chem. Soc.*, Adv. Chem. Ser., vol. 211, Washington, DC, 1986, p. 59.
789. M. Suess, J. Kressler, H. W. Kammer, Polymer, **28**, 957 (1987).
790. M. Suess, J. Kressler, H. W. Kammer, K. Heinemann, Polym. Bull., **16**, 371 (1986).
791. J. Sun, I. Cabasso, Macromolecules, **24**, 3603 (1991).
792. J. Sun, H. L. Frisch, I. Cabasso, J. Polym. Sci., Polym. Phys. Ed., **27**, 2657 (1989).
793. Z. Sun, H. Li, Y. Zhuang, M. Ding, Z. Feng, Polym. Bull., **26**, 557 (1991).
794. L. Suspene, T. M. Lam, J. P. Pascault, J. Appl. Polym. Sci., **39**, 1347 (1990).
795. T. Suzuki, E. M. Pearce, T. K. Kwei, Polymer, **33**, 198 (1992).
796. B. T. Swinyard, J. A. Barrie, D. J. Walsh, Polym. Commun., **28**, 331 (1987).
797. A. A. Tager, L. V. Adamova, V. S. Blinov, N. Ye. Klimova, Ye. N. Racheiskova, T. Ye. Chudinovskikh, T. V. Litvinova, L. A. Mazyrina, Vysokomol. Soedin., Ser. A, **29**, 2327 (1987); Polym. Sci. USSR, **29**, 2557 (1987).
798. Y. Takagi, T. Ougizawa, T. Inoue, Polymer, **28**, 103 (1987).
799. M. Takayanagi, H. Harima, Y. Iwata, Mem. Faculty Eng. Kyushu Univ., **23**, 1 (1963).
800. H. Tanaka, A. J. Lovinger, Macromolecules, **20**, 2638 (1987).
801. R. Tannanbaum, M. Rutkowska, A. Eisenberg, Am. Chem. Soc., Polym. Prepr., **27** (1), 345 (1986).
802. G. ten Brinke, E. Roerdink, G. Challa, in: K. Solc (Ed.), "Polymer Compatibility and Incompatibility: Principles and Practices", MMI Press, Harwood, 1982, p. 77.
803. G. ten Brinke, E. Rubinstein, F. E. Karasz, W. J. MacKnight, R. Vukovic, J. Appl. Phys., **56**, 2440 (1984).
804. E. V. Thompson, in: D. Klempner, K. C. Frisch (Eds.), "Polymer Alloys II: Blends, Blcoks, Grafts and IPN's", Plenum, New York, 1980, p. 1.
805. G. Thyagarajan, V. Janarthanan, Polymer, **30**, 1797 (1989).
806. S. P. Ting, B. J. Bulkin, E. M. Pearce, T. K. Kwei, J. Polym. Sci., Polym. Chem., **19**, 1451 (1981).
807. S. P. Ting, E. M. Pearce, T. K. Kwei, J. Polym. Sci., Polym. Lett. Ed., **18**, 201 (1980).
808. P. E. Tomlins, J. S. Higgins, J. Chem. Phys., **90**, 6691 (1989).

809. H. Tomura, H. Saito, T. Inoue, Macromolecules, **25**, 1161 (1992).
810. C. A. Trask, C. M. Roland, Polym. Commun., **29**, 332 (1988).
811. C. A. Trask, C. M. Roland, Macromolecules, **22**, 256 (1989).
812. C. Tremblay, R. E. Prud'homme, J. Polym. Sci., Polym. Phys. Ed., **22**, 1857 (1984).
813. Y. Tsujita, T. Higaki, A. Takizawa, T. Kiroshita, J. Polym. Sci., Polym. Chem. Ed., **23**, 1255 (1985).
814. Y. Tsujita, K. Iwakiri, T. Kiroshita, A. Takizawa, W. J. MacKnight, J. Polym. Sci., Polym. Phys. Ed., **25**, 415 (1987).
815. J. M. Ubrich, F. B. C. Larbi, J. L. Halary, L. Monnerie, B. Bauer, C. C. Han, Macromolecules, **19**, 810 (1986).
816. H. Ueda, F. E. Karasz, Macromolecules, **18**, 2719 (1985).
817. H. Ueda, F. E. Karasz, J. Macromol. Sci., Chem. A, **27**, 1693 (1990).
818. H. Ueda, F. E. Karasz, Polym. J., **24**, 1363 (1992).
819. C. Uriarte, I. J. Eguiazabal, M. Llanos, J. I. Iribarren, J. J. Iruin, Macromolecules, **20**, 3038 (1987).
820. C. Uriarte, J. J. Iruin, M. J. Fernandez-Berridi, J. M. Elorza, Polymer, **30**, 1155 (1989).
821. P. van Ende, G. Groeninckx, H. Reynaers, C. Samyn, Polymer, **33**, 3598 (1992).
822. Van Tam Bui, D. Baril, Thanh Long Vu, Makromol. Chem., Macromol. Symp., **16**, 267 (1988).
823. D. L. Vanderhart, G. C. Campbell, R. M. Briber, Macromolecules, **25**, 4734 (1992).
824. J. Vanderschueren, A. Janssens, M. Ladang, J. Niezette, Polymer, **23**, 395 (1982).
825. D. F. Varnell, J. P. Runt, M. M. Coleman, Polymer, **24**, 37 (1983).
826. K. T. Varughese, J. Appl. Polym. Sci., **39**, 205 (1990).
827. K. T. Varughese, P. P. De, S. K. Sanyal, S. K. De, J. Appl. Polym. Sci., **37**, 2537 (1989).
828. K. T. Varughese, G. B. Nando, P. P. De, S. K. De, J. Mater. Sci., **23**, 3894 (1986).
829. H. Vazquez-Torres, C. A. Cruz-Ramos, J. Appl. Polym. Sci., **32**, 6095 (1986).
830. A. Verma, B. L. Deopura, A. K. Sengupta, J. Appl. Polym. Sci., **31**, 747 (1986).
831. J. M. Vion, R. Jerome, Ph. Teyssie, M. Aubin, R. E. Prud'homme, Macromolecules, **19**, 1828 (1986).
832. M. Vivas de Meftahi, J. M. J. Frechet, Polymer, **29**, 477 (1988).
833. E. J. Voremkamp, G. Challa, Polymer, **29**, 86 (1988).
834. R. Vukovic, G. Bogdanic, V. Kuresevic, F. E. Karasz, W. J. MacKnight, Eur. Polym. J., **24**, 123 (1988).
835. R. Vukovic, F. E. Karasz, W. J. MacKnight, Polymer, **24**, 529 (1983).
836. R. Vukovic, F. E. Karasz, W. J. MacKnight, J. Appl. Polym. Sci., **28**, 219 (1983).
837. R. Vukovic, V. Kuresevic, G. Bogdanic, D. Fles, Thermochim. Acta, **195**, 351 (1992).
838. R. Vukovic, V. Kuresevic, G. Bogdanic, N. Segudovic, Polym. Bull., **28**, 473 (1992).
839. R. Vukovic, V. Kuresevic, F. E. Karasz, W. J. MacKnight, Thermochim. Acta, **54**, 349 (1982).
840. R. Vukovic, V. Kuresevic, F. E. Karasz, W. J. MacKnight, Proc. 7th Int. Conf. Thermal Analysis, **2**, 1078 (1982) [Chem. Abstr., **99**, 176649b (1983)].
841. R. Vukovic, V. Kuresevic, F. E. Karasz, W. J. MacKnight, J. Appl. Polym. Sci., **30**, 317 (1985).
842. R. Vukovic, V. Kuresevic, F. E. Karasz, W. J. MacKnight, Proc. 2nd Eur. Symp. Thermal Analysis, 243 (1981) [Chem. Abstr., **96**, 36064v (1982)].
843. R. Vukovic, V. Kuresevic, N. Segudovic, F. E. Karasz, W. J. MacKnight, J. Appl. Polym. Sci., **28**, 1379 (1983).
844. Wahrmund, D. C., R. E. Bernstein, J. W. Barlow, D. R. Paul, Polym. Eng. Sci., **18**, 677 (1978).
845. D. J. Walsh, G. L. Cheng, Polymer, **23**, 1965 (1982).
846. D. J. Walsh, G. L. Cheng, Polymer, **25**, 495 (1984).
847. D. J. Walsh, G. T. Dee, J. T. Halary, Macromolecules, **22**, 3385 (1989).
848. D. J. Walsh, J. S. Higgins, Z. Chai, Polymer, **23**, 336 (1982).
849. D. J. Walsh, J. S. Higgins, S. Rostami, Macromolecules, **16**, 388 (1983).
850. D. J. Walsh, J. G. McKeown, Polymer, **21**, 1330 (1980).
851. D. J. Walsh, C. K. Sham, Macromolecules, **22**, 3799 (1989).
852. D. J. Walsh, L. Shi, Z. Chai, Polymer, **22**, 1005 (1981).
853. D. J. Walsh, V. B. Singh, Makromol. Chem., **185**, 1979 (1984).
854. M. H. Walters, D. N. Keyte, Rubber Chem. Technol., **38**, 62 (1965).
855. C. B. Wang, S. L. Cooper, J. Polym. Sci., Polym. Phys. Ed., **21**, 11 (1983).
856. H. L. Wang, L. Toppare, J. E. Fernandez, Macromolecules, **23**, 1053 (1990).
857. L. F. Wang, E. M. Pearce, T. K. Kwei, J. Polym. Sci., Polym. Phys. Ed., **29**, 619 (1991).
858. X. Wang, H. Saito, T. Inoue, Kobunshi Ronbushu, **48**, 443 (1991).
859. Y. Wang, H. Morawetz, Macromolecules, **22**, 164 (1989).
860. Y. Y. Wang, S. A. Chen, Polym. Eng., Sci., **21**, 47 (1981).
861. R. A. Weiss, C. Beretta, S. Sasongko, A. Garton, J. Appl. Polym. Sci., **41**, 91 (1990).
862. C. Wendland, S. Klotz, V. Krieger, H. J. Cantow, Am. Chem. Soc., Polym. Prepr., **30** (2), 109 (1989).
863. R. E. Wetton, W. J. MacKnight, J. R. Fried, F. E. Karasz, Macromolecules, **11**, 158 (1978).
864. R. E. Wetton, P. J. Williams, Polym. Mater. Sci. Eng., **59**, 846 (1988).
865. J. M. Widmaier, G. Mignard, Eur. Polym. J., **23**, 989 (1987).
866. K. A. Wolf, J. Polym. Sci., Part C, **4**, 1626 (1963).
867. H. Wolff, Plast. Kautsch., **4**, 244 (1957).
868. E. M. Woo, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **28**, 1347 (1983).
869. E. M. Woo, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **29**, 3837 (1984).

870. E. M. Woo, J. W. Barlow, D. R. Paul, J. Polym. Sci., Polym. Symp., **71**, 137 (1984).
871. E. M. Woo, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **30**, 4243 (1985).
872. E. M. Woo, J. W. Barlow, D. R. Paul, Polymer, **26**, 763 (1985).
873. E. M. Woo, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **32**, 3889 (1986).
874. S. Xie, W. J. MacKnight, F. E. Karasz, J. Appl. Polym. Sci., **29**, 2679 (1984).
875. X. Xu, L. Zhang, H. Li, Polym. Eng. Sci., **27**, 398 (1987).
876. Y. Xu, J. F. Graf, P. C. Painter, M. M. Coleman, Polymer, **31**, 3103 (1991).
877. H. Yang, G. Hadziioannou, R. S. Stein, J. Polym. Sci., Polym. Phys. Ed., **21**, 159 (1983).
878. H. Yang, S. Ricci, M. Collins, Macromolecules, **24**, 5218 (1991).
879. H. Yang, S. Richard, C. C. Han, B. J. Bauer, E. J. Kramer, Polym. Commun., **27**, 132 (1986).
880. H. Yang, M. Shibayama, R. S. Stein, N. Shimizu, T. Hashimoto, Macromolecules, **19**, 1667 (1986).
881. T. P. Yang, E. M. Pearce, T. K. Kwei, N. L. Yang, Macromolecules, **22**, 1813 (1989).
882. N. Yoshimura, K. Fujimoto, Nippon Gomu Kyokaishi, **41**, 161 (1968); Rubber Chem. Technol., **42**, 1009 (1968).
883. D. Yu, Y. Zheng, Z. Wu, X. Tang, B. Jiang, J. Appl. Polym. Sci., **41**, 2649 (1990).
884. D. Yu, E. H. Hellmann, G. P. Hellmann, Makromol. Chem., **192**, 2749 (1991).
885. G. A. Zakrzewski, Polymer, **14**, 347 (1973).
886. J. Zelinger, E. Volfova, H. Zahraknikova, Z. Pelzbauer, Int. J. Polym. Mater., **5**, 99 (1976).
887. H. Zhang, R. E. Prud'homme, J. Polym. Sci., Polym. Phys. Ed., **25**, 723 (1987).
888. X. Zhang, K. Hikichi, Macromolecules, **25**, 2336 (1992).
889. X. Zhang, K. Takegoshi, K. Hikichi, Macromolecules, **24**, 5756 (1991).
890. X. Zhang, K. Takegoshi, K. Hikichi, Polymer, **33**, 712 (1992).
891. X. Zhang, K. Takegoshi, K. Hikichi, Polym. J., **24**, 1403 (1992).
892. Y. Zhao, D. Ma, R. E. Prud'homme, Polymer, **32**, 791 (1991).
893. Y. Zhao, R. E. Prud'homme, Macromolecules, **23**, 713 (1990).
894. Y. Zhao, R. E. Prud'homme, Polym. Bull., **26**, 101 (1991).
895. J. Q. Zheng, T. Kyu, Polym. Eng. Sci., **32**, 1004 (1992).
896. Z. L. Zhou, A. Eisenberg, J. Polym. Sci., Polym. Phys. Ed., **21**, 595 (1983).
897. K. J. Zhu, S. F. Chen, T. Ho, E. M. Pearce, T. K. Kwei, Macromolecules, **23**, 150 (1990).
898. P. Zhuang, T. Kyu, J. L. White, Polym. Eng. Sci., **28**, 1095 (1988).
899. J. Zimmerman, E. M. Pearce, I. K. Miller, J. A. Muzzio, I. E. Epstein, E. A. Hosegood, J. Appl. Polym. Sci., **17**, 849 (1973).
900. J. J. Ziska, J. W. Barlow, D. R. Paul, Polymer, **22**, 918 (1981).
901. L. Y. Zlatkevich, V. G. Nikolskii, Rubber Chem. Technol., **46**, 1210 (1973).
902. P. Zucco, P. C. Painter, Am. Chem. Soc., Polym. Prepr., **29** (1), 341 (1988).
903. H. Abe, Y. Doi, M. M. Satkowski, I. Noda, Macromolecules, **27**, 50 (1994).
904. D. Acierno, C. Naddeo, Polymer, **35**, 1994 (1994).
905. T. O. Ahn, H. S. Eum, H. M. Jeong, J. Y. Park, Polym. Bull., **30**, 461 (1993).
906. T. O. Ahn, M. Lee, H. M. Jeong, K. Cho, Pollimo, **17**, 242 (1993).
907. T. O. Ahn, M. Lee, H. M. Jeong, K. Cho, J. Polym. Sci., Part B: Polym. Phys., **33**, 327 (1995).
908. T. O. Ahn, C. Y. Ryu, J. H. Sunwoo, H. M. Jeong, Macromol. Rapid Commun., **15**, 265 (1994).
909. I. Akiba, S. Akiyama, Polym. J., **26**, 873 (1994).
910. M. Alberdi, E. Espi, M. J. Fernandez-Berridi, J. J. Iruin, Polym. J., **26**, 1037 (1994).
911. A. Alegria, I. Telleria, J. Colmenero, J. Non-Cryst. Solid, **172**, 961 (1994).
912. E. Andresen, H. G. Zechmann, Colloid Polym. Sci., **272**, 1352 (1994).
913. K. Aoi, A. Takasu, M. Okada, Macromol. Rapid Commun., **16**, 757 (1995).
914. K. Aoi, A. Takasu, M. Okada, Macromol. Rapid Commun., **16**, 53 (1995).
915. A. Asano, K. Takegoshi, K. Hikichi, Polymer, **35**, 5630 (1994).
916. A. Avramova, Polymer, **36**, 801 (1995).
917. C. G. Bazuin, L. Rancourt, S. Villeneuve, A. Soldera, J. Polym. Sci., Part B: Polym. Phys., **31**, 1431 (1993).
918. G. Beaucage, R. S. Stein, Macromolecules, **26**, 1617 (1993).
919. L. A. Belfiore, C. Qin, E. Ueda, A. T. N. Pires, J. Polym. Sci., Part B: Polym. Phys., **31**, 409 (1993).
920. S. Y. Bell, J. M. G. Cowie, I. J. McEwen, Polymer, **35**, 786 (1994).
921. Z. Benabdelghani, R. Belkhiri, S. Djadoun, Polym. Bull., **35**, 329 (1995).
922. D. E. Bhagwagar, P. C. Painter, M. M. Coleman, Macromolecules, **27**, 7139 (1994).
923. E. Blumm, A. J. Owen, Polymer, **36**, 4077 (1995).
924. R. H. Bott, J. A. Kuphal, L. M. Robeson, D. Sagl, J. Appl. Polym. Sci., **58**, 1593 (1995).
925. D. Braun, D. Leiss, M. J. Bergmann, G. P. Hellmann, Eur. Polym. J., **29**, 225 (1993).
926. W. Brostow, K. H. Seo, J. B. Back, J. C. Lim, Polym. Eng. Sci., **35**, 1016 (1995).
927. C. M. Buchanan, S. C. Gedon, B. G. Pearcy, A. W. White, M. D. Wood, Macromolecules, **26**, 5704 (1993).
928. A. B. Burdin, A. A. Tager, Vysokomol. Soedin, Ser. A., **35**, 1290 (1993).
929. T. A. Callaghan, D. R. Paul, Macromolecules, **26**, 2439 (1993).

930. T. A. Callaghan, D. R. Paul, J. Polym. Sci., Part B: Polym. Phys., **32**, 1813 (1994).
931. T. A. Callaghan, D. R. Paul, J. Polym. Sci., Part B: Polym. Phys., **32**, 1847 (1994).
932. T. A. Callaghan, K. Takakuwa, D. R. Paul, A. R. Padwa, Polymer, **34**, 3796 (1993).
933. G. Ceccorulli, M. Pizzoli, M. Scandola, Macromolecules, **26**, 6722 (1993).
934. L. C. Cesteros, J. R. Isasi, I. Katime, Macromolecules, **26**, 7256 (1993).
935. L. C. Cesteros, J. R. Isasi, I. Katime, J. Polym. Sci., Part B: Polym. Phys., **32**, 223 (1994).
936. L. C. Cesteros, J. R. Isasi, I. Katime, Macromolecules, **27**, 7887 (1994).
937. L. C. Cesteros, E. Meaurio, I. Katime, Macromolecules, **26**, 2323 (1993).
938. L. C. Cesteros, E. Meaurio, I. Katime, Polym. Int., **34**, 97 (1994).
939. L. C. Cesteros, J. L. Velada, I. Katime, Polymer, **36**, 3183 (1995).
940. Y. T. Cha, E. T. Kim, T. K. Ahn, S. Choe, Polym. J., **26**, 1227 (1994).
941. S. S. Chakraborty, N. Maiti, B. M. Mandal, S. N. Bhattacharyya, Polymer, **34**, 111 (1993).
942. H. L. Chen, Macromolecules, **28**, 2845 (1995).
943. J. Chen, S. H. Goh, S. Y. Lee, K. S. Siow, Polymer, **35**, 1477 (1994).
944. J. Chen, S. H. Goh, S. Y. Lee, K. S. Siow, J. Appl. Polym. Sci., **56**, 761 (1995).
945. S. S. Chen, T. Kyu, J. Macromol. Sci. B: Phys., **32**, 63 (1993).
946. H. L. Chen, R. S. Porter, J. Polym. Sci., Part B: Polym. Phys., **31**, 1845 (1993).
947. H. Cherdron, M. Haubs, F. Herold, A. Schneller, O. Herrmann-Schonherr, R. Wagener, J. Appl. Polym. Sci., **53**, 507 (1994).
948. Y. W. Cheung, R. S. Stein, Macromolecules, **27**, 2512 (1994).
949. Y. W. Cheung, R. S. Stein, J. S. Lin, G. D. Wignall, Macromolecules, **27**, 2520 (1994).
950. Y. W. Cheung, R. S. Stein, G. D. Wignall, H. E. Yang, Macromolecules, **26**, 5365 (1993).
951. J. W. Cho, J. Tasaka, S. Miyata, Polym. J., **25**, 1267 (1993).
952. S. Choe, Y. J. Cha, H. S. Lee, J. S. Yoon, H. J. Choi, Polymer, **36**, 4977 (1995).
953. C. H. Chu, B. Berner, J. Appl. Polym. Sci., **47**, 1083 (1993).
954. S. Cimmino, E. Dipace, E. Martuscelli, C. Silvestre, Polymer, **34**, 2799 (1993).
955. K. J. Chu, W. Ji, Makromol. Chem. Phys., **196**, 479 (1995).
956. S. Classen, M. Vogt, J. H. Wendorff, Polym. Adv. Technol., **6**, 616 (1995).
957. M. R. Coleman, R. Kohn, W. J. Koros, J. Appl. Polym. Sci., **50**, 1059 (1993).
958. M. M. Colemen, Y. Xu, S. R. Macio, P. C. Painter, Macromolecules, **26**, 3457 (1993).
959. M. M. Colemen, X. Yang, P. C. Painter, Y. H. Kim, J. Polym. Sci., Part A: Polym. Chem., **32**, 1817 (1994).
960. M. M. Colemen, X. Yang, H. Zhang, P. C. Painter, J. Macromol. Sci.-Phys., **32**, 295 (1993).
961. M. M. Colemen, X. Yang, H. Zhang, P. C. Painter, K. V. Scherer. Jr, J. Polym. Sci., Part A: Polym. Chem., **31**, 2039 (1993).
962. P. Cote, J. Brisson, Macromolecules, **27**, 7329 (1994).
963. J. M. G. Cowie, A. Demaude, Polym. Adv. Technol., **5**, 178 (1994).
964. J. M. G. Cowie, B. G. Devlin, I. J. McEwen, Polymer, **34**, 501 (1993).
965. J. M. G. Cowie, R. Ferguson, I. J. McEwen, V. M. C. Reid, Polymer, **35**, 1473 (1994).
966. J. M. G. Cowie, J. H. Harris, Eur. Polym. J., **30**, 707 (1994).
967. J. M. G. Cowie, G. Li, I. J. McEwen, Polymer, **35**, 5518 (1994).
968. J. M. G. Cowie, A. A. N. Reilly, Eur. Polym. J., **29**, 455 (1993).
969. J. M. G. Cowie, A. A. N. Reilly, J. Appl. Polym. Sci., **47**, 1155 (1993).
970. J. M. G. Cowie, L. M. Watson, I. J. McEwen, Polym. Bull., **32**, 729 (1993).
971. M. D. Dadmun, C. C. Han, Mat. Res. Soc. Symp. Proc., **305**, 171 (1993).
972. Y. K. Dai, E. Y. Chu, Z. S. Xu, E. M. Pearce, Y. Okamoto, T. K. Kwei, J. Polym. Sci., Part A: Polym. Chem., **32**, 397 (1994).
973. J. Dai, S. H. Goh, S. Y. Lee, K. S. Siow, Polymer, **34**, 4314 (1993).
974. J. Dai, S. H. Goh, S. Y. Lee, K. S. Siow, Polymer, **35**, 2174 (1994).
975. J. Dai, S. H. Goh, S. Y. Lee, K. S. Siow, J. Appl. Polym. Sci., **53**, 837 (1994).
976. J. Dai, S. H. Goh, S. Y. Lee, K. S. Siow, Polym. J., **26**, 905 (1994).
977. J. Dai, S. H. Goh, S. Y. Lee, K. S. Siow, Polym. J., **27**, 558 (1995).
978. J. Dai, S. H. Goh, S. Y. Lee, K. S. Siow, J. Polym. Res., **2**, 209 (1995).
979. K. Darragas, G. Groeninckx, H. Reynaers, C. Samyn, Eur. Polym. J., **30**, 1165 (1994).
980. J. Datta, A. K. Nandi, Polymer, **35**, 4804 (1994).
981. A. M. de Ilarduya, J. L. Eguiburu, E. Espi, J. J. Iruin, M. J. Fernandez-Berridi, Makro. Chem., **194**, 501 (1993).
982. R. De Juann, A. Etxeberria, M. Cortazar, J. J. Iruin, Macromolecules, **27**, 1395 (1994).
983. H. de los Santos Jones, Y. Liu, P. T. Inglefield, A. A. Jones, C. K. Kim, D. R. Paul, Polymer, **35**, 57 (1994).
984. W. de Oliveira, W. G. Glasser, J. Appl. Polym. Sci., **51**, 563 (1994).
985. S. Dadoun, F E. Karasz, F. Metref, Maromol. Symp., **78**, 155 (1994).
986. L. Dong, D. J. T. Hill, A. K. Whittaker, K. P. Ghiggino, Macromolecules, **27**, 5912 (1994).
987. E. Dubini Paglia, P. L. Beltrame, M. Canetti, A. Seves, B. Marcandalli, E. Martuscelli, Polymer, **34**, 996 (1993).

988. S. Dutta, S. S. Chakraborty, B. M. Mandal, S. N. Bhattacharyya, Polymer, **34**, 3499 (1993).
989. J. L. Eguiburu,, A. M. de Ilarduya, J. J. Iruin, M. J. Fernandez-Berridi, Polym. Int., **33**, 393 (1994).
990. C. D. Eisenbach, J. Hofmann, K. Fischer, Macromol. Rapid Commun., **15**, 117 (1994).
991. C. D. Eisenbach, J. Hofmann, W. J. MacKnight, Macromolecules, **27**, 3162 (1994).
992. T. S. Ellis, J. Polym. Sci., Part B: Polym. Phys., **31**, 1109 (1993).
993. T. S. Ellis, Polymer, **36**, 3919 (1995).
994. A. Etxeberria, J. M. Elorza, J. J. Iruin, C. Macro, M. A. Gomez, J. G. Faton, Eur. Polym. J., **29**, 1483 (1993).
995. A. Etxeberria, M. Iriarte, C. Uriarte, J. J. Iruin, Macromolecules, **28**, 7188 (1995).
996. A. Etxeberria, C. Uriarte, M. J. Fernandez-Berridi, J. J. Iruin, Polymer, **35**, 2128 (1994).
997. L. Fambri, A. Penati, J. Kolarik, Angew. Makromol. Chem., **209**, 119 (1993).
998. H. Feng, Z. Feng, L. Shen, Polymer, **34**, 2516 (1993).
999. H. Q. Feng, L. F. Shen, Z. L. Feng, Eur. Polym. J., **31**, 243 (1995).
1000. H. Feng, C. Ye, P. Zhang, Z. Sun, Z. Feng, Macromol. Chem. Phys., **196**, 2587 (1995).
1001. F. Feraz, A. S. H. Hamou, S. Djadoun, Eur. Polym. J., **31**, 665 (1995).
1002. M. J. Feranadez-Berridi, M. Valero, A. M. de Ilarduya, E. Espi, J. J. Iruin, Polymer, **34**, 38 (1993).
1003. M. L. Franzen, J. R. Elliott, Jr., T. Kyu, Macromolecules, **28**, 5147 (1995).
1004. M. Gada, R. A. Gross, S. P. McCarthy, Stud. Polym. Sci., **12**, 177 (1994).
1005. K. P. Gallagher, X. Zhang, J. P. Runt, G. Huynh-ba, J. S. Lin, Macromolecules, **26**, 588 (1993).
1006. M. E. Galvin, S. Heffner, K. I. Winey, Macromolecules, **27**, 3520 (1994).
1007. P. P. Gan, D. R. Paul, Polymer, **35**, 3513 (1994).
1008. P. P. Gan, D. R. Paul, J. Appl. Polym. Sci., **54**, 317 (1994).
1009. P. P. Gan, D. R. Paul, J. Polym. Sci., Part B: Polym. Phys., **33**, 1693 (1995).
1010. P. P. Gan, D. R. Paul, A. R. Padwa, Polymer, **35**, 1487 (1994).
1011. P. P. Gan, D. R. Paul, A. R. Padwa, Polymer, **35**, 3351 (1994).
1012. F. O. Garces, K. Sivadasan, P. Somasundaran, N. J. Turro, Macromolecules, **27**, 272 (1994).
1013. Z. G. Gardlund, Polymer, **34**, 1850 (1993).
1014. L. Gargallo, N. Gatica, D. Radic, Int. J. Polym. Mater., **27**, 107 (1994).
1015. M. Gaztelumendi, J. Nazabal, J. Polym. Sci., Part B: Polym. Phys., **33**, 603 (1995).
1016. S. H. Goh, S. Y. Lee, A. S. H. Ho, Bull. Singapore Natl. Inst. Chem., **22**, 47 (1994).
1017. S. H. Goh, S. Y. Lee, S. M. Low, Macromolecules, **27**, 6736 (1994).
1018. S. H. Goh, S. Y. Lee, K. S. Siow, J. Chen, Polymer, **34**, 2898 (1993).
1019. J. M. Gomez-Elvira, J. L. Halary, L. Monnerie, L. J. Fetters, Macromolecules, **27**, 3370 (1994).
1020. D. Goncalves, A. Waddon, F. E. Karasz, L. Akcelrud, Synth. Metal., **74**, 197 (1995).
1021. K. R. Gorda, D. J. Peiffer, J. Appl. Polym. Sci., **50**, 1977 (1993).
1022. A. Gottschalk, K. Muhlbach, F. Seitz, R. Stadler, C. Auschra, Macromol. Symp., **83**, 127 (1994).
1023. Q. Guo, Polymer, **34**, 70 (1993).
1024. Q. Guo, Macromol. Rapid Commun., **16**, 785 (1995).
1025. Q. Guo, K. Sun, T. Fang, Y. Qi, Z. Feng, J. Appl. Polym. Sci., **48**, 547 (1993).
1026. Q. H. Guo, X. J. Yuan, L. K. Chen, N. Sun, Cellul. Chem. Technol., **27**, 369 (1993).
1027. M. Guo, H. G. Zachmann, Polymer, **34**, 2503 (1993).
1028. C. S. Ha, W. J. Cho, J. H. Ryou, R. J. Roe, Polymer, **34**, 505 (1993).
1029. J. T. Haponiuk, A. Balas, J. Therm. Anal., **43**, 215 (1995).
1030. H. Hayashi, M. Kakimoto, Y. Imai, Polymer, **36**, 2797 (1995).
1031. H. Hayashi, S. Nakata, M. Kakimoto, Y. Imai, J. Appl. Polym. Sci., **49**, 1241 (1993).
1032. S. A. Heffner, P. A. Mirau, Macromolecules, **27**, 7283 (1994).
1033. K. S. Ho, K. Levon, J. Mao, W. Y. Zheng, J. Laakso, Synth. Met., **57**, 3591 (1993).
1034. J. Hong, S. H. Goh, S. Y. Lee, K. S. Siow, Polymer, **36**, 143 (1995).
1035. J. Hong, S. H. Goh, S. Y. Lee, K. S. Siow, Polym. Networks Blends, **5**, 101 (1995).
1036. D. T. Hseih, D. G. Peiffer, J. Appl. Polym. Sci., **47**, 1469 (1993).
1037. B. S. Hsiao, B. B. Sauer, J. Polym. Sci., Part B: Polym. Phys., **31**, 901 (1993).
1038. J. Huang, A. Prasad, H. Marand, Polymer, **35**, 1896 (1994).
1039. C. C. Hung, W. G. Carson, S. P. Bohan, J. Polym. Sci., Part B: Polym. Phys., **32**, 141 (1994).
1040. P. P. Huo, P. Cebe, M. Capel, Macromolecules, **26**, 4275 (1993).
1041. K. Ikawa, S. Tanaka, K. Ueno, Polym. Networks Blends, **3**, 31 (1993).
1042. C. T. Imrie, B. J. A. Paterson, Macromolecules, **27**, 6673 (1994).
1043. P. Iriondo, J. J. Iruin, M. J. Fernandez-Berridi, Polymer, **36**, 3235 (1995).
1044. J. R. Isasi, L. C. Cesteros, I. Katime, Polymer, **34**, 2374 (1993).
1045. J. R. Isasi, L. C. Cesteros, I. Katime, Macromol. Rapid Commun., **15**, 903 (1994).
1046. J. R. Isasi, L. C. Cesteros, I. Katime, Polymer, **36**, 1235 (1995).
1047. V. Janarthanan, J. Kresssler, F. E. Karasz, W. J. MacKnight, J. Polym. Sci., B: Polym. Phys., **31**, 1013 (1993).

1048. S. C. Jeong, H. M. Jeong, M. H. Oh, B. K. Kim, Polym. Bull., **33**, 237 (1994).
1049. M. Jiang, T. Huang, J. Xie, Macromol. Chem. Phys., **196**, 803 (1995).
1050. M. Jiang, X. Qiu, W. Qin, L. Fei, Macromolecules, **28**, 730 (1995).
1051. M. Jiang, C. Zhou,, Z. Zhang, Polym. Bull., **30**, 449 (1993).
1052. W. H. Jo, J. Y. Kim, M. S. Lee, J. Polym. Sci., Part B: Polym. Phys., **32**, 1321 (1994).
1053. W. H. Jo, J. Y. Kim, M. S. Lee, Polym. J., **26**, 465 (1994).
1054. W. H. Jo, M. R. Lee, B. G. Min, M. S. Lee, Polym. Bull., **33**, 113 (1994).
1055. L. Jong, E. M. Pearce, T. K. Kwei, Polymer, **34**, 48 (1993).
1056. A. K. Kalkar, P. S. Parkhi, J. Appl. Polym. Sci., **57**, 233 (1995).
1057. A. K. Kalkar, N. K. Roy, Eur. Polym. J., **29**, 1391 (1993).
1058. R. P. Kambour, W. L. Nachlis, J. D. Carbeck, Polymer, **35**, 209 (1994).
1059. Y. Kano, S. Akiyama, T. Kasemura, Int. J. Adhesion and Adhesives, **15**, 219 (1995).
1060. Y. Kano, S. Kawahara, S. Akiyama, J. Adhes., **41**, 25 (1993).
1061. R. J. Karcha, R. S. Porter, J. Polym. Sci., Part B: Polym. Phys., **31**, 821 (1993).
1062. S. Kawahara, S. Akiyama, Macromolecules, **26**, 2428 (1993).
1063. S. Kawahara, K. Sato, S. Akiyama, J. Polym. Sci., Part B: Polym. Phys., **32**, 15 (1994).
1064. L. W. Kelts, C. J. T. Landry, D. M. Teegarden, Macromolecules, **29**, 2941 (1993).
1065. B. K. Kim, I. H. Kim, Polym. Plast. Technol. Eng., **32**, 167 (1993).
1066. K. J. Kim, G. B. Kim, S. H. Han, J. Appl. Polym. Sci., **49**, 7 (1993).
1067. S. J. Kim, B. K. Kim, H. M. Jeong, J. Appl. Polym. Sci., **51**, 2187 (1994).
1068. C. K. Kim, J. J. Lim, D. R. Paul, Polym. Eng. Sci., **34**, 1788 (1994).
1069. J. H. Kim, C. J. Lim, K. E. Min, Pollimo, **17**, 654 (1993).
1070. C. K. Kim, D. R. Paul, Polym. Eng. Sci., **34**, 24 (1994).
1071. S. H. Kim, E. M. Pearce, T. K. Kwei, J. Polym. Sci., Part A: Polym. Chem., **31**, 3167 (1993).
1072. N. Kinami, T. Okuyama, M. Okamoto, T. Inone, Polymer, **36**, 4449 (1995).
1073. S. Kobayashi, J. W. Cho, S. Miyata, Polym. J., **26**, 235 (1994).
1074. J. S. Kollodge, R. S. Porter, Polymer, **34**, 4990 (1993).
1075. T. Kondo, C. Sawatari, R. St. J. Manley, D. G. Gray, Macromolecules, **27**, 210 (1994).
1076. Y. Kosaka, T. Uryu, Macromolecules, **27**, 6286 (1994).
1077. Y. Kosaka, T. Uryu, J. Polym. Sci., Part A: Polym. Chem., **33**, 2221 (1995).
1078. T. Kovacic, B. Baric, I. Klaric, Thermochim. Acta, **231**, 215 (1994).
1079. N. Koyama, Y. Doi, Can. J. Microbiol., **41** (Suppl. 1), 316 (1995).
1080. C. Kummerlowe, H. W. Kammer, M. Malinconico, E. Martuscelli, Polymer, **34**, 1677 (1993).
1081. P. P. Kundu, D. K. Tripathy, B. K. Samanthroy, Polym. Networks Blends, **5**, 11 (1995).
1082. S. Y. Kwak, J. Appl. Polym. Sci., **53**, 1823 (1994) 94-81.
1083. T. K. Kwei, Y. K. Dai, X. Lu, R. A. Weiss, Macromolecules, **26**, 6583 (1993).
1084. T. Kyu, S. S. Chen, J. Macromol. Sci. Phys. B, **32**, 55 (1993).
1085. T. Kyu, C. C. Ko, D. S. Lim, S. D. Smith, I. Noda, J. Polym. Sci., Part B: Polym. Phys., **31**, 1641 (1993).
1086. C. J. T. Landry, B. K. Coltrain, D. M. Teegarden, W. T. Ferrar, Macromolecules, **26**, 5543 (1993).
1087. C. J. T. Landry, W. T. Ferrar, D. M. Teegarden, B. K. Coltrain, Macromolecules, **26**, 35 (1993).
1088. M. R. Landry, D. J. Massa, C. J. T. Landry, D. M. Teegarden, R. H. Colby, T. E. Long, P. M. Henrichs, J. Appl. Polym. Sci., **54**, 991 (1994).
1089. C. J. T. Landry, D. J. Massa, D. M. Teegarden, M. R. Landry, P. M. Henrichs, R. H. Colby, T. E. Long, Macromolecules, **26**, 6299 (1993).
1090. D. Lath, E. Lathova, Chem. Papers, **48**, 226 (1994).
1091. D. Lath, E. Lathova, J. M. G. Cowie, Makromol. Chem., **194**, 3087 (1993).
1092. W. K. Lee, C. S. Ha, Pollimo, **18**, 935 (1994).
1093. J. S. Lee, H. J. Kim, D. S. Lee, Polym. Bull., **30**, 229 (1993).
1094. L. M. Leung, K. K. C. Lo, Polym. Int., **36**, 339 (1995).
1095. K. Levon, E. Chu, K. S. Ho, T. K. Kwei, J. Mao, W. Y. Zheng, J. Laakso, J. Polym. Sci., Part B: Polym. Phys., **33**, 537 (1995).
1096. E. G. Lezcano, M. G. Prolongo, C. S. Coll, Polymer, **36**, 595 (1995).
1097. W. Li, R. E. Prudhomne, J. Polym. Sci., Part B: Polym. Phys., **31**, 719 (1993).
1098. J. C. Lim, J. K. Park, H. Y. Song, J. Polym. Sci. Polym. Phys., **32**, 29 (1994).
1099. Y. G. Lin, H. W. Lee, H. H. Winter, S. Dashevsky, K. S. Kim, Polymer, **34**, 4703 (1993).
1100. J. Liu, Y. C. Jean, H. Yang, Macromolecules, **28**, 5774 (1995).
1101. D. J. Lohse, L. J. Fetters, M. J. Doyle, H. C. Wang, C. Kow, Macromolecules, **26**, 3444 (1993).
1102. J. Lokaj, A. Sikora, Z. Pientka, M. Bleha, J. Appl. Polym. Sci., **58**, 1485 (1995).
1103. N. Lotti, M. Pizzoli, G. Ceccorulli, M. Scandola, Polymer, **34**, 4935 (1993).
1104. S. M. Low, S. Y. Lee, S. H. Goh, Eur. Polym. J., **29**, 909 (1993).
1105. S. M. Low, S. Y. Lee, S. H. Goh, Eur. Polym. J., **29**, 1075 (1993).
1106. S. M. Low, S. Y. Lee, S. H. Goh, Eur. Polym. J., **29**, 1543 (1993).
1107. S. M. Low, S. Y. Lee, S. H. Goh, Macromolecules, **26**, 2631 (1993).
1108. S. M. Low, S. Y. Lee, S. H. Goh, Eur. Polym. J., **30**, 139 (1994).

1109. S. M. Low, S. H. Goh, S. Y. Lee, M. K. Neo, Polym. Bull., **32**, 187 (1994).
1110. S. M. Low, S. H. Goh, S. Y. Lee, Macromolecules, **27**, 3764 (1994).
1111. S. M. Low, S. H. Goh, S. Y. Lee, Polymer, **35**, 3290 (1994).
1112. S. Lu, E. M. Pearce, T. K. Kwei, J. Polym. Sci., Part A: Polym. Chem., **32**, 2607 (1994).
1113. S. Lu, E. M. Pearce, T. K. Kwei, Polymer, **36**, 2435 (1995).
1114. S. Lu, E. M. Pearce, T. K. Kwei, Polym. Eng. Sci., **35**, 1113 (1995).
1115. X. Lu, R. A. Weiss, Macromolecules, **28**, 3022 (1995).
1116. D. Luo, T. K. Kwei, E. M. Pearce, Macromol. Symp., **98**, 769 (1995).
1117. D. Luo, E. M. Pearce, T. K. Kwei, Macromolecules, **26**, 6220 (1993).
1118. X. Luo, S. Zheng, D. Ma, K. Hu, Chin. J. Polym. Sci., **13**, 144 (1995).
1119. X. Luo, S. Zheng, N. Zhang, D. Ma, Polymer, **35**, 2619 (1994).
1120. D. Ma, R. Zhang, X. Luo, Polymer, **36**, 3963 (1995).
1121. T. R. Maier, A. M. Jamieson, R. Simha, J. Appl. Polym. Sci., **51**, 1053 (1994).
1122. P. Maiti, A. K. Nandi, Macromolecules, **28**, 8511 (1995).
1123. B. Majumdar, H. Keskkula, D. R. Paul, N. G. Harvey, Polymer, **35**, 4263 (1994).
1124. A. A. Mansour, S. A. Madbouly, Polym. Int., **36**, 269 (1995).
1125. D. J. Massa, K. A. Shriner, S. R. Turner, B. I. Voit, Macromolecules, **28**, 3214 (1995).
1126. C. Marco, M. A. Gomez, J. G. Fatou, A. Etxeberria, M. M. Elorza, J. J. Iruin, Eur. Polym. J., **29**, 1477 (1993).
1127. J. M. Martinez, J. I. Eguiazabal, J. Nazabal, J. Appl. Polym. Sci., **48**, 935 (1993).
1128. S. L. Maunu, J. Kinnunen, K. Soljamo, F. Sundholm, Polymer, **34**, 1141 (1993).
1129. K. G. McGrath, C. M. Roland, Rubber Chem. Technol., **67**, 629 (1994).
1130. N. Mekhilef, P. Hadjiandreou, Polymer, **36**, 2165 (1995).
1131. F. Metref, S. Djadoun, Polym. Bull., **34**, 485 (1995).
1132. S. P. Mishra, P. Venkidusamy, J. Appl. Polym. Sci., **58**, 2229 (1995).
1133. Y. Miyashita, Y. Yamada, N. Kimura, Y. Nishio, H. Suzuki, Seni Gakkaishi, **51**, 396 (1995).
1134. S. Mohanty, S. Roy, R. N. Santra, G. B. Nando, J. Appl. Polym. Sci., **58**, 1947 (1995).
1135. A. Molnar, A. Eisenberg, Polymer, **34**, 1918 (1993).
1136. S. Nakata, M. Kakimoto, Y. Imai, Polym. J., **25**, 569 (1993).
1137. M. K. Neo, S. H. Goh, Polym. Networks Blends, **3**, 131 (1993).
1138. C. W. A. Ng, M. A. Bellinger, W. J. MacKnight, Macromolecules, **27**, 6942 (1994).
1139. M. Nishimoto, Y. Takami, A. Tohara, H. Kasahara, Polymer, **36**, 1441 (1995).
1140. Y. Nishio, H. Suzuki, K. Morisaki, Polym. Int., **31**, 15 (1993).
1141. A. Opazo, L. Gargallo, D. Radic, Int. J. Polym. Mater., **27**, 117 (1994).
1142. S. J. Organ, Polymer, **35**, 86 (1994).
1143. S. J. Organ, P. J. Barham, Polymer, **34**, 459 (1993).
1144. A. A. C. M. Oudhuis, G. ten Brinke, F. E. Karasz, Polymer, **34**, 1991 (1993).
1145. A. A. C. M. Oudhuis, H. J. Thiewes, P. F. van Hutten, G. ten Brinke, Polymer, **35**, 3936 (1994).
1146. T. Ougizawa, D. J. Walsh, Polym. J., **25**, 1315 (1993).
1147. S. Palsule, J. M. G. Cowie, Polym. Bull., **33**, 241 (1994).
1148. R. A. Pearson, A. F. Yee, J. Appl. Polym. Sci., **48**, 1051 (1993).
1149. P. Pedrosa, J. A. Pomposo, E. Calahorra, M. Cortazar, Polymer, **36**, 3889 (1995).
1150. P. Pedrosa, J. A. Pomposo, E. Calahorra, M. Cortazar, Macromolecules, **27**, 102 (1994).
1151. M. V. Pellow-Jarman, P. J. Hendra, M. J. J. Hetem, Spectrochim. Acta, Part A, **51A**, 2107 (1995).
1152. J. Peng, S. H. Goh, S. Y. Lee, K. S. Siow, Polymer, **34**, 4930 (1993).
1153. J. Peng, S. H. Goh, S. Y. Lee, K. S. Siow, Polymer, **35**, 1482 (1994).
1154. J. Peng, S. H. Goh, S. Y. Lee, K. S. Siow, Polym. Networks Blends, **4**, 139 (1994).
1155. J. Peng, S. H. Goh, S. Y. Lee, K. S. Siow, Polym. Networks Blends, **5**, 19 (1995).
1156. S. Percec, L. Melamud, J. Appl. Polym. Sci., **47**, 723 (1993).
1157. P. Perrin, R. E. Prud'homme, Acta Polym., **44**, 307 (1993).
1158. D. M. Petrovic-Djakov, J. M. Filipovic, Lj. P. Vrhovac, J. S. Velickovic, J. Therm. Anal., **40**, 741 (1993).
1159. J. Pionteck, V. Reid, W. J. MacKnight, Acta Polym., **46**, 156 (1995).
1160. M. C. Piton, A. Natansohn, Macromolecules, **28**, 1197 (1995).
1161. J. A. Pomposo, E. Calahorra, I. Eguiazabal, M. Cortazar, Macromolecules, **26**, 2104 (1993).
1162. J. A. Pomposo, M. Cortazar, E. Calahorra, Macromolecules, **27**, 245 (1994).
1163. J. A. Pomposo, M. Cortazar, E. Calahorra, Macromolecules, **27**, 252 (1994).
1164. J. A. Pomposo, I. Eguiazabal, E. Calahorra, M. Cortazar, Polymer, **34**, 95 (1993).
1165. X. Qiu, M. Jiang, Polymer, **35**, 5084 (1994).
1166. X. Qiu, M. Jiang, Polymer, **36**, 3601 (1995).
1167. M. Rabeony, D. B. Siano, D. G. Peiffer, E. Siakali-Koulafa, N. Hadjichristidis, Polymer, **35**, 1033 (1994).
1168. P. Rajagopalan, J. S. Kim, H. P. Brack, X. Lu, A. Eisenberg, R. A. Weiss, W. M. Risen. Jr, J. Polym. Sci., Part B: Polym. Phys., **33**, 495 (1995).
1169. P. Ramesh, S. K. De, J. Appl. Polym. Sci., **50**, 1369 (1993).
1170. P. Ramesh, D. Khastgir, S. K. De, Plast. Rubber Compos. Process Appl., **20**, 35 (1993).
1171. D. Rana, B. M. Mandal, S. N. Bhattacharyya, Polymer, **34**, 1454 (1993).
1172. I. Ray, S. Roy, D. Khastgir, Polym. Bull., **30**, 685 (1993).

1173. J. Runt, L. Jin, S. Talibuddin, C. R. Davis, Macromolecules, **28**, 2781 (1995).
1174. J. Runt, D. M. Miley, K. P. Gallagher, X. Zhang, C. A. Barron, S. K. Kumar, Polym. Adv. Technol., **5**, 333 (1994).
1175. J. H. Ryou, C. S. Ha, W. J. Cho, J. Polym. Sci., Part A: Polym. Chem., **31**, 325 (1993).
1176. K. Sakurai, E. Douglas, W. J. MacKnight, Macromolecules, **26**, 208 (1993).
1177. A. Sanchis, M. G. Prolongo, R. G. Rubio, R. M. Masegosa, Polym. J., **27**, 10 (1995).
1178. B. B. Sauer, B. S. Hsiao, J. Polym. Sci., Part B: Polym. Phys., **31**, 917 (93).
1179. B. B. Sauer, B. S. Hsiao, Polymer, **34**, 3315 (1993).
1180. M. Sankarapandian, K. Kishore, Macromolecules, **27**, 7278 (1994).
1181. R. N. Santra, T. K. Chaki, S. Roy, G. B. Nando, Angew. Makromol. Chem., **213**, 7 (1993).
1182. R. N. Santra, S. Roy, A. K. Bhowmick, G. B. Nando, Polym. Eng. Sci., **33**, 1352 (1993).
1183. M. Scandola, Can. J. Microbiol., **41** (Suppl. 1), 310 (1995).
1184. H. A. Schneider, F. Glatz, R. Mulhaupt, New Polym. Mater., **4**, 289 (1995).
1185. K. Schulze, J. Kressler, H. W. Kammer, Polymer, **34**, 3704 (1993).
1186. A. K. Sen, G. S. Mukherjee, Polymer, **34**, 2386 (1993).
1187. M. Shibayama, K. Uenoyama, J. Oura, S. Nomura, T. Iwamoto, Polymer, **36**, 4811 (1995).
1188. M. Shiyematsu, M. Morita, I. Sakata, Macromol. Chem. Phys., **195**, 2827 (1994).
1189. A. Siciliano, A. Seves, T. De Marco, S. Cimmino, E. Martuscelli, C. Silvestre, Macromolecules, **28**, 8065 (1995).
1190. P. Smith, A. Eisenberg, Macromolecules, **27**, 545 (1994).
1191. D. D. Sotiropoulou, O. E. Avramidou, N. K. Kalfoglou, Polymer, **34**, 2297 (1993).
1192. J. Straka, P. Schmidt, J. Dybal, B. Schneider, J. Speracek, Polymer, **36**, 1147 (1995).
1193. P. Svoboda, J. Kressler, T. Chiba, T. Inoue, H. W. Kammer, Macromolecules, **27**, 1154 (1994).
1194. K. Takakuwa, S. Gupta, D. R. Paul, J. Polym. Sci., Part B: Polym. Phys., **32**, 1719 (1994).
1195. K. Takegoshi, K. Hikichi, Polym. J., **26**, 1377 (1994).
1196. K. Takegoshi, K. Tsuchiya, K. Hikichi, Polym. J., **27**, 284 (1995).
1197. T. Tanigami, Y. Shirai, K. Yamaura, S. Matsuzawa, Polymer, **35**, 1970 (1994).
1198. R. E. Taylor-Smith, R. A. Register, Macromolecules, **26**, 2802 (1993).
1199. A. Tezuka, K. Takegoshi, K. Hikichi, J. Mol. Struct., **355**, 9 (1995).
1200. H. Tomita, R. A. Register, Macromolecules, **26**, 2796 (1993).
1201. M. Tsukada, M. Romano, A. Seves, Acta Polym., **43**, 327 (1992).
1202. Y. Tsukahara, J. Inoue, Y. Ohta, S. Kohjiya, Polymer, **35**, 5785 (1994).
1203. Q. Tran-Cong, H. Nakano, J. Okinaka, R. Kawakubo, Polymer, **35**, 1242 (1994).
1204. H. Ueda, F. E. Karasz, Polym. J., **26**, 771 (1994).
1205. C. Uriarte, J. Alfageme, A. Etxeberria, J. J. Iruin, Eur. Polym. J., **31**, 609 (1995).
1206. M. M. Vaidya, K. Levon, E. M. Pearce, J. Polym. Sci., Part B: Polym. Phys., **33**, 2093 (1995).
1207. D. L. VanderHart, Macromolecules, **27**, 2837 (1994).
1208. D. L. VanderHart, R. St. J. Manley, J. D. Barnes, Macromolecules, **27**, 2826 (1994).
1209. M. Vanneste, G. Groeninckx, Polymer, **35**, 1051 (1994).
1210. M. Vanneste, G. Groeninekx, Polymer, **36**, 4253 (1995).
1211. H. Vazquez-Torres, J. V. Canich-Rodriguez, C. A. Cruz-Ramos, J. Appl. Polym. Sci., **50**, 777 (1993).
1212. J. L. Velada, L. C. Cesteros, E. Meaurio, I. Katime, Polymer, **36**, 2765 (1995).
1213. I. M. Vermeesch, G. Groeninckx, Polymer, **36**, 1039 (1995).
1214. I. M. Vermeesch, G. Groeninckx, M. M. Coleman, Macromolecules, **26**, 6643 (1993).
1215. R. Vukovic, G. Bogdanic, V. Kuresevic, N. Segudovic, F. E. Karasz, W. J. MacKnight, Polymer, **35**, 3055 (1994).
1216. R. Vukovic, G. Bogdanic, V. Kuresevic, V. Srica, F. E. Karasz, W. J. MacKnight, J. Appl. Polym. Sci., **52**, 1499 (1994).
1217. R. Vukovic, G. Bogdanic, V. Kuresevic, M. Tomaskovic, F. E. Karasz, W. J. MacKnight, J. Polym. Sci., Part B: Polym. Phys., **32**, 1079 (1994).
1218. R. Vukovic, G. Bogdanic, N. Segudovic, D. Fles, F. E. Karasz, W. J. MacKnight, Thermochim. Acta, **264**, 125 (1995).
1219. R. Vukovic, V. Kureseric, G. Bogdanic, D. Fles, Thermochim. Acta, **233**, 75 (1994).
1220. R. Vukovic, M. Zuanic, G. Bogdanic, F. E. Karasz, W. J. MacKnight, Polymer, **34**, 1449 (1993).
1221. H. L. Wang, J. E. Fernandez, Macromolecules, **26**, 3336 (1993).
1222. J. M. Warakomski, R. P. Dion, J. Appl. Polym. Sci., **46**, 1057 (1992).
1223. R. A. Weiss, X. Lu, Polymer, **35**, 1963 (1994).
1224. J. L. White, P. A. Mirau, Macromolecules, **27**, 1648 (1994).
1225. G. D. Wignall, J. D. Londono, J. S. Lin, L. G. Alamo, M. J. Galante, L. Mandelkern, Macromolecules, **28**, 3156 (1995).
1226. H. Witteler, G. Lieser, M. Droscher, Makro. Chem. Rapid Commun., **14**, 401 (1993).
1227. A. Wrzyszczynski, X. Qu, L. Szosland, E. Adamczak, L. A. Linden, J. F. Rabek, Polym. Bull., **34**, 493 (1995).
1228. S. Xie, A. Natansohn, P. Rochon, Macromolecules, **27**, 1489 (1994).
1229. F. Y. Xu, J. C. W. Chien, Macromolecules, **27**, 6589 (1994).
1230. H. Yang, T. K. Kwei, Y. Dai, Macromolecules, **26**, 842 (1993).
1231. H. Yang, E. Y. Ni, W. McKenna, Composite Interfaces, **1**, 439 (1993).
1232. H. Yang, W. Yetter, Polymer, **35**, 2417 (1994).

1233. Y. T. Yeo, S. Y. Lee, S. H. Goh, Eur. Polym. J., **30**, 495 (1994).
1234. Y. T. Yeo, S. Y. Lee, S. H. Goh, Eur. Polym. J., **30**, 1117 (1994).
1235. Y. T. Yeo, S. Y. Lee, S. H. Goh, Eur. Polym. J., **31**, 39 (1995).
1236. Y. T. Yeo, S. Y. Lee, S. H. Goh, J. Appl. Polym. Sci., **55**, 1651 (1995).
1237. J. S. Yoon. C. S. Choi, S. J. Maing, H. J. Choi, H. S. Lee, S. J. Choi, Eur. Polym. J., **29**, 1359 (1993).
1238. K. H. Yoon, S. C. Lee, O. O. Park, Polym. J., **26**, 816 (1994).
1239. N. Yoshie, Y. Azuma, M. Sakurai, Y. Inoue, J. Appl. Polym. Sci., **56**, 17 (1995).
1240. H. Zhang, D. E. Bhagwagar, J. F. Graf, P. C. Painter, M. M. Coleman, Polymer, **35**, 5379 (1994).
1241. L. Zhang, C. Xiong, X. Deng, J. Appl. Polym. Sci., **56**, 105 (1995).
1242. R. Zhang, X. Luo, D. Ma, J. Appl. Polym. Sci., **55**, 455 (1995).
1243. R. Zhang, X. Luo, D. Ma, Eur. Polym. J., **31**, 1011 (1995).
1244. P. Zhang, Z. Sun, G. Li, Y. Zhuang, M. Ding, Z. Feng, Makromol. Chem., **194**, 1871 (1993).
1245. W. Zhao, L. Yu, X. Zhong, Y. Zhang, J. Sun, J. Macromol. Sci. B: Phys., **34**, 231 (1995).
1246. W. Zheng, K. Levon, T. Taka, J. Lankso, J. E. Osterholm, J. Polym. Sci., Part B: Polym. Phys., **33**, 1289 (1995).
1247. K. J. Zhu, L. Wang, J. Wang, S. Yang, Macromol. Chem. Phys., **195**, 1965 (1994).
1248. K. J. Zhu, L. Wang, S. Yang, Polym. Int., **36**, 293 (1995).
1249. H. Zhuang, E. M. Pearce, T. K. Kwei, Macromolecules, **27**, 6398 (1994).
1250. H. Zhuang, E. M. Pearce, T. K. Kwei, Polymer, **36**, 2237 (1995).
1251. M. D. Zipper, G. P. Simon, M. R. Tant, J. D. Small, G. M. Stack, A. J. Hill, Polym. Int., **36**, 127 (1995).
1252. W. Guo, G. R. Mitchell, Polymer, **35**, 3706 (1994).
1253. A. M. de Ilarduya, J. J. Iruin, M. J. Fernandez-Berridi, Macromolecules, **28**, 3707 (1995).
1254. M. Jiang, T. Huang, J. Xie, Macromol. Chem. Phys., **196**, 787 (1995).
1255. X. Zhang, K. Takegoshi, K. Hikichi, Polym. J., **23**, 79 (1991).
1256. X. Zhang, K. Takegoshi, K. Hikichi, Polym. J., **23**, 87 (1991).
1257. X. Zhang, K. Takegoshi, K. Hikichi, Polymer, **33**, 718 (1992).
1258. W. W. Y. Lau, Y. Q. Jiang, P. P. K. Tan, Polym. Int., **31**, 163 (1993).

Heat Capacities of High Polymers*,**

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† General Note: The letters in parenthesis (after polymer names) correspond to the author's file number in Ref. 40, and are not necessarily the accepted abbreviation.

A. INTRODUCTION

The molar heat capacity at constant pressure, C_p, is defined as the amount of heat, Q, needed to raise the temperature, T, of one mole of substance at constant (atmospheric) pressure by one kelvin in the absence of latent heat effects, such as heats of fusion, crystallization,

reaction, etc. In differential form, one writes: $C_p \equiv dQ/dT = (\partial H/\partial T)_{p,n}$, [= heat capacity at constant pressure p, and number of moles, n (for polymers usually expressed per mole of repeating unit, *i.e.* for poly(propylene) per 0.04208 kg); H = enthalpy = $U + pV$ (U = total energy, V = molar volume)]. Frequently, one makes use of the specific heat capacity, c_p, which refers commonly to the heat capacity of one gram of material ($c_p \equiv C_p/M$, where M represents the molar mass of the repeating unit in Da).

The heat capacity is not only of theoretical interest, but has also considerable importance for practical applications, *e.g.*, for calculation of enthalpy differences between two temperatures or as the basis for interpretation of differential scanning calorimetry (DSC) curves. Furthermore, C_p is a basic quantity of thermodynamics. All integral functions can be derived from the data on C_p: the enthalpy, $H = \int C_p\,dT$, equal to the total heat content when integrated from 0 K to the temperature of interest and adding all intervening heats of transition, ΔH, at their equilibrium transition temperatures (energy and enthalpy are conserved as expressed by the first law of thermodynamics); the entropy, $S = \int (C_p/T)\,dT$, a measure of the degree of disorder of the sample, also to be integrated from 0 K to the temperature of interest and to be corrected by the intervening entropies of transition, ΔS ($S = 0$ J/K/mol for equilibrium crystals at 0 K and positive for all other temperatures and states–third law of thermodynamics; and the free enthalpy, $G = H - TS$, a measure of the stability of the sample (the state with the lowest free enthalpy is most stable – second law of thermodynamics). Figure 1 displays these integral functions for the example of polyethylene (PE, see Table 4) (34).

A representative experimental heat capacity measured by differential scanning calorimetry (DSC) is given in Fig. 2 for semicrystalline poly(oxyethyleneoxyterephthaloyl) (PET, see Table 37) (35). It can be seen that the heat capacity, C_p, given in J/K/mol, is not a single-valued function of temperature. It depends on the state of the polymer [crystalline, amorphous (solid or liquid), or semicrystalline]. The common semicrystalline samples, characterized by their weight fraction (or percentage) crystallinity w_c, are not in equilibrium, and their heat capacity depends on the thermal history since different cooling or annealing conditions may lead to different values of w_c. In addition, particularly in the melting region, the method of measurement, such as DSC or temperature-modulated DSC (TMDSC), may also yield different experimental answers (35). The TMDSC (heavy curve) is a measure of the limited reversibility of the melting of polymers, while the DSC curve is a measure of the breadth of the melting which arises from crystallite size-distributions and reorganization during melting (36).

In case of amorphous polymers (or the amorphous portion of a semicrystalline polymer) the heat capacity changes at the glass transition temperature, T_g, by ΔC_p, proportional to the weight fraction of amorphous material present, as shown in Fig. 3. In addition, the value of ΔC_p is often proportional to the number of flexible parts (beads) into which the repeating unit can be divided [≈ 11 J/K (mol of beads)$^{-1}$ (34)]. The detailed apparent heat-capacity response of the sample in the glass-transition region is dependent on the thermal history of the substance under investigation and also on the heating rate of measurement. A maximum of heat capacity may be caused by enthalpy relaxation and may easily be confused with a melting process, as illustrated in Fig. 3, for atactic poly(styrene) (PS, see Table 20.). A number of other non-equilibrium processes that change the appearance of the heat capacity are (34): minima (exotherms) caused by stress relaxation that may be confused with a crystallization process, crystal perfection, annealing or fusion that add to or subtract from the heat capacity. Positive (endothermic) effects are usually indicative of disordering, negative (exothermic) effects are usually indicative of ordering (36).

Plotting ΔC_p versus ΔH allows extrapolation of data of semicrystalline samples to the completely amorphous and crystalline states, as shown in Fig. 4 (37). Negative deviations from the straight line exist for many stiffer polymers and indicate the presence of a rigid–amorphous fraction, *i.e.* a third phase in a semicrystalline sample that does not become mobile at the glass transition, but is not crystalline (three-phase model with the third phase most likely placed at the surfaces of the crystals where molecules continue into the amorphous phase). The effect of a rigid–amorphous phase on the heat capacity of poly(oxymethylene) is shown in Fig. 5 (38).

The heat capacity tables of this chapter contain mean values of the best data available in the literature. To make the tables universal, effects of broad glass transitions, thermal history, rigid–amorphous fractions, and effects of fusion, annealing, or crystallization were eliminated by appropriate extrapolation. Data for the "solid" refer to crystals as well as glasses. Only in the temperature range from 0 to perhaps 70 K are the heat capacities of glasses slightly higher than that for crystals. For semicrystalline samples between glass transition and melting, the crystallinity must be known. The heat capacity is then the sum of the heat capacities of the crystalline weight fraction and the melt weight fraction, as shown in Fig. 6. Fusion, annealing and crystallization, as well as glass transition broadening, must be added from additional experimental information, as illustrated above.

The tables given in this chapter are based on the ATHAS Data Bank of experimental heat capacities and its update for 1990 [U. Gaur, H. C. Shu, A. Mehta, S. F. Lau, B. B. Wunderlich, M. Varma-Nair, and B. Wunderlich, J. Phys. Chem. Ref. Data, **10**, 89, 119, 1001, 1051, (1981), **11**, 313, 1065, (1982), **12**, 29, 65, 91 (1983), and **20**, 349, (1991)]. The term "NA" in the value-column indicates that no recommendation of experimental data can be made at present; "NA" in the index-column is a reminder that the data point is gained by extrapolation of the experimental data due to transition effect.

The experimental data bank, as presented in this chapter, has been interpreted in terms of the molecular motion of the

polymers (39). The results of this extended data bank can be accessed over the World Wide Web (40). It contains data tables for over 200 polymer and polymer-related materials, including all poly(amino acids) and some proteins. In addition to the experimental data as shown in Tables 1–50, the ATHAS Data Bank contains a link of the solid heat capacity to approximate vibrational spectra as illustrated in Fig. 7 for poly(ethylene). Also, empirical addition schemes for the heat capacity of liquid and solid polymers have been developed, based on the molecular structure (41). An example is given in Fig. 8, for a series of aliphatic nylons (see also Tables 38–40).

B. EXPERIMENTAL CURVES

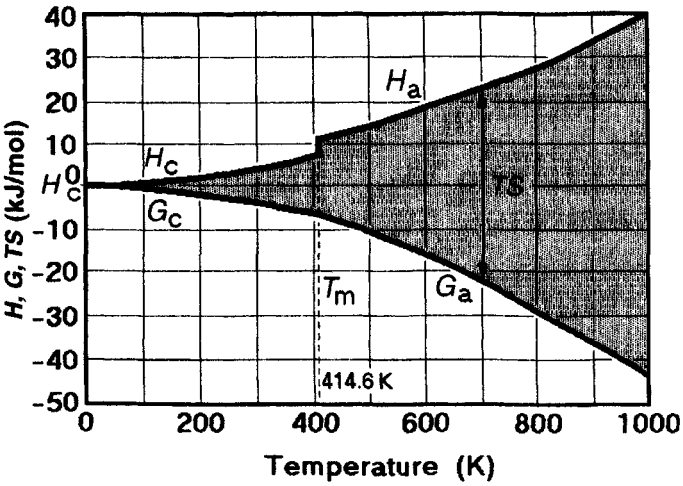


Figure 1. The integral thermodynamic functions H, G, and $TS = H - G$ for PE. The subscripts "a" refer to the amorphous, "c" to the crystalline samples; the equilibrium melting occurs at T_m; H_c^0 is set arbitrarily to zero, it can also be adjusted to the heat of formation $\Delta H_f(T)$ by adding the integral of C_p from 298.15 to 0 K to ΔH_f (298.15) (Ref. 34).

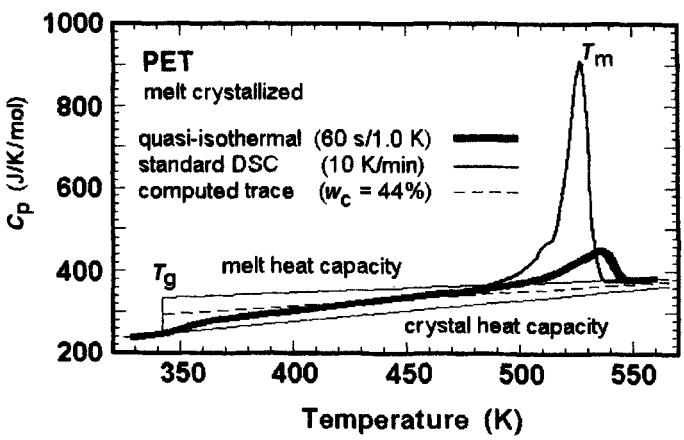


Figure 2. Experimental heat capacities of PET. The glass-transition region is marked by T_g, the melting region by T_m. The solid, thin lines indicate the data of Table 37, the dashed line indicates the crystallinity-corrected heat capacity for the semi-crystalline sample, 44% crystalline, 56% amorphous. The quasi-isothermal data were measured by TMDSC at 2 K intervals with a sinusoidal temperature modulation of ± 1.0 K and a modulation period of 60 s (Ref. 35).

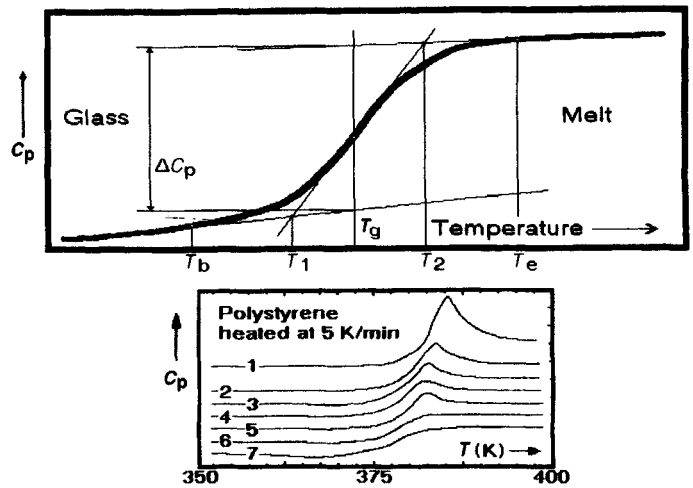


Figure 3. Schematic of the change of the heat capacity at the glass transition (top) and experimental data by DSC for polystyrene after cooling at different rates (bottom). The five characteristic temperatures of the glass transition (T_b, T_1, T_g, T_2, T_e) are marked and apply to a given time scale only. The experimental data display an enthalpy relaxation (endothermic peak if the cooling rate that sets the thermal history is less than the heating rate). The samples 1–7 were cooled at 0.084, 0.20, 0.52, 1.10, 2.50, 5.00, and 30.0 K/min, respectively.

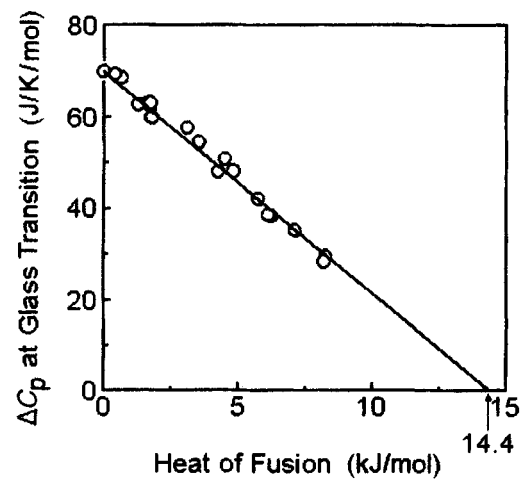


Figure 4. Change of the heat capacity of poly(oxymethylene-carbonyloxyethylene) (poly-*p*-dioxanone) at the glass transition with heat of fusion. (Ref. 37).

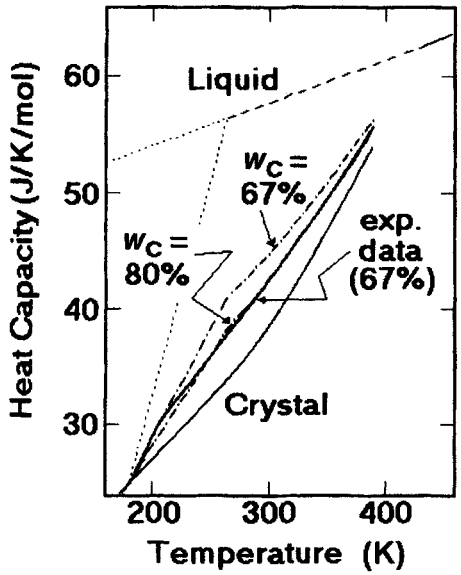


Figure 5. Effect of a rigid-amorphous fraction on the heat capacity of semicrystalline poly(oxymethylene). The measured C_p of a 67% crystalline sample (as determined from its heat of fusion) shows the heat capacity expected for an 80% crystalline sample. The 13% difference from w_c is attributed to the rigid-amorphous phase (Ref. 38).

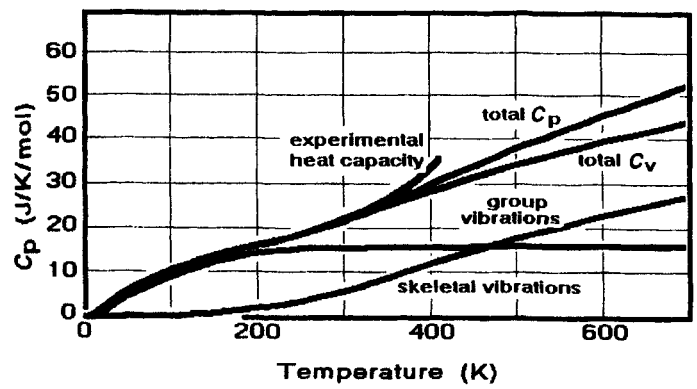


Figure 7. The contributions of skeletal and group vibrations to the heat capacity of polyethylene at constant volume, C_v. Note that, once established, the vibrational heat capacity can be extended safely beyond the range of measurement and deviations from the calculated heat capacity can be interpreted in terms of large-amplitude motion and ordering or disordering.

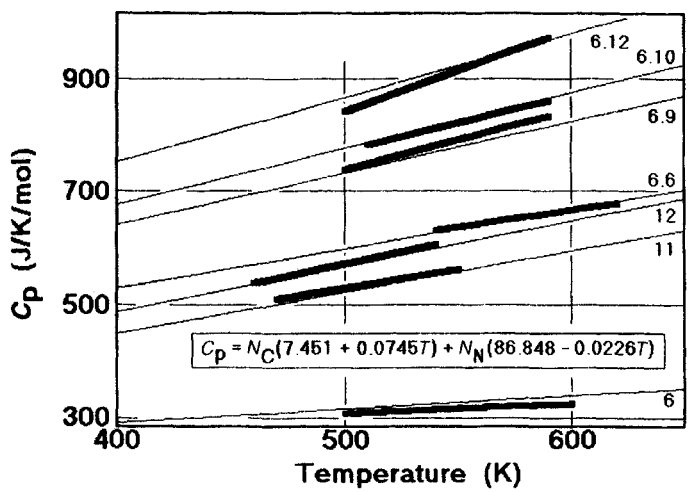


Figure 8. Heat capacity of various aliphatic nylons as measured (heavy lines) and computed from the equation (thin lines) (N_C = number of carbon atoms, N_N number of nitrogen atoms per repeating unit).

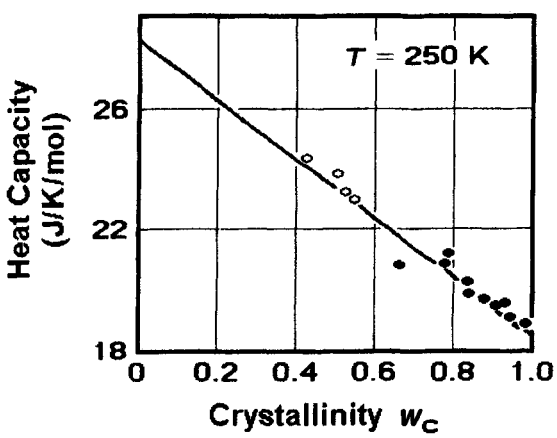
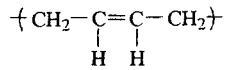


Figure 6. Heat capacity of poly(ethylene) as a function of crystallinity above the glass transition temperature (T_g = 237 K). The extrapolated data for zero crystallinity are entered into Table 4 for the amorphous (melt) state, for crystallinity one, for the solid.

C. DATA TABLES FOR SOLIDS AND LIQUIDS

TABLE 1. *cis*-1,4-POLY(BUTADIENE) (PBD*c*)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	1.18	NA	NA
20	NA	6.52	NA	NA
30		11.79	NA	NA
40		16.15	NA	NA
50		19.98	NA	NA
60		23.18	NA	NA
70		26.16	NA	NA
80		29.07	NA	NA
90		31.87	NA	NA

TABLE 1. *cont'd*

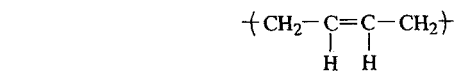
	Solid		Melt	
T (K)	Index	Value	Index	Value
100		34.63	NA	NA
110		37.34	NA	NA
120		40.05	NA	NA
130		42.82	NA	NA
140		45.62	NA	NA
150		48.50	NA	NA
160		53.93	NA	NA
170	NA	NA	NA	NA
180	NA	NA	NA	NA
190	NA	NA	NA	NA
200	NA	NA	NA	NA
210	NA	NA	NA	NA
220	NA	NA	NA	NA
230	NA	NA	NA	NA
240	NA	NA	NA	NA
250	NA	NA	NA	NA
260	NA	NA	NA	NA
270	NA	NA	NA	NA
280	NA	NA	NA	NA
290	NA	NA		104.20
300	NA	NA		106.00
310	NA	NA		107.70
320	NA	NA		109.50
330	NA	NA		111.30
340	NA	NA		113.10
350	NA	NA		114.90
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

References: 1. U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 41 (1983).
2. J. Grebowicz, W. Aycock, B. Wunderlich, Polymer, **27**, 575 (1986).

T_g: 171.0 K (ΔC_p = 29.10)
T_m: 284.0 K (C_p = 86.78–103.1)
Solid: 10–20 K Calculated C_p value by fitting experimental data above 22 K to the Debye function (Data Bank, 1980)
30–160 K Experimental C_p data (Data Bank, 1980)
170–600 K C_p data are not available
Melt: 10–280 K Melt unstable
290–350 K Experimental C_p data (Data Bank, 1985)
360–600 K C_p data are not available

TABLE 2. ***trans*-1,4-POLY(BUTADIENE) (PBD*t*)**



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	0.80	NA	NA
20	NA	4.82	NA	NA
30		9.70	NA	NA
40		14.39	NA	NA
50		18.74	NA	NA
60		22.56	NA	NA
70		25.89	NA	NA
80		29.00	NA	NA
90		32.02	NA	NA
100		34.98	NA	NA
110		37.86	NA	NA
120		40.70	NA	NA
130		43.53	NA	NA
140		46.26	NA	NA
150		48.99	NA	NA
160		51.71	NA	NA
170		54.52	NA	NA
180		57.88	NA	NA
190	NA	NA	NA	NA
200	NA	NA	NA	NA
210	NA	NA	NA	NA
220	NA	NA	NA	NA
230	NA	NA	NA	NA
240	NA	NA	NA	NA
250	NA	NA	NA	NA
260	NA	NA	NA	NA
270	NA	NA	NA	NA
280	NA	NA	NA	NA
290	NA	NA	NA	NA
300	NA	NA	NA	NA
310	NA	NA	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	116.60
370	NA	NA	NA	118.40
380	NA	NA	NA	120.20
390	NA	NA	NA	122.00
400	NA	NA	NA	123.70
410	NA	NA	NA	125.50
420	NA	NA	NA	127.30
430	NA	NA	NA	129.10
440	NA	NA		130.90
450	NA	NA		132.60
460	NA	NA		134.40
470	NA	NA		136.20
480	NA	NA		138.00
490	NA	NA		139.00
500	NA	NA		141.50
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

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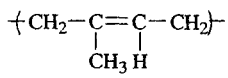
TABLE 2. *cont'd*

Notes

References: 1. U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Data, **12** (1), 41 (1983).
2. J. Grebowicz, W. Ayock, B. Wunderlich, Polymer, **27**, 575 (1986).

T_g:	180.0 K	(ΔC_p = 28.20)
T_m:	356.0 K	(C_p = 107.5–115.9) (Crystal → Condis Crystal)
	437.0 K	(Condis Crystal → Melt)
Solid:	10–20 K	Calculated C_p value by fitting C_p data above 22 K into the Debye function (Data Bank, 1980)
	30–180 K	Experimental C_p data (Data Bank, 1980)
	190–600 K	C_p data are not available
Melt:	10–350 K	Melt unstable
	360–430 K	Extrapolated data from C_p in the region of 440–500 K (Condis Cryst. Region; Ref. 2)
	440–500 K	Experimental C_p data (Data Bank, 1985)
	510–600 K	C_p data are not available

TABLE 3. ***cis*-1,4-POLY(2-METHYLBUTADIENE) (PMBD)**



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		2.41	NA	NA
20		8.32	NA	NA
30		14.24	NA	NA
40		19.54	NA	NA
50		24.34	NA	NA
60		28.83	NA	NA
70		33.06	NA	NA
80		37.07	NA	NA
90		40.90	NA	NA
100		44.58	NA	NA
110		48.12	NA	NA
120		51.57	NA	NA
130		54.93	NA	NA
140		58.23	NA	NA
150		61.48	NA	NA
160		64.70	NA	NA
170		67.89	NA	NA
180		71.06	NA	NA
190		74.85	NA	NA
200	NA	78.03	NA	108.90
210	NA	NA		111.00
220	NA	NA		112.80
230	NA	NA		114.70
240	NA	NA		116.70
250	NA	NA		118.70
260	NA	NA		120.90
270	NA	NA		123.20
280	NA	NA		125.40
290	NA	NA		127.80
300	NA	NA		130.20
310	NA	NA		132.60
320	NA	NA		135.00
330	NA	NA		137.50
340	NA	NA		139.90
350	NA	NA		142.40
360	NA	NA		144.80
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA

TABLE 3. *cont'd*

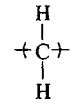
	Solid		Melt	
T (K)	Index	Value	Index	Value
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

References: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 41 (1983).

T_g:	200.0 K	(ΔC_p = 78.03–108.9)
Solid:	10–190 K	Experimental C_p data (Data Bank, 1980)
	200 K	Extrapolated value from C_p in the region of 170–190 K (Data Bank, 1980)
	210–600 K	C_p data are not available
Melt:	10–190 K	Melt unstable
	200 K	Extrapolated data from C_p in the region of 210–360 K (Data Bank, 1980)
	210–360 K	Experimental C_p data (Data Bank, 1980)
	370–600 K	C_p data are not available

TABLE 4. **POLY(ETHYLENE) (PE)**



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		0.10	NA	NA
20		0.73	NA	NA
30		1.84	NA	NA
40		3.19	NA	NA
50		4.52	NA	NA
60		5.74	NA	NA
70		6.83	NA	NA
80		7.79	NA	NA
90		8.66	NA	NA
100		9.45	NA	NA
110		10.20	NA	NA
120		10.92	NA	NA
130		11.60	NA	NA

TABLE 4. *cont'd*

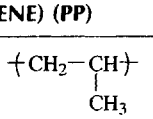
	Solid		Melt	
T (K)	Index	Value	Index	Value
140		12.25	NA	NA
150		12.87	NA	NA
160		13.45	NA	NA
170		14.00	NA	NA
180		14.53	NA	NA
190		15.05	NA	NA
200		15.57	NA	NA
210		16.10	NA	NA
220		16.66	NA	NA
230		17.25	NA	NA
240		17.88		28.30
250		18.55		28.73
260		19.23		29.16
270		19.91	NA	29.60
280		20.58	NA	30.03
290		21.21	NA	30.46
300		21.81	NA	30.89
310		22.38	NA	31.33
320		22.95	NA	31.76
330		23.56	NA	32.19
340		24.30	NA	32.62
350		25.25	NA	33.06
360		26.51	NA	33.49
370		28.13	NA	33.92
380		30.12	NA	34.35
390		32.36		34.79
400		34.19		35.22
410		36.19		35.65
420		38.27		36.08
430		40.44		36.52
440		42.68		36.95
450		44.98		37.38
460		47.35		37.81
470	NA	NA		38.25
480	NA	NA		38.68
490	NA	NA		39.11
500	NA	NA		39.54
510	NA	NA		39.98
520	NA	NA		40.41
530	NA	NA		40.84
540	NA	NA		41.27
550	NA	NA		41.71
560	NA	NA		42.14
570	NA	NA		42.57
580	NA	NA		43.00
590	NA	NA		43.44
600	NA	NA		43.87

Notes

References: U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10** (1), 119 (1981).

T_g:	252.0 K	(C_p = 18.69–28.79)
T_m:	414.6 K	(C_p = 37.13–35.85)
Glass:	10–70 K	Experimental glass C_p data shows a slightly higher value than cryst. C_p data from 10 K to 70 K, respectively: 0.371, 1.40, 2.60, 3.90, 5.14, 6.26, 7.25 (Data Bank, 1980)
Melt:	10–230 K	Melt unstable
	240–260 K	C_p obtained from crystallinity extrapolation and molten PE, match at 252 K (Data Bank, 1980)
	270–380 K	Extrapolated value from C_p in the region of 390–600 K
	390–600 K	Experimental C_p data (Data Bank, 1980)
Crystal:	10–460 K	Experimental C_p data (Data Bank, 1980)
	470–600 K	C_p data are not available

TABLE 5. POLY(PROPYLENE) (PP)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		0.55	NA	NA
20		3.57	NA	NA
30		7.07	NA	NA
40		10.20	NA	NA
50		13.02	NA	NA
60		15.67	NA	NA
70		18.27	NA	NA
80		20.90	NA	NA
90		23.65	NA	NA
100		26.25	NA	NA
110		28.69	NA	NA
120		31.03	NA	NA
130		33.28	NA	NA
140		35.45	NA	NA
150		37.57	NA	NA
160		39.64	NA	NA
170		41.68	NA	NA
180		43.68	NA	NA
190		45.66	NA	NA
200		47.63	NA	NA
210		49.59	NA	NA
220		51.54	NA	NA
230		53.50	NA	NA
240		55.16	NA	NA
250		56.83	NA	NA
260		58.75	NA	82.29
270		60.87	NA	83.80
280		63.18		85.32
290		65.64		86.83
300		68.24		88.34
310		70.95	NA	89.86
320		73.77	NA	91.37
330		76.68	NA	92.88
340		79.67	NA	94.39
350		82.73	NA	95.91
360		85.86	NA	97.42
370		89.04	NA	98.93
380		92.27	NA	100.45
390		95.55	NA	101.96
400		98.97	NA	103.47
410		102.20	NA	104.99
420		105.60		106.50
430		109.00		108.01
440		112.50		109.52
450		116.00		111.04
460		119.50		112.55
470	NA	NA		114.06
480	NA	NA		115.58
490	NA	NA		117.09
500	NA	NA		118.60
510	NA	NA		120.11
520	NA	NA		121.63
530	NA	NA		123.14
540	NA	NA		124.65
550	NA	NA		126.17
560	NA	NA		127.68
570	NA	NA		129.19
580	NA	NA		130.70
590	NA	NA		132.22
600	NA	NA		133.73

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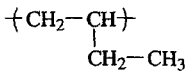
TABLE 5. *cont'd*

Notes

References: U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10** (4), 1051 (1981).

T_g:	260.0 K	(C_p = 64.92–82.29)
T_m:	460.7 K	(C_p = 119.7–112.7)
Glass:	10–250 K	Experimental glass C_p data shows a slightly higher value than cryst. C_p data from 10 K to 250 K, respectively: 1.876, 5.077, 8.721, 11.89, 15.01, 18.36, 20.91, 23.54, 26.07, 28.51, 30.85, 33.11, 35.30, 37.42, 39.48, 41.48, 43.43, 45.35, 47.44, 49.63, 51.92, 54.31, 56.80, 59.40, 62.10. (Data Bank, 1980)
Melt:	10–250 K	Melt unstable
	260–270 K	Extrapolated value from C_p in the region of 280–300 K and region of 420–600 K (Data Bank, 1980)
	280–300 K	C_p of amorphous PP obtained from crystallinity extrapolation of experimental C_p data of semicrystalline samples (Data Bank, 1980)
	310–410 K	Extrapolated value from C_p in the region of 420–600 K (Data Bank, 1980)
	420–600 K	Experimental C_p data (Data Bank, 1980)
Crystal:	10–460 K	Experimental C_p data (Data Bank, 1980)
	470–600 K	C_p data are not available

TABLE 6. POLY(1-BUTENE) (PB)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		0.96	NA	NA
20		5.42	NA	NA
30		10.20	NA	NA
40		14.86	NA	NA
50		19.26	NA	NA
60		23.27	NA	NA
70		27.01	NA	NA
80		30.69	NA	NA
90		34.30	NA	NA
100		37.78	NA	NA
110		41.16	NA	NA
120		44.35	NA	NA
130		47.47	NA	NA
140		50.48	NA	NA
150		53.38	NA	NA
160		56.22	NA	NA
170		59.08	NA	NA
180		62.06	NA	NA
190		65.14	NA	NA
200		68.40	NA	NA
210		71.88	NA	NA
220	NA	75.05	NA	NA
230	NA	78.32	NA	NA
240	NA	81.59	NA	NA
250	NA	NA	NA	107.81
260	NA	NA	NA	109.65
270	NA	NA	NA	111.49
280	NA	NA	NA	113.33
290	NA	NA	NA	115.18
300	NA	NA	NA	117.02
310	NA	NA	NA	118.86
320	NA	NA	NA	120.71
330	NA	NA	NA	122.55

TABLE 6. *cont'd*

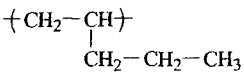
	Solid		Melt	
T (K)	Index	Value	Index	Value
340	NA	NA	NA	124.39
350	NA	NA	NA	126.24
360	NA	NA	NA	128.08
370	NA	NA	NA	129.02
380	NA	NA	NA	131.76
390	NA	NA		133.61
400	NA	NA		135.45
410	NA	NA		137.29
420	NA	NA		139.14
430	NA	NA		140.98
440	NA	NA		142.82
450	NA	NA		144.67
460	NA	NA		146.51
470	NA	NA		148.35
480	NA	NA		150.19
490	NA	NA		152.04
500	NA	NA		153.88
510	NA	NA		155.72
520	NA	NA		157.57
530	NA	NA		159.41
540	NA	NA		161.25
550	NA	NA		163.10
560	NA	NA		164.94
570	NA	NA		166.78
580	NA	NA		168.62
590	NA	NA		170.47
600	NA	NA		172.31

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 34 (1983).

T_g:	249.0 K	(C_p = 84.54–107.6)
T_m:	411.2 K	
Solid:	10–210 K	Experimental C_p data of semicrystalline sample (Data Bank, 1980)
	220–240 K	Extrapolated value from C_p in the region of 180–210 K (Data Bank, 1980)
	250–600 K	C_p data are not available
Melt:	10–230 K	Melt unstable
	240–380 K	Extrapolated value from C_p in the region of 390–630 K
	390–600 K	Experimental C_p data (Data Bank, 1980)

TABLE 7. POLY(1-PENTENE) (PPE)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA
90	NA	NA	NA	NA
100	NA	NA	NA	NA

TABLE 7. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
110	NA	NA	NA	NA
120	NA	NA	NA	NA
130	NA	NA	NA	NA
140	NA	NA	NA	NA
150	NA	NA	NA	NA
160	NA	NA	NA	NA
170	NA	NA	NA	NA
180	NA	NA	NA	NA
190	NA	NA	NA	NA
200		87.90	NA	NA
210		90.86	NA	NA
220		93.82	NA	NA
230		96.78	NA	NA
240	NA	NA		126.70
250	NA	NA		129.64
260	NA	NA		132.58
270	NA	NA		135.52
280	NA	NA		138.46
290	NA	NA		141.40
300	NA	NA		144.34
310	NA	NA		147.28
320	NA	NA		150.22
330	NA	NA		153.16
340	NA	NA		156.10
350	NA	NA		159.04
360	NA	NA		161.98
370	NA	NA		164.92
380	NA	NA		167.86
390	NA	NA		170.80
400	NA	NA		173.74
410	NA	NA		176.68
420	NA	NA		179.62
430	NA	NA		182.56
440	NA	NA		185.50
450	NA	NA		188.44
460	NA	NA		191.38
470	NA	NA		194.32
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 36 (1983).

T_g: 233.0 K ($C_p = 97.67-124.7$)

T_m: 403.2 K

Solid: 10–190 K C_p data are not available

200–230 K Experimental C_p data of semicrystalline isotactic sample agree with amorphous atactic sample (Data Bank, 1980)

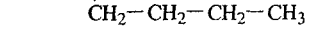
240–600 K C_p data are not available

Melt: 10–230 K Melt unstable

240–470 K Experimental C_p data from atactic sample above T_g and isotatic sample above T_m (Data Bank, 1980)

480–600 K C_p data are not available

TABLE 8. POLY(1-HEXENE) (PHE)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20		9.09	NA	NA
30		16.42	NA	NA
40		24.00	NA	NA
50		30.16	NA	NA
60		36.71	NA	NA
70		42.67	NA	NA
80		48.33	NA	NA
90		53.74	NA	NA
100		59.08	NA	NA
110		64.30	NA	NA
120		69.61	NA	NA
130		74.68	NA	NA
140		79.88	NA	NA
150		84.96	NA	NA
160		89.69	NA	NA
170		95.11	NA	NA
180		100.90	NA	NA
190		106.45	NA	NA
200	NA	112.09	NA	NA
210	NA	117.74	NA	NA
220	NA	123.38	NA	NA
230	NA	NA		152.77
240	NA	NA		156.47
250	NA	NA		160.18
260	NA	NA		163.88
270	NA	NA		167.58
280	NA	NA		171.28
290	NA	NA		174.98
300	NA	NA	NA	NA
310	NA	NA	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

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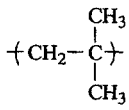
TABLE 8. *cont'd*

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 37 (1983).

T_g:	233.0 K	(C_p = 125.1–150.2)
Solid:	20–190 K	Experimental C_p data of total amorphous sample (Data Bank, 1980)
	200–220 K	Extrapolated value from C_p in the region of 170–190 K (Data Bank, 1980)
	230–600 K	C_p data are not available
Melt:	10–220 K	Melt unstable
	230–290 K	Experimental C_p data from amorphous sample above T_g (Data Bank, 1980)
	300–600 K	C_p data are not available outside the region of 223–290 K

TABLE 9. POLY(ISOBUTENE) (PIB)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20		4.66	NA	NA
30		7.84	NA	NA
40		10.76	NA	NA
50		13.69	NA	NA
60		16.78	NA	NA
70		20.10	NA	NA
80		23.65	NA	NA
90		27.34	NA	NA
100		30.97	NA	NA
110		34.58	NA	NA
120		38.15	NA	NA
130		41.72	NA	NA
140		45.20	NA	NA
150		48.59	NA	NA
160		51.82	NA	NA
170		55.01	NA	NA
180		58.07	NA	NA
190	NA	61.22	NA	NA
200	NA	63.34	NA	85.63
210	NA	NA		88.08
220	NA	NA		90.52
230	NA	NA		92.97
240	NA	NA		95.41
250	NA	NA		97.86
260	NA	NA		100.31
270	NA	NA		102.75
280	NA	NA		105.20
290	NA	NA		107.64
300	NA	NA		110.09
310	NA	NA		112.54
320	NA	NA		114.98
330	NA	NA		117.43
340	NA	NA		119.87
350	NA	NA		122.32
360	NA	NA		124.77
370	NA	NA		127.21
380	NA	NA		129.66
390	NA	NA	NA	NA

TABLE 9. *cont'd*

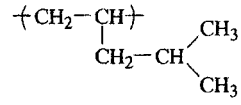
	Solid		Melt	
T (K)	Index	Value	Index	Value
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 38 (1983).

T_g:	200.0 K	(C_p = 63.34–85.63)
T_m:	317.0 K	
Solid:	20–180 K	Experimental C_p data of glassy amorphous sample (Data Bank, 1980)
	190–200 K	Extrapolated value from C_p in the region of 160–180 K (Data Bank, 1980)
	210–600 K	C_p data are not available
Melt:	10–190 K	Melt unstable
	200 K	Extrapolated value from C_p in the region of 210–380 K
	210–380 K	Experimental C_p data from amorphous sample above T_g (Data Bank, 1980)
	390–600 K	C_p data are not available

TABLE 10. POLY(4-METHYL-1-PENTENE) (P4MPE)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80		47.21	NA	NA
90		52.54	NA	NA

TABLE 10. *cont'd*

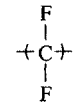
	Solid		Melt	
T (K)	Index	Value	Index	Value
100		57.52	NA	NA
110		62.22	NA	NA
120		66.72	NA	NA
130		71.06	NA	NA
140		75.29	NA	NA
150		79.45	NA	NA
160		83.57	NA	NA
170		87.66	NA	NA
180		91.75	NA	NA
190		95.87	NA	NA
200		100.00	NA	NA
210		104.20	NA	NA
220		108.40	NA	NA
230		112.80	NA	NA
240		117.10	NA	NA
250		121.60	NA	NA
260		126.20	NA	NA
270		130.80	NA	NA
280		135.60	NA	NA
290		140.40	NA	NA
300		145.40	NA	NA
310	NA	NA	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 39 (1983).

T_g:	303.0 K	(C_p = 147.0–180.7)
T_m:	523.2 K	
Solid:	10–70 K	C_p data are not available
	80–300 K	Experimental C_p data (Data Bank, 1980)
	310–600 K	C_p data are not available
Melt:	10–600 K	C_p data are not available

TABLE 11. POLY(TETRAFLUOROETHYLENE) (PTFE)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		1.23	NA	NA
20		3.92	NA	NA
30		6.30	NA	NA
40		8.44	NA	NA
50		10.31	NA	NA
60		12.27	NA	NA
70		14.17	NA	NA
80		16.02	NA	NA
90		17.75	NA	NA
100		19.37	NA	NA
110		21.20	NA	NA
120		22.90	NA	NA
130		24.53	NA	NA
140		26.13	NA	NA
150		27.65	NA	NA
160		29.13	NA	NA
170		30.62	NA	NA
180		31.98	NA	NA
190		33.24	NA	NA
200		34.47	NA	NA
210		35.67	NA	NA
220		36.82	NA	NA
230		37.94	NA	NA
240		39.03	NA	46.85
250		40.10	NA	47.61
260		41.13	NA	48.36
270		42.15	NA	49.12
280		43.15	NA	49.89
290		44.13	NA	50.65
300		45.09	NA	51.42
310		46.02	NA	52.19
320		46.95	NA	52.96
330		47.86	NA	53.74
340		48.76	NA	54.52
350		49.63	NA	55.30
360		50.49	NA	56.08
370		51.35	NA	56.87
380		52.17	NA	57.66
390		52.99	NA	58.45
400		53.81	NA	59.24
410		54.60	NA	60.04
420		55.39	NA	60.84
430		56.15	NA	61.64
440		56.91	NA	62.44
450		57.65	NA	63.25
460		58.39	NA	64.06
470		59.10	NA	64.87
480		59.82		65.39
490		60.51		65.72
500		61.20		66.05
510		61.88		66.38
520		62.55		66.72
530		63.20		67.05
540		63.86		67.38
550		64.48		67.71
560		65.12		68.04
570		65.75		68.38
580		66.36		68.71

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TABLE 11. *cont'd*

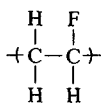
	Solid		Melt	
T (K)	Index	Value	Index	Value
590		66.96		69.04
600		67.58		69.37

Notes

Reference: S. F. Lau, H. Suzuki, B. Wunderlich, J. Polym. Sci., Polym. Phys. Ed., **22**, 379 (1984).

T_g:	240.0 K	(C_p = 39.03–46.85)
T_m:	605.0 K	(C_p = 67.88–69.54)
Solid:	10–600 K	All data are taken from the experimental C_p data of cryst. PTFE (Recommended Data, 1984)
Melt:	10–230 K	Melt unstable
	240–470 K	Extrapolated value from C_p in the region of 480–700 K
	480–600 K	Experimental C_p data, preliminary recommended data, 1984, based on experiments from 560 to 700 K and crystallinity extrapolation at low temperature (Eqs. 4 and 4′ of reference)

TABLE 12. POLY(VINYL FLUORIDE) (PVF)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	0.47	NA	NA
20	NA	2.80	NA	NA
30	NA	5.70	NA	NA
40	NA	8.50	NA	NA
50	NA	11.80	NA	NA
60	NA	14.50	NA	NA
70	NA	17.10	NA	NA
80		20.27	NA	NA
90		22.18	NA	NA
100		23.96	NA	NA
110		25.64	NA	NA
120		27.25	NA	NA
130		28.83	NA	NA
140		30.39	NA	NA
150		31.95	NA	NA
160		33.51	NA	NA
170		35.09	NA	NA
180		36.70	NA	NA
190		38.34	NA	NA
200		40.02	NA	NA
210		41.75	NA	NA
220		43.52	NA	NA
230		45.34	NA	NA
240		47.23	NA	NA
250		49.17	NA	NA
260		51.18	NA	NA
270		53.25	NA	NA
280		55.40	NA	NA
290		57.62	NA	NA
300		59.91	NA	NA
310		62.29	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA

TABLE 12. *cont'd*

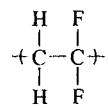
	Solid		Melt	
T (K)	Index	Value	Index	Value
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 43 (1983).

T_g:	314.0 K	(C_p = 63.27–81.07)
T_m:	503.2 K	
Solid:	10–70 K	C_p data are calculated from the Tarasov eq. (Data Bank, 1980)
	80–310 K	Experimental C_p data (Data Bank, 1980)
	320–600 K	C_p data are not available
Melt:	10–600 K	C_p data are not available

TABLE 13. POLY(VINYLIDENE FLUORIDE) (PVF2)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		1.28	NA	NA
20		5.25	NA	NA
30		9.10	NA	NA
40		12.20	NA	NA
50		15.00	NA	NA
60		17.90	NA	NA

TABLE 13. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
70		20.90	NA	NA
80		23.80	NA	NA
90		26.10	NA	NA
100		28.40	NA	NA
110		30.90	NA	NA
120		33.00	NA	NA
130		35.20	NA	NA
140		37.50	NA	NA
150		39.60	NA	NA
160		41.80	NA	NA
170		44.00	NA	NA
180		46.10	NA	NA
190		48.30	NA	NA
200		50.30	NA	NA
210		52.80	NA	NA
220		54.90	NA	NA
230		57.10	NA	NA
240	NA	NA	NA	NA
250	NA	NA	NA	NA
260	NA	NA	NA	NA
270	NA	NA	NA	NA
280	NA	NA	NA	NA
290	NA	NA	NA	NA
300	NA	NA	NA	NA
310	NA	NA	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 45 (1983).

T_g: 233.0 K (C_p = 57.80–80.60)

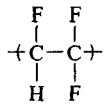
T_m: 483.2 K

Solid: 10–230 K Experimental C_p data (Data Bank, 1980)

240–600 K C_p data are not available

Melt: 10–600 K C_p data are not available

TABLE 14. POLY(TRIFLUOROETHYLENE) (P3FE)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30		12.29	NA	NA
40		15.24	NA	NA
50		17.61	NA	NA
60		20.28	NA	NA
70		23.65	NA	NA
80		26.43	NA	NA
90		29.99	NA	NA
100		33.21	NA	NA
110		36.15	NA	NA
120		38.89	NA	NA
130		41.48	NA	NA
140		43.97	NA	NA
150		46.39	NA	NA
160		48.78	NA	NA
170		51.16	NA	NA
180		53.56	NA	NA
190		55.99	NA	NA
200		58.46	NA	NA
210		61.00	NA	NA
220		63.61	NA	NA
230		66.31	NA	NA
240		69.11	NA	NA
250		72.01	NA	NA
260		75.02	NA	NA
270		78.16	NA	NA
280		81.43	NA	NA
290		84.84	NA	NA
300		88.40	NA	NA
310	NA	NA	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA

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TABLE 14. *cont'd*

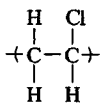
	Solid		Melt	
T (K)	Index	Value	Index	Value
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 46 (1983).

T_g:	304.0 K	(C_p = 89.87–110.87)
T_m:	495.2 K	
Solid:	30–300 K	Experimental C_p data (Data Bank, 1980)
	310–600 K	C_p data are not available
Melt:	10–600 K	C_p data are not available

TABLE 15. POLY(VINYL CHLORIDE) (PVC)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		1.81	NA	NA
20		5.91	NA	NA
30		9.95	NA	NA
40		13.38	NA	NA
50		16.28	NA	NA
60		18.77	NA	NA
70		20.99	NA	NA
80		23.05	NA	NA
90		24.98	NA	NA
100		26.82	NA	NA
110		28.59	NA	NA
120		30.29	NA	NA
130		31.94	NA	NA
140		33.56	NA	NA
150		35.15	NA	NA
160		36.73	NA	NA
170		38.31	NA	NA
180		39.88	NA	NA
190		41.45	NA	NA
200		43.03	NA	NA
210		44.61	NA	NA
220		46.21	NA	NA
230		47.79	NA	NA
240		49.37	NA	NA
250		50.99	NA	NA
260		52.63	NA	NA
270		54.29	NA	NA
280		55.95	NA	NA
290		57.64	NA	NA
300		59.35	NA	NA
310	NA	61.11	NA	NA
320	NA	62.94	NA	NA
330	NA	64.88	NA	NA
340	NA	66.96	NA	NA
350	NA	68.85	NA	NA
360	NA	NA		91.08
370	NA	NA		94.56
380	NA	NA		98.05

TABLE 15. *cont'd*

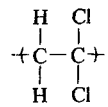
	Solid		Melt	
T (K)	Index	Value	Index	Value
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 49 (1983).

T_g:	354.0 K	(C_p = 69.62–88.99)
T_m:	546.0 K	
Solid:	10–350 K	All data are taken from the experimental C_p of glass (Data Bank, 1980)
	360–600 K	C_p data are not available
Melt:	10–350 K	Melt unstable
	360–380 K	Experimental C_p data (Data Bank, 1980)
	390–600 K	C_p data are not available

TABLE 16. POLY(VINYLIDENE CHLORIDE) (PVC2)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	7.32	NA	NA
40	NA	12.53	NA	NA
50	NA	17.38	NA	NA
60		21.49	NA	NA
70		26.08	NA	NA
80		30.01	NA	NA
90		33.38	NA	NA
100		36.31	NA	NA
110		38.91	NA	NA
120		41.26	NA	NA
130		43.45	NA	NA
140		45.53	NA	NA
150		47.53	NA	NA

TABLE 16. *cont'd*

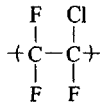
	Solid		Melt	
T (K)	Index	Value	Index	Value
160		49.50	NA	NA
170		51.46	NA	NA
180		53.44	NA	NA
190		55.45	NA	NA
200		57.51	NA	NA
210		59.64	NA	NA
220		61.84	NA	NA
230		64.12	NA	NA
240		66.50	NA	NA
250		69.98	NA	NA
260	NA	NA	NA	NA
270	NA	NA	NA	NA
280	NA	NA	NA	NA
290	NA	NA	NA	NA
300	NA	NA	NA	NA
310	NA	NA	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 52 (1983).

T_g:	255.0 K	(C_p = 70.26)
T_m:	463.0 K	
Solid:	30–50 K	C_p data are calculated from the Tarasov eq. (Data Bank, 1980)
	60–250 K	Experimental C_p data (Data Bank, 1980)
	260–600 K	C_p data are not available
Melt:	30–600 K	C_p data are not available

TABLE 17. POLY(CHLOROTRIFLUOROETHYLENE) (PC3FE)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80		32.46	NA	NA
90		36.77	NA	NA
100		40.81	NA	NA
110		44.63	NA	NA
120		48.24	NA	NA
130		51.69	NA	NA
140		55.00	NA	NA
150		58.18	NA	NA
160		61.26	NA	NA
170		64.26	NA	NA
180		67.18	NA	NA
190		70.05	NA	NA
200		72.87	NA	NA
210		75.66	NA	NA
220		78.41	NA	NA
230		81.15	NA	NA
240		83.87	NA	NA
250		86.57	NA	NA
260		89.27	NA	NA
270		91.98	NA	NA
280		94.68	NA	NA
290		97.39	NA	NA
300		100.10	NA	NA
310	NA	102.90	NA	NA
320	NA	105.60	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA

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TABLE 17. *cont'd*

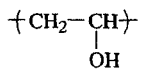
	Solid		Melt	
T (K)	Index	Value	Index	Value
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 54 (1983).

T_g:	325.0 K	(C_p = 107.0)
T_m:	493.0 K	
Solid:	10–70 K	C_p data are not available
	80–300 K	Experimental C_p data (Data Bank, 1980)
	310–320 K	Extrapolated value from C_p in the region of 80–300 K (Data Bank, 1980)
	330–600 K	C_p data are not available
Melt:	30–600 K	C_p data are not available

TABLE 18. POLY(VINYL ALCOHOL) (PVA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60		11.78	NA	NA
70		15.05	NA	NA
80		17.94	NA	NA
90		20.46	NA	NA
100		22.69	NA	NA
110		24.69	NA	NA
120		26.54	NA	NA
130		28.29	NA	NA
140		29.98	NA	NA
150		31.66	NA	NA
160		33.34	NA	NA
170		35.06	NA	NA
180		36.84	NA	NA
190		38.69	NA	NA
200		40.64	NA	NA
210		42.69	NA	NA
220		44.85	NA	NA
230		47.16	NA	NA
240		49.60	NA	NA
250		52.21	NA	NA
260		54.99	NA	NA
270		57.95	NA	NA
280		61.12	NA	NA
290		64.50	NA	NA
300		68.11	NA	NA
310	NA	NA	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA
350	NA	NA	NA	NA

TABLE 18. *cont'd*

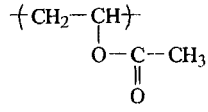
	Solid		Melt	
T (K)	Index	Value	Index	Value
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 56 (1983).

Solid:	10–50 K	C_p data are not available
	60–300 K	Experimental C_p data (Data Bank, 1980)
	310–600 K	C_p data are not available
Melt:	30–600 K	C_p data are not available

TABLE 19. POLY(VINYL ACETATE) (PVAc)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80		27.81	NA	NA
90		31.17	NA	NA
100		34.54	NA	NA
110		37.91	NA	NA
120		41.27	NA	NA
130		44.64	NA	NA

TABLE 19. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
140		48.00	NA	NA
150		51.37	NA	NA
160		54.74	NA	NA
170		58.10	NA	NA
180		61.47	NA	NA
190		64.83	NA	NA
200		68.20	NA	NA
210		71.57	NA	NA
220		74.93	NA	NA
230		78.30	NA	NA
240		81.66	NA	NA
250		85.03	NA	NA
260		88.40	NA	NA
270		91.76	NA	NA
280		95.13	NA	NA
290		98.49	NA	NA
300		101.86	NA	NA
310	NA	NA		157.50
320	NA	NA		158.48
330	NA	NA		159.45
340	NA	NA		160.43
350	NA	NA		161.41
360	NA	NA		162.39
370	NA	NA		163.37
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 56 (1983).

T_g:	304.0 K	(C_p = 103.2–156.9)
Solid:	10–70 K	C_p data are not available
	80–300 K	Experimental C_p data (Data Bank, 1980)
	310–600 K	C_p data are not available
Melt:	10–300 K	Melt unstable
	310–370 K	Experimental C_p data (Data Bank, 1980)
	380–600 K	C_p data are not available

TABLE 20. POLY(STYRENE) (PS)

$$\left(CH_2-\underset{\displaystyle C_6H_5}{\underset{|}{CH}}\right)$$

	Solid		Melt	
T (K)	Index	Value	Index	Value
10		3.33	NA	NA
20		10.77	NA	NA
30		17.47	NA	NA
40		23.13	NA	NA
50		28.01	NA	NA
60		32.37	NA	NA
70		36.38	NA	NA
80		40.17	NA	NA
90		43.81	NA	NA
100		47.37	NA	NA
110		50.88	NA	NA
120		57.38	NA	NA
130		57.89	NA	NA
140		61.43	NA	NA
150		65.01	NA	NA
160		68.65	NA	NA
170		72.36	NA	NA
180		76.14	NA	NA
190		80.01	NA	NA
200		83.97	NA	NA
210		88.03	NA	NA
220		92.19	NA	NA
230		96.01	NA	NA
240		100.16	NA	NA
250		104.45	NA	NA
260		108.85	NA	NA
270		113.36	NA	NA
280		117.96	NA	NA
290		122.64	NA	NA
300		127.38	NA	NA
310		132.18	NA	NA
320		137.02	NA	NA
330		141.92	NA	NA
340		146.85	NA	NA
350		151.82	NA	NA
360		156.81	NA	NA
370		161.84	NA	NA
380	NA	NA		195.93
390	NA	NA		198.59
400	NA	NA		201.24
410	NA	NA		203.89
420	NA	NA		206.55
430	NA	NA		209.20
440	NA	NA		211.85
450	NA	NA		214.51
460	NA	NA		217.16
470	NA	NA		219.81
480	NA	NA		222.46
490	NA	NA		225.12
500	NA	NA		227.77
510	NA	NA		230.42
520	NA	NA		233.08
530	NA	NA		235.73
540	NA	NA		238.38
550	NA	NA		241.04
560	NA	NA		243.69
570	NA	NA		246.34
580	NA	NA		248.99
590	NA	NA		251.65
600	NA	NA		254.30

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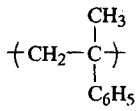
TABLE 20. *cont'd*

Notes

Reference: U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (2), 313 (1982).

T_g:	373.0 K	(C_p = 163.4–194.1)
T_m:	516.2 K	
Solid:	10–70 K	Experimental C_p data of glassy samples (Data Bank, 1980)
	80–370 K	C_p data of glassy samples agree with C_p data of semi-crystalline samples. (Data Bank, 1980)
	380–600 K	C_p data are not available
Melt:	10–370 K	Melt unstable
	380–600 K	Experimental C_p data (Data Bank, 1980)

TABLE 21. POLY(α-METHYLSTYRENE) (PMS)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	0.18	NA	NA
20	NA	1.47	NA	NA
30	NA	4.61	NA	NA
40	NA	11.10	NA	NA
50	NA	19.50	NA	NA
60		28.97	NA	NA
70		38.52	NA	NA
80		45.22	NA	NA
90		51.84	NA	NA
100		55.69	NA	NA
110		61.13	NA	NA
120		66.16	NA	NA
130		70.89	NA	NA
140		75.09	NA	NA
150		79.38	NA	NA
160		83.77	NA	NA
170		88.23	NA	NA
180		92.76	NA	NA
190		97.36	NA	NA
200		102.01	NA	NA
210		106.71	NA	NA
220		111.46	NA	NA
230		116.26	NA	NA
240		121.09	NA	NA
250		125.95	NA	NA
260		130.85	NA	NA
270		135.78	NA	NA
280		140.73	NA	NA
290		145.70	NA	NA
300		150.70	NA	NA
310		155.71	NA	NA
320		160.74	NA	NA
330		165.79	NA	NA
340		170.84	NA	NA
350		175.91	NA	NA
360		180.98	NA	NA
370		186.06	NA	NA
380		191.15	NA	NA
390		196.24	NA	NA
400		201.33	NA	NA
410		206.42	NA	NA
420		211.52	NA	NA

TABLE 21. *cont'd*

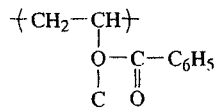
	Solid		Melt	
T (K)	Index	Value	Index	Value
430		216.61	NA	NA
440		221.69	NA	NA
450	NA	NA		252.68
460	NA	NA		258.44
470	NA	NA		264.20
480	NA	NA		269.96
490	NA	NA		275.72
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 58 (1983).

T_g:	441.0 K	(C_p = 222.2–247.5)
Solid:	10–50 K	C_p data are calculated from the Tarasov equation (Data Bank, 1980)
	60–440 K	Experimental C_p data of glassy samples (Data Bank, 1980)
	450–600 K	C_p data are not available
Melt:	10–440 K	Melt unstable
	450–490 K	Experimental C_p data (Data Bank, 1980)
	500–600 K	C_p data are not available

TABLE 22. POLY(VINYL BENZOATE) (PVBZ)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA
90	NA	NA	NA	NA
100	NA	NA	NA	NA
110	NA	NA	NA	NA
120	NA	NA	NA	NA
130	NA	NA	NA	NA
140	NA	NA	NA	NA
150	NA	NA	NA	NA
160	NA	NA	NA	NA
170	NA	NA	NA	NA
180	NA	NA	NA	NA

TABLE 22. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
190		106.39	NA	NA
200		111.57	NA	NA
210		116.75	NA	NA
220		121.93	NA	NA
230		127.10	NA	NA
240		132.28	NA	NA
250		137.46	NA	NA
260		142.64	NA	NA
270		147.82	NA	NA
280		152.99	NA	NA
290		158.17	NA	NA
300		163.35	NA	NA
310	NA	168.53	NA	NA
320	NA	173.71	NA	NA
330	NA	178.88	NA	NA
340	NA	184.06	NA	NA
350	NA	NA	NA	258.08
360	NA	NA	NA	260.96
370	NA	NA		263.84
380	NA	NA		266.71
390	NA	NA		269.59
400	NA	NA		272.47
410	NA	NA		275.35
420	NA	NA		278.23
430	NA	NA		281.10
440	NA	NA		283.98
450	NA	NA		286.86
460	NA	NA		289.74
470	NA	NA		292.62
480	NA	NA		295.49
490	NA	NA		298.37
500	NA	NA		301.25
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

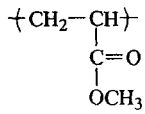
Reference: U. Gaur, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12** (1), 59 (1983).

T_g: 347.0 K

Solid:
- 10–180 K: C_p data are not available
- 190–300 K: Experimental C_p data (Data Bank, 1980)
- 310–340 K: Extrapolated value from C_p in the region of 190–300 K
- 350–600 K: C_p data are not available

Melt:
- 10–340 K: Melt unstable
- 350–360 K: Extrapolated value from C_p in the region of 370–500 K
- 370–500 K: Experimental C_p data (Data Bank, 1980)
- 510–600 K: C_p data are not available

TABLE 23. POLY(METHYL ACRYLATE) (PMA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	0.46	NA	NA
20	NA	4.64	NA	NA
30	NA	11.70	NA	NA
40	NA	19.35	NA	NA
50		26.57	NA	NA
60		33.09	NA	NA
70		38.90	NA	NA
80		44.08	NA	NA
90		48.75	NA	NA
100		52.98	NA	NA
110		56.88	NA	NA
120		60.50	NA	NA
130		63.91	NA	NA
140		67.14	NA	NA
150		70.23	NA	NA
160		73.21	NA	NA
170		76.12	NA	NA
180		78.95	NA	NA
190		81.75	NA	NA
200		84.51	NA	NA
210		87.25	NA	NA
220		89.99	NA	NA
230		92.72	NA	NA
240		95.46	NA	NA
250		98.21	NA	NA
260		100.98	NA	NA
270		103.77	NA	NA
280	NA	NA		148.74
290	NA	NA		150.37
300	NA	NA		151.99
310	NA	NA		153.62
320	NA	NA		155.24
330	NA	NA		156.87
340	NA	NA		158.49
350	NA	NA		160.12
360	NA	NA		161.74
370	NA	NA		163.37
380	NA	NA		164.99
390	NA	NA		166.62
400	NA	NA		168.24
410	NA	NA		169.87
420	NA	NA		171.49
430	NA	NA		173.12
440	NA	NA		174.14
450	NA	NA		176.37
460	NA	NA		177.99
470	NA	NA		179.62
480	NA	NA		181.24
490	NA	NA		182.87
500	NA	NA		184.49
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA

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TABLE 23. *cont'd*

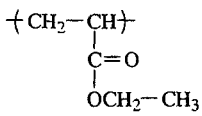
	Solid		Melt	
T (K)	Index	Value	Index	Value
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

T_g:	279.0 K	($\Delta C_p = 42.30$)
Solid:	10–40 K	Extrapolated value from C_p in the region of 50–270 K (Data Bank, 1982)
	50–270 K	Experimental C_p data (Data Bank, 1982)
	280–600 K	C_p data are not available
Melt:	10–270 K	Melt unstable
	280–500 K	Experimental C_p data (Data Bank, 1982)
	510–600 K	C_p data are not available

TABLE 24. POLY(ETHYL ACRYLATE) (PEA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA
90		57.99	NA	NA
100		62.50	NA	NA
110		66.83	NA	NA
120		71.06	NA	NA
130		75.20	NA	NA
140		79.28	NA	NA
150		83.31	NA	NA
160		87.32	NA	NA
170		91.30	NA	NA
180		95.26	NA	NA
190		99.20	NA	NA
200		103.13	NA	NA
210		107.05	NA	NA
220		110.96	NA	NA
230		114.87	NA	NA
240		118.76	NA	NA
250	NA	NA		168.06
260	NA	NA		170.22
270	NA	NA		172.39
280	NA	NA		174.55
290	NA	NA		176.72
300	NA	NA		178.88
310	NA	NA		181.04
320	NA	NA		183.21
330	NA	NA		185.37

TABLE 24. *cont'd*

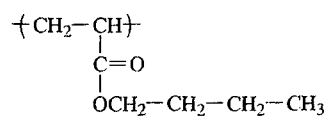
	Solid		Melt	
T (K)	Index	Value	Index	Value
340	NA	NA		187.54
350	NA	NA		189.70
360	NA	NA		191.86
370	NA	NA		194.03
380	NA	NA		196.19
390	NA	NA		198.36
400	NA	NA		200.52
410	NA	NA		202.68
420	NA	NA		204.85
430	NA	NA		207.01
440	NA	NA		209.18
450	NA	NA		211.34
460	NA	NA		213.50
470	NA	NA		215.67
480	NA	NA		217.83
490	NA	NA		220.00
500	NA	NA		222.16
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

T_g:	249.0 K	($\Delta C_p = 45.60$)
Solid:	10–80 K	C_p data are not available
	90–240 K	Experimental C_p data (Data Bank, 1982)
	250–600 K	C_p data are not available
Melt:	10–240 K	Melt unstable
	250–500 K	Experimental C_p data (Data Bank, 1982)
	510–600 K	C_p data are not available

TABLE 25. POLY(*n*-BUTYL ACRYLATE) (P*n*BA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80		71.75	NA	NA
90		78.20	NA	NA
100		84.65	NA	NA
110		91.10	NA	NA

TABLE 25. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
120		97.55	NA	NA
130		104.01	NA	NA
140		110.46	NA	NA
150		116.91	NA	NA
160		123.36	NA	NA
170		129.81	NA	NA
180		136.27	NA	NA
190		142.72	NA	NA
200		149.17	NA	NA
210		155.62	NA	NA
220	NA	NA		206.90
230	NA	NA		210.19
240	NA	NA		213.49
250	NA	NA		216.79
260	NA	NA		220.09
270	NA	NA		223.39
280	NA	NA		226.68
290	NA	NA		229.98
300	NA	NA		233.28
310	NA	NA		236.58
320	NA	NA		239.88
330	NA	NA		243.17
340	NA	NA		246.47
350	NA	NA		249.77
360	NA	NA		253.07
370	NA	NA		256.37
380	NA	NA		259.66
390	NA	NA		262.96
400	NA	NA		266.26
410	NA	NA		269.56
420	NA	NA		272.86
430	NA	NA		276.15
440	NA	NA		279.45
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

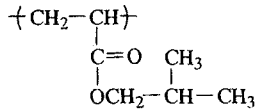
Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

T_g: 218.0 K ($\Delta C_p = 45.40$)

Solid: 10–70 K C_p data are not available
80–210 K Experimental C_p data (Data Bank, 1982)
220–600 K C_p data are not available

Melt: 10–210 K Melt unstable
220–440 K Experimental C_p data (Data Bank, 1982)
450–600 K C_p data are not available

TABLE 26. POLY(ISO-BUTYL ACRYLATE) (PIBA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA
90	NA	NA	NA	NA
100	NA	NA	NA	NA
110	NA	NA	NA	NA
120	NA	NA	NA	NA
130	NA	NA	NA	NA
140	NA	NA	NA	NA
150	NA	NA	NA	NA
160	NA	NA	NA	NA
170	NA	NA	NA	NA
180	NA	NA	NA	NA
190	NA	NA	NA	NA
200	NA	NA	NA	NA
210	NA	NA	NA	NA
220		155.80	NA	NA
230		163.50	NA	NA
240		171.30	NA	NA
250	NA	NA		215.17
260	NA	NA		218.55
270	NA	NA		221.94
280	NA	NA		225.32
290	NA	NA		228.71
300	NA	NA		232.09
310	NA	NA		235.47
320	NA	NA		238.86
330	NA	NA		242.24
340	NA	NA		245.63
350	NA	NA		248.01
360	NA	NA		252.39
370	NA	NA		255.78
380	NA	NA		259.16
390	NA	NA		262.55
400	NA	NA		265.93
410	NA	NA		269.31
420	NA	NA		272.70
430	NA	NA		276.08
440	NA	NA		279.47
450	NA	NA		282.85
460	NA	NA		286.23
470	NA	NA		289.62
480	NA	NA		293.00
490	NA	NA		296.39
500	NA	NA		299.77
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA

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TABLE 26. *cont'd*

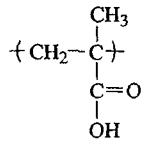
	Solid		Melt	
T (K)	Index	Value	Index	Value
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

T_g:	249.0 K	(ΔC_p = 36.60)
Solid:	10–210 K	C_p data are not available
	220–240 K	Experimental C_p data (Data Bank, 1982)
	250–600 K	C_p data are not available
Melt:	10–240 K	Melt unstable
	250–500 K	Experimental C_p data (Data Bank, 1982)
	510–600 K	C_p data are not available

TABLE 27. POLY(METHACRYLIC ACID) (PMAA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		1.23	NA	NA
20		5.50	NA	NA
30		10.85	NA	NA
40		16.38	NA	NA
50		21.76	NA	NA
60		26.90	NA	NA
70		31.80	NA	NA
80		36.46	NA	NA
90		40.91	NA	NA
100		45.18	NA	NA
110		49.27	NA	NA
120		53.23	NA	NA
130		57.06	NA	NA
140		60.77	NA	NA
150		64.39	NA	NA
160		67.93	NA	NA
170		71.39	NA	NA
180		74.78	NA	NA
190		78.12	NA	NA
200		81.41	NA	NA
210		84.65	NA	NA
220		87.85	NA	NA
230		91.02	NA	NA
240		94.16	NA	NA
250		97.27	NA	NA
260		100.40	NA	NA
270		103.40	NA	NA
280		106.50	NA	NA
290		109.50	NA	NA
300		112.50	NA	NA
310	NA	NA	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA

TABLE 27. *cont'd*

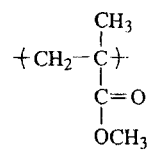
	Solid		Melt	
T (K)	Index	Value	Index	Value
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

Solid:	10–300 K	Experimental C_p data (Data Bank, 1982)
	310–600 K	C_p data are not available
Melt:	10–600 K	C_p data are not available

TABLE 28. POLY(METHYL METHACRYLATE) (PMMA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		1.67	NA	NA
20		7.97	NA	NA
30		14.67	NA	NA
40		21.38	NA	NA
50		27.67	NA	NA
60		34.53	NA	NA
70		41.30	NA	NA
80		47.29	NA	NA
90		52.64	NA	NA
100		57.49	NA	NA
110		61.97	NA	NA
120		66.18	NA	NA
130		70.20	NA	NA
140		74.10	NA	NA

TABLE 29. *cont'd*

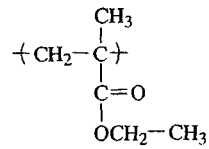
	Solid		Melt	
T (K)	Index	Value	Index	Value
150		77.94	NA	NA
160		81.74	NA	NA
170		85.55	NA	NA
180		89.40	NA	NA
190		93.52	NA	NA
200		97.52	NA	NA
210		101.52	NA	NA
220		105.53	NA	NA
230		109.54	NA	NA
240		113.56	NA	NA
250		117.58	NA	NA
260		121.60	NA	NA
270		125.63	NA	NA
280		129.66	NA	NA
290		133.69	NA	NA
300		137.72	NA	NA
310		141.75	NA	NA
320		145.78	NA	NA
330		149.82	NA	NA
340		153.85	NA	NA
350		157.89	NA	NA
360		161.93	NA	NA
370		165.97	NA	NA
380	NA	NA		203.16
390	NA	NA		205.54
400	NA	NA		207.91
410	NA	NA		210.28
420	NA	NA		212.66
430	NA	NA		215.03
440	NA	NA		217.41
450	NA	NA		219.78
460	NA	NA		222.15
470	NA	NA		224.53
480	NA	NA		226.90
490	NA	NA		229.28
500	NA	NA		231.65
510	NA	NA		234.02
520	NA	NA		236.40
530	NA	NA		238.77
540	NA	NA		241.15
550	NA	NA		243.52
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1068 (1982).

T_g:	378.0 K	(C_p = 169.2–202.7)
Solid:	10–370 K	Experimental C_p data (Data Bank, 1982)
	380–600 K	C_p data are not available
Melt:	10–370 K	Melt unstable
	380–550 K	Experimental C_p data (Data Bank, 1982)
	560–600 K	C_p data are not available

TABLE 29. POLY(ETHYL METHACRYLATE) (PEMA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80		58.84	NA	NA
90		63.60	NA	NA
100		68.42	NA	NA
110		73.28	NA	NA
120		78.17	NA	NA
130		83.08	NA	NA
140		88.01	NA	NA
150		92.94	NA	NA
160		97.88	NA	NA
170		102.83	NA	NA
180		107.79	NA	NA
190		112.75	NA	NA
200		117.71	NA	NA
210		122.67	NA	NA
220		127.64	NA	NA
230		132.61	NA	NA
240		137.58	NA	NA
250		142.55	NA	NA
260		147.52	NA	NA
270		152.49	NA	NA
280		157.47	NA	NA
290		162.44	NA	NA
300		167.42	NA	NA
310		172.39	NA	NA
320		177.37	NA	NA
330		182.34	NA	NA
340	NA	NA		218.77
350	NA	NA		222.47
360	NA	NA		226.17
370	NA	NA		229.87
380	NA	NA		233.57
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA

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TABLE 29. *cont'd*

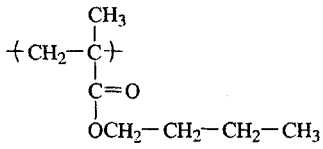
	Solid		Melt	
T (K)	Index	Value	Index	Value
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

T_g:	338.0 K	($\Delta C_p = 31.70$)
Solid:	10–70 K	C_p data are not available
	80–330 K	Experimental C_p data (Data Bank, 1982)
	340–600 K	C_p data are not available
Melt:	10–330 K	Melt unstable
	340–380 K	Experimental C_p data (Data Bank, 1982)
	390–600 K	C_p data are not available

TABLE 30. POLY(*n*-BUTYL METHACRYLATE) (P*n*BMA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80		77.81	NA	NA
90		84.85	NA	NA
100		91.96	NA	NA
110		99.13	NA	NA
120		106.32	NA	NA
130		113.54	NA	NA
140		120.77	NA	NA
150		128.01	NA	NA
160		135.27	NA	NA
170		142.53	NA	NA
180		149.80	NA	NA
190		157.07	NA	NA
200		164.34	NA	NA
210		171.62	NA	NA
220		178.90	NA	NA
230		186.18	NA	NA
240		193.47	NA	NA
250		200.75	NA	NA
260		208.04	NA	NA
270		215.33	NA	NA
280		222.62	NA	NA
290		229.91	NA	NA
300	NA	NA		263.41
310	NA	NA		268.29
320	NA	NA		273.17

TABLE 30. *cont'd*

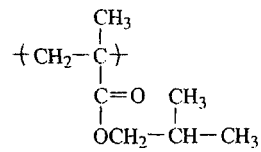
	Solid		Melt	
T (K)	Index	Value	Index	Value
330	NA	NA		278.05
340	NA	NA		282.93
350	NA	NA		287.82
360	NA	NA		292.70
370	NA	NA		297.58
380	NA	NA		302.46
390	NA	NA		307.34
400	NA	NA		312.22
410	NA	NA		317.10
420	NA	NA		321.98
430	NA	NA		326.86
440	NA	NA		331.74
450	NA	NA		336.63
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

T_g:	293.0 K	($\Delta C_p = 29.70$)
Solid:	10–70 K	C_p data are not available
	80–290 K	Experimental C_p data (Data Bank, 1982)
	300–600 K	C_p data are not available
Melt:	10–290 K	Melt unstable
	300–450 K	Experimental C_p data (Data Bank, 1982)
	460–600 K	C_p data are not available

TABLE 31. POLY(ISO-BUTYL-METHACRYLATE) (PIBMA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA

TABLE 32. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
90	NA	NA	NA	NA
100	NA	NA	NA	NA
110	NA	NA	NA	NA
120	NA	NA	NA	NA
130	NA	NA	NA	NA
140	NA	NA	NA	NA
150	NA	NA	NA	NA
160	NA	NA	NA	NA
170	NA	NA	NA	NA
180	NA	NA	NA	NA
190	NA	NA	NA	NA
200	NA	NA	NA	NA
210	NA	NA	NA	NA
220	NA	NA	NA	NA
230		173.90	NA	NA
240		181.00	NA	NA
250		188.00	NA	NA
260		195.10	NA	NA
270		202.20	NA	NA
280		209.20	NA	NA
290		216.30	NA	NA
300		223.40	NA	NA
310		230.40	NA	NA
320		237.50	NA	NA
330	NA	NA		281.77
340	NA	NA		284.43
350	NA	NA		287.10
360	NA	NA		289.77
370	NA	NA		292.43
380	NA	NA		295.10
390	NA	NA		297.76
400	NA	NA		300.43
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

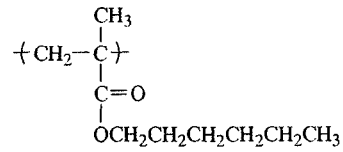
Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

T_g: 326.0 K ($\Delta C_p = 39.00$)

Solid: 10–220 K C_p data are not available
230–320 K Experimental C_p data (Data Bank, 1982)
330–600 K C_p data are not available

Melt: 10–320 K Melt unstable
330–400 K Experimental C_p data (Data Bank, 1982)
410–600 K C_p data are not available

TABLE 32. POLY(HEXYL METHACRYLATE) (PHMA)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA
90	NA	NA	NA	NA
100	NA	NA	NA	NA
110	NA	NA	NA	NA
120	NA	NA	NA	NA
130	NA	NA	NA	NA
140	NA	NA	NA	NA
150	NA	NA	NA	NA
160	NA	NA	NA	NA
170	NA	NA	NA	NA
180	NA	NA	NA	NA
190	NA	NA	NA	NA
200	NA	NA	NA	NA
210	NA	NA	NA	NA
220	NA	NA	NA	NA
230	NA	NA	NA	NA
240	NA	NA	NA	NA
250	NA	NA	NA	NA
260	NA	NA	NA	NA
270	NA	NA		310.77
280	NA	NA		315.46
290	NA	NA		320.14
300	NA	NA		324.83
310	NA	NA		329.52
320	NA	NA		334.20
330	NA	NA		338.89
340	NA	NA		343.57
350	NA	NA		348.26
360	NA	NA		352.95
370	NA	NA		357.63
380	NA	NA		362.32
390	NA	NA		367.00
400	NA	NA		371.69
410	NA	NA		376.38
420	NA	NA		381.06
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA

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TABLE 32. *cont'd*

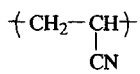
	Solid		Melt	
T (K)	Index	Value	Index	Value
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

Solid:	10–600 K	C_p data are not available
Melt:	10–260 K	Melt unstable
	270–420 K	Experimental C_p data (Data Bank, 1982)
	430–600 K	C_p data are not available

TABLE 33. POLY(ACRYLONITRILE) (PAN)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	1.17	NA	NA
20	NA	1.93	NA	NA
30	NA	4.94	NA	NA
40	NA	9.55	NA	NA
50	NA	13.77	NA	NA
60		18.09	NA	NA
70		21.69	NA	NA
80		24.87	NA	NA
90		27.69	NA	NA
100		30.22	NA	NA
110		32.54	NA	NA
120		34.67	NA	NA
130		36.68	NA	NA
140		38.59	NA	NA
150		40.44	NA	NA
160		42.23	NA	NA
170		44.00	NA	NA
180		45.75	NA	NA
190		47.51	NA	NA
200		49.27	NA	NA
210		51.04	NA	NA
220		52.85	NA	NA
230		54.68	NA	NA
240		56.56	NA	NA
250		58.47	NA	NA
260		60.43	NA	NA
270		62.45	NA	NA
280		64.51	NA	NA
290		66.64	NA	NA
300		68.83	NA	NA
310		71.08	NA	NA
320		73.41	NA	NA
330		75.80	NA	NA
340		78.27	NA	NA
350		80.82	NA	NA
360		83.45	NA	NA
370		86.16	NA	NA
380	NA	NA	NA	NA

TABLE 33. *cont'd*

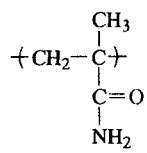
	Solid		Melt	
T (K)	Index	Value	Index	Value
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

T_g:	378.0 K	
T_m:	593.2 K	
Solid:	10–50 K	C_p value are calculated from the Tarasov equation (Data Bank, 1982)
	60–370 K	Experimental C_p data (Data Bank, 1982)
	380–600 K	C_p data are not available
Melt:	10–600 K	C_p data are not available

TABLE 34. POLY(METHACRYLAMIDE) (PMAM)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	0.21	NA	NA
20	NA	2.81	NA	NA
30	NA	8.14	NA	NA
40	NA	14.70	NA	NA
50	NA	21.50	NA	NA
60		28.07	NA	NA
70		34.25	NA	NA
80		39.99	NA	NA
90		45.31	NA	NA
100		50.25	NA	NA
110		54.86	NA	NA
120		59.20	NA	NA
130		63.29	NA	NA
140		67.18	NA	NA
150		70.90	NA	NA

TABLE 34. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
160		74.48	NA	NA
170		77.94	NA	NA
180		81.30	NA	NA
190		84.59	NA	NA
200		87.81	NA	NA
210		90.98	NA	NA
220		94.11	NA	NA
230		97.21	NA	NA
240		100.30	NA	NA
250		103.30	NA	NA
260		106.40	NA	NA
270		109.50	NA	NA
280		112.50	NA	NA
290		115.60	NA	NA
300		118.70	NA	NA
310	NA	NA	NA	NA
320	NA	NA	NA	NA
330	NA	NA	NA	NA
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **11** (4), 1074 (1982).

Solid: 10–50 K C_p value are calculated from the Tarasov equation (Data Bank, 1982)

60–300 K Experimental C_p data (Data Bank, 1982)

310–600 K C_p data are not available

Melt: 10–600 K C_p data are not available

TABLE 35. POLY(OXY-1-OXOETHYLENE) (PGL)

	Solid		Melt	
T (K)	Index	Value	Index	Value
10		0.78	NA	NA
20		3.98	NA	NA
30		8.02	NA	NA
40		12.12	NA	NA
50		16.04	NA	NA
60		19.57	NA	NA
70		22.72	NA	NA
80		25.55	NA	NA
90		28.11	NA	NA
100		30.47	NA	NA
110		32.67	NA	NA
120		34.74	NA	NA
130		36.69	NA	NA
140		38.53	NA	NA
150		40.28	NA	NA
160		41.96	NA	NA
170		43.58	NA	NA
180		45.15	NA	NA
190		46.69	NA	NA
200		48.22	NA	NA
210		49.99	NA	NA
220		51.71	NA	NA
230		53.42	NA	NA
240		55.14	NA	NA
250		56.85	NA	NA
260		58.57	NA	NA
270		60.28	NA	NA
280		62.00	NA	NA
290		63.71	NA	NA
300	NA	65.42	NA	NA
310	NA	67.14	NA	NA
320	NA	68.85	NA	NA
330	NA	70.57	NA	NA
340	NA	72.28	NA	NA
350	NA	74.00	NA	114.13
360	NA	75.71	NA	114.51
370	NA	77.43	NA	114.90
380	NA	79.14	NA	115.28
390	NA	80.86	NA	115.66
400	NA	82.57	NA	116.04
410	NA	84.29	NA	116.42
420	NA	86.00	NA	116.80
430	NA	87.72	NA	117.18
440	NA	89.43	NA	117.56
450	NA	91.15	NA	117.94
460	NA	92.86	NA	118.32
470	NA	94.58	NA	118.70
480	NA	96.29	NA	119.09
490	NA	98.01	NA	119.47
500	NA	99.72	NA	119.85
510	NA	NA		120.23
520	NA	NA		120.61
530	NA	NA		120.99
540	NA	NA		121.37
550	NA	NA		121.75
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

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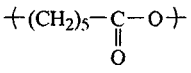
TABLE 35. *cont'd*

Notes

References: 1. B. Lebedev, A. Yevstropov, Makromol. Chem., 185 (6), 1235 (1984).
2. R. Pan, Master Thesis, Dept. of Chemistry, Rensselaer Polytechnic Institute, Dec. 1986.

T_g:	318.0 K	(ΔC_p = 44.7)
T_m:	501.0 K	(ΔC_p = 20.0)
Solid:	10–290 K	Experimental C_p data interpolated by Spline function (Data Bank, 1985)
	300–500 K	Linear extrapolated data from the region of 200–290 K (Data Bank, 1985)
	510–600 K	C_p data are not available
Melt:	10–310 K	Melt unstable
	350–500 K	Linear extrapolated data from the region of 500–550 K (Data Bank, 1985)
	510–550 K	Experimental C_p data (Data Bank, 1985)
	560–600 K	C_p data are not available

TABLE 36. POLY(OXY-1-OXOHEXAMETHYLENE) (PCL)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		2.42	NA	NA
20		10.72	NA	NA
30		20.66	NA	NA
40		30.17	NA	NA
50		37.95	NA	NA
60		45.74	NA	NA
70		53.02	NA	NA
80		59.61	NA	NA
90		65.62	NA	NA
100		71.14	NA	NA
110		76.25	NA	NA
120		81.02	NA	NA
130		85.55	NA	NA
140		89.91	NA	NA
150		94.19	NA	NA
160		98.73	NA	NA
170		103.28	NA	NA
180		107.83	NA	NA
190		112.38	NA	NA
200		116.92	NA	NA
210		121.47	NA	180.81
220		126.02	NA	183.72
230		130.57	NA	186.64
240		135.12	NA	189.55
250		139.67	NA	192.47
260		144.22	NA	195.38
270		148.77	NA	198.30
280		153.32	NA	201.21
290	NA	157.87	NA	204.13
300	NA	162.42	NA	207.04
310	NA	166.97	NA	209.96
320	NA	171.51	NA	212.87
330	NA	176.06	NA	215.79
340	NA	180.61		218.70
350	NA	NA		221.62
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA

TABLE 36. *cont'd*

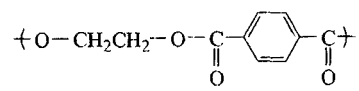
	Solid		Melt	
T (K)	Index	Value	Index	Value
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

References: 1. B. Lebedev, A. Yevstropov, Makromol. Chem., 185 (6), 1235 (1984).
2. R. Pan, Master Thesis, Dept. of Chemistry, Rensselaer Polytechnic Institute, Dec. 1986.

T_g:	209.0 K	(ΔC_p = 60.0)
T_m:	342.0 K	(ΔC_p = 38.0)
Solid:	10–200 K	Experimental C_p data interpolated by Spline function (Data Bank, 1985)
	210–340 K	Linear extrapolated data from the region of 150–200 K (Data Bank, 1985)
	350–600 K	C_p data are not available
Melt:	10–200 K	Melt unstable
	210–300 K	Linear extrapolated data from the region of 342–350 K (Data Bank, 1985)
	340–350 K	Experimental C_p data (Data Bank, 1985)
	360–600 K	C_p data are not available

TABLE 37. POLY(OXYETHYLENEOXYTEREPHTHALOYL) (PET)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		12.43	NA	NA
20	NA	NA	NA	NA
30		22.91	NA	NA
40		36.18	NA	NA
50		48.24	NA	NA
60		56.68	NA	NA
70		63.92	NA	NA
80		69.95	NA	NA
90		77.18	NA	NA
100		84.42	NA	NA
110		91.70	NA	NA
120		100.10	NA	NA
130		105.90	NA	NA
140		112.60	NA	NA

TABLE 37. *cont'd*

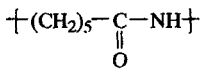
	Solid		Melt	
T (K)	Index	Value	Index	Value
150		119.40	NA	NA
160		126.20	NA	NA
170		133.10	NA	NA
180		140.10	NA	NA
190		147.10	NA	NA
200		154.10	NA	NA
210		161.20	NA	NA
220		168.20	NA	NA
230		175.30	NA	NA
240		182.40	NA	NA
250		189.50	NA	NA
260		196.70	NA	NA
270		203.80	NA	NA
280		210.90	NA	NA
290		218.10	NA	NA
300		225.20	NA	NA
310		232.40	NA	NA
320		239.60	NA	NA
330		246.70	NA	NA
340		253.90	NA	NA
350	NA	NA		335.70
360	NA	NA		338.50
370	NA	NA		341.30
380	NA	NA		344.10
390	NA	NA		346.90
400	NA	NA		349.80
410	NA	NA		352.60
420	NA	NA		355.40
430	NA	NA		358.20
440	NA	NA		361.00
450	NA	NA		363.90
460	NA	NA		366.70
470	NA	NA		369.50
480	NA	NA		372.30
490	NA	NA		375.10
500	NA	NA		378.00
510	NA	NA		380.80
520	NA	NA		383.60
530	NA	NA		386.40
540	NA	NA		389.20
550	NA	NA		392.10
560	NA	NA		394.90
570	NA	NA		397.70
580	NA	NA		400.50
590	NA	NA		403.30
600	NA	NA	NA	406.15

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12**, (1), 70 (1983).

T_g:	342.0 K	(C_p = 255.6–333.4)
T_m:	553.0 K	
Solid:	10–340 K	Experimental C_p (Data Bank, 1983)
	350–600 K	C_p data are not available
Melt:	10–340 K	Melt unstable
	350–590 K	Experimental C_p data (Data Bank, 1983)
	600 K	Linear extrapolated data from C_p in the region of 350–590 K (Data Bank, 1985)

TABLE 38. POLY(IMINO-(1-OXOHEXAMETHYLENE)) (NYLON 6)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70		49.78	NA	NA
80		56.40	NA	NA
90		62.45	NA	NA
100		68.06	NA	NA
110		73.34	NA	NA
120		78.38	NA	NA
130		83.26	NA	NA
140		88.01	NA	NA
150		92.70	NA	NA
160		97.35	NA	NA
170		102.00	NA	NA
180		106.70	NA	NA
190		111.40	NA	NA
200		116.20	NA	NA
210		121.00	NA	NA
220		126.00	NA	NA
230		131.00	NA	NA
240		136.20	NA	NA
250		141.40	NA	NA
260		146.90	NA	NA
270		152.40	NA	NA
280		158.10	NA	NA
290		164.00	NA	NA
300		170.00	NA	NA
310		176.20	NA	NA
320	NA	NA	NA	272.80
330	NA	NA	NA	274.30
340	NA	NA	NA	275.80
350	NA	NA	NA	277.40
360	NA	NA	NA	278.90
370	NA	NA	NA	280.40
380	NA	NA	NA	281.90
390	NA	NA	NA	283.50
400	NA	NA	NA	285.00
410	NA	NA	NA	286.50
420	NA	NA	NA	288.00
430	NA	NA	NA	289.60
440	NA	NA	NA	291.10
450	NA	NA	NA	292.60
460	NA	NA	NA	294.10
470	NA	NA	NA	295.70
480	NA	NA	NA	297.20
490	NA	NA	NA	298.70
500	NA	NA		300.20
510	NA	NA		301.80
520	NA	NA		303.30
530	NA	NA		304.80
540	NA	NA		306.30
550	NA	NA		307.90
560	NA	NA		309.40
570	NA	NA		310.90
580	NA	NA		312.40
590	NA	NA		314.00
600	NA	NA		315.50

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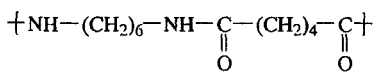
TABLE 38. *cont'd*

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12**, (1), 83 (1983).

T_g:	313.0 K	(C_p = 178.1–271.7)
T_m:	553.0 K	
Solid:	10–60 K	C_p data are not available
	70–310 K	Experimental C_p data (Data Bank, 1983)
	320–600 K	C_p data are not available
Melt:	10–310 K	Melt unstable
	320–600 K	Experimental C_p data (Data Bank, 1983)

TABLE 39. POLY(IMINOADIPOYLIMINOHEXAMETHYLENE) (NYLON 66)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA
90	NA	NA	NA	NA
100	NA	NA	NA	NA
110	NA	NA	NA	NA
120	NA	NA	NA	NA
130	NA	NA	NA	NA
140	NA	NA	NA	NA
150	NA	NA	NA	NA
160	NA	NA	NA	NA
170	NA	NA	NA	NA
180	NA	NA	NA	NA
190	NA	NA	NA	NA
200	NA	NA	NA	NA
210	NA	NA	NA	NA
220	NA	NA	NA	NA
230		252.10	NA	NA
240		263.40	NA	NA
250		274.70	NA	NA
260		286.00	NA	NA
270		297.30	NA	NA
280		308.60	NA	NA
290		320.00	NA	NA
300		331.30	NA	NA
310		342.60	NA	NA
320		354.00	NA	NA
330	NA	NA		505.70
340	NA	NA		510.40
350	NA	NA		515.00
360	NA	NA		519.70
370	NA	NA		524.40
380	NA	NA		529.70
390	NA	NA		533.80
400	NA	NA		538.50
410	NA	NA		543.10
420	NA	NA		547.80
430	NA	NA		552.50
440	NA	NA		557.20

TABLE 39. *cont'd*

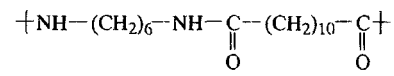
	Solid		Melt	
T (K)	Index	Value	Index	Value
450	NA	NA		561.90
460	NA	NA		566.50
470	NA	NA		571.20
480	NA	NA		575.90
490	NA	NA		580.60
500	NA	NA		585.30
510	NA	NA		590.00
520	NA	NA		594.60
530	NA	NA		599.30
540	NA	NA		604.00
550	NA	NA		608.70
560	NA	NA		613.40
570	NA	NA		618.10
580	NA	NA		622.70
590	NA	NA		627.40
600	NA	NA		632.10

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12**, (1), 84 (1983).

T_g:	323.0 K	(C_p = 357.4–502.4)
T_m:	553.0 K	
Solid:	10–220 K	C_p data are not available
	230–320 K	Experimental C_p data (Data Bank, 1983)
	330–600 K	C_p data are not available
Melt:	10–320 K	Melt unstable
	330–600 K	Experimental C_p data (Data Bank, 1983)

TABLE 40. POLY(IMINOADIPOYLIMINODODECAMETHYLENE) (NYLON 612)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA
90	NA	NA	NA	NA
100	NA	NA	NA	NA
110	NA	NA	NA	NA
120	NA	NA	NA	NA
130	NA	NA	NA	NA
140	NA	NA	NA	NA
150	NA	NA	NA	NA
160	NA	NA	NA	NA
170	NA	NA	NA	NA
180	NA	NA	NA	NA
190	NA	NA	NA	NA
200	NA	NA	NA	NA
210	NA	NA	NA	NA
220	NA	NA	NA	NA
230		362.40	NA	NA
240		378.70	NA	NA

TABLE 40. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
250		395.00	NA	NA
260		411.20	NA	NA
270		427.50	NA	NA
280		443.80	NA	NA
290		460.10	NA	NA
300		476.30	NA	NA
310		492.60	NA	NA
320	NA	NA		723.00
330	NA	NA		729.10
340	NA	NA		735.10
350	NA	NA		741.10
360	NA	NA		747.20
370	NA	NA		753.20
380	NA	NA		759.30
390	NA	NA		765.30
400	NA	NA		771.30
410	NA	NA		777.40
420	NA	NA		783.40
430	NA	NA		789.50
440	NA	NA		795.50
450	NA	NA		801.50
460	NA	NA		807.60
470	NA	NA		831.60
480	NA	NA		819.60
490	NA	NA		825.70
500	NA	NA		831.70
510	NA	NA		837.80
520	NA	NA		843.80
530	NA	NA		849.80
540	NA	NA		855.90
550	NA	NA		861.90
560	NA	NA		867.90
570	NA	NA		874.00
580	NA	NA		880.00
590	NA	NA		886.10
600	NA	NA		892.10

Notes

Reference: U. Gaur, S. F. Lau, B. B. Wunderlich, B. Wunderlich, J. Phys. Chem. Ref. Data, **12**, (1), 87 (1983).

T_g:	319.0 K	(C_p = 507.3–722.1)
Solid:	10–220 K	C_p data are not available
	230–310 K	Experimental C_p data (Data Bank, 1983)
	330–600 K	C_p data are not available
Melt:	10–320 K	Melt unstable
	320–600 K	Experimental C_p data (Data Bank, 1983)

TABLE 41. POLY(OXYMETHYLENE) (POM)

$$-(O-CH_2)-$$

	Solid		Melt	
T (K)	Index	Value	Index	Value
10		0.42	NA	NA
20		2.46	NA	NA
30		5.41	NA	NA
40		7.97	NA	NA
50		10.10	NA	NA
60		11.85	NA	NA
70		13.30	NA	NA
80		14.54	NA	NA
90		15.65	NA	NA
100		16.68	NA	NA

TABLE 41. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
110		17.68	NA	NA
120		18.68	NA	NA
130		19.71	NA	NA
140		20.75	NA	NA
150		21.82	NA	NA
160		22.89	NA	NA
170		23.97	NA	NA
180		25.05	NA	NA
190		26.11	NA	53.58
200		27.15	NA	53.95
210		28.18	NA	54.32
220		29.20	NA	54.69
230		30.21	NA	55.07
240		31.24	NA	55.44
250		32.29	NA	55.81
260		33.39	NA	56.18
270		34.55	NA	56.55
280		35.79	NA	56.93
290		37.11	NA	57.30
300		38.52	NA	57.67
310		40.02	NA	58.04
320		41.61	NA	58.41
330		43.27	NA	58.79
340		45.00	NA	59.16
350		46.77	NA	59.53
360		48.57	NA	59.90
370		50.38	NA	60.27
380		52.20	NA	60.65
390		54.00	NA	61.02
400	NA	NA	NA	61.39
410	NA	NA	NA	61.76
420	NA	NA	NA	62.13
430	NA	NA		62.51
440	NA	NA		62.88
450	NA	NA		63.25
460	NA	NA		63.62
470	NA	NA		63.99
480	NA	NA		64.37
490	NA	NA		64.74
500	NA	NA		65.11
510	NA	NA		65.48
520	NA	NA		65.85
530	NA	NA		66.23
540	NA	NA		66.60
550	NA	NA	NA	66.97
560	NA	NA	NA	67.34
570	NA	NA	NA	67.71
580	NA	NA	NA	68.09
590	NA	NA	NA	68.46
600	NA	NA	NA	68.83

Notes

References: 1. U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10** (4), 1005 (1981).
2. H. Suzuki, B. Wunderlich, J. Polym. Sci., Polym. Phys. Ed., **23**, 1671 (1985).

T_g:	190.0 K	(C_p = 26.11–53.58)
T_m:	457.2 K	
Solid:	10–390 K	Experimental C_p data (Data Bank, 1981)
	400–600 K	C_p data are not available
Melt:	10–180 K	C_p data are not available
	190–420 K	Extrapolated value from C_p in the region of 430–540 K (Data Bank, 1985)
	430–540 K	Experimental C_p data (Data Bank, 1985)
	550–600 K	Extrapolated value from C_p in the region of 430–540 K (Data Bank, 1985)

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TABLE 42. POLY(OXYETHYLENE)

$-\{O-CH_2CH_2\}-$

	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	0.51	NA	NA
20	NA	3.18	NA	NA
30	NA	7.24	NA	NA
40	NA	11.16	NA	NA
50	NA	14.62	NA	NA
60	NA	17.60	NA	NA
70	NA	20.13	NA	NA
80	NA	22.33	NA	NA
90		24.90	NA	NA
100		26.93	NA	NA
110		28.78	NA	NA
120		30.44	NA	NA
130		32.10	NA	NA
140		33.57	NA	NA
150		35.05	NA	NA
160		36.53	NA	NA
170		37.80	NA	NA
180		39.11	NA	NA
190		40.40	NA	NA
200		41.88	NA	NA
210		43.17	NA	81.88
220		44.64	NA	82.55
230	NA	45.91	NA	83.21
240	NA	47.26	NA	83.88
250	NA	48.61	NA	84.55
260	NA	49.96	NA	85.22
270	NA	51.31	NA	85.89
280	NA	52.66	NA	86.55
290	NA	54.01	NA	87.22
300	NA	55.36	NA	87.89
310	NA	56.71	NA	88.56
320	NA	58.06	NA	89.23
330	NA	59.41		89.89
340	NA	60.76		90.56
350	NA	NA		91.23
360	NA	NA		91.90
370	NA	NA		92.57
380	NA	NA		93.23
390	NA	NA		93.90
400	NA	NA		94.57
410	NA	NA		95.24
420	NA	NA		95.91
430	NA	NA		96.57
440	NA	NA	NA	97.24
450	NA	NA	NA	97.91
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

TABLE 42. *cont'd*

Notes

References:
1. U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10** (4), 1010 (1981).
2. H. Suzuki, B. Wunderlich, J. Polym. Sci., Polym. Phys. Ed., **23**, 1671 (1985).

T_g:	260.0 K	(C_p = 42.65–81.61)
T_m:	342.0 K	(C_p = 61.03–90.70)
Solid:	10–80 K	Value estimated by additivity (Data Bank, 1981)
	90–220 K	Experimental C_p data (Data Bank, 1981)
	230–340 K	Extrapolated value from C_p in the region of 160–220 K (Data Bank, 1981)
	350–600 K	C_p data are not available
Melt:	10–200 K	C_p data are not available
	210–320 K	Extrapolated value from C_p in the region of 430–540 K (Data Bank, 1985)
	330–450 K	Experimental C_p data (Data Bank, 1985)
	460–600 K	C_p data are not available

TABLE 43. POLY(OXYTRIMETHYLENE) (PO3M)

$-\{O-(CH_2)_3\}-$

	Solid		Melt	
T (K)	Index	Value	Index	Value
10		0.86	NA	NA
20		4.68	NA	NA
30		9.31	NA	NA
40		13.65	NA	NA
50		17.64	NA	NA
60		21.37	NA	NA
70		25.82	NA	NA
80		29.59	NA	NA
90		32.90	NA	NA
100		35.87	NA	NA
110		38.58	NA	NA
120		41.11	NA	NA
130		43.51	NA	NA
140		45.83	NA	NA
150		48.11	NA	NA
160		50.37	NA	NA
170		52.65	NA	NA
180		54.97	NA	NA
190	NA	57.33	NA	NA
200	NA	59.01		109.24
210	NA	61.08		110.25
220	NA	63.15		111.26
230	NA	65.23		112.27
240	NA	67.30		113.28
250	NA	69.37		114.29
260	NA	71.44		115.30
270	NA	73.51		116.31
280	NA	75.59		117.32
290	NA	77.66		118.33
300	NA	79.73		119.34
310	NA	NA		120.35
320	NA	NA		121.36
330	NA	NA		122.37
340	NA	NA	NA	NA
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA

TABLE 43. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10** (4), 1015 (1981).

T_g:	195.0 K	(C_p = 57.97–108.7)
T_m:	308.0 K	(C_p = 81.39–120.1)
Solid:	10–180 K	Experimental C_p data (Data Bank, 1981)
	190–300 K	Extrapolated value from C_p in the region of 150–180 K (Data Bank, 1981)
	310–600 K	C_p data are not available
Melt:	10–190 K	Melt unstable
	200–330 K	Experimental C_p data (Data Bank, 1981)
	340–600 K	C_p data are not available

TABLE 44. POLY(OXYTETRAMETHYLENE) (PO4M)

$$-\{O-(CH_2)_4\}-$$

	Solid		Melt	
T (K)	Index	Value	Index	Value
10		1.47	NA	NA
20		6.59	NA	NA
30		12.42	NA	NA
40		18.32	NA	NA
50		24.63	NA	NA
60		29.74	NA	NA
70		34.61	NA	NA
80		39.41	NA	NA
90		43.67	NA	NA
100		47.27	NA	NA
110		50.72	NA	NA
120		54.70	NA	NA
130		57.99	NA	NA
140		61.12	NA	NA
150		64.46	NA	NA
160		67.96	NA	NA
170	NA	71.20	NA	NA
180	NA	74.52	NA	NA
190	NA	NA	NA	124.20
200	NA	NA	NA	125.92
210	NA	NA	NA	127.64
220	NA	NA	NA	129.36
230	NA	NA	NA	131.09
240	NA	NA	NA	132.81
250	NA	NA	NA	134.53

TABLE 44. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
260	NA	NA	NA	136.26
270	NA	NA	NA	137.98
280	NA	NA		139.70
290	NA	NA		141.42
300	NA	NA		143.15
310	NA	NA		144.87
320	NA	NA		146.59
330	NA	NA		148.32
340	NA	NA		150.04
350	NA	NA	NA	NA
360	NA	NA	NA	NA
370	NA	NA	NA	NA
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10** (4), 1023 (1981).

T_g:	189.0 K	(C_p = 77.51–124.0)
T_m:	330.0 K	
Solid:	10–160 K	Experimental C_p data (Data Bank, 1981)
	170–180 K	Extrapolated value from C_p in the region of 130–160 K (Data Bank, 1981)
	190–600 K	C_p data are not available
Melt:	10–180 K	Melt unstable
	190–270 K	Extrapolated value from C_p in the region of 280–340 K (Data Bank, 1981)
	280–340 K	Experimental C_p data (Data Bank, 1981)
	350–600 K	C_p data are not available

TABLE 45. POLY(OXYPROPYLENE) (POPr)

$$-\{O-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}\}-$$

	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA

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TABLE 45. *cont'd*

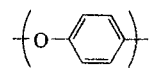
	Solid		Melt	
T (K)	Index	Value	Index	Value
70	NA	NA	NA	NA
80	NA	31.21	NA	NA
90	NA	34.33	NA	NA
100	NA	37.37	NA	NA
110	NA	40.34	NA	NA
120	NA	43.22	NA	NA
130	NA	46.03	NA	NA
140	NA	48.76	NA	NA
150	NA	51.41	NA	NA
160	NA	53.98	NA	NA
170	NA	56.48	NA	NA
180	NA	58.89	NA	NA
190	NA	61.23	NA	NA
200	NA	NA		95.46
210	NA	NA		97.04
220	NA	NA		98.61
230	NA	NA		100.19
240	NA	NA		101.77
250	NA	NA		103.35
260	NA	NA		104.92
270	NA	NA		106.50
280	NA	NA		108.08
290	NA	NA		109.65
300	NA	NA		111.23
310	NA	NA		112.81
320	NA	NA		114.38
330	NA	NA		115.96
340	NA	NA		117.54
350	NA	NA		119.12
360	NA	NA		120.69
370	NA	NA		122.27
380	NA	NA	NA	NA
390	NA	NA	NA	NA
400	NA	NA	NA	NA
410	NA	NA	NA	NA
420	NA	NA	NA	NA
430	NA	NA	NA	NA
440	NA	NA	NA	NA
450	NA	NA	NA	NA
460	NA	NA	NA	NA
470	NA	NA	NA	NA
480	NA	NA	NA	NA
490	NA	NA	NA	NA
500	NA	NA	NA	NA
510	NA	NA	NA	NA
520	NA	NA	NA	NA
530	NA	NA	NA	NA
540	NA	NA	NA	NA
550	NA	NA	NA	NA
560	NA	NA	NA	NA
570	NA	NA	NA	NA
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

Reference: U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10** (4), 1033 (1981).

T_g:	198.0 K	(C_p = 63.05–95.20)
T_m:	348.0 K	
Solid:	10–70 K	C_p data are not available
	80–190 K	Experimental C_p data (Data Bank, 1981)
	200–600 K	C_p data are not available
Melt:	10–190 K	Melt unstable
	200–370 K	Experimental C_p data (Data Bank, 1981)
	380–600 K	C_p data are not available

TABLE 46. POLY(OXY-1,4-PHENYLENE) (POPh)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA
90	NA	NA	NA	NA
100	NA	NA	NA	NA
110	NA	NA	NA	NA
120	NA	NA	NA	NA
130	NA	NA	NA	NA
140	NA	NA	NA	NA
150	NA	NA	NA	NA
160	NA	NA	NA	NA
170	NA	NA	NA	NA
180	NA	NA	NA	NA
190	NA	NA	NA	NA
200	NA	NA	NA	NA
210	NA	NA	NA	NA
220	NA	NA	NA	NA
230	NA	NA	NA	NA
240	NA	NA	NA	NA
250	NA	NA	NA	NA
260	NA	NA	NA	NA
270	NA	NA	NA	NA
280	NA	NA	NA	NA
290	NA	NA	NA	NA
300		109.10	NA	NA
310		112.40	NA	NA
320		115.80	NA	NA
330		119.20	NA	NA
340		122.50	NA	NA
350		125.90	NA	NA
360	NA	NA		150.30
370	NA	NA		151.70
380	NA	NA		153.10
390	NA	NA	NA	154.60
400	NA	NA	NA	156.00
410	NA	NA	NA	157.40
420	NA	NA	NA	158.80
430	NA	NA	NA	160.30
440	NA	NA	NA	161.70
450	NA	NA	NA	163.10
460	NA	NA	NA	164.50
470	NA	NA	NA	166.00
480	NA	NA	NA	167.40
490	NA	NA	NA	168.80
500	NA	NA	NA	170.20
510	NA	NA	NA	171.70
520	NA	NA	NA	173.10
530	NA	NA	NA	174.50
540	NA	NA	NA	175.90
550	NA	NA	NA	177.40
560	NA	NA	NA	178.80
570	NA	NA	NA	180.20
580	NA	NA	NA	181.60
590	NA	NA		183.10
600	NA	NA		184.50

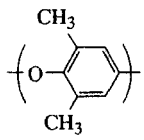
TABLE 46. *cont'd*

Notes

Reference: U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10** (4), 1033 (1981).

T_g:	358.0 K	(C_p = 128.6–150.0)
T_m:	535.0 K	
Solid:	10–290 K	C_p data are not available
	300–350 K	Experimental C_p data (Data Bank, 1981)
	360–600 K	C_p data are not available
Melt:	10–350 K	Melt unstable
	360–380 K	Experimental C_p data (Data Bank, 1981)
	390–580 K	Extrapolated value from C_p in the region of 360–380 K & 590–620 K (Data Bank, 1981)
	590–600 K	Experimental C_p data (Data Bank, 1981)

TABLE 47. POLY(OXY-2,6-DIMETHYL-1,4-PHENYLENE) (PPO)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		4.05	NA	NA
20		11.52	NA	NA
30		19.12	NA	NA
40		26.45	NA	NA
50		33.44	NA	NA
60		40.12	NA	NA
70		46.49	NA	NA
80		52.50	NA	NA
90		57.80	NA	NA
100		63.01	NA	NA
110		68.07	NA	NA
120		72.91	NA	NA
130		77.45	NA	NA
140		82.04	NA	NA
150		86.42	NA	NA
160		89.74	NA	NA
170		94.14	NA	NA
180		98.48	NA	NA
190		102.80	NA	NA
200		107.00	NA	NA
210		111.20	NA	NA
220		115.33	NA	NA
230		119.57	NA	NA
240		123.81	NA	NA
250		128.04	NA	NA
260		132.33	NA	NA
270		136.60	NA	NA
280		140.91	NA	NA
290		145.27	NA	NA
300		149.66	NA	NA
310		154.08	NA	NA
320		158.58	NA	NA
330		162.10	NA	NA
340		166.20	NA	NA
350		170.20	NA	NA
360		174.50	NA	NA
370		178.60	NA	NA
380		182.70	NA	NA
390		186.80	NA	NA
400		190.50	NA	NA
410		194.50	NA	NA

TABLE 47. *cont'd*

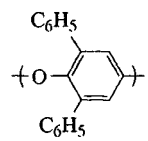
	Solid		Melt	
T (K)	Index	Value	Index	Value
420		198.20	NA	NA
430		201.30	NA	NA
440		204.70	NA	NA
450		208.10	NA	NA
460		211.60	NA	NA
470		215.00	NA	NA
480		218.40	NA	NA
490	NA	NA		252.80
500	NA	NA		255.10
510	NA	NA		257.30
520	NA	NA		259.60
530	NA	NA		261.90
540	NA	NA		264.20
550	NA	NA		266.50
560	NA	NA		268.70
570	NA	NA		271.00
580	NA	NA	NA	NA
590	NA	NA	NA	NA
600	NA	NA	NA	NA

Notes

References: 1. U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10**, 1036 (1981).
2. B. V. Lebedev, V. G. Vasil'ev, T. A. Bykova, E. G. Kiparisova, and B. Wunderlich, Vysokomol. Soedin. A, **38**, 216; Engl. Translat., Polym. Sci. Ser. A, **38** (2), 104 (1996)

T_g:	482.0 K	(C_p = 219.1–251.0)
T_m:	545.0 K	
Solid:	10–70 K	Experimental C_p data from Ref. 2
	80–320 K	Average experimental C_p data from Ref. 1 and 2
	320–480 K	Experimental C_p data from Ref. 1
	490–600 K	C_p data are not available
Melt:	10–480 K	Melt metastable
	490–570 K	Experimental C_p data from Ref. 1
	580–600 K	C_p data are not available

TABLE 48. POLY(OXY-2,6-DIPHENYL-1,4-PHENYLENE) (POPPO)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10	NA	NA	NA	NA
20	NA	NA	NA	NA
30	NA	NA	NA	NA
40	NA	NA	NA	NA
50	NA	NA	NA	NA
60	NA	NA	NA	NA
70	NA	NA	NA	NA
80	NA	NA	NA	NA
90	NA	NA	NA	NA
100	NA	NA	NA	NA
110	NA	NA	NA	NA
120	NA	NA	NA	NA
130	NA	NA	NA	NA
140	NA	NA	NA	NA
150	NA	NA	NA	NA

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TABLE 48. *cont'd*

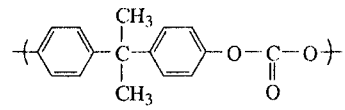
	Solid		Melt	
T (K)	Index	Value	Index	Value
160	NA	NA	NA	NA
170	NA	NA	NA	NA
180		180.60	NA	NA
190		189.20	NA	NA
200		198.20	NA	NA
210		207.30	NA	NA
220		215.90	NA	NA
230		223.10	NA	NA
240		229.40	NA	NA
250		235.90	NA	NA
260		243.30	NA	NA
270		251.20	NA	NA
280		259.30	NA	NA
290		267.00	NA	NA
300		274.10	NA	NA
310		280.50	NA	NA
320		287.00	NA	NA
330		294.50	NA	NA
340		302.50	NA	NA
350		309.50	NA	NA
360		315.00	NA	NA
370		320.20	NA	NA
380		326.30	NA	NA
390		333.00	NA	NA
400		339.40	NA	NA
410		344.80	NA	NA
420		350.10	NA	NA
430		356.30	NA	NA
440		363.00	NA	NA
450		369.50	NA	NA
460		375.20	NA	NA
470		380.30	NA	NA
480	NA	387.20	NA	NA
490	NA	393.30	NA	NA
500	NA	NA		476.50
510	NA	NA		483.40
520	NA	NA		490.30
530	NA	NA		497.30
540	NA	NA	NA	504.10
550	NA	NA	NA	510.90
560	NA	NA	NA	517.80
570	NA	NA	NA	524.70
580	NA	NA	NA	531.60
590	NA	NA	NA	538.50
600	NA	NA	NA	545.40

Notes

Reference: U. Gaur, B. Wunderlich, J. Phys. Chem. Ref. Data, **10**, (4) 1036 (1981).

T_g:	493.0 K	(C_p = 395.1–471.7)
T_m:	735.0 K	
Solid:	10–170 K	C_p data are not available
	180–470 K	Experimental C_p data (Data Bank, 1981)
	480–490 K	Extrapolated value from C_p in the region of 420–470 K (Data Bank, 1981)
	500–600 K	C_p data are not available
Melt:	10–490 K	Melt unstable
	500–530 K	Experimental C_p data (Data Bank, 1981)
	540–600 K	Extrapolated value from C_p in the region of 500–530 K & 780–820 K

TABLE 49. POLY(OXYCARBONYLOXY-1,4-PHENYLENE-ISOPROPYLIDENE-1,4-PHENYLENE) (PC)



	Solid		Melt	
T (K)	Index	Value	Index	Value
10		5.95	NA	NA
20		25.38	NA	NA
30		39.85	NA	NA
40		52.46	NA	NA
50		63.75	NA	NA
60		73.72	NA	NA
70		82.62	NA	NA
80		91.39	NA	NA
90		100.40	NA	NA
100		109.70	NA	NA
110		118.80	NA	NA
120		128.90	NA	NA
130		139.00	NA	NA
140		148.90	NA	NA
150		158.80	NA	NA
160		168.70	NA	NA
170		178.50	NA	NA
180		188.30	NA	NA
190		198.00	NA	NA
200		207.80	NA	NA
210		217.60	NA	NA
220		227.30	NA	NA
230		237.10	NA	NA
240		247.00	NA	NA
250		256.80	NA	NA
260		266.70	NA	NA
270		276.70	NA	NA
280		286.70	NA	NA
290		296.70	NA	NA
300		306.80	NA	NA
310		316.90	NA	NA
320		327.10	NA	NA
330		337.40	NA	NA
340		347.70	NA	NA
350		358.10	NA	NA
360		368.50	NA	NA
370		379.10	NA	NA
380		389.70	NA	NA
390		400.30	NA	NA
400		411.10	NA	NA
410		421.90	NA	NA
420		NA		480.30
430		NA		486.00
440		NA		491.80
450		NA		497.60
460		NA		503.40
470		NA		509.20
480		NA		515.00
490		NA		520.80
500		NA		526.60
510		NA		532.40
520		NA		538.20
530		NA		543.90
540		NA		549.70
550		NA		555.50
560		NA		561.30

TABLE 49. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
570		NA	NA	NA
580		NA	NA	NA
590		NA	NA	NA
600		NA	NA	NA

Notes

Reference: U. Gaur, S. F. Lau, B. Wunderlich, J. Phys. Chem. Ref. Data, **12**, (1) 93 (1983).

T_g:	418.0 K	(C_p = 430.6–479.1)
T_m:	608.2 K	
Solid:	10–410 K	Experimental C_p data of glassy sample (Data Bank, 1983)
	420–600 K	C_p data are not available
Melt:	10–410 K	Melt unstable
	420–560 K	Experimental C_p data (Data Bank, 1983)
	570–600 K	C_p data are not available

TABLE 50. TRIGONAL SELENIUM (SeT)

—(Se)—

	Solid		Melt	
T (K)	Index	Value	Index	Value
10		0.64	NA	NA
20		3.45	NA	NA
30		6.34	NA	NA
40		8.85	NA	NA
50		10.98	NA	NA
60		12.80	NA	NA
70		14.39	NA	NA
80		15.80	NA	NA
90		17.10	NA	NA
100		18.19	NA	NA
110		19.10	NA	NA
120		19.84	NA	NA
130		20.49	NA	NA
140		21.07	NA	NA
150		21.58	NA	NA
160		22.03	NA	NA
170		22.41	NA	NA
180		22.75	NA	NA
190		23.05	NA	NA
200		23.32	NA	NA
210		23.56	NA	NA

TABLE 50. *cont'd*

	Solid		Melt	
T (K)	Index	Value	Index	Value
220		23.78	NA	NA
230		23.99	NA	NA
240		24.18	NA	NA
250		24.36	NA	NA
260		24.52	NA	NA
270		24.68	NA	NA
280		24.83	NA	NA
290		24.97	NA	NA
300		25.11	NA	NA
310		25.25		40.11
320		25.38		39.82
330		25.51		39.54
340		25.63		39.26
350		25.75		38.98
360		25.88		38.72
370		26.00		38.46
380		26.12		38.21
390		26.24		37.96
400		26.36		37.72
410		26.48		37.49
420		26.60		37.26
430		26.72		37.04
440		26.84		36.82
450		26.96		36.62
460		27.08		36.42
470		27.20		36.22
480		27.33		36.03
490		27.45		35.85
500		27.58		35.68
510	NA	NA		35.51
520	NA	NA		35.35
530	NA	NA		35.19
540	NA	NA		35.04
550	NA	NA		34.90
560	NA	NA		34.76
570	NA	NA		34.64
580	NA	NA		34.51
590	NA	NA		34.40
600	NA	NA		34.29

Notes

Reference: U. Gaur, H. C. Shu, A. Mehta, B. Wunderlich, J. Phys. Chem. Ref. Data, **10**, (1) 89 (1981).

T_g:	303.4 K	(C_p = 27.02–40.31)
T_m:	492.2 K	(C_p = 27.51–35.78)
Solid:	10–500 K	Experimental C_p data (Data Bank, 1981)
	510–600 K	C_p data are not available
Melt:	10–300 K	Melt unstable
	310–600 K	Experimental C_p data (Data Bank, 1981)

D. REFERENCES

1. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **12**, 297 (1962).
2. Idem, **12**, 277 (1962).
3. M. Dole, W. P. Hettinger, N. R. Larson, J. A. Wethington, J. Chem. Phys., **20**, 718 (1952).
4. B. Wunderlich, M. Dole, J. Polym. Sci., **24**, 201 (1957).
5. G. T. Furukawa, M. L. Reilly, J. Res. Nat. Bur. Std., **56**, 285 (1956).
6. N. Bekkedahl, H. Matheson, J. Res. Nat. Bur. Std., **15**, 503 (1935).
7. S. S. Chang, A. B. Bestul, J. Res. Nat. Bur. Std., **75A**, 113 (1971).
8. L. A. Wood, N. Bekkedahl, Polym. Lett., **5**, 169 (1967).
9. E. Y. Roinishvili, N. N. Tavkhelidze, V. B. Akopyan, Vysokomol. Soedin. B, **9**, 254 (1967).
10. C. W. Smith, M. Dole, J. Polym. Sci., **20**, 37 (1956).
11. K. H. Hellwege, W. Knappe, W. Wetzel, Kolloid-Z., **180**, 126 (1962).
12. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **3**, 263, (1962).

13. H. Wilski, Kolloid-Z. -Z. Polym., **248**, 867 (1971).
14. F. S. Dainton, D. M. Evans, F. E. Hoare, T. P. Melia, Polymer, **12**, 286 (1962).
15. E. Passaglia, H. K. Kevorkian, J. Appl. Phys., **34**, 90 (1963).
16. F. E. Karash, H. E. Bair, J. M. O'Reilly, J. Phys. Chem., **69**, 2657 (1965).
17. S. Ichihara, A. Komatsu, T. Hata, Polym. J., **2**, 644 (1971).
18. G. T. Furukawa, R. E. Mc Coskey, G. J. King, J. Res. Nat. Bur. Std., **50**, 357 (1953).
19. G. T. Furukawa, R. E. Mc Coskey, M. L. Reilly, J. Res. Nat. Bur. Std., **55**, 127 (1955).
20. B. V. Lebedev, I. B. Rabinowich, V. A. Budarina, Vysokomol. Soedin. A, **9**, 488 (1967).
21. Th. Grewer, H. Wilski, Kolloid-Z. -Z. Polym., **226**, 46 (1968).
22. P. Marx, M. Dole, J. Am. Chem. Soc., **77**, 4771 (1955).
23. G. T. Furukawa, R. E. Mc Coskey, G. J. King, J. Res. Nat. Bur. Std., **43**, 273 (1952).
24. T. B. Douglas, A. W. Harman, J. Res. Nat. Bur. Std., **69A**, 149 (1965).
25. P. Marx, M. Dole, J. Am. Chem. Soc., **77**, 4771 (1955).
26. S. Yoshida, M. Sakiyama, S. Seki, Polym. J., **1**, 573 (1970).
27. J. M. O'Reilly, F. E. Karasz, H. E. Bair, J. Polym. Sci. C, **6**, 109 (1963).
28. F. E. Karasz, H. E. Bair, J. M. O'Reilly, J. Polym. Sci. Part A: **2**, 1141 (1968).
29. T. Hatakeyama, K. Nakamura, H. Hatakeyama, Polymer, **23**, 1801 (1982).
30. V. P. Kolesov, I. E. Paylov, S. M. Skutarov, Zh. Fiz. Khim., **36**, 770 (1962).
31. D. R. Gee, T. P. Melia, Polymer, **11**, 192 (1970).
32. P. Marx, C. W. Smith, A. E. Worthington, M. Dole, J. Phys. Chem., **59**, 1015 (1955).
33. R. C. Wilhoit, M. Dole, J. Phys. Chem., **57**, 14 (1953).
34. B. Wunderlich, in: Academic Press, New York, 1990; E. Turi (Ed.), "Thermal Characterization of Polymeric Materials." Academic Press, New York, 1997.
35. Okazaki, B. Wunderlich, Macromolecules, **30**, 1758 (1997).
36. B. Wunderlich, "Macromolecular Physics, vol. 3, Crystal Melting." Academic Press, Boston, MA, 1980.
37. K. Ishikiriyama, M. Pyda, G. Zhang, T. Forschner, J. Grebowicz, B. Wunderlich, J. Macromol. Sci. B: Phys., **37**, 27 (1998).
38. H. Suzuki, J. Grebowicz, B. Wunderlich, Brit. Polymer J., **17**, 1 (1985).
39. B. Wunderlich, Pure and Applied Chem., **67**, 1919 (1995).
40. World Wide Web, URL: http://funnelweb.utcc.utk.edu/~athas.
41. R. Pan, M-Y. Cao, B. Wunderlich, J. Thermal Anal., **31**, 1319 (1986); U. Gaur, M.-Y. Cao, R. Pan, B. Wunderlich, **31**, 421 (1986).

Surface and Interfacial Tensions of Polymers, Oligomers, Plasticizers, and Organic Pigments

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A. INTRODUCTION

1. Definition

The surface (interfacial) tension is the reversible work required to create a unit surface (interfacial) area at constant temperature (T), pressure (P), and composition (n) (1), i.e.,

$$\gamma = (\partial G/\partial A)_{T,P,n} \qquad \text{(A1)}$$

where γ is the surface (interfacial) tension, G the Gibbs free energy of the system, and A the surface (interfacial) area.

The specific surface free energy f_s is the free energy per unit surface area (1), i.e.,

$$f_s = \gamma + (C_1\mu_1 + C_2\mu_2) \tag{A2}$$

where C_j is the surface concentration (the number of moles per unit area) of component j, and μ_j is the chemical potential of component j. Thus $\gamma = f_s - (C_1\mu_1 + C_2\mu_2)$, which means that the surface tension is the *excess* specific surface free energy. The *excess* means in excess of the bulk phase. This distinguishes the surface tension and the specific surface free energy (1).

2. Temperature Dependence

The surface tension of polymers, just as with small-molecule liquids, varies with temperature (1) by

$$\gamma = \gamma_0(1 - T/T_c)^{11/9} \tag{A3}$$

where γ_0 is the surface tension at $T = 0\,K$, and T_c is the critical temperature. The temperature coefficient of surface tension is thus given (1) by

$$-d\gamma/dT = (11/9)(\gamma_0/T_c)(1 - T/T_c)^{2/9} \tag{A4}$$

Since $T_c \simeq 1000\,K$ for most polymers (1), $-d\gamma/dT$ is therefore practically constant at ordinary temperatures, i.e., $T \ll T_c$.

For polymer pairs that have an upper consolute curve and an UCST (upper critical solution temperature), the interfacial tension decreases with increasing temperature and vanishes at the upper consolute temperature. This is the case for most pairs of immiscible polymers. On the other hand, for polymers that have a lower consolute curve and an LCST (lower critical solution temperature), the interfacial tension vanishes at the lower consolute point and increases with increasing temperature in the immiscible region. A few polymer pairs appear to behave this way (24).

3. Macleod's Relation

The surface tension varies with density ρ according to Macleod's relation (1)

$$\gamma = \gamma^0\rho^\beta \tag{A5}$$

where γ^0 and β are constants, independent of temperature. β is known as the Macleod's exponent, and usually has a value of 3.0–4.5 for polymers (1). Thus the surface tension is "solely" determined by the density for a given chemical composition.

γ^0 is, to a close approximation, dependent only on the chemical constitution, and is related to Sugden's parachor (1) by

$$\gamma^0 = \left(\sum P_r\right)/M_r \tag{A6}$$

where $\sum P_r$ is the parachor of the repeat unit and M_r its formula weight. Equation (A6) neglects the end group effect (1). The parachor is a group additive quantity, independent of temperature, and its values have been tabulated elsewhere (104,105).

Macleod's relation provides a number of important relationships for analyzing the effects of molecular weight, glass transition, phase transition (crystallization), and chemical constitution on surface tension, as discussed below.

4. Molecular Weight Dependence

The surface tension varies with the number-average molecular weight M_n (1) by

$$\gamma^{1/4} = \gamma_\infty^{1/4} - k_1/M_n \tag{A7}$$

where γ_∞ is the surface tension at infinite molecular weight and k_1 is a constant. Equation (A7) follows directly from Macleod's relation (1). Alternatively, an empirical relation is given (1,81) as

$$\gamma = \gamma_\infty - k_2/M_n^{2/3} \tag{A8}$$

where k_2 is a constant. Equations (A7) and (A8) fit the experimental data equally well.

Usually, the surface tension increases with increasing molecular weight. When the molecular weight is greater than about 2000–3000, the surface tension reaches within about 1 mN/m (dyne/cm) of the value at infinite molecular weight. Therefore, the effect of molecular weight may usually be neglected for polymers, except for oligomers. Several exceptions have however been observed and explained in terms of end-group effects (1,4,103).

Relationships similar to Eqs. (A7) and (A8) may be written for the interfacial tension (1,81), i.e.,

$$\gamma^{1/4} = k_0 - k_3/M_{n1} - k_4/M_{n2} \tag{A9}$$

and

$$\gamma_{12} = k_0' - k_3'/M_{n1}^{2/3} - k_4'/M_{n2}^{2/3} \tag{A10}$$

where $k_0, k_3, k_4, k_0', k_3'$, and k_4' are constants, as discussed elsewhere (1).

5. Effects of Glass and Crystal-Melt Transitions

Macleod's relation also provides relationships for the effects of glass and crystal-melt transitions. At the glass transition, the surface tension is continuous, but its temperature coefficient is discontinuous. $(d\gamma/dT)_g$ below the glass temperature, T_g, is related to $(d\gamma/dT)_r$ above T_g (1) by

$$(d\gamma/dT)_g = (\alpha_g/\alpha_r)(d\gamma/dT)_r \tag{A11}$$

where α_g is the isobaric volumetric thermal expansion coefficient in the glassy region and α_r is that in the rubbery region. Since α_g is usually smaller than α_r, the surface tension varies less with temperature in the glassy region than in the rubbery region. Since $-d\gamma/dT$ is usually small, this effect is often neglected when the temperature range of interest is small.

On the other hand, both the surface tension and its temperature coefficient are discontinuous at the crystal-melt transition. The surface tension of crystalline surface, γ_c, is related to that of amorphous surface γ_a (1) by

$$\gamma_c = (\rho_c/\rho_a)^{\beta}\gamma_a \qquad \text{(A12)}$$

where ρ_c is the crystalline density, and ρ_a the amorphous density. Since $\beta = 3.0$–4.5, and usually $\rho_c > \rho_a$, the crystalline surface can have a much higher surface tension than the amorphous surface. For example, polyethylene has $\beta = 3.2$, $\rho_a = 0.855$ g/cc, $\rho_c = 1.0$ g/cc, and $\gamma_a = 35.7$ mN/m at 20°C. Thus its crystalline surface tension is given by

$$\gamma_c = (35.7)(1/0.855)^{3.2} = 58.9 \text{ mN/m}$$

Note that the value given above refers to a truly crystalline surface. Most "melt-crystallized" semicrystalline polymers, however, have amorphous surface. Only when they are nucleated in the melt against certain nucleating surfaces, or their single crystals are grown from solutions, are crystalline surfaces obtained.

6. Effect of Surface Chemical Constitution

The Macleod–Sugden equation shows that the surface tension depends on surface chemical constitution and density. Since the density is roughly a function of chemical constitution (105), it is expected that a given surface constitution will have a certain "characteristic" surface tension. Some of these values are listed in Table 1. These values should, however, be used only qualitatively.

7. Dispersion (Nonpolar) and Polar Components

The surface tension can be separated into dispersion (nonpolar) and polar components (1) i.e.,

$$\gamma = \gamma^d + \gamma^p \qquad \text{(A13)}$$

where γ^d is the dispersion component (arising from dispersion-force interactions) and γ^p the polar component (arising from various dipolar and specific interactions). The polarity x^p is defined as

$$x^p = \gamma^p/\gamma \qquad \text{(A14)}$$

which is independent of temperature (1), i.e.,

$$dx^p/dT = 0 \qquad \text{(A15)}$$

The interfacial tension is related to the surface tension and the polarity of the two contiguous phases by the harmonic-mean equation (1),

$$\gamma_{12} = \gamma_1 + \gamma_2 - \frac{4\gamma_1^d\gamma_2^d}{\gamma_1^d + \gamma_2^d} - \frac{4\gamma_1^p\gamma_2^p}{\gamma_1^p + \gamma_2^p} \qquad \text{(A16)}$$

or the geometric-mean equation (1,54,108,109),

$$\gamma_{12} = \gamma_1 + \gamma_2 - 2(\gamma_1^d\gamma_2^d)^{1/2} - 2(\gamma_1^p\gamma_2^p)^{1/2} \qquad \text{(A17)}$$

where the subscripts 1 and 2 refer to the two individual phases. The harmonic-mean equation has been shown to predict the interfacial tension between polymers adequately, whereas the geometric-mean equation is often less satisfactory (1,4).

The γ^d and γ^p components can be determined either from the interfacial tension or the contact angles by using Eq. (A16), the harmonic-mean equation, or less satisfactorily by Eq. (A17), the geometric-mean equation, as discussed elsewhere (1).

The polarity may also be defined in terms of the cohesive energy density (1), i.e.,

$$x^p = (\Delta E^p/\Delta E) = (\delta^p/\delta)^2 \qquad \text{(A18)}$$

where ΔE is the cohesive energy density, ΔE^p its polar component, δ the solubility parameter, and δ^p its polar component. The polarity values determined from interfacial properties have been found to agree with those determined from cohesive energy densities for a number of polymers (1). It should be cautioned, however, that Eq. (A14) defines the surface polarity, while Eq. (A18) defines the bulk polarity. These two values may not be the same, if the polymer has sufficiently long surface-active segments, which tend to preferentially adsorb on the surfaces.

B. TABLES

Table 1 lists typical values of surface tension for some surface chemical constitutions.

Table 2 lists the surface tension, its temperature coefficient, polarity, and Macleod's exponent for amorphous surfaces. The surface tension for crystalline surface may be calculated from the amorphous value by using the Macleod's exponent. All data were obtained by direct reversible measurements in the melt. The values for the solid were obtained by extrapolation from the melts. These are the most reliable and preferred. On the other hand, some values were obtained by the contact angle method. These are denoted with the symbol (c). The two values usually agree well.

Table 3 lists the interfacial tension and its temperature coefficient for amorphous interfaces. All data were obtained by direct reversible measurements in the melt. The values for the solid were obtained by extrapolation from the melt and have been shown to be reliable. The interfacial tension

may also be estimated from the surface tension and the polarity of the two phases by using the harmonic-mean equation (Eq. (A16)).

TABLE 1. SURFACE CHEMICAL CONSTITUTION AND SURFACE TENSION

Surface chemical constitution	Surface tension at 20°C (mN/m ≡ dyn/cm)
1.1. HYDROCARBON SURFACES	
$-CH_3$	30
$-CH_2$	36
$-CH_2-$ and $\urcorner CH\urcorner$	43
$\urcorner CH\urcorner$ (phenyl ring edge)	45
1.2. FLUOROCARBON SURFACES	
$-CF_3$	15
$-CF_2H$	26
$-CF_3$ and $-CF_2-$	17
$-CF_2-$	23
$-CH_2CF_3$	23
$-CF_2-CFH-$	30
$-CF_2-CH_2-$	33
$-CFH-CH_2-$	37
1.3. CHLOROCARBON SURFACES	
$-CHCl-CH_2-$	42
$-CCl_2-CH_2-$	45
$=CCl_2$	50
1.4. SILICONE SURFACES	
$-O-Si(CH_3)_2-O-$	20
$-O-Si(CH_3)(C_6H_5)-O-$	26

TABLE 2. SURFACE TENSION, POLARITY, AND MACLEOD'S EXPONENT FOR AMORPHOUS SURFACES

Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma/dT$ (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
2.1. HYDROCARBON POLYMERS							
Poly(acetylene)							
cis-isomer	51 (c)	–	–	–	0.08	–	5
trans-isomer	52 (c)	–	–	–	0.05	–	5
Poly(butadiene), $M_n = 5400$							
carboxyl acid end group	48.6	29.9	22.7	0.1440	–	–	70
methyl ester end group	43.1	28.8	23.3	0.1098	–	–	70
Poly(butadiene-*stat*-acrylonitrile), carboxyl end group							
18% by weight AN $M_n = 5300$	51.3	40.0	35.7	0.0866	–	–	70
27% by weight AN $M_n = 5900$	52.6	40.2	35.4	0.0955	–	–	70
Poly(ethylene), branched							
$M_n = 7000$	35.3	26.6	23.3	0.067	0	3.3	1–3,7–14
$M_n = 2000$	33.7	25.9	22.9	0.060	0	–	15
Poly(ethylene), linear							
$M_n = \infty$	36.8	29.4	26.6	0.056	0	3.2	9
$M_w = 67000$	35.7	28.2	25.4	0.057	0	3.2	1–3,7,11 14,15
Poly(ethylene-*stat*-acrylic acid)							
mol% acrylic acid 20	59 (c)	–	–	–	–	–	19
14	55 (c)	–	–	–	–	–	19
8	44 (c)	–	–	–	–	–	19
3	41 (c)	–	–	–	–	–	19
Poly(ethylene-*stat*-propylene)							
mol% PP M_w							
34 37000	33.8	25.7	22.6	0.062	0	3.0	13
36 33000	33.1	25.4	22.4	0.059	0	3.0	13
44 15000	32.0	24.1	21.0	0.061	0	3.3	13
50 63000	31.0	23.5	20.6	0.058	0	3.2	13
60 58000	30.7	23.2	20.3	0.058	0	2.8	13
Poly(ethylene-*stat*-propylene-*stat*-hexadiene), EPDM, E/PP/HD/ = 69.5/26.5/4.0 weight ratio comonomers $M_n = 17000$, $M_w = 70000$	34.5	26.2	23.0	0.064	0	–	20

TABLE 2. *cont'd*

Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma/dT$ (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
Poly(ethylene-*stat*-vinyl acetate)							
E/VA weight ratio 75/25	35.5	26.8	23.5	0.067	0.050	–	12
82.3/17.7	34.1	27.1	24.4	0.054	0.025	–	21–24
73.4/26.6	31.3	26.5	24.7	0.037	0.064	–	21–24
69.1/30.9	30.6	26.4	24.7	0.033	0.083	–	21–24
61.3/38.7	33.0	26.9	24.6	0.047	0.108	–	21–24
Poly(isobutene)							
$M_n = 2700$	33.6	25.3	22.1	0.064	–	4.1	7,11,17,26
Poly(isoprene)							
cis-isomer	32 (c)	–	–	–	–	–	6
cyclized	34 (c)	–	–	–	–	–	6
trans isomer	31 (c)	–	–	–	–	–	6
Poly(4-methyl-1-pentene)	25 (c)	–	–	–	–	–	6
Poly(propylene)							
atactic	29.4	22.1	19.3	0.056	0	3.2	4,11,13,16
isotactic	29.4	22.1	19.3	0.056	0	3.2	4,13
isotactic and atactic mixture	30.1	22.5	19.6	0.058	0.020	3.2	17,18
Poly(styrene) and copolymers, see Styrene Polymers							
Poly(xylylenes), see Styrene Polymers							
2.2. STYRENE POLYMERS							
Poly(2-methylstyrene)							
$M_n = 3000$	38.7	31.1	28.2	0.058	–	4.0	4,17,21
Poly(styrene)							
$M_v = 44000$	40.7	31.4	27.8	0.072	0.168	4.4	8,28–30
$M_n = 9300$	39.4	31.0	27.7	0.065	–	–	31
$M_n = 1700$	39.3	29.2	25.4	0.077	–	–	31
Poly(styrene-*stat*-acrylonitrile)							
mol% acrylonitrile 32.7	43 (c)	–	–	–	–	–	32
27.6	42 (c)	–	–	–	–	–	32
16.8	40 (c)	–	–	–	–	–	32
6.9	37 (c)	–	–	–	–	–	32
Poly(styrene-*stat*-2,2,3,3-tetrafluoropropyl methacrylate							
mol% styrene 98	31 (c)	–	–	–	–	–	33
95	30 (c)	–	–	–	–	–	33
90	29 (c)	–	–	–	–	–	33
80	28 (c)	–	–	–	–	–	33
70	27 (c)	–	–	–	–	–	33
60	26 (c)	–	–	–	–	–	33
50	25 (c)	–	–	–	–	–	33
40	25 (c)	–	–	–	–	–	33
30	21 (c)	–	–	–	–	–	33
20	20 (c)	–	–	–	–	–	33
10	19 (c)	–	–	–	–	–	33
2	19 (c)	–	–	–	–	–	33
1	18 (c)	–	–	–	–	–	33
Poly(xylylene), see also Halogenated Hydrocarbons	46.3 (c)	–	–	–	0.013	–	27
2.3. HALOGENATED HYDROCARBON POLYMERS							
Poly(2-chloro-1,4-xylylene), poly((3-chloro-1,4-phenylene)ethylene)	47.1 (c)	–	–	–	0.019	–	27
Poly(chloroprene) $M_v = 30000$	43.6	32.3	28.0	0.086	0.108	4.2	2,3
Poly(chlorotrifluoroethylene)	30.9	22.2	18.9	0.067	0.282	–	1–3,37
Poly(chlorotrifluoroethylene-*stat*-tetrafluoroethylene) Wt.% TFE							
60	25.2 (c)	–	–	–	0.348	–	1–3
80	21.3 (c)	–	–	–	0.354	–	1–3
Poly(dichloro-1,4-xylylene), poly((dichloro-1,4-phenylene)ethylene)	46.3 (c)	–	–	–	0.009	–	27

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TABLE 2. *cont'd*

Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma/dT$ (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
Poly(hexachlorobutadiene)	41.5 (c)	–	–	–	0.019	–	34
Poly(hexafluoropropylene)	17.0 (c)	–	–	–	0.120	–	1–3
	12.8 (c)	–	–	–	0.065	–	1–3
Poly(tetrafluoroethylene)							
$M = \infty$	23.9	16.3	13.4	0.058	0.089	–	9
$C_{21}F_{44}$ ($M = 1088$)	21.5	13.1	9.8	0.065	–	–	9,38,39
Poly(tetrafluoroethylene-*stat*-ethylene)							
50/50 weight ratio TFE/E	27.6 (c)	–	–	–	0.286	–	1–3
Poly(trifluoroethylene)	27.3 (c)	–	–	–	0.118	–	1–3
Poly(vinyl chloride)	41.9 (c)	–	–	–	0.146	–	1–3
Poly(vinyl fluoride)	38.4 (c)	–	–	–	0.292	–	1–3
Poly(vinylidene chloride)	45.4 (c)	–	–	–	0.196	–	1–3
Poly(vinylidene fluoride)	33.2 (c)	–	–	–	0.376	–	1–3
2.4. VINYL POLYMERS – ESTERS							
Poly(ethylene-*stat*-vinyl acetate), see Hydrocarbon Polymers							
Poly(vinyl acetate) $M_w = 11000–120000$	36.5	27.9	24.6	0.066	0.329	3.4	7.12
Poly(vinyl butyrate)	31.1	22.0	18.5	0.070	0.172	3.5	4,40
Poly(vinyl decanoate)	28.9	20.9	17.8	0.062	0.061	3.3	4,40
Poly(vinyl dodecanoate)	29.1	20.9	17.8	0.063	0.044	3.2	4,40
Poly(vinyl hexadecanoate)	30.9	22.5	19.0	0.066	0.035	3.2	4,40
Poly(vinyl hexanoate)	29.4	21.2	18.1	0.063	0.128	3.4	4,40
Poly(vinyl propionate)	34.0	24.7	21.1	0.072	0.221	3.6	4,40
Poly(vinyl octanoate)	28.7	20.7	17.7	0.061	0.078	3.3	4,40
2.5. VINYL POLYMERS – OTHERS							
Poly(acrylamide)	52.3 (c)	–	–	–	0.493	–	41
Poly(acrylonitrile)	50 (c)	–	–	–	–	–	1
Poly(heptafluoroisopropoxy)ethylene)	15 (c)	–	–	–	–	–	42
Poly(1-(heptafluoroisopropoxy)methyl) propylene-*stat*-maleic acid)	19 (c)	–	–	–	–	–	43
Poly(1-heptafluoroisopropoxy)propylene)	21 (c)	–	–	–	–	–	42
Poly(methacrylonitrile)	39 (c)	–	–	–	–	–	1
Poly(styrene-*stat*-acrylonitrile), see Styrene Polymers							
Poly(vinyl alcohol)	37 (c)	–	–	–	–	–	44
Poly(vinyl butyral)	38 (c)	–	–	–	–	–	1
Poly(vinyl ethyl ether), poly(ethoxyethylene)	36 (c)	–	–	–	–	–	45
Poly(vinyl methyl ether), poly(methoxyethylene)							
$M_n = 46500$, $M_w = 99000$	31.8	22.1	18.3	0.075	–	–	70
Poly(vinyl formal)	39 (c)	–	–	–	–	–	35
2.6. ACRYLIC POLYMERS – NONFLUORINATED							
Poly(acrylamide) see Vinyl Polymers (nonesters)							
Poly(acrylonitrile), see Vinyl Polymers (nonesters)							
Poly(*n*-butyl acrylate)							
$M_n = 32000$	33.7	24.6	21.1	0.070	0.098	–	1
	30.7	22.8	19.7	0.061	0.127	–	46,107
Poly(ethyl acrylate)							
$M_n = 28000$	37.0	27.0	23.2	0.070	0.174	–	1
	33.7	26.4	23.6	0.056	0.205	–	46,107
Poly(ethylene-*stat*-acrylic acid), see Hydrocarbon Polymers							
Poly(2-ethylhexyl acrylate)							
$M_n = 34000$	30.2	21.1	17.6	0.070	0.028	–	1
	29.2	21.1	18.0	0.062	0.076	–	46,107
Poly(methyl acrylate)							
$M_n = 25000$	41.0	31.0	27.2	0.070	0.248	–	1
	42.7	31.0	26.5	0.090	0.210	–	46,107

TABLE 2. *cont'd*

Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma/dT$ (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
2.7. ACRYLIC POLYMERS - FLUORINATED							
Poly((1-chlorodifluoromethyl)tetrafluoroethyl acrylate)	19 (c)	–	–	–	–	–	47
Poly(di(chlorodifluoromethyl) fluoromethyl acrylate)	20 (c)	–	–	–	–	–	47
Poly(1,1-dihydroheptafluorobutyl acrylate)	15 (c)	–	–	–	–	–	47
Poly(1,1-dihydropentafluoroisopropyl acrylate)	15 (c)	–	–	–	–	–	47
Poly(1,1-dihydropentadecafluorooctyl acrylate)	10 (c)	–	–	–	–	–	47
Poly(heptafluoroisopropyl acrylate)	14 (c)	–	–	–	–	–	47
Poly(5-(heptafluoroisopropoxy)pentyl acrylate)	18 (c)	–	–	–	–	–	47
Poly(11-(heptafluoroisopropoxy)undecyl acrylate)	20 (c)	–	–	–	–	–	47
Poly(2-heptafluoropropoxy)ethyl acrylate)	16 (c)	–	–	–	–	–	47
Poly(nonafluoroisobutyl acrylate)	14 (c)	–	–	–	–	–	47
2.8. METHACRYLIC POLYMERS - NONFLUORINATED							
Poly(benzyl methacrylate)	36 (c)	–	–	–	–	–	49
Poly(*n*-butyl methacrylate)							
$M_v = 37000$	31.2	23.5	20.6	0.059	0.158	4.2	8
Poly(isobutyl methacrylate)							
$M_v = 35000$	30.9	23.1	20.1	0.060	0.139	–	2
Poly(*t*-butyl methacrylate)							
$M_v = 6000$	30.4	22.7	19.8	0.059	0.120	–	2
Poly(*t*-butylaminoethyl methacrylate)	34 (c)	–	–	–	–	–	1,35
Poly(dodecyl methacrylate), poly(lauryl methacrylate)	32.8 (c)	–	–	–	–	–	1,35
Poly(ethyl methacrylate)							
$M_v = 5200$	35.9	26.8	23.3	0.070	–	–	1,35
Poly(2-ethylhexyl methacrylate)							
$M_v = 64000$	28.8	20.8	17.7	0.062	–	–	1,35
Poly(*n*-hexyl methacrylate)							
$M_v = 52000$	30.0	22.0	18.9	0.062	–	–	1,35
Poly(dimethylaminoethyl methacrylate)	36 (c)	–	–	–	–	–	1,35
Poly(hydroxyethyl methacrylate), see also Methacrylic Hydrogels	37 (c)	–	–	–	–	–1,	5
Poly(lauryl methacrylate), see Poly(dodecyl methacrylate)							
Poly(methyl methacrylate), see also Methacrylic Hydrogels							
$M = 3000$	41.1	31.2	27.4	0.076	0.281	4.2	8
Poly(phenyl methacrylate)	35 (c)	–	–	–	–	–	49
Poly(*n*-propyl methacrylate)							
$M = 8500$	33.3	24.7	21.5	0.065	–	–	1,35
Poly(stearyl methacrylate), poly(octadecyl methacrylate)	36.3 (c)	–	–	–	–	–	47
2.9. METHACRYLIC POLYMERS - FLUORINATED							
Poly(1,1-dihydropentadecafluorooctyl methacrylate)	10.5 (c)	–	–	–	0.048	–	50,53,54
Poly(heptafluoroisopropyl methacrylate)	15 (c)	–	–	–	–	–	47
Poly(heptadecafluorooctyl methacrylate)	15.3 (c)	–	–	–	0.094	–	51,52
Poly(1-hydrotetrafluoroethyl methacrylate)	15 (c)	–	–	–	–	–	47
Poly(1,1-dihydrotetrafluoropropyl methacrylate)	19 (c)	–	–	–	–	–	33
Poly(1-hydrohexafluoroisopropyl methacrylate)	15 (c)	–	–	–	–	–	47,55
Poly(*t*-nonafluorobutyl methacrylate)	15 (c)	–	–	–	–	–	48
Poly(styrene-*stat*-2,2,3,3-tetrafluoropropyl methacrylate), see Styrene Polymers							
2.10. METHACRYLIC HYDROGELS							
Poly(hydroxyethyl methacrylate)							
dry	37 (c)	–	–	–	–	–	1,35
40 wt.% water	69.0 (c)	–	–	–	–	–	56
Poly(hydroxyethyl methacrylate-*stat*-dimethylaminoethyl methacrylate)							
weight ratio wt.% water							
99.9/0.1 40	68.2 (c)	–	–	–	0.701	–	56
97/3 59	65.3 (c)	–	–	–	0.657	–	56

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TABLE 2. *cont'd*

Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma/dT$ (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
Poly(hydroxyethyl methacrylate-*stat*-methoxyethyl methacrylate)							
weight ratio wt.% water							
75/25 33	69.1 (c)	–	–	–	0.670	–	56
50/50 23	67.8 (c)	–	–	–	0.608	–	56
25/70 19	68.0 (c)	–	–	–	0.496	–	56
Poly(hydroxyethyl methacrylate-*stat*-methacrylic acid)							
weight ratio wt.% water							
99.1/0.1 40	68.7 (c)	–	–	–	0.700	–	56
97/3 39	67.6 (c)	–	–	–	0.704	–	56
Poly(hydroxyethyl methacrylate-*stat*-methyl methacrylate)							
weight ratio wt.% water							
95/5 43	68.6 (c)	–	–	–	0.697	–	56
75/25 43	67.0 (c)	–	–	–	0.640	–	56
50/50 21	65.7 (c)	–	–	–	0.543	–	56
25/75 7	65.6 (c)	–	–	–	0.410	–	56
5/95 3	54.4 (c)	–	–	–	0.393	–	56
Poly(methyl methacrylate), see also Methacrylic Polymers for Dry Surfaces							
3 wt.% water	60.9 (c)	–	–	–	0.282	–	56
Poly(methyoxyethyl methacrylate)							
2 wt.% water	71.1 (c)	–	–	–	0.301	–	56

2.11. POLY(ETHERS) (see also POLY(ETHER) COPOLYMERS, POLY(SULFONE), POLY(CARBONATE), PHENOXY RESIN, EPOXY RESIN, VINYL POLYMERS)

Poly(chloral)	37.9 (c)	–	–	–	0.037	–	34
Poly(epichlorohydrin)							
$M = 1500$	43.2 (25°C)	–	–	–	–	–	57
Poly(oxybutene)-diol	38.2	28.0	24.0	0.079	0.003	3.7	58
Poly(oxyisobutene)-diol							
$M = 30000$	27.5	19.0	15.7	0.066	–	3.8	61
Poly(oxydecamethylene)	36.1	27.2	23.8	0.068	–	2.8	58
Poly(oxyethylene)-diol							
$M = -17000$	42.9	30.1	25.2	0.098	0.284	3.0	9,59–61
Poly(oxyethylene)-diol							
$M = 6000$	42.9	33.0	29.2	0.076	0.284	3.0	9,11,12,
$M = 6000$	42.5	30.1	25.4	0.095	0.165	2.9	9,62
Poly(oxyethylene)-dimethylether							
$M = 114$	28.6	16.0	11.1	0.097	–	4.4	9,59–61
$M = 148$	31.1	18.6	13.8	0.096	–	4.4	9,59–61
$M = 182$	32.9	20.5	15.8	0.095	–	4.0	9,59–61
$M = 600$	37.5	26.1	21.7	0.088	–	3.5	9,59–61
$M = 5000$	44.1	32.7	28.3	0.088	–	3.5	9,59–61
$M = 100000$	44.2	32.8	28.4	0.088	–	3.5	9,56–61
Poly(oxyhexamethylene)-diol	37.4	27.5	23.7	0.076	–	3.2	58
Poly(oxymethylene)	44.6 (c)	–	–	–	0.316	–	1,35
Poly(oxypropylene)-diol							
$M = 2025$	31.5	21.1	17.1	0.080	–	3.4	13
$M = ?$	31.7	20.6	16.4	0.085	0.014	3.4	62
$M = 3000$	31.2	20.9	17.0	0.079	–	3.4	13,57,60,61
$M = 400–4100$	31.1	21.6	17.9	0.073	–	3.4	57,60,61
Poly(oxypropylene)-dimethylether							
$M = 3000$	30.7	18.3	13.6	0.095	–	3.6	57,60,61
Poly(oxytetramethylene)							
$M = 43000$	31.9	24.0	20.9	0.061	0142	–	2,12
$M = 2500$	38.2	27.9	24.0	0.079	–	3.7	13

TABLE 2. *cont'd*

Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma/dT$ (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
2.12. POLY(ETHER) COPOLYMERS							
Poly(oxyethylene)-*block*-poly(oxypropylene)-*block*-poly(oxyethylene)							
wt.% OP / DP of blocks: OE / OP / OE							
19.0 / 46 / 16 / 46	36.8	27.8	24.3	0.069	0.129	2.8	57,60,62
48.9 / 11 / 16 / 11	34.2	25.0	21.5	0.071	0.105	3.0	57,60,62
56.9 / 8 / 16 / 8	33.7	24.3	20.7	0.072	0.094	3.1	57,60,62
84.1 / 2 / 16 / 2	32.9	22.8	18.9	0.078	0.047	3.4	57,60,62
19.0 / 75 / 30 / 75	34.7	26.6	23.5	0.062	0.096	2.5	57,60,62
55.1 / 13 / 30 / 13	33.4	24.7	21.3	0.067	–	2.7	57,60,62
59.6 / 10 / 30 / 10	32.7	23.7	20.2	0.069	–	2.8	57,60,62
85.5 / 3 / 30 / 3	31.8	22.1	18.4	0.074	–	3.3	57,60,62
18.9 / 110 / 39 / 110	33.8	26.3	23.4	0.058	0.095	2.6	57,60,62
20.2 / 122 / 47 / 122	33.2	26.0	23.2	0.056	0.074	2.4	57,60,62
20.2 / 139 / 56 / 139	32.3	25.5	22.9	0.052	0.069	2.5	57,60,62
38.3 / 56 / 56 / 56	32.4	25.0	22.1	0.057	–	2.6	57,60,62
57.7 / 28 / 56 / 28	32.1	24.4	21.4	0.059	–	2.7	57,60,62
66.7 / 19 / 56 / 19	31.9	23.8	20.7	0.062	–	2.8	57,60,62
86.7 / 6 / 56 / 6	31.7	23.2	20.0.0	0.065	–	2.9	57,60,62
Poly(oxyethylene-*stat*-oxypropylene) mol% oxypropylene							
17	40.5	28.1	23.4	0.095	0.139	3.6	57,62
30	39.3	26.9	22.2	0.095	0.127	3.6	57,62
41	37.5	25.5	20.9	0.092	0.112	3.6	57,62
56	34.9	23.2	18.7	0.090	0.061	3.6	57,62
76	33.4	22.3	18.0	0.086	0.040	3.3	57,62
85	32.4	21.3	17.0	0.086	0.031	3.4	57,62
Poly(oxyethylene-*stat*-oxypropylene)-*block*-poly(oxydimethylsilylene)-*block*-poly(oxyethylene-*stat*-oxypropylene)]; i.e., Poly(oxyethylene-*stat*-oxypropylene) has a comonomer weight ratio of 50/50, $M_n = 2200$							
wt.% of polysiloxane / DP of polysiloxane							
0 / 0	34.8 (25°C)	–	–	–	–	–	63
4.6 / 4.9	28.0 (25°C)	–	–	–	–	–	63
7.5 / 7.9	25.4 (25°C)	–	–	–	–	–	63
9.9 / 10.4	21.4 (25°C)	–	–	–	–	–	63
19.8 / 20.8	20.6 (25°C)	–	–	–	–	–	63
32.6 / 34.3	20.5 (25°C)	–	–	–	–	–	63
Poly(oxytetramethylene-*stat*-oxypropylene)							
mol% oxypropylene / M_w							
25 / 2100	36.4	26.0	22.0	0.080	–	3.3	13
35 / 1200	36.0	25.7	21.8	0.079	–	3.2	13
46 / 1900	34.9	24.3	20.2	0.082	–	3.5	13
52 / 2250	34.3	23.8	19.7	0.081	–	3.5	13
67 / 2200	33.1	22.8	18.9	0.079	–	3.4	13
85 / 1700	31.9	21.6	17.7	0.079	–	3.5	13
2.13. POLY(ESTERS)							
Poly(oxydecamethyleneoxyisophthaloyl), poly(decamethylene isophthalate)	42.7	33.3	29.7	0.072	0.192	3.0	98
Poly(oxydodecamethyleneoxyisophthaloyl), poly(dodecamethylene isophthalate)	40.0	30.9	27.4	0.070	0.149	3.0	98

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TABLE 2. *cont'd*

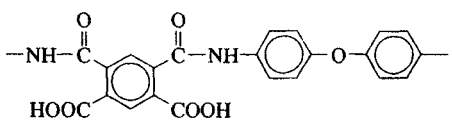
Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma/dT$ (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
Poly(oxyethyleneoxyterephthaloyl), poly(ethylene terephthalate) $M_n = 16000$ $M_w = 37000$	44.6	36.2 27.7 (280°C)	32.9 24.8 (325°C)	0.065	0.221	–	1,14,20,64
Poly(oxyhexamethyleneoxyisophthaloyl), poly hexamethylene isophthalate)	45.6	35.6	31.7	0.077	0.242	3.3	98
Poly(oxytetramethyleneoxyisophthaloyl), poly(butylene isophthalate)	47.8	37.4	33.4	0.080	0.270	3.2	98
Poly(oxytrimethyleneoxyisophthaloyl), poly(propylene isophthalate)	49.3	38.5	34.3	0.083	0.288	3.0	98
2.14. POLY(CARBONATES)							
Poly(oxycarbonyloxy-1, 4-phenyleneisopropylidene-1,4-phenylene), polycarbonate of bisphenol-A	42.9	35.1	32.1	0.060	0.246	–	35
	45 (c)	–	–	–	–	–	65
2.15. POLY(SULFONES)							
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenyleneisoproypylidene-1,4-phenylene), polysulfone	46.6 (c)	–	–	–	0.347	–	35
2.16. PHENOXY RESINS							
Poly(oxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy-2-hydroxytrimethylene), phenoxy resin $M = 30000$	43.0	–	–	–	–	–	35
2.17. EPOXY RESINS							
Diglycidyl ether of bisphenol-A, 2,3-di(glycidoxy-1,4-phenylene) propane pure, $M = 340$	47.2	–	–	–	–	–	67
	45.6	–	–	–	0.432	–	35,68
Diglycidyl ether of bisphenol-A, 2,3-(diglycidoxy-1,4-phenylene)propane chain extended with bisphenol-A	51.2(23 °C)	–	–	–	–	–	85
Diglycidyl ether of bisphenol-A with 6 wt.% of *N-N*-diethylaminopropylamine							
fresh, mixture, not cured	33.7	–	–	–	–	–	67,68
cured	46.8 (c)	–	–	–	–	–	35
Diglycidyl ether of bisphenol-A with stoichiometric amount of triethylenetetramine cured	39.1 (c)	–	–	–	0.166	–	35
Diglycidyl ether of bisphenol-A with 0.5 wt.% of *N*-methylaminopropyltrimethoxysilane fresh, mixture, not cured	39.1	–	–	–	–	–	68
Diglycidyl ether of bisphenol-A with 0.5 wt.% of γ-glycidoxypropyltrimethoxysilane fresh, mixture, not cured	32.9	–	–	–	–	–	68
Curing agent: γ-aminopropyltrimethoxysilane $M = 179$	25.5	–	–	–	–	–	68
Curing agent: *N,N*-dibutylaminopropylamine $M = 186$	25.3	–	–	–	–	–	68
Curing agent: *N,N*-diethylaminopropylamine $M = 130$	24.6	–	–	–	–	–	68
Curing agent: *N,N*-dimethylaminopropylamine $M = 102$	23.9	–	–	–	–	–	68
Curing agent: *N*-methylaminopropyltrimethoxysilane $M = 193$	27.0	–	–	–	–	–	68
Curing agent: γ-glycidoxypropyltrimethoxysilane $M = 236$	23.9	–	–	–	–	–	68
2.18. POLY(AMIDES)							
Poly(glycine), (nylon 2), see Poly(peptides)							
Poly(imino(1-oxodecamethylene))(nylon 11)	–	22.6 (225°C)	–	–	–	–	72

TABLE 2. *cont'd*

Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	−dγ/d*T* (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
Poly(imino(1-oxododecamethylene) (nylon 12)	35.8 (c)	–	–	–	0.154	–	73
Poly(imino(1-oxohexamethylene)) (nylon 6)	38.4 (c)	–	–	–	0.188	–	35
	–	–	36.1 (265°C)		–	–	72
Poly(imino(1-oxotetramethylene)) (nylon 4)	48.5 (c)	–	–	–	0.427	–	52,73
Poly(imino-1,4-phenyleneiminoterephthaloyl), poly(1,4-phenylene terephthalamide)							
fiber (unsized)	48.5 (c)	–	–	–	0.216	–	74
cast sheet (H-bonding parallel to surface)	33.7 (c)	–	–	–	0.206	–	75
cast sheet (H-bonding vertical to surface)	31.3 (c)	–	–	–	0.202	–	75
Poly(iminoadipoyliminohexamethylene) (nylon 66)	46.5	38.1	34.8	0.065	0.344	–	1,14,20,
$M_n = 17000$, $M_w = 35000$		29.6 (280°C)	26.7 (325°C)				
Poly(iminoazelaoyliminononamethylene) (nylon 99)	36(c)	–	–	–	–	–	71
Poly(iminopimeloyliminoheptamethylene) (nylon 77)	43 (c)	–	–	–	–	–	71
Poly(iminosebacoyliminodecamethylene) (nylon 1010)	32 (c)	–	–	–	–	–	71
Poly(iminosebacoyliminohexamethylene) (nylon 610)	–	–	37.0 (265°C)		–	–	72
Poly(iminosuberoyliminooctamethylene) (nylon 88)	34 (c)	–	–	–	–	–	71

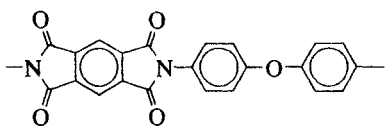
2.19. POLY(IMIDES)

Poly(iminocarbonyl-(4,6-dicarboxy-1,3-phenylene)-carbonylimino-1,4-phenyleneoxy-1,4-phenylene), KAPTON H film



	41.0 (c)	–	–	–	0.358	–	35,76

Poly[(1,3,5,7-tetraoxo-2,3,6,7-tetrahydro-1H,5H-benzo[1,2-c:4,5-c] dipyrrol-2,6-diyl-1,4-phenyleneoxy-1,4-phenylene), cured KAPTON H film



	37.7 (c)	–	–	–	0.223	–	35,76

2.20. POLY(IMINES)

Poly((benzoylimino)ethylene)	26 (c)	–	–	–	–	–	77
Poly((butyrylimino)ethylene)	25 (c)	–	–	–	–	–	77
Poly((dodecanoylimino)ethylene)	22 (c)	–	–	–	–	–	78
Poly((dodecanoylimino)ethylene-*stat*-(acetylimino) trimethylene)	22 (c)	–	–	–	–	–	78
Poly((heptanoylimino)ethylene)	22 (c)	–	–	–	–	–	77
Poly((hexanoylimino)ethylene)	23 (c)	–	–	–	–	–	77
Poly(((3-methyl)butyrylimino)ethylene)	24 (c)	–	–	–	–	–	77
Poly((pentadecafluorooctadecanoylimino)ethylene)	11 (c)	–	–	–	–	–	77
Poly((pentanoylimino)ethylene)	23 (c)	–	–	–	–	–	77

2.21. POLY(URETHANES)

Poly(methylenediphenyldiisocyanate-*alt*-(butanediol poly(oxytetramethylene)diol) (ESTANE 5714)	39 (c)	–	–	–	0.179	–	79
Poly(hexamethylene diisocyanate-*alt*-triethylene glycol)	38.6 (c)	–	–	–	0.194	–	73
Poly(4-methyl-1,3-phenylene diisocyanate-*alt*-tripropylene glycol)	36.3 (c)	–	–	–	0.344	–	73

2.22. POLY(SILOXANES) (see also ORGANOSILANES)

Poly(oxydiethylsilylene)							
158 cS	25.7	16.2	12.5	0.073	–	–	80
Poly(oxydimethylsilylene)							
$M = 75000$	20.39	14.15	11.75	0.048	0.042	3.5	2.9
	21.17	–	–	–	–	–	26
	21.01	13.47	10.57	0.058	–	–	83

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TABLE 2. *cont'd*

Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	−dγ/dT (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
$M = 3900$	20.17	13.93	11.53	0.048	0.042	3.5	2,9
	20.59	–	–	–		–	26
	20.47	12.93	10.03	0.058	–	–	83
60000 cS	19.8	13.5	11.1	0.048	0.042	3.5	2,11,14, 80,82
$M = 1274$	19.9	–	–	–	–	3.5	80–82
$M = 607$	18.8	–	–	–	–	3.7	80–82
$M = 310$	17.6	–	–	–	–	3.7	80–82
$M = 162$	15.7	–	–	–	–	4.0	80–82
Poly(oxydimethylsilylene), α,ω-difunctional [R–(Si(CH$_3$)$_2$–O–)$_n$-Si(CH$_3$)$_2$–R] R = (CH$_2$)$_3$NH$_2$							
$M_n = 2086$	20.88	14.12	11.52	0.052	–	–	70
$M_n = 7916$	19.86	12.97	10.32	0.053	–	–	70
Poly(oxydimethylsilylene), α,ω-difunctional R–(Si(CH$_3$)$_2$–O–)$_n$-Si(CH$_3$)$_2$–R							
R = (CH$_2$)$_5$NH$_2$; $M_n = 1132$	21.00	14.37	11.82	0.051	–	–	70
R = (CH$_2$)$_3$COOH; $M_n = 2194$	20.54	13.52	10.82	0.054	–	–	70
Poly(oxydimethylsilylene) block copolymers, see Poly(ester) Copolymers							
Poly(oxymethylphenylsilylene)							
102 cS	26.1	11.8	6.3	0.11	–	–	80

2.23. ORGANOSILANES – HYDROLYZED AND CONDENSED FILMS (see also POLYSILOXANES and EPOXY RESINS)

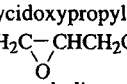
2.23.1. ALKYL AND ALKYLENE SILANES

Ethyltriethoxysilane, CH$_3$CH$_2$Si(OC$_2$H$_5$)$_3$							
on silica (no catalyst)	36.7 (c)	–	–	–	–	–	1,35
on silica (acetic acid catalyst)	34.5 (c)	–	–	–	–	–	1,35
on silica (propionic acid catalyst)	27.8 (c)	–	–	–	–	–	1,35
on silica (piperidine catalyst)	33.4 (c)	–	–	–	–	–	1,35
γ-Methacryloxypropyltrimethoxysilane, CH$_2$=C(CH$_3$)–C(=O)O(CH$_2$)$_3$Si(OCH$_3$)$_3$							
on soda-lime glass	44.8 (c)	–	–	–	–	–	1,35
Methyltrimethoxysilane, CH$_3$Si(OCH$_3$)$_3$							
on soda-lime glass	28.0 (c)	–	–	–	–	–	1,35
Phenyltrimethoxysilane, C$_6$H$_5$Si(OCH$_3$)$_3$							
on soda-lime glass	43.0 (c)	–	–	–	–	–	1,35
Vinyltriethoxysilane, CH$_2$=CHSi(OC$_2$H$_5$)$_3$							
on silica	33.4 (c)	–	–	–	–	–	1,35
Vinytrimethoxysilane, CH$_2$=CHSi(OCH$_3$)$_3$							
on soda-lime glass	28.6 (c)	–	–	–	–	–	1,35

2.23.2. AMINO SILANES

N-β-(aminoethyl)-γ-aminopropyltrimethoxysilane, NH$_2$(CH$_2$)$_2$NH(CH$_2$)$_3$Si(OCH$_3$)$_3$							
on soda-lime glass	33.7 (c)	–	–	–	–	–	1,35
γ-Aminopropyltriethoxysilane, NH$_2$CH$_2$CH$_2$CH$_2$Si(OC$_2$H$_5$)$_3$							
on soda-lime glass	35.7 (c)	–	–	–	–	–	1,35

2.23.3. EPOXY SILANE

Glycidoxypropyltrimethoxysilane, 							
on soda-lime glass (no catalyst)	49.4 (c)	–	–	–	–	–	1,35
on soda-lime glass (acetic acid catalyst, pH = 4)	66.9 (c)	–	–	–	–	–	1,35

2.23.4. HALOGENATED ALKYL SILANES

p-Bromophenyltrimethoxysilane, BrC$_6$H$_5$Si(OCH$_3$)$_3$							
on soda-lime glass	49.3 (c)	–	–	–	–	–	1,35
p-Chlorophenylethyltrimethoxysilane, ClC$_6$H$_4$CH$_2$CH$_2$Si(OCH$_3$)$_3$							
on pyrex glass	51.0 (c)	–	–	–	–	–	1,35

TABLE 2. *cont'd*

Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	−dγ/d*T* (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
on silica	49.9 (c)	–	–	–	–	–	1,35
on stainless steel	53.3 (c)	–	–	–	–	–	1,35
on α-alumina	49.9 (c)	–	–	–	–	–	1,35
γ-Chloropropyltrimethoxysilane, $Cl(CH_2)_3Si(OCH_3)_3$							
on soda-lime glass	40.7 (c)						
on pyrex glass	48.8 (c)	–	–	–	–	–	1,35
on stainless steel	49.9 (c)	–	–	–	–	–	1,35
3-(1,1-Dihydroperfluorooctoxy)propyltriethoxysilane, $CF_3(CF_2)_6CH_2O(CH_2)_3Si(OC_2H_5)_3$							
on gold	21.5 (c)	–	–	–	–	–	1,35
on pyrex glass	18.8 (c)	–	–	–	–	–	1,35
on stainless steel	18.8 (c)	–	–	–	–	–	1,35
γ-Perfluoroisopropoxypropyltrimethoxysilane, $(CF_3)_2CFO(CH_2)_3Si(OCH_3)_3$							
on pyrex glass	24.2 (c)	–	–	–	–	–	1,35
on silica	22.8 (c)	–	–	–	–	–	1,35
on stainless steel	18.8 (c)	–	–	–	–	–	1,35
2.23.5. MERCAPTO SILANE							
γ-Mercaptopropyltrimethoxysilane, $HS(CH_2)_3Si(OCH_3)_3$							
on soda-lime glass	41.9 (c)	–	–	–	–	–	1,35
2.24. CELLULOSICS							
Amylose	37 (c)	–	–	–	–	–	84
Amylopectin	35 (c)	–	–	–	–	–	84
Cellophane	45.4 (c)	–	–	–	0.344	–	51,52
Cellulose							
regenerated from cotton	42 (c)	–	–	–	–	–	86
regenerated from wood pulp	36–42 (c)	–	–	–	–	–	86
Cellulose acetate	45.9 (c)	–	–	–	0.296	–	51,52,87,88
Cellulose acetate butyrate	34 (c)	–	–	–	–	–	1
Ethyl cellulose	32 (c)	–	–	–	–	–	1
Hemicellulose							
arabinogalactan	33 (c)	–	–	–	–	–	86
galactoglucomannan	37 (c)	–	–	–	–	–	86
hardwood xylan	33–37 (c)	–	–	–	–	–	86
softwood xylan	35 (c)	–	–	–	–	–	86
Nitrocellulose	38 (c)	–	–	–	–	–	1
Starch	39 (c)	–	–	–	–	–	89
2.25. POLY(PEPTIDES)							
Casein	43 (c)	–	–	–	–	–	90
Poly(L-alanine)	45.2 (c)	–	–	–	0.203	–	52,91
Poly(γ-benyzl L-glutamate)	39.2 (c)	–	–	–	0.224	–	52,92
Poly(glycine), (nylon2)	50.1 (c)	–	–	–	0.443	–	52,92
Poly(γ-methyl L-glutamate)							
α-helix random tangle	44.0 (c)	–	–	–	0.375	–	52,92
α-sheet	48.0 (c)	–	–	–	0.192	–	41
α-helix on water	44.4 (c)	–	–	–	0.435	–	52,92
β-extended chain	35.5 (c)	–	–	–	0.191	–	52,92
β-sheet	37.8 (c)	–	–	–	0.119	–	41
β-random chain	45.4 (c)	–	–	–	0.440	–	52,92
Mixed α and β sheet	42.1 (c)	–	–	–	0.359	–	52,92
Wool	45 (c)	–	–	–	–	–	90
2.26. MISCELLANEOUS							
2.26.1. HYDROCARBONS							
Eicosane, $C_{20}H_{44}$							
$M = 282$	28.9	25.6 (60°C)	–	0.0833	0	–	1,9,62,102
Hexatriacontane, $C_{36}H_{74}$							
$M = 507$	31.4	26.1 (100°C)	–	0.066	0	–	1,9,94,102
	24 (c)	–	–	–	–	–	94

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TABLE 2. *cont'd*

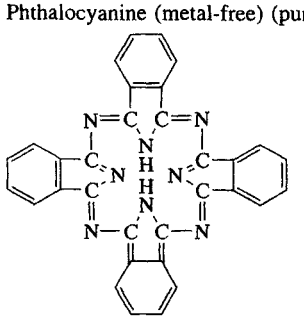
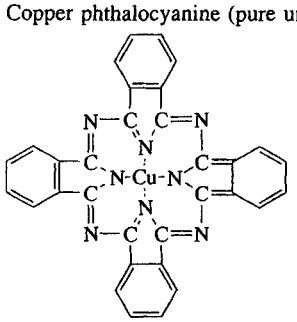
Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma/t$ (mN/m	Polarity (x^P)	Macleod's exponent (β)	Refs.
Paraffin wax (mp 65°C)	34.7	32.0 (65°C)	26.9 (150°C)	0.06	0	–	1,94,95
	31.0 (c)	–	–	–	–	–	1
	25.5 (c)	–	–	–	–	–	108
2.26.2. ALKANE-DIOLS							
1,4-Butanediol, $M = 90$	44.2	37.6 (90°C)	–	0.094	0.330	3.6	58
1,10-Decanediol, $M = 174$	39.3	34.2 (90°C)	–	0.073	0.137	2.9	58
1,6-Hexanediol, $M = 118$	42.2	36.5 (90°C)	–	0.081	0.238	3.3	58
1,5-Pentanediol, $M = 104$	41.6	36.7 (90°C)	–	0.070	0.281	2.8	58
1,3-Propanediol, $M = 76$	45.9	38.9 (90°C)	–	0.100	0.375	3.6	58
2.26.3. ADHESIVES							
Natural rubber + ester rosin pressure sensitive adhesive	36 (c)	–	–	–	–	–	45,101
Poly(isobutene)-based pressure sensitive adhesive	30 (c)	–	–	–	–	–	101
Phenol-resorcinol adhesive	52 (c)	–	–	–	–	–	93
Resorcinol adhesive	51 (c)	–	–	–	–	–	90
Urea-formaldehyde adhesive	61 (c)	–	–	–	–	–	90
2.26.4. SULFUR							
"Monomeric" sulfur, below the floor temperature of 160°C							
	$\gamma = 72.95 - 0.1019T$, for $T = 120-160$°C						97
"Polymeric" sulfur, above the floor temperature of 160°C							
	$\gamma = 65.31 - 0.0553T$, for $T = 160-440$°C						97
Solid sulfur	128 (c)	–	–	–	–	–	96
2.26.5. PLASTICIZERS							
Di(*n*-butyl)-*o*-phthalate $M = 278$	33.1	23.1	19.2	0.077	0.051	3.4	100
Di(*n*-dodecyl)-*o*-phthalate $M = 502$	30.8	22.6	19.5	0.063	0.029	2.9	100
Di(ethyl)-*o*-phthalate $M = 222$	36.6	24.6	20.0	0.092	0.115	3.8	100
Di(*n*-heptyl)-*o*-phthalate $M = 362$	30.0	20.6	17.0	0.072	0.047	3.5	100
Di(methyl)-*o*-phthalate $M = 194$	39.5	26.9	22.0	0.097	0.190	3.7	100
Di(*n*-nonyl)-*o*-phthalate $M = 418$	28.9	19.9	16.5	0.069	0.028	3.4	100
Di(*n*-propyl)-*o*-phthalate $M = 250$	34.4	23.3	19.1	0.085	0.067	3.3	100
Tricresyl phosphate	40.9	–	–	–	0.027	–	1
2.26.6 ORGANIC PIGMENTS							
Phthalocyanine (metal-free) (pure untreated surface) 	52.8 (c)	–	–	–	0.240	–	1,99
Copper phthalocyanine (pure untreated surface) 	46.9 (c)	–	–	–	0.147	–	1,99

TABLE 2. *cont'd*

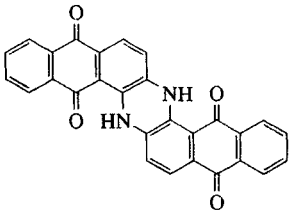
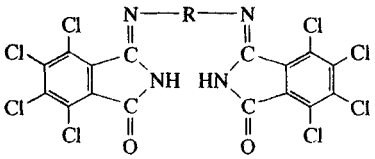
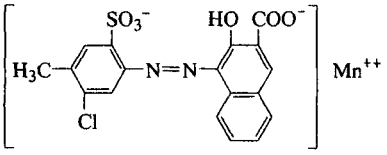
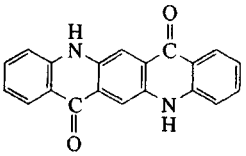
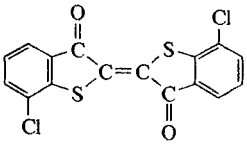
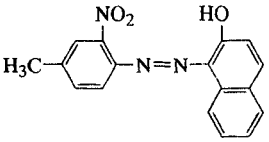
Polymer	Surface tension γ (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma/dT$ (mN/m/K)	Polarity (x^P)	Macleod's exponent (β)	Refs.
Chlorinated copper phthalocyanine (pure untreated surface)	42.0 (c)	–	–	–	0.148	–	1,99
Idanthrone (pure untreated surface) 	63.2 (c)	–	–	–	0.475	–	1,99
Isoindolinone (pure untreated surface) 	47.2 (c)	–	–	–	0.318	–	1,99
Manganese β-oxynaphthoic acid derivative (pure untreated surface) 	51.9 (c)	–	–	–	0.464	–	1,99
γ-Quinacridone (pure untreated surface) 	49.1 (c)	–	–	–	0.273	–	1,99
Thioindigoid red (pure untreated surface) 	51.4 (c)	–	–	–	0.317	–	1,99
Toluidine red (pure untreated surface) 	53.0 (c)	–	–	–	0.251	–	1,99

TABLE 3. INTERFACIAL TENSION FOR AMORPHOUS INTERFACES

Polymer	Interfacial tension γ_{12} (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma_{12}/dT$ (mN/m/k)	Refs.
3.1. HYDROCARBON POLYMERS vs. OTHERS					
3.1.1. POLY(BUTADIENE) vs. OTHERS					
Poly(butadiene) vs. poly(oxydimethylsilylene) PBD $M_n = 960$, PDMS $M_n = 3900$	2.48	2.07	1.91	0.00322	103

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TABLE 3. *cont'd*

Polymer	Interfacial tension γ_{12} (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma_{12}/dT$ (mN/m/k)	Refs.
Poly(butadiene) vs. poly(oxydimethylsilylene) PBD $M_n = 2350$, PDMS $M_n = 3900$	3.86	2.70	2.25	0.00895	103
Poly(butadiene) vs. poly(oxydimethylsilylene) PBD $M_n = 2350$ PDMS $M_n = 5200$	3.98	2.85	2.42	0.00865	103
Poly(butadiene) vs. poly(oxydiethylsilylene) PBD $M_n = 960$, PDMS $M_n = 5200$	2.58	2.25	2.13	0.00250	103
Poly(butadiene) vs. epoxy resin (diglycidyl ether of bisphenol-A, chain extended with bisphenol-A)	1.77 (23°C)	1.40 (55°C)	–	0.0105	70
3.1.2. POLY(BUTADIENE-*stat*-ACRYLONITRILE) vs. OTHERS					
Poly(butadiene-*stat*-acrylonitrile) 18 wt.% AN vs. epoxy resin (same as above)	1.23	0.57 (55°C)	–	0.0198	70
Poly(butadiene-*stat*-acrylonitrile) 26 wt.% AN vs. epoxy resin (same as above)	–	0.58 (55°C)	–	–	70
3.1.3. POLY(ETHYLENE) vs. OTHERS					
Poly(ethylene) (branched) vs. poly(propylene) (atactic)	–	1.1	–	–	17
Poly(ethylene) (linear) vs. poly(styrene)	8.3	5.7	4.7	0.020	8
	–	–	4.4	–	106
Poly(ethylene) (branched) vs. poly(styrene)	5.6	5.0	4.8	0.0046	46
Poly(ethylene) (branched) vs. poly(chloroprene)	4.6	3.6	3.2	0.0075	2
Poly(ethylene) (linear) vs. poly(methyl acrylate)	10.6	8.2	7.3	0.018	1,35
Poly(ethylene) (branched) vs. poly(methyl acrylate)	–	6.5	–	–	46
Poly(ethylene) (linear) vs. poly(ethyl acrylate)	7.5	5.4	4.6	0.016	1,35
Poly(ethylene) (branched) vs. poly(ethyl acrylate)	–	5.5	–	–	46
Poly(ethylene) (linear) vs. poly(*n*-butyl acrylate)	5.0	3.3	2.7	0.014	1,35
Poly(ethylene) (branched) vs. poly(*n*-butyl acrylate)	5.5	3.8	3.2	0.013	46,107
Poly(ethylene) (linear) vs. poly(2-ethylhexyl acrylate)	3.1	1.8	1.3	0.010	1,35
Poly(ethylene) (branched) vs. poly(2-ethylhexyl acrylate)	3.5	2.5	2.1	0.0078	46,107
Poly(ethylene) (linear) vs. poly(methyl methacrylate)	11.8	9.5	8.6	0.018	8
Poly(ethylene) (linear) vs. poly(*n*-butyl methacrylate)	7.1	5.1	4.4	0.015	8
Poly(ethylene) (branched) vs. poly(iso-butyl methacrylate)	5.5	4.2	3.7	0.010	2
Poly(ethylene) (branched) vs. poly(*tert*-butyl methacrylate)	5.9	4.7	4.3	0.009	2
Poly(ethylene) (linear) vs. poly(poly(vinyl acetate)	14.6	11.0	9.7	0.027	7,12
Poly(ethylene) (branched) vs. poly(vinyl acetate)	–	10.5	–	–	18
Poly(ethylene) (branched) vs. poly(ethylene-*stat*-vinyl acetate), 25 wt.% VAc	2.0	1.3	1.1	0.005	12,18
Poly(ethylene) vs. poly(ethylene-*stat*-vinyl acetate), 12 wt.% VAc	–	0.3	–	–	23
Poly(ethylene) vs. poly(ethylene-*stat*-vinyl acetate), 17.7 wt.% VAc	–	0.8	–	–	23
Poly(ethylene) vs. poly(ethylene-*stat*-vinyl acetate), 25 wt.% VAc	–	1.5	–	–	23
Poly(ethylene) vs. poly(ethylene-*stat*-vinyl acetate), 26.6 wt.% VAc	–	1.7	–	–	23
Poly(ethylene) vs. poly(ethylene-*stat*-vinyl acetate), 30.9 wt.% VAc	–	2.2	–	–	23
Poly(ethylene) vs. poly(ethylene-*stat*-vinyl acetate), 38.7 wt.% VAc	–	2.9	–	–	23
Poly(ethylene) vs. poly(ethylene-*stat*-vinyl acetate), 50 wt.% VAc	–	4.1	–	–	35
Poly(ethylene) vs. poly(ethylene-*stat*-vinyl acetate), 75 wt.% VAc	–	7.3	–	–	35
Poly(ethylene) (linear) vs. poly(oxyethyleneoxyterephthaloyl)	9.4	6.5 (280°C)	6.0 (325°C)	–	35
Poly(ethylene) (branched) vs. poly(oxyethyleneoxyterephthaloyl)	13.7	9.8 (280°C)	9.2 (325°C)	–	14
Poly(ethylene) (branched) vs. poly(oxytrimethyleneoxy-isophthaloyl)	15.4	11.5	10.0	0.030	98
Poly(ethylene) (branched) vs. poly(oxytetramethyleneoxy-isophthaloyl)	13.5	10.1	8.8	0.026	98
Poly(ethylene) (branched vs. poly(oxyhexamethyleneoxy-isophthaloyl)	11.1	8.4	7.3	0.021	98
Poly(ethylene) (branched) vs. poly(oxyisophthaloyloxydeca-methylene)	8.3	6.4	5.6	0.015	98
Poly(ethylene) (branched) vs. poly(oxyisophthaloyloxydodeca-methylene)	5.9	4.5	3.9	0.011	98
Poly(ethylene) (linear) vs. poly(iminohexamethylene-iminoadipoyl) (nylon 66)	14.9	10.2 (280°C)	9.4 (325°C)	–	35
Poly(ethylene) (branched) vs. poly(iminohexamethylene-iminoadipoyl) (nylon 66)	–	17.0 (280°C)	15.2 (325°C)	0.041	14
Poly(ethylene) vs. poly(imino(1-oxohexamethylene)) (nylon 6)	–	–	10.7 (250°C)	–	106
Poly(ethylene) (branched) vs. poly(oxyethylene)	11.6	9.5	8.7	0.016	12

TABLE 3. *cont'd*

Polymer	Interfacial tension γ_{12} (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma_{12}/dT$ (mN/m/k)	Refs.
Poly(ethylene) (branched) vs. poly(oxytetramethylene)	5.1	4.1	3.8	0.007	2,12
Poly(ethylene) (branched) vs. poly(oxydimethylsilylene)	5.3	5.1	5.0	0.002	212,82
3.1.4. POLY(ETHYLENE-*stat*-PROPYLENE-*stat*-HEXADIENE), (EPDM) vs. OTHERS					
EPDM (E/P/HD 69.5/26.5/4.0 weight ratio) vs. poly-(oxyethyleneoxyterephthaloyl), poly(ethylene terephthalate)	8.6	6.2 (280°C)	5.7 (325°C)	0.0093	20,35
EPDM (E/P/HD 69.5/6.5/4.0 weight ratio) vs. poly(imino-hexamethyleneiminoadipoyl) (nylon 66)	14.7	9.7 (280°C)	8.8 (325°C)	0.0193	20,35
Poly(ethylene-*stat*-vinyl acetate) vs. poly(ethylene), see reverse					
Poly(ethylene-*stat*-vinyl acetate) vs. poly(vinyl acetate), see reverse					
Poly(ethylene-*stat*-vinyl acetate) vs. poly(ethylene *stat*-vinyl acetate)					
E/VAc (25 wt.% VAc) vs. E/VAc (12 wt.% VAc)	–	0.45	–	–	23
E/VAc (25 wt.% VAc) vs. E/VAc (17.7 wt.% VAc)	–	0.13	–	–	23
E/VAc (25 wt.% VAc) vs. E/VAc (38.7 wt.% VAc)	–	0.17	–	–	23
E/VAc (25 wt.% VAc) vs. E/VAc (50 wt.% VAc)	–	1.0	–	–	35
E/VAc (25 wt.% VAc) vs. E/VAc (75 wt.% VAc)	–	3.6	–	–	35
3.1.5. POLY(ISOBUTENE) vs. OTHERS					
Poly(isobutene) vs. poly(vinyl acetate)	9.9	7.3	6.3	0.020	7
Poly(isobutene) vs. poly(oxydimethylsilylene)	4.9	4.1	3.8	0.006	82
3.1.6. POLY(PROPYLENE) vs. OTHERS					
Poly(propylene) vs. poly(ethylene), see reverse					
Poly(propylene) vs. poly(styrene)					
Poly(propylene) vs. poly(oxydimethylsilylene)	3.2	2.9	2.8	0.002	17
3.1.7. HALOGENATED HYDROCARBON POLYMERS vs. OTHERS					
Poly(chloroprene) vs. poly(ethylene), see reverse					
Poly(chloroprene) vs. poly(*n*-butyl methacrylate)	2.2	1.5	1.3	0.0047	2
Poly(chloroprene) vs. poly(styrene), see reverse					
Poly(chloroprene) vs. poly(oxydimethylsilylene)	7.1	6.4	6.2	0.0050	2
3.2. STYRENE POLYMERS vs. OTHERS					
Poly(styrene) vs. poly(ethylene), see reverse					
Poly(styrene) vs. poly(chloroprene)	0.7	0.5	0.4	0.0014	2
Poly(styrene) vs. poly(methyl acrylate)	–	3.2	–	–	46
Poly(styrene) vs. poly(ethyl acrylate)	–	1.4	–	–	46
Poly(styrene) vs. poly(*n*-butyl acrylate)	–	1.4	–	–	46
Poly(styrene) vs. poly(2-ethylhexyl acrylate)	–	2.5	–	–	46
Poly(styrene) vs. poly(vinyl acetate)	4.2	3.7	3.4	0.0044	2.24
Poly(styrene) vs. poly(methyl methacrylate)	3.2	1.5	0.8	0.013	8
	–	–	1.3	–	106
Poly(styrene) vs. poly(oxydimethylsilylene)	6.1	6.1	6.1	~0	82
Poly(styrene) vs. poly(ethylene-*stat*-vinyl acetate), 17.7 wt.% VAc	1.8	2.8	3.2	−0.0074	24
Poly(styrene) vs. poly(ethylene-*stat*-vinyl acetate), 30.9 wt.% VAc	0.3	2.1	2.8	−0.0136	24
Poly(styrene) vs. poly(ethylene-*stat*-vinyl acetate), 38.7 wt.% VAc	2.3	2.0	1.9	0.0020	24
Poly(styrene) vs. poly(ethylene-*stat*-vinyl acetate), 26.6 wt.% VAc	3.3	2.6	2.3	0.0050	24
3.3. VINYL POLYMERS vs. OTHERS					
Poly(ethylene-*stat*-vinyl acetate) vs. poly(ethylene), see reverse					
Poly(ethylene-*stat*-vinyl acetate) vs. polystyrene, see reverse					
Poly(ethylene-*stat*-vinyl acetate) vs. self, see Hydrocarbon Polymers vs. Others					
3.3.1. POLY VINYL ACETATE vs. OTHERS					
Poly(vinyl acetate) vs. poly(isobutene), see reverse					
Poly(vinyl acetate) vs. poly(ethylene-*stat*-vinyl acetate), 12 wt.% VAc	–	8.3	–	–	23
Poly(vinyl acetate) vs. poly(ethylene-*stat*-vinyl acetate), 17.7 wt.% VAc	–	7.5	–	–	23.
Poly(vinyl acetate) vs. poly(ethylene-*stat*-vinyl acetate), 25 wt.% VAc	6.3	5.8	5.5	0.0043	12
	–	6.5	–	–	23

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TABLE 3. *cont'd*

Polymer	Interfacial tension γ_{12} (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma_{12}/dT$ (mN/m/k)	Refs.
Poly(vinyl acetate) vs. poly(ethylene-*stat*-vinyl acetate), 26.6 wt.% VAc	–	6.1	–	–	23
Poly(vinyl acetate) vs. poly(ethylene-*stat*-vinyl acetate), 30.9 wt.% VAc	–	5.6	–	–	23
Poly(vinyl acetate) vs. poly(ethylene-*stat*-vinyl acetate), 38.7 wt.% VAc	–	4.7	–	–	23
Poly(vinyl acetate) vs. poly(ethylene-*stat*-vinyl acetate), 50 wt.% VAc	–	3.5	–	35	
Poly(vinyl acetate) vs. poly(ethylene-*stat*-vinyl acetate), 75 wt.% VAc	–	1.2	–	–	35
Poly(vinyl acetate) vs. poly(*n*-butyl methacrylate)	4.2	2.8	2.3	0.010	8
Poly(vinyl acetate) vs. poly(oxytetramethylene)	5.5	4.5	4.1	0.0081	12
Poly(vinyl acetate) vs. poly(oxydimethylsilylene)	8.4	7.3	6.9	0.0081	2,12,82
Poly(vinyl acetate) vs. eicosane, $C_{20}H_{44}$	7.1	4.6	3.6	0.0195	40
3.3.2. POLY(VINYL BUTYRATE) vs. OTHERS					
Poly(vinyl butyrate) vs. eicosane, $C_{20}H_{44}$	1.9	0.9	0.5	0.0078	40
3.3.3. POLY(VINYL PROPIONATE) vs. OTHERS					
Poly(vinyl propionate) vs. eicosane, $C_{20}H_{44}$	3.5	2.5	2.1	0.0080	40
3.4. ACRYLIC POLYMERS vs. OTHERS					
3.4.1. POLY(METHYL ACRYLATE) vs. OTHERS					
Poly(methyl acrylate) vs. poly(ethylene), see reverse					
Poly(methyl acrylate) vs. poly(styrene), see reverse					
Poly(methyl acrylate) vs. poly(ethyl acrylate)	2.4	13	0.9	0.008	1,35,46
Poly(methyl acrylate) vs. poly(*n*-butyl acrylate)	4.0	3.0	2.6	0.008	1,35,46
Poly(methyl acrylate) vs. poly(2-ethylhexyl acrylate)	6.6	5.7	5.4	0.007	1,35,46
3.4.2. POLY(ETHYL ACRYLATE) vs. OTHERS					
Poly(ethyl acrylate) vs. poly(ethylene), see reverse					
Poly(ethyl acrylate) vs. poly(styrene), see reverse					
Poly(ethyl acrylate) vs. poly(methyl acrylate), see reverse					
Poly(ethyl acrylate) vs. poly(*n*-butyl acrylate)	2.1	1.3	1.0	0.006	1,35,46
Poly(ethyl acrylate) vs. poly(2-ethylhexyl acrylate)	3.9	3.2	3.0	0.005	1,35,46
3.4.3. POLY(*n*-BUTYL ACRYLATE) vs. OTHERS					
Poly(*n*-butyl acrylate) vs. poly(ethylene), see reverse					
Poly(*n*-butyl acrylate) vs. poly(styrene), see reverse					
Poly(*n*-butyl acrylate) vs. poly(methyl acrylate), see reverse					
Poly(*n*-butyl acrylate) vs. poly(ethyl acrylate), see reverse					
Poly(*n*-butyl acrylate) vs. poly(2-ethylhexyl acrylate),	1.8	1.2	1.0	0.005	1,35,46
3.4.4. POLY(2-ETHYLHEXYL ACRYLATE) vs. OTHERS					
Poly(2-ethylhexyl acrylate) vs. poly(ethylene), see reverse					
Poly(2-ethylhexyl acrylate) vs. poly(styrene), see reverse					
Poly(2-ethylhexyl acrylate) vs. poly(methyl acrylate), see reverse					
Poly(2-ethylhexyl acrylate) vs. poly(ethyl acrylate), see reverse					
Poly(2-ethylhexyl acrylate) vs. poly(*n*-butyl acrylate), see reverse					
3.5. METHACRYLIC POLYMERS vs. OTHERS					
3.5.1. POLY(METHYL METHACRYLATE) vs. OTHERS					
Poly(methyl methacrylate) vs. poly(ethylene), see reverse					
Poly(methyl methacrylate) vs. poly(styrene), see reverse					
Poly(methyl methacrylate) vs. poly(*n*-butyl methacrylate)	3.4	1.9	1.3	0.012	8
Poly(methyl methacrylate) vs. poly(*tert*-butyl methacrylate)	3.0	2.2	1.9	0.006	2
3.5.2. POLY(*n*-BUTYL METHACRYLATE) vs. OTHERS					
Poly(*n*-butyl methacrylate) vs. poly(ethylene), see reverse					
Poly(*n*-butyl methacrylate) vs. poly(vinyl acetate), see reverse					

TABLE 3. *cont'd*

Polymer	Interfacial tension γ_{12} (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma_{12}/dT$ (mN/m/k)	Refs.
Poly(*n*-butyl methacrylate) vs. poly(chloroprene), see reverse					
Poly(*n*-butyl methacrylate) vs. poly(oxydimethylsilylene), see reverse					
3.5.3. POLY(*iso*-BUTYL METHACRYLATE) vs. OTHERS					
Poly(iso-butyl methacylate) vs. poly(ethylene), see reverse					
3.5.4. POLY(*tert*-BUTYL METHACRYLATE) vs. OTHERS					
Poly(*t*-butyl methacrylate) vs. poly(ethylene), see reverse					
Poly(*t*-butyl methacrylate) vs. poly(oxydimethylsilylene), see reverse					
3.6. POLY(ETHERS) vs. OTHERS					
3.6.1. POLY(OXYETHYLENE) vs. OTHERS					
Poly(oxyethylene) vs. poly(ethylene), see reverse					
Poly(oxyethylene) vs. poly(ethylene-*stat*-vinyl acetate), 25 wt.% VAc	6.4	5.8	5.5	0.0045	12
Poly(oxyethylene) vs. poly(2-ethylhexyl, acrylate)	6.6	4.8	4.1	0.014	107
Poly(oxyethylene) vs. poly(*n*-butyl acrylate)	2.8	2.0	1.7	0.0065	107
Poly(oxyethylene) vs. poly(oxytetramethylene)	4.5	3.8	3.6	0.0051	12
Poly(oxyethylene) vs. poly(oxydimethylsilylene), see reverse					
Poly(oxyethylene) vs. eicosane, $C_{20}H_{44}$	–	9.3 (90°C)	–	0.040	58
3.6.2. POLY(OXYTETRAMETHYLENE) vs. OTHERS					
Poly(oxytetramethylene) vs. poly(oxyethylene), see reverse					
Poly(oxytetramethylene) vs. poly(ethylene), see reverse					
Poly(oxytetramethylene) vs. poly(vinyl acetate), see reverse					
Poly(oxytetramethylene) vs. poly(ethylene-*stat*-vinyl acetate), 25 wt.% VAc	1.5	1.2	1.1	0.0023	12
Poly(oxytetramethylene) vs. poly(oxydimethylsiloxane), see reverse					
Poly(oxytetramethylene) vs. eicosane, $C_{20}H_{44}$	–	1.1 (90°C)	–	0.013	58
3.7. POLY(ESTERS) vs. OTHERS					
Poly(oxyethyleneoxyterephthaloyl) vs. polyethylene, see reverse					
Poly(oxyethyleneoxyterephthaloyl) vs. poly(ethylene-*stat*-propylene-*stat*-hexadiene), see reverse					
Poly(oxytrimethyleneoxyisophthaloyl) vs. poly(ethylene), see reverse					
Poly(oxytetramethyleneoxyisophthaloyl) vs. poly(ethylene), see reverse					
Poly(oxyhexamethyleneoxyisophthaloyl) vs. poly(ethylene), see reverse					
Poly(oxyisophthaloyloxydecamethylene) vs. poly(ethylene), see reverse					
Poly(oxyisophthaloyloxydodecamethylene) vs. poly(ethylene), see reverse					
3.8. POLY(AMIDES) vs. OTHERS					
Poly(iminohexamethyleneiminoadipoyl) vs. polyethylene, see reverse					
Poly(iminohexamethyleneiminoadipoyl) vs. poly(ethylene-*stat*-propylene-*stat*-hexadiene), see reverse					
Poly(imino(1-oxohexamethylene)) vs. poly(ethylene), see reverse					
3.9. EPOXY RESIN vs. OTHERS					
Epoxy resin vs. poly(butadiene), see reverse					
Epoxy resin vs. poly(butadiene-*stat*-acrylonitrile), see reverse					
3.10. POLY(SILOXANES) vs. OTHERS					
Poly(oxydimethylsilylene) vs. poly(ethylene), see reverse					
Poly(oxydimethylsilylene) vs. poly(styrene), see reverse					
Poly(oxydimethylsilylene) vs. poly(propylene), see reverse					
Poly(oxydimethylsilylene) vs. poly(isobutene), see reverse					
Poly(oxydimethylsilylene) vs. poly(chloroprene), see reverse					
Poly(oxydimethylsilylene) vs. poly(vinyl acetate), see reverse					
Poly(oxydimethylsilylene) vs. poly(butadiene), see reverse					
Poly(oxydimethylsilylene) vs. poly(*n*-butyl methacrylate)	4.2	3.8	3.6	0.0037	2
Poly(oxydimethylsilylene) vs. poly(*t*-butyl methacrylate)	3.6	3.3	3.2	0.0025	2
Poly(oxydimethylsilylene) vs. poly(oxyethylene)	10.8	9.8	9.4	0.0078	12
Poly(oxydimethylsilylene) vs. poly(oxytetramethylene)	6.4	6.3	6.2	0.0012	2,12

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TABLE 3. *cont'd*

Polymer	Interfacial tension γ_{12} (mN/m ≡ dyn/cm) 20°C	150°C	200°C	$-d\gamma_{12}/dT$ (mN/m/k)	Refs.
3.11. MISCELLANEOUS					
Eicosane ($C_{20}H_{44}$) vs. poly(oxyethylene), see reverse					
Eicosane ($C_{20}H_{44}$) vs. poly(oxytetramethylene), see reverse					
Eicosane ($C_{20}H_{44}$) vs. poly(vinyl acetate), see reverse					
Eicosane ($C_{20}H_{44}$) vs. poly(vinyl propionate), see reverse					
Eicosane ($C_{20}H_{44}$) vs. poly(vinyl butyrate), see reverse					
Eicosane ($C_{20}H_{44}$) vs. 1,3-propanediol 25 wt.% VAc	45.9	38.9 (90°C)	–	0.100	58
Eicosane ($C_{20}H_{44}$) vs. 1,4-butanediol	44.2	37.6 (90°C)	–	0.094	58
Eicosane ($C_{20}H_{44}$) vs. 1,5-pentanediol	41.6	36.7 (90°C)	–	0.070	58
Eicosane ($C_{20}H_{44}$) vs. 1,6-hexanediol	42.2	36.5 (90°C)	–	0.081	58
Eicosane ($C_{20}H_{44}$) vs. 1,10-decanediol	39.3	34.2 (90°C)	–	0.073	58

C. REFERENCES

1. S. Wu, "Polymer Interface and Adhesion", Marcel Dekker, New York, 1982.
2. S. Wu, J. Polym. Sci. C, **34**, 19 (1971).
3. S. Wu, J. Adhesion, **5**, 39 (1973).
4. T. Hata, T. Kasemura, in: L. H. Lee (Ed.), "Adhesion and Adsorption of Polymers", Vol. 12a, Plenum, New York, 1980, p. 15.
5. H. Schonhorn, G. L. Baker, F. S. Bates, J. Polym. Sci., Polym. Phys. Ed., **23**, 1555 (1985).
6. L. H. Lee, J. Polym. Sci. A-2, **5**, 1103 (1967).
7. S. Wu, J. Colloid Interface Sci., **31**, 153 (1969).
8. S. Wu, J. Phys. Chem., **74**, 632 (1970).
9. S. Wu, J. Macromol. Sci., **C10**, 1 (1974).
10. S. Wu, in: D. R. Paul, S. Newman, (Eds.), "Polymer Blends", Vol. 1, Academic, New York, 1978, p. 243.
11. R. J. Roe, J. Phys. Chem. **72**, 2013 (1968).
12. R. J. Roe, J. Colloid Interface Sci., **31**, 228 (1969).
13. T. Kasemura, N. Yamashita, K. Suzuki, T. Kondo, T. Hata, Kobunshi Ronbunshu, **35**, 263 (1978).
14. W. Berger, K. Olbricht, H. W. Kammer, Faserforsch. Textiltech., **27** (1), 9 (1976); **27** (1), 15 (1976).
15. R. H. Dettre, R. E. Johnon, Jr., J. Colloid Interface Sci., **21**, 367 (1966).
16. H. Schonhorn, L. H. Sharpe, J. Polym. Sci. A, **3**, 569 (1965).
17. Y. Oda, T. Hata, Preprints, 17th Annual Meeting High Polymer Society, Japan, 1968, p. 167.
18. Y. Oda, T. Hata, Preprints, Sixth Symp. Adhesion and Adhesives, The Adhesion Society of Japan, 1968, p. 69.
19. W. H. Smarook, S. Bonotto, Polym. Eng. Sci., **8**, 41 (1968).
20. S. Wu, Polym. Eng. Sci., **27**, 335 (1987).
21. T. Hata, Kobunshi, **17**, 594 (1968).
22. T. Hata, Hyomen, **6**, 659 (1968).
23. U. Uraki, T. Hata, Preprints, Thirteenth Symp. Adhesion and Adhesives, The Adhesion Society of Japan, 1975.
24. Y. Kitazaki, H. Yamauchi, T. Hata, Preprints, Ninth Symp. Adhesion and Adhesives, Japan, 1971, p. 51.
25. H. Edwards, J. Appl. Polym. Sci., **12**, 2213 (1968).
26. D. G. LeGrand, G. L. Gaines, Jr., J. Colloid Interface Sci., **31**, 162 (1969).
27. T. E. Nowlin, D. F. Smith, Jr., J. Appl. Polym. Sci., **25**, 1619 (1980).
28. W. Y. Lau, C. M. Burns, Surface Sci., **30**, 478 (1972).
29. W. Y. Lay, C. M. Burns, J. Colloid Interface Sci., **45**, 295 (1973).
30. W. Y. Lay, C. M. Burns, J. Polym. Sci., Polym. Phys. Ed., **12**, 431 (1974).
31. D. G. LeGrand, G. L. Gaines, Jr., J. Colloid Interface Sci., **42**, 181 (1973).
32. L. H. Lee, Adv. Chem. Ser., **87**, 85 (1968).
33. K. Tamaribuchi, Polym. Preprints, Am. Chem. Soc., **8**, 631 (1967).
34. D. H. Kaelble, P. J. Dyne, L. Maus, J. Adhesion, **6**, 239 (1974). Collected or estimated from various sources.
35. Collected or estimated from various sources.
36. M. K. Bernett, W. A. Zisman, J. Phys. Chem., **64**, 1292 (1960).
37. H. Schonhorn, F. W. Ryan, L. H. Sharpe, J. Polym. Sci. A-2, **4**, 538 (1966).
38. R. H. Dettre, R. E. Johnson, Jr., J. Colloid Interface Sci., **31**, 568 (1969).
39. R. H. Dettre, R. E. Johnson, Jr., J. Phys. Chem., **71**, 1529 (1967).
40. T. Kasemura, F. Uzi, T. Kondo, T. Hata, Kobunshi Ronbunshu, **36**, 337 (1979).
41. Y. Kitazaki, T. Hata, J. Adhesion Soc. Japan, **8**, 131, (1972).
42. A. G. Pittman, B. A. Ludwig, D. L. Sharp, J. Polym. Sci. A-1, **6**, 1741 (1968).
43. W. L. Wasley, A. G. Pittman, Polym. Lett., **10**, 279 (1972).
44. W. A. Zisman, Adv. Chem. Ser., **43**, 1 (1964).
45. Y. Kitazaki, A. Watanabe, M. Toyama, Preprints, fourth Symp. Adhesion and Adhesives Japan, 1966 p. 31.
46. H. Tozaki, T. Hata, Preprints, 26th Annual Meeting of the Society of Polymer Science, Japan, 1977.
47. A. G. Pittman, D. L. Sharp, B. A. Ludwig, J. Polym. Sci. A-1, **6**, 1729 (1968).

48. J. N. Roitman, A. G. Pittman, Polym. Lett., **10**, 499 (1972).
49. H. W. Fox, W. A. Zisman, J. Colloid Sci., **7**, 428 (1952).
50. M. K. Bernett, W. A. Zisman, J. Phys. Chem., **66** 1207 (1962).
51. V. L. Gott, R. E. Baier, Public Document PH43-68-84-4-2, National Technical Information Service, Springfield, Virginia, 22151, May 1973.
52. D. H. Kaelble, J. Moacanin, Polymer, **18**, 475 (1977).
53. M. K. Bernett, H. Ravener, J. Adhesion, **9**, 157 (1978).
54. D. H. Kaelble, "Physical Chemistry of Adhesion", Wiley, New York, 1971.
55. A. G. Pittman, in: L. A. Wall (Ed.), "Fluoropolymers", Vol. 25, Wiley, New York, 1972, p. 419.
56. R. N. King, J. D. Andrade, S. M. Ma, D. E. Gregonis, L. R. Brostrom, J. Colloid Interface Sci., **103**, 62 (1985).
57. A. K. Rastogi, L. E. St. Pierre, J. Colloid Interface Sci., **31**, 168 (1969).
58. T. Kasemura, N. Yamashita, K. Suzuki, T. Kondo, T. Hata, Kobunshi Ronbunshu, **35**, 215 (1978).
59. G. W. Bender, D. G. LeGrand, G. L. Gaines, Jr., Macromol., **2**, 681 (1969).
60. A. K. Rastogi, L. E. St. Pierre, J. Colloid Interface Sci., **35**, 16 (1971).
61. T. Kasemura, T. Hata, Kobunshi Ronbunshu, **33**, 192 (1976).
62. T. Kasemura, K. Suzuki, F. Uzi, T. Kondo, T. Hata, Kobunshi Ronbunshu, **35**, 779 (1978).
63. T. C. Kendrick, B. M. Kingston, N. C. Lyoyd, M. J. Owen, J. Colloid Interface Sci., **24**, 135 (1967).
64. H. T. Patterson, K. H. Hu, T. H. Grindtaff, J. Polym. Sci. C, **34**, 31 (1971).
65. R. E. Baier, "Surface Properties of Materials for Prosthetic Implants", Calspan Corporation, CAL Report VH-2801-p-2, February 1970.
66. A. Herczeg, G. S. Ronay, W. C. Simpson, Proc. Nat. Soc. Aerospace Mat. Proc. Eng. Conf., Dallas, Texas, **2**, 221 (1970).
67. L. H. Sharpe, H. Schonhorn, Adv. Chem. Ser., **43**, 189 (1964).
68. R. T. Foister, J. Colloid Interface Sci., **99**, 568 (1984).
69. L. S. Penn, E. R. Bowler, Surface Interface Anal., **3** (4), 161 (1981).
70. J. T. Koberstein, Chemical Engineering Department, Princeton University, Princeton, New Jersey, Private Communication, 1986.
71. T. Fort, Jr., Adv. Chem. Ser., **43**, 302 (1964).
72. F. J. Hybart, T. R. White, J. Appl. Polym. Sci., **3**, 118 (1960).
73. E. Nyilas, W. A. Morton, R. D. Cumming, D. M. Lederman, T. H. Chiu, Polym. Preprints, Am. Chem. Soc., **16** (2), 165 (1975).
74. L. S. Penn, T. K. Liao, Composites Technol. Rev., **6** (3), 133 (1984).
75. M. Hayashi, R. Katou, T. Sakai, Seni Gakkaishi, **38** (4), T-147 (1982).
76. E. Sacher, J. Appl. Polym. Sci., **22**, 2137 (1978).
77. M. Litt, J. Herz. J. Colloid Interface Sci., **31**, 248 (1969).
78. M. Litt, J. Herz. Polym. Preprints, Am. Chem. Soc., **10**, 905 (1969).
79. H. J. Busscher, A. W. J. Van Pelt, H. P. de Jong, J. Arends, J. Colloid Interface Sci., **95**, 23 (1983).
80. H. W. Fox, P. W. Taylor, W. A. Zisman, Ind. Eng. Chem., **39**, 1401 (1947).
81. D. G. LeGrand, G. L. Gaines, Jr., J. Polym. Sci. C, **34**, 45 (1971).; J. Colloid Interface Sci., **50**, 272 (1975).
82. Y. Kitazaki, T. Hata, Preprints, 18th Annual Meeting High Polymer Society Japan, 1969, p. 478.
83. Q. S. Bhatia, J. K. Chen, J. T. Koberstein, J. E. Sohn, J. A. Emerson, J. Colloid Interface Sci., **106**, 353 (1985).
84. B. R. Ray, J. R. Anderson, J. J. Scholz, J. Phys. Chem., **62**, 1220 (1958).
85. J. E. Sohn, J. A. Emerson, J. K. Chen, A. F. Siegel, J. T. Kobersein, Preprints, Polym. Mater. Sci. Eng., Am. Chem. Soc., **49**, 447 (1983).
86. P. Luner, M. Sandell, J. Polym. Sci. C, **28**, 115 (1969).
87. W. C. Hamilton, J. Colloid Interface Sci., **40**, 219 (1972).
88. H. J. Busscher, J. Arends, J. Colloid Interface Sci., **81**, 75 (1981).
89. B. R. Ray, W. L. Wasley, J. J. Scholz, J. Phys. Chem., **62**, 1220 (1970).
90. H. D. Feltman, J. R. McPhee, Textile Res. J., **34**, 634 (1964).
91. R. E. Baier, C. Akers, S. Perlmutter, V. L. Gott, J. O'Riodan, Calspan Report WO6-EB-5307-M-18, National Institute of Health, Maryland, March 1976.
92. R. E. Baier, W. A. Zisman, Macromol., **3**, 70 (1970).
93. V. R. Gray, in "Wetting", The Society of Chemical Industry Monograph No. 25, London, 1967, p. 99.
94. S. Wu, J. Colloid Interface Sci., **71**, 605 (1979); **73**, 590 (1980).
95. J. F. Padday, Proc. 2nd Intern, Congr. Surface Activity, **3**, 136 (1957).
96. E. Chibowski, B. Bilinski, A. Waksmundzki, W. Wojcik, J. Colloid Interface Sci., **86**, 559 (1982).
97. R. Fanelli, J. Am. Chem. Soc., **72**, 4016 (1950).
98. T. Kasemura, T. Kondo, T. Hata, Kobunshi Ronbunshu, **36**, 815 (1979).
99. S. Wu, K. J. Brzozowski, J. Colloid Interface Sci., **37**, 686 (1971).
100. T. Kasemura, N. Yamashita, T. Hata, Kobunshi Ronbunshu, **33**, 703 (1976).
101. M. Toyama, T. Ito, Polym-Plast. Technol. Eng., **2** (2), 161 (1973).
102. J. J. Kasper, E. V. Kring, J. Phys. Chem **59**, 1019 (1955); J. J. Jasper, E. R. Kerr, F. Gregorich, J. Am. Chem. Soc., **75**, 5252 (1953).
103. S. H. Anastasiadis, J. K. Chen, J. T. Koberstein, J. E. Sohn, J. A. Emerson, Polym. Eng. Sci., **26**, 1410 (1986).
104. O. R. Quayle, Chem. Rev., **53**, 439 (1953).
105. D. W. van Krevelen, "Properties of Polymers", Elsevier, Amsterdam, 1976.
106. J. J. Elmendorp, G. de Vos, Polym. Eng. Sci., **26**, 415 (1986).
107. T. Kasemura, M. Inagaki, T. Hata, Kobunshi Ronbunshu, **44**, 131 (1987).
108. F. M. Fowkes, in "Chemistry and Physics of Interfaces", American Chemical Society, Washington DC, 1968, p. 1.
109. D. K. Owens and R. C. Wendt, J. Appl. Polym. Sci., **13**, 1741 (1969).

Permeability and Diffusion Data

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A. INTRODUCTION

The transmission of molecules through polymer films is named "permeability". There are many dimensions and units found in the literature for the general expression "permeability". In this paper the permeability coefficient is used. It has the dimension

$$P = \frac{(\text{quantity of permeant}) \times (\text{film thickness})}{(\text{area}) \times (\text{time}) \times (\text{pressure drop across the film})}$$

and is the best definition for permeability.

The permeability coefficient, in a strict sense, is not only a function of the chemical structure of the polymer. It also varies with the morphology of the polymer and depends on many physical factors such as density, crystallinity, and orientation. However, the chemical structure of a polymer can be considered to be the predominant factor which controls the magnitude of the permeability coefficient.

The following general trends in permeability, as related to some influencing factors, may be useful for the proper interpretation of the tables:

Density can be regarded as a measure of the free volume between the molecules of the polymer structure. In general, the higher the density, the lower is the permeability.

Crystallinity of a semicrystalline polymer reduces the permeability significantly compared to the value of the corresponding amorphous polymer; i.e., the higher the degree of crystallinity, the lower the permeability. The crystallinity and the density of a polymer are strongly related. The higher the crystallinity the higher is the density of a given polymer.

Molecular mass of a polymer has been found to have little effect on the permeability of polymers, except at a very low range of molecular masses.

Orientation of polymer molecules reduces the permeability.

Crosslinking decreases the permeability, especially for higher degrees of crosslinking and for large molecular size permeants.

The method and degree of vulcanization has a significant effect on the permeability of elastomers.

Plasticizers increase the permeability.

Humidity increases the permeability of some hydrophilic polymers.

Liquid permeants have the same permeabilities as their corresponding saturated vapors, though higher permeabilities may occur especially if parts of the polymer are being dissolved.

Solution cast films have variable permeabilities depending upon the kind of solvent used and the drying technique. Poor solvents tend to yield films of higher permeability.

Fillers, usually inorganic fillers, decrease the permeability; however, the effect is complicated by the type, shape, and amount of filler and its interaction with the polymer.

Thickness of film does not, in principle, affect the permeability coefficient, the diffusion coefficient, and the solubility coefficient. In practice, different values may be obtained from films of variable thickness, which in turn may be due to differences in drawing, orientation, and crystallinity.

If a permeant does not interact with the polymer under investigation, the permeability coefficient is characteristic for the permeant–polymer system. This is the case with the permeation of many gases, such as H_2, He, N_2, O_2, and CO_2, through many polymers. On the other hand, if a permeant interacts with the polymer, the permeation coefficient is no longer a constant, and may depend on the special conditions of the measurement and the history of the polymer film. In such cases, a single value of the permeability coefficient does not represent the characteristic permeability of the polymer, and it is necessary to know the dependency of the permeability coefficient on all possible variables in order to obtain the complete profile of the permeability of the polymer.

In these cases, the transmission rate, which has the dimension

$$Q = \frac{(\text{quantity of permeant})}{(\text{area}) \times (\text{time})}$$

is often used for practical purposes. Since the transmission rate, Q, includes neither the pressure of the permeant nor the thickness of the polymer in its dimension, it is necessary to know either the pressure or the concentration of the permeant and the thickness of the polymer under the conditions of measurement.

For both P and Q, the quantity of permeant can be expressed by mass, moles, or gaseous volume at standard temperature and pressure. These can readily be converted from one unit into another.

The preferred SI unit of the permeability coefficient used in this book is

$$\text{unit of } P: \frac{\text{cm}^3(273.15\,\text{K};\ 1.013 \times 10^5\,\text{Pa}) \times \text{cm}}{\text{cm}^3 \times \text{s} \times \text{Pa}}$$

where (273.15 K; 1.013×10^5 Pa) means standard temperature and pressure (STP). Therefore permeability coefficients given in this paper are in the range of 10^{-11} – 10^{-16} $\text{cm}^3 \times \text{cm}/\text{cm}^2 \times \text{s} \times \text{Pa}$ for many polymers and permeants.

The mostly used units and their conversion factors are listed in Table B.

The permeation of molecules through flawless polymers occurs by the steps of dissolution of a permeant in the polymer and diffusion of the dissolved permeant. The product of the diffusion coefficient D and the solubility coefficient S is referred to as the permeability coefficient.

$$P = D \times S$$

Units of D and S used in these tables are

$$\text{unit of } D: \frac{\text{cm}^2}{\text{s}}$$

$$\text{unit of } S: \frac{\text{cm}^2(273.15\,\text{K};\ 1.013 \times 10^5\,\text{Pa})}{\text{cm}^3 \times \text{Pa}}$$

The solubility coefficients cited in the following tables are often calculated by

$$S = \frac{P}{D}$$

The temperature dependence of the permeability coefficient P, the diffusion coefficient D, and the solubility coefficient S can be represented by

$$P = P_0 \times \exp(-E_P/RT)$$
$$D = D_0 \times \exp(-E_D/RT)$$

and

$$S = S_0 \times \exp(-E_S/RT)$$

Consequently

$$E_P = E_D + E_S$$

where E_P is the activation energy of permeation, E_D the activation energy of diffusion, and E_S the heat of solution. P_0, D_0, and S_0 are the pre-exponential factors. Values of E_P, E_D, and E_S are given in kJ/mol in the following tables. R is the gas constant (8.3144×10^{-3} kJ/K mol); T is the temperature in K. In the following tables the temperature range is given in which P_0, E_P, E_D, and E_S are relevant, as far as the authors have reported it. The permeability coefficient can be determined for a given temperature by means of the pre-exponential factor P_0 and the activation energy of permeation E_P. In the following tables, permeability, diffusion, and solubility coefficients are listed for many polymers. The pre-exponential factor P_0, the activation energy of permeation E_P, the activation energy of diffusion E_D, and the heat of solution E_S are also given. The pre-exponential factors D_0 and S_0 can be determined by the following equations:

$$D_0 = D \times \exp(E_D/RT)$$
$$S_0 = S \times \exp(E_S/RT)$$

B. CONVERSION FACTORS FOR VARIOUS UNITS OF THE PERMEABILITY COEFFICIENT

From	Multiplication factors to obtain P in: $\frac{[cm^3][cm]}{[cm^2][s][cm\ Hg]}$	$\frac{[cm^3][cm]}{[cm^2][s][Pa]}$	$\frac{[cm^3][cm]}{[m^2][day][atm]}$
$\frac{[cm^3][cm]}{[cm^2][s][cm\ Hg]}$	1	7.5×10^{-4}	6.57×10^{10}
$\frac{[cm^3][mm]}{[cm^2][s][cm\ Hg]}$	10^{-1}	7.5×10^{-5}	6.57×10^{9}
$\frac{[cm^3][cm]}{[cm^2][s][atm]}$	1.32×10^{-2}	9.87×10^{-6}	8.64×10^{8}
$\frac{[cm^3][mil]}{[cm^2][day][atm]}$	3.87×10^{-14}	2.90×10^{-17}	2.54×10^{-3}
$\frac{[in^3][mil]}{[100in^2][day][atm]}$	9.82×10^{-12}	7.37×10^{-15}	6.45×10^{-1}
$\frac{[cm^3][cm]}{[m^2][day][atm]}$	1.52×10^{-11}	1.14×10^{-14}	1
$\frac{[cm^3][cm]}{[m^2][day][bar]}$	1.54×10^{-11}	1.16×10^{-14}	1.01
$\frac{[cm^3][cm]}{[cm^2][s][Pa]}$	1.33×10^{3}	1	8.75×10^{13}

C. TABLES

TABLE 1. PERMEABILITY COEFFICIENTS, DIFFUSION COEFFICIENTS, AND SOLUBILITY COEFFICIENTS OF POLYMERS

Units used in the tables are as follows: P, P_0 in cm^3 (273.15 K; 1.013×10^5 Pa) × cm/(cm^2 × s × Pa); D in cm^2/s; S in cm^3(273.15 K; 1.013×10^5 Pa)/(cm^3 × Pa); E_P, E_D, E_S in kJ/mol; T in °C; (273.15 K; 1.013×10^5 Pa) means standard temperature and pressure (STP).

Polymer	Permeant	T	P (×10^{13})	D (×10^6)	S (×10^6)	Temp. range	P_0 (×10^7)	E_P	E_D	E_S	Refs.
1.1. POLY(ALKANES)											
Poly(ethylene) LLDPE	C_2H_4O	30	68.9	0.031	225						80
Poly(ethylene)	H_2	25	7.4	0.474	1.58						49
density 0.914 g/cm^3, LDPE	D_2	25	6.6	0.476	1.38						49
	He	25	3.7	6.8	0.0544	5–60	4.62	34.8	24.7	10.1	16
	O_2	25	2.2	0.46	0.472	5–60	66.5	42.7	40.2	2.5	16
	Ar	25	2.1	0.36	0.571	5–60	174	45.2	42.3	2.9	16
	Ne	25	0.48	2.42	0.020						49
	Kr	25	2.15	0.169	1.28						49
	Xe	25	4.01	0.069	5.82						49
	CO_2	25	9.5	0.372	2.54	5–60	62.0	38.9	38.5	0.4	16

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Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
	CO	25	1.1	0.332	0.336	5–60	154	46.5	39.8	6.7	16
	N_2	25	0.73	0.320	0.228	5–60	329	49.4	41.5	7.9	16
	CH_4	25	2.2	0.193	1.13	5–60	425	47.3	45.6	1.7	16
	C_2H_6	25	5.1	0.068	7.55	5–60	985	47.3	53.6	−6.3	16
	C_3H_4	25	32	0.105	30.2	5–60	209	38.9	49.8	−10.9	16
	C_3H_6	25	11	0.058	18.8	5–60	459	43.5	52.3	−8.8	16
	C_3H_8	25	7.1	0.0322	21.3	5–60	1170	46.9	55.7	−8.8	16
	SF_6	25	0.13	0.0135	0.951	5–60	4050	59.9	62.0	−2.1	16
	H_2S	20	27.0								5
	NH_3	20	21.0								5
	CH_3Br	20	110								37
	C_2H_4O	30	75			0–60					37
	H_2O	25	68			10–90	48.8	33.5			21
density 0.964 g/cm^3, HDPE	He	25	0.86	3.07	0.028	5–60	0.137	29.7	23.4	6.3	16
	O_2	25	0.30	0.17	0.18	5–60	0.423	35.1	36.8	−1.7	16
	Ar	25	1.30	0.12	1.1	5–60	5.23	37.7	38.9	−1.2	16
	CO_2	25	0.27	0.12	0.22	5–60	0.0506	30.1	35.6	−5.5	16
	CO	25	0.15	0.096	0.15	5–60	1.15	39.3	36.8	2.5	16
	N_2	25	0.11	0.093	0.15	5–60	0.991	39.7	37.7	2.0	16
	CH_4	25	0.29	0.057	0.51	5–60	3.76	40.6	43.5	−2.9	16
	C_2H_6	25	0.44	0.015	3.0	5–60	13.3	42.7	52.3	−9.6	16
	C_3H_4	25	3.0	0.025	12	5–60	1.89	33.1	47.3	−14.2	16
	C_3H_6	25	0.87	0.011	8.2	5–60	5.68	38.9	52.3	−13.4	16
	C_3H_8	25	0.404	0.0049	8.3	5–60	28.5	44.8	56.9	−12.1	16
	n-C_4H_{10}	40		0.02					62.3		64
	n-C_5H_{12}	25		0.0033					76.6		64
	n-C_6H_{14}	0		0.000075							64
	SF_6	25	0.0063	0.0016	0.39	5–60	29.5	55.2	62.8	−7.6	16
	H_2S	20	6.5								5
	NH_3	20	8.0								5
	H_2O	25	9.0								21
Poly(ethylene) HDPE	C_2H_4O	30	21.4	0.021	97.5						80
Poly(ethylene) (Hizex 7000F)											
Isotropic density 0.943 g/cm^3	He	25	1.33	3.50	0.0379						74
draw ratio 8 0.958	He	25	1.46	3.20	0.0457						74
9 0.957	He	25	1.44	3.55	0.0406						74
10 0.958	He	25	0.885	2.38	0.0372						74
12 0.960	He	25	0.407	1.69	0.0241						74
14.5 0.962	He	25	0.278	1.14	0.0244						74
20.5 0.966	He	25	0.129	0.632	0.0204						74
Poly(ethylene) (Hizex 7000F)											
draw ratio 1.0 density 0.945 g/cm^3	N_2	25	0.2	0.094	0.217						99
	O_2	25	0.76	0.196	0.395						99
	CO_2	25	3.2	0.148	2.16						99
6.6 0.954	He	25	1.1	3.3	0.0346						99
6.4	N_2	25	0.11	0.075	0.138						99
	O_2	25	0.34	0.099	0.346						99
	CO_2	25	1.4	0.080	1.82						99
11.8 0.959	He	25	0.37	1.5	0.0247						99
	O_2	25	0.074	0.032	0.237						99
	CO_2	25	0.28	0.02	1.41						99
12.0 0.959	He	25	0.3	1.7	0.0188						99
	O_2	25	0.10	0.0459	0.197						99
17.6	He	25	0.12	0.54	0.0217						99
	O_2	25	0.012	0.0055	0.217						99
	CO_2	25	0.045	0.0035	1.29						99
20.2	He	25	0.11	0.48	0.0237						99
	O_2	25	0.15	0.0065	0.168						99
Poly(ethylene-hexene-1) copolymer (Rigidex 002-55)											
Isotropic density 0.943 g/cm^3	He	25	1.11	3.27	0.0340						74
draw ratio 7 0.955	He	25	0.23	3.32	0.0278						74
8.5 0.958	He	25	0.878								74
9.5 0.959	He	25	0.773								74
10 0.960	He	25	0.695								74
11 0.961	He	25	0.515								74
13 0.963	He	25	0.257								74
16.5 0.964	He	25	0.206								74
Poly(ethylene-hexene-1) copolymer (Rigidex 002-55)											
draw ratio 1.0 density 0.945 g/cm^3	N_2	25	0.179	0.0945	0.188						99
	O_2	25	0.662	0.185	0.357						99
	CO_2	25	3	0.132	2.27						99
6.5	N_2	25	0.059	0.0411	0.148						99
	O_2	25	0.0522	0.0696	0.289						99
	CO_2	25	1.0	0.056	1.78						99

Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
11.7	He	25	0.242	1.2	0.0197						99
	O_2	25	0.044	0.0203	0.218						99
	CO_2	25	0.18	0.0113	1.58						99
14.7	He	25	0.14	0.713	0.0197						99
	O_2	25	0.025	0.0148	0.167						99
17.6	He	25	0.097	0.57	0.0168						99
	O_2	25	0.0098	0.0048	0.207						99
	CO_2	25	0.034	0.0025	1.32						99
Poly(ethylene-*co*-propylene) 40/60 amorphous	He	24.4	24.0	16.5	0.146	0–50	1.56	27.6	15.0	12.6	25
31.0 CH_3-groups per 100 carbons	Ne	24.4	9.0	4.5	0.20	0–50	2.48	31.8	28.5	3.3	25
	Ar	24.4	9.8	1.06	0.93	0–50	46.6	39.6	37.1	2.5	25
	N_2	24.4	3.7	0.67	0.545	0–50	167	43.8	45.8	−2.0	25
26.5 CH_3-groups per 100 carbons	He	24.1	21.8	17.9	0.121	0–50	1.93	28.1	24.8	3.3	25
	Ne	24.1	8.5	5.3	0.16	0–50	4.89	32.7	31.9	0.8	25
	Ar	24.1	11.0	1.3	0.83	0–50	46.7	37.8	42.0	−4.2	25
	N_2	24.1	4.1	0.94	0.44	0–50	97.5	42.1	42.9	−0.8	25
21.5 CH_3-groups per 100 carbons	He	26.6	16.0	9.4	0.17	0–50	2.37	29.7	27.0	2.7	25
	Ne	26.6	6.9	3.5	0.20	0–50	5.67	34.1	31.9	2.2	25
	Ar	26.6	9.2	1.06	0.87	0–50	27.2	37.2	39.6	−2.4	25
	N_2	26.6	3.7	0.79	0.46	0–50	54.2	41.2	44.0	−2.8	25
Poly(propylene)	He	20	0.28	19.5		20–70	0.153	32.2			13
density 0.907 g/cm^3	H_2	20	31.0	2.1		20–70	224	38.5			13
crystallinity 50%	N_2	30	0.33			20–70	1280	55.7			19
	O_2	30	1.7			20–70	278	47.7			19
	CO_2	30	6.9			20–70	24.0	38.1			19
	H_2O	30	51.0			10–90	900	42.3			19,21
	CH_3Br	30	15.0			0–50	1.56×10^6	64.2			19
	H_2S	20	2.4								5
	NH_3	20	6.9								5
(Trespaphan GND)	H_2O	23	15.8			10–50	0.249	23.8			73
density 0.8871 g/cm^3	He	33	10.6			30–55	0.631	28			47
crystallinity 43%	O_2	33	1.68			30–55	117	46			47
	N_2	33	0.424			30–55	1010	55			47
	CO_2	33	5.43			30–55	173	44			47
density 0.8931 g/cm^3	He	33	8.98			30–55	0.534	28			47
crystallinity 50%	O_2	33	1.48			30–55	103	46			47
	N_2	33	0.355			30–55	11500	55			47
	CO_2	33	4.64			30–55	148	44			47
density 0.8998 g/cm^3	He	33	7.7			30–55	0.458	28			47
crystallinity 58%	O_2	33	1.28			30–55	89.4	46			47
	N_2	33	0.306			30–55	732	55			47
	CO_2	33	4.05			30–55	129	44			47
density 0.9016 g/cm^3	He	33	8.69			30–55	0.236	26			47
crystallinity 60%	O_2	33	1.78			30–55	11.8	40			47
	N_2	33	0.494			30–55	34.5	46			47
Poly(propylene)	C_2H_4O	30	0.9	0.0022	45						80
1.2. POLY(STYRENES)											
Poly(styrene)	O_2	25	1.9								42
	H_2O	25	1350								42
biaxially oriented	He	25	14.0								41
	H_2	25	17.0								41
	O_2	25	2.0								41
	N_2	25	0.59								41
	CO_2	25	7.9								41
	H_2O	25	840	0.14							27
	H_2O	23	717			10–50	0.000028	−7.79			73
ultradrawn											
draw ratio 1.0	CO_2	25	6.0	0.0024	248						72
1.8	CO_2	25	4.35	0.0019	233						72
3.1	CO_2	25	2.18	0.00075	293						72
4.4	CO_2	25	1.13								72
5.0	CO_2	25	0.75	0.00046	158						72
Poly(styrene-*co*-styrenesulfonic acid) Na^+-polysalt											
sulfonated with 0 mol% SO_3^-	H_2	23	18.5								83
(Polystyrene)	N_2	23	0.368	0.055	0.67						83
	O_2	23	2.18	0.12	2.0						83
	CO_2	23	10.58	0.054	19.5						83
	CH_4	23	0.585	0.011	5.3						83
sulfonated with 15.2 mol% SO_3^-	H_2	23	10.13								83
	N_2	23	0.0975	0.0042	2.3						83
	O_2	23	0.743	0.021	3.6						83
	CO_2	23	3.53	0.0091	39.8						83
	CH_4	23	0.150	0.0026	5.8						83

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Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
sulfonated with 27.5 mol% SO_3^-	H_2	23	5.85								83
	N_2	23	0.0465	0.0012	4.1						83
	O_2	23	0.443	0.0080	5.5						83
	CO_2	23	2.18	0.0027	80.6						83
	CH_4	23	0.0443	0.00096	4.7						83
Poly(styrene-*co*-styrenesulfonic acid) Mg^{2+}-polysalt											
sulfonated with 15.2 mol% SO_3^-	H_2	23	7.20								83
	N_2	23	0.0563	0.0069	0.83						83
	O_2	23	0.510	0.028	1.8						83
	CO_2	23	2.25	0.0078	28.6						83
	CH_4	23	0.0525	0.0025	2.1						83
sulfonated with 27.5 mol% SO_3^-	H_2	23	5.18								83
	N_2	23	0.0270	0.0051	0.53						83
	O_2	23	0.315	0.017	1.9						83
	CO_2	23	1.65	0.0037	44.6						83
	CH_4	23	0.0255	0.0026	0.98						83
1.3. POLY(METHACRYLATES)											
Poly(methyl methacrylate)	H_2	35	2.78								88
	He	35	5.70								88
	Ar	35	0.0158								88
	N_2	35	6.15								88
	O_2	34	0.116								51
	O_2	35	0.0653								88
	CO_2	35	2.33								88
	CH_4	35	0.00263								88
	H_2O	23	480								73
Poly(methyl methacrylate) + (cellulose acetate butyrate) + (silicone) – Blend 50/40/10	O_2	34	4.52								51
Poly(methyl methacrylate) + (Styrene/acrylonitrile) – Blend											
80/20	CO_2	35	0.435								88
50/50	CO_2	35	0.938								88
20/80	CO_2	35	1.58								88
Poly(methyl acrylate) + Poly(epichlorohydrin) – Blend	He	35	7.95	7.88	0.101						77
100/0	Ar	35	0.383	0.0513	0.747						77
	N_2	35	0.140	0.0385	0.365						77
	CH_4	35	0.176	0.0121	1.46						77
75/25	He	35	6.78	7.15	0.0945						77
	Ar	35	0.410	0.0595	0.689						77
	N_2	35	0.144	0.0442	0.326						77
	CH_4	35	0.221	0.0167	1.32						77
50/50	He	35	5.59	6.54	0.0855						77
	Ar	35	0.443	0.0677	0.655						77
	N_2	35	0.159	0.0508	0.313						77
	CH_4	35	0.302	0.0242	1.25						77
25/75	He	35	4.61								77
	Ar	35	0.495	0.0792	0.625						77
	N_2	35	0.178	0.0595	0.299						77
	CH_4	35	0.411	0.0333	1.23						77
0/100	He	35	3.62	4.88	0.0743						77
	Ar	35	0.533	0.0918	0.580						77
	N_2	35	0.180	0.0655	0.275						77
	CH_4	35	0.493	0.0422	1.17						77
Poly(ethyl methacrylate)	He	25	5.18	42.5	0.0122	25–80	0.255	26.8	15.5	11.3	29
	Ne	25	2.28	1.5	0.152	25–80	0.242	28.7	23.9	4.8	29
	O_2	25	0.889	0.106	0.839	25–80	2.1	36.4	31.8	4.6	29
	Ar	25	0.433	0.0208	2.08	25–80	2.03	38.1	43.1	−5.0	29
	N_2	25	0.170	0.0301	0.565	25–80	2.18	40.6	42.7	−2.1	29
	CO_2	25	3.79	0.0336	11.3	25–80	0.435	28.9	33.1	−4.2	29
	Kr	25	0.287	0.0065	4.40	25–80	10.1	43.1	46.1	−3.0	29
	H_2S	25	2.88	0.0046	62.6	25–80	22.8	39.4	47.7	−8.3	29
	SF_6	25	0.0127	0.00023	5.62	25–80	0.885	44.8	64.5	−19.7	29
	H_2O	25	2380	0.102	2333	25–65	0.00555	2.1	36.4	−34.3	29
		65	2880	0.729	395	65–80	1.58	18.0	63.2	−45.2	29
Poly(2-hydroxyethyl-methacrylate) highly crosslinked	O_2[a]	34	2.72								51
	O_2[b]	34	5.43								51

Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
1.4. POLY(NITRILES)											
Poly(acrylonitrile)	O_2	25	0.00015								42
	CO_2	25	0.00060								42
	H_2O	25	230								42
(Barex)	O_2	25	0.0041								42
		38	0.014								42
	CO_2	25	0.012								42
		38	0.022								42
	H_2O	25	490								42
		38	520								42
Poly(acrylonitrile-*co*-styrene) 86/14	O_2	25	0.0032								42
	CO_2	25	0.011								42
	H_2O	25	640								42
66/34	O_2	25	0.036								42
	CO_2	25	0.16								42
	H_2O	25	1500								42
57/43	O_2	25	0.14								42
	CO_2	25	0.27								42
	H_2O	25	1800								42
39/61	O_2	25	0.35								42
	CO_2	25	1.0								42
	H_2O	25	1900								42
Poly(acrylonitrile-*co*-methyl acrylate-*co*-butadiene) 79/15/6	O_2	25	0.0041								42
	CO_2	25	0.012								42
	H_2O	25	970								42
Poly(methacrylonitrile)	O_2	25	0.00090	0.0003	0.3						42
	CO_2	25	0.0024	0.0002	1.2						42
	H_2O	25	310								42
(Lopac)	O_2	25	0.0026								42
		38	0.0053								42
	CO_2	25	0.0081								42
		38	0.016								42
	H_2O	25	260								42
		38	270								42
Poly(methacrylonitrile-*co*-styrene) 97/3	O_2	25	0.0018								42
	CO_2	25	0.0059								42
	H_2O	25	370								42
82/18	O_2	25	0.0098	0.0026	0.38						42
	CO_2	25	0.038	0.0018	2.1						42
	H_2O	25	700								42
61/39	O_2	25	0.068								42
	CO_2	25	0.21								42
	H_2O	25	1300								42
53/47	O_2	25	0.12								42
	CO_2	25	0.38								42
	H_2O	25	1400								42
38/62	O_2	25	0.29								42
	CO_2	25	0.88								42
	H_2O	25	1600								42
18/82	O_2	25	0.81								42
	CO_2	25	2.4								42
	H_2O	25	1500								42
Poly(methacrylonitrile-*co*-styrene-*co*-butadiene) 88/7/5	O_2	25	0.0036								42
	CO_2	25	0.011								42
	H_2O	25	450								42
83/7/10	O_2	25	0.0052								42
	CO_2	25	0.015								42
	H_2O	25	500								42
78/7/15	O_2	25	0.0072								42
	CO_2	25	0.024								42
	H_2O	25	580								42
1.5. POLY(VINYLS)											
Poly(vinyl acetate)	He	10	4.95	6.46	0.0784	0–20	0.00134	13.2	17.4	−4.2	14
		30	9.44	9.55	0.101	25–40	2.23	31.2	22.4	8.8	14
	H_2	10	2.99	1.32	0.237	0–20	0.0023	15.7	21.6	−5.9	14
		30	6.84	2.63	0.254	25–40	10.4	41.7	31.4	10.3	14
	Ne	10	0.838	0.794	0.106	0–20	0.000111	11.5	30.8	−19.3	14
		30	1.97	1.66	0.118	25–40	14.1	39.8	35.4	4.4	14
	O_2	10	0.136	0.0178	0.766	0–20	0.00074	20.2	46.4	−26.2	14
		30	0.367	0.0562	0.637	25–40	1680	56.1	60.7	−4.6	14
	Ar	10	0.0569	0.00479	1.11	0–20	0.0471	32.1	47.6	−15.5	14
		30	0.143	0.0162	0.943	25–40	4950	61.2	69.1	−7.9	14
	Kr	10	0.0172	0.000602	2.78	0–20	0.00826	30.8	60.7	−29.9	15
		30	0.0582	0.00295	1.96	25–40	7.71×10^{5}	76.2	81.2	−5.0	15

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Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
	CH_4	25	0.0237	0.0017	1.39	25–40	0.72×10^{7}	82.7	80.8	1.9	15
	CO_2	35	1.52	0.00747	20.4						86
Poly(vinyl alcohol)	He[c]	20	0.00075								10
	H_2[c]	25	0.00672								28
	D_2[c]	100	0.0134								57
	N_2[c]	14	0.0001								11
	N_2[d]	14	0.248	0.045	0.53						11
	O_2[c]	25	0.00665								28
	O_2[c]	23	0.00005								59
	CO_2[c]	25	0.00924								28
	CO_2[d]	25	65.0								28
	H_2S[c]	25	0.0052								28
	C_2H_4O[c]	0	0.0015								37
	H_2O	25	5.25								98
Poly(vinyl benzoate)	H_2	25	6.55	1.93	0.34	20–60	0.0055	16.7	21.9	−5.2	65
		70	18.3	6.39	0.29	60–85	0.251	27.2	24.5	2.7	65
	He	25	6.67	6.17	0.11	20–60	0.00345	15.5	16.9	−1.3	65
		70	17.1	16.1	0.11	60–85	0.458	29.1	22.4	6.7	65
	Ne	25	1.51	1.08	0.14	20–60	0.00137	16.9	23.2	−6.3	65
		70	4.28	3.83	0.11	60–85	0.758	34.5	28.6	5.9	65
	Ar	25	0.358	0.033	1.08	20–60	0.0020	21.4	31.6	−10.1	65
		35	0.459	0.0491	0.93						86
		70	1.24	0.181	0.69	60–85	0.863	38.4	38.1	0.29	65
	Kr	25	0.192	0.0059	3.25	20–60	0.00188	22.8	38.4	−15.5	65
		70	0.712	0.049	1.45	60–85	2.565	43.1	50.7	−7.7	65
	Xe	25	0.071	0.00064	11.1	20–60	0.00723	28.6	46.0	−17.5	65
		70	0.43								65
	N_2	25	0.124	0.026	0.48	20–60	0.00104	22.4	33.0	−10.6	65
		70	0.463	0.158	0.29	60–85	1.26	42.3	45.2	−2.9	65
	O_2	25	0.715	0.079	0.91	20–60	0.00060	16.7	25.1	−8.5	65
		70	2.11	0.309	0.68	60–85	2.76	40.2	30.5	9.8	65
	CO_2	25	4.17	0.020	20.9	20–60	0.00020	9.6	32.9	−23.4	65
		35	4.74	0.030	16.2						86
		70	8.06	0.117	6.9	60–85	0.734	32.6	45.2	−12.5	65
Poly(vinyl butyral)	H_2O	25	608								98
Poly(vinyl-*p*-isopropylbenzoate)	He	25	8.93	15.5	0.0578	20–67	0.590	27.5	18.4	9.1	86
		80	49.9	51.9	0.096	67–95	0.537	27.3	23.5	3.7	86
	Ne	25	3.88	2.72	0.0585	20–67	0.00251	16.0	17.1	−1.1	86
		80	13.9	8.71	0.160	67–95	1.09	33.1	23.4	9.6	86
	Ar	25	1.90	0.165	1.15	20–67	0.00131	16.2	27.5	−11.3	86
		80	7.67	1.11	0.691	67–95	11.6	41.8	39.7	2.1	86
	Kr	25	0.940	0.0388	3.91	20–67	0.000248	13.8	32.2	−18.3	86
		80	6.25	0.393	1.61	67–95	40.6	46.1	47.3	−1.3	86
	Xe	25	0.00942	0.00579	16.5	20–67	0.00000289	14.2	36.1	−22.0	86
		80	4.62	0.0898	5.14	67–95	534.8	54.5	61.5	−7.0	86
	O_2	25	2.44	0.279	0.0546	20–67	0.00180	16.4	26.8	−10.4	86
		80	12.4	1.948	0.637	67–95	6.92	38.9	37.4	1.4	86
	N_2	25	0.647	0.117	0.550	20–67	0.000878	17.9	29.3	−11.4	86
		80	2.99	0.874	0.348	67–95	19.4	46.1	42.4	3.6	86
	CO_2	25	15.7	0.0760	20.6	20–67	0.000110	4.8	30.4	25.5	86
		80	32.9	0.638	5.17	67–95	3.57	34.0	43.9	−9.9	86
	CH_4	25	1.32	0.0462	2.84	20–67	0.000512	14.8	33.1	−18.3	86
		80	5.65	0.478	1.163	67–95	58.7	47.4	48.1	−0.6	86
Poly(vinyl-*m*-methylbenzoate)	Ar	35	0.642	0.0738	0.870						86
	CO_2	35	5.05	0.0361	14.0						86
Poly(vinyl chloride)	He	25	1.5	2.8	0.055	25–80	0.212	29.9	20.7	9.2	32
unplasticized	H_2	25	1.3	0.5	0.26	25–80	1.46	34.5	34.5	0	32
	Ne	25	0.29	0.25	0.12	25–80	0.274	34.1	31.5	2.6	32
	N_2	25	0.0089	0.0038	0.23	25–80	9380	69.0	61.9	7.1	32
	Ar	25	0.0086	0.0012	0.75	25–80	6.28	57.8	51.5	6.3	32
	Kr	25	0.0060	0.00041	0.54	25–70	24.2	55.3	62.8	−7.5	32
		80	0.14			75–90	8.85×10^{5}	86.6	128.5	−41.9	32
	O_2	25	0.034	0.012	0.29	25–80	179	55.8	54.6	1.2	32
	CO_2	25	0.12	0.0025	4.7	25–90	930	56.8	64.6	−7.8	32
	CH_4	25	0.021	0.0013	1.7	25–85	7950	66.2	70.3	−4.1	32
	H_2O	25	206	0.024	870	25–80	2.04	22.9	41.8	−18.9	32
	H_2S	20	0.14	0.000046	310						5
	NH_3	20	3.7								5
	C_2H_4O	30	4.4	0.0013	330						80
unplasticized	He	35	2.52								75
(Pentaform Type TH 170/01)	N_2	35	0.00709	0.00188							75
	Ar	35	0.0277	0.00339							75
	CO_2	35	0.182	0.00145							75
	CH_4	35	0.00810								75

Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
unplasticized	He	35	2.24								75
(CS 5760)	N_2	35	0.00501	0.00137							75
	Ar	35	0.0208	0.00242							75
	CO_2	35	0.138	0.00145							75
	CH_4	35	0.00638	0.000331							75
plasticized with tricresyl triphospate %											
0	H_2	27	1.8	0.48							45
5	H_2	27	1.4	0.456							45
10.2	H_2	27	1.3	0.422							45
15	H_2	27	1.3	0.444							45
20.1	H_2	27	1.6	0.473							45
30.8	H_2	27	2.1	1.56							45
40.0	H_2	27	2.7	2.25							45
0	CO	27	0.019	0.0023							45
5	CO	27	0.012	0.0021							45
10.2	CO	27	0.013	0.0016							45
15	CO	27	0.014	0.0025							45
20.1	CO	27	0.021	0.0029							45
30.8	CO	27	0.079	0.0054							45
40.0	CO	27	0.281	0.029							45
Poly(vinyl chloride)/Oligo(dimethylsiloxane)											
100/0	N_2	25	0.0226								85
	O_2	25	0.103								85
90/10	N_2	25	0.0281								85
	O_2	25	0.120								85
80/20	N_2	25	0.0316								85
	O_2	25	0.151								85
75/25	N_2	25	0.0359								85
	O_2	25	0.164								85
60/40	N_2	25	0.0656								85
	O_2	25	0.295								85
50/50	N_2	25	0.153								85
	O_2	25	0.678								85
40/60	N_2	25	0.320								85
	O_2	25	1.74								85
30/70	N_2	25	1.51								85
	O_2	25	3.83								85
Poly(vinyl chloride)-*g*-oligo(dimethylsiloxane)/Oligo(dimethylsiloxane)											
100/0	N_2	25	0.0269								85
	O_2	25	0.164								85
80/20	N_2	25	0.0271								85
	O_2	25	0.165								85
70/30	N_2	25	0.0431								85
	O_2	25	0.254								85
60/40	N_2	25	0.0609								85
	O_2	25	0.413								85
50/50	N_2	25	0.120								85
	O_2	25	0.795								85
40/60	N_2	25	0.417								85
	O_2	25	2.43								85
30/70	N_2	25	1.01								85
	O_2	25	4.99								85
Poly(vinyl flouride)	H_2	35	0.41								96
	He	35	0.98								96
	N_2	35	0.0068								96
	O_2	35	0.030								96
	CO_2	35	0.20								96
	CH_4	35	0.0075								96
Poly(vinylidene chloride)	O_2	35	0.00038								96
Poly(vinylidene chloride) (Saran)	He	34	0.233								10
	N_2	30	0.000706			0–90	900	70.3			37
	O_2	30	0.00383			0–90	825	66.6			37
	CO_2	30	0.0218			0–90	24.8	51.5			37
	CH_3Br	30	0.00218								37
	H_2S	30	0.027	9.6×10^{-5}	27.6	30–75	1.58×10^{5}	74.5			8
	H_2O	25	7.0			10–60	863	46.1			21
Poly(vinylidene chloride-*co*-vinyl chloride) 88/12 plasticized with acetyltributyl citrate (%)											
0.5	H_2O	30	1.38								50
2.7	H_2O	30	3.07								50

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Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
4.9	H_2O	30	4.77								50
7.2	H_2O	30	7.38								50
Poly(vinyl cyclohexanecarboxylate)	He	25	9.69	7.33	0.148	15–52	0.0158	18.3	16.3	0.25	81
	He	70	33.5	20.3	0.166	52–85	0.547	27.7	23.7	4.4	81
	O_2	25	1.30	0.122	1.08	15–52	0.00302	19.2	30.9	−11.7	81
	O_2	70	6.50	0.772	0.835	52–85	7.26	39.7	41.1	−0.34	81
	Ar	25	0.851	0.0670	1.27	15–52	0.0115	23.6	34.6	−11.1	81
	Ar	35	1.17	0.106	1.10						86
	Ar	70	4.75	0.570	0.934	52–85	20.2	43.5	43.5	−0.011	81
	N_2	25	0.256	0.0421	0.609	15–52	0.00690	25.3	35.6	-11.1	81
	N_2	70	1.82	0.368	0.492	52–85	60.4	49.4	46.9	1.8	81
	CO_2	25	5.05	0.0254	20.0	15–52	0.00233	15.2	37.5	−22.3	81
	CO_2	70	21.4	0.270	8.36	52–85	6.68	36.1	47.7	−11.7	81
1.6. *FLUORINE CONTAINING POLYMERS*											
Poly(tetrafluoroethylene)	H_2	25	7.4			20–130	0.0415	21.4			23
	H_2	25	7.25	0.147	4.94						49
	N_2	25	1.0	0.088	1.2	20–130	0.0202	24.4	29.9	−5.5	23
	O_2	25	3.2	0.0152	2.1	20–130	0.00698	19.1	26.3	−7.2	23
	CO_2	25	7.5	0.095	9.2	20–130	0.00248	14.0	28.6	−14.6	23
	NO_2	25	12.0	0.0037	320	20–130	0.00087	10.7	58.6	−47.9	23
	N_2O_4	25	28.0	0.025	110	20–130	9.7×10^{-6}	−2.1			23
	D_2	25	7.48	0.116	6.42						49
	T_2	25	6.19	0.052	11.9						49
	CH_4	25	0.56	0.023	2.46						49
	He	25	9.0	0.811	1.09						49
	Ne	25	2.69	0.251	1.09						49
	Ar	25	1.50	0.056	2.66						49
	Kr	25	0.73	0.015	4.83						49
	Xe	25	0.30	0.004	7.40						49
	H_2O	38	6								98
	CH_4	90	3.85	0.221	1.74	40–195	0.273	33.7	34.7	−1.0	53
	C_2H_6	90	2.15	0.0763	2.81	40–195	0.251	35.2	36.2	−1.0	53
	C_3H_8	90	0.96	0.0251	3.87	40–195	0.923	41.6	44.8	−3.2	53
	n-C_4H_{10}	90	0.878	0.0187	4.69	40–195	1.22	42.7	51.5	−8.8	53
	i-C_4H_{10}	90	0.345	0.0086	4.03	40–195	4.64	49.6	58.1	−8.5	53
(Hostaflon PFA)	He	23	33.0			20–100	0.0119	14.5			73
	O_2	22	3.17			20–90	0.0386	23.1			73
	N_2	23	0.92			20–100	0.105	28.7			73
	CO_2	24	10.0			20–100	0.0236	19.2			73
	CH_4	25	0.79			20–100	0.292	31.8			73
	SO_2	24	2.11			5–100	0.0425	24.5			73
	Cl_2	23	1.35			20–100	0.0259	24.3			73
	HCl	22	2.04			20–80	0.122	27.0			73
	H_2O	23	13.4			10–80	0.0797	21.4			73
Poly(tetrafluorethylene-*co*-hexafluoropropene) (Teflon FEP)	N_2	25	1.19	0.0948	1.25	20–80	0.273	30.6	38.5	−7.9	24
	O_2	25	3.68	0.184	2.02	20–80	0.109	25.5	34.7	−9.2	24,67
	CO_2	25	9.53	0.105	9.08	20–80	0.0443	20.9	36.6	−15.7	24
	CH_4	90	9.98	0.491	2.15	40–195	1.09	35.0	41.3	−6.3	53,55
	C_2H_6	90	6.23	0.140	4.37	40–195	1.22	36.8	53.8	−17.0	53,55
	C_3H_8	90	3.11	0.0489	6.38	40–195	6.30	43.9	59.5	−15.6	53,55
	n-C_4H_{10}	90	2.18	0.0278	7.8	40–195	134	54.2	65.0	−10.8	53,55
	i-C_4H_{10}	90	0.855	0.0146	5.86	40–195	4200	67.4	77.8	−10.4	53,55
	CH_3Cl	25	0.763	0.00667	11.5	40–120	0.945	34.8	50.1	−15.3	55
	C_6H_6	25	0.263	2.45×10^{-4}	107	40–120	122	49.5	69.1	−19.6	55
	H_2O	25	12.8			25–60	0.494	26.2	56.4	−30.2	56,66
		70	37.3			60–85	1.58×10^{-7}	−15.6	16.5	−32.1	56,66
Poly(tetrafluorethylene-*co*-ethylene) (Hostaflon ET)	He	23	4.23								54
	O_2	23	0.40								54
	N_2	23	0.137								54
	CO_2	23	1.49								54
	H_2O	23	2.8								54
	He	25	5.67			20–100	0.0557	22.8			73
	O_2	22	0.78			20–100	0.371	32.1			73
	N_2	23	0.25			5–100	0.0790	31.2			73
	CO_2	23	4.39			20–100	0.0562	23.3			73
	CH_4	23	0.23			20–100	0.326	34.9			73
	Cl_2	23	0.807			20–100	0.0349	26.3			73
	HCl	23	2.2			20–100	0.394	29.8			73
	SO_2	23	1.96			5–100	0.0137	21.8			73
	H_2O	25	74			20–55	0.000121	1.2			73
		80	115			55–95	0.0387	17.1			73
Poly(trifluorochloroethylene)	He	20	5.1					23.4			10
unplasticized	H_2	20	0.705					30.2			10
Kel-F 300 film of about	N_2	25	0.00375			25–75	150	59.9			20
30% crystallinity	O_2	40	0.030			0–80	4.35	46.9			20
	CO_2	40	0.158			40–80	24.8	49.4			20
	H_2O	25	0.218								37

Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
	C_2H_2O	60	3.3								37
	CH_3Br	60	3.45								37
crystallinity 80%	N_2	25	0.00225			25–75	1.05	49.8			20
	O_2	40	0.0188			40–60	0.698	45.6			20
	CO_2	40	0.036			40–80	2.03	46.5			20
crystallinity 30%	N_2	50	0.413								20
plasticized[c]	O_2	30	0.42								20
	CO_2	50	5.63								20
Poly(trifluorochloroethylene-*co*-ethylene) (Halar)	He	23	4.0								54
	O_2	23	0.114								54
	N_2	23	0.0457								54
	CO_2	23	0.457								54
	H_2O	23	2.8								54
Poly(vinyl fluoride) (Tedlar)	He	23	0.347								54
	N_2	23	0.0012								54
	O_2	23	0.0139								54
	CO_2	23	0.069								54
	H_2O	23	10.1								54
Poly(vinyl fluoride) (Kynar)	He	35	1.29	1.87							78
	H_2	35	0.404	0.336							78
	Ar	35	0.0323	0.00985	0.394						78
	N_2	35	0.0167	0.00716	0.234						78
	O_2	35	0.0620	0.0172							78
	CH_4	35	0.0150	0.00294	0.457						78
	CO_2	35	0.388	0.00566	7.21						78
1.7. POLY(DIENES)											
Poly(butadiene)	H_2	25	31.6	9.6	0.326	25–50	2.15	27.6	21.3	6.3	35
	N_2	25	4.84	1.1	0.444	25–50	4.91	34.3	30.1	4.2	35
	O_2	25	14.3	1.5	0.957	25–50	2.27	29.7	28.5	1.2	35
	CO_2	25	104	1.05	9.87	25–50	0.683	21.8	30.6	−8.8	35
cis-1.4	He	24	24.5	15.7	0.156	0–45	0.0855	20.3	17.3	2.9	25
	Ne	24	14.4	6.55	0.220	0–45	0.096	21.8	17.4	4.4	25
	Ar	24	30.8	4.06	0.758	0–45	0.084	19.4	21.3	−1.9	25
	N_2	24	14.4	2.96	0.488	0–45	0.078	21.3	25.0	−3.7	25
Poly(butadiene-*co*-acrylonitrile)	He	25	12.7	15.5	0.0819	25–50	1.24	28.5	17.6	10.9	35
80/20 (Perbunan 18)	H_2	25	18.9	6.43	0.296	25–50	3.52	30.1	26.0	4.1	35
	N_2	25	1.89	0.51	0.375	25–50	34.9	41.5	35.6	5.9	35
	O_2	25	6.15	0.79	0.77	25–50	230	36.0	33.9	2.1	35
	CO_2	25	47.6	0.425	11.2	25–50	6.43	29.3	38.5	−9.2	35
73/27 (Perbunan, German)	He	25	9.2	11.7	0.0789	25–50	1.24	29.3	21.8	7.5	35
	H_2	25	11.9	4.5	0.267	25–50	7.43	33.1	28.9	4.2	35
	N_2	25	0.8	0.25	0.316	25–50	180	47.7	43.5	4.2	35
	O_2	25	2.9	0.43	0.671	25–50	37.3	40.6	38.5	2.1	35
	CO_2	25	23.2	0.19	12.2	25–50	20.0	33.9	44.8	−10.9	35
	C_2H_2	25	18.7	0.0764	24.5	25–50	282	41.0	51.5	−10.5	35
	C_3H_8	50	58.5	0.141	41.5						35
68/32 (Hycar OR 25)	He	25	7.4	11.2	0.0661	25–50	1.9	30.9	21.8	9.1	35
	H_2	25	8.85	3.85	0.227	25–50	8.97	34.3	29.3	5.0	35
	N_2	25	0.454	0.152	0.296	25–50	473	51.5	49.0	2.5	35
	O_2	25	1.76	0.28	0.632	25–50	85.6	43.9	43.1	0.8	35
	CO_2	25	13.9	0.107	13.0	25–50	55.5	37.7	50.2	−12.5	35
61/39 (Hycar OR 15)	He	25	5.13	7.92	0.0592	25–50	2.23	32.2	23.0	9.2	35
	H_2	25	5.35	2.43	0.217	25–50	14.9	36.8	31.8	5.0	35
	N_2	25	0.177	0.064	0.276	25–50	2330	57.8	53.2	4.6	35
	O_2	25	0.721	13.6	0.533	25–50	444	50.2	45.6	4.6	35
	CO_2	25	5.59	0.038	14.7	25–50	272	43.9	56.1	−12.2	35
Poly(butadiene-*co*-styrene)	He	24	17.2	15.7	0.110	0–50	4.79	26.8	23.7	3.1	25
92/8 (Ameripol 1502)	Ne	24	7.28	5.47	0.133	0–50	0.908	29.2	29.4	−0.2	25
	Ar	24	9.53	1.39	0.686	0–50	7.88	33.7	33.3	0.4	25
	N_2	24	3.83	1.05	0.365	0–50	8.63	36.3	36.0	0.3	25
80/20 (Hycar 2001)	He	24	10.1	16.0	0.0628	0–50	1.41	29.5	18.0	11.5	25
	Ne	24	3.76	4.23	0.0885	0–50	0.56	29.5	25.7	3.8	25
	Ar	24	3.37	58.0	0.581	0–50	19.2	42.2	41.3	0.9	25
	N_2	24	1.28	42.8	0.30	0–50	23.8	41.6	42.2	−0.6	25
Poly(butadiene) hydrogenated crystallinity 29%, Hydropol density 0.894 g/cm^3	He	25	11.8	15.1	0.0781	5–60	17.2	35.2	24.7	10.5	16
	O_2	25	8.49	1.2	0.708	5–60	128	41.0	38.9	2.1	16
	Ar	25	8.29	0.96	0.864	5–60	292	43.1	40.2	2.9	16
	CO_2	25	36.3	0.91	3.99	5–60	85.8	36.4	36.8	−0.4	16
	CO	25	4.64	0.82	0.566	5–60	324	44.8	37.3	7.5	16
	N_2	25	3.00	0.74	0.405	5–60	489	46.9	39.4	7.5	16
	CH_4	25	9.77	0.54	1.81	5–60	942	45.6	44.0	1.6	16
	C_2H_6	25	25.0	0.24	10.4	5–60	543	41.9	49.4	−7.5	16
	C_3H_4	25	169	0.31	54.5	5–60	340	36.0	45.2	−9.2	16
	C_3H_6	25	62.2	0.20	31.1	5–60	680	40.2	48.2	−8.0	16
	C_3H_8	25	40.5	0.12	33.8	5–60	1680	43.5	52.3	−8.8	16
	SF_6	25	0.849	0.056	1.52	5–60	7790	56.9	60.3	−3.4	16

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Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
Poly(chloroprene) (Neoprene G)	H_2	25	10.2	3.8	0.286	15–50	8.80	33.9	27.6	6.3	33,36
	O_2	25	2.96	0.38	0.740	15–50	54.7	41.5	39.4	2.1	33,36
	N_2	25	0.88	0.24	0.355	15–50	224	44.4	43.1	1.3	33,36
	Ar	25	2.84	0.16							36
	CO_2	25	19.2	0.23	8.19	15–50	2499	35.6	45.2	−9.6	33,36
	CH_4	25	2.47								33
	H_2O	25	683								31
	NH_3	25			86.8						33
	SO_2	25			176						33
(Neoprene W)	H_2O	23	837			23–60	4.04×10^{11}	83.3			52
	H_2O[f]	23	5130			23–60	2.4×10^{6}	49.2			52
(Neoprene)	He	35	15.0	60.4	0.0247						90
	H_2	35	20.7	9.33	0.221						90
	D_2	35	20.0	10.7	0.188						90
	O_2	35	6.62	1.01	0.656						90
	N_2	35	2.23	0.533	0.419						90
Poly(dimethylbutadiene)	He	25	10.8			25–50	0.735	27.6			35,58
(Methyl rubber)	H_2	25	12.8	3.9	0.326	25–50	9.40	33.5	31.4	2.1	35
	N_2	25	0.355	0.079	0.454	25–50	2010	55.7	51.9	3.8	35
	O_2	25	1.58	0.14	1.13	25–50	303	47.3	46.5	0.8	35
	CO_2	25	5.63	0.063	8.98	25–50	917	46.9	53.6	−6.7	35,33
	CH_4	25	0.622			25–50	50300	62.3			58
Poly(ethylene-*co*-propylene-*co*-diene)	H_2O	23	157			23–60	6.84×10^{8}	71.7			52
Poly(isobutene-*co*-isoprene) 98/2	H_2O[f]	23	367			23–60	745000	52.8			52
(Butyl rubber)	He	25	6.32	5.93	0.109	25–50	2.34	31.8	24.3	7.5	35
	H_2	25	5.43	1.52	0.355	25–50	12.8	36.4	33.9	2.5	35
	N_2	25	0.243	0.045	0.543	25–50	349	52.3	50.7	1.6	35
	O_2	25	0.977	0.081	1.20	25–50	68.3	44.8	49.8	−5.0	35
	CO_2	25	3.89	0.0578	6.71	25–50	71.9	41.5	50.2	−8.7	35
	C_2H_2	25	1.26	0.0203	6.22	25–50	407	48.6	53.6	−5.0	35
	C_3H_8	50	10.3	0.0224	46.4						35
	H_2O	37.5	82.5								12
Poly(isoprene), amorphous	He	25		21.6					19.7		16
(Natural rubber)	O_2	25	17.6	1.73	1.02	20–50	2.38	29.3	33.5	−4.2	16,35
	Ar	25	17.2	1.36	1.26	20–50	9.15	32.7	33.1	−0.4	16
	CO_2	25	115	1.25	9.20	20–50	0.755	21.8	34.3	−12.5	16,35
	CO	25	11.8	1.35	0.874	20–50	3.16	31.0	31.0	0	16
Poly(isoprene)	N_2	25	7.11	1.17	0.608	20–50	12.2	35.6	33.5	2.1	16,35
	CH_4	25	22.7	0.89	2.55	20–50	6.08	31.0	36.4	−5.4	16
	C_2H_6	25		0.40					42.7		16
	C_3H_4	25	415	0.50	83.5	20–50	3.76	22.6	37.7	−15.1	16
	C_3H_6	25	154	0.31	49.7	20–50	17.7	28.9	42.7	−13.8	16
	C_3H_8	25	126	0.21	60.0	20–50	1.34	23.0	46.5	−23.5	16,35
	SF_6	25	2.70	0.115	2.35	20–50	4.62	35.6	50.2	−14.6	16
	C_2H_2	25	74.5	0.467	16.0	25–50	17.0	30.6	39.8	−9.2	35
	n-C_4H_{10}	50	526								58
	n-C_5H_{12}	50	1240								58
	H_2O	25	1720								31, 98
trans-	H_2	25	10.9	5.0	0.227	15–40	296	42.3	31.8	13.0	34
(Vulcanized purified gutta percha)		50	106	21	0.493	50–70	0.582	23.0	26.4	−0.4	34
	O_2	25	4.64	0.7	0.661	15–40	2560	49.8	38.5	13.8	34
		50	56.3	5.3	1.05	50–70	0.197	21.8	26.8	−2.1	34
	N_2	25	1.62	0.5	0.326	15–40	1560	52.3	41.0	13.8	34
		50	23.7	4.2	0.562	50–70	0.316	25.5	28.1	0.4	34
	CO_2	25	26.7	0.47	5.52	15–40	217	39.3	41.0	0.8	34
		50	281	4.1	6.81	50–70	0.0671	14.7	28.1	−10.5	34
	H_2O	25	379								31
Poly(isoprene-*co*-acrylonitrile) 74/26	He	25	5.85	8.01	0.0731	25–50	2.17	31.8	20.5	11.3	35
	H_2	25	5.58	2.47	0.227	25–50	26.2	38.1	31.0	7.1	35
	N_2	25	0.136	0.045	0.306	25–50	13500	62.8	60.7	2.1	35
	O_2	25	0.642	0.092	0.701	25–50	1330	53.2	50.7	2.5	35
	CO_2	25	3.26	0.031	10.6	25–50	4690	52.3	60.3	−8.0	35
Poly(isoprene-*co*-methacrylonitrile) 74/26	H_2	25	10.2	3.55	0.286	25–50	8.80	33.9	28.9	5.0	35
	N_2	25	0.449	0.123	0.365	25–50	1090	53.6	48.6	5.0	35
	O_2	25	1.77	0.24	0.740	25–50	209	46.1	40.2	5.9	35
	CO_2	25	10.7	0.091	11.8	25–50	273	42.3	51.1	−8.8	35
Poly(isoprene) hydrochloride	He	20	14.3								10
(Pliofilm FM-1)	N_2	30	0.113								37
	O_2	30	40.5								37
	CO_2	30	0.91			10–60	1.43	36.0			37,43
	H_2S	30	0.075	0.0002	38.5	30–60	403000	73.7			8
	C_2H_4O	0	548								37
	CH_3Br	60	25.5								37

Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
(Pliofilm NO)	H_2O	25		4100		25–60			58.6		39
Poly(methyl-1-pentenylene)	He	25	75.8								41
	H_2	25	102								41
	N_2	25	5.87	0.55	1.07						41
	O_2	25	24.2	1.01	2.40						41
	CO_2	25	69.5	684	0.0102						41
Poly(2-methyl-1,3-pentadiene-*co*-4-methyl-1,3-pentadiene 85/15	H_2	25	32.0			25–50	5.08	29.7			35
	N_2	25	2.06	0.30	0.691	25–50	336	46.9	46.5	0.4	35
	O_2	25	7.52	0.555	1.36	25–50	59.6	39.4	41.0	−1.6	35
	CO_2	25	33.8			25–50	41.9	34.8			35
1.8. POLY(XYLYLENES)											
Poly(*p*-2,6-dichloroxylylene)	H_2O	25	4.97	0.00594	84.6	30–80	0.0129	19.5	36.0	−16.5	48
Poly(chloro-*p*-xylylene) (parylene C)	He	25	1.41	0.103	0.391	10–80		21.7	14.6	7.1	84
	H_2	25	0.717	0.0675	0.670	10–80		24.1	18.5	5.6	84
	N_2	25	0.00626	0.00178	0.0863	10–80		31.3	30.1	1.2	84
	O_2	25	0.0358	0.00351	0.567	10–80		29.5	32.5	−3.0	84
	CO_2	25	0.105	0.00164	4.62	10–80		27.8	51.3	−23.4	84
1.9. POLY(OXIDES)											
Poly(oxymethylene)	CO_2	25	1.35	0.014	9.64	25–100	0.462	31.4	49.0		38
	H_2O	25	683	0.027	2530	25–100	0.129	13.0			38
grafted with 2.8% butadiene	CO_2	25	3.83	0.044	8.69	25–100	0.095	25.1	38.5		38
	H_2O	25	998	0.070	1430	25–100	0.0126	6.3			38
(Hostaform)	O_2	23	0.057								73
	CO_2	23	1.28								73
	H_2O	23	422			23–60	0.129	14.1			73
Poly(oxy-2,6-dimethyl-1,4-phenylene)	He	25	58.5								41
	H_2	25	84.6								41
	N_2	25	2.86								41
	O_2	25	11.9								41
	CO_2	25	56.8	0.0601	94.6						41
	H_2O	25	3045	0.17	1797						40
Poly(phenylene oxide) brominated											
0% Br[a]	CO_2	35	38	0.17	21.7						100
	CH_4	35	2.2	0.021	10.6						100
36% Br	CO_2	35	38	0.16	23.7						100
	CH_4	35	2.0	0.016	11.8						100
91% Br	CO_2	35	51	0.20	25.4						100
	CH_4	35	2.6	0.02	13.1						100
106% Br	CO_2	35	81	0.30	27.2						100
	CH_4	35	4.7	0.033	14.3						100
1.10. POLY(ESTERS), POLY(CARBONATES)											
Poly(oxybutyleneoxyterephthaloyl), amorphous	He	25	1.28	0.13	0.888						44
	Ne	25	0.237	0.022	0.997						44
	Ar	25	0.078	0.0040	1.97						44
	CO_2	25	0.217	0.0062	3.46						44
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene) (Lexan)	He	25	7.5			25–125	0.0248	20.1			22
	H_2	25	9.0	0.64	1.38	25–175	0.0816	22.6	20.9	1.7	22
	Ne	100	10.9								22
	Ar	25	0.6	0.015	4.15	25–125	0.00544	22.6	25.1	−2.5	22
		175	32.3	0.53	6.12	125–175	70.2	54.4			22
	O_2	25	1.05	0.021	5.03	0–125	0.00252	19.3	32.2	−12.9	22
		175	22.5			125–175	0.0919	31.0			22
	CO_2	25	6.0	0.0048	124	25–100	0.00365	15.9	37.7	−21.8	22
	SF_6	25	4.88×10^{-6}	1.0×10^{-7}	4.94	25–125	0.0118	53.6	83.7	−30.1	22
		175	0.075			125–175	3.21×10^{7}	125.6			22
	H_2O	25	1050	0.68					26.0		22
	N_2	25	0.225			0–125	0.00559	25.1			22
		175	14.3			125–175	1.09	41.9			22
Poly(oxyethyleneoxyisophthaloyl)	O_2	30	0.012	0.0029	0.41						60
Poly(oxyethyleneoxyterephthaloyl), amorphous	He	25	2.37	3.1	0.077	20–60	0.0199	21.3	19.2	−1.1	17,18
	Ar	25	0.0217	0.0024	0.79	20–60	0.00319	28.9	48.2	−15.1	17,18
	N_2	25	0.0108	0.0019	0.55	20–60	0.00025	26.4	47.7	−23.9	17,18
	O_2	25	0.0444	0.0045	0.98	20–60	0.227	37.7	48.6	−14.7	17,18
	CH_4	25	0.0070	0.00030	2.3	20–60	0.00021	24.7	51.1	−27.2	17,18
	CO_2	25	0.227	0.00083	28.0	20–60	0.0232	27.6	52.3	−28.5	17,18
crystallinity 40%	He	25	1.09	2.0	0.052	20–80	0.00308	19.7	20.1	0.9	17,18
		90	6.3			80–130	0.0266	25.5	27.6	0.9	17,18
	N_2	25	0.00513	0.0013	0.45	20–80	0.00275	32.7	44.0	−18.4	17,18
		90	0.068			80–130	20.6	59.0	58.6	0	17,18

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Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
	O_2	25	0.0257	0.0035	0.72	20–80	0.0122	32.2	46.1	−13.0	17,18
		90	0.31			80–130	0.744	44.4	50.7	0	17,18
	CH_4	25	0.00257	0.00013	2.0	20–80	0.00719	36.8	52.3	−22.2	17,18
		90	0.091			80–130	63.2	61.5	67.0	−12.9	17,18
	CO_2	25	0.118	0.00057	20.0	20–80	0.000197	18.4	50.2	−31.4	17,18
		90	1.0			80–130	22.2	51.1	64.5	−19.7	17,18
(Mylar A)	H_2	25	0.448								57
	D_2	25	0.073								57
	N_2	25	0.0038			−25–80	0.000945	30.8			37
	O_2	25	0.030			−25–80	0.000026	16.8			37
	CH_3Br	25	0.013			25–90	0.0163	34.8			37
	C_2H_4O	25	0.084			25–90	0.000534	21.7			37
	H_2O	25	98			10–60	0.00032	2.9			21
(Hostaphan)	H_2S	20	0.14	0.000071	195						5
	NH_3	20	1.1								5
	N_2	20	0.0034								61
	O_2	23	0.014								72
	O_2	30	0.042	0.0065	0.62						60
	H_2O	23	114			10–40	0.00012	0.1			73
Poly(ethylene terephthalate)	H_2O	25	113								98
Poly(ethylene terephthalate-*co*-isophthalate) 50/50	O_2	30	0.019	0.0035	0.49						60
Poly(ethylene terephthalate-*co*-2,6-naphthalene dicarboxylate) 50/50	O_2	30	0.031	0.0048	0.61						60
Poly(oxyterephthaloyloxymethylene-1,4-cyclohexylenemethylene)	O_2	30	0.19	0.014	1.3						60
Poly(tetramethylene adipate) diol	O_2	35	0.0975	0.076	0.1275						76
molecular weight 1050	CO_2	35	0.525	0.027	1.95						76
density 1.20 g/cm^3	CO	35	0.0278	0.039	0.07125						76
	N_2	35									76
	CH_4	35	0.0353	0.027	0.1275						76
	H_2O	23	3900	0.041	9525						76
molecular weight 1510	O_2	35	0.563	0.16	0.3525						76
density 1.19 g/cm^3	CO_2	35	4.8	0.057	8.25						76
	CO	35	0.195	0.074	0.2625						76
	N_2	35	0.113	0.053	0.2100						76
	CH_4	35	0.383	0.058	0.6600						76
	H_2O	23	5925	0.055	10875						76
molecular weight 2100	O_2	35	1.5	0.39	0.375						76
density 1.18 g/cm^3	CO_2	35	15	0.13	6.53						76
	CO	35	0.9	0.11	0.750						76
	N_2	35	0.6	0.22	0.270						76
	CH_4	35	1.2	0.17	0.750						76
	H_2O	23	7425	0.078	9525						76
Poly(carbonate) (Lexan)	H_2O	25	1050								98
Poly(carbonate)	CO_2	35	4.5	0.0209	14.5						100
	CH_4	35	0.20	0.0046	4.0						100
Poly(carbonate-sulfone) (diphenyl carbonate + 1,10-decanediol)	CO_2	25	9.5								95
(diphenyl carbonate + 1,4-bis(3-hydroxypropylsulfonyl)butane	CO_2	25	0.035								95
Poly(tetrabutylene carbonate)	CO_2	35	3.17	0.0166	19.1						100
	CH_4	35	0.0945	0.0013	7.24						100
Poly(tetrachlorine carbonate)	CO_2	35	5.00	0.0253	19.6						100
	CH_4	35	0.168	0.0022	7.61						100
Poly(tetramethylene hexafluoro carbonate)	CO_2	35	82.5	0.24	32.6						100
	CH_4	35	3.45	0.0348	9.58						100
Poly(tetramethylene carbonate)	CO_2	35	13.2	0.061	21.5						100
	CH_4	35	0.60	0.0081	7.37						100
Poly(tetramethylene carbonate) diol	O_2	35	0.105	0.029	0.368						76
molecular weight 1150	CO_2	35	0.675	0.012	5.40						76
density 1.29 g/cm^3	CO	35	0.0525	0.010	0.525						76
	N_2	35	0.0308	0.019	0.165						76
	CH_4	35	0.0375	0.008	0.473						76
	H_2O	23	825	0.012	7223						76

Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
molecular weight 1570	O_2	35	0.143	0.054	0.263						76
density 1.27 g/cm^3	CO_2	35	1.35	0.017	7.50						76
	CO	35	0.105	0.033	0.315						76
	N_2	35	0.0660	0.050	0.105						76
	CH_4	35	0.105	0.019	0.540						76
	H_2O	23	2100	0.020	10575						76
molecular weight 3070	O_2	35	0.308	0.072	0.473						76
density 1.21 g/cm^3	CO_2	35	2.48	0.026	9.75						76
	CO	35	0.143	0.043	0.345						76
	N_2	35	0.0900	0.026	0.345						76
	CH_4	35	0.0975	0.015	0.675						76
	H_2O	23	1875	0.030	6143						76
Poly(hexamethylene carbonate) diol	O_2	35	0.188	0.022	0.825						76
molecular weight 1040	CO_2	35	1.05	0.022	6.98						76
density 1.29 g/cm^3	CO	35	0.0450	0.015	0.300						76
	N_2	35	0.0450	0.017	0.263						76
	CH_4	35	0.0375	0.0041	0.825						76
	H_2O	23	825	0.013	6825						76
molecular weight 1320	O_2	35	0.225	0.026	0.825						76
density 1.25 g/cm^3	CO_2	35	2.03	0.029	7.05						76
	CO	35	0.150	0.015	0.375						76
	N_2	35	0.0825	0.037	0.225						76
	CH_4	35	0.0675	0.012	0.563						76
	H_2O	23	1050	0.015	7230						76
molecular weight 1500	O_2	35	0.623	0.12	0.525						76
density 1.20 g/cm^3	CO_2	35	2.85	0.038	7.50						76
	CO	35	0.225	0.056	0.450						76
	N_2	35	0.173	0.076	0.225						76
	CH_4	35	0.165	0.019	0.900						76
	H_2O	23	1575	0.017	923						76
molecular weight 2380	O_2	35	1.65	0.19	0.825						76
density 1.18 g/cm^3	CO_2	35	11.3	0.21	5.33						76
	CO	35	0.750	0.14	0.525						76
	N_2	35	0.570	0.17	0.338						76
	CH_4	35	0.825	0.11	0.750						76
	H_2O	23	3450	0.070	4973						76
molecular weight 2860	O_2	35	2.18	0.22	0.975						76
density 1.18 g/cm^3	CO_2	35	9.00	0.27	3.38						76
	CO	35	0.975	0.23	0.450						76
	N_2	35	0.900	0.24	0.360						76
	CH_4	35	1.20	0.12	0.975						76
	H_2O	23	4200	0.076	5595						76
Poly(pentamethylene-hexamethylene carbonate) diol	O_2	35	0.825	0.079	1.05						76
molecular weight 2170	CO_2	35	7.50	0.089	8.25						76
density 1.18 g/cm^3	CO	35	0.443	0.074	0.600						76
	N_2	35	0.278	0.079	0.353						76
	CH_4	35	0.480	0.038	1.28						76
	H_2O	23	2475	0.034	7215						76
Poly(dipropylene glycol carbonate) diol	O_2	35	0.398	0.081	0.488						76
molecular weight 3010	CO_2	35	2.70	0.035	7.50						76
density 1.19 g/cm^3	CO	35	0.135	0.033	0.405						76
	N_2	35	0.0825	0.040	0.210						76
	CH_4	35	0.105	0.021	0.495						76
	H_2O	23	2025	0.019	10800						76
Poly(dipropylene glycol)poly(propylene glycol carbonate) diol	O_2	35	0.975	0.23	0.420						76
molecular weight 2020	CO_2	35	7.28	0.061	12.0						76
density 1.16 g/cm^3	CO	35	0.585	0.16	0.375						76
	N_2	35	0.278	0.13	0.218						76
	CH_4	35	0.570	0.048	1.20						76
	H_2O	23	3000	0.020	15000						76
Poly(triethylene glycol carbonate) diol	O_2	35	0.263	0.079	0.330						76
molecular weight 2580	CO_2	35	3.15	0.050	6.30						76
density 1.20 g/cm^3	CO	35	0.143	0.063	0.225						76
	N_2	35	0.0825	0.074	0.113						76
	CH_4	35	0.150	0.037	0.405						76
	H_2O	23	9750	0.034	25500						76

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Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
Poly(hexamethylene carbonate pentamethylene ester) diol	O_2	35	1.50	0.24	0.630						76
molecular weight 2470	CO_2	35	20.3	0.16	12.8						76
density 1.17 g/cm^3	CO	35	0.750	0.24	0.315						76
	N_2	35	0.600	0.18	0.330						76
	CH_4	35	1.28	0.14	0.900						76
	H_2O	23	655	0.11	59250						76
1.11. POLY(SILOXANES)											
Poly(oxydimethylsilylene) with 10%	He	0	175	41	0.424	−40–25	0.0946	14.3	8.0	6.3	2,41
filler Scantocel CS,	Ne	0	143	16.1	0.888	−40–0	0.0543	13.5	8.4	5.1	2
vulcanized silicon rubber	Ar	0	413	12.2	3.36	−40–0	0.0606	9.8	10.3	−0.5	2
	Kr	0	765	8.0	9.67	−40–0	0.0134	6.5	12.2	−5.7	2
	Xe	0	1910	4.8	39.4	−40–0	0.00781	3.2	14.0	−10.8	2
	H_2	0	348	47.0	0.73	−40–25	0.145	13.7	9.3	4.4	2,41
	N_2	0	170	8.5	2.00	−40–25	0.0197	10.8	11.3	−0.5	2,41
	O_2	0	367	12.0	3.06	−40–25	0.0169	8.7	9.0	−0.3	2,41
	CO_2	25	2430								41
	C_3H_8	25	319	12.0	2.4	0–70	0.00401	6.3	11.3	−5.0	46
	n-C_4H_{10}	0	14600	2.4	608	−40–0			14.2		2
	H_2O	35	32300			35–65	0.0303				3
Poly(dimethyl siloxane)	H_2	35	705								82
	He	35	420								82,94
	N_2	35	353								82
	O_2	35	695								82,94
	CO_2	35	3489								82,94
	CH_4	35	1002								82,94
	C_2H_4	35	2738								82
	C_2H_6	35	3150								82
	C_3H_8	35	6338								82
	H_2S	35	6413								82
	NH_3	35	9015								82
Poly(methyl ethyl siloxane)	He	35	172								94
	O_2	35	234								94
	CO_2	35	1130								94
	CH_4	35	356								94
Poly(methyl propyl siloxane)	He	35	191								94
	O_2	35	287								94
	CO_2	35	1727								94
	CH_4	35	428								94
Poly(methyl octyl siloxane)	H_2	35	233								82
	He	35	99.8								82,94
	N_2	35	65.3								82
	O_2	35	144								82,94
	CO_2	35	695								82,94
	CH_4	35	236								82,94
	C_2H_4	35	608								82
	C_2H_6	35	720								82
	C_3H_8	35	1493								82
	H_2S	35	1470								82
	NH_3	35	2213								82
Poly(triflouropropyl siloxane)	H_2O	25	16050								98
Poly(triflouropropyl methyl siloxane)	He	35	185								94
	O_2	35	163								94
	CO_2	35	1041								94
	CH_4	35	151								94
Poly(phenyl methyl siloxane)	He	35	25.5								94
	O_2	35	24.0								94
	CO_2	35	179								94
	CH_4	35	27.0								94
Poly(dimethyl silmethylene)	He	35	73.5								94
	O_2	35	75.8								94
	CO_2	35	412								94
	CH_4	35	97.5								94
Poly(silethylene siloxane)	He	35	204								94
	O_2	35	296								94
	CO_2	35	1412								94
	CH_4	35	448								94
Poly(silhexylene siloxane)	He	35	146								94
	O_2	35	199								94
	CO_2	35	980								94
	CH_4	35	296								94

Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
Poly(siloctylene siloxane)	He	35	131								94
	O_2	35	178								94
	CO_2	35	899								94
	CH_4	35	270								94
Poly(meta-silphenylene siloxane)	He	35	63.0								94
	O_2	35	55.5								94
	CO_2	35	389								94
	CH_4	35	79.5								94
Poly(para-silphenylene siloxane)	He	35	24.5								82, 94
	N_2	35	2.40								82
	O_2	35	8.21								82,94
	CO_2	35	38.3								82,94
	CH_4	35	7.65								82,94
	C_2H_4	35	15.0								82
	C_2H_6	35	11.3								82
	C_3H_8	35	19.5								82
	H_2S	35	141								82
	NH_3	35	223								82
1.12. POLY(AMIDES), POLY(IMIDES)											
Poly(iminoadipoyliminohexamethylene) (Nylon 66)											
undrawn	CO_2	5	0.018	0.00018	9.97						4
	CO_2	25	0.052	0.00083	6.32						4
drawn	CO_2	5	0.023	0.00018	12.8						4
	CO_2	25	0.071	0.00048	14.8						4
Poly(imino-1-oxohexamethylene) (Nylon 6)	He	20	0.398					36.4			10
	N_2	30	0.00713			0–90	1.05	46.9			37
	O_2	30	0.0285			0–90	0.975	43.5			37
	CO_2	20	0.066			0–90	1.2	40.6			37
	H_2O	25	0.139								20
	H_2S	30	0.255	0.00047	55.7	0–80	2400	58.2			8
	NH_3	20	0.878								5
	CH_3Br	60	0.63			60–80	137	53.2			37
Poly(imino-1-oxoundecamethylene) (Nylon 11)	He	30	1.46	3.5	0.0424	10–50	0.0827	27.6	22.2	5.4	1
	H_2	30	1.34	0.984	0.136	20–60	0.342	31.4	29.7	1.7	1
	Ne	30	0.26	0.437	0.0592	20–60	0.30	35.2	32.7	2.5	1
	Ar	40	0.143	0.0372	0.385	40–60	1.38	41.9	44.8	−2.9	1
	CO_2	40	0.754	0.0191	3.94	40–60	0.338	33.9	51.9	−18.0	1
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′]dipyrrole-2,6[1H,3H]-diyl-1,4-phenyleneoxy-1,4-phenylene) (Kapton)	O_2	160	1.7	0.168		135–240	0.00028	18.4	31.0	−12.6	67
	H_2O	30	431	0.0018	21400	30–80	0.00026	−1.28	42.0	−43.2	68
Poly(2,2′-bis(*p*-trimellitoxyphenyl) propane dianhydride-*co*-3,3′-dimethyl-benzidine)	H_2	30	4.85	0.396	1.22						104
	N_2	30	0.0384	0.00222	1.73						104
	O_2	30	0.304	0.0106	2.87						104
Poly(2,2′-bis(*p*-trimellitoxyphenyl) propane dianhydride-*co*-4,4′-diaminodiphenyl ether)	H_2	30	3.62	0.329	1.10						104
	N_2	30	0.0363	0.00206	1.76						104
	O_2	30	0.233	0.00878	2.65						104
Poly(2,2′-bis(*p*-trimellitoxyphenyl) propane dianhydride-*co*-4,4′-diaminodiphenyl methylene)	H_2	30	3.49	0.289	1.21						104
	N_2	30	0.0343	0.00187	1.83						104
	O_2	30	0.220	0.00736	2.99						104
Poly(2,2′-bis(*p*-trimellitoxyphenyl) propane dianhydride-*co*-benzidine)	H_2	30	0.542	0.0536	1.01						104
	N_2	30	0.00485	0.000327	1.49						104
	O_2	30	0.0344	0.00167	2.06						104
Poly(2,2′-bis(*p*-trimellitoxyphenyl) propane dianhydride-*co*-3,3′,4,4′-tetraaminodiphenyl ether)	H_2	30	5.52	0.402	1.37						104
	N_2	30	0.0439	0.00223	2.32						104
	O_2	30	0.326	0.0102	3.19						104
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*m*,*m*′-diaminodiphenyl methylene)	O_2	25	0.0235								105
	CO_2	25	0.0340								105
	H_2O	25	291								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*m*,*p*′-diaminodiphenyl methylene)	O_2	25	0.0578								105
	H_2O	25	573								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*p*,*p*′-diaminodiphenyl methylene)	O_2	25	0.140								105
	CO_2	25	0.324								105
	H_2O	25	984								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*p*,*p*′-diaminodiphenyl ether)	O_2	25	0.0927								105
	CO_2	25	0.357								105
	H_2O	25	719								105

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Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-benzidine)	O_2	25	0.00668								105
	CO_2	25	0.0234								105
	H_2O	25	94.4								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*p*-phenylenediamine)	O_2	25	0.0103								105
	H_2O	25	231								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*m*-phenylenediamine)	O_2	25	0.0308								105
	CO_2	25	0.0882								105
	H_2O	25	804								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-diaminodiphenyl sulfide)	O_2	25	0.0992								105
	CO_2	25	0.307								105
	H_2O	25	787								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*m,m*′-diaminobenzophenone)	O_2	25	0.0219								105
	CO_2	25	0.0807								105
	H_2O	25	428								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*m,p*′-diaminobenzophenone)	O_2	25	0.0450								105
	CO_2	25	0.0887								105
	H_2O	25	521								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*p,p*′-diaminobenzophenone)	O_2	25	0.0451								105
	CO_2	25	0.132								105
	H_2O	25	599								105
Poly(3,3′,4,4′-benzophenone tetracarboxylic dianhydride-*co*-*p,p*′-diaminostilbene)	O_2	25	0.00214								105
	CO_2	25	0.00692								105
Poly(pyromellitic dianhydride-*co*-benzidine)	O_2	25	0.00451								105
	CO_2	25	0.00790								105
	H_2O	25	131								105
Poly(pyromellitic dianhydride -*co*-*m,m*′-diaminobenzophenone)	O_2	25	0.0260								105
	H_2O	25	727								105
Poly(pyromellitic dianhydride -*co*-p,p′-diaminobenzophenone)	O_2	25	0.110								105
	CO_2	25	0.135								105
	H_2O	25	1009								105
Poly(pyromellitic dianhydride -*co*-*p,p*′-diaminodiphenyl ether)	O_2	25	0.242								105
	CO_2	25	1.36								105
	H_2O	25	1882								105
Poly(pyromellitic dianhydride -*co*-*p,p*′-diaminodiphenyl sulfide)	O_2	25	0.336								105
	H_2O	25	1882								105
Poly(pyromellitic dianhydride -*co*-*p,p*′-diaminodiphenyl methylene)	O_2	25	0.368								105
	CO_2	25	1.32								105
	H_2O	25	1582								105
1.13. POLY(URETHANES)											
Poly(2-methyl-2-dimethylaminomethylpropylenehexa methylene dicarbamate) (Poly(NPM-HMDI))	N_2	25	0.00975								91
	O_2	25	0.0645								91
Poly(2-methyl-2-dimethylaminomethyl-1,3-propylene-diphenylmethane-4,4′-dicarbamate) (Poly(NPM-MDI))	N_2	25	0.0105								91
	O_2	25	0.0675								91
Poly(urethane-sulfone) (tolylene diisocyanate + 1,10-decanediol)	CO_2	25	0.23								95
(tolylene diisocyanate + 1,3-bis(3-hydroxypropylsulfonyl)propane)	CO_2	25	0.0064								95
(tolylene diisocyanate + 1,4-bis(3-hydroxypropylsulfonyl)butane)	CO_2	25	0.0038								95
1.14. POLY(SULFONES)											
Poly(sulfone)	CO_2	35	4.20	0.020	20.7						100
	CH_4	35	0.19	0.0034	5.60						100
Poly(tetramethylsulfone)	CO_2	35	15.8	0.064	24.7						100
	CH_4	35	0.71	0.0079	9.14						100

Polymer	Permeant	T	P (×10[13])	D (×10[6])	S (×10[6])	Temp. range	P_0 (×10[7])	E_P	E_D	E_S	Refs.
Poly(dimethylsulfone)	CO_2	35	1.58	0.0094	16.8						100
	CH_4	35	0.0525	0.0010	5.09						100
Poly(tetramethylhexafluorosulfone)	CO_2	35	54	0.14	39.5						100
	CH_4	35	2.25	0.016	14.6						100
1.15. POLY(ARYL ETHER ETHER KETONE)											
Poly(aryl ether ether ketone)	He	25	5.24			25-75	17.2				101
		55	9.90								101
		75	14.0								101
	O_2	25	0.41			25-75	18.7				101
		55	0.82								101
		75	1.21								101
	CO_2	25	2.03			25-75	13.8				101
		55	3.40								101
		75	4.54								101
semicrystalline (30%)	H_2O	5		0.0017							102
		20		0.0053							102
		35		0.0082							103
		40		0.0152							102
		50		0.0190							103
		60		0.0356							102
		65		0.0390							103
		80		0.0660							103
		95		0.133							103
1.16. CELLULOSE AND DERIVATIVES											
Cellulose hydrate	He[c]	20	0.000375					41.4			10
(Cellophane)	H_2[c]	25	0.00472								28
	N_2[c]	25	0.0024								28
	O_2[c]	25	0.0016								28
	CO_2[c]	25	0.00353								28
	SO_2[c]	25	0.00129								28
	H_2S[c]	25	0.0086								28
	H_2S	45	0.0045	1.6×10^{-6}	276	45-60	2.1×10^6	89.6			8
	NH_3[c]	25	0.0118								28
	H_2O	25	18900								7
	H_2[h]	25	0.012								28
	N_2[h]	25	0.00507								28
	O_2[h]	25	0.00536								28
	CO_2[h]	25	0.00974								28
	H_2[i]	25	0.0244								28
	N_2[i]	25	0.00559								28
	O_2[i]	25	0.00665								28
	CO_2[i]	25	0.0539								28
	H_2[d]	25	0.0598								28
	N_2[d]	25	0.0138								28
	O_2[d]	25	0.0087								28
	CO_2[d]	25	0.192								28
	NH_3[d]	25	133								28
	SO_2[d]	25	28.4								28
	H_2S[d]		0.425								28
Cellulose acetate	He	20	10.2								10
	H_2	20	2.63								10
		35	11.2								87
	N_2[j]	30	0.21			- 25-60	0.00975	27.1			37
		35	0.173								87
	O_2[j]	30	0.585			- 25-60	0.00375	20.9			37
		35	0.788								87
	CO_2[j]	30	17.3				0.0218	18.0			37
		35	4.47								87
	H_2O	25	4130			10-60	0.0053	0			21
	H_2O[j]	25	5500			10-60	0.0053	0			21
	H_2S	30	2.63	0.0010	266	0-60	0.000668	20.5			8
	H_2S[j]	30	4.58	0.0022	207	0-60	0.00188	21.4			8
	CH_4	35	0.154								87
	C_3H_4O[j]	30	30.0								37
	CH_3Br[j]	30	4.2								37
Cellulose acetobutyrate	O_2	34	3.56								51
Cellulose nitrate	He	25	5.18	0.431	1.2						9
	H_2	20	1.5					23.8			10
	N_2	25	0.087	0.0193	0.45						9
	O_2	25	1.46	0.15	0.975						9
	Ar	25	0.0825	0.00753	1.05						9
	CO_2	25	1.59	0.0221	7.2						9
	SO_2	25	1.32	0.0018	73.3						9
	NH_3	25	42.8	0.00786	544						9
	H_2O	25	4720	0.0262	18000						9

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Polymer	Permeant	T	P ($\times 10^{13}$)	D ($\times 10^{6}$)	S ($\times 10^{6}$)	Temp. range	P_0 ($\times 10^{7}$)	E_P	E_D	E_S	Refs.
	C_2H_6	25	0.0473	0.00013	36.2						9
	C_3H_8	25	0.0063	0.000021	29.6						9
Ethyl cellulose	He	25	40.1	2.21	1.8						9
	H_2	20	65.3								10
	N_2	25	3.32	0.233	1.43						9
	O_2	25	11.0	0.639	1.73						9
	Ar	25	7.65	0.403	1.88						9
	CO_2	25	84.8	0.565	15.0						9
	SO_2	25	198	0.0734	270						9
	NH_3	25	529	0.146	363						9
	H_2O	25	6700	0.0286	23400						9
	C_2H_6	25	6.9	0.019	36.3						9
	C_2H_6	30	10.0								79
	C_2H_6	40	10.3								79
	C_2H_6	50	12.2								79
	C_2H_6	60	14.9								79
	C_2H_6	70	20.1								79
	C_3H_8	25	2.78	0.00293	94.7						9
	C_3H_8	30	4.73								79
	C_3H_8	40	5.32								79
	C_3H_8	50	6.36								79
	C_3H_8	60	9.05								79
	C_3H_8	70	10.8								79
	n-C_4H_{10}	25	2.9	0.00146	199						9
	n-C_4H_{10}	30	4.20								79
	n-C_4H_{10}	40	6.30								79
	n-C_4H_{10}	50	7.43								79
	n-C_4H_{10}	60	9.77								79
	n-C_4H_{10}	70	11.4								79
	n-C_5H_{12}	25	2.78	0.00138	201						9
	n-C_6H_{14}	25	5.75	0.00124	462						9
(Ethocel 610)	C_2H_4O[j]	30	308								37

[a] Thickness 0.29 mm.
[b] Thickness 0.84 mm.
[c] Relative humidity 0%.
[d] Relative humidity 100%.
[e] Plasticizer was low-molecular-weight polytrifluorochloroethylene.
[f] 3.5% aqueous sodium chloride solution (simulated sea water).
[g] The % bromination refers to 100 times the number of bromine atoms per repeat unit.
[h] Relative humidity 43%.
[i] Relative humidity 76%.
[j] Plasticized.

TABLE 2. PERMEABILTY COEFFICIENTS OF SIX DIFFERENT FLUORINATED HYDROCARBONS THROUGH POLYMERS[a]

Polymer	$CFCl_3$ (24.5°C)	CF_2Cl_2 (20°C)	CF_3Cl (20°C)	$CHFCl_2$ (20°C)	CHF_2Cl (20°C)	CF_2Cl–CF_2Cl (20°C)	N_2 (20°C)
Poly(ethylene) low density	37.5	152	2.03	1.88	0.795	0.212	0.914
high density	9.75	30	0.673	0.374	0.106	0.0683	0.219
high density, drawn	3.38	4.35	0.326	0.161	0.0266	0.0336	0.133
Poly(propylene)	4.88	137	0.132	0.00338	0.0018	0.00728	0.151
Poly(vinyl chloride) (Genotherm UG), unplasticized	0.697	0.00668	0.00585	0.0016	0.0016	0.00503	0.0247
Poly(vinyl chloride) (Guttagena T52), plasticized	656	63.8	1.88	0.359	0.185	0.054	0.0649
Poly(oxyethyleneoxyterephthaloyl) [Poly(ethylene terephthalate)] (Hostaphan)	0.00188	0.00135	0.0016	0.00135	0.00165	0.0039	0.00343
Poly(imino-1-oxohexamethylene) (Supronyl S)	0.0315	0.0329	0.0111	0.00623	0.00803	0.00923	0.0075
Poly(imino-1-oxohexamethylene) (Supronyl N), plasticized	52.5	3.6	0.254	0.00473	0.00143	0.00225	0.0133

[a] Ref. 61; P in $cm^3 \times cm/(cm^2 \times s \times Pa) \times 10^{-13}$.

TABLE 3. PERMEABILITY COEFFICIENTS OF VARIOUS ORGANIC COMPOUNDS THROUGH LOW-DENSITY POLY(ETHYLENE)[a]

Permeant	0°C	21.1°C	54.4°C	73.9°C
Acetaldehyde	1.34	2.36		
Acetanilide		0.008	0.16	1.34
Acetic acid	0.14	1.22	25.9	119

Permeant	0°C	21.1°C	54.4°C	73.9°C
Acetic anhydride	0.051	0.32	11.6	83.7
Acetone	0.55	2.67	72.3	295
Allyl alcohol	0.063	0.57	9.04	54.2
Amyl acetate	0.22	3.42	106	430
i-Amyl alcohol		0.078		
i-Amyl propionate		3.9		
n-Amyl propionate		11.8		
Aniline	0.098	0.67	23.4	116
Benzaldehyde	0.15	2.67	81.0	417
Benzene	17.7	173	1770	5400
Benzoic acid		0.028	2.24	14.2
Benzyl alcohol		0.16		
n-Butyl acetate		5.9		
n-Butyl alcohol	0.04	0.18	8.02	59
s-Butyl alcohol	0.05	0.24	14.8	109
t-Butyl alcohol	0.02	0.10	10.6	92.8
Butyraldehyde	0.35	3.93	229	393
n-Butyric acid	0.075	1.89	39.3	228
Camphor		0.12	8.5	63.7
Carbon tetrachloride	20	240	3080	17700
Chloroacetic acid		0.12	5.9	23.6
m-Chloroaniline		0.63		
Chlorobenzene	22.7	179	1730	8300
Chloromaleic anhydride		0.28		
p-Chlorotoluene	11.8	119	1640	7300
Cyclohexane	12.4	98.7	1470	5900
Decane	3.7	28.0	480	1620
1,2-Dibromoethane		55.0		
Dibutyl ether	4.4	33.6	582	1230
Dibutyl phthalate			5.7	10.9
o-Dichlorobenzene		60.9		
Diethylene glycol				0.30
Diethyl ether	18.9	123	3140	11800
Diethyl oxalate		0.28		
Dipentene	5.9	50.3	798	3190
Ethyl acetate	0.75	6.5	149	669
Ethyl alcohol		0.28		
Ethylene glycol				0.50
Ethylene glycol monobutyl ether		0.26	13.8	70.8
Ethyl formate		5.9		
Ethyl mercaptan	10.2	157	3931	15700
Ethyl propionate		9.8		
Formaldehyde		0.21		
Formamide		0.20		
Formic acid	0.10	0.26	4.7	20.0
Furfuryl alcohol		0.09		
Glycerin			0.14	0.63
n-Heptane	19.1	106	1040	3200
n-Heptene		106		
n-Heptyl acetate		7.1		
Heptyl alcohol		0.39		
n-Hexane	18.9	138	3540	11800
Hydroquinone			0.02	0.14
Methyl acetate		5.5		
Methyl alcohol	0.1	0.48	10.9	39.3
Methyl ethyl ketone	1.45	4.95	128	550
Methylcyclohexane		108		
Nitrobenzene	0.15	1.93	39.3	165
Nitroethane	0.38	1.06	25.1	113
Nitromethane		0.83		
Octadecane		0.71		
Octyl alcohol	0.039	0.20	10.1	73.9
Oxalic acid			0.08	0.09
i-Pentane	18.9	106	1970	7860
n-Pentane	38.1	206	5900	23600
Phenol	0.04	0.2	9.4	47.1

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Permeant	0°C	21.1°C	54.4°C	73.9°C
n-Propyl alcohol	0.03	0.2	8.8	66.0
i-Propylamine	0.90	16.0	275	1260
Tetradecane	0.67	5.7	159	393
Tetradecene		4.7		
Toluene	22.7	199	2270	11300
1,1,1-Trichloroethane		102		
o-Xylene	14.2	101	1420	6530
p-Xylene	33.7	191	1890	6410

[a] Ref. 62; P in g × mm/(m² × d).

TABLE 4. PERMEABILITY COEFFICIENTS AND DIFFUSION COEFFICIENTS OF AN EQUIMOLAR MIXTURE OF VARIOUS COMPOUNDS (1.25 M EACH) THROUGH HIGH-DENSITY POLY(ETHYLENE)[a]

	P		D ($\times 10^6$)	
Permeant	22°C	50°C	22°C	50°C
1,1,2-Trichlorethane (0.167 g/ml)	2.4	8.8	0.0142	0.0833
1,2-Dichloroethane (0.124 g/ml)	3.3	14.6	0.0225	0.114
Methylene chloride (0.106 g/ml)	5.3	5.9	0.0889	
Styrene (0.130 g/ml)	2.7	10.0	0.0500	0.167
Benzene (0.098 g/ml)	3.1	10.0	0.0417	0.361
Toluene (0.098 g/ml)	3.3	10.2	0.0722	0.417
Naphthalene (0.160 g/ml)	0.73	4.4		0.0333
2-Methyl naphthalene (0.178 g/ml)	0.49	2.7		0.0203

[a] Ref. 92; P in g × mm/(m² × d) and D in cm²/s.

TABLE 5. PERMEABILITY COEFFICIENTS OF VARIOUS ORGANIC COMPOUNDS THROUGH HIGH-DENSITY POLY(ETHYLENE) AND POLY(PROPYLENE)[a]

	High-density poly(ethylene)		Poly(propylene)	
Permeant	22.8°C	48.9°C	22.8°C	48.9°C
Acetic acid	0.39	2.36		
Acetone	0.67	3.9	0.28	8.84
n-Butyl ether	6.6	43.6		
Carbon tetrachloride	32	260	181	1100
n-Decane	7.3	39	22	153
Dipentene	7.6	54	29.5	161
Ethyl acetate	1.7	18.5	5.8	55
Methanol	0.1	2.0	0.08	1.5
Toluene	39	196	142	770

[a] Ref. 62; P in g × mm/(m² × d).

TABLE 6. PERMEABILITY COEFFICIENTS OF VARIOUS ORGANIC COMPOUNDS THROUGH IRRADIATION CROSSLINKED LOW-DENSITY POLY(ETHYLENE)[a]

	Poly(ethylene) unirradiated (crystallinity 58%)					Poly(ethylene) irradiated with 260 kGy (electron irradiation)				
Permeant	T (°C)	P $\left(\frac{g \times mm}{m^2 \times d}\right)$	Temp. range (°C)	P_0 $\left(\frac{g \times mm}{m^2 \times d}\right)$	E_P $\left(\frac{kJ}{mol}\right)$	T (°C)	P $\left(\frac{g \times mm}{m^2 \times d}\right)$	Temp. range (°C)	P_0 $\left(\frac{g \times mm}{m^2 \times d}\right)$	E_P $\left(\frac{kJ}{mol}\right)$
Acetic acid	21.1	1.27	20–75	1.108×10^{13}	72.9	21.1	1.55	20–75	2.783×10^{12}	68.8
Amyl acetate	21.1	3.80	20–75	2.016×10^{15}	83.0	21.1	5.03	20–75	3.269×10^{14}	77.9
Aniline	21.1	0.70	20–75	6.088×10^{14}	84.0	21.1	0.81	20–75	2.315×10^{14}	81.1
Benzaldehyde	21.1	2.76	20–75	6.375×10^{14}	80.7	21.1	3.36	20–75	4.513×10^{14}	79.6
Benzene	21.1	181	20–75	1.083×10^{13}	60.8	21.1	207	20–75	1.677×10^{12}	55.2

Permeant	Poly(ethylene) unirradiated (crystallinity 58%)					Poly(ethylene) irradiated with 260 kGy (electron irradiation)				
	T (°C)	P $\left(\frac{g \times mm}{m^2 \times d}\right)$	Temp. range (°C)	P_0 $\left(\frac{g \times mm}{m^2 \times d}\right)$	E_P $\left(\frac{kJ}{mol}\right)$	T (°C)	P $\left(\frac{g \times mm}{m^2 \times d}\right)$	Temp. range (°C)	P_0 $\left(\frac{g \times mm}{m^2 \times d}\right)$	E_P $\left(\frac{kJ}{mol}\right)$
Dibutyl ether	21.1	37.1	20–75	3.051×10^{13}	67.0	21.1	45.2	20–75	7.153×10^{12}	63.1
Ethyl acetate	21.1	9.39	20–75	6.831×10^{14}	78.2	21.1	11.4	20–75	1.797×10^{14}	74.4
n-Heptane	21.1	110	20–75	1.756×10^{12}	57.5	21.1	120	20–75	5.55×10^{11}	54.4
Methyl alcohol	21.1	0.43	20–75	1.160×10^{13}	75.7	21.1	0.49	20–75	7.843×10^{12}	74.0
Methyl ethyl ketone	21.1	5.03	20–75	3.195×10^{14}	77.7	21.1	5.86	20–75	4.310×10^{14}	78.1
n-Propyl alcohol	21.1	0.20	20–75	3.195×10^{15}	91.2	21.1	0.36	20–75	1.639×10^{14}	82.6
o-Xylene	21.1	116	20–75	8.599×10^{13}	67.0	21.1	120	20–75	6.676×10^{12}	60.4

[a] Ref. 63.

TABLE 7. PERMEABILITY COEFFICIENTS OF GASES THROUGH IRRADIATION CROSSLINKED LOW-DENSITY POLY(ETHYLENE)[a]

Gas	Irradiation Dose[b] (kGy)	T (°C)	D ($\times 10^6$) (cm^2/s)	P ($\times 10^{13}$) ($cm^3 \times cm/cm^2 \times s \times Pa$)	E_D (kJ/mol)	E_P (kJ/mol)
He	0	40	3.5	82.5	20.7	26.0
	200	40			22.5	27.7
	500	40		78.8	26.3	28.1
N_2	0	40	0.52	18.0		28.1
	100	40		17.3		
	200	40		15.0		30.1
	500	40		12.8		31.3
CH_4	0	40	0.35	47.3	28.5	40.3
	200	40	0.33	44.3	30.1	40.7
	500	40	0.31	36.8	33.5	41.1
C_2H_6	0	40	0.18	82.5	34.5	41.0
	200	40	0.16	78.0	37.6	43.4
	500	40	0.15	69.0	39.8	45.9
C_3H_8	0	40	0.05	142	50.0	45.3
	200	40	0.04	112	54.7	48.1
	500	40	0.03	90	55.6	50.3

[a] Ref. 71.
[b] Films were irradiated in acetylene atmosphere; dose rate was 6.15 kGy/h.

TABLE 8. PERMEABILITY COEFFICIENTS OF CHEMICALLY CROSSLINKED POLY(OXYPROPYLENE)[a]

Polymer	Molecular mass of poly(oxypropylene) between crosslinks	Permeability coefficient for 23°C ($\times 10^{13}$)	
		H_2	CO
Poly(oxypropylene), crosslinked with stoichiometric quantities of tris (*p*-isocyanatophenylthiophosphate) (Desmodur RF)	425	2.86	0.0608
	725	7.26	0.65
	1025	18.80	2.93
	2000	28.73	7.65
	3000	44.10	12.60

[a] Refs. 69 and 70; P in $cm^3 \times cm/(cm^2 \times s \times Pa)$.

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TABLE 9. PERMEABILITY COEFFICIENTS OF GASES THROUGH VARIOUS ELASTOMERS[a]

Elastomer		Permeation coefficient									
		N_2		O_2		CO_2		H_2		He	
		20°C	80°C	20°C	80°C	20°C	80°C	20°C	80°C	20°C	80°C
NBR	shore 70	0.172	5.39	0.639	10.8	6.11	61.1	3.25	27.2	2.56	25.6
NBR	shore 60	0.161	6.11	0.75	16.1	6.11	83.3	3.75	47.2	2.33	33.1
NBR	shore 50	0.231	6.39	0.722	13.6	4.08	28.6	4.58	38.3	2.31	20.6
NEM	shore 70	0.275	4.78								
SBR	shore 52	0.583	10.8	2.14	24.2	13.1	77.8	6.67	48.6	4.31	28.9
NR	shore 66	1.72	33.9	5.06	55.6	35.6	239	9.17	86.1	4.53	34.7
CR	shore 42	0.611	12.8	1.31	30.6	6.94	75.0	5.00	61.1	3.42	26.4
VMQ	shore 50	128	331	286	528			225	583	156	444
ECO	shore 65	0.169	3.25								
FKM	shore 70	0.172	4.75	0.306	7.22	3.53	61.1	2.58	41.7	4.89	48.1
ACM	shore 70	0.694	13.3	2.58	18.6	27.8	52.8	10.6	80.6	6.39	43.1
EPDM	shore 68	2.08	22.5	6.11	42.2	12.8	48.1	15.6	86.1	10.0	61.1
CIIR	shore 68	0.119	2.22	0.242	4.08	1.42	24.4	1.92	26.1	2.28	20.0
CSM	shore 70	0.144	3.92	0.494	7.78	3.47	16.7	2.94	20.8	2.97	26.4
AU	shore 94	0.286	4.72								

[a] Ref. 89; P in 10^{-13} cm^3 × cm/(cm^2 × s × Pa).

TABLE 10. PERMEABILITY COEFFICIENTS OF GASES THROUGH VARIOUS COMMERCIAL ELASTOMERS AT 35°C[a]

Elastomer tradename	He			H_2			D_2			O_2			N_2		
	P^b	D^c	S^d	P^b	D^c	S^d	P^b	D^c	S^d	P^b	D^c	S^d	P^b	D^c	S^d
Hypalon® 40	6.6	6.31	0.10	8.25	2.65	0.31	7.65	2.32	0.33	1.78	2.04	0.87	0.473	0.10	0.48
Hypalon® 45	8.63	10.3	0.08	11.7	4.84	0.24	11.2	5.35	0.21	3.15	4.86	0.65	0.960	0.34	0.28
Viton® E60	22.9	8.10	0.28	7.95	1.87	0.42	750	1.79	0.42	1.71	1.39	1.23	0.638	0.06	0.99
Viton® GF	32.9	18.3	0.18	16.4	3.45	0.47	15.7	2.78	0.56	3.15	3.86	0.82	1.60	0.13	1.24
Hydrin® 100 w/filler	1.13	0.73	0.15	1.05	0.23	0.47	1.20	0.18	0.67	0.233	0.03	0.73	0.113	0.008	1.43
Hydrin® 100	4.13	5.26	0.08	6.68	2.92	0.23	6.60	2.93	0.23	0.788	0.17	0.47	0.330	0.09	0.36
Kraton® G1652	34.0	19.9	0.17	51.5	12.3	0.42	48.2	10.1	0.48	20.0	1.92	1.04	7.35	1.18	0.62
Kraton® FG	34.7	22.4	0.15	50.2	11.6	0.43	44.9	9.68	0.46	18.4	1.43	1.29	6.47	0.98	0.66
Kraton® KG VTEOS	33.8	20.5	0.16	52.3	13.3	0.39	47.7	9.93	0.48	19.1	1.81	1.06	7.43	1.14	0.65

[a] Ref. 90.
[b] P in 10^{-13} cm^3 × cm/(cm^2 × s × Pa).
[c] D in 10^{-6} cm^2/s.
[d] S in 10^{-6} cm^3/(cm^3 × Pa).

TABLE 11. PERMEABILITY, DIFFUSION AND SOLUBILITY COEFFICIENTS OF ALKANES THROUGH SANTOPRENE™ (BLEND OF ETHYLENE–PROPYLENE COPOLYMER AND ISOTACTIC POLY(PROPYLENE))[a]

Alkane	T^b	P^c	D^d	S^e	Temp. range	E_P[f]	E_D	E_S
n-Pentane	25	1.51	6.07	0.25				
n-Hexane	25	1.11	2.88	0.39	25–70	13.86	12.87	0.99
	40	1.48	3.83	0.39				
	55	1.85	4.63	0.40				
n-Heptane	25	1.39	3.22	0.43	25–70	8.64	8.23	0.41
	40	1.67	3.74	0.45				
	55	2.13	4.72	0.45				
	70	2.12	4.82	0.44				
n-Octane	25	1.03	1.83	0.56	25–70	14.10	12.89	1.21
	40	1.38	2.36	0.58				
	55	1.73	2.88	0.60				
	70	2.18	3.64	0.60				

Alkane	T^b	P^c	D^d	S^e	Temp. range	$E_P{}^f$	E_D	E_S
n-Nonane	25	0.82	1.69	0.49	25–70	16.65	15.12	1.53
	40	1.13	2.23	0.51				
	55	1.50	2.87	0.52				
	70	1.99	3.79	0.53				
n-Decane	25	0.67	1.34	0.50	25–70	14.86	12.67	2.19
	40	0.87	1.78	0.49				
	55	1.13	2.10	0.54				
	70	1.47	2.67	0.55				
n-Dodecane	25	0.37	0.78	0.47	25–70	20.80	18.30	2.51
	40	0.52	1.04	0.50				
	55	0.76	1.44	0.53				
	70	1.10	2.05	0.54				
n-Tetradecane	25	0.28	0.62	0.45	25–70	19.48	17.41	2.07
	40	0.34	0.69	0.49				
	55	0.55	1.08	0.51				
	70	0.75	1.49	0.50				
n-Hexadecane	25	0.17	0.34	0.50	25–70	20.04	19.43	0.61
	40	0.27	0.55	0.49				
	55	0.38	0.74	0.51				
	70	0.49	0.97	0.51				
2,2,4-Trimethylpentane	25	0.56	1.66	0.34	25–70	12.67	11.19	1.48
	40	0.76	2.07	0.37				
	55	0.89	2.40	0.37				
	70	1.12	3.05	0.37				
Cyclohexane	25	0.85	0.90	0.94	25–70	17.43	14.70	2.74
	40	1.27	1.27	1.00				
	55	1.82	1.74	1.05				
	70	2.08	1.90	1.09				
1,2,3,4-Tetrahydronaphthalene	25	0.49	0.56	0.88	25–70	17.81	13.69	4.12
	40	0.74	0.80	0.93				
	55	0.92	0.93	0.99				
	70	1.29	1.19	1.08				

[a] Ref. 93.
[b] T and Temp. range in °C.
[c] P in 10^{-6} cm^3 × cm/(cm^2 × s).
[d] D in 10^{-6} cm^2/s.
[e] S in cm^3/cm^3.
[f] E_P, E_D and E_S in kJ/mol.

TABLE 12. PERMEABILITY, DIFFUSION AND SOLUBILITY COEFFICIENTS OF ESTERS THROUGH POLY(EPICHLOROHYDRIN) (ECO)[a]

Alkane	T^b	P^c	D^d	S^e	Temp. range	$E_P{}^f$	E_D	E_S
Methyl acetate	25	0.998	0.712	1.40	25–50	16.25	18.21	−1.94
	40	1.450	1.069	1.36				
	50	1.640	1.246	1.32				
(Ethyl acetate)	25	0.765	0.561	1.36	25–60	12.48	14.78	−2.30
	44	0.985	0.779	1.26				
	60	1.304	1.052	1.24				
(*n*-Butyl acetate)	25	0.346	0.295	1.17	25–60	14.34	15.02	−0.68
	44	0.480	0.418	1.15				
	60	0.636	0.559	1.14				
(Iso-amyl acetate)	25	0.317	0.267	1.19	25–60	15.99	17.34	−1.35
	44	0.486	0.420	1.16				
	60	0.621	0.554	1.12				
(Methyl salicylate)	25	0.146	0.071	2.06	25–60	31.15	31.15	0.10
	44	0.327	0.155	2.11				
	60	0.546	0.263	2.08				

[a] Ref. 97.
[b] T and Temp. range in °C.
[c] P in 10^{-6} cm^3 × cm/(cm^2 × s).
[d] D in 10^{-6} cm^2/s.
[e] S in cm^3/cm^3.
[f] E_P, E_D and E_S in kJ/mol.

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D. REFERENCES

1. R. Ash, R. M. Barrer, D. G. Palmer, Polymer, **11**, 421 (1970).
2. R. M. Barrer, H. T. Chio, J. Polym. Sci. C, **10**, 111 (1966).
3. J. A. Barrie, B. Plott, Polymer, **4**, 303 (1963).
4. W. W. Brandt, J. Polym. Sci., **41**, 415 (1959).
5. V. L. Braunisch, H. Lenhart, Kolloid-Z., **177**, 24 (1961).
6. J. Crank, C. Robinson, Proc. Roy. Soc. (London) A, **204**, 549 (1951).
7. P. M. Hauser, A. D. McLaren, Ind. Eng. Chem. Int. Ed., **40**, 112 (1948).
8. W. Heilman, V. Tammela, J. A. Meyer, V. Stannett, M. Szwarc, Ind. Eng. Chem., **48**, 821 (1956).
9. P. Y. Hsieh, J. Appl. Polym. Sci., **7**, 1743 (1963).
10. Y. Ito, Kobunshi Kagaku, **18**, 124 (1961).
11. Y. Ito, Kobunshi Kagaku, **18**, 158 (1961).
12. Y. Iyengar, J. Poly. Sci. B, **3**, 663 (1965).
13. D. Jeschke, Ph. D. Thesis, Mainz, 1960.
14. P. Mears, J. Am. Chem. Soc., **76**, 3415 (1954).
15. P. Mears, Trans. Faraday Soc., **53**, 101 (1957).
16. A. S. Michaels, H. J. Bixler, J. Polym. Sci., **50**, 413 (1961).
17. A. S. Michaels, W. R. Vieth, J. A. Barrie, J. Appl. Phys., **34**, 1 (1963).
18. A. S. Michaels, W. R. Vieth, J. A. Barrie, J. Appl. Phys., **34**, 13 (1963).
19. A. W. Myers, V. Stannett, M. Szwarc, J. Polym. Sci., **35**, 285 (1959).
20. A. W. Myers, V. Tammela, V. Stannett, M. Szwarc, Mod. Plastics, **37** (10), 139 (1960).
21. A. W. Myers, J. A. Meyer, C. E. Rogers, V. Stannett, M. Szwarc, Tappi, **44**, 58 (1961).
22. F. J. Norton, J. Appl. Polym. Sci., **7**, 1649 (1963).
23. R. A. Pasternak, M. V. Christenson, J. Heller, Macromolecules, **3**, 366 (1970).
24. R. A. Pasternak, G. L. Burns, J. Heller, Macromolecules, **4**, 470 (1971).
25. D. R. Paul, A. T. DiBenedetto, J. Polym. Sci. C, **10**, 17 (1965).
26. R. W. Roberts, K. Kammermeyer, J. Appl. Polym. Sci. **7**, 2183 (1963).
27. G. Rust, F. Herrero, Materialprüfung, **11** (5), 166 (1969).
28. V. L. Simril, A. Hershberger, Modern Plastics, **27** (11), 95 (1950).
29. V. Stannett, J. L. Williams, J. Polym. Sci. C, **10**, 45 (1966).
30. V. Stannett, North Carolina State University unpublished data.
31. R. L. Taylor, D. B. Hermann, A. R. Kemp, Ind. Eng. Chem. Int. Ed., **28**, 1255 (1936).
32. B. P. Tikhomirov, H. B. Hopfenberg, V. T. Stannett, J. L. Williams, Makromol. Chem., **118**, 177 (1968).
33. G. J. Amerongen, J. Appl. Phys., **17**, 972 (1946).
34. G. J. Amerongen, J. Polym. Sci., **2**, 381 (1947).
35. G. J. Amerongen, J. Polym. Sci., **5**, 307 (1950).
36. G. J. Amerongen, Rubber Chem. Technol., **37**, 1065 (1964).
37. R. Waack, N. H. Alex, H. L. Frisch, V. Stannett, M. Szwarc, Ind. Eng. Chem., **47**, 2524 (1955).
38. J. L. Williams, V. Stannett, J. Appl. Polym. Sci., **14**, 1949 (1970).
39. H. Yasuda, V. Stannett, J. Polym. Sci., **57**, 907 (1962).
40. H. Yasuda, V. Stannett, J. Macromol. Sci. B, **3** (4), 589 (1969).
41. H. Yasuda, Kj. Rosengren, J. Appl. Polym. Sci., **14**, 2839 (1970).
42. M. Salame, J. Polym. Sci. Symp., **41**, 1 (1973).
43. H. Platenius, Modern Packaging, **17**, 2103 (1943).
44. L. Phan Thuy, A. Lukaschov, J. Springer, Colloid Polym. Sci., **261**, 973 (1983).
45. M. Sefcik, J. Schaefer, F. May, D. Raucher, J. Polym. Sci., **21**, 1041 (1983).
46. N. Minoura, S. Tani, T. Nakagawa, J. Polym. Sci., **22**, 833 (1978).
47. S. Sczi, J. Springer, Colloid Polym. Sci., **259**, 1170 (1981).
48. E. Sacher, J. Polym. Sci., **28**, 1535 (1983).
49. H. Miyake, M. Matsuyama, K. Ashida, K. Watanabe, J. Vac. Sci. Technol. A-1, **3**, 1447 (1983).
50. P. T. Delassus, D. J. Grieser, J. Vinyl Technol. **2** (3), 195 (1980).
51. W. H. Yang, V. F. Smolen, N. A. Peppas, J. Membr. Sci., **9**, 53 (1981).
52. P. E. Cassidy, T. M. Aminabhavi, J. C. Brunson, Rubber Chem. Technol., **56**, 357 (1983).
53. N. Y. Yan, R. M. Felder, W. J. Koros, J. Polym. Sci., **25**, 1755 (1980).
54. H. Fitz, Kunststoffe, **70**, 27 (1980).
55. T. Duncan, W. J. Koros, R. M. Felder, J. Polym. Sci., **28**, 209 (1983).
56. E. Sacher, J. R. Susko, J. Polym. Sci., **27**, 3893 (1982).
57. D. H. W. Carstens, E. P. Ehart, J. Polym. Sci., **29**, 261 (1984).
58. H. Kenbishi, Int. Polym. Sci. Technol., **8** (4), (1981).
59. H. Schuch, Kautschuk, Gummi, Kunststoffe, **33** (9), (1980).
60. R. Light, R. Seymour, Soc. Plast. Eng., **29**, 417 (1983).
61. H. Braunisch, B. Hoffmann, Kältetechnik, **13** (2), 59 (1961).
62. M. Solame, S. P. E. Trans., **1**, 153 (1961).
63. H. Bent, J. Polym. Sci., **24**, 387 (1957).
64. J. Flynn, Polymer, **23**, 1325 (1982).
65. T. Hirose, K. Mizoguchi, Y. Kamiya, J. Polym. Sci., **30**, 401 (1985).
66. E. Sacher, J. Susko, J. Polym. Sci., **24**, 1997 (1979).
67. W. Koros, J. Wang, R. Felder, J. Polym. Sci., **26**, 2805 (1981).
68. E. Sacher, R. Susko, J. Polym. Sci., **23**, 2355 (1979).
69. A. L. Andrady, M. D. Sefcik, J. Polym. Sci., **21**, 2453 (1983).
70. A. L. Andrady, M. D. Sefcik, J. Polym. Sci., **22**, 237 (1984).
71. R. W. Macdonald, R. Y. M. Huang, J. Polym. Sci., **26**, 2239 (1981).
72. L. H. Wang, R. S. Porter, J. Polym. Sci., **22**, 1645 (1984).
73. S. Pauly, HOECHST AG, Werk Kalle, unpublished data.
74. P. S. Holden, G. A. J. Orchard, I. M. Ward, J. Polym. Sci., **23**, 709 (1985).

75. M. J. El-Hibri, D. R. Paul, J. Appl. Polym. Sci., **30**, 3649 (1985).
76. N. Cao, M. Pegoraro, F. Bianchi, L. Di Landro, L. Zanderighi, J. Appl. Polym. Sci., **48**, 1831 (1993).
77. J. S. Chiou, J. W. Barlow, D. R. Paul, J. Appl. Polym. Sci., **30**, 1173 (1985).
78. M. J. El-Hibri, D. R. Paul, J. Appl. Polym. Sci., **31**, 2533 (1986).
79. E. Casur, T. G. Smith, J. Appl. Polym. Sci., **31**, 1441 (1986).
80. A. Phatak, C. M. Burns, R. Y. M. Huang, J. Appl. Polym. Sci., **34**, 1835 (1987).
81. T. Hirose, K. Mizoguchi, Y. Kamiya, J. Appl. Polym. Sci., **35**, 517 (1988).
82. S. A. Stern, B. D. Bhide, J. Appl. Polym. Sci., **38**, 2131 (1989).
83. W.-J. Chen, C. R. Martin, J. Membr. Sci., **95**, 51 (1994).
84. A. Tanioka, N. Fukushima, K. Hasegawa, K. Miyasaka, J. Appl. Polym. Sci., **54**, 219 (1994).
85. M. S. Lerma, K. Iwamoto, M. Seno, J. Appl. Polym. Sci., **33**, 625 (1987).
86. T. Hirose, K. Mizoguchi, K. Terada, J. Appl. Polym. Sci., **58**, 1031 (1995).
87. J. Li, K. Nagai, T. Nakagawa, S. Wang, J. Appl. Polym. Sci., **58**, 1455 (1995).
88. J. S. Chiou, D. R. Paul, J. Appl. Polym. Sci., **34**, 1037 (1987).
89. W. Beckmann, Kautschuk, Gummi, Kunststoffe, **44**, 323 (1991).
90. M. W. Fitch, W. J. Koros, R. L. Nolen, J. R. Carnes, J. Appl. Polym. Sci., **47**, 1033 (1993).
91. T. Tsuchida, Y. Arimatsu, K. Yoda, Polym. Comm., **29**, 246 (1988).
92. L. N. Britton, R. B. Ashman, T. M. Aminabhavi, P. E. Cassidy, J. Appl. Polym. Sci., **38**, 227 (1989).
93. T. M. Aminabhavi, H. T. S. Phayde, J. Appl. Polym. Sci., **55**, 17 (1995).
94. S. A. Stern, V. M. Shah, B. J. Hardy, J. Polym. Sci. B, **25**, 1263 (1987).
95. T. Zhang, M. H. Litt, C. E. Rogers, J. Polym. Sci. B, **32**, 1671 (1994).
96. J. M. Mohr, D. R. Paul, J. Appl. Polym. Sci., **42**, 1711 (1991).
97. T. Aminabhavi, R. S. Munnolli, J. Polym. Eng., **14**, 53 (1995).
98. H. Fu, L. Jia, Chin. J. Polym. Sci., **13**, 189 (1995).
99. J. A. Webb, D. I. Bower, I. M. Ward, P. T. Cardew, J. Polym. Sci. B, **31**, 743 (1993).
100. K. Gosal, B. D. Freeman, Polym. Advanced Techn., **5**, 673 (1994).
101. G. Golemme, E. Drioli, F. Lufrano, Polym. Sci., **36**, 1647 (1994).
102. G. Mensitieri, A. Apicella, J. M. Kenny, L. Nicolais, J. Appl. Polym. Sci., **37**, 381 (1989).
103. M. A. Grayson, C. J. Wolf, J. Polym. Sci. B, **25**, 31 (1987).
104. G. Xuesong, L. Fengcai, Polymer, **36**, 1035 (1995).
105. G. F. Sykes, A. K. St. Clair, J. Appl. Polym. Sci., **32**, 3725 (1986).

Refractive Indices of Polymers

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A. INTRODUCTION

The refractive indices of polymers provide fundamental physical property information that can be used in characterization, processing, and usage analyses. In-depth description of refractive indices based on classic electromagnetic theory can be found in basic texts (1,2). The purpose of this monograph is to provide a convenient reference of values and some general remarks in approximating and utilizing refractive indices of polymers.

The refractive index, reported as the velocity of light in a vacuum relative to the velocity in the polymer, depends on the wavelength of light. All values reported in this chapter refer to the sodium spectral line $\lambda = 5893$Å, referred to as n_D in this literature. However, for simplicity, the subscript D is omitted here in the subsequent discussion and tables. In general, it should be kept in mind that for any polymer, the refractive index represents an average property of its aggregate state (3,4). Orientation, conformation, as well as phase changes may all contribute to variations in the index of refraction of a polymer with a given chemical structure. Thus, the refractive index may not be expected to be equal along all axes of the molecule (5). Consequently, the measured refractive index is an average value, with contributions from the refractions of the molecule's different chemical bonds. It can be shown from classic electromagnetic theory that the square of the refractive index (n^2) is the appropriate form to use in molecular descriptions of refractive indices since it is related to the dielectric tensor, having a magnitude as well as a directional description (1,4). However, since $1.3 < n < 1.7$ for most polymers, correlations proposed for n^2 almost always find a simplified version with n (3,4).

B. MOLAR REFRACTION

The molar refraction of a substance, a product of its refractive index, molecular weight, and density, may be assumed to be a fundamental physical property of materials (1). Furthermore, the molar refraction may be treated (as established by Denbigh) as the sum of the bond refractions that make up the molecule (6). These two postulates make possible, the calculation of the refractive indices of polymers almost from first principles (7,8). Two useful relations for the molar refraction, R, are provided here (1):
The Lorentz–Lorentz equation

$$R = \frac{n^2 - 1}{n^2 + 2}\frac{M}{\rho} \tag{B1}$$

The Gladstone–Dale equation

$$R = (n - 1)\frac{M}{\rho} \tag{B2}$$

where ρ is the density of the polymer and M its molecular weight. The molar refraction may be assumed to be the sum of the refractions of all the bonds or groups that make up the polymer system, i.e., $R = \sum_i R_i$ (6,7). Chemical group refractivities that can be used in the calculation of refractive index of polymers have been recently summarized by Van Krevelen (8). For example, for a homopolymer whose repeat monomer unit refractivity, R_M, is known, with $R = \mathrm{DP}\,R_M$ (where DP is the degree of polymerization) and $M = \mathrm{DP}\,M_M$ (M_M = molecular weight of the monomer unit), then the refractive index of the polymer by the Gladstone–Dale expression (Eq. B2) becomes

$$\mathrm{DP}\,R_M = (n - 1)\frac{\mathrm{DP}\,M_M}{\rho} \quad \text{or} \quad n = 1 + \frac{M_m}{R_m}\rho \tag{B3}$$

This may be calculated only from a knowledge of the monomer unit molar refraction and molecular weight. The

use of bond or group refractions in the calculation of refractive indices of polymers has been demonstrated to be in good agreement with the measured values for homogeneous, unoriented polymers (8).

C. REFRACTIVE INDICES OF HETEROGENEOUS POLYMERS

For polymeric systems whose phase dimensions are below the wavelength of the incident light, the application of the Maxwell relations is valid and the refractive index of the polymer may be expressed in terms of the dielectric constants of the constituent phases (1). The dielectric constant, k, is equivalent to the square of the refractive index ($k = n^2$), and it can be demonstrated that an upper bound of the refractive index of the polymer may be obtained as the sum of dielectric constants of the constituent phases (4,5):

$$n^2 = \sum_i v_i n_i^2 \tag{C1}$$

where n_i and $v_i (\sum v_i = 1)$ are the refractive indices and volume fractions of the constituent phases of the components that make up the polymer. Similarly, the lower bound of a polymer's refractive index may be computed as the sum of the inverse of the dielectric constants of the constituent phases (4):

$$\frac{1}{n^2} = \sum_i \frac{v_i}{n_i^2} \tag{C2}$$

For most heterogeneous polymers, however, whose constituent refractive indices differ by less than 0.2, i.e., $(n_i - n_j \leq 0.2)$, the refractive index of the polymer may be computed by the Gladstone–Dale relation, assuming additivity of the refractive indices of the constituent phases (3,4):

$$n = \sum_i v_i n_i \tag{C3}$$

The calculation methods described above may be used for multiphase polymers such as blends, block or graft copolymers, composites, and semicrystalline polymers.

D. OPTICAL ANISOTROPY*

Small as well as long polymer molecules exhibit different refractive indices parallel and perpendicular to the molecular chain axis. Thus, all polymers should be viewed as intrinsically anisotropic. The degree of anisotropy exhibited in bulk property behavior will depend on the molecular orientation imposed by deformation during processing or testing. Only in its unoriented state can a polymer be characterized by a single refractive index, since its optical properties will be identical in all directions. Consequently, optical anisotropy is one of the simplest and often used methods of studying orientation in polymers (5). By polarizing light in a specific direction, the refractive index of a polymer can be measured in that direction. Consequently, all three principal refractive indices of a polymer, n_x, n_y, n_z, corresponding to an x, y, and z direction, can be measured. For example, a refractometer fitted with a polarized eye piece has been successfully used to measure refractive indices along the major axes of polymer films (9). The average refractive indices of a polymer may be computed from the refractive indices along the axes of the polymer:

$$n = (n_x + n_y + n_z)/3 \tag{D1}$$

For the unoriented (isotropic) condition ($n_x = n_y = n_z$), a single value of the refractive index provides a unique optical description ($n = n_x = n_y = n_z$). For oriented polymers, the difference in refractive indices along two orthogonal directions can be used as the measure of optical anisotropy and, consequently, orientation of the polymer. The difference in refractive indices is defined as the birefringence or double refraction. According to Stein (5), there are three birefringences, Δ_{ij}'s that may be defined for a cartesian coordinate system, with two of them being independent. They are

$$\Delta n_{xy} = n_x - n_y \tag{D2}$$

$$\Delta n_{xz} = n_x - n_z \tag{D3}$$

$$\Delta n_{yz} = n_y - n_z = \Delta n_{xz} - \Delta n_{xy} \tag{D4}$$

For a polymer exhibiting transverse isotropic symmetry, as in the case of uniaxial orientation (i.e., deformation along one axis), a single birefringence is enough to characterize orientation and anisotropy. For example, for a polymer oriented uniaxially along the x direction, $\Delta n_{xy} = \Delta n_{xz}$ and $\Delta n_{yz} = 0$. For a fully oriented polymer, where the molecular chain axis is aligned with the deformation axis (perfect orientation) a limiting value of birefringence is reached, which is defined as the intrinsic birefringence of the polymer (10,11) and denoted as

$$\Delta n_{xz} = n_{xy}^0 f \tag{D5}$$

where f is the Hermans orientation function

$$f = \frac{\overline{3\cos^2\theta - 1}}{2} \tag{D6}$$

defined on the average angle θ that the molecular chain axis makes with the deformation axis. The Hermans orientation function (D6), $-1/2 \leq f \leq 1$ assumes the value of $-1/2$ when the molecular chain axis is perpendicular to the deformation axis; 0 for random orientation (isotropy); and -1 for perfect alignment of the chains with the deforma-

*See also corresponding table in this Handbook.

tion axis. The use of the refractive index in the calculation of birefringence, rather than the difference of some function of the dielectric constant (n^2 or $1/n^2$), is an approximation to optical anisotropy. Thus, if the birefringence is used as a measure of orientation, as was proposed by Stein, form effects should be considered (5,7,11)

$$\Delta n_{xz} = n_{xy} f + \Delta_{\text{form}} \tag{D7}$$

The importance of form may be estimated from bond polarizabilities for a given polymer (4,7). Thus, to characterize completely the optical properties of a given polymer, its average refractive index as well as the intrinsic and form birefringences must be specified. To date, very few polymers have been optically characterized in such detail. However, as more refined and specific performance is demanded from polymers, this information is forthcoming. In order to demonstrate the usefulness of complete characterization and to summarize the data available till date, an example for semicrystalline polymers is provided, along with a table showing reported values (for a few polymers) of average refractive indices and intrinsic birefringences.

E. APPLICATIONS

Using Eq. (B2), the unoriented refractive index of a semicrystalline polymer may be calculated from a knowledge of its intrinsic crystalline and noncrystalline densities and refractive indices. First, the volume fraction crystallinity, v_{cr}, is calculated from the densities:

$$v_{cr} = \frac{\rho - \rho_{nc}}{\rho_{cr} - \rho_{nc}} \tag{E1}$$

where ρ is the bulk measured density, and ρ_{cr} and ρ_{nc} are the crystalline and noncrystalline densities. Then the average refractive index is

$$n = v_{cr} n_{cr} + (1 - v_{cr}) n_{nc} \tag{E2}$$

From Stein's equation of birefringence additivity (neglecting form birefringence) (5,11)

$$\Delta n_{xy} = v_{cr}\, \Delta n^0_{xy_{cr}} f_{cr} + (1 - v_{cr}) \Delta n^0_{xy_{nc}} f_{nc} \tag{E3}$$

where $\Delta n^0_{xy_{cr}}$ and $\Delta n^0_{xy_{nc}}$ are the crystalline and noncrystalline intrinsic birefringences, and f_{cr} and f_{nc} are the corresponding crystalline and noncrystalline Hermans orientation functions.

If the intrinsic properties of a semicrystalline polymer are known, Eq. (E2) can be used to calculate the volume fraction crystallinity by simply measuring the polymer's refractive index, while Eq. (E3) can be used to obtain the value of the noncrystalline orientation function by measuring birefringence and crystalline orientation by an independent technique [e.g., wide angle X-ray (5)]. For a single-phase polymer like polystyrene, the orientation function can be directly calculated from the measured birefringence value and the known intrinsic birefringence:

$$f = \frac{\Delta n_{xy}}{\Delta n^0_{xy}} \tag{E4}$$

Once these quantities are known, it becomes possible to completely characterize and predict the optical anisotropy of a given polymer from its unoriented to its oriented state (3,4).

Table 1 is provided to demonstrate that such basic optical parameters may be obtained from the literature (3,4,9,12–16). The recognition of polymers as anisotropic heterogeneous systems continues to expand and their refractive indices provide useful insight into their process–structure–property interrelations (18–20). It is hoped that continuous investigations in existing as well as new polymers will expand this compilation.

Finally, Table 2 provides average refractive indices of polymers arranged in order of increasing values, while Table 3 provides the same list in alphabetical order. Some additional values from the recent literature (21–23) have been added to the earlier compilation (17) to form the material for the present edition.

TABLE 1. INTRINSIC OPTICAL PROPERTIES OF SELECTED POLYMERS

Polymer	Crystalline density (ρ_{cr})	Noncrystalline density (ρ_{nc})	Phase average refractive indices		Intrinsic birefringence ($\times 10^3$)	
			n_{cr}	n_{nc}	Δn^0_{cr}	Δn^0_{nc}
Poly(propylene)	0.936	0.850	1.519–1.524	1.471	23–30	60–86
Poly(ethylene)	1.000	0.855	1.560–1.565	1.476–1.483	58–66	
Low orientation						120
High orientation						194
Poly(oxyethylene-oxyterephthaloyl)	1.455	1.335	1.639–1.643	1.570–1.577	220–230	220–275
Poly(styrene)	Amorphous	1.06	Amorphous	1.590–1.592	Amorphous	195

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TABLE 2. AVERAGE REFRACTIVE INDICES OF POLYMERS (IN ORDER OF INCREASING n)[a]

Polymer	n	T (°C)
Poly(tetrafluoroethylene-*co*-hexafluoropropylene)	1.338	
Poly(pentadecafluorooctyl acrylate)	1.339	25
Poly(tetrafluoro-3-(heptafluoropropoxy)propyl acrylate)	1.346	25
Poly(tetrafluoro-3-(pentafluoroethoxy)propyl acrylate)	1.348	25
Poly(tetrafluoroethylene)	1.35–1.38	
Poly(undecafluorohexyl acrylate)	1.356	
Poly(nonafluoropentyl acrylate)	1.360	25
Poly(tetrafluoro-3-(trifluoromethoxy)propyl acrylate)	1.360	25
Poly(pentafluorovinyl propionate)	1.364	25
Poly(heptafluorobutyl acrylate)	1.367	25
Poly(trifluorovinyl acetate)	1.375	25
Poly(octafluoropentyl acrylate)	1.380	25
Poly(pentafluoropropyl acrylate)	1.385	25
Poly(2-heptafluorobutoxy)ethyl acrylate)	1.390	25
Poly(2,2,3,4,4,4-hexafluorobutyl acrylate)	1.392	25
Poly(trifluoroethyl acrylate)	1.407	25
Poly(2-(1,1,2,2-tetrafluoroethoxy)ethyl acrylate)	1.412	25
Poly(trifluoroisopropyl methacrylate)	1.4177	20
Poly(2,2,2-trifluoro-1-methylethyl methacrylate)	1.4185	
Poly(2-trifluoroethoxy)ethyl acrylate	1.419	25
Poly(trifluorochloroethylene)	1.42–1.43	
Poly(vinylidene fluoride)	1.42	25
Poly(oxydimethylsilylene) (poly(dimethyl siloxane))	1.43	
Poly(trifluoroethyl methacrylate)	1.437	
Poly(oxypropylene)	1.4495	
Poly(vinyl isobutyl ether)	1.4507	30
Poly(vinyl ethyl ether)	1.4540	30
Poly(oxyethylene)	1.4563	30
Poly(vinyl butyl ether)	1.4563	30
Poly(vinyl pentyl ether)	1.4581	30
Poly(vinyl hexyl ether)	1.4591	30
Poly(4-methyl-1-pentene)	1.459–1.465	
Cellulose acetate butyrate	1.46–1.49	
Poly(4-fluoro-2-trifluoromethylstyrene)	1.46	
Poly(vinyl octyl ether)	1.4613	30
Poly(vinyl-2-ethylhexyl ether)	1.4626	30
Poly(vinyl decyl ether)	1.4628	30
Poly(2-methoxyethyl acrylate)	1.463	25
Poly(butyl acrylate)	1.4631	30
	1.466	20
Poly(*tert*-butyl methacrylate)	1.4638	20
Poly(vinyl dodecyl ether)	1.4640	30
Poly(3-ethoxypropyl acrylate)	1.465	25
Poly(oxy-1-oxopentamethylene)	1.465	50
Poly(vinyl propionate)	1.4665	20
Poly(vinyl acetate)	1.4665	20
Poly(vinyl methyl ether)	1.467	20
Poly(ethyl acrylate)	1.4685	20
Poly(ethylene-*co*-vinyl acetate) (80%–20% vinyl acetate)	1.47–1.50	
Cellulose propionate	1.47–1.49	
Cellulose acetate propionate	1.47	
Benzyl cellulose	1.47–1.58	
Phenol–formaldehyde resins	1.47–1.70	
Cellulose triacetate	1.47–1.48	
Poly(vinyl methyl ether) (isotactic)	1.4700	30
Poly(3-methoxypropyl acrylate)	1.471	25
Poly(2-ethoxyethyl acrylate)	1.471	25
Poly(methyl acrylate)	1.472–1.480	
Poly(isopropyl methacrylate)	1.4728	20
Poly(1-decene)	1.4730	
Poly(propylene) (atactic, density 0.8575 g/cm^3)	1.4735	20
Poly(vinyl *sec*-butyl ether) (isotactic)	1.4740	30
Poly(dodecyl methacrylate)	1.4740	20
Poly(oxyethyleneoxysuccinoyl) (poly(ethylene succinate))	1.4744	25
Poly(tetradecyl methacrylate)	1.4746	30

TABLE 2. *cont'd*

Polymer	*n*	*T* (°C)
Poly(ethylene-*co*-propylene) (EPR-rubber)	1.4748–1.48	
Poly(hexadecyl methacrylate)	1.4750	30
Poly(vinyl formate)	1.4757	20
Poly(2-fluoroethyl methacrylate)	1.4768	20
Poly(isobutyl methacrylate)	1.477	20
Ethyl cellulose	1.479	21
Poly(vinyl acetal)	1.48–1.50	
Cellulose acetate	1.48–1.50	
Cellulose acetate (39.8% acetyl content)	1.4700	25
Cellulose tripropionate	1.48–1.49	
Poly(oxymethylene)	1.48	20
Poly(vinyl butyral)	1.48–1.49	
Poly(*n*-hexyl methacrylate)	1.4813	20
Poly(*n*-butyl methacrylate)	1.483	20–25
Poly(ethylidene dimethacrylate)	1.4831	20
Poly(2-ethoxyethyl methacrylate)	1.4833	20
Poly(oxyethyleneoxymaleoyl) (poly(ethylene maleate))	1.4840	25
Poly(*n*-propyl methacrylate)	1.484	25
Poly(3,3,5-trimethylcyclohexyl methacrylate)	1.485	20
Poly(ethyl methacrylate)	1.485	20–25
Poly(2-nitro-2-methylpropyl methacrylate)	1.4868	20
Poly(triethylcarbinyl methacrylate)	1.4889	20
Poly(1,1-diethylpropyl methacrylate)	1.4889	20
Poly(methyl methacrylate)	1.4893	23
	1.490	20
Poly(2-decyl-1,3-butadiene)	1.4899	20.5
Poly(vinyl alcohol)	1.49–1.53	
	1.5214	20
Poly(ethyl glycolate methacrylate)	1.4903	20
Poly(3-methylcyclohexyl methacrylate)	1.4947	20
Poly(cyclohexyl α-ethoxyacrylate)	1.4969	20
Methyl cellulose (low viscosity)	1.497	25
Poly(4-methylcyclohexyl methacrylate)	1.4975	20
Poly(decamethylene glycol dimethacrylate)	1.4990	
Poly(urethanes)	1.5–1.6	
Poly(1,2-butadiene)	1.5000	20
Poly(vinyl formal)	1.50	
Poly(2-bromo-4-trifluoromethylstyrene)	1.5	25
Cellulose nitrate	1.5–1.514	
Poly(*sec*-butyl α-chloroacrylate)	1.500	25
Poly(2-heptyl-1,3-butadiene)	1.5000	
Poly(ethyl α-chloroacrylate)	1.502	25
Poly(2-isopropyl-1,3-butadiene)	1.5028	30
Poly(2-methylcyclohexyl methacrylate)	1.5028	20
Poly(propylene) (density 0.9075 g/cm^3)	1.5030	20
Poly(isobutene)	1.505–1.51	
Poly(bornyl methacrylate)	1.5059	20
Poly(2-*tert*-butyl-1,3-butadiene)	1.5060	24.6
Poly(ethylene glycol dimethacrylate)	1.5063	20
Poly(cyclohexyl methacrylate)	1.5066	20
Poly(cyclohexanediol-1,4-dimethacrylate)	1.5067	20
Butyl rubber(unvulcanized)	1.508	
Poly(tetrahydrofurfuryl methacrylate)	1.5096	20
Gutta percha (β)	1.509	
Poly(ethylene) ionomer	1.51	
Poly(oxyethylene) (high molecular weight)	1.51–1.54	
Poly(ethylene) (density 0.914 g/cm^3)	1.51	20
(density 0.94–0.945 g/cm^3)	1.52–1.53	20
(density 0.965 g/cm^3)	1.545	20
Poly(1-methylcyclohexyl methacrylate)	1.5111	20
Poly(2-hydroxyethyl methacrylate)	1.5119	20
Poly(vinyl chloroacetate)	1.512	25
Poly(butene) (isotactic)	1.5125	
Poly(vinyl methacrylate)	1.5129	20
Poly(*N*-butyl methacrylamide)	1.5135	20

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TABLE 2. *cont'd*

Polymer	n	T (°C)
Gutta percha (α)	1.514	50
Terpene resin	1.515	25
Poly(1,3-butadiene)	1.5154	25
Shellac	1.51	1.53
Poly(methyl α-chloroacrylate)	1.517	20
Poly(2-chloroethyl methacrylate)	1.517	20
Poly(2-diethylaminoethyl methacrylate)	1.5174	20
Poly(2-chlorocyclohexyl methacrylate)	1.5179	20
Poly(1,3-butadiene) (35% *cis*; 56% *trans*; 7% 1,2-content)	1.5180	
Natural rubber	1.519–1.52	
Poly(allyl methacrylate)	1.5196	20
Poly(vinyl chloride) + 40% dioctyl phthalate	1.52	
Poly(acrylonitrile)	1.52	
	1.5187	25
Poly(methacrylonitrile)	1.52	
Poly(1,3-butadiene) (high *cis*-type)	1.52	
Poly(butadiene-*co*-acrylonitrile)	1.52	
Poly(methyl isopropenyl ketone)	1.5200	20
Poly(isoprene)	1.521	20
Poly(ester) resin, rigid (ca. 50% styrene)	1.523–1.54	
Poly((*N*-2-methoxyethyl)methacrylamide)	1.5246	20
Poly(2,3-dimethylbutadiene) (methyl rubber)	1.525	
Poly(vinyl chloride-*co*-vinyl acetate) (95/5–90/10)	1.525–1.535	
Poly(acrylic acid)	1.527	25
Poly(1,3-dichloropropyl methacrylate)	1.5270	20
Poly(2-chloro-1-(chloromethyl)ethyl methacrylate)	1.5270	20
Poly(acrolein)	1.529	
Poly(1-vinyl-2-pyrrolidone)	1.53	
Poly(*N*-methylmaleimide-*alt*-isobutene)	1.53	
Hydrochlorinated rubber	1.53–1.55	
Poly(imino(1-oxohexamethylene)) (nylon 6)	1.53	
Poly(iminoadipoyliminohexamethylene) (nylon 6,6)	1.53	
Poly(iminoadipoylimino tetramethylene) (nylon moulding 6,10)	1.53	
oriented fibres	1.515–1.565	
Poly(butadiene-*co*-styrene) (– 30% styrene) block copolymer	1.53	
Poly(cyclohexyl α-chloroacrylate)	1.532	25
Poly(2-chloroethyl α-chloroacrylate)	1.533	25
Poly(butadiene-*co*-styrene) (~75/25)	1.535	
Poly(2-aminoethyl methacrylate)	1.537	20
Poly(furfuryl methacrylate)	1.5381	
Proteins	1.539–1.541	
Poly(butylmercaptyl methacrylate)	1.5390	20
Poly(1-phenyl-*n*-amyl methacrylate)	1.5396	20
Poly(*N*-methyl-methacrylamide)	1.5398	20
Cellulose	1.54	
Poly(vinyl chloride)	1.54–1.55	
Urea–formaldehyde resin	1.54–1.56	
Poly(*sec*-butyl α-bromoacrylate)	1.542	25
Poly(cyclohexyl α-bromoacrylate)	1.542	25
Poly(2-bromoethyl methacrylate)	1.5426	20
Poly(dihydroabietic acid)	1.544	
Poly(abietic acid)	1.546	25
Poly(ethylmercaptyl methacrylate)	1.547	20
Poly(*N*-allyl methacrylamide)	1.5476	20
Poly(1-phenylethyl methacrylate)	1.5487	20
Poly(vinylfuran)	1.55	20
Poly(2-vinyltetrahydrofuran)	1.55	20
Poly(vinyl chloride) + 40% tricresyl phosphate	1.55	
Epoxy resins	1.55–1.60	
Poly(*p*-methoxybenzyl methacrylate)	1.552	20
Poly(isopropyl methacrylate)	1.552	
Poly(*p*-isopropylstyrene)	1.554	20
Poly(chloroprene)	1.554–1.558	
Poly(oxyethylene)-α-benzoate-ω-methacrylate)	1.555	20
Poly(*p,p*-xylylenyl dimethacrylate)	1.5559	20

TABLE 2. *cont'd*

Polymer	*n*	*T* (°C)
Poly(1-phenylallyl methacrylate)	1.5573	20
Poly(*p*-cyclohexylphenyl methacrylate)	1.5575	20
Poly(2-phenylethyl methacrylate)	1.5592	20
Poly(oxycarbonyloxy-1,4-phenylene-1-propylbutylidene-1,4-phenylene)	1.5602	
Poly(1-(*o*-chlorophenyl)ethyl methacrylate)	1.5624	20
Poly(styrene-*co*-maleic anhydride)	1.564	21
Poly(1-phenylcyclohexyl methacrylate)	1.5645	20
Poly(oxycarbonyloxy-1,4-phenylene-1,3-dimethylbutylidene-1,4-phenylene)	1.5671	
Poly(methyl α-bromoacrylate)	1.5672	20
Poly(benzyl methacrylate)	1.5680	20
Poly(2-(phenylsulfonyl)ethyl methacrylate)	1.5682	
Poly(*m*-cresyl methacrylate)	1.5683	20
Poly(styrene-*co*-acrylonitrile) (– 75/25)	1.57	
Poly(oxycarbonyloxy-1,4-phenyleneisobutylidene-1,4-phenylene)	1.5702	
Poly(*o*-methoxyphenyl methacrylate)	1.5705	20
Poly(phenyl methacrylate)	1.5706	20
Poly(*o*-cresyl methacrylate)	1.5707	20
Poly(diallyl phthalate)	1.572	20
Poly(2,3-dibromopropyl methacrylate)	1.5739	20
Poly(oxycarbonyloxy-1,4-phenylene-1-methyl-butylidene-1,4-phenylene)	1.5745	
Poly(oxy-2,6-dimethylphenylene)	1.575	
Poly(oxyethyleneoxyterephthaloyl) (amorphous) (poly(ethylene terephthalate))	1.5750	20
Poly(vinyl benzoate)	1.5775	20
Poly(oxycarbonyloxy-1,4-phenylenebutylidene-1,4-phenylene)	1.5792	
Poly(1,2-diphenylethyl methacrylate)	1.5816	20
Poly(*o*-chlorobenzyl methacrylate)	1.5823	20
Poly(oxycarbonyloxy-1,4-phenylene-*sec*-butylidene-1,4-phenylene)	1.5827	
Poly(oxypentaerythritoloxyphthaloyl)	1.584	20
Poly(*m*-nitrobenzyl methacrylate)	1.5845	20
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)	1.5850	
Poly(*N*-(2-phenylethyl)methacrylamide)	1.5857	20
Poly(4-methoxy-2-methylstyrene)	1.5868	20
Poly(*o*-methylstyrene)	1.5874	20
Poly(styrene)	1.59–1.592	20
Poly(oxycarbonyloxy-1,4-phenylenecyclohexylidene-1,4-phenylene)	1.5900	20
Poly(*o*-methoxystyrene)	1.5932	20
Poly(diphenylmethyl methacrylate)	1.5933	20
Poly(oxycarbonyloxy-1,4-phenyleneethylidene-1,4-phenylene)	1.5937	
Poly(*p*-bromophenyl methacrylate)	1.5964	20
Poly(*N*-benzyl methacrylamide)	1.5965	20
Poly(*p*-methoxystyrene)	1.5967	20
Hard rubber (32% S)	1.6	
Poly(vinylidene chloride)	1.60–1.63	
Poly(sulfides) (Thiokol)	1.6–1.7	
Poly(*o*-chlorodiphenylmethyl methacrylate)	1.6040	
Poly(oxycarbonyloxy-1,4-(2,6-dichloro)phenylene-isopropylidene-1,4-(2,6-dichloro)phenylene)	1.6056	
Poly(oxycarbonyloxybis(1,4-(3,5-dichlorophenylene)))	1.6056	
Poly(pentachlorophenyl methacrylate)	1.608	20
Poly(*o*-chlorostyrene)	1.6098	20
Poly(phenyl α-bromoacrylate)	1.612	25
Poly(*p*-divinylbenzene)	1.6150	20
Poly(*N*-vinylphthalimide)	1.6200	20
Poly(2,6-dichlorostyrene)	1.6248	20
Poly(β-naphthyl methacrylate)	1.6298	20
Poly(α-naphthyl carbinyl methacrylate)	1.63	
Poly(sulfone)	1.633	
Poly(2-vinylthiophene)	1.6376	20
Poly(α-naphthyl methacrylate)	1.6410	20
Poly(oxycarbonyloxy-1,4-phenylenediphenyl-methylene-1,4-phenylene)	1.6539	
Poly(vinyl phenyl sulfide)	1.6568	20
Butylphenol formaldehyde resin	1.66	
Urea–thiourea–formaldehyde resin	1.660	25
Poly(vinylnaphthalene)	1.6818	20

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TABLE 2. *cont'd*

Polymer	*n*	*T* (°C)
Poly(vinylcarbazole)	1.683	20
Naphthalene–formaldehyde resin	1.696	
Phenol–formaldehyde resin	1.70	
Poly(pentabromophenyl methacrylate)	1.71	

[a]Temperature in °C are listed when available. In general, n may be assumed to decrease with increasing temperature. Approximate change may be calculated as $dn/dT = -1 \times 10^{-4}$ to 2×10^{-4}/°C for glassy polymers and $dn/dT = -3 \times 10^{-4}$ to 5×10^{-4}/°C for amorphous unoriented polymers above the glass transition.

TABLE 3. AVERAGE REFRACTIVE INDICES OF POLYMERS (IN ALPHABETICAL ORDER)

Polymer	*n*	*T* (°C)
Benzyl cellulose	1.47–1.58	
Butylphenol formaldehyde resin	1.66	
Butyl rubber(unvulcanized)	1.508	
Cellulose	1.54	
Cellulose acetate	1.48–1.50	
Cellulose acetate (39.8% acetyl content)	1.4700	25
Cellulose acetate butyrate	1.46–1.49	
Cellulose acetate propionate	1.47–1.48	
Cellulose nitrate	1.5–1.514	
Cellulose propionate	1.47–1.49	
Cellulose triacetate	1.47–1.48	
Cellulose tripropionate	1.48–1.49	
Epoxy resins	1.55–1.60	
Ethyl cellulose	1.479	21
Gutta percha (α)	1.514	50
Gutta percha (β)	1.509	
Hard rubber (32% S)	1.6	
Hydrochlorinated rubber	1.53–1.55	
Methyl cellulose (low viscosity)	1.497	25
Naphthalene–formaldehyde resin	1.696	
Natural rubber	1.519–1.52	
Phenol–formaldehyde resin	1.70	
Phenol–formaldehyde resins	1.47–1.70	
Poly(abietic acid)	1.546	25
Poly(acrolein)	1.529	
Poly(acrylic acid)	1.527	25
Poly(acrylonitrile)	1.52	
	1.5187	25
Poly(allyl methacrylate)	1.5196	20
Poly(2-aminoethyl methacrylate)	1.537	20
Poly(*N*-allyl methacrylamide)	1.5476	20
Poly(*N*-benzyl methacrylamide)	1.5965	20
Poly(*n*-butyl methacrylate)	1.483	20–25
Poly(benzyl methacrylate)	1.5680	20
Poly(bornyl methacrylate)	1.5059	20
Poly(2-bromoethyl methacrylate)	1.5426	20
Poly(*p*-bromophenyl methacrylate)	1.5964	20
Poly(2-bromo-4-trifluoromethylstyrene)	1.5	25
Poly(1,2-butadiene)	1.5000	20
Poly(1,3-butadiene)	1.5154	25
Poly(1,3-butadiene) (high *cis*-type)	1.52	
Poly(1,3-butadiene) (35% *cis*; 56% *trans*; 7% 1,2-content)	1.5180	
Poly(butadiene-*co*-acrylonitrile)	1.52	
Poly(butadiene-*co*-styrene) (~75/25)	1.535	
Poly(butadiene-*co*-styrene) (– 30% styrene) block copolymer	1.53	
Poly(butene) (isotactic)	1.5125	
Poly(butyl acrylate)	1.4631	30
	1.466	20
Poly(*sec*-butyl α-bromoacrylate)	1.542	25
Poly(*sec*-butyl α-chloroacrylate)	1.500	25
Poly(2-*tert*-butyl-1,3-butadiene)	1.5060	24.6

TABLE 3. *cont'd*

Polymer	n	T (°C)
Poly(*tert*-butyl methacrylate)	1.4638	20
Poly(butylmercaptyl methacrylate)	1.5390	20
Poly(*N*-butyl methacrylamide)	1.5135	20
Poly(*o*-chlorobenzyl methacrylate)	1.5823	20
Poly(2-chloro-1-(chloromethyl)ethyl methacrylate)	1.5270	20
Poly(2-chlorocyclohexyl methacrylate)	1.5179	20
Poly(*o*-chlorodiphenylmethyl methacrylate)	1.6040	
Poly(2-chloroethyl α-chloroacrylate)	1.533	25
Poly(2-chloroethyl methacrylate)	1.517	20
Poly(1-(*o*-chlorophenyl)ethyl methacrylate)	1.5624	20
Poly(chloroprene)	1.554–1.558	
Poly(*o*-chlorostyrene)	1.6098	20
Poly(*m*-cresyl methacrylate)	1.5683	20
Poly(*o*-cresyl methacrylate)	1.5707	20
Poly(cyclohexanediol-1,4-dimethacrylate)	1.5067	20
Poly(cyclohexyl α-bromoacrylate)	1.542	25
Poly(cyclohexyl α-chloroacrylate)	1.532	25
Poly(cyclohexyl α-ethoxyacrylate)	1.4969	20
Poly(cyclohexyl methacrylate)	1.5066	20
Poly(*p*-cyclohexylphenyl methacrylate)	1.5575	20
Poly(decamethylene glycol dimethacrylate)	1.4990	
Poly(1-decene)	1.4730	
Poly(2-decyl-1,3-butadiene)	1.4899	20.5
Poly(diallyl phthalate)	1.572	20
Poly(2,3-dibromopropyl methacrylate)	1.5739	20
Poly(1,3-dichloropropyl methacrylate)	1.5270	20
Poly(2,6-dichlorostyrene)	1.6248	20
Poly(2-diethylaminoethyl methacrylate)	1.5174	20
Poly(1,1-diethylpropyl methacrylate)	1.4889	20
Poly(dihydroabietic acid)	1.544	
Poly(2,3-dimethylbutadiene) (methyl rubber)	1.525	
Poly(1,2-diphenylethyl methacrylate)	1.5816	20
Poly(diphenylmethyl methacrylate)	1.5933	20
Poly(*p*-divinylbenzene)	1.6150	20
Poly(dodecyl methacrylate)	1.4740	20
Poly(ester) resin, rigid (ca. 50% styrene)	1.523–1.54	
Poly(2-ethoxyethyl acrylate)	1.471	25
Poly(2-ethoxyethyl methacrylate)	1.4833	20
Poly(3-ethoxypropyl acrylate)	1.465	25
Poly(ethyl acrylate)	1.4685	20
Poly(ethyl α-chloroacrylate)	1.502	25
Poly(ethyl glycolate methacrylate)	1.4903	20
Poly(ethyl methacrylate)	1.485	20–25
Poly(ethylene) (density 0.914 g/cm^3)	1.51	20
(density 0.94–0.945 g/cm^3)	1.52–1.53	20
(density 0.965 g/cm^3)	1.545	20
Poly(ethylene) ionomer	1.51	
Poly(ethylene-*co*-propylene) (EPR-rubber)	1.4748–1.48	
Poly(ethylene-*co*-vinyl acetate) (80%–20% vinyl acetate)	1.47–1.50	
Poly(ethylene glycol dimethacrylate)	1.5063	20
Poly(ethylidene dimethacrylate)	1.4831	20
Poly(ethylmercaptyl methacrylate)	1.547	20
Poly(2-fluoroethyl methacrylate)	1.4768	20
Poly(4-fluoro-2-trifluoromethylstyrene)	1.46	
Poly(furfuryl methacrylate)	1.5381	
Poly(2-heptafluorobutoxy)ethyl acrylate)	1.390	25
Poly(heptafluorobutyl acrylate)	1.367	25
Poly(2-heptyl-1,3-butadiene)	1.5000	
Poly(hexadecyl methacrylate)	1.4750	30
Poly(2,2,3,4,4,4-hexafluorobutyl acrylate)	1.392	25
Poly(*n*-hexyl methacrylate)	1.4813	20
Poly(2-hydroxyethyl methacrylate)	1.5119	20
Poly(iminoadipoyliminohexamethylene) (nylon 6,6)	1.53	
Poly(iminoadipoylimino tetramethylene) (nylon moulding 6,10)	1.53	
oriented fibers	1.515–1.565	

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TABLE 3. *cont'd*

Polymer	*n*	*T* (°C)
Poly(imino(1-oxohexamethylene)) (nylon 6)	1.53	
Poly(isobutene)	1.505–1.51	
Poly(isobutyl methacrylate)	1.477	20
Poly(isoprene)	1.521	20
Poly(2-isopropyl-1,3-butadiene)	1.5028	30
Poly(isopropyl methacrylate)	1.4728	20
	1.552	
Poly(*p*-isopropylstyrene)	1.554	20
Poly(methacrylonitrile)	1.52	
Poly(*p*-methoxybenzyl methacrylate)	1.552	20
Poly(2-methoxyethyl acrylate)	1.463	25
Poly((*N*-2-methoxyethyl)methacrylamide)	1.5246	20
Poly(4-methoxy-2-methylstyrene)	1.5868	20
Poly(*o*-methoxyphenyl methacrylate)	1.5705	20
Poly(3-methoxypropyl acrylate)	1.471	25
Poly(*o*-methoxystyrene)	1.5932	20
Poly(*p*-methoxystyrene)	1.5967	20
Poly(methyl acrylate)	1.472–1.480	
Poly(methyl α-bromoacrylate)	1.5672	20
Poly(methyl α-chloroacrylate)	1.517	20
Poly(methyl isopropenyl ketone)	1.5200	20
Poly(methyl methacrylate)	1.4893	23
	1.490	20
Poly(*N*-methyl-methacrylamide)	1.5398	20
Poly(4-methyl-1-pentene)	1.459–1.465	
Poly(1-methylcyclohexyl methacrylate)	1.5111	20
Poly(2-methylcyclohexyl methacrylate)	1.5028	20
Poly(3-methylcyclohexyl methacrylate)	1.4947	20
Poly(4-methylcyclohexyl methacrylate)	1.4975	20
Poly(*N*-methylmaleimide-alt-isobutene)	1.53	
Poly(*o*-methylstyrene)	1.5874	20
Poly(α-naphthyl carbinyl methacrylate)	1.63	
Poly(α-naphthyl methacrylate)	1.6410	20
Poly(β-naphthyl methacrylate)	1.6298	20
Poly(*m*-nitrobenzyl methacrylate)	1.5845	20
Poly(2-nitro-2-methylpropyl methacrylate)	1.4868	20
Poly(nonafluoropentyl acrylate)	1.360	25
Poly(octafluoropentyl acrylate)	1.380	25
Poly(oxycarbonyloxybis(1,4-(3,5-dichlorophenylene))	1.6056	
Poly(oxycarbonyloxy-1,4-(2,6-dichloro)phenylene-isopropylidene-1,4-(2,6-dichloro)phenylene))	1.6056	
Poly(oxycarbonyloxy-1,4-phenylene-1,3-dimethylbutylidene-1,4-phenylene)	1.5671	
Poly(oxycarbonyloxy-1,4-phenylene-1-methyl-butylidene-1,4-phenylene)	1.5745	
Poly(oxycarbonyloxy-1,4-phenylene-1-propylbutylidene-1,4-phenylene)	1.5602	
Poly(oxycarbonyloxy-1,4-phenylene-*sec*-butylidene-1,4-phenylene)	1.5827	
Poly(oxycarbonyloxy-1,4-phenylenebutylidene-1,4-phenylene)	1.5792	
Poly(oxycarbonyloxy-1,4-phenylenecyclohexylidene-1,4-phenylene)	1.5900	20
Poly(oxycarbonyloxy-1,4-phenylenediphenyl-methylene-1,4-phenylene)	1.6539	
Poly(oxycarbonyloxy-1,4-phenyleneethylidene-1,4-phenylene)	1.5937	
Poly(oxycarbonyloxy-1,4-phenyleneisobutylidene-1,4-phenylene)	1.5702	
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)	1.5850	
Poly(oxy-2,6-dimethylphenylene)	1.575	
Poly(oxydimethylsilylene) (poly(dimethyl siloxane))	1.43	
Poly(oxyethylene) (high molecular weight)	1.4563	30
	1.51–1.54	
Poly(oxyethylene)-α-benzoate-ω-methacrylate)	1.555	20
Poly(oxyethyleneoxymaleoyl) (poly(ethylene maleate))	1.4840	25
Poly(oxyethyleneoxysuccinoyl) (poly(ethylene succinate))	1.4744	25
Poly(oxyethyleneoxyterephthaloyl) (amorphous) (poly(ethylene terephthalate))	1.5750	20
Poly(oxymethylene)	1.48	20
Poly(oxy-1-oxopentamethylene)	1.465	50
Poly(oxypentaerythritoloxyphthaloyl)	1.584	20
Poly(oxypropylene)	1.4495	
Poly(pentabromophenyl methacrylate)	1.71	
Poly(pentachlorophenyl methacrylate)	1.608	20
Poly(pentadecafluorooctyl acrylate)	1.339	25

TABLE 3. *cont'd*

Polymer	n	T (°C)
Poly(pentafluoropropyl acrylate)	1.385	25
Poly(pentafluorovinyl propionate)	1.364	25
Poly(1-phenyl-*n*-amyl methacrylate)	1.5396	20
Poly(phenyl α-bromoacrylate)	1.612	25
Poly(phenyl methacrylate)	1.5706	20
Poly(1-phenylallyl methacrylate)	1.5573	20
Poly(1-phenylcyclohexyl methacrylate)	1.5645	20
Poly(1-phenylethyl methacrylate)	1.5487	20
Poly(2-phenylethyl methacrylate)	1.5592	20
Poly(*N*-(2-phenylethyl)methacrylamide)	1.5857	20
Poly(2-(phenylsulfonyl)ethyl methacrylate)	1.5682	
Poly(*n*-propyl methacrylate)	1.484	25
Poly(propylene) (density 0.9075 g/cm^3)	1.5030	20
Poly(propylene) (atactic, density 0.8575 g/cm^3)	1.4735	20
Poly(styrene)	1.59–1.592	20
Poly(styrene-*co*-acrylonitrile) (– 75/25)	1.57	
Poly(styrene-*co*-maleic anhydride)	1.564	21
Poly(sulfides) (Thiokol)	1.6–1.7	
Poly(sulfone)	1.633	
Poly(tetradecyl methacrylate)	1.4746	30
Poly(tetrafluoro-3-(heptafluoropropoxy)propyl acrylate)	1.346	25
Poly(tetrafluoro-3-(pentafluoroethoxy)propyl acrylate)	1.348	25
Poly(tetrafluoro-3-(trifluoromethoxy)propyl acrylate)	1.360	25
Poly(2-(1,1,2,2-tetrafluoroethoxy)ethyl acrylate)	1.412	25
Poly(tetrafluoroethylene)	1.35(– 1.38)	
Poly(tetrafluoroethylene-*co*-hexafluoropropylene)	1.338	
Poly(tetrahydrofurfuryl methacrylate)	1.5096	20
Poly(triethylcarbinyl methacrylate)	1.4889	20
Poly(2,2,2-trifluoro-1-methylethyl methacrylate)	1.4185	
Poly(trifluorochloroethylene)	1.42–1.43	
Poly(2-trifluoroethoxy)ethyl acrylate)	1.419	25
Poly(trifluoroethyl acrylate)	1.407	25
Poly(trifluoroethyl methacrylate)	1.437	
Poly(trifluoroisopropyl methacrylate)	1.4177	20
Poly(trifluorovinyl acetate)	1.375	25
Poly(3,3,5-trimethylcyclohexyl methacrylate)	1.485	20
Poly(undecafluorohexyl acrylate)	1.356	
Poly(urethanes)	1.5–1.6	
Poly(vinyl acetal)	1.48–1.50	
Poly(vinyl acetate)	1.4665	20
Poly(vinyl alcohol)	1.49–1.53	
	1.5214	20
Poly(vinyl benzoate)	1.5775	20
Poly(vinyl butyl ether)	1.4563	30
Poly(vinyl *sec*-butyl ether) (isotactic)	1.4740	30
Poly(vinyl butyral)	1.48–1.49	
Poly(vinyl chloride)	1.54–1.55	
Poly(vinyl chloride) + 40% dioctyl phthalate	1.52	
Poly(vinyl chloride) + 40% tricresyl phosphate	1.55	
Poly(vinyl chloride-*co*-vinyl acetate) (95/5–90/10)	1.525–1.535	
Poly(vinyl chloroacetate)	1.512	25
Poly(vinyl decyl ether)	1.4628	30
Poly(vinyl dodecyl ether)	1.4640	30
Poly(vinyl ethyl ether)	1.4540	30
Poly(vinyl-2-ethylhexyl ether)	1.4626	30
Poly(vinyl formal)	1.50	
Poly(vinyl formate)	1.4757	20
Poly(vinyl hexyl ether)	1.4591	30
Poly(vinyl isobutyl ether)	1.4507	30
Poly(vinyl methacrylate)	1.5129	20
Poly(vinyl methyl ether)	1.467	20
Poly(vinyl methyl ether) (isotactic)	1.4700	30
Poly(vinyl octyl ether)	1.4613	30
Poly(vinyl pentyl ether)	1.4581	30
Poly(vinyl phenyl sulfide)	1.6568	20

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TABLE 3. *cont'd*

Polymer	n	T (°C)
Poly(vinyl propionate)	1.4665	20
Poly(vinylcarbazole)	1.683	20
Poly(vinylfuran)	1.55	20
Poly(vinylidene chloride)	1.60–1.63	
Poly(vinylidene fluoride)	1.42	25
Poly(vinylnaphthalene)	1.6818	20
Poly(1-vinyl-2-pyrrolidone)	1.53	
Poly(*N*-vinylphthalimide)	1.6200	20
Poly(2-vinyltetrahydrofuran)	1.55	20
Poly(2-vinylthiophene)	1.6376	20
Proteins	1.539–1.541	
Shellac	1.51–1.53	
Terpene resin	1.515	25
Urea–formaldehyde resin	1.54–1.56	
Urea–thiourea–formaldehyde resin	1.660	25
Poly(*p*,*p*-xylylenyl dimethacrylate)	1.5559	20

F. REFERENCES

1. J. R. Partington, "An Advanced Treatise of Physical Chemistry: Physico-Chemical Optics", vol. IV, Longmans, Green and Co., London, 1960.
2. J. F. Nye, "Physical Properties of Crystals", Oxford University Press, London, 1960.
3. J. C. Seferis, R. J. Samuels, Polym. Eng. Sci., **19**, 975 (1979).
4. A. R. Wedgewood, J. C. Seferis, Polym. Eng. Sci., **24**, 328 (1984).
5. R. S. Stein, G. L. Wilkes, in: I. M. Ward, (Ed.) "Structure and Properties of Oriented Polymers", Wiley, New York, 1975.
6. K. G. Denbigh, Trans. Faraday Soc., **36**, 936 (1940).
7. R. L. Rowell, R. S. Stein, J. Chem. Phys., **47**, 2985 (1967).
8. D. W. Krevelen, "Properties of Polymers – Part III: Properties of Polymers in Fields of Force", 3rd ed., Elsevier, New York, 1990.
9. R. J. Samuels, J. Appl. Polym. Sci., **26**, 1383 (1981).
10. P. H. Hermans, "Contribution to the Physics of Cellulose Fibers", Elsevier, New York, 1946.
11. R. S. Stein, J. Polym. Sci. A-2, **7**, 1021 (1969).
12. A. J. deVries, C. Bonnebat, J. Polym. Sci., Polym. Symp., **58**, 109 (1977).
13. A. J. deVries, Pure Appl. Chem., **53**, 1011 (1981).
14. R. S. Stein, J. Appl. Phys., **32**, 1280 (1961).
15. G. W. Shael, J. Appl. Polym. Sci., **8**, 2717 (1964).
16. G. W. Shael, J. Appl. Polym. Sci., **12**, 903 (1968).
17. J. C. Seferis, in: J. Brandrup, E. H. Immergut (Eds.), "Polymer Handbook", 3rd ed., Wiley, New York, 1989, p. VI/451.
18. C. Y. Cha, R. J. Samuels, J. Polym. Sci. B: Polym. Phys., **33**, 259 (1995).
19. S. S. Hardaker, R. J. Samuels, J. Polym. Sci. B: Polym. Phys., **35**, 777 (1997).
20. C. Cha, R. Samuels, S. Hardaker, R. Gregory, Synthetic metals, **84**, 743 (1997).
21. S. Hayashi, K. Asada, S. Hovike, H. Furahata, T. Hirai, J. Colloid Interface Sci., **176**, 370 (1995).
22. D. F. Stamatialis, M. Wessling, M. Sanopoulou, H. Strathmann, J. H. Petropoulos, J. Membrane Sci., **130**, 75 (1997).
23. T. Doi, S. Yukioka, H. Inoue, A. Akimoto, J. Appl. Pol. Sci., **63**, 925 (1997).

Radiation Resistance of Plastics and Elastomers

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A. INTRODUCTION

1. General Comments

The data on radiation resistance have been compiled mainly for practical polymer materials that have been classified into four groups: thermoplastics, elastomers, all aromatic polymers, and composite materials. The values of radiation resistance were determined by the change in mechanical properties. Radiation-induced degradation of polymer materials depends greatly on the irradiation conditions, as indicated in the tables; hence, the reader is advised to use these data with caution, considering all the factors.

2. Criterion for Radiation Resistance

Radiation resistance is characterized by the half-value-dose of significant changes in mechanical properties. For elastomers and thermoplastics, with large ultimate elongation by tensile test, the resistance is characterized by 100% elongation at break. The half-value-dose means the absorbed dose that reduces a property to 50% of its initial value under defined environments. The SI unit gray (Gy; 1 Gy = 1 J/kg = 100 rad) is used, and values in tables are in units of MGy (10^6 Gy) when there is no notation. In cases where the absorbed dose in the experiment was terminated before reaching the criterion, the dose is given with the sign ">".

The properties can be affected differently, therefore more than one relevant property should be considered. In most cases, the properties used to characterize radiation resistance are the elongation at break for elastomers and flexible plastics, and the flexural strength for rigid plastics and composite materials. However, the elongation at break sometimes increases with elastomers deterioration, but in that case, the strength decreases greatly.

3. Factors of Influence and their Consideration in Tables

Radiation resistance depends on polymer formulation as well as on the conditions of radiation exposure, such as the environmental atmosphere, temperature, dose rate, mechanical stress, etc. The most effective factor is the oxidation induced by radiation. Radiation resistance is indicated in two categories, with and without oxidation, together with the irradiation conditions employed.

3.1. Type of Polymer and Formulation The polymer samples reported were generally formulated by mixing with chemical agents, such as of stabilizers and fillers. Elastomers were normally crosslinked. The formulation and/or crosslinking have many variations for a given type of polymer. The formulation is determined by the suppliers based on the application (which may need, improve the mechanical or thermal properties), and the contents are usually not disclosed. Radiation resistance is influenced by formulation and crosslinking. In general, radiation resistance can be increased with inorganic fillers, with good adhesion between the polymer and reinforcing fillers, with

antioxidants, and with aromatic compounds. Organic fillers usually decrease radiation resistance. Some of the aromatic polymers reported may contain fillers of glass or carbon fibers.

The first column of the tables give the polymer name, along with the formulation or cross-linking. The second column gives the form of the test sample: either sheets (formed by heat press) or films (formed by extrusion or inflation), and their thickness (in mm).

3.2. Type of Radiation and Dosimetry The overall radiation effect does not depend on the type of radiation (such as X rays, gamma rays, electron beams, nuclear reactor irradiation, or other accelerated particle radiation), but only on the absorbed dose. There are some uncertainties about the given dose, especially to different types of radiation. In the following tables, the accuracy of the dose is estimated to be ±10% for gamma-rays and ±20% for electron beams or accelerated particles.

3.3. Dose Rate and the Atmosphere In an inert gas atmosphere or under vacuum, the dependence of dose rate on polymeric degradation is negligible at ambient temperature (1). If the other stresses, such as thermal degradation or light exposure, are eliminated during irradiation, the degradation should depend on the accumulated dose. If oxygen is present, the induced oxidation will decrease radiation resistance. The overall oxidation is determined by the rate of oxygen consumption by radiation-induced oxidation in the polymer matrix, and by the rate of oxygen supply into the matrix. The thickness of the polymer oxidation layer (L_{ox}) is a function of radiation dose rate (I), oxygen partial pressure (P_{o_2}) at one atmosphere total pressure, oxygen permeability coefficient (P_c) into the polymer matrix, and oxygen consumption factor (Φ), and is given by the following equation. (Ref. 2,3).

$$L_{ox} = (P_c P_{o_2} / I \Phi)^{1/2} \quad (1)$$

When the test material is a thick sheet, the oxidation layer occurs only at the surface, and the inside is not oxidized. The oxidation layer increases with decreasing dose rate. For most polymers, the degradation is more progressive with oxidation than without it. The dose rate affects the ratio of oxidation layer to material thickness. The degradation with oxidation can be observed by irradiation of a thin film; here, oxidation proceeds throughout the film. Oxidation throughout a thick film requires irradiation with oxygen under pressure rather than decreasing the dose rate (4). In the following tables, radiation resistance with oxidation implies that oxidation takes place throughout the material.

The other dose-rate dependent degradation, chain-reaction oxidation, has been reported at low dose-rate irradiations over long time periods. We believe that it is difficult to distinguish these chain reactions from thermal oxidation. In the tables, the data for very low dose rate have been excluded.

3.4. Temperature The irradiation temperature is ambient temperature, i.e. about 20–30 °C. The data for low-temperature irradiation is listed for few materials. Several experiments performed at higher irradiation temperatures are also listed. As the resistance depends on irradiation temperature, it is supposed that chemical reactions increase greatly due to the increased molecular motions during irradiation.

3.5. Other Stresses The tables do not consider mechanical stress and deformation during irradiation. These factors can have a large influence on degradation.

B. LIST OF SYMBOLS USED

σ	Tensile strength at break (ultimate strength)
$\sigma_{1/2}$	Half of tensile strength at break
ε	Elongation at break (ultimate elongation)
$\varepsilon_{1/2}$	Half of elongation at break
$\varepsilon_{100\%}$	100% of elongation at break
σ_F	Flexural strength
$\sigma_{F1/2}$	Half of flexural strength
IL	Inter Lamella Shear Strength (ILSS)
$IL_{1/2}$	Half of ILSS
γ	^{60}Co- γ-rays
EB	Electron beams, ( ) energy
n	Neutron, ( ) energy
H^+	Proton, ( ) energy
RT	Room temperature
Vac	Vaccum (including seal after vacuum)
Air	Air atmosphere
He	Helium-gas atmosphere
N_2	Nitrogen-gas atmosphere
O_2	Oxygen gas, ( ) indicates pressure

C. TABLES OF RADIATION RESISTANCE

TABLE 1. THERMOPLASTICS

Polymer	Form	Property: Initial value	Property: Criterions	Radiation resistance: Without oxidation (surface oxidation): Dose (MGy)	Irradiation conditions	Refs.	With oxidation: Dose (MGy)	Irradiation condition	Refs.	Remarks
Poly(ethylene) low-density (LDPE), non formulated	Sheet, 1 mm	σ : 18 MPa ε : 500%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>1.5 >1.5	γ, RT, vac, 5 kGy/h	5	0.25 0.4	γ, RT, O_2 (1.0 MPa), 5 kGy/h	5	
low-density (LDPE), non formulted	Film, 0.1 mm	σ : 45 MPa ε : 800%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>2 2.0	EB (2 MeV), RT, vac, 2.5 kGy/s or H^+(8 MeV), RT, vac, 0.7 kGy/s	6				
cross-linked (XLPE), formulated, cable insulator	Sheet, 1 mm	σ : 17 MPa ε : 500%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	3 3	γ, RT, vac, 6 kGy/h	7	1.0 2.6	γ, RT, O_2 (0.4 MPa), 6 kGy/h	7	
	Tube, 5 mm ϕ	σ : 20 MPa ε : 700%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	1.0 0.8	γ, RT, air, 10 kGy/h	8	0.4 0.4	γ, RT, O_2 (1.0 MPa), 2.8 kGy/h or air, 50 Gy/h	8	
UHMW-PE (ultra high molecular weight-PE)	Fiber, 0.011 mm ϕ	σ : 3.7 GPa	$\sigma_{1/2}$				0.1 – 0.2	γ, RT, O_2 (0.03 MPa), 10 kGy/h	9	
Poly(propylene) (PP), isotactic PP	Sheet, 0.5 mm	σ : 32 MPa ε : 210%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	0.5 0.04	γ, RT, vac, 10 kGy/h	10	0.15	γ, RT, O_2 (0.1 MPa), 10 kGy/h	10	
	Film, 0.1 mm	ε : 1200%	$\varepsilon_{100\%}$	0.02	EB (2 MeV), RT, vac, 2.5 kGy/s or H^+(8 MeV), RT, vac, 0.7 kGy/s	6				
Poly(vinyl chloride) (PVC), formulated	Sheet, 1.0 mm	σ : 28 MPa ε : 370%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>3 1.7	γ, RT, vac, 6 kGy/h	7	0.3 0.5	γ, RT, O_2 (0.4 MPa), 6 kGy/h	7	
Poly(vinyl chloride) (PVC), cable jacket	Tube, OD 18 mm ϕ 1.8 mm t	σ : 20 MPa ε : 190%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>1.4 1.0	γ, RT, air, 10 kGy/h	8	0.6 0.1	γ, RT, O_2 (1 MPa), 2.8 kGy/h, or air, 50 Gy/h	8	Low dose rate in air has the same effect as high dose rate in O_2 under pressure
Poly (12-aminolauric acid), Nylon-12	Sheet, 0.5 mm	σ : 80 MPa ε : 430%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>3 2.3	γ, RT, vac, 5 kGy/h	11	0.15 0.18	γ, RT, O_2 (0.7 MPa), 1.2 kGy/h	11	
Poly (ε-caprolactam), Nylon-6	Sheet, 0.5 mm	σ : 130 MPa ε : 400%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>3 >3	γ, RT, vac, 5 kGy/h	11	0.06 0.07	γ, RT, O_2 (0.7 MPa), 1.2 kGy/h	11	Inner part is not oxidized by oxygen under pressure
Poly(ethylene-terephthalate) (PET)	Film, 0.1 mm	σ : 220 MPa ε : 75%	$\sigma_{1/2}$ $\varepsilon_{1/2}$ $\sigma_{1/2}$	2.5 1.8 2.5	γ, RT, vac, 10 kGy/h H^+(8 MeV), RT, vac, 0.7 kGy/s	12 6	1.8 1.4	γ, RT, air, 8.5 kGy/h, or O_2 (2 MPa), 8.5 kGy/h	12	Oxidation proceeds throughout film in air at a relatively high dose rate
Poly(ethylene), 2,6-naphthalate (PEN)	Film, 0.075 mm	σ : 280 MPa ε : 30%	$\sigma_{1/2}$ $\varepsilon_{1/2}$	42 12	γ, RT, vac, 10 kGy/h	12	7.8 4.5	γ, RT, O_2 (0.6 MPa), 8.5 kGy/h	12	Cross-linking under vacuum irradiation
Poly(methylmethacrylate) (PMMA)	Sheet, 3 mm	σ_F: 130 MPa	$\sigma_{F1/2}$ $\sigma_{F1/2}$	0.15 1.5	γ, RT, vac, 10 kGy/h γ, 77 K, N_2, 30 kGy/h H^+(30, 45 MeV), RT, vac, 30 Gy/s	 13 14				
Poly(styrene) (PS), atactic PS	Sheet, 0.5 mm	σ : 60 MPa	$\sigma_{1/2}$ $\sigma_{1/2}$	>10 2	γ, RT, vac, 10 kGy/h σ, 80°C, vac, 10 kGy/h	15 15				
Poly(styrene) (PS), syndiotactic PS	Sheet, 0.5 mm	σ : 60 MPa	$\sigma_{1/2}$ $\sigma_{1/2}$	>30 10–20	γ, RT, vac, 10 kGy/h γ, 80°C, vac, 10 kGy/h	15 15				
Poly(tetrafluoroethylene) (PTFE)	Sheet, 0.5 mm	σ : 55 MPa ε : 400%	$\sigma_{1/2}$ $\varepsilon_{100\%}$ $\varepsilon_{100\%}$	0.0035 0.015 0.1	γ, RT, vac, 10 kGy/h γ, 77 K, N_2, 30 kGy/h	16 13	0.0015 0.002	γ, RT, air, 10 kGy/h	16	In air, oxidation proceeds throughout sheet
Cross-linked PTFE (XL-PTFE)	Sheet, 0.5 mm	σ : 25 MPa ε : 270%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	2.0 1.2	γ, RT, vac, 10 Gy/h	16	0.45 0.25	γ, RT, air, 10 kGy/h	16	In air, oxidation proceeds throughout sheet
Poly(vinylidene fluoride) (PVDF)	Sheet, 1 mm		$\sigma_{1/2}$ $\varepsilon_{100\%}$	1.0 0.3	γ, RT, air, 10 Gy/h	17	0.6 0.2	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	
	Film, 0.1 mm	σ : 110 MPa ε : 800%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	0.4 0.6	H^+(8 MeV), RT, vac, 0.7 kGy/s	6				
Poly(ethylene-tetrafluoro ethylene-*co*-polymer) (ETFE), Tefzel®	Film, 0.1 mm	σ : 65 MPa ε : 650%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>2.5 2.5	EB (2 MeV), RT, vac, 2.5 kGy/s or H^+ (8 MeV), RT, vac, 0.7 kGy/s	6				

TABLE 2. ELASTOMERS

Polymer	Form	Property: Initial value	Property: Criterion	Radiation resistance: Without oxidation (surface oxidation): Dose (MGy)	Irradiation conditions	Refs.	With oxidation: Dose (MGy)	Irradiation conditions	Refs.	Remarks
Ethyl acrylate rubber formulated, commercial	Sheet, 1 mm	σ : 11 MPa ε : 270%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	3 1.0	γ, RT, air, 10 kGy/h	17	3 >2	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Oxidation proceeds throughout in O_2 under pressure
Chlorosulphonated polyethylene (Hypalon™) formulated	Sheet, 1 mm	σ : 21 MPa ε : 390%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>3 1.0	γ, RT, air, 10 kGy/h	17	1.2 0.6	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Oxidation proceeds throughout in O_2 under pressure
Chlorosulphonated polyethylene (Hypalon™) cable jacket	Tube, 20 mm ϕ 2 mm t	σ : 12 MPa ε : 520%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>2 1.5	γ, RT, air, 10 kGy/h	8	>2 1.5	γ, RT, air, 50 Gy/h, or O_2 (1.0 MPa), 2.8 kGy/h	8	Oxidation proceeds throughout at very low dose rate or in O_2 under pressure
Ethylene-propylene-copolymer (EPR) formulated, commercial	Sheet, 1 mm	σ : 11 MPa ε : 270%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	3 1.0	γ, RT, air, 10 kGy/h	17	3 >2	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	In O_2 oxidation proceeds throughout sheet
Ethylene-propylene-ter-polymer (EPDM) formulated, commercial	Sheet, 1 mm	σ : 12 MPa ε : 800%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>1.5 >1.5	γ, RT, air, 10 kGy/h	17	>1.5 >1.5	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Oxidation is only on the surface at high dose rate, and proceeds throughout sheet in O_2 under pressure
	Tube, 5 mm ϕ Cable insulator	σ : 8.5 MPa ε : 590%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>10 >4	γ, RT, air, 10 kGy/h	8				
Ethylene-vinylacetate-rubber (EVM or EVA) formulated, commercial	Sheet, 1 mm	σ : 11 MPa ε : 270%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	1.5 >3	γ, RT, air, 10 kGy/h	17	>3 >3	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
	Film, 0.1 mm	σ : 30 MPa ε : 110%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>2.5 2.5	EB (2 MeV), RT, vac, 2.5 kGy/s or H^+(8 MeV), RT, vac, 0.7 kGy/s	6				
Poly(vinylidene fluoride-*co*-hexafluoropropylene) (Viton™) commercial	Sheet, 1 mm	σ : MPa ε : %	$\sigma_{1/2}$ $\varepsilon_{100\%}$	0.7 0.8	γ, RT, air, 10 kGy/h	17	0.3 0.3	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
Poly(tetrafluoroethylene-*co*-propylene) (Aflus™) commercial	Sheet, 1 mm	σ : MPa ε : %	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>3 0.5	γ, RT, air, 10 kGy/h	17	0.1 1.5	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
Poly-(*cis*-isoprene) (Natural rubber) formulated, commercial	Sheet, 1 mm	σ : 22 MPa ε : 580%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	1.0 2.5	γ, RT, air, 10 kGy/h	17	0.3 0.7	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
Poly(butadiene-*co*-acrylonitrile) (Nitrile/butadiene rubber), formulated, commercial	Sheet, 1 mm	σ : 17 MPa ε : 670%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	2.5 1.5	γ, RT, air, 10 kGy/h	17	2.5 3	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
Poly(isobutene-*co*-isoprene) (Butyl rubber) formulated, commercial	Sheet, 1 mm	σ : MPa ε : %	$\sigma_{1/2}$ $\varepsilon_{100\%}$	0.06 0.25	γ, RT, air, 10 kGy/h	17	0.06 0.2	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
Silicone rubber (Q) di-methylsilicone, commercial	Sheet, 1 mm	σ : MPa ϵ : %	$\sigma_{1/2}$ $\varepsilon_{100\%}$	0.8 0.5	γ, RT, air, 10 kGy/h	17	1.2 0.8	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
Silicone rubber di-methyl silicone/phenyl silicone (30 phr), commercial	Sheet, 1 mm	σ : MPa ε : %	$\sigma_{1/2}$ $\varepsilon_{100\%}$	2.5 >3	γ, RT, air, 10 kGy/h	17	2.3 2.3	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
Poly(butadiene-*co*-styrene) (styrene-butadiene rubber (SBR), commercial	Sheet, 1 mm	σ : MPa ε : %	$\sigma_{1/2}$ $\varepsilon_{100\%}$	>3 2	γ, RT, air, 10 kGy/h	17	1.2 0.7	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
Polyurethane rubber (AU), commercial	Sheet, 1 mm	σ : 19 MPa ε : 420%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	1.4 3.0	γ, RT, air, 10 kGy/h	17	0.4 3	γ, RT, O_2 (0.5 MPa), 5 kGy/h	17	Same as above
Polyurethane–polyester, commercial	Sheet, 1–3 mm	σ : 35–60 MPa ε : 600%	$\sigma_{1/2}$ $\varepsilon_{100\%}$	0.5–1.0 3–5	Reactor, RT, air, 200 kGy/h	18	0.2 >0.5	X-rays, RT, air, 100 kGy/h	18	Same as above

TABLE 3. AROMATIC POLYMERS

Polymer	Form	Property		Radiation resistance						Remarks
				Without oxidation (surface oxidation)			With oxidation			
		Initial value	Criterion	Dose (MGy)	Irradiation conditions	Refs.	Dose (MGy)	Irradiation conditions	Refs.	
Poly(phenyl-amide), Nomex®	Film, 0.05 mm	σ : 100 MPa	$\sigma_{1/2}$	15	γ, RT, air, 10 kGy/h	19	6	γ, RT, O_2 (0.5 MPa), 5 kGy/h	19	Oxidation of surface in air irradiation
		ε : 9%	$\varepsilon_{1/2}$	10			3			
Poly-arylene-ether-nitrile, BEK® (Idemitsu Kosan)	Sheet, 2 mm	σ : 82 MPa	$\sigma_{1/2}$	370	Reactor, RT, air, 750 kGy/h	20				Dose in reactor irradiation: 95% γ-rays and 5% neutron; temperature of material is estimated to be heated up to 100 °C or more
		ε : 21%	$\varepsilon_{1/2}$	200						
		σ_F : 107 MPa	$\sigma_{F/2}$	>370						
Poly(ether-ether-ketone) (PEEK) amorphous	Film, 0.125 mm	σ : 135 MPa	$\sigma_{1/2}$	>300	EB (2 MeV), RT, He, 6.4 kGy/s	21	10	γ, RT, O_2 (0.5 MPa), 5 kGy/h	19	
		ε : 210%	$\varepsilon_{1/2}$	100			>10			
	Film, 0.050 mm	σ : 120 MPa	$\sigma_{1/2}$	>300	γ, RT, N_2, 16 kGy/h	22	20	γ, RT, air, 16 kGy/h	22	Inner part is not oxidized in air at high dose-rate
		ε : 220%	$\varepsilon_{1/2}$	125			8			
Poly(ether-ether-ketone) (PEEK) crystalline	Film, 0.125 mm	σ : 190 MPa	$\sigma_{1/2}$	>300	EB (2 MeV), RT, He, 6.4 kGy/s	21	10	γ, RT, O_2 (0.5 MPa), 5 kGy/h	19	
		ε : 190%	$\varepsilon_{1/2}$	85			8			
	Film, 0.050 mm	σ : 105 MPa		>300	γ, RT, N_2, 16 kGy/h	22	12	γ, RT, air, 16 kGy/h	22	
		ε : 240%		45			5			
Poly(ether-ether-imide), ULTEM®	Film, 0.05 mm	σ : 140 MPa	$\sigma_{1/2}$	165	γ, RT, N_2, 16 kGy/h	22	12	γ, RT, air, 16 kGy/h	22	
		ε : 160%	$\varepsilon_{1/2}$	15			5			
Poly(imide), Kapton®	Film, 0.125 mm	σ : 240 MPa	$\sigma_{1/2}$	>300	EB (2 MeV), RT, He, 6.4 kGy/s	21	12	γ, RT, O_2 (0.7 MPa), 5 kGy/h	23	
		ε : 90%	$\varepsilon_{1/2}$	140			3	γ, RT, O_2 (0.6 MPa), 5 kGy/h	22	
	Film, 0.05 mm	σ : 240 MPa	$\sigma_{1/2}$	>300	γ, RT, N_2, 16 kGy/h	22	7			
		ε : 90–120%	$\varepsilon_{1/2}$	160			3			
Poly(imide), Upilex-R®	Film, 0.125 mm	σ : 260 MPa	$\sigma_{1/2}$	>300	EB (2 MeV), RT, He, 6.4 kGy/s	21				
		ε : 130–100%	$\varepsilon_{1/2}$	60						
	Film, 0.05 mm	σ : 320 MPa	$\sigma_{1/2}$	>300	γ, RT, N_2, 16 kGy/h	22	>20	γ, RT, O_2 (0.7 MPa), 5 kGy/h	19	
		ε : 100%	$\varepsilon_{1/2}$	180			20			
			$\sigma_{1/2}$				36	γ, RT, air, 16 kGy/h	22	Inner part is not oxidized in air irradiation
			$\varepsilon_{1/2}$				30			
Poly(imide), Upilex-S®	Film, 0.125 mm	σ : 390 MPa	$\sigma_{1/2}$	>300	EB (2 MeV), RT, He, 6.4 kGy/s	21				
		ε : 30–50%	$\varepsilon_{1/2}$	>300						
	Film, 0.05 mm	σ : 380 MPa	$\sigma_{1/2}$	>300	γ, RT, N_2, 16 kGy/h	22	60	γ, RT, air, 16 kGy/h	22	Inner part is not oxidized in air irradiation
		ε : 36%	$\varepsilon_{1/2}$	230			40			
Poly(imide), Thermo-plastics, Regulus®	Film, 0.100 mm	σ : 120 MPa	$\sigma_{1/2}$	>300	EB (2 MeV), RT, He, 6.4 kGy/s	21				
		ε : 110%	$\varepsilon_{1/2}$	70						
	Sheet, 2 mm	σ : 78 MPa	$\sigma_{1/2}$	>120	EB (2 MeV), RT, air, 5 kGy/s	24				
		ε : 60%	$\varepsilon_{1/2}$	90						
Poly(ethersulfone), (PES)	Film, 0.10 mm	σ : 85 MPa	$\sigma_{1/2}$	>2	EB (2 MeV), RT, vac, 2.5 kGy/s	6				
		ε : 190%	$\varepsilon_{1/2}$	1						

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TABLE 4. ORGANIC COMPOSITE MATERIALS[a]

Polymer	Form	Property: Initial value	Property: Criterion	Radiation resistance without oxidation (surface oxidation): Dose (MGy)	Irradiation conditions	Refs.	Remarks
GFRP(epoxy resin), G-10 CR®	Sheet, 2 mm	σ_F : 600 MPa	$\sigma_{F1/2}$	2	γ, RT, N_2, 10 kGy/h	13	σ_F : 1.2 GPa at 77 K
			$\sigma_{F1/2}$	25	γ, 77 K, N_2, 30 kGy/h		
			$\sigma_{F1/2}$	2	H^+ (30, 45 MeV), RT, vac, 5 kGy/h	14	H^+ was irradiated by beam scanning throughout sheet
GFRP(epoxy resin), G-11 CR®	Sheet, 2 mm	σ_F : 630 MPa	$\sigma_{F1/2}$	40	γ, RT, air, 10 kGy/h	25	σ_F : 1.2 GPa at 77 K
			$\sigma_{F1/2}$				In air, oxidation is limited only to the surface layer in air at high dose-rate
GFRP(epoxy resin), (TGDDM/DDM), Vf: 40%	Sheet, 2 mm	σ_F : 0.5–.7 GPa	$\sigma_{F1/2}$	>120	EB (2 MeV), RT, air, 5 kGy/s	26	σ_F : 1.2 GPa at 77 K
		IL : 50 MPa	$IL_{1/2}$	70–90			In air, oxidation is limited only to the surface layer in air at high dose-rate
GFRP(epoxy resin), M50J/2500®	Sheet, 2 mm	σ_F : 0.8 GPa	$\sigma_{F1/2}$	>60	EB (2 MeV), RT, air, 5 kGy/s	27	Same as above
		IL : 60 MPa	$IL_{1/2}$	>60			
GFRP(silicon resin)	Sheet, 4–6 mm	σ_F : 1.5 GPa	$\sigma_{F1/2}$	50	Reactor, RT, air, 1–2 MGy/h	28	In air oxidation is limited only to the surface layer in air at high dose-rate
GFRP(polyester), Vf: 40%	Sheet, 4–6 mm	σ_F : 2–3 GPa	$\sigma_{F1/2}$	10–50	Reactor, RT, air, 1–2 MGy/h	28	Same as above
GFRP(Kerimide)	Sheet, 4–6 mm	σ_F : 5 GPa	$\sigma_{F1/2}$	80	Reactor, RT, air, 1–2 MGy/h	28	Same as above
CFRP(epoxy resin), T300/3601, IM-6/R6376®	Sheet, 2 mm	σ_F : 2.0 GPa	$\sigma_{F1/2}$	>120	EB (2 MeV), RT, air, 5 kGy/s	29	Same as above
CFRP(PEEK), T6142/PEEK	Sheet, 2 mm	σ_F : 700 MPa	$\sigma_{F1/2}$	>120	EB (2 MeV), RT, air, 5 kGy/s	30	Same as above
CFRP(polyimide), T800H/PMR-15®	Sheet, 2 mm	σ_F : 1.2 GPa	$\sigma_{F1/2}$	>60	EB (2 MeV), RT, air, 5 kGy/s	27	Same as above
		IL : 80 MPa	$IL_{1/2}$	>60			
AFRP(epoxy resin), Kevlar 49/DGEBA, (FiBRA)®	Fabric rod, 3–20 mm ϕ	σ : 1.6 GPa	σ : 90%	5	EB (2 MeV), RT, air, 5 kGy/h	31	Same as above
		σ_F : 150 MPa	$\sigma_{F1/2}$	30			
		IL : 53 MPa	$IL_{1/2}$	60			
AFRP(epoxy resin), Kevlar 49/TGDDM/StO, (New FiBRA)®	Fabric rod, 3–20 mm ϕ	σ : 1.6 GPa	σ : 90%	>1205	EB (2 MeV), RT, air, 5 kGy/h	31	Same as above
		σ_F : 150 MPa	$\sigma_{F1/2}$	>120			
		IL : 45 MPa	$IL_{1/2}$	>120			

[a] GFRP: Glass fiber reinforced plastic; CFRP: Carbon fiber reinforced plastic; ARFP: Aramid fiber reinforced plastic.

D. REFERENCES

1. Y. Haruyama, Y. Morita, T. Seguchi, T. Tanaka, T. Kanazawa, K. Yotsumoto, K. Yoshida, Report of Japan Atomic Energy Research Institute, JAERI-M 88-197 (1988).
2. T. Seguchi, S. Hashimoto, K. Arakawa, N. Hayakawa, W. Kawakami, I. Kuriyama, Radiat. Phys. Chem., **17**, 195 (1981).
3. H. Kashiwabara, T. Seguchi, Ch. 11 in: A. Singh, J. Siverman (Eds.), "Radiation Processing of Polymers", Hanser Publishers, 1991.
4. T. Seguchi, K. Arakawa, N. Hayakawa, S. Machi, Radiat. Phys. Chem., **18**, 671 (1981).
5. T. Seguchi, K. Arakawa, M. Ito, N. Hayakawa, S. Machi, Radiat. Phys. Chem., **21**, 495 (1983).
6. T. Sasuga, S. Kawanishi, T. Seguchi, I. Kohno, Polymer, **30**, 2054 (1989).
7. M. Sugiyama, M. Nitta, T. Yagi, T. Seguchi, Inst. Elect. Eng. Japan, Papers of Technical Meeting on Electrical Insulating Materials, DEI-94-88 (1994).
8. T. Seguchi, K. Yoshida, Inst. Elect. Eng. Japan, Papers of Technical Meeting on Electrical Insulating Materials, EIM-82-113 (1982).
9. Y. Izumi, M. Nishii, T. Seguchi, K. Ema, T. Yamamoto, Radiat. Phys. Chem., **37**, 213 (1991).
10. E. A. Hegazy, T. Seguchi, K. Arakawa, S. Machi, J. Appl. Polym. Soc., **26**, 1361 (1981).
11. Y. Morita, T. Seguchi, Inst. Elect. Eng. Japan, Papers of Technical Meeting on Electrical Insulating Materials, EIM-83-131 (1983).
12. T. Saito, T. Seguchi, S. Nagano, I. Omae, JAERI-M 89-096.
13. H. Kudoh, N. Kasai, T. Sasuga, T. Seguchi, Radiat. Phys. Chem., **43**, 329 (1994).
14. H. Kudoh, T. Sasuga, T. Seguchi, in: R. L. Clough, S. W. Shalaby (Eds.), "Irradiation of Polymers", ACS Sympo. Series 620, 1 (1996).
15. Y. Tabata, A. Oshima, K. Takashika, T. Seguchi, Radiat. Phys. Chem., **48**, 563 (1996).
16. A. Oshima, S. Ikeda, T. Seguchi, Y. Tabata, Radiat. Phys. Chem., **49**, 279 (1997).
17. T. Seguchi, T. Sasuga, K. Yoshida, H. Yagyu, Y. Yamamoto, Inst. Elect. Eng. Japan, Papers of Technical Meeting on Electrical Insulating Materials, EIM-82-114 (1982).
18. H. Shonbacher, M. Tablet, CERN-89-12 (1989).
19. T. Seguchi, K. Arakawa, K. Yoshida, Y. Yamamoto, H. Yagyu, Inst. Elect. Eng. Japan., Papers of Technical Meeting on Electrical Insulating Materials, EIM-83-132 (1983).

20. K. Arigane, A. Otomo, H. Sasajima, H. Takahashi, T. Seguchi, S. Murakami, JAERI-Tech 94-004 (1994).
21. T. Kawamata, M. Nagai, Y. Morita, T. Yagi, T. Seguchi, Inst. Elect. Eng. Japan., Papers of Technical Meeting on Electrical Insulating Materials, DEI-94-91 (1994).
22. Y. Morita, K. Watanabe, H. Yagyu, Inst. Elect. Eng. Japan., Papers of Technical Meeting on Electrical Insulating Materials, EIM-88-129 (1988) and their additional data.
23. T. Sasuga, Polymer, **29**, 1562 (1989).
24. T. Hirade, Y. Hama, T. Sasuga, T. Seguchi, Polymer, **32**, 2499 (1991).
25. S. Egusa, H. Nakajima, M. Oshikiri, M. Hagiwara, S. Shimamoto, J. Nucl. Mater., **137**, 173 (1985).
26. K. Sonoda, Y. Yamamoto, O. Hashimoto, T. Ono, H. Tomita, A. Udagawa, S. Egusa, T. Sasuga, T. Seguchi, N. Tamura, Japan. J. Appl. Phys., **28**, 1950 (1989).
27. A. Udagawa, K. Yudate, H. Kudoh, T. Sasuga, Y. Morino, T. Seguchi, JAERI-Tech 95-007 (1995).
28. H. Schönbacher, A. Stolarz-Izycka, CERN 79-08 (1979).
29. T. Hirade, N. Takeda, A. Udagawa, T. Sasuga, T. Seguchi, Y. Hama, Adv. Composite Mater., **1**, 321 (1991).
30. T. Sasuga, T. Seguchi, H. Sakai, T. Nakamura, M. Masutani, J. Mater. Sci., **24**, 1570 (1989).
31. A. Udagawa, T. Moriya, S. Matsubara, T. Seguchi, Proc. Atomic Energy Soc. Japan. Conf., Sendai, September (1996).

PVT Relationships and Equations of State of Polymers

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A. INTRODUCTION

Pressure–volume–temperature (PVT) behavior of polymeric materials are important both from the practical and scientific viewpoint. On the practical side, a general knowledge of PVT properties are often required for processing polymers. On the scientific side, PVT properties are often required as input information on molecular models of polymer solutions. Information on molecular packing is also reflected in PVT properties. Numerous equations of state (EOS) have been developed to describe polymers. One can divide them into two categories: phenomenological and molecular.

Equations of state in the first category are for the empirical representation of volumetric data. The majority of equations in this group are the so-called isothermal compressibility equations (1–4). The purely empirical, original Tait equation and its modification by Tammon have been almost exclusively used for organics, including polymers (2,3,5,6).

Sanchez et al. re-examined polymer bulk data in a rigorous classical thermodynamic analysis (7,8). A new principle of temperature–pressure (T–P) superposition of compression response was found. Stated briefly, a dimensionless pressure variable is used to superpose compression data as a function of temperature into a universal curve. The governing parameter of compression is the first pressure coefficient $B_1(=\partial B/\partial P)_{P=0}$, of the bulk modulus B. It is related to the asymmetry of the free energy around its minimum, between dilation and compression. For polymers, B_1 is around 10, and universal. A new isothermal equation of state was formulated through a Padé analysis of the pressure dependence of the bulk modulus (8). Both the Tait and the Padé equations describe almost perfectly the isothermal pressure dependence of volume. Both can be used to smoothen experimental PVT data.

Equations of state in the second category arise from molecular models and the development of a proper partition function (9). Most earlier works on this group are based on the lattice or quasi-lattice description of a fluid. Three types of models have played a dominant role in the past several decades. These are the cell, hole, and lattice fluid models.

In Prigogine's cell model (10–12), any monomer is considered to be enclosed by the cell of surrounding molecules. The configurational potential is separated into the mean potential (between the center of the cell and surroundings), and the cell potential (generated by the movement of the central monomer within the cell). The cell potential was further simplified to be athermal. The Simha–Somcynsky theory (13) incorporated the lattice imperfection (hole) into the simple cell model by Prigogine et al. Flory et al. (14) borrowed some ideas from the cell model (separation of internal and external degrees of freedom) but used a Tonk's gas-type description for the repulsive part of the intermolecular potential. Unlike the cell model, the molecules are not localized in space, and a van der Waals mean field energy is employed. Meanwhile, the lattice fluid model (15,16) introduces lattice vacancies as pseudo-solvent molecules into the incompressible Flory–Huggins model. In this model, an incompressible polymer solution is converted into a compressible bulk polymer liquid.

All theoretical equations of state suggest a corresponding states behavior of polymer *PVT* properties that requires three scaling parameters such as characteristic temperature (T^*), pressure (P^*), and volume (V^*).

The derivation of the partition function can in principle be done in continuum space. This approach yields the so called integral equation theory (9). The self consistent integral equations of the interparticle distribution functions are to be determined. For the simplest hard sphere chains, several analytical equations of state were suggested by Chiew (17–22), Wertheim (23–27) and Chapman (28,29). Chiew's equation of state is based on Baxter's solution of the Percus–Yevick integral equation for the adhesive hard sphere (AHS) system. The AHS has a strong, but extremely short ranged attractive interaction at its surface. This peculiar interaction is treated as chemical bonds. Wertheim and Chapmann considered this adhesive attraction as a perturbation to the hard sphere potential in the framework of graph theory. A different and unique work on this subject was performed by Hall and coworkers (30) without recourse to integral equation theory. A particle insertion method was generalized to hard chains in continuum space.

To apply these continuum models to real systems, an energetic contribution to pressure should be added on the basis of a perturbation theory. In the simplest approximation a mean field (van der Waals) energy is added (28,31). The incorporation of the energy term turns these off-lattice models into 3-parameter (V^*, T^*, P^*) corresponding states models.

There are empirical or semiempirical equations of state that can be included partly in the two categories. The Sanchez–Cho (32), and the Hartmann–Haque (33) equations are considered the most important in applications. These two equations of state also suggest a corresponding states behavior.

This article only deals with liquid data (above T_g or T_m). The mathematical expression of the liquid *PVT* data with either the Padé or the Tait equations are tabulated here for extracting the 3 characteristic parameters (V^*, T^*, P^*) for given corresponding states equations of state.

The characteristic parameters are essential in the application of a given equation of state to real systems. We present, therefore, a compendium of such parameters for the 3-parameter equations of state described above. These parameters have their own physical meaning according to theories at hand. However, these are treated as adjustable parameters in a general nonlinear regression routine to best represent the smoothed experimental values regenerated from the Sanchez–Cho or the Tait equations. We used the volume data over the whole experimental *T–P* range. All model parameters are determined from the same data sets.

B. ISOTHERMAL COMPRESSIBILITY EQUATIONS

The recently found principle of temperature–pressure (*T–P*) superposition (7,8) implies that the $\ln(V/V_0)$, is a function of the scaled pressure, P/B_0, as is compression, given below:

$$\ln(V/V_0) = f\left(\frac{P}{B_0}\right) = -\frac{P}{B_0} + \frac{B_1}{2!}\left(\frac{P}{B_0}\right)^2 - \frac{1}{3!}(2B_1^2 - B_0B_2)\left(\frac{P}{B_0}\right)^3 + \cdots \quad \text{(A1)}$$

where V_0 and B_0 are volume and the bulk modulus at zero pressure, respectively. The right hand side of Eq. (A1) is a rigorous power series expression of $\ln(V/V_0)$. The additional contributing numbers $B_1 (i \geq 0)$ are the coefficients in the series expansion of the bulk modulus B in pressure:

$$B = B_0 + B_1P + \frac{1}{2}B_2P^2 + \cdots \quad \text{(A2)}$$

A tractable approximation to the infinite series in Eq. (A1) was suggested by Sanchez et al. They utilized the empirical observation that the bulk modulus B in Eq. (A2) is linear in pressure to high pressures. Expecting that the modulus B diverges as $P^{\omega}(\omega < 1)$, a Padé analysis and the successive integration of B yielded the Padé equation (8) given below:

$$\ln(V/V_0) = \frac{\omega}{(1-\omega)B_1}\left[1 - \left(1 + \frac{B_1}{\omega}\frac{P}{B_0}\right)^{1-\omega}\right] \quad \text{(A3)}$$

where the curvature parameter, ω, is defined as

$$\omega = B_1^2/(B_1^2 - B_0B_2) \quad \text{(A4)}$$

A pair of universal values of B_1 and ω ($B_1 = 10.2$, $\omega = 0.9$) is observed to be excellent in correlating the bulk volumetric data of various polymers.

The historically well known Tait equation falls into the general statement of the T–P superposition in Eq. (A1):

$$\ln(V/V_0) = \ln\left[1 - C\ln\left(1 + \frac{1}{CB_0}P\right)\right] \qquad \text{(A5)}$$

where C is universally 0.0894 (Cutler's constant). Equation (A5) is the modified version, by Tammon, of the original Tait equation, which was empirically formulated for water. One should notice that the product, CB_0, comprises the so-called Tait parameter in the literature. Equations (A3) and (A5) yield an accurate representation (within experimental error) of isothermal, volume–pressure relationships.

The zero pressure volume, V_0, and modulus, B_0, are additionally required to complete Eqs. (A3) and (A5). The compilation of these parameters are shown in Table 1. In this table, the abbreviation of polymer names are used. The explanation of the abbreviated names and experimental T–P range are listed in Table 2. Several types of fitting functions are used for V_0. Typically used are the polynomial and exponential expressions. For the bulk modulus, the exponential dependence on temperature is almost exclusively used. Some other polymer PVT data that are not used to obtain the characteristic parameters can be found in the literature, including blends and liquid crystalline polymers (38,50,55,59–67).

C. EMPIRICAL OR SEMIEMPIRICAL 3-PARAMETER EQUATIONS OF STATE

Equations in this section are in between the compressibility equations and succeeding model equations of state: (a) they are formulated on the basis of phenomenological arguments; (b) they suggest a corresponding behavior or a universal relationship between dimensionless V, T and P. The Sanchez–Cho and the Hartmann–Haque equations of state are considered the most important equations in this category.

Sanchez–Cho (S–C) Equation of State The Sanchez–Cho equation of state (32) was formulated by combining the empirically observed temperature dependence of the zero pressure volume V_0 and modulus B_0 with the isothermal Padé equation, Eq. (A3):

$$\tilde{V} = \frac{1}{(1-\tilde{T})}\exp\left[\frac{\omega}{(1-\omega)B_1}\left\{1 - \left(1 + \frac{B_1\tilde{P}e^{\tilde{b}\tilde{T}}}{\omega}\right)^{1-\omega}\right\}\right] \qquad \text{(A6)}$$

where the required universal numbers are $B_1 = 10.2$, $\omega = 0.9$, and $\tilde{b} = 9$. The dimensionless variables are defined as:

$$\tilde{V} \equiv V/V^*, \tilde{T} \equiv T/T^*, \tilde{P} \equiv P/P^* \qquad \text{(A7)}$$

The V^* and P^* are related to the van der Waals volume and cohesive energy density (32). The Sanchez–Cho equation provides volume as an explicit function of temperature and pressure, as is shown in Eq. (A6). Equation (A6) is to date the most accurate 3-parameter equation of state. The comparison of Eq. (A6) with the empirical Tait or Padé equation over the whole experimental T–P range gives the average deviation in volume less than 0.0004 cm^3/g. The Sanchez–Cho equation is useful in extrapolating data to high pressures and investigating the negative pressure region. The three characteristic parameters for the Sanchez–Cho equation are given in Table 4.

Hartmann–Haque (H–H) Equation of State The Hartmann–Haque equation of state (33) has the following functional form:

$$\tilde{P}\tilde{V} = \tilde{T}^{3/2} - \ln\tilde{V} \qquad \text{(A8)}$$

where the dimensionless variables $\tilde{P}, \tilde{V}, \tilde{T}$ are defined as in Eq. (A7). The P^* is identified as the isothermal bulk modulus extrapolated to zero temperature and pressure. The zero pressure isobar in Eq. (A8) originates in the Simha–Somcynsky hole model (13) in the next section. The three characteristic parameters for the Hartmann–Haque equation are given in Table 5. The average deviation between Eq. (A8) and regenerated bulk PVT data (data smoothed using the Tait Eq.) is 0.0009 cm^3/g or more than a factor of 2 worse than the S–C equation.

D. LATTICE OR QUASI LATTICE MODELS

The most frequently used model equations of state based on the lattice or quasi lattice phase space are visited in this section. These are various cell models of Prigogine et al., the hole model, and the lattice fluid model. The common feature of the models in this section is the separation of pressure into the thermal (P_{th}) and internal (P_i) pressures:

$$\frac{\tilde{P}}{\tilde{T}} = \frac{\tilde{P}_{th}(\tilde{V})}{\tilde{T}} + \frac{\tilde{P}_i(\tilde{V})}{\tilde{T}} \qquad \text{(A9)}$$

where the dimensionless variables are defined as before Eq. (A7). The physical meaning of the characteristic parameters P^*, V^*, T^* vary among the models. But in general, V^* is related to the van der Waals volume, T^* is proportional to monomer–monomer interaction energy, and P^* is related to the cohesive energy density.

Simple Cell Model of Prigogine et al. The cell model by Prigogine et al. (10–12) is an extension of the cell model for small molecules by Lennard–Jones and Devonshire (68) to polymers. Each monomer in the system is considered to be trapped in the cell created by the surroundings. The general cell potential, generated by the surroundings, is simplified to be athermal. This turns the simple cell model into a free volume theory. The mean potential between the centers of different cells are described by the Lennard–Jones 6–12 potential. The dimensionless equation of state has the following form:

$$\frac{\tilde{P}}{\tilde{T}} = \frac{1}{\tilde{V}^{2/3}(\tilde{V}^{1/3} - 2^{-1/6})} - \frac{2}{\tilde{T}}\left(\frac{1.2045}{\tilde{V}^3} - \frac{1.011}{\tilde{V}^5}\right) \quad \text{(A10)}$$

The factor $2^{-1/6}$ originates in the choice of the hexagonal close packing lattice as a cell geometry. The factors 1.2045 and 1.011 correct the effects of higher coordination shells on the internal energy.

Table 6 gives the three characteristic parameters P^*, V^*, T^* for the Prigogine cell model for the polymers in Tables 1 and 2. The average deviation between Eq. (A10) and regenerated *PVT* data is 0.0008 cm^3/g.

Cell Model by Flory, Orwoll, and Vrij (FOV) The cell model by Flory et al. (14) has the simpler mathematical form than the Prigogine cell model:

$$\frac{\tilde{P}}{\tilde{T}} = \frac{1}{\tilde{V}^{2/3}(\tilde{V}^{1/3} - 1)} - \frac{1}{\tilde{T}\tilde{V}^2} \quad \text{(A11)}$$

The Lennard–Jones energy is replaced with a uniform background (van der Waals) energy.

The characteristic parameters P^*, V^*, T^* for the FOV theory are shown in Table 7. The average deviation between Eq. (A11) and regenerated *PVT* data is 0.0022 cm^3/g. It has often been pointed out that the fitting performance of the FOV theory is less accurate, especially when covering a large pressure range. However, we present the parameters for the whole pressure range. This will provide a guidance when one fits the *PVT* data for smaller pressure ranges of interest.

Hole Model by Simha and Somcynsky (S–S) The simple cell model by Prigogine et al. is modified by Simha and Somcynsky (13) to allow for lattice imperfections (holes). The model can be written as the following equation of state:

$$\frac{\tilde{P}}{\tilde{T}} = \frac{(y\tilde{V})^{1/3}}{\tilde{V}[(y\tilde{V})^{1/3} - 2^{-1/6}y]} - \frac{2y}{\tilde{T}\tilde{V}}\left(\frac{1.2045}{(y\tilde{V})^2} - \frac{1.011}{(y\tilde{V})^4}\right) \quad \text{(A12)}$$

where the factor "y" is the fraction of occupied lattice sites and satisfies

$$(s/3c)[1 + \ln(1-y)/y] = -\frac{[(y\tilde{V}^{1/3}/3 - 2^{-1/6}y]}{[(y\tilde{V})^{1/3} - 2^{-1/6}y]} - \frac{y}{6\tilde{T}}\left(\frac{2.409}{(y\tilde{V})^2} - \frac{3.033}{(y\tilde{V})^4}\right) \quad \text{(A13)}$$

The factor $s/3c$ in Eq. (A13) is called the flexibility ratio, which is usually set to 1. Other factors are the same as those in the Prigogine cell model.

In order to calculate volume at given temperatures and pressures, one should solve Eqs. (A12) and (A13) simultaneously. The calculational complexity is a drawback of this model. Simha and Somcynsky suggested the approximate representation of the S–S equation of state:

$$\tilde{V} = 0.9018\exp(23.835\tilde{T}^{3/2})[1 - 0.089\ln\{1 + 1.0472\tilde{P}\exp(49.22\tilde{T})\}] \quad \text{(A14)}$$

The characteristic parameters P^*, V^*, T^* for the S–S theory are shown in Table 8. The parameters are obtained by fitting the original S–S equation of state in Eq. (A12) with Eq. (A13), rather than fitting Eq. (A14). The average deviation between Eq. (A12) and regenerated *PVT* data is 0.0007 cm^3/g.

Lattice Fluid Model The Lattice Fluid model developed by Sanchez and Lacombe (15,16) introduces vacancies into the classical incompressible Flory–Huggins model. The lattice vacancy is treated as a pseudo particle in the system. The free energy of an incompressible binary polymer solution is then converted to that of the bulk polymer. The equation of state for a polymer is given below:

$$\frac{\tilde{P}}{\tilde{T}} = -\left[\ln\left(1 - \frac{1}{\tilde{V}}\right) + \frac{1}{\tilde{V}}\right] - \frac{1}{\tilde{T}\tilde{V}^2} \quad \text{(A15)}$$

Table 9 gives the three characteristic parameters P^*, V^*, T^* for the lattice fluid (LF) theory. As was with the FOV model, the LF model does not correlate *PVT* data well over large pressure ranges. The average deviation between Eq. (A15) and regenerated bulk *PVT* data is 0.0033 cm^3/g.

E. CONTINUUM MODELS

The widely used models presented in the last section are derived by considering the partition function of a chain system on a lattice or quasi lattice. The lattice or quasi lattice phase space can be further generalized to be a continuum counterpart. This approach yields, in general, the integral equation theory, which focuses on solving the integral equations of the particle–particle distribution functions (9). The system of hard chains has been treated the most, because it is the simplest and provides the possibility of analytical equation of state in some cases.

One approach of this category is to solve the integral equations using the Percus–Yevick closure for the system of adhesive hard sphere (AHS) mixtures (17–22). An adhesive hard sphere is a hard sphere that has attractive sites at surface. The attractive interaction on these attractive sites is infinitely strong and infinitesimally short ranged. The Percus–Yevick closure yields an analytical solution for such systems. The adhesive attraction, which resembles the chemical bonding, is used to build up chains by employing the proper connectivity constraints.

A different approach of this category is to manipulate the graphs in the cluster expansion of the system of hard spheres with multiple attractive sites (23–29). The attractive interaction is the same as that of the AHS system, and is treated as a perturbation to the reference hard

sphere potential. The steric incompatibility is taken into account by Wertheim to approximate the graph expansion for distribution functions. This approach is called the thermodynamic perturbation theory (TPT) (23–27) or the statistically associating fluid theory (SAFT) (28,29).

A unique approach to the hard chain equation of state has been developed by Hall and coworkers (30). Flory's on-lattice particle insertion analysis is generalized to the off-lattice calculation. This theory is called the generalized Flory (GF) model. The enhanced evaluation of the insertion probability is further manipulated to yield the equation of state.

Since the developed equations of state are for athermal chain systems, an energetic contribution to pressure is required in order for these models to be applicable to real systems. The simplest model adds a van der Waals (vdW) term as a perturbation to the reference pressure (28,31):

$$\frac{\tilde{P}}{\tilde{T}} = \left(\frac{\tilde{P}}{\tilde{T}}\right)_0 - \frac{1}{\tilde{T}\tilde{V}^2} \tag{A16}$$

where the term with the subscript 0 implies the athermal reference pressure. The dimensionless variables are defined as usual (Eq. (A7)).

The specific mathematical forms of the reference equation of state are as follows:

$$\left(\frac{\tilde{P}}{\tilde{T}}\right)_0 = \frac{3}{2}\frac{1+1/\tilde{V}}{\tilde{V}^2(1-1/\tilde{V})^3} \quad \text{(AHS)} \tag{A17}$$

$$\left(\frac{\tilde{P}}{\tilde{T}}\right)_0 = \frac{3}{\tilde{V}^2(1-1/\tilde{V})^3}\left[1 - \frac{(1-1/\tilde{V}^2]}{(2+1/\tilde{V})}\right] \quad \text{(TPT or SAFT)} \tag{A18}$$

$$\left(\frac{\tilde{P}}{\tilde{T}}\right)_0 = 1.31659\frac{2-1/\tilde{V}}{\tilde{V}^2(1-1/\tilde{V})^3} \quad \text{(GF)} \tag{A19}$$

The three characteristic parameters P^*, V^*, T^* for the AHS + vdW equation of state are shown in Table 10. The average deviation between the AHS + vdW equation and regenerated bulk data is 0.0012 cm^3/g. The parameters for other two equations are relatively close to those for the AHS + vdW equation, and are not tabulated here.

F. TABLES

TABLE 1. ZERO PRESSURE VOLUME ($V_0(T)$) AND BULK MODULUS ($B_0(T)$)[a]

Polymer	$V_0(T)$ (cm^3/g)	$B_0(T)$ (MPa)
PDMS	$1.0079\exp(9.121\times10^{-4}t)$	$1000.00\exp(-5.701\times10^{-3}t)$
PS	$0.9287\exp(5.131\times10^{-4}t)$	$2426.17\exp(-3.319\times10^{-3}t)$
PoMS	$0.9396\exp(5.306\times10^{-4}t)$	$2929.53\exp(-4.114\times10^{-3}t)$
PMMA	$0.8254+2.8383\times10^{-4}t+7.792\times10^{-7}t^2$	$3215.88\exp(-4.146\times10^{-3}t)$
PnBMA	$0.9341+5.5254\times10^{-4}t+6.5803\times10^{-7}t^2+1.5691\times10^{-10}t^3$	$2535.79\exp(-5.344\times10^{-3}t)$
PCHMA	$0.8793+4.0504\times10^{-4}t+7.774\times10^{-7}t^2-7.7534\times10^{-10}t^3$	$3302.01\exp(-5.220\times10^{-3}t)$
PEA	$0.8756\exp(7.241\times10^{-4}t)$	$2161.07\exp(-4.839\times10^{-3}t)$
PEMA	$0.8614\exp(7.468\times10^{-4}t)$	$2918.34\exp(-5.356\times10^{-3}t)$
PMA	$0.8365\exp(6.795\times10^{-4}t)$	$2637.58\exp(-4.493\times10^{-3}t)$
PVAc	$0.8250+5.820\times10^{-4}t+2.940\times10^{-7}t^2$	$2\,291.95\exp(-4.346\times10^{-3}t)$
LPE	$0.9172\exp(7.806\times10^{-4}t)$	$1976.51\exp(-4.661\times10^{-3}t)$
BPE	$0.9399\exp(7.341\times10^{-4}t)$	$1980.98\exp(-4.699\times10^{-3}t)$
LDPE-A	$1.1484\exp(6.950\times10^{-4}t)$	$2157.72\exp(-4.701\times10^{-3}t)$
LDPE-B	$1.1524\exp(6.700\times10^{-4}t)$	$2199.11\exp(-4.601\times10^{-3}t)$
LDPE-C	$1.1516\exp(6.730\times10^{-4}t)$	$2088.37\exp(-4.391\times10^{-3}t)$
PBD	$1.0970\exp(6.600\times10^{-4}t)$	$1987.70\exp(-3.593\times10^{-3}t)$
PBD8	$0.9352+4.9988\times10^{-4}t+1.268\times10^{-7}t^2$	$10915.90\exp(-5.344\times10^{-3}t)$
PBD24	$0.9352+4.9988\times10^{-4}t+1.268\times10^{-7}t^2$	$10915.90\exp(-5.344\times10^{-3}t)$
PBD40	$0.9352+4.9988\times10^{-4}t+1.268\times10^{-7}t^2$	$10915.90\exp(-5.344\times10^{-3}t)$
PBD50	$0.9352+4.9988\times10^{-4}t+1.268\times10^{-7}t^2$	$10915.90\exp(-5.344\times10^{-3}t)$
PBD87	$0.9352+4.9988\times10^{-4}t+1.268\times10^{-7}t^2$	$10915.90\exp(-5.344\times10^{-3}t)$
i-PB	$1.1417\exp(6.751\times10^{-4}t)$	$1873.60\exp(-4.533\times10^{-3}t)$
PAr	$0.73381\exp(1.626\times10^{-5}T^{3/2})$	$3321.03\exp(-3.375\times10^{-3}t)$
PCL	$0.9049\exp(6.392\times10^{-4}t)$	$2114.09\exp(-3.931\times10^{-3}t)$
PC	$0.73565\exp(1.859\times10^{-5}T^{3/2})$	$3467.56\exp(-4.078\times10^{-3}t)$
BCPC	$0.6737+3.634\times10^{-4}t+2.370\times10^{-7}t^2$	$4064.88\exp(-4.921\times10^{-3}t)$
HFPC	$0.6111+4.898\times10^{-4}t+1.730\times10^{-7}t^2$	$2646.53\exp(-5.156\times10^{-3}t)$
TMPC	$0.8497+5.073\times10^{-4}t+3.832\times10^{-7}t^2$	$2588.37\exp(-4.242\times10^{-3}t)$
PET	$0.6883+5.90\times10^{-4}t$	$4135.35\exp(-4.150\times10^{-3}t)$
PIB	$1.0750\exp(5.651\times10^{-4}t)$	$2240.49\exp(-4.329\times10^{-3}t)$
PI8	$0.9352+4.9988\times10^{-4}t+1.268\times10^{-7}t^2$	$10915.90\exp(-5.344\times10^{-3}t)$
PI14	$0.9352+4.9988\times10^{-4}t+1.268\times10^{-7}t^2$	$10915.90\exp(-5.344\times10^{-3}t)$
PI41	$0.9352+4.9988\times10^{-4}t^21.268\times10^{-7}t^2$	$10915.90\exp(-5.344\times10^{-3}t)$

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TABLE 1. *cont'd*

Polymer	$V_0(T)$ (cm^3/g)	$B_0(T)$ (MPa)
PI56	$0.9352 + 4.9988 \times 10^{-4}t + 1.268 \times 10^{-7}t^2$	$10915.90 \exp(-5.344 \times 10^{-3}t)$
i-PP	$1.1606 \exp(6.700 \times 10^{-4}t)$	$1667.79 \exp(-4.177 \times 10^{-3}t)$
a-PP	$1.1841 - 1.091 \times 10^{-4}t + 5.286 \times 10^{-6}t^2$	$1813.20 \exp(-6.604 \times 10^{-3}t)$
Phenoxy	$0.76644 \exp(1.921 \times 10^{-5}T^{3/2})$	$4025.73 \exp(-4.378 \times 10^{-3}t)$
PSO	$0.7644 + 3.419 \times 10^{-4}t + 3.126 \times 10^{-7}t^2$	$4092.84 \exp(-3.757 \times 10^{-3}t)$
PEO	$0.8766 \exp(7.087 \times 10^{-4}t)$	$2323.27 \exp(-3.947 \times 10^{-3}t)$
PVME	$0.9585 \exp(6.653 \times 10^{-4}t)$	$2413.87 \exp(-4.588 \times 10^{-3}t)$
PEEK	$0.7158 \exp(6.690 \times 10^{-4}t)$	$4340.04 \exp(-4.124 \times 10^{-3}t)$
PTFE	$0.3200 - 9.5862 \times 10^{-4}t$	$4756.15 \exp(-9.380 \times 10^{-3}t)$
PTHF	$1.0043 \exp(6.691 \times 10^{-4}t)$	$1997.76 \exp(-4.233 \times 10^{-3}t)$
PMP	$1.4075 - 9.905 \times 10^{-4}t + 3.497 \times 10^{-6}t^2$	$376.7 + 2.134t - 7.0445 \times 10^{-3}t^2$
PA6	$0.7597 \exp(4.701 \times 10^{-4}t)$	$4213.65 \exp(-4.660 \times 10^{-3}t)$
PA66	$0.7657 \exp(6.600 \times 10^{-4}t)$	$3539.15 \exp(-5.040 \times 10^{-3}t)$
PECH	$0.7216 \exp(5.825 \times 10^{-4}t)$	$2665.55 \exp(-4.171 \times 10^{-3}t)$
PVC	$0.7196 + 5.581 \times 10^{-5}t + 1.468 \times 10^{-6}t^2$	$3290.83 \exp(-5.321 \times 10^{-3}t)$
PPO	$0.78075 \exp(2.151 \times 10^{-5}T^{3/2})$	$2548.10 \exp(-4.290 \times 10^{-3}t)$
E/P50	$1.2291 + 5.799 \times 10^{-5}t + 1.964 \times 10^{-6}t^2$	$5447.43 \exp(-8.103 \times 10^{-3}t)$
EVA18	$1.02391 \exp(2.173 \times 10^{-5}T^{3/2})$	$2105.15 \exp(-4.537 \times 10^{-3}t)$
EVA25	$1.00416 \exp(2.244 \times 10^{-5}T^{3/2})$	$2062.64 \exp(-4.734 \times 10^{-3}t)$
EVA28	$1.00832 \exp(2.241 \times 10^{-5}T^{3/2})$	$2052.57 \exp(-4.457 \times 10^{-3}t)$
EVA40	$1.06332 \exp(2.288 \times 10^{-5}T^{3/2})$	$2294.18 \exp(-4.989 \times 10^{-3}t)$
SAN3	$0.9233 + 3.936 \times 10^{-4}t + 5.685 \times 10^{-7}t^2$	$2682.33 \exp(-4.376 \times 10^{-3}t)$
SAN6	$0.9211 + 4.370 \times 10^{-4}t + 5.846 \times 10^{-7}t^2$	$2538.03 \exp(-4.286 \times 10^{-3}t)$
SAN15	$0.9044 + 4.207 \times 10^{-4}t + 4.077 \times 10^{-7}t^2$	$2666.67 \exp(-3.943 \times 10^{-3}t)$
SAN18	$0.9016 + 4.036 \times 10^{-4}t + 4.206 \times 10^{-7}t^2$	$2689.04 \exp(-3.858 \times 10^{-3}t)$
SAN4	$0.8871 + 3.406 \times 10^{-4}t + 4.938 \times 10^{-7}t^2$	$3236.02 \exp(-4.431 \times 10^{-3}t)$
SAN70	$0.8528 + 3.616 \times 10^{-4}t + 2.634 \times 10^{-7}t^2$	$3751.68 \exp(-3.923 \times 10^{-3}t)$
SMMA20	$0.9063 + 3.570 \times 10^{-4}t + 6.532 \times 10^{-7}t^2$	$2595.08 \exp(-4.143 \times 10^{-3}t)$
SMMA60	$0.8610 + 3.350 \times 10^{-4}t + 6.98010^{-7}t^2$	$2919.46 \exp(-4.611 \times 10^{-3}t)$

[a] In the mathematical expression of volume, t and T are in °C and K, respectively. See Table 2 for references.

TABLE 2. ABBREVIATIONS OF POLYMER NAMES AND THE EXPERIMENTAL TEMPERATURE–PRESSURE RANGE FOR POLYMER LIQUIDS[a]

Polymer abbreviation	Monomer or source-based Name	T (°C)	P (MPa)	Refs.
PDMS	Dimethyl siloxane	25–70	0–100	34,35,36
PS	Styrene	115–196	0–200	37
PoMS[b]	*o*-Methyl styrene	139–198	0–180	37
PMMA	Methyl methacrylate	114–159	0–200	38
PBMA[b]	*n*-Butyl methacrylate	34–200	0–200	38
PCHMA[b]	Cyclo hexyl methacrylate	123–198	0–200	38
PEA	Ethyl acrylate	37–210	0–196	34
PEMA[b]	Ethyl methacrylate	113–161	0–196	34
PMA	Methyl acrylate	37–220	0–196	34
PVAC	Vinyl acetate	35–100	0–80	39
LPE[b]	Linear polyethylene (PE)	142–200	0–200	38
BPE[b]	Branched PE	125–198	0–200	38
LDPE-A	Low density PE-A	112–225	0–196	40
LDPE-B	Low density PE-B	112–225	0–196	40
LDPE-C	Low density PE-C	112–225	0–196	40
PBD[b]	Butadiene	4–55	0–283.5	41
PBD8[b]	Butadiene with 8% 1,2 content	25–200	0–200	42
PBD24[b]	Butadiene with 24% 1,2 content	25–200	0–200	42
PBD40[b]	Butadiene with 40% 1,2 content	25–200	0–200	42
PBD50[b]	Butadiene with 50% 1,2 content	25–200	0–200	42
PBD87[b]	Butadiene with 87% 1,2 content	25–200	0–200	42

TABLE 2. *cont'd*

Polymer abbrevation	Monomer or source-based Name	T (°C)	P (MPa)	Refs.
PB	1-Butene	133–246	0–196	43
PAr[b]	Arylate	177–310	0–176.5	44
PCL	Caprolactone	100–148	0–200	34
PC	Carbonate (PC)	151–340	0–176.5	44
BCPC[b]	Bisphenol chloral PC	220–280	0–50	45
HFPC[b]	Hexafluoro bisphenol-A PC	220–280	0–50	45
TMPC[b]	Tetramethyl bisphenol-A PC	210–290	0–50	45
PET	Ethylene terephthalate	274–342	0–196	46
PIB	Isobutylene	53–110	0–00	36,47
PI8	Isoprene with 8% 3,4 content	25–200	0–200	42
PI14	Isoprene with 14% 3,4 content	25–200	0–200	42
PI41	Isoprene with 41% 3,4 content	25–200	0–200	42
PI56	Isoprene with 56% 3,4 content	25–200	0–200	42
i-PP	Isotactic polypropylene	170–297	0–196	43
a-PP	Atactic polypropylene	80–120	0–100	34
Phenoxy[b]	Phenoxy	68–300	0–176.5	44
PSO	Sulfone	202–371	0–196	48
PEO	Ethylene oxide	88–224	0–68.5	49
PVME	Vinyl methyl ether	30–198	0–200	34,50
PEEK	Ether ether ketone	346–398	0–200	51
PTFE	Tetrafluoro ethylene	330–372	0–39	52
PTHF[b]	Tetrahydro furane	62–166	0–78.5	49
PMP	4-Methyl-1-pentene	241–319	0–196	53
PA6	Amide 6	236–296	0–196	54

TABLE 2. *cont'd*

Polymer abbrevation	Monomer or source-name	*T* (°C)	*P* (MPa)	Refs.
PA66	Amide 66	246–298	0–196	54
PECH[b]	Epichlorohydrine	60–140	0–200	34
PVC	Vinyl chloride	100–150	0–200	34
PPO	Phenylene oxide	203–320	0–176.5	55
E/P50	Ethylene/propylene 50%	140–250	0–62.5	56
EVA18	Ethylene/vinyl acetate 18%	112–219	0–176.5	57
EVA25	Ethylene/vinyl acetate 25%	94–233	0–176.5	57
EVA28	Ethylene/vinyl acetate 28%	94–235	0–176.5	57
EVA40	Ethylene/vinyl acetate 40%	75–235	0–176.5	57
SAN3	Styrene/acrylonitrile 2.7%	105–266	0–200	45
SAN6	Styrene/acrylonitrile 5.7%	96–267	0–200	45
SAN15	Styrene/acrylonitrile 15.3%	132–262	0–196	45
SAN18	Styrene/acrylonitrile 18.0%	104–255	0–196	45
SAN40	Styrene/acrylonitrile 40%	100–270	0–196	45
SAN70	Styrene/acrylonitrile 70%	100–271	0–196	58
SMMA20[a]	Styrene/methyl methacrylate 20%	110–270	0–196	58
SMMA60[a]	Styrene/methyl methacrylate 60%	110–270	0–200	58

[a] See Abbreviations Table in Part VIII of this Handbook.
[b] Abbreviation may not be recognized by international organization.

TABLE 3. *PVT* PROPERTIES OF OTHER POLYMERS[a]

Polymer	*T* (°C)	*P* (MPa)	Note	Refs.
HDPE	140–203	0–196.0		59
LDPE	121–175	0–196.0		59
HMLPE	137–200	0–200.0		38
i-Poly(1-pentene)	140–172	0.1		60
Poly(vinylidene fluoride)	175–240	0.1		61
Poly(methylene oxide)	155–185	18.3–150.9	Figure only	62
Poly(butylene terephthalate)	215–280	20.7–103.4	Figure only	63
Qiana nylon	270–320	20.7–103.4	Figure only	63
Natural rubber vulcanizated with 10–28% sulfur	10–82	0–1013.0		64
Hevea rubber uncrosslinked crosslinked (6% sulfur) crosslinked (11.5% sulfur)	10–85	0–80.0	Simha–Somcynsky parameters only	65
PS/PPO blends	30–350	0–176.5		55
PS/PVME blends	30–198	0–200.0		50
Main chain LCPs			Figure only up	66
Azomethine ether LCP	20–300	0–200.0	to nematic	67

[a] This table lists polymers not mentioned in the earlier tables.

TABLE 4. CHARACTERISTIC PARAMETERS FOR THE SANCHEZ–CHO EQUATION OF STATE

Polymer	V^* (cm^3/g)	T^* (K)	P^* (MPa)
PDMS	0.8071	1375.1	6212.9
PS	0.8165	2277.2	7867.6
PoMS	0.8368	2380.6	7772.1
PMMA	0.7139	2184.2	9873.0
PBMA	0.7963	1855.9	9025.3
PCHMA	0.7710	2195.1	8515.1

TABLE 4. *cont'd*

Polymer	V^* (cm^3/g)	T^* (K)	P^* (MPa)
PEA	0.7398	1747.4	9131.8
PEMA	0.7332	1771.2	11257.0
PMA	0.7121	1829.0	10623.9
PVAc	0.6918	1696.9	10399.9
LPE	0.9491	1655.0	9943.5
BPE	0.9723	1751.9	8630.7
LDPE-A	0.9852	1865.4	8214.5
LDPE-B	0.9937	1923.8	7983.8
LDPE-C	0.9869	1880.5	8225.3
PBD	0.9115	1633.8	9443.2
PBD8	0.9308	1798.6	9136.3
PBD24	0.9359	1819.0	8708.0
PBD40	0.9357	1842.9	8352.0
PBD50	0.9408	1892.0	7757.3
PBD87	0.9498	1905.6	7241.8
PB	0.9854	1924.1	6891.1
PAr	0.6839	2243.9	11557.3
PCL	0.7671	1849.0	9530.1
PC	0.6871	2070.3	12106.1
BCPC	0.5971	2249.1	9931.3
HFPC	0.5264	1788.2	10173.1
TMPC	0.7261	1908.0	10573.0
PET	0.6199	2022.2	15278.8
PIB	0.9382	2130.2	7045.3
PI 8	0.9453	1921.0	7669.6
PI 14	0.9366	1911.3	8219.9
PI 41	0.9370	1912.7	8112.3
PI 56	0.9330	1854.5	8600.1
i-PP	1.0116	1991.5	6118.1
a-PP	0.9690	1776.2	6202.8
Phenoxy	0.7242	2103.4	12728.1
PSO	0.6655	2232.2	13286.4
PEO	0.7441	1789.1	10805.4
PVME	0.8187	1861.3	9270.5
PEEK	0.6395	2126.5	14335.5
PTFE	0.3638	1400.7	9757.5
PTHF	0.8561	1843.0	8160.2
PMP	1.0089	1885.0	6452.5
PA6	0.7130	3140.3	5718.4
PA66	0.6887	2195.2	8391.9
PECH	0.6269	2068.9	8899.3
PVC	0.6252	2395.4	7551.2
PPO	0.7181	1810.4	11776.9
E/P50	1.0582	2384.7	6421.0
EVA18	0.9585	1878.9	8107.5
EVA25	0.9338	1848.3	7957.5
EVA28	0.9241	1812.5	8622.1
EVA40	0.8864	1856.1	8441.9
SAN3	0.8030	2185.7	7878.5
SAN6	0.7896	2010.7	8896.8
SAN15	0.7860	2170.4	8602.0
SAN18	0.7854	2208.4	8495.6
SAN40	0.7853	2435.2	7772.6
SAN70	0.7616	2546.6	9177.0
SMMA20	0.7789	2105.6	8531.3
SMMA60	0.7408	2099.5	8858.9

TABLE 5. CHARACTERISTIC PARAMETERS FOR THE HARTMANN–HAQUE EQUATION OF STATE

Polymer	V^* (cm^3/g)	T^* (K)	P^* (MPa)
PDMS	0.8795	1106	1837
PS	0.8754	1603	2956

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TABLE 5. *cont'd*

Polymer	V^* (cm^3/g)	T^* (K)	P^* (MPa)
PoMS	0.8891	1608	3096
PMMA	0.7582	1467	3819
PBMA	0.8552	1309	3268
PCHMA	0.8249	1517	3038
PEA	0.8038	1284	3076
PEMA	0.7976	1298	3654
PMA	0.7712	1334	3680
PVAc	0.7368	1151	3817
LPE	1.0381	1211	2804
BPE	1.0677	1313	2510
LDPE-A	1.0769	1387	2516
LDPE-B	1.0842	1426	2509
LDPE-C	1.0781	1396	2538
PBD	0.9855	1199	3333
PBD8	1.0065	1301	3109
PBD24	1.0118	1317	2989
PBD40	1.0113	1336	2891
PBD50	1.0157	1370	2742
PBD87	1.0264	1387	2557
PB	1.0818	1452	2066
PAr	0.7408	1614	3709
PCL	0.8418	1411	3013
PC	0.7474	1502	3644
BCPC	0.5356	1520	3641
HFPC	0.5758	1306	2920
TMPC	0.8010	1430	2907
PET	0.6802	1484	4071
PIB	0.9935	1422	2976
PI8	1.0199	1389	2733
PI14	1.0098	1375	2925
PI41	1.0103	1377	2889
PI56	1.0076	1339	2990
i-PP	1.1066	1475	1852
a-PP	1.0292	1197	2383
Phenoxy	0.7784	1481	4357
PSO	0.7246	1623	3972
PEO	0.8005	1254	3504
PVME	0.8836	1342	3245
PEEK	0.7044	1568	3668
PTFE	0.3683	900	3251
PTHF	0.9165	1276	2894
PMP	1.1228	1449	1670
PA6	0.7654	2255	2264
PA66	0.7559	1643	2453
PECH	0.6734	1456	3775
PVC	0.6559	1532	3595
PPO	0.7906	1339	3079
E/P50	1.1112	1516	2247
EVA18	1.0443	1380	2521
EVA25	1.0171	1356	2474
EVA28	1.0077	1333	2646
EVA40	0.9604	1339	2716
SAN3	0.8615	1534	2840
SAN6	0.8545	1448	3010
SAN15	0.8495	1564	2957
SAN18	0.8448	1566	3024
SAN40	0.8326	1630	3138
SAN70	0.8140	1776	3578
SMMA20	0.8453	1539	2812
SMMA60	0.8037	1530	2912

TABLE 6. CHARACTERISTIC PARAMETERS FOR THE SIMPLE CELL MODEL EQUATION OF STATE

Polymer	V^* (cm^3/g)	T^* (K)	P^* (MPa)
PDMS	0.9180	3254	388.0
PS	0.9148	5167	599.0
PoMS	0.9302	5214	622.7
PMMA	0.7941	4783	765.0
PBMA	0.8931	4237	683.0
PCHMA	0.8906	4873	630.1
PEA	0.8394	4167	652.7
PEMA	0.8321	4190	772.8
PMA	0.8054	4318	768.3
PVAc	0.7731	3781	755.9
LPE	1.0978	4124	588.1
BPE	1.1179	4309	535.2
LDPE-A	1.1195	4430	549.8
LDPE-B	1.1267	4537	544.6
LDPE-C	1.1210	4461	552.3
PBD	1.0269	3811	681.5
PBD8	1.0488	4171	655.5
PBD24	1.0539	4212	629.9
PBD40	1.0532	4263	608.3
PBD50	1.0573	4357	575.4
PBD87	1.0679	4399	537.9
i-PB	1.1216	4600	460.6
PAr	0.7745	5257	776.0
PCL	0.8758	4478	632.0
PC	0.7835	4969	774.7
BCPC	0.6676	5046	762.1
HFPC	0.6116	4525	620.7
TMPC	0.8473	4875	609.4
PET	0.7193	5070	868.8
PIB	1.0393	4593	580.0
PI8	1.0617	4415	572.1
PI14	1.0516	4382	611.4
PI41	1.0521	4386	604.1
PI56	1.0495	4277	628.0
i-PP	1.1470	4712	425.8
a-PP	1.0758	3884	493.4
Phenoxy	0.8145	4837	903.2
PSO	0.7593	5358	844.5
PEO	0.8447	4226	696.8
PVME	0.9210	4298	677.0
PEEK	0.7465	5418	797.0
PTFE	0.4829	4488	323.9
PTHF	0.9609	4185	576.8
PMP	1.1652	4721	400.0
PA6	0.7956	7048	465.4
PA66	0.7841	5231	550.9
PECH	0.6997	4662	757.7
PVC	0.6855	4938	719.2
PPO	0.8355	4585	668.5
E/P50	1.1653	4982	453.5
EVA18	1.0874	4434	544.5
EVA25	1.0592	4368	537.6
EVA28	1.0509	4319	572.9
EVA4	1.0012	4326	584.9
SAN3	0.8995	4969	603.2
SAN6	0.8950	4761	640.8
SAN15	0.8875	5069	618.8
SAN18	0.8826	5070	630.4
SAN40	0.8696	5273	653.6
SAN70	0.8493	5676	723.1
SMMA20	0.8795	4908	600.7
SMMA60	0.8362	4884	622.9

TABLE 7. CHARACTERISTIC PARAMETERS FOR THE FLORY, ORWOLL, AND VRIJ EQUATION OF STATE

Polymer	V^* (cm^3/g)	T^* (K)	P^* (MPa)
PDMS	0.8264	5184	326.9
PS	0.8277	8118	405.2
PoMS	0.8457	8463	441.5
PMMA	0.7204	7717	568.8
PBMA	0.8087	6794	509.6
PCHMA	0.7772	7700	461.4
PEA	0.7563	6599	511.5
PEMA	0.7503	6703	641.3
PMA	0.7277	6894	599.2
PVAc	0.7090	6449	599.7
LPE	0.9818	6548	537.6
BPE	0.9992	6710	453.1
LDPE-A	0.9963	6774	469.5
LDPE-B	1.0025	6896	456.4
LDPE-C	0.9974	6809	471.0
PBD	0.9173	5521	454.4
PBD8	0.9404	6418	505.5
PBD24	0.9438	6432	480.9
PBD40	0.9419	6453	460.7
PBD50	0.9443	6514	425.0
PBD87	0.9511	6474	395.9
PB	0.9867	6838	403.9
PAr	0.6991	8470	651.2
PCL	0.7830	6754	486.2
PC	0.7070	8039	671.0
BCPC	0.6065	8287	611.0
HFPC	0.5521	7360	542.7
TMPC	0.7720	8156	514.2
PET	0.6452	8215	851.0
PIB	0.9455	7396	396.0
PI8	0.9480	6573	419.3
PI14	0.9415	6622	450.1
PI41	0.9415	6613	444.0
PI56	0.9400	6514	473.7
i-PP	1.0072	7011	397.4
a-PP	0.9755	6351	405.9
Phenoxy	0.7389	7869	713.2
PSO	0.6847	8664	738.2
PEO	0.7719	7147	601.6
PVME	0.8266	6607	512.8
PEEK	0.6642	8667	832.9
PTFE	0.4215	7088	404.9
PTHF	0.8774	7006	459.8
PMP	1.0203	7079	395.0
PA6	0.6896	9182	411.0
PA66	0.6885	7865	558.3
PECH	0.6321	7192	524.6
PVC	0.6210	7752	501.8
PPO	0.7472	7360	650.9
E/P50	1.0650	8377	360.2
EVA18	0.9724	6870	453.6
EVA25	0.9475	6770	444.8
EVA28	0.9416	6759	480.5
EVA40	0.8985	6766	475.5
SAN3	0.8129	7897	453.3
SAN6	0.8087	7622	503.1
SAN15	0.8020	8053	465.0
SAN18	0.7974	7999	461.0
SAN40	0.7869	8369	474.7
SAN70	0.7648	8701	500.9
SMMA20	0.7865	7521	475.8
SMMA60	0.7494	7558	491.3

TABLE 8. CHARACTERISTIC PARAMETERS FOR THE SIMHA-SOMCYNSKY EQUATION OF STATE

Polymer	V^* (cm^3/g)	T^* (K)	P^* (MPa)
PDMS	0.9592	7864	501.4
PS	0.9634	12840	715.9
PoMS	0.9814	13080	746.1
PMMA	0.8369	11940	926.4
PBMA	0.9358	10310	856.0
PCHMA	0.9047	12030	772.2
PEA	0.8773	11040	830.8
PEMA	0.8710	10190	987.0
PMA	0.8431	10460	969.1
PVAc	0.8126	9348	947.4
LPE	1.1406	9793	786.4
BPE	1.1674	10390	692.3
LDPE-A	1.1664	10580	716.2
LDPE-B	1.1734	10860	703.6
LDPE-C	1.1679	10660	718.8
PBD	1.0766	9225	815.0
PBD8	1.0960	10029	827.0
PBD24	1.1013	10122	792.3
PBD40	1.1003	10229	763.7
PBD50	1.1049	10450	716.8
PBD87	1.1149	10502	671.2
PB	1.1666	10920	603.7
PAr	0.8091	12390	1003.0
PCL	0.9173	10870	784.5
PC	0.8156	11830	1020.0
BCPC	0.6975	12190	987.8
HFPC	0.6317	10550	851.0
TMPC	0.8794	11540	819.2
PET	0.7426	11800	1194.0
PIB	1.0940	11360	686.6
PI8	1.1094	10578	711.5
PI14	1.0997	10541	760.1
PI41	1.1000	10547	750.7
PI56	1.0970	10284	786.9
i-PP	1.1884	11060	573.0
a-PP	1.1274	9494	627.7
Phenoxy	0.8529	11730	1139.0
PSO	0.7903	12770	1116.0
PEO	0.8812	10150	914.5
PVME	0.9632	10360	848.1
PEEK	0.7705	12580	1086.0
PTFE	0.4339	8126	658.1
PTHF	1.0087	10280	725.5
PMP	1.2050	11030	545.3
PA6	0.8327	16870	549.9
PA66	0.8195	12640	706.9
PECH	0.7343	11370	913.1
PVC	0.7320	12350	849.5
PPO	0.8602	10580	929.4
E/P50	1.2227	12220	572.0
EVA18	1.1341	10630	705.6
EVA25	1.1040	10440	697.8
EVA28	1.0949	10310	747.2
EVA40	1.0446	10360	753.9
SAN3	0.9416	12070	764.2
SAN6	0.9352	11490	823.8
SAN15	0.9299	12360	779.2
SAN18	0.9255	12380	785.3
SAN40	0.9124	12900	811.8
SAN70	0.8906	13790	874.7
SMMA20	0.9186	11800	764.0
SMMA60	0.8739	11780	791.1

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TABLE 9. CHARACTERISTIC PARAMETERS FOR THE SANCHEZ–LACOMBE EQUATION OF STATE

Polymer	V^* (cm^3/g)	T^* (K)	P^* (MPa)
PDMS	0.9022	466	288.5
PS	0.8929	688	371.5
PoMS	0.9142	725	405.7
PMMA	0.7805	668	516.9
PBMA	0.8789	596	442.1
PCHMA	0.8444	675	410.0
PEA	0.8216	581	450.6
PEMA	0.8189	602	567.5
PMA	0.7908	606	521.9
PVAc	0.7768	582	501.3
LPE	1.0734	596	479.8
BPE	1.0878	601	411.3
LDPE-A	1.0827	603	429.9
LDPE-B	1.0878	610	421.4
LDPE-C	1.0835	606	431.8
PBD	0.9851	462	440.2
PBD8	1.0167	554	456.0
PBD24	1.0194	553	437.1
PBD40	1.0161	552	422.5
PBD50	1.0169	553	394.8
PBD87	1.0224	547	373.6
PB	1.0739	609	377.5
PAr	0.7632	760	568.7
PCL	0.8477	589	453.4
PC	0.7737	728	574.4
BCPC	0.6651	753	513.6
HFPC	0.6077	680	455.4
TMPC	0.8493	752	432.4
PET	0.7102	761	726.1
PIB	1.0213	623	350.4
PI8	1.0196	555	392.3
PI14	1.0141	562	415.6
PI41	1.0139	561	410.7
PI56	1.0141	557	433.1
i-PP	1.0958	633	366.4
a-PP	1.0658	570	354.2
Phenoxy	0.8032	690	607.4
PSO	0.7496	787	635.3
PEO	0.8492	656	492.2
PVME	0.8930	567	463.0
PEEK	0.7310	804	713.7
PTFE	0.4515	630	357.2
PTHF	0.9587	626	385.6
PMP	1.1153	650	355.7
PA6	0.7406	785	422.5
PA66	0.7502	713	513.7
PECH	0.6803	606	482.4
PVC	0.6686	656	469.0
PPO	0.8220	681	554.1
E/P50	1.1652	753	301.4
EVA18	1.0580	612	407.7
EVA25	1.0307	602	397.0
EVA28	1.0259	604	424.1
EVA40	0.9772	600	419.8
SAN3	0.8860	701	391.6
SAN6	0.8841	686	429.7
SAN15	0.8733	712	404.1
SAN18	0.8672	703	400.1
SAN40	0.8556	734	412.1
SAN70	0.8222	726	466.4
SMMA20	0.8520	658	434.7
SMMA60	0.8125	662	446.0

TABLE 10. CHARACTERISTIC PARAMETERS FOR THE AHS + vdW EQUATION OF STATE

Polymer	V^* (cm^3/g)	T^* (K)	P^* (MPa)
PDMS	0.5629	7331	707.2
PS	0.6080	16542	731.7
PoMS	0.6222	17194	731.6
PMMA	0.5264	15173	986.3
PBMA	0.5820	12106	868.5
PCHMA	0.5719	15573	793.6
PEA	0.5294	10289	933.1
PEMA	0.5143	9659	1252.0
PMA	0.5111	10989	1105.2
PVAc	0.4891	9713	1246.9
LPE	0.6471	7955	1136.2
BPE	0.6801	9336	924.9
LDPE-A	0.6995	10596	845.6
LDPE-B	0.7117	11372	802.4
LDPE-C	0.7017	10761	846.0
PBD	0.6793	11913	867.5
PBD8	0.6764	11417	900.7
PBD24	0.6832	11794	843.1
PBD40	0.6865	12248	792.9
PBD50	0.6962	13112	710.5
PBD87	0.7069	13574	642.2
i-PB	0.7029	11078	680.3
PAr	0.4779	12060	1319.1
PCL	0.5249	8788	943.2
PC	0.4691	10088	1439.2
BCPC	0.4016	10381	1248.0
HFPC	0.3627	8598	946.0
TMPC	0.4638	7446	1496.6
PET	0.4041	8344	1984.3
PIB	0.7079	16891	687.8
PI 8	0.7030	13661	689.7
PI 14	0.6936	13326	756.7
PI 41	0.6943	13369	744.3
PI 56	0.6850	12371	817.0
i-P	0.6987	10082	691.1
a-PP	0.6935	10648	672.6
Phenoxy	0.515	212153	1377.3
PSO	0.4541	10866	1582.8
PEO	0.4974	8139	1439.8
PVME	0.5991	12258	900.3
PEEK	0.4164	8666	1801.4
PTFE	0.1807	3156	2185.6
PTHF	0.6026	10303	965.0
PMP	0.6857	8842	718.6
PA6	0.5321	23165	533.4
PA66	0.4818	11705	896.1
PECH	0.4707	15781	820.9
PVC	0.4747	19538	673.1
PPO	0.4664	7365	1538.8
E/P50	0.7592	14530	718.7
EVA18	0.6792	10625	852.7
EVA25	0.6669	10405	831.9
EVA28	0.6589	9791	934.8
EVA40	0.7130	10665	877.9
SAN3	0.5894	14537	737.9
SAN6	0.5662	11971	902.0
SAN15	0.5685	13512	858.1
SAN18	0.5763	14760	816.5
SAN40	0.5911	18577	678.8
SAN70	0.5745	19851	827.2
SMMA20	0.5622	12948	850.6
SMMA60	0.5344	12855	883.0

G. REFERENCES

1. J. R. Macdonald, Rev. Mod. Phys., **41**, 316 (1969).
2. P. G. Tait, Phys. Chem., **2**, 1 (1888).
3. G. Tammon, Z. Phys. Chem., **17**, 620 (1895).
4. F. D. Murnaghan, Proc. Nat. Acad. Sci., **30**, 244 (1944).
5. A. T. Hayward, Brit. J. Appl. Phys., **18**, 976 (1967).
6. W. G. Cutler, R. H. McMickle, W. Webb, J. Chem. Phys., **29**, 727 (1958).
7. I. C. Sanchez, J. Cho, W.-J. Chen, J. Phys. Chem., **97**, 6120 (1993).
8. I. C. Sanchez, J. Cho, W.-J. Chen, Macromolecules, **26**, 4234 (1993).
9. D. A. McQuarrie, "Statistical Mechanics", Harper & Row, New York, 1976.
10. I. Prigogine, V. Mathot, J. Chem. Phys., **20**, 49 (1952).
11. I. Prigogine, N. Trappeniers, V. Mathot, Disc. Faraday Soc., **15**, 93 (1953).
12. I. Prigogine, A. Bellemans, V. Mathot, "The Molecular Theory of Solutions", North Holland, Amsterdam, Netherlands, 1957.
13. R. Simha, T. Somcynsky, Macromolecules, **2**, 342 (1969).
14. P. J. Flory, R. A. Orwoll, A. Vrij, J. Am. Chem. Soc., **86**, 3515 (1964).
15. I. C. Sanchez, R. H. Lacombe, J. Phys. Chem., **80**, 2352 (1976).
16. I. C. Sanchez, R. H. Lacombe, J. Polym. Sci., Polym. Lett. Ed., **15**, 71 (1977).
17. R. J. Baxter, J. Chem. Phys., **49**, 2770 (1968).
18. R. J. Baxter, J. Chem. Phys., **52**, 2559 (1970).
19. B. Barboy, Chem. Phys., **11**, 357 (1975).
20. Y. C. Chiew, Molec. Phys., **70**, 129 (1990).
21. A. O. Malakhov, E. B. Brun, Macromolecules, **25**, 6262 (1992).
22. V. S. Mitlin, I. C. Sanchez, J. Chem. Phys., **99**, 533 (1993).
23. M. S. Wertheim, J. Stat. Phys., **35**, 19 (1984).
24. M. S. Wertheim, J. Stat. Phys., **35**, 35 (1984).
25. M. S. Wertheim, J. Stat. Phys., **42**, 459 (1986).
26. M. S. Wertheim, J. Stat. Phys., **42**, 477 (1986).
27. M. S. Wertheim, J. Chem. Phys., **87**, 7323 (1987).
28. W. G. Chapman, G. Jackson, K. E. Gubbins, Mol. Phys., **65**, 1057 (1988).
29. W. G. Chapman, K. E. Gubbins, G. Jackson, M. Radosz, Ind. Eng. Chem. Res., **29**, 1709 (1990).
30. R. Dickman, C. K. Hall, J. Chem. Phys., **85**, 4108 (1986).
31. G. R. Brannock, I. C. Sanchez, Macromolecules, **26**, 4970 (1993).
32. I. C. Sanchez, J. Cho, Polymer, **36**, 2929 (1995).
33. B. Hartmann, M. A. Haque, J. Appl. Polym. Sci., **30**, 1553 (1985).
34. P. A. Rodgers, J. Appl. Polym. Sci., **48**, 1061 (1993).
35. H. Shih, P. J. Flory, Macromolecules, **5**, 758 (1972).
36. S. Beret, J. M. Prausnitz, Macromolecules, **8**, 536 (1975).
37. A. Quach, R. Simha, J. Appl. Phys., **42**, 4592 (1971).
38. O. Olabisi, R. Simha, Macromolecules, **8**, 206 (1975).
39. J. E. McKinney, M. Goldstein, J. Res. Nat. Bur. Stand., **78A**, 331 (1974).
40. P. Zoller, J. Appl. Polym. Sci., **23**, 1051 (1979).
41. J. W. Barlow, Polym. Eng. Sci., **18**, 238 (1978).
42. Y. X. Yi, P. Zoller, J. Polym. Sci., Polym. Phys. Ed., **31**, 779 (1993).
43. P. Zoller, J. Appl. Polym. Sci., **23**, 1057 (1979).
44. P. Zoller, J. Polym. Sci., Polym. Phys. Ed., **20**, 1453 (1982).
45. C. K. Kim, PhD, Dissertation, The University of Texas at Austin, 1992.
46. P. Zoller, P. Bolli, J. Macromol. Sci. B: Phys., **18**, 555 (1968).
47. B. E. Eichinger, P. J. Flory, Macromolecules, **1**, 285 (1975).
48. P. Zoller, J. Polym. Sci., Polym. Phys. Ed., **16**, 1261 (1978).
49. R. K. Jain, R. Simha, J. Polym. Sci., Polym. Phys. Ed., **17**, 1929 (1979).
50. T. Ougizawa, G. T. Dee, D. J. Walsh, Macromolecules, **24**, 3834 (1991).
51. P. Zoller, T. A. Kehl, H. W. Starkweather Jr., G. A. Jones, J. Polym. Sci., Polym. Phys. Ed., **27**, 993 (1989).
52. P. Zoller, J. Appl. Polym. Sci., **22**, 633 (1978).
53. P. Zoller, J. Appl. Polym. Sci., **21**, 3129 (1977).
54. Y. Z. Wang, W. J. Chia, K. H. Hsieh, H. C. Tseng, J. Appl. Polym. Sci., **44**, 1731 (1992).
55. P. Zoller, H. H. Hoehn, J. Polym. Sci., Polym. Phys. Ed., **20**, 1385 (1982).
56. R. G. Griskey, N. Waldman, Mod. Plast., **43**, 245 (1966).
57. P. Zoller, R. K. Jain, R. Simha, J. Polym. Sci., Polym. Phys. Ed., **24**, 687 (1968).
58. C. K. Kim, D. R. Paul, Polymer, **33**, 2089 (1992).
59. K. H. Hellwege, W. Knappe, P. Lehmann, Kolloid-Z. Z. Polym., **183**, 110 (1962).
60. F. Danusso, G. Gianotti, Makromol. Chem., **61**, 164 (1963).
61. K. Nakagawa, Y. Ishida, Kolloid-Z. Z. Polym., **251**, 103 (1973).
62. V.-H. Karl, F. Asmussen, K. Ueberreiter, Makromol. Chem., **178**, 2037 (1977).
63. K.-Y. Wei, J. A. Cuculo, D. W. Ihm, J. Polym. Sci. Polym. Phys. Ed., **21**, 1091 (1983).
64. C. E. Weir, J. Res. Nat. Bur. Stand., **50**, 153 (1953).
65. O. Olabisi, R. Simha, Macromolecules, **8**, 211 (1975).
66. D. J. Walsh, G. T. Dee, P. W. Wojtkowski, Polymer, **30**, 1467 (1989).
67. Y. A. Fakhreddine, P. Zoller, J. Polym. Sci., Part B: Polym. Phys., **32**, 2445 (1994).
68. J. E. Lennard-Jones, A. F. Devonshire, Proc. R. Soc., London Ser. A, **163**, 53 (1937).

SECTION VII

SOLUTION PROPERTIES

Viscosity – Molecular Weight Relationships and Unperturbed Dimensions of Linear Chain Molecules

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A. INTRODUCTION

1. The Viscosity – Molecular Weight Relationship

The limiting viscosity number $[\eta]$ of a solution – which has long been called the intrinsic viscosity – is defined as

$$[\eta] = \lim_{c \to 0} \frac{\eta - \eta_0}{\eta_0 c} \qquad \text{(A1)}$$

in terms of the solvent viscosity, η_0, the solution viscosity, η, and the solute concentration, c. The concentration, c, is expressed in grams of solute per milliliter of solution or, more frequently, in grams of solute per 100 milliliters of solution, the limiting viscosity number being given in the reciprocal of these units, *i.e.*, in milliliters per gram or in deciliters per gram. Here, following the IUPAC 1952-recommendations (1), we adopt the former unit. The quantity $[\eta]$ of a polymer solution is a measure of the capacity of a polymer molecule to enhance the viscosity, which depends on the size and the shape of the polymer molecule. Within a given series of polymer homologs, $[\eta]$ increases with the molecular weight M; hence it is a measure of M.

Section C gives the limiting viscosity number – molecular weight relationships for polymers, in various solvents and at various temperatures. The table contains the constants of the equation

$$[\eta] = KM^a \qquad \text{(A2)}$$

which is known as the Mark–Houwink–Sakurada equation. It is now well established that for linear, flexible polymers, under special conditions of temperature or solvent, (usually known as the Flory "theta" temperature or solvent, (2)), the above equation becomes

$$[\eta]_\theta = K_\theta M^{0.50} \qquad \text{(A3)}$$

In the tables, the θ sign in parenthesis (next to the temperature data) indicates that the viscosity constants were obtained under the theta condition. Since Eq. (A3) is approximately valid over the whole molecular weight range, K_θ and $a = 0.50$ may be used without modification, outside of the molecular weight range in which they were determined. However, it must be noted that $[\eta]$ is rather sensitive to temperature in the vicinity of θ, especially when M is higher than 5×10^5.

In ordinary good solvents, the constants K and a obtained are valid only within a rather limited range of M (3,4). It is therefore, quite probable that the tabulated relationships are in error outside the indicated range of M (see eighth column in the table). As for the effect of temperature, however, both K and a mostly become insensitive to the temperature when a exceeds about 0.70, and they may be used in a 10-degree range on either side of temperature at which the constants were determined.

The method of determination of the molecular weight and the number of fractionated samples (Fr.) or whole polymer samples (WP) used to determine the $[\eta]$–M relationship are also given in the ninth and the sixth or seventh columns, respectively. The abbreviations used are as follows:

(A) Methods Yielding the Number-Average Molecular Weight (M_n)

CR	cryoscopy
EG	end-group titration
VOS	vapor pressure osmometry
EB	ebullioscopy
OS	osmotic pressure

(B) Methods Yielding the Weight-Average Molecular Weight (M_w)

LS	light scattering
LLS	low-angle laser light scattering
SE	sedimentation equilibrium
SEC, GPC	size exclusion chromatography gel permeation chromatography
SA	approach to the sedimentation equilibrium (Archibald's method)

(C) Empirical or Semi-Empirical Methods

EM	electron microscopy
GPC	gel permeation chromatography
LV	limiting viscosity number – molecular weight relationship
PR	analysis of polymerization rate (yielding M_n)
DV	diffusion and viscosity
MV	melt viscosity – molecular weight relationship
SD	sedimentation and diffusion
SV	sedimentation and viscosity

Thus, for example, the constants tabulated are for the $[\eta]$–M relationships, expressed in terms of M_n or M_w, if the method is specified as OS or LS, respectively; *i.e.*,

$$[\eta] = K_n M_n^a \qquad \text{(A4)}$$

or

$$[\eta] = K_w M_w^a \qquad \text{(A5)}$$

The values of K_n and K_w, especially the former, are greatly influenced by the molecular weight distribution (MWD) of the polymer samples, and caution must be taken in using these relationships.

To illustrate this effect, let us assume that

(i) Equation (A2) is applicable to the molecule i with molecular weight M_i over the whole range of M; *i.e.*,

$$[\eta]_i = KM_i^a \tag{A6}$$

(ii) The weight fraction w_i of the molecules i in a given sample can be represented by a continuous exponential function,

$$w_i(M_i) = [y^{h+1}/\Gamma(h+1)]\, M_i^h \exp(-yM_i) \tag{A7}$$

$$y = h/M_n = (h+1)/M_w \tag{A8}$$

or by the log-normal function,

$$w_i(M_i) = AM_i \exp[-p^2(\ln M_i/M_0)^2] \tag{A9}$$

where h, A, p, and M_0 are constants, and Γ represents the gamma function.
Then, since $[\eta] = \sum_i w_i[\eta]_i$, we obtain

$$K_n = K\Gamma(a+h+1)/h^a\,\Gamma(h+1) \tag{A10}$$

$$K_w = K\Gamma(a+h+1)/(h+1)^a\,\Gamma(h+1) \tag{A11}$$

for the exponential MWD, and

$$K_n = K(M_w/M_n)^{0.5a(a+1)} \tag{A12}$$

$$K_w = K(M_w/M_n)^{0.5a(a-1)} \tag{A13}$$

for the log-normal MWD (5). The values of K_n/K and K_w/K calculated by these equations are shown in Section B. This table may be used for estimating an error due to MWD in the determination of M.

As an example, let us assume that a given polymer sample has the exponential MWD with $M_w/M_n = 2.0$, while an available $[\eta]$–M_n equation has been obtained for samples with a narrow MWD, *e.g.*, $M_w/M_n = 1.1$. Further, let a be 0.70. Then, to find the correct value of M_n of the given sample from $[\eta]$, we must use Eq. (A4) with $K_n = 1.54K$, instead of the available equation with $K_n = 1.06K$. Use of the latter would lead to an overestimate M_n' which is related to the correct M_n by

$$[\eta] = 1.54KM_n^{0.70} = 1.06KM_n'^{0.70} \tag{A14}$$

The error amounts to about 70%, *i.e.*, $M_n' = 1.7M_n$. Thus, application of the viscosity equation written in M_n is to be restricted to within a narrow class of samples, unless an appropriate correction is made. On the other hand, if an $[\eta]$–M_w equation is available for the same pair of working and reference samples as above, we have

$$[\eta] = 0.951KM_w^{0.70} = 0.991KM_w'^{0.70} \tag{A15}$$

instead of Eq. (A14). Hence, the error M_w amounts to only 6% ($M_w' = 0.94M_w$), which will be negligible for more practical purposes.

Based on the above consideration, we classify the heterogeneity of polymers into four classes, A–D, as shown in the last column of the table in Section B, and indicate it in the tenth column of the tables in Section C, as a measure of the heterogeneity of the reference samples used.

It is desirable that readers select their own relationship by inspecting these data on heterogeneity as well as those on the number of samples and the molecular weight range. Generally speaking, a "good" $[\eta]$–M relationship is one that has been obtained on the basis of M_w for at least four samples of classes A and B (exceptionally C) or on the basis of M_n for those of class A (exceptionally B), whose molecular weights range over at least one half orders of magnitude.

In the "Remarks" column of Section C, we have occasionally indicated by the letter R, a "recommended" relationship for the convenience of readers. In the range of low molecular weight (mostly less than 10^4), the constant a becomes 0.50 irrespective of the solvent. This type of relationship cannot be used, even approximately, at higher molecular weights. This case is noted by the letter L. High conversion polymers are also marked by the letter H, where the $[\eta]$–M relationships are less reproducible due to chain branching than are ordinary ones. The abbreviations used are as follows:

A	Narrow MWD polymers, or well-fractionated polymers, $M_w/M_n \leq 1.25$
B	Ordinary fractionated polymers, $1.30 \leq M_w/M_n \leq 1.75$
C	Poorly fractionated polymers or most probable MWD polymers, $1.8 \leq M_w/M_n \leq 2.4$
D	Wide MWD polymers, $M_w/M_n \geq 2.5$
H	High conversion polymers, including branches
L	Limited to low-molecular-weight polymers
R	Recommended relationship

In Section C, polymers are arranged according to their structure in subgroups. Within each subgroup, the polymers are, in principle, given in alphabetical order. Within each polymer, the solvents are also arranged in alphabetical order, followed by the mixed solvents.

Chain configurational data are occasionally given in the first column. The data given in parentheses refer to only one set of viscosity constants listed in the same row, while the data given without parentheses refer to a series of sets listed in the same and succeeding rows. Thus, for example, the data "N content, 13.9 wt.%" are effective only for the sixth row of cellulose trinitrate, and the data "95%-*cis*, 1%-*trans*, 4%-1,2" are effective for the fourth to eighth rows of polybutadiene.

The tables in Section C are essentially based on the table published by Kurata and Stockmeyer (3). Data were also taken from tables published by Peterlin (7), Meyerhoff (8), Elias (9), and Krause (10), the last one including a number of unpublished data on acrylic and methacrylic polymers. We are also grateful to these authors. Thanks are also due to J. Brandrup and K. Kamide for their help with this compilation.

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2. Unperturbed Dimensions of Linear Chain Molecules

The mean-square end-to-end distance $\langle r^2 \rangle$ of a linear chain molecule in solution is usually expressed in terms of two basic quantities, the unperturbed mean-square end-to-end distance $\langle r^2 \rangle_0$ and the expansion factor α; *i.e.*,

$$\langle r^2 \rangle = \langle r^2 \rangle_0 \alpha^2 \tag{A16}$$

The latter quantity α represents the effect of "long-range interactions" which can be described as an osmotic swelling of the chain by the solvent–polymer interactions, while the unperturbed dimension $\langle r^2 \rangle_0$ represents the effect of "short-range interactions" such as bond angle restrictions and steric hindrances to internal rotation. The steric hindrances are also influenced by the torques exerted on the chain by solvent molecules, but the effect is rather small in many cases (11).

For sufficiently long chain, $\langle r^2 \rangle_0$ becomes proportional to $\sum_i n_i l_i^2$ where n_i is the number of the ith-kind bond of length l_i. The quantity C_∞, defined by

$$C_\infty = \lim_{n \to \infty} \langle r^2 \rangle_0 \Big/ \sum_i n_i l_i^2 \tag{A17}$$

is often called the characteristic ratio and it serves as a measure of the effect of short-range interactions.

The freely rotating state is a hypothetical state of the chain in which the bond angle restrictions are retained, but the steric hindrances to internal rotation are released. The mean-square end-to-end distance of the freely rotating chain $\langle r^2 \rangle_{0f}$ can be readily calculated from the given basic structure of the chain. For instance, if the chain consists of only one kind of bond of length l, we obtain

$$\langle r^2 \rangle_{0f} = nl^2[(1 + \cos\theta)/(1 - \cos\theta)] \tag{A18}$$

where n is the number of bonds and θ is the supplement of the valence bond angle. For vinyl polymer chains, $l = 0.154$ [nm], $\cos\theta = 1/3$, and $n = M/m = 2M/M_u$; and hence

$$(\langle r^2 \rangle_{0f}/M)^{1/2} = 0.308/M_u^{1/2} = 0.218/m^{1/2}\ [\text{nm}] \tag{A19}$$

where M_u is the molecular weight of the repeating unit and m is the average molecular weight per skeletal link. Similar expressions for r_{0f} $(= \langle r^2 \rangle_{0f}^{1/2})$ can be also obtained for more complicated chains. The results are summarized in Section D.

The ratio of $\langle r^2 \rangle_0$ to $\langle r^2 \rangle_{0f}$, then, represents the effect of steric hindrance on the average chain dimensions:

$$\sigma = r_0/r_{0f} = (\langle r^2 \rangle_0/\langle r^2 \rangle_{0f})^{1/2} \tag{A20}$$

The quantity σ is independent of n. Section E gives a list of the unperturbed dimensions of linear chain molecules which were obtained under various conditions of solvent and temperature. The values of $r_0/M^{1/2}$, $r_{0f}/M^{1/2}$, σ and C_∞ are given, together with the experimental values of $S_{0z}/M_w^{1/2}$, a_p, or K_0 from which r_0 was computed. S_{0z} which is the abbreviation of $\langle S^2 \rangle_{0z}^{1/2}$ is the z-average value of the unperturbed radius of gyration, a_p is the persistence length, and K_0 is the viscosity constant corresponding to K_θ in Eq. (A3). The methods used to determine these quantities are also indicated in the tenth column of the tables by using the following abbreviations:

(A) Light scattering
- LT Zimm's plot in a theta solvent yielding $S_{0z}/M_w^{1/2}$. After a heterogeneity correction is made, the tabulated value of $r_0/M^{1/2}$ $(= 6^{1/2} S_{0w}/M_w^{1/2})$ is obtained.
- LD dissymmetry method in a theta solvent. Less reliable for heterogeneous samples than the former method.
- LG Zimm's plot in good solvents yielding $S_z/M_w^{1/2}$. After corrections for the excluded volume effect and heterogeneity are made, the tabulated value of $r_0/M^{1/2}$ is obtained (3,12).

(B) X-ray small-angle scattering
- XS the persistence length a_p is obtained irrespective of the solvent nature. The tabulated values of $r_0/M^{1/2}$ are the asymptotic values for infinitely high molecular weight (13,14).

(C) Limiting viscosity number
- VT viscosity – molecular weight relationship in a theta solvent. Equation (A3) $r_0/M^{1/2}$ is calculated by the Flory and Fox relation, $K_0 = \Phi_0 (r_0/M^{1/2})^3$. The following values of Φ_0 were used:
  - 2.7×10^{23} for well-fractionated polymers (class A in Section C)
  - 2.5×10^{23} for ordinary fractionated polymers (class B)
  - 2.1×10^{23} for poorly fractionated or unfractionated polymers (class C or D)
- VG viscosity – molecular weight relationship in good solvents. K_0 was estimated by using the Kurata–Stockmeyer–Fixman plot (3,4) or other analogous plots (12).
- VWC viscosity analyzed by the wormlike cylinder model.
- VA viscosity in good solvents. The correction of excluded volume effect is made by using the Flory–Krigbaum–Orofino theory of the second virial coefficient A_2 or other analogous theories (12).

(D) Method yielding the temperature dependence of r_0.
ST stress – temperature coefficient of undiluted or swollen samples.
VTe viscosity – temperature coefficient of the intrinsic viscosity

The polymers are arranged in Section E in the same order as in Section C. For each polymer, smoothed values of $r_0/M^{1/2}$, σ, and C_∞, which were mostly obtained by VT or VG, are given in the first line, followed by some typical values obtained by more direct methods such as LT or XS. The listed values of $r_0/M^{1/2}$ sometimes scatter appreciably, reflecting the difficulty involved, both experimental and theoretical, in the determination of this quantity. Especially in the case of cellulose chains, the right magnitude of r_0 is yet in controversy (542, 549,3,691,696,688,678,686,12). In recent papers, emphasis has often been put on the effect of temperature or solvent on the unperturbed dimensions. These data are put together at the end of the tabulation for each polymer. Section E is also based on the tables published by Kurata and Stockmeyer (3).

B. EFFECT OF MOLECULAR WEIGHT DISTRIBUTION ON THE VISCOSITY CONSTANT K

	$a = 0.5$		$a = 0.6$		$a = 0.7$		$a = 0.8$		$a = 0.9$		$a = 1.0$		
M_w/M_n	K_n/K	K_w/K	K_n/K	K_w/K	K_n/K	K_w/K	K_n/K	K_w/K	K_n/K	K_w/K	K_n/K	K_w/K	Class
MOLECULAR WEIGHT DISTRIBUTION: EXPONENTIAL TYPE (Eq. (A7))													
30	4.87	0.890	6.91	0.897	9.85	0.911	14.18	0.933	20.56	0.963	30	1	D
15	3.46	0.893	4.57	0.900	6.08	0.914	8.16	0.935	11.02	0.964	15	1	D
10	2.83	0.896	3.59	0.903	4.59	0.917	5.91	0.937	7.67	0.965	10	1	D
5	2.03	0.907	2.40	0.913	2.85	0.925	3.42	0.943	4.12	0.968	5	1	D
3	1.60	0.921	1.79	0.926	2.02	0.936	2.29	0.952	2.62	0.973	3	1	D
2	1.33	0.940	1.43	0.943	1.54	0.951	1.68	0.963	1.83	0.979	2	1	C
1.75	1.25	0.948	1.33	0.951	1.42	0.958	1.51	0.968	1.63	0.982	1.75	1	B
1.50	1.18	0.959	1.23	0.961	1.28	0.967	1.35	0.975	1.42	0.986	1.50	1	B
1.25	1.09	0.975	1.12	0.977	1.15	0.980	1.18	0.985	1.21	0.991	1.25	1	A
1.10	1.04	0.989	1.05	0.989	1.06	0.991	1.07	0.993	1.09	0.996	1.10	1	A
MOLECULAR WEIGHT DISTRIBUTION: NORMAL TYPE (Eq. (A9))													
30	3.58	0.654	5.12	0.665	7.57	0.700	11.58	0.762	18.32	0.858	30	1	D
15	2.76	0.713	3.67	0.723	5.01	0.753	7.03	0.805	10.13	0.885	15	1	D
10	2.37	0.750	3.02	0.759	3.94	0.785	5.25	0.832	7.16	0.902	10	1	D
5	1.83	0.818	2.17	0.824	2.61	0.845	3.19	0.879	3.96	0.930	5	1	D
3	1.51	0.872	1.69	0.877	1.92	0.891	2.21	0.916	2.56	0.952	3	1	D
2	1.30	0.917	1.39	0.920	1.51	0.930	1.65	0.946	1.81	0.969	2	1	C
1.75	1.23	0.932	1.31	0.935	1.40	0.943	1.50	0.956	1.61	0.975	1.75	1	B
1.50	1.16	0.951	1.21	0.953	1.27	0.958	1.34	0.968	1.41	0.982	1.50	1	B
1.25	1.09	0.973	1.11	0.974	1.14	0.977	1.17	0.982	1.21	0.990	1.25	1	A
1.10	1.04	0.988	1.05	0.989	1.06	0.990	1.07	0.992	1.08	0.996	1.10	1	A

C. TABLES OF VISCOSITY – MOLECULAR WEIGHT RELATIONSHIPS, $[\eta] = KM^a$

TABLE 1. MAIN-CHAIN ACYCLIC CARBON POLYMERS

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
1.1. POLY(DIENES)										
Poly(butadiene)										
linear	Dioxane	12.1	139	0.52	4	–	2.1–23.2	LG	B–C	889
	Toluene	35	16.9	0.765	6	–	2.1–23.2	LG	B–C	889
ring	Toluene	35	11.8–10.6	0.765	3	–	3.8–6.1	LG	B–C	889
98%-*cis*, 2%-1,2	Benzene	30 (θ)	33.7	0.715	9	–	5–50	OS	A,R	15
	Isobutyl acetate	20.5	185	0.50	6	–	5–50	OS	A	15
	Toluene	30	30.5	0.725	9	–	5–50	OS	A	15
95%-*cis*, 1%-*trans*, 4%-1,2	Benzene	30	8.5	0.78	4	–	15–50	LS	A	16
	Cyclohexane	30	11.2	0.75	4	–	15–50	LS	A	16
	5-Methyl-2-hexanone	12.6 (θ)	150	0.50	4	–	15–35	LS	B	17

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TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	3-Pentanone	10.3 (θ)	152	0.50	4	–	10–25	LS	B	17
	Toluene	30	33.9	0.688	8	–	10–65	OS	A	18
94%-*cis*, 4%-*trans*, 2%-1,2	Benzene	25	41.4	0.70	8	–	9–120	OS	A	19
	Dioxane	20.2 (θ)	205	0.50	8	–	9–120	OS	A	19
92%-*cis*, 3%-*trans*, 5%-1,2	Benzene	32	10	0.77	13	–	10–160	LS	B,R	20
51%-*trans*, 43%-*cis*, 6%-1,2	Toluene	30	39	0.713	6	–	11–25	OS	A	21
57%-*trans*, 36%-*cis*, 7%-1,2	Tetrahydrofuran	30	25.6	0.74	2	–	1–57	LS	A	733
71%-*trans*, 4%-*cis*, 25%-1,2	Cyclohexane	25	12	0.77	8	–	230–880	LS	C	22
79%-*trans*, 21%-*cis*	Cyclohexane	20	36	0.70	12	–	23–130	LS	B,R	23
97%-*trans*, 3%-1,2	Cyclohexane	40	28.2	0.70	7	–	4–17	LS	B	24
	Toluene	30	29.4	0.753	6	–	5–16	OS	A	25
ca. 100%-*cis*	Benzene	32	14.5	0.76	8	–	18–50	LS	A	26
	Heptane/hexane (1/1, v/v)	20	138	0.53	5	–	?	SD	A	27
65%-1,2, 25%-*trans*, 10%-*cis*	Toluene	25	110	0.62	8	–	7–70	OS	B	28
5°C-emulsion, randomly branched	3-Pentanone	24 (θ)	$M^{2/3}[\eta] = 7.15 + 3.47M$		10	–	10–100	OS	C	29
50°C-emulsion, randomly branched	Benzene	5 (θ)	$M^{2/3}[\eta]^{4/3} = 4.61 + 0.3287M$		16		5–124	OS	C	29
Poly(butadiene-*co*-acrylonitrile), Buna-N rubber										
	Acetone	25	50	0.64	5	–	2.5–10	OS	B	28
	Benzene	25	13	0.55	5	–	2.5–10	OS	B	28
	Chloroform	25	54	0.68	5	–	2.5–10	OS	B	28
	Toluene	25	49	0.64	7	–	2.5–40	OS	B	28
Poly(butadiene-*co*-styrene), Buna-S, GR-S, or SBR rubber										
	Benzene	25	52.5	0.66	24	–	1–160	OS		45
		25	54	0.66	8	–	1–165	OS	B	46
	Cyclohexane	30	31.6	0.70	6	–	5–25	OS	A	47
	2-Pentanone	21 (θ)	185	0.50	6	–	5–25	OS	A	47
	Toluene	25	52.5	0.667	25	–	2.5–50	OS	B	28
		30	16.5	0.78	–	9	3–35	OS		48
		30	37.9	0.71	6	–	5–25	OS	A	47
linear fraction	Toluene	30	21.4	0.74	15	–	3–20	OS	A,R	41
branched fraction	Toluene	30	535	0.48	20	–	20–100	OS	B	41
Poly(1-butenylene-*co*-vinylethylene)										
43%-1,2	1,4-Dioxane	15.7 (θ)	173	0.50	6	–	0.88–22	LLS, SEC, OS	A	890
	Tetrahydrofuran	30	32.3	0.72	6	–	0.88–22	LLS, SEC, OS	A	890
Poly(2-*tert*-butylbutadiene)	Benzene	21	4.2	0.80	–	8	6–90	SD	A	30
	Octane	21	4.2	0.80	–	7	6–35	SD	A	30
Poly(chloroprene)										
Neoprene CG	Benzene	25	2.02	0.89	10	–	6–150	OS	B	31
Neoprene GN	Benzene	25	14.6	0.73	16	–	2–96	OS	B	32
Neoprene W	Benzene	25	15.5	0.71	8	–	5–100	OS	B	33
		25	15.5	0.72	9	–	5–80	LS	B,R	34
	Butanone	25 (θ)	113	0.50	7	–	15–300	LS	A	35
	Butyl acetate	25	37.8	0.62	7	–	15–300	LS	A	35
	Carbon tetrachloride	25	22.1	0.69	7	–	15–300	LS	A	35
	Cyclohexane	45.5 (θ)	107	0.50	7	–	15–70	LS	B	34
type, unspecified	Toluene	25	50	0.615	13	–	4–120	OS	B	28
Poly(isoprene)										
natural rubber	Benzene	30	18.5	0.74	–	4	8–28	OS	C	37
	Cyclohexane	27	30	0.70	–	1	ca. 185	LS, SD	C	38
	4-Methyl-2-pentanone	35	60.7	0.57			5–100	LS	B	698
	2-Pentanone	14.5 (θ)	119	0.50	–	4	8–28	OS	C	37
	Toluene	25	50.2	0.667	20	–	7–100	OS	B,R	39
		35	17.4	0.74			5–100	LS	B	698
synthetic *cis*	Hexane	20	68.4	0.58	5	–	5–80	SD	A	40
	Toluene	30	8.51	0.77	5	–	20–100	LS	A	41
85–91%-*cis*	Toluene	30	20.0	0.728	–	12	14–580	LS	A,R	42
		30	15	0.74	–	16	2–15	PR	A	43
	2,2,4-Trimethylpentane	30	22.2	0.683	–	8	23–580	LS	A	42
	Heptane/propanol (78/22, v/v)	30	37	0.63	–	6	43–580	LS	A	42

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
84%-*cis*, 14%-*trans*, 2%-1,2	Benzene	25	13.3	0.78	20	–	2–80	OS	B	44
		25	11.2	0.78	25	–	2–60	OS	B	44
	Dioxane	34 (θ)	145	0.50	30	–	2–50	OS	B	44
71%-*cis*, 22%-*trans*, 7%-3,4	Cyclohexane	35	20.2	0.73	5	–	2–32	OS	A	786
	Dioxane	35	94	0.53	5	–	2–32	OS	A	786
	4-Methyl-2-pentanone	35	76	0.56	5	–	2–32	OS	A	786
	Toluene	35	17.5	0.74	5	–	2–32	OS	A	786
gutta percha	Benzene	25	35.5	0.71	9	–	0.2–5	OS	A,R	19
	Dioxane	47.7 (θ)	191	0.50	9	–	0.2–5	OS	A	19
	Propyl acetate	60 (θ)	232	0.50	–	3	10–20	OS	C	37
synthetic *trans*	Benzene	32	43.7	0.65	24	–	8–140	LS	C	26
synthetic *trans* (98%)	Benzene	30	18.1	0.722			14–77	LS	B	699
	Cyclohexane	30	16.2	0.736	8	–	14–77	LS	B	699
	Hexane	30	13.8	0.711	7	–	14–77	LS	B	699
	Toluene	30	17.6	0.729			14–77	LS	B	699
98% 1,4-*trans*	Benzene	30	18.1	0.722	6	–	14–69	LS	B	887
	Cyclohexane	30	16.2	0.736	8	–	16–130	LS	B	887
	n-Hexane	30	13.8	0.711	7	–	16–105	LS	B	887
	Toluene	30	17.6	0.729	5	–	14–77	LS	B	887
70% *cis*-1,4, 23% *trans*-1,4 7% 3,4	Cyclohexane	25	20.5	0.730	11	–	1.5–342	LS	A	891
76% *cis*-1,4, 19% *trans*-1,4 5% 3,4 star type	Cyclohexane	25	18.0	0.74	5	–	33–724	LS	A	892
70% *cis*-1,4, 23% *trans*-1,4 7% 3,4 (mol. wt. of arm = various)										
3 arm	Cyclohexane	25	12.5	0.757	–	–	–	LS	–	893
4 arm	Cyclohexane	25	11.6	0.753	–	–	–	LS	–	893
8 arm	1,4-Dioxane	34	75.5	0.493	–	–	–	LS	–	893
	Toluene	34	9.47	0.726	–	–	–	LS	–	893
12 arm	1,4-Dioxane	34	53.9	0.501	–	–	–	LS	–	893
	Toluene	34	5.70	0.741	–	–	–	LS	–	893
16 arm	1,4-Dioxane	34	37.4	0.502	–	–	–	LS	–	893
	Toluene	34	3.04	0.764	–	–	–	LS	–	893
trans 1,4	Benzene	30	18.1	0.722	6	–	14–70	LS	B	894
	Cyclohexane	30	16.2	0.736	6	–	14–70	LS	B	894
	Hexane	30	13.8	0.711	6	–	14–70	LS	B	894
	Toluene	30	17.6	0.729	4	–	14–70	LS	B	894
Poly(isoprene)-*block*-poly(styrene), A_k-B_n										
k/*n*, (50/50, w/w)	Cyclohexane	35	21.8	0.68	5	–	3–54	OS	A	786
	Dioxane	35	32.6	0.63	5	–	3–54	OS	A	786
	4-Methyl-2-pentanone	35	53.1	0.57	5	–	3–54	OS	A	786
	Toluene	35	14.6	0.73	5	–	3–54	OS	A	786
k/*n*, (25/75, w/w)	Cyclohexane	35	39.3	0.60	6	–	3–48	OS	A	786
	Dioxane	35	24.8	0.65	6	–	3–48	OS	A	786
	4-Methyl-2-pentanone	35	62.8	0.55	6	–	3–48	OS	A	786
	Toluene	35	14.3	0.72	6	–	3–48	OS	A	786
Poly(isopropenylethylene-*co*-1-methyl-1-vinylethylene) 20–25%-1,2,70–75%-3,4										
	2-Octanol	30.5 (θ)	102	0.50	7	–	1.3–27	LLS, SEC, OS	A	890
	Tetrahydrofuran	30	11.6	0.77	7	–	1.3–27	LLS, SEC, OS	A	890
Poly(1-methyl-1-butylene-*co*-isopropenylethylene) 51%-1,4,49%-3,4										
	2-Octanol	41.3	78	0.52	8	–	1.1–14	LLS, SEC, OS	A	890
	Tetrahydrofuran	30	14.6	0.75	8	–	1.1–14	LLS, SEC, OS	A	890
Poly(1,1,2-trichlorobutadiene)	Benzene	25	31.6	0.66	11	–	25–130	LS		36
1.2. POLY(ALKENES), POLY(ACETYLENES)										
Poly(alkene) C_{10}–C_{18}	Toluene	25	12.7	1.04	12	–	2–18	LS	B	86
Poly(alkene) C_{12}–C_{18}	Cetane	38	21	0.61	10	–	4–700	LS	B	87
Poly(1-butene)	2-Octanol	23.6	60.5	0.52	6	–	2.7–55	LLS, SEC	A	896
	Tetrahydrofuran	25	8.24	0.76	6	–	2.7–55	LLS, SEC	A	896
atactic	Anisole	86.2 (θ)	123	0.50	3	–	10–130	LS	C	81
	Benzene	30	22.4	0.72	11	–	0.03–0.5	EG	B,L	82

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TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	Ethylcyclohexane	70	7.34	0.80	5	–	4–130	LS	C	81
	Phenyl ether	141 (θ)	104	0.50	5	–	2–66	OS	A	700
isotactic	Anisole	89 (θ)	111	0.50	5	–	4–57	OS	B	700
	Decalin	115	9.49	0.73	6	–	4.5–90	LS		83
	Ethylcyclohexane	70	7.34	0.80	4	–	8–94	LS	A	81
	Heptane	35	4.73	0.80	6	–	4.5–90	LS		83
		60	15.0	0.69	6	–	4.5–90	LS		83
	Nonane	80	5.85	0.80	4	–	11–94	LS	A	81
	Phenetole	64.5 (θ)	113	0.50	5	–	4–57	OS	B	700
	Phenyl ether	148 (θ)	103	0.50	5	–	4–57	OS	B	700
	1,2,4,-Trichlorobenzene	135	11.8	0.729				GPC		701
	Cyclohexane/propanol (80/20, v/v)	35	102	0.59	6	–	3–73	LS	B–C	702
	(70/30, v/v)	35	253	0.51	6	–	3–73	LS	B–C	702
	(65/35, v/v)	35	497	0.44	6	–	3–73	LS	B–C	702
Poly(ethylene)										
low pressure	Biphenyl	127.5 (θ)	323	0.50	4	–	2–30	LV	B	58
		130 (θ)	302	0.50	5	–	5.7–27	LS, GPC		703
	1-Chloronaphthalene	125	138	0.58	?	?	?	LS	?	59
		125	18.4	0.78	10	–	5–100	LS		60
		125	43	0.67	10	–	5–100	LS	C,D	61
		129	27.1	0.71	26	–	5–100	LS	D	62
	Decalin	135	67.7	0.67	–	>10	3–100	LS	D	63
		135	46	0.73	23	–	3–64	LS		64
		135	62	0.70	7	–	2–105	LS	B,R	65,66
		135	58.5	0.725	9	–	0.4–50	OS	B	67,68
		135	62	0.70	7	–	3–120	GPC, LS	B	704
	Decanol	153.3 (θ)	302	0.50	?	–	2–105	LV	B	58
	Diphenyl ether	161.4 (θ)	295	0.50	6	–	2–105	LS	B	65
	Diphenylmethane	142.2 (θ)	315	0.50	?	–	2–105	LV	B	58
	Dodecanol	137.3 (θ)	307	0.50	5	–	2–105	LV	B	58
		138 (θ)	316	0.50	–	8	8–32	LS	F	69
		144.5		0.61	6	–	0.8–59	LS		705
	Octanol	180.1 (θ)	286	0.50	?	–	2–105	LV	B	58
	Tetralin	105	16.2	0.83	4	–	13–57	LS	C	70
		120	23.6	0.78	36	–	5–100	LS		60
		120	32.6	0.77	20	–	0.3–50	LS	B	71
		130	43.5	0.76	6	–	2–30	OS	B	71
		130	51	0.725	9	–	0.4–50	OS	B,R	72
		130	37.8	0.72	–	10	8–17	LS	D	73
	1,2,4-Trichlorobenzene	135	95.4	0.64	4	–	3–45	LS	A	706
		135	51	0.706	19	–	0.8–123	GPC, LS	B	704
		135	51.6	0.691	4	–		GPC		701
	3,5,5-Trimethylhexyl acetate	121		0.55	6	–	1–59	LS		705
	p-Xylene	105	16.5	0.83	4	–	13–50	LS	C	70
		105	17.6	0.83	8	–	1–18	OS	C	74
		105	51	0.725	?	–	0.4–50	LV	B,R	75
	Paraffin wax ($M_n = 390 \pm 10$)									
		150	(42)	(0.65)	9	–	0.04–11	LS	D	76
high pressure	Decalin	70	38.73	0.738	8	–	0.2–3.5	OS	B	77
	p-Xylene	75	135	0.63	–	22	0.2–7.6	OS	D	78
		81	105	0.63	7	–	1–10	OS	D	79
(normal paraffin)	Carbon tetrachloride	20	$[\eta] = -1.14 + 0.104\,M$		–	7	0.024–0.048	CR	A	80
Poly(ethylene-*alt*-tetrafluoroethylene)										
	Diisobutyl adipate	240	2.3	0.71	–	3	54–116	LS	B	898
Poly(ethylene-*co*-isopropylethylene)										
	n-Hexyl acetate	65	171	0.51	6	–	0.92–23	LLS, SEC, OS	A	890
	Tetrahydrofuran	30	71.3	0.64	6	–	0.92–23	LLS, SEC, OS	A	890
Poly(ethylene-*co*-propylene)	Benzene	19.0	201	0.502	6	–	2.8–39	LS	A	897
		21.4	136	0.543	6	–	2.8–39	LS	A	897
	n-Decyl acetate	5.0	162	0.523	6	–	2.8–39	LS	A	897
	n-Heptyl acetate	38.0	156	0.522	6	–	2.8–39	LS	A	897

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(ethylene-*co*-propylene-*co*-diene), EPDM rubber										
	Cyclohexane	40	53.1	0.75	20	–	3–30	OS	A	41
Poly(isobutene)	Anisole	105 (θ)	91	0.50	–	–	18–188	LV	B	49
	Benzene	24 (θ)	107	0.50	15	–	18–188	LV	B	49
		25	83	0.53	9	–	0.05–126	OS, CR	B,R	50
		30	61	0.56	9	–	0.05–126	OS, CR	B	50
		40	43	0.60	9	–	0.05–126	OS, CR	B	50
		60	26	0.66	9	–	0.05–126	OS, CR	B	50
	Benzene	25	100	0.504	–	9	1.35–148	SEC, LLS	A–B	1010
	Carbon tetrachloride	30	29	0.68	12	–	0.05–126	OS, CR	B	50
	Chloroform	25	71	0.55	12	–	5.7–490	LS	A(?)	899
	Cyclohexane	10	25.6	0.639	–	6	2.6–110	LV	A–B	901
		20	24.2	0.697	–	6	2.6–110	LV	A–B	901
		30	23.3	0.699	–	6	2.6–110	LV	A–B	901
		40	22.9	0.701	–	6	2.6–110	LV	A–B	901
		50	22.4	0.702	–	6	2.6–110	LV	A–B	901
	Cyclohexane	25	40	0.72	6	–	14–34	OS	B	51
		30	27.6	0.69	7	–	4–71	OS	A,R	52
		30	26.5	0.69	12	–	0.05–126	OS, CR	B	50
	Cyclohexane	25	13.5	0.740	–	18	6.4–161	SEC, LLS	A–B	1010
	Decalin	25	22	0.70	6	–	530–1680	LS	A–B	53,54
	Diisobutylene	20	36	0.64	23	–	1–130	OS	A,R	55,52
		25	130	0.50	5	–	0.4–2.5	OS	A,L	56
	n-Heptane	25	15.8	0.697	–	7	0.84–148	SEC, LLS	A–B	1010
	Isooctane	10	38.1	0.624	–	6	2.6–110	LV	A–B	901
		20	36.8	0.626	–	6	2.6–110	LV	A–B	901
		30	36.2	0.627	–	6	2.6–110	LV	A–B	901
		40	34.6	0.631	–	6	2.6–110	LV	A–B	901
		50	33.6	0.633	–	6	2.6–110	LV	A–B	901
	Phenetole	86 (θ)	91	0.50	4	–	5–188	LV	B	49
	Toluene	0	40	0.60	8	–	1–146	LV	B	50
		15	24	0.65	6	–	1–146	LV	B	50
		25	87	0.56	6	–	14–34	OS	B	51
		30	20	0.67	5	–	1–146	LV	B,R	50
		50	20	0.68	6	–	1–146	LV	B	50
		60	13.5	0.71	4	–	11–146	LV	B	50
		90	12.6	0.72	3	–	46–146	LV	B	50
oligomer–polymer	Benzene	25 (θ)	($M_w > 10^5$)	0.50	19	–	0.011–179	LS	A	904
			($M_w < 10^5$)	Not const.						
	Isoamyl isovalerate	25 (θ)	($M_w > 10^5$)	0.50	19	–	0.011–179	LS	A	904
			($M_w < 10^5$)	Not const.						
Poly(isobutene-*co*-isoprene), butyl rubber	Benzene	22.8 (θ)	115	0.50	5	–	15–72	LS	A	787
	Carbon tetrachloride	25	10.7	0.78	6	–	10–30	OS	A	57
	5-Methyl-3-heptanone	55.5 (θ)	109	0.50	5	–	15–72	LS	A	787
	Toluene	25	66	0.60	5	–	15–30	OS	A	57
		30	21.4	0.678	8	–	10–30	OS	A	57
Poly(methylbutylene)	*n*-Hexyl acetate	60.9	169	0.51	8	–	0.84–60	LLS, SEC, OS	A	890
	Tetrahydrofuran	30	42.2	0.68	8	–	0.84–60	LLS, SEC, OS	A	890
Poly(isopropylethylene-*co*-1-methyl-1-ethylethylene)										
	2-Octanol	26.2	92	0.49	7	–	1.5–28	LLS, SEC, OS	A	890
	Tetrahydrofuran	30	10.3	0.75	7	–	1.5–28	LLS, SEC, OS	A	890
Poly(3-methyl-1-butene)	Diisobutylene	20	42	0.63	6	–	1–20	LS	A	85
Poly(1-methylbutylene-*co*-isopropylethylene)										
	2-Octanol	53.3	74	0.52	8	–	1.1–43	LLS, SEC, OS	A	890
	Tetrahydrofuran	30	29.8	0.68	8	–	1.1–43	LLS, SEC, OS	A	890
Poly(4-methyl-1-pentene)	Biphenyl	194.6 (θ)	152	0.50	7	–	6–30	OS	B	707
	Decalin	130	19.5	0.75	5	–	6–30	OS	B	707
	Diphenyl ether	210.0 (θ)	158	0.50	6	–	6–30	OS	B	707
	Diphenylmethane	176.6 (θ)	160	0.50	6	–	6–30	OS	B	707
Poly(1-octene)	Bromobenzene	25	2.90	0.78	5	–	25–400	LS	A	84
	Cyclohexane	30	5.75	0.78	6	–	25–400	LS	A	84
	Phenetole	50.4 (θ)	65.5	0.50	4	–	60–400	LS	A	84

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(pentenamer)										
80–85%-*trans*,	Cyclohexane	30	56.9	0.68	5	–	3.6–63	LS	B	788
19–12%-*cis*	Isopentyl acetate	38 (θ)	234	0.50	8	–	3.6–63	LS	B	788
	Toluene	30	52.1	0.69	10	–	3.6–63	LS	B	788
Poly(1-phenyl-1-propyne)	Cyclohexane	36 (θ) (wormlike behavior)	–	0.5–0.8	12	–	1.6–145	LS, SE	A	902
	Toluene	25 (wormlike behavior)	–	Not const.	12	–	1.6–145	LS, SE	A	902
Poly(propylene)										
	n-Octyl acetate	27.0	175	0.512	6	–	2.8–39	LS	A	897
	3-Nonanol	5.0	113	0.536	7	–	2.3–42	LS, SEC	A	897
	1-Octanol	77.0	110	0.504	7	–	2.3–42	LS, SEC	A	897
atactic	Benzene	25	27.0	0.71	6	–	6–31	OS	A	88
		30	33.8	0.67	6	–	2–34	OS	A	89
	Biphenyl	129 (θ)	128.3	0.50	3		4–71	LV	B	708
	1-Chloronaphthalene	74 (θ)	182	0.50	3	–	4–33	OS	A	90
	Cyclohexane	25	16.0	0.80	6	–	6–31	OS	A	88
		30	20.9	0.76	6	–	2–34	OS	A	89
	Cyclohexanone	92 (θ)	172	0.50	4	–	1.5–33	OS	A	90
	Decalin	135	15.8	0.77	6	–	2–39	OS	A	91
		135	11.0	0.80	6	–	2–62	LS	A,R	88
		135	54.3	0.65	–	10	2–72	LS	D	92
	Isobutyl acetate	58 (θ)	158.5	0.50	3	–	4–71	LV	B	708
	Isopentyl acetate	34 (θ)	168.5	0.50	6	–	2–34	OS	A	89
	Phenyl ether	145	192	0.47	3	–	3.7–21	OS	A	90
		153 (θ)	120	0.50	3	–	3.7–21	OS	A	90
	Tetralin	130	1.24	0.96	–	–	?	?		93
	Toluene	30	21.8	0.725	7	–	2–34	OS	A	89
isotactic	Biphenyl	125.1 (θ)	152	0.50	4	–	5–42	LV	A	94
		125 (θ)	141.0	0.50	5	–	5–50	OS	B	708
	1-Chloronaphthalene	139	21.5	0.67	11	–	10–170	LS		95
		145	4.9	0.80	9	–	5–63	LS	A,R	96
	Decalin	135	11.0	0.80	6	–	2–62	LS	A,R	88
		135	10.0	0.80	4	–	10–100	LS	A,R	97
	Dibenzyl ether	183.2 (θ)	106	0.50	4	–	5–42	LV	A	94
	Diphenyl ether	142.8 (θ)	137	0.50	4	–	5–42	LV	A	94
		145 (θ)	132	0.50	4	–	3–48	OS	A	90
		153	112	0.54	4	–	3–48	OS	A	90
	Tetralin	135	2.5	1.0	5	–	2–11	OS		98
		135	9.17	0.80	9	–	4–54	OS	A,R	96
		135	19.3	0.74	5	–	5–50	OS	B	708
	p-Xylene	85	96	0.63	12	–	?	OS		99
syndiotactic head-to-head	Heptane	30	31.2	0.71	5	–	9–45	LS	A	100
94%-*trans*, 6%-1,2	Cyclohexane	30	493	0.39	5	–	0.2–1.1	VOS, OS	L	709
		30	4.16	0.86	5	–	2.1–4.2	VOS, OS		709
89%-*trans*, 11%-1,2	Cyclohexane	30	295	0.43	5	–	0.3–0.8	VOS, OS	L	709
		30	3.82	0.90	3	–	1.7–3.2	VOS, OS		709
1.3. POLY(ACRYLIC ACID) AND DERIVATIVES										
Poly(acrylamide)	Water	30	6.31	0.80	7	–	2–50	SD	B	101
		30	68	0.66	–	21	1–20	PR	C	102
		30	6.5	0.82	7	–	4–127	OS	B	710
	Aq. NaCl (1 N)	25	19.1	0.71	–	5	49–320	LS	D	905
	Aq. NaCl (0.12 M)	30	5.31	0.79	7	–	15–153	LS	B	972
Poly(*N*-acryloyl-*m*-aminobenzoic acid)										
	N,*N*-Dimethylacetamide	30	9.1	0.84	8	–	1.3–5.5	VOS, GPC	A–B	908
	N,*N*-Dimethylformamide	30	8.9	0.83	8	–	1.3–5.5	VOS, GPC	A–B	908
	Dimethyl sulfoxide	30	10.6	0.80	8	–	1.3–5.5	VOS, GPC	A–B	908
Poly(*N*-acryloyl-*o*-aminobenzoic acid)										
	N,*N*-Dimethylformamide	25	3.74	0.93	7	–	0.7–2.9	VOS, GPC	A–B	909
	Dimethyl sulfoxide	25	3.83	0.92	7	–	0.7–2.9	COS, GPC	A–B	909
	Tetrahydrofuran	25	3.39	0.89	7	–	0.7–2.9	COS, GPC	A–B	909

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(acrylamide-*co*-*N*,*N*,*N*-trimethyl aminoethyl chloride acrylate)										
70% acrylamide	Aq. NaCl (1 N)	25	10.5	0.73	–	6	45–270	LS	D	905
Poly(acrylic acid)	1,4-Dioxane	30 (θ)	76	0.50	–	4	13–82	OS	B	104
–, sodium salt	Aq. NaOH (2 M)	25	42.2	0.64	12	–	4–50	OS	C	105
	Aq. NaCl									
	(0.012 M)	20	–	0.93	7	–	7–180	LV	B	106
	(1 M)	25	15.47	0.90	12	–	4–50	OS	C	105
Poly(acrylic acid)										
–, sodium salt	Aq. NaCl									
	(0.05 M)	25	7.35	0.88	4	–	9.4–60	LLS, GPC	B–C	906
	(0.1 M)	25	14.6	0.80	4	–	9.4–60	LLS, GPC	B–C	906
	(0.3 M)	25	16.9	0.75	4	–	9.4–60	LLS, GPC	B–C	906
	(0.5 M)	25	18.6	0.72	4	–	9.4–60	LLS, GPC	B–C	906
	(1.0 M)	25	41.5	0.63	4	–	9.4–60	LLS, GPC	B–C	906
	Aq. NaBr									
	(1.5 M)	15	145	0.50	4	–	9.4–60	LLS, GPC	B–C	906
	(1.5 M)	15 (θ)	165	0.50	5	–	6–64	LV	C	107
		15 (θ)	124	0.50	4	–	12–83	LS	C	108
	(0.5 M)	15	52.7	0.628	7	–	1–50	LV	C	109
		25	50.6	0.656	7	–	2–80	LV	C,R	110
	(0.1 M)	15	25.4	0.755	7	–	1–50	LV	C	109
		25	31.2	0.755	7	–	2–80	LV	C	110
	(0.05 M)	15	28.1	0.77	7	–	1–50	LV	C	109
	(0.025 M)	15	16.3	0.84	7	–	1–50	LV	C	109
		25	17.6	0.85	7	–	2–80	LV	C	110
	(0.01 M)	15	13.6	0.89	7	–	1–50	LV	C	109
		25	13.2	0.91	7	–	2–80	LV	C	110
	(0.005 M)	15	(44.2)	0.83	7	–	1–50	LV	C	109
	(0.0025 M)	15	(24.9)	0.89	7	–	1–50	LV	C	109
	Aq. NaSCN	30 (θ)	154	0.50	5	–	6–64	LV	C	107
	(1.12 M)	30 (θ)	121	0.50	4	–	12–83	LS	C	111
Poly(acrylonitrile)	γ-Butyrolactone	20	34.3	0.730	5	–	4–40	LV (LS)	A,R	134
(polymerized at –30°C)		30	57.2	0.67	6	–	4–30	SA	B	135
(polymerized at 60°C)		30	34.2	0.70	5	–	6–30	SA	B	135
		30	40.0	0.69	–	5	15–53	LS	D	136
		50	28.7	0.740	5	–	4–40	LS	A	134
	Dimethylformamide	20	17.7	0.78	5	–	7–30	LS	B	137
		25	16.6	0.81	5	–	5–27	SD	B	138
		25	24.3	0.75	–	4	3–25	LS	C	139
		25	39.2	0.75	–	16	3–100	OS	C	140
		25	52.0	0.690	7	–	5–52	LS	B,R	711
	(Deionized DMF)	25	15.5	0.80	3	5	3–10	LS, SD	B–C	141
		25	57.4	0.73	–	8	0.3–1.5	EG	L	142
		25	39.6	0.75	–	7	4–30	OS	C	143
		25	44.3	0.70	–	7	2–20	LS	C	143
		25	69.8	0.65	–	21	8–140	LS	C	144
(polymerized at –30°C)		30	29.6	0.74	7	–	4–30	SA	B	135
(polymerized at 60°C)		30	20.9	0.75	7	–	6–30	SA	B	135
		30	33.5	0.72	–	6	16–48	LS	D	136
		35	27.8	0.76	9	–	3–58	DV	B	145
		35	31.7	0.746	12	–	9–76	LS	A,R	134
		50	30.0	0.752	22	–	4–102	LV	A	134
	Dimethylacetamide	20	30.7	0.761	6	–	2–40	LV	A	134
		35	27.5	0.767	6	–	2–40	LV	A	134
		50	27.4	0.764	6	–	2–40	LV	A	134
	Dimethyl sulfoxide	20	32.1	0.750	9	–	9–40	LV	A	134
		50	28.3	0.758	9	–	9–40	LV	A	134
		140	20.9	0.75	–	6	4–40	LS		146
	Ethylene carbonate	50	29.5	0.718	13	–	7–40	LV	A	134
	Ethylene carbonate/water (85/15, w/w)	25	263	0.49	7	–	5.2–52	LS, OS	A–B	951

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	Hydroxyacetonitrile	20	40.9	0.697	8	–	4–34	LV	A	134
		50	35.4	0.707	8	–	4–34	LV	A	134
	Aq. HNO_3 (60%)	0	33.9	0.740	6	–	2–40	LV	A	134
		20	30.7	0.747	5	–	4–40	LV	A	134
Poly(benzyl acrylate)	Butanone	35	0.587	0.883	?		?	OS		337
Poly(butyl acrylate)	Acetone	25	6.85	0.75	–	6	5–27	LS	C	112
Poly(*tert*-butyl acrylate)	Acetone	25	4.7	0.75	5	–	7–31	LS	B	712
	Butanone	25	3.2	0.80	5	–	7–31	LS	B	712
	Hexane	24.2 (θ)	49.0	0.50	5	–	7–31	LS	B	712
	Methanol	25	16.0	0.61	5	–	7–31	LS	B	712
	Pentane	25	22.0	0.57	5	–	7–31	LS	B	712
Poly(1,1-dihydroperfluorobutyl acrylate)										
	Benzofluoride	26.6	13	0.56	7	3	20–200	LS	B	113
	Methyl perfluorobutyrate	26.6	12	0.60	7	3	20–200	LS	B	113
Poly(*N*,*N*-dimethylacrylamide)	Methanol	25	17.5	0.68	–	8	5–122	LS	C	103
	Water	25	23.2	0.81	–	6	5–122	LS	C	103
		40	20.0	0.65	–	4	11–122	LS	C	103
Poly(ethyl acrylate)	Acetone	25	51	0.59	7	–	35–450	LS	B,R	114
		30	20.0	0.66	5	–	16–50	OS	B,R	115
	Benzene	30	27.7	0.67	–	7	5–67	OS	C	116
	Butanone	30	2.68	0.80	5	–	48–700	LS	B–C	117
	Chloroform	30	31.4	0.68	–	5	9–54	OS	C	116
	Ethyl acetate	30	26.0	0.66	–	5	9–54	OS	C	116
Poly(2-ferrocenylethyl acrylate)	Benzene	25	4.68	0.70	–	3	1.4–2.7	VOS, GPC	D	713
Poly(ferrocenylmethyl acrylate)	Benzene	25	6.84	0.75	–	8	0.7–2	VOS, GPC	C–D	714
Poly(hexadecyl acrylate)	Methanol	30	48.7	0.55	–	6	6–70	OS	C	116
	Heptane	20	1.74	0.82	6	–	1–10	LS	B	118
Poly(*N*-isopropylacrylamide)	Tetrahydrofuran	27	9.59	0.65	8	–	10–100	OS	A	975
	Water	20 (θ)	145	0.50	8	–	10–100	OS	A	975
Poly(isopropyl acrylate)	Acetone	30	13.0	0.69	6	–	6–30	LS	B	119
	Benzene	25	14.9	0.70	9	–	7–70	OS	B	120
		25	12.4	0.701	20	–	4–100	LS	B,R	121
		30	11.8	0.71	4	–	7–20	LS	B	119
	Bromobenzene	25	11.3	0.704	20	–	4–100	LS	B	121
		60	11.6	0.698	20	–	4–100	LS	B	121
	Chloroform	30	14.1	0.72	5	–	7–30	LS	B	122
isotactic	2,2,3,3-tetrafluoro-propanol	25	19.7	0.697	7	–	10–65	LS	B	121
atactic		25	17.3	0.703	6	–	8–110	LS	B	121
syndiotactic		25	15.9	0.708	6	–	20–110	LS	B	121
isotactic		60	17.9	0.693	4	–	10–65	LS	B	121
atactic and syndiotactic		60	14.7	0.704	6	–	20–110	LS	B	121
Poly(methyl acrylate)	Acetone	20	(7.40)	(0.76)	–	4	7–32	OS		123
		25	5.5	0.77	8	–	28–160	LS	B,R	124
		25	19.8	0.66	9	–	30–250	LS	B	125
		25	5.20	0.77	11	–	4–183	LS	B	715
		30	28.2	0.52	7	–	4–45	OS	B	126
	Benzene	25	2.58	0.85	4	–	20–130	OS		127
		30	4.5	0.78	7	7	7–160	LS		128
		30	3.56	0.798	6	–	25–190	LS	B,R	129
		30	4.59	0.795	6	–	15–140	OS	B	129
		35	12.8	0.71	–	5	5–30	OS	C	130
	Butanone	20	3.5	0.81	13	–	6–240	LS	A–B,R	128
		25	14.1	0.67	4	–	17–68	LS	B	131
		30	3.97	0.772	6	–	25–190	LS	B	129
		35	(34)	(0.61)	–	3	5–47	LV	C	132
	Dimethyl malonate	30	3.51	0.793	4	–	50–190	LS	B	129
	Ethyl acetate	35	11	0.69	–	8	24–148	LS	A	133
	Isopentyl acetate	62.5 (θ)	68	0.50	6	–	20–160	LS	B	129
	2-methylcyclohexanol	56.0 (θ)	68	0.50	4	–	40–105	LS	B	129
	Toluene	30	7.79	0.697	6	–	25–190	LS	B	129
		35	21	0.60	–	7	12–69	LS	A	133

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	Butanone–2-propanol									
	(42/58, v/v)	20 (θ)	81	0.50	5	–	29–140	LS	B	124
	(1/1, v/v)	27.5 (θ)	54.4	0.50	4	–	14–83	LS	C	108
		30 (θ)	72	0.50	4	–	50–190	LS	B	129
branched	(42/58 v/v)	20 (θ)	290	0.40	6	–	37–250	LS	B	125
Poly(1-methylphenyl acrylate)	Butyl acetate	25	14.7	0.63	8	–	2–110	SD	A	346
Poly(morpholinocarbonylethylene)										
	Dimethylformamide	25	18	0.65	?		?	LS	C	338
	Aqueous NaCl (0.1 M)	20	64	0.68	?		?	LS	C	338
Poly(piperidinocarbonylethylene)	Dimethylformamide	25	32	0.56	?		?	LS	C	338
Poly(propyl acrylate)	Butanone	30	15.0	0.687	4	–	71–181	LS	A	117
1.4. POLY(α-SUBSTITUTED ACRYLIC ACID) AND DERIVATIVES										
Poly(benzyl methacrylate)	Benzene	30	1.03	0.82	–	9	17–120	LS		339
Poly(butyl methacrylate)	Acetone	25	18.4	0.62	5	–	100–600	LS	A	150
	Benzene	30	(4.0)	(0.77)	–	3	8–300	LS		151
	Benzene	25	3.82	0.774	6	–	1270–2070	–	–	912
	Butanone	23	1.56	0.81	10	–	25–260	LS	B	152
		25	9.7	0.68	5	–	11–670	LS	A	150
		30	(1.15)	(0.89)	3	–	67–132	OS	C	153
	Chloroform	20	2.9	0.78	8	–	4–800	LS	B,R	154
		25	4.37	0.80	6	–	8–80	OS		155
	Cyclohexane	25	21.0	0.648	6	–	1230–2450	–	–	912
	Dioxane	25	23.7	0.630	6	–	1330–2070	–	–	912
	Methyl ethyl ketone	25	6.13	0.726	10	–	959–2160	–	–	912
	2-Propanol	21.5 (θ)	29.5	0.50	8	–	30–260	LS	B	152
		21.5 (θ)	38	0.50	9	–	4–800	LS	B,R	154
		23.7 (θ)	36.6	0.50	5	–	40–170	LS	B	156
Poly(*tert*-butyl methacrylate)	Butyl acetate	25	22.0	0.63	6	–	46–870	LS	A	157
anionic	Butanone	25	12.0	0.675	6	–	2.8–107	LLS, SEC	A	917
	Cyclohexane	10.0	62.0	0.499	6	–	2.8–107	LLS, SEC	A	917
	Tetrahydrofuran	30	15.2	0.66	6	–	2.8–107	LLS, SEC	A	917
free radical	Butanone	25	5.91	0.73	4	–	14–155	LLS, SEC	A	917
	Cyclohexane	10.0	46.0	0.505	5	–	6.8–155	LLS, SEC	A	917
	n-Heptane	64.0	62.1	0.476	7	–	3.4–155	LLS, SEC	A	917
	Tetrahydrofuran	30	5.84	0.76	6	–	3.4–155	LLS, SEC	A	917
Poly(2-*tert*-butylphenyl methacrylate)										
	Benzene	25	7.8	0.68	8	–	4–113	LS	B	771
	Butanone	25	9.0	0.64	8	–	4–113	LS	B	771
	Cyclohexane	18.4 (θ)	35.5	0.50	8	–	4–113	LS	B	771
Poly(4-*tert*-butylphenyl methacrylate)										
	Acetone	20	5.75	0.68	15	–	6–350	LS		340
		25	16	0.60	6	–	11–204	LS	B	716
	Bromobenzene	20	4.1	0.71	7	–	15–2500	LS		341
	Carbon tetrachloride	20	4.1	0.71	7	–	20–2500	LS		341
	Chloroform	20	2.4	0.78	15	–	6–300	LS	A–B	342
	Cyclohexane	25	47	0.49	6	–	11–204	LS	B	716
	Tetrahydrofuran	25	9.4	0.68	6	–	7–88	L	B	716
Poly[1-(*N*-carbethoxyphenyl)-methacrylamide]										
	Acetone	Unc.	0.00115	1.35	4	–	26–74	LS		369
	Dimethylformamide	Unc.	This relation not followed		5	–	48–140	LS		369
	Ethyl acetate	Unc.	0.00446	1.25	5	–	26–11	LS		369
Poly(2-chloroethyl methacrylate)	*o*-Dichlorobenzene	35.7 (θ)	47.4	0.50	6	–	3.2–54	LLS, SEC	A–B	914
	Tetrahydrofuran	30	6.83	0.72	6	–	3.2–54	LLS, SEC	A–B	914
Poly(4-chlorophenyl methacrylate)										
	Benzene		9.2	0.66	8	–	10–610	LS	A	343
	Carbon tetrachloride		20.0	0.58	8	–	10–610	LS	A	343
	Dioxane		6.1	0.70	8	–	10–610	LS	A	343
Poly(cyclobutyl methacrylate)	1-Butanol	37.5	49.2	0.494	6	–	4.8–31		A–B	915
Poly(cyclododecyl methacrylate)	Cyclohexane	30	8.84	0.651	7	–	5.1–475		B–C	915
	n-Hexyl acetate	35	35.4	0.494	6	–	5.1–475		B–C	915
	Toluene	30	6.03	0.687	7	–	5.1–475		B–C	915

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TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(cyclohexyl methacrylate)	Benzene	30	8.4	0.69	5	–	80–200	LS		344
		25	3.54	0.77	9	–	10–419	LS	A	717
	Butanol	23 (θ)	33.7	0.50	5	–	57–445	LS	B	345
		22.5 (θ)	45.2	0.50			10–125	LS	A	718
		25	31.8	0.533			10–125	LS	A	718
	Butanone	25	5.79	0.68	6	–	57–560	LS	B	345
		30	7.0	0.66	5	–	80–200	LS		344
	Cyclohexane	25	8.8	0.67	9	–	10–419	LS	A	717
Poly(cyclohexylthiolmethacrylate)										
	Cyclohexane	25	8.65	0.63	8	–	4–60	LS	B	719
	Tetrahydrofuran	35	4.07	0.74	8	–	4–60	LS	B	719
Poly(cyclooctyl methacrylate)	2-Butanol	45	43.2	0.488	7	–	6.6–418		A–C	915
	1,4-Dioxane	30	26.0	0.549	8	–	3.1–418		A–C	915
	Toluene	30	8.70	0.677	8	–	3.1–418		A–C	915
Poly(cyclopentyl methacrylate)	Cyclohexane	36	54.8	0.487	8	–	4.6–358		A–B	915
	1,4-Dioxane	30	10.9	0.659	8	–	2.9–358		A–B	915
	Ethyl acetate	30	19.4	0.588	9	–	2.9–358		A–B	915
	Methyl ethyl ketone	30	13.6	0.631	9	–	2.9–358		A–B	915
	Toluene	30	8.38	0.696	9	–	2.9–358		A–B	915
Poly(2,6-dimethylphenyl methacrylate)										
	Chlorobenzene	25	18.4	0.65	4	–	3.4–33	OS, SEC	A	923
	Tetrahydrofuran	25	33.1	0.59	8	–	3.4–82	OS, SEC	A	923
	Tetrahydrofuran/water	25 (θ)	75.0	0.50	4	–	5.7–33	OS, SEC	A	923
	Toluene	25 (θ)	78.0	0.50	9	–	3.4–82	OS, SEC	A	923
Poly(diphenylmethyl methacrylate)										
	3-Heptanone	45	38.6	0.485	9	–	8.7–575	LLS, SEC	A–B	911
	Toluene	25	3.61	0.712	7	–	42–575	LLS, SEC	A–B	911
Poly(decyl methacrylate)	Ethyl acetate	11 (θ)	34.7	0.50	10	–	2.9–92	LS, OS, SEC	B	913
	Tetrahydrofuran	30	4.56	0.73	10	–	2.9–92	LS, OS, SEC	B	913
Poly(dodecyl methacrylate)	Amyl acetate	31	36.6	0.53	10	–	10–135	LS, OS, SEC	B	913
	Tetrahydrofuran	30	1.05	0.64	10	–	10–135	LS, OS, SEC	B	913
Poly(dodecyl methacrylate)	Butyl acetate	23	8.64	0.64	8	–	26–360	LS	A	158
	Isopropyl acetate	13 (θ)	32.2	0.50	7	–	26–360	LS	A	158
	Pentanol	29.5 (θ)	34.8	0.50	7	–	27–240	LS	A	159
Poly(2-ethylbutyl methacrylate)	Butanone	25	2.21	0.77	8	–	48–332	LS	A	160
	2-Propanol	27.4 (θ)	33.7	0.50	8	–	48–332	LS	A	160
Poly(ethyl methacrylate)	Butanone	23	2.83	0.79	10	–	20–263	LS	A	161
	Ethyl acetate	35	8.6	0.71	–	11	65–1200	LS	C	162
	2-Propanol	36.9 (θ)	47.5	0.50	4	–	22–130	LS	B	156
	Butanone-2-propanol (1/7, v/v)	23 (θ)	47.3	0.50	10	–	20–263	LS	A	161
	Ethyl acetate/ethanol (2/9, v/v)	35	47.6	0.53	6	–	78–500	LS	A	162
	(1/6, v/v)	35 (θ)	56.4	0.50	6	–	60–420	LS	A	162
Poly(2-ferrocenylethyl methacrylate)										
	Benzene		3.12	0.76	–	6	2–9	VOS, GPC	C,D	713
Poly(ferrocenylmethyl methacrylate)										
	Benzene	25	27.8	0.58	–	6	0.6–3.6		C,D	714
Poly(hexadecyl methacrylate)	Benzene	21	5.9	0.71	3	–	130–440	SD	B	163
	Carbon tetrachloride	21	2.37	0.78	5	–	130–440	SD	B	163
	Heptane	21	3.92	0.75	5	–	130–440	SD	B	163
		25	35.1	0.56	9	–	20–110	LS		164
Poly[4-(4-hexadecyloxybenzoyloxy)-phenyl methacrylate]										
	Carbon tetrachloride		33.1	0.5	20	–	10–2000	SD		730
Poly(hexyl methacrylate)	Butanone	23	2.12	0.78	8	–	6–41	LS	A	165
	2-Propanol	32.6 (θ)	43.0	0.50	8	–	6–41	LS	A	165
Poly(2-hydroxyethyl methacrylate)										
	Dimethylformamide	30	10.6	0.70	6	–	4–52	LS	A,B	720
	Dimethyl sulfoxide	30	12.9	0.69	5	–	4–52	LS	A,B	720
	Methanol	30	52.4	0.51	7	–	4–52	LS	A,B	720
Poly[1-(2-hydroxyethyl) pyridiniumbenzene sulfonate methacrylate]										
	Aq. KCL (0.5 M)	25	2.62	0.71	–	14	19–950	LS		721

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly[1(2-hydroxyethyl) trimethylammoniumbenzene sulfonate methacrylate]										
	Aq. KCl (0.5 M)	25	4.04	0.70	–	8	15–400	LS		721
Poly[*N*-(2-hydroxypropyl) methacrylamide]										
	Aqueous KCl (0.1 M)	25	11.0	0.67	12	–	2–500	LS		722
Poly(D,L-isobornyl methacrylate)	1-Octanol	39.6 (θ)	31.7	0.50	6	–	9–120	GPC		723
	Tetrahydrofuran	30	3.68	0.73	6	–	9–120	GPC		723
Poly(isobutyl methacrylate)	Acetone	25	0.199	0.94	6	–	300–1100	LS	C	166
		25	23.4	0.66	6	–	50–116	OS		724
		60	18.2	0.68	6	–	50–116	OS		724
	Acetone	25	8.41	0.66	6	–	1020–3460	–	–	919
	Benzene	25	3.88	0.74	5	–	1020–3460	–	–	919
	Benzene	25	7.03	0.77	6	–	50–116	OS		724
	Butanone	20	5.56	0.73	6	–	300–1100	LS	C	166
		25	8.61	0.70	7	–	300–1100	LS	C	166
		30	7.47	0.71	6	–	300–1100	LS	C	166
		44	2.18	0.79	6	–	300–1100	LS	C	166
	Carbon tetrachloride	25	4.88	0.72	6	–	1020–3460	–	–	919
	1,4-Dioxane	25	6.89	0.68	6	–	1020–3460	–	–	919
	2-Hydroxymethyl-tetrahydrofuran	25	85.8	0.56	6	–	50–116	OS		724
Poly(5-*p*-methyl methacrylate)	Benzene	25	9.6	0.67	6	–	12–230	LS	B	771
	Cyclohexane	25	11.5	0.65	6	–	12–230	LS	B	771
	2-Pentanone	25 (θ)	43.6	0.50	6	–	12–230	LS	B	771
	Tetrahydrofuran	25	11.5	0.65	6	–	12–230	LS	B	771
Poly(methacrolein)	Dimethylformamide	20	2.8	0.97	–	?	0.5–2	OS, CR	?	204
Poly(methacrylic acid)	*N,N*-Dimethylformamide/ 1,4-dioxane (5/7, v/v)	26.9 (θ)	103	0.50	7	–	27.5–101	LS	–	907
	Methanol	26	242	0.51	6	–	4–20	OS	B	147
	Aq. HCl (0.002 M)	30	66	0.50	7	–	10–90	LV	C	148
	Aq. $NaNO_3$ (2 M)	25	44.9	0.65	6	–	8–70	OS	B	149
Poly(methacrylonitrile)	Acetone	20	95.5	0.56	–	4	35–100	OS	C	202
	Dimethylformamide	29.2	306	0.503	–	15	0.6–8	LV	C,H	203
Poly(2-methoxyethyl methacrylate)										
	Butanone	25	7.34	0.71	12	–	4–220	LS	A–B	725
	Tetrahydrofuran	25	7.57	0.71	12	–	4–220	LS	A–B	725
Poly(methyl butacrylate)	Butanol	13 (θ)	57.0	0.50	4	–	6–60	LS	A	168
	Butanone	30	5.43	0.73	10	–	7–430	LS	A	168
Poly(methyl α-chloroacrylate)	Chloroform	30	3.08	0.78	8	–	20–780	LS	D	726
Poly(methyl ethacrylate)	Benzene	30	2.35	0.82	6	–	16–110	LS	A	168
	Butanone	30	4.29	0.75	10	–	4–200	LS	A	168
	2,6-Dimethyl-4-heptanone	11.4 (θ)	67.6	0.50	10	–	4–200	LS	A	168
Poly(methyl methacrylate) atactic	Acetone	20	5.5	0.73	7	–	7–700	SD	A–B,R	169
		20	3.90	0.76	7	–	7–700	SD	A–B	169
		25	7.5	0.70	9	–	8–137	LS	B	170
		25	6.76	0.71	10	–	3–700	SD	A–B	171
		25	7.5	0.70	14	–	2–740	LS, SD	A–B	172
		25	5.3	0.73	7	–	2–780	LS	A–B,R	173
		25	9.6	0.69	4	–	180–350	LS	A–B	174
		25	7.5	0.70	4	6	3–98	LS	B–C	175
		25	2.45	0.80	9	–	6–210	OS	B–C	176
		25	6.59	0.71	6	–	5–41	OS	B	177
		30	7.7	0.70	6	–	6–263	LS	A–B	178
		39	6.40	0.72	6	–	5–41	OS	B	177
		46	6.18	0.72	6	–	5–41	OS	B	177
	Acetonitrile	30	39.3	0.50	6	–	10–86	LV	A–B	178
		45 (θ)	48	0.50	6	–	10–260	LV	A–B,R	179
		50	29	0.54	6	–	10–260	LV	A–B	180
		65	9.8	0.64	5	–	10–260	LV	A–B	180
	Benzene	20	8.35	0.73	7	–	7–700	SD	A–B	169
		20	15.1	0.70	7	–	8–90	SD		181
		25	7.24	0.76	10	–	6–100	OS	B	182
		25	5.5	0.76	11	–	2–740	LS	A–B,R	173

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TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
		25	3.80	0.79	5	–	24–450	LS		183
		25	83	0.52	7	–	0.03–1	EB	A,L	184
		30	5.2	0.76	9	–	6–250	LS	A–B,R	178
		30	6.27	0.76	5	–	4–73	OS	A	185
		30	104	0.50	9	–	0.02–2	OS	A,L	185
		30	195	0.41	5	–	0.3–2	LS	A–B,L	178
		39	6.74	0.75	6	–	5–41	OS	B	177
		53	6.52	0.76	6	–	5–41	OS	B	177
	Butanone	25	6.8	0.72	9	–	8–137	LS	B,R	170
		25	7.1	0.72	7	–	41–330	LS	A–B	174
		25	6.8	0.72	4	6	3–98	LS	B–C	175
		25	9.39	0.68	15	–	16–910	LS	A–B	186
plasma initiated		25	11 ± 2	0.69			1610–3000	LS		727
	Butyl chloride	35.4 (θ)	50.5	0.50	4	–	13–68	SA	A–B	187
	Chloroform	20	9.6	0.78	18	–	1.4–60	OS		188
		20	4.88	0.82	8	–	6–100	OS	B	182
		20	4.85	0.80	9	–	8–200	SD	A–B,R	169
		20	6.0	0.79	12	–	3–780	LS	A–B	173,189
		25	4.8	0.80	9	–	8–137	LS	B	170
		25	3.4	0.83	6	–	40–330	LS	A–B	174
		25	5.81	0.79	6	–	5–41	OS	B	177
living type		30	4.3	0.80	–	8	13–263	LS	A–B	178
		39	5.02	0.80	6	–	5–41	OS	B	177
		53	3.90	0.79	6	–	5–41	OS	B	177
		Unc.	5.1	0.79	13	–	7–400	LS	B	190
	p-Cymene	159.7 (θ)	57.5	0.50	4	–	6.6–171	LV	A–B	191
	1,2-Dichloroethane	25	17.0	0.68	4	6	3–98	LS	B–C	175
		30	5.3	0.77	–	7	6–263	LS	A–B,R	178
	Ethyl acetate	20	21.1	0.64	8	34	6–110	SD		192
	3-Heptanone	33.7 (θ)	63.1	0.50	4	–	6.6–171	LV	A–B	191
	4-Heptanone	33.8 (θ)	48	0.50	5	–	1–172	LS	A–B,R	179
	Methyl isobutyrate	30	9.9	0.67	6	–	19–260	LV	A–B	178
	Methyl methacrylate	20	16.2	0.65	9	–	1.8–160	LS	A	918
		40	11.3	0.68	9	–	1.8–160	LS	A	918
		60	21.2	0.64	9	–	1.8–160	LS	A	918
	Methyl methacrylate	30	6.75	0.72	3	–	13–170	LV	A–B	178
	Nitroethane	25	5.70	0.74	2	6	10–200	LS	C	193
	3-Octane	72 (θ)	50	0.50	3	–	13–260	LV	A–B	179
	Propanol	84.4 (θ)	67.9	0.50	4	–	6.6–171	LV	A–B	191
	Tetrachloroethane	25	12.8	0.73	6	–	5–41	OS	B	177
		53	12.2	0.73	6	–	5–41	OS	B	177
	2,2,3,3-Tetrafluoro-propanol	25	7.2	0.79	7	–	7–95	LV	A	194
	Tetrahydrofuran	25	7.5	0.72				LS	A	732
	Toluene	25	7.1	0.73	7	–	4–330	LS	A–B	174
		25	8.12	0.71	6	–	5–41	OS	B	177
		25	78	0.50	10	–	0.2–7	OS	A,L	195
		30	7.0	0.71	6	–	19–263	LV	A–B	178
		39	7.24	0.72	6	–	5–41	OS	B	177
		53	6.63	0.73	6	–	5–41	OS	B	177
	Butanone/2-propanol (55/45, v/v)	23	47.0	0.55	6	–	40–300	LS	A–B	174
	(50/50, v/v)	25 (θ)	59.2	0.50	7	–	30–280	LS	A–B	196
		25 (θ)	42.8	0.50	5	–	77–490	LS	A–B	186
	Methanol/toluene (9/5, v/v)	26.2 (θ)	55.9	0.50	3	–	60–300	LS	A–B	156
oligomer-polymer										
atactic, $f_{racem} = 0.79$	Acetonitrile	44.0	–	Not const.	22	–	0.032–283	LS	A	916
	Benzene	30.0	–	Not const.	15	–	0.032–75.8	LS	A	916
	n-Butyl chloride	40.8	–	Not const.	21	–	0.032–283	LS	A	916
isotactic	Acetone	30	23.0	0.63	7	–	5–128	LS	A–B	199
	Acetonitrile	20	130	0.448	5	–	3–19	LV	A	198
		27.6 (θ)	75.5	0.500	5	–	3–19	LV	A	198

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
		35	46	0.546	5	–	3–19	LV	A	198
		50	26.2	0.602	5	–	3–19	LV	A	198
		27.5 (θ)	76.2	0.50	6	–	13–68	LS	B	728
	Benzene	30	5.2	0.76	5	–	5–128	LS	A–B	199
	p-Cymene	152.1 (θ)	56.6	0.50	4	–	7–131	LV	A–B	191
	3-Heptanone	40.0 (θ)	87.0	0.50	4	–	7–131	LV	A–B	191
	Propanol	75.9 (θ)	76.1	0.50	4	–	7–131	LV	A–B	191
	2,2,3,3-Tetrafluoro-propanol	25	7.05	0.78	11	–	2–100	LV	B	194
	Butanone/2-propanol (1/1, v/v)	30.3 (θ)	90.0	0.50	4	–	7–131	LV	A–B	191
5%-iso, 51%-hetero, 43%-syndiotactic	Acetonitrile/chlorobenzene									
	(4.4/95.6, v/v)	25 (θ)	68.3	0.50	4	–	73–232	LS	A	785
	(91.2/8.8, v/v)	25 (θ)	63.5	0.50	4	–	73–232	LS	A	785
Poly(α-methyleneglutaronitrile)	Dimethylformamide	25	31.6	0.65	–	8		LS	H	731
Poly(methyl phenylacrylate), see 1.9 Other Compounds poly(1-methoxycarbonyl-1-phenylethylene)										
Poly(β-naphthyl methacrylate)	Benzene	20	27.2	0.55	6	–	17–262	LS	B	729
	Tetralin	20 (θ)	47.5	0.50	4	–	57–262	LS	B	729
Poly[4-(4-nonyloxy-phenyl methacrylate], Poly(phenylmethacrylic ester of nonyloxybenzoic acid)										
	Carbon tetrachloride		24.3	0.5	20	–	20–220	SD		730
Poly(octadecyl methacrylate)	Tetrahydrofuran	30	2.5	0.75	–	4	20–170	LS	C,H	200
Poly(octyl methacrylate)	Butanol	16.8 (θ)	26.8	0.50	10	–	33–1250	LS	B	201
	Butanone	23	4.47	0.69	10	–	33–1250	LS	B	201
	Butyl acetone	10.5 (θ)	–	0.45	4	–	62–320	LS	C	920
		20	–	0.53	4	–	62–320	LS	C	920
		30	–	0.54	4	–	62–320	LS	C	920
Poly(*N*-phenyl methacrylamide)	Acetone	20	28.2	0.75	8	–	10–320	LS		370
	n-Propyl acetate	36 (θ)	37.1	0.50	11	–	1.5–93	LS, OS, SEC	B	913
	Tetrahydrofuran	30	8.95	0.67	11	–	1.5–93	LS, OS, SEC	B	913
Poly(2-selenolylmethyl methacrylate)										
	Chlorobenzene	25	36.3	0.56	5	–	3.6–21	VOS, SEC	A–B	922
	Tetrahydrofuran	25	18.1	0.65	5	–	3.6–21	VOS, SEC	A–B	922
Poly(stearyl methacrylate)	Tetrahydrofuran	30	9.0	0.67	12	–	1.5–94	LS	B	733,751
Poly(tetrahydrofurfuryl methacrylate)										
	Acetone	30	24.0	0.66	7	–	16–62	OS	B	734
Poly(tetrahydro-4H-pyranyl 2-methacrylate)										
	Isobutanol	30.4 (θ)	31.9	0.50	6	–	4–85	GPC		735
Poly(2-thiophenmethyl methacrylate)										
	Chlorobenzene	25 (θ)	4.35	0.50	6	–	48–59	SEC	A–B	921
	Tetrahydrofuran	25	6.95	0.72	6	–	48–59	SEC	A–B	921
	Thiophen	25	9.00	0.66	6	–	48–59	SEC	A–B	921
Poly(*N,N,N*-trimethyl aminoethyl chloride acrylate)										
	Aq. NaCl (1 N)	25	2.3	0.82	–	4	85–510	LS	D	905
Poly(tridecyl methacrylate)	Ethyl acetate	27 (θ)	32.2	0.50	10	–	8.2–138	LS, OS, SEC	B	913
	Tetrahydrofuran	30	4.74	0.71	10	–	8.2–138	LS, OS, SEC	B	913
	Tetrahydrofuran	30	2.93	0.76	6	–	4–85	GPC		735
Poly[2-(triphenylmethoxy)ethyl methacrylate]										
	Mesitylene	47 (θ)	30	0.50				LV		736
1.5. POLY(VINYL ETHERS)										
Poly[(hexadecyloxy)ethylene]	Heptane	21	70.8	0.50	6	–	0.5–3	SD	B,L	205
Poly(methoxyethylene)	Benzene	30	76	0.60	13	–	1–45	LS	B	206
	Butanone	30	137	0.56	13	–	1–45	LS	B	206
Poly[(octadecyloxy)ethylene]	Benzene	25	170	0.47	–	7	0.1–1.5	LS	D,H	200
	Tetrahydrofuran	30	224	0.35	–	7	9.4–11	LS	D,H	200
Poly(vinyl methyl ether), see Poly(methoxyethylene)										
1.6. POLY(VINYL ALCOHOL), POLY(VINYL HALIDES)										
Poly(chlorotrifluoroethylene)	2,5-Dichlorobenzo-trifluoride	130	6.15	0.74	7	–	7–51	OS	B	234

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TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(tetrafluoroethylene-*alt*-ethylene), see also Poly(ethylene-*alt*-tetra-fluoroethylene) in group 1.2										
Poly(vinyl alcohol)	Water	25	20	0.76	6	–	0.6–2.1	OS	B	208
		25	300	0.50	4	–	0.9–17	SD		209
		25	140	0.60	3	–	1–7	SD	B	210
		30	66.6	0.64	8	–	0.6–16	OS	B	212
		30	42.8	0.64	–	14	1–80	LS	C	213
		30	45.3	0.64	–	–	1–80	LS	A,R	213
		30	73.4	0.63	7	–	3–12	LV	B	737
		80	94	0.56	–	5	10–46	LS	B	214
	Phenol/water (85/15, v/v)	30	24.6	0.80	–	21	3–12	LV	B	215
	Water/dimethylsulfoxide (100/0, v/v)	30	74.3	0.63	7	–	DP = 815–2830	VG	C	924
	(90/10, v/v)	30	–	Not clear	5	–	DP = 815–3400	VG	C	924
Poly(vinyl bromide)	Cyclohexanone	25	32.8	0.55	7	–	2–10	LS	B	217
	Tetrahydrofuran	25	15.9	0.64	7	–	2–10	LS	B	217
	Methanol/tetrahydrofuran (17/83, v/v)	20	38.8	0.50	7	–	2–10	LS	B	218
Poly(vinyl chloride)	Benzyl alcohol	155.4 (θ)	156	0.50	9	–	4–35	LS	B	219
	Chlorobenzene	30	71.2	0.59	7	–	3–19	SA	B	220
	Cyclohexanone	20	11.6	0.85	–	6	2–10	OS	C	221
		20	13.7	1.0	7	5	7–13	OS	C,D	222
		20	112.5	0.63	5	3	9–15	OS	D,H	222
		25	12.3	0.83	11	–	2–14	OS		223
		25	24	0.77	13	–	3–14	OS		224
		25	204	0.56	?	–	2–15	OS	C	225
		25	174	0.55	6	–	6–22	LS	C	226
		25	8.5	0.75	5	–	4–20	LS	B	227
		25	13.8	0.78	28	–	1–12	LS	A,B,R	228
		30	16.3	0.77	6	–	3–19	SA	B	220
	Tetrahydrofuran	20	3.63	0.92	20	–	2–17	OS	B	229
		25	15.0	0.77	22	–	1–12	LS	A,B	228
		25	16.3	0.766	23	–	2–30	LS	A,B,R	230
		25	49.8	0.69	5	–	4–40	LS	A–B	231
		30	63.8	0.65	9	–	3–32	LS		232
		30	83.3	0.83	7	–	3–19	SA	B	220
		30	219	0.54	16	–	5–30	LS		233
Poly(vinyl fluoride)	Dimethylformamide	90	6.42	0.80	–	9	14–66	SV	D	235
Poly(vinylidene chloride)	Hexamethylphosphoramide	25	25.8	0.65	–	7	0.8–12	LS	C	738
	1-Methyl-2-pyrrolidone	25	13.1	0.69	–	7	0.8–12	LS	C	738
	Tetramethylene sulfoxide	25	13.9	0.69	–	7	0.8–12	LS	C	738
Poly(vinylidene fluoride)	*N*,*N*-Dimethylacetamide, *N*,*N*-dimethylformamide, *N*-methylpyrrolidone, *N*,*N*-dimethyl-*N*,*N*-trimethylene urea	25	45	0.70	–	7	–	LS	–	925
1.7. POLY(VINYL ESTERS)										
Poly(allyl acetate)	Benzene	27	66	0.53	8	–	0.1–0.3	CR		216
Poly(vinyl acetate)	Acetone	6	$[\eta] = 0.104 M^{0.50} + 0.00725 M^{0.90}$		21	–	0.3–150	LS	A	236
		18	24.5	0.67	6	–	4–34	OS	B	237
		20	15.8	0.69	6	–	19–72	LS		238
		25	21.4	0.68	6	–	4–34	OS	B	237
		25	18.8	0.69	?	?	?	LS		239
		25	14.6	0.72	–	6	0.7–1.3	EG	C,L	240
		25	10.8	0.72	10	–	0.9–2.5	EG	B,L	240
		30	17.6	0.68	16	–	2–163	OS	A–B	241
		30	8.6	0.74	8	–	8–66	LS	A–B	242
		30	17.4	0.70	?	–	7–68	OS		243
		30	10.2	0.72	–	8	3–126	LS	C	244
		30	10.1	0.73	11	–	6–150	LS	A	236

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
		30	$[\eta] = 0.097 M^{0.50} + 0.00723 M^{0.90}$		22	–	0.3–150	LS	A	236
		46	13.8	0.71	6	–	4–34	OS	A	236
	Acetonitrile	25	16.2	0.71	–	–	24–215	LS	B	246
		30	41.5	0.62	4	–	97–153	LS	A–B	247
	Benzene	30	22	0.65	5	–	34–102	LS	A–B	248
		30	56.3	0.62	24	–	3–86	OS	B	249
		30	56.3	0.62	12	–	7–54	LS	B	250
		35	21.6	0.675	14	–	5–40	LS	A–B	251
	Butanone	25	13.4	0.71	6	–	25–346	LS	A	252
		25	42	0.62	15	–	2–120	SD,LS	A,B	253
		30	10.7	0.71	–	13	3–120	LS	C	244
	Chlorobenzene	25	110	0.50	9	–	0.15–7	OS	A	195
		25	94.4	0.56	6	–	4–34	OS	A	236
		53	53.7	0.60	6	–	4–34	OS	A	236
		67	28.9	0.65	6	–	4–34	OS	A	236
	Chloroform	20	15.8	0.74	?	?	7–68	OS		243
		25	20.3	0.72	5	–	4–34	OS	A	236
		53	14.7	0.74	5	–	4–34	OS	A	236
	Dioxane	25	11.4	0.74	5	–	4–34	OS	B	237
		53, 60	10.2	0.75	5	–	4–34	OS	B	237
	Ethanol	56.9 (θ)	90	0.50	5	–	4–150	OS, LS	A	236
	Ethyl formate	30	32	0.65	4	–	16–154	LS	A–B	247
	3-Heptanone	26.8 (θ)	82.0	0.50	5	–	4–150	OS, LS	A	236
		29 (θ)	92.9	0.50	18	–	5–83	LS	A–B	255
	Methanol	6 (θ)	101	0.50		–	0.3–150	OS, LS, VOS		236,245
		25	38.0	0.59	5	–	4–22	OS	B	237
		30	31.4	0.60	–	13	3–120	LS	C	244
		53	36.6	0.59	5	–	4–22	OS	B	237
	6-Methyl-3-heptanone	66 (θ)	82.0	0.50	9	–	14–83	LS	A–B	255
		66 (θ)	78.0	0.50	3	–	9–150	OS, LS	A	236
	4-Methyl-2-pentanone	30	44.9	0.60	5	–	12–69	LS		247
	Tetrahydrofuran	25	16	0.70	11	–	5–50	GPC	B,C	739
		35	15.6	0.708	9	–	1.7–117	GPC	B	740
	Toluene	25	108	0.53	4	–	4–15	OS	B	237
		67	156	0.49	4	–	4–15	OS	B	237
	1,2,4-Trichlorobenzene	35	33.0	0.623			5–40	LS		251
	Heptane/3-methyl-2-butanone (27.3/72.7, v/v)	25	92	0.50	6	–	25–287	LS	C	244
Poly(vinyl benzoate)	Xylene	32.5 (θ)	62.0	0.50	5	–	10–24	OS	B	334
Poly(vinyl butyrate)	Benzene	30	11.15	0.735	–	4	3–15	OS	C	256
Poly(vinyl caproate)	Benzene	30	15.47	0.689	–	4	3–126	OS	C	256
Poly(vinyl 4-chlorobenzoate)	Water	30	64.0	0.64	7	–	6–35	LV	B	336
	Butanol/butanone (47/53, v/v)	60 (θ)	73	0.50	7	–	6–35	LV	B	336
Poly(vinyl formate)	Acetone	30	29.3	0.63	–	9	3–41	LV	C	257
	Acetonitrile	30	14.1	0.717	–	9	3–41	LV	C	257
	Dioxane	30	20.7	0.68	–	8	3–41	LV	C	257
	Methyl acetate	30	37.6	0.61	–	7	3–24	LV	C	257
	Methyl formate	30	14.1	0.722	–	7	3–24	LV	C	257
Poly(vinyl isobutyrate)	Benzene	30	11.05	0.711	–	4	5–20	OS	C	256
Poly(vinyl isocaproate)	Benzene	30	51.0	0.575	–	4	3–17	OS	C	256
Poly(vinyl pivalate)	Acetone	25	2.88	0.77	4	–	40–217	LS	C	258
	Butanone–methanol (0.897 g/ml)	20	53	0.50	2	–	222–344	LS	C	258
Poly(vinyl sulfate)	Aq. NaCl (0.5 M)	20	0.55	1.06	6	–	1–6	LV	C	261
1.8. POLY(STYRENE) AND DERIVATIVES										
Poly(4-acetoxystyrene)	Butyl acetate	26.8 (θ)	52.0	0.50	13	–	15–219	LS	A–B	937
	Dioxane	25	16.7	0.65	22	–	0.91–352	LS	A–B	937
	Isopropyl acetate	19.7 (θ)	52.0	0.50	13	–	15–219	LS	A–B	937
	Tetrahydrofuran	25	17.5	0.64	20	–	0.91–352	LS	A–B	937

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TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(4-bromostyrene)	Benzene	20 (θ)	95.5	0.53	10	–	3–30	OS	B	347
		26.3 (θ)	50.0	0.50	5	–	84–250	LS	A,R	348
	Chlorobenzene	30	7.43	0.69	5	–	59–400	LS	A	348
	Toluene	30	18.2	0.57	5	–	63–400	LS	A	349
Poly(*p-tert*-butylstyrene)	Benzene	35	7.1	0.74	4	–	1.8–640	LS	A	741
	Benzene	35	6.58	0.739	7	–	1.3–174	LLS, SEC, OS	A	934
	Cyclohexane	25	8.52	0.717	10	–	1.3–240	LLS, SEC, OS	A	934
	Cyclohexane	35	9.9	0.71	4	–	1.8–640	LS	A	741
	1-Hexanol	65.0	64.7	0.480	6	–	16–240	LLS, SEC, OS	A	934
	Cyclohexane	35.0	11.1	0.694	7	–	2.7–45.5	OS, LLS	A	938
	1,4-Dioxane	25.0	21.4	0.604	4	–	9.3–45.5	OS, LLS	A	938
		38.0	14.3	0.644	4	–	9.3–45.5	OS, LLS	A	938
	1-Nitropropane	31.0	60.0	0.489	7	–	2.7–45.5	OS, LLS	A	938
	1-Nitropropane	31 (θ)	61	0.49	4	–	1.8–640	LS	A	741
	3-Nonanol	10.9	66.5	0.490	6	–	16–240	LLS, SEC, OS	A	934
	2-Octanol	32.7	60.8	0.492	6	–	16–240	LLS, SEC, OS	A	934
	2-Octanol	32.7	61.0	0.489	7	–	2.7–45.5	OS, LLS	A	938
	Tetrahydrofuran	30.0	10.4	0.70	7	–	2.7–45.5	OS, LLS	A	938
Poly(2-chlorostyrene)	Butanone	24.5 (θ)	46.8	0.50	7	–	20–80	LS	A,R	742
		25 (θ)	46.0	0.50	5	–	20–100	LS	A,B	743
	Toluene	25	11.5	0.66	6	–	14–101	LS	A,B	743
		30	14.3	0.65	10	–	23–143	LS	A	350
Poly(4-chlorostyrene)	Benzene	30	30.6	0.56	–	8	10–200	LS	C	351
		26.7	29.3	0.56	5	–	34–180	LV	B	744
	Benzene–methanol (4.5/1, v/v)	41.6 (θ)	56.8	0.50	5	–	34–180	LV	B	744
	Butanone	25	29	0.59	7	–	3–140	LS	B,R	352
		30	3.52	0.75	6	–	17–270	OS	B	353
	Chlorobenzene	30	2.19	0.80	6	–	17–270	OS	B	353
	Chloroform	30	14.8	0.65	–	8	10–200	LS	C	351
	Dioxane	30	17.6	0.62	–	8	10–200	LS	C	351
	Ethylbenzene	30	21.8	0.60	6	–	10–180	LS	B	745
	Toluene	20	24.1	0.605	–	7	2–40	LS	B	354
		25	13.2	0.645	–	7	1–244	LS	B	355
		30	13.0	0.64	6	–	3–140	LS	B,R	352
		30	11.8	0.65	7	–	21–140	LS	A	349
		30	5.37	0.71	7	–	17–270	OS	B	353
Poly(4-cyclohexylstyrene)	Heptane	30	32.3	0.54	6	–	4–30	OS	A–B	266
	Toluene	30	10.6	0.69	7	–	2–30	OS	A–B	266
Poly(2,5-dichlorostyrene)	Toluene	21	12.6	0.69	9	–	7–66	LS		356
	Ethanol/ethyl acetate (1/15, w/w)	30.5 (θ)	35.5	0.50	8	–	50–130	LS		357
Poly(3,4-dichlorostyrene)	Chlorobenzene	30	4.39	0.72	7	–	8–51	OS	A	358
	o-Dichlorobenzene	30	4.11	0.73	7	–	8–51	OS	A	358
	Butanol/butyl acetate (1/13, w/w)	32.9 (θ)		0.50	8	–	40–540	LS		359
Poly(2,4-dimethylstyrene)	Benzene	20	3.8	0.79	6	–	3–22	LS	B,C	746
	Butyl acetate	20	10.2	0.68	8	–	3–22	LS	B,C	746
	Cyclohexane	20	14.8	0.65	7	–	3–22	LS	B,C	746
	Toluene	30	9.52	0.70	–	9	5–120	LS	C	333
Poly(1,4-divinylbenzene)	*trans*-Decalin	25.0 (θ)	286	0.32	7	–	0.7–180	LS, SE	Branching	895
Poly(3-fluorostyrene)	Benzene	25	15.3	0.69						747
	Butanone	25	13.8	0.70						747
	Carbon tetrachloride	25	65.6	0.53			0.5–8	LS	C	747
	Chloroform	25	12.8	0.70	–	8	1–15	LS	C	747
Poly(4-fluorostyrene)	Benzene	25	40.8	0.58						747
	Butanone	25	11.1	0.73						747
	Carbon tetrachloride	25	82.8	0.50						747
	Chloroform	25	16.1	0.69	–	8	0.2–13	LS	C	747
Poly(4-hydroxystyrene)	Dioxane	25	20.3	0.66	11	–	3.8–583	LS	–	940
	Ethyl propionate	25	63.7	0.52	11	–	3.8–583	LS	–	940
	Isobutyl acetate	25	59.4	0.53	11	–	3.8–583	LS	–	940
	Tetrahydrofuran	25	37.2	0.60	10	–	3.8–583	LS	–	940
	1-Chloro-*n*-hexane	5.7	91.6	0.468	9	–	5.9–90.3	LS, SEC, OS	A	888

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(4-iodostyrene)	Dioxane	20	33	0.51	10	6	10–118	LV	B-C	360
Poly((*p*-isopropyl-α-methylstyrene)										
	Tetrahydrofuran	25	45	0.55	8	–	1–63	LS	A-D	748
Poly(*p*-isopropylstyrene)	Toluene	25	12.3	0.69	–	5	14–75	LS	B,C	265
Poly(*o*-methoxystyrene)	Butanone	30	18.6	0.59	5	–	13–35	LS	A–B	362
	Toluene	30	6.40	0.71	5	–	13–35	LS	A–B	362
	Methanol/toluene (25/75, v/v)	30 (θ)	57.5	0.50	4	–	15–30	LS	A–B	362
Poly(*p*-methoxystyrene)	Butanone	30	3.75	0.73	5	–	13–75	LS	A–B	362
		35	8.6	0.68	6	–	1–100	LS	B	352
	Chlorocyclohexane	25	17.7	0.63	16	–	22–220	LS	A	363
	Pentyl acetate	25	55	0.52	16	–	22–220	LS	A	363
	Toluene	25	10.5	0.70	16	–	22–220	LS	A	363
		30	5.28	0.73	5	–	13–75	LS	B	362
		30	18.0	0.62	6	–	1–100	LS	B	352
	Methanol/toluene (28.1/71.9, v/v)	30 (θ)	62.1	0.50	5	–	7–180	LS	B	362
Poly(α-methylstyrene) anionic, (ca. 50%-hetero, ca. 40%-syndio)										
	Benzene	30	10.3	0.72	–	9	4–170	LS	A	319
		30	9.15	0.726	–	6	37.5–685	LS	A	873
	Cyclohexane	34.5 (θ)	73	0.50	–	10	4–750	LS, OS	A	320
		37 (θ)	78	0.50	–	9	9–400	LS	A	321
		38 (θ)	76	0.50	–	6	2–66	LS	A	322
		38.6 (θ)	76.0	0.50	–	9	4–170	LS	A	323
		39	71.3	0.51	–	9	3–140	LS	A	324
	trans-Decalin	9.5 (θ)	67	0.50	–	9	8–750	LS, OS	A	320
	Toluene	25	7.06	0.744	–	9	8–750	LS, OS	A	320
		25	7.81	0.73	–	6	3–60	SD	A	325
		30	10.8	0.71	–	13	2–66	LS	A	322,326
cationic	Benzene	30	24.9	0.647	4	–	14–91	OS	B	327
(10%-hetero, 90%-syndio)	Cyclohexane	32.5 (θ)	66.0	0.50	5	–	2–370	LS	B	328
(19%-hetero, 80%-syndio)		33.3 (θ)	72.7	0.50	8	–	2–18	LS	B	328
	Toluene	30 (θ)	2.2	0.80	6	–	1–100	LS	B	329
	Benzene/methanol (79.4/20.6, v/v)	30	76.8	0.50	4	–	14–91	OS	B	327
		10.0	71.8	0.491	9	–	5.9–90.3	LLS, SEC, OS	A	888
	1-Chloro-*n*-heptane	20.0	94.9	0.459	9	–	5.9–90.3	LLS, SEC, OS	A	888
		27.0	79.6	0.477	9	–	5.9–90.3	LLS, SEC, OS	A	888
	1-Chloro-*n*-octane	43.0	94.8	0.461	9	–	5.9–90.3	LLS, SEC, OS	A	888
		53.0	65.1	0.497	9	–	5.9–90.3	LLS, SEC, OS	A	888
		80.0	74.6	0.484	9	–	5.9–90.3	LLS, SEC, OS	A	888
	n-Hexyl acetate	85.0	86.4	0.474	9	–	5.9–90.3	LLS, SEC, OS	A	888
	Cyclohexane	34.5	74.0	0.493	11	–	2.01–90.3	LLS, SEC, OS	A	888
	trans-Decalin	10.0	83.0	0.480	11	–	2.01–90.3	LLS, SEC, OS	A	888
	n-Butyl chloride	35.5	40.0	0.56	16	–	0.2–130	OS, GPC	B	927
	Carbon tetrachloride	45	19.4	0.67	16	–	0.2–130	OS, GPC	B	927
	Chloroform	25	13.3	0.70	16	–	0.2–130	OS, GPC	B	927
	Tetrahydrofuran	25	11.1	0.69	16	–	0.2–130	OS, GPC	B	927
	Toluene	25	10.1	0.71	16	–	0.2–130	OS, GPC	B	927
	n-Butyl chloride	5	33.6	0.570	8	–	6.0–107	LLS, LS	A	935
		25	27.0	0.590	15	–	5.4–354	LLS, LS	A	935
		50	26.5	0.594	8	–	6.0–107	LLS, LS	A	935
Poly(*m*-methylstyrene)	Benzene	30	7.36	0.76	9	–	8–115	OS	A	330
	Cyclohexane	30	11.76	0.70	7	–	15–83	OS	A	330
	Ethyl acetate	30	17.42	0.64	7	–	15–83	OS	A	330
Poly(*p*-methylstyrene)	Diethyl succinate	16.4 (θ)	70	0.50	6	–	16–200	LS	A	331
	Toluene	30	8.86	0.74	9	–	19–180	LS	A	331
Poly(methylstyrene), position of substituent, unspecified										
	Cyclohexane	20	22	0.68	6	–	11–133	SV	A	332
Poly[(2,3,4,5,6-pentafluorostyrene)]										
	4-Methyl-2-pentanone	20	4.37	0.736	–	21	10–260	OS	C	364
Poly(styrene)	Benzene	20	6.3	0.78	18	–	1–300	SD	A	270
atactic		20	12.3	0.72	7	–	0.6–520	SD	A,R	271

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
		25	22.7	0.72	–	7	0.2–0.8	CR	C,L	272
		25	41.7	0.60	9	–	0.1–1	CR	B,L	272
		25	34.0	0.65	11	–	0.04–0.8	EG	A,L	273
		25	9.52	0.744	6	–	3–61	OS	A	274
		25	9.18	0.743	6	–	3–70	LS	A	275
		25	11.3	0.73	10	–	7–180	OS	A	276
		34	9.8	0.737	10	–	8–80	DV	A	277
	Butanone	25	39	0.58	16	–	1–180	LS	A,R	278
		25	30.5	0.60	5	–	7–150	OS	A	276
		25	19.5	0.635	7	–	12–280	LS	A	279
		30	23	0.62	7	–	40–370	LS	B	280
		34	28.9	0.60	10	–	8–80	DV	A	281,282
		35	17.1	0.64	13	–	4–640	LS	A	752
	Butyl chloride	40.8	15.1	0.659	5	–	29–106	LS	B	283
	n-Butyl chloride	25	17.4	0.662 ($M_w > 3.7 \times 10^4$)	8	–	0.97–67.5	LLS, SEC	A	933
		50	16.2	0.665 ($M_w > 3.7 \times 10^4$)	8	–	0.97–67.5	LLS, SEC	A	933
	Carbon tetrachloride	10	12.6	0.717	–	6	1.8–180	LV	A–B	901
		20	12.0	0.720	–	6	1.8–180	LV	A–B	901
		30	11.4	0.724	–	6	1.8–180	LV	A–B	901
		40	11.2	0.725	–	6	1.8–180	LV	A–B	901
		50	11.0	0.726	–	6	1.8–180	LV	A–B	901
	Chlorobenzene	25.7	7.4	0.749	4	–	62–424	LS	B	283
	Chloroform	25	7.16	0.76	8	–	12–280	LS	A	279
		25	11.2	0.73	5	–	7–150	OS	A	276
		30	4.9	0.794	4	–	19–373	OS	B	284
	Cyclohexane	28	108.0	0.479	7	–	0.6–69	OS	A	285
		34 (θ)	82	0.50	15	–	1–70	LV	A	274
		34 (θ)	90.2	0.503	9	–	0.6–69	OS	A	285
		34.5 (θ)	84.6	0.50	8	–	14–200	LS	A,R	286
		35 (θ)	80	0.50	3	–	8–42	LS	A	287
		35 (θ)	70	0.50	8	–	3–200	SD	B	288
		35 (θ)	76	0.50	10	–	4–137	LS	B	283
		40	41.6	0.554	10	–	4–137	LS	B	283
		45	34.7	0.575	10	–	4–137	LS	B	283
		50	26.9	0.599	10	–	4–137	LS	B	283
		50	36.4	0.584	7	–	4–52	LS	A	289
	cis-Decalin	25	40	0.574	8	–	4.8–2360	LS	A	936
	Decalin (100%-*trans*)	20	149	0.44	7	–	14–200	LS	A	290
		23	98	0.48	7	–	14–200	LS	A	290
		23.8 (θ)	–	0.50	–	–		LS	A	290
		25	67	0.52	7	–	14–200	LS	A	290
		30	61	0.53	6	–	14–200	LS	A	290
		60	22	0.63	4	–	14–200	LS	A	290
	Decalin (73%-*trans*)	18 (θ)	77	0.50	4	–	14–140	LS	A	290
		30	36	0.58	4	–	14–140	LS	A	290
		40	37	0.58	4	–	14–140	LS	A	290
		60	22	0.64	4	–	14–140	LS	A	290
		100	15.7	0.67	6	–	14–200	LS	A	290
	Dichloroethane	25	21.0	0.66	7	–	1–180	LS	A	278
		35	14.3	0.69	11	–	10–500	LS	A	689
	1,2-Dichloroethane	35	14.3	0.69	11	–	9–540	LS	A	752
	Diethyl malonate	34.2 (θ)	71.8	0.50	3	–	39–400	LV	B	291
	Diethyl oxalate	55.8 (θ)	73.0	0.50	3	–	39–400	LV	B	291
	Dimethylformamide	35	31.8	0.603	5	–	0.4–87	LS	A	754
	Dioxane	34	15.0	0.694	10	–	8–80	DV	A	282
	Ethyl acetate/cyclohexane (100/0–10/90)	–	18.2–43.8	0.54–0.67	5	–	11–96	–	A	939
		10	11.0	0.733	–	6	1.8–180	LV	A–B	901
		25	9.32	0.738	–	6	1.8–180	LV	A–B	901
		40	9.07	0.739	–	6	1.8–180	LV	A–B	901
		55	8.79	0.742	–	6	1.8–180	LV	A–B	901
		65	8.90	0.740	–	6	1.8–180	LV	A–B	901

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	Ethylbenzene	25	17.6	0.68	5	–	7–150	OS	A	276
	Ethylcyclohexane	70 (θ)	75	0.50	2	–	36–127	LV	B	292
	Methylcyclohexane	70 (θ)	76	0.50	?	?	?	?		293
		70.5 (θ)	69.6	0.50	3	–	39–400	LV	B	291
	4-Methyl-2-pentanone	35	61.9	0.53			5–100	LS	B	698
	Tetrahydrofuran	25	11.0	0.725	–	7	1–100	GPC	A	749
		25	14	0.70				LS	C	756
	Toluene	20	4.16	0.788	10	–	4–137	LS	B	283
		25	7.5	0.75	8	–	12–280	LS	A	279
		25	8.48	0.748	7	–	4–52	LS	A	289
		25	10.5	0.73	6	–	16–100	LS	A,R	294
		25	17	0.69	9	–	1–160	LS	A	278
		25	7.54	0.783	?	?	5–80	OS		295
		25	13.4	0.71	5	–	7–150	OS	A	276
		25	44	0.65	–	9	0.5–4.5	OS		296
		25	(a increases with M)		10	–	0.08–3.7	CR	L	297
		25	100	0.50	8	–	0.05–0.5	CR	A,R,L	298
		30	9.2	0.72	9	–	4–146	LS	A	299
		30	12.0	0.71	8	–	40–370	LS	B	280
		30	11.0	0.725	7	–	8–85	OS	A–B	300
		34	9.7	0.733	10	–	8–80	DV	A	282
		35	12.6	0.71	16	–	3–650	LS	A	752
		35	12.9	0.71			5–100	LS	B	698
	Trichloro benzene	135	1.75	0.67						697
	Benzene/methanol (74/26, v/v)	34 (θ)	89	0.50	10	–	8–80	DV	A	277
	Butanone/methanol									
	(97.5/2.5, v/v)	25	22.4	0.62	8	–	12–280	LS	A	279
	(95.9/5.0, v/v)	25	26.3	0.60	8	–	12–280	LS	A	279
	(92.5/7.5, v/v)	25	35.7	0.57	8	–	12–280	LS	A	279
	(89/11, v/v)	25 (θ)	73	0.50	8	–	12–280	LS	A	279
	Butanone/2-propanol									
	(6/1, v/v)	23 (θ)	73	0.50	9	–	4–146	LS	A	299
	(82.6/17.4, v/v)	34 (θ)	71.8	0.50	10	–	8–80	DV	A	282
	Chloroform/methanol									
	(90/10, v/v)	25	7.7	0.75	8	–	12–280	LS	A	279,278
	(80/20, v/v)	25	12	0.68	8	–	12–280	LS	A	279,278
	(75/25, v/v)	25	46	0.54	8	–	12–280	LS	A	279,278
	(74.7/24.3, v/v)	25 (θ)	73	0.50	8	–	12–280	LS	A	279,278
	Dioxane/methanol									
	(65.1/34.9, v/v)	34 (θ)	72.6	0.50	10	–	8–80	DV	A	282
	Toluene/methanol									
	(90/10, v/v)	25	10.4	0.715	8	–	12–280	LS	A	279
	(80/20, v/v)	25	26	0.612	8	–	12–280	LS	A	279
	(76.9/24.8, v/v)	25	92	0.50	12	–	0.07–3.5	DV	A,L	298,297
	(75.2/24.8, v/v)	34 (θ)	88	0.50	10	–	8–80	DV	A	282
atactic, anionic	Benzene	25	100	0.50	–	7	0.04–1	VOS, EB	A,L	301
		25	7.8	0.75	17	–	40–6000	LS	A,R	750
		30	8.5	0.75	–	12	2.5–150	VOS	A	301
		30	11.5	0.73	–	5	25–300	LS	A	302
		30	9.50	0.74	–	6	31–500	LS	A	649
	Cyclohexane	34 (θ)	74.5	0.50	–	?	?	LS	B	304
		34.5 (θ)	85	0.50	–	12	0.04–150	LS	A,R	301,303
		34.5 (θ)	88	0.50	–	9	31–970	LS	A	649
		34.5 (θ)	88	0.50	17	–	1–6000	LS	A	750
		34.6 (θ)	91	0.50	–	4	25–300	LS	A	302
		35 (θ)	86	0.50	–	7	2–50	LS	A	305
	Cyclohexene	25	16.3	0.68	–	3	20–107	LS	A	306
	Decalin (66%-*cis*)	12.2 (θ)	80	0.50	–	6	2–50	LS	A	303
	Decalin (99%-*trans*)	20.4 (θ)	81	0.50	–	8	31–760	LS	A	649
	Dichloroethane	30	8.38	0.74	–	5	25–300	LS	A	302
	Dioctyl phthalate	22.0 (θ)	80	0.50	–	4	40–160	LS	A	303
	Tetrahydrofuran	25	13.63	0.714	13	–	2–4000	LS, OS, SD	A,R	753

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TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	Thiophenol	25	14.6	0.70	6	–	42–842	LS	A	778
	Toluene	20	10.69	0.724	20	–	3–4000	LS, OS, SD	A,R	753
		20	11.2	0.72	–	6	3–24	SD		307
		25	9.77	0.73	–	12	1–104	SD	A,R	308
		25	34.5	0.62	–	25	0.4–230	SD	B	309
		30	8.81	0.75	–	5	25–300	LS	A	302
		30.3	10.4	0.73	–	15	2.6–50	OS, LS	A	310
isotactic	Benzene	30	9.5	0.77	6	–	4–75	OS		311
		30	10.6	0.735	7	–	4–37	OS	A–B,R	312
	Chloroform	30	25.9	0.734	3	–	9–32	OS	C-D	284
	o-Dichlorobenzene	25	17.9	0.677	5	–	2–100	LV	C	313
	Toluene	30	11.0	0.725	7	–	3–37	OS	A–B	312
		30	9.3	0.72	5	–	15–71	LS	A–B,R	314
branched, random type	Butanone	25	(a decreases with M)		5	–	30–200	LS	B-C	315
	Cyclohexane	35 (θ)	(a decreases with M)		9	–	8–300	LS	A	316
	Toluene	30	(a decreases with M)		9	–	8–300	LS	A	316
head-to-head	Tetrahydrofuran	25	53	0.61	8	–		LS	C	756
		35 (not θ)	–	0.465	12	–	0.69–43.6	GPC	B	926
		40	–	0.5	3	–	16.1–43.6	GPC	B	926
		35	–	0.67	12	–	0.69–43.6	GPC	B	926
	Cyclohexane	34.5	53	0.50	8	–	3–27	LLS, GPC	A–B	928
		35	–	0.52	10	–	4.4–27.6	LS	A–B	930
	Tetrahydrofuran	30	12.0	0.68	8	–	3–27	LLS, GPC	A–B	928
		25	–	–	–	–	?–18	LS	A–B	930
	Toluene	25	6.38	0.734	8	–	3–27	LLS, GPC	A–B	928
ring	Cyclohexane	40.0	55.6	0.50	10	–	0.7–44	LS		755
	Toluene	35.0	12.4	0.67	10	–	0.7–44	LS		755
star type										
three branches	Benzene	25	5.7	0.76	5	–	3–96	OS, GPC	A	757
	Cyclohexane	35	36	0.55	5	–	3–96	OS, GPC	A	757
four branches	Cyclohexane	35 (θ)	63.1	0.50	9	–	5–140	OS, LS	A	758
		50	26.5	0.58	9	–	5–140	OS, LS	A	758
	Toluene	35	7.4	0.73	9	–	5–140	OS, LS	A	758
regular H-shaped	Cyclohexane	35 (θ)	66.5	0.50	6	–	11–170	LS	A	759
	Toluene	35	74.7	0.73	6	–	11–170	LS	A	759
star type 3 arm	Cyclohexane	16.6–30.1 (θ) (depend on M_w)	–	–	–	–	–	LS	–	931
	Toluene	35	–	Not const.	–	–	–	LS	–	931
star type 15 arm	Cyclohexane	34.5 (not θ)	30	0.5	4	–	86–302	LS, OS, GPC	A–B	941
20 arm	Cyclohexane	34.5 (not θ)	24	0.5	3	–	26–148	LS, OS, GPC	A–B	941
star type, anionic	Cyclohexane	34 (θ)	$g' = 0.94$ (3 branches)*							304
			$g' = 0.82$ (4 branches)*							304
	Decalin	15	$g' = 0.48$ (9 branches)*							318
	Toluene	25	$g' = 0.90$ (3 branches)*							304
		34	$g' = 0.84$ (4 branches)*							304
Poly(styrenesulfonic acid)	Aq. HCl (0.52 M)	25	(0.344)	(1.0)	3	–	18–46	LV		365
	Aq. NaCl (0.52 M)	25	(0.312)	(1.0)	3	–	18–46	LV		365
–, sodium salt	Aq. NaCl									
	(4.17 M)	25 (θ)	20.4	0.50	4	–	49–228	LS	B	366
	(0.5 M)	25	18.6	0.64	6	–	39–234	LS	B,R	366
	(0.1 M)	25	17.8	0.68	6	–	39–234	LS	B	366
	(0.05 M)	25	13.9	0.72	6	–	39–234	LS	B	366
	(0.02 M)	25	10.1	0.78	6	–	39–234	LS	B	366
	(0.01 M)	25	2.8	0.89	5	–	39–234	LS	B	366
	(0.05 M)	25	2.3	0.93	5	–	49–234	LS	B	366
Aq. KCl	(3.1 M)	25	20.4	0.50	4	–	49–234	LS	B	366

[a] $g' = [\eta]$ of branched molecules/$[\eta]$ of linear molecules with same mol. wt.

1.9. OTHER COMPOUNDS

Poly(2-acrylamino-2-methylpropanesulfonamide)										
	Water	25	12.3	0.757	10	–	85–122	OS, LS	B	1011
	Water/1,4-dioxane (81.5/18.5, v/v)	25 (θ)	172	0.50	10	–	85–122	OS, LS	B	1011

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(allylammonium chloride)	Aqueous NaCl (0.05 M)	25	2.40	0.975	9	–	4.1–18	LS	–	900
	(0.2 M)	25	7.18	0.815	7	–	2.8–18	LS	–	900
	(0.5 M)	25	7.19	0.794	6	–	2.8–18	LS	–	900
	(1.0 M)	25	13.9	0.714	9	–	4.1–18	LS	–	900
Poly[(biphenyl-4-yl)-ethylene]	Benzene	20	21.4	0.619	5	–	7–170	LS	B	264
		30	29.5	0.59	6	–	1–110	LV	B	264
		75	27.7	0.589	5	–	7–170	LS	B	264
Poly(*tert*-butyl crotonate)	Butyl chloride	25	7.7	0.82	–	11	0.2–32	OS, GPC	A	760
	Toluene	25	7.7	0.82	10	–	0.6–35	OS, GPC	A	760
Poly(carbanilinoxyethylene), (Poly(vinyl carbanilate))	Dioxane	20	13.7	0.68	11	–	6–200	LS	A	335
	Dioxane/methanol (28/72, v/v)	20 (θ)	64.5	0.51	5	–	6–200	LS	A	335
Poly(dibutyl itaconate)	Toluene	25	5.70	0.70	8	–	20–105	LS	B	762
Poly(dicyclohexyl itaconate)	Tetrahydrofuran	25	23.3	0.58	5	–	5–56	OS	B	761
	Toluene	25	13.1	0.623	10	–	5–56	OS	B	761
Poly(didecyl itaconate)	Toluene	25	8.01	0.64	11	–	13–82	LS	B	762
Poly(diethyl fumarate)	Benzene	30	1.62	0.87	10	–	?–20	OS, GPC	A–B	1009
Poly(didodecyl itaconate)	Toluene	25	11.7	0.59	–	–	–	LS	–	945
Poly(diethyl itaconate)	Toluene	25	1.48	0.80	–	8	5–61	LS	B	762
Poly[di(ethylcyclohexyl) itaconate]										
	Toluene	25	2.62	0.73			16–170	LS	B	763
Poly(dihexadecyl itaconate)	Toluene	25	11.4	0.60	–	–	–	LS	–	945
Poly(dihexyl itaconate)	Toluene	25	3.71	0.72	–	8	12–122	LS	B	762
Poly(diicosayl itaconate)	Toluene	25	14.3	0.56	–	–	–	LS	–	945
Poly(diisopropyl fumarate)	Benzene	30	0.753	0.98	11	–	4.7–30	OS, GPC	A–B	1009
		(Wormlike behavior)								
Poly(dimethyl itaconate)	Benzene	25	5.15	0.68	–	8	4–120	LS	B	762
Poly[di(methylcyclohexyl) itaconate]										
	Toluene	25	2.02	0.76			13–102	LS	B	763
Poly(dioctyl itaconate)	Toluene	25	3.67	0.71	–	8	11–163	LS		762
Poly(diphenylmethylene)	Benzene		218	0.328	?		1–90	?		267
Poly(diphenyl itaconate)	Toluene	25	4.47	0.69	9	–	5.7–57	LS	–	943
Poly(dioctadecyl itaconate)	Toluene	25	13.1	0.59	–	–	–	LS	–	945
Poly(dipropyl itaconate)	Toluene	25	1.62	0.78	–	6	13–109	LS		762
Poly[di(propylcyclohexyl itaconate]										
	Pentyl acetate	25	14.0	0.56			16–91	LS	B	763
	Toluene	25	2.23	0.73			16–91	LS	B	763
Poly(ditetradecyl itaconate)	Toluene	25	9.91	0.61	–	–	–	LS	–	945
Poly(diundecyl itaconate)	Toluene	25	10.01	0.61	–	4	9–250	LS		762
Poly(1-methoxycarbonyl-1-phenylethylene)										
	Benzene	30	35.6	0.566	8	–	6–40	LS	A	361
	Chloroform	30	12.7	0.661	8	–	6–40	LS	A	361
	Ethylbenzene	15 (θ)	51.4	0.507	8	–	6–40	LS	A	361
Poly(monodecyl itaconate)	Tetrahydrofuran	25	–	Not const.	11	–	3.3–72.4	LS, SEC	A	1006
		(Wormlike behavior)								
Poly(monododecyl itaconate)	Tetrahydrofuran	25	11.2	0.65	9	–	1.7–233	LS, SEC	A	949
		(Wormlike behavior)								
Poly[bis(phenylethyl) itaconate]	Toluene	25	5.26	0.66	8	–	9.0–45	LS	–	943
Poly[bis(phenyl-*n*-propyl) itaconate]										
	Toluene	25	5.97	0.65	12	–	6.5–52	LS	–	943
Poly(9-vinyladenine)	Aq. NaCl (0.1 M)/ $NaAs(CH_3)_2$(0.1 M); pH7	26	241	0.35	3	–	9–51	OS		830
		40	29.2	0.53	3	–	9–51	OS		830
Poly(1-vinyl-3-benzylimidazolium chloride)										
	Aq. NaCl (0.2 M)	25	5.89	0.74	4	–	18–130	LS	B	813
	Methanol/$N(CH_3)_4$ Br(0.01 M)	25	13.4	0.63	5	–	18–160	LS	B	813
Poly(4-vinylbenzyl trimethylammonium chloride)										
	Aq. NaCl (0.5 M)	25	19.6	0.67	4	–	1–21	OS, GPC		764
	(0.1 M)	25	18.6	0.70	4	–	1–21	OS, GPC		764
	(0.01 M)	25	7.87	0.85	4	–	1–21	OS, GPC		764
	(0.002 M)	25	5.77	0.88	4	–	1–21	OS, GPC		764

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TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(vinylcarbazole)	Benzene	25	30.5	0.58	11	–	0.7–45	LS	A	367
	Bromobenzene	25	5.14	0.76	7	–	7–49	GPC	A–B	765
	Chlorobenzene	30	5.23	0.755	6	–	7–57	GPC	A–B	766
		45	4.44	0.776	6	–	7–57	GPC	A–B	766
	Chloroform	25	5.93	0.74	7	–	7–49	GPC	A–B	765
		25	13.6	0.67	8	–	3–45	LS	A	367
	Cyclohexanone	25	20.0	0.61	9	–	2–45	LS	A	367
	1,2-Dichlorobenzene	25	11.0	0.68	7	–	4–44	GPC	A–B	767
	1,3-Dichlorobenzene	25	5.6	0.75	6	–	7–49	GPC	A–B	767
	Nitrobenzene	25	9.25	0.69	6	–	7–49	GPC	A–B	765
		30	7.19	0.716	6	–	7–57	GPC	A–B	766
		45	6.03	0.739	6	–	7–57	GPC	A–B	766
	Tetrachloroethane	25	12.9	0.68	9	–	2–45	LS	A	367
	Tetrahydrofuran	25	14.4	0.65	10	–	1–45	LS	A	367
	Toluene	37 (θ)	76.2	0.50	7	–	4–107	OS	A	368
Poly(*N*-vinyl-3,6-dibromo carbazole)										
	p-Chloro-*m*-cresol	112.9 (θ)	30.2	0.50	7	–	4.8–125	–	–	942
	o-Chlorophenol	60.0 (θ)	27.5	0.50	8	–	4.8–125	–	–	942
Poly(1-vinylimidazole)	Aq. NaCl (0.1 M)	25	122	0.51	5	–	9–90	LS	B	813
	(5.0 M)	25 (θ)	121	0.50	8	–	5–90	LS	B	813
	Aq. NaSCN (0.1 M)	25 (θ)	105	0.50	5	–	9–90	LS	B	813
	Methanol/$N(CH_3)_4$ Br(0.01 M)	25	48.5	0.63	6	–	9–130	LS	B	813
protonated	Aq. HCl (0.1 M)/ NaCl (1 M)	25 (θ)	169	0.50	5	–	9–90	LS	B	813
Poly(5-vinyl-2-methylpyridine)	Butanone	25	13.9	0.65	5	–	13–88	LS	A	375
		25	19	0.64	15	–	6–100	LS	A	376
	Dimethylformamide	25	13.0	0.76	6	–	4–40	OS	A–B	377
	Methanol	25	18.0	0.83	8	–	4–40	OS	A–B	377
		25	18.6	0.70	9	–	7–80	LS	A	376
		25	8.0	0.76	9	–	13–88	LS	A	375
Poly(1-vinylnaphthalene)	Benzene	20	2.20	0.82	4	–	4–17	LS	B	264
		75	1.03	0.88	4	–	4–17	LS	B	264
Poly(2-vinylnaphthalene)	Benzene	17	1.7	0.80	11	–	10–100	LS		268
		20	6.90	0.719	6	–	6–68	LS	B	264
		75	8.69	0.695	6	–	6–69	LS	B	264
	Decalin/toluene (13/10, w/w)	30.2 (θ)		0.50	8	–	10–100	LS		269
Poly(3-vinylpyrene)	Chloroform	25 (θ)	51.0	0.500	9	–	3–50	LS	A	768
	1,2-Dichlorobenzene	25	11.7	0.655	9	–	3–50	LS	A	768
	Tetrahydrofuran	25	31.8	0.547	9	–	3–50	LS	A	768
Poly(2-vinylpyridine)	Benzene	25	6.6	0.72	3	–	3–11	LS		769
		25	17.0	0.64	14	–	3–93	LS	B,C	371
	Benzene	10	151	0.445	5	–	9.4–196	LS, OS	A	944
	Benzene	10	149	0.43	5	–	9.4–196	LS, OS	A	952
		11.4 (θ)	81	0.50	5	–	9.4–196	LS, OS	A	952
		12	56.0	0.53	5	–	9.4–196	LS, OS	A	952
		15	31.8	0.59	5	–	9.4–196	LS, OS	A	952
	Butanone	25	97.2	0.47	14	–	3–93	LS	B,C	207
	Dimethylformamide	25	14.7	0.67	14	–	3–93	LS	B,C	371
	Dioxane	25	30.9	0.58	14	–	3–93	LS	B,C	371
	Methanol	25	11.3	0.73	14	–	3–93	LS	B,C	371
	Methyl ethyl ketone	25	93.3	0.480	5	–	9.4–196	LS, OS	A	944
	Pyridine	25	9.84	0.727	5	–	9.4–196	LS, OS	A	944
	2-Propanol	25	18.4	0.67	5	–	9.4–196	LS, OS	A	952
	Pyridine	25	9.9	0.73	4	–	9.4–196	LS, OS	A	952
	Pyridine	25	13.8	0.69	14	–	3–93	LS	B,C	207
	Ethanol/water (92/8, w/w)	25	12.2	0.73	14	–	3–93	LS	B,C	371
Poly(2-vinylpyridine 1-oxide)	Benzyl alcohol	25	9.35	0.73	14	–	1.3–34	LS	A–B	946
	1-Butanol	25	7.78	0.72	14	–	1.3–34	LS	A–B	946
	Chloroform	25	15.8	0.64	14	–	1.3–34	LS	A–B	946
	Methanol	25	7.55	0.72	14	–	1.3–34	LS	A–B	946

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	1-Propanol	25	10.3	0.70	14	–	1.3–34	LS	A–B	946
	2-Propanol	25	12.0	0.68	14	–	1.3–34	LS	A–B	946
Poly(4-vinylpyridine)	Ethanol	25	(1.51)	(0.52)	–	3	1–4	SD	C	372
		25	25.0	0.68	8	–	10–185	LS	A–B	373
	Water	25	22.0	0.687	8	–	10–185	LS	A–B	373
	Butanone/2-propanol	25	38.0	0.57	7	–	7–224	LS	B	374
	Ethanol/water (92/8, w/w)	25	12.0	0.73	7	–	7–224	LS	B	374
Poly(vinylpyrrolidone)	Chloroform	25	19.4	0.64	4	2	2–23	LS	B	378
	Methanol	30	23	0.65	–	6	2–23	LS	B	378
	Water	20	64	0.58	3	–	1–9	SD	B	379
		25	67.6	0.55	15	–	0.7–10	LS	B,R	378
		25	4.1	0.85	–	5	1–4	SD	C,D	211
		30	14	0.70	9	–	1–20	SD	B	381
		30	39.3	0.59	6	–	8–110	OS	A,R	383
	Acetone/water (66.8/33.2, v/v)	25 (θ)	75.0	0.50	–	3	1.2–108	LS	B	384
	Aq. Na_2SO_4 (0.55 M)	25 (θ)	58.0	0.500	5	–	9–46	OS	B	770
Poly(1-vinyl-2-pyrrolidone)										
high M	Aq. sodium acetate (0.1 M)	25	8.86	0.74 (Mixed)			4–220	–	–	948
low M	Aq. sodium acetate (0.1 M)	25	64	0.53 (Mixed)			0.2–4	–	–	948
Poly(vinylsulfonic acid)	Aq. KBr (0.347 M)	5.7 (θ)	68.8	0.50	5	–	4–39	LS	B	259
		15	30.8	0.61	5	–	8–39	LS	B	259
		30	24.5	0.75	5	–	8–39	LS	B	259
		50	26.6	0.76	5	–	8–39	LS	B	259
	Aq. KCl	5.5 (θ)	68.2	0.50	5	–	4–39	LS	B	259
	(0.349 M)	25	16.7	0.79	5	–	4–39	LS	B	259
	(0.650 M)	26.0 (θ)	79.5	0.50	5	–	4–39	LS	B	259
	(1.001 M)	44.5 (θ)	80.3	0.50	5	–	4–39	LS	B	259
	Aq. NaBr	−0.6 (θ)	95.5	0.50	5	–	4–39	LS	B	259
	(0.346 M)	10	26.8	0.73	5	–	8–39	LS	B	259
		20	25.1	0.76	5	–	8–39	LS	B	259
		30	22.0	0.79	5	–	8–39	LS	B	259
	(1.008 M)	40.1 (θ)	94.5	0.50	5	–	4–39	LS	B	259
	Aq. NaCl									
	(1.003 M)	32.4 (θ)	96.1	0.50	5	–	4–39	LS	B	259
	(0.5 M)	20	21.5	0.65	–	6	0.3–3	SD	C	260
Poly(vinyltrimethylsilane)	Cyclohexane	25	8.2	0.71	5	–	59–213	LS	B	610
1.10. COPOLYMERS										
Poly(acrylonitrile-*co*-butadiene), see also Poly(butadiene-*co*-acrylonitrile) in group 1.1										
(18/82, w/w, random)	Toluene	25	251	0.50	7	–	0.06–1.26	OS	A	590
(26/74, w/w, random)	Toluene	25	260	0.50	5	–	0.15–0.40	OS	A	590
Poly(acrylonitrile-*co*-glycidyl methacrylate)										
	Dimethylformamide	30	175	0.65	?	?	?	?		591
Poly(acrylonitrile-*co*-methyl acrylate)										
	Dimethylformamide	20	17.9	0.79	6	–	2–21	LS	B	592
Poly(acrylonitrile-*stat*-methyl acrylate), (91.5/8.5, w/w)										
	Dimethylformamide	25	21.3	0.743	11	–	2–53	LS		772
	Ethylene carbonate/water (82.5/17.5, w/w)	25 (θ)	152	0.502	8	–				772
Poly(acrylonitrile-*co*-styrene)										
(38.3/61.7, mol/mol,)	Butanone	30	36	0.62	16	–	15–120	LS	B	593
	Tetrahydrofuran	25	21.5	0.68	4	–	10–78	LS	B	594
azeotropic (62.6/37.4, mol/mol, random)	Butanone	30	53	0.61	11	–	19–56	LS	B	595
	Dimethylformamide	30	12	0.77	11	–	19–56	LS	B	595
Poly(acrylonitrile-*stat*-styrene) (27.4/72.6, mol/mol)	Dimethylformamide	30	12.0	0.74	6	–	14–58	LS	B	773

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TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
(38.5/61.5, mol/mol)	Dimethylformamide	30	16.2	0.73	6	–	22–106	LS	B	773
(47.5/52.5, mol/mol)	Dimethylformamide	30	17.2	0.73	6	–	14–78	LS	B	773
Poly(acrylonitrile-*alt*-styrene) (1/1, mol/mol)										
	Butanone	30	24.3	0.67	10	–	5–100	LS	B	774
	Dimethylformamide	30	7.63	0.76	9	–	8–100	LS	B	774
	Butanone/methanol (63.6/36.4, v/v)	30 (θ)	140	0.50	10	–	5–100	LS	B	774
Poly(acrylonitrile-*co*-vinylidene chloride) 58 wt.% AN										
	γ-Butyrolactone	25	112	0.576	8	–	4.2–50	LS	–	961
	Dimethylacetamide	25	99.9	0.603	8	–	4.2–50	LS	–	961
	Dimethylformamide	25	102	0.591	8	–	4.2–50	LS	–	961
	N-methyl-2-pyrrolidone	25	114	0.595	8	–	4.2–50	LS	–	961
	70 wt.% HNO_3	25	142	0.521	8	–	4.2–50	LS	–	961
Poly(*p*-aminobenzoic acid-*stat*-6-aminohexanoic acid) (1/1, mol/mol)										
	Dichloroacetic acid	30	23.7	1.13	4	–	12–60	LS		775
	Trifluoroacetic acid	30	132.5	0.99	4	–	12–60	LS		775
Poly(butadiene-*co*-methacrylamide) (90/10, w/w, random)										
	Toluene	25	437	0.50	5	–	0.09–0.11	OS	A	590
Poly(butadiene-*co*-2-methyl-5-vinylpyridine)										
	Toluene	25	309	0.50	5	–	0.08–1.04	OS	A	590
Poly(butadiene-*co*-styrene), see also Poly(butadiene-*co*-styrene) in group 1.1 (84/16 mol/mol, random)										
	Benzene	25	39.4	0.70	4	–	2–51	OS	A	596
	Dibutyl phthalate	56	472	0.40	6	–	2–51	OS	A	596
	2-Pentanone	23.8 (θ)	167	0.50	5	–	7–51	OS	A	596
Poly(butyl itaconate-*co*-dibutyl itaconate), (40/60, mol/mol, random)										
	Acetone	25	575	0.32	6	–	9–70	LS	B	597
	Methanol	25	354	0.32	7	–	11–110	LS	B	597
	m-Xylene	25	1040	0.21	7	–	11–110	LS	B	597
Poly(butyl methacrylate-*alt*-styrene)										
	Butanone	25	5.3	0.76	10	–	32–320	LS	B	776
Poly(butyl methacrylate-*stat*-styrene) (1/1, mol/mol)										
	Butanone	25	4.9	0.77	10	–	15–208	LS	B	776
		35	5.98	0.75	8	–	22–56	LS	B	777
	Cyclohexane	35	7.88	0.70	6	–	34–50	LS	B	777
Poly(*tert*-butylphenyl methacrylate)-*stat*-vinylpyrrolidone)										
(33.7/66.3, mol/mol)	Benzene	25	118	0.53	5	–	33–96	OS	C	779
	Chloroform	25	105	0.57	5	–	33–96	OS	C	779
(60.4/39.6, mol/mol)	Benzene	25	130	0.55	4	–		OS	C	779
	Chloroform	25	247	0.51	4	–		OS	C	779
Poly(*p*-*tert*-butylstyrene)-*block*-poly(dimethyl-siloxane)-*block*-poly(*p*-*tert*-butylstyrene) 28 wt.% PBS										
	Methyl ethyl ketone	10	252	0.39	4	–	12–129	OS, GPC	C–D	957
		15	58.9	0.50	4	–	12–129	OS, GPC	C–D	957
		20	40.5	0.54	4	–	12–129	OS, GPC	C–D	957
		31	19.3	0.61	4	–	12–129	OS, GPC	C–D	957
		38	10.7	0.66	4	–	12–129	OS, GPC	C–D	957
	Benzene	35	15.4	0.66	4	–	9.3–128	OS, LS	B	960
	Methyl ethyl ketone	15 (θ)	58.9	0.50	4	–	9.3–128	OS, LS	B	960
		20	40.5	0.54	4	–	9.3–128	OS, LS	B	960
		31	19.3	0.61	4	–	9.3–128	OS, LS	B	960
Poly(*p*-chlorostyrene-*stat*-methylmethacrylate), (52/48, mol/mol)										
	Benzene	27	7.94	0.72	9	–	15–120	LS	C	598
	Benzene/hexane (60/40, v/v)	22.3 (θ)	64	0.50	8	–	15–120	LS	C	598
Poly(4-chlorostyrene)-*block*-poly(styrene)-*block*-poly(4-chlorostyrene), A_k–B_n–A_k										
$2k/n$, (33/67, mol/mol)	Carbon tetrachloride	40	6.24	0.76	–	5	14–61	OS	A	780
	Cumene	40	5.88	0.76	–	5	14–61	OS	A	780
$2k/n$, (50/50, mol/mol)	Carbon tetrachloride	40	6.63	0.74	–	10	19–74	OS	A	780
	Cumene	40	5.76	0.75	–	10	19–74	OS	A	780
Poly(diethyl fumarate-*co*-isobutene)										
50/50, mol/mol	Benzene/petrol ether	20	340	0.44	4	–	1–14	SD	C	599
Poly(diethyl fumarate-*co*-vinylcarbazole)										
	Benzene	25	14.6	0.67	6	–	2–13	LS	B	789

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(*p*-diethylphosphonomethylstyrene-*co*-styrene), (1/4, mol/mol, random)										
	Benzene	20	1.95	0.90	5	–	15–50	SV	B	
	Tetrachloroethane	20	0.0836	1.18	11	–	9–51	SV	B	
Poly(dimethyl itaconate-*co*-styrene)										
(75/25, w/w)	Toluene	25	6.6	0.68	6	–	6–22	LS	A	683
(67/33, w/w)		25	9.0	0.67	6	–	4–19	LS	A	683
(59/41, w/w)		25	9.7	0.67	8	–	5–38	LS	A	683
(49/51, w/w)		25	11.7	0.67	7	–	6–24	LS	A	683
(29.5/70.5, w/w)		25	12.8	0.67	8	–	7–36	LS	A	683
(27/73, w/w)		25	10.9	0.69	8	–	6–40	LS	A	683
(0/100, w/w)		25	11.45	0.712	12	–	3–58	LS	A	683
Poly(dimethyl siloxane-*co*-diphenylsiloxane)										
(54/45, mol/mol)	Benzene	25	40.7	0.60	5	–	7–57	OS	A	596
	Dimethyl phthalate	82.5	512	0.31	5	–	11–57	OS	A	596
	Ethanol/toluene (37/63, w/w)	29.5 (θ)	78	0.50	5	–	7–35	OS	A	596
(66/34, mol/mol)	Benzene	25	15.6	0.68	4	–	3.7–100	LS	A	596
	Hexane	36 (θ)	141	0.44	4	–	3.7–100	LS	A	596
	Benzene/2-propanol (44/56, w/w)	42 (θ)	74	0.50	4	–	3.7–100	LS	A	596
Poly(divinylstyrene-*co*-styrene), see also Poly(styrene), branched, random type, in group 1.8.										
	Benzene	25	37.2	0.70	5	–	5–80	SD		602
	Octane	21	162	0.50	6	–	5–80	SD		602
Poly(ethyl acrylate-*stat*-methyl methacrylate), (80/20, mol/mol)										
	Acetone	25	62	0.57	10	–	65–800	LS	B	114
Poly(ethyl methacrylate-*alt*-styrene)										
	Butanone	25	9.3	0.72	9	–	19–470	LS	B	776
Poly(ethyl methacrylate-*stat*-styrene)										
(1/1, mol/mol)	Butanone	25	12.0	0.70	10	–	3–185	LS	B	776
Poly(ethylene-*co*-α-methyl-styrene), $[(ET)_m(MS)_n]_p$										
$m/n = 3/4$	Cyclohexane	30	92	0.56	5	–	0.7–6	SA	B	604
	Dioxane	30	76	0.58	5	–	0.7–6	SA	B	604
	Toluene	30	32	0.68	5	–	0.7–6	SA	B	604
	Butanone/cyclohexane (60/40, v/v)	30 (θ)	135	0.50	5	–	0.7–6	SA	B	604
$m/n = 5/4$	Cyclohexane	30	65	0.60	5	–	0.8–7	SA	B	604
	Dioxane	30	89	0.56	5	–	0.8–7	SA	B	604
	Toluene	30	37	0.66	5	–	0.8–7	SA	B	604
	Butanone/cyclohexane (75/25, v/v)	30 (θ)	140	0.50	5	–	0.8–7	SA	B	604
$m/n = 5/7$	Cyclohexane	30 (θ)	112	0.50	4	–	1.5–7	SA	B	604
	Dioxane	30	123	0.49	4	–	1.5–7	SA	B	604
	Toluene	30	56	0.58	4	–	1.5–7	SA	B	604
Poly(hexadecyl methacrylate-*co*-methyl methacrylate)										
(25/75, mol/mol, random)	Heptane	25	85.0	0.38	8	–	4–47	LS	A	605
	Propyl acetate	25	17.1	0.62	8	–	4–47	LS	A	605
(38/62, mol/mol, random)	Chloroform	25	36.3	0.57	8	–	16–195	LS	A	605
	Heptane	25	6.9	0.65	8	–	4–47	LS	A	605
	Propyl acetate	25	53.6	0.50	8	–	4–47	LS	A	605
(50/50, mol/mol, random)	Chloroform	25	53	0.54	8	–	10–154	LS	A	605
	Heptane	25	32.0	0.52	8	–	4–47	LS	A	605
	Propyl acetate	25	91.3	0.43	8	–	4–47	LS	A	605
Poly(1-hexene-*co*-sulfur dioxide), see Poly[sulfonyl(butylethylene) in group 3.10.										
Poly(isobutene-*co*-isoprene), see group 1.2.										
Poly(isoprene)-*block*-poly(styrene), see group 1.1.										
Poly(maleic acid-*alt*-styrene)										
DN[b] ≈ 9.5	Aq. NaCl (0.02 M)	25	395	0.59	7	–	29–130	LV	B	842
0.50		25	37.3	0.67	7	–	29–130	LV	B	842
0.20		25	0.51	0.87	7	–	29–130	LV	B	842
0.10		25	0.01	0.98	7	–	29–130	LV	B	842
Poly(methacrylic acid-*co*-methyl)methacrylate) (7.4/92.6, w/w)										
	Acetone	20	3.4	0.74	9	–	26–105	LS		607

[b] DN–Degree of neutralization

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(4-methoxystyrene-*stat*-styrene)										
(24.4/75.6, mol/mol)	Toluene	25	7.0	0.75	9	–	4–42	OS	B	781
(26.4/73.6, mol/mol)	Butanone	25,50	40	0.585	8	–	7–80	LV	B	782
	Toluene	25	18.6	0.68	8	–	7–80	LV	B	782
(46.2/53.8, mol/mol)	Toluene	25	7.3	0.755	13	–	3.5–70	OS	B	781
(53.0/47.0, mol/mol)	Butanone	25 (θ)	100	0.49	7	–	7–80	LV	B	782
	Toluene	25	37	0.615	7	–	7–80	LV	B	782
(74.0/26.0, mol/mol)	Toluene	25	8.2	0.755	9	–	5–35	OS	B	781
(75.6/34.4, mol/mol)	*tert*-Butylbenzene	25 (θ)	83	0.49	6	–	7–80	LV	B	782
	Toluene	25	16.9	0.673	7	–	7–80	LV	B	782
Poly(methyl acrylate-*stat*-methylmethacrylate)										
(17/83, w/w)	Butanone	25	11.7	0.70	4	–	56–208	LS	B	131
(29/71, w/w)		25	11.1	0.63	4	–	37–137	LS	B	131
(67/33, w/w)		25	36.6	0.60	4	–	71–187	LS	B	131
Poly(methyl acrylate-*co*-styrene)										
(22/78, mol/mol, random)	Benzene	30	8.93	0.744	16	–	2.6–80	LS	A	129
	Butanone	30	21.1	0.640	12	–	2.6–80	LS	A	129
	2-Methylcyclohexanol	43.5 (θ)	77	0.50	8	–	2.6–80	LS	A	129
(33/67, mol/mol, random)	Benzene	30	7.18	0.759	9	–	6.6–36	LS	A	129
	Butanone	30	11.4	0.696	4	–	6.6–36	LS	A	129
	2-Methylcyclohexanol	35.0 (θ)	76	0.50	7	–	6.6–36	LS	A	129
(47/53, mol/mol, random)	Butanone	30	10.7	0.724	9	–	6.7–24.4	OS	A	129
(50/50, mol/mol, random)	Ethyl acetate	35	41.6	0.57	9	–	18–116	LS	A	133
(59/41, mol/mol, random)	Benzene	30	6.15	0.780	6	–	7–40	LS	A	129
	Butanone	30	11.3	0.703	4	–	12–40	LS	A	129
	2-Methylcyclohexanol	36.6 (θ)	76	0.50	6	–	7–40	LS	A	129
(76/24, mol/mol, random)	Benzene	30	7.42	0.766	6	–	7.2–28	LS	A	129
	Butanone	30	9.16	0.728	5	–	8.9–28	LS	A	129
	2-Methylcyclohexanol	29.4 (θ)	75	0.50	5	–	6.5–24	LS	A	129
Poly(methyl methacrylate-*co*-*p*-isopropylstyrene), 2/3, mol/mol, *graft*										
	Butanone	25	0.021	1.11	–	6	31–65	LS		611
Poly(methyl methacrylate-*co*-2-methyl-5-vinylpyridine) (85/15, mol/mol, random)										
	Acetic acid	25	170	0.51	3	–	37–150	LV	B	612
Poly(methyl methacrylate-*co*-styrene)										
(10/90, mol/mol, random)	1-Chlorobutane	40.8	16.6	0.609	5	–	20–82	LS	B	613
(30/70, mol/mol, random)	1-Chlorobutane	30	17.6	0.67	9	–	5–55	LS	B	614
	Cyclohexanol	64.0 (θ)	71.6	0.51	4	–	5–55	LS	B	614
	Toluene	30	8.32	0.75	10	–	5–55	LS	B	614
(44/56, mol/mol, random)	1-Chlorobutane	30	24.9	0.63	10	–	5–81	LS	B	614
	Cyclohexanol	64.0 (θ)	70.0	0.51	4	–	10–81	LS	B	614
	Toluene	30	13.2	0.71	11	–	4.8–81	LS	B	614
(50/50, mol/mol, random)	Butanone	25	15.4	0.675	11	–	5–227	LS	B	615
(52/48, mol/mol, random)	1-Chlorobutane	40.8	49.0	0.575	5	–	18–115	LS	B	613
(71/29, mol/mol, random)	1-Chlorobutane	30	24.9	0.63	10	–	4.8–81	LS	B	614
	Cyclohexanol	68.0 (θ)	97.3	0.47	5	–	15–106	LS	B	614
	Toluene	30	11.4	0.70	8	–	7–106	LS	B	614
(94/6, mol/mol, random)	1-Chlorobutane	40.8	27.6	0.617	5	–	20–100	LS	B	613
nearly equimolar, three blocks (MSM) PS% 86.1–90.4, *graft* (S on *M*; $M_{PS} = 0.7–1.0 \times 10^4$)	Cyclohexanol	81.0 (θ)	$\lim_{M_w \to 0}[\eta]_\theta / M^{1/2} = 63$		–	7	3.4–147	LS	B	616
	Benzene	25	$[\eta] = 0.00918 M_{PS}^{0.743} g^{0.77}$				83–121	LV		617
	Butanone	25	$[\eta] = 0.0390 M_{PS}^{0.58} g^{0.59}$				53–285	LV		617
graft	Bromoform	–	56	0.6	6	–	180–320	LV		618
Poly(methyl methacrylate)-*block*-poly(styrene), A_k–B_n, k/n, (54/46, w/w)										
	Butanone	30	9.4	0.69	6	–	5.8–93	LS	A	783
	1-Chlorobutane	30	22.4	0.60	6	–	5.8–93	LS	A	783
	Toluene	30	7.3	0.73	6	–	5.8–93	LS	A	783
Poly(2-methyl-1-pentene-*co*-sulfur dioxide), see Poly[sulfonyl(1-methyl-1-propylethylene)], group 3.10.										
Poly(α-methylstyrene-*co*-styrene)										
	Benzene	20	14.4	0.69	5	–	18–80	SV	A	332
Poly(1-octadecane-*alt*-maleic acid)										
	Ethyl acetate	25	55	0.48	6	–	2.4–9.9	LS	–	956

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(octyl methacrylate-*alt*-styrene)										
	Butanone	25	7.4	0.72	11	–	12–450	LS	B	776
Poly(octyl methacrylate-*stat*-styrene), (1/1, mol/mol)										
	Butanone	25	7.8	0.71	10	–	10–198	LS	B	776
Poly(styrene)-*block*-poly(butadiene)-*block*-poly(styrene) 50 wt.%-PS										
	Toluene	30	71.3	0.62	7	–	5.7–140	OS, LS	A–B	958
Poly(styrene)-*block*-poly(4-chloro-styrene)-*block*-poly(styrene), $B_n-A_k-B_n$										
$k/2n$, (33/67, mol/mol)	Carbon tetrachloride	40	6.10	0.77	–	4	20–82	OS	A	780
	Cumene	40	3.99	0.79	–	4	20–82	OS	A	780
$k/2n$, (50/50, mol/mol)	Butanone	30	8.14	0.72	–	8	31–89	OS	A	780
	Carbon tetrachloride	40	5.58	0.76	–	8	31–89	OS	A	780
	Cumene	40	6.84	0.74	–	8	31–89	OS	A	780
	Toluene	30	4.62	0.79	–	8	31–89	OS	A	780
$k/2n$, (66/34, mol/mol)	Carbon tetrachloride	40	5.94	0.74	–	4	15–54	OS	A	780
	Cumene	40	5.21	0.75	–	4	15–54	OS	A	780
Poly(styrene-*co*-sulfur dioxide), see Poly[sulfonyl(phenylethylene)], group 3.10.										
Poly(styrene)-*graft*-poly(methyl methacrylate), MMA content, 13 wt.%										
	Bromoform		(16.5)	(0.58)	5	–	17–196	SD	A	784
Poly(styrene)-*block*-poly(dimethylsiloxane) 12-62% PS										
	Methyl ethyl ketone	25	Various	Various	Various	–	Various	OS, GPC	B-C	955
Poly(styrene-*co*-monoethyl maleate)										
	Acetone	26.4 (θ)	51.1	0.50	9	–	20–180	LS	A	317
	Dioxane	25	11.2	0.702	9	–	20–180	LS	A	317
	Tetrahydrofuran	25	7.50	0.695	9	–	20–180	LS	A	317
–, sodium salt	Aq. NaCl									
	(0.005 M)	25	5.8	0.87	4	–	40–130	LS	A	317
	(0.01 M)	25	5.5	0.85	5	–	40–180	LS	A	317
	(0.03 M)	25	6.3	0.80	5	–	40–180	LS	A	317
	(0.05 M)	25	11	0.73	5	–	40–180	LS	A	317
	(0.075 M)	25	10	0.71	5	–	40–180	LS	A	317
	(0.15 M)	25	15	0.65	5	–	40–180	LS	A	317
	(0.3 M)	25	21	0.60	5	–	40–180	LS	A	317
	(0.6 M)	25	55	0.50	5	–	40–180	LS	A	317
Poly(styrene)-*block*-poly(2-vinylpyridine) 50 wt.%-PS										
	Benzene	10	35.2	0.605	4	–	6.5–153	LS, OS	A	944
	Methyl ethyl ketone	25	31.9	0.587	4	–	6.5–153	LS, OS	A	944
	Pyridine	25	9.02	0.734	4	–	6.5–153	LS, OS	A	944

TABLE 2. MAIN-CHAIN CARBOCYCLIC POLYMERS

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^4$)	Method	Remarks	Refs.
Poly(acenaphthenylene)	Benzene	25	30.04	0.594	11	–	2–100	OS	B	262
		25	2.82	0.74	4	–	4–100	LS	A,B	263
	Ethylene chloride	25	20.0	0.54	6	–	6–125	LS	A,B	263
	Dioxane	25	11.5	0.61	7	–	6–145	LS	A,B	263
	Methylene chloride	25	6.92	0.66	5	–	6–145	LS	A,B	263
	Toluene	25	6.76	0.66	17	–	3–175	LS	A,B	263
Poly(4,7-dimethylindene)	Benzene	25	3.5	0.77	9	–	5.8–140	LS	C,D	811
Poly(6-methylindene)	Benzene	25	34	0.60	13	–	30–1320	LS	C,D	811

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TABLE 3. MAIN-CHAIN HETEROATOM POLYMERS

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
3.1. POLY(OXIDES), POLY(ETHERS)										
Poly(butene oxide), see Poly[oxy(ethylethylene)]										
Poly(ethylene oxide), see Poly(oxyethylene)										
Poly[oxy(*tert*-butyl-ethylene)]	Benzene	25	39.7	0.686	9	–	8–520	LS	A–B	385
isotactic	Xylene	80	1.48	0.56	8	–	67–640	LS		790
Poly(1,2-cyclohexylene)	Toluene	35	3.5	0.83	22	–	2–50	OS	B	472
Poly(oxydecamethylene)	Benzene	35	195	0.53	7	–	0.1–0.9	SE	B	386
	Chloroform	30	172	0.56	9	–	0.05–0.9	SE	B	386
Poly(2,6-dichlorophenylene oxide)										
from 2,4,6-trichlorophenol	Toluene	37	150	0.327	–	6(?)	1.6–25	LS	Branching	964
from 4-bromo-2,6-dichlorophenol	Toluene	37	207	0.320	–	6(?)	1.6–25	LS	Branching	964
Poly(oxy-2,6-dimethyl-1,4-phenylene)										
	Benzene	25	26.0	0.69	8	–	3–17	LS	B	473
	Carbon tetrachloride	25	75.5	0.585	5	–	7–17	LS	B	473
	Chlorobenzene	25	37.8	0.66	7	–	2–42	LS	B	474
		90	51.4	0.63	7	–	3–18	LS	B	473
	Chloroform	25	48.3	0.64	8	–	2–42	LS	B	474
	Toluene	25	28.5	0.68	15	–	2–42	LS	B	474
Poly(dioxolane), see Poly(oxymethyleneoxyethylene)										
Poly[2,2′-(*p*,*p*-oxydi-*p*-phenylene)6,6′-oxybis(4-phenylquinoline)]										
	Chloroform	25	250	0.57	9	–	7.6–32.5	LLS, SEC	B	962
Poly(oxy-2,6-diphenyl-1,4-phenylene)										
	Benzene	50	19.72	0.638	–	10	1–110	LS	A	792
	Chlorobenzene	25	13.9	0.68	–	10	4–145	LS	C	473
		90	15.5	0.67	–	10	4–145	LS	C	473
	Chloroform	25	32.1	0.609	–	15	1–110	LS	A	792
	Toluene	25	21.4	0.635	–	10	4–145	LS	C	473
Poly(oxyethylene)	Acetone	25	32	0.67	5	–	7–100	LV	A,R	387
		25	156	0.50	7	–	0.02–0.3	EG	A,L	388
	Benzene	20	48	0.68	12	–	0.01–1.9	EG	A	389
		25	39.7	0.686	9	–	8–520	LS	A,R	385
		25	129	0.50	12	–	0.02–0.8	EG	A,L	388
	Carbon tetrachloride	20	69	0.61	9	–	0.02–1.1	EG	A	389
		25	62	0.64	5	–	7–100	LV	A	387
	Chloroform	25	206	0.50	6	–	0.02–0.15	EG	A,L	388
	Cyclohexane	20	$[\eta] = 0.5 + 0.035 M^{0.64}$		11	–	0.006–1.1	EG	A	389
	Diethylene glycol diethyl ether	50	140	0.51	5	–	7–100	LV	A	387
	Dimethylformamide	25	$[\eta] = 2.0 + 0.024 M^{0.73}$		10	–	0.1–3	LS, SD	A	390
	Dioxane	20	$[\eta] = 0.75 + 0.035 M^{0.71}$		13	–	0.006–1.1	EG	A	389
		25	138	0.50	7	–	0.02–0.15	EG	A,L	388
	Methanol	20	$[\eta] = 2.0 + 0.033 M^{0.72}$		12	–	0.006–1.9	EG	A	389
		25	85.2	0.57	?		?	LS, SD	A	391
	4-Methylpentan-2-one	50	120	0.52	5	–	7–100	LV	A	387
	Toluene	35	14.5	0.70	–	4	0.04–0.4	EG	C,L	392
	Water	20	$[\eta] = 2.0 + 0.016 M^{0.76}$		11	–	0.006–1.1	EG	A	391
		25	156	0.50	5	–	0.019–0.1	EG	A,L	393
		30	12.5	0.78	–	6	2–500	LS, SD	C	394
		35	6.4	0.82	–	5	3–700	LV	C,R	395
		35	16.6	0.82	–	4	0.04–0.4	EG	C,L	392
		45	6.9	0.81	–	5	3–700	LV	C	395
	Water	3	60.2	0.67	–	6	0.31–3.0	GPC	A–B	963
		5	61.7	0.66	–	6	0.31–3.0	GPC	A–B	963
		10	56.8	0.67	–	6	0.31–3.0	GPC	A–B	963
		15	50.5	0.68	–	6	0.31–3.0	GPC	A–B	963
		25	49.9	0.67	–	6	0.31–3.0	GPC	A–B	963
		35	48.8	0.66	–	6	0.31–3.0	GPC	A–B	963
		55	77.7	0.61	–	6	0.31–3.0	GPC	A–B	963
		75	76.3	0.57	–	6	0.31–3.0	GPC	A–B	963
		85	118	0.54	–	6	0.31–3.0	GPC	A–B	963

TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	Aq. KOH (0.96 M)	25	144	0.53	–	6	0.31–3.0	GPC	A–B	963
	(1.10 M)	25	165	0.51	–	6	0.31–3.0	GPC	A–B	963
	(1.24 M)	25	185	0.50	–	6	0.31–3.0	GPC	A–B	963
	(1.48 M)	25	309	0.43	–	6	0.31–3.0	GPC	A–B	963
	Aq. K_2SO_4	35 (θ)	130	0.50	–	5	3–700	LV	C	395
	(0.45 M)	35	280	0.45	–	5	7–100	LV	A	387
	Aq. $MgSO_4$ (0.39 M)	45 (θ)	100	0.50	–	5	3–700	LV	C	395
Poly[oxy(ethylethylene)]	Benzene	25	15.9	0.75	10	–	5–120	LS	B–C	396
		30	3.39	0.84	9	–	20–210	LS	B	397
	Butanol	25	19.6	0.69	10	–	5–120	LS	B–C	396
	Butanone	30	4.08	0.79	9	–	20–210	LS	B	397
	Hexane	25	14.3	0.73	10	–	5–120	LS	B–C	396
	2-Propanol	30 (θ)	86.5	0.50	9	–	20–210	LS	B	397
		30 (θ)	111	0.50	9	–	5–120	LS	B–C	396
Poly[oxy(hexamethylene)]	Benzene	25	86.9	0.62	1	8	0.01–1.5	SE, CR	C	398
	Dioxane	25	131	0.55	1	10	0.01–1.5	SE, CR	C	398
Poly(oxymethylene)	Dimethylformamide	130	22.4	0.71	7	–	0.15–1.5	EG	B	399
		140	18.1	0.73	7	–	0.15–1.5	EG	B	399
	Hexafluoroacetone-sesquihydrate (1/1.7, mol/mol, with triethylamine 1%, v/v)	25	46.0	0.74	7	–	0.15–1.5	EG	B	399
		25	87	0.69	–	5	2–16	LS	C	402
	Phenol/tetrachloroethane (1/3, w/w)	90	27.5	0.80	–	18	0.8–10	EG	C,D	400
	(1/3, v/v)	90	5.22	0.93	?		?	OS		403
Poly(oxymethyleneoxyethylene)	Chlorobenzene	25	200	0.50	4	14	9–100	LS	D	404
	p-Chlorophenol	60	41.3	0.724	–	3	7–13	LS	C	405
	1H,1H,5H-octafluoro-pentanol-1	110	13.35	0.810	–	3	7–13	LS	C	405
	Tetrahydrofuran	25	17	0.73	4	–	3–11	OS		791
Poly[oxy(phenylethylene)]	Benzene	30	92.2	0.758	10	–	1.4–81	LS	B,C	385
	Toluene	25	67.9	0.766	10	–	1.4–81	LS	B,C	385
Poly(oxy-1,4-phenylene-oxy-1,4-phenylene-carbonyl-1,4-phenylene), (PEEK)										
	p-Chlorophenol/ *o*-dichloro-benzene (60/40, w/w)	35	58.6	0.702	7	–	0.86–8	LS	B	993
	Methanesulfonic acid	30	22.6	0.818	8	–	0.46–8	LS	B	993
Poly(oxypropylene)	Acetone	25	75.5	0.56	5	–	0.1–0.4	LS	A	406
	Benzene	20	11.1	0.79	5	–	0.07–0.33	SE	A	407
		25	11.2	0.77	3	–	3–70	LS	A–B	408
		25	14	0.8	?		?	?		409
isotactic		25	38.5	0.73	–	8	0.5–92	LV	C	410
		25	41.3	0.64	11	–	1–8	LS	A	411
		25	41.5	0.65	5	–	0.05–0.4	LS	A	406
	Hexane	46	19.7	0.67	6	10	3.4–367	LS	A–B	408
	Methanol	20	40.6	0.64	6	–	0.05–0.33	SE	A	407
		25	76.9	0.55	10	–	1–7	LS		411
	Tetrahydrofuran	20	55.0	0.62	6	–	0.05–0.33	SE	A	407
	Toluene	20	20.8	0.72	5	–	0.07–0.33	SE	A	407
		25	12.9	0.75	3	–	3–70	LS	A–B	408
	Toluene/2,2,4-trimethyl-pentane (5/7, v/v)	39.5 (θ)	107.5	0.50	7	–	1–7	LS	A	411
oligomer	Acetone	20	75.5	0.56	5	–	0.1–0.4		L	793
	Benzene	20	41.5	0.65	5	–	0.04–0.4		L	793
Poly(oxytetramethylene)	Benzene	30	131	0.60	–	12	2.6–113	LS	A	412
	Ethyl acetate	30	42.2	0.65	–	12	2.6–113	LS	A	412
	Toluene	28	25.1	0.78	10	–	3–12	OS	A–B	413
	Ethyl acetate/hexane (22.7/77.3, w/w)	31.8 (θ)	206	0.49	–	11	2.6–113	LS	A	412
Poly(oxytrimethylene)	Acetone	30	76.0	0.59	–	7	2.8–20	LS	A	414
	Benzene	30	21.9	0.78	–	15	2.8–30	LS	A	414
	Carbon tetrachloride	30	26.7	0.75	–	11	2.8–25	LS	A	414

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TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(phenyl ether-*alt*-phenyl ketone), (PEK)										
	Sulfuric acid (96.4%)	?	56	0.73	–	6	1.2–5	LS, SEC	C–D	1008
Poly(propylene oxide), see Poly(oxypropylene)										
Poly(tetrahydrofuran), see Poly(oxytetramethylene)										
3.2. POLY(ESTERS), POLY(CARBONATES)										
Poly(butylene isophthalate)	Chloroform	?	3.3	0.92	6	–	1.4–3.7	VOS, EG	–	966
Poly(butylene adipate-*co*-butylene isophthalate)										
(100 mol% BADP)	Chloroform	30	73.1	0.69	6	–	0.67–1.45	VOS	C	967
(80 mol% BADP)	Chloroform	30	47.4	0.73	4	–	0.92–1.9	VOS	C	967
(50 mol% BADP)	Chloroform	30	26.7	0.78	4	–	0.80–1.8	VOS	C	967
(20 mol% BADP)	Chloroform	30	11.2	0.80	4	–	0.88–2.3	VOS	C	967
Bisphenol A poly(carbonates), see Poly[oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene]										
Poly(*N*-decanoyldehydroalanine methyl ester)										
	Tetrahydrofuran	25	26.3	0.63	6	–	10–254	LLS	–	968
Poly(ethylene terephthalate), see Poly(oxyethyleneoxyterephthaloyl)										
Poly(D-β-hydroxybutyrate), see Poly(D-oxy-1-oxo-3-methyl-trimethylene)										
Poly(4,4′-isopropylidene-diphenyl carbonate)-*block*-poly(styrene)										
26 wt.%-PS	Chloroform	25	31.1	0.74	9	–	6–17	OS, LS	A	959
	Diethyl carbonate	25	228	0.54	9	–	6–17	OS, LS	A	959
	1,4-Dioxane	25	60.8	0.68	9	–	6–17	OS, LS	A	959
	Methylene chloride	25	21.2	0.77	9	–	6–17	OS, LS	A	959
	Tetrahydrofuran	25	136	0.57	9	–	6–17	OS, LS	A	959
48 wt.%-PS	Chloroform	25	65.1	0.71	8	–	3–11	OS, LS	A	959
	Diethyl carbonate	25	378	0.53	8	–	3–11	OS, LS	A	959
	1,4-Dioxane	25	80.5	0.68	8	–	3–11	OS, LS	A	959
	Methylene chloride	25	28.6	0.79	8	–	3–11	OS, LS	A	959
	Tetrahydrofuran	25	352	0.54	8	–	3–11	OS, LS	A	959
50 wt.%-PS	Chloroform	25	62.7	0.71	6	–	6–27	OS, LS	A	959
	Diethyl carbonate	25	316	0.55	6	–	6–27	OS, LS	A	959
	1,4-Dioxane	25	79.8	0.68	6	–	6–27	OS, LS	A	959
	Methylene chloride	25	20.4	0.81	6	–	6–27	OS, LS	A	959
	Tetrahydrofuran	25	243	0.56	6	–	6–27	OS, LS	A	959
73 wt.%-PS	Chloroform	25	52.2	0.74	9	–	6–17	OS, LS	A	959
	Diethyl carbonate	25	347	0.55	9	–	6–17	OS, LS	A	959
	1,4-Dioxane	25	117	0.67	9	–	6–17	OS, LS	A	959
	Methylene chloride	25	31.5	0.80	9	–	6–17	OS, LS	A	959
	Tetrahydrofuran	25	143	0.65	9	–	6–17	OS, LS	A	959
Poly(D,L-β-methyl-β-propiolactone), see Poly(D,L-β-methyl-β-propiolactone), see Poly(D,L-oxy-1-oxo-3-methyl-trimethylene)										
Poly(oxyadipoyloxydecamethylene)										
	Chlorobenzene	25	11.7	0.84	–	7	0.3–3	LV	C	415
	Diethyl succinate	79	5.8	0.86	–	12	1–3	LV	C	415
Poly(oxybutynedioyl-oxyhexamethylene)	Benzene	20	151	0.55	?	–	0.1–0.5	OS	B	416
	Chloroform	20	91	0.61	?	–	0.1–0.5	OS	A	416
Poly[oxycarbonyl(bicyclo-[2,2,2]octan-2,5-dion)carbonyloxyhexamethylene]										
	Chloroform	20	–	–	–	4	1.4–3.9	OS	C	475
Poly[oxycarbonyloxy-1,4-phenyleneisopropylidene1,4-phenylene]										
	Butyl benzyl ether	170 (θ)	210	0.50	8	–	4–31	LS	B	476
	Chloroform	20	277	0.50			1.5–6	LS		477
		25	12.0	0.82	8	–	1–7	LS	A	478
		25	30.1	0.74	12	–	0.4–47	LS	A,R	796
	Dichloromethane	25	29.9	0.74	12	–	0.4–47	LS	A	796
	Dioxane/cyclohexane (68/32)	25 (θ)	2.5–2.8	Various	9	–	–	SEC	B	965
	Ethylene chloride	25	20.4	0.76	8	–	1–7	LS	A	478
	Methylene chloride	25	11.1	0.82	6	–	1–27	SD	B	479
		25	11.9	0.80	12	–	1–76	LS	B,R	476
		25	38.9	0.70	8	–	1–7	LS	A	478
	Tetrachloroethane	25	13.4	0.82	8	–	1–7	LS	A	478
	Tetrahydrofuran	25	38.9	0.70	8	–	1–7	LS	A	478
		25	39.9	0.70	–	6	1–27	SD	C,R	479
		25	37.8	0.71	12	–	0.4–47	LS	A	796

TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	Tetrahydrofuran	Room temp.	31.4	0.70	9	–	–	SEC	B	965
	Cyclohexane/dioxane (36.1/63.9, w/w)	25	210	0.50	4	–	30–75	LS	B	476
Poly(oxy-1,4-cyclohexyleneoxysebacoyl)										
cis	Chloroform	20	27.8	0.78	–	5	2.1–4.6	OS	C	480
trans	Chloroform	20	18.3	0.86	–	9	1.1–3.7	OS	C	480
Poly[oxy-1,10-di-oxoperfluorodecamethyleneoxydecamethylene)										
	Chloroform	25	15.5	0.70	7	–	0.8–8	LS		795
	s-Tetrachloroethane	25 (θ)	88	0.50	7	–	0.8–8	LS		795
Poly(oxyethyleneoxycarbonyl-1,4-phenyleneoxyethyleneoxy-1,4-phenylenecarbonyl)										
	o-Chlorophenol	35	145	0.59	9	–	1.3–11	OS		801
	Phenol–tetrachloroethane (1/2, w/w)	35	112	0.63	9	–	1.3–11	OS		801
Poly(oxyethyleneoxyterephthaloyl)										
	o-Chlorophenol	25	17	0.83	–	7	0.8–2.0	EG	C	481
		25	19	0.81	6	–	1.5–3.8	EG	B	482
		25	30	0.77	–	34	1.1–2.9	EG	C	483
		25	42.5	0.69	7	–	2–15	SD	S	484
		25	65.6	0.73	–	5	1.2–2.5	OS	C	485
		55	26	0.77	6	–	1.5–3.8	EG	B	482
	m-Cresol	25	0.77	0.95	–	5	0.04–1.2	EG	A,L	486
	Dichloroacetic acid	45	400	0.50	7	–	1.5–3.8	EG	B	482
	Tetrachloroethane	50	13.8	0.87	–	6	0.04–0.1	EG	A,L	487
	Trifluoroacetic acid	25	140	0.64	7	–	1.5–3.8	EG	B	482
		30	43.3	0.68	–	9	2.5–12	LS	C	488
		35	130	0.66	7	–	1.5–3.8	EG	B	482
		55	105	0.69	6	–	1.5–3.8	EG	B	482
	Dichloroethane/phenol (6/4, v/v)		9.2	0.8				EG		489
	Phenol/tetrachloroethane	25	140	0.64	6	–	1.5–3.8	EG	B	482
	(40/60, w/w)	35	125	0.65	6	–	1.5–3.8	EG	B	482
	(3/5, v/v)	30	22.9	0.73	–	9	2.5–12	LS	C	488
	(50/50, v/v)	20	75.5	0.685	–	38	0.3–3	EG	C	490
		25	21	0.82	–	9	0.5–3	EG	C	491
		25	12.7	0.86						492
	Phenol/tetrachlorophenol	25	46.8	0.68				LS		493
	Phenol/trichlorophenol (10/7, v/v)	29.8	28.0	0.775	–	4	0.1–0.4	EG	C	494
		30	630	0.47	–	8	1.1–4	OS	C	495
Poly(oxyfumaroyloxyhexamethylene)										
	Chloroform	20	27.1	0.80	5	–	2–4.3	OS	B	417
Poly[oxy(hexahydro-3,6-endomethylenephthaloyl)-oxyhexamethylene]										
cis	Benzene	20	4.64	0.86	13	–	2.3–7.5	OS	B	496
	Chloroform	20	9.33	0.83	13	–	2.3–7.5	OS	B	496
trans	Benzene	20	17.4	0.75	10	–	3.3–11	OS	B	496
	Chloroform	20	17.9	0.77	11	–	3.3–15	OS	B	496
Poly[oxy(hexahydroterephthaloyl)oxyoctamethylene]										
cis	Chloroform	20	22.9	0.79	6	–	3.3–5.5	OS	B	480
trans	Chloroform	20	18.9	0.84	6	–	2.4–4.4	OS	B	480
Poly(oxyhexamethyleneoxy-2,9-dibutylsebacoyl)										
	Benzene	20	37.4	0.74	?	–	0.9–2.4	OS	B	418
Poly(oxyhexamethyleneoxysebacoyl)										
	Benzene	20	62.7	0.69	9	–	0.6–1.8	OS	B	418
	Chloroform	20	72.5	0.70	9	–	2–10	OS	B	419
Poly(oxymaleoyloxyhexamethylene)										
	Benzene	20	76.3	0.60	7	–	1.3–6.6	OS	B	417
	Chloroform	20	36.2	0.73	7	–	1.3–6.6	OS	B	417
	Tetrahydrofuran	20	43.7	0.66	7	–	1.3–6.6	OS	B	417
Poly(oxy-1-oxo-dimethyltrimethylene)										
	Trifluoroacetic acid	20	24.2	0.74	11	–	5–54	GPC	B, C	800
Poly(oxy-1-oxohexamethylene)	Benzene	25	72	0.63	6	–	3.4–12	LS	A	808
		30	9.94	0.82	9	–	1.4–15	SV	B	447

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TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	Dimethylformamide	30	19.1	0.73	9	–	1.4–15	SV	B	447
	Dioxane	25	90	0.61	6	–	3.4–12	LS	A	808
	Ethyl acetate	25	180	0.52	6	–	3.4–12	LS	A	808
Poly(oxy-1-oxo-3-methyl-trimethylene)	Chloroform	30	7.7	0.82	–	5	2–78	SD	C, D	660
	2,2,2-Trifluoroethanol	30	25.1	0.74	–	6	2–101	LS	C, D	661
D-	Chloroform	30	11.8	0.78	9	–	11–164	LS	B, C	797
D,L-	Ethylene dichloride	30	9.18	0.78	9	–	11–164	LS	B, C	797
	Trifluoroethanol	25	12.5	0.80	9	–	11–164	LS	B, C	797
	Butyl chloride	13 (θ)	100	0.50	4	–	2–15	LS	A	798
	Chloroform	30	16.6	0.76	5	–	2–15	LS	A	798
	1-Chloronaphthalene	40	39.6	0.62	4	–	2–15	LS	A	798
	Ethylene dichloride	30	16.8	0.74	6	–	2–15	LS	A	798
	Trifluoroethanol	25	22.2	0.76	8	–	2–15	LS	A	798
		30	17.5	0.78	8	–	2–15	LS	A	798
Poly(oxysebacoyloxyhexadecamethylene)										
	Chloroform	20	74.7	0.70	4	–	2–10	OS	B	419
Poly(oxysuccinyloxyhexamethylene)										
	Benzene	20	43.3	0.70	22	–	1.5–5	OS	B	417
	Chloroform	20	24.4	0.79	18	–	1.5–5	OS	B	417
	Tetrahydrofuran	20	44.3	0.69	13	–	1.5–5	OS	B	417
Poly[oxytetra(ethyleneoxy)carbonyl(1-methylethyleneoxy)thio(2-methylethylene(carbonyl)]										
	Chloroform	?	34.7	0.714	–	?	< 1.5	EG	?	421
Poly[oxy-(1,1,3-trimethyl-3-phenylindan-5,4′-dicarbonyl)oxy-1,4-phenyleneisopropylidene-1,4-phenylene]										
	Tetrahydrofuran	25	30.5	0.691	7	–	2.6–30	LS	A	794
Poly(oxyundecanoyl)	Chloroform	20	21.4	0.60	7	–	3–49	OS	B	419
		25	36.3	0.82	–	6	0.5–1.3	EG	C	420
Poly(phenylhydroquinone-*co*-terephthalic acid)										
	o-Dichlorobenzene/*p*-chlorophenol (1/1, w/w)									
		25	30	0.869	5	–	5.0–16	LS	–	969
		(Wormlike behavior)								
Poly(β-propiolactone)	Chloroform	30	28.2	0.71	13	–	0.68–86	LS	B–D	991
	1,4-Dioxane	30	17	0.71	–	–	0.68–86	LS	B–D	991
	2,2,2,-Trifluoroethanol	30	17.2	0.78	13	–	0.68–86	LS	B–D	991
3.3. POLY(AMIDES) (FOR COPOLY(AMIDES), see 3.10 COPOLYMERS)										
Poly[(butylimino)carbonyl], [Poly(butyl isocyanate)]										
	Benzene	20	1.10	1.11	–	7	1.8–21	SD	A,R	441
	Carbon tetrachloride	–	31.6	1.2	9	–	1–30	SD	B	823
		22	7.81×10^{-5}	2.0	8	–	2.5–14	OS, LS	A	825
	Tetrachloromethane	20			–	2	6.6–16	SD	D	442
	Tetrahydrofuran	20	0.457	1.18	–	7	1.8–21	SD	A	441
		40	–	1.8–2.0	4	–	< 10	SE		821
		40	–	0.26–0.37	4	–	> 100	SE		821
Poly[(chlorohexylimino)-carbonyl], [Poly(chlorohexyl isocyanate)]										
	Carbon tetrachloride	21	0.22	1.21	10	–	1.4–31	SD	B,C	826
Poly(*trans*-2,5-dimethylpiperazo-1,3-phthaloyl)										
	m-Cresol	25	–	0.75	9	–	6–237	LS		828
	Trifluoroethanol	25	–	0.75	10	–	3–237	LS		828
Poly[(hexyliminocarbonyl], [Poly(hexyl isocyanate)]										
	Carbon tetrachloride	25	0.0152	1.10	10	–	3.8–43	LS	A	822
	Chloroform	25	0.0499	0.97	10	–	3.8–43	LS	A	822
	Hexane	25	1.0	1.2	8	–	6.8–39	LS	A	824
		25	88	0.77	8	–	$> 10^6$	LS	A	824
		25	5.4	0.97	6	–	9.1–230	LS	A, B	827
	Tetrahydrofuran	25	0.0220	1.06	10	–	3.8–43	LS	A	822
	Toluene	25	0.0248	1.05	11	–	3.8–43	LS	A	822
Poly(hexyl isocyanate)	Butyl chloride	25	(Wormlike behavior)		31	–	0.44–724	LS, SE	A	970
	Dichloromethane	20	–	Not const.	13	–	0.44–91	LS, OS, SE, LV	A	984
			(Wormlike behavior)							
	Toluene	10	–	Not const.	13	–	0.44–91	LS, OS, SE, LV	A	984
			(Wormlike behavior)							
		25	–	Not const.	13	–	0.44–91	LS, OS, SE, LV	A	984
			(Wormlike behavior)							

TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
		40	– (Wormlike behavior)	Not const	13	–	0.44–91	LS, OS, SE, LV	A	984
Poly(iminoadipoyliminohexamethylene), (Nylon 66)										
	o-Chlorophenol	25	168	0.62	–	2	1.4–5	LS, EG	C	443
	m-Cresol	25	240	0.61	–	2	1.4–5	LS, G	C	443
		25	$[\eta] = 0.5 + 0.0353 M^{0.792}$		13	–	0.015–5	LS,EG	B	444
	Dichloroacetic acid	25	$[\eta] = 0.5 + 0.352 M^{0.551}$		13	–	0.015–5	LS,EG	B	444
	2,2,3,3-Tetrafluoropropanol/CF_3COONa (0.1 M)	25	114	0.66	–	2	1.4–5	LV	C	443
	Aq. HCOOH (90%, v/v)	25	35.3	0.786	3	11	0.6–6.5	LS, EG	C	443
		25	110	0.72	–	20	0.5–2.5	EG	C	446
		25	$[\eta] = 2.5 + 0.0132 M^{0.873}$		13	–	0.015–5	LS,EG	B	444
	Aq. HCOOH (90% vol)/HCOONa (0.1 M)	25	32.8	0.74	–	19	1–5	EG	C,R	445
		25	87.7	0.65	–	2	1.4–5	LS, EG	C	443
		25	$[\eta] = 1.0 + 0.0516 M^{0.687}$		6	–	0.015–5	LS,EG	B, R	444
	Aq. HCOOH (90%, v/v)/KCl (2.3 M)	25 (θ)	227	0.50	–	2	1.4–5	LS, EG	C	443
		25 (θ)	253	0.50	7	–	0.015–5	LS, EG	B-C	444
	Aq. H_2SO_4 (95%, v/v)	25	$[\eta] = 2.5 + 0.0249 M^{0.832}$		12	–	0.015–5	LS,EG	B	444
	(96%, v/v)	25	115	0.67	–	2	1.4–5	LV	C	443
Poly(iminocarbonyl-1,4-phenylene), Poly(*p*-benzamide)										
	Sulfuric acid (96%)	20	1.6	1.7	–	10	0.7–2.3	DV		807
Poly(imino-4,6-carboxyisophthaloylimino-1,4-phenylene)										
	Dimethylformamide	30	252	0.56	–	14	0.9–2.5	LS		802
Poly(iminohexamethyleneiminosebacoyl), (Nylon 6-10)										
	m-Cresol	25	13.5	0.96	–	5	0.8–2.4	SD	B	454
Poly(iminohexamethyleneiminoterephthaloyloxyhexamethyleneoxyterephthaloyl)										
	Phenol/tetrachloroethane (1/1, w/w)	25	36.4	0.75	–	8	1.6–7.4	GPC	D	806
Poly(iminohexamethyleneiminothiocarbonylhexamethylenethiocarbonyl)										
	Dimethylformamide/ LiCl (5%)	20	22.8	0.76	12	–	0.7–4.8	EG		805
Poly(iminohexamethyleneiminothiocarbonyltetramethylenethiocarbonyl)										
	Dimethylformamide/ LiCl (5%)	20	99.1	0.65	12	–	0.8–2.2	EG		805
Poly[imino(1-oxododecamethylene)], (Nylon 12)										
	m-Cresol	25	81	0.74	38	–	0.3–13	LS	B,R	803
		25	46.3	0.75	16	5	1–13	LS	B	812
	Sulfuric acid (96%)	25	69.4	0.64	15	5	1–13	LS	B	812
Poly[imino(1-oxohexamethylene)], (Nylon 6)										
	m-Cresol	25	320	0.62	6	–	0.05–0.5	EG	B	448
	Trifluoroethanol	−20	53.3	0.74	5	–	1.3–10	LS	B	450
		25	53.6	0.75	5	–	1.3–10	LS	B	450
		50	58.2	0.73	5	–	1.3–10	LS	B	450
	Aq. HCOOH (85%)	−10	26.8	0.82	6	–	0.7–12	LS	B	450
		0	24.8	0.82	6	–	0.7–12	LS	B	450
		10	23.4	0.82	6	–	0.7–12	LS	B	450
		20	75	0.70			0.45–1.6	EG		451
		25	22.6	0.82	11	–	0.7–12	LS	B,R	450
	Aq. HCOOH (65%)	25	229	0.50	5	–	0.7–12	LS	B	450
	Aq. H_2SO_4 (40%)	25	59.2	0.69			0.3–1.3	EG		452
ring oligomer	*m*-Cresol	25	2100	0.22	–	4	0.02–0.06	VOS	A	449
	Ethylene chlorohydrin	25	870	0.27	–	3	0.03–0.06	VOS	A	449
Monochain, polymerized with stearic acid										
	Conc. H_2SO_4	25	63	0.76	–	7	0.2–1.4	EG	B	453
Dichain, polymerized with sebacic acid										
	Conc. H_2SO_4	25	42	0.79	–	14	0.2–2.3	EG	B	453

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TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Tetrachain, polymerized with a tetrabasic acid										
	Conc. H_2SO_4	25	55	0.74	–	11	0.2–1.9	EG	B	453
Octachain, polymerized with a octabasic acid										
	Conc. H_2SO_4	25	13.5	0.86	–	5	0.4–2.6	EG	B	453
Poly(iminoterephthaloylhydrazocarbonyl-1,4-phenylene), [Poly(terephthaloyl-*p*-aminobenzhydrazide)]										
	Dimethyl sulfoxide	25	–	1.3–0.9	15	–	0.2–5.8	LS	B	809
		25	1.05	0.795	–	8	1.2–65	LS	B	810
Poly(iminoterephthaloylimino-1,4-phenylenefluoren-9-ylidene-1,4-phenylene)										
	Dimethylformamide	25	110	0.66	7	–	1.5–7.8	LS	B	503
Poly(iminoterephthaloylimino-1,4-phenylene-phthalidylidene-1,4-phenylene)										
	Dimethylformamide	25	277	0.59	8	–	1.4–6.9	LS	B	503
Poly[iminoterephthaloylimino(trimethylhexamethylene)], 2,4,4-/2,2,4-trimethyl (1/1, mol/mol)										
	Dimethylformamide	25	27.4	0.71	17	–	1.2–28	LS, SA	B	804
Poly(isophthaloyl-*trans*-2,5-dimethylpiperazine)										
	N-methyl-2-pyrrolidone	25	– (Wormlike behavior)	Not const	18	–	0.26–237	LS, SE	A	971
Poly[(octylimino)carbonyl], [Poly(octyl isocyanate)]										
	Carbon tetrachloride	22	2.02×10^{-5}	2.0	4	–	5.9–20.0	OS, LS	A	825
Poly(1,4-phenyleneterephthalamide), (Kevlar)										
	96% Sulfuric acid	30	1.09	1.25	–	3	1.6–3.8	LS	–	973
Poly(terephthalamide-*p*-benzohydrazide)										
	Dimethyl sulfoxide	25	(Wormlike behavior)		6	–	0.94–5.8	LS, SE	A	974

3.4. POLY(AMINO ACIDS)

Poly(β-benzyl-L-aspartate), see Poly(iminocarbonyl-L-benzyloxycarbonylethylidene)										
Poly(γ-benzyl-L-glutamate), see Poly(iminocarbonyl-L-benzyloxycarbonylpropylidene)										
Poly[(benzylimino)carbonylethylene], (Poly(*N*-benzyl-β-alanine))										
	Dichloroacetic acid	25	120	0.525	–	6	0.15–1.8	EG	B,L	455
Poly(iminocarbonyl-L-benzyloxycarbonylethylidene), (Poly(β-benzyl-L-aspartate))										
	m-Cresol	15	–	1.15	5	–	0.8–24	LS	B	456
		70	–	0.74	5	–	0.8–24	LS	B	456
	Hexamethylphosphoramide	25	–	0.80	4	–	2–24	LS	B	456
	Chloroform/dichloroacetic acid (98/2, v/v)	25	–	1.30	5	–	0.8–24	LS	B	456
Poly(iminocarbonyl-L-benzyloxycarbonylpropylidene),										
	Dichloroacetic acid	25	2.78	0.87	–	6	2–34	LS	C	457
		25	8.8	0.77	8	–	6–	LS	B	814
(Poly(γ-benzyl-L-glutamate))	Dimethylformamide	25	0.0056	1.45	8	–	6–	LS	B	814
		25	0.00029	1.70	–	5	7–34	LS	C	457
	Dichloroacetic acid/heptane									
	(55/45, v/v)	21	116	0.53	–	4	1.5–10	LS	C	458
	(90/10, v/v)	21	25.4	0.68	–	4	1.5–10	LS	C	458
D,L-glutamate	Dichloroacetic acid	25	2.85	0.85	–	6	1.5–10	LS	C	459
	Dimethylformamide	25	37.7	0.55	–	6	1.5–10	LS	C	459
Poly(iminocarbonyl-L-hydroxycarbonylpropylidene), [Poly(α-L-glutamic acid)]										
	Sodium phosphate buffer (1.01 M, pH, 6.8)	25.5	1.55	0.96	5	–	4–15	LS	C	815
Poly[iminocarbonyl-L-(*N*-hydroxy-ethylcarbamoylpropylidene)], {Poly[N^5-(2-hydroxyethyl)-L-glutamine]}										
	Sodium phosphate buffer (0.05 M, pH, 7.4)	25	26.2	0.65	6	–	0.8–15	SE	C	816
	Water	30	–	0.55	4	–	1.7–7.2	OS	C	817
Poly[iminocarbonyl-L-(*N*-hydroxypropylcarbamoylpropylidene)], (Poly(N^5-(3-hydroxypropyl)-L-glutamine))										
	Methanol	25	–	1.6	4	–	20–40	LS	B	460
	Water	25	–	0.6–1.0	5	–	2–33	LS	B	460
Poly(iminocarbonyl-L-methoxycarbonylpropylidene), [Poly(γ-methyl-L-glutamate)]										
	m-Cresol	25		>1			3–21	OS	C	463
	Dichloroacetic acid	25	29	0.74	–	6	3–21	OS	C	463
D,L-glutamate	*m*-Cresol	25	11	0.78	–	6	3.2–8.2	OS	C	464
	Dichloroacetic acid	25	5.9	0.85	–	6	3.2–8.2	OS	C	464
Poly(iminocarbonyl-L-methoxyethylideneiminocarbonyl-L-hydroxyethylideneiminocarbonylmethylene), [Poly(Asp(OCH_3)-Ser(H)-Gly)]										
	Dichloroacetic acid	30	868	0.367	–	9	0.25–1.1	SA	B,C	467

TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(iminocarbonyl-L-methylthiopropylidene), [Poly(L-methionine)]	acid (1/1, mol)									
	Dichloroacetic acid		3.0	0.87	–	5	3–50	LS	B	820
	Dichloroacetic acid/trifluoroacetic									
		36	0.62	–	4	3–50	LS	B	820	
Poly(iminocarbonyl-L-*p*-nitrobenzyloxycarbonylpropylidene), [Poly(γ-*p*-nitrobenzyl-L-glutamate)]										
	Dichloroacetic acid	25	11.5	0.72	–	10	1–5	LS	B	608
	Dimethylformamide	25	0.0170	1.36	–	10	1–5	LS	B	608
Poly(iminocarbonyl-phenylethylidene), [Poly(phenylalanine)]										
L-	Chloroform	25	0.00346	1.48	–	11	2.2–14	LS	B	465
D,L	Chloroform/dichloro-acetic acid (2/3, v/v)	21	118	0.55				LS	B	466
Poly(γ-hydroxy-L-proline)	Water	30	92	0.68	3	–	1.6–3.5	GPC	A	818
Poly(α,ε-L-lysine)	Dimethylformamide/ LiCl (1%)	20	2500	0			0.05–23		A	819
Poly[(methylimino)carbonylmethylene], (Poly[sarcosine)]										
	Water	20	56	0.88	–	5	0.7–1.6	EG	C	468
Poly(1-proline), see Poly[(L-1,2-pyrrolidindiyl)carbonyl] in group 3.9.										
3.5. POLY(UREAS), POLY(URETHANES), POLY(IMINES)										
Poly(ethylene imine), see also Poly(iminoethylene)										
	Methanol	25	10.4	0.95	8	–	0.24–2.8	LS	–	976
Poly(iminoethylene)	Water	25	$[\eta] = 2.14\,P^{0.35}$ P: number of N atoms		4	–	$P = 4–13$	CR	D	470
Poly(oxytetramethyleneoxycarbonylimino-2,4-tolyleneiminocarbonyl)										
	Dimethylformamide	30	54	0.74	–	5	0.35–1.6	LS	C	497
Poly[oxytetramethyleneoxycarbonylimino-(6-pentyloxy-1,3-phenylene)-iminocarbonyl]										
	Dimethylformamide	20	8.1	0.86	–	5	0.9–4.3	SV	C	498
Poly(oxytetramethyleneoxycarbonylimino-[6-(αH,ωH,ωH-perfluoroalkylene)oxy-1,3-phenylene]-iminocarbonyl)										
number of F atoms										
4	Acetone	20	7.1	0.81	–	5	0.5–4	SV	C	498
8	Acetone	20	4.3	0.785	–	5	2–16	SV	C	498
12	Acetone	20	13.5	0.67	–	5	1.7–28	SV	C	498
16	Acetone	20	25.6	0.615	–	5	0.9–9	SV	C	498
Poly(terephthalic acid-4-aminobenzohydrazide)										
	Dimethyl sulfoxide	25	(Wormlike behavior)		11	–	0.38–26	LS	–	977
Poly(urethane)with methylene *bis*(4-phenyl isocyanate), tetramethylene glycol and ethylenediamine										
	Dimethylacetamide	25	101	0.59	8	–	3.2–35	LS	–	978
Poly(ureyleneheptamethylene)	Dichloroacetic acid	46	338	0.505	–	10	0.3–2.4	LS	C	471
	Sulfuric acid (90%)	25	50.0	0.714	–	14	0.13–2.4	LS	C	471
		46	223	0.506	–	7	0.06–2.4	LS	C	471
	(96%)	25	37.5	0.757	–	5	0.4–2.4	LS	C	471
	(98%)	46	240	0.53	–	7	0.2–2.4	LS	C	471
3.6. POLY(SULFIDES)										
Poly(thiopropylene)	Benzene	20	3.3	0.86	7	–	3.8–20.4	LS	B	436
		30	5.0	0.78	–	10	3.6–73	LS		839
Poly(trimethylene sulfide)	Chloroform	25	31.0	0.76	6	–	1–4.7	OS	B	979
3.7. POLY(PHOSPHATES)										
Lithium–potassium copoly(phosphate)										
Li/K (9/1, mol/mol)	Aq. NaBr (0.035 N)		0.073	1.17	–	6	0.4–1	EG	–	980
(8/2, mol/mol)	Aq. NaBr (0.035 N)		0.63	0.93	–	6	0.4–1	EG	–	980
(7/3, mol/mol)	Aq. NaBr (0.035 N)		7.3	0.89	–	6	0.4–1	EG	–	980
(6/4, mol/mol)	Aq. NaBr (0.035 N)		27	0.75	–	5	0.4–1	EG	–	980
(5/5, mol/mol)	Aq. NaBr (0.035 N)		25	0.74	–	6	0.4–1	EG	–	980
Poly[oxy(hydroxyphosphinylidene)]										
	Aq. NaBr (0.35 M)	25	6.5	0.69	–	16	1–125	LS	C	422
	(0.415 M)	25 (θ)	49.4	0.50	–	9	1–125	LS	C	422
sodium salt	Aq. NaBr (0.035 M)	25.5	69	0.61	–	5	0.09–1	EG	C	423

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TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(phosphoric acid), see Poly[oxy(hydroxyphosphinylidene)]										
Sodium–potassium copoly(phosphate)										
Na/K (9/1, mol/mol)	Aq. NaBr (0.035 N)		3.6	0.93	–	4	0.4–0.8	EG	–	980
(8/2, mol/mol)	Aq. NaBr (0.035 N)		11	0.81	–	4	0.4–0.8	EG	–	980
(7/3, mol/mol)	Aq. NaBr (0.035 N)		12	0.80	–	4	0.4–0.8	EG	–	980
(6/4, mol/mol)	Aq. NaBr (0.035 N)		57	0.60	–	4	0.4–0.8	EG	–	980
(5/5, mol/mol)	Aq. NaBr (0.035 N)		73	0.57	–	4	0.4–0.8	EG	–	980
(4/6, mol/mol)	Aq. NaBr (0.035 N)		150	0.50	–	4	0.4–0.8	EG	–	980
3.8. POLY(SILOXANES), POLY(SILSESQUIOXANES)										
Poly(dimethyl siloxane), see Poly[oxy(dimethylsilylene)]										
Poly[(1-isobutyl-3-phenylsilsesquioxane)]										
	Benzene	21	1.4	0.90	?	–	1.2–15	SD	?	619
		21	110	0.54	?	–	20–230	SD	?	619
	Butyl acetate	24	Same as above two data							619
Poly(3-methylbutenesilsesquioxane)										
	Benzene	21	5.4	0.88	5	–	9–60	SD	B	603
		21	1.6	0.90	?	–	0.35–74	SD	B	619
	Butyl acetate	24	5.4	0.88	13	–	9–60	SD	B	603
Poly(methylphenylsiloxane)	Cyclohexane	25	27.3	0.60	4	–	12–46	LS	C	983
		50	15.6	0.65	4	–	12–46	LS	C	983
	Methylcyclohexane	20	30.6	0.58	4	–	12–46	LS	C	983
	Tetrahydrofuran	25	16.5	0.69	4	–	12–46	LS	C	983
Poly[oxy(dimethylsilylene)]	Benzene	20	12	0.68	4	–	5.5–12	LV	A–B	424
	Bromobenzene	78.7 (θ)	76	0.50	3	–	8–106	LS	A	425
	Bromocyclohexane	28 (θ)	78	0.50	5	–	10–92	SD	A	426
		28 (θ)	79.5	0.50	6	–	4–90	LS	A	838
		29.0 (θ)	74	0.50	5	–	3.3–106	LS	A,R	425
	Bromocyclohexane	37 (θ)	78	0.5	7	–	4.5–32	LS	–	982
	Bromocyclohexane/ phenetole (6/7, v/v)	36.3 (θ)	75.5	0.50	4	–	4.5–106	LS	A	425
	Butanone	20	64	0.525	6	–	4–90	LS	A	838
		20 (θ)	81	0.50	5	–	5–66	OS	A	427
		30	48	0.55	8	–	5–66	OS	A	427
	$C_8F_{18}/C_2Cl_4F_2$ (33.17/66.83, w/w), low cohesive energy density mixture	22.5 (θ)	106	0.50	4	–	55–120	LS	A–B	424
	Chlorobenzene/dimethyl phthalate (45/6, v/v)	57.5 (θ)	76	0.50	3	–	8–106	LS	A	425
	Cyclohexane	35	10.2	0.735	8	–	1.7–90	LS	A	838
	Ethyl iodide	2.1 (θ)	70	0.50	2	–	34–106	LS	A	425
	Phenetole	83 (θ)	79	0.50	–	2	5–66	OS	A	427
		89.5 (θ)	73	0.50	4	–	4.5–106	LS	A,R	425
	Toluene	20	20.0	0.66	–	?	0.3–20	OS, LS	C	428
		25	2.43	0.84	–	7	1.9–13	LS	C	429
		25	8.28	0.72	5	–	10–92	SD	A	426
		25	21.5	0.65	–	?	2–130	OS		430
		25	75	0.50	5	–	0.2–1.0	OS		56
	Toluene	25	13.6	0.69	13	–	0.26–63	OS, LS	B	981
	Toluene	25	18.7	0.66	18	–	0.2–30	LS, SA	B	837
		35	12.5	0.703	8	–	1.7–90	LS	A	838
star type, 3 branches	Toluene	20	23.9	0.64	10	–	4–35	LS	A	431
star type, 4 branches	Toluene	20	64.5	0.54	10	–	0.8–25	LS	A	431
Poly[oxy(dimethylsilylene)1,4-phenylenedimethylsilylene]										
	Toluene	25	11.2	0.75	6	–	7–40	LS	B	499
Poly[oxy(dipropylsilylene)]	2-Pentanone	76 (θ)	87.1	0.50	4	–	2.5–27	OS	A	433
	Toluene	10 (θ)	109	0.50	6	–	2.5–30	OS	A	433
		25	43.5	0.58	15	–	1.7–43	OS	A	433
Poly[oxy(methylsilylene)]										
Me/Si = 1.5	Chlorobenzene	20 (θ)	326	0.21	12	–	0.1–500	LS		432
Me/Si = 1.8	Chlorobenzene/dimethyl phthalate (90.7/9.3, w/w)	20 (θ)	240	0.28	3	–	5–100	LS		432

TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly[oxy(methylphenylsilylene)]	Cyclohexane	25	5.52	0.72	13	–	6–124	LS	A	434
	Diisobutylamine	30.4 (θ)	51.5	0.50	9	–	6–124	LS	A	434
	Toluene	25	3.90	0.78	20	–	6–124	LS	A	434
Poly[oxy(γ-trifluoropropylmethylsilylene)]										
	Cyclohexyl acetate	25.0 (θ)	41.0	0.50	12	–	12–451	LS	A	435
	Ethyl acetate	25	5.92	0.70	9	–	20–451	LS	A	435
	Methyl hexanoate	72.8 (θ)	44.5	0.50	7	–	44–451	LS	A	435
Poly(phenylsilsesquioxane)	Benzene	–	–	0.92	–	–		LS		500
		21	0.77	0.90	7	–	1.7–6.1	SD	B	501
		21	2.38	0.85	14	–	0.4–88	SD	B	502
		21	0.13	1.10	8	–	3.7–15	SD	B,L	501,603
		21	7.6	0.70	5	–	10–31	SD	B	603
		25	Not linear					LS		843
	Bromoform	21	0.13	1.09	8	–	3.7–15	SD	B	501
		21	2.38	0.85	12	–	3.6–88	SD	B	502
	Benzene/bromoform	21 (θ)	2.38	0.85	5	–	14–71	SD	B	502
	(60/40, w/w)	21 (θ)	220	0.50	?	–	60–340	SD	?	619
cis-syndiotactic	1,2-Dichloroethane	50.5 (θ)	2.12	0.87	4	–	5–30	OS		601
3.9. POLY(HETEROCYCLICS)										
Poly(2,6-benzothiazole)	Methanesulfonic acid	30.0	12.6	1.0	6	–	0.36–11	LLS	–	989
Poly(benzobisthiazole-2,6-diyl-1,4-phenylene)										
	Methanesulfonic acid	25	0.014		1.8		1.5–3.7	LS		833
Poly[2,5-benzoxazole)	Methanesulfonic acid	30.0	10.9	1.02	4	–	0.49–11	LLS	–	989
Poly[bis(*m*-chlorophenoxy)phosphazene]										
	Chloroform	25	–	0.67	10	–	59–668	LS		836
Poly[6,6′-bis(3-phenylquinoxaline)-2,2′-diyl-1,4-phenylene)]										
	m-Cresol	40	86	0.63	–	8	1–35	LS		845
Poly[(1,3-dihydro-3-oxoisobenzofuran-1-ylidene)-1,4-phenyleneiminoterephthaloylimino-1,4-phenylene]										
	Dimethylformamide	25	277	0.59	7	–	1.4–5.5	LS	B	503
Poly[(1,3-dihydro-3-oxo-2-phenylisoindole-1-ylidene)-1,4-phenyleneoxyterephthaloyloxy-1,4-phenylene]										
	Tetrachloroethane	20	41.0	0.684	10	–	0.9–3	LS	B	514
	Tetrahydrofuran	20	259	0.488	5	–	1–3	LS	B	514
Poly[(5,7-dihydro-1,3,5,7-tetraoxobenzo[1,2-c:4,5-c′]-dipyrrole-2,6(1H,3H)diyl)-1,4-phenylene-(1,3-dihydro-3-oxoisobenzofuran-1-ylidene)-1,4-phenylene]										
	Dimethylformamide	20	328	0.516	26	–	0.4–17	LS	B	515
Poly(3,3′-dimethylthietane)	Tetrahydrofuran	25	11.0	0.79	7	–	3.8–12.9	OS	B	986
Poly(1-isobutyl-2,5-oxopyrrolidine-3,4-diyl)										
	Butyl acetate	21	22	0.65	13	–	19–340	SD	A	512
Poly[2-methoxy-4,6-di(*p*,*p*′-isopropylidene diphenyloxy)-*s*-triazine]										
	Chloroform	30	16.8	0.74	6	–	9–55.2	OS, LS	B–C	985
	1,2-Dichloroethane	30	21.6	0.71	4	–	14–55.2	OS, LS	B–C	985
	1,2-Dichloroethane/ *n*-butanol (80/15, v/v)	30	27.9	0.68	4	–	14–55.2	OS, LS	B–C	985
Poly(methylene-*N*,*N*′-dimethylpiperidinium chloride)										
	Aq. NaCl (0.1 M)	20	39.8	0.660	5	–	8–195	LS	A	829
	(0.4 M)	20	13.9	0.542	5	–	8–195	LS	A	829
Poly[*N*-(*n*-octadecyl)maleimide]	Benzene	25	5.38	0.74	17	–	1.2–33	OS, VOS, GPC	B	987
	1-Chloronaphthalene	25	2.89	0.77	7	–	1.5–12	OS, VOS, GPC	B	987
	o-Dichlorobenzene	25	4.03	0.75	7	–	1.8–12	OS, VOS, GPC	B	987
	Tetrahydrofuran	25	9.61	0.67	15	–	1.3–33	OS, VOS, GPC	B	987
	Toluene	25	6.04	0.70	8	–	1.2–17	OS, VOS, GPC	B	987
	1-Decanol	39.4 (θ)	18.3	0.50	6	–	2–19	LV	B	988
	1-Hexanol	79.0 (θ)	37.0	0.50	6	–	1.4–9	LV	B	988
	1-Octanol	53.7 (θ)	23.5	0.50	6	–	1.1–12	LV	B	988
Poly(*N*,*N*′-(*p*-phenylene)-3,3′,4,4′-biphenyltetracarboxylic acid diamine]										
	N-methylpyrrolidinone	25	9.5	0.90	–	4	0.59–7.5	LLS, SEC	–	990
	Dimethyl sulfoxide	25	38	0.77	–	6	0.58–5.6	LLS, SEC	–	990
Poly(4-phenylquinoline-2,6-diyloxy-4-phenylguinoline-2,6-diyl-1,4-phenyleneoxy-1,4-phenylene)										
	Chloroform	25	90	0.66	3	–	4–8	OS		832
Poly[(4-phenyl-1,2,4-triazol-3,5-diyl)-1,3(or 1,4)-phenylene]										
	Phenol/water (90/10, w/w)	–	845	0.56	–	5	1.3–2.7	OS	–	516
Poly[(L-1,2-pyrrolidindiyl)-carbonyl]										
	Water, acetic acid	25	No simple relation		–	5	1–5	OS	C	507

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TABLE 3. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	Mol. wt. range M ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Poly(quinoline-2,4-diyl-1,4-phenyleneoxy-1,4-phenylene-quinoline-4,2-diyl-1,4-phenyleneoxy-1,4-phenylene)										
	Chloroform	20	550	0.44	3	–	3–8	OS		831
Poly(1-*p*-tolyl-2,5-oxopyrrolidine-3,4-diyl)										
	Dimethylformamide	21	15.5	0.7	6	–	4–56	SD	B	513
3.10. COPOLYMERS (MALEIC ANHYDRIDE, SULFONES)										
Poly[3,9-bis(ethylidone-2,4,8,10-tetraoxaspiro(5,5)undecane)-*ran*-(*trans*-cyclohexane-dimethanol)/(1,6-hexanediol)]										
trans-CDM/1,6-ND (100/0)	Tetrahydrofuran	30	49.2	0.57	?	?	?	LLS	–	954
(70/30)	Tetrahydrofuran	30	22	0.65	?	?	?	LLS	–	954
(50/50)	Tetrahydrofuran	30	17.2	0.68	?	?	?	LLS	–	954
(29/71)	Tetrahydrofuran	30	11.9	0.74	?	?	?	LLS	–	954
(0/100)	Tetrahydrofuran	30	3.78	0.81	?	?	?	LLS	–	954
Poly[(tetrahydro-2,5-dioxo-3,4-furandiyl)(1-isobutyloxyethylene)]										
	Acetone	30	124.7	0.506	5	–	21–111	LS	B	504
	Butanone	30	119.4	0.512	5	–	21–111	LS	B	504
	Tetrahydrofuran	30	75.6	0.552	5	–	21–111	LS	B	504
Poly[(tetrahydro-2,5-dioxo-3,4-furandiyl)(1-methoxycarbonyl-1-methylethylene)]										
	Acetone	30	12.4	0.69	6	–	20–71	LS	B	505
	Dimethylsulfoxide	30	7.5	0.77	6	–	20–71	LS	B	505
	Dioxane	30	26.1	0.64	6	–	20–71	LS	B	505
	Tetrahydrofuran	30	13.4	0.69	6	–	20–71	LS	B	505
Poly[(tetrahydro-2,5-dioxo-3,4-furandiyl)(1-phenylethylene)]										
	Acetone	30	8.69	0.74	6	–	13–75	OS	A	506
	Tetrahydrofuran	30	5.07	0.81	6	–	13–75	OS	A	506
Poly[sulfonyl(bicycloheptene)]	Chloroform	25	1.58	0.79	4	–	27–227	LS	D	840
	Cyclohexane	25	3.05	0.73	4	–	27–227	LS	D	840
Poly[sulfonyl(butylethylene)]	Acetone	20	5.9	0.74	7	–	5–60	LS, SD	B	437
	Benzene	25	8.9	0.70	5	–	9–107	OS	A,R	438
	Chloroform	25	5.8	0.75	6	–	7–54	OS	A,R	439
	Dioxane	25	6.2	0.76	5	–	9–107	OS	A	438
	Hexylchloride	13 (θ)	33	0.55	5	–	10–60	LS, SD	B	437
	Butanone/2-propanol (29.8/70.2, v/v)	8 (θ)	53	0.50	6	–	7–54	OS	A	439
	(37/63, v/v)	24 (θ)	53	0.50	6	–	7–54	OS	A	439
	Dioxane/hexane (40/60, v/v)	20 (θ)	65	0.50	7	–	9–107	OS	A	438
Poly[sulfonyl(1-methyl-1-propylethylene)]										
	Chloroform	20	5.9	0.81	6	–	4–50	OS	A	439
	Butanone/2-propanol (39.5/60.5, v/v)	22.5 (θ)	91	0.50	6	–	4–50	OS	A	439
	Butanone/hexane (35.4/64.6, v/v)	11.5 (θ)	91	0.50	6	–	4–50	OS	A	439
Poly[sulfonyl(phenyleneoxyphenylene)]										
	Dimethylformamide	25	33	0.64	6	–	3–25	LS	C	841
Poly[sulfonyl(phenylethylene)]	Tetrahydrofuran	30	3.89	0.78	5	–	15–40	OS	A	440
3.11. OTHER COMPOUNDS										
Poly(di-*n*-hexylsilane)	Tetrahydrofuran	25	17.0	0.672	7	–	18–960	LS	B–D	995
Poly(oxy-1,4-phenylene-oxy-1,4-phenylene-*N*-phenylimine-1,4-phenylene)										
	Tetrahydrofuran	25	63.4	0.617	8	–	0.58–10	LS	B	993
Poly[oxythiocarbonyloxy-1,4-phenylene(methyl)phenylmethylene-1,4-phenylene]										
	Chloroform	25	71.0	0.60	5	–	1.5–14.2	OS, SEC	C	992
	Dichloromethane	25	141	0.55	5	–	1.5–14.2	OS, SEC	C	992
	1,4-Dioxane	25	153	0.50	5	–	1.5–14.2	OS, SEC	C	992
	Tetrahydrofuran	25	61.7	0.61	5	–	1.5–14.2	OS, SEC	C	992
Poly[*trans*-bis(tributylphosphine)platinum-1,4-butadiynediyl]										
	Heptane	25	Wormlike Cylinder							834
	Tetrahydrofuran	25	6.5×10^{-7}	1.7	5	–	4–10	GPC	B	835
Poly[*trans*-bis(tributylphosphine)platinum-1,4-phenylenediethynylene]										
	Tetrahydrofuran	25	2.0×10^{-6}	1.7				GPC	B	835
Poly(hexamethylenedimethylammonium bromide)										
	Aq. KBr (0.4 M)	25	62.2	0.58	5	–	1.3–5	LS	C	844
Poly(hexamethylenedimethylammonium chloride)										
	Aq. KCl (0.4 M)	25	1.02	1.00	4	–	1.3–3	LS	C	844

TABLE 4. CELLULOSE AND DERIVATIVES, POLY(SACCHARIDES) (See also table "Properties of Cellulose Materials")

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
(Acetoxypropyl) cellulose	Dimethyl phthalate	25	–	0.88	6	–	2.2–43	SEC, LLS	A–C	1001
		105	–	0.75	5	–	7.6–43	SEC, LLS	A–C	1001
		150	–	0.57	6	–	2.2–43	SEC, LLS	A–C	1001
			(Wormlike behavior)							
Amylose	Dimethyl sulfoxide	20	3.97	0.82	–	14	2–217	LS	C	517
		25	1.25	0.87	9	–	22–310	LS	B	518
		25	15.1	0.70			8–180	LS	B	519
		25	30.6	0.64	6	–	27–220	LS	B	520
	Ethylenediamine	25	15.5	0.70	6	–	31–310	LS	B	518
	Formamide	20	22.6	0.67	–	12	2–157	LS	C	517
		25	30.5	0.62	14	–	8–180	LS	B	519
	Water	20	13.2	0.68	–	12	36–217	LS	C	517
	Acetone/dimethyl sulfoxide (43.5/56.5, v/v)	20 (θ)	83.1	0.51	–	10	2–157	LS	C	517
	Aq. KCl (0.33 M)	22.5	33.9	0.59	5	–	16–230	LS	B	521
		25 (θ)	112	0.50	5	–	16–230	LS	B	522
		25 (θ)	115	0.50	6	–	27–220	LS	B	520
	(0.50 M)	25 (θ)	61.1	0.50					B	523
	Aq. KOH (0.15 M)	25	8.36	0.77	?	–	8–180	LS	B	519
	(0.2 M)	25	6.92	0.78	5	–	16–229	LS	B	522
	(0.5 M)	25	8.50	0.76	6	–	27–220	LS	B	520
	(1 M)	25	1.18	0.89	5	–	31–310	LS	B	518
	Aq. NaOH (0.5 M)	20	3.65	0.85	–	16	2–217	LS	C	517
Amylose triacetate	Chloroform	30	1.06	0.92	12	–	12–480	LS	B	524
		30	4.90	0.85	4	–	21–102	LS	A	525
		50	5.20	0.83	4	–	21–102	LS	A	525
	Methyl acetate	25	5.60	0.80	–	3	7–19	SD	D	526
	Nitromethane	22.5	8.50	0.73	12	–	14–310	LS	B	519,527
		25	3.58	0.78	6	–	4–100	LS	B,C	846
		30	2.18	0.82	6	–	4–100	LS	B,C	846
		30	9.93	0.76	4	–	21–102	LS	A	525
		50	8.71	0.76	4	–	21–102	LS	A	525
	Chloroform/cyclohexane (80/20, v/v)	30	4.64	0.85	4	–	21–102	LS	A	525
	(50/50, v/v)	30	7.41	0.79	4	–	21–102	LS	A	525
	Methanol/nitromethane (70.7/29.3, v/v)	30 (θ)	98.4	0.51	4	–	21–102	LS	A	525
	(50/50, v/v)	30	6.49	0.75	4	–	21–102	LS	A	525
	(25/75, v/v)	30	10.23	0.76	4	–	21–102	LS	A	525
	Nitromethane/propanol (43.3/56.7, v/v)	25 (θ)	91.6	0.50	12	–	14–310	LS	B	519
	(50/50, v/v)	25	17.0	0.66	12	–	14–310	LS	B	519
Amylose tributyrate	Butanone	25	3.39	0.77	7	–	7–160	LS	B	847
	Carbon tetrachloride	25	3.46	0.79	7	–	7–160	LS	B	847
	Ethyl acetate	25	5.59	0.74	7	–	7–160	LS	B	847
	Tetrahydrofuran	25	1.66	0.86	7	–	7–160	LS	B	847
Amylose tricarbanilate	Acetone	20	0.814	0.90	–	26	4–490	LS	B	528
	Dioxane	20	0.906	0.92	–	25	4–360	LS	B	528
	Dioxane	–	(Wormlike behavior)		7	–	21–290	LS	A	997
	Pyridine	20	0.589	0.92	–	20	4–360	LS	B	528
		25	2.00	0.83	11	–	30–244	LS	B	848,849
Amylose tricarbethoxymethylcarbamate	Acetone	20	27.6	0.63	13	–	9–380	LV	B	529
Amylose tripropionate	Ethyl acetate	25	31.3	0.58	8	–	1–61	OS, GPC	B	850
	Tetrahydrofuran	25	24.8	0.61	6	–	1–61	OS, GPC	B	850
Amylose, carboxymethyl –, sodium salt	Aq. NaCl (0.35 M)	37.5	25.2	0.64				LS	A	609
	(0.5 M, pH 8)	35	209	0.53	6	–	5–21	OS	B	530
	(0.78 M; 0.02% NaN_3)	35	37.1	0.61	6	–	7–29	LS	B	531
Amylose, diethylaminoethyl –, hydrochloride	Aq. NaCl (0.78 M; 0.02% NaN_3)	35	82.8	0.55	5	–	4–23	LS	B	531
Arginic acid, sodium salt	Aq. NaCl (0.2 M)	25	7.97	1.0	–	7	5–19	OS	C	532

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TABLE 4. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Cellulose, see also table "Properties of Cellulose Materials"										
	Cadoxen	25	33.8	0.77	5	–	20–100	SD	C,R	533
		25	38.5	0.76	4	–	1.0–3.4	SE, LS	B–C,R	534
	Cuprammonium	20	105	0.66	9	–	2–25	OS	C	535
		25	8.5	0.81	–	5	8–96	OS	C	536
	Cupriethylene	25	13.3	0.905	32	–	1–54	OS	B-C	537
	N,N-Dimethylactamide/ 9% LiCl	30	0.128	1.19	9	–	12.5–70	LLS	–	1000
Cellulose acetate	*N,N*-Dimethylacetamide	25	191	0.60	10	–	4–15	LS	B	851
DS[a] 0.49	Dimethyl sulfoxide	25	171	0.61	10	–	4–15	LS	B	851
	Formamide	25	20.9	0.60	10	–	4–15	LS	B	851
	Water	25	20.9	0.60	10	–	4–15	LS	B	851
DS[a] 1.75	*N,N*-Dimethylacetamide	25	95.8	0.65	7	–	3–14	LS	B	852
Cellulose acetate butyrate	Acetic acid	25	14.6	0.83	–	5	1–21	OS	B–C	538
	Acetone	25	13.7	0.85	–	11	1–21	OS	B–C	538
	Ethyl acetate	30	11.3	0.83	6	–	3–14	LS		853
	Nitromethane	30	16.5	0.78	6	–	3–14	LS		853
Cellulose diacetate	Acetone	25	133	0.616	9	–	6–27	LS	A	854
Ac. acid content, 55.5 wt.%	Dimethylacetamide	25	39.5	0.738	5	–	6–19	LS	A	855
	N,N-Dimethylactamide	30	11.0	0.848	6	–	3.58–17.0	SE	A–B	1007
			(Wormlike behavior)							
	Tetrahydrofuran	25	51.3	0.688	6	–	7–30	LS	A	854
	Trifluoroacetic acid	25	52.7	0.696	5	–	6–19	LS	A	855
Cellulose nitrate	Tetrahydrofuran	25	–	0.80–0.84	–	9	2.6–27	LLS, OS	D	1004
Cellulose triacetate	Acetone	20	2.38	1.0	5	–	2–14	SD	B	539
		25	14.9	0.82	8	–	2–39	OS	A(?)	540
		25	8.97	0.90	14	–	1–18	OS	B,R	541
		25	33.0	0.760	9	–	2–30	OS	C	535
		25	28.9	0.725	9	–	8–69	LS	A	855
	Chloroform	20	2.2	0.95	13	–	1–13	LS	A	856
		25	45.4	0.649	7	–	8–69	LS	A	855
		30	14.4	0.800	6	–	10–18	LS	A	857
		30	4.5	0.9	5	–	3–18	LV	C	542
	o-Cresol	30	6.15	0.9	5	–	3–18	LV	C	542
	Dichloromethane	20	24.7	0.704	7	–	6–69	LS	A	855
	Dimethylacetamide	25	26.4	0.750	10	–	6–69	LS	A	855
	Tetrachloroethane	25	39.3	0.662	10	–	6–69	LS	A	855
	Trifluoroacetic acid	25	39.6	0.706	8	–	6–69	LS	A	855
	Acetone/water (80/20, v/v)	20	2.65	1.0	9	–	2–11	SD	B	539
		25	21.0	0.803		–	2–30	OS	C	535
	Ethanol/methylene chloride (20/80, v/v)	13.9	0.834		–	2	~30	OS	C	5315
Cellulose tributyrate	Butanone	30	4.3	0.87	7	–	6–32	LS	B,R	543
		30	18.2	0.80	7	–	8–32	OS	C–D	544,2
	Tributyrin	0	5.3	0.87	4	–	6–32	LS	B	543
		25	5.6	0.85	4	–	6–32	LS	B	543
		50	6.1	0.82	4	–	6–32	LS	B	543
		70	6.2	0.80	4	–	6–32	LS	B	543
	Dodecane/tetralin (75/25, v/v)	130 (θ)	82	0.50	3	–	11–21	OS	C–D	544,2
Cellulose tricarbanilate	Acetone	0	1.10	0.93	6	–	31–220	LS	B–C	545
		20	4.66	0.84	–	16	7–270	LS	B	528
		25	1.43	0.91	6	–	31–220	LS	B–C,R	545
		35	1.31	0.90	6	–	31–220	LS	B–C	545
	Acetone	25	–	Not const.	9	–	22–770	LS, GPC	B–C	996
			(Wormlike behavior)							
	Anisole	94 (θ)	130	0.50	4	–	31–220	LS	B–C	546
	Butanone	15	2.02	0.89	–	10	21–110	LS	D	858
		30	3.42	0.84	–	10	21–110	LS	D	858
	Cyclohexanol	73 (θ)	156	0.51	–	5	21–110	LS	D	858
	Cyclohexanone	25	1.91	0.86	5	–	31–220	LS	B–C	545
		35	2.02	0.85	5	–	31–220	LS	B–C	545

TABLE 4. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
	Dioxane	20	4.20	0.88	–	15	7–270	LS	B	528
		25	0.813	0.97	5	–	31–220	LS	B–C	545
		35	0.865	0.96	5	–	31–220	LS	B–C	545
		50	0.849	0.95	4	15	31–94	LS	B–C	545
		15	0.89	0.97	–	10	21–110	LS	D	858
		21		0.95	11	–	1–11	SD	B	859
		30	1.04	0.95	–	10	21–110	LS	D	858
		70	2.57	0.86	–	10	21–110	LS	D	858
	Pyridine	20	3.46	0.86	–	12	7–270	LS	B	528
	Tetrahydrofuran	25	–	Not const.	12	–	2.7–370	LS, GPC	B–C	996
			(Wormlike behavior)							
Cellulose trihexanoate	Dimethylformamide	41 (θ)	245	0.50	7		6–130	LS	C–D	547
	Dioxane	35	125	0.57	7		4–130	LS	C–D	547
Cellulose trinitrate	Acetone	20	2.80	1.00	13		1–250	SD	B	548
		25	1.69	1.00	11	–	8–265	LS	B–C	549
		25	1.66	0.86	6	–	68–250	LS	C	550
		25	10.8	0.89	4	–	4–32	LS	C–D	551
(N content, 12.9 wt.%)		25	5.70	0.90	4	–	15–200	LS	A,R	552
(N content, 13.9 wt.%)		25	6.93	0.91	6	–	8–400	LS	A,R	552
		25	7.00	0.933	9	–	5–50	OS	B–C	535
		25	11.0	0.91	33	–	3–100	OS	B–C	537
		25	23.5	0.78	6	–	7–26	OS	B–C	553
	Butyl acetate	25	5.68	0.969	9	–	5–50	OS	B–C	535
	Butyl formate	25	23	0.81	6	–	7–26	OS	B–C	553
	Cyclohexanone	25	2.24	0.810	6	–	7–22	OS	B–C	554
	Ethyl acetate	25	3.8	1.03	33	–	3–100	OS	B–C	537
		25	8.3	0.90	6	–	7–26	OS	B–C	553
		25	1.66	0.86	7	–	68–250	LS	C	550
		30	2.50	1.01	6	–	4–57	LS	B–C	555
	Ethyl butyrate	25	3.64	1.0	?	–	5–50	OS	B–C	556
	Ethyl formate	25	30	0.79	6	–	7–26	OS	B–C	553
	Ethyl lactate	25	12.2	0.92	10	–	3–65	OS	B–C	537
	2-Heptanone	25	5.0	0.93	6	–	7–26	OS	B–C	553
	Methyl acetate	25	18.3	0.835	6	–	7–22	OS	B–C	554
	Nitrobenzene	25	6.1	0.945	6	–	7–22	OS	B–C	554
	Pentyl acetate	25	1.1	1.04	6	–	7–26	OS	B–C	553
Cellulose trioctanoate	Dimethylformamide	140 (θ)	113	0.50	3	–	10–32	OS	B–C	544
	γ-Phenylpropanol	48 (θ)	129	0.50	3	–	8–32	OS	B–C	544
	Toluene	30	17.3	0.70	6	–	8–35	OS	B–C	544
(Cyanoethyl)(hydroxypropyl) cellulose	Tetrahydrofuran	25	1.14	0.99	8	–	5.4–27	LLS, SEC	C–D	1002
			(Wormlike behavior)							
Ethyl cellulose	Acetone	20	1.51	1.05	5	–	1.1–8	SD	A	557
	Benzene	20	1.34	1.07	5	–	1.1–8	SD	A	557
		25	29.2	0.81	6	–	4–14	OS	B–C	558
		60	35.8	0.78	6	–	4–14	OS	B–C	558
	Butanone	25	18.2	0.84	6	–	4–14	OS	B–C	558
		60	26.7	0.79	6	–	4–14	OS	B–C	558
	Butyl acetate	25	14.0	0.87	6	–	4–14	OS	B–C	558
		60	18.1	0.83	6	–	4–14	OS	B–C	558
	Chloroform	25	11.8	0.89	6	–	4–14	OS	B–C	558
		46	9.3	0.90	6	–	4–14	OS	B–C	558
	Ethyl acetate	25	10.7	0.89	6	–	4–14	OS	B–C	558
		60	14.0	0.85	6	–	4–14	OS	B–C	558
	Methanol	25	52.3	0.65	6	–	10–41	LS	B–C	559
	Nitroethane	25	4.2	0.96	6	–	4–14	OS	B–C	558
		60	22.6	0.79	6	–	4–14	OS	B–C	558
Ethyl hydroxyethyl cellulose	Water	25	37	0.80	4	–	5–18	SD, LS	B	560
Hydroxyethyl cellulose	Cadoxen	25	17.4	0.79	4	–	8–61	LS	B	561
	Water	25	9.53	0.87	5	–	8–63	LS	B	561
DS[a] 0.88	Aq. HCl (4 M)	25	$[\eta] = 1.2\ DP_w^{0.87}$ (DP_w; weight-average degree of polymerization)						B	562
Methyl cellulose										
DS[a] 1.74	Water	25	316	0.55	–	5	12–57	LS	C–D	563
	Aq. HCl (4 M)	25	$[\eta] = 1.6\ DP_w^{0.86}$					LS	B	562

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TABLE 4. *cont'd*

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples W.P.	Mol. wt. range ($\times 10^{-4}$)	Method(s)	Remarks	Refs.
Sodium carboxymethylcellulose	Aq. NaOH (0.78 M)/NaN_3(0.1 M)	35	37	0.61	6	–	7–30	OS, LS	B	531
DS[a] 0.2–1.0	Cadoxen	25	33.4	0.73	–	5	5–106	LS	C	564
DS[a] 0.96	Aq. HCl (4 M)	25	$[\eta] = 0.97\ DP_w^{0.83}$						B	562
DS[a] 0.62–0.74	Aq. NaCl (0.001 M)	25	0.100	1.40	8	–	4.5–35	SD	C–D	565
	(0.01 M)	25	0.646	1.20	3	–	4.5–35	SD	C–D	565
	(0.1 M)	25	12.3	0.91	8	–	4.5–35	SD	C–D	565
DS[a] 1.06	(0.005 M)	25	7.2	0.95	4	–	14–106	LS	C–D	566
	(0.01 M)	25	8.1	0.92	4	–	14–106	LS	C–D	566
	(0.05 M)	25	19	0.82	4	–	14–106	LS	C–D	566
	(0.2 M)	25	43	0.74	4	–	14–106	LS	C–D	566
Sodium cellulose xanthate	Aq. NaOH (1 M)	0	$[\eta] = 1.67\ DP_w^{0.82} + 0.62\ DS_{2,3} - 0.20\ DS_6 DP_w^{0.94}$						C	567
			DS_i, Degree of substitution at the i (= 2, 3 or 6) positions in glucose unit							
Dextran										
linear fraction	Formamide	25	165	0.49	5	–	0.2–3.2	OS	C	568
	Water	25	97.8	0.50	10	–	2–10	LS	C,R	569
		25	49.3	0.60	10	–	0.04–4.5	EA	C	570
		50	39.3	0.61	6	–	0.2–3.2	OS	C	568
branched fraction	Methanol/water (40.5/59.5, v/v)	34 (θ)	10.3	0.25	–	6	7–51	LS	C	860
	Water	25	–	0.20	9	–	80	LS	C	569
		32.7			6	–				571
Guaran triacetate	Acetonitrile	25	2.62	0.87	4	–	7–85	LS	A	572
		25	311	0.52	5	–	206–534	LS	B	572
Hyaluronic acid	Aq. HCl (0.1 M)	25	27.9	0.763	5	–	7–103	LS	A	573
	Aq. NaCl (0.2 M)	25	22.8	0.816	8	–	7–103	LS	A	573
	(0.5 M)	25	31.8	0.777	5	–	11–103	LS	A	573
Pullulan	Water	25	22.1	0.66	9	–	11–236	LS	A	998
		25	–	0.65	8	–	0.53–102	SE	A	1003
Salep glucomannan triacetate	Nitroethane	30	This relation not followed		11	–	0.06–0.4	LS	A–B	574
Shizophyllan (β-1,3-D-glucan)	Dimethyl sulfoxide	25	–	0.69	6	–	3–190	SE, LS	A	852,863
	Aq. NaOH	25	–	1.8	4	–	10–25	LS	A	863
		25	–	1.1	3	–	300–570	LS	A	863
Xanthan	Cadoxen	25	–	0.87			3–15	LS		865
		25		0.70			20–360	LS		865
	Aq. NaCl (0.1 M)	25	0.17	1.14	5	–	30–700	LS		866
		25	Double strand helix				3–360	LS		865

[a] DS: Degree of substitution.

TABLE 5. MISCELLANEOUS

Polymer	Solvent	Temp. (°C)	K ($\times 10^3$) (ml/g)	a	No. of samples Fr.	No. of samples WP	M ($\times 10^{-4}$)	Mol. wt. range Method	Remarks	Ref.
Imogolite	Aqueous acetic acid/ 0.02 wt.% NaN_3	30	$[\eta] = 9.35 \times 10^{-10} M^2/\ln M$ (sonic wave)		6 –		59–208	LS	A	1005

D. CALCULATED UNPERTURBED DIMENSIONS OF FREELY ROTATING CHAINS

Chain type	$r_{of}/M^{1/2}$ (nm mol$^{1/2}$/g$^{1/2}$)		Refs.
Poly(methylene) chain	$0.308/M_u^{1/2}$	$0.218/m^{1/2}$	2
Amylosic chain	$0.426/M_u^{1/2}$	$0.191/m^{1/2}$	518
Cellulosic chain	$0.790/M_u^{1/2}$	$0.353/m^{1/2}$	620
Gutta-percha (*trans* polydiene)	$0.580/M_u^{1/2}$	$0.290/m^{1/2}$	620,621
Natural rubber (*cis* polydiene)	$0.383/M_u^{1/2}$	$0.201/m^{1/2}$	620
Poly(peptide)	$0.383/M_u^{1/2}$	$0.221/m^{1/2}$	620

E. UNPERTURBED DIMENSIONS OF LINEAR POLYMER MOLECULES

TABLE 6. MAIN-CHAIN ACYCLIC CARBON POLYMERS

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
6.1. POLY(DIENES)										
Poly(butadiene)										
100%-*cis*	Dioxane	20.2	–	205	920	547	1.68	5.15	VT	19
98%-*cis*, 2%-1,2	Isobutyl acetate	20.5	–	185	880	547	1.61	4.75	VT	15
95%-*cis*, 4%-1,2	2-Pentanone	59.7	–	157	835	546	1.53	4.3	VT	17
	3-Pentanone	10.3	–	152	825	546	1.51	4.2	VT	17
92%-*cis*, 5%-1,2	Benzene	32	–	150 ± 20	820 ± 40	545	1.50 ± 0.08	4.15	VG	3(20)
71%-*trans*, 25%-1,2	Cyclohexane	25	–	300 ± 40	1030 ± 50	702	1.45 ± 0.08	7.3	VG	3(22)
79%-*trans*, 21%-1,2	Cyclohexane	20	–	280 ± 25	1010 ± 30	742	1.36 ± 0.05	6.9	VG	3(23)
97%-*trans*, 3%-1,2	Cyclohexane	40	–	200 ± 30	935 ± 40	768	1.22 ± 0.07	5.4	VG	24
100%-*cis*	Various solvents	50					1.63	4.9	VT	622
	Undiluted	50–90	$d \ln r_0^2/dT = 0.4 \times 10^{-3}$ (deg^{-1})						ST	622
100%-*trans*	Decalin	55					1.23	5.8	VA	623
	Undiluted		$d \ln r_0^2/dT = -0.6 \times 10^{-3}$ (deg^{-1})						ST	623
1,4-*trans*	Oligomeric 1,4-Poly(butadiene)	25.0–105.8	$d \ln r_0^2/dT = -0.10 \times 10^{-3}$ (deg^{-1})						VG	888
linear	Dioxane	12.1	–	155	–	–	–	6.15	VG	889
	Toluene	35	–	150	–	–	–	–	VG	889
Poly(1-butenylene-*co*-vinylethylene) 43%-1,2										
	1,4-Dioxane	15.7	–	173	–	–	–	6.0	VT	890
Poly(chloroprene) 85%-*trans*	Benzene	25	–	115 ± 20	750 ± 80	535	1.40 ± 0.15	5.6	VG	3(32,33)
	Butanone	25	–	113	750	535	1.40	5.6	VT	35
	Cyclohexane	45.5	–	107	755	535	1.41	5.65	VT	34
	Butanone	25	313	–	750	535	1.40	5.6	LT	624
Poly(2,3-dimethylbutadiene) 85%-*trans*-1,4, 3%-1,2										
	Cyclohexane/*n*-propanol	25	–	226	–	–	–	8.59 ± 0.14	VG,VT	883
Poly(isoprene) 100%-*cis*	Benzene; 2-pentanone	~20	–	130 ± 20	810 ± 45	485	1.67 ± 0.09	5.0	VT,VG	3,37
	Diisopropyl ether	22	0.76	–	847	485	1.74	5.5	XS	625
	2-Pentanone	14.5		119				4.7	VT	626
	Undiluted	−10 to 70	$d \ln r_0^2/dT = 0.41 \times 10^{-3}$ (deg^{-1})						ST	626
		30–70	$d \ln r_0^2/dT = 0.56 \times 10^{-3}$ (deg^{-1})						ST,VT	627
100%-*trans*	Propyl acetate	60	–	232	970	703	1.38	7.2	VT	37
	Dioxane	47.7		191	910	703	1.30	6.35	VT	19
	Undiluted	~60	$d \ln r_0^2/dT = -0.27 \times 10^{-3}$ (deg^{-1})						ST	623
trans-1,4	Benzene, cyclohexane, Toluene	30	–	198	884	–	1.26	5.99	VG	894
	Hexane	30	–	135	778	–	1.11	4.64	VG	894
71%-*cis*, 22%-*trans*, 7%-3,4	Cyclohexane; dioxane; 4-methyl-2-pentanone; toluene	35	–	126	766	–	–	–	VG	786
Poly(isopropenylethylene-*co*-1-methyl-1-vinylethylene) 20–25%-1,2,70–75%-3,4										
	2-Octanol	30.5	–	102	–	–	–	7.9	VT	890
Poly[isoprene-*block*-poly(styrene)], A_k–B_n										
k/*n*, (75/25, w/w)	4-Methyl-2-pentanone; toluene	30	–	128	782	–	–	–	VG	867
k/*n*, (50/50, w/w)	Cyclohexane; dioxane; 4-methyl-2-pentanone; toluene	35	–	110	741	–	–	–	VG	786
	Butanone	44	–	116	757	–	–	–	VT	867
	Cyclohexane	46	–	116	756	–	–	–	VG	867
	4-Methyl-2-pentanone	30	–	124	773	–	–	–	VG	867
	Toluene	30	–	115	754	–	–	–	VG	867
k/*n*, (25/75, w/w)	Cyclohexane; dioxane; 4-methyl-2-pentanone; toluene	35	–	97	711	–	–	–	VG	786
Poly(methylbutylene) 43%-1,2	*n*-Hexyl acetate	60.9	–	189	–	–	–	6.2	VG	890
Poly(1-methyl-1-butylene-*co*-isopropenylethylene) 51%-1,4,49%-3,4										
	2-Octanol	41.3	–	96	–	–	–	5.3	VG	890

* References in parentheses include data used for the calculations of end to end distance in Ref. 3.

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TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_θ ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0t}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0t}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
6.2. POLY(ALKENES), POLY(ACETYLENES)										
Poly(1-butene)	2-Octanol	23.6	–	74	–	–	–	5.3	VG	896
	Tetrahydrofuran	25	–	74	–	–	–	5.3	VG	896
atactic	Anisole; ethylcyclohexane	~70	–	123 ± 10	775 ± 25	427	1.82 ± 0.05	6.6	VT,VG	3(81)
	Nonane	35	590 ± 50	–	1180 ± 70	427	2.76 ± 0.20	15.1	LT	81
	Phenyl ether	141	–	104	747	–	–	–	VT	700
	Undiluted	140–200	$d \ln r_0^2/dT = (0.5 \pm 0.04) \times 10^{-3}$ (deg^{-1})						ST	633
isotactic	Anisole	89	–	111	763	–	–	–	VT	700
	Nonane	80	510 ± 50	–	1290 ± 90	427	3.00 ± 0.20	18.0	LT	81
	Phenetole	64.5	–	113	767	–	–	–	VT	700
	Phenyl ether	148	–	103	744	–	–	–	VT	700
	Undiluted	140–200	$d \ln r_0^2/dT = (0.09 \pm 0.07) \times 10^{-3}$ (deg^{-1})						ST	634
		160	$d \ln r_0^2/dT = -0.1 \times 10^{-3}$ (deg^{-1})						ST	634
Poly(ethylene)	1-Chloronaphthalene; tetralin; *p*-Xylene	~100	–	230	950 ± 40	582	1.63 ± 0.08	5.3	VG	3(61,67, 74)
	Decalin	140	–	–	1070	582	1.84	6.8	VA	65
	Bis-2-ethylhexyl adipate	145	–	225	940 ± 40	582	1.61	5.2	VT	630
		145	690 ± 100	–	1320 ± 150	582	2.27 ± 0.26	10.3	LT	630
	Biphenyl	127.5	–	323	1085	582	1.87	7.0	VT	58
	Dodecanol	137.3	–	307	1070	582	1.84	6.8	VT	58
		138	–	316 ± 7	1080	582	1.86	6.9	VT	69
	Diphenylmethane	142.2	–	315	1080	582	1.86	6.9	VT	58
	Decanol	153.3	–	302	1065	582	1.83	7.6	VT	65
	Diphenyl ether	161.4	–	295	1060	582	1.79	6.4	VT	58
	Octanol	180.1	–	286	1040	582	1.79	6.4	VT	58
	Biphenyl	127.5	–	330	1095	582	1.88	7.1	VT	631
	Diphenylmethane	142.2	–	322	1085	582	1.87	7.0	VT	631
	Diphenyl ether	163.9	–	309	1070	582	1.84	6.8	VT	631
	Undiluted; diluted with triacontane; dotriacontane	140–190	$d \ln r_0^2/dT = -(1.15 \pm 0.1) \times 10^{-3}$ (deg^{-1})						ST	629
	Hexadecane	140	$d \ln r_0^2/dT = -1.2 \times 10^{-3}$ (deg^{-1})						VA	632
low pressure	Biphenyl	130	–	302	1130	–	–	–	VT	703
	Decalin	120	–	360	1200	–	–	–	VG	703
	1,2,4-Trichlorobenzene	135	–	360	1200	–	–	–	VG	703
Poly(ethylene-*co*-isopropylethylene)										
	n-Hexyl acetate	65	–	188	–	–	–	6.3	VG	890
Poly(ethylene-*co*-propylene)	Benzene	19.0	–	206	938	–	–	6.85	VG	897
		21.4	–	203	933	–	–	6.78	VG	897
	n-Decyl acetate	5.0	–	203	933	–	–	6.78	VG	897
	n-Heptyl acetate	38.0	–	194	919	–	–	6.57	VG	897
	n-Octyl acetate	27.0	–	198	925	–	–	6.66	VG	897
Poly(isobutene)	Anisole	105	–	91 ± 5	700 ± 20	412	1.70 ± 0.05	5.8	VT	49
	Benzene	24	–	107 ± 5	740 ± 20	412	1.80 ± 0.05	6.5	VT	49
	Benzene	25	–	105	749	–	–	6.7	VG	1010
	Phenetole	86	–	91 ± 5	700 ± 20	412	1.70 ± 0.05	5.8	VT	49
	Heptane/propanol (80/20, v/v)	25	390	166	780	412	1.9	7.2	LT	628
	Undiluted; diluted with hexadecane	60	$d \ln r_0^2/dT = -(0.1 \pm 0.05) \times 10^{-3}$ (deg^{-1})						ST	629
Poly(isopropylethylene-*co*-1-methyl-1-ethylethylene)										
	2-Octanol	26.2	–	83	–	–	–	7.2	VG	890
Poly(1-methylbutylene-*co*-isopropylethylene)										
	2-Octanol	53.3	–	91	–	–	–	5.0	VG	890
Poly(1-octene)	Bromobenzene	25	–	50	570 ± 50	291	1.96 ± 0.15	7.7	VG	84
	Cyclohexane	30	–	100	710 ± 60	291	2.44 ± 0.20	11.9	VG	84
	Phenetole	50.4	–	65	625 ± 30	291	2.14 ± 0.10	9.1	VT	84
Poly(pentenamer) 80–85%-*trans*, 19–12%-*cis*	Isopentyl acetate	38	–	234	991	–	1.26	–	VT	788

TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_θ ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
Poly(1-pentene)										
atactic	*Undiluted*	40–140	$d\ln r_0^2/dT = (0.53 \pm 0.05) \times 10^{-3}$ (deg^{-1})						ST	633
isotactic	2-Pentanol	62.4	–	121	790	368	2.14	9.2	VT	633
	Undiluted	80–140	$d\ln r_0^2/dT = (0.34 \pm 0.04) \times 10^{-3}$ (deg^{-1})						ST	633
		~90	$d\ln r_0^2/dT = -0.3 \times 10^{-3}$ (deg^{-1})						ST	634
		~60	$d\ln r_0^2/dT = -0.2 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(propylene)										
	3-Nonanol	5.0	–	158	857	–	–	6.6	VG	897
	1-Octanol	77.0	–	115	773	–	–	5.4	VG	897
atactic	Isoamyl acetate; benzene; cyclohexane; toluene	30	–	156 ± 15	835 ± 25	475	1.76 ± 0.05	6.2	VT,VG	3(88,89)
	Decalin	135	–	125 ± 20	755 ± 35	475	1.63 ± 0.08	5.3	VG	3(88,91)
	1-Chloronaphthalene	74	–	182	880	475	1.85	6.85	VT	90
	Cyclohexanone	92	–	172	870	475	1.83	6.7	VT	90
	Diphenyl ether	153	–	120	765	475	1.61	5.2	VT	90
	Diphenyl	129	–	128.3	820	–	–	–	VT	708
	Isoamyl acetate	34	–	168.5	900	–	–	–	VT	708
	Isobutyl acetate	58	–	158.5	880	–	–	–	VT	708
	Phenyl ether	146	–	125.6	820	–	–	–	VT	708
	Above 4 solvents	–	$d\ln r_0^2/dT = 1.8 \times 10^{-3}$ (deg^{-1})			–	–	–	VT	708
isotactic	1-Chloronaphthalene; decalin; tetralin	~140	–	120 ± 20	765 ± 40	475	1.61 ± 0.08	5.2	VG	3(88,96)
	Diphenyl ether	145	–	132	790	475	1.66	5.5	VT	90
		145	–	94	710	475	1.49	4.45	VT	630
		145	370 ± 30	–	685 ± 30	475	1.44 ± 0.07	4.15	LT	630
	Biphenyl	125.1	–	152	809	475	1.70	5.8	VT	94
	Diphenyl ether	142.8	–	137	782	475	1.62	5.25	VT	94
	Dibenzyl ether	183.2	–	106	718	475	1.51	4.55	VT	94
	Diphenyl	125	–	141.0	850	–	–	–	VT	708
	Phenyl ether	143	–	130.4	830	–	–	–	VT	708
	Above 2 solvents	–	$d\ln r_0^2/dT = 3 \times 10^{-3}$ (deg^{-1})			–	–	–	VT	708
syndiotactic	Heptane	30	–	164	830	475	1.75	6.1	VG	100
	Isoamyl acetate	45	–	172	843	475	1.77	6.25	VT	100
head-to-head										
94%-*trans*, 6%-1,2	Cyclohexane	30	–	80	674	–	1.42	4.0	VG	709
89%-*trans*, 11%-1,2	Cyclohexane	30	–	70	646	–	1.36	3.7	VG	709
Poly(vinylethylene)										
atactic	2-Octanol	32.8	–	120	–	–	–	7.0	VT	896
6.3. POLY(ACRYLIC ACID) AND DERIVATIVES										
Poly(acrylamide)	Water	30	–	260 ± 40	100 ± 50	367	2.72 ± 0.10	14.8	VG	3(101)
		30	–	–	862	–	2.36	11.14	VG	710
Poly(acrylic acid)	1,4-Dioxane	30	–	76	665	363	1.83	6.7	VT	104
50% neutralization	Aq. NaCl $C_s = 0.02$ mol/l	30	–	95	692	–	–	–	VG	868
	0.10 mol/l	30	–	121	750	–	–	–	VG	868
	0.50 mol/l	30	–	114	735	–	–	–	VG	868
100%-neutralization	Aq. NaCl, $C_s = 0.02$ mol/l	30	–	112	730	–	–	–	VG	868
	0.10 mol/l	30	–	121	750	–	–	–	VG	868
	0.50 mol/l	30	–	133	774	–	–	–	VG	868
	1.50 mol/l	30	–	120	750	–	–	–	VG	868
–, sodium salt	Aq. NaBr (1.5 M)	15	–	124	756	318	2.38	11.3	VT	108,109
		15	–	–	1030	318	3.24	21	LD	108
	Aq. NaSCN (1.25 M)	30	–	121	752	318	2.36	11.1	VT	111
Poly(acrylonitrile)	Dimethylformamide	25	–	210 ± 15	930 ± 20	422	2.20 ± 0.05	9.7	VG	3(138,139)
(polymd. at – 30°C)	γ-Butyrolactone;	30	–	250	970	422	2.30	10.6	VG	135
	dimethyl formamide	30	–	250	900	422	2.13	9.1	VG	135
(polymd. at 60°C)										
	Ethylene carbonate/water (85/15, w/w)	25	–	–	950	–	2.25	10.1	Various	951
	Dimethylformamide/ [illegible]	25	–	–	1060	–	2.51	12.6	Various	951

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TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
	55 wt.% HNO_3	25	–	–	1060	–	2.51	12.6	Various	951
	67 wt.% HNO_3	25	–	–	1290	–	3.06	18.5	Various	951
Poly(*N*-acryloyl-*m*-aminobenzoic acid)										
	N,*N*-dimethylacetamide, *N*,*N*-dimethylformamide, Dimethyl sulfoxide	30	–	85	670	223	3.0	18.1	VG	908
Poly(*N*-acryloyl-*o*-aminobenzoic acid)										
	N,*N*-dimethylformamide, Dimethyl sulfoxide, Tetrahydrofran	25	–	85	660	223	2.96	18.3	VG	909
Poly(butyl acrylate)	Undiluted	60	$d\ln r_0^2/dT = -0.2 \times 10^{-3}$ (deg^{-1})						ST	635
		76	$d\ln r_0^2/dT = 0$ (deg^{-1})						ST	634
Poly(*sec*-butyl acrylate)	Undiluted	60	$d\ln r_0^2/dT = -0.2 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(*tert*-butyl acrylate)	Undiluted	60	$d\ln r_0^2/dT = -0.2 \times 10^{-3}$ (deg^{-1})						ST	635
	Acetone; butanone; methanol; pentane	25	–	47 ± 1	607 ± 3	–	$2.2 \sim 2.3$	–	VG	712
Poly(*N*,*N'*-dimethylacrylamide)	Methanol; water	25	–	78 ± 15	670 ± 40	309	2.17 ± 0.14	9.15	VG	3(103)
Poly(dodecyl acrylate)	Undiluted	60	$d\ln r_0^2/dT = 1.0 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(ethyl acrylate)	Acetone; methanol	30	–	90 ± 10	720 ± 30	308	2.34 ± 0.10	10.9	VG	3(115,116)
	Acetone	25	–	–	856	308	2.78	15.4	VG	114
	Undiluted	60	$d\ln r_0^2/dT = -0.2 \times 10^{-3}$ (deg^{-1})						ST	635
		76	$d\ln r_0^2/dT = -0.4 \times 10^{-3}$ (deg^{-1})						ST	634
Poly(hexyl acrylate)	Undiluted	60	$d\ln r_0^2/dT = -0.3 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(isopentyl acrylate)	Undiluted	60	$d\ln r_0^2/dT = -0.2 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(isopropyl acrylate)	Benzene	25	–	–	540 ± 25	287	1.88 ± 0.08	7.1	VA	120
	Bromobenzene	60	–	–	540 ± 25	287	1.88 ± 0.08	7.1	VA	120
	2,2,3,3-Tetrafluoro-propanol	25	–	92	700 ± 30	287	2.42 ± 0.10	11.7	VG	121
	Undiluted	60	$d\ln r_0^2/dT = -0.3 \times 10^{-3}$ (deg^{-1})						ST	635
isotactic	Bromobenzene	60	–	–	630 ± 30	287	2.20 ± 0.10	9.7	VA	120
syndiotactic	Bromobenzene	60	–	–	546 ± 35	287	1.90 ± 0.12	7.2	VA	120
Poly(methyl acrylate)	Various solvents	30	–	81 ± 10	680 ± 30	332	2.05 ± 0.10	8.4	VG	3(123,132)
	Isopentyl acetate	62.5	–	68	650	332	1.96	7.7	VT	129
	2-Methylcyclohexanol	56.0	–	68	650	332	1.96	7.7	VT	129
	Butanone/2-propanol (42/58, v/v)	20	–	81	680	332	2.05	8.4	VT	124
	(50/50, v/v)	30	–	72	665	332	2.00	8.0	VT	129
		27.5	–	–	720	332	2.17	9.4	LD	108
	Undiluted	60	$d\ln r_0^2/dT = -0.2 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(octyl acrylate)	Undiluted	60	$d\ln r_0^2/dT = -0.2 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(morpholinocarbonylethylene)										
	Dimethylformamide	25	–	70 ± 10	630 ± 40	260	2.42 ± 0.15	11.7	VG	3(338)
Poly(piperidinocarbonylethylene)										
	Dimethylformamide	25	–	58 ± 10	600 ± 40	261	2.30 ± 0.15	10.6	VG	3(338)
Poly(propyl acrylate)	Undiluted	60	$d\ln r_0^2/dT = -0.3 \times 10^{-3}$ (deg^{-1})						ST	635
6.4. POLY(α-SUBSTITUTED ACRYLIC ACID) AND DERIVATIVES										
Poly(butyl methacrylate)	Butanone; 2-propanol	23	–	34 ± 5	510 ± 20	258	1.98 ± 0.10	7.85	VT,VG	3(152) (154,156)
	Dimethylformamide	23.6	309	–	659	–	–	–	LT	910
	2-Propanol	23	–	–	530	258	2.06	8.5	LD	152
	Undiluted	60	$d\ln r_0^2/dT = 2.5 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(*sec*-butyl methacrylate)	Undiluted	60	$d\ln r_0^2/dT = -0.2 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(*tert*-butyl methacrylate)	Butyl acetate	25	–	–	894	258	3.45	23.8	VA	157
anionic	Cyclohexane	10.0	–	60.8	–	–	–	11.8	VG	917
free radical	Cyclohexane	10.0	–	48.9	–	–	–	10.2	VG	917
	n-Heptane	64.0	–	48.6	–	–	–	10.2	VG	917
	Butanone, cyclohexane, *n*-Heptane, Tetrahydrofuran	25–64	$d\ln r_0^2/dT = 0$ (deg^{-1})			–	–	–	VTe	917
Poly(*tert*-butylphenyl methacrylate)										
	Acetone	20	–	35 ± 5	515 ± 20	208	2.48 ± 0.10		VG	3(340)

TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_θ ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
Poly(2-*tert*-butylphenyl methacrylate)										
	Benzene; butanone	25	–	35	550	–	2.6	–	VG	771
	Cyclohexane	18.4	–	35	550	–	2.6	–	VG	771
Poly(4-*tert*-butylphenyl methacrylate)										
	Acetone; cyclohexane; tetrahydrofuran	25	–	45	565	–	–	–	VG	716
Poly(2-chloroethyl methacrylate)	*o*-Dichlorobenzene	35.7	–	47.5	–	–	–	10.5	VT	914
Poly(cyclobutyl methacrylate)	1-Butanol	37.5	–	47.7	–	–	–	10.0	VG	915
Poly(cyclohexyl methacrylate)	Butanol	23	–	34	510	237	2.15	9.25	VT	345
	Benzene; butanol; butanone; cyclohexane; dioxane	25	–	45	590	240	2.5	–	VG	717
Poly(cyclododecyl methacrylate)	Cyclohexane,toluene	30	–	33.0	–	–	–	14.2	VG	915
	Butanol	22.5	224	45.2	549	236	2.33	–	VT	718
	n-Hexyl acetate	35	–	33.0	–	–	–	14.2	VG	915
		21–27	228	–	558	236	2.37	–	LG	718
Poly(cyclohexyl thiolmethacrylate)										
	Cyclohexane	25	–	30.0	490	230	2.13	–	VG	719
	Tetrahydrofuran	35	–	30.0	490	230	2.13	–	VG	719
Poly(cyclooctyl methacrylate)	2-Butanol	45	–	37.7	–	–	–	12.1	VG	915
	1,4-Dioxane, toluene	30	–	37.7	–	–	–	12.1	VG	915
Poly(cyclopentyl methacrylate)	Cyclohexane	36	–	47.8	–	–	–	11.2	VG	915
	1,4-Dioxane,ethyl acetate, methyl ethyl ketone,									
	Toluene	30	–	47.8	–	–	–	11.2	VG	915
Poly(decahydro-β-naphthyl methacrylate)										
	–	–	–	43	600	210	2.9	–	–	717
Poly(decyl methacrylate)	Ethyl acetate	11	–	35.7	–	–	–	13.0	VT	913
Poly(2,6-dimethylphenyl methacrylate)										
	Chlorobenzene, tetrahydrofuran	25	–	77	675	223	3.02	18.3	VG	923
	Tetrahydrofuran/water, toluene	25	–	77	675	223	3.02	18.3	VT	923
Poly(diphenylmethyl methacrylate)										
	3-Heptanone	45	–	33.1	510	–	–	14.0	VG	911
Poly(dodecyl methacrylate)	Isopropyl acetate	13	–	32.2	490	193	2.54	12.9	VT	158
	Pentanol	29.5	–	34.8	500	193	2.59	13.4	VT	159
		29.5	222	–	500	193	2.59	13.4	LD	159
	Undiluted	60	$d\ln r_0^2/dT = 2.6 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(docosyl methacrylate)	Amyl acetate, tetrahydrofuran	31,30	–	37.9	–	–	–	23.6	VG	913
Poly(2-ethylbutyl methacrylate)	Butanone; 2-propanol	25	–	36 ± 5	510 ± 30	236	2.16 ± 0.13	9.3	VG,VT	3(160)
	2-Propanol	27.4	–	33.7	500	236	2.12	8.95	VT	160
Poly(ethyl methacrylate)	Butanone	23	–	49.3	565 ± 15	288	1.96 ± 0.05	7.7	VG	3(161)
	2-Propanol	36.5	–	47.5	575	288	2.00	8.0	VT	156
	Butanone/2-propanol (1/7, v/v)	23	–	47.3	560	288	1.94	7.55	VT	161
		23	–	–	560	288	1.94	7.55	LD	161
Poly(hexadecyl methacrylate)	Heptane	21	–	60	620	175	3.54	25.1	VG	163
Poly(hexyl methacrylate)	Butanone	30	–	41 ± 4	530 ± 20	236	2.25 ± 0.08	10.1	VT,VG	3(165)
	2-Propanol	32.6	–	43	540	236	2.29	10.5	VT	165
		32.6	–	–	580	236	2.46	12.1	LD	165
	Undiluted	60	$d\ln r_0^2/dT = 2.2 \times 10^{-3}$ (deg^{-1})						ST	635
Poly[*N*-(2-hydroxypropyl) methacrylamide]										
	Aq. NaCl (10%); water	50	–	52	590	–	–	–	VG	722
	Ethanol	10	–	52	590	–	–	–	VG	722
	Dimethylformamide	25	–	52	590	–	–	–	VG	722
Poly(isobutyl methacrylate)	2-Hydroxymethyl-tetrahydrofuran	19.5	–	26.5	501	–	–	–	VT	724
	Acetone	25	–	89.8	290	–	–	–	VG	919
	Benzene	25	–	113	313	–	–	–	VG	919

TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
	Carbon tetrachloride	25	–	105	305	–	–	–	VG	919
	1,4-Dioxane	25	–	98.4	299	–	–	–	VG	919
Poly(isopentyl methacrylate)	Undiluted	60	$d\ln r_0^2/dT = 1.4 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(isopropyl methacrylate)	Undiluted	60	$d\ln r_0^2/dT = 2.5 \times 10^{-3}$ (deg^{-1})						ST	635
Poly(5-*p*-menthyl methacrylate)	Benzene; cyclohexane; 2-pentanone; tetrahydrofuran	25	–	44.0	630	–	3.1	–	VT,VG	771
Poly(methacrylic acid)	Aq. NaCl	25	–	200	900	334	2.7	14.6	VG	110
	N,N-Dimethylformamide/ 1,4-Dioxane(5/7, v/v)	26.9 (θ)	284	103	–	–	2.13	–	LT,VT	907
Poly(methyl butacrylate)	Butanol	13	–	57	590	258	2.28	10.4	VT	168
Poly(methyl ethacrylate)	2,6-Dimethyl-4-heptanone	11.4	–	67.6	620	288	2.15	9.25	VT	168
Poly((methyl methacrylate) atactic	Various solvents	25	–	70 ± 20	640 ± 60	308	2.08 ± 0.20	8.65	VT,VG	3(170, 173, 174,181, 193,196)
	Acetonitrile	45	–	49 ± 5	555 ± 15	308	1.80 ± 0.05	6.5	VT	179
	Butyl chloride	35.4	219	–	537	308	1.74	6.05	LT	636
		40.8	292 ± 6	–	620 ± 15	308	2.01 ± 0.05	8.1	LT	283
	Benzene, toluene	21	0.72 ± 0.05	–	653 ± 25	308	2.12 ± 0.08	9.0	XS	637
	Benzene; chloroform; etc. (various)	Various	–	46 ± 9	–	–	–	–	VG	869
	2-Methyl-4-pentanone	−42	–	36.0	500	308	1.62	5.25	VT	636
	Methyl isovaleriate	−37	–	41.5	525	308	1.70	5.8	VT	636
	Butyl acetate	−20	–	40.6	520	308	1.69	5.7	VT	636
	Butanone/2-propanol (58.2/41.8, v/v)	4.0	–	47.2	550	308	1.78	6.35	VT	636
	(55/45, v/v)	12.8	–	49.8	560	308	1.82	6.65	VT	636
	(50/50, v/v)	22.8	–	50.4	610	308	1.98	7.85	VT	636
	(46.8/53.2, v/v)	28.5	–	50.8	610	308	1.99	7.95	VT	636
	Butyl chloride	35.4	–	52.6	620	308	2.00	8.0	VT	636
	4-Heptanone	40.4	–	53.2	620	308	2.01	8.1	VT	636
	Isoamyl acetate	57.5	–	53.5	620	308	2.01	8.1	VT	636
	4-Heptanone	33	–	47 ± 4	550 ± 15	308	1.78 ± 0.05	6.35	VT	179
	3-Octanone	72	–	50 ± 3	560 ± 10	308	1.82 ± 0.03	6.65	VT	179
	Undiluted	168	$d\ln r_0^2/dT = 0.1 \times 10^{-3}$ (deg^{-1})						ST	634
isotactic	Acetonitrile	27.6	–	75.5	670	308	2.17	9.4	VT	198
		27.5	273	76.2	669	–	–	–	VT	728
	Butanone/2-propanol (50/50, v/v)	30.3	–	90	715	308	2.32	10.8	VT	191
	3-Heptanone	40	–	87	710	308	2.30	10.6	VT	191
	Propanol	75.9	–	76.1	680	308	2.21	9.75	VT	191
	p-Cymene	152.1	–	56.6	610	308	1.98	7.85	VT	191
Poly(methyl methacrylate)-*block*-poly(α-methylstyrene)-*block*-poly(methylmethacrylate)										
0.0 wt.% PMS	Chloroform, toluene, tetrahydrofuran, carbon tetrachloride	25, 45	660	–	–	–	–	2.08	VA	927
6.7 wt.% PMS	*ibid.*	25,45	600–690	–	–	–	–	1.98–2.28	VA	927
8.3 wt.% PMS	*ibid.*	25,45	670–770	–	–	–	–	2.24–2.54	VA	927
21.0 wt.% PMS	*ibid.*	25,45	630–720	–	–	–	–	2.17–2.45	VA	927
31.8 wt.% PMS	*ibid.*	25,45	680–780	–	–	–	–	2.27–2.82	VA	927
78.1 wt.% PMS	*ibid.*	25,45	710–810	–	–	–	–	2.48–2.80	VA	927
100 wt.% PMS	*ibid.*	25,45	653	–	–	–	–	2.30	VA	927
Poly(methyl-phenylacrylate), (see group 1.7 Poly(methoxycarbonyl-1-phenylethylene)										
Poly(β-naphthyl methacrylate)	Benzene	Various	–	47.2	574	–	–	–	VG	729
	Tetralin	Various	–	47.5	575	–	–	–	VG	729
	Toluene	Various	–	39.0	538	–	–	–	VG	729
	–	–	–	54	660	210	3.1	–	–	717
Poly(octadecyl methacrylate)	Butyl acetone	10.5,20,30	–	–	–	–	–	14.5	VG	920
Poly(octyl methacrylate)	Butanol; butanone	20	–	30 ± 5	480 ± 20	219	2.19 ± 0.09	9.6	VT, VG	3(201)
	Butanol	16.8	–	–	500	219	2.28	10.4	LD	201
	Undiluted	60	$d\ln r_0^2/dT = 2.2 \times 10^{-3}$ (deg^{-1})						ST	635

TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
Poly(*N*-phenylmethacrylamide)	Acetone	20	–	38 ± 9	520 ± 40	242	2.15 ± 0.16	9.25	VG	3(370)
Poly(phenyl methacrylate)	Acetone; benzene; butanone; dioxane	25	–	57	670	240	2.8	–	VG	717
Poly(2-selenolylmethyl methacrylate)										
	Chlorobenzene	25	–	56	580	204	2.84	–	VG	922
	Tetrahydrofuran	25	–	65	607	204	2.98	18.0	VG	922
Poly(stearyl methacrylate)	*n*-Propyl acetate	36	–	38.2	–	–	–	20.4	VT	913
Poly(tetrahydrofurfuryl)methacrylate)										
	2-Hydroxymethyl-tetrahydrofuran	31.2	–	34.9	573	–	–	–	VT	734
Poly(2-thiophenmethyl methacrylate)										
	Chlorobenzene	25	–	45.0	594	228	2.45	10.8	VT	921
	Tetrahydrofuran, thiophene	25	–	45.0	594	228	2.45	10.8	VG	921
Poly(tridecyl methacrylate)	Ethyl acetate	27	–	33.3	–	–	–	14.7	VT	913
Poly[2-(triphenylmethoxy)ethyl methacrylate]										
	Mesitylene	47	–	30	487	–	–	18–19	VT	736
6.5. POLY(VINYL ETHERS), POLY(VINYL ALCOHOL), POLY(VINYL ESTERS), POLY(VINYL HALIDES)										
Poly(chlorotrifluoroethylene)	2,5-Dichlorobenzo-trifluoride	130	–	52 ± 3	580 ± 15	286	2.03 ± 0.07	8.25	VG	3(234)
Poly(methoxyethylene)	Benzene; butanone	30	–	195 ± 30	900 ± 50	404	2.23 ± 0.13	9.95	VG	3(206)
Poly(tetrafluoroethylene)	Mixed digomers of poly-(chlorotrifluoroethylene)	340	330 ± 50	In this poor solvent				7 ± 2	LT	885,886
	Perfluorokerosene	300	–	~300	1070	308	~3.5	~24	VG	690
Poly(vinyl acetate)	Various solvents	25	–	93 ± 10	705 ± 10	332	2.12 ± 0.09	9.0	VT, VG	3(242, 244, 252,255)
	3-Heptanone	29	0.95 ± 0.05	–	790 ± 20	332	2.38 ± 0.07	11.3	XS	641
	Heptane/3-methyl-2-butanone (26.8/73.2, v/v)	25	318 ± 10	–	745 ± 20	332	2.24 ± 0.07	10.0	LT	252
	Methanol	6	–	101	720	332	2.17	9.4	VT	236
	3-Heptanone	26.8	–	82.0	670	332	2.02	8.15	VT	236
	Ethanol	56.9	–	90	690	332	2.08	8.65	VT	236
	6-Methyl-3-heptanone	66	–	78	660	332	1.99	7.9	VT	236
	Tetrahydrofuran	35	–	116 ± 9	774 ± 20	–	–	–	VG	740
Poly(vinyl alcohol)	Water	30	–	222 ± 25	950 ± 40	464	2.04 ± 0.10	8.3	VG	3(208,210, 212)
	Undiluted	50	$d \ln r_0^2/dT = 0.0$ (deg^{-1})						ST	638
		90	$d \ln r_0^2/dT = 0.5 \times 10^{-3}$ (deg^{-1})						ST	639
Poly(vinyl benzoate)	Xylene	32.5	–	62 ± 8	620 ± 25	252	2.46 ± 0.10	12.1	VT	334
Poly(vinyl bromide)	Cyclohexane; tetrahydro-furan; methanol/THF (17/83, v/v)	20	–	40 ± 5	540 ± 20	298	1.82 ± 0.07	6.6	VT, VG	3(217, 218)
	1-Methylnaphthalene	~20	1.09	–	763	298	2.56	13.1	XS	640
Poly(vinyl butyrate)	Benzene	30	–	80 ± 10	670 ± 35	288	2.32 ± 0.12		VG	3(256)
Poly(vinyl chloride)	Cyclohexanone; tetrahydrofuran	~25	–	100 ± 30	393		1.83 ± 0.15	6.7	VG	3(221,223, 224,229)
	Benzyl alcohol	155.4	–	156	820	393	2.08	8.65	VT	219
	Tetrahydrofuran	25	–	103	717	390	1.84	–	VG	879
Poly(vinyl 4-chlorobenzoate)	Butanol/butanone (47/53, v/v)	60	–	73	665	228	2.92	17.1	VT	336
Poly(vinyl fluoride)	Dimethylformamide	90	–	128	787	457	1.72	5.9	VG	235
Poly(vinyl hexanoate)	Benzene	30	–	91 ± 10	700 ± 30	258	2.71 ± 0.12		VG	3(256)
Poly(vinyl isobutyrate)	Benzene	30	–	80 ± 10	670 ± 35	288	2.32 ± 0.12		VG	3(256)
Poly(vinyl methyl ether), see Poly(methoxyethylene)										
Poly(vinyl pivalate)	Butanone/methanol (0.897 g/ml)	20	–	53 ± 5	580 ± 20	253	2.29 ± 0.08		VT	258
Poly(vinylidene chloride)	1-Methyl-2-pyrrolidone; Tetramethylene sulfoxide	25	–	60 ± 10	620 ± 30	–	–	8 ± 1	VG	738

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TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
Poly(vinylidene fluoride)	*N,N*-dimethylacetamide, *N,N*-dimethylformamide, *N*-methylpyrrolidone, *N,N*-dimethy-*N,N*-trimethylene urea	25	330	–	–	–	–	8.9	LV	925
6.6. POLY(STYRENE) AND DERIVATIVES										
Poly(4-acetoxystyrene)	Butyl acetate,	26.8								
	Isopropyl acetate,	19.7	–	54	593	–	2.37	11.2	VT	937
	Dioxane, tetrahydrofuran	25	–	54	593	–	2.37	11.2	VG	937
Poly(4-bromostyrene)	Benzene	26.3	–	50	570	228	2.50	12.5	VT	348
		30	–	28.5	–	–	2.05	–	VG	870
	Chloroform	30	–	46.0	–	–	2.41	–	VG	870
	Dimethylformamide	30	–	44.0	–	–	2.37	–	VG	870
	Dioxane	30	–	30.0	–	–	2.09	–	VG	870
	Toluene	30	–	45	564	228	2.48	12.3	VG	349
		30	237	–	554	228	2.43	11.8	LG	349
Poly(*p*-*tert*-butylstyrene)	Benzene; cyclohexane	35	–	62	612	–	–	–	VG	741
	1-Nitropropane	31	255	61 ± 5	610 ± 15	–	–	13.4 ± 0.7	VT	741
		31	286	–	–	–	–	–	LT	741
	Benzene	35	–	58.2	–	–	–	–	VG	934
	Cyclohexane	25	–	57.7	–	–	–	–	VG	934
	1-Hexanol	65.0	–	51.1	–	–	–	11.9	VG	934
	3-Nonanol	10.9	–	59.5	–	–	–	13.1	VG	934
	2-Octanol	32.7	–	55.7	–	–	–	12.6	VG	934
	1-Nitropropane	31.0								
	2-Octanol	32.7	–	54.2	–	–	–	12.3	VG	938
Poly(2-chlorostyrene)	Butanone	24.5	237	46.8	580	–	2.22	–	VT	742
		24.5	236	–	580	–	2.22	–	LT	742
Poly(4-chlorostyrene)	Butanone; chlorobenzene; toluene	30	–	50 ± 5	560 ± 20	261	2.15 ± 0.07	9.25	VG	3(352,355), 353
	Benzene; benzene/methanol (5.5/1, v/v)	26.7	–	59	590	–	2.25	–	VG	744
	Toluene	30	–	58	615	261	2.36	11.1	VG	349
		30	272	–	615	261	2.36	11.1	LG	349
Poly(4-cyclohexylstyrene)	Heptane; toluene	30	–	53 ± 3	570 ± 20	226	2.52 ± 0.07	12.7	VG	266
	Toluene	60	–	51 ± 3	560 ± 20	226	2.48 ± 0.07	12.2	VG	266
Poly(2,5-dichlorostyrene)	Ethanol/ethyl acetate (1/15, w/w)	30.5	–	35.5	510	234	2.18	9.5	VT	3(357)
Poly3,4-dichlorostyrene)	Butanol/butyl acetate (1/13, w/w)	32.9	–	71	640	234	2.7	14.6	VT	359
	Chlorobenzene; *o*-dichlorobenzene	30	–	38 ± 5	510 ± 20	234	2.18 ± 0.08	9.5	VG	358
Poly(2,4-dimethylstyrene)	Cyclohexane; 1,2-dichloroethane; dioxane	20	–	59–60	635–640	–	–	VG	746	
	n-Decane; *n*-heptane	50	–	48–49	595–600	–	–	–	VG	746
	p-Dichlorobenzene	60	–	48.5	595	–	–	–	VG	746
	Toluene	30	–	60 ± 5	630 ± 15	268	2.35 ± 0.07	11.0	VG	333
Poly(4-hydroxystyrene)	Dioxane	25	–	82.7	–	–	2.38	11.3	VG	940
	Ethyl propionate	25	–	90.0	–	–	2.45	12.0	VG	940
	Isobutyl acetate	25	–	90.0	–	–	2.45	12.0	VG	940
	Tetrahydrofuran	25	–	83.7	–	–	2.39	11.4	VG	940
Poly(*o*-methoxystyrene)	Methanol/toluene (25/75, v/v)	30	–	57.5	600	266	2.26	10.2	VT	362
Poly(*p*-methoxystyrene)	Amyl acetate	75.0	–	69	635 ± 15	–	2.40 ± 0.06	–	VT	871
	Amyl acetate; cyclohexane; toluene	23.4	–	71	640	–	–	–	VG	363
	Butanone; chlorocyclohexane; toluene	20, 50, 75	–	66–73	–	–	–	–	VG	782
	Methanol/toluene (28.1/71.9, v/v)	30	–	62.1	630	266	2.37	11.2	VT	362

TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
	4-Methyl-2-pentanone	23.4	–	64	620 ± 15	–	2.33 ± 0.07	–	VT	871
	2-Methyl-2-phenylpropane	52.2	–	74	650 ± 20	–	2.45 ± 0.06	–	VT	871
	Toluene	25	–	74	650 ± 20	–	2.45 ± 0.06	–	VT	871
Poly(methyl methacrylate)-*block*-poly(α-methylstyrene)-*block*-	poly(methyl methacrylate), see also group 1.4									
Poly(α-methylstyrene)	1-Chloro-*n*-hexane	5.7	–	66.0	–	–	–	10.4	VG	888
		10.0	–	65.2	–	–	–	10.3	VG	888
	1-Chloro-*n*-heptane	20.0	–	62.5	–	–	–	10.0	VG	888
		27.0	–	62.7	–	–	–	10.0	VG	888
	1-Chloro-*n*-octane	43.0	–	63.8	–	–	–	10.1	VG	888
		53.0	–	63.1	–	–	–	10.1	VG	888
		80.0	–	63.8	–	–	–	10.1	VG	888
	n-Hexyl acetate	85.0	–	66.5	–	–	–	10.4	VG	888
	Cyclohexane	34.5	–	68.9	–	–	–	10.7	VG	888
	trans-Decalin	10.0	–	68.0	–	–	–	10.6	VG	888
anionic (atactic)	Benzene; cyclohexane	~30	–	76 ± 5	650 ± 15	284	2.29 ± 0.05	10.5	VT, VG	319,323
	trans-Decalin	9.5	–	67 ± 2	625 ± 5	284	2.20 ± 0.02	9.7	VT	320
	Cyclohexane	~38	–	76 ± 2	650 ± 10	284	2.29 ± 0.03	10.5	VT	320,321 322,323
ca. 50%-hetero, ca. 40%-syndio		34.5	290	–	710	–	–	–	LT	872
		34.5	–	73	645	–	–	–	VT	872
		34.9	263	–	645	–	–	–	LT	873
		34.9	–	74.2	650	–	–	–	VT	873
	trans-Decalin (100%)	9.5	283	–	690	–	–	–	LT	872
		9.5	–	67	630	–	–	–	VT	872
		9.6	250	–	615	–	–	–	LT	873
		9.6	–	62.5	615	–	–	–	VT	873
cationic (syndiotactic)	Toluene; benzene/methanol (79.4/20.6, v/v)	30	–	74 ± 10	670 ± 25	284	2.36 ± 0.10	11.1	VT, VG	3(329, 327)
	Cyclohexane	~33	–	69 ± 3	655 ± 10	284	2.31 ± 0.03	10.7	VT	328
Poly(*m*-methylstyrene)	Benzene; cyclohexane; ethyl acetate	30	–	84.0	664	284	2.34	11.0	VG	330
		40	–	86.8	671	284	2.37	11.2	VG	330
		50	–	89.7	678	284	2.39	11.4	VG	330
Poly(*p*-methylstyrene)	Butanone; cyclohexane; toluene	30	–	68 ± 5	620 ± 15	284	2.18 ± 0.05	9.5	VG	651,331
	Diethyl succinate	16.4	–	70	655	284	2.31	10.7	VT	331
		16.4	291	–	680	284	2.39	11.4	LT	331
Poly(styrene)										
atactic	Various solvents	~30	–	82 ± 5	670 ± 15	302	2.22 ± 0.05	9.85	VT, VG	3(274, 277, 279,282 286,299, 301,304)
	n-Butyl chloride	25, 50	–	77	–	–	–	–	VG	933
	Cyclohexane	34.5	297 ± 3	–	–	–	–	–	LT	875
	trans-Decalin	16–80	–	81.4	–	–	–	–	Monte Carlo, G	876
	Ethylcyclohexane; methylcyclohexane	~70	–	75 ± 5	650 ± 15	302	2.15 ± 0.05	9.25		3(292,293)
	Cyclohexane	34	282 ± 5	–	690 ± 10	302	2.28 ± 0.04	10.4	LT	643
		35	306	–	730	302	2.42	11.7	LT	278
		35	300	–	670	302	2.22	9.85	LT	644
		35	0.92 ± 0.03	–	705 ± 15	302	2.33 ± 0.05	10.9	XS	645
	Benzene; toluene	25	0.91 ± 0.02	–	700 ± 15	302	2.32 ± 0.04	10.8	XS	645
	Benzene/ethanol (71.5/28.5, v/v)	25	296	–	645	302	2.14	9.15	LT	628
	Butanone/2-propanol (87/13, v/v)	67	317	–	757	302	2.50	12.5	LT	278
	1-Chloroundecane	32.8	–	–	775	302	2.56	13.1	VA	646
	Cyclohexane	34.8	–	–	768	302	2.54	12.9	VA	646
	Dimethyl malonate	35.9	–	–	762	302	2.52	12.7	VA	646

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TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_θ ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
	73%-*trans*-Decalin	18	–	77	655	302	2.17	9.4	VT	290
	100%-*trans*-Decalin	24	–	82	670	302	2.22	9.85	VT	290
	Butyl formate	−9	–	77.4	655	302	2.17	9.4	VT	647
	Hexyl-*m*-xylol	12.5	–	77.0	655	302	2.17	9.4	VT	647
	Decalin	29.5	–	77.9	655	302	2.17	9.4	VT	647
	Diethyl malonate	31	–	70.5	635	302	2.10	8.8	VT	647
	Cyclohexane	34	–	79.5	660	302	2.18	9.5	VT	647
	Diethyl oxalate	51.5	–	72.2	640	302	2.12	9.0	VT	647
	Methylcyclohexane	68	–	78.0	655	302	2.17	9.4	VT	647
	Cyclohexanol	83.5	–	50.8	575	302	1.90	7.2	VT	647
	1-Chlorodecane	6.6	–	78.0	655	302	2.17	9.4	VT	648
	1-Chloroundecane	32.8	–	78.7	660	302	2.18	9.5	VT	648
	1-Chlorododecane	58.6	–	80.7	665	302	2.20	9.7	VT	648
	Cyclohexane/methylcyclohexane									
	(1/0, v/v)	34.5	–	77.9	655	302	2.17	9.4	VT	291
	(2/1, v/v)	43.0	–	77.6	655	302	2.17	9.4	VT	291
	(1/1, v/v)	48.0	–	74.8	650	302	2.15	9.25	VT	291
	(1/2, v/v)	54.0	–	73.0	645	302	2.14	9.15	VT	291
	(0/1, v/v)	70.5	–	69.6	635	302	2.10	8.8	VT	291
	Diethyl malonate	34.2	–	71.8	640	302	2.12	9.0	VT	291
	Diethyl oxalate	55.8	–	73.0	645	302	2.14	9.15	VT	291
	Undiluted	150	$d \ln r_0^2/dT = 0.4 \times 10^{-3}$ (deg^{-1})						ST	634
atactic, anionic	Benzene/isopropyl alcohol									
	(64/36, v/v)	25.5	312	–	764	–	–	–	LT	874
		25.5	–	107	–	–	–	–	VT	874
	Cyclohexane	34.5	288	–	706	–	–	–	LV	877
		34.5	–	89.7	–	–	–	–	VT	877
		34.5	280	–	685	–	–	–	LT	878
		~34.5	–	88 ± 3	685 ± 10	302	2.27 ± 0.03	10.3	VT	301,302. 303,305,649
	Decalin; dioctyl phthalate	12–22	–	80 ± 1	665 ± 5	302	2.20 ± 0.02	9.7	VT	303,649
isotactic	Benzene; toluene	30	–	90 ± 10	695 ± 25	302	2.30 ± 0.08	10.5	VG	3(311,312, 314)
	Chlorobenzene	25.3	–	176	890	302	2.94	17.3	VG	650
head-to-head	Tetrahydrofuran	25	–	135	805	–	–	–	VG	756
head-to-tail	Tetrahydrofuran	25	–	78	660	–	–	–	VG	756
star type										
four branches	Cyclohexane	35	–	63.1	661	–	–	–	VT	758
		34.5	280	–	685	–	–	–	LT	878
six branches	Cyclohexane	34.5	190	–	465	–	–	–	LT	878
star type 12.3 arm	Cyclohexane	34.6	$<S^2>_{0z}/<S^2>_{0z,lin} = 0.276$		–	–	–	–	LT	929
16.8 arm	Cyclohexane	34.6	$<S^2>_{0z}/<S^2>_{0z,lin} = 0.228$		–	–	–	–	LT	929
regular H-shaped	Cyclohexane	35	–	66.5	–	–	–	–	VT	759
		35	235	–	–	–	–	–	LT	759
Poly(styrene-*p*-sulfonic acid) –, sodium salt	Aq. NaCl (4.17 M)	25	–	20.4	425	214	1.98	7.85	VT	366
6.7. OTHER COMPOUNDS										
Poly(2-acrylamino-2-methyl-propanesulfonamide)										
	Water/1,4-dioxane (81.5/18.5, v/v)	25	–	172	880	–	4.0	32	VT	1011
Poly[(biphenyl-4-yl)ethylene]	Benzene	30	–	63.0	605	230	2.63	13.8	VG	264
		20–75	$d \ln r_0^2/dT = (0.23 \pm 0.01) \times 10^{-3}$ (deg^{-1})						VG	264
Poly(carbanilinoxyethylene)	Dioxane/methanol (28/72, v/v)	20	–	75	680	241	2.82	15.9	VT	335
Poly[1-(9-carbazyl)-2,3-(diethoxycarbonyl)-butane-1,4-diyl]										
	Benzene	25	–	54 ± 2	573	228	2.51	–	VG	789
Poly(dibutyl itaconate)	Toluene	25	–	35.4	508	198	2.60	–	VG	762
Poly(dicyclohexyl itaconate)	Tetrahydrofuran; toluene	25	–	44 ± 2	–	–	2.99	–	VG	761
Poly(didecyl itaconate)	Toluene	25	–	33.8	503	152	3.30	–	VG	762

TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
Poly(didodecyl itaconate)	Toluene	25	–	33.6	–	–	3.6	25	VG	945
Poly(diethyl itaconate)	Toluene	25	–	22.3	447	226	1.98	–	VG	762
Poly(diethylcyclohexyl itaconate)	Toluene	25	–	27.5	–	–	2.83	–	VG	763
Poly(dihexadecyl itaconate)	Toluene	25	–	33.5	–	–	3.9	30	VG	945
Poly(dihexyl itaconate)	Toluene	25	–	34.8	528	179	2.90	–	VG	762
Poly(diicosayl itaconate)	Toluene	25	–	26.7	–	–	3.9	31	VG	945
Poly(dimethyl itaconate)	Benzene	25	–	30.2	494	245	2.02	–	VG	762
Poly(dimethylcyclohexyl itaconate)	Toluene	25	–	27.5	–	–	2.83	–	VG	763
Poly(dioctadecyl itaconate)	Toluene	25	–	30.7	–	–	4.0	31	VG	945
Poly(diphenyl itaconate)	Toluene	25	–	28.4	–	–	2.69	14.6	VG	943
Poly(dipropyl itaconate)	Toluene	25	–	27.3	478	211	2.27	–	VG	762
Poly(dipropylcyclohexyl itaconate)	Amyl acetate	25	–	27.0	–	–	2.93	–	VG	763
	Toluene	25	–	24.8	–	–	2.84	–	VG	763
Poly(ditetradecyl itaconate)	Toluene	25	–	33.6	–	–	3.7	27	VG	945
Poly(diundecyl itaconate)	Toluene	25	–	33.9	514	147	3.50	–	VG	762
Poly(1-methoxycarbonyl-1-phenylethylene)	Benzene; chloroform	30	–	54 ± 1	585 ± 5	242	2.42 ± 0.02	11.7	VG	361
	Ethylbenzene	15	–	54 ± 1	585 ± 5	242	2.42 ± 0.02	11.7	VT	361
Poly[bis(phenylethyl)itaconate]	Toluene	25	–	28.1	–	–	2.78	14.8	VG	943
Poly[bis(phenyl-*n*-propyl)itaconate]	Toluene	25	–	29.7	–	–	2.91	15.6	VG	943
Poly(*N*-vinyl-3,6-dibromo carbazole)	*p*-Chloro-*m*-cresol	112.9	–	30.2	470	–	2.89	16.7	VT	942
	o-Chlorophenol	60.0	–	27.5	481	–	2.96	17.5	VT	942
Poly(vinylcarbazole)	Benzene; chloroform; tetrachloroethane; tetrahydrofuran	25	–	68 ± 2	619	222	2.82	15.9	VG	367
	Bromobenzene; chloroform; nitrobenzene	25	–	59.2	590	222	2.66	–	VG	765
	Chlorobenzene; nitrobenzene	30–45	–	56.8	583	–	–	–	VG	766
	1,2-Dichlorobenzene	25	–	58.5	589	–	–	–	VG	767
	1,2-Dichloroethane	25	–	59.6	592	–	–	–	VG	767
	Toluene	37	–	76.2 ± 5	633	222	2.85	16.2	VT	368
Poly(1-vinylimidazole)	Aq. NaCl (0.1 M); (5 M); aqueous NaSCN (0.1 M)	25	–	116 ± 10	774 ± 22	–	–	12–15	VT	813
protonated	Acetate (0.1 M)	25	–	414	1180	–	–	15–28	VT	813
	Acetate (0.1 M)/NaCl (1 M)	25	–	183	900	–	–	15–28	VT	813
	Aq. HCl (0.1 M)/ NaCl (1 M)	25	–	169	880	–	–	15–28	VT	813
	Propionate (0.1 M)	25	–	392	1160	–	–	15–28	VT	813
Poly(1-vinyl-3-benzyl-imidazoliumchloride)	Aq. NaCl (0.01 M); (0.02 M); (0.05 M); acetate (0.1 M); aqueous NaH_2PO_4 (0.1 M)	25	–	63	–	–	–	18	VG	813
Poly(1-vinylnaphthalene)	Benzene	30	–	24.2	435	248	1.76	6.2	VG	264
		75	–	–	405	248	1.63	5.3	VG	264
		20–75	$d \ln r_0^2/dT = -(1.87 \pm 0.04) \times 10^{-3}$ (deg^{-1})						VG	264
Poly(2-vinylnaphthalene)	Benzene	30	–	64.7	610	248	2.45	12.0	VG	264
		65	–	–	595	248	2.40	11.5	VG	264
		20–75	$d \ln r_0^2/dT = -(0.83 \pm 0.03) \times 10^{-3}$ (deg^{-1})						VG	264
	Decalin/toluene (13/10, w/w)	30.2	–	–	–	248	ca. 3.1	ca. 19.2	VT	269
Poly(3-vinylpyrene)	Chloroform; 1,2-dichlorobenzene; tetrahydrofuran	25	–	52.0	577	–	2.84	–	VG	768

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TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
Poly(2-vinylpyridine)	Various solvents	25	–	82 ± 10	660 ± 30	300	2.20 ± 0.10	9.7	VG	371
	Benzene	15	–	72	635	300	2.12	9.0	VG	652
		25	–	71	633	300	2.11	8.9	VG	652
		30	–	59	595	300	1.98	7.85	VG	652
		40	–	52	570	300	1.90	7.2	VG	652
		50	–	57	590	300	1.96	7.65	VG	652
		60	–	62	605	300	2.02	8.15	VG	652
	Benzene; chloroform; etc. (various)	Various	–	72 ± 11	660 ± 30	–	–	–	VG	869
	Chloroform	0	–	88	690	300	2.24	10.0	VG	652
		25	–	87.5	689	300	2.24	10.0	VG	652
	Benzene, 2-propanol, pyridine	10–15 25	–	81	–	300	2.3	–	VT,VG	952
Poly(2-vinylpyridine 1-oxide)	Benzyl alcohol, 1-butanol, chloroform, methanol, 1-propanol, 2-propanol	25	–	55	576	278	2.07	8.6	VG	946
Poly(4-vinylpyridine)	Ethanol; water	25	–	94 ± 10	710 ± 30	300	2.37 ± 0.10	11.2	VG	3(373,374)
Poly(5-vinyl-2-methylpyridine)	Butanone; methanol	25	–	69 ± 5	652 ± 15	282	2.31 ± 0.05	10.6	VG	375
	Butyl acetate	21.8	–	83	675	282	2.39	11.4	VT	381
	4-Methyl-2-pentanone	37.4	–	83	675	282	2.39	11.4	VT	381
	Pentyl acetate	48.2	–	80	665	282	2.36	11.1	VT	381
Poly(vinylpyrrolidone)	Water	~25	–	100 ± 15	720 ± 40	292	2.48 ± 0.12	12.3	VG	3(380,382)
	Acetone/water (66.8/33.2, v/v)	25	–	75	650	292	2.22	9.85	VT	384
	Butanone/2-propanol (94/4, v/v)	25	–	61	630	292	2.16	9.3	VT	653
	Aq. Na_2SO_4 (0.55 M); 2-propanol	25	–	58.0	650	290	2.24	–	VT,VG	770
	Chloroform	25	–	30.0	520	290	1.79	–	VG	770
	Chloroform/acetone (40/60, v/v)	25	–	66.0	680	290	2.34	–	VG	770
	Water; water/acetone (80/20, v/v)	25	–	23.0	480	290	1.66	–	VG	770
	Water/acetone (33.2/66.8, v/v)	25	–	74.0	710	290	2.45	–	VT	770
Poly(1-vinyl-2-pyrrolidone)	Methanol/3-heptanone, ethanol/3-heptanone, propyl alcohol/ 3-heptanone	25	–	61	–	–	–	–	VG	947
Poly(vinyl sulfate)	Aq. NaCl (0.5 M)	20	–	25 ± 15	460 ± 80	278	1.65 ± 0.30	5.45	VG	3(261)
Poly(vinylsulfonic acid)	Aq. KBr (0.349 M)	5.7	–	68.8	650(788)†	296	2.19(2.66)†	9.6	VT	259,642
	Aq. KCl (0.349 M)	5.5	–	68.2	650(786)†	296	2.19(2.66)†	9.6	VT	259,642
	(0.650 M)	26.0	–	79.5	685(830)†	296	2.31(2.80)†	10.6	VT	259,642
	(1.001 M)	44.5	–	80.3	690(832)†	296	2.33(2.81)†	10.8	VT	259,642
	Aq. NaBr (0.347 M)	−0.6	–	95.5	730(882)†	296	2.46(2.98)†	12.1	VT	259,642
	Aq. NaCl (1.003 M)	32.4	–	96.1	730(880)†	296	2.46(2.97)†	12.1	VT	259,642
	Aq. NaBr (1.008 M)	40.1	–	94.5	725(875)†	296	2.45(2.96)†	12.0	VT	259,642
6.8. COPOLYMERS										
Poly(acrylonitrile-*stat*-α-methylstyrene) ca. (46/54, mol/mol)	Tetrahydrofuran	–	409	–	–	–	–	–	LG	878
Poly(acrylonitrile-*co*-styrene)										
(38.3/61.7, mol/mol) azeotropic	Butanone	30	–	124	770	335	2.30 ± 0.05	10.6	VG	593,595
(62.6/37.4, mol/mol, random)	Butanone; dimethylformamide	30	–	170	840	362	2.32	10.8	VG	595
(38.5/61.5, mol/mol)	Butanone	–	354	–	–	–	–	–	LG	878
azeotropic	Tetrahydrofuran	–	390	–	–	–	–	–	LG	878

† The values of $r_0/M^{1/2}$ and σ given in parentheses were obtained by using $\Phi_0 = 1.39 \times 10^{23}$, while for the others $\Phi_0 = 2.5 \times 10^{23}$ was used.

TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
(1/1, mol/mol, alternating)	Butanone; dichloroethane; dimethylformamide	30	–	139	785	–	2.25	–	VG	774
	Butanone/methanol (63.6/36.4, v/v)	30	–	140	785	–	2.25	–	VT	774
Poly(acrylonitrile-*co*-vinylidene chloride)	γ-Butyrolactone	25	–	–	940	–	2.46	13.2	Various	961
	Dimethylacetamide	25	–	–	980	–	2.57	14.4	Various	961
58 wt.%-AN	Dimethylformamide	25	–	–	950	–	2.49	13.5	Various	961
	N-Methyl-2-pyrrolidone	25	–	–	1000	–	2.62	15.0	Various	961
	70 wt.% HNO_3	25	–	–	860	–	2.31	10.8	Various	961
Poly(butadiene-*stat*-styrene) (84/16, mol/mol)										
	2-Pentanone	23.8	–	23.8	460				VT	596
Poly(butyl itaconate-*stat*-dibutyl) itaconate) (40.5/59.5, mol/mol)										
	Acetone	25	–	83			3.2		VG	682
	Methanol/*m*-xylene									
	(100/0, v/v)	25	–	51			2.7		VG	682
	(80/20, v/v)	25	–	62			2.8		VG	682
	(65/35, v/v)	25	–	78			3.1		VG	682
	(50/50, v/v)	25	–	98			3.4		VG	682
	(30/70, v/v)	25	–	101			3.4		VG	682
	(10/90, v/v)	25	–	82			3.2		VG	682
	(0/100, v/v)	25	–	42			2.5		VG	682
Poly(*p*-chlorostyrene-*stat*-methyl methacrylate) (51.6/48.4, mol/mol)										
	Benzene/hexane (960/40, v/v)	22.3	–	64	660				VT	598
poly(dimethyl itaconate-*co*-styrene)										
(100/0, w/w)	Benzene	25	–	30.4	495	245	2.02		VT	683
(75/25, w/w)	Toluene	25	–	35.4	508	260	1.96		VT	683
(67/33, w/w)		25	–	40.3	544	266	2.04		VT	683
(59/41, w/w)		25	–	45.7	553	270	2.05		VT	683
(49/51, w/w)		25	–	55.3	590	278	2.12		VT	683
(29.5/70.5, w/w)		25	–	63.4	617	287	2.15		VT	683
(27/73, w/w)		25	–	63.7	618	288	2.15		VT	683
(0/100, w/w)		25	–	78.0	661	302	2.17		VT	683
Poly(ethyl acrylate-*stat*-methyl methacrylate) (80/20, mol/mol)										
	Acetone	25	–	–	823	308	2.67		VG	114
Poly(ethylene-*co*-α-methyl-styrene), $[(ET)_m(MS)_n]_p$										
$m/n = 3/4$	Butanone/cyclohexane (60/40, v/v)	30	–	135	820	345	2.38		VT	604
$m/n = 5/4$	Butanone/cyclohexane (75/25, v/v)	30	–	140	830	373	2.22		VT	604
$m/n = 5/7$	Cyclohexane	30	–	112	770	343	2.24		VT	604
Poly(4-methoxystyrene-*stat*-styrene)										
(24.4/75.6, mol/mol)	Toluene	25	–	81	–	–	–	–	VG	781
(26.4/73.6, mol/mol)	Butanone; chlorocyclohexane; toluene	25, 50, 75	–	95–100	–	–	–	–	VG	782
(46.2/53.8, mol/mol)	Toluene	25	–	109	–	–	–	–	VG	781
(53.0/47.0, mol/mol)	Butanone; chlorocyclohexane; toluene	25, 50, 75	–	86–109	–	–	–	–	VG	782
(74.0/26.0, mol/mol)	Toluene	25	–	98	–	–	–	–	VG	781
(75.6/24.4, mol/mol)	Butanone; chlorocyclohexane; toluene	25, 50, 75	–	75–83	–	–	–	–	VG	782
Poly(methyl acrylate-*stat*-styrene)										
(50/50, mol/mol)	Ethyl acetate	35	–	104 ± 2	1010 ± 20	314	3.22		VG	133
(22/78, mol/mol)	Various solvents	~30	–	75	650				VT,VG	129
(33–76, mol/mol)		~30	–	76	650				VT	129
(47/53, mol/mol)		~30	–	77	655				VG	129
(59/41, mol/mol)		~30	–	76	650				VT,VG	129
(76/24, mol/mol)		~30	–	75	650				VT	129
Poly(methyl methacrylate-*co*-styrene)										
(100/0, mol/mol, random)	1-Chlorobutane	40.8	–	50	583	308	1.89		VT	613
(94/6, mol/mol, random)		40.8	–	59	616	308	2.00		VG	613
(58/42, mol/mol, random)		40.8	–	95	728	305	2.39		VG	613

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TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.*
(10/90, mol/mol, random)		40.8	–	89	707	302	2.34		VG	613
(0/100, mol/mol, random)		40.8	–	80	685	302	2.27		VG	613
(71/29, mol/mol, random)	Various solvents	~30	–	66 ± 2	625 ± 95		2.05		VG,VT	614
(44/56, mol/mol, random)	Various solvents	~30	–	75 ± 2	655 ± 95		2.15		VG,VT	614
(30/70, mol/mol, random)	Various solvents	~30	–	77 ± 2	660 ± 95		2.17		VG,VT	614
three blocks (MSM) nearly equimolar	Cyclohexanol	81	–	63	617	305	2.04		VT	616
Poly(styrene)-*block*-poly(butadiene)-*block*-poly(styrene) 50 wt.%-PS										
	Toluene	30	–	–	970–986	–	–	–	VG	958
Poly(styrene-*stat*-vinylpyrrolidone)										
(87/13, w/w)	Butanone	25	–	96					VT	684
	Butanone/2-propanol (75/25, v/v)	25	–	76					VT	684
(13/87, w/w)	Butanone/2-propanol (97/3, v/v)	25	341	75					LT,VT	684
Poly(styrene-*co*-monomethyl maleate)										
	Acetone	26.4	–	51.1	575	285	2.02	8.15	VT	317
	Aq. NaCl (0.6 M)	25	–	55	585	285	2.05	8.4	VT	317
Poly(trifluoronitrosomethane-*co*-tetrafluoroethylene)										
		35	–	38	510 ± 25	304	1.68 ± 0.08		VT	3(685)
Poly(vinylchloride-*stat*-diphenylpropene)										
	Tetrahydrofuran	25	–	127	768	320	2.40	–	VG	8792
(77.3/22.7, mol/mol)	Tetrahydrofuran	25	–	123	761	290	2.62	–	VG	879
(62/38, mol/mol)	Tetrahydrofuran	25	–	100	707	250	2.82	–	VG	879
(33/67, mol/mol)	Tetrahydrofuran	25	–	70	630	233	2.70	–	VG	879
(11/89, mol/mol)	Tetrahydrofuran	25	–	62	605	230	2.63	–	VG	879
(4/96, mol/mol)	Tetrahydrofuran	25	–	56	585	224	2.61	–	VG	879
(0/100, mol/mol)	Tetrahydrofuran	25	–	50	563	221	2.55	–	VG	879

TABLE 7. MAIN-CHAIN CARBOCYCLIC POLYMERS

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	Method	Ref.
Poly(1,2-acenaphthenylene)									
trans	Various solvents	25	–	36 ± 3	520 ± 20	354	1.47 ± 0.05	VG	263

TABLE 8. MAIN-CHAIN HETEROATOM POLYMERS

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
8.1. POLY(OXIDES), POLY(ETHERS)										
Poly(butene oxide), see Poly[oxy(ethylethylene)]										
Poly(ethylene oxide), see Poly(oxyethylene)										
Poly[oxy(*tert*-butylethylene)]	Benzene	25	–	230	930	377*	2.47	13.6	VG	385
isotactic	Xylene	80	–	–	1040	–	–	15.9	VG	790
Poly(oxy-1,2-cyclohexylene)	Toluene	35	–	53	592	359	1.65		VG	472
Poly(oxydecamethylene)	Benzene; chloroform	~30	–	240	960	570	1.68	7.5	VG	386
Poly[oxy(2,6-dimethyl-1,4-phenylene)]										
	Chlorobenzene; toluene	25	–	166 ± 5	833 ± 10	715	1.16 ± 0.02	2.7	VG	474
	Benzene; carbon tetrachloride	25	–	175 ± 8	850 ± 10	715	1.13 ± 0.02	2.6	VG	473

TABLE 8. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
Poly[oxy(2,6-diphenyl-1,4-phenylene)]										
	Chlorobenzene; toluene	25	–	80 ± 5	660 ± 20	500	1.32 ± 0.04	3.5	VG	473
Poly(oxyethylene)	Various solvents	~20	–	110 ± 10	750 ± 30	541	1.38 ± 0.06	3.8	VG	3(389,390)
	Aq. K_2SO_4 (0.45 M); aq. $MgSO_4$ (0.39 M)	~40	–	115 ± 15	775 ± 30	541	1.43 ± 0.06	4.1	VT	395,655
	Benzene	25	–	129	790	541	1.46	4.25	VG	656,388
	Acetone	25	–	170	840	541	1.55	4.8	VG	387
	Various poor solvents	50	–	170	840	541	1.55	4.8	VG	387
	Undiluted	60	$d\ln r_0^2/dT = (0.23 \pm 0.02) \times 10^{-3}$ (deg^{-1})						ST	655
	Water	25	–	175	888	566	1.57	–	VG	963
		3–85	–	201–156	–	–	–	–	VG	963
	Aq. KOH (1.24 M)	25	–	177	888	566	1.57	–	VG	963
	(0.96–1.48 M)	25	–	183–180	–	–	–	–	VG	963
Poly[oxy(ethylethylene)]	Benzene; butanone; 2-propanol	30	–	87	700	423	1.66 ± 0.05	5.5	VG,VT	397
	2-Propanol	30	–	110	730	427	1.71	5.85	VT	396
Poly(oxyhexamethylene)	Benzene; dioxane	25	–	185	910	565*	1.61	5.15	VG	398
Poly(oxymethylene)	Hexafluoroacetone sesquihydrate (1/1.7, mol/mol)	25	–	430 ± 40	1200 ± 80	522	2.3 ± 0.2	10.5	VG	402
Poly[oxy(2-methyl-6-phenyl-1,4-phenylene)]										
	Dioxane/methylcyclohexane (1/1, v/v)	25	–	110 ± 5	790 ± 20	580	1.36 ± 0.04	3.7	VT	673
Poly[oxy(phenylethylene)]	Toluene	25	–	75	640	344*	1.85	6.85	VG	385
Poly(oxypropylene)	Benzene; methanol	25	–	115 ± 10	750 ± 25	472	1.59 ± 0.05	5.05	VG	3(411)
	Toluene/2,2-trimethylpentane (5/7, v/v)	39.5	–	107.5	735	472	1.56	4.85	VT	411
	2,2,4-Trimethylpentane	50	375	–	800	494*	1.62	5.75	LT	654
Poly(oxytetramethylene)	Diethyl malonate	33.5	–	243	945	556	1.70	5.8	VT	658
	Ethyl acetate	30	–	180 ± 20	860 ± 30	556	1.55 ± 0.05	4.8	VG	412
	Ethyl acetate/hexane (22.7/77.3, w/w)	30.4	–	267	975	556	1.75	6.1	VT	658
		31.8	–	210	900 ± 20	556	1.62 ± 0.03	5.25	VT	412
	2-Propanol	44.6	–	231	930	556	1.67	5.6	VT	658
	Undiluted	60	$d\ln r_0^2/dT = -1.33 \times 10^{-3}$ (deg^{-1})						ST	659
Poly(oxytrimethylene)	Acetone; benzene; carbontetrachloride	30	–	126	795	550	1.45	4.2	VG	414
Poly(propyleneoxide), see Poly(oxypropylene)										
Poly(tetrahydrofuran), see Poly(oxytetramethylene)										

8.2. POLY(ESTERS), POLY(CARBONATES)

Poly(butylene adipate-*co*-butylene isophthalate)										
(100 mol% BADP)	Chloroform	30	–	247	950	–	–	–	VG	967
(80 mol% BADP)	Chloroform	30	–	225	923	–	–	–	VG	967
(50 mol% BADP)	Chloroform	30	–	160	823	–	–	–	VG	967
(20 mol% BADP)	Chloroform	30	–	99	700	–	–	–	VG	967
Bisphenol A poly(carbonate), see Poly(oxycarbonyl-1,4-phenyleneisopropylidene-1,4-phenylene)										
	Poly(ethylene terephthalate), see Poly(oxyethyleneoxyterephthaloyl)									
	Chlorobenzene	25	–	100 ± 10	720 ± 25	540	1.33 ± 0.05	3.55	VT	3(415)
Poly(β-hydroxybutyrate)	Chloroform/*n*-heptane, (50.2/49.8, v/v)	25	–	157	860	–	–	–	VT,LT	950
	2,2,2-Trifluoroethanol/water (52.6/47.4, v/v)	25	–	149	840	–	–	–	VT,LT	950
	2,2,2-Trifluoroethanol	25	380–390	145–160	830–960	–	–	–	VG,LG	950
Poly(4,4′-isopropylidenediphenyl carbonate)-*block*-poly(styrene)										
26 wt.%-PS	Chloroform, diethyl carbonate, 1,4-dioxane, methylene chloride, tetrahydrofuran	25	–	–	910–1040	–	–	–	VG	959

* These values of r_{0f} of poly(epoxide) chains were calculated by $0.377/M_u^{1/2}$ [nm.mol$^{1/2}$/gram$^{1/2}$], while those given without asterisk were calculated by $0.360/M_u^{1/2}$. The former is due to Allen et al. (Ref. 695). The latter is based on the assumption that all valence angles of skeleton are tetrahedral.

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TABLE 8. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
48 wt.%-PS	*ibid.*	25			1000–1200	–	–	–	VG	959
50 wt.%-PS	*ibid.*	25			1040–1200	–	–	–	VG	959
73 wt.%-PS	*ibid.*	25			1120–1270	–	–	–	VG	959
Poly(oxybutynedioyloxyhexamethylene)										
	Benzene; chloroform	20	–	180 ± 20	870 ± 30	627	1.39 ± 0.05	3.9	VG	3(416)
Poly[oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene]										
	Methylene chloride; tetrahydrofuran	25	–	180 ± 20	880 ± 20	796	1.10 ± 0.05		VG	3(479)
	Butyl benzyl ether	170	–	210	940	796	1.18		VT	476
	Cyclohexane/dioxane (36.1/63.9, w/w)	25	–	210	940	796	1.18		VT	476
	Chloroform; tetrahydrofuran	25	–	150 ± 13	840	796	1.05		VG	478
	Hexane/tetrachloroethane (54/46, v/v)	30	–	230	930	796	1.16		VT	694
	Various	25	–	210 ± 17	920 ± 50	–	1.35	–	VG	880
Poly(oxy-1,4-cyclohexyleneoxysebacoyl)										
cis	Chloroform	20	–	140 ± 20	800 ± 30	495	1.62 ± 0.05		VG	3(480)
trans	Chloroform	20	–	160 ± 20	840 ± 30	633	1.33 ± 0.05		VG	3(480)
Poly(oxyethyleneoxyterephthaloyl)										
	Phenol/tetrachloroethane (1/1, v/v)	25	–	160 ± 15	840 ± 25	687	1.22 ± 0.03	3.15	VG	3(491)
	o-Chlorophenol	25	–	210	910	687	1.33	3.7	VG	484
	Trifluoroacetic acid	30	–	242	975	687	1.42	4.25	VG	488
Poly(oxyfumaroyloxyhexamethylene)										
	Chloroform	20–50	–	180 ± 20	870 ± 30	592	1.47 ± 0.05	4.3	VG	3(417)
Poly[oxy(hexahydroterephthaloyl)oxyoctamethylene]										
cis	Chloroform	20	–	140 ± 20	800 ± 30	495	1.62 ± 0.05		VG	3(480)
trans	Chloroform	20	–	160 ± 20	840 ± 30	633	1.33 ± 0.05		VG	3(480)
Poly(oxyhexamethyleneoxy-2,9-dibutylsebacoyl)										
	Benzene	20	–	155 ± 25	835 ± 70	457	1.82 ± 0.15	6.6		3(418)
Poly(oxyhexamethyleneoxysebacoyl)										
	Benzene; chloroform	20	–	215 ± 60	910 ± 100	540	1.70 ± 0.17	5.8		3(418,419)
Poly[oxyisophthaloyloxy-1,4-phenylene (fluoren-9-yildene)-1,4-phenylene]										
	Tetrachloroethane; tetrahydrofuran	20	–	210	902				VG	674
Poly(oxymaleoyloxyhexamethylene)										
	Benzene; chloroform; tetrahydrofuran	20–50	–	135 ± 15	790 ± 30	510	1.55 ± 0.05	4.8	VG	3(417)
Poly(oxy-1-oxo-dimethyltrimethylene)										
	Trifluoroacetic acid	20	–	238	950	–	–	–	VG	800
Poly(oxy-1-oxo-hexamethylene)	Benzene; dioxane; ethyl acetate	25	–	–	–	–	–	5.9 ± 0.3	VG	808
		–	–	–	–	–	–	5.6	–	881
Poly(oxy-1-oxo-3-methyltrimethylene), D,L										
	Chloroform; ethylene dichloride	30	–	133	875	–	–	–	VG	799
	Trifluoroethanol	25	–	133	875	–	–	–	VG	799
Poly(oxyperfluorosebacoyloxydecamethylene)										
	Chloroform; *s*-tetrachloroethane	25	–	–	–	–	–	6.3 ± 0.5	VG,VT	795
Poly(oxysebacoyloxyhexadecamethylene)										
	Chloroform	20	–	270 ± 40	100 ± 50	555	1.80 ± 0.10	6.5	VG	3(419)
Poly(oxysuccinyloxyhexamethylene)										
	Benzene; chloroform; tetrahydrofuran	20–60	–	165 ± 30	850 ± 60	522	1.62 ± 0.14	5.25	VG	3(417)
Poly[oxy-(1,1,3-trimethyl-3-phenylindan-5,4′-dicarbonyl)–oxy-1,4-phenyleneisopropylidene-1,4-phenylene]										
	Tetrahydrofuran	25	–	156 ± 2	850 ± 10	–	–	–	VG	794,882
Poly(oxyundecanoyl)	Chloroform	20	–	185 ± 60	880 ± 100	550	1.60 ± 0.16	5.1	VG	3(419,420)
Poly(β-propiolactone)	Chloroform, 1,4-dioxane, 2,2,2,-trifluoroethanol	30	–	146	–	–	–	–	VG	991

TABLE 8. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
8.3. POLY(AMIDES)										
Poly(iminoadipoyliminohexamethylene), (Nylon 66)										
	Aq. HCOOH (90%, v/v)	25	–	190 ± 20	890 ± 40	545	1.63 ± 0.08	5.3	VG	3(445,446)
	Aqueous HCOOH (90%, v/v)									
	KCl (2.3 M)	25	–	253	1010	545	1.85	6.85	VT	444
		25	–	192	935	545	1.72	5.93	VT	668,669
Poly(iminohexamethyleneiminoterephthaloyloxyhexamethyleneoxyterephthaloyl)										
	Phenol/tetrachloroethane (1/1, w/w)	25	–	250	1000	–	–	4.76	VG	806
Poly(iminohexamethyleneiminothiocarbonylhexamethylenethiocarbonyl)										
	Dimethylformamide/ LiCl (5%)	20	–	142	–	–	–	–	VG	805
Poly(iminohexamethyleneiminothiocarbonyltetramethylenethiocarbonyl)										
	Dimethylformamide/ LiCl (5%)	20	–	279	–	–	–	–	VG	805
Poly[imino(1-oxododecamethylene)], (Nylon 12)										
	m-Cresol	25	–	266	–	–	–	6.77	VG	803
		25	–	238	970	–	6.29	–	VG	812
	Sulfuric acid (96%)	25	–	222	949	–	6.01	–	VG	812
Poly[imino(1-oxohexamethylene)], (Nylon 6)										
	Sulfuric acid	25	–	190 ± 10	890 ± 20	545	1.63 ± 0.04	5.3	VG	3(453)
	Aq. HCOOH (65–85%)	25	–	229	970	545	1.78	6.35	VT,VG	450
Poly[iminoterephthaloylimino-1,4-phenylene(fluoren-9-ylidene)-1,4-phenylene]										
	Dimethylformamide	25	–	409	~1200				VG	503
Poly[iminoterephthaloylimino(trimethylhexamethylene)], 2,4,4-/2,2,4-trimethyl (1/1, mol/mol)										
	Aniline	142	–	148	850	640	1.33	–	VG	804
	Dimethylformamide	25	–	174	–	–	–	–	VG	804
	Pyrroline	62	–	151	850	640	1.33	–	VG	804
Poly(isophthaloyl-*trans*-2,5-dimethylpiperazine)										
	N-methyl-2-pyrrolidone	25	355	–	–	–	–	–	VG	971
8.4. POLY(AMINO ACIDS)										
Poly(β-benzyl-L-aspartate), see Poly(iminocarbonyl-L-benzyloxycarbonylethylidene)										
Poly(γ-benzylglutamate), see Poly(iminocarbonyl-benzyloxycarbonylpropylidene)										
Poly(iminocarbonyl-L-benzyloxycarbonylethylidene)										
	m-Cresol	100	–	–	600	268	2.24 ± 0.1	9.6	VA	670
Poly(iminocarbonyl-benzyloxycarbonylpropylidene)										
L	Dichloroacetic acid	25	–	58 ± 5	600 ± 20	259	2.32 ± 0.08	10.3	VG	3(457)
		25	–				2.14	8.8	VA	670(457)
D,L	Dichloroacetic acid; dimethylformamide	25	–	58 ± 5	600 ± 20	259	2.32 ± 0.08	10.3	VG	3(459)
Poly(iminocarbonyl-L-carboxypropylidene), [Poly(L-glutamic acid)]										
	Phosphate buffer (Na^+, 0.3 M; pH 7.85)	17	–	–	720	337	2.14 ± 0.1	8.8	VA	670
Poly(iminocarbonyl-L-hydrocarbonylpropylidene), [Poly(α-L-glutamic acid)]										
	Sodium phosphate buffer (pH, ca. 7)	25.5	–	98.4	775	–	–	–	VG	815
Poly(iminocarbonyl-L-(*N*-hydroxyethylcarbamoylpropylidene)], {Poly[N^5-(2-hydroxyethyl)-L-glutamine]}										
	Water	30	–	–	–	–	–	10 ± 1	Complex	817
Poly(γ-hydroxy-L-proline)	Water	30	–	–	–	–	–	15.8	VA	818
Poly[(methylimino)carbonylmethylene], [Poly(sarcosine)]										
	Water	20	–	50 ± 20	570 ± 90	455	1.25 ± 0.20	3.0	VG	3(468)
8.5. POLY(URETHANES)										
Poly(oxytetramethyleneoxycarbonylimino-2,4-tolyleneiminocarbonyl)										
	Dimethylformamide	30	–	–	1030	515	2.0		VG	497
8.6. POLY(SULFIDES)										
Poly(thiopropylene)	Benzene	20	–	60	600				VG	436
Poly(trimethylene sulfide)	Chloroform	25	–	150	783	–	–	4.0	VG	979

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TABLE 8. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ ($\times 10^4$ nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
8.7. POLY(PHOSPHATES)										
Poly[oxy(hydroxyphosphinylidene)]										
	Aq. NaBr (0.35–0.415 M)	25	–	50 ± 3	560 ± 20	370	1.51 ± 0.04	6.6	VT,VG	3(422)
	Aq. LiBr									662
	Aq. CsCl (0.96 M)	30	–				3.93	7.1	LT	663
	Aq. LiCl (2.9 M)	30	–				2.25		LT	663
	Aq. NaCl (0.96 M)	30	–				2.79		LT	663
8.8. POLY(SILOXANES), POLY(SILSESQUIOXANES), POLY(SILMETHYLENES)										
Poly(dimethyl siloxane), see Poly[oxy(dimethylsilylene)]										
Poly(dimethylsilmethylene)	Heptane/propanol (68.8/31.2, v/v)	25	540	–	988	450	2.2	–	LT	671
Poly(dimethylsiltrimethylene)	Heptane	25	–	–	1200	480	2.5	–	LG	671
Poly(diphenylsiltrimethylene)	Cyclohexanol/toluene (63.5/36.5, v/v)	25	580	–	1160	322	3.6	–	LT	671
Poly[oxy(dimethylsilylene)	Butanone, toluene	~25	–	80 ± 5	670 ± 20	482	1.39 ± 0.05	6.25	VT,VG	3(427, 428) 664
	Various theta solvents	2–90	266 ± 10	–	612 ± 13	482	1.27 ± 0.03	5.2	LT	425
	$C_8F_{18}/C_2Cl_4F_2$ (33/67, w/w)	22.5	–	106	740	482	1.54	7.6	VT	424
	Ethyl iodide	2	–	70	640	482	1.33	5.7	VT	425
	Bromocyclohexane	29	–	74	655	482	1.36	6.0	VT	425
		28	–	79.5	[655]	–	–	–	VT	838
		28	294 ± 14	–	–	–	–	–	LT	838
	Bromocyclohexane	37	–	78	685	–	–	–	VT	982
	Bromocyclohexane/ phenetole (6/7, v/v)	36	–	75	660	482	1.37	6.1	VT	425
	Chlorobenzene/dimethyl phthalate (45/6, v/v)	57.5	–	76	660	482	1.37	6.1	VT	425
	Bromobenzene	78.5	–	76	660	482	1.37	6.1	VT	425
	Phenetole	89.5	–	73	650	482	1.35	5.9	VT	425
	Toluene	25	–	75	670	–	–	–	VG	981
		30	–	78	678	–	–	–	VG	953
	Undiluted	40–100	$d \ln r_0^2/dT = -(0.78 \pm 0.06) \times 10^{-3}$ (deg^{-1})						ST	665
	Diluted with liquid silicon	30–105	$d \ln r_0^2/dT = -(0.71 \pm 0.13) \times 10^{-3}$ (deg^{-1})						VT	665
Poly[oxy(dipropylsilylene)]	2-Pentanone	76	–	87.1	703	372	1.89	12.0	VT	433
	Toluene	10	–	109	759	372	2.04	14.0	VT	433
Poly[oxy(methylphenylsilylene)]	Diisobutylamine	30.4	–	51.5	575	363	1.58	8.35	VT	434
	Cyclohexane	25	–	71.5	–	–	–	10.8	VG	983
		50	–	71.5	–	–	–	10.8	VG	983
	Methylcyclohexane	20	–	65.0	–	–	–	10.2	VG	983
	Tetrahydrofuran	25	–	96.0	–	–	–	13.4	VG	983
Poly[oxy(γ-trifluoropropylmethylsilylene)]										
	Cyclohexyl acetate	25.0	–	41.0	550(648)*	341	1.61(1.90)*	6.3	VT	435
	Methyl hexanoate	72.8	–	44.5	565(667)*	341	1.66(1.96)*	6.65	VT	435
Poly(phenylsilsesquioxane)	1,2-Dichloroethane	50.5	–	–	1160	–	–		VT	675
8.9. POLY(HETEROCYCLICS)										
Poly(3,3′-dimethylthietane)	Tetrahydrofuran	25	–	120	744	–	–	–	VG	986
Poly[2-methoxy-4,6-di(*p,p′*-isopropylidene diphenyloxy)-*s*-triazine]										
	Chloroform, 1,2-dichloroethane, 1,2-dichloroethane/*n*-butanol (80/15, v/v)	30	–	16.5–17.5	850–863	778	1.1	–	VG	985
Poly[*N*-(*n*-octadecyl)maleimide]	Benzene	25	–	28	–	–	–	16.6	VG	987
	1-Chloronaphthalene	25	–	27	–	–	–	16.2	VG	987
	o-Dichlorobenzene	25	–	25	–	–	–	15.4	VG	987

* The values of $r_0/M^{1/2}$ and σ given in parentheses were obtained by using $\Phi_0 = 1.39 \times 10^{23}$, while for the others $\Phi_0 = 2.5 \times 10^{23}$ was used.

TABLE 8. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_θ ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
	Tetrahydrofuran	25	–	37	–	–	–	20.0	VG	987
	Toluene	25	–	24	–	–	–	15.0	VG	987
	1-Decanol	39.4	–	18.3	–	–	–	11.7	VT	988
	1-Hexanol	79.0	–	37.0	–	–	–	18.8	VT	988
	1-Octanol	53.7	–	23.5	–	–	–	13.8	VT	988
Poly[6,6′-bis(3-phenylquinoxaline)-2,2′-diyl-1,4-phenylene]										
	m-Cresol	40	–	213	1000	–	–	–	VG	849
Poly[(1,3-dihydro-3-oxo-isobenzofuran-1-ylidene)-1,4-phenyleneoxyisophthaloyloxy-1,4-phenylene]										
	Tetrachloroethane; tetrahydrofuran	20	–	195	880	–	–		VG	674
Poly[(1,3-dihydro-3-oxo-isobenzofuran-1-ylidene)-1,4-phenyleneiminoterephthaloylimino-1,4-phenylene]										
	Dimethylformamide	25	–	558	~1500				VG	503
Poly[(D,L-1,2-pyrrolidindiyl)carbonyl]										
	Water	25	–	ca. 25	ca. 570	390	ca. 1.5		VT	3(507)
Poly(1-isobutyl-2,5-dioxopyrrolidin-3,4-diyl)										
	Butyl acetate	21	–	132	790	–	–		VG	512
Poly(1-*p*-tolyl-2,5-dioxopyrrolidin-3,4-diyl)										
	Dimethylformamide	21	–	75	670	–	–		VG	513
8.10. COPOLYMERS (MALEIC ANHYDRIDE, SULFONES, SILOXANES)										
8.10.1. MALEIC ANHYDRIDE COMPOLYMERS										
Poly[3,9-bis(ethylidone-2,4,8,10-tetraoxaspiro(5,5)undecane)-*ran*-*trans*-cyclohexane-dimethanol)/(1,6-hexanediol)]										
trans-CDM/1,6-ND (100/0)	Tetrahydrofuran	30	–	87.0	–	–	–	–	LS	954
(70/30)	Tetrahydrofuran	30	–	67.0	–	–	–	–	LS	954
(50/50)	Tetrahydrofuran	30	–	61.5	–	–	–	–	LS	954
(29/71)	Tetrahydrofuran	30	–	50.0	–	–	–	–	LS	954
(0/100)	Tetrahydrofuran	30	–	30.0	–	–	–	–	LS	954
Poly[(tetrahydro-2,5-dioxo-3,4-furandiyl)-1-phenylethylene]										
	Tetrahydrofuran	30	–	82.6	732	383	1.91		VG	506
8.10.2. SULFONES										
Poly(sulfonylbicycloheptene)	Chloroform	25, 40	–	29.0	488	–	–	4.2	VG	840
	Cyclohexane	25	–	29.0	488	–	–	4.2	VG	840
Poly[sulfonyl(butylethylene)]	Hexyl chloride	13	–	67 ± 2	625 ± 10	350	1.79 ± 0.02		VT	666
		13	296 ± 9	–	725 ± 22	350	2.07 ± 0.07		LT	666
	Benzene/cyclohexane (43/57, v/v)	25	–	66 ± 2	642 ± 7	350	1.84 ± 0.02		VG	348
	Butanone/2-propanol (30–37/70, v/v)	8–24	–	53	580	350	1.66		VT	439
	Dioxane/hexane (40/60, v/v)	20	–	65 ± 2	638 ± 7	350	1.83 ± 0.02		VT	438
Poly[sulfonyl(1-methyl-1-propylethylene)]										
	Butanone/hexane (35.4/64.6, v/v)	11.5	–	91	700				VT	439
Poly(sulfonyl-*p*-phenoxyphenylene)										
	Dimethylformamide	25	–	106	753 ± 20	771	0.98 ± 0.10	–	VG	841
Poly[sulfonyl(phenylethylene)]	Tetrahydrofuran	30	–	57.4	649	425	1.53		VG	440,667
8.10.3. SILOXANES										
Poly(*p*-*tert*-butylstyrene)-*block*-polydimethylsiloxane)-*block*-poly(*p*-*tert*-butylstyrene) 28 wt.% PBS										
	Methyl ethyl ketone	15	–	58.9	620	–	–	–	VT	957
	Benzene	35	–	68	650	–	–	–	VG	960
	Methyl ethyl ketone	15	–	59	620	–	–	–	VT	960
		20	–	62	630	–	–	–	VG	960
		31	–	62	630	–	–	–	VG	960
Poly(dimethyl siloxane-*co*-diphenyl siloxane)										
(95/5, mol/mol)	Bromobenzene	9	–	100	742	–	1.56		VT	596
(66/34, mol/mol)	Benzene/2-propanol (44/56, w/w)	42	–	74	670	–	1.70		VT	596

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TABLE 8. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
(55/45, mol/mol)	Dimethylphthalate	82.5	–	78	675	–	1.81		VT	596
	Ethanol/toluene (37/63, w/w)	29.5	–	78	675	–	1.81		VT	596
Poly(styrene)-*block*-poly(dimethyl siloxane) 68% PS										
	Butanone	30	–	81	687		–	–	VG	953
	Decalin	30	–	80	687	–	–	–	VG	953
	Toluene	30	–	82	687	–	–	–	VG	953
Poly(styrene)-*block*-poly(dimethyl siloxane)-*block*-poly(styrene) 68% PS										
	Toluene	30	–	80	687	–	–	–	VG	953
8.11. OTHER COMPOUNDS										
Poly(2-acrylamino-2-methylpropanesulfonic acid)										
cesium salt	Aq. CsCl (1 N)	25	–	16	–	–	–	11	VG	864
sodium salt	Aq. CsCl (1 N)	25	–	32	–	–	–	12	VG	864
	Aq. NaCl	25	–	30	–	–	–	12	VG	864
tetrabutylammonium salt	Aq. NaCl (0.1 N; 1 N; 5 N)	25	–	10	–	–	–	11	VG	864
	Organic solvents	25	–	23	–	–	–	19	VG	864
	Tetrabutylammonium bromide (0.5 N)	25	–	30	–	–	–	23	VG	864
Poly(dihexoxyphosphazene)	Tetrahydrofuran	25	–	30.8	–	–	–	$C_n = 13.2$	VG	994
Poly(*di-n*-hexylsilane)	Tetrahydrofuran	25	–	–	740	–	–	19	VG	995
Poly(hexamethylenedimethylammonium bromide)										
	Aq. KBr (0.4 M)	25	–	117.4	825	–	–	–	VG	844
Poly[oxythiocarbonyloxy-1,4-phenylene(methyl)phenylmethylene-1,4-phenylene]										
	Chloroform	25	–	190	913	–	–	5.26	VG	992
	Dichloromethane	25	–	225	965	–	–	5.89	VG	992
	1,4-Dioxane	25	–	175	888	–	–	4.98	VG	992
	Tetrahydrofuran	25	–	165	871	–	–	4.79	VG	992

TABLE 9. CELLULOSE AND DERIVATIVES, POLY(SACCHARIDES)

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
Amylose	Dimethyl sulfoxide; ethylene diamine	25	–	56 ± 12	600 ± 50	335	1.79 ± 0.15		VG	3(518)
	Various solvents	25	–	–	700	335	2.08		VT,VG	517
	Aq. KCl (0.33 M); dimethyl sulfoxide	25	–	110 ± 5	750 ± 25	335	2.24 ± 0.08		VT,VG	3(520)
	Aq. KCl (0.5 M)	25	–	61	625	335	1.87		VT	523
	Aq. KOH (0.15 M)	25	–	164					VG	519
	Aq. KCl (0.33 M)	25	–	–	–	–	–	5.2[a]	VT	530
	Nitromethane	22.5	–	–	920	335	2.75		LG	527
	Water/dimethyl sulfoxide (40/60, v/v)	25	–	–	–	–	–	4.1	VA,LG	884
	Water	24	–	–	–	–	–	11.7	VG,LG	999
Amylose acetate	Nitromethane	25	–	51	589	–	2.22	–	VG	846
Amylose triacetate	Chloroform; nitromethane	30	–	47 ± 10	580 ± 60	250	2.32 ± 0.24		VG	3(524)
		30	–	48	580	250	2.32		VG	676
		30	–		800 ± 15	250	3.2 ± 0.06		VG	677
Amylose tributyrate	Butanone	25	–	–	557	–	–	–	LG	847
	Butanone; carbon tetrachloride	25	–	–	674	–	–	6.83	VG	847
	Ethyl acetate	25	–	–	595	–	–	–	LG	847

TABLE 9. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_θ ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
	Ethyl acetate; tetrahydrofuran	25	–	–	611	–	–	5.63	VG	847
	–	–	–	–	–	–	2.21–2.24	–	–	861
Amylose tricarbanilate	Acetone; dioxane; pyridine	20	–	27 ± 5	470 ± 30	187	2.51 ± 0.16		VG	3(528)
	Dioxane/methanol (49/51, v/v)	20	–	–	2180	–	11.7		LT	678
	Pyridine	25	471–478	–	–	–	–	–	LG	848
Amylose tripropionate	Ethyl acetate	25	–	57	583	–	2.32	6.21	VT?	850
Carboxymethyl amylose, sodium salt	Aq. NaCl (0.65 M)	37.5					2.62	7.9[a]	VT	609
	Aq. NaCl (0.5 M; pH 8)	35					2.95	10.0[a]	VA	530
	Aq. NaCl (0.78 M; 0.02% NaN_3)	35					2.15	5.3[a]	VA	531
Diethylaminoethyl amylose hydrochloride	Aq. NaCl (0.78 M; 0.02% NaN_3)	35						6.4[a]	VA	531
Cellulose	Cupriethylene diamine	25	–	180 ± 80	900 ± 150	620	1.45 ± 0.25		VG	3(537)
	Cadoxen	25	0.24	495	1250	620	2.0		VG	691,534
	N,N-dimethylacetamide/ 9% LiCl	30	–	–	–	–	6.71	91.9	LG	1000
Cellulose diacetate										
Ac. acid content, 55.5 wt.%	Acetone	25	–	–	–	–	3.59	26.3	LS	854
	Tetrahydrofuran	25	–	–	–	–	2.56	13.5	LS	854
Cellulose triacetate	Acetone; chloroform; *o*-cresol	25–30	–	108 ± 10	750 ± 30	465	1.61 ± 0.07		VG	3(541,542)
	Ethylacetate; dioxane; methyl acetate; tetrahydrofuran	25			730–740	465	1.57–1.59		VG	680
	Chloroform	30	–	–	1180	–	–	–	VG	857
Cellulose tributyrate	Butanone	30	–	97 ± 15	730 ± 40	408	1.79 ± 0.10		VG	3(544)
	Dodecane/tetralin (75/25, v/v)	130	–	82	690	408	1.69		VT	544
Cellulose tricarbanilate	Acetone; dioxane; pyridine	20	–	130 ± 30	810 ± 70	346	2.34 ± 0.20		VG	3(528)
	Acetone	~25	–	65 ± 3	635	346	1.83		VG	693
	Acetone	25	–	–	1450	–	–	–	VWC	996
	Cyclohexane	~25	–	83.5 ± 3	690	346	1.99		VG	693
	Dioxane	~25	–	44 ± 3	560	346	1.61		VG	693
	Dioxane/methanol (42.5/57.5, v/v)	20	–	–	1120	346	3.24		LT	678
	Anisol	94	ca. 1000	130	805	346	2.32		LT,VT	546
	Cyclohexanol	73	ca. 1050	–	875	346	2.52		LT	546
		73	–	156	1090	–	–	–	VT	858
	Dioxane	15	–	76	860	–	–	–	VG	858
		30	–	68	830	–	–	–	VG	858
		45	–	61	800	–	–	–	VG	858
		60	–	56	780	–	–	–	VG	858
		70	–	51	750	–	–	–	VG	858
	Tetrahydrofuran	25	–	–	1410	–	–	–	VWC	996
Cellulose trihexanoate	Dimethylformamide	41	–	240	980	370	2.65		VT	547
	Dioxane/water (100/7, v/v)	63	–	224	960	370	2.60		VT	547
Cellulose trinitrate	Acetone; ethyl acetate	~25	–	130 ± 30	810 ± 50	458	1.77 ± 0.11		VG	3(535,537 548,549)
	Acetone	25	360	–	720	458	1.57		LG	3(549)
		22	0.26 ± 0.01	–	930 ± 15	458	2.03 ± 0.03		XS	641
		~20	0.40–0.70	–	1180–1590	458	2.58–3.46		XS	679
		20	0.22–0.28	–	850–975	458	1.85–2.12		XS	679
(N%, 13.9)	Acetone	25	970	–	2410	462	4.7		LG	686
(N%, 12.9)	Acetone	25	530	–	1780	475	3.4		LG	686
(N%, 13.5)	Ethyl acetate	30	700	–	2050	467	3.9		LG	555
(N%, 12.0)	Acetone	20	0.48	–		486	2.5		XS	687

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TABLE 9. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
Cellulose trioctanoate	Toluene	30	–	127 ± 15	800 ± 40	340	2.35 ± 0.12		VG	3(544)
	Dimethylformamide	140	–	113	770	340	2.27		VT	544
	γ-Phenylpropanol	48	–	129	805	340	2.37		VT	544
(Cyanoethyl)(hydroxypropyl) cellulose	Tetrahydrofuran	25	$d \ln r_0^2/dT = 3.8 \times 10^{-3}$ (deg^{-1})	–	–	–	–	–	VTe	1002
Ethyl cellulose	Methanol	25	–	232 ± 10	970 ± 20	520	1.87 ± 0.03		VG	3(559)
Ethyl hydroxyethyl cellulose	Water	25	–	550 ± 80	1300 ± 50	545	2.38 ± 0.09		VG	3(560)
		25	–	335	1100	545	2.0		VG	691,560, 692
Hydroxyethyl cellulose	Cadoxen; water	25	–	250	1000	514	1.9		VG	691,561
Methyl cellulose	Water	25	–	920 ± 100	1500 ± 60	581	2.58 ± 0.10		VG	3(563)
Pullulan	Water	25	360	91	–	–	–	–	VG	1003
Sodium carboxymethyl cellulose	Cadoxen	25	–	357	1130	563	2.0		VG	691,565
	Aq. NaCl (0.005–0.2 M)	25	–	420	1190	502	2.4		VG	691,565
Sodium cellulose xanthate	Aq. NaOH (1 M)	30	–	290	1380				VG	681
Hyaluronic acid	Aq. HCl (0.1 M)	25	–	–	–	–	2.5	–	VA	688

[a] These values of the characteristic ratio C_∞ of cellulosic chains were obtained by $C_\infty = r_0^2/\mathrm{DP}\,l^2$, where DP is the degree of polymerization and $l = 0.425$ (nm).

F. REFERENCES

1. "Report on Nomenclature in the Field of Macromolecules, International Union of Pure and Applied Chemistry", J. Polym. Sci., **8**, 257 (1952).
2. P. J. Flory, "Principles of Polymer Chemistry", Cornell University Press, Ithaca, New York, 1953.
3. M. Kurata, W. H. Stockmayer, Fortschr. Hochpolymer. Forsch., **3**, 196 (1963).
4. W. H. Stockmayer, M. Fixman, J. Polym. Sci. C, **1**, 137 (1963).
5. R. Koningsveld, C. A. Tuijnman, Makromol. Chem., **38**, 39,44 (1960).
6. "A Structure-Based Nomenclature for Linear Polymers", Macromolecules, **1**, 193 (1968).
7. A. Peterlin, "Viskosität und Form", in: H. A. Stuart (Ed.), "Die Physik der Hochpolymeren", vol. II, Springer, Berlin, 1953.
8. G. Meyerhoff, Fortschr. Hochpolymer. Forsch., **3**, 59 (1961).
9. H. -G. Elias, Kunststoffe-Plastics, **4**, 1 (1961).
10. S. Krause, "Dilute Solution Properties of Acrylic and Methacrylic Polymers", Part I, Revision 1, Rohm & Haas, Philadelphia, Pennsylvania, 1961.
11. P. J. Flory, "Statistical Mechanics of Chain Molecules", Interscience, Wiley, New York, 1969.
12. H. Yamakawa, "Modern Theory of Polymer Solutions", Harper & Row, New York, 1971.
13. A. Peterlin, J. Polym. Sci., **47**, 403 (1960).
14. S. Heine, O. Kratky, G. Porod, P. J. Schmitz, Makromol. Chem., **44/46**, 682 (1961); S. Heine, *ibid.*, **48**, 205 (1961); S. Heine, O. Kratky, J. Roppert, *ibid.*, **56**, 150 (1962).
15. F. Danusso, G. Moraglio, G. Gianott, J. Polym. Sci., **51**, 475 (1961).
16. H. Fujita, N. Takeguchi, K. Kawahara, M. Abe, H. Utiyama, M. Kurata, Paper presented the Twelfth Polymer Symposium, Nagoya, Japan, Nov., 1963.
17. M. Abe, Y. Murakami, H. Fujita, J. Appl. Polym. Sci., **9**, 2549 (1965); see also Ref. 254.
18. R. Endo, Nippon Gomu Kyokaishi (J. Rubber Ind. Japan), **34**, 522 (1961). See also, M. Takeda, R. Endo, Rept. Progr. Polym. Phys. Japan, **6**, 37 (1963).
19. I. Ya. Poddubnyi, Ye. G. Erenberg, M. A. Yeremina, Vysokomol. Soedin. A, **10**, 1381 (1968).
20. W. G. Cooper, G. Vaughan, D. E. Eaves, R. W. Madden, J. Polym. Sci., **50**, 159 (1961).
21. R. Endo, Nippon Gomu Kyokaishi (J. Rubber Ind. Japan), **35**, 658 (1962).
22. R. L. Cleland, J. Polym. Sci., **27**, 349 (1958); see also Ref. 26.
23. P. L. Ribeyrolles, A. Guyot, H. Benoit, J. Chim. Phys., **56**, 377 (1959); see also Ref. 26.
24. M. Kurata, H. Utiyama, K. Kajitani, T. Koyama, H. Fujita, Paper presented at the Twelfth Polymer Symposium, Nagoya, Japan, Nov., 1963.
25. R. Endo, Nippon Gomu Kyokaishi (J. Rubber Ind. Japan), **34**, 527 (1961).
26. W. Cooper, D. E. Eaves, G. Vaughan, J. Polym. Sci., **59**, 241 (1962).
27. I. Ya. Poddubnyi, V. A. Grechanovskii, Vysokomol. Soedin., **6**, 64 (1964).
28. R. L. Scott, W. C. Carter, M. Magat, J. Am. Chem. Soc., **71**, 220 (1949).
29. D. J. Pollock, L. J. Elyash, T. W. Dewitt, J. Polym. Sci., **15**, 335 (1955).
30. V. S. Skazka, M. Kozhokaru, G. A. Fomin, L. F. Roguleva, Vysokomol. Soedin. A, **9**, 177 (1967).

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TABLE 9. *cont'd*

Polymer	Solvent	Temp. (°C)	$S_{0z}/M_w^{1/2}$ ($\times 10^4$) or a_p (nm)	K_0 ($\times 10^3$) (ml/g)	$r_0/M^{1/2}$ $\times 10^4$ (nm)	$r_{0f}/M^{1/2}$ ($\times 10^4$) (nm)	$\sigma = r_0/r_{0f}$	$C_\infty = r_0^2/nl^2$	Method(s)	Refs.
Cellulose trioctanoate	Toluene	30	–	127 ± 15	800 ± 40	340	2.35 ± 0.12		VG	3(544)
	Dimethylformamide	140	–	113	770	340	2.27		VT	544
	γ-Phenylpropanol	48	–	129	805	340	2.37		VT	544
(Cyanoethyl)(hydroxypropyl) cellulose	Tetrahydrofuran	25	$d \ln r_0^2/dT = 3.8 \times 10^{-3}$ (deg^{-1})	–	–	–	–	–	VTe	1002
Ethyl cellulose	Methanol	25	–	232 ± 10	970 ± 20	520	1.87 ± 0.03		VG	3(559)
Ethyl hydroxyethyl cellulose	Water	25	–	550 ± 80	1300 ± 50	545	2.38 ± 0.09		VG	3(560)
		25	–	335	1100	545	2.0		VG	691,560, 692
Hydroxyethyl cellulose	Cadoxen; water	25	–	250	1000	514	1.9		VG	691,561
Methyl cellulose	Water	25	–	920 ± 100	1500 ± 60	581	2.58 ± 0.10		VG	3(563)
Pullulan	Water	25	360	91	–	–	–	–	VG	1003
Sodium carboxymethyl cellulose	Cadoxen	25	–	357	1130	563	2.0		VG	691,565
	Aq. NaCl (0.005–0.2 M)	25	–	420	1190	502	2.4		VG	691,565
Sodium cellulose xanthate	Aq. NaOH (1 M)	30	–	290	1380				VG	681
Hyaluronic acid	Aq. HCl (0.1 M)	25	–	–	–	–	2.5	–	VA	688

[a] These values of the characteristic ratio C_∞ of cellulosic chains were obtained by $C_\infty = r_0^2/DP\,l^2$, where DP is the degree of polymerization and $l = 0.425$ (nm).

F. REFERENCES

1. "Report on Nomenclature in the Field of Macromolecules, International Union of Pure and Applied Chemistry", J. Polym. Sci., **8**, 257 (1952).
2. P. J. Flory, "Principles of Polymer Chemistry", Cornell University Press, Ithaca, New York, 1953.
3. M. Kurata, W. H. Stockmayer, Fortschr. Hochpolymer. Forsch., **3**, 196 (1963).
4. W. H. Stockmayer, M. Fixman, J. Polym. Sci. C, **1**, 137 (1963).
5. R. Koningsveld, C. A. Tuijnman, Makromol. Chem., **38**, 39,44 (1960).
6. "A Structure-Based Nomenclature for Linear Polymers", Macromolecules, **1**, 193 (1968).
7. A. Peterlin, "Viskosität und Form", in: H. A. Stuart (Ed.), "Die Physik der Hochpolymeren", vol. II, Springer, Berlin, 1953.
8. G. Meyerhoff, Fortschr. Hochpolymer. Forsch., **3**, 59 (1961).
9. H. -G. Elias, Kunststoffe-Plastics, **4**, 1 (1961).
10. S. Krause, "Dilute Solution Properties of Acrylic and Methacrylic Polymers", Part I, Revision 1, Rohm & Haas, Philadelphia, Pennsylvania, 1961.
11. P. J. Flory, "Statistical Mechanics of Chain Molecules", Interscience, Wiley, New York, 1969.
12. H. Yamakawa, "Modern Theory of Polymer Solutions", Harper & Row, New York, 1971.
13. A. Peterlin, J. Polym. Sci., **47**, 403 (1960).
14. S. Heine, O. Kratky, G. Porod, P. J. Schmitz, Makromol. Chem., **44/46**, 682 (1961); S. Heine, *ibid.*, **48**, 205 (1961); S. Heine, O. Kratky, J. Roppert, *ibid.*, **56**, 150 (1962).
15. F. Danusso, G. Moraglio, G. Gianott, J. Polym. Sci., **51**, 475 (1961).
16. H. Fujita, N. Takeguchi, K. Kawahara, M. Abe, H. Utiyama, M. Kurata, Paper presented the Twelfth Polymer Symposium, Nagoya, Japan, Nov., 1963.
17. M. Abe, Y. Murakami, H. Fujita, J. Appl. Polym. Sci., **9**, 2549 (1965); see also Ref. 254.
18. R. Endo, Nippon Gomu Kyokaishi (J. Rubber Ind. Japan), **34**, 522 (1961). See also, M. Takeda, R. Endo, Rept. Progr. Polym. Phys. Japan, **6**, 37 (1963).
19. I. Ya. Poddubnyi, Ye. G. Erenberg, M. A. Yeremina, Vysokomol. Soedin. A, **10**, 1381 (1968).
20. W. G. Cooper, G. Vaughan, D. E. Eaves, R. W. Madden, J. Polym. Sci., **50**, 159 (1961).
21. R. Endo, Nippon Gomu Kyokaishi (J. Rubber Ind. Japan), **35**, 658 (1962).
22. R. L. Cleland, J. Polym. Sci., **27**, 349 (1958); see also Ref. 26.
23. P. L. Ribeyrolles, A. Guyot, H. Benoit, J. Chim. Phys., **56**, 377 (1959); see also Ref. 26.
24. M. Kurata, H. Utiyama, K. Kajitani, T. Koyama, H. Fujita, Paper presented at the Twelfth Polymer Symposium, Nagoya, Japan, Nov., 1963.
25. R. Endo, Nippon Gomu Kyokaishi (J. Rubber Ind. Japan), **34**, 527 (1961).
26. W. Cooper, D. E. Eaves, G. Vaughan, J. Polym. Sci., **59**, 241 (1962).
27. I. Ya. Poddubnyi, V. A. Grechanovskii, Vysokomol. Soedin., **6**, 64 (1964).
28. R. L. Scott, W. C. Carter, M. Magat, J. Am. Chem. Soc., **71**, 220 (1949).
29. D. J. Pollock, L. J. Elyash, T. W. Dewitt, J. Polym. Sci., **15**, 335 (1955).
30. V. S. Skazka, M. Kozhokaru, G. A. Fomin, L. F. Roguleva, Vysokomol. Soedin. A, **9**, 177 (1967).

31. W. E. Mochel, J. B. Nichols, J. Am. Chem. Soc., **71**, 3435 (1949).
32. W. E. Mochel, J. B. Nichols, C. J. Mighton, J. Am. Chem. Soc., **70**, 2185 (1948).
33. W. E. Mochel, J. B. Nichols, Ind. Eng. Chem., **43**, 154 (1951).
34. K. Hanafusa, A. Teramoto, H. Fujita, J. Phys. Chem., **70**, 4004 (1966).
35. K. Kawahara, T. Norisuye, H. Fujita, J. Chem. Phys., **49**, 4339 (1968).
36. S. A. Pavlova, T. A. Soboleva, A. P. Supran, Vysokomol. Soedin., **6**, 122 (1964).
37. H. L. Wagner, P. J. Flory, J. Am. Chem. Soc., **74**, 195 (1952).
38. K. Altgelt, G. V. Schulz, Makromol. Chem., **36**, 209 (1960).
39. W. C. Carter, R. L. Scott, M. Magat, J. Am. Chem. Soc., **68**, 1480 (1946).
40. I. Ya. Poddubnyi, V. A. Grechanovskii, A. V. Podalinskii, Vysokomol. Soedin., **5**, 1588 (1964).
41. M. Abe, M. Iwama, T. Homma, Kogyo Kagaku Zasshi (J. Chem. Soc. Japan, Ind. Chem. Sec.), **72**, 2313 (1969).
42. W. H. Beattie, C. Booth, J. Appl. Polym. Sci., **7**, 507 (1963).
43. H. Brody, M. Ladaeki, R. Milkovitch, M. Szwarc, J. Polym. Sci., **25**, 221 (1959).
44. I. Y. Poddubnyi, E. G. Ehrenberg, J. Polym. Sci., **57**, 545 (1962). [K = 13.3 in Benzene is for Li-type polymers and 11.2 for Ziegler-types. No difference is found in dioxane.]
45. H. C. Tingey, R. H. Ewart, G. E. Hulse, unpublished work; cited in Ref. 28; see also, D. M. French, R. H. Ewart, Anal. Chem., **19**, 165 (1947).
46. J. A. Yanko, J. Polym. Sci., **3**, 576 (1958).
47. T. Homma, H. Fujita, J. Appl. Polym. Sci., **9**, 1701 (1965). [Bound styrene, 24%.]
48. J.-Y. Chien, W. Chin, Y.-S. Cheng, Kolloidn. Zh., **19**, 515 (1957).
49. T. G. Fox, P. J. Flory, J. Am. Chem. Soc., **73**, 1909 (1951).
50. T. G. Fox, P. J. Flory, J. Phys. Colloid Chem., **53**, 197 (1949).
51. C. E. H. Bawn, E. S. Hill, M. A. Wajid, Trans. Faraday Soc., **52**, 1651 (1956).
52. W. R. Krigbaum, P. J. Flory, J. Am. Chem. Soc., **75**, 1775 (1953).
53. A. Ram, Thesis, Mass. Inst. Tech., Cambridge, Mass., USA, 1961.
54. F. R. Cottrell, E. W. Merrill, K. A. Smith, J. Polym. Sci. A-2, **7**, 1415 (1969).
55. P. J. Flory, J. Am. Chem. Soc., **65**, 372 (1943).
56. U. Bianchi, M. Dalpiaz, E. Patrone, Makromol. Chem., **80**, 112 (1964).
57. N. M. Tret'yakova, L. V. Kosmodem'yanskii, R. G. Romanova, E. G. Lazaryants, Vysokomol. Soedin. A, **12**, 2754 (1970).
58. R. Chiang, J. Phys. Chem., **70**, 2348 (1966).
59. R. W. Wheatcraft, cited in H. Wesslau, Makromol. Chem., **20**, 111 (1956).
60. E. Duch, L. Kuechler, Z. Elektrochem., **60**, 218 (1956).
61. J. T. Atkins, L. T. Muus, C. W. Smith, E. T. Pieski, J. Am. Chem. Soc., **79**, 5089 (1957).
62. A. Kotera, T. Saito, K. Takamizawa, Y. Miyazawa, Rept. Progr. Polymer Phys. Japan, **3**, 58 (1960).
63. P. S. Francis, R. Cooke, Jr., J. H. Elliott, J. Polym. Sci., **31**, 453 (1957).
64. P. M. Henry, J. Polym. Sci., **36**, 3 (1959).
65. R. Chiang, J. Phys. Chem., **69**, 1645 (1965).
66. R. Chiang, J. Polym. Sci., **36**, 91 (1959).
67. L. H. Tung, J. Polym. Sci., **24**, 333 (1957).
68. L. H. Tung, J. Polym. Sci., **36**, 287 (1959).
69. C. J. Stacy, R. L. Arnett, J. Phys. Chem., **69**, 3109 (1965). See also Ref. 73.
70. Q. A. Trementozzi, J. Polym. Sci., **36**, 113 (1959).
71. H. Wesslau, Makromol. Chem., **26**, 96 (1958).
72. H. S. Kaufmann, E. K. Walsh, J. Polym. Sci., **26**, 124 (L) (1957).
73. C. J. Stacy, R. L. Arnett, J. Polym. Sci. A, **2**, 167 (1964).
74. W. R. Krigbaum, Q. A. Trementozzi, J. Polym. Sci., **28**, 295 (1958).
75. R. A. Mendelson, E. E. Drott, J. Polym. Sci. B, **11**, 795 (1968).
76. W. F. Busse, R. Longworth, J. Polym. Sci., **58**, 49 (1962).
77. K. Ueberreiter, H.-J. Orthman, S. Sorge, Makromol. Chem., **8**, 21 (1952).
78. I. Harris, J. Polym. Sci., **8**, 353 (1952).
79. Q. A. Trementozzi, J. Polym. Sci., **23**, 887 (1957).
80. K. H. Meyer, A. van der Wyk, Helv. Chim. Acta, **18**, 1067 (1935).
81. W. R. Krigbaum, J. E. Kurz, P. Smith, J. Phys. Chem., **65**, 1984 (1961).
82. R. Endo, K. Iimura, M. Takeda, Bull. Chem. Soc. Japan, **37**, 950 (1964).
83. S. S. Stivala, R. J. Valles, D. W. Levi, J. Appl. Polym. Sci., **7**, 97 (1963).
84. J. B. Kinsinger, L. E. Ballard, J. Polym. Sci. A, **3**, 3963 (1965).
85. I. H. Billick, J. P. Kennedy, Polymer, **6**, 175 (1965).
86. J. L. Jungnickel, F. T. Weiss, J. Polym. Sci., **49**, 437 (1961).
87. D. L. Flowers, W. A. Hewett, R. D. Mullineaux, J. Polym. Sci. A, **2**, 2305 (1964).
88. J. B. Kinsinger, R. E. Hughes, J. Phys. Chem., **63**, 2002 (1959).
89. F. Danusso, G. Moraglio, Rend. Acad. Naz. Lincei, **25**, 509 (1958).
90. J. B. Kinsinger, R. E. Hughes, J. Phys. Chem., **67**, 1922 (1963).
91. F. Danusso, G. Moraglio, Makromol. Chem., **28**, 250 (1958).
92. L. Westerman, J. Polym. Sci. A, **1**, 411 (1963).
93. H. J. L. Schuurmans, R. A. Mendelson, unpublished work; cited in E. Kohn, H. J. L. Schuurmans, J. V. Cavender, R. A. Mendelson, J. Polym. Sci., **58**, 681 (1962).
94. A. Nakajima, A. Saijo, J. Polym. Sci. A-2, **6**, 735 (1968).
95. A. Kotera, K. Takamizawa, T. Kamata, H. Kawaguchi, Rept. Progr. Polym. Phys. Japan, **4**, 131 (1961).

96. P. Parrini, F. Sebastiano, G. Messina, Makromol. Chem., **38**, 27 (1960).
97. R. Chiang, J. Polym. Sci., **38**, 235 (1958).
98. G. Ciampa, Chim. Ind. (Milan), **38**, 298 (1956).
99. F. Ang, H. Mark, Monatsh. Chem., **88**, 427 (1957).
100. H. Inagaki, T. Miyamoto, S. Ohta, J. Phys. Chem, **70**, 3420 (1966).
101. W. Scholtan, Makromol. Chem., **14**, 169 (1954).
102. E. Collinson, F. S. Dainton, G. S. McNaughton, Trans. Faraday Soc., **53**, 489 (1957).
103. L. Trossarelli, M. Meirone, J. Polym. Sci., **57**, 445 (1962).
104. S. Newman, W. R. Krigbaum, C. Laugier, P. J. Flory, J. Polym. Sci., **14**, 451 (1954).
105. A. Takahashi, N. Hayashi, I. Kagawa, Kogyo Kagaku Zasshi (J. Chem. Soc. Japan, Ind. Chem. Sec.), **60**, 1059 (1957). See also Ref. 126.
106. V. N. Tsvetkov, S. Ya. Lyubina, T. V. Barskaya, Vysokomol. Soedin., **6**, 806 (1964); V. N. Tsvetkov, V. S. Skazka, G. V. Tarasova, V. M. Yamshchikov, S. Ya Lyubina, *ibid.* Ser. A, **10**, 74 (1968).
107. A. Takahashi, S. Yamori, I. Kagawa, Nippon Kagaku Zasshi (J. Chem. Soc. Japan, Pure Chem. Sec.), **83**, 11 (1962).
108. A. Takahashi, T. Kamei, I. Kagawa, Nippon Kagaku Zasshi (J. Chem. Soc. Japan, Pure Chem. Sec.), **83**, 14 (1962).
109. A. Takahashi, M. Nagasawa, J. Am. Chem. Soc., **86**, 543 (1964).
110. I. Noda, T. Tsuge, M. Nagasawa, J. Phys. Chem., **74**, 710 (1970).
111. A. Soda, I. Kagawa Nippon Kagaku Zasshi (J. Chem. Soc. Japan, Pure Chem. Sec.), **83**, 412 (1962).
112. G. Saini, L. Trossarelli, Atti Accad. Sci. Torino, Classe Sci. Fis., Mat. Nat., **90**, 410 (1955–56). cf. Ref. 10.
113. G. B. Rathmann, F. A. Bovey, J. Polym. Sci., **15**, 544 (1955).
114. M. Giurgea, C. Ghita, I. Baltog, A. Lupu, J. Polym. Sci. A-2, **4**, 529 (1966).
115. H. Sumitomo, Y. Hachihama, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **10**, 544 (1953). See also Y. Hachihama, H. Sumitomo, Technol. Rept. Osaka Univ., **3**, 385 (1953).
116. H. Sumitomo, Y. Hachihama, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **12**, 479 (1955). See also Y. Hachihama, H. Sumitomo, Technol. Rept. Osaka Univ., **5**, 485 (1956).
117. D. Mangaraj, S. K. Patra, Makromol. Chem., **107**, 230 (1967).
118. I. G. Soboleva, N. V. Makletsova, S. S. Medvedev, Dokl. Akad. Nauk. SSSR, **94**, 289 (1954).
119. E. S. Cohn, T. A. Orofino, I. L. Scogna, unpublished work; cited in Ref. 10.
120. J. E. Mark, R. A. Wessling, R. E. Hughes, J. Phys. Chem., **70**, 1895 (1966).
121. R. A. Wessling, J. E. Mark, E. Hamori, J. E. Mark, J. Phys. Chem., **70**, 1903 (1966).
122. S. Krause, unpublished work; cited in Ref. 10.
123. H. Staudinger, H. Warth, Z. Prakt. Chem., **155**, 261 (1940).
124. L. Trossareli, G. Saini, Atti Accad. Sci. Torino, Classe Sci. Fis., Mat. Nat., **90**, 419 (1955–56); cited in Ref. 10.
125. L. Trossareli, G. Saini, Atti Accad. Sci. Torino, Classe Sci. Fis., Mat. Nat., **90**, 431 (1955–56); cited in Ref. 10.
126. H. Ito, S. Shimizu, S. Suzuki, Kogyo Kagaku Zasshi (J. Chem. Soc. Japan, Ind. Chem. Sec.), **59**, 930 (1956).
127. G. M. Guzman, Anales Real Soc. Espan. Fis. Quim. (Madrid) Ser. B, **52**, 377 (1956); cited in Ref. 10.
128. Rohm and Haas, old data; reported in Ref. 10.
129. H. Matsuda, K. Yamano, H. Inagaki, J. Polym. Sci. A-2, **7**, 609 (1969).
130. J. N. Sen, S. R. Chatterjee, S. R. Palit, J. Sci. Ind. Res. B, (India), **11**, 90 (1952); cited in Ref. 10.
131. A. Kotera, T. Saito, Y. Watanabe, M. Ohama, Makromol. Chem., **87**, 195 (1965).
132. N. T. Srinivasan, M. Santappa, Makromol. Chem., **27**, 61 (1958).
133. K. Karunakaran, M. Santappa, J. Polym. Sci. A-2, **6**, 713 (1968).
134. Y. Fujisaki, H. Kobayashi, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **19**, 73, 81 (1962).
135. H. Inagaki, K. Hayashi, T. Matsuo, Makromol. Chem., **84**, 80 (1965).
136. T. Shibukawa, M. Sone, A. Uchida, K. Iwahori, J. Polym. Sci. A-1, **6**, 147 (1968).
137. W. Scholtan, H. Marzolph, Makromol. Chem., **57**, 52 (1962).
138. J. Bisschops, J. Polym. Sci., **17**, 81 (1955).
139. R. L. Cleland, W. H. Stockmayer, J. Polym. Sci., **17**, 473 (1955).
140. R. F. Onyon, J. Polym. Sci., **22**, 13 (1956).
141. W. R. Krigbaum, A. M. Kotliar, J. Polym. Sci., **32**, 323 (1958).
142. C. H. Bamford, A. D. Jenkins, R. Johnston, E. F. T. White, Trans. Faraday Soc., **55**, 168 (1959).
143. P. F. Onyon, J. Polym. Sci., **37**, 315 (1959).
144. L. H. Peebles, J. Polym. Sci. A, **3**, 361 (1965).
145. H. Kobayashi, J. Polym. Sci., **39**, 369 (1959).
146. R. Chiang, J. C. Stauffer, J. Polym. Sci. A-2, **5**, 101 (1967).
147. N. M. Wiederhorn, A. R. Brown, J. Polym. Sci., **8**, 651 (1952).
148. A. Katchalsky, H. Eisenberg, J. Polym. Sci., **6**, 145 (1951).
149. R. Arnold, S. R. Caplan, Trans. Faraday Soc., **51**, 857 (1955).
150. R. van Leemput, R. Stein, J. Polym. Sci. A, **1**, 985 (1963). [*K* values are given for monodisperse polymers.]
151. Rohm and Haas, old data; reported in Ref. 10.
152. S. N. Chinai, R. A. Guzzi, J. Polym. Sci., **21**, 417 (1956).
153. A. S. Nair, M. S. Muthana, Makromol. Chem., **47**, 114 (1961).
154. V. N. Tsvetkov, S. I. Klenin, Zh. Tekhn. Fiz., **29**, 1393 (1959).
155. Z. Menčik, Chem. Listy, **46**, 407 (1952).
156. S. N. Chinai, R. J. Valles, J. Polym. Sci., **39**, 363 (1959).
157. M. Kozhokaryu, V. S. Skazka, K. G. Berdnikova, Vysokomol. Soedin., **8**, 1063 (1967).
158. S. N. Chinai, R. A. Guzzi, J. Polym. Sci., **41**, 475 (1959).

159. H. T. Lee, D. W. Levi, J. Polym. Sci., **47**, 449 (1960).
160. F. E. Didot, S. N. Chinai, D. W. Levi, J. Polym. Sci., **43**, 557 (1960).
161. S. N. Chinai, R. J. Samuels, J. Polym. Sci., **19**, 463 (1956).
162. K. Karunakaran, M. Santappa, Makromol. Chem., **111**, 20 (1968).
163. V. N. Tsvetkov, D. Khardi, I. N. Shtennikova, Ye. V. Komeyeva, G. F. Pirogova, K. Nitrai, Vysokomol. Soedin. Ser. A, **11**, 349 (1969).
164. J.-Y. Chieng, M.-H. Shih, Z. Physik. Chem. (Leipzig), **207**, 60 (1957).
165. S. N. Chinai, J. Polym. Sci., **25**, 413 (1957).
166. R. J. Valles, J. Polym. Sci. A, **3**, 3853 (1965); see also R. J. Valles, E. C. Schramm, *ibid.*, **3**, 3664 (1965); and Ref. 167.
167. R. J. Valles, Makromol. Chem., **100**, 167 (1967).
168. M. Iwama, H. Utiyama, M. Kurata, J. Makromol. Chem., **1**, 701 (1966).
169. G. Meyerhoff, G. V. Schulz, Makromol. Chem., **7**, 294 (1951).
170. J. Bischoff, V. Desreux, Bull Soc. Chim. Belges, **61**, 10 (1952).
171. G. V. Schulz, G. Meyerhoff, Z. Elektrochem., **56**, 904 (1952).
172. G. V. Schulz, H.-J. Cantow, G. Meyerhoff, J. Polym. Sci., **10**, 79 (1953).
173. H.-J. Cantow, G. V. Schulz, Z. Physik. Chem. (Frankfurt), **2**, 117 (1954); see also H.-J. Cantow, G. V. Schulz, *ibid.*, **1**, 365 (1954).
174. S. N. Chinai, J. D. Matlack, A. L. Resnick, R. J. Samuels, J. Polym. Sci., **17**, 391 (1955).
175. F. W. Billmeyer, Jr., C. B. de Than, J. Am. Chem. Soc., **77**, 4763 (1955).
176. S. L. Kapur, J. Sci, Ind. Res. B (India), **15**, 239 (1956).
177. W. R. Moore, R. J. Fort, J. Polym. Sci. A, **1**, 929 (1963).
178. E. Cohn-Ginsberg, T. G. Fox, H. F. Mason, Polymer, **3**, 97 (1962).
179. T. G. Fox, Polymer, **3**, 111 (1962).
180. E. S. Cohn, T. G. Fox, unpublished work; cited in Ref. 10.
181. A. F. V. Eriksson, Acta Chem. Scand., **7**, 623 (1953).
182. J. H. Baxendale, S. Bywater, M. G. Evans, J. Polym. Sci., **1**, 237 (1946).
183. J.-Y. Chien, L.-H. Shin, K.-I. Shin, H. H. Pao, Acta Chim. Sinica., **23**, 215 (1957). cf. Ref. 10.
184. K. G. Schoen, G. V. Schulz, Z. Physik. Chem. (Frankfurt), **2**, 197 (1954).
185. T. G. Fox, J. B. Kinsinger, H. F. Mason, E. M. Schuele, Polymer, **3**, 71 (1962).
186. J. Hakozaki, Nippon Kagaku Zasshi (J. Chem. Soc. Japan, Pure Chem. Sec.), **82**, 155 (1961).
187. H. Inagaki, S. Kawai, Makromol. Chem., **79**, 42 (1964). See also Ref. 197.
188. G. V. Schulz, A. Dinglinger, Z. Prakt, Chem., **158**, 137 (1941).
189. H. J. Cantow, J. Pouget, C. Wippler, Makromol. Chem., **14**, 110 (1954).
190. V. N. Tsvetkov, S. I. Klenin, J. Polym. Sci., **30**, 187 (1958).
191. I. Sakurada, A. Nakajima, O. Yoshizaki, K. Nakamae, Kolloid-Z., **186**, 41 (1962).
192. A. F. V. Eriksson, Acta Chem. Scand., **10**, 378 (1956).
193. E. F. Casassa, W. H. Stockmayer, Polymer, **3**, 53 (1962).
194. E. Hamori, L. R. Prusinowski, P. G. Sparks, R. E. Hughes, J. Phys. Chem., **69**, 1101 (1965).
195. E. Patrone, E. Bianchi, Makromol. Chem., **94**, 52 (1966).
196. S. N. Chinai, C. W. Bondurant, J. Polym. Sci., **22**, 555 (1956).
197. G. V. Schulz, R. Kirste, Z. Phys. Chem. (Frankfurt), **30**, 171 (1961).
198. S. Krause, E. Cohn-Ginsberg, J. Phys. Chem., **67**, 1479 (1963).
199. S. Krause, E. Cohn-Ginsberg, Polymer, **3**, 565 (1962).
200. J. G. Fee, W. S. Port, L. P. Whitnauer, J. Polym. Sci., **33**, 95 (1958).
201. S. N. Chinai, A. L. Resnick, H. T. Lee, J. Polym. Sci., **33**, 471 (1958).
202. N. Fuhrman, R. B. Mesrobian, J. Am. Chem. Soc., **76**, 3281 (1954).
203. C. G. Overberger, E. M. Pearce, N. Mayes, J. Polym. Sci., **34**, 109 (1959).
204. R. C. Schulz, S. Suzuki, H. Cherdron, W. Kern, Makromol. Chem., **53**, 145 (1962).
205. Ye. V. Korneyeva, V. N. Tsvetkov, P. N. Lavrenko, Vysokomol. Soedin. A, **12**, 1369 (1970).
206. J. A. Manson, G. J. Arquette, Makromol. Chem., **37**, 187 (1960).
207. S. Arichi, J. Sci. Hiroshima Univ. Ser. A-II, **29**, 97 (1965).
208. P. J. Flory, F. Leutner, J. Polym. Sci., **3**, 880 (1948).
209. K. Dialer, K. Vogler, F. Patat, Helv. Chim. Acta, **35**, 869 (1952).
210. H. A. Dieu, J. Polym. Sci., **12**, 417 (1954).
211. L. E. Miller, F. A. Hamm, J. Phys. Chem., **57**, 110 (1953).
212. A. Nakajima, K. Furutate, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **6**, 460 (1949).
213. M. Matsumoto, Y. Ohyanagi, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **17**, 191 (1960). [The second line is the constant corrected for monodisperse sample.]
214. T. Matsuo, H. Inagaki, Makromol. Chem., **55**, 151 (1962).
215. M. Matsumoto, K. Imai, J. Polym. Sci., **24**, 125 (1957).
216. M. Litt, F. R. Eirich, J. Polym. Sci., **45**, 379 (1960).
217. A. C. Ciferri, M. Kryszewski, G. Weil, J. Polym. Sci., **27**, 167 (1958).
218. A. Ciferri, M. Lauretti, Ann. Chim. (Rome), **48**, 198 (1958).
219. M. Sato, Y. Koshiishi, M. Asahina, J. Polym. Sci. B, **1**, 233 (1963).
220. H. Inagaki, J. Nakazawa, private communication.
221. J. W. Breitenbach, E. L. Forster, A. J. Renner, Kolloid-Z., **127**, 1 (1952).
222. C. Bier, H. Kramer, Makromol. Chem., **18/19**, 151 (1955).
223. G. Ciampa, H. Schwindt, Makromol. Chem., **21**, 169 (1954).
224. F. Danusso, G. Moraglio, S. Cazzera, Chim. Ind. (Milan), **36**, 883 (1954).
225. Z. Mencik, Collection Czech. Chem. Commun., **21**, 517 (1956).

226. W. R. Moore, R. J. Hutchinson, Nature, **200**, 1095 (1963).
227. J. Vavra, J. Lapcik, J. Sabados, J. Polym. Sci. A-2, **5**, 1305 (1967).
228. M. Bohdanecky, K. Solc, P. Kratochvil, M. Kolinsky, M. Pyska, D. Lim, J. Polym. Sci. A-2, **5**, 343 (1967).
229. H. Batzer, A. Nisch, Makromol. Chem., **22**, 131 (1957).
230. M. Freeman, P. P. Manning, J. Polym. Sci. A, **2**, 2017 (1964).
231. A. Takahashi, M. Ohara, I. Kagawa, Kogyo Kagaku Zasshi (J. Chem. Soc. Japan, Ind. Chem. Sec.), **66**, 960 (1963).
232. M. Sato, Y. Koshiishi, M. Asahina, cited in: A. Nakajima, K. Kato, Makromol. Chem., **95**, 52 (1966).
233. T. Kobayashi, Bull. Chem. Soc. Japan, **35**, 726 (1962).
234. E. K. Walsh, H. S. Kaufman, J. Polym. Sci., **26**, 1 (1957).
235. M. L. Wallach, M. A. Kabayama, J. Polym. Sci. A-1, **4**, 2667 (1966).
236. M. Ueda, K. Kajitani, Makromol. Chem., **108**, 138 (1967), and private communication.
237. W. R. Moore, M. Murphy, J. Polym. Sci., **56**, 519 (1962).
238. V. N. Tsvetkov, S. Ya. Kotlyar, Zh. Fiz. Khim., **30**, 1100 (1956).
239. G. Saini, L. Maldifassi, L. Trossarelli, Ann. Chim. (Rome), **44**, 1954 (1954); cited in H.-G. Elias, Kunststoffe-Plastics, **4**, 8 (1961).
240. G. S. Misra, V. P. Gupta, Makromol. Chem., **71**, 110 (1964).
241. R. H. Wagner, J. Polym. Sci., **2**, 21 (1947).
242. S. N. Chinai, P. C. Scherer, D. W. Levi, J. Polym. Sci., **17**, 117 (1955).
243. K. Z. Fattakhov, E. S. Pisarenko, L. N. Verkotina, Kolloidn. Zh., **18**, 101 (1956).
244. M. Matsumoto, Y. Ohyanagi, J. Polym. Sci., **46**, 441 (1960). Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **17**, 1 (1960).
245. R. Naito, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **16**, 7 (1959).
246. Bevak, Thesis, Mass. Inst. Tech., Cambridge, Mass., USA, 1955.
247. V. Kalpagam, R. Rao, J. Polym. Sci. A, **1**, 233 (1963).
248. M. R. Rao, V. Kalpagam, J. Polym. Sci., **49**, S 14 (1961).
249. A. Nakajima, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **11**, 142 (1954).
250. V. V. Varadiah, J. Polym. Sci., **19**, 477 (1956).
251. G. C. Berry, L. M. Hobbs, V. C. Long, Polymer, **5**, 31 (1964).
252. A. R. Schulz, J. Am. Chem. Soc., **76**, 3423 (1954); see also R. O. Howard, Thesis, Mass. Inst. Tech., Cambridge, MA, 1952.
253. H.-G. Elias, F. Patat, Makromol. Chem., **25**, 13 (1957).
254. M. Abe, H. Fujita, J. Phys. Chem., **69**, 3263 (1965).
255. M. Matsumoto, Y. Ohyanagi, J. Polym. Sci., **50**, S1 (1961); see also Ref. 244.
256. C. J. Kurian, M. S. Muthana, Makromol. Chem., **29**, 1 (1959).
257. K. Fujii, S. Imoto, J. Ukida, M. Matsumoto, Kobunshi Kagaku (Chem. High Polym., Tokyo), **19**, 581 (1962).
258. H. Hopff, J. Dohany, Makromol. Chem., **69**, 131 (1963).
259. H. Eisenberg, D. Woodside, J. Chem. Phys., **36**, 1844 (1962).
260. K. Dialer, R. Kerber, Makromol. Chem., **17**, 56 (1955).
261. F. Patat, K. Vogler, Helv. Chim. Acta, **35**, 128 (1952).
262. J. Springer, K. Ueberreiter, R. Wenzel, Makromol. Chem., **96**, 122 (1966).
263. J. M. Barrales-Rienda, D. C. Pepper, Polymer, **8**, 337 (1967).
264. L. A. Utracki, R. Simha, Makromol. Chem., **117**, 94 (1968). See also, L. A. Utracki, R. Simha, N. Eliezer, Polymer, **10**, 43 (1969).
265. F. S. Holahan, S. S. Stivala, D. W. Levi, J. Polym. Sci. A, **3**, 3987 (1955).
266. N. Kuwahara, K. Ogino, M. Konuma, N. Iida, M. Kaneko, J. Polym. Sci. A-2, **4**, 173 (1966).
267. V. V. Korshak, S. L. Sosin, V. P. Alekseeva, J. Polym. Sci., **52**, 213 (1961).
268. V. E. Eskin, O. Z. Korotkina, Vysokomol. Soedin., **1**, 1580 (1959).
269. V. E. Eskin, O. Z. Korotkina, Vysokomol. Soedin., **2**, 272 (1960).
270. M. Cantow, G. Meyerhoff, G. V. Schulz, Makromol. Chem., **49**, 1 (1961).
271. G. Meyerhoff, Z. Phys. Chem. (Frankfurt), **4**, 335 (1955).
272. D. C. Pepper, J. Polym. Sci., **7**, 347 (1951). See also D. C. Pepper, Proc. Roy. Dublin Soc., **25**, 239 (1951).
273. G. S. Misra, R. C. Rastogi, V. P. Gupta, Makromol. Chem., **50**, 72 (1961).
274. W. R. Krigbaum, P. J. Flory, J. Polym. Sci., **11**, 37 (1953).
275. T. A. Orofino, F. Wenger, J. Phys. Chem., **67**, 566 (1963).
276. C. E. H. Bawn, C. Freeman, A. Kamaliddin, Trans. Faraday Soc., **46**, 1107 (1950).
277. C. Rossi, Chimica delle Macromolecole (Sept. 1961); Consiglio Nazionalle delle Ricerche, Roma (1963) p. 153.
278. P. Outer, C. I. Carr, B. H. Zimm, J. Chem. Phys., **18**, 830 (1950).
279. J. Oth, V. Desreux, Bull. Soc. Chim. Belges, **63**, 285 (1954).
280. T. Oyama, K. Kawahara, M. Ueda, Nippon Kagaku Zasshi (J. Chem. Soc. Japan, Pure Chem. Sec.), **79**, 727 (1958).
281. U. Bianchi, V. Magnasso, C. Rossi, Ric. Sci., **58**, 1412 (1958).
282. U. Bianchi, V. Magnasso, C. Rossi, Chim. Ind. (Milan), **40**, 263 (1958); See also, Ref. 276.
283. H. Utiyama, Thesis, Kyoto Univ., Kyoto, Japan, 1962.
284. R. Endo, M. Takeda, J. Polym. Sci., **56**, 28 (1962). [The tabulated constants are calculated from the original figure.]
285. L. Utracki, R. Simha, J. Phys. Chem., **67**, 1052 (1963).
286. H. Inagaki, H. Suzuki, M. Fujii, T. Matsuo, J. Phys. Chem., **70**, 1718 (1966). See also H. Inagaki, H. Suzuki, M. Kurata, J. Polym. Sci. C, **15**, 409 (1966).
287. T. Homma, K. Kawahara, H. Fujita, M. Ueda, Makromol. Chem., **67**, 132 (1963).
288. H. J. Cantow, Makromol. Chem., **30**, 169 (1959).
289. L. A. Papazian, Polymer, **10**, 399 (1969).
290. H. Inagaki, private communication [based on the data given in Ref. 286, and also on unpublished data].
291. M. Abe, H. Fujita, J. Phys. Chem., **69**, 3263 (1965).

292. T. G. Fox, P. J. Flory, J. Am. Chem. Soc., **73**, 1915 (1951).
293. E. T. Dimitru, L. H. Cragg, unpublished work; cited in: P. J. Flory, "Principles of Polymer Chemistry", Cornell Univ. Press, Ithaca, New York, 1953, p. 615, Table XXVII.
294. J. W. Breitenbach, H. Gabler, O. F. Olaj, Makromol. Chem., **81**, 32 (1964).
295. R. H. Ewart, H. C. Tingey, M. Wales, unpublished work; cited in W. V. Smith, J. Am. Chem. Soc., **68**, 2063 (1946).
296. C. H. Bamford, M. J. S. Dewar, Proc. Roy. Soc. London A, **192**, 329 (1948).
297. H. Marzolph, G. V. Schulz, Makromol. Chem., **13**, 120 (1954).
298. C. Rossi, U. Bianchi, E. Bianchi, Makromol. Chem., **41**, 31 (1960); see also, Ref. 301.
299. S. N. Chinai, P. C. Scherer, C. W. Bondurat, D. W. Levi, J. Polym. Sci., **22**, 527 (1956).
300. F. Danusso, G. Moraglio, J. Polym. Sci., **24**, 161 (1957).
301. T. Altares, D. P. Wyman, V. R. Allen, J. Polym. Sci. A, **2**, 4533 (1964).
302. A. Yamamoto, M. Fujii, G. Tanaka, H. Yamakawa, Polym. J., **2**, 799 (1971).
303. G. C. Berry, J. Chem. Phys., **46**, 1338 (1967).
304. M. Morton, T. E. Helminiak, S. D. Gadkary, F. Bueche, J. Polym. Sci., **57**, 471 (1962).
305. J. M. G. Cowie, S. Bywater, Polymer, **6**, 197 (1965).
306. J. M. G. Cowie, E. L. Cussler, J. Chem. Phys., **46**, 4886 (1967).
307. G. Meyerhoff, Z. Physik. Chem. (Frankfurt), **23**, 100 (1960).
308. H. W. McCormick, J. Polym. Sci., **36**, 341 (1959).
309. R. N. Mukherjea, P. Remmp, J. Chim. Phys., **56**, 94 (1959).
310. J. M. G. Cowie, D. J. Worsfold, S. Bywater, Trans. Faraday Soc., **57**, 537 (1961).
311. F. Ang, J. Polym. Sci., **25**, 126 (1957).
312. G. Natta, F. Danusso, G. Moraglio, Makromol. Chem., **20**, 37 (1956). See also Ref. 300.
313. W. R. Krigbaum, D. K. Carpenter, S. Newman, J. Phys. Chem., **62**, 1586 (1958).
314. L. Trossarelli, E. Campi, G. Saini, J. Polym. Sci., **35**, 205 (1959).
315. C. D. Thurmond, B. H. Zimm, J. Polym. Sci., **8**, 477 (1952).
316. M. Kurata, M. Abe, M. Iwama, M. Matsushima, Polym. J., **3**, 729 (1972).
317. S. Minatono, Thesis, Kyoto Univ., Kyoto, Japan, 1966.
318. Y. Mitsuda, K. Osaki, L. Schrag, J. D. Ferry, Polym. J., **4**, 24 (1973).
319. K. Sakato, M. Kurata, Polym. J., **1**, 260 (1970).
320. I. Noda, K. Mizutani, T. Kato, T. Fujimoto, M. Nagasawa, Macromolecules, **3**, 787 (1970).
321. J. M. G. Cowie, S. Bywater, D. J. Worsfold, Polymer, **8**, 105 (1967).
322. A. F. Sirianni, D. J. Worsfold, S. Bywater, Trans. Faraday Soc., **55**, 2124 (1959).
323. M. Abe, K. Sakato, T. Kageyama, M. Fukatsu, M. Kurata, Bull. Chem. Soc. Japan, **41**, 2330 (1968).
324. I. Noda, S. Saito, T. Fujimoto, M. Nagasawa, J. Phys. Chem., **71**, 4048 (1967).
325. H. W. McCormick, J. Polym. Sci., **41**, 327 (1959).
326. B. J. Cottam, J. M. G. Cowie, S. Bywater, Makromol. Chem., **86**, 116 (1965).
327. S. Okamura, T. Higashimura, Y. Imanishi, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **16**, 244 (1959).
328. J. M. G. Cowie, S. Bywater, J. Polym. Sci. A-2, **6**, 499 (1968).
329. A. Kotera, T. Saito, H. Matsuda, R. Kamata, Rept. Progr. Polym. Phys. Japan, **3**, 51 (1960).
330. A. K. Chaudhuri, D. K. Sarkar, S. Palit, Makromol. Chem., **111**, 36 (1968).
331. G. Tanaka, S. Imai, H. Yamakawa, J. Chem. Phys., **52**, 2639 (1970).
332. V. P. Budtov, V. M. Belyayev, Vysokomol. Soedin. A, **12**, 1909 (1970).
333. C. S. H. Chen, R. F. Stamm, J. Polym. Sci., **58**, 369 (1962).
334. I. Sakurada, Y. Sakaguchi, S. Kokuryo, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **17**, 227 (1960).
335. W. Burchard, M. Nosseir, Makromol. Chem., **82**, 109 (1964).
336. Y. Sakaguchi, J. Nishino, K. Tsugawa, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **20**, 661 (1963).
337. S. K. Patra, Thesis, I. I. T. Kharagpur, India 1965; cited in S. K. Patra, D. Mangaraj, Makromol. Chem., **111**, 168 (1968).
338. J. Parrod, J. Elles, J. Polym. Sci., **29**, 411 (1958).
339. E. S. Cohn, unpublished work; cited in Ref. 10.
340. V. N. Tsvetkov, O. V. Kalisov, Zh. Fiz. Khim., **33**, 710 (1959).
341. O. V. Kalisov, I. N. Shtennikova, Vysokomol. Soedin., **1**, 842 (1959).
342. V. N. Tsvetkov, S. I. Klenin, J. Polym. Sci., **30**, 187 (1958).
343. M. Tricot, V. Desreux, Makromol. Chem., **149**, 185 (1971).
344. E. S. Cohn, I. L. Scogna, T. A. Orofino, unpublished work; cited in Ref. 10.
345. J. Hakozaki, Nippon Kagaku Zasshi (J. Chem. Soc. Japan, Pure Chem. Sec.), **82**, 158 (1961).
346. V. N. Tsvetkov, V. S. Skazka, N. A. Nikitin, I. B. Stepanenko, Vysokomol. Soedin., **6**, 69 (1964).
347. W. Kern, D. Brawn, Makromol. Chem., **27**, 23 (1958).
348. K. Takashima, G. Tanaka, H. Yamakawa, Polym. J., **2**, 245 (1971).
349. Y. Noguchi, A. Aoki, G. Tanaka, H. Yamakawa, J. Chem. Phys., **52**, 2651 (1970).
350. K. Matsumura, Makromol. Chem., **124**, 204 (1969).
351. R. B. Mohite, S. Gundiah, S. L. Kapur, Makromol. Chem., **116**, 280 (1968).
352. A. Kotera, T. Saito, H. Matsuda, R. Kamata, Rept. Progr. Polym. Phys. Japan, **3**, 51 (1960).
353. N. Kuwahara, K. Ogino, A. Kasai, S. Ueno, M. Kaneko, J. Polym. Sci. A, **3**, 985 (1965).
354. G. Greber, J. Tolle, W. Burchard, Makromol. Chem., **71**, 47 (1964).
355. J. E. Davis, Thesis, Mass. Ins. Tech., Cambridge, Mass., USA, 1960.
356. E. F. Frisman, L. F. Shalaeva, Dokl. Akad. Nauk. SSSR, **101**, 970 (1955).

357. V. E. Eskin, K. Z. Gumargalieva, Vysokomol. Soedin., **2**, 265 (1960).
358. N. Kuwahara, K. Ogino, M. Konuma, N. Iida, M. Kaneko, J. Polym. Sci. A-2, **4**, 173 (1966).
359. V. E. Eskin, L. N. Andreeva, Vysokomol. Soedin., **3**, 435 (1961).
360. D. Braun, T.-O. Ahn, W. Kern, Makromol. Chem., **53**, 154 (1962).
361. S. Minatono, Thesis, Kyoto Univ., Kyoto, Japan, 1967.
362. Y. Imanishi, T. Higashimura, S. Okamura, Kobunshi Kagaku (Chem. High Polym., Japan, Tokyo), **22**, 241 (1965).
363. G. Ceccorulli, M. Pizzoli, G. Stea, Makromol. Chem., **142**, 153 (1971).
364. W. A. Pryor, T.-L. Huang, Macromolecules, **2**, 70 (1969).
365. M. Kato, T. Nakagawa, H. Akamatsu, Bull. Chem. Soc. Japan, **33**, 322 (1960).
366. A. Takahashi, T. Kato, M. Nagasawa, J. Phys. Chem., **71**, 2001 (1967).
367. G. Sitaramaiah, D. Jacobs, Polymer, **11**, 165 (1970).
368. N. Kuwahara, S. Higashide, M. Nakata, M. Kaneko, J. Polym. Sci. A-2, **7**, 285 (1969).
369. G. M. Chetyrkina, V. G. Aldoshin, S. Y. Frenkel, Vysokomol. Soedin., **1**, 1133 (1959).
370. V. N. Tsvetkov, V. G. Aldoshin, Zh. Fiz. Khim., **33**, 2767 (1959).
371. S. Arichi, Bull. Chem. Soc. Japan, **39**, 439 (1966).
372. D. O. Jordan, A. R. Mathieson, M. R. Porter, J. Polym. Sci., **21**, 473 (1956).
373. J. B. Berkowitz, M. Yamin, R. M. Fuoss, J. Polym. Sci., **28**, 69 (1958).
374. A. G. Boyes, U-P. Strauss, J. Polym. Sci., **22**, 463 (1956).
375. M. Miura, Y. Kubota, T. Masuzukawa, Bull. Chem. Soc. Japan, **38**, 316 (1965).
376. C. Garbuglio, L. Crescentini, A. Mula, G. B. Gechele, Makromol. Chem., **97**, 97 (1966); see also Ref. 381.
377. H. Sato, T. Yamamoto, Nippon Kagaku Zasshi (J. Chem. Soc. Japan, Pure Chem. Sec.), **80**, 1393 (1959).
378. G. B. Levy, H. P. Frank, J. Polym. Sci., **17**, 247 (1955); see also Ref. 380.
379. K. Dialer, K. Vogler, Makromol. Chem., **6**, 191 (1951).
380. G. B. Levy, H. P. Frank, J. Polym. Sci., **10**, 371 (1953).
381. G. B. Gechele, L. Crescentini, J. Polym. Sci. A, **3**, 3599 (1965).
382. W. Scholtan, Makromol. Chem., **7**, 209 (1951).
383. L. C. Cerney, T. E. Helminiak, J. F. Meier, J. Polym. Sci., **44**, 539 (1960).
384. H.-G. Elias, Makromol. Chem., **50**, 1 (1961).
385. G. Allen, C. Booth, S. J. Hurst, M. N. Jones, C. Price, Polymer, **8**, 391 (1967).
386. K. Yamamoto, H. Fujita, Polymer, **8**, 517 (1967).
387. D. R. Beech, C. Booth, J. Polym. Sci. A-2, **7**, 575 (1969).
388. C. Rossi, C. Cuniberti, J. Polym. Sci. B, **2**, 681 (1964).
389. C. Sadron, P. Rempp, J. Polym. Sci., **29**, 127 (1958).
390. T. A. Ritscher, H.-G. Elias, Makromol. Chem., **30**, 48 (1959).
391. H.-G. Elias, Kunststoffe-Plastics, **4**, 1 (1961).
392. D. K. Thomas, A. Charlesby, J. Polym. Sci., **42**, 195 (1960).
393. C. Rossi, E. Bianchi, G. Conio, Chim. Ind. (Milan), **45**, 1498 (1963).
394. F. E. Bailey, Jr., J. L. Kucera, L. G. Imhof, J. Polym. Sci., **32**, 517 (1958).
395. F. E. Bailey, Jr., R. W. Callard, J. Appl. Polym. Sci., **1**, 56 (1959).
396. C. Booth, R. Orme, Polymer, **11**, 626 (1970).
397. M. Matsushima, M. Fukatsu, M. Kurata, Bull. Chem. Soc. Japan, **41**, 2570 (1968).
398. K. Yamamoto, H. Fujita, Polymer, **7**, 557 (1966).
399. L. Hoehr, V. Jaacks, H. Cherdron, S. Iwabuchi, W. Kern, Makromol. Chem., **103**, 279 (1967).
400. J. Majer, Makromol. Chem., **86**, 253 (1965).
401. V. Kokle, F. W. Billmeyer, Jr., J. Polym. Sci. B, **3**, 47 (1965).
402. W. H. Stockmayer, L.-L. Chen, J. Polym. Sci. A-2, **4**, 437 (1966).
403. J. Majer, unpublished; cited in M. Kučera, E. Spousta, Makromol. Chem., **76**, 183 (1964).
404. N. A. Pravikova, Ye. B. Berman, Ye. B. Lyudvig, A. G. Davtyan, Vysokomol. Soedin. A, **12**, 580 (1970).
405. H. L. Wagner, K. F. Wissbrun, Makromol. Chem., **81**, 14 (1964).
406. G. Meyerhoff, U. Moritz, Makromol. Chem., **109**, 143 (1967).
407. W. Scholtan, S. Y. Lie, Makromol. Chem., **108**, 104 (1967).
408. G. Alen, A. C. Booth, M. M. Jones, Polymer, **5**, 195 (1964).
409. R. E. Hughes, J. Richards, unpublished work; cited in P. E. Ebert, C. C. Price, J. Polym. Sci., **34**, 157 (1959).
410. R. J. Valles, Makromol. Chem., **113**, 147 (1968). [The constants are expressed in terms of M_n].
411. J. Moacanin, J. Appl. Polym. Sci., **1**, 272 (1959).
412. M. Kurata, H. Utiyama, K. Kamada, Makromol. Chem., **88**, 281 (1965).
413. S. M. Ali, M. B. Huglin, Makromol. Chem., **84**, 117 (1965).
414. K. Yamamoto, A. Teramoto, H. Fujita, Polymer, **7**, 267 (1966).
415. P. J. Flory, P. B. Stickney, J. Am. Chem. Soc., **62**, 3032 (1940).
416. H. Batzer, G. Weisenberger, Makromol. Chem., **11**, 83 (1953).
417. H. Batzer, B. Mohr. Makromol. Chem., **8**, 217 (1952).
418. H. Batzer, Makromol. Chem., **10**, 13 (1953).
419. H. Batzer, Makromol. Chem., **5**, 5 (1950).
420. W. O. Baker, C. S. Fuller, J. H. Heiss, Jr., J. Am. Chem. Soc., **63**, 3316 (1941).
421. J. G. Erickson, J. Polym. Sci. A-1, **4**, 519 (1966).
422. U. P. Strauss, P. L. Wineman, J. Am. Chem. Soc., **80**, 2366 (1958).
423. M. Nakagaki, S. Ohashi, F. Minato, Bull. Chem. Soc. Japan, **36**, 341 (1963).
424. V. Crescenzi, P. J. Flory, J. Am. Chem. Soc., **86**, 141 (1964).
425. G. V. Schulz, A. Haug, Z. Phys. Chem. (Frankfurt), **34**, 328 (1962).
426. A. Haug, G. Meyerhoff, Makromol. Chem., **53**, 91 (1962).

427. P. J. Flory, L. Mandelkern, J. B. Kinsinger, W. B. Schulz, J. Am. Chem. Soc., **74**, 3364 (1952).
428. A. J. Barry, J. Appl. Phys., **17**, 1020 (1946).
429. H. H. Takimoto, C. T. Forbes, R. K. Laudenslager, J. Appl. Polym. Sci., **5**, 153 (1961).
430. A. Ya. Korolev, K. A. Andrianov, L. S. Utesheva, T. E. Vredenskaya, Dokl. Akad. Nauk. SSSR, **89**, 65 (1953).
431. K. A. Andrianov, B. G. Zavin, N. V. Pertsova, Vysokomol. Soedin. A, **10**, 46 (1968).
432. F. P. Price, S. G. Martin, J. P. Bianchi, J. Polym. Sci., **22**, 41 (1956).
433. C.-L. Lee, F. A. Emerson, J. Polym. Sci. A-2, **5**, 829 (1967).
434. R. R. Buch, H. M. Klimisch, O. K. Johannson, J. Polym. Sci. A-2, **8**, 541 (1970).
435. R. R. Buch, H. M. Klimisch, O. K. Johannson, J. Polym. Sci. A-2, **7**, 563 (1969).
436. V. Ye. Eskin, A. Ye. Nesterov, Vysokomol. Soedin., **8**, 141 (1966).
437. K. J. Ivin, H. A. Ende, G. Meyerhoff, Polymer, **3**, 129 (1962).
438. T. W. Bates, K. J. Ivin, Polymer, **8**, 263 (1967).
439. T. W. Bates, J. Biggins, K. J. Ivin, Makromol. Chem., **87**, 180 (1965).
440. R. Endo, T. Manago, M. Takeda, Bull. Chem. Soc. japan, **39**, 733 (1966).
441. W. Burchard, Makromol. Chem., **67**, 182 (1963).
442. V. N. Tsvetkov, I. N. Shtennikova, Ye. I. Ryumtsev, L. N. Andreyeva, Yu. P. Getmanchuk, Yu. L. Spirin, R. I. Dryagileva, Vysokomol. Soedin. A, **10**, 2132 (1968); see also V. N. Tsvetkov, Eur. Polym. J. Suppl., 237 (1969).
443. J. J. Burke, T. A. Orofino, J. Polym. Sci. A-2, **7**, 1 (1969).
444. H.-G. Elias, R. Schumacher, Makromol. Chem., **76**, 23 (1964).
445. G. J. Howard, J. Polym. Sci., **37**, 310 (1959).
446. G. B. Taylor, J. Am. Chem. Soc., **69**, 635 (1947).
447. J. V. Koleske, R. D. Lundberg, J. Polym. Sci. A-2, **7**, 897 (1967).
448. K. Hoshino, M. Watanabe, Nippon Kagaku Zasshi (J. Chem. Soc. Japan, Pure Chem. Sec.), **70**, 24 (1949).
449. R. Okada, T. Fukumura, H. Tanzawa, J. Polym. Sci. B, **4**, 971 (1966).
450. A. Mattiussi, G. B. Gechele, R. Francesconi, J. Polym. Sci. A-2, **7**, 411 (1969); see also G. B. Gechele, A. Mattiussi, Eur. Polym. J., **1**, 47 (1965); A. Mattiussi, G. B. Gechele, N. Pizzoli, *ibid.*, **2**, 283 (1966).
451. R. Bennewitz, Faserforsch. Textiltech., **5**, 155 (1954).
452. J.-J. Chien, L.-H. Shih, K.-I. Shih, Acta Chem. Sinica, **21**, 50 (1955).
453. J. R. Schaefgen, P. J. Flory, J. Am. Chem. Soc., **70**, 2709 (1948).
454. P. W. Morgan, S. L. Kwolek, J. Polym. Sci. A, **1**, 1147 (1963).
455. A. Zilkha, Y. Burstein, Biopolymers, **2**, 147 (1964).
456. Y. Hayashi, A. Teramoto, K. Kawahara, H. Fujita, Biopolymers, **8**, 403 (1969).
457. P. Doty, J. H. Bradbury, A. M. Holtzer, J. Am. Chem. Soc., **78**, 947 (1956).
458. E. Marchal, C. Dufour, J. Polym. Sci. C, **30**, 77 (1970).
459. G. Spach, Compt. Rend., **249**, 543 (1959).
460. K. Okita, A. Teramoto, H. Fujita, Biopolymers, **9**, 717 (1970).
461. N. Lupu-Lotan, A. Yaron, A. Berger, M. Sela, Biopolymers, **3**, 625 (1965).
462. A. Nakajima, T. Hayashi, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **24**, 230 (1967).
463. S. Tanaka, Thesis, Kyoto Univ. Kyoto, Japan, 1972.
464. A. Nakajima, S. Tanaka, K. Itoh, Polym. J., **3**, 398 (1972).
465. J. Marchal, C. Lapp, J. Chim. Phys., **60**, 756 (1963).
466. C. Lapp, J. Marchal, J. Chim. Phys., **62**, 1032 (1965).
467. D. F. Detar, F. F. Rogers, Jr., H. Bach, J. Am. Chem. Soc., **89**, 3039 (1967).
468. J. H. Fessler, A. G. Ogston, Trans. Faraday Soc., **47**, 667 (1951).
469. C. Tanford, K. Kawahara, S. Lapanje, J. Am. Chem. Soc., **89**, 729 (1967).
470. G. Thomson, S. A. Rice, M. Nagasawa, J. Am. Chem. Soc., **85**, 2537 (1963).
471. J. Feisst, H.-G. Elias, Makromol. Chem., **82**, 78 (1964).
472. A. Mele, R. Rufo, L. Santonato, Polymer, **10**, 233 (1969).
473. P. J. Akers, G. Allen, M. J. Bethell, Polymer, **9**, 575 (1968).
474. J. M. Barrales-Rienda, D. C. Pepper, J. Polym. Sci. B, **4**, 939 (1966).
475. H. Batzer, G. Benzing, Makromol. Chem., **62**, 66 (1963).
476. G. C. Berry, H. Nomura, K. G. Mayhan, J. Polym. Sci. A-2, **5**, 1 (1967).
477. A. D. Chirico, Chim. Ind. (Milan), **42**, 248 (1960).
478. G. Sitaramaiah, J. Polym. Sci. A, **3**, 2743 (1965).
479. G. V. Schulz, A. Horbach, Makromol. Chem., **29**, 93 (1959).
480. H. Batzer, G. Fritz, Makromol. Chem., **14**, 179 (1955).
481. D. A. S. Ravens, I. M. Ward, Trans. Faraday Soc., **57**, 150 (1961).
482. W. R. Moore, D. Sanderson, Polymer, **9**, 153 (1968).
483. I. M. Ward, Nature, **180**, 141 (1957).
484. G. Meyerhoff, S. Shimotsuma, Makromol. Chem., **135**, 195 (1970).
485. I. Marshall, A. Todd, Trans. Faraday Soc., **49**, 67 (1953).
486. H. Zahn, C. Borstlap, G. Valk, Makromol. Chem., **64**, 18 (1963).
487. B. Seidel, J. Polym. Sci., **55**, 411 (1961).
488. M. L. Wallach, Makromol. Chem., **103**, 19 (1967).
489. E. V. Kuznetsov, A. O. Wisel, I. M. Shermergorn, C. C. Tjulenjeu, Vysokomol. Soedin., **2**, 205 (1960).
490. H. M. Koepp, H. Werner, Makromol. Chem., **32**, 79 (1959).
491. A. Conix, Makromol. Chem., **26**, 226 (1958).
492. W. Griehl, S. Neue, Faserforsch. Textiltech., **5**, 423 (1954).
493. L. D. Moore, ACS Meeting, Cleveland, Polym. Preprints, **1**, 234 (1960).
494. N. G. Gaylord, S. Rosenbaum, J. Polym. Sci., **39**, 545 (1959).
495. W. A. Lanke, reported in W. R. Krigbaum, J. Polym. Sci., **28**, 213 (1958).

496. H. Batzer, H. Juergen, Makromol. Chem., **44/49**, 179 (1961).
497. B. F. Malichenko, O. N. Tsypina, A. Ye. Nesterov, A. S. Shevlyakov, Yu. S. Lipatov, Vysokomol. Soedin. A, **9**, 2624 (1967).
498. B. F. Malichenko, A. V. Yazlovitskii, A. Ye. Nesterov, Vysokomol. Soedin. A, **12**, 1700 (1970).
499. R. J. Merker, M. J. Scott, J. Polym. Sci. A, **2**, 15 (1964).
500. J. F. Brown, Jr., L. H. Vogt, Jr., A. Katchman, J. W. Eustance, K. M. Kiser, K. W. Krantz, J. Am. Chem. Soc., **82**, 6194 (1960).
501. V. N. Tsvetkov, K. A. Andrianov, G. I. Okhrimenko, I. N. Shtenikova, G. A. Fomin, M. G. Vitovskaya, V. I. Pakhomov, A. A. Yarosh, D. N. Andreyev, Vysokomol. Soedin. A, **12**, 1892 (1970).
502. V. N. Tsvetkov, K. A. Andrianov, I. N. Shtennikova, G. I. Okhrimenko, L. N. Andreyeva, G. A. Fomin, V. I. Pakhomov, Vysokomol. Soedin. A, **10**, 547 (1968).
503. V. V. Rode, P. N. Gribkova, V. V. Korshak, Vysokomol. Soedin. A, **11**, 57 (1969).
504. R. Endo, H. Iimura, M. Takeda, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **29**, 44 (1972).
505. R. Endo, M. Hattori, M. Takeda, Kobunshi Kagaku (Chem. High Polym. (Tokyo)), **29**, 48 (1972).
506. R. Endo, T. Hinokuma, M. Takeda, J. Polym. Sci. A-2, **6**, 665 (1968).
507. I. Z. Steinberg, W. F. Harrington, A. Berger, M. Sela, E. Katchalski, J. Am. Chem. Soc., **82**, 5263 (1960).
512. V. N. Tsvetkov, G. A. Fomin, P. N. Lavrenko, I. N. Shtennikova, T. V. Sheremeteva, I. I. Godunova, Vysokomol. Soedin. A, **10**, 903 (1968).
513. V. N. Tsvetkov, N. N. Kupriyanova, G. V. Tarasova, P. N. Lavrenko, I. I. Migunova, Vysokomol. Soedin. A, **12**, 1974 (1970).
514. L. V. Dubrovina, S. A. Pavlova, V. V. Korshak, Vysokomol. Soedin., **8**, 752 (1966).
515. V. V. Korshak, S. A. Pavlova, L. V. Boiko, T. M. Babchinitser, S. V. Vinogradova, Ya. S. Vygodskii, N. A. Golubeva, Vysokomol. Soedin. A, **12**, 56 (1970).
516. J. R. Holsten, M. R. Lilyquist, J. Polym. Sci. A, **3**, 3905 (1965).
517. W. Burchard, Makromol. Chem., **64**, 110 (1963).
518. J. M. G. Cowie, Makromol. Chem., **42**, 230 (1961).
519. W. Banks, C. T. Greenwood, Polymer, **10**, 257 (1969).
520. W. W. Everett, J. F. Foster, J. Am. Chem. Soc., **81**, 3459, 3464 (1959).
521. W. Banks, C. T. Greenwood, Makromol. Chem., **67**, 49 (1963).
522. W. Banks, C. T. Greenwood, Eur. Polym. J., **5**, 649 (1969).
523. J. M. G. Cowie, Makromol. Chem., **53**, 13 (1962).
524. J. M. G. Cowie, J. Polym. Sci., **49**, 455 (1961).
525. R. S. Patel, R. D. Patel, Makromol. Chem., **90**, 262 (1966).
526. B. A. Dombrow, C. O. Beckmann, J. Phys. Colloid Chem., **51**, 107 (1947).
527. W. Banks, C. T. Greenwood, D. J. Hourston, Trans. Faraday Soc., **64**, 363 (1968); see also Makromol. Chem., **111**, 226 (1968).
528. W. Burchard, E. Husemann, Makromol. Chem., **44/46**, 358 (1961).
529. H. Husemann, R. Resz, R. Wermer, Makromol. Chem., **47**, 48 (1961).
530. D. A. Brant, B. K. Min, Macromolecules, **2**, 1 (1969).
531. K. D. Goebel, D. A. Brant, Macromolecules, **3**, 634 (1970).
532. F. G. Donnan, R. C. Rose, Can. J. Research, B, **28**, 105 (1950).
533. D. Henley, Arkiv Kemi, **18**, 327 (1961).
534. W. Brown, R. Wilkstroem, Eur. Polym. J., **1**, 1 (1965).
535. W. G. Harland, in: Honeyman, Ed., "Recent Advances in Chemistry of Cellulose and Starch" Interscience, New York, 1959, p. 265.
536. N. Gralen, T. Svedberg, Nature, **152**, 625 (1943).
537. E. H. Immergut, B. G. Ranby, H. Mark, Ind. Eng. Chem., **45**, 2383 (1953).
538. J. W. Tamblyn, D. R. Morey, R. H. Wagner, Ind. Eng. Chem., **37**, 573 (1945).
539. F. H. Holmes, D. J. Smith, Trans. Faraday Soc., **53**, 67 (1957).
540. W. J. B. Badgley, H. Mark, J. Phys. Colloid Chem., **51**, 58 (1947).
541. H. J. Philipp, C. F. Bjork, J. Polym. Sci., **6**, 549 (1951).
542. P. J. Flory, O. K. Spurr, Jr., D. K. Carpenter, J. Polym. Sci., **27**, 231 (1958).
543. L. Huppenthal, S. Claesson, Roczniki Chem., **39**, 1867 (1965).
544. L. Mandelkern, P. J. Flory, J. Am. Chem. Soc., **74**, 2517 (1952).
545. V. P. Shanbhag, Arkiv Kemi, **29**, 1 (1968).
546. V. P. Shanbhag, J. Oehman, Arkiv Kemi, **29**, 163 (1968).
547. W. R. Krigbaum, L. H. Sperling, J. Phys. Chem., **64**, 99 (1960).
548. G. Meyerhoff, J. Polym. Sci., **29**, 399 (1958).
549. A. M. Holtzer, H. Benoit, P. Doty, J. Phys. Chem., **58**, 624 (1954).
550. M. M. Huque, D. A. I. Goring, S. G. Mason, Can. J. Chem., **36**, 952 (1958).
551. R. M. Badger, R. H. Blaker, J. Phys. Colloid Chem., **53**, 1056 (1949).
552. E. Penzel, G. V. Shulz, Makromol. Chem., **113**, 64 (1968).
553. W. R. Moore, J. A. Epstein, J. Appl. Chem. (London), **6**, 168 (1956).
554. W. R. Moore, G. D. Edge, J. Polym. Sci., **47**, 469 (1960).
555. M. L. Hunt, S. Newman, H. A. Scheraga, P. J. Flory, J. Phys. Chem., **60**, 1278 (1956).
556. W. G. Harland, Nature, **170**, 667 (1952).
557. G. Meyerhoff, N. Suetterlin, Makromol. Chem., **87**, 258 (1965).
558. W. R. Moore, A. M. Brown, J. Colloid Sci., **14**, 1343 (1959).
559. V. N. Tsvetkov, S. Ya. Kotlyar, Zh. Fiz. Khim., **30**, 1100 (1956).
560. R. St. J. Manley, Arkiv Kemi, **9**, 519 (1956).
561. W. Brown, D. Henley, J. Oehman, Makromol. Chem., **64**, 49 (1963); see also W. Brown, Arkiv Kemi, **18**, 227 (1961).
562. H. Vink, Makromol. Chem., **94**, 1 (1966).

563. W. B. Neeley, J. Polym. Sci. A, **1**, 311 (1963).
564. W. Brown, D. Henley, J. Oehman, Makromol. Chem., **62**, 164 (1963).
565. G. Sitaramaiah, D. A. I. Goring J. Polym. Sci., **58**, 1107 (1962).
566. W. Brown, D. Henley, Makromol. Chem., **79**, 68 (1964).
567. H. Elmgren, Arkiv Kemi, **24**, 237 (1965).
568. K. Gekko, Makromol. Chem., **148**, 229 (1971).
569. F. R. Senti, N. N. Hellman, N. H. Ludwig, G. E. Babcock, R. Tobin, C. A. Class, B. L. Lamberts, J. Polym. Sci., **17**, 527 (1955); see also Ref. 8.
570. K. Gekko, H. Noguchi, Biopolymers, **10**, 1513 (1971).
571. L. H. Arond, H. P. Frank, J. Phys. Chem., **58**, 953 (1954).
572. J. V. Koleska, S. F. Kurath, J. Polym. Sci. A, **2**, 4123 (1964).
573. R. L. Cleland, J. L. Wang, Biopolymers, **9**, 799 (1970).
574. D. H. Juers, H. A. Swenson, S. F. Kurath, J. Polym. Sci. A-2, **5**, 361 (1967).
590. N. M. Tret'yakova, L. V. Kosmodem'yanoskii, R. G. Romanova, E. G. Lazaryants, Vysokomol. Soedin. A, **12**, 2754 (1970).
591. Houtz, cited in Y. Iwakura, Y. Kurosaki, N. Nakabayashi, Makromol. Chem., **44/46**, 570 (1961).
592. W. Scholtan, H. Marzolph, Makromol. Chem., **57**, 52 (1962).
593. Y. Shimura, I. Mita, H. Kambe, J. Polym. Sci. B, **2**, 403 (1964).
594. H. Gerrens, H. Ohlinger, R. Fricker, Makromol. Chem., **87**, 209 (1965).
595. Y. Shimura, J. Polym. Sci., A-2, **4**, 423 (1966).
596. Ye. G. Erenburg, G. G. Kartasheva, M. A. Yeremina, I. Ya. Poddubnyi, Vysokomol. Soedin. A, **9**, 2709 (1967).
597. Ye. A. Bekturov, L. A. Bimendina, S. V. Bereza, Vysokomol. Soedin. A, **12**, 2179 (1970).
598. R. B. Mohite, S. Gundiah, S. I. Kapur, Makromol. Chem., **108**, 52 (1967).
599. H. Hopff, D. Starck, Makromol. Chem., **48**, 50 (1961).
600. S. R. Rafikov, S. A. Pavlova, Sh. Shayakhmotov, Vysokomol. Soedin. A, **11**, 1990 (1969).
601. T. E. Helminiak, C. L. Benner, W. E. Gibbs, Polymer Preprints, **8**, 284 (1967); see also CA, 1967, 116045k; cited by W. H. Stockmayer, IUPAC Symposium, Leiden, Netherlands, 1970.
602. I. Ya. Poddubnyi, V. A. Grechanovskii, M. I. Mosevitskii, Vysokomol. Soedin., **5**, 1042 (1964).
603. V. N. Tsvetkov, K. A. Andrianov, G. I. Okhrimenko, M. G. Vitovskaya, Eur. Polym. J., **7**, 1215 (1971).
604. H. Tanzawa, T. Tanaka, A. Soda, J. Polym. Sci. A-2, **7**, 929 (1969).
605. S. V. Bereza, Ye. A. Bekturov, S. R. Rafikov, Vysokomol. Soedin. A, **11**, 1681 (1969).
606. S. V. Bereza, S. R. Rafikov, Ye. A. Bekturov, R. Ye. Legkunets, Vysokomol. Soedin. A, **10**, 2536 (1968).
607. V. A. Myagchenkov, E. V. Kuznetsov, O. A. Ishhakov, V. M. Lichkina, Vysokomol. Soedin., **5**, 724 (1963).
608. J. Feyen, K. deGroot, A. C. deVisser, A. Bantjes, Biopolymers, **10**, 2509 (1971).
609. J. R. Patel, C. K. Patel, R. D. Patel, Stärke, **19**, 330 (1967); cited in: J. R. Patel, R. D. Patel, Biopolymers, **10**, 839 (1971).
610. A. B. Duchkova, N. S. Nametkin, V. S. Khotimskii, L. Ye. Gusel'nikov, S. G. Durgar'yan, Vysokomol. Soedin., **8**, 1814 (1966).
611. J. D. Matlack, S. N. Chinai, R. A. Guzzi, D. W. Levi, J. Polym. Sci., **49**, 533 (1961).
612. S. M. Kochergin, V. P. Barabanov, I. G. Fedorova, Vysokomol. Soedin., **8**, 916 (1966).
613. H. Utiyama, K. Kajitani, M. Kurata, unpublished work. Cf. Rept. Progr. Polym. Phys. Japan, **6**, 29 (1963).
614. T. Kotaka, Y. Murakami, H. Inagaki, J. Phys. Chem., **72**, 829 (1968); see also T. Kotaka, H. Ohnuma, Y. Murakami, *ibid.*, **70**, 4099 (1966).
615. W. H. Stockmayer, L. D. Moore, Jr., M. Fixman, B. N. Epstein, J. Polym. Sci., **16**, 517 (1955).
616. T. Kotaka, H. Ohnuma, H. Inagaki, Polymers, **10**, 517 (1969); see also H. Ohnuma, H. Inagaki, *ibid.*, **10**, 501 (1969).
617. S. Polowinskii, J. Polym. Sci. A-2, **5**, 891 (1967).
618. V. N. Tsvetkov, S. Ya. Magarik, T. Kadyrov, G. A. Andreyeva, Vysokomol. Soedin. A, **10**, 943 (1968).
619. V. N. Tsvetkov, Makromol. Chem., **160**, 1 (1972).
620. H. Benoit, J. Polym. Sci., **3**, 376 (1948).
621. H. Markovitz, J. Chem. Phys., **20**, 868 (1952).
622. J. E. Mark, J. Am. Chem. Soc., **88**, 4354 (1966).
623. J. E. Mark, J. Am. Chem. Soc., **89**, 6829 (1967).
624. T. Norisuye, K. Kawahara, A. Teramoto, H. Fujita, J. Chem. Phys., **49**, 4330 (1968).
625. O. Kratky, H. Sand, Kolloid-Z., **172**, 18 (1960).
626. J. E. Mark, Ref. 622; based on the stress-temperature data of L. A. Wood and F. L. Roth, J. Appl. Phys., **15**, 781 (1944).
627. J. E. Mark, Ref. 622; based on the stress-temperature data of A. Ciferri, Makromol. Chem., **43**, 152 (1961).
628. E. D. Kunst, Rec. Trav. Chim., **69**, 125 (1950).
629. A. Ciferri, C. A. J. Hoeve, P. J. Flory, J. Am. Chem. Soc., **83**, 1015 (1961); A. Ciferri, J. Polym. Sci., **54**, 149 (1961).
630. A. Kotera, H. Matsuda, A. Wada, Rept. Progr. Polym. Phys. Japan, **8**, 5 (1965).
631. A. Nakajima, F. Hamada, S. Hayashi, J. Polym. Sci. C, **15**, 285 (1966).
632. P. J. Flory, A. Ciferri, R. Chiang, J. Am. Chem. Soc., **83**, 1023 (1961).
633. J. E. Mark, P. J. Flory, J. Am. Chem. Soc., **87**, 1423 (1965).
634. Unpublished data from the Chemstrand Research Center; cited in A. Ciferri, J. Polym. Sci. A, **2**, 3089 (1964).
635. A. V. Tobolsky, D. Carlson, N. Indicator, J. Polym. Sci., **54**, 175 (1961).
636. R. Kirste, G. V. Schulz, Z. Phys. Chem. (Frankfurt), **30**, 171 (1961).
637. R. Kirste, O. Kratky, Z. Phys. Chem. (Frankfurt), **31**, 363 (1962).
638. A. Nakajima, H. Yamakawa, J. Phys. Chem., **67**, 654 (1963).
639. H. Abe, W. Prins, J. Polym. Sci. C, **2**, 527 (1963).

640. O. Kratky, G. Porod, Rec. Trav. Chim., **68**, 1106 (1949).
641. E. Wada, K. Okano, Rept. Progr. Polym. Phys. Japan, **6**, 1 (1963).
642. E. Eisenberg, E. F. Casassa, J. Polym. Sci., **47**, 29 (1960).
643. N. T. Notley, P. J. W. Debye, J. Polym. Sci., **17**, 99 (1955).
644. W. R. Krigbaum, D. K. Carpenter, J. Phys. Chem., **59**, 1166 (1955).
645. E. Wada, K. Okano, Rept. Progr. Polym. Phys. Japan, **7**, 19 (1964).
646. T. A. Orofino, J. W. Mickey, Jr., J. Chem. Phys., **38**, 2513 (1963).
647. G. V. Schulz, H. Baumann, Makromol. Chem., **60**, 120 (1963).
648. T. A. Orofino, A. Ciferri, J. Phys. Chem., **68**, 3136 (1964).
649. M. Fukuda, M. Fukutomi, Y. Kato, T. Hashimoto, [Private communication.]
650. H. Utiyama, J. Phys. Chem., **69**, 4138 (1965).
651. N. Kuwahara, K. Ogino, A. Kasai, S. Ueno, M. Kaneko, J. Polym. Sci. A, **3**, 985 (1965).
652. A. Dondos, H. Benoit, Makromol. Chem., **135**, 181 (1970).
653. V. Ye. Eskin, O. Z. Korotkina, Vysokomol. Soedin. A, **12**, 2216 (1970).
654. G. Allen, C. Booth, C. Price, Polymer, **8**, 397,414 (1967).
655. J. E. Mark, P. J. Flory, J. Am. Chem. Soc., **87**, 1415 (1965).
656. C. Rossi, A. Perico, J. Chem. Phys., **53**, 1223 (1970).
657. W. Silberszyc, J. Polym. Sci. B, **1**, 577 (1963).
658. J. M. Evans, M. B. Huglin, Makromol. Chem., **127**, 141 (1969).
659. K. Bak, E. Elefante, J. E. Mark, J. Phys. Chem., **71**, 4007 (1967).
660. R. H. Marchessault, K. Okamura, C. J. Su, Macromolecules, **3**, 735 (1970).
661. J. Cornibert, R. H. Marchessault, H. Benoit, G. Weill, Macromolecules, **3**, 741 (1970).
662. U. P. Strauss, P. Ander, J. Phys. Chem., **66**, 2235 (1962).
663. J. K. Peterson, Thesis, Ohio State Univ., Ohio, USA, 1961.
664. Ye. G. Erenburg, G. G. Kartasheva, M. A. Yeremina, I. Ya. Poddubnyi, Vysokomol. Soedin. A, **9**, 2709 (1967).
665. J. E. Mark, P. J. Flory, J. Am. Chem. Soc., **86**, 138 (1964).
666. K. J. Ivin, H. A. Ende, J. Polym. Sci., **54**, S17 (1961).
667. R. Endo, T. Hinokuma, M. Takeda, J. Polym. Sci. A-2, **6**, 665 (1968).
668. P. R. Saunders, J. Polym. Sci. A, **2**, 3765 (1964).
669. P. J. Flory, A. D. Williams, J. Polym. Sci. A-2, **5**, 399 (1967).
670. D. A. Brant, P. J. Flory, J. Am. Chem. Soc., **87**, 2788 (1965).
671. A. Yu. Koshevnik, M. M. Kusakov, E. A. Razumovskaya, Vysokomol. Soedin. A, **10**, 2795 (1968).
672. A. E. Tonelli, P. J. Flory, Macromolecules, **2**, 225 (1969).
673. A. R. Schulz, J. Polym. Sci. A-2, **8**, 883 (1970).
674. S. A. Pavlova, V. V. Korshak, L. V. Dubrovina, G. I. Timofeyeva, Vysokomol. Soedin. A, **9**, 2624 (1967).
675. W. H. Stockmayer, IUPAC Symposium, Leiden, Netherlands, 1970. [This estimate of $r_0/M^{1/2}$ was based on the viscosity data given in Ref. 601.]
676. J. M. G. Cowie, P. M. Toporowski, Polymer, **5**, 601 (1964).
677. R. S. Patel, R. D. Patel, J. Polym. Sci. A, **3**, 2123 (1965); A-2, **4**, 835 (1966).
678. W. Burchard, Makromol. Chem., **88**, 11 (1965).
679. S. Heine, O. Kratky, G. Porod, P. Schmitz, Makromol. Chem., **44/46**, 682 (1961).
680. A. M. Rijke, J. Polym. Sci. B, **4**, 131 (1966).
681. B. Das, A. K. Ray, P. K. Choudhary, J. Phys. Chem., **73**, 3413 (1969); see also B. Das, P. K. Choudhary, J. Polym. Sci. A-1, **5**, 769 (1967).
682. Ye. A. Bekturov, L. A. Bimendina, S. V. Bareza, Vysokomol. Soedin. A, **12**, 2179 (1970).
683. J. Velickovic, D. Jovanovic, J. Vukajlovic, Makromol. Chem., **129**, 203 (1969).
684. V. Ye. Eskin, O. Z. Korotkina, Vysokomol. Soedin. A, **12**, 2216 (1970).
685. G. A. Morneau, P. I. Roth, A. R. Schulz, J. Polym. Sci., **55**, 609 (1961).
686. G. V. Schulz, E. Penzel, Makromol. Chem., **112**, 260 (1968).
687. O. Kratky, H. Leopold, G. Puchwein, Kolloid-Z. -Z. Polymere, **216/217**, 255 (1967).
688. R. L. Cleland, Biopolymers, **10**, 1925 (1971).
689. M. Nakata, Makromol. Chem., **149**, 99 (1971).
690. T. W. Bates, W. H. Stockmayer, J. Chem. Phys., **45**, 2321 (1966); Macromolecules, **1**, 12 (1968).
691. W. Brown, D. Henley, Makromol. Chem., **75**, 179 (1964).
692. R. St. John Manley, Svensk Papperstidn. **61**, 96 (1958).
693. V. P. Shanbhag, Arkiv Kemi, **29**, 139 (1968).
694. S. Teramachi, A. Takahashi, I. Kagawa, Kogyo Kagaku Zasshi (J. Chem. Soc. Japan, Ind. Chem. Sec.), **69**, 685 (1966).
695. G. Allen, C. Booth, C. Price, Polymer, **8**, 414 (1967).
696. P. J. Flory, Makromol. Chem., **98**, 128 (1966).
697. Recommendation of IUPAC Macromolecular Division working party, "Molecular Characterization of Commercial Polymers", Strassbourg, 12.3.1971.
698. N. Corbin, J. Prud'homme, J. Polym. Sci., Polym. Phys. Ed., **15**, 1937 (1977).
699. P. N. Chaturvedi, C. K. Patel, J. Polym. Sci., Polym. Phys. Ed., **23**, 1255 (1985).
700. G. Moraglio, G. Gianotti, F. Zoppi, U. Bonicelli, Eur. Polym. J., **7**, 303 (1971).
701. D. Constantin, Eur. Polym. J., **13**, 907 (1977).
702. I. Katime, P. Garro, J. M. Teijon, Eur. Polym. J., **11**, 881 (1975).
703. G. R. Williamson, A. Cervenka, Eur. Polym. J., **10**, 295 (1974).
704. E. P. Otocka, R. J. Roe, M. Y. Hellman, P. M. Muglia, Macromolecules, **4**, 507 (1971).
705. H. L. Wagner, C. A. J. Hoeve, J. Polym. Sci., Polym. Symp., **54**, 327 (1976).
706. G. R. Williamson, A. Cervenka, Eur. Polym. J., **8**, 1009 (1972).
707. S. Tani, F. Hamada, A. Nakajima, Polym. J., **5**, 86 (1973).
708. G. Moraglio, G. Gianotti, U. Bonicelli, Eur. Polym. J., **9**, 623 (1973).

709. S. Arichi, M. Y. Pedram, J. M. G. Cowie, Eur. Polym. J., **15**, 113 (1979).
710. G. S. Misra, S. H. Bhattacharya, Eur. Polym. J., **15**, 125 (1979).
711. K. Kamide, Y. Miyazaki, H. Kobayashi, Polym. J., **17**, 607 (1985).
712. R. Jerome, V. Desreux, Eur. Polym. J., **6**, 411 (1970).
713. C. U. Pittman, Jr., R. L. Voges, W. R. Jones, Macromolecules, **4**, 291 (1971).
714. C. U. Pittman, Jr., J. C. Lai, D. P. Vanderpool, M. Good, R. Prado, Macromolecules, **3**, 746 (1970).
715. M. Guaita, O. Chiantore, Makromol. Chem., **176**, 185 (1975).
716. L. Gargallo, D. Radic, I. Katime, Eur. Polym. J., **16**, 383 (1980).
717. N. Hadjichristidis, M. Devaleriola, V. Desreux, Eur. Polym. J., **8**, 1193 (1972).
718. I. A. Katime Amashta, G. Sanchez, Eur. Polym. J., **11**, 223 (1975).
719. N. Hadjichristidis, C. Touloupis, L. J. Fetters, Macromolecules, **14**, 128 (1981).
720. R. J. Fort, T. M. Polyzoidis, Eur. Polym. J., **12**, 685 (1976).
721. J. Stejskal, M. J. Benes, P. Krotochvil, J. Peska, J. Polym. Sci., Polym. Phys. Ed., **11**, 1803 (1973).
722. M. Bohdanecky, H. Bazilova, J. Kopecek, Eur. Polym. J., **10**, 405 (1974).
723. N. Hadjichristidis, J. Mays, W. Ferry, L. J. Fetters, J. Polym. Sci., Polym. Phys. Ed., **22**, 1745 (1984).
724. M. M. Zafar, R. Mahmood, S. A. Wadood, Makromol. Chem., **160**, 313 (1972).
725. J. Stejskal, J. Janca, P. Kratochvil, Polym. J., **8**, 549 (1976).
726. J. N. Helbert, C.-Y. Chen, C. U. Pittman, Jr., G. L. Hagnauer, Macromolecules, **11**, 1104 (1978).
727. Y. Kashiwagi, Y. Einaga, H. Fujita, Polym. J., **12**, 271 (1980).
728. I. Katime, A. Roig, L. M. Leon, S. Montero, Eur. Polym. J., **13**, 59 (1977).
729. J. Niezette, N. Hadjichristidis, V. Desreux, Eur. Polym. J., **13**, 41 (1977).
730. V. N. Tsvetkov, E. I. Riumtsev, I. N. Shtennikova, E. V. Korneeva, B. A. Krenstel, Yu. B. Amerik, Eur. Polym. J., **9**, 481 (1973).
731. M. Galin, J.-C. Galin, Eur. Polym. J., **9**, 1149 (1973).
732. Z. Grubisic, P. Rempp, H. Benoit, J. Polym. Sci. B, **5**, 753 (1967).
733. X. Zhongde, S. Mingshi, N. Hadjichristidis, L. J. Fetters, Macromolecules, **14**, 1591 (1981).
734. M. M. Zafar, R. Mahmood, Makromol. Chem., **175**, 903 (1974).
735. N. Hadjichristidis, J. Mays, R. D. Vargo, L. J. Fetters, J. Polym. Sci., Polym. Phys. Ed., **21**, 189 (1983).
736. M. Dolezalova, V. Petrus, Z. Tuzar, M. Bohdanecky, Eur. Polym. J., **12**, 701 (1976).
737. K. Yamaura, K. Hirata, S. Tamura, S. Matsuzawa, J. Polym. Sci., Polym. Phys. Ed., **23**, 1703 (1985).
738. K. Matsuo, W. H. Stockmayer, Macromolecules, **8**, 660 (1975).
739. F. Cane, T. Capaccioli, Eur. Polym. J., **14**, 185 (1978).
740. C. M. L. Atkinson, R. Dietz, Eur. Polym. J., **15**, 21 (1979).
741. Z. Kucukyavuz, S. Kucukyavuz, Eur. Polym. J., **14**, 867 (1978).
742. I. Katime, A. Roig, Eur. Polym. J., **11**, 603 (1975).
743. K. Matsumura, Polym. J., **1**, 332 (1970).
744. I. Hernandez-Fuentes, M. G. Prolongo, Eur. Polym. J., **15**, 571 (1979).
745. Y. Izumi, Y. Miyake, Polym. J., **3**, 647 (1972).
746. V. E. Eskin, T. N. Nekrasova, U. Juraev, Eur. Polym. J., **11**, 341 (1975).
747. M. Matsuo, W. H. Stockmayer, J. Polym. Sci., Polym. Phys. Ed., **11**, 43 (1973).
748. S. L. Malhotra, J. Leonard, E. Ahad, Eur. Polym. J., **14**, 257 (1978).
749. A. L. Spatorico, B. Coulter, J. Polym. Sci., Polym. Phys. Ed., **11**, 1139 (1973).
750. Y. Einaga, Y. Miyaki, H. Fujita, J. Polym. Sci., Polym. Phys. Ed., **17**, 2103 (1979).
751. Z. Xu, N. Hadjichristidis, L. J. Fetters, cited in Ref. 733.
752. M. Nakata, Makromol. Chem., **149**, 99 (1971).
753. G. Meyerhoff, B. Appelt, Macromolecules, **12**, 968 (1979), **13**, 657 (1980).
754. C. W. Tsimpris, B. Suryanrayanan, K. G. Mayhan, J. Polym. Sci. A-2, **10**, 1837 (1972).
755. J. Roovers, J. Polym. Sci., Polym. Phys. Ed., **23**, 1117 (1985).
756. C. Strazielle, H. Benoit, O. Vogl, Eur. Polym. J., **14**, 331 (1978).
757. J. Herz, M. Hert, C. Strazielle, Makromol. Chem., **160**, 213 (1972).
758. J. E. L. Roovers, S. Bywater, Macromolecules, **5**, 384 (1972).
759. J. Roovers, P. M. Toporowski, Macromolecules, **14**, 1174 (1981).
760. I. Noda, Y. Yamamoto, T. Kitano, M. Nagasawa, Macromolecules, **14**, 1306 (1981).
761. J. Veličkovič, S. Ćoseva, R. J. Fort, Eur. Polym. J., **11**, 377 (1975).
762. J. Veličkovič, S. Vasovič Makromol. Chem., **153**, 207 (1972).
763. J. Veličkovič, J. Filipović, S. Ćoseva, Eur. Polym. J., **15**, 521 (1979).
764. Y. Higo, Y. Kato, M. Itoh, N. Kozuka, I. Noda, M. Nagasawa, Polym. J., **14**, 809 (1982).
765. L. M. León, I. Katime, M. Gonzalez, J. Figueruelo, Eur. Polym. J., **14** 671 (1978).
766. L. M. León, J. Galaz, L. M. Garcia, M. S. Anasagasti, Eur. Polym. J., **16**, 921 (1980).
767. L. M. León, I. Katime, M. Rodriguez, Eur. Polym. J., **15**, 29 (1979).
768. G. Sitaramaiah, D. Jacobs, Makromol. Chem., **164**, 237 (1973).
769. G. Fourche, C. Tourenne, Eur. Polym. J., **12**, 663 (1976).
770. R. Meza, L. Gargallo, Eur. Polym. J., **13**, 235 (1977).
771. M. Tricot, J. Bleus, J.-P. Riga, V. Desreux, Makromol. Chem., **175**, 913 (1974).

772. K. Kamide, Y. Miyazaki, H. Kobayashi, Polym. J., **14**, 591 (1982).
773. C. R. Reddy, V. Kalpagam, J. Polym. Sci., Polym. Phys. Ed., **14**, 749 (1976)
774. C. Cincu, Eur. Polym. J., **10**, 29 (1974).
775. S. Ali, Polym. J., **15**, 685 (1979).
776. J. Podešva, M. Bohdanecký, P. Kratochvil, J. Polym. Sci., Polym. Phys. Ed., **15**, 1521 (1977).
777. K. S. V. Srinivasan, M. Santappa, J. Polym. Sci., Polym. Phys. Ed., **11**, 331 (1973).
778. N. Nemoto, T. Inoue, Y. Tsunashima, M. Kurata, Bull. Inst. Chem. Res., Kyoto Univ., **62**, 177 (1984).
779. D. Radic, L. Gargallo, J. Polym. Sci., Polym. Phys. Ed., **16**, 977 (1978).
780. M. Shima, E. Ogawa, S. Ban, M. Sato, J. Polym. Sci., Polym. Phys. Ed., **15**, 1999 (1977).
781. M. Pizzoli, G. Ceccorulli, Eur. Polym. J., **8**, 769 (1972).
782. M. Pizzoli, G. Ceccorulli, G. Stea, Makromol. Chem., **164**, 273 (1973).
783. T. Tanaka, T. Kotaka, K. Ban, M. Hattori, H. Inagaki, Macromolecules, **10**, 960 (1977).
784. S. Ya. Magarik, G. M. Pavlov, G. A. Fomin, Macromolecules, **11**, 294 (1978).
785. M. G. Prolongo, R. M. Masegosa, I. Hernandez-Fuentes, A. Horta, Macromolecules, **14**, 1526 (1981).
786. J. Prud'homme, J. E. L. Roovers, S. Bywater, Eur. Polym. J., **8**, 901 (1972).
787. T. Tsuji, H. Fujita, Polym. J., **4**, 409 (1973).
788. G. Gianotti, U. Bonicelli, D. Borghi, Makromol. Chem., **166**, 235 (1973).
789. Cr. Simionescu, J. Cobianu, Eur. Polym. J., **15**, 173 (1979).
790. N. Doddi, W. C. Forsman, C. C. Price, J. Polym. Sci., Polym. Phys. Ed., **12**, 1395 (1974).
791. R. Alamo, J. G. Fatou, A. Bello, Polym. J., **15**, 491 (1983).
792. A. R. Schulz, A. L. Bridgman, E. M. Hadsell, C. R. McCullough, J. Polym. Sci. A-2, **10**, 273 (1972).
793. G. Meyerhoff, Makromol. Chem., **145**, 189 (1971).
794. A. L. Spatorico, J. Polym. Sci., Polym. Phys. Ed., **12**, 159 (1974).
795. K. Matsuo, W. H. Stockmayer, G. F. Needham, J. Polym. Sci., Polym. Sypm., **71**, 95 (1984).
796. T. Tsuji, T. Norisuye, H. Fujita, Polym. J., **7**, 558 (1975).
797. S. A. Akita, Y. Einaga, Y. Miyaki, H. Fujita, Macromolecules, **9**, 774 (1976).
798. T. Hirose, Y. Einaga, H. Fujita, Polym. J., **11**, 819 (1979).
799. Y. Miyaki, Y. Einaga, T. Hirose, H. Fujita, Macromolecules, **10**, 1356 (1977).
800. T. Capaccioli, L. Zotteri, Eur. Polym. J., **11**, 729 (1975).
801. K. Kamide, Y. Miyazaki, H. Kobayashi, Polym. J., **9**, 317 (1977).
802. C. W. Tsimpris, K. G. Mayhan, J. Polym. Sci., Polym. Phys. Ed., **11**, 1151 (1973).
803. B. Lánská, M. Bohdanecký, J. Šebenda, Z. Tuzar, Eur. Polym. J., **14**, 807 (1978).
804. J. Herold, G. Meyerhoff, Eur. Polym. J., **15**, 525 (1979).
805. J. C. Gressier, G. Levesque, Eur. Polym. J., **16**, 1167 (1980).
806. A. DeChirico, Eur. Polym. J., **13**, 537 (1977).
807. V. N. Tsvetkov, G. I. Kudryavtsev, I. N. Shennikova, T. V. Pekar, E. N. Zakkarova, V. D. Kalmykova, A. V. Volokhina, Eur. Polym. J., **12**, 517 (1976).
808. A. A. Jones, W. H. Stockmayer, R. J. Molinari, J. Polym. Sci., Polym. Symp., **54**, 227 (1976).
809. K. Sakurai, K. Ochi, T. Norisuye, H. Fujita, Polym. J., **16**, 559 (1984).
810. E. Bianchi, A. Ciferri, J. Preston, W. R. Krigbaum, J. Polym. Sci., Polym. Phys. Ed., **19**, 863 (1981).
811. H. Garreau, E. Marechal, Eur. Polym. J., **9**, 263 (1973).
812. Z. Tuzar, M. Bohdanecký, R. Puffr, J. Sebenda, Eur. Polym. J., **11**, 851 (1975).
813. J. S. Tan, A. R. Sochor, Macromolecules, **14**, 1700 (1981).
814. S. Itoh, N. Nishioka, T. Norisuye, A. Teramoto, Macromolecules, **14**, 904 (1981).
815. R. B. Howkins, A. Holtzer, Macromolecules, **5**, 294 (1972).
816. M. Oka, T. Hayashi, A. Nakajima, Polym. J., **17**, 621 (1985).
817. W. L. Mattice, J.-T. Lo, Macromolecules, **5**, 734 (1972).
818. D. S. Clark, W. L. Mattice, Macromolecules, **10**, 369 (1977).
819. S. M. Aharoni, C. R. Crosby III, E. K. Walsh, Macromolecules, **15**, 1093 (1982).
820. C. Fougnot, J. Jozefovicz, J. Polym. Sci., Polym. Symp., **39**, 199 (1972).
821. M. R. Ambler, D. McIntyre, L. J. Fetters, Macromolecules **11**, 300 (1978).
822. M. N. Berger, B. M. Tidswell, J. Polym. Sci., Polym. Symp., **42**, 1063 (1973).
823. V. N. Tsvetkov, I. N. Shtennikova, E. I. Rjumtsev, Yu. P. Getmanchuk, Eur. Polym. J., **7**, 767 (1971).
824. H. Murakami, T. Norisuye, H. Fujita, Macromolecules, **13**, 345 (1980).
825. A. J. Bur, L. J. Fetters, Macromolecules, **6**, 874 (1973).
826. V. N. Tsvetkov, E. I. Rjumtsev, N. V. Pogodina, I. N. S. Shtennikova, Eur. Polym. J., **11**, 37 (1975).
827. D. N. Rubingh, H. Yu, Macromolecules, **9**, 681 (1976).
828. J. Sadanobu, T. Norisuye, H. Fujita, Polym. J., **13**, 75 (1981).
829. S. Ioan, E. Dumitrium S. Maxim, A. Carpov, Eur. Polym. J., **13**, 109 (1977).
830. H. Kaye, H. J. Chou, J. Polym. Sci., Polym. Phys. Ed., **13**, 477 (1975).
831. J. F. Wolfe, J. K. Stille, Macromolecules, **9**, 489 (1976).
832. S. O. Norris, J. K. Stille, Macromolecules, **9**, 496 (1976).
833. S.-G. Chu, S. Venkatraman, G. C. Berry, Y. Einaga, Macromolecules, **14**, 939 (1981).
834. M. Motowoka, T. Norisuye, A. Teramoto, H. Fujita, Polym. J., **11**, 665 (1979).
835. S. Takahashi, M. Kariya, T. Yatake, K. Sonogashira, N. Hagihara, Macromolecules, **11**, 1063 (1978).
836. G. L. Hagnauer, B. R. Laliberte, J. Polym. Sci., Polym. Phys. Ed., **14**, 367 (1976).
837. J. Brzezinski, Z. Czlonkowska-Kohutnicka, B. Czarnecka, A. Kornas-Calka, Eur. Polym. J., **9**, 733 (1973).
838. J. G. Zilliox, J. E. L. Roovers, S. Bywater, Macromolecules, **8**, 573 (1975).

839. A. Stokes, Eur. Polym. J., **6**, 719 (1970).
840. A. C. Ouano, E. Gipstein, W. Kaye, B. Dawson, Macromolecules, **8**, 558 (1975).
841. G. Allen, J. McAnish, Eur. Polym. J., **6**, 1635 (1970).
842. N. Ohno, K. Nitta, S. Makino, S. Sugai, J. Polym. Sci., Polym. Phys. Ed., **11**, 413 (1973).
843. J. Kovář, L. Mrkvičková-Vaculová, M. Bohdanecký, Makromol. Chem., **176**, 1829 (1975).
844. D. Casson, A. Rembaum, Macromolecules, **5**, 75 (1972).
845. G. L. Hagnauer, G. D. Mulligan, Macromolecules, **6**, 477 (1973).
846. J. M. G. Cowie, A. Maconnachie, Eur. Polym. J., **11**, 79 (1975).
847. J. M. G. Cowie, A. Maconnachie, Eur. Polym. J., **10**, 367, 375 (1974).
848. W. Banks, C. T. Greenwood, Makromol. Chem., **144**, 135 (1971).
849. W. Banks, C. T. Greenwood, Eur. Polym. J., **7**, 263 (1971).
850. J. M. G. Cowie, A. Maconnachie, J. Polym. Sci., Polym. Phys. Ed., **12**, 407 (1974).
851. K. Kamide, M. Saito, T. Abe, Polym. J., **13**, 421 (1981).
852. M. Saito, Polym. J., **15**, 249 (1983).
853. N. G. Vyas, S. Shashikant, C. K. Patel, R. D. Patel, J. Polym. Sci., Polym. Phys. Ed., **17**, 2021 (1979).
854. K. Kamide, T. Terakawa, Y. Miyazaki, Polym. J., **11**, 285 (1979).
855. K. Kamide, Y. Miyazaki, T. Abe, Polym. J., **11**, 523 (1979).
856. N. P. Dymarchuk, K. P. Mischenko, T. V. Formina, Zh. Prikl. Khim. (Leningrad), **37**, 2263 (1964), cited in Ref. 855.
857. P. R. M. Nair, R. M. Gohil, K. C. Patel, R. D. Patel, Eur. Polym. J., **13**, 273 (1977).
858. J. T. Guthrie, M. B. Huglin, R. W. Richards, V. I. Shah, A. H. Simpson, Eur. Polym. J., **11**, 527 (1975).
859. V. N. Tsvetkov, E. I. Rjumtsev, L. N. Andreeva, N. V. Pogodina, P. N. Lavrenko, L. Kutsenko, Eur. Polym. J., **10**, 563 (1974).
860. L. C. Cerney, J. McTiernan, D. M. Stasiw, J. Polym. Sci., Polym. Symp., **42**, 1455 (1973).
861. J. M. G. Cowie, A. Maconnachie, Data cited in Ref. 850.
862. T. Norisuye, T. Yanaki, H. Fujita, J. Polym. Sci., Polym. Phys. Ed., **18**, 547 (1980).
863. Y. Kashiwagi, T. Norisuye, H. Fujita, Macromolecules, **14**, 1220 (1981).
864. L. W. Fisher, A. R. Sochor, J. S. Tan, Macromolecules, **10**, 955 (1977).
865. T. Sato, T. Norisuye, H. Fujita, Polym. J., **16**, 341 (1984).
866. M. Milas, M. Rinaudo, B. Tinland, Polym. Bull., **14**, 157 (1985).
867. J. R. Urwin, M. Girolamo, Makromol. Chem., **160**, 183 (1972).
868. P. J. Kay, F. E. Treloar, Makromol. Chem., **175**, 3207 (1974).
869. A. Dondos, H. Benoit, Macromolecules, **4**, 279 (1971).
870. P. Karayannidis, A. Dondos, Makromol. Chem., **147**, 135 (1971).
871. M. Pezzoli, C. Stea, G. Ceccorulli, G. B. Gechele, Eur. Polym. J., **6**, 1219 (1970).
872. T. Kato, K. Miyaso, I. Noda, T. Fujimoto, M. Nagasawa, Macromolecules, **3**, 777 (1970).
873. Y. Tsunashima, Dactoral Thesis, Kyoto Univ., 1972.
874. K. Takashima, K. Nakae, M. Shibata, H. Yamakawa, Macromolecules, **7**, 641 (1974).
875. Y. Miyaki, Y. Einaga, H. Fujita, Macromolecules, **11**, 1180 (1978).
876. P. Munk, B. O. Gutierrez, Macromolecules, **12**, 467 (1979).
877. Y. Mikayi, Y. Einaga, H. Fujita, M. Fukuda, Macromolecules, **13**, 588 (1980).
878. G. Glöckner, Eur. Polym. J., **15**, 727 (1979).
879. A. Dondos, Eur. Polym. J., **7**, 405 (1971).
880. V. R. Moore, M. A. Uddin, Eur. Polym. J., **6**, 121 (1970).
881. M. R. Knecht, H.-G. Elias, Makromol. Chem., **157**, 1 (1972).
882. F. S. Tan, J. Polym. Sci., Polym. Phys. Ed., **12**, 175 (1974).
883. J. Ansorena, J. J. Iruin, G. M. Guzmán, J. Polym. Sci., Polym. Phys. Ed., **18**, 173 (1980).
884. R. C. Jordan, D. A. Brant, Macromolecules, **13**, 491 (1980).
885. B. Chu, C. Wu, J. Zuo, Macromolecules, **20**, 700 (1987).
886. W. H. Stockmayer, in: M. Nagasawa (Ed.), "Molecular Conformation and Dynamics of Macromolecules in Condensed Systems", Elsevier, Amsterdam, 1988, p. 1.
887. P. N. Chaturvedi, C. K. Patel, J. Polym. Sci., Polym. Phys. Ed., **23**, 1255(1985).
888. J. W. Mays, N. Hadjichristidis, W. W. Graessley. L. J. Fetters, J. Polym. Sci., Polym. Phys. Ed., **24**, 2553 (1986).
889. J. Roovers, P. M. Tororowski, J. Polym. Sci., Polym. Phys. Ed., **26**, 1251 (1988).
890. J. Mays, N. Hadjichristidis, L. J. Fetters, Macromolecules, **17**, 2723 (1984).
891. N. S. Davidson, L. J. Fetters, W. G. Funk, N. Hadjichristidis, W. W. Graessley, Macromolecules, **20**, 2614 (1987).
892. Y. Tsunashima, M. Hirata, N. Nemoto, M. Kurata, Macromolecules, **21**, 1107 (1988).
893. B. J. Baucr, L. J. Fetters, W. W. Graessley, N. Hadjichristidis, G. F. Quack, Macromolecules, **22**, 2337 (1989).
894. P. N. Chaturvedi, C. K. Patel, Makromol. Chem., **186**, 2341 (1985).
895. K. Kajiwara, H. Suzuki, H. Inagaki, M. Maeda, T. Tsuruta, Makromol. Chem., **187**, 2257 (1986).
896. X. Zhongde, N. Hadjichristidis, J. M. Carella, L. J. Fetters, Macromolecules, **16**, 925 (1983).
897. J. W. Mays, L. J. Fetters, Macromolecules, **22**, 921 (1989).
898. Z. Wang, A. Tontisakis, W. H. Tuminello, W. Buck, B. Chu, Macromolecules, **23**, 1444 (1990).
899. W. Brown, P. Zhou, Macromolecules, **24**, 5151 (1991).
900. H. Ochiai, M. Handa, H. Matsumoto, T. Moriga, I. Murakami, Makromol. Chem., **186**, 2547 (1985).
901. F. Gundert, B. A. Wolf, Makromol. Chem., **187**, 2969 (1986).
902. T. Hirao, A. Teramoto, T. Sato, T. Norisuye, T. Masuda, T. Higashimura, Polym. J., **23**, 925 (1991).
903. Y. Tsunashima, H. Hara, H. Narita, F. Horii, ICR Annual Report, **2**, 28 (1995).

904. F. Abe, Y. Einaga, H. Yamakawa, Macromolecules, **24**, 4423 (1991).
905. F. Mabire, R. Audebert, C. Quivoron, Polym., **25**, 1317 (1984).
906. T. Kato, T. Tokuya, T. Nozaki, A. Takahashi, Polym., **25**, 218 (1984).
907. K. Sivadasan, S. Gundiah, Polym., **28**, 1426 (1987).
908. T. Desai, P. Shah, B. Suthar, J. Polym. Sci., Polym. Phys. Ed., **26**, 491 (1988).
909. K. Patal, P. P. Shah, B. Suthar, J. Polym. Sci., Polym. Phys. Ed., **26**, 491 (1988).
910. B. C. Simionescu, S. Ioan, C. I. Simionescu, J. Polym. Sci., Polym. Phys. Ed., **25**, 829 (1987); C. I. Simionescu, B. C. Simionescu, S. Ioan, J. Polym. Sci., Polym. Lett. Ed., **23**, 121 (1985).
911. J. W. Mays, N. Hadjichristidis, J. S. Lindner, J. Polym. Sci., Polym. Phys. Ed., **28**, 1881 (1990).
912. C. I. Simionescu, S. Ioan, B. C. Simionescu, Europ. Polym. J., **23**, 69 (1987).
913. X. Zhongde, N. Hadjichristidis, L. J. Fetters, Macromolecules, **17**, 2303 (1984).
914. J. W. Mays, W. Ferry, N. Hadjichristidis, L. J. Fetters, Macromolecules, **18**, 2330 (1985).
915. E. Siakali-Kioulafa, N. Hadjichristidis, J. W. Mays, Macromolecules, **22**, 2059 (1989).
916. Y. Fujii, Y. Tamai, T. Konishi, H. Yamakawa, Macromolecules, **24**, 1608 (1991).
917. A. Karandinos, S. Nan, J. W. Mays, Macromolecules, **24**, 2007 (1991).
918. M. Stickler, D. Panke, W. Wunderlich, Makromol. Chem., **188**, 2651 (1987).
919. C. I. Simionescu, S. Ioan, A. Flondor, B. C. Simionescu, Makromol. Chem., **189**, 2331 (1988).
920. M. Ricker, M. Schmidt, Makromol. Chem., **192**, 679 (1991).
921. L. Gargallo, M. I. Munoz, D. Radic, Polym., **27**, 1416 (1986).
922. L. Gargallo, M. I. Munoz, F. R. Dias, D. Radic, Polym., **29**, 1898 (1988).
923. L. Gargallo, N. Hamidi, D. Radic, Polym., **31**, 924 (1990).
924. K. Yamaura, K. Hirata, S. Tamura, S. Matsuzawa, J. Polym. Sci., Polym. Phys. Ed., **23**, 1703 (1985).
925. G. Lutringer, G. Weill, Polym., **32**, 877 (1991).
926. J. Roovers, J. Polym. Sci., Polym. Phys. Ed., **23**, 1117 (1985).
927. T. Corner, F. W. Peaker, J. Polym. Sci., Polym. Phys. Ed., **22**, 2105 (1984).
928. H. Zhiduau, Y. Meina, Z. Xuanqi, W. Wiaobing, J. Xiaoming, H. Jiaxian, L. Chunrong, W. Linfu, Europ. Polym. J., **22**, 597 (1986).
929. J. Roovers, N. Hadjichristidis, L. J. Fetters, Macromolecules, **16**, 214 (1983).
930. G. B. McKenna, G. Hadziioannou, P. Lutz, G. Hild, C. Strazielle, C. Straupe, P. Reimpp, A. Kovacs, Macromolecules, **20**, 498 (1987).
931. N. Khasat, R. W. Pennisi, N. Hadjichristidis, L. J. Fetters, Macromolecules, **21**, 1100 (1988).
932. Y. Einaga, H. Koyama, T. Konishi, H. Yamakawa, Macromolecules, **22**, 3419 (1989).
933. M. E. Lewis, S. Nan, J. W. Mays, Macromolecules, **24**, 197 (1991).
934. J. W. Mays, S. Nan, D. Whitfield, Macromolecules, **24**, 315 (1991).
935. J. W. Mays, S. Nan, M. E. Lewis, Macromolecules, **24**, 4857 (1991).
936. W. -M, Kulicke, M. Prescher, Makromol. Chem., **185**, 2619 (1984).
937. S. Arichi, N. Sakamoto, S. Himuro, M. Miki, M. Yoshida, Polymer, **26**, 1175 (1985).
938. J. W. Mays, W. F. Ferry, N. Hadjichristidis, W. G. Funk, L. J. Fetters, Polym.er **27**, 129 (1986).
939. A. A. Abdel-Azim, S. A. Moustafa, M. M. E. Dessouky, F. Abdel-Rehim, S. A. Hassan, Polymer, **27**, 1406 (1986).
940. S. Arichi, N. Sakamoto, M. Yoshida, S. Himuro, Polymer, **27**, 1761 (1986).
941. T. Masuda, Y. Ohta, T. Yamauchi, S. Onogi, Polym. J., **16**, 273 (1984).
942. J. M. Barrales-Rienda, P. A. G. Gomez, Europ. Polym. J., **20**, 1213 (1984).
943. J. Velickovic, J. Filipovic, Macromolecules, **17**, 611 (1984).
944. Y. Matsushita, Y. Nakao, K. Shimizu, I. Noda, M. Nagasawa, Macromolecules, **21**, 2790 (1988).
945. J. Velickovic, J. Filipovic, Makromol. Chem., **185**, 569 (1984).
946. N. Sakamoto, H. Tanaka, K. Yamaguchi, M. Yasuda, S. Arichi, Makromol. Chem., **186**, 1881 (1985).
947. R. Gavara, A. Campos, J. E. Figueruelo, Makromol. Chem., **191**, 1899 (1990).
948. G. M. Pavlov, E. F. Panarin, E. V. Korneeva, C. V. Kurochkin, V. E. Baikov, V. N. Ushakova, Makromol. Chem., **191**, 2889 (1990).
949. A. Leon, L. Gargallo, D. Radic, A. Horta, Polym., **32**, 761 (1991).
950. M. B. Huglin, M. A. Radwan, Polym., **32**, 1293 (1991).
951. K. Kamide, Y. Miyazaki, H. Kobayashi, Polym. J., **17**, 607 (1985).
952. Y. Matsushita, K. Shimizu, Y. Nakao, H. Choshi, I. Noda, M. Nagasawa, Polym. J., **18**, 361 (1986).
953. M. Benchabane, R. Van Leemput, J. Polym. Sci., Polym. Phys. Ed., **23**, 991 (1985).
954. D. B. Cotts, J. Polym. Sci., Polym. Phys. Ed., **23**, 2217 (1985).
955. S. L. Malhotra, T. L. Bluhm, Y. Deslandes, Europ. Polym. J., **22**, 391 (1986).
956. K. Matsuo, W. H. Stockmayer, F. Bangerter, Macromolecules, **18**, 1346 (1985).
957. Z. Kucukyavuz, S. Kucukyavuz, Makromol. Chem., **187**, 2469 (1986).
958. Z. Kucukyavuz, Makromol. Chem, **191**, 2205 (1990).
959. A. Dems, G. Strobin, Makromol, Chem., **192**, 131 (1991).
960. S. Kucukyavus, Z. Kucukyavus, G. Erdogan, Polym., **31**, 379 (1990).
961. K. Kamide, Y. Miyazaki, H. Yamazaki, Polym. J., **18**, 645 (1986).

962. P. M. Cotts, J. Polym. Sci., Polym. Phys. Ed., **24**, 1493 (1986).
963. P. Gregory, M. B. Huglin, Makromol. Chem., **187**, 1745 (1986).
964. D. Kisakurek, N. Binboga, J. F. Harrod, Polym., **28**, 1767 (1987).
965. J. I. Eguiazabal, J. Areizaga, J. J. Iruin, G. M. Guzman, Europ. Polym. J., **21**, 711 (1985).
966. F. Pilati, A. Munari, P. Manaresi, G. Miloni, V. Bonora, Europ. Polym. J., **23**, 265 (1987).
967. A. Munari, P. Manaresi, E. Chiorboli, A. Chiolle, Europ. Polym. J., **28**, 101 (1992).
968. L. J. Mathias, R. E. Hermes, Macromolecules, **19**, 1536 (1986).
969. W. R. Krigbaum, T. Tanaka, Macromolecules, **21**, 743 (1988).
970. M. Kuwata, H. Murakami, T. Norisuye, H. Fujita, Macromolecules, **17**, 2731 (1984).
971. T. Kitagawa, J. Sadanobu, T. Norisuye, Macromolecules, **23**, 602 (1990).
972. R. A. Kulkarni, S. Gundiah, Makromol. Chem., **185**, 549 (1984).
973. Q. Ying, B. Chu, R. Qian, J. Bao, J. Zhang, C, Xu, Polymer, **26**, 1401 (1985).
974. K. Sakurai, K. Ochi, T. Norisuye, H. Fujita, Polym. J., **16**, 559 (1984).
975. S. Fujishige, Polym. J., **19**, 297 (1987).
976. K. F. Weyts, E. J. Goethals, W. H. Bunge, C. J. B. Treslong, Europ. Polym. J., **26**, 445 (1990).
977. W. R. Krigbaum, G. Brelsford, Macromolecules, **21**, 2502 (1988).
978. K. Kamide, A. Kiguchi, Y. Miyazaki, Polym. J., **18**, 919 (1986).
979. A. Sanchez, A. Bello, C. Marco, J. G. Fatou, Makromol. Chem., **189**, 399 (1988).
980. H. N. Bhargava, C. B. Sharma, R. R. Srivastava, Polmer J., **18**, 619 (1986); H. N. Bhargava, R. R. Srivastava, M. Singh, Polym. J., **19**, 1285 (1987).
981. A. Lapp, J. Herz, C. Strazielle, Makromol. Chem., **186**, 1919 (1985).
982. J. Horska, V. Petrus, M. Bohdanecky, Polym., **28**, 1207 (1987).
983. C. Salom, J. J. Freire, I. Hernandes-Fuentas, Polym., **30**, 615 (1989).
984. T. Itoh, H. Chikiri, A. Teramoto, S. M. Aharoni, Polym. J., **20**, 143 (1988).
985. P. P. Shah, Europ. Polym. J., **20**, 519 (1984).
986. S. Lazcano, A. Bello, C. Marco, J. G. Fatou, Europ. Polym. J., **24**, 985 (1988).
987. J. M. Barrales-Rienda, C. R. Galicia, J. J. Freire, A. Horta, Macromolecules, **16**, 940 (1983).
988. J. M. Barrales-Rienda, C. R. Galicia, J. J. Freire, A, Horta, Macromolecules, **16**, 1707 (1983).
989. A. W. Chow, S. P. Bitler, P. E. Penwell, D. J. Osborne, J. F. Wolfe, Macromolecules, **22**, 3514 (1989).
990. S. A. Swanson, P. M. Cotts, R. Siemens, S. H. Kim, Macromolecules, **24**, 1352 (1991).
991. D. Rosenvasser, R. V. Figini, Makromol. Chem., **185**, 1361 (1984).
992. L. Gargallo, E. Soto, L. H. Tagle, D. Radic, Europ. Polym. J., **24**, 1119 (1988).
993. J. Roovers, J. D. Cooney, P. M. Toporowski, Macromolecules, **23**, 1611 (1990).
994. J. Bravo, M. P. Tarazona, E. Saiz, Macromolecules, **24**, 4089 (1991).
995. P. M. Cotts, S. Ferline, G. Dagli, D. S. Pearson, Macromolecules, **24**, 6730 (1991).
996. J. Danhelka, M. Netopilik, M. Bohdanecky, J. Polym. Sci., Polym. Phys. Ed., **25**, 1801 (1987).
997. B. Pfannemuller, M. Schmidt, G. Ziegast, K. Matsuo, Macromolecules, **17**, 710 (1984).
998. T. Kato, T. Katsuki, A. Takahashi, Macromolecules, **17**, 1726 (1984).
999. S. G. Ring, K. J. l'Anson, V. J. Morris, Macromolecules, **18**, 182 (1985).
1000. C. L. McCormick, P. A. Callais, B. H. Hutchinson, Jr., Macromolecules, **18**, 2394 (1985).
1001. G. V. Laivins, D. G. Gray, Macromolecules, **18**, 1746 (1985).
1002. J. W. Mays, Macromolecules, **21**, 3179 (1988).
1003. K. Nishinari, K. Kohyama, P. A. Williams, G. O. Phillips, W. Burchard, K. Ogino, Macromolecules, **24**, 5590 (1991).
1004. A. Wirscn, Makromol. Chem., **189**, 833 (1988).
1005. N. Donkai, H. Inagaki, K. Kajiwara, H. Urakawa, M. Schmidt, Makromol. Chem., **186**, 2623 (1985).
1006. A. Leon, L. Gargallo, A. Horta, D. Radic, J. Polym. Sci., Polym. Phys. Ed., **27**, 2337 (1989).
1007. H. Kawanishi, Y. Tsunashima, F. Horii, S. Okada, Polym. preprints Japan **45**, 3387 (1996).
1008. C. Wei-Berk, G. C. Berry, J. Polym. Sci., Polym. Phys. Ed., **28**, 1873 (1990).
1009. A. Matsumoto, T. Tarui, T. Otsu, Macromolecules, **23**, 5102 (1990).
1010. L. J. Fetters, N. Hadjichristidis, J. S. Lindner, J. W. Mays, W. W. Wilson, Macromolecules, **24**, 3127(1991).
1011. S. R. Gooda, M. B. Huglin, Macromolecules, **25**, 4215(1992).

Note: References 508–511 and 575–589 are for biological polymers such as collagen, gelatin and poly(nucleotides), which do not appear in the present tables.

Sedimentation Coefficients, Diffusion Coefficients, Partial Specific Volumes, Frictional Ratios, and Second Virial Coefficients of Polymers in Solution*,**

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* Based on a table in the second edition, by P. E. O. Klärner, and H. A. Ende, BASF AG, Ludwigshafen, Germany.

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A. SEDIMENTATION COEFFICIENTS, DIFFUSION COEFFICIENTS, PARTIAL SPECIFIC VOLUMES, AND FRICTIONAL RATIOS OF POLYMERS IN SOLUTION

(2, 349–357, 799, 800)

1. Introduction

1.1. Sedimentation Coefficient The sedimentation coefficient s is defined as the sedimentation velocity in a unit force field

$$s = (\mathrm{d}r/\mathrm{d}t)/(\omega^2 r) \qquad \text{(A1)}$$

where r is the distance from center of rotation, and ω, the angular velocity. For a given polymer–solvent system, the sedimentation coefficient is dependent on temperature T, pressure p, and polymer concentration C. Experimentally, sedimentation coefficients can be determined only with an ultracentrifuge.

Concentration Dependence

For polymer solutions studied by ultracentrifugation, the following concentration dependence of the sedimentation coefficient holds:

$$1/s = (1/s_0)\,(1 + k_s C + k_s' C^2 + \cdots) \qquad \text{(A2)}$$

where s_0 is the sedimentation coefficient at zero concentration. In Section B the concentration dependence of s according to Eq. (A2) is listed. In some cases, special extrapolation procedures are used. The tables in Section B also refer to the appropriate reference. For special treatments, see, e.g., Gehatia (358,359).

Sector shaped cells are normally used in an ultracentrifuge. In these cells, the square-dilution rule has to be taken into account (360):

$$C_t/C_0 = (r_0/r_t)^2 \qquad \text{(A3)}$$

where r_0 and r_t are the distances of the meniscus and of the boundary from the axis of rotation and C_0 and C_t the concentrations at zero time and at time t at the boundary, respectively.

Averages of the Sedimentation Coefficient

The average of a function $X(M)$ is defined as

$$X_\beta = \left(\int_0^\infty W(M) M^{\beta-1} X(M)\,\mathrm{d}M\right) \Big/ \left(\int_0^\infty W(M) M^{\beta-1}\,\mathrm{d}M\right) \qquad \text{(A4)}$$

with $\int_0^\infty W(M)\,\mathrm{d}M = 1$, where $W(M)$ denotes the probability density function (in our case the molar mass distribution). The average with $\beta = 0$ is called the number average X_n, with $\beta = 1$, the mass average X_w, and with $\beta = 2$, the z-average X_z.

In principle it is possible to evaluate the various averages of the sedimentation coefficient from the distribution of the concentration gradient, $\partial C/\partial r$, in the cell:

$$s_n = M_n \int_0^\infty W(M) M^{-1} s(M)\,\mathrm{d}M$$

$$s_w = \int_0^\infty W(M) s(M)\,\mathrm{d}M$$

$$s_z = (1/M_w) \int_0^\infty W(M) M s(M)\,\mathrm{d}M \qquad \text{(A5)}$$

where $W(M) = C(M)/\int_0^\infty C(M)\,\mathrm{d}M = C(M)/C_0$ is the mass distribution of the polydisperse polymer in the cell and s_n, s_w, and s_z the number average, mass average and z-average of the sedimentation coefficient. In many cases the migration of the maximum of the quantity $\partial C/\partial r$ is determined, which leads to a sedimentation coefficient s_t, that in turn is related, in a complicated manner, to the averages of s in Eq. (A5). Another method of evaluating the sedimentation coefficient is by observing the median, that is the line dividing the gradient curve in two equal areas, yielding a value s_m which is also a rather complicated average. Provided the skewness of the molar mass distribution is not very pronounced, s_w can be calculated by (Ref. 259)

$$s_w = (3s_m - s_t)/2 \qquad \text{(A6)}$$

Most ultracentrifuges measure the refractive index difference between polymer and solvent, $\mathrm{d}n$, or the refractive index gradient $\partial n/\partial r$ versus r (Schlieren optics). With certain assumptions $\partial n/\partial r$ can be related to the concentration gradient (Ref. 14)

$$\partial n/\partial r = R(\partial C/\partial r) \qquad \text{(A7)}$$

where R is called the specific refractive index increment. The interference optics measure the deflection of the parallel interference lines in the solution. This deflection is directly proportional to the concentration of the polymer.

The absorption optics measure the optical density of the system as a function of the rotor distance, which, according to the Lambert–Beers law is proportional to the concentration of the polymer. The intensity is measured with a photoelectric scanner.

A multiplexer is used in order to measure several concentrations during one run (869–871).

Sedimentation Coefficient – Molar Mass Relationship
The dependence of s on the molar mass can be given by a power law expression:

$$s = K_s M^{a_s} \tag{A8}$$

where K_s and a_s are empirical constants determined for each polymer–solvent system at given values of temperature and pressure. In Section B, relationship (A8) is listed whenever quoted in the literature, in preference to single s values,

Temperature Dependence, Pressure Dependence
Svedberg and Pedersen (2) and Mosimann and Signer (361) derived expressions for the temperature and pressure dependence of the sedimentation coefficient with restricted validity:

$$s/s^\theta = (\eta_1^\theta/\eta_1)\,(1 - v_2^\theta \rho_1^\theta)/(1 - v_2\rho_1) \tag{A9}$$

where s^θ, η_1^θ, v_2^θ and ρ_1^θ denotes the values at the reference temperature and/or pressure. Equation (A9) holds in the case of an incompressible medium and if no changes of size, shape and solvation state of the dissolved molecules in the temperature and pressure region take place. From these assumptions it may be deduced that Eq. (A9) can be applied with sufficient accuracy only on proteins in aqueous solvents.

Due to the high centrifugal fields applied in the ultracentrifuge, pressure gradients with pressures up to 200 bars are encountered in the cell and change appreciably the viscosity and the density of the solvent and the partial specific volume of the polymer. Thus the sedimentation coefficient, s, measured at a pressure p differs from the sedimentation coefficient, s^0, measured at 0 bar. Application of Eq. (A9) for the calculation of s requires the knowledge of η_1 (362), ρ_1 (given in a number of Handbooks) and v_2 (148). More precise equations for the calculation of the pressure influence on the sedimentation have been worked out by several authors (363,364,365,800). The equation of Oth and Desreux (Ref. 363)

$$s = s^0(1 - \mu p) \tag{A10}$$

has very often been used for the calculation of s^0 (800). The coefficient μ depends on the polymer–solvent system and varies only slightly with pressure.

In Section B, the intrinsic sedimentation coefficient, $[s^0]$, instead of s^0, has been occasionally listed. $[s^0]$ is related to Eq. (9): $[s_0] = s_0\eta_1/(1 - v_2\rho_1)$.

1.2. Diffusion Coefficient The translational diffusion coefficient D is defined by Ficks' first law:

$$J = -D\nabla C \tag{A11}$$

where J is the flow of substance (total number of particles transported in unit time across unit surface) and ∇C the concentration gradient. For one dimensional diffusion, Eq. (A11) reduces to

$$(1/A)\,(\partial n/\partial t) = -D(\partial C/\partial r) \tag{A12}$$

where A denotes the area and n the number of moles. The negative sign in Eqs. (A11) and (A12) indicate that diffusion takes place in the direction of decreasing concentration. For practical reasons, Ficks second law is prefered

$$(\partial C/\partial t) = D\Delta C \tag{A13}$$

where Δ denotes the Laplace operator. Eq. (A13) reduces to

$$(\partial C/\partial t) = D(\partial^2 C/\partial r^2) \tag{A14}$$

for the one dimensional case. Eqs. (A13) and (A14) allow the determination of D when solved with consideration to certain initial and boundary conditions.

Experimentally translational diffusion coefficients can be determined with an ultracentrifuge (see Section 1.3) in special diffusion cells using Eqs. (A11)–(A14) (800) and with dynamic light scattering (DLS). Since 1970, when the dynamic light scattering became available, nearly all diffusion coefficients have been measured by this method. Based on the theory of Pecora (853,854), the experimentally measured autocorrelation function $g_2(t)$ is linked with the diffusion coefficient in polymer solutions (Refs. 730,855):

$$\lim_{c\to 0} g_2(t) = A + B\left(M_w^{-1}\int_0^\infty MW(M)p(q,M)\exp(-q^2D(M)t)\mathrm{d}M\right)^n \tag{A15}$$

where $n = 1$ for the heterodyne procedure and $n = 2$ for the homodyne one. Eq. (A15) reduces, in the case of a monodisperse polymer solution, to

$$\lim_{c\to 0} g_2(t) = A + B(p(q,M)\exp(-q^2D(M)t))^n \tag{A16}$$

A and B are measurable constants. Extrapolation of q against zero yields $P(q,M) = 1$. Note that $g_2(t)$ is the z-average of the function, $\exp(-q_2 D(M)t)$. In this way one obtains the direct measurable quantity D_z from dynamic light scattering on polydisperse polymer solutions.

For polydisperse polymer solutions, Eq. (A16) may be expanded in a series (Ref. 856):

$$\lim_{\substack{C\to 0\\ q\to 0}} g(t) = \exp(-q^2 D_z t)\,(1 + (\mu_2/2!)t^2 - (\mu_3/3!)t^3 + \cdots) \tag{A17}$$

where the moments, μ_i, are related to the molar mass distribution of the polymer (671).

More general methods for the determination of the distribution of the diffusion coefficient and of the molar mass are given elsewhere (857,858).

As has been pointed out, only the translational diffusion coefficient is listed in the tables. For higher modes of the autocorrelation function (e.g. rotation, vibration) see the appropriate literature (853,854).

The entire description of the translational diffusion of a polymer molecule in solution requires the determination of the mutual or cooperative diffusion coefficient, D_m, which characterizes the relaxation of a concentration gradient and the self-diffusion coefficient, D_s, which describes the Brownian or random motions of the polymer molecule (860–863).

$$D_m = (RT/f)\,(1 - v_2\rho_1)\,(1 + 2A_2MC + 3A_3MC^2 + \cdots)$$
$$D_s = RT/f$$

At infinite dilution, $D_m = D_s \times D_m$ is normally measured with classical diffusion cells or dynamic light scattering whereas D_s can be studied by measuring the migration of labeled solute molecules (864), by pulsed field gradient NMR (865), and by forced Rayleigh scattering (866). Normally, D_m is given in the tables. In some cases, especially where D_s is extrapolated to zero concentration, $D_s = D_m$ is given in the tables.

Concentration Dependence

The concentration dependence of the diffusion coefficient may be described as

$$D = D_0(1 + k_D C + k'_D C^2 + \cdots) \tag{A18}$$

where D_0 is the diffusion coefficient at zero concentration. In Section B a concentration dependence of D according to Eq. (A18) is listed. In some cases, special extrapolation procedures are used. In these cases Section B refers to the appropriate references.

Averages of the Diffusion Coefficient

Similar considerations as made on sedimentation coefficients hold for diffusion coefficients. From measurements in a diffusion cell, one obtains the various averages of the diffusion coefficient from the distribution of the concentration gradient, $\partial C/\partial r$, in the cell:

$$D_n = M_n \int_0^\infty W(M)M^{-1}D(M)\,dM$$
$$D_w = \int_0^\infty W(M)D(M)\,dM$$
$$D_z = (1/M_w)\int_0^\infty W(M)MD(M)\,dM \tag{A19}$$

where $W(M) = C(M)/C_0$.

The diffusion cell may be combined in the same manner as in the case of the ultracentrifuge with Schlieren optics, interference optics, and absorbtion optics. In most cases, the mass average and the area average of the diffusion coefficient are determined. They can be calculated with the help of Eqs. (A14) and (A19):

$$D_w = \left(\int_0^\infty r^2(\partial C/\partial r)\,dr\right) \Big/ \left(2t\int_0^\infty (\partial C/\partial r)\,dr\right) \tag{A20}$$

$$D_A = \int_0^\infty ((\partial C/\partial r)\,dr)^2/(4\pi t(\partial C/\partial r)_{max})^2 \tag{A21}$$

$(\partial C/\partial r)_{max}$ is the maximum height of $\partial C/\partial r$. The integral in Eq. (A21) is the area of the curve.

As has been pointed out, dynamic light scattering measurements normally yield the z-average of D. Nevertheless it might be possible to determine averages other than D_z with the help of Eqs. (A19) and (A21). In Section B the different averages of D are quoted. In some cases unusual averages are listed. In these cases Section B refers to the appropriate references.

Diffusion Coefficient – Molar Mass Relationship

The diffusion coefficient – molar mass dependence frequently takes the form

$$D = K_D M^{-a_D} \tag{A22}$$

where K_D and a_D are constants for each polymer–solvent system at given values of temperature and pressure. In Section B relationship A(22) is listed whenever quoted in the literature, in preference to single D values.

Temperature Dependence, Pressure Dependence

A similar expression as Eq. (A9) holds under the same restricted conditions for the temperature and pressure dependence of the diffusion coefficient:

$$(D/T)/(D^\theta/T_0) = \eta_1^\theta/\eta_1 \tag{A23}$$

where D^θ and η_1^θ denote the values at the reference temperature T_0 and/or pressure p_0. As has been pointed out in Eq. (A9), Eq. (A23) can be applied with sufficient accuracy on proteins in aqueous solvents only.

The temperature dependence of the diffusion coefficient is often described by the following exponential function

$$D = D_\infty \exp(-E_D/RT) \tag{A24}$$

where E_D denotes the apparent activation energy of the diffusion and D_∞ the diffusion coefficient at the limit $T = \infty$. In Section B we have listed the temperature and pressure dependence of D whenever quoted in the literature.

1.3. Molar Mass Averages Determined from Sedimentation and Diffusion Coefficients For the experimental determination of the molar mass from sedimentation and diffusion measurements, the Svedberg Eq. (A2) is used:

$$M = (s_0/(1/v_2\rho_1))/(D_0/RT) \tag{A25}$$

Here s_0 and D_0 are the corrected and standardized coefficients for zero polymer concentration. For polydisperse polymers the various averages of the sedimentation and diffusion coefficient s and D are inserted, obtaining certain molar mass averages $M_{\beta,\gamma}$. Thus Eq. (A25) acquires the more general form:

$$M_{\beta,\gamma} = (s_{0,\beta}/(1 - v_2\rho_1))/(D_{0,\gamma}/RT) \tag{A26}$$

For β = n, w, z, ... and γ = n, w, z, ... Eq. (A26) defines several different molar mass averages $M_{n,n}$, $M_{w,n}$, $M_{w,w}$ etc. The averages, $M_{n,n}$, and $M_{w,w}$, are different from M_n and M_w respectively (859). The coefficients with β = n and z and γ = n are determinable only with large error. Instead, the coefficients with $\beta = t$ and m (see under Section 1.1.) and γ = A and w (see under Section 1.2.) are usually evaluated. Thus, rather peculiar averages, e.g. $M_{t,A}$, $M_{m,w}$, and $M_{m,A}$, result. The more straightforward molar mass averages $M_{n,w}$, $M_{w,w}$, $M_{z,w}$, etc., in relation to the familiar averages, M_n, M_w, and M_z, are found elsewhere in the literature (369–371,859).

1.4. Partial Specific Volumes The volume, V^{id}, of an ideal two component system can be expressed in terms of the masses m_1 and m_2 and the specific volumes, v_1^0 and v_2^0, of the two components by the equation

$$V^{id} = m_1 v_1^0 + m_2 v_2^0 \tag{A27}$$

Most components do not behave ideally upon mixing; i.e. they react with each other in a way so that the total volume deviates from V^{id}. The total volume can then be written as

$$V = m_1 v_1 + m_2 v_2; \quad v_i = (\partial V/\partial m_i)_{p,T,m_{j\neq i}} \tag{A28}$$

where v_i $(i = 1, 2)$ is the partial specific volume of component i. When component 2 is polymer and component 1 solvent, then, for practical reasons it is convenient to introduce the so-called apparent partial specific volume v_2^* which is defined by

$$V = m_1 v_1^0 + m_2 v_2^* \tag{A29}$$

where v_1^0 denotes the specific volume of the solvent. The quantity, v_2^*, now contains the parameter of nonideal mixing of both the solvent and the polymer. In practice, however v_2^* differs not much from v_2 if the polymer concentration is kept low (up to 1%). Dividing Eq. (A29) by the total mass $m_1 + m_2$ leads to

$$v = w_1 v_1^0 + w_2 v_2^* \tag{A30}$$

where v is the specific volume of the solution and $w_i = m_i/\sum m_i$ are the mass fractions of the solvent $(i = 1)$ and the polymer $(i = 2)$.

With $v = 1/\rho$ and $v_1^0 = 1/\rho_1$, ρ and ρ_1 being the densities of the solution and the solvent, respectively, it is readily found that

$$v_2^* = (1/w_2)\,(\rho_1 - w_1\rho)/(\rho\rho_1) \tag{A31}$$

Eq. (A31) demonstrates that v_2^* can be determined by measuring ρ_1 and ρ. Numerous methods for determining densities are described in the literature (268,372–379).

In order to determine the partial specific volume from the apparent specific volume, Eq. (A29) yields

$$(\partial(m_2 v_2^*)/\partial m_2)_{m_1} = m_2(\partial v_2^*/\partial m_2)_{m_1} + v_2^* = (\partial V/\partial m_2)_{m_1} \tag{A32}$$

where $(\partial V/\partial m_2)_{m_1}$ is, according to definition, equal to v_2. In terms of mass fractions, Eq. (A32) can finally be written as

$$v_2 = v_2^* + w_1 w_2(\partial v_2^*/\partial w_2) \tag{A33}$$

Most values reported in the literature are v_2^*-values rather than v_2-values, since extrapolation according to Eq. (A33) is usually omitted. The differences between v_2^* and v_2 are, however, often small.

1.5. Frictional Ratios The molar frictional coefficient, f_{sp}, of an unsolvated spherical molecule may be computed by the formula based on Stokes law:

$$f_{sp} = 6\pi\eta R_H N_A = 6\pi\eta[3N_A M v_2/(4\pi)]^{1/3} \tag{A34}$$

where η is the viscosity of the solvent and N_A is Avogadro's number. When the shape of a molecule deviates from a sphere, or when it is solvated, then the frictional coefficient, f_0, of such a molecule is larger than that of the spherical molecule. The frictional ratio, f_0/f_{sp}, thus permits to draw conclusions concerning either solvation or shape of the

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molecule. It is possible to calculate the dimensions of the nonspherical molecule, provided a particular model (ellipsoid, cylinder, etc.) for the molecule is adopted and either the degree of solvation is known or assumed to be negligible. The molar frictional coefficient can be determined either from sedimentation velocity data, provided the molar mass is known from independent measurements according to the Svedberg equation

$$f_0 s_0 = M(1 - v_2\rho_1) \qquad \text{(A35)}$$

or from diffusion measurements, using the relation

$$f_0 = RT/D_0 \qquad \text{(A36)}$$

Eqs. (A34)–(A36) are most frequently used for calculating the frictional ratio. f_0/f_{sp} is listed in the tables. Other relationships for the determination of f_0/f_{sp} are quoted in the literature (2). In these cases special reference to the literature is made in the tables.

In few cases, only the molar frictional coefficient f_0, rather than f_0/f_{sp}, was quoted in the literature. These values are inserted in the same column as the values for the frictional ratio.

1.6. List of Symbols and Abbreviations

Symbols

C	Concentration
C_0	Initial polymer concentration
C_t	Polymer concentration at time t
D	Diffusion coefficient
D_A	Area average of the diffusion coefficient
D_n, D_w, D_z	Number average, mass average, and z-average of the diffusion coefficient
D	Diffusion coefficient at reference temperature and/or pressure
D_0	Diffusion coefficient at zero concentration
η_1	Viscosity of solvent
η_1^θ	Viscosity of solvent at reference temperature and/or pressure
f_{sp}	Molar frictional coefficient of a spherical molecule
f_0	Molar frictional coefficient at zero concentration
k_s, k_s	Concentration coefficients defined by Eq. (A2)
k_D, k_D	Concentration coefficients defined by Eq. (A18)
M	Molar mass of the polymer
M_n, M_w, M_z	Number average, mass average and z-average of the molar mass of the polymer
M	Molar mass determined by Mark–Houwink equation
M_{MW}	Molar mass determined by equations from Mandelkern and Flory, or Wales and Van Holde (see Section 1.7)
$M_{\beta,\gamma}$	Molar mass average determined by Eq. (A26) from $s_{0,\beta}$ and $D_{0,\gamma}$
$M_{s,w}$, $M_{s,A}$, $M_{s,D}$	Molar mass average determined from undefined sedimentation coefficient s and D_w, D_A and undefined diffusion coefficient D, respectively
n	Refractive index or number of moles
ω	Angular velocity
r	Distance from center of rotation
r_0	Distance of meniscus from center of rotation
r_t	Distance of boundary from center of rotation at time t
ρ	Density of solution
ρ_1	Density of the solvent
R	Gas constant
s	Sedimentation coefficient
s_n, s_w, s_z	Number average, mass average, and z-average of the sedimentation coefficient
s_θ	Sedimentation coefficient at reference temperature and/or pressure
s^0	Sedimentation coefficient at pressure $p = 0$ bar
s_0	Sedimentation coefficient at zero concentration
$[s_0]$	Intrinsic sedimentation coefficient (see under Section 1.1)
s_m	Sedimentation coefficient determined from migration of gradient curve median
s_t	Sedimentation coefficient determined from migration of gradient curve maximum
T	Temperature
θ	Theta temperature ($A_2 = 0$)
t	Time
v_i	Partial specific volume of component i ($i = 1$, solvent; $i = 2$, solute)
v_1^0	Specific volume of component i
v_2^*	Apparent specific volume of solute

Abbreviations

A	Archibald method
approx.	approximately
CLS	Classical light scattering method
DLS	Dynamic light scattering method
NBS	National Bureau of Standards, USA (now National Institute of Standard and Technology, NIST)
OS	Osmometry
PCC	Pressure Chemical Co., Pittsburg, PA, USA
RT	Room temperature
SE	Sedimentation equilibrium
SRM	Standard reference material
SV	Sedimentation velocity
TS	Toyo Soda Co., Japan
V	Number of single values given in Ref. cited

1.7. Miscellaneous With certain assumptions it is possible to determine the molar mass from a combination of the intrinsic viscosity $[\eta]$ and the limiting sedimentation coefficient s_0. In this way, Mandelkern and Flory (128) and Wales and Van Holde (134) derived an expression. An expression similar to the equation between M, $[\eta]$, and s_0 was derived between M, $[\eta]$, and D_0 (128). The molar masses determined from either one of these relationships are refered to as MW.

B. TABLES OF SEDIMENTATION COEFFICIENTS, DIFFUSION COEFFICIENTS, PARTIAL SPECIFIC VOLUMES, AND FRICTIONAL RATIOS OF POLYMERS IN SOLUTION

TABLE 1. POLY(ALKENES)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Poly(ethylene)	1-Bromonaphthalene	110	4.7	1.42						M_v	38
			14.2	2.03							
			21.8	2.39							
			55.0	3.39							
linear	1-Bromonaphthalene	110	3.5–274	$s_0 = 7.74 \times 10^{-15} M^{0.344}$						M_v	38
		120	69	−4.09						M_{MW}	509
			136	−5.14							
branched		120	88	−5.18						M_{MW}	509
linear	1-Chloronaphthalene	120		−1.67						$[\eta] = 51$ cm^3/g	27
				−2.37						90	
				−3.31						169	
branched, fractions branches per molec.											
1.9				−1.06						$[\eta] = 18$ cm^3/g	27
2.6				−1.91						33	
3.0				−2.85						56	
2.9				−2.77						60	
4.6				−3.57						66	
5.7				−4.11						74	
6.4				−4.52						88	
13.7				−7.30						118	
16.1				−7.87						129	
26.3				−11.37						139	
		110							[a]		532
	Diphenyl	123							[a]		
		130	26			9.80				D_A, M_w, D_w given in Ref.	593
			60			6.42					
			79			6.42					
			99			4.73					
			261			3.13					
			493			2.15					
branched	Tetralin	80	25.7			0.87				$D_{0,z}$; $M_w/M_n \cong 3$	619
			503			0.20				$D_{0,z}$; $M_w/M_n \cong 5$	
Poly(1-butene)	Ethyl octanoate	22 ($T=\theta$)	800	1.30	55.5				1.106[b]	M_{MW};	409,410
Poly(isobutene)	Cyclohexane	34 ($T=\theta$)		6.4[c]	721 ± 159					M_v; Oppanol B 100	576
									1.091	M_n	32,621,622
		20	30.9	0.925	54						
			86.7	1.49	128						
			172	1.94	191						
			672	3.33	510						

			1420	4.45	836					
		25	21			6.32	4.8		M_w; $M_w/M_n = 3.2$	620
			78			2.99	41		M_w; $M_w/M_n = 2.1$	
			122			2.26	53		M_w; $M_w/M_n = 2.0$	
			176			1.84	79		M_w; $M_w/M_n = 2.7$	
			490			1.01	167		M_w; $M_w/M_n = 4.5$	
			698			0.79	379		M_w; $M_w/M_n = 5.1$	
			1800			0.43	348		M_w; $M_w/M_n = 10.8$	
	Ethyloctanoate	22	750	1.3	55.5			1.106	s_t	237
	n-Heptane	25	145						M_w; C from 0.53 to 2.95 g/dm^3; M_n, D_A given in Ref.	511
		25	65.2–308			$D_0 = 5.01 \times 10^4$ M$^{0.555}$			M_w; CLS	449
	Octane	20.9	52					1.072	M	148
		23.2	1600					1.069		
		20.7	1600					1.075		
		25	1900	5.8					M_w, s^0, $C = 3.461$ g/dm^3; 25V; C from 3.461 to 16.293 g/dm$_3$ p from 0 to 46×10^5 Pa	623
	Tetralin	24	388			0.68			Oppanol B 50; $M_w/M_n = 3.0$	624
	Toluene	24	388			2.7			Oppanol B 50; $M_w/M_n = 3.0$	624
Poly(isobutylene)	Benzene	24.5				$D_0 = 1.27 \times 10^{-8}\ M_w^{-0.48}$;			$M_w/M_n = 1.2$	989
	Chloroform	25				$D_0 = 6.22 \times 10^{-8}\ M_w^{-0.60}$			$M_w/M_n = 1.2$	
	Cyclohexane	25				$D_0 = 1.78 \times 10^{-4}\ M_w^{-0.572}$	CLS and DLS		990	
						$1.33 \times 10^5 \le M_w \le 14.8 \times 10^5$ g/mol				
	Isoamyl isovalerate (IAIV)	25 ($T = \theta$)	18.1			5.1	−8.7		$M_w/M_n = 1.08$	1050
			48.5			3.1	−16		$M_w/M_n = 1.06$	
			139			1.9	−24		$M_w/M_n = 1.07$	
			422			1.1	−47		$M_w/M_n = 1.09$	
			634			0.86	−64		$M_w/M_n = 1.06$	
			819			0.75	−80		$M_w/M_n = 1.09$	
			1760			0.52	−110		$M_w/M_n = 1.09$	
	n-Heptane	25	18.1			16.3	−1.5		M_w	1050
			48.5			9.5	3.5		M_w	
			139			5.5	13		M_w	
			422			2.8	56		M_w	
			634			2.3	68		M_w	
			819			2.0	88		M_w	
			1760			1.3	180		M_w	
		25				$D_0 = 4.10 \times 10^{-4}\ M_w^{-0.560}$;			CLS and DLS	990
						$0.64 \times 10^5 \le M_w \le 14.8 \times 10^5$ g/mol				
	Isoamyl isovalerate (IAIV)	22.1				$D_0 = 5.94 \times 10^{-5}\ M_w^{-0.493}$;			CLS and DLS	990
						$0.64 \times 10^5 \le M_w \le 14.8 \times 10^5$ g/mol				
		25 ($T = \theta$)	420			1.06			$[\eta] = 70.9$ cm^3/g	1051
			634			0.86			$[\eta] = 88.2$ cm^3/g	
			819			0.75			$[\eta] = 99.8$ cm^3/g	
			1760						$[\eta] = 146$ cm^3/g	

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	υ_2 (cm^3/g)	Remarks	Refs.
Poly(1-hexene)	Acetone	20	46.8	11.5		16.2				s_f; D_A; $M_{f,A}$	53
			222	21.5		6.4					
			571	33.8		3.0					
		20							0.793		54
Poly(tetrafluoroethylene)	Perfluorotetracosane	325	260			3.66	−35.2				1004
			2100			1.21	−130.0				

[a] See Ref.for various methods applied and values obtained.
[b] Corrected for pressure influence.
[c] See Ref. for def. of s.

TABLE 2. POLY(DIENES)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	υ_2 (cm^3/g)	Remarks	Refs.
Poly(butadiene)	Diethyl ketone	10.3	60	1.76						M_{MW}	590
			187	2.76							
			350	3.45							
			436	4.28							
			778	4.52							
			1380	5.15							
cis-4	Dodecane	80				23.8					625
cis-4	Hexafluorobenzene	80				16.3					625
95%, 1,4 *cis*	Diethylketone	10.3 ($T=\theta$)		$s_0 = 0.53 \times 10^{-15}\ M^{0.5}$						s_n, s_w, s_f, $s_{m,0}$ given in Ref.	574
90%, 1,4 *cis* (TiJ_4)	Hexane/heptane (1/1)	20	55–1080	$s_0 = 2.80 \times 10^{-15}\ M^{0.48}$						$M_{s,D}$	323
90%, 1,4 *cis* (CoCl2)	Hexane/heptane (1/1)		34.7–1040	$s_0 = 2.33 \times 10^{-15}\ M^{0.50}$						$M_{s,D}$	323
cis-4	Hexatriacontan	80				4.78					625
linear	Cyclohexane	25				$D_0 = 1.45 \times 10^{-4}\ M_w^{-0.561}$; $11000 \le M_w \le 760000$ g/mol					987
	Dioxane	26.5 ($T=\theta$)				$D_0 = 6.34 \times 10^{-5}\ M_w^{-0.496}$; $11000 \le M_w \le 760000$ g/mol					
18-armed star	Cyclohexane	25				$D_0 = 2.39 \times 10^{-4}\ M_w^{-0570}$; $99000 \le M_w \le 1900000$ g/mol					
	Dioxane	26.5 ($T=\theta$)				$D_0 = 8.87 \times 10^{-5}\ M_w^{-0.497}$; $99000 \le M_w \le 1900000$ g/mol					
Poly(butadiene-*stat*-acrylonitrile)	Methyl ethyl ketone/cyclohexane (47.5/52.5)	22 ($T=\theta$)	34.8–780	$s_0 = 1.28 \times 10^{-2}\ M^{0.5}$		$D_0 = 1.38 \times 10^{-4}\ M^{-0.5}$;				D_A; $M_{s,D}$, k_s given in Ref.	552
	Methyl ethyl ketone/isopropanol (60/40)	20 ($T=\theta$)	475	7.58	19.7					D_A; $M_{s,D}$	552
			205	4.65	16.0						
			101	3.43	20.6						
Poly(butadiene-*co*-styrene)											
branched	Benzene	25	81.9–168	1.21–4.63				0.995 ± 0.005		See Ref.	406
	Methyl *n*-propyl ketone	21 ($T=\theta$)	49–514	$s_0 = 1.04 \times 10^{-15}\ M^{0.5}$							573
21 mass% styrene	Methyl *n*-propyl ketone	21 ($T=\theta$)	49–514	$s_0 = 0.83 \times 10^{-15}\ M^{0.5}$						M_n	344
Poly(chloroprene)	Methyl ethyl ketone	25	56–380	$s_0 = 3.1 \times 10^{-15}\ M^{0.5}$				0.773		M_w; OS; SE	444

Poly(isoprene)									
linear	Carbon tetrachloride	5	5		$D_0 = 10^{-7.73\pm0.08}\ M^{-0.54\pm0.04}$			$M_w/M_n = 1.05-1.11$	626
	Carbon tetrachloride	50	5		$D_0 = 10^{-7.47\pm0.05}\ M^{-0.61\pm0.01}$			$M_w/M_n = 1.02-1.07$	
4-armed star	Carbon tetrachloride	50	21.2–122		$D_0 = 10^{-7.53\pm0.15}\ M^{-0.57\pm0.03}$			M_w	
18-armed star	Carbon tetrachloride	50	61.7–630		$D_0 = 10^{-7.22\pm0.12}\ M^{-0.61\pm0.03}$			M_w	
3-armed star	Cyclohexane	–	137		2.4	–		$[\eta] = 0.958$ dL/g	988
4-armed star			173		1.87	76.5		$[\eta] = 0.973$ dL/g	
			380		1.20	138		$[\eta] = 1.74$ dL/g	
8-armed star			795		0.91	235		$[\eta] = 1.83$ dL/g	
			1760		0.52	250		$[\eta] = 3.30$ dL/g	
			4400		–	–		$[\eta] = 6.37$ dL/g	
12-armed star			96		–	–		$[\eta] = 0.294$ dL/g	
			810		0.89	–		–	
			3900		0.37	560		$[\eta] = 4.33$ dL/g	
			5280		0.305	331		$[\eta] = 5.63$ dL/g	
18-armed star			93		–	–		$[\eta] = 0.182$ dL/g	
			218		2.4	–		–	
			384		1.88	35		–	
			800		1.06	100		$[\eta] = 1.02$ dL/g	
			1480		0.71	169		$[\eta] = 1.58$ dL/g	
			3570		0.455	303		$[\eta] = 2.90$ dL/g	
natural rubber, crepe	Chloroform	20	270	15.5	2.24		3.32	$M_{s,D}$; Determ. of f_0/f_{sp} see Ref.	337
			485	15.5	1.26		5.10		
			930	27.5	1.16		5.26		
			125		2.63		3.82	M_w; OS; Determ. of f_0/f_{sp} see Ref.	337
			275		1.64		4.71		
			330		1.55		4.67		
			450		1.41		4.64		
			760		1.44		3.82		
	Cyclohexane	20	1600	4.6	0.48			$M_{s,D}$	
			1750	4.76	0.46				
			1800	4.35	0.41				
			1600	4.6	0.48		1.10	s_t; $M_{t,A}$	29
natural rubber	Hexane	20	270	9	3			$M_{s,D}$	337
			1660	21	1.01		1.54		
1,4-*cis*	Hexane	20		$s_0 = 5.01 \times 10^{-15}\ M^{0.45}$	$D_0 = 3.98 \times 10^{-2}\ M^{-0.55}$			$M_{s,D}$	324
1,4-*cis* ($TiCl_4$/AL $(C_7H_{15})_3$)	Hexane	25	460–970	$s_0 = 7.76 \times 10^{-2}\ M^{0.42}$				$M_w/M_n = 1.8-2.9$	627

TABLE 3. ACRYLIC POLYMERS

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Poly(acrylamide)	Water	20	19.4–534	$s_t = 8.17 \times 10^{-15} M^{0.31}$		$D_A = 8.46 \times 10^{-4} M^{-0.69}$			0.769	D measured in H_2O/KCl (0.2 mol/dm^3); $M_{t,A}$	266,154
			130–8200	$s_w = 1.07 \times 10^{-15} M_w^{0.48\pm0.01}$		$D_{z,0} = 1.24 \times 10^{-4} M_w^{-0.53\pm0.01}$			0.708	SV; 15V; M_w/M_n = 1.15–1.4	628
		20	5000						0.693	C = 0.06, 1.2 g/dm^3	629
			250, 610						0.696	M_w; M_w/M_n = 1.5; C = 5, 10 g/dm^3	630
	Water/NaCl (0.2 mol/dm^3)	20	250, 610						0.687	M_w; M_w/M_n = 1.5; C = 5, 10 g/dm^3	630
		20		$s_0 = 8.17 \times 10^{-15} M_{s,D}^{0.31}$		$D_0 = 8.46 \times 10^{-4} M^{-0.69}$					632
	Water/NaCl (0.5 mol/dm^3)	20	490–5150	$s_0 = 9 \times 10^{-15} M_w^{0.32}$						M_w/M_n = 2.5	633
		25	8400	$s_0 = 4.9 \times 10^{-15} M_w^{0.37}$	1570	0.3	88		0.697	M_w; $D_{z,0}$	631
									0.710		790
	Water/NaCl										
	Water/NaCl (0.1 mol/dm^3)		100–8000	$s_0 = 1.07 \times 10^{-15} M^{0.48\pm0.01}$		$D_0 = 1.24 \times 10^{-4} M^{-0.53\pm0.01}$					634, 635
	Water/NaCl (0.1 mol/dm^3)		8200							M_w; see Ref. for s_0 and k_s	667
H-acrylamide	Water/NaCl (1 M)	25	2330			0.505					1008
2,3,3-D_3- acrylamide		1710				0.630					
	Water/NaCl (2 g/dm^3)	20	8000			1.6	97			M_w; $D_{z,0}$	634
	Water/NaCl (20 g/dm^3)	20	160			0.14	1370			M_w; $D_{z,0}$	634
	Water/$NaNO_3$ (0.5 mol/dm^3)	25	260			1.29 ± 0.05	72			M_w; $D_{z,0}$	636
			460			1.15 ± 0.10	277			M_w; $D_{z,0}$	636
partially hydrolyzed Pusher 700 (Dow Chemical)	Water	26				0.280				D_z; C = 0.5 g/dm^3	637
	Water/NaCl (0.1 mol/dm^3)					0.562				D_z; C = 0.5 g/dm^3; 4V	
	Water/NaCl (0.2 mol/dm^3)					0.383				D_z; C = 0.5 g/dm^3; 4V	
Poly(*p*-ethoxycarbonyl phenylmethacrylamide)	Acetone		479			7.05				M_w; D_A	267
			575			7.59					
			955			3.90					
			1060			2.52					
	Dimethylformamide		1000			2.18				M_w; D_A	267
	Dimethylformamide/ formamide (90/10)		740			1.16				M_w; D_A	267
			1100			1.11					
	Ethyl acetate		260–740			$D_A = 2.8 \times 10^{-4} M_w^{-0.69}$					267
Poly(*N*-isopropyl-acrylamide)	Water	15	290		99[a]			M_n 0.887	chloroform		498
		20		2.48							
		25	1000	3.04					0.896	M_w in chloroform	
		27.5		3.55							
		30		4.17							
		32		4.76							
		33		5.08						0.902	
Poly(*N*-isopropyl methacrylamide)	Water	20				$D_0 = 2.02 \times 10^{-4} M^{-0.57}$					1005

Poly(*N*-isopropylmethacrylamide-*co*-acrylamide)						
IPAM-100	Deionized water	–	3000	0.490	CLS, DLS	1006
IPAM-85			3100	0.439		
IPAM-70			4500	0.394		
IPAM-55			3900	0.421		
IPAM-40			2200	0.524		
IPAM			5200	0.335		
Poly(acrylamide-*co*-sodium-3-acrylamide-3-methylbutanoate-*co*-(2-(acrylamido)-2-methylpropyl) trimethylammonium chloride)					Terpolymer composition mol% AM: NaAMB: AMPTAC	1023
AM	NaAMB		AMPTAC			
ATABAM 0.5–0.5	Water/NaCl (1 M)	25	3430	0.498	99 : 0.5 : 0.5	
ATABAM 2.5–2.5			13900	0.313	89.1 : 5.1 : 5.8	
ATABAM 5.0–5.0			3450	0.476	85.9 : 7.9 : 6.2	
ATABAM 10–10			10800	0.264	80.5 : 9.7 : 9.8	
ATABAM 15–15			12200	0.313	74.6 : 11.5 : 14.0	
ATABAM 5.0–10			11200	0.361	75.7 : 13.6 : 10.7	
ATABAM 10–5.0			19400	0.389	85.3 : 4.5 : 10.2	
Poly(acrylamide-*co*-3-(2-acrylamido-2-methylpropane dimethylammonio)-1-propane-sulfonate) (AM-*co*-AMPDAPS)					AMPDAPS in the copolymer mol%	
DAPSAM-1	Water/1 M NaCl	25	3200	0.505	1	1010
DAPSAM-5			12400	0.316	5	
DAPSAM-10-2			700	0.401	9.6	
DAPSAM-10-3			21500	0.320	10	
DAPSAM-25-2			8200	0.397	27.7	
DAPSAM-40-2			15100	0.362	38.9	
DAPSAM-40-3			17500	0.324	40	
DAPSAM-60-2			5400	0.360	58.8	
DAPSAM-75-2			6200	0.490	70.1	
DAPSAM-100			3000	0.628	100	
Poly(acrylamide-*co*-NaAMPS-*co*-AMPTAC)						
ATASAM 0.5–0.5	Water/1 M NaCl	25	6030	0.423	$dn/dC = 0.1426\,cm^3/g$	1022
ATASAM 2.5–2.5			2780	0.577	$dn/dC = 0.1481\,cm^3/g$	
ATASAM 5–5			5070	0.381	$dn/dC = 0.1510\,cm^3/g$	
ATASAM 10–10			5870	0.449	$dn/dC = 0.1395\,cm^3/g$	
ATASAM 15–15			6770	0.433	$dn/dC = 0.1344\,cm^3/g$	
ATASAM 10–5			3880	0.413	$dn/dC = 0.1376\,cm^3/g$	
Poly(NaAMPS-*co*-AMPTAC)					AMPTAC in the copolymer mol%	1021
ATAS-0	Water/1 M NaCl	25	6120	0.387	0.0	
ATAS-10-2			8220	0.419	14.7	
ATAS-25-2			6200	0.473	36.1	
ATAS-40-2			7920	0.385	45.4	
ATAS-50-2			7680	0.463	51.7	
ATAS-70-2			4980	0.406	65.7	
ATAS-100			1470	0.637	100.0	
Poly(1,4-benzamide)	Dimethylacetamide/ 3% (g/mol) LiCl	25	55	$D_0 = 4.38 \times 10^{-4}\, M_w^{-0.82}$		1007
Poly(1,4-phenylene terephthalamide)	96 wt.% H_2SO_4	30	48.3	0.072		
			38.4	0.088		
			22.7	0.113		
			15.9	0.159		

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TABLE 3. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			14.5			0.196					
						$D_{z,0} = 2.72 \times 10^{-5}\ M_w^{-0.77}$					
	96 wt.% H_2SO_4/	30	42.6			0.074					
	0.05 M K_2SO_4		29.8			0.093					
						$D_{z,0} = 1.51 \times 10^{-5}\ M_w^{-0.72}$					
Poly(*m*-phenylene isophthalamide)	Dimethylacetamide	26	4.2–230	$s_0 = 1.72 \times 10^{-15}\ M_w^{0.44}$		$D_0 = 1.13 \times 10^{-4}\ M^{-0.56}$			0.66	SV	1082
	Dimethylacetamide/ 3% LiCl	25		$s_0 = 0.68 \times 10^{-15}\ M_w^{0.44}$		$D_0 = 0.55 \times 10^{-4}\ M^{-0.56}$			0.71		
Poly(acrylonitrile)	Dimethylformamide	20	78.7	1.72						M_V	569
			67–1030	1.65–4.68						See Ref. for s and M_{MW}	533
		25	48	1.52		3.6			0.83	s_t; D_w, $M_{t,w}$	115
			113	2.08		2.1					
			155	2.36		1.75					
			270	2.95		1.25					
		25	105	2.04					0.830	s_t; M_w	74
			210	2.48							
			16			7.14				D_w; D_n given in Ref.	206
			28			5.72					
			63			5.08					
			193			4.03					
			214			3.91					
		35	28–575			$D_0 = 2.19 \times 10^{-4}\ M^{-0.58}$	$k_D = 2.31 \times 10^{-9} \times D_0^{-1.29}$				201
			980	4.6						$C = 1.5$–10 g/dm^3	792
	Dimethylformamide/ benzene 30%	RT	320	3.08						$M_{s,n}$	462
	Dimethylformamide/ benzene 37%	RT		7000	138					$M_{s,n}$ calculated from $s_0 = 0.025$	
	Dimethylformamide/ tetrahydrofuran (1/1)		1200	5.78[b]						M_n; $C = 0.648$ g/ dm^3; 5V	638
			210	2.94[b]						M_n; $C = 1.323$ g/ dm^3; 12V	
	Dimethylformamide/methanol		1200	4.03[b]						M_n; $C = 1.270$ g/ dm^3; 4V	
Poly(acrylonitrile-*stat*-styrene)	Butanone		330	12.1						$C = 1.5$–10 g/dm^3	792
Poly(acrylonitrile-*alt*-styrene)	Butanone		170	10.2						$C = 1.5$–10 g/dm^3	792
Poly(amidobenzimidazole)	Dimethylacetamide	25	74.7	1.33	537						1046
			72.3	1.32	525						
			63.1	1.28	550						
			55.7	1.25	428						
			46.9	1.11	285						
			39.6	0.99	183						
			27.3	0.93	142						
			10.1	0.80	51						

	Dimethylacetamide/ 0.3% LiCl	25	79.3	1.45	900					
			60.6	1.33	890					
			41.1	1.12	720					
			29.3	0.88	460					
Poly(benzyl acrylate-*co*-methyl acrylate) different comp.	Methyl ethyl ketone	20							v_2 and M given	506
Poly(*n*-butyl-acrylate)	Methyl isobutyl ketone	23.3	46					0.944	M_v	148
Poly(*tert*-butyl-acrylate)	Methyl ethyl ketone	25	306	17.7	0.94				M_w; s_0	512
	Acetone			0.207	0.84					
	Methanol			0.237	0.53					
	Pentane			0.266	0.22					
	Methyl ethyl ketone	25	237	0.160	0.88				M_w; s_0	
	Acetone			0.194	0.74					
	Methanol			0.230	0.49					
	Pentane			0.244	0.21					
	Acetone	25	180	0.182	0.64				M_w; s_0	
	Methanol			0.203	0.43					
	Pentane			0.228	0.20					
	Methyl ethyl ketone	25	149	0.139	0.65				M_w; s_0	
	Acetone			0.162	0.55					
	Methanol				0.39					
	Pentane			0.191	0.19					
	Methyl ethyl ketone	25	75	0.109	0.59				M_w; s_0	
	Acetone			0.117	0.32					
	Methanol			0.127	0.22					
	Pentane			0.137	0.13					
Poly(*n*-butyl methacrylate)	Methyl isobutyl ketone	21.2	820					0.922	M_v	148
	Propan-2-ol	21.5	40–8000			$6.3 \times 10^{-5} M^{-0.50}$				399
Microgels	Ethyl acetate	20	254			0.174			M_w, $C = 0.1$–0.5 g/dm^3; see Ref.	646
			254			0.145			M_w, $C = 0.1$–0.5 g/dm^3; see Ref.	
	Tetrahydrofuran		340			0.181			M_w, $C = 0.1$–0.5 g/dm^3; see Ref.	
			380			0.191			M_w, $C = 0.1$–0.5 g/dm^3; see Ref.	
		25		$s_0 = 0.12 M^{0.31}$						1052
Poly(*tert*-butyl methacrylate)	Butyl acetate	20		$s_0 = 3.02 \times 10^{-5} M_w^{0.41}$					$C = 0.5$ g/dm^3	551
		21				$D_0 = 2.51 \times 10^{-4} M_w^{-0.55}$			D_A; $M_{s,D}$ given in Ref.	
Poly(cetyl methacrylate)	Heptane	21		$s_0 = 1.42 \times 10^{-14} \times M^{0.36}$		$D_0 = 1.74 \times 10^{-3} M^{-0.64}$			$M_{s,D}$; D_A	560
Poly(cholesteryl acrylate)	Benzene	21	140–780	$s_0 = 1.9 \times 10^{-15} M^{0.46}$		$D_0 = 3.2 \times 10^{-4} M^{-0.54}$			$M_{s,D}$	580
Poly(2.6-diisopropyl-phenylmethacrylate)	Toluene	25						0.789		1053
Poly(2.6-dimethyl-phenylmethacrylate)	Toluene	25						0.894		1053
Poly(acrylic acid)	Water/NaCl (0.2 mol/dm^3)	20–22[c]	1100		5.6		4.77	0.73	$M_{s,D}$;	555
	(0.02 mol/dm^3)					35.0	20.0			
	(0.012 mol/dm^3)			7.3 ± 0.5	81.0	0.65 ± 0.02[d]	28.2			
	(0.006 mol/dm^3)				114.0		49.0			

TABLE 3. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
	(1.0 mol/dm^3)		420		2.4		0.2			$M_{s,D}$	
	(0.2 mol/dm^3)				4.2		1.0				
	(0.012 mol/dm^3)			6.5 ± 0.5	32.0	1.35 ± 0.05[d]	10.0				555
	(0.006 mol/dm^3)				60.0		27.4				
	(0.0012 mol/dm^3)						146				
	(1.0 mol/dm^3)		70				22.3			$M_{s,D}$	
	(0.2 mol/dm^3)				0.0		22.3				
	(0.012 mol/dm^3)			2.4 ± 0.1	2.75	3.2 ± 0.1[d]	150				
	(0.006 mol/dm^3)				8.0		286				
Poly(ethyl acrylate)	Methyl isobutyl ketone	24.3	34						0.951	M_v	148
	n-Propanol	28	18–155	$s_0 = 1.05 \times 10^{-2} M_n^{0.47}$						M_{MW}	456
Poly(ethyl acrylate-*co*-acrylic-acid) sodium salt	Water/NaCl (0.05 mol/dm^3)		185–1300			$D_{o,z} = 5.76 \times 10^{-4} M^{-0.65}$				$M_w/M_n = 1.8$–2.3	637
	(0.1 mol/dm^3)					$D_{o,z} = 8.55 \times 10^{-4} M^{-0.67}$					
	(0.3 mol/dm^3)					$D_{o,z} = 3.73 \times 10^{-4} M^{-0.59}$					
	(0.5 mol/dm^3)					$D_{o,z} = 1.54 \times 10^{-4} M^{-0.52}$					
	(0.7 mol/dm^3)					$D_{o,z} = 1.41 \times 10^{-4} M^{-0.51}$					
	(1.0 mol/dm^3)					$D_{o,z} = 1.24 \times 10^{-4} M^{-0.50}$					
	(1.2 mol/dm^3)					$D_{o,z} = 1.10 \times 10^{-4} M^{-0.49}$					
	2-Heptanone					$D_{o,z} = 1.35 \times 10^{-4} M^{-0.51}$					
Poly(ethyl acrylate-*co*-methylacrylate) different comp.	Methyl ethyl ketone	20								v_2 and M given	506
Poly(ethyl methacrylate)	Acetone	20	77–7440	$s_0 \sim M^{0.43}$		$D_0 \sim M^{-0.57}$		$f_{sp} \sim M^{0.57}$	0.798	$M_{f,A}$	59
			34.1–9350	8.95–120		17.4–0.85			0.798	$M_{f,A}$; s_f; See Ref.	18
									0.7942	extrapolated value	268
			24.6–7440	7.45–107	23–60	20.0–0.95	0–63			$M_{f,w}$; s_f; D_A, See Ref.	52
	Methyl isobutyl ketone	22.9	410						0.863	M_v	148
Poly(isooctyl methacrylate)	(theta-solvent)[e]	25	1036	63.0–67.5					0.900	A[e]	427
			875							$C = 2.5$ g/cm^3; s_w (Trautmann)	
Poly(phenyl methacrylate)	Toluene	25							0.789		1053
Poly(sulfoalkyl methacrylate) alkyl: hexyl	0.1 M NaCl	25	306			1.22	1.54×10^{-4}			$M_w/M_n = 1.5$; $dn/dC = 0.135$ cm^3/g	100
octyl			494			1.02	1.72×10^{-4}			$M_w/M_n = 2.0$; $dn/dC = 0.140$ cm^3/g	
decyl			701			1.00	1.99×10^{-4}			$M_w/M_n = 1.8$; $dn/dC = 0.149$ cm^3/g	
ethyl			734			0.975	2.07×10^{-4}			$M_w/M_n = 2.1$; $dn/dC = 0.170$ cm^3/g	
Poly(methacrylic acid)	Dimethylsulfoxide	25	34	0.46						M_{MW}	638
			130	1.02							
			260	1.27							
			429	1.66							
			432	1.85							
			381	1.60							

			777	2.80							
			1404	3.02							
Poly(methacrylic acid)	Sodium salt										
	Water/NaCl, $Y^f = 4$	20	237	2.90		1.150				s_t; apparent M	124
	5		339	3.87		1.072					
	7		442	4.36		0.923					
	9		530	4.73		0.838					
	14		690	5.46		0.742					
	19		768	5.66		0.692					
	29		880	6.04		0.643					
	39		943	6.28		0.621					
	49		986	6.43		0.611					
	59		1014	6.50		0.602					
	∞		1112	7.22				3.45		M of uncharged polymer	
	Water/HCl (pH = 2.0)	21.3	62						0.712	M_v	148
	Water/NaOH (pH = 12.0)	21.4	78						0.385	M_v	148
	Water/NaCl (0.01 mol/dm^3)/ NaOH(pH = 12.0)	21.8	78						0.412	M_v	148
Poly(methyl acrylate-*co*-styrene) different comp.	Methyl ethyl ketone	20								v_2 and M given	506
Poly(methyl methacrylate)	Acetone	20	115				10.25	1.19×10^{-5} cm^g		M_{MW}	460
			201	20.3	72	6.8	18	1.97×10^{-5} cm^g	0.789	$M_{t,A}$; s_t; D_A	15
			1380	46.1	224	2.2		5.35×10^{-5} cm^g		$M_{t,A}$; s_t; D_A	15
			6350	88.5	614	0.92		14.1×10^{-5} cm^g		$M_{t,A}$; s_t; D_A	15
		20	65–1980			$D_{0,z} = (8.09 \pm 0.68) \times 10^{-4}\ M_w^{-0.576 \pm 0.006}$				$M_w/M_n = 1.20$–1.44	639
		20	65–1980				-0.41–73.05			$M_w/M_n = 1.20$–1.44; see Ref.	639
		20	16			27.06 ± 0.70	-6.97			$D_{0,z}$; $M_w/M_n = 1.18$	639
		20	29			17.20 ± 0.17	-4.77			$D_{0,z}$; $M_w/M_n = 1.29$	
		25							0.798		1054
		30							0.825		494
	Acetonitrile	38		$s_0 = 18.6 \times 10^{-15}\ M^{0.51}$						See Ref. for range of M_w	570
		44	1.16			101	-0.7			M_w	1057
			1.80			82.1	-2.1			$M_w/M_n = 1.08$ for all	
			2.95			68.0	-3.7				
			4.18			57.5	-3.1				
			7.55			40.8	-3.7				
			10.9			36.4	-4.1				
			20.2			25.6	-5.1				
			35.3			21.0	-8.1				
			74.0			14.3	-11.4				
			119			11.6	-13.2				
			218			8.9	-24.2				
	Acetonitrile-d_3		1.1			74.1				NMR-selfdiffusion	1080
	Benzene	20	115				5.05	1.25×10^{-5} cm^g		M_{MW}	460
		25	1.9–77						0.8273 – 0.8077	M_{MW}; (AIBN) See Ref.	268
		25							0.8069 + 39.4/M	(B_2O_2) See Ref.	268
									0.818		211
		30							0.879		494
	Benzene-d_6		1.1			42.8				NMR-self diffusion	1080

TABLE 3. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_θ ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	ν_2 (cm^3/g)	Remarks	Refs.
	Butanone	25							0.799		1054
	n-Butyl acetate	25								Value for infinite molar mass	208
	n-Butyl chloride	34.5	65–1980			$D_{0,z} = (3.08 \pm 0.68) \times 10^{-4}\ M_w^{-0.498 \pm 0.007}$				$M_w/M_n = 1.20$–1.44	639
		34.5	65–1980							$M_w/M_n = 1.20$–1.44; see Ref.	639
		34.5	16			22.38 ± 0.01	-6.2			$D_{0,z}$; $M_w/M_n = 1.18$	639
		34.5	29			14.92 ± 0.05	-6.52			$D_{0,z}$; $M_w/M_n = 1.29$	
		35 ($T = \theta$)		40.1	251 ± 53					See Ref. for def. of s	
		35.6	200	15.7	22	7.18	-30	1.65×10^{-5} cm^g	0.82	$M_{t,A}$; s_t; D_A	6,15
			1300	39.1	41	2.91	-55	4.09×10^{-5} cm^g			
			6550	86.0	107	1.15	-35	9.38×10^{-5} cm^g			
		40.8	1.80			62.4	-3.0			M_w	1057
			2.95			51.5	-4.0			$M_w/M_n \approx 1.08$ for all	
			4.18			44.0	-3.8				
			7.55			32.7	-5.1				
			10.9			28.1	-6.1				
			20.2			21.8	-9.0				
			35.3			16.4	-10.7				
			74.0			11.2	-14.6				
			119			9.1	-18.8				
			218			6.8	-22.4				
	Carbon tetrachloride								0.8038	Value for infinite molar mass	268
	Chlorobenzene	25							0.8118	Value for infinite molar mass	268
	Chloroform	25							0.7942	Value for infinite molar mass	268
	Chloroform-d		1.1			46.7				NMR-self diffusion	1080
	Cyclohexanone	25–65	31							See Ref. for D_0 and k_D-values	640
	o-Dichlorobenzene	25							8 0.8187	Value for infinite molar mass	268
	Diethyl ketone								0.8015	Value for infinite molar mass	268
	1,4-Dioxane	25							0.8181	Value for infinite molar mass	268
		25	200	2.11	134			2.17×10^{-5} cm^g		$M_{t,A}$; s_t	15
			1300	4.5	412			6.35×10^{-5} cm^g			15
			6550	8.55	1040			17.6×10^{-5} cm^g			15
	Ethyl acetate								0.7963	Value for infinite molar mass	268
		20	79	7.6	2.5	7.21	0		0.787	$M_{s,0}$; D_w	211
			127	8.8	0.8	5.66	0			$s_0 = -0.06 + 0.026\ M^{0.5}$	
	Ethyl acetate		207	11.7	1.8	4.36					211
			315	14.1	1.9	3.73					

			381	16.3	2.4	3.79					
			473	19.4	5.2	3.38					
			701	21.5	5.2	2.54					
			931	25.0	6.0	2.34					
			180	9.86	0.59					$M_{r,w}$	212
			1965	34.0	3.1						
			18–1965	$s_0 = 1.51 \times 10^{-15}\ M^{0.48}$							
	Methyl ethyl ketone								0.7993	Value for infinite molar mass	268
	Methyl isobutyl ketone	21.8	480						0.816	M_v	148
	Methyl methacrylate		370			1.95	943			M_w; D_z; $M_w/M_n = 2$, $C = 1.34\ g/dm^3$	641
			454			1.51				M_w; D_z; $M_w/M_n = 2$, $C = 0.859\ g/dm^3$	641
	Tetrahydrofuran								0.8085	Value for infinite molar mass	268
			45–448			$D_{0,z} = 2.27 \times 10^{-5}\ M^{0.333}$					642
	Thiophenol	25	107	1.15	87					$s_0 = 1.17 \times 10^{-13}\ M_w^{0.40}$	1056
			260	1.63	126					$k_s = 2.50 \times 10^{-2}\ M_w^{0.69}$	
			265	1.66	150						
			401	2.08	201						
			844	2.56	315						
			1100	3.06	375						
			2140	3.94	615						
			4050	4.88	955						
	Toluene	20	115			5.80		$1.19 \times 10^{-5}\ cm^g$		M_n	460
									0.807	Value for infinite molar mass	268
									0.8101	Value for infinite molar mass	
		25	1400	20.62						M_v	645
	m-Xylene	17	190						0.8063	M_v	148
		22.15	190						0.8019	M_v	
		29.5	190						0.8190	M_v	
		45.0	190						0.8143	M_v	
atactic	Acetone	25	1.83			69.2	−1.6			M_w	1058
			4.18			48.2	−1.8			$M_w/M_n \approx 1.08$ for all	
			7.55			36.5	−2.9				
			20.2			22.4	−1.1				
			50.4			14.3	1.8				
			74.0			11.5	1.9				
			119			9.1	7.4				
			204			6.7	19				
			312			5.5	13				
			482			4.3	29				
			952			2.9	64				
	Acetonitrile	44 ($T = \theta$)	1.83			78.0	−1.7			M_w; $M_w/M_n \approx 1.08$ for all	1058
			4.18			57.5	−3.1				
			7.55			40.8	−3.7				
			20.2			26.9	−6.1				
			50.4			17.6	−8.0				

TABLE 3. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			74.0			14.3	−11				
			119			11.6	−13				
			204			8.9	−16				
			312			7.3	−15				
			482			5.9	−33				
			952			4.2	−38				
isotactic	Acetone	25	3.04			52.4	−1.5			M_w; $M_w/M_n \approx 1.06$ for all	1058
			7.27			34.5	−2.5				
			30.9			16.6	−2.8				
			70.1			10.9	−1.5				
			154			7.1	2.3				
			379			4.3	15				
			686			3.2	37				
			978			2.6	57				
	Acetonitrile	28 ($T=\theta$)	3.04			49.9	−2.6			M_w; $M_w/M_n \approx 1.06$ for all	1058
			7.27			33.0	−4.9				
	Acetonitrile		30.9			16.2	−9.8				
			70.1			10.6	−14				
			154			7.2	−21				
			379			4.6	−36				
			686			3.4	−52				
			978			2.9	−66				
Poly(methacrylic acid)/Poly(vinylpyrrolidone)											
complex	Water	25	1000	24.4						M_{MW}	638
			1384	30.3							
			1324	28.5							
			1313	30.3							
			1262	28.5							
			1335	33.3							
			1136	25.0							
			1138	26.6							
	Dimethylsulfoxide	25	1065	2.63						M_{MW}	638
			936	2.50							
			746	2.30							
			692	2.20							
			536	2.10							
			428	1.79							
			311	1.60							
			130	1.08							
Poly(methyl methacrylate-*co*-ethylene dimethylacrylate)											
randomly branched	Ethyl acetate	20								M_w; see Ref. for polymerization conditions, s_0, k_s and M values	549
	Mesitylene	47 ($T=\theta$)	380	20.7					0.680	M_w (measured in dioxane); $v_2\eta_1$	643
			800	30.8							
			1050	36.0							
			2630	58.6							

		47 ($T=\theta$)	380–2630	$s_0 = 3.1 \times 10^{13}\ M^{0.5}$				M_w	644
	Toluene	25	1000	20.57				M_w	645
			1350	26.39				M_w	
			170	12.6			0.702	M_w; (measured in dioxane); $v_2\eta_1$	643
			51	6.7					
			380	16.6					
			800	24.3					
			1050	26.5					
			2630	42.7					

[a] For all temperatures.
[b] See Ref.
[c] See Ref.
[d] See Ref. for def. of D
[e] See Ref.
[f] Y = total concentration (mol/dm^3) of Na-ions divided by concentration (mol/dm^3) of poly(methacrylic acid).
[g] $f_0' = kT/(D_0\eta_1)$.

TABLE 4. VINYL POLYMERS

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Poly(hydroxyether) composed of 1,4-anhydroerythritol and diglycidyl ether/bisphenol A											
		20	3.1			16				$[\eta] = 8.7\,cm^3/g$	1085
			2.5			13				$[\eta] = 9.9\,cm^3/g$	
			4.9			9.6				$[\eta] = 10.0\,cm^3/g$	
			6.95			8.0				–	
			8.26			7.5				$[\eta] = 11.1\,cm^3/g$	
			11.8			6.7				$[\eta] = 11.9\,cm^3/g$	
			12.3			5.8				$[\eta] = 12.2\,cm^3/g$	
			14.2			5.5				–	
Poly(methoxy ethylene)	Amyl alcohol	23.5	15						0.983	M_v	148
Poly(vinyl acetal)	Toluene	20.9							0.848		148
Poly(vinyl acetate)	Cyclohexanone	23.8	48						0.849	M_v	148
	Diethyl ketone	13							0.821		60
		40							0.833		
	Methyl ethyl ketone	20	17–1200	$s_0 = 9.8 \times 10^{-15}\ M_w^{0.38}$		$D_0 = 7.8 \times 10^{-4}\ M_w^{0.63}$				$M_{f,A}$; s_f; D_A	110,122
	1,2,3-Trichloropropane	13							0.83		60
		40							0.841		
microgels	Methanol	25	2360–17500			$D_{0,z} = 9.78 \times 10^{-5}\ M_w^{-0.478}$				M_w	647
	Methanol	25	31700–422900			0.16				M_w	647
Poly(vinyl alcohol)	Water	20	22.4						0.765		129
			13	0.96		7.46				$M_{f,w}$; s_f	129
			34	1.26		3.77					
			74	1.96		2.68					
			90	1.86		2.16					
				$s_0 = 4.4 \times 10^{-15}\ M^{0.32}$		$D_0 = 5.5 \times 10^{-10}\ M^{-0.68}$			0.765		130
		21.6	21						0.750	M_v	148

TABLE 4. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
		25	8.5	0.61		6.89	228				
			23.3	0.96		3.97	282				
			64.7	1.54		2.3	347	$1.04 \times M^{-0.166}$			119
			167	2.52		1.45	400				
		25	107			3.06	−8			D_A; $M_w/M_n = 1.01$	648
		25	9	1.10		7.56				$M_{s,D}$; s_t; $C = 2$–10 g/dm^3	649
			28	2.13		5.45		4.02			
			130	3.22		2.65		4.39			
			492	4.35		1.75		4.62			
		25	94.5						0.754		650
		25	18			12.3				M_v; $C = 10$ g/dm^3	658
		25	43			9.8				M_v; $C = 10$ g/dm^3	
		30	6.4						0.769	M_v	151
			19.8						0.766		
			54						0.763		
	Water/borax	25	94.5						0.767		650
	(0.0060 mol/dm^3)										
	(0.0100 mol/dm^3)								0.768		
	(0.0149 mol/dm^3)								0.775		
	(0.0299 mol/dm^3)								0.771		
	(0.0498 mol/dm^3)								0.777		
	Water/NaCl	25		2.14					0.79		152
	(0.2 mol/dm^3)										
	D_2O/NaCl			1.22							
	(0.2 mol/dm^3)										
	Water/*n*-Propanol	25	9	0.90		5.45				$M_{s,D}$; s_t; $C = 2$–10 g/dm^3	649
	(10 mass%)	20	1.14	4.10		3.82					
			63	1.82		3.55		3.45			
			200	2.33		2.45		3.22			
	Water/*n*-Propanol	25	13	0.63		4.00				$M_{s,D}$; s_t; $C = 2$–10 g/dm^3	649
	(20 mass%)		30	0.95		2.15		4.00			
			74	1.25		1.65		5.08			
			230	1.45		1.05		5.30			
	Water/*n*-Propanol	25	10	0.62		3.25				$M_{s,D}$; s_t; $C = 2$–10 g/dm^3	649
	(30 mass%)		21	0.92		2.85		3.45			
			74	1.05		1.55		4.25			
			200	1.15		0.85		4.70			
	Water/*n*-Propanol	25 ($T = \theta$)	14	0.82		3.25				$M_{s,D}$; s_t; $C = 2$–10 g/dm^3	649
	(35 mass%)		21	0.92		2.80		3.42			
			76	1.32		1.55		4.15			
			217	1.64		1.05		4.23			
Poly(4-vinylbiphenyl)		25							0.82	See Ref. for solvent	480
Poly(vinyl butyral)	Amyl alcohol	20							0.883	$[\eta] = 122$ cm^3/g	148
Poly(4-vinyl-*N*-butyl-pyridinium bromide)	Water/NaCl	20	115	3.0		2.8		2.38	0.77	$C = 1$ g/dm^3	135
	(0.2 mol/dm^3)	25	25	1.10							131

Poly(*N*-vinylcarbazole)	Benzene	20	4550	33.3			0.786	CLS	457
			847	16.3				CLS	
	Chlorobenzene	23.1					0.793	$[\eta] = 122\,cm^3/g$	148
Poly(vinyl chloride)	Cyclohexanone	23.4	940				0.711	M_v	148
		25					0.6926		602
			19	0.622			0.705	M_n	75
			53	1.075					
			66	1.18					
			120	1.56					
			29–110			[a]			640
		120	100–180	$s_0 = 2.195 \times 10^{-15}\,M^{0.49}$				M_w	651
	1-Dioxane		39			[a]			640
	Tetrahydrofuran	20	97–105	7.69–5.32	247–132			M_w; M_{MW} given in Ref.	544
		25					0.7429		181
							0.6951		602
Fractionated			90	6.8				M_w	602
			46	5.41					
			20.2–183	$s_0 = 6.50 \times 10^{-2}\,M^{0.41}$	55			M_v	596
Poly(vinyl chloride-*co*-vinyl acetate)	Cyclohexanone	23.5					0.723	$[\eta] = 72.2\,cm^3/g$	148
Poly(vinyl chloride-*co*-vinylidene chloride)	Cyclohexanone	23.5					0.703	$[\eta] = 77.4\,cm^3/g$	148
Poly(vinyl fluoride)	Dimethylformamide/ LiBr (0.1 mol/dm^3)	100	114–521	$s_0 = 0.101 \times M^{0.40}$				M_n; M_{MW}	421
			143–654	$s_0 = 9.29 \times 10^{-2}\,M^{0.4}$			0.72	s_0 from $s_0 = s(1 + 1.66[\eta]C)$; M_{MW}	441
Poly(vinyl methyl ether)	See Poly(methoxy ethylene)								
Poly(1-vinyl-naphthalene)	Toluene	25					0.82		480
Poly(2-vinyl-naphthalene)	Toluene	25					0.84		480
Poly(vinylpyridine)	Ethanol	25	97	3.6	179		0.682	M_{MW}; Ref. 134	132
			430	6.45	629		0.689		
Poly(vinyl sulfonic acid)	Sodium salt					21.3	0.410	s_l; See Ref. for M and D	116
	Water/NaCl (0.5 mol/dm^3)	20	3	1.57		15.0	0.428		
			5	1.96		11.0	0.430		
			9	2.32		9.75	0.424		
			12.5	2.94		5.95	0.408		
			29.5	4.27		6.40	0.424		
			29.5	4.46					
Poly(vinyl trimethylsilane)	Cyclohexane	RT	500			1.0 (1.04)		$M_w/M_n = 1.4–2.1$; $C = 0.5\,g/dm^3$	652
		25		$s = 1.02 \times 10^{-15}\,M_w^{0.43}$					761
						$D = 2.4 \times 10^{-4}\,M_w^{-0.57}$			
Poly(vinylpyrrolidone)	Methanol	25	23	3.41		9.60	0.730	$M_{l,D}$; SV	150
			20				0.776		144
	Water	20	500				0.783		144
			12–74	$s_0 = 8.81 \times 10^{-16}\,M^{0.50}$	$D_0 = 1.0 \times 10^{-4}\,M^{0.50}$			$M_{l,A}$; D_A	143,144

TABLE 4. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			13–41.5	0.82–1.42		7.55–4.14		1.70–2.11	0.802	s_t; 7V; see Ref. for M and D v_2 at 24.85°C; D_A given in Ref.	40
			94.5	10	1.17	1.17		1.66		$M_{t,A}$; s_t; D_A	153
			2700	12.20	0.50	0.50				$M_{t,A}$	
			10.6	0.85	8.66	8.66				$M_{s,D}$; $C = 50\%$	178
			42	1.66	4.31	4.31		2.12		$M_{s,D}$; $C = 5\%$	178
			86	1.94	2.44	2.44		2.94		$M_{s,D}$; $C = 5\%$	178
unfractionated									0.775		178
		23.1	29						0.781	M_v	148
		25							0.820		150
			ca. 32						0.780	M_{MW}	487
			30	20					0.784		144
			50						0.801		144
	Water/CaBr (1 mol/dm^3)	20	24.5			5.5				$C = 10$ g/dm^3	611,613
	Water/KBr (1 mol/dm^3)		24.5			4.78				$C = 10$ g/dm^3	611,613
	Water/LiBr (1 mol/dm^3)		24.5			4.06				$C = 10$ g/dm^3	611,613
	Water/$MgBr_2$ (1 mol/dm^3)		24.5			4.01				$C = 10.03$ g/dm^3	611,613
	Water/NaCl (0.2 mol/dm^3)		12	1.0				1.46		$M_{t,A}$	144
			25	1.43	68						144
			39.2	1.59	85			2.09			144
			55.6	1.76				2.38			144
			310	4.77	391						144
			584	7.13				2.81			144
			1129	7.13	720			4.36			144
			1900	12.5	1310						144
			3420	16.7	1630						144
				$s_0 = 8.81 \times 10^{-16} M^{0.5}$							144
(Luviskol K 30)	Phosphate buffer (0.1 mol/dm^3; pH = 7)	20	945	7.15						$M_{t,A}$; s_t	153
	Water/NaSCN (0.6 mol/dm^3)	25							0.772	M_{MW}	487
	Water/Na_2SO_4 (0.2 mol/dm^3)								0.771	M_{MW}	487
	Water/NaAc (0.1 M)	25		$s_0 = 2.14 \times 10^{-15} M^{0.42}$		$D_0 = 2.42 \times 10^{-4} M^{-0.58}$				$M_w > 4 \times 10^4$ g/mol	1059
				$s_0 = 1.15 \times 10^{-15} M^{0.48}$		$D_0 = 1.29 \times 10^{-4} M^{-0.52}$				$M_w \leq 4 \times 10^4$ g/mol	

[a] See Ref.

TABLE 5. STYRENE POLYMERS

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Poly(styrene)-poly(dimethylsiloxane)											1064
	THF	25	46.8			7.28	19			PS : $M_w = 32100$ g/mol	
	7.5 vol% PDMS/THF	25	46.8			1.32	36			PS : $M_w = 32100$ g/mol	
	15 vol% PDMS/THF	25	46.8			0.72	8.4			PS : $M_w = 32100$ g/mol	
Poly(styrene)-*b*-poly(ethylene/propylene)											1067
	2-Pentanol	25	105			7.03				$[\eta] = 16.2$ cm^3/g	
	5% 2-Pentanol/5-methyl-2-hexanone	25	105			7.10				$[\eta] = 15.5$ cm^3/g	
	10% 2-Pentanol/5-methyl-2-hexanone	25	105			6.96				$[\eta] = 13.6$ cm^3/g	
	15% 2-Pentanol/5-methyl-2-hexanone	25	105			6.73				$[\eta] = 12.9$ cm^3/g	
Poly(*o*-chlorostyrene)	Benzene	30							0.792		585
Poly(*p*-chlorostyrene)	Benzene	30							0.797		585
Poly(*p*-isopropyl-α-methylstyrene)	Toluene	21.5	18–625			$D_z = (5 \pm 2) \times 10^8 M_w^{(-0.61 \pm 0.04)}$				M_w; $M_w/M_n = 1.1$–3.1; $C = 0.01$–0.3 g/dm^3	653
		21.5	18–625			$D_z/K^{LN} = (4 \pm 2) \times 10^{-8} M_w^{(-0.57 \pm 0.03)}$				M_w; corrected for polydispersitiy (Log–Norm distribution)	653
Poly(α-methylstyrene) narrow distribution	Benzene	30	48.4–1670	$s_0 = 3.61 \times 10^{-15} M^{0.43}$						M_w from SE,LS; k_s, D_0, k_D given in Ref.	522
		30	8.64			19.88 ± 0.28				M_w (CLS); $M_w/M_n = 1.59$; $D_{z,0}$	654
			32.7			10.65 ± 0.11	4.67 ± 1.98			M_w (CLS); $M_w/M_n = 1.22$; $D_{z,0}$	
			91.7			6.02 ± 0.02	14.5 ± 0.4			M_w (CLS); $M_w/M_n = 1.03$; $D_{z,0}$	
			145.3			4.80 ± 0.05	20.8 ± 1.4			M_w (CLS); $M_w/M_n = 1.04$; $D_{z,0}$	
			394.6			2.74 ± 0.12	52.2 ± 1.0			M_w (CLS); $M_w/M_n = 1.26$; $D_{z,0}$	
			976.4			1.58 ± 0.07	101.4 ± 1.6			M_w (CLS); $M_w/M_n = 1.10$; $D_{z,0}$	
			100–1000			$D_{z,0} = (5.24 \pm 0.48) \times 10^{-4} M^{(0.59 \pm 0.01)}$				M_w (CLS); $M_w/M_n = 1.1$–1.6	654
	Chloroform	21	500			3.1				M_w; D_w	558
Poly(α-methylstyrene)	Cyclohexane	20				$D_z = 4.78 \times 10^{-5} M_w^{-0.425}$				$1.03 \leq M_w/M_n \leq 1.15$	1070
		28				$D_z = 6.49 \times 10^{-5} M_w^{-0.425}$				$1.03 \leq M_w/M_n \leq 1.15$	1070
		32	110–3280	$s_0 = 1.67 \times 10^{-15} M_w^{0.5}$		$D_0 = 1.36 \times 10^{-8} M_w^{-0.5}$				$M_z/M_w = 1.04$–1.2	657
		34.5	204			3.00					655
			204	8.06	20			$f_0 = 1.78 \times 10^{-5}$ cm		M_w; $f_0\ \eta_1$	656
			444	11.7	36			2.60×10^{-5} cm			
			768	16.5	38			3.41×10^{-5} cm			
			1190	21.6	91			3.87×10^{-5} cm			
			1820	24.3	68			5.21×10^{-5} cm			
			3300	33.0	119			7.03×10^{-5} cm			

TABLE 5. STYRENE POLYMERS

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			7470	49.8	134			10.6×10^{-5} cm			
			204–7470	$s_0 = 1.8 \times 10^{-15} M_w^{0.5}$					0.886		656
narrow distribution	Cyclohexane	39	48.4–1670	$s_0 = 2.0 \times 10^{-15} M^{0.5}$						M_w (SE); k_s given in Ref.	522
		35 ($T=\theta$)	37–627	$s_0 = 1.72 \times 10^{-15} M^{0.49}$						M_w; A; M_n given in Ref.; s_{in}	321
			730	13.89						s_0	616
			1520	19.47						M_w	
		37		$s_0 = 1.86 \times 10^{-15} M_w^{0.5}$				$f_0 = 4.1 \times 10^{-8} M_w^{0.5}$ cm		M_w; D_0 given in Ref.	68
		38.2		$s_0 = 1.93 \times 10^{-15} M_w^{0.5}$					0.889	M_w dep. of k_s given in Ref.	594
		35–125	6500							$C = 16.4$ g/dm^3; see Ref. for s_0-values	550
		39	42.7–1430	$s_0 = 2.0 \times 10^{-15} M^{0.5}$	$k_s = 6.2 \times 10^4 M^{0.5}$					M_w; pressure dep. of s given in Ref.	601
		39	42.5–1170	$s_0 = 2.04 \times 10^{-15} M_w^{0.50}$					1.886	M_v; CLS; k_s given in Ref.	472
	trans-Decalin	9.5 ($T=\theta$)	204	1.74	29			$f_0 = 1.62 \times 10^{-5}$ cm		M_w; f_0 η_1	656
			444	2.42	27			2.47×10^{-5} cm			
			768	3.40	38			3.12×10^{-5} cm			
			1190	4.41	62			3.72×10^{-5} cm			
			1820	5.15	57			4.82×10^{-5} cm			
	trans-Decalin	3300	6.85	89				$f_0 = 6.65 \times 10^{-5}$ cm			
			7470	10.6	106			9.73×10^{-5} cm			
			204–7470	$s_0 = 3.9 \times 10^{-16} M_w^{0.5}$					0.894	M_w; s_0	656
		20	1190	4.98	109			$f_0 = 4.19 \times 10^{-5}$ cm		M_w; f_0 η_1	656
			1820	5.71	137			5.53×10^{-5} cm			
			3300	7.41	178			7.82×10^{-5} cm			
			7470	11.4	353			11.5×10^{-5} cm			
									0.892		
		40	204	2.71	45			$f_0 = 1.99 \times 10^{-5}$ cm		M_w; f_0 η_1	656
			444	3.38	82			2.95×10^{-5} cm			
			768	5.26	134			3.68×10^{-5} cm			
			1190	6.41	167			4.91×10^{-5} cm			
			1820	7.41	207			6.43×10^{-5} cm			
			3300	10.3	356			8.74×10^{-5} cm			
			7470	14.3	628			13.8×10^{-5} cm			
									0.892		
	Toluene	25	42.5–1170	$s_0 = 3.01 \times 10^{-15} M^{0.44}$					0.873	M_n; A; LS	472
		35	19.8	11.37						M_v	616
			40.1	17.78						M_v	
		25–120	6500	2.38–8.0						$C = 16.4$ g/dm^3; see Ref. for s_0-values	550
		37	20	$s_0 = 4.02 \times 10^{-15} M_w^{0.43}$						M_w; D_0 and f_0 given in Ref.	68
		25	8.64			19.55±0.18	1.12 ± 0.30			M_w; (CLS) $M_w/M_n = 1.59$	654
			32.7			10.67 ± 0.05	4.63 ± 0.30			M_w; (CLS) $M_w/M_n = 1.22$	

			91.7			6.10 ± 0.02	13.5 ± 0.4		M_w; (CLS) $M_w/M_n = 1.03$	
			145.3			4.76 ± 0.07	19.7 ± 2.4		M_w; (CLS) $M_w/M_n = 1.04$	
			394.6			2.77 ± 0.02	46.0 ± 1.8		M_w; (CLS) $M_w/M_n = 1.26$	
			792.9			1.80 ± 0.02	88.9 ± 4.9		M_w; (CLS) $M_w/M_n = 1.20$	
			976.4			1.61 ± 0.02	105.3 ± 7.2		M_w; (CLS) $M_w/M_n = 1.10$	
			100–1000			$D_{z,0} = (5.24 \pm 0.48) \times 10^{-4}\ M^{-0.59 \pm 0.01}$				654
			204	6.76	91		$f_0 = 2.25 \times 10^{-5}$ cm		M_w; f_0 η_1	656
			444	9.26	167		3.49×10^{-5} cm			
			768	12.0	247		4.78×10^{-5} cm			
			1190	15.1	353		6.04×10^{-5} cm			
			1820	17.2	457		7.81×10^{-5} cm			
			3300	22.7	743		10.9×10^{-5} cm			
			7470	33.8	1450		16.5×10^{-5} cm			
			204–7470	$s_0 = 3.6 \times 10^{-15}\ M_w^{0.43}$						
Poly(α-methylstyrene-*block*-butadiene)	Toluene							0.873		656
Poly(α-methylstyrene-*star*-butadiene) different comp.	Cyclohexane	37							s_0 and M_v given in Ref.	567
Poly(styrene)	Acetone	20	1200–2450	11.9–16.7		1.18–0.80			s_1; D_w; $M_{t,w}$	50
		25						0.899	Extrapolated value	263
	Benzene	20	150					0.911	M_n; $M_w = 450.10$	606
		25							See Ref. for v_2	532
		25	9.2							480
			1.32			27.9				180
			1.95			25.0				
			2.8			21.1				
			3.9			17.2				
		25	10.6			11.7			SV; M_w	150
			67	2.5		4.1				
			606	7.14		1.50				
		25						0.91		150
	Benzene	25	∞					0.9175		268
			1.18–520	$s_0 \sim M^{0.47}$		$D_0 \sim M^{0.53}$				43
			2.3–222					0.9078–0.9177	M_{MW}; $v_2 = v_{2,\infty} - 23/M$; B_2O_2	268
NBS 705		35	179			3.29	22		M_w; D_A	577
NBS 419			411			1.57	59		M_w; D_A	577
Styron 683-7			277			3.26	45		M_w; D_A	577
						$D_0 = 3.05 \times 10^{-4}\ M^{-0.55}$				
		40	4000	19.89	100.4			0.917	M_w; CLS	172
PCC		25	392, 1700						M_w; see Ref. for s_0 and k_s	793
four branched star		25	312, 2210						M_w; see Ref. for s_0 and k_s	793
			140	4.3					$C = 1.5–10$ g/dm^3	792
		25	24.8			12.4 ± 0.1			D_z; $C = 1.5–89$ g/dm^3	661
			86.6			5.7 ± 0.1			D_z	
			162			3.6 ± 0.1			D_z	
			411			2.5 ± 0.1			D_z	
			1110			1.1 ± 0.1			D_z	

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			4330			0.6 ± 0.1				D_z	
			88700			0.7 ± 0.1				D_z	
		30	79.8			5.18 ± 0.02	15.44			M_w; $D_{z,0}$	659
		30	775			1.41	120			$M_w/M_n = 1.01$	660
			2420			0.745	310			$M_w/M_n = 1.04$	
			5530			0.512	490			$M_w/M_n = 1.04$	
			9700			0.353	640			$M_w/M_n = 1.03$	
			13400			0.284	740				
four branched star	Benzene	20	24–3800			$D_{z,0} = (2.18 \pm 0.32) \times 10^{-4} M_z^{-(0.55 \pm 0.02)}$				M_w; $M_w/M_n = 1.13$	662
		25	785			1.30	150			M_w; $M_w/M_n = 1.01$	663
			24–24000			$D_{z,0} = (2.18 \pm 0.32) \times 10^{-4} M_z^{-(0.55 \pm 0.02)}$				M_w; $M_w/M_n = 0.02$–1.2; k_s given in Ref.	664
PCC, 4b		20	117			4.17				M_w; D_z; $C = 4$ g/dm^3	
PCC, 1c			186			3.26					
PCC, 3b			372			2.29					
PCC, 13a			596			1.92					
PCC, 4b	Benzene/cyclohexane	20	117			4.23				M_w; D_z; $C = 4$ g/dm^3	665
PCC, 1c	(90/10)		186			3.42					
PCC, 3b			372			2.30					
PCC, 13a			596			1.84					
PCC, 4b	Benzene/cyclohexane		117			4.14				M_w; D_z; $C = 4$ g/dm^3	665
PCC, 1c	(80/20)		186			3.33					
PCC, 3b			372			2.27					
PCC, 13a			596			1.80					
PCC, 4b	Benzene/cyclohexane		117			4.15				M_w; D_z; $C = 4$ g/dm^3	665
PCC, 1c	(70/30)		186			3.26					
PCC, 3b			372			2.24					
PCC, 13a			596			1.74					
PCC, 4b	Benzene/cyclohexane		117			4.15				M_w; D_z; $C = 4$ g/dm^3	665
PCC, 1c	(60/40)		186			3.37					
PCC, 3b			372			2.26					
PCC, 13a			596			1.82					
PCC, 4b	Benzene/cyclohexane		117			4.18				M_w; D_z; $C = 4$ g/dm^3	665
PCC, 1c	(50/50)		186			3.24					
PCC, 3b			372			2.15					
PCC, 13a			596			1.69					
PCC, 4b	Benzene/cyclohexane		117			3.96				M_w; D_z; $C = 4$ g/dm^3	665
PCC, 1c	(40/60)		186			3.12					
PCC, 3b			372			2.25					
PCC, 13a	Benzene/cyclohexane (40/60)		596			1.75					
PCC, 4b	Benzene/cyclohexane		117			3.86				M_w; D_z; $C = 4$ g/dm^3	665
PCC, 1c	(30/70)		186			2.96					

PCC, 3b			372		2.00					
PCC, 13a			596		1.75					
PCC, 4b	Benzene/cyclohexane		117		3.64				M_w; D_z; $C = 4$ g/dm^3	665
PCC, 1c	(20/80)		186		2.88					
PCC, 3b			372		1.99					
PCC, 13a			596		1.57					
PCC, 4b	Benzene/cyclohexane		117		3.39				M_w; D_z; $C = 4$ g/dm^3	665
PCC, 1c	(10/90)		186		2.64					
PCC, 3b			372		1.88					
PCC, 13a			596		1.48					
	Benzene/methanol (1/3.27 vol)	25						0.660		150
	Benzene/propane-2-ol (64/36)	25.5 ($T = \theta$)	33		8.2				M_w; $D_{z,0}$	666
			111		4.14	−21.0				
			180		3.367 ± 0.012	−31.6 ± 1.2				
			392		2.266 ± 0.006	−34.4 ± 0.6				
			670		1.751 ± 0.007	−45.0 ± 1.0				
			860		1.512 ± 0.014	−48.7 ± 2.6				
			1000		1.411 ± 0.001	−5.03 ± 0.8				
			1800		1.065 ± 0.004	−58.4 ± 1.0				
			2050		0.942 ± 0.006	−64.6 ± 1.7				
			33–2050		$D_{z,0} \sim M^{-0.497 \pm 0.006}$				M_w; $D_{z,0}$	666
Poly(styrene)	Benzene	25	5.0		10.4				DLS	1038
			10.0		7.5	2			DLS	
			17.5		5.1	3			DLS	
			35.5		3.5	7			DLS	
			50.0		2.8	10			DLS	
Polyscience	Bromobenzene		10.000						M_w; see Ref. for s_0 and k_s	667
Waters			3800						M_w; see Ref. for s_0 and k_s	667
	Butanone	20	110	7.13					CLS; M: av. of 6 obs.	11
			331	13.41					CLS; M: av. of 6 obs.	
			985	26.0					CLS; M: av. of 6 obs.	
			720	23.8					CLS; M: av. of 6 obs.	
			528	18.4	3.25				M_w; s_t; D_A	3
			250	12.6	73					125
			550	18.2	130					
			800	21.3	170					
			1300	26.0	230					
			90	12		12	1.38		$M_{w,w}$; s_w; D_w	126
			180	21		6.4	2.05			
			750	21		2.6	3.18			
			960	22		2.04	3.75			
			1700	31		1.70	3.71			
			2800	45		1.48	3.59			
			3400	30		0.83	6.05			

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			5000	45		0.84	5.24				
			5500	48		0.81	5.25				
			450	17.9		3.4				D_A; av. of 4 obs.	4
						3.75				D_w;	4
			528					$f_0 = 3.12 \times 10^{-5}$ cm		$f_0 = kT/(D_0 \eta_1)$	3
		23.4	1800			1.56				CLS	568
	Butanone	25				$D_0 = 3.1 \times 10^{-4} M^{-0.53}$				CLS	568
	2-Butanone	25				$D_z = 4.58 \times 10^{-4} M_w^{-0.55}$				$3.7 \leq M_w \times 10^{-4} \leq 302$	1071
										$1.03 \leq M_w/M_n \leq 1.15$	1071
		30	238			5.13	5.89			DLS	995
			426			3.83	13.4			DLS	
			1210			2.21	55.3			DLS	
			3850			1.19	140			DLS	
			5470			0.976	140			DLS	
PCC, 4b, 13a, 61208	Butanone		110, 670, 950							M_w; See Ref. for D_0 and k_D	897
		25	20.8–1700			$D_{z,0} = (3.1 \pm 0.2) \times 10^{-4} M^{(-0.53 \pm 0.02)}$				M_w	669
PCC		20	100–1800			$D_{z,0} = 4.1 \times 10^{-4} M^{-0.54}$				M_w; $M_w/M_n \leq 1.06$; 4V	670
PCC-1c		21–2	200			5.86 ± 0.06				M_w; $M_w/M_n = 1.04$; $D_{z,0}$	671
NBS 705			190			5.66 ± 0.09				M_w (SE); $M_w/M_n \leq 1.11$; $D_{z,0}$	
PCC-50124			254			4.89 ± 0.06				M_w; $M_w/M_n \leq 1.17$; $D_{z,0}$	
NBS 706			288			4.37 ± 0.03				M_w (SE); $M_w/M_n \leq 2.1$; $D_{z,0}$	
blend			233			5.22 ± 0.06				M_w; (DLS); $M_w/M_n \leq 1.4$; $D_{z,0}$	
		25	5400–15300			$D_{w,0} = 5.5 \times 10^{-4} M_w^{-0.561}$				M_w; $M_w/M_n = 1.3$–1.6; 4V	672
PCC, 2b		25	20.8			22.5 ± 1.0	-25 ± 5			M_w; $M_w/M_n = 1.06$;	673
PCC, 7b			33			16.0 ± 0.6	-26 ± 5			M_w; $M_w/M_n = 1.06$; $D_{z,0}$	
PCC, 4b			111			8.1 ± 0.3	-11 ± 4			M_w; $M_w/M_n = 1.06$; $D_{z,0}$	
PCC, 1c			200			5.95 ± 0.1	-2 ± 3			M_w; $M_w/M_n = 1.06$; $D_{z,0}$	
PCC, 3b			392			3.95 ± 0.15	9 ± 4			M_w; $M_w/M_n = 1.1$; $D_{z,0}$	
PCC, 5a			507			3.30 ± 0.10	19 ± 4			M_w; $M_w/M_n = 1.2$; $D_{z,0}$	
PCC, 13a			670			2.9 ± 0.1	27 ± 4			M_w; $M_w/M_n = 1.1$; $D_{z,0}$	
PCC, 14b			2700			1.30 ± 0.03	72 ± 5			M_w; $M_w/M_n = 1.3$; $D_{z,0}$	
JVC, 7M			8700			0.70 ± 0.03				M_w; $M_w/M_n = 1.3$; $D_{z,0}$	
		25	20.8–8700			$D_{z,0} = (5.5 \pm 0.2) \times 10^{-4} M_w^{-0.561 \pm 0.005}$				M_w; $M_w/M_n = 1.06$–1.3	673
PCC, 2b		25	21			22.5				M_w; $M_w/M_n = 1.06$	674
PCC, 4b			110			7.90				M_w; $M_w/M_n = 1.06$	
PCC, 1c			200			5.96				M_w; $M_w/M_n = 1.06$	
PCC, 13a			670			2.95				M_w; $M_w/M_n = 1.15$	
PCC, 14b			270			1.37				M_w; $M_w/M_n = 1.3$	
		25	10000			0.62 ± 0.01				M_w	675
		25	∞						0.9078	Extrapolated value	268
			325–2800	$\log s_0 = 17.845 + 0.455 \log M_w$						$[s_0] = 1.385 \times 10^{-2} s_0^0$	169

NBS 706		25	257.8			3.88			M_w; CLS	571
NBS 705			179.3			5.4			M_w; CLS	571
Waters sample W-61970			2145			1.62			M_w; CLS	571
						$D_0 = 1.96 \times 10^{-4}\ M^{-0.49}$				571
		25 ($T = \theta$)	127	35.76–37.82					For solvents See Ref.; s_w; A; $C = 2$ g/dm^3	427
		27	82	9.8	43	9.4	−86		M_n; D_w; $D_w = k\,M_{f,w}^{-0.53}$	12
			140	13.3	48	5.14	−12			
		25		$s_0 = 4.62 \times 10^{-15}\ M^{0.452}$	$0.107\ M^{0.55}$				$M_w/M_n = 1.6$–2.3	745
					$k_s = 8.08 \times 10^{-2}\ M^{0.70}$					
		25	107			7.44	5		M_w; $M_w/M_n = 1.01$; D_A; $C = 15.1$ g/dm^3	648
			45–448			$D_{z,0} = 5.85 \times 10^{-4}\ M^{-0.58}$			M_w; DLS	642
						50			M_w; $D_{z,0}$; $C = 7.9$ g/dm^3	716
			2.2			6.18			M_w; $D_{z,0}$; $C = 1.7$ g/dm^3	
	Butanone	27	480	20.0	111	3.32	49			12
			770			2.51	0			
			1100	25.0	175	2.64	40			
	Butanone/butanol (74/26)	25	200							169
			372.5	18.05						
			830	26.6	123				M_w; CLS	
			1065	32.0	162					
			2300	47.1	272					
	Butyl acetate	25	∞					0.9162		268
	Carbon tetrachloride	23	600			1.0			M_w; $D_{z,0}$; k_D given in Ref.	677
		24	2880			0.7			DLS; $M_w/M_n = 1.09$	991
		25	10000			0.179 ± 0.012			M_w; $D_{z,0}$	676
		25						0.931	A. Paar Digital Densitometer	1060
		25	2000			0.63			M_w; $M_w/M_n < 1.3$; $C = 20$ g/dm^3; 3V	799
		27	82			4.43	-1.31^a		M_n; D_w	12
			140			2.27	0^a			
			480			1.41	0^a			
			1100			1.04	0.5^a			
						$D_w = k\,M_{f,w}^{-0.59}$		0.9087	Extrapolated value	268
		30	79.8			3.70 ± 0.2	38.92			12
	Chlorobenzene	25	∞						M_w; $D_{z,0}$	659
		25						0.9207	Extrapolated value	268
	Chloroform	20	250	9.2	180			0.912		1060
										125
		25						0.908		5
			550	13.1	320			0.924		1060
			800	15.4	400					
			1300	17.8	500					
			540	−15.1		(1.84)	$f_0 = 3.86 \times 10^{-5}$ cm		s_f; M_w; $f_0 = kT/(D_{0,1})$	3
standard sample no. 706		21	278			5.8			M_w; D_w	558
		25						0.9100	Extrapolated value	268
			115–2800	$\log s_0 = 17.992 + 0.415 \log M_w$					$[s_0] = -1.595 \times 10^{-2}\ s_0$	268

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
PCC, 4b		22	110			6.1				M_w; $M_w/M_n = 1.06$	678
		21	500	−12.30	330					M_w; flotation	748
			5000	−20.80	690						
			13300	−27.10	950						
			16400	−33.3	1420						
	Cyclohexane	25	∞						0.9293	Extrapolated value	268
			181–1040	$s_0 = 3.85 \times 10^{-15} M_w^{0.42}$						$M_{s,d}$	605
		28	180			ca. 2.2				M_n	414
		30	180			ca. 2.5				M_n	414
			90			4.0				$C = 17.8$ g/dm^3	587
			237			ca. 3.1				$C = 1.47$–17.9 g/dm^3	587
		34 ($T=\theta$)	1150	16.9[b]	72 22					M_v	576
		34.5 ($T=\theta$)	0.37			67.4	−1.03			$M_w/M_n \approx 1.02$	1069
			0.58			58.2	−1.67			$M_w/M_n \approx 1.02$	
			0.90			43.1	−1.91			$M_w/M_n \approx 1.02$	
			1.78			32.9	−3.45			$M_w/M_n \approx 1.02$	
			3.48			21.7	−3.78			$M_w/M_n \approx 1.02$	
			10.1			13.0	−6.07			$M_w/M_n \approx 1.02$	
			40.0			6.58	−10.0			$M_w/M_n \approx 1.02$	
			191.0			3.04	−21.7			$M_w/M_n \approx 1.02$	
			732.0			1.57	−38.9			$M_w/M_n \approx 1.02$	
			1320.0			1.17	−68.4			$M_w/M_n \approx 1.02$	
		35.1 ($T=\theta$)	702			1.69				DLS	1061
			2300			0.93				DLS	1061
	Cyclohexane	35	32–1640							$M_w f_0 = kT/(D_0\eta_1)$ given in Ref.	6
			540	12.1		1.92		$f_0 = 2.95 \times 10^{-5}$ cm		M_w; $f_0 = kT/(D_0\eta_1)$	3
									0.934		5
			11.4–1040	$s_0 = 1.69 \times 10^{-15} M^{0.48}$						M_w; A	107
				$s_0 = 1.35 \times 10^{-15} M_w^{0.51}$		$D_0 = 1.21 \times 10^{-4} M_w^{-0.49}$					96
			4000	28.1	5.06				0.928	M_w; CLS	172
										See Ref. for v_2	532
Styron 683–7			125			3.86	−7			M_w; M_n given in Ref.; D_w	592
						2.71					592
NBS 705						3.19	5			D_w	592
						2.88				D_A	592
			309	8.43						M_n	486
			975	16.2					0.943		
			4900	34.5							
Styron 683-7		35	277			3.86	−6			M_w; D_A	577
NBS 705			179			3.19	−4			M_w; D_A	
NBS 419			411			2.67	−4			M_w; D_A	
				$s_0 = 1.50 \times 10^{-15} M^{0.5}$		$D_0 = 1.21 \times 10^{-4} M^{-0.5}$				M_w; dep. of k_s given in Ref.	594
			194–929	$s_0 = 1.50 \times 10^{-15} M^{0.5}$		0.928				$M_{s,D}$	605
			150						0.933	M_n; $M_w = 450.000$	606

Waters sample no. 25170			13	51	2.28	−4.3		M_{MW}; $M_w = 51.000$, $M_n = 49.000$	543
		16		2.42	−1.4				
		22		2.61	−0.8				
		28		2.88	5.8				
		35		3.26	12.1				
		32	1500	16.8	24			M_{MW}	
		34		17.2	46				
		36		17.6	71				
		38		18.0	86				
		40		18.6	105				
		42		19.0	125				
		28–40	180			ca. 2.9		M_n	414
		45	4000	29.0	1570		0.934	M_w (CLS)	172
		55	4000	34.0	3570		0.936	M_w (CLS)	172
		35	79.8			4.61 ± 0.2	−17.62		659
		35	498			1.5		M_w; D_z; see Ref. M_n	679
		35	179, 1050					M_w; D_z given in Ref.	679
		35	179			3.23 ± 0.02		M_w; $M_w/M_n = 1.06$; $D_{z,0}$; k_D	680
		38				3.37 ± 0.05		Given in Ref.	680
		40.4				3.64 ± 0.01			
		45				3.83 ± 0.04			
		55				4.61 ± 0.08			
		45	900			1.49 ± 0.02			
		55				1.83 ± 0.03			
		35 ($T = \theta$)	4100–48000			$D_{z,0} = 1.39 \times 10^{-4}\ M^{-0.5}$		M_w	681
		40	200			2.9		M_w C = 10 g/dm^3	682
NBS 705		35	179.3			2.62		M_w; $M_w/M_n = 1.07$; D_z	684
Waters sample no. 25167				867			1.20	M_w; $M_w/M_n = 1.12$; D_z	
Waters sample no. 61970				2145			0.80	M_w; $M_w/M_n = 1.21$; D_z	
			4500			0.34		M_w; D_z; $C = 5.4$ g/dm^3	
PCC, 4a		34.5 (T = θ)	97	4.64 ± 0.04	19 ± 1.5			M_w; $M_w/M_n = 1.06$	685
PCC, 1a			160	5.93 ± 0.03	21.5 ± 1			M_w; $M_w/M_n = 1.06$	
PCC, 1c	Cyclohexane		190	6.44 ± 0.03	26 ± 1			M_w; $M_w/M_n = 1.06$	685
PCC, 3a			410	9.68 ± 0.08	38 ± 4			M_w; $M_w/M_n = 1.06$	
PCC, 6a			860	14.05 ± 0.08	58 ± 4			M_w; $M_w/M_n = 1.15$	
			1800					$M_w/M_n = 1.20$	
			97–1800	$s_0 = (1.51 \pm 0.02) \times 10^{-15}\ M^{0.5}$					
				$s_0 = (0.062 \pm 0.003)\ M^{0.5}$					
								M_w; $M_w/M_n = 1.15$	
		35.4	233	7.3				M_w	
TS-6			422	9.7				M_w	686
PCC, 60914			600	11.6				M_w; $M_w/M_n = 1.04$	
TS-8			1260	18.2				M_w; $M_w/M_n = 1.1$	
TS-34			3840	29.4				M_w; $M_w/M_n = 1.05$	
TS-31			8420	40.8				M_w; $M_w/M_n = 1.05$	
TS-33			20600	66.7				M_w; $M_w/M_n = 1.19$	
			233–20600	$s_0 = (1.50 \pm 0.01) \times 10^{-15}\ M^{0.5}$				M_w	686
				$s_0 = (0.052 \pm 0.010)\ M^{0.5}$					
PCC, 4b, 13a, PS-705; F-4		35	110–4100			$D_{z,0} = (1.38 \pm 0.04) \times 10^{-4}\ M^{-0.5}$		M_w; M_w/M_n = 1.06–1.15, k_s given in Ref.	687

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
		35 ± 0.2	110–4100			$D_{z,0} = (1.38 \pm 0.04) \times 10^{-4}\ M^{-0.5}$				M_w; M_w/M_n = 1.06–1.15, k_s given in Ref.	688
PCC		35.4	230	7.3 ± 0.2						M_w; $C = 2.5$ g/dm^3; s_0 given in Ref.	689
		28	230	7.2 ± 0.2						M_w	689
SRM 1478		35	37.4					0.9320 ± 0.0007			690
		35	51 ± 2.5			2.07 ± 0.02				M_w; $M_z/M_w < 1.1$; D_z	691
			410 ± 20			5.86 ± 0.22				M_w; $M_z/M_w < 1.1$; D_z	
			186 ± 10			2.56 ± 0.07				M_w; $M_z/M_w = 1.87$; D_z	
			254 ± 13			2.56 ± 0.03				M_w; $M_z/M_w = 1.24$; 0.05; D_z	
PCC		10	37			3.69				M_w; M_w/M_n = 1.06; $D_{z,0}$	692
		15				4.08					
		20				4.51					
		30				5.44					
		35 ($T = \theta$)				5.01					
		40				6.36					
		50				7.27					
PCC		15	110			ca. 2.51				M_w; M_w/M_n = 1.06; $D_{z,0}$	692
		20				2.78					
		30				3.38					
		35 ($T = \theta$)				3.69					
		40				3.97					
		50				4.57					
		60				5.17					
		70				5.91					
		80				6.67					
		90				7.47					
		100				8.28					
PCC		20	350			1.72				M_w; M_w/M_n = 1.1; $D_{z,0}$	692
		30				1.99					
		35 ($T = \theta$)				2.12					
		40				2.26					
		50				2.61					
		60				2.96					
		70				3.36					
		80				3.80					
		90				4.25					
		100				4.77					
PCC		35 ($T = \theta$)	37–350			$D_{z,0} = (1.24 \pm 0.05) \times 10^{-8}\ M_z^{-0.498 \pm 0.02}$				M_w; M_w/M_n = 1.06–1.1; $D_{z,0}$	692
PCC	Cyclohexane	35	670			1.66				M_w; $D_{z,0}$; C = 4.2–79.9 g/dm^3	795
PCC		35	107–775	$s_0 = 1.49 \times 10^{-15}\ M^{0.5}$		$D_{z,0} = 1.40 \times 10^{-4}\ M^{-0.5}$				M_w; C = 0.9–8.08	694

								g/dm^3; 7V; corrected for polydispersity	
		35 ($T=\theta$)	2.41 ± 0.20			2.43 ± 0.05		M_w; M_z/M_w = 1.61 ± 0.05; D; C = 2 g/dm^3	695
			2.45 ± 0.26			2.46 ± 0.09		M_w; M_z/M_w = 1.59 ± 0.19; D; C = 2 g/dm^3	
PCC, 4b		20	117			3.14		M_w	665
		37	250			2.25		M_w; D_z; C = 10 g/dm^3	693
			269			1.95		M_w; D_z; C = 10 g/dm^3	
		35	670					D_z given in Ref.	796
Styron 683-7		35	340	7.14	24	3.69	− 8.4	M_w; M_n = 111000; $D_{w,0}$; see Ref.	797
			265	7.76		3.23	− 5.4	M_w; M_n = 182000; $D_{w,0}$; see Ref.	
			425	9.23		3.07	− 7.1	M_w; M_n = 202000; $D_{w,0}$; see Ref.	
			365	8.25		3.35	− 9.4	M_w; M_n = 168000; $D_{w,0}$; see Ref.	
		35 ± 0.1	4100–48000			$D_{z,0} = 1.39 \times 10^{-4}\ M^{-0.5}$		M_w; M_w/M_n = 1.02–1.17	696
						$D_z = D_{z,0} - 7.76 \times 10^9\ C$			696
		34.5	98	4.64 ± 0.04				M_w; $M_w/M_n \leq 1.06$; pressure dependence of s_0 given in Ref.	697
			160	5.93 ± 0.03					
			190	6.44 ± 0.03				M_w; $M_w/M_n \leq 1.06$	
			411	9.68 ± 0.08				M_w; $M_w/M_n \leq 1.06$	
			860	14.05 ± 0.08				M_w; $M_w/M_n \leq 1.15$	
		35						0.934	743
		34.5 ($T=\theta$)	238			2.74	− 22.0	DLS	995
			426			2.09	− 26.6	DLS	
			1210			1.24	− 48.6	DLS	
			3850			0.67	− 50.0	DLS	
			5470			0.58	− 85.6	DLS	
		44.5	238			3.22	− 10.6	DLS	
			426			2.46	− 8.3	DLS	
			1210			1.42	1.21	DLS	
			3850			0.75	8.67	DLS	
			5470			0.612	43.5	DLS	
TS-32	Cyclohexane	35.6	6770	38.02				M_w; M_w/M_n = 1.14	698
		38.7		38.06					
TS-31		35.6	8420	41.19				M_w; M_w/M_n = 1.17	
		36.6		40.79					
		39.0		40.70					
TS-33		35.6	20600	64.36				M_w	
		36.6		64.75					
		38.0		63.19					
		39.0		61.75					
		35.4	100–20600	$s_0 = 1.50 \times 10^{-15}\ M^{0.50}$				M_w	

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
NBS-705		35	179							M_w;$M_w/M_n = 1.07$; $D_{z,0}$, k_D given in Ref.	700
PCC, 8a		35	10.3							M_w; $M_w/M_n = 1.06$; $D_{z,0}$, k_D given in Ref.	700
		30	37			6.4 ± 0.02				M_w; $M_w/M_n = 1.03$; $D_{z,0}$; DLS	701
		20.3 ($T = T_c$)	85	3.00					0.9214	M_w; $M_w/M_n \leq 1.06$; T_c critical temp.	702
		34.5 ($T = \theta$)		3.7					0.9285		
		24.3 ($T = T_c$)	140	4.96		0.9280			0.9225		
		34.5 ($T = \theta$)		5.0		0.9219					
		27.0 ($T = T_c$)	320	7.60		0.9260					
		34.5 ($T = \theta$)		8.8		0.9195					
		30.0 ($T = T_c$)	700	12.60		0.9220					
		34.5 ($T = \theta$)		12.0		0.9179					
		31.0 ($T = T_c$)	1100	13.20		0.9200					
		34.5 ($T = \theta$)		14.2						M_w	702
		34.5 ($T = \theta$)	85–1100	$s_0 = 1.58 \times 10^{-15} M^{0.49}$		4.33	−4			M_w; $M_w/M_n = 1.01$; D_A; $C = 15.3$ g/dm^3	648
F 10		35	107			2.81	−8			M_w; $M_w/M_n \approx 3$; D_A; $C = 13$ g/dm^3	
Styron 683-7		35	302			6.10				M_w; $M_w/M_n \leq 1.06$; s_0	703
PCC, 7a		34.5	51	3.28		4.50					
Waters, 41995			98.2	4.66		4.03			0.936		
PCC, 4b			110	5.09		3.36					
NBS-705			183	6–33		2.33			0.936		
PCC, 3b			390	9.26		2.26					
Waters, 25166N			470	9.34		2.20					
Water, 25166			411	9.62		2.05					
PCC, 5a			498	11.16		1.64					
Waters, 25167			867	13.64		1.12				M_w; $M_w/M_n = 1.2$; s_0	703
PCC, 14a		34.5	1800	19.41		1.09					
Waters, 61970			2145	19.41		$D_{z,0} = (1.388 \pm 0.42) \times 10^{-4} M^{-0.5}$				M_w; $M_w/M_n = 1.2$	704
		34.5	51–2145	$s_0 = (1.514 \pm 0.077) \times 10^{-15} M^{0.5}$							
					$k_s = (0.65 \pm 0.17)$						
		34.5	1.2			35.0	−2.0			DLS	993
			3.1			22.3	−2.8			DLS	993
			4.0			19.45	−3.9			DLS	993
			10.7			12.30	−5.7			DLS	993
			30.2			7.31	−10.8			DLS	993
PCC, 2b		35	21			8.97	−9			M_w; $M_w/M_n < 0.06$; DLS	674
PCC, 4b			110			4.20	−18			M_w; $M_w/M_n < 0.06$; DLS	
PCC, 1c			200			3.05	−16			M_w; $M_w/M_n < 0.06$; DLS	
PCC, 13a			670			1.66	−34			M_w; $M_w/M_n < 0.15$; DLS	
PCC, 14b			2700			0.84	−56			M_w; $M_w/M_n < 0.3$; DLS	
			21–2700			$D_{z,0} = (1.3 \pm 0.2) \times 10^{-4} \times M_w^{-0.497 \pm 0.006}$				M_w; $M_w/M_n = 1.6$–2.3; DLS	674

		21–2700	$s_0 = 1.46 \times 10^{-15}\ M^{0.501}$			$k_D = (4.1 \pm 0.3) \times 10^{-4} \times M_w^{0.48 \pm 0.04}$	
				$k_s = 5.69 \times 10^{-2}\ M^{0.50}$			
	35	500	11.50	40			M_w; $M_w/M_n = 1.6$–2.3 745
Cyclohexane		5000	24.40	104			M_w 748
		16400	50.00	150			
linear, S102	35	84	4.29				M_w; s_D corrected for pressure influence 683
linear, PS100f2		116	4.94				
linear, S108		238	7.66				
linear, PS800f2		675	12.2				
linear, PS1000f2		1700	20.15				
star	35	960			1.66 ± 0.04	-4100 ± 400	M_w; $D_{z,0}$; DLS 719
branched	34.5	980			1.63		$M_w/M_n = 1.17$ 994
branched		1730			1.28		$M_w/M_n = 1.24$
branched		2390			1.09		$M_w/M_n = 1.19$
branched		4780			0.767		$M_w/M_n = 1.06$
branched		14300			0.509		$M_w/M_n = 1.03$
branched	44.5	980			1.82		994
branched		1730			1.37		
branched		2390			1.18		
branched		4780			0.860		
branched		14300			0.569		
branched, 4-armed stars, S131A	35	50.7	3.38				M_w; s_0 683
branched, 4-armed stars, S121A		93.5	4.72				
branched, 4-armed stars, S111A		154	6.17				
branched, 4-armed stars, S161A		351	9.34				
branched, 4-armed stars, S141AA		521	12.0				
branched, 4-armed stars, S181A		1027	16.2				
branched, 4-armed stars, S191A		1390	18.7				
branched, 6-armed stars, HS061A	35	110	5.51				M_w; s_0 683
branched, 6-armed stars, HS051A		1090	17.4				
Comb, C612	35	475	10.8				M_w; s_0 683
Comb, C622		624	13.7				
Comb, C632		913	17.4				
Comb, C642		1630	25.0				
Comb, C652		3130	35.6				
Comb, C732		1530	20.6				
Comb, C742		2230	26.5				
Comb, C752		3620	36.3				
	35		$s_0 = 1.49 \times 10^{-15}\ M_w^{0.5}$				s_0 683
PCC, 3b	35	392					M_w; $M_w/M_n \leq 1.10$ s_0 and k_s given in Ref. 793
PCC, 14a		1700					M_w; $M_w/M_n \leq 1.20$ 793
branched, 4-armed stars		312					M_w; $M_w/M_n \leq 1.15$ 793
		2210					M_w; $M_w/M_n \leq 1.2$
branched, 3-armed stars		12.4			11.73		M_w;D_z 705
		920			1.47		
		1190			1.28		
branched, 12-armed stars		55			6.61		
		149			4.04		
		467			2.42		

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TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			808			1.85					
			1110			1.61					
			1690			1.35					
branched, 18-armed stars			8770			0.621					
branched, 9-armed stars		35	950			1.66 ± 0.04	-41 ± 4			M_w; $D_{z,0}$	706
PCC	Cyclohexanone	25	200			5.2				M_w; $M_w/M_n \leq 1.06$	624
PCC, 4b	Cyclopentane	5	111	8.07	2.0 g/g					M_w; $M_w/M_n \leq 1.06$	707
		11		8.40	10.9 g/g						
		15		8.84	16.1 g/g						
		20		8.90	17.1 g/g						
		30		9.94	22.9 g/g						
		40		11.03	27.0 g/g						
PCC, 2b		20 ($T=\theta$)	20.8	3.97	9.7 g/g					M_w; $M_w/M_n \leq 1.06$	707
PCC, 3b		20 ($T=\theta$)	392	16.12	36.5 g/g					M_w; $M_w/M_n \leq 1.10$	707
PCC, 61208		20 ($T=\theta$)	900	26.60	64.9 g/g					M_w; $M_w/M_n \leq 1.10$	707
		20 ($T=\theta$)	20.8–900	$s_0 = 8.7 \times 10^{-14}\, M_w^{0.50}$							707
					$k_s = 1.9\, M_w^{0.50}$						707
TS-12		20 ($T=\theta$)	4480	34.85						M_w; $M_w/M_n \approx 1.14$; $C = 4.448$ g/dm^3; 4V	708
PCC		20.4 ($T=\theta$)	110			5.91				M_w; $M_w/M_n \approx 1.06$; D_z $C = 0.55$ mass%; 6V	756
		24.8				4.54				D_z; $C = 5.95$ mass%; 3V	
		29.0				5.27				D_z; $C = 5.95$ mass%; 3V	
		34.0				6.13				D_z; $C = 5.95$ mass%; 3V	
Polym. Soc. (Japan), stand. sample	*trans*-Decalin	18.2	150			0.85				M_n; D_A; D_w; M_w; M_{MW} given in Ref.	502
		30				1.16					
		40				0.93					
		50				1.43					
		60				1.52					
		70				1.82					
		80				1.96					
		90				2.69					
		100				3.30					
		110				3.59					
		18.2	450			0.69				M_n; D_A; D_w; M_w; M_{MW} given in Ref.	502
		40				0.76					
		60				1.10					
		80				1.45					
		110				2.23					
		27	770	2.73	73	0.50	-15			M_n; D_w	12
TS, F80		20.4 ($T=\theta$)	775			0.5223	-38.8 ± 2.4			M_w	709
TS, FF33			2420			0.3035	-82.2 ± 7.3				
TS, FF35			5530			0.1935	-104 ± 11				
TS, FF36			9700			0.1458	-119 ± 19		0.13	M_w	
		20.4	775–9700			$D_0 = (4.60 \pm 0.08) \times 10^{-5}\, M^{-0.5}$			0.43 ± 0.02	M_w	710
		25 ($T=\theta$)	1800	5.43						M_w; $M_w/M_n \approx 1.20$	711

		20	390	2.34	27.3			M_w; $M_w/M_n \leq 1.10$	712
		25		2.60	58.8				
		30		2.78	65.6				
		35		3.06	79.1				
		40		3.34	94.9				
NBS-705		20	179.3			1.11	1.72×10^{-5}	M_w; $M_w/M_n = 1.07$, $D_{z,0}$, DLS; f_0/η_0	713
		30				1.33	1.77×10^{-5}		
		40				1.59	1.82×10^{-5}		
		20.5 ($T = \theta$)	37			6.95		M_w; $M_w/M_n = 1.06$; $D_{z,0}$; DLS	714
			180			3.13		M_w; $M_w/M_n = 1.06$; $D_{z,0}$; DLS	
			670			1.65		M_w; $M_w/M_n = 1.15$; $D_{z,0}$; DLS	
			1260			1.172		M_w; $M_w/M_n = 1.05$; $D_{z,0}$; DLS	
	trans-Decalin		2880			0.736		M_w; $M_w/M_n = 1.09$; $D_{z,0}$; DLS	714
			5050			0.566		M_w; $M_w/M_n = 1.02$; $D_{z,0}$; DLS	
		20.5 ($T = \theta$)	37–5050			$D_{z,0} = 1.309 \times 10^{-4} M_w^{-0.5}$		M_w; M_w/M_n = 1.02–1.15; DLS	
PCC, 3b		20	390			0.70	0.9259	M_w; 10V; $C = 0$–84.5 g/dm^3	724,725
		25					0.9282		
		30				0.86	0.9308	M_w; 9V; $C = 0$–86.7 g/dm^3	
		35					0.9336		
		40				1.01	0.9357	M_w; 9V; $C = 0$–83.0 g/dm^3	
	Dibutyl phthalate	25	680					M_w; $D_{z,0}$; DLS	715
		120							
	o-Dichlorobenzene	25	∞				0.9289	Extrapolated value	268
	Dichloroethane	21	500	−4.26	292			M_w; flotation	748
			5000	−8.35	880				
			13300	−13.30	1330				
			16400	−16.70	1960				
	Diethyl ketone	25	∞				0.9106	Extrapolated value	268
	Dimethylformamide	21	500	3.60	103			M_w	748
			5000	6.70	167				
			13300	7.70	190				
	Dioxane	25					0.930		1060
	1,4-Dioxane	50	2			27.7		Taylor dispersion method	1062
			4			19.8			
			9			14.9			
			17.5			9.9			
			50			5.8			
			100			4.1			
		75	2			41.8			
			4			30.0			
			9			21.3			
			17.5			14.1			

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			50			8.2					
			100			5.9					
		95	2			52.7					
			4			39.1					
			9			28.0					
			17.5			18.4					
	1,4-Dioxane	95	50			10.8				Taylor dispersion method	1062
			100			7.6					
		105	2			59.5					
			4			44.1					
			9			32.5					
			17.5			20.9					
			50			12.4					
			100			8.7					
	Dioxane	25	∞						0.9270	Extrapolated value	268
		30	79.8			3.10 ± 0.2	13.7			M_w; $D_{z,0}$; DLS	659
	Epichlorohydrin	21	500	−1.97	145					M_w; flotation	748
			5000	−3.85	460						
			13300	−7.15	1160						
			16400	−11.10	2170						
	Epichlorohydrin/ dimethyl maleate (1.3/1.0 mol/mol)		500	−1.35	36					M_w; flotation	748
			5000	−2.63	97						
			13300	−3.03	168						
			16400	−5.26	278						
	Ethyl acetate	25	∞						0.9132	Extrapolated value	268
		27				2.64	12			M_n; D_w	12
PCC, 4b		20	117			6.23				M_w; D_z	665
PCC, 1c			186			5.05					
PCC, 3b			372			3.31					
PCC, 13a			596			2.45					
	Ethyl acetate/ cyclohexane (90/10)	20	117			6.05				M_w; D_z	665
			186			4.54					
			372			3.21					
			596			2.38					
	Ethyl acetate/ cyclohexane (80/20)	20	117			5.81				M_w; D_z	665
			186			4.55					
			372			3.05					
			596			2.45					
	Ethyl acetate/ cyclohexane (70/30)	20	117			5.73				M_w; D_z	665
			186			4.36					
			372			2.89					
			596			2.20					

	Ethyl acetate/ cyclohexane (60/40)	20	117			5.28			M_w; D_z 665
			186			4.11			
			372			2.76			
			596			2.15			
	Ethyl acetate/ cyclohexane (50/50)	20	117			4.95			M_w; D_z 665
			186			3.92			
			372			2.53			
			596			1.99			
	Ethyl acetate/ cyclohexane (40/60)	20	117			4.72			M_w; D_z 665
			186			3.68			
			372			2.39			
			596			1.92			
	Ethyl acetate/ cyclohexane (30/70)	20	117			4.27			M_w; D_z 665
			186			3.54			
			372			2.31			
			596			1.81			
	Ethyl acetate/ cyclohexane (20/80)	20	117			4.08			M_w; D_z 665
			186			3.09			
			372			2.10			
			596			1.53			
	Ethyl acetate/ cyclohexane (10/90)	20	117			3.69			M_w; D_z 665
			186			2.73			
			372			1.79			
			596			1.49			
	Ethyl benzene	27	770	9.5	3.55	0.96	130		M_n; D_w 12
		30	42.5–1430	$s_0 = 3.28 \times 10^{-15}\ M^{0.42}$					M_v; 12V; k_s 612 given in Ref.
		130							D given in Ref. 794
	Ethyl *n*-butyrate	25						0.919	1060
	Mesitylene	25						0.9260	Extrapolated value 268
	Methylethylketone	25						0.919	1060
	Methyl isopropyl ketone	20	94–1200	7.2–34		6.95–2.5			7V; s_t; $D_A = D_w$; $M_{t,w}$ 66
	Nitrobenzene	25						0.904	1060
	Tetrahydrofuran	25	3840			0.682	571		$M_w/M_n = 1.04$ 992
			5480			0.531	1114		$M_w/M_n = 1.15$ 992
			8420			0.432	1071		$M_w/M_n = 1.17$
			20000			0.241	1501		$M_w/M_n = 1.20$
	Tetrahydrofuran	25						0.9102	Extrapolated value 268
linear		25	10.5			15.8	8		$M_w/M_n = 1.15$ 996
			45.0			8.1	20		$M_w/M_n = 1.19$
			79.5			6.15	38		$M_w/M_n = 1.08$
			182.0			3.95	46		$M_w/M_n = 1.10$
						$D_0^{linear} = (1.40 \pm 0.20) \times 10^{-4}\ M_w^{-0.48 \pm 0.02}$			
cyclic			11.1			6			$M_w/M_n = 1.20$
			37.8			14			$M_w/M_n = 1.21$
			79.4			20			$M_w/M_n = 1.14$

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	υ_2 (cm^3/g)	Remarks	Refs.
			181.5			27				$M_w/M_n = 1.11$	
						$D_0^{cyclic} = (2.40 \pm 0.17) \times 10^{-4}\ M_w^{-0.52\pm0.02}$					
		30	198			13.41				M_w; $D_{z,0}$; k_D given in Ref.; DLS	718
			370			8.81					
			570			8.02					
			110			4.92					
			390			2.47					
			411			2.51					
			3000			0.813					
			10000			0.435					
linear		30	180			3.25	36 ± 8			M_w; $D_{z,0}$; DLS	719,722
			390			2.48	96 ± 7				
			3000			0.813	390 ± 90				
			10000			0.435	620 ± 30				
star			960			1.71 ± 0.04	5100 ± 500			M_w; $D_{z,0}$; DLS	719,722
linear		30	180–10000			$D_{z,0} = (3.2 \pm 0.1) \times 10^{-4}\ M_w^{-0.56\pm0.02}$				M_w; DLS	720,722
		30	19.8			13.41	76			M_v; $M_w/M_n = 1.06$; $D_{z,0}$; DLS	721
			37			8.81	24				
			51			8.02	14.9				
			110			4.92	32				
			411			2.51	75				
		30	20–411			$D_{z,0} = (3.0 \pm 0.4) \times 10^{-4}\ M^{-0.549\pm0.013}$				DLS	721
		24	20.4			12.7	7			M_w; $D_{z,0}$; DLS	723
			51			7.65	16				
			97.2			5.43	28				
			160			3.99	44				
			411			2.35	91				
			860			1.59	136				
			1800			1.01	264				
			20–1800			$D_{z,0} = (3.45 \pm 0.18) \times 10^{-4}\ M_w^{-0.564\pm0.004}$				M_w; DLS	723
atactic	4-*tert*-Butyltoluene	50	5.38			19.2	−2.69			DLS	1068
			20.2			9.31	−2.44			DLS	
			359			2.01	20.4			DLS	
			732			1.38	57.0			DLS	
			1270			1.02	77.1			DLS	
			3840			0.53	129			DLS	
atactic	Toluene	15	0.92			52.0	−1.22			$M_w/M_n = 1.01$	1068
			5.38			20.0	0.525			$M_w/M_n = 1.03$	
			9.98			14.1	2.38			$M_w/M_n = 1.03$	
			40.0			6.85	10.7			$M_w/M_n = 1.02$	
			97.3			4.15	21.8			$M_w/M_n = 1.02$	
			191			2.82	40.6			$M_w/M_n = 1.02$	
			359			1.96	71.2			$M_w/M_n = 1.01$	
	Toluene		732			1.33	130			$M_w/M_n = 1.03$	1068
			1270			0.978	206			$M_w/M_n = 1.03$	
			3840			0.502	319			$M_w/M_n = 1.05$	

	20	536	8.75		1.95	$f_0 = 3.55 \times 10^{-5}$ cm		M_w; s_t; D_A; $f_0' = kT/(D_0\eta_1)$	3
		450	7.92		2.02			Av. of 3 obs.	4
							0.910		5
		150					0.917	M_n; $M_w = 450000$	606
		1.2–520	$s_0 \sim M^{0.47}$		$D \sim M^{-0.53}$				43
		300	6.25		2.50			M_w; s_t; D_A	1
		45	7.20		2.85			M_n; s_w; D_w	
		140	4.75		4.03			$M_{t,A}$; s_t; D_A	16
		960	9.65		1.17				
		2850	18.1		0.74				
		250–1300	5.9–11.2					4V; k_s given in Ref.	125
Szwarc "living polymer"			57.5–226	3.7–5.20		8.05–3.29		4V; $M_{t,A}$ $C = 1$ g/dm^3	44
	22.7	190					0.918	M_v	148
	23.5		3.24	36.4				s_t	237
		75.0	3.24	36.4			0.917	M_{MW}	409
	25						0.916	Extrapolated value	268
					$D = 2.15 \times 10^{-4}\ M_w^{-0.53}$				591
								v_2 given in Ref.	532
		74			4.13			D_A; $C = 0.79$ g/dm^3	511
					3.78			D_A; $C = 1.19$ g/dm^3	511
	25	1260			1.265			DLS	1078
		2950			0.81			DLS	1078
		8000			0.43			DLS	1078
		15000			0.31			DLS	1078
	27						0.917		57
		770			1.22	96		M_n (Ref. 13); D_w	12
		1.2	0.66					s_t	17
		2	0.44						17
		40					0.917		1,237
	30	13.3					0.948	M_n	467
PCC	25	200			3.4			M_w; $M_w/M_n < 1.06$	624
TS-12	25	4480	14.01					M_w; $M_w/M_n = 1.14$; $C = 0.429$ g/dm^3; 12V ($C = 0.429$–37.02 g/dm^3	708
TS-33	25	20600	15.47					M_w; $C = 0.487$ g/dm^3; 6V ($C = 0.487$–4.445 g/dm^3)	708
	30	250			2.57			M_w; $M_n = 100000$; $D_{z,0}$; DLS	721
		269			2.37			M_w; $M_n = 59000$; $D_{z,0}$; DLS	
		411			1.982			M_w; $M_n = 388000$; $D_{z,0}$; DLA	
	30	238			3.19	56.7		DLS	995
		426			2.32	74.2		DLS	995
		1210			1.28	190		DLS	995
		3850			0.68	361		DLS	995
		5470			0.55	546		DLS	995
	30	79.8			6.32 ± 0.2	20.57		M_w; $D_{z,0}$; DLS	659
		93.46			5.22 ± 0.2	10.476			
		347.22			2.20 ± 0.1	61.36			
		1136.36			1.20 ± 0.1	173.3			
		3802.28			0.85 ± 0.05	1552.94			

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
		30	37			12.6 ± 3.8				M_w; $M_w/M_n = 1.03$; $D_{z,0}$; DLS	701
PCC		20	670							M_w; D_z given in Ref.; DLS	796
PCC		20	670			1.58				M_w; $D_{z,0}$ C = 4.2–79.9 g/dm^3; DLS	795
	Toluene	23	179			3.59 ± 0.25	85.9 ± 73.7			M_w; $M_w/M_n = 1.06$; $D_{z,0}$; $k_D = k_D D_{z,0}$; DLS	680
		55	179			5.09 ± 0.03	198.1 ± 10.1			M_w; $M_w/M_n = 1.06$; $D_{z,0}$; $k_D = k_D D_{z,0}$; DLS	680
PCC, 4b		10	110			2.3				M_w; $M_w/M_n = 1.06$; D_z; C = 28.7 g/dm^3, DLS	678
		20	99.5			4.45	19			M_w; $M_w/M_n = 1.07$; $D_{z,0}$; DLS	727
F 80			783			1.34	113			M_w; $D_{z,0}$; DLS	
F 126			1280			1.01	173			M_w; $D_{z,0}$; DLS	
F 288			2890			0.53	432			M_w; $D_{z,0}$; DLS	
		20	18800			0.207				M_w; $M_w/M_n = 1.16$; $D_{z,0}$; C = 0.0715–0.3511 g/dm^3, DLS	730
		20	37			8.027				M_w; $M_w/M_n = 1.03$; $D_{z,0}$; DLS	731
			110			4.180				$M_w/M_n = 1.03$	
			200			3.175				$M_w/M_n = 1.03$	
			422			1.900				$M_w/M_n = 1.04$	
			775			1.360				$M_w/M_n = 1.01$	
			1230			1.083				$M_w/M_n = 1.02$	
			2420			0.720				$M_w/M_n = 1.04$	
	Toluene	20	1.2			40.1	−0.8			DLS	993
			3.1			28.5				DLS	
			4.0			25.8				DLS	
			10.7			15.7	1.7			DLS	
			30.2			8.82	5.0			DLS	
			48.5			6.78	10.0			DLS	
			97.0			4.39	21.0			DLS	
			424			1.96	70.9			DLS	
		25	25.6			2.56				M_w; $M_w/M_n = 1.1$; $D_{z,0}$; DLS	744
			38.6			1.95					
			64.9			1.32					
			238			0.80					
				$s_0 = 3.43 \times 10^{-15} M^{0.413}$						$M_w/M_n = 1.6–2.3$	745
				$k_s = 8.08 \times 10^{-2} M^{0.74}$							
linear	Toluene	25							0.916		1075
4-ring-shaped		25	30	2.554					0.922	SV	1075
			78.5	3.869						SV	1075
			123	4.808						SV	1075

			272	6.909			SV	1075
			3.1			28.5	DLS	1076
			4.0			25.8	DLS	1076
			10.7			15.1	DLS	1076
		21	500	7.30	270		M_w	748
			5000	15.20	840			
			13300	18.50	1000			
			16400	25.00	1490			
			4590			0.497	$M_w/M_n = 1.05$	731
			9500			0.310	$M_w/M_n = 1.03$	
			16100			0.205	$M_w/M_n = 1.16$	
		20	37–16100			$D_0 = 3.32 \times 10^{-4}\ M^{0.574}$	M_w; extrapolated for $C = 0$ and $M_w/M_n = 1.0$; DLS	728,729, 731
		20	110			4.25 ± 0.10	M_w; $D_{z,0}$; DLS	732
			200			3.25 ± 0.10	M_w; $D_{z,0}$; DLS	732
		25	111			3.954 ± 0.002 25.8 ± 0.003	M_w	733
microgel		24	410			2.07	Cross link ratio 1:20	1066
	Toluene							
NBS 706		25	260			$D_1 = 3.38$; $D_2 = 2.96$; $D_3 = 2.86$	M_w; $D_1 \le D_2 \le D_3$; $C < 0.45$ g/dm^3	734
PCC, 4b		25	110	4.00		4.25, 4.50	M_w; 23V; $C = 0$–120 g/dm^3	724
TS, F870			9500				M_w; $M_w/M_n = 0.03$; D given in Ref.	735
TS, F225			2420				M_w; $M_w/M_n = 0.04$; D given in Ref.	
TS, F510			5530				M_w; $M_w/M_n = 0.04$; D given in Ref.	
	Toluene	25	17.4			14.4	Taylor dispersion technique	1074
			110			4.42		
			900			1.34		
		25	111			4.35 30 ± 3	M_w; $M_w/M_n = 1.06$; $D_{z,0}$; $p = 1$ bar	726
						2.48 22 ± 6	M_w; $M_w/M_n = 1.06$; $D_{z,0}$; $p = 960$ bar	
						1.52 20 ± 6	M_w; $M_w/M_n = 1.06$; $D_{z,0}$; $p = 1940$ bar	
						0.92 20 ± 5	M_w; $M_w/M_n = 1.06$; $D_{z,0}$; $p = 2900$ bar	
						0.56 20 ± 6	M_w; $M_w/M_n = 1.06$; $D_{z,0}$; $p = 3870$ bar	
						0.35 23 ± 10	M_w; $M_w/M_n = 1.06$; $D_{z,0}$; $p = 4840$ bar	
PCC, 7a		25	51	2.98		7.20	M_w; $M_w/M_n = 1.06$; s_0; $D_1 = D_A$	736
Waters, 41995			98.2	3.94		5.31		
PCC, 4b			110.0	4.32		4.58		
NBS 705			183	5.12		3.48		

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
NBS 705	Toluene		191			3.49				DLS	1019
NBS 1479			939			1.44				DLS	1019
TS 550	Toluene		5850			0.62				DLS	1019
TS 2000			15900			0.31				DLS	1019
PCC, 3b			390	7.15		2.46					736
Waters, 25166 N			470	7.20		2.33					
Waters, 25166			411	7.32		2.24					
PCC, 5a			498	7.93		2.05					
Waters, 25167			867	9.92		1.52					
PCC, 14a			1800	13.70		1.00				M_w; $M_w/M_n = 1.2$; s_0; $D_1 = D_A$	736
Waters 61970	Toluene		2145	13.53		1.00			0.917	M_w	736
		25	51–2145		$k_s/[\eta] = 1.65 \pm 0.15$					M_w	736
		25	51–2145	$s_0 = 2.84 \times 10^{-15} M_{s,D}^{0.43}$							736
		RT	1000			1.3				M_w; D_z; $C = 0.05\%$; DLS	717
			13000			0.6				M_w; D_z; $C = 0.05\%$; DLS	717
linear		35	116	4.80						M_w; s_0	683
			216	6.22							
			275	7.06							
			675	9.97							
			860	11.1							
			1700	15.0							
branched	Toluene	30	980			1.88					994
			1730			1.39					994
			2390			1.28					994
			4780			0.895					994
			14300			0.595					994
branched, 4-armed stars		35	50.7	3.60						M_w; s_0	683
			93.5	4.59							
			154	5.81							
			351	8.33							
			521	9.75							
			1027	13.2							
			1390	14.5							
branched, 6-armed stars		35	110	5.41						M_w; s_0	683
			1090	14.6							
comb		35	475	9.91						M_w; s_0	683
			624	11.8							
			913	14.6							
			1630	20.4							
			3130	29.0							
			1050	12.9							
			1190	14.6							
			1530	16.4							

			2230	21.3				
			3620	27.6				
		21	500	7.30	270			748
			5000	15.20	840		M_w; D_z; $C = 0.05\%$; DLS	
			13300	18.50	1000			
			16400	25.00	1490			
	Toluene	25				0.919		1060
	Toluene/butanone (3/1)	20	528	9.9		2.11	$f_0 = 3.65 \times 10^{-5}$ cm	s_A; D_A; $f_0 = kT/(D_0\eta_1)$ 3
	(1/1)		553	12.9		2.45	$f_0 = 3.43 \times 10^{-5}$ cm	s_A; D_A; $f_0 = kT/(D_0\eta_1)$ 3
	(1/3)		554	15.6		2.82	$f_0 = 3.29 \times 10^{-5}$ cm	s_A; D_A; $f_0 = kT/(D_0\eta_1)$ 3
	m-Xylene	20						
	p-Xylene	25				0.9205	Extrapolated value	268
halomethylated, containing 5.66% P	Tetrachloroethane	20	90.7–507	$s_0 = 9.77 \times 10^{-15}\ M^{0.266}$			v_2 given in Ref.	532
latex	Water						M_{MW}	561
latex, DCC, 481 nm		20				0.081	See Ref.	737
latex		20				0.475	D_z; DLS	727
latex, 481 nm		RT				0.41	D_z; DLS	738
latex, 109 nm		RT				0.089	D_z; DLS; $C = 0.05$ g/dm^3	739
latex, 109 nm		20				0.415	D_z; DLS; $C = 0.05$ g/dm^3	739
latex		24.7				0.431	D_z; DLS	740
		40.2				0.637	D_z; DLS	723
		55.2				0.865		
		70.4				1.15		
latex		25				0.439	D_z; DLS; $C = 0.11\%$	684
latex		25				0.652 ± 0.007	D_z; DLS; $C = 0.004\%$	741
latex		25				0.654 ± 0.007	D_z; DLS; $C = 0.022\%$	741
latex	Water/NaCl (0.01 mol/dm^3)	25				0.687 ± 0.007	D_z; DLS; $C = 0.004\%$	741
						0.696 ± 0.008	D_z; DLS; $C = 0.023\%$	741
latex		20				0.856 ± 0.043	$D_{z,0}$; DLS	746
	PMMA/benzene	30						
	$C_{PMMA} = 0$ g/cm^3		8420			0.38		
	$C_{PMMA} = 0.91$ g/cm^3		8420			0.0594	$M_{PMMA} = 4.05 \times 10^6$ g/mol	1077
	$C_{PMMA} = 1.29$ g/cm^3		8420			0.0330		
	$C_{PMMA} = 2.72$ g/cm^3		8420			0.0055		
	$C_{PMMA} = 3.65$ g/cm^3		8420			0.0021		
	$C_{PMMA} = 0.91$ g/cm^3		8420			0.125	$M_{PMMA} = 1.93 \times 10^6$ g/mol	1077
	$C_{PMMA} = 1.28$ g/cm^3		8420			0.0837		
	$C_{PMMA} = 2.69$ g/cm^3		8420			0.0281		
	$C_{PMMA} = 3.67$ g/cm^3		8420			0.0148		
	$C_{PMMA} = 3.70$ g/cm^3		8420			0.0258	$M_{PMMA} = 1.09 \times 10^6$ g/mol	1077
	$C_{PMMA} = 3.67$ g/cm^3		8420			0.0364	$M_{PMMA} = 8.5 \times 10^5$ g/mol	
	$C_{PMMA} = 0$ g/cm^3		1260			1.10		
	$C_{PMMA} = 3.65$ g/cm^3		1260			0.0347	$M_{PMMA} = 4.05 \times 10^6$ g/mol	1077
	$C_{PMMA} = 3.65$ g/cm^3		1260			0.0727	$M_{PMMA} = 1.93 \times 10^6$ g/mol	

TABLE 5. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm³/g)	D_0 ($\times 10^{7}$) (cm²/s)	k_D (cm³/g)	f_0/f_{sp}	v_2 (cm³/g)	Remarks	Refs.
	$C_{PMMA} = 3.65$ g/cm³		1260			0.111				$M_{PMMA} = 1.09 \times 10^6$ g/mol	1077
	$C_{PMMA} = 3.70$ g/cm³		1260			0.137				$M_{PMMA} = 8.5 \times 105$ g/mol	
Poly(styrene-*co*-butadiene)	Methyl *n*-propyl ketone	21	125–985	$s_0 = 3.31 \times 10^{-16} M^{0.62}$						M_n	590
24 mass% St.		21 ($T = \theta$)	49–514	$s_0 = 8.3 \times 10^{-16} M^{0.5}$							
Poly(styrene)-*block*-(dimethylsiloxane)	THF	25	46.8			7.280	19			Tracer diff. coef.	1064
	7.5 vol% PDMS/THF	25	46.8			1.320	−36				
	15 vol% PDMS/THF	25	46.8			0.715	−8.4				
Poly(styrene)-*block*-(hydrogenated polybutadiene)-*block*-polystyrene											
	1,4-Dioxane/0–30 vol% *n*-heptane	25	4800				4.02			0 vol% *n*-Heptane	1018
			4440				6.63			10 vol% *n*-Heptane	1018
Poly(styrene)-*block*-(hydrogenated polybutadiene)-*block*-polystyrene											
	1,4-Dioxane/0–30 vol% *n*-heptane		4200				7.54			20 vol% *n*-Heptane	1018
			3770				11.95			30 vol% *n*-Heptane	1018
Poly(styrene-*co*-divinylbenzene)											
30/70	*n*-Octane	21	65–710	$s_0 = 1.59 \times 10^{-15} M^{0.50}$		$D_0 = 1.49 \times 10^{-4} M^{-0.50}$			1.05	$M_{s,D}$	322
Poly(styrene-*co*-isooctyl methacrylate)											
20/80	Theta-solvent (see Ref.)	25 ($T = \theta$)	59.6 53.5	25.87–28.42					0.761	s_w; A $C = 1.67$ g/dm³	427
Poly(styrene-*block*-methyl methacrylate)											
	Butyl acetate	RT	600			1.6				M_w; D_z; $C = 0.05\%$; DLS	717
Poly(α-methylstyrene)	Toluene	25				$D_{z,0} = 3.14 \times 10^{-4} M_w^{-0.556}$				$7000 \leq M_w \leq 1140000$ g/mol	998
	n-Butylchloride	25	53.8			11.4	−24.8			$[\eta] = 17.0$ cm³/g; $D_0 = 3.81 \times 10^{-4} M_w^{-0.531}$	997
			74.3			10.1	−22.3			$[\eta] = 20.5$ cm³/g; $D_0 = 3.81 \times 10^{-4} M_w^{-0.531}$	997
			171			6.38	−24.3			$[\eta] = 32.5$ cm³/g; $D_0 = 3.81 \times 10^{-4} M_w^{-0.531}$	997
			303			4.56	−23.2			$[\eta] = 46.4$ cm³/g; $D_0 = 3.81 \times 10^{-4} M_w^{-0.531}$	997
			666			3.04	−26.1			$[\eta] = 74.1$ cm³/g; $D_0 = 3.81 \times 10^{-4} M_w^{-0.531}$	997
			881			2.67	−25.4			$[\eta] = 86.2$ cm³/g; $D_0 = 3.81 \times 10^{-4} M_w^{-0.531}$	997
			3540			1.26	−21.9			$[\eta] = 198$ cm³/g; $D_0 = 3.81 \times 10^{-4} M_w^{-0.531}$	997
Poly(styrenesulfonic acid)	Water/KCl (0.2 mol/dm³)	20	670–1150	8.6–11.5						M_{MW}; single data given in Ref.	127
	Water/NaCl (0.005–0.2 mol/dm³)	25	350–2650	4.95–16.4						M_{MW}; single data given in Ref.	452
PCC	Water/NaCl (0.01 mol/dm³)		400			0.90				M_w; $M_w/M_n < 1.1$; $D_{z,0}$; 10V $C = 0.01$–17.5 g/dm³	742

PCC			650	0.75	M_w; M_w/M_n<1.1; $D_{z,0}$; 9V C=0.049–26.4 g/dm^3	742
PCC			1200	0.40	M_w; M_w/M_n<1.1; $D_{z,0}$; 12V C=0.01–27.0 g/dm^3	742
PCC	Water/NaCl (0.01 mol/dm^3)		650	0.727 ± 0.007	M_w; D_z; C=0.05 g/dm^3; 9V; DLS	747
PCC	Water/NaCl (0.025 mol/dm^3)		650	0.864 ± 0.09	M_w; D_z; C=0.075 g/dm^3; 14V; DLS	
PCC	Water/NaCl (0.05 mol/dm^3)		650	0.92 ± 0.01	M_w; D_z; C=0.28 g/dm^3; 9V; DLS	
PCC	Water/NaCl (0.10 mol/dm^3)		650	1.01 ± 0.01	M_w; D_z; C=0.05 g/dm^3; 11V; DLS	
	Water/NaCl (0.1 mol/dm^3)	12	340	1.05	M_w; D_z; C=13.1 g/dm^3; DLS	756
		15		1.20	M_w; D_z; C=13.1 g/dm^3; DLS	
		20		1.65	M_w; D_z; C=13.1 g/dm^3; DLS	
		25		2.7	M_w; D_z; C=13.1 g/dm^3; DLS	
Sulfonated Poly(styrene) Ionomers						
	DMF/1 × 10^{-4} M salt	25	400	1.63 828	Ion content: 2.6%	1089
	DMF/1 × 10^{-3} M salt			1.58 237		
	DMF/1 × 10^{-2} M salt			1.68 21		
	DMF/1 × 10^{-1} M salt			1.77 8		
Poly(sodium-styrene-sulfonate-*co*-2-vinylnaphthalene)				$D_{z,0} = (1.5 \pm 0.8) \times 10^{-3}\ M_w^{-(0.75 \pm 0.06)}$	1073	
Poly(styrene)	PMPS	50	2	0.415 ± 0.041 105	PC-spectroscopy; PMPS: M_w = 2600 g/mol; M_w/M_n = 1.5	1079
			10.2	0.321 ± 0.028 46		
			12.2	0.289 ± 0.026 94		
			17.5	0.265 ± 0.032 122		
		65	2	0.728 ± 0.041 121		
			10.2	0.554 ± 0.033 49		
			12.2	0.448 ± 0.059 80		
			17.5	0.419 ± 0.045 121		

[a] See Ref.
[b] See Ref. for def. of s.

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TABLE 6. (O,C)-HETEROCHAIN POLYMERS (POLY(ETHERS), POLY(ESTERS), POLY(CARBONATES))

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Poly(carbonate)											
PMC	1,4-Dioxane	25	41.8						0.804	$M_w/M_n = 1.10$	1045
PMC	Methylene chloride	20	41.8						0.8999	$M_w/M_n = 1.10$	
PMC	Tetrahydrofuran	20	41.8						0.7738	$M_w/M_n = 1.10$	
PPhC	Benzene	25	47.0						0.762	$M_w/M_n = 1.09$	
Poly(thiocarbonate)											
PHTC	Benzene	25	19.1						0.753	$M_w/M_n = 1.23$	
PHTC	1,4-Dioxane		19.1						0.795	$M_w/M_n = 1.23$	
PMTC-1			9.4						0.790	$M_w/M_n = 1.34$	
PMTC-2			63.0						0.808	$M_w/M_n = 1.30$	
PetTC	Benzene		14.0						0.790	$M_w/M_n = 1.40$	
PiBuTC			40.0						0.834	$M_w/M_n = 1.24$	
PPhTC			93.0						0.780	$M_w/M_n = 2.07$	
P3ClPhTC-1			5.8						0.719	PAAR DMA 55	1044
P3ClPhTC-2			34.0						0.718		
P4ClPhTC-1			5.0						0.738		
		35	5.0						0.724		
		50	5.0						0.729		
P4ClPhTC-2		25	31.0						0.732		
P34ClPhTC-1			4.5						0.700		
	1,4-Dioxane		4.5						0.704		
	Chlorobenzene		4.5						0.705		
P34ClPhTC-2	Benzene		28.0						0.694		
Poly(ethylene terephthalate) See Poly(oxyethyleneoxyterephthaloyl)											
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)											
	Methylene chloride	20							0.8099		45
	Tetrahydrofuran	20	8.5–266	$s_0 = 1.33 \times 10^{-14}\ M_{s,D}^{0.362}$		$1/D_0 = 1.569 \times 10^{-3}\ M_{s,D}^{0.638}$				$M_{t,w}$; s_t	45
Poly(oxycarbonyl-propylene)	Chloroform	20	780	-9.95					0.812	M_w from Scheraga–Mandelkern-Eq. (Ref. 536)	535,537
			370	-8.16							
			156	-4.96							
			83.5	-3.50							
			21.1	-2.00							
Poly(oxyethylene)	Acetone	25	4.3			37.8				M_w; CLS	184
	Formamide	25	4.3			5.76				M_w; CLS	184
	Methanol	25	4.3			23.5				M_w; CLS	184
			23.8			10.7					
			19.2	3.18		10.7			0.785	$M_{t,D}$; s_t	150
			40			8.23				$T_\theta = 16.1$°C; $[\eta] = 79.5 \times 10^{-3}\ M^{0.58}$ for $T = 25$°C	1003
			74			5.92				$T_\theta = 17.2$°C	
			145			4.05				$T_\theta = 16.7$°C	
			594			1.63				$T_\theta = 17.0$°C	
			1200			1.10				$T_\theta = 16.5$°C	
	Water	20	0.29			37.0				D_z	177
			0.625			29.2					

			1.25		24.0			
			3.3		13.3			
			5.8		11.6			
			8.8		10.3			
			10.6		7.2			
	Water	20	1.426		23.6			180
			1.470		22.6			
			1.778		19.7			
			1.822		20.1			
		22.8	470			0.834	M_v	148
		23	0.24			0.846		
		25	4.3		11.5			184
		25	177		1.33		$M_w/M_n = 1.04$	1002
			300		1.08		$M_w/M_n = 1.05$	
			679		0.79		$M_w/M_n = 1.10$	
			1190		0.63		$M_w/M_n = 1.12$	
			20			0.9058	For $p = 1$ MPa	1001
						0.8689	For $p = 100$ Mpa	
						0.8420	For $p = 200$ Mpa	
			12	0.59	7.35			184
			23.8	0.93	4.85			
			37.3	1.09	3.95		M_w; CLS	150
TS, SE 15		25	17.7		1.33		M_w; $M_w/M_n = 1.04$; D_z; $C = 1.003$ g/dm^3; DLS	744
TS, SE 30			30.0		1.08		M_w; $M_w/M_n = 1.05$; D_z; $C = 1.003$ g/dm^3; DLS	
TS, SE 70			67.9		0.79		M_w; $M_w/M_n = 1.10$; D_z; $C = 1.003$ g/dm^3; DLS	
TS, SE 150			119		0.63		M_w; $M_w/M_n = 1.12$; D_z; $C = 1.003$ g/dm^3; DLS	
			17.7–119		$D_z = K_D M^{-0.39}$		M_w; $M_w/M_n = 1.12$; $C = 1.003$ g/dm^3; DLS	744
		25 ± 0.02	1130	3.73	1.58	26.8	M_w; $D_{0,A}$; $M_{s,D}$; M_w given in Ref.; $f_0 = RT/D_{0,A}$	749
			1470	4.16	1.33	29.1		
			1490	4.18	1.36	28.4		
			1900	4.67	1.15	30.9		
			2000	4.78	1.12	31.2		
			2610	5.39	0.93	34.4		
			2630	5.42	0.93	34.2		
			2670	5.34	0.94	33.8		
			1130–2670	$s_0 = 6.40 \times 10^{-16} M_s^{0.456}$	$D_0 = 1.86 \times 10^{-4} M_D^{-0.67}$		M_w	749
TS, SE 8, SE 30, SE 150		25	5.7–1200				M_w; $M_w/M_n = 1.12$; D given in Ref.	750
		25	320	2.15	1.0		$M_{s,D}$; $M_{s,D}/M_n = 1.1$	751
Poly(ethylene glycol)	D_2O	5			$D_0 = k M^{-0.597 \pm 0.020}$			1088
		25			$D_0 = k M^{-0.534 \pm 0.021}$			
	D_2O/diethylene glycol 1/1	25			$D_0 = k M^{-0.542 \pm 0.058}$			
Poly(oxy-2,6-dimethyl-1,4-phenylene)	Toluene	40–54			0.0206–0.0251		$C = 30$ mass%	765
		55–65			0.0463–0.0509		$C = 40$ mass%	765
		69–78			0.0556–0.0624		$C = 50$ mass%	765

TABLE 6. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Poly(oxyethyleneoxyterephthaloyl)(polyethylene terephthalate)											
	m-cresol	25	111	0.103		0.120				$M_{s,D}$	526
			77.3	0.153		0.155				A_2 given in Ref.	
			61.0	0.138		0.186					
			41.3	0.132		0.263					
			36.8	0.125	180	0.280					
			23.0			0.365					
			60.2			0.205					
unfractionated			87.3			0.130					
				$s_0 = K_s M^{0.26}$		$D_0 = K_D M^{-0.74}$					
		50	111	0.435					0.678	$M_{s,D}$	526
			77.3	0.405							
			61.0	0.391							
			41.3	0.372	140	0.992	80				
			36.8	0.338							
			23.0	0.439							
unfractionated			60.2	0.408		0.381					
			87.3	$s_0 = K_s M^{0.22}$							
Poly(oxypropylene)	Acetone	20	0.074–3.375			$D_0 = K_D M^{-0.52}$		$f_0 = 4.4 \times 10^{-7} M^{0.52}$		D_A; $f_0 = kT/(D_0\eta_1)$	465
	Benzene	20	0.134–3.375			$D_0 = K_D M^{-0.55}$		$f_0 = 3.2 \times 10^{-8} M^{0.55}$		D_A; M from condensation and boiling temperature elevation	465
TS, SE 5	Water	25	40			3.73				M_w; $M_w/M_n = 1.03$; D_z; $C = 14$ g/dm^3; DLS	765
TS, SE 8			73			2.09				M_w; $M_w/M_n = 1.02$; D_z; $C = 4.8$ g/dm^3; DLS	
TS, SE 15			148			1.66				M_w; $M_w/M_n = 1.04$; D_z; $C = 3.2$ g/dm^3; DLS	
TS, SE 30			278			1.72				M_w; $M_w/M_n = 1.05$; D_z; $C = 10.6$ g/dm^3; DLS	
TS, SE 70			661			1.07				M_w; $M_w/M_n = 1.10$; D_z; $C = 3.6$ g/dm^3; DLS	
TS, SE 15		15	148			1.73				M_w; $M_w/M_n = 1.04$; D_z; $C = 10.5$ g/dm^3; DLS	
		34				2.47					
		43				2.88					
TS, SE 15		25	148			2.63				M_w; $M_w/M_n = 1.0$; D_z; $C = 25.6$ g/dm^3; DLS	
		30.3				2.72					
		35.4				3.08					
		40.5				3.25					
		45.5				3.49					

TABLE 7. (N,C)-AND (O,N,C)-HETEROCHAIN POLYMERS (POLY(AMIDES), POLY(UREAS), POLY(URETHANES))

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm³/g)	D_0 ($\times 10^{7}$) (cm²/s)	k_D (cm³/g)	f_0/f_{sp}	v_2 (cm³/g)	Remarks	Refs.
Poly(ethylenimine)											
linear	100 mM NaCl	25	41.5			2.48	2.3			DLS	1090
	25 mM NaCl		41.5			1.92	9.2				
	10 mM NaCl		41.5			1.39	25				
	25 mM NaCl		88.2			1.12	15				
Poly(*n*-butyliminocarbonyl) (Poly(butylisocyanate))											
	Tetrahydrofuran	20	18–211	$s_0 = 3.05 \times 10^{-14}\ M^{0.16}$		$D_0 = 1.69 \times 10^{-4}\ M^{-0.85}$				$M_{s,D}$	325
Poly(hexyliminocarbonyl) (Poly(hexylisocyanate))											
	Hexane	25	52.7			8.11	27.3			M_w; $D_{0,z}$;	752
			202			3.17	−263			$C = 0.2$–6.0 g/kg	
			411			1.85	−242				
Poly(iminohexamethyleneiminoadipoyl) (Nylon 66)											
	Formic acid (98–100%)	22.4	16						0.890	M_v	148
Poly(*iminoisophthaloylimino*-1,3-phenylene)											
	Dimethylformamide	25	30.2–156	$s_0 = 1.9 \times 10^{-15}\ M^{0.44}$					0.678		589
	LiCl (2.5 g/dm³ + 96% H_2SO_4) in dimethylformamide		20.7–142	$s_0 = 2.8 \times 10^{-15}\ M^{0.39}$					0.682		589
Poly(iminoisophthaloylimino-1,3-phenylene-sulfonyl-1,3-phenylene) (Poly(3,3′-diphenylsulfonisophthalamide))											
		25	79	2.2						M_w	764
			82	2.5							
			110	2.6							
			130	2.8							
			175	3.3							
			240	3.5							
			360	4.4							
		25	79–360	$s_0 = 2.56 \times 10^{-15}\ M^{0.40}$						$M_w/M_n \approx 2.6$	764
Poly(iminoterephthaloylimino-1,4-phenylene-sulfonyl-1,3-phenylene) (Poly(4,4-diphenylsulfonterephthalamide))											
		20	25.0	1.50						M_w	763,764
			33.4	1.71							
			50.0	2.00							
			68.5	2.40							
			72.0	2.57							
			90.0	2.70							
			98.0	2.77							
			109.0	2.86						$M_w/M_n \approx 1.5$	763
			25–109	$s_0 = 2.34 \times 10^{-15}\ M^{0.42}$						From dn/dc (see Ref.)	532
Poly(iminoterephthaloylimino-1,4-phenyleneanthron-9-ylidene-1,4-phenylene)											
	Dimethylformamide	25		$s_0 = 3.0 \times 10^{-15}\ M^{0.40}$					0.747		579
Poly(l-lysine) branched by benzyl aspartate and benzyl glutamate											
	Dimethylformamide	20								s_0, k_s, D_A as function of composition and M given	448
Poly(4,4′-oxydiphenylene) pyromellitamic acid)											
	Dimethylformamide/ LiBr (0.1 mol/dm³)	25		$s_0 = 3.9 \times 10^{-15}\ M^{0.39}$					0.578		753

TABLE 7. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Poly(4,4′-phenylene) pyromellitamic acid)											
	Dimethylformamide/ LiBr (0.1 mol/dm^3)	25								s_0 given in Ref.	753
Poly(oxy-1,4-phenyleneiminocarbonyl-2,5-dicarboxy-1,4-phenylenecarbonylimino-1,4-phenylene)											
	Dimethylacetamide/	15	9.9–266	0.99–3.22						17V; CLS	455
	LiBr (0.1 mol/dm^3)	15	18.4–45.4	$s_0 = 3.72 \times 10^{-15} M_w^{0.36}$						M_n	403
Poly(iminoisophthaloylimino-1,3-phenylene)											
	Dimethylformamide/ LiCl (0.25%)	25		(3.30–3.70)	0.18					$C = 0.55–1.10$ g/dm^3; See Ref.	754
	Dimethylformamide/	25	4.3	0.33		6.19				$M_{s,D}$	755
	LiCl (3%)		5.9	0.33		4.47					755
			7.7	0.39	49	4.03					
			10.6	0.43		3.24					
			11.2	0.37		2.65					
			15.5	0.57		2.95					
			20	0.66	107	2.65					
			24	0.53		1.78					
			25	0.76		2.47					
			27	0.76	160	2.00					
			37	0.86		1.85					
			38	0.77		1.62					
			41	0.86	210	1.67					
			43	0.73	90	1.37					
			47	0.78		1.33					
			48	0.81	190	1.35					
			50	0.82		1.31					
			55	0.88		1.28					
			57	0.89		1.26					
			65	0.92		1.13					
			69	0.90		1.04					
			70	0.95		1.08					
			72	0.95		1.05					
			74	0.96	283	1.04					
			75	0.98		1.05					
			80	0.98	270	0.98					
			82	1.08		1.06					
			96	1.14	250	0.95					
			112	1.15	315	0.82					
unfractionated			112	1.18		0.84					
			113	1.24		0.88					
			113	1.24	290	0.88					
			118	1.19		0.81					
			124	1.18		0.76					
			133	1.25		0.75					
			143	1.23		0.69					
			147	1.36	465	0.74					
			154	1.33		0.69					

			163	1.47		0.72			
			184	1.54	575	0.67			
			209	1.67		0.64			
			233	1.95	667	0.67			
			237	1.81		0.61			
			296	1.82	875	0.49			
			305	2.06	870	0.54	227		
		25	4.3–305	$s_0 = 0.88 \times 10^{-15}\ M^{0.43}$		$D_0 = 0.71 \times 10^{-8}\ M^{-0.57}$		0.71 ± 0.02	$M_{s,D}$	755
	Dimethylformamide/ LiCl (0.005 g/dm^3)	25	112			1.91			$M_{s,D}$; $C = 0.3–0.4$ g/dm^3	755
	Dimethylformamide/ LiCl (0.010 g/dm^3)	25	237			0.88				
			124			1.20				
	Dimethylformamide/ LiCl (0.030 g/dm^3)	25	305			0.50			$M_{s,D}$;	755
			237			0.68			$C = 0.3–0.4$ g/dm^3	
			124			1.08				
			112			1.28				
Poly(iminoisophthaloylimino-1,3-phenylene)										
	Dimethylformamide/ LiCl (0.040 g/dm^3)	25	305			0.40			$M_{s,D}$;	755
			124			0.71			$C = 0.3–0.4$ g/dm^3	
	Dimethylformamide/ LiCl (0.050 g/dm^3)	25	305			0.30			$M_{s,D}$;	755
			112			0.97			$C = 0.3–0.4$ g/dm^3	
Poly(naphthoylene imidobenzimidazole)										
	sulfuric acid 96%	30				$D_0 = 3.9 \times 10^{-6}\ M^{-0.55}$			$M \geq 50 \times 10^3$ g/mol	1043
						$D_0 = 3.4 \times 10^{-5}\ M^{-0.74}$			$M \leq 80 \times 10^3$ g/mol	
Poly(urethane) from Poly(oxycaproyl) (A) diphenylmethanediisocyanate, ethylenediamine										
linear										
A:$M_n = 1.3 \times 10^3$	Dimethylformamide	25	26.2–128	$s_0 = 1.36 \times 10^{-15}\ M^{0.44}$				0.848	M_{MW}	607,608
A:$M_n = 2.8 \times 10^3$			266–306	$s_0 = 9.86 \times 10^{-16}\ M^{0.46}$				0.8745	7V; k_s given in Ref.	
A:$M_n = 1.8 \times 10^3$			146.4–17.9	$s_0 = 1.18 \times 10^{-15}\ M^{0.45}$				0.8581	7V; k_s given in Ref.	
branched			200	2.91	292				M_{MW}	
			96.7	2.10	176				M_{MW}	
branched			137	2.50	243				M_{MW}	
			71.2	1.85	159					
Poly(ureylenehepta-methylene)	Dimethylformamide	55						0.955		183
		90						0.949		
	Formic acid (97%)	25						0.915		183
	Sulfonic acid (96%)	25	23.5					0.974	M_w; CLS	183
		60						0.965		
Poly(ureylene-2,4,4-trimethylpentamethylene)										
	Dimethylformamide	55						0.955		183
		90						0.949		
	Formic acid (97%)	25						0.915		

TABLE 8. OTHER SYNTHETIC POLYMERS

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Poly(bibenzoate):											
Poly(heptamethylene p,p'-bibenzoate)											
	Cl_3CH	25							0.814		1086
	Cl_2CH_2								0.820		
	$Cl_4C_2H_2$								0.835		
Poly(oxybis(ethylene oxide) p,p'-bibenzoate)											
	Cl_3CH	25							0.733		1086
	$Cl_4C_2H_2$								0.768		
Poly(oxytris(ethylene oxide) p,p'-bibenzoate)											
	Cl_3CH	25							0.745		1086
	$Cl_4C_2H_2$								0.761		
Poly(oxytetraquis(ethylene oxide) p,p'-bibenzoate)											
	Cl_3CH	25							0.759		1086
	$Cl_4C_2H_2$								0.783		
Phenol novolac	Acetone	RT	0.770			70.2				M_n; D_w; M_w; D_A given in Ref.	762
			1.300			63.7					
			1.930			53.5					
			2.280			43.6					
			4.030			34.2					
Phenol resol	Acetone	RT	0.370			106.8				M_n; D_w; M_w; D_A given in Ref.	762
			0.740			88.3					
			1.500			71.1					
			2.800			46.0					
			7.530			27.1					
Poly(terephthalic acid-4-aminobenzohydrazide)											
	Dimethylsulfoxide	23	3.78			4.02				CLS, DLS	1017
			5.67			2.97					
			7.06			2.28					
			9.16			2.25					
			9.79			1.75					
			16.6			1.32	−16.2				
			28.5			1.00	−31.0				
			47.1			0.806	−101				
			83.9			0.665	−192				
			155			0.393	−349				
			260			0.255	−588				
Poly(acenaphthylene)	Benzene	25							0.77		480
Poly(*o*-methyl-phenylene)	Toluene	20	5.6	1.10		18.5		1.68	0.841	$M_{t,w}$; D_w; v_2 independ. of M	140
			9.3	1.36		12.5		1.94		D_w	
			12.6	1.55		11.1		2.04		D_A	
			15.8	1.73		11.0		2.06		$M_{t,w}$; D_w	
			16.8	2.20		12.1				D_w	
			23.4	2.35		9.41		2.04		$M_{t,w}$; D_A	
			36.7	2.87		8.38		2.26		D_w	
			77.2	4.40		5.34		2.42		$M_{t,w}$; D_w	
			270	8.25		3.52		2.97		D_w	

Poly(L-*p*-tolyl-2,5-dioxopyrrolidine-3,4-diyl) (Poly-*p*-tolyl-maleimide)										
	Dimethylformamide	21	40–560	$s_0 = 2.76 \times 10^{-14}\ M^{0.42}$	$D_0 = 3.47 \times 10^{-3}\ M^{-0.58}$			0.74	$M_{s,D}$	563
Poly(*N*-phenylmaleimide-*co*-styrene)										
	Tetrahydrofuran	25						0.9064	$w_{NPI} = 0$	1047
								0.8040	$w_{NPI} = 0.430$	
								0.7768	$w_{NPI} = 0.528$	
								0.7584	$w_{NPI} = 0.569$	
								0.7400	$w_{NPI} = 0.702$	
								0.6988	$w_{NPI} = 1.000$	

TABLE 9. INORGANIC POLYMERS

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Poly(bis(*m*-chlorophenoxy)phosphazene)											
	Chloroform	25 ± 0.01	3450 ± 240			1.40 ± 0.01	116			$D_{0,z}$	757
						$D_0 = (2.08 \pm 0.06) \times 10^{-4}\ M^{-0.57}$					757
Poly(metaphosphoric acid)											
magnesium complex	Water/NaCl (0.4 mol/dm^3)	38	1000				1.39	2.8	0.370		139
potassium salt	Water/NaCl (0.1 mol/dm^3)	14	480				1.05	5.2	0.311	$M_{r,w}$; s_r; D_w	139
	Water/NaCl (0.4 mol/dm^3)	32	1100				1.08	3.8	0.316		
	Water/NaCl (0.1 mol/dm^3)/ethanol (5%)	20	570				1.25	4.1	0.315		
	Water/NaCl (0.1 mol/dm^3)/ethanol (8%)	25	650				1.36	3.6	0.318		
Poly(3-methyl-1-butene-1-silsesquioxane), cyclolinear											
	Butylacetate	24	92–600	$s_0 = 1.1 \times 10^{-14}\ M^{0.31}$		$D_0 = 1.1 \times 10^{-3}\ M^{-0.69}$			0.863 ± 0.003	$M_{s,D}$; s (13V); D (13V)	583
Poly(oxydimethylsilylene) (polydimethyl siloxane)											
	Bromocyclohexane	28	101	−2.40		1.66				$M_{s,D}$	326
			165	−2.70		1.15					
			550	−4.60		0.59					
									1.025		326
linear		28 ($T = \theta$)	0.312			28.2				M_n; $M_w/M_n = 1.0$; D; $C = 0.69$ mass%	758
			0.386			23.5				M_n; $M_w/M_n = 1.0$; D; $C = 0.77$ mass%	
			0.700			17.4				M_n; $M_w/M_n = 1.01$; D; $C = 0.80$ mass%	
			1.130			13.3				M_n; $M_w/M_n = 1.04$; D; $C = 0.76$ mass%	
			2.130			9.69				M_n; $M_w/M_n = 1.11$; D; $C = 0.72$ mass%	
			5.280			5.96				M_n; $M_w/M_n = 1.03$; D; $C = 0.98$ mass%	

TABLE 9. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			10.240			4.32				M_n; $M_w/M_n = 1.20$; D; $C = 0.71$ mass%	
			14.980			3.27				M_n; $M_w/M_n = 1.14$; D; $C = 0.80$ mass%	
		28 ($T = \theta$)	4.030			9.61				M_n; $M_w/M_n = 1.08$; D; $C = 0.68$ mass%	758
			10.240			5.75				M_n; $M_w/M_n = 1.20$; D; $C = 0.71$ mass%	
			14.980			4.47				M_n; $M_w/M_n = 1.12$; D; $C = 0.63$ mass%	
						$D_0 = K_D M^{-0.54\pm0.02}$					758
cyclic		15	0.222			33.7				M_n; $M_w/M_n = 1.0$; D; $C = 0.74$ mass%	
			0.296			27.4				M_n; $M_w/M_n = 1.0$; D; $C = 0.78$ mass%	
			0.520			21.7				M_n; $M_w/M_n = 1.02$; D; $C = 0.76$ mass%	
			1.070			15.0				M_n; $M_w/M_n = 1.04$; D; $C = 0.79$ mass%	
			2.080			10.9				M_n; $M_w/M_n = 1.01$; D; $C = 0.76$ mass%	
			3.710			8.30				M_n; $M_w/M_n = 1.04$; D; $C = 0.73$ mass%	
			8.480			5.56				M_n; $M_w/M_n = 1.14$; D; $C = 0.73$ mass%	
			17.380			3.53				M_n; $M_w/M_n = 1.08$; D; $C = 0.73$ mass%	
		28	3.710			11.5				M_n; $M_w/M_n = 1.04$; D; $C = 0.77$ mass%	758
			8.480			7.67				M_n; $M_w/M_n = 1.06$; D; $C = 0.72$ mass%	
			17.380			4.81				M_n; $M_w/M_n = 1.08$; D; $C = 0.73$ mass%	
	Butanone	10	16						1.018		148
		20							1.0282	M_v	
		35							1.0468		
	Toluene	20	301	4.09		3.00				$M_{s,D}$	326
			920	6.07		1.60					
		22.4	9.4						1.024	M_v	148
linear		25	0.3122			124.00				M_n; $M_w/M_n = 1.0$; D; $C = 1.405$ mass%	759
			0.3863			107.10				M_n; $M_w/M_n = 1.0$; D; $C = 1.375$ mass%	
			0.6086			80.00				M_n; $M_w/M_n = 1.03$; D; $C = 1.388$ mass%	
			1.053			58.92				M_n; $M_w/M_n = 1.02$; D; $C = 1.333$ mass%	

			1.442		49.42			M_n; $M_w/M_n = 1.02$; D; $C = 1.371$ mass%	
Poly(oxydimethylsilylene) (polydimethyl siloxane)									
linear	Toluene		3.573		33.52			M_n; $M_w/M_n = 1.04$; D; $C = 1.243$ mass%	
			6.018		26.65			M_n; $M_w/M_n = 1.05$; D; $C = 1.337$ mass%	
			14.460		15.03			M_n; $M_w/M_n = 1.07$; D; $C = 1.249$ mass%	
Poly(oxydimethylsilylene) (polydimethyl siloxane)									
			23.570		10.88			M_n; $M_w/M_n = 1.08$; D; $C = 1.750$ mass%	
			0.3–23.6		$D_0 = K_D\, M^{-0.436 \pm 0.002}$			M_n; $M_w/M_n \approx 1.08$	759
cyclic	Toluene		0.2223		145.20			M_n; $M_w/M_n = 1.0$; D; $C = 1.483$ mass%	759
			0.2964		123.30			M_n; $M_w/M_n = 1.0$; D; $C = 1.294$ mass%	
			0.7410		78.00			M_n; $M_w/M_n = 1.07$; D; $C = 1.180$ mass%	
			1.189		64.73			M_n; $M_w/M_n = 1.08$; D; $C = 1.512$ mass%	
			1.534		57.13			M_n; $M_w/M_n = 1.05$; D; $C = 1.180$ mass%	
			3.171		40.95			M_n; $M_w/M_n = 1.06$; D; $C = 1.504$ mass%	
			4.172		37.07			M_n; $M_w/M_n = 1.04$; D; $C = 1.351$ mass%	
			9.077		24.22			M_n; $M_w/M_n = 1.06$; D; $C = 1.237$ mass%	
			17.673		16.26			M_n; $M_w/M_n = 1.07$; D; $C = 1.432$ mass%	
			0.2–17.7		$D \sim M^{-0.43 \pm 0.002}$			M_n; $M_w/M_n \approx 1.08$	759
linear		25	23.30		11.15	8.18		M_w; $D_{0,z}$	760
			12.40		17.35	3.36			
			0.627		82.05	−2.08			
cyclic			23.20		13.05	2.33		M_w; $D_{0,z}$	760
			11.50		21.75	−2.00			
			0.296		125.4	−1.503			
Poly(oxydiphenylsilylene) (Poly(phenyl siloxane))									
	Benzene	21	4–3350	$s_0 = 9.16 \times 10^{-15}\, M^{0.37}$	$D_0 = 6.16 \times 10^{-4}\, M^{-0.63}$		0.807	$M_{s,D}$; D_A	556
Poly(oxymethylphenylsilylene)									
	Toluene	22					0.006	$[\eta] = 132\,cm^3/g$	148

TABLE 10. POLY(SACCHARIDES)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Amylose	1,4-dioxane	20	2.8	0.303		11.4	−3			M_w; s_0; $D_{z,0}$	768
			5.8	0.642		10.8	0				
			9.2	0.805		8.2	1				
			17.3	1.27		6.9	−4				
			32.9	1.44		4.1	10				
			49.0	1.86	62	3.56	5				
			83.5	1.92	33	2.15	17			$M_{s,D}$; s_0; $D_{z,0}$	768
			121	2.29	41	1.77	22				
			356	3.13	77	0.825	36				
			860	4.16	141	0.452	52				
			1130	4.16	160	0.344	142				
			1280	4.54	183	0.332	166				
			2050	5.40	280	0.247	346				
			2730	5.85	284	0.20	447				
				$s_0 = 6.6 \times 10^{-15}\ M_{s,D}^{0.30}$		$D_0 = 6.6 \times 10^{-4}\ M_{s,D}^{-0.70}$					768
Carboxymethyl cellulose	Water	RT				1.2				$C = 0.1$ mass%	766
						2.8				$C = 0.5$ mass%	
						3.7				$C = 1.0$ mass%	
						12.0				$C = 2.0$ mass%	
sodium salt	Water/NaOH(0.01 mol/dm^3; pH = 12.0)	20.6	700						0.505	M_v	148
	Water/NaCl(0.001 mol/dm^3)	25	44.7	2.40							314
			46	2.30							
			90.6	2.50							
			94.2	2.58							
			163	2.65							
			194	2.62							
			280	2.85							
			346	2.96							
	Water/NaCl(0.005 mol/dm^3)		70	2.54						M_n	277,278
			104	3.00							
			175	3.14							
			290	3.97							
	Water/NaCl(0.01 mol/dm^3)		70	2.76						M_n	277,278
			104	3.18							
			175	4.25							
			290	4.28							
			44.7	2.70							314
			163	3.11							
			346	4.53							
	Water/NaCl					51.5					767
Cellulose											
absorbent cotton	Cuprammonium	20		4.28		2.64				s_l; $C = 1.0$ g/dm^3	63
aged, alkali cell.	hydroxide		150	3.8	110	0.84	30	5.3	0.508	$M_{l,w}$; s_l; D_w	14
overaged			44	1.9	20	1.54	45	4.6	0.508		14
cotton (Georgia)			1750	10.4	900	0.2	350	9.8	0.508		14
Holo cellulose (spruce)			340	7.2	390	0.72	145	4.7	0.508		14

linters, unbleached			1500	10.3	600	0.23	320	9.0	0.508		14
linters, bleached			490	7.2	520	0.43	130	6.1	0.508		14
			1200	9.2	690	0.26	250	8.6	0.508		14
linters, cotton				3.94		2.83					63
filter paper											
Cellulose											
flax fiber			5900	17.5	1700	0.1	1200	13.1	0.508	s_t; $C = 1$ g/dm^3	14
nettle fiber			1900	14.0	1000	0.25	320	7.7	0.508	$M_{t,w}$; s_t; D_w	14
ramie fiber			2000	10.8	860	0.18	250	10.4	0.508		14
sulfate cell.			400	6.5	390	0.54	105	5.9	0.508	$M_{t,w}$; s_t; D_w	14
sulfate cell.,			570	6.3		0.37	190	7.7	0.508		14
acetylated											
sulfate cell., precipit.			320	6.3		0.66	165	5.2	0.508		14
sulfite cell.			500	6.3	380	0.45	115				14
sulfite cell., bleached				4.28		2.72				s_t; constituent A	63
			2.73			3.67				s_t; constituent B	63
viscose staple fiber (sulfate cellulose)											
–, (spruce)			75	2.6	50	1.04	0	4.8	0.508	$M_{t,w}$; s_t; D_w	14
–, (pine)			230	4.1	170	0.54	75	6.4	0.508		14
wood cellulose (Cross				4.28		2.70				s_t; constituent A	63
and Bevan aspen)				2.73		3.53				s_t; constituent B	63
–Cellulose, cotton			53	4.6		5.4			0.642	s_t	62
linters											
–Cellulose, holo			680	6.5	320	0.32	295	8.4	0.508	$M_{t,w}$; s_t; D_w	14
cellulose											
β Cellulose			27	1.6	50	1.97	50	4.0	0.508		14
Cellulose	Cupriethylenediamine	25	175	5.5		1.2				$M_{s,w}$; D_w	234
			9.5	8.3		0.95					
									0.65		97
	Cadmium	12	33.6	1.25						M_w; SE	425
	ethylenediamine		24.5	1.13							
			18.8	1.04							
			10.1	0.74							
linters hydrol.		25							0.50		97
			43	1.80	153	1.77				$M_{t,w}$; s_t; D_w	97
			155	2.75	370	0.75					97
			210	3.17	465	0.64					97
			275	3.80	710	0.58					97
			435	4.53	1000	0.44					97
			745	5.49	1410	0.31					97
sulfite pulp			115	2.87	367	1.06					97
	Iron–tartaric acid										
	Complex Na salt	20	111	1.75		0.89			0.654	$M_{s,A}$; s_t; D_A	235
			201.4	2.17		0.60					
			364	2.98		0.46					
			539	3.35		0.35					
Cellulose acetate	Acetone		100			5.4	0			D_w	13
		20	10.35	4.07	600	20.7				$M_{t,w}$; s_t; D_A	39
			51	7.5	5600	7.7					
			143	10.9	14300	4.0					

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TABLE 10. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			194	12.5	19100	3.4					
			10.4–194	$s_0 = 1.22 \times 10^{-14}\ M^{0.38}$					0.68	$M_{t,w}$; $M_w/M_n = 1.27$	39,769
from cotton linters	Acetone	25	150	16.7					0.687	M_n; OS; M_{MW} given in Ref.	554,557
			136	16.0					0.687		
			66	10.5					0.692		
			145	14.7					0.698		
			110	12.8					0.697		
			26	7.0					0.690		
			24.2–249	$s_0 = 1.2 \times 10^{-14}\ M^{0.39}$						$M_w/M_n = 1.15$–1.99	557
Cellulose acetate											
from wood cellulose	Acetone			10.0					0.700		
			40	8.4					0.700		557
		20	21.2–122	$s_0 = 2.21 \times 10^{-14}\ M^{0.33}$						$M_{s,D}$; $M_w/M_n =$ 1.12–2.44	770
degree of substitution: 2.46		25	24.2	5.28	24.3					M_w	769
			37.6	6.26	43.8						
			59.0	7.31	65.1						
			75.0	7.67	68.3						
			111	9.80	104						
			159	11.60	169						
			249	12.50	188						
			24.2–249	$s_0 = 1.02 \times 10^{-14}\ M^{0.39}$						M_w; $M_w/M_n = 1.2$–1.3	
degree of substitution: 2.92											
	N,N-Dimethylacetamide	25	36.7	2.27	71.7					M_w	769
			82.2	2.83	101						
			137	3.32	131						
			308	4.04	205						
			500	4.17	228						
Cellulose triacetate	sym-Tetrachloroethane	21.9	250						0.744	M_v	148
Cellulose acetate-*co*-butyrate	sym-Tetrachloroethane	22.7	97						0.777	M_v	148
Cellulose carbanilate	Acetone		3930	45.4							792
	Dioxane	20	1.35–2580	$s_0 = 2.35 \times 10^{-15}\ M^{0.36}$					0.72	$M_{s,D}$; D_A, k_s and k_D given in Ref.	68
		23	14							$M_{s,D}$	771
			24			4.7					
			28			4.9					
			34			4.1					
			39			3.4					
			46			3.6					
			44			3.2					
			66			2.6					
			61			2.8					
			84			2.5					
			107			2.3					

			14–107			$D_0 = 9.6 \times 10^{-6}\ Z_{s,D}^{-0.68}$				Z = degree of polymerization	771
	Dioxane	20	4.43	0.520		11.0	−5			M_w; s_0; $D_{z,0}$; DLS	768
			8.70	0.743		8.0	0				
			11.9	0.823		6.5	0				
			17.8	0.95		5.0	5				
			42.0	1.25		2.80	2				
			80.8	1.51	177	1.75	10				
			141	1.88	145	1.25	8				
			215	2.18	150	0.95	60				
			278	2.34	164	0.79	64				
			1040	4.00	380	0.36	188				
			1570	4.35	574	0.26	262				
			2580	5.50	863	0.20	400				
		20	4.43–2580	$s_0 = 2.35 \times 10^{-15}\ M_{s,D}^{0.37}$		$D_0 = 2.2 \times 10^{-4}\ M_{s,D}^{-0.63}$					768
	Ethyl acetate	23	14	3.9		19.4			0.72	$M_{s,D}$	771
			24	4.7		13.8					
			28	4.8		12.1					
			34	5.3		10.9					
			39	5.4		9.6					
Cellulose carbanilate	Ethylacetate		46	6.1		9.3					771
			44	5.5		8.8					
			66	6.9		7.3					
			61	7.0		8.0					
			84	7.0		5.8					
			107	8.3		5.4					
			14–107	$s_0 = 1.3 \times 10^{-13}\ Z^{0.32}$		$D_0 = 2.2 \times 10^{-5}\ Z_{s,D}^{-0.68}$			0.72	Z = degree of polymerization	771
Cellulose glycolate sodium salt	Water/NaCl(0.2 mol/dm^3)	20	55			3.08	185			$M_{t,w}$; D_w	14
	Water/NaCl(0.5 mol/dm^3)				3.30	215				D_w	14
	Water/NaCl(1.0 mol/dm^3)		55	3.4	1.3	3.22		3.2	0.53	$M_{t,w}$; s_t; D_w	14
Methyl cellulose	Water	25	19	3.44	78	4.34				$[\eta] = 67\ cm^3/g$	1083
			31	2.62		3.80				$[\eta] = 110\ cm^3/g$	
			39	2.57	140	3.12				$[\eta] = 138\ cm^3/g$	
			58	2.50		2.44				$[\eta] = 242\ cm^3/g$	
			91	2.07		1.63				$[\eta] = 238\ cm^3/g$	
			89	1.90	270	1.75				$[\eta] = 260\ cm^3/g$	
			87	1.80						$[\eta] = 270\ cm^3/g$	
			102	1.71		1.75				$[\eta] = 310\ cm^3/g$	
			163	1.62	540	1.33				$[\eta] = 450\ cm^3/g$	
			190	1.40		1.17				$[\eta] = 500\ cm^3/g$	
			252	1.36	620	0.90				$[\eta] = 760\ cm^3/g$	
			408	0.94	740	0.73				$[\eta] = 830\ cm^3/g$	
Cellulose nitrate 13.8% N	Acetone	20	6.2					1.7	0.51	$M_{s,D}$	223
			30					3.2		SE	
			80.2	12.0				4.6		SE	
			199					5.4		$M_{s,D}$	
			613	30.0				6.9		$M_{s,D}$	
			100	13.9		6.5		3.74		$M_{s,D}$	276
									0.57		64
			190	14.0		3.29				s_t; $D_A = D_w$	65

TABLE 10. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			242	14.2		2.60				$M_{t,w}$	
			13.8	6.5		20.9				$M_{w,w}$; s_w; D_w	259
			16.3	7.1		19.3			0.572		
			27.8	8.3		13.3			0.570		
			100	12.6		5.6			0.566		
			153	15.3		4.5					
			187	14.5		3.5					
			267	16.8		2.8					
			495	25.3		2.3					
			1175	26.2		1.0					
cotton (13.8% N)			13	7.2		25.0				$M_{t,w}$; s_t; $D_A = D_w$	64
			23	8.8		16.5					
			46	11.1		10.8					
			74	10.8		6.45					
			90	12.8		6.30					
			178	15.2		3.80					
			262	17.1		2.90					
			456	20.5		2.00					
			630	21.5		1.50					
			2490	37.5		0.66					58,64
cotton (13.5–14% N)			74	10.8		6.45				$M_{t,w}$; s_t; $D_A = D_w$	49,58
	Acetone		178	15.2		3.80					58
			262	17.1		2.90					
			456	20.5		2.00					
			593	20.0		1.50					
			1130	30.5		1.20					
holo cellulose (spruce)			340	21.6	420	2.57	175	6.3	0.51	$M_{t,w}$; s_t; D_w	14
			780	13.4	210	2.4	65				14
holo cellulose (-cellulose)			230					7.7	0.51	$M_{t,w}$; s_t; D_w	14
linters			250	16		2.8				$M_{t,w}$; s_t; $D_A = D_w$	65
			344	15.8		2.05					
			780	19.0	420	1.0	210	12.2	0.510	$M_{t,w}$; s_t; D_w	14
linters, bleached			400	14.0	280	1.44	120	10.6	0.510		14
			600	19.0	420	1.11	225				
			680					11.5	0.510		
linters, bleached			6	5.2		35				$M_{s,D}$	223
and unbleached			30	8.7						SE	
			80	12.0						SE	
			199	18.0		3.7				$M_{s,D}$	
			613	30.0		2.0				$M_{s,D}$	
rayon pulp			24	17.6	480	3.01	235	6.0	0.51	$M_{t,w}$; s_t; D_w	14
			300	15.5	260	2.15	100	7.9	0.51		
			350	16.7	360	1.94		8.2	0.51		
			510	18.3	480	1.48		9.6	0.51		
			570	23.3	1150	1.66		8.2	0.51		
sulfate pulp			410	16.2	350	1.56	280	9.6	0.51	$M_{t,w}$; s_t; D_w	14
sulfite pulp			430	16.4	360	1.56	265	9.6	0.51	$M_{t,w}$; s_t; D_w	14
13.9% N			533			1.46				M_w; $D_{z,0}$; DLS	772

			870			1.03			
			1750			0.69			
			3600			0.43			
			430–3600			$D_0 = K_D M^{-0.72}$		M_w; DLS	772
	Butyl acetate						0.584		259
	Cyclohexanone		80.2	1.1			0.59	SE; $[s_0]$ given in Ref.	223
			613	2.0				$M_{s,D}$	
	Ethyl acetate	21.3	350				0.556	M_v	148
chemical cotton		20	370	11.4		1.49		$M_{f,w}$; s_f; D_w	91
			690	13.1		0.91			
raw cotton (Delfos variety)				1500			0.47		91
linters			195	9.3		2.40		$M_{f,w}$; s_f; $D_A = D_w$	65
viscose rayon			93	9.26		4.77		$M_{f,w}$; s_f; D_w	91
		20	17.5				0.545	M_n	91
		20	$Z_w = 3300$ (CLS)			0.59		$Z_n = 2600$ (OS); $D_{w,0}$; DLS; Z = degree of polymerization	773
			$Z_w = 2800$ (CLS)			0.77		$Z_n = 2100$ (OS); D_w,0; DLS	
			$Z_w = 2400$ (CLS)			0.97		$Z_n = 1600$ (OS); D_w,0; DLS	
purified chemical		30	41.3	6.34				M_w; $M_n = 34600$	90
linters			76	7.10				M_w; $M_n = 60600$	
			128	8.65				M_w; $M_n = 89000$	
			110	9.62				M_n	
			248	11.13				M_w	
			573	12.66				M_w; $M_n = 257000$	
			41.3–573	$s_0 = 3.04 \times 10^{-14} M_w^{0.29}$					90
	Methanol	20	80.2				0.54		223
	Pentyl acetate	20	27			13.3		$M_{w,w}$	259
			91			5.8			
			100			5.6			
			153			4.5			
			186			3.5			
			282			2.9			
			80.2	4.0			0.55	SE	223
			613	6.5				$M_{s,D}$	
unfractionated	Water	21	335	10.2	540			$M_{s,D}$	774
fractionated			63	5.8	100			$M_{s,D}$	774
			106	6.4	120				
			143	7.2	220				
			191	8.2	320				
			228	8.7	400				
			260	9.5	470				
			300	9.7	500				
			342	9.8	610				
			444	11.1	690				
			600	11.8	870				
			658	12.4	870				
			822	13.9	1250				

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TABLE 10. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
			983	15.1	1220						
		21							0.59		774
Cellulose xanthate	Water/NaCl (1 mol/dm^3)	0	$Z = 1460$	2.10				$f_0 = 1.16 \times 10^6$ g s^{-1} mol		See Ref. for degree of subst.	437
			$Z = 1060$	2.25				$f_0 = 1.01 \times 10^6$ g s^{-1} mol		Z = degree of polymerization	
			$Z = 590$	1.68				$f_0 = 0.73 \times 10^6$ g s^{-1} mol			
			$Z = 450$	1.81				$f_0 = 0.57 \times 10^6$ g s^{-1} mol			
			$Z = 360$	1.22				$f_0 = 0.52 \times 10^6$ g s^{-1} mol			
			$Z = 350$	1.05				$f_0 = 0.49 \times 10^6$ g s^{-1} mol			
sodium salt	Water/NaOH (0.2 mol/dm^3)/NaCl (1.0 mol/dm^3)	20	35–126	2.3–4.4	70–280	3.35–1.83	20–0	3.4–4.3	0.530	$M_{t,w}$; D_w	14
α-Cyclodextrin	Water	25 ± 0.01				32.0					775
Pullulan	Water + 0.02% (m/m)										
	Sodium azide	25	167			2.36	−32.0			$[\eta] = 42.5$ cm^3/g	1087
		40				3.20	−37.1				
		60				4.30	−35.3				
		80				6.70	−46.3				
		25	303			1.80	0			$[\eta] = 65.3$ cm^3/g	
		40				2.50	0				
		60				4.10	5.1				
		80				5.00	12.8				
		25	513			1.40	23.6			$[\eta] = 95.4$ cm^3/g	
		40				2.00	28.0				
		60				3.00	26.7				
		80				3.80	27.5				
		25	1000			1.00	55.0			$[\eta] = 154.2$ cm^3/g	
		40				1.40	65.7				
		60				1.90	84.2				
		80				2.95	54.9				
	Water	25	107			2.56	−20.1			$[\eta] = 44$ cm^3/g	1013
			188			1.94	−22.7			$[\eta] = 70$ cm^3/g	
			384			1.34	−30.5			$[\eta] = 102$ cm^3/g	
			572			1.12	−40.2			$[\eta] = 145$ cm^3/g	
			811			0.92	−36.4			$[\eta] = 175$ cm^3/g	
			1050			0.787	−35.0			$[\eta] = 225$ cm^3/g	
			1490			0.630	76.2			$[\eta] = 250$ cm^3/g	
			2100			0.567	30.2			$[\eta] = 310$ cm^3/g	
			2360			0.530	114			$[\eta] = 338$ cm^3/g	
Dextran											
T 500, linear	Ethanediol	5	154.3			4.40				M_w; $M_w/M_n = 1.26$; $C = 1$ g/dm^3	779
			191.4			4.26				M_w; $M_w/M_n = 1.17$; $C = 1$ g/dm^3	
			242.1			4.34				M_w; $M_w/M_n = 1.17$; $C = 1$ g/dm^3	
			309.6			3.66				M_w; $M_w/M_n = 1.25$; $C = 1$ g/dm^3	

		367.7	3.34	M_w; $M_w/M_n = 1.23$; $C = 1$ g/dm^3	
		434.1	3.48	M_w; $M_w/M_n = 1.32$; $C = 1$ g/dm^3	
linear		28.9	6.01	M_w; $M_w/M_n = 1.22$; $C = 1$ g/dm^3	779
		57.3	4.90	M_w; $M_w/M_n = 1.16$; $C = 1$ g/dm^3	
		92.5	4.72	M_w; $M_w/M_n = 1.16$; $C = 1$ g/dm^3	
		142.9	4.12	M_w; $M_w/M_n = 1.17$; $C = 1$ g/dm^3	
		185.6	2.86	M_w; $M_w/M_n = 1.13$; $C = 1$ g/dm^3	
		272.0	3.20	M_w; $M_w/M_n = 1.24$; $C = 1$ g/dm^3	
		433.4	2.80	M_w; $M_w/M_n = 1.31$; $C = 1$ g/dm^3	
branched		380.5	2.88	M_w; $M_w/M_n = 1.20$; $C = 1$ g/dm^3	779
		443.1	2.79	M_w; $M_w/M_n = 1.24$; $C = 1$ g/dm^3	
		493.3	2.77	M_w; $M_w/M_n = 1.29$; $C = 1$ g/dm^3	
		564.6	2.29	M_w; $M_w/M_n = 1.38$; $C = 1$ g/dm^3	
		617.7	2.38	M_w; $M_w/M_n = 1.35$; $C = 1$ g/dm^3	
		782.5	2.53	M_w; $M_w/M_n = 1.50$; $C = 1$ g/dm^3	
branched		280.8	3.66	M_w; $M_w/M_n = 1.34$; $C = 1$ g/dm^3	779
		324.6	3.49	M_w; $M_w/M_n = 1.28$; $C = 1$ g/dm^3	
		360.0	2.77	M_w; $M_w/M_n = 1.20$; $C = 1$ g/dm^3	
		401.6	2.76	M_w; $M_w/M_n = 1.28$; $C = 1$ g/dm^3	
		424.0	2.77	M_w; $M_w/M_n = 1.28$; $C = 1$ g/dm^3	
		472.5	2.49	M_w; $M_w/M_n = 1.28$; $C = 1$ g/dm^3	
		580.7	2.93	M_w; $M_w/M_n = 1.35$; $C = 1$ g/dm^3	
T 500, linear	5	154–434	$D_z = K_D M^{-0.28}$	M_w; $M_w/M_n < 1.3$; $C = 1$ g/dm^3	779
linear		29–434	$D_z = K_D M^{-0.30}$	M_w; $M_w/M_n < 1.3$; $C = 1$ g/dm^3	779
branched		380–783	$D_z = K_D M^{-0.25}$	M_w; $M_w/M_n < 1.5$; $C = 1$ g/dm^3	779
branched		280–580	$D_z = K_D M^{-0.40}$	M_w; $M_w/M_n < 1.4$; $C = 1$ g/dm^3	779

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TABLE 10. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
	Water	20	10–200	1.1–6.0						s_t vs. M diagram	23
			20	2.02						M determ. by Ogston's method (Ref. 24,25)	
			44	3.04						s_t	24
			65	3.47							
			150	5.47							
			460	9.6							
		20.9	160						0.60	M_v	148
		20	18–400	$s_0 = 2.45 \times 10^{-16} M_w^{0.44}$							26
			13–200	$s_0 = 2.36 \times 10^{-16} M_n^{0.45}$							26
T 10		20	6			8.9				M_w; $D_{z,0}$; M_w/M_n = 1.68; DLS	776
T 20			16.5			6.7				M_w; $D_{z,0}$; $M_w/M_n = 1.24$; DLS	
T 70			39.5			3.3				M_w; $D_{z,0}$; $M_w/M_n = 1.76$; DLS	
T 150			86.0			2.2				M_w; $D_{z,0}$; $M_w/M_n = 1.80$; DLS	
Dextran											
FDR 7783			120.3			2.2				M_w; $D_{z,0}$; $M_w/M_n = 1.32$; DLS	
			6–120.3			$D_{z,0} = (1.17 \pm 0.3) \times 10^{-4} M_z^{0.52 \pm 0.021}$				M_w/M_n 1.8; DLS; see Ref.	776
		25								$C = 2.72$ mass%; self diffusion; 5V	777
				3.24 ± 0.2						$C = 0.36$ mass%; 19V	777
								0.643 ± 0.002			777
		75				7.3 ± 0.4				$C = 2.72$ mass%; self diffusion; 9V	
Leuconostoc mesenteroides		20	9.8 ± 0.5			7.3 ± 0.3				M_w(CLS); M_w/M_n = 1.6; $D_{z,0}$; DLS	
			20			5.8 ± 0.4				M_w(CLS); $M_w/M_n = 1.2$; $D_{z,0}$; DLS	
			43 ± 4			3.9 ± 0.1				M_w(CLS); $M_w/M_n = 1.3$; $D_{z,0}$; DLS	
			75 ± 4			3.1 ± 0.1				M_w(CLS); $M_w/M_n = 1.6$; $D_{z,0}$; DLS	
			110 ± 5			2.8 ± 0.1				M_w(CLS); $M_w/M_n = 1.4$; $D_{z,0}$; DLS	
			150 ± 7			2.3 ± 0.1				M_w(CLS); $M_w/M_n = 1.8$; $D_{z,0}$; DLS	
			230 ± 10			1.90 ± 0.05				M_w(CLS); $M_w/M_n = 2.1$; $D_{z,0}$; DLS	
			490 ± 30			1.25 ± 0.05				M_w(CLS); $M_w/M_n = 2.3$; $D_{z,0}$; DLS	
		20	10–500			$D_{z,0} = 2.25 \times 10^{-4} M^{-0.45}$				D in mm^2/s, M in kg/mol	778

T 500 linear	Water	25	154.3	2.40	M_w; $M_w/M_n = 1.26$; $C = 1$ g/dm^3	779
			191.3	2.22	M_w; $M_w/M_n = 1.17$; $C = 1$ g/dm^3	
			242.1	2.08	M_w; $M_w/M_n = 1.17$; $C = 1$ g/dm^3	
			309.6	1.93	M_w; $M_w/M_n = 1.25$; $C = 1$ g/dm^3	
			367.7	1.75	M_w; $M_w/M_n = 1.23$; $C = 1$ g/dm^3	
			434.1	1.55	M_w; $M_w/M_n = 1.32$; $C = 1$ g/dm^3	
linear			28.9	3.17	M_w; $M_w/M_n = 1.22$; $C = 1$ g/dm^3	779
			57.3	3.08	M_w; $M_w/M_n = 1.16$; $C = 1$ g/dm^3	
			92.5	2.52	M_w; $M_w/M_n = 1.16$; $C = 1$ g/dm^3	
			142.9	2.51	M_w; $M_w/M_n = 1.17$; $C = 1$ g/dm^3	
			185.6	2.13	M_w; $M_w/M_n = 1.13$; $C = 1$ g/dm^3	
			272.0	1.82	M_w; $M_w/M_n = 1.24$; $C = 1$ g/dm^3	
			433.4	1.46	M_w; $M_w/M_n = 1.31$; $C = 1$ g/dm^3	
branched			380.5	1.68	M_w; $M_w/M_n = 1.20$; $C = 1$ g/dm^3	779
			443.1	1.53	M_w; $M_w/M_n = 1.24$; $C = 1$ g/dm^3	
			493.3	1.49	M_w; $M_w/M_n = 1.29$; $C = 1$ g/dm^3	
			564.6	1.42	M_w; $M_w/M_n = 1.38$; $C = 1$ g/dm^3	
			617.7	1.35	M_w; $M_w/M_n = 1.35$; $C = 1$ g/dm^3	
			782.5	1.24	M_w; $M_w/M_n = 1.50$; $C = 1$ g/dm^3	
			280.8	1.95	M_w; $M_w/M_n = 1.34$; $C = 1$ g/dm^3	779
			324.6	1.83	M_w; $M_w/M_n = 1.28$; $C = 1$ g/dm^3	
			360.0	1.72	M_w; $M_w/M_n = 1.20$; $C = 1$ g/dm^3	
			401.6	1.67	M_w; $M_w/M_n = 1.28$; $C = 1$ g/dm^3	
			424.0	1.62	M_w; $M_w/M_n = 1.28$; $C = 1$ g/dm^3	
			472.5	1.53	M_w; $M_w/M_n = 1.28$; $C = 1$ g/dm^3	
			580.7	1.41	M_w; $M_w/M_n = 1.35$; $C = 1$ g/dm^3	

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TABLE 10. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
T 500, linear		25	154–434			$D_z = K_D M^{-0.40}$				M_w; $M_w/M_n = 1.3$; $C = 1$ g/dm^3	779
linear			29–434			$D_z = K_D M^{-0.29}$				M_w; $M_w/M_n = 1.3$; $C = 1$ g/dm^3	779
branched			380–783			$D_z = K_D M^{-0.41}$				M_w; $M_w/M_n = 1.5$; $C = 1$ g/dm^3	779
branched			280–580			$D_z = K_D M^{-0.44}$				M_w; $M_w/M_n = 1.4$; $C = 1$ g/dm^3	779
T 250				3.3						Extrapolated value ($C = 0$); 25V	
									0.598	$C = 0.0903$–0.1508 mass%	
			11; 19; 100; 510							D given in Ref.	750
	Water	25	9.9			9.10	~10			$M_w/M_n = 1.62$	999
			39.7			4.39	~10			$M_w/M_n = 1.60$	
			65.0			3.32	~10			$M_w/M_n = 1.85$	
			481			1.40	~20			$M_w/M_n = 2.69$	
Dextran	Water		2270			0.74	~60				
			3250			0.66	~90				
PL 10	0.05 g/cL sodium azide	20	7.5			9.00				$[\eta] = 7.42$ cm^3/g	1015
PL 15			14.5			6.31				$[\eta] = 10.53$ cm^3/g	
PL 40			41.6			4.14				$[\eta] = 18.85$ cm^3/g	
PL 100			120.7			2.57				$[\eta] = 29.77$ cm^3/g	
PL 400			374			1.30				$[\eta] = 50.57$ cm^3/g	
PL 900			1270			0.78				$[\eta] = 64.20$ cm^3/g	
	Water/Na_2HPO_4	20	14	1.64		7.50		1.88		$M_{f,A}$; s_f; D_A	92
	(0.025 mol/dm^3)/NaH_2PO_4		36	2.64		4.69		2.19			
	(0.025 mol/dm^3)		84	4.1		3.12		2.49			
			141	5.3		2.40		2.79			
			240	6.55		1.73		3.15			
			1990	23.3		0.75		3.61			
			3050	26.4		0.55		4.25			
			38000	150		0.25		4.03			
p/bar	Water										
1		30	1000			1.04	81.5			DLS	1084
2000						1.00	71.4				
1		70	1000			1.58	124.1				
2000						1.46	100.7				
1		100	1000			2.60	105.8				
2000						2.26	112.4				
1		30	7400			0.58	63.3				
2000						0.53	59.2				
1		70	7400			1.42	32.9				
2000						1.19	29.0				
1		100	7400			2.00	30.8				
2000						1.66	35.5				
1		30	17500			0.42	−49.6				

2000					0.38	−45.2					
l		50	17500		0.61	−51.0					
2000					0.54	−46.3					
l		100	17500		1.48	−49.3					
2000					1.23	−47.1					
l		30	21100		0.98	−29.4					
2000					0.35	−31.6					
l		50	21100		0.85	−37.9					
2000					0.73	−33.4					
l		70	21100		0.60	−38.2					
2000					0.52	−28.1					
l		100	21100		1.45	−53.7					
2000					1.15	−54.0					
Leuconostoc mesenteroides											
ca. 5% branched	Water/Na_2HPO_4 (0.05 mol/dm^3)/NaH_2PO_4 (0.05 mol/dm^3)	20	11.2–437	1.57–9.82	8.80–1.40		1.74–3.22	0.611	$M_{t,A}$; s_t; D_A see Ref. for f_0/f_{sp} and D_0	167
				$s_t = 1.45 \times 10^{-15}\ M_{t,A}^{0.50}$						
highly branched	Water/Na_2HPO_4 (0.05 mol/dm^3)/NaH_2PO_4 (0.05 mol/dm^3)	20	28.8					0.603	$M_{t,A}$; s_t; D_A see Ref. for f_0/f_{sp} and D_0	167
Dextran										
				$s_t = 3.43 \times 10^{-15}\ M_{t,A}^{0.42}$						167
2-(Diethylamino)ethyl-dextran										
DEAE PL I	0.8 M sodium nitrate	20	1.58		17.6				$[\eta] = 3.34$ cm^3/g	1015
DEAE PL 10			11.30		7.8				$[\eta] = 8.72$ cm^3/g	
DEAE PL 15			21		5.52				$[\eta] = 11.59$ cm^3/g	
DEAE PL 40			45.7		3.56				$[\eta] = 19.45$ cm^3/g	
DEAE PL 100			131		2.04				$[\eta] = 31.49$ cm^3/g	
DEAE PL 400			429		1.13				$[\eta] = 52.15$ cm^3/g	
DEAE PL 900			1474		0.709				$[\eta] = 65.73$ cm^3/g	
β-Dextrin	Water	20	1.134	0.48				0.627	$C = 1$ g/dm^3	171
Ethyl cellulose	Acetone	20	78.8	5.53	5.7			0.885	$M_{s,D}$; D_A	412
			52.1	4.81	7.5					
			35.9	4.33	9.8					
			19.4	3.48	14.6					
			11.45	3.02	21.4					
			20		3.46				$C = 6.1$ g/dm^3; see Ref. for D	21
			53		1.43				$C = 5.0$ g/dm^3	
			90		1.05				$C = 5.16$ g/dm^3	
	sym-tetrachloroethane	21.9	190					0.864	M_v	148
Ethyl hydroxyethyl cellulose										
ca. 18% ethoxyl	Water	20	92	2.0	1.9	30	3.8	0.721	$M_{t,w}$; s_t; D_w; D_A given in Ref.	94
ca. 30% ethoxyl + hydroxyethyl			140	2.32	1.4	30	4.1	0.711		
			200	2.63	1.1	30	5.1	0.703		
			300	3.0	0.8	30	6.0	0.705		
			92–300	$s = 3.1 \times 10^{-15}\ M^{0.36}$	$D_w = 10^{-4}\ M^{-0.7}$				MA,w; s_t	94

TABLE 10. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Ficoll (Pharmacia)	Water	25	400	4.37						M_w; s_0; 11V; $C = 5.7–298$ g/dm^3	782
Galactomannan											
Carob	Water	20	319	4.65		0.97			0.629	$M_{w,w}$; $D_{w,0}$; $M_{s,D}$;	781
Guar			657	5.33		0.54			0.635	$D_{A,0}$ given in Ref.	
Lotus I			3.7	1.35		24.2			0.628		
Lotus II			332	5.43		1.09			0.630		
Lucerne			239	7.69		2.14			0.636		
Redclover			416	4.81		0.77			0.630		
Galactomannan											
S. japonica			152	3.02		1.32			0.629		
Soybean			16.6	1.72		6.90			0.634		
										k_s, k_D, a_s, a_D given in Ref.	781
Glucose carbanilate	1,4-dioxane	20	0.76	0.190		23.5	−50			M_w; $D_{z,0}$; DLS	768
Hydroxyethyl	Water	19.2	470						0.678	M_v	148
		25	125	2.82	385	1.86		3.96	0.701	$M_{t,w}$; s_t; v_2 at 20°C	98
			330	4.45	1020	1.11		4.78			
			420	4.90	1190	0.96	5.12				
			545	5.65	1580	0.85		5.29			
Hydroxyethyl											
			125–545	$s = 1.0 \times 10^{-15} M_{t,w}^{0.46}$		$D = 8.2 \times 10^{-5} M^{-0.54}$				$M_{t,w}$; s_t	98
Natrosol 250G		25	150	2.89						$M_{s,D}$; s_0; 9V; $C = 0.4–48$ g/dm^3	782
Hydroxypropyl cellulose										$M_{s,D}$; s_0; 11V; $C = 1.1–98$ g/dm^3	782
Klucel-L	Water	25	73	1.72							
		25	73	1.87	101.4				0.773	$M_{s,D}$; $M_{s,D}/M_n < 2$; s_0	783
		30	73	2.17	96.8						
		34	73	2.28	74.8						
		37	73	2.36	59.2						
	Water	40 ($T = \theta$)	73	2.54	44.4	5.0				M_w; $C = 10$ g/dm^3	784
			11			5.0					
			17			3.0					
			60			2.7					
			110			1.9					
			1000			0.57				M_w; s_0; $D_{z,0}$; DLS	768
Maltotriose carbanilate	1,4-Dioxane	20	1.8	0.332		17.3	−6				
Methyl cellulose											
22.6% methoxyl	Water	20	14	0.83				3.0	0.68	M_w; SE	222
27.05% methoxyl			70.6	1.29		1.49				$M_{t,A}$; s_t; D_A	19
27.44% methoxyl			151	1.67		0.92			0.704		
28.45% methoxyl			302	2.50		0.71			0.710		
28.8% methoxyl			22.6	0.79		3.05			0.683		19,20
			24.3	0.79				3.9	0.717	M_w; SE	222
31.7% methoxyl			34.3	0.89		2.34			0.720	$M_{t,A}$; s_t; D_A	19,20
31.8% methoxyl			38.1	0.89				4.5	0.73	M_w; SE	222

ca. 28% methoxyl							$v_2 = 0.0560 + 0.0542\ C_{OCH_3}$ (C in%)			19,20
				$s_0 = 0.85 \times 10^{-15}\ M_{I,A}^{0.45}$		$D_0 = 7.9 \times 10^{-5} M_{I,A}^{-0.56}$				
	Water/NaCl (0.2 mol/dm^3)	20	14.1			4.45			av. of M not specified; see Ref. for D	21
			24.3			3.05			$C = 5.0$ g/dm^3; see Ref. for D	
			38.1			2.47			See Ref. for D	
Xanthan	Water (deionized)	RT	2000			0.242			$D_{z,0}$; DLS	785
			2000						M_w; $D_{z,0}$; DLS	785
	Water/NaCl (0.3 mol/dm^3)		2000			0.194			$D_{z,0}$; DLS	785
	Water/NaCl (1 mol/dm^3)		2000			0.155				785
	Water/urea (4 mol/dm^3) (heat treated)		2000			0.280				785
	Water 0.1 M NaCl	25	9000			0.166			M_w; $D_{z,0}$; DLS; $C = 0.025$–0.1%	786,787
native			2940			0.155	873		DLS	1011,1012
pyruvate-free			1370			0.260	296			
acetate-free			1770			0.260	231			

TABLE 11. OTHER BIOPOLYMERS [PROTEINS, POLY(NUCLEOTIDES)]

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
β-Alanine	Water	25				101.4				D; $C = 1$ mass%	788
		36				147.1					
Aldolase			40			2.65	$D_0 k_D = -0.07$			$D_{0,z}$; $D_0\eta_0/T =$ 13.04	789,718
Poly(γ-benzyl-L-glutamate)	Dichloroacetic acid										
	0 to 100 vol%[a]	25	530	3.8–4.2						M_v; s	566
	N,N-Dimethylformamide	21	59–364	$s_c = 2.8 \times 10^{-14}\ M^{0.2}$		$D_c = 2.8 \times 10^{-3}\ M^{-0.8}$			0.786	$M_{s,D}$; for s_c and D_c see Ref.	559
		25	300		100	1.1			0.779	CLS	581
			550		180	0.75				CLS	
		25	51–410	1.62–3.91	9–141					for s_0 and k_s see Ref.	434
			36.4–4350	1.77–3.95	15–141					CLS; for s_0, k_s and M_A see Ref.	450

[a] helix to coil transition between 60 and 80 vol%

TABLE 11. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
		RT	83	2.2						M_v; D_z	717
			220	3.0		1.30					
			370	3.5		0.84					
			38				2.0			M_w; $M_n = 29000$; $D_{0,z}$ given in Ref.	668
			73				6.8			M_w; $M_n = 43000$; $D_{0,z}$ given in Ref.	
			80				3.9			M_w; $M_n = 48000$; $D_{0,z}$ given in Ref.	
			95				4.9			M_w; $M_n = 55000$; $D_{0,z}$ given in Ref.	
			200				15.0			M_w; $M_n = 101000$; $D_{0,z}$ given in Ref.	
			210				18.6			M_w; $M_n = 128000$; $D_{0,z}$ given in Ref.	
			560				33.5			M_w; $M_n = 190000$; $D_{0,z}$ given in Ref.	
		25	320	3.70						M_w; 10V ($C = 0.58$–15.08 g/dm^3)	782
		30	300			1.33	11				1016
Poly(γ-benzyl-α,L-glutamate)	Benzylalcohol	70	318			0.4763					1042
Poly(L-lysine)-hydrobromide		25									
	Water/NaBr (3.02×10^{-1} mol/dm^3)			84.8		$c_p = 3 \times 10^{-3}$ g/cm^3; $D = 2.6 \times 10^{-7}$ cm^2/s $c_p = 1 \times 10^{-3}$ g/cm^3; $D = 2.2 \times 10^{-7}$ cm^2/s $c_p = 5 \times 10^{-4}$ g/cm^3; $D = 2.0 \times 10^{-7}$ cm^2/s					1041
	Water/NaBr (7.41×10^{-2} mol/dm^3)		84.8			$c_p = 3 \times 10^{-3}$ g/cm^3; $D = 3.0 \times 10^{-7}$ cm^2/s $c_p = 1 \times 10^{-3}$ g/cm^3; $D = 2.4 \times 10^{-7}$ cm^2/s $c_p = 5 \times 10^{-3}$ g/cm^3; $D = 2.2 \times 10^{-7}$ cm^2/s					
	Water/NaBr (5.75×10^{-3} mol/dm^3)		84.8			$c_p = 5 \times 10^{-3}$ g/cm^3; $D = 18 \times 10^{-7}$ cm^2/s $c_p = 1 \times 10^{-3}$ g/cm^3; $D = 8 \times 10^{-7}$ cm^2/s $c_p = 1 \times 10^{-4}$ g/cm^3; $D = 1.7 \times 10^{-7}$ cm^2/s					
	Water/NaBr (2.00×10^{-3} mol/dm^3)		84.8			$c_p = 5 \times 10^{-3}$ g/cm^3; $D = 32 \times 10^{-7}$ cm^2/s $c_p = 1 \times 10^{-3}$ g/cm^3; $D = 11 \times 10^{-7}$ cm^2/s $c_p = 5 \times 10^{-5}$ g/cm^3; $D = 1.6 \times 10^{-7}$ cm^2/s					
	Water/NaBr (7.41×10^{-4} mol/dm^3)		84.8			$c_p = 3 \times 10^{-3}$ g/cm^3; $D = 30 \times 10^{-7}$ cm^2/s $c_p = 7.5 \times 10^{-4}$ g/cm^3; $D = 12 \times 10^{-7}$ cm^2/s $c_p = 8 \times 10^{-5}$ g/cm^3; $D = 1.9 \times 10^{-7}$ cm^2/s					
	Water/NaBr (6.92×10^{-5} mol/dm^3)		84.8			$c_p = 5 \times 10^{-3}$ g/cm^3; $D = 40 \times 10^{-7}$ cm^2/s $c_p = 3 \times 10^{-3}$ g/cm^3; $D = 42 \times 10^{-7}$ cm^2/s $c_p = 1.5 \times 10^{-3}$ g/cm^3; $D = 30 \times 10^{-7}$ cm^2/s					
Poly(adenylic acid)											
	Water/NaCl (0.1 mol/dm^3)	20	3500			0.84				D_z; $C = 4.8$ g/dm^3; DLS	883
	Water/NaCl (0.01 mol/dm^3)		3500			0.38				D_z; $C = 2.4$ g/dm^3; DLS	

	Water/cacodylate (pH = 7.4)		3500		0.10		D_z; $C = 0.12$ g/dm^3; DLS	
Bovine plasma albumin								
	Water/NaCl (0.2 mol/dm^3)	18–22	68		5.8 ± 0.1		M_w; $D_{D,w}$; DLS	876
	Water/NaCl (0.1 mol/dm^3)/phosphate (0.1 mol/dm^3; pH = 7)	20			D_z, $M_w = 4.0 \times 10^{-2}$ cm/s g/mol		$C = 25\%$	852
Bovine serum albumin								
	Water (pH = 5.2)	20			6.70		D_{app}	872
	Water	25			9.03		D; $C = 1$ mass%	788
		36			16.5		D; $C = 1$ mass%	788
	Water/NaCl (0.10–0.15 mol/dm^3)	22			5.65 ± 0.35		$D_{0,z}$; DLS	879
	Water/Tris/HCl (0.5 mmol/dm^3; pH = 7.4)	20			6.00		D_{app}	740
	(10.0 mmol/dm^3; pH = 7.4)				5.50		D_{app}	740
	(100.0 mmol/dm^3; pH = 7.0)				6.1		D_{app}	873
	(120.0 mmol/dm^3; pH = 4.7)				5.90		D_{app}	874
	(1000.0 mmol/dm^3; pH = 7.8)				5.00		D_{app}	875
	Water/Gdn-HCl	30	67		1.907	$D_0 k_D = (3.26 \times 10^{-6}$ cm^5/g/s	DLS; Gdn = Guanidin	718
	Water/phosphate buffer	20		4.5			$s_{0,w}$; SV	877
	Water/phosphate buffer (pH = 6)	20			5.76 ± 0.05		$D_{0,z}$; DLS	878
denaturated	Water/Gdn-HCl	25	69		1.907	$D_0 k_D = 3.28 \times 10^{-6}$ cm^5/g/s	$D_{0,z}$; DLS; Gdn = Guanidin	789
globular		25	69		6.69	$D_0 k_D = -4.52 \times 10^{-6}$ cm^5/g/s	$D_{0,z}$; DLS	789
Casein micelles								
α_{S1}-casein	0.2 M phosphate buffer, pH = 6.7	35	3400		0.45 ± 0.03			1048
Chromatin subunit								
	Water/NaCl (0.1–2 mol/dm^3)				3.4–1.8		$M_w/M_n = 1.07$; see Ref.	880
Chymotrypsinogen	Water/Gdn-HCl	30	5.7		3.15	$D_0 k_D = -0.70 \times 10^{-6}$ cm^5/g/s	DLS; Gdn = Guanidin	718
		25	25.7		3.15	$D_0 k_D = -0.70 \times 10^{-6}$ cm^5/g/s	DLS	789
	Water/NaCl (0.2 mol/dm^3)	18–22	29		7.4 ± 0.3		M_w; $D_{0,w}$; DLS	876
Collagen	Water	5			4.36		D_z; $C = 158$ g/dm^3; DLS	881
		12			4.32		D_z; $C = 157$ g/dm^3; DLS	
		20			4.12		D_z; $C = 147$ g/dm^3; DLS	
native		20			0.86 ± 0.02		D_z; $C = 0.4$ mass%; DLS	
DNA	Water/NaCl (0.2 mol/dm^3)/Na_2EDTA (0.01 mol/dm^3)	20			0.223 ± 0.06		D_z; DLS; pH = 6.5	884
calf thymus	Water/NaCl (0.1 mol/dm^3)	20	14500	22.0	0.080		D_z; DLS	885,887
		20	156		2.06		CLS and DLS	1014

TABLE 11. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
	Water/NaCl (0.5 mol/dm^3)	20	10000	20.0		0.114					
	Water/NH_4Ac (0.4 mol/dm^3)	20	11700	17.1		0.096					
	Water/LiCl (5.5 mol/dm^3)	20	10300	9.3		0.130–0.193				D_z; DLS; see Ref.	885,887
	Water/NaCl (0.14 mol/dm^3)/Tris (0.01 mol/dm^3)	25				0.05				D_z; DLS; pH = 8	886
	Water/NaCl (1.0 mol/dm^3)	20				0.118				D_z; DLS; pH = 7	887
	Water/NaCl (1.0 mol/dm^3)/EDTA	22–23				0.15				D_z; C = 0.04 g/dm^3; DLS; pH = 9.6	888,887
	Water/NaCl (1.0 mol/dm^3)	22–23				0.14				D_z; C = 0.04 g/dm^3; DLS; pH = 10.3	
histone (1:1)	Water/NaCl (0.8 mol/dm^3)	20	1500			0.265 ± 0.008				D_z; DLS; urea (2 mol/dm^3); phosphate buffer	889
	Water/NaCl (1.0 mol/dm^3)					0.234 ± 0.009					
	Water/NaCl (1.2 mol/dm^3)					0.183 ± 0.005					
	Water/NaCl (1.4 mol/dm^3)					0.187 ± 0.005					
	Water/NaCl (1.5 mol/dm^3)					0.186 ± 0.005					
pure	Water/NaCl (0.8 mol/dm^3)	20	1500			0.155 ± 0.003				D_z; DLS; C = 0.7 g/dm^3	889
	Water/NaCl (0.8 mol/dm^3)	20	1500			0.166 ± 0.012				D_z; DLS; C = 0.35 g/dm^3	
	Water	5				4.36				D_z; C = 158 g/dm^3; DLS	881
histone F_1 (1:1)		20									
	Water/NaCl (0.8 mol/dm^3)	20	1500			0.103 ± 0.004				D_z; DLS	889
histone F_2A (1:1)	Water/NaCl (0.8 mol/dm^3)	20	1500			0.259 ± 0.009				D_z; DLS	
E. coli phage 14	Water	20	(1.9 ± 0.2) 10^5			0.25 × 0.03				$D_{0,z}$; DLS	738
			(1.925 ± 0.066) 10^5			0.295 ± 0.003				$D_{0,z}$; DLS	895
Gelatin	Formamide	25	292			0.358	35			M_w/M_n = 2.3	1000
			346			0.324	50			M_w/M_n = 2.7	
			371			0.311	63			M_w/M_n = 1.8	
Glycoprotein ovomucoid	Water					4.03	$D_0k_D = 7.25 \times 10^{-6}$ cm^5/g/s			$D_{0,z}$; DLS	789
Glycoproteins											
Blood-group-specific glycoprotein (human)		–						0.731		Nonprotein component 2%	1049
Butyrylcholinesterase (horse)									0.707	20%	
Cervial (bovine)									0.635	74%	
Fibrinogen (human)									0.725	2.5%	
Prothrombin (human)									0.713	10%	
Submaxillary (ovine)									0.685	41%	

Submaxillary (porcine)						0.66	54%	
Tamm-Horsfall (human)						0.692	28%	
Lipoproteins								
HDL (human)						0.884	50%	1049
HDL_2 (human)						0.897	59%	
LDL_1 (porcine)						0.952	72%	
LpB (human)						0.953	78.5%	
$VHDL_1$ (human)						0.83	38%	
Nucleoproteins								
Alfalfa mosaic virus						0.703	16%; RNA	1049
Bacteriophage fd (*E. coli*)						0.71	12%; DNA	
Bacteriophage R 17 (*E. coli*)						0.681	30%; RNA	
Bacteriophage T 7 (*E. coli*)						0.639	51%; DNA	
Bacteriophage T 2 (*E. coli*)						0.66	55%; DNA	
Fog virus 3						0.713	10% lipide	
Nucleosome (chicken)						0.654	45%; DNA	
30 S Ribosome (*E. coli*)						0.601	70%; RNA	
Southern bean mosaic virus (cowpeas)						0.693	21%; RNA	
Tobacco mosaic virus						0.728	5%; RNA	
Metalloprotein								
Ferritin (horse spleen)						0.589	38% iron hydroxide	1049
haemocyanin (*M. trunculus*)	Water/NaCl (0.1 mol/dm^3)/ acetate buffer (0.1 mol/dm^3)		9200		$(1.03 \pm 0.01)-(1.08 \pm 0.01)$		$D_{0,z}$; DLS; pH = 5.7	878
Haemoglobin	Water	13			5.58		D_z; DLS; $C = 1.25$ mass%	890
		15			6.46		D_z for $C = 17$ mass% given in Ref.	
		18			6.81			
		21			7.38			
		24			7.82			
		27			8.13			
		30			8.33			
		33			8.50			
		35			8.59			
		37			8.61			
Histone (lysin rich)	Water/NaCl (0.14 mol/ dm^3)/Tris (0.01 mol/dm^3)	25	11		8.5		$D_{0,z}$; DLS; pH = 8	891
Lysozyme	Water	20			10		D_z; $C = 20$ g/dm^3; DLS	892
	Water/NaCl (0.1 mol/ dm^3)/phosphate buffer	20			10.6 ± 0.02		$D_{0,z}$; DLS; pH = 6.0	878
Nitrochitin	Dimethylformamide	26		$s_0 = 1.12 \times 10^{-14}\ M_w^{0.29}$ $10000 \leq M_w \leq 170000$ g/mol	$D_0 = 7.02 \times 10^{-4}\ M_w^{-0.70}$ $10000 \leq M_w \leq 170000$ g/mol			1081
Polynucleosome								
	Water/Tris–HCl (30.0 mmol/dm^3)	20			0.65		D_z; DLS; pH = 7.4	740
	Water/Tris–HCl (10.0 mmol/dm^3)				0.61		D_z; DLS; pH = 7.4	
	Water/Tris–HCl (0.5 mmol/dm^3)				0.53		D_z; DLS; pH = 7.4	
	Water/Tris–HCl (0.1 mmol/dm^3)				0.39		D_z; DLS; pH = 7.4	

TABLE 11. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	s_0 ($\times 10^{13}$) (s)	k_s (cm^3/g)	D_0 ($\times 10^7$) (cm^2/s)	k_D (cm^3/g)	f_0/f_{sp}	v_2 (cm^3/g)	Remarks	Refs.
Pepsinogen	Water	25	40			2.52	$D_0k_D = 3.95 \times 10^{-6}$ cm^5/g/s			$D_{0,z}$; DLS	789,718
Phosphatidyl choline vesicles											
	D_2O/CsCl (0.001 mol/dm^3)	25				0.341 1.24				D_z; DLS; $C = 13$ g/dm^3	893
Serum albumin	Water/NaCl (0.2 mol/dm^3)/NaF (0.005 mol/dm^3)	20	69	4.0		6.1 ± 0.1				$D_{0,z}$; D_w and D_s given in Ref.	896
Sucrose	Water	25				52.3				$C = 1$ mass%	788
		36				65.8				$C = 1$ mass%	
Urease	Water/NaCl (0.2 mol/dm^3)	18–22	60			3.4 ± 0.1				M_w; $D_{0,w}$	876
Virus R 17	Water	20	4020 ± 170			1.534 ± 0.015				$D_{0,z}$; DLS	894
Qβ			4550 ± 160			1.423 ± 0.014				$D_{0,z}$; DLS	
BSV			8810 ± 170			1.246 ± 0.013				$D_{0,z}$; DLS	
PM2			47900 ± 1700			0.650 ± 0.007				$D_{0,z}$; DLS	
T7			50900 ± 1100			0.644 ± 0.007				$D_{0,z}$; DLS	
Proteins	Water/Gdn HCl (6 mol/dm^3)/mercaptoethanol (0.1 mol/dm^3)	25				$D_{0,z} = 6.96 \times 10^{-6}\ n^{-0.560}$				DLS; n = number of chain segments; Gdn = Guanidine	789

C. SECOND VIRIAL COEFFICIENTS OF POLYMERS IN SOLUTION

1. Introduction

The behavior of polymers in solution can be described by the colligative properties introducing the virial coefficients A_2, A_3, etc.

According to the theory of McMillan and Mayer (866–868), the second virial coefficient of a monodisperse polymer solution is defined as

$$A_2 = (N_A/(2VM^2)) \int (F_1(1)F_1(2) - F_2(1,2))\,\mathrm{d}(1,2) \tag{C1}$$

where N_A is Avogadro's number, V is the volume of the system, M is the molar mass of the dissoluted polymer, $F_1(1)$, $F_1(2)$, and F_2 (1,2) are the molecular distribution functions of species 1 (solvent molecule) and species 2 (solute molecule), and d(1,2) is the differential volume element in hyperspace. The integral in Eq. (C1) is called the cluster integral. The distribution functions for the molecules, F_1 and F_2, are to be evaluated at zero concentration.

There are a couple of methods for the experimental determination of the second virial coefficient.

1.1. Colligative Properties The vapor pressure depression of the solvent in a polymer solution is expressed as

$$(p_1^0 - p_1)/p_1 = \Delta p_1/p_1 = V_1^0 C(1/M_n + A_2 C + A_3 C^2 + \cdots) \tag{C2}$$

where p_1^0 and p_1 are the vapor pressure of the pure solvent and the solvent in the solution, V_1^0 is the molar volume of the pure solvent, C is the mass concentration of the polymer, M_n is the number average of the molar mass, and A_2, A_3, ... are the virial coefficients. For theta conditions, the virial coefficents are zero.

The freezing point depression, $T_{F,1}$, of a polymer solution is

$$T_1^0 - T_1 = \Delta T_{F,1} = E_F C(1/M_n + A_2 C + A_3 C^2 + \cdots) \tag{C3}$$

and the boiling point increase, $T_{B,1}$, is

$$T_1 - T_1^0 = \Delta T_{B,1} = E_B C(1/M_n + A_2 C + A_3 C^2 + \cdots) \tag{C4}$$

with T_1^0 and T_1 the boiling point (freezing point) of the pure solvent and of the solvent in the solution, and E_F and E_B are the cryoscopic and ebullioscopic constants.

The osmotic pressure of a polymer solution can be described as

$$\pi = RTC(1/M_n + A_2 C + A_3 C^2 + \cdots) \tag{C5}$$

where R and T are the gas constant and the absolute temperature. In some cases $B = RTA_2$ is given (instead of A_2).

With certain assumptions it is possible to express the third virial coefficent, A_3, in terms of the second virial coefficient (379–381). Especially when A_2 is small, it is found that, for example, Eq. (C5) can be approximately expressed as (Ref. 382)

$$\pi = (RTC/M_n)(1 + (A_2 M_n/2)C)^2 \tag{C6}$$

if $A_3 = A_2^2 M_n/4$.

Following Eqs. (C2)–(C6), the dimension of A_2 is given in $\mathrm{mol\,cm^3/g^2}$ or $\mathrm{mol\,m^3/kg^2}$.

1.2. Scattering Methods (Classical Light Scattering, X-ray Scattering, Neutron Scattering) Based on the fluctuation theory of Smoluchowski (383) and Einstein (384), which links osmotic pressure with classical light scattering, Debye (385) established the fundamental equation for the light scattering of polymers in solution:

$$KC/R(q) = (M_w p_z(q))^{-1} + 2A_2 Q(q)C + \cdots \tag{C7}$$

with $q = (4\pi/\lambda)\sin\theta/2$ and

$$p(q) = (2/N^2)\int_0^N (N-n)\sin q h_n/q h_n\,\mathrm{d}n$$

Equation (C7) makes it possible to determine the virial coefficients from classical light scattering measurements and can also be applied on X-ray and neutron scattering. In Eq. (C7), K is a constant and $R(q)$ is the excess intensity of the scattered beam at the value q, at unit distance and unit primary intensity. M_w denotes the mass average of the molar mass and $P_z(q)$ is the z-mean of the scattering function from which the dimensions of the dissoluted polymer can be calculated (h_n is the distance between scattering points 0 and n). The $Q(q)$-function corresponds to the q-dependence of A_2. $P(q)$ denotes the scattering function of a monodisperse polymer.

1.3. Sedimentation Velocity The determination of sedimentation and diffusion coefficient, s and D, yields the virial coefficients of a polymer solution (Ref. 800):

$$(D/s)(1 - v_2\rho_1) = RT(1/M + 2A_2 C + 3A_3 C^2 + \cdots) \tag{C8}$$

where v_2 is the partial specific volume of the solute and ρ_1 is the density of the solvent.

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1.4. Sedimentation Equilibrium The sedimentation equilibrium in an ultracentrifuge gives (Ref. 800)

$$\omega^2 r(1 - v_2\rho_1)/(\partial \ln C/\partial r) = RT(1/M + 2A_2 C + 3A_3 C^2 + \cdots) \quad \text{(C9)}$$

where ω is the angular velocity and r is the distance from the center of rotation.

1.5. p–V–T Measurements The virial coefficients are related to thermodynamic quantities

$$A_2 + A_3 C + \cdots = -\mu_1^E/(RTC^2 V_1) \quad \text{(C10)}$$

where V_1 is the partial molar volume of the solvent and μ_1^E is the excess chemical potential of the solvent in the solution.

$$\mu_1^E = (\partial G_m^E/\partial x_1)_{p,T,x_2};\ G_m = G_m^E - G_m^{id} \quad \text{(C11)}$$

G_m^E is the molar excess Gibbs energy, G_m is the molar Gibbs energy, and G_m^{id} is the molar Gibbs energy of an ideal solution. With the help of the equation

$$(\partial G_m/\partial p)_{T,x_j} = V_m^E \quad \text{(C12)}$$

μ_1^E, and consequently the virial coefficients, can be calculated from p–V–T measurements.

1.6. Averages of the Second Virial Coefficient The average of A_2 for heterogeneous polymers is rather complicated and depends on the measuring method. If A_2 is obtained from measurements based on colligative methods, it will give the average (Ref. 860)

$$A_2^{\pi} = \int_{i=0}^{\infty}\int_{j=0}^{\infty} A_{ij}\, w(M_i)w(M_j)\, dM_i\, dM_j = (A_2)_{w,w} \quad \text{(C13)}$$

This is a double mass average of A_2.

Classical light scattering measurements will give the average (Ref. 860)

$$A_2^{LS} = (1/M_w)^2 \int_0^{\infty}\int_0^{\infty} A_{ij} M_i M_j\, w(M_i)w(M_j)\, dM_i\, dM_j = (A_2)_{z,z} \quad \text{(C14)}$$

This is a double z-average of A_2. A_{ij} is defined as

$$A_{ij} = (N_A/(2VM_iM_j)) \int (F_1(i)F_1(j) - F_2(i,j))d(i,j) \quad \text{(C15)}$$

where i stands for a molecule of the ith species and j for one of the jth species.

1.7. Second Virial Coefficient – Molar Mass Relationship The dependence of A_2 on the molar mass can be given by a power expression

$$A_2 = K_A M^{a_A} \quad \text{(C16)}$$

where K_A and a_A are empirical constants determined for each polymer solvent system at given values of temperature and pressure.

In Section C, relationship (C16) is listed whenever quoted in the literature, in preferences to single A_2 values.

1.8. Temperature Dependence, Pressure Dependence There is no general equation concerning the temperature and pressure dependence. The equation (Ref. 860)

$$A_2 = (N_A/2)(\beta_0/M_0)(1 - \theta/T) \quad \text{(C17)}$$

where β_0 is for the independent part of the excluded volume, M_0 is the molar mass of a segment, and θ is the theta temperature, has only a restricted validity.

1.9. Abbreviations

A	Archibald method
approx.	Approximately
CLS	Classical light scattering
DLS	Dynamic light scattering
Cont.	Continued
GPC	Gel permeation chromatography
LALLS	Low angle laser light scattering
OS	Osmotic pressure osmometry
RT	Room temperature
SANS	Small angle neutron scattering
SAXS	Small angle X-ray scattering
SE	Sedimentation equilibrium
SV	Sedimentation velocity
V	Number of single values given in ref. cited
VOS	Vapor pressure osmometry

1.10. Miscellaneous For solvent mixtures, the volume fraction $\varphi_i = v_i/\sum v_i$ is used.

In some cases the second virial coefficient is replaced by $B = RTA_2$.

Several attempts have been made to find theoretical explanations for the nonideal behavior of polymer solutions. There exist numerous excellent books on this topic (272,387,389–398,618,860).

The following tables give second virial coefficients for different polymer solutions. The polymers are arranged according to the hierarchial system in this book. In order to indicate the number of values reported in the original reference, a number, e.g., 5V is given in the column "Remarks". Values for solvent mixtures of different ratios, data for different pH etc., are treated separately in some cases. The user is asked to refer to the original references to find the detailed data and conditions.

D. TABLE OF SECOND VIRIAL COEFFICIENTS OF POLYMERS IN SOLUTION

TABLE 12. POLY(ALKENES), POLY(ALKYNES)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(L-butene)						
atactic	*n*-Nonane	35	44.1–1300	2.4	CLS	109
isotactic		80	105–935	6.05–1.05		
atactic	Toluene	45	26.3–558	10.8–4.10	OS	109
isotactic		45	90.1–775	6.57–3.73		
atactic	Toluene	45		$M^{-0.32}$	OS	316
isotactic				$M^{-0.25}$		
Poly(ethylene)	1-Chloronaphthalene	125	115–2160	12.0–0.78	9V; CLS; A_2M; (cm^3/g)	291
		135	107	3.5	CLS	820
			113	3.9		
			120	3.0		
			141	5.6		
		135	13.63	15.9	CLS	821
			32.1	10.9		
			119.6	10.3		
	Diphenylmethane	142	82–89	−0.25 to 0.93	CLS	820
	1,2,4-Trichlorobenzene	135	55	21.0	CLS	820
			91	19.0		
			99	18.8		
			131	23.5		
			91	19.0		
	1,2,4-Trichlorobenzene	135	570	15	CLS	905
	Trichlorobenzene	135	89	11.4	CLS	904
			415	9.1		
NBS standard		140	11.4–28.9	41.1–45.2	4V; VOS $M_w/M_n = 1.11$–4.56	822
linear			1.75–2.06	20.11–34.37	3V; VOS $M_w/M_n = 1.39$	822
			3.53	7.59	VOS	822
			3.72	6.70	$M_w/M_n = 1.03$	
			22.95, 24.8	4.17	VOS $M_w/M_n = 1.08$	822
			3.77	4.36	VOS $M_w/M_n = 1.39$	
branched			17.6 (GPC)	2.64	VOS $M_w/M_n = 22.96$	822
			22.7 (GPC)	2.17	VOS $M_w/M_n = 9.24$	822
			26.6 (GPC)	3.22	VOS $M_w/M_n = 18.59$	822
low pressure			121–526	10.5 (approx.)	8V; CLS	296
Marlex			310	11.6	CLS	296
			625	10.6	CLS	296
high pressure			144–720			
linear, fractions		135	20–1000	$6.3 \times 10^{-3}\ M^{-0.15}$		231a
	n-Decane	115	145–700	5.9–1.6	2V; CLS; M_n given in Ref.	231
	1,2,3,4-Tetrahydro-naphthalene	105	145–690	21.8–6.6	2V; CLS; M_n given in Ref.	231
		125	105–219	26.8–1.68	8V; CLS; A_2M; (cm^3/g)	291
branched, varying degree		81	38–735	3.6–0.9	6V; CLS	292
			2600–6100	0 (approx.)		
high pressure		81	573	0.92	CLS	293
			1980	0.82		
low pressure		105	125–465	23.1–15.9	3V; CLS	293
high pressure, fractionated	*p*-Xylene	81	9–215	27.8–6.54	12V; OS	292
low pressure, fractionated		105	11.2–160	33.5–26.1	4V; OS	292
				$26.2 \times 10^{-3}\ M^{-0.24}$		292a
HDPE, linear	*n*-Decane	130	600	4.8	CLS, $A_{2,1} = 4.70 \times 10^{-3}\ M_w^{-0.17}$	906
			520	5.0		
			310	5.4		
LDPE, branched	*n*-Decane	130	520	2.0	CLS	906
			310	2.4		
Lupolen 1840 D	*n*-Decane	130	600	1.9	CLS	906
Poly(ethylene-*co*-tetrafluoroethylene)	Diisobutyladipate	240	540	1.97	CLS	1026
			1160	1.02		
			3210 ± 290	0.511 ± 0.054		

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TABLE 12. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(isobutene)	Benzene	24–30	1260	78.1×10^{-4}	$(1-(298/T))$; CLS; M_v	308
						311
		30	556	45.0	A_2M; (cm^3/g)	160
			722	0.52		
		40		$A_2 = 0.75 \times 10^{-3}\ M^{-0.12}$	OS	317
	Chloroform	25	–	$A_2 = 3.45 \times 10^{-3}\ M^{-0.204}$	$M_w/M_n \approx 1.2$	989
	Cyclohexane	23	27–16100	$A_2 = 6.88 \times 10^{-3}\ M_w^{-0.203}$	CLS	990
				$A_2 = 6.88 \times 10^{-3}\ M_w^{-0.203}$	CLS	898
		25	550	260.0	CLS	202
	Cyclohexane	25	26.1	10.6	CLS; A_3 reported	901
			42.9	9.55	CLS; A_3 reported; $M_w/M_n = 1.08$	
			77.7	8.40	CLS; A_3 reported; $M_w/M_n = 1.10$	
			181	7.05	CLS; A_3 reported; $M_w/M_n = 1.09$	
			488	5.30	CLS; A_3 reported; $M_w/M_n = 1.10$	
			878	4.55	CLS; A_3 reported; $M_w/M_n = 1.09$	
			1580	3.85	CLS; A_3 reported	
			3390	3.13	CLS; A_3 reported	
			7830	2.68	CLS; A_3 reported	
			14100	2.54	CLS; A_3 reported	
			2400	5.0	CLS	454
		–	391–4700	4.39	CLS	843
	Dibutyl ether	0	9000 ± 1000	1.0 (approx.)	CLS; graphical representation of T-dependence given in Ref.	438
	Ethyl caprylate	25	674	41.0	CLS	202
	Ethyl heptanoate	25–75	56	0.430–1.62	6V; CLS; $M_w/M_n = 1.13$	934
		25–75	122	−0.234 to 1.69	6V; CLS; $M_w/M_n = 1.20$	
		25–75	230	−0.363 to 1.14	6V; CLS; $M_w/M_n = 1.19$	
		26–70	612	−0.358 to 1.093	6V; CLS; $M_w/M_n = 1.24$	
		25–75	1100	−0.270 to 1.012	6V; CLS; $M_w/M_n = 1.30$	
	n-Heptane	23	167–16100	$A_2 = 3.08 \times 10^{-3}\ M^{-0.188}$	CLS	990
		23		$A_2 = 3.08 \times 10^{-3}\ M^{-0.188}$	CLS; $1.14 \leq M_w/M_n \leq 2.28$	898
		25	27.1	4.96	CLS; A_3 reported	900
			42.5	4.45		
			81.3	3.94		
			179	3.22		
			487	2.65		
			890	2.42		
			1580	2.08		
			3310	1.78		
			7750	1.51		
			50–22	$A_2 = 1.49 \times 10^{-2}\ M^{-0.28}$	17V; CLS; different methods of polymer preparation; see Ref.	600
			400–1900	2.85–2.75	3V; CLS	56
		60	1900	1.75		56
	Heptane/propyl alcohol					
	90/10	25	1900	1.95	CLS	56
	80/20		1900	0		56
		60	1900	0.85	CLS	56
	72/28	60	1900	−0.25		56
	Isoamyl isovalerate	21	80.4	−0,23	CLS; see Ref. for A_3	899
			178	−0.31		
			482	−0.30		
			872	−0.33		
			1600	−0.40		
		23	80.4	−0.14	CLS; see Ref. for A_3	899
			178	−0.20		
			482	−0.20		
			872	−0.19		
			1600	−0.20		
		25	80.4	−0.06	CLS; see Ref. for A_3	899
			178	−0.09		
			482	−0.10		
			872	−0.08		
			1600	−0.10		

TABLE 12. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
		27	80.4	00	CLS; see Ref. for A_3	899
			178	0.0		
			482	0.0		
			872	0.0		
			1600	0.0		
		29	80.4	0.10	CLS; see Ref. for A_3	899
			178	0.10		
			482	0.10		
			872	0.09		
			1600	0.08		
		31	80.4	0.20	CLS; see Ref. for A_3	899
			178	0.18		
			482	0.18		
			872	0.15		
			1600	0.13		
		33	80.4	0.30	CLS; see Ref. for A_3	899
			178	0.25		
			482	0.25		
			872	0.23		
			1600	0.19		
		35	80.4	0.35	CLS; see Ref. for A_3	899
			178	0.30		
	Isoamyl isovalerate		482	0.28		899
		35	872	0.28		
			1600	0.24		
		37	80.4	0.40	CLS; see Ref. for A_3	
			178	0.35		
			482	0.35		
			872	0.33		
			1600	0.30		
	Isooctane	25	5	7.96	CLS	902
			6.2	8.03		
			8.9	6.9		
			11.8	6.61		
			15.8	6.2		
			22.9	5.56		
			26.1	5.30		
			33.9	5.00		
			39.0	4.76		
			56.4	4.42		
			61.4	4.23		
			80.5	4.09		
			87.0	3.93		
			95.0	4.08		
			122	3.84		
			140	3.72		
			180	3.67		
			230	3.57		
			270	3.51		
			393	3.34		
			612	3.15		
			832	3.14		
			1100	3.00		
			1360	2.94		
	Isooctane	25	2400	1.1	CLS	454
	n-Pentane	0	9000 ± 1000	1.2 (approx.)	CLS; graphical representation of T-dependence given in Ref.	438
Poly(1-octadecene/maleic anhydride)	Ethylacetate	25	28	3.0	CLS	981
			37	2.9		
			51	2.8		
			68	2.5		
			99	2.1		
Poly(1-octene)	Bromobenzene	25	250–4000	3.12–1.62	CLS	420

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TABLE 12. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(propylene)	1,2,4-Trichlorobenzene	140	23.0–28.0	12.0–16.9	5V; VOS; $M_w/M_n = 10.00$	822
			44.5–49.8	12.5–15.0	3V; VOS; $M_w/M_n = 5.70$	822
			34.7–37.1	23.8–22.4	2V; VOS; $M_w/M_n = 8.60$	822
isotactic	α-Chloronaphthalene	147 ± 1	484 (approx.)	4.3 (approx.)	3V; CLS; three different wavelengths used	446
			116 (approx.)	3.65 (approx.)	3V; CLS; three different wavelengths used	
atactic	Benzene	25		$3.2 \times 10^{-3} M^{-0.20}$	CLS	318
	Bromobenzene	100	2.16	1.21	OS	518,520
				10.8	CLS	518,520
		37–130	2.16	−2–+46.5	4V; OS; B; (bar cm^6/g^2)	518,520
	Carbon tetrachloride	37	1.25–2.50	54–69.5	6V; OS; B; (bar cm^6/g^2); values for four more solvents given in Ref.	518,520
isotactic	1-Chloronaphthalene	125		$4.3 \times 10^{-3} M^{-0.16}$	CLS	318
		135		$16.5 \times 10^{-3} M^{-0.27}$	CLS	318
		140	107–1110	3.9 (approx.)	4V; CLS	315
atactic	Cyclohexane	25		$20 \times 10^{-3} M^{-0.26}$		318
Poly(1-pentene)						
(80% *trans*)	Hexane	25	105–17200	1.8–3.2	11V; CLS; $M_z/M_w = 1.1–1.9$	835
isotactic	Toluene	30	549–25.2	0.86–2.96	7V; OS; (cm^4/g)	604
Poly(tetrafluoroethylene)	Poly(chlorotrifluoro-ethylene) mixed oligomeres	340	290 ± 20	0.67 ± 0.13		1039
	Perfluorotetracosane	325	260 ± 15%	0.19 ± 21%	CLS	1004
			2100 ± 15%	0.10 ± 20%	CLS	
Poly(1-phenyl-1-propene)	Toluene	25	1450	2.91	CLS	1034
			1320	3.78		
			613	3.58		
			372	4.31		
			258	3.84		
			142	4.82		
			140	4.77		
			100	5.46		
			57	6.05		
			55.9	6.75	SE; $M_z/M_w = 1.06$	
Poly(1-phenyl-1-propene)	Toluene	25	43.9	9.50	SE; $M_z/M_w = 1.08$	1034
			28.2	9.88	SE; $M_z/M_w = 1.06$	
			22.1	10.0	SE; $M_z/M_w = 1.09$	
			15.5	15.7	SE; $M_z/M_w = 1.06$	

TABLE 13. POLY(DIENES)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(butadiene)						
1,4-*cis*	Benzene	28.6	60–293	15.3 (approx.)	11V; OS	297
			138	27.9	OS	
	Cyclohexane	25	20	16.4 (14.9)	OS, (CLS)	903
			46	25.0		
			70.9	13.2 (12.7)		
			76	13.6		
			130	11.6 (10.3)		
			355	9.2 (8.5)		
			548	(7.9)		
			925	(6.7)		
			2950	(4.9)		
			3740	(4.6)		
			16500	(3.6)		
linear		25	11–760	$A_2 = 1.32 \times 10^{-2} M_w^{-0.218}$	CLS	987
18-armed-star			99–1900	$A_2 = 5.0 \times 10^{-3} M^{-0.22}$	CLS	
unfractionated		28.6	840–4350	2.92 (approx.)	11V; CLS	297

TABLE 13. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
fractionated		28.6	143–1640	7.50 to −1.63	7V; CLS	297
1,4-*trans* 69%		28.6	1070	1.67	CLS	297
60–70% unfractionated		25	5400–19000	0.52 (approx.)	7V; CLS	288
fractionated		25	3400–18000	2.6 to −0.4	12V; CLS	288
			2300–11000	2.67 (approx.)	5V; CLS	
		34.2	17.4–370	18.3–9.48	4V; OS	497
	Dioxane	34.2	17.9–434	3.81–1.49	4V; OS	497
	Toluene	34.2	18.2–466	19.4–11.5	7V; OS	497
Poly(butadiene-*co*-styrene)						
25-30% styrene	Benzene	28.6	58–112	13.5 (approx.)	4V; OS	297
25–30%	Cyclohexane	28.6	800–26700	0.9 (approx.)	4V; CLS	297
24%		30	49–514	1.7–1.0	8V; OS	344
diblock, 32.9–3.59 mass% PST						
		34.2	42.6–625	10.0–7.55	4V; OS	497
	Dioxane	34.2	43.6–610	5.8–1.64	4V; OS	
	Toluene	34.2	43.0–625	12.6–8.17	4V; OS	
triblock, 45.9–4.05 mass% PST	Cyclohexane	34.2	35.8–517	10.1–8.69	8V; OS	497
	Dioxane	34.2	34–170	6.63–2.16	6V; OS	
	Toluene	34.2	34.6–516	12.5–9.57	8V; OS	
Poly(chloroprene)	Butanone	25	147–2920	0	7V; CLS	493
	n-Butyl acetate	25	149–3020	2.52–1.24	7V; CLS	493
	Carbon tetrachloride	25	156–3180	4.59–2.03	7V; CLS	493
	trans-Decalin	0.4–48.7	587	−0.10 to +1.49	12V; CLS	493
		1.1–48.7	865	−0.074 to +1.43	13V; CLS	
		0.4–48.7	1660	−0.05 to +1.12	12V; CLS	
Chlorinated rubber	Butanone	29	156	3.60	OS	478
		35	4410	−0.18	CLS	478
Chlorinated rubber-*co*-poly(ethyl methacrylate)						
20.72%	Butanone	29	197	4.80	OS	478
Poly(isoprene)						
cis	Cyclohexane	20	1600	6.5	$M_{s,D}$; B; (bar cm^6/g^2)	29
		25	620 ± 10%	5.0 ± 20%	OS	508
	Carbon tetrachloride	25	32.4	0.123	CLS; M_w/M_n given	927
			71.95	0.114		
			97.2	0.109		
			295	0.104		
			452	0.088		
natural rubber		7	1700	14.2	CLS; B; (bar cm^6/g^2)	30
		27		14.3		
		7	1300	11.7	CLS; B; (bar cm^6/g^2)	28
		27	1300	12.7		
		25	300	6.2	OS; values for CLS in Ref.	508
epoxidized 10 mole%	Chloroform	25	430	3.30	CLS	980
nitrile rubber 100% hydrogenated						
	Chloroform	25	150	3.34	CLS	980
brominated, 0% bromine	Cyclohexane	20	1800	14.3	CLS; B; (bar cm^6/g^2)	31
5.8% bromine			1850	9		
7.3% bromine			1900	4.5		
25.8% bromine			2900	2.6		
41.5% bromine			3100	0.4		
43.8% bromine			3800	0		
Poly(isoprene)						
long-arm-stars						
functionality 2.9–4.0	Cyclohexane	–	35.2–410	7.0–12.6	16V; CLS	988
functionality 4.9–18.1	Toluene	–	41–6800	1.6–9.4	28V; CLS	988
Poly(isoprene-*co*-styrene)						
block copolymer	Cyclohexane	30	221–370	5.51–4.40	CLS; values for OS given in Ref.	495
	Methyl isobutyl ketone	30	221–370	1.25–1.74	CLS; values for OS given in Ref.	495
	Toluene	30	221–370	9.04–10.2	CLS; values for OS given in Ref.	495

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TABLE 13. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(isoprene)/Poly(ethylenepropylene)						
star polymers	Cyclohexane	–	61	8.06	CLS	928
	Methylcyclohexane	–	13.5	2.91	CLS	
	Tetrahydrofuran	–	4.2	11.1	CLS	
	Methylcyclohexane-d_{14}	–	61	8.24	SANS	928
		–	13.5	3.03	SANS	
		–	4.2	5.5	SANS	

TABLE 14. POLY(ACRYLICS)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(acrylamide)	Dimethylsulfoxide	25	1100	1.3	CLS	827
	Ethanolamine	25	1500	7.3	CLS	827
	Water	25–60	390	217 × (1-(235/T))	CLS	308
		25	4700	0.64	CLS; (J cm^6/g^2)	93
			1100–2000	4.9–5.4	5V; CLS	827
		25	1400	1.8	5V; CLS; $M_w/M_n = 2.5$ p dependence given in Ref.	832
	Aq. NaCl 0.1 M	4	33	3.3	CLS	974
		9.5		4.5		
		18		5.2		
		25		6.1		
		28		6.0		
		39		6.5		
		49		6.8		
		60		6.8		
		4.5	220	2.0	CLS; $M_w/M_n = 1.7$	
		14		2.4		
		24.5		3.0		
		25		2.8		
		34.5		3.0		
		44		3.1		
		55.5		3.2		
		2.5	6700	0.3	CLS; $M_w/M_n = 2.2$	
		6.5		0.5		
		10		0.7		
		18.5		1.1		
		26.4		1.3		
		30		1.8		
	Aq. NaCl 0.2 M	20	255–2750	3.06–2.81	$M_z/M_w = 1.44$–1.73	833
	Aq. NaCl 0.5 M	25	900–1400	5.3–7.7	6V; CLS	827
	Aq. NaCl 1 M	25	2330	3.66		1008
			1710	3.14	Deuterated monomer	
			2130	0.43	Microemulsion	
			3670	0.29	Microemulsion, deuterated	
Poly(acrylamide-*co*-3-(2-acrylamido-2-methylpropanedimethylammonio)-1-propanesulfonate) (AMPDAPS)						
AMPDAPS 1 mol%	Aq. NaCl 1 M	25	3200	2.67	CLS	1010
5 mol%			12400	1.76		
9.6 mol%			7000	1.49		
10 mol%			21500	1.69		
27.7 mol%			8200	1.33		
38.9 mol%			15100	0.75		
40 mol%			17500	1.21		
58.8 mol%			5400	0.49		
70.1 mol%			6200	0.48		
100 mol%			3000	0.25		
Poly(acrylamide-*co*-sodium acrylate) (AM)						
100 mol% AM	Water	25	5000	5.1	CLS	1036
				3.7	LALLS	
	Aq. Na_2SO_4 0.1 M		5000	2.5	CLS	
				2.4	LALLS	
83 mol% AM			5290	3.8	CLS	

TABLE 14. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
70 mol% AM			5550	4.8		
55 mol% AM			5960	8.1		
36 mol% AM			6110	12.3		
				9.7	LALLS	
9.4 mol% AM			6450	8.9	CLS	
				7.7	LALLS	
Poly(acrylamide-*co*-sodium acrylate)						
5.6 mol% AM			6160	5.5	CLS	1036
				6.8	LALLS	
3.7 mol% AM	NaBr 0.5 M		6500	2.8	CLS	
Poly(acrylamide-*co*-sodium-3-methacrylamido-3-methylbutanoate)						
(AM)	(NaMAMB)					
0 mol% AM	0.1 M NaCl	25	2600	3.5	CLS	1025
31.7 mol% AM			2000	3.2		
42.7 mol% AM			4200	3.3		
58.1 mol% AM			6600	3.5		
70.4 mol% AM			6900	3.1		
87.9 mol% AM			7200	3.2		
100 mol% AM			8900	3.5		
Poly(sodium-2-(acrylamido)-2-methylpropanesulfonate-*co*-(2-(acrylamido)-2-methylpropyl)trimethylammonium chloride)						
	(NaAMPS)		(AMPTAC)			
0 mol% AMPTAC	Aq. NaCl 1 M	25	6120	0.92	CLS	1021
14.7 mol% AMPTAC			8220	0.49		
36.1 mol% AMPTAC			62000	−0.02		
45.4 mol% AMPTAC			7920	0.59		
51.7 mol% AMPTAC			7680	0.76		
65.7 mol% AMPTAC			4980	0.34		
100 mol% AMPTAC			1470	2.11		
Poly(acrylamide-*co*-sodium-2-(acrylamido)-2-methylpropanesulfonate-*co*-(2-(acrylamido)-2-methylpropyl)trimethyl ammonium chloride)						
	(NaAMPS)			(AMPTAC)		
mol% (NaAMPS)/mol% (AMPTAC)						
0.5/0.5	Aq. NaCl 1 M	25	6030	2.95	CLS	1022
2.5/2.5			2780	1.65		
5/5			5070	2.00		
10/10			5870	1.57		
15/15			6770	1.31		
10/5			3880	1.56		
Poly(acrylamide-*co*-sodium-3-acrylamide-3-methylbutanoate-*co*-(2-(acrylamido)-2-methylpropyl)trimethylammonium chloride)						
	(NaAMB)		(AMPTAC)			
mol% (NaAMB)/mol% (AMPTAC)						
0.5/0.5	Aq. NaCl 1 M	25	3430	3.61	CLS	1023
5.1/5.8			13900	1.66		
7.9/6.2			3450	1.63		
9.7/9.8			10800	2.06		
11.5/14.0			12200	1.75		
13.6/10.7			19400	1.85		
4.5/10.2			11200	1.84		
Poly(acrylamide-co-maleic anhydride)	Aq. Na NO_3	21.5	24.4	4.0	CLS	828
Poly(acrylic acid)	Aq. NaCl 1 M	25	500	2.8	CLS	829
	0.1 M			13		
	0.01 M			88		
	IS: 0.102–0.994	27.5	770	5.95–196	6V; CLS; IS = ionic strength	320
	Dioxane	–	230	−0.1	CLS	970
			420	0.1		
			890	0.2		
			1510	0.2		
			2440	0.0		
			5600	0.3		
	1,4-Dioxane	25	134–1220	28.0–18	5V; CLS; A_2M; (cm^3/g)	202
	0.2 M Hydrogen chloride	20–68	1100	49.9 × (1-(287/T))	CLS	308
sodium salt	Aq. NaCl 0.01 M	25	64	7.98	OS; B; (bar cm^6/g^2)	254
	0.05 M		58.7	1.22	OS;	
	0.1 M		30–644	0.68 (approx.)	6V;	
	1.0 M		58	0.07		

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TABLE 14. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(acrylic acid-*co*-acrylamide-*co*-3-((2-acrylamido-2-methylpropyl)dimethylammonio)-1-propanesulfonate)						
(AA)	(AM)		(AMPDAPS)			
AA/AM/AMPDAPS mol% (in feed)						
90/5/5			3000	2.77	LALLS	1035
80/10/10			5400	2.95		
50/25/25			7900	2.23		
20/40/40			4700	2.23		
90/10/0			4000	5.76		
75/25/0			1900	9.49		
90/0/10			7000	1.49		
75/0/25			8200	1.33		
Poly(*n*-butyl acrylate)	Acetone	20	32–920	5.0–2.9	7V; CLS	525
	Tetrahydrofuran	–	29	4.8	CLS	970
			76	4.5		
			110	5.0		
			200	3.9		
			790	3.4		
Poly(*n*-butyl acrylate)	Tetrahydrofuran		1480	3.3		970
		25	118–245	0.0146–0.0197	4V; CLS	1037
Poly(*n*-butyl acrylate-*co*-methyl methacrylate)						
mass% 79.5/20.5	Acetone	20	22–740	8.0–2.6	7V; CLS	524
56/44			24–710	11.0–4.0	7V; CLS	
29.5/70.5			27–800	9.5–3.6	7V; CLS	
Poly(*tert*-butyl acrylate)	Butanone	25	306–75	2.37–4.27	5V; CLS	512
	Tetrahydrofuran	–	110	4.7	CLS	970
			180	3.7		
			350	3.9		
			700	3.5		
			820	2.5		
Poly(*n*-butyl methacrylate)	Acetone	20	57–490	2.3–1.5	5V; CLS	525
		RT	108–582	3.61–1.69	5V; OS	280
	Carbontetrachloride	20	21633 ± 1700	$A_2 = 1.760 \times 10^{-3}\ M_w^{-0.18}$	CLS	910
			116–613	3.04–1.33	5V; CLS; A_3 reported	
	Butanone	23	250–2580	2.08–1.69	7V; CLS	165,205
			117–666	3.65–1.95	5V; CLS; A_3 reported	280
	Isopropyl alcohol	20	108–506	– 0.7 to 0.39	4V; CLS	279
		23.7	107–650	– 0.15 to 0.1	5V	
		25.0	107–653	0.03 to 0.2	5V	
		31.0	107–653	0.54 (approx.)	5V	
		45.0	109–262	1.44 (approx.)	3V	
	Methyl ethylketone	25	21633 ± 1700	$A_2 = 2.585 \times 10^{-3}\ M_w^{-0.21}$	CLS	910
Poly(*tert*-butyl methacrylate)						
	Butanone	25	27.7	3.81	CLS; $A_2 = 1.25 \times 10^{-3}\ M_w^{-0.115}$; anionic polymerization	912
			66.0	3.55		
			190	3.14		
			414	2.73		
			835	2.61		
			1070	2.56		
			34.2	5.63	CLS; $A_2 = 7.55 \times 10^{-3}\ M_w^{-0.25}$; free-radical polymerization	912
			68.3	5.23		
			138	4.53		
			242	3.34		
			342	3.01		
			611	2.79		
			1550	2.44		
Poly(butyl methacrylate-*co*-methyl methacrylate)						
50/50 mass%	Acetone	20	70–430	3.2–2.2	5V; CLS	525
Poly(*tert*-butylphenyl methacrylate)						
Poly(carbethoxyphenyl-methacrylamide)	Dimethylformamide	20	995	2.2	CLS	267
	DMF/formamide 90/10			1.2		
	Ethyl acetate			6.2		
Poly(cetyl methacrylate)	Heptane	25	200–1070	3.38–1.9	9V; CLS	263

TABLE 14. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(1,1-dihydroperfluorobutyl acrylate)						
	Benzotrifluoride	25	2000–38000	0.2 (approx.)	10V; CLS	203
	Methyl perfluorobutyrate		5200–14100	0.5	2V; CLS	203
Poly(ethyl acrylate)	Acetone	20	55–860	5.0–3.1	6V; CLS	525
		28	320–8000	10.52 (approx.)	10V; OS; B; (bar cm^6/g^2)	208
		30	145–691	14.6 (approx.)	13V; OS; B; (bar cm^6/g^2)	233
			132–691	18.1–14.1	7V; OS; B; (bar cm^6/g^2)	271
			145–711	13.3 (approx.)	5V; OS; B; (bar cm^6/g^2)	270
	Tetrahydrofuran	–	69	8.1	CLS	970
			180	4.7		
			230	5.3		
			420	4.1		
			810	3.8		
			1040	4.3		
Poly(ethyl acrylate-*co*-methyl methacrylate)						
50/50 mass%	Acetone	20	220–660	4.3–3.5	5V; CLS	525
	Dimethylformamide	20		see Ref.	CLS	515
	Butanone	20		see Ref.	OS	515
Poly(2-ethylhexyl acrylate)	Tetrahydrofuran	–	210	2.6	CLS	970
			280	2.4		
			400	2.1		
			490	2.4		
			880	1.7		
			1340	2.0		
	Toluene	37	660–72	$A_2 = 0.0045\ M^{-0.25}$	4V; OS	586
Poly(ethyl methacrylate)	Ethyl acetate	35	926–11620	1.45–1.01	9V; CLS	471
				1.38–1.20	Calculated acc. to Orofino, Flory	
	Butanone	23	200–2630	4.9–1.83	9V; CLS	162
Poly(*n*-hexyl acrylate)	Tetrahydrofuran	–	100	4.7	CLS	970
			170	4.5		
			320	3.4		
			590	3.2		
			1040	2.4		
Poly(*n*-hexyl methacrylate)	Butanone	23	640–4050	1.77–1.17	8V; CLS	295
Poly(*i*-butyl acrylate)	Tetrahydrofuran	–	200	4.5	CLS	970
			490	3.6		
			780	3.0		
			1000	3.0		
Poly(isopropyl acrylate)						
highly isotactic	Bromobenzene	60	74.3–509	6.21–3.52	6V; OS	429
moderately isotactic	Bromobenzene	60	109–46	5.83–4.12	4V; OS	
syndiotactic	Bromobenzene	60	142–564	7.06–3.71	4V; OS	
atactic	Benzene	25	71.7–703	7.27–3.60	5V; OS	
Poly(isopropylacrylamide (IPAM)/Poly(acrylamide) copolymer						
IPAM = 100 mol%	Water	–	3000	2.94	CLS	1006
IPAM = 85 mol%			3100	3.01		
IPAM = 70 mol%			4500	3.31		
IPAM = 55 mol%			3900	3.46		
IPAM = 40 mol%			2200	3.62		
IPAM = 0 mol%			5200	3.78		
Poly(*N*-isopropyl-methacrylamide)	Water	20	–	$A_2 = 5.9 \times 10^{-3}\ M_w^{-0.25}$	CLS	1005
Poly(*n*-lauryl acrylate)	Tetrahydrofuran	–	95	6.5	CLS	970
			260	2.3		
			600	1.6		
			1000	0.9		
Poly(lauryl methacrylate)	*n*-Butyl acetate	23	260–3600	1.10–0.47	8V; CLS	205
Poly(methacrylamide)	Water	22–56	320	4.49 (1 – (279/T))	CLS	308
Poly(methacrylic acid)	Hydrochloric acid	27–53	590	14.8 ((329/T) – 1)	CLS; T in K	308
	0.02 M					
	0.1 M		146–682	1.76 (approx.)	3V; OS	240
	0.114 M	25	200–427	16.9 (approx.)	5V; CLS	240
	1.0 M		84–169	2.2–0.4	3V; OS	240
	LiBr in ethanol 0.125 M	25	42–383	5.2–2.3	4V; CLS	249
		30	34–421	4.9–1.8	5V; OS	249

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TABLE 14. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
	Methanol	20–40	517 (approx.)	2.42 (approx.)	3V; CLS	346
	0.05 M NaCl	20–60	505 (approx.)	1.465–0	4V; CLS	346
	0.5 M NaCl	20–40	510	0.96–0.1	3V; CLS	
	Water + x M NaCl					
	$x = 0.18$–0.0007	25	196	0–780	5V; CLS	355
Poly(methyl acrylate)	Acetone	20	77–880	4.5–2.8	5V; CLS	525
		25	280–2500	4.2–2.4	17V; CLS	246,247
	Ethyl acetate	35	362–1480	1.92 (approx.)	5V; CLS	482
	Butanone/isopropanol 58/42	20	290–1720	0.1–0.06	8V; CLS	246
	Tetrahydrofuran	–	49	4.1	CLS	970
			69	4.7		
			100	3.8		
			260	3.7		
			430	3.0		
			570	4.1		
Poly(methyl acrylate-*co*-chlorinated rubber) 86.9/13.1 *graft* copolymer	Butanone	35	33940	0.079	CLS	478
Poly(methyl acrylate-*co*-methyl methacrylate) 50/50 mass%	Acetone	20	80–525	5.5–3.5	10V; CLS	525
Poly(methyl butacrylate)	Butanone	30	75.2–4280	2.77–1.00	10V; CLS	445
Poly(methyl ethacrylate)	Butanone	30	46.1–1910	4.78–1.72	10V; CLS	445
Poly(methyl methacrylate)	Acetone	20		See Ref.	Small angle X-ray scat. varying tacticities measured	
			0.2–510	30–1.5	10V; small angle X-ray scat.	492
			23–5300	12.2–1.55	10V; SE; B; (bar cm^6/g^2)	52
			200–6550	2.25–0.68	3V; SE	15
		22	550	1.9	CLS; identical samples;	469
			510	1.5	Small angle X-ray scat.	469
		25	0.203–0.385	−24.2 to 113.0	4V; B; (bar cm^6/g^2); M_n	408
			0.691–2.480	627–47.5	3V; B; (bar cm^6/g^2); M_n	408
		–	341–7800	12	CLS	842
	Acetonitrile	20	0.613–20	−4.4 to 6.5	8V; CLS	915
		25	0.613–50.2	−3.6 to 8.0	9V; CLS	
		30	0.613–496	−3.0 to 9.5	11V; CLS	
		35		−2.3 to 11	11V; CLS	
		40		−1.9 to 12	11V; CLS	
		44 (θ)		−1.5 to 13	11V; CLS	
		47		−1 to 14	11V; CLS	
		50		−0.5 to 15	11V; CLS	
Poly(methylmethacrylate)	Acetonitrile	55		0–16.5	11V; CLS	915
		28–42	20–496	−1 to −0.08	17V; CLS	
	Acetonitrile/1-butanol $\varphi_{MeCN} = 0.3$–0.9	25	169	−0.89 to 2.24	7V; CLS MeCN = acetonitrile	916
	Acetonitrile/pentyl acetate $\varphi_{MeCN} = 0$–0.95			−0.76 to 2.58	11V; CLS	
	Acetonitrile/1-propanol $\varphi_{MeCN} = 0.3$–0.9			−0.16 to 2.30	7V; CLS	
	Acetonitrile/methanol $\varphi_{MeCN} = 0.5$–1			−0.97 to 0.38	7V; CLS	
	Acetonitrile/carbon tetrachloride	25	120		10V; CLS MeCN = acetonitrile	823
	$\varphi_{MeCN} = 0\%$			0.6		
	$\varphi_{MeCN} = 40\%$			5.9		
	$\varphi_{MeCN} = 100\%$			−0.65		
atactic			51 ± 4.3	8.2 ± 0.61	CLS; B; (bar cm^6/g^2)	419
			3670 ± 50	2.12 ± 0.05	CLS; B; (bar cm^6/g^2)	419
				$2.9 \times 10^{-3}\ M^{-0.22}$	CLS	273
			68–1370	$(7.1$–$3.5) \times 10^3$	7V; CLS; 981(cm^{12}/g)	232
			240–4470	2.3–1.1	5V; CLS	264
			27.7–7800	0.975–0.29	12V; CLS; (J cm^3/g^2)	99
			1450	19	CLS	202
			2.69–2500	0.97–390	13V	275
			86–1050	2.90–1.58	7V; OS	202
			92–798	1.89–1.52	9V; CLS	202

TABLE 14. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
			486–6620	2.38–0.82	4V; CLS	55
		30		$2.63 \times 10^{-3} M^{0.78}$	OS; A_2M; (cm^3/g)	274
			161	1.85	OS	494
atactic	Acetonitrile	44	0.594–204	−1.5 to 12.5	10V; CLS	917
		47	312	0.100	CLS	
			482	0.114		
			765	0.107		
		50	312	0.175	CLS	
			482	0.184		
			765	0.177		
		55	312	0.256	CLS	
			482	0.279		
			765	0.292		
	Acetone	25	0.594–1580	1.25–47.3	19V; CLS	
	Chloroform	25	204	5.66	CLS	
			312	5.06		
			482	4.39		
			765	3.91		
			1580	3.31		
	Nitroethane	30	204	3.32	CLS	
			312	2.97		
			482	2.61		
			765	2.32		
			1580	1.99		
isotactic	Acetone	25	0.789–1930	1.23–17.2	16V; CLS; M_w/M_n see Ref.	918
	Acetonitrile	35	688	0.20	CLS; M_w/M_n see Ref.	
		45		0.40		
		55		0.57		
	Chloroform	25	155	6.92	CLS; M_w/M_n see Ref.	
			313	6.12		
			380	5.57		
			688	4.91		
			946	4.39		
			1930	3.80		
	Nitroethane	30	155	4.33	CLS; M_w/M_n see Ref.	
			313	3.66		
			380	3.47		
			688	3.03		
			946	2.64		
			1930	2.33		
	Tetrahydrofuran	–	90	5.0	CLS; M_w/M_n given	956
		–	305	3.9		
		–	506	3.2		
unfractionated	Acetonitrile	30.5	162–647	28.8–88	12V; OS; A_2 M; (cm^3/g)	274
fractionated			64–726	12.3–95	5V; OS; A_2 M; (cm^3/g)	274
	Acetonitrile	40	10	−0.1	SAXS	492
		43.3	10	0	SAXS	492
		50	10	0.2	SAXS	492
	Benzene	25	0.188–0.935	−222.2 to 8.9	8V; B; (bar cm^6/g^2); M_n	408
		30	433	2.63	OS	494
	Butanone	25	5–52.8	7.38–2.18	6V; CLS	418
	Butanone/isopropanol		75.5	2.70	CLS	418
	58.2/41.8	4–36	210	0.001–0.369	9V; CLS; (J cm^3/g^2)	13
	55/45	4–36	210	−0.128 to 0.293	9V; CLS; (J cm^3/g^2)	13
	50/50	8–40	210	−0.365 to 0.237	9V; CLS; (J cm^3/g^2)	13
fractionated	Butanone/isopropanol					
	46.8/53.2	18–40	210	−0.248 to 0.174	7V; CLS; (J cm^3/g^2)	13
	Butyl acetate	25	486–6620	0.96–0.36	4V; CLS	55
		−18 to 60	210	0–1.20	6V; CLS	210,215
	Butyl chloride	14–48	210	−0.242 to 0.088	10V; CLS; (J cm^3/g^2)	13
	Carbon tetrachloride	25	0.202–0.755	40.0–6.7	3V; B; (bar cm^6/g^2); M_n	408
			1.160–4.625	23.2–0.0	4V; B; (bar cm^6/g^2); M_n	408
	Chlorobutane	4–48	30	−1.6 to 0.365	12V; CLS	210,215
		28–48	1100	−0.3 to 0.325	7V; CLS	210,215
		28–48	4600	−0.23 to 0.31	7V; CLS	210,215
		35.6	200–6550	−0.24 to 0.07	3V; SD	15

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TABLE 14. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
	1-Chlorobutane/2-butanol $\varphi_{bu} = 0$–0.7991	40	2400–2690	0–1.28	12V; CLS	909
	1-Chlorobutane/2-methoxy-ethanol	40 ± 0.2	2330–2640	0–116	14V; CLS	913
	Chloroform	25	830–1370	(12–10) × 10^3	2V; CLS; 981 (cm^3/g)	232
			0.204–0.379	− 53.9 to 81.0	4V; B; (bar cm^6/g^2); M_n	408
			0.710–5.240	49.6–21.9	3V; B; (bar cm^6/g^2); M_n	408
	Chloroform/butyl acetate % butyl acetate	25			CLS; apparent values for M_w	914
	0		172	0.75		
	20		260	0.30		
	40		270	0.31		
	60		294	0.53		
	80		227	0.33		
	90		212	0.51		
	100		172	0.05		
	Dimethylformamide		32.2 ± 0.3	10.9–0.16	OS; B; (bar cm^6/g^2); M_n	419
			2180 60	6.2–0.08	OS; B; (bar cm^6/g^2); M_n	419
	Dioxane		486–6620	3.24–1.43	4V; CLS	5
			1220–2000	3.1–3.0	2V; CLS	294
		38	5690	1.82	CLS	908
	α-Chloronaphthalene/toluene $\varphi_{CNA} = 0.72$	38	5690	1.75	CLS	908
	α-Chloronaphthalene/polystyrene ($C_{PS} = 3.76$ g/l)	38	5690	− 0.11	CLS	908
	Ethyl acetate		172–3100	1.9–1.2	3V; OS	196
	4-heptanone	16–62	2100	− 0.76 to 0.52	10V; CLS	210,215
	Isoamyl acetate	20–60	210	− 0.276 to 0.014	11V; CLS; (J cm^3/g^2)	13
	Isopentyl acetate	20–60	2100	− 1.13–0.055	12V; CLS	210,215
unfractionated	Butanone	25	20–1730	(4.02–1.39) × 10^{-2}	7V; CLS; A_2M; (cm^3/g^2); A_3 reported	273
			130–2760	(3.04–1.07) × 10^{-2}	4V; CLS	273
			760–1400	(5.4–4.2) × 10^3	4V; CLS; 981 (cm^{12}/g)	232
			1220	1.6	CLS	294
			34–9800	2.78–1.46	10V; CLS	253
			410–3260	1.92–1.66	7V; CLS	163
	Butanone/isopropyl alcohol					
	55/45	23	410–3260	0.08–0.02	6V; CLS	163
	58.2/41.8	4–36	210	0.005–1.44	9V; CLS	210,215
	55/45	4–36	210	− 0.57 to 1.14	10V	210,215
	50/50	8–50	210	− 1.56 to 0.91	10V	210,215
	46.8/53.2	8–40	210	− 1.02 to 0.68	8V	210,215
linear	Butanone	40	45.9–940	4.43–2.34	8V; CLS; OS results also given in Ref.	570
randomly branched			48.9–4490	5.02–1.95	12V; CLS; OS results also given in Ref.	570
			1470–6990	1.44–0.61	3V; CLS; OS results also given in Ref.	570
unfractionated	Nitroethane	25	22–1710	5.51–1.68	6V; CLS; A_3 reported	273
			114–564	3.38–1.73	2V; CLS; A_3 reported	273
				0.67 × 10^{-3} $M^{-0.257}$	CLS	273
			1220–2000	2.6 (approx.)	2V; CLS	294
	Tetrahydrofuran		47–9300	5.3–1.1	6V; CLS	121
	Toluene	30.5	12.3–20	5.7–8.5	3V; OS	274
Poly(methyl methacrylate-*co*-chlorinated rubber) (45.9/54.1)						
graft polymer	Butanone	35	8150	0.36	CLS	478
Poly(methyl methacrylate-*co*-ethylene dimethacrylate)						
randomly branched	Butanone	40		See Ref.	11V	549
Poly(methyl methacrylate)/90% poly(styrene) (side chains)						
graft polymer	Benzene	–	1800	0.52	1V; CLS	824
Poly(*n*-octyl methacrylate)	Butanone	23	3260–12520	0.77–0.35	7V; CLS	209
Poly(*N*-phenylmethacrylamide)	Acetone	20	100–3200	4.1–3.2	CLS	229
Poly(phenyl acrylate)	Ethyl lactate	11.55	1460	0.0	CLS; $M_n = 0.90 \times 10^6$ g/mol	919
		13.65		6.0		

TABLE 14. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
		16.65		29.2		
		22.75		51.7		
		31.65		68.7		
		45.45		94.0		
		59.45		117		
		74.45		125		
		89.85		141		
		106.15		150		
Poly(*n*-propyl acrylate)	Tetrahydrofuran	–	110	4.0	CLS	970
			200	5.3		
			440	3.6		
			630	3.3		
			800	3.0		
			980	2.9		
			1080	2.9		
Poly(sulfodecylmethacrylate)	Aq. NaCl 0.1 M	25	734	1.91	CLS; $M_w/M_n = 2.1$	1009
Poly(sulfoethylmethacrylate)	Aq. NaCl 0.1 M	25	306	6.41	CLS; $M_w/M_n = 1.5$	
Poly(sulfohexylmethacrylate)	Aq. NaCl 0.1 M	25	494	5.61	CLS; $M_w/M_n = 2.0$	
Poly(sulfooctylmethacrylate)	Aq. NaCl 0.1 M	25	701	2.03	CLS; $M_w/M_n = 1.8$	

TABLE 15. POLY(VINYLS)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(acrylonitrile)	Dimethylformamide	20	1041	4.217	CLS	971
			1132	4.064		
			1235	3.988		
			1371	3.589		
			1778	3.631		
			2367	3.456		
			2973	3.291		
			3078	3.311		
				$A_2 = 9.44 \times 10^{-3}\ M_w^{-0.219}$		
		20	98–120	22.9–21.4	CLS	603
				$4.94 \times 10^{-2}\ M^{-0.24}$	2 Different wavelengths used	
		20	9–69	32.2–7.0	5V; OS; (cm^4/g)	248
			91–762	$2.43 \times 10^{-2}\ M^{-22}$	CLS	95
		25	43–298	21 (approx.)	7V; OS	199
			27–159	16–20	OS	200
		25	103	18.3	CLS	938
		25–40	33–101	19.1 (approx.)	13V; OS	74
		35	28–575	$2.74 \times 10^{-2}\ M^{-0.24}$	9V; M	201
highly isotactic, 73.4% iso triades						
	HNO_3; 60–70 mass%	25	150–174	0.073–0.254	4V; CLS	973
	HNO_3; 70 mass%	25	158–263	0.252–0.420	5V; CLS	
	Dimethylsulfoxide	25	155–2620	–0.002 to 0.153	4V; CLS	
Poly(acrylonitrile)/vinylidene chloride copolymer, 59 wt.% AN	Dimethylacetamide	25	42	0.132	CLS; $A_2 = 5.77 \times 10^{-3}\ M_w^{-0.14}$	972
			62	0.121		
			90	0.120		
			140	0.120		
			204	0.115		
			270	0.110		
			348	0.095		
			500	0.090		
	Butyrolactone	25	200	0.119	CLS	
	Dimethylformamide	25	196	0.137	CLS	
	70 mass% HNO_3	25	–	0.0075	CLS	
Poly(*n*-butoxyethylene)	Butanone	25	215–861	2.3–0.7	2V; CLS	149
Poly(ethoxyethylene)	Butanone	25	347–409	3.4–2.3	2V; CLS	149
Poly(isopropoxyethylene)	Butanone	25	536–894	1.9–1.7	2V; CLS	149
Poly(methoxyethylene)	Butanone	25	13–450	3.89 (approx.)	12V; CLS	149

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TABLE 15. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(2-methyl-5-vinylpyridine)	Butanone	25	67–1200	2.8–1.2	13V; CLS	432
Poly(1-trimethylsilyl ethylene)	Cyclohexane	25	2130–590	4.85–12.00	6V; CLS	553
Poly(vinyl acetate)	Acetone	30 (approx.)	27–845	8.80–3.34	9V; CLS	330
			343–722	3.66–3.50	CLS	331
			78–660	6.5–2.5	18V; CLS	289
	Methanol	25	2360–422900	7.50–0.172	9V; DLS	831
branched	Benzene	30	258–45.2	4.76 (approx.)	10V; OS	453
			5203–331	0.208–2.066	8V; CLS	
	Methanol	30 (approx.)	370–620	3.5 (approx.)	2V; CLS	331
linear		30 (approx.)	1620–1900	1.14–1.3	2V; CLS	333
branched		36 (approx.)	2920	0.99	3V; CLS	333
		36 (approx.)	3580	1.06		
	Butanone	25	870–3420	3.51–2.43	11V; CLS	111
			58–367	10 (approx.)	8V; OS; B; (bar cm^6/g^2)	112
			93–1100	5 (approx.)	CLS; B; (bar cm^6/g^2)	112
		30	40–1268	7.00–2.95	12V; CLS	330
			300–772	3.93–2.90	CLS	331
linear	Butanone	36 (approx.)	440–3580	3.2–2.74	10V; CLS	333
branched		36 (approx.)	430–12.60	2.99–0.66	9V; CLS	333
	Tetrahydrofuran	25	13.8–47.7	0.160–0.280	3V; CLS	1037
linear	Trichlorobenzene	36 (approx.)	1380–3760	2.29–1.75	12V; CLS	333
				$5.94 \times 10^{-3}\ M^{-0.230}$		333
branched	Trichlorobenzene	36 (approx.)	1610–4110	1.69–1.18	19V; CLS	333
Poly(vinyl alcohol)	Water	30	180–196	3.9–5.2	2 V; CLS	282,283
		73.5	245	1.119	CLS	282,283
		80	100–458	2.189–0.775	5V; CLS	282,283
Poly(vinyl bromide)	Tetrahydrofuran	RT	6–76	6–0	14V; CLS	287
			104–270	2.7–2.0	2V; CLS	
Poly(vinyl butyral)	Dioxane	37	541–89.5	5.53–10.3	8V; OS	597
			208–73.9	9.15–11.3	5V; OS	
			181–57.5	9.45–12.2	7V; OS	
Poly(vinyl-*iso*-butyral)			462–207	6.9–9.6	6V; OS	
Poly(vinyl caprolactam)	Dimethylformamide	20	377–1670	0.337–0.662	8V; CLS	1024
Poly(vinyl carbanilate)	Cyclohexanol	40–90	1100	– 1.2 to 1.2 (approx.)	8V; CLS	504
	Cyclohexanone	15–90		2.2–1.9 (approx.)	6V; CLS	504
	Diethylene glycol/ diethyl ether	5–95		2.7–1.5 (approx.)	7V; CLS	504
	Diethyl ketone	20–90		– 0.5–1.8 (approx.)	8V; CLS	
	Dioxane	20	7.2–183	$(3.4–1.71) \times 10^{-4}$	11V; CLS; (cm^2/g^2)	401
		5–95	1100	2.7–2.2 (approx.)	7V; CLS	504
	Pyridine	10–90		2.5–2.0 (approx.)	4V; CLS	504
Poly(*N*-vinylcarbazole)	Benzene	25	4550–136	0.57–1.32	CLS	457
			1630	0.61	CLS	499
			2290	0.54	CLS	499
	Dioxane	37	42.4–1070	2.35–0.91	8V	499
				$4.26 \times 10^{-3}\ M^{-0.27}$		
	Toluene	37–43	298	0–0.27	3V; OS	499
		37–43	522	0.01–0.26	3V; OS	499
Poly(vinyl chloride)	Cyclohexanone	25	118 (119)	11	A; (CLS)	602
			59 (50)	12 (18)	A; (CLS)	602
		30	32.3–97.4	3.7×10^2	OS; (cm^4/g)	108
		30–70	90	$3.9 \times (1 + (1613/T))$	CLS; M_v	308,309
		25	48.8	12.8×10^{-4}	OS;	451
			44.9	16.2×10^{-4}	OS; more values for seven different PVC-preparations given in Ref.	451
			24.8	6.85	OS; (J cm^3/g^2)	75
			37–110	2.5 (approx.)	5V; OS; (J cm^3/g^2)	75
			122	15	A	602
			56 (53)	14 (17)	A; (CLS)	602
		29	136	9.03	OS	478
Poly(vinyl chloride-*co*-butylacrylate) comonomer content (mol%)						
16.6	Tetrahydrofuran	25	173–302	0.0518–0.0649	4V; CLS	1037
40.0			205–632	0.0156–0.0244	3V; CLS	

TABLE 15. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(vinyl chloride-*co*-*n*-butyl ether)						
comonomer content (mol%)						
9.2	Tetrahydrofuran	25	30–98	0.077–0.115	3V; CLS	1037
Poly(vinyl chloride-*co*-methylacrylate)						
comonomer content (mol%)						
25.5	Tetrahydrofuran	25	221–476	0.0503–0.0570	3V; CLS	1037
41.6			121–891	0.0316–0.0803	4V; CLS	
64.0			379–2510	0.014–0.033	4V; CLS	
Poly(vinyl chloride-*co*-vinyl acetate)						
comonomer content (mol%)						
11.4	Tetrahydrofuran	25	29.5–96.7	0.089–0.119	4V; CLS	1037
25.0			21–185	0.050–0.148	4V; CLS	
40.0			43–179	0.064–0.118	4V; CLS	
66.0			100–708	0.012–0.028	4V; CLS	
Poly(vinyl chloride-*co*-vinylidene chloride)						
comonomer content (mol%)						
15	Tetrahydrofuran	25	50–223	0.0449–0.0910	4V; CLS	1037
42			79–187	0.0260–0.0396	4V; CLS	
Poly(vinylchloride-*co*-poly(ethyl methacrylate))						
84.55%/15.45%	Tetrahydrofuran	29	161	4.80	OS	478
Poly(vinyl chloride-*co*-poly(methyl methacrylate))						
45.1%/54.9% *graft* copolymer	Tetrahydrofuran	29	302	9.55	OS	478
Poly(vinyl methyl ether) see Poly(methoxyethylene)						
Poly(2-vinylpiridine)	Aq. H_2SO_4; 0.01 M	25	288	11	CLS	977
	0.1 M			2.2		
	0.25 M			2.1		
	0.5 M			1.7		
	0.75 M			1.3		
	1.0 M			1.4		
	5.1 M			3.0		
	10.3 M			3.0		
Poly(4-vinylpiridine)	Ethyl alcohol	RT	102–1850	4.13 (approx.)	10V; CLS	286
Poly(vinylpyrrolidone)	Cyclohexanone	25	24.5–37.9	2.6 (approx.)	3V; OS	191
	Ethanol	20	63–500	1.15–5.47	8V; CLS	1024
	Water	20	310	4.2	$M_w/M_n = 1.5$	985
		25	11.6–75.4	3.39 (approx.)	12V; CLS	191
			19.5–933	64.7–2.52	6V; CLS	
			310	4.2	CLS; $M_w/M_n = 1.5$	942
Poly(vinylpyrrolidone-*co*-vinylcaprolactam)						
1:1 stat.	Ethanol	20	140–557	0.457–1.89	8V; CLS	1024

TABLE 16. POLY(STYRENES)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(*p*-bromostyrene)	Toluene	30	3860–630	0.076–0.113	5V; CLS	516
Poly(*p*-*tert*-butylstyrene)-*b*-poly(dimethylsiloxane)-*b*-poly(*p*-*tert*-butylstyrene)						
anionically synthetized						
diblock	Toluene	30	92.7	4.23	CLS; M_w/M_n given	966
triblock			115	3.24		
			291	2.56		
			829	2.10		
Poly(*m*-chlorostyrene)	Butanone	30	1064–322	1.44–1.78	5V; CLS	595
				$A_2 = 1.54 \times 10^{-3} M^{-0.17}$		
	Toluene	35	1075–145	2.42–5.08	6V; CLS	
				$A_2 = 26.5 \times 10^{-3} M^{-0.33}$		
Poly(*o*-chlorostyrene)	*n*-Butyl acetate	5	415	0.570	OS	585
		30	403	0.449	OS	585
		55	401	0.337	OS	585
	Chlorobenzene	30	406	1.72	OS	585

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TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
	Toluene	25	1005–142	1.78–2.70	6V; CLS	527
		30	407	1.73	OS	585
		35	1087–435	2.18–2.70	6V; CLS	500
			1429–2350	1.96–3.00	4V; CLS	500
				$5.22 \times 10^{-3} M^{-0.23}$		
Poly(*p*-chlorostyrene)	Butanone	21	480–4220	1.29–0.752	10V; CLS	300
	n-Butyl acetate	5	180	2.96	OS	585
		30	176	2.39	OS	585
		50	165	1.90	OS	585
	Chlorobenzene	30	316	2.17	OS	585
	Toluene	30	472–192	0.96–1.69	5V; OS	609
				$A_2 = \mathrm{K} M^{-0.6}$	A_3 given in Ref.	
			1370–214	0.570–0.895	6V; CLS	516
			206	0.825	CLS	516
			463	0.625	OS	585
Poly(2,5-dichlorostyrene)	Dioxane	20	590–19600	0.88–0.20	7V; CLS	250
Poly(*p*-methoxystyrene)	Butanone	25	870–220	$A_2 = \mathrm{K} M^{-0.20}$	CLS; 6V	610
	Toluene	30	850–220	$A_2 = \mathrm{K} M^{-0.40}$	CLS; 6V	
Poly(α-methylstyrene)	Benzene	25	1.45	17.4	CLS; B; (bar cm^6/g^2)	407
			6.76	13.7	CLS; B; (bar cm^6/g^2)	407
		30		See Ref.		522
	n-Butylchloride	–	6.9–3540	0.60–2.97	15V; CLS; M_w/M_n, M_z/M_w given	997
				$A_2 = 3.11 \times 10^{-3} M_w^{-0.255}$	$M_w/M_n = 1.03–1.15$	
Poly(α-methylstyrene)	Cyclohexane	20	60–313	$A_2 = 2.4 \times 10^{-7} M_w^{0.50}$	CLS	935
		24	61–1020	$A_2 = 6.0 \times 10^{-9} M_w^{0.72}$	CLS	
		30	180–1080	$A_2 = 5.5 \times 10^{-10} M_w^{0.84}$	CLS	
monodisperse		40	1220	0.218		484
		50	1220	0.509		484
		50–32	310	0.575 to −0.142	7V; CLS; A_2-values for mixtures of these samples given in Ref.	484
			17.3	0.461 to −0.116	7V; CLS; A_2-values for mixtures of these samples given in Ref.	484
			3000	0.406 to −0.109	7V; CLS; A_2-values for mixtures of these samples given in Ref.	484
Poly(α-methylstyrene)	Toluene	25	1.45	49	CLS; B; (bar cm^6/g^2)	407
			6.76	34	$M_{s,D}$; SE; B; (bar cm^6/g^2)	407
			20.1	8.05	CLS	998
			24.7	7.03		
			59.0	5.54		
			63.1	5.29		
			75.9	5.67		
			118	4.63		
			177	4.55		
			305	3.73		
			313	3.56		
			665	2.90		
			903	2.71		
			1140	2.56		
				$A_2 = 1.15 \times 10^{-2} M_w^{-0.273}$		
		34.5	204–7470	2.08	CLS	845
		37	278–46.9	4.2–5.9	8V; OS	584
Poly(α-methylstyrene-*co*-acrylonitrile)						
46 mol% acrylonitrile	Tetrahydrofuran	–	45–417	7.70 (mean)	19V; CLS	819
Poly(α-methylstyrene-*co*-styrene)						
50/50	Toluene	37	253–106	5.9–5.2	4V; OS	584
Poly(*o*-methylstyrene)	Toluene	30	430–102	2.64–4.80	7V; CLS	588
				$A_2 = 3.89 \times 10^{-2} M^{-0.38}$		
Poly(*p*-methylstyrene)	Butanone	30	1210–391	1.00–1.28	4V; CLS	517
	Cyclohexane	30	847–263	1.56–1.92	3V; CLS	517
	Dichlororethane	30	1180–500	1.70–2.28	3V; CLS	517

TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
	Diethyl succinate	60–16	1970	0.589 to −0.026	8V; CLS	517
			896	0.819 to −0.001	8V; CLS	
			762	0.900 to −0.013	8V; CLS	
			684	0.830 to −0.046	8V; CLS	
	Toluene	30	1800–192	2.18–4.37	9V; CLS	517
Poly(styrene)	Benzene	20	1330–7100	3.6–3.3	4V; CLS; A_2M; (cm^3/g)	514
			2700	3.4	CLS; A_2M; (cm^3/g); A_3 given in Ref.	514
		–	0.207–3100	54–3.1	9V; CLS	801
		20	150	3.95	CLS;	804
			35.6–1750	7.24–2.22	$M_z/M_w = 1.01–1.55$	833
		25	16700	2.23	CLS; A_2M; (cm^3/g)	805
			12500–19200	1.52–1.12	4V; CLS; $A_2 = 2.9 \times 10^{-3} M_w^{-0.59}$	807
			8760–56800	1.82–1.09	6V; CLS	808
				$A_2 = 1.05 \times 10^{-2} M_w^{-0.276}$	CLS	1038
		30	589	3.56, 3.30	CLS	818
		25	10.3	11.3	CLS; A_3 given in Ref.	922
			19.8	8.80		
			97.9	5.78		
			200	4.72		
			845	3.33		
			1380	2.85		
			3150	2.29		
			8560	1.80		
			14500	1.49		
			20100	1.38		
	Bromobenzene	20	35.5–1750	6.38–2.15	$M_z/M_w = 1.01–1.55$	833
	Butanone	20	150	0.0127	CLS	804
		25	–	0.58	CLS; A_2M; (cm^3/g)	806
	2-Butanone	25	50	6.90	CLS	926
			233	1.80	CLS	
			670	1.53	CLS	
			1800	1.46	CLS	
			mixtures			
			1800*/670; $w^a = 0.15–0.85$	1.390–1.496	6V; CLS	926
			1800*/233; $w^a = 0.20–0.63$	1.540–1.740	4V; CLS	
			1800*/50; $w^a = 0.13–0.91$	0.63–1.36	6V; CLS	
		30	233–5840	$A_2 = 4.37 \times 10^{-3} M_w^{-0.277}$	CLS; $M_w/M_n = 1.04–1.16$	995
	Carbon tetrachloride	20	150	3.58	CLS	804
			17.5–1800	2.23–7.69	5V; CLS; $M_w/M_n = 1.04–1.30$	934
		24	2880	2.58	CLS	991
		25	35	6.47	CLS	927
			45	5.73		
			61	5.57		
			186	4.83		
			422	3.52		
			775	2.08		
	Chloroform	25		6.56	CLS	804
	Cyclohexane	15	0.37–40	−2.75 to 10	11V; CLS; M_w/M_n given in Ref.	920
		20		−2.00 to 13	11V; CLS; M_w/M_n given in Ref.	
		25		−1.20 to 15	11V; CLS; M_w/M_n given in Ref.	
		30		−0.45 to 18	11V; CLS; M_w/M_n given in Ref.	
		34.5 (θ)		0.03–21	14V; CLS; M_w/M_n given in Ref.	
			20.7–3150	0–0.1	6V; A_3 given in Ref.	923
		40	0.37–40	0.40–22	11V; CLS; M_w/M_n given in Ref.	920
		44.5	233–5840	$A_2 = 6.30 \times 10^{-4} M_w^{-0.173}$	CLS; $M_w/M_n = 1.04–1.16$	995
		45		0.77–24	11V; CLS; M_w/M_n given in Ref.	920
		50		1.10–26	11V; CLS; M_w/M_n given in Ref.	
	Cyclopentane	23	160	28	CLS; M_w/M_n given in Ref.	924
			260	23		
			400	20		
	1,4 Dioxane	20	150	2.75	CLS	804

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TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
	Decalin (*trans*)	20	390	−0.06	CLS	815
		25		0.25	CLS	815
		30		0.58	CLS	815
		35		0.85	CLS	815
		40		1.11	CLS	815
	Dimethylformamide	25	704	1.9	CLS	938
	Dioctylphtalate	12 (θ)	5000	0	Extrapolated value	930
		22	2050	−0.036	CLS; $M_w/M_n \leq 1.30$	921
	Ethylbenzene	20	50–1800	3.10–8.41	5V; CLS; $M_w/M_n = 1.06$–1.30	934
		25		$A_2 = (0.011 \pm 0.005)\, M_w^{-0.27\pm0.02}$	CLS; $0.1106 \leq M_w \leq 8.48 \times 10^6$	929
	Methyl acetate	30	179.3	−0.235	DLS; $M_w/M_n = 1.07$	811
		43		0	DLS	811
		55		0.22	DLS	811
		70		0.39	DLS	811
		100		0.21	DLS	811
		114		0	DLS	811
		125		−0.35	DLS	811
	1-Phenyldecane	21.6	390	−3.22	CLS	813
		27.0		−0.83		
		31.3		1.54		
		35.0		2.04		
		41.3		3.97		
	Tetrahydrofuran	25	90–20000	1.41–5.91	9V; CLS; $M_w/M_n = 1.04$–1.20	992
			25.75 ± 1.10	10.52 ± 0.20	LALLS; $M_w/M_n = 1.03$	1031
			32.09 ± 1.28	9.07 ± 0.15	LALLS; $M_w/M_n = 1.02$	
		30		$A_2 = (0.0194 \pm 0.004)\, M_w^{-0.286\pm0.02}$	CLS; $0.110^6 \leq M_w \leq 8.48 \times 10^6$	929
	Tetrahydrofuran/15 vol% Poly(dimethylsiloxane)	25	25.75 ± 1.10	-14.7 ± 3.9	LALLS; $M_w/M_n = 1.03$	1031
			32.09 ± 1.28	-17.1 ± 9.1	LALLS; $M_w/M_n = 1.02$	
	Toluene	12	622–4400	2.27	CLS	841
		20	12300	2.18	CLS	810
			23400	1.82		
			24400	2.32		
			29700	1.75		
			35600	1.48		
			40200	1.37		
			150	0.0127	CLS	804
			1.2–424	3.5–21.7	8V; CLS	993
			21.6	7.3	SANS	993
		23	160	450	CLS; M_w/M_n given in Ref.	924
			260	510		
			400	540		
	Toluene	25	20.5	0 (approx.)	CLS; microgel containing; see Ref.	802
			95.4	10		
			9700	159		
			22300	336		
			2960	2.24	CLS; $M_w/M_n = 1.05$	803
			10800–22200	1.90–1.28	9V; CLS; $A_2 = 4.79 \times 10^{-3}\, M_w^{-0.63}$	807
			50–20000	5.94–1.4	11V; CLS; $M_w/M_n = 1.06$–1.30 values for A_3 given in Ref.	814
	Toluene		1810	2.14	CLS	812
			125–2960	3.29	CLS	847
		30	1290	2.35	CLS	809
			2580	1.99		
			6120	1.64		
			11100	1.34		
			15900	1.21		
			586	3.05	CLS	818
		30	6,7–533	3.04–74	10V; OS; M_w/M_n given in Ref. $A_2 = 0.964\, M^{-0.5} - 4.36 \times 10^{-3}$	925
			233–5840	$A_2 = 1.58 \times 10^{-2}\, M_w^{-0.286}$	CLS; $M_w/M_n = 1.04$–1.16	995
	Xylene	20	107 ± 5	4.8 ± 0.1	CLS	1032

TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
linear	Tetrahydrofuran	25	10.5–182	4.62–11.0	4V; CLS; $M_w/M_n = 1.08–1.19$	996
cyclic			11.1–181.5	4.37–9.33	4V; CLS; $M_w/M_n = 1.08–1.20$	
	Xylene	24	120 ± 5	7.0 ± 1	SANS	937
linear	Toluene (deuterated > 99%)	22 ± 5	132	9.1	SANS	931
			143	9.2		
			163	7.7		
			196	6.8		
			215	8.0		
cyclic			127	7.8	SANS	931
			143	7.7		
			163	7.1		
			184	6.6		
			194	7.0		
linear	Toluene	–	10–390	3.9–24.4	8V; CLS	932
	Methyl ethylketone	–		1.48–19.5	7V; CLS	
	Cyclohexane	30-40	110	– 0.368 to 0.446	3V; CLS	
cyclic	Toluene	–	15–227	3.87–31.7	11V; CLS	
	Methyl ethylketone	–		1.69–14.8	11V; CLS	
	Cyclohexane	25–40	40	– 1.27 to 2.0	4V; CLS	932
			105	– 1.17 to 0.932	4V; CLS	
			132	– 1.93 to 0.599	4V; CLS	
			227	– 2.82 to 0.631	4V; CLS	
Mixtures A/B	Toluene	21	36 (A)		4V	851
mass% B: 0–1			1600 (B)	9.3–2.6		
Mixtures C/B			4.9 (C)		8V	851
mass% B: 0–1			1600 (B)	20.0–2.6		
Mixtures D/B			2.3 (D)		7V	851
mass% B: 0–1			1600 (B)	25–2.6		
NBS 705 standard	*trans*-Decalin	20	179.3	– 0.036	CLS; $M_w/M_n = 1.07$	830
		30		0.593		
		40		1.17		
	Toluene	–	179	6.00	CLS; $M_w/M_n = 1.05$	1019,1020
NBS 1479 standard	Toluene	–	50	4.00	CLS; $M_w/M_n = 1.1$	
Toyo Soda 550	Toluene	–	5500	2.09	CLS; $M_w/M_n = 1.2$	
Toyo Soda 2000	Toluene	–	20000	1.61	CLS; $M_w/M_n = 1.3$	
IUPAC-sample			79	6.2	OS	11
IUPAC-sample			233	5.0	OS	11
IUPAC-sample			630	5.1	OS	11
			3.38 ± 0.08	23 ± 3	M from isothermal distillation	463
branched			170	4.9	CLS	312
linear				5.7		312
branched			257	3.3		312
linear				5.4		312
		25	109–5000	3.83 (approx.)	8V; CLS	56
			3200–62	2.9–6.0	5V; CLS	454
atactic			680	4.9×10^{-4}	M_v; (cm^4/g)	562
		40	4000	3.2	CLS	172
		60		2.30		56
	Benzene/ethanol					
	80/20	25	1750	1.75	CLS	56
		60		1.55		56
	70/30	25		0		56
		60		1.1		56
	Benzene/heptane					
	Heptane 0–56 vol.%	35.0 ± 0.2	23.4–2370	3.06–0	9V; CLS; A_2-values corrected for preferential adsorption given in Ref.	440
	Benzene/hexane					
	50/50	25	1750	1.0		56
		60		0.9		56
	40/60	25		0		56
		60		0.65		56
	30/70	60		0		56

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TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
	Benzene/isopropanol					
	Isopropanol 0–39 vol.%	35.0 ± 0.2	2340–4170	3.06–0	8V; CLS	440
	n-Butyl acetate	5	411	1.88	OS	585
		30	404	2.08		
		55	402	2.00		
	Carbon tetrachloride	25	920	2.6	CLS	294
	Chlorobenzene	25	98.4	5.41	OS	404
		30	388	3.77	OS	585
	Chloroform	20	372–2800	8.0–6.6	4V; CLS	176
	Chloroform/methanol					
	100/0	25	330	7.5	CLS	176
	90/10			5.4		
	80/20			1.9		
	Chloroform	30	100	5.4–5.1	5V; CLS; $A_{2,S}$, $A_{2,H}$ given in Ref.	528
isotactic	1-Chloro-4-methylbenzene	RT	410–1050	2.4 (approx.)	2V; CLS	328
	1-Chloronaphthalene	130	34.5 (approx.)	1.33×10^2 (approx.)	OS; $A_2 M$/(cm^3/g); see Ref. for various extrapolation methods	523
isotactic	*o*-Dichlorobenzene	RT	196–264	2.9–1.9	3V; OS	328
atactic	*n*-Butylchloride	15	360	2.48	CLS	967
			739	2.00		
			1270	1.81		
			3920	1.35		
	4-*tert*-Butyltoluene	50	5.37	5.12	CLS	967
			20.2	4.18		
			364	2.03		
			743	1.74		
			1280	1.55		
			3950	1.17		
	1-Chloro-*n*-undecane	27.96–43.99	406	−6.0 to 9.0	9V; CLS; $A_2 M$; (cm^3/g); M_n given in Ref.	133
	Cyclohexane	23–45	68.7	−0.410 to 0.226	6V; CLS; (J cm^3/g^2)	105
		27–41.5	1610	−0.37 to 0.258	5V; CLS	166
		30	50–203	−0.276 to −0.445	4V; OS	106
		40		0.634–0.445	4V	106
		50		1.33–0.901	4V	106
		34.2	78–210	−0.061 to 0.071	3V; OS	497
		35.5–55	See Ref.	−0.153 to 0.538	5V; CLS	329
		35–55	3200	0–0.54	3V; CLS	334
			4000	0–0.88	3V; CLS	172
		31.94–43.99	406	−131 to 23.1	5V; CLS; $A_2 M$; (cm^3/g)	133
		29.92–44.03		−16.3 to 22.7	5V; CLS; $A_2 M$; (cm^3/g)	133
		28	500	0.07	CLS	454
		33		0		
		50		0.5		
		40	100	0.4–0.25	3V; CLS; p dependence in Ref.	528
		45		0.79–0.87	3V; CLS; p dependence in Ref.	528
		50		1.11–1.19	3V; CLS; p dependence in Ref.	528
		50	513–45.9	0.95–1.29	7V; CLS	503
	Cyclohexane	40	370	0.320	CLS	967
			744	0.304		
			1270	0.321		
		45	370	0.526		
			743	0.537		
			1280	0.509		
		50	5.48	2.50		
			10.3	1.62		
			20.5	1.20		
			370	0.764		
			746	0.738		
			1280	0.674		
		55	370	0.887		
			746	0.854		
			1280	0.797		

TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
	Cyclohexane	34.5 (θ)	0.162	100	CLS; $M_w/M_n = 1.00$	968
			0.266	43	CLS; $M_w/M_n = 1.00$	
			0.370	33	CLS; $M_w/M_n = 1.00$	
			1.23	4.10	CLS; $M_w/M_n = 1.03$	
			2.27	1.33	CLS; $M_w/M_n = 1.05$	
			5.38	0.41	CLS; $M_w/M_n = 1.03$	
			10.1	0.28	CLS; $M_w/M_n = 1.03$	
			20.6	0.10	CLS; $M_w/M_n = 1.02$	
	Decalin		500	0.8×10^{-4}	M_v; (cm^4/g)	562
		25	1800	0.19 to −0.01	4V; CLS; p dependence in Ref.	528
		30		0.40–0.20	5V; CLS; p dependence in Ref.	528
		40		0.72–0.45	5V; CLS; p dependence in Ref.	528
atactic	Dichloroethane	22	3–17	7–2.54	6V; CLS	166
		25	3	6.8	CLS	166
		35	439–5360	See Ref.	11V; CLS	564
		67	16	2.7	CLS	166
	Dichloroethane/cyclohexane					
	65/35	22	1610	3.98	CLS	166
		67		3.32		166
	35/65	22		2.76		166
		67		2.70		166
	6/94	22		0.78		166
		67		1.27		166
	Dioxane	25	900–920	2.9–2.8	2V; CLS	294
			97.3	3.08	OS; A_3 given in Ref.	404
		34.2	77.6–196	5.11–4.48	3V; OS	497
	Dioxane/heptane					
	Heptane 0–58.5 vol.%	35 ± 0.2	2240–2630	2.51–0	9V; CLS	440
	Ethyl acetate	25	243	0.7	OS	196
	Ethylbenzene		680	4.7×10^{-4}	M_v; (cm^4/g)	562
	Hexyl-*m*-xylene	16	570–6000	0	3V; CLS; SAXS	492
		20	14.9–39.8	0.2–0.05	2V; CLS; SAXS	492
atactic	Poly(isobutene)/ cyclohexane	25–40	84.2	3.59	5V; CLS	599
		30	38.8	1.81		599
		35		0.93		599
		40		0.46		599
	Butanone	7.5–45	68.7	0.586	CLS; (J cm^3/g^2)	105
	Toluene	15	5.37	12.2	CLS	967
			20.0	8.80		
			191	4.36		
			357	3.72		
			723	3.12		
			1290	2.81		
			3940	2.07		
	Toluene	15	0.578	44	CLS; $M_w/M_n = 1.00$	968
			0.680	43	CLS; $M_w/M_n = 1.00$	
			0.904	37	CLS; $M_w/M_n = 1.01$	
			1.23	25	CLS; $M_w/M_n = 1.03$	
			2.27	17	CLS; $M_w/M_n = 1.05$	
			3.48	14.6	CLS; $M_w/M_n = 1.07$	
			5.38	12.2	CLS; $M_w/M_n = 1.03$	
			20.0	8.80	CLS; $M_w/M_n = 1.02$	
			39.4	6.45	CLS	
			191	4.36		
			357	3.72		
			723	3.12		
			1290	2.81		
			3940	2.07		
IUPAC-sample		RT	1030	1.29	CLS	253
		20	115–2800	2.2–0.72	8V; CLS	176
		22	2–1770	4.3–0.86	9V; CLS	166
		25	3.2–980	3.5–1.05	8V; CLS	166
			245–540	1.2–1.0	2V; OS	196

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TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
			96.3–785	166–5.3	5V; OS	176
			170–257	1.9	2V; CLS; OS	312
			900	1.3	CLS	294
		35	45.1–6350	See Ref.	13V; CLS	564
		67	517–980	0.89–0.81	3V; CLS	166
	Butanone/isopropyl alcohol					
	87/13	22	16.3–17.4	(0.2 to −0.3)	2V; CLS	166
		67		0	2V; CLS	166
	Tetrahydrofuran	30	24.4–5700	8.32–2.105	6V; CLS; $A_{2,H}$, $A_{2,S}$ given in Ref.	481
	Tetralin	25	680	4.0×10^{-4}	M_v; (cm^4/g)	562
	Toluene	12	54–4400	30–792	8V; CLS; A_2M; (cm^3/g)	548
branched	Cyclohexane	34.5	980–14300	−0.08–0.03	5V; CLS; $M_w/M_n = 1.03–1.24$	994
		44.5		0.02–0.18	5V; CLS; $M_w/M_n = 1.03–1.24$	
	Toluene	30		0.38–2.12	5V; CLS; $M_w/M_n = 1.03–1.24$	
star branched			384	115	CLS; A_2M; (cm^3/g)	548
			1870	380	CLS; A_2M; (cm^3/g)	548
comb branched			2320	404	CLS; A_2M; (cm^3/g)	548
			3880	520	CLS; A_2M; (cm^3/g)	548
		15	1800	2.24–2.39 (approx.)	5V; CLS	528
		20	5.85–125	9.4–4.5	5V; SAXS	492
		25	1800	2.27–2.41	5V; CLS	528
			209–3920	4.20–1.84	4V; CLS	492
			680	5.0×10^{-4}	M_v; (cm^4/g)	562
			99–952	6.5–3.5	7V; CLS; OS	176
			237	1.23	See Ref.; (J cm^3/g^2)	56
comb branched	Toluene		72–1800	$(11–7.0) \times 10^3$	10V; CLS; (cm^4/g)	245
		30	100	4.6	5V; CLS; $A_{2,H}$; $A_{2,S}$ given in Ref.	585
			373	3.45	OS	585
			13.3	4.49	OS	467
			418–37.6	4.54–5.88	7V; OS	503
			30.9–612	$7.41 \times 10^{-3} M^{0.75}$	A_2M; (cm^3/g)	169
			155–710	4.17–1.56	4V; CLS	290
				$A_2 = k M^{-0.22}$	OS	317
				$= 4.6 \times 10^{-3} M^{-0.22}$	OS	319
				$= 2.1 \times 10^{-3} M^{-0.16}$	OS	319
		30.27	56–452	29–145	5V; OS; A_2M; (cm^3/g)	327
		34.2	77.8–196	$(5.43–4.46) \times 10^{-4}$	3V; OS	497
		35	33.1–6460	See Ref.	16V; CLS	564
		37	366–50.9	5.5–6.9	8V; OS	584
		40	209–1600	4.28–2.46	3V; CLS	121
		22–67	1610	3.12–2.43	2V; CLS	166
		28–67	980	$20.4 \times [(479/T) - 1]$	M_v; CLS; A_2M; (cm^3/g)	308,310
Poly(styrene)-*b*-(Polyethylene/Polypropylene)						
	4-Methyl-2-pentanone/ 2-Chlorobutane (mixtures)					
	100%/0%	25	50000	0.058	CLS; micelles	1033
	90%/10%		48000	0.061		
	80%/20%		35000	0.059		
	70%/30%		29000	0.062		
	60%/40%	20	20000	0.054		
	50%/50%	25	100	2.7	CLS; unassociated	
Poly(styrenesulfonate)						
sodium salt; degree of sulfonation = 0.84	$NaNO_3$-solutions; ionic strength 0.005–2.0	10	350	0.08–34.2	9V; CLS; $M_w/M_n \approx 1.2$	936
		20		0.3–43.2	9V; CLS; $M_w/M_n \approx 1.2$	
		30		0.3–64.8	9V; CLS; $M_w/M_n \approx 1.2$	
	Xylene + 2% methanol	24	110 ± 10	3.7 ± 1	SANS; sulfonation level 1.39 mol%	937
Poly(styrene)/Poly(acrylonitrile)						
mass ratio 3:1	Dimethylformamide	25	538	0.0051	CLS	938

TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(styrene)(PST)/Poly(*p*-chlorostyrene)						
mixtures, mass frac. PST						
1.000	Butanone	30	341	1.15	OS	969
0.786			–	1.51		
0.593			–	1.70		
0.434			–	1.78		
0.272			–	1.76		
0.133			–	1.82		
0.000			330	1.72		
1.000	Cumene	55	341	2.90	OS	969
0.786			–	2.38		
0.593			–	1.92		
0.434			–	1.39		
0.272			–	0.84		
0.133			–	0.37		
0.000			330	−0.20		
1.000	Toluene	30	341	3.73	OS	969
0.786			–	3.41		
0.593			–	2.99		
0.434			–	2.64		
0.272			–	2.30		
0.133			–	1.75		
0.000			330	1.20		
block-copolymer						
mass frac. PST						
0.650	Butanone	30	379	1.68	OS	969
0.467			325	1.78		
0.372			368	1.75		
0.655			374	1.68		
0.509			356	1.82		
0.311			299	1.85		
0.650	Cumene	55	379	2.19	OS	969
0.467			325	1.65		
0.372			368	1.37		
0.655			374	2.27		
0.509			356	1.85		
0.311			299	1.52		
0.650	Toluene	30	379	3.02	OS	969
0.467			325	2.62		
0.372			368	2.23		
0.655			374	3.00		
0.509			356	2.73		
0.311			299	2.14		
Poly(styrene)/poly (isobutylene)	Toluene	21	194–2400 (PS) 156–2440 (PIB)	−9.80 to 8.36	16V; CLS; concentration and angle dependence given in Ref.	817
Poly(styrene)-*block*-Poly(ethylene/propylene)						
35 ± 3 mass% PS	Methyl ethylketone	25	105	420	Micelles; $M_w = 47.6 \times 10^{-6}$ g/mol	933
	Methyl propylketone			310	Micelles; $M_w = 18.2 \times 10^{-6}$ g/mol	
	Methyl isobutylketone			680	Micelles; $M_w = 42.2 \times 10^{-6}$ g/mol	
	Diethylketone			960	Micelles; $M_w = 43.0 \times 10^{-6}$ g/mol	
	5-Methyl-2-hexanone			780	Micelles; $M_w = 43.5 \times 10^{-6}$ g/mol	
Poly(styrene-*co*-maleic anhydride)/poly(ethyl acrylate)						
A	B					
graft copolymer						
A	B					
56.5/43.5	Acetone	25	53.4	2.1×10^{-4}	OS; (cm^4/g)	489
77/23			71.2	2.1×10^{-4}		
0/100			350	5.7×10^{-4}		
Poly(styrene-*co*-*p*-methoxystyrene)						
random						
mol% *p*-methoxystyrene						
	Toluene	25	71–350	5.0–3.0	7V; OS	617
53.8			35.5–701	4.1–3.2	12V; OS	617

TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
75.6			44–421	5.9–3.5	8V; OS	617
26.4			61–665	5.34–4.29	9V; CLS	617
				$A_2 = 2.0 \times 10^{-3} M^{-0.115}$		
53.0			66–1783	5.46–3.41	11V; CLS	617
				$A_2 = 2.4 \times 10^{-3} M^{-0.135}$		
75.6			79–1717	4.54–3.00	8V; CLS	617
				$A_2 = 2.3 \times 10^{-3} M^{-0.145}$		
Poly(styrene-*co*-methyl acrylate)						
	Dimethylformamide	20			See Ref. OS	515
	Butanone	20			See Ref. CLS	515
Poly(styrene-*co*-methyl methacrylate)						
approx. 50/50	Carbon tetrachloride	25	1330–1900	2.3–2.4	2V; CLS	294
	Dioxane		1330–1900	3.8–3.6	2V; CLS	294
random, mol% St. 29.3	2-Ethoxyethanol	40.0 (68.0)	354	0	OS	443
56.2	(Cyclohexanol)	58.4 (61.3)	350	0		
70.2		72.8 (63.0)	342	0		
block copolymer						
mol% St. 27	1,4-Dioxane			11.0	CLS	816
33				9.98		
46				11.8		
56				10.9		
100				7.37		
mol% St. 49.6	2-Ethoxyethanol	81.0	317	0	OS	443
	(Cyclohexanol)	81.3	317	0		
73.2		84.0	392	0		
85.1		84.0	193	0		
100		81.8	206	0		
mass% St. 38	Benzene	30	1530	1.48	CLS	818
				1.25		
	Toluene			1.65		
		69		4.92		
fractionated	Butanone	25	49–2270	3.2–1.45	11V; CLS	294
unfractionated			970	1.55		294
	Isoamyl acetate				See Ref.	507
	Nitroethane		1330–1900	1.5–1.2	2V; CLS	294
	Toluene				See Ref.	507
Poly(styrene)-*block*-(hydrogenated Polybutadiene)-*block*-Poly(styrene)						
	1,4-Dioxane/*n*-heptane					
	n-Heptane 0 vol%	25	4800	0.021	CLS, micelles	1018
	n-Heptane 10 vol%		4440	0.042		
	n-Heptane 20 vol%		4200	0.075		
	n-Heptane 30 vol%		3770	0.127		

[a] w = mass fraction of component *.

TABLE 17. (O, C)-HETEROCHAIN POLYMERS (POLY(ETHERS), POLY(ESTERS), POLY(CARBONATES))

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(carbonate) see Poly(oxycarbonyloxy.....)						
Poly(3,3′-dimethyl oxetane)		25	18.5	0.159	OS	976
	Cyclohexane					
			37.0	0.162		
			52.0	0.156		
			55.0	0.154		
			67.5	0.148		
			130	0.155		
Poly(ethylene terephthalate) see Poly(oxyethyleneoxyterephthaloyl)						
Poly(formaldehyde) see Poly(oxymethylene)						
Poly(oxybutene)	Butanone	30	2100–343	2.3–1.8	6V; CLS	598

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TABLE 16. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
75.6			44–421	5.9–3.5	8V; OS	617
26.4			61–665	5.34–4.29	9V; CLS	617
				$A_2 = 2.0 \times 10^{-3} M^{-0.115}$		
53.0			66–1783	5.46–3.41	11V; CLS	617
				$A_2 = 2.4 \times 10^{-3} M^{-0.135}$		
75.6			79–1717	4.54–3.00	8V; CLS	617
				$A_2 = 2.3 \times 10^{-3} M^{-0.145}$		
Poly(styrene-*co*-methyl acrylate)						
	Dimethylformamide	20			See Ref. OS	515
	Butanone	20			See Ref. CLS	515
Poly(styrene-*co*-methyl methacrylate)						
approx. 50/50	Carbon tetrachloride	25	1330–1900	2.3–2.4	2V; CLS	294
	Dioxane		1330–1900	3.8–3.6	2V; CLS	294
random, mol% St. 29.3	2-Ethoxyethanol	40.0 (68.0)	354	0	OS	443
56.2	(Cyclohexanol)	58.4 (61.3)	350	0		
70.2		72.8 (63.0)	342	0		
block copolymer						
mol% St. 27	1,4-Dioxane			11.0	CLS	816
33				9.98		
46				11.8		
56				10.9		
100				7.37		
mol% St. 49.6	2-Ethoxyethanol	81.0	317	0	OS	443
	(Cyclohexanol)	81.3	317	0		
73.2		84.0	392	0		
85.1		84.0	193	0		
100		81.8	206	0		
mass% St. 38	Benzene	30	1530	1.48	CLS	818
				1.25		
	Toluene			1.65		
		69		4.92		
fractionated	Butanone	25	49–2270	3.2–1.45	11V; CLS	294
unfractionated			970	1.55		294
	Isoamyl acetate				See Ref.	507
	Nitroethane		1330–1900	1.5–1.2	2V; CLS	294
	Toluene				See Ref.	507
Poly(styrene)-*block*-(hydrogenated Polybutadiene)-*block*-Poly(styrene)						
	1,4-Dioxane/*n*-heptane					
	n-Heptane 0 vol%	25	4800	0.021	CLS, micelles	1018
	n-Heptane 10 vol%		4440	0.042		
	n-Heptane 20 vol%		4200	0.075		
	n-Heptane 30 vol%		3770	0.127		

[a] w = mass fraction of component *.

TABLE 17. (O, C)-HETEROCHAIN POLYMERS (POLY(ETHERS), POLY(ESTERS), POLY(CARBONATES))

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(carbonate) see Poly(oxycarbonyloxy.....)						
Poly(3,3′-dimethyl oxetane)		25	18.5	0.159	OS	976
	Cyclohexane					
			37.0	0.162		
			52.0	0.156		
			55.0	0.154		
			67.5	0.148		
			130	0.155		
Poly(ethylene terephthalate) see Poly(oxyethyleneoxyterephthaloyl)						
Poly(formaldehyde) see Poly(oxymethylene)						
Poly(oxybutene)	Butanone	30	2100–343	2.3–1.8	6V; CLS	598

TABLE 17. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)						
	Chloroform	25	76.9–28.5	11.9–14.4	6V; OS	496
			10.2–463	24–8.91	14V; CLS; two different wavelengths M_w/M_n given in Ref.	850
		30	88–37.3	12.5–18.0	6V; OS	496
	Cyclohexanone	25	76.9–31.6	8.8–10.1	5V; OS	496
	Dioxane			10.9–12.4	5V; OS	496
	Dioxane/cyclohexanone		76.9–28.5	0	4V; OS	496
	Ethylene chloride		76.9–31.6	10.6–12.7	5V; OS	496
	Ethylene dichloride	20	56.4–97.7	11.0–9.0	3V; OS; M_n given	850
		25	359	7.43	CLS	850
		30	146–401	7.83–5.75	4V; OS; M_n given	850
	Methylene chloride		76.9–28.5	12.1–14.4	4V; OS	496
			61–122	3.6–2.5	2V; CLS; (J cm^3/g^2)	45
		27	23–182	3.9–2.9	6V; OS; (J cm^3/g^2)	45
	Tetrahydrofuran	20	8–266	4–1.29	6V; $M_{s,D}$; (J cm^3/g^2)	45
		25	76.9–31.6	11.0–13.0	5V; OS	496
			59.2–469	8.7–5.23	7V; CLS	850
		27	23–92	2.5–1.7	6V; OS; (J cm^3/g^2)	45
Poly(oxyethylene)	Benzene	25	7.70	27.4	CLS; B; (bar cm^6/g^2)	407
			3.79	78	OS; B; (bar cm^6/g^2)	407
	Dimethylformamide		3.79	30	OS; B; (bar cm^6/g^2)	407
		25–120	3.5 (approx.)	37–47	4V; OS; B; (bar cm^6/g^2)	188
	Methanol	25	7.70	66.0	OS; B; (bar cm^6/g^2)	407
			3.79	56.0	OS; B; (bar cm^6/g^2)	407
			0.316–6.75	18.0–16.4	3V; OS	518
			See Ref.	170–34.8	3V; CLS	423
			0.062–37.3	1220–46	CLS; B; (bar cm^6/g^2); data for seven solvents and for OS; SE given in Ref.	423
			1–31	102.5–39	11V; CLS; B; (bar cm^6/g^2)	192
			1–10	84.5–47.5	4V; CLS; B; (bar cm^6/g^2)	193
			4–23	87–46	3V; CLS; B; (bar cm^6/g^2)	184
			3–48	48–27.5	2V; CLS; B; (bar cm^6/g^2)	185
	Methanol	25	$68.5 \leq M_w \leq 852$	$A_2 = 0.0329\ M_w^{-0.28\pm0.06}$	CLS	907
			40	29.5	CLS; $M_w/M_n = 1.03$	1003
			74	28.3	CLS; $M_w/M_n = 1.02$	
			145	23.9	CLS; $M_w/M_n = 1.04$	
			594	18.7	CLS; $M_w/M_n = 1.10$	
			1200	17.6	CLS; $M_w/M_n = 1.12$	
	Water	5	6.95	0.465	LALLS; $M_w/M_n = 1.04$	975
			35.7	0.318	LALLS; $M_w/M_n = 1.04$	
		20	1.59	0.2963	CLS	965
			2.88	0.2611	CLS	
		25	10.9–800	116–30.4	5V; B; (bar cm^6/g^2)	189
			10.1	62	CLS; B; (bar cm^6/g^2)	184
			6.95	0.289	LALLS; $M_w/M_n = 1.04$	975
			20.5	0.243	LALLS; $M_w/M_n = 1.03$	
			35.7	0.232	LALLS; $M_w/M_n = 1.04$	
			177	12.17	CLS; $M_w/M_n = 1.04$	1002
			300	9.25	CLS; $M_w/M_n = 1.05$	
			1190	1.36	CLS; $M_w/M_n = 1.12$	
		40	6.95	0.267	LALLS; $M_w/M_n = 1.04$	975
			20.5	0.196	LALLS; $M_w/M_n = 1.03$	
			35.7	0.195	LALLS; $M_w/M_n = 1.04$	
	Water (D_2O)	25	20 ± 2	24 ± 2	SANS; $p = 0$ MPa; $M_w/M_n < 1.1$	1001
			18 ± 2	16 ± 2	SANS; $p = 100$ MPa; $M_w/M_n < 1.1$	
			19	13 ± 2	SANS; $p = 200$ MPa; $M_w/M_n < 1.1$	
Poly(oxyethylene)/Poly(oxypropylene) copolymer star structure molar fraction oxyethylene 0.724–0.866	Water	25	30.1–36	0.06–0.215	4V; CLS	986

TABLE 17. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(oxyethyleneoxyterephthaloyl)						
fractionated	*o*-Chlorophenol	25	107–18.1	14–25.5	8V; OS	526
				40 (approx.)	Different types of osmometers used; see Ref.	
unfractionated		25	39	26	Different types of osmometers used; see Ref.	
fractionated		50	74–26.2	15 (approx.)	4V; OS	
	o-Chlorophenol/tetrachloroethylene 60/40	25	12.2	19	OS	542
	Hexafluoro isopropanol	25	33	39 ± 1.8	CLS	911
Poly(oxyhexamethyleneoxydipoyl)	Chloroform	25	17.2	53	OS	542
Poly(oxyhexamethyleneoxyterephthaloyl)	*o*-Chlorophenol/tetrachloroethylene 60/40	25	11.3	35	OS	542
Poly(oxymethylene)	Hexafluoroacetone hydrate	25	23–170	$(5.6–2.0) \times 10^{-3}$	5V; CLS; see Ref.	430
–; diacetate	Hexafluoroacetone	25	23–185	56–25	5V; CLS	236
unfractionated	*p*-Chlorophenol	70	41	32	OS	510
fractionated			86.5–27	35 av.	5V; OS	
unfractionate			35–58	31–28	3V; OS	
fractionated			46–66.5	32–31	2V	
Poly(oxymethylene-*co*-oxyethylene)	1H, 1H, 5H-Octafluoropentanol-1	110	34–56	See Ref.	OS; (cm^3/mol); CLS-measurements also given in Ref.	402
Poly(oxypentamethyleneoxyterephthaloyl)	*o*-Chlorophenol/tetrachloroethylene 60/40	25	10.6	47	OS	542
Poly(oxypropylene)	Acetone	25	0.067	–90	CLS	475
			0.125	0	CLS	
			0.45–3.85	15.2 (approx.)	6V; CLS	
	Hexane	46	783–901	0.46–4.50	5V; CLS	428
	Methanol	20	0.535–3.31	$(10.75–0.95) \times 10^{-3}$	6V; A; A_2; (cm^3/g); see Ref.	470
	Iso-octane	48–85	901	0–1.58	9V; CLS	428
		50–89	783	–0.25–1.72	7V; CLS	
	Hexane	46	34.2–4410	3.16–0.523	16V; CLS	207
Poly(oxytetramethylene), Poly(tetrahydrofuran)	Ethyl acetate	30	34.6–1030	6.14–2.47	10V; CLS	413
Poly(oxytetramethyleneoxyadipoyl	Chloroform	25	14.7	40	OS	542
Poly(oxytetramethyleneoxyisophthaloyl)	*o*-Chlorophenol/tetrachloroethylene 60/40	25	12.8	27	OS	542
Poly(oxytetramethyleneoxyterephthaloyl)	*o*-Chlorophenol/tetrachloroethylene 60/40	25	13.4	30	OS	542
Poly(β-propiolactone)	2,2,2-trifluoroethanol	30	$6.80 \times 10^3 \leq M_w \leq 861 \times 10^3$			
				$A_2 = (0.025 \pm 0.003)\, M_w^{-0.19 \pm 0.01}$	CLS	978
Poly(terephthalic acid-4-aminobenzohydrazide)	Dimethyl sulfoxide	23	3.78	0.787	CLS	1017
			5.67	0.529		
			7.06	0.636		
			9.16	0.572		
			9.79	0.406		
			16.6	0.442		
			28.5	0.594		
			47.1	0.559		
			83.9	0.498		
			155	0.171		
			260	0.327		

TABLE 18. (N, C)- AND (O, N, C)-HETEROCHAIN POLYMERS (POLYAMIDES, POLYUREAS, POLYURETHANES)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(allylammonium hydrochloride)	Aq. NaCl 0.05–3.0 M	25	180	2.4–33.7	7V	1028
Poly(1,4-benzamide)	Dimethylacetamide, 3% LiCl	–	55	3.06	CLS	1007
Poly((*n*-butylimino)carbonyl) (Poly(*n*-butyl isocyanate))	Chloroform	22	61–10000	25.1	31V; CLS	565
	Toluene	37	23 ± 0.5 – 530 ± 60		7V; OS	433
				21.2 ± 2.7		433
		37	23–523	21.3	2V; OS	565
Poly(dimethyldiallyl-ammonium chloride)	Aq. NaCl 1.0 M (pH = 5.5)	23 ± 2	89	0.278	CLS	1029
			114	0.247		
			281	0.15		
			352	0.122		
			418	0.127		
			473	0.222		
Poly(ethylene imine)	Aq.. NaCl 1 M	35	16.6–5320	$A_2 = 0.135\ M_w^{-0.53 \pm 0.02}$	CLS; $M_w/M_n = 1.8$–2.2	1030
Poly(*N*-(3-hydroxypropyl)-L-glutamine)	Methanol	35	366 ± 5 – 100 ± 5		5V; CLS	534
				$2.40–4.1 \pm 0.6$		534
			55 ± 1	3.3	CLS	
Poly(imino-1,4(2-cyano-phenylene)iminoterephthaloyl)	Sulfuric acid 96%	25	4.1–20.8	0.15–1.11	9V; CLS	849
Poly(imino(2,5-dicarboxy-terephthaloyl)imino-1,4-phenyleneoxy-1,4-phenylene)	Dimethylacetamide					
	0.1 M LiBr	15	22.5–266	$(21.1–14.3) \times 10^{-4}$	13V; CLS; A_2M; (cm^3/g); OS	403
Poly(iminohexamethyleneiminoadipoyl) (Nylon 66)	*m*-Cresol	60	18	183	OS; B; (bar cm^6/g^2)	217
	Formic acid (90%)	25	18	840	OS; B; (bar cm^6/g^2)	217
	Formic acid (90%)/KCl 0.2–2.5 M KCl	25	31	59.2 to –7.0	7V; CLS; B; (bar cm^6/g^2)	219
	Formic acid (90%)/KCl 2.3 M KCl	25	31	0		219
	Formic acid/2 M KCl 82.5–40% Acid	25	31	–9.4 to 36.5	5V; CLS; B; (bar cm^6/g^2)	219
	Formic acid (90%)/ 2 M KCl	25	2–52	312–10.1	20V; CLS; B; (bar cm^6/g^2)	218
	Formic acid (75–98%) 0.5 M Sodium formide					
	2,2,3,3-Tetrafluoropropanol	25	32	1.0–4.0 (approx.)	6V; CLS	405
	0.1 M Sodium trifluoroacetate/2,2,3,3-tetrafluoropropanol	25	62	57.1	CLS; B; (bar cm^6/g^2)	220,221
Poly(*n*-hexyliminocarbonyl)	Toluene	37	61–606	20 (approx.)	3V; OS; see Ref. for measurements in THF	565
Poly(*n*-hexyl isocyanate)	Dichloromethane	20	111	6.10	CLS	983
			180	6.24		
			253	6.49		
			342	6.28		
			481	6.46		
			724	5.97		
			1060	6.23		
Poly((*n*-octylimino)carbonyl)	Toluene	37	68–200	20 (approx.)	3V; OS; see Ref. for measurements in THF	565
Poly(ureyleneheptamethylene)	Dichloroacetic acid	25	28.5	0	CLS; B; (bar cm^6/g^2)	400
		45	17.5	6.5	CLS; B; (bar cm^6/g^2)	400
		73	17.8	38	CLS; B; (bar cm^6/g^2)	400
		98	13.2	52	CLS; B; (bar cm^6/g^2)	400
		45	3.1–17.8	32–6.2	6V; CLS; B; (bar cm^6/g^2)	183
		73	2.6–23.5	82–8.0	6V; CLS; B; (bar cm^6/g^2)	183
		25–98	17.5 (approx.)	0–52	4V; CLS; B; (bar cm^6/g^2)	183
	Formic acid 98%/KCl 96/4	45	0.5	5	CLS; B; (bar cm^6/g^2)	183

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TABLE 18. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(ureyleneheptamethylene)	Formic acid 98%/KCl 96/4	45	3–13.9	104–59	6V; CLS; B; (bar cm^6/g^2)	183
	Formic acid 85.8–99.5%/ KCl 3.0–10.0%	73	5.7–12.5	512–30	8V; CLS; B; (bar cm^6/g^2)	400
		83	19.5–26.3	83–0.0	6V; CLS; B; (bar cm^6/g^2)	400
	Formic acid 98%/KCl 4%	25	10.6	−190	CLS; B; (bar cm^6/g^2)	400

TABLE 19. OTHER SYNTHETIC POLYMERS

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(acenaphthylene)	Toluene	25	1720–33	0.26–2.42	17V; CLS	458
			1140–23	0.28–2.10	17V; OS	
Poly(*tert*-butyl crotonate)	*n*-Butylchloride	25	60.6–351	7.75–5.81	4V; CLS	838
	Ethylacetate	25	60.6–351	2.11–1.46	4V; CLS	838
Poly(trimethylene sulfide)	Chloroform	25	19.3	0.2047	OS	1027
			21.5	0.1808		
			27.3	0.1750		
			33.4	0.1634		
			47.3	0.1651		

TABLE 20. INORGANIC POLYMERS

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(dihexoxyphosphazene)	Tetrahydrofuran	18	2340	0.758	CLS	982
		25	26.3	3.98		
			245	1.67		
			917	1.27		
			2280	0.655		
		32	2260	0.576		
Poly(di-*N*-hexylsilane)	Hexane	20	6200	0.76	CLS; M_w/M_n given	953
	Tetrahydrofuran	–	100–10000	$A_2 = 0.0019\ M_w^{-0.19}$	CLS	955
Poly(ferrocenylsilane) symmetrically substituted	Tetrahydrofuran	20	229	1.3	CLS	954
Poly(oxydimethylsilylene) (Poly(dimethyl siloxane))	Toluene	–	3.6	10.3	CLS; M_w/M_n given	956
		–	7.2	8.6		
		–	25.5	6.3		
		–	79.2	4.4		
		–	282	3.6		
		–	625	3.0		
linear	Toluene	25	1.89–30.1	13.6–8.04	8V; CLS	836
cyclic			1.98–21.8	13.0–6.76	9V; CLS	836
linear	Bromocyclohexane	28	0.93	8.42	6V; CLS	836
			3.03	2.62		
			10.6–32.6	0		
	Bromobenzene	59.1–110.8	80	−2.025 to 1.750	9V; CLS	216
		70–108.3	340	−0.587 to 0.890	8V; CLS	
		73.7–111.1	1060	−0.272 to 0.772	7V; CLS	
	Bromocyclohexane	16.7–60.2	33	−1.3 to 2.260	8V; CLS	216
		18.2–60.2	85	−0.782 to 2.012	8V; CLS	
		29.2–60.2	340	−0.136 to 1.329	8V; CLS	
		24.1–90.3	780	−0.257 to 1.985	10V; CLS	
		25.5–60.2	1060	−0.190 to 0.985	6V; CLS	
			15400	32		437
			14000	37		437
			5800	36		437

TABLE 20. *cont'd*

Polymer	Solvent	T (°C)	M (× 10⁻³) (g/mol)	A_2 (× 10⁴) (mol cm³/g²)	Remarks	Refs.
			5400	43		437
			4400	38		437
			4300	50		437
	Bromocyclohexane/	23.9–88.8	80	−1.23 to 2.14	10V; CLS	216
	phenetol (6/7)	26.8–93.9	340	−0.536 to 1.490	10V; CLS	
		31.8–90	1060	−0.239 to 1.280	8V; CLS	
	Chlorobenzene	20	107	1.88	OS	501
			232	1.70	OS	501
	n-Dodecane (DOD)/	20	68	0.08–4.32	11V; CLS	957
	butanone $\varphi_{DOD} = 0.00–1.00$		80.5	0.92–4.29	10V; CLS	
			125	0.09–3.47	11V; CLS	
			150	0.10–3.08		
	n-Hexadecane (HED)/	20	80.5	0.01–2.49	11V; CLS	
	butanone $\varphi_{HED} = 0.00–1.00$		125	0.00–2.14		
			150	0.00–2.09		
	n-Undecane (UND)/	20	31.5	0.04–6.08	9V; CLS	958
	butanone $\varphi_{UND} = 0.00–1.00$		80.5	0.12–5.00		
			125	0.093–4.65	9V; CLS; LALLS	958,963
			150	0.096–4.50	9V; CLS; LALLS	958,963
Poly(oxydimethylsilylene) linear	(Poly(dimethyl siloxane))					
	n-Hexane (HEX)/	20	31.5	0.04–9.55	10V; LALLS	963
	butanone $\varphi_{HEX} = 0–1$		80.5	0.12–7.60	10V; LALLS	
			125	0.093–6.78	10V; LALLS	
			150	0.096–6.47	10V; LALLS	
	n-Heptane (HEP)/	20	31.5	0.04–8.95	11V; LALLS	963
	butanone $\varphi_{HEP} = 0–1$		80.5	0.12–7.00	11V; LALLS	
			125	0.093–6.36	11V; LALLS	
			150	0.096–6.03	11V; LALLS	
	n-Nonane (NON)/	20	31.5	0.04–7.92	10V; LALLS	963
	butanone $\varphi_{NON} = 0–1$		80.5	0.12–6.38	10V; LALLS	
			125	0.093–5.78	10V; LALLS	
			150	0.096–5.43	10V; LALLS	
	n-Decane (DEC)/	20	68	0.08–5.35	11V; LALLS	963
	butanone $\varphi_{DEC} = 0–1$		80.5	0.12–5.20	11V; LALLS	
			125	0.09–4.74	11V; LALLS	
			150	0.10–4.64	11V; LALLS	
	Phenetol	80.8–122.8	45	−2.15 to 3.823	8V; CLS	216
		81.7–123.1	80	−0.93 to 3.23	7V	
		79.5–122.8	340	−0.54 to 1.796	7V	
		90–112.8	1060	0.015–1.466	6V	
	Styrene	14	23.5	−2.19	CLS; A_3 given	962
		28		−0.96		
		35		0		
		60		1.37		
		25	860	−0.77	CLS; A_3 given	
		28		−0.42		
		30		−0.36		
		33		−0.22		
		35		−0.04		
		40		0.26		
		50		0.59		
		25	130	−1.06	CLS	961
		30		−0.86		
		35 (θ)		0		
		40		0.17		
		50		0.68		
	Toluene	20	100	3.53	OS	501
			235	3.18	OS	501
		25	0.66–2.015	25.0–4.05	3V; OS	518
				−2.66 to 10.6	3V; CLS	519
Poly(oxymethyl phenylsilylene) (Poly(methylphenylsiloxane))	Acetone	25	257	1.6	CLS; $M_w/M_n = 1.65$	959
	Cyclohexane		270	1.2	CLS; $M_w/M_n = 1.65$	

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TABLE 20. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly(phenyl silsesquioxane)	Toluene	37	488.5–26.4	1.123–2.434	15V; OS	459
Poly(tetramethyl-*p*-silphenylene-siloxane)	Toluene	25	382	0.118	OS	960
		40		0.118		
		55		0.118		
		25	360	0.112	OS	960
		32.5		0.113		

TABLE 21. POLY(SACCHARIDES)

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Amylose acetate	Nitromethane	–	148–3110	2.43	CLS	846
Cellulose						
hydrolized linters	Cadmium ethylenediamine	25	225–945	16.1 (approx.)	5V; CLS	97
sulfite pulp			215	12.1	CLS	
regenerated	Aq. NaOH, 8 wt.%	3.5	80	4.79	CLS	941
		10		3.75		
		26		3.33		
		35		2.75		
		41		−0.58		
		45		−3.06		
	Aq. LiOH, 6 wt.%	25	121	11.0	CLS	
Cellulose acetate	Acetone	RT	60–173	9.4–5.8	4V; CLS	302
Cellulose acetate phthalate acetyl content: 19.5% phthalyl content: 21.4%	0.5 N $NaHCO_3$	30		$A_2 = 2.34 \times 10^{-3} M_w^{1.11}$	CLS; $1.68 \leq M_w \leq 13.44$ ($\times 10^6$)	939
Cellulose diacetate	Acetone	12.1	94	4.1	CLS	825
		25.4		3.8		
		37.7		3.6		
		45.0		3.5		
		50.3		3.4		
Cellulose diacetate	Butanone	30	71	−0.5	CLS	826
		40		−0.25	$(dA_2/dT) = 0.25 \times 10^{-5}$ [cm^3/(g K)]	
		50		0		
		60		0.25		
		40	92	−0.25		
		50		0		
			141	−2.1		
Cellulose carbanilate	Dioxane	20	2580–80.8	2.45–11.5	7V	68
			42.0–1.35	7.5–20.0	6V	68
Cellulose nitrate						
(13.9% N)	Acetone	25	81–3850	10.8–8.2	6V; CLS	477
(12.9% N)		25	141–1700	13.3–12.5	4V; CLS	
(ca. 13.55% N)		RT	61.6–2482	0.24 (approx.)	14V; OS; B; (bar L^2/g^2)	61
		25	77–2640	6.10 (approx.)	11V; CLS	22
		–	780	11.2	CLS	824
from row cotton (ca. 13.66% N)		15	22.8–417	0.24 (approx.)	8V; OS; B; (bar L^2/g^2)	195
from cotton		20	31–661	0.28 (approx.)	11V; OS; B; (bar L^2/g^2)	194
	Butyl acetate	20	150–400	1.0–0.5	CLS; B; (bar cm^6/g^2)	239
		25	30–360	3.5–0.3	OS; B; (bar cm^6/g^2)	239
from viscose rayon	Ethyl acetate	29.7	71.5	44.1	OS; A_2M; (cm^3/g)	91
from chemical cotton (12.4% N)			295–450	28.5–25.7	2V; OS; A_2M; (cm^3/g)	91
	Butanone	25	130	10.8	OS	196

TABLE 21. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Cellulose tricarbanilate	Tetrahydrofuran	25	105	7.0	CLS; M_w/M_n given	940
			200	7.2		
			294	5.7		
			385	5.0		
			508	5.6		
			578	4.2		
			787	4.2		
			1050	4.3		
			1330	4.2		
			1820	4.0		
			3700	3.8		
			86	5.5	CLS; M_w/M_n given	940
			128	4.5		
			185	8.0		
			225	6.7		
			267	3.7		
			400	4.8		
			513	4.8		
			885	4.5		
			1720	3.3		
Dextran	0.05 g/100 ml sodium azide	20	7.5	9.80	CLS	1015
			14.5	1.00		
			41.6	7.26		
			120.7	5.87		
			374	2.30		
			1270	1.05		
	Water	20	36.96	0.0616	CLS	965
			67.94	0.0421		
			113.07	0.0352		
			530.61	0.0302		
			485	4.6	$M_w/M_n = 1.95$	985
		25	485	4.6	CLS; $M_w/M_n = 1.95$	942
			9.9	13.8	CLS	999
			39.7	9.2		
			65.0	77		
			481	3.9		
			2270	0.9		
			3250	0.7		
(2-(Diethylamino)ethyl) dextran	Aq. sodium nitrate 0.8 M	20	1.58	1.34	CLS	1015
			11.3	11.6		
			21.0	9.5		
			45.7	4.6		
			131	3.31		
			405	2.1		
			429	2.28		
			495	1.95		
			640	1.4		
			1474	1.15		
Pullulan	Water	25	107	2.50	CLS; M_w/M_n given	1013
			188	2.20		
			384	1.60		
			572	1.60		
			811	1.38		
			1050	1.35		
			1490	1.50		
			2100	1.20		
			2360	1.46		
				$A_2 = 5.42 \times 10^{-3}\ M_w^{-0.26}$		
hydrophobized		25	760	2.60	CLS	943
Tetramethylammonium gellan gum						
	Tetramethylammonium chloride					
	$C = 0.0200$ mol/dm^3	28	97	0.71	OS	984
	$C = 0.0500$ mol/dm^3		98	0.45		
	$C = 0.0750$ mol/dm^3		95	0.39		

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TABLE 21. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
	$C = 0.2000$ mol/dm^3		98	0.30		
	$C = 0.5000$ mol/dm^3		100	0.24		
	$C = 0.0125$ mol/dm^3	40	49	0.95	OS	984
	$C = 0.0200$ mol/dm^3		50	0.71		
	$C = 0.0500$ mol/dm^3		48	0.44		
	$C = 0.0750$ mol/dm^3		52	0.34		
	$C = 0.2000$ mol/dm^3		52	0.21		
	$C = 0.5000$ mol/dm^3		48	0.12		
Xanthan, A	Synth. seawater brine	–	4200	2.0	CLS	946
			5000	2.3		
			5100	2.6		
			5400	2.4		
			8100	2.7		
			8200	2.5		
			10100	2.5		
Xanthan B			4100	0.95		
Xanthan C			12200	1.5		
Xanthan D			5200	2.1		
Xanthan A	0.1 M NaCl	–	5000	3.6	CLS; 0.22 μm filter used	
			5100	4.0	CLS	
			9600	4.2	CLS	
Xanthan D			5800	3.5	CLS; 0.22 μm filter used	
powdered	0.1 M NaCl	–	7400	3.2	CLS	
broth	0.085 M NaCl	–	1800	6–8	CLS	
double stranded helix	0.1 N NaCl	25	74	6.80	CLS	944
			112	6.18		
			164	5.64		
			209	5.34		
			240	5.02		
			362	4.77		
			603	4.35		
			994	3.79		
			1420	3.99		
			2560	3.08		
			3940	2.91		
			7400	3.17		
sodium salt	0.005 M NaCl	25	233	20(± 4)	CLS; A_3 given	945
	0.007 M		225	17.5(± 3.5)		
	0.01 M		229	16(± 1)		
	0.02 M		238	8.0(± 1.0)		
	0.03 M		226	7.5(± 0.5)		
	0.06 M		222	5.3(± 0.5)		
	0.1 M		220	4.8(± 0.3)		
	0.2 M		221	2.95(± 0.05)		
	0.6 M		228	2.1(± 0.1)		
	1.0 M		227	1.7 (± 0.5)		
native	0.1 M NaCl	20	2940	4.94	CLS	1011,1012
acetyl-free			1770	6.00	CLS	
pyruvate-free			1370	5.22	CLS	

TABLE 22. OTHER BIOPOLYMERS (PROTEINS, POLY(NUCLEOTIDES))

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Calf thymus desoxyribonucleic acid	Aq. NaCl 0.2 M	20	156 ± 5	0.0072 ± 0.0002	CLS	1014
Gelatin (30.2% C, 8.2% H, 18.5% N)	0.5 M KSCN	25	400	2.0	CLS	979
	Formamide		380	2.6	CLS	979
			292	3.7	CLS; M_w/M_n given	1000
			346	4.5		
			371	6.2		

TABLE 22. *cont'd*

Polymer	Solvent	T (°C)	M ($\times 10^{-3}$) (g/mol)	A_2 ($\times 10^4$) (mol cm^3/g^2)	Remarks	Refs.
Poly((γ-benzyl-L-glutamate)	Chlorobenzene	25	920	5.1	CLS	488
	Chloroform		130–358	5.0–4.7	4V; CLS	299
	Chloroform/formamide		392	2.0		435
			408	2.2		435
	Cyclohexanone		230	16	CLS	488
	Dochloroacetic acid		21.4–336	15.7–7.5	3V; CLS	299
	Dichloroethane		490	5.0	CLS	488
			555	23.2	CLS	476
	Dimethylformamide	5	179	3.52	CLS	949
		15	60	5.36	CLS	
			149	3.94	CLS	
			179	4.14	CLS	
		15–55	27.4	$-6 \pm 1.3 - 2.5 \pm 0.27$	5V; OS	521
		20	179	4.43	CLS	949
		20–55	140	$-0.9 \pm 0.44 - 2.6 \pm 0.15$	5V; OS	521
		25	18.5–414	2.5 (approx.)	13V; CLS	435
			350	4.7	CLS	488
			132–417	See Ref.	CLS	450
			179	3.46	CLS	949
		25–70	.6	16.0	3V; OS; CLS; M_w; B; (bar cm^6/g^2)	476
			33.4	9.1	5V; B; (bar cm^6/g^2)	476
			154	9.3	4V; B; (bar cm^6/g^2)	476
			222	9.1	4V; B; (bar cm^6/g^2); for measurements in five more solvents see Ref.	476
		30	277	3.35	CLS	949
			179 ± 1	3.1 ± 0.1		1016
		37	69–250	3.5–2.8	OS	450
			76–176	3.8–2.0	5V; OS	435
		40	60	5.38	CLS	949
			149	3.35	CLS	
			179	3.86	CLS	
		45	27.4–490	2.2 (approx.)	8V; OS	521
		75	60	5.70	CLS	949
			149	4.77	CLS	
			179	3.80	CLS	
			277	3.45	CLS	
		–	102	3.9	CLS	950
Poly(γ-benzyl-L-glutamate) (PBG)	Dimethylformamide	25	25–1100	1.1–3.3	10V; CLS	951
PBG-S-S-PBG			0.4–16	1.8–13.7	11V; CLS	
PBG-N=N-PBG			8–24	1.9–5.1	4V; CLS	
PBG-$(CH_2)_{10}$-PBG			19–44	1.2–2.6	6V; CLS	
	Dioxane	25	400	5.3	CLS	488
	Tetrahydrofuran		440	5.8	CLS	488
Desoxyribonucleic acid	Aq. NaCl, 1 M	25	105 ± 0.03	0.30 ± 0.02	CLS; A_3, A_4 given	947
			1.00 ± 0.05	0.26 ± 0.06		
	0.1 M		109 ± 0.03	0.69 ± 0.05		
	0.01 M		107 ± 0.05	2.9 ± 0.3		
fragments	Aq. NaCl, 0.002 M	25	117 ± 18	181 ± 4	CLS; theoretical values given	948
	0.005 M			61 ± 2		
	0.01 M			29 ± 2		
	0.02 M			20 ± 2		
	0.04 M			10 ± 1.5		
	0.1 M			6 ± 1.5		
	0.5 M			2 ± 1.5		
Poly(L-glutamic acid)	Aq. NaCl, 0.02 M	–	112	0.88	CLS	952
	0.05 M	–		0.48	CLS	
	0.1 M	–		0.38	CLS	
	0.05 M	–	104	0.50	CLS	
	0.2 M	–	104	0.30	CLS	

E. REFERENCES

1. G. Meyerhoff, Z. Electrochem., **61**, 1250 (1957).
2. T. Svedberg, K. O. Pederson, "Die Ultrazentrifuge", Th. Steinkopf, Dresden und Leipzig, 1940.
3. G. Meyerhoff, Makromol. Chem., **37**, 97 (1960).
4. H. P. Frank, H. F. Mark, J. Polym. Sci., **17**, 1 (1955).
5. G. Meyerhoff, in "Conference on the Ultracentrifuge", Academic Press, New York, 1963.
6. G. Meyerhoff, Makromol. Chem., **72**, 214 (1964).
7. J. G. Kirkwood, J. Riseman, J. Chem. Phys., **16**, 565 (1948).
8. J. Riseman, J. G. Kirkwood, in: F. R. Eirich, (Ed.), "Rheology, Theory and Application", vol. I, Academic Press, New York, 1956.
9. M. Kurata, H. Yamakawa, J. Chem. Phys., **29**, 311 (1958).
10. H. Kuhn, A. Silberberg, J. Polym. Sci., **14**, 193 (1954).
11. International Union of Pure and Applied Chemistry, J. Polym. Sci., **10**, 129 (1953).
12. A. F. Schick, S. J. Singer, J. Phys. Chem., **54**, 1028 (1950).
13. G. V. Schulz, H. Inagaki, R. Kirste, Z. Phys. Chem. N. F., **24**, 390 (1960).
14. N. Gralèn, Dissertation, Uppsala, 1944.
15. H. Luetge, G. Meyerhoff, Makromol. Chem., **68**, 180 (1963).
16. K. Nachtigall, G. Meyerhoff, Z. Phys. Chem. (Frankfurt), **30**, 35 (1961).
17. G. Meyerhoff, Makromol. Chem., **15**, 68 (1955).
18. G. Meyerhoff, Makromol. Chem., **12**, 45 (1954).
19. K. Uda, G. Meyerhoff, Makromol. Chem., **67**, 168 (1961).
20. R. Signer, J. Liecht., Helv. Chim. Acta, **21**, 530 (1938).
21. A. Polson, Kolloid-Z., **83**, 172 (1938).
22. A. M. Holtzer, H. Benoit, M. Doty, J. Phys. Chem., **58**, 624 (1954).
23. J. W. Williams, W. M. Saunders, J. Phys. Chem., **58**, 854 (1954).
24. A. G. Ogston, E. F. Woods, Trans. Faraday Soc., **50**, 635 (1954).
25. A. G. Ogston, Trans. Faraday Soc., **49**, 1481 (1953).
26. F. R. Senti, N. N. Hellmann, N. H. Ludwig, G. Babcock, R. Tobin, C. A. Glass, B. L. Lamberts, J. Polym. Sci., **17**, 527 (1955).
27. L. D. Moore, Jr., G. R. Greear, J. O. Sharp, J. Polym. Sci., **59**, 339 (1962).
28. G. V. Schulz, K. Altgelt, H.-J. Cantow, Makromol. Chem., **21**, 13 (1956).
29. K. Altgelt, G. V. Schulz, Makromol. Chem., **32**, 66 (1959).
30. K. Altgelt, G. V. Schulz, Makromol. Chem., **36**, 209 (1960).
31. G. V. Schulz, A. Mula, Proc. Nat. Rubber, Res. Conf., Kuala Lumpur, 1960, p. 602.
32. L. Mandelkern, W. R. Krigbaum, H. A. Scheraga, P. J. Flory, J. Chem. Phys., **20**, 1392 (1952).
33. K. H. Schachman, M. A. Lauffer, J. Am. Chem. Soc., **71**, 536 (1949).
34. M. A. Lauffer, J. Am. Chem. Soc., **66**, 1188 (1944).
35. H. Neurath, Chem. Rev., **30**, 357 (1942).
36. M. A. Lauffer, J. Am. Chem. Soc., **66**, 1195 (1944).
37. H. K. Schachman, W. J. Kauzman, J. Phys. Chem., **53**, 150 (1949).
38. H. W. McCormick, J. Polym. Sci. A, **1**, 103 (1963).
39. S. J. Singer, J. Chem. Phys., **15**, 341 (1947).
40. L. E. Miller, F. A. Hamm, J. Phys. Chem., **57**, 110 (1953).
41. H. Ende, G. Meyerhoff, G. V. Schulz, Z. Naturforsch., **13b**, 713 (1958).
42. S. Shulman, J. Am. Chem. Soc., **75**, 5846 (1953).
43. G. Meyerhoff, Z. Physik. Chem. (Frankfurt), **4**, 335 (1955).
44. G. Meyerhoff, Z. Physik. Chem. (Frankfurt), **23**, 100 (1960).
45. V. Schulz, A. Horbach, Makromol. Chem., **29**, 93 (1959).
46. G. V. Schulz, D. Laue, O. Bodmann, Makromol. Chem., **31**, 75 (1959).
47. R. Cecil, A. G. Ogston, Biochem. J., **42**, 229 (1948).
48. G. V. Schulz, H. A. Ende, Z. Physik. Chem. (Frankfurt), **36**, 82 (1963).
49. G. V. Schulz, M. Marx, Makromol. Chem., **14**, 52 (1954).
50. M. Cantow, G. Meyerhoff, G. V. Schulz, Makromol. Chem., **49**, 1 (1961).
51. O. Bodmann, D. Kranz, G. V. Schulz, Makromol. Chem., **41**, 225 (1960).
52. G. V. Schulz, G. Meyerhoff, Z. Elektrochem., **56**, 545 (1952).
53. K. J. Ivin, H. A. Ende, G. Meyerhoff, Polymer, **3**, 129 (1962).
54. K. J. Ivin, J. Polym. Sci., **25**, 228 (1957).
55. G. V. Schulz, H. Doll, Z. Elektrochem., **63**, 301 (1959).
56. E. D. Kunst, Dissertation, Groningen, 1950.
57. G. V. Schulz, K. V. Guenner, H. Gerrens, Z. Physik. Chem. (Frankfurt), **4**, 192 (1955).
58. G. Meyerhoff, Naturwissenschaften, **41**, 13 (1954).
59. G. Meyerhoff, G. V. Schulz, Makromol. Chem., **7**, 294 (1951).
60. G. V. Browning, J. D. Ferry, J. Chem. Phys., **17**, 1107 (1949).
61. A. Muenster, J. Polym. Sci., **8**, 633 (1952).
62. J. Stamm, J. Am. Chem. Soc., **52**, 3047 (1930).
63. J. Stamm, J. Am Chem. Soc., **52**, 3062 (1930).
64. G. Meyerhoff, J. Polym. Sci., **29**, 399 (1958).
65. G. Meyerhoff, Makromol. Chem., **32**, 249 (1959).
66. J. Hengstenberg, G. V. Schulz, Makromol. Chem., **2**, 5 (1948).
67. A. Polson, Kolloid-Z., **87**, 149 (1939).
68. J. M. G. Cowie, S. Bywater, J. Polym. Sci. C, **30**, 85 (1970).
69. T. Svedberg, K. O. Pedersen, The Ultracentrifuge, The Clarendon Press, Oxford, 1940.
70. A. G. Polson, Dissertation University of Stellenbosch, South Africa, 1937.
71. R. O. Carter, J. Am. Chem. Soc., **63**, 1960 (1941).
72. H. G. Tennent, C. F. Vilbrandt, J. Am. Chem. Soc., **65**, 424 (1943).
73. K. O. Petersen, J. J. I. Andersson, unpubl. results, cited. in Ref. 67.

74. W. R. Krigbaum, A. M. Kotliar, J. Polym. Sci., **32**, 323 (1958).
75. A. Oth, Ind. Chim. Belge, **20**, 423 (1955).
76. T. Svedberg, I.-B. Eriksson-Quensel, J. Am. Chem. Soc., **56**, 1700 (1934).
77. L. Krejci, T. Svedberg, J. Am. Chem. Soc., **56**, 1706 (1934).
78. T. Svedberg, I.-B. Eriksson, J. Am. Chem. Soc., **55**, 2834 (1933).
79. W. F. H. M. Mommaertz, R. G. Parrish, J. Biol. Chem., **188**, 545 (1951).
80. P. Johnson, R. Landolt, Nature, **165**, 430 (1950).
81. G. L. Miller, W. C. Price, Arch. Biochem. Biophys., **10**, 467 (1946).
82. W. F. H. M. Mommaertz, Arkiv Kemi Mineral. Geol., **19A**, No. 17 (1945).
83. N. W. Pirie, Advan. Enzymol., **5**, 1 (1945).
84. M. A. Lauffer, J. Phys. Chem., **44**, 1137 (1940).
85. M. A. Lauffer, W. M. Stanley, J. Biol. Chem., **135**, 463 (1940).
86. A. G. Ogston, Biochem. J., **37**, 78 (1943).
87. A. S. McFarlane, R. A. Kekwick, Biochem. J., **32**, 1607 (1938).
88. F. C. Bawden, N. W. Pirie, unpubl. (cit. in Ref. 83).
89. M. M. Huque, D. A. Goring, S. G. Mason, Can. J. Chem., **36**, 952 (1958).
90. L. Hunt, S. Newman, H. A. Scheraga, P. J. Flory, J. Phys. Chem., **60**, 1278 (1956).
91. S. Newman, L. Loeb, C. M. Conrad, J. Polym. Sci., **10**, 463 (1953).
92. B. Ingelman, M. S. Halling, Arkiv Kemi, **1**, 61 (1949).
93. H.-J. Cantow, Z. Naturforsch., **7b**, 485 (1952).
94. R. S. J. Manley, Arkiv Kemi, **9**, 519 (1956).
95. Y. Fujisaki, H. Kobayashi, Chem. High Polym. (Japan), **19**, 81 (1962).
96. H.-J. Cantow, Makromol. Chem., **30**, 169 (1959).
97. D. Henley, Arkiv Kemi, **18**, 327 (1961).
98. W. Brown, Arkiv Kemi, **18**, 227 (1961).
99. H.-J. Cantow, G. V. Schulz, Z. Phys. Chem. (Frankfurt), **2**, 117 (1954).
100. D. J. Bell, H. Gutfreund, R. Cecil, A. G. Ogston, Biochem. J., **42**, 405 (1945).
101. R. W. G. Wyckoff, J. Biol. Chem., **121**, 219 (1937).
102. H. Neurath, A. M. Saum, J. Biol. Chem., **126**, 435 (1938).
103. F. C. Bawden, N. W. Pirie, Proc. Roy. Soc. (London), Ser. B, **123**, 274 (1937).
104. W. M. Stanley, J. Phys. Chem., **42**, 55 (1938).
105. H.-J. Cantow, Z. Physik. Chem. (Frankfurt), **7**, 58 (1956).
106. W. R. Krigbaum, J. Am. Chem. Soc., **76**, 3758 (1954).
107. H. W. McCormick, J. Polym. Sci., **36**, 341 (1959).
108. J. W. Breitenbach, E. L. Forster, A. J. Renner, Kolloid-Z., **127**, 1 (1952).
109. W. R. Krigbaum, J. E. Kurz, P. Smith, J. Phys. Chem., **65**, 1984 (1961).
110. H.-G. Elias, F. Patat, J. Polym. Sci., **29**, 141 (1958).
111. A. R. Shultz, J. Am. Chem. Soc., **76**, 3422 (1954).
112. H.-G. Elias, F. Patat, Makromol. Chem., **25**, 13 (1958).
113. T. Svedberg, I.-B. Eriksson, J. Am. Chem. Soc., **54**, 3998 (1933).
114. L. Krejci, T. Svedberg, J. Am. Chem. Soc., **57**, 946 (1935).
115. J. Bisschops, J. Polym. Sci., **17**, 81 (1955).
116. K. Dialer, R. Kerber, Makromol. Chem., **17**, 56 (1955).
117. A. Oth, Bull. Soc., Schim. Belges, **64**, 484 (1955).
118. A. R. Peacocke, H. R. Schachmann, Biochim. Biophys. Acta, **15**, 198 (1954).
119. K. Dialer, K. Vogler, F. Patat, Helv. Chim. Acta, **35**, 869 (1952).
120. J. L. Oncley, G. Scatchard, A. Brown, J. Phys. Chem., **51**, 184 (1947).
121. G. V. Schulz, H. Baumann, R. Darskus, J. Phys. Chem., **70**, 3647 (1966).
122. R. A. Kekwick, K. O. Pedersen, Biochem. J., **30**, 2201 (1936).
123. M. Deidelberger, K. O. Pedersen, J. Gen. Physiol., **19**, 95 (1935).
124. G. J. Howard, D. O. Jordan, J. Polym. Sci., **12**, 209 (1954).
125. S. Newman, F. Eirich, J. Colloid Sci., **5**, 541 (1950).
126. N. Gralèn, G. Lagermalm, J. Phys. Chem., **56**, 514 (1952).
127. J. A. V. Butler, A. B. Robins, K. V. Shooter, Proc. Roy. Soc. (London) Ser. A, **241**, 299 (1951).
128. L. Mandelkern, P. J. Flory, J. Chem. Phys., **20**, 212 (1952).
129. H. A. Dieu, J. Polym. Sci., **12**, 417 (1957).
130. L. Freund, M. Daune, J. Polym. Sci., **29**, 161 (1958).
131. D. O. Jordan, A. R. Mathieson, M. R. Porter, J. Polym. Sci., **21**, 463 (1956).
132. D. O. Jordan, A. R. Mathieson, M. R. Porter, J. Polym. Sci., **21**, 473 (1956).
133. T. A. Orofino, J. W. Mickey, Jr., J. Chem. Phys., **38**, 2512 (1963).
134. M. Wales, K. E. Van Holde, J. Polym. Sci., **14**, 81 (1954).
135. B. Rosen, P. Kamath, F. Eirich, Disc. Faraday Soc., **11**, 135 (1951).
136. D. A. I. Goring, C. Chepeswick, J. Colloid Sci., **10**, 440 (1955).
137. L. P. Witnauer, F. R. Senti, M. D. Stern, J. Polym. Sci., **16**, 1 (1955).
138. R. L. Baldwin, Biochem. J., **55**, 644 (1953).
139. H. Malmgren, Acta Chem. Scand., **6**, 1 (1952).
140. R. Gehm, Acta Chem. Scand., **5**, 270 (1951).
141. J. H. Fessler, A. G. Ogston, Trans. Faraday Soc., **47**, 667 (1951).
142. L. W. Nichol, A. B. Roy, Biochemistry, **4**, 386 (1965).
143. W. Scholtan, Makromol. Chem., **36**, 162 (1960).
144. W. Scholtan, Makromol. Chem., **7**, 209 (1952).
145. C. G. Holmberg, Arkiv Kemi, Mineral. Geol., **17A**, No. 28 (1944).
146. V. L. Koenig, Arch. Biochem., **25**, 241 (1950).
147. V. L. Koenig, J. D. Perrings, Arch. Biochem. Biophys., **41**, 367 (1952).
148. G. R. Andersson, Arkiv Kemi, **20**, 513 (1963).
149. J. A. Manson, G. J. Arquette, Makromol. Chem., **37**, 187 (1960).
150. H.-G. Elias, Makromol. Chem., **50**, 1 (1961).

151. K. Nakanishi, M. Kurata, Bull. Chem. Soc. (Japan), **33**, 152 (1960).
152. W. G. Martin, W. H. Cook, C. A. Winkler, Can. J. Chem., **34**, 809 (1956).
153. W. Scholtan, Makromol. Chem., **23**, 128 (1957).
154. V. L. Koenig, J. D. Perrings, Arch. Biochem. Biophys., **36**, 147 (1952).
155. V. L. Koenig, J. D. Perrings, Arch. Biochem. Biophys., **40**, 218 (1952).
156. V. L. Koenig, K. O. Pedersen, Arch. Biochem., **25**, 97 (1950).
157. I. B. Eriksson-Quensel, Biochem. J., **32**, 585 (1938).
158. A. Tiselius, D. Gross, Kolloid-Z., **66**, 11 (1934).
159. K. O. Pedersen, Biochem. J., **30**, 961 (1936).
160. W. R. Krigbaum, P. J. Flory, J. Am. Chem. Soc., **75**, 1775 (1953).
161. M. D. Dayhoff, G. E. Perlmann, D. A. MacInnes, J. Am. Chem. Soc., **74**, 2515 (1952).
162. S. N. Chinai, R. J. Samuels, J. Polym. Sci., **19**, 463 (1956).
163. S. N. Chinai, J. D. Matlack, A. L. Resnick, R. J. Samuels, J. Polym. Sci., **17**, 391 (1955).
165. S. N. Chinai, R. A. Guzzi, J. Polym. Sci., **21**, 417 (1956).
166. P. Outer, C. I. Carr, B. H. Zimm, J. Chem. Phys., **18**, 830 (1950).
167. K. A. Granath, J. Colloid Sci., **13**, 308 (1958).
168. O. Lamm, A. Polson, Biochem. J., **30**, 528 (1936).
169. J. Oth, V. Desreux, Bull. Soc. Chem. Belges, **66**, 303 (1957).
170. W. B. Bridgman, J. Am Chem. Soc., **68**, 857 (1946).
171. H. K. Schachman, W. F. Harrington, J. Polym. Sci., **12**, 379, (1954).
172. D. McIntyre, A. Wims, L. C. Williams, L. Mandelkern, J. Phys. Chem., **66**, 1932 (1962).
173. H. K. Schachman, Protein and Function, Brookhaven Symposia in Biology: No. 13, 49 (1960)
174. H. K. Schachman, J. Am. Chem. Soc., **73**, 4453 (1951).
175. I.-B. Eriksson-Quensel, T. Svedberg, Biol. Bull., **71**, 498 (1936).
176. J. Oth, V. Desreux, Bull. Soc. Chim. Belges, **63**, 285 (1954).
177. P. J. Rempp, J. Chim. Phys., **54**, 432 (1957).
178. K. Dialer, K. Vogler, Makromol. Chem., **6**, 191 (1951).
180. C. Rosse, U. Bianchi, V. Mangasco, J. Polym. Sci., **30**, 175 (1958).
181. G. Kegeles, S. M. Klainer, W. J. Salem, J. Phys. Chem., **61**, 1286 (1957).
182. G. W. Schwert, S. Kaufmann, J. Biol. Chem., **179**, 655 (1949).
183. J. Feisst, H.-G. Elias, Makromol. Chem., **82**, 78 (1965).
184. H.-G. Elias, Z. Physik. Chem. (Frankfurt), **28**, 303 (1961).
185. H.-G. Elias, Chem. Ing.-Tech., **33**, 359 (1961).
186. K. Froembling, F. Patat, Makromol. Chem., **25**, 41 (1958).
187. H.-G. Elias, Makromol. Chem., **27**, 192 (1958).
188. H.-G. Elias, E. Maenner, Makromol. Chem., **40**, 207 (1960).
189. H.-G. Elias, Angew. Chem., **73**, 209 (1961).
190. F. Patat, H.-G. Elias, Naturwissenschaften, **46**, 322 (1959).
191. J. Hengstenberg, E. Schuch, Makromol. Chem., **7**, 236 (1952).
192. T. A. Ritscher, H.-G. Elias, Makromol. Chem., **30**, 48 (1959).
193. C. Sadron, P. Rempp, J. Polym. Sci., **29**, 127 (1958).
194. A. Muenster, Z. Physik. Chem. (Leipzig), **197**, 17 (1951).
195. H. Diener, A. Muenster, Z. Physik. Chem. (Frankfurt), **13**, 202 (1957).
196. G. Jacobsson, Acta Chem. Scand., **8**, 1843 (1954).
197. W. Burchard, Z. Physik. Chem. (Frankfurt), **42**, 293 (1964).
198. W. Burchard, Makromol. Chem., **59**, 16 (1963).
199. P. E. Onyon, J. Polym. Sci., **37**, 315 (1959).
200. P. E. Onyon, J. Polym. Sci., **22**, 13 (1956).
201. H. Kobayashi, J. Polym. Sci., **39**, 369 (1959).
202. S. Newman, W. R. Krigbaum, C. Laugier, P. J. Flory, J. Polym. Sci., **14**, 451 (1954).
203. G. B. Rathmann, F. A. Bovey, J. Polym. Sci., **15**, 544 (1955).
204. I. J. O'Donell, E. F. Woods, J. Polym. Sci., **21**, 397 (1956).
205. S. N. Chinai, R. A. Guzzi, J. Polym. Sci., **41**, 475 (1956).
206. H. Kobayashi, J. Polym. Sci., **26**, 230 (1957).
207. G. Allen, C. Booth, M. N. Jones, Polymer (London), **5**, 195 (1964).
208. J. E. Hansen, M. G. McCarthy, T. J. Dietz, J. Polym. Sci., **7**, 77 (1951).
209. S. N. Chinai, A. L. Resnick, H. T. Lee, J. Polym. Sci., **33**, 471 (1958).
210. R. Kirste, G. V. Schulz, Z. Physik. Chem. (Frankfurt), **27**, 301 (1961)
211. A. F. V. Eriksson, Acta Chem. Scand., **7**, 623 (1953).
212. A. F. V. Eriksson, Acta Chem. Scand., **10**, 378 (1956).
213. A. Ehrenberg, Acta Chem. Scand., **11**, 1257 (1957).
214. R. V. Webber, J. Am. Chem. Soc., **78**, 536 (1956).
215. G. V. Schulz, R. Kirste, Z. Physik. Chem. (Frankfurt), **30**, 171 (1961).
216. G. V. Schulz, A. Haug, R. Kirste, Z. Physik. Chem., **38**, 1 (1963).
217. R. Schumacher, H.-G. Elias, Makromol. Chem., **76**, 12 (1964).
218. H.-G. Elias, R. Schumacher, Makromol. Chem., **76**, 23 (1964).
219. P. R. Saunders, J. Polym. Sci., **57**, 131 (1961).
220. D. W. Carlson, Dissertation, University Delaware, June 1959.
221. H. C. Beachell, D. W. Carlson, J. Polym. Sci., **40**, 543 (1959).
222. R. Signer, P. V. Tavel, Helv. Chim. Acta, **21**, 535 (1938).
223. H. Mosimann, Helv. Chim. Acta, **26**, 61 (1943).
224. B. H. Soerbo, Acta Chem. Scand., **7**, 1129 (1953).
225. A. Ehrenberg, H. Theorell, Acta Chem. Scand., **9**, 1193 (1949).
226. H. Theorell, R. Bonnichsen, Acta Chem. Scand., **5**, 1105 (1951).
227. H. G. Boman, Acta Chem. Scand., **5**, 1311 (1951).
228. A. Ehrenberg, S. Palèus, Acta Chem. Scand., **9**, 538 (1955).

229. V. N. Tsvetkov, V. G. Alsoshin, Zh. Fiz. Khim., **33**, 2767 (1959). (Russ. J. Phys. Chem., **33**, 619 (1959).
230. R. Chiang, J. Phys. Chem., **69**, 1645 (1965).
231. V. Kokle, F. W. Billmeyer, Jr., L. T. Muus, E. J. Newitt, J. Polym. Sci., **62**, 251 (1962).
232. J. Bischoff, V. Desreux, Bull. Soc. Chim. Belges, **61**, 10 (1952).
233. Y. Hachihama, H. Sumitomo, Technol. Rept. Osaka Univ., **3**, 385 (1953).
234. H. Vink, Arkiv. Kemi, **14**, 29 (1957).
235. S. Claesson, W. Bergmann, G. Jayme, Svensk Papperstid, **62**, 141 (1959).
236. W. H. Stockmayer, L. L. Chan, ACS Meeting, Detroit, 1965, Polymer Preprints, **6**, 333 (1965).
237. M. J. R. Cantow, R. S. Porter, J. F. Johnson, ACS Meeting, Detroit, 1965, Polymer Preprints, **6**, 338 (1965).
238. G. M. Guzman, Anales Real Soc. Espan, Fis. Quim. B (Madrid), **52**, 377 (1956).
239. J. Schurz, Papier, **15**, 530 (1961).
240. R. Arnold, S. R. Caplan, Trans. Faraday Soc., **51**, 857 (1955).
241. A. Ehrenberg, K. Dalziel, Acta Chem. Scand., **11**, 398 (1957).
242. A. Ehrenberg, K. Agner, Acta Chem. Scand., **12**, 95 (1958).
243. K. Z. Fattakhov, V. N. Tsvetkov, O. V. K. Kallistov, Zh. Eksperim. i. Teor. Fiz., **26**, 351 (1954).
244. M. L. Wagner, H. A. Scheraga, J. Phys. Chem., **60**, 1066 (1956).
245. C. E. H. Bawn, R. F. Freeman, A. R. Kamaliddin, Trans. Faraday Soc., **46**, 862 (1950).
246. L. Trossarelli, G. Saini, Atti Accad. Sci. Torino: Classe Sci. Fis. Mat. Nat., **90**, 419 (1955-56).
247. G. Saini, L. Trossarelli, Atti Accad. Sci. Torino: Classe Sci. Fis. Mat. Nat., **90**, 431 (1955-56).
248. H. Frind, Faserforsch. Textiltech., **5**, 540 (1954).
249. Q. A. Trementozzi, J. Am Chem. Soc., **76**, 5273 (1954).
250. V. E. Eskin, T. I. Vokov, Vysokomol. Soedin., **5**, 614 (1963).
251. L. G. Longsworth, J. Am. Chem. Soc., **75**, 5705 (1953).
252. L. G. Longsworth, H. Phys. Chem., **58**, 770 (1954).
253. F. W. Billmeyer, Jr., C. B. DeThan, J. Am. Chem. Soc., **77**, 4763 (1955).
254. A. Takahashi, J. Hayashi, I. Kagawa, Kogyo Kagaku Zasshi, **60**, 1059 (1957).
255. H. H. Weber, Biochim. Biophys. Acta, **4**, 12 (1950).
256. O. Snellman, T. Erdoes, Biochim. Biophys. Acta, **2**, 650 (1948).
257. E. J. Cohn, J. L. Oncley, L. E. Strong, W. L. Hughes, Jr., S. H. Armstrong, Jr., J. Clin, Invest., **23**, 417 (1944).
259. I. Jullander, Arkiv Kemi, **21A**, 8 (1945).
260. D. S. Feingold, M. Gehatia, J. Polym. Sci., **22**, 783 (1957).
261. G. L. Borgin, J. Am. Chem. Soc., **71**, 2247 (1949).
262. J. Y. Chien, L. H. Shih, S. C. Yu, J. Polym. Sci., **29**, 117 (1958).
263. J. Y. Chien, L. H. Shih, Z. Physik. Chem., **207**, 60 (1957).
264. J. Y. Chien, L. H. Shih, K. T. Shih, Hua Hsueh Hsueh Pao, **23**, 215 (1957).
265. H.-G. Elias, H. H. Scshluhbach, Ann. Chem., **627**, 126 (1959).
266. W. Scholtan, Makromol. Chem., **14**, 169 (1954).
267. G. M. Chetyrkina, V. G. Alsoshin, S. Y. Frenkel, Vysokomol. Soedin., **1**, 1133 (1959).
268. G. V. Schulz, M. Hoffmann, Makromol. Chem., **23**, 220 (1957).
269. H. Mosiman, T. Svedberg, Kolloid-Z., **100**, 99 (1942).
270. H. Sumitomo, Y. Hachihama, Chem. High Polymers (Japan), **10**, 544 (1953).
271. H. Sumitomo, Y. Yatsuhama, Chem. High Polymers (Japan), **11**, 65 (1954).
272. H.-G. Elias, Makromolekuele, Huethig and Wepf, Basel-Heidelberg (1971)
273. E. F. Casassa, W. H. Stockmayer, Polymer, **3**, 53 (1962).
274. T. G. Fox, J. B. Kinsinger, H. F. Mason, E. M. Schuele, Polymer, **3**, 71 (1962).
275. E. Cohn-Ginsberg, T. G. Fox, H. F. Mason, Polymer, **3**, 97 (1962).
276. H. Campbell, P. Johnson, Trans. Faraday Soc., **40**, 221 (1944).
277. W. Brown, D. Henley, Makromol. Chem., **79**, 68 (1964).
278. W. Brown, D. Henley, J. Oehman, Arkiv Kemi, **22**, 189 (1964).
279. R. Van Leemput, R. Stein, J. Polym. Sci. A, **2**, 4039 (1964).
280. R. Van Leemput, R. Stein, J. Polym. Sci. A, **1**, 985 (1963).
281. W. Burchard, E. Husemann, Makromol. Chem., **44/46**, 358 (1961).
282. T. Matsuo, H. Inagaki, Makromol. Chem., **55**, 150 (1962).
283. T. Matsuo, H. Inagaki, Makromol. Chem., **53**, 130 (1962).
284. W. Brown, D. Henley, J. Oehman, Makromol. Chem., **62**, 164 (1963).
285. W. Brown, D. Henley, J. Oehman, Makromol. Chem., **64**, 49 (1963).
286. J. B. Berkowitz, M. Yamin, R. M. Fuoss, J. Polym. Sci., **28**, 69 (1958).
287. A. Ciferri, M. Kryszewski, G. Weill, J. Polym. Sci., **27**, 167 (1958).
288. R. L. Cleland, J. Polym. Sci., **27**, 349 (1958).
289. S. N. Chinai, P. C. Scherer, D. W. Lewi, J. Polym. Sci., **17**, 117 (1955).
290. L. Trossarelli, E. Campi, G. Saini, J. Polym. Sci., **35**, 205 (1959).
291. L. H. Tung, J. Polym. Sci. A, **2**, 4875 (1964).
292. Q. A. Trementozzi, J. Polym. Sci., **23**, 887 (1957).
292a. W. R. Krigbaum, Q. A. Trementozzi, J. Polym. Sci., **28**, 295, (1958).
293. Q. A. Trementozzi, J. Polym. Sci., **36**, 113 (1959).
294. W. H. Stockmayer, L. D. Moore, Jr., M. Fixman, B. N. Epstein, J. Polym. Sci., **16**, 517 (1955).
295. S. N. Chinai, J. Polym. Sci., **25**, 413 (1957).
296. L. H. Tung, J. Polym. Sci., **36**, 287 (1959).
297. W. Cooper, G. Vaughan, D. E. Eaves, R. W. Madden, J. Polym. Sci., **50**, 159 (1961).
298. P. Dreizehn, R. W. Noble, D. F. Waugh, J. Am. Chem. Soc., **84**, 4938 (1962).

299. P. Doty, J. H. Bradburg, A. M. Holtzer, J. Am. Chem. Soc., **78**, 947 (1956).
300. S. Mao, E. V. Frisman, Vysokomol. Soedin., **4**, 1839 (1962).
301. H. Boedtker, P. Doty, J. Am. Chem. Soc., **78**, 4267 (1956).
302. R. S. Stein, P. Doty, J. Am. Chem. Soc., **68**, 159 (1946).
303. U. P. Strauss, P. L. Wineman, J. Am. Chem. Soc., **80**, 2366 (1958).
304. H. Boedtker, S. Simmons, J. Am. Chem. Soc., **80**, 2550 (1958).
305. A. Holtzner, S. Lowey, J. Am. Chem. Soc., **81**, 1370 (1959).
306. R. Bjorklund, S. Katz, J. Am. Chem. Soc., **78**, 2123 (1956).
307. V. N. Tsetkov, O. V. Kalistov, Zh. Fiz. Khim., **33**, 710 (1959).
308. A. Silberberg, J. Elissaf, A. K. Katschalsky, J. Polym. Sci., **23**, 259 (1957).
309. P. Doty, F. Mishuck, J. Am. Chem. Soc., **69**, 1631 (1947).
310. P. Doty, M. Brownstein, W. Schlener, J. Phys. Colloid, Chem., **53**, 213 (1949).
311. W. R. Krigbaum, P. J. Flory, J. Am. Chem. Soc., **75**, 5254 (1953).
312. M. Morton, T. E. Helminiak, S. D. Gadkary, F. Bueche, J. Polym. Sci., **57**, 471 (1962).
313. J. V. Koleske, S. F. Kurath, J. Polym. Sci. A, **2**, 4123 (1964).
314. G. Sitaramaiah, D. A. I. Goring, J. Polym. Sci., **58**, 1107 (1962).
315. R. Chiang, J. Polym. Sci., **28**, 235 (1958).
316. W. R. Krigbaum, J. E. Kurz, P. Smith, J. Phys. Chem., **65**, 1984 (1961).
317. P. J. Flory, J. Am. Chem. Soc., **65**, 372 (1943).
318. J. B. Kinsinger, R. E. Hughes, J. Phys. Chem., **63**, 2002 (1959). From R. Chiang, Chapter in: B. Ke (Ed.), "Newer Methods of Polymer Characterization", Interscience, New York, 1964, p. 471.
319. F. Danusso, G. Moraglio, Makromol. Chem., **28**, 250 (1958).
320. T. A. Orofino, P. J. Flory, J. Phys. Chem., **63**, 283 (1959).
321. H. W. McCormick, J. Polym. Sci., **41**, 327 (1959).
322. I. Y. Poddubnyi, V. A. Grechanovskii, M. I. Mosevitskii, Vysokomol. Soedin., **5**, 1042 (1964).
323. I. Y. Poddubnyi, V. A. Grechanovskii, M. I. Mosevitskii, Vysokomol. Soedin., **5**, 1049 (1964).
324. I. Y. Poddubnyi, V. A. Grechanovskii, A. V. Podalinskii, Vysokomol. Soedin., **5**, 1588 (1964).
325. W. Burchard, Makromol. Chem., **67**, 182 (1963).
326. A. von Haug, G. Meyerhoff, Makromol. Chem., **53**, 91 (1962).
327. J. M. G. Cowie, D. J. Worsfold, S. Bywater, Trans. Faraday Soc., **57**, 705 (1961).
328. W. R. Krigbaum, D. K. Carpenter, S. Newman, J. Phys. Chem., **62**, 1586 (1958).
329. W. R. Krigbaum, D. K. Carpenter, J. Phys. Chem., **59**, 1166 (1955).
330. M. Matsumoto, Y. Ohyanagi, J. Polym. Sci., **46**, 441 (1960).
331. Y. Ohyanagi, M. Matsumoto, Chem. High Polymer (Japan), **16**, 296 (1959).
332. J. Moacanin, J. Appl. Polym. Sci., **1**, 272 (1959).
333. G. C. Berry, L. M. Hobbs, V. C. Long, Polymer (London), **5**, 31 (1964).
334. W. R. Krigbaum, L. H. Sperling, J. Phys. Chem., **64**, 99 (1960).
335. H. J. L. Trap, J. J. Hermans, J. Phys. Chem., **58**, 757 (1954).
336. N. S. Schneider, P. Doty, J. Phys. Chem., **58**, 762 (1954).
337. S. Bywater, P. Johnson, Trans. Faraday Soc., **47**, 195 (1951).
338. E. V. Gouinlock, Jr., P. J. Flory, H. A. Scheraga, J. Polym. Sci., **16**, 383 (1955).
339. K. Krishnamurti, T. Svedberg, J. Am. Chem. Soc., **52**, 2897 (1930).
340. E. O. Kraemer, J. Phys. Chem., **45**, 660 (1941).
341. E. O. Kraemer, J. Phys. Chem., **46**, 177 (1942).
342. L. H. Arond, H. P. Frank, J. Phys. Chem., **58**, 953 (1954).
343. H. Boedtker, P. Doty, J. Phys. Chem., **58**, 968 (1954).
344. T. Homma,. Fujita, J. Appl. Polym. Sci., **9**, 1701 (1965).
345. B. P. Brand, D. A. I. Goring, P. Johnson, Trans. Faraday Soc., **51**, 872 (1955).
346. E. A. Kanevskaya, P. I. Zubov, L. V. Ivanova, Yu. S. Liptaov, Vysokomol. Soedin., **6**, 981 (1964).
347. S. Lowey, C. Cohen, J. Mol. Biol., **4**, 293 (1962).
348. M. J. Kronman, M. D. Stern, J. Phys. Chem., **59**, 969 (1955).
349. G. Meyerhoff, "Bestimmung des Molekulargewichtes durch Diffusions-und Sedimentationsmessungen (Ultrazentrifuge)". In: Houben-Weyl "Methoden der organischen Chemie", vol. III, Part 1, Georg Thieme Verlag, Stuttgart, 1955, p. 390.
350. J. Hengstenberg, in: H. A. Stuart: "Sedimentation und Diffusion von Makromolekuelen", Physik der Hochpolymeren, vol. II, Springer, Berlin, Goettingen, Heidelberg, 1953, p. 411.
351. A. L. Geddes, "Determination of Diffusivity", in: A. Weissberger: Technique of Organic Chemistry, vol. I, Part 1, Interscience, New York, 1949, p. 551.
352. J. B. Nichols, E. D. Bailey, "Determinations with the Ultracentrifuge", in: A. Weissberger: Technique of Organic Chemistry, vol. I, Part 1, Interscience, New York, 1949, p. 621.
353. P. O. Kinell, B. G. Roanby, Advan. Colloid, Sci., **3**, 182 (1950).
354. H. K. Schachman, "Ultracentrifugation in Biochemistry", Academic Press, New York and London, 1959.
355. R. L. Baldwin, K. E. Van Holde, Fortschr. Hochpolymer-Forsch., **1**, 451 (1960).
356. H. Fujita, "Mathematical Theory of Sedimentation Analysis", Academic Press, New York, 1962.
357. J. W. Williams, "Ultracentrifugal Analysis in Theory and Experiment", Academic Press, New York, London, 1963.
358. M. Gehatia, J. Polym. Sci., **57**, 241 (1962).
359. M. Gehatia, Naturwissenschaften, **48**, 598 (1961).
360. T. Svedberg, H. Rinde, J. Am. Chem. Soc., **46**, 2677 (1924).
361. H. Mosimann, R. Signer, Helv. Chim. Acta, **27**, 1123 (1944).
362. P. W. Bridgman, Proc. Am. Acad. Arts Sci., **61**, 57 (1926).
363. J. Oth, V. Desreux, Bull. Soc. Chim. Belges, **63**, 133 (1954).

364. M. Wales, J. Am. Chem. Soc., **81**, 4758 (1959).
365. J. Fujita, J. Chem. Phys., **24**, 1084 (1956).
366. H.-G. Elias, Makromol. Chem., **29**, 30 (1959).
367. H. Mark, In: Saenger, R., "Der feste Koerper", Hirzel Verlag, Zuerich, 1938, p. 65.
368. Houwink, R. R., J. Prakt. Chem., **157**, 15 (1940).
369. I. Jullander, Arkiv Kemi, Mineral, Geol., **19B**, No. 4 (1944).
370. I. Jullander, "Ueber die Berechnung von Molekulargewichten aus Messungen von Sedimentationsgeschwindigkeit und Diffusion", in: "The Svedberg", Jubilee Volume, Almquist and Wiksells Boktryckeri AB, Uppsala, 1944.
371. S. J. Singer, J. Polym. Sci., **1**, 445 (1946).
372. N. Bauer, "Determination of Density", in: A. Weissberger: "Technique of Organic Chemistry", vol. I, Part 1, Interscience, New York, 1949, p. 253.
373. H. Kienitz, "Bestimmung der Dichte", in: Houben-Weyl: "Methoden der organischen Chemie", vol. III, Part 1, Georg Thieme Verlag, Stuttgart, 1955, p. 163.
374. W. Geffcken, C. Beckmann, A. Kruis, Z. Physik. Chem. B, **20**, 398 (1933).
375. A. B. Lamb, R. E. Lee, J. Am. Chem. Soc., **35**, 1666 (1913).
376. D. A. MacInnes, M. O. Dayhoff, B. R. Ray, Rev. Sci. Instr., **22**, 642 (1951).
377. D. A. MacInnes, M. O. Dayhoff, J. Am. Chem. Soc., **74**, 1017 (1952).
378. J. Stauff, G. Ruemmler, Kolloid-Z., **166**, 152 (1959).
379. P. J. Flory, W. R. Krigbaum, J. Chem. Phys., **18**, 1086 (1950).
380. P. J. Flory, J. Chem. Phys., **17**, 1347 (1949).
381. W. H. Stockmayer, E. F. Casassa, J. Chem. Phys., **20**, 1560 (1952).
382. W. R. Krigbaum, P. J. Flory, J. Polym. Sci., **9**, 503 (1952).
383. M. v. Smoluchowski, Ann. Physik., **25**, 205 (1908).
384. A. Einstein, Ann. Physik., **33**, 1275 (1910).
385. P. Debye, J. Appl. Physics, **15**, 338 (1944).
386. G. V. Schulz, Z. Physik. Chem., **193**, 168 (1944).
387. P. J. Flory, "Principles of Polymer Chemistry", Cornell University Press, Ithaca, New York, 1953.
388. L. Mandelkern, P. J. Flory, J. Chem. Phys., **19**, 984 (1951).
389. C. Tanford, "Physical Chemistry of Macromolecules", Wiley, New York, 1961.
390. M. V. Volkenstein, "Configurational Statistics of Polymeric Chains", Interscience, New York-London, 1963.
391. P. J. Flory, W. R. Krigbaum, Ann. Rev. Phys. Chem., **2**, 383 (1951).
392. F. T. Wall, L. A. Hiller, Jr., Ann. Rev. Phys. Chem., **5**, 267 (1954).
393. J. J. Hermans, Ann. Rev. Phys. Chem., **8**, 179 (1957).
394. E. F. Casassa, Ann. Rev. Phys. Chem., **11**, 477 (1960).
395. W. H. Stockmayer, Makromol. Chem., **35**, 54 (1960).
396. T. B. Grimley, "The Theory of High Polymer Solutions", in: J. C. Robb, F. W. Peaker (Eds.), "Progress in High Polymers", vol. I, Academic Press, New York, 1961.
397. R. E. Hughes, C. A. V. Frankenberg, Ann. Rev. Phys. Chem., **14**, 291 (1963).
398. P. J. Flory, J. Chem. Phys., **10**, 51 (1942).
399. M. L. Huggins, J. Phys. Chem., **46**, 151 (1942).
400. J. Feisst, H.-G. Elias, Makromol. Chem., **82**, 78 (1965).
401. W. Burchard, M. Nosslir, Makromol. Chem., **82**, 109 (1965).
402. H. L. Wagner, K. F. Wissbrun, Makromol. Chem., **81**, 14 (1965).
403. M. L. Wallach, Polymer Preprints, **6/1**, 53 (1965).
404. J. Leonard, H. Daoust, J. Phys. Chem., **69**, 1174 (1965).
405. P. R. Sunders, J. Polym. Sci. A-3, 1221 (1965).
406. J. Blanchford, R. F. Robertson, J. Polym. Sci. A, **3**, 1289 (1965).
407. H.-G. Elias, H. Schlumpf, Makromol. Chem., **85**, 118 (1965).
408. J. Springer et al., Ber. Bunsenges. Physik. Chem., **69**, 494 (1965).
409. M. J. R. Cantow et al. Polymer Preprints, **6/1**, 338 (1965).
410. M. J. R. Contow et al. Makromol. Chem., **87**, 248 (1965).
411. D. Kranz et al. Biochim. Biophys. Acta, **102**, 514 (1965).
412. G. Meyerhoff, N. Suetterlin, Makromol. Chem., **87**, 258 (1965).
413. M. Kurata, H. Utiyama, K. Kamada, Makromol. Chem., **88**, 281 (1965).
414. G. Rehage et al. Makromol. Chem., **88**, 232 (1965).
415. R. Kirste, W. Wunderlich, J. Polym. Sci. B, **3**, 851 (1965).
416. A. Chatterjee, J. Am. Chem. Soc., **86**, 3640 (1964).
417. T. Lindahl et al., J. Am. Chem. Soc., **87**, 4961 (1965).
418. C. Rossi, E. Bianchi, E. Pedemonte, Makromol. Chem., **89**, 95 (1965).
419. H.-G. Elias, O. Etter, Makromol. Chem., 228 (1965).
420. J. B. Kinsinger, L. E. Ballard, J. Polym. Sci. A, **3**, 3963 (1965).
421. M. L. Wallach, M. A. Kabayama, Polymer Preprints, **7/1**, 237 (1966).
422. B. R. Jennings, H. G. Jerrard, J. Chem. Phys., **44**, 1291 (1966).
423. H.-G. Elias, H. P. Lys, Makromol. Chem., **92**, 1 (1966).
424. P. W. Heaps et al., Nature, **209**, 397 (1966).
425. W. Brown, R. Wirkstroem, Eur. Polym. J., **1**, 1 (1965).
426. O. Wetter, Z. Naturforschung, **19b**, 60 (1964).
427. M. Kalfus, J. Mitus, J. Polym. Sci. A, **1/4**, 953 (1966).
428. G. Allen, C. Booth, C. Price, Polymer, **7**, 167 (1966).
429. J. E. Mark, R. A. Wessling, R. E. Hughes, J. Phys. Chem., **70**, 1895 (1966).
430. W. H. Stockmayer, L. L. Chan, J. Polym. Sci. A, **4**, 437 (1966).
431. G. C. Berry, J. Chem. Phys., **44**, 4550 (1966).
432. C. Garbuglio et al., Makromol. Chem., **97**, 97 (1966).
433. L. J. Fetters, H. Yu, Polymer Preprints, **7/2**, 443 (1966).
434. H. Fujita et al., Biopolymers, **4**, 781 (1966).
435. H. Fujita et al., Biopolymers, **4**, 769 (1966).
436. J. Rosenbloom, E. C. Cox, Biopolymers, **4**, 747 (1966).
437. H. Elmgren, Arkiv Kemi, **24/3**, 237 (1965).
438. G. Delmas, D. Patterson, Polymer, **7**, 513 (1966).
439. M. J. Hunter, J. Phys. Chem., **70**, 3285 (1966).

440. J. M. G. Cowie, S. Bywater, J. Macromol. Chem., **1**, 851 (1966).
441. M. L. Wallach, M. A. Kabayama, J. Polym. Sci. A-1, **4**, 2667 (1966).
442. W. Scholtan, S. Y. Lie, Makromol. Chem., **98**, 204 (1966).
443. T. Kataka et al., J. Phys. Chem., **70**, 4099 (1966).
444. K. Hanafusa et al., J. Phys. Chem., **70**, 4004 (1966).
445. M. Iwama et al., J. Macromol. Chem., **1**, 701 (1966).
446. D. K. Carpenter, J. Polym. Sci. A, **2**, 923 (1966).
447. C. Tanford et al., J. Am Chem. Soc., **89**, 729 (1967).
448. N. Eliezer, Biopolymers, **5**, 105 (1967).
449. B. Porsch, M. Kubin, J. Polym. Sci. C, **16**, 515 (1967).
450. A. Teramoto, T. Yamashita, H. Fujita, J. Chem. Phys., **46**, 1919 (1967).
451. M. Bohdanecky et al., J. Polym. Sci. A-2, **5**, 343 (1967).
452. M. Nagasawa, Y. Eguchi, J. Phys., **71**, 880 (1967).
453. S. Gundiah, Makromol. Chem., **104**, 196 (1967).
454. A. A. Tager, V. M. Andreeva, J. Polym. Sci. C, **16**, 1145 (1967).
455. M. L. Wallach, J. Polym. Sci. A-2, **5**, 653 (1967).
456. J. Llopis, A. Albert, P. Usobiaga, European Polym. J., **3**, 259 (1967).
457. J. Naghizadeh, J. Springer, Kolloid-Z., **215**, 21 (1967).
458. J. M. Barrales-Rienda, D. C. Pepper, Polymer, **8**, 337 (1967).
459. T. E. Helminiak, C. L. Brenner, W. E. Gibbs, Polymer Preprints, **8**, 284 (1967).
460. G. Meyerhoff, J. Polymer Sci. C, **16**, 1579 (1967).
461. D. H. Juers, H. A. Swenson, S. F. Kurath, J. Polym. Sci. A-2, **5**, 361 (1967).
462. R. B. Beevers, Polymer, **8**, 419 (1967).
463. P. Callot, A. Banderet, J. Chim. Phys., **64**, 1260 (1967).
464. S. Lapanje, C. Tanford, J. Am. Chem. Soc., **89**, 5030 (1967).
465. G. Meyerhoff, Makromol. Chem., **107**, 124 (1967).
466. J. M. Barrales-Rienda, D. C. Pepper, European Polym. J., **3**, 535 (1967).
467. N. Kuwahara, T. Okazawa, M. Kaneko, J. Chem. Phys., **47**, 3357 (1967).
468. W. J. Closs, J. G. Jerrard, B. R. Jennings, Nature, **215**, 1196 (1967).
469. W. Wunderlich, Makromol. Chem., **108**, 315 (1967).
470. W. Scholtan, S. Y. Lie, Makromol. Chem., **108**, 104 (1967).
471. K. Karunakaran, M. Santappa, Makromol. Chem., **111**, 20 (1967).
472. J. Noda et al., J. Phys. Chem., **71**, 4048 (1967).
473. W. Banks, C. F. Greenwood, D. J. Hourston, Trans. Faraday Soc., **64**, 363 (1968); Makromol. Chem., **111**, 226 (1968).
474. R. G. Kirste, W. Wunderlich, Z. Physik. Chem., **58**, 133 (1968).
475. G. Meyerhoff, U. Moritz, Makromol. Chem., **109**, 143 (1968).
476. H.-G. Elias, J. Gerber, Makromol. Chem., **112**, 142 (1967).
477. E. Penzel, G. V. Schulz, Makromol. Chem., **112**, 260 (1968).
478. S. P. Rao, N. Santappa, J. Polym. Sci. A-1, **6**, 95 (1968).
479. T. Kotaka, Y. Murakami, H. Inagaki, J. Phys. Chem., **72**, 829 (1968).
480. J. Moakanin et al., J. Macromol. Sci. A: Chem., **1/8**, 1497 (1967).
481. G. V. Schulz, H. Baumann, Makromol. Chem., **114**, 122 (1968).
482. K. Karunakaran, M. Santappa, J. Polym. Sci. A-2, **6**, 713 (1968).
483. H. Triebel, Biopolymers, **6**, 449 (1968).
484. T. Kato, K. Miyaso, M. Nagasawa, J. Phys. Chem., **72**, 2161 (1968).
485. G. Cohen, H. Eisenberg, Biopolymers, **6**, 1077 (1968).
486. T. Kotak, N. Donkai, J. Polym. Sci. A-2, **6**, 1457 (1968).
487. J. Goldfarb, S. Rodriguez, Makromol. Chem., **116**, 96 (1968).
488. J. Lilie, J. Springer, K. Ueberreiter, Kolloid-Z., **226**, 138 (1968).
489. J. Danon, J. Polym. Sci. C, **16**, 4071 (1968).
490. J. M. G. Cowie, Polymer, **9**, 587 (1968).
491. Y. Arnaud, J. Polym. Sci. C., **16**, 4103 (1968).
492. R. G. Kirste, G. Wild, Makromol. Chem., **121**, 174 (1969).
493. T. Norisuye et al., J. Chem. Phys., **49**, 4330 (1968).
494. N. Kuwahara, T. Oikawa, M. Kaneko, J. Chem. Phys., **49**, 4972 (1968).
495. D. N. Cramod, J. R. Urwin, Europ. Polym. J., **5**, 35 and 45 (1969).
496. W. R. Moore, M. Uddin, European Polym. J., **5**, 185 (1969).
497. L. A. Utracki, R. Simha, J. Fetters, J. Polym. Sci. A-2, **6**, 2051 (1968).
498. M. Heskins, J. E. Guillet, J. Macromol. Sci. A, **2/8**, 1441 (1968).
499. N. Kuwahara et al., J. Polym. Sci. A-2, **7**, 285 (1969).
500. K. Matsumura, Makromol. Chem., **124**, 204 (1969).
501. N. Kuwahara et al., J. Polym. Sci. C, **23**, 543 (1968).
502. A. Kotera et al., J. Polym. Sci. C, **23**, 619 (1968).
503. L. A. Papazian, Polymer, **10**, 399 (1969).
504. W. Burchard, Polymer, **10**, 467 (1969).
505. A. H. Rosenberg, W. Studier, Biopolymers, **7**, 765 (1969).
506. W. Maechtle, H. Fischer, Angew. Makromol. Chem., **7**, 147 (1969).
507. J. Danon, R. Derai, European Polym. J., **5**, 659 (1969).
508. T. S. Ng, G. V. Schulz, Makromol. Chem., **127**, 165 (1969).
509. L. H. Tung, G. W. Knight, J. Polym. Sci. A-2, **7**, 1623 (1969).
510. R. Dick, D. Roussel, H. Benoit, J. Chim. Phys., **66**, 1612 (1969).
511. M. Kubin, B. Porsch, European Polym. J., **6**, 97 (1970).
512. R. Jerome, V. Desreux, European Polym. J., **6**, 411 (1970).
513. A. Takahashi et al., J. Phys. Chem., **74**, 944 (1970).
514. H. Dautzenberg, Faserforsch. Textiltechn., **21**, 117 (1970).
515. W. Maechtle, Angew. Makromol. Chem., **10**, 1 (1970).
516. Y. Noguchi et al., J. Chem. Phys., **52**, 2651 (1970).
517. G. Tanaka, S. Imai, H. Yamakawa, J. Chem. Phys., **52**, 2639 (1970).

518. K. Kamide, K. Sugamiya, C. Nakayama, Makromol. Chem., **132**, 75 (1970).
519. G. Adank, H.-G. Elias, Makromol. Chem., **102**, 151 (1967).
520. H.-G. Elias, H. Dietschy, Makromol. Chem., **105**, 102 (1967).
521. K. D. Goebel, W. G. Miller, Macromolecules, **3**, 64 (1970).
522. K. Sakato, M. Kurata, Polymer J., **1**, 260 (1970).
523. R. L. Arnett, R. Q. Gregg, J. Phys. Chem., **74**, 1593 (1970).
524. W. Wunderlich, Angew. Chem., **11**, 73 (1970).
525. W. Wunderlich, Angew. Makromol. Chem., **11**, 189 (1970).
526. G. Meyerhoff, S. Shimotsuma, Makromol. Chem., **135**, 195 (1970).
527. K. Matsumura, Polymer J., **1**, 322 (1970).
528. M. Lechner, G. V. Schulz, Eur. Polym. J., **6**, 945 (1970).
529. R. L. Cleland, J. L. Wang, Biopolymers, **9**, 799 (1970).
530. T. C. Laurent, M. Ryan, A. Pietruszkiewicz, Biochim. Biophys. Acta, **42**, 476 (1960).
531. W. Banks, C. T. Greenwood, J. Sloss, Makromol. Chem., **140**, 109 (1970).
532. B. J. Rietveld, J. Polym. Sci. A-2, **8**, 1837 (1970).
533. K.-J. Linow, B. Philipp, Faserforsch. Textiltechn., **21**, 483 (1970).
534. K. Okita, A. Teramoto, H. Fujita, Polym. J., **1**, 582 (1970).
535. R. H. Marchessault, K. Okamura, C. J. Su, Macromolecules, **3**, 735 (1970).
536. H. A. Scheraga, L. Mandelkern, J. Am. Chem. Soc., **75**, 179 (1953).
537. J. Cornibert et al., Macromolecules, **3**, 741 (1970).
538. A. Krasna, J. R. Dawson, J. A. Harpst, Biopolymers, **9**, 1017 (1970).
539. A. Krasna, Biopolymers, **9**, 1029 (1970).
540. K. E. Reinert, J. Strassburger, H. Triebel, Biopolymers, **10**, 285 (1971).
541. K. E. Reinert, Biopolymers, **10**, 275 (1971).
542. M. Gilbert, F. J. Hybart, J. Polym. Sci. A-1, **9**, 227 (1971).
543. V. Petrus, J. Danihel, M. Bohdanecky, Europ. Polym. J., **7**, 143 (1971).
544. W. J. Bengough, G. F. Grant, Eur. Polym. J., **7**, 203 (1971).
545. W. Banks, C. T. Greenwood, J. Sloss, Eur. Polym. J., **7**, 263 (1971).
546. J. R. Urwin, M. Girolamo, Makromol. Chem., **142**, 161 (1971).
547. W. Banks, C. T. Greenwood, Makromol. Chem., **144**, 135 (1971).
548. G. C. Berry, J. Polym. Sci. A-2,, **9**, 687 (1971).
549. K. Kamada, H. Sato, Polym. J., **2**, 489 (1971).
550. P. F. Mijnlieff, W. J. M. Jaspers, Trans. Faraday Soc., **67/6**, 1837 (1971).
551. M. Kozhokaryu, V. S. Skazka, K. G. Berdnikova, Vysokomol. Soedin., **8**, 1063 (1966).
552. J. J. Poddubnyi, A. V. Podalinskii, V. A. Grechanovskii, Vysokomol. Soedin., **8**, 1556 (1966).
553. A. B. Duckova et al., Vysokomol. Soedin., **8**, 1814 (1966).
554. V. M. Golubev, S. Ya. Frenkel, Vysokomol. Soedin. A, **9**, 1847 (1967).
555. V. N. Tsvetkov et al., Vysokomol. Soedin. A, **10**, 74 (1968).
556. V. N. Tsvetkov et al., Vysokomol. Soedin. A, **10**, 547 (1968).
557. V. M. Golubev, S. Ya. Frenkel, Vysokomol. Soedin. A, **10**, 750 (1968).
558. Yu. Ye. Eizner, A. N. Cherkasov, S. J. Klenin, Vysokomol. Soedin. A, **10**, 1971 (1968).
559. V. N. Tsvetkov et al., Vysokomol. Soedin., **7**, 1103 (1965).
560. V. N. Tsvetkov et al., Vysokomol. Soedin. A, **11**, 349 (1970).
561. S. R. Rafikov, Vysokomol. Soedin. A, **11**, 1990 (1969).
562. A. A. Tager et al., Vysokomol. Soedin. A, **12**, 1320 (1970).
563. V. N. Tsvetkov et al., Vysokomol. Soedin. A, **12**, 1974 (1970).
564. N. Nakata, Makromol. Chem., **149**, 99 (1971).
565. L. J. Fetters, H. Yu, Macromolecules, **4**, 385 (1971).
566. K. de Groot et al., Kolloid-Z., **246**, 578 (1971).
567. K. F. Elgert, E. Seiler, Makromol. Chem., **151**, 83 (1972).
568. R. Gabler, N. C. Ford, Jr., F. E. Karasz, Polymer Preprints, **12**, 776 (1971).
569. K.-J. Linow, B. Philipp, Faserforsch. Textiltechn., **22**, 444 (1971).
570. K. Kamada, H. Sato, Polymer J., **2**, 593 (1971).
571. O. Kramer, J. E. Frederick, Macromolecules, **4**, 613 (1971).
572. W. L. Petikolas, Adv. Polym. Sci., **9**, 285 (1972).
573. T. Homma, H. Fujita, J. Appl. Polym. Sci., **9**, 1701 (1965).
574. M. Abe et al., J. Appl Polym. Sci., **8**, 2549 (1965).
575. L. C. Cerny, R. C. Graham, H. James, Jr., J. Appl. Polym. Sci., **11**, 1941 (1967).
576. W. J. Closs, B. R. Jennings, H. G. Jerrard, Europ. Polym. J., **4**, 639 (1968).
577. H. Matsuda, H. Aonuma, S. Kuroiwa, J. Appl. Polym. Sci., **14**, 335 (1970).
578. M. G. Wirick, H. Waldmann, J. Appl. Polym. Sci., **14**, 579 (1970).
579. J. K. Nekrasov et al., Vysokomol. Soedin. A, **14**, 866 (1972).
580. V. N. Tsvetkov et al., Vysokomol. Soedin. A, **14**, 427 (1972).
581. A. Wada, N. C. Ford, F. E. Karasz, Polymer Preprints, **13**, 191 (1972).
582. K. Lederer, Polymer Preprints, **13**, 203 (1972).
583. V. N. Tsvetkov et al., Vysokomol. Soedin. A, **14**, 369 (1972).
584. D. J. Goldwasser, D. J. Williams, Polymer Preprints, **12**, 539 (1971).
585. K. Kubo, K. Ogino, Bull. Chem. Soc. (Japan), **44**, 997 (1971).
586. T. Lucas, Compt. Rend. C, **270**, 1377 (1970).
587. M. Gorden, S. Polowinski, British Polymer J. , **2**, 182 (1970).
588. K. Matsumura, Bull. Chem. Soc. (Japan), **43**, 1303 (1970).
589. J. K. Nekrasov, Vysokomol. Soedin. A, **13**, 1707 (1971).
590. M. Abe, M. Iwama, T. Homma, J. Chem. Soc. (Japan) Ind. Chem. Sect., **72**, 2313 (1969).
591. V. Bugdahl, Kautschuk, Gummi, Kunststoffe, **22**, 486 (1969).

592. H. Matsuda, H. Aonuma, S. Kuroiwa, Rep. Prog. Polym. Phys. Jpn., **11**, 25 (1968).
593. A. Kotera, H. Matsuda, K. Takemura, Rep. Prog. Polym. Phys. Jpn., **11**, 61 (1968).
594. A. Kotera, T. Hamada, Rep. Prog. Polym. Phys. Jpn., **11**, 57 (1968).
595. K. Matsumura, Bull. Chem. Soc. (Japan), **42**, 1874 (1969).
596. G. Moraglio, F. Zoppi, F. Danusso, Chim. Ind. (Milan), **51**, 117 (1969).
597. H. Matsuda, K. Yamano, H. Inagaki, J. Chem. Soc. (Japan) Ind. Chem. Sect., **73**, 390 (1970).
598. M. Matsushima, M. Fukatsu, M, Kurata, Bull. Chem. Soc. (Japan), **41**, 2570 (1968).
599. A. J. Hyde, A. G. Tanner, J. Colloid Interf. Sci., **28**, 179 (1968).
600. Cervenka et al., Collection Czech. Chem. Commun., **33**, 4248 (1968).
601. M. Abe et al., Bull. Chem. Soc. (Japan), **41**, 2330 (1968).
602. V. Petrus, Collection Czech. Chem. Commun., **33**, 119 (1968).
603. K. Kamide et al., Chem. High Polymers (Tokyo), **24**, 679 (1967).
604. G. Moraglio, N. Oddo, Chim. Ind., **48**, 224 (1966).
605. J. M. G. Cowie, E. L. Cussler, J. Chem. Phys., **46**, 4886 (1967).
606. K. F. Elgert, K. Cammann, Z. Anal. Chem., **226**, 193 (1967).
607. H. Sato, Bull. Chem. Soc. (Japan), **39**, 2335 (1966).
608. H. Sato, Bull. Chem. Soc. (Japan), **39**, 2340 (1966).
609. K. Takamizawa, Bull. Chem. Soc. (Japan), **39**, 1168 (1966).
610. A. Kotera et al. Bull. Chem. Soc. (Japan), **39**, 1192 (1966).
611. C. Trevisoi, R. Francesconi, Chim. Ind., **47**, 1184 (1965).
612. G. Moraglio, J. Masini, Chim. Ind., **47**, 1050 (1965).
613. C. Trevissoi, R. Francesconi, Chim. Ind., **47**, 744 (1965).
614. J. Eigner, P. Doty, J. Mol. Biol., **12**, 549 (1965).
615. R. Jaenicke, M. Gehatia, Biochim. Biophys. Acta, **45**, 217 (1960).
616. A. Soda, T. Fujimoto, M. Nagasawa, J. Phys. Chem., **71**, 4274 (1967).
617. M. Pizzoli, G. Ceccorulli, Europ. Polym. J., **8**, 769 (1972).
618. M. B. Huglin, "Light Scattering from Polymer Solutions", Academic Press, London, New York, 1972.
619. M. Helmstedt, G. Fleischer, H. K. Roth, Plaste Kautsch., **29**, 628
620. G. Patlmamanoharan, H. d. Hek, A. Vrij, Colloid Polym. Sci., **261**, 786 (1983).
621. G. M. Kok, A. Rudin, J. Appl. Polym. Sci., **27**, 3357 (1982).
622. H. K. Mahabadi, A. Rudin, Makromol. Chem., **179**, 2977 (1978).
623. B. Nystroem, J. Roots, Makromol. Chem., **180**, 2419 (1979).
624. B. Chitrangad, H. R. Osmers, J. Polym. Sci., Polym. Phys. Ed., **18**, 1219 (1980).
625. R. D. Ferguson, E. von Meerwall, J. Polym. Sci., Polym. Phys. Ed., **18**, 1285 (1980).
626. C. Xuexin, X. Zhongde, E. von Meerwall, N. Seunge, N. Hadjichristidis, L. J. Fetters, Macromolecules, **17**, 1343 (1984).
627. A. A. Berg, Vysokomol. Soedin., Ser. A, **22**, 543 (1980).
628. T. Schwartz, J. Francois, G. Weill, Polymer, **21**, 247 (1980).
629. W.-M. Kulicke, R. Kniewske, J. Klein, Prog. Polym. Sci., **8**, 373 (1982).
630. P. Munk, T. M. Aminabhavi, P. Williams, D. E. Hoffmann, M. Chmelir, Macro-molecules, **13**, 871 (1980).
631. W. Maechtle, Makromol. Chem., **183**, 2515 (1982).
632. W. Scholtan, Makromol. Chem., **14**, 169 (1954).
633. J. Klein, K.-D. Conrad, Makromol. Chem., **181**, 227 (1980).
634. J. Francois, T. Schwartz, G. Weill, Macromolecules, **13**, 564 (1980).
635. J. Francois, D. Sarazin, T. Schwartz, G. Weill, Polymer, **20**, 696 (1979).
636. A. DiNapoli, B. Chu, C. Cha, Macromolecules, **15**, 1174 (1982).
637. S. Wunder, N. C. Ford, F. E. Karasz, J. Tan, J. Colloid Interface Sci., **63**, 290 (1978).
638. L. A. Bimendina, V. V. Roganov, E. A. Bekturov, J. Polym. Sci., Polym. Symp., **44**, 65 (1974).
639. H.-U. ter Meer, W. Burchard, W. Wunderlich, Colloid Polym. Sci., **258**, 675 (1980).
640. G. Rosskopf, J. Polawka, G. Rehage, W. Borchard, Ber. Bunsenges. Phys. Chem., **83**, 360 (1979).
641. B. Chu, D. Lee, Macromolecules, **17**, 926 (1984).
642. R. B. Flippen, Soc. Plast. Eng. Tech. Pap., **29**, 550 (1983).
643. M. Dolezalova, V. Petrus, Z. Tuzar, M. Bohdanecky, Eur. Polym. J., **12**, 701 (1976).
644. M. Dolezalova, V. Petrus, M. Bohdanecky, Collect. Czech. Chem.. Commun., **42**, 1960 (1977).
645. B. Nystroem, L. O. Sundeloef, M. Bohdanecky, V. Petrus, J. Polym. Sci., Polym. Lett. Ed., **17**, 543 (1979).
646. D. Kunz, A. Thurn, W. Burchard, Colloid Polym. Sci., **261**, 635 (1983).
647. M. Schmidt, D. Nerger, W. Burchard, Polymer, **20**, 582 (1979).
648. M. Okabe, K. Shibuya, H. Matsuda, Kobunshi Ronbushu, **38**, 15 (1981).
649. I. Dekany, S. Sipos, F. Szanto, Coll. Polym. Sci., **251**, 752 (1973).
650. N. Ochiai, V. Fujina, Y. Tadokoro, I. Murakami, Polym. J., **13**, 351 (1981).
651. A. Malick, Polimery (Warsaw), **24**, 166 (1979).
652. N. S. Nametkin, O. B. Semenov, S. G. Durgaryan, V. S. Skazka, V. G. Filippova, V. Ya. Nikolayev, Polym. Sci. (USSR) A, **17**, 1115 (1975).
653. D. Landheer, S.-L. Malhatra, J. Macromol. Sci., Chem., **16**, 1349 (1981).
654. J. C. Selser, Macromolecules, **14**, 346 (1981).
655. I. Noda, S. Goto, M. Nagasawa, Polym. J., **8**, 507 (1976).
656. I. Noda, K. Mizatani, K. Tadaya, Macromolecules, **10**, 618 (1977).

657. P. M. Lavrenko, A. A. Bojkov, L. N. Andraeva, E. V. Beljaeva, A. F. Podolskij, Vysokomol. Soedin., Ser. A, **23**, 1937 (1981).
658. P. König, F. Wolf, Plaste Kautsch., **25**, 21 (1978).
659. E. Ozdemir, R. W. Richards, Polymer, **24**, 1097 (1983).
660. N. Nemoto, Y. Makita, Y. Tsunashima, M. Kurata, Macromolecules, **17**, 425 (1984).
661. J. Hadgraft, A. J. Hyde, R. W. Richards, J. Chem. Soc., Faraday Trans., **2**, 75, 1495 (1979).
662. M. Adam., M. Delsanti, J. de Physique, **37**, 1045 (1976).
663. N. Nemoto, Y. Tsunashima, M. Kurata, Polym. J., **13**, 827 (1981).
664. M. Adam, M. Delsanti, Macromolecules, **10**, 1229 (1977).
665. T. M. Aminabhavi, P. Munk, Macromolecules, **12**, 1194 (1979).
666. R. A. Frost, D. Caroline, Macromolecules, **10**, 616 (1977).
667. G. Pouyet, F. Francois, J. Dayantis, G. Weill, Macromolecules, **13**, 176 (1980).
668. M. Schmitt, Macromolecules, **17**, 553 (1984).
669. N. C. Ford, F. E. Karasz, J. E. Owen, Farad. Disc. Chem. Soc., **49**, 228 (1970).
670. C. M. Kok, F. R. Hallett, A. Rudin, Eur. Polym. J., **21**, 33 (1985).
671. J. C. Selser, Macromolecules, **12**, 909 (1979).
672. T. A. King, M. F. Treadaway, J. Chem. Soc., Faraday Trans., **2**, 72, 1473 (1976).
673. T. A. King, A. Knox, J. D. G. McAdam, Polymer, **14**, 293 (1973).
674. T. A. King, A. Knox, W. J. Lee, J. D. G. McAdam, Polymer, **14**, 151 (1973).
675. I. Nishio, A. Wada, Polym. J., **12**, 145 (1980).
676. Y.-H. Lin, B. Chu, Macromolecules, **14**, 385 (1981).
677. G. D. Patterson, J. P. Jarry, C. P. Lindsey, Macromolecules, **13**, 668 (1980).
678. W. Brown, R. M. Johnsen, P. Stilbs, Polym. Bull., **9**, 305 (1983).
679. E. J. Amis, C. C. Han, Y. Matsushita, Polymer, **25**, 650 (1984).
680. C. C. Han, A. Z. Akcasu, Polymer, **22**, 1165 (1981).
681. C. C. Han, A. Z. Akcasu, Amer. Chem. Soc., Div. Polym. Chem., Polym. Prepr., **22**, 68 (1981).
682. G. Allen, P. Vasudevan, E. Y. Hawkins, T. A. King, J. Chem. Soc., Faraday Trans., **2**, 73, 449 (1977).
683. J. Roovers, P. M. Toporowski, J. Polym. Sci., Polym. Phys. Ed., **18**, 1907 (1980).
684. T. F. Reed, J. E. Frederick, Macromolecules, **4**, 72 (1971).
685. J. J. Mulderije, Macromolecules, **13**, 1207 (1980).
686. P. Vidakovic, C. Allain, F. Rondelez, Macromolecules, **15**, 1571 (1982).
687. C. C. Han, F. L. McCrackin, Polymer, **20**, 4278 (1979).
688. D. C. Han, Polymer, **20**, 259 (1979).
689. C. Destor, F. Rondelez, Polymer, **22**, 67 (1981).
690. F. W. Wang, F. L. McCrackin, Polymer, **24**, 1541 (1983).
691. J. C. Brown, P. N. Pusey, R. Dietz, J. Chem. Phys., **62**, 1136 (1975)
692. V. J. Novotny, J. Chem. Phys., **78**, 183 (1982).
693. A. M. Jamieson, M. E. McDonnell, Adv. Chem. Ser., **174**, 163 (1979).
694. M. Okabe, J. Polym. Sci., **22**, 477 (1984).
695. T. A. King, M. F. Treadaway, J. Chem. Soc., Faraday Trans., **2**, 73, 1616 (1977).
696. C. C. Han, A. Z. Akcasu, Macromolecules, **14**, 1080 (1981).
697. J. J. Mulderije, Macromolecules, **15**, 506 (1982).
698. P. Vidakovic, F. Rondelez, Macromolecules, **16**, 253 (1983).
699. P. Vidakovic, C. Allain, F. Rondelez, J. Phys. (Paris), **42**, 323 (1981).
700. E. Gulari, E. Gulari, T. Chu, B. Chu, Polymer, **20**, 347 (1979).
701. T. Cosgrove, J. M. Sutherland, Polymer, **24**, 534 (1983).
702. G. Ponyet, F. Candau, J. Dayantis, Makromol. Chem., **177**, 2973 (1976).
703. V. Petrus, B. Porsch, B. Nystroem, L. O. Sundeloef, Makromol. Chem., **184**, 295 (1983).
704. M. Bohdanecky, V. Petrus, B. Porsch, L. O. Sundeloef, Makromol. Chem., **184**, 309 (1983).
705. H. Huber, W. Burchard, L. J. Fetters, Macromolecules, **17**, 541 (1984).
706. T. L. Yu, H. Reiharian, A. M. Jamieson, Macromolecules, **13**, 1590 (1980).
707. B. Nystroem, J. Roots, R. Bergman, Polymer, **20**, 157 (1979).
708. J. Roots, B. Nystroem, J. Polym. Sci., Polym. Phys. Ed., **19**, 479 (1981).
709. Y. Tsunashima, N. Nemoto, M. Kurata, Macromolecules, **16**, 1184 (1983).
710. Y. Tsunashima, N. Nemoto, Macromolecules, **16**, 1941 (1983).
711. B. Nystroem, B. Porsch, L. O. Sundeloef, Eur. Polym. J., **13**, 683 (1977).
712. B. Nystroem, J. Roots, Eur. Polym. J., **14**, 551 (1978).
713. T. Nose, B. Chu, Macromolecules, **12**, 590 (1979).
714. M. J. Pritchard, D. Caroline, Macromolecules, **14**, 424 (1981).
715. S. J. Candau, I. Butler, T. A. King, Polymer, **24**, 1601 (1983).
716. A. C. Quano, J. Colloid Interface Sci., **63**, 275 (1978).
717. V. S. Skazka, Russ. Chem. Rev., **53**, 506 (1984); V. S. Skazka, Vysokomol. Soed. A, **17**, 2607 (1975).
718. M. E. McDonnell, A. M. Jamieson, J. Polym. Sci., Polym. Phys. Ed., **18**, 1781 (1980).
719. T. L. Yu, H. Reihanian, A. M. Jamieson, Macromolecules, **13**, 1590 (1980).
720. H. Reihanian, T. L. Yu, A. M. Jamieson, Am. Chem. Soc., Div. Polym. Chem. Polym. Prepr., **22**, 78 (1981).
721. M. E. McDonnell, A. M. Jamieson, J. Macromol. Sci., Phys., **13**, 67 (1977).
722. T. L. Yu, H. Reihanian, J. G. Southwick, A. M. Jamieson, J. Macromol. Sci., Phys., **18**, 777 (1981).
723. W. Mandema, H. Zeldenrust, Polymer, **18**, 835 (1977).
724. J. Roots, B. Nystroem, L. O. Sundeloef, B. Porsch, Polymer, **20**, 337 (1979).
725. B. Nystroem, J. Roots, Polymer, **18**, 1289 (1977).

726. J. Roots, B. Nystroem, Macromolecules, **15**, 553 (1982).
727. S. Bantle, M. Schmidt, W. Burchard, Macromolecules, **15**, 1604 (1982).
728. J. Raczek, G. Meyerhoff, Macromolecules, **13**, 1251 (1980).
729. J. Raczek, Eur. Polym. J., **19**, 607 (1983).
730. J. Raczek, Eur. Polym. J., **18**, 847 (1982).
731. J. Raczek, Eur. Polym. J., **18**, 393 (1982).
732. P. N. Pusey, J. M. Vaughan, G. Williams, J. Chem. Soc., Faraday Trans., **2**, 70, 1696 (1974).
733. T. M. Aminabhavi, J. Macromol. Sci., Chem., **19**, 687 (1983).
734. B. Porsch, L. O. Sundeloef, Collect. Czech. Chem. Commun., **46**, 3018 (1981).
735. J. Hendrix, B. Saleh, K. Gnädig, L. de Maeyer, Polymer, **18**, 10 (1977).
736. V. Petrus, B. Porsch, B. Nystroem, L. O. Sundeloef, Macromolecular Chemistry and Physics (Base), **183**, 1279 (1982).
737. M. H. Birnboim, H. Weiss, J. Polym. Sci., Polym. Phys. Ed., **17**, 2225 (1979).
738. K. Ohbayashi, M. Minoda, M. Utiyama, Macromolecules, **14**, 1031 (1981).
739. M. Schmidt, W. Burchard, N. C. Ford, Macromolecules, **11**, 452 (1978).
740. K. S. Schmitz, M. Lu, J. Gauntt, J. Chem. Phys., **78**, 5059 (1983).
741. C. M. Trotter, D. N. Pinder, J. Chem. Phys., **75**, 118 (1981).
742. R. S. Koene, M. Mandel, Macromolecules, **16**, 220 (1983).
743. H. Matsuda, I. Yamada, M. Okabe, S. Kuroiwa, Polym. J., **9**, 527 (1977).
744. Y. Kambe, C. Honda, Polym. Commun., **24**, 208 (1983).
745. B. Appelt, G. Meyerhoff, Macromolecules, **13**, 657 (1980).
746. J. C. Brown, P. N. Pusey, J. W. Goodwin, R. H. Otterwill, J. Phys. A, **8**, 664 (1975).
747. R. S. Koene, T. Nicolai, M. Mandel, Macromolecules, **16**, 227 (1983).
748. V. S. Skazka, G. V. Tarasova, Polym. Sci. USSR, **17**, 2062 (1975).
749. E. Bortel, A. Kochunowski, Macromol. Chem., Rapid Commun., **1**, 205 (1980).
750. W. Brown, P. Stilbs, Polymer, **24**, 188 (1983).
751. B. Nystroem, S. Boileau, P. Hemery, J. Roots, Eur. Polym. J., **17**, 249 (1981).
752. K. Kubota, B. Chu, Macromolecules, **16**, 105 (1983).
753. O. V. Kallistov, J. E. Svetlov, I. G. Silinskaya, V. P. Sklizkova, V. V. Kudriavtsev, M. M. Koton, Eur. Polym. J., **18**, 1103 (1982).
754. I. K. Nekrusov, Vysokomol. Soedin. A, **19**, 910 (1977).
755. V. N. Tovetkov, P. N. Lavrenko, G. M. Pavlov, S. V. Busin, E. P. Astapenko, A. A. Bojkov, N. A. Shildyayeva, S. A. Didenko, B. F. Malicnenko, Vysokomol. Soedin. A, **24**, 2343 (1982). Polym. Sci. USSR 24, 2689 (1982).
756. W. Brown, Macromolecules, **17**, 66 (1984).
757. B. Chu, E. Gulari, Macromolecules, **12**, 445 (1979).
758. C. J. Edwards, R. F. Stapto, J. A. Semlyen, Polymer, **23**, 865 (1982).
759. C. J. Edwards, R. F. Stapto, J. A. Semlyen, Polymer, **21**, 781 (1980).
760. C. J. Edwards, S. Bantle, W. Burchard, R. F. Stapto, J. A. Semlyen Polymer, **23**, 873 (1982).
761. N. S. Nametkin, O. B. Semonov, S. G. Durgaryan, V. S. Skazka, V. G. Filippova, Polym. Sci. USSR, **17**, 1115 (1975).
762. S.-J. Ishida, T. Kitagawa, Y. Nakamoto, K. Kaneko, Polym. Bull., **10**, 533 (1983).
763. G. E. Prozorova, A. V. Pavlov, A. M. Scetinin, M. M. Iovleva, S. P. Papkov, Vysokumol. Soedin. B, **19**, 750 (1977).
764. E. S. Bilimova, V. M. Golubev, Vysokomol. Soedin. B, **21**, 854 (1979).
765. R. T. van Emmerik, C. A. Smolders, Eur. Polym. J., **9**, 931 (1973).
766. D. Singh, K. D. Nigam, J. Appl. Polym. Sci., **23**, 3021 (1979).
767. R. G. Griskey, G. H. Sedahmed, J. Appl. Polym. Sci., **17**, 3761 (1973).
768. W. Sutter, W. Burchard, Makromol. Chem., **179**, 1961 (1978).
769. S. J. Ishida, H. Komatsu, H. Katoh, M. Saito, Y. Miyazaki, K. Kamide, Makromol. Chem., **183**, 3075 (1982).
770. F. H. Holmes, D. I. Smith, J. Chem. Soc., Faraday Trans., **53**, 69 (1957)
771. V. N. Tsvetkov, E. J. Rjumtsev, L. N. Andreeva, N. V. Pogodina, P. N. Lavrenko, L. J. Kutsenko, Eur. Polym. J., **10**, 563 (1974).
772. G. Meyerhoff, G. Büldt, Makromol. Chem., **175**, 675 (1974).
773. D. B. Sellen, Polymer, **16**, 169 (1975).
774. G. M. Pavlov, S. J. Frenkel, Vysokomol. Soedin. A, **25**, 1015 (1983).
775. T. Uemara, T. Moro, J. Komiyama, T. Lijima, Macromolecules, **12**, 737 (1979).
776. B. N. Preston, W. D. Comper, A. E. Hughes, I. Snook, W. van Hagen, J. Chem. Soc., Faraday Trans., **1**, 78, 1209 (1982).
777. W. Brown, P. Stilbs, R. M. Johnsen, J. Polym. Sci., Polym. Phys. Ed., **20**, 1771 (1982).
778. D. B. Sellen, Polymer, **16**, 561 (1975).
779. E. M. Fahner, G. H. Großmann, K. H. Ebert, Makromol. Chem., **185**, 2205 (1984).
780. H. Elmgren, J. Polym. Sci., Polym. Lett. Ed., **20**, 389 (1982).
781. W. R. Sharman, E. L. Richards, G. N. Malcolm, Biopolymers, **17**, 2817 (1978).
782. L. D. Sundeloef, B. Nystroem, J. Polym. Sci., Polym. Lett. Ed., **15**, 377 (1977).
783. B. Nystroem, R. Bergman, Eur. Polym. J., **14**, 431 (1978).
784. S. A. Chung, D. G. Gray, J. Colloid Interface Sci., **67**, 255 (1978).
785. J. G. Southwick, A. M. Jamieson, J. Blackwell, Appl. Polym. Symp., **37**, 385 (1983).
786. J. G. Southwick, M. E. McDonnell, A. M. Jamieson, J. Blackwell, Macromolecules, **12**, 305 (1979).
787. G. Holzwarth, Carbohydr. Res., **66**, 173 (1978).

788. B. Nystroem, J. Roots, Eur. Polym. J., **16**, 201 (1980).
789. M. E. McDonnell, A. M. Jamieson, Biopolymers, **15**, 1283 (1976).
790. K. J. Linow, O. Okatova, B. Philipp, Acta Polym., **34**, 376 (1983).
791. G. Muller, Polym. Bull., **11**, 391 (1984).
792. K. J. Linow, B. Philipp, Faserforsch. Textiltech., **29**, 452 (1978).
793. B. Nystroem, J. Roots, Polymer, **21**, 183 (1980).
794. J. L. Duda, Y. C. Ni, J. S. Vrentas, Macromolecules, **12**, 459 (1979).
795. G. Büldt, G. Meyerhoff, Makromol. Chem., Suppl., **1**, 359 (1975).
796. G. Büldt, Eur. Polym. J., **12**, 239 (1976).
797. M. Okabe, H. Matsuda, J. Appl. Polym. Sci., **28**, 2325 (1983).
798. A. M. Jamieson, C. T. Presley, Int. J. Polym. Mater., **4**, 161 (1976).
799. P. T. Callaghan, D. N. Pinder, Macromolecules, **13**, 1085 (1980).
800. H. Fujita, Foundations of Ultracentrifugal Analysis, Wiley, New York, 1975
801. V. E. Eskin, I. A. Baranovskaja, Vysokomol. Soedin. A, **19**, 533 (1977).
802. O. Prochazka, Z. Tuzar, P. Kartochvil, Makromol. Chem., **184**, 2097 (1983).
803. M. D. Lechner, Eur. Polym. J., **17**, 3 (1981).
804. L. Wolinski, K. Witkowski, Z. Turzynski, Makromol. Chem., **180**, 2399 (1979).
805. H. Suzuki, Brit. Polymer J., **11**, 35 (1979).
806. H. Suzuki, Brit. Polymer J., **11**, 41 (1979).
807. V. Chrastova, D. Mikulasova, J. Lacok, P. Citovicky, Polymer, **22**, 1054 (1981).
808. Y. Miyaki et al., Macromolecules, **11**, 1180 (1978).
809. H. Utiyama et al., Macromolecules, **11**, 506 (1978).
810. B. Appelt, G. Meyerhoff, Macromolecules, **13**, 657 (1980).
811. K. Kubota, K. M. Abbey, B. Chu, Macromolecules, **16**, 137 (1983).
812. M. Hara, A. Nakajima, Polym. J., **12**, 711 (1980).
813. H. Geerissen, B. A. Wolf, Makromol. Chem., Rapid Comm.,**3**, 17 (1982).
814. R. Kniewske, W. M. Kulicke, Makromol. Chem., **184**, 2173 (1983).
815. J. Roots, B. Nyström, Eur. Polym. J., **14**, 773 (1978).
816. S. F. Sun, Makromol. Chem., Rapid Comm., **4**, 203 (1983).
817. M. W. van den Esker, J. Laven, A. Broeckman, A. Vrij, J. Polym. Sci., Polym. Phys. Ed., **14**, 1953 (1976).
818. H. Utiyama, K. Takenaka, M. Mizumori, M. Fukuda, Macromolecules, **7**, 28 (1974).
819. G. Glöckner et al., Acta Polymerica, **30**, 10 (1979).
820. J. Stejskal, J. Horska, P. Kratochvil, J. Appl. Polym. Sci., **27**, 3929 (1982).
821. C. C. Han, P. H. Verdier, H. L. Wagner, J. of Research, **83**, 185 (1978).
822. F. M. Mirabella jr., J. Appl. Polym. Sci., **25**, 1775 (1980).
823. J. Vazquez, L. de Blas, M. G. Prolongo, R. M. Masegosa, I. Hernandez-Fuentes, A. Horta, Makromol. Chem., **185**, 797 (1984).
824. H. Dautzenberg, J. Polym. Sci., Polym. Symp., **61**, 83 (1977).
825. H. Suzuki, Y. Miyazaki, K. Kamide, Eur. Polym. J., **16**, 703 (1980).
826. H. Suzuki, et al., Eur. Polym. J., **18**, 831 (1982).
827. J. Stejskal, J. Horska, Makromol. Chem., **183**, 2527 (1982).
828. L. M. Ionescu, I. Petrea, I. I. Bujor, I. Demetrescu, Makromol. Chem., **184**, 1005 (1983).
829. M. Hara, A. Nakajima, Polym. J., **12**, 701 (1980).
830. T. Nose, B. Chu, Macromolecules, **12**, 590 (1979).
831. M. Schmidt, D. Nerger, W. Burchard, Polymer, **20**, 582 (1979).
832. M. D. Lechner, D. G. Steinmeier, Makromol. Chem., Rapid Commun., **2**, 421 (1981).
833. T. M. Aminabhavi, R. C. Patel, K. Bridger, E. S. Jayadevappa, J. Macromol. Sci. Chem. A, **17**, 1283 (1982).
834. K. Gekko, J. Am. Chem. Soc., **103**, 415 (1981).
835. A. L. Izjumnikov, G. R. Polkjaova, A. R. Gautmacher, Vysokomol. Soedin. A, **25**, 2341 (1983).
836. C. J. Edwards, R. F. Stepto, J. A. Semlyen, Polymer, **23**, 869 (1982).
837. V. N. Tsvetkov, S. O. Tsepelevich, Eur. Polym. J., **19**, 267 (1983).
838. I. Noda, T. Imai, T. Kitano, M. Nagasawa, Macromolecules, **14**, 1303 (1981).
839. T. Sato, T. Norisuye, H. Fujita, Polym. J., **16**, 341 (1984).
840. G. V. Schulz, E. Penzel, Makromol. Chem., **112**, 260 (1968).
841. G. C. Berry, J. Chem. Phys., **44**, 4550 (1966).
842. H. J. Cantow, G. V. Schulz, Z. Phys. Chem. N.F., **2**, 117 (1954).
843. T. Matsumoto, N. Nishioka, H. Fujita, J. Polym. Sci. A-2, **10**, 23 (1972).
844. T. Norisuye, K. Kawahara, A. Teramoto, H. Fujita, J. Chem. Phys., **49**, 4330 (1968).
845. T. Kato, K. Miyaso, I. Noda, T. Fujimoto, M. Nagasawa, Macromolecules, **3**, 777 (1970).
846. W. Banks, C. T. Greenwood, Eur. Polym. J., **4**, 245 (1968).
847. D. Rahlwes, R. G. Kirste, Makromol. Chem., **178**, 1793 (1977).
848. G. Glöckner, Eur. Polym. J., **15**, 727 (1979).
849. A. L. Izyumnikov et al., Vysokomol. Soedin. A, **25**, 582 (1983).
850. T. Tsuji, T. Norisuye, H. Fujita, Polym. J., **7**, 558 (1975).
851. T. L. Welzen, Brit. Polym. J., **12**, 95 (1980).
852. D. B. Sellen, Polymer, **16**, 773 (1975).
853. B. J. Berne, R. Pecora, "Dynamic Light Scattering", Wiley, New York, 1976.
854. B. Chu, "Laser Light Scattering", Academic Press, New York, 1974.
855. J. Raczek, Eur. Polym. J., **19**, 607 (1983).
856. D. E. Koppel, J. Chem. Phys., **57**, 4814 (1972).
857. G. S. Greschner, Eur. Polym. J., **20**, 475 (1984).

858. J. Raczek, Eur. Polym. J., **18**, 393 (1982).
859. H.-G. Elias, R. Bareiss, J. G. Watterson, Advan. Polym. Sci., **11**, 111 (1973).
860. H. Yamakawa, "Modern Theory of Polymer Solutions", Harper and Row, New York, 1971.
861. J. A. Marquese, J. M. Deutch, J. Chem. Phys., **73**, 5396 (1980).
862. B. Nyström, J. Roots, J. Macromol. Sci., Rev. Macromol. Chem. C, **19**, 35 (1980).
863. W. Brown, P. Stilbs, R. M. Johnsen, J. Polym. Sci., Polym. Phys. Ed., **20**, 1771 (1982).
864. R. G. Kitchen, B. N. Preston, J. D. Wells, J. Polym. Sci., Polym. Symp., **55**, 39 (1976).
865. J. E. Tanner, K.-J. Liu, J. E. Andersson, Macromolecules, **4**, 586 (1971); H. Hervet, L. Leger, F. Rondelez, Phys. Rev. Lett., **42**, 1681 (1979).
866. W. G. McMillan, J. E. Mayer, J. Chem. Phys., **13**, 276 (1945).
867. B. H. Zimm, J. Chem. Phys., **14**, 164 (1946).
868. A. C. Albrecht, J. Chem. Phys., **27**, 1002 (1957).
869. C. H. Paul, D. A. Yphantis, Anal. Biochem., **48**, 588 (1972).
870. J. Flossdorf, H. Schilling, K. P. Schindler, Makromol. Chem., **179**, 1617 (1978).
871. W. Mächtle, U. Klodwig, Makromol. Chem., **180**, 2507 (1979).
872. R. Giordano, G. Maisano, F. Mellamace, N. Nicoli, F. Wanderlengh, J. Chem. Phys., **75**, 4770 (1981).
873. J. D. Harrey, R. Geddes, P. R. Wills, Biopolymers, **18**, 2249 (1979).
874. Y. S. Oh, C. S. Johnson Jr., J. Chem. Phys., **74**, 2717 (1981).
875. N. A. Weissmann, R. C. Pan, A. R. Ware, J. Chem. Phys., **70**, 2897 (1979).
876. D. B. Sellen, Polymer, **14**, 359 (1973).
877. D. Langevin, F. Rondelez, Polymer, **19**, 875 (1978).
878. R. Ford, E. Jakeman, C. J. Oliver, E. R. Pike, R. J. Blagrove, E. Wood, A. R. Peacocke, Nature, **227**, 242 (1970).
879. B. D. Fair, D. Y. Chao, A. M. Jamieson, J. Colloid Interface Sci., **66**, 323 (1978).
880. E. Hantz, A. Cao, E. Taillandier, P. Tivant, M. Drifford, N. Defer, J. Kruh, Int. J. Biol. Macromol., **5**, 130 (1983).
881. E. J. Amis, P. A. Janmey, J. D. Ferry, H. Yu, Polym. Bull., **6**, 13 (1981).
882. G. C. Fletcher, Biopolymers, **15**, 2201 (1976).
883. H. Reihanian, A. M. Jamieson, Macromolecules, **12**, 684 (1979).
884. J. Douglas, H. Eisenberg, Biopolymers, **15**, 61 (1976).
885. N. Parthasarathy, K. S. Schmitz, M. K. Cowman, Biopolymers, **19**, 1137 (1980).
886. F. C. Chen, B. Chu, J. Chem. Phys., **68**, 2235 (1977).
887. K. S. Schmitz, Biopolymers, **18**, 479 (1979).
888. S.-C. Lin, J. M. Schurr, Biopolymers, **17**, 425 (1978).
889. W. L. Wun, W. Prins, Biopolymers, **14**, 111 (1975).
890. W. B. Veldkamp, I. R. Votano, Biopolymers, **19**, 111 (1980).
891. F. C. Chen, A. Yeh, B. Chu, J. Chem. Phys., **65**, 4508 (1977).
892. D. F. Nicoli, G. B. Benedek, Biopolymers, **15**, 2421 (1976).
893. F. C. Chen, A. Chrzeszczyk, B. Chu, J. Chem. Phys., **64**, 3403 (1976).
894. R. D. Camerini-Otero, P. N. Pusey, D. E. Koppel, D. W. Schaefer, R. M. Franklin, Biochemistry, **13**, 960 (1974).
895. S. B. Dubin, G. B. Benedek, F. C. Bancroft, D. J. Freifelder, Mol. Biol., **54**, 547 (1970).
896. R. G. Kitchen, B. N. Preston, J. D. Wells, J. Polym. Sci., Polym. Symp., **55**, 39 (1976).
897. J. Carlfors, R. Rymden, B. Nystroem, Polym. Commun., **24**, 263 (1983).
898. L. J. Fetters, N. Hadjichristidis,J. S. Lindner,J.W. Mays, W.W. Wilson, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **32** (1), 640–641 (1991).
899. K. Akasaka, Y. Nakamura, T. Norisuye, A. Teramoto, Polym. J., **26** (3), 363 (1994).
900. K. Akasaka, Y. Nakamura, T. Norisuye, A. Teramoto, Polym. J., **26** (12), 1387 (1994).
901. Y. Nakamura, K. Akasaka, K. Katayama, T. Norisuye, A. Teramoto, Macromolecules, **25** (3), 1134 (1992).
902. H. Geerissen, P. Schuetzeichel, B. A. Wolf, Macromolecules, **24** (1), 304 (1991).
903. R. H. Colby, L. J. Fetters, W. W. Groessly, Macromolecules, **20** (9), 2226 (1987).
904. C. Wu, D. Lilge, J. Appl. Polym. Sci., **50** (10), 1753 (1993).
905. J. W. Pope, B. Chu, Macromolecules, **17** (12), 2633 (1984).
906. E. Nordmeier, U. Lanver, M. D. Lechner, Macromolecules, **23** (4), 1072 (1990).
907. S. Kinugasa, H. Nakahara, J. I. Kawahara, Y. Koga, H. Takaya, J. Polym. Sci., Part B: Polym. Phys., **34** (3), 583 (1996).
908. B. Chu, D. Wu, Macromolecules, **20** (7), 1606 (1987).
909. M. Nakata, Y. Nakano, K. Kawate, Macromolecules, **21** (8), 2509 (1988).
910. B.C. Simionescu, S. Ioan, C.I. Simionescu, J. Polym. Sci., Polym. Phys. Ed., **25** (4), 829 (1987).
911. M. Naoki, I. H. Park, S. L. Wunder, B. Chu, J. Polym. Sci., Polym. Phys. Ed., **23** (12), 2567 (1985).
912. A. Karandinos, S. Nan, J. W. Mays, N. Hadjichristidis, Macromolecules, **24** (8), 2007 (1991).
913. M. Nakata, N. Numasawa, Macromolecules, **18** (9), 1736 (1985).
914. I. Katime, R. B. Valenciano, M. S. Anasgasti, D. Yoldi, Eur. Polym. J., **21** (5), 485 (1985).
915. F. Abe, Y. Einaga, H. Yamakawa, Macromolecules, **28** (3), 694 (1995).
916. R. M. Masegosa, M. G. Prolongo, I. Hernandez-Fuentes, A. Horta, Macromolecules, **17** (6), 1181 (1984).
917. F. Abe, Y. Einaga, H. Yamakawa, Macromolecules, **27** (12), 3262 (1994).
918. M. Kamijo, F. Abe, Y. Einaga, H. Yamakawa, Macromolecules, **28** (12), 4159 (1995).
919. M. B. Huglin, M. K. Khorasani, P. M. Sasia, Polymer, **29** (9), 1590 (1988).
920. H. Yamakawa, F. Abe, Y. Einaga, Macromolecules, **27** (20), 5704 (1994).
921. M. J. Solomon, S. J. Muller, J. Polym. Sci., Part B: Polym. Phys., **34** (1), 181 (1996).

922. Y. Nakamura, T. Norisuye, A. Teramoto, J. Polym. Sci., Polym. Phys. Ed., **29** (2), 153 (1991).
923. Y. Nakamura, T. Norisuye, A. Teramoto, Macromolecules, **24** (17), 4904 (1991).
924. L. J. Vagberg, K. A. Cogan, A. P. Gast, Macromolecules, **24** (7), 1670 (1991).
925. L. Zhang, D. Qin, R. Qian, Polym. J., **17** (4), 657 (1985).
926. C. M. Kok, A. Rudin, Eur. Polym. J., **22** (2), 107 (1986).
927. S. Allorio, S. Pispas, E. Siakali-Kioulafa, N. Hadjichristidis, J. Polym. Sci., Part B: Polym. Phys., **33** (16), 2229 (1995).
928. L. Willner, O. Jucknischke, D. Richter, L. J. Fetters, J. S. Huang, B. Farago, Makromol. Chem., Macromol. Symp., **61**, 122 (1992).
929. K. Venkataswamy, A. M. Jamieson, R. G. Petshek, Org. Coat. Appl. Polym. Sci. Proc., **55**, 339 (1986).
930. T. Nicolai, W. Brown, Macromolecules, **29** (5), 1698 (1996).
931. M. Ragnetti, D. Geiser, H. Hoecker, R. C. Oberthuer, Macromol. Chem. Phys., Suppl., **186** (8), 1701 (1985).
932. H. Zhang, Z. He, Polym. Commun., **32** (8), 239 (1991).
933. J. R. Quintana, M. Villacampa, I. A. Katime, Polymer, **34** (11), 2380 (1993).
934. F. Gundert, B. A. Wolf, J. Chem. Phys., **87** (10), 6156 (1987).
935. J. Li, J. Mays, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **35** (1), 130 (1994).
936. E. Nordmeier, W. Dauwe, Polym. J., **24** (3), 229 (1992).
937. A. M. Pedley, J. S. Higgins, D. G. Peiffer, A. R. Rennie, Macromolecules, **23** (9), 2494 (1990).
938. B. Gantchev, M. Mihailov, Polym. Bull., **17** (2), 181 (1987).
939. C. P. Patel, H. C. Trivedi, K. C. Patel, R. D. Patel, J. Polym. Mater., **8** (3), 231 (1991).
940. J. Danhelka, M. Netopilik, M. Bohdanecky, J. Polym. Sci., Polym. Phys. Ed., **25** (9), 1801 (1987).
941. K. Kamide, M. Saito, K. Kowsaka, Polym. J., **19** (10), 1173 (1987).
942. J. Desbrieres, R. Borsali, M. Rinaudo, M. Milas, Macromolecules, **26** (10), 2592 (1993).
943. K. Akiyoshi, S. Deguchi, N. Moriguchi, A. Yamaguchi, J. Sunamoto, Macromolecules, **26** (12), 3062 (1993).
944. T. Sato, T. Norisuye, H. Fujita, Macromolecules, **17** (12), 2696 (1984).
945. K. Kawakami, T. Norisuye, Macromolecules, **24** (17), 4898 (1991).
946. E.A. Lange, Org. Coat. Appl. Polym. Sci. Proc., **55**, 567 (1986).
947. P: Wissenburg, T. Odijk, P. Cirkel, M. Mandel, Macromolecules, **28** (7), 2315 (1995).
948. T. Nicolai, M. Mandel, Macromolecules, **22** (1), 438 (1989).
949. L. M. DeLong, P. S. Russo, Macromolecules, **24** (23), 6139 (1991).
950. M. A. Tracy, R. Pecora, Macromolecules, **25** (1), 337 (1992).
951. G. D. Rudkovskaja, B. M. Sabsels, I. A. Baranovskaja, N. N. Uljanova, u.a., Vysokomol. Soedin., A, **31** (1), 133 (1989).
952. A. Kidera, A. Nakajima, Macromolecules, **17** (4), 659 (1984).
953. P. Metzger Cotts, Org. Coat. Appl. Polym. Sci. Proc., **53**, 336 (1985).
954. D. A. Foucher, R. Ziembinski, B.-Z. Tang, P. M. Macdonald, J. Massey, Jaeger, Macromolecules, **26** (11), 2878 (1993).
955. P. M. Cotts, S. Ferline, G. Dagli, D. S. Pearson, Macromolecules, **24** (25), 6730 (1991).
956. M. Anasagasti, I. Katime, C. Strazielle, Macromol. Chem. Phys., Suppl., **188** (1), 201 (1987).
957. B. Celda, A. Campos, R. Gavara, J. E. Figueruelo, Polymer, **27** (8), 1247 (1986).
958. A. Campos, B. Celda, J. Mora, J. E. Figueruelo, Polymer, **25** (10), 1479 (1984).
959. M. R. Gomez-Anton, A. Horta, I. Hernandez-Fuentes, Polym. Commun., **27** (1), 5 (1986).
960. S. Wang, J. E. Mark, Polym. Bull., **31** (2), 205 (1993).
961. A. Lapp, C. Strazielle, Makromol. Chem., Rapid Commun., **6** (9), 591 (1985).
962. A. Lapp, C. Strazielle, Macromol. Chem. Phys., Suppl., **188** (12), 2921 (1987).
963. A. Campos, B. Celda, J. Mora, J. E. Figueruelo, Eur. Polym. J., **20** (12), 1187 (1984).
964. B. Celda, A. Campos, R. Tejero, J. E. Figueruelo, Eur. Polym. J., **22** (2), 129 (1986).
965. F. Doebert, A. Pfennig, M. Stumpf, Macromolecules, **28** (23), 7860 (1995).
966. S. Kuecuekyavuz, Z. Kuecuekyavuz, G. Erdogan, Polymer, **31** (2), 379 (1990).
967. H. Yamakawa, F. Abe, Y. Einaga, Macromolecules, **26** (8), 1898 (1993).
968. Y. Einaga, F. Abe, H. Yamakawa, Macromolecules, **26** (23), 6243 (1993).
969. E. Ogawa, N. Yamaguchi, M. Shima, Polym. J., **18** (12), 903 (1986).
970. E. Penzel, N. Goetz, Angew. Makromol. Chem., **178**, 191 (1990).
971. B. C. Simionescu, S. Ioan, M. Bercea, C. I. Simionescu, Eur. Polym. J., **27** (6), 553 (1991).
972. K. Kamide, Y. Miyazaki, H. Yamazaki, Polym. J., **18** (9), 645 (1986).
973. H. Yamazaki, Y. Miyazaki, M. Saito, K. Kamide, Polym. J., **23** (6), 725 (1991).
974. A. Kanda, M. Duval, D. Sarazin, J. Francois, Polymer, **26** (3), 406 (1985).
975. H. Hasse, H.-P. Kany, R. Tintinger, G. Maurer, Macromolecules, **28** (10), 3540 (1995).
976. E. Perez, Bello, J.G. Fatou, Eur. Polym. J., **24** (5), 431 (1988).
977. M. Yoshida, N. Sakamoto, K. Ikemi, S. Arichi, Bull. Chem. Soc. Jpn., **66** (6), 1598 (1993).
978. D. Rosenvasser, R. V. Figini, Macromol. Chem. Phys., Suppl., **185** (7), 1361 (1984).
979. J. Stejskal, D. Strakova, P. Kratochvil, Macromol. Chem. Phys., Suppl., **188** (4), 855 (1987).

980. S. Roy, S. Bhattacharjee, A. K. Bhowmick, B. R. Gupta, R. A. Kulkarni, Macromol. Rep. A, **30** (3/4), 301 (1993).
981. K. Matsuo, W. H. Stockmayer, F. Bangerter, Macromolecules, **18** (6), 1346 (1985).
982. J. Bravo, M. P. Tarazona, E. Saiz, Macromolecules, **24** (14), 4089 (1991).
983. Y. Jinbo, T. Sato, A. Teramoto, Macromolecules, **27** (21), 6080 (1994).
984. E. Ogawa, Polym. J., **27** (6), 567 (1995).
985. B. Tinland, R. Borsali, Macromolecules, **27** (8), 2141 (1994).
986. A. Louai, D. Sarazin, G. Pollet, F. Moreaux, Polymer, **32** (4), 703 (1991).
987. J. Roovers, J. E. Martin, J. Polym. Sci., Polym. Phys. Ed., **27** (12), 2513 (1989).
988. B. J. Bauer, L. J. Fetters, W. W. Graessley, N. Hadjichristidis, G. F. Quack, Macromolecules, **22** (5), 2337 (1989).
989. W. Brown, P. Zhou, Macromolecules, **24** (18), 5151 (1991).
990. L. J. Fetters, N. Hadjichristidis, J. S. Lindner, J. W. Mays, W. W. Wilson, Macromolecules, **24** (11), 3127 (1991).
991. C. H. Wang, X. Q. Zhang, Macromolecules, **28** (7), 2288 (1995).
992. M. Bhatt, A. M. Jamieson, Macromolecules, **21** (10), 3015 (1988).
993. K. Huber, S. Bantle, P. Lutz, W. Burchard, Macromolecules, **18** (7), 1461 (1985).
994. M. Murayama, M. Okada, T. Fukutomi, T. Nose, Macromol. Chem. Phys., Suppl., **188** (4), 829 (1987).
995. B. K. Varma, Y. Fujita, M. Takahashi, T. Nose, J. Polym. Sci., Polym. Phys. Ed., **22** (10), 1781 (1984).
996. M. Duval, P. Lutz, C. Strazielle, Makromol. Chem., Rapid Commun., **6** (2), 71 (1985).
997. J. W. Mays, S. Nan, M. E. Lewis, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **32** (1), S 74 (1991).
998. J. S. Lindner, W. W. Wilson, J. W. Mays, Macromolecules, **21** (11), 3304 (1988).
999. C. Wu, Macromolecules, **26** (15), 3821 (1993).
1000. C. Wu, J. Polym. Sci., Part B: Polym. Phys., **32** (5), 803 (1994).
1001. N. Vennemann, M. D. Lechner, R. C. Oberthuer, Polymer, **28** (10), 1738 (1987).
1002. Y. Kambe, C. Honda, Polym. Commun., **25** (5), 154 (1984).
1003. P. Zhou, W. Brown, Macromolecules, **23** (4), 1131 (1990).
1004. B. Chu, C. Wu, W. Buck, Macromolecules, **22** (2), 831 (1989).
1005. K. Kubota, K. Hamano, N. Kuwahara, S. Fujishige, I. Ando, Polym. J., **22** (12), 1051 (1990).
1006. P. S. Mumick, C. L. McCormick, Polym. Eng. Sci., **34** (18), 1419 (1994).
1007. Q. Ying, J. Wang, C. Long, B. Chu, Org. Coat. Appl. Polym. Sci. Proc., **54**, 546 (1986).
1008. Q. Ying, G. Wu, B. Chu, R. Farinato, L. Jackson, Macromolecules, **29** (13), 4646 (1996).
1009. W. Peiqiang, M. Siddiq, C. Huiying, Q. Di, C. Wu, Macromolecules, **29** (1), 277 (1996).
1010. C. L. McCormick, L. C. Salazar, Polymer, **33** (21), 4617 (1992).
1011. T. Coviello, W. Burchard, M. Dentini, V. Crescenzi, Macromolecules, **20** (5), 1102 (1987).
1012. T. Coviello, K. Kajiwara, W. Burchard, M. Dentini, V. Crescenzi, Macromolecules, **19** (11), 2826 (1986).
1013. T. Kato, T. Katsuki, A. Takahashi, Macromolecules, **17** (9), 1726 (1984).
1014. H. T. Goinga, R. Pecora, Macromolecules, **24** (23), 6128 (1991).
1015. R. Hanselmann, W. Burchard, R. Lemmes, D. Schwengers, Macromol. Chem. Phys., **196** (7), 2259 (1995).
1016. P. S. Russo, F. E. Karasz, K. H. Langley, J. Chem. Phys., **80** (10), 5312 (1984).
1017. W. R. Krigbaum, G. Brelsford, Macromolecules, **21** (8), 2502 (1988).
1018. C. Konak, Z. Tuzar, P. Stepanek, G. Sedlacek, P. Kratochvil, Prog. Colloid Polym. Sci., **71**, 15 (1985).
1019. T. A. Seery, E. J. Amis, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **29** (2), 389 (1988).
1020. T. A. Seery, J. A. Shorter, E. J. Amis, Polymer, **30** (7), 1197 (1989).
1021. C. L. McCormick, L. C. Salazar, Macromolecules, **25** (7), 1896 (1992).
1022. C. L. McCormick, L. C. Salazar, Polymer, **33** (20), 4384 (1992).
1023. C. L. McCormick, L. C. Salazar, J. Appl. Polym. Sci., **48** (6), 1115 (1993).
1024. E. Vrinov, F. T. Abdullaev, U. M. Mirzaev, Vysokomol. Soedin. A, **31** (3), 602 (1989).
1025. C. L. McCormick, K. P. Blackmon, Macromolecules, **19** (6), 1512 (1986).
1026. B. Chu, C. Wu, W. Buck, Macromolecules, **22** (1), 371 (1989).
1027. A. Sanchez, A. Bello, C. Marco, J. G. Fatou, Macromol. Chem. Phys., Suppl., **189** (2), 399 (1988).
1028. H. Ochiai, M. Handa, H. Matsumoto, T. Moriga, I. Murakami, Macromol. Chem. Phys., Suppl., **186** (12), 2547 (1985).
1029. C. W. Burkhardt, K. J. McCarthy, D. P. Parazak, J. Polym. Sci., Polym. Lett. Ed., **25** (5), 209 (1987).
1030. I. H. Park, E.-J. Choi, Polymer, **37** (2), 313 (1996).
1031. M. R. Aven, C. Cohen, Macromolecules, **23** (2), 476 (1990).
1032. A. M. Pedley, J. S. Higgins, D. G. Peiffer, W. Burchard, Macromolecules, **23** (5), 1434 (1990).
1033. M. Villacampa, E. D. de Apodaca, J. R. Quintana, I. Katime, Macromolecules, **28** (12), 4144 (1995).
1034. T. Hirao, A. Teramoto, T. Sato, T. Norisuye, T. Masuda, T. Higashimura, Polym. J., **23** (8), 925 (1991).
1035. E. E. Kathmann, A. D. de Davis, C. L. McCormick, Macromolecules, **27** (12), 3156 (1994).
1036. W. M. Kulicke, H. H. Hoerl, Colloid Polym. Sci., **263** (7), 530 (1985).
1037. K. S. Minsker, R. B. Pancheshnikova, Yu. B. Monakov, G. E. Zaikov, Eur. Polym. J., **21** (11), 981 (1985).
1038. J. Li, Y. Wan, Z. Xu, J.W. Mays, Macromolecules, **28** (15), 5347 (1995).
1039. B. Chu, C. Wu, et al., Macromolecules, **20** (3), 700 (1987).
1040. Q. Ying, B. Chu, Macromolecules, **19** (6), 1580 (1986).

1041. N. Nemoto, H. Matsuda, Y. Tsunashima, M. Kurata, Macromolecules, **17** (9), 1731 (1984).
1042. P. Shukla, M. Muthukumar, K. H. Langley, J. Appl. Polym. Sci., **44** (12), 2115 (1992).
1043. P. N. Lavrenko, O. V. Okatova, S. V. Vinogradova, A. L. Rusanov, Ponomarev, I. I. Eur. Polym. J., **26** (12), 1337 (1990).
1044. J. Mogin, C. Abradelo, M. F. Rey-Stolle, L. H. Tagle, I. Hernandez-Fuentes, Eur. Polym. J., **28** (2), 187 (1992).
1045. M. J. Fabre, L. H. Tagle, L. Gargallo, D. Radic, I. Hernandez-Fuentes, Eur. Polym. J., **25** (12), 1315 (1989).
1046. E. B. Tarabukina, Z. N. Slavina, S. Ya. Frenkel, Acta Polym., **42** (6), 266 (1991).
1047. F. Francuskiewicz, G. Gloeckner, P. Kratochvil, Angew. Makromol. Chem., **206** (Mar), 121 (1993).
1048. A. Thurn, W. Burchard, R. Niki, Colloid Polym. Sci., **265** (10), 897 (1987).
1049. H. Durchschlag, Colloid Polym. Sci., **267** (12), 1139 (1989).
1050. M Osa, F. Abe, T. Yoshizaki, Y. Einaga, H. Yamakawa, Macromolecules, **29** (6), 2302 (1996).
1051. T. Konishi, T. Yoshizaki, H. Yamakawa, Macromolecules, **24** (20), 5614 (1991).
1052. W. Maechtle, G. Ley, J. Streib, Prog. Colloid Polym. Sci., **99**, 144 (1995).
1053. L. Gargallo, D. Radic, N. Hamidi, L. Cardenas, A. Horta, J. Hernandez-Fuentes, J. Appl. Polym. Sci., **36** (6), 1495 (1988).
1054. H. Ochiai, K. Kamata, I. Murakami, Polym. Commun., **25** (5), 158 (1984).
1055. E. Morales, M. R. Gomez-Anton, I. F. Pierola, J. Appl. Polym. Sci., **58** (11), 1943 (1995).
1056. N. Nemoto, S. Okada, T. Inoue, M. Kurata, Macromolecules, **21** (5), 1502 (1988).
1057. K. Dehara, T. Yoshizaki, H. Yamakawa, Macromolecules, **26** (19), 5137 (1993).
1058. T. Arai, N. Sawatari, T. Yoshizaki, Y. Einaga, H. Yamakawa, Macromolecules, **29** (6), 2309 (1996).
1059. G. M. Pavlov, E. F. Panarin, E. V. Korneeva, K. V. Kurockin, V. E. Bajkov, Vysokomol. Soedin. A, **32** (6), 1190 (1990).
1060. M. B. Huglin, S. J. O'Donohue, M. A. Radwan, Eur. Polym. J., **25** (6), 543 (1989).
1061. G. Vansco, Eur. Polym. J., **26** (3), 345 (1990).
1062. A. Barooah, C. K. Sun, S. H. Chen, J. Polym. Sci., Polym. Phys. Ed., **24** (4), 817 (1986).
1063. C. M. Kok, F. R. Hallett, A. Rudin, Eur. Polym. J., **21** (1), 33 (1985).
1064. J. W. Klein, C. Cohen, Polymer, **34** (20), 4186 (1993).
1065. R. Mieczkowski, L. Huppenthal, Polimery (Warsaw), **32** (10), 397 (1987).
1066. P. Rossmanith, W. Koehler, Macromolecules, **29** (9), 3203 (1996).
1067. J. R. Quintana, M. D. Janez, M. Villacampa, I. Katime, Macromolecules, **28** (12), 4139 (1995).
1068. T. Arai, F. Abe, T. Yoshizaki, Y. Einaga, H. Yamakawa, Macromolecules, **28** (10), 3609 (1995).
1069. T. Yamada, T. Yoshizaki, H. Yamakawa, Macromolecules, **25** (1), 377 (1992).
1070. N. Hadjichristidis, J. S. Lindner, J. W. Mays, W. W. Wilson, Macromolecules, **24** (25), 6725 (1991).
1071. M. E. Lewis, S. Nan, W. Yunan, J. Li, J. W. Mays, N. Hadjichristidis, Macromole-cules, **24** (25), 6686 (1991).
1072. Y. Tsunashima, Macromolecules, **23** (11), 2963 (1990).
1073. G. J. Vancso, B. R. White, J. E. Guilet, Eur. Polym. J., **29** (5), 751 (1993).
1074. J. S. Vrentan, C. H. Chu, J. Polym. Sci., Polym. Phys. Ed., **27** (2), 465 (1989).
1075. J. Huang, C. Li, B. He, Macromol. Chem. Phys., Suppl., **187** (1), 149 (1986).
1076. K. Huber, W. Burchard, A. U. Akcasu, Macromolecules, **18** (12), 2743 (1985).
1077. N. Numasawa, K. Kuwamoto, T. Nose, Macromolecules, **19** (10), 2593 (1986).
1078. W. Brown, R. Rymden, Macromolecules, **21** (3), 840 (1988).
1079. M. M. Rodrigo, C. Cohen, Macromolecules, **21** (7), 2091 (1988).
1080. F. D. Blum, T. L. Maver, R. Raghavan, Polym. Prepr. (Am. Chem. Soc., Div. Polym. Chem.), **28** (1), 340, (1987).
1081. G. M. Pavlov, N. V. Pogodina, S. V. Busin, A. B. Melnikov, E. B. Lysenko, u.a., Vysokomol. Soedin. A, **28** (4), 688 (1986).
1082. P. N. Lavrenko, E. P. Astapenko, S. V. Busin, O. V. Okatova, Vysokomol. Soedin. A, **28** (8), 1669 (1986).
1083. G. Pavlov, N. Michailova, E. Tarabukina, E. Korneeva, Prog. Colloid Polym. Sci., **99**, 109 (1995).
1084. R. I. Stankovic, S. Jovanovic, L. Ilic, E. Nordmeier, M. D. Lechner, Polymer, **32** (2), 235 (1991).
1085. E. Wachenfeld-Eisele, W. Burchard, Polymer, **29** (3), 471 (1988).
1086. M. J. Gonzalez-Tejera, J. M. Perena, A. Bello, I. Hernandez-Fuentes, Polym. Bull., **31** (1), 111 (1993).
1087. R. I. Stankovic, L. Ilic, E. Nordmeier, S. Jovanovic, M. D. Lechner, Polym. Bull., **27** (3), 337 (1991).
1088. R. A. Waggoner, F. D. Blum, J. C. Lang, Macromolecules, **28** (8), 2658 (1995).
1089. J.-L. Wu, M. Hara, Macromolecules, **27** (4), 923 (1994).
1090. R. G. Smits, M. E. Kuil, M. Mandel, Macromolecules, **26** (25), 6808 (1993).
1091. S. S. Sorlie, Macromolecules, **23** (2), 487 (1990).
1092. S. J. Rathbone, C. A. Haynes, H. W. Blanch, J. M. Prausnitz, Macromolecules, **23** (17), 3944 (1990).

Polymolecularity Correction Factors

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A. LIST OF SYMBOLS USED

a_D	exponent in $D = K_D M^{a_D}$
a_s	exponent in $s = K_s M^{a_s}$
$a_{[\eta]}$	exponent in $[\eta] = K_{[\eta]} M^{a_{[\eta]}}$
A_2	second virial coefficient
$\bar{A}_2^{(OP)}$, $\bar{A}_2^{(LS)}$	A_2 from osmotic pressure, light scattering, resp.
B, B', B''	constants
D	diffusion coefficient
$\bar{D}_A$	"area average" diffusion coefficient $\bar{D}_A = \left(\sum_i w_i D_i^{-0.5}\right)^{-2}$
D_i	diffusion coefficient of component i
$\bar{D}_w$	weight-average diffusion coefficient $\bar{D}_w = \sum_i w_i D_i$
$\bar{D}_z$	z-average diffusion coefficient $\bar{D}_z = \left(\sum_i w_i M_i D_i\right)/\left(\sum_i w_i M_i\right)$
$[f]_\Theta$	intrinsic frictional coefficient in the theta state
i	integers 1,2,..., i
k	degree of coupling in Schulz–Zimm distributions of the molecular weight $k = (\bar{M}_w/\bar{M}_n - 1)^{-1}$
K_D	constant in $D = K_D M^{a_D}$
K_0	constant for the determination of unperturbed dimensions $K_0 = \Phi_\Theta(\langle r^2\rangle_0/M)^{3/2}$
K_s	constant in $s = K_s M^{a_s}$
$K_{\langle S^2\rangle}$	constant in $\langle S^2\rangle = K_{\langle S^2\rangle} M^{1+\varepsilon}$
$K_{[\eta]}$	constant in $[\eta] = K_{[\eta]} M^{a_{[\eta]}}$
L	reduced contour length $L = L'/\lambda^{-1}$
L'	contour length
$\bar{L}_n$, $\bar{L}_w$, $\bar{L}_z$	number-, weight-, z-average L
M	molecular weight[1]
$\bar{M}_{av}$	any average of the molecular weight
$\bar{M}_D$	molecular weight from D
M_i	molecular weight of component i
M_j	molecular weight of compound j
$\bar{M}_m$	molecular weight of median in logarithmic normal distributions
$\bar{M}_n$	number-average molecular weight $\bar{M}_n = 1/\sum_i w_i M_i^{-1}$
$\bar{M}_s$	molecular weight from s
$\bar{M}_{\langle S^2\rangle}$	molecular weight from $\langle S^2\rangle$
$\bar{M}_{sD}$	molecular weight from s and D
$\bar{M}_v$	viscosity-average molecular weight $\bar{M}_v = \left(\sum_i w_i M_i^{a_{[\eta]}}\right)^{1/a_{[\eta]}}$
$\bar{M}_w$	weight-average molecular weight $\bar{M}_w = \sum_i w_i M_i$
$\bar{M}_{s[\eta]}$	molecular weight from s and $[\eta]$
$\bar{M}_{[\eta]D}$	molecular weight from $[\eta]$ and D
n_i	mole fraction of component i
N_L	Avogadro number
P_0	constant in $[f]_\Theta = P_0\langle r^2\rangle_0^{1/2}$
q	polymolecularity correction factor
q_B	q for Baumann–Stockmayer–Fixman procedure
q_{BSF}	q for Burchard–Stockmayer–Fixman equation

[1] According to IUPAC recommendations (1,2), "molecular weight" is understood to be the relative molecular mass, the numerical values of which are identical to those of the molar mass in g/mol and therefore molecular weight is denoted here with the same symbol as the molar mass, viz. M.

q_{BKS}	q for Baumann–Kurata–Stockmayer procedure
$q_{CB,1}, q_{CB,2}$	q's for Cowie–Bywater procedure
q_{Diff}	q for $D = K_D M^{a_D}$
q_{FF}	q for Fox–Flory equation
q_{Gyr}	q for $\langle S^2 \rangle = K_{\langle S^2 \rangle} M^{1+\varepsilon}$
q_{KMSH}	q for $[\eta] = K_{[\eta]} M^{a_{[\eta]}}$
$q(\bar{L}_w)$	q for stiff polymer chains
$q_{\langle r^2 \rangle_n}$, $q_{\langle r^2 \rangle_w}$, $q_{\langle r^2 \rangle_z}$	q's for conversion of $[\langle r^2 \rangle^{3/2}]_n^{2/3}$ into $\langle r^2 \rangle_n$, $\langle r^2 \rangle_w$ and $\langle r^2 \rangle_z$, respectively
$q_{\langle S^2 \rangle_w}$, $q_{\langle S^2 \rangle_n}$	q's for conversion of $\langle S^2 \rangle_z$ into $\langle S^2 \rangle_w$ and $\langle S^2 \rangle_n$, respectively
q_{Sed}	q for $s = K_s M^{a_s}$
$\langle r^2 \rangle$	mean-square end-to-end distance
$\langle r_i^2 \rangle$	$\langle r^2 \rangle$ of component i
$\langle r^2 \rangle_{av}$	any average of $\langle r^2 \rangle$
$\langle r^2 \rangle_0$	mean-square unperturbed end-to-end distance
$[\overline{\langle r^2 \rangle^{3/2}}]_n$	1.5th moment of number distribution of mean-square end-to-end distance $[\overline{\langle r^2 \rangle^{3/2}}]_n = \sum_i n_i \langle r_i^2 \rangle^{3/2}$
$\langle r^2 \rangle_n$	number-average mean-square end-to-end distance $\langle r^2 \rangle_n = \sum_i n_i \langle r_i^2 \rangle$
$\langle r^2 \rangle_w$	weight-average mean-square end-to-end distance $\langle r^2 \rangle_w = \sum_i w_i \langle r_i^2 \rangle$
$\langle r^2 \rangle_z$	z-average mean-square end-to-end distance $\langle r^2 \rangle_z = (\sum_i w_i M_i r_i^2)/(\sum_i w_i M_i)$
s	sedimentation coefficient
$\bar{s}_{max}$	sedimentation coefficient determined from the maximum of the concentration gradient curve
$\bar{s}_w$	weight-average sedimentation coefficient
$\langle S^2 \rangle$	mean-square radius of gyration
$\langle S^2 \rangle_0$	unperturbed mean-square radius of gyration
$\langle S_i^2 \rangle$	$\langle S^2 \rangle$ of component i
$\langle S^2 \rangle_{av}$	any average of $\langle S^2 \rangle$
$\langle S^2 \rangle_n$	number-average mean-square radius of gyration $\langle S^2 \rangle_n = \sum_i n_i \langle S_i^2 \rangle$
$\langle S^2 \rangle_w$	weight-average mean-square radius of gyration $\langle S^2 \rangle_w = \sum_i w_i \langle S_i^2 \rangle$
$\langle S^2 \rangle_z$	z-average mean-square radius of gyration $\langle S^2 \rangle_z = (\sum_i w_i M_i \langle S_i^2 \rangle)/(\sum_i w_i M_i)$
$\langle S^2 \rangle_{0,z}$	unperturbed $\langle S^2 \rangle_z$
v_2	partial specific volume of the solute (index 2)
w_i	mass fraction of component i
w_j	mass fraction of component j
ε	exponent in $\langle S^2 \rangle = K_{\langle S^2 \rangle} M^{1+\varepsilon}$
η	viscosity (index 1: solvent)
$[\eta]$	intrinsic viscosity
$[\eta]_i$	intrinsic viscosity of component i
$[\bar{\eta}]_w$	weight-average intrinsic viscosity $[\bar{\eta}]_w \equiv [\eta] = \sum_i w_i [\eta]_i$
λ^{-1}	length of Kuhn statistical segment
π	pi = 3.14159
ρ	density (index 1: solvent)
σ	standard deviation of logarithmic normal distribution
Θ	index theta state
$\sum_i$	summation over all components i
$\Phi(\varepsilon)$	viscosity function; $\Phi_\Theta = \Phi(\varepsilon = 0) = 2.5 \times 10^{23}\ mol^{-1}$ for end-to-end distance and $[\eta]$ in cm^3/g is also called Flory constant

B. INTRODUCTION

Relationships between molecular weight M (1,2),[1] and properties dependent on M, for polymers exhibiting a molecular weight distribution (MWD, polymolecularity[2]) require the use of corresponding averages ("principle of corresponding averages" (3)). For example, in Staudinger's viscosity law (4), modified by Kuhn (5), Mark (6), Sakurada (7) and Houwink (8,9) (KMSH), the experimentally determined intrinsic viscosity $[\eta]$ is the weight-average intrinsic viscosity (10–12) but the corresponding molecular weight is the viscosity-average molecular weight (13,14) $\bar{M}_v$:

$$[\eta] = K_{[\eta]} \bar{M}_v^{a_{[\eta]}}$$

where

$$[\eta] = \sum_i w_i [\eta]_i \quad \text{and} \quad \bar{M}_v = \left(\sum_i w_i M_i^{a_{[\eta]}} \right)^{1/a_{[\eta]}}$$

A direct experimental determination of $\bar{M}_v$ is not possible (exception: the intrinsic viscosity $[\bar{\eta}]_\Theta$ and the weight-average sedimentation coefficient $\bar{s}_w$, both for the theta state, inserted in the Flory–Krigbaum–Mandelkern–Scheraga equation (15–17) result in the viscosity-average molecular weight for the theta state, $\bar{M}_{v,\Theta}$ (18)). Therefore, for the determination of the numerical values of the constant $K_{[\eta]}$ and the exponent $a_{[\eta]}$ in Eq. (B1) different molecular weight averages, $\bar{M}_{av}$, are usually applied, e.g. the weight-average molecular weight ($\bar{M}_{av} = \bar{M}_w$), the number-average molecular weight ($\bar{M}_{av} = \bar{M}_n$), or the molecular weight averages obtained with Svedberg's equation[3] (19) $\bar{M}_{sD}$ or the equations of Flory, Krigbaum, Mandelkern and Scheraga (15–17) $\bar{M}_{s[\eta]}$ and $\bar{M}_{[\eta]D}$. From $\bar{M}_{av}$, the viscosity-average molecular weight $\bar{M}_v$ can be calculated, provided the type and width of the molecular weight distribution of the polymer samples or fractions are known (14).

1. Alternatively, a polymolecularity correction factor q can be defined which corrects Eq. (B1) for the use of $\bar{M}_{av}$ instead of $\bar{M}_v$:

$$[\eta] = K_{[\eta]} \bar{M}_v^{a_{[\eta]}} = K_{[\eta]} q_{KMSH} \bar{M}_{av}^{a_{[\eta]}} \qquad \text{(B1)}$$

$$q_{KMSH} = (\bar{M}_v / \bar{M}_{av})^{a_{[\eta]}} \qquad \text{(B2)}$$

Equation (B2) reads, for $\bar{M}_{av} \equiv \bar{M}_w$:

$$q_{KMSH,\bar{M}_w} = \left(\sum_i w_i M_i^{a_{[\eta]}} \right) \Big/ \left(\sum_i w_i M_i \right)^{a_{[\eta]}} \qquad \text{(B3)}$$

(see Tables 2 and 3).

[2] Polymolecularity is understood to describe the phenomenon that most polymers are nonuniform (2) with respect to relative molecular mass.
[3] See this Handbook page VII–85.

For $\bar{M}_{av} \equiv \bar{M}_n$:

$$q_{KMSH,\bar{M}_n} = \left(\sum_i w_i M_i^{a_{[\eta]}}\right) \Big/ \left(\sum_i w_i M_i\right)^{a_{[\eta]}} \quad \text{(B4)}$$

(see Tables 4 and 5).

For $\bar{M}_{av} \equiv \bar{M}_{\bar{s}_w \bar{D}_w}$:

$$q_{KMSH,\bar{s}_w \bar{D}_w} = \left(\sum_i w_i M_i^{a_{[\eta]}}\right)\left(\sum_i w_i M_i^{(2-a_{[\eta]})/3}\right)^{-a_{[\eta]}} \times \left(\sum_i w_i M_i^{-(1+a_{[\eta]})/3}\right)^{a_{[\eta]}} \quad \text{(B5)}$$

(see Tables 6 and 7).

For $\bar{M}_{av} \equiv \bar{M}_{\bar{s}_w \bar{D}_A}$:

$$q_{KMSH,\bar{s}_w \bar{D}_A} = \left(\sum_i w_i M_i^{a_{[\eta]}}\right)\left(\sum_i w_i M_i^{(2-a_{[\eta]})/3}\right)^{-a_{[\eta]}} \times \left(\sum_i w_i M_i^{(1+a_{[\eta]})/6}\right)^{-2a_{[\eta]}} \quad \text{(B6)}$$

(see Tables 8 and 9).

For $\bar{M}_{av} \equiv \bar{M}_{\bar{s}_w \bar{D}_z}$:

$$q_{KMSH,\bar{s}_w \bar{D}_z} = \left(\sum_i w_i M_i^{a_{[\eta]}}\right)\left(\sum_i w_i M_i\right)^{-a_{[\eta]}} \quad \text{(B7)}$$

(see Tables 10 and 11).

For $\bar{M}_{av} \equiv \bar{M}_{\bar{s}_w [\bar{\eta}]_w}$:

$$q_{KMSH,\bar{s}_w [\bar{\eta}]_w} = \left(\sum_i w_i M_i^{a_{[\eta]}}\right)^{1-0.5a_{[\eta]}} \times \left(\sum_i w_i M_i^{(2-a_{[\eta]})/3}\right)^{-1.5a_{[\eta]}} \quad \text{(B8)}$$

(see Tables 12 and 13).

For $\bar{M}_{av} \equiv \bar{M}_{[\bar{\eta}]_w \bar{D}_w}$:

$$q_{KMSH,[\bar{\eta}]_w \bar{D}_w} = \left(\sum_i w_i M_i^{a_{[\eta]}}\right)^{a_{[\eta]}+1}\left(\sum_i w_i M_i^{-(1+a_{[\eta]})/3}\right)^{3a_{[\eta]}} \quad \text{(B9)}$$

(see Tables 14 and 15).

For $\bar{M}_{av} \equiv \bar{M}_{[\bar{\eta}]_w \bar{D}_A}$:

$$q_{KMSH,[\bar{\eta}]_w \bar{D}_A} = \left(\sum_i w_i M_i^{a_{[\eta]}}\right)^{1+a_{[\eta]}}\left(\sum_i w_i M_i^{(1+a_{[\eta]})/6}\right)^{-6a_{[\eta]}} \quad \text{(B10)}$$

(see Tables 16 and 17).

For $\bar{M}_{av} \equiv \bar{M}_{[\bar{\eta}]_w \bar{D}_z}$:

$$q_{KMSH,[\bar{\eta}]_w \bar{D}_z} = \left(\sum_i w_i M_i^{a_{[\eta]}}\right)^{1+a_{[\eta]}}\left(\sum_i w_i M_i^{(2-a_{[\eta]})/3}\right)^{3a_{[\eta]}} \times \left(\sum_i w_i M_i\right)^{-3a_{[\eta]}} \quad \text{(B11)}$$

(see Tables 18 and 19).

2. Correspondingly, polymolecularity correction factors are useful for the relationship of sedimentation coefficient versus molecular weight,

$$s = K_s \bar{M}_s^{a_s} = K_s q_{Sed} \bar{M}_{av}^{a_s} \quad \text{(B12)}$$

the relationship of diffusion coefficient versus molecular weight,

$$D = K_D \bar{M}_D^{a_D} = K_D q_{Diff} \bar{M}_{av}^{a_D} \quad \text{(B13)}$$

and the relationship of mean-square radius of gyration[4] versus molecular weight

$$\langle S^2 \rangle = K_{\langle S^2 \rangle} \bar{M}_{\langle S^2 \rangle}^{1+\varepsilon} = K_{\langle S^2 \rangle} q_{Gyr} \bar{M}_{av}^{1+\varepsilon} \quad \text{(B14)}$$

if other than the corresponding molecular weight averages $\bar{M}_s$, $\bar{M}_D$, and $\bar{M}_{\langle S^2 \rangle}$, respectively (3), are to be used for the numerical evaluation of the constants K_s, K_D, $K_{\langle S^2 \rangle}$ and the exponents a_s, a_D, ε, respectively. Analogous to Eq. (B1) the polymolecularity correction factors for Eqs. (B12), (B13), and (B14) are found to be:

$$q_{Sed} = (\bar{M}_s / \bar{M}_{av})^{a_s} \quad \text{(B15)}$$

$$q_{Diff} = (\bar{M}_D / \bar{M}_{av})^{a_D} \quad \text{(B16)}$$

$$q_{Gyr} = (\bar{M}_{\langle S^2 \rangle} / \bar{M}_{av})^{1+\varepsilon} \quad \text{(B17)}$$

Eq. (B15) reads for $s \equiv \bar{s}_w$, $\bar{M}_{av} \equiv \bar{M}_w$:

$$q_{Sed,\bar{s}_w \bar{M}_w} = \left(\sum_i w_i M_i^{a_s}\right) \Big/ \left(\sum_i w_i M_i\right)^{a_s} \quad \text{(B18)}$$

(see Tables 20 and 21).

For $s \equiv \bar{s}_w$, $\bar{M}_{av} = \bar{M}_n$:

$$q_{Sed,\bar{s}_w \bar{M}_n} = \left(\sum_i w_i M_i^{a_s}\right)\left(\sum_i w_i / M_i\right)^{a_s} \quad \text{(B19)}$$

(see Tables 22 and 23).

[4] The symbol for the radius of gyration is a *capital S* – contrary to the recommendations by IUPAC (2) – in order to distinguish it from the symbol for the sedimentation coefficient *s*.

For $s \equiv \bar{s}_w$, $\bar{M}_{av} \equiv \bar{M}_{\bar{s}_w \bar{D}_w}$ (20):

$$q_{\mathrm{Sed},\bar{s}_w \bar{M}_{\bar{s}_w \bar{D}_w}} = \left(\sum_i w_i M_i^{-(a_{|\eta|}+1)/3} \right)^{(2-a_{|\eta|})/3} \div \left(\sum_i w_i / M_i^{(2-a_{|\eta|})/3} \right)^{-(1+a_{|\eta|})/3} \quad \text{(B20)}$$

(see Tables 24 and 25).
For calculating $\bar{s}_w$ from experimentally determined $\bar{s}_{max}$, see Tables 26 and 27 (21).

3. Equation (B16) reads for $D \equiv \bar{D}_A$, $\bar{M}_{av} \equiv \bar{M}_n$:

$$q_{\mathrm{Diff},\bar{D}_A \bar{M}_n} = \left(\sum_i w_i M_i^{-a_D/2} \right)^{-2} \left(\sum_i w_i M_i^{-1} \right)^{a_D} \quad \text{(B21)}$$

(see Tables 28 and 29).
For $D \equiv \bar{D}_A$, $\bar{M}_{av} \equiv \bar{M}_{\bar{s}_w \bar{D}_A}$ (20):

$$q_{\mathrm{Diff},\bar{D}_A \bar{M}_{\bar{s}_w \bar{D}_A}} = \left(\sum_i w_i M_i^{(2-a_{|\eta|})/3} \right)^{(a_{|\eta|}+1)/3} \times \left(\sum_i w_i M_i^{(a_{|\eta|}+1)/6} \right)^{2(a_{|\eta|}-2)/3} \quad \text{(B22)}$$

(see Tables 30 and 31).
For $D \equiv \bar{D}_w$, $\bar{M}_{av} \equiv \bar{M}_n$:

$$q_{\mathrm{Diff},\bar{D}_w \bar{M}_n} = \left(\sum_i w_i M_i^{a_D} \right) \left(\sum_i w_i M_i^{-1} \right)^{a_D} \quad \text{(B23)}$$

(see Tables 32 and 33).
For $D \equiv \bar{D}_z$, $\bar{M}_{av} \equiv \bar{M}_n$:

$$q_{\mathrm{Diff},\bar{D}_z \bar{M}_n} = \left(\sum_i w_i M_i^{1+a_D} \right) \left(\sum_i w_i M_i \right)^{-1} \times \left(\sum_i w_i M_i^{-1} \right)^{a_D} \quad \text{(B24)}$$

(see Tables 34 and 35).
For $D \equiv \bar{D}_A$, $\bar{M}_{av} \equiv \bar{M}_w$:

$$q_{\mathrm{Diff},\bar{D}_A \bar{M}_w} = \left(\sum_i w_i M_i^{-a_D/2} \right)^{-2} \left(\sum_i w_i M_i \right)^{-a_D} \quad \text{(B25)}$$

(see Tables 36 and 37).
For $D \equiv \bar{D}_w$, $\bar{M}_{av} \equiv \bar{M}_w$:

$$q_{\mathrm{Diff},\bar{D}_w \bar{M}_w} = \left(\sum_i w_i M_i^{a_D} \right) \left(\sum_i w_i M_i \right)^{-a_D} \quad \text{(B26)}$$

(see Tables 38 and 39).

For $D \equiv \bar{D}_z$, $\bar{M}_{av} \equiv \bar{M}_w$:

$$q_{\mathrm{Diff},\bar{D}_z \bar{M}_w} = \left(\sum_i w_i M_i^{1+a_D} \right) \left(\sum_i w_i M_i \right)^{-(1+a_D)} \quad \text{(B27)}$$

(see Tables 40 and 41).

4. Equation (B17) reads, for $\langle S^2 \rangle \equiv \langle S^2 \rangle_z$, $\bar{M}_{av} \equiv \bar{M}_w$:

$$q_{\mathrm{Gyr},\langle S^2 \rangle_z \bar{M}_w} = \left(\sum_i w_i M_i^{\varepsilon+2} \right) \left(\sum_i w_i M_i \right)^{-(\varepsilon+2)} \quad \text{(B28)}$$

(see Tables 42 and 43).
For $\langle S^2 \rangle \equiv \langle S^2 \rangle_z$, $\bar{M}_{av} \equiv \bar{M}_z$:

$$q_{\mathrm{Gyr},\langle S^2 \rangle_z \bar{M}_z} = \left(\sum_i w_i M_i^{\varepsilon+2} \right) \left(\sum_i w_i M_i \right)^{\varepsilon} \times \left(\sum_i w_i M_i^2 \right)^{-(\varepsilon+1)} \quad \text{(B29)}$$

(see Tables 44 and 45).
For $\langle S^2 \rangle \equiv \langle S^2 \rangle_z$, $\bar{M}_{av} \equiv \bar{M}_n$:

$$q_{\mathrm{Gyr},\langle S^2 \rangle_z \bar{M}_n} = \left(\sum_i w_i M_i^{\varepsilon+2} \right) \left(\sum_i w_i M_i \right)^{-1} \times \left(\sum_i w_i M_i^{-1} \right)^{\varepsilon+1} \quad \text{(B30)}$$

(see Tables 46 and 47).

5. In order to obtain the correct numerical values of constant K_{A_2} and exponent a_{A_2} in the exponential relationship between the second virial coefficient A_2 and molecular weight M

$$A_2 = K_{A_2} M^{a_{A_2}} \quad \text{(B31)}$$

it is necessary to determine the corresponding averages $\bar{A}_{2,av}$ and $\bar{M}_{av}$ with respect to Eq. (B31):

$$\bar{A}_{2,av} = K_{A_2} \bar{M}_{av}^{a_{A_2}} \quad \text{(B32)}$$

$\bar{A}_{2,av}$ is the measured second virial coefficient of a nonuniform (2) polymer sample, while $\bar{M}_{av}$ is the value of the molecular weight which fulfils Eq. (B32) if K_{A_2} and a_{A_2} are valid for strictly uniform (2) samples i:

$$A_{2,i} = K_{A_2} M_i^{a_{A_2}} \quad \text{(B33)}$$

In paper (22), the corresponding averages of the molecular weight $\bar{M}_{av,\bar{A}_2^{(OP)}}$ and $\bar{M}_{av,\bar{A}_2^{(LS)}}$ are calculated for the average second virial coefficients $\bar{A}_2^{(OP)}$ and $\bar{A}_2^{(LS)}$ obtained from

osmotic pressure (OP) and light scattering (LS) measurements, respectively, according to Flory and Krigbaum (23)

$$\bar{A}_2^{(\mathrm{OP})} = \sum_i \sum_j w_i w_j (A_2)_{ij} \tag{B34}$$

$$\bar{A}_2^{(\mathrm{LS})} = (1/\bar{M}_\mathrm{w})^2 \left[\sum_i \sum_j w_i M_i w_j M_j (A_2)_{ij} \right] \tag{B35}$$

where w_i is the mass fraction of uniform polymer species i, w_j the mass fraction of uniform polymer species j, M_i and M_j their molecular weights, respectively, and $(A_2)_{ij}$ is the second virial coefficient effective between species i and j. Since $\bar{M}_{\mathrm{av},\bar{A}_2^{(\mathrm{OP})}}$ and $\bar{M}_{\mathrm{av},\bar{A}_2^{(\mathrm{LS})}}$ are not directly experimentally accessible, polymolecularity correction factors (2) $q_{\bar{M}_\mathrm{n},\bar{A}_2^{(\mathrm{OP})}}$ and $q_{\bar{M}_\mathrm{w},\bar{A}_2^{(\mathrm{LS})}}$ are defined so that $\bar{M}_{\mathrm{av},\bar{A}_2^{(\mathrm{OP})}}$ and $\bar{M}_{\mathrm{av},\bar{A}_2^{(\mathrm{LS})}}$ can be calculated from experimentally determined $\bar{M}_\mathrm{n}$ and $\bar{M}_\mathrm{w}$, respectively:

$$\bar{A}_2^{(\mathrm{OP})} = K_{A_2} \bar{M}_{\mathrm{av},\bar{A}_2^{(\mathrm{OP})}}^{a_{A_2}} = K_{A_2} \bar{M}_\mathrm{n}^{a_{A_2}} q_{\bar{M}_\mathrm{n},\bar{A}_2^{(\mathrm{OP})}} \tag{B36}$$

$$\bar{A}_2^{(\mathrm{LS})} = K_{A_2} \bar{M}_{\mathrm{av},\bar{A}_2^{(\mathrm{LS})}}^{a_{A_2}} = K_{A_2} \bar{M}_\mathrm{w}^{a_{A_2}} q_{\bar{M}_\mathrm{w},\bar{A}_2^{(\mathrm{LS})}} \tag{B37}$$

Thus the respective polymolecularity correction factors

$$q_{\bar{M}_\mathrm{n},\bar{A}_2^{(\mathrm{OP})}} = (\bar{M}_{\mathrm{av},\bar{A}_2^{(\mathrm{OP})}} / \bar{M}_\mathrm{n})^{a_{A_2}} \tag{B38}$$

$$q_{\bar{M}_\mathrm{w},\bar{A}_2^{(\mathrm{LS})}} = (\bar{M}_{\mathrm{av},\bar{A}_2^{(\mathrm{LS})}} / \bar{M}_\mathrm{w})^{a_{A_2}} \tag{B39}$$

can be obtained. The numerical values of $q_{\bar{M}_n,\bar{A}_2^{(\mathrm{OP})}}$ and $q_{\bar{M}_w,\bar{A}_2^{(\mathrm{LS})}}$ are calculated numerically and listed for Schulz–Zimm (SZ), i.e. $q^{(\mathrm{SZ})}_{\bar{M}_\mathrm{n},\bar{A}_2^{(\mathrm{OP})}}$ and $q^{(\mathrm{SZ})}_{\bar{M}_\mathrm{w},\bar{A}_2^{(\mathrm{LS})}}$, and logarithmic normal (LN), i.e. $q^{(\mathrm{LN})}_{\bar{M}_\mathrm{n},\bar{A}_2^{(\mathrm{OP})}}$ and $q^{(\mathrm{LN})}_{\bar{M}_\mathrm{w},\bar{A}_2^{(\mathrm{LS})}}$, distributions of the molecular weight (see Tables 48–51).

6. Similarly, application of the principle of corresponding averages (3,9,24–26) to the relationship of Fox and Flory

$$[\eta] = \Phi(\varepsilon) \frac{\langle r^2 \rangle^{3/2}}{M} \tag{B40}$$

and defining a polymolecularity correction factor q_FF

$$[\eta] = \Phi(\varepsilon) q_\mathrm{FF} \frac{\langle r^2 \rangle_\mathrm{av}^{3/2}}{\bar{M}_\mathrm{av}} \tag{B41}$$

for the use of the z-average mean-square end-to-end distance $\langle r^2 \rangle_z$ and $\bar{M}_\mathrm{av} \equiv \bar{M}_\mathrm{n}$ in Eq. (B41) leads to

$$q_{\mathrm{FF},\langle r^2\rangle_z \bar{M}_\mathrm{n}} = \left(\sum_i w_i M_i^{-1} \right)^{-1} \left(\sum_i w_i M_i^{a_{[\eta]}} \right) \times \left(\sum_i w_i M_i^{(2a_{[\eta]}+5)/3} \right)^{-3/2} \left(\sum_i w_i M_i \right)^{3/2} \tag{B42}$$

(see Tables 52 and 53), whereas for $\langle r^2 \rangle_z$ and $\bar{M}_\mathrm{av} = \bar{M}_\mathrm{w}$ Eq. (B41) gives

$$q_{\mathrm{FF},\langle r^2\rangle_z \bar{M}_\mathrm{w}} = \left(\sum_i w_i M_i \right)^{5/2} \left(\sum_i w_i M_i^{(2a_{[\eta]}+5)/3} \right)^{-3/2} \times \left(\sum_i w_i M_i^{a_{[\eta]}} \right) \tag{B43}$$

(see Tables 54 and 55).

7. For converting the experimentally determined z-average mean-square radius of gyration $\langle S^2 \rangle_z$ into the weight-average or the number-average mean-square radii of gyration $\langle S^2 \rangle_\mathrm{w}$ or $\langle S^2 \rangle_\mathrm{n}$, respectively

$$\langle S^2 \rangle_\mathrm{w} = q_{\langle S^2 \rangle_\mathrm{w}} \langle S^2 \rangle_z \tag{B44}$$

$$\langle S^2 \rangle_\mathrm{n} = q_{\langle S^2 \rangle_\mathrm{n}} \langle S^2 \rangle_z \tag{B45}$$

the polymolecularity correction factors $q_{\langle S^2 \rangle_\mathrm{w}}$ and/or $q_{\langle S^2 \rangle_\mathrm{n}}$

$$q_{\langle S^2 \rangle_\mathrm{w}} = \left(\sum_i w_i M_i^{1+\varepsilon} \right) \left(\sum_i w_i M_i \right) \left(\sum_i w_i M_i^{\varepsilon+2} \right)^{-1} \tag{B46}$$

$$q_{\langle S^2 \rangle_\mathrm{n}} = \left(\sum_i w_i M_i^{\varepsilon} \right) \left(\sum_i w_i M_i^{-1} \right)^{-1} \left(\sum_i w_i M_i^{\varepsilon+2} \right)^{-1} \times \left(\sum_i w_i M_i \right) \tag{B47}$$

are to be used (26) (see Tables 56–59).

8. Correspondingly, in order to convert the 1.5th moment of the number distribution of the mean-square end-to-end distance $\overline{[\langle r^2 \rangle^{3/2}]}_\mathrm{n}$,† resulting from Eq. (B40) if $\bar{M}_\mathrm{n}$ is used (27), into $\langle r^2 \rangle_\mathrm{n}$, $\langle r^2 \rangle_\mathrm{w}$ or $\langle r^2 \rangle_z$

$$\langle r^2 \rangle_\mathrm{n} = q_{\langle r^2 \rangle_\mathrm{n}} \overline{[\langle r^2 \rangle^{3/2}]}_\mathrm{n}^{2/3} \tag{B48}$$

$$\langle r^2 \rangle_\mathrm{w} = q_{\langle r^2 \rangle_\mathrm{w}} \overline{[\langle r^2 \rangle^{3/2}]}_\mathrm{n}^{2/3} \tag{B49}$$

$$\langle r^2 \rangle_z = q_{\langle r^2 \rangle_z} \overline{[\langle r^2 \rangle^{3/2}]}_\mathrm{n}^{2/3} \tag{B50}$$

the polymolecularity correction factors $q_{\langle r^2 \rangle_\mathrm{n}}$, $q_{\langle r^2 \rangle_\mathrm{w}}$, and $q_{\langle r^2 \rangle_z}$ are to be used:

$$q_{\langle r^2 \rangle_\mathrm{n}} = \left(\sum_i w_i M_i^{(2a_{[\eta]}-1)/3} \right) \left(\sum_i w_i M_i^{-1} \right)^{-1/3} \times \left(\sum_i w_i M_i^{a_{[\eta]}} \right)^{-2/3} \tag{B51}$$

† $\overline{[\langle r^2 \rangle^{3/2}]}_\mathrm{n} = \sum_i n_i \langle r_i^2 \rangle^{3/2} \neq \langle r^2 \rangle_\mathrm{n}^{3/2} = \left(\sum_i n_i \langle r_i^2 \rangle \right)^{3/2}$

$$q_{\langle r^2\rangle_w} = \left(\sum_i w_i M_i^{(2a_{[\eta]}+2)/3}\right)\left(\sum_i w_i M_i^{-1}\right)^{2/3} \times \left(\sum_i w_i M_i^{a_{[\eta]}}\right)^{-2/3} \quad \text{(B52)}$$

$$q_{\langle r^2\rangle_z} = \left(\sum_i w_i M_i^{(2a_{[\eta]}+5)/3}\right)\left(\sum_i w_i M_i\right)^{-1} \times \left(\sum_i w_i M_i^{-1}\right)^{2/3}\left(\sum_i w_i M_i^{a_{[\eta]}}\right)^{-2/3} \quad \text{(B53)}$$

(see Tables 60–65).

9. The experimentally obtained ratio of the z-average unperturbed mean-square radius of gyration and the weight-average molecular weight $\langle S^2\rangle_{0,z}/\bar{M}_w$ of a polymolecular sample of stiff polymer chains can be transformed into $\langle S^2\rangle_0/M$ for a monomolecular polymer, the molecular weight M of which is equal to $\bar{M}_w$ of the polymolecular sample, according to:

$$\langle S^2\rangle_0/M = (\langle S^2\rangle_{0,z}/\bar{M}_w)(\bar{M}_w/\bar{M}_z)q(\bar{L}_w) \quad \text{(B54)}$$

where $q(\bar{L}_w)$ is the polymolecularity correction factor (28) (see Tables 66 and 67).

10. Application of the principle of corresponding averages to the various relationships for the experimental determination of the unperturbed dimensions of polymers in thermodynamically good solvents leads unambiguously to the correct polymolecularity correction factors (29).

10.1. The original (30) and modified (31) procedures of Burchard (32), Stockmayer and Fixman (30) can be handled with polymolecularity correction factors q_{BSF} depending upon which average of the molecular weight ($\bar{M}_w$ or $\bar{M}_n$) is to be used (33–35):

$$[\eta]/(q_{BSF,\bar{M}_w}\bar{M}_w^{1/2}) = K_0 + B'\bar{M}_w^{1/2}/q_{BSF,\bar{M}_w} \quad \text{(B55)}$$

$$[\eta]/({}^1q_{BSF,\bar{M}_n}\bar{M}_n^{1/2}) = K_0 + B'\bar{M}_n^{1/2}/{}^2q_{BSF,\bar{M}_n} \quad \text{(B56)}$$

The general formulae for $q_{BSF,\bar{M}_w}$ (33) and for ${}^1q_{BSF,\bar{M}_n}$ and ${}^2q_{BSF,\bar{M}_n}$ (34) were found to be

$$q_{BSF,\bar{M}_w} = \left(\sum_i w_i M_i^{1/2}\right)\left(\sum_i w_i M_i\right)^{-1/2} \quad \text{(B57)}$$

$${}^1q_{BSF,\bar{M}_n} = \left(\sum_i w_i M_i^{1/2}\right)\left(\sum_i w_i M_i^{-1}\right)^{1/2} \quad \text{(B58)}$$

$${}^2q_{BSF,\bar{M}_n} = \left(\sum_i w_i M_i^{1/2}\right)\left(\sum_i w_i M_i^{-1}\right)^{-1/2} \times \left(\sum_i w_i M_i\right)^{-1} \quad \text{(B59)}$$

(see Tables 68–70).

10.2. In the equation of Cowie and Bywater (36) also, two polymolecularity correction factors are necessary. If $\bar{M}_w$ and $\bar{s}_w$ are to be used, one obtains (14,33):

$$(\bar{M}_w^{1/2}/\bar{s}_w)(1-\tilde{v}_2\rho_1)/(\eta_1 N_L) = P_0 q_{CB,1} + q_{CB,2}B''\bar{M}_w^{1/2} \quad \text{(B60)}$$

with

$$q_{CB,1} = \left(\sum_i w_i M_i^{1-0.5\varepsilon}\right)\left(\sum_i w_i M_i\right)^{-1/2} \times \left(\sum_i w_i M_i^{0.5(1-\varepsilon)}\right)^{-1} \quad \text{(B61)}$$

$$q_{CB,2} = \left(\sum_i w_i M_i^{1.5-0.5\varepsilon}\right)\left(\sum_i w_i M_i\right)^{-1} \times \left(\sum_i w_i M_i^{0.5(1-\varepsilon)}\right)^{-1} \quad \text{(B62)}$$

(see Tables 71–74).

10.3. Baumann (37) and later Banks and Greenwood (38) describe procedures for the evaluation of unperturbed dimensions from the determination of the molecular weight and the radius of gyration for a series of polymer homologues. The Baumann–Stockmayer–Fixman procedure uses the theory of Stockmayer and Fixman (30) and with the two polymolecularity correction factors $q_{B,\bar{M}_{av}}$ (for the molecular weight) and $q_{B,\langle S^2\rangle_{av}}$ (for the radius of gyration) reads (39):

$$[\langle S^2\rangle_{av}q_{B,\langle S^2\rangle_{av}}/(\bar{M}_{av}q_{B,\bar{M}_{av}})]^{3/2} = K_0^{3/2} + B_{BSF}(\bar{M}_{av}q_{B,\bar{M}_{av}})^{1/2} \quad \text{(B63)}$$

For $\bar{M}_{av} \equiv \bar{M}_w$:

$$q_{B,\bar{M}_w} = \left(\sum_i w_i M_i^3\right)^2\left(\sum_i w_i M_i\right)^{-1}\left(\sum_i w_i M_i^{5/2}\right)^{-2} \quad \text{(B64)}$$

and for $\langle S^2\rangle_{av} \equiv \langle S^2\rangle_z$:

$$q_{B,\langle S^2\rangle_z} = \left(\sum_i w_i M_i^3\right)^2\left(\sum_i w_i M_i\right)\left(\sum_i w_i M_i^{2.5+1.5\varepsilon}\right)^{2/3} \times \left(\sum_i w_i M_i^{5/2}\right)^{-8/3}\left(\sum_i w_i M_i^{2+\varepsilon}\right)^{-1} \quad \text{(B65)}$$

(see Tables 75–78).

10.4. Application of the theory of Kurata and Stockmayer (40) leads to the Baumann–Kurata–Stockmayer procedure (41)

$$\langle S^2\rangle_{\mathrm{av}} q_{\mathrm{BKS},\langle S^2\rangle_{\mathrm{av}}}/(\bar{M}_{\mathrm{av}} q_{\mathrm{BKS},\bar{M}_{\mathrm{av}}}) = K_0 + B_{\mathrm{BKS}}\bar{M}_{\mathrm{av}} q_{\mathrm{BKS},\bar{M}_{\mathrm{av}}}/(\langle S^2\rangle_{\mathrm{av}} q_{\mathrm{BKS},\langle S^2\rangle_{\mathrm{av}}})^{1/2} \quad \text{(B66)}$$

For $\bar{M}_{\mathrm{av}} \equiv \bar{M}_{\mathrm{w}}$:

$$q_{\mathrm{BKS},\bar{M}_{\mathrm{w}}} = \left(\sum_i w_i M_i\right)^{-1}\left(\sum_i w_i M_i^2\right)^{-3}\left(\sum_i w_i M_i^{2+\varepsilon}\right) \times \left(\sum_i w_i M_i^{2.5-0.5\varepsilon}\right)^2 \quad \text{(B67)}$$

and for $\langle S^2\rangle_{\mathrm{av}} \equiv \langle S^2\rangle_{\mathrm{z}}$:

$$q_{\mathrm{BKS},\langle S^2\rangle_{\mathrm{z}}} = \left(\sum_i w_i M_i\right)\left(\sum_i w_i M_i^2\right)^{-4}\left(\sum_i w_i M_i^{2+\varepsilon}\right) \times \left(\sum_i w_i M_i^{2.5-0.5\varepsilon}\right)^2 \quad \text{(B68)}$$

(see Tables 79–82).

The numerical values of the polymolecularity correction factors Eqs. (B3–B11, B18–B30, B42, B43, B46, B47, B51–B53, B57–B59, B61, B62, B64, B65, B67, B68) are tabulated on the following pages for Schulz–Zimm (SZ) distributions of the molecular weight (42,43)

$$w_i(M) = [(k/\bar{M}_{\mathrm{n}})^{k+1}/\Gamma(k+1)]M^k \exp[-(k/\bar{M}_{\mathrm{n}})M] \quad \text{(B69)}$$

and for logarithmic normal (LN) distributions of the molecular weight (44–47)

$$w_i(M) = [1/(\sigma\sqrt{2\pi})]M^{-1}\exp\{-[1/(2\sigma^2)][\log_{\mathrm{e}}(M/\bar{M}_{\mathrm{m}})]^2\} \quad \text{(B70)}$$

as a function of the width of the respective distribution expressed as $\bar{M}_{\mathrm{w}}/\bar{M}_{\mathrm{n}}$ and where relevant of the exponent $a_{[\eta]}$ of Eq. (B1). The calculations have been conducted by inserting Eq. (B69) or Eq. (B70) for SZ or LN, respectively, into the general expressions Eqs. (B3–B11, B18–B30, B42, B43, B46, B47, B51–B53, B57–B59, B61, B62, B64, B65, B67, B68), replacing the summations by integrals and solution of the integrals (48). The resulting formulae of the polymolecularity correction factors $q^{(\mathrm{SZ})}$ and $q^{(\mathrm{LN})}$ are given on the titles of the tables.

Analogously, it is possible to calculate the corresponding formulae for the polymolecularity correctiom factors from the general expressions, Eqs. (B3–B11, B18–B30, B42, B43, B46, B47, B51–B53, B57–B59, B61, B62, B64, B65, B67, B68), for any molecular weight distribution (MWD) of which a closed mathematical expression can be formulated and for which the resulting integrals can be solved. In cases where the MWD is known but no closed mathematical expression can be formulated, or if the resulting integrals cannot be solved analytically, the polymolecularity correction factors nevertheless can be calculated numerically (22,28), i.e. because the general polymolecularity correction formulae represent moments (14) or products of moments of the MWD, they can in principle be taken from any experimentally determined (e.g. by gel-permeation chromatography) molecular weight distribution curve of the polymers or polymer fractions investigated.

In case the type of MWD of the polymers or polymer fractions investigated is not known the numerical values of the following tables still give an appropriate impression of the sensitivity of the methods listed towards type and width of the MWD as compared with the range of the experimental errors. Furthermore, as long as the difference between the numerical values of $q^{(\mathrm{SZ})}$ and $q^{(\mathrm{LN})}$ is not significant, the sensitivity of the method considered toward the type of MWD most probably is not strong, in other words any MWD other than SZ or LN will not exhibit a q value much different from $q^{(\mathrm{SZ})}$ or $q^{(\mathrm{LN})}$. When corresponding numerical values for $q^{(\mathrm{SZ})}$ and $q^{(\mathrm{LN})}$ differ substantially (cf. Tables 75 and 77) and/or $q \gg 1$ (cf. e.g. Table 47) or $q \ll 1$ (cf. e.g. Table 53), it is preferable to look for another experimental method which provides more adequate correction. A closer look upon the numerical values of the following tables also reveals that there are methods which are not sensitive towards polymolecularity (such as setting up $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ with $\bar{M}_{\bar{s}_{\mathrm{w}}\bar{D}_{\mathrm{A}}}$ (Tables 8 and 9) whereas others are very sensitive towards polymolecularity (such as correlating $\bar{D}_{\mathrm{z}}$ and $\bar{M}_{\mathrm{n}}$ in $D = K_D M^{a_D}$ (Tables 34 and 35)).

The exponents $a_{[\eta]}$, a_s, a_D, ε, and a_{A_2} of Eqs. B1, B12–B14 and B31 are interconnected with each other (49–55) by the "Exponents' Rule" (50,14):

$$a_{[\eta]} = 2 - 3a_s = -(1 + 3a_D) = 1.5\varepsilon + 0.5 = a_{A_2} + 1$$

In the following tables the numerical values for $\bar{M}_{\mathrm{w}}/\bar{M}_{\mathrm{n}}$ range from 1.1 to 20.0, those for $a_{[\eta]}$ from 0.4 to 0.9, in order to cover also limiting cases which occur in practical work (e.g., $a_{[\eta]} < 0.5; \varepsilon < 0$). The intervals for both parameters $\bar{M}_{\mathrm{w}}/\bar{M}_{\mathrm{n}}$ and $a_{[\eta]}$ as well as the number of digits behind the decimal sign for the polymolecularity correction factors are chosen so that for any value of $\bar{M}_{\mathrm{w}}/\bar{M}_{\mathrm{n}}$ and $a_{[\eta]}$ within the above-mentioned limits the numerical value of the polymolecularity correction factor can be found by interpolation of the numerical values given in the tables with an accuracy of at least about 1% except, of course, when extremely low numerical values are involved. In order to enable the user to calculate the numerical values of the polymolecularity correction factors for any $\bar{M}_{\mathrm{w}}/\bar{M}_{\mathrm{n}}$ and $a_{[\eta]}$ (or a_s, a_D, ε, and a_{A_2}, respectively) the exact formulae for all polymolecularity correction factors tabulated are given on top of the corresponding tables; for this purpose a table of the numerical values of the gamma function $\Gamma(x)$ for $1.00 \le x \le 2.00$ is included

(Table 1). For $x<1$ and $x>2$, the following formula can be used

$$\Gamma(x+1) = x\Gamma(x)$$

TABLE 1. NUMERICAL VALUES OF GAMMA FUNCTION $\Gamma(x)$

x	$\Gamma(x)$	x	$\Gamma(x)$	x	$\Gamma(x)$	x	$\Gamma(x)$
1.00	1.00000	1.25	0.90640	1.50	0.88623	1.75	0.91906
1.01	0.99433	1.26	0.90440	1.51	0.88659	1.76	0.92137
1.02	0.98884	1.27	0.90250	1.52	0.88704	1.77	0.92376
1.03	0.98355	1.28	0.90072	1.53	0.88757	1.78	0.92623
1.04	0.97844	1.29	0.89904	1.54	0.88818	1.79	0.92877
1.05	0.97350	1.30	0.89747	1.55	0.88887	1.80	0.93138
1.06	0.96874	1.31	0.89600	1.56	0.88964	1.81	0.93408
1.07	0.96415	1.32	0.89464	1.57	0.89049	1.82	0.93685
1.08	0.95973	1.33	0.89338	1.58	0.89142	1.83	0.93969
1.09	0.95546	1.34	0.89222	1.59	0.89243	1.84	0.94261
1.10	0.95135	1.35	0.89115	1.60	0.89352	1.85	0.94561
1.11	0.94739	1.36	0.89018	1.61	0.89468	1.86	0.94869
1.12	0.94359	1.37	0.88931	1.62	0.89592	1.87	0.95184
1.13	0.93993	1.38	0.88854	1.63	0.89724	1.88	0.95507
1.14	0.93642	1.39	0.88785	1.64	0.89864	1.89	0.95838
1.15	0.93304	1.40	0.88726	1.65	0.90012	1.90	0.96177
1.16	0.92980	1.41	0.88676	1.66	0.90167	1.91	0.96523
1.17	0.92670	1.42	0.88636	1.67	0.90330	1.92	0.96878
1.18	0.92373	1.43	0.88604	1.68	0.90500	1.93	0.97240
1.19	0.92088	1.44	0.88580	1.69	0.90678	1.94	0.97610
1.20	0.91817	1.45	0.88565	1.70	0.90864	1.95	0.97988
1.21	0.91558	1.46	0.88560	1.71	0.91057	1.96	0.98374
1.22	0.91311	1.47	0.88563	1.72	0.91258	1.97	0.98768
1.23	0.91075	1.48	0.88575	1.73	0.91466	1.98	0.99171
1.24	0.90852	1.49	0.88595	1.74	0.91683	1.99	0.99581
						2.00	1.00000

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C. INTRINSIC VISCOSITY VERSUS MOLECULAR WEIGHT RELATIONSHIP

TABLE 2. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{KMSH,\bar{M}_w}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING WEIGHT-AVERAGE MOLECULAR WEIGHT $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{KMSH,\bar{M}_w} = \frac{\Gamma(k+a_{[\eta]}+1)}{(k+1)^{a_{[\eta]}}\Gamma(k+1)} \qquad [\eta] = K_{[\eta]}\bar{M}_w^{a_{[\eta]}} q^{(SZ)}_{KMSH,\bar{M}_w}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	**0.40**	**0.50**	**0.70**	**0.90**
1.1	0.9891	0.9887	0.9906	0.9960
1.3	0.9725	0.9716	0.9765	0.9900
1.5	0.9605	0.9594	0.9665	0.9858
1.7	0.9515	0.9502	0.9590	0.9827
2.0	0.9414	0.9400	0.9509	0.9793
2.5	0.9301	0.9287	0.9419	0.9755
3.0	0.9228	0.9213	0.9360	0.9731
4.0	0.9137	0.9123	0.9289	0.9701
5.0	0.9083	0.9069	0.9247	0.9684
10.0	0.8977	0.8964	0.9166	0.9650
20.0	0.8924	0.8913	0.9126	0.9634

TABLE 3. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{KMSH,\bar{M}_w}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING WEIGHT-AVERAGE MOLECULAR WEIGHT $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{KMSH,\bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}-a_{[\eta]})/2} \qquad [\eta] = K_{[\eta]}\bar{M}_w^{a_{[\eta]}} q^{(LN)}_{KMSH,\bar{M}_w}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	**0.40**	**0.50**	**0.70**	**0.90**
1.1	0.9886	0.9882	0.9900	0.9957
1.3	0.9690	0.9677	0.9728	0.9883
1.5	0.9525	0.9506	0.9583	0.9819
1.7	0.9383	0.9358	0.9458	0.9764
2.0	0.9202	0.9170	0.9298	0.9693
2.5	0.8959	0.8918	0.9083	0.9596
3.0	0.8765	0.8717	0.8911	0.9518
4.0	0.8467	0.8409	0.8645	0.9395
5.0	0.8244	0.8178	0.8445	0.9301
6.0	0.8065	0.7993	0.8285	0.9225
8.0	0.7792	0.7711	0.8039	0.9107
10.0	0.7586	0.7499	0.7852	0.9016
20.0	0.6980	0.6877	0.7301	0.8739

TABLE 4. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{KMSH,\bar{M}_n}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING NUMBER-AVERAGE MOLECULAR WEIGHT $\bar{M}_n$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{KMSH,\bar{M}_n} = \frac{\Gamma(k+a_{[\eta]}+1)}{k^{a_{[\eta]}}\Gamma(k+1)} \qquad [\eta] = K_{[\eta]}\bar{M}_n^{a_{[\eta]}} q^{(SZ)}_{KMSH,\bar{M}_n}$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	**0.40**	**0.50**	**0.60**	**0.70**	**0.80**	**0.90**
1.1	1.0276	1.0370	1.0474	1.0589	1.0715	1.0852
1.3	1.0802	1.1078	1.1388	1.1733	1.2115	1.2537
1.5	1.1297	1.1750	1.2262	1.2837	1.3480	1.4199
2.0	1.2422	1.3293	1.4296	1.5447	1.6765	1.8274
2.5	1.3419	1.4684	1.6162	1.7888	1.9901	2.2253
3.0	1.4320	1.5958	1.7898	2.0197	2.2921	2.6155
4.0	1.5908	1.8245	2.1078	2.4514	2.8692	3.3782
5.0	1.7290	2.0279	2.3968	2.8529	3.4186	4.1222
6.0	1.8525	2.2129	2.6643	3.2317	3.9469	4.8516
8.0	2.0683	2.5429	3.1523	3.9380	4.9549	6.2762
10.0	2.2548	2.8348	3.5944	4.5937	5.9138	7.6655
15.0	2.6415	3.4586	4.5684	6.0836	8.1624	11.03
20.0	2.9579	3.9860	5.4197	7.4299	10.26	14.28

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TABLE 5. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{KMSH,\bar{M}_n}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING NUMBER-AVERAGE MOLECULAR WEIGHT $\bar{M}_n$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{KMSH,\bar{M}_n} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}+a_{[\eta]})/2} \qquad [\eta] = K_{[\eta]}\bar{M}_n^{a_{[\eta]}} q^{(LN)}_{KMSH,\bar{M}_n}$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	1.0270	1.0364	1.0468	1.0583	1.0710	1.0849
1.3	1.0762	1.1034	1.1342	1.1690	1.2079	1.2515
1.5	1.1202	1.1642	1.2149	1.2728	1.3390	1.4144
1.7	1.1602	1.2202	1.2901	1.3713	1.4653	1.5741
2.0	1.2142	1.2968	1.3947	1.5105	1.6472	1.8088
2.5	1.2925	1.4100	1.5524	1.7249	1.9343	2.1890
3.0	1.3602	1.5098	1.6944	1.9226	2.2056	2.5582
4.0	1.4743	1.6818	1.9453	2.2815	2.7132	3.2716
5.0	1.5693	1.8286	2.1652	2.6055	3.1861	3.9593
6.0	1.6515	1.9580	2.3633	2.9040	3.6330	4.6272
8.0	1.7901	2.1810	2.7132	3.4462	4.4691	5.9176
10.0	1.9055	2.3714	3.0200	3.9355	5.2481	7.1614
20.0	2.3136	3.0753	4.2121	5.9445	8.6445	12.953

TABLE 6. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{KMSH,\bar{s}_w\bar{D}_w}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $\bar{s}_w$ AND $\bar{D}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{KMSH,\bar{s}_w\bar{D}_w} = \left(\frac{3}{3k - a_{[\eta]} + 2}\right)^{a_{[\eta]}} \frac{\Gamma(k + a_{[\eta]} + 1)}{\Gamma(k+1)}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{\bar{s}_w\bar{D}_w} q^{(SZ)}_{KMSH,\bar{s}_w\bar{D}_w}$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	1.0064	1.0120	1.0191	1.0279	1.0384	1.0506
1.3	1.0179	1.0331	1.0527	1.0771	1.1065	1.1413
1.5	1.0277	1.0509	1.0812	1.1190	1.1651	1.2203
1.7	1.0362	1.0662	1.1056	1.1551	1.2160	1.2898
2.0	1.0470	1.0854	1.1361	1.2006	1.2809	1.3795
2.5	1.0608	1.1100	1.1755	1.2598	1.3664	1.5000
3.0	1.0711	1.1284	1.2051	1.3048	1.4323	1.5943
4.0	1.0855	1.1539	1.2465	1.3684	1.5269	1.7326
5.0	1.0950	1.1708	1.2741	1.4113	1.5918	1.8291
6.0	1.1017	1.1828	1.2938	1.4421	1.6389	1.9003
8.0	1.1106	1.1987	1.3200	1.4836	1.7030	1.9983
10.0	1.1162	1.2088	1.3367	1.5102	1.7445	2.0626
20.0	1.1281	1.2301	1.3724	1.5676	1.8353	2.2059

TABLE 7. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{KMSH,\bar{s}_w\bar{D}_w}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $\bar{s}_w$ AND $\bar{D}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{KMSH,\bar{s}_w\bar{D}_w} = (\bar{M}_w/\bar{M}_n)^{(5a^2_{[\eta]}-a_{[\eta]})/6} \qquad [\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{\bar{s}_w\bar{D}_w} q^{(LN)}_{KMSH,\bar{s}_w\bar{D}_w}$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	1.0064	1.0120	1.0192	1.0282	1.0389	1.0513
1.3	1.0176	1.0333	1.0539	1.0795	1.1107	1.1477

TABLE 7. *cont'd*

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.5	1.0274	1.0520	1.0845	1.1255	1.1761	1.2372
1.7	1.0360	1.0686	1.1120	1.1674	1.2365	1.3213
2.0	1.0473	1.0905	1.1487	1.2241	1.3195	1.4389
2.5	1.0630	1.1214	1.2011	1.3064	1.4427	1.6178
3.0	1.0760	1.1472	1.2457	1.3777	1.5518	1.7803
4.0	1.0968	1.1892	1.3195	1.4983	1.7411	2.0705
5.0	1.1133	1.2228	1.3797	1.5991	1.9037	2.3279
6.0	1.1269	1.2510	1.4310	1.6864	2.0477	2.5617
8.0	1.1487	1.2968	1.5157	1.8340	2.2974	2.9794
10.0	1.1659	1.3335	1.5849	1.9573	2.5119	3.3497
20.0	1.2211	1.4542	1.8206	2.3959	3.3145	4.8199

TABLE 8. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{KMSH,\bar{s}_w\bar{D}_A}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $\bar{s}_w$ AND $\bar{D}_A$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{KMSH,\bar{s}_w\bar{D}_A} = \frac{\Gamma(k + a_{[\eta]} + 1)[\Gamma(k+1)]^{3a_{[\eta]}-1}}{[\Gamma((3k - a_{[\eta]} + 5)/3)]^{a_{[\eta]}}[\Gamma((6k + a_{[\eta]} + 7)/6]^{2a_{[\eta]}}}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{\bar{s}_w\bar{D}_A} q^{(SZ)}_{KMSH,\bar{s}_w\bar{D}_A}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	1.0001	1.0029	1.0114	1.0236
1.5	1.0007	1.0113	1.0439	1.0907
2.0	1.0012	1.0176	1.0678	1.1404
3.0	1.0018	1.0243	1.0929	1.1926
5.0	1.0024	1.0299	1.1137	1.2363
10.0	1.0029	1.0342	1.1297	1.2700
20.0	1.0032	1.0364	1.1378	1.2871

TABLE 9. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{KMSH,\bar{s}_w\bar{D}_A}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $\bar{s}_w$ AND $\bar{D}_A$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{KMSH,\bar{s}_w\bar{D}_A} = (\bar{M}_w/\bar{M}_n)^{(8a^2_{[\eta]}-3a_{[\eta]}-a^3_{[\eta]})/12}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{\bar{s}_w\bar{D}_A} q^{(LN)}_{KMSH,\bar{s}_w\bar{D}_A}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	1.0001	1.0030	1.0118	1.0245
1.5	1.0005	1.0128	1.0512	1.1086
2.0	1.0009	1.0219	1.0891	1.1927
3.0	1.0015	1.0349	1.1448	1.3222
5.0	1.0021	1.0516	1.2191	1.5056
10.0	1.0031	1.0746	1.3276	1.7958
20.0	1.0040	1.0981	1.4459	2.1418

TABLE 10. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{KMSH,\bar{s}_w\bar{D}_z}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $\bar{s}_w$ AND $\bar{D}_z$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{KMSH,\bar{s}_w\bar{D}_z} = \frac{\Gamma(k + a_{[\eta]} + 1)}{(k+1)^{a_{[\eta]}}\Gamma(k+1)} \qquad [\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{\bar{s}_w\bar{D}_z} q^{(SZ)}_{KMSH,\bar{s}_w\bar{D}_z}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	0.9891	0.9887	0.9906	0.9960
1.3	0.9725	0.9716	0.9765	0.9900
1.5	0.9605	0.9594	0.9665	0.9858
1.7	0.9515	0.9502	0.9590	0.9827
2.0	0.9414	0.9400	0.9509	0.9793
2.5	0.9301	0.9287	0.9419	0.9755
3.0	0.9228	0.9213	0.9360	0.9731
4.0	0.9137	0.9123	0.9289	0.9701
5.0	0.9083	0.9069	0.9247	0.9684
10.0	0.8977	0.8964	0.9166	0.9650
20.0	0.8924	0.8913	0.9126	0.9634

TABLE 11. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{KMSH,\bar{s}_w\bar{D}_z}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $\bar{s}_w$ AND $\bar{D}_z$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{KMSH,\bar{s}_w\bar{D}_z} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]} - a_{[\eta]})/2} \qquad [\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{\bar{s}_w\bar{D}_z} q^{(LN)}_{KMSH,\bar{s}_w\bar{D}_z}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	0.9886	0.9882	0.9900	0.9957
1.3	0.9690	0.9677	0.9728	0.9883
1.5	0.9525	0.9506	0.9583	0.9819
1.7	0.9383	0.9358	0.9458	0.9764
2.0	0.9202	0.9170	0.9298	0.9693
2.5	0.8959	0.8918	0.9083	0.9596
3.0	0.8765	0.8717	0.8911	0.9518
4.0	0.8467	0.8409	0.8645	0.9395
5.0	0.8244	0.8178	0.8445	0.9301
6.0	0.8065	0.7993	0.8285	0.9225
8.0	0.7792	0.7711	0.8039	0.9107
10.0	0.7586	0.7499	0.7852	0.9016
20.0	0.6980	0.6877	0.7301	0.8739

TABLE 12. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{KMSH,\bar{s}_w[\bar{\eta}]_w}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $\bar{s}_w$ AND $[\eta] \equiv [\bar{\eta}]_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{KMSH,\bar{s}_w[\bar{\eta}]_w} = \frac{1}{[\Gamma((3k - a_{[\eta]} + 5)/3)]^{1.5a_{[\eta]}}} \frac{\Gamma(k + a_{[\eta]} + 1)]^{1-0.5a_{[\eta]}}}{[\Gamma(k+1)]^{1-2a_{[\eta]}}}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{\bar{s}_w[\bar{\eta}]_w} q^{(SZ)}_{KMSH,\bar{s}_w[\bar{\eta}]_w}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50[a]	0.70	0.90
1.1	0.9980	1.0000	1.0056	1.0122
1.5	0.9925	1.0000	1.0211	1.0459
2.0	0.9885	1.0000	1.0323	1.0702
5.0	0.9811	1.0000	1.0530	1.1153
10.0	0.9786	1.0000	1.0600	1.1307
20.0	0.9773	1.0000	1.0635	1.1385

[a] See Ref. 18.

TABLE 13. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{KMSH,\bar{s}_w[\bar{\eta}]_w}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $\bar{s}_w$ AND $[\eta] \equiv [\bar{\eta}]_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{KMSH,\bar{s}_w[\bar{\eta}]_w} = (\bar{M}_w/\bar{M}_n)^{(5a^2_{[\eta]} - 2a^3_{[\eta]} - 2a_{[\eta]})/6}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{\bar{s}_w[\bar{\eta}]_w} q^{(LN)}_{KMSH,\bar{s}_w[\bar{\eta}]_w}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50[a]	0.70	0.90
1.1	0.9980	1.0000	1.0058	1.0127
1.5	0.9914	1.0000	1.0249	1.0550
2.0	0.9853	1.0000	1.0429	1.0958
5.0	0.9662	1.0000	1.1026	1.2367
10.0	0.9521	1.0000	1.1499	1.3552
20.0	0.9381	1.0000	1.1993	1.4850

[a] See Ref. 18.

TABLE 14. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{KMSH,[\bar{\eta}]_w\bar{D}_w}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $[\eta] \equiv [\bar{\eta}]_w$ AND $\bar{D}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{KMSH,[\bar{\eta}]_w\bar{D}_w} = \frac{[\Gamma(k + a_{[\eta]} + 1)]^{a_{[\eta]}+1}[\Gamma((3k - a_{[\eta]} + 2)/3]^{3a_{[\eta]}}}{[\Gamma(k+1)]^{4a_{[\eta]}+1}}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{[\bar{\eta}]_w\bar{D}_w} q^{(SZ)}_{KMSH,[\bar{\eta}]_w\bar{D}_w}$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	1.0234	1.0363	1.0530	1.0741	1.1001	1.1318
1.3	1.0655	1.1025	1.1512	1.2143	1.2949	1.3973
1.5	1.1020	1.1607	1.2395	1.3438	1.4809	1.6611
1.7	1.1338	1.2122	1.3189	1.4629	1.6570	1.9195
2.0	1.1744	1.2787	1.4236	1.6240	1.9022	2.2923
2.5	1.2275	1.3676	1.5672	1.8519	1.2628	2.8666
3.0	1.2680	1.4367	1.6815	2.0391	2.5706	3.3795
4.0	1.3253	1.5365	1.8511	2.3264	3.0620	4.2389
5.0	1.3639	1.6050	1.9705	2.5351	3.4334	4.9193
8.0	1.4286	1.7224	2.1810	2.9160	4.1406	6.2842
10.0	1.4522	1.7661	2.2611	3.0653	4.4280	6.8637
20.0	1.5030	1.8614	2.4394	3.4059	5.1043	8.2812

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TABLE 15. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{KMSH,[\bar{\eta}]_w\bar{D}_w}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $[\eta] \equiv [\bar{\eta}]_w$ AND $\bar{D}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{KMSH,[\bar{\eta}]_w\bar{D}_w} = (\bar{M}_w/\bar{M}_n)^{(4a^3_{[\eta]}+5a^2_{[\eta]}+a_{[\eta]})/6}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{[\bar{\eta}]_w\bar{D}_w} q^{(LN)}_{KMSH,[\bar{\eta}]_w\bar{D}_w}$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	1.0234	1.0364	1.0532	1.0745	1.1008	1.1331
1.3	1.0657	1.1034	1.1534	1.2186	1.3027	1.4105
1.5	1.1034	1.1642	1.2468	1.3574	1.5049	1.7016
1.7	1.1374	1.2202	1.3346	1.4917	1.7072	2.0050
2.0	1.1832	1.2968	1.4580	1.6861	2.0111	2.4811
2.5	1.2490	1.4100	1.6462	1.9949	2.5184	3.3243
3.0	1.3055	1.5098	1.8178	2.2887	3.0265	4.2219
4.0	1.3999	1.6818	2.1258	2.8428	4.0446	6.1560
5.0	1.4778	1.8286	2.4002	3.3635	5.0648	8.2480
8.0	1.6563	2.1810	3.0994	4.7932	8.1342	15.27
10.0	1.7485	2.3714	3.4995	5.6711	10.19	20.46
20.0	2.0688	3.0753	5.1022	9.5619	20.49	50.78

TABLE 16. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{KMSH,[\bar{\eta}]_w\bar{D}_A}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $[\eta] \equiv [\bar{\eta}]_w$ AND $\bar{D}_A$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{KMSH,[\bar{\eta}]_w\bar{D}_A} = \frac{[\Gamma(k+1)]^{5a_{[\eta]}-1}[\Gamma(k+a_{[\eta]}+1)]^{a_{[\eta]}+1}}{[\Gamma((6k+a_{[\eta]}+7)/6]^{6a_{[\eta]}}}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{[\bar{\eta}]_w\bar{D}_A} q^{(SZ)}_{KMSH,[\bar{\eta}]_w\bar{D}_A}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	1.0044	1.0088	1.0231	1.0468
1.5	1.0172	1.0344	1.0909	1.1861
2.0	1.0270	1.0538	1.1426	1.2949
3.0	1.0375	1.0747	1.1983	1.4147
5.0	1.0465	1.0924	1.2458	1.5189
10.0	1.0535	1.1061	1.2831	1.6020
20.0	1.0570	1.1132	1.3022	1.6452

TABLE 17. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{KMSH,[\bar{\eta}]_w\bar{D}_A}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $[\eta] \equiv [\bar{\eta}]_w$ AND $\bar{D}_A$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{KMSH,[\bar{\eta}]_w\bar{D}_A} = (\bar{M}_w/\bar{M}_n)^{(5a^3_{[\eta]}+4a^2_{[\eta]}-a_{[\eta]})/12}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{[\bar{\eta}]_w\bar{D}_A} q^{(LN)}_{KMSH,[\bar{\eta}]_w\bar{D}_A}$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	1.0045	1.0090	1.0154	1.0239	1.0349	1.0487
1.5	1.0191	1.0387	1.0670	1.1057	1.1572	1.2241
2.0	1.0329	1.0671	1.1173	1.1875	1.2834	1.4130

TABLE 17. *cont'd*

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
3.0	1.0526	1.1085	1.1922	1.3131	1.4851	1.7297
5.0	1.0780	1.1629	1.2937	1.4903	1.7850	2.2316
10.0	1.1134	1.2409	1.4454	1.7698	2.2909	3.1532
20.0	1.1500	1.3243	1.6150	2.1016	2.9402	4.4554

TABLE 18. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{KMSH,[\bar{\eta}]_w\bar{D}_z}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $[\eta] \equiv [\bar{\eta}]_w$ AND $\bar{D}_z$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{KMSH,[\bar{\eta}]_w\bar{D}_z} = \frac{[\Gamma(k+a_{[\eta]}+1)]^{a_{[\eta]}+1}[\Gamma((3k-a_{[\eta]}+5)/3)]^{3a_{[\eta]}}}{(k+1)^{3a_{[\eta]}}[\Gamma(k+1)]^{4a_{[\eta]}+1}}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{[\bar{\eta}]_w\bar{D}_z} q^{(SZ)}_{KMSH,[\bar{\eta}]_w\bar{D}_z}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	0.9715	0.9665	0.9612	0.9644
1.5	0.8997	0.8830	0.8657	0.8756
2.0	0.8538	0.8305	0.8086	0.8200
3.0	0.8108	0.7820	0.7529	0.7684
4.0	0.7904	0.7592	0.7277	0.7442
5.0	0.7784	0.7459	0.7132	0.7301
10.0	0.7553	0.7204	0.6853	0.7029
20.0	0.7442	0.7081	0.6719	0.6899

TABLE 19. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{KMSH,[\bar{\eta}]_w\bar{D}_z}$ FOR $[\eta] = K_{[\eta]}M^{a_{[\eta]}}$ USING MOLECULAR WEIGHT FROM $[\eta] \equiv [\bar{\eta}]_w$ AND $\bar{D}_z$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{KMSH,[\bar{\eta}]_w\bar{D}_z} = (\bar{M}_w/\bar{M}_n)^{(4a^3_{[\eta]}-a^2_{[\eta]}-5a_{[\eta]})/6}$$

$$[\eta] = K_{[\eta]}\bar{M}^{a_{[\eta]}}_{[\bar{\eta}]_w\bar{D}_z} q^{(LN)}_{KMSH,[\bar{\eta}]_w\bar{D}_z}$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	0.9702	0.9649	0.9611	0.9593	0.9597	0.9627
1.5	0.8793	0.8589	0.8448	0.8378	0.8393	0.8506
2.0	0.8026	0.7711	0.7495	0.7390	0.7412	0.7584
3.0	0.7057	0.6623	0.6332	0.6192	0.6221	0.6451
4.0	0.6441	0.5946	0.5617	0.5461	0.5494	0.5751
5.0	0.6001	0.5469	0.5120	0.4955	0.4989	0.5262
6.0	0.5663	0.5107	0.4746	0.4576	0.4611	0.4892
8.0	0.5169	0.4585	0.4210	0.4036	0.4073	0.4362
10.0	0.4816	0.4217	0.3837	0.3662	0.3698	0.3990
20.0	0.3865	0.3252	0.2876	0.2706	0.2741	0.3026

D. SEDIMENTATION COEFFICIENT VERSUS MOLECULAR WEIGHT RELATIONSHIP

TABLE 20. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Sed,\bar{s}_w\bar{M}_w}$ FOR $s = K_s M^{a_s}$ USING $\bar{s}_w$ AND $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Sed,\bar{s}_w\bar{M}_w} = \frac{\Gamma(k+1+a_s)}{(k+1)^{a_s}\Gamma(k+1)} = \frac{\Gamma((3k-a_{[\eta]}+5)/3)}{(k+1)^{(2-a_{[\eta]})/3}\Gamma(k+1)}$$

$$\bar{s}_w = K_s \bar{M}_w^{a_s} q^{(SZ)}_{Sed,\bar{s}_w\bar{M}_w}$$

	$a_{[\eta]}$ and (a_s)			
$\bar{M}_w/\bar{M}_n$	0.40 (0.53)	0.50 (0.50)	0.70 (0.43)	0.90 (0.37)
1.1	0.9888	0.9887	0.9889	0.9895
1.5	0.9597	0.9594	0.9598	0.9616
2.0	0.9406	0.9400	0.9404	0.9429
3.0	0.9222	0.9213	0.9216	0.9246
4.0	0.9133	0.9123	0.9124	0.9157
5.0	0.9080	0.9069	0.9070	0.9104
8.0	0.9003	0.8990	0.8990	0.9025
20.0	0.8927	0.8913	0.8911	0.8947

TABLE 21. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Sed,\bar{s}_w\bar{M}_w}$ FOR $s = K_s M^{a_s}$ USING $\bar{s}_w$ AND $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Sed,\bar{s}_w\bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{(a_s^2-a_s)/2} = (\bar{M}_w/\bar{M}_n)^{(a_{[\eta]}^2-a_{[\eta]}-2)/18}$$

$$\bar{s}_w = K_s \bar{M}_w^{a_s} q^{(LN)}_{Sed,\bar{s}_w\bar{M}_w}$$

	$a_{[\eta]}$ and (a_s)		
$\bar{M}_w/\bar{M}_n$	0.40 (0.53)	0.50 (0.50)	0.90 (0.37)
1.1	0.9882	0.9882	0.9890
1.5	0.9508	0.9506	0.9540
2.0	0.9174	0.9170	0.9227
3.0	0.8722	0.8717	0.8802
5.0	0.8185	0.8178	0.8295
7.0	0.7849	0.7841	0.7978
10.0	0.7509	0.7499	0.7654
15.0	0.7139	0.7128	0.7302
20.0	0.6888	0.6877	0.7062

TABLE 22. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Sed,\bar{s}_w\bar{M}_n}$ FOR $s = K_s M^{a_s}$ USING $\bar{s}_w$ AND $\bar{M}_n$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Sed,\bar{s}_w\bar{M}_n} = \frac{\Gamma(k+1+a_s)}{k^{a_s}\Gamma(k+1)} = \frac{\Gamma((3k-a_{[\eta]}+5)/3)}{k^{(2-a_{[\eta]})/3}\Gamma(k+1)} \qquad \bar{s}_w = K_s \bar{M}_n^{a_s} q^{(SZ)}_{Sed,\bar{s}_w\bar{M}_n}$$

	$a_{[\eta]}$ and (a_s)					
$\bar{M}_w/\bar{M}_n$	0.40 (0.53)	0.50 (0.50)	0.60 (0.47)	0.70 (0.43)	0.80 (0.40)	0.90 (0.37)
1.1	1.0403	1.0370	1.0337	1.0306	1.0276	1.0247
1.5	1.1914	1.1750	1.1592	1.1441	1.1297	1.1158
2.0	1.3612	1.3293	1.2989	1.2699	1.2422	1.2158
3.0	1.6568	1.5958	1.5381	1.4835	1.4320	1.3833
4.0	1.9129	1.8245	1.7416	1.6638	1.5908	1.5223
5.0	2.1423	2.0279	1.9213	1.8218	1.7290	1.6425
6.0	2.3521	2.2129	2.0837	1.9638	1.8525	1.7493
8.0	2.7292	2.5429	2.3714	2.2136	2.0683	1.9345
10.0	3.0654	2.8348	2.6240	2.4312	2.2548	2.0934
20.0	4.4117	3.9860	3.6050	3.2638	2.9579	2.6836

TABLE 23. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Sed,\bar{s}_w\bar{M}_n}$ FOR $s = K_s M^{a_s}$ USING $\bar{s}_w$ AND $\bar{M}_n$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Sed,\bar{s}_w\bar{M}_n} = (\bar{M}_w/\bar{M}_n)^{(a_s^2+a_s)/2} = (\bar{M}_w/\bar{M}_n)^{(a_{[\eta]}^2-7a_{[\eta]}+10)/18} \qquad \bar{s}_w = K_s \bar{M}_n^{a_s} q^{(LN)}_{Sed,\bar{s}_w\bar{M}_n}$$

	$a_{[\eta]}$ and (a_s)					
$\bar{M}_w/\bar{M}_n$	0.40 (0.53)	0.50 (0.50)	0.60 (0.47)	0.70 (0.43)	0.80 (0.40)	0.90 (0.37)
1.1	1.0397	1.0364	1.0332	1.0300	1.0270	1.0242
1.5	1.1803	1.1642	1.1488	1.1342	1.1202	1.1069
2.0	1.3277	1.2968	1.2677	1.2402	1.2142	1.1897
3.0	1.5671	1.5098	1.4564	1.4066	1.3602	1.3169
4.0	1.7627	1.6818	1.6071	1.5381	1.4743	1.4153
5.0	1.9311	1.8286	1.7346	1.6484	1.5693	1.4967
6.0	2.0805	1.9580	1.8463	1.7445	1.6515	1.5666
8.0	2.3403	2.1810	2.0373	1.9075	1.7901	1.6837
10.0	2.5638	2.3714	2.1990	2.0444	1.9055	1.7806
20.0	3.4039	3.0753	2.7877	2.5354	2.3136	2.1183

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TABLE 24. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Sed,\bar{s}_w\bar{M}_{\bar{s}_w\bar{D}_w}}$ FOR $s = K_sM^{a_s}$ USING $\bar{s}_w$ AND $\bar{M}_{\bar{s}_w\bar{D}_w}$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Sed,\bar{s}_w\bar{M}_{\bar{s}_w\bar{D}_w}} = \frac{[\Gamma((3k+5-a_{[\eta]})/3)]^{(a_{[\eta]}+1)/3}[\Gamma((3k+2-a_{[\eta]})/3)]^{(2-a_{[\eta]})/3}}{\Gamma(k+1)} \qquad \bar{s}_w = K_s\bar{M}^{a_s}_{\bar{s}_w\bar{D}_w}q^{(SZ)}_{Sed,\bar{s}_w\bar{M}_{\bar{s}_w\bar{D}_w}}$$

	$a_{[\eta]}$ and (a_s)					
$\bar{M}_w/\bar{M}_n$	0.40 (0.53)	0.50 (0.50)	0.60 (0.47)	0.70 (0.43)	0.80 (0.40)	0.90 (0.37)
1.1	1.0119	1.0120	1.0119	1.0118	1.0116	1.0112
1.5	1.0503	1.0509	1.0512	1.0509	1.0502	1.0490
2.0	1.0837	1.0854	1.0863	1.0864	1.0857	1.0842
3.0	1.1250	1.1284	1.1307	1.1320	1.1319	1.1306
5.0	1.1648	1.1704	1.1746	1.1773	1.1783	1.1774
8.0	1.1897	1.1966	1.2018	1.2053	1.2066	1.2057
10.0	1.1982	1.2055	1.2111	1.2147	1.2161	1.2151
20.0	1.2153	1.2234	1.2295	1.2334	1.2348	1.2335

TABLE 25. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Sed,\bar{s}_w\bar{M}_{\bar{s}_w\bar{D}_w}}$ FOR $s = K_sM^{a_s}$ USING $\bar{s}_w$ AND $\bar{M}_{\bar{s}_w\bar{D}_w}$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT[a]

$$q^{(LN)}_{Sed,\bar{s}_w\bar{M}_{\bar{s}_w\bar{D}_w}} = (\bar{M}_w/\bar{M}_n)^{(a_{[\eta]}-a^2_{[\eta]}+2)/18} \qquad \bar{s}_w = K_s\bar{M}^{a_s}_{\bar{s}_w\bar{D}_w}q^{(LN)}_{Sed,\bar{s}_w\bar{M}_{\bar{s}_w\bar{D}_w}}$$

	$a_{[\eta]}$ and (a_s)					
$\bar{M}_w/\bar{M}_n$	0.40 (0.53)	0.50 (0.50)	0.60 (0.47)	0.70 (0.43)	0.80 (0.40)	0.90 (0.37)
1.1	1.0119	1.0120	1.0119	1.0118	1.0115	1.0111
1.5	1.0518	1.0520	1.0518	1.0510	1.0499	1.0482
2.0	1.0901	1.0905	1.0901	1.0888	1.0867	1.0838
3.0	1.1465	1.1472	1.1465	1.1444	1.1409	1.1361
5.0	1.2218	1.2228	1.2218	1.2185	1.2130	1.2055
6.0	1.2498	1.2510	1.2498	1.2461	1.2399	1.2313
8.0	1.2953	1.2968	1.2953	1.2909	1.2834	1.2731
10.0	1.3318	1.3335	1.3318	1.3267	1.3183	1.3065
20.0	1.4518	1.4542	1.4518	1.4446	1.4326	1.4160

[a] Unfortunately there is a printing error in Ref. 20. The exponent $(a_{[\eta]} - a^2_{[\eta]} + 2)/18$ given here is correct.

TABLE 26. VALUES OF $\bar{s}_w/\bar{s}_{max}$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT, CALCULATED ACCORDING TO

$$\left(\frac{\bar{s}_w}{\bar{s}_{max}}\right)_{SZ} = \frac{\Gamma(k+a_s+1)}{(k+1-a_s)^{a_s}\Gamma(k+1)} = \frac{\Gamma(k-a_{[\eta]}/3+5/3)}{(k+1/3+a_{[\eta]}/3)^{(2-a_{[\eta]})/3}\Gamma(k+1)}$$

	$a_{[\eta]}$ and (a_s)					
$\bar{M}_w/\bar{M}_n$	0.40 (0.53)	0.50 (0.50)	0.60 (0.47)	0.70 (0.43)	0.80 (0.40)	0.90 (0.37)
1.1	1.0153	1.0120	1.0090	1.0063	1.0039	1.0018
1.2	1.0294	1.0230	1.0172	1.0120	1.0075	1.0037
1.5	1.0653	1.0509	1.0382	1.0269	1.0171	1.0087
2.0	1.1097	1.0854	1.0640	1.0454	1.0293	1.0156
3.0	1.1657	1.1284	1.0961	1.0683	1.0446	1.0247
5.0	1.2217	1.1708	1.1275	1.0907	1.0598	1.0340
10.0	1.2724	1.2088	1.1553	1.1105	1.0731	1.0422
20.0	1.3013	1.2301	1.1708	1.1214	1.0805	1.0468

TABLE 27. VALUES OF $\bar{s}_w/\bar{s}_{max}$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT, CALCULATED ACCORDING TO

$$\left(\frac{\bar{s}_w}{\bar{s}_{max}}\right)_{LN} = \frac{\bar{M}_w^{1.5a_s^2}}{\bar{M}_n} = \frac{\bar{M}^{a_{[\eta]}^2-4a_{[\eta]}+4)/6}}{\bar{M}_n}$$

	$a_{[\eta]}$ and (a_s)					
$\bar{M}_w/\bar{M}_n$	0.40 (0.53)	0.50 (0.50)	0.60 (0.47)	0.70 (0.43)	0.80 (0.40)	0.90 (0.37)
1.1	1.0415	1.0364	1.0316	1.0272	1.0231	1.0194
1.2	1.0809	1.0708	1.0614	1.0527	1.0447	1.0375
1.5	1.1887	1.1642	1.1416	1.1210	1.1022	1.0852
2.0	1.3441	1.2968	1.2541	1.2156	1.1810	1.1500
3.0	1.5980	1.5098	1.4317	1.3626	1.3017	1.2480
5.0	1.9871	1.8286	1.6917	1.5735	1.4715	1.3834
10.0	2.6710	2.3714	2.1216	1.9128	1.7378	1.5910
20.0	3.5901	3.0753	2.6607	2.3252	2.0523	1.8297

E. DIFFUSION COEFFICIENT VERSUS MOLECULAR WEIGHT RELATIONSHIP

TABLE 28. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Diff,\bar{D}_A\bar{M}_n}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_A$ AND $\bar{M}_n$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Diff,\bar{D}_A\bar{M}_n} = \frac{1}{k^{a_D}}\left(\frac{\Gamma(k+1)}{\Gamma((2k-a_D+2)/2)}\right)^2 = \frac{1}{k^{-(a_{[\eta]}+1)/3}}\left(\frac{\Gamma(k+1)}{\Gamma((6k+a_{[\eta]}+7)/6)}\right)^2$$

$$\bar{D}_A = K_D \bar{M}_n^{a_D} q^{(SZ)}_{Diff,\bar{D}_A\bar{M}_n}$$

	$a_{[\eta]}$ and (a_D)		
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.90 (−0.63)
1.1	0.9723	0.9700	0.9602
1.5	0.8798	0.8705	0.8324
2.0	0.7939	0.7790	0.7200
2.5	0.7291	0.7107	0.6392
3.0	0.6783	0.6575	0.5781
4.0	0.6026	0.5789	0.4908
5.0	0.5483	0.5230	0.4309
6.0	0.5068	0.4806	0.3868
8.0	0.4467	0.4197	03253
10.0	0.4045	0.3773	0.2840
20.0	0.2955	0.2695	0.1851

TABLE 29. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Diff,\bar{D}_A\bar{M}_n}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_A$ AND $\bar{M}_n$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Diff,\bar{D}_A\bar{M}_n} = (\bar{M}_w/\bar{M}_n)^{-(a_D^2-2a_D)/4} = (\bar{M}_w/\bar{M}_n)^{-(a_{[\eta]}^2+8a_{[\eta]}+7)/36}$$

$$\bar{D}_A = K_D \bar{M}_n^{a_D} q^{(LN)}_{Diff,\bar{D}_A\bar{M}_n}$$

	$a_{[\eta]}$ and (a_D)		
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.90 (−0.63)
1.1	0.9729	0.9707	0.9610
1.5	0.8899	0.8810	0.8445
2.0	0.8192	0.8052	0.7490
3.0	0.7289	0.7094	0.6325
5.0	0.6293	0.6047	0.5112
10.0	0.5155	0.4870	0.3829
20.0	0.4223	0.3921	0.2868

TABLE 30. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Diff,\bar{D}_A\bar{M}_{\bar{s}_w\bar{D}_A}}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_A$ AND $\bar{M}_{\bar{s}_w\bar{D}_A}$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Diff,\bar{D}_A\bar{M}_{\bar{s}_w\bar{D}_A}} = \frac{[\Gamma((3k+5-a_{[\eta]})/3)]^{(a_{[\eta]}+1)/3}[\Gamma((6k+7+a_{[\eta]})/6)]^{(2a_{[\eta]}-4)/3}}{[\Gamma(k+1)]^{a_{[\eta]}-1}} \qquad \bar{D}_A = K_D \bar{M}^{a_D}_{\bar{s}_w\bar{D}_A} q^{(SZ)}_{Diff,\bar{D}_A\bar{M}_{\bar{s}_w\bar{D}_A}}$$

	$a_{[\eta]}$ and (a_D)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.60 (−0.53)	0.70 (−0.57)	0.80 (−0.60)	0.90 (−0.63)
1.1	1.0035	1.0029	1.0023	1.0017	1.0011	1.0005
2.0	1.0210	1.0176	1.0140	1.0104	1.0068	1.0033
4.0	1.0331	1.0278	1.0221	1.0164	1.0107	1.0052
8.0	1.0395	1.0318	1.0264	1.0196	1.0128	1.0062
20.0	1.0434	1.0364	1.0290	1.0215	1.0140	1.0068

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TABLE 31. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Diff,\bar{D}_A\bar{M}_{3_w\bar{D}_A}}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_A$ AND $\bar{M}_{\bar{3}_w\bar{D}_A}$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Diff,\bar{D}_A\bar{M}_{3_w\bar{D}_A}} = (\bar{M}_w/\bar{M}_n)^{(a^3_{[\eta]}-2a^2_{[\eta]}-a_{[\eta]}+2)/36} \qquad \bar{D}_A = K_D\bar{M}^{a_D}_{\bar{3}_w\bar{D}_A} q^{(LN)}_{Diff,\bar{D}_A\bar{M}_{3_w\bar{D}_A}}$$

	$a_{[\eta]}$ and (a_D)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.60 (−0.53)	0.70 (−0.57)	0.80 (−0.60)	0.90 (−0.63)
1.1	1.0036	1.0030	1.0024	1.0018	1.0011	1.0006
1.5	1.0153	1.0128	1.0101	1.0075	1.0049	1.0024
2.0	1.0262	1.0219	1.0174	1.0128	1.0084	1.0040
3.0	1.0419	1.0349	1.0277	1.0204	1.0133	1.0064
5.0	1.0619	1.0516	1.0409	1.0301	1.0195	1.0094
8.0	1.0807	1.0671	1.0531	1.0390	1.0253	1.0121
20.0	1.1183	1.0981	1.0774	1.0567	1.0366	1.0175

TABLE 32. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Diff,\bar{D}_w\bar{M}_n}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_w$ AND $\bar{M}_n$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Diff,\bar{D}_w\bar{M}_n} = \frac{1}{k^{a_D}}\frac{\Gamma(k+a_D+1)}{\Gamma(k+1)} = \frac{1}{k^{-(a_{[\eta]}+1)/3}}\frac{\Gamma((3k-a_{[\eta]}+2)/3)}{\Gamma(k+1)}$$

$$\bar{D}_w = K_D\bar{M}^{a_D}_n q^{(SZ)}_{Diff,\bar{D}_w\bar{M}_n}$$

	$a_{[\eta]}$ and (a_D)		
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.90 (−0.63)
1.1	0.9877	0.9876	0.9884
1.5	0.9406	0.9400	0.9429
2.0	0.8878	0.8862	0.8896
3.0	0.8017	0.7979	0.7981
5.0	0.6837	0.6760	0.6659
7.0	0.6064	0.5959	0.5768
10.0	0.5285	0.5154	0.4868
15.0	0.4478	0.4323	0.3945
20.0	0.3963	0.3796	0.3369

TABLE 33. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Diff,\bar{D}_w\bar{M}_n}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_w$ AND $\bar{M}_n$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Diff,\bar{D}_w\bar{M}_n} = (\bar{M}_w/\bar{M}_n)^{(a^2_D+a_D)/2} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}-a_{[\eta]}-2)/18}$$

$$\bar{D}_w = K_D\bar{M}^{a_D}_n q^{(LN)}_{Diff,\bar{D}_w\bar{M}_n}$$

	$a_{[\eta]}$ and (a_D)		
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.90 (−0.63)
1.1	0.9882	0.9882	0.9890
1.5	0.9508	0.9506	0.9540
2.0	0.9174	0.9170	0.9227
3.0	0.8722	0.8717	0.8802
5.0	0.8185	0.8178	0.8295
7.0	0.7849	0.7841	0.7978
10.0	0.7509	0.7499	0.7654
15.0	0.7139	0.7128	0.7302
20.0	0.6888	0.6877	0.7062

TABLE 34. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Diff,\bar{D}_z\bar{M}_n}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_z$ AND $\bar{M}_n$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Diff,\bar{D}_z\bar{M}_n} = \frac{1}{k^{a_D}(k+1)}\frac{\Gamma(k+a_D+2)}{\Gamma(k+1)} = \frac{k^{(a_{[\eta]}+1)/3}}{k+1}\frac{\Gamma((3k-a_{[\eta]}+5)/3)}{\Gamma(k+1)}$$

$$\bar{D}_z = K_D\bar{M}^{a_D}_n q^{(SZ)}_{Diff,\bar{D}_z\bar{M}_n}$$

	$a_{[\eta]}$ and (a_D)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.70 (−0.57)	0.90 (−0.63)
1.1	0.9458	0.9427	0.9369	0.9315
1.5	0.7942	0.7833	0.7628	0.7439
2.0	0.6806	0.6647	0.6349	0.6079
3.0	0.5523	0.5319	0.4945	0.4611
4.0	0.4782	0.4561	0.4159	0.3806
5.0	0.4285	0.4056	0.3644	0.3285
6.0	0.3920	0.3688	0.3273	0.2915
8.0	0.3412	0.3179	0.2767	0.2418
10.0	0.3065	0.2835	0.2431	0.2093
20.0	0.2206	0.1993	0.1632	0.1342

TABLE 35. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Diff,\bar{D}_z\bar{M}_n}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_z$ AND $\bar{M}_n$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Diff,\bar{D}_z\bar{M}_n} = (\bar{M}_w/\bar{M}_n)^{(a^2_D+3a_D)/2} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}-7a_{[\eta]}-8)/18}$$

$$\bar{D}_z = K_D\bar{M}^{a_D}_n q^{(LN)}_{Diff,\bar{D}_z\bar{M}_n}$$

	$a_{[\eta]}$ and (a_D)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.70 (−0.57)	0.90 (−0.63)
1.1	0.9452	0.9422	0.9364	0.9311
1.5	0.7869	0.7761	0.7561	0.7380
2.0	0.6638	0.6484	0.6201	0.5948
3.0	0.5224	0.5033	0.4689	0.4390
4.0	0.4407	0.4204	0.3845	0.3538
5.0	0.3862	0.3657	0.3297	0.2993
6.0	0.3468	0.3263	0.2907	0.2611
8.0	0.2925	0.2726	0.2384	0.2105
10.0	0.2564	0.2371	0.2044	0.1781
20.0	0.1702	0.1538	0.1268	0.1059

TABLE 36. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Diff,\bar{D}_A\bar{M}_w}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_A$ AND $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Diff,\bar{D}_A\bar{M}_w} = \frac{[\Gamma(k+1)]^2(k+1)^{-a_D}}{[\Gamma(k-a_D/2+1)]^2} = \frac{[\Gamma(k+1)]^2(k+1)^{(a_{[\eta]}+1)/3}}{[\Gamma((6k+a_{[\eta]}+7)/6)]^2}$$

$$\bar{D}_A = K_D \bar{M}_w^{a_D} q^{(SZ)}_{Diff,\bar{D}_A\bar{M}_w}$$

	$a_{[\eta]}$ and (a_D)		
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.90 (−0.63)
1.1	1.0165	1.0173	1.0200
1.5	1.0631	1.0661	1.0760
2.0	1.0971	1.1017	1.1168
3.0	1.1326	1.1388	1.1592
5.0	1.1620	1.1695	1.1941
10.0	1.1846	1.1931	1.2209
20.0	1.1960	1.2051	1.2345

TABLE 37. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Diff,\bar{D}_A\bar{M}_w}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_A$ AND $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Diff,\bar{D}_A\bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{-(a_D^2+2a_D)/4} = (\bar{M}_w/\bar{M}_n)^{-(a^2_{[\eta]}-4a_{[\eta]}-5)/36}$$

$$\bar{D}_A = K_D \bar{M}_w^{a_D} q^{(LN)}_{Diff,\bar{D}_A\bar{M}_w}$$

	$a_{[\eta]}$ and (a_D)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.70 (−0.57)	0.90 (−0.63)
1.1	1.0172	1.0180	1.0195	1.0208
1.3	1.0481	1.0504	1.0547	1.0584
1.5	1.0752	1.0790	1.0858	1.0917
1.7	1.0996	1.1046	1.1138	1.1217
2.0	1.1320	1.1388	1.1511	1.1618
2.5	1.1781	1.1874	1.2045	1.2193
3.0	1.2172	1.2287	1.2499	1.2684
4.0	1.2815	1.2968	1.3251	1.3498
5.0	1.3336	1.3522	1.3865	1.4166
8.0	1.4506	1.4768	1.5254	1.5683
10.0	1.5097	1.5399	1.5961	1.6458
20.0	1.7090	1.7537	1.8373	1.9122

TABLE 38. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Diff,\bar{D}_w\bar{M}_w}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_w$ AND $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Diff,\bar{D}_w\bar{M}_w} = \frac{(k+1)^{-a_D}\Gamma(k+a_D+1)}{\Gamma(k+1)} = \frac{(k+1)^{(a_{[\eta]}+1)/3}\Gamma((3k-a_{[\eta]}+2)/3)}{\Gamma(k+1)} \qquad \bar{D}_w = K_D \bar{M}_w^{a_D} q^{(SZ)}_{Diff,\bar{D}_w\bar{M}_w}$$

	$a_{[\eta]}$ and (a_D)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.60 (−0.53)	0.70 (−0.57)	0.80 (−0.60)	0.90 (−0.63)
1.1	1.0326	1.0358	1.0391	1.0426	1.0462	1.0499
1.3	1.0891	1.0984	1.1081	1.1182	1.1288	1.1400
1.5	1.1365	1.1512	1.1668	1.1833	1.2007	1.2190
1.7	1.1767	1.1965	1.2175	1.2399	1.2637	1.2889
2.0	1.2268	1.2533	1.2817	1.3122	1.3448	1.3799
2.5	1.2909	1.3267	1.3655	1.4076	1.4533	1.5031
3.0	1.3386	1.3820	1.4293	1.4811	1.5379	1.6003
4.0	1.4050	1.4596	1.5200	1.5868	1.6612	1.7441
5.0	1.4490	1.5115	1.5812	1.6592	1.7467	1.8453
8.0	1.5216	1.5983	1.6849	1.7831	1.8953	2.0242
10.0	1.5479	1.6299	1.7230	1.8293	1.9514	2.0927
20.0	1.6037	1.6977	1.8056	1.9302	2.0754	2.2461

TABLE 39. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Diff,\bar{D}_w\bar{M}_w}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_w$ AND $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Diff,\bar{D}_w\bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{(a_D^2-a_D)/2} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}+5a_{[\eta]}+4)/18} \qquad \bar{D}_w = K_D \bar{M}_w^{a_D} q^{(LN)}_{Diff,\bar{D}_w\bar{M}_w}$$

	$a_{[\eta]}$ and (a_D)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.60 (−0.53)	0.70 (−0.57)	0.80 (−0.60)	0.90 (−0.63)
1.1	1.0332	1.0364	1.0397	1.0432	1.0468	1.0505
1.3	1.0939	1.1034	1.1132	1.1235	1.1342	1.1453
1.5	1.1488	1.1642	1.1803	1.1972	1.2149	1.2333
1.7	1.1991	1.2202	1.2423	1.2656	1.2901	1.3158
2.0	1.2677	1.2968	1.3277	1.3603	1.3947	1.4312
2.5	1.3683	1.4100	1.4545	1.5019	1.5524	1.6063
3.0	1.4564	1.5098	1.5671	1.6285	1.6944	1.7651

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TABLE 39. *cont'd*

	$a_{[\eta]}$ and (a_D)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.60 (−0.53)	0.70 (−0.57)	0.80 (−0.60)	0.90 (−0.63)
4.0	1.6071	1.6818	1.7627	1.8503	1.9453	2.0483
5.0	1.7346	1.8286	1.9311	2.0430	2.1652	2.2989
8.0	2.0373	2.1810	2.3403	2.5169	2.7132	2.9316
10.0	2.1990	2.3714	2.5638	2.7790	3.0200	3.2902
20.0	2.7877	3.0753	3.4039	3.7802	4.2121	4.7089

TABLE 40. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Diff,\bar{D}_z\bar{M}_w}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_z$ AND $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Diff,\bar{D}_z\bar{M}_w} = \frac{1}{(k+1)^{1+a_D}} \frac{\Gamma(k+a_D+2)}{\Gamma(k+1)} = \frac{1}{(k+1)^{(2-a_{[\eta]})/3}} \frac{\Gamma((3k-a_{[\eta]}+5)/3)}{\Gamma(k+1)}$$

$$\bar{D}_z = K_D \bar{M}_w^{a_D} q^{(SZ)}_{Diff,\bar{D}_z\bar{M}_w}$$

	$a_{[\eta]}$ and (a_D)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.60 (−0.53)	0.90 (−0.63)
1.1	0.9888	0.9887	0.9887	0.9895
1.5	0.9597	0.9594	0.9594	0.9616
2.0	0.9406	0.9400	0.9399	0.9429
3.0	0.9222	0.9213	0.9211	0.9246
4.0	0.9133	0.9123	0.9120	0.9157
5.0	0.9080	0.9069	0.9066	0.9104
6.0	0.9046	0.9034	0.9030	0.9068
8.0	0.9003	0.8990	0.8986	0.9025
10.0	0.8978	0.8964	0.8960	0.8999
20.0	0.8927	0.8913	0.8908	0.8947

TABLE 41. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Diff,\bar{D}_z\bar{M}_w}$ FOR $D = K_D M^{a_D}$ USING $\bar{D}_z$ AND $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Diff,\bar{D}_z\bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{(a^2_D+a_D)/2} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}-a_{[\eta]}-2)/18}$$

$$\bar{D}_z = K_D \bar{M}_w^{a_D} q^{(LN)}_{Diff,\bar{D}_z\bar{M}_w}$$

	$a_{[\eta]}$ and (a_D)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.47)	0.50 (−0.50)	0.70 (−0.57)	0.90 (−0.63)
1.1	0.9882	0.9882	0.9884	0.9890
1.5	0.9508	0.9506	0.9514	0.9540
2.0	0.9174	0.9170	0.9184	0.9227
3.0	0.8722	0.8717	0.8738	0.8802
4.0	0.8415	0.8409	0.8435	0.8513
5.0	0.8185	0.8178	0.8207	0.8295
6.0	0.8001	0.7993	0.8025	0.8122
8.0	0.7720	0.7711	0.7747	0.7855
10.0	0.7509	0.7499	0.7537	0.7654
20.0	0.6888	0.6877	0.6922	0.7062

F. MEAN-SQUARE RADIUS OF GYRATION VERSUS MOLECULAR WEIGHT RELATIONSHIP

TABLE 42. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Gyr,\langle S^2\rangle_z\bar{M}_w}$ FOR $\langle S^2\rangle = K_{\langle S^2\rangle} M^{1+\varepsilon}$ USING $\langle S^2\rangle_z$ AND $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT (57)

$$q^{(SZ)}_{Gyr,\langle S^2\rangle_z\bar{M}_w} = \frac{\Gamma(k+\varepsilon+3)}{(k+1)^{\varepsilon+2}\Gamma(k+1)} = \frac{\Gamma((3k+2a_{[\eta]}+8)/3)}{(k+1)^{(2a_{[\eta]}+5)/3}\Gamma(k+1)} \qquad \langle S^2\rangle_z = K_{\langle S^2\rangle} \bar{M}_w^{1+\varepsilon} q^{(SZ)}_{Gyr,\langle S^2\rangle_z\bar{M}_w}$$

	$a_{[\eta]}$ and (ε)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.70 (0.13)	0.80 (0.20)	0.90 (0.27)
1.1	1.0818	1.0909	1.1005	1.1105	1.1210	1.1320
1.5	1.2982	1.3333	1.3707	1.4105	1.4529	1.4979
2.0	1.4457	1.5000	1.5583	1.6210	1.6881	1.7602
3.0	1.5921	1.6667	1.7473	1.8345	1.9287	2.0305
4.0	1.6650	1.7500	1.8422	1.9423	2.0508	2.1685
5.0	1.7086	1.8000	1.8993	2.0073	2.1246	2.2521
6.0	1.7376	1.8333	1.9375	2.0508	2.1741	2.3083
8.0	1.7739	1.8750	1.9852	2.1053	2.2361	2.3788
10.0	1.7956	1.9000	2.0138	2.1380	2.2735	2.4213
20.0	1.8390	1.9500	2.0713	2.2037	2.6291	2.5069

TABLE 43. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Gyr,\langle S^2\rangle_z \bar{M}_w}$ FOR $\langle S^2\rangle = K_{\langle S^2\rangle} M^{1+\varepsilon}$ USING $\langle S^2\rangle_z$ AND $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Gyr,\langle S^2\rangle_z \bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{(\varepsilon^2+3\varepsilon+2)/2} = (\bar{M}_w/\bar{M}_n)^{(2a^2_{[\eta]}+7a_{[\eta]}+5)/9} \qquad \langle S^2\rangle_z = K_{\langle S^2\rangle} \bar{M}_w^{1+\varepsilon} q^{(LN)}_{Gyr,\langle S^2\rangle_z \bar{M}_w}$$

	$a_{[\eta]}$ and (ε)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.70 (0.13)	0.80 (0.20)	0.90 (0.27)
1.1	1.0898	1.1000	1.1108	1.1221	1.1341	1.1466
1.5	1.4417	1.5000	1.5635	1.6326	1.7078	1.7897
2.0	1.8689	2.0000	2.1469	2.3116	2.4967	2.7049
3.0	2.6944	3.0000	3.3566	3.7739	4.2638	4.8410
4.0	3.4929	4.0000	4.6090	5.3435	6.2333	7.3163
5.0	4.2720	5.0000	5.8941	6.9981	8.3684	10.08
6.0	5.0358	6.0000	7.2060	8.7237	10.65	13.09
8.0	6.5281	8.0000	9.8948	12.35	15.56	19.79
10.0	7.9840	10.00	12.65	16.18	20.89	27.26
20.0	14.92	20.00	27.17	37.39	52.16	73.74

TABLE 44. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Gyr,\langle S^2\rangle_z \bar{M}_z}$ FOR $\langle S^2\rangle = K_{\langle S^2\rangle} M^{1+\varepsilon}$ USING $\langle S^2\rangle_z$ AND $\bar{M}_z$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Gyr,\langle S^2\rangle_z \bar{M}_z} = \frac{\Gamma(k+\varepsilon+3)}{(k+1)(k+2)^{\varepsilon+1}\Gamma(k+1)} = \frac{\Gamma((3k+2a_{[\eta]}+8)/3)}{(k+1)(k+2)^{(2a_{[\eta]}+2)/3}\Gamma(k+1)} \qquad \langle S^2\rangle_z = K_{\langle S^2\rangle} \bar{M}_z^{1+\varepsilon} q^{(SZ)}_{Gyr,\langle S^2\rangle_z \bar{M}_z}$$

	$a_{[\eta]}$ and (ε)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.70 (0.13)	0.80 (0.20)	0.90 (0.27)
1.1	0.9974	1.0000	1.0029	1.0062	1.0099	1.0139
1.5	0.9925	1.0000	1.0085	1.0181	1.0287	1.0405
2.0	0.9902	1.0000	1.0112	1.0238	1.0378	1.0532
3.0	0.9883	1.0000	1.0133	1.0282	1.0448	1.0632
5.0	0.9872	1.0000	1.0146	1.0311	1.0494	1.0697
20.0	0.9860	1.0000	1.0159	1.0338	1.0538	1.0758

TABLE 45. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Gyr,\langle S^2\rangle_z \bar{M}_z}$ FOR $\langle S^2\rangle = K_{\langle S^2\rangle} M^{1+\varepsilon}$ USING $\langle S^2\rangle_z$ AND $\bar{M}_z$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Gyr,\langle S^2\rangle_z \bar{M}_z} = (\bar{M}_w/\bar{M}_n)^{(\varepsilon^2+\varepsilon)/2} = (\bar{M}_w/\bar{M}_n)^{(2a^2_{[\eta]}+a_{[\eta]}-1)/9} \qquad \langle S^2\rangle_z = K_{\langle S^2\rangle} \bar{M}_z^{1+\varepsilon} q^{(LN)}_{Gyr,\langle S^2\rangle_z \bar{M}_z}$$

	$a_{[\eta]}$ and (ε)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.70 (0.13)	0.80 (0.20)	0.90 (0.27)
1.1	0.9970	1.0000	1.0034	1.0072	1.0115	1.0162
1.3	0.9919	1.0000	1.0094	1.0200	1.0320	1.0453
1.5	0.9875	1.0000	1.0145	1.0311	1.0499	1.0709
1.7	0.9836	1.0000	1.0190	1.0409	1.0657	1.0938
2.0	0.9787	1.0000	1.0250	1.0538	1.0867	1.1242
2.5	0.9719	1.0000	1.0331	1.0717	1.1162	1.1674
3.0	0.9664	1.0000	1.0398	1.0865	1.1409	1.2039
4.0	0.9578	1.0000	1.0505	1.1104	1.1810	1.2638
5.0	0.9512	1.0000	1.0589	1.1293	1.2130	1.3123
6.0	0.9458	1.0000	1.0658	1.1450	1.2399	1.3534
7.0	0.9413	1.0000	1.0716	1.1584	1.2630	1.3891
10.0	0.9309	1.0000	1.0853	1.1900	1.3183	1.4753
20.0	0.9110	1.0000	1.1124	1.2540	1.4326	1.6586

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TABLE 46. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{Gyr,\langle S^2\rangle_z \bar{M}_n}$ FOR $\langle S^2\rangle = K_{\langle S^2\rangle} M^{1+\varepsilon}$ USING $\langle S^2\rangle_z$ AND $\bar{M}_n$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{Gyr,\langle S^2\rangle_z \bar{M}_n} = \frac{\Gamma(k+\varepsilon+3)}{k^{\varepsilon+1}(k+1)\Gamma(k+1)} = \frac{\Gamma((3k+2a_{[\eta]}+8)/3)}{k^{(2a_{[\eta]}+2)/3}(k+1)\Gamma(k+1)} \qquad \langle S^2\rangle_z = K_{\langle S^2\rangle}\bar{M}_n^{1+\varepsilon} q^{(SZ)}_{Gyr,\langle S^2\rangle_z \bar{M}_n}$$

	$a_{[\eta]}$ and (ε)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.70 (0.07)	0.90 (0.27)
1.1	1.1825	1.2000	1.2372	1.2773
1.3	1.5419	1.6000	1.7268	1.8693
1.5	1.8954	2.0000	2.2333	2.5034
1.7	2.2444	2.4000	2.7537	3.1732
2.0	2.7608	3.0000	3.5558	4.2352
2.5	3.6069	4.0000	4.9399	6.1333
3.0	4.4390	5.0000	6.3715	8.1651
4.0	6.0720	7.0000	9.3464	12.55
5.0	7.6738	9.0000	12.44	17.30
8.0	12.35	15.00	22.22	33.13
10.0	15.40	19.00	29.06	44.74
20.0	30.12	39.00	65.71	111.46

TABLE 47. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{Gyr,\langle S^2\rangle_z \bar{M}_n}$ FOR $\langle S^2\rangle = K_{\langle S^2\rangle} M^{1+\varepsilon}$ USING $\langle S^2\rangle_z$ AND $\bar{M}_n$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{Gyr,\langle S^2\rangle_z \bar{M}_n} = (\bar{M}_w/\bar{M}_n)^{(\varepsilon^2+5\varepsilon+4)/2} = (\bar{M}_w/\bar{M}_n)^{(2a^2_{[\eta]}+13a_{[\eta]}+11)/9} \qquad \langle S^2\rangle_z = K_{\langle S^2\rangle}\bar{M}_n^{1+\varepsilon} q^{(LN)}_{Gyr,\langle S^2\rangle_z \bar{M}_n}$$

	$a_{[\eta]}$ and (ε)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.70 (0.13)	0.80 (0.20)	0.90 (0.27)
1.1	1.1912	1.2100	1.2296	1.2501	1.2715	1.2938
1.3	1.6186	1.6900	1.7666	1.8488	1.9370	2.0319
1.5	2.1049	2.2500	2.4095	2.5849	2.7781	2.9911
1.7	2.6485	2.8900	3.1610	3.4655	3.8083	4.1949
2.0	3.5691	4.0000	4.4968	5.0708	5.7358	6.5080
2.5	5.3758	6.2500	7.2960	8.5519	10.06	11.89
3.0	7.5125	9.0000	10.83	13.11	15.93	19.47
4.0	12.74	16.00	20.22	25.71	32.90	42.35
5.0	19.19	25.00	32.81	43.37	57.73	77.41
8.0	45.46	64.00	90.93	130.39	188.71	275.64
10.0	68.48	100.00	147.53	219.90	331.13	503.76

G. SECOND VIRIAL COEFFICIENT VERSUS MOLECULAR WEIGHT RELATIONSHIP

TABLE 48. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{\bar{M}_n,\bar{A}_2^{(OP)}}$ FOR $\bar{A}_2^{(OP)} = K_{A_2} M^{a_{A_2}}$ USING $\bar{M}_n$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{\bar{M}_n,\bar{A}_2^{(OP)}} = (\bar{M}_{av,\bar{A}_2^{(OP)}}/\bar{M}_n)^{a_{A_2}} = (\bar{M}_{av,\bar{A}_2^{(OP)}}/\bar{M}_n)^{a_{[\eta]}-1}$$

	$a_{[\eta]}$ and (a_{A_2})		
$\bar{M}_w/\bar{M}_n$	0.65 (−0.35)	0.80 (−0.20)	0.95 (−0.05)
1.1	1.0492	1.0267	1.0063
1.2	1.0606	1.0318	1.0073

TABLE 48. *cont'd*

	$a_{[\eta]}$ and (a_{A_2})		
$\bar{M}_w/\bar{M}_n$	0.65 (−0.35)	0.80 (−0.20)	0.95 (−0.05)
1.3	1.0653	1.0333	1.0074
1.4	1.0666	1.0329	1.0070
1.5	1.0658	1.0314	1.0064
1.7	1.0603	1.0264	1.0048
2.0	1.0469	1.0164	1.0018
3.0	0.9886	0.9775	0.9907
5.0	0.8949	0.9167	0.9735

TABLE 49. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{\bar{M}_n,\bar{A}_2^{(OP)}}$ FOR $\bar{A}_2^{(OP)} = K_{A_2}M^{a_{A_2}}$ USING $\bar{M}_n$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{\bar{M}_n,\bar{A}_2^{(OP)}} = (\bar{M}_{av,\bar{A}_2^{(OP)}}/\bar{M}_n)^{a_{A_2}} = (\bar{M}_{av,\bar{A}_2^{(OP)}}/\bar{M}_n)^{a_{[\eta]}-1}$$

	$a_{[\eta]}$ and (a_{A_2})		
$\bar{M}_w/\bar{M}_n$	0.65 (−0.35)	0.80 (−0.20)	0.95 (−0.05)
1.1	1.0495	1.0270	1.0064
1.2	1.0619	1.0330	1.0077
1.3	1.0685	1.0358	1.0081
1.4	1.0723	1.0371	1.0083
1.5	1.0745	1.0375	1.0082
1.7	1.0761	1.0371	1.0078
2.0	1.0750	1.0347	1.0068
3.0	1.0630	1.0237	1.0030
5.0	1.0378	1.0042	0.9969
10.0	0.9924	0.9713	0.9867
20.0	0.9318	0.9280	0.9733

TABLE 50. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{\bar{M}_w,\bar{A}_2^{(LS)}}$ FOR $\bar{A}_2^{(LS)} = K_{A_2}M^{a_{A_2}}$ USING $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{\bar{M}_w,\bar{A}_2^{(LS)}} = (\bar{M}_{av,\bar{A}_2^{(LS)}}/\bar{M}_w)^{a_{A_2}} = (\bar{M}_{av,\bar{A}_2^{(LS)}}/\bar{M}_w)^{a_{[\eta]}-1}$$

	$a_{[\eta]}$ and (a_{A_2})		
$\bar{M}_w/\bar{M}_n$	0.65 (−0.35)	0.80 (−0.20)	0.95 (−0.05)
1.1	1.0477	1.0260	1.0063
1.2	1.0579	1.0308	1.0072
1.3	1.0626	1.0326	1.0075

TABLE 50. *cont'd*

	$a_{[\eta]}$ and (a_{A_2})		
$\bar{M}_w/\bar{M}_n$	0.65 (−0.35)	0.80 (−0.20)	0.95 (−0.05)
1.4	1.0649	1.0333	1.0075
1.5	1.0661	1.0334	1.0074
1.7	1.0667	1.0329	1.0071
2.0	1.0657	1.0314	1.0065
3.0	1.0600	1.0267	1.0050
5.0	1.0531	1.0217	0.0035

TABLE 51. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{\bar{M}_w,\bar{A}_2^{(LS)}}$ FOR $\bar{A}_2^{(LS)} = K_{A_2}M^{a_{A_2}}$ USING $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{\bar{M}_w,\bar{A}_2^{(LS)}} = (\bar{M}_{av,\bar{A}_2^{(LS)}}/\bar{M}_w)^{a_{A_2}} = (\bar{M}_{av,\bar{A}_2^{(LS)}}/\bar{M}_n)^{a_{[\eta]}-1}$$

	$a_{[\eta]}$ and (a_{A_2})		
$\bar{M}_w/\bar{M}_n$	0.65 (−0.35)	0.80 (−0.20)	0.95 (−0.05)
1.1	1.0496	1.0271	1.0065
1.2	1.0620	1.0331	1.0078
1.3	1.0686	1.0359	1.0082
1.4	1.0723	1.0371	1.0084
1.5	1.0742	1.0375	1.0083
1.7	1.0764	1.0374	1.0082
2.0	1.0753	1.0350	1.0071
3.0	1.0634	1.0241	1.0035
5.0	1.0391	1.0057	0.9985
10.0	0.9968	0.9752	0.9887
20.0	0.9604	0.9513	0.9816

H. DETERMINATION OF THE DIMENSIONS OF MACROMOLECULES

1. Fox–Flory Relationship

TABLE 52. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{FF,\langle r^2\rangle_z,\bar{M}_n}$ FOR FOX–FLORY RELATIONSHIP $[\eta] = \Phi(\varepsilon)\langle r^2\rangle^{3/2}/M$ USING $\langle r^2\rangle_z$ AND $\bar{M}_n$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{FF,\langle r^2\rangle_z,\bar{M}_n} = k(k+1)^{3/2}[\Gamma(k+1)]^{1/2}\frac{\Gamma((2k+3\varepsilon+3)/2)}{[\Gamma(k+\varepsilon+3)]^{3/2}} = k(k+1)^{3/2}[\Gamma(k+1)]^{1/2}\frac{\Gamma(k+a_{[\eta]}+1)}{[\Gamma((3k+2a_{[\eta]}+8)/3)]^{3/2}} \qquad [\eta] = \Phi(\varepsilon)\frac{\langle r^2\rangle_z^{3/2}}{\bar{M}_n}q^{(SZ)}_{FF,\langle r^2\rangle_z,\bar{M}_n}$$

	$a_{[\eta]}$ and (ε)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.70 (0.13)	0.80 (0.20)	0.90 (0.27)
1.1	0.7991	0.7888	0.7790	0.7695	0.7604	0.7518
1.3	0.5642	0.5474	0.5317	0.5171	0.5034	0.4905
1.5	0.4329	0.4154	0.3994	0.3846	0.3710	0.3585
2.0	0.2708	0.2558	0.2424	0.2304	0.2195	0.2097
3.0	0.1531	0.1427	0.1336	0.1256	0.1185	0.1121
4.0	0.1063	0.0985	0.0917	0.0858	0.0806	0.0760
5.0	0.0813	0.0751	0.0697	0.0650	0.0609	0.0573
10.0	0.0373	0.0342	0.0316	0.0293	0.0273	0.0256
20.0	0.0179	0.0164	0.0151	0.0139	0.0130	0.0121

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TABLE 53. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{FF,\langle r^2\rangle_z \bar{M}_n}$ FOR FOX–FLORY RELATIONSHIP $[\eta] = \Phi(\varepsilon)\langle r^2\rangle^{3/2}/M$ USING $\langle r^2\rangle_z$ AND $\bar{M}_n$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{FF,\langle r^2\rangle_z \bar{M}_n} = (\bar{M}_w/\bar{M}_n)^{(3\varepsilon^2-18\varepsilon-21)/8} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}-10a_{[\eta]}-11)/6} \qquad [\eta] = \Phi(\varepsilon)\frac{\langle r^2\rangle_z^{3/2}}{\bar{M}_n} q^{(LN)}_{FF,\langle r^2\rangle_z \bar{M}_n}$$

	$a_{[\eta]}$ and (ε)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.70 (0.13)	0.80 (0.20)	0.90 (0.27)
1.1	0.7900	0.7787	0.7677	0.7572	0.7470	0.7372
1.5	0.3668	0.3450	0.3248	0.3063	0.2892	0.2734
2.0	0.1801	0.1621	0.1463	0.1323	0.1199	0.1089
3.0	0.0661	0.0559	0.0475	0.0405	0.0347	0.0298
4.0	0.0324	0.0263	0.0214	0.0175	0.0144	0.0119
5.0	0.0187	0.0146	0.0115	0.0091	0.0073	0.0058
6.0	0.0119	0.0091	0.0069	0.0054	0.0042	0.0032
8.0	0.0058	0.0043	0.0031	0.0023	0.0017	0.0013
10.0	0.0034	0.0024	0.0017	0.0012	0.0009	0.0006

TABLE 54. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{FF,\langle r^2\rangle_z \bar{M}_w}$ FOR FOX–FLORY RELATIONSHIP $[\eta] = \Phi(\varepsilon)\langle r^2\rangle^{3/2}/M$ USING $\langle r^2\rangle_z$ AND $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{FF,\langle r^2\rangle_z \bar{M}_w} = (k+1)^{5/2}[\Gamma(k+1)]^{1/2}\frac{\Gamma((2k+3\varepsilon+3)/2)}{[\Gamma(k+\varepsilon+3)]^{3/2}}$$

$$= (k+1)^{5/2}[\Gamma(k+1)]^{1/2}\frac{\Gamma(k+a_{[\eta]}+1)}{[\Gamma((3k+2a_{[\eta]}+8)/3)]^{3/2}}$$

$$[\eta] = \Phi(\varepsilon)\frac{\langle r^2\rangle_z^{3/2}}{\bar{M}_w} q^{(SZ)}_{FF,\langle r^2\rangle_z \bar{M}_w}$$

	$a_{[\eta]}$ and (ε)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.70 (0.13)	0.90 (0.27)
1.1	0.8791	0.8677	0.8465	0.8269
1.5	0.6494	0.6231	0.5769	0.5377
2.0	0.5416	0.5117	0.4607	0.4193
3.0	0.4593	0.4282	0.3767	0.3363
4.0	0.4253	0.3941	0.3432	0.3038
5.0	0.4067	0.3755	0.3252	0.2865
20.0	0.3578	0.3273	0.2789	0.2427

TABLE 55. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{FF,\langle r^2\rangle_z \bar{M}_w}$ FOR FOX–FLORY RELATIONSHIP $[\eta] = \Phi(\varepsilon)\langle r^2\rangle^{3/2}/M$ USING $\langle r^2\rangle_z$ AND $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{FF,\langle r^2\rangle_z \bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{(3\varepsilon^2-18\varepsilon-13)/8} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}-10a_{[\eta]}-5)/6}$$

$$[\eta] = \Phi(\varepsilon)\frac{\langle r^2\rangle_z^{3/2}}{\bar{M}_w} q^{(LN)}_{FF,\langle r^2\rangle_z \bar{M}_w}$$

	$a_{[\eta]}$ and (ε)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.70 (0.13)	0.90 (0.27)
1.1	0.8690	0.8565	0.8329	0.8110
1.5	0.5502	0.5174	0.4594	0.4101
2.0	0.3601	0.3242	0.2646	0.2179
3.0	0.1982	0.1678	0.1215	0.0894
4.0	0.1297	0.1051	0.0700	0.0475
5.0	0.0934	0.0731	0.0456	0.0291
10.0	0.0336	0.0237	0.0121	0.0063
20.0	0.0121	0.0077	0.0032	0.0014

2. Interconversion of Different Averages of the Mean-Square Radius of Gyration

TABLE 56. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{\langle S^2\rangle_w}$ FOR THE CALCULATION OF $\langle S^2\rangle_w$ FROM $\langle S^2\rangle_z$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{\langle S^2\rangle_w} = \frac{k+1}{(3k+2a_{[\eta]}+5)/3} \qquad \langle S^2\rangle_w = q^{(SZ)}_{\langle S^2\rangle_w}\langle S^2\rangle_z$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	0.9218	0.9167	0.9116	0.9066	0.9016	0.8967
1.2	0.8654	0.8571	0.8491	0.8411	0.8333	0.8257
1.3	0.8228	0.8125	0.8025	0.7927	0.7831	0.7738
1.4	0.7895	0.7778	0.7664	0.7554	0.7447	0.7343
1.5	0.7627	0.7500	0.7377	0.7258	0.7143	0.7031
1.7	0.7224	0.7083	0.6948	0.6818	0.6693	0.6572

TABLE 56. *cont'd*

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	**0.40**	**0.50**	**0.60**	**0.70**	**0.80**	**0.90**
1.9	0.6934	0.6786	0.6643	0.6507	0.6376	0.6250
2.0	0.6818	0.6667	0.6522	0.6383	0.6250	0.6122
2.5	0.6410	0.6250	0.6098	0.5952	0.5814	0.5682
3.0	0.6164	0.6000	0.5844	0.5696	0.5556	0.5422
4.0	0.5882	0.5714	0.5556	0.5405	0.5263	0.5128
6.0	0.5625	0.5455	0.5294	0.5143	0.5000	0.4865
10.0	0.5435	0.5263	0.5102	0.4950	0.4808	0.4673
20.0	0.5300	0.5128	0.4967	0.4815	0.4673	0.4539

TABLE 57. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{\langle S^2\rangle_w}$ FOR THE CALCULATION OF $\langle S^2\rangle_w$ FROM $\langle S^2\rangle_z$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{\langle S^2\rangle_w} = (\bar{M}_w/\bar{M}_n)^{-(2a_{[\eta]}+2)/3} \qquad \langle S^2\rangle_w = q^{(LN)}_{\langle S^2\rangle_w}\langle S^2\rangle_z$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	**0.40**	**0.50**	**0.60**	**0.70**	**0.80**	**0.90**
1.1	0.9149	0.9091	0.9033	0.8976	0.8919	0.8863
1.2	0.8435	0.8333	0.8233	0.8133	0.8035	0.7938
1.3	0.7828	0.7692	0.7559	0.7428	0.7299	0.7173
1.4	0.7305	0.7143	0.6984	0.6829	0.6678	0.6530
1.5	0.6849	0.6667	0.6489	0.6316	0.6147	0.5983
1.7	0.6094	0.5882	0.5678	0.5481	0.5290	0.5106
1.9	0.5493	0.5263	0.5043	0.4831	0.4629	0.4435
2.0	0.5236	0.5000	0.4774	0.4559	0.4353	0.4156
2.5	0.4252	0.4000	0.3763	0.3540	0.3330	0.3133
3.0	0.3587	0.3333	0.3098	0.2879	0.2676	0.2487
4.0	0.2742	0.2500	0.2279	0.2078	0.1895	0.1727
5.0	0.2227	0.2000	0.1797	0.1614	0.1450	0.1302
6.0	0.1878	0.1667	0.1479	0.1312	0.1165	0.1034
8.0	0.1436	0.1250	0.1088	0.0947	0.0825	0.0718
10.0	0.1166	0.1000	0.0858	0.0736	0.0631	0.0541
20.0	0.0611	0.0500	0.0409	0.0335	0.0275	0.0225

TABLE 58. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{\langle S^2\rangle_n}$ FOR THE CALCULATION OF $\langle S^2\rangle_n$ FROM $\langle S^2\rangle_z$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{\langle S^2\rangle_n} = \frac{k(k+1)}{[(3k+2a_{[\eta]}+2)/3](3k+2a_{[\eta]}+5)/3)} \qquad \langle S^2\rangle_n = q^{(SZ)}_{\langle S^2\rangle_n}\langle S^2\rangle_z$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	**0.40**	**0.50**	**0.60**	**0.70**	**0.80**	**0.90**
1.1	0.8431	0.8333	0.8237	0.8143	0.8050	0.7959
1.2	0.7293	0.7143	0.6998	0.6857	0.6720	0.6588
1.3	0.6428	0.6250	0.6079	0.5916	0.5758	0.5607
1.4	0.5749	0.5556	0.5372	0.5198	0.5032	0.4873
1.5	0.5200	0.5000	0.4811	0.4633	0.4464	0.4305
1.7	0.4369	0.4167	0.3978	0.3802	0.3637	0.3483
1.9	0.3769	0.3571	0.3389	0.3221	0.3065	0.2921
2.0	0.3527	0.3333	0.3156	0.2992	0.2841	0.2701
2.5	0.2671	0.2500	0.2345	0.2205	0.2076	0.1959
3.0	0.2150	0.2000	0.1865	0.1744	0.1634	0.1534
4.0	0.1548	0.1429	0.1323	0.1229	0.1144	0.1068
5.0	0.1210	0.1111	0.1024	0.0948	0.0880	0.0819
6.0	0.0993	0.0909	0.0836	0.0771	0.0714	0.0663
10.0	0.0578	0.0526	0.0481	0.0442	0.0407	0.0377
20.0	0.0283	0.0256	0.0234	0.0214	0.0196	0.0181

TABLE 59. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{\langle S^2\rangle_n}$ FOR THE CALCULATION OF $\langle S^2\rangle_n$ FROM $\langle S^2\rangle_z$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{\langle S^2\rangle_n} = (\bar{M}_w/\bar{M}_n)^{-(4a_{[\eta]}+4)/3} \qquad \langle S^2\rangle_n = q^{(LN)}_{\langle S^2\rangle_n}\langle S^2\rangle_z$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	0.8370	0.8264	0.8160	0.8057	0.7955	0.7855
1.2	0.7115	0.6944	0.6778	0.6615	0.6456	0.6301
1.3	0.6128	0.5917	0.5714	0.5517	0.5328	0.5145
1.4	0.5336	0.5102	0.4878	0.4664	0.4460	0.4264
1.5	0.4691	0.4444	0.4211	0.3989	0.3779	0.3580
1.7	0.3714	0.3460	0.3224	0.3004	0.2798	0.2607
1.9	0.3018	0.2770	0.2543	0.2334	0.2143	0.1967
2.0	0.2742	0.2500	0.2279	0.2078	0.1895	0.1727
2.5	0.1808	0.1600	0.1416	0.1253	0.1109	0.0981
3.0	0.1286	0.1111	0.0960	0.0829	0.0716	0.0618
4.0	0.0752	0.0625	0.0520	0.0432	0.0359	0.0298
5.0	0.0496	0.0400	0.0323	0.0260	0.0210	0.0170
6.0	0.0353	0.0278	0.0219	0.0172	0.0136	0.0107
10.0	0.136	0.0100	0.0074	0.0054	0.0040	0.0029

3. Interconversion of Different Averages of the Mean-Square End-to-End Distance

TABLE 60. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{\langle r^2\rangle_n}$ FOR THE CALCULATION OF $\langle r^2\rangle_n$ FROM $[\overline{\langle r^2\rangle^{3/2}}]_n \equiv \sum_i n_i \langle r_i^2\rangle^{3/2}$ AS OBTAINED FROM FOX–FLORY RELATIONSHIP WITH $\bar{M}_n$ ACCORDING TO EQ. (B40) FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{\langle r^2\rangle_n} = \left(\frac{k}{\Gamma(k+1)}\right)^{1/3} \frac{\Gamma((3k+2a_{[\eta]}+2)/3)}{[\Gamma(k+1+a_{[\eta]})]^{2/3}}$$

$$\langle r^2\rangle_n = q^{(SZ)}_{\langle r^2\rangle_n}[\overline{\langle r^2\rangle^{3/2}}]_n^{2/3}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	0.9790	0.9761	0.9697	0.9627
1.5	0.9087	0.8981	0.8760	0.8531
2.0	0.8426	0.8271	0.7962	0.7654
3.0	0.7513	0.7323	0.6954	0.6600
4.0	0.6898	0.6697	0.6315	0.5957
5.0	0.6443	0.6242	0.5861	0.5508
6.0	0.6089	0.5889	0.5514	0.5170
8.0	0.5561	0.5368	0.5009	0.4681
10.0	0.5178	0.4993	0.4649	0.4337
20.0	0.4136	0.3978	0.3688	0.3429

TABLE 61. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{\langle r^2\rangle_n}$ FOR THE CALCULATION OF $\langle r^2\rangle_n$ FROM $[\overline{\langle r^2\rangle^{3/2}}]_n \equiv \sum_i n_i \langle r_i^2\rangle^{3/2}$ AS OBTAINED FROM FOX–FLORY RELATIONSHIP WITH $\bar{M}_n$ ACCORDING TO EQ. (B40) FOR POLYMERS WITH A LOGARITHMIC NORMAL DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{\langle r^2\rangle_n} = (\bar{M}_w/\bar{M}_n)^{-(a^2_{[\eta]}+2a_{[\eta]}+1)/9} \qquad \langle r^2\rangle_n = q^{(LN)}_{\langle r^2\rangle_n}[\overline{\langle r^2\rangle^{3/2}}]_n^{2/3}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	0.9795	0.9765	0.9699	0.9625
1.5	0.9155	0.9036	0.8779	0.8499
2.0	0.8599	0.8409	0.8005	0.7573
3.0	0.7872	0.7598	0.7027	0.6436
4.0	0.7394	0.7071	0.6407	0.5735
5.0	0.7043	0.6687	0.5964	0.5244
6.0	0.6769	0.6389	0.5625	0.4874
8.0	0.6358	0.5946	0.5129	0.4343
10.0	0.6057	0.5623	0.4774	0.3971
20.0	0.5208	0.4729	0.3821	0.3007

TABLE 62. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{\langle r^2\rangle_w}$ FOR THE CALCULATION OF $\langle r^2\rangle_w$ FROM $[\overline{\langle r^2\rangle^{3/2}}]_n \equiv \sum_i n_i \langle r_i^2\rangle^{3/2}$ AS OBTAINED FROM FOX–FLORY RELATIONSHIP WITH $\bar{M}_n$ ACCORDING TO EQ. (B40) FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{\langle r^2\rangle_w} = \frac{\Gamma((3k + 2a_{[\eta]} + 5)/3)}{k^{2/3}[\Gamma(k+1)]^{1/3}[\Gamma(k + a_{[\eta]} + 1)]^{2/3}}$$

$$\langle r^2\rangle_w = q^{(SZ)}_{\langle r^2\rangle_w}[\overline{\langle r^2\rangle^{3/2}}]_n^{2/3}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	1.0704	1.0737	1.0796	1.0846
1.3	1.2051	1.2142	1.2305	1.2441
1.5	1.3328	1.3471	1.3724	1.3933
1.7	1.4548	1.4738	1.5071	1.5346
2.0	1.6290	1.6543	1.6985	1.7348
2.5	1.9005	1.9351	1.9954	2.0445
3.0	2.1538	2.1969	2.2715	2.3320
4.0	2.6211	2.6789	2.7788	2.8594
5.0	3.0498	3.1208	3.2432	3.3416
8.0	4.1891	4.2941	4.4744	4.6189
10.0	4.8677	4.9925	5.2066	5.3779
20.0	7.7480	7.9556	8.3107	8.5941

TABLE 63. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{\langle r^2\rangle_w}$ FOR THE CALCULATION OF $\langle r^2\rangle_w$ FROM $[\overline{\langle r^2\rangle^{3/2}}]_n \equiv \sum_i n_i \langle r_i^2\rangle^{3/2}$ AS OBTAINED FROM FOX–FLORY RELATIONSHIP WITH $\bar{M}_n$ ACCORDING TO EQ. (B40) FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{\langle r^2\rangle_w} = (\bar{M}_w/\bar{M}_n)^{-(a^2_{[\eta]} - 4a_{[\eta]} - 5)/9} \qquad \langle r^2\rangle_w = q^{(LN)}_{\langle r^2\rangle_w}[\overline{\langle r^2\rangle^{3/2}}]_n^{2/3}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	1.0706	1.0741	1.0805	1.0860
1.3	1.2065	1.2175	1.2375	1.2549
1.5	1.3366	1.3554	1.3900	1.4204
1.7	1.4618	1.4888	1.5388	1.5829
2.0	1.6421	1.6818	1.7559	1.8220
2.5	1.9264	1.9882	2.1048	2.2102
3.0	2.1949	2.2795	2.4408	2.5881
4.0	2.6965	2.8284	3.0832	3.3198
5.0	3.1634	3.3437	3.6959	4.0271
6.0	3.6042	3.8337	4.2858	4.7156
7.0	4.0245	4.3035	4.8574	5.3886
8.0	4.4280	4.7568	5.4139	6.0489
9.0	4.8174	5.1962	5.9574	6.6981
10.0	5.1946	5.6234	6.4897	7.3376
15.0	6.9432	7.6220	9.0208	10.423
20.0	8.5302	9.4574	11.395	13.369

TABLE 64. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{\langle r^2\rangle_z}$ FOR THE CALCULATION OF $\langle r^2\rangle_z$ FROM $[\overline{\langle r^2\rangle^{3/2}}]_n \equiv \sum_i n_i \langle r_i^2\rangle^{3/2}$ AS OBTAINED FROM FOX–FLORY RELATIONSHIP WITH $\bar{M}_n$ ACCORDING TO EQ. (B40) FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ}_{\langle r^2\rangle_z}) = \frac{\Gamma((3k + 2a_{[\eta]} + 8)/3)}{k^{2/3}(k+1)[\Gamma(k+1)]^{1/3}[\Gamma(k + a_{[\eta]} + 1)]^{2/3}}$$

$$\langle r^2\rangle_z = q^{(SZ)}_{\langle r^2\rangle_z}[\overline{\langle r^2\rangle^{3/2}}]_n^{2/3}$$

	$a_{[\eta]}$			
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.70	0.90
1.1	1.1612	1.1713	1.1908	1.2095
1.3	1.4646	1.4944	1.5523	1.6078
1.5	1.7475	1.7961	1.8908	1.9816
1.7	2.0139	2.0806	2.2104	2.3350
2.0	2.3891	2.4814	2.6610	2.8335
2.5	2.9647	3.0962	3.3523	3.5984
3.0	3.4940	3.6615	3.9877	4.3012
4.0	4.4558	4.6881	5.1408	5.5758
5.0	5.3269	5.6175	6.1837	6.7278
8.0	7.6102	8.0515	8.9116	9.7381
10.0	8.9565	9.4858	10.52	11.51
20.0	14.62	15.51	17.26	18.94

TABLE 65. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{\langle r^2\rangle_z}$ FOR THE CALCULATION OF $\langle r^2\rangle_z$ FROM $[\overline{\langle r^2\rangle^{3/2}}]_n \equiv \sum_i n_i \langle r_i^2\rangle^{3/2}$ AS OBTAINED FROM FOX–FLORY RELATIONSHIP WITH $\bar{M}_n$ ACCORDING TO EQ. (B40) FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(LN)}_{\langle r^2\rangle_z} = (\bar{M}_w/\bar{M}_n)^{-(a^2_{[\eta]} - 10a_{[\eta]} - 11)/9} \qquad \langle r^2\rangle_z = q^{(LN)}_{\langle r^2\rangle_z}[\overline{\langle r^2\rangle^{3/2}}]_n^{2/3}$$

	$a_{[\eta]}$					
$\bar{M}_w/\bar{M}_n$	0.40	0.50	0.60	0.70	0.80	0.90
1.1	1.1702	1.1815	1.1927	1.2037	1.2146	1.2253
1.3	1.5413	1.5827	1.6243	1.6660	1.7078	1.7497
1.5	1.9514	2.0331	2.1163	2.2009	2.2868	2.3739
1.7	2.3987	2.5310	2.6673	2.8077	2.9520	3.1000
2.0	3.1359	3.3636	3.6022	3.8519	4.1125	4.3839
2.5	4.5306	4.9704	5.4419	5.9459	6.4833	7.0550
3.0	6.1196	6.8385	7.6233	8.4775	9.4043	10.41
4.0	9.8340	11.31	12.98	14.84	16.91	19.22
5.0	14.21	16.72	19.60	22.90	26.66	30.93
6.0	19.19	23.00	27.46	32.65	38.67	45.62
7.0	24.74	30.12	36.52	44.07	52.97	63.38
8.0	30.84	38.05	46.74	57.15	69.55	84.25
10.0	44.55	56.23	70.61	88.22	109.65	135.59
15.0	86.94	114.33	149.44	194.16	250.74	321.88

4. The Effect of Polymolecularity on the Radius of Gyration of Stiff Polymer Chains

TABLE 66. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}(\bar{L}_w)$ FOR STIFF POLYMER CHAINS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF CHAIN LENGTHS L

	$q^{(SZ)}(\bar{L}_w)$ for $\bar{L}_w$						
$\bar{L}_w/\bar{L}_n$†	1.5	2	4	10	20	40	100
1.1	0.8986	0.8720	0.8901	0.9523	0.9759	0.9879	0.9952
1.2	0.7714	0.8335	0.8902	0.9537	0.9768	0.9883	0.9953
1.5	0.8098	0.8669	0.9100	0.9626	0.9812	0.9905	0.9962
2	0.7592	0.8227	0.8838	0.9507	0.9751	0.9875	0.9950
3	0.7720	0.7899	0.8637	0.9414	0.9703	0.9851	0.9940
5	0.6985	0.7692	0.8509	0.9355	0.9671	0.9835	0.9933
10	0.6836	0.7558	0.8425	0.9316	0.9651	0.9824	0.9928

† $\bar{L}_w/\bar{L}_n = \bar{M}_w/\bar{M}_n$.

TABLE 67. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}(\bar{L}_w)$ FOR STIFF POLYMER CHAINS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF CHAIN LENGTHS L

	$q^{(LN)}(\bar{L}_w)$ for $\bar{L}_w$						
$\bar{L}_w/\bar{L}_n$†	1.5	2	4	10	20	40	100
1.1	0.919	0.9636	0.9651	0.9861	0.9932	0.9966	0.9986
1.2	0.864	0.913	0.9377	0.9746	0.9875	0.9937	0.9975
1.5	0.749	0.817	0.8822	0.9505	0.9751	0.9875	0.9950
2	0.663	0.737	0.8325	0.9276	0.9631	0.9814	0.9924
3	0.57	0.665	0.7876	0.9056	0.9513	0.9753	0.9900
5	0.52	0.612	0.7537	0.8887	0.9422	0.9705	0.9881
10	0.51	0.589	0.7303	0.8764	0.9354	0.9670	0.9866

† $\bar{L}_w/\bar{L}_n = \bar{M}_w/\bar{M}_n$.

I. DETERMINATION OF UNPERTURBED DIMENSIONS OF MACROMOLECULES

1. Burchard–Stockmayer–Fixman Procedure

TABLE 68. POLYMOLECULARITY CORRECTION FACTORS $q_{BSF,\bar{M}_w}$ FOR BURCHARD–STOCKMAYER–FIXMAN PROCEDURE (EQ. B55) USING $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT AND FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$$q^{(SZ)}_{BSF,\bar{M}_w} = \frac{1}{(k+1)^{1/2}} \frac{\Gamma(k+1.5)}{\Gamma(k+1)} \qquad [\eta]/(\bar{M}_w^{1/2} q^{(SZ)}_{BSF,\bar{M}_w}) = K_0 + B'\bar{M}_w^{1/2}/q^{(SZ)}_{BSF,\bar{M}_w}$$

$$q^{(LN)}_{BSF,\bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{-1/8} \qquad [\eta]/(\bar{M}_w^{1/2} q^{(LN)}_{BSF,\bar{M}_w}) = K_0 + B'\bar{M}_w^{1/2}/q^{(LN)}_{BSF,\bar{M}_w}$$

$\bar{M}_w/\bar{M}_n$	1.1	1.5	2.0	3.0	5.0	10.0	20.0
$q^{(SZ)}_{BSF,\bar{M}_w}$	0.9887	0.9594	0.9400	0.9213	0.9069	0.8964	0.8913
$q^{(LN)}_{BSF,\bar{M}_w}$	0.9882	0.9506	0.9170	0.8717	0.8178	0.7499	0.6877

TABLE 69. POLYMOLECULARITY CORRECTION FACTORS ${}^1q^{(SZ)}_{BSF,\bar{M}_n}$ AND ${}^2q^{(SZ)}_{BSF,\bar{M}_n}$ FOR BURCHARD–STOCKMAYER–FIXMAN PROCEDURE (EQ. B56) USING $\bar{M}_n$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT

$${}^1q^{(SZ)}_{BSF,\bar{M}_n} = \frac{\Gamma(k+1.5)}{k^{0.5}\Gamma(k+1)} \qquad {}^2q^{(SZ)}_{BSF,\bar{M}_n} = \frac{k^{0.5}\Gamma(k+1.5)}{(k+1)\Gamma(k+1)} \qquad [\eta]/({}^1q^{(SZ)}_{BSF,\bar{M}_n}\bar{M}_n^{1/2}) = K_0 + B'\bar{M}_n^{1/2}/{}^2q^{(SZ)}_{BSF,\bar{M}_n}$$

$\bar{M}_w/\bar{M}_n$	1.1	1.3	1.5	1.7	2.0	2.5	3.0	4.0	5.0	6.0	8.0	10.0	15.0	20.0
${}^1q^{(SZ)}_{BSF,\bar{M}_n}$	1.0370	1.1078	1.1750	1.2389	1.3293	1.4684	1.5958	1.8245	2.0279	2.2129	2.5429	2.8348	3.4586	3.9860
${}^2q^{(SZ)}_{BSF,\bar{M}_n}$	0.9427	0.8522	0.7833	0.7287	0.6647	0.5874	0.5319	0.4561	0.4056	0.3688	0.3179	0.2835	0.2306	0.1993

TABLE 70. POLYMOLECULARITY CORRECTION FACTORS ${}^1q^{(LN)}_{BSF,\bar{M}_n}$ AND ${}^2q^{(LN)}_{BSF,\bar{M}_n}$ FOR BURCHARD-STOCKMAYER-FIXMAN PROCEDURE (EQ. B56) USING $\bar{M}_n$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT

$${}^1q^{(LN)}_{BSF,\bar{M}_n} = (\bar{M}_w/\bar{M}_n)^{3/8} \qquad {}^2q^{(LN)}_{BSF,\bar{M}_n} = (\bar{M}_w/\bar{M}_n)^{-5/8} \qquad [\eta]/({}^1q^{(LN)}_{BSF,\bar{M}_n}\bar{M}_n^{1/2}) = K_0 + B'\bar{M}_n^{1/2}/{}^2q^{(LN)}_{BSF,\bar{M}_n}$$

$\bar{M}_w/\bar{M}_n$	1.1	1.3	1.5	1.7	2.0	2.5	3.0	4.0	5.0	6.0	8.0	10.0	15.0	20.0
${}^1q^{(LN)}_{BSF,\bar{M}_n}$	1.0364	1.1034	1.1642	1.2202	1.2968	1.4100	1.5098	1.6818	1.8286	1.9580	2.1810	2.3714	2.7608	3.0753
${}^2q^{(LN)}_{BSF,\bar{M}_n}$	0.9422	0.8488	0.7761	0.7177	0.6484	0.5640	0.5033	0.4204	0.3657	0.3263	0.2726	0.2371	0.1841	0.1538

2. Cowie-Bywater Procedure

TABLE 71. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{CB,1}$ FOR THE COWIE-BYWATER PROCEDURE (EQ. B60) USING $\bar{M}_w$ AND $\bar{s}_w$ FOR POLYMERS WITH A SCHULZ-ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 72)

$$q^{(SZ)}_{CB,1} = \frac{\Gamma(k-0.5\varepsilon+2)}{(k+1)^{0.5}\Gamma(k-0.5\varepsilon+1.5)}$$

$$(\bar{M}_w^{1/2}/\bar{s}_w)(1-\tilde{\nu}_2\rho_1)/(\eta_1 N_L) = P_0 q^{(SZ)}_{CB,1} + B''\bar{M}_w^{1/2}q^{(SZ)}_{CB,2}$$

	$a_{[\eta]}$ and (ε)		
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.90 (0.27)
1.1	1.0129	1.0114	1.0054
1.5	1.0477	1.0424	1.0209
2.0	1.0716	1.0638	1.0322
3.0	1.0955	1.0854	1.0441
5.0	1.1145	1.1026	1.0538
10.0	1.1287	1.1155	1.0612
20.0	1.1358	1.1220	1.0650

TABLE 72. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{CB,2}$ FOR THE COWIE-BYWATER PROCEDURE (EQ. B60) USING $\bar{M}_w$ AND $\bar{s}_w$ FOR POLYMERS WITH A SCHULZ-ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 71)

$$q^{(SZ)}_{CB,2} = \frac{k-0.5\varepsilon+1.5}{k+1}$$

$$(\bar{M}_w^{1/2}/\bar{s}_w)(1-\tilde{\nu}_2\rho_1)/(\eta_1 N_L) = P_0 q^{(SZ)}_{CB,1} + B''\bar{M}_w^{1/2}q^{(SZ)}_{CB,2}$$

	$a_{[\eta]}$ and (ε)		
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.90 (0.27)
1.1	1.0485	1.0455	1.0333
1.5	1.1778	1.1667	1.1222
2.0	1.2667	1.2500	1.1833
3.0	1.3556	1.3333	1.2444
5.0	1.4267	1.4000	1.2933
10.0	1.4800	1.4500	1.3300
20.0	1.5067	1.4750	1.3483

TABLE 73. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{CB,1}$ FOR THE COWIE-BYWATER PROCEDURE (EQ. B60) USING $\bar{M}_w$ AND $\bar{s}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 74)

$$q^{(LN)}_{CB,1} = (\bar{M}_w/\bar{M}_n)^{(1-2\varepsilon)/8}$$

$$(\bar{M}_w^{1/2}/\bar{s}_w)(1-\tilde{\nu}_2\rho_1)/(\eta_1 N_L) = P_0 q^{(LN)}_{CB,1} + B''\bar{M}_w^{1/2}q^{(LN)}_{CB,2}$$

	$a_{[\eta]}$ and (ε)		
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.90 (0.27)
1.1	1.0136	1.0120	1.0056
1.5	1.0591	1.0520	1.0239
2.0	1.1032	1.0905	1.0413
3.0	1.1684	1.1472	1.0662
5.0	1.2561	1.2228	1.0984
10.0	1.3857	1.3335	1.1438
15.0	1.4676	1.4029	1.1711
20.0	1.5287	1.4542	1.1909

TABLE 74. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{CB,2}$ FOR THE COWIE-BYWATER PROCEDURE (EQ. B60) USING $\bar{M}_w$ AND $\bar{s}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 73)

$$q^{(LN)}_{CB,2} = (\bar{M}_w/\bar{M}_n)^{(1-\epsilon)/2}$$

$$(\bar{M}_w^{1/2}/\bar{s}_w)(1-\tilde{\nu}_2\rho_1)/(\eta_1 N_L) = P_0 q^{(LN)}_{CB,1} + B''\bar{M}_w^{1/2}q^{(LN)}_{CB,2}$$

	$a_{[\eta]}$ and (ε)		
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.90 (0.27)
1.1	1.0521	1.0488	1.0356
1.5	1.2414	1.2247	1.1603
2.0	1.4473	1.4142	1.2894
3.0	1.7967	1.7321	1.4960
5.0	2.3593	2.2361	1.8042
10.0	3.4145	3.1623	2.3263
12.0	3.7633	3.4641	2.4871
16.0	4.3873	4.0000	2.7638
20.0	4.9418	4.4721	2.9995

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3. Baumann–Stockmayer–Fixman Procedure

TABLE 75. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{B,\bar{M}_w}$ FOR THE BAUMANN–STOCKMAYER–FIXMAN PROCEDURE (EQ. B63) USING $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 76)

$$q^{(SZ)}_{B,\bar{M}_w} = (k+1)(k+2)^2(k+3)^2\frac{[\Gamma(k+1)]^2}{[\Gamma(k+3.5)]^2}$$

$$[\langle S^2\rangle_{av} q^{(SZ)}_{B,\langle S^2\rangle_{av}}/(\bar{M}_w q^{(SZ)}_{B,\bar{M}_w})]^{3/2} = K_0^{3/2} + B_{BSF}(\bar{M}_w q^{(SZ)}_{B,\bar{M}_w})^{1/2}$$

$\bar{M}_w/\bar{M}_n$	$q^{(SZ)}_{B,\bar{M}_w}$
1.1	1.2048
1.3	1.5203
1.5	1.7520
1.7	1.9292
2.0	2.1286
2.5	2.3547
3.0	2.5055
5.0	2.8071
10.0	3.0333
20.0	3.1464

TABLE 76. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{B,\langle S^2\rangle_z}$ FOR THE BAUMANN–STOCKMAYER–FIXMAN PROCEDURE (EQ. B63) USING $\langle S^2\rangle_z$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 75)

$$q^{(SZ)}_{B,\langle S^2\rangle_z} = (k+1)^3(k+2)^2(k+3)^2 \times \frac{[\Gamma(k+1)]^3}{[\Gamma(k+3.5)]^{8/3}}\frac{[\Gamma(k+1.5\varepsilon+3.5)]^{2/3}}{\Gamma(k+\varepsilon+3)}$$

$$[\langle S^2\rangle_z q^{(SZ)}_{B,\langle S^2\rangle_z}/(\bar{M}_{av} q^{(SZ)}_{B,\bar{M}_{av}})]^{3/2} = K_0^{3/2} + B_{BSF}(\bar{M}_{av} q^{(SZ)}_{B,\bar{M}_{av}})^{1/2}$$

	$a_{[\eta]}$ and (ε)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.70 (0.13)	0.90 (0.27)
1.1	1.1016	1.1044	1.1105	1.1173
1.3	1.2287	1.2353	1.2496	1.2654
1.5	1.3050	1.3140	1.3334	1.3549
1.7	1.3560	1.3665	1.3894	1.4147
2.0	1.4069	1.4191	1.4456	1.4747
3.0	1.4884	1.5033	1.5355	1.5709
5.0	1.5427	1.5595	1.5956	1.6352
10.0	1.5785	1.5965	1.6352	1.6776
20.0	1.5950	1.6135	1.6535	1.6971

TABLE 77. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{B,\bar{M}_w}$ FOR THE BAUMANN–STOCKMAYER–FIXMAN PROCEDURE (EQ. B63) USING $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 78)

$$q^{(LN)}_{B,\bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{2.25}$$

$$[\langle S^2\rangle_{av} q^{(LN)}_{B,\langle S^2\rangle_{av}}/(\bar{M}_w q^{(LN)}_{B,\bar{M}_w})]^{3/2} = K_0^{3/2} + B_{BSF}(\bar{M}_w q^{(LN)}_{B,\bar{M}_w})^{1/2}$$

$\bar{M}_w/\bar{M}_n$	$q^{(LN)}_{B,\bar{M}_w}$
1.1	1.2392
1.3	1.8046
1.5	2.4900
1.7	3.3000
2.0	4.7568
2.5	7.8590
3.0	11.845
4.0	22.627
5.0	37.384
6.0	56.343
7.0	79.702
8.0	107.63
9.0	140.30
10.0	177.83

TABLE 78. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{B,\langle S^2\rangle_z}$ FOR THE BAUMANN–STOCKMAYER–FIXMAN PROCEDURE (EQ. B63) USING $\langle S^2\rangle_z$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 77)

$$q^{(LN)}_{B,\langle S^2\rangle_z} = (\bar{M}_w/\bar{M}_n)^{0.25\varepsilon^2+0.5\varepsilon+1.25}$$

$$[\langle S^2\rangle_z q^{(LN)}_{B,\langle S^2\rangle_z}/(\bar{M}_{av} q^{(LN)}_{B,\bar{M}_{av}})]^{3/2} = K_0^{3/2} + B_{BSF}(\bar{M}_{av} q^{(LN)}_{B,\bar{M}_{av}})^{1/2}$$

	$a_{[\eta]}$ and (ε)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.70 (0.13)	0.90 (0.27)
1.1	1.1231	1.1265	1.1342	1.1429
1.3	1.3764	1.3881	1.4143	1.4443
1.5	1.6385	1.6600	1.7086	1.7649
1.7	1.9083	1.9412	2.0158	2.1032
2.0	2.3259	2.3784	2.4986	2.6410
3.0	3.8109	3.9482	4.2690	4.6612
4.0	5.4097	5.6569	6.2429	6.9751
5.0	7.0989	7.4767	8.3833	9.5353
6.0	8.8637	9.3905	10.67	12.31
8.0	12.58	13.45	15.60	18.42
10.0	16.51	17.78	20.95	25.18
12.0	20.62	22.33	26.65	32.51
14.0	24.87	27.08	32.67	40.35
16.0	29.27	32.00	38.97	48.65
20.0	38.40	42.29	52.34	66.51

4. Baumann–Kurata–Stockmayer Procedure

TABLE 79. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{BKS,\bar{M}_w}$ FOR THE BAUMANN–KURATA–STOCKMAYER PROCEDURE (EQ. B66) USING $\bar{M}_w$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 80)

$$q^{(SZ)}_{BKS,\bar{M}_w} = \frac{\Gamma(k+\varepsilon+3)[\Gamma(k-0.5\varepsilon+3.5)]^2}{(k+1)^4(k+2)^3[\Gamma(k+1)]^3} = \frac{\Gamma((3k+2a_{[\eta]}+8)/3)[\Gamma((3k-a_{[\eta]}+11)/3)]^2}{(k+1)^4(k+2)^3[\Gamma(k+1)]^3}$$

$$\langle S^2\rangle_{av} q^{(SZ)}_{BKS,\langle S^2\rangle_{av}}/(\bar{M}_w q^{(SZ)}_{BKS,\bar{M}_w}) = K_0 + B_{BKS}\bar{M}_w q^{(SZ)}_{BKS,\bar{M}_w}/(\langle S^2\rangle_{av} q^{(SZ)}_{BKS,\langle S^2\rangle_{av}})^{1/2}$$

	$a_{[\eta]}$ and (ε)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.90 (0.27)
1.1	1.1627	1.1593	1.1566	1.1521
1.3	1.4135	1.4050	1.3981	1.3868
1.5	1.5978	1.5855	1.5756	1.5593
1.7	1.7388	1.7236	1.7114	1.6913
2.0	1.8976	1.8791	1.8643	1.8399
2.5	2.0776	2.0554	2.0376	2.0085
3.0	2.1976	2.1730	2.1532	2.1209
5.0	2.4377	2.4082	2.3845	2.3458
10.0	2.6179	2.5847	2.5580	2.5146
20.0	2.7080	2.6729	2.6448	2.5990

TABLE 80. POLYMOLECULARITY CORRECTION FACTORS $q^{(SZ)}_{BKS,\langle S^2\rangle_z}$ FOR THE BAUMANN–KURATA–STOCKMAYER PROCEDURE (EQ. B66) USING $\langle S^2\rangle_z$ FOR POLYMERS WITH A SCHULZ–ZIMM (SZ) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 79)

$$q^{(SZ)}_{BKS,\langle S^2\rangle_z} = \frac{\Gamma(k+\varepsilon+3)[\Gamma(k-0.5\varepsilon+3.5)]^2}{(k+1)^3(k+2)^4[\Gamma(k+1)]^3} = \frac{\Gamma((3k+2a_{[\eta]}+8)/3)[\Gamma((3k-a_{[\eta]}+11)/3)]^2}{(k+1)^3(k+2)^4[\Gamma(k+1)]^3}$$

$$\langle S^2\rangle_z q^{(SZ)}_{BKS,\langle S^2\rangle_z}/(\bar{M}_{av} q^{(SZ)}_{BKS,\bar{M}_{av}}) = K_0 + B_{BKS}\bar{M}_{av} q^{(SZ)}_{BKS,\bar{M}_{av}}/(\langle S^2\rangle_z q^{(SZ)}_{BKS,\langle S^2\rangle_z})^{1/2}$$

	$a_{[\eta]}$ and (ε)			
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.90 (0.27)
1.1	1.0658	1.0627	1.0602	1.0561
1.3	1.1485	1.1416	1.1360	1.1268
1.5	1.1983	1.1891	1.1817	1.1694
1.7	1.2317	1.2209	1.2122	1.1980
2.0	1.2650	1.2528	1.2428	1.2266
2.5	1.2985	1.2846	1.2735	1.2553
3.0	1.3185	1.3038	1.2919	1.2725
5.0	1.3543	1.3379	1.3247	1.3032
10.0	1.3778	1.3604	1.3463	1.3235
20.0	1.3887	1.3707	1.3563	1.3328

TABLE 81. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{BKS,\bar{M}_w}$ FOR THE BAUMANN–KURATA–STOCKMAYER PROCEDURE (EQ. B66) USING $\bar{M}_w$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 82)

$$q^{(LN)}_{BKS,\bar{M}_w} = (\bar{M}_w/\bar{M}_n)^{0.75\varepsilon^2-0.5\varepsilon+1.75} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}-2a_{[\eta]}+6)/3} \qquad \langle S^2\rangle_{av} q^{(LN)}_{BKS,\langle S^2\rangle_{av}}/(\bar{M}_w q^{(LN)}_{BKS,\bar{M}_w}) = K_0 + B_{BKS}\bar{M}_w q^{(LN)}_{BKS,\bar{M}_w}/(\langle S^2\rangle_{av} q^{(LN)}_{BKS,\langle S^2\rangle_{av}})^{1/2}$$

	$a_{[\eta]}$ and (ε)					
$\bar{M}_w/\bar{M}_n$	0.40 (−0.07)	0.50 (0.00)	0.60 (0.07)	0.70 (0.13)	0.80 (0.20)	0.90 (0.27)
1.1	1.1856	1.1815	1.1781	1.1755	1.1737	1.1725
1.3	1.5980	1.5827	1.5703	1.5607	1.5539	1.5498
1.5	2.0636	2.0331	2.0085	1.9896	1.9762	1.9682
1.7	2.5807	2.5310	2.4910	2.4603	2.4387	2.4258
2.0	3.4502	3.3636	3.2944	3.2415	3.2043	3.1821
2.5	5.1403	4.9704	4.8357	4.7334	4.6616	4.6191
3.0	7.1196	6.8385	6.6168	6.4493	6.3323	6.2631
4.0	11.90	11.31	10.85	10.51	10.27	10.13
5.0	17.73	16.72	15.93	15.34	14.94	14.70
8.0	41.07	38.05	35.75	34.06	32.90	32.22
10.0	61.19	56.23	52.48	49.74	47.86	46.77
20.0	211.11	189.15	172.89	161.22	153.37	148.84

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TABLE 82. POLYMOLECULARITY CORRECTION FACTORS $q^{(LN)}_{BKS,\langle S^2\rangle_z}$ FOR THE BAUMANN–KURATA–STOCKMAYER PROCEDURE (EQ. B66) USING $\langle S^2\rangle_z$ FOR POLYMERS WITH A LOGARITHMIC NORMAL (LN) DISTRIBUTION OF THE MOLECULAR WEIGHT (SEE ALSO TABLE 81)

$q^{(LN)}_{BKS,\langle S^2\rangle_z} = (\bar{M}_w/\bar{M}_n)^{0.75\varepsilon^2-0.5\varepsilon+0.75} = (\bar{M}_w/\bar{M}_n)^{(a^2_{[\eta]}-2a_{[\eta]}+3)/3}$ $\quad \langle S^2\rangle_z q^{(LN)}_{BKS,\langle S^2\rangle_z}/(\bar{M}_{av} q^{(LN)}_{BKS,\bar{M}_{av}}) = K_0 + B_{BKS}\bar{M}_{av} q^{(LN)}_{BKS,\bar{M}_{av}}/(\langle S^2\rangle_z q^{(LN)}_{BKS,\langle S^2\rangle_z})^{1/2}$

	$a_{[\eta]}$ and (ε)					
$\bar{M}_w/\bar{M}_n$	**0.40 (−0.07)**	**0.50 (0.00)**	**0.60 (0.07)**	**0.70 (0.13)**	**0.80 (0.20)**	**0.90 (0.27)**
1.1	1.0779	1.0741	1.0710	1.0687	1.0670	1.0659
1.3	1.2292	1.2175	1.2079	1.2006	1.1953	1.1922
1.5	1.3757	1.3554	1.3390	1.3264	1.3175	1.3121
1.7	1.5181	1.4888	1.4653	1.4473	1.4345	1.4269
2.0	1.7251	1.6818	1.6472	1.6208	1.6021	1.5911
2.5	2.0561	1.9882	1.9343	1.8934	1.8647	1.8477
3.0	2.3732	2.2795	2.2056	2.1498	2.1108	2.0877
4.0	2.9759	2.8284	2.7132	2.6268	2.5669	2.5315
5.0	3.5470	3.3437	3.1861	3.0687	2.9874	2.9397
8.0	5.1337	4.7568	4.4691	4.2575	4.1125	4.0278
10.0	6.1188	5.6234	5.2481	4.9736	4.7863	4.6774
20.0	10.5555	9.4574	8.6445	8.0609	7.6683	7.4420

J. REFERENCES

1. "Quantities, Units and Symbols in Physical Chemistry", second edition, Blackwell Scientific Publications, Oxford, 1993.
2. "Compendium of Macromolecular Nomenclature", Blackwell Scientific Publications, Oxford, 1991.
3. R. E. Bareiss, Makromol. Chem., **184**, 1509 (1983).
4. H. Staudinger, W. Heuer, Ber. Dtsch. Chem. Ges. B, **63**, 222 (1930).
5. W. Kuhn, Kolloid-Z., **68**, 2 (1934).
6. H. Mark, in "Der Feste Koerper (Vortraege zum 50jaehrigen Bestehen der Physikalischen Gesellschaft Zuerich 1937)", R. Saenger (Ed.), Hirzel, Leipzig, 1938.
7. I. Sakurada, Kasen-Koenshu, **5**, 33 (1940), Nippon Kagaku Senni Kenkyujo (Institute of Chemical Fiber, Japan), 5th meeting, Oct. 8, 1940, coll. vol. of lectures.
8. R. Houwink, J. Prakt. Chem., **157**, 15 (1941).
9. R. E. Bareiss, Chem. Ztg., **105**, 97 (1981).
10. H. G. Bungenberg de Jong, W.A.L. Dekker, Biochem. Z., **212**, 318 (1929).
11. W. Philippoff, "Viskositaet der Kolloide", in "Handbuch der Kolloidwissenschaft in Einzeldarstellungen", Vol. IX, Th. Steinkopff, Dresden-Leipzig, 1942, p. 333.
12. G. Meyerhoff, Adv. Polym. Sci., **3**, 59 (1961).
13. P. J. Flory, J. Am. Chem. Soc., **65**, 372 (1943).
14. H.-G. Elias, R. (R. E.) Bareiss, J. G. Watterson, Adv. Polym. Sci., **11**, 111 (1973).
15. L. Mandelkern, P. J. Flory, J. Chem. Phys., **20**, 212 (1952).
16. L. Mandelkern, W. R. Krigbaum, H. A. Scheraga, P. J. Flory, J. Chem. Phys., **20**, 1392 (1952).
17. H. A. Scheraga, L. Mandelkern, J. Am. Chem. Soc., **75**, 179 (1953).
18. H.-G. Elias, R. (R. E.) Bareiss, J. Macromol. Sci., Chem., **1**, 1377 (1967).
19. T. Svedberg, Kolloid-Z., Erg. -Bd., **36**, 53 (1925).
20. R. E. Bareiss, Vysokomol. Soedin. Ser. A, **32**, 171 (1990).
21. V. Petrus, M. Bohdanecký, R. E. Bareiss, Polym. Commun., **31**, 68 (1990).
22. R. E. Bareiss, B. Jung, A. R. Khokhlov, Macromol. Theory Simul., **5**, 157 (1996).
23. P. J. Flory, W. R. Krigbaum, J. Chem. Phys., **18**, 1086 (1950).
24. R. E. Bareiss, Makromol. Chem., **182**, 1761 (1981).
25. R. E. Bareiss, Makromol. Chem., Rapid Commun., **2**, 523 (1981).
26. R. E. Bareiss, Eur. Polym. J., **19**, 847 (1983).
27. S. Newman, W. R. Krigbaum, C. Laugier, P. J. Flory, J. Polym. Sci., **14**, 451 (1954).
28. V. Petrus, M. Bohdanecký, R. E. Bareiss, Makromol. Chem., **188**, 2447 (1987).
29. R. E. Bareiss, Makromol. Chem., **187**, 955, 1335 (1986).
30. W. H. Stockmayer, M. Fixman, J. Polym. Sci., Part C: Polym. Symp., **1**, 137 (1963).
31. H. Yamakawa, "Modern Theory of Polymer Solutions", Harper & Row, New York, 1971.
32. W. Burchard, Makromol. Chem., **50**, 20 (1961).
33. W. Sutter, A. Kuppel, Makromol. Chem., **149**, 271 (1971).
34. M. Guaita, R. E. Bareiss, Makromol. Chem., **180**, 2735 (1979).
35. V. Petrus, M. Bohdanecký, R. E. Bareiss, Makromol. Chem., **190**, 1453 (1989).
36. J. M. G. Cowie, S. Bywater, Polymer, **6**, 197 (1965).
37. H. Baumann, J. Polym. Sci. B, **3**, 1069 (1965).
38. W. Banks, C. T. Greenwood, Eur. Polym. J., **4**, 249 (1968).
39. R. E. Bareiss, Makromol. Chem., **170**, 251 (1973).
40. M. Kurata, W. H. Stockmayer, Adv. Polym. Sci., **3**, 196 (1963).

41. R. E. Bareiss, J. Bellido, Makromol. Chem., **177**, 3571 (1976).
42. G. V. Schulz, Z. Phys. Chem., Abt. B, **43**, 25 (1939).
43. B. H. Zimm, J. Chem. Phys., **16**, 1099 (1948).
44. W. D. Lansing, E. O. Kraemer, J. Am. Chem. Soc., **57**, 1369 (1935).
45. H. Wesslau, Makromol. Chem., **20**, 111 (1956).
46. E. P. Honig, J. Phys. Chem., **69**, 4418 (1965).
47. J. G. Watterson, J. Macromol. Sci., Chem., **5**, 459 (1971).
48. See e.g. Appendices in Refs. 24 and 41.
49. N. F. Gralen, Dissertation, Uppsala, 1944.
50. B. Vollmert, Angew. Chem., **71**, 119 (1959).
51. N. Yamada, H. Matsuda, H. Nakamura, Rep. Prog. Polym. Phys. Jpn., **4**, 81 (1961).
52. K. G. Berdnikova, G. V. Tarasova, V. S. Skazka, N. A. Nikitin, G. V. Dyuzhev, Vysokomol. Soedin., **6**, 2057 (1964); Eng. translation in Polym. Sci. USSR, **6**, 2280 (1965).
53. D. M. Crothers, B. H. Zimm, J. Mol. Biol., **12**, 524 (1965).
54. H.-G. Elias, Makromol. Chem., **122**, 264 (1969).
55. R. E. Bareiss, Makromol. Chem. **185**, 1623 (1984).
56. R. Koningsveld, C. A. F. Tuijnman, Makromol. Chem., **38**, 39 (1960).
57. R. Ch. Oberthuer, Makromol. Chem., **179**, 2693 (1978).

Polymer–Solvent Interaction Parameters

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A. INTRODUCTION

The classical thermodynamics of binary polymer–solvent systems was developed independently by P. J. Flory (1–3) and M. L. Huggins (4–6). It is based on the well-known lattice model qualitatively formulated by K. H. Meyer (7), who pointed out the effect of the differences in molecular size of polymer and solvent molecules on the entropy of mixing. The quantitative calculation of the entropy of mixing led to the introduction of a dimensionless quantity, the so-called Flory–Huggins interaction parameter χ, for the thermodynamic description of polymer solutions.

It was evident from the beginning that χ is a function of T; however, at first χ was considered to be independent of the concentration of the polymer. Subsequent experiments have shown the necessity of treating χ as a function of composition (8,9). Furthermore, χ also depends on M_2 (the molecular weight of the polymer), not only at high dilution, but – according to recent experiments (10) – even at high concentrations of polymers.

B. CONCENTRATION VARIABLES

Since the numerical value of the interaction parameter depends on the concentration variable chosen, one has to specify it in each case. The following three quantities are most frequently used, where the subscript 1 stands for the solvent and the subscript 2 for the polymer:

1. Volume fractions φ_i:

$$\varphi_2 = \frac{n_2 V_2}{n_1 V_1 + n_2 V_2} \tag{B1}$$

n_i moles of component i
V_i molar volume of component i

2. Weight fractions w_i:

$$w_2 = \frac{m_2}{m_1 + m_2} \tag{B2}$$

m_i weight of component i

3. Segment fractions s_i (hard core volume fractions):

$$s_2 = \frac{n_2 V_2^*}{n_1 V_1^* + n_2 V_2^*} \tag{B3}$$

V_i^* molar hard core volume (characteristic volumes) of components related to reduced volumes $\tilde{V}_i$ by $\tilde{V}_i = V_i/V_i^*$ (11).

C. CONVERSION OF CONCENTRATION VARIABLES

	φ_2	w_2	s_2
φ_2	–	$w_2 \frac{\rho}{\rho_2}$	$s_2 \frac{\tilde{V}_2}{\tilde{V}}$
w_2	$\varphi_2 \frac{\rho_2}{\rho}$	–	$s_2 \frac{\rho_2^*}{\rho^*}$
s_2	$\varphi_2 \frac{\tilde{V}}{\tilde{V}_2}$	$w_2 \frac{\rho^*}{\rho_2^*}$	–

ρ, ρ_i are the densities of the solution and of the pure component i, $1/\rho = w_1/\rho_1 + w_2/\rho_2$;
ρ^*, ρ_i^* are the hard core (characteristic) densities of the solution and of the component i, $\rho_i^* = M_i/V_i^*$ where M_i is the molar mass, $1/\rho^* = w_1/\rho_1^* + w_2/\rho_2^*$;
$\tilde{V}$, $\tilde{V}_i$ are the reduced volumes of the solution and of the component i.

D. BASIC EQUATIONS

The reduced molar Gibbs energy of mixing, $\Delta G_M/RT$, is given by the following equation:

$$\frac{\Delta G_M}{RT} = x_1 \ln \varphi_1 + x_2 \ln \varphi_2 + g x_1 \varphi_2 \tag{D1}$$

where x_i is the mole fraction of component i, R is the gas constant, T is the absolute temperature and g stands for the integral polymer–solvent interaction parameter; g refers to the interaction (mixing of the pure liquid compounds) of a solvent molecule with a polymer segment, the size of which is defined by the molar volume of the solvent. This definition implies some difficulties as soon as mixed solvents or solvents with more than one segment are considered.

Alternatively, the Gibbs energy of mixing can be normalized to the volume, so that it reads

$$\frac{\Delta G_M}{V} = \frac{RT}{V_1}\varphi_1 \ln \varphi_1 + \frac{RT}{V_2}\varphi_2 \ln \varphi_2 + B\varphi_1\varphi_2 \qquad \text{(D2)}$$

where B is an interaction energy–density parameter that no longer depends on the definition of a segment, and relates to g and molar volume of the segment V_{seg} by

$$B = RT\frac{g}{V_{seg}} \qquad \text{(D3)}$$

The sum

$$x_1 \ln \varphi_1 + x_2 \ln \varphi_2 = \frac{\Delta G_M^{comb}}{RT} \qquad \text{(D4)}$$

of Eq. (D1) is the combinatorial part of ΔG_M. The remaining term

$$gx_1\varphi_2 = \frac{\Delta G_M^R}{RT} \qquad \text{(D5)}$$

is the so-called reduced residual Gibbs energy of mixing.

Partial differentiation of $\Delta G/RT$ with respect to n_1, the number of moles of solvent, or n_2, the number of moles of polymer, respectively, yields the reduced partial molar Gibbs energy of mixing of the solvent $\Delta G_1/RT = \Delta\mu_1/RT$, or the reduced partial molar Gibbs energy of mixing of the polymer, $\Delta G_2/RT = \Delta\mu_2/RT$:

$$\frac{\Delta G_1}{RT} = \frac{\Delta\mu_1}{RT} = \ln(1-\varphi_2) + \left(1 - \frac{1}{N}\right)\varphi_2 + \chi\varphi_2^2 \qquad \text{(D6)}$$

$$\frac{\Delta G_2}{RT} = \frac{\Delta\mu_2}{RT} = \ln(1-\varphi_1) + (1-N)\varphi_1 + \xi N\varphi_1^2 \qquad \text{(D7)}$$

$N = V_2/V_1$ (where $V_1 = V_{seg}$), the ratio of the molar volumes of the components is, in general, different from the degree of polymerization but proportional to it; χ and ξ are the Flory–Huggins interaction parameters in the differential equation for the solvent and for the polymer, respectively.

Separating ΔG_1 and ΔG_2 like ΔG_M (Eqs. D1–D5) into a combinatorial and a residual contribution yields the following expressions for the different interaction parameters, which demonstrates that they measure the deviation of a real polymer solution from purely combinatorial behavior:

$$g = \frac{\Delta G_M^R}{RTx_1\varphi_2} \qquad \text{(D8)}$$

$$\chi = \frac{\Delta G_1^R}{RT\varphi_2^2} \qquad \text{(D9)}$$

$$\xi = \frac{\Delta G_2^R}{RTN\varphi_1^2} \qquad \text{(D10)}$$

From phenomenological thermodynamics, one obtains the interrelations between these parameters, given in Eqs. (D11–D13):

$$g = \varphi_2\chi + (1-\varphi_2)\xi = \frac{1}{\varphi_1}\int_0^{\varphi_1} \chi d\varphi_1 \qquad \text{(D11)}$$

$$\chi = g + \varphi_1\frac{\partial g}{\partial\varphi_1} = g - (1-\varphi_2)\frac{\partial g}{\partial\varphi_2} \qquad \text{(D12)}$$

$$\xi = g + \varphi_2\frac{\partial g}{\partial\varphi_2} \qquad \text{(D13)}$$

Since it is the chemical potential of the solvent that is normally measured, χ is practically the only interaction parameter that can be directly obtained. In the following text, the term polymer–solvent interaction parameter (Flory–Huggins parameter) is, therefore, exclusively applied to χ.

E. METHODS OF DETERMINATION

Rearrangement of Eq. (D6) yields

$$\chi = \frac{\Delta G_1^R}{RT\varphi_2^2} = \frac{\Delta G_1}{RT\varphi_2^2} - \frac{1}{\varphi_2^2}\left(\ln(1-\varphi_2) + \left(1 - \frac{1}{N}\right)\varphi_2\right) \qquad \text{(E1)}$$

METHODS FOR THE DETERMINATION OF POLYMER–SOLVENT INTERACTION PARAMETERS

Typical range of concentration	Methods	Abbreviation in text	Refs.
$\varphi_2 \to 0$	Scattering methods	LS SAXS SANS	20 129
$0 < \varphi_2 < 0.3$	Osmosis	OS	21–25
	Light scattering	LS	
	Critical miscibility (only applicable with poor solvents)	CM	
	Differential vapor–pressure methods	ΔVP	26
$0.3 < \varphi_2 < 0.8$	Vapor–pressure methods	VP	3
$\varphi_2 \to 1$	Inverse gas chromatography (gas liquid chromatography)	IGC	27–30

Since $\Delta G_1 = RT \ln a_1$, the interaction parameter χ is accessible from the measured activity of the solvent. In order to get the information for the entire range of concentrations, a combination of several methods is necessary. The table given on the previous page shows the concentration ranges for the applicability of the most commonly used methods, the abbreviations used in the text to specify the method, and some pertinent references. Information stemming from the following sources is not incorporated in the tables: melting point depression of polymer (3,12,13), freezing point depression of solvents (8), swelling equilibria (3,14–17), and intrinsic viscosities (18,19).

F. TEMPERATURE DEPENDENCE

The temperature dependence of χ can be described for most systems exhibiting UCSTs by means of the following equation:

$$\chi(T) = a + \frac{b}{T} \tag{F1}$$

Knowing the temperature dependence of χ, it is possible to split this quantity in its enthalpic and entropic part (χ_H and χ_S).

$$\chi = \chi_H + \chi_S \tag{F2}$$

where

$$\chi_H = -T\left(\frac{\partial \chi}{\partial T}\right)_{p,\varphi_2} = \frac{\Delta H_1^R}{RT\varphi_2^2} \tag{F3}$$

$$\chi_S = \left(\frac{\partial (T\chi)}{\partial T}\right)_{p,\varphi_2} = \frac{\Delta S_1^R}{R\varphi_2^2} \tag{F4}$$

G. CONCENTRATION DEPENDENCE

The following expression for the Flory–Huggins parameter was deduced on the basis of the lattice theory (25):

$$\chi = \alpha + \frac{\beta(1-\gamma)}{(1-\gamma\varphi_2)^2} \tag{G1}$$

where α and γ are constants within a certain temperature range and β varies with T ($\beta = \beta_0 + \beta_1/T$).

For the integral interaction parameter, g, the equation reads:

$$g = \alpha + \frac{\beta}{(1-\gamma\varphi_2)} \tag{G2}$$

For an analytical representation of the experimentally found concentration dependence, the following power series with $n = 2$ is normally adequate:

$$\chi = \sum_{i=0}^{n} \chi_i \varphi_2^i \tag{G3}$$

$$g = \sum_{i=0}^{n} \frac{\chi_i}{i+1} \frac{1-\varphi_2^{i+1}}{1-\varphi_2} \tag{G4}$$

According to experimental information, three characteristic behaviors can be distinguished:

1. Very often, χ increases greatly with polymer concentration, particularly in the case of poor solvents. The systems benzene/polyisobutylene (PIB) (21,31,32), *n*-pentane/polyisobutylene (33,34), and cyclohexane/polystyrene (22) (Figs. 1 and 2) constitute examples.

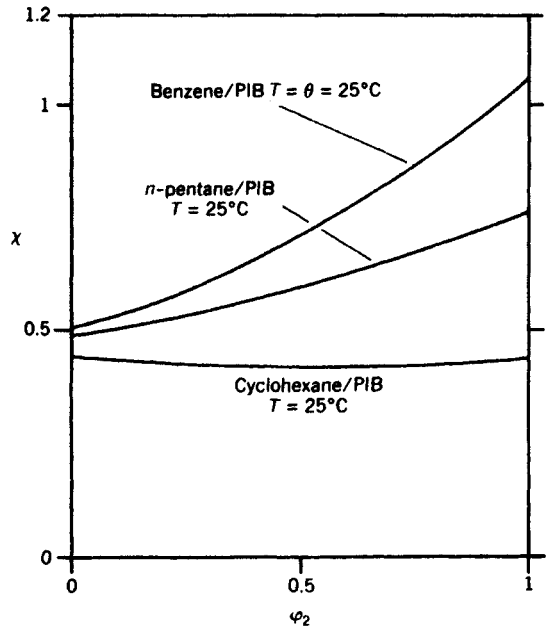

Figure 1. Examples for the variation of the Flory–Huggins interaction parameter, χ (Eq. D14), with the volume fraction φ_2 of polymer for the systems benzene/poly(isobutylene) (PIB), *n*-pentane/PIB and cyclohexane/PIB at the indicated temperatures.

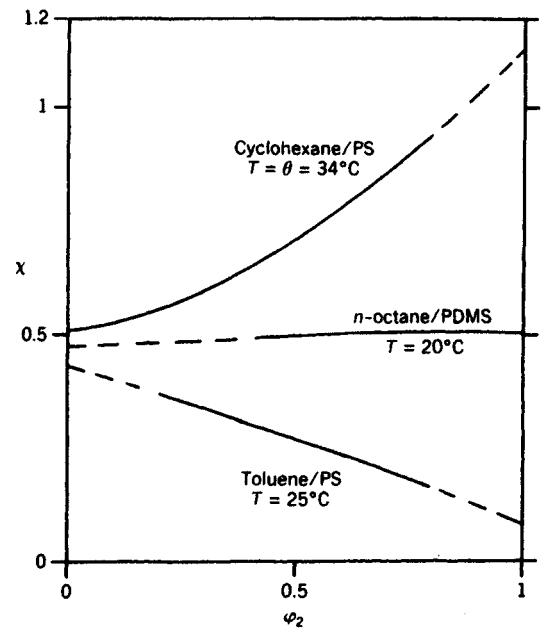

Figure 2. Examples for the variation of the Flory–Huggins interaction parameter, χ (Eq. D14), with the volume fraction φ_2 of polymer for the systems cyclohexane/polystyrene (PS), *n*-octane/poly(dimethyl siloxane) (PDMS) and toluene/PS at the indicated temperatures.

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COEFFICIENTS χ_0, χ_1, χ_2 (Eq. G3) OF SYSTEMS SELECTED FOR GRAPHICAL REPRESENTATION (Figs. 1 AND 2)

System	T/(°C)	χ_0	χ_1	χ_2	Refs.
Benzene/PIB	25	0.506	0.229	0.317	21,31,32
n-Pentane/PIB	25	0.484	0.184	0.080	33,34
Cyclohexane/PIB	25	0.440	−0.083	0.068	30,31,35
Cyclohexane/PS	34	0.507	0.149	0.464	22
n-Octane/PDMS	20	0.501	−0.035	0.042	36
Toluene/PS	25	0.431	−0.311	−0.036	37

2. In some cases, χ seems to be independent of composition, as proposed by the original Flory–Huggins theory. These findings mainly concern good solvents; examples are the systems cyclohexane/polyisobutylene (30,31,35) and *n*-octane/poly(dimethyl siloxane) (PDMS) (36) (Figs. 1 and 2).
3. In a few cases, mostly with highly exothermal systems, χ decreases with concentration, as with toluene/polystyrene (37) (Fig. 2).

The lines drawn in Figs. 1 and 2 have been calculated by means of Eq. (G3) and the coefficients χ_i, collected in the table given above.

H. MOLECULAR WEIGHT DEPENDENCE

Second osmotic virial coefficients, as obtained from osmotic pressure or light scattering measurements with dilute polymer solutions, depend markedly on M_2; since this quantity is closely related (cf. Eq. I1) to χ, its dependence on chain length in this concentration regime is obvious. Recent experiments (10) have shown that χ may depend on the molar mass of the polymer even for concentrated solutions. Furthermore, there is some evidence (given in Ref. 60, resulting from a common valuation of the data given in Refs. 83,58,33,30 and 90, and from a comparison of the results reported in Refs. 99 and 110) that even χ_∞, the χ parameter for an infinitely large excess of polymer, may depend on chain length. According to the few presently available experimental data, a reduction of χ upon an increase of M_2 according to some scaling law seems to be the normal case.

I. POLYMER–SOLVENT INTERACTION PARAMETERS, χ

Values of the polymer–solvent interaction parameters, defined by Eq. (E1) in terms of volume fractions, have been collected. They are tabulated as a function of polymer concentration for various temperatures. Data given on the basis of segment fractions, s_2, or weight fractions, w_2, were converted into volume fractions φ_2.

If needed, χ_0 can be calculated from the second osmotic virial coefficient A_2 (see corresponding chapter in this Handbook by means of the following equation:

$$\chi_0 = 0.5 - \rho_2^2 V_1 A_2 \qquad (I1)$$

Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
Cellulose acetate 2.5										
Acetone	30	104		0.30	0.51				VP	38
Dioxane	30	104		0.31	0.51				VP	38
Methyl acetate	30	104		0.43	0.59				VP	38
Pyridine	30	104		0.07	0.09				VP	38
Cellulose acetate 3.0										
Dichloromethane	25	157		0.38	0.45	0.49			VP	39
Trichloromethane	30	157		0.36	0.45	0.51			VP	39
Cellulose acetate *x*										
Dimethylformamide	50	36				0.67	0.55		VP	40
Dimethylformamide	70	36				0.39	0.31		VP	40
Cellulose nitrate 2.6										
Acetone	20	100		0.14	0.06	−0.37	−1.24		VP	41
Acetonitrile	20	100			0.59	0.42	0.12	−0.1	VP	41
Cyclopentanone	20	100		0.42	0.07	−0.71	−2.4		VP	41
Dioxane	20	100			1.2	−0.25	−1.7		VP	41
Ethyl acetate	20	130		0.04	−0.43	−1.35			VP	42
Ethyl formate	20	130		−0.08	−0.14	−0.42	−3.2		VP	42
Isopentyl acetate	20	130		−0.89	−1.8	−3.3			VP	42
Isopropyl ketone	20	100		0.62	−0.08	−1.7			VP	41
Methyl isopropyl ketone	20	100		−0.5	−0.52	−1.6			VP	41
Methyl *tert*-butyl ketone	20	100		0.16	−1.5	−2.8	−3.7		VP	41
Nitromethane	20	100		0.66	0.64	0.6	0.45		VP	41
n-Propyl acetate	20	130		−0.38	−0.83	−2.0	−4.1		VP	42

Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
Cellulose, hydroxypropyl										
Acetone	25	100				0.52	0.67		VP	43
Ethanol	25	100				0.35	0.44		VP	43
Tetrahydrofuran	25	100				0.55	0.51		VP	43
Water	25	100				0.74	1.02		VP	44
Poly(acrylonitrile)										
Dimethylformamide	70	30.6					1.28		VP	40
Poly(α-methyl styrene)										
Cumene	65	17		0.49	0.47	0.51	0.6		VP	45
α-Methylstyrene	65	17			0.46	0.47	0.47		VP	45
Toluene	25				0.51	0.6				46
	25	53		0.41					VP	47
Poly(arylate)										
Chlorobenzene	140–200	7						0.54–0.30	IGC	48
o-Dichlorobenzene	140–200	7						0.59–0.27	IGC	48
Diethylene glycol diethyl ether	140–200	7						0.08–0.56	IGC	48
n-Dodecane	140–200	7						1.34–1.24	IGC	48
Ethylbenzene	140–200	7						0.82–0.48	IGC	48
Tetralin	140–200	7						0.75–0.57	IGC	48
p-Xylene	140–200	7						0.84–0.39	IGC	48
Poly(butadiene)										
Acetone	40–100	13.1						1.64–1.32	IGC	49
Acetonitrile	40–100	13.1						2.86–2.36	IGC	49
Benzene	23.5	65.2				0.06	0.02		VP	50
	40–100	13.1						0.26–0.19	IGC	49
1-Butanol	40–100	13.1						2.76–1.64	IGC	49
Butyl acetate	40–100	13.1						0.52–0.45	IGC	49
Butyronitrile	40–100	13.1						1.90–1.57	IGC	49
1-Chlorobutane	40–100	13.1						0.33–0.25	IGC	49
1-Chloropentane	40–100	13.1						0.37–0.32	IGC	49
1-Chloropropane	40–100	13.1						0.36–0.30	IGC	49
Cyclohexane	23.5	65.2					0.26		VP	50
	40–100	13.1						0.22–0.15	IGC	49
Dibutylether	40–100	12.2						0.37–0.34	IGC	49
Dichloromethane	23.5	65.2				0.27	0.12		VP	50
Diethylether	40–100	13.1						0.43–0.40	IGC	49
Diethylketone	40–100	13.1						0.82–0.67	IGC	49
Dipropylether	40–100	13.1						0.51–2.26	IGC	49
Ethanol	40–100	13.1						3.41–2.30	IGC	49
Ethyl acetate	40–100	13.1						0.86–0.69	IGC	49
Ethylbenzene	80	200				0.31	0.37		VP	51
Ethylcyclohexane	40–100	13.1						0.09–0.04	IGC	49
n-Heptane	40–100	13.1						0.36–0.31	IGC	49
1-Heptene	40–100	13.1						0.25–0.23	IGC	49
n-Hexane	40–100	13.1						0.37–0.38	IGC	49
1-Hexene	40–100	13.1						0.32–0.28	IGC	49
Methanol	40–100	13.1						3.85–2.70	IGC	49
Methyl acetate	40–100	13.1						1.14–0.92	IGC	49
Methyl cyclohexane	40–100	13.1						0.17–0.13	IGC	49
Methyl ethyl ketone	40–100	13.1						1.19–0.97	IGC	49
Methyl isobutyl ketone	40–100	13.1						0.95–0.72	IGC	49
n-Octane	40–100	13.1						0.34–0.28	IGC	49
1-Octene	40–100	13.1						0.23–0.20	IGC	49
n-Pentane	40–100	13.1						0.44–0.36	IGC	49
1-Propanol	40–100	13.1						2.97–1.97	IGC	49
2-Propanol	40–100	13.1						2.94–1.86	IGC	49
Propionitrile	40–100	13.1						2.25–1.82	IGC	49
Propyl acetate	40–100	13.1						0.67–0.55	IGC	49
Tetrachloromethane	23.5	65.2				−0.04	−0.12		VP	50
Tetrachloromethane	40–100	13.1						0.10–0.15	IGC	49
Tetrahydrofuran	40–100	13.1						0.35–0.29	IGC	49

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Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
Toluene	40–100	13.1						0.14–0.12	IGC	49
Trichloromethane	23.5	65.2				−0.2	−0.34		VP	50
Trichloromethane	40–100	13.1						0.05–0.07	IGC	49
Poly(butene-1)										
n-Decane	115–135	10						0.30	IGC	52
n-Heptane	115–135	10						0.38	IGC	52
n-Nonane	115–135	10						0.32	IGC	52
n-Octane	115–135	10						0.36	IGC	52
Poly(decene)										
Toluene	30	213.9		0.49	0.44	0.39	0.35	0.32	VP	53
Poly(dimethyl siloxane)										
Acetone	100							1.33	IGC	54
Benzene	20	50			0.64	0.71	0.77	0.85	VP	36
	25	1.54			0.56	0.65	0.72	(0.76)	VP	55
	25	4.17				0.66	0.8		VP	55
	25	10				0.66	0.73	0.79	OS, VP	56
	25	15.7		0.56	0.63	0.76	0.82		VP	57
	25	50				0.66	0.73	0.79	VP, IGC	58
	25	50						0.81	IGC	59
	25–70	50						0.79–0.74	VP, IGC	58
	25–70	50						0.81–0.75	IGC	59
	30	6.65					0.71	0.74	VP	60
	30	15.65					0.71	0.75	VP	60
	30	26					0.72	0.74	VP	60
	30	89					0.45		VP	61
	30	89					0.74	0.77	VP	60
	40	1.54				0.85	0.81	(0.72)	VP	55
	40	4.17				0.75	0.73	(0.83)	VP	55
Cycloheptane	25–70	50						0.56–0.53	VP, IGC	58
Cyclohexane	20	50	0.41	0.44					OS	62
	25	10		0.46	0.48	0.50		0.47	OS, VP	56
	25	50				0.50		0.47	VP, IGC	58
	25–70	50						0.47–0.48	VP, IGC	58
	30				0.43	0.43	0.42	0.42	VP	63
	30	89					0.46	0.45	VP	64
Cyclooctane	25–70	50						0.66–0.61	VP, IGC	58
Cyclopentane	25–70	50						0.42–0.46	VP, IGC	58
n-Decane	20	30						0.64		65
Diisobutyl ketone	35	80		0.49					OS	66
1,4-Dioxane	25–70	50						1.32–1.18	VP, IGC	58
	100							1.06	IGC	54
Ethyl *n*-butyl ketone	35	80		0.52					OS	66
Ethylbenzene	23.5	50			0.54	0.63	0.7		VP	36
	25–70	50						0.77–0.73	VP, IGC	58
	25–70	50						0.83–0.78	IGC	59
n-Heptane	20	30						0.50		65
	20	50			0.45	0.45	0.46	0.48	VP	36
	25–70	50						0.48–0.50	VP,IGC	58
	25–70	50						0.50–0.49	IGC	59
	30	89					0.43	0.45	VP	64
Hexamethyl disiloxane	23.2	50			0.3	0.28	0.26	0.25	VP	36
	25	1.54			0.27	0.25	0.24		VP	55
	25	4.17			0.21	0.22	0.25		VP	55
	25	50						0.28	VP, IGC	58
	25–70	50						0.28–0.34	VP, IGC	58
n-Hexane	20		0.40	0.40				0.45	VP	40
	20	30						0.50		65
	25	89					0.41	0.43	VP	61
	25–70	50						0.46–0.47	VP, IGC	58
	25–70	50						0.45	IGC	59
	30	6.65					0.37	0.39	VP	67
	30	15.65					0.38	0.40	VP	67
	30	26					0.38	0.41	VP	67

Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
	30	89					0.41	0.43	VP	61
	30	89					0.38	0.40	VP	64
	30	89					0.40	0.41	VP	67
	35	89					0.41	0.42	VP	61
Mesitylene	25–70	50						0.95–0.86	VP, IGC	58
Methyl ethyl ketone	20	80		0.57					OS	66
	25	280	0.50						OS	68
	30	50			0.51	0.64	0.79		VP	69
	35	80		0.57					OS	66
	50	80		0.56					OS	66
Methyl isobutyl ketone	20	80		0.54					OS	66
2-Methylbutane	25	50						0.39	IGC	59
	100							1.10	IGC	54
2-Methylheptane	25–70	50						0.49–0.47	VP, IGC	58
	25–70	50						0.52–0.50	IGC	59
2-Methylhexane	25–70	50						0.45–0.46	VP, IGC	58
	25–70	50						0.46	IGC	59
3-Methylhexane	25–70	50						0.45–0.43	VP, IGC	58
2-Methylpentane	25–70	50						0.45–0.43	IGC	59
Octamethylcyclo-tetra siloxane	140	10.1			0.51	0.38	0.25		VP	70
Octamethyltrisiloxane	23.2	50			0.22	0.19	0.17	0.14	VP	36
n-Octane	20	30						0.53		65
	20	50			0.50	0.51	0.52	0.55	VP	36
	20	50		0.42	0.50	0.49	0.50	0.51	OS	62
	25	4.17			0.36	0.40	0.44		VP	55
	25	50						0.52	VP, IGC	58
	25–70	50						0.56–0.52	IGC	59
	35	50						0.54	IGC	59
	35	144		0.42					OS	71
	40	4.17				0.40	0.40	(0.44)	VP	55
n-Pentane	25–70	50						0.45–0.49	VP, IGC	58
	25–70	50						0.41–0.43	IGC	59
	20	30						0.48		65
	20	50			0.43	0.41	0.40	0.40	VP	36
	25	50						0.45	VP, IGC	58
	100							0.31	IGC	54
Tetrachloromethane	100							0.36	IGC	54
Toluene	20	50	0.45	0.50	0.59	0.64	0.72	0.82	VP	36
	25	1.54			0.53	0.58	0.65	(0.74)	VP	55
	25	4.17				0.59	0.65		VP	55
	25	50						0.75	VP, IGC	58
	25–70	50						0.75–0.71	VP, IGC	58
	25–70	50						0.80–0.75	IGC	59
	40	4.17				0.63	0.68	(0.69)	VP	55
	100							0.59	IGC	54
Trichloromethane	30	89					0.64	0.65	VP	72
	100							0.60	IGC	54
2,2,4-Trimethylpentane	20	30						0.43		65
	25	4.17			0.35	0.35	0.38		VP	55
	25–70	50						0.45–0.43	VP, IGC	58
p-Xylene	25	50				0.58	0.67	0.78	VP	36
	25	50						0.75	VP, IGC	58
	25–70	50						0.80–0.77	IGC	59
m-Xylene	25–70	50						0.82–0.76	VP, IGC	58
o-Xylene	25–70	50						0.86–0.80	VP, IGC	58
Poly(dodecene)										
Toluene	30	94.9			0.34	0.47	0.61	0.74	VP	53
Poly(ε-caprolactam)										
Tetrachloromethane	65	10.7			0.54	0.46	0.39		VP	73
Poly(epichlorohydrin)										
Acetone	100							0.28	IGC	54
Benzene	100							0.25	IGC	54

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Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
n-Butane	100							1.65	IGC	54
1-Butanol	100							1.12	IGC	54
Butyl acetate	100							0.36	IGC	54
Chlorobenzene	100							0.24	IGC	54
1-Chlorobutane	100							0.62	IGC	54
Chloromethane	100							0.36	IGC	54
1-Chloropentane	100							0.65	IGC	54
Cycloheptane	100							1.19	IGC	54
Cyclohexane	100							1.25	IGC	54
Cyclohexene	100							0.84	IGC	54
Cyclooctane	100							1.20	IGC	54
Cyclopentane	100							1.16	IGC	54
n-Decane	100							2.12	IGC	54
1,1-Dichloroethane	100							0.33	IGC	54
1,2-Dichloroethane	100							0.20	IGC	54
Dichloromethane	100							0.18	IGC	54
1,4-Dioxane	100							0.04	IGC	54
Ethanol	100							1.58	IGC	54
Ethyl acetate	100							0.35	IGC	54
Ethyl benzene	100							0.44	IGC	54
n-Heptane	100							1.79	IGC	54
n-Hexane	100							1.72	IGC	54
Methyl acetate	100							0.39	IGC	54
Methyl ethyl ketone	100							0.20	IGC	54
n-Nonane	100							1.99	IGC	54
n-Octane	100							1.89	IGC	54
n-Pentane	100							1.64	IGC	54
1-Pentanol	100							1.03	IGC	54
Propane	100							1.71	IGC	54
1-Propanol	100							1.22	IGC	54
Propyl acetate	100							0.33	IGC	54
Tetrachloromethane	100							0.69	IGC	54
Tetrahydrofuran	100							0.01	IGC	54
Toluene	100							0.31	IGC	54
1,1,1-Trichloroethane	100							0.46	IGC	54
Trichloroethylene	100							0.53	IGC	54
Trichloromethane	100							0.25	IGC	54
n-Undecane	100							2.24	IGC	54
Poly(ethyl methacrylate)										
Dichloromethane	23.5	144				−0.52	−0.66		VP	74
Tetrachloromethane	23.5	144.4				−0.01	−0.12		VP	74
Toluene	23.5	144				0.19	0.15		VP	74
Trichloromethane	23.5	144			−0.94	−1.05			VP	74
Poly(ethylene glycol)										
Ethylbenzene	70	2				0.87	0.83		VP	75
	70	3.272			0.38	0.42	0.45		VP	40
	70	4				0.85	0.82		VP	75
	70	6				0.71	0.69		VP	75
Poly(ethylene oxide)										
Benzene	50.3	6.1		0.18	0.14	0.1			VP	76
	70	100				0.23	0.22		VP	73
	70	6.1		0.19	0.14	0.12	0.09		VP	76
	100							0.13	IGC	54
Cyclohexane	100							1.23	IGC	54
n-Decane	100							2.10	IGC	54
Ethanol	100							0.70	IGC	54
n-Octane	100							1.85	IGC	54
Toluene	99.8	5.2			0.41	0.3			VP	40
	100							0.26	IGC	54
Trichloromethane	100							−0.55	IGC	54
Poly(ethylene)HD										
cis-Decalin	149	105						0.07	IGC	77

Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
trans-Decalin	149	105						0.05	IGC	77
n-Decane	145–190							0.18	IGC	28
	149	105						0.31	IGC	77
	185	105						0.12	IGC	78
2,4-Dimethyl hexane	149	105						0.38	IGC	77
3,4-Dimethyl hexane	149	105						0.31	IGC	77
n-Dodecane	149							0.28	IGC	77
Ethylbenzene	149	105						0.37	IGC	77
Mesitylene	149	105						0.28	IGC	77
2-Methylheptane	149	105						0.39	IGC	77
3-Methylhexane	149	105						0.40	IGC	77
n-Nonane	149	105						0.34	IGC	77
n-Octane	149	105						0.36	IGC	77
Tetralin	149	105						0.32	IGC	77
Toluene	149	105						0.39	IGC	77
2,2,4-Trimethyl hexane	149	105						0.35	IGC	77
2,2,4-Trimethyl pentane	149	105						0.40	IGC	77
m-Xylene	149	105						0.34	IGC	77
p-Xylene	149	105						0.32	IGC	77
Poly(ethylene)L D										
Benzene	125–135							0.43–0.36	IGC	79
1-Butanol	135							1.38	IGC	79
Chlorobenzene	135							0.34	IGC	79
Cyclohexane	125–135							0.18	IGC	79
Cyclohexanol	135							1.22	IGC	79
cis-Decalin	120–145	82						0.03		77
trans-Decalin	120–145	82						0.02–0.00		77
n-Decane	120–145	82						0.25–0.26		77
2,4-Dimethyl hexane	120–145	82						0.34–0.32		77
2,5-Dimethyl hexane	120–145	82						0.35		77
3,4-Dimethyl hexane	120–145	82						0.25		77
n-Dodecane	110–145							0.18		28
	120–145	82						0.23–0.24		77
Ethylbenzene	120–145	82						0.33		77
n-Heptane	108.9	0.45		0.29	0.31	0.34	0.2		VP	80
Mesitylene	120–145	82						0.24–0.25		77
3-Methyl heptane	120–145	82						0.31–0.30		77
3-Methyl hexane	120–145	82						0.33–0.34		77
n-Nonane	120–145	82						0.28		77
n-Octane	120–145	82						0.31–0.30		77
Phenol	135							1.5	IGC	79
Tetrachloromethane	135							0.24	IGC	79
Tetralin	120–145	82						0.29–0.28		77
Toluene	120–145	82						0.34		77
Trichloromethane	135							0.41	IGC	79
2,2,4-Trimethyl hexane	120–145	82						0.29–0.28		77
2,2,4-Trimethyl pentane	120–145	82						0.34		77
m-Xylene	120–145	82						0.29		77
p-Xylene	120–145	82						0.27–0.28		77
Poly(heptene)										
Toluene	30	224.1			0.48	0.45	0.43		VP	53
Poly(isobutylene)										
Benzene	10	40			0.67	0.78	0.92		VP	32
	25							0.91	IGC	30
	25	40		0.57	0.66	0.75	0.88	1.06	VP	32
	25	40						0.87	IGC	81
	25	40						0.86	IGC	83
	25	50				0.76	0.82		VP	82
	25	90	0.50	0.57					OS	21
	25–50	66						0.91–0.79	IGC	30
	25–65	40						0.86–0.61	IGC	83
	26.9					0.73	0.82	1.07	VP	84
	39.6	40				0.70	0.80		VP	32

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Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
	40	50				0.69	0.75		VP	82
	40	50				0.65	0.74	0.83	VP	85
	40–65	53						0.87–0.77		86
	60	50				0.52	0.60	0.69	VP	82
	65	50				0.73	0.74	0.75	VP	82
	65	100				0.73	0.74	0.74	VP	82
	80	50				0.4	0.49	0.58	VP	85
n-Butane	100							0.65	IGC	87
Chlorobenzene	100							0.70	IGC	87
Cyclohexane	25	26.1	0.403						OS	88
	25	40			0.42	0.42	0.41	0.43	VP	35
	25	40						0.56	IGC	81
	25	40						0.54	IGC	83
	25	50				0.33	0.37		VP	82
	25	50	0.44						OS	89
	25	66						0.43	IGC	30
	25–65	40						0.54–0.42	IGC	83
	25–50	66						0.43–0.40	IGC	30
	40–65	53						0.47–0.42		86
	40	50				0.30	0.38		VP	82
	40	50					0.45	0.52	VP	82
	60	50					0.41	0.48	VP	85
	65	50				0.33	0.38		VP	82
	100							0.39	IGC	87
2,2-Dimethylpropane	25–46	1000						0.82–0.87	VP	90
	35	1000					0.82		VP	90
Ethylbenzene	25	50				0.34	0.55		VP	82
	40	50				0.4	0.5		VP	82
n-Heptane	25	40						0.57	IGC	81
	25–65	40						0.57–0.47	IGC	83
n-Hexane	25	40						0.65	IGC	81
	25	50				0.4	0.4		VP	82
	25–65	40						0.63–0.50	IGC	83
Methyl ethyl ketone	100							1.55	IGC	87
2-Methylbutane	25–35	1000					0.65		VP	90
	35–46	1000						0.65–0.68	VP	90
n-Nonane	25	40						0.49	IGC	81
n-Octane	25	40						0.52	IGC	81
	25	40		0.44	0.48				OS	91
	25–65	40						0.52–0.43	IGC	83
n-Pentane	25	40			0.57	0.62	0.68	0.75	VP	34
	25	40						0.73	IGC	81
	25	40						0.72	IGC	83
	25	50						0.28	VP, IGC	58
	25	62	0.48	0.53	0.57	0.62	0.63	0.68	VP	33
	25	66						0.70	IGC	30
	25	1000	0.48	0.53	0.57	0.62	0.63	0.68	VP	90
	25–50	66						0.70–0.60	IGC	30
	25–65	40						0.72–0.57	IGC	83
	35	62			0.58	0.63	0.63		VP	33
	35	1000			0.62	0.62	0.62	0.62	VP	90
	40	62		0.53	0.57				VP	33
	45	62			0.56	0.62	0.63		VP	33
	55	62				0.63	0.63		VP	33
Tetrachloromethane	100							0.48	IGC	87
Toluene	100							0.60	IGC	87
Trichloromethane	100							0.78	IGC	87
Poly(isoprene)										
Acetone	25						1.27	1.8	VP	92
Benzene	10	100				0.42	0.43		VP	93
	25		0.40	0.41	0.41	0.41	0.41	0.41	OS, VP	94
	25							0.46	IGC	96
	25					0.43	0.43		VP	95
	25	100	0.39	0.40	0.41	0.41	0.42	0.43	VP	93

Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
	25–55							0.46–0.43	IGC	96
	40						0.41	0.44	IGC	96
	40	100					0.41	0.44	VP	93
Ethyl acetate	25					0.82	0.96	1.24	VP	97
	25				0.69	0.81	0.96	1.23	VP	92
	50				0.68	0.77	0.91		VP	97
	50				0.68	0.79	0.92		VP	92
Ethylbenzene	25–55							0.34–0.30	IGC	96
n-Heptane	25–55							0.51–0.49	IGC	96
n-Hexane	25–55							0.54–0.50	IGC	96
Methyl ethyl ketone	25					0.86	1.05	1.43	VP	92
	45					0.83	0.99	1.20	VP	92
2-Methylheptane	25–55							0.50–0.47	IGC	96
2-Methylhexane	25–55							0.52–0.50	IGC	96
2-Methylpentane	25–55							0.56–0.52	IGC	96
n-Octane	25–55							0.49–0.46	IGC	96
n-Pentane	25–55							0.61–0.53	IGC	96
Tetrachloromethane	23.5	100				0.20	0.10		VP	50
Toluene	25–55							0.36–0.32	IGC	96
	30	270			0.46	0.42	0.38		VP	98
Trichloromethane	23.5	100				0.25	0.16		VP	50
2,2,4-Trimethylpentane	25–55							0.49–0.46	IGC	96
p-Xylene	25–55							0.28–0.26	IGC	96
Poly(methyl acrylate)										
Acetone	100							0.4	IGC	54
Benzene	90–110	200						0.51–0.46	IGC	99
n-Butane	100							1.86	IGC	54
1-Butanol	100							0.79	IGC	54
n-Butylbenzene	90–110	200						1.15–1.05	IGC	99
tert-Butylbenzene	90–110	200						1.03–0.95	IGC	99
n-Butylcyclohexane	90–110	200						2.34–2.13	IGC	99
Cyclohexane	90–110	200						1.72–1.51	IGC	99
cis-Decalin	90–110	200						2.07–1.84	IGC	99
trans-Decalin	90–110	200						2.13–1.90	IGC	99
n-Decane	88–110	200						2.68–2.43	IGC	99
Dichloromethane	23.5	63.2				−0.22	−0.43		VP	74
1,4-Dioxane	100							0.20	IGC	54
n-Dodecane	90–110	200						3.00–2.75	IGC	99
Ethanol	100							1.01	IGC	54
Ethyl acetate	100							0.43	IGC	54
Ethylbenzene	90–110	200						0.83–0.75	IGC	99
n-Heptane	100							2.10	IGC	54
n-Hexane	100							2.08	IGC	54
Methyl ethyl ketone	100							0.40	IGC	54
Naphthalene	100–110	200						0.49–0.47	IGC	99
n-Nonane	100							2.4	IGC	54
n-Octane	90–110	200						2.38–2.19	IGC	99
n-Pentane	100							1.92	IGC	54
Propane	100							2.5	IGC	54
n-Tetradecane	90–110	200						3.36–3.06	IGC	99
Tetrahydrofuran	100							0.34	IGC	54
Tetralin	90–110	200						1.04–0.95	IGC	99
3,3,4,4-Tetramethyl-hexane	90–110	200						2.17–1.95	IGC	99
Toluene	90–110	200						0.67–0.62	IGC	99
Trichloromethane	23.5	63.2				−0.53	−0.58		VP	74
3,4,5-Trimethylheptane	90–110	200						2.44–2.20	IGC	99
2,2,5-Trimethylhexane	90–110	200						2.48–2.21	IGC	99
2,2,4-Trimethylpentane	90–110	200						2.35–2.07	IGC	99
Poly(methyl methacrylate)										
1,2-Dichloroethane	23.5	33.2			−0.63	−0.7	−0.74		VP	74
	50	5.5			0.25	0.17	0.32		VP	40
	50	8.43			0.14	0.21	0.33		VP	40

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Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
	70	5.5			0.06	0.21	0.33		VP	40
	70	8.43				0.31	0.28		VP	40
3,3-Dimethyl-2-butanone	50	3.15			0.59	0.45	0.34		VP	100
	50.2	8.56			0.55	0.55	0.6		VP	100
	69.3	3.15				0.63	0.46		VP	100
	70.3	8.56				0.57	0.74		VP	100
Mesitylene	130	8.43				0.85	0.99		VP	40
Tetrachloromethane	50	3.16			0.56	0.56			VP	100
	50.2	8.56				0.6	0.65		VP	100
	69.7	3.15				0.57	0.62		VP	100
	70.7	8.56				0.53	0.66		VP	100
Toluene	23.5	33.2					0.45		VP	74
	48.5	19.77			0.62	0.53	0.45		VP	101
	49.8	3.16				0.60	0.69		VP	100
	50	36.41				0.50	0.51		VP	40
	50	72				0.51	0.53		VP	40
	50.1	8.56				0.58	0.73		VP	100
	69.6	3.16			0.53	0.53	0.63		VP	100
	70	36.41				0.45	0.51		VP	40
	70	72				0.44	0.50		VP	100
	70.4	8.56				0.51	0.71		VP	100
	160	110						0.89	VP	102
p-Xylene	50	8.43			0.71	0.67	0.89		VP	100
	70	8.43				0.89	0.84		VP	100
	100	8.46				0.77	0.73		VP	100
	136.2	8.46				0.46	0.63		VP	100
Poly(*n*-butyl acrylate)										
Benzene	23.5	33				0.17	0.02		VP	74
Toluene	23.5	33			0.12	0.05			VP	74
Trichloromethane	23.5	33			−0.32	−0.39			VP	74
Poly(*n*-butyl methacrylate)										
Acetone	120	35						0.93	IGC	103
Acetonitrile	120	35						1.40	IGC	103
Benzene	50.5	36.8			0.21	0.14			VP	40
1,2-Dichloroethane	50.8	36.8			0.41	0.37			VP	40
Ethanol	40	8.9	0.509	0.57	0.68	0.86	1.09	1.38	VP	104
	60	8.9	0.484	0.55	0.66	0.8	0.99	1.19	VP	104
	80	8.9	0.477	0.53	0.63	0.76	0.89	1.01	VP	104
Ethyl acetate	120	35						0.67	IGC	103
n-Hexane	120	35						0.78	IGC	103
Mesitylene	100	36.8				0.41			VP	40
Methyl ethyl ketone	50.5	36.8				0.3	0.44		VP	40
	120	35						0.62	IGC	103
1-Propanol	120	35						1.01	IGC	103
2-Propanol	27	8.9	0.492	0.54	0.65	0.82	1.03	1.29	VP	104
	56	8.9	0.432	0.49	0.59	0.72	0.88	1.11	VP	104
	86	8.9	0.399	0.46	0.54	0.64	0.77	0.95	VP	104
Toluene	50.2	36.8			0.24	0.17			VP	40
Trichloromethane	50.5	36.8			0.01				VP	40
	120	35						0.10	IGC	103
m-Xylene	70.8	36.8				0.48	0.4		VP	40
o-Xylene	71.3	36.8				0.26	0.27		VP	40
Poly(*o*-chlorostyrene)										
Benzene	25	607			0.56	0.58	0.60	0.62	VP	105
	40	607				0.56	0.62	0.67	VP	105
Methyl ethyl ketone	40	607					0.83	1.02	VP	105
Poly(octadecene)										
Toluene	30	193.5		0.50	0.51	0.74			VP	53
Poly(*p*-bromostyrene)										
Toluene	20	10			0.64	0.65			VP	106

Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
Poly(*p*-chlorostyrene)										
Toluene	22	10		0.53	0.58	0.55			VP	106
Poly(*p*-methyl styrene)										
Toluene	20	10		0.41	0.4	0.35			VP	106
Poly(propylene glycol)										
Benzene	47.2	500		0.16	0.03	−0.03	−0.07		VP	107
	74.7	500		0.15	0.08	0.02	−0.02		VP	107
Methanol	25	3.35		0.45	0.48				VP	108
Poly(propylene)										
Acetone	100							1.72	IGC	87
Benzene	100							0.51	IGC	87
Cyclohexane	100							0.22	IGC	87
2,4-Dimethyl-3-pentanone	25	20		0.55	0.61	0.7	0.86	1.11	VP	109
	45	20		0.57	0.57	0.64	0.79	1.01	VP	109
n-Hexane	80							0.18		29
n-Octane	100							0.22	IGC	87
3-Pentanone	25	20			0.70	0.82	1.08	1.66	VP	109
	45	20			0.70	0.71	0.97	1.55	VP	109
Tetrachloromethane	100							0.29	IGC	87
Tetrahydrofuran	100							0.55	IGC	87
Toluene	100							0.43	IGC	87
Trichloromethane	100							0.61	IGC	87
Poly(styrene)										
Acetic acid	162–229	20						3.00–2.15	IGC	110
Acetone	25	15.7				0.81	0.94	1.1	VP	111
	50	15.7				0.80	0.92		VP	111
	150	17.5						0.65	IGC	112
	162–229	20						1.30–0.56	IGC	110
Acetonitrile	162–229	20						2.02–0.93	IGC	110
Aniline	162–229	20						1.11–0.68	IGC	110
Benzaldehyde	162–229	20						1.22–0.80	IGC	110
Benzene	20	20			0.57	0.39	0.21		VP	113
	20	500			0.49	0.46	0.44		VP	113
	23.5	600				0.30	0.19		VP	114
	25	90		0.42	0.37	0.32	0.26		VP	37
	30	19.2				0.41	0.3		VP	115
	30	103.8				0.54	0.35		VP	115
	40	218				0.26	0.28		VP	116
	50	19.2				0.45	0.26		VP	115
	50	103.8				0.53	0.21		VP	115
	60	218				0.25	0.31		VP	116
	70	19.2				0.32	0.27		VP	115
	70	103.8				0.40	0.35		VP	115
	80	218				0.14	0.23		VP	116
	120							0.33	IGC	78
	133–210	200						0.32–0.20	IGC	117
	162–229	20						0.66–0.13	IGC	110
	140–200	76						0.59–0.60	IGC	118
	150–200	97						0.38	IGC	30
	130–155	275						0.31–0.28	VP	102
	160–180	120						0.29–0.24	IGC	99
Benzyl alcohol	162–229	20						1.42–0.65	IGC	110
1-Butanol	162–229	20						1.47–0.82	IGC	110
2-Butanol	162–229	20						1.71–0.81	IGC	110
n-Butyl acetate	162–229	20						1.01–0.45	IGC	110
tert-Butyl acetate	10	11	0.50	0.54	0.64				CM, LS	120
	30	11	0.50	0.54	0.64				LS, CM	119
	50	11	0.49	0.53	0.63				CM, LS	120
	143–183	11						0.73–0.52	CM	120
n-Butylbenzene	183–203	120						0.38–0.35	IGC	99
n-Butylcyclohexane	160–180	120						0.77–0.71	IGC	99

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Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
Chlorobenzene	162–229	20						0.68–0.28	IGC	110
Cyclohexane	20	500				0.68	0.92		VP	113
	30			0.57					CM	25
	30	19.2				0.78	0.77		VP	115
	30	100				0.79	0.98		VP	121
	30	103.8				0.87	0.89		VP	115
	30	150		0.47	0.65	0.71			VP	122
	30	218				0.69	0.92	1.14	VP	123
	34	25.1	0.50	0.57	0.64	0.75	0.93		OS, VP	22
	35	51		0.56					LS	23
	35	233	0.50	0.56	0.64	0.75	0.91	(1.15)	HSGC	124
	40	100				0.75	0.92		VP	121
	40	218				0.76	0.92		VP	116
	44	25.1	0.49	0.57	0.63	0.72	0.93		OS, VP	22
	45	51		0.55					LS	23
	45	233	0.49	0.55	0.63	0.73	0.85	(1.02)	HSGC	124
	50	100				0.66	0.82		VP	121
	50	103.8				0.52	0.73		VP	115
	50	150		0.34	0.55	0.72			VP	122
	55	233	0.48	0.53	0.59	0.66	0.73	(0.82)	HSGC	124
	60	100				0.62	0.78		VP	121
	60	218				0.72	0.89		VP	116
	65	51		0.53					LS	23
	65	233	0.48	0.51	0.55	0.60	0.65	(0.70)	HSGC	124
	70	19.2				0.65	0.65		VP	115
	70	103.8				0.57	0.71		VP	115
	80	218				0.68	0.8	0.92	VP	116
	150	17.5						0.47	IGC	112
	140–200	76						1.05–0.85	IGC	118
	162–229	20						1.11–0.46	IGC	110
	160–180	120						0.62–0.53	IGC	99
Cyclohexanone	162–229	20						1.08–0.49	IGC	110
cis-Decalin	183–203	120						0.47–0.42	IGC	99
trans-Decalin	183–203	120						0.52–0.46	IGC	99
n-Decane	140–200	76						1.51–1.13	IGC	118
	183–203	120						1.01–0.94	IGC	99
Di(*n*-propyl)ether	20	500			0.71	0.81	0.91		VP	113
o-Dichlorobenzene	133–210	200						0.59–0.30	IGC	117
1,2-Dichloroethane	162–229	20						0.85–0.22	IGC	110
Dichloromethane	23.5	600				0.24	0.10		VP	114
	162–229	20						0.62–0.21	IGC	110
Diethyl ether	162–229	20						0.78–0.71	IGC	110
Diisopropyl ether	162–229	20						1.42–0.41	IGC	110
Dimethylformamide	140–200	76						1.29–1.16	IGC	118
1,4-Dioxane	20	200				0.35	0.35		VP	113
	20	500				0.44	0.44		VP	113
	50	10.3			0.67	0.61	0.78	1.10	VP	101
	133–171	200						0.52–0.39	IGC	117
	162–229	20						0.95–0.42	IGC	110
n-Dodecane	183–203	120						1.09–1.00	IGC	99
Ethanol	162–229	20						1.80–0.43	IGC	110
Ethyl acetate	150	17.5						0.46	IGC	112
	162–229	20						1.14–0.35	IGC	110
Ethylbenzene	30	230			0.40	0.40			VP	123
	50	19.2			0.88	0.66	0.43		VP	125
	50	103.8				0.75	0.91		VP	125
	70	19.2				0.74	0.65		VP	125
	70	103.8				0.71	0.83		VP	125
	130	275		0.23	0.28	0.33	0.39	0.43	VP	102
	140	275						0.46	VP	102
	150	17.5						0.27	IGC	112
	120–185							0.22–0.14	IGC	29
	133–210	200						0.40–0.23	IGC	117
	140–200	76						0.57–0.58	IGC	118
	160–180	275						0.32–0.25	VP	102
Ethylene glycol	162–229	20						3.77–2.23	IGC	110

Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
Formamide	162–229	20						4.11–3.16	IGC	110
n-Heptane	140–200	76						1.49–1.14	IGC	118
	162–229	20						1.33–0.25	IGC	110
n-Hexadecane	183–203	120						1.22–1.14	IGC	99
n-Hexane	162–229	20						1.35–0.03	IGC	110
Methanol	162–229	20						2.19–0.44	IGC	110
Methyl cyclohexane	72	110	0.49	0.58	0.67				OS	126
Methyl ethyl ketone	25	290			0.63	0.69	0.77		VP	127
	48.5	10.3			0.63	0.67	0.73		VP	101
	70	290				0.63	0.72		VP	127
	120–200	35						1.13–0.9	IGC	128
	133–171	200						0.80–0.67	IGC	117
	162–229	20						1.16–0.36	IGC	110
Naphthalene	183–203	120						0.13–0.11	IGC	99
Nitrobenzene	162–229	20						1.18–0.72	IGC	110
n-Nonane	130–175	53.7						1.00–0.97	VP	51
n-Octane	150	17.5						1.24	IGC	112
n-Octane	162–229	20						2.19–0.80	IGC	110
i-Octane	162–229	20						1.72–0.35	IGC	110
1-Octanol	162–229	20						1.14–0.55	IGC	110
n-Pentane	162–229	20						1.12–0.83	IGC	110
1-Pentanol	162–229	20						1.75–0.86	IGC	110
3-Pentanone	20	200			0.63	0.65	0.95		VP	113
	20	500				0.65	0.87		VP	113
1-Propanol	120–200	9						1.54/1.38/1.4	IGC	128
	162–229	20						1.71–0.27	IGC	110
2-Propanol	162–229	20						1.74–0.15	IGC	110
Propyl acetate	25	290			0.66	0.66	0.66		VP	111
	70	290			0.61	0.6	0.59		VP	111
Pyridine	162–229	20						1.02–0.23	IGC	110
Styrene	133–210	200						0.23/0.28/0.23	IGC	117
Tetrachloromethane	20	500				0.37	0.37		VP	113
	23.5	600					0.31		VP	114
	133–171	200						0.49–0.19	IGC	117
	162–229	20						0.90–0.26	IGC	110
n-Tetradecane	183–203	120						1.14–1.08	IGC	99
Tetrahydrofuran	162–229	20						0.70–0.16	IGC	110
Tetralin	183–203	120						0.20–0.11	IGC	99
3,3,4,4-Tetramethyl-hexane	160–180	120						0.90–0.76	IGC	99
Toluene	20	200				0.52	0.34		VP	113
	20	500				0.41	0.33		VP	113
	22	10		0.40	0.42	0.39			VP	106
	23.5	600				0.39	0.25		VP	114
	25	51		0.37					LS	23
	25	90		0.37	0.30	0.23	0.16		VP	37
	25	100	0.44	0.41	0.40				SAXS	129
	25	170	0.43						OS	130
	25	290			0.42	0.37	0.31		VP	127
	40	218				0.36	0.37		VP	116
	45	51		0.37					LS	23
	48.5	10.3		0.35	0.34	0.33	0.32		VP	101
	49.8	45				0.45	0.53		VP	40
	60	218				0.35	0.37		VP	116
	65	51		0.37					LS	23
	69.5	45				0.57	0.57		VP	40
	80	290			0.40	0.35			VP	127
	100	103.8				0.48	0.3		VP	131
	110	275		0.52	0.40	0.37	0.35		VP	102
	130	275					0.21	0.18	VP	102
	133–210	200						0.32–0.21	IGC	117
	162–229	20						0.67–0.04	IGC	110
	140–200	76						0.57–0.56	IGC	118
Trichloroethane	23.5	600				0.07	−0.1		VP	114
Trichloroethylene	162–229	20						0.69–0.12	IGC	110

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Solvent	T (°C)	M_2 (kg/mol)	χ (0.0)	χ (0.2)	χ (0.4)	χ (0.6)	χ (0.8)	χ (1.0)	Method	Refs.
Trichloromethane	25	290		0.52	0.32	0.23	0.17		VP	111
	50	290		0.45	0.34	0.23	0.14		VP	111
	120–200	9						1.4–1.48	IGC	128
	162–229	20						0.43–0.01	IGC	110
Water	162–229	20						4.40–3.10	IGC	110
m-Xylene	100	19.2				0.61	0.52		VP	40
o-Xylene	100	19.2				0.48	0.43		VP	40
	100	103.8				0.53	0.43		VP	40
	100	1538				0.52	0.39		VP	40
	133–210	200						0.38–0.25	IGC	117
	162–229	20						0.72–0.26	IGC	110
m-Xylene	130–175	53.7						0.39–0.35	VP	51
Poly(*tert*-butyl methacrylate)										
Benzene	50	12				0.39	0.36		VP	40
	69.5	12				0.31	0.36		VP	40
Ethylbenzene	69.5	12				0.62	0.69		VP	40
Mesitylene	100	12				0.70	0.71		VP	40
Methyl ethyl ketone	50	12				0.33	0.52		VP	40
	69.5	12				0.43	0.62		VP	40
3-Pentanone	50	12				0.63	0.58		VP	40
Tetrachloromethane	50	12				0.42	0.42		VP	40
Toluene	100	12				0.47	0.44		VP	40
	50	12				0.44	0.44		VP	40
	69.6	12				0.37	0.44		VP	40
Trichloromethane	50	12			0.25	0.03			VP	40
m-Xylene	69.5	12				0.67	0.62		VP	40
o-Xylene	69.5	12				0.55	0.63		VP	40
p-Xylene	69.5	12				0.42	0.57		VP	40
Poly(vinyl acetate)										
Acetone	25		0.454						SAXS	132
	30	8.6				0.38	0.48		VP	133
	30	109.1			0.31	0.38	0.55		VP	133
	30	170					0.33	0.31	VP	134
	30	1041				0.41	0.55		VP	133
	40	170					0.34	0.35	VP	134
Vinyl acetate	30	110			0.41	0.29	0.22		VP	135
Poly (vinyl methyl ether)										
Acetone	150	68.5						0.54	IGC	112
Cyclohexane	150	68.5						0.47	IGC	112
Ethyl acetate	150	68.5						0.21	IGC	112
Ethylbenzene	150	68.5						0.10	IGC	112
n-Octane	150	68.5						0.97	IGC	112
Poly (vinyl methyl ketone)										
n-Decane	83–163	500						3.91–3.54	IGC	136
n-Dodecane	113–173	500						3.89–3.16	IGC	136
n-Heptane	73–163	500						3.56–3.05	IGC	136
n-Nonane	73–163	500						3.48–3.20	IGC	136
n-Octane	73–163	500						3.55–3.11	IGC	136
n-Undecane	83–163	500						3.86–3.68	IGC	136

J. REFERENCES

1. P. J. Flory, J. Chem. Phys., **10**, 51 (1942).
2. P. J. Flory, J. Chem. Phys., **12**, 425 (1944).
3. P. J. Flory, "Principles of Polymer Chemistry", Cornell University Press, Ithaca, 1953.
4. M. L. Huggins, J. Chem. Phys., **46**, 151 (1942).
5. M. L. Huggins, J. Am. Chem. Soc., **64**, 1712 (1942).
6. M. L. Huggins, "Physical Chemistry of High Polymers", Wiley, New York, 1958.
7. K. H. Meyer, Z. Phys. Chem., **44**, 383 (1939).
8. R. A. Orwoll, Rubber Chem. Technol., **50**, 451 (1977).
9. A. F. M. Barton, "Handbook of Solubility Parameters and Dher Cohesion Parameters", CRC Press, Boca Raton, 1983.

10. H. M. Petri, N. Schuld, B. A. Wolf, Macromolecules, **28**, 4975 (1995).
11. P. J. Flory, Disc. Faraday Soc., **49**, 7 (1970).
12. P. J. Flory, J. Chem. Phys., **17**, 223 (1949).
13. C. E. Anagnostopoulos, A. Y. Coran, H. R. Gamrath, J. Appl. Polym. Sci., **4**, 181 (1960).
14. P. J. Flory, Rehner, Jr., J. Chem. Phys., **11**, 521 (1943).
15. P. J. Flory, J. Chem. Phys., **18**, 108 (1950).
16. P. Doty, H. S. Zable, J. Polym. Sci., **1**, 90 (1946).
17. A. F. Blanchard, P. M. Wootton, J. Polym. Sci., **34**, 627 (1959).
18. G. M. Bristow, W. F. Watson, Trans. Faraday Soc., **54**, 1742 (1958).
19. C. M. Kok, A. Rudin, J. Appl. Polym. Sci., **27**, 353 (1982).
20. M. B. Huglin (Ed.), "Light Scattering from Polymer Solutions", Academic Press, London (1972).
21. P. J. Flory, H. Daoust, J. Polym. Sci., **25**, 429 (1957).
22. W. R. Krigbaum, D. O. Geymer, J. Am. Chem. Soc., **81**, 1859 (1959).
23. T. G. Scholte, Eur. Polym. J., **6**, 1063 (1970).
24. R. Koningsveld, L. A. Kleintjens, A. R. Shultz, J. Polym. Sci. Part A-2, **8**, 1261 (1970).
25. R. Koningsveld, L. A. Kleintjens, Macromolecules, **4**, 637 (1971).
26. C. A. Haynes, R. A. Beyon, R. S. King, H. W. Blanch, J. M. Prausnitz, J. Chem., Phys., **93**, 5612 (1989).
27. O. Smidsrod, J. E. Guillet, Macromolecules, **2**, 272 (1969).
28. D. Patterson, Y. B. Tewari, H. P. Schreiber, J. E. Guillet, Macromolecules, **4**, 356 (1971).
29. N. F. Brockmeier, R. W. McCoy, J. A. Meyer, Macromolecules, **5**, 464 (1972).
30. R. D. Newman, J. M. Prausnitz, J. Chem. Phys., **76**, 1492 (1972).
31. P. J. Flory, J. Am. Chem. Soc., **65**, 372 (1943).
32. B. E. Eichinger, P. J. Flory, Trans. Faraday Soc., **64**, 2053 (1968).
33. C. H. Baker, W. B. Brown, G. Gee, J. S. Rowlinson, D. Stubley, Y. E. Yeadon, Polymer, **3**, 215 (1962).
34. B. E. Eichinger, P. J. Flory, Trans. Faraday Soc., **64**, 2066 (1968).
35. B. E. Eichinger, P. J. Flory, Trans. Faraday Soc., **64**, 2061 (1968).
36. R. S. Chahal, W.-P. Kao, D. Patterson, J. Chem. Soc., Faraday Trans. 1, **69**, 1834 (1973).
37. I. Noda, Y. Higo, N. Ueno, T. Fujimoto, Macromolecules, **17**, 1055 (1984).
38. W. R. Moore, R. Shuttleworth, J. Polym. Sci., Part A, **1**, 733 (1963).
39. W. R. Moore, R. Shuttleworth, J. Polym. Sci., Part A, **1**, 1985 (1963).
40. C. Wohlfarth, Physical Sciences Data Vol. 44, Elsevier (1990).
41. E. C. Baughan, A. L. Jones, K. Stewart, Proc. Roy. Soc., London, Ser. A, **225**, 478 (1954).
42. A. L. Jones, Trans. Faraday Soc., **52**, 1408 (1956).
43. J. S. Aspler, D. G. Gray, Polymer, **23**, 43 (1982).
44. J. S. Aspler, D. G. Gray, Macromolecules, **12**, 562 (1979).
45. S. G. Canagrata, D. Margerison, J. P. Newport, Trans. Faraday Soc., **63**, 3058 (1966).
46. I. Noda, N. Kato, T. Kitano, et al., Macromolecules, **14**, 668 (1981).
47. A. Hamdouni, J. Leonard, V. T. Bui, Polymer Commun., **31**, 258 (1990).
48. J. I. Eguiazabal, M. J. Fernandez-Berridi, J. J. Iruin, J. M. Elorza, Polym. Bull., **13**, 463 (1985).
49. P. Alessi, A. Cortessi, P. Sacomani, E. Vallés, Macromolecules, **26**, 6175 (1993).
50. S. Saeki, J. C. Holste, D. C. Bonner, J. Polym. Sci., Polym. Phys. Ed., **20**, 793 (1982).
51. Y. Iwai, Y. Arai, J. Chem. Eng., Japan, **22**, 155 (1989).
52. G. Charlet, R. Ducasse, G. Delmas, Polymer, **22**, 1190 (1981).
53. P. J. T. Tait, P. J. Livesey, Polymer, **11**, 359 (1970).
54. P. Munk, P. Hattam, Q. Du, et. al., J. Appl. Polym. Sci.: Appl. Polym. Symp., **45**, 289 (1990).
55. E. Dolch, M. Glaser, A. Heintz, H. Wagner, R. N. Lichtenthaler, Ber. Bunsenges. Phys. Chem., **88**, 479 (1984).
56. P. J. Flory, H. Shih, Macromolecules, **5**, 761 (1972).
57. M. J. Newing, Trans. Faraday Soc., **46**, 613 (1950).
58. R. N. Lichtenthaler, D. D. Liu, J. M. Prausnitz, Ber. Bunsengesell., **78**, 470 (1974).
59. W. R. Summers, Y. B. Tewari, H. P. Schreiber, Macromolecules, **5**, 12 (1972).
60. A. J. Ashworth, G. J. Price, Macromolecules, **19**, 358 (1986).
61. A. J. Ashworth, G. J. Price, Thermochim. Acta, **82**, 161 (1984).
62. N. Kuwahara, T. Okazawa, M. Keneko, J. Polym. Sci., Part C, **23**, 543 (1968).
63. R. W. Brotzman, B. E. Eichinger, Macromolecules, **15**, 531 (1982).
64. A. J. Ashworth, et. al., Macromolecules, **17**, 1090 (1984).
65. W. E. Hammers, C. L. Ligny, J. Polym. Sci., Polym. Phys. Ed., **12**, 2065 (1974).
66. T. Shiomi, Z. Izumi, F. Hamada, A. Nakajima, Macromolecules, **13**, 1149 (1980).
67. A. J. Ashworth, G. J. Price, J. Chem. Soc., Faraday Trans. I, **81**, 473 (1985).
68. D. W. Scott, J. Am. Chem. Soc., **68**, 1877 (1946).
69. A. Muramoto, Polymer, **23**, 1311 (1982).
70. R. C. Osthoff, W. T. Grubb, J. Am. Chem. Soc., **76**, 399 (1954).
71. K. Sugamiya, N. Kuwahara, M. Kaneko, Macromolecules, **7**, 66 (1974).
72. A. J. Ashworth, G. J. Price, Macromolecules, **19**, 362 (1986).
73. C. Panayiotou, J. H. Vera, Polym. J., **16**, 89 (1984).
74. S. Saeki, J. Holste, D. C. Bonner, J. Polym. Sci. Polym. Phys, Ed., **21**, 2049 (1983).
75. C. Wohlfarth, W. Zschoch, M. T. Rätzsch, Acta Polymerica, **32**, 616 (1981).
76. C. Booth, C. J. Devoy, Polymer, **12**, 309 (1971).
77. H. P. Schreiber, Y. B. Tewari, D. Patterson. J. Polym. Sci., Polym. Phys. Ed., **11**, 15 (1973).
78. N. F. Brockmeier, R. W. McCoy, J. A. Meyer, Macromolecules, **5**, 130 (1972).

79. G. DiPaola-Baranyi, J. E. Guillet, H.-E. Jeberien, J. Klein, Makromol. Chem., **181**, 215 (1980).
80. J. H. van der Waails, J. J. Hermans, Rec. Trav. Chim. Pays Bas., **69**, 971 (1950).
81. Y.-K. Leung, B. E. Eichinger, J. Phys. Chem., **78**, 60 (1974).
82. H. Masuoka, N. Murashige, M. Yorizane, Fluid Phase Equil., **18**, 155 (1984).
83. Y.-K. Leung, B. E. Eichinger, Macromolecules, **7**, 685 (1974).
84. R. S. Jessup, J. Res. Nat. Bur. Stand., **60**, 47 (1958).
85. N.-H. Wang, S. Takashima, H. Masuoka, Kagaku Kogaku Ronbunshu, **15**, 313 (1989).
86. R. N. Lichtenthaler, D. D. Liu, J. M. Prausnitz, Macromolecules, **7**, 565 (1974).
87. Q. Du, P. Hattam, P. Munk, J. Chem. Eng. Data, **35**, 367 (1990).
88. Y. Nakamura, K. Akasaka, K. Katayama, T. Norisuye, A. Teramoto, Macromolecules, **25**, 1134 (1992).
89. P. J. Flory, J. Am. Chem. Soc., **65**, 372 (1943).
90. S. Prager, E. Bagley, F. A. Long, J. Am. Chem. Soc., **75**, 2742 (1953).
91. P. J. Flory, J. L. Ellenson, B. E. Eichinger, Macromolecules, **1**, 279 (1968).
92. C. Booth, G. Gee. G. Holden, G. R. Williamson, Polymer, **5**, 343 (1964).
93. B. E. Eichinger, P. J. Flory, Trans. Faraday Soc., **64**, 2035 (1968).
94. G. Gee. J. Chem. Soc., **5**, 280 (1947).
95. G. Gee, J. B. M. Herbert, R. C. Roberts, Polymer, **6**, 541 (1965).
96. Y. B. Tewari, H. P. Schreiber, Macromolecules, **5**, 329 (1972).
97. C. Booth, G. Gee, G. R. Williamson, J. Polym. Sci., **23**, 3 (1957).
98. K. H. Meyer, E. Wolff, C. G. Boissonnas, Helv. Chim. Acta, **23**, 430 (1940).
99. G. DiPaola-Baranyi, J. Guillet, Macromolecules, **11**, 228 (1978).
100. E. Regener, C. Wohlfarth, M. T. Rätzsch, S. Höring, Acta Polymerica, **39**, 618 (1988).
101. P. J. T. Tait, A. M. Abushihada, Polymer, **18**, 810 (1977).
102. J. S. Vrentas, J. L. Duda, S. T. Hsieh, Ind. Eng. Chem., Process Des. Dev., **22**, 326 (1983).
103. D. J. Walsh, J. G. McKeown, Polymer, **21**, 1330 (1980).
104. S. P. Nunes, B. A. Wolf, Macromolecules, **20**, 1948 (1987).
105. K. Gekko, K. Matsumura, Bull. Chem. Soc. Japan, **46**, 1554 (1973).
106. R. Corneliussen, S. A. Rice, H. Yamakawa, J. Chem. Phys., **38**, 1768 (1963).
107. C. Booth, C. J. Devoy, Polymer, **12**, 320 (1971).
108. M. L. Lakhanpal, B. E. Conway. J. Polym. Sci., **46**, 75 (1960).
109. W. B. Brown, G. Gee, W. D: Taylor, Polymer, **5**, 362 (1964).
110. S. Gündüz, S. Dincer, Polymer, **21**, 1041 (1980).
111. C. E. H. Bawn, M. A. Wajid, Trans. Faraday Soc., **52**, 1658 (1956).
112. S. Klotz, R. H. Schuster, H.-J. Cantow, Makromol. Chem., **187**, 1491 (1986).
113. E. C. Baughan, Trans. Faraday Soc., **44**, 495 (1948).
114. S. Saeki, J. C. Holste, D. C. Bonner, J. Polym. Sci., Poly. Phys. Ed., **19**, 307 (1981).
115. M. T. Rätzsch, M. Opel, C. Wohlfarth, Acta Polymerica, **31**, 218 (1980).
116. N.-H. Wang, S. Takashima, H. Matsuoka, Kagaku Kogaku Ronbunshu, **15**, 795 (1989).
117. F. H. Covitz, J. King, J. Polym. Sci. A, **10**, 689 (1972).
118. M. Galin, M. C. Rupprecht, Polymer, **19**, 506 (1978).
119. B. A. Wolf, H. J. Adam, J. Chem. Phys., **75**, 4121 (1981).
120. K. Schotsch, B. A. Wolf, H.-E. Jeberien, J. Klein, Makromol. Chem., **185**, 2169 (1984).
121. Y. C. Bae, J. J. Shim, D. S. Soane, J. M. Prausnitz, J. Polym. Sci., **47**, 1193 (1993).
122. K. Schmoll, E. Jeckel, Z. Elektrochemie, **60**, 756 (1956).
123. T. Katayama, K. Matsumura, Y. Urahama, Kagaju Kogaku, **35**, 1012 (1971).
124. H. M. Petri, B. A. Wolf. Macromolecules, **27**, 2714 (1994).
125. M. T. Rätzsch, G. Illig, C. Wohlfarth, Acta Polymerica, **33**, 89 (1982).
126. K. Kamide, K. Sugamiya, T. Kawai, Y. Miyazaki, Polym. J., **12**, 67 (1980).
127. C. E. H. Bawn, R. F. J . Freeman, A. R. Kamaliddin, Trans. Faraday Soc., **46**, 677 (1950).
128. C. Uriate, M. J. Fdez.-Berride, J. M. Elorza, J. J. Iruin, L. Kleintjens, Polymer, **30**, 1493 (1989).
129. S. Kinugasa, H. Hayashi, F. Hamada, A. Nakajima, Macromolecules, **18**, 582 (1985).
130. C. E. Bawn, E. S. Hill and M. A. Wajid, Trans. Faraday Soc., **52**, 1651 (1958).
131. G. Illig, Dissertation, TH Leuna-Merseburg, 1981.
132. A. M. Hecht, F. Horkay, E.Geissler, J. P. Benoit, Macromolecules, **24**, 4183 (1991).
133. K. Mathsumara, T. Katayama, Kagaku Kogaku, **38**, 388 (1974).
134. R. J. Kokes, A. R. DiPietro, F. A. Long, J. Am. Chem. Soc., **75**, 6319 (1953).
135. A. Nakajima, H. Yamakawa, I. Sakurada, J. Polym. Sci., **35**, 489 (1959).
136. Zeki Y. Al-Saigh, Poly. Int., **40**, 25 (1996).

Concentration Dependence of the Viscosity of Dilute Polymer Solutions: Huggins and Schulz–Blaschke Constants

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A. INTRODUCTION

The dependence of the viscosity of dilute polymer solutions on concentration can be described by a power series:

$$\eta = \eta_0(1 + a_1c + a_2c^2 + a_3c^3 + \cdots) \qquad \text{(A1)}$$

where η is the viscosity of the solution and η_0 is the viscosity of the solvent. This equation is usually presented in another form:

$$\eta = \eta_0(1 + [\eta]c + k_1[\eta]^2c^2 + k_2[\eta]^3c^3 + \cdots) \qquad \text{(A2)}$$

where $[\eta]$ is the limiting viscosity number and $k_i(k_1, k_2, \cdots)$ is a dimensionless constant. If terms with the third and higher powers of concentration are neglected, then Eq. (A2) becomes

$$\eta = \eta_0(1 + [\eta]c + k_1[\eta]^2c^2) \qquad \text{(A3)}$$

which, since $\eta/\eta_0 = \eta_{rel}$ (viscosity ratio) and $\eta_{rel} - 1 = \eta_{sp}$ (specific viscosity), can be put in the form

$$\eta_{sp} = [\eta]c + k_1[\eta]^2c^2 \qquad \text{(A4)}$$

which becomes the Huggins equation (1)

$$\eta_{sp}/c = [\eta](1 + k_h[\eta]c) \qquad \text{(A5)}$$

where k_h is the Huggins viscosity constant that is the most commonly used dilute solution viscosity number or index. It is easily determined from the slope of a plot of η_{sp}/c versus c.

The Huggins constant can be thought of as a measure of the "goodness" of the solvent for the polymer with values around 0.3 in good solvents and 0.5–1 in theta solvents. It should be noted that these generalizations break down with extremely dilute solutions (2) and with chain aggregation (3). In both cases, values tend to be higher. The Huggins constant is independent of molecular weight for many polymers, but is dependent on molecular weight for polymers that associate easily in solution (3).

The Huggins equation and Huggins constants can be used to determine values for the limiting viscosity number, $[\eta]$. For extrapolation procedures related to $[\eta]$ and k_h, see (Ref. 4). Applications of the Huggins equation to measure $[\eta]$ are given in Refs. (5,6). The latter gives an equation for $[\eta]$:

$$[\eta] = \frac{(1 + 4k_h\eta_{sp})^{1/2} - 1}{2k_hc} \qquad \text{(A6)}$$

which can be used for single point determinations, i.e., calculating $[\eta]$ from a single viscosity measurement at a known concentration. Computer programs relating solution viscosity and molecular parameters are found in Refs. (7,8).

Other semiempirical equations, such as those of Kraemer and Schulz and Blaschke, also are used for processing viscosity data. The Kraemer equation (Ref. 9) is

$$\ln \eta_{rel}/c = [\eta](1 - k_k[\eta]c) \qquad \text{(A7)}$$

where k_k is the Kraemer constant. Sometimes, the term $k_k[\eta]c$ is written as positive and the constants are listed as negative numbers. The result is the same. The Schulz–Blaschke equation (10) is

$$\eta_{sp}/c = [\eta](1 + k_{sb}\eta_{sp}) \quad \text{(A8)}$$

Although few Schulz–Blaschke constants have been published, some investigators contend that the Schulz–Blaschke equation is superior because it works with a wider range of concentrations and is not bound to a constant specific viscosity region (11). On the other hand, it is not recommended for single point determinations because of large errors (6). Huggins' and Kraemer's equations, used singly or in combined form, are better (6).

In addition to the polymer–solvent system, k_h and k_{sb} depend on concentration (whether semidilute, dilute, or extremely dilute), molecular weight, degree of branching, shear rate of measurement, and whether there can be preferential solvation (copolymers and mixed solvents). With polymers in good solvents, measured values for the Huggins constant are between 0.3 and 0.4. In some cases, they are even lower (2). With decreasing goodness of the solvent, the Huggins constant increases steadily, reaching 0.55 or more for theta conditions.

Regarding concentration, deGennes (12) has defined a critical concentration, c^*, above which solutions are semidilute and below which they are dilute. Most k_h and k_{sb} values have been measured in the dilute region. Dondos (2) has defined another critical concentration, c^{**}, below which the solution is called extremely dilute. Values for k_h measured below c^{**} tend to be higher than those measured between c^* and c^{**}. It is interesting that Peterson and Fixman (13) predicted higher values for Huggins constants except for those below c^{**}. It may be that the low values in the dilute region are due to polymer coil compression because of lack of sufficient volume (2).

The Huggins constant is not supposed to be dependent on the molecular weight of the polymer. In many cases, it is not (14–18), or the dependence is small (14,19,20), but other results show definite molecular weight dependence (3,21). It appears that the Huggins constant is related to molecular weight in polymers that associate in solution, either by the effect of strong ionic or polar interactions, or by the effect of hydrogen bonds (3).

There is a relationship between k_h and the coil expansion factor or coefficient of expansion, α, which is a measure of the extent of the expansion of the polymer coil in a particular solvent. In good solvents, the coil is more extended than in poor solvents and α is correspondingly larger. The Huggins constant seems to vary with α^3 (3,14,22,23). One expression for this is (Ref. 23)

$$k_h = a + b\alpha^3 + c \quad \text{(A9)}$$

Although there is no agreement in the literature as regards the influence of branching on viscosity constants, most results indicate that k_h and k_{sb} increase with higher degrees of branching (24–26). There is no agreement on the effect of shearing on viscosity constants either, the literature indicating both increases (27) and decreases (28) with increasing shear rate. Shearing that produced chain scission lowered the limiting viscosity number $[\eta]$ and raised k_h, and did so more at higher shear rates, although there was a tendency toward stabilization at the highest shear rates (29). The influence of shear rate is usually neglected below 1500 sec^{-1} (11).

Huggins constants also can be determined for dilute suspensions. The k_h value for such a suspension is sensitive to the degree and nature of the polydispersity of the specimen (30).

The following tables contain Huggins and Schulz–Blaschke constants taken from the literature, along with the relevant references. The values are listed by limiting viscosity number $[\eta]$, the units of which are ml/g. Values marked with an asterisk (*) are interpolated values from the literature (mostly mean values calculated by Stickler and Sutterlin (11)). For information on molecular weight and other factors, as well as conditions of measurement not given in the tables, the reader may refer to the original publications.

B. TABLES

TABLE 1. HUGGINS CONSTANTS

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
1.1. POLY(DIENES) AND POLY(ALKENES)						
Poly(1,3-butadiene), *cis*	Toluene	25				31
	Phenyloctane	25		451	0.46	32
		50		447	0.46	
	Dioctyl phthalate	16.5		147	0.95	
		21.5		162	0.77	
Poly(isoprene) *cis*	Toluene	25				31
Poly(isoprene) stars	Cyclohexane	25	Three-armed stars	35.4	0.33	33
			Four-armed stars	37.6	0.35	
			Three-armed stars	185	0.4	
			Four-armed stars	195	0.46	

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
	Toluene	34	Chlorosilane-linked stars	14–780	0.35–0.88	33
	1,4-Dioxane	34		14–164	0.63–9.2	
	Toluene	34	Divinylbenzene-linked stars	82–195	0.50–1.02	33
	1,4-Dioxane	34		30–70.5	0.71–1.19	
Natural rubber	Benzene	30		354	0.32	34
	n-Hexane	30		170	0.35	34
Natural rubber (epoxidized)	Tetrahydrofuran	25?	1% Epoxy	415	0.84	35
			9.5% Epoxy	218	0.43	
			18% Epoxy	30	4.27	
Gutta percha	Toluene	25				31
Poly(chloroprene)	Benzene	25		10	0.58*	36
				30–120	0.34 ± 0.06*	
				150	0.43	
				200	0.52*	
Poly(ethylene)	Tetralin	80				37
	p-Xylene	100	Low pressure	38	0.71	38,39
				97	0.45	
				327	0.69	
	Tetralin	80	Low pressure			40
	p-Xylene	81	Low pressure	35	0.38	38,39
				71	0.27	
				141	0.39	
				153	0.65	
	p-Xylene	81	High pressure	159	0.89	38,39
Poly(isobutene)	Benzene	105	High pressure			39,41
		20		24	0.54	14
				107	0.87	
		24		128	0.49*	42
		25		27	0.49	14
				100	0.59	
				183	0.7	
		30		30	0.49*	14
				100	0.57*	
				240	0.70*	
		40		30–320	0.42* ± 0.02	14
Poly(isobutene)	Carbon tetrachloride	30		14–42	0.36	43
	Chlorobenzene	25		155	0.36	14
	Cyclohexane	25		19	0.38 ± 0.02	44
				54	0.35 ± 0.04	
				384	0.30 ± 0.01	
	Cyclohexane	30		309	0.32	45
				484	0.19	
	Cyclohexane	30	Ref. also includes data for 10	29	0.41	46
			20, 40, 50 C	84	0.34	
				261	0.33	
				389	0.31	
	Dibutyl ether	25		165	0.33	14
	Diisobutylene	25	$M_v = (16–926) \times 10^3$		0.40 ± 0.05	47
	Heptane	25		185	0.3	14
	Isooctane	30	Ref. also includes data for 10,	22	0.57	46
			20, 40, 50 C; only small	56	0.43	
			variation in k across range	157	0.37	
				219	0.42	
		30	Ref. also includes data for	8.4	0.94	48
			20, 40 C; only small	16.7	0.58	
			variation in k across range	35.7	0.47	
	Tetrachloroethane	25		276	0.27	14
	Tetradecane	25		189	0.33	14
	Toluene	25		171	0.36	14
Poly(propylene)	Benzene	30	Atactic	30	0.43	49
				95	0.42	50
				150	0.42	49
	Carbon tetrachloride	50	Atactic	253	0.25	50
	Chlorobenzene	30	Atactic	165	0.33	50
	Chloroform	30	Atactic	196	0.31	50

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TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
	α-Chloronaphthalene	139	Atactic			51
	Cyclohexane	30	Atactic	45–330	0.27 ± 0.02	49
				276	0.24	50
		60–70	Isotactic		0.36	52
	Decalin	30	Atactic	277	0.25	50
		135	Isotactic			51
	2-Ethyl hexyl acetate	30	Atactic	120	0.38	50
	n-Heptane	30	Atactic	205	0.34	50
	Isoamyl acetate	34	Atactic	26	0.45	49
				65	0.65	
				98	0.83	
	Isopropyl ether	30	Atactic	156	0.36	50
	Tetrahydrofuran	30	Atactic	176	0.32	50
	Tetralin	30	Atactic	155	0.36	50
		130	Atactic			51
	Toluene	30	Atactic	30–120	0.36* ± 0.02	49
				190–220	0.40* ± 0.03	
				180	0.35	51
	p-Xylene	125	Atactic			51
1.2. POLY(ACRYLIC ACID) AND POLY(METHACRYLIC ACID) DERIVATIVES						
Poly(acrylic acid)	Water (0.1 M NaCl)	25				53
	Water (0.25 M NaCl)	25	Degree of neutr. = 0	20	0.35	54
			Degree of neutr. = 0.3	150	0.25	
			Degree of neutr. = 0.76; data also for 5, 15, 35, 45 C	240	0.25	
Poly(acrylonitrile)	Dimethyl formamide	25				55
		30		70–190	0.3	56,57
				157	0.4	55,58
				106	0.28	59
				134–296	0.31	
				170–245	0.31*	60
				309–342	0.34 ± 0.1	59
				239–711	0.33	56
				117	0.35	61
		40, 50, 60, 70				60
		60				62
Poly(ethyl acrylate)	Acetone	30		46–177	0.35	63,64
		30	Telomers			65
Poly(decyl methacrylate)	Isooctane	25	Ref. also includes data for 50, 75, 100 C	34	0.55	66
				36	0.45	
				38	0.35	
				40	0.25	
Poly(ethyl methacrylate)	Benzene	35		145	0.29	67
		52		148	0.29	
		60		145	0.36	
	n-Butanol	45		47	0.7	67
		52		55	0.54	
		60		64	0.45	
		70		72	0.4	
	n-Butyl bromide	42		160	0.24	67
		50		149	0.4	
		58		159	0.3	
		65		166	0.25	
	n-Butyl chloride	40		142	0.32	67
		47		144	0.3	
		61		147	0.26	
	Methyl acetate	20–53	Conformational changes occurred in this range	16–50	~0.5 to ~−0.3	68
	Ethyl acetate	35		133	0.31	67
				120–970	0.25–0.27	69
		45		134	0.28	67
		55				
		65				

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
		20–65	Conformational changes occurred in this range	12–48	~0.45 to ~−0.1	68
	Isobutyl acetate	20–65	Conformational changes occurred in this range	13–45	~0.5 to ~−0.15	68
	Isoamyl acetate	50		145	0.39	67
		70		152	0.37	
		80		156	0.32	
	Isopropanol	37		88	1.06	67
		44		129	0.27	
		51		151	0.39	
		60		183	0.43	
	Methyl ethyl ketone	23–45		157–158	0.34–0.35	67
	Methyl *n*-propyl ketone	40		196	0.2	67
		50–70		183–189	0.33 ± 0.01	
	m-Xylene	47		131	0.39	67
		54		134	0.33	
		61		137	0.33	
		70		139	0.29	
Poly(methacrylic acid)	Water (0.1 M NaCl)	25				53
	Water (0.25 M NaCl)	25	Degree of neutr. = 0	30	2.33	54
			Degree of neutr. = 0.1	35	0.65	
			Degree of neutr. = 0.3	240	0.33	
			Degree of neutr. = 0.81; also data for 5, 15, 35, 45 C	575	0.3	
Poly(methacrylonitrile)	Dimethyl formamide	29	Anionic	26	0.4	70
				60–143	0.34* ± 0.03	
Poly(methyl acrylate)	Acetone	30		95–298	0.38	71
				333	0.37	72
		40		314	0.38	72
	Benzene	25		52–343	0.37	71
		30		383	0.31	72
		35		50–335	0.38	71
		40		382	0.38	72
		45, 55		48–330	0.38	71
	Carbon tetrachloride/ methanol	35	Vol. fraction of methanol: 0.33 ([η] maximum, k minimum)	25–214	2.94 ± 0.06	73
	Chlorobenzene	25		53–305	0.4	71
	Chloroform	30		220	0.41	74
		35		215	0.39	
		40		222	0.36	
	Ethyl acetate	30, 35, 40		115–120	0.50 ± 0.01	74
	Methyl ethyl ketone	25		45	0.65	75
				85	0.56	
				116	0.39	
		30		108	0.45	73
				302	0.43	72
		35, 40		104	0.44	73
	Toluene	25		34–147	0.59	71
		30		61	0.59	73
				201	0.33	72
		35		34–160	0.52	71
		40		69		73
		45		35–169		71
		55		35–175		71
Poly(methyl methacrylate)	Acetone	25				76
		25?			0.35	82
	Benzene	20		85	0.25	77
		25		4	1.65*	78
				9.5	0.69*	
				14	0.5	
				22	0.35	
				26–40	0.30*	
				68–331	0.25 ± 0.01	
				667	0.18	

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TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
			90% Isotactic	41	0.39	14
				49–61	0.33	
				73–125	0.3* ± 0.03	
	Butyl acetate	25		7	0.9	78
				29	0.67	
				64	0.58	
				110	0.47	
			90% Isotactic	22	0.67	14
				39	0.61	
				52	0.57	
	n-Butyl bromide	35		47	0.5	67
		50		63	0.41	
		58		70	0.35	
	Chloroform	25		19	0.32	
				146	0.25	
				223–275	0.38	79
				390	0.22*	
		30, 60		870	0.18	80
	Cyclohexanone	40, 60		138, 152	0.44	67
		50		146	0.45	
		70		136	0.4	
	Dimethyl formamide	25		56	0.48	14
					0.38	82
	Ethyl acetate	25		53	0.41	14
	Isoamyl acetate	50		36	0.71	67
		65		44	0.63	
		80		47	0.5	
	Methyl ethyl ketone	25		7	0.9	78
				27	0.49	
				56	0.4	14
				137	0.33	78
				151	0.4	75
				172	0.49	
		30, 60				80
	Di-*n*-propyl ketone	33.8	Star-branched polymer	11.4	1.66*	81
				17	1.48	
		35–50				78
	Tetrachloroethane	25		101	0.29	14
	Tetrahydrofuran	30	Star-branched polymer	24	0.161*	81
	Toluene	25		40	0.49*	78
				9	0.63	
				24	0.46	
				55	0.41	14
				83	0.33	78
				229	0.25	
				414	0.23	
		30		37	0.43 ± 0.04	44,80
				80	0.41 ± 0.03	
		60				80
	o-Xylene	40		68	0.34	67
		50		72	0.21	
		60		76	0.32	
		70		81	0.27	
	m-Xylene	40		55	0.48	67
		50		62	0.67	
		60		71	0.55	
		70		80	0.35	
	p-Xylene	40		42	1.31	67
		50		48	1.26	
		60		59	0.93	
		70		64	0.47	
Poly(hexyl methacrylate)	*n*-Heptane	30		31	0.5	83
Poly(heptyl methacrylate)	*n*-Heptane	30		45	0.44	83
Poly(octyl methacrylate)	*n*-Heptane	30		64	0.45	83
Poly(tetradecyl methacrylate)	*n*-Heptane	30		58	0.32	83

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
Poly(cyclobutyl methyl methacrylate)	1-Butanol	40		13	1.49	84
				20	1.16	
				36	1.33	
	Tetrahydrofuran	30		22	0.36	84
				57	0.31	
				142	0.3	
Poly(cyclohexyl methyl methacrylate)	1-Butanol	45		10	0.74	84
				17	0.81	
				41	0.62	
	Cyclohexane	30		17	0.18	84
				54	0.43	
				91	0.31	
	Tetrahydrofuran	30		16	0.52	84
				35	0.35	
				105	0.36	
Poly(cyclooctyl methacrylate)	1-Hexanol	38		10	0.94	84
				16	1.46	
				23	1.77	
	Cyclohexane	30		17	0.26	84
				38	0.39	
	Tetrahydrofuran	30		25	0.4	84
				76	0.37	
Poly(cycloundecyl methacrylate)						
	Cyclohexane	30		9	0.69	84
				18	0.49	
				36	0.38	
	Tetrahydrofuran	30		9	0.7	84
				17	0.55	
				36	0.37	
Poly(diphenyl methyl methacrylate)	Toluene	25		19	0.4	85
				95	0.34	
				230	0.35	
	3-Heptanone	45		13	0.76	85
				39	0.73	
				73	0.61	
Poly(2,6-diisopropyl phenyl methacrylate)	Tetrahydrofuran	25		22.6–98.0	0.42	224
	Toluene	25		34.0–88.4	0.41	
	Tetrahydrofuran/water (90.9/9.1, v/v)	25	Theta solvent	33.9–54.1	0.61	
Poly(2,3-epoxypropyl methacrylate)	Tetrahydrofuran	25		11.6	0.79	86
				24.7	0.47	
				28.2	0.52	
	1,4-Dioxane	25		13.5	0.32	86
				16.5	0.55	
				18.5	0.45	
				23	0.41	
				62.7	0.37	
Poly(pentachlorophenyl methacrylate)	Benzene	40		13–42	1.25	87
	Toluene	25		22–87	0.48	
	Ethyl benzene	25		14–23	1.17	
	o-Xylene	25		29–155	0.42	
	Chlorobenzene	25		29–148	0.43	
	o-Dichlorobenzene	25		33–178	0.36	
Poly(2-biphenyl methacrylate)	Chloroform	25		11–111	0.34	88
	Tetrahydrofuran	25		11.0–99	0.33	
	Benzene	25		11.0–98	0.35	
	1,4-Dioxane	25		10.0–91	0.35	
Poly(4-biphenyl methacrylate)	Chloroform	25		11–133	0.29	89
	Tetrahydrofuran	25		11–111	0.3	
	Benzene	25		9.0–64	0.4	
	1,4-Dioxane	25		10.0–87	0.33	

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TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
Poly(di-isobornyl methacrylate)	Tetrahydrofuran	30		25.3–104	0.23 ± 0.01	90
	1-Octanol	39.6		13.1	1.29	
				23	1.13	
Poly(acrylamide)	Water	25		1300–1700	0.35 av.	6
				274	0.16	91
				1353	0.5	
	Water (1 M NaCl)	25		1200–1900	0.33 av.	6
	Water (pH 7)	30		21	0.46	92
1.3. VINYL POLYMERS						
Poly(4-isopropyl styrene)	Toluene	25		44–139	0.34 ± 0.04	93
Poly(4-isopropyl α-methyl styrene)	Toluene	21.5		16.5	0.52	94
				25.5	0.3	
				44	0.42	
				60	0.33	
Poly(styrene)	Benzene	25		105	0.37	95
		25	Atactic	73.5	0.36	96
		25	Atactic	1.7	0.79	109
				5.5	0.71	
				10.2	0.54	
	Benzene/isopropanol	25	Theta temperature, atactic	40.4	0.54	
	Butyl acetate	25		43	0.38	14
				126	0.35	
				150	0.33	
	4-*tert*-Butyltoluene	50	Atactic	7, 13.7	0.57	97
				82.9, 132	0.38	
				187, 432	0.36	
	Carbon tetrachloride	25				98
		30	Ref. also includes data for 10, 20, 40, 50 C	14.5	0.55	46
				43	0.41	
				180	0.33	
				386	0.29	
	Chloroform	30, 60				81
	Cyclohexane	30	Ref. also includes data for 15, 20, 25, 32, 40, 50 C	10.7–60.7	0.84 ± 0.03	46
		$34.5 = \Theta$		10.8–114	0.77 ± 0.06	
		31, 95	Trifunctional star molecule	42	0.6	95
		34		16	0.5	99
				48	0.52	
		34.5	$c > c^{**}$		0.48	2
			$c < c^{**}$		0.78	
		34.5		178	0.61	100
		34.5	$k = 0.5$ for MW $> 2.1 \times 10^5$; $k > 0.5$ for MW $< 2.1 \times 10^5$			101
		34.5		1.5	1.3	109
				2.5	0.91	
				3.7	0.96	
				5.4	0.93	
				6.6	0.81	
				9	0.67	
		34	Branched	29–31	$0.86^* \pm 0.02$	102
		34.3, 34.8	Trifunctional star molecule	43, 44	0.59	95
		34.5	Atactic	97	0.59	97
		35				103
		36		152	0.51	14
		40		174	0.49	14
		40, 22	Trifunctional star	47	0.53	95
		45	$c < c^{**}$		1.28	2
		50		55	0.50 ± 0.005	99
	cis-Decaline	20		20	$0.41^* \pm 0.02$	14
				60	$0.43^* \pm 0.02$	
				120	$0.45^* \pm 0.02$	
	trans-Decaline	20		38	0.6	14
				80	0.71	

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
		25		50	0.49* ± 0.01	14
				130	0.51* ± 0.02	
		30		47	0.4	14
				117	0.42	
	Diethyl malonate	34	Theta temperature	45	0.47	14
				82	0.5	
		40		45	0.5	14
				83	0.47	
	Dimethyl formamide	25		19	0.5	14
				36	0.44	
				76–172	0.39	
		30				57
		35		12.2	0.48	104
				77.2	0.4	
				188	0.41	
		95		20–700	0.48	105
	Dioctyl phthalate	25		49	0.59	14
				89	0.58	
		35		51	0.57	14
				86	0.5	
		45		45	0.56	14
				54–87	0.5	
	1,4-Dioxane	25	$c < c^{**}$		0.81	2
			$c > c^{**}$		0.35	
	Ethyl benzene	25	Ref. also includes data for	13.4	0.503	46
			10, 40, 55, 65 C	26.1	0.423	
				41.7	0.365	
				384	0.291	
	Methyl ethyl ketone	25				98
		30				57,81
		40				64
		40.3		44–151	0.54 ± 0.09	106
		60				81
	1-Methyl naphthalene	25				107
	Toluene	15	Atactic	292	0.34	97
		21		40–125	0.35 ± 0.03	99
				360	0.30 ± 0.05	
				460	0.25 ± 0.08	
		25				98,107
		25	Anionic	8	0.51	39,108
				14	0.44	
				36–272	0.34 ± 0.03	
		25	Thermal	41–123	0.34 ± 0.02	39,108
			Branched	56	0.41	102
				71	0.38	
			Linear	63–84	0.31	102
		30		14	0.46 ± 0.02	39,81,83
				73–234	0.38 ± 0.015	106
				28	0.41 ± 0.02	83
				98	0.39	95
				203	0.34 ± 0.01	83
		30		12.6	0.56	104
				28.1	0.43	
				64	0.38	
				362	0.32	
		60				81
	Toluene/methanol (77/23)	25	Theta temperature	28–80	0.51–0.52	23
Poly(vinyl acetate)	Acetone	18		68	0.33	110
		25		30	0.46	39,64,110
				42–66	0.35*	
				91	0.37	
				134	0.41	
		25		3.7	1.54	111,112
				6.6	1.13	
				8.4	0.79	

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TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
				11.6, 12.9	0.53	
				20.8	0.38	
				27.1	0.34	
		30		70	0.35* ± 0.02	113
				250	0.37* ± 0.02	
				450	0.39 ± 0.02	
		32, 39, 46		65, 64, 63	0.41, 0.42, 0.44	110
	Benzene	30		126	0.17	28
				50–450	0.37 ± 0.03	113
				147	0.25	28
				197	0.3	
				262	0.45	
				281	0.65	
		30	Zero shear plot	124–381	0.34	
			Emulsion polymer	77	1.42	113
				133	1.21	
				229	1.05	
		35	Branched	45	0.28	115
				72–200	0.35* ± 0.02	
				300	0.37* ± 0.02	
				500	0.39* ± 0.02	
				700	0.42* ± 0.02	
		35	Linear	137–177	0.33* ± 0.01	115
				196–250	0.34* ± 0.01	
				281–570	0.35* ± 0.01	
		40		204	0.25	28
				243	0.28	
				268	0.36	
				354	0.44	
			Zero shear plot	197–413	0.42	
	Chlorobenzene	18		71	0.44	110
		25		37–45	0.35* ± 0.03	39,110
				66–69	0.44* ± 0.01	
				99–121	0.35* ± 0.04	
		32, 39, 46		68, 67, 65	0.43, 0.44, 0.4	110
	2-Chloroethylene	25				64
	Chloroform	18		101	0.34	110
		25		45	0.31	39,110
				62–92	0.37* ± 0.02	
				146	0.28	
		32, 39, 46		99, 95, 92	0.37, 0.39, 0.41	110
	Dioxane	18		77	0.27	110
		25		75	0.29	39,110
		32, 39, 46		74, 73, 72	0.31, 0.32, 0.34	110
	Methanol	18		41	0.61	110
		25		30	0.41* ± 0.04	39,110
				40	0.49* ± 0.03	
				50	0.57* ± 0.03	
		30				116
		32, 39, 46		44, 45, 47	0.52, 0.41, 0.39	110
	Toluene	18		46	0.57	110
		25		46	0.55	39,110
		30				116
		32, 39, 46		47, 47, 48	0.56, 0.56, 0.54	110
	Water	25	Partially alcoholized			117
			72.5% Hydrolyzed, unfract.	21	0.67	118
			Middle fractions	20	0.98	
			77.9% Hydrolyzed, unfract.	9.5	2.27	
			Middle fractions	10	3.87	
			73.3% Hydrolyzed, unfract.	20.2	1.03	
			Middle fractions	20.6	1.2	
Poly(vinyl alcohol)	Dimethyl sulfoxide	30				119
	DMSO/water (4/1, w/w)	30	No cooling	122	0.43	225
			Precooled to form clusters	77–99	0.75 ± 0.03	

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
			Precooled to form clusters	100	1.1	
			Precooled to form clusters	132	1.8	
	Water	25			0.5	120,139
	Water	30		30	0.57*	119,121
		30(?)		42.1	0.59	122
				71, 77, 86.5	0.69	
				96	0.71	
				165	0.81	
		30		40	0.59*	119,121
		30		40	0.9	123
		30		50	0.60*	121
				60	0.66*	
				70–80	0.68*	
				100	0.74*	
				170	0.83*	
	Water (dil. NaOH)	30	0.042 M, 0.084 M NaOH	88, 85	0.65, 0.7	114
			0.167 M NaOH	78	1.6	
	Water (boric acid/	30	0.005 M Boric acid	22	25	114
	0.167 M NaOH)		0.01 M	28	16	
			0.02 M	42	5.5	
			0.06 M	84	1.8	
	Water	60		32	1.26	123
				59	0.93	
Poly(vinyl chloride)	Acetone	20		36	0.42	14
	Anisole	25		34	0.46	14
	Bromobenzene	25		40	0.4	14
	Butyl acetate	25		43	0.38	14
	γ-Butyrolactone	25		34	0.43	14
	Chlorobenzene	25		39	0.42	14
	Chloroform	20		38	0.49	14
	Cyclohexanone	20				64,124
		25		90–130	0.34	125,126
		30		100–121	0.35	127,128,129
		20–60		23–132		130
		80	Samples with different tacticity	94	0.36	131
				108	0.43	
				287	0.57	
				338	0.64	
				438	0.48	
	Cyclopentanone	20		46	0.33	14
	1,2-Dichloroethane	25		41	0.4	14
	Dimethyl formamide	25		40	0.3	14
	Dioxane	25		33	0.5	14
				40	0.48*	
				50	0.44*	
				60	0.41*	
				70	0.38*	
	Ethyl acetate	25		39	0.43	14
	Ethyl acetyl acetate	25		34	0.43	14
	Methyl ethyl ketone	25		26	0.34	14
				32	0.35	
				47–83	0.38 ± 0.01	
	Morpholine	30		71	0.64	132
				89	0.57	
				108	0.49	
		50		64	0.81	132
				79	0.67	
				98	0.53	
		69		58	0.85	132
				70	0.7	
				87	0.55	
	Nitrobenzene	25		41	0.33	14
	Tetrachloroethane	25		29	0.42	14
				36–76	0.37* ± 0.01	
				93	0.35	

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TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
	Tetrahydrofuran	20	100% Conversion	20–143	0.28* ± 0.02	133
				49	0.35	14
				171	0.31	133
				201–220	0.36* ± 0.02	
		30		76–91	0.44* ± 0.01	132
				107	0.47	
	Tetrahydrofurfuryl alcohol	25		44	0.35	14
	Tetramethylene sulfoxide	25		46	0.46	14
Poly(vinyl cyanoethyl ether)	Water	30				121
	Acetone	30				121
Poly(vinyl biphenyl)	Benzene	20		7	0.55	134
				22–38	0.51* ± 0.04	
				93–134	0.34* ± 0.01	
		30		7	0.56	134
				21–37	0.51* ± 0.03	
				92–133	0.35* ± 0.01	
		45		21–37	0.54* ± 0.02	134
				91–129	0.36* ± 0.01	
		65		7	0.51	134
				21–36	0.51* ± 0.12	
				91–126	0.36* ± 0.01	
		75		21–36	0.43* ± 0.01	134
				90–122	0.36* ± 0.01	
Poly(vinyl butyral)	Methanol	25		30	2.69	135
	Ethanol	25		63.2	0.49	135
	n-Propanol	25		75.6	0.58	135
	Isopropanol	25		53.9	1.01	135
	n-Butanol	25		85.2	0.201	135
	Isobutanol	25		57.2	2.29	135
	Cyclohexanone	25		75.1	0.43	135
Poly(vinyl formate)	Acetone	30		50	0.34	136
				113	0.54	
	Acetonitrile	30		48	0.3	136
				162	0.51	
	Dioxane	30		44	0.3	136
				134	0.55	
	Methyl acetate	30		62	0.42	136
				74	0.46	
	Methyl formate	30		48	0.3	136
				107	0.46	
Poly(*N*-vinyl imidazole)	Water (0.2 M NaBr)	25	Degree of neutr. = 0	106	0.63	137
			Degree of neutr. = 0.3	131	0.5	
			Degree of neutr. = 0.8	206	0.36	
Poly(1-vinyl naphthalene)	Benzene	20		16	0.46	134
				27	0.45	
				37	0.41* ± 0.03	
		30		15	0.47	134
				27	0.48	
				37	0.43* ± 0.02	
		45		15	0.61	134
				26	0.57	
				36	0.43* ± 0.02	
		65		14	0.75	134
				25	0.46	
				36	0.44* ± 0.01	
		75		14	0.78	134
				25	0.6	
				35	0.49* ± 0.01	
Poly(2-vinyl naphthalene)	Benzene	20		18	0.5	134
				46	0.41	
				67–107	0.35* ± 0.01	
		30		18	0.5	134
				45	0.43	
				86	0.38	
				106	0.36	

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
		45		17	0.58	134
				43	0.45	
				63–103	0.39* ± 0.01	
		65		17	0.51	134
				20	0.43	
				43–101	0.38* ± 0.06	
		75		17	0.48	134
				43–100	0.38* ± 0.06	
Poly(vinyl phenylketone)	Benzene	25				112
Poly(4-vinylpyridine)	Ethanol	25				138
Poly(vinyl pyrollidone)	Dimethyl formamide	25			0.45	139
	Methanol	25		33		140
	Methanol + 0.33 N GC	25	GC; guanidine hydrochloride	30.7		
	Ethanol	25		34		
				28		
	Propanol	25		34		
				29		
	Water	15		17.2	0.45	141
		25		15	0.62	
		35		13.3	0.77	
		45		11.5	0.81	
		55		10.6	0.95	
		65		1.25	1.3	
	Water (salt soln.)	25	1.2 M NaH_2PO_4	15.5	0.97	226
		35	Theta temperature	14.7	1.99	
		25	0.5 M Na_2SO_4	16.9	0.85	
		36	Theta temperature	12.9	2.16	
		25	0.5 M Na_2CO_3	16.2	1.07	
		39	Theta temperature; Ref. contains data for seven other salts	15.5	2.6	
Poly(vinyltrimethylsilane)	Cyclohexane	25		96–154	0.35 ± 0.03	142
Poly(vinyl 4-trimethylsilyl benzoate)	Toluene	25		136	0.19	143
Poly(vinyl 3-trimethylsilyl benzoate)	Toluene	25		214	0.54	143
Poly(vinylidene fluoride)	Dimethylacetamide	25		76	0.35	144
				116	0.37	
				143	0.32	
				189	0.27	
				325		0.42
		25	$[\eta]$ corr. for aggregation	108	0.33	145
				124	0.37	
				132	0.42	
				174	0.47	
	Dimethyl formamide	25		74	0.31	144
				128	0.36	
				175	0.39	
Poly(ethylene glycol monovinyl ether)	Water	25		~75	0.54	146
	Ethanol	25		?	0.47	
	Dimethyl formamide	25		~65	0.4	
1.4. POLY(OXIDES)						
Poly(oxyethylene)	Benzene	20		2	~3.7	147
				48	0.4	
	Chloroform	20		3	~0.9	147
				82	0.4	
	Dimethyl formamide	20		3	~0.6	147
				45	0.4	
	Toluene	30		3	~2.5	147
				39	0.4	
		35		1.1	8.16	148
				1.9	3.41	
				2.4	2.06	
				5.5	2.3	

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
	Water	10.2	$MW = 1.0 \times 10^5$	98.3	0.63	227
		18.1		91.4	0.71	
		24		84.9	0.87	
		32		77.5	0.97	
	Water	20		3	1.1	147
				5	0.4	
		25		37.7	0.39	149
		30		28.1	0.39	150
				36.5	0.46	149
				41.1	0.22	150
		35		1.7	4.95	148
				3	2.42	
				5	0.93	
				10.8	0.44	
	Water (2.40 M NaCl)	25		26.1	0.82	150
		54	Theta temperature	17	1.64	
	Water (1.5 M Na acetate)	25		28.9	0.56	149
		55	Theta temperature	20.8	0.9	
	Water (1.5 M Na formate)	25		29.5	0.57	149
		52	Theta temperature	21.2	0.98	
	Water (0.80 M KF)	25		29.5	0.56	151
		49	Theta temperature	22.7	0.76	
	Water (2.40 M KNO_3)	25		26	0.55	149
		53	Theta temperature; Refs. 149–151 include data on a number of other salts	20	0.87	
	Water (urea)	10.2		97	0.53	227
		20.1		89	0.57	
		32		80	0.72	
		10.2		116	0.43	
		16–32		109–96	0.47 ± 0.03	
Poly(oxymethylene)	*p*-Chlorophenol	60		135–270	0.3	152
	Dimethyl formamide	140		69	0.23	152
				102	0.21	
		150				152
Poly(oxypropylene)	Benzene	35				153
1.5. POLY(ESTERS)						
Poly(oxyethyleneoxy adipoyl)	Cyclohexanone	25				154
	Dimethyl formamide	25				154
Poly(bis(oxyethylene) oxyadipoyl)	Methyl ethyl ketone	25	Oligomers with different functionality			155
			Nonfunctional	3.1	0.39, 0.42	155
				3.6	0.46	
				4.3	0.42	
			Monofunctional	3.3	0.3	155
				4.2	0.34	
				4.5	0.42	
				5.2	0.32	
			Bifunctional	3.5	0.25	155
				4.1	0.32	
				4.8	0.26	
				5.3	0.22	
				5.7	0.3	
				8.7	0.25	
Poly(oxyethyleneoxy sebacoyl)	Cyclohexanone	25				154
Poly(oxyethyleneoxy terephthaloyl)	*m*-Cresol/phenol (1/1)	25				156
Poly(mono-ethyl itaconate)	Methanol (0.1 M KI)	25		40.2	0.25	157
				67.1	0.34	
				86	0.47	
	2-Ethoxyethanol	25		41.3	0.2	157
				55	0.22	
				71	0.3	

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	[η]	k_h	Refs.
1.6. POLYAMIDES						
Poly(iminoadipoylimino hexamethylene) (Nylon 66)	*m*-Cresol	20				158
	Formic acid (90%)	25		83	0.2	159
				100	0.22* ± 0.01	
				120	0.24* ± 0.02	
				140	0.27* ± 0.02	
				200	0.29* ± 0.01	
Poly(iminoisophthaloyl imino-1,2-phenylene)	Dimethyl formamide (1.18 N LiCl)	25	1,3- and 1,4-phenylene polymers also in this Ref. (in DMF/LiCl and sulfuric acid/LiCl)			160
Poly[imino(1-oxohexa methylene)] (Nylon 6)	Tricresol	25	Polymers with different end groups			161
	m-Cresol	?	Branched polymers with different end groups	27–220	2.75–0.26	162
		20–60				163
	Tricresol	25	(0.019 vol.% water)	50	1.1	161,164
				109	0.49	
				335	0.29	
	Formic acid (85%)	25		70	0.32* ± 0.08	165
				80	0.22* ± 0.06	
				85	0.18* ± 0.02	
				48	0.24	164
				86	0.12	
				460	0.13	
	(68.8%)	25		32	0.38	164
				200	0.32	
				240	0.34	166
				558	0.56	
				965	0.89	
				1340	1.31	
				73	0.24	166
				101	0.66	
				155	1.36	
				305	3.36	
	Hexafluoroisopropanol	25	(0.05 M CF_3COONa)	200–500	0.23	167
	Sulfuric acid (96%)	25		94–104	0.30* ± 0.07	165
	(93%)	25		63	0.42	164
				125	0.26	
				490	0.23	
				502	0.19	
	(80%)	25		48	0.5	164
				90	0.34	
				325	0.3	
	(62%)	25		34	1.06	164
				60	0.74	
				195	0.36	
	(60%)	25		52	0.52* ± 0.1	165
				54	0.44* ± 0.1	
				57	0.34* ± 0.06	
	(52.5%)	25		30	1.37	164
				50	0.96	
				135	0.44	
				152	0.39	
	(33%)	25				165
Poly[imino(1-oxo-undeca-methylene)]	Phenol: CCl_4 (1 : 1)	25				168
	Nylon 12	45				
		75				
Poly(iminoterephthaloyl imino-1,2-phenylene)	Sulfuric acid	25	1,3- and 1,4-phenylene polymers also in this Ref. (in sulfuric acid)			160
	Dimethyl formamide (1.18 N LiCl)	25				160
Poly(amidobenzimidazole) Aramid fibers	Sulfuric acid	25	Virgin fibers	440	0.5	169
		25	Heated to 140°C	460	0.54	
		25	Heated to 160°C	440	0.65	
		25	Heated to 200°C	500	0.6	
		25	Heated to 250°C	520	0.6	

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TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
Poly(*p*-benzanalide terephthalamide)	Sulfuric acid	25	Rod-like polymer, high MW	>100	~0.5	170
			Low MW	<10	>1.0	
Poly(caprolactam)	Sulfuric acid (80%)		Fibers	121–143	0.29* ± 0.02	171
1.7. OTHER COMPOUNDS						
Al-isopropoxypoly(vinyl butyral)	Methanol	25		32.4	1.87	135
	Ethanol	25		46.7	0.95	
	n-Propanol	25		51.9	0.87	
	Isopropanol	25		52.7	0.41	
	n-Butanol	25		58.6	0.54	
	Isobutanol	25		45.1	0.95	
	Cyclohexanone	25		51.4	0.91	
Poly(2-acrylamido-2-methyl propanesulfonic acid)	Water	25		68	0.53	228
	Water			218	0.27	
Poly(2-acrylamido-2-methyl propane sulfonamide	Water			178	0.31	228
	Water	25		480	0.25	229
Poly(epichlorohydrin)	Tetrahydrofuran	25		44–555	0.36* ± 0.03	172
	Cyclohexanone	25		51–755	0.36* ± 0.01	
Poly(*N,N'*-dimethyl(acryl amidopropyl) ammomium propane sulfonate	Water (0.5 M salt)	30	NH_4Cl	32	0.9	173
			LiCl	37	0.29	
			NaCl	39	0.16	
			KCl	40	0.15	
			$MgCl_2$	50	0.45	
			$CaCl_2$	55	0.48	
			$SrCl_2$	54	0.53	
Poly(dimethylaminoethyl methacrylate dimethyl sulfate)	Water (0.1 M salt)	30	KF	386	0.21	174
			KCl	243	0.23	
			KBr	224	0.25	
			KI	170	0.28	
			LiCl	280	0.11	
			NaCl	253	0.21	
			KCl	243	0.23	
			CsCl	247	0.5	
Poly[3-dimethyl(methacryl oyloxyethyl)ammonium propane sulfonate]	Water (KCl)	30	0.25 M	49	0.7	175
			0.30 M	55	0.55	
			0.40 M	62	0.41	
			0.50 M	74	0.41	
			0.8 M	80	0.34	
			1.50 M	93	0.29	
	Water (0.5 M)	30	LiCl	57	0.56	175
			NaCl	68	0.5	
			KCl	74	0.41	
			KBr	80	0.39	
			KI	98	0.38	
			$KClO_4$	106	0.37	
			$CaCl_2$	77	0.37	
Poly(ether imide)	Dichloromethane	10		43	0.43	176
		25		37	0.66	
	Pyridine	10		25	2.13	176
		25		23	2.44	
	N-Methyl pyrollidone	10		51	0.48	176
		25		45	0.59	
	NMP/water (96/9)	10		33	1	176
		25		34	0.8	
		50		29	1.16	
	NMP/water (95/5)	25		25	1.93	176
		50		23	2.23	
Poly(ethylene imine)	Water (1 M NaCl)	35		10.4–64.4	0.76* ± 0.02	230
Poly(ethylene oxide)urethanes	Tetrahydrofuran	25	Different hydrophobic end caps	19, 18.5, 24	0.4	177
Na-polyphosphate	Water (NaBr)	25		50–276	0.29* ± 0.04	178

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
Poly(di-*n*-hexylsilane)	Tetrahydrofuran	25		58	0.33	179
				87	0.36	
				133	0.34	
				143	0.4	
				192	0.44	
				515	0.38	
				831	0.66	
Poly(oxydimethylsilylene)	Benzene	25		251	0.44	180
	Carbon tetrachloride	25		312	0.51	180
	Chlorobenzene	25		219	0.61	180
	Cyclohexane	25		324	0.37	180
	n-Heptane	25		305	0.45	180
	Methyl ethyl ketone	25		151	0.8	180
	Tetrahydrofuran	25		319	0.39	180
	Toluene	25		62–241	0.55* ± 0.02	180
				146	0.51	181
				577–727	0.46* ± 0.01	182
				10–210	0.35–0.50	182
Poly(methylene *N,N*-dimethylpiperidinium chloride)	Water (NaCl)	20–60	Ionic strengths ranging 0.1–0.5 M			183
Poly[bis(2-phenylethoxy) phosphazene]	Tetrahydrofuran	25		8.4	0.42	231
				11.7	0.62	
				12.9	0.32	
Phenolphthalein poly(aryl ether sulfone)	Dimethyl formamide	25	$MW = 0.8 \times 10^5$	29	0.49	Bo
			$MW = 2.20 \times 10^5$	53.6	0.62	
			$MW = 4.90 \times 10^5$	89.3	0.51	
	Chloroform	25	Reference includes data from additional solvents, molecular weights	28.5	0.74	Bo
				50.5	0.89	
				82.7	0.78	
Poly(phenyl quinoxaline)	Chloroform	25?		65	0.53	184
	Chloroform/toluene (98/2)			75	0.33	
	Chloroform/toluene (95/5)			62	0.31	
	Chloroform/ethanol (95/5)			78	0.06	
Poly(sodium acrylate)	Water (NaCl)	25	0.4% NaCl	256	0.38	185
			1.30%	153	0.42	
			5.00%	77	0.51	
			10.00%	59	0.65	
	Water (0.4% NaCl)	25	3% C8 substitution	276	0.34	
	10% NaCl			78	0.48	
	0.40% NaCl		3% C12 substitution	278	0.37	
	1.30% NaCl			165	0.6	
	5.00% NaCl			64	2.2	
	10.00% NaCl			34	6.3	
Poly(styrene sulfonic acid), potassium salt	Water (KCl)	25	Ionic str., $I = 1.00$ M	8.2	0.75	186,187
			$I = 4.0 \times 10^{-1}$ M, MW 41.5	15.7	1.14	
			$I = 1.0 \times 10^{-2}$ M	27.5	3.73	
			$I = 3.0 \times 10^{-1}$ M	39	12.4	
			$I = 1.0 \times 10^{-1}$ M	65.7	20	
			$I = 1.00$ M, MW 109.5	15.7	0.76	
			$I = 1.0 \times 10^{-1}$ M	39.9	0.49	
			$I = 7.0 \times 10^{-1}$ M	83.5	2.93	
			$I = 3.0 \times 10^{-1}$ M	105	6.62	
			$I = 1.0 \times 10^{-1}$ M	148	18.7	
			$I = 3.1$ M, MW 1170	16.6	4.92	
			$I = 1.0$ M	115	0.37	
			$I = 1.0 \times 10^{-1}$ M	361	0.26	
			$I = 1.0 \times 10^{-3}$ M	3353	0.73	
			$I = 3.0 \times 10^{-4}$ M	5786	1.28	
			$I = 3.0 \times 10^{-5}$ M	9661	7.91	
Sodium poly(styrene-sulfonate)-diallyldimethyl ammonium chloride			$I = 10^{-4}$ M	4480	22.9	233
			$I = 10^{-3}$ M	2750	2.51	
			$I = 10^{-2}$ M	1150	0.39	
			$I = 0.5$ M	202	0.29	

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TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
			Ref. also gives data for copolymers with acrylamide			
Poly(thiopropylene)	Benzene	20				188
1.8. CELLULOSE, CELLULOSE DERIVATIVES, AND POLYSACCHARIDES						
Alginate(sodium)	Water ($NaCl/CuCl_2$)	30	Near gelation point	180	0.27	234
Amylose	Water (KOH)	25				189
	Water (NaOH)	25				189
	Water (KOH–NaCl)	25				189
	Water (1 N KOH)	25		139	0.19	190
		30		129	0.2	190
		35		121	0.48	190
		40		107	0.88	190
Arabinoglucuronoxylan	Water	25		41.6 ± 2.2	3.6 ± 0.6	235
Cellulose	Cd-ethylenediamine (Cadoxen)	20		339–1395	0.50 ± 0.01	191
		25		426–510	0.46* ± 0.01	192
	Cu-ethylenediamine (Cuoxen)	25		440	0.54* ± 0.02	192
				466	0.59	
				510	0.65	
	Zn-ethylenediamine (Zincoxen)	25		248–383	0.47* ± 0.09	192
	LiCl/dimethylacetamide	30	MW = 4.2 to 4.5 × 10^5	641–682	0.43–0.59	236
Cellulose acetate	Acetone	30				193
	Formic acid	30				193
	Pyridine	30				193
Cellulose diacetate		25		295	0.56	194
				196	0.8	
				185	1.3	
				155	1	
		30				
	Dimethyl formamide	25?	Subst. = 2.34 as-is reprecipitated	158	0.48	195
				145	0.87	
			Subst. = 2.34 as-is reprecipitated	230	0.32	
				200	0.72	
			Subst. = 2.55 as-is reprecipitated	199	0.29	
				195	0.35	
Cellulose triacetate	Acetic acid	25		72	0.42	196
				117	0.57	
	Chloroform	25		43	0.45	196
				72	0.43	
	Methylene chloride	15–30	Θ temp. = 27°C			197
	Tetrachloroethane	25		50	0.34	196
				82	0.38	
Cellulose acetate phthalate	Water (NaCl solns.)	35	Salt conc. = 0.002 M	90	8.64	198
			Salt conc. = 0.006 M	71	3.36	
			Salt conc. = 0.0.1 M	68	1.69	
Cellulose nitrate	Acetone	20		695–1195	0.53* ± 0.08	191
	Diethyl adipate	25				199
	Isoamyl acetate	25				200
Methyl cellulose	Water	20		400	1.2	237
Ethyl cellulose	Ethyl acetate	20		41–188	0.65 ± 0.15	191
Ethyl hydroxyethyl cellulose	Water	20		440	1	238
			0.5 mM/g SDS	175	11.3	
			1.0 mM/g	135	18	
			2.0 mM/g	125	13	
			5.0 mM/g	140	7.5	
Hydroxyethyl cellulose	Water	0		423	0.57	201
				845	0.74	
				1052	0.8	
				1218	0.87	
		15		402	0.43	201
				794	0.59	
				977	0.68	
				1130	0.75	

TABLE 1. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_h	Refs.
		20		273	1.22	237
		25		128	0.31	239
		25		381	0.3	201
				754	0.47	
				895	0.62	
				1062	0.67	
		35		370	0.22	201
				725	0.4	
				875	0.49	
				1010	0.57	
		40		100	0.75	239
		60		86	0.85	
	Water	25	pH 2.2	622	4.04	202
			pH 3.0	624	3.09	
			pH 5.0	610	2.94	
			pH 7.0	600	3.1	
			pH 9.0	674	2.39	
			pH 11.0	740	1.48	
			pH 12.0	792	1.23	
	Water (LiCl)	25	1×10^{-1} to 1×10^{-3} M	900* ± 42	0.98* ± 0.16	202
	Water (KCl	25		928* ± 14	0.90* ± 0.03	202
	Water ($CuCl_2$)	25		951* ± 1	0.86* ± 0.01	202
	Water ($MnCl_2$)	25		930* ± 15	0.98* ± 0.08	202
Degraded hydroxyethyl cellulose	Water	25	Thermal degradation in water or salt soln. at 60, 80, 100°C			202
Hydrophobically modified hydroxyethyl cellulose	Water	25		110	1.45	239
		40		91	2.22	
		60		69	3.75	
Hydroxypropyl cellulose	Water	20		134	0.83	237
Hydroxy propylmethyl cellulose	Water	20		740	0.54	
Na-carboxymethyl cellulose	Water (NaCl)	25	Subst. 0.75; 0.01, 0.1, and 0.2 M NaCl			198
	Water (1 M NaCl)	25	Subst. 0.5–2.5			203
	Water (1 N NaOH)	25	Subst. 0.5–2.5			203
	Water (NaCl)	25				204
Acidic hetero polysaccharides	0.5 M NaCl/10% cadox.	30	from *Auricularia auricula judae*	72	0.2	205
	0.5 M K, NaH phosphates			101	0.36	
β-D-Glucan	Water	30	from *Auricularia auricula judae*	4900	0.49	205
	10% Cadoxen			4050	0.41	205
	Water			265	0.21	205
	10% Cadoxen			353	0.23	205
	88% Formic acid			750	0.49	205
Lyzozyme	Water	5	Enzyme from egg-whites (glycoside hydrolase)	3.05	1.35	240
		15		2.83	1.42	
		25		2.66	1.48	
		40		2.49	1.54	
		55		2.41	1.58	
Microbial polysaccharide	Water (NaCl)	25?	Kelco S-657; 0.001 M NaCl	752	0.99	206
				838	0.62	
				1452	0.73	
				2194	0.81	
			0.1 M NaCl	680	0.7	206
				750	0.43	
				1380	0.59	
				2097	0.6	
Welan gum	Water (NaCl)	25?	Kelco S-130; 0.001 M NaCl	738	0.94	206
				1078	0.99	
				1537	1	
				2471	0.86	
			0.1 M NaCl	641	0.46	206
				984	0.55	
				1510	0.51	
				2370	0.62	
Xanthan gum	Water (0.1 M NaCl)	25	Native	10300	0.45	207
			Renatured	13200	0.46	

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TABLE 2. SCHULZ–BLASCHKE CONSTANTS

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_{sb}	Refs.
Poly(isoprene)						
Gutta percha	Toluene[a]	25				208
Poly(chloroprene)	Benzene	25		19–208	0.32	209
Poly(methylene)	*m*-Xylene[a]	120				208
Poly(ethylene)	Decalin	135	Low pressure	121–1535	0.29	210
Poly(1-butene)	Decalin	115	Stereoregular	$\eta_{sp}<1$	0.48	211
Poly(isobutene)	Diisobutylene	20				208
Poly(acrylonitrile)	Dimethyl formamide	25				208
Poly(methyl methacrylate)	Acetone	25				208
				$\eta_{sp}<1$	0.5	211,212
		32		$1/k_{sb}=6.09\log(\eta_{sp}+5)-2.05$		213
	Benzene[a]	25				208,212
	Butyl acetate	25				212
	Chloroform	20		25–480	0.3	10
	Chloroform[a]	25				208
		27.3				214
		32		$1/k_{sb}=5.41\log(\eta_{sp}+5)-1.15$		213
	Methyl ethyl ketone	25		$\eta_{sp}<1$	0.5	211
	Toluene[a]	25				208
Poly(ethyl methacrylate)	Methyl ethyl ketone	23		$\eta_{sp}<1$	0.33	211
Poly(*n*-butyl methacrylate)	Isopropanol	21.5	Theta temperature	$\eta_{sp}<1$	1	211
	Methyl ethyl ketone	23		$\eta_{sp}<1$	0.36	211
Poly(1-ethylbutyl methacrylate	Isopropanol	27.4	Theta temperature	$\eta_{sp}<1$	0.89	211
	Methyl ethyl ketone	25		$\eta_{sp}<1$	0.3	211
Poly(*n*-hexyl methacrylate)	Isopropanol	32.6	Theta temperature	$\eta_{sp}<1$	0.7	211
	Methyl ethyl ketone	23		$\eta_{sp}<1$	0.36	211
Poly(*n*-octyl methacrylate)	*n*-Butanol	16.8	Theta temperature	$\eta_{sp}<1$	0.8	211
	Methyl ethyl ketone	23		$\eta_{sp}<1$	0.46	211
Poly(*n*-lauryl methacrylate)	*n*-Butyl acetate	23		$\eta_{sp}<1$	0.38	211
	Isopropyl acetate	13	Theta temperature	$\eta_{sp}<1$	0.59	211
Poly(acrylamide)	Water	25		1400–1800	$0.20^{*}\pm0.06$	6
	Water (1 M NaCl)	25		1200–2000	$0.20^{*}\pm0.07$	6
Poly(vinyl alcohol)	Water	30		42.5–178	$0.46^{*}\pm0.02$	122
Poly(vinyl chloride)	Cyclohexanone	25	MW 69000–106000		0.28 ± 0.01	215
			reprecipitated		0.32 ± 0.03	
	Cyclohexanone[a]	25				208
		20–60		21–139		130
Poly(vinyl acetate)	Benzene	30		$1/k_{sb}=5.224\log(\eta_{sp}+4)-0.65$		213
	Toluene	25				208
Poly(styrene)	Benzene	25		1		208
		30		$1/k_{sb}=10.534\log(\eta_{sp}+3)-2.65$		213
	Butanone[a]	25				208
	Chloroform	25				208
	Cyclohexane	34				208
	Ethyl acetate	25				208
	Methyl ethyl ketone	30				214
	Toluene[a]	25				208
		30				214
	m-Xylene[a]	25				208
Poly(α-methyl styrene)	Cyclohexane	30		28	0.82	216
				34	0.99	
				52	1.05	
				68	1.02	
				78	0.77	
	Cyclohexane/ ethylacetate (50/50)	30		37	0.4	216
				56	0.29	
				88	0.37	
				119	0.38	
				164	0.39	
Poly(2-methyl-5-vinyl tetrazole)	Chloroform	20		26	0.4	217
				37	0.26	
				56	0.41	
				86	0.26	

TABLE 2. *cont'd*

Polymer	Solvent	T (°C)	Remarks	$[\eta]$	k_{sb}	Refs.
				186	0.19	
				270	0.19	
	Dimethyl formamide	20		20	0.59	217
				39	0.38	
				64	0.33	
				128	0.23	
				236	0.22	
	Methylene chloride	20		27	0.26	217
				34	0.19	
				55	0.29	
				89	0.25	
				168	0.3	
				244	0.35	
Poly(oxyethylene)	Acetonitrile	25		$M_v > 10^4$	0.22	218
	Benzene	10		$M_v > 10^4$	0.21	218
		37		$M_v > 10^4$	0.23	
	Benzyl alcohol	25		$M_v > 10^4$	0.19	218
	Carbon tetrachloride	25		$M_v > 10^4$	0.3	218
	Dimethylformamide	25		$M_v > 10^4$	0.27	218
	Dioxane	25		$M_v > 10^4$	0.27	218
	Methanol	25		$M_v > 10^4$	0.26	218
	Methyl acetate	25		$M_v > 10^4$	0.27	218
	Water	25		$M_v > 10^4$	0.17	218
Poly(oxydimethylsilylene)	Butanone[a]	25				208
	Chloroform[a]	25				208
Poly(iminoadipoylimino hexamethylene) (Nylon 66)	Formic acid (90%)	25		83	0.2	159
				100	$0.22^* \pm 0.02$	
				120	$0.24^* \pm 0.02$	
				140	$0.26^* \pm 0.02$	
				200	$0.28^* \pm 0.01$	
Poly(amide imide)	*N*-Methyl pyrollidone	30		19–87	0.3	219
Cellulose	Cadoxen	20		339–1395	0.29 ± 0.08	191
	Cuoxam	20		946–1125	0.29	220
	Cuoxen	20		190–2800	0.29	221
Cellulose acetate	Acetone	25				64
Cellulose nitrate	Acetone	20	(13.7% N)	180–500	0.29	222
				500–1000	0.34	
				1000–1400	0.37	
				1400–2000	0.4	
				695–1995	0.30 ± 0.03	191
		27				64
	Ethyl acetate	20	(13.7% N)	200–600	0.26	222
				600–1200	0.3	
				1200–1800	0.33	
				1800–2400	0.37	
Ethyl cellulose	Acetone	20, 30, 40	Subst. = 2.73	24–180	0.35, 0.33, 0.32	223
	Benzene	20, 30	Subst. = 2.73	26–450	0.36, 0.35	223
		40	Subst. = 2.73	362–418	0.34	
	Isobutanol	20, 30, 40	Subst. = 2.73	25–420	0.34, 0.33, 0.32	223
	n-Butyl chloride	20, 30, 40	Subst. = 2.73	25–470	0.33, 0.33, 0.32	223
	Ethyl acetate	20, 30, 40	Subst. = 2.73	25–450	0.29, 0.28, 0.27	223
		20		41–1888	0.31 ± 0.05	191

[a] k_{sb} is roughly calculable from the Arrhenius constant.

C. REFERENCES

1. M. L. Huggins, J. Am. Chem. Soc., **64**, 2716 (1942).
2. A. Dondos, C. Tsitsilianis, Polym. Int., **28**, 151 (1992).
3. I. Dort, Polymer (London), **29**, 490 (1988).
4. T. Sakai, J. Polym. Sci. A-2, **6**, 1659 (1968).
5. P. M. Reilly, B. M. E. van der Hoff, M. Ziogas, J. Appl. Polym. Sci., **24**, 2087 (1979).
6. M. M. Rafi'ee Fanood, M. H. George, Polymer, **28**, 2244 (1987).
7. A. Penati et al., J. Appl. Polym. Sci., **19**, 2583 (1975).
8. A. Varada Rajulu, A. Padma, J. Polym. Mater., **6**, 15 (1989).
9. E. O. Kraemer, Ind. Eng. Chem., **30**, 1200 (1938).
10. G. V. Schulz, F. Blaschke, J. Prakt. Chem., **158**, 130 (1941), **159**, 146 (1941).

11. M. Stickler, N. Sütterlin, in: J. Brandrup, E. H. Immergut (Eds.) "Polymer Handbook", 3rd Ed., Wiley-Interscience, Wiley, New York, 1989, p. VII-183.
12. P. G. de Gennes, "Scaling Concepts in Polymer Physics", Cornell Univ. Press, Ithaca, NY, 1979.
13. J. M. Peterson, M. Fixman, J. Chem. Phys., **39**, 2516 (1963).
14. M. Bohdanecký, Collect. Czech. Chem. Commun., **35**, 1972 (1970).
15. K. K. Chee, J. Appl. Polymer. Sci., **23**, 1639 (1979).
16. M. Bohdanecký, Z. Tuzar, Collect, Czech. Chem. Commun., **34**, 3318 (1969).
17. C. G. Seefried et al., J. Polym. Sci., Polym. Phys. Edn., **18**, 817 (1980).
18. J. Moacanin, J. Appl. Polym. Sci., **1**, 272 (1959).
19. N. Howard, M. B. Huglin, R. W. Richards, J. Appl. Polymer Sci., **16**, 1525 (1972).
20. P. Munk, M. T. Abijaoude, M. E. Halbrook, J. Polym. Sci., Polym. Physics Edn., **16**, 105 (1978).
21. B. Lánská et al., Eur. Polym. J., **14**, 807 (1978).
22. R. Berger, Makromol. Chem., **102**, 24 (1967).
23. A. A. Abdel-Azim, Macromol. Rapid Commun., **15**, 183 (1994).
24. C. E. H. Bawn, M. B. Huglin, Polymer (London), **3**, 615 (1962).
25. J. Blackford, R. F. Robertson, J. Polym. Sci. A, **3**, 1289, 1303 (1965).
26. J. A. Manson, L. H. Cragg, Can. J. Chem., **30**, 482 (1952); **36**, 858 (1958).
27. L. M. Hobbs, V. C. Long, Polymer (London), **4**, 479 (1963).
28. S. L. Kapur, S. Gundiah, J. Polym. Sci., **26**, 89 (1957).
29. M. Moan, A. Omari, Polym. Degrad. and Stab., **35**, 277 (1992).
30. R. B. Jones, Physica A (Amsterdam), **212**, 43 (1994).
31. G. M. Bristow, J. Polym. Sci., **62**, 168 (1962).
32. R. H. Colby et al., Macromolec., **24**, 3873 (1991).
33. B. J. Bauer et al., Macromolec., **22**, 2337 (1989).
34. S. L. Kapur, S. Gundiah, Makromol. Chem., **26**, 119 (1958).
35. N. V. Bac, L. Terlemezyan, M. Mihailov, J. Appl. Polym. Sci., **50**, 845 (1993).
36. J. Polácek, Collect. Czech. Chem. Commun., **25**, 2103 (1960).
37. L. Nicolas, J. Polym. Sci., **29**, 191 (1958).
38. Q. A. Trementozzi, J. Polym. Sci., **23**, 887 (1957).
39. O. F. Solomon, I. Z. Ciuta, Bul. Inst. Politeh. "Gheorghe Gheorgie-Dej". Bucuresti, **30**, 87 (1968).
40. E. Willert, R. Berger, G. Langhammer, Plaste Kautschuk, **14**, 303 (1967).
41. H. J. Schuurmans, J. Polym. Sci., **57**, 557 (1962).
42. V. P. Budtov, Vysokomol. Soedin., Ser. A, **9**, 765, (1967).
43. W. F. Hadden, Jr., R. S. Porter, J. F. Johnson, J. Appl. Polym. Sci., **8**, 1371 (1964).
44. L.-Y. Chou, J. L. Zakin, J. Colloid Interface Sci., **25**, 547 (1967).
45. A. Penati, G. Pagani, E. Beacco, Ann. Chim. (Rome), **64**, 99 (1974).
46. F. Gundert, B. A. Wolf, Makromol. Chem., **187**, 2969 (1986).
47. P. H. Plesch, P. P. Rutherford, Polymer, **1**, 271 (1960).
48. H. Geerissen, P. Schützeichel, B. A. Wolf, Macromolec., **24**, 304 (1991).
49. F. Danuso, G. Moraglio, Atti Accad. Nazl. Lincei, Rend., Classe Sci. Fis. Mat. Nat., **25**, 509 (1958).
50. G. Moraglio, F. Danusso, Ann. Chim. (Rome), **49**, 902 (1959).
51. K. Kamido, Kobunshi Kagaku, **21**, 152 (1964).
52. P. Xie, Q. Ying, L. Shi, Gaofenzi Xuebao, 257 (1990); Chem. Abstr., **114**, 123693h (1991).
53. G. Barone et al., Ric. Sci., **36**, 477 (1966).
54. M. Sakurai et al., Polym. J. (Tokyo), **25**, 1247 (1993).
55. N. M. Beder, A. B. Pakshver, Khim, Volokna, **3**, 21 (1961).
56. T. Shibukawa, K. Nakaguchi, Kobunshi Kagaku, **14**, 353 (1957).
57. Y. Shimura, J. Polym. Sci. A-2, **4**, 423 (1966).
58. S. Uchiyama, Sen-i Gakaishi, **22**, 373 (1966).
59. T. Shibukawa, K. Nakaguchi, Kobunshi Kagaku, **14**, 203 (1957).
60. M. Katayama, T. Ogoshi, Kobunshi Kagaku, **13**, 114, 148 (1958).
61. N. S. Batty, J. T. Guthrie, Polymer, **19**, 1145 (1978).
62. T. Shibukawa, K. Nakaguchi, Kobunshi Kagaku, **14**, 291, 294 (1957).
63. H. Somitomo, Y. Yatsukama, Kobunshi Kagaku, **11**, 65 (1954).
64. M. L. Huggins, "Physical Chemistry of High Polymers", Wiley, New York, 1958, p. 82.
65. Y. Piétrasantra, G. Rigal, P. Schaeffner, Makromol. Chem., **182**, 1371 (1981).
66. N. Schott, B. Will, B. A. Wolf, Makromol. Chem., **189**, 2067 (1988).
67. P. Vasudevan, M. Santappa, Makromol. Chem., **137**, 262 (1970).
68. J. Y. Olayemi, Makromol. Chem., **183**, 2547 (1982).
69. K. Karunakaran, M. Santappa, Current Sci. (India), **33**, 551 (1964).
70. J. Herz et al., Compt. Rend., **261**, 1319 (1965).
71. N. Yokomichi, K. Ogino, T. Nakagawa, Nippon Kagaku Zasshi, **87**, 233 (1996).
72. S. Gundiah, N. V. Viswanathan, S. L. Kapur, J. Sci. Ind. Res., **19B**, 191 (1960).
73. B. Maitra, A. K. Nandi, Polymer, **34**, 1260 (1993).
74. N. T. Srinivasan, M. Santappa, Makromol. Chem., **27**, 61 (1958).
75. A. Kotera et al., Makromol. Chem., **87**, 195 (1965).
76. A. Blumstein, F. W. Billmeyer, Jr., J. Polym. Sci. A-2, **4**, 465 (1966).
77. G. S. Kolesnikov, Tsing Han-Ming, Vysokomol. Soedin., **3**, 1210 (1961).
78. M. Bohdanecký, Coll. Czech. Chem. Commun., **31**, 4095 (1966).
79. A. S. Badran et al., J. Appl. Polym. Sci., **45**, 333 (1992).
80. S. Morimoto, A. Machi, Kogyo Gijutsuin Sen I Kogyo Shikensho Kenkyn Hokoku, **67**, 1 (1964).

81. V. Efstratiadis et al., Polym. Int., **33**, 171 (1994).
82. G. N. Tsidvintseva, Deposited Doc., **1978**, VINITI 1704–78; Chem. Abstr., **91**, 193784u (1979).
83. Z. Li, W. Zhang, K. Feng, Zhongshan Daxue Xuebao, Ziran Kexueban, **3**, 27, (1986).
84. D. Pateropoulou, E. Siakali-Kioulafa, N. Hadjichristidis, Macromol. Chem. Phys., **195**, 173 (1994).
85. J. W. Mays, N. Hadjichristidis, J. S. Linder, J. Polym. Sci. B: Polym. Phys., **28**, 1181 (1990).
86. B. Bednar et al., Sb. Vys. Sk. Chem. Technol. Praze, **53**, 241 (1980).
87. M. Becerra, D. Radic, L. Gargallo, Makromol. Chem., **179**, 2241 (1978).
88. J. B. Alexopoulos, N. Hadjichristidis, Makromol. Chem., **180**, 455 (1979).
89. J. B. Alexopoulos, N. Hajichristidis, A. Vassiliadis, Polymer, **16**, 386 (1975).
90. N. Hadjichristidis et al., J. Polym. Sci., Polym. Phys. Ed., **22**, 1745 (1984).
91. Rafi'ee Fanood, M. H. George, Polymer, **28**, 2241 (1987).
92. H. Kamogawa, T. Sekiya, Kogyo Kagaku Zasshi, **63**, 1631 (1960).
93. F. S. Holahan, S. S. Stivala, D. W. Levi, J. Polym. Sci. A, **3**, 3987 (1965).
94. D. Landheer, S.-L. Malhotra, J. Macromol. Sci.-Chem. A, **16**, 1349 (1981).
95. T. A. Orofino, F. Wenger, J. Phys. Chem., **67**, 566 (1963).
96. M. Pulat, N. Uyysal, Cemil Senvar, Marmara Univ. Fen Blimleri Derg., **5**, 125 (1988).
97. T. Arai et al., Macromolec., **28**, 3609 (1995).
98. J. R. Urwin, J. M. Stearne, Makromol. Chem., **78**, 204 (1964).
99. V. P. Budtov, Vysokomol. Soedin., Ser. A, **9**, 765 (1967).
100. J. Schurz, H. Pippan, Monatsch. Chem., **94**, 859 (1963).
101. K. K. Chee, J. Appl. Polym. Sci., **27**, 1675 (1982).
102. M. Morton et al., J. Polym. Sci., **57**, 471 (1962).
103. H.-G. Elias, O. Etter, Makromol. Chem., **66**, 56 (1963).
104. J. L. Zakin, R. Wu, H. Luh, K. G. Mayan, J. Polym. Sci., Polym. Phys. Ed., **14**, 299 (1976).
105. V. A. Myagchenkov, E. V. Kuzentsov, V. Ya. Kitkevich, Vysokomol. Soedin., **6**, 1366 (1964).
106. A. I. Goldberg, W. P. Hohenstein, H. Mark, J. Polym. Sci., **2**, 503 (1947).
107. U. Lohmander, R. Stroemberg, Makromol. Chem., **72**, 143 (1964).
108. H. W. McCormick, J. Colloid Sci., **16**, 635 (1961).
109. Y. Einaga et al., Macromolec, **22**, 3419 (1989).
110. W. E. Moore, M. Murphy, J. Polym. Sci., **56**, 519 (1962).
111. A.-M. Meffroy-Biget, A. Unanue, P. Stevenot, C. R. Hebd. Seances Acad. Sci., Ser. C., **281**, 179 (1975).
112. A.-M. Meffroy-Biget et al., C. R. Hebd. Seances Acad. Sci., Serv. C, **281**, 429 (1975).
113. K. Imai, U. Maeda, M. Matsumoto, Kobunshi Kagaku, **14**, 419 (1957).
114. M. Shibayama et al., Macromolec., **29**, 885 (1986).
115. L. M. Hobbs, V. C. Long, Polymer, **4**, 479 (1963).
116. M. Matsumoto, U. Maeda, K. Imai, Kobunshi Kagaku, **14**, 425 (1957).
117. A. Beresniewicz, J. Polym. Sci., **39**, 63 (1959).
118. F. Lerner, M. Alon, J. Polym. Sci. A: Polym. Chem., **25**, 181 (1987).
119. M. Negishi et al., Kobunshi Kagaku, **14**, 239 (1957).
120. L. Z. Vilenchik et al., Chromatographia, **24**, 633 (1987).
121. M. Matsumota, K. Imai, Kobunshi Kagaku, **12**, 402 (1955).
122. T. Ikaya, K. Imai, Kobunshi Ronbunshu, **36**, 329 (1979).
123. Y. Sakaguchi, N. Yasuhira, Kobunshi Kagaku, **13**, 437 (1956).
124. J. W. Breitenbach, E. L. Forster, A. J. Renner, Kolloid-A., **127**, 1 (1952).
125. F. Dunusso, G. Moraglio, S. Gazzeva, Chim. Ind. (Milan), **36**, 883 (1964).
126. L. C. Grotz, J. Polym. Sci. B, **2**, 883 (1964).
127. G. M. Guzman, J. M. G. Faton, Anales Real Soc. Espan. Frs. Quim. (Madrid), **54-B**, 263 (1958).
128. K. Goto, Y. Ono, S. Furusawa, Kobunshi Kagaku, **11**, 437 (1954).
129. H. Sobue, Y. Tabata, Y. Tajima, Kogyo Kagaku Zasshi, **61**, 1328 (1958).
130. L. Lapcik, V. Roubal, Zb. Pr. Chemikotechnol. Fak. SVST, **1972**, 15 (1974).
131. E. L. Madruga, J. Millan, J. Polym. Sci., Polym. Chem. Ed., **12**, 2111 (1974).
132. Y. Nakamura, M. Saito, Kobunshi Kagaku, **17**, 718 (1960).
133. F. Krasovec, J. Stefan, Inst. Reports (Ljubljana), **3**, 203 (1956).
134. L. A. Utracki, R. Simha, Makromol. Chem., **177**, 94 (1968).
135. R. Hippe, Bull. Acad. Pol. Sci., Ser. Sci. Chim., **23**, 363 (1975).
136. K. Fujii et al., Kobunshi Kagaku, **19**, 581 (1962).
137. M. Sakurai et al., Polymer J. (Tokyo), **26**, 658 (1994).
138. D. O. Jordan, A. R. Mathieson, M. R. Porter, J. Polym. Sci., **21**, 473 (1956).
139. L. Z. Vilenchik et al., Vysokomol. Soedin, Ser. B, **31** (2), 114 (1989).
140. G. Liang, X. Liang, R. Feng, Zhongshan Daxue Xuebao, Ziran Kexueban, **3**, 126, (1980).
141. N. Ahmad, B. Ahmad, J. Chem. Soc. Pak., **12**, 246 (1990).
142. A. B. Duchkova et al., Vysokomol. Soedin., **8**, 1814 (1966).
143. H. Hopff, M. A. Osman, Makromol. Chem., **135**, 175 (1970).
144. G. Lutringer, G. Weill, Polymer, **32**, 877 (1991).
145. G. Lutringer, B. Meurer, G. Weill, Polymer, **32**, 884 (1991).
146. G. S. Nurkeeva et al., Vysokomol. Soedin., Ser. B, **28**, 649 (1986).
147. G. Mueh, Kolloid-Z., **196**, 64,c 140 (1964).
148. D. K. Thomas, A. Charlesby, J. Polym. Sci., **42**, 195 (1960).
149. M. Ataman, J. Macromol. Sci.-Chem. A, **24**, 967 (1987).
150. E. A. Boucher, P. M. Hines, J. Polym. Sci., Polym. Phys. Ed., **16**, 501 (1978).
151. M. Ataman and E. A. Boucher, J. Polym. Sci.: Polym. Phys. Ed., **20**, 1585 (1982).
152. W. Tummler, Plaste Kautschuk, **12**, 582 (1965).
153. K. Okazaki, Kogyo Kagaku Zasshi, **64**, 342 (1961).

154. A. Orszagh, F. Fejgin, Polimery, **8**, 233 (1963).
155. N. N. Filatova et al., Polym. Sci. USSR, **24**, 1144 (1982).
156. H. Sobue, A. Kajiura, Kogyo Kagaku Zasshi, **62**, 1908 (1959).
157. L. Gargallo et al., Eur. Polym. J., **29**, 609 (1993).
158. G. Prati, Ann. Chim. (Rome), **47**, 51 (1957).
159. E. Heim, Faserforsch. u. Textiltech., **11**, 513 (1960).
160. V. Z. Nikonov, L. B. Sokolov, Vysokomol. Soedin., **8**, 1529 (1966).
161. J. Šebanda, J. Králicek, Coll. Czech. Chem. Commun., **31**, 2534 (1966).
162. J. Králicek, J. Kondeliková, V. Kubánek, Sb. Vys. Sk. Chem.-Technol. Praze, Org. Chem. Technol. C, **18**, 61 (1973).
163. L. Huppenthal, R. Szczepanska, Polimery (Warsaw), **21**, 16 (1976).
164. R. Rybnikár, Faserforsch. U. Textiltech., **9**, 500 (1958).
165. W. Sbrolli, T. Capaccioli, Chim. Ind. (Milan), **42**, 243 (1960).
166. O. Quadrat, M. Bohdanecký, Coll. Czech. Chem. Commun., **29**, 2469 (1964).
167. A. V. Pavlov, S. E. Bresler, S. R. Rafikov, Vysokomol. Soedin., **6**, 2068 (1964).
168. M. K. Baloch, Polym. Bull. (Berlin), **34**, 469 (1995).
169. L. V. Avrorova et al., Khim. Volokna, 20, (1988).
170. S. M. Aharoni, Macromolec., **20**, 2010 (1987).
171. D. Pishev, N. Angelova, Przegl. Wlok. Tech. Wlok., **46**, 5 (1992).
172. H. Li et al., Chemical Journal of Chinese Universities, **10**, 1065 (1989).
173. W.-F. Lee, C.-C. Tsai, Polymer, **36**, 357 (1992).
174. D.-J. Liaw, S. J. Shian, K. R. Lee, J. Appl. Polym. Sci., **45**, 61 (1992).
175. D.-J. Liaw et al., J. Appl. Polym. Sci., **34**, 999 (1987).
176. A. Viallat, R. Pedro Bom, J. P. Cohen Addad, Polymer, **33**, 4379 (1992).
177. C. Maechling-Strasser et al., Polymer, **33**, 627 (1992).
178. U.-P. Strauss, J. Polym. Sci., **33**, 291 (1958).
179. P. M. Cotts et al., Macromolec., **24**, 6730 (1991).
180. M. Takeda, A. Yamada, T. Saito, Kobunshi Kagaku, **14**, 265 (1957).
181. M. Takeda, A. Yamada, Kobunshi Kagaku, **14**, 247 (1957).
182. J. Brzezinski, Polimery (Warsaw), **18**, 67 (1973).
183. S. Maxim et al., Eur. Polym. J., **13**, 105 (1977).
184. M. S. Matevosyan, A. A. Askadskii, G. L. Slonimskii, Vysokomol. Soedin., Ser. A, **29**, 761 (1987).
185. B. Magny, I. Ilioponlos, R. Audebert, Polym. Comm., **32**, 456 (1991).
186. R. M. Davis, W. B. Russel, Proc. IUPAC Macromol. Symp., **28**, 902 (1982).
187. R. M. Davis, W. B. Russel, Macromolec., **20**, 518 (1987).
188. V. E. Eskin, A. E. Nesterrov, Vysokomol. Soedin., **8**, 141 (1966).
189. S. R. Erilander, R. M. Purvinas, Stärke, **20**, 37 (1968).
190. M. K. S. Morsi, C. Sterling, J. Appl. Polym. Sci., **10**, 925 (1966).
191. L. S. Bolotnikova, T. I. Samsonova, Zh. Prikl. Khim., **38**, 2299 (1965).
192. M. W. Pandit, R. S. Singh, Indian J. Technol., **6**, 172 (1968).
193. V. P. Kharitonova, A. B. Pakshver, Kolloid. Zhur., **20**, 110 (1958).
194. G. Khamrakulov et al., Vysokomol. Soedin, Ser. A, **35**, 705 (1993).
195. L. A. Knypikova, T. A. Savitskaya, D. D. Grinshpan, Khim. Drev., **8**, (1993).
196. P. Haward, R. S. Parikh, J. Polym. Sci. A-1, **6**, 537 (1968).
197. I. I. Ryskina et al., Polym. Sci. USSR, **18**, 2854 (1976).
198. C. Patel et al., J. Polym. Mater., **4**, 35 (1987).
199. U. Lohmander, R. Stroemberg, Makromol. Chem., **72**, 143 (1964).
200. F. Higashide, J. Nakajima, Kogyo Kagaku Zasshi, **69**, 1329 (1966).
201. W. Brown, Arkiv Kemi, **18**, 227 (1961).
202. A. Blazková, J. Hrivíková, L. Lapcík, Chem. Pap., **44**, 289 (1990).
203. K. Watanabe, M. Nakamura, Kogyo Kaguku Zassshi, **69**, 1329 (1966).
204. D. A. I. Goring, S. Sitaramaiah, Polymer, **4**, 7 (1963).
205. L. Zhang et al., Carbohydr. Res., **270**, 1 (1995).
206. R. Urbani, D. A. Brant, Carbohydr. Polym., **11**, 169 (1989).
207. M. Milas et al., Macromolec., **23**, 2506 (1990).
208. M. Hoffman, Makromol. Chem., **24**, 222, 245 (1957).
209. J. Polácek, J. Polym. Sci., **39**, 469 (1959).
210. H. Wesslau, Makromol. Chem., **20**, 111 (1956).
211. R. J. Valles, M. C. Otzinger, D. W. Levi, J. Appl. Polym. Sci., **4**, 92 (1960).
212. H. Craubner, Makromol. Chem., **93**, 24 (1966).
213. V. V. Varadaiah, R. Rao, Polym. Sci., **19**, 379 (1956).
214. V. E. Hart, J. Polym. Sci., **17**, 215 (1955).
215. H. Anders, H. Zimmerman, H. Behnisch, Acta Polym., **37**, 63 (1986).
216. L. D. Choo, L. M. Goo, Nonmunjip – Sanop Kwahak Kisul Yonguso (Inha Taehakkyo), **17**, 247 (1989).
217. T. I. Bychkova et al., Vysokomol. Soedin., Ser. B, **28**, 456 (1986).
218. H.-G. Elias, Makromol. Chem., **99**, 219 (1966).
219. N. Tsubokawa, Y. Sone, Kobunshi Ronbunshu, **43**, 71 (1986).
220. E. Husemann, G. V. Schulz, J. Makromol. Chem., **1**, 197 (1943).
221. M. Marx, Makromol. Chem., **16**, 157 (1955); Papier, **10**, 135 (1956).
222. M. Marx, G. V. Schulz, Makromol. Chem., **31**, 140 (1959); see also M. Marx-Figini, Makromol. Chem., **36**, 220 (1960).
223. G. Meyerhoff, N. Sütterlin in Ref., **11**, p. VII-200.
224. L. Gargallo, N. Hamidi, D. Radic, Polym. Int., **24**, 1 (1991).
225. T. Takigawa, K. Urayama, T. Masuda, J. Chem. Phys., **93**, 7310 (1990).
226. A. Güner, J. Appl. Polym. Sci., **62**, 78 (1996).
227. S. I. Jeon, G. D. Chang, J. Korean Chem. Soc., **40**, 748 (1996).
228. D. Dragan, Angew. Makromol. Chem., **236**, 85 (1996).

229. S. R. Gooda, M. B. Huglin, Macromolec., **25**, 4215 (1992).
230. I. H. Park, E.-J. Choi, Polymer, **37**, 313 (1996).
231. J. A. Perdigón, M. P. Tarazona, E. Saiz, An. Quim. Int. Ed., **92**, 385 (1996).
232. S.-Q. Bo et al., Chem. J. Chinese Univ. (Gaodeng Xuexiao Huaxue Xuebao), **17**, 820 (1996).
233. F. Brand, H. Dautzenberg, Langmuir, **13**, 2905 (1997).
234. H. Zheng et al., J. Chem. Phys., **105**, 7746 (1996).
235. R. Dhami et al., Carbohydr. Polym., **28**, 113 (1995).
236. T. Wang et al., Xianweisu Kexue Yu Jishu, **4**, 7 (1996).
237. S. Nilsson, L.-O. Sundelöf, B. Porsch, Carbohydr. Polym., **28**, 265 (1995).
238. C. Holmberg, Colloid Polym. Sci., **274**, 836 (1996).
239. L. Picton, G. Muller, Progr. Colloid Polym. Sci., **102**, 26 (1996).
240. K. Monkos, Biochim. Biophys. Acta, **1339**, 304 (1997).

Theta Solvents

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A. INTRODUCTION

1. Fundamentals

1.1 Thermodynamics The theta state is defined as that state of a polymer solution at which the excess chemical potential, and correspondingly, the excess Gibbs energy of dilution is zero. For a given polymer–solvent system, this state is obtained at a certain characteristic temperature, the theta temperature Θ. A solvent at this temperature is called a theta solvent.

Since Flory (110) was the first to show the importance of the theta state for a better understanding of physical structures and properties of polymers, the theta temperature is sometimes also called "Flory temperature". The name "van't Hoff temperature" has also been suggested (127). However, neither "Flory temperature" nor "van't Hoff temperature" has gained acceptance.

The exact definition of the theta state is given by chemical thermodynamics. The chemical potential of a solvent 1, $\Delta\mu_1$, can be split into an ideal term and an excess term:

$$\Delta\mu_1 = \Delta\mu_1^{id} + \Delta\mu_1^{exc} \qquad (A1)$$

where the excess chemical potential at the thermodynamic temperature T is given by the enthalpy of dilution, ΔH_1, and the excess entropy of dilution, ΔS_1^{exc}:

$$\Delta\mu_1^{exc} = \Delta H_1 - T\Delta S_1^{exc} \qquad (A2)$$

A zero excess chemical potential does not imply that both the enthalpy of solution and the excess entropy of solution are zero as in the case of true ideal solutions where $\Delta H_1 \equiv 0$ and $\Delta S_1^{exc} = 0$. Rather, a theta state means only that the terms on the right side of Eq. (A2) compensate each other at the theta temperature Θ, i.e., $\Delta H_1 \equiv \Theta\Delta S_1^{exc}$ for $\Delta\mu_1^{exc}$ at $T = \Theta$. Solutions in the theta state are thus not thermodynamically ideal but thermodynamically pseudo-ideal (306).

The chemical potential cannot be measured directly. It may be expressed, however, by the product of the osmotic pressure Π of the solution and the partial specific volume ${}^*v_1$ of the solvent:

$$-\Delta\mu_1 = {}^*v_1\Pi \qquad (A3)$$

The concentration dependence of the osmotic pressure of non-electrolyte solutions can be written as a power series

$$\Pi = A_1c_2 + A_2c_2^2 + A_3c_2^3 + \cdots \qquad (A4)$$

where $c_2 = m_2/V$ is the mass concentration of the polymer (mass m_2 of the polymer 2 per total volume V of the solution *after* mixing polymer and solvent) and A_1, A_2, A_3,... are first, second, third,... virial coefficients, respectively. The second virial coefficient is a measure of the interaction between two bodies, the third virial coefficient a measure of interactions between three bodies, etc. Second and third virial coefficients depend on the chemical structure and the molar mass of the polymer, the chemical structure of the solvent (and thus its interaction with the polymer), and the temperature.

The first virial coefficient is given by

$$A_1 = RT/\overline{M}_n \tag{A5}$$

where R is the general gas constant ($R = 8.314510$ J/K/mol) and $\overline{M}_n$ is the number-average molar mass of the polymer. Eq. (A4) is thus often written as

$$\frac{1}{(\overline{M}_n)_{app}} \equiv \frac{\Pi}{RTc_2} = \frac{1}{\overline{M}_n} + Bc_2 + Cc_2^2 + \cdots \tag{A6}$$

where $(\overline{M}_n)_{app}$ is the apparent number-average molar mass, $B = A_2/RT$, and $C = A_3/RT$.

The third term of the right-hand side of Eq. (A4) is often negligible at low polymer concentrations. Combinations of Eqs. (A1), (A3) and (A4) then yields,

$$A_2 = -\Delta\mu_1^{exc}/({}^*v_1 c_2^2) \tag{A7}$$

Since by definition the excess chemical potential is zero at the theta temperature, the second virial coefficient (but not necessarily the third virial coefficient) is also zero ($A_2 = 0$ at $T = \Theta$). The concentration dependence of the reduced osmotic pressure, Π/c_2, is thus zero at the theta temperature, albeit at low concentrations only. The same reasoning applies to other colligative methods (cryoscopy, ebullioscopy, vapor phase osmometry) and also to those physical methods that measure concentration dependencies of apparent molar masses (static light scattering, sedimentation equilibrium, combination of sedimentation and diffusion coefficient, etc.). An example is static light scattering which yields at the scattering angle $\vartheta \to 0°$:

$$\frac{1}{(\overline{M}_w)_{app}} \equiv \frac{Kc_2}{R_0} = \frac{1}{\overline{M}_w} + 2Bc_2 + 3Cc_2^2 + \cdots \tag{A8}$$

where $(\overline{M}_w)_{app}$ = apparent mass-average molar mass, K = optical constant at zero angle, and R_0 = Rayleigh ratio at zero scattering angle. The virial coefficients B and C of light scattering average differently over interactions than those of colligative measurements; B and C of polymolecular ("polydisperse") polymers are numerically different for osmotic pressure and light scattering measurements.

The theta temperature may also be phenomenologically defined as the critical miscibility temperature at the limit of infinite molar mass (110). Since a solution may exhibit two critical miscibility temperatures, a polymer solution may display two theta temperatures. In the case of endothermic solutions, a lowering of temperature leads to less positive second virial coefficients and the theta temperature corresponds to an upper critical solution temperature (UCST) in the limit of infinite molar mass. The theta temperature corresponds to a lower critical solution temperature (LCST) in the case of exothermic solutions (see, e.g., (92)).

1.2. Unperturbed Dimensions The thermodynamic behavior described above results from the fact that long-range interactions between polymer segments are absent in the theta state. Long-range interactions are intramolecular interactions between segments or groups of one and the same polymer molecule that are separated by many chemical bonds along the chain; they are remote in sequence but adjacent in space. Short-range interactions are interactions between chain groups that are near to each other in sequence. They are determined by the length of chain bonds, the valence angles between chain atoms, and the hindrance to rotation around chain bonds (usually due to a pair (4 chain bonds) or two pairs (5 chain bonds) of conformers in the chain).

In a thermodynamically good solvent ($A_2 \gg 0$), polymer–solvent interactions are much larger than polymer–polymer and solvent–solvent interactions. The polymer chain expands in order to minimize polymer–polymer contacts; the dimensions of these chains are said to be perturbed. In such perturbed chains, the space occupied by one polymer segment is excluded for all other segments of the same polymer chain, i.e., even at infinite dilution.

In a theta solvent ($A_2 = 0$), polymer–solvent interactions are just balanced by polymer–polymer and solvent–solvent interactions. Long-range interactions disappear and the polymer chain assumes its so-called unperturbed dimensions which manifest themselves for linear chains by a dependence of the root-mean square radii of gyration s, on the square root of molar masses:

$$\langle s^2 \rangle_0^{1/2} = K_{s,0} M^{1/2} \tag{A9}$$

Intrinsic viscosities $[\eta]$ are also controlled by the space requirements of polymer molecules. For unperturbed coils of linear chains, one can write (see textbooks of polymer science)

$$[\eta]_0 = \Phi_0(\langle s^2 \rangle_0 / M)^{3/2} M^{1/2} = K_0 M^{a_v} \tag{A10}$$

where Φ_0 is usually assumed to be a universal constant (but see Ref. 498). The dependence of the radius of gyration, $\langle s^2 \rangle^{1/2}$, or the intrinsic viscosity, $[\eta]$, on the square-root of the molar mass is generally taken as a manifestation of unperturbed dimensions and thus also of the presence of the theta state of the polymer.

The assumption that theta conditions lead to unperturbed dimensions is often fulfilled within limits of error for most of the investigated polymers but it is not true in the general

sense. The thermodynamic definition of the theta state refers to the *global* properties of polymer solutions whereas the definition of unperturbed dimensions relates to the *local* properties of the chain. Both definitions agree *exactly* only if infinitely thin chains are surrounded by a locally and globally homogeneous continuum of solvent molecules.

The ability of a theta solvent to generate unperturbed dimensions thus depends on the constitution and configuration of the polymer, the solvent, and the temperature [83] since all these factors influence long-range interactions (excluded volume effects and interactions with the solvent). Small molar mass dependencies of theta temperatures are sometimes detectable for single solvents (230), especially at low molar masses when the effects of end groups or small helical segments (525) become noticeable.

However, if the macromolecules have long side chains and if the main chain and the side chains are constitutionally different, then the theta state of a polymer solution may not be identical with the unperturbed state of polymer coils, even for single solvents. An example is the solution of poly(octadecyl methacrylate) in butyl acetate which is in the theta state at 10.5°C ($A_2 = 0$). At this temperature, the exponent a_v of the intrinsic viscosity/molar mass relationship of Eq. (A10) was found as $a_v = 0.44$ instead of 0.50, due to a contraction of the long side chains (406). A similar discrepancy between the theta state of the polymer–solvent system and the unperturbed dimensions of the polymer coils has been found for cyclic poly(styrene) in cyclohexane where $\Theta = 28.5$°C according to thermodynamics but $a_v = 0.40$ at the same temperature; $a_v = 1/2$ was observed for the considerably higher temperature of $T = 40$°C (366). In both cases, intrinsic viscosities were assumed to reflect unperturbed dimensions. This is not necessarily true, however, since atactic poly(methyl methacrylate) exhibited the same reduced mean-square radius of gyration, $\langle s^2 \rangle_0 / \overline{M}_w$, in the two theta solvents acetonitrile ($\Theta = 44$°C) and *n*-butyl chloride ($\Theta = 40.5$°C) whereas the corresponding viscosity quantities, $[\eta]_\Theta / \overline{M}_w^{1/2}$, definitely differed from each other despite the almost identical theta temperatures (512).

In mixed solvents, these effects may be enhanced by a possible distribution of solvent components, depending on the polarities of the polymer 2 and liquids 1 and 3; 1 is a solvent and liquid 3 usually a non-solvent (precipitant). It seems that in the absence of large variations of theta temperatures, the unperturbed dimension of a given apolar polymer is approximately the same in various apolar solvents. However, unperturbed dimensions (as determined from intrinsic viscosities) may vary widely for polar polymers in various polar theta mixtures (31,95,154). For a review of various methods for the determination of unperturbed dimensions in mixed solvents see (526).

2. Methods to Determine Theta Solvents

Methods for the determination of theta temperatures are discussed in greater detail in two reviews (527,528). As mentioned above, theta conditions do not necessarily lead to unperturbed dimensions of coils (and *vice versa*).

2.1. Phase Equilibrium (PE) Binary mixtures of a polymer in a single solvent phase-separate at various temperatures, T_{sep}, depending on the volume fraction ϕ_2 of the polymer. The maximum of the $T_{sep} = f(\phi_2)$-function is called the critical solution temperature T_{cst}. The experiment is repeated for a series of dilute solutions of polymers of the same constitution and configuration but of different molar mass. The relation between the critical solution temperature and the molar mass of the polymer is based on the Flory–Huggins lattice theory which predicts that

$$\frac{1}{T_{cst}} = \frac{1}{\Theta}\left[1 + \frac{1}{\psi_1}\left(\frac{1}{r^{1/2}} + \frac{1}{2r}\right)\right] \qquad \text{(A11)}$$

where ψ_1 is the Flory residual-entropy parameter and r is the ratio of the molecular volume of the polymer to that of the solvent. Since r is proportional to the (number-average) molar mass of the polymer, Eq. (A11) can be written as

$$\frac{1}{T_{cst}} = \frac{1}{\Theta}\left[1 + \frac{b}{(\overline{M}_n)^{1/2}}\right] \qquad \text{(A12)}$$

for polymers of sufficiently high r (or $\overline{M}_n$) (Shultz–Flory method) (110,529). Extrapolation of the inverse critical solution temperatures to infinite molar mass delivers the inverse theta temperature Θ.

The data must be extrapolated not only to infinite molar mass but also to infinite dilution if used with mixed solvents (191). Furthermore, the method can be used only for phase separations into two liquid phases. Care must therefore be taken with crystalline polymers.

2.2. Second Virial Coefficient (A) All absolute methods for the determination of molar masses may be used; most commonly applied are membrane osmometry and static light scattering. Less frequently used are vapor phase osmometry and sedimentation equilibrium. Absolute methods that depend on specific solvent behaviour cannot be applied; examples are cryoscopy and ebullioscopy. Measurements must be performed at sufficiently low polymer concentrations in order to avoid effects of third virial coefficients.

The concentration dependence of reciprocal apparent molar masses is determined for a given polymer–solvent pair at different temperatures. Second virial coefficients are calculated from the slopes of these curves (see Eqs. (A6) and (A8)) and plotted against temperature (110). The dependence of the second virial coefficients on temperature is linear only in the proximity of the theta temperature. Care must be taken, therefore, to work close to the theta temperature. Alternatively, the second virial coefficient may be measured at a constant temperature, using different solvent–nonsolvent ratios. However, appropriate precautions and corrections have to be used if mixed solvents are

used (Donnan equilibria in membrane osmometry, effects of differences in refractive indices in static light scattering, etc).

2.3. Cloud-Point Titration (CP) Dilute polymer solutions of different concentrations are titrated at constant temperature with a nonsolvent until the first sign of cloudiness. The volume fraction $\phi_{3,cp}$ of the nonsolvent 3 at the cloud point is then plotted against the logarithm of the volume fraction $\phi_{2,cp}$ of the polymer at the cloud point (89,90,531) and extrapolated to pure polymer:

$$\phi_{3,cp} = \phi_{3,\Theta} - B_{cp} \ln \phi_{2,cp} \qquad \text{(A13)}$$

where B_{cp} is the slope of the $\phi_{3,cp} = f(\ln \phi_{2,cp})$-curve and $\phi_{3,\Theta}$ is the volume fraction of the nonsolvent in the theta mixture solvent–nonsolvent. At high molar masses (small B_{cp}) and polymer densities $\rho_2 \approx 1$ g/ml, mass-concentrations $c_{2,cp} = \rho_2 \phi_{2,cp}$ (in g/ml) can be used instead of volume fractions $\phi_{2,cp}$ (93); neglect of these conditions may lead to errors (58). This "Elias method" has been shown to have theoretical justification from the Flory–Huggins lattice model (532,533); a comprehensive review of this method is available (93).

2.4. Cloud Temperature Titration (CT) In a variation of the cloud point titration method, cloud temperatures may be determined by cooling or heating of dilute polymer solutions of various concentrations until cloudiness appears (58). The inverse cloud point temperature T_{ct} is then plotted against the logarithm of the volume fraction $\phi_{2,ct}$ of the polymer 2 at the cloud point and extrapolated to pure polymer:

$$\frac{1}{T_{ct}} = \frac{1}{\Theta} - B_{ct} \ln \phi_{2,ct} \qquad \text{(A14)}$$

This method is also based on the Flory–Huggins lattice model (522,533). In a modification of the method, inverse cloud temperatures are plotted against $1/r^{0.6}$ (530), again based on the Flory–Huggins theory:

$$\frac{1}{T_{ct}} = \frac{1}{\Theta}\left(1 + \frac{B_r}{r^{0.6}}\right) \qquad \text{(A15)}$$

where r is the ratio of molecular volumes of polymer and solvent.

2.5. Unperturbed Dimensions (RGM, VM, DM, SM) Root-mean-square radii of gyration, $\langle s^2 \rangle^{1/2}$, depend on the a_s power of the molar mass M (110):

$$\langle s^2 \rangle^{1/2} = K_s M^{a_s} \qquad \text{(A16)}$$

The exponent becomes $a_s = 1/2$ for the unperturbed state where $\langle s^2 \rangle = \langle s^2 \rangle_0$. The temperature dependence of the function $\langle s^2 \rangle = f(M)$ can thus be used to determine the temperature T_0 at which $a_s = 1/2$ (RGM method).

This method is cumbersome; more often used is the temperature dependence of the intrinsic viscosity/molar mass relationship

$$[\eta] = K_v M^{a_v} \qquad \text{(A17)}$$

because $a_v = 1/2$ at the unperturbed state (at T_0) (VM method). Less often used are the corresponding relationships between molar masses and other hydrodynamic properties such as sedimentation coefficients (SM method) or diffusion coefficients (DM method).

Unperturbed radii of gyration (from scattering experiments) always seem to indicate theta conditions (cf. Ref. 512) but the database is too small to allow definite proof. Unperturbed dimensions from hydrodynamic measurements in single solvents sometimes do not coincide with the theta state (cf. Refs. 366,466,512); see also Section 1.2. Hydrodynamic measurements in mixed solvents very often deliver different unperturbed dimensions for theta conditions (cf. Refs. 31,83,95,154).

2.6. Other Methods (IG, PL, R) The theta state and/or unperturbed dimensions have occasionally been determined by various other methods: refractometry (R) (7,24), dielectric measurements (E) (5), polarized luminescence (PL) (399), inverse gas chromatography (IG) (412), and dynamic light scattering (LS) (376).

Common methods for the determination of theta temperatures Θ, theta compositions $\phi_{i,\Theta}$ of mixed solvents, and/or unperturbed dimensions are compared in the following table:

Method		Required for the application			
Abbreviation	**Name**	**Minimum number of polymers**	**Knowledge of molar mass**	**Type of solvent**	**Determined parameter**
A	Second virial coefficient	3	Yes	Preferably single	Θ
CP	Cloud point titration	1	No	Mixed	$\phi_{3,\Theta}$
CT	Cloud point temperature	1	No	Preferably single	Θ
PE	Phase equilibria (critical solution temperature)	3	Yes	Preferably single	Θ
VM	Intrinsic viscosity/molar mass	3	Yes	Single or mixed	$\Theta, \phi_{3,\Theta}$

B. TABLES OF THETA SOLVENTS FOR POLYMERS

In the following tables, theta solvents and theta temperatures have been compiled for various polymers from the literature. The polymers are subdivided into groups of chemically similar compounds (see beginning of chapter for Contents). Within each group, polymers are arranged alphabetically, mostly according to the polymer names used in the original reference; no attempt has been made to use either IUPAC names (systematic names) or poly(monomer) names exclusively. Monomer units in alternating and random copolymers are listed alphabetically, regardless of their proportions; their relative amounts are given in parentheses on a mole/mole basis. Segments of block and graft copolymers are usually noted in the sequence reported by authors. The following abbreviations are used: *alt* (alternating), *block* (self-explanatory), *co* (statistical or random (or unspecified)), *graft* (self-explanatory).

Polymers are further characterized by their tacticity (if any), provided it has been reported in the literature. Polymers by free radical polymerizations are usually assumed by authors to be "atactic". Tacticities are usually given as mole fractions of isotactic diads (x_i), syndiotactic diads (x_s), isotactic triads (x_{ii}), syndiotactic triads (x_{ss}), heterotactic triads (x_{is}), *cis*-configurations (x_{cis}), *trans*-configurations (x_{trans}). If no quantitative characterization was reported, tacticities are described as it (predominantly isotactic), st (predominantly syndiotactic), at (neither predominantly iso- nor syndiotactic), cis (predominantly cis-tactic), or trans (predominantly trans-tactic).

For each polymer, theta solvents are given in alphabetical order; no attempt was made to convert commonly used solvent names into systematic names (e.g. dioxane = 1,4-dioxane). In mixed solvents, the liquid listed first may thus be a solvent or a nonsolvent. Compositions of mixed solvents are given in vol./vol. unless otherwise noted (ex: w/w indicates weight per weight).

A recent (smaller) list (496) of "Theta Temperatures" is mainly concerned with unperturbed dimensions. Since these dimensions are often evaluated from measurements at nontheta temperatures using various theories of intrinsic viscosity/molar mass, or (far less frequent) radius of gyration/molar mass relationships, reported "theta temperatures" are often not theta temperatures *per se* but the temperatures at which measurements in thermodynamically *good* solvents have been performed. A comprehensive list of unperturbed dimensions of various polymers can be found in Section VII/1 of this Handbook and a list of characteristic ratios of polymethacrylates in (497).

TABLE 1. HOMOCHAIN POLYMERS

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
1.1. POLY(ALKANES)				
Poly(acenaphthylene)	Ethylene dichloride (1,2-dichloro ethane)	35	A, VM	238
		30	A, VM	237
		20	PE, VM	16
Poly(1-butene), atactic	*i*-Amyl acetate	23	PE, VM	245,246
	Anisole	83	VM	245
		83	PE, VM	246
		86.2	PE	190
	Diphenyl ether	141	PE, VM	246
	Phenetole	61	VM	245
		61	PE, VM	246
	Toluene	−46	PE, VM	246
–, isotactic	Anisole	89.1	PE	190
		89	PE, VM	246
	Cyclohexane/*n*-propanol (69/31)	35.0	VM	304
	Diphenyl ether	148	PE, VM	246
	Phenetole	64.5	PE, VM	246
Poly(butene-*co*-ethylene), 40 branches per 100 backbone carbon atoms	3-Octanol	5.0	VM	514
	2-Octanol	29.0	VM	514
	1-Octanol	59.0	VM	514
Poly(ethylene)	*t*-Amyl alcohol	199.2	PE	251
	Anisole	153.5	PE	251
	Benzyl phenyl ether	191.5	PE	251
	Bis(2-ethylhexyl) adipate	170	PE	324
		145	VM	351
		145	PE	240
	Bis(2-ethylhexyl sebacate)	150	PE	324
	Carbon disulfide	−73	PE	510
	n-Decanol	153.3	PE	251
	Dibutyl phthalate	>200	PE	324
	Dioctyl adipate	145	VM	508
	Diphenyl	128	PE	353

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TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
		127.5	PE, VM	485
		127.5	VM	492
		125	PE	324
		118	CT	355
		118	PE	462
	Diphenyl ether	165	A	177
		163.9	PE, VM	251,485
		161.4	PE	46
	Diphenyl methane	142.2	PE, VM	251,485
	Diphenylene oxide	118	PE	324
	n-Dodecanol	143.4	PE	353
		138	PE, VM	324
		137.3	PE	251
	n-Heptane	173.9	PE	132
	n-Hexane	133.3	PE	132
		133	PE	252
	n-Hexanol/*p*-xylene (70/30)	155	PE	8
	n-Hexanol/xylene (70/30)	170	CT	136
	Nitrobenzene	>200	PE	324
	p-Nonyl phenol	162.4	PE	251
	n-Octane	210	PE	132
	n-Octanol	180.1	PE	251
	p-Octyl phenol	174.5	PE	251
	n-Pentane	85	PE	252
		80.0	PE	132
	3,5,5-Trimethylhexyl acetate	126	A	353
		121	PE	353
Poly(ethylene-*alt*-propylene)	Benzene	19–21.4	CP	453
	n-Decyl acetate	5.0	CP	453
	n-Heptyl acetate	38.0	CP	453
	n-Hexyl acetate	60.9	CP	453
	n-Octyl acetate	27.0	CP	453
Poly(ethylene-*co*-propylene)	Methanol/toluene (50/50)			346
Poly(ethyl ethylene)	2-Octanol	23.5	A, VM	458
		23.5	VM	514
Poly(1-hexene)	Phenetole	61.3	VM	210
	Butanone/*n*-hexane (29.8/70.2)	8	PE	18
	Butanone/*i*-propanol (37/63)	24	PE	18
		23.5	A, PE	154
	(41.5/58.5)	4	A, PE	154
	Dioxane/*n*-hexane (40/60)	20.5	VM	19
	n-Hexyl chloride	13	A, PE	154
		13	A	18
Poly(isobutene)	*i*-Amyl benzyl ether	23.7	A, VM	339
	i-Amyl butyrate	28	A, VM	339
	i-Amyl *i*-valerate	22.1	A	183
		25.0	DM, VM	427
		25.0	A	517
		27	A	499
	i-Amyl *n*-valerate	22.1	SM, VM	221
		21.0	A	339
	n-Amyl butyrate	22.0	A	339
	Anisole	105.5	PE, VM	113
	Benzene	25.0	A	517
		25		44
		24.0	PE, VM	113
		24	VM	320
		22.8	A, VM	339
	n-Butanol/*n*-hexane (23.6/76.4)	25.0	CP	100
	n-Butanol/methylcyclohexane (29.2/70.8)	25.0	CP	100
	(42.1/57.9)	49.0	CP	100
	Butanone/carbon tetrachloride (33.7/66.3)	25.0	CP	100
	Butanone/cyclohexane (36.8/63.2)	25.0	CP	100
	Butanone/*n*-hexane (36.6/63.4)	25.0	CP	100
	n-Butyl *n*-butyrate	46.2	A	339

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	Carbon tetrachloride/dioxane (63.8/36.2)	25.0	CP	100
	Chlorobenzene/*n*-propanol (67.5/37.5)	49.0	CP	100
	(76.0./24.0)	25.0	CP	100
	(79.7/20.3)	14.0	CP	100
	Chloroform/*n*-propanol (57.9/42.1)	49.0	CP	100
	(77.1/22.9)	25.0	CP	100
	(79.5/20.5)	14.0	CP	100
	Cyclohexane/dioxane (45.1/54.9)	25.0	CP	100
	Cyclohexane/*n*-propanol	35	VM	305
	Cyclohexanol/toluene (29.3/70.7)	25.0	CP	100
	Cyclopentane	188		536
	n-Decanol/*n*-hexane (41.1/58.9)	25.0	CP	100
	n-Decanol/methylcyclohexane (47.5/52.5)	25.0	CP	100
	Dibuty ether	204	A, RGM, VM	76
	Dioxane/*n*-hexane (48.2/51.8)	25.0	CP	100
	Dioxane/methylcyclohexane (51.0/49.0)	25.0	CP	100
	Diphenyl ether	148	PE	113
	Diphenyl ether/ethylbenzene (50/50)	76.0	PE	113
	(25/75)	26.8	PE	113
	Ethylbenzene	−24.0	PE	113
	Ethyl caproate	57	PE	113
	Ethyl caprylate	22	PE	263
	Ethyl heptanoate	33		44
		33	PE	113
	Ethyl hexanoate	57		44
	Ethyl octanoate	22		44
	3-Ethylpentane	185		536
	n-Heptanol/*n*-hexane (37.4/62.6)	25.0	CP	100
	n-Heptanol/methylcyclohexane (39.5/60.5)	25.0	CP	100
	n-Hexane/*n*-hexanol (68.3/31.7)	25.0	CP	100
	n-Hexane/3-methylbutanone (57.6/43.4)	25.0	CP	100
	n-Hexane/*n*-octanol (63.7/36.3)	25.0	CP	100
	n-Hexane/*n*-pentanol (71.7/28.3)	25.0	CP	100
	n-Hexane/*n*-propanol (80.3/19.7)	25.0	CP	100
	2-Methylbutane (M = 23000, 150000)	52 ± 2		536
	–, (M = 760000)	45		536
	Methyl capronate	72.7	A	183
	Methyl caprylate	24.0	A	183
	Methylcycohexane/*n*-octanol (56.0/44.0)	25.0	CP	100
	Methylcyclohexane/*n*-pentanol (65.2/34.8)	25.0	CP	100
	Methylcyclohexane/*n*-propanol (74.2/25.8)	25.0	CP	100
	2-Methylhexane	153		536
	Methyl heptanoate	47.7	A	183
	3-Methylheptanone-5	55.5	A, VM	339
	Methyl pelargonate	5.9	A	183
	2-Methylpentane	103		536
	Pentane	76.0	A, RGM, VM	76
	Phenetole	87		44
		86.0	PE, VM	113
	Toluene	−13.0	PE, VM	113
Poly(2-methyl-2-butylene) (= hydrogenated 1,4-poly(isoprene))	*n*-Hexyl acetate	60.9	A	363
Poly(methylethylene)	2-Octanol	37.6	VM	454
Poly([S]-4-methyl-1-hexene), $x_i > 0.95$	*o*-Chloronaphthalene	165	VM	262
	o-Dichlorobenzene	133	VM	262
Poly(4-methyl-1-pentene, $0.90 \leq x_i \leq 0.94$	Diphenyl	194.6	PE, VM	334
	Diphenyl ether	210.0	PE, VM	334
	Diphenylmethane	176.6	PE, VM	334
Poly[1-(4-methylpentyl)-1-butylene-*co*-1-(1,5-dimethylhexyl)ethylene (= hydrogenated poly(myrcene)	2-Octanol	51.0	VM	514
Poly(2-methylpentene-1-sulfone)	Butanone/*n*-hexane (35.4/64.6)	11.5	PE	18
	Butanone/*i*-propanol	22.5	PE	18
Poly(1-octene)	*n*-Pentane	162.5	PE	169
	Phenetole	50.4	PE	170

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TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
Poly(1-pentene), atactic	Anisole	85	PE, VM	472
	i-Butyl acetate	32.5	PE, VM	472
	Diphenylmethane	121.0	PE, VM	472
	Phenetole	64.0	PE, VM	472
	Phenetole	48.3	A	192
	Phenyl ether	149	PE, VM	472
–, isotactic	*i*-Amyl acetate	31.5	VM	243
	Anisole	85	PE, VM	472
	i-Butyl acetate	32.5	PE, VM	472
	Diphenylmethane	121.0	PE, VM	472
	2-Pentanol	62.4	PE	219
	Phenetole	55.8	A	192
		64.0	PE, VM	472
	Phenyl ether	149	PE, VM	472
Poly(propylene), atactic	*i*-Amyl acetate	34	VM	69
		34	VM	244
	n-Amyl acetate	36.6	A	333
	n-Butanol/carbon tetrachloride (33/67)	25.0	CP	78
	n-Butanol/*n*-hexane (32/68)	25.0	CP	78
	n-Butanol/methylcyclohexane (34/66)	25.0	CP	78
	i-Butyl acetate	65.5	A	333
		58.0	VM	244
	n-Butyl acetate	58.5	A	332
	Carbon tetrachloride/*n*-propanol (74/26)	25.0	CP	78
	1-Chloronaphthalene	74	PE	171,521
		68	A	314
	Cyclohexanone	92	PE	171,521
	Diphenyl	129	VM	244
	Diphenyl ether	153.3	PE	172
		146	VM	244
	n-Hexane/*n*-propanol	25	CP	78
	Methylcyclohexane/*n*-propanol (69/31)	25	CP	78
	1-Octanol	77	CP	453
	2-Octanol	37.6	CP	453
	3-Octanol	5.0	CP	453
	n-Propyl acetate	85.5	A	333
–, isotactic	*i*-Amyl acetate	70	A	152
		34		69
	i-Amyl benzylether	111	PE	448
	p-*t*-Amyl phenol	140.8	PE	448
	Benzyl phenyl ether	181.8	PE	448
	Benzyl propionate	132	PE	448
	n-Butyl alcohol	147.2	PE	448
	t-Butyl alcohol	122	PE	448
	p-*t*-Butyl phenol	166.0	PE	448
	1-Chloronaphthalene	74	A	521
	p-Cresol	206	PE	448
	Dibenzyl ether	183.2	PE	255
		183.2	PE	448
	Diphenyl	125.1	PE	255
		125.1	PE	448
		125.0	VM	244
	Diphenyl ether	146.2	PE	172
		145	PE, VM	171,521
		143	VM	244
		142.8	PE	255
		142.8	PE	448
	Diphenyl methane	116.5	PE	448
	p-Ethyl phenol	184	PE	448
	i-Octyl phenol	115	PE	448
	p-Octyl phenol	106	PE	448
–, syndiotactic	*i*-Amyl acetate	45	A	152
Poly(propylene), head-to-head	*i*-Amylacetate	43	PE	471
		56.8	VM	514
Poly(vinyl ethylene), at	2-Octanol	32.8	A, VM	458

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
1.2. POLY(ALKENES)				
1,4-Poly(butadiene), 97.0% cis	*n*-Heptane	−1	PE	242
	n-Propyl acetate	35.5	A, VM	242
95.0% cis	Diethyl ketone	10.3	VM	362
94.6% cis	5-Methyl-2-hexanone/2-pentanone (1/3)	46.2	PE	2
	(1/1)	32.7	PE	2
	(3/1)	22.3	PE	2
	3-Pentanone	10.3	PE	2
	3-Pentanone/2-pentanone (3/2)	30.0	PE	2
94.0% cis	Diethyl ketone	213	PE	66
	Ethyl propyl ketone	240	PE	66
	Propylene oxide	146	PE	66
93.0% cis	Diethyl ketone	208	PE	66
		14	PE	66
	Ethyl propyl ketone	237	PE	66
		−22	PE	66
	Propylene oxide	141	PE	66
		35	PE	66
90.0% cis	*i*-Butyl acetate	20.5	A, VM	70
	n-Heptane/*n*-hexane (25/75)	20	PE	282
	(50/50)	5	PE	282
	5-Methyl-2-hexanone	12.6	PE	1
	2-Pentanone	59.7	PE	1
	3-Pentanone	10.6	PE	1
–, 70% cis-1,4; 23% trans-1,4; 7% 3,4	Dioxane	22.3	VM	447
–, 43% 1,2 units; trans/cis = 2.3/1	Dioxane	15.7	A	363
–, 36% cis; 57% trans; 7% 1,2-vinyl	Dioxane	26.5	A, VM	480
–, –, hydrogenated	*n*-Hexyl acetate	65.0	A	363
1,4-Poly(butadiene), 9% cyclization	Cyclohexanone/dioxane (12/88)	30	A, VM	358
31% cyclization	Cyclohexanone/dioxane (17/83)	30	A, VM	358
46% cyclization	Cyclohexanone/dioxane (21/79)	30	A, VM	358
63% cyclization	Cyclohexanone/dioxane (30/70)	30	A, VM	358
81% cyclization	Cyclohexanone/dioxane (39/61)	30	A, VM	358
Poly(butadiene-*co*-styrene) (76.1/23.9)	Methyl *i*-butyl ketone	46	PE	145
	2-Pentanone	21	PE	145
(75/25)	*n*-Octane	21	VM	284
(70/30)	*n*-Octane	21	PE, DM, SM, VM	283
1,4-Poly(chloroprene), cis	Butanone	25	A	133
		25	VM	455,495
	Cyclohexane	45.5	PE	133
	Cyclopentane	56.3	PE	133
	Decalin, *trans*	2	A	455
1,4-Poly(2,3-dimethylbutadiene)				
>85% trans-1,4; 3% 1,2-vinyl	Cyclohexane/*n*-propanol (81.3/18.7)	25	CP	483
1,4-Poly(isoprene),				
Natural rubber	Butanone	25	VM	88
	n-Hexane/*i*-Propanol (1/1)	21	VM	284
	2-Pentanone	14.5	PE	352
96% cis	Dioxane	31.2	PE	11
	n-Heptane/*n*-Propanol (69.5/30.5; w/w)	25	A, CP, VM	74
94% cis	Methyl isobutyl ketone	16.5	A	42
	Methyl propyl ketone	33	A	42
trans (gutta percha)	*n*-Propyl acetate	60	PE	352
trans (other)	Dioxane	47.7	PE	352
	i-Propanol/toluene (36.2/63.8)	50	A, CP	11
	(35.5/64.5)	45	A, CP	11
	(34.2/65.8)	40	A, CP	11
	(33.5/66.5)	35	A, CP	11
	(32.4/67.6)	30	A, CP	11
–, 51% 1,4-units (mainly trans; 49% 3,4-units, 1-2% 1,2-units	2-Octanol	41.3	A	363
–, –, hydrogenated	2-Octanol	55.3	A	363
–, <5% 1,4-units, 20–25% 1,2-units, 70–75% 3,4-units	2-Octanol	30.5	A	363

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TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
–, –, hydrogenated	2-Octanol	26.2	A	363
1,4-Poly(isoprene), star	Dioxane	34.1	A	130
4-arm star	Dioxane	33.4	A	130
6-arm star	Dioxane	33.5	A	130
8-arm star	Dioxane	32.8	A	20
12-arm star	Dioxane	32.9	A	20
1,4-Poly(isoprene), 94% cis, linear	Methyl propyl ketone	33	A	42
3 branches	Methyl propyl ketone	33	A	42
11 branches	Methyl propyl ketone	27.8	A	42
22 branches	Methyl propyl ketone	23.5	A	42
22 branches	Methyl isobutyl ketone	15	A	42
1,2-*co*-3,4-Poly(isoprene) (35/65)	Benzene/*i*-propanol (55/45)	20	CP	98
1,4-Poly(isoprene), brominated (30% of double bonds)	Cyclohexane	20	A	310
Poly(isoprene-*block*-styrene)	Butanone	44	VM	347
Poly(1-methyl-1-butenylene-*co*-isopropenylethylene)	2-Octanol	41.3	A	363
Poly(2-methyl-1,3-pentadiene) (= Poly(1,3-dimethyl-1-butenylene))	Dioxane	61.2	VM	454
	2-Octanol	28.9	VM	454
Poly[1-(4-methylpentyl)-1-butylene-*co*-1-(1,5-dimethylhexyl)ethylene]	2-Octanol	51.0	A	363
Poly(myrcene), 90% 1,4, 10% 3,4-	2-Octanol	35.9	VM	514
Poly(pentenamer), 80–85% trans	*i*-Amyl acetate	38	VM	347,125
85% trans	*i*-Amyl acetate	38	VM	361
Poly(1-phenyl-1-propyne)	Cyclohexane	36.0	A	491
1.3. POLY(STYRENES)				
Poly(4-acetoxystyrene)	Butyl acetate	26.8	VM	367
	i-Propyl acetate	19.7	VM	367
Poly(*p*-bromostyrene)	Benzene	20	VM	168
Poly(*p*-*t*-butylstyrene)	1-Hexanol	65.0	A, PE	515
	1-Nitropropane	31.0	A, VM	197,465
		31.0	A, PE	515
		31	VM	370
	3-Nonanol	10.9	A, PE	515
	2-Octanol	32.7	A, PE	515
	2-Octanol	32.7	VM	370,473
Poly(*p*-*t*-butylstyrene)-*block*-poly(dimethylsiloxane)-*block*-poly(*p*-*t*-butylstyrene); 28 wt.% butylstyrene units	1-Nitropropane	157	PE	365
	Butanone	31	PE	365
Poly(*o*-chlorostyrene)	Butanone	25	PE, VM	224
Poly(*p*-chlorostyrene)	Benzene	8	PE, VM	224
	Benzene/methanol (5.5/1)	26.7	PE	139
	(5.0/1)	32.4	PE	139
	(4.5/1)	41.6	PE	139
	n-Butyl acetate	33.7	VM	195
	t-Butyl acetate	65.4	PE	156
	n-Butyl carbitol	50.1	PE	156
	Carbon tetrachloride	58.9	PE	196
		52.3	PE	158
		50.7	PE	156
	Carbon tetrachloride/toluene (1/1)	13.4	PE	196
	(5/2)	32.0	PE	196
	Cumene	60.9	PE	158
		59.0	PE	157
	Ethyl acetate	11.7	VM	195
	Ethylbenzene	−7.5	PE	158
		−14.7	PE	156
	Ethyl chloroacetate	−1.3	PE	158
		−1.8	PE	156
	Methyl chloroacetate	65.0	PE	158
		64.6	PE	156
	i-Propyl acetate	75.7	PE	156
		65.0	PE	158

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	i-Propyl benzene	59.0	PE	156
	i-Propyl chloroacetate	−8.2	PE	156
		−8.2	PE	158
	Tetrachloroethylene	44.4	PE	156
Poly(*p*-chlorostyrene-*co*-methyl methacrylate) (51.6/48.4)	*trans*-Decalin/trichloroethylene (78/22)	22.3	VM	239
Poly(*p*-decylstyrene)	Butanone/ethylbutyl ketone (1/1.24)	21.0	A	464
		18.1	PE	464
Poly(2,5-dichlorostyrene)	Ethanol/ethyl acetate (6.3/93.7; w/w)	30.5	A	103
Poly(3,4-dichlorostyrene)	*n*-Butanol/*n*-butyl acetate (7.1/92.9; w/w)	32.9	A	102
Poly(*p*-hexylstyrene)	Butanone	30.0	A	464
		29.3	PE	464
Poly(*p*-methoxystyrene)	*i*-Amyl acetate	75.0	PE	281
	Benzene/cyclohexanol (61.2/38.8)	25	A, CP	226
	Benzene/*n*-heptane (72.5/27.5)	25	CP	295
	Benzene/methanol (73.7/26.3)	25	A, CP	226
	Butanone/*n*-heptane (80.0/20.0)	25	A, CP	226
	Butanone/*n*-propanol (80.6/19.4)	25	A, CP	226
	t-Butylbenzene	52.2	PE	281
	Chloroform/cyclohexane (63.9/36.1)	25	A, CP	226
	Chloroform/methanol (66.5/33.5)	25	A, CP	226
	Cyclohexanol/toluene (37.2/62.8)	25	A, CP	226
	Dichlorodecane	92.6	PE	281
	Methyl isobutyl ketone	23.4	CP	281
Poly(*p*-methoxystyrene-*co*-styrene)				
26.4/73.6	*i*-Amyl formate	30.5	CT	280
53.0/47.0	*i*-Amyl formate	22.8	CT	280
	i-Amyl propionate	18.1	CT	280
	Hexyl methyl ketone	27.0	CT	280
75.6/24.4	*i*-Amyl formate	37.0	CT	280
	i-Amyl propionate	>50.0	CT	280
	t-Butylbenzene	12.5	CT	280
	Hexyl methyl ketone	>30	CT	280
Poly(α-methylstyrene) $x_s = 0.95$; $x_{ss} = 0.90$	Cyclohexane	32.3	A	63
$x_s = 0.91$; $x_{ss} = 0.81$	Cyclohexane	33.1	A	63
$x_s = 0.83$; $x_{ss} = 0.68$	Cyclohexane	34.3	A	63
$x_s = 0.65$; $x_{ss} = 0.44$	Cyclohexane	36.8	A	63
$x_{ss} = 0.44$	Cyclohexane	37.0	VM	64
$x_s = 0.64$	Cyclohexane	39	A, VM	3
atactic	Benzene/methanol (79.4/20.6)	30		272
	Butanone	31		217
	n-Butyl chloride	−10		536
		139		536
	1-Chlorodecane	80.0	VM	470
	1-Chloroheptane	20, 27	VM	479
	1-Chlorohexane	5.7, 10	VM	479
	1-Chlorooctane	43, 53	VM	479
	Cyclohexane	34.5	VM	479
		38	VM	59
	trans-Decalin	9.5	VM	266
		10.0	VM	479
	Diethyl succinate	16.4	A, VM	494
	n-Hexyl acetate	85.0	VM	479
isotactic	Butanone	45		217
Poly(*p*,*i*-propylstyrene)	Dioxane/*i*-propanol (35/65)	20	CP	98
Poly(*p*,*i*-propylstyrene-*co*-styrene)				
(88/12)	Dioxane/*i*-propanol (37/63)	20	CP	98
(56/44)	Dioxane/*i*-propanol (44/56)	20	CP	98
(33/67)	Dioxane/*i*-propanol (48/52)	20	CP	98
Poly(styrene)	*i*-Amyl acetate	220	PE	296
		−49	PE	296
	Benzene	249.8	PE	460
	Benzene/*i*-Propanol (58/42)	35	A	62
	Benzene/cyclohexanol (38.4/61.6)	25	A, CP	95
	Benzene/*n*-heptane (44/56)	35	A	62

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TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	Benzene/*n*-hexane (38/61)	20	CP	128
	(34.7/65.3)	25	A, CP	95
	Benzene/methanol (74.7/25.3)	35	A	62
	(74/26)	34	VM	31
	(77.8/22.2)	25	A, CP	95
	(78/22)	23.5	A	42
	(78/22)	21.5	PE	42
	Benzene/*i*-propanol (66/34)	20	CP	98
	(64.2/35.8)	25	A, CP	95
	(69/31)	35	A	62
	Bicyclohexyl	61	PE	522
	Butanone	148.8	PE	460
	Butanone/methanol (89/11)	25	A, VM	273
	(89.7/11.3)	25	A, CP	95
	(88.9/11.1)	30	PE	315
	Butanone/*i*-propanol (85.7/14.3)	23	A	53
	(82.6/17.4)	34	VM	31
	(87/13)	67	A	115
	(87/13)	67	A	274
	i-Butyl acetate	−46	PE	296
	t-Butyl acetate	−34.7	PE	374
		109.3	PE	374
	Butylchloride	35.6	SM	233
	n-Butyl formate	−9	A	307
		−7.5	CP	375
	Carbon tetrachloride/*n*-butanol (65/35)	35	A	62
	Carbon tetrachloride/heptane (53/47)	35	A	62
	Carbon tetrachloride/methanol (81.7/28.3)	25	A, CP	95
	Chlorobenzene/di-*i*-propyl ether (32/68)	25	A, CP	95
	1-Chlorodecane	6	CT	22
		6.6	A	269
		8.5	VM	456
	1-Chlorodecane/3-methylcyclohexane			
	(21.8/78.2)	22.8	CT	22
	(25.0/75.0)	35.0	CT	22
	(50.5/49.5)	59.6	CT	22
	(90.0/10.0)	87.8	CT	22
	1-Chlorododecane	58.6	A	269
		58.6	VM	456
	Chloroform/methanol (74.7/25.3)	25	A, VM	273
	(75.2/24.8)	25	A, CP	95
	1-Chloroundecane	32.8	A	269
		32.8	A	270
		32.8	VM	456
	Cyclodecane	16	PE	445
	Cycloheptane	17	PE	445
		19.0	VM	456
	Cyclohexane	213	PE	371
		34.0	VM	31
		34.0	PE	189
		34.0	A	307
		34.0	PE	439
		34.4	A, VM	188
		34.5	VM	2, 476
		34.5	VM	488
		34.5	A	42
		34.5	VM	290
		34.5	DM, VM	427
		34.5	VM	456
		34.8	A	270
		35.0	A	274
		35.0	DM, SM	43,308
		35.0	DM	232
		35.0	A	229
		35.0	SM	144

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
		35.0		328
		35.0	A	439
	Cyclohexane, deuterated	40.2		328
		40.0	A	439
		38	PE	438
	Cyclohexane/methylcyclohexane (2/1)	43	VM	2
	(1/1)	48	VM	2
	(1/2)	54	VM	2
	Cyclohexane/toluene (86.9/13.1)	15	PE	315
	Cyclohexanol	79	CT	135
		83.5	A	118
		83.5	VM	290
		83.5	A	307
		86	PE	461
		87	PE	8
		87.8	CT	22
	Cyclopentane	19.6	PE	203
		19.6	A, VM	368
		19.8	PE	460
		20		536
		20.5	VM	456
		20.6	PE	204
		154		536
		154.5	A, VM	368
		154.2	PE	203
	Cyclooctane	12	PE	445
		20.5	VM	456
	Decalin ($w_{cis} = 0$)	23.8	A	153
	($w_{cis} = 0$)	22.8	CT	22
	($w_{cis} = 0$)	21.2	A	26
	($w_{cis} = 0$)	21.0	DM, VM	427
	($w_{cis} = 0$)	20.4	PE, VM	119
	($w_{cis} = 0.20$)	21.2	A	26
	($w_{cis} = 0.23$)	19.3	PE	268
	($w_{cis} = 0.50$)	15.5	A	26
	($w_{cis} = 0.54$)	16.0		160
	($w_{cis} = 0.60$)	15.0	A	26
	($w_{cis} = 0.66$)	15.2	A	451
	($w_{cis} = 0.80$)	14.0	A	26
	($w_{cis} = 1.00$)	12.5	A	26
	($w_{cis} = 1.00$)	12.2	A	451
	($w_{cis} = 1.00$)	10.8	CP	523
	cis/trans	19.3	PE	271
	cis/trans	14.5	A	42
	unknown	31	A	289
	unknown	29.5	A	153
	unknown	18.2	DM	182
	Diethyl ether/dimethoxy ethane (0.7/0.3)	−5	A	356
		−27	A	356
	Di(2-ethylhexyl) phthalate	22		160
	Diethyl malonate	31	A	307
		34.2	VM	2
		34.5	VM	456
		35.6	PE	42
		35.9	A	270
		36.0	VM	503
	Diethyl malonate/diethyl oxalate (4/1)	40	VM	2
	(1/1)	47	VM	2
	(1/4)	52.6	VM	2
	Diethyl oxalate	51.5	A	307
		55.8	VM	2
		58.2	VM	456
		59.6	CT	22
	Dimethyl succinate	67.6	VM	456
	Dioctylphthalate	22		26

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TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	Dioxane/*n*-heptane (41.5/58.5)	35	A	62
	Dioxane/*n*-hexane (38/62)	20	CP	128
	Dioxane/methanol (71.4/28.6)	25	A, CP	95
	(65.1/34.9)	34	VM	31
	(66.5/33.5)	35	A	62
	(66.5/33.5)	35	A, VM	82
	Dioxane/*i*-propanol (55/45)	20	CP	98
	(51.5/48.5)	35	A	62
	Ethyl acetate	−44.3	LS	376
		−44	PE	296
		−43.5	CP	375
		139	PE	296
	Ethyl acetoacetate	108.5	CP	147
	Ethylcyclohexane	70	PE	114
		75.0	VM	456
	Heptane/nitropropane (58/42)	35	A	62
	Heptane/toluene (52.4/47.6)	30	PE	315
	Hexane/3-methylbutanone (48/52)	20	CP	99
	Hexyl *m*-xylene	12.5	A	307
	Methanol/tetrahydrofuran (28.7/71.3)	25	A, CP	95
	Methanol/toluene (20/80)	25	A, VM	273
	(23/77)	25.0	CP	505
	(23.1/76.6)	25.0	VM	490
	(24.8/75.2)	34	VM	31,490
	(27.2/72.8)	45.0	VM	490
	Methyl acetate	43	PE	296
		114	PE	296
	Methyl cyclohexane	60	VM	290
		67.2	A	184
		68	A	307
		68.0	VM	456
		69.3	A	438
		70.5	VM	2
		70.5	VM	85
	3-Methyl cyclohexanol	98.0	CT	22
		98.0	VM	296
	Methyl cyclopentane	75	PE	440
		144	PE	440
	D,L-Menthol	115	VM	290
	1-Phenyldecane	28.0	A	124
		30.6	PE	124
	i-Propyl acetate	−27	PE	296
		107	PE	296
	n-Propyl acetate	−80	PE	296
		178	PE	296
	D,L-Terpineol	78.5	VM	290
	Tetrahydrofuran/water (92.3/7.7)	25	CP	322
	Toluene (estimated theta temperature)	−154		375
Poly(styrene), deuterated	Cyclohexane	30		328
		30	A, PE	439
	Cyclohexane, deuterated	36		328
		36	A	439
		35	PE	439
	Carbondisulfide	36	RGM	60
Poly(styrene), H-shaped	Cyclohexane	34.8	VM	372
Poly(styrene), head-to-head	Cyclohexane	19	A	329
		16		369
Poly(styrene), combs (theta temperatures depend strongly on number and length of branches)	Benzene/methanol (78/22)	16–24	PE	42
		16–24	A, VM	502
	Cyclohexane	18–33	PE	42
		18–33	A, VM	502
		20–36	A, CT	75
		28–33	A	294

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	Decalin (cis-trans)	4.5	PE	42
		4.5	A, VM	502
	Diethyl malonate	22	PE	42
		22	A, VM	502
Poly(styrene), 3-star	Cyclohexane	34.8	A	493
Poly(styrene), 9.4-star	Cyclohexane	22	VM	486
Poly(styrene), (6–15.5) stars	Cyclohexane	18–33	A	524
Poly(styrene), cyclic	Cyclohexane	40	VM	366
		28.5	A	373
Poly(1,4-divinylbenzene) (soluble; by anionic polymerization)	Decalin (trans)	25	A	364
1.4. POLY(VINYLS)				
Poly(2,5-dimethyl-4-vinylethynylpiperidol-4)	Dimethylformamide	25	VM	444
Poly(divinyl ether-*co*-maleic anhydride methyl ester) (1/2)	Tetrahydrofuran	30	VM	302
Poly(isobutyl vinyl ether-*alt*-maleic anhydride)	Acetone	29.8	VM	235
	NaCl/water (1 mol/l)	25	VM	235
Poly(maleic acid-*alt*-vinyl pyrrolidone)	Acetone/dimethylsulfoxide (65.4/34.6)	25	VM	434
	Na_2SO_4/water (0.986 mol/l)	25	VM	434
Poly(2-methyl-5-vinylpyridine)	*i*-Amyl acetate	53.2	PE	123
	n-Amyl acetate	48.2	VM	120
		48.4	PE	123
	i-Butyl acetate	49.0	PE	123
	n-Butyl acetate	21.8	VM	120
		21.8	PE	123
	i-Butyl methyl ketone	37.4	VM	120
		38.7	PE	123
	Ethyl *n*-butyrate	50.0	PE	123
	Ethyl propionate	25.4	PE	123
	Propionitrile	−3.6	PE	123
	n-Propyl acetate	19.3	PE, VM	123
	n-Propyl propionate	58.0	PE	123
	Tetrahydronaphthalene	49.5	PE	123
Poly(9-vinyl adenine)	Sodium cacodylate/water (0.1 mol/l)	26	A	167
	Sodium chloride/water (0.1 mol/l)	40	A	167
Poly(vinyl acetate)	Acetone/*i*-propanol (23/77)	30	PE	338
	Butanone/*i*-propanol (73.2/26.8)	25	A,PE	316
	n-Butyl ethyl ketone	29.0	VM	223
	Cetyl alcohol	123	PE	28
	Di-*i*-butyl ketone	136.5	CP	147
	Ethanol	19	A	42
	Ethanol/methanol (80/20)	17	PE	250
	(60/40)	26.5	PE	250
	(50/50)	34	PE	338
	(40/60)	36	PE	250
	n-Heptane/3-methylbutanone (26.8/73.2)	25	A,PE	316
	(27.3/72.7)	30	A,PE	222
	3-Heptanone	29	A,PE	223
		29	A	42
	Methanol	6	PE	250
		6	PE	345
	6-Methyl-3-heptanone	66	A,PE	223
	i-Propanol	52.6	CP	380
	i-Propanol, deuterated	51.8	CP	380
–, branched	Ethanol	12–15	A	42
	3-Heptanone	26	A	42
–, grafted by styrene	Di(*i*-butyl) ketone	136.5	CP	147
	Ethyl acetoacetate	108.5	CP	147
Poly(vinyl alcohol)	*t*-Butanol/water (32.0/68.0; w/w)	25	CP	357
	Ethanol/water (41.5/58.5; w/w)	25	CP	357
	Methanol/water (41.7/58.3; w/w)	25	CP	357
	i-Propanol/water (39.4/60.6; w/w)	25	CP	357
	n-Propanol/water (35.1/64.9; w/w)	25	CP	357
	NaCl/water (2 mol/l)	25	CP	258
		27 ± 3		536

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	Water	97	PE	79
–, urethanized 4.9%	*n*-Propanol/water (30/40)	30	CT	299
	(40/50)	60	CT	299
8.1%	*n*-Propanol/water (30/40)	30	CT	299
	(40/50)	60	CT	299
11.5%	*n*-Propanol/water (30/40)	30	CT	299
	(40/50)	60	CT	299
Poly(vinyl benzoate)	Xylene	32.5	A	301
Poly(4-vinyl biphenyl)	2-Methoxyethanol/dimethoxyethane (70.6/29.4)	30	A,VM	237
Poly(vinyl bromide)	Methanol/tetrahydrofuran (17/83)	20	A	56
Poly(vinyl carbanilate)	Acetone/water (92.2/7.8)	20	A	41
	Butanone/water (44/56)	20	A	41
	Diethyl ketone	35	A	41
	Dioxane/methanol (28/72)	20	A	41
	Methanol/tetrahydrofuran (71.5/28.5)	20	A	41
Poly(*N*-vinyl carbazole)	Chlorobenzene	−37.4	VM	207
	Chlorobenzene/methanol (85.9/14.1)	25	CP,VM	126
	1,3-Dichlorobenzene/methanol (85.0/15.0)	25	CP,VM	126
	Nitrobenzene	−21.4	VM	207
	Toluene	37	A,VM	200
		39	A	249
Poly(vinyl chloride)	Benzyl alcohol	155.4	PE	303
	n-Butanol/cyclohexanone (15.8/100)	59	E	4
	(41.5/100)	72	E	4
	Cyclohexane/dimethylformamide (100/12.8)	40.5	E	4
	Cyclohexanone	22	A	321
		51	E	4
	Dimethylformamide	36.5	E	4
	Methanol/tetrahydrofuran (42/58)	22	A,CP	137
	Phenetol	88	VM	377
	Tetrahydrofuran/water (100/5)	17	E	4
	(91.5/8.5)	25	A	186
	(100/11.9)	30	CT	253
	(100/11.9)	30	E	4
	(100/9.5)	30	CT	253
	o-Xylene	84	VM	377
–, isotactic and syndiotactic			E	6
Poly(vinyl *p*-chlorobenzoate)	*n*-Butanol/butanone (47/53)	60	PE,VM	298
Poly(vinyl cyclohexane)	Tetrahydrofuran	25	A	97
Poly(vinylethylene) (98% 1,2-, 2% 1,4-)	1-Hexanol	66	VM	514
Poly(*N*-vinylimidazole)	*n*-Butanol	25		337
Poly(vinyl methyl ether)	Cyclohexane	51	A	520
Poly(β-vinyl naphthalene)	Decalin/toluene (56.5/43.5)	30.2	A,VM	104
Poly(vinyl pivalate)	Acetone/methanol (38.2/61.8)	20	CP	80
	Benzene/methanol (33.3/66.7)	20	CP	80
	Butanone/methanol (24.6/75.4)	20	CP	80
Poly(3-vinylpyrene)	Chloroform	25	VM	319
	Methanol/tetrahydrofuran (8/92)	25	CP,VM	319
Poly(2-vinylpyridine)	Benzene	11.4	VM	413
		15	VM	81
		16.2		378
	n-Heptane/*n*-propanol (59.6/40.4; w/w)	25.0	PE	149
Poly(*N*-vinylpyrrolidone)	Acetone/water (66.8/33.2)	25.0	CP,VM	90
	Dioxane	−10	A	39
	NaCl/water (0.55 mol/l)	25	PE	234
	Water	130.5	CT	381
Poly(vinyl stearate)	*n*-Octane	17	VM	313
Poly(*N*-vinyl succinamic acid)	NaCl/water (0.2 mol/l)	25	VM	265
Poly(vinyl sulfonic acid)	KBr/water (0.347 mol/l)	5.7	PE	87
	KCl/water (0.349 mol/l)	5.5	PE	87
	(0.65 mol/l)	26.0	PE	87
	(1.001 mol/l)	44.5	PE	87
	NaBr/water (0.346 mol/l)	−0.6	PE	87
	(1.008 mol/l)	40.1	PE	87
	NaCl/water (1.003 mol/l)	32.4	PE	87

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
Poly(1-vinyl-1,2,4-triazole)	Dimethylformamide	25	VM	430
	Dimethylformamide/dioxane (76.6/23.4)	25	CP	91
		25	CP	211
1.5. POLY(ACRYLICS) AND RELATED COMPOUNDS				
Poly(acrylamide)	Methanol/water (2/3)	20	VM	33
	(41/59)	25	A, VM	463
	NaCl in water (0.1 mol/l)/methanol (59/41)	21	A, RGM	442
Poly(2-acrylamido-2-methylpropane-sulfone amide)	Dioxane/water (18.5/81.5)	25	CP	391
Poly(acrylic acid)	Dioxane	30	A, PE	263
	Water/0.2 mol/l HCl	14 ± 5		536
Poly(acrylonitrile)	Dimethylformamide/methanol (81/19)	20	A, VM	387
	Ethylene carbonate	60	VM	163
	Ethylene carbonate/water (85/15; w/w)	29.3	VM	384
	(85/15; v/v)	44	VM	468
	Nitric acid/water (55/45; w/w)	25	VM	384
Poly(acrylonitrile-*co*-butadiene) (35/65)	Butanone/cyclohexane (47.5/52.5)	25	SM, VM	306
(40/60)	Acetonitrile/butanone (79/21)	19–22	DM, SM	285
	Benzene/cyclohexane (85.5/14.5)	19–22	DM, SM	285
	Butanone/cyclohexane (47.4/52.5)	22	DM, SM	285
	Butanone/*i*-propanol (60/40)	20	DM, SM	285
	Cyclohexane/ethyl acetate (32.7/67.3)	19–22	DM, SM	285
Poly(acrylonitrile-*co*-methyl acrylate) (95/5)	Ethylene carbonate/water (82.5/17.5; w/w)	24.8	VM	385
	Nitric acid/water (51/49; w/w)	24.8	VM	385
Poly(acrylonitrile-*co*-methyl methacrylate)				
$x_{AN} = 0.236$	Benzene	19.8	VM	379
$x_{AN} = 0.5$	Benzene/dimethylformamide (6.5/1)	30		379
$x_{AN} = 0.74$	Benzene/dimethylformamide (5/3)	30		379
Poly(acrylonitrile-*co*-styrene) (1/2)	Benzene/methanol (66.7/33.3)	25	CP	91
	Dimethylformamide/methanol (44.7/55.3)	25	CP	91
(51/49)	Ethyl acetate	40–43	PE	216
Poly(*t*-butyl acrylate)	Benzene/methanol (52/48)	25		213
	n-Hexane	24.2	VM	159
Poly(*t*-butyl acrylate-*co*-vinylidene chloride) (16.7/83.3)	Benzyl alcohol	44	PE	12
Poly(cetyl acrylate)	Butyl acetate	15.2	VM	341
Poly(cyclohexyl methacrylate)	*n*-Butanol	23.0	VM	450
Poly(decyl acrylate)	*i*-Amyl alcohol	−8.5	A, PE	340
	n-Butyl alcohol	−24.5	A, PE	340
	Ethyl acetate	9–11	A, PE	340
	n-Hexanol	34	A	388
		43.2	PE	388
	n-Pentanol	70	A	388
		75.6	PE	388
Poly(ethyl acrylate)	*n*-Butanol	44.9	PE	212
	Ethanol	37.4	PE	212
	Methanol	20.5	PE	212
	n-Propanol	39.5	PE	212
		39.5	VM	323
Poly(ethyl acrylate-*co*-vinylidene chloride) (14.9/85.1)	Ethyl acetoacetate	49.6	PE	12
Poly(hexyl acrylate-*co*-vinylidene chloride) (14.5/85.5)	Benzyl alcohol	56.8	PE	12
Poly(hexadecyl acrylate)	*n*-Heptanol	20.5	A	388
		30.1	PE	388
	n-Hexanol	52.2	A	388
		60.2	PE	388
Poly(phenyl acrylamide)	Water	−8	A	441
Poly(*N*-*i*-propyl acrylamide)	Dioxane	20		389
	Water	31	PE	142
		30.6	VM	390
		30.6	LS	534
		20	VM	382

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TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
Poly(*i*-propyl acrylate), atactic	1,2-Butanediol/1,3-butanediol (68.4/31.6)	121.0	PE	218
	n-Decane	166.6	PE	218
isotactic	1,2-Butanediol/1,3-butanediol (68.4/31.6)	123.5	PE	354
	n-Decane	178.0	PE	354
syndiotactic	*n*-Decane	168.3	PE	218
Poly(*n*-octyl acrylate-*co*-vinylidene chloride) (15.5/84.4)	Benzyl alcohol	77.9	PE	12
Poly(phenyl acrylate)	Ethyl lactate	11.7	A, CP	146
		13.8	VT	383
Sodium poly(acrylate)	NaBr/water (1.504 mol/l)	15.0	A	452
1.6. POLY(METHACRYLICS) AND RELATED COMPOUNDS				
Poly(benzyl methacrylate)	Cyclohexanone	73.2	PE,VM	291
	Cyclopentanol	83.5	A	292
Poly(*p*-biphenyl methacrylate)	Benzene	10	VM	129
Poly(DL-*i*-bornyl methacrylate)	1-Octanol	39.6	A	396
Poly(4-*t*-butylcyclohexyl methacrylate)	Cyclohexane	−133		288
	n-Butanol	40		288
Poly(*n*-butyl methacrylate)	*i*-Butanol	10.7	VM	205
	n-Decane	84.8	VM	205
	Dimethylformamide	23.6	PE	254
	n-Dodecane	101	VM	205
	Ethanol	≈45	IG	412
	n-Hexadecane	119.8	VM	205
	Methyl carbitol	87.9	PE	254
	Methyl cellosolve	59.6	PE	254
	n-Octane	68.9	VM	205
	i-Propanol	20.9	VM	205
		21.5	A,VM	50
		23	IG	412
		23.7	A, PE	54
		25	A	348
Poly(*i*-butyl methacrylate), isotactic	Dimethylformamide	48.1	PE	254
	Methyl carbitol	115.6	PE	254
	Methyl cellosolve	74.9	PE	254
Poly(*t*-butyl methacrylate)	Cyclohexane	10	VM	466
		10	PE	518
	n-Heptane	64	PE	518
Poly(2-*t*-butylphenyl methacrylate)	Cyclohexane	18.5	PE,VM	336
Poly(*p*-*t*-butylphenyl methacrylate)	Cyclohexane	25	VM	121
Poly(2-chloroethyl methacrylate)	*o*-Dichlorobenzene	35.7	A	397
Poly(cyclobutyl methacrylate)	*n*-Butanol	37.5	PE	404
Poly(cyclobutylmethyl methacrylate)	*n*-Butanol	40.0	VM	504
Poly(cyclododecyl methacrylate)	Hexyl acetate	35	PE	404
Poly(cyclohexyl methacrylate)	*n*-Butanol	22.5	A,VM	166
		22.5	VM	131
		22.5	VM	312
	n-Decane	93.6	VM	312
	n-Decanol	23.0	VM	312
	n-Dodecane	97.5	VM	312
	n-Hexanol	9.2	VM	312
	n-Nonanol	20.2	VM	312
	n-Octane	83.4	VM	312
	n-Octanol	17.9	VM	312
	n-Propanol	39.5	VM	312
Poly(cyclooctyl methacrylate)	2-Butanol	45	PE	404
Poly(cyclopentyl methacrylate)	Cyclohexane	36	A	404
Poly(decyl methacrylate)	Ethyl acetate	11	A	401,457
	n-Pentanol	9.6	PE	140
Poly(2,6-dimethylphenyl methacrylate)	Toluene	25.0	A	411
	Tetrahydrofuran/water (90.6/9.4)	25.0	A	411
Poly(2,6-diisopropylphenyl methacrylate)	Tetrahydrofuran/water (90.1/9.1)	25	VM	467

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
Poly(diphenylmethyl methacrylate)	3-Heptanone	45	PE,VM	405,478
Poly(docosyl methacrylate)	Amyl acetate	31	A	401,457
Poly(dodecyl methacrylate)	*n*-Amyl alcohol	29.5	A,VM	247
Poly(ethylbutyl methacrylate)	*i*-Propanol	27.4	A, PE	77
Poly(ethyl methacrylate)	Butanone/*i*-propanol (12.5/87.5)	23	A, PE	52
	n-Butyl bromide	0	VM	349
	Methyl *n*-propyl ketone	− 1	VM	349
	i-Propanol	36.9	A, PE	54
		36.9	VM	474
	m-Xylene	− 3	VM	349
Poly(glycidyl methacrylate-*co*-lauryl methacrylate) (3/97)	Ethanol/*n*-heptane (69/31)	− 24	CP	257
	(62/38)	1	CP	257
	(56/44)	24	CP	257
	(49/51)	40	CP	257
	n-Heptane/*n*-propanol (56/44)	1	CP	257
	(70/30)	24	CP	257
Poly(*n*-hexyl methacrylate)	*i*-Propanol	32.6	A, PE	47
		37	VM	34
	n-Propanol	20.0	A	388
		21.3	PE	388
	Urea/water (4 mol/l)	10	VM	86
	(6 mol/l)	27.2	VM	86
	(8 mol/l)	52.5	VM	86
Poly(2-hydroxyethyl methacrylate), it	2-Butanol	3.7	VM	484
	Ethanol	15.8	VM	484
	1-Propanol	32.1	VM	484
	2-Propanol	14.0	VM	484
	Water	15.3	VM	484
Poly(isobutyl methacrylate)	2-Hydroxymethyltetrahydrofuran	19.5	PE	507
Poly(*n*-lauryl methacrylate)	*i*-Propyl acetate	13	A, PE	48
		13	PE,VM	489
	n-Pentanol	29.5	A, PE	206
		29.5	VM	208
Poly(5-*p*-menthyl methacrylate)	2-Pentanone	25	VM	336
Poly(methacrylic acid)	Dioxane/dimethylformamide (7/5)	26.9	VM	392
	HCl/water (0.002 mol/l)	30	VM	165
	Methanol	151	PE	393
		26	VM	394
	NaCl/water (0.5 mol/l)	43	A	164
	(0.05 mol/l)	68	A	164
Poly(methacrylonitrile)	Dimethylformamide	29.2	VM	275
Poly(hexyleneoxyphenylenecarboxy-phenyleneoxymethylene methacrylate)	Toluene	15.5	A	402
	Toluene-d_8	6.5	A	402
Poly(methyl *n*-butacrylate)	*n*-Butanol	13	A	155
Poly(methyl ethacrylate)	Di(*i*-butyl) ketone	11.4	A	155
Poly(methyl methacrylate)	Acetone	− 126		288
		− (55 ± 10)	VM	112
	Acetone/ethanol (47.7/52.3)	25	A, CP	96
	Acetone/methanol (78.1/21.9)	25	VM	278
	Acetonitrile	28	VM	278
		30	A,VM	57
		30.5	RGM	414
		32.3	A,VM	409
		38	VM	162
		44	A	407,410,519
		44	DM,VM	427
		44.0	RGM	512,513
		45 ± 10	VM	112
		60	PE,VM	112
	Acetonitrile/butanol (93/7)	25	A	220
	Acetonitrile/methanol (90/10)	25	A	220
	Acetonitrile/pentyl acetate (93/7)	25	A	220
	Acetonitrile/*n*-propanol	25	A	220

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TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	i-Amyl acetate	57.5	A	174
	n-Amyl acetate	41	VM	112
	n-Amyl methyl ketone	15	VM	293
	Benzene	−223	VM	112
	Benzene/*n*-hexane (70/30)	20	CP	128
	2-Butanol/1-chlorobutane (80/20)	39.2	A	403
	Butanone	−98	VM	112
	Butanone/cyclohexane (59.5/40.5)	25	A, CP	96
	Butanone/*n*-hexane (70.7/29.3)	25	A, CP	96
	Butanone/*i*-propanol (58.2/41.8)	4	A	174
	(55/45)	7.0	A	311
	(55/45)	12.8	A	174
	(38/62)	20	CP	128
	(50/50)	22.8	A	174
	(50/50)	25	PE	300
	(50/50)	25	VM	32
	(50/50)	25	A	49
	(58.2/41.8)	25	A	96
	(46.8/53.2)	28.5	A	174
	2-Butoxyethanol	75.0	VM	278
		75.0	VM	443
	n-Butyl acetate	−20	A	174
	n-Butyl chloride	32.6	A	173
		34.0	VM	278
		35.0	A	311
		35.0	RGM	414
		35.0		509
		35.4	A	174
		35.4	VM	400
		40.6	A	407
		40.8	DM,VM	427
		40.8	RGM	512,513
		40.8	A	519
	Carbon tetrachloride	27	VM	112
	Carbon tetrachloride/*n*-hexane (99.4/0.6)	25	CP	96
	Carbon tetrachloride/methanol (53.3/46.7)	25	CP	96
	Chloroform	−273	VM	112
	Cyclohexane/dioxane (53/47)	20	A,VM	82
	(53.4/46.6)	25	A,CP	96
	Cyclohexanol	77.6		117
	Cyclohexanone/*i*-propanol (51.6/48.4)	21–22	VM	279
	Decalin (trans)	23.5	VM	277
	Dichloroethane	−233	VM	112
	2,2-Dimethyl-4-pentanone	35	PE	112
	2,4-Dimethyl-3-pentanone	41	VM	32
		46	PE	112
	Dioxane/*n*-hexane (59/41)	20	CP	98
	Dioxane/water (85/15)	25	A	343
	2-Ethoxyethanol	27.1	VM	443
		30.1	VM	278
		37.8	A,VM	409
	Ethyl acetate	−98	VM	112
	Ethyl *n*-butyraldehyde	22	PE	112
	2-Heptanone	11	VM	112
		11.5	RGM	414
	3-Heptanone	25	VM	112
		33.7	PE	300
	4-Heptanone	33.8	PE	112,516
		34 ± 10	VM	112
		40.4	A	174
	n-Hexane/3-methyl-2-butanone (17/83)	20	CP	99
	n-Hexane/toluene (18.8/81.2)	25	CP	96
	Methanol/toluene (64.3/35.7)	26.2	A, PE	54

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	2-Methoxyethanol	−2	VM	278
		−2	VM	443
	Methyl *i*-butyrate	−(53 ± 10)	VM	112
	Methyl methacrylate	−163	VM	112
	2-Methyl-4-pentanone	−42	A	174
	Methyl *i*-valerate	−37	A	174
	Nitrobenzene/*i*-propanol (51.6/48.4; w/w)	18	VM	227
	Nitromethane	32.0	RGM	414
	2-Octanone	60	PE	112
	3-Octanone	72	PE	112,516
	Tetralin	74	CT	147
	Toluene	−(65 ± 10)	VM	112
		−178	PE	408
	m-Xylene	24	VM	277
$x_i = 0.07$	*p*-Cymene	159.7	PE	300
	n-Propanol	84.4	PE	300
$x_i = 0.08$	Acetonitrile	43.0	A	511
$x_i = 0.09$	Butanone/*i*-propanol (50/50)	30.3	PE	300
	p-Cumene	152.1	PE	300
	3-Heptanone	40.0	PE	300
	n-Propanol	75.9	PE	300
$x_i = 0.94$	Acetonitrile	27.6	PE	187
	Butanone/*i*-propanol (55/45)	25.0	A	311
	n-Butyl chloride	26.5	A	311
$x_i = 1.00$	Butanone/*i*-propanol (55/45)	8.0	A	311
	n-Butyl chloride	26.5	A	311
	n-Propanol	85.2	PE	300
Poly(methyl methacrylate-*co*-9,10-anthrylenedimethyl dimethacrylate (99.86/0.14)	Toluene/octane (78/22)	25	PL	399
Poly(methyl methacrylate-*block*-α-methylstyrene)	1-Chlorohexane	93	PE	179
	3-Heptanone	84	PE	179
	2-Methyl cyclohexanol	62	PE	179
	2-Octanone	94	PE	179
Poly(methyl methacrylate-*alt*-styrene) (50/50)	Cyclohexanol	60.8	A,VM	181
Poly(methyl methacrylate-*block*-styrene) (14.9/85.1)	Cyclohexanol	84	A,VM	181
(15.4/84.6)		84	A	267
(26.8/73.2)		84	A,VM	180
(27.6/72.4)		84	A	267
(various)		82	PE	179
(50/50)		81.6	A	118
(50/50)		81.1	A,VM	181
(50.4/49.6)		81.3	A,VM	180
(50.4/49.6)	2-Ethoxyethanol	81.0	A,VM	180
(51.4/48.6)		81.0	A	267
(51.4/48.6)	Cyclohexanol	81.3	A	267
(64.1/35.9)		80.5	A	267
(64.1/35.9)	2-Ethoxyethanol	69.5	A	267
Poly(methyl methacrylate-*co*-styrene) (23.7/76.3)	Benzene/*n*-hexane (44/56)	20	CP	99
	Benzene/*i*-propanol (57/43)	20	CP	99
	n-Hexane/3-methylbutane (40/60)	20	CP	99
(29.8/70.2)	Cyclohexanol	63.0	A,VM	180
	2-Ethoxyethanol	72.8	A,VM	180
(30.6/69.4)	Cyclohexanol	63.0	A	267
	2-Ethoxyethanol	72.8	A	267
(41.9/58.1)	Benzene/*n*-hexane (51/49)	20.0	CP	99
	Benzene/*i*-propanol (51/49)	20.0	CP	99
	n-Hexane/3-methylbutanone (34/66)	20.0	CP	99
(43.8/56.2)	Cyclohexanol	61.3	A,VM	180
	2-Ethoxyethanol	58.4	A,VM	180
(44.8/55.2)	Cyclohexanol	61.3	A	267
	2-Ethoxyethanol	58.4	A	267

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TABLE 1. *cont'd*

Polymer (mol/mol)		Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	(48.0/52.0)	Cyclohexanol	61.6	A,VM	181
	(50.0/50.0)	Cyclohexanol	68.6	A	118
	(51.4/48.6)	Cyclohexanol	81.3	A	178
		2-Ethoxyethanol	81.0		178
	(57.7/42.3)	Benzene/*n*-hexane (59/41)	20	CP	99
		Benzene/*i*-propanol (48/52)	20	CP	99
		n-Hexane/3-methylbutanone (29/71)	20	CP	99
	(64.1/35.9)	Cyclohexanol	80.5	A	178
		2-Ethoxyethanol	69.5	A	178
	(70.7/29.3)	Cyclohexanol	68.0	A,VM	180
		2-Ethoxyethanol	40	A,VM	180
	(71.5/28.5)	Cyclohexanol	68.2	A	267
		2-Ethoxyethanol	40	A	267
	(73.9/26.1)	Benzene/*n*-hexane (62/38)	20	CP	99
		Benzene/*n*-propanol (41/59)	20	CP	99
		n-Hexane/3-methylbutanone (24/76)	20	CP	99
Poly(β-naphthyl methacrylate)		Tetralin	20	VM	264
		Toluene	52	VM	264
Poly(octadecyl methacrylate)		Butyl acetate	10.5	A	406
Poly(*n*-octyl methacrylate)		*n*-Butanol	16.8	A, PE	51
Poly(*i*-octyl methacrylate)		Acetone/*n*-heptane (64.1/35.9)	25	A, CP	161
Poly(*i*-octyl methacrylate-*co*-styrene) (20/80)		Butanone/methanol (99.4/0.6)	25	A, CP	161
Poly(pentachlorophenyl methacrylate)		Benzene	37	PE	287
			40	R,VM	24
		Benzene/ethylbenzene (5/5)	30	PE	287
		Benzene/toluene (1/9)	10	PE	287
		(5/5)	14	PE	287
		Chloroform	35	PE	287
		o-Dichlorobenzene	−126		288
		Ethylbenzene	25	PE	287
			25		288
			25	R,VM	24
Poly(4-phenylbutyl-1-methacrylate)		1-Chloroundecane	23.0	VM	477
Poly(2-phenylethyl-1-methacrylate)		1-Chloroheptane	27.5	VM	477
Poly(phenyl thiolmethacrylate)		Butanone	25	VM	176
Poly(stearyl methacrylate)		Propyl acetate	36	A	401,457
Poly(tetrahydrofurfuryl methacrylate)		2-Hydroxymethyltetrahydrofuran/methanol (80/20)	31.2	PE,VM	359
Poly(tetrahydro-4*H*-pyranyl-2-methacrylate)		*i*-Butanol	30.5	A	481
Poly(2,4,5-trichlorophenyl methacrylate)		Benzene	21.8	VM	395
			25.0	VM	398
Poly(tridecyl methacrylate)		Ethyl acetate	27	A	401,457

1.7. OTHER CARBON CHAINS

Poly[di(*n*-amyl) itaconate]	Acetone/methanol (99/1)	30	CP	350
	Acetone/water (98/2)	30	CP	350
	Benzene/methanol (60/40)	30	CP	350
	Butanone/methanol (63/37)	30	CP	350
	Chloroform/methanol (52/48)	30	CP	350
	Ethyl acetate/methanol (65/31)	30	CP	350
	Methanol/toluene (43/57)	30	CP	350
Poly(dibenzyl itaconate)	Tetrahydrofuran/water (87.4/12.6)	25.0	VM	470
Poly[di(*n*-butyl) itaconate]	Acetone/methanol (99/1)	30	CP	350
	Acetone/water (97/3)	30	CP	350
	Benzene/methanol (42/58)	30	CP	350
	Butanone/methanol (38/62)	30	CP	350
	Chloroform/methanol (40/60)	30	CP	350
	Ethyl acetate/methanol (40/60)	30	CP	350
	Methanol/toluene (54/46)	30	CP	350
Poly[di(*n*-decyl) itaconate]	Benzene/methanol (82/18)	30	CP	350
	Chloroform/methanol (75/25)	30	CP	350
	Methanol/toluene (21/79)	30	CP	350
Poly(diethyl itaconate)	Acetone/water (91/9)	30	CP	350
	Benzene/*n*-heptane (53/47)	30	CP	350

TABLE 1. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	Butanone/mineral spirits (34/66)	30	CP	350
	Ethanol/water (91/9)	30	CP	350
Poly[di(*n*-heptyl) itaconate]	Benzene/methanol (42/58)	30	CP	350
	Butanone/methanol (88/12)	30	CP	350
Poly[di(*n*-hexyl) itaconate]	Benzene/methanol (64/36)	30	CP	350
	Butanone/methanol (68/32)	30	CP	350
	Chloroform/methanol (64/36)	30	CP	350
	Ethyl acetate/methanol (78/22)	30	CP	350
	Methanol/toluene (28/72)	30	CP	350
Poly(dimethyl itaconate)	Acetone/methanol (71/29)	30	CP	350
	Acetone/water (80/20)	30	CP	350
	Benzene/*n*-hexane (93/7)	30	CP	350
	Benzene/methanol (41/59)	30	CP	350
	Butanone/cyclohexane (86/14)	30	CP	350
	Butanone/*n*-hexane (90/10)	30	CP	350
	Butanone/methanol (62/38)	30	CP	350
	Chloroform/methanol (32/68)	30	CP	350
	Ethanol/water (95/5)	30	CP	350
Poly(*N*-2,4-dimethyl phenylmaleimide)	Butyl acetate	21	DM,VM	342
Poly[di(*n*-nonyl) itaconate]	Ethanol/water (5/95)	30	CP	350
Poly[di(*n*-octyl) itaconate]	Benzene/methanol (80/20)	30	CP	350
	Butanone/methanol (99/1)	30	CP	350
	Chloroform/methanol (71/29)	30	CP	350
	Ethyl acetate/methanol (99/1)	30	CP	350
	Methanol/toluene (22/78)	30	CP	350
Poly[di(*n*-propyl itaconate]	Acetone/water (94/6)	30	CP	350
	Ethanol/water (8/2)	30	CP	350
Poly[di(*n*-undecyl) itaconate]	Benzene/methanol (83/17)	30	CP	350
	Chloroform/methanol (77/23)	30	CP	350
Poly[*N*-(*n*-octadecyl) maleimide]	*n*-Butanol	120.5	CP,VM	17
	n-Decanol	39.3	CP,VM	17
	n-Hexanol	78.9	CP,VM	17
	n-Octanol	53.6	CP,VM	17
Poly(mono(*n*-octyl) itaconate)	2-Propanol	30	VM	386
Poly(tetrafluoroethylene-*co*-trifluoronitrosomethane (1/1)	Trichlorotrifluoroethane	35	VM	248

TABLE 2. HETEROCHAIN POLYMERS

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
2.1. POLY(ACETALS) AND POLY(ETHERS)				
Poly(oxy-2,6-dimethyl-1,4-phenylene)	Methylene chloride	89.2	VM	317
Poly(oxyethylene); poly(ethylene oxide)	Acetonitrile/*i*-propyl ether (45/55)	20	CP	98
	Benzene/*i*-octane (100/48)	70.8	CT	134
		71	PE	8
	$CaCl_2$/water (2 mol/l)	82	CT	260
		86	CT	36
	Chloroform/*n*-hexane (54/46)	20	CP	98
	(47.4/52.6)	20	CT,VM	90
	CsCl/water (2 mol/l)	60	CT	260
	Diethylene glycol diethylether	50	VM	25
	KCl/water (2 mol/l)	54	CT	36
		57	CT	260
	KF/water (0.30 mol/l)	78.0	PE	482
	(0.50 mol/l)	66.0	PE	482
	(0.80 mol/l)	49.0	PE	482
	(1.00 mol/l)	38.0	PE	482
	KNO_3/water (2 mol/l)	65	CT	36
	KOH/water (1.24 mol/l)	25	PE,VM	416
	K_3PO_4/water (0.20 mol/l)	63.0	PE	482

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Table 2. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	(0.30 mol/l)	45.0	PE	482
	(0.35 mol/l)	38.0	PE	482
	(0.40 mol/l)	26.0	PE	482
	K_2SO_4/water (0.45 mol/l)	34	CT	259
		34	CT	36
		35	PE	13
		35	VM	25
	LiCl/water (2 mol/l)	90	CT	36
		90	CT	260
	Methyl *i*-butyl ketone	50	VM	25
	$MgCl_2$/water (2 mol/l)	80	CT	260
		90	CT	36
	$MgSO_4$/water (0.39 mol/l)	42	CT	259
		42	CP	36
		42		536
		45	PE	13
	NaCl/water (2 mol/l)	60	CT	260
		61	CT	260
	Na acetate/water (0.75 mol/l)	75.0	PE	482
	(1.50 mol/l)	55.0	PE	482
	(2.00 mol/l)	45.0	PE	482
	(3.00 mol/l)	31.0	PE	482
	$Na_2S_2O_3$ (0.20 mol/l)	74.0	PE	482
	(0.30 mol/l)	63.0	PE	482
	(0.40 mol/l)	52.0	PE	482
	(0.50 mol/l)	41.0	PE	482
	NH_4Cl/water (2 mol/l)	76	CT	260
		77	CT	36
	Nitroethane/*i*-propyl ether (45/55)	20	CP	98
	RbCl/water (2 mol/l)	56	CT	260
	$SrCl_2$water (2 mol/l)	73	CT	260
		82	CT	36
	$ZnSO_4$/water (0.30 mol/l)	54.0	PE	482
	(0.35 mol/l)	48.0	PE	482
	(0.45 mol/l)	35.0	PE	482
	21 salts in water (various concentrations)	97		417
	Water	5.1	DM,VM	45
		96	CT	36
		102	A	418
		103	VM	416
		108.3	VM	419
		117	CT	260
		92 - 109	VT	383
Poly(oxyethylethylene), poly(butylene oxide)	*i*-Propanol	29.8	VM	225
		30	VM	35
Poly(oxy(2-hydroxytrimethylene)oxy-1,4-phenylene-isopropylidene-1,4-phenylene)	Cyclohexanone/toluene (32.5/67.5)	22	VM	194
	Tetrahydrofuran/toluene (2/1)	30	SM	37
Poly(oxypropylene); poly(propylene oxide)	*i*-Octane	50	A	10
Poly(oxytetrafluoroethylene)-*co*-poly(oxydifluoromethylene) (40/60)	Freon TF	25	VM	415
Poly(oxytetramethylene); poly(tetrahydrofuran)	Acetonitrile/benzene (61.5/38.5)	25	CP	94
	Acetonitrile/butanone (38.3/61.7)	25	CP	94
	Acetonitrile/carbon tetrachloride (50.2/49.8)	25	CP	94
	Acetonitrile/chlorobenzene (60.1/39.9)	25	CP	94
	Acetonitrile/tetrahydrofuran (58.7/41.3)	25	CP	94
	Acetonitrile/toluene (61/39)	25	CP	94
	n-Butanol	5	PE	107
	Butanone	25	VM	214
	Chlorobenzene	25	VM	214
	Chlorobenzene/*n*-octane (25.0/75.0)	10	A, CP	106
	(21.5/78.5)	25.8	A, CP	106

Table 2. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	(14.5/85.5)	46.0	A, CP	106
	(13.0/87.0)	63.0	A, CP	106
	(10.9/89.1)	80.0	A, CP	106
	Cyclohexane/*n*-heptane	26	VM	214,228
	Diethyl malonate	33.5	PE	105
	Ethyl acetate/*n*-hexane (22.7/77.3)	30.4	PE	105
		33	A	199
	i-Propanol	44.6	PE	105
	Toluene	28.3	A	9
Poly(oxytrimethylene)	Cyclohexane	27	PE	55
2.2. POLY(ESTERS)				
Poly(bisphenol A-*co*-(isophthalic acid + terephthalic acid; 1:1)	Cyclohexane/dioxane (32/68)	25	CP	437
Poly(decamethylene perfluorosebacate)	*s*-Tetrachloroethane	25	VM	446
Poly(diphenylphthalide dicarboxylic acid-*alt*-*p*,*p*′-dihydroxydiphenyl)	Dichloroethane	25	A	420
Poly(β-hydroxybutyrate)	Chloroform/heptane (50.2/49.8)	25	CP	421
	2,2,2-Trifluoroethanol/water (52.6/47.4)	25	CP	421
Poly(1-(6-[4-(4-methoxyphenoxycarbonyl)-phenoxy]hexyloxycarbonyl)-1-methyl-ethylene)	Toluene	17.0	A,VM	506
Poly(oxyadipoyloxyhexamethylene); poly(hexamethylene adipate)	Cyclohexanol	64	PE	360
Poly(oxyadipoly-poly(oxyethylene) mol. wt. of PEOX = 150	Chloroform/*n*-hexane (65/35)	20	CP	98
200	(63/37)	20	CP	99
300	(62/38)	20	CP	98
387	(60/40)	20	CP	98
600	(59/41)	20	CP	98
2580	(57/43)	20	CP	98
7000	(56/44)	20	CP	98
Poly(oxyadipoyloxytetramethylene); poly(tetramethylene adipate)	Benzene/*n*-hexane (83.3/16.7)	25	CP	175
	Cyclohexanol	77	PE	360
	Chlorobenzene/*n*-hexane (75.9/24.1)	25	CP	175
	n-Hexane/nitrobenzene (43.3/56.7)	25	CP	175
Poly(oxycaproyl); poly(ε-caprolactone)	Benzene/*n*-hexane (85.1/14.9)	25	CP	175
	Chlorobenzene/*n*-hexane (70.2/29.8)	25	CP	175
	Nitrobenzene/*n*-hexane (67.0/33.0)	25	CP	175
Poly(oxycarbonyloxy-1,4-phenylene-isopropylidene-1,4-phenylene); bisphenol A polycarbonate	*n*-Butyl benzyl ether	170	A	27
	Chloroform	20	A	73
	Chloroform/ethanol (74.5/25.5)	18	CP	193
	Cyclohexane/dioxane (36.1/63.9; w/w)	25	VM	241
Poly(oxyethyleneoxyadipoyl)	Cyclohexanol	114.5	PE	360
Poly(oxyethyleneoxysuberoyl)	Cyclohexanol	88	PE	360
Poly(oxyethyleneoxyterephthaloyl); poly(ethylene terephthalate)	Cyclohexane/dichloroacetic acid (6/4)	25	A,VM	344
Poly(oxyoctamethyleneoctasebacoyl)	Benzene/*n*-hexane (48.3/51.7)	25	CP	175
	Bromobenzene/*n*-hexane (45.0/55.0)	25	CP	175
	Chlorobenzene/*n*-hexane (37.5/62.5)	25	CP	175
DL-Poly(oxy(1-oxo-3-methyltrimethylene)) poly(DL-β-methyl-propiolactone)	Butyl chloride	13	A,VM	143
Poly(oxysebacoyloxy-poly(oxyethylene)); mol. wt. of PEOX = 387	Acetonitrile/*i*-propyl ether (35.8/64.2)	20	CP	98
	Chloroform/*n*-hexane (56/44)	20	CP	98
Poly(12-oxystearoyl); poly(12-hydroxystearic acid)	*n*-Butanol/*n*-heptane (81/19)	1	CP	257
	Ethanol/*n*-heptane (39/61)	1	CP	257
	(51/49)	24	CP	257
	(59/41)	40	CP	257
	n-Heptane/*n*-propanol (35/65)	1	CP	257
	(16/84)	24	CP	257

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Table 2. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
Poly(oxysuberoyl-poly(oxyethylene)) mol. wt. of PEOX = 387	Acetonitrile/*i*-propyl ether (38.6/61.4)	20	CP	98
	Chloroform/*n*-hexane (58/42)	20	CP	98
Poly(oxysebacoyloxy-poly(oxyethylene)); mol. wt. of PEOX = 387	Acetonitrile/*i*-propyl ether (35.8/64.2)	20	CP	98
	Chloroform/*n*-hexane (61/39)	20	CP	98
Poly(phenolphthalein-*alt*-terephthalic acid)	Tetrahydrofuran	19	A	420
Poly[2,2-propane-bis(4-phenylthiocarbonate)]	Chloroform/cyclohexane (76.5/23.5)	25	VM	469
2.3. POLY(AMIDES)				
Poly(2-acrylamido-2-methylpropane sulfonamide)	Dioxane/water (18.5/81.5)	25	A,VM	475
Poly(γ-benzyl-L-glutamate)	Dichloroethane/diethylene glycol (80/20)	25	A	209
Poly([*n*-hexylimino)carbonyl]; poly(*n*-hexyl isocyanate)	Carbon tetrachloride/methanol (81.5/18.5)	25	CP	29
	Methanol/toluene (19.5/80.5)	25	CP	29
Poly(iminoadipoyliminohexamethylene); poly(hexamethylene adipamide)	Carbon tetrachloride/*m*-cresol/cyclohexane	20	SM	335
	Formic acid/KCl/water	25	A,VM	101
Poly(imino(1-oxohexamethylene)); poly(ε-caprolactam)	$CaCl_2$/methanol (15 g/dl)	35	VM	256
	Chloroform/perfluoro alcohol (7/3)	25	PE	198
	m-Cresol/decalin (25/75)	30	CT	148
	KCl in formic acid (2.3 mol/l)/water (90/10)	25	VM	422
Poly(iminoterephthaloylimino-2,2,4/2,4,4-trimethylhexamethylene)	Aniline	142	CT, PE	141
	Pyrroline	62	PE	141
Poly[*N*-methylimino(1-oxodecamethylene)]; poly(*N*-methyldodecane lactam)	Dioxane	30.5	A,VM	286
Poly(L-proline)	Trifluoroethanol	130	A	227
	Water	100	A	227
Poly(pyromellitic acid-*co*-4-amino-1,4-phenyleneoxy-1,4-phenyleneamine)	Dimethylacetamide/dioxane (1/2.5)	21	A,VM	449
Poly(pyromellitic acid-*co*-4-amino-1,4-phenyleneoxy-1,4-phenyleneoxy-1,4-phenyleneamine)	Dimethylacetamide/dioxane (1/7.5)	21	A,VM	449
2.4. POLYUREAS, POLYURETHANES AND OTHER NITROGEN CHAINS				
Poly(1,4-butylene glycol-*co*-4,4′-diphenyl-methanediisocyanate-*co*-poly(oxycaprolyl))	Dimethylformamide	82	VM	318
Poly(ethylene glycol-*co*-4,4′-diphenyl-methane diisocyanate)	Acetone/dimethylformamide (21/79)	25	VM	23
Poly[2-methoxy-4,6-di(*p*,*p*′-isopropylidene diphenyloxy) *s*-triazine]	Butanol/1,2-dichloroethane (38/62)	30	VM	435
Poly(ureylene heptamethylene); Poly(heptamethylene urea)	Dichloroacetic acid	46	VM	109
	Sulfuric acid (90%)	46	VM	109
Polyurethane from polypropylene glycol and 2,4,4-tolylene diisocyanate	2,4,4-Trimethyl pentane	39.5	VM	236
2.5. POLYSACCHARIDES				
Amylopectin	Water	25	A	325
Amylose	Acetone/dimethyl sulfoxide (43.5/56.5)	20	VM	40
	Dimethyl sulfoxide/methanol (49/51)	25.0	VM	61
	Dimethyl sulfoxide/0.5 mol/l KCl in water (25/75)	25	VM	61
	Guanidinium chloride in water (4.2 mol/l)	25	VM	424
	0.33 mol/l KCl in water	22.5	VM	14
		25	VM	108
	Sodium acetate buffer	25	VM	61
Amylose tricarbanilate	Pyridine/water (86.7/13.3)	25	A	15
Cellulose diacetate	Acetone	155	PE	330
		159.8	PE	426
	Butanone	37	PE	331
		49	PE	426

Table 2. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
		50	A	332
		158.8	PE	426
		170	PE	331
Cellulose triacetate	Acetone	27	PE	65
		181	PE	65
Cellulose tributyrate	Tetrachloroethane	56.6	A	150
Cellulose tricaproate	Dimethylformamide	41	A	191
Cellulose tricaprylate	Dimethylformamide	140	PE	215
	3-Phenylpropanol	48	PE	215
Cellulose tricarbanilate	Anisole	94	A	425
	Cyclohexanol	73	A	425
Dextran	Water	101.8	A	535
Hydroxyethyl cellulose	Water	−73	A	423
2.6. CARBON–SULFUR CHAINS				
Poly(arylate-arylenesulfonoxide)	Dioxane	23.5		501
Poly(arylate-*block*-arylenesulfonoxide)	Dioxane	23.5		501
Poly(propylene sulfide)	*n*-Heptane/toluene (31.5/68.5)	25	CP	261
Poly(sulfonyl-1,4-phenyleneoxy-1,4-phenylene-dimethylmethylene-1,4-phenylene-oxy)	Dimethylsulfoxide	105.5	VM	428
Poly(arylene sulfonate)	Dioxane	26.8	PE	436
Poly(diphenylether sulfone)	Dimethylformamide/methanol (83/17)	25.0	VM	500
2.7. SILICON–OXYGEN CHAINS				
Poly(oxydimethylsilylene) poly(dimethyl siloxane)	Acetone/toluene (6/5)	29.5	A	68
	Benzene	−7	A	202
		−5.2	VM	201
		−3	A	202
	Benzyl alcohol/carbon tetrachloride (21.2/78.8)	25	CP	94
	Benzyl alcohol/chloroform (12.8/87.2)	25	CP	94
	Benzyl alcohol/ethyl benzene (14.6/85.4)	25	CP	94
	Benzyl alcohol/methylene chloride (22.7/77.3)	25	CP	94
	Benzyl alcohol/trichloroethylene (21.7/78.3)	25	CP	94
	Benzyl alcohol/toluene (16.2/83.8)	25	CP	94
	Bromobenzene	78.3	VM	201
		78.7	A	309
	Bromocyclohexane	28	VM	433,459
		36.8	VM	201
		39.0	A	209
	Bromocyclohexane/phenetole (85.7/14.3)	36.3	A	309
	Butanone	19.8	PE	432
		19.8	VM	201
		20.0	A	67
		20.0	PE	111
		20.0	A	202
	n-Butyl acetate	−38.2	VM	201
	Chlorobenzene	−19.2	VM	201
		68	A	202
	Chlorobenzene/dimethyl phthalate (88.2/11.8)	56.3	A	309
	Cyclohexane	−81	A	202
		−68.2	VM	201
	Cyclohexanol/toluene (34/66)	25	CP	94
	Dichlorobenzene	87	PE	71
	Ethanol/*n*-heptane (49/51)	63	CT	72
		65	PE	72
		67 ± 2		536
	Ethyl acetate	4.8	VM	201
		18	A	202
	Ethyl iodide	2.1	A	309
	n-Heptane	−173.2	VM	201
	n-Hexane	−173.2	VM	201
	Methylcyclohexane	−113.2	VM	201

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Table 2. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	Methylcyclopentane	−98.2	VM	201
	Nitroethane/toluene (25.9/74.1)	25	CP	94
	Nitromethane/toluene (18.9/81.1)	25	CP	94
	Nitropropane/toluene (36/64)	25	CP	94
	n-Nonane	−113.2	VM	201
	n-Octane	−143.2	VM	201
	Perfluorooctane/tetrachlorodifluoroethane (33.2/66.8; w/w)	22.5	A, PE	67
	Phenetole	83.0	PE	111
		83.0	A	202
		89.3	VM	201
		89.5	A	309
	n-Propyl acetate	−28.2	VM	201
	Styrene	35	A	429
	Toluene	−30	A	202
		−33.2	VM	201
	Xylene	−47	A	202
		−48.2	VM	201
Poly(methylphenylsiloxane)	Di-*i*-butylamine	30.4	VM	487
Poly(phenylsilasesquioxane)	1,2-Dichloroethane	50.5	VM	138,431
		56		276
	Mesitylene/toluene (30/70)	25	A	185
Poly(γ-trifluoropropyl methyl siloxamer)	Cyclohexyl acetate	25	VM	38
	Methyl hexanoate	72.8	VM	38
2.8. PHOSPHORUS–OXYGEN CHAINS				
Poly(phosphate)	LiBr/water (1.80 mol/l)	25	A	326
	LiCl/water (0.40 mol/l)	20	A	297
	NaBr/water (0.415 mol/l)	25	A	327

C. REFERENCES

1. M. Abe, H. Fujita, Rep. Progr. Polym. Phys. Japan, **7**, 42 (1964).
2. M. Abe, H. Fujita, J. Phys. Chem., **69**, 3263 (1965).
3. M. Abe, K. Sakato, T. Kageyama, M. Fukatsu, M. Kurata, Bull. Chem. Soc. Japan, **41**, 2330 (1968).
4. P. Adamski, Vysokomol. Soedin. A, **13**, 709 (1971); Polym. Sci. USSR, **13**, 803 (1973).
5. P. Adamski, Vysokomol. Soedin. B, **13**, 666 (1971).
6. P. Adamski, Zesz. Nauk Politech. Lodz, **23**, 5 (1971).
7. P. Adamski, M. Kryshevski, IUPAC Symp. Macromolecules, Toronto, 1968.
8. A. M. Afifi-Effat, J. N. Hay, M. Wiles, J. Polym. Sci. B, **11**, 87 (1973).
9. S. M. Ali, M. B. Huglin, Makromol. Chem., **84**, 117 (1965).
10. G. Allen, C. Booth, C. Price, Polymer, **7**, 167 (1966).
11. F. J. Ansorene, L. M. Revuelta, G. M. Guzmán, J. J. Iruin, Europ. Polym. J., **18**, 19 (1982).
12. M. Asahina, M. Sato, Bull. Chem. Soc. Japan, **35**, 630 (1962).
13. F. E. Bailey, Jr., R. W. Callard, J. Appl. Polym. Sci., **1**, 56 (1959).
14. W. Banks, C. T. Greenwood, Makromol. Chem., **67**, 49 (1963); Carbohydrate Res., **7**, 349 (1968).
15. W. Banks, C. T. Greenwood, J. Sloss, Makromol. Chem., **140**, 109 (1970).
16. J. M. Barrales-Rienda, D. C. Pepper, Polymer, **8**, 337 (1967).
17. J. M. Barrales-Rienda, C. Romero Galicia, J. F. Freire, A. Horta, Macromolecules, **16**, 337 (1983).
18. T. W. Bates, J. Biggins, K. J. Ivin, Makromol. Chem., **87**, 180 (1965).
19. T. W. Bates, K. J. Ivin, Polymer, **8**, 263 (1967).
20. B. J. Bauer, N. Hadjichristidis, L. J. Fetters, J. E. L. Roovers, J. Am. Chem. Soc., **102**, 2410 (1980).
21. U. Baumann, H. Schreiber, K. Tessmar, Makromol. Chem., **36**, 81 (1960).
22. A. Bazuaye, M. B. Huglin, Polymer, **20**, 44 (1979).
23. H. C. Beachell, J. C. Peterson, J. Polym. Sci. A-1, **7**, 2021 (1969).
24. M. Becerra, D. Radić, L. Gargallo, Makromol. Chem. **179**, 2241 (1978).
25. D. R. Beach, C. Booth, J. Polym. Sci. A-2, **7**, 575 (1969).
26. G. C. Berry, J. Chem. Phys., **44**, 4550 (1966).
27. G. C. Berry, H. Nomura, K. G. Mayhan, J. Polym. Sci. A-2, **5**, 1 (1967).
28. G. C. Berry, H. Nakayasu, T. G. Fox, J. Polym. Sci., Polym. Phys. Ed., **17**, 1825 (1979).
29. M. N. Berger, B. M. Tidswell, J. Polym. Sci.-Symp., **42**, 1063 (1973).
30. U. Bianchi, V. Magnasco, C. Rossi, Ric. Sci., **28**, 1412 (1958).
31. U. Bianchi, V. Magnasco, C. Rossi, Chim. Ind. (Milan), **40**, 263 (1958).

32. M. Bohdanecký, Coll. Czech. Chem. Commun., **29**, 876 (1964).
33. M. Bohdanecký, V. Petrus, B. Sedláček, Makromol. Chem., **184**, 2061 (1983).
34. M. Bohdanecký, Z. Tuzar, Coll. Czech. Chem. Commun., **34**, 3318 (1969).
35. C. Booth, R. Orme, Polymer, **11**, 626 (1970).
36. E. A. Boucher, P. M. Hines, J. Polym. Sci., Polym. Phys. Ed., **16**, 501 (1978).
37. J. Brezinski, B. Czamecka, Polimery Tworzywa, **14**, 117 (1969).
38. R. R. Buch, H. M. Klimisch, O. K. Johannson, J. Polym. Sci. A-2, **7**, 563 (1969).
39. W. Burchard, Habilitation Thesis, University of Freiburg, Germany.
40. W. Burchard, Makromol. Chem., **64**, 110 (1963).
41. W. Burchard, Z. Physik. Chem. (Frankfurt/M.), **43**, 265 (1964).
42. F. Candau, C. Strazielle, H. Benoit, Makromol. Chem., **170**, 165 (1973).
43. H. -J. Cantow, Makromol. Chem., **30**, 169 (1959).
44. D. K. Carpenter, J. H. Hsieh, ACS Polymer Preprints, **13**, 981 (1972).
45. B. Chew, A. Couper, J. Chem. Soc. Faraday Soc., [1], **72**, 382 (1976).
46. R. Chiang, J. Phys. Chem., **69**, 1645 (1965).
47. S. N. Chinai, J. Polym. Sci., **25**, 413 (1957).
48. S. N. Chinai, J. Polym. Sci., **44**, 475 (1959).
49. S. N. Chinai, C. W. Bondurant, J. Polym. Sci., **22**, 555 (1956).
50. S. N. Chinai, R. A. Guzzi, J. Polym. Sci., **21**, 417 (1956).
51. S. N. Chinai, A. L. Resnick, H. T. Lee, J. Polym. Sci., **33**, 471 (1958).
52. S. N. Chinai, R. J. Samuels, J. Polym. Sci., **19**, 463 (1956).
53. S. N. Chinai, P. C. Scherer, C. W. Bondurant, D. W. Levi, J. Polym. Sci., **22**, 527, (1956).
54. S. N. Chinai, R. J. Valles, J. Polym. **39**, 363 (1959).
55. D. S. Chiu, Y. Takahashi, J. E. Mark, Polymer, **17**, 670 (1976).
56. A. Ciferri, M. Lauretti, Ann. Chim. (Rome), **48**, 198 (1958).
57. E. Cohn-Ginsberg, T. G. Fox, H. F. Mason, Polymer, **3**, 97 (1962).
58. C. F. Cornet, H. van Ballegooijen, Polymer, **7**, 293 (1966).
59. B. J. Cottam, J. M. G. Cowie, S. Bywater, Makromol. Chem., **86**, 116 (1965).
60. J. P. Cotton, D. Decker, H. Benoit, B. Farnoux, J. Higgins, G. Jannink, R. Ober, J. des Cloizeaux, Macromolecules, **7**, 863 (1974).
61. J. M. G. Cowie, Makromol. Chem., **59**, 189 (1963).
62. J. M. G. Cowie, J. Polymer. Sci. C, **23**, 267 (1968).
63. J. M. G. Cowie, S. Bywater, J. Polym. Sci. A-2, **6**, 499 (1968).
64. J. M. G. Cowie, S. Bywater, D. J. Worsfold, Polymer, **8**, 105 (1967).
65. J. M. G. Cowie, A. Maconnachie, R. J. Ranson, Macromolecules, **4**, 57 (1971).
66. J. M. G. Cowie, I. J. McEwen, Polymer, **16**, 933 (1975).
67. V. Crescenci, P. J. Flory, J. Am. Chem. Soc., **86**, 141 (1964).
68. A. Czuppon, IUPAC Symp. Macromolecules, Prague 1965, Preprint 459.
69. F. Danusso, G. Moraglio, Atti Accad. Nazl. Lincei Rend. Classe Sci. Fis. Mat. Nat., [8], **25**, 509 (1958).
70. F. Danusso, G. Moraglio, G. Gianotti, J. Polym. Sci., **51**, 475 (1961).
71. J. V. Dawkins, M. Hemming, Makromol. Chem., **155**, 75 (1972).
72. J. V. Dawkins, G. Taylor, Colloid Polym. Sci., **258**, 79 (1980).
73. A. de Chirico, Chim. Ind. (Milan), **42**, 248 (1960).
74. A. de Chirico, Chim. Ind. (Milan), **46**, 53 (1964).
75. D. Decker, Makromol. Chem., **125**, 136 (1969).
76. G. Delmas, D. Patterson, Polymer, **7**, 513 (1966).
77. F. E. Didot, S. N. Chinai, D. W. Levi, J. Polym. Sci., **43**, 557 (1960).
78. H. Dietschy, Ph.D. Thesis, ETH Zürich 1966.
79. H. A. Dieu, J. Polym. Sci., **12**, 417 (1954).
80. J. Dohany, Ph.D. Thesis, ETH Zürich 1963.
81. A. Dondos, Makromol. Chem., **135**. 181 (1970).
82. A. Dondos, H. Benoit, J. Polym. Sci. B, **7**, 279 (1969).
83. A. Dondos, H. Benoit, Macromolecules, **4**, 279 (1971).
84. P. Dreyfus, M. P. Dreyfus, Adv. Polym. Sci., **4**, 528 (1967).
85. E. T. Dumitru, L. H. Cragg, Cited in ref. [110], p. 615.
86. K. Dušek, M. Bohdanecký, E. Prokopová, Eur. Polym. J., **10**, 239 (1974).
87. H. Eisenberg, D. Woodside, J. Polym. Phys. **36**, 1844 (1962).
88. K. H. Elgert, PhD Thesis, Techn. Univ. of Munich, 1963.
89. H.-G. Elias, Makromol. Chem., **33**, 140 (1959).
90. H.-G. Elias, Makromol. Chem., **50**, 1 (1961).
91. H.-G. Elias, Makromol. Chem., **54**, 78 (1962).
92. H.-G. Elias, "Makromoleküle", Hüthig and Wepf, Basle, Fourth Ed. 1981, p. 200; "Macromolecules", Plenum, New York, Second Ed. 1984, Vol. I., p. 232.
93. H.-G. Elias, in: L. H. Tung, (Ed.), "Fractionation of Synthetic Polymers", Dekker, New York, 1977.
94. H.-G. Elias, G. Adank, Makromol. Chem., **69**, 241 (1963).
95. H.-G. Elias, O. Etter, Makromol. Chem., **66**, 56 (1963).
96. H.-G. Elias, O. Etter, Makromol. Chem., **89**, 228 (1965).
97. H.-G. Elias, O. Etter, J. Makromol. Chem., **1**, 431 (1966).
98. H.-G. Elias, U. Gruber, Makromol. Chem., **78**, 72 (1964).
99. U. Gruber, H. -G. Elias, Makromol. Chem., **86**, 168 (1965).
100. H. -G. Elias, F. W. Ibrahim, Makromol. Chem., **89**, 12 (1965).
101. H. -G. Elias, R. Schumacher, Makromol. Chem., **76**, 23 (1964).
102. V. E. Eskin, L. N. Andreeva, Vysokomol. Soedin., **3**, 435 (1961); cited in J. Polym. Sci., **56**, S 39 (1962).
103. V. E. Eskin, K. Z. Gumargalieva, Vysokomol. Soedin., **2**, 265 (1960).
104. V. E. Eskin, O. Z. Korothina, Vysokomol. Soedin., **2**, 272 (1960); J. Polym. Sci. USSR, **2**, 247 (1961).
105. J. M. Evans, M. B. Huglin, Makromol. Chem., **127**, 141 (1969).
106. J. M. Evans, M. B. Huglin, Europ. Polym. J., **6**, 1161 (1970).

107. J. M. Evans, M. B. Huglin, cited in [84].
108. W. W. Everett, J. F. Foster, J. Am. Chem. Soc., **81**, 3459, 3464 (1959).
109. J. Feisst, H.-G. Elias, Makromol. Chem., **82**, 78 (1965).
110. P. J. Flory, "Principles of Polymer Chemistry", Cornell Univ. Press, Ithaca, NY, 1953.
111. P. J. Flory, L. Mandelkern, J. B. Kinsinger, W. B. Shultz, J. Am. Chem. Soc., **74**, 3364 (1952).
112. T. G. Fox, Polymer, **3**, 111 (1962).
113. T. G. Fox, P. J. Flory, J. Am. Chem. Soc., **73**, 1909 (1959).
114. T. G. Fox, P. J. Flory, J. Am. Chem. Soc., **73**, 1915 (1959).
115. T. G. Fox, J. B. Kinsinger, H. F. Mason, E. M. Schuele, Polymer, **3**, 71 (1962).
116. D. M. French, A. W. Casey, C. I. Collins, P. Kirchner, ACS Polymer Preprints, **7**, 447 (1966).
117. D. Froelich, Ph.D. Thesis, Univ. of Strasbourg, 1966; cited in [83].
118. D. Froelich, H. Benoit, Makromol. Chem., **92**, 224 (1966).
119. M. Fukuda, M. Fukutomi, Y. Kato, T. Hashimoto, J. Polym. Sci.-Polym. Chem. Ed., **12**, 871 (1974).
120. C. Garbuglio, L. Crescentini, A. Mula, G. B. Gechele, Makromol. Chem., **97**, 97 (1966).
121. L. Gargallo, J. Niezzette, V. Desreux, Bull. Soc. Roy. Sci. Liege, **1–2**, 82 (1977); cited in [122].
122. L. Gargallo, D. Radić, I. Katime, Europ. Polym. J., **16**, 383 (1980).
123. G. B. Gechele, L. Crescentini, J. Polym. Sci. A, **3**, 3599 (1965).
124. H. Geerissen, B. A. Wolf, Makromol. Chem., Rapid Commun., **3**, 17 (1982).
125. G. Gianotti, U. Bonicelli, D. Borghi, Makromol. Chem., **166**, 235 (1973).
126. C. González, F. Zamora, G. M. Guzmán, L. M. León, Makromol. Chem., **185**, 339 (1984).
127. T. B. Grimley, Proc. Roy. Soc. [London] A, **212**, 339 (1952).
128. U. Gruber, Ph.D. Thesis, ETH Zürich, 1964.
129. N. Hadjichristidis, Polymer, **16**, 848 (1975).
130. N. Hadjichristidis, J. E. L. Roovers, J. Polym. Sci., Polym. Phys. Ed., **12**, 2521 (1974).
131. J. Hakozaki, J. Chem. Soc. Japan, Pure Chem. Sect. (Nippon Kagaku Zasshi), **82**, 158 (1961).
132. F. Hamada, K. Fujisawa, A. Nakajima, Polym. J., **4**, 316 (1973).
133. K. Hanafusa, A. Teramoto, H. Fujita, J. Chem. Phys., **70**, 4004 (1966).
134. J. N. Hay, A. M. Afifi-Effat, Brit. Polym. J., **9**, 1 (1977).
135. J. N. Hay, P. A. Fitzgerald, J. Polym. Sci., Polym. Chem. Ed., **18**, 1079 (1980).
136. J. N. Hay, M. Wiles, Brit. Polym. J., **10**, 181 (1978).
137. W. Hayduk, H. A. Bromfield, J. Appl. Polym. Sci., **22**, 149 (1978).
138. T. E. Helminiak, C. L. Benner, W. E. Gibbs, ACS Polymer Preprints, **8**, 284 (1967).
139. I. Hernandez-Fuentes, M. G. Prolongo, Europ. Polym. J., **15**, 571 (1979).
140. F. K. Herold, B. A. Wolf, Makromol. Chem., **184**, 2539 (1981).
141. J. Herold, G. Meyerhoff, Europ. Polym. J., **15**, 525 (1979).
142. M. Heskins, J. E. Guillet, J. Macromol. Sci. A, **2**, 1441 (1968).
143. T. Hirosye, Y. Einaga, H. Fujita, Polym. J., **11**, 819 (1979).
144. T. Homma, K. Kawahara, H. Fujita, M. Ueda, Makromol. Chem., **67**, 132 (1963).
145. T. Homma, K. Kawahara, H. Fujita, J. Appl. Polym. Sci., **8**, 2853 (1964).
146. F. Hooshmand-Mozaffar, M. K. Hoseinalizadeh-Khorasani, M. B. Huglin, Polymer, **21**, 413 (1980).
147. F. Horii, Y. Ikada, I. Sakurada, J. Polym. Sci., Polym. Chem. Ed., **12**, 323 (1974).
148. N. Howard, M. B. Huglin, R. W. Richards, J. Appl. Polym. Sci., **16**, 1525 (1972).
149. A. J. Hyde, R. B. Taylor, Polymer, **4**, 1 (1963).
150. T. Ikeda, H. Kawaguchi, Rep. Progr. Polym. Phys. Japan, **9**, 23 (1966).
151. H. Inagaki, A. Nakazawa, T. Kotaka, Bull. Inst. Chem. Res. Kyoto Univ., **43**, 135 (1965).
152. H. Inagaki, T. Miyamoto, S. Ohta, J. Phys. Chem., **70**, 3420 (1966).
153. H. Inagaki, H. Suzuki, M. Fujii, T. Matsuo, J. Phys. Chem., **70**, 1718 (1966).
154. K. J. Ivin, H. A. Ende, G. Meyerhoff, Polymer, **3**, 129 (1962).
155. M. Iwama, H. Utiyama, M. Kurata, J. Macromol. Chem., **1**, 701 (1966).
156. Y. Izumi, Y. Miyake, Polym. J., **3**, 647 (1972).
157. Y. Izumi, Y. Miyake, Polym. J., **4**, 205 (1973).
158. Y. Izumi, S. Otsuka, Y. Miyake, Rep. Progr. Polym. Phys. Japan, **12**, 13 (1969).
159. R. Jerome, V. Desreux, Europ. Polym. J., **6**, 411 (1970).
160. R. M. Johnson, J. L. Schrag, J. D. Ferry, Polym. J., **1**, 742 (1970).
161. M. Kalfus, J. Mitus, J. Polym. Sci. A-1, **4**, 953 (1966).
162. K. Kamada, H. Sato, Polym. J., **2**, 593 (1971).
163. K. Kamide, T. Terekawa, Makromol. Chem., **155**, 25 (1972).
164. Ye. A. Kanevskaya, P. I. Zubov, L. V. Ivanova, Yu. S. Lipatov, Vysokomol. Soedin., **6**, 981 (1964); Polym. Sci. USSR, **6**, 1080 (1964).
165. A. Katchalsky, H. Eisenberg, J. Polym. Sci., **6**, 145 (1951).
166. I. A. Katime Amashta, G. Sanchez, Europ. Polym. J., **11**, 223 (1975).
167. H. Kaye, H. J. Cou, J. Polym. Sci., Polym. Phys. Ed., **13**, 477 (1975).
168. W. Kern, D. Braun, Makromol. Chem., **27**, 23 (1958).
169. J. B. Kinsinger, L. E. Ballard, J. Polym. Sci. B, **2**, 879 (1964).
170. J. B. Kinsinger, L. E. Ballard, J. Polym. Sci. A, **3**, 3963 (1965).
171. J. B. Kinsinger, R. E. Hughes, J. Phys. Chem., **67**, 1922 (1963).
172. J. B. Kinsinger, R. A. Wessling, J. Am. Chem. Soc., **81**, 2908 (1959).
173. R. Kirste, G. V. Schulz, Z. Physik. Chem. (Frankfurt/M.), **27**, 310 (1960).
174. R. G. Kirste, G. V. Schulz, Z. Physik. Chem. (Frankfurt/M.), **30**, 171 (1961).

175. M. R. Knecht, H.-G. Elias, Makromol. Chem., **157**, 1 (1972).
176. A. Kokkiaris, C. Touloupis, N. Hadjichristidis, Polymer, **22**, 63 (1981).
177. R. Koningsveld, Th. G. Schoolte, private communication.
178. T. Kotaka, H. Ohnuma, H. Inagaki, Polymer, **10**, 517 (1969).
179. T. Kotaka, H. Ohnuma, H. Inagaki, ACS Polymer Preprints, **11**, 660 (1970).
180. T. Kotaka, H. Ohnuma, Y. Murakami, J. Phys. Chem., **70**, 4099 (1966).
181. T. Kotaka, T. Tanaka, H. Ohnuma, Y. Murakami, H. Inagaki, Polymer J., **1**, 245 (1970).
182. A. Kotera, H. Matsuda, K. Konishi, K. Takemura, J. Polym. Sci. C, **23**, 619 (1968).
183. A. Kotera, N. Onda, T. Saito, Rep. Progr. Polym. Phys. Japan, **17**, 25 (1974).
184. A. Kotera, T. Saito, N. Yamaguchi, Annual Meeting Chem. Soc. Japan, 1962; cited in A. Kotera, N. Ise, A. Senuma, T. Hamada, IUPAC Symp. Macromol., Prague 1965, Preprint 232.
185. J. Kovář, L. Mrkvičková-Vaculová, M. Bohdanecký, Makromol. Chem., **176**, 1829 (1975).
186. P. Kratochvil, D. Strtaková, Makromol. Chem., **154**, 325 (1972).
187. S. Krause, E. Cohn-Ginsburg, J. Phys. Chem., **67**, 1479 (1963).
188. W. R. Krigbaum, D. K. Carpenter, J. Phys. Chem., **59**, 1166 (1955).
189. W. R. Krigbaum, P. J. Flory, J. Polym. Sci., **11**, 37 (1953).
190. W. R. Krigbaum, J. E. Kurz, P. Smith, J. Phys. Chem., **65**, 1984 (1961).
191. W. R. Krigbaum, L. H. Sperling, J. Phys. Chem., **64**, 99 (1960).
192. W. R. Krigbaum, J. D. Woods, J. Polym. Sci. A, **2**, 3075 (1964).
193. S. Krozer, Plaste Kautschuk, **15**, 175 (1968).
194. S. Krozer, A. Pastusiak, IUPAC Symp. Macromol., Prague 1965, Preprint 609.
195. K. Kubo, K. Ogino, Rep. Progr. Polym. Phys. Japan, **10**, 111 (1967).
196. K. Kubo, K. Ogino, T. Nakagawa, Rep. Progr. Polym. Phys. Japan, **10**, 7 (1967).
197. Z. Küçükyavuz, A. Küçükyavuz, Europ. Polym. J., **14**, 867 (1978).
198. A. M. Kulakova, I. K. Nekrasov, Vysokomol. Soedin. A, **18**, 127 (1976); Polym. Sci. USSR, **18**, 146 (1976).
199. M. Kurata, H. Utiyama, K. Kamada, Makromol. Chem., **88**, 281 (1965).
200. N. Kuwahara, S. Higashida, M. Nakata, K. Kaneko, J. Polym. Sci. A-2, **7**, 285 (1969).
201. N. Kuwahara, M. Kaneko, Makromol. Chem., **82** 205 (1965).
202. N. Kuwahara, T. Miyake, M. Kaneko, J. Furuichi, Rep. Progr. Polym. Phys. Japan, **5**, 1 (1962).
203. N. Kuwahara, S. Saeki, S. Konno, M. Kaneko, Macromolecules, **6**, 589 (1973).
204. N. Kuwahara, S. Saeki, S. Konno, Polymer, **15**, 66 (1974).
205. D. Lath, M. Bohdanecký, J. Polym. Sci.-Letters Ed., **15**, 555 (1977).
206. H. T. Lee, D. W. Levi, J. Polym. Sci., **47**, 449 (1960).
207. L. M. León, J. Galaz, L. M. Garcia, M. S. Anasagasti, Europ. Polym. J., **16**, 921 (1980).
208. D. W. Levi, H. T. Lee, R. J. Valles, J. Polym. Sci., **62**, S163 (1962).
209. J. Lilie, J. Springer, K. Ueberreiter, Kolloid-Z., **226**, 138 (1968).
210. F. C. Lin, S. S. Stivala, J. A. Biesenberger, J. Appl. Polym. Sci., **17**, 3465 (1973).
211. M. Lippay. Ph.D. Thesis, ETH Zürich, 1963.
212. J. Llopis, A. Albert, P. Usobiaga, Europ. Polym. J., **3**, 259 (1967).
213. T. Lucas, Bull. Liaison Lab. (Lab. Prof. Peint)., **20**, 15 (1967).
214. N. V. Makletsova, I. V. Epel'baum, B. A. Rozenberg, Ye. B. Lyudvig, Vysokomol. Soedin., **7**, 70 (1965); Polym. Sci. USSR, **7**, 73 (1966).
215. L. Mandelkern, P. J. Flory, J. Am. Chem. Soc., **74**, 2517 (1952).
216. P. V. Mangalam, V. P. Kalpagam, J. Polym. Sci.-Polym. Phys. Ed., **20**, 773 (1982).
217. V. P. Mardykin, K. B. Martinovich, A. I. Volozhin, N. M. Denisyuk, Vysokomol. Soedin. B, **16**, 23 (1974); CA **81**: 13900r (1974).
218. J. E. Mark, Diss. Abstr., **23**, 1205 (1962).
219. J. E. Mark, P. J. Flory, J. Am. Chem. Soc., **87**, 1423 (1965).
220. R. M. Masegosa, M. G. Prolongo, I. Hernandez-Fuentes, A. Horta, Macromolecules, **17**, 1181 (1984).
221. T. Matsumoto, N. Nishioka, H. Fujita, J. Polym. Sci. A-2, **10**, 23 (1972).
222. M. Matsumoto, Y. Ohyanagi, J. Polym. Sci., **46**, 441 (1960).
223. M. Matsumoto, Y. Ohyanagi, J. Polym. Sci., **50**, S1 (1961).
224. K. Matsumura, Polym. J., **1** 322 (1970).
225. M. Matsushima, M. Fukatsu, M. Kurata, Bull. Chem. Soc. Japan, **41**, 2570 (1968).
226. A. Mattiussi, E. Conto, G. B. Gechele, Europ. Polym. J., **8**, 429 (1972).
227. W. L. Mattice, L. Mandelkern, Macromolecules, **4**, 271 (1971).
228. R. M. May, R. E. Wetton, cited in [84].
229. H. W. McCormick, J. Polym. Sci., **36**, 341 (1959).
230. D. McIntyre, J. H. O'Mara, B. C. Konouck, J. Am. Chem. Soc., **81**, 3498 (1959).
231. R. A. Mendelson, J. Polym. Sci., **46**, 493 (1960).
232. G. Meyerhoff, Makromol. Chem., **37**, 97 (1960).
233. G. Meyerhoff, in: J. W. Williams (Ed.), "Ultracentrifugal Analysis in Theory and Practice". Academic Press, New York, 1963.
234. R. Meza, L. Gargallo, Europ. Polym. J., **13**, 235 (1977).
235. S. Miyamoto, Y. Ishii, H. Ohnuma, Makromol. Chem., **182**, 483 (1981).
236. J. Moacanin, J. Appl. Polym. Sci., **1**, 272 (1959).
237. J. Moacanin, A. Rembaum, R. K. Laudenslager, ACS Polymer Preprints, **4/2**, 179 (1963).

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Table 2. *cont'd*

Polymer (mol/mol)	Theta solvent (v/v)	Theta temp. (°C)	Method	Refs.
	Methylcyclopentane	−98.2	VM	201
	Nitroethane/toluene (25.9/74.1)	25	CP	94
	Nitromethane/toluene (18.9/81.1)	25	CP	94
	Nitropropane/toluene (36/64)	25	CP	94
	n-Nonane	−113.2	VM	201
	n-Octane	−143.2	VM	201
	Perfluorooctane/tetrachlorodifluoroethane (33.2/66.8; w/w)	22.5	A, PE	67
	Phenetole	83.0	PE	111
		83.0	A	202
		89.3	VM	201
		89.5	A	309
	n-Propyl acetate	−28.2	VM	201
	Styrene	35	A	429
	Toluene	−30	A	202
		−33.2	VM	201
	Xylene	−47	A	202
		−48.2	VM	201
Poly(methylphenylsiloxane)	Di-*i*-butylamine	30.4	VM	487
Poly(phenylsilasesquioxane)	1,2-Dichloroethane	50.5	VM	138,431
		56		276
	Mesitylene/toluene (30/70)	25	A	185
Poly(γ-trifluoropropyl methyl siloxamer)	Cyclohexyl acetate	25	VM	38
	Methyl hexanoate	72.8	VM	38
2.8. PHOSPHORUS–OXYGEN CHAINS				
Poly(phosphate)	LiBr/water (1.80 mol/l)	25	A	326
	LiCl/water (0.40 mol/l)	20	A	297
	NaBr/water (0.415 mol/l)	25	A	327

C. REFERENCES

1. M. Abe, H. Fujita, Rep. Progr. Polym. Phys. Japan, **7**, 42 (1964).
2. M. Abe, H. Fujita, J. Phys. Chem., **69**, 3263 (1965).
3. M. Abe, K. Sakato, T. Kageyama, M. Fukatsu, M. Kurata, Bull. Chem. Soc. Japan, **41**, 2330 (1968).
4. P. Adamski, Vysokomol. Soedin. A, **13**, 709 (1971); Polym. Sci. USSR, **13**, 803 (1973).
5. P. Adamski, Vysokomol. Soedin. B, **13**, 666 (1971).
6. P. Adamski, Zesz. Nauk Politech. Lodz, **23**, 5 (1971).
7. P. Adamski, M. Kryshevski, IUPAC Symp. Macromolecules, Toronto, 1968.
8. A. M. Afifi-Effat, J. N. Hay, M. Wiles, J. Polym. Sci. B, **11**, 87 (1973).
9. S. M. Ali, M. B. Huglin, Makromol. Chem., **84**, 117 (1965).
10. G. Allen, C. Booth, C. Price, Polymer, **7**, 167 (1966).
11. F. J. Ansorene, L. M. Revuelta, G. M. Guzmán, J. J. Iruin, Europ. Polym. J., **18**, 19 (1982).
12. M. Asahina, M. Sato, Bull. Chem. Soc. Japan, **35**, 630 (1962).
13. F. E. Bailey, Jr., R. W. Callard, J. Appl. Polym. Sci., **1**, 56 (1959).
14. W. Banks, C. T. Greenwood, Makromol. Chem., **67**, 49 (1963); Carbohydrate Res., **7**, 349 (1968).
15. W. Banks, C. T. Greenwood, J. Sloss, Makromol. Chem., **140**, 109 (1970).
16. J. M. Barrales-Rienda, D. C. Pepper, Polymer, **8**, 337 (1967).
17. J. M. Barrales-Rienda, C. Romero Galicia, J. F. Freire, A. Horta, Macromolecules, **16**, 337 (1983).
18. T. W. Bates, J. Biggins, K. J. Ivin, Makromol. Chem., **87**, 180 (1965).
19. T. W. Bates, K. J. Ivin, Polymer, **8**, 263 (1967).
20. B. J. Bauer, N. Hadjichristidis, L. J. Fetters, J. E. L. Roovers, J. Am. Chem. Soc., **102**, 2410 (1980).
21. U. Baumann, H. Schreiber, K. Tessmar, Makromol. Chem., **36**, 81 (1960).
22. A. Bazuaye, M. B. Huglin, Polymer, **20**, 44 (1979).
23. H. C. Beachell, J. C. Peterson, J. Polym. Sci. A-1, **7**, 2021 (1969).
24. M. Becerra, D. Radić, L. Gargallo, Makromol. Chem. **179**, 2241 (1978).
25. D. R. Beach, C. Booth, J. Polym. Sci. A-2, **7**, 575 (1969).
26. G. C. Berry, J. Chem. Phys., **44**, 4550 (1966).
27. G. C. Berry, H. Nomura, K. G. Mayhan, J. Polym. Sci. A-2, **5**, 1 (1967).
28. G. C. Berry, H. Nakayasu, T. G. Fox, J. Polym. Sci., Polym. Phys. Ed., **17**, 1825 (1979).
29. M. N. Berger, B. M. Tidswell, J. Polym. Sci.-Symp., **42**, 1063 (1973).
30. U. Bianchi, V. Magnasco, C. Rossi, Ric. Sci., **28**, 1412 (1958).
31. U. Bianchi, V. Magnasco, C. Rossi, Chim. Ind. (Milan), **40**, 263 (1958).

32. M. Bohdanecký, Coll. Czech. Chem. Commun., **29**, 876 (1964).
33. M. Bohdanecký, V. Petrus, B. Sedláček, Makromol. Chem., **184**, 2061 (1983).
34. M. Bohdanecký, Z. Tuzar, Coll. Czech. Chem. Commun., **34**, 3318 (1969).
35. C. Booth, R. Orme, Polymer, **11**, 626 (1970).
36. E. A. Boucher, P. M. Hines, J. Polym. Sci., Polym. Phys. Ed., **16**, 501 (1978).
37. J. Brezinski, B. Czamecka, Polimery Tworzywa, **14**, 117 (1969).
38. R. R. Buch, H. M. Klimisch, O. K. Johannson, J. Polym. Sci. A-2, **7**, 563 (1969).
39. W. Burchard, Habilitation Thesis, University of Freiburg, Germany.
40. W. Burchard, Makromol. Chem., **64**, 110 (1963).
41. W. Burchard, Z. Physik. Chem. (Frankfurt/M.), **43**, 265 (1964).
42. F. Candau, C. Strazielle, H. Benoit, Makromol. Chem., **170**, 165 (1973).
43. H. -J. Cantow, Makromol. Chem., **30**, 169 (1959).
44. D. K. Carpenter, J. H. Hsieh, ACS Polymer Preprints, **13**, 981 (1972).
45. B. Chew, A. Couper, J. Chem. Soc. Faraday Soc., [1], **72**, 382 (1976).
46. R. Chiang, J. Phys. Chem., **69**, 1645 (1965).
47. S. N. Chinai, J. Polym. Sci., **25**, 413 (1957).
48. S. N. Chinai, J. Polym. Sci., **44**, 475 (1959).
49. S. N. Chinai, C. W. Bondurant, J. Polym. Sci., **22**, 555 (1956).
50. S. N. Chinai, R. A. Guzzi, J. Polym. Sci., **21**, 417 (1956).
51. S. N. Chinai, A. L. Resnick, H. T. Lee, J. Polym. Sci., **33**, 471 (1958).
52. S. N. Chinai, R. J. Samuels, J. Polym. Sci., **19**, 463 (1956).
53. S. N. Chinai, P. C. Scherer, C. W. Bondurant, D. W. Levi, J. Polym. Sci., **22**, 527, (1956).
54. S. N. Chinai, R. J. Valles, J. Polym. **39**, 363 (1959).
55. D. S. Chiu, Y. Takahashi, J. E. Mark, Polymer, **17**, 670 (1976).
56. A. Ciferri, M. Lauretti, Ann. Chim. (Rome), **48**, 198 (1958).
57. E. Cohn-Ginsberg, T. G. Fox, H. F. Mason, Polymer, **3**, 97 (1962).
58. C. F. Cornet, H. van Ballegooijen, Polymer, **7**, 293 (1966).
59. B. J. Cottam, J. M. G. Cowie, S. Bywater, Makromol. Chem., **86**, 116 (1965).
60. J. P. Cotton, D. Decker, H. Benoit, B. Farnoux, J. Higgins, G. Jannink, R. Ober, J. des Cloizeaux, Macromolecules, **7**, 863 (1974).
61. J. M. G. Cowie, Makromol. Chem., **59**, 189 (1963).
62. J. M. G. Cowie, J. Polymer. Sci. C, **23**, 267 (1968).
63. J. M. G. Cowie, S. Bywater, J. Polym. Sci. A-2, **6**, 499 (1968).
64. J. M. G. Cowie, S. Bywater, D. J. Worsfold, Polymer, **8**, 105 (1967).
65. J. M. G. Cowie, A. Maconnachie, R. J. Ranson, Macromolecules, **4**, 57 (1971).
66. J. M. G. Cowie, I. J. McEwen, Polymer, **16**, 933 (1975).
67. V. Crescenci, P. J. Flory, J. Am. Chem. Soc., **86**, 141 (1964).
68. A. Czuppon, IUPAC Symp. Macromolecules, Prague 1965, Preprint 459.
69. F. Danusso, G. Moraglio, Atti Accad. Nazl. Lincei Rend. Classe Sci. Fis. Mat. Nat., [8], **25**, 509 (1958).
70. F. Danusso, G. Moraglio, G. Gianotti, J. Polym. Sci., **51**, 475 (1961).
71. J. V. Dawkins, M. Hemming, Makromol. Chem., **155**, 75 (1972).
72. J. V. Dawkins, G. Taylor, Colloid Polym. Sci., **258**, 79 (1980).
73. A. de Chirico, Chim. Ind. (Milan), **42**, 248 (1960).
74. A. de Chirico, Chim. Ind. (Milan), **46**, 53 (1964).
75. D. Decker, Makromol. Chem., **125**, 136 (1969).
76. G. Delmas, D. Patterson, Polymer, **7**, 513 (1966).
77. F. E. Didot, S. N. Chinai, D. W. Levi, J. Polym. Sci., **43**, 557 (1960).
78. H. Dietschy, Ph.D. Thesis, ETH Zürich 1966.
79. H. A. Dieu, J. Polym. Sci., **12**, 417 (1954).
80. J. Dohany, Ph.D. Thesis, ETH Zürich 1963.
81. A. Dondos, Makromol. Chem., **135**. 181 (1970).
82. A. Dondos, H. Benoit, J. Polym. Sci. B, **7**, 279 (1969).
83. A. Dondos, H. Benoit, Macromolecules, **4**, 279 (1971).
84. P. Dreyfus, M. P. Dreyfus, Adv. Polym. Sci., **4**, 528 (1967).
85. E. T. Dumitru, L. H. Cragg, Cited in ref. [110], p. 615.
86. K. Dušek, M. Bohdanecký, E. Prokopová, Eur. Polym. J., **10**, 239 (1974).
87. H. Eisenberg, D. Woodside, J. Polym. Phys. **36**, 1844 (1962).
88. K. H. Elgert, PhD Thesis, Techn. Univ. of Munich, 1963.
89. H.-G. Elias, Makromol. Chem., **33**, 140 (1959).
90. H.-G. Elias, Makromol. Chem., **50**, 1 (1961).
91. H.-G. Elias, Makromol. Chem., **54**, 78 (1962).
92. H.-G. Elias, "Makromoleküle", Hüthig and Wepf, Basle, Fourth Ed. 1981, p. 200; "Macromolecules", Plenum, New York, Second Ed. 1984, Vol. I., p. 232.
93. H.-G. Elias, in: L. H. Tung, (Ed.), "Fractionation of Synthetic Polymers", Dekker, New York, 1977.
94. H.-G. Elias, G. Adank, Makromol. Chem., **69**, 241 (1963).
95. H.-G. Elias, O. Etter, Makromol. Chem., **66**, 56 (1963).
96. H.-G. Elias, O. Etter, Makromol. Chem., **89**, 228 (1965).
97. H.-G. Elias, O. Etter, J. Makromol. Chem., **1**, 431 (1966).
98. H.-G. Elias, U. Gruber, Makromol. Chem., **78**, 72 (1964).
99. U. Gruber, H. -G. Elias, Makromol. Chem., **86**, 168 (1965).
100. H. -G. Elias, F. W. Ibrahim, Makromol. Chem., **89**, 12 (1965).
101. H. -G. Elias, R. Schumacher, Makromol. Chem., **76**, 23 (1964).
102. V. E. Eskin, L. N. Andreeva, Vysokomol. Soedin., **3**, 435 (1961); cited in J. Polym. Sci., **56**, S 39 (1962).
103. V. E. Eskin, K. Z. Gumargalieva, Vysokomol. Soedin., **2**, 265 (1960).
104. V. E. Eskin, O. Z. Korothina, Vysokomol. Soedin., **2**, 272 (1960); J. Polym. Sci. USSR, **2**, 247 (1961).
105. J. M. Evans, M. B. Huglin, Makromol. Chem., **127**, 141 (1969).
106. J. M. Evans, M. B. Huglin, Europ. Polym. J., **6**, 1161 (1970).

107. J. M. Evans, M. B. Huglin, cited in [84].
108. W. W. Everett, J. F. Foster, J. Am. Chem. Soc., **81**, 3459, 3464 (1959).
109. J. Feisst, H.-G. Elias, Makromol. Chem., **82**, 78 (1965).
110. P. J. Flory, "Principles of Polymer Chemistry", Cornell Univ. Press, Ithaca, NY, 1953.
111. P. J. Flory, L. Mandelkern, J. B. Kinsinger, W. B. Shultz, J. Am. Chem. Soc., **74**, 3364 (1952).
112. T. G. Fox, Polymer, **3**, 111 (1962).
113. T. G. Fox, P. J. Flory, J. Am. Chem. Soc., **73**, 1909 (1959).
114. T. G. Fox, P. J. Flory, J. Am. Chem. Soc., **73**, 1915 (1959).
115. T. G. Fox, J. B. Kinsinger, H. F. Mason, E. M. Schuele, Polymer, **3**, 71 (1962).
116. D. M. French, A. W. Casey, C. I. Collins, P. Kirchner, ACS Polymer Preprints, **7**, 447 (1966).
117. D. Froelich, Ph.D. Thesis, Univ. of Strasbourg, 1966; cited in [83].
118. D. Froelich, H. Benoit, Makromol. Chem., **92**, 224 (1966).
119. M. Fukuda, M. Fukutomi, Y. Kato, T. Hashimoto, J. Polym. Sci.-Polym. Chem. Ed., **12**, 871 (1974).
120. C. Garbuglio, L. Crescentini, A. Mula, G. B. Gechele, Makromol. Chem., **97**, 97 (1966).
121. L. Gargallo, J. Niezzette, V. Desreux, Bull. Soc. Roy. Sci. Liege, **1–2**, 82 (1977); cited in [122].
122. L. Gargallo, D. Radić, I. Katime, Europ. Polym. J., **16**, 383 (1980).
123. G. B. Gechele, L. Crescentini, J. Polym. Sci. A, **3**, 3599 (1965).
124. H. Geerissen, B. A. Wolf, Makromol. Chem., Rapid Commun., **3**, 17 (1982).
125. G. Gianotti, U. Bonicelli, D. Borghi, Makromol. Chem., **166**, 235 (1973).
126. C. González, F. Zamora, G. M. Guzmán, L. M. León, Makromol. Chem., **185**, 339 (1984).
127. T. B. Grimley, Proc. Roy. Soc. [London] A, **212**, 339 (1952).
128. U. Gruber, Ph.D. Thesis, ETH Zürich, 1964.
129. N. Hadjichristidis, Polymer, **16**, 848 (1975).
130. N. Hadjichristidis, J. E. L. Roovers, J. Polym. Sci., Polym. Phys. Ed., **12**, 2521 (1974).
131. J. Hakozaki, J. Chem. Soc. Japan, Pure Chem. Sect. (Nippon Kagaku Zasshi), **82**, 158 (1961).
132. F. Hamada, K. Fujisawa, A. Nakajima, Polym. J., **4**, 316 (1973).
133. K. Hanafusa, A. Teramoto, H. Fujita, J. Chem. Phys., **70**, 4004 (1966).
134. J. N. Hay, A. M. Afifi-Effat, Brit. Polym. J., **9**, 1 (1977).
135. J. N. Hay, P. A. Fitzgerald, J. Polym. Sci., Polym. Chem. Ed., **18**, 1079 (1980).
136. J. N. Hay, M. Wiles, Brit. Polym. J., **10**, 181 (1978).
137. W. Hayduk, H. A. Bromfield, J. Appl. Polym. Sci., **22**, 149 (1978).
138. T. E. Helminiak, C. L. Benner, W. E. Gibbs, ACS Polymer Preprints, **8**, 284 (1967).
139. I. Hernandez-Fuentes, M. G. Prolongo, Europ. Polym. J., **15**, 571 (1979).
140. F. K. Herold, B. A. Wolf, Makromol. Chem., **184**, 2539 (1981).
141. J. Herold, G. Meyerhoff, Europ. Polym. J., **15**, 525 (1979).
142. M. Heskins, J. E. Guillet, J. Macromol. Sci. A, **2**, 1441 (1968).
143. T. Hirosye, Y. Einaga, H. Fujita, Polym. J., **11**, 819 (1979).
144. T. Homma, K. Kawahara, H. Fujita, M. Ueda, Makromol. Chem., **67**, 132 (1963).
145. T. Homma, K. Kawahara, H. Fujita, J. Appl. Polym. Sci., **8**, 2853 (1964).
146. F. Hooshmand-Mozaffar, M. K. Hoseinalizadeh-Khorasani, M. B. Huglin, Polymer, **21**, 413 (1980).
147. F. Horii, Y. Ikada, I. Sakurada, J. Polym. Sci., Polym. Chem. Ed., **12**, 323 (1974).
148. N. Howard, M. B. Huglin, R. W. Richards, J. Appl. Polym. Sci., **16**, 1525 (1972).
149. A. J. Hyde, R. B. Taylor, Polymer, **4**, 1 (1963).
150. T. Ikeda, H. Kawaguchi, Rep. Progr. Polym. Phys. Japan, **9**, 23 (1966).
151. H. Inagaki, A. Nakazawa, T. Kotaka, Bull. Inst. Chem. Res. Kyoto Univ., **43**, 135 (1965).
152. H. Inagaki, T. Miyamoto, S. Ohta, J. Phys. Chem., **70**, 3420 (1966).
153. H. Inagaki, H. Suzuki, M. Fujii, T. Matsuo, J. Phys. Chem., **70**, 1718 (1966).
154. K. J. Ivin, H. A. Ende, G. Meyerhoff, Polymer, **3**, 129 (1962).
155. M. Iwama, H. Utiyama, M. Kurata, J. Macromol. Chem., **1**, 701 (1966).
156. Y. Izumi, Y. Miyake, Polym. J., **3**, 647 (1972).
157. Y. Izumi, Y. Miyake, Polym. J., **4**, 205 (1973).
158. Y. Izumi, S. Otsuka, Y. Miyake, Rep. Progr. Polym. Phys. Japan, **12**, 13 (1969).
159. R. Jerome, V. Desreux, Europ. Polym. J., **6**, 411 (1970).
160. R. M. Johnson, J. L. Schrag, J. D. Ferry, Polym. J., **1**, 742 (1970).
161. M. Kalfus, J. Mitus, J. Polym. Sci. A-1, **4**, 953 (1966).
162. K. Kamada, H. Sato, Polym. J., **2**, 593 (1971).
163. K. Kamide, T. Terekawa, Makromol. Chem., **155**, 25 (1972).
164. Ye. A. Kanevskaya, P. I. Zubov, L. V. Ivanova, Yu. S. Lipatov, Vysokomol. Soedin., **6**, 981 (1964); Polym. Sci. USSR, **6**, 1080 (1964).
165. A. Katchalsky, H. Eisenberg, J. Polym. Sci., **6**, 145 (1951).
166. I. A. Katime Amashta, G. Sanchez, Europ. Polym. J., **11**, 223 (1975).
167. H. Kaye, H. J. Cou, J. Polym. Sci., Polym. Phys. Ed., **13**, 477 (1975).
168. W. Kern, D. Braun, Makromol. Chem., **27**, 23 (1958).
169. J. B. Kinsinger, L. E. Ballard, J. Polym. Sci. B, **2**, 879 (1964).
170. J. B. Kinsinger, L. E. Ballard, J. Polym. Sci. A, **3**, 3963 (1965).
171. J. B. Kinsinger, R. E. Hughes, J. Phys. Chem., **67**, 1922 (1963).
172. J. B. Kinsinger, R. A. Wessling, J. Am. Chem. Soc., **81**, 2908 (1959).
173. R. Kirste, G. V. Schulz, Z. Physik. Chem. (Frankfurt/M.), **27**, 310 (1960).
174. R. G. Kirste, G. V. Schulz, Z. Physik. Chem. (Frankfurt/M.), **30**, 171 (1961).

175. M. R. Knecht, H.-G. Elias, Makromol. Chem., **157**, 1 (1972).
176. A. Kokkiaris, C. Touloupis, N. Hadjichristidis, Polymer, **22**, 63 (1981).
177. R. Koningsveld, Th. G. Schoolte, private communication.
178. T. Kotaka, H. Ohnuma, H. Inagaki, Polymer, **10**, 517 (1969).
179. T. Kotaka, H. Ohnuma, H. Inagaki, ACS Polymer Preprints, **11**, 660 (1970).
180. T. Kotaka, H. Ohnuma, Y. Murakami, J. Phys. Chem., **70**, 4099 (1966).
181. T. Kotaka, T. Tanaka, H. Ohnuma, Y. Murakami, H. Inagaki, Polymer J., **1**, 245 (1970).
182. A. Kotera, H. Matsuda, K. Konishi, K. Takemura, J. Polym. Sci. C, **23**, 619 (1968).
183. A. Kotera, N. Onda, T. Saito, Rep. Progr. Polym. Phys. Japan, **17**, 25 (1974).
184. A. Kotera, T. Saito, N. Yamaguchi, Annual Meeting Chem. Soc. Japan, 1962; cited in A. Kotera, N. Ise, A. Senuma, T. Hamada, IUPAC Symp. Macromol., Prague 1965, Preprint 232.
185. J. Kovář, L. Mrkvičková-Vaculová, M. Bohdanecký, Makromol. Chem., **176**, 1829 (1975).
186. P. Kratochvil, D. Strtaková, Makromol. Chem., **154**, 325 (1972).
187. S. Krause, E. Cohn-Ginsburg, J. Phys. Chem., **67**, 1479 (1963).
188. W. R. Krigbaum, D. K. Carpenter, J. Phys. Chem., **59**, 1166 (1955).
189. W. R. Krigbaum, P. J. Flory, J. Polym. Sci., **11**, 37 (1953).
190. W. R. Krigbaum, J. E. Kurz, P. Smith, J. Phys. Chem., **65**, 1984 (1961).
191. W. R. Krigbaum, L. H. Sperling, J. Phys. Chem., **64**, 99 (1960).
192. W. R. Krigbaum, J. D. Woods, J. Polym. Sci. A, **2**, 3075 (1964).
193. S. Krozer, Plaste Kautschuk, **15**, 175 (1968).
194. S. Krozer, A. Pastusiak, IUPAC Symp. Macromol., Prague 1965, Preprint 609.
195. K. Kubo, K. Ogino, Rep. Progr. Polym. Phys. Japan, **10**, 111 (1967).
196. K. Kubo, K. Ogino, T. Nakagawa, Rep. Progr. Polym. Phys. Japan, **10**, 7 (1967).
197. Z. Küçükyavuz, A. Küçükyavuz, Europ. Polym. J., **14**, 867 (1978).
198. A. M. Kulakova, I. K. Nekrasov, Vysokomol. Soedin. A, **18**, 127 (1976); Polym. Sci. USSR, **18**, 146 (1976).
199. M. Kurata, H. Utiyama, K. Kamada, Makromol. Chem., **88**, 281 (1965).
200. N. Kuwahara, S. Higashida, M. Nakata, K. Kaneko, J. Polym. Sci. A-2, **7**, 285 (1969).
201. N. Kuwahara, M. Kaneko, Makromol. Chem., **82** 205 (1965).
202. N. Kuwahara, T. Miyake, M. Kaneko, J. Furuichi, Rep. Progr. Polym. Phys. Japan, **5**, 1 (1962).
203. N. Kuwahara, S. Saeki, S. Konno, M. Kaneko, Macromolecules, **6**, 589 (1973).
204. N. Kuwahara, S. Saeki, S. Konno, Polymer, **15**, 66 (1974).
205. D. Lath, M. Bohdanecký, J. Polym. Sci.-Letters Ed., **15**, 555 (1977).
206. H. T. Lee, D. W. Levi, J. Polym. Sci., **47**, 449 (1960).
207. L. M. León, J. Galaz, L. M. Garcia, M. S. Anasagasti, Europ. Polym. J., **16**, 921 (1980).
208. D. W. Levi, H. T. Lee, R. J. Valles, J. Polym. Sci., **62**, S163 (1962).
209. J. Lilie, J. Springer, K. Ueberreiter, Kolloid-Z., **226**, 138 (1968).
210. F. C. Lin, S. S. Stivala, J. A. Biesenberger, J. Appl. Polym. Sci., **17**, 3465 (1973).
211. M. Lippay. Ph.D. Thesis, ETH Zürich, 1963.
212. J. Llopis, A. Albert, P. Usobiaga, Europ. Polym. J., **3**, 259 (1967).
213. T. Lucas, Bull. Liaison Lab. (Lab. Prof. Peint)., **20**, 15 (1967).
214. N. V. Makletsova, I. V. Epel'baum, B. A. Rozenberg, Ye. B. Lyudvig, Vysokomol. Soedin., **7**, 70 (1965); Polym. Sci. USSR, **7**, 73 (1966).
215. L. Mandelkern, P. J. Flory, J. Am. Chem. Soc., **74**, 2517 (1952).
216. P. V. Mangalam, V. P. Kalpagam, J. Polym. Sci.-Polym. Phys. Ed., **20**, 773 (1982).
217. V. P. Mardykin, K. B. Martinovich, A. I. Volozhin, N. M. Denisyuk, Vysokomol. Soedin. B, **16**, 23 (1974); CA **81**: 13900r (1974).
218. J. E. Mark, Diss. Abstr., **23**, 1205 (1962).
219. J. E. Mark, P. J. Flory, J. Am. Chem. Soc., **87**, 1423 (1965).
220. R. M. Masegosa, M. G. Prolongo, I. Hernandez-Fuentes, A. Horta, Macromolecules, **17**, 1181 (1984).
221. T. Matsumoto, N. Nishioka, H. Fujita, J. Polym. Sci. A-2, **10**, 23 (1972).
222. M. Matsumoto, Y. Ohyanagi, J. Polym. Sci., **46**, 441 (1960).
223. M. Matsumoto, Y. Ohyanagi, J. Polym. Sci., **50**, S1 (1961).
224. K. Matsumura, Polym. J., **1** 322 (1970).
225. M. Matsushima, M. Fukatsu, M. Kurata, Bull. Chem. Soc. Japan, **41**, 2570 (1968).
226. A. Mattiussi, E. Conto, G. B. Gechele, Europ. Polym. J., **8**, 429 (1972).
227. W. L. Mattice, L. Mandelkern, Macromolecules, **4**, 271 (1971).
228. R. M. May, R. E. Wetton, cited in [84].
229. H. W. McCormick, J. Polym. Sci., **36**, 341 (1959).
230. D. McIntyre, J. H. O'Mara, B. C. Konouck, J. Am. Chem. Soc., **81**, 3498 (1959).
231. R. A. Mendelson, J. Polym. Sci., **46**, 493 (1960).
232. G. Meyerhoff, Makromol. Chem., **37**, 97 (1960).
233. G. Meyerhoff, in: J. W. Williams (Ed.), "Ultracentrifugal Analysis in Theory and Practice". Academic Press, New York, 1963.
234. R. Meza, L. Gargallo, Europ. Polym. J., **13**, 235 (1977).
235. S. Miyamoto, Y. Ishii, H. Ohnuma, Makromol. Chem., **182**, 483 (1981).
236. J. Moacanin, J. Appl. Polym. Sci., **1**, 272 (1959).
237. J. Moacanin, A. Rembaum, R. K. Laudenslager, ACS Polymer Preprints, **4/2**, 179 (1963).

238. J. Moacanin, A. Rembaum, R. K. Laudenslager, R. Adler, J. Macromol. Sci., Chem., A: **1**, 1497 (1967).
239. R. B. Mohite, S. Gundiah, S. L. Kapur, Makromol. Chem., **108**, 52 (1967).
240. L. D. Moore, Jr., J. Polym. Sci., **36**, 155 (1959).
241. W. R. Moore, M. Uddin, Europ. Polym. J., **5**, 185 (1969).
242. G. Moraglio, Europ. Polym. J., **1**, 103 (1965).
243. G. Moraglio, J. Brzenzainski, J. Polym. Sci. B, **2**, 1105 (1964).
244. G. Moraglio, G. Gianotti, U. Bonicelli, Europ. Polym. J., **9**, 623 (1973).
245. G. Moraglio, G. Gianotti, F. Danusso, Europ. Polym. J., **3**, 251 (1967).
246. G. Moraglio, G. Gianotti, F. Zoppi, U. Bonicelli, Europ. Polym. J., **7**, 303 (1971).
247. U. Moritz, G. Meyerhoff, Makromol. Chem., **139**, 23 (1970).
248. G. A. Morneau, P. I. Roth, A. R. Shultz, J. Polym. Sci., **55**, 609 (1961).
249. J. Naghizadeh, J. Springer, Kolloid-Z. Z. Polym., **215**, 21 (1967).
250. R. Naito, Chem. High Polym. Japan (Kobunshi Kagaku), **16**, 7 (1959).
251. A. Nakajima, H. Fujiwara, F. Hamada, J. Polym. Sci. A-2, **4**, 507 (1966).
252. A. Nakajima, F. Hamada, Rep. Progr. Polym. Phys. Japan, **9**, 41 (1966).
253. A. Nakajima, K. Kato, Makromol. Chem., **95**, 52 (1966).
254. A. Nakajima, K. Okazaki, Chem. High Polym. Japan (Kobunshi Kagaku), **22**, 791 (1965).
255. A. Nakajima, H. Saijyo, J. Polym. Sci. A-2, **6**, 735 (1968).
256. A. Nakajima, K. Tanaami, Polym. J., **5**, 248 (1973).
257. D. H. Napper, Trans. Faraday Soc., **64**, 1701 (1968).
258. D. H. Napper, Kolloid-Z., **234**, 1149 (1969).
259. D. H. Napper, J. Colloid Interface Sci., **32**, 106 (1970).
260. D. H. Napper, J. Colloid Interface Sci., **33**, 384 (1970).
261. D. W. Nash, C. C. Pepper, Polymer, **16**, 105 (1975).
262. P. Neuenschwander, P. Pino, Makromol. Chem., **181**, 737 (1980).
263. S. Newman, W. R. Krigbaum, C. Laugier, P. J. Flory, J. Polym. Sci., **14**, 451 (1954).
264. J. Niezette, N. Hadjichristidis, V. Desreux, Europ. Polym. J., **13**, 41 (1977).
265. A. Nikolaev, N. V. Daniel, A. M. Torotseva, I. Varga, N. V. Ivanova, Vysokomol. Soedin., **6**, 292 (1964); Polym. Sci. USSR, **6**, 336 (1964).
266. I. Noda, K. Mizutani, T. Kato, T. Fujimoto, N. Nagasawa, Macromolecules, **3**, 787 (1970).
267. H. Ohnuma, Y. Murakami, T. Kotaka, Rep. Progr. Polym. Phys. Japan, **10**, 63 (1967).
268. R. Okada, Y. Toyoshima, H. Fujita, Makromol. Chem., **59**, 137 (1963).
269. T. A. Orofino, A. Ciferri, J. Phys. Chem., **68**, 3136 (1964).
270. T. A. Orofino, J. W. Mickey, Jr., J. Chem. Phys., **38**, 2512 (1963).
271. R. Okada, Y. Toyoshima, H. Fujita, Makromol. Chem., **59**, 137 (1963).
272. S. Okamura, T. Higashimura, Y. Imanishi, Chem. High Polym. Japan (Kobunshi Kagaku), **16**, 244 (1959).
273. J. Oth, V. Desreux, Bull. Soc. Chim. Belge, **63**, 285 (1954).
274. P. Outer, C. I. Carr, B. H. Zimm, J. Chem. Phys., **18**, 830 (1950).
275. C. G. Overberger, E. M. Pearce, N. Mayes, J. Polym. Sci., **34**, 109 (1959).
276. O. Quadrat, Coll. Czech. Chem. Commun., **35**, 2564 (1970).
277. O. Quadrat, M. Bohdanecký, J. Polym. Sci. A-2, **5**, 1309 (1967).
278. O. Quadrat, M. Bohdanecký, L. Mrkvičková, Makromol. Chem., **182**, 445 (1981).
279. O. Quadrat, M. Bohdanecký, P. Munk, J. Polym. Sci. C, **16**, 95 (1967).
280. M. Pissoli, G. Ceccorulli, G. Stea, Makromol. Chem., **164**, 273 (1973).
281. M. Pizzoli, G. Stera, G. Ceccorulli, G. B. Gechele, Europ. Polym. J., **6**, 1219 (1970).
282. I. Ya. Poddubnyi, V. A. Grechanovskii, M. I. Mosevitskii, Vysokomol. Soedin, **5**, 1049 (1963); Polym Sci. USSR, **5**, 105 (1964).
283. I. Ya. Poddubnyi, V. A. Grechanovskii, M. I. Mosevitskii, A. V. Podalinskii, Vysokomol. Soedin, **5**, 1042 (1963); Polym. Sci. USSR, **5**, 97 (1964).
284. I. Ya. Poddubnyi, V. A. Grechanovskii, A. V. Podalinskii, J. Polym. Sci. C, **16**, 3109 (1968).
285. I. Ya. Poddubnyi, A. V. Podalinskii, V. A. Grechanovskii, Polym. Sci. USSR, **8**, 1715 (1966).
286. R. Puffr, Z. Tuzar, L. Mrkvičková, J. Šebenda, Makromol. Chem., **184**, 1957 (1983).
287. D. Radić, L. Gargallo, Makromol. Chem., **180**, 1329 (1979).
288. D. Radić, L. Gargallo, Polymer, **22**, 410 (1981).
289. C. Reiss, H. Benoit, Compt. Rend. Acad. Sci. (Paris), **253**, 268 (1962).
290. C. Reiss, H. Benoit, J. Polym. Sci. C, **16**, 3079 (1968).
291. R. W. Richards, Polymer, **18**, 114 (1977).
292. R. W. Richards, G. Disselhoff, Angew. Makromol. Chem., **66**, 221 (1978).
293. A. Roig, I. Katime, V. C. Ramiro, L. M. Leon, An. Quim., **69**, 965 (1973).
294. J. Roovers, Polymer, **20**, 843 (1979).
295. R. Rossi, A. Mattiussi, G. B. Gechele, D. Damiani, Chim. Ind. (Milan), **53**, 350 (1971).
296. S. Saeki, S. Konno, N. Kuwahara, M. Nakata, M. Kaneko, Macromolecules, **7**, 521 (1974).
297. G. Saini, L. Trossarelli, J. Polym. Sci., **23**, 563 (1957).
298. Y. Sakaguchi, J. Nishino, K. Tsugawa, Chem. High Polym., Japan (Kobunshi Kagaku), **20**, 661 (1963).
299. I. Sakurada, A. Nakajima, K. Shibetani, J. Polym. Sci. A, **2**, 3345 (1964).
300. I. Sakurada, A. Nakajima, O. Yoshizaki, K. Nakamae, Kolloid-Z., **186**, 41 (1962).
301. I. Sakurada, Y. Sakaguchi, S. Kokuryo, Chem. High. Polym. Japan (Kobunshi Kagaku), **17**, 227 (1960).
302. R. J. Samuels, Polymer, **18**, 452 (1977).
303. M. Sato, Y. Koshishi, M. Asahina, J. Polym. Sci. B, **1**, 233 (1963).

304. K. S. Sastry, R. D. Patel, Europ. Polym. J., **5**, 79 (1969).
305. P. R. Saunders, J. Polym. Sci., **57**, 131 (1961).
306. W. Scholtan, H. Lange, R. Casper, U. Pohl, D. Wendisch, R. Mayer-Mader, Angew. Makromol. Chem., **27**, 1 (1972).
307. G. V. Schulz, H. Baumann, Makromol. Chem., **60**, 120 (1963).
308. G. V. Schulz, H. J. Cantow, Z. Elektrochem., **60**, 517 (1956).
309. G. V. Schulz, A. Haug, Z. Physik. Chem. (Frankfurt/M.), **34**, 328 (1962).
310. G. V. Schulz, A. Mula, Makromol. Chem., **44/46**, 479 (1961).
311. G. V. Schulz, W. Wunderlich, R. Kirste, Makromol. Chem., **75**, 22 (1964).
312. K. Sedlak, D. Lath, Makrotest sb, Prednasek, Celustatni Kof., **5**, 71 (1978); CA **89**, 180549e (1978).
313. V. S. Shazka, G. A. Fomin, G. V. Tarasova, I. G. Kirillova, V. M. Yamshchikov, A. Ye. Grishchenko, I. A. Shefer, I. S. Alekseyeva, Polym. Sci. USSR, **15**, 2903 (1973).
314. Y. Shiokawa, H. Takeo, K. Takamizawa, T. Oyama, Rep. Progr. Polym. Phys. Japan, **7**, 49 (1964).
315. A. R. Shultz, Ph.D. Thesis, Cornell Univ., cited in [110].
316. A. R. Shultz, J. Am. Chem. Soc., **76**, 3422 (1954).
317. A. R. Shultz, C. R. McCullough, J. Polym. Sci. A-2, **7**, 1577 (1969).
318. L. Šimek, J. Hrnčiřik, M. Mládek, Angew. Makromol. Chem., **92**, 33 (1980).
319. G. Sitaramaiah, D. Jacobs, Makromol. Chem., **164**, 237 (1973).
320. V. S. Skazka, R. A. Zobov, A. Moshpauenko, Vysokomol. Soedin., **4**, 1257 (1962).
321. E. M. Sörvik, J. Appl. Polym. Sci., **21**, 2769 (1977).
322. T. Spychaj, D. Lath, D. Berek, Polymer, **20**, 437 (1979).
323. K. S. V. Srinivasan, M. Santappa, Polymer, **14**, 5 (1973).
324. C. J. Stacy, R. L. Arnett, J. Phys. Chem., **69**, 3109 (1965).
325. C. J. Stacy, J. F. Foster, J. Polym. Sci., **20**, 56 (1956).
326. U. P. Strauss, P. Ander, J. Phys. Chem., **66**, 2235 (1962).
327. U. P. Strauss, P. L. Wineman, J. Am. Chem. Soc., **80**, 3266 (1958).
328. C. Strazielle et al., cited in J. P. Cotton, D. Decker, H. Benoit, B. Farnoux, J. Higgins, G. Jannink, R. Ober, C. Picot, J. des Cloizeaux, Macromolecules, **7**, 863 (1974).
329. C. Strazielle, H. Benoit, O. Vogl, Europ. Polym. J., **14**, 331 (1978).
330. H. Suzuki, K. Kamide, M. Saitoh, Europ. Polym. J., **18**, 123 (1982).
331. H. Suzuki, Y. Muraoka, M. Saitoh, K. Kamide, Brit. Polym. J., **14**, 23 (1982).
332. H. Suzuki, Y. Muraoka, M. Saitoh, K. Kamide, Europ. Polym. J., **18**, 831 (1982).
333. K. Takamizawa, Y. Kambara, T. Oyama, Rep. Progr. Polym. Phys. Japan, **10**, 85 (1967).
334. S. Tani, F. Hamada, A. Nakajima, Polym. J., **5**, 86 (1973).
335. J. O. Threkeld, H. A. Ende, J. Polym. Sci. A-2, **4**, 663 (1966).
336. M. Tricot, J. P. Bleus, J.-P. Riga, V. Desreux, Makromol. Chem., **175**, 913 (1974).
337. M. Tricot, C. Mertens, P. Collete, V. Desreux, Bull. Roy. Soc. Liege, **43**, 502 (1974).
338. S. Tsuchiya, Y. Sakaguchi, I. Sakurada, Chem. High. Polym. Japan (Kobunshi Kagaku), **18**, 346 (1961).
339. T. Tsuji, H. Fujita, Polym. J., **4**, 409 (1973).
340. V. N. Tsvetkov, L. N. Andreyeva, Ye. V. Korneyeva, P. N. Lavrenko, N. A. Platé, V. P. Shibayev, B. S. Petrukhin, Vysokomol. Soedin. A, **14**, 1944 (1972).
341. V. N. Tsvetkov, L. N. Andreyeva, Ye. V. Komeyeva, P. N. Lavrenko, N. A. Platé, V. P. Shibayev, B. S. Petrukhin, Vysokomol. Soedin. A, **13**, 2226 (1971); Polym. Sci. USSR, **13**, 2501 (1971).
342. V. N. Tsvetkov, G. V. Tarasova, Ye. L. Vinogradov, N. Kuprianova, V. M. Yamshchikov, V. S. Kozka, V. S. Ivanov, V. K. Smirnova, I. I. Migunova, Vysokomol. Soedin. A, **13**, 620 (1971); Polym. Sci. USSR, **13**, 705 (1971).
343. Z. Tuzar, P. Kratochvil, Coll. Czech. Chem. Comm., **32**, 3358 (1967).
344. Z. Tuzar, V. Vošický, M. Bohdanecký, J. Jalovecký, Makromol. Chem., **180**, 1399 (1979).
345. M. Ueda, K. Kajitani, Makromol. Chem., **108**, 138 (1967)
346. M. Uhniat, M. Rubaj, T. Marciniak, J. Bankiewicz, Polimery, **17**, 141 (1972); CA **77**: 153493c (1972).
347. I. R. Urwin, M. Girolamo, Makromol. Chem., **150**, 179 (1971).
348. R. van Leemput, R. Stein, J. Polym. Sci. A, **2**, 4039 (1964).
349. P. Vasudevan, M. Santappa, Makromol. Chem., **137**, 261 (1970).
350. J. Valičković, V. Juraničová, J. Filipović, Angew. Makromol. Chem., **24**, 77 (1972).
351. H. Völker, F.-J. Luig, Angew. Makromol. Chem., **12**, 43 (1970).
352. H. L. Wagner, P. J. Flory, J. Am. Chem. Soc., **74**, 195 (1952).
353. H. L. Wagner, C. A. J. Hoeve, J. Polym. Sci.-Symp., **54**, 327 (1976).
354. R. A. Wessling, Diss. Abstr., **23**, 1536 (1962).
355. G. R. Williamson, A. Cervenko, Europ. Polym. J., **10**, 295 (1974).
356. B. A. Wolf, H. F. Bieringer, J. W. Breitenbach, Polymer, **17**, 605 (1976).
357. E. Wolfram, Kolloid-Z. Z. Polym., **227**, 86 (1968).
358. K. Yamamoto, N. Bessho, T. Shiibashi, E. Maekawa, Polym. J., **13**, 555 (1981).
359. M. M. Zafar, R. Mahmood, Makromol. Chem., **175**, 903 (1974).
360. E. A. Zavaglia, F. W. Billmeyer, Jr., Official Digest, **36**, 221 (1964).
361. G. Gianotti, U. Bonicelli, D. Borghi, Makromol. Chem., **166**, 235 (1973).
362. M. Abe, Y. Murakami, H. Fujita, J. Appl. Polym. Sci., **9**, 2549 (1965).
363. J. Mays, H. Hadjichristidis, L. J. Fetters, Macromolecules, **17**, 2723 (1984).
364. K. Kajiwara, H. Suzuki, H. Inagaki, M. Maeda, T. Tsuruta, Makromol. Chem., **187**, 2251 (1986).
365. Z. Küçükyavuz, S. Küçükyavuz, Makromol. Chem., **187** 2469 (1986).
366. J. Roovers, J. Polym. Sci., Polym. Phys. Ed., **23**, 1117 (1985).

367. S. Arichi, N. Sakamoto, S. Himura, M. Miki, M. Yoshida, Polymer, **26**, 1175 (1985).
368. G. C. Berry, E. F. Casassa, P.-Y. Liu, J. Polym. Sci., Polym. Phys. B, **25**, 673 (1987).
369. O. Vogl, M. Malanga, W. Berger, in W. J. Bailey, T. Tsuruta, Eds., "Contemporary Topics in Polymer Science", Vol. 4, Plenum, New York 1984, p. 35.
370. J. W. Mays, W. M. Ferry, N. Hadjichristidis, W. G. Funk, L. J. Fetters, Polymer, **27**, 129 (1986).
371. R. W. Richards, A. Maconnachie, G. Allen, Polymer, **22**, 153 (1981).
372. J. L. Roovers, P. M. Toporowski, Macromolecules, **14**, 1174 (1981).
373. J. L. Roovers, P. M. Toporowski, Macromolecules, **16**, 843 (1983).
374. B. A. Wolf, H. J. Adam, J. Chem. Phys., **75**, 4121 (1981).
375. P. Vidaković, F. Rondelez, Macromolecules, **18**, 700 (1985).
376. W. Brown, P. Štěpánek, Macromolecules, **21**, 1791 (1988).
377. H. Geerissen, J. Roos, B. A. Wolf, Makromol. Chem., **186**, 753 (1985).
378. S. Arichi, Bull. Chem. Soc. Japan, **39**, 439 (1966).
379. P. V. Mangalam, V. P. Kalpagam, Polymer, **23**, 991 (1982).
380. F. Horkay, H. B. Stanley, E. Geissler, S. M. King, Macromolecules, **28**, 678 (1995).
381. P. Sakellariou, Polymer, **33**, 1339 (1992).
382. S. Fujishige, Polymer J., **19**, 297 (1987).
383. P. Gregory, M. B. Huglin, M. K. H. Khorasani, P. M. Sasia, Brit. Polym. J., **20**, 1 (1988).
384. K. Kamide, Y. Miyazaki, H. Kobayashi, Polymer J., **17**, 607 (1985).
385. K. Kamide, Y. Miyazaki, H. Kobayashi, Polymer J., **14**, 591 (1982).
386. L. Gargallo, D. Radić, A. Leon, Makromol. Chem., **186**, 1289 (1985).
387. W. Scholtan, H. Marzolph, Makromol. Chem., **57**, 52 (1962).
388. V. P. Shibaev, Habilitation Thesis, Moscow State University, 1974; cited by N. A. Platé, V. P. Shibaev, "Comb-Shaped Polymers and Liquid Crystals", Plenum, New York, 1987.
389. M. P. Breton, Diss. Abstracts International B, **42/11**, 4446 (1982).
390. K. Kubota, S. Fujishige, I. Ando, Polymer J., **22**, 15 (1990).
391. S. R. Gooda, M. B. Huglin, Macromolecules, **25**, 4215 (1992).
392. K. Sivadasan, S. Gundiah, Polymer, **28**, 1426 (1987).
393. K. Nomiama, M. Kaneko, Rep. Progr. Polym. Phys. Japan, **24**, 15 (1981).
394. N. M. Wiederborn, A. R. Brown, J. Polym. Sci., **8**, 651 (1952).
395. N. Hadjichristidis, C. Touloupis, Makromol. Chem., **183**, 611 (1982).
396. N. Hadjichristidis, J. Mays, W. Ferry, L. J. Fetters, J. Polym. Sci.-Polym. Phys. Ed., **22**, 1745 (1984).
397. J. W. Mays, W. Ferry, N. Hadjichristidis, L. J. Fetters, Macromolecules, **18**, 2330 (1985).
398. N. Hadjichristidis, Makromol. Chem., **178**, 1463 (1977).
399. M. Krakovyak, T. Ananieva, E. Anufrieva, T. Nekrasova, G. Sinitsina, Makromol. Chem., **186**, 1075 (1985).
400. H. Inagaki, S. Kawai, Makromol. Chem., **79**, 42 (1964).
401. X. Zhongde, N. Hadjichristidis, L. J. Fetters, Macromolecules, **17**, 2303 (1984).
402. H. G. Ohm, R. G. Kirste, R. C. Oberthür, Makromol. Chem., **189**, 1387 (1988).
403. M. Nakata, Y. Nakano, K. Kawate, Macromolecules, **21**, 2509 (1988).
404. E. Siakali-Kioulafa, N. Hadjichristidis, J. W. Mays, Macromolecules, **22**, 2059 (1989).
405. J. W. Mays, N. Hadjichristidis, J. S. Lindner, ACS Polymer Preprints, **32/1**, 148 (1991).
406. M. Ricker, M. Schmidt, ACS Polymer Preprints, **31/2**, 122 (1990) Makromol. Chem. **192**, 679 (1991).
407. Y. Tamai, T. Konishi, Y. Einaga, M. Fujii, H. Yamakawa, Macromolecules, **23**, 4067 (1990).
408. G. M. Burnett, P. Meares, C. Paton, Trans. Faraday, Soc., **58**, 737 (1962).
409. M. Guaita, O. Chiantore, W. Burchard, Makromol. Chem., **192**, 2333 (1991).
410. T. Yoshizaki, H. Hayashi, H. Yamakawa, Macromolecules, **26**, 4037 (1993).
411. L. Gargallo, N. Hamidi, D. Radić, Polymer, **31**, 924 (1990).
412. S. P. Nunes, B. A. Wolf, H. E. Jeberien, Macromolecules, **20**, 1948 (1987).
413. Y. Matsushita, K. Shimizu, Y. Nakao, H. Chashi, I. Noda, M. Nagasawa, Polym. J., **18**, 361 (1986).
414. I. Katime, J. R. Quintana, Polym. Bull., **6**, 455 (1982).
415. M. J. R. Cantow, R. B. Larrabee, E. M. Barrall II, R. S. Butner, P. Cotts, F. Levy, T. Y. Ting, Makromol. Chem., **187**, 2475 (1986).
416. P. Gregory, M. B. Huglin, Makromol. Chem., **187**, 1745 (1986).
417. M. Ataman, Colloid Polym. Sci., **265**, 19 (1987).
418. W. F. Polik, W. Burchard, Macromolecules, **16**, 978 (1983).
419. T. Amu, Polymer, **23**, 1775 (1982).
420. Ye. A. Gladkova, S.-S. A. Pavlova, L. V. Dubrovina, V. V. Korshak, Vysokomol. soyed. A, **26**, 53 (1984); Polymer Sci. USSR, **26**, 59 (1984).
421. M. B. Huglin, M. A. Radwan, Polymer, **32**, 1293 (1991).
422. P. R. Saunders, J. Polym. Sci. A, **2**, 3765 (1964).
423. W. Brown, Arkiv Kemi, **18**, 227 (1961).
424. H. L. Griffin, S. R. Erlander, F. R. Senti, Die Stärke, **1**, 8 (1967).
425. V. P. Shanbhag, J. Öhman, Arkiv Kemi, **29**, 163 (1968).
426. K. Kamide, S. Matsuda; cited in K. Kamide, M. Saito, Adv. Polym. Sci., **83**, 1 (1987).
427. T. Konishi, Y. Yoshizaki, H. Yamakawa, Macromolecules, **24**, 5614 (1991).
428. G. Allen, J. McAnish, C. Strazielle, Europ. Polym. J., **5**, 319 (1969).
429. A. Lapp, C. Strazielle, Makromol. Chem.-Rapid Comm., **6**, 591 (1985); Makromol. Chem., **188**, 2921 (1987).
430. L. A. Tartarova, T. G. Ermakova, N. F. Kedreena, B. A. Kasekeen, D. D. Novekov, B. A. Lopirev, Vysokomol. Soedin., **24**, 697 (1982).
431. T. E. Helminiak, C. L. Benner, W. E. Gibbs, ACS Polymer Preprints, **8/1**, 284 (1967).

432. P. J. Flory, L. Mandelkern, J. B. Kinsinger, W. B. Shultz, J. Am. Chem. Soc., **74**, 3364 (1952).
433. A. Haug, G. Meyerhoff, Makromol. Chem., **53**, 91 (1962).
434. H. Rios, L. Gargallo, D. Radić, J. Polym. Sci., B: Polym. Phys. **24**, 2421 (1986).
435. P. P. Shah, J. Macromol. Sci. A: Chem. **21**, 547 (1984).
436. L. V. Dubrovina, S.-S. A. Pavlova, M. A. Ponomareva, Vysokomol. Soyed. A, **27**, 780 (1985); Polym. Sci. USSR, **27**, 875 (1985).
437. J. I. Egaiazábal, J. Areizaga, J. J. Iruin, G. M. Guzmán, Europ. Polym. J., **21**, 711 (1985).
438. I. H. Park, J.-H. Kim, T. Chang, Macromolecules, **25**, 7300 (1992).
439. C. Strazielle, H. Benoit, Macromolecules, **8**, 203 (1975).
440. J. M. G. Cowie, I. J. McEwen, Polymer, **25**, 1107 (1984).
441. A. Kanda, M. Duval, D. Sarazin, J. François, Polymer, **26**, 406 (1985).
442. T. Schwartz, J. Sabbadin, J. François, Polymer, **22**, 609 (1981).
443. O. Quadrat, L. Mrkvičková, Europ. Polym. J., **17**, 1155 (1981).
444. E. A. Bekturov, S. L. Kudaibergenov, J. Macromol. Sci. B: Phys. **25**, 133 (1986).
445. J. M. G. Cowie, I. J. McEwen, Brit. Polym. J., **18**, 387 (1986).
446. K. Matsuo, W. H. Stockmayer, G. F. Needham, J. Polym. Sci., Polym. Symp., **71**, 95 (1984).
447. N. Hadjichristidis, Z. Zhongde, L. J. Fetters, J. Roovers, J. Polym. Sci., Polym. Phys. Ed., **20**, 743 (1982).
448. A. Nakajima, H. Fujiwara, J. Polym. Sci. A-2, **6**, 723 (1968).
449. T. M. Birshtein, Europ. Polym. J., **13**, 375 (1977).
450. N. Hadjichristidis, M. Devaleriola, V. Desreux, Europ. Polym. J., **8**, 1193 (1972).
451. G. C. Berry, J. Chem. Phys., **46**, 1338 (1967).
452. A. Takahashi, S. Yamori, I. Kagawa, J. Chem. Soc. Japan, Ind. Chem. Section., **83**, 11 (1962); cited in A. Takahashi, M. Nagasawa, J. Am. Chem. Soc., **86**, 543 (1964).
453. J. W. Mays, L. J. Fetters, Macromolecules, **22**, 921 (1989).
454. Xu Zhongde, J. Mays, C. Xuexin, N. Hadjichristidis, F. C. Schilling, H. E. Bair, D. S. Pearson, L. J. Fetters, Macromolecules, **18**, 2560 (1985).
455. T. Norisuye, K. Kawahara, A. Teramoto, H. Fujita, J. Chem. Phys., **49**, 4330 (1968).
456. J. W. Mays, N. Hadjichristidis, L. J. Fetters, Macromolecules, **18**, 2231 (1985).
457. Xu Zhongde, N. Hadjichristidis, L. J. Fetters, Macromolecules, **17**, 2303 (1984).
458. Xu Zhongde, N. Hadjichristidis, J. M. Carella, L. J. Fetters, Macromolecules, **16**, 925 (1983).
459. J. G. Zilliox, J. E. L. Roovers, S. Bywater, Macromolecules, **8**, 573 (1975).
460. S. Saeki, N. Kuwahara, S. Konno, M. Kaneko, Macromolecules, **6**, 589 (1973).
461. A. M. Afifi-Effat, J. N. Hay, Brit. Polym. J., **8**, 91 (1976).
462. W. R. A. D. Moore, W. Millns, Brit. Polym. J., **1**, 81 (1969).
463. A. L. Izyumnikov, L. V. Mineyev, V. A. Maslennikov, L. S. Sidorina, O. S. Samsonova, A. D. Abkin, Polym. Sci. USSR, **30**, 1062 (1988).
464. S. Ya. Magarik, A. P. Filippov, N. V. D'yakonova, Polym. Sci. USSR, **29**, 772 (1987).
465. S. Küçuükyavuz, Z. Küçuükyavuz, Polym. Commun., **31**, 35 (1990).
466. A. Karandinos, J. W. Mays, H. Hadjichristidis, Polym. Bull., **24**, 251 (1990).
467. L. Gargallo, N. Hamidi, D. Radić, Polym. Internat., **24**, 1 (1991).
468. B. Simionescu, S. Ioan, M. Bercea, C. I. Simionescu, Europ. Polym. J., **27**, 589 (1991).
469. L. Gargallo, E. Soto, F. R. Diaz, L. H. Tagle, D. Radić, Europ. Polym. J., **23**, 571 (1987).
470. M. Yazdani-Pedram, L. Gargallo, D. Radić, Europ. Polym. J., **21**, 461 (1985).
471. S. Arichi, M. Y. Pedram, J. M. G. Cowie, Europ. Polym. J., **15**, 113 (1979).
472. G. Moraglio, G. Gianotti, Europ. Polym. J., **5**, 781 (1969).
473. A. George, W. W. Wilson, J. S. Lindner, J. W. Mays, Polymer, **35**, 600 (1994).
474. A. Kuntman, B. M. Baysal, Polymer, **34**, 3723 (1993).
475. S. R. Gooda, M. B. Huglin, Polymer, **34**, 1913 (1993).
476. A.-A. A. Abdel-Azim, M. B. Huglin, Polymer, **24**, 1429 (1983).
477. Y. -Z. Chen, J. W. Mays, N. Hadjichristidis, J. Polym. Sci., Polym. Phys. Ed., **32**, 715 (1994).
478. J. W. Mays, N. Hadjichristidis, J. S. Lindner, J. Polym. Sci., Polym. Phys. Ed., **28**, 1881 (1990).
479. J. W. Mays, N. Hadjichristidis, W. W. Graessley, L. J. Fetters, J. Polym. Sci., Polym. Phys. Ed., **24**, 2553 (1986).
480. N. Hadjichristidis, Xu Zhongde, L. J. Fetters, J. Roovers, J. Polym. Sci., Polym. Phys. Ed., **20**, 743 (1982).
481. N. Hadjichristidis, J. W. Mays, R. D. Vargo, L. J. Fetters, J. Polym. Sci., Polym. Phys. Ed., **21**, 189 (1983).
482. M. Ataman, E. A. Boucher, J. Polym. Sci., Polym. Phys. Ed., **20**, 1585 (1982).
483. J. Ansorena, J. J. Iruin, G. M. Guzmán, J. Polym. Sci., Polym. Phys. Ed., **18**, 173 (1980).
484. S. H. Oh, M. S. John, J. Polym. Sci., Polym. Chem. Ed., **27**, 1731 (1989).
485. A. Nakajima, F. Hamada, S. Hayashi, J. Polym. Sci. C, **15**, 285 (1966).
486. C. Tsitslianis, E. Pierri, A. Dondos, J. Polym. Sci., Polym. Letters Ed., **21**, 685 (1983).
487. R. R. Buch, H. M. Klimisch, O. K. Johannoson, J. Polym. Sci. A-2, **8**, 541 (1970).
488. T. Altares, D. P. Wyman, V. R. Allen, J. Polym. Sci. A, **2**, 4533 (1964).
489. S. N. Chinai, R. A. Guzzi, J. Polym. Sci., **41**, 475 (1959).
490. U. Bianchi, V. Magnasco, J. Polym. Sci., **41**, 177 (1959).
491. T. Hirao, A. Teramoto, T. Sato, T. Norisuye, T. Masuda, T. Higashimura, Polym. J., **23**, 925 (1991).
492. R. Chiang, J. Phys. Chem., **70**, 2348 (1966).
493. T. A. Orofino, F. Wenger, J. Phys. Chem., **67**, 566 (1963).

494. G. Tanaka, S. Imai, H. Yamakawa, J. Chem. Phys., **52**, 2639 (1970).
495. K. Kawahara, T. Norisuye, H. Fujita, J. Chem. Phys., **49**, 4339 (1968).
496. P. R. Sundarajan, in: J. E. Mark, (Ed.), "Physical Properties of Polymers Handbook", AIP Press, American Institute of Physics, Woodbury. NY, 1996, p. 197.
497. J. W. Mays, N. Hadjichristidis, J. Macromol. Sci.-Rev. Macromol. Chem. Phys. C, **28**, 371 (1988).
498. H. -G. Elias, Polymere, Hüthig and Wepf, Zug (Switzerland) 1996, p. 204; Polymers, VCH Publ., Weinheim (Germany) 1997, p. 209.
499. K. Akasaka, Y. Nakamura, T. Norisuye, A. Teramoto, Polym. J., **26**, 363 (1994).
500. Y. H. Park, D. C. Lee, Polymer (Korea), **12**, 749 (1988); cited in (496).
501. I. A. Ronova, M. A. Ponomareva, L. V. Dubrovina, et al., Int. Polym. Sci. Techn., **13**, T 90 (1986); cited in (496).
502. I. Noda, T. Horikawa, T. Kato, T. Fujimoto, M. Nagasawa, Macromolecules, **3**, 795 (1970).
503. M. Bohdanecký, D. Berek, Macromol. Chem., Rapid Commun., **6**, 275 (1985).
504. D. Pateropoulou, E. Siakali-Kioulafa, N. Hadjichristidis, Macromol. Chem. Phys., **195**, 173 (1994).
505. A.-A. A. Abdel-Azim, Macromol. Chem. Phys., **195**, 159 (1994).
506. L. Fritz, J. Springer, Makromol. Chem., **194**, 2047 (1993).
507. M. M. Zafar, R. Mahmood, S. A. Wadood, Makromol. Chem., **160**, 313 (1972).
508. T. Hama, K. Yamaguchi, T. Suzuki, Makromol. Chem., **155**, 283 (1972).
509. H. Lütje, G. Meyerhoff, Makromol. Chem., **68**, 180 (1963).
510. H. M. Tan, A. Hiltner, E. Baer, Macromolecules, **16**, 28 (1983).
511. T. Yoshizaki, K. Hayashi, H. Yamakawa, Macromolecules, **27**, 4259 (1994).
512. F. Abe, K. Horita, Y. Einaga, H. Yamakawa, Macromolecules, **27**, 725 (1994).
513. Y. Takaeda, T. Yoshizaki, H. Yamakawa, Macromolecules, **26**, 3742 (1993).
514. P. Hattam, S. Gauntlett, J. W. Mays, N. Hadjichristidis, R. N. Young, L. J. Fetters, Macromolecules, **24**, 6199 (1991).
515. J. W. Mays, S. Nan, D. Whitfield, Macromolecules, **24**, 315 (1991).
516. J. W. Mays, S. Nan, W. Yunan, J. Li, N. Hadjichristidis, Macromolecules, **24**, 4469 (1991).
517. F. Abe, Y. Einaga, H. Yamakawa, Macromolecules, **24**, 4423 (1991).
518. A. Karandinos, S. Nan, J. W. Mays, N. Hadjichristidis, Macromolecules, **24**, 2007 (1991).
519. F. Fujii, Y. Tamai, T. Konishi, H. Yamakawa, Macromolecules, **24**, 1608 (1991).
520. H. M. Petri, N. Schuld, B. A. Wolf, Macromolecules, **28**, 4975 (1995).
521. J. B. Kinsinger, R. E. Hughes, J. Phys. Chem., **63**, 2002 (1959).
522. F. Strehle, Th. Dorfmüller, D. Samios, Macromolecules, **25**, 3569 (1992); N. Pesce da Silveira, D. Samios, F. Strehle, Th. Dorfmüller, Macromol. Chem. Phys., **197**, 1945 (1996).
523. W.-M. Kulicke, M. Prescher, Makromol. Chem., **185**, 2619 (1984).
524. J.-G. Zilliox, Makromol. Chem., **156**, 121 (1972).
525. F. Abe, Y. Einaga, H. Yamakawa, Macromolecules, **27**, 3262 (1994).
526. A.-A. A. Abdel-Azim, M. B. Huglin, Europ. Polym. J., **18**, 735 (1982).
527. R. Jenkins, R. S. Porter, Adv. Polym. Sci., **36**, 1 (1980).
528. J. W. Mays, N. Hadjichristidis, J. Macromol. Sci. Rev.-Macromol. Phys. C, **28**, 371 (1988).
529. A. R. Shultz, P. J. Flory, J. Am. Chem. Soc., **74**, 4760 (1952).
530. G. Talamini, G. Bidotto, Makromol. Chem., **110**, 111 (1967).
531. H.-G. Elias, M. Dobler, H.-R. Wyss, J. Polym. Sci., **46**, 264 (1960).
532. D. H. Napper, Polymer, **10**, 181 (1969).
533. D. H. Napper, Polymeric Stabilization of Colloidal Dispersions, Academic Press, London 1983.
534. S. Q. Zhou, S. Y. Fan, S. T. F. An-yeung, C. Wu, Polymer, **36**, 1341 (1995).
535. F. Döbert, A. Pfennig, M. Stumpf, Macromolecules, **28**, 7860 (1995).
536. D. H. Napper, in: C. A. Finch, (Ed.), Chemistry and Technology of Water-Soluble, Polymers, Plenum, New York 1983.

Fractionation of Polymers

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A. PRINCIPLES OF POLYMER FRACTIONATION

As a general rule, the composition of a polymeric substance is not homogeneous. The differences between the macromolecules of such a substance may be classified according to three main properties: (a) molecular weight, (b) chemical composition, and (c) molecular configuration and structure. Fractionation of a polymeric substance means the separation of that substance into its different molecular species, using a suitable experimental technique, in order to obtain homogeneous fractions. The molecular weight distribution is a general feature for practically all synthetic polymers; it is a consequence of the particular nature of the polymerization process by which they are made. Natural polymers usually have molecular weight distributions as a result of the degradation processes suffered by the substance during

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isolation from living tissues. Additional causes, such as more or less accidental degradation during processing, improper handling, or routine use, may contribute substantially to an increase in the width of the natural molecular weight distribution already present in the sample. The existence of molecular weight heterogeneity in macromolecular substances is therefore quite general. It is one of their fundamental properties and is directly responsible for the necessity of using several molecular weight averages for their description. It also exerts a permanent influence on all the properties of the substance, both in solution and in the solid state.

Differences in chemical composition in polymers originate from those reaction which offer several possibilities of substitution along the backbone of the macromolecules, for instance, the synthesis of random, block and graft copolymers and then any partial chemical transformations to which the substances can be subjected.

The third kind of heterogeneity mentioned above refers to differences in the physical configuration of the macromolecules, such as those between linear and branched polymers, and also to differences in the tacticity of the several molecular species present in the mixture, which is usually reflected in varying amounts of amorphous and crystalline materials in the substance.

Most of the experimental techniques developed so far to fractionate polymers refer to fractionation according to molecular weight. Chemical composition and physical structure differences are handled by more or less sophisticated modifications of the solubility method, such as varying the nature of solvent/nonsolvent mixture or the temperature of extraction, or by using an appropriate active support (adsorbent). The experimental techniques referred to in the tables are mostly based on the variation of some properties directly related to the molecular size. It is common to classify the fractionation method according to their preparative analytical character. The latter methods do not isolate fraction; they are mainly intended to explore the molecular weight distribution of the polymer. The classification of fractionation methods together with the basic idea of each experimental technique is briefly described below. Reviews of polymer fractionation have been published by Cragg and Hammerschlag (660), Desreux and Oth (745), Schulz (2902), Conrad (613), Hall (1198), Channen (544), Guzmán (1172,1174), Bello, Barrales-Rienda and Guzmán (276,277), Fuchs and Leugering (936), Kaesbauer and Schuch (1584), Screaton (2924), Schneider (2878), Cantow (506), Johnson, Porter and Cantow (1506), Moll (2231), Tung (3284,3286), Francuskiewicz (918). Reviews dealing with some specific aspects of fractionation have been published by Schurz (2915), Samsonov (2826), Schneider (2880), Giddings (1047, 1050–1052,1147). Reviews concerning theoretical aspects of polymer fractionation have been given in the book of Tompa (3217), and by Voorn (3397), Huggins and Okamoto (1392), and Koningsveld (1749).

In recent years, gel permeation chromatography (GPC) has become a technique of widespread use not only for molecular weight distribution but also for preparative fractionation. There exists much information on this technique in recent literature. General reviews on GPC have been published by Cantow, Porter and Johnson (505), Heitz and Kern (1262), Altgelt and Moore (71), Johnson and Porter (1507,1508), Altgelt (72), Determann (748), Altgelt and Segal (73), Lambert (1854), Ouano, Barrall II and Johnson (2492), Dawkins and Yeadon (717), Janča (1471), Ouano (2494), Barrales-Rienda (225), Bruessau (452), Hamielec and Styring (1202), Dawkins (718) and books by Yau, Kirkland and Bly (3552), Belenkii and Vilenchik (275), Janča (1474). Also, nonexclusion liquid chromatographies and cross fractionations for copolymers have been reviewed by Glöckner (1071,1076,1080), Vela Estrada and Hamielec (3366) and Mori (2283); temperature rising elution fractionation (TREF) by Wild (1710,3475, 3476); and field flow fractionation (FFF) by Janča (1476).

B. FRACTIONATION METHODS

1. Fractionation by Solubility

(a) Fractional Precipitation

Addition of Nonsolvent Successive precipitation of polymer species from a solution by addition of a miscible nonsolvent. The larger molecules precipitate first.

Lowering the Temperature Successive precipitation of polymer species from a solution by controlled cooling. The larger molecules precipitate first.

Solvent Volatilization Successive precipitation of polymer species from a solution of the polymer in a solvent/nonsolvent mixture by controlled evaporation of the more volatile solvent. The larger molecules precipitate first.

Pressure Variation at Lower Critical Solution Temperature Successive separation is carried out at a lower critical solution temperature (LCST) under isothermal conditions by changing the pressure of the system. Low molecular weight species are precipitated first. The efficiency of the method is poor in low molecular weight ranges.

(b) Turbidimetric Titration Continuous precipitation of polymer species from a very dilute solution by progressive addition of nonsolvent. In the absence of coagulation, the amount of polymer precipitated can be measured by the increase in optical density of the solution. The larger molecules precipitate first. This is an analytical method and can also be reversed, i.e; the polymer is first precipitated completely, and then redissolved by progressive addition of solvent.

(c) Summative Precipitation Simultaneous precipitation of polymer species from several solutions of the same sample by addition of increasing amounts of nonsolvent to the solution. The sum of all the precipitates constitutes a cumulative weight distribution. This is an analytical method.

(d) Cumulative Volume of Precipitate Successive precipitation of polymer species from a solution by addition of increasing amounts of nonsolvent. Fractions are not isolated

and the cumulative volume of precipitate is observed and determined after each nonsolvent increment. This is an analytical method.

(e) Fractional Solution

Direct Extraction Small polymer molecules are brought in direct contact with the solvent to extract low-molecular weight species from the swollen polymer. Subsequent fractions can be obtained either by different solvent mixtures of increasing dissolution power, or by raising the temperature stepwise with the same solvent. Smaller molecules are extracted first.

Film Extraction

CONTINUOUS OPERATION (CONTINUOUS FILM EXTRACTION) (CFE) The polymer solution is applied as a thin coating on both sides of a slowly moving belt. On passing through a drying region the solvent evaporates; the thin film on the belt is extracted continuously in a series of tubes containing solvent/nonsolvent mixtures of increasing dissolution power kept at constant temperatures.

BATCH OPERATION (FILM EXTRACTION) It works by using support material to obtain a very thin film with a large surface area. A metal foil is coated thinly with a polymer solution and the solvent is stripped out; after drying, the foil is cut into strips and extracted successively with solvent/nonsolvent mixtures of increasing solvent power, or by raising the temperature stepwise with the same solvent. Smaller molecules are extracted first.

Column Extraction The polymer is distributed on an inert support (small glass beads or sea sand) packed in a column which will subsequently have solvent passed through it either in an upward or downward direction. Successive elutions then take place in steps by the use of different solvent mixtures or by different extraction temperatures.

Temperature-Rising Elution Fractionation (TREF) The sample is loaded by injection in hot solution into a column packed with an inert support. The column is held in a hot bath. The loaded column is then cooled down to room temperature at a rate as slow as possible. The polymer to be fractionated must be deposited on the inert support by programmed cooling slow enough for crystallization, in such a way that it facilitates and promotes the crystallization of the different branched species in layers one upon the other. This step favours the effective separation by elution with increasing temperatures, which may be carried out stepwise or continuously. Therefore, polymer species are eluted with a solvent from the column as the temperature rises at a selected rate of heating. Molecules are eluted first according to their short-chain branches, i.e., with respect to degree of branching.

Coacervate Extraction A liquid polymer-rich gel phase (coacervate) is extracted by a solvent, forming a sol phase. Both the sol phase (or extracting agent) and the gel phase (coacervate) contain the same solvent and nonsolvent and are formed only together with the polymer fractionated (i.e., solvent and nonsolvent are completely miscible without the polymer). Thus, this procedure is a liquid–liquid extraction. Successive extraction of polymer species can be done from the coacervate. Smaller molecules are extracted first.

Continuous Polymer Fractionation (CPF) It represents a special variety of a counter-current extraction that functions in the following way: A comparatively concentrated solution of the polymer in a given solvent [it can also be a solvent/non-solvent mixture (the feed, FD)] is introduced into a pulsed sieve-bottom column at one end, and the same solvent or solvent/nonsolvent mixture but free of polymer (extracting agent, EA) is added at the other end. Both phases are homogeneous as they enter the column, but the CPF splits the polymer into two fractions by demixing inside the column in such a way that the molecules of different mass distribute between the two phases. This treatment really forms a coacervate due to the limited solubility of the polymer in the solvent/nonsolvent system. The coacervate is continuously extracted inside a counter-current column by the liquid extracting agent whereby the polymer molecules will be distributed over the counter-current phases according to their MW. The high molecular weight material remains in the phase resulting from the feed and leaves the column as the gel phase, whereas the low molecular weight material is extracted with the sol phase, so these two fractions leave the column as gel and sol phases, respectively. Thus, the original polymer can be split into two fractions at a desired MW in one continuous run.

(f) Partition Between Immiscible Solvents

Polymer species are distributed according to molecular size between two immiscible liquids of different solvent power. Counter-current extraction is particularly suitable for this method.

(g) Fractionation with Demixing Solvents

It is possible to separate chemically different polymers dissolved in a suitable homogeneous solvent mixture quantitatively, if the solvents are miscible at high temperatures but show phase separation at a lower temperature. Demixing is reached by slow-cooling the system to the desired separation (equilibrium) temperature. Stirring of the system is advantageous. Details of the demixing procedure depend on polymer species. Blends of polymers and copolymers, including block-, graft- and statistical copolymers, can be separated in one fractionation step with respect to their chemical nature. Multistep fractionation methods using demixing solvents (cumulative, successive, semicumulative, and head–tail type of fractionation) have been used to determine the chemical distribution of binary copolymers as well as block- and graft-copolymers.

(h) Fractional Crystallization

Successive separation of species from polymer solutions by crystallization at different temperatures. The crystallization can be carried out under stagnant conditions, but considerable improvement in the efficiency is obtained if the crystallization is induced by fast stirring (stirring crystallization). Under these conditions, the high molecular weight constituent crystallizes first, settling on the stirrer as long thin fibrillar crystals. The technique can be coupled with fractional

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extraction at increasing solution temperatures. The method is poorly reproducible.

Crystallization Analysis Fractionation (CRYSTAF) It is based on the segregation of crystals of different morphology or comonomer content by crystallization. In CRYSTAF, the separation and analysis are performed in a single step (the crystallization cycle), where concentration of the polymer solution is intermittently sampled and analyzed as the temperature goes down. The temperature–concentration data which are obtained correspond, in the cases of branched poly(olefins) or their copolymers with a-olefins, to the cumulative curve of the side chain branching distribution (SCBD). The last point at the lowest temperature of the experiment is the soluble or noncrystallizable fraction.

2. Fractionation by Chromatography

(a) Adsorption Chromatography (Chromatography on an active support) The adsorption of polymer species on an active support depends on the molecular weight.

Frontal Analysis The active support is packed in a column. A solution of polymer passes down the column and is collected when leaving it. The concentration in the effluent changes with the volume collected, and presents successive fronts due to the differential adsorption of molecular species on the active support.

Elution Analysis

COLUMN A small quantity of polymer is adsorbed on the upper portion of the support packed in the column. Elution with a suitable solvent takes place; each component moves down the column at a different rate and is completely displaced at some time and collect in the effluent; in the gradient elution method, the eluent is a liquid of increasing solvent power.

THIN LAYER CHROMATOGRAPHY (TLC) A thin layer chromatographic system is represented by the following three elements: a stationary phase which is a thin layer formed usually on a glass plate with an adsorbent like silica gel (the thin layer is activated by drying); a mobile phase which is any solvent or mixture used as a developer, and a sample. Separation is effected because the migration rate of each component relative to that of the mobile phase is retarded by the stationary phase to different degrees. A spot of the polymer solution is put on the plate, and after drying, the plate is placed vertically in a tank containing a suitable solvent or mixture. As the solvent moves upwards, it carries along each component at different rates. Separated, colorless species can be made visible by appropriate methods. When TLC is used in combination with a concentration gradient method, the composition of the developer (solvent or mixture) changes with the development time. In general, the TLC of a polymer may be classified into four types depending on the separation mechanism: adsorption–desorption, partition, molecular sieving (size exclusion), and ion-exchange process. However, only two of the above mechanisms, adsorption and molecular sieving, are applicable to separation of high polymers.

(b) Competitive Selective Adsorption This method is applied to separate mixtures of tactic forms into their components. The principle of separation consists of a competitive adsorption of two different tactic polymers from the solution onto solid surface. The two tactic forms compete for occupying active sites on the solid adsorbent; one of them will be selectively adsorbed on the adsorbent, while the second one will be unadsorbed and remain in solution. In practice, the separation is carried out by adding the solid adsorbent to the solution of the tactic mixture. The solvent must be selected in such a way that the stereocomplex formation between tactic forms rarely occurs.

(c) "Precipitation" Chromatography (Baker–Williams Method) (Chromatography on an inactive support) The support is an inert material packed in a column. On top of the column, a small amount of polymer is placed, as in elution chromatography. A temperature gradient is set along the column, the upper part being at higher temperature than the bottom; then the elution takes place with a solvent mixture of increasing solvent power. Polymer species move down the column being in a continuous exchange between a precipitated phase and a saturated solution. This distribution depends on the molecular size. Precipitation chromatography can also be carried out in the absence of a temperature gradient. It then becomes essentially a continuous fractionation by fractional solution. (see under Fractional Solution Column Extraction).

(d) Normal-Phase and Reversed-Phase Liquid Chromatography Copolymer separation by non-steric liquid chromatography (NELC) can be achieved in practice by the difference of adsorption power between the copolymers and the stationary phase. In both NPLC and RPLC, the mobile phase should be a good solvent for the copolymers during the entire elution.

In NPLC, stationary phases are polar, for example, silica gel, silica-CN, silica-NH_2 and poly(acrylonitrile) gel; retention times increases with sample polarity and mobile phases are less polar. Gradient elution is performed to increase the content of a polar solvent in the mobile phase, i.e. the polarity of the eluent increases in the course of a run.

RPLC is performed on non polar columns, i.e. stationary phases are hydrophobic (e.g. packed with crosslinked poly(styrene) (PS), or an alkyl branched phase materials such as silica-ODS), and the retention times decrease with polarity. Mobile phases are polar. RP gradient elution is performed to increase the content of less polar solvent in the mobile phase, i.e. the polarity of the eluent decreases in the course of a run.

(e) Gel Permeation Chromatography (GPC) (Chromatography on a porous support) Gel permeation chromatography, sometimes referred to as size-exclusion chromatography (SEC), is an analytical or preparative technique in which solute molecules are separated according to their effective hydrodynamic volumes in solution. A

gel permeation chromatographic system is composed of three fundamental elements. The stationary phase is an expanded and highly crosslinked polymer gel network (expanded silica gel, gel of porous glass, or some of these, but modified by chemical or physical treatments). The gel is packed into a chromatographic column; a mobile phase which may be any solvent or mixture for the polymer and a good swelling agent for the gel. The polymer sample in dilute solution is injected onto a column which contains porous particles with solvent and which is continuously fluxed with eluent. The separation occurs as the polymer molecules in the eluent solvent (mobile phase) percolate through the porous bed (stationary phase). As a result of the restrictions imposed by the size of the pores on the larger molecules, there is a greater pore volume available to the low molecular weight species giving them in effect a longer path length and they, therefore, are more strongly retarded during elution. In other words, small molecules are able to penetrate into the pores of the beads more readily than the larger ones. This effectively increases the retention volume as molecular size decreases. Hence an inverse relationship between retention volume and molecular weight of the polymer exists.

Since from an ideal point of view polymer solubility and adsorption are not occurring in this chromatographic process, the volume of eluent required for the elution of any macromolecules species must essentially be dependent on chain length and appears to be insensitive to structure (universal calibration). However, this seems to be not always the case (non-exclusion effects).

(f) Surface Exclusion Chromatography (Hydrodynamic exclusion chromatography) Conventional gel permeation chromatography (GPC) fails when applied to the characterization of very large macromolecules. Such polymers are easily oriented or deformed by hydrodynamic forces occurring in columns. Hence, steric pore exclusion mechanisms are disturbed, separation efficiency is poor, and the relationship between elution volume and molecular weight is no longer found valid for the highest molecular weight species.

Surface exclusion chromatography (hydrodynamic exclusion chromatography) has been developed for rigid particle size determinations. This technique has been shown to be efficient for size separation of proteins and latexes. It uses a non-porous packing as a stationary phase.

(g) Partition Chromatography Polymer species are distributed between two liquid phases, one of them mobile and the other fixed by absorption on a support. The support consists of strips or sheets of porous paper. The immobile phase is packed in a column.

(h) Phase Distribution Chromatography (PDC) Separation of polymer components in a PDC column is based on thermodynamic and kinetic interactions between the mobile phase (the polymer to be analyzed is dissolved in a Q-solvent) and a gel (stationary phase) of the same linear high polymer situated as a non-crosslinked coating on the surface of small glass bead. It is important to keep the temperature always below the Q-temperature of the sample. The separation efficiency increases sharply with decreasing temperature since the higher molecular species dissolve in the gel phase to a higher extent than the lower molecular species, the former leave the column later. It is a powerful method for the correct determination of narrow molecular weight distributions.

(i) Field Flow Fractionation (FFF) Field flow fractionation resembles chromatography in both the experimental and dynamical aspects of its operation, and it can therefore be thought of as a chromatographic method. It requires neither a stationary phase in the classical sense, nor the packing to support as is characteristic of chromatography. For this reason FFF has been described as one phase chromatography. Like chromatography, it is an elution technique used to separate components as they are transported through a column by a stream of solvent. The method is based on the difference between the flow rates in the center and near the column walls which obey Poiseuille's law. Retention is caused by an external field or gradient which forces the solute into narrow layers occupying the slow flow regions near the wall of an empty channel. Several types of FFF based on the type of force field applied have been developed.

Classical FFF

THERMAL FFF Thermal field flow fractionation (ThFFF) is a high-resolution separation method for a wide range of relatively non-polar polymers in suitable organic solvents. Certain polar macromolecules can also be fractionated by ThFFF in totally aqueous systems. ThFFF separations are carried out with a single mobile phase in a thin, open channel by applying a large thermal gradient across highly polished parallel plates between which a spacer is clamped. It is usually composed of two metallic blocks with highly polished surfaces. ThFFF separations are governed by two transport mechanisms acting in opposition: thermal diffusion, which is the movement of mass in response to a temperature gradient, and ordinary (Fickian) diffusion, which is the movement of mass in response to the concentration gradient established by thermal diffusion.

SEDIMENTATION FFF Either natural gravitational or centrifugal forces in the centrifuge serve as an effective field. The channel is usually coiled around the interior basket of a centrifuge rotor. This technique separates components on the basis of their mass differences.

ELECTRICAL FFF The altogether homogeneous field is induced by electrical current across the channel. Charged macromolecules interact with the field and are separated according to the ratio of electrophoretic mobility to diffusion coefficient.

FLOW FFF Flow field-flow fractionation separates according to differences in diffusion coefficients.

PRESSURE FFF Pressure field-flow fractionation (pressure FFF) and flow FFF are by nature very similar. In both

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cases, it is the cross flow that represents the lateral field. The principal difference is that in flow FFF, the flow field is applied externally across the channel formed between two parallel plane membranes, whereas in pressure FFF, the lateral flow across the wall of a circular-cross-section capillary is initiated by an internal pressure drop in a liquid pumped along the semipermeable capillary.

MAGNETIC FFF Magnetic field-flow fractionation has been studied in only a few cases dealing with theoretical principles of the separation and retention of bovine serum albumin in the presence of nickel (II) ions in a magnetic field of 400 G and retention of metal oxides.

CONCENTRATION FFF Concentration field flow fractionation (concentration FFF) is the only FFF technique that could make use of a concentration gradient of a mixed solvent across the channel in order to induce effective chemical forces or chemical field.

SHEAR FFF Shear field-flow fractionation (shear FFF) has been proposed as a technique in which shear forces are responsible for migration perpendicular to flow. An internal shear force field is induced in the annular space between two concentric cylinders that are in relative rotational motion.

Focusing FFF

SEDIMENTATION-FLOTATION FOCUSING FFF In SFFFFF solute particles or macromolecules "sediment" or float in the density gradient of a liquid phase according to the difference between the local density of the liquid and that of the particle or macromolecule. At quasi-equilibrium, the solute species are focused in a thin layer at the position where the density of the environment is the same as the solute density.

ISOELECTRIC FOCUSING FFF The electrophoretic mobility of amphoteric macromolecules is a function of pH and is zero at the isoelectric point. This means that if a stable pH gradient is formed in the FFF channel due to an applied electrical field, the amphoteric solute under the influence of this field is focused into the position of the isoelectric point. In a steady state, this transport is balanced by ordinary diffusion.

(j) Ion-Exchange Chromatography The support is an ion-exchange resin, constituting an immobile phase, through which the solution of polymer is passed. This method is appropriate for polymer species bearing electric charges; the molecules are distributed between the liquid phase and the interface, according to their ionic adsorption forces, which depend on the electric charge and the size of the macromolecules.

(k) Hydrodynamic Chromatography Hydrodynamic chromatography (HDC) is a rather new technique for separating solutes or dispersed materials at high dilution in the micron range, according to decreasing size. It is based on the effect of a radial flow gradient inducing different mobilities according to size. Larger particles are located preferentially in the axis of capillaries where the flow rate is maximum; whereas smaller ones are closer to the walls where the flow rate is minimum. Packed and open capillary columns can separate particles below and above one mm, respectively. It involves injection of a dilute latex sample into a carrier stream (the aqueous eluent) which is pumped through the system at a constant flow rate. The column is packed with non-porous beads. Larger particles exit the column before smaller ones and are detected using a suitable system of detection. It is a complementary method to field-flow fractionation (FFF), steric exclusion chromatography (SEC) and superfluid chromatography.

3. Cross Fractionation

Most of the copolymers and polymers obtained by chemical modification reactions generally possess a two-dimensional distribution: MWD (Molecular Weight Distribution) and CCD (Distribution of Chemical Composition). An analogous problem exists with polyolefins: MWD and distribution of Short-Chain Branching (SCB). It is normally impossible that they can be adequately fractionated by both molecular weight (MWD) and by chemical composition (CCD). The problem is overcome by cross fractionation. The term cross-fractionation refers to any combination of fractionation methods capable of evaluating the distribution in size and composition.

(a) Classical Cross Fractionation Cross fractionation requires two solvent/nonsolvent systems to separate in two different directions. A solvent/nonsolvent combination fractionating solely by MW would be appropriate for the evaluation of the MWD, another one separating by CC would be suited for measuring the CCD of the material. Fractions of the starting material obtained by one precipitant are redissolved and fractionated once more with the help of the complementary precipitant yielding the final fractions. Owing to its high expenditure of time and materials, classical cross-fractionation is seldom performed. However, this procedure is irreplaceable for the production of narrow molecular weight distribution fractions in both MW and CCD on a preparative scale.

(b) Non-Classical Cross Fractionation This technique of cross fractionation can be performed off-line or on-line. Typical cross-fractionations include partition chromatography-SEC, gradient elution-SEC, SEC-turbidimetric titration, gradient elution-TREF, TREF-SEC and demixing fractionation-demixing fractionation.

(c) Orthogonal Chromatography This technique is a special case of non-classical cross fractionation in which two coupled SEC techniques are used. SEC in different eluents can separate a copolymer in two diverging directions. Two SEC instruments are coupled together so that the eluent from the first one flows through the injection valve of the second one.

(d) Cross Fractionation by Bidimensional TLC Combined types of TLC are often for the separation of polymers, copolymers, etc. This technique allows separation according to chemical composition, molecular weight, and stereoregularity. Copolymers, LD and LLD poly(ethy-

lenes) and chemically modified polymers are extremely multicomponent materials which often exhibit simultaneous distributions of molecular weight, composition, and monomer sequence length. Interference of the remaining two distributions while attempting to elucidate one of them greatly complicates analysis. Cross fractionation (bidimensional) TLC is a method to elucidate such problems.

4. Fractionation by Sedimentation

(a) Sedimentation Velocity Sedimentation velocity of polymer species in a high centrifugal field is a function of molecular size. This is usually an analytical method unless a velocity ultracentrifuge of the preparative type is used.

(b) Sedimentation Equilibrium At lower rotational speeds of the ultracentrifuge, it becomes possible to create an equilibrium situation such that the rate of sedimentation is exactly equal to the rate of back-diffusion of the macromolecules. Larger polymer molecules are then found closer to the bottom of the cell than smaller ones.

(c) Density Gradient Technique In a performed or self-generated density gradient, the polymer collects around the position of its own density in a band. The width of this band is dependent on the molecular weight of the polymer. A band of polydisperse macromolecules contains more high-molecular-weight polymer near its maximum than near its "tails". This technique is also capable of separating according to composition.

5. Fractionation by Diffusion

(a) Thermal Diffusion A polymer solution is placed between two surfaces, and a high temperature gradient is established between them. The solution is in contact with an upper and a lower reservoir. The temperature gradient gives rise to a thermal circulation of the molecules, producing a separation of polymer species, which migrate towards the lower reservoir. Thermal diffusion is more pronounced for the larger molecules than for the smaller.

(b) Brownian Diffusion Polymer molecules diffuse at different rates from a solution into a solvent by a boundary, depending on their molecular weights. The translational diffusion constants can be determined by special optical means and related to the inhomogeneity of the sample. This is an analytical method.

6. Fractionation by Ultrafiltration Through Porous Membranes

A polymer solution is submitted to ultrafiltration through a series of membranes of different porosity. The rate of diffusion depends on the molecular size and the degree of permeability of the membranes. It is possible to isolate fractions with varying molecular weights at different times.

7. Fractionation by Zone Melting

A solid solvent is packed in a column. A small amount of polymer is put on top of the solid solvent and dissolved by heating a narrow zone. After solidification, a lower zone is heated, melted, and resolidified. The full column is treated in this way. Polymer species move down the column at different rates, depending on their molecular size during the molten stage. At the end, the polymer is distributed throughout the entire column and recovered by cutting the solid into several portions and sublimating the solvent.

8. Electron Microscopic Counting Method

With most glassy amorphous polymers, it is possible by means of the electron microscope, to observe single spherical molecules, if very dilute solutions in a solvent/nonsolvent mixture are sprayed onto a thin substrate. Consequently, the weight- and number-average molecular weights and molecular weight distributions can be obtained. The method can be recommended and applied only for polymers with molecular weights higher than 5×10^5.

C. TABLES OF FRACTIONATION SYSTEMS FOR DIFFERENT POLYMERS

TABLE 1. MAIN-CHAIN ACYLIC CARBON POLYMERS

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
1.1. POLY(DIENES)				
Poly(diacetylene)	Chromatography	Chloroform	SEC, styragel	2640,3450
			SEC, ultrastyragel	1677
		Methylene dichloride	SEC, μ-styragel	2620
Poly(butadiene)	Chromatography	1,2,4-Trichlorobenzene	SEC	1380,2073,2074
		Benzene/ethanol	Precipitation chromatogr.	624,626
		Benzene/methanol	Precipitation chromatogr., 35–28°C	2636
		Butyl acetate/isopropanol	Precipitation chromatogr., 25–55°C, 1,4-*cis*	2756

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Carbon tetrachloride followed by amyl chloride	Two dimension TLC, silica gel, 1,4-*trans*,1,4-*cis* and 1,2-vinyl	779
		Chloroform	SEC	2925,2927
			SEC, carboxy + hydroxy term.	1879
		Chloroform, tetrahydrofuran	Three-arm star	2608
		Diisobutene/isooctane	Precipitation chromatogr.	1397
		Diisobutene/*n*-propanol	Precipitation chromatogr., 1,4-*cis*	1397
		Ethyl acetate	Hydroxytelechelic; SEC, styragel	740
		Methanol, ethanol	Column elution, carboxy term.	1880
		Methylene chloride	Hydroxytelechelic; SEC, Porasil	740
			SEC, 1,4,-*trans*, 1,4,-*cis* and 1,2,-vinyl (modified)	1919
		n-Heptane/isooctane	Precipitation chromatogr. 50–40°C	1272
		o-Dichlorobenzene	SEC, 130°C	1251,1252
			SEC, 140°C, 1,2-	141,142
			SEC, 150°C, 1,4-*cis*	3031
			SEC, 25°C, 1,4-*cis*	1214
		Tetrahydrofuran	SEC	12,379,380,381,547, 886,956,1824,1852, 2667,2677,3314,3346
			SEC, 25°C	2555
			SEC, deuterated	146
			SEC, low MW	1842,1878
			SEC, modified	2900,2901
			SEC, phenogel, telechelic	592
			SEC, star branched	1196,1783
			SEC, μ-styragel, 30°C, three arms	2561
			SEC, carboxy term	2925,2926,2927
			SEC, cyclized	366
		Toluene	SEC	7,2073,3241
		Toluene/isopropyl alcohol	Precipitation chromatogr., 44–24°C	1379
		Toluene/*n*-propanol	Precipitation chromatogr., 1,4,-*cis*.	1397
		Trichlorobenzene	SEC, 135°C, hydrogenated	2555
		Trichloroethylene	SEC, 23°C	3195
	Coacervation	Benzene/pentane		1214
	Extraction	Benzene/acetone	Branched polymer	3364
		Benzene/acetone-methanol (9/1)	28°C	625
	Field flow	Water (0.1% w/w FL-70 detergent + 0.02% sodium azide	Sedimentation, latexes	352
	Fractional crystallization	Heptane	Several temperatures, 1,4,-*trans*	3241,3242
	Fractional precipitation	Benzene/acetone:dioxane followed by acetone : methanol		1001
		Benzene/methanol		428,430
			25°C	882
			25°, 1,4,-*cis*	2373
			25°C, Hydroxy term.	1396
			Low. temp., 1,4,-*cis*	7
		Benzene/*n*-butanol		1726
		Heptane/acetone	1,4,-*cis*	2260
		n-pentane–dioxane/methanol	Hydrogenated	3510
		Pentane/methanol	30°C, carboxy term.	922
		Tetrahydrofuran/water	Chlorinated	3136,3346
		Toluene/ethanol	1,4,-*cis*	442
		Toluene/heptane	1,4,-*trans*	3416
		Toluene/methanol	1,4,-*cis*	813,3597
			28°C	110
			Cyclic, 1,4,-*cis*-tri- and tetrachain	2666,3586
		Toluene/*n*-butanol		1726
	Fractional solution	Benzene/methanol	Cyclopolymer	3524
		Chloroform, chloroform–ethanol, ethanol, carbon tetrachloride, carbon tetrachloride-chloroform	Column elution, carboxy + hydroxy term.	1879

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Chloroform/methanol	Column elution	2925,2926
		Ethyl ether	Extraction	2371
	Sedimentation velocity	Diethyl ketone	Ultracentrifuge, 10.3°C	3333
		n-heptane/isooctane	Ultracentrifuge, 20.5°C 1,4,-*cis*	2259
		n-Hexane/*n*-heptane (1/1)	Ultracentrifuge, strereoreg.	59,2624
		Octane	Ultracentrifuge	429,430
	Turbidimetric titration	Carbon tetrachloride/*n*-butanol		1332,1332,3559
Poly[(3,5-di-*tert*-butyl-4-hydroxyphenyl)acetylene]				
	Chromatography	Tetrahydrofuran	SEC, styragel	2401
Poly(*tert*-butylacetylene)	Chromatography	Chloroform	SEC, μ-styragel	2071
		Chloroform	SEC, styragel	2356
Poly(1-chloro-1-decyne)	Chromatography	Chloroform	SEC, styragel	3568
Poly(1-chloro-1-hexadecyne)	Chromatography	Chloroform	SEC, styragel	3568
Poly(1-chloro-1-hexyne)	Chromatography	Chloroform	SEC, styragel	3568
Poly(1-chloro-1-octyne)	Chromatography	Chloroform	SEC, styragel	3568
Poly(1,1,2-trichlorobutadiene)	Fractional precipitation	Benzene/petroleum ether		2546,3027
Poly(chloroprene)	Chromatography	Benzene	SEC, 25°C	1984
		Benzene/methanol	Precipitation chromatogr., 28–35°C	2636
		Butanone	SEC, 25°C	1610
		Butanone : methyl alcohol (0.887/0.113)	SEC, 25°C	1610
		Chloroform	SEC	2890
		Tetrahydrofuran	SEC	649
	Fractional precipitation	Benzene/acetone		1233
		Benzene/methanol		1206,2221, 2222,2223
			1,4,-*trans*	2847
		Butanone/methanol	25°C	682
	Fractional solution	Benzene/isopropanol	30°C	344
		Benzene/methanol	Neoprene, 30°C cont, film ext.	345,2890
	Sedimentation velocity	Butanone	20°C	2890
	Thermal diffusion	Benzene		1725,1789
	Turbidimetric titration	Benzene/methanol		2129
Poly(1-ferrocenyl-1,3-butadiene)				
	Chromatography	Tetrahydrofuran	SEC	3347
Poly[*o*-(trifluoromethyl)phenyl)acetylene]				
	Chromatography	Chloroform	SEC, styragel	2067,2070
Poly(2,4-hexadiyn-1,6-ylene-asbacate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1814
Poly(1,4-bis(homoallyl)cubane)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	557
Poly(5-hydroxyoctylene)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2700
Poly(isoprene)	Chromatography	Benzene/methanol	Precipitation chromatogr.	365,1964, 2332,2636
			Precipitation chromatogr., glass beads	436
		Chloroform	30°C, SEC, styragel	297,708,716
			SEC, ultrastyragel	3553
		Cyclohexane	SEC	716,1962
		Cyclohexanone	Partition on paper	195,309
		Methylene chloride	SEC μ-styragel	1919
		o-Dichlorobenzene	SEC, 135°C	709
		Tetrahydrofuran	SEC, μ-styragel, 25°C, amino-capped	700
			SEC, μ-styragel, 30°C, three and four arms	2561
			SEC, styragel, several structures	22,81,112,427, 495,886,1166, 1184,1185,1243, 1308,2105,2677, 3396
		Tetrahydrofuran, toluene/methanol (95/5)	SEC, μ-styragel and sodium salt carboxylated	888
		Toluene	SEC, preparative styragel	551
		Toluene/isopropyl alcohol	Precipitation chromatogr.	1379
		Trichlorobenzene	SEC, 130°C	1215,1380

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Field flow	Tetrahydrofuran	Thermal	1146
	Fractional precipitation	Benzene/acetone	Havea	1233
		Benzene/ethanol	Gutta percha, balata, synthetic *trans*-poly(isoprene)	625
		Benzene/isopropanol	Low temp.	359
		Benzene/methanol	*cis*	10
			Low temp.	359,551,1039,1158
		Benzene/*n*-butanol		2640
		Chloroform/acetone	Pale crepe	484
		Dichloroethane/butanone		1963
		Hexane/1-propanol	*cis*, 26°C	2597
		Toluene–ethanol (4 : 1)/ethanol	Rubber	3230,3438
		Toluene/boiling methanol	Chlorinated natural rubber	65
		Toluene/methanol		87,88,112,256, 386,470,883, 1828,2332, 2759,3406
			trans,1-4,*cis*,1-4	436,3507
		Toluene/*n*-butanol	30°C	2639
	Fractional solution	Acetone	Have a extraction	1505
		Acetone, *n*-hexane	Guayule, extraction	1231,1232,2123
		Amyl acetate/2-ethoxyethanol mixtures	Column extraction	1828
		Benzene/methanol	25°C, column extraction, natural rubber	437,3455
	Thermal diffusion	Tetrahydrofuran	Natural rubber	1904
	Trubidimetric titration	Carbon tetrachloride/*n*-butanol		1332
		Toluene–ethanol (4 : 1)/ethanol	25°C	2643
Poly(1-isopropylidenedicyclopentadiene)				
	Fractional precipitation	Toluene/methanol		533
Poly(1,4-(2,3-dimethylbutadiene)				
	Fractional crystallization	Toluene/methanol	28°C	111
Poly(1,4-*cis*-(2-methyl-1,3-pentadiene)				
	Fractional solution	Diethyl ether, heptane, toluene	Direct extraction	485
Poly(norbornadiene)	Extraction	Toluene, *o*-dichlorobenzene		2218
Poly(1,3-pentadiene)	Fractional solution	Benzene/butanone	Amorph/cryst., 1,4-*cis*, isotactic	2370
Poly(perfluorobutadiene)	Fractional solution	Hexane, hexafluorobenzene	Extraction	3220,3221
Poly(1-phenylbutadiene)	Chromatography	Tetrahydrofuran	SEC, styragel	3106
Poly(phenylacetylene)	Chromatography	Tetrahydrofuran	SEC, 25°C, styragel	2932
Poly(spiro-2,4-hepta-4,6-diene)	Fractional precipitation	Benzene/isopropanol		1247
1.2. POLY(ALKENES)				
Poly(α-olefins) (General)	Chromatography	Benzene/ethanol	Precipitation chromatogr.	843
Poly(1-butene) See Poly(ethylethylene)				
Poly(butylethylene)	Chromatography	1,2,4-Trichlorobenzene	80°C, styragel, SEC	3418
		Tetrahydrofuran	SEC, ultrastyragel, 30°C	160
	Fractional precipitation	Cyclohexane/acetone		1931
		Toluene/*n*-butanol		3223
		Xylene/triethylene glycol	130°C	2974
Poly(cyclohexylethylene)	Fractional solution	Benzene	Extraction	2497
Poly(cyclopentylethylene)	Fractional solution	Ethyl ether	Extraction	2497
Poly(cyclopropylethylene)	Fractional solution	Benzene	Extraction	2497
		Ethyl ether, *n*-heptane, octane, nonane	Extraction	2501
Poly(ethylene)	Chromatography	1,2,4-Trichlorobenzene	SEC, 135°C	2520
			SEC, μ-styragel, 145°C, HDPE, LLDPE, LDPE	731
			SEC, styragel, 140°C	3456
			SEC, styragel, 145°C	1925
			SEC, ultrastyragel, 135°C	824
		Chlorobenzene	SEC, 100°C merckogel, and styragel	1266
		Diphenyl ether, tetralin/triethylene glycol	Continuous fractionation (counter current extraction) > 130°C	1018

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Ligroin/2-(2-butyl)ethanol	Precipitation chromatogr.	3015
		o-Dichlorobenzene	SEC	3318
			SEC, Low MW HDPE	3376
			SEC, styragel	2355
			SEC, styragel, 130°C, 135°C, 138°C	190,235,236,617, 709,712,963,1045, 1411,1646,1981, 1982,2347,3030, 3479
		Perchloroethylene	SEC, 110°C	2783,3394
		Propane	Supercritical fluid fractionation	2127
		Tetralin	SEC, 125°C	1984
		Tetralin/2-butoxyethanol	Precipitation chromatogr., 110–160°C	611,1143
		Toluene	SEC, 25°C, low MW	81,126
		Trichlorobenzene	SEC	379,381,509,607, 614,615,639,723,751, 785,786,1006,1311,1643, 1644,1704,1989,2076, 2139,2346,2347,2348, 2349,2482,2509,2590, 2661,2697,2698,2820, 2821,2868,2881,3282, 3377,3403,3433,3434, 3435,3441,3458,3462, 3469,3470,3471,3472, 3480
			SEC, styragel, 135–140°C, branched	1823
	Coacervation	Dibutyl phthalate/decalin (60/40)		3480
		o-Dichlorobenzene/triethylene glycol	138°C	709
		Xylene/poly(oxyethylene)		1839,2451,2919
		Xylene/triethylene glycol	Branched	1823
		Xylene : ethyl cellosolve–poly(oxyethylene) (50/50)	120°C	2661
	Cross-fractionation	*o*-Dichlorobenzene	SEC (135°C), LLDPE	3602
			TREF; SEC, 140°C, LLDPE	3318
		Trichlorobenzene	TREF-Chromosorb, SEC 140°C	2183
		Xylene	TREF (silica)	3602
	Extraction	Phenyl ether, tetralin/triethylene glycol		1014
	Fractional crystallization	1,2,4-Trichlorobenzene	CRYSTAF, LLDPE, 95–30°C	1675,2237
			CRYSTAF, LLPDE, 95–30°C	2235
			CRYSTAF, LLPDE, VLDPE, 95–30°C	2236
			CRYSTAF, metallocene resins blend	2236
		Xylene		1287,1625,1627,1645, 1748,2076,2559,2560, 2665
			UHMWPE	2119
		Xylene, perchloroethylene or 1-Chloronaphthalene	Lowering temp. stirred	2560
	Fractional demixing	Dimethylformamide/methylcyclohexane	Chlorinated PE	1817
	Fractional precipitation	2,4-Dimethylpentane	At a lower critical solution, temp. is sensitive to the side-group content	212
		2-Ethylhexanol/decalin (85/15)	Solvent volatilization., low temp.	1619,3451, 3452,3453
		Amyl acetate	133°C, lowering temp.	21

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Benzene/poly(oxyethylene)	75°C	2364
		Epichlorhydrine/*n*-hexane	35°C	1566
		Liquid ethylene at 130 atm. and 80°C	Releasing pressure	3298
		p-Xylene/*n*-hexanol	Turbidimetric by lowering temp.	16
		Tetralin/2-ethoxyethanol (1/3)	Lowering temp., branched	1990
		Tetralin/benzyl alcohol (3/2)	Lowering temp. 105–165°C	1578,1709
		Toluene/methanol		697
		Toluene/*n*-butanol	115–100°C Lowering temp.	1815
		Toluene/*n*-propanol	Lowering temp.	2364,3299
		Toluene/poly(oxyethylene)	80°C	2392
		Toluene/triethylene glycol	107°C	1578
		Xylene/*n*-butanol	100–115°C	1815
		Xylene/*n*-propanol	90°C	125
		Xylene/poly(oxyethylene)	75°C, 80°C, 130°C	1555,2314,2364, 2450,2451
		Xylene/triethylene glycol	90°C	18,278,279,280, 281,342,1628, 1630,2339,2701, 2974,3163,3279, 3280,3281
	Fractional solution	1,2,4-Trichlorobenzene	LLDPE	2895,3478
		2-Ethoxy ethyl acetate (cellosolve)/decalin	130°C, glass beads	3480
		Anhydric carbonic, pentane	Supercritical	2884
		Carbon dioxide, propane, propane–modified carbon dioxide	SCFF, Low MW HDPE	3376
		Cyclohexanone	Rising temp.	2896
		Decalin	150°C	1872
		Decalin/2-ethoxy-ethyl acetate	Column elution	1126
		Decalin/ethylene glycol–hexanol	139°C, column elution	1735
		Diisopropyl ether, benzene, xylene	Extraction, waxes	1722
		Ethyl benzene/cellosolve	120°C, column elution	963
		Mesitylene/2-butoxyethanol	Column elution, 126°C	3224
		o-Dichlorobenzene/dimethyl phthalate	127°C, glass beads column elution	2933,2934
		Petroleum ether	Column elution, low MW	1025
		Tetralin	Quartz sand, column extraction, variable temp. gradient	467
		Tetralin/2-butoxyethanol	126°C, column elution analytical and preparative, branched	3224
		Trimethylbenzene/butylcellosolve	Column elution, glass beads, 130°C	1954
		Toluene	Column elution	743,744,2364
		Toluene/poly(oxyethylene)	80°C	2390,2391
		Xylene		1235,1628,1629,1642
			128°C, column extraction	2466
			Rising temp.	305,615
		Xylene, chloronaphthalene, perchloroethylene	Increasing temp.	2560
		Xylene/butyl cellosolve	125°C, Column elution	2808,3220
		Xylene/2-butoxyethanol	Column elution	574,641,871,1224, 1246,2338,2340, 2344,3224
		Xylene/2-ethoxy-ethyl acetate (cellosolves)	Column elution, 125°C	2466
		Xylene/butyl cellosolve	Column elution, 127°C	2343
		Xylene/diethylene glycol monomethyl ether	Column elution 126°C	917,1200,1276,1342, 1359,1632,1668,1669, 1832,2076,2138,2246, 2354,2804,2820,2821, 2867,3452,3453,3469, 3470,3519

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Xylene/ethyl cellosolve	Column elution, 120°C, LLDPE	1367
		Xylene/*n*-butanol	Film extraction	935
		Xylene/trichloroethylene	Boiling mixt., 46–86°C	1642
		Xylene/triethylene glycol	Column elution	279,3279
	Sedimentation equilibrium	Biphenyl	Ultracentrifuge, 123.2°C	2893
	Sedimentation velocity	1-Bromonaphthalene	Ultracentrifuge, 110°C	2117
		1-Chloronaphthalene	Ultracentrifuge, 120°C	2244
	Stirring-induced fractional precipitation	Decalin	100–78°C	1293
	Thermal field flow	*o*-Dichlorobenzene	90–140°C	2520
		Water/ethylene glycol or silicone oil, water/glycerol	90–130°C	2520
	TREF	1,2,4-Trichlorobenzene	Chromosorb-P, TREF room – 20°C/hour	3241
			Glass beads, LLDPE	2896
			LLDPE	728,1514,1849
		Butyl cellosolve	Glass beads, deuterated	1954
		Chloronaphthalene	HP-LPDE, HDPE, LLDPE	1648
		o-Dichlorobenzene	Chromosorb-P, analytical, low density and low linear density	1710
			Direction of separation SCB	3318
			SEC	3603
			TREF (glass beads)	2355
		Trichlorobenzene	20°C/h	3474
			Chromosorb-P, analytical + preparative, very low density	1581
			HP-LDPE/LLDPE blends	3026
			LLDPE, 140°C	2183
			Steel shot, analytical + preparative, low linear density	1255
		Xylene	25–120°C	1367
			Chromosorb P, branched	1823
			Chromosorb-P, analytical + preparative, low density	3473
			LLDPE, sylanated glass beads	186
			Sand, preparative, low density	2973
	Turbidimetric titration	α-Chloronaphthalene/20% dimethyl phthalate		1413,3162
		Xylene/*n*-hexanol		1238
Poly(ethylethylene) *Poly(1-butene)*				
	Chromatography	Benzene/butanone	Precipitation chromatogr., 18–50°C	1179
		Dichlorobenzene	SEC, 125°C	1982
		Trichlorobenzene	SEC, styragel, 80°C, 135°C	2758,3418
	Extraction	Boiling ether	Tacticity	480
	Fractional precipitation	Cyclohexane/acetone	35°C	635,2841
		Cyclohexane/cyclohexanol-glycol (3/1)	115°C	3083
		Decane	Lowering temp.	1589
		o-Dichlorobenzene/ dimethylformamide		548,2263
		Petroleum ether/acetone		735
		Toluene/methanol	30°C, atactic	2261,2263
		Xylene/triethylene glycol	130°C	2974
	Fractional solution	Benzene/methanol	Extraction	843
Poly(hexadecylethylene) *(poly(1-octadecene)*				
	Chromatography	Trichlorobenzene	80°C	3415
	Fractional precipitation	Xylene/triethylene glycol	130°C	2974
Poly(1-hexene) see Poly(butylethylene)				
Poly(hexylethylene) *poly(1-octene)*				
	Chromatography	Trichlorobenzene	SEC, styragel, 80°C	3418
	Fractional precipitation	Cyclohexane/acetone		1682
		Xylene/triethylene glycol	130°C	2974

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(isobutene) see Poly(1,1-dimethylethylene)				
Poly(3-methyl-1-hexene)	Fractional solution	Acetone, ethyl acetate, diethyl ether, cyclohexane	Extraction	3387
Poly(4-methyl-1-hexene)	Fractional solution	Acetone, ethylacetate, diethyl ether, cyclohexane	Extraction	3387
Poly(3,7-dimethyl-1-octene) see Poly(1,5-dimethylhexyl-ethylene)				
Poly(3-methyl-1-pentene) see Poly(1-methylpropyl)ethylene)				
Poly(4-methyl-1-pentene) see Poly(2-methylpropyl)ethylene)				
Poly(endo, exo-5,6-dimethylbicyclo[2.2.1]hept-2-ene)				
	Chromatography	Tetrahydrofuran	SEC, Shodex	3095
Poly(2-methylbutyl)ethylene) *(poly-4-methyl-1-hexene)*				
	Chromatography	Alcohols, aromatic hydrocarbons, ethers	Column elution on active support	2603
		Methylcyclohexane	SEC, styragel, 60°C	2382
	Fractional precipitation	Benzene–nitrobenzene	Lowering temp. 61–25°C	1973,2382
	Fractional solution	Benzene and ethyl ether	Stereoisomers, extraction	2382
		Boiling (acetone, ethyl acetate, ethyl ether, isopropyl ether and cyclohexane)	Stereoisomers, extraction	331
Poly(1,1-dimethylethylene) *Poly(isobutylene)*				
	Chromatography	2-Methylheptane	Column elution Charcoal and colloidal silica	1959
		Benzene/acetone	Precipitation chromatogr., glass beads, 28–50°C	500,501,2648
		Carbon tetrachloride	SEC	620
		Chloroform	SEC, μ-styragel, 38°C	772
		Diisobutene/*n*-butanol	Precipitation chromatogr., 10–60°C	2513
		Heptane	SEC, styragel	2657
		Hexane/ethanol	Precipitation chromatogr., 10–60°C	2513
		Tetrachloromethane, chloroform, ethanol–chloroform	Column elution, 30°C silica gel	3583
		Tetrahydrofuran		965
			SEC μ-styragel	1389,1657,1663,2387,2679
			SEC, ultrastyragel	1595
			SEC, styragel	431,504,966,1655,2229, 2294,2398,2399,3570
			SEC, ultrastyragel	164
		Trichlorobenzene	SEC, 150°C	2650
		Xylene/*n*-propanol	Precipitation chromatogr., 25–70°C glass bead	2229
			Precipitation chromatogr., 10–60°C	2513
	Fractional precipitation	2,4,4-Trimethylpentene/ *n*-butanol		1959
		Benzene/acetone		322,323,644,645, 678,903,913,914, 928,1796,2043,2044
		Benzene/methanol	28°C	785,909,1492
		Chloroform–methylene chloride/acetone		2833
		Cyclohexane/butanone	25°C	2524
		Heptane/ethanol		373
		Liquid ethylene at 130 atm. releasing pressure	80°C	3298
		Toluene/methanol		184,2664,3126
	Fractional solution	Benzene (theta technique)	Column elution, 24.5°C	2086
		Benzene, benzene/methanol	Column elution	2353
		Hexane/ethanol	Column elution, 25°C	3583
		Toluene, methyl ethyl ketone	Continous operation	1016
		Toluene/acetone mixtures	Extraction, 30°C	322
		Toluene/butanone, heptane/butanone	Continous operation	1017
		Toluene/methanol	Column elution, charcoal	324,1857

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Sedimentation velocity	2,2,4-Trimethylpentane		3410
	Thermal diffusion	*n*-heptane	Gradient 40–50°C	1833
	Turbidimetric titration	Butyl acetate, butyl ether, methoxymethanol, methyl propyl ketone, diethyl ketone, ethyl amyl ketone, heptane/*n*-butanol	Lowering temp.	2234
Poly(1,5-dimethylhexylethylene)				
	Chromatography	Acetone/ethyl ether, benzene, ethyl ether	20–78°C polymeric active support	2603
	Extraction	Boiling (acetone, ethyl acetate, ethyl ether, diisopropyl ether, cyclohexane)		595
Poly(2-methylpropene)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	568
Poly(1-methylpropylethylene)	Chromatography	Acetone/ethyl ether, benzene	Column elution, 18–20°C, active support, acetone soluble fraction	2603
	Fractional solution	Boiling (acetone, ethyl acetate, and benzene)	Extraction	2602
Poly(2-methylpropylethylene)	Fractional precipitation	Benzene/nitrobenzene (3/1)	Lowering temp.	2382
		Decalin/ethyl cellosolve		3150
	Fractional solution	Benzene and ethyl ether	Stereoisomers, extraction	2382
Poly(nonylethylene) *Poly(1-undecene)*				
	Chromatography	Trichlorobenzene	SEC, styragel, 80°C	3418
Poly(octadecylethylene)	Fractional solution	Boiling ethyl ether	Extraction	1996
Poly(1-octenylene) *Poly(octenamer)*				
	Chromatography	Tetrahydrofuran	SEC, styragel, spherosil	126,1066
Poly(1-pentenylene) *Poly(pentenamer)*				
	Chromatography	*o*-Dichlorobenzene	SEC, styragel, 135°C	3608
		Tetrahydrofuran	SEC, styragel, spherosil	126,2832,3490
	Fractional precipitation	Toluene/methanol	40/20 (°C)	1044,3490
Poly(pentylethylene) *Poly(1-heptene)*				
	Chromatography	Trichlorobenzene	SEC, styragel, 80°C	3418
Poly(propylene)	Chromatography	1,2,4-Trichlorobenzene		608
			SEC	3541
			SEC, μ-styragel, 135°C	1907
			SEC, μ-styragel, 145°C	1925
			SEC, ultrastyragel, 140°C	2670
			SEC, ultrastyragel, 135°C	824
		1,2-Dichlorobenzene	SEC, μ-styragel, 135°C	185
		Cyclohexane	SEC, μ-styragel, 70°C	3561
		Isopropyl ether	High MW and crystallinity	2367
		Ligroin/2-butoxyethanol	Precipitation chromatogr., 75–170°C	3015
		o-Dichlorobenzene	SEC	1634
			SEC, styragel, 135°C	767,768,769,771, 1588,1638,1946, 1982,1994,2434, 2760,2789, 3365,3386
			SEC, styragel, 140°C	2216
			SEC, styragel, preparative 135°C	148
		Petroleum ether/methanol	Silica gel, 0–48°C, low MW	1026
		Tetralin	Column elution increasing temp. 20–150°C	612
			SEC, 130–145°C	1853
		Tetralin/2-butoxyethanol	Precipitation chromatogr, 140–180°C	611
		Tetralin/diethylene glycol monobutyl ether (mixtures)	Column elution, 165°C according to MW	612
		Tetralin/diethylene glycol monomethyl ethyl ether	Column elution, 145°C	2265
		Trichlorobenzene	SEC, 135°C,130°C,150°C	607,673,1774, 2487,2488,3377

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Trichlorobenzene/dimethyl phthalate	Precipitation chromatogr. temp. grad. 55–125°C glass beads	169
	Coacervation	*o*-Dichlorobenzene/triethylene glycol	Isotactic	191
		Tetralin/poly(oxyethylene)	140°C	2720
	Extraction		17 Hydrocarbon fractions with increasing b.p. from 35 to 130°C	2337
		Acetone, ethyl ether, *n*-pentane, *n*-hexane, *n*-heptane, *n*-octane	Isotactic	1956,2201,2372,2797
		Boiling (acetone, ethyl ether, *n*-hexane, *n*-heptane	Amorphous and crystalline	958
		Boiling (heptane, xylene)	Tacticity	536
		Boiling diethyl ether, hexane, heptane	Tacticity	479
		Boiling heptane, octane	Tacticity	213
		Boiling hexane, heptane, toluene	Syndiotactic	185
		Boiling xylene	Tacticity	536
		Boiling xylene, octane	Tacticity	535
		Boiling (ethyl ether, *n*-heptane, methyl cyclohexane)	Amorphous and crystalline	3351
		Different solvents	Different temp.	1539
		Ethyl ether, *n*-hexane, *n*-heptane	Syndiotactic	2438
		Heptane	Boiling	1970,3032,3252
			Hot and cold	892
		Heptane, diethyl ether, hexane/heptane	Kumagawa extraction	478
		Heptane, *n*-octane, xylene	Tacticity	2536
		Hexane	Boiling	2751
		n-Octane, *n*-heptane, *n*-hexane, *n*-pentane	Stereoregular separation	2539
		Xylene/ethylene glycol monoethyl ether		2674
	Fractional crystallization	1,2,4-trichlorobenzene	CRYSTAF, isotactic + sindiotactic mixture	2236
		n-Heptane	Stirred, opt. act. (amorphous and crystalline)	2500
		o-Xylene		341
		Octane	Lowering temp.	2539
		Perchloroethylene	70–90°C, isotactic	2560
		Xylene	Stereoregular separation	1550
	Fractional precipitation	Benzene/acetone		906
		Benzene/methanol	Atactic	293,1679
		Cyclohexane/acetone	Atactic	1681,2252
		Cyclohexane/isopropanol		124
		Cyclohexanone/ethylene glycol or dimethyl phthalate	Isotactic, 130–135°C	1681
		Decalin/phenol–decalin/ propylene glycol	130°C, nitrogen atm.	1870
		o-Dichlorobenzene/benzyl alcohol		2264
		Trichlorobenzene/dimethyl phthalate	140°C	169
		Xylene/poly(oxyethylene)	112°C,134°C	704,1557
		Xylene/triethylene glycol	130°C	2974
	Fractional solution	17 Hydrocarbon fractions with increasing b.p. from 35 to 130°C		
		Acetone, diethyl ether, *n*-pentane, *n*-hexane, *n*-heptane	Extraction	1940

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Acetone, diethyl ether, *n*-pentane, *n*-hexane, *n*-heptane	Extraction, anisotactic	2752
		Butyl cellosolve/diethyleneglycol monobutyl ether	165°C	1360
		Decalin	Rising temp.	2435
		Decalin/2-ethoxyethanol	Column extraction,110°C	1416
		Decalin/acetophenone	Column elution,146.5°C	2556
		Decalin/benzyl alcohol	Column elution,152°C	2556
		Decalin/diethylene glycol monobutyl ether	Column elution,161°C	2434
		Decalin/ethyl carbitol	Column elution,160°C	2435
		Diethyl ether, pentane heptane, aliphatic hydrocarbons, decalin	Successive extraction	1907
		Ether, hexane, heptane		609
		Ethylene glycol monobutyl ether/diethylene glycol monobutyl ether	Gradient extraction several temp.	1907
		Hexane, *n*-decane		1684
		Kerosene/diethylene glycol monomethyl ether	Column extraction,150°C	3350,3467
		Kerosene/diethyleneglycol monobutyl ether	Column elution,155°C,160°C	1420,2433
		Kerosene/ethylene glycol (10%) in 2-butoxyethanol	Column extraction 156°C, isotactic	673,872,2983
		Nine solvents of increasing solvent power and boiling point	According to crystallinity	2385
		o-Dichlorobenzene/diethylene glycol	Column elution, 168°C, 170°C and 172°C	704,1329,2137,2649
		o-Dichlorobenzene/diethylene glycol monobutyl ether	Column elution,166°C	3517,3518
		o-Dichlorobenzene/dimethyl phthalate	Column elution,150°C	2933
		Tetralin	Column extraction, increasing temp.	3016
		Tetralin/2-butoxyethanol	Column extraction	3146
		Tetralin/2-ethoxyethanol	Column extraction,170°C	2519
		Tetralin/diethylene glycol monomethyl ether–ethylene glycol (75%–25%)	Column extraction,180°C, isotactic	1312
		Tetralin/dimethyl phthalate	Column extraction,155–178°C	2161
		Xylene/2-ethoxyethanol	165°C	1359
		Xylene/ethylene glycol monoethyl ether	Gradient extraction, several temp.	1907
		Xylene/ethylene glycol monoethyl ether, ethylene glycol monobutyl ether/diethyleneglycol monobutyl ether	Temperature (121–128°C) and (159–161°C)	536
		Xylene/methanol	Column extraction, 56°C, atactic	2649,2983
		Xylene/poly(oxyethylene)	Coacervate extraction,134°C	704
	TREF	Trichlorobenzene	140–30°C	3026
		Xylene	20–130°C, column extraction	1536
			25–130°C, glass beads	2216
	Turbidimetric titration	Benzene/methanol	30°C, atactic	1522
		Chloroform/isopropanol	30°C, atactic	1522
		Tetralin/butyl cellosolve	110°C	3145,3146
Poly(propylethylene)	Fractional precipitation	Toluene/methanol	30°C, isotactic	696,2262
Poly(tetradecylethylene) *Poly(1-hexadecene)*	Chromatography	Trichlorobenzene	SEC, styragel, 80°C	3418
Poly(undecylethylene) *Poly(1-tridecene)*	Chromatography	Trichlorobenzene	SEC, styragel, 80°C	3418
Poly(vinylcyclohexane) see Poly(cyclohexylethylene)				

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
1.3. POLY(ACRYLIC ACID) AND DERIVATIVES				
Poly(acrylamide)	Chromatography	Sodium sulfate 0.5 M	SEC, aquagel-OH	2121
	Electron microscopy	Water/*n*-propanol (dispersant agent)		2686
	Fractional precipitation	Formamide	SEC	2465
		Formamide/water (1/15)	SEC, surface modified silica	336
		Sodium nitrate (water) 0.5 N	SEC	752
		Sodium sulfate (water) 0.1 N	SEC	1703
		Water	SEC, TSK-gel, Sephadex	1224,2361
		Water/acetone–methanol		3500
		Water/isopropyl alcohol		3375
		Water/methanol	30°C, 25°C	252,253,336,805, 2199,2465,2910
			Hydrolyzed, decreasing temp.	2920
	Sedimentation velocity	Water		1993
Poly(acrylic acid)	Chromatography	Dioxane	Continuous fractionation (counter current extraction) > 40°C	1177
		Sodium chloride (water) 0.3 N	SEC, TSK-gel sodium salt	1616
		Water	SEC	1636
	Electrophoresis	Sodium acetate buffer, pH = 5.0		560
	Extraction	Water/isopropanol	Partially neutralizated	868
	Extraction (CPF)	Dioxane		1177
	Fractional precipitation	Methanol/ether		1680,1708
		Methanol/ethyl acetate	20°C	3255,3263
		Methanol/water	Θ-point phase as a sodium salt	2176
		n-Propanol/1,2-dichloroethane		3046
	Fractional solution	Dioxane	CPF	1177
		Dioxane–water	Extraction	1848
Poly(acrylic acid), sodium salt	Field flow	Water + 0.02 M triethanolamine + 0.1 M potassium chloride		1046,1047,1048,1049, 1050,1051,1052
	Fractional precipitation	Sodium hydroxide 0.4 N (water)/ methanol–water–sodium hydroxide 2.2 N	Stepwise lowering temp. 40–20°C	2809
Poly(acrylic anhydride)	Chromatography	Tetrahydrofuran	SEC, cyclocopolymer	2083
Poly(*N*-acryloyl-*o*-aminobenzoic acid)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2532
	Fractional precipitation	Tetrahydrofuran/petroleum ether	30°C	2532
Poly(*N*-acryloyl-*N'*-methyl piperazine)				
	Chromatography	Tetrahydrofuran		3506
Poly[4-(acryloyloxy)propoxy)-4'-cyanobiphenyl]				
	Chromatography	Chloroform	SEC, styragel	2839
Poly(benzyl acrylate) chromium tricarbonyl				
	Chromatography	Tetrahydrofuran	SEC	2611
Poly(*n*-butyl acrylate)	Chromatography	Butyl acetate	SEC, styragel, 70°C	3501
		Methylene chloride	SEC	1928
		Tetrahydrofuran	SEC	826
			SEC, ultrastyragel	1980
	Fractional solution	Nitrous oxide	Supercritical	2882
Poly(*tert*-butyl acrylate)	Chromatography	Tetrahydrofuran	SEC, star-shaped molecules	3245
	Fractional precipitation	Methanol/methanol–water	25°C	1498
Poly(*n*-butyl α-bromoacrylate)	Chromatography	Tetrahydrofuran		2852
Poly(4-(butylphenylazoxy)phenyl acrylate) *Poly(n-butyl-p-acryloyl azoxy benzene)*				
	Fractional precipitation	Chloroform/*n*-heptane		2544
Poly(cholesteryl acrylate)	Fractional precipitation	Benzene/acetone		3271
Poly(*n*-decyl acrylate)	Fractional precipitation	*n*-heptane/acetone		857,3269
Poly(diacetone acrylamide) *Poly(N-(1,1-dimethyl-3-oxobutyl)acrylamide)*				
	Fractional precipitation	Methanol/water		1502
Poly(ethyl acrylate)	Field flow	Water + FL-70 tensoactive	Sedimentation, cyclical-field, spheres	1903
	Fractional precipitation	Acetone/methanol		2019
		Acetone/water–methanol (1:5)	30°C	1941,1942,1943
		Butanone/methanol–water	30°C	3056
	Turbidimetric titration	Chloroform/*n*-hexane		1991
Poly(ethyl-α-bromoacrylate)	Chromatography	Tetrahydrofuran		2852

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(2-ethyl-*n*-hexyl acrylate)	Fractional precipitation	Tetrahydrofuran/ethanol-*n*-hexane (2/1)	25°C	3442
Poly[(*N*-ethylcarbazol-3-yl) methyl acrylate]				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3000,3001
Poly(ferrocenyl ethyl acrylate)	Chromatography	Tetrahydrofuran	SEC	2612
Poly(ferrocenyl methyl acrylate)				
	Chromatography	Tetrahydrofuran	SEC	2609,2610
Poly(fluoroalkyl-2-trifluoromethyl acrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2362
Poly(*N*-formyl acrylamide)	Chromatography	Dimethylformamide	SEC, vinyl acetate copolymer gel (Merckgel)	3225
Poly(1,1-dihydroperfluorobutyl) acrylate				
	Fractional precipitation	Benzenetrifluoride/methanol		2714
Poly(2-hydroxyethyl acrylate)	Chromatography	Tetrahydrofuran	SEC, styragel, telomer	402
Poly(*N*-isoproplyacrylamide)	Chromatography	Methanol		2124
	Fractional precipitation	Ethanol/diethyl ether		3207
Poly(isopropyl acrylate)	Fractional precipitation	Benzene/*n*-hexane	25°C, atactic	2042
		Chlorobenzene/*n*-hexadecane	110–115°C, stereoregular	2042
Poly(isopropyl-α-bromoacrylate)				
	Chromatography	Tetrahydrofuran		2852
Poly(*N*-isopropylacrylamide)	Chromatography	Dimethylpirrolidone	SEC, μ-bondagel	2864
	Fractional precipitation	Acetone/*n*-hexane		950
	Phase separation	Water-fed acetone/*n*-hexane		2445
Poly[4′-methoxyphenyl-4-((acryloxyloxy)propoxy)benzoate)]				
	Chromatography	Chloroform	SEC, styragel	2839
			SEC, styragel, analytical, preparative	1408
Poly(*trans*-11{4-[5-(4-methoxyphenyl)-1,3-dioxan-2-yl]-2,6-dimethylphenoxy}undecyl acrylate]				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	1381
Poly(methyl acrylate)	Fractional precipitation	Acetone/petroleum ether		1136
		Acetone/water–methanol		1167,1958
		Benzene/methanol		2059
		Benzene/petroleum ether		2001
		Butanone/methanol		1773
		Liquid ethylene at 130 atm and 80°C	Releasing pressure	3298
		Tetrahydrofuran/methanol–water (1 : 1)	25°C	3442
	Fractional solution	Chlorodifluoromethane	Supercritical fluid fractionation	2127
	Turbidimetric titration	Toluene/methanol		3398
Poly(methyl α-bromoacrylate)	Chromatography	Tetrahydrofuran	SEC, styragel	159,2852
Poly(methyl α-chloroacrylate)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	159
	Fractional precipitation	Acetonitrile/methanol		1267
Poly(*N*-(1,1,-dimethyl-3-imidazolylpropyl acrylamide)				
	Fractional precipitation	Methanol/mesitylene		1207
Poly(methyl α-(trifluoromethyl)acrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 30°C, 40°C	1446
Poly(*N*,*N*-dimethylacrylamide)	Chromatography	Dimethylformamide + lithium bromide 0.1%		3042
		Tetrahydrofuran	SEC, ultrastyragel, 25°C	3506
Poly(2-methylsulfinyl ethylacrylate)				
	Fractional precipitation	Chloroform/toluene	30°C	1335
Poly(1-(2-methylsulfinyl) ethoxycarbonyl)ethylene)				
Poly(*n*-octadecyl acrylate)	Chromatography	Tetrahydrofuran		2964
Poly(1-oxyl-2,2,6,6-tetramethyl-4-piperidyl) acrylamide				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 22°C	1582
Poly(1-oxyl-2,2,6,6-tetramethyl-4-piperidyl) acrylate				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 22°C	1582
Poly(1,2,2,6,6-pentamethyl-4-piperidyl) acrylate				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 22°C	1582
Poly(*n*-pentyl α-bromoacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2852
Poly(phenyl acrylate)	Fractional precipitation	Toluene/methanol	28°C	1347

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(4-phenylazoxyphenyl acrylate)				
	Fractional precipitation	Chloroform/*n*-heptane		2544
Poly(*n*-propyl acrylate)	Fractional precipitation	Acetone/methanol		2019
Poly(*n*-propyl α-bromoacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2852
Poly(3-sulfonylpropane methacrylamide)				
	Fractional precipitation	Ethanol/*n*-heptane		894
1.4. POLY(METHACRYLIC ACID) AND DERIVATIVES				
Poly(alkylphenyl methacrylates)s (General)				
	Fractional precipitation	Benzene/methanol		2690
Poly(6-(anisyloxycarbonylphenoxy)hexylmethacrylate)				
	Fractional precipitation	Trichloromethane/methanol		3053
Poly(benzyl methacrylate)	Fractional precipitation	Butanone/methanol		2749
Poly(2-biphenyl methacrylate)	Fractional precipitation	Benzene/methanol		38
Poly(*p-tert*-butylcyclohexyl methacrylate)				
	Fractional precipitation	Dioxane/methanol		990
Poly(*n*-butyl methacrylate)	Chromatography	Benzene	SEC, styragel	1263
		Tetrahydrofuran	SEC	690
			SEC, styragel	2824
		Tetralin	SEC, styragel, 75°C	2975
		Toluene		261
	Fractional precipitation	Acetone/methanol		393,1489,1605, 2628,2994
			Lowering temp.	581
		Acetone/water		3348
		Benzene/methanol	SEC, styragel	1519,2419
		Toluene/methanol		2993,2997
	Thermal diffusion	Benzene		1724
Poly(*tert*-butyl methacrylate) *Poly(1-tert-butoxycarbonyl)-1-methylethylene*				
	Chromatography	Tetrahydrofuran	SEC	3109
			SEC, styragel	1685
			SEC, μ styragel 30°C	3358
			SEC, ultrastyragel	1577
	Fractional precipitation	Acetone/methanol		1777
		Toluene/methanol, toluene/methanol + water		1577
	Sedimentation velocity	Butyl acetate	20–21°C	292
Poly(2-*tert*-butylphenyl methacrylate) *Poly(1-(2-tert-butyl phenoxyl carbonyl)-1-methylethylene))*				
	Fractional precipitation	Benzene/methanol		3232
Poly(*p-tert*-butylphenyl methacrylate)				
	Fractional precipitation	Acetone/2-propanol		990,991
Poly(carbethoxyphenyl methacrylamide) *[Poly(1-(Carbethoxyphenylimino carbonyl)-1-methylethylene)]*				
	Sedimentation velocity	Ethyl acetate	20°C	923
Poly(4-cetyloxycarbonylphenyl methacrylate)				
	Fractional precipitation	Benzene/methanol		3275
Poly(chloresteryl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC	1377
Poly(2-chlorophenyl methacrylate)				
	Fractional precipitation	Benzene/isopropanol		2396
Poly(4-chlorophenylmethacrylate) *[Poly(1-4-chlorophenyloxycarbonyl) 1-methylethylene]*				
	Fractional precipitation	Chloroform/methanol		3231
Poly(cyclohexyl methacrylate)	Chromatography	Toluene	SEC, styragel, 80°C	1180
	Fractional precipitation	Dioxane/methanol	25°C	1180
			Lowering temp.	2144
		Tetrahydrofuran/methanol		1605
Poly(1-(2-*N*-cyclohexyl-*N*-methyl-carbamoyloxy) ethyl) methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 28°C	2668
Poly(cyclohexyl thiolmethacrylate)				
	Fractional precipitation	Toluene/methanol		1186
Poly(decahydro-β-naphthyl methacrylate)				
	Fractional precipitation	Dioxane/methanol	25°C	1181
Poly(*n*-decyl methacrylate)	Fractional precipitation	Toluene/methanol		3599

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(*n*-docosyl methacrylate)	Fractional precipitation	Toluene/methanol		3599
Poly(dodecyl methacrylate)	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	2285
	Fractional precipitation	Toluene/methanol		2285
Poly(2,3-epoxypropyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, silica gel, 25°C	2295
	Fractional precipitation	Benzene/methanol		3232
	Sedimentation velocity	Heptane–acetone	Theta-solvent	1544
Poly(2-(*N*,*N*-diethyl carbamoyloxy)-2-methyl-ethyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2668
Poly(ethyl methacrylate) *[Poly-(1-(ethoxycarbonyl)-1-methyl ethylene)]*				
	Chromatography	Tetrahydrofuran	SEC, styragel	2824
	Fractional precipitation	Acetone/acetone + water system		351,1608
		Acetone/acetone–water (4 : 1)	Lowering temp.	582,1583
		Benzene/methanol		2459
		Benzene/*n*-hexane		3362,3363
		Benzene/petroleum ether		2628
		Methanol/water, chloroform/methanol		2460
Poly(ethyl methacrylate)-d_{10}	Chromatography	Tetrahydrofuran	SEC, styragel predeuterated	1226
Poly(1-(2-(*N*,*N*-diethylcarbamoyloxy)ethylmethacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 28°C	2668
Poly(*N*-ethylcarbazol-3-yl)methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3000
Poly(ethylene glycol methacrylate)				
	Fractional precipitation	Ethanol/methanol (0.7/1), *n*-heptane–benzene (1 : 1)	30°C	369,372,799
		Ethanol/*n*-heptane–benzene (1 : 1)	30°C	369
Poly(ferrocenylethyl methacrylate) *(Poly-(1-(ferrocenylethoxycarbonyl)-1-methylethylene))*				
	Chromatography	Tetrahydrofuran	SEC	2612
Poly(ferrocenylmethyl methacrylate) *(Poly-(1-(ferrocenylmethoxycarbonyl)-1-methylethylene))*				
	Chromatography	Dimethylformamide	SEC, styragel	2615
		Tetrahydrofuran	SEC	2609,2610,2614, 2615,3247
Poly(11-(hexadecyldimethylammonio)-undecyl methacrylate)				
	Chromatography	Chloroform + 0.005 M tetrabutylammonium bromide	SEC, μ-styragel and ultrastyragel	376
Poly(*n*-hexyl methacrylate) *(Poly(1-(hexyl-oxycarbonyl)-1-methylethylene)*				
	Fractional precipitation	Acetone/ethanol	Lowering temp.	583
Poly(2-hydroxyethyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	1122
	Fractional precipitation	Benzene : ethanol (1 : 1)/*n*-heptane	30°C	910
		Methanol/*n*-heptane–benzene (1 : 1)		1496
Poly(*N*-(2-hydroxyethyl)phthalimido methacrylate) *[Poly(1-(phthalimedoethoxycarbonyl)-1-methylethylene)]*				
	Fractional precipitation	Chloroform/petroleum ether	20°C	3595
Poly(*N*-(2-hydroxypropyl) methacrylamide)				
	Chromatography	Tris buffer solution (pH = 8.3 in 0.5 M sodium chloride)	SEC, Sepharose	3088
	Fractional precipitation	Methanol/acetone	40°C	374
Poly[1-(5-indanyloxy carbonyl)-1-methylethylene)] *Poly(5-indanyl methacrylate)*				
	Fractional precipitation	Toluene/ethanol		998
Poly(isobornyl methacrylate)	Fractional precipitation	Toluene/methanol		1187
Poly(isobutyl methacrylate) *(Poly-(1-(isobutoxycarbonyl)-1-methylethylene))*				
	Fractional precipitation	Acetone/methanol	30°C	3334,3335,3574
		Toluene/methanol		2998
Poly(isooctyl methacrylate) *(Poly-(1-(isooctyloxycarbonyl)-1-methylethylene))*				
	Sedimentation velocity	Heptane/acetone	Theta-solvent	1544
Poly(5-*p*-menthyl methacrylate) *(Poly(1-(5-p-menthyloxycarbonyl)-1-methylethylene))*				
	Fractional precipitation	Benzene/methanol		3232
Poly(methacrylamide)	Fractional precipitation	Formamide/methanol	25°C	3211
Poly(methacrylic acid) *[poly(1-carboxy-1-methylethylene)]*				
	Electrophoresis	Water (sodium phosphate, dibasic)	Stereoregular separation	162

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Fractional precipitation	Methanol/butyl acetate–2-pentanone (10 : 1)		1361
		Methanol/diethyl ether		3012
		Methanol/ethyl acetate + acetic acid		115,335,929, 2377,2378
		Methanol/ethyl ether	25°C	904,1598,2153
		Methanol/methyl isobutyl ketone		3464
	Sedimentation equilibrium	Methanol		517
Poly-*N*-[4-(methacryloyloxy)phenyl]-2-(4-methoxy phenyl)acetamide				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 25°C	971
Poly(2-methoxyethyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, silica gel	3388
	Fractional precipitation	Toluene/ethanol	27°C	3074
Poly-*N*-[4-[[4-methoxyphenyl)acetyl]oxyphenyl]methacrylamide]				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 20°C	971
Poly[*trans*-11{4-[5-(4-methoxyphenyl)-1,3-dioxan-2-yl]-2,6-dimethylphenoxy}undecyl methacrylate]				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	1381
Poly(4-methoxyphenyl methacrylate) *(Poly(1-(methoxyphenoxycarbonyl)-1-methylethylene)*				
	Chromatography	GPC		3454
Poly(2,2-dimethyl-1,3-dioxolan-4-yl)methyl methacrylate				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	2274
Poly(methyl methacrylate) *Poly(1-(methoxycarboonyl)-1-methylethylene)*				
	Chromatography	1,2,4-Trichlorobenzene	SEC, styragel, 145°C	572
		2,2,2-Trifluoroethanol	SEC, 50°C	2671,2672
		2-Ethoxyethanol	SEC, silica gel	577
		Acetone	Activated carbon + super cel	601
			SEC	2158,2159
			TLC	2210
		Acetone/methanol	Precipitation chromatogr. silanated glass beads	1951
			Precipitation chromatogr., periodic temp. changes	2634,2637
		Acetonitrile/*n*-butyl chloride		91
		Benzene/cyclohexane, heptane/ethylacetate, cyclohexane/ethylacetate, benzene/cyclohexane/ethylacetate	ATLC, oligomers	356
		Benzene/methanol	Precipitation chromatogr.	215,710,1175
		Butanone/butanone–70% ethanol		3440
		Chloroform	SEC, preparative	3138
			SEC, silica gel, 25°C	2211
			SEC, styragel, 33°C	60,188,707,2094
		Chloroform/methanol	Precipitation chromatogr.	397
			TLC	1419
		Dimethylformamide	SEC, styragel, 70°C	357,3402
		Dioxane	ThFFF	1686
		Ethyl acetate	TLC	2210
		Methylene chloride	SEC, μ-styragel, 32°C	60
		Petroleum ether : tetrahydrofuran	HPLC, gradient	356
		Tetrahydrofuran		730,2811
			Perdeuterated, SEC, Zorbax	974
			SEC	59,60,167,229, 284,298,357,441, 758,1383,1387, 2103,2150,2151, 2601,2671,2873, 2874,3024,3109,3426
			SEC, silanated silica gel, 25°C	3079
			SEC, styragel, poreglass, 25°C	2801
			SEC, styragel, 25°C telechelic	2442
			SEC, μ-styragel, 25°C	1602
			SEC, μ-styragel, 25°C, isotactic and syndiotactic samples	1601

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
			SEC, μ-styragel, Isotactic (99%), syndiotactic (92%)	1606
			Syndiotactic	3439
		Tetrahydrofuran : triethylamino (9 : 1)	SEC, μ-styragel, 35°C	670
		Trichloroethylene	SEC, 23°C	3195
		Xylene	SEC, syndiotactic	304
	Field flow	Tetrahydrofuran	Thermal	1146
	Fractional precipitation	Acetone/Aq. acetone		251
		Acetone/methanol		368,373,605,916, 1467,1491,1603,1605, 2873,2999,3049,3521
		Acetone/methanol–water (1 : 1)		605,915
		Acetone/*n*-heptane		1318
		Acetone/*n*-hexane	Summative method	331,1718,2062
		Acetone/petroleum ether	20°C	851
		Acetone/shellsolve	Low MW	605,915
		Acetone/water		273,580,605,915, 949,1415,2905,3607
		Aq. acetone solutions	Lowering temp.	251
		Benzene–chloroform (3 : 1)/petroleum ether		2148
		Benzene/cyclohexane		850,851,1205,1679
		Benzene/isopropyl alcohol		1599,1600
		Benzene/methanol		166,303,727,953, 1175,1466,1547,1604, 2317,2419,2533,2535, 2600,2770,2771,2772, 2773,3138,3606
			Isotactic	2036
			Isotactic (99%), syndiotactic (92%)	1606
			Isotactic and syndiotactic samples	1601
			Isotactic atactic	1602
		Benzene/*n*-hexane	25°C	74,909,949,1205, 1363,1785,1992,2250, 3256,3362,3363
		Benzene/petroleum ether	25°C	367,368,662,663, 2685,3137,3253
		Benzene/shellsolve		605,915,1786
		Butanone–ethanol/cyclohexane		3440
		Butanone/methanol		518,1773
		Butanone/*n*-hexane		518
		Carbon tetrachloride/shellsolve		605,915,916
		Chloroform/*n*-heptane	Lowering temp.	2987
		Chloroform/petroleum ether		132,3238
		Dioxane/*n*-heptane	30°C, imidized	3141
		Methyl ethyl ketone/methanol		589
		Toluene/methanol–water		2109
		Toluene/petroleum ether		640
	Fractional solution	Acetone/methanol	Column extraction	742,2635,2638,2906
		Acetone/*n*-hexane	Column extraction, 20°C glass beads, analytical and preparative	2302
		Benzene/cyclohexane	Column extraction	2906
		Benzene/methanol		2421
			Column extraction	215
		Butanone/cyclohexane : methanol		3440
		Chloroform/methanol	Column extraction	397
		Chloroform/petroleum ether	Film extraction	935
		Toluene/methanol, benzene/methanol		2350
		Toluene/petroleum ether		831
	Sedimentation velocity	Acetone	Ultracentrifuge	1680
		Ethyl acetate	Ultracentrifuge, 20°C	852,853

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Ethyl acetate/acetone	Ultracentrifuge, 20°C	1679
		Nitromethane	Ultracentrifuge, Θ-temp. 28°C	2769
	Thermal diffusion	Acetone		931
		Benzene	Plate type apparatus	1726,1727
		Toluene		1863,1868
	Turbidimetric titration	Acetone/water		58,1012,1219
		Benzene/cyclohexane		664
		Benzene/methanol		1991
		Butanone/water–methanol (3 : 1)		58
		Chloroform/methanol		1991
Poly(2-(4-methyl-1-piperazinylcarbonyloxy)ethylmethacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 26°C	2668
Poly(2-(4-methyl-1-piperazinylcarbonyloxy)-2-methylethylmethacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 28°C	2668
Poly(dimethylamino) ethyl methacrylate				
	Chromatography	Tetrahydrofuran : triethylamino (9 : 1)	SEC, μ-styragel, 35°C	670
Poly(1-(2-(*N*,*N*-dimethylcarbamoyloxy)ethylmethacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 28°C	2668
Poly(2-(*N*,*N*-dimethylcarbamoyloxy)-2-methylethyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 28°C	2668
Poly(*p*-methylphenyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC	3454
Poly(2,5-dimethylpyrazolyl methacrylate)				
	Fractional precipitation	Acetone/*n*-hexane	20°C	2960
Poly(2-morpholinocarbonyloxy)ethylmethacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 26°C	2668
Poly(1-naphthyl methacrylate)	Chromatography	*n*-Butyl acetate	SEC, bioglass	143
		Tetrahydrofuran	SEC, 40°C	2402
Poly(2-[(3,5-dinitrobenzoyl)oxy]ethyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3000,3001
Poly(1-(1-nonyloxy-4-phenoxycarbonyl)phenyl methacrylate)				
	Fractional precipitation	Benzene/acetone		476
Poly(octadecyl methacrylate)	Fractional precipitation	Toluene/methanol		2750
Poly(*n*-octyl methacrylate)	Chromatography	Tetrahydrofuran	SEC, styragel	2824
	Fractional precipitation	Toluene/ethanol		2628
Poly(3-(8-oxyl-7,7,9,9-tetramethyl-2,4-dioxo-1,3,8-triazaspiro(4,5)-dec-3-yl)propyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 22°C	1582
Poly(pentachlorophenyl methacrylate)				
	Fractional precipitation	Toluene/petroleum ether		259
Poly(1,2,2,6,6-pentamethyl-4-piperidyl methacrylate) *Poly(1-methyl-1-(1,2,2,6,6-pentamethyl-4-piperidyl)oxycarbonyl ethylene)*				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 22°C	1582
Poly(phenyl methacrylate)	Chromatography	Tetrahydrofuran/water	SEC, styragel	989
		Toluene	SEC, styragel, 80°C	1180
	Fractional precipitation	Acetone/isopropanol	25°C	1180
Poly(*N*-phenylmethacrylamide) *(Poly-1-(phenyliminocarbonyl)-1-methylethylene))*				
	Fractional precipitation	Acetone/benzene		2853
Poly(diphenylmethyl methacrylate)				
	Chromatography	Chloroform	SEC, μ-styragel	766
Poly(propionyloxyethyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	1496
Poly(propyl methacrylate)	Fractional precipitation	Sodium chloride 1.0 N (water)/ methanol–isopropanol (1 : 1) Sodium sulfonate salt		3143
Poly(*n*-stearyl methacrylate)	Fractional precipitation	Toluene/methanol		3598,3599
Poly(tetrahydrofuran methacrylate)				
	Chromatography	Tetrahydrofuran	SEC	954
Poly(tetrahydrofurfuryl methacrylate)				
	Fractional precipitation	Acetone/benzene	25°C	3575
Poly(4-(1,1,3,3-tetramethylbutyl)phenylmethacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	989
	Fractional precipitation	Benzene/methanol		2446
Poly(2-thiophene methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	993

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Fractional precipitation	Benzene/methanol		993
Poly(2,4,5-trichlorophenyl methacrylate) *Poly(1(2,4,5-trichlorophenoxycarbonyl)-1-methylethylene)*				
	Fractional precipitation	Benzene/methanol		1182,2396
Poly(*n*-tridecyl methacrylate)	Fractional precipitation	Toluene/methanol		3599
Poly(2-trimethylammonium ethyl methacrylate chloride)				
	Chromatography	Nitric acid (0.1 M) + sodium nitrate (0.1 M)/water		3080,3081
	Sedimentation velocity	Sodium chloride 0.1 N	25°C	3081
Poly(triphenylmethyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2453
Poly(trityl methacrylate)	Fractional precipitation	Hexamethyl phosphotriamide–benzene (40 : 60)/petroleum ether		2394
1.5. OTHER α- AND β-SUBSTITUTED POLY(ACRYLICS) AND POLY(METHACRYLICS)				
Poly(di-*n*-alkyl itaconates)	Fractional precipitation	Benzene/methanol	Poly dimethyl, diethyl, di-propyl itaconates	755
Poly(di-*n*-amyl itaconate) *[Poly(1-(n-amyl-oxycarbonylmethyl)-1-(n-amyloxycarbonyl)ethylene]*				
	Turbidimetric titration	Acetone/methanol	30°C	3368
		Benzene/methanol	30°C	3368
		Butanone/methanol	30°C	3368
		Chloroform/methanol	30°C	3368
		Ethyl acetate/methanol	30°C	3368
		Toluene/methanol	30°C	3368
Poly(di-benzyl itaconate)	Chromatography	Benzene/methanol	Precipitation chromatogr., glass beads, grad. temp.	3373
	Fractional precipitation	Isobutyl methyl ketone/methanol	25°C	3554
Poly(*tert*-butyl crotonate)	Chromatography	1,1,3,3,3,-hexafluoro-2-propanol	SEC, μ-styraged, analytical	3324
		Carbon dioxide/methylene dichloride	SFC, non-bonded silica gel, 160°C, 20 MPa.	3324
		Chloroform	SEC, styragel, preparative, oligomers	3324
	Fractional precipitation	Toluene/methanol		2408
Poly(di-*n*-butyl itaconate)	Chromatography	Benzene/methanol	Precipitation chromatogr., glass beads, grad. temp.	3370
		Tetrahydrofuran	SEC, ultrastyragel	1365
	Fractional precipitation	Benzene/methanol		1365
	Turbidimetric titration	Acetone/methanol	30°C	3368
		Acetone/water	30°C	3368
		Benzene/methanol	30°C	3368
		Butanone/methanol	30°C	3368
		Chloroform/methanol	30°C	3368
		Ethyl acetate/methanol	30°C	3368
		Toluene/methanol	30°C	3368
Cholesteryl vinyl succinate	Chromatography	Tetrahydrofuran	SEC	2424
Poly(dicyclohexyl itaconate)	Fractional precipitation	Toluene/methanol	25°C	3371
Poly(di-*n*-decyl itaconate) *Poly(1-(n-decyloxycarbonylmethyl)-1-n-decyloxycarbonyl)ethylene)*				
	Chromatography	Benzene/methanol	Precipitation chromatogr.	3370
	Turbidimetric titration	Benzene/methanol	30°C	3368
		Chloroform/methanol	30°C	3368
		Toluene/methanol	30°C	3368
Poly(mono-*n*-decyl itaconate)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2691
	Fractional precipitation	Butanol/methanol:water mixtures (2 : 1)		1912,2691
Poly[1-(decyloxycarbonylmethyl)-1-(methoxycarbonyl)ethylene] *Poly(methyl decyl itaconate)*				
	Fractional solution	Chloroform/ethanol:methanol (1 : 1)		1914
Poly(di-*n*-dodecyl itaconate) *Poly(1-(n-dodecyloxycarbonyl methyl)-1-n-decyl-oxycarbonyl)ethylene*				
	Fractional precipitation	Benzene/methanol		3371
Poly(mono-*n*-dodecyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2691

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Fractional precipitation	Butanol/methanol–water mixture (2 : 1)		1913
		Butanol/methanol : water (2 : 1)		2691
Poly(1,1-(diethoxycarbonyl)ethylene)				
	Chromatography	Chloroform	SEC	1460
	Fractional precipitation	Tetrahydrofuran/water	50°C	1460
Poly(ethyl cyanoacrylate)	Fractional precipitation	Methylene chloride/methanol		781
Poly(di-*n*-ethyl itaconate)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	1365
	Fractional precipitation	Benzene/methanol		1365
	Turbidimetric titration	Acetone/water, benzene/*n*-heptane, butanone/*n*-heptane, butanone/mineral spirits, ethanol/water	30°C	3368
Poly(mono-*n*-ethyl itaconate)	Fractional precipitation	Methanol–acetone (1 : 1)/diethyl ether		997
Poly(di-*n*-heptyl itaconate)	Turbidimetric titration	Benzene/methanol, butanone/methanol	30°C	3368
Poly(di-*n*-hexadecyl itaconate) *Poly(1-(n-hexadecyloxycarbonylmethyl-1-(n-hexadecyloxycarbonyl)ethylene*				
	Fractional precipitation	Benzene/methanol		3374
	Turbidimetric titration	Benzene/methanol, butanone/methanol, chloroform/methanol, ethyl acetate/methanol, toluene/methanol	30°C	3368
Poly(1-(*n*-diicosayloxy-carbonylmethyl)1-(diicosayloxycarbonyl)ethylene *Poly(diicosayl itaconate)*				
	Fractional solution	Benzene/methanol	Column elution	3374
Poly(methyl butacrylate)	Fractional solution	Acetone/methanol, benzene/*n*-hexane	Column elution	1456
Poly(methyl-*n*-decyl itaconate)	Chromatography	Tetrachloroethane–phenol (3:1)/*n*-hexanol	SEC, μ-styragel	2691
	Fractional precipitation	Butanol/methanol:water (2 : 1)		2691
Poly(methyl-*n*-dodecyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2691
	Fractional precipitation	Butanol/methanol:water (2 : 1)		2691
Poly(di-*n*-methyl itaconate)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	1365
	Fractional precipitation	Acetone/methanol		2674
		Benzene/methanol		1365
	Turbidimetric titration	Acetone/methanol	30°C	3368
		Acetone/water	30°C	3368
		Benzene/water	30°C	3368
		Benzene/*n*-hexane	30°C	3368
		Butanone/cyclohexane	30°C	3368
		Butanone/methanol	30°C	3368
		Butanone/*n*-hexane	30°C	3368
		Chloroform/methanol	30°C	3368
Poly(di-*n*-octadecyl itaconate) *Poly(1-(n-octadecyloxycarbonylmethyl-1-octadecyloxycarbonyl)ethylene)*				
	Fractional precipitation	Benzene/methanol		3374
Poly(di-*n*-octyl itaconate)	Turbidimetric titration	Benzene/methanol, butanone/methanol, chloroform/methanol, ethyl acetate/methanol, toluene/methanol	30°C	3368
Poly(di-*n*-phenyl ethyl itaconate)				
	Chromatography	Benzene/methanol	Precipitation chromatogr., glass beads, temp. grad. 18–40°C	3373
Poly(diphenyl itaconate)	Chromatography	Benzene/methanol	Precipitation chromatogr., glass beads, temp. grad. 20–40°C	3372
Poly(di(phenyl-*n*-propyl) itaconate) *Poly(1-phenylpropyloxycarbonylmethyl)-1-(phenylpropyloxycarbonyl)ethylene)*				
	Chromatography	Benzene/methanol	Precipitation chromatogr., glass beads, temp. grad. 18–40°C	3373
Poly(di-*n*-propyl itaconate)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	1365
	Fractional precipitation	Benzene/methanol		1365
	Turbidimetric titration	Acetone/water	30°C	3368

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Ethanol/water	30°C	3348
Poly(di-*n*-tetradecyl itaconate) *Poly(1-n-tetradecyloxycarbonylmethyl)-1-(tetradecyloxycarbonyl)ethylene)*				
	Fractional precipitation	Benzene/methanol		3374
Poly(di-*n*-undecyl itaconate)	Turbidimetric titration	Benzene/methanol	30°C	3368
		Chloroform/methanol	30°C	3368
1.6. POLY(VINYL ETHERS)				
Poly[5-[[[2′,5′-bis[4″-*n*-alkoxybenzoyl]oxy]benzyl]oxy]carbonyl]bicyclo[2.2.1]hept-2-enes				
	Fractional precipitation	Tetrahydrofuran/hexane		2676
Poly(allyloxyethylene) *Poly(allyl vinyl ether)*				
	Chromatography	Tetrahydrofuran	SEC μ-bondagel	2989
Poly(butoxyethylene)	Fractional precipitation	Benzene/methanol		2977
Poly(2-chloro-4-methylpenthyl 4′[[8-vinyl oxy]octyl]oxy]biphenyl-4-carboxylete)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C, (r,s)	2569
Poly(2-chloroethoxyethylene)	Chromatography	Chloroform	SEC, styragel	1300
Poly(hexadecyloxyethylene)	Fractional precipitation	*n*-Heptane/acetone		1760
Poly(isobutoxyethylene)	Chromatography	Chloroform	SEC, styragel	2212
		Tetrahydrofuran	SEC μ-styragel	2856
			SEC, styragel	1244,1827
	Extraction	Acetone, benzene, ethyl ether, methylene chloride	Oppanol C	1972
	Fractional precipitation	Toluene-butanone (1 : 1)/ethanol	Lowering temp.	2865
Poly(isobutyl propenyl ether) *[Poly(1-isobutoxy-2-methylethylene)]*				
	Chromatography	Chloroform	SEC, styragel	1299
Poly(isobutyl vinyl ether)	Chromatography	Chloroform		1597,1692
			SEC, styragel	1569
		Toluene, chloroform	SEC, styragel, analytical and preparative	584
Poly(isopropyloxyethylene)	Extraction	Acetone, benzene, ethyl ether, methylene chloride		1972
Poly 5-[[[*n*-[(4′-methoxy-4-biphenyl)oxy]alkyl]-oxy]carbonyl]bicyclo [2.2.1]hept-2-enes				
	Chromatography	Tetrahydrofuran	SEC, styragel	1744
Poly(methoxyethylene)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2857
	Fractional precipitation	Benzene/*n*-hexane, *n*-heptane or decane		2025
		Chloroform/heptane		247
		Methanol	Lowering temp. 65 to 25°C and to 3°C	3352
		Toluene/heptane		246
		Water	Raising temp.	2025
	Fractional solution	Toluene/petroleum ether	Continous extraction (CPF)	2587
Poly(vinyl isobutyl ether) see Poly(isobutoxyethylene)				
Poly(vinyl methyl ether) see Poly (methoxyethylene)				
1.7. POLY(VINYL ALCOHOL), POLY(VINYL KETONES), POLY(VINYL HALIDES), POLY(VINYL NITRILES)				
Poly(acrylonitrile)	Chromatography	Chloroform	SEC, styragel	1781
		Dimethylacetamide	SEC, styragel	330,1965
		Dimethylformamide	SEC	985,1846
			SEC, Merck gel	1461
			SEC, Sephadex	1781
			SEC, styragel, 70°C	1142
	Fractional precipitation	Dimethyl sulfoxide–aq. hydrochloric acid/toluene	35°C	1715
		Dimethyl sulfoxide/toluene	35°C	946,947,1414, 1554,1558,1716
		Dimethylformamide	110–90°C	2888
		Dimethylformamide/ethanol	pH 6.0, ageing period	2173
		Dimethylformamide/lauric alcohol		3071
		Dimethylformamide/ligroin	50–60°C	1111
		Dimethylformamide/*n*-heptane	60°C	267,339,383,594, 945,948,1371, 1626,2167

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(divinylbenzene-*co*-methacrylate)				
	Fractional precipitation	Benzene/methanol		563
Poly(divinylbenzene-*co*-vinylpyridene)				
	Chromatography	Benzene	SEC, styragel	2327
Poly(4-vinylpyridine-*co*-*tert*-butyl methacrylate)				
	Chromatography	Dimethylformamide/triethylamine/ pyridine (8 : 1 : 1)		669
Poly(vinylpyrrolidone-*co*-diamyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2692
Poly(vinylpyrrolidone-*co*-diethyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2692
Poly(vinylpyrrolidone-*co*-dimethyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2692
Poly(vinylpyrrolidone-*co*-dipropyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2692
Poly(vinylpyrrolidone-*co*-monomethyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2692
Poly(vinylpyrrolidone-*co*-monoamyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2692
Poly(vinylpyrrolidone-*co*-monoethyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2692
Poly(vinylpyrrolidone-*co*-monopropyl itaconate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2692
1.12. BLOCK COPOLYMERS				
Poly(acrylamide)-*block*-poly(methyl methacrylate)				
	Fractional precipitation	Water/methanol followed by water		3020
Poly(acrylamido-2-methylpropane-sulfonic acid)-*block*-poly(styrene)-*block*-poly(2-acrylamido-2-methyl propane-sulfonic acid)				
	Extraction	Hot cyclohexane, methanol–ethyl ether (5 : 3), benzene	Triblock	1752
Poly(acrylate)-*block*-poly(methyl methacrylate)				
	Fractional precipitation	Chloroform/methanol	25°C	25
Poly(acrylonitrile)-*block*-poly(methyl methacrylate)				
	Extraction	Acetone, dimethylformamide		2958
Poly(acrylonitrile)-*block*-poly(styrene)				
	Chromatography	Benzene	TLC, homopolymer sep.	2379
		Chloroform/dimethylformamide (18/2)	TLC, homopolymer sep.	2379
		Dimethylformamide	SEC, macroporous glass	2379
			SEC, styragel	2379
Poly(acrylonitrile)-*block*-poly(ethylene glycol)-*block*-poly(acrylonitrile)				
	Fractional precipitation	Dimethylformamide/water	Chemical separation	2326
Poly(γ-benzyl-L-glutamate)-*block*-poly(styrene)				
	Extraction	Cyclohexane, butanone–isopropanol (2 : 8)	Scrambled type	2011
		Dimethylformamide/formic acid, dimethylformamide/ cyclohexane	Homopolymer	783
	Fractional precipitation	Dimethylformamide–ethyl acetate–cyclohexane (1 : 2 : 3)/methanol		783
Poly(butadiene)-*block*-poly(acrylonitrile)				
	Fractional precipitation	Isooctane-isoamyl alcohol (1 : 1)	Lowering temp.	2955
Poly(butadiene)-*block*-poly(ethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1239
Poly(1,3-butadiene)-*block*-poly(2-isoprenylnaphthalene)				
	Chromatography	Tetrahydrofuran	SEC	846
Poly(1,2-butadiene)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, phenogel, 25°C	1022
Poly(1,4-butadiene)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, phenogel, 25°C	1022
Poly(butadiene)-*block*-poly(styrene)				
	Chromatography	Benzene	TLC, to separate homopolymer	2379
		Chloroform	SEC, styragel	3427

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Chloroform/carbon tetrachloride, tetrahydrofuran/methanol	TLC, silica gel gradient elution	1769
		Cyclohexane–chloroform	TLC, concentration gradient	780
		Methanol, acetone, butanone, ethyl acetate, tetrahydrofuran, chloroform, toluene, etc.	TLC, silica gel	1769
		o-Dichlorobenzene	SEC	83
			SEC, styragel, 135°C	2365
		Tetrahydrofuran	SEC	542,2795
			SEC, macroporous glass	2379,2834,3287
			SEC, styragel	1770,2379,2541
		Toluene	HP SEC	237
			SEC, 45°C	1788
			SEC, macroporous glass	2379
			SEC, styragel	2379
	Fractional precipitation	Benzene/butanone and cyclohexane/isooctane	Cross fractionation	3172
		Benzene/methanol		3558
	Fractional solution	Benzene	Extraction	2472
	Sedimentation velocity	*n*-Heptane	25°C	3047
		n-Octane/cyclohexane (3/1)	21°C	2627
Poly(butadiene)-*block*-poly(styrene-*co*-acrylonitrile)				
	Chromatography	Tetrahydrofuran	SEC	1801
Poly(1-butene)-*block*-poly(propylene)				
	Fractional solution	Tetraline	Column extraction, increasing temp.	630
Poly(butyl acrylate)-*block*-poly(methyl methacrylate)				
	Fractional solution	*n*-Butanol/methanol		1150
Poly(*n*-butyl acrylate)-*block*-poly(vinyl chloride)				
	Chromatography	Dimethylformamide	SEC, porous silica, 45°C	3442
	Fractional precipitation	Tetrahydrofuran/ethanol	25°C	3442
Poly(*tert*-butyl methacrylate)-*block*-poly(dimethylamino)ethyl methacrylate				
	Chromatography	Tetrahydrofuran:triethylamine (9 : 1)	SEC, μ-styragel, 35°C	670
Poly(*tert*-butyl methacrylate)-*block*-poly(oxyethylene)				
	Chromatography	Tetrahydrofuran	SEC	3109
			SEC, styragel, 25°C	3107
Poly(*n*-butyl methacrylate)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2011
	Extraction	Isopropanol, cyclohexane	Scrambled type	2011
Poly(*tert*-butyl methacrylate)-*block*-poly(styrene)				
	Chromatography	Trichloroethylene	35°C	714
Poly(*n*-butyl methacrylate)-*block*-poly(vinyl chloride)				
	Fractional precipitation	Tetrahydrofuran/methanol–acetone (9 : 1)	25°C	3442
Poly(*tert*-butyl-4-vinylbenzoate)-*block*-poly(*tert*-butyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC	3419
Poly(*tert*-butylacrylate)-*block*-poly(styrene)-*block*-poly(*tert*-butylacrylate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 30°C	2099
Poly(*p*-*tert*-butylstyrene)-*block*-poly(isobutylene)-*block*-poly(*tert*-butylstyrene)				
	Chromatography	Tetrahydrofuran	SEC	1667
Poly(*p*-chlorostyrene)-*block*-poly(*n*-hexadecyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2012
Poly(*p*-chlorostyrene)-*block*-poly(styrene)				
	Fractional precipitation	Benzene/methanol		2440
Poly(*p*-chlorostyrene)-*block*-poly(styrene)-*block*-poly(*p*-chlorostyrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2967
Poly(4-cyanostyrene)-*block*-poly(*tert*-butyl methacrylate)-*block*-poly(4-cyanostyrene)				
	Chromatography	Dimethylformamide	SEC	1437
Poly(cyclohexyl methacrylate)-*block*-poly(styrene)				
	Extraction	Acetone, cyclohexane	Scrambled type	2011
Poly(*n*-dodecyl methacrylate)-*block*-poly(ethyl methacrylate)				
	Fractional precipitation	Tetrahydrofuran/methanol		2952
		Tetrahydrofuran/methanol-2-propanol		2952
Poly(ethyl acrylate)-*block*-poly(vinyl chloride)				
	Fractional precipitation	Tetrahydrofuran/methanol		3442

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(ethyl methacrylate)-*block*-poly(*n*-hexyl methacrylate)				
	Fractional precipitation	Tetrahydrofuran/methanol–water		2952
Poly(ethyl methacrylate)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2011
	Extraction	Butanone/isopropanol (2 : 8), cyclohexane		2011
Poly(ethyl methacrylate)-*block*-poly(*N*-vinylcarbazole)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2012
Poly(ethylene)-*block*-poly(propylene)				
	Fractional precipitation	Cyclohexane/isopropyl alcohol		2554
Poly(ethylene-*co*-propylene)-*block*-poly(vinylcyclohexane)				
	Chromatography	Tetrahydrofuran	SEC, phenogel, 25°C	1022
Poly(ethylene)-*block*-poly(vinylcyclohexane)				
	Chromatography	Tetrahydrofuran	SEC, phenogel, 25°C	1022
Poly(ethylene-*co*-propylene)-*block*-poly(propylene)				
	Chromatography	*o*-Dichlorobenzene	SEC, shodex, 140°C	770
Poly(2-ethylhexyl acrylate)-*block*-poly(vinyl chloride)				
	Fractional precipitation	Tetrahydrofuran/ethanol–*n*-hexane (2 : 1)	25°C	3442
Poly(ferrocenylmethyl methacrylate)-*block*-poly(acrylonitrile)				
	Chromatography	Dimethylformamide, tetrahydrofuran	SEC, styragel	2615
Poly(ferrocenylmethyl methacrylate)-*block*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran, dimethylformamide	SEC, styragel	2615
Poly(hexadecyl methacrylate)-*block*-poly(*p*-methoxystyrene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2012
Poly(*n*-hexyl methacrylate)-*block*-poly(lauryl methacrylate)				
	Fractional precipitation	Benzene/methanol		29
Poly(*n*-hexyl methacrylate)-*block*-poly(methyl methacrylate)				
	Fractional precipitation	Benzene/methanol	20°C	29
Poly(dihydronaphthalene)-*block*-poly(styrene)				
	Extraction	Ethyl acetate, butanone		538
Poly(isobutene)-*block*-poly(indene)				
	Fractional solution	Butanone, *n*-hexane, methylene dichloride, dioxane	Extraction	2386
Poly(isobutene)-*block*-poly(indene)-*block*-poly(isobutene)				
	Fractional precipitation	Benzene-*n*-pentane/butanone		3495
Poly(isobutene)-*block*-poly(α-methylstyrene)				
	Chromatography	Tetrahydrofuran,	SEC, styragel	1655
	Extraction	*n*-Hexane, 1-nitropropane, *n*-pentane	Tris-stars	1659
Poly(isobutene)-*block*-poly(styrene)				
	Chromatography	Benzene/isopropanol, cyclohexane/iso-propanol	Precipitation chromatogr., thermal gradient	695
		Tetrahydrofuran	SEC, styragel	1653
	Extraction	Butanone, *n*-pentane, acetone		1655
		Butanone, *n*-pentane, *n*-heptane		1064
	Fractional precipitation	Benzene/isopropanol, cyclohexane/*n*-propanol		695
Poly(isobutyl methacrylate)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2011
	Extraction	Acetone, cyclohexane	Scrambled type	2011
Poly(isobutyl vinyl ether)-*block*-poly(isobutylene)				
	Chromatography	Tetrahydrofuran	SEC	1188
Poly(isobutyl vinyl ether)-*block*-poly(*p*-methoxystyrene)				
	Chromatography	Butanone	SEC	1297
Poly(isobutylene)-*block*-poly(amide)				
	Chromatography	Tetrahydrofuran	SEC	3584
Poly(isobutylene)-*block*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC	3129
Poly(isobutylene)-*block*-poly(α-methylstyrene)				
	Chromatography	Tetrahydrofuran	SEC	1927

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(isodecyl methacrylate)-*block*-poly(styrene)				
	Extraction	Isopropanol–butanone (8 : 2), cyclohexane	Scrambled type	2011
Poly(isoprene)-*block*-poly(*tert*-butylacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	3312
Poly(isoprene)-*block*-poly(2,6-dihydroxypropyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, 40°C	2273
Poly(isoprene)-*block*-poly(methyl methacrylate)				
	Fractional precipitation	Benzene/methanol		1163
Poly(isoprene)-*block*-poly(α-methylstyrene)-*block*-poly(isoprene)				
	Chromatography	Chloroform/*n*-hexane (7/10)	TLC	2379
		Toluene	SEC, styragel	2379
Poly(isoprene)-*block*-(novolac resins)				
	Turbidimetric titration	Acetone/water		1775
Poly(1,4-isoprene)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	*cis*, SEC, phenogel, 25°C	1022
Poly(isoprene)-*block*-poly(styrene)-*block*-poly(isoprene)				
	Chromatography	Tetrahydrofuran, chloroform	SEC, μ-styragel, end functionalized	2606
Poly(isoprene)-*block*-poly(styrene)-*block*-poly(isoprene)-*block*-poly(4-vinylbenzyldimethylamine)-*block*-poly(isoprene)				
	Chromatography	Tetrahydrofuran (5% *N,N*-dimethylbenzylamine)	SEC, styragel	2209
Poly(isoprene)-*block*-poly(4-vinyl pyridine)				
	Chromatography	Toluene	SEC, 70°C	1269
Poly(2-isoprophenylbenzoxazole)-*block*-poly(*tert*-butyl methacrylate)-*block*-poly(2-isoprophenylbenzoxazole)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	1438
Poly(2-isoprophenylbenzoxazole)-*block*-poly(styrene)-*block*-poly(2-isoprophenylbenzoxazole)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	1438
Poly(isopropyl methacrylate)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel scrambled type	2011
	Extraction	Cyclohexanol, cyclohexane, or acetonitrile, isopropanol, methyl isobutyl ketone, methyl isobutyl ketone/acetone		2011
Poly(4,4′-isopropylidenediphenyl carbonate)-*block*-poly(styrene)				
	Fractional precipitation	Chloroform/heptane, benzene/cyclohexane		736
		Chloroform/heptane, benzene/heptane	Multiblock	737
Poly(lauryl methacrylate)-*block*-poly(methyl methacrylate)				
	Fractional precipitation	Benzene/methanol, benzene/methanol–isopropanol (1 : 1), benzene/methanol–*n*-heptane (1 : 1)	20°C	29
Poly(methacrylic acid)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	1596
	Extraction	Benzene, followed by methanol		1596
Poly(*p*-methoxystyrene)-*block*-poly(cyclohexene oxide)				
	Chromatography	Carbon tetrachloride/tetrahydrofuran	TLC, silica gel, gradient	1152
	Turbidimetric titration	Benzene, butanone (3 : 1)/ethanol		1152
Poly(methyl acrylate)-*block*-poly(vinyl chloride)				
	Fractional precipitation	Tetrahydrofuran/methanol–water (1 : 1)	25°C	3442
Poly(methyl methacrylate)-*block*-poly(*tert*-butyl acrylate)				
	Chromatography	Tetrahydrofuran		3357
Poly(methyl methacrylate)-*block*-poly(*tert*-butyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC	2503
Poly(methyl methacrylate)-*block*-poly(α-chloroacrylate)				
	Fractional precipitation	Chloroform–acetone (1 : 2)/methanol		2462
	Fractional solution	Chloroform/acetone mixtures	Selective solution pf homopolymers, extraction	2462

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Turbidimetric titration	Dimethylformamide/*n*-dibutyl ether	20°C	2462
Poly(methyl methacrylate)-*block*-poly((dimethylamino)ethyl methacrylate)				
	Chromatography	Tetrahydrofuran : triethylamine (9 : 1)	SEC, μ-styragel, 35°C	670
Poly(methyl methacrylate)-*block*-poly(ethylene)-*block*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	3107
Poly(methyl methacrylate)-*block*-poly(α-methylstyrene)-*block*-poly(methyl methacrylate)				
	Chromatography	Chloroform	SEC, styragel, preparative	633
Poly(methyl methacrylate)-*block*-poly(*n*-octadecyl methacrylate)				
	Fractional precipitation	Benzene/methanol	20°C	29
Poly(methyl methacrylate)-*block*-poly(styrene)				
	Chromatography	Acetone/water	Precipitation chromatogr.	248
		Benzene, followed by chloroform–methanol (6 : 13)	TLC	2379
		Benzene, methanol	TLC, silica gel	3578
		Benzene/methanol	Precipitation chromatogr.	2233
			TLC	1570
		Carbon tetrachloride, butanone	TLC	1771
		Nitroethane/acetone	TLC, silica gel	1572
		Tetrahydrofuran	SEC	250,2011,2443
			SEC, silanized glass beads	193
	Extraction	Acetone, tetralin		2233
		Acetone, tetralin, cyclohexane, acetonitrile		3316
		Butanone/isopropanol (3 : 7), cyclohexane	Scrambled type	2011
		Cyclohexane, acetonitrile, benzene, methanol–acetonitrile (1 : 1)		2307
	Field flow	Cyclohexane, acetonitrile		2958
	Fractional precipitation	1-Chlorobutane/*n*-hexane		1771
		Benzene–acetone (1 : 1)/petroleum ether		3316
		Benzene–chlorobenzene (1 : 1)/petroleum ether		926
		Benzene–chlorobenzene (1 : 1)/methanol		165
		Benzene/cyclohexane		1355
		Benzene/methanol		1355,1415,2233
			Column extraction	472
		Butanone/diisopropyl ether	30°C	1784
		Chloroform/methanol		3020
	Turbidimetric titration	Acetone/water		2131
		Benzene/methanol	25°C	1415
		Butanone/isopropanol		1355
		Butanone/water–methanol (3 : 1)		2233
Poly(methyl methacrylate)-*block*-poly(styrene)-*block*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	777
Poly(methyl methacrylate)-*block*-poly(tetrafluoroethylene)				
	Extraction	Butanone, butanone/ethyl ether (1 : 5)		3444
	Turbidimetric titration	Acetone, water		2131
Poly(methyl methacrylate)-*block*-poly(vinyl acetate)				
	Turbidimetric titration	Acetone/water		2552
Poly(methyl methacrylate)-*block*-poly(vinyl chloride)				
	Chromatography	Dimethylformamide	SEC, porous silica, 45°C	3442
	Fractional precipitation	Tetrahydrofuran/methanol–acetone (9 : 1)	25°C	3442
Poly(2-methyl-2-oxazoline)-*block*-poly(styrene)				
	Extraction	Water, methanol, ethanol, diethyl ether, tetrahydrofuran, chloroform		2954

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(methyl vinyl ether)-*block*-poly(isobutyl vinyl ether)-*block*-poly(methyl vinyl ether)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2857
Poly(*p*-*N*,*N*-dimethylaminostyrene)-*block*-poly(methacrylic acid)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2284
Poly(2-*N*,*N*-dimethylcarbamoyloxyethyl methacrylate)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 28°C	2668
Poly(α-methylstyrene)-*block*-poly((dimethylamino)ethyl methacrylate)				
	Chromatography	Tetrahydrofuran : triethylamine (9 : 1)	SEC, μ-styragel, 35°C	670
Poly(α-methylstyrene)-*block*-poly(isobutyl vinyl ether)-*block*-poly(α-methylstyrene)				
	Extraction	2-Propanol		2855
Poly(α-methylstyrene)-*block*-poly(isobutylene)-poly(α-methylstyrene)				
	Chromatography	Tetrahydrofuran	SEC	1926
			SEC, μ-styragel	1656
	Extraction	*n*-Hexane, 1-nitropropane		1659
Poly(α-methylstyrene)-*block*-poly(isoprene)-*block*-poly(α-methylstyrene)				
	Chromatography	Toluene	SEC, 70°C	885
	Sedimentation analysis	Cyclohexane		885
		Tetrahydrofuran		885
Poly(α-methylstyrene)-*block*-poly(propylene sulfide)-*block*-poly(α-methylstyrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 45°C	2289
Poly(α-methylstyrene)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, ultra styragel, 25°C	1676
	Density gradient technique	Bromoform/benzene	Ultracentrifuge	2695
	Fractional precipitation	Benzene/methanol		1787
		Butanone–benzene (6 : 1)/methanol + 0.01% calcium chloride		165
Poly(4-octylstyrene)-*block*-poly(2,6-dihydroxypropyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, 40°C	2273
Poly(oxyethylene)-*block*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	3107
Poly(oxyethylene)-*block*-poly(styrene)				
	Chromatography	Ethyl acetate/methanol	TLC, cellulose F	3306
		Tetrahydrofuran	SEC, styragel, 40°C	2352,3306
Poly(phenyl methacrylate)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel scrambled type	2011
Poly(diphenyl methyl methacrylate)-*block*-poly(trityl methacrylate)				
	Chromatography	Chloroform	SEC, μ-styragel	766
Poly(propylene)-*block*-poly(ethylene-*co*-propylene)-*block*-poly(propylene)				
	Chromatography	*o*-Dichlorobenzene	SEC, shodex, 140°C	770
Poly(styrene)-*block*-poly(butadiene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, star shaped	1503
			SEC, 40°C	3572
	Cross-Fractionation	Tetrahydrofuran	SEC	1331
		Tetrahydrofuran/methanol	Turbidimetric titration	1331
	Fractional precipitation	Benzene/methanol	Star copolymers A2B	1405
Poly(styrene)-*block*-poly(butadiene)-*block*-poly(styrene)				
	Chromatography	GPC, spherical silica		3296
		Methylene chloride	SEC, μ-styragel	1919
		Tetrahydrofuran	SEC, silica gel	1031
			SEC, styragel	1166
		Toluene	SEC, 45°C	1788,2941
	Sedimentation analysis	Tetrahydrofuran/allyl alcohol	30°C	3295
Poly(styrene)-*block*-poly(butadiene)-*block*-(urea)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C star branched	84,318,2669
	Sedimentation velocity	Tetrahydrofuran/ethanol	25°C, several compositions	2669
Poly(styrene)-*block*-poly(butadiene)-*block*-poly(4-vinylpyridine)				
	Chromatography	Tetrahydrofuran	SEC, styragel	117
Poly(styrene)-*block*-poly(*tert*-butyl acrylate)				
	Chromatography	Tetrahydrofuran	SEC	3389
Poly(styrene)-*block*-poly(*tert*-butyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, phenogel, 25°C	2703
			SEC, styragel, 25°C	3006,3007
	Fractional precipitation	Tetrahydrofuran/methanol		3007

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(styrene)-*block*-poly(*tert*-butylacrylate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 30°C	2099
Poly(styrene)-*block*-poly(*tert*-butylmethacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	2703
Poly(styrene)-*block*-poly(*tert*-butylstyrene)-*block*-poly(styrene)				
	Chromatography	*n*-Hexane, butanone, dimethylformamide	TLC	327
Poly(styrene)-*block*-poly(*p*-chlorostyrene)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2967
Poly(styrene)-*block*-poly(2-cinnamoylethyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	1151
			SEC, styragel	3155
Poly(styrene)-poly(2,3-dihydroxypropyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, 40°C	2273
Poly(styrene)-*block*-poly(ethene-*co*-butene)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, μ-bondagel	650
	Fractional precipitation	Cyclohexane/1-propanol	25°C	650
Poly(styrene)-*block*-poly(ethylene-*co*-butene)				
	Fractional precipitation	Toluene : cyclohexane (3 : 1)/methanol		1195
Poly(styrene)-*block*-poly(ethylene/butylene)-*block*-poly(styrene)				
	Chromatography	4-methyl-2-pentane	SEC	3380
Poly(styrene)-*block*-poly(*p*-hydroxystyrene)				
	Chromatography	Tetrahydrofuran		3594
Poly(styrene)-*block*-poly(isoprene)				
	Chromatography	1,2-Dichloroethane	SEC, 25°C	2630
		Base lubricating oil	Hydrogenated, SEC, styragel, 50°C	2664
		Methylene chloride	Modified, SEC, α-styragel	1919
		Tetrahydrofuran	SEC, μ-styragel, 30°C	2605
			SEC, 42°C	661
			SEC, styragel	1062,2102
			Star copolymers, SEC, styragel	318
		Toluene	SEC (multiblocks)	1320,2266
			SEC, styragel, 35°C (multiblocks)	632
	Fractional demixing	Dimethylformamide/ methylcyclohexane	Hydrogenated, diblock	2629
	Fractional precipitation	Benzene/methanol	25°C	650
			One, two, three and four arms	2663
			Star copolymers A2B	1405
	Sedimentation analysis	Carbon tetrachloride/cyclohexane	Multiblocks	2598
	Turbidimetric titration	Acetone/methanol		1775
		Toluene/methanol		2870
Poly(styrene)-*block*-poly(isoprene)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 30°C	2605
			SEC, styragel	1166,1728
		Tetrahydrofuran, chloroform	SEC, μ-styragel, end functionalized	2606
		Toluene	Hydrogenated, SEC, styragel	2941
Poly(styrene)-*block*-poly(isoprene)-*block*-poly(4-vinylpyridine)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1977
Poly(styrene)-*block*-poly(2-isoprophenylbenzoxazole)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	1438
Poly(styrene)-*block*-poly(methacrylic acid)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	2703
	Fractional precipitation	Benzene/methanol		2016
Poly(styrene)-*block*-poly(metal methacrylate)				
	Fractional precipitation	Benzene/methanol	Ionomers	741
Poly(styrene)-*block*-poly(methacrylic acid)				
	Chromatography	Tetrahydrofuran	SEC, phenogel, 25°C	2703
	Sedimentation velocity	Dioxane : water (80 : 20)		2682
Poly(styrene)-*block*-poly([6-[4-(4-methoxyphenyl) phenoxy] hexylmethacrylate]				
	Chromatography	Tetrahydrofuran	SEC	3513
Poly(styrene)-*block*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, 40°C	3572

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Extraction	Cyclohexane, acetonitrile, benzene		2600
Poly(styrene)-*block*-poly(methyl methacrylate)-*block*-poly(*n*-butyl methacrylate)				
	Fractional precipitation	Tetrahydrofuran/petroleum ether		1253
Poly(styrene)-*block*-poly(methyl methacrylate)-*block*-poly(styrene)				
	Chromatography	Benzene/butanone	TLC, silica gel	1572
		Carbon tetrachloride/methyl acetate	TLC, silica gel	1572
		Dimethylformamide	SEC, macroporous glass	2379
		Nitroethane/acetone	TLC, silica gel	1572
		Toluene	SEC, styragel	2379
	Fractional precipitation	Benzene, chlorobenzene/ petroleum ether		1194
Poly(styrene)-*block*-poly((dimethylamino)ethylmethacrylate)				
	Chromatography	Tetrahydrofuran : triethylamine (9 : 1)	SEC, μ-styragel, 35°C	670
Poly(styrene)-*block*-poly(dimethylsiloxane)				
	Chromatography	Tetrachloroethylene	SEC, μ-styragel	802
	Fractional precipitation	Tetrahydrofuran/methanol	Macrocycles	3560
Poly(styrene)-*block*-poly(oxyadipoyloxy-1,4-dichlorophenyleneisopropylidene-1,4-dichlorophenylene)				
	Fractional precipitation	Methylene dichloride/*n*-hexane		1873
Poly(styrene)-*block*-poly(vinyl chloride)				
	Fractional precipitation	Tetrahydrofuran/methanol		1892,3442
Poly(styrene)-*block*-poly(4-vinyl-*N*-ethyl pyridinium bromide)				
	Sedimentation	Water–methanol–0.01 M lithium bromide	25°C	2945
	Sedimentation velocity	Water 0.1 M potassium bromide		2944
Poly(styrene)-*block*-poly[(3-vinyl phenyl)methoxy(-*tert*-butyldimethylsilane)]-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC	1310
Poly(styrene)-*block*-poly(*N*-vinylcarbazole)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2012
	Extraction	Cyclohexane, toluene	Scrambled type	2012
Poly(styrene)-*block*-poly(2-vinylpyridene)-*block*-poly(butadiene)				
	Chromatography	Tetrahydrofuran	SEC	3429
Poly(styrene)-*block*-poly(2-vinylpyridine)				
	Chromatography	Tetrahydrofuran		2909
	Fractional precipitation	Benzene/*n*-heptane		1131,1132
Poly(styrene)-*block*-poly(4-vinylpyridine)				
	Chromatography	Tetrahydrofuran	SEC, 40°C	1441,1442
		Tetrahydrofuran, dimethylformamide	SEC, styragel, 60°C (THF), 70°C (DMF)	113
	Fractional precipitation	Benzene/*n*-heptane		1133
	Thermal field flow	Tetrahydrofuran, dimethylformamide		114
Poly(styrene-*co*-maleic anhydride)-*block*-poly(butadiene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	1802
Poly(2-vinyl naphthalene)-*block*-poly(vinyldiphenyl anthracene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3043
Poly(vinylcyclohexane)-*block*-poly(ethyl ethylene)				
	Chromatography	Tetrahydrofuran	SEC, phenogel, 25°C	1022
Poly[2-(4-vinylphenyl)-4,4-dimethyl-2-oxazoline]-*block*-poly(styrene)-*block*-poly[2-(4-vinylphenyl)-4,4-dimethyl-2-oxazoline]				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	1309
Poly(4-vinylpyridine)-*block*-poly(styrene)-*block*-poly(4-vinylpyridine)				
	Chromatography	Tetrahydrofuran	SEC, 40°C	1441,1442
Poly(4-vinylpyridine)-*block*-poly(2-vinylpyridine)				
	Fractional precipitation	Tetrahydrofuran–methanol/ *n*-heptane	20°C	1134

1.13. GRAFT COPOLYMERS

Poly(acrylamide)-*graft*-poly(vinyl alcohol)				
	Turbidimetric titration	Formic acid/butanone		1173
		Water/ethanol		1173
Poly(acrylic acid)-*graft*-poly(tetramethylene oxide)				
	Chromatography	Tetrahydrofuran	SEC	3196

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(acrylonitrile)-*graft*-poly(methyl methacrylate)				
	Extraction	Benzene		2949
Poly(acrylonitrile)-*graft*-poly(styrene)				
	Fractional solution	Toluene–dimethylformamide		1494
Poly(*N*-acryloyl pyrrolidone-*co*-acrylamide)-*graft*-poly(2-pyrrolidone)				
	Extraction	Boiling water		3540
Poly(*N*-acryloyl pyrrolidone-*co*-acrylonitrile)-*graft*-poly(2-pyrrolidone)				
	Extraction	Boiling water		3540
Poly(*N*-acryloyl pyrrolidone-*co*-methyl methacrylate)-*graft*-poly(2-pyrrolidone)				
	Extraction	Boiling water		3540
Poly(β-alanine)-*graft*-poly(acrylic acid)				
	Turbidimetric titration	Formic acid 85%/ethanol		1173
Poly(butadiene)-*graft*-poly(acrylate)				
	Chromatography	Tetrahydrofuran	SEC	1386
Poly(butadiene)-*graft*-poly(methacrylate)				
	Chromatography	Tetrahydrofuran	*cis*	1386
Poly(butadiene)-*graft*-poly(styrene)				
	Chromatography	Tetrahydrofuran	*cis*	1386
			SEC, styragel	492,3285
		Toluene	SEC, styragel	2360
	Extraction	Butanone		3565
		Cyclohexane–acetone, toluene		1390
	Turbidimetric titration	Benzene/acetone–methanol (9 : 1)		192
Poly(butadiene)-*graft*-poly(styrene-*co*-acrylonitrile)				
	Chromatography	Tetrahydrofuran		1250
Poly(butadiene)-*graft*-poly(vinyl chloride)				
	Chromatography	Tetrahydrofuran	SEC	3,1891
	Extraction	*n*-Pentane		3197
Poly(*n*-butyl acrylate)-*graft*-poly(methyl methacrylate)				
	Extraction	Acetonitrile		2427
Poly((*n*-butyl acrylate-*co*-styrene)-*graft*-poly(oxyethylene))				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, macromer	2726
Poly(butyl acrylate)-*co*-poly(styrene)-*graft*-poly(oxyethylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, macromer	2726
Poly(butyl methacrylate)-*graft*-poly(methacrylate)				
	Fractional precipitation	Acetone/methanol–water (1 : 1)		1249
Poly(butyl methacrylate)-*graft*-poly(styrene)				
	Fractional precipitation	Acetone/water–methanol (50 : 50)		1249
		Benzene/methanol		1249
Chlorinated poly(cyclopentadiene)-*graft*-poly(butadiene)				
	Chromatography	Tetrahydrofuran	SEC, lichrospher, 40°C	1290
Chlorinated rubber-*graft*-poly(ethyl methacrylate)				
	Fractional precipitation	Butanone/methanol		2707
Chlorinated rubber-*graft*-poly(methyl acrylate)				
	Fractional precipitation	Butanone/methanol		2707
Chlorinated rubber-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Butanone/methanol		2707
Poly(chloroprene)-*graft*-poly(isobutene)				
	Fractional precipitation	Benzene–butanone (1 : 4)/methanol		1721
Poly(chloroprene)-*graft*-(poly(isobutene-*block*-α-methylstyrene))				
	Extraction	*n*-Pentane, and tetrahydrofuran–ethyl acetate		1660
		n-Pentane, ethyl ether, butanone, tetrahydrofuran/ethyl ether, tetrahydrofuran/butanone		1664
Poly(chloroprene)-*graft*-poly(isobutyl vinyl ether)				
	Extraction	*n*-Pentane		1654
	Fractional precipitation	Tetrahydrofuran/butanone		1654
Poly(chloroprene)-*graft*-poly(α-methylstyrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1662
	Extraction	Butanone, acetone/*n*-pentane (6 : 4), 1-nitropropane		1662

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(*p*-chlorostyrene)-*graft*-poly(styrene)				
	Fractional precipitation	Benzene/methanol		2206
Chlorosulfonated poly(ethylene)-*graft*-poly(methyl methacrylate)				
	Extraction	Acetonitrile		2848
Poly(4-(1,1-dicyanoethylazo)benzyl methacrylate)-*graft*-poly(methacrylonitrile)				
	Chromatography	Dimethylformamide	SEC, fractogel	2423
Poly(4-(1,1-dicyanoethylazo)benzyl methacrylate-*co*-methyl methacrylate)-*graft*-poly(styrene)				
	Chromatography	Dimethylformamide	SEC, fractogel	2423
Poly(*N*-eosin vinylamine hydrochloride)-*graft*-poly(acrylamide)				
	Fractional precipitation	Water/acetone		3020
Poly(*N*-eosin vinylamine hydrochloride)-*graft*-poly(acrylic acid)				
	Fractional precipitation	Water/acetone		3020
Poly(*N*-eosin vinylamine hydrochloride)-*graft*-poly(acrylonitrile)				
	Fractional precipitation	Water/acetone		3020
Poly(*N*-eosin vinylamine hydrochloride)-*graft*-poly(styrene)				
	Fractional precipitation	Water/acetic acid		3020
Poly(ethyl-α-chloroacrylate)-*graft*-poly(vinyl acetate)				
	Fractional precipitation	Acetone/ethanol		3019
		Butanone/methanol–water (1 : 1)		3019
Poly(ethylene)-*graft*-poly(acrylamide)				
	Extraction	Water		2651
Poly(ethylene)-*graft*-poly(acrylic acid)				
	Extraction	Water		262
	Fractional precipitation	Xylene/methanol	130°C	2753
Poly(ethylene)-*graft*-poly(1,2-epoxy-3-butene)				
	Extraction	Chloroform		2310
Poly(ethylene-*co*-vinylacetate)-*graft*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2258
Poly(ethylene)-*graft*-poly(2,3,4,5,6-pentafluorostyrene)				
	Extraction	Acetone		3483
Poly(ethylene-*co*-propylene)-*graft*-poly(butadiene)				
	Chromatography	Toluene	SEC	78
			SEC, styragel	78
Poly(ethylene-*co*-propylene-*co*-1,4-hexadiene)-*graft*-poly(isobutene)				
	Extraction	*n*-Pentane, 3-pentanone		1650
Poly(ethylene-*co*-propylene-*co*-hexadiene)-*graft*-poly(styrene-*graft*-poly(α-methylstyrene))				
	Extraction	Acetone, *n*-pentane		1650
Poly(ethylene-*co*-propylene)-*graft*-poly(styrene)				
	Chromatography	Toluene	SEC, styragel	78
Poly(ethylene-*co*-propylene)-*graft*-poly(styrene-*co*-acrylonitrile)				
	Fractional precipitation	Tetrahydrofuran		772
Poly(ethylene)-*graft*-poly(styrene)				
	Extraction	Benzene		1553,3483
	Fractional precipitation	Toluene/methanol	90°C	559
Poly(ethylene-*co*-vinyl acetate)-*graft*-poly(styrene)				
	Fractionation by solvent mixing	Dimethylformamide/ methylcyclohexane		1818
Poly(ethylene-*co*-vinyl acetate)-*graft*-poly(styrene-*co*-acrylonitrile)				
	Cross fractionation by solvent mixing	Perchloroethylene/propylene carbonate (0.6 : 0.4)	Three layers	1818
	Extraction	Dimethylformamide/ methylcyclohexane	Lowering temp. and solvent demixing	1818
Poly(ethylene-*co*-propane-*co*-diene)-*graft*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC	151
Poly(ethylene-*co*-propene-*co*-diene)-*graft*-poly(methyl methacrylate)				
	Chromatography	Isooctane/tetrahydrofuran		151
		Tetrahydrofuran	SEC, μ-styragel	151
Poly(fluoroalkyl acrylate)-*graft*-(poly(methyl methacrylate))				
	Extraction	Ethyl ether, methanol	Macromer	3534
Poly(hydroxyethyl methacrylate-*graft*-methyl methacrylate)				
	Extraction	Ethyl ether, methanol	Macromer	3534
Poly(isobutene oxide)-*graft*-poly(styrene)				
	Extraction	Benzene		2463

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(isobutene)-*graft*-poly(styrene)				
	Chromatography	Benzene–cyclohexane/isopropanol	Precipitation chromatogr.	694
		Cyclohexane/*n*-propanol	Precipitation chromatogr.	545
	Extraction	Butanone, cyclohexane		395
		n-Hexane, cyclohexane		2930
	Fractional precipitation	Cyclohexane/*n*-propanol		545
	Turbidimetric titration	Toluene/methanol		1778
Poly(isobutylene-*co*-isoprene)-*graft*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC	593
Poly(isoprene)-*graft*-poly(styrene)				
	Chromatography	Toluene	SEC, styragel, 82°C, 35°C	489,492,1183
Poly(1,3-diisopropenylbenzene)-*graft*-poly(styrene)				
	Fractional precipitation	Benzene/methanol		1976
Poly(*p*-isopropylstyrene)-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Benzene/methanol		2077
Poly(methyl acrylate)-*graft*-poly(methyl methacrylate-d 5 sub)				
	Chromatography	Chloroform	SEC, styragel	2094
Poly(methyl acrylate)-*graft*-poly(styrene)				
	Fractional precipitation	Chloroform/methanol		3021
Poly(methyl acrylate-*co*-styrene-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Chloroform/methanol		3022
Poly(methyl acrylate)-*graft*-poly(vinyl chloride)				
	Fractional precipitation	Benzene/methanol		2707
Poly(methyl methacrylate)-*graft*-poly(butadiene)				
	Extraction	Petroleum ether, cyclohexane, acetone		490
Poly(methyl methacrylate)-*graft*-poly(β-hydroxybutyrate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 35°C	1776
Poly(methyl methacrylate)-*graft*-poly(isobutene)				
	Chromatography	Benzene	SEC	1665
	Extraction	Butanone, cyclohexane		395
Poly(methyl methacrylate)-*graft*-poly(isoprene)				
	Turbidimetric titration	Methylene dichloride + HCl/methanol + HCl		1333
Poly(methyl methacrylate)-*graft*-poly(isopropylstyrene)				
	Fractional precipitation	Benzene/isopropanol		833
Poly(methyl methacrylate)-*graft*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, permagel + ultrastyragel, 30–45°C	3005
	Extraction	Ethyl ether, methanol	Macromer	3534
Poly(methyl methacrylate)-*graft*-poly(oxypropylene)				
	Fractional precipitation	Toluene/*n*-heptane	25°C	2041
Poly(methyl methacrylate)-*graft*-poly(stearyl methacrylate)				
	Chromatography	Chloroform	Macromer, SEC, styragel	1447
Poly(methyl methacrylate)-*graft*-poly(styrene)				
	Chromatography	Tetrahydrofuran/acetonitrile	Reversed phase, silica-C18(subindice), gradient	3189
			SEC, μ-styragel	3190
		Tetrahydrofuran/*n*-heptane	HPLC reverse and normal phase mode	3147
		Tetrahydrofuran : acetonitrile gradient	HPLC (octadecyl modified silica), adsorption, composition	3190
	Extraction	Ethyl ether, acetonitrile, benzene		2205
	Fractional precipitation	Benzene–chlorobenzene (1 : 1)/methanol		570
		Benzene–acetone (1 : 1)/methanol + 0.1 M calcium chloride		109
		Benzene–chlorobenzene (1 : 1)/petroleum ether		109,973,1997
		Benzene/methanol		1249
		Benzene/petroleum ether		1249

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Butanone/methanol–water (1 : 1)		2722,3019
		Chloroform/methanol		3017,3021
	Turbidimetric titration	Benzene–chlorobenzene (1 : 1)/ petroleum ether		2200
Poly(methyl methacrylate)-*graft*-poly(vinyl acetate)				
	Chromatography	Acetone/acetone–water	Precipitation chromatogr., 45–15°C	1212
	Fractional precipitation	Acetone/methanol		1249,3017,3019
		Acetone/water–methanol (50 : 50)		1249
	Fractional solution	Acetone/water	Column extraction	57
Poly(methyl methacrylate)-*graft*-poly(vinyl chloride)				
	Fractional precipitation	Dioxane/methanol		3017
Poly(dimethyl vinyl siloxane)-*graft*-poly(acrylonitrile)				
	Extraction	Dimethylformamide, boiling benzene		268
Poly(α-methylstyrene)-*graft*-chlorobutyl rubber				
	Chromatography	Tetrahydrofuran	SEC, styragel	1662
	Extraction	*n*-Pentane, butanone, nitropropane		1662
Poly(α-methylstyrene-*co*-1,3-diisopropenylbenzene)-*graft*-poly(styrene)				
	Fractional precipitation	Benzene/methanol		1976
Poly(phenyl vinyl sulfide)-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Benzene/methanol		1747
Poly(phthalimidoacrylate-*co*-styrene)-*graft*-poly(oxyethylene)				
	Chromatography	Water–methanol	SEC	308
Poly(propylene)-*graft*-poly(butyl acrylate)				
	Extraction	Acetone		1462
Poly(propylene)-*graft*-poly(*N*,*N*-dimethylaminoethyl methacrylate)				
	Chromatography	Xylene/*n*-octanol	Column extraction, 126°C	1462
	Extraction	Acetone		1462
Poly(propylene)-*graft*-poly(styrene)				
	Chromatography	Benzene/methanol	Precipitation chromatogr. temp. gradient, glass beads	900,901
		Chloroform/isopropanol	Column elution	900,901,1521
	Extraction	Benzene		738
	Turbidimetric titration	Benzene/methanol		900,1522
Rubber-*graft*-poly(methyl methacrylate)				
	Extraction	Acetone		194
		Petroleum ether, acetone		108
	Fractional precipitation	Acetone/*n*-hexane		1718
Poly(styrene-*co*-maleic anhydride)-*graft*-poly(acrylonitrile)				
	Fractional solution	Aq. acetone, tetrahydrofuran, dimethylformamide	Extraction	2730
Poly(styrene)-*graft*-poly(ethyl acrylate)				
	Extraction	Cyclohexane, ethyl ether, acetonitrile		1393
Poly(styrene-*alt*-maleic anhydride)-*graft*-poly(ethylene oxide)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	818
Poly(styrene-*co*-maleic anhydride)-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Acetone/methanol		2730
Poly(styrene-*co*-maleic anhydride)-*graft*-poly(vinyl chloride)				
	Extraction	Butanone, ammonium hydroxide		1888
Poly(styrene)-*graft*-poly(methyl methacrylate)				
	Extraction	Cyclohexane, acetonitrile, benzene		3512
Poly(styrene)-*graft*-poly(*tert*-oxyethylene)				
	Chromatography	Tetrahydrofuran	SEC	1480
Poly(styrene-*co*-1-phenyl-2-(3-vinylphenylthiodiazene)-*graft*-poly(acrylonitrile)				
	Extraction	Chloroform, nitromethane/water, dimethylformamide		2422
Poly(styrene)-*graft*-poly(styrene-*co*-methyl methacrylate)				
	Fractional precipitation	Benzene/methanol		94
Poly(styrene)-*graft*-poly(vinyl acetate)				
	Fractional precipitation	Butanone/methanol		3017

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(styrene-*co*-vinyl benzyl chloride)-*graft*-poly(styrene)				
	Fractional precipitation	Benzene/methanol		497
Poly(styrene-*co*-vinylbenzophenone)-*graft*-poly(acrylonitrile)				
	Chromatography	Dimethylformamide	SEC	957
Poly(styrene-*co*-vinylbenzophenone)-*graft*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	957
Poly(tetrafluoroethylene-*co*-perfluorovinyl ether)-*graft*-poly(acrylic acid)				
	Extraction	Water		1261
Poly(tetrahydrofuran-*block*-ε-caprolactone)				
	Chromatography	Tetrahydrofuran	SEC	2413
Poly(tetrahydrofuran)-*graft*-poly(styrene)				
	Fractional precipitation	Tetrahydrofuran/isopropanol	Lowering temp.	2447
Poly(vinyl acetate)-*graft*-poly(methyl methacrylate)				
	Chromatography	Butanone/carbon tetrachloride (8/2)	TLC, silica gel	1357
		Methanol/water (9/1)	TLC, silica gel	1357
	Fractional precipitation	Acetone/methanol–water (1 : 2)		3017
Poly(vinyl acetate)-*graft*-poly(styrene)				
	Chromatography	Methanol–water (9 : 1)/chloroform		1357
	Fractional precipitation	Benzene/petroleum ether		1249
Poly(vinyl alcohol)-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Benzene/*n*-butanol		2814
Poly(vinyl alcohol)-*graft*-poly(styrene)				
	Chromatography	Benzene/methanol/butanone	Silica gel, column adsorption	1356
Poly(vinyl benzoate)-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Acetone/methanol		3018
Poly(vinyl benzoate)-*graft*-poly(vinyl acetate)				
	Fractional precipitation	Acetone/methanol–water		3018
Poly(vinyl chloride)-*graft*-poly(ethyl methacrylate)				
	Extraction	Acetone		546
	Fractional precipitation	Butanone/methanol		2707
		Dioxane/methanol		3017
		Tetrahydrofuran–acetone (1 : 3)/ methanol–water (4 : 1)		360
Poly(vinyl chloride)-*graft*-poly(glycidyl methacrylate)				
	Extraction	Butanone, isopropanol		2715
Poly(vinyl chloride)-*graft*-poly(isobutene)				
	Chromatography	Tetrahydrofuran	SEC	3
	Extraction	*n*-Pentane		1156,3197
Poly(vinyl chloride)-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Butanone/methanol		2707
Poly(vinyl chloride)-*graft*-poly(α-methylstyrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3237
	Extraction	Benzene		3237
Poly(vinyl chloride)-*graft*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1652
	Extraction	Benzene, toluene toluene/dichloroethane		546
	Fractional precipitation	Benzene/methanol		973,2206
		Tetrahydrofuran/methanol		1869
		Tetrahydrofuran/petroleum ether		1652,1869
		Tetrahydrofuran/water		1869
Poly(vinyl chloride)-*graft*-poly(styrene-*co*-acrylonitrile)				
	Extraction	Toluene/dichloroethane mixtures		546
Poly(*N*-vinylcarbazole)-*graft*-poly(isoprene)				
	Chromatography	Tetrahydrofuran	SEC	1434
Poly(2-vinylpyridine)-*graft*-poly(acrylonitrile)				
	Fractional solution	Ethanol	Extraction	1493
		Ethanol/dimethylformamide		1493
Poly(2-vinylpyridine)-*graft*-poly(vinyl acetate)				
	Extraction	Benzene, water		3161
Poly(2-vinylpyridine)-*graft*-poly(vinyl alcohol)				
	Extraction	Benzene, water		3161

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
1.14. MIXTURE OF POLYMERS				
Poly(butadiene) + poly(α-methylstyrene)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.34 : 0.66)	25°C	1816
Poly(butadiene) + poly(styrene-*co*-acrylonitrile)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.43 : 0.57)	25°C	1816
Poly(butadiene) + poly(vinyl chloride)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.4 : 0.6), (0.3 : 0.7)		1816
Poly(1-butene) + poly(propylene)				
	Fractional solution	Tetralin	Column elution increasing temp.	3016
Chlorinated poly(vinyl chloride) + poly(vinylchloride)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.4 : 0.6)	25°C, < 35% Chlorinated	1816
Poly(ethylene) + poly(ethylene)	TREF	Chloronaphthalene	HP-LDPE/LLDPE (ethylene-*co*-1-butene)	1648
Poly(ethylene) + poly(ethylene-*co*-propylene-*co*-diene)				
	TREF	Chloronaphthalene	EPDM	1648
Poly(ethylene) + poly(ethylene-*co*-vinylacetate)				
	TREF	Chloronaphthalene	LLDPE(ethylene-*co*-1-octene) + HP-EVA	1648
Poly(ethylene) + poly(isobutylene)				
	TREF	Chloronaphthalene		1648
Poly(ethylene) + poly(oxyethylene)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.2 : 0.8)		1816
Poly(ethylene) + poly(propylene)				
	Chromatography	1,2,4-Trichlorobenzene	SEC, 145°C	572
	Fractional crystallization	Perchloroethylene	Stirring	2560
	Fractional solution	Tetralin	Column elution increasing temp.	3016
Poly(ethylene) + poly(vinyl acetate)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.3 : 0.7)	25°C, LLDPE	1816
Poly(ethylene-*co*-propylene)-*block*-poly(ethylene) + poly(ethylene-*co*-propylene)-*block*-poly(ethylene)-*block*-poly(ethylene-*co*-propylene) mixture				
	Chromatography	Tetrahydrofuran	SEC, phenogel, 25°C	1021
	Fractional precipitation	Cyclohexanone + toluene/methanol		1021
Poly(ethylene-*co*-propylene) + poly(styrene-*co*-butylacrylate)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.2 : 0.8)		1816
Poly(*p*-iodostyrene) + poly(vinyl acetate)				
	Extraction	Benzene		750
Poly(isobutylene) + poly(α-methylstyrene)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.32 : 0.68)		1816
Poly(isobutylene) + poly(oxyethylene)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.2 : 0.8)	25°C	1816
Poly(isobutylene) + poly(styrene-*co*-butylacrylate)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.2 : 0.8)	25°C, 72% S	1816
Poly(isoprene)-*block*-poly(*tert*-butylacrylate) + poly(isoprene)				
	Fractional precipitation	Toluene/methanol	20°C	3312
Poly(*p*-methoxystyrene) + poly(cyclohexane)				
	Chromatography	Carbon tetrachloride/ tetrahydrofuran	TLC, silica gel, gradient	1152
	Turbidimetric titration	Benzene : butanone (3 : 1)/ethanol		1152
Poly(methyl methacrylate) + poly(iminoadipoyliminohexamethylene) *(Nylon 66)*				
	Density gradient technique	Dichloroacetic acid/methanol		1350
	Extraction	Chloroform		1350

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TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Fractional precipitation	Dichloroacetic acid/methanol		1350
		Formic acid/methanol		1350
		m-Cresol/methanol		1350
	Turbidimetric titration	*m*-Cresol/methanol		1350
Poly(methyl methacrylate) + poly(lauryl methacrylate)				
	Fractional precipitation	Benzene/methanol	20°C	29
Poly(methyl methacrylate) + poly(*n*-hexyl methacrylate)				
	Fractional precipitation	Benzene/methanol	20°C	29
Poly(dimethylsiloxane) + poly(ethylene-*co*-vinylacetate)				
	Fractional demixing	Methanol : heptane (0.48 : 0.52)	25°C, 70% VA	1816
Poly(dimethylsiloxane) + poly(α-methylstyrene)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.4 : 0.6)	25°C	1816
Poly(dimethylsiloxane) + poly(oxyethylene)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.2 : 0.8), methanol : heptane (0.28 : 0.72)	25°C	1816
Poly(dimethylsiloxane) + poly(vinyl chloride)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.4 : 0.6), (0.3 : 0.7)	25°C	1816
Poly(methacrylic acid)-*block*-poly(2-vinyl pyridine) + poly(2-vinyl pyridine)				
	Extraction	Methanol	Homopolymer	3082
Poly(methacrylic acid)-*block*-poly(2-vinyl pyridine) + poly(methacrylic acid)				
	Extraction	Acetonitrile	Homopolymer	3082
Poly(propylene) + poly(ethylene-propylene) rubber				
	Fractional precipitation	1,2,4-Trichlorobenzene	TREF	2185
Poly(propylene) + poly(ethylene-*co*-propylene) + poly(ethylene)				
	TREF	Trichlorobenzene	Chromosorb P, analytical and preparative, impact-resistane PP	2184
Poly(styrene) + poly(butadiene)	Chromatography	Tetrahydrofuran		2795
Poly(styrene) + poly(dimethylsiloxane)				
	Chromatography	Toluene	SEC, 20°C	701
Poly(styrene) + poly(ethylene)	Chromatography	1,2,4-Trichlorobenzene	SEC, styragel, 145°C	572
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.24 : 76), (0.25 : 0.75)		1816
Poly(styrene) + poly(ethylene-*co*-propylene)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.2 : 0.8)	25°C (53% PE-47% pp)	1816
Poly(styrene) + poly(ethylene-*co*-propylene-*co*-dicyclo pentadiene)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.2 : 0.8)	25°C (50% PE-43% PP-7% DCP)	1816
Poly(styrene) + poly(ethylene-*co*-vinylacetate)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.3 : 0.7)	25°C 8.5% VA	1816
Poly(styrene) + poly(isobutylene)				
	Fractional demixing	Dimethylformamide : methylcyclohexane (0.24 : 0.76)	45°C	1816
Poly(styrene)-*block*-poly(isoprene) + poly(styrene)				
	Fractional precipitation	Cyclohexane/1-propanol, dioxane/1-propanol		650
Poly(styrene) + poly(methyl methacrylate)				
	Chromatography	Toulene	SEC + adsorption (separation mixtures)	1477
		Toulene, tetrahydrofuran	SEC/adsorption, (separation) mixtures	1478
		Trichloroethylene	SEC, 23°C	3195
	Fractional precipitation	Benzene/methanol	30°C	176
		Tetrahydrofuran	SEC	1898
	Turbidimetric titration	Benzene/methanol	25°C	1415
Poly(styrene) + poly(oxyethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	348
Poly(styrene) + paraffin	Fractional solution	Chloroform	Column extraction	1732
Poly(styrene) + poly(vinyl acetate)				
	Extraction	Methanol		750

TABLE 1. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(styrene) + poly(vinyl chlroide)				
	Extraction	Cyclohexane		750
Poly(vinyl acetate) + poly(vinyl chloride)				
	Extraction	Tetrahydrofuran		750
Poly(vinyl alkyl ether) + poly(cyclic ethers)				
	Extraction	*n*-Heptane, acetone		2644

TABLE 2. MAIN-CHAIN CARBOCYLIC POLYMERS

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
2.1. POLY(PHENYLENES)				
Poly(2,5-dialkoxy phenylene)	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	3305
Poly(anthracene)	Extraction	Benzene, acetone, etc.		1304
	Fractional precipitation	Benzene/methanol		307,1128
Poly(benzene-*co*-anthracene)	Extraction	Benzene,bromobenzene		307
Poly(bromophenylenes) oligomers				
	Fractional solution	Dioxane/water		2914
Poly(*p*-dichloromethylbenzene-*co*-benzene)				
	Chromatography	Chloroform	SEC	1117
Poly(*p*-dichloromethylbenzene-*co*-diphenylmethane)				
	Chromatography	Chloroform	SEC	1118
Poly(cumarone-*co*-indene)	Fractional precipitation	Benzene–ethyl acetate/methonal		3587
Poly(ethynylene-phenyleneethynylene)				
	Chromatography	*o*-Dichlorobenzene	SEC, 120°C	3459
Poly(7,16-dihydroheptacenes)	Chromatography	Tetrahydrofuran	SEC, analytical and preparative	2840
Poly(indene)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3202
Poly(bis-indenylene)	Chromatography	Benzene	SEC	2393
		Tetrahydrofuan	SEC, μ-styragel	2678
Poly(1,3-diisopropenylbenzene)	Chromatography	Tetrahydrofuran	SEC, styragel	1976
Poly(1,4-diisopropenylbenzene)	Chromatography	Tetrahydrofuran	SEC, styragel	1976
Poly[(2-methoxycarbonyl)phenylene-1,4-diyl]				
	Chromatography	Tetrahydrofuran	SEC	552
Poly(dimethylbiphenylene)	Chromatography	Chloroform	SEC, styragel	1797
Poly(phenylenes)	Fractional precipitation	Toluene/cyclohexane		537
Poly(1,4-phenylene(methylmethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	541
Poly(diphenylmethylene)	Fractional precipitation	Benzene/methanol		3332
Poly(divinylbenzenes)	Chromatography	Tetrahydrofuran	SEC, TSK gel	2403
2.2. FORMALDEHYDE RESINS				
Phenolic resin (General)	Fractional precipitation	Acetone, tetrahydrofuran/water		3091,3092
Poly(ureas) – General	Fractional precipitation	Phenol–tetrachloroethane (1 : 1)/ethanol-*n*-heptane		2550
Poly(dibenzothiophenolformaldehyde)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1061
p-tert-Butylphenolformaldehyde				
	Chromatography	Tetrahydrofuran	SEC, styragel	2416
p-Cresol–formaldehyde	Chromatography	Tetrahydrofuran	SEC, Merck gel	414
	Fractional precipitation	Benzene/methanol, Benzene/petroleum ether		1670
		Dimethylformamide/water		3213
		Dioxane, dichloroethane, trichloroethylene/methanol, ethanol,*n*-propanol, *n*-butanol, ethyl ether, formamide		3390

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TABLE 2. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Tetradydrofuran–methanol(1 : 2) water		1670
		Tetrahydrofuran/petroleum ether		1670
		Toluene/petroleum ether		1670
Formaldehyde – sodium napthalene sulfonate				
	Fractional precipitation	Water/acetone		2572
Poly(*p*-hydroxybenzoic acid-*co*-*p*-bromophenol-*co*-formaldehyde)				
	Fractional precipitation	Methanol/water	Random copolymer	550
Melamine – formaldehyde resins	Chromatography	Dimethylformamide	SEC	415
			SEC, Merck gel	413,417
Methylphenol derivatives – formaldehyde				
	Chromatography	Benzene/acetic acid/water (80 : 24 : 3)	Paper chromatogr., 15°C	891
Phenol–formaldehyde	Chromatography	Benzene – acetic acid – water	Paper partition	891,2558
		Tetrahydrofuran	SEC, μ-styragel	175,788,3098
			SEC, Merck gel	414
			SEC, styragel	1432,1562,2239, 2790,3407
	Coacervation	Ethanol or dioxane/salts in water		481
		Methanol, ethanol, *n*-propanol, novolac, dioxane/carbon dioxide, water solution		482
	Fractional precipitation	Acetone–methanol/petroleum ether		527
		Acetone/petroleum ether	30°C	1562
		Acetone/water		
		Dioxane, dichloroethane, trichloroethylene/methanol, ethanol, *n*-propanol, *n*-butanol, ethyl ether, formamide		527,3212,3213,
		Methanol/water		481
	Turbidimetric titration	Acetone/diluted sulfuric acid		526
		Methanol/diluted sulfuric acid		526
		Toluene/*n*-hexane, toluene/methanol		11
Phenolated phenanthrene – formaldehyde				
	Chromatography	Dichloroethane	SEC, styragel, analytical and preparative	1277
		Tetrahydrofuran	SEC, styragel, analytical and preparative	1277
Poly(phenol)s	Chromatography	Water, methanol	SEC	3485
Poly(diphenyl ether-*co*-formaldehyde)				
	Fractional precipitation	Benzene/methanol	30°C	2397
Resorcinol–formaldehyde	Chromatography	Tetrahydrofuran	SEC, oligomers	1348
Urea–formaldehyde	Chromatography	Tetrahydrofuran	SEC	1349,1617
	Fractional precipitation	Diemthyl sulfoxide/acetone		1617
		Ethanol–water(1 : 1)/methanol		3391

TABLE 3. MAIN-CHAIN HETEROATOM POLYMERS

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
3.1. POLY(OXIDES)				
Poly(acetaldehyde), see Poly(oxyethylidene)				
Poly(acrolein)	Chromatography	Tetrahydrofuran	SEC,Styragl	1144
Poly(aldehydes) general	Extraction	Boiling acetone, diisopropyl ether, benzene	Amorphous + crystalline	2368

TABLE 2. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Tetradydrofuran–methanol(1 : 2) water		1670
		Tetrahydrofuran/petroleum ether		1670
		Toluene/petroleum ether		1670
Formaldehyde – sodium napthalene sulfonate				
	Fractional precipitation	Water/acetone		2572
Poly(*p*-hydroxybenzoic acid-*co*-*p*-bromophenol-*co*-formaldehyde)				
	Fractional precipitation	Methanol/water	Random copolymer	550
Melamine – formaldehyde resins	Chromatography	Dimethylformamide	SEC	415
			SEC, Merck gel	413,417
Methylphenol derivatives – formaldehyde				
	Chromatography	Benzene/acetic acid/water (80 : 24 : 3)	Paper chromatogr., 15°C	891
Phenol–formaldehyde	Chromatography	Benzene – acetic acid – water	Paper partition	891,2558
		Tetrahydrofuran	SEC, μ-styragel	175,788,3098
			SEC, Merck gel	414
			SEC, styragel	1432,1562,2239, 2790,3407
	Coacervation	Ethanol or dioxane/salts in water		481
		Methanol, ethanol, *n*-propanol, novolac, dioxane/carbon dioxide, water solution		482
	Fractional precipitation	Acetone–methanol/petroleum ether		527
		Acetone/petroleum ether	30°C	1562
		Acetone/water		
		Dioxane, dichloroethane, trichloroethylene/methanol, ethanol, *n*-propanol, *n*-butanol, ethyl ether, formamide		527,3212,3213,
		Methanol/water		481
	Turbidimetric titration	Acetone/diluted sulfuric acid		526
		Methanol/diluted sulfuric acid		526
		Toluene/*n*-hexane, toluene/methanol		11
Phenolated phenanthrene – formaldehyde				
	Chromatography	Dichloroethane	SEC, styragel, analytical and preparative	1277
		Tetrahydrofuran	SEC, styragel, analytical and preparative	1277
Poly(phenol)s	Chromatography	Water, methanol	SEC	3485
Poly(diphenyl ether-*co*-formaldehyde)				
	Fractional precipitation	Benzene/methanol	30°C	2397
Resorcinol–formaldehyde	Chromatography	Tetrahydrofuran	SEC, oligomers	1348
Urea–formaldehyde	Chromatography	Tetrahydrofuran	SEC	1349,1617
	Fractional precipitation	Diemthyl sulfoxide/acetone		1617
		Ethanol–water(1 : 1)/methanol		3391

TABLE 3. MAIN-CHAIN HETEROATOM POLYMERS

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
3.1. POLY(OXIDES)				
Poly(acetaldehyde), see Poly(oxyethylidene)				
Poly(acrolein)	Chromatography	Tetrahydrofuran	SEC,Styragl	1144
Poly(aldehydes) general	Extraction	Boiling acetone, diisopropyl ether, benzene	Amorphous + crystalline	2368

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Methanol, chloroform	Amorphous + crystalline	938
		Propio-,butyr, isobutyr-, and isovaleraldehydes	Amorphous + crystalline	2368
Poly(β-allyloxypropionaldehyde)				
	Chromatography	Tetrahydrofuran	SEC, styragel	137
Poly(aryl ether isoquinoline)s	Chromatography	Chloroform	SEC, μ-styragel	3004
Poly(1-butene oxide), see Poly(oxy(ethylethylene)				
Poly(epichlorohydrin), see Poly(oxy(chloromethyl)ethylene)				
Poly(epoxides)	Chromatography	Benzene	SEC, styragel	1263
Poly(ether ether ketimine)	Chromatography	Tetrahydrofuran	SEC,μ-styragel, 35°C	2778
	Fractional precipitation	Benzene/*n*-hexane	20–35°C	2778
Poly(ethylene oxide), see Poly(oxyethylene)				
Poly(glutaraldehyde)	Chromatography	Dimethylformamide	SEC, styragel	2038
Methyl [poly(ether ether ketone)]				
	Fractional precipitation	*N*-Methyl-2-pyrrolidone + chloroform/methanol	Hot dissolved in NMP and diluted in chloroform	3417
Poly(2-methyl-1,4,6-trioxa-spiro-(4,4)-nonane)				
	Chromatography	Tetrahydrofuran	SEC,μ-styragel	3226
Poly(*cis*-1,4-bis((methylsfulfonyl)methyl) cyclohexane),(*trans*-1,4-bis(methylsulfonyl)methyl)cyclohexane)				
	Chromatography	Chloroform	SEC, styragel, 40°C	2566
Poly[1,3-bis(1-naphthoxy)benzene]				
	Chromatography	Chloroform	SEC, styragel, 40°C	2567
Poly[2,2′-bis(1-naphthoxy)biphenyl]				
	Chromatography	Chloroform	SEC, styragel, 40°C	2567
Poly[4,4′-bis(1-naphthoxy)biphenyl]				
	Chromatography	Chloroform	SEC, styragel, 40°C	2567
Poly(oxadiazo-3,5-diyl-perfluoroethylidene-*block*-oxyperfluoropropylene)				
	Chromatography	1,1,2-Trichloro-1,2,2-trifluoroethane (Halocarbon 113)	SEC, porous silica	1765
Poly(oxy-2-butenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 30°C	2324
Poly(oxy(*tert*-butyl)ethylene)	Fractional crystallization	Benzene	80°C, lowering temp.	53
	Fractional precipitation	Benzene/methanol		764
Poly(oxy(4-chlorobutyl)ethylene)				
	Cromatography	Tetrahydrofuran	SEC, styragel	2966
Poly(oxy(2-chloroethyl)ethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2966
Poly(oxy(chloromethyl)ethylene)				
	Extraction; precipitation	Acetone(cold), acetone/methanol, methanol/water		1431
Poly(oxy(3-chloropropyl)oxirane)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2966
Poly(oxy-2-cyanoethyloxymethylene)				
	Fractional solution	Acetone, dimethylformamide		3097
Poly(oxy-1,2-cyclohexylene)	Chromatography	Tetrahydrofuran		1240,1241
	Fractional precipitation	Benzene/methanol	25°C	2128
Poly(oxy-1,4-cyclohexylene) (*poly(1,4-cyclohexane oxide)*				
	Chromatography	*m*-Cresol-chlorobenzene (1 : 1)	SEC, 25°C	3073
		Tetrahydrofuran	SEC,μ-styragel	319
Poly(oxy-3,5-dimethyl-1,4-cyclohexylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1754
Poly(oxy-2,6-dimethyl-1,4-phenylene)				
	Chromatography	Carbon tetrachloride/methanol (98/2)	TLC, silica gel	2785
		Chloroform	SEC, styragel	120,3219
		Dimethyl sulfoxide	SEC	1932
		Methylene dichloride	TLC, silica gel	2785
		o-Dichlorobenzene	SEC, 80°C	1391
			SEC, styragel, 140°C	3320
		Tetrahydrofuran	SEC	2724,3460
			SEC, μ-styragel	2564
			SEC, μ-styragel, 40°C	230
		Toluene	SEC	31
	Fractional precipitation	Benzene/methanol		120

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Chloroform/methanol	20°C, preparative	217,1871,2504
		Ethylene dichloride/nitromethane		3061
		Toluene/methanol	20°C, preparative	217,220,1871
		Toluene/petroleum ether		152
	Fractional solution	Chloroform/acetone		3320
		Chloroform/methanol	Column extraction	456
	Stirring fractionation	Dioxane		31
	Turbidimetric titration	Chloroform/ethanol	25°C	456
Poly(oxy-1,1-dimethyltrimethylene) *[(Poly(2,2-dimethyloxacyclobutane)]*				
	Chromatography	Tetrahydrofuran		1758
Poly(oxy-2,2′-dimethytrimethylene) *Poly(3,3′-dimethyloxetane)*				
	Fractional precipitation	Chlorobenzene/ethanol		1099
		Cyclohexane/ethanol	30°C	2577,2579
Poly(oxy-2,6-diphenyl-1,4-phenylene)				
	Chromatography	Benzene	50°C	2981
		Methylene dichloride	25°C	1517
		Tetrahydrofuran		1087
Poly(oxy(ethylethylene) *poly(1-butene oxide)*				
	Chromatography	Tetrahydrofuran	SEC, styragel	27,1889,2009
		Trichlorobenzene	SEC, 90°C	389
	Fractional precipitation	Benzene/methanol		389
Poly(oxy(2-hydroxytrimethylene)oxy-1,4-phenyleneiopropylidene-1,4-phenylene)				
	Chromatography	Chloroform	SEC,μ-styragel	455
		Chloroform/ethanol (2/1)	SEC, Sephadex	245
		Tetrahydrofuran	SEC, butyl methacrylate gel	1843
			SEC, styragel	819,847,1843
	Fractional solution	Chloroform/*n*-hexane	Column extraction	2322
Poly(oxy-3-methyl-3-chloromethyltrimethylene)				
	Sedimentation velocity	Butanone	21°C	36
Poly(oxy-2-methyl-1,4-cyclohexylene)				
	Chromatography	Tetrahydrofuran	SEC	1754,1755
Poly(oxy(1-methyl)trimethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1756,1757,2441
Poly(oxy(2-methyl)trimethylene)				
	Fractional precipitation	Chloroform/methanol	25°C	2739
Poly(oxy-4-phenylazo-1,2-phenylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2312
Poly(oxydecamethylene)	Fractional precipitation	Benzene/methanol		3523
Poly(oxyethylene)	Chromatography	Acetone	SEC, Sephadex, 50°C	1520
		Benzene	SEC, styragel	1263
		Butanone/water (1/1)	TLC, kieselgel	465
		Chloroform/ethanol (10/1)	TLC, alumina	978
		Chloroform/pyridine(5/7)	TLC, silica gel	978
		Dimethylacetamide	SEC, 74°C	1813,1822
		Dimethylacetamide	GLC of trimethylsilyl ether derivatives, low MW	898
		Ethylene glycol/methanol (4/1)	TLC, gradient const.	2480
		Methanol followed by methanol/dimethylformamide (4/1)	Gradient elution, TLC	2480
		o-Dichlorobenzene	SEC	3437
			SEC, 130°C	1251,1252
		Pyridene–water (0.1 : 10)	TLC, silica gel	978
		Tetrahydrofuran	SEC	27,1264,1265
			SEC, 24.9°C	566
			SEC, butyl methacrylate-*co*-triethylene glycol methacrylate gel	2003
			SEC, PS-PSOE network	269
			SEC, styragel, 25°C	667,1683, 2049,3034
			Tosylated, SEC, ultrastyragel	3112
			Tosylated, SEC, μ-spherogel	3112

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
			Tosylated, SEC, μ-styragel	3112
		Toluene	SEC, butyl methacrylate-*co*-triethylene glycol methacrylate gel	2003
			SEC, TSK gel	1224
		Trichloroethylene	SEC, 23°C	3195
		Water	SEC, TSK gel	1224,3557
			SEC, aquagel-OH	2121
			SEC, modified glass beads	1611,1612
		Water, water/methanol (80:20), (60:40), dimethylformamide	SEC, poly(acrylamide gel)	719
	Extraction	Water/chloroform–benzene	Distribution between two immiscible liquid countercurrent	520
		Water/*n*-hexane	Distribution between two immiscible liquid countercurrent	520,521
	Field flow	Methanol		1689
	Fractional crytallization	Benzene/isooctane		388
		Ethanol		1237
	Fractional precipitation	Benzene/ethyl ether	20°C	1528,1529
		Benzene/isooctane		16,52,264,265, 266,353,388,391, 1820,1986,2290, 3062,3063,3064, 3065,3066
			37°C	3086
			65°C, low MW	734
		Benzene/isooctane (68/32)	Lowering temp.	2731
		Benzene/isooctane (7/3)	Lowering temp.	263
		Benzene/*n*-heptane		1237,2757,2807
		Benzene/*n*-hexane	Low temp.	180
		Chloroform/*n*-hexane		172
		Isopropanol/*n*-heptane		2757
		Toluene/isooctane		1987
		Touene/*n*-hexane	60°C	720
		Water(0.375 M sodium percarbonate)	LCST, raisng temp., star	3557
		Water, potassium sulfate	Lowering temp.	792
	Fractional solution	Benzene/*n*-heptane	Low MW	721
	Sedimentation equilibrium	Aq. 0.1–hydrochloric acid	Ultracentrifuge, 25°C	398
	Sedimentation velocity	Aq. 0.1–hydrochloric acid	Ultracentrifuge, 25°C	398
	Turbidimetric titration	Benzene/isooctane		16
		Benzene/*n*-heptane	35°C	1236
		Chloroform/*n*-hexane		833
		Isooctane/benzene (100/48)	Lowering temp.	1237
Poly[oxyethyleneoxyethyleneoxyethyleneoxy(1,4′-biphenylyl)-2,4-phenyl] butane				
	Chromatography	Tetrahydrofuran	SEC, styragel,40°C	2571
Poly(oxyhexamethylene)	Fractional precipitation	Benzene/methanol	40°C	2032
		Butanone	30°C, phase separation at 28°C	3522
		Dioxane/methanol	54°C	2031
		Tetrahydrofuran/methanol	40°C	2028,2031,2033,2734
Poly(oxymethylene(2-hydroxy-5-methyl-1,3-phenylene) methylene)				
	Fractional precipitation	Benzene/petroleum ether	35°C	1524
Poly(oxymethylene) *Poly(formaldehyde)*				
	Chromatography	1,1,1,3,3,3-Hexafluoro-2-propanol	SEC	1222
		Dimethylacetamide	SEC, styragel, 150°C diol, diacetate and diethyl ether ends	610
		Dimethylformamide	SEC	419
	Extraction	Dimethylformamide	Lowering temp. fractional	757,2461
		o-Dichlorobenzene	150–164°C	2002
	Fractional crystallization	*p*-Chlorophenol	Stirred, 60°C	1712,1713
		Phenol	Lowering temp. diacetate	3072
	Fractional precipitation	Dimethylformamide	Lowering temp.	757
			Methoxy capped, 123.5–106°C	1327

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Fractional solution	Phenol/ethyl cellosolve	Column extraction, Kieselite	1535,1713
		Tetrachloroethane–phenol (3 : 1)/*n*-hexanol	120°C	1129
			Column extraction, 130°C	1129
	Turbidimetric titration	Dimethylformamide	Lowering temp.	3359
Poly(oxymethylene oxy tetramethylene) *Poly(1,3-dioxepane)*				
	Fractional precipitation	Benzene/methanol	30°C	2035
Poly(oxymethyleneoxyethylene) *Poly(1,3-dioxolane)*				
	Chromatography	Benzene	SEC	1640
		Tetrahydrofuran	SEC, styragel	1807
		Water and water + Sodium chloride 0.1N	SEC	290
	Fractionation precipitation	Benzene/*n*-hexane–ethanol		34
		Chlorobenzene	Lowering temp.	2735
		Chloroform/*n*-hexane–methanol	25°C	32,33
		Dichloroethane/methanol	Lowering temp.	121
Poly(oxymethyleneoxymethylene-1,4-phenymethylene)				
	Fractional precipitation	Chlorofrom/methanol		2737
Poly(oxymethyleneoxytetramethylene)				
	Fractional separation	Benzene/methanol		1002
Poly(oxyoctadecylethylene)	Extraction	Boiling methylene dichloride		107
		Fractional precipitation dichloride/*n*-hexanol	Methylene	107
Poly(oxyoctamethylene)	Fractional precipitation	Benzene/methanol		2029,2030,2574
Poly(oxyphenylethylene)	Fractional precipitation	Benzene/isooctane	25°C	53
		Toluene/*n*-hexane	30°C	2091
Poly(oxypropylene) *Poly(propylene oxide)*				
	Chromatography	Acetone	TLC	3330
		Aq. 0.3 M sodium hydroxide	SEC, 25°C Polyurethane gel	3326
		Aq. 1.0 M sodium chloride	SEC, 25°C PMMA gel	3326
		Benzene	Membrane chromatogr.	2155
			SEC, styragel	3307
		Benzene,*n*-heptane	Extraction from an active support	961
		Butanone	TLC	3330
		Chlorobenzene	Membrane chromatogr.	2155
		Chloroform	SEC, styragel, preparative	1435
		Chloroform	SEC, phenogel, oligomers	3227
		Ethyl acetate	TLC	3330
		Ethyl acetate + water	TLC, silica gel	3326
		Ethyl acetate/water, ethyl acetate/butanone	Column, elution, silica gel	3326
		Isooctane	SEC, Merck gel, 95–96°C	77
		Methanol	Column elution, silica gel	3326
		Methanol/water	Precipitation chromatogr. 20–60°C	2889
		Tetrahydrofuran	SEC	3156
			SEC, styragel	27,1435,1780, 2150,2161,2540, 2541,2889, 3336,3337
			SEC, styragel, 40°C, stereoregular-*trans*	76
		Toluene	SEC	2889
	Distribution between immiscible liquids	Water/chloroform–benzene	Countercrurrent	520
		Water/*n*-hexane	Countercurrent	5,507,-521
	Extraction	Acetone	0°C(D,L)	3250
			Tacticity	1307
		Acetone(cold), acetone/methanol, methanol/water	Amorphous + crystalline	1431
		Acetone (0°C), *n*-hexane (– 78°C)	Optically active	1422,1423, 1424,1425
		n-Hexane	– 78°C(D-L)	1423

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Fractional crystallization	Acetone, *n*-hexane	Lowering temp.	3307
		Isooctane	40°C	51
			Lowering temp.	49,385,387,2655
			Lowering temp.	76
	Fractional precipitation	Acetone	Lowering temp. 0°C to −78°C crystalline + amorphous(D, L comp.)	960 960
		Isooctane	Lowering temp.	3311
	Fractional solution	Acetone	0°C, crystalline + amorphous	962
		Acetone/isopropyl ether	Column extraction, alumina	2288
		Ethanol/water	Column extraction	627
		Isooctane		390,606
			60°C, lowering temp.	49
		Isooctane/octamethyl tetrasiloxane	Amorphous, extracted with isooctane at 40°C, crystalline fractionated at 70,74°C	51
		Isopropanol/water	70°C, 74°C	50
		Methanol/water	25°C	50
Poly(oxytetramethylene)	Chromatography	Acetone/water		1844,2431
		Butanone	SEC, styragel	3125
		Dimethyformamide	SEC, Merckogel PVA gel	1790
		Methanol–water mixtures	Dicarbamates, HPLC, 35°C	92
		Tetrahydrofuran	SEC (Low MW)	1220
			SEC, styragel	92,672,1746, 1840,1841,3125
			SEC, ultrastyragel, 30°C	567
		Toluene	SEC, Styragel	329
		Toluene, ethyl benzene, ethyl acetate, dioxane, tetrahydrofuran, cyclohexane, methyl ethyl ketone, benzene	ThFFF	3339
	Distribution between immiscibe liquids	Cyclohexane–toluene (9 : 1)/ water–methanol		1509
	Extraction	Isopropanol/water	Continuous extraction	345
		Water/acetone	Mixtures	1844
	Fractional precipitation	Acetone	Lowering temp.	2786
		Benzene/methanol		43,861,862
		Benzene/*n*-hexane	60°C	180
			Lowering temp.	179
		Toluene/methanol		2005
	Fractional solution	Butanone	Column, elution, silica gel	1840,1841
		Ethyl ether/petroleum ether	22°C	1396
		Hydrocarbons, alcohols, water		2296
		Isopropanol/water	50 C, low MW	344
	Sedimentation velocity	Ethyl acetate–*n*-hexane (22.3 : 77.7)	30°C	3531
	Turbidimetric titration	Ethanol/water		2005
Poly(oxytrifluoropropylene) *Poly(trifluoropropylene oxide)*				
	Coacervation	Acetone/toluene–water (1 : 25)	30°C	1691
Poly(oxytrimethylene)	Chromatography	Chloroform	SEC, μ-styragel, 35°C	2687
	Fractional precipitation	Acetone/water		2578,3134
Perfluoropoly(ether) copolymer (General)				
	Fractional precipitation	CFC 113/methanol		2831
Perfluoropoly(oxyalklenes)	Chromatography	Delifrene-LS/methanol	Oligomers	1923
	Fractional precipitation	Delifrene-LS/methanol	Oligomers	1923
Poly(2-phenoxymethyl-1,4,6-trioxa-spiro-4,4-nonane)(polymer derived from :)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3226
Poly(phenyl glycidyl ether)	Chromatography	Butanone,butanone + *n*-heptane	Adsorption chromatogr. silica gel, 20°C	1834
		Tetrahydrofuran	SEC, styragel	1834
Poly(2-phenyl-1,4,6-trioxa-spiro-(4,4)-nonane)(polymer from :)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3226

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(prophyrin oxide)s	Chromatography	Chloroform	SEC, μ-styragel	2859
Poly(tetrahydrofuran)	Chromatography	Chloroform	SEC, ultrastyragel	3553
Poly(thio-2,2′-dimethyltrimethylene)				
	Fractional precipitation	Benzene/methanol	25°C	1882,1883
Poly(triazine-2,4 diyl-perfluoroethylidene-block-oxyperfluoropropylene)				
	Chromatography	1,1,2-Trichloro-1,2,2,-trifluo roethane(Halocarbon 113)	SEC, porous silica	1765

3.2. POLY(CARBONATES)

Poly(carbonates) (general) poly(oxycarbonyloxy .....)				
	Fractional precipitation	Methylene chloride/*n*-heptane		3216
	Turbidimetric titration	Chloroform/methanol		1804
Poly(carbon dioxide)-*alt*-(oxypropylene); poly(oxypropylene)				
	Chromatography	Tetrahydrofuran		28
Poly(2,4-hexadiyn-1,6-ylene carbonate)				
	Chromatography	Terahydrofuran	SEC, styragel	1814
Poly(bis(4-hydroxyphenyl)isopropylidene; di(4-hydroxyphenyl)ethylene)-*alt*-phosgene)				
	Turbidimetric titration	Methylene dichloride/methanol		1740
Poly(bis(4- hydroxyphenyl)isopropylidene; diethylene glycol)-*alt*-phosgene)				
	Chromatography	Tetraydrofuran	SEC, μ-styragel	1999
Poly(bis(4-hdroxyphenyl)isopropylidene; (4-hydroxy-3,5-dichlorophenyl) isopropylidene)-*alt*-phosgene)				
	Turbidimetric titration	Methylene dichloride/methanol		1740
Poly(bis(hydroxyphenyl)isopropylidene; di(4-hydroxyphenyl)sulfone-*alt*-phosgene)				
	Turbidimetric titration	Methylene dichloride		1740
Poly(bis(4-hydroxyphenyl)isopropylidene terephthalic acid)-*alt*-phosgene				
	Chromatography	Tetrahydrofuran	SEC, zorbax	2045
Poly(bis(4-hydroxyphenyl)isopropylidene;terephthalic acid)-*alt*-phosgene)				
	Chromatography	*o*-Dichlorobenzene	SEC, styragel	2662
Poly(oxybutane oxyadipoyl)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, oligomers	3121
Poly(oxycarbonyloxy-1,4-phenylene isopropylidene-1,4-phenylene)				
	Chromatography	Methylene chloride/diethylene glycol	Continous fractionation (counter current extraction) > 25°C	3443
		Methylene dichloride	SEC, ultrastyragel	173
			SEC, μ-styragel	1906
			SEC, styragel	4,1321,3562
		Methylene dichloride/ *n*-heptane		3291,3292,3293
		Tetrahydrofuran	SEC, μ-styragel	65,174,177, 231,458,683,806, 854,2171,3443
			SEC, ultrastyragel, 35°C	824
	Extraction	Hot benzene	Low MW	1906
	Extraction (CPF)	Methylene chloride–diethylene glycol	Continous operation	3443
	Fractional precipitation	1-2-Dichloroethane/isooctane	50°C	231
		Chloroform/methanol	23°C, 25°C	539,1068,2254, 2256,2257
		Chloroform/*n*-octane		1068
		Methylene chloride–cresol/ petroleum ether	−20°C	1975
		Methylene chloride/methanol	0, 20, 25, 30°C	1068,1353,1975,3009
		Methylene chloride/*n*-heptane		312
		Methylene chloride/*n*-octane	25°C, 30°C	1068
		Methylene chloride/petroleum ether	25°C	1975
		Methylene dichloride/methanol		173
		Tetrachloroethane/paraffin oils	25°C	1975
	Fractional solution	Methylene dichloride/*n*-hexane		1069
	Turbidimetric titration	Dioxane–tetralin (7 : 10)/cyclohexane/decalin(1 : 1))		2297
		Methylene dichloride/methanol		1069
Poly(oxycarbonyloxy-1,4-phenylene-1-cyclohexylene-1,4-phenylene)				
	Fractional precipitation	Methylene dichloride/methanol		999

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(oxycarbonyloxytrimethylene) *Poly(trimethylene carbonate)*				
	Chromatography	Tetrahydrofuran	SEC	1795
Poly[oxythiocarbonyloxy-1,4-phenylene (methyl) phenylmethylene-1,4-phenylene]				
	Fractional precipitation	Chloroform/methanol		996
Poly[2,2-propane-bis(4-phenylthiocarbonate)]				
	Fractional precipitation	Chloroform/methanol		994
3.3. POLY(ESTERS)				
Poly(esters) (general)	Chromatography	Chloroform/ethanol, ethyl ether/water	Paper chromatogr., oligomers from aliphatic and armoatic diacids and glycols	122
		Ethanol-cyclohexane(3 : 7)/ cyclohexane	Precipitation chromatogr.	2646
Poly(adipic acid-*co*-ethylene glycol), see Poly(oxyethyleneoxyadipoyl)				
Poly(adipic acid-co-poly(ethylene glycol)				
	Fractional precipitation	Acetonitrile/isopropyl ether		833
		Chloroform/*n*-hexane		833
	Fractional solution	Butanone	Column extraction, silica gel	3559
Poly(adipic acid-*co*-glycerol)	Fractional precipitation	Benzene/petroleum ether		527
Poly(adipic acid-*alt*-(propylene glycol, ethylene glycol)				
	Chromatography	Toluene/ethanol (93 : 7), toluene/ethanol (90 : 10)		3331
Alkyd resins	Turbidimetric titration	Acetone/water		1454
		Butanone/methanol		2768
		Dioxane/water		1454
Poly(benzene-1,3,5-triacetic acid-*co*-decamethylene glycol)				
	Chromatography	Benzene	SEC	1541
	Sedimentation velocity	Butanone	Ultracentrifuge	3105
Poly(ω-caprolactone), see Poly(oxy(1-oxohexamethylene)				
Poly(diethylene glycol-1,4-*trans*-cyclohexanedicarboxylate)				
	Fractional precipitation	Chloroform/methanol	35°C	2745
Poly(ethylene terephthalate), see Poly(oxyethyleneoxyterephthaloyl)				
Poly(2,4-hexadiyn-1,6-ylene adipate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1814
Poly(β-hydroxyalkanoate)s	Chromatography	Chloroform	SEC, styragel + ultrastyragel	1678
			SEC, ultrastyragel	1254
Poly(1,2-dihydroxybenzene phthalate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	3384
Poly(1,3-dihydroxybenzene phthalate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	3384
Poly(1,4-dihydroxybenzene phthalate)				
	Chromatography	Tetrafydrofuran	SEC, μ-styragel, 25°C	3384
Poly(4-hydroxybutyrate)	Chromatography	Chloroform	SEC, ultrastyragel, 25°C	2
Poly(β-hydroxybutyrate)	Chromatography	Chloroform	SEC, ultrastyragel, (S)-hydroxybutyrate	3592
			SEC, μ-styragel, 40°C, [(R)-3-hydroxybutyrate]	1458,1549
	Extraction	Acetone	Syndiotactic, (R, S)-hydroxybutyrate	1326
	Fractional precipitation	Chloroform/methanol	(S)-hydroxybutyrate	3592
		Methylene chloride/diethyl ether	β-D,L-hydroxybutyrate	1794
Poly(11-hydroxyundecanoic acid) see Poly(oxy (1-oxoundecamethylene))				
Poly((isophthalic acid maleic anhydride)-*alt*-propylene glycol)				
	Chromatography	Acetone/*n*-heptane	Precipitation chromatogr., 27–50°C	502
Poly((isophthalic terephthalic acid (1 : 1))-*alt*-bis(4-hydroxyphenyl)isopropylidene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	1843
Poly(lauric acid-*co*-glycerol)	Fractional precipitation	Toluene/*n*-heptane	Alkyd resin	2329
Poly((maleic anhydride; phthalic anhydride)-*alt*-(ethylene glycol; cyclohexanol))				
	Fractional precipitation	Acetone/petroleum ether		1593
Poly(1-(6-[4-(4-methoxyphenoxycarbonyl)phenoxy]hexyloxycarbonyl)-1-methylethylene))				
	Chromatography	Tetrahydrofuran	SEC	930
	Fractional precipitation	Chloroform/methanol		2380

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(oxy-2(4-(carbazol-9-yl)butyl)trimethyleneoxyadipoyl)				
	Fractional precipitation	Tetrahydrofuran/*n*-hexane		3166
Poly(oxy-2(4-(carbazol-9-yl)butyl)trimethyleneoxysebacoyl)				
	Fractional precipitation	Tetrahydrofuran/*n*-hexane		3166
Poly(oxy-2(4-(carbazol-9-yl)butyl)trimethyleneoxysuccinyl)				
	Fractional precipitation	Tetrahydrofuran/*n*-hexane		3166
Poly((oxy-2′,5′ -dihexyl-*p*-terphenyl-1,4-oxy-2-2-dihexyl terephthaloyl)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 25°C	1546
Poly(oxy(1,4-dimethyltetramethylene)oxysebacoyl)				
	Chromatography	Chloroform	SEC, styragel	2426
Poly(oxy(2,2′-methylethyltrimethylene)oxysebacoyl)				
	Chromatography	Chloroform	SEC, styragel	2426
Poly(di-(4-oxy-3-methylphenyl-2,2-propane oxy terephthaloyl))				
	Fractional precipitation	Phenol : tetrachloroethylene (3 : 1)/heptane		2549
Poly(oxy(1-oxy2,2-dimethyl-3-isopropylidenyl)trimethylene)				
	Chromatography	Chloroform	SEC, styragel	2493
		Tetrahydrofuran	SEC, styragal	2493
Poly(oxy(1-oxo-2-dimethyletylene)				
	Chromatography	Hexafluoroisopropanol	SEC, μ-styragel	2616
Poly(oxy(1-oxo-2,2-dimethyltrimethylene) *(Poly(pivalolactone))*				
	Chromatography	*m*-Cresol	SEC, 120°C, 112°C	396
Poly(oxy(1-oxo-3-ethyltrimethylene) *Poly(β-ethyl-β-propiolactone)*				
	Fractional precipitation	Chloroform/ethyl ether	Crystalline/amorphous	3191
Poly(oxy(1-oxo-2-methylethylene)) *Poly(D,L-lactic acid)*				
	Chromatography	Tetrahydrofuran	SEC, 30°C	3342
Poly(oxy(1-oxo-3-methyltrimethylene) *Poly(α-methyl-β-propiolactone), Poly(α-butyrolactone)*				
	Chromatography	Tetrahydrofuran	SEC, styragel	3548,3549
	Fractional precipitation	Chloroform/acetone	Crystalline /amorphous	3191
		Chloroform/chloroform–propylene glycol (1 : 2)		2208
Poly(oxy-1-oxo(2-(1,2,3,-tetrahydro-5-methyl-2,4-dioxo-1-pyrimidinyl)methyltrimethylene)				
	Chromatography	Dimethylformamide	SEC, Sephadex	1228
Poly(oxy(1-oxoethylene) *Poly(glycolate)*				
	Chromatography	Hexafluoroisopropanol	SEC, μ-styragel	2616
Poly(oxy(1-oxohexamethylene)	Chromatography	Methylene chloride	SEC, ultrastyragel, 25°C	1793
		o-Dichlorobenzene	SEC, 135°C	2713
		Tetrahydrofuran	SEC, shodex	863
			SEC, μ-styragel	2869
			SEC, styragel	860,2024
			SEC, ultrastyragel	794,795,796
	Fractional precipitation	Benzene/isooctane	34°C	2583
		Benzene/*n*-heptane		1738
		Benzene/petroleum ether	25°C	668
			Oligomers	1160
		Butanone/*n*-hexane	Oligomers	1160
	Fractional solution	Benzene/isooctane mixtures	Column extraction, temp.	2583
	Sedimentation velocity	Benzene	30°C	437
	Turbidimetric titration	Benzene/petroleum ether	Oligomers	1160
Poly(oxy(1-oxopentamethylene) *Poly(β-valerolactone)*				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2711
Poly(oxy(1-oxotrimethylene) *Poly(β-propiolactone)*				
	Chromatography	*m*-Cresol	SEC, styragel, 100°C	3430
	Fractional precipitation	Chloroform/petroleum ether	30°C	668
		Chloroform/toluene	Lowering temp.	2781
Poly(oxy(1-oxoundecamethylene)				
	Fractional precipitation	Benzene/methanol	55°C	1953
Poly(oxy-1-oxo(2(1,2,3,4-tetrahydro-2,4-dioxo-1-pyrimidinyl)methyltrimethylene)				
	Chromatography	Dimethylformamide	SEC, Sephadex	1228
Poly(oxy(1,1,2,2-tetramethyl(ethylene-oxysebacoyl)				
	Chromatography	Chloroform	SEC, styragel	2426
Poly(oxy-2″,3″,5″-triphenyl-*p*-quinquophenyl-4,4′′′′-oxy-2,5-diphenyl terephthaloyl)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 25°C	1546
Poly(oxy-2″,3″,5″-triphenyl-*p*-terphenyl-4,4′′′′-oxy-2,5-diphenyl terephthaloyl)				
	Chromatography	Tetrahydrofuran	SEC, ultastyragel, 25°C	1546

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(oxyadipoyloxydecamethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1512
Poly(oxybutyleneoxy adipoyl) *Poly(butylene adipate)*				
	Chromatography	Tetrahydrofuran	SEC	2241
			SEC, ultrastyragel, oligomers	3121
Poly(oxycarbonyl-1,2-phenylene) *Poly(2-hydroxy-benzoate)*				
	Chromatography	*m*-Cresol/chlorofrom (15/85)	SEC, styragel, 38°C	2806
Poly(oxycarbonyl-1,3-phenylene) *Poly(3-hydroxybenzoate)*				
	Chromatography	Chloroform	SEC, Styragel	1791
Poly(oxycarbonyl-1,4-phenylene-*co*-oxycarbonyl-2,6-naphthylene)				
	Chromatography	3,5-bis(Trifluoromethyl)/decalin	SEC, 60°C	1819
	Fractional precipitation	3,5-bis(Trifluoromethyl)phenol/ decalin	90°C	1803,1819
Poly(oxycarbonyl-1,4-phenylene-isopropylidene-1,4-phenyleneoxycarbonyl-1,4-phenylene-1,6-indanylene)				
	Chromatography	Cyclohexane/dioxane	Precipitation chromatogr., thermal gradient, acid wahed glass beads	3045
Poly(oxycarbonyl-1,4-phenyleneoxytherephthaloyloxy-1,4-phenylenecarbonyloxydecamethylene)				
	Fractional precipitation	Trifluoroacetic acid/*o*-xylene		3277
	Sedimentation velocity	Trifluoroacetic acid	25°C	3277
Poly[2,2-bis(4-oxycyclohexyl)propane adipate]				
	Chromatography	Methylene dichloride		1003
Poly(oxydecamethyleneoxysebacoyl)				
	Fractional solution	Benzene/ethanol	Column elution	3157
		Tetrachloroethane/*n*-heptane	Column elution	3157
Poly(di(oxethylene)oxyadipoyl)	Chromatography	*n*-Hexane–butanone–ethanol	Column extraction, silica gel	890
	Extraction	Butanone/*n*-hexane	Oligomers, column extraction	1159
	Turbidimetric titration	Butanone/petroleum ether		1159
Poly(oxyethylene oxyethyleneethyleneoxyadipoyl)				
	Chromatography	Tetrahydrofuran	SEC, ultratyragel, oligomers	3121
Poly(di(oxyethylene)oxyterephthaloyl)				
	Fractional precipitation	Chloroform/ethanol	30°C	1944,2738,2744
Poly(oxyethyleneoxyadipoyl)	Fractional crytallization	Ethanol	Hot	1324
	Fractional precipitation	Benzene/petroleum ether		2693
		Chloroform/petroleum ether		1594
Poly(di(oxethylene)oxycarbonyl-1,4-cyclohexylenecarbonyl)*trans*				
	Chromatography	Methylene dichloride	SEC, styragel	893
Poly(oxyethyleneoxycarbonyl-1,4-phenyleneoxyethylenoxy-1,4-phenylene carbonyl)				
	Fractional precipitation	1,1,2,2-Tetrachloroethane/*n*-heptane, *o*-chlorophenol(3 : 2)	20°C	1560
Poly(di(oxyethylene)oxyisophthaloyl)				
	Fractional precipitation	Chloroform/methanol	30°C	2740
Poly(oxyethyleneoxyisophthaloyl)				
	Chromatography	Tetrahydrofuran	SEC, 37°C	333
	Fractional precipitation	Tetrachloroethane/*n*-heptane		3515
Poly(oxyethyleneoxyphthaloyl)	Fractional precipitation	Ethylene dichloride/petroleum ether		1303
Poly(oxyethyleneoxysebacoyl)	Chromatography	Tetrahydrofuran	SEC, 37°C	333
	Fractional precipitation	Hot methanol		1324
Poly(oxyethyleneoxysuberoyl)	Chromatography	Tetrahydrofuran	SEC, 37°C	333
	Fractional precipitation	Hot methanol		1324
Poly(oxyethyleneoxysuccinoyl)	Chromatography	*m*-Cresol	SEC, styragel, 100°C	3431
Poly(oxyethyleneoxyterephthaloyl) *Poly(ethylene terephthalate)*				
	Chromatography	Chloroform/ethanol (94/6)	TLC, Kiesselgel oligomers	761
		m-Cresol	SEC	804
			SEC, styragel, 125°C	1278
		Methylene dichloride/ hexafluoroispropanol (70/30)	SEC, styragel	1894
		Nitrobenzene/tetrachloroethylene (5/95)		2521
			SEC, styragel 100°C	3310
		o-tert-Butylphenol	SEC, styragel, 110°C	2792
		Tetrahydrofuran	Samples were dissolved in phenol–tetrachloroethane- (3 : 2) SEC, 37°C	333

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
			SEC, μ-styragel, 40°C, PET oligomers	2972
			SEC, oligomers	2972
			SEC, styragel, 40°C, analytical and preparative, oligomers	2972
		Trifluoroacetic, acid/chloroform(10 : 90)eluent *o*-Chlorophenol	SEC, 50°C	2162
		Trifluoroacetic acid chloroform(20 : 80) solvent	Charcoal	450
	Coacervation	Phenol–tetrachloroethane/*n*-heptane		3290
	Distribution between immiscible liquids	*m*-Cresol/petroleum ether		3300
	Extraction	Hot chloroform	Oligomers	1906
		Methylene chloride–ethanol (4 : 6)		475
	Fractional precipitation	1,1,2,2-Tetrachloroethane/*n*-heptane, *o*-chlorophenol (3 : 2)	20°C	1560
		Dimethylformamide	Lowering temp.	3300
		m-Cresol/ligroin (b.p. 100°C)	50°C	3300
		o-Chlorophenol/*n*-heptane		2160,2162,2242, 2247,2255
		Phenol–1,2,4-trichlorobenzene/*n*-heptane		3140
		Phenol–tetrachloroethane(1 : 1)/ligroin	25°C	3400
			65°C	1111
		Phenol–tetrachloroethane/*n*-heptane		1023,1024,1278,2712
		Phenol/cyclohexane	70°C	1723
	Fractional solution	Phenol–tetrachloroethane(3 : 2)/*n*-nonane	Film extraction, 95°C	935
		Phenol/*n*-heptane	56°C	2506
	Turbidimetric titration	Phenol–tetrachloroethane/*n*-heptane		2006
Poly(tri(oxyethylene)oxyterephthaloyl)				
	Fractional precipitation	Chloroform/methanol	30°C	2741
Poly(oxyhexamethylene oxy-4,4′-biphenyleneoyl)				
	Chromatography	Phenol + tetrachloroethane	SEC, 50°C	3214
Poly(oxyhexamethyleneoxy)(α,α-dibutyl)sebacoyl)				
	Fractional precipitation	Benzene/methanol		241,3481
	Fractional precipitation	Benzene/methanol		241,3481
Poly(oxyhexamethyleneoxysebacoyl)				
	Chromatography	Chloroform	SEC, styragel	2426
	Fractional solution	Benzene/methanol, tetrachloroethane/*n*-heptane	Column elution	3157
	Turbidimetric titration	Carbon tetrachloride/*n*-heptane	20°C	1763
Poly(oxyhexane oxyadipoyl)	Chromatography	Tetrahydrofuran	SEC, styragel, oligomers	3121
Poly(oxyisophthaloyloxy-1,4-phenylene-(2-phenyl)phthalimidine-1,4-phenylene)				
	Fractional precipitation	Tetrachloroethane/tetrahydrofuran		790
	Sedimentation velocity	Tetrahydrofuran	20°C	790
Poly(oxyisophthaloyloxy-1,4-phenylene-phthalidylidene-1,4-phenylene)				
	Chromatography	Tetrachloroethane/ethanol	Precipitation chromatogr., 15–65°C	789
	Extraction	Phenol–tetrachloroethane(1 : 3)/*n*-heptane		789
	Turbidimetric titration	Tetrachloroethane/ethanol		789
Poly(oxyneopentyleneoxyadipoyl)				
	Fractional precipitation	Chloroform/hexane		2742
Poly(oxypropyleneoxyisophthaloyl)				
	Fractional precipitation	Chloroform/methanol	30°C	2576
Poly(oxypropyleneoxyterephthaloyl)				
	Chromatography	Chloroform	SEC, styragel or bio-beads	337
		Tetrahydrofuran	SEC, styragel or bio-beads	337
	Extraction	Methylene dichloride/ethanol (4/6)	Cyclic oligomers	475
	Fractional precipitation	Chloroform/methanol	30°C	2576
Poly(oxysebacoyloxydecamethylene)				
	Turbidimetric titration	Carbon tetrachloride/*n*-heptane	20°C	1763

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(oxysuccinoyloxyhexamethylene)				
	Chromatography	Butanone–cyclohexane	Precipitation chromatogr., 70–27°C and 70–40°C	1210
		Butanone–cyclohexane (3 : 1)/cyclohexane		1210
		Butanone (1 : 4)	Precipitation chromatogr., 27–70°C	1210 2321
Poly(oxytartaroyloxyhexamethylene)				
	Chromatography	Benzene	Urea	1671
Poly(oxyterephthaloyloxy(3-methylphenylene)isopropylidene-1,4-(3-methylphenylene) *Poly(arylate)*				
	Fractional precipitation	Dichloroethane/*n*-hexane		3244
Poly(oxyterephthaloyloxy-1,4-chlorophenyleneisopropylidene-1,4-chloro/phenylene)				
	Chromatography	Chloroform		343
		Tetrahydrofuran	SEC, styragel, 25°C	1843
	Fractional precipitation	Dichloroethane/*n*-hexane		3244
Poly(oxyterephthaloyloxy-1,4-phenylenefluoren-9-ylidene-1,4-phenylene) *Poly(arylate)*				
	Fractional precipitation	Carbon tetrachloride–phenol (3 : 1)/*n*-heptane		791
		Dichloroethane/*n*-hexane		3025
Poly(oxytetramethyleneoxyadipoyl) *Poly(tetramethylene adipate)*				
	Chromatography	Benezene–ethanol (3 : 1)	TLC, silica gel	3327
		Tetrahydrofuran	SEC, 37°C	333
Poly(oxytetramethyleneoxyterephthaloyl) *Poly(tetramethylene terephthalate)*				
	Chromatography	Chloroform–*n*-hexane(10 : 5) and chloroform–*n*-hexane(10 : 3)	Cyclic oligomers extracted with *o*-dichlorobenzene, TLC, silica gel	812
		Hexafluoroisopropanol		3014
		n-Hexane–dioxane (7 : 3)	HPLC, lichrosorb, cyclic oligomer, extracted with *o*-dichlorobenzene	812
	Extraction	Methylene dichloride–ethanol (4 : 6)	Cyclic oligomers	475
	Fractional precipitation	1,1,2,2-Tetrachloroethane–phenol/*n*–heptane		1354,2303
Poly(oxytrimethyleneoxysuccinyl) *Poly(trimethylene succinate)*				
	Chromatography	Tetrahydrofuran	SEC, styragel	1512
Poly(phenylhydroquinone-*co*-terephthalic acid)				
	Fractional solution	*o*-Dichlorobenzene : *p*-chlorobenzene (50 : 50)/methanol	23°C	1798
Poly(phthalic acid-*alt*-pentaerythrytol)				
	Fractional precipitation	Acetone	Lowering temp.	364
Poly(phthalic anhydride-*alt*-(coconut oil;glycerol))				
	Fractional precipitation	Acetone/water	Alkyd resin	896
Poly(phthalic anhydride-*alt*-glycerol)				
	Fractional precipitation	Acetone/water	Alkyd resin	897
		Butanone–acetone/methanol		362
		Butanone–acetone/methanol–water		362
Poly(phthalic anhydride; lauric acid)-*alt*-glycerol)				
	Fractional precipitation	Toluene/*n*-heptane		2339
Poly(phthalidylidene-1,4-phenylene oxy-1,4-phenylene carbonyl-1,4-phenyleneoxy-1,4-phenylene)				
	Fractional precipitation	*N*,*N*-Dimethyformamide/ethanol		3499
Poly(pivalolactone) see Poly(oxy(1-oxo-2,2-dimethyltrimethylene)				
Poly(sebacic acid-*alt*-(hexanediol;hexanetriol)				
	Fractional precipitation	Benzene/methanol		242
Poly(sebacic, adipic, tartaric acid)-*alt*-(1,6-hexanediol-1,10-decanediol)				
	Fractional precipitation	Benzene/methanol		240,3481
	Fractional solution	Benzene/methanol		240,3481
Poly(succinic acid-*co*-hexanediol) see Poly(oxysuccinoyloxyhexamethylene)				
	Fractional precipitation		Benzene/methanol	242
Poly((succinico;pimelic acid)-*alt*-(hexanediol)				
Poly(terephthalic acid-*co*-ethylene glycol) see Poly(oxyethyleneoxyterephthaloyl)				
Poly(ditrimethylene glycol terephthalate)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 30°C	3068
	Fractional precipitation	Toluene/methanol		1102

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
3.4. POLY(URETHANES) AND POLY(UREAS)				
Poly(urethanes) (General)	Fractional precipitation	Acetone, benzene, ethyl acetate, chloroform and their mixtures with DMF, dimethylformamide/di-*n*-pentyl ether		2992
Poly(poly(caprolactone)-*co*-4,4′diphenylmethane diisocyanate-*co*-2,4-butanediol)				
	Fractional precipitation	Dimethylacetamide/diphenyl ether		2991
Poly(poly(caprolactone)-*co*-4,4′diphenylmethane diisocyanate-*co*-ethylene diamine)				
	Fractional precipitation	Dimethylacetamide/*n*-heptane–ethyl ether		2842
Poly((caprolactone)-*co*-trimethylol ethane-*co*-4,4′-diphenylmethane diisocyanate-*co*-ethylene diamine)				
	Fractional precipitation	Dimethylacetamide/*n*-heptane–ethyl ether		2842
Poly(poly(caprolactone)diol-*co*-4,4′-diphenylmethane diisocyanate-*co*-1,5-pentanediol)				
	Fractional precipitation	Dimethylformamide/dimethylformamide–water(4 : 1)	30°C	2935
Poly((ethylene diamine-*co*-2,4-toluenediisocyanate)-*co*-poly(tetramethylene oxide))				
	Chromatography	Dimethylformamide + 0.01 M lithium bromide		2505
Poly(ethylene diamine-*co*-2,4-toluenediisocyanate-*co*-poly-(oxytetramethyleneoxyadipoyl)				
	Chromatography	Dimethylformamide + 0.01 M lithium bromide	SEC	2505
Poly[4,4′-diisocyanodiphenylmethane-*co*-2,4-hexadene-1,6-diol-*co*-poly(propylene glycol)]				
	Chromatography	Tetrahydrofuran	SEC, styragel	1382
Poly(oxycarbonyloxy-1,4-phenyleneureylene-1,4-phenyleneureylene-1,4-phenylene)				
	Turbidimetric titration	Dimethylformamide		1739
Poly(oxycarbonyloxy-1,4-phenyleneureylene-1,5-naphthyleneureylene-1,4-phenylene)				
	Turbidimetric titration	Dimethylformamide		1739
Poly(oxycarbonyloxy-1,4-phenyleneureylene-1,4-phenylenemethylene-1,4-phenyleneureylene-1,4-phenylene)				
	Turbidimetric titration	Dimethylformamide		1739
Poly(oxycarbonyloxy-1,4-phenyleneureylene-1,4-toluylene ureylene 1,4-phenylene)				
	Turbidimetric titration	Dimethylformamide		1739
Poly(oxycarbonyloxy-1,4-phenyleneureylenehexamethyleneureylene-1,4-phenylene)				
	Turbidimetric titration	Dimethylformamide		1739
Poly(oxycarbonyl(phenylimino)ethylene)(Poly(*N*-phenylethylenimine-*alt*-carbon dioxide)				
	Extraction	Methanol	Homopolymer	3029
Poly(poly(oxyethyleneoxyadipoyl)-*co*-4,4′-diphenylmethane diisocyanate-*co*-1,4-butanediol)				
	Fractional precipitation	Dimethylformamide/dibutyl ether		2991
Poly(oxyethyleneoxycarbonylimino-1,4-phenylene-methylene-1,4-phenylene)				
	Extraction	Dimethylformamide/acetone	30°C	254
	Fractional solution	Dimethylformamide/acetone		255
Poly(poly(oxypropylene)-*co*-toluene diisocyanate)				
	Fractional solution	Benzene/isooctane	Column extraction, 34°C	1421
Poly(oxytetramethylene oxycarbonylimino hexamethyleneiminocarbonyl)				
	Fractional precipitation	Benzene/2,2,4-trimethylpentane	25°C	42
Poly(poly(oxytetramethyleneoxyadipoyl)-*co*-4,4′-diphenylmethanediisocyanate-*co*-1,4-butenediol)				
	Fractional precipitation	Dimethylformamide/dibutyl ether		2991
Poly(poly(oxytetramethyleneoxyadipoyl)-*co*-toluylenediisocyanate)				
	Chromatography	Ethyl acetate	SEC, styragel, 20°C	3328
Poly(oxytetramethyleneoxycarbonylimino-2,4-toloyleneiminocarbonyl)				
	Chromatography	Dimethylformamide + lithium bromide (0.01%)	SEC, μ-styragel	2704
Poly(oxytetramethyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)				
	Chromatography	Tetrahydrofuran	SEC, styragel	488
Poly(oxytetramethyleneoxycarbonyliminohexamethyleneiminocarbonyl)				
	Chromatography	Formic acid/water (88%)	Paper partition chromatogr.	157
	Fractional solution	*m*-Cresol/*n*-hexane		2400
Poly(ureylene(1-ethoxycarbonyl)pentamethylene) *Poly(urea of ethyl ester* L-*lysine)*				
	Coacervation	Trichloroethane–phenol/*n*-heptane		1764

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
3.5. POLY(AMIDES) AND POLY(IMINES)				
Poly(amido amine)	Chromatography	Dimethylformamide		2130
Poly((6-aminocaproic acid; adipic acid)-*alt*-hexamethylenediamine)				
	Fractional precipitation	Phenol/water	70–90°C	243
Poly(butylimino carbonyl)	Fractional precipitation	Carbon tetrachloride/methanol		1227
Poly((*N-tert*-butylimino)ethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 20°C	1089
Poly(butyliminocarbonyl) *[Poly(butylisocyanate)]*				
	Chromatography	Benzene	SEC	468
	Fractional precipitation	Cabon tetrachloride/methanol		3264
		Tetrachloroethane/methanol		2978,3274
	Sedimentation analysis	Carbon tetrachloride		468
Poly(ε-caprolactam) see Poly(imino(1-oxohexamethylene))				
Poly(caprolactam-*co*-hexamethylenediamine-*co*-adipic acid-*co*-azelaic acid)				
	Extraction	Methanol/water mixtures	35°C	2543
	Turbidimetric titration	Methanol/water mixtures		2543
Poly(ethylimino-*trans*-fumaroylethylamino-1,4-(2-methyl)phenylene-methylene-1,4-(2-methyl)phenylene)				
	Fractional precipitation	Dimethylformamide/water		876
		Methanol/water		876
	Turbidimetric titration	Dimethylformamide/water		875
Poly((R)-1-deuterio-*n*-hexyl isocyanate)				
	Chromatography	Chloroform	SEC, styragel, 40°C analytical and preparative	2452
Poly(*n*-hexyl isocyanate)	Fractional precipitation	Chloroform/methanol		787
	Sedimentation equilibrium	Hot hexane, hot benzene, hot hexafluoroisopropanol, *m*-cresol	Ultracentrifuge, 25°C	2851
Poly(hexyliminocarbonyl) *Poly(hexylisocyanate)*				
	Chromatography	Chloroform	SEC, styragel, 40°C	1451
	Coacervation	Tetrachloromethane/methanol		358
		Toluene/methanol	25°C	300,301
	Fractional precipitation	Benzene/methanol		1500
		Carbon tetrachloride/methanol		1451,1810,2309, 2787,3068
	Sedimentation equilibrium	Hexane	25°C	1451
Poly(imino(1-oxodecamethylene)) *[Nylon 10]*				
	Distribution between immiscible liquids	Phenol/water	70°C	1368
Poly(imino(1-oxododecamethylene)) *[Nylon 2]*				
	Chromatography	Benzyl/alcohol	SEC	2522
		Hexamethylphosphotriamide	SEC, styragel, 85°C	2507
		Methylene dichloride	SEC, μ-styragel	1465
		Tetrahydrofuran	SEC	834
	Turbidimetric titration	*m*-Cresol/*n*-heptane		2287
Poly(imino(1-oxohexamethylene)) *[Nylon 6]*				
	Chromatography	Benzyl alcohol	SEC, lichrospher, 100°C	2523
			SEC, 100°C	2522
		Formic acid/water	Paper partition, 20°C	157
		Hexamethylphosphortriamide	SEC, 85°C	2507
		Hydrochloric and 0.1M	SEC, biogel	320
		m-Cresol	SEC, 105°C, 133°C	235,236
			SEC, styragel, 112°C	616
		Methylene chloride	SEC, μ-styragel	1465
		Phenol–ethanol–water(1 : 2.6 : 1.6)	SEC, Sephadex	3422
		Phenol–*n*-propanol (7 : 3)	SEC, 65°C	2523
		Trifluoroethanol	SEC, styragel	2098
	Distribution between immiscible liquids	Phenol/water	Continuous	807
	Fractional crystallization	Phenol–tetrachloroethane (1 : 1)/ethanol–*n*-heptane	25°C, stirred	328
	Fractional precipitation	Formic acid–water (85 : 15)–5% lithium chloride/water–5% lithium chloride	Amorphous	325

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		m-Cresol/cyclohexane		1516
		m-Cresol/ethyl ether	26°C	1192
		m-Cresol/ligroin		1111,1125, 2548,3294
		Phenol–tetrachloroethylene (1 : 1)/*n*-heptane	25°C	2694
		Phenol/methanol–water		1821
		Phenol/water	70°C	3482
	Fractional solution	Conc. formic acid	20–90°C	666
		Formic acid/propyl acetate	Film extraction	935
		Phenol/ethylene glycol/water	Coacervation	3289
		Phenol/water mixtures	Column extraction, 70°C	1010,2096
		Water–methanol (81.5 : 18.5)/phenol	Column extraction, linear and branched	2097
	Sedimentation equilibrium	Formic acid–water (85%) + 2M potassium chloride	25°C	2893
	Sedimentation velocity	Hexafluoroisopropanol		2542
	Summative precipitation	Formic acid (85%) water		638
	Thermal diffusion	Formic acid–water 90%		1863
	Turbidimetric titration	Formic acid/dibutyl ether		1274
		Hydrochloric acid/Aq. ammonium sulfate		2642
		m-Cresol–isobutanol–petroleum ether (1 : 9) petroleum ether–tetrachloroethane (9 : 1)		1110
		m-Cresol/*n*-heptane		2548
		m-Cresol/petroleum ether		665
		Phenol 94%/50% methanol		1011
		Phenol–water–96% ethanol (1 : 1 : 1)/46% ethanol		214
Poly(imino(1-oxooctamethylene)) *[Nylon 8]*				
	Chromatography	Methylene dichloride	SEC, μ-styragel	1465
	Fractional solution	Phenol/water–methanol	Coacervation, 25°C	525
Poly(imino(1-oxotetramethylene) *[Nylon 4]*				
	Chromatography	*m*-Cresol	SEC, 90°C	1385
		Methylene dichloride	SEC, μ-styragel	1465
	Fractional precipitation	*m*-Cresol–benzene(6 : 1)/*n*-heptane		2764
Poly(imino(1-oxoundecamethylene) *[Nylon 11]*				
	Chromatography	Benzyl alcohol	SEC, μ-styragel	2953
		Hexamethyl phosphotriamide	SEC, 85°C	2507
		Methylene dichloride	SEC, μ-styragel	1465
	Fractional precipitation	*m*-Cresol/ethyl alochol or *n*-heptane		3192
Poly(iminoperchloroterephthaloyliminohexamethylene) *Poly(hexamethylene perchloroterephthalamide)*				
	Fractional precipitation	*m*-Cresol/cyclohexane		1875
Poly(imino-1,4-phenyleneimino-terephthaloyl)				
	Chromatography	Sulfuric acid 90%	Preparative SEC, silica gel	128
		Sulfuric acid 96%	SEC	130
		Tetrahydrofuran	SEC, Shodex	2432
Poly(imino-1,3-phenyleneimino-isopthaloyl) *Poly(m-phenylene isophthalamide)*				
	Fractional precipitation	Dimethylformamide + 5% lithium chloride/dioxane		3276,3385
	Sedimentation velocity	Dimethylformamide + lithium chloride	Ultracentrifuge	2375
Poly(imino-1,4-phenylenecarbonylhydroazoterephthaloyl) *Poly(terephthalamide-p-benzohydrazide)*				
	Fractional precipitation (50 : 30–50 : 70)/dimethyl	Dimethylsulfoxide : dichloro-methane sulfoxide	25°C, the precipitate is redisolved by temp. lowering	2816
Poly(iminoadipoyl iminodecamethylene)*[Nylon 610]*				
	Distribution between immiscible liquids	Phenol/water	Continuous, Nylon 610	807
Poly(iminoadipoyl iminohexamethylene) *[Nylon 66]*				
	Chromatography	Formic acid–water (88%)	Partition chromatogr. 20°C	157
		Hexafluoroisopropanol	SEC, styragel	1395
		Methylene dichloride	SEC, μ-styragel	1465

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Distribution between immiscible liquids	Phenol/water	Continous, Nylon 66	807
	Fractional precipitation	*m*-Cresol/cyclohexane		1375,1516
		Phenol/water	Nylon 66	3165
	Sedimentation gradient	Carbon tetrachloride, *m*-cresol, cyclohexane	Ultracentrifuge	3204
	Turbidimetric titration	*m*-Cresol/cyclohexane	Nylon 66	1376
		m-Cresol/*n*-heptane		2287
Poly(iminocarbonyl(3-carboxybiphenyl)4,4′-ylenecarbonylimino-1,4-phenylenemethylene-1,4-phenylene)				
	Fractional precipitation	*m*-Cresol/cyclohexane		3128
Poly(iminocarbonyl-1,4-(2,5-methoxycarbonyl)phenylenecarbonylimino-1,4-(3-phthalidyl)phenylene-methylene-1,4-phenylene)				
	Fractional solution	Trichloroethylene-phenol-*n*-heptane		2232
Poly (iminocarbonyl-1,4-phenylene) [Poly(*p*-benzamide)]				
	Chromatography	Sulfuric acid 92%	SEC, preparative, silica gel	129,130
Poly(iminocarbonyl-1,4-phenyleneiminocarbonyl tetramethylene)				
	Fractional precipitation	Dimethylacetamide/Lithium chloride/propionic acid		39
Poly(iminoethylene)	Chromatography	Water	SEC, Sephadex	2673
		Water 0.109 N sodium chloride	SEC, biogel	1369
	Curtain electrophoresis	Acetic acid–water		2584
Poly(iminohexamethylene iminosebacoyl)				
	Chromatography	Formic acid–water (88%)	Paper partition	157
		Methylene dichloride	SEC, μ-styragel	1465
	Mechanical fractionation	Tyler screens	Interfacial condensation	2272
Poly(iminoisophthaloyl iminodecamethylene)				
	Turbidimetric titration	*p*-Cresol/acetone		3033
Poly(imino(1-oxohexamethylene)*co*-imino(1-oxododecamethylene) *Poly(6-caprolactam-co-12-dodecanelactam)*				
	Fractional precipitation	*m*-Cresol/*n*-heptane		44
Poly(iminoterepthaloylimino-2-methyl-4,4-dimethylhexamethylene)				
	Fractional precipitation	Dimethylformamide/ethyl acetate		1283
Poly(iminoterepthaloylimino-3-methyl-4,4-dimethylhexamethylene)				
	Chromatography	Methylene dichloride	SEC, μ-styragel	1465
Poly(iminoterepthaloylimino-2-phenylenefluorene-9-ylidene-1,4-phenylene)				
	Fractional precipitation	Dimethylformamide/acetone		2765
		Dimethylformamide/acetone–ethanol (2 : 1)		2765
Poly(iminoterephthaloylimino-1,4-phenylenephthalidylidene-1,4-phenylene)				
	Coacervation	Dimethylformamide (3% HCl 1 N)/ethanol		2374
		Dimethylformamide (acidified HCl)/ethanol		2376
	Fractional precipitation	Dimethylformamide/acetone		2765
		Dimethylformamide/acetone–ethanol (2 : 1)		2765
Poly(iminotetramethylene iminoisophthaloyl)				
	Turbidimetric titration	Cresol/acetone		3033
Poly(isopropylmethacrylamide)	Fractional precipitation	Ethanol/diethyl ether		3207
Poly(*trans*-2,5-dimethyl-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)				
	Chromatography	Dichloroethylene/cyclohexane	Column elution, 30°C	2240
Poly(4,4′-methylenedianiline-*co*-1-chloro 2,3-epoxypropane) *(epoxy resin)*				
	Chromatography	Acetonitrile–water gradient silica gel, preparative	HPLC reverse phase, graft	1145
		Tetrahydrofuran	SEC, styragel, preparative	1145
Poly[(dimethylimino) decamethylene bromide]				
	Fractional solution	Ethylene glycol monoethyl ether/diisopropyl ether		2899
Poly(*trans*-2,5-dimethylpiperazinediylisophthaloyl)				
	Fractional precipitation	Chloroform–1–5%phenol/*n*-hexane		2292
		Methylene chloride–ethanol, (10%) *n*-hexane		1694,2802
		Methylene chloride-methanol (85 : 15) *n*-hexane)		2291
	Sedimentation equilibrium	*N*-methylpyrralidone	25°C	1694

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(oxysebacoyloxyethylene iminoterephthaloyliminoethylene)				
	Chromatography	Methylene dichloride	SEC,μ-styragel	1465
Poly(oxysuccinoyloxyethylene iminoterephthaloyliminoethylene)				
	Chromatography	Methylene dichloride	SEC,μ-styragel	1465
Poly(oxyterephthaloyloxyethylene iminoterephthaloyl iminoethylene)				
	Chromatography	Methylene dichloride	SEC, μ-styragel	1465
Poly(oxytetramethyleneoxysuccinoyl iminohexamethylene iminosuccinoyl)				
	Chromatography	Methylene dichloride	SEC, μ-styragel	1465
Poly(oxytetramethyleneoxyuccinoyl iminododecamethylene iminosuccinoyl)				
	Chromatography	Methylene dichloride	SEC, μ-styragel	1465
Poly(phenylcarbamoyl)iminoethylene) *Poly(1-(N-phenylcarbamoyl) aziridine)*				
	Chromatography	Tetrahydrofuran	SEC, styragel	119
Poly[*N*,*N*′-(*p*-phenylene)-3,3′,4,4′-biphenyl tetracarboxylic acid diamide]				
	Chromatography	Dimethylpyrrolidone	SEC, μ-styragel, 60°C	3114
Poly(*m*-phenyleneisophthalamide)				
	Chromatography	*N*-methyl-2-pyrrolidone,*N*,*N*, dimethylacetamide		1847
Poly(*N*-phenylnorbornene-5,6-dicarboximides)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	144
Poly(piperazine-1,4-diyl-ethylene) *Poly(ethylene-co-piperazine)*				
	Fractional precipitation	Water/acetone		2675
Poly(piperazine-2,4-diyl-sebacoyl)				
	Mechanical fractionation	Tyler screens	Interfacial condensation	2272
Poly(quaterthienyl-*N*-dodecyl pyrrole)				
	Chromatography	Tetrahydrofuran		2515
Poly(terephthalic acid-4-iminobenzohydrazide)				
	Fractional solution	Dimethylsulfoxide/ethyl acetate		1799
Poly(1,2,6,7-tetrahydrobenzo(1,2-C : 4,5-C′) dipyrrole-2,6-diyl-terephthaloyl)				
	Chromatography	*m*-Cresol	SEC, styragel, 95°C	763
Poly(1,2,6,7-tetradhydrobenzo(1,2-C : 4,5-C′) dipyrrole-2,6-diyl-sebacoyl)				
	Chromatography	*m*-Cresol	SEC, styragel, 95°C	763
Poly(1,2,6,7-tetrahydrobenzo(1,2-C : 4,5-C′)dipyrrole-2,6-diyl-carbonyl-1,4-phenyleneoxy-1,4-phenylenecarbonyl)				
	Chromatography	*m*-Cresol	SEC, styragel	763
Poly(1,2,6,7-tetrahydrobenzo(1,2-C : 4,5-C′)dipyrrole-2,6-diyl-adipoyl)				
	Chromatography	*m*-Cresol	SEC, styragel, 95°C	763
Poly(1,2,6,7-tetrahydrobenzo(1,2-C : 4,5-C′)dipyrrole-2,6-diyl-isophthaloyl)				
	Chromatography	*m*-Cresol	SEC, styragel, 95°C	763
Poly(*tert*-thienyl-*N*-dodecyl pyrole)				
	Chromatography	Tetrahydrofuran		2515
3.6. POLY(AMINO ACIDS)				
Poly(γ-benzyl-L-glutamate) *poly(imino(1-(2-benzyloxycarbonylethyl)-2-oxo-ethylene)*				
	Chromatography	Dimethylformamide	SEC	2330
		m-Cresol	SEC, 100°C	1030
		m-Cresol-chlorobenzene/benzoic acid	SEC	405
		Methylene dichloride/methanol	Precipitation chromatogr.	636,637,1085
		Tetrahydrofuran	SEC, styragel	3528
	Dialysis	Dioxane		2203
	Fractional precipitation	Dichloroethane/petroleum ether		3257,3258,3229
		Dioxane/ethanol		88
		Methylene dichloride/methanol		406
	Fractional solution	Formic acid, ethanol (hot)	Extraction	2203
Poly(γ-benzyl-D-glutamate-*co*-benzyl-L-glutamate)				
	Fractional precipitation	Methylene dichloride/methanol		1248
	Sedimentation equilibrium	Dimethylformamide	Ultracentrifuge	1248
Poly(γ-benzyl-L-glutamate-*N*-carboxy anhydride)				
	Fractional precipitation	Methylene dichloride/methanol	25°C	3528
Poly(*N*-carbobenzoxy-L-lysine)-*co*-poly(1-lysine-γ-methyl-L-glutamate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2812
	Fractional precipitation	Dimethylformamide/ethyl ether		2812
Poly(carbobenzoxy-L-lysine-*co*-styrene)				
	Fractional precipitation	Chloroform–hexamethylphosphor-triamide/methanol		334

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Casein-*graft*-poly(vinyl acetate)	Extraction	Acetone	cambiar de grupo?	2225
Deoxyribonucleic acid	Sedimentation velocity	Sodium dodecyl sulfate	Ultracentrifuge	2976
Fibrinogen	Sedimentation	Buffer solution	20°C	841
		Phosphate buffer		840
Gelatin	Coacervation	Water/ethanol		2652,3067
Gelatin-*graft*-poly(methyl acrylate)				
	Extraction	Tetrahydrofuran		2325
Poly(glutamate, sodium salt)	Chromatography	Sodium nitrate (water) 0.1 M	SEC, silica gel	2762
Poly(L-glutamic acid)	Fractional precipitation	Dimethyl sulfoxide/2-propanol		1673
Glycogen	Sedimentation	Water		434
Heparin	Fractional precipitation	Water/ethanol		821,987
Poly(L-histidine) *Poly(imino-1-(4-imidazolylmethyl)-2-oxo-ethylene)*				
	Chromatography	Lithium chloride 0.5 M + 1 M HCl pH = 2.35 buffer	Gel filtration, Dextran gel	2215
Poly(*N*-super-5-(4-hydroxbutyl)-L-glutamine-L-lysine)				
	Fractional precipitation	Methanol/ethyl ether		811
Poly(hydroxybutylglutamine)	Fractional precipitation	Methanol/ethyl ether		2780
Poly(hydroxybutylglutamine-*co*-L-arginine)				
	Fractional precipitation	Methanol–1 M aq. sodium chloride (9 : 1)/ethyl ether		1751
Poly(hydroxybutylglutamine-*co*-glycine)				
	Fractional precipitation	Methanol/ethyl ether		89
Poly(hydroxybutylglutamine-*co*-L-serine)				
	Fractional precipitation	Methanol/ethyl ether		1394
Poly(hydroxybutylglutamine-*ran*-L-leucine)				
	Fractional precipitation	Methanol/ethyl ether		69
Poly(*N*-super-5-omegahydroxyethyl-L-glutamine)				
	Chromatography	Water	SEC, Sephadex, preparative	2095
Poly(*N*-super-5-(3-hydroxypropyl)-L-glutamine				
	Chromatography	Water	SEC	2454
Poly(hydroxypropyl glutamine-*co*-L-alanine)				
	Fractional precipitation	Methanol/ethyl ether		2621
Poly(hydroxypropyl glutamine-*co*-hydroxybutylglutamine)				
	Fractional precipitation	Methanol/ethyl ether		3395
Poly(hydroxypropyl glutamine-*co*-L-leucine)				
	Fractional precipitation	Methanol/ethyl ether		69
Poly(hydroxypropyl glutamine)	Fractional precipitation	Methanol/ethyl ether		3395
Poly(imino(1-(2-*N*-benzylcarbamoylethyl)-2-oxoethylene-*block*-poly(butadiene)				
	Extraction	Methanol–water (1 : 1)		2580
	Fractional precipitation	Hexamethylphosphortriamide/ acetone		2580
Lactoglobulin	Salt fractionation	Ammonium sulfate solution 2.19 M		3215
Poly(α epsilon-l-lysine)	Chromatography	Dimethylformamide + 1% lithium chloride		24
Poly(lysine-*co*-lactic acid)	Chromatography	Chloroform	SEC	228
Poly(ε-C-L-lysyl-γ-benzyl-L-glutamate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2464
Poly(L-lysyl-L-valyl-L-lysine)	Chromatography	Water, 0.1 M potassium phosphate dibasic and monobasic		597
Poly(γ-methyl-D-glutamate-*co*-γ-methyl-L-glutamate)				
	Fractional precipitation	Methylene dichloride/methanol		1248
	Sedimentation equilibrium	*m*-Cresol		1248
Poly(γ-methyl-D,L-methylglutamate-*block*-poly(butadiene)-*block*-poly(γ-methyl-D,L-glutamate)				
	Fractional precipitation	*n*-Hexane–chloroform/ethanol		1248
Polypeptides from (L-,D-Lysine + L-tyrosine)				
	Fractional precipitation	Water + 0.1 *N* chloride acid/0.1 M sodium citrate		2165
Poly(L-proline)	Chromatography	Water	SEC	980
	Sedimentation velocity	Propionic acid	20°C	515
Proteins	Chromatography	Hexafluoroisopropanol	Silk	1464
	Fractional precipitation	Ethanol	Blood plasma, large scale	1921
Poly(sarcosine)	Centrifugation	Dioxane/water	Precipitation chromatogr.	511

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Silk fibroin	Chromatography	Formic acid 99.3% + 4.5% LiCl	SEC, bioglass, 30°C	3229
	Fractional precipitation	LiBr 9.3 M/aq. rivanol (6,9-diamino-2-ethoxy-acridine lactate)	4°C	3229
		Lithium sulfocyanide/water		2986
Poly(*N*-tosyl-L-tyrosine-formal)	Fractional precipitation	Acetone/water		2946
3.7. POLY(SULFIDES), POLY(SULFONES), POLY(SULFONAMIDES)				
Poly(1-(*n*-alkylthio)-1-propynes)				
	Chromatography	Chloroform	SEC, styragel	2068
Poly(1-butenesulfone) see Poly (sulfonylethylethylene)				
Poly(3-*n*-dodecylthiophene-2,5-diyl)				
	Extraction	Hexane, acetone, methylene dichloride		3526
Poly(1,2-epithiopropane)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3436
Poly(ether sulfone)	Chromatography	*N*-methyl-2-pyrrolidine	SEC	1995
Poly(2,5-ethylene (thiene [3,2-b] thiophenediyl) ethylene)				
	Chromatography	Chloroform	SEC, styragel	2798
Poly(2,5-ethylenethiophenediylethylene)				
	Chromatography	Chloroform	SEC, styragel	2798
Poly(5,5′-ethynylene-2,2′-bithiophenediylethynylene)				
	Chromatography	Chloroform	SEC, styragel	2798
Poly(2,5-ethynylene-3-butylthiophenediylethynylene)				
	Chromatography	Chloroform	SEC, styragel	2798
Poly(2,5-ethynylene-3-hexylthiophenediylethynylene)				
	Chromatography	Chloroform	SEC, styragel	2798
Poly(2,5-ethynylene-3-methylthiophenediylethynylene)				
	Chromatography	Chloroform	SEC, styragel	2798
Poly(2,5-ethynylene-3-octylthiophenediylethynylene)				
	Chromatography	Chloroform	SEC, styragel	2798
Poly(2,5-ethynylene-3-phenylthiophenediylethynylene)				
	Chromatography	Chloroform	SEC, styragel	2798
Poly(3-hexylthiophene)	Chromatography	Tetrahydrofuran	SEC	1260,1336
Poly(3-*n*-hexylthiophene-2,5-diyl)				
	Extraction	Hexane, acetone, methylene dichloride, tetrahydrofuran		3526
Poly[2,5-(3-methyl thieno [3,2-b] thiophenylene)]				
	Chromatography	Chloroform	SEC, styragel	2798
Poly(3-octylthiophene)	Chromatography	Tetrahydrofuran		1939
Poly(oxy-1,4-phenylene sulfonyl-1,4-phenylene)				
	Chromatography	Chloroform	SEC, styragel	3571
		N-methypyrrolidone + lithium bromide 0.1 M	SEC, Shodex, 25°C	549
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-(2-chloromethyl)-phenylene-isopropylidene-1,4-phenylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2564
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-(2-vinyl)-phenyleneisopropylidene-1,4-phenylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2564
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)				
	Chromatography	Chloroform	SEC, μ-styragel	455,1280,2927
		Dimethylacetamide, dimethylformamide	SEC	2927
		Tetrahydrofuran	SEC, μ-styragel	2564,2565
			SEC, styragel	2927
	Fractional precipitation	Cholorobenzene/isopropanol		1157
		Dimethylsulfoxide	80°C, lowering temp.	54
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-(2-ethynyl)phenyleneisopropylidene-1,4-phenylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2564
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylene(1-methylethylidene)-1,4-phenylene)				
	Chromatography	Tetrahydrofuran	SEC	3468
Poly(oxyethylenethioethylene)	Fractional precipitation	Chloroform/*n*-hexane		2436
Poly(phenylene sulfide)	Chromatography	1-Chloronaphthalene	SEC, μ-styragel, 210°C	1370
		1-Cycolhexyl-2-pyrrolidone + 20 nM lithium chloride, 1-chloronaphthalene	SEC, styragel, 210°C	
		α-Chloronaphthalene	SEC, 210°C	3248

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(styrene sulfonic acid)	Chromatography	Water	SEC	691
Poly(sulfide rubber)s	Chromatography	Dioxane	Column elution, silica gel	1037
Poly(sulfonyl-1,2-cyclohexylene)	Fractional precipitation	Benzene/methanol		873
Poly(sulfonyl-1,2-cyclopentylene)	Fractional precipitation	Dioxane/methanol		873
Poly(sulfonyl-1-methyl-1-propylethylene)	Fractional precipitation	Acetone/methanol	Lowering temp.	238
Poly(sulfonyl-1,4-phenylenesulfonylimino-(ethoxycarbonyl)pentamethyleneimino)	Fractional precipitation	Dimethylformamide/ethanol		258
Poly(sulfonyl-1,4-phenyleoxy-1,3-propylenoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, 25°C	3525
Poly(sulfonyl-phenylethylene)	Fractional precipitation	Tetrahydrofuran/methanol	40°C	844
Poly(sulfonylbutylethylene) *Poly(hexene-1-sulphone)*	Fractional precipitation	Acetone/methanol	Lowering temp.	238,239,1455
Poly(sulfonylethylethylene)	Chromatography	Tetrahydrofuran	SEC 40°C	446
	Fractional precipitation	Acetone/methanol	Lowering temp.	447
Poly(tetrathioethylene)	Chromatography	Dioxane		1029
Poly(2,5-thieno [3,2-b]thiophenylene)	Chromatography	Chloroform	SEC, styragel	2798
Poly(1,4-bis(2-thienyl)-2,5-didodecyloxyphenylene)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2794
Poly(1,4-bis(2-thienyl)-2,5-dieicosinoxyphenylene)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2794
Poly(1,4-bis(2-thienyl)-2,5-diheptoxyphenylene)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2794
Poly(1,4-bis(2-thienyl)-2,5-dihexadecyloxyphenylene)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2794
Poly(1,4-bis(2-thienyl)-2,5-dihexylphenylene)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2794
Poly(1,4-bis(2-thienyl)-2-dodecyloxy-5-methoxyphenylene)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2794
Poly(1,4-bis(2-thienyl)-2-heptoxy-5-methoxyphenylene)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2794
Poly(thio-*tert*-butylethylene)	Fractional precipitation	Benzene/methanol		800
Poly(thio-1,2-cyclohexylene)	Chromatography	Tetrahydrofuran	SEC, styragel	1090
Poly(thio-2,5-dimethyl-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, 25°C	3525
Poly(thio(1-methyltrimethylene)	Chromatography	Benzene	SEC, 45°C	1573
Poly(thio-1,4-phenyleneoxy-1,10-decyleneoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, 25°C	3525
Poly(thio-1,4-phenyleneoxy-1,7-heptyleneoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, 25°C	3525
Poly(thio-1,4-phenyleneoxy-1,6-hexyleneoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, 25°C	3525
Poly(thio-1,4-phenyleneoxy-1,8-octyleneoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, 25°C	3525
Poly(thio-1,4-phenyleneoxy-1,5-pentyleneoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, 25°C	3525
Poly(thio-1,4-phenylene)	Chromatography	1-Chloronaphthalene	SEC, 210°C, porasil and μ-bondagel	3060
		Dimethylsulfoxide	SEC, Sephadex only DMSO soluble fraction	1366
		N-Methyl-2-pyrrolidone	SEC, Asahipac, 25°C	3525
	Fractional precipitation	Toluene, chloronapthalene	Rising temp.	1961
Poly(thio-1,4-phenyleneoxy-1,4-bytyleneoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, 25°C	3525
Poly(thio-1,4-phenyleneoxy,1,2-ethyleneoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, 25°C	3525
Poly(thio-1,4-phenyleneoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC, Asahipac, 25°C	3525
Poly(thio-1,4-phenyleneoxy-1,3-propyleneoxy-1,4-phenylene)	Chromatography	*N*-Methyl-2-pyrrolidone	SEC	

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(thio-2,3,5,6-tetramethyl 1,4-phenylene)				
	Chromatography	*N*-methyl-2-pyrrolidone	SEC, Asahipac, 25°C	3525
Poly(thiohexamethylene)	Fractional precipitation	Benzene/methanol	40°C	2575,2829
Poly(thiomethylethylethylene)	Chromatography	Tetrahydrofuran	SEC, styragel	2696
Poly(thiophene)	Chromatography	Chloroform	Chloroform soluble fraction SEC, Ja-gel	507
		Chloroform–*n*-hexane–acetone (20 : 5 : 2)		
		n-Hexane–chloroform–Butanone (30 : 6 : 11)	Chloroform soluble fraction, TLC, silica gel coate with poly(acrylamide)	
Poly(thiophene-,5-diylvinylene)	*Poly(2,5-thienylenevinylene)*			
	Chromatography	Chloroform	SEC	725
Poly(thiopropylene)	Chromatography	Methylene dichloride	SEC, poragel	3002
	Extraction	Ethyl ether		801
	Fractional precipitation	Benzene/methanol		855,2363
		Carbon tetrachloride/*n*-hexane		2363
Poly(thiotetramethylene)	Fractional pecipitation	Benzene/methanol		2034
Poly(thiotrimethylene)	Fractional precipitation	Chlroform/hexane	50°C	2827,2830
Poly(vinyl phenyl sulfoxide)	Chromatography	Tetrahydrofuran	SEC, TSK gel	1620

3.8. POLY(SILANES) AND POLY(SILOXANES)

Poly(di-*n*-alkylsiloxane)s	Chromatography	Toluene	SEC, μ-styragel	2495
Poly(2,3-benzo-1-silabutenes)	Chromatography	Tetrahydrofuran	SEC, μ-styragel, analytical and preparative	3199
Poly[(3,4-benzo-1-silacyclo pent-3-en-1-ylene)-*cis* and *trans*-but-2-en-1,4-ylene]				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 20°C	3420
Poly(benzyldimethylvinylsilane)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2471
Poly(di-*n*-butylsiloxane)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2230
Poly[[(2,2-dichloro-1-oxo-3,4-cyclobutanediyl)]-*cis*-bis(methylene)](methylphenylsilylene)]				
	Chromatography	Tetrahydrofuran	SEC	1899
Poly(chlorophenyl silsesquioxane)	*Poly(chlorophenyl siloxane)*			
	Fractional precipitation	Benzene/*n*-heptane	Ladder	95
Poly(dichlorophenyl silsesquioxane)				
	Fractional precipitation	Benzene/*n*-hexane	Ladder	95
Poly(chromium potassium sulfate-*co*-phenyl sodiumoxydihydroxysilane)				
	Fractional precipitation	Carbon tetrachloride/petroleum ether		156
Poly(di-*n*-decylsiloxane)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2230
Poly(di-*n*-ethylsiloxane)	Chromatography	Toluene	SEC	1729
Poly(ferrooxyhalophenylsilylene)				
	Fractional precipitation	Benzene/petroleum ether		483
Poly(ferrocenyl di-*n*-butylsilane)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel SEC, ultrastyragel	911 911
Poly(ferrocenyl dimethylsilane)	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel	911
Poly(ferrocenyl diethylsilane)	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel	911
Poly(ferrocenyl di-*n*-hexyl silane)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel	911
Poly(ferrocenyl methyl, propylen 3-trifluoro silene)				
	Chromatography	Tetrahydrofuran + 0.1% *tert-n*-butylammonium bromide	SEC, ultrastyragel	912
Poly(ferrocenyl methyl ferrocenylsilane)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel	912
Poly(ferrocenyl methyl 5-norobornylsilane)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel	912

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(ferrocenyl methyl, *n*-octadecyl silane)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel	912
Poly(ferrocenyl methyl, phenyl silane)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel	912
Poly(ferrocenyl methyl, vinyl silane)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel	912
Poly(ferrocenyl methylsilane)	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, ultrastyragel	912
Poly(di-*n*-heptylsiloxane)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2230
Poly(hexamethylcyclotrisiloxane)				
	Chromatography	Chloroform	SEC, 25°C	585
Poly(1,3,5,7,9,11-hexaphenyl-1,3,5,7,9,11-hexacyclohexasiloxane)				
	Fractional precipitation	Benzene/methanol		102
Poly(di-*n*-hexylsilane)	Chromatography	Tetrahydrofuran	SEC	646,2763
Poly(di-*n*-hexylsiloxane)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2230
Poly(methyl cyclohexylsilylene-*co*-cyclohexylenesilylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3591
Poly(2,2-dimethyl-5,5-diphenyl-1-oxa-2,5-disilacyclopentane) *Poly(oxydimethylsilyleneethylene-diphenylsilylene)*				
	Chromatography	Tetrahydrofuran	SEC, porosity glass	3102
Poly(2,5-dimethyl-2,5-diphenyl-1-oxa-,5-disilacyclopentane) *Poly(oxymethylphenylsilylenemethylenemethylpenylsilylene)*				
	Chromatography	Tetrahydrofuran	SEC, porosity glass	3102
Poly(1,1-dimethyl-1-germa-*cis*-pent-3-ene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	3588
Poly(methyl-*n*-hexylsilylane)	Chromatography	Tetrahydrofuran	SEC	1468
	Fractional precipitation	Cyclohexane/isopropanol		749
Poly(methyl-β-phenethyl-*co*-methylphenyl silylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3591
Poly(methyl-β-phenethylsilylene-*co*-methyl-*p*-tolylsilylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3591
Poly(methyl(2-phenylethyl)silylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3591
Poly(1-methyl-1-sila-*cis*-pent-3-ene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3588
			SEC, ultrastyragel, 20°C	3601
Poly(methyl-3,3,3-trifluoropropylmethylsiloxane)				
	Fractional precipitation	Ethyl acetate/toluene		459
Poly(5-methyl-2,2,5-triphenyl-1-oxa-disilacyclopentane) *Poly(oxydiphenylsilyleneethylenemethylphenylsilyeene)*				
	Chromatography	Tetrahydrofuran	SEC, porosity glass	3102
Poly(1-methyl-1-vinyl-1-silabutene)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 20°C	1930
Poly((3-methylbutenyl)1-silsesquioxane)				
	Fractional precipitation	Benzene/methanol		3270
		Cyclohexane/acetone–methanol (1 : 5)		3270
Poly[(methylchlorosilylene)methylene]				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	161
Poly(methylcyclohexylsilylene)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3591
	Fractional precipitation	Cyclohexane/isopropanol		749
Poly(methylcyclohexylsilylene-*co*-methyl-*p*-tolylsilylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3591
Poly(methycylclohexylsilylene-*co*-methylphenylethylsilylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3591
Poly[(methyldeuteriosilylene)methylene]				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	161
Poly(methylethylsiloxane)	Fractional precipitation	Benzene/methanol		1358
Poly(methylphenylsilane)	Fractional precipitation	Toluene/isopropanol		749
Poly(methylphenylsiloxane)	Fractional precipitation	Toluene/methanol		2819
Poly(dimethylsilmethylene)	Chromatography	Toluene	SEC	1924
	Fractional precipitation	Benzene/methanol		1711
Poly(dimethylsiloxane)	Chromatography	Chloroform	SEC, preparative	3514
		Tetrahydrofuran	SEC	3099
			SEC, μ-styragel, 35°C	778

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Fractional precipitation	Benzene/methanol		1358
		Toluene/methanol	25°C	2060
	Fractional precipitation	Andrici carbonic	Supercritical	3593
		Methy ethyl ketone/methanol		3514
		Supercritical	Nitrous oxide	2883
		Toluene/acetone		2143
Poly[(di-methylsilylene)methylene]				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	161
Poly[(methylsilylene)methylene]				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	161
Poly(di-*n*-nonylsiloxane)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2230
Poly(di-*n*-octylsiloxane)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2230
Poly(oxy(ditrimethylsiloxy)silylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1533
Poly(oxydichlorophenyl)titano-tetra-(oxyhlorophenylsilylene)				
	Fractional precipitation	Carbon tetrachloride/petroleum ether		155
Poly(1,4-bis-(oxydimethylsilyl)phenylenedimethyl-silane) *Poly(oxydimethylsilyleneoxydimethylsilylene1,4-phenylenedimethylsilylene)*				
	Chromatography	Tetrahydrofuran	SEC, styragel, 30°C	2613
Poly(oxydimethylsilylene) *Poly(dimethyl siloxane)*				
	Chromatography	Benzene	SEC, 55°C	967
		Chloroform	SEC, 35°C, 30°C	713,2174
			SEC, styragel, 30°C	2789
		Ethanol/methanol, ethyl ether/ethanol, and ethyl ether	Gradient elution, carbon	197
		n-Heptane/butanone	SEC, μ-styragel, 20°C	494
		n-Hexane/butanone	SEC, μ-styragel, 20°C	494
		o-Dichlorobenzene	SEC, styragel, 87°C	713
		Tetrahydrofuran	SEC, 25°C	1905
		Toluene	Oligomers, SEC	285
			SEC	1328,1649
			SEC, μ-styragel	1890,1896
			SEC, difunctio.	1081
			SEC, octadecyltrichlorosilane modified silica gel	1257
			SEC, styragel, 37°C	1897,3496
			SEC, styragel, preparative	765
		Trichloroethylene	Precipitation chromatogr., temp. gradient, quartz sand, branched polymer	99
			SEC, 25°C, 23°C	2961,3195
	Distribution between immiscible liquids	Methanol/carbon tetrachloride in counter current with cyclohexane		208
	Foam fractionation	Benzene		967
	Fractional precipitation	Benzene/methanol	25°C	322,323,457, 679,1257,1649, 1835,1837,3094
		Cyclohexane–carbon tetrachloride (3 : 1)/methanol		104
		Ethyl acetate/methanol		711,1257,1449,2174
		Toluene/methanol		153
	Fractional solution	Benzene/methanol	Trifunctional star	100
		Cyclohexane–carbon tetrachloride (3 : 1)/methanol	25°C, extraction	967,2585
	Sedimentation velocity	*n*-heptane	20°C	2585
	Turbidimetric titration	Butanone/methanol		2174
		Methyl ethyl ketone/methanol	25°C	706
Poly(oxydimethylsilylene-1,4-phenyleneoxy-1,4-phenylenedimethylsilylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	808
Poly(oxydimethylsilylene-1,4-phenylenedimethylsilylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	808
Poly(oxydiphenylsilylene)	Fractional precipitation	Benzene/methanol		3261,3265

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(oxydiphenyltitano-tetraoxydiphenylsilylene)				
	Fractional precipitation	Carbon tetrachloride/petroleum ether		155
Poly(oxyhydroxyaluminooxydichlorophenylsilylene)				
	Fractional precipitation	Carbon tetrachloride/petroleum ether		154
Poly(oxyhydroxyaluminooxydiphenylsilylene)				
	Fractional precipitation	Carbon tetrachloride/petroleum ether		154
Poly(oxymethyl-3,3,3-trifluoropropylsilylene)				
	Fractional precipitation	Ethyl acetate/toluene		460
Poly(oxymethylphenylsilylene)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 25°C	2000
	Fractional precipitation	Cyclohexane–carbon tetrachloride (3 : 1)/methanol		104
		Toluene/isopropanol		461,2000
	Fractional solution	Cyclohexane–carbon tetrachloride (3 : 1)/methanol		105
Poly(*n*-pentyl-*n*-alkylsilanes)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	1705
Poly(di-*n*-pentylsiloxane)	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2230
Poly(di-*n*-pentylsilyene)	Chromatography	Cyclohexane	SEC, μ-styragel, chiral	925
Poly(1,1-diphenyl-2,2,3,3,4,4-hexamethylcyclotetrasiloxane)				
	Fractional precipitation	Benzene/methanol		97
Poly(1,1-diphenyl-1-sila-*cis*-pent-3-ene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	3588
Poly(phenyl siloxane)	Fractional precipitation	Benzene/methanol	20°C	2547
Poly(1-phenyl-1-vinyl-1-silabutene)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 20°C	1930
Poly(diphenylene siloxane)	Fractional precipitation	Benzene/methanol		2604
Poly(*m*-phenylene siloxane)	Fractional precipitation	Benzene/methanol		2604
Poly(*p*-phenylene siloxane)	Fractional precipitation	Benzene/methanol		2604
Poly(*p*-phenylenetetramethyldisiloxane) *Poly(oxydimethylsilyene-1,4-phenylenedimethylsilylene)*				
	Chromatography	Tetrahydrofuran	SEC, styragel, 30°C	2613
Poly(phenylmethylsiloxane)	Chromatography	Methanol/tetrahydrofuran	Liquid absorption (HPLC), silica-Csubindice 18, cyclic	602
		Toluene	SEC, styragel, preparative	602
Poly(diphenylsilane)	Chromatography	Tetrahydrofuran	SEC	1510
Poly(phenylsilsesquioxane)	Fractional precipitation	Benzene/methanol		106,1270,2684,3272
		Benzene/*n*-heptane		106
		Benzene/*n*-hexane		95
Poly(1,3-diphenyltetradisiloxane)				
	Fractional precipitation	Benzene/methanol		102
Poly(dipropyl siloxane)	Fractional precipitation	Toluene/methanol		1895
Poly(2,2,5,5-tetramethyl-1-oxa-2,5-disilacyclopentane) *Poly(oxydimethylsilyleneethylene-dimethylsilylene)*				
	Chromatography	Tetrahydrofuran	SEC, porasil	3103
Poly(2,2,5-tetramethyl-5-phenyl-1-oxa-2,5-disilacyclopentane) *Poly(oxydimethylsilyleneethylene-methylphenylsilylene)*				
	Chromatography	Tetrahydrofuran	SEC, porosity glass	3102
Poly(tetramethyl-*p*-silphenylene siloxane)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2456
Poly(tetramethyl-*p*-silphenylene-siloxane)				
	Fractional precipitation	Benzene/methanol–water		3421
Poly(2,2,5,5-tetraphenyl]-1-oxa-disiiacyclopentane) *Poly(oxydiphenylsilylenethylene-diphenylsilylene)*				
	Chromatography	Tetrahydrofuran	SEC, porosity glass	3102
Poly(1,3,5,7-tetraphenyl-1,3,5,7-tetrachlorocyclotetrasiloxane)				
	Fractional precipitation	Benzene/methanol		102
Poly(1,3,5,7-tetrapheylhexatetrasiloxane)				
	Fractional precipitation	Benzene/methanol		102
Poly[4-([1-trifluoromethyl)alkoxoly(*n*) carbonyl)phenyl 4′-alkoxy(*m*) biphenyl-4-carboxylate siloxane)]				
	Chromatography	Tetrahydrofuran	SEC, styragel	629
Poly[(3,3,3-trifluoropropyl)methyl]silane				
	Chromatography	Tetrahydrofuran	SEC, styragel	940
Poly(1,1,3-trimethyl-1-sila-*cis*-pent-3-ene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	3588
			SEC, ultrastyragel, 20°C	2517

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(*p*[bis(trimethylsilyl)methyl]*iso* propenylbenzene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2328
Poly[(*o*-trimethylsilyl)phenyl acetylene]				
	Chromatography	Chloroform	SEC, styragel	2069
Poly(1,3,5-triphenylpentatrisiloxane)				
	Fractional precipitation	Benzene/methanol		102
Poly(bis(undecenyloxy)methylsilane)				
	Chromatography	Tetrahydrofuran	SEC	2110
Poly(1,1-divinyl-1-vinyl-1-silabutene)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 20°C	1930
Poly(divinyltetramethyldisiloxane)				
	Fractional precipitation	Butanone/methanol	25°C	1161

3.9. POLY(PHOSPHAZENES) AND RELATED POLYMERS

Poly(aminocarbophosphazenes)	Chromatography	Tetrahydrofuran + 0.1wt.% tetra *n*-butylammonium bromide	SEC, phenogel	47
Poly[bis-(4-biphyloxy)phosphazene)]				
	Chromatography	Tetrahydrofuran + 0.1 wt.% tetra *n*-butylammonium bromide	SEC, μ-styragel, 35°C	2299
Poly(*bis-p*-bromophenyleneoxy)phosphazene				
	Chromatography	Tetrahydrofuran + 0.1 wt.% tetra-*n*-butylmammonium bromide	SEC, μ-styragel, 25°C	2299
Poly(*trans*-2,4-dicarboxy phenoxy-3,4,6,6-tetraphenoxy)cyclotriphosphazene-*alt*-oxy-1,4-phenyleneisopropylidine-1,4-phenyleoxy)				
	Chromatography	Cresol : chloroform (20 : 80)	SEC	2214
Poly(di(*m*-chlorophenoxy)phosphazene)				
	Chromatography	Chloroform	SEC, styragel	1191
Poly(di-(*p*-chlorophenoxy)phosphazene)				
	Fractional precipitation	Dimethylformamide	Lowering temp.	55
Poly(bis-*p*-chlorophenyleneoxy)phosphazene				
	Chromatography	Tetrahydrofuran + 0.1 wt.% tetra-*n*-butylammonium bromide	SEC, μ-styragel, 25°C	2299
Poly(diphenoxy phosphazene)	Chromatography	Tetrahydrofuran + 0.1 wt.% tetra-*n*-butylammonium bromide	SEC, μ-styragel, 35°C	2299
Poly(fluoroalkoxyphosphazene)	Fractional solution	1,1,2-Trichloro-1,2,2-trifluoroethane/acetone	Extraction	1190
Poly(dihexoxyphosphazene)	Fractional precipitation	Tetrahydrofuran/ethanol		421
Poly(di-(*p*-isopropylphenoxy)phosphazene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1094
Poly(methyl phenyl phosphazene)				
	Chromatography	Tetrahydrofuran + 0.1 % *n*-tetrabutylammonuium bromide	SEC, μ-styragel, 30°C	3489
Poly(di-(*p*-methylanilino)phosphazene)				
	Fractional precipitation	Tetrahydrofuran/diethylene glycol		2593
Poly(dimethyloxy phosphazene)				
	Chromatography	Tetrahydrofuran + 0.1 wt.% tetra-*n*-butylammonium bromide	SEC, μ-styragel 35°C	2299
Poly(di *p*-methylphenylenoxy phosphazene)				
	Chromatography	Tetrahydrofuran + 0.1 wt.% tetra-*n*-butylammonium bromide	SEC, μ-styragel, 35°C	2299
Poly[bis(β-naphthoxy) phosphazene)				
	Chromatography	Tetrahydrofuran + 0.1 wt.% tetra-*n*-butylammonium bromide	SEC, μ-styragel, 35°C	2299
		Tetrahydrofuran + 0.1 % tetrakis (*n*-butyl ammonium bromide)	SEC, styragel	422
	Fractional precipitation	Tetrahydrofuran/hexane		422

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(octafluroropentoxy-phosphazene)				
	Fractional precipitation	Toluene–tetrahydrofuran (30 : 15)/petroleum ether		874
Poly(diphenoxyphosphazene)	Chromatography	Tetrahydrofuran	SEC, pore glass beads	45
Poly((*p*-phenylazoanilino)phosphazene)				
	Chromatography	Aq. 95% methanol	SEC, styragel	46
Poly((1-phenylazona-phthalene-2-oxy-methylamino)phosphazene)				
	Chromatography	Aq. 95% methanol	SEC, styragel	46
Poly((1-phenylazonapthalene-2-oxytrifluoroethyoxy)phosphazene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	46
Poly(di-(*p*-phenylphenoxy)phosphazene)				
	Fractional precipitation	Dimethylformamide	Lowering temp.	55
Polymer from chromium (II) acetate and potassium diphenylphosphinate				
	Chromatography	Tetrahydrofuran	SEC	733
Poly((2-trifluoroethoxy)(1-difluoro-2,3,4-hexafluoropentoxy) phosphazene) *Phosphonitrilic fluoroelastomers*				
	Chromatography	Dimethylformamide	SEC, styragel	514
	Fractional precipitation	Acetonitrile/benzene		514
Poly(bis-2,2,2-trifluoroethoxy)phenoxy phosphazenes)				
	Chromatography	Tetrahydrofuran + wt.% tetra-*n*-butylammonium bromide	SEC, μ-styragel, 35°C	2299
		Tetrahydrofuran + 0.875 g tetra-*n*-butylammonium bromide/l	SEC, styragel	48
	Fractional precipitation	Tetrahydrofuran/methanol		48
Poly(2,2,2-trifluoroethoxy)phosphazene				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	1937
	Fractional precipitation	Acetone/cycloxhexane	25°C	55
		Methanol	Lowering temp. 40°C to 25°C	55
Poly (urylene-1,4-phenyleneureylenephenylphosphinylidene)				
	Fractional precipitation	Dimemthyl sulfoxide/carbon tetrachloride	25°C	516

3.10. OTHER COMPOUNDS

Poly(2-acrylamido-2-methylpropanesulfonic acid)				
	Fractional precipitation	$Ca\ Cl_2 \cdot 6\ H_2O$ 3.0 M/2-propanol		1105
Poly(5-(1-adamantyloxy-2H-pyrrole-2-one)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3484
Poly(aniline)-*co*-poly(*N*-butyl aniline)				
	Chromatography	*N*-methylpyrrolidone	SEC, trimethylsilyl-branched phase, 25°C	302
Poly(arylate) from : (terephthalic acid chloride + 2-phenyl-bis-(4-hydroxyphenyl)-phthalimidine)				
	Coacervation	Tetrachloroethane–phenol (3 : 1)/ *n*-heptane		867
Poly(dibenzimidazole(2,3-a)(6,7-a)-1,8-dioxo-1,2,7,8-tetrahydro-4,5-benzoph enantroline)				
	Chromatography	Methanesulfonic acid, 96% sulfuric acid	SEC, porous silica	313
Poly(benzo(1,2-d : 5,4-d′) bisoxazole-2,6-diyl-1,4-phenylene)				
	Chromatography	Methanesulfonic acid	SEC	648
Poly(5,5′-dibenzoimidazoie-2,2′-diyl-1,-3-phenylene)				
	Fractional precipitation	Dimethylacetamide/*n*-hexane	61°C	1271
	Sedimentaion equilibrium	Dimethylacetamide	Ultracentrifuge, 40°C	1019
		Dimethylacetamide/lithium chloride/propionic acid	Ultracentrifuge	3466
Poly(*N*-butyl aniline)	Chromatography	*N*-Methylpyrrolidone	SEC, trimethylsilyl-branched-phase, 25°C	573
Poly(chalcones) (General)	Chromatography	Dimethyl sulfoxide	SEC	2014
Poly(cyclodiborozane)s	Chromatography	Tetrahydrofuran	SEC, styragel	591
Poly{*N*-[3,4,5-tris(*n*-dodecan-1-yloxy) phenyl]maleimide}				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	2570
Poly(*N*-dodecyl aniline)	Chromatography	*N*-Methylpyrrolidone	SEC, trimethylsilyl-branched-phase, 25°C	573
Epoxy	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	2275
Poly(ether ester)s	Chromatography	Methylene dichloride	SEC, ultrastyragel	3206
Poly(*N*-ethyl aniline)	Chromatography	*N*-Methylpyrrolidone	SEC, trimethylsilyl-branched-phase, 25°C	573

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(ferrocenediylfurfurylidene)				
	Fractional precipitation	Dioxane/isopropanol–water		2383
Poly(ferrocenylgermanes)	Chromatography	Tetrahydrofuran + 0.1 % tetra-*n*-butylammonium bromide	SEC, ultrastyragel	2557
Poly(5,7-dihydro-1,3,5,7-tetraoxobenzo-(1,2-c : 4,5-c′)dipyrrole-2,6-(1H, 3H)-diyl-1,4-phenylenephthalidylidene-1,4-phenylene)				
	Fractional precipitation	Nitrobenzene/dichloroethane	50°C and 63°C	1762
Poly(2,3-dihydrobenzofuran-2,3-diyl)				
	Fractional precipitation	Benzene/methanol	Optically active	866
Poly(1-isobutyl-2,5-dioxopyrrolidin-3,4-diyl)				
	Fractional precipitation	Benzene/methanol		3262
Poly(2-methoxy-1,3,5-triazine-4,6-diyloxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy)				
	Chromatography	Chloroform	SEC, styragel, 30°C	2957
	Fractional precipitation	1,2-Dichloroethane/methanol	30°C	2957
Poly(*N*-methylaniline)	Chromatography	*N*-Methylpyrrolidone	SEC, trimethylsilyl-branched-phase, 25°C	573
Poly(1,2,4-dimethylphenyl-2,5-dioxopyrrolidin-3,4-diyl)				
	Fractional precipitation	Ethyl acetate/methanol	21°C	3267
Poly(nitrilo-2-methyl-1,4-phenylene nitrilomethylidyne-1,4-phenylenemethylidyne)				
	Chromatography	Sulfuric acid 96%	SEC	2169
Poly(1-octadecyl-2,5-dioxopyrrolidin-3,4-diyl)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	224
	Fractional precipitation	Toluene/methanol	20°C	224
Poly(1,3-dioxa-2-propyl-4,6-cyclohexylene-methylene) *Poly (vinyl butyral)*				
	Chromatography	Tetrahydrofuran	Acetylated, SEC, spherosil	688
	Fractional precipitation	Isopropanol/water		1313
Poly(oxadiazoledily-1,4-phenylenephthalidylidene-1,4-phenylene)				
	Fractional precipitation	Tetrachloroethane–phenol/*n*-heptane		2510,3209
	Sedimentation analysis	Dimethylformamide		2510
Poly(2-oxo-1,3-dioxolan-4,5-diyl) *Poly vinylcarbonato*				
	Fractional precipitation	Dimethylformamide/methanol		2912
		Dimethylformamide/*n*-butanol		2912
Poly(2,5-dioxopyrrolidine-1,3-diylthio-4,4-biphenylenethio-2,5-dioxopyrrolidine-1,4-phenylenemethylene-1,4-phenylene)				
	Chromatography	Dimethylformamide + lithium nitrate	SEC, silica gel	3461
Poly(2,5-dioxopyrrolidine-1,3-diylthio-1,4-phenyleneoxy-1,4-phenyleneoxy-1,4-phenylenethio-2,5-dioxopyrrolidine-1,4-phenylenemethylene-1,4-phenylene)				
	Chromatography	Dimethylformamide + lithium nitrate	SEC, silica gel	3461
Poly(4,4′dioxy-2,2′-dimethylazoxybenzenedodecanedioyl)				
	Chromatography	Chloroform	SEC, styragel, 40°C	2335
Poly(oxybenzoxazole-5,2-diyl-1,4-phenylenephthalidylindene-1,4-phenylenebenzoxazole-2,5-diyl)				
	Fractional precipitation	Tetrachloroethane–phenol/*n*-heptane		2309
Poly(oxyphthalimide-5,2-diyl-1,4-phenylenephthalidylidene-1,4-phenylenephthalimide-2,5-diyl)				
	Fractional precipitation	Tetrachloroethane–phenol/*n*-heptane		3209
Poly(phenol formaldehyde)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	578
Phenol novocal	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	2275
Poly(1-phenyl-2,5-dioxopyrrolidin-3,4-diyl) *Poly[N-phenylmaleimide]*				
	Chromatography	Tetrahydrofuran	SEC, styragel, 38°C	1189
Poly(2,2-(1,4-phenyl)-7,7-hydroxy-bis-(3-pheylquinoxaline)				
	Fractional precipitation	Carbon tetrachloride–phenol/*n*-heptane		3211
Poly(1,4-phenylenemethylene-3,5-pyrazole)				
	Fractional precipitation	Dimethylformamide/benzene	25°C	3233
Poly(*p*-phenylene)s	Chromatography	Chloroform	SEC, styragel, 40°C	2568
Poly(3-phenylquinoxaline-2,7-diyloxy-3-phenylquinoxaline-7,2-diyl-1,4-phenylene)				
	Fractional precipitation	Tetrachloroethane–phenol (1 : 1)/*n*-heptane		3210
Poly(1,4-piperazinediyl-2,5-dioxopyrrolidine-1,3-diyl-hexamethylene-2,5-dioxopyrrolidine-1,3-diyl)				
	Fractional precipitation	Chloroform/acetone		1000
Poly(*N*-propyl aniline)	Chromatography	*N*-Methylpyrrolidone	SEC, trimethylsilyl-branched-phase, 25°C	573
Poly(propyleneglycol)-*co*-di(4-isocyanotophenyl)methane-*co*-pyromellitic dianhydride) *Poly(etherimide)*				
	Chromatography	Dimethylformamide	SEC, styragel	724
Poly(2,4-quinolinediyl-1,4-phenyleneoxy-1,4-phenylene)				
	Fractional precipitation	Chloroform/methanol		3492

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly[1-(3-sulfonatepropyl)-pyridinium-2-yl]-ethylene)] *Poly(2-vinylpyridinium sulfopropylbetaine)*				
	Fractional precipitation	Water/methanol		2238
Poly(1-tolyl-2,5-dioxopyrrolidin-3,4-diyl)				
	Fractional precipitation	Chloroform/methanol	20°C	3266
Poly[(4,4′-trimethylene-bis-(1-methyl piperidine)-*N*-(2,3-butene)di-tetrafluoroborate]				
	Chromatography	Acetonitrile + acetic acid 0.5 M (70 : 30)	SEC, HEMA	2898
	Fractional solution	Ethylene glycol monomethyl ether/diisopropyl ether	Column extraction, CPF	2898
Poly[(4,4′-trimethylene-bis-(1-methyl piperidine)-*N*-(2,3-butene)dibromide]				
	Chromatography	Acetonitrile + acetic acid 0.5 M, (70 : 30)	SEC, HEMA	2898
	Fractional solution	Ethylene glycol monomethyl ether/diisopropyl ether	Column extraction, CPF	2898
Wool-*graft*-poly(*n*-butyl acrylate)				
	Extraction	Benzene or acetone		3356
Wool-*graft*-poly(ethyl acrylate)	Extraction	Benzene		2196
		Benzene or acetone		3356
Wool-*graft*-poly(methyl acrylate)				
	Extraction	Benzene		2192,2195,2197
		Benzene or acetone		3356
Wool-*graft*-poly(methyl methacrylate)				
	Extraction	Benzene		6,2193
		Benzene or acetone		828,3356
Wool-*graft*-poly(methyl acrylate-co-vinyl acetate)				
	Extraction	Benzene		2197
Wool-*graft*-poly(vinyl acetate)	Extraction	Benzene		2189,2195,2197

3.11. RANDOM COPOLYMERS

Poly(acetaldehyde-*co*-chloral)	Fractional precipitation	Chloroform/methanol		1457
Poly(acrolein-*co*-thiophenolmercaptal)				
	Fractional precipitation	Benzene/methanol		2911
Poly(acrylonitrile-*co*-α-hydroxyisobutyric acid anhydrosulfite)				
	Turbidimetric titration	Dimethylformamide/chloroform		1426
Poly(5-(1-adamantyloxy)-2H-pyrrole-2-one-*co*-methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3484
Poly(5(1-adamantyloxy)-2H-pyrrole-2-one-*co*-styrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3484
Poly(*N*-alaninylmaleimide-*co*-methacrylic acid)				
	Chromatography	Dimethylformamide	SEC, μ-styragel, 40°C	1901
Poly(*N*-alaninylmaleimide-*co*-vinyl acetate)				
	Chromatography	Dimethylformamide	SEC, μ-styragel, 40°C	1901
Poly(di-*n*-alkylsiloxane-*co*-di-*n*-hexylsiloxane)				
	Chromatography	Toluene	SEC	2496
Poly(β-allyloxypropionaldehyde-*co*-acrylonitrile)				
	Chromatography	Tetrahydrofuran	SEC, styragel	137
Poly(β-allyloxypropionaldehyde-*co*-styrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	137
Poly(butyleneglycol-*alt*-(terephthalic acid-sebacic acid))				
	Chromatography	*o*-Chlorophenol		2047
Poly(carbon monoxide-*co*-ethylene)				
	Chromatography	Trichlorobenzene	SEC, styragel, 135°C	1108
Poly(carbon monoxide-*co*-vinyl chloride)				
	Chromatography	Carbon tetrachloride–ethyl alcohol(1 : 1) tetrahydrofuran	TLC, gradient elution	1633
Poly(1,3-bis(p-carboxyphenoxy)propane-*co*-sebacic acid)				
	Chromatography	Chloroform	SEC, styragel	685
Poly(3,3-bis(chloromethyl)oxacyclobutane-*co*-β-propiolactone)				
	Chromatography	Chlorobenzene	SEC, styragel	3465
	Fractional precipitation	Ethylene dichloride/ethanol		3124
Poly(cyclohexane oxide-*co*-nitroethylene)				
	Extraction	Benzene		1429

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly[1-(3-sulfonatepropyl)-pyridinium-2-yl]-ethylene)] *Poly(2-vinylpyridinium sulfopropylbetaine)*				
	Fractional precipitation	Water/methanol		2238
Poly(1-tolyl-2,5-dioxopyrrolidin-3,4-diyl)				
	Fractional precipitation	Chloroform/methanol	20°C	3266
Poly[(4,4′-trimethylene-bis-(1-methyl piperidine)-*N*-(2,3-butene)di-tetrafluoroborate]				
	Chromatography	Acetonitrile + acetic acid 0.5 M (70 : 30)	SEC, HEMA	2898
	Fractional solution	Ethylene glycol monomethyl ether/diisopropyl ether	Column extraction, CPF	2898
Poly[(4,4′-trimethylene-bis-(1-methyl piperidine)-*N*-(2,3-butene)dibromide]				
	Chromatography	Acetonitrile + acetic acid 0.5 M, (70 : 30)	SEC, HEMA	2898
	Fractional solution	Ethylene glycol monomethyl ether/diisopropyl ether	Column extraction, CPF	2898
Wool-*graft*-poly(*n*-butyl acrylate)				
	Extraction	Benzene or acetone		3356
Wool-*graft*-poly(ethyl acrylate)	Extraction	Benzene		2196
		Benzene or acetone		3356
Wool-*graft*-poly(methyl acrylate)				
	Extraction	Benzene		2192,2195,2197
		Benzene or acetone		3356
Wool-*graft*-poly(methyl methacrylate)				
	Extraction	Benzene		6,2193
		Benzene or acetone		828,3356
Wool-*graft*-poly(methyl acrylate-co-vinyl acetate)				
	Extraction	Benzene		2197
Wool-*graft*-poly(vinyl acetate)	Extraction	Benzene		2189,2195,2197

3.11. RANDOM COPOLYMERS

Poly(acetaldehyde-*co*-chloral)	Fractional precipitation	Chloroform/methanol		1457
Poly(acrolein-*co*-thiophenolmercaptal)				
	Fractional precipitation	Benzene/methanol		2911
Poly(acrylonitrile-*co*-α-hydroxyisobutyric acid anhydrosulfite)				
	Turbidimetric titration	Dimethylformamide/chloroform		1426
Poly(5-(1-adamantyloxy)-2H-pyrrole-2-one-*co*-methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3484
Poly(5(1-adamantyloxy)-2H-pyrrole-2-one-*co*-styrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3484
Poly(*N*-alaninylmaleimide-*co*-methacrylic acid)				
	Chromatography	Dimethylformamide	SEC, μ-styragel, 40°C	1901
Poly(*N*-alaninylmaleimide-*co*-vinyl acetate)				
	Chromatography	Dimethylformamide	SEC, μ-styragel, 40°C	1901
Poly(di-*n*-alkylsiloxane-*co*-di-*n*-hexylsiloxane)				
	Chromatography	Toluene	SEC	2496
Poly(β-allyloxypropionaldehyde-*co*-acrylonitrile)				
	Chromatography	Tetrahydrofuran	SEC, styragel	137
Poly(β-allyloxypropionaldehyde-*co*-styrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	137
Poly(butyleneglycol-*alt*-(terephthalic acid-sebacic acid))				
	Chromatography	*o*-Chlorophenol		2047
Poly(carbon monoxide-*co*-ethylene)				
	Chromatography	Trichlorobenzene	SEC, styragel, 135°C	1108
Poly(carbon monoxide-*co*-vinyl chloride)				
	Chromatography	Carbon tetrachloride–ethyl alcohol(1 : 1) tetrahydrofuran	TLC, gradient elution	1633
Poly(1,3-bis(p-carboxyphenoxy)propane-*co*-sebacic acid)				
	Chromatography	Chloroform	SEC, styragel	685
Poly(3,3-bis(chloromethyl)oxacyclobutane-*co*-β-propiolactone)				
	Chromatography	Chlorobenzene	SEC, styragel	3465
	Fractional precipitation	Ethylene dichloride/ethanol		3124
Poly(cyclohexane oxide-*co*-nitroethylene)				
	Extraction	Benzene		1429

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(cyclohexene-*alt*-*N*-ethylmaleimide)				
	Chromatography	Ethanol/cyclohexane	Alternating, SEC *n*-octylporasil, 25°C	3486
Poly(cyclohexene carbonate-*co*-cyclohexene ether)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 35°C	3101
Poly(diketene-*co*-tetrahydrofuran)				
	Extraction	Methanol, benzene		3123
Poly(epichlorohydrin-*co*-styrene)				
	Turbidimetric titration	Benzene/methanol		2177
Poly(ether ketone)s	Chromatography	Tetrahydrofuran	SEC, styragel	1792
Poly(diethylazocarboxylate-*alt*-*N*-vinylcarbazole)				
	Extraction	Acetone or carbon tetrachloride		3288
Poly(ethylene-*co*-e-caprolactone)				
	Chromatography	Tetrahydrofuran	SEC, Shodex	863
Poly(ethylene glycol-*co*-2,2-butyl-hydroxyphenyl)propane-*alt*-terephthalic acid)				
	Extraction	Boiling, ethyl ether, acetone, 1,2-dichloropropane, tetrahydrofuran, 1,2-dichloroethane		1201
Poly(etheylene terephthalate-*co*-caprolactone)				
	Fractional precipitation	Chloroform/methanol		3497
Poly(ferrocenylsilene-*co*-ferrocenylgermane)s				
	Chromatography	Tetrahydrofuran + 0.1 wt.% tetra-*n*-butylammonium bromide	SEC, ultrastyragel	2557
Poly(formaldehyde-*co*-dioxolane)				
	Fractional precipitation	Tetrachloroethane–phenol/cyclohexane		1433
Poly(1,6-hexamethylene diol-*co*-2-methyl-2-ethyl-1,3-propanediol-*alt*-sebacic acid)				
	Chromatography	Chloroform	SEC, styragel	2426
Poly(3-hydroxybutyrate-*co*-hydroxyvalerate)				
	Fractional precipitation	Chloroform/heptane		3567
Poly(3-hydroxybutyrate-*co*-3-hydroxypropianate)				
	Chromatography	Chloroform	SEC, styragel	2968
			SEC, ultrastyragel, 40°C	1826
	Fractional precipitation	Chloroform/*n*-heptane		508
Poly(3-hydroxybutyrate-*co*-4-hydroxypropianate)				
	Chromatography	Chloroform	SEC, ultrastyragel, 40°C	1826
Poly(3-hydroxybutyrate-*co*-3-hydroxyvalerate)				
	Fractional precipitation	Acetone/water		2204
Poly[(*N*-(4-hydroxyphenyl)maleimide)-*alt*-(*p*-trimethylsilyl-α-methylstyrene)]				
	Chromatography	Tetrahydrofuran	SEC, styragel, 40°C	575
Poly(D,L-lactic acid-*co*-glycolic acid)				
	Chromatography	Methylene chloride	SEC	2793
Poly(methyl-2-ethyl-2-thiirane-*co*-thiopropylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2696
Poly(methyl methacrylate-*co*-*p*-cresyl methacrylate)				
	Fractional precipitation	Benzene/methanol	Crosslinked with *p*-cresol formaldehyde resins	2641
Poly(methyl methacrylate-*co*-dithio-3-chlorophenylene dithiophenylene methylenephenylene)				
	Extraction	Acetone and chloroform		1199
Poly(methyl methacrylate-*co*-oxyethylenedithioethylene)				
	Extraction	Acetone, carbon disulfide		1199
Poly(1,4-bis-(di-methylhydroxysilyl)phenylene-*co*-dimethylsiloxane)				
	Fractional precipitation	Benzene/methanol	20°C	98
Poly(methylphenyl phosphazene-*co*-methyl-3-amino phenyl phosphazene)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, μ-styragel, 30°C	3489
Poly(methylphenyl phosphazene-*co*-methyl-3-imino carbonyl ethylene)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, μ-styragel 30°C	3489
Poly(methylphenyl phosphazene-*co*-methyl-3-nitro phenyl phosphazene)				
	Chromatography	Tetrahydrofuran + 0.1% *n*-tetrabutylammonium bromide	SEC, μ-styragel, 25°C	3489
Poly(dimethylsilapropylene-*co*-diphenylsilapropynylenes)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	400

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(dimethylsilylene-*co*-diphenylsilylenes)				
	Fractional precipitation	Tetrahydrofuran/2-propanol		3590
Poly(dimethylsilylene-*co*-methyl-β-phenethylsilylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3590
	Fractional precipitation	Tetrahydrofuran/2-propanol		3590
Poly(dimethylsilylene-*co*-metylcyclohexylsilylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3590
	Fractional precipitation	Tetrahydrofuran/2-propanol		3590
Poly(dimethylsiloxane-*co*-isophorone diisocyanate)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel, 40°C	3604
Poly(dimethylethylsilylene-*co*-phenylmethylsylylene)				
	Chromatography	Tetrahydrofuran	SEC, Shodex	145
Poly(octamethylcyclotetrasiloxane-*co*-octaphenylcyclotetrasiloxane)				
	Chromatography	Benzene	SEC, styragel	103
Poly(octaphenylcyclotetrasiloxane-*co*-octa methylcyclotetrasiloxane)				
	Chromatography	Benzene	SEC, styragel	101
Poly(dioxolane-*co*-tetrahydrofuran)				
	Fractional precipitation	Chloroform/*n*-hexane		3563
Poly(oxyethylene-*co*-oxypropylene)				
	Chromatography	Butanone, water saturated	TLC, silica gel	3329
		Dioxane–water (10 : 1)	TLC, silica gel	3329
		Hexane–butanone methanol (6 : 2 : 1)	TLC, silica gel	3329
	Turbidimetric titration	Dimethylformamide/water		1056
		Methanol/water		1056
Poly(phenyl methacrylate-*co*-methyl methacrylate)				
	Fractional precipitation	Benzene/methanol		995
Poly(*n*-propylmethylsilane-*co*-cyclohexylmethylsilane)				
	Chromatography	Chloroform	SEC, μ-styragel	2858
	Fractional precipitation	Toluene/ethyl acetate	25°C	2858
Poly(*n*-propylmethysilene-*co*-isopropylmethylsilane)				
	Chromatography	Chloroform	SEC, μ-styragel	2858
	Fractional precipitation	Toluene/ethyl acetate	25°C	2858
Poly(*n*-propylmethylsilene-*co*-phenylmethylsilene)				
	Chromatography	Chloroform	SEC, μ-styragel	2858
	Fractional precipitation	Toluene/ethyl acetate	25°C	2858
Poly(silarylene)-*alt*-poly(siloxane)s				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	809
Poly(styrene-*co*-ε-caprolactam)	Chromatography	Tetrahydrofuran	SEC	3533
			SEC, styragel	3532
	Extraction	Benzene and trifluoroethanol		1279
	Fractional precipitation	Benzene/methanol		3532,3533
Poly(styrene-*co*-oxyethylenedithioethylene)				
	Extraction	Cyclohexane and carbon disulfide		1199
Poly(tetrahydrofuran-*co*-1,2-butene oxide)				
	Chromatography	Tetrahydrofuran	SEC	1204
Poly(tetrahydrofuran-*co*-oxyproylene)				
	Chromatography	Tetrahydrofuran	SEC	346,347,349, 350,1204
Poly(tetrahydrofuran-*co*-*n*-propyl glycidyl ether)				
	Chromatography	Tetrahydrofuran	SEC	1204
Poly(tetrahydrofuran-*co*-vinyl butyl ether)				
	Extraction	*n*-Heptane, acetone		2644
Poly(dithiotetramethylene-*co*-methyl methacrylate)				
	Extraction	Acetone and carbon tetrachloride		1199
Poly(dithiotetramethylene-*co*-styrene)				
	Extraction	Cyclohexane and carbon disulfide		1199
Poly(3,3,3-trifluoropropylmethylsilane-*co*-methylpropylsilane)				
	Chromatography	Tetrahydrofuran	SEC	941

3.12. BLOCK COPOLYMERS

Poly(acetaldehyde)-*block*-poly(oxypropylene)				
	Fractional solution	Methanol, chlorofom	Extraction	939

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(5-(*p*-acrylamidophenyl)-10,15,20-triphenylporphin)-*block*-poly(styrene)				
	Chromatography	Benzene	SEC, bioglass	1223
Poly(acrylonitrile)-*block*-poly(butandiene)-*block*-epoxy resins				
	Turbidimetric titration	Chloroform/methanol		1579
Poly(acrylonitrile)-*block*-poly(oxyethylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	3110
Poly(acrylonitrile)-*block*-poly(oxymethylene)				
	Extraction	Acetonitrile		969
Poly(γ-benzyl-L-glutamate)-*block*-poly(butadiene)				
	Extraction	Ethyl acetate–methanol (85 : 15); dimethylformamide		2580
	Fractional precipitation	Chloroform–*n*-heptane (1 : 1)/ methanol		2580
		Chloroform/hexane		2345
Poly(γ-benzyl-D,L-glutamate)-*block*-poly(butadiene)-*block*-poly(γ-benzyl-D,L-glutamate)				
	Fractional precipitation	*n*-Hexane–chloroform/ethanol		1248
Poly(butadiene)-*block*-poly(ε-caprolactam)				
	Extraction	Tetramethylsilane, trifluoroethanol		1279
	Fractional precipitation	*m*-Cresol/cyclohexane		1279
Poly(butadiene)-*block*-poly(methyl-2-ethyl-2-thiirane)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2696
Poly(butadiene)-*block*-poly(oxy-1,4-phenylenesulfonyl-1,4-phenylene-1,4-phenylene-isopropylidene-1,4-phenylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	2653
Poly(butadiene)-*block*-poly(oxymethylene)				
	Extraction	Toluene, acetone, methanol, water		2414
Poly(butadiene-*co*-styrene)-*block*-(epoxy resins)				
	Turbidmetric titration	Chloroform/ethanol		1579
Poly(*n*-butyl isocyanate)-*block*-poly(styrene)				
	Chromatography	Falta disolvent	SEC	1082,1083
Poly(β-butyrolactone)-*block*-poly(oxyethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3550
Poly(β-butyrolactone)-*block*-poly(oxypropylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3550
Poly(β-butyrolactone)-*block*-poly(propiolactone)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3550
Poly(ε-caprolactam)-*block*-poly(1,4-butyleneglycol-*co*-4,4-diphenylmethanediisocyanate)				
	Fractional precipitation	Dimethylformamide/diamyl ether		2990
Poly(ε-caprolactam)-*block*-poly(isobutene)-*block*-poly(ε-caprolactam)				
	Extraction	*n*-Hexane, 2,2,2 trifluoroethanol		3494
Poly(ε-caprolactam)-*block*-poly(styrene)				
	Fractional precipitation	*m*-Cresol/cyclohexane		1279
	Turbidimetric titration	Toluene/*n*-hexane		1109
Poly(ε-caprolactone)-*block*-poly(lactide)				
	Extraction	Acetone, chloroform		881
Poly(ε-caprolactone)-*block*-poly(D,L-lactide)-*block*-poly(ε-caprolactone)				
	Chromatography	Dimethylformamide	SEC, styragel, 45°C	3036
Poly(ε-caprolactone)-*block*-poly(oxyethylene)				
	Fractional precipitation	Benzene–methanol/isooctane		2581,2582
		Benzene/isooctane		2581,2582
Poly(ε-caprolactone)-*block*-poly(β-propiolactone)				
	Fractional precipitation	Toluene/petroleum ether		1203
Poly(ε-caprolactone)-*block*-(tetrahydrofuran)-*block*-(ε-caprolactone)				
	Chromatography	Tetrahydrofuran	SEC	2412
Expoxide resin-*block*-poly(butadiene-*co*-styrene)				
	Turbidimetric titration	Chloroform/methanol		1580
Poly(ethylene glycol)-*block*-Poly(L-lysine)				
	Centrifugation	Tetrahydrofuran	SEC	1211
Poly(ethylene terepthalate)-*block*-poly(tetramethylene ether)				
	Cross fractional precipitation	Tetrachloroethane–methanol, trichloroethylene–*n*-heptane	20°C	3600
Poly(2,2-bis(4-hydroxyphenyl)propane-*block*-epxoy resin)				
	Chromatography	Tetrahydrofuran		686

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(2,2-bis(4-hydroxyphenyl)propane-*co*-bis(4-hydroxy-2,5-dichlorophenyl)methane-*co*-phosgene), (*block* copolymer)				
	Chromatography	Chloroform	SEC, styragel	2493
		Tetrahydrofuran	SEC, styragel	2493
Poly(isoprene)-*block*-poly(oxytetramethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1
Poly(isoprene)-*block*-(2,4-toluene diisocyanate)				
	Chromatography	Toluene, tetrahydrofuran	SEC	1082
Poly(D,L lactide)-*block*-poly(ethyleneglycol)				
	Chromatography	Tetrahydrofuran	SEC	739
Poly(methyl-2-ethyl-2-thiirane)-*block*-poly(propylene sulfide)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2696
Poly(methyl glycidyl ether)-*block*-poly-(oxytetramethylene)				
	Fractional precipitation	Methanol/water		2645
Poly(methyl methacrylate)-*block*-poly-(oxytetramethylene)-*block*-poly(methylmethacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	3108
	Fractional precipitation	Choroform/methanol		3108
Poly(methyl methacrylate)-*block*-(2,4-toluenediisocyanate)				
	Chromatography	Toluene, tetrahydrofuran	SEC	1082
Poly(dimethyl siloxane)-*block*-poly(styrene)				
	Chromatography	Toluene	SEC, 20°C	701
Poly(dimethylsilylene)-*block*-poly(styrene)				
	Chromatography	Tetrahydrofuran	SEC	287,554
		Toluene	SEC, 20°C	701
Poly(dimethylsilylene)-*block*-poly(styrene)-*block*-poly(dimethylsilylene)				
	Chromatography	Toluene	SEC, μ-styragel, 25°C	178
			SEC, styragel	1511
Poly(α-methylstyrene)-*block*-poly-(oxytetramethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2747
Poly(α-methylstyrene)-*block*-poly(propylenesulfide)-*block*-poly(α-methylstyrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	2747
Poly(1,3-dioxolane)-*block*-poly(p-methoxystyrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1410
Poly(oxybutylene)-*block*-poly(oxyethyylene)-*block*-poly(oxybutylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3542
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)-*alt*-poly(isobutene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	1929
Poly(oxydimethylsilylene)-*block*-poly(oxymethylsilylene)				
	Chromatography	Ethyl acetate	SEC, styragel, 30°C	93
Poly(oxyethylene)-*block*-poly(ethyl methacylate)				
	Fractional precipitation	Benzene/2,2,4-trimethylpentane		2951
Poly(oxyethylene)-*block*-poly(isoprene)-*block*-poly(ethylene oxide)				
	Sedimentation equilibrium	Water, chloroform	20°C	2271
	Sedimentation velocity	Water, chloroform	20°C	2271
Poly(oxyethylene)-*block*-poly(isoprene)-*block*-poly(styrene)-*block*-poly(isoprene)-*block*-poly(oxyethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 30°C	1728
Poly(oxyethylene)-*block*-poly(methacrylonitrile)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	3110
Poly(oxyethylene)-*block*-poly(methyl methacrylate)-*block*-poly(oxyethylene)				
	Chromatography	Tetrahydrofuran	SEC	3109
Poly(oxyethlene)-*block*-poly(α-methylstyrene)				
	Chromatography	Tetrahydrofuran	SEC	1631
Poly(oxyethylene)-*block*-poly(oxybutylene)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3543
Poly(oxyethylene)-*block*-poly(oxyethylene)-*co*-bis(4-isocyanotophenyl)methane-*co*-poly(oxyethyleneoxyadipoyl)-*block*-poly(oxethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	970
Poly(oxyethylene)-*block*-poly(oxyethyleneoxyadipoyl)				
	Fractional precipitation	Benzene/isooctane	35°C	970
Poly(oxyethylene)-*block*-poly(oxyethyleneoxyadipoyl)-1-methyl-2,4-disocyanato-benzene-poly(oxy-ethyleneoxyadipoyl-*block*-polyoxethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 25°C	970
Poly(oxyethylene)-*block*-poly(oxypropylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	26
Poly(oxyethylene)-*block*-poly(oxypropylene)-*block*-poly(oxyethylene)				
	Chromatography	Dimethylformamide + lithium chloride	SEC, silica-Ti-fluoropolymer, 30°C	3424

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TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(oxyethylene)-*block*-poly(isoprene)-poly(styrene)-*block*-poly(styrene)-*block*-poly(oxyethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 30°C	1728
Poly(oxyethylene)-*block*-poly(2-vinylpyridine)				
	Chromatography	Dimethylformamide	SEC, styragel	2039
	Fractional precipitation	Benzene/trimethyl-2,2, 4-pentane(isooctane)	35°C	2039
Poly(oxyethyleneoxyadipoyl)-*block*-poly(styrene)-*block*-poly(methyl methacrylate)				
	Fractional precipitation	Tetrahydrofuran/water		1253
Poly(oxymethylene)-*block*-poly(isoprene)				
	Extraction	Toluene, acetone, methanol water		2414
Poly(oxymethylene)-*block*-poly(2-methyl-5-vinylpyiidine)				
	Extraction water	Toluene, acetone, methanol,		2414
Poly(oxymethylene)-*block*-poly(α-methylstyrene)				
	Extraction	Toluene, acetone, methanol		2414
Poly(oxymethylene)-*block*-poly(styrene)				
	Extraction	Toluene, acetone, methanol,water		2414
Poly(oxypropylene)-*block*-poly(oxyethylene)-*block*-poly(oxypropylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3383
Poly(oxypropylene)-*block*-poly-(oxytetramethylene)				
	Chromatography	Tetrahydrofuran	SEC, styragel, 30°C	1761
Poly(oxypropylene)-*block*-poly(styrene)				
	Fractional precipitation	Benzene/methanol		2179
Poly(styrene)-*block*-poly(isoprene)-*block*-poly(ethylene sulfide)				
	Chromatography	Tetrahydrofuran	SEC, styragel	628
Poly(styrene)-*co*-5,10,15,20-tetra(α,α,α,α-*o*-methacrylamidophenyl)porphin)				
	Chromatography	Benzene	SEC, bioglass	1223
Poly(styrene)-*block*-(1-methyl-2,4-diisocyanate cyclohexane)				
	Chromatography	Toluene, tetrahydrofuran		1082
Poly(styrene)-*block*-poly(oxyethylene)				
	Chromatography	Dimethylformamide	SEC, Merck gel	3415
		Tetrahydrofuran	Star polymer	3246
Poly(styrene)-*block*-poly(oxypropylene)				
	Chromatography	Dimethylformamide	SEC, Merck gel	3415
Poly(styrene)-*block*-poly(oxytetramethylene)				
	Chromatography	Dimethylformamide	SEC, Merck gel	3415
		Tetrahydrofuran	SEC, styragel	1,2746,3530
Poly(styrene)-*block*-(2,4-toluene diisocyanate)				
	Chromatography	Toluene, tetrahydrofuran	SEC	1082,1083
Poly(tetrahydrofuran)-*block*-(ε-caprolactone)				
	Chromatography	Tetrahydrofuran	SEC	2411
Poly[γ-(vinyloxy)ethyl cinnamate]-*block*-poly(isobutyl vinyl ether)				
	Chromatography	Ethyl acetate	SEC, styragel	1938
3.13. GRAFT COPOLYMERS				
Poly(acrylic acid)-*graft*-poly(β-alanine)				
	Turbidimetric titration	Formic acid–water (85 : 15)/ethanol		1173
Poly(acrylonitrile)-*graft*-poly(imino-1,3-phenyleneiminoisophthaioyl)				
	Extraction	Dimethylformamide		1587
		Nitromethane/water (5%)		1587
	Fractional precipitation	Dimethylformamide/methanol		1587
Poly(acrylonitrile)-*graft*-poly(iminoadipoyliminohexamethylene)				
	Extraction	Dimethylformamide		1587
Poly(epoxide)-*graft*-poly(organosiloxanes)				
	Turbidimetric titration	Dioxane/water		96
Poly(α-hydroxyisobutyric acid anhydrosulfite)-*graft*-poly(acrylonitrile)				
	Turbidimetric titration	Dimethylformamide/chloroform		1426
Poly(bis(4-isopropylphenoxy)phosphazene)-*graft*-poly(styrene)				
	Fractional precipitation	Benzene/acetone		1067
Poly(methyl methacrylate)-*graft*-poly(imino-1-oxohexamethylene)				
	Extraction	Benzene or acetone		1259
Poly(methyl methacrylate)-*graft*-poly(iminoadipoyliminohexamethylene)				
	Fractional precipitation	Formic acid/methanol		1350
		m-Cresol/methanol		1350

TABLE 3. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Poly(methyl methacrylate)-*graft*-poly(oxy-1-oxo-hexamethylene)				
	Extraction	Benzene		2301
Poly(dimethylsiloxane)-*graft*-poly(methyl methacrylate)				
	Extraction	Dimethylsulfoxide/tetrachlororethane		2630,3077
	Fractional demixing	Dimethyl sulfoxide/ tetrachloroethane		3077
Poly(2-methyl-2-oxazoline)-*graft*-poly(vinyl chloride)				
	Extraction	Methanol		3236
Poly(oxyethylene)-*graft*-poly(*n*-butyl methacrylate)				
	Fractional solution	Benzene/2,2,4-trimethylpentane	Column extraction	3200
Poly(oxyethylene)-*graft*-poly(iminoadipoyliminohexamethylene)				
	Extraction	Methanol		1586
Poly(oxyethylene)-*graft*-poly(methyl methacrylate)				
	Extraction	Ethylacetate, methanol		420
	Fractional precipitation	Chloroform/ethyl ether		3297
Poly(oxyethylene)-*graft*-poly(*n*-propylmethacrylate)				
	Fractional solution	Benzene/2,2,4-trimethylpentane	Column extraction	3200
Poly(oxyethyleneoxycarbonylimino-1,4-phenylenemethyleneiminocarbonyl)-*graft*-poly(acrylonitrile)				
	Extraction	Dimethylformamide/ethanol		14
Poly(oxyethyleneoxyterephthaloyl)-*graft*-poly(vinyl acetate)				
	Extraction	Acetone		870
Poly(oxyethyleneoxyterephthaloyl)-*graft*-poly(oxyethyleneoxysuccinoyl)				
	Chromatography	*o-tert*-Butylphenol	SEC, styragel, 110°C	2792
Poly(oxymethyleneoxyethylene)-*graft*-poly(*p*-methoxystyrene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	1410
Poly(phenyl methyl silylene)-*graft*-poly(tetrahydrofuran)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	1378
Poly(pivalolactone)-*graft*-poly(ethylene-*co*-propylene-*co*-1,4-hexadiene)				
	Chromatography	Tetrahydrofuran	SEC, styragel	3198
	Extraction	Hot hexane, hot benzene, hot hexafluoroisopropanol, *m*-cresol		3198
Poly(propylene)-*graft*-poly(2(2-hydroxy-5-vinylphenyl)2H-benzotriazole)				
	Chromatography	Tetrahydrofuran	SEC	2658
Poly(siloxane)-*graft*-poly(pivalolactone)				
	Chromatography	Methylene chloride/ hexafluoroisopropyl alcohol		3423
Poly(styrene)-*graft*-poly(imino-1-oxohexamethylene)				
	Extraction	Benzene		3483
			$n=4, 6, 10$	877
Poly(styrene)-*graft*-poly(oxyfumaroyloxy-(methylene)$_n$				
	Extraction	Benzene		3483
Poly(styrene)-*graft*-poly(oxymaleoyloxyhexamethylene)				
	Fractional precipitation	Benzene/methanol		2410
		Benzene/petroleum ether		2410
Poly(styrene)-*graft*-poly(siloxane)				
	Turbidimetric titration	Benzene/methanol–acetone (3 : 1)		2178
Poly(vinylpyrrolidone)-*graft*-poly(oxy-(*cis*-2-butenylene)oxycarbonylimino-1, 4-phenylenemethylene-1,4-phenyleneimino-carbonyl)				
	Chromatography	Dimethylformamide	SEC, styragel	820
Poly(vinylprrolidone)-*graft*-poly(oxypropyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl)				
	Chromatography	Dimethylformamide	SEC, styragel	820
Poly(vinylpyrrolidone)-*graft*-poly-(oxytetramethyleneoxycarbonylimino-1,4-phenylenemethylene 1,4-phenyleneiminocarbonyl)				
	Chromatography	Dimethylformamide	SEC, styragel	820

TABLE 4. POLY(SACCHARIDES)

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
4.1. POLY(SACCHARIDES)				
Acetoxy propyl cellulose	Fractional precipitation	Benzene/heptane		2979
Alginates	Centrifugation	Water	Ethylene diammonium alginates	623

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TABLE 4. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
	Fractional precipitation	Water/manganous chloride + calcium chloride	Sodium alginate	2118
		Water/sodium chloride	Sodium alginate	1230
Amilose tributyrate	Fractional precipitation	Toulene/petroleum ether	25°C	657
Amylopectin	Chromatography	Dimethyl sulfoxide	SEC, silica porous gel, 60°C	2817
Amylopectin xanthate	Fractional precipitation	Water/manganous chloride + calcium chloride	Sodium salt	2022
Amylose	Chromatography	Dimethyl sulfoxide	SEC, μ-bondagel	1900
			SEC, Porasil, 30°C	3341
	Fractional precipitation	Acetone/acetone–water (1 : 1)	Acetate	1400
		Acetone/water	Carbanilate	469
		Dimethyl sulfoxide/acetone		1013
		Dimethyl sulfoxide/ethanol		197,652,1013
		Diemthyl sulfoxide/*n*-butanol	25°C	198,199,200,201,202
		Dimethylformamide/methanol + 10% $CaCl_2$	Carbanilate	1401
		Nitromethane/methanol	Acetate	653
		Water/methanol	Glucoamicoethyl	3457
Amylose acetate	Fractional precipitation	Dimethyl sulfoxide/*n*-butanol	25°C	198,199,202, 203,653,654
		Nitromethane/methanol	25°C	653,654
Amylose benzoate	Fractional solution	Nitromethane/methanol	31°C	2525,2526,2527
Amylose triacetate	Fractional precipitation	Nitromethane/methanol		659
Amylose tricarbanilate	Chromatography	Ethanol/water	25°C	2594,2595
	Fractional precipitation	Acetone/methanol	25°C	204,205,206,207
		Dimethylformamide/methanol + 10% $CaCl_2$	20°C	677,1401,1402,2595
	Sedimentation velocity	Dioxane		3104
Amylose tripropianate	Fractional precipitation	Ethyl acetate/methanol	25°C	658
Amylose xanthane	Fractional precipitation	Water 1N sodium hydroxide/methanol	Sodium salt	2021
Arabinogalactam	Chromatography	Water	SEC, Sephadex G-75	3118
Arabinose	Chromatography	Chloroform/acetone, pyridine, glacial acetic acid	Araban acetate, charcoal	1107
	Fractional precipitation	Acetone/petroleum ether	Araban acetate	1107
Carboxymethyl amylose	Chromatography	Sodium chloride 0.5M aq. solution		408,1084
	Fractional precipitation	Sodium chloride 0.5M /acetone		2023
		Water–acetone (7 : 3) + 0.5 % sodium chloride/acetone	30°C, sodium salt	2529,2530,2531
Carboxymethyl cellulose	Fractional precipitation	Water/acetone	Sodium salt	449
		Water/ethanol		1112
Cellulose	Centrifugal sedimentation	Water suspension	Crystals suspension micel sol	1483
	Chromatography	Cadoxen	SEC	294
	Extraction	Iron–sodium tartrate	Active charcoal	1484
			Regenerated fibers	2538
		Water, 0.25% ammonium oxalate and 80% ethanol		1881
	Fractional precipitation	Acetone/acetone–water		2072
		Cadoxen–water (1 : 1)/propanol–water(3 : 1)		2573,2732,2733
		Cadoxen–water/propanol–water	25°C	1176
		Cadoxen/water–glycerol(1 : 1)		377
		Cuene/propanol, cuoxam/propanol	Varying temp.	2988
		Iron-sodium tartrate/mannitol		1487,1488
		Sodium hydroxide 2.0 M/methanol	Rayon summative method	631
		Sulfuric acid (60–65%)/water	− 15°C	2995
	Fractional solution	Cadoxen		1485,1486
		Cuene	Extraction	2306
		Cuoxam		2982
		Iron–sodium tartrate solution/ sodiumchloride 5 *N*	Summative method	1481,1482
Cellulose acetate	Chromatography	1,2-Dichloroethane	SEC, Porasil A, triacetate	3120
		Chloroform–acetone (1 : 1)	Triacetate, calcium silicate as absorbent	2333

TABLE 4. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Methylene dichloride–acetone/methanol	TLC, silica gel for MW distribution	1559
		Methylene dichloride–tetrahydrofuran/methanol	TLC, silica gel for MW distribution	1559
		Methylene dichloride/methanol	TLC, silica gel for acetyl group distribution	1559
		Tetrahydrofuran	SEC	433,2304,3149
			SEC, styragel triacetate	466
	Diffusion into porous charcoal	Acetone, dioxane	Countercurrent	3115,3116
	Fractional precipitation	1-Chloro-2,3-epoxypropane/*n*-hexane	35°C	1565
		Acetone–ethanol (5 : 3)/*n*-hexane	Triacetate	909,2246
		Acetone–water (3 : 1)/water		3070
		Acetone/95% ethanol	Lowering temp.	162,1566,2596, 3037,3113
		Acetone/ethanol	23°C	1564,1568,1750,3153
		Acetone/*n*-heptane–acetone (1 : 1)	Diacetate	1097
		Acetone/water		1097,1169
		Acetone/water followed by acetone/*n*-pentane	Cross fractionation	2780
		Aq. acetone/*n*-heptane–acetone (3 : 1)		2863
		Chloroform–acetone (1 : 1)/petroleum ether	Triacetate, 25°C	1373,1374
		Chloroform–acetone/petroleum ether	25°C	1372
		Chloroform/ethanol	Diacetate	762
		Tetrachloroethylene/ethanol		762
	Fractional solution	Acetone	Adsorption on to activated carbon	2170
		Acetone/methanol	Column extraction	742,743
		Dioxane/*n*-hexane mixtures/dioxane/heptane reflux	Triacetate	656
		Water/water–acetone	Summative	1750
	Sedimentation velocity	Acetone	25°C, diacetate	1097,3410,3411
	Thermal diffusion	Dichloroethane	Triacetate	1862,1863,1868
	Turbidimetric titration	Acetic acid/water		2180
		Acetone/ethanol		1845
		Butanol–ethanol(4 : 1)/95% ethanol		338
		Dimethylformamide/dibutyl ether		1059
Cellulose acetate–butyrate	Fractional precipitation	Acetone/isopropyl alcohol		2269
		Acetone/water–acetone (1 : 1) + 2% salt and then water alone + 2% salt		3139
	Turbidimetric titration	Acetone/ethanol–water (3 : 1)		2267
Cellulose cyanoethyl	Fractional precipitation	Methylene dichloride–methanol (10 : 1)/methanol	21°C	3278
	Sedimentation velocity	Acetone	26°C	3278
Cellulose nitrate	Chromatography	Acetone	Activated carbon	601
			SEC	2158,2159
			SEC, Porasil, 12–13% *N*	438
		Acetone–methanol–chloroform (10 : 10 : 5)	TLC, silica gel, 25°C, TLC, kiseselgur, 25°C	1563
		Acetone/cyclohexane followed by acetone/*n*-methanol	Elution chromatography, starch	439
		Dioxane–methanol–2-propanol (15 : 10 : 1)	TLC, silica gel, 25°C, TLC, Kieselgur, 25°C	1563
		Ethyl acetate	SEC, styragel	1341
		Nitromethane–methanol (2 : 8)	TLC, kieselgur, 25°C	1563
			TLC, silica gel, 25°C, TLC, alumina, 25°C	1563
		Tetrahydrofuran	SEC	2150,2151,2158, 2304,2305,2491
			SEC, μ-styragel	2058

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TABLE 4. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
			SEC, μ-styragel,styragel, 23°C	3488
			SEC, porous silica gel	2057
			SEC, styragel	37,810,2154, 2755,2936,2937, 2938,2939 2940, 3404,3432
	Diffiusion into porous charcoal	Acetone and dioxane	Countercurrent	3116
	Distribution between immiscible liquids	Cellulose triacetate gel in methyl acetate, water, ethanol, butyl acetate		439
		Fractional precipitation	Acetone–9% water/water	2202,2782,3201
		Acetone–ethanol (5 : 3)/*n*-hexane	Room temp.	885,2245,2248
		Acetone–ethanol (5 : 3)/shellsolv	High MW	1398
		Acetone/acetone–water (1 : 1)		2334,3580,3581,3582
		Acetone/acetone–water (4 : 1) until pure water	0°C, dark	2052,2053,2054,2897
		Acetone/acetone–water mixtures	12–13% *N*	438
		Acetone/ligroin		140
		Acetone/*n*-heptane	Lowering temp.	2987
		Acetone/*n*-hexane		477,703,2633
			25°C	1563
			Low MW	1398
		Acetone/*n*-hexane(1/1)/*n*-hexane		1213
		Acetone/petroleum ether		2334,3399
		Acetone/water	Triangle method	20,703,2051,2125, 2126,3208,3425
		Acetone/water followed by acetone/*n*-heptane	Cross fractionation	15
		Acetone/water–acetone		2056,2562,2908
		Acetone/water–acetone (5 : 1)		2055
		Ethyl acetate/*n*-heptane		2861
	Fractional solution	Acetone/methanol–water	Column extraction	743
		Ethyl acetate/95% ethanol		838,1292
	Membrane diffusion	Acetone		3381
	Sedimentation velocity	Acetone	Ultracentrifuge	2479
	Turbidimetric titration	Acetone/acetone–water	Dinitrate	2448
		Acetone/methanol–water (9 : 1)		2313
		Acetone/water	Dinitrate	2313
Cellulose propionate	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	519
Cellulose tributyrate	Fractional precipitation	Acetone/water		1856
Cellulose tricarbanilate	Chromatography	Acetone	SEC, spherosil	687
		Acetone/water	Sand, gradual increase in temp.	687
		Tetrahydrofuran	SEC	432,2298,3193
			SEC, spherosil	687
	Fractional precipitation	Acetone/1% aq. sodium chloride		432
		Acetone/acetone–water (1 : 1)		1403
		Acetone/methanol		118
		Benzene/methanol		3273
	Fractional solution	Acetone/water		3104
Celllulose tripropionate	Chromatography	Tetrahydrofuran	SEC	433
	Fractional precipitation	Acetone/petroleum ether		869
Cellulose xanthate	Fractional precipitation	Sodium chloride–10% aq. methanol	Viscose fraction	1040
Cyanoethyl (hydroxypropyl)cellulose				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel, 25°C	2107
	Fractional precipitation	Acetone/*n*-heptane		2107
Dextran	Chromatography	Dimethyl sulfoxide/methanol (15%) + ammonium acetate	SEC	2818
		Sodium chloride water 0.5%	SEC, porous glass	232
		Sodium nitrate 0.1 M	SEC, spherosil	2762
		Water	Activated carbon	601

TABLE 4. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
			SEC	1116,1884
			SEC, controlled porous glass	234,816
			SEC, aguapak	378
			SEC, aquagel-OH	2121
			SEC, Porasil	3028
			SEC, porous glass beads bonded with hydrophilic compounds	1611,1612
			SEC, Sephadex, preparative	232,233,234
			SEC, sepharose	233,234
			SEC, sepheron	1070,2617
			SEC, TSK gel	1224
		Water(10 mM sodium chloride)		3343
		Water, cadoxen	SEC	294
		Water/1% diethylene glycol	SEC, porous silica beads	1885
		Water/ethanol	Precipitation chromatogr.	814,815
	Fractional precipitation	Water/95% Ethanol	Acid hydrolyzed	2950
		Water/manganous chloride + calcium chloride	Low MW	127,222, 1028,3046,3605
	Fractional solution	Water–ethanol		2381
	Sedimentation velocity	Water	Ultracentrifuge, 20°C	814
	Turbidimetric titration	Water/ethanol		1130
	Ultrafiltration	Water		2084
			Amicom-XM-4B and UM-1 diaflomembranes	183
Ethyl cellulose	Chromatography	Methylene dichloride	SEC	440
	Fractional precipitation	Acetone/water		2152,3093
		Ethyl acetate –acetone (1 : 4)/water		2862
Poly(2-(diethylamino)ethyl)dextran				
	Chromatography	Water, 0.8M sodium nitrate		1209
Diethylamino amylose hydrochloride				
	Chromatography	Aq. sodium chloride 0.5M	SEC	1084
Poly(2-D-glucopyranosyl-3-oxymethyl)-acrylic acid ethyl ester)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2632
Poly(D-glucose)	Fractional precipitation	Water/methanol	Lowering temp.	2164
Guaran triacetate	Fractional precipitation	Chloroform/ethanol	22.5°C	1737
	Sedimentation velocity	Acetonitrile	25°C	1337
Gum arabic	Fractional precipitation	Aq. sodium chloride 0.5M/acetone		3117
Hemicellulose	Fractional precipitation	KOH 5%/methanol		3309
Hemicellulolse acetate 3309	Fractional precipitation	Chloroform-ethanol (9 : 1)–/*n*-hexane		
Tri-*o*-heptylcellulose	Chromatography	Chlroroform	SEC, preparative	3130
	Fractional precipitation	Benzene/methanol		3130
Hydroxyethyl cellulose	Chromatography	Sodium chloride 0.25M/sodium phosphate 0.005M		2121
	Fractional precipitation	Water/acetone	25°C	448,836
Hydroxyethyl cellulose acetate	Chromatography	Tetrahydrofuran	SEC	3498
2-Hydroxypropyl cellulose, benzoic acid ester				
	Fractional precipitation	Acetone/methanol–water (85 : 15)		316
Hydroxypropy; cellulose	Chromatography	Tetrahydrofuran	SEC, 25°C	3487
	Extraction	Ethanol(anhydrous)–heptane mixtures	13	
	Fractional precipitation	Ethanol/heptane		1759,2761
		Ethanol/*n*-heptane	30°C	3487
o-(2,3-Dihyproxypropyl)cellulose				
	Chromatography	Acetic acid solution (water)	SEC, Tosoh, 25°C	2850
2-Hydroxypropyl cellulose propionic acid ester				
	Fractional precipitation	Benzene/*n*-heptane		3240
Hydroxypropyl lignin	Fractional precipitation	Acetone/hexane		1647
Levan	Chromatography	Dimethyl sulfoxide/methanol (15%) + ammonium acetate	SEC	2818
Lignin	Chromatography	Dimethyl sulfoxide	SEC	3041
			SEC, Sephadex	2111
		Sodium hydroxide 0.10M	SEC, Sephadex	2835

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TABLE 4. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
		Sodium hydroxide 0.10M (carbonate-free)	SEC, analytical and preparative	2836
	Fractional precipitation	Sulfuric acid	By pH change and centrifugation	2837
Ligninsulfonic acid, sodium salt				
	Chromatography	Sodium chloride 0.0001M Aq.	SEC, Sephadex G-50	1153,1154
		Water	SEC, 44°C	378
	Extraction	Water/ethanol		879
	Fractional precipitation	Water/ethanol		879
	Fractional solution	Aq. sodium choride/95% ethanol	Brownian diffusion	2219
Poly(3-D-methacryloyl glucomic acid)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2632
Methyl cellulose	Thermal diffusion	Water		2779
Pectins	Fractional precipitation	Ethylene diamine	Amylopectin temp.	3058
	Fractional precipitation	Dioxane/petroleum ether	Propianate, lowering	2502
	Membrane diffusion	Water		931
Pullulan	Field flow	Water		1688
	Fractional precipitation	Water/isopropyl alcohol	Extracellular	1615
Schizophyllan	Chromatography	Sodium hydroxide water 0.01M	SEC, TSK gel	1450
	Fractional precipitation	Water/ethanol, water/acetone		1450
Scleroglucan	Chromatography	Dimethyl sulfoxide/methanol (15%) + ammonium acetate	SEC	2818
Starch	Alkaline leaching	Sodium hydroxide 0.5 M (water)		196
	Aqueous leaching	Water	Several temp. and times	196,650,1121
	Centrifugation	Dilute alkali		249
	Chromatography	Dimethylsulfoxide + 0.03 M aqueous sodium nitrate	SEC, μ-styragel, 80°C	590
		Dimethyl sulfoxide/methanol (15%) + ammonium acetate	SEC	2818
	Extraction	Aq. sodium hydroxide/*n*-butanol		196,248
	Fractional precipitation	Thymol/*n*-butanol		651
Tribenzyl glucan	Sedimentation velocity	Benzene	8°C benzylated lineat α-, 1, 6 glucans	2140
Poly(vinyl galactose ketone)	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2632
Poly(vinyl isopropylidene arabinose ketone)				
	Chromatography	Tetrahydrofuran	SEC, ultrastyragel	2632
Xanthan	Chromatography	Aq. sodium sulfate 0.5M	SEC, preparative	2514
		Sodium nitrate 0.1M + 1% ethylene glycol	SEC, silica gel	1855
		Water + sodium chloride 5 g/l	Surface exclusion chromatography SiC particles, 30°C	1893
	Fractional precipitation	Acetone/0.5M sodium chloride		1178
Xylan	Fractional precipitation	Dimethyl sulfoxide/ethanol		1887

4.2. GRAFT COPOLYMERS

Acetyl cellulose-*graft*-poly(acrylamide)				
	Extraction	Water, acetone		3160
Amylopectine-*graft*-poly(methyl methacrylate)				
	Extraction	1,2-Dichloroethane		1100
Amylose triacetate-*graft*-poly-(4,4′-diphenylmethane diisocyanate-*co*-poly(butadiene))				
	Chromatography	Chloroform	SEC, μ-bondagel	1978
Amylose-*graft*-poly(4,4′-diphenylmethane diisocyanate-*co*-poly(ethylene-*co*-poly(ethylene-*co*-propylene adipate))				
	Chromatography	Dimethyl sulfoxide	SEC, μ-bondagel	1900
Amylose-*graft*-poly(4,4′-diphenylmethane diisocyanate-*co*-poly(ethylene terephthalate))				
	Chromatography	Dimethyl sulfoxide	SEC, μ-bondagel	1900
Amylose-*graft*-poly(methyl acrylate)				
	Fractional precipitation	Dimethyl sulfoxide/*n*-butanol		2534
Amylose-*graft*-poly(methyl methacrylate)				
	Extraction	1,2-Dichloroethane		1100
Amylose triacetate-*graft*-poly-(4,4′-diphenylmethane diisocyanate-*co*-propylene glycol)				
	Chromatography	Chloroform	SEC, μ-bondagel	1978
Amylose triacetate-*graft*-poly-(2,4-toluene diisocyanate-*co*-poly(butadiene))				
	Chromatography	Chloroform	SEC, μ-bondagel	1978

TABLE 4. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Amylose triacetate-*graft*-poly(2,4-toluene diisocyanate-*co*-poly(oxypropylene)				
	Chromatography	Chloroform	SEC, μ-bondagel	1978
Benzyl cellulose-*graft*-poly(styrene)				
	Fractional precipitation	Benzene/methanol		530
Cellulose-*graft*-poly(acrylamide)				
	Extraction	Dimethylformamide,water cuprammonium	Cellophane	1008
Cellulose-*graft*-poly(acrylonitrile)		Dimethylformamide, water cuprammonium	Cellophane	1008
Cellulose-*graft*-poly(ethyl acrylate)				
	Extraction	Acetone	55°C	1534
		Benzene		2191
Cellulose-*graft*-poly(methyl methacrylate)				
	Extraction	Acetone		1007,1284,1809
		Acetone or benzene	Viscose	827
		Benzene		1041
		Dichloroethane		2187
		Xanthation process		1041
Cellulose-*graft*-poly(styrene)	Extraction	Boiling benzene		2175
		Tetrahydrofuran–water (98 : 2)		1162
	Fractional precipitation	Dioxane/methanol	20°C	2914
Cellulose-*graft*-poly(2-vinylpyridine)				
	Chromatography	Benzene/methanol	Precipitation chromatography, celite, 15–60°C	3447
	Extraction	Benzene		2803
Cellulose nitrate-*graft*-poly(methyl methacrylate)				
	Chromatography	Tetrahydrofuran	SEC, μ-styragel	3057
	Extraction	Benzene, methanol		3057
Cellulose acetate- *graft*-poly(acrylamide)				
	Extraction	Acetone		1008
Cellulose- acetate-*graft*-poly(acrylonitrile)				
	Extraction	Dimethylformamide	Acetylated	2748
		Dimethylformamide–dioxane (3.5 : 2)	Cotton	2748
	Fractional precipitation	Dimethylformamide/chloroform		2748
Cellulose-acetate-*graft*-poly(butyl acrylate)				
	Extraction;precipitation	Acetone/water		1406
		Benzene		1406
	Turbidimetric titration	Actone–*n*-methanol (6 : 4)/ methanol		1406
Cellulose acetate-*graft*-poly(ethyl acrylate)				
	Extraction;precipitation	Acetone/water		1406
		Benzene		1406
	Turbidimetric titration	Acetone–methanol(6 : 4)/methanol		1406
Cellulose-acetate-*graft*-poly(methyl acrylate)				
	Extraction;precipitation	Acetone/water		1406
		Benzene		1406
	Turbidimetric titration	Acetone–methanol (6 : 4)/methanol		1406
Cellulose-acetate-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Pyridine, acetone (95%/–5%)/water		342
Cellulose acetate-*graft*-poly(2-methyl-5-vinylpyridine)				
	Extraction	Acetone, methylene chloride, methanol		1523
Cellulose acetate-*graft*-poly(oxymethylene)				
	Fractional precipitation	Dimethylformamide/ethyl ether		209
Cellulose acetate-*graft*-poly(styrene)				
	Chromatography	Benzene/ethanol	Precipitation chromatogr., celite, 15–60°C	3447
		Chloroform	TLC, silica gel	1357
		Methylene dichloride–methanol (1 : 1)	TLC, silica gel	1357
	Fractional precipitation	Chloroform/*n*-heptane	Triacetylated viscose	2748
		Pyridine/Aq. hydrochloric acid		3547

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TABLE 4. *cont'd*

Polymer	Method of fractionation	Solvent or solvent/nonsolvent mixture	Remarks	Refs.
Cellulose acetate-*graft*-poly(vinyl acetate)				
	Extraction;precipitation	Methanol; benzene/methanol		509
Cellulose acetate-*graft*-poly(3-vinylpyridine)				
	Extraction	Methanol;acetone–dichloroethane (1 : 4)		3158
Cellulose acetate-*graft*-poly-(*N*-vinylpyrrolidone)				
	Extraction	Water, acetone		3159
Cotton-*graft*-poly(acrylamide)	Extraction	Water		3414
Cotton-*graft*-poly(glycidyl methacrylate)				
	Extraction	Butanone		3414
Cotton-*graft*-poly(methacrylic acid)				
	Extraction	Water		3414
Dextran-*graft*-poly(acrylamide)	Chromatography	Water (0.5M potassium phthalate)	SEC	2112,3505
Dextran-*graft*-poly(acrylamide-*co*-sodium-2-acrylamido-2-methyl propane sulfonate)				
	Chromatography	Water 0.4M Sodium sulfate	SEC, porous glass	2113
Dextran-*graft*-poly(methyl methacrylate)				
	Extraction	Several solvents		2467
Dextran-*graft*-poly-(*N*-vinylpyrrolidone)				
	Turbidimetric titration	Water/acetone		2962
Ethyl cellulose-*graft*-poly(acrylamide)				
	Extraction	Hot methanol		1008
Ethyl cellulose-*graft*-poly(methyl methacrylate)				
	Fractional precipitation	Acetone/methanol		530
Methyl cellulose-*graft*-poly(acrylonitrile)				
	Extraction	Water	Hot	530
Methylated xylan-*graft*-poly(styrene)				
	Fractional precipitation	Benzene/petroleum ether		2425
Oxycellulose-*graft*-poly(methyl methacrylate)				
	Extraction	Benzene		1041
		Xanthation process		1041
Starch-*graft*-poly(acrylamide)	Extraction	Etahn, 1,2-diol and acetic acid		3003
Starcg-*graft*-poly(acrylonitrile)	Chromatography	Dimethylformamide	SEC, 75°C	1141
	Extraction	Dimethylformamide		1142
		Dimethylformamide, γ butyrolactone, dimethysulfoxide		865
Starch-*graft*-poly(methyl methacrylate)				
	Extraction	Acetone		2190
	Extraction;precipitation	Benzene; Aq. dimethylfomamide/methanol		530
Starch-*graft*-poly(styrene)	Chromatography	Tetrahydrofuran		1273
Starch-*graft*-poly(vinyl acetate)	Extraction	Water at 100°C;benzene		2194
Tricarbanilated (cellulose)-*graft*-poly(styrene)				
	Fractional precipitation	Dioxane/water		2917
4.3. MIXTURES OF POLYMERS				
Cellulose acetate + poly(oxyethylene)				
	Fractional precipitation	Dimethylformamide/ethyl ether		209

D. REFERENCES

1. M. J. M. Abadie, F. Schue, T. Sovel, D. B. Hartley, D. H. Richards, Polymer, **23**, 445 (1982).
2. R. Abate, A. Ballistreri, G. Montaudo, E. Impallomeni, Macromolecules, **27**, 332 (1994).
3. K. B. Abbås, N. G. Thame, J. Polym. Sci., Polym. Chem. Ed., **13**, 59 (1975).
4. K. B. Abbås, Polymer, **21**, 936 (1980).
5. A. H. Abdel-Alim, A. E. Hamielec, J. Appl. Polym. Sci., **17**, 3033 (1973).
6. S. Abdel-Fattah, I. Geczy, Polymer J., **6**, 542 (1974).
7. M. Abe, Y. Murakami, H. Fujita, J. Appl. Polym. Sci., **9**, 2549 (1965).
8. Z. Abe, H. Tanaka, M. Sumimoto, J. Polym. Sci., Polym. Chem. Ed., **16**, 305 (1978).
9. G. J. K. Acres, F. L. Dalton, J. Polym. Sci. A, **1**, 2419 (1963).
10. K. Adachi, T. Kotaka, Macromolecules, **20**, 2018 (1987).
11. H. E. Adams, P. O. Power, Ind. Eng. Chem., Anal. Chem., **15**, 711 (1943).

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2. R. Abate, A. Ballistreri, G. Montaudo, E. Impallomeni, Macromolecules, **27**, 332 (1994).
3. K. B. Abbås, N. G. Thame, J. Polym. Sci., Polym. Chem. Ed., **13**, 59 (1975).
4. K. B. Abbås, Polymer, **21**, 936 (1980).
5. A. H. Abdel-Alim, A. E. Hamielec, J. Appl. Polym. Sci., **17**, 3033 (1973).
6. S. Abdel-Fattah, I. Geczy, Polymer J., **6**, 542 (1974).
7. M. Abe, Y. Murakami, H. Fujita, J. Appl. Polym. Sci., **9**, 2549 (1965).
8. Z. Abe, H. Tanaka, M. Sumimoto, J. Polym. Sci., Polym. Chem. Ed., **16**, 305 (1978).
9. G. J. K. Acres, F. L. Dalton, J. Polym. Sci. A, **1**, 2419 (1963).
10. K. Adachi, T. Kotaka, Macromolecules, **20**, 2018 (1987).
11. H. E. Adams, P. O. Power, Ind. Eng. Chem., Anal. Chem., **15**, 711 (1943).

12. H. E. Adams, E. Ahad, M. S. Chang, D. B. Davis, D. M. French, J. Appl. Polym. Sci., **17**, 269 (1973).
13. M. A. Aden, E. Bianchi, A. Cifferi, G. Conio, A. Tealdi, Macromolecules, **17**, 2010 (1984).
14. K. Adibi, M. H. George, J. A. Barrie, J. Polym. Sci., Poly. Chem. Ed., **19**, 57 (1981).
15. K. Aejmelaeus, Ann. Acad. Sci., Fennicae., Serie A, **2**, 63 (1950).
16. A. M. Afifi-Effat, J. N. Hay, M. Wiles, J. Polym. Sci. B, **11**, 87 (1973).
17. A. M. Afifi-Effat, J. N. Hay, Br. Polym. J., **8**, 91 (1976).
18. A. B. Agasaryan, G. P. Belov, S. P. Davtyan, M. L. Eritsyan, Eur. Polym. J., **11**, 549 (1975).
19. H. Agbaje, J. Springer, Eur. Polym. J., **22**, 943 (1986).
20. J. C. Aggarwal, J. L. Mc Carthy, J. Indian Chem. Soc., **26**, 11 (1949).
21. S. L. Aggarwal, L. Marker, M. J. Carrano, J. Appl. Polym. Sci., **3**, 78 (1960).
22. R. K. Agnihotri, D. Falcon, E. C. Fredericks, J. Polym. Sci., Polym. Chem. Ed., **10**, 1839 (1972).
23. E. Ahad, Y. Sicotte, J. Polym. Sci. C, **30**, 163 (1970).
24. S. M. Aharoni, C. R. Crosby III, E. K. Walsh, Macromolecules, **15**, 1093 (1982).
25. T. O. Ahn, J. Lee, H. M. Jeong, K. Cho, Eur. Polym. J., **30**, 353 (1994).
26. T. Aida, S. Inoue, Makromol. Chem., Rapid. Commun., **1**, 677 (1980).
27. T. Aida, S. Inoue, Macromolecules, **14**, 1162 (1981).
28. T. Aida, S. Inoue, Macromolecules, **15**, 682 (1982).
29. H. Ailhaud, Y. Gallot, A. Skoulios, Makromol. Chem., **140**, 179 (1970).
30. L. Akcelrod, A. S. Gomes, J. Polym. Sci., Polym. Chem. Ed., **16**, 2431 (1978).
31. P. J. Akers, G. Allen, M. J. Bethell, Polymer, **9**, 575 (1968).
32. R. Alamo, J. G. Fatou, J. Guzmán, Polymer, **23**, 374 (1982).
33. R. Alamo, J. G. Fatou, J. Guzmán, Polymer, **23**, 379 (1982).
34. R. Alamo, J. G. Fatou, A. Bello, Polym. J., **15**, 491 (1983).
35. A. Albert, D. C. Pepper, Proc. Roy. Soc. (London) A, **263**, 75 (1961).
36. G. P. Aleksink, V. V. Shamanin, A. F. Podolsky, L. V. Alferova, V. A. Kropachev, Polym. J., **13**, 33 (1981).
37. W. J. Alexander, Th. E. Muller, J. Polym. Sci., C, **36**, 87 (1971).
38. J. Alexopoulos, N. Hadjichristidis, Makromol. Chem., **180**, 455 (1979).
39. S. Ali, Br. Polym. J., **10**, 108 (1978).
40. S. Ali, A. K. Raina, Makromol. Chem., **179**, 2925 (1978).
41. S. Ali, N. Ahmad, Br. Polym. J., **14**, 113 (1982).
42. S. I. Ali, J. Appl. Polym. Sci., Appl. Polym. Symp., **51**, 293 (1992).
43. S. M. Ali, M. B. Huglin, Makromol. Chem., **84**, 117 (1965).
44. R. Alijev, Z. Tuzar, J. Kondelíková, H. Verlová, J. Králiček, M. Buděšinký, Angew. Makromol. Chem., **105**, 99 (1982).
45. H. R. Allcock, G. Y. Moore, W. J. Cook, Macromolecules, **7**, 571 (1974).
46. H. R. Allcock, S. D. Wright, K. M. Kosyder, Macromolecules, **11**, 357 (1978).
47. H. R. Allcock, S. M. Coley, Ch. T. Morrissey, Macromolecules, **27**, 2904 (1994).
48. H. R. Allcock, Y.-B. Kim, Macromolecules **27**, 3933 (1994).
49. G. Allen, C. Booth, M. N. Jones, Polymer, **5**, 257 (1964).
50. G. Allen, C. Booth, M. N. Jones, Polymer, **5**, 195 (1964).
51. G. Allen, C. Booth, C. Price, Polymer, **8**, 397 (1967).
52. G. Allen, C. Booth, S. J. Hurst, M. N. Jones, C. Price, Polymer, **8**, 391 (1967).
53. G. Allen, C. Booth, S. J. Hurst, Polymer, **8**, 385 (1967).
54. G. Allen, J. Mc Ainsh, C. Strazielle, Eur. Polym. J., **5**, 319 (1969).
55. G. Allen, C. J. Lewis, S. M. Todd, Polymer, **11**, 44 (1970).
56. P. E. M. Allen, J. M. Downer, G. W. Hastings, H. W. Melville, P. Molyneux, J. R. Urwin, Nature (London), **177**, 910 (1956).
57. P. E. M. Allen, G. M. Burnett, J. M. Downer, H. W. Melville, Makromol. Chem., **38**, 72 (1960).
58. P. E. M. Allen, R. Hardy, J. R. Mayer, P. Molyneux, Makromol. Chem. **38**, 52 (1960).
59. P. E. M. Allen, R. P. Chaplin, D. O. Jordan, Eur. Polym. J., **8**, 271 (1972).
60. P. E. M. Allen, C. Mair, Eur. Polym. J., **20**, 697 (1984).
61. V. R. Allen, S. R. Turner, J. Macromol. Sci., Part A: Chem., **5**, 229 (1971).
62. D. F. Alliet, J. M. Pacco, Polym. Preprints, **8** (2), 1288 (1967).
63. D. F. Alliet, J. M. Pacco, J. Polym. Sci., C, **21**, 199 (1968).
64. D. F. Alliet, J. Appl. Polym. Symp., **8**, 39 (1969).
65. R. Allirot, Compt. Rend., **231**, 1065 (1950).
66. Y. Almog, M. Levy, J. Polym. Sci. Polym. Chem. Ed., **18**, 1 (1980).
67. T. Altares Jr., D. P. Wyman, V. R. Allen, J. Polym. Sci. A-1, **2**, 4533 (1964).
68. T. Altares Jr., D. P. Wyman, V. R. Allen, K. Meyersen, J. Polym. Sci. A-1, **3**, 4131 (1965).
69. J. E. Alter, G. T. Taylor, H. A. Scheraga, Macromolecules, **5**, 739 (1972).
70. K. H. Altgelt, J. Appl. Polym. Sci., **9**, 3389 (1965).
71. K. H. Altgelt, J. C. Moore, in: M. J. R. Cantow (Ed.), "Polymer Fractionation" Academic Press, London, 1967.
72. K. H. Altgelt, in: J. C. Giddings, R. A. Keller (Eds.), "Advances in Chromatography", vol 7, Marcel Dekker, New York. 1968.
73. K. H. Altgelt, L. Segal Eds., "Gel Permeation Chromatography", Marcel Dekker, New York, 1971.
74. J. M. Alvariño, A. Bello, G. M. Guzmán, Anales Quím. (Madrid), **66**, 105 (1970).
75. J. M. Alvariño, M. L. Hernandez, L. Lain, M. De Renobales, A. Torre, Eur. Polym. J., **14**, 991 (1978).
76. K. Alyürük, T. Özden, N. Colak, Polymer, **27**, 2009 (1986).
77. K. Alyürük, K. Hartani, Polymer, **30**, 2328 (1989).
78. A. J. Amass, E. W. Duck, J. R. Hawkins, J. M. Locke, Eur. Polym. J., **8**, 781 (1972).
79. M. R. Ambler, R. D. Mate, J. Polym. Sci., Polym. Chem. Ed., **10**, 2677 (1972).

80. M. R. Ambler, D. Mc Intyre, J. Appl. Polym. Sci., **21**, 3237 (1977).
81. M. R. Ambler, J. Polym. Sci., Polym. Lett. Ed., **14**, 683 (1978).
82. M. R. Ambler, W. G. Stevenson, L. J. Fetters, Polymer, **19**, 48 (1978).
83. R. J. Ambrose, J. Appl. Polym. Sci., **15**, 1297 (1971).
84. R. J. Ambrose, W. L. Hergenrother, J. Polym. Sci. Polym. Lett. Ed., **14**, 603 (1976).
85. Y. Amerik, J. E. Guillet, Macromolecules, **4**, 375 (1971).
86. P. Ammendola, T. Taucredi, A. Zambelli, Macromolecules, **19**, 307 (1986).
87. K. Anandakumaran, C. C. Kuo, S, Mukherji and A. E. Woodward, J. Polym. Sci., Polym. Chem. Ed., **20**, 1669 (1982).
88. K. Anandakumaran, W. Herman, A. E. Woodward, Macromolecules, **16**, 563 (1983).
89. V. S. Ananthanarajanan, R. H. Andreatta, D. Poland, H. A. Scheraga, Macromolecules, **4**, 417 (1971).
90. K. B. Anderson, A. Holmström, E. M. Sörvik, Makromol. Chem., **166**, 247 (1978).
91. G. D. Andrews, A. Vatvars, Macromolecules, **14**, 1603 (1981).
92. G. D. Andrews, A. Vatvars, G. Pruckmayr, Macromolecules, **15**, 1580 (1982).
93. A. P. Andreyev, I. A. Vakhtina, O. G. Tarakanov, Polym. Sci. USSR, **22**, 2877 (1980); Vysokomol. Soedin. A, **22**, 2627 (1980).
94. G. A. Andreyeva, S. P. Mitsengendler, K. I. Sokolova, A. A. Korotkov, Polym. Sci. USSR, **8**, 2391 (1966); Vysokomol. Soedin., **8**, 2159 (1966).
95. L. N. Andreyeva, S. V. Bushin, A. I. Mashoshin, Ye. B. Lysenko, K. P. Smirnov, V. N. Yemel'yanov, V. N. Tsvetkov, Polym. Sci. USSR, **24**, 2405 (1982); Vysokomol. Soedin. A, **24**, 2101 (1982).
96. K. A. Andrianov et al., Vysokomol. Soedin., **3**, 1692 (1961).
97. K. A. Andrianov, S. Ye. Yakushkina, L. N. Guniava, Poly. Sci. USSR, **8**, 2398 (1966); Vysokomol. Soedin., **8**, 2166 (1966).
98. K. A. Andrianov, V. I. Pakhomov, V. M. Gel'perina, D. N. Mukhina, Polym. Sci. USSR, **8**, 1787 (1966); Vysokomol. Soedin., **8**, 1618 (1966).
99. K. A. Andrianov, B. G. Zavin, N. V. Pertsova, Polym. Sci. USSR, **10**, 49 (1968); Vysokomol. Soedin. A, **10**, 46 (1968).
100. K. A. Andrianov, I. I. Tverdokhlebova, S. A. Pavlova, B. G. Zavin, J. Polym. Sci. C, **22**, 741 (1969).
101. K. A. Andrianov, B. G. Zavin, G. F. Sablina, Polym. Sci. USSR, **14**, 1294 (1972); Vysokomol. Soedin. A, **14**, 1156 (1972).
102. K. A. Andrianov, S. A. Paulova, I. I. Tverdokhlebova, V. N. Yemel'yanov, T. A. Larina, A. Yu. Rabkina, Polym. Sci. USSR, **14**, 2628 (1972); Vysokomol. Soedin. A, **14**, 2246 (1972).
103. K. A. Andrianov, A. A. Zhdanov, B. G. Zavin, G. F. Sablina, Polymer Sci. USSR, **14**, 2078 (1972); Vysokomol. Soedin. A, **14**, 1855 (1972).
104. K. A. Andrianov, S. A. Pavlova, I. I. Tverdokhlebova, N. V. Pertsova, V. A. Temnikovskii, Polym. Sci. USSR, **14**, 2035 (1972); Vysokomol. Soedin. A, **14**, 1816 (1972).
105. K. A. Andrianov, S.-S. A. Pavlova, I. P. Tverdokhlebova, N. V. Pertsova, V. A. Temnikovskii, L. N. Pronina, Polym. Sci. USSR, **19**, 532 (1977); Vysokomol. Soedin. A, **19**, 466 (1977).
106. K. A. Andrianov, I. I. Tverdokhlebova, I. I. Mamayeva, A. Yu. Rabkina, B. G. Zavin, V. M. Men'shov, A. K. Chekalov, S.-S. A. Pavlova, Polym. Sci. USSR, **20**, 1857 (1976); Vysokomol. Soedin. A, **20**, 1646 (1978).
107. F. Andruzzi, D. Lupinacci, P. L. Magagnini, Macromolecules, **13**, 15 (1980).
108. D. J. Angier, D. T. Turner, J. Polym. Sci., **28**, 265 (1958).
109. D. J. Angier, R. J. Cereza, W. F. Watson, J. Polym. Sci., **34**, 699 (1959).
110. F. J. Ansorena, J. J. Iruin, G. M. Guzmán, Eur. Polym. J., **16**, 165 (1980).
111. F. J. Ansorena, J. J. Iruin, G. M. Guzmán, J. Polym. Sci. Polym. Chem. Ed., **18**, 173 (1980).
112. F. J. Ansorena, L. M. Revuelta, G. M. Guzmán, J. J. Iruin, Eur. Poly J., **18**, 19 (1982).
113. M. Antonietti, S. Heinz, M. Schmidt, Ch. Rosenauer, Macromolecules, **27**, 3276 (1994).
114. M. Antonietti, A. Briel, C. Tank, Acta Polym., **46**, 254 (1995).
115. E. V. Anufrieva, T. M. Birshtein, T. N. Nekrasova, O. B. Ptitsyn, T. V. Sheveleva, J. Polym. Sci. C, **16**, 3519 (1968).
116. S. Aoshima, T. Higashimura, Macromolecules, **22**, 1010 (1989).
117. K. Arai, T. Kotaka, Y. Kitano, K. Yoshimura, Macromolecules, **13**, 1670 (1980).
118. K. Araki, Y. Imamura, Makromol. Chem. **183**, 1334 (1982).
119. T. Araki, F. Nogami, H. Tsukube, K. Nagata, S. Iyoshi, J. Polym. Chem. Ed., **16**, 1037 (1978).
120. M. A. de Araujo, R. Stadler, Makromol. Chem., **189**, 2169 (1988).
121. P. Archambault, R. E. Prud'homme, J. Polym. Sci. Polym. Phys. Ed., **18**, 35 (1980).
122. I. Arendt, H. J. Schenck, Kunststoffe, **48**, 111 (1958).
123. S. Arichi, S. Mitsuta, N. Sakamoto, H. Murata, Bull. Chem. Soc. Japan, **39**, 428 (1966).
124. S. Arichi, M. J. Pedram, J. M. G. Cowie, Eur. Polym. J., **5**, 107 (1979).
125. R. S. Aries, A. P. Sachs, J. Polym. Sci., **21**, 551 (1956).
126. J.-P. Arlie, Y. Chauvin, D. Commereve, J.-P. Soufflet, Makromol. Chem., **175**, 861 (1974).
127. L. H. Arond, H. P. Frank, J. Phys. Chem, **58**, 953 (1954).
128. M. Arpin, C. Strazielle, Makromol. Chem., **177**, 293 (1976).
129. M. Arpin, C. Strazielle, Makromol. Chem., **177**, 581 (1976).
130. M. Arpin, C. Strazielle, Polymer, **18**, 591 (1977).
131. F. Arranz, M. Galin, J. C. Galin, P. Rempp, IUPAC Boston, vol. 1, 432 (1971).
132. F. Arranz, J. C. Galin, Makromol. Chem., **152**, 185 (1972).
133. F. Arranz, M. Sánchez-Chaves, A. Molinero, R. Martinez, Makromol. Chem., Rapid. Commun., **4**, 29 (1983).

134. R. Arshady, B. S. R. Reddy, M. H. George, Polymer, **25**, 1161 (1984).
135. S. M. Arulsamy, M. Santappa, Makromol. Chem., **178**, 2451 (1977).
136. N. Asada, H. Hosozawa, A. Toyoda, H. Sato, Rubber Chem. Technol., **63**, 181 (1990).
137. J. Asakura, Y. Matsubara, M. Yoshihara, T. Maeshima, J. Macromol. Sci. Part A: Chem., **14**, 803 (1980).
138. R. Asami, M. Gyi, M. Takaki, T. Ikuta, Polym. J., **10**, 301 (1978).
139. R. Asami, M. Takaki, K. Kita, E. Asakura, Polym. Bull., **2**, 713 (1980).
140. H. Asaoka, A. Suzuki, J. Soc. Text. Cell Ind. (Japan), **11**, 32 (1955).
141. H. Ashitaka, H. Ichikawa, H. Ueno, A. Nagasaka, J. Polym. Sci. Polym. Chem. Ed., **21**, 1853 (1983).
142. H. Ashitaka, K. Jinda, H. Ueno, J. Polym. Sci., Polym. Chem. Ed., **21**, 1951 (1983).
143. J. S. Aspler, C. E. Hoyle, J. E. Guillet, Macromolecules, **11**, 925 (1978).
144. J. Asrar, Macromolecules, **27**, 4036 (1994).
145. T. Asuke, Y. Chien-Hua, R. West, Macromolecules, **27**, 3023 (1994).
146. E. L. Atkin, L. A. Kleintjens, R. Koningsveld, L. J. Fetters, Makromol. Chem., **184**, 377 (1984).
147. C. M. L. Atkinson, R. Dietz, Brit. Polym. J., **6**, 133 (1974).
148. C. M. L. Atkinson, R. Dietz, Makromol. Chem. 177, 213 (1976).
149. C. M. L. Atkinson, R. Dietz,Eur. Poly. J., **15**, 21 (1979).
150. C. M. L. Atkinson, R. Dietz, M. A. Francis, Polymer, **21**, 891 (1980).
151. M. Augenstein, M. Stickler, Makromol. Chem., **191**, 415 (1990).
152. C. Auschra, R. Stadler, I. G. Voight-Martin, Polymer, **34**, 2081 (1993).
153. M. R. Aven, C. Cohen, Makromol. Chem., **189**, 881 (1988).
154. T. P. Avilova, V. T. Bykov, G. Ya. Zolotar, Polym. Sci. USSR, **7**, 917 (1965); Vysokomol. Soedin., **7**, 831 (1965).
155. T. P. Avilova, V. T. Bykov, V. P Marinin, N. P. Shapkin, Polym. Sci. USSR, **7**, 2377 (1965); Vysokomol. Soedin., **7**, 2168 (1965).
156. T. P. Avilova, V. T. Bykov, L. A. Kondratenko, Polym. Sci. USSR, **8**, 12 (1966), Vysokomol. Soedin., **8**, 14 (1966).
157. C. W. Ayers, Analyst, **78**, 382 (1953).
158. M. A. Azzen, U. S. Yousef, G. Pierre, Eur. Polym. J., **32**, 111 (1996).
159. G. N. Babu, A. Narula, S. L. Hsu, J. C. W. Chien, Macromolecules, **17**, 2749 (1984).
160. G. N. Babu, R. A. Newmark, J. C. W. Chien, Macromolecules, **27**, 3383 (1994).
161. E. Bacqué, J.-P. Pillto, M. Birot, J. Dunogués, Macromolecules, **21**, 34 (1988).
162. W. J. Badgley, H. F. Mark, J. Phys. Chem., **51**, 58 (1947).
163. H. J. Bae, T. Miyashita, M. Iino, M. Matsuda, Macromolecules, **21**, 26 (1988).
164. X. H. Bae, Z. Fodor, R. Faust, Macromolecules, **30**, 198 (1997).
165. M. Baer, J. Polym. Sci. A, **2**, 417 (1964).
166. G. Bagby, R. S. Lehrle, J. C. Robb. Makromol. Chem., **119**, 122 (1968).
167. G. Bagby, R. S. Lehrle, J. C. Robb. Polymer, **9**, 284 (1968).
168. F. Bai, S. E. Webber, Macromolecules, **21**, 625 (1988).
169. M. D. Baijal, R. M. Diller, F. R. Pool, Polym. Preprints, **10** (2), 1464 (1969).
170. M. D. Baijal, K. M. Kanppila, Polym. Eng. Sci., **11**, 182 (1971).
171. D. Bailey, D. Tirrell, C. Pinazzi, O. Vogl, Macromolecules, **11**, 312 (1978).
172. F. E. Bailey, G. M. Powell, K. L. Smith, Ind. Eng. Chem., **50**, 8 (1958).
173. C. Bailly, D. Daoust, R. Legras, J. P. Mercier, A. Lapp, CV. Sterazielle, Polymer, **27**, 1410 (1986).
174. Ch. Bailly, M. Daumerie, R. Legras, J. P. Mercier, J. Polym. Sci., Polym. Phys. Ed., **23**, 493 (1985).
175. D. R. Bain, J. D. Wagner, Polymer, **25**, 403 (1984).
176. F. C. Baines, J. C. Bevington, Eur. Polym. J., **3**, 593 (1967).
177. H. E. Bair, D. R. Falcone, M. Y. Hellman, G. E. Johnson, P. G. Kelleher, J. Appl. Polym. Sci., **26**, 1777 (1981).
178. P. Bajaj, S. K. Varshney, A. Misra, J. Polym. Sci., Polym. Chem. Ed., **18**, 295 (1980).
179. K. Bak, G. Elefante, J. E. Mark, J. Phys. Chem., **71**, 4007 (1967).
180. K. Bak, G. Elefante, J. E. Mark, Polym. Preprints, **8** (2), 998 (1967).
181. B. Baker, P. J. T. Tait, Polymer, **8**, 225 (1967).
182. C. A. Baker, R. J. P. Williams, J. Chem. Soc., 2352 (1956).
183. R. W. Baker, J. Appl. Polym. Sci., **13**, 369 (1969).
184. N. F. Bakeyev, I. S. Lakoba, Polym. Sci. USSR, **14**, 2849 (1972); Vysokomol. Soedin. A, **14**, 2443 (1972).
185. G. Balbontin, D. Dainelli, M. Galimberti, G. Paganetto, Makromol. Chem., **193**, 693 (1992).
186. G. Balbontin, I. Camurati, T. Dall'occo, A. Finotti, R. Franzese, G. Vecellio, Angew. Makromol. Chem., **219**, 139 (1994).
187. S. T. Balke, R. Patel, 29th Canadian Chemical Engineering Conference, Sarnia, Ontario, 1979.
188. D. G. H. Ballard, J. V. Dawkins, Makromol. Chem., **148**, 195 (1971).
189. D. G. H. Ballard, J. V. Dawkins, J. M. Key, P. W. Van Lienden, Makromol. Che., **165**, 173 (1973).
190. D. G. H. Ballard, J. V. Dawkins,, Eur. Polym. J., **10**, 829 (1974).
191. D. G. H. Ballard, P. Cheshire, G. W. Longman, J. Schelten, Polymer, **19**, 379 (1978).
192. G. D. Ballova, V. M. Bulatova, K. A. Vylegzhanina, Ye. I. Yegorova, L. L. Sul'zhenko, G. P. Fratkina, Polym. Sci. USSR, **11**, 2080 (1969); Vysokomol, Soedin. A, **11**, 1827 (1969).
193. C. H. Bamford, G. C. Eastmond, J. Woo, D. H. Richards, Polymer, **23**, 643 (1982).
194. A. Banderet, W. Kobryner, Compt. Rend., **244**, 604 (1957).
195. T. F. Banigan, Science, **117**, 249 (1953).
196. W. Banks, C. T. Greenwood, J. Thomson, Makromol. Chem., **31**, 197 (1959).

197. W. Banks, C. T. Greenwood, Makromol. Chem., **67**, 49 (1963).
198. W. Banks, C. T. Greenwood, Eur. Polym. J., **4**, 249 (1968).
199. W. Banks, C. T. Greenwood, Eur. Polym. J., **4**, 377 (1968).
200. W. Banks, C. T. Greenwood, Makromol. Chem., **114**, 245 (1968).
201. W. Banks, C. T. Greenwood, D. J. Hourston, Makromol. Chem., **111**, 226 (1968).
202. W. Banks, C. T. Greenwood, D. J. Hourston, Trans. Faraday Soc., **64**, 363 (1968).
203. W. Banks, C. T. Greenwood, Polymer, **10**, 257 (1969).
204. W. Banks, C. T. Greenwood, J. Sloss, Makromol. Chem., **140**, 109 (1970).
205. W. Banks, C. T. Greenwood, J. Sloss, Makromol. Chem., **140**, 119 (1970).
206. W. Banks, C. T. Greenwood, J. Sloss, Eur. Polym. J., **7**, 263 (1971).
207. W. Banks, C. T. Greenwood, Makromol. Chem., **144**, 135 (1971).
208. D. W. Bannister, C. S. G. Phillips, R. J. P. Williams, Anal. Chem., **26**, 1451 (1954).
209. A. Bar-Ilan, A. Zilkha, Eur. Polym. J., **6**, 403 (1970).
210. I. A. Baranovskaya, V. Ye. Eskin, Polym. Sci. USSR, **7**, 373 (1965); Vysokomol. Soedin., **7**, 339 (1964).
211. K. Baranwal, H. Jacobs, J. Appl. Polym. Sci., **13**, 797 (1969).
212. A. Barbalata, T. Bohossian, G. Delmas, J. Appl. Polym. Sci., **46**, 411 (1992).
213. P. C. Barbé, L. Noristi, G. Baruzzi, Makromol. Chem., **193**, 229 (1992).
214. G. M. Bardina, A. A. Speranskii, V. K. Murav'ev, V. M. Kharitonov, Khim. Volokna., 26 (1968).
215. A. Barlow, R. S. Lehrle, J. C. Robb, D. Sunderland, Polymer, **8**, 537 (1967).
216. W. K. R. Barnikol, G. V. Schulz, Z. Physik. Chem. N. F. (Frankfurt), **47**, 89 (1965).
217. J. M. Barrales-Rienda, D. C. Pepper, Eur. Polym. J., **3**, 535 (1967).
218. J. M. Barrales-Rienda, D. C. Pepper, Polymer, **8**, 337 (1967).
219. J. M. Barrales-Rienda, G. R. Brown, D. C. Pepper, Polymer, **10**, 327 (1969).
220. J. M. Barrales-Rienda, Anales Quím. (Madrid), **66**, 767 (1970).
221. J. M. Barrales-Rienda, J. Gonzales Ramos, M. V. Dabrio Bañuls, Anales Quím. (Madrid), **67**, 651 (1971).
222. J. M. Barrales-Rienda, A. Horta, E. Sáiz, J. Macromol. Sci., Part A: Chem., **8**, 715 (1974).
223. J. M. Barrales-Rienda, P. A. Galera Gómez, J. Gonzales Ramos, E. Otero Aenlle, Anales Quím. (Madrid), **76**, 49 (1980).
224. J. M. Barrales-Rienda, C. Romero Galicia, A. Horta, Macromolecules, **16**, 932 (1983).
225. J. M. Barrales-Rienda, Rev. Plást. Mod. (Madrid), **324**, 605 (1983).
226. J. M. Barrales-Rienda, P. A. Galera Gómez, Eur. Polym. J., **20**, 1213 (1984).
227. J. M. Barrales-Rienda, P. A. Galera Gonez, A. Horta, E. Sáiz, Macromolecules, **18**, 2572 (1985).
228. D. Barrera, E. Zylstra, P. T. Lausbury, R. Langer, Macromolecules, **28**, 425 (1995).
229. C. A. Barson, J. C. Robb, Brit. Polym. J., **3**, 53 (1971).
230. M. Bartmann, U. Kowalczik, Makromol. Chem., **189**, 2285 (1988).
231. R. L. Bartosiewicz, C. Booth, A. Marshall, Eur. Polym. J., **10**, 783 (1974).
232. A. M. Basedow, K. H. Ebert, H. Ederer, H. Hunger, Makromol. Chem., **177**, 1501 (1976).
233. A. M. Basedow, K. H. Ebert, U. Ruland, Makromol. Chem., **179**, 1351 (1978).
234. A. M. Basedow, K. H. Ebert, W. Feigenbutz, Makromol. Chem., **181**, 1071 (1980).
235. G. L. Bata, J. E. Hazell, L. A. Prince, IUPAC Toronto, Preprint A, **4**, 14 (1968).
236. G. L. Bata, J. E. Hazell, L. A. Prince, J. Polym. Sci. C., **30**, 157 (1970).
237. F. S. Bates, R. E. Cohen, C. V. Barney, Macromolecules, **15**, 589 (1982).
238. T. W. Bates, J. Biggins, K. J. Ivin, Makromol. Chem., **87**, 180 (1965).
239. T. W. Bates, K. J. Ivin, Polymer, **8**, 263 (1967).
240. H. Batzer, Makromol. Chem., **5**. 66 (1950).
241. H. Batzer, F. Wiloth, Makromol, Chem., **8**, 41 (1952).
242. H. Batzer, Makromol. Chem., **12**, 145 (1954).
243. H. Batzer, A. Moeschle, Makromol. Chem., **22**, 195 (1957).
244. H. Batzer, A. Nisch, Makromol. Chem., **22**, 131 (1957).
245. H. Batzer, S. A. Zahir, J. Appl. Polym. Sci., **21**, 1843 (1977).
246. B. J. Bauer, R. M. Briber, C. C. Han, Macromolecules, **22**, 940 (1989).
247. B. J. Bauer, B. Hanley, Y, Muroga, Polymer Commun., **30** 19 (1989).
248. H. Baum, G. A. Gilbert, H. L. Wood, J. Chem. Soc., 4047 (1955).
249. H . Baum. G. A. Gilbert, J. Colloid. Sci., **11**, 428 (1956).
250. G. F. Bauman, S. Steingiser, J. Polym. Sci. A, **1**, 3396 (1963).
251. J. H. Baxendale, S. Bywater, M. G. Evans, Trans. Faraday Soc., **42**, 675 (1946).
252. B. Baysal, G. Adler, D. Ballantine, A. Glines, J. Polym. Sci. B, **1**, 257 (1963).
253. B. Baysal, J. Polym. Sci. C, **4**, 935 (1963).
254. H. C. Beachell, J. C. Peterson, Polym. Preprints, **8** (1), 456 (1967).
255. H. C. Beachell, J. C. Peterson, J. Polym. Sci. A-1, **7**, 2021 (1969).
256. W. H. Beattie, C. Booth, J. Appl. Polym. Sci., **7**, 508 (1963).
257. W. H. Beattie, J. Polym. Sci. A-1, **3**, 527 (1965).
258. J. Beaumais, G. Muller, J.-C. Fenyo, Polym. Bull., **2**, 449 (1980).
259. M. Becerra, D. Radic, L. Gargallo, Makromol. Chem., **179**, 2241 (1978).
260. G. Beck, J. Kiwi, D. Lindenau, W. Schnabel. Eur. Polym. J., **10**, 1069 (1974).

261. G. P. van der Beck, M. A. Cohen Stuart, G. J. Fleer, Macromolecules, **24**, 3553 (1991).
262. C. G. Beddows, H. Gil, J. T. Guthrie, Polym. Bull., **3**, 645 (1980).
263. D. R. Beech, C. Booth, J. Polym. Sci. A-2, **7**, 575 (1969).
264. D. R. Beech, C. Booth, I. H. Hillier, C. J. Pickles, Eur. Polym. J., **8**, 799 (1972).
265. D. R. Beech, C. Booth, D. V. Dodgson, R. R. Sharpe, J. R. S. Waring, Polymer, **13**, 73 (1972).
266. D. R. Beech, C. Booth, Polymer, **13**, 355 (1972).
267. R. B. Beevers, J. Polym. Sci. A-1, **2**, 5257 (1964).
268. S. I. Beilin, Y. L. Vollershtein, B. A. Dolgoplosk, V. A. Kargin, M. N. Shvarts, T. V. Fremel, Polym. Sci. USSR, **10**, 1 (1968); Vysokomol. Soedin. A, **10**, 3 (1968).
269. G. Beinert, P. Chaumont, P. Rempp, J. Herz, Eur. Polym. J., **20**, 837 (1984).
270. Ye. A. Bekturov, L. A. Bimendina, S. V. Bereza, Polym. Sci. USSR, **10**, 2468 (1970); Vysokomol. Soedin. A, **12**, 2179 (1970).
271. Ye. A. Bekturov, R. Ye. Legkunets, Polym. Sci. USSR, **12**, 702 (1970), Vysokomol. Soedin. A, **12**, 626 (1970).
272. E. A. Bekturov, Sh. Sh. Shajakhmetov, S. E. Kudaibergenov, Polymer, **21**, 787 (1980).
273. M. Bel'govskii, V. M. Gol'dberg, I. A. Krasotkina, D. Ya. Toptygin, Polym. Sci. USSR, **13**, 758 (1971); Vysokomol. Soedin. A, **13**, 666 (1971).
274. M. Belbachir, B. Boutevin, Y. Pietrasanta, G. Rigal, Makromol. Chem., **183**, 2347 (1982).
275. B. G. Belenkii, L. Z. Vilenchik, "Modern Liquid Chromatography of Macromolecules", Elsevier, Amsterdam, 1983.
276. A. Bello, J. M. Barrales-Rienda, G. M. Guzmán, in: J. Brandrup, E. H. Immergut, (Eds.), "Polymer Handbook". 2nd. ed., Wiley, New York, 1989.
277. A. Bello, J. M. Barrales-Rienda, G. M. Guzmán, in: J. Brandrup, E. H. Immergut, (Eds.), "Polymer Handbook". 3rd. ed., Wiley, New York, 1989.
278. G. P. Belov, S. K. Kaltochikhina, L. I. Atanasova, Polym. Sci. USSR, **8**, 2046 (1966); Vysokomol. Soedin., **8**, 1852 (1966).
279. G. P. Belov, A. P. Lisitskaya, N. M. Chirkov, V. I. Tsvetkova, Vysokomol. Soedin. A, **9**, 1269 (1967); Polymer Sci. USSR, **9**, 1417 (1967).
280. G. P. Belov, A. P. Lisitskaya, T. I. Solovyeva, N. M. Chirkov, Eur. Polym. J., **6**, 29 (1970).
281. G. P. Belov, V. I. Kuznetsov, T. I. Solovyeva, N. M. Chirkov, S. S. Ivanchev, Makromol. Chem., **140**, 213 (1970).
282. G. P. Belov, V. N. Belova, L. N. Raspopov, Y. V. Kissin, K.-M. A. Brikenshtein, N. M. Chirkov, Polym. J., **3**, 681 (1972).
283. L. Beltzung, C. Strazielle, Makromol. Chem., **185**, 1145 (1984).
284. L. Beltzung, C. Strazielle, Makromol. Chem., **185**, 1155 (1982).
285. M. Beltzung, C. Picot, P. Rempp, J. Herz, Macromolecules, **15** 1594 (1982).
286. V. M. Belyayev, V. P. Budtov, S. Ya. Frenkel', N. V. Daniel, Polym. Sci. USSR., **14**, 2724 (1972); Vysokomol. Soedin. A, **14**, 2335 (1972).
287. M. Benchanabe, R. Van Leemput, J. Polym. Sci. Polym. Phys. Ed., **23**, 991 (1985).
288. W. I. Bengough, G. F. Grant, Eur. Polym. J., **7**, 203 (1971).
289. M. A. Benincasa, J. C. Gidding, Anal. Chem., **64**, 790 (1992).
290. A. Benkhira, E. Franta, M. Rawiso, J. François, Macromolecules, **27**, 3963 (1994).
291. H. Benoît, Z. Grübisic, P. Rempp, D. Decker, J.-G. Zilliox, J. Chim. Phys., **63**, 1507 (1966).
292. K. G. Berdnikova, G. V. Tarasova, V. S. Skazka, N. A. Nikitin, G. V. Dyuzhev, Polym. Sci. USSR, **6**, 2280 (1964); Vysokomol. Soedin. **6**, 2057 (1964).
293. D. Berek, D. Lath, V. Durdovic, J. Polym. Sci. C, **16**, 659 (1967).
294. D. Berek, G. Katuscakova, I. Novak, Brit. Polym. J., **9**, 62 (1977).
295. A. Beresniewicz, J. Polym. Sci., **35**, 321 (1959).
296. S. V. Bereza, Ye. A. Bekturov, S. R. Rafikov, Polym. Sci. USSR, **11**, 1907 (1969); Vysokomol. Soedin. A, **11**, 1681 (1969).
297. A. A. Berg. V. G. Kozlov, V. P. Budtov, Yu. B. Monakov, S. R. Rafikov, Polym. Sci. USSR, **22**, 601 (1980); Vysokomol. Soedin., A, **22**, 543 (1980).
298. K. C. Berger, G. V. Schulz, Makromol. Chem., **136**, 221 (1970).
299. K. C. Berger, Makromol. Chem., **175**, 2121 (1974).
300. M. N. Berger, B. M. Tidswell, IUPAC Helsinki, vol. III, Preprint II-63 (1972).
301. M. N. Berger, A. Cervenka, Eur. Polym. J., **10**, 205 (1974).
302. J.-Y. Bergeron, L. H. Dao, Macromolecules, **25**, 3332 (1992).
303. H. Berghmans, G. Smets, Makromol. Chem., **115**, 187 (1968).
304. H. Berghmans, A. Donkers, L. Frenay, W. Stoks, F. E. De Schryver, P. Moldenars, J. Mewis, Polymer, **28**, 97 (1987).
305. C. Bergstrom, E. Avela, J. Appl. Polym. Sci., **23**, 163 (1979).
306. J. B. Berkowitz, M. Yamin, R. M. Fuoss, J. Polym. Sci., **28**, 69 (1958).
307. A. A. Berlin, V. A. Grigorovskaya, O. G. Sel'skaya, Polym. Sci. USSR, **12**, 2879 (1970); Vysokomol. Soedin. 12-A, 2541 (1970).
308. I. V. Berlinova, a., Amzil, N. G. Vladimirov, J. Polym. Sci. Part. A: Polym. Chem., **33**, 1751 (1995).
309. D. E. Bernal, Rev. General Caoutchouc, **32**, 889 (1955).
310. G. C. Berry, L. M. Hobbs, V. C. Long, Polymer, **5**, 31 (1964).
311. G. C. Berry, R. G. Craig, Polymer, **5**, 19 (1964).
312. G. C. Berry, H. Nomura, K. G. Mayhan, J. Polym. Sci. A-2, **5**, 1 (1967).
313. G. C. Berry, S. P. Yen, Adv. Chem. Ser., **91**, 734 (1969).
314. G. C. Berry, E. F. Casassa, P.-Y Liu, J. Polym. Sci. Part B: Polym. Phys., **25**, 673 (1987).
315. J. C. Bevington, G. M. Guzmán, H. W. Melville, Proc. Roy. Soc. (London) A, **221**, 437 (1954).
316. S. N. Bhadani, S. L. Tseng, D. G. Gray, Makromol. Chem., **184**, 1727 (1983).

317. G. S. Bhargava, H. U. Khan, K. K. Bhattacharyya, J. Appl. Polym. Sci., **23**, 1181 (1979).
318. L.-K. Bi, L. J. Fetters, Macromolecules, **9**, 732 (1976).
319. Y. Bi, D. C. Neckers, Macromolecules, **27**, 3683 (1994).
320. E. Biagini, B. Pedemonte, E. Pedemonte, S. Russo, A. Turturro, Makromol. Chem., **183**, 2131 (1982).
321. U. Bianchi, V. Magnasco, J. Polym. Sci., **41**, 177 (1959).
322. U. Bianchi, M. Dalpiaz, E. Patrone, Makromol. Chem., **80**, 112 (1964).
323. U. Bianchi, E. Patrone, M. Dalpiaz, Makromol. Chem., **84**, 230 (1965).
324. U. Bianchi, A. Peterlin, J. Polym. Sci. A-2, **6**, 1011 (1968).
325. U. Bianchi, M. Bruzzone, M. Mormino, J. Polym. Sci. Polym. Lett. Ed., **15**, 345 (1977).
326. J. P. Bianchi, F. P. Price, B. H. Zimm, J. Polym. Sci., **25**, 27 (1957).
327. E. Bigdeli, R. W. Lenz, B. Oster, R. D. Lundberg, J. Polym. Sci., Polym. Chem. Ed., **16**, 469 (1978).
328. P. Biernacki, S. Chrzczonowicz, M. Wlodarczyk, Eur. Polym. J., **7**, 739 (1971).
329. R. Bill, M. Droesher, G. Wegner, Makromol. Chem., **182**, 1033 (1981).
330. N. C. Billingham, L. M. Boxall, A. D. Jenkins, P. D. Lees, Eur. Polym. J., **10**, 981 (1974).
331. F. W. Billmeyer, W. H. Stockmayer, J. Polym. Sci., **5**, 121 (1950).
332. F. W. Billmever, R. N. Kelley, Polym. Preprints, **8** (2), 1259 (1967).
333. F. W. Billmeyer, I. Katz, Macromolecules, **2**, 105 (1969).
334. J. P. Billot, A. Douy, B. Gallot, Makromol. Chem., **178** 1641 (1977).
335. L. A. Bimendina, V. V. Roganov, Ye. A. Kekturov, Polym. Sci. USSR, **16**, 3274 (1974); Vysokomol. Sodin. A, **16**, 2810 (1974).
336. R. Biran, J. V. Dawkins, Eur. Polym. J., **20**, 129 (1984).
337. A. W. Birley, J. V. Kawkins, D. Kyriacos, Polymer, **21**, 632 (1980).
338. J. Bischoff, V. Desreux, Bull. Soc. Chim. Belges, **60**, 137 (1951).
339. J. Bischops, J. Polym. Sci., **17**, 81 (1955).
340. B. Biyani, A. J. Campagna, D. Daruwalla, C. M. Srivastava, P. Erlich, J. Macromol. Sci., Part A: Chem., 9, 327 (1975).
341. D. A. Blackadder, G. J. Le Poidevin, Polymer, **18**, 547 (1977).
342. W. R. Blackmore, W. Alexander, Can. J. Chem., **39**, 1888 (1961).
343. I. V. Blagodatskikh, L. V. Dubrovina, S.-S. A. Pavlova, Polym. Sci. USSR, **24**, 400 (1982); Vysokomol. Soedin. A, **24**, 367 (1982).
344. D. E. Blair, J. Appl. Polym. Sci., **14**, 2469 (1970).
345. D. E. Blair, Polym. Preprints, **11** (2), 967 (1970).
346. L. P. Blanchard, M. D. Baijal, Polym. preprints, **7** (2), 944 (1966).
347. L. P. Blanchard, M. D. Baijal, J. Polym. Sci. A, **15**, 2045 (1967).
348. L.-P. Blanchard, V. Hornof, H.-H. Lam, S. L. Malhotra, Eur. Polym. J., **10**, 1057 (1974).
349. L. P. Blanchard, G. G. Gabra, V. Hornof, S. L. Malhotra, J. Polym. Sci., Polym. Chem. Ed., **13**, 271 (1975).
350. L. P. Blanchard, G. G. Gabra, S. L. Malhotra, J. Polym. Sci., Polym. Chem. Ed., **13**, 1619 (1975).
351. M. D. Blanco, J. M. Teijón, I. Katime, Eur. Polym. J., **26**, 249 (1990).
352. M. Blanda, P.-L. Reschiglian, F. Dondi, R. Beckett, Polym. Int., **33**, 61 (1994).
353. M. J. Blandamer, M. F. Fox, E. Powell, J. W. Stafford, Makromol. Chem., **124**, 222 (1969).
354. F. Blasco, E. Riande, J. Polym. Sci., Polym. Phys. Ed., **21**, 835 (1983).
355. A. Bledzki, M. I. Prosinska, D. Berek, J. Polym. Sci., Polym. Chem. Ed., **16**, 107 (1978).
356. A. Bledzki, H. Balard, D. Braun, Makromol. Chem., **189**, 2807 (1988).
357. T. Bleha, T. Spychaj, R. Vondra, D. Berek, J. Polym. Sci., Polym. Phys. Ed., **21**, 1903 (1983).
358. H. Block, J. J. Gosling, S. M. Walker, Polymer, **25**, 1465 (1984).
359. G. F. Bloomfield, Rubber Chem. Technol., **24**, 737 (1951).
360. Z. Blumczynski, T. Skowzonski, W. Laskawski, Angew. Makromol. Chem., **97**, 115 (1981).
361. D. D. Bly, Polym. Preprints, **8** (2), 1234 (1967).
362. E. G. Bobalek, E. R. Moore, S. S. Levy, C. C. Lee, J. Appl. Polym. Sci., **8**, 625 (1964).
363. L. L. Boehm, R. H. Casper, G. V. Schulz, J. Polym. Sci., Polym. Phys. Ed., 12, 239 (1974).
364. Ye. P. Bogomolova, A. A. Trapeznikov, V. N. Fedotov, Vysokomol. Soedin. A, **12**, 1216 (1970); Polym. Sci. USSR, **12**, 1375 (1970).
365. V. Boháčková, J. Polácek, Z. Grubisic, H. Benoît, J. Chim. Phys., **66**, 207 (1969).
366. V. Boháčková, E. Fišerová, Z. Gallot-Grubišic, J. Polácek, M. Stolka, J. Chim. Phys., **67**, 777 (1970).
367. M. Bohdanecký, Coll. Czech. Chem. Commun., **29**, 876 (1964).
368. M. Bohdanecký, Coll. Czech. Chem. Commun., **31**, 4095 (1966).
369. M. Bohdanecký, Z. Tuzar, M. Štoll, R. Chromecek, Coll. Czech. Commun., **33**, 4104 (1968).
370. M. Bohdanecký, V. Petrus, P. Kratochvíl, Coll. Czech. Chem. Commun., **33**, 4089 (1968).
371. M. Bohdanecký, V. Petrus, P. Kratochvil, Coll. Czech. Chem. Commun., **34**, 1168 (1969).
372. M. Bohdanecký, Z. Tuzar, Coll. Czech. Chem. Commun., **34**, 3318 (1969).
373. M. Bohdanecký, Coll. Czech. Chem. Commun., **35**, 1972 (1970).
374. M. Bohdanecký, H. Bazilova, J. Kopeček, Eur. Polym. J., **10**, 405 (1974).
375. G. Boileux-Steffan, Q. T. Pham, Makromol. Chem., **185**, 877 (1984).
376. D. Bolikal, S. L. Regen, Macromolecules, **17**, 1287 (1984).
377. L. S. Bolotnikova, T. I. Samsonova, Polym. Sci. USSR, **6**, 590 (1964); Vysokomol. Soedin., **6**, 533 (1964).

378. K. J. Bombaugh, W. A. Dark, R. N. King, J. Polym. Sci. C-21, 131 (1968).
379. K. A. Boni, F. A. Sliemers, P. B. Stickney, Polym. Preprints., **8** (1), 446 (1967).
380. K. A. Boni, F. A. Sliemers, P. B. Stickney, J. Polym. Sci. A-2, **6**, 1567 (1968).
381. K. A. Boni, F. A. Sliemers, P. B. Stiekney, J. Polym. Sci. A-2, **6**, 1579 (1968).
382. G. Bontá, B. M. Gallo, S. Russo, C. Uliana, Polymer, **17**, 217 (1976).
383. C. Booth, L. R. Beason, J. Polym. Sci., **42**, 108 (1960).
384. C. Booth, L. R. Beason, J. Polym. Sci., **42**, 99 (1960).
385. C. Booth, M. N. Jones, E. Powell, Nature (London), **196**, 772 (1962).
386. C. Booth, Polymer, **4**, 471 (1963).
387. C. Booth, W. C. E. Higginson, E. Powell, Polymer, **5**, 479 (1964).
388. C. Booth, C. Price, Polymer, **7**, 85 (1966).
389. C. Booth, R. Orme, Polymer, **11**, 626 (1970).
390. C. Booth, D. V. Dodgson, I. H. Hillier, Polymer, **11**, 11 (1970).
391. C. Booth, J. M. Bruce, M. Buggy, Polymer, **13**, 475 (1972).
392. C. Booth, J.-L. Forget, I. Georgii, W. S. Li, Price, Eur. Polym. J., **16**, 255 (1980).
393. O. V. Boronina, V. A. Kasaikin, M. B. Lachinov, V. P. Zubov, V. A. Kavanov, Polym. Sci. USSR, **22**, 2028 (1980); Vysokomol. Soedin. A, **22**, 1850 (1980).
394. J. Borrajo, C. Cordon, J. M. Carella, S. Toso, G. Goizueta, J. Polym. Sci. Part B: Polym. Phys., **33**, 1627 (1995).
395. P. Borrell, G. Riess, A. Banderet, Bull. Soc. Chim. France, **2**, 354 (1961).
396. C. Borri, S. Brueckner, V. Crescenzi, G. Della Fortuna, A. Mariano, P. Scarazzato, Eur. Polym. J., **7**, 1515 (1971).
397. E. Borsig, M. Lazár, M. Capla, S. Florián, Angew. Makromol. Chem., **9**, 89 (1969).
398. E. Bortel, A. Kochanowski, Makromol. Chem. Rapid Commun., **1**, 205 (1980).
399. E. Bortel, M. Stysto, Makromol. Chem., **189**, 1155 (1988).
400. R. Bortolin, S. S. D. Brown, B. Parbhoo, Macromolecules, **23**, 2465 (1990).
401. J. J. Bourguignon, H. Bellisent, J. C. Galin, Polymer, **18**, 937 (1977).
402. B. Boutevin, M. Maliszewicz, Y. Pietrasanta, Makromol. Chem., **183**, 2333 (1982).
403. T. N. Bowner, E. Reichmams, C. W. Wilkins Jr., M. Y. Hellman, J. Polym. Sci., Polym. Chem. Ed., **20**, 2661 (1982).
404. R. F. Boyer, J. Polym. Sci., **8**, 73, 197 (1952).
405. E. M. Bradbury, C. Crane-Robinson, H. W. E. Rattle, Polymer, **11**, 277 (1970).
406. E. M. Bradbury, C. Crane-Robinson, P. G. Hartman, Polymer, **14**, 543 (1973).
407. J. G. Braks, R. Y. M. Huang, Makromol. Chem., **185**, 317 (1984).
408. D. A. Brant, B. Kwon Min, Macromolecules, **2**, 1 (1969).
409. D Braun, W. Fischer, Makromol. Chem., **85**, 155 (1965).
410. D. Braun, F.-J. Quesada Lucas, W. Neumann, Makromol. Chem., **127**, 253 (1969).
411. D. Braun, D. Chaudhari, Makromol. Chem., **133**, 241 (1970).
412. D. Braun, D. Chaudhari, W. Maechtle, Angew. Makromol. Chem., **15** 83 (1971).
413. D. Braun, V. Legradic, Angew. Makromol. Chem., **25**, 193 (1972).
414. D. Braun, J. Arndt, Angew. Makromol. Chem., **73**, 143 (1978).
415. D. Braun, W. Pandjojo, Angew. Makromol. Chem., **80**, 195 (1979).
416. D. Braun, G. Holzer, R. I. Noriega, Angew. Makromol. Chem., **89**, 111 (1980).
417. D. Braun, P. Guenther, W. Pandjojo, Angew. Makromol. Chem., **102**, 147 (1982).
418. D. Braun, E. Bezdadea, M. Dethloff, Angew. Makromol. Chem., **147**, 49 (1987).
419. D. Braun, S. Ross, Angew. Makromol. Chem., **228**, 185 (1995).
420. D. Braun, C. Heittig-Rupaner, Angew. Makromol. Chem., **225**, 83 (1995).
421. J. Bravo, M. P. Tarazona, E. Saiz, Macromolecules, **24**, 4089 (1991).
422. J. Bravo, M. P. Tarazona, E. Saiz, Macromolecules, **25**, 5625 (1992).
423. J. W. Breitenbach, H. G. Burger, Makromol. Chem., **54**, 60 (1962).
424. J. W. Breitenbach, H. G. Burger, A. Schindler, Monatsh. Chem., **93**, 160 (1962).
425. J. W. Breitenbach, H. Gabler, O. F. Olaj, Makromol. Chem., **81**, 32 (1965).
426. J. W. Breitenbach, B. A. Wolf, Makromol. Chem., **108**, 263 (1967).
427. P. Bres, D. H. Richards, M. J. Stewart, M. Viguier, Brit. Polym, J., **16**, 1 (1984).
428. S. E. Bresler, I. Ya. Poddubnyi, S. Ya. Frenkel, Zh. Tekhn. Fiz., **23**, 1521 (1953).
429. S. E. Bresler, S. Ya. Frenkel, Zh. Tekhn. Fiz., **23**, 1502 (1953).
430. S. E. Bresler, I. Y. Poddubnyi, S. Ya. Frenkel, Rubber Chem. Technol., **30**, 507 (1957).
431. P. I. Brewer, Polymer, **9**, 545 (1968).
432. R. J. Brewer, L. J. Tanghe, S. Bailey, J. T. Burr, J. Polym. Sci. A, **16**, 1697 (1968).
433. R. J. Brewer, L. J. Tanghe, S. Bailey, J. Polym. Sci. A, **17**, 1635 (1969).
434. W. B. Bridgman, J. Am. Chem. Soc., **64**, 2349 (1942).
435. S. L. Brimhall, M. N. Myers, K. D. Caldwell, J. C. Giddings, J. Polym. Sci., Polym. Lett. Ed., **22**, 339 (1984).
436. G. M. Bristow, B. Westall, Polymer, **8**, 609 (1967).
437. G. L. Borde, J. V. Koleske, J. Macromol. Sci., Part A: Chem., **6**, 1109 (1972).
438. B. W. Brodman, M. P. Devine, M. J. Gurbarg, J. Macromol. Sci., Part A: Chem., **8**, 837 (1974).
439. M. C. Brooks, R. M. Badger, J. Am. Chem., Soc., **72**, 1705, 4384 (1950).

440. A. P. Brookshaw, D. E. Hillman, J. I. Paul, Brit. Polym. J., **5**, 229 (1973).
441. J.-C. Brosse, J.-C. Lenain, J.-M. Gauthier, Makromol. Chem., **183**, 2685 (1982).
442. R. W. Brotzman, P. J. Flory, Macromolecules, **20**, 351 (1987).
443. L. de Brouckere, E. Bidaine, A. Van der Heyden, Bull. Soc. Chim. Belges, **58**, 418 (1949).
444. F. M. Brower, H. W. Mc Cormick, J. Polym. Sci., **36**, 341 (1959).
445. F. M. Brower, H. W. Mc Cormick, J. Polym. Sci. A-1, 1749 (1963).
446. J. R. Borwn, J. H. O'Donnell, Macromolecules, **5**, 109 (1972).
447. J. R. Borwn, J. H. O'Donnell, J. Macromol. Sci., Part A: Chem., **6**, 1411 (1972).
448. W. Brown, Arkiv Kemi, **18**, 227 (1961).
449. W. Brown, D. Henley, J. Oehman, Arkiy Kemi, **22**, 189 (1964).
450. S. D. Bruck, J. Polym. Sci., **32**, 519 (1958).
451. S. Brueckner, G. Allegra, G. Gianotti, E. Moraglio, Eur. Polym. J., **10**, 347 (1974).
452. R. Bruessau, Analytiker-Taschenbuch, **4**, 415 (1984).
453. W. A. J. Bryce, I. G. Meldrum, J. Polym. Sci., C, **16**, 3321 (1968).
454. W. A. J. Bryce, G. Mc Gibbon, I. G. Meldrum, Polymer, **11**, 394 (1970).
455. L. E. Brydia, O. M. Garty, J. Polym. Sci., Polym. Chem. Ed., **18**, 1577 (1980).
456. J. Brzezinski, Z. Czlonkowska-Kohutnicka, O. Kallistov, Sz. Krozer, Plaste Kautschuk, **15**, 403 (1968).
457. J. Brzezinski, Z. Czlonkowska-Kohutnicka, B. Czarnecka, A. Kornas-Calka, Eur. Polym. J., **9**, 733 (1973).
458. L. Brzezinski, Z. Dobkowski, Eur. Polym. J., **16**, 85 (1980).
459. R. R. Buch, H. M. Klimisch, O. K. Johannson, Polym. Preprints, **6** (2), 1022 (1965).
460. R. R. Buch, H. M. Klimisch, O. K. Johanson, J. Polym. Sci. A-2, **7**, 563 (1969).
461. R. R. Buch, H. M. Klimisch, O. K. Johannson, J. Polym. Sci. A-2, **8**, 541 (1970).
462. V. P. Budtov, N. G. Podosenova, V. M. Gal'perin, G. A. Otradina, V. G. Rupyshev, N. M. Domareva, Ye. O. Krivchenko, Polym. Sci. USSR, **18**, 1742 (1976); Vysokomol. Soedin. A, **18**, 1522 (1976).
463. V. P. Budtov, S. S. Ivanchev, U. M. Belyayev, O. S. Romanova, G. A. Otradina, Polym. Sci. USSR, **18**, 2662 (1976); Vysokomol. Soedin. A, **18**, 2328 (1976).
464. R. Buening, R. Minke, D. Schulze, Angew. Makromol. Chem., **25**, 27 (1972).
465. K. Buerger, Z. Anal. Chem., **196**, 259 (1963).
466. A. Buleon, H. Chanzy, P. Froment, J. Polym. Sci., Polym. Phys. Ed., **20**, 1081 (1982).
467. A. A. Buniyat-Zade, Ye. A. Osipov, A. Bazimova, Polym. Sci. USSR, **14**, 808 (1972); Vysokomol. Soedin. A, **14**, 722 (1972).
468. A. J. Bur, L. J. Fetters, Macromolecules, **6**, 874 (1973).
469. W. Burchard, B. Fritz, R. Lippert, B. Pfannemuller, E. Husemann, Makromol. Chem., **44–46**, 358 (1961).
470. D. R. Burfield, L. C. Chew, S. N. Gan, Polymer, **17**, 713 (1976).
471. D. E. Burge, D. B. Bruss, J. Polym. Sci. A-1, 1927 (1963).
472. G. M. Burnett, P. Meares, C. Paton, Trans. Faraday Soc., **58**, 723 (1962).
473. H. Burns, D. K. Carpenter, Macromolecules, **1**, 384 (1968).
474. F. W. Burns, B. Mc Carthy, R. M. O'Connor, D. C. Pepper, IUPAC Helsinki, vol 1, Preprint, 1/30 (1972).
475. K. Burzin, W. Holtrup, R. Feinauer, Angew. Makromol Chem., **74**, 93 (1978).
476. S. V. Bushin, Ye. V. Korneyeva, I. I. Konstantinov, Yu. B. Amerik, S. A. Didenko, I. N. Shtennikova, V. N. Tsvetkov, Polym. Sci. USSR, **24**, 1671 (1982); Vysokomol. Soedin. A, **24**, 1469 (1982).
477. S. V. Bushin, Ye. B. Lysenko, V. A. Cherkasov, K. P. Smironov, S. A. Didenko, G. N. Marchenko, V. N. Tsvetkov, Polym. Sci. USSR, **25**, 2212 (1983); Vysokomol. Sodin. A, **25**, 1899 (1983).
478. V. Busico, P. Corradini, L. De Martino, F. Graziano, A. Iadicicco, Makromol. Chem., **192**, 49 (1991).
479. V. Busico, P. Corradini, R. De Biasio, M. Trifuoggi, Makromol. Chem., **193**, 1765 (1992).
480. V. Busico, P. Corradini, R. De Biasio, Makromol. Chem., **193**, 897 (1992).
481. A. Buzagh, K. Udvarhelyi, F. Horkay, Kolloid Z., **154**, 130 (1957).
482. A. Buzagh, K. Udvarhelyi, F. Horkay, Kolloid Z., **157**, 53 (1958).
483. V. T. Bykov, T. P. Avilova, N. P. Shapkin, Polym. Sci. USSR, **12**, 813 (1970); Vysokomol. Soedin. A, **12**, 724 (1970).
484. S. Bywater, P. Johnson, Trans. Faraday Soc., **47**, 195 (1951).
485. F. Cabassi, W. Porzio, G. Ricci, S. Brückner, S. V. Meille, Makromol. Chem., **189**, 2135 (1988).
486. R. E. Cais, J. M. Kometani, Macromolecules, **17**, 1932 (1984).
487. K. D. Caldwell, S. L. Brimmhall, J. Cao, J. C. Giddings, J. Appl. Polym. Sci., **36**, 7.3 (1988).
488. Y. Camberlin, J. P. Pascault, M. Letoffe, P. Clardy, J. Polym. Sci., Polym. Chem. Ed., **20**, 383 (1982).
489. G. G. Cameron, M. Y. Qureshi, J. Polym. Sci., Polym. Chem. Ed., **18**, 2143 (1980).
490. G. G. Cameron, M. Y. Qureshi, J. Polym. Sci., Polym. Chem. Ed., **18**, 3149 (1980).
491. H. Campbell, P. O. Kane, I. G. Ottewill, J. Polym. Sci., **12**, 611 (1954).
492. D. S. Campbell, A. J. Tinker, Polymer, **25**, 1146 (1984).
493. A. Campos, L. Borque, J. E. Figueruelo, Anales Quim. (Madrid), B, **74**, 701 (1978).
494. A. Compos, B. Celda, R. Tejero, J. E. Figueruelo, Eur. Polym. J., **20**, 447 (1984).
495. E. Compos-López, J. L. Angulo-Sánchez, J. Polym. Sci., Polym. Lett. Ed., **14**, 649 (1976).
496. F. Can, T. Capaccioli, Eur. Polym. J., **14**, 185 (1978).
497. F. Candau, P. Rempp, Makromol. Chem., **122**, 15 (1969).
498. H. J. Cantow, Makromol. Chem., **30**, 81 (1959).

499. M. J. R. Cantow, R. S. Porter, J. F. Johnson, Makromol. Chem., **49**, 1 (1961).
500. M. J. R. Cantow, R. S. Porter, J. F. Johnson, Nature (London), **192**, 752 (1962).
501. M. J. R. Cantow, R. S. Porter, J. F. Johnson, J. Polym. Sci., C, **1**, 187 (1963).
502. M. J. R. Cantow, R. S. Porter, J. F. Johnson, J. Appl. Polym. Sci., **8**, 2963 (1964).
503. M. J. R. Cantow, R. S. Porter, J. F. Johnson, Polym. Preprints, **6** (1), 338 (1965).
504. M. J. R. Cantow, R. S. Porter, J. F. Johnson, J. Polym. Sci. B, **4**, 707 (1966).
505. M. J. R. Cantow, R. S. Porter, J. F. Johnson, J. Macromol. Sci: Rev. Macromol. Chem., **1**, 393 (1966).
506. M. J. R. Cantow (Ed.) "Polymer Fractionation", Academic Press, New York, 1966.
507. Y. Cao, Q. Wu, K. Guo, R. Qian, Makromol. Chem., **185**, 389 (1984).
508. A. Cao, M. Ichikawa, K-i. Kasuya, N. Yoshio, N. Asakawa, Y. Inoue, Y. Doi, H. Abe, Polym. J., **28**, 1096 (1996).
509. G. Capaccio, I. M. Ward, J. Polym. Sci., Polym. Phys. Ed., **20**, 1107 (1982).
510. T. Capaccioli, L. Zotteri, Eur. Polym. J., **11**, 729 (1975).
511. S. R. Caplan, J. Polym. Sci., **35**, 409 (1959).
512. K. R. Cardunen, R. O. Carter III, M. Zimbo, J. L. Gerlock, D. R. Bauer, Macromolecules, **21**, 1598 (1988).
513. C. Carlini, J. Polym. Sci., Polym. Chem. Ed., **18**, 799 (1980).
514. D. W. Carlson, E. O'Rourke, J. K. Valaitis, A. G. Altenau, J. Polym. Sci., Polym. Chem. Ed., **14**, 1379 (1978).
515. S. H. Carr, E. Baer, A. G. Walton, J. Macromol. Sci., Part B: Phys., **6**, 15 (1972).
516. C. E. Carraher Jr., T. W. Brandt, Makromol. Chem., **130**, 166 (1970).
517. B. Carton, V. Bottiglione, M. Morcellet, C. Loucheux, Makromol. Chem., **179**, 2931 (1978).
518. E. F. Casassa, W. H. Stockmayer, Polymer, **3**, 53 (1962).
519. G. A. Casay, A. George, N. Hadjichristidis, J. S. Lindner, J. Polym. Sci., Part B: Polym. Phys., **33**, 1537 (1995).
520. L. C. Case, J. Phys. Chem., **62**, 895 (1958).
521. L. C. Case, Makromol. Chem., **41**, 61 (1960).
522. R. H. Casper, G. V. Schulz, J. Polym. Sci. A-2, **8**, 833 (1970).
523. R. H. Casper, G. V. Schulz, Sep., **6**, 321 (1971).
524. G. Ceccorulli, M. Pizzoli, G. Stea, Makromol. Chem., **142**, 153 (1971).
525. G. Ceccorulli, F. Manescalchi, Makromol. Chem., **168**, 303 (1973).
526. J. Čepelák, Plaste Kautschuk, **2**, 34 (1955).
527. J. Čepelák, Chem. Prumysl., **6**, 106 (1956).
528. R. J. Ceresa, Polymer, **1**, 480 (1960).
529. R. J. Ceresa, Polymer, **1**, 488 (1960).
530. R. J. Ceresa, Polymer, **2**, 213 (1961).
531. L. C. Cerny, T. E. Helminiak, J. F. Meier, J. Polym. Sci., **44**, 539 (1960).
532. S. Cesca, A. Roggero, T. Salvatori, A. De Chirico, G. Santi, Makromol. Chem., **133**, 161 (1970).
533. S. Cesca, A. Priola, A. De Chirico, G. Santi, Makromol. Chem., **143**, 211 (1971).
534. S. Cesca, P. V. Duranti, A. Roggero, G. Santi, M. Bruzzone, J. Polym. Sci., Polym. Chem. Ed., **12**, 1209 (1974).
535. J. C. Chadwick, A. Miedema, B. J. Ruisch, O. Sudmeijer, Makromol. Chem., **193**, 1463 (1992).
536. J. C. Chadwick, A. Miedema, O. Sudmeijer, Makromol. Chem. Phys., **195** 167 (1994).
537. A. J. Chalk, A. R. Gilbert, J. Polym. Sci., Polym. Chem. Ed., **10**, 2033 (1972).
538. G. Champetier, M. Fontanille, A.-C. Korn, P. Sigwalt, J. Polym. Sci., **58**, 911 (1962).
539. J. V. Champion, R. A. Desson, G. H. Neeten, Polymer, **15**, 301 (1974).
540. R. K. S. Chan, Polym. Eng. Sci., **11**, 152 (1971).
541. J. E. Chandler, B. H. Johnson, R. W. Lenz, Macromolecules, **13**, 377 (1980).
542. F. S. C. Chang, Polym. Preprints, **12** (2), 835 (1971).
543. E. P., Chang, R. Salovey, J. Polym. Sci., Polym. Chem. Ed., **12**, 2977 (1974).
544. E. W. Channen, Rev. Pure Appl. Chem., **9**, 225 (1959).
545. A. Chapiro, P. Cordier, J. Josefowicz, J. Sebban-Danon, J. Polym. Sci., C, **4**, 491 (1963).
546. A Chapiro, G. Palma, Eur. Polym. J., **3**, 151 (1967).
547. R. P. Chaplin, W. Ching, Chromatographia, **11**, 600 (1978).
548. G. Charlet, R. Ducasse, G. Delmas, Polymer, **22**, 1190 (1981).
549. Y. Charlier, P. Godard, D. Daoust, C. Strazielle, Macromolecules, **27**, 3604 (1994).
550. S. K. Chatterjee, N. Dattagupta, Brit. Polym. J., **6**, 293 (1974).
551. P. N. Chaturvedi, C. K. Patel, J. Polym. Sci., Polym. Chem. Ed., **23**, 1255 (1985).
552. V. Chatuverdi, S. Tanaka, K. Kaeriyama, Macromolecules, **26**, 2607 (1993).
553. A. K. Chaudhuri, D. K. Sarkar, S. R. Palit, Makromol. Chem., **111**, 36 (1968).
554. P. Chaumont, G. Beinert, J. Herz, P. Rempp, Polymer, **22**, 663 (1981).
555. P. Chaumont, G. Beinert, J. Herz, Eur. Polym. J., **18**, 875 (1982).
556. P. Chaumont, G. Beinert, J. E. Herz, P. Rempp, Makromol. Chem., **183**, 1181 (1982).
557. Y. Chauvin, L. Saussine, Macromolecules, **29**, 1163 (1996).
558. A. S. Chawla, R. Y. M. Huang, J. Polym. Sci., Polym. Chem. Ed., **13**, 1271 (1975).
559. W. K. W. Chen, H. Z. Friedlander, J. Polym. Sci. C, **4**, 1195 (1963).
560. J.-L. Chen, H. Morawetz, Macromolecules, **15**, 1185 (1982).
561. C.-Y. Chen, M. Igbal. C. U. Pittman Jr., J. N. Helbert, Makromol. Chem., **179**, 2109 (1978).
562. C.-Y. Chen, C. U. Pittman Jr, J. N. Helbert, J. Polym. Sci., Polym. Chem. Ed., **18**, 169 (1980).
563. H. Chen, K. Ishizu, T. Fukutomi, T. Kakurai, J. Polym. Sci., Polym. Chem. Ed., **22**, 2123 (1984).

564. H. R. Chen, L. P. Blanchard, IUPAC Leiden Vol 1, 37 (1970).
565. R.-S. Cheng, X.-H. Yan, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 123 (1991).
566. S. Z. D. Cheng, A. Zhang, J. Cheu, D. P. Heberer, J. Polym. Sci. Part B: Polym. Phys., **29**, 287 (1991).
567. A. Chenite, F. Brisse, Macromolecules, **25**, 776 (1992).
568. H. Cheradame, J.-C. Habimana, E. Rousset, F. J. Chen, Marcomolecules, **27**, 631 (1994).
569. H. Cheradame, P. P. Kisilitsa, O. G. Sel'skaya, A. A. Berlin, Polym. Sci. USSR, **10**, 223 (1968); Vysokomol. Soedin. A, **10**, 196 (1968).
570. A. N. Cherkasov, S. I. Klenin, G. A. Andreyeva, Polym. Sci. USSR, **12**, 1382 (1970); Vysokomol. Soedin. A, **12**, 1223 (1970).
571. A. N. Cherkasov, S. I. Klenin, M. Z. El'Sabei, Polym. Sci. USSR, **14**, 1486 (1972); Vysokomol. Soedin. A, **14**, 1326 (1972).
572. P. Cheung, S. T. Balke, T. C. Schunk, T. H. Mourey, J. Appl. Polym. Sci., Appl. Polym. Symp., **52**, 105 (1993).
573. J.-W. Chevalier, J.-Y. Bergeron, L. H. Dao, Macromolecules, **25**, 3325 (1992).
574. R. Chiang, J. Phys. Chem., **69**, 1645 (1965).
575. W.-Y. Chiang, J.-Y. Lu, Angew. Makromol. Chem., **205**, 75 (1993).
576. O. Chiantore, S. Pokorný, M. Bleha, J. Janča, Angew. Makromol. Chem., **105**, 61 (1982).
577. O. Chiantore, M. Guaita, J. Liquid. Chromatog., **8**, 1413 (1985).
578. O. Chiantore, M. Guaita, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 431 (1991).
579. O. Chianotore, P. Cinquina, M. Guaita, Eur. Polym. J., **30**, 1043 (1994).
580. S. N. China, J. D. Matlack, A. L. Resnick, R. L. Samuels, J. Polym. Sci., **17**, 391 (1955).
581. S. N. China, R. A. Guzzi, J. Polym. Sci., **21**, 417 (1956).
582. S. N. China, R. J. Samuels, J. Polym. Sci., **19**, 463 (1956).
583. S. N. China J. Polym. Sci., **25**, 413 (1957).
584. W. O. Choi, M. Sawamoto, T. Higashimura, Macromolecules, **23**, 48 (1990).
585. J. Chojnowski, L. Wilczek, Makromol. Chem., **180**, 117 (1979).
586. M. S. Choudhary, I. K. Varma, Angew. Mackromol. Chem., **87**, 75 (1980).
587. M. S. Choudhary, I. K. Varma, J. Macromol. Sci., Part A: Chem., **20**, 941 (1983).
588. R. C. L. Chow, C. S. Marvel, J. Polym. Sci. A-1, **5**, 2949 (1967).
589. B. Chu, D-quing Wu, Macromolecules, **20**, 1606 (1987).
590. J.-Y. Chuang, J. Appl. Polym. Sci., Appl. Polym. Symp., **45**, 227 (1990).
591. Y. Chujo, I. Tomita, T. Saegusa, Macromolecules, **27**, 6714 (1994).
592. T. C. Chung, M. Chasmawala, Macromolecules, **25**, 5137 (1992).
593. T. C. Chung, W. Janvikul, R. Bernard, R. Hu, Polymer, **36**, 3565 (1995).
594. G. Ciampa, H. Schwindt, Chim. Ind., **37**, 169 (1955).
595. F. Ciardelli, G. Montagnoli, D. Pini, O. Pieroni, C. Carlini, E. Benedetti, Makromol. Chem., **147**, 53 (1971).
596. F. Ciardelli, P. Locatelli, M. Marchetti, A. Zambelli, Makromol. Chem., **175**, 923 (1974).
597. R. Ciaschi, M. D'Alagni, G. Mignucci, Makromol. Chem., **180**, 2883 (1979).
598. C. Cincu, Eur. Polym. J., **10**, 29 (1974).
599. C. Cincu, C. V. Uglea, Makromol. Chem., **177**, 2457 (1976).
600. P. Cinquina, G. Gianotti, D. Borghi, Polymer Commun., **31**, 30 (1990).
601. S. Claesson, Disc. Faraday Soc., **7**, 321 (1949).
602. S. J. Clarson, J. A. Semlyen, Polymer, **27**, 1633 (1986).
603. D. Cleverdon, D. Laker, J. Appl. Chem., **1**, 6 (1951).
604. I. E. Climie, E. F. T. White, J. Polym. Sci., **47**, 149 (1960).
605. E. Cohn-Ginsberg, T. G. Fox, H. F. Mason, Polymer, **3**, 97 (1962).
606. N. Colak, K. Alyürük, Macromolecules, **30**, 1709 (1989).
607. H. Coll, D. K. Gilding, J. Polym. Sci. A-2, **8**, 89 (1970).
608. J. W. Collete, C. W. Tullock, N. Mac Donald, W. H. Buck, A. C. L. Su, J. R. Harrell, R. Mulhaught, B. C. Anderson, Macromolecules, **22**, 3851 (1989).
609. J. W. Collete, D. W. Ovenall, W. H. Buck, R. C. Fergurson, Macromolecules, **22**, 3858 (1989).
610. P. Colombo, S. Custro, P. Radici, J. Polym. Sci., Polym. Chem. Ed., **18**, 681 (1980).
611. R. L. Combs, D. F. Slonaker, J. T. Summers, Org. Coatings Plastics Preprints, **21**, 249 (1961).
612. R. L. Combs, D. F. Slonaker, F. B. Joyner, H. W. Coover Jr., J. Polym. Sci. A-1, **5**, 215 (1967).
613. C. M. Conrad, Ind. Eng. Chem., **45**, 2511 (1953).
614. D. Constantin, Eur. Polym. J., **13**, 907 (1977).
615. D. Constantin, M. Hert, J.-P. Machon, Makromol. Chem., **179**, 1581 (1978).
616. W. Conti, E. Sorta, Eur. Polym. J., **8**, 475 (1972).
617. M. Cooper, R. St. John Manley, J. Macromol. Sci., Part B: Phys., **10**, 59 (1974).
618. A. R. Cooper, J. F. Johnson, A. R. Bruzzone, Polym. Preprints, **10** (2), 1455 (1969).
619. A. R. Cooper, A. R. Bruzzone, J. F. Johnson, Makromol. Chem., **138**, 279 (1970).
620. A. R. Cooper, J. F. Johnson, Polym. Preprints, **12** (2), 738 (1971).
621. A. R. Cooper, J. Polym. Sci., Polym. Phys. Ed., **12**, 1969 (1974).
622. A. R. Cooper, A. J. Hughes, J. F. Johnson, Chromatograhia, **8**, 136 (1975).
623. R. E. Cooper, J. Upadhyay, A. Wassermann, J. Polym. Sci., **60**, S46 (1962).
624. W. Cooper, G. Vaughan, D. E. Eaves, R. W. Madden, J. Polym. Sci., **50**, 159 (1961).
625. W. Cooper, D. E. Eaves, G. Vaughan, J. Polym. Sci., **59**, 241 (1962).
626. W. Cooper, G. Vaughan, J. Yardles, J. Polym. Sci., **59**, S2 (1962).
627. W. Cooper, G. A. Pope, G. Vaughan, Eur. Polym. J., **4**, 207 (1968).

628. W. Cooper, P. T. Hale, J. S. Walker, Polymer, **15**, 175 (1974).
629. N. F. Cooray, M. Kakimoto, Y. Imai, Y. Suzuki, Macromolecules, **27**, 1592 (1994).
630. H. W. Coover, Jr., R. L. Mc Connell, F. B. Joyner, D. F. Slonaker, R. L. Combs, J. Polym. Sci. A-1, **4**, 2563 (1966).
631. S. Coppick, O. A. Battista, M. R. Lytton, Ind. Eng. Chem., **42**, 2533 (1950).
632. N. Corbin, J. Prud'homme, J. Polym. Sci., Polym. Chem. Ed., **14**, 1645 (1976).
633. T. Corner, F. W. Peaker, J. Polym. Sci., Polym. Chem. Ed., **22**, 2119 (1984).
634. C. F. Cornet, Polymer, **9**, 7 (1968).
635. M. Cortazar, G. M. Guzmán, Makromol. Chem., **183**, 721 (1982).
636. A. Cosani, E. Peggion, E. Scoffone, A. S. Verdini, J. Polym. Sci., B, **4**, 55 (1965).
637. A. Cosani, E. Peggion, E. Scoffone, A. S. Verdini, Makromol. Chem., **97** 113 (1966).
638. G. Costa, E. Pedemonte, S. Russo, E. Sava, Polymer, **20**, 713 (1979).
639. J. A. Cote, M. Shida, J. Polym. Sci. A-2, **9**, 421 (1971).
640. B. J. Cottam, D. M. Wiles, S. Bywater, Can. J. Chem., **41**, 1905 (1963).
641. B. J. Cottam, J. Appl. Polym. Sci., **9**, 1853 (1965).
642. B. J. Cottam, J. M. G. Cowie, S. Bywater, Makromol. Chem., **86**, 116 (1965).
643. G. R. Cotton, W. C. Schneider, J. Appl. Polym. Sci., **7**, 1243 (1963).
644. F. R. Cottrell, E. W. Merrill, K. A. Smith, IUPAC Toronto, Preprint A, **8**, 1 (1968).
645. F. R. Cottrell, E. W. Merrili, K. A. Smith, J. Polym. Sci. A-2, **7**, 1415 (1969).
646. P. M. Cotts, S. Ferline, G. Dagli, D. S. Pearson, Macromolecules, **24**, 6730 (1991).
647. P. M. Cotts, Macromolecules, **27**, 6487 (1994).
648. D. B. Cotts, G. C. Berry, Macromolecules, **14**, 930 (1981).
649. J. Čoupek, K. Bouchal, Makromol. Chem., **135**, 69 (1970).
650. J. M. G. Cowie, C. T. Greenwood, J. Chem. Soc., 2862 (1957).
651. J. M. G. Cowie, C. T. Greenwood, J. Chem. Soc., 4640 (1957).
652. J. M. G. Cowie, Makromol. Chem., **42**, 230 (1960).
653. J. M. G. Cowie, J. Polym. Sci., **49**, 455 (1961).
654. J. M. G. Cowie, P. M. Toporowski, Polymer, **5**, 601 (1964).
655. J. M. G. Cowie, S. Bywater, J. Polym. Sci. A-2, **6**, 499 (1968).
656. J. M. G. Cowie, R. J. Ranson, Makromol. Chem., **143**, 105 (1971).
657. J. M. G. Cowie, A. Macconnachie, Eur. Polym. J., **10**, 367 (1974).
658. J. M. G. Cowie, A. Macconnachie, J. Polym. Sci., Polm. Phys. Ed., **12**, 407 (1974).
659. J. M. G. Cowie, A. Macconnachie, Eur. Polym. J., **11**, 79 (1975).
660. L. H. Cragg, H. Hammerschlag, Chem. Rev., **39**, 79 (1946).
661. D. N. Cramond, J. M. Hammond, J. R. Urwin, Eur. Polym. J., **4**, 451 (1968).
662. H. Craubner, Makromol. Chem., **78**. 121 (1964).
663. H. Craubner, Makromol. Chem., **89**, 105 (1965).
664. H. Craubner, Ber. Bunsenges. Phys. Chem., **74**, 1262 (1970).
665. H. Craubner, J. Polym. Sci., C, **42**, 889 (1973).
666. H. Craubner, Makromol. Chem., **175**, 2461 (1974).
667. J. R. Craven, R. H. Mobbs, C. Booth, J. R. M. Giles, Makromol. Chem. Rapid Commun, **7**, 81 (1986).
668. V. Crescenzi, G. Manzini, G. Calzolari, C. Borri, Eur. Polym. J., **8**, 449 (1972).
669. S. Creutz, P. Teyssé, R. Jérôme, Macromolecules, **30**, 1 (1997).
670. S. Creutz, P. Teyssié, R. Jérôme, Macromolecules, **30**, 6 (1997).
671. M. A. Crook, F. S. Walker, Nature (London), **198**, 1163 (1963).
672. T. G. Croucher, R. E. Wetton, Polymer, **17**, 205 (1976).
673. P. Crouzet, F. Fine, P. Mangin, J. Appl. Polym. Sci., **13**, 205 (1969).
674. C. Crouzet, J. Marchal, Makromol. Chem., **177**, 2819 (1976).
675. E. Csakvari, M. Azori, F. Tüdos, Polym. Bull., **5**, 413 (1981).
676. P. Cukor, M. Rubner, J. Polym. Sci., Polym. Chem. Ed., **18**, 909 (1980).
677. R. J. E. Cumberbich, W. G. Harland, J. Textile Inst., **49**, T679 (1958).
678. C. Cuniberti, U. Bianchi, Polymer, **7**, 151 (1966).
679. C. Cuniberti, J. Polym. Sci. A-2, **8**, 2051 (1970).
680. C. Cuniberti, U. Bianchi, Polymer, **15**, 346 (1974).
681. C. Cuniberti, A. Perico, Eur. Polym. J., **16**, 887 (1980).
682. J. Curchod, T. Ve, J. Appl. Polym. Sci., **9**, 3541 (1965).
683. Z. Czlonkowska-Kohutnicka, Z. Dobkowski, Eur. Polym. J., **18**, 911 (1982).
684. G. F. D'Alelio, A. B. Finestone, L. Taft, T. J. Miranda, J. Polym. Sci., **45**, 83 (1960).
685. A. D'Emanuele, J. Kost, J. L. Hill, R. Langer, Macromolecules, **25**, 511 (1992).
686. E. Dan, M. B. Roller, J. Polym. Sci., Polym. Lett. Ed., **21**, 875 (1983).
687. J. Daňhelka, I. Koessler, V. Boháčková, J. Polym. Sci., Polym. Chem. Ed., **14**, 287 (1976).
688. J. Daňhelka, L. Mrkvičková, V. Štěphan, I. Koessler, Angew. Makromol. Chem., **122**, 77 (1984).
689. M. Danielwicz, M. Kubin, J. Appl. Polym. Sci., **26**, 951 (1981).
690. P. O. Danis, D. E. Karr, W. J. Simonsick, Macromolecules, **28**, 1229 (1995).
691. P. O. Danis, D. E. Karr, Macromolecules, **28**, 8548 (1995).
692. J. Danon, J. Jozefovicz, Eur. Polym. J., **5**, 405 (1969).
693. J. Danon, R. Derai, Eur. Polym. J., **5**, 659 (1969).
694. J. Danon, J. Jozefowicz, J. Polym. Sci. C, **16**, 4523 (1969).
695. J. Danon, J. Jozefovicz, Eur. Polym. J., **6**, 199 (1970).
696. F. Danusso, G. Gianotti, Makromol. Chem., **6**, 164 (1963).

697. B. Das, B. B. Konar, J. Macromol. Sci., Part A: Chem. **19**, 601 (1983).
698. C. David, D. Baeyens-Volant, G. Delaunois, O. L. Vinh, W. Piret, G. Geuskens, Eur. Polym. J., **14**, 501 (1978).
699. M. G. Davidson, W. M. Deen, Macomolecules, **21**, 3474 (1988).
700. N. S. Davidson, L. J. Fetters, W. G. Funk, W. W. Graessley, N. Hadjichristidis, Macromolecules, **21**, 112 (1988).
701. W. G. Davies, D. P. Jones, Polym. Preprints, **11** (2), 447 (1970).
702. D. H. Davies, D. C. Phillips, J. D. B. Smith, J. Polym. Sci., Polym. Chem. Ed., **17**, 1153 (1979).
703. W. E. Davis, J. Am. Chem. Soc., **69**, 1453 (1947).
704. T. E. Davis, R. L. Tobias, J. Polym. Sci., **50**, 227 (1961).
705. T. P. Davis, K. F. O'Driscoll, M. C. Piton, M. A. Winnik, Macromolecules, **23**, 2113 (1990).
706. V. P. Davyodova, N. A. Pravikova, T. A. Yakushina, V. I. Yakovleva, Polym. Sci. USSR, **8**, 479 (1966); Vysokomol. Soedin., **8**, 436 (1966).
707. J. V. Dawkins, J. Macromol. Sci. B, **2**, 623 (1968).
708. J. V. Dawkins, R. Denyer, J. W. Maddock, Polymer, **10**, 154 (1969).
709. J. V. Dawkins, Eur. Polym. J., **6**, 831 (1970).
710. J. V. Dawkins, F. W. Peaker, Eur. Polym. J., **6**, 209 (1970).
711. J. V. Dawkins, J. W. Maddock, D. Coupe, J. Polym. Sci. A-2, **8**, 1803 (1970).
712. J. V. Dawkins, J. W. Maddock, Eur. Polym. J., **7**, 1537 (1971).
713. J. V. Dawkins, M. Hemming, Makromol. Chem., **155**, 75 (1972).
714. J. V. Dawkins, M. Hemming, J. Appl. Polym. Sci., **19**, 3107 (1975).
715. J. V. Dawkins, G. Taylor, J. Polym. Sci., Poly. Lett. Ed., **13**, 29 (1975).
716. J. V. Dawkins, M. Hemming, Makromol. Chem., **176**, 1777 (1975).
717. J. V. Dawkins, G. Yeadon, "High Performance Gel Permeation Chromatography" in Development in Polymer Characterization. vol 1, Chapter 23, Applied Science Publishers Ltd., London, 1978.
718. J. V. Dawkins in C. Booth, in: C. Price (Ed.), "Comprehensive Polymer Science", vol. 1, Pergamon, Oxford, (1989), p. 231.
719. J. V. Dawkins, N. P. Gabbott, A. M. C. Montenegro, L. L. Lloyd, F. P. Warner, J. Appl. Polym. Sci., Appl. Polym. Symp., **45**, 103 (1990).
720. F. De Candia, Makromol. Chem., **141**, 177 (1971).
721. F. De Candia, V. Vittoria, U. Bianchi, E. Patrone, Macromolecules, **5**, 493 (1972).
722. A. De Chirico, S. Arrighetti, M. Bruzzone, Polymer, **22**, 529 (1981).
723. A. C. De Kok, A. C. Oomens, J. Liq. Chromatog., **5**, 807 (1982).
724. A. C. De Visser, A. A. Driessen, J. G. C. Wolke, Makromol. Chem., Rapid Commun., **1**, 177 (1980).
725. M. De Wit, G. Verreyt, Y. Bao, H. J. Geise, Macromolecules, **28**, 5144 (1995).
726. P. C. Deb, S. R. Palit, Makromol. Chem., **128**, 123 (1969).
727. P. C. Deb, S. R. Palit, Makromol. Chem., **166**, 227 (1973).
728. F. Defoor, G. Groeninckx, P. Shouterden, B. Van der Heijden, Polymer, **33**, 3878 (1992).
729. F. Defoor, G. Groeninckx, H. Reynaers, P. Schouterden, B. V. Hejiden, J. Appl. Polym. Sci., **47**, 1839 (1993).
730. C. Degoulet, T. Nicolai, D. Durand, J. P. Busnel, Macromolecules, **28**, 6819 (1995).
731. A. W. Degroot, J. Appl. Polym. Sci., Appl. Polym. Symp., **43**, 85 (1989).
732. A. D. Delman, B. B. Slimms, A. E. Ruff, J. Polym. Sci., **45**, 415 (1960).
733. A. D. Delman, J. Kelly, J. Mironov, B. B. Simms, J. Polym. Sci. A-1, **4**, 1277 (1966).
734. G. Delmas, J. Appl. Polym. Sci., **12**, 839 (1968).
735. G. Delmas, P. Tancrede, Eur. Polym. J., **9**, 199 (1973).
736. A. Dems, G. Strobin, Makromol. Chem., **192**, 131 (1991).
737. A. Dems, G. Strobin, Makromol. Chem., **192**, 115 (1991).
738. F. Denes, V. Percec, M. Totolin, J. P. Kennedy, Polym. Bull., **2**, 499 (1980).
739. X. M. Deng, C. D. Xiong, L. M. Cheng, H. H. Huang, R. P. Xu, J. Appl. Polym. Sci., **55**, 1193 (1995).
740. I. Deschères. O. Païssé, J. N. Colonna-Ceccaldi, Makromol. Chem., **188**, 583 (1987).
741. A. Desjardins, A. Eisenberg, Macromolecules, **24**, 5779 (1991).
742. V. Desreux, Bull. Soc. Chim. Belges, **51**, 416 (1948).
743. V. Desreux, Rec. Trav. Chim., **68**, 789 (1949).
744. V. Desreux, M. C. Spiegels, Bull. Soc. Chim. Belges, **59**, 476 (1950).
745. V. Desreux, A. Oth, Chem. Weekbl., **48**, 247 (1952).
746. H. Determann, G. Lueben, Th. Wieland, Makromol. Chem., **73**, 168 (1964).
747. H. Determann, M. Kriewen, Th. Wieland, Makromol. Chem., **114**, 256 (1968).
748. H. Determann, "Gel Chromatography", 2nd ed., Springer, New York, 1969.
749. J. Devaux, J. Sledz, F. Schúe, L. Giral, H. Naarmann, Eur. Poly. J., **25**, 263 (1989).
750. H. Dexheimer, O. Fuchs, Makromol. Chem., **96**, 172 (1966).
751. H. Dexheimer, E. Helmes, H. J. Leugering, IUPAC Toronto, Preprint A, **8**, 1 (1968).
752. A. Di Napoli, B. Chu, C. Cha, Macromolecules, **15**, 1174 (1982).
753. K. Dialer, K. Vogler, Makromol. Chem. **6**, 191 (1951).
754. K. Dialer, K. Vogler, F. Patat, Helv. Chim. Acta, **35**, 869 (1952).
755. R. Díaz Calleja, L. Gargallo, D. Radic, Macromolecules, **28**, 6963 (1995).
756. A. Diaz-Barrios, A. Rengel, J. Polym. Sci., Polym. Chem. Ed., **22**, 519 (1984).
757. R. Dick, H. Sack, H. Benoît, J. Polym. Sci. C, **16**, 4597 (1969).
758. B. Dickens, J. W. Martin, D. Waksman, Polymer, **25**, 706 (1984).
759. H. A. Dieu, J. Polym. Sci., **12**, 417 (1954).

760. V. Dimonie, M. S. El-Aasser, A. Klein, J. W. Vanderhoff, J. Polym. Sci., Polym. Chem. Ed., **22**, 2197 (1984).
761. K. Dimov, E. Terlemezyan, J. Polym. Sci., Polym. Chem. Ed., **10**, 3133 (1972).
762. A. Dobry, J. Chim. Phys., **35**, 387 (1938).
763. N. Doddi, J. Polym. Sci., Polym. Chem. Ed., **12**, 761 (1974).
764. N. Doddi, W. C. Forsman, C. C. Price, J. Polym. Sci., Polym. Phys. Ed., **12**, 1395 (1974).
765. K. Dodgson, D. Sympson, J. A. Semlyen, Polymer, **19**, 1285 (1978).
766. M. A. Doherty, T. E. Hogen-Esch, Makromol. Chem., **187**, 61 (1986).
767. Y. Doi, M. Takada, T. Keii, Makromol. Chem., **180**, 57 (1979).
768. Y. Doi, S. Ueki, T. Keii, Makromol. Chem., **180**, 1359 (1979).
769. Y. Doi, A. Morinaga, T. Keii, Makromol. Chem. Rapid Commun., **1**, 193 (1980).
770. Y. Doi, S. Ueki, T. Keii, Makromol. Chem. Rapid Commun., **3**, 225 (1982).
771. Y. Doi, S. Ueki, S. Tamura, S. Nagahara, T. Keii, Polymer, **23**, 258 (1982).
772. T. Doi, A. Akimoto, A. Matsumoto, T. Otsu, J. Polym. Sci. Part A: Polym. Chem., **34**, 367 (1996).
773. A. Domard, M. Rinaudo, C. Rochas, J. Polym. Sci., Polym. Phys. Ed., **17**, 673 (1979).
774. Ye, S. Domnina, G. G. Skvortsova, Polym. Sci. USSR, **8**, 1394 (1966); Vysokomol. Soedin., **8**, 1268 (1966).
775. A. Dondos, H. Benoît, Makromol. Chem., **129**, 35 (1969).
776. A. Dondos, Makromol. Chem., **135**, 181 (1970).
777. A. Dondos, P. Rempp, H. Benoît, Makromol. Chem., **15**, 1659 (1974).
778. X. Dong, A. Proctor, D. M. Hercules, Macromolecules, **30**, 63 (1997).
779. N. Donkai, N. Murayama, T. Miyamoto, H. Inagaki, Makromol. Chem., **175**, 187 (1974).
780. N. Donkai, T. Miyamoto, H. Inagaki, Polym. J., **7**, 577 (1975).
781. E. F. Donnelly, D. C. Pepper, Makromol. Chem., Rapid Commun., **2**, 439 (1981).
782. P. Doty, H. Wagner, S. Singer, J. Phys. Colloid. Chem., **51**, 322 (1947).
783. A. Douy, B. Gallot, Polymer, **23**, 1039 (1982).
784. S. Drăgan, S. Ioan, M. Dima, Polym. Bull., **7**,473 (1982).
785. E. E. Drott, R. A. Mendelson, J. Polym. Sci. A-2, **8**, 1373 (1970).
786. E. E. Drott, R. A. Mendelson, Polym. Preprints, **12** (1), 277 (1971).
787. D. B. Dupré, H. Wang, Macromolecules, **25**, 7155 (1992).
788. P. L. Dubin, M. Kronstadt, J. Polym. Sci., Polym. Chem. Ed., **20**, 1709 (1982).
789. L. V. Dubrovina, G. I. Timofeyeva, V. V. Korshak, S. S. Pavlova, Polym. Sci. USSR, **6**, 2225 (1964); Vysokomol. Soedin., **6**, 2011 (1964).
790. L. V. Dubrovina, S. A. Pavlova, V. V. Korshak, Polym. Sci. USSR, **8**, 2171 (1966); Vysokomol. Soedin., **8**, 1965 (1966).
791. L. V. Dubrovina, S. A. Pavlova, V. A. Vasnev, S. V. Vinogradova, V. V. Korshak, Polym. Sci. USSR, **12**, 1484 (1970); Vysokomol. Soedin. A, **12**, 1308 (1970).
792. S. A. Dubrovskii, I. V. Kumpanenko, V. M. Gol'dberg, K. S. Kazanskii, Polym. Sci. USSR, **17**, 3141 (1975); Vysokomol. Soedin. A, **17**, 2733 (1975).
793. A. B. Duchkova, N. S. Nametkin, V. S. Khotimskii, L. Ye. Gusel'nikov, S. G. Durgar'yan, Polym. Sci. USSR, **8**, 2002 (1966); Vysokomol. Soedin., **8**, 1814 (1966).
794. A. Duda, Z. Florjanczyk, A. Hofman, S. Slomkowski, S. Penczek, Macromolecules, **23**, 1640 (1990).
795. A. Duda, Macromolecules, **27**, 576 (1994).
796. A. Duda, Macromolecules, **29**, 1399 (1996).
797. J. H. Duerksen, A. E. Hamielec, Polym. Preprints, **8** (2), 1337 (1967).
798. J. H. Duerksen, A. E. Hamielec, J. Polym. Sci. C, **21**, 83 (1968).
799. K. Dušek, Coll. Czech. Chem. Commun., **34**, 3309 (1969).
800. P. Dumas, N. Spassky, P. Sigwalt, J. Polym. Sci., Polym. Chem. Ed., **12**, 1001 (1974).
801. P. Dumas, N. Spassky, P. Sigwalt, J. Polym. Sci., Polym. Chem. Ed., **14**, 1015 (1976).
802. T. Dumelow, S. R. Holding, L. J. Maisey, J. V. Dawkins, Polymer, **27**, 1170 (1986).
803. A. S. Dunn, B. D. Stead, H. W. Melville, Trans. Faraday Soc., **50**, 279 (1954).
804. D. T. Duong, J. P. Bell, J. Polym. Sci., Polym. Phys. Ed., **13**, 765 (1975).
805. M. Duval, J. Francois, D. Sarazin, Polymer, **26**, 397 (1985).
806. I. Duvdevani, J. A. Biesenberger, M. Tan, J. Polym. Sci. B, **9**, 429 (1971).
807. N. Duveau, A. Piguet, J. Polym. Sci., **57**, 357 (1962).
808. P. R. Dvornic, R. W. Lenz, J. Polym. Sci., Polym. Chem. Ed., **20**, 593 (1982).
809. P. R. Dvornic, R. W. Lenz, Macromolecules, **27**, 5833 (1994).
810. J. Dyer, L. H. Phifer, J. Polym. Sci. C, **36**, 103 (1971).
811. M. K. Dygert, G. T. Taylor, F. Cardinaux, H. A. Scheraga, Macromolecules, **9**, 794 (1976).
812. G. C. East, A. M. Girshab, Polymer, **23**, 323 (1982).
813. K. C. Eberly, B. L. Johnson, J. Polym. Sci., **3**, 283 (1948).
814. K. H. Ebert, M. Brosche, K. F. Elgert, Makromol. Chem., **72**, 191 (1964).
815. K. H. Ebert, G. Schenk, G. Rupprecht, M. Brosche, H. W. Weng, D. Heinicke, Makromol. Chem., **96**, 206 (1966).
816. K. H. Ebert, H. J. Ederer, J. Emmert, Polym. Bull., **6**, 471 (1982).
817. S. Ebner, H. Keul, H. Höcker, Macromolecules, **29**, 553 (1996).
818. A. R. Eckert, S. E. Webber, Macromolecules, **29**, 560 (1996).
819. G. D. Edwards, J. Appl. Polym. Sci., **9**, 3845 (1965).
820. H. S. Egboh, M. H. Goerge, J. A. Barrie, D. J. Walsh, J. Polym. Sci., Polym. Chem. Ed., **20**, 2879 (1982).
821. J. Ehrlich, S. S. Stivala, Polymer, **15**, 204 (1974).
822. B. S. Ehrlich, W. V. Smith, Polym. Preprints, **12** (2), 847 (1971).

823. A. Eigel, H. Buenemann, N. Dattagupta, Makromol. Chem., **175**, 1847 (1974).
824. J. L. Ekmanis, R. A. Skinner, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 57 (1991).
825. A. A. El'Saied, S. Ya. Mirlina, V. A. Kargin, Polym. Sci. USSR, **11**, 314 (1969); Vysokomol. Soedin. A, **11**, 282 (1969).
826. M. S. El-Aasser, T. Makgawinata, J. W. Vanderhoff, J. Polym. Sci., Polym. Chem. Ed., **21**, 2363 (1983).
827. M. H. El-Rafie, A. I. Waly, A. Hebeish, J. Polym. Sci., Polym. Chem. Ed., **14**, 2903 (1976).
828. M. H. El-Rafie, S. H. Abdel-Fatth, E. M. Khalil, A. Hebeish, Angew. Makromol. Chem., **87**, 63 (1980).
829. M. Z. El-Sabee, A. H. Ahmed, S. Mawaziny, Eur. Polym. J., **10**, 1149 (1974).
830. K. F. Elgert, R. Wohlschiess, Angew. Makromol. Chem., **57**, 87 (1977).
831. E. J. Elgood, N. S. Heath, B. J. O'Brien, D. H. Solomon, J. Appl. Polym. Sci., **8**, 881 (1964).
832. H.-G. Elias, F. Patat, Makromol. Chem., **25**, 13 (1957).
833. H.-G. Elias, V. Gruber, Makromol. Chem., **78**, 72 (1964).
834. H.-G. Elias, A. Fritz, Makromol. Chem., **114**, 31 (1968).
835. H.-G. Elias, V. S. Kamat, Makromol. Chem., **117**, 61 (1968).
836. H. Elmgren, S. Norrby, Makromol. Chem., **123**, 265 (1969).
837. W. L. Elsdon, J. M. Goldwasser, A. Rudin, J. Polym. Sci., Polym. Phys. Ed., **19**, 483 (1981).
838. C. Emery, W. E. Cohen, Aust. J. Appl. Sci., **2**, 473 (1951).
839. A. H. Emery, H. G. Drickamer, J. Chem. Phys., **3**, 2252 (1953).
840. H. A. Ende, G. V. Schulz, Z. Physk. Chem. N. F. (Frankfurt), **33**, 143 (1962).
841. H. A. Ende, J. M. Peterson, G. V. Schulz, Z. Physk. Chem. N. F. (Frankfurt), **41**, 224 (1964).
842. R. Endo, Chem. High Polym. (Japan), **19**, 39 (1962).
843. R. Endo, K. Iimura, M. Takeda, Bull. Chem. Soc. Japan, **37**, 950 (1964).
844. R. Endo, T. Manago, M. Takeda, Bull. Chem. Soc. Japan, **39**, 733 (1966).
845. R. Endo, T. Hinokuma, M. Takeda, J. Polym. Sci. A-2, **6**, 665 (1968).
846. D. Engel, R. C. Schulz, Eur. Polym. J., **19**, 967 (1983).
847. N. S. Enikolopyan, M. A. Markevitch, L. S. Sakhonenko, S. Z. Rogovina, V. G. Oshmyan, J. Polym. Sci., Polym. Chem. Ed., **20**, 1231 (1982).
848. Ye. G. Erenburg, L. V. Pavlova, Ye. O. Osipchuk, I. M. Dolgopol'skii, A. I. Konshin, R. L. Rabinovich, I. Ya. Poddubnyi, Polym. Sci. USSR, **20**, 431 (1978); Vysokomol. Soedin. A, **20**, 382 (1978).
849. Ye. G. Erenburg, L. V. Pavlova, Y. O. Osipchuk, I. M. Dolgopol'skii, A. I. Konshin, A. M. Greis, I. Y. Poddubnyi, Polym. Sci. USSR, **21**, 2496 (1979).
850. A. F. V. Eriksson, Acta Chem. Scand., **3**, 1 (1949).
851. A. F. V. Eriksson, Acta Chem. Scand., **7**, 377 (1953).
852. A. F. V. Eriksson, Acta Chem. Scand., **7**, 623 (1953).
853. A. F. V. Eriksson, Acta Chem. Scand., **10**, 360, 378 (1953).
854. B. Erman, D. C. Marvin, P. A. Irvine, P. J. Flory, Macromolecules, **15**, 664 (1982).
855. V. Ye. Eskin, A. Ye. Nesterov, Polym. Sci. USSR, **8**, 152 (1966); Vysokomol. Soedin., **8**, 141 (1966).
856. V. E. Eskin. I. N. Serdjuk, J. Polym. Sci. C, **23**, 309 (1968).
857. V. Ye. Eskin, A. I. Kipper, Polym. Sci. USSR., **16**, 2157 (1974); Vysokomol. Soedin. A, **16**, 1862 (1974).
858. V. Ye. Eskin, U. Zhurayev, T. N. Nekrasova, Polym. Sci. USSR, **18**, 2890 (1976); Vysokomol. Soedin. A, **18**, 2529 (1976).
859. V. Ye. Eskin, S. Ya. Magarik, U. B. Zhurayev, G. D. Rudkovskaya, Polym. Sci. USSR, **20**, 2494 (1978); Vysokomol. Soedin. A, **20**, 2219 (1978).
860. G. A. Estrina, A. I. Kuzayev, S. P. Davtyan, B. A. Rozenberg, Polym. Sci. USSR, **22**, 1421 (1980); Vysokomol. Soedin. A, **22**, 1294 (1980).
861. J. M. Evans, M. B. Huglin, Makromol. Chem. **127**, 141 (1969).
862. J. M. Evans, M. B. Huglin, R. F. T. Stepto, Makromol. Chem., **146**, 91 (1971).
863. W. J. Evans, H. Katsumata, Macromolecules, **27**, 4011 (1994).
864. J. Exner, M. Bohdanecký, Coll. Czech. Chem. Commun., **31**, 3985 (1966).
865. G. F. Fanta, R. C. Burr, C. R. Russel, C. E. Rist, J. Appl. Polym. Sci., **10**, 929 (1966).
866. M. Farina, G. Bressan, Makromol. Chem., **61**, 79 (1963).
867. Z. M. Farmazyan, S.-S. Pavlova, L. V. Dubrovina, Polym. Sci. USSR, **22**, 1735 (1980); Vysokomol. Soedin. A, **22**, 1582 (1980).
868. D. Farrar, M. Hawe, Patent Ep 85.308260 A2 860625.
869. G. R. Farvardin, P. Howard, Polymer, **20**, 667 (1979).
870. S. A. Faterpeker, S. P. Potnis, Angew. Makromol. Chem., **90**, 69 (1980).
871. J. G. Fatou, L. Mandelkern, J. Phys. Chem., **69**, 417 (1965).
872. J. M. G. Fatou, Eur. Polym. J., **7**, 1057 (1971).
873. A. H. Fawcett, K. J. Ivin, Polymer, **13**, 439 (1972).
874. S. G. Federov, G. S. Gol'din, G. S. Nikitina, A. V. Kisin, V. M. Nosova, Polym. Sci. USSR, **22**, 2270 (1980); Vysokomol. Soedin. A, **22**, 2070 (1980).
875. O. Ya. Fedotova et al. Vysokomol. Soedin. **5**, 900 (1963).
876. O. Ya. Fedotova, M. I. Shtil'man, Polym. Sci. USSR, **8**, 1332 (1966); Vysokomol. Soedin., **8**, 1209 (1966).
877. R. Feinauer, W. Funke, K. Hamann, Makromol. Chem., **82**, 123 (1965).
878. D. Feldman, C. Uglea, N. Simionescu, J. Polym. Sci. A-1, **7**, 439 (1969).
879. V. F. Felicetta, A. Ahola, J. L. Mc Carthy, J. Am. Chem. Soc., **78**, 1899 (1956).
880. R. E. Felter, E. S. Moyer, L. N. Ray Jr., J. Polym. Sci., B, **7**, 529 (1969).
881. X. D. Feng, C. X. Song, W. Y. Chen, J. Polym. Sci., Polym. Phys. Ed., **21**, 593 (1983).
882. S. Fernandez Bermudez, J. G. Fatou, J. Royo, Anales Quím. (Madrid) 66, 789 (1970).
883. M. J. Fernandez-Berridi, F. J. Ansorena, J. J. Iruin, G. M. Guzmán, Macromolecules, **13**, 190 (1980).

884. J. D. Ferry, D. C. Udy, F. C. Wu, G. E. Heckler, D. B. Fordyce, J. Colloid. Sci., **6**, 429 (1951).
885. L. J. Fetters, M. Morton, Macromolecules, **2**, 453 (1969).
886. L. J. Fetters, M. Morton, Macromolecules, **7**, 552 (1974).
887. L. J. Fetters, D. Mc Intyre, Polymer, **19**, 463 (1978).
888. L. J. Fetters, W. W. Graessley, N. Hadjichristidis, A. D. Kiss, D. S. Pearson. L. B. Younghouse, Macromolecules, **21**, 1644 (1988).
889. R. V. Figini, H. Hostalka, K. Hurm, G. Loehr, G. V. Schulz, Z. Physik. Chem. N. F. (Frankfurt), **45**, 269 (1965).
890. N. N. Filatova, D. Ya. Rossina, V. V. Yevreinov, S. G. Entelis, Polym. Sci. USSR., **20**, 2663 (1978); Vysokomol. Soedin. A, **20**, 2367 (1978).
891. S. R. Finn, J. W. James, J. Appl. Chem., **6**, 466 (1956).
892. A. P. Firsov, I. V. Yeremina, N. M. Chirkov, Polym. Sci. USSR, **6**, 417 (1964); Vysokomol. Soedin., **6**, 377 (1964).
893. L. W. Fisher, G. H. Pearson. P. D. Yacobucci, J. Polym. Sci., Polym. Phys. Ed., **18**, 1455 (1980).
894. L. W. Fisher, A. R. Sochor, J. S. Tan, Macromolecules, **10**, 949 (1977).
895. E. B. Fitzgerald, R. M. Fuoss, Ind. Eng. Chem., **42**, 1603 (1950).
896. J. R. Fletcher, L. Polgar, D. H. Solomom, J. Appl. Polym. Sci., **8**, 659 (1964).
897. J. R. Fletcher, L. Polgar, D. H. Solomom, J. Appl. Polym. Sci., **8**, 663 (1964).
898. J. R. Fletcher, H. E. Persinger, J. Polym. Sci. A-1, **6**, 1025 (1968).
899. A. Flondor, S. Maxim, S. Ioan, Eur. Polym. J., **22**, 683 (1986).
900. Š. Florián, D. Lath, V. Durdôvič, Angew. Mokromol. Chem., **10**, 189 (1970).
901. Š. Florián, D. Lath, Z. Máňásek, Angew. Makromol. Chem., **13**, 43 (1970).
902. Š. Florián, E. Lathová, P. Mach, Eur. Polym. J., **30**, 449 (1994).
903. P. J. Flory, J. Am. Chem. Soc., **65**, 372 (1943).
904. P. J. Flory, J. Am. Chem. Soc., **65**, 375 (1943).
905. D. L. Flowers, W. A. Hewett, R. D. Mullineaux, J. Polym. Sci. A-1, **2**, 2305 (1964).
906. C. M. Fontan, J. Phys. Chem., **63**, 1167 (1959).
907. J. Fontán, J. Millán Rodriguez, Anales Quím. (Madrid), **60**, 411 (1964).
908. J. L. Forget, C. Booth, P. H. Canham, M. Duggleby, T. A. King, C. Price, J. Polym. Sci., Polym. Phys. Ed., **17**, 1403 (1979).
909. R. J. Fort, R. J. Hutchinson, W. R. Moore, M. Murphy, Polymer, **4**, 35 (1963).
910. R. J. Fort, T. M. Polyzoidis, Eur. Polym. J., **12**, 685 (1976).
911. D. A. Foucher, B. Z. Tang, I. Manners, J. Am. Chem. Soc., **114**, 6246 (1992).
912. D. Foucher, R. Ziembinki, R. Petersen, J. Pudelski, M. Edwars, Y. Ni, J. Massey, C. R. Jaeger, G. J. Vancso, I. Manners, Macromolecules, **27**, 3992 (1994).
913. T. G. Fox, P. J. Flory, J. Am. Chem. Soc., **70**, 2384 (1984).
914. T. G. Fox, P. J. Flory, J. Phys. Chem., **53**, 197 (1949).
915. T. G. Fox, J. B. Kinsinger, H. F. Mason, E. M. Schuele, Polymer, **3**, 71 (1962).
916. T. G. Fox, Polymer, **3**, 111 (1962).
917. P. S. Francis, R. C. Cooke Jr., J. H. Elliot, J. Polym. Sci., **31**, 453 (1958).
918. F. Francuskiewicz, "Polymer Fractionation", Springer, Berlin, 1994.
919. H. P. Frank, J. W. Breitenbach, J. Polym. Sci., **6**, 609 (1951).
920. J. M. J. Fréchet, S. Matuszczak, B. Reck, H. D. H. Stöver, C. G. Willson, Macromolecules, **24**, 1746 (1991).
921. M. Freeman, P. P. Manning, J. Polym. Sci. A-1, **2**, 2017 (1964).
922. D. M. French, A. W. Casey, C. I. Collins, P. Kirchner, Polym. Preprints, **7** (1), 447 (1966).
923. S. Ya. Frenkel, S. I. Klenin, Polym. Sci. USSR, **6**, 1571 (1964); Vysokomol. Soedin., **6**, 1420 (1964).
924. L. Freund, M. Daune, J. Polym. Sci., **29**, 161 (1958).
925. H. Frey, Möller, K. Matyjaszewski, Macromolecules, **27**, 1814 (1994).
926. D. Freyss, P. Rempp, H. Benoît, J. Polym. Sci., B, **2**, 217 (1964).
927. H. L. Frisch, Q. Xu, Macromolecules, **25**, 5145 (1992).
928. E. V. Frisman, M. A. Sibileva, M. A. Chebishyan, Polym. Sci. USSR, **9**, 1191 (1967); Vysokomol. Soedin., **9** 1071 (1967).
929. E. V. Frisman, M. A. Sibileva, N. Tkhi Kim Ngan, T. N. Nekrasova, Polym. Sci. USSR, **10**, 2125 (1968); Vysokomol. Soedin. A, **10**, 1834 (1968).
930. L. Fritz, J. Springer, Makromol. Chem. 194, 2047 (1993).
931. H. Fritzemeier, J. J. Hermans, Bull. Soc. Chem. Belges, **57**, 136 (1984).
932. P. E. Froehlnig, D. M. Koenhen, A. Bantjes, C. A. Smolders, Polymer, **17**, 835 (1976).
933. O. Fuchs, Makromol. Chem. **5**, 245 (1951).
934. O. Fuchs, Makromol. Chem. **7**, 259 (1952).
935. O. Fuchs, Z. Elektrochem. **60**, 229 (1956).
936. O. Fuchs, H. J. Leugering, in: R. Nitsche, K. A. Wolf (Eds.), "Kunststoffe" vol. 1, Springer, Berlin, p. 118 1962.
937. O. Fuchs, Makromol. Chem. **58**, 65 (1962).
938. H. Fujii, J. Furukawa, T. Saegusa, A. Kawasaki, Makromol. Chem., **40**, 226 (1960).
939. H. Fujii, T. Fujii, T. Saegusa, J. Furukawa, Makromol. Chem., **63**, 147 (1963).
940. M. Fujino, T. Hisaki, M. Fujiki, N. Matsumoto, Macromolecules, **25**, 1079 (1992).
941. M. Fujimo, T. Hisaki, N. Matsumoto, J. Polym. Sci. Part A, Polym. Chem., **33**, 2279 (1995).
942. T. Fujimoto, N. Ozaki, M. Nagasawa, J. Polym. Sci. A-1, **3**, 2259 (1965).
943. T. Fujimoto, H. Narukawa, M. Nagasawa, Macromolecules, **3**, 57 (1970).
944. T. Fujimoto, S. Tani, K. Takano, M. Ogawa, M. Nagasawa, Macromolecules, **11**, 673 (1978).
945. Y. Fujisaki, H. Kobayashi, Chem. High Polym. (Japan), **18**, 305 (1961).
946. Y. Fujisaki, H. Kobayashi, Chem. High Polym. (Japan), **19**, 73 (1962).

947. Y. Fujisaki, H. Kobayashi, Chem. High Polym. (Japan), **19**, 49 (1962).
948. Y. Fujisaki, High Polym. (Japan), **19**, 64 (1962).
949. S. Fujisaki, H.-G. Elias, Makromol. Chem., **155**, 127 (1972).
950. S. Fujishige, Polym. J., **19**, 297 (1987).
951. M. Fukuda, M. Fukutomi, Y. Kato, T. Hashimoto, J. Polym. Sci., Polym. Phys. Ed., **12**, 871 (1974).
952. T. Fukuda, Y. Ma, H. Inagaki, Polym. J., **14**, 705 (1982).
953. T. Fukutomi, A. Yokota, K. Ishizu, J. Polym. Sci., Polym. Chem. Ed., **22**, 2983 (1984).
954. A. P. Full, J. E. Puig, L. U. Gron, E. W. Kaler, J. R. Minter, T. H. Mourey, J. Texter, Macromolecules, **25**, 5157 (1992).
955. B. L. Funt, D. R. Richardson, J. Polym. Sci. A-1, **8**, 1055 (1970).
956. B. L. Funt, V. Hornof, J. Appl. Polym. Sci., **15**, 2439 (1971).
957. B. L. Funt, L. C. Hsu, J. Polym. Sci., Polym. Chem. Ed., **18**, 1957 (1980).
958. J. Furukawa, T. Saegusa, T. Tsuruta, S. Anzai, T. Narumiya, A. Kawasaki, Makromol. Chem., **41**, 17 (1960).
959. J. Furukawa, T. Saegusa, N. Mise, A. Kawasaki, Makromol. Chem., **39**, 243 (1960).
960. J. Furukawa, S. Akutsu, T. Saegusa, Makromol. Chem., **81**, 100 (1965).
961. J. Furukawa, S. Akustsu, T. Saegusa, Makromol. Chem., **94**, 68 (1966).
962. J. Furukawa, T. Saegusa, S. Yasui, S. Akutsu, Makromol. Chem., **94**, 74 (1966).
963. E. A. Fushman, L. F. Borisova, Ye. I. Vizen, N. M. Chirkov, Polym. Sci. USSR, **16**, 2282 (1974); Vysokomol. Soedin. A, **16**, 1972 (1974).
964. A. Gabert, H. Seide, G. Langhammer, J. Polym. Sci. C, **16**, 3547 (1968).
965. A. C. Gadkari, M. Zsuga, J. P. Kennedy, Polym. Bull., **18**, 317 (1987).
966. D. Gaeckle, D. Patterson, Macromolecules, **5**, 136 (1972).
967. G. L. Gaines Jr., D. G. Le Grand, J. Polym. Sci. B, **6**, 625 (1968).
968. V. M. Gal'perin, N. A. Kuznetzov, A. L. Smolyanskii, Polym. Sci. USSR, **14**, 3122 (1972); Vysokomol. Soedin. A, **14**, 2684 (1972).
969. J. C. Galin, Makromol. Chem., **124**, 118 (1969).
970. M. Galin, J. C. Galin, Makromol. Chem. 160, 321 (1972).
971. A. Gallardo, J. San Román, Macromolecules, **25**, 5836 (1992).
972. A. Gallo, Chim. Ind. (Milan), **35**, 487 (1953).
973. Y. Gallot, P. Rempp, J. Parrod, J. Polym. Sci. B, **1**, 329 (1963).
974. M. E. Galvin, S. Heffner, K. I. Winey, Macromolecules, **27**, 3520 (1994).
975. L. W. Gamble, W. T. Wipke, T. Lane, Poly. Preprints, **4** (2), 162 (1963).
976. L. W. Gamble, W. T. Wipke, T. Lane, J. Appl. Polym. Sci., **9**, 1503 (1965).
977. S. I. Ganicheva, V. M. Belyayev, V. V. Krivobokov, O. V. Kallistov, S. Ya. Frenkel, Polym. Sci. USSR, **24**, 1734 (1982); Vysokomol. Soedin. A, **24**, 1523 (1982).
978. E. S. Gankina, M. D. Val'chikina, B. G. Belen'kii, Polym. Sci. USSR, **18**, 1345 (1976); Vysokomol, Soedin. A, **18**, 1170 (1976).
979. E. S. Gankina, S. I. Ganicheva, Y. M. Belyayev, Y. Y. Kever, B. G. Belen'kii, Polym. Sci. USSR, **24**, 909 (1982); Vysokomol. Soedin. A, **24**, 814 (1982).
980. V. Ganser, J. Engel, D. Winklmair, G. Krause, Biopolymers, **9**, 329 (1970).
981. C. Garbuglio, G. Pezzin, C. Badoni, Chim. Ind. (Milan), **46**, 797 (1964).
982. C. Garbuglio, L. Crescentini, A. Mula, G. B. Gechele, Makromol. Chem., **97**, 97 (1966).
983. C. Garbuglio, A. Mula, L. Chinellato, J. Polym. Sci. C, **16**, 1529 (1967).
984. M. García-Martí, K. H. Reichert, Makromol. Chem., **144**, 17 (1971).
985. L. H. García-Rubio, A. E. Hamielec, J. Appl. Polym. Sci., **23**, 1397 (1979).
986. L. H. García-Rubio, A. E. Hamielec, J. F. Mc Gregor, ACS Symposium Series in: C. D. Craver (Ed.), "Polymer Characterization: Spectroscopic, Chromatographic and Physical Instrumental Methods", ACS Advances in Chemistry Series, **203**, 311 (1983).
987. S. K. Garg, S. S. Stivala, Polymer, **19**, 243 (1978).
988. S. K. Garg, S. S. Stivala, Polymer, **23**, 514 (1982).
989. L. Gargallo, H. Rios, D. Radić, Polym. Bull., **11**, 525 (1984).
990. L. Gargallo, Makromol. Chem., **177**, 233 (1976).
991. L. Gargallo, D. Radić, I. Katime, Eur. Polym. J., **16**, 383 (1980).
992. L. Gargallo, C. L. Güemes, D. Radić, Eur. Polym. J., **20**, 483 (1984).
993. L. Gargallo, M. I. Muñoz, D. Radić, Polymer, **27**, 1416 (1986).
994. L. Gargallo, E. Soto, F. R. Diaz, L. H. Tagle, D. Radić, Eur. Polym. J., **23**, 571 (1987).
995. L. Gargallo, D. Radić, L. H. Tagle, Polym. Bull., **17**, 15 (1987).
996. L. Gargallo, E. Soto, L. H. Tagle, D. Radić, Eur. Polym. J., **24**, 1119 (1988).
997. L. Gargallo, M. Yazdani-Pedram, D. Radić, A. Horta, J. Bravo, Eur. Polym. J., **29**, 609 (1993).
998. L. Gargallo, F. Martínez-Piña, A. Leiva, D. Radić, Eur. Polym. J., **32**, 1303 (1996).
999. T. I. Garmonova, M. G. Vitovskaya, P. N. Laurenko, V. N. Tsvetkov, Ye. V. Korovina, Polym. Sci. USSR, **13**, 996 (1971); Vysokomol. Soedin. A, **13**, 884 (1971).
1000. T. I. Garmonova, M. G. Vitovskaya, S. V. Bushin, T. V. Sheremeteva, A. P. Zaitseva, Polym. Sci. USSR, **16**, 306 (1974); Vysokomol. Soedin. A, **16**, 265 (1974).
1001. V. Garten, W. Becker, Makromol. Chem., **3**, 78 (1949).
1002. J. Garza, C. Marco, J. G. Fatou, A. Bello, Polymer, **22**, 477 (1981).
1003. R. G. Gaughan, H. W. Hill, J. E. Inda, J. Polym. Sci. :Part A, Polym. Chem., **24**, 419 (1986).
1004. G. Gavoret, J. Duclaax, J. Chim. Phys., **42**, 41 (1945).
1005. G. Gavoret, M. Magat, J. Chim. Phys., **44**, 90 (1947).

1006. N. G. Gaylord, A. Takahashi, S. Kikuchi, R. A. Guzzi, J. Polym. Sci., Polym. Lett. Ed., **10**, 95 (1972).
1007. N. G. Gaylord, S. Maiti, S. S. Dixit, J. Polym. Sci. Polym. Lett. Ed., **10**, 855 (1972).
1008. N. Geacintov, V. Stannett, E. W. Abrahamson, J. J. Hermans, J. Appl. Polym. Sci., **3**, 54 (1960).
1009. B. Gebus, R. Knoesel, Mackromol. Chem., **175**, 1439 (1974).
1010. G. B. Gechele, A. Mattiussi, Eur. Polym. J., **1**, 47 (1965).
1011. I. Geczy, G. Revai, Kolor Ert., **12**, 186 (1970).
1012. I. Geczy, S. H. Abdel-Fattah, Angew. Makromol. Chem., **43**, 75 (1975).
1013. R. Geddes, C. T. Greenwood, A. W. Mac Gregor, Makromol. Chem., **79**, 189 (1964).
1014. H. Geeriseen, P. Schuetzeichel, B. H. Wolf, Makromol. Chem. **191**, 659 (1970).
1015. H. Geeriseen, J. Roos, B. A. Wolf, Makromol. Chem., **186**, 735 (1985).
1016. H. Geerissen, J. Ross, P. Schützeichel, B. A. Wolf, J. Appl. Polym. Sci., **34**, 271 (1987).
1017. H. Geerissen, P. Schützeichel, B. A. Wolf, J. Appl. Polym. Sci., **34**, 287 (1987).
1018. H. Geerissen, P. Schützeichel, B. A. Wolf, Makromol. Chem., **191**, 659 (1990).
1019. M. Gehatia, D. R. Wiff, IUPAC Boston, Vol. **11**, 1045 (1971).
1020. M. Gehatia, D. R. Wiff, Eur. Polym. J., **8**, 585 (1972).
1021. M. D. Gehlsen, K. Almdal, F. S. Bates, Macromolecules, **25**, 939 (1992).
1022. M. D. Gehlsen, F. S. Bates, Macromolecules, **27**, 3611 (1994).
1023. K. Gehrke, G. Reinisch, IUPAC Praga, Preprint, 444 (1965).
1024. K. Gehrke, G. Reinisch, J. Polym. Sci. A-1, **7**, 1571 (1969).
1025. G. Geiseler, H. Herold, F. Runge, Erdoel Kohle, **7**, 357 (1954).
1026. G. Geiseler, H. P. Baumann, Z. Elektrochem., **62**, 209 (1958).
1027. D. Geiser, H. Hoecker, Polym. Bull. **2**, 591 (1980).
1028. K. Gekko, Makromol. Chem., **148**, 229 (1971).
1029. A. N. Genkin, T. P. Nasonova, I. Y. Poddubnyi, Vysokomol. Soedin., **4**, 1088 (1962).
1030. J. Gerber, H.-G. Elias, Makromol. Chem., **112**, 142 (1968).
1031. K. Gerberding, G. Heinz, W. Helkmann, Makromol. Chem. Rapid Commun., **1**, 221 (1980).
1032. H. Geerissen, J. Roos, B. A. Wolf, Makromol. Chem., **186**, 735–801 (1987).
1033. R. German, R. Hank, G. Vaughan, Kautschuk Gummi, Kunststoffe, **19**, 67 (1966).
1034. H. Gerrens, H. Ohlinger, R. Fricker, Makromol. Chem., **87**, 209 (1965).
1035. C. Gerth, H. H. Meyer, Angew. Makromol. Chem., **74**, 81 (1978).
1036. J. A. Gervasi, A. B. Gosnell, V. Stannett, Polym. Preprints, **8** (1), 785 (1967).
1037. A. N. Geukin et al., Vysokomol. Soedin., **4**, 1088 (1962).
1038. A. V. Gevorkyan, Ye. S. Yegiyan, N. G. Oganeyan, R. V. Bagdasaryan, Polym. Sci. USSR, **12**, 1218 (1970); Vysokomol. Soedin. A, **12**, 1078 (1970).
1039. A. V. Gevorkyan, L. Kh. Simonyan, K. A. Torosyan, A. Sh. Safarov, Polym. Sci. USSR, **14**, 448 (1974); Vysokomol. Soedin. A, **14**, 385 (1974).
1040. K. K. Ghosh, P. K. Choudhury, Makromol. Chem., **102**, 217 (1967).
1041. P. Ghosh, M. Banerjee, T. K. Ghosh, J. Macromol. Sci. A: Chem. **18**, 607 (1982).
1042. H.-B. Gia, R. Jérôme, P. Teyssié, J. Polym. Sci., Polym. Chem. Ed., **18**, 3488 (1980).
1043. E. Giannetti, P. Mazzocchi, E. Albizzati, T. Fiorani, F. Milani, Makromol. Chem., **185**, 2133 (1984).
1044. G. Gianotti, U. Bonicelli, D. Borghi, Makromol. Chem., **166**, 235 (1973).
1045. G. Gianotti, A. Cicuta, D. Romanini, Polymer, **21**, 1087 (1980).
1046. J. C. Giddings, M. N. Myers, G.-C. Lin, M. Martin, J. Chromatog., **142**, 23 (1977).
1047. J. C. Giddings, S. R. Fisher, M. N. Myers, Am. Laboratory, **10**, 15 (1978).
1048. J. C. Giddings, M. Martin, M. N. Myers, J. Chromatog., **158**, 419 (1978).
1049. J. C. Giddings, G.-C. Lin, M. N. Myers, J. Liq. Chromatog., **1**, 1 (1978).
1050. J. C. Giddings, M. N. Myers, K. D. Caldwell, S. R. Fisher, Methods of Biochem. Anal., **26**, 79 (1980).
1051. J. C. Giddings Pure. Appl. Chem., **51**, 1459 (1979).
1052. J. C. Giddings, Anal. Chem., **53**, 1170A (1981).
1053. J. C. Giddings, M. Martin, M. N. Myers, J. Polym. Sci., Polym. Sci. Phys. Ed., **19**, 815 (1981).
1054. J. C. Giddings. V. Kumar, P. S. Williams, M. N. Myers, in "Polymer Characterization" Adv. Chem. Series, **227**, 3 (1990).
1055. J. C. Giddings. Y. Xu, M. N. Myers, Anal. Chem., **66**, 3047 (1994).
1056. H. Giesekus, unpublished report to Farbenfabriken Bayer 1964, quoted by H. Giesekus, in: M. J. R. Cantow, (Ed.), "Polymer Fractionation", Academic Press, New York, 1967, p. 191.
1057. H. Giesekus, in: M. J. R. Cantow, (Ed.), "Polymer Fractionation", Academic Press, New York, 1967, p. 191.
1058. H. Giesekus, unpublished report to Farbenfabriken Bayer 1954, quoted by H. Giesekus, in: M. J. R. Cantow, (Ed.), "Polymer Fractionation", Academic Press, New York, 1967, p. 191.
1059. H. Giesekus, unpublished report to Farbenfabriken Bayer 1955, quoted by H. Giesekus, in: M. J. R. Cantow (Ed.), "Polymer Fractionation" Academic Press, New York, 1967, p. 191.
1060. L. Gilet, M.-F. Grenier-Loustalot, J. Bounoure, Polymer, **33**, 4605 (1992).
1061. E. Gipstein, A. C. Ouano, W. A. Hewett, Macromolecules, **5**, 249 (1972).
1062. M. Girolamo, J. R. Urwin, Eur. Polym. J., **8**, 1159 (1972).
1063. M. Giurgea. C. Ghita, I. Baltog, A. Lupu, J. Polym. Sci. A-2, **4**, 529 (1966).

1064. P. Giusti, Polym. J., **12**, 555 (1980).
1065. J. L. Glajch, D. C. Warren, M. A. Kaiser, L. B. Rogers, Anal. Chem., **50**, 1962 (1978).
1066. W. Glenz, W. Holtrup, F. W. Kuepper, H. H. Meyer, Angew Makromol. Chem., **37**, 97 (1974).
1067. M. Gleria, A. Bolognesi, W. Porzio, M. Castellani, S. Destri, G. Audisio, Macromolecules, **20**, 469 (1987).
1068. G. Glöckner, Plaste Kautschuk, **10**, 154 (1963).
1069. G. Glöckner, Z. Physik. Chem. (Leipzig), **229**, 98 (1965).
1070. G. Glöckner, J. Chromatog., **191**, 279 (1980).
1071. G. Glöckner, "Polymer charakterisienrong durch Flüssigkeitschromatographie" Hüthing Heidelberg (1982).
1072. G. Glöckner, J. H. M. Van den Berg, N. L. J. Mejerink, T. G. Scholte, R. Koningsveld, Macromolecules, **17**, 962 (1984).
1073. G. Glöckner, V. Albrecht, F. Francuskiewicz, D. Ilchmann, Angew. Makromol. Angew. Makromol. Chem., **130**, 41 (1985).
1074. G. Glöckner, J. H. M. van der Berg, J. Chromatog., **352**, 511 (1986).
1075. G. Glöckner, J. H. M. van der Berg, J. Chromatog., **384**, 135 (1987).
1076. G. Glöckner, in: Elsevier (Ed.), "Polymer Characterization by liquid Chromatography", Amsterdam, 1987.
1077. G. Glöckner, M. Stickler, W. Wunderlich, J. Appl. Polym. Sci., **37**, 3147 (1989).
1078. G. Glöckner, J. Appl. Polym. Sci., Appl. Polym. Symp., **43**, 39 (1989).
1079. G. Glöckner, J. Stejskal, P. Kratochvil, Makromol. Chem., **190**, 427 (1989).
1080. G. Glöckner, "Gradient HPLC of Copolymer and Chromatographic Cross-fractionation", Springer Berlin, 1991.
1081. Y. Gnanov, P. Rempp, Makromol. Chem., **189**, 1997 (1988).
1082. R. A. Godfrey, G. W. Miller, J. Polym. Sci. A-1, **7**, 2387 (1969).
1083. R. A. Godfrey, G. W. Miller, Polym. Preprints **10** (1), 218 (1969).
1084. K. D. Goebel, D. A. Brant, Macromolecules, **3**, 634 (1970).
1085. K. D. Goebel, W. G. Miller, Macromolecules, **3**, 64 (1970).
1086. R. Goeden, D. D. Davis, M. Y. Hellman, A. J. Lovinger, F. H. Winslow, Macromolecules, **21**, 1212 (1988).
1087. D. J. Goedhart, A. Opschoor, IUPAC Leiden, vol. 1, 163 (1970).
1088. D. J. Goedhart, A. Opschoor, J. Polym. Sci. A-2, **8**, 1227 (1970).
1089. E. J. Goethals, A. Munir, R. Devaux, L. Vandenberghe, Makromol. Chem., Rapid Commun., **3**, 515 (1982).
1090. E. J. Goethals, H. Nuytten, R. P. Simonds, Polym. Bull., **7**, 89 (1982).
1091. A. L. Goff, I. P. Yakovlev, V. M. Zhulin, M. G. Gonikberg, Polym. Sci. USSR, **11**, 1487 (1969); Vysokomol. Soedin. A, **11**, 1309 (1969).
1092. S. H. Goh, S. Y. Lee, S. M. Low, Macromolecules, **27**, 6736 (1994).
1093. A. I. Goldberg, W. P. Hohenstein, H. F. Mark, J. Polym. Sci., **2**, 503 (1947).
1094. L. Goldfarb, N. D. Hann, R. L. Dieck, D. C. Messersmith, J. Polym. Sci., Polym. Chem. Ed., **16**, 1505 (1978).
1095. F. J. Golemba, J. E. Guillet, Macromolecules, **5**, 212 (1972).
1096. M. A. Golub, J. Polym. Sci., **11**, 281 (1953).
1097. V. M. Golubev, S. Ya. Frenkel, Polym. Sci. USSR, **10**, 869 (1968); Vysokomol. Soedin. A, **10**, 750 (1968).
1098. M. T. Gomes Freire, L. Danicher, M. Lambla, Makromol. Chem., **189**, 67 (1988).
1099. M. A. Gómez, J. G. Fatou, A. Bello, Eur. Polym. J., **22**, 661 (1986).
1100. I. Goñi, M. D. Gurruchaga, M. Valero, G. M. Guzmán, J. Polym. Sci., Polym, Chem. Ed., **22**, 1327 (1984).
1101. C. Gonzalez, F. Zamora, G. M. Guzmán, L. M. León, Makromol. Chem., **185**, 339 (1984).
1102. C. C. González, E. Riande, A. Bello, J. M. Pereña, Macromolecules, **21**, 3230 (1988).
1103. M. C. González, F. Zamora, Eur. Polym. J., **26**, 431 (1990).
1104. G. Gooberman, J. Polym. Sci., **40**, 469 (1959).
1105. S. R. Gooda, M. B. Huglin, Macromolecules, **25**, 4215 (1992).
1106. S. R. Gooda, M. B. Huglin, Polymer, **34**, 1913 (1993).
1107. A. E. Goodban, H. S. Owens, J. Polym. Sci., **23**, 825 (1957).
1108. R. Gooden, M. Y. Hellman. R. S. Hutton, F. H. Winslow, Macromolecules, **17**, 2830 (1984).
1109. M. Goodman, L. Scarso, Polym. Preprints, **2**, 116 (1961).
1110. A. Gorcienko, W. Griehl, H. Siebert, Faserforsch. Textiltech., **6**, 105 (1955).
1111. A. Gorcienko, Faserforsch. Textiltech., **4**, 499 (1953).
1112. D. A. I. Goring, G. Sitaramaiah, Polymer, **4**, 7 (1963).
1113. A. B. Gosnell, J. A. Gervasi, A. Schindler, J. Polym. Sci. A-1, **4**, 1401 (1966).
1114. N. Gospodinova, L. Terlemerzyan, M. Milhailov, V. Men Han, K. Ben Du, Eur. Polym. J., **28**, 961 (1992).
1115. N. Gralen, G. Lagermalm, J. Phys. Chem., **56**, 514 (1952).
1116. K. A. Granath, P. Flodin, Makromol. Chem., **48**, 160 (1961).
1117. N. Grassie, I. G. Meldrum, Eur. Polym. J., **6**, 499 (1970).
1118. N. Grassie, I. G. Meldrum, Eur. Polym. J., **6**, 513 (1970).
1119. J. H. S. Green, H. T. Hookway, M. F. Vaughan, Chem. Ind., 862 (1958).
1120. J. H. S. Green, M. F. Vaughan, Chem. Ind., 29 (1958).
1121. C. T. Greenwood, Adv. Carbohydr. Chem., **11**, 335 (1956).
1122. D. E. Gregonis, G. A. Russell, J. D. Andrade, A. C. De Visser, Polymer, **19**, 1279 (1978).
1123. G. S. Greschner, Makromol. Chem., **180**, 2551 (1979).
1124. G. S. Greschner, Makromol. Chem., **183**, 2823 (1982).
1125. W. Griehl, H. Lueckert, J. Polym. Sci., **30**, 399 (1958).
1126. B. M. Grieveson, Makromol. Chem., **84**, 93 (1965).
1127. C. H. Griffiths, A. Van Laeken, J. Polym. Sci., Polym. Phys. Ed., **14**, 1433 (1976).
1128. V. A. Grigorovskaya, V. Ye. Basin, L. S. Lyubchenko, D. K. Khakimova, Ye. S. Cherepanova, A. A. Berlin, Polym. Sci. USSR, **18**, 2960 (1976); Vysokomol, Soedin. A, **18**, 2590 (1976).
1129. H. Grohn, H. Friedrich, J. Polym. Sci. C, **16**, 3737 (1968).
1130. A. Gronawall, H. Hint, B. Ingelman, G. Wallenius, O. Wilander, Scand. J. Chem. Lab. Invest., **4**, 363 (1952).
1131. P. Grosius, Y. Gallot, A. Skoulios, Makromol. Chem., **127**, 94 (1969).

1132. P. Grosius, Y. Gallot, A. Skoulios, Eur. Polym. J., **6**, 355 (1970).
1133. P. Grosius, Y. Gallot, A. Skoulios, Makromol. Chem, **132**, 35 (1970).
1134. P. Grosius, Y. Gallot, A. Skoulios, Makromol. Chem., **136**, 191 (1970).
1135. R. I. Gruz, T. Yu. Verkhoglyadova, Ye. F. Panarin, S. N. Ushakov, Polym. Sci. USSR, **13**, 736 (1971); Vysokomol. Soedin. A, **13**, 647 (1971).
1136. M. Guaita, O. Chiantore, Makromol. Chem., **176**, 185 (1975).
1137. J. M. Guenet, Z. Gallot, C. Picot, H. Benoît. J. Appl. Polym. Sci., **21**, 2181 (1977).
1138. J. M. Guenet, Macromolecules, **20**, 2874 (1987).
1139. B. Guenther Soares, A. de S. Gomes, Polym. Bull., **22**, 207 (1989).
1140. J. A. Guest, K. Matsuo, W. H. Stockmayer, Macromolecules, **13**, 560 (1980).
1141. L. A. Gugliemelli, M. O. Weaver, C. R. Russell, C. E. Rist, J. Polym. Sci. B, **9**, 151 (1971).
1142. L. A. Gugliemelli, C. L. Swanson, F. L. Baker, W. M. Doane, C. R. Russell, J. Polym. Sci., Polym. Chem. Ed., **12**, 2683 (1974).
1143. J. E. Guillet, R. L. Combs, D. F. Slonaker, H. W. Coover, J. Polym. Sci., **47**, 307 (1960).
1144. D. Gulino, J. Cole, J. P. Pascault, Makromol. Chem., **182**, 1215 (1981).
1145. D. Gulino, J. Galy, J. P. Pascault, Q. T. Pham. Makromol. Chem., **185**, 297 (1984).
1146. J. J. Gunderson, J. C. Giddings, Macromolecules, **19**, 2618 (1986).
1147. J. J. Gunderson, J. C. Giddings, C. Booth, C. Price, (Eds.), "Comprensive Polymer Science", vol. 1, Pergamon, Oxford, 1989, p. 279.
1148. S. Gundiah, N. V. Visvanathan, S. L. Kapur, Indian J. Chem., **4**, 344 (1966).
1149. S. Gundiah, Makromol. Chem., **104**, (1967).
1150. B. Gunesin, I. Piirma, J. Appl. Polym. Sci., **26**, 3103 (1981).
1151. A. Guo, G. Liu, J. Tao, Macromolecules, **29**, 2487 (1996).
1152. H.-Q. Guo, A. Kajiwara, Y. Morishima, M. Kamachi, Macromolecules, **29**, 2354 (1996).
1153. P. R. Gupta, J. L. Mc Carthy, Macromolecules, **1**, 236 (1968).
1154. P. R. Gupta, J. L. Mc Carthy, Macromolecules, **1**, 495 (1968).
1155. I. Gupta, S. N. Gupta, D. C. Neckers, J. Polym. Sci., Polym. Chem. Ed., **20**, 147 (1982).
1156. S. N. Gupta, J. P. Kennedy, Polym. Bull., **1**, 253 (1979).
1157. V. V. Gur'yanova, O. F. Alkayeva, S. Ye. Melamed, Reitburd, A. L. Narkon, B. M. Arshava, A. V. Pavlov, D. D. Novikov, Polym. Sci. USSR, **25**, 438 (1983); Vysokomol. Soedin. A, **25**, 375 (1983).
1158. A. A. Guryleva, M. P. Dianov, Polym Sci. USSR, **10**, 1661 (1968); Vysokomol. Soedin. A, **10**, 1431 (1968).
1159. A. A. Guryleva, B. Ya. Teitel'baum. R. A. Shlyakhter, L. P. Moskevich, Polym. Sci. USSR, **14**, 1370 (1972); Vysokomol. Soedin. A, **14**, 1221 (1972).
1160. A. A. Guryleva, R. A. Shlyakhter, B. Ya. Teitel'baum. Polym. Sci. USSR, **16**, 1427 (1974); Vysokomol. Soedin. A, **16**, 1235 (1974).
1161. L. Ye. Gusel'nikov, A. Yu. Koshevnik, I. M. Kosheleva, M. M. Kusakov, Polym. Sci. USSR, **7**, 860 (1965); Vysokomol. Soedin., **7**, 950 (1965).
1162. J. T. Guthrie, M. Ryder, F. I. Abdel-Hay, Polym. Bull., **1**, 501 (1979).
1163. J. M. Guyon-Gelliu, J. Golé, J. P. Pascault, J. Appl. Polym. Sci., **19**, 3173 (1975).
1164. A. Guyot, P. Quang-Tho, J. Chim. Phys., **63**, 742 (1966).
1165. A. Guyot, P. Rocaniere, J. Appl. Polym. Sci., **13**, 2019 (1969).
1166. P. Guyot, J. C. Favier, M. Fontanille, P. Sigwalt, Polymer, **23**, 73 (1982).
1167. G. M. Guzmán, Anales Real Soc. Esp. Fis. Quím. B, **50**, 631 (1954).
1168. G. M. Guzmán, J. Polym. Sci., **19**, 519 (1956).
1169. G. M. Guzmán, J. M. G. Fatou, Anales Real Soc. Esp. Fis. Quím. B, **53**, 669 (1957).
1170. G. M. Guzmán, J. M. G. Fatou, Anales Real Soc. Esp. Fis. Quím. B, **54**, 609 (1958).
1171. G. M. Guzmán, J. M. G. Fatou, Anales Real Soc. Esp. Fis. Quím. B, **54**, 263 (1958).
1172. G. M. Guzmán, in: J. C. Robb, F. W. Peaker, (Eds.), "Progress in High Polymers", Heywood, London, 1961.
1173. G. M. Guzmán, F. Arranz, Anales Real Soc. Esp. Fis. Quím. B, **59**, 445 (1963).
1174. G. M. Guzmán, in: J. Brandrup, E. H. Immergut, (Eds.), "Polymer Handbook" 1st ed., Interscience, New York, 1966.
1175. G. M. Guzmán, A. Bello, Makromol. Chem., **107**, 46 (1967).
1176. G. M. Guzmán, E. Riande, J. M. Pereña, A. G. Ureña, Eur. Polym. J., **10**, 537 (1974).
1177. M. Haberer, B. A. Wolf, Angew. Makromol. Chem., **228**, 179 (1995).
1178. L. S. Hacche, G. E. Washington, D. A. Brant, Macromolecules, **20**, 2180 (1987).
1179. W. F. Haddon, R. S. Porter, J. F. Johnson, J. Appl. Polym. Sci., **8**, 1371 (1964).
1180. N. Hadjichristidis, M. Devaleriola, V. Desreux, Eur. Polym. J., **8**, 1193 (1972).
1181. N. Hadjichristidis, V. Desreux, J. Macromol. Sci. Part A: Chem., **6**, 1227 (1972).
1182. N. Hadjichristidis, Makromol. Chem., **178**, 1463 (1977).
1183. H. Hadjichristidis, J. Roovers, J. Polym. Sci., Polym. Phys. Ed., **16**, 851 (1978).
1184. H. Hadjichristidis, L. J. Fetters, Macromolecules, **11**, 668 (1978).
1185. H. Hadjichristidis, A. Guyot, L. J. Fetters, Macromolecules, **11**, 668 (1978).
1186. H. Hadjichristidis, C. Touloupis, L. J. Fetters, Macromolecules, **14**, 128 (1981).
1187. H. Hadjichristidis, J. Mays, W. Ferry, L. J. Fetters, J. Polym. Sci., Polym. Chem. Ed., **22**, 1745 (1984).
1188. S. Hadjikyriacon, R. Faust, Macromolecules, **28**, 7893 (1995).

Previous Page

1189. T. Hagiwara, T. Shimizu, T. Someno, T. Yamagishi, H. Hamana, T. Narita, Macromolecules, **21**, 3324 (1988).
1190. G. L. Hagnauer, N. S. Schneider, J. Polym. Sci. A-2, **10**, 699 (1972).
1191. G. L. Hahnauer, B. R. Laliberte, J. Polym. Sci., Polym. Phys. Ed., **14**, 367 (1976).
1192. P. Hague, M. B. Huglin, B. L. Johnson, J. Smith, J. Appl. Polym. Sci., **12**, 2105 (1968).
1193. W. Hahn, W. Muller, R. W. Webber, Makromol. Chem., **21**, 131 (1956).
1194. H. Haïda, J.-P. Lingelser, Y. Gallot, M. Duval, Makromol. Chem., **192**, 270 (1991).
1195. D. A. Hajduk, S. M. Gruner, P. Rangarajan, R. A. Register, L. J. Fetters, Ch. Honeker, R. J. Albalak, E. L. Thomas, Macromolecules, **27**, 490 (1994).
1196. A. F. Halasa, H. E. Adams, J. Polym. Sci. C, **30**, 169 (1970).
1197. A. Halbwachs, Z. Grubisie-Gallot, Makromol. Chem. Rapid Commun., **7**, 709 (1986).
1198. R. W. Hall, in: P. W. Allen, (Ed.), "Techniques of Polymer Characterization", Butterworths, London, 1959.
1199. M. L. Hallensleben, Makromol. Chem., **177**, 3213 (1976).
1200. T. Hama, K. Yamaguchi, T. Suzaki, Makromol. Chem., **155**, 283 (1972).
1201. F. L. Hamb, J. Polym. Sci., Polym. Chem. Ed., **10**, 3217 (1972).
1202. A. E. Hamielec, M. Styring, Pure Appl. Chem., **57**, 955 (1985).
1203. A. Hamiton, R. Jerôme, Ph. Tyssié, J. Polym. Sci., Polym. Chem. Ed., **15**, 1035 (1977).
1204. J. M. Hammond, J. F. Hooper, W. G. P. Robertson, J. Polym. Sci. A-1, **9**, 281 (1971).
1205. E. Hamori, L. R. Prusinowski, P. G. Sparks, R. E. Hughes, J. Phys. Chem., **69**, 1101 (1965).
1206. K. Hanafusa, A. Teramoto, H. Fujita, J. Phys. Chem., **70**, 4004 (1966).
1207. T. M. Handel, I. S. Ponticello, J. S. Tan, Macromolecules, **20**, 264 (1987).
1208. J. Hannus, G. Smets, Bull. Soc. Chim. Belges, **60**, 76 (1951).
1209. R. Hanselmann, W. Buchard, R. Lemmes, D. Schwengers, Makromol. Chem. Phys., **196**, 2259 (1995).
1210. C. M. Hansen, G. A. Sather, J. Appl. Polym. Sci., **8**, 2479 (1964).
1211. A. Harada, K. Kataoka, Macromolecules, **28**, 5294 (1995).
1212. R. Hardy, P. E. M. Allen, Makromol. Chem., **42**, 38 (1960).
1213. W. G. Harland, J. Textile Inst., **46**, 483 (1955).
1214. D. J. Harmon, J. Polym. Sci. C, **8**, 243 (1965).
1215. D. J. Harmon, H. L. Jacobs, J. Appl. Polym. Sci., **10**, 253 (1966).
1216. D. J. Harmon, J. Appl. Polym. Sci., **11**, 1333 (1967).
1217. R. E. Harrington, B. H. Zimm, Polym. Preprints, **6** (1), 346 (1965).
1218. R. E. Harrington, P. G. Pegoraro, J. Polym. Sci. A-1, **4**, 475 (1966).
1219. I. Harris, R. G. J. Miller, J. Polym. Sci., **7**, 377 (1951).
1220. C. A. Harrison, Th. H. Mourey, J. Appl. Polym. Sci., **56**, 211 (1995).
1221. N. Hartler, Acta Chem. Scand., **11**, 1162 (1957).
1222. M. Hasagawa, K. Yamamoto, T. Shiwaku, T. Hashimoto, Macromolecules, **23**, 2629 (1990).
1223. E. Hasegawa, J. Nemoto, T. Kanayama, E. Tsuchida, Eur. Polym. J., **14**, 123 (1978).
1224. T. Hashimoto, H. Sasaki, M. Aiura, Y. Kato, J. Polym. Sci., Polym. Phys. Ed., **16**, 1789 (1978).
1225. G. W. Hastings, D. W. Ovenall, F. W. Peaker, Nature (London) 1091 (1956).
1226. K. Hatada, T. Kitayama, S. Okakata, H. Yuki, Polym. J., **14**, 971 (1982).
1227. D. Hatman, B. Chu, Macromolecules, **17**, 1537 (1984).
1228. M. Hattori, H. Takai, M. Kinoshita, Makromol. Chem., **179** 905 (1978).
1229. S. Hattori, H. Endoh, H. Nakahara, T. Kamata, M. Hamashima, Polym. J., **10**, 173 (1978).
1230. A. Haugh, Acta Chem. Scand., **13**, 601 (1959).
1231. E. A. Hauser, D. S. Le Beau, Indian Rubber J., **110**, 601 (1946).
1232. E. A. Hauser, D. S. Le Beau, Rubber Chem. Tech., **20**, 70 (1947).
1233. E. A. Hauser, D. S. Le Beau, J. Phys. Chem., **54**, 256 (1950).
1234. A. Hauss, Angew, Makromol. Chem., **8**, 73 (1969).
1235. S. W. Hawkins, H. Smith, J. Polym. Sci., **28**, 341 (1958).
1236. J. N. Hay, M. Sabir, R. L. T. Steven, Polymer, **10**, 187 (1969).
1237. J. N. Hay, A. M. Afifi-Effat, Brit. Polym. J., **1**, 9 (1977).
1238. J. N. Hay, M. Wiles, Brit. Polym. J., **10**, 181 (1978).
1239. J. N. Hay, J. F. Mc Cabe, J. Polym. Sci., Polym. Chem. Ed., **16**, 2893 (1978).
1240. S. Hayase, T. Ito, S. Suzuki, M. Wada, J. Polym. Sci., Polym. Chem. Ed., **19**, 2541 (1981).
1241. S. Hayase, Y. Onishi, K. Yoshikiyo, S. Suzuki, M. Wada, J. Polym. Sci., Polym. Chem. Ed., **20**, 3155 (1982).
1242. K. Hayashi, D. C. Pepper, Polym. J., **8**, 1 (1976).
1243. K. Hayshi, M. Tachibana, S. Okamura, J. Polym. Sci., Polym. Chem. Ed., **18**, 3381 (1980).
1244. K. Hayshi, J. Polym. Sci., Polym. Chem. Ed., **18**, 179 (1980).
1245. K. Hayshi, N. Kotani, J. Polym. Sci., Polym. Chem. Ed., **18**, 191 (1980).
1246. S. Hayshi, F. Hamada, A. Saijoo, A. Nakajima, Kobunshi Kagaku Japan, **24**, 769 (1967).
1247. S. Hayshi, S. Ikuma, K. Okuyama, T. Kajiyama, M. Takayanagy, J. Polym. Sci., Polym. Phys. Ed., **17**, 1995 (1979).
1248. T. Hayshi, G. W. Chen, A. Nakjima, Polym. J., **16**, 739 (1984).
1249. P. Hayden, R. Roberts, Int. J. Appl. Radiation Isotopes, **5**, 269 (1959).
1250. R. Hayes, S. Futamura, J. Polym. Sci., Polym. Chem. Ed., **19**, 985 (1981).
1251. J. E. Hazell, L. A. Prince, H. E. Stapelfeldt, Polym. Preprints, **8** (2), 1303 (1967).
1252. J. E. Hazell, L. A. Prince, H. E. Stapelfeldt, Polym. Sci. C, **21**, 43 (1968).
1253. B. Hazer, Angew. Makromol. Chem., **129**, 31 (1985).

1254. B. Hazer, R. W. Lenz, R. C. Fuller, Macromolecules, **27**, 45 (1994).
1255. L. A. Hazlitt, D. G. Moldovan, US Patent., **4**, 798.081 (1989).
1256. L. G. Hazlitt, J. Appl. Polym. Sci., Appl. Polym. Symp., **45**, 25 (1990).
1257. Q.-J. He, X.-Z. Li, D.-Y. Meng, J. Liquid. Chromatog., **5**, 1223 (1982).
1258. Z.-D. He, X.-C. Zhang, R.-S. Cheng, J. Liquid. Chromatog., **5**, 1209 (1982).
1259. A. Hebeish, M. H. El-Rafie, A. I. Waly, J. Polym. Sci., Polym. Chem. Ed., **14**, 2895 (1976).
1260. G. W. Heffner, D. S. Pearson, Macromolecules, **24**, 6295 (1991).
1261. E. A. Hegazy, N. B. El-Assy, A.-G. M. Rabie, I. Ishigaki, J. Okamoto, J. Polym. Sci., Polym. Chem. Ed., **22**, 597 (1984).
1262. W. Heitz, W. Kern, Angew. Makromol. Chem., **1**, 150 (1967).
1263. W. Heitz, B. Boemer, H. Ullner, Makromol. Chem., **121**, 102 (1969).
1264. W. Heitz, Angew. Makromol. Chem., **10**, 115 (1970).
1265. W. Heitz, Makromol. Chem., **145**, 141 (1971).
1266. W. Heitz, T. Wirth, R. Peters, G. Strobl, E. W. Fischer, Makromol. Chem., **162**, 63 (1972).
1267. J. N. Helbert, C.-Y. Chen, C. V. Pittman Jr., G. L. Hagnauer, Macromolecules, **11**, 1104 (1978).
1268. J. Heller, J. Moacanin, J. Polym. Sci. B, **6**, 595 (1968).
1269. J. Heller, J. F. Schimscheimer, R. A. Pasternak, C. B. Kingsley, J. Moacanin, J. Polym. Sci. A-1, **7**, 73 (1969).
1270. T. E. Helminiak, C. L. Benner, W. E. Gibbs, Polym. Preprints, **8** (1), 284 (1967).
1271. T. E. Helminiak, C. L. Benner, W. E. Gibbs, Polym. Preprints, **11** (1), 291 (1970).
1272. J. F. Henderson, J. M. Hulme, J. Appl. Polym. Sci., **11**, 2349 (1967).
1273. A. M. Henderson, A. Rudin, J. Polym. Sci., Polym. Chem. Ed., **19**, 1707 (1981).
1274. J. Hengstenberg, Angew. Chem., **65**, 350 (1953).
1275. J. Hengstenberg, Z. Elektrochem., **60**, 236 (1956).
1276. P. M. Henry, J. Polym. Sci., **36**, 3 (1959).
1277. L. D. Herbert, J. G. Robinson, G. Marr, Brit. Polym. J., **14**, 180 (1982).
1278. W. L. Hergenrother, C. J. Nelson, J. Polym. Sci., Polym. Chem. Ed., **12**, 2905 (1974).
1279. W. L. Hergenrother, R. J. Ambrose, J. Polym. Sci., Polym. Chem. Ed., **12**, 2613 (1974).
1280. P. M. Hergenrother, J. Polymer Sci., Polym. Chem. Ed., **20**, 1313 (1982).
1281. J.J. Hermans, H. A. Ende, J. Polym. Sci. C, **1**, 161 (1963).
1282. I. Hernández-Fuentes, M. G. Prolongo, Eur. Polym. J., **15**, 671 (1979).
1283. J. Hérold, G. Meyerhoff, Eur. Polym. J., **15**, 525 (1979).
1284. R. Hérold, J.-P. Fouassier, Angew. Makromol. Chem., **86**, 123 (1980).
1285. P. Herrent, J. Polym. Sci., **8**, 346 (1952).
1286. A. Herrera de Mola, M. Marx-Figini, R. V. Figini, Makromol. Chem., **176**, 2455 (1975).
1287. M. Hert, J. Rault, E. Robelin, Makromol. Chem., **184**, 225 (1983).
1288. J. Herz, D. Decker-Freyss, P. Rempp, J. Polym. Sci. C, **16**, 4035 (1968).
1289. J. Herz, M. Hert, C. Strazielle, Makromol. Chem., **160**, 213 (1972).
1290. G. Heublein, W. Freitag, W. Mock, Makromol. Chem., **181**, 267 (1980).
1291. G. Heufer, D. Braun, J. Polym. Sci. B, **3**, 495 (1965).
1292. E. Heuser, Wim. Shockley, R. Kjellgreen, Tappi, **33**, 92 (1950).
1293. M. E. S. Hibibe, M. C. A. Esperidião, J. Polym. Sci. Part B, Polym. Phys. Ed., **33**, 759 (1995).
1294. T. Higashimura, T. Matsuda, S. Okamura, J. Polym. Sci. A-1, **8**, 483 (1970).
1295. T. Higashimura, O. Kishiro, J. Polym. Sci., Polym. Chem. Ed., **12**, 967 (1974).
1296. T. Higashimura, O. Kishiro, T. Takeda, J. Polym. Sci., Polym. Chem. Ed., **14**, 1089 (1976).
1297. T. Higashimura, M. Mitsuhashi, M. Sawamoto, Macromolecules, **12**, 178 (1979).
1298. T. Higashimura, H. Teranishi, M. Sawamoto, Polym. J., **12**, 393 (1980).
1299. T. Higashimura, A. Tavizaki, M. Sawamoto, J. Polym. Sci., Polym. Sci., Polym. Chem. Ed., **22**, 3173 (1984).
1300. T. Higashimura, Y. M. Law, M. Sawamoto, Polym. J., **16**, 401 (1984).
1301. Y. Higo, Y. Kato, M. Itoh, N. Kozuka, I. Noda, M. Nagasawa, Polym. J., **14**, 809 (1982).
1302. G. Hild, A. Kohler, P. Rempp, Eur. Polym. J., **16**, 525 (1980).
1303. A. Hilt, K. H. Reichert, K. Hamann, Makromol. Chem., **101**, 246 (1967).
1304. O. Hinterhofer, Makromol. Chem., **181**, 67 (1980).
1305. T. Hirabayashi, K. Yokota, Polym. J., **14**, 789 (1982).
1306. A. Hirai, K. Yamaguchi, K. Takenaka, K. Suzuki, S. Nakahama, N. Yamazaki, Makromol. Chem., Rapid Commun., **3**, 941 (1982).
1307. T. Hirano, P. H. Khanh, T. Tsuruta, Makromol. Chem., **153**, 331 (1972).
1308. A. Hirao,, I. Hattori, T. Sasagawa, K. Yamaguchi, S. Nakahama, N. Yamazaki, Makromol. Chem., Rapid Commun., **3**, 59 (1982).
1309. A. Hirao, Y. Ishino, S. Nakahama, Macromolecules, **21**, 561 (1988).
1310. A. Hirao, K. Kitamura, K. Takenata, S. Nakahama, Macromolecules, **26**, 4995 (1993).
1311. S. Hirata, H. Hasegawa, A. Kishimoto, J. Appl. Polym. Sci., **14**, 2025 (1970).
1312. M. Hirooka, H. Kanda, K. Nakaguchi, J. Polym. Sci. B, **1**, 701 (1963).
1313. S. M. Hirshfield, E. R. Allen, J. Polym. Sci., **39**, 554 (1959).
1314. T. Hjerberg, E. M. Soervik, J. Polym. Sci., Polym. Chem. Ed., **16**, 645 (1978).

1315. T. Hjerberg, E. M. Soervik, J. Macromol. Sci., Part A: Chem., **17**, 983 (1982).
1316. M. Hloušek, J. Lániková, Chem. Prumsysl, **16** (14), 160 (1966).
1317. M. Hloušek, J. Lániková, J. Polym. Sci. C, **16**, 935 (1967).
1318. N. Ho-Duc, H. Daoust, Can. J. Chem., **46**, 2456 (1968).
1319. N. Ho-Duc, H. Daoust, A. Gourdenne, Polym. Preprints, **12** (1), 639 (1971).
1320. N. Ho-Duc, J. Prud'homme, Macromolecules, **6**, 472 (1973).
1321. H. C. Hoare, D. E. Hillman, Brit. Polym. J., **3**, 259 (1971).
1322. L. M. Hobbs, S. C. Kothari, V. C. Long, G. C. Sutaria, J. Polym. Sci., **22**, 123 (1956).
1323. L. M. Hobbs, V. C. Long, Polymer, **4**, 479 (1963).
1324. S. J. Hobbs, F. W. Billmeyer Jr., J. Polym. Sci. A-2, **8**, 1387 (1970).
1325. H. Höcker, K. Saeki, Makromol. Chem., **148**, 107 (1971).
1326. P. J. Hocking, R. H. Marchessault, Polym. Bull., **30**, 163 (1993).
1327. L. Hoehr, V. Jaacks, H. Cherdron, S. Iwabuchi, W. Kern, Makromol. Chem., **103**, 279 (1967).
1328. H. J. Hoelle, B. R. Lehnen, Eur. Polym. J., **11**, 663 (1975).
1329. A. S. Hoffman, B. A. Fries, P. C. Condit, J. Polym. Sci. C, **4**, 109 (1963).
1330. J. D. Hoffman, B. H. Zimm, J. Polym. Sci., **15**, 405 (1955).
1331. M. Hoffman, H. Urban, Makromol. Chem., **178**, 2683 (1977).
1332. M. Hoffmann, Makromol. Chem., **57**, 96 (1962).
1333. M. Hoffmann, unpublished report to Farbenfabriken Bayer, 1959, quoted by H. Giesekus, in: M. J. R. Cantow, (Ed.), "Polymer Fractionation", Academic Press, 1967, p. 191.
1334. M. Hoffmann, Makromol. Chem., **177**, 1021 (1976).
1335. V. Hofmann, H. Ringsdorf, E. Schaumloeffel, Makromol. Chem., **181**, 351 (1980).
1336. S. Holdcrof, Macromolecules, **24**, 4834 (1991).
1337. G. E. Holdcroft, P. H. Plesch, Makromol. Chem., **185**, 27 (1984).
1338. P. J. Holdsworth, A. Keller, I. M. Ward, T. Williams, Makromol. Chem., **125**, 82 (1969).
1339. P. J. Holdsworth, A. Keller, I. M. Ward, T. Williams, Makromol. Chem., **125**, 70 (1969).
1340. G. M. Holob, P. Erlich, J. Polym. Sci., Polym. Phys. Ed., **15**, 627 (1977).
1341. C. Holt, W. Mackie, D. B. Sellen, Polymer, **19**, 1421 (1978).
1342. W. Holtrup, Makromol. Chem., **178**, 2335 (1977).
1343. R. B. Homer, M. Shinitzky, Macromolecules, **1**, 469 (1968).
1344. T. Homma, K. Kawahara, H. Fujita, M. Ueda, Makromol. Chem., **67**, 132 (1963).
1345. T. Homma, K. Kawahara, H. Fujita, J. Appl. Polym. Sci., **8**, 2853 (1964).
1346. T. Homma, H. Fujita, J. Appl. Polym. Sci., **9**, 1701 (1965).
1347. F. Hooshmand-Mozaffar, M. K. Hoseinalizadeh-Khorasani, M. B. Huglin, Polymer, **21**, 413 (1980).
1348. P. Hope, R. Anderson, A. S. Bloss, Brit. Polym. J., **5**, 67 (1973).
1349. P. Hope, B. P. Stark, Brit. Polym. J., **5**, 363 (1973).
1350. H. Hopff, M. De Tomasi, H.-G. Elias, Makromol. Chem., **96**, 41 (1966).
1351. H. Hopff, M. A. Osman, Makromol. Chem., **143**, 289 (1971).
1352. J. Höpken, M. Möller, Macromolecules, **25**, 1461 (1992).
1353. A. Horbach, H. Vernaleken, K. Weirauch, Makromol. Chem., **181**, 111 (1980).
1354. A. Horbach, R. Binsack, H. Mueller, Angew. Makromol. Chem., **98**, 35 (1981).
1355. K. Horie, D. Mikulášová, Makromol. Chem., **175**, 2091 (1974).
1356. F. Horii, Y. Ikada, J. Polym. Sci., Polym. Lett. Ed., **12**, 27 (1974).
1357. F. Horii, Y. Ikada, I. Sakurada, J. Polym. Sci., Polym. Chem. Ed., **13**, 755 (1975).
1358. H. Horiuchi, S. Irie, T. Nose, Polymer, **32**, 1970 (1991).
1359. R H. Horowitz, Polym. Preprints, **3**, 167 (1962).
1360. R. H. Horowitz, Polym. Preprints, **4**, 689 (1963).
1361. J. Horský, V. Petrus, M. Bohdneckỳ, Makromol. Chem., **187**, 2621 (1986).
1362. J. Horský, M. Bohdneckỳ, Eur. Polym. J., **26**, 1109 (1990).
1363. A. Horta, I. A. Katime, Macromolecules, **17**, 2734 (1984).
1364. A. Horta, E. Saiz, J. M. Barrales-Rienda, P. A. Galera Gómez, Polymer, **27**, 139 (1986).
1365. A. Horta, L. Gargallo, D. Radić, Macromolecules, **23**, 5320 (1990).
1366. B. Hortling, J. J. Lindberg, Makromol. Chem., **179**, 1707 (1978).
1367. S. Hosoda, Polym. J., **20**, 383 (1988).
1368. K. Hoshino, M. Watanabe, J. Am. Chem. Soc., **73**, 4816 (1951).
1369. R. E. Hostetler, J. W. Swanson, J. Polym. Sci., Polym, Chem, Ed., **12**, 29 (1974).
1370. T. Housaki, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 75 (1991).
1371. R. C. Houtz, Textile Res. J., **20**, 786 (1950).
1372. P. Howard, R. S. Parikh, IUPAC Praga, Preprint P 265 (1965).
1373. P. Howard, R. S. Parikh, J. Polym. Sci. A-1, **4**, 407 (1966).
1374. P. Howard, R. S. Parikh, J. Polym. Sci. C, **30**, 17 (1970).
1375. G. J. Howard, J. Polym. Sci., **37**, 310 (1959).
1376. G. J. Howard, J. Polym. Sci. A, **1**, 2667 (1963).
1377. C. E. Hoyle, C. P. Chawla, Macromolecules, **28**, 1946 (1995).
1378. J. S. Hrkach, K. Matyjaszewski, Macromolecules, **23**, 4042 (1990).
1379. H. L. Hsieh, J. Polym. Sci. A-1, **3**, 191 (1965).
1380. H. L. Hsieh, O. F. Mc Kinney, J. Polym. Sci. B, **4**, 843 (1966).
1381. C. S. Hsu, V. Percec, Makromol. Chem., **189**, 1141 (1988).
1382. X. Hu, J. L. Stanford, R. J. Day, R. J. Young, Macromolecules, **25**, 672 (1992).
1383. R. Y. Hunag, G. Mayer, IUPAC Budapest, Vol. III, Preprint 6/03 (1969).
1384. R. Y. M. Huang, J. F. Westlake, S. C. Sharma, J. Polym. Sci. A-1, **7**, 1729 (1969).

1385. R. Y. M. Huang, U. Y. Kim, J. M. Dickson, D. R. Lloyd, C. Y. Peng, J. Appl. Polym. Sci., **26**, 1135 (1981).

1386. N. Huang, D. C. Suudberg, J. Polym. Sci. Part A: Polym. Chem., **33**, 2551 (1995).

1387. R. Y. M. Huang, P. Chandramouli, J. Polym. Sci. A-1, **7**, 1393 (1969).

1388. R. Y. M. Huang, J. F. Westlake, IUPAC Budapest, Vol. III, **8**, 49 (1970).

1389. C. Huber, K. H. Lederer, J. Polym. Sci., Polym. Lett. Ed., **18**, 535 (1980).

1390. P. Hubin-Eschger, Angew. Makromol. Chem., **26**, 107 (1972).

1391. P. Hudec, Z. Slama, H. Stepankova, Makromol. Chem., **180**, 351 (1979).

1392. M. L. Huggins, H. Okamoto, in: M. J. R. Cantow (Ed.), "Polymer Fractionation", Academic Press, New York, 1996.

1393. L. J. Hughes, G. L. Brown. J. Appl. Polym. Sci., **7**, 59 (1963).

1394. L. J. Hughes, R. H. Andreatta, H. A. Scheraga, Macromolecules, **5**, 187 (1972).

1395. A. J. Hughes, J. P. Bell, J. Polym. Sci., Polym. Phys. Ed., **16**, 201 (1978).

1396. M. B. Huglin. D. H. Whitehurst, D. Sims, J. Appl. Polym. Sci., **12**, 1889 (1968).

1397. J. M. Hulme, L. A. Mc Leod, Polymer, **3**, 153 (1962).

1398. M. L. Hunt, S. Newman, H. A. Scheraga, P. J. Flory, J. Phys. Chem., **60**, 1278 (1956).

1399. B. K. Hunter, K. E. Russell, M. Y. Scammell, S. L. Thompson, J. Polym. Sci., Polym. Chem. Ed., **22**, 1383 (1984).

1400. E. Husemann, Makromol. Chem., **26**, p. 181, p. 199 (1958).

1401. E. Husemann, B. Pfannemuller, Makromol. Chem., **49**, 214 (1961).

1402. E. Husemann, B. Pfannemuller, Makromol. Chem., **69**, 74 (1963).

1403. E. Husemann, R. Werner, Makromol. Chem., **59**, 43 (1963).

1404. A. J. Hyde, R. B. Taylor, Polymer, **4**, 1 (1963).

1405. H. Iatron, E. Siakali-Kionlafa, N. Hadjichristidis, J. Roovers, J. Polym. Sci. Part B, Polym. Phys., **33**, 1925 (1995).

1406. F. Ide, R. Handa, K. Nakatsuka, Chem. High Polym. (Japan), **21**, 57 (1964).

1407. Ye. V. Ignatova, M. B. Lachinov, V. P. Zubov, Ye. S. Garina, V. A. Kabanov, Polym. Sci. USSR, **22**, 3057 (1980); Vysokomol. Soedin. A, **22**, 2787 (1980).

1408. T. Ikeda, S. Horiuchi, D. B. Karaujit, S. Kurihara, S. Tazuke, Macromolecules, **23**, 36 (1990).

1409. T. Ikeda, S. Kurihara, D. B. Karaujit, S. Tazuke, Macromolecules, **23**, 3938 (1990).

1410. I. Ikeda, K. Suzuki, T. Maekawa, T. Higashimura, Polym. J., **12**, 423 (1980).

1411. K.-H. Illers, Makromol. Chem., **118**, 88 (1968).

1412. K. Imai, M. Matsumoto, Bull. Chem. Soc. Japan, **36**, 455 (1963).

1413. L. G. Imhof, J. Appl. Polym. Sci., **10**, 1137 (1966).

1414. H. Inagaki, K. Hayashi, T. Matsuo, Makromol. Chem., **84**, 80 (1965).

1415. H. Inagaki, T. Miyamoto, Makromol. Chem., **87**, 166 (1965).

1416. H. Inagaki, T. Miyamoto, S. Ohta, J. Phys. Chem., **70**, 3420 (1966).

1417. H. Inagaki, H. Suzuki, M. Fujii, T. Matsuo, J. Phys. Chem., **70**, 1718 (1966).

1418. H. Inagaki, H. Matsuda, F. Kamiyama, Macromolecules, **1**, 520 (1968).

1419. H. Inagaki, F. Kamiyama, T. Yagi, Macromolecules, **4**, 133 (1971).

1420. H. Inamura, H. Ochiai, H. Yamamura, J. Polym. Sci., Polym. Phys. Ed., **14**, 1221 (1976).

1421. J. D. Ingham, N. S. Rapp, J. Polym. Sci., A-1, **2**, 4941 (1964).

1422. S. Inoue, T. Tsuruta, N. Yoshida, Kôgyô Kagaku Zassi, **67**, 1439 (1964).

1423. S. Inoue, T. Tsuruta, N. Yoshida, Makromol. Chem., **79**, 34 (1964).

1424. S. Inoue, Y. Yokota, N. Yoshida, T. Tsuruta, Makromol. Chem., **90**, 131 (1966).

1425. S. Inoue, T. Tsukuma, M. Kawaguchi, T. Tsuruta, Makromol. Chem., **103**, 151 (1967).

1426. S. Inoue, K. Tsubaki, T. Yamada, T. Tsuruta, Makromol. Chem., **125**, 181 (1969).

1427. T. Inoue, S. Tazuke, J. Polym. Sci., Polym. Chem. Ed., **19**, 2861 (1981).

1428. K. Irabo, S. Anzai, A. Onishi, Bull. Chem. Soc. Japan, **41**, 501 (1968).

1429. M. Irie, S. Tomimoto, K. Hayashi, J. Polym. Sci., Polym. Chem. Ed., **11**, 1859 (1973).

1430. J. J. Iruin, G. M. Guzmán, J. Ansorena, Makromol. Chem., **182**, 2789 (1981).

1431. S. Ishida. Bull. Chem. Soc. Japan, **33**, 727 (1960).

1432. S. Ishida, Y. Tsutsumi, K. Kaneko, J. Polym. Sci., Polym. Chem. Ed., **19**, 1609 (1981).

1433. I. Ishigaki, N. Morishita, Y. Morita, A. Ito, J. Appl. Polym. Sci., **21**, 749 (1977).

1434. S. Ishikausa, Makromol. Chem. Phys., **196**, 485 (1995).

1435. M. Ishimori, T. Hagiwara, T. Tsuruta, Makromol. Chem., **179**, 2337 (1978).

1436. T. Ishizone, J. Tsuchiya, A. Hirao, S. Nakahama, Macromolecules, **25**, 4840 (1992).

1437. T. Ishizone, A. Hirao, S. Nakahama, Macromolecules, **26**, 6964 (1993).

1438. T. Ishizone, N. Oka, A. Hirao, S. Nakahama, Macromolecules, **29**, 528 (1996).

1439. K. Ishizu, K. Ino, Y. Koshiro, T. Fukutomi, T. Kakurai, J. Polym. Sci., Polym. Chem. Ed., **20**, 2545 (1982).

1440. K. Ishizu, A. Ohtaka, T. Fukutomi, T. Kakurai, J. Polym. Sci., Polym. Chem. Ed., **20**, 3199 (1982).

1441. K. Ishizu, Y. Kashi, T. Fukutomi, T. Kakurai, Makromol. Chem., **183**, 3099 (1982).

1442. K. Ishizu, K. Inagaki, K. Bessho, T. Fukutomi, Makromol. Chem., **185**, 1169 (1984).

1443. K. Ishizu, T. Fukuyama, Macromolecules, **22**, 244 (1989).

1444. Y. Isono, T. Fujimoto, H. Inagaki, M. Shishido, M. Nagasawa, Polym. J., **12**, 131 (1980).

1445. H. Itagaki, A. Okamoto, K. Horie, I. Mita, Eur. Polym. J., **18**, 885 (1982).
1446. H. Ito, D. C. Miller, C. G. Willson, Macromolecules, **15**, 915 (1982).
1447. K. Ito, N. Usami, Y. Yamashita, Macromolecules, **13**, 216 (1980).
1448. K. Ito, Y. Tomi, S. Kawaguchi, Macromolecules, **25**, 1538 (1992).
1449. Y. Ito, S. Shuishido, J. Polym. Sci., Polym. Phys. Ed., **10**, 2239 (1972).
1450. T. Itou, A. Teramoto, Polym. J., **16**, 779 (1984).
1451. T. Itou, M. Chikiri, A. Taremoto, S. M. Aharoni, Polym. J., **20**, 143 (1988).
1452. S. S. Ivanchev, N. I. Solomko, Polym. Sci. USSR, **8**, 353 (1966); Vysokomol. Soedin., **8**, 322 (1966).
1453. S. S. Ivanchev, L. V. Shumnyi, V. V. Konovalenko, Polym. Sci. USSR, **22**, 3000 (1980); Vysokomol. Soedin. A, **22**, 2735 (1980).
1454. J. Ivanfi, E. Ady, J. Paint. Technol., **43**, 199 (1970).
1455. K. J. Ivin, H. A. Ende, G. Meyerhoff, Polymer, **3**, 129 (1962).
1456. M. Iwama, H. Utiyama, M. Kurata, J. Macromol. Sci. Chem., **1**, 701 (1966).
1457. T. Iwata, T. Saegusa, H. Fujii, J. Furukawa, Makromol. Chem., **97**, 49 (1966).
1458. T. Iwata, Y. Doi, K. Kasuya, Y. Inoue, Macromolecules, **30**, 833 (1997).
1459. Y. Izumi, Y. Miyake, Polym. J., **3**, 647 (1972).
1460. V. Jaacks, G. Franzmann, Makromol. Chem., **143**, 283 (1971).
1461. V. Jaacks, C. D. Eisenbach, W. Kern, Makromol. Chem., **161**, 139 (1972).
1462. H. Jabloner, R. H. Mumma, J. Polym. Sci., Polym. Chem. Ed., **10**, 763 (1972).
1463. C. Jackson, D. J. Frater, J. W. Mays, J. Polym. Sci. B, Polym. Phys., **33**, 2155 (1995).
1464. C. Jackson, J. P. O'Brien, Macromolecules, **28**, 5975 (1955).
1465. E. Jacobi, H. Schuttenberg, R. C. Schulz, Makromol. Chem. Rapid. Commun., **1**, 397 (1980).
1466. J. Jaffe, C. Berliner, J. Chim. Phys., **63**, 389 (1966).
1467. J. Jaffe, C. Berliner, M. Lambert, J. Chim. Phys., **64**, 499 (1967).
1468. B. Jambe, J. Marchand.-Brynaert, J. Devaux, J. Polym. Sci. :Part A, Polym. Chem., **33**, 1283 (1995).
1469. J. Janča, S. Pokorný, J. Chromatog., **170**, 319 (1979).
1470. J. Janča, M. Kolínský, J. Appl. Polym. Sci., **21**, 83 (1977).
1471. J. Janča, Chem. Listy, **73**, 476 (1979).
1472. J. Janča, S. Pokorný, M. Kolínský, J. Appl. Polym. Sci., **23**, 1811 (1979).
1473. J. Janča, S. Pokorný, L. Z. Vilenchik, B. G. Belenkii, J. Chromatog., **211**, 39 (1981).
1474. J. Janča (Ed.), "Chromatographic Science Series", vol. 25, Marcel Dekker, New York, 1984.
1475. J. Janča, D. Pribylová, K. Bouchal, V. Tyráckova, J. Liquid. Chromatogr., **9**, 2059 (1986).
1476. J. Janča, "Field-Flow Fractionation – Analysis of Macromolecules and Particles", Chromatographic Science Series, vol 39, Marcel Dekker, New York, 1988.
1477. M. Jancó, T. Prudskova, D. Bereck, J. Appl. Polym. Sci., **55**, 393 (1995).
1478. M. Jancó, D. Bereck, T. Prudskova, Polymer, **36**, 3295 (1995).
1479. M. Janic, M. J. Beneš, J. Pěska, Makromol. Chem., **138**, 99 (1970).
1480. P. Jannasch, B. Wesslen, J. Polym. Sci. Part A, Polym. Chem., **33**, 1465 (1995).
1481. G. Jayme, K. Kringstad, Angew. Chem., **73**, 219 (1961).
1482. G. Jayme, P. Kleppe, A. Kunschner, Makromol. Chem., **48**, 144 (1961).
1483. G. Jayme, H. Knolle, Makromol. Chem., **82**, 190 (1965).
1484. G. Jayme, J. Troeften, Naturwissenschaften, **52**, 496 (1965).
1485. G. Jayme, P. Kleppe, Papier, **15**, 6 (1961).
1486. G. Jayme, K. K. Hasvold, Papier, **20**, 657 (1966).
1487. G. Jayme, G. El-Kodsi, Papier, **24**, p. 125, p. 679 (1970).
1488. G. Jayme, G. El-Kodsi, Papier, **25**, 589, 627 (1971).
1489. J. M. Jelich, S. P. Nunes, E. Paul, B. A. Wolf, Macromolecules, **20**, 1943 (1987).
1490. H. H. G. Jellinek, G. White, J. Polym. Sci., **6**, 757 (1951).
1491. H. H. G. Jellinek, M. D. Lun, Polym. Preprints, **8** (1), 533 (1967).
1492. H. H. G. Jellinek, J. J. Lichorat, Polym. J., **12**, 347 (1980).
1493. A. M. Jendrychowska-Bonamour, Eur. Polym. J., **6**, 1491 (1970).
1494. A. M. Jendrychowska-Bonamour, C. Michot, Eur. Polym. J., **13**, 251 (1977).
1495. R. E. Jentoft, T. H. Gouw, J. Polym. Sci., B, **7**, 811 (1969).
1496. S. I. Jeon, M. S. Jhon, J. Polym. Sci., Polym. Chem. Ed., **22**, 3555 (1984).
1497. S. I. Jeon, D. W. Lee, J. Polym. Sci. B, Polym. Phys., **33**, 411 (1995).
1498. R. Jerome, V. Desreux, Eur. Polym. J., **6**, 411 (1970).
1499. M. Jiang, T. Huang, J. Xie, Makromol. Chem. Phys., **196**, 803 (1995).
1500. Y. Jinbo, T. Sato, A. Teramoto, Macromolecules, **27**, 6080 (1994).
1501. B. Jirgensons, J. Polym. Sci., **8** 519 (1952).
1502. T. Jochsberger, N. Indictor, N. Güdüllüdglu, C. J. Shahani, N. S. Baer, J. Polym. Sci., Polym. Chem. Ed., **16**, 309 (1978).
1503. C. Johann, P. Kila, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 111 (1991).
1504. R. M. Johnsen, Chem. Scripta, **1**, 81 (1971).
1505. B. L. Johnson, Ind. Eng. Chem., **40**, 351 (1948).
1506. J. F. Johnson, R. S. Porter, M. J. R. Cantow, in: H. F. Mark, N. G. Gaylord, N. M. Bikales, (Eds.), "Fractionation" in Encyclopedia Polym. Sci., Technol. Interscience, New York, **7**, 231 (1967).
1507. J. F. Johnson, R. S. Porter (Eds.), "Analytical Gel Permeation Chromatography", J. Polym. Sci. C, **21**, Interscience, New York, 1968.

1508. J. F. Johnson, R. S. Porter, "Gel Permeation Chromatography" in: A. D. Jenkins, (Ed.), "Progress in Polymer Science", vol. 2, Pergamon Press, 201 (1970).
1509. P. R. Johnston, J. Appl. Polym. Sci., **9**, 467 (1965).
1510. R. G. Jones, R. E. Benfield, R. H. Cragg, A. C. Swain, S. J. Webb, Macromolecules, **26**, 4878 (1993).
1511. F. R. Jones, Eur. Polym. J., **10**, 249 (1974).
1512. F. R. Jones, L. E. Scales, J. A. Semiyen, Polymer, **15**, 738 (1974).
1513. E. F. Jordan Jr., A. N. Wrigley, J. Polym. Sci. A-1, **2**, 3909 (1964).
1514. P. L. Joskowicz, A. Muñoz, J. Barrera, A. Müller, Makromol. Chem. Phys., **196**, 385 (1995).
1515. S. Jovanović, J. Romatowski, G. V. Schulz, Makromol. Chem., **85**, 187 (1965).
1516. J. Juilfs, Kolloid. Z., **141**, 88 (1955).
1517. P. C. Juliano, J. V. Crivello, D. E. Floryan, Polym. Preprints, **12** (1), 632 (1971).
1518. J. L. Jungnickel, F. T. Weiss, J. Polym. Sci., **49**, 437 (1961).
1519. J. Junquera, N. Cardona, J. E. Figueruelo, Makromol. Chem., **160**, 159 (1972).
1520. G. R. Juppien, Angew. Makromol. Chem., **70**, 199 (1978).
1521. V. Juranićová, Š. Florián, D. Berek, Eur. Polym. J., **6**, 63 (1970).
1522. V. Juranićová, Š. Florián, D. Berek, Eur. Polym. J., **6**, 57 (1970).
1523. R. I. Kabalyunas, G. D. Shershneva, R. M. Livshits, Z. A. Rogovin, Polym. Sci. USSR, **8**, 260 (1966); Vysokomol. Soedin., **8**, 240 (1966).
1524. H. Kaemmerer, W. Kern, W. Heuser, J. Polym. Sci., **28**, 331 (1958).
1525. H. Kaemmerer, F. Rocaboy, Compt. Rend., **256**, 4440 (1963).
1526. H. Kaemmerer, F. Rocaboy, Makromol. Chem., **72**, 76 (1964).
1527. H. Kaemmerer, K.-G. Steinfort, W. Kern, Makromol. Chem., **70**, 173 (1964).
1528. H. Kaemmerer, P. N. Grover, Makromol. Chem., **96**, 270 (1966).
1529. H. Kaemmerer, P. N. Grover, Makromol. Chem., **99**, 49 (1966).
1530. H. Kaemmerer, K. F. Mueck, Makromol. Chem., **136**, 161 (1970).
1531. H. Kaemmerer, K. F. Mueck, Makromol. Chem., **136**, 179 (1970).
1532. K. Kagawa, T. Oishi, K. Matsusaki, M. Fujimoto, Polymer, **36**, 941 (1995).
1533. M. Kajiwara, N. Yamamoto, Brit. Polym. J., **16**, 148 (1984).
1534. Y. Kakamura, O. Hinojosa, J. C. Arthur Jr., J. Appl. Polym. Sci., **15**, 391 (1971).
1535. H. Kakiuchi, W. Fukuda, Kôgyô Kagaku Zasshi, **66**, 964 (1963).
1536. M. Kakugo, T. Miyatake, Y. Naito, K. Mizunuma, Macromolecules, **21**, 314 (1988).
1537. M. Kakugo, T. Miyatake, K. Mizunuma, Y. Kawai, Macromolecules, **21**, 2309 (1988).
1538. M. Kakugo, Y. Naito, K. Mizunuma, T. Miyatake, Makromol. Chem., **190**, 849 (1989).
1539. M. Kakugo, T. Miyatake, Y. Naiko, K. Mizunuma, Makromol. Chem., **190**, 505 (1989).
1540. M. Kakugo, T. Miyatake, K. Mizunuma, Macromolecules, **24**, 1469 (1991).
1541. J. Kálal, M. Gordon, C. Devoy, Makromol. Chem., **152**, 233 (1972).
1542. J. Kálal, V. Marousek, F. Švec, Angew. Makromol. Chem., **38**, 45 (1974).
1543. M. Kalfus, E. Kopytovski, Z. Skupinska, S. Lesniak, Polym. Sci. USSR, **7**, 1830 (1965); Vysokomol. Soedin., **7**, 1655 (1965).
1544. M. Kalfus. J. Mitus, J. Polym. Sci. A-1, **4**, 953 (1966).
1545. R. Kalir, A. Zilkha, Eur. Polym. J., **14**, 557 (1978).
1546. J. K. Kallitsis, F. Kakali, K. G. Gravalos, Macromolecules, **27**, 4509 (1994).
1547. K. K. Kalnin'sh, A. N. Krasowskii, B. G. Belen'kii, G. A. Andreyeva, Polym. Sci., USSR, **18**, 2636 (1976); Vysokomol. Soedin. A, **18**, 2304 (1976).
1548. K. Kamada, H. Sato, Polym. J., **2**, 593 (1971).
1549. Y. Kamaguchi, Y. Doi, Macromolecules, **25**, 2324 (1992).
1550. P. M. Kamath, L. Wild, Polym. Eng. Sci., **6**, 213 (1966).
1551. Y. Kambe, C. Honda, Angew. Makromol. Chem., **25**, 163 (1972).
1552. Y. Kambe, C. Honda, Angew. Makromol. Chem., **48**, 175 (1975).
1553. I. Kamel, S. Machi, J. Silverman, J. Polym. Sci., Polym. Chem. Ed., **10**, 1019 (1972).
1554. K. Kamide, K. Fujii, H. Kobayashi, Makromol. Chem., **117**, 190 (1968).
1555. K. Kamide, K. Ohno, T. Kawai, Makromol. Chem., **137**, 1 (1970).
1556. K. Kamide, K. Sugamiya, T. Ogawa, C. Nakayama, N. Baba, Makromol. Chem., **135**, 23 (1970).
1557. K. Kamide, K. Yamaguchi, Makromol. Chem., **162**, 205 (1972).
1558. K. Kamide, T. Terakawa, Makromol. Chem., **155**, 25 (1972).
1559. K. Kamide, S. Manabe, E. Osafune, Makromol. Chem., **168**, 173 (1973).
1560. K. Kamide, Y. Miyazaki, H. Kobayashi, Polym. J., **9**, 317 (1977).
1561. K. Kamide, Y. Miyazaki, T. Abe, Polym. J., **9**, 395 (1977).
1562. K. Kamide, Y. Miyakawa, Makromol. Chem., **179**, 359 (1978).
1563. K. Kamide, T. Okada, T. Terakawa, K. Kaneko, Polym. J., **10**, 547 (1978).
1564. K. Kamide, T. Terakawa, Y. Miyazaki, Polym. J., **11**, 285 (1979).
1565. K. Kamide, Y. Miyazaki, T. Abe, Polym. J., **11**, 523 (1979).
1566. K. Kamide, Y. Miyazaki, T. Abe, Brit. Polym. J., **13**, 168 (1981).
1567. K. Kamide, Y. Miyazaki, H. Kobayashi, Polym. J., **14**, 591 (1982).
1568. K. Kamide, M. Saito, Eur. Polym. J., **20**, 903 (1984).

1569. M. Kamigaito, K. Yamaoka, M. Sawamoto, T. Higashimura, Macromolecules, **25**, 6400 (1992).
1570. F. Kamiyama, H. Matsuda, H. Inagaki, Makromol. Chem., **125**, 286 (1969).
1571. F. Kamiyama, H. Matsuda, H. Inagaki, Polym. J., **1**, 518 (1970).
1572. F. Kamiyama, H. Inagaki, T. Kotaka, Polym. J., **3**, 470 (1972).
1573. R. Kammereck, L. J. Fetters, M. Morton, Polym. Preprints, **11** (1), 72 (1970).
1574. B. K. Kang, K. F. O'Driscoll, Macromolecules, **7**, 886 (1974).
1575. P. J. Kangle, E. Pacsu, J. Polym. Sci., **54**, 301 (1961).
1576. S. L. Kapur, S. Gundiah, J. Polym. Sci., **26**, 89 (1957).
1577. A. Karandinos, S. Nan, J. W. Mays, N. Hadjichristidis, Macromolecules, **24**, 2007 (1991).
1578. A. N. Karasev, I. N. Andreyeva, N. M. Domareva, K. I. Kosmatykh, M. G. Karaseva, N. A. Domnicheva, Polym. Sci. USSR, **12**, 1275 (1970); Vysokomol. Soedin. A, **12**, 1127 (1970).
1579. V. A. Kargin, N. A. Platé, A. S. Dobrynina, Colloid. J. USSR, **20**, 315 (1958).
1580. V. A. Kargin, N. A. Platé, A. S. Dobrynina, kolliod. Zh., **20**, 332 (1958).
1581. S. Karoglaniza, I. R. Harrison, Am. Chem. Soc. Proceedings. Polym. Mat. Sci. and Eng., **61**, 748 (1989).
1582. F. E. Karrer, Makromol. Chem., **181**, 595 (1980).
1583. K. Karuwakaran, M. Santappa, Makromol. Chem., **111**, 20 (1968).
1584. F. Käsbauer, E. Schuch in: K. A. Wolf (Ed.), "Kunststoffe", Vol. 1, Springer, Berlin, 1962.
1585. T. Kase, M. Imahori, T. Kazama, Y. Isono, T. Fujimoto, Macromolecules, **24**, 1714 (1991).
1586. H. A. Kashani, M. H. George, J. A. Barrie, J. Polym. Sci., Polym. Lett., **15**, 565 (1977).
1587. H. A. Kashani, J. A. Barrie, M. H. George, J. Polym. Sci., Polym. Chem. Ed., **16**, 533 (1978).
1588. N. Kashiwa, J. Yoshitake, Makromol. Chem., Rapid Commun., **3**, 211 (1982).
1589. N. Kashiwa, J. Yoshitake, A. Mizuno, T. Tsutmi, Polymer, **28**, 1227 (1987).
1590. A. K. Kashyap, V. Kalpagam, C. Rami Reddy, Polymer, **18**, 878 (1977).
1591. A. K. Kashyap, V. Kalpagam, J. Polym. Sci., Polym. Chem. Ed., **17**, 225 (1979).
1592. A. K. Kashyap, V. Kalpagam, Makromol. Chem., **180**, 1243 (1979).
1593. A. Kaštánek, J. Zelenka, K. Hajek, J. Appl. Polym. Sci., **26**, 4117 (1981).
1594. A. Kaštánek, O. Dufka, I. Dosoudilová, J. Appl. Polym. Sci., **38**, 1883 (1989).
1595. G. Kaszas, J. E. Puskas, C. C. Chen, J. P. Kennedy, Macromolecules, **23**, 3909 (1990).
1596. S. Kataoka, T. Ando, Polymer, **25**, 507 (1984).
1597. H. Katayama, M. Kamigaito, M. Sawamoto, T. Higashimura, Macromolecules, **28**, 3747 (1995).
1598. A. Katchalski, H. Eisenberg, J. Polym. Sci., **6**, 145 (1951).
1599. I. Katime, A. Roig, L. M. León, Anales Quím. (Madrid), **67**, 811 (1971).
1600. I. Katime, A. Roig, P. Gutierrez Cabanas, Eur. Poly. J., **10**, 897 (1974).
1601. I. A. Katime, J. R. Quintana, C. Strazielle, Makromol. Chem., **187**, 1441 (1986).
1602. I. Katime, J. R. Quintana, R. Valenciano, C. Strazielle, Polymer, **27**, 742 (1986).
1603. I. A. Katime, M. B. Huglin, P. M. Sasia, Eur. Polym. J., **24**, 561 (1988).
1604. I. A. Katime, J. R. Quintana, Eur. Polym. J., **24**, 775 (1988).
1605. I. A. Katime, P. M. Sasia, B. Eguia, Eur. Polym. J., **24**, 1159 (1988).
1606. I. A. Katime, J. R. Quintana, Makromol. Chem., **189**, 1373 (1988).
1607. I. A. Katime, J. R. Quintana, L. C. Cesteros, M. C. Peleteiro, Polym. Bull., **21**, 69 (1989).
1608. I. Katime, D. Blanco, J. M. Teijón, Polym. Bull., **21**, 89 (1989).
1609. Y. Kato, S. Kido, T. Hashimoto, J. Polym. Phys. Ed., **11**, 2329 (1973).
1610. Y. Kato, T. Hashimoto, J. Polym. Sci., Polym. Phys. Ed., **12**, 813 (1974).
1611. Y. Kato, K. Kamiya, H. Sasaki, T. Hashimoto, J. Chromatogr. **193**, 311 (1980).
1612. Y. Kato, K. Kamiya, H. Sasaki, T. Hashimoto, J. Chromatogr. **193**, 297 (1980).
1613. T. Kato, K. Miyaso, M. Nagasawa, J. Phys. Chem., **72**, 2161 (1968).
1614. T. Kato, T. Hashimoto, T. Fujimoto, M. Nagasawa, J. Polym. Sci., Polym. Phys. Ed., **13**, 1849 (1975).
1615. T. Kato, T. Katsuki, A. Takahashi, Macromolecules, **17**, 1726 (1984).
1616. T. Kato, T. Tokuya, T. Nozaki, A. Takahashi, Polymer, **25**, 218 (1984)
1617. S. Katuscak, M. Tomas, O. Schiessl, J. Appl. Polym. Sci., **26**, 381 (1981).
1618. H. S. Kaufman, E. Solomon, Ind. Eng. Chem., **45**, 1779 (1953).
1619. H. S. Kaufman, E. K. Walsh, J. Polym. Sci., **26**, 124 (1957).
1620. R. S. Kauga, T. Hogen-Esch, E. Raudrianalimanana, A. Soum, M. Fontanille, Macromolecules, **23**, 4235 (1990).
1621. S. K. Kaumar, V. W. Suter, R. C. Reid, Fluid Phas. Equilib., **29**, 373 (1986).
1622. K. Kawahara, Chem. High Polym. (Japan), **18**, 687 (1961).
1623. K. Kawahara, R. Okada, J. Polym. Sci., **56**, S7 (1962).
1624. K. Kawahara, Makromol. Chem., **73**, 1 (1964).
1625. T. Kawai, A. Keller, J. Polym. Sci. B, **2**, 333 (1964).
1626. T. Kawai, E. Ida, Kolloid. Z.-Z. Polym., **194**, 40 (1964).
1627. T. Kawai, Makromol. Chem., **102**, 125 (1967).
1628. T. Kawai, T. Goto, H. Maeda, Kolloid. Z.-Z. Polym., **223**, 117 (1968).
1629. T. Kawai, K. Ehara, H. Sasano, K. Kamide, Makromol Chem., **111**, 271 (1968).
1630. T. Kawai, T. Hama, K. Ehara, Makromol. Chem., **113**, 282 (1968).

1631. T. Kawai, S. Shiozaki, S. Sonoda. H. Nakagawa, T. Matsumoto, H. Maeda, Makromol. Chem., **128**, 252 (1969).

1632. T. Kawai, M. Hosoi, K. Kamide, Makromol. Chem., **146**, 55 (1971).

1633. W. Kawai, T. Ichihashi, J. Polym. Sci., Polym. Chem. Ed., **12**, 201 (1974).

1634. H. Kawamura, T. Hayashi, Y. Inone, R. Chûjô, Macromolecules, **22**, 2181 (1989).

1635. A. Kawasaki, J. Furukawa, T. Saegusa, M. Mise, T. Tsuruta, Makromol. Chem., **42**, 25 (1960).

1636. P. J. Kay, F. E. Treloar, Makromol. Chem., **175**, 3207 (1974).

1637. H. Kaye, H. J. Chou, J. Polym. Sci., Polym. Phys. Ed., **13**, 477 (1975).

1638. T. Keii, Y. Doi, E. Suzuki, M. Tamura, M. Murata, K. Soga, Makromol. Chem., **185**, 1537 (1984).

1639. G. I. Keim, D. L. Christman, L. R. Kangas, S. K. Keahey, Macromolecules, **5**, 217 (1972).

1640. T. Kelen, D. Schlotterbeck, V. Jaacks, IUPAC Boston, Preprints vol. 11, 649 (1971).

1641. T. Kelen, B. Iván, T. T. Nagy, B. Turcsanyi, F. Tüdös, J. P. Kennedy, Polym. Bull., **1**, 79 (1978).

1642. A. Keller, A. O'Connor, Polymer, **1**, 163 (1960).

1643. A. Keller, Kolloid. Z.-Z. Polym., **231**, 386 (1969).

1644. A. Keller, E. Martuscelli, D. J. Priest, Y. Udagawa, IUPAC Leiden, Preprints vol. **11**, 843 (1970).

1645. A. Keller, F. M. Willmouth, J. Polym. Sci. A-2, **8**, 1443 (1970).

1646. A. Keller, F. M. Willmouth, J. Macromol. Sci., Part B: Phy., **6**, 493 (1972).

1647. S. S. Kelley, T. C. Ward, T. G. Rials, W. G. Glasser, J. Appl. Polym. Sci., **37**, 2961 (1989).

1648. E. C. Kelusky, C. T. Elston, R. E. Murray, Polym. Eng. Sci. **27**, 1562 (1987).

1649. T. C. Kendrick, J. Polym. Sci. A-2, **7**, 297 (1969).

1650. J. P. Kennedy, A. Vidal, J. Polym. Sci., Polym. Chem. Ed., **13**, 1765 (1975).

1651. J. P. Kennedy, M. Nakao, J. Macromol. Sci., Part A: Chem., **11**, 1621 (1977).

1652. J. P. Kennedy, M. Nakao, J. Macromol. Sci., Part A: Chem., A, **12**, 197 (1978).

1653. J. P. Kennedy, S. C. Feinberg, S. Y. Huang, J. Polym. Sci., Polym. Chem. Ed., **16**, 243 (1978).

1654. J. P. Kennedy, S. S. Plamthottam, J. Macromol. Sci., Part A: Chem., **14**, 729 (1980).

1655. J. P. Kennedy, S. Y. Huang, R. A. Smith, J. Macromol. Sci., Part A: Chem., **14**, 1085 (1980).

1656. J. P. Kennedy, R. A. Smith, J. Polym. Sci., Polym. Chem. Ed., **18**, 1539 (1980).

1657. J. P. Kennedy, R. A. Smith, J. Polym. Sci., Polym. Chem. Ed., **18**, 1523 (1980).

1658. J. P. Kennedy, D. Y. L. Chung, A. Guyot, J. Polym. Sci., Polymer Chem. Ed., **19**, 2737 (1981).

1659. J. P. Kennedy, S. C. Guhaniyogi, L. R. Ross, J. Macromol. Sci., Part A: Chem., **18**, 119 (1982).

1660. J. P. Kennedy, S. S. Plamthottam, B. Ivan, J. Macromol. Sci., Part A: Chem., **17**, 637 (1982).

1661. J. P. Kennedy, R. T. Chou, J. Macromol. Sci., Part A: Chem., **18**, 11 (1982).

1662. J. P. Kennedy, S. C. Guhaniyogi, J. Macromol. Sci., Part A: Chem., **18**, 103 (1982).

1663. J. P. Kennedy, R. T. Chou, V. S. C. Chang, J. Macromol. Sci., Part A Chem., **18**, 39 (1982).

1664. J. P. Kennedy, S. S. Plamthottam, Polym. Bull., **7**, 337 (1982).

1665. J. P. Kennedy, M. Hiza, J. Polym. Sci., Polym. Chem. Ed., **21**, 1033 (1983).

1666. J. P. Kennedy, J. Kurian, Macromolecules, **23**, 3736 (1990).

1667. J. P. Kennedy, N. Meguriya, B. Kesler, Macromolecules, **24**, 6572 (1991).

1668. A. S. Kenyon, I. O. Salyer, J. Polym. Sci., **43**, 427 (1960).

1669. A. S. Kenyon, I. O. Salyer, J. E. Kurz, D. R. Brown, J. Polym. Sci. C, **8**, 205 (1965).

1670. W. Kern, H. Kämmerer, G. Dall'Asta, R. Dieck, Makromol. Chem., **6**, 206 (1951).

1671. W. Kern, H. Schmidt, H. S. Von Steinwehr, Makromol. Chem., **16**, 74 (1955).

1672. F. F. Khodzhevanov, N. S. Nametkin, S. G. Durgaryan, I. Ye. Chernyakov, Polym. Sci. USSR, **14**, 1022 (1972); Vysokomol. Soedin. A, **14**, 920 (1972).

1673. A. Kidera, A. Nakajima, Macromolecules, **17**, 659 (1984).

1674. K. Kikukawa, S. Nozakura, S. Murahashi, Polym. J., **3**, 52 (1972).

1675. J. D. Kim. J. B. Soares, G. L. Rempel, B. Monrabal, Control of Molecular Weight and Chemical Composition Distyribution of Polyolefins with supported Metallocene Catalyst. Present at the MetCon-97, USA.

1676. J. K. Kim, C. D. Han, Macromolecules, **25**, 271 (1992).

1677. W. H. Kim, N. B. Kodali, J. Kumar, S. K. Tripathy, Macromolecules, **27**, 1819 (1994).

1678. Y. B. Kim, R. W. Lenz, R. C. Fuller, Macromolecules, **25**, 1852 (1992).

1679. P. O. Kinell, Acta Chem. Scand., **1**, 832 (1947).

1680. P. O. Kinell, Acta Chem. Scand., **1**, 335 (1947).

1681. J. B. Kinsinger, R. E. Hughes, J. Phys. Chem., **63**, 2002 (1959).

1682. J. B. Kinsinger, L. E. Ballard, J. Polym. Sci. A-1, **3**, 3963 (1965).

1683. S. Kinugasa, S. Hattori, K. Ishikawa, J. Appl. Polym. Sci., Appl. Polym. Symp., **52**, 205 (1993).

1684. M. Kioka, H. Makio, A. Mizuno, N. Kashiwa, Polymer, **35**, 580 (1994).

1685. E. Kiran, J. K. Gillham, E. Gipstein, J. Macromol. Sci., Part B: Phys., **9**, 341 (1974).

1686. J. J. Kirkland, S. W. Rementer, W. W. Yau, Anal. Chem., **60**, 610 (1988).

1687. J. J. Kirkland, S. W. Rementer, W. W. Yau, Macromolecules, **38**, 1383 (1989).

1688. J. J. Kirkland, C. H. Dilks Jr., S. W. Rementer, Anal. Chem., **64**, 1295 (1992).

1689. J. J. Kirkland, C. H. Dilks, Anal. Chem., **64**, 2836 (1992).

1690. J. E. Kirsh, T. A. Soos, T. M. Karaputadze, Eur. Polym. J., **15**, 223 (1979).

1691. L. A. Kiseleva, V. M. Men'shov, I. A. Zadorozhnyi, V. A. Ponomarenko, Polym. Sci. USSR, **18**, 1426 (1976); Vysokomol. Soedin. A, **18**, 1243 (1976).
1692. Y. Kishimoto, S. Aoshima, T. Higashimura, Macromolecules, **22**, 3877 (1989).
1693. J. V. Kissin, J. Polym. Sci. Part A, Polym. Chem., **33**, 227 (1995).
1694. T. Kitagawa, J. Sadanosu, T. Norisuye, Macromolecules, **23**, 602 (1990).
1695. J. Klein, E. Killmann, Makromol. Chem., **96**, 193 (1966).
1696. J. Klein, F. Patat, Makromol. Chem., **97**, 189 (1966).
1697. J. Klein, U. Wittenberger, Makromol. Chem., **122**, 1 (1969).
1698. J. Klein, G. Weinhold, Angew. Makromol. Chem., **10**, 49 (1970).
1699. J. Klein, H. Friedel, J. Appl. Polym. Sci., **14**, 1927 (1970).
1700. J. Klein, R. Heitzmann, Makromol. Chem., **179**, 1895 (1978).
1701. J. Klein, U. Standt, W. Kaminsky, Polym. Bull., **1**, 287 (1979).
1702. J. Klein, U. D. Standt, Colloid Polym. Sci., **259**, 977 (1981).
1703. J. Klein, A. Westerkamp, J. Polym. Sci., Polym. Chem. Ed., **19**, 707 (1981).
1704. L. A. Kleintjens, R. Koningsveld, W. H. Stockmayer, Brit. Polym. J., **8**, 144 (1976).
1705. B. M. Klemann, R. West, J. A. Koutsky, Macromolecules, **29**, 198 (1996).
1706. V. I. Klenin, G. G. Uglanova, Polym. Sci. USSR, **11**, 2587 (1969); Vysokomol. Soedin. A, **11**, 2273 (1969).
1707. E. Klespers, W. Hartmann, J. Polymer Sci., Polym. Lett. Ed., **15**, 707, 713 (1977).
1708. N. Th. M. Klooster, F. Van der Touw, M. Mandel, Macromolecules, **17**, 2070 (1984).
1709. A. M. Knebel'man, L. A. Kantor, D. F. Kagan, Polym. Sci USSR, **12**, 3117 (1970); Vysokomol. Soedin. A, **12**, 2745 (1970).
1710. D. C. Knobeloch, L. Wild, SPE Poliolefins IV Conference Preprints, 427 (1984).
1711. J. H. Ko, J. E. Mark, Macromolecules, **8**, 869 (1975).
1712. E. Kobayashi, T. Higashimura, S. Okamura, J. Macromol. Sci., Part A Chem., **1**, 1519 (1967).
1713. E. Kobayashi, S. Okamura, S. Signer, J. Appl. Polym. Sci., **12**, 1661 (1968).
1714. H. Kobayashi, J. Polym. Sci., **26**, 230 (1957).
1715. H. Kobayashi, Y. Fujisaki, J. Polym. Sci. B, **1**, 15 (1963).
1716. H. Kobayashi, K. Sasaguri, Y. Fujisaki, T. Amano, J. Polym. Sci. A-1, **2**, 313 (1964).
1717. T. Kobayashi, Bull. Chem. Soc. Japan, **35**, 726 (1962).
1718. W. Kobryner, A. Benderet, Compt. Rend., **245**, 689 (1957).
1719. R. Kocher, Ch. Sadron, Makromol. Chem., **10**, 172 (1953).
1720. S. M. Kochergin, V. P. Barabanov, I. G. Fedorova, Polym. Sci. USSR, **8**, 1006 (1966); Vysokomol. Soedin., **8**, 916 (1966).
1721. G. Kockelbergh, G. Smets, J. Polym. Sci., **33**, 227 (1958).
1722. H. Koelbel, W. H. E. Mueller, H. Hammer, Makromol. Chem., **70**, 1 (1964).
1723. H. M. Koepp, H. Werrer, Makromol. Chem., **32**, 309 (1959).
1724. I. Koessler, J. Krejsa, J. Polym. Sci., **29**, 69 (1958).
1725. I. Koessler, M. Stolka, J. Polym. Sci., **44**, 213 (1960).
1726. I. Koessler, J. Krejsa, J. Polym. Sci., **57**, 509 (1962).
1727. I. Koesslerr, H. Krauserova, J. Polym. Sci. A-1, **4**, 1329 (1966).
1728. D. W. Koetsier, A. Bantjes, J. Feijen, D. J. Lyman, J. Polym. Sci., Polym. Chem. Ed., **16**, 511 (1978).
1729. G. Kögler, A. Hasenhindlm, M. Möller, Macromolecules, **22**, 490 (1989).
1730. A. Kohler, J. G. Zilliox, P. Rempp, J. Polacek, I. Koessler, Eur. Polym. J., **8**, 627 (1972).
1731. S. Kohjiya, K. Iwata, S. Yamashita, T. Miyamoto, H. Inagaki, Polym. J., **15**, 869 (1983).
1732. H. W. Kohlschuetter, K. Unger, K. Vogel, Makromol. Chem., **93**, 1 (1966).
1733. J. Koivumäk, J. V. Seppäla, Polym. Bull., **31**, 441 (1993).
1734. K. Kojima, M. Sawamoto, T. Higashimura, Macromolecules, **23**, 948 (1990).
1735. V. Kokle, F. W. Billmeyer Jr., J. Polym. Sci. C, **8**, 217 (1965).
1736. E. Kokufuta, Polymer, **21**, 177 (1980).
1737. J. V. Koleske, S. F. Kurath, J. Polym. Sci. A-1, **2**, 4123 (1964).
1738. J. V. Koleske, R. D. Lundberg, J. Polym. Sci. A-2, **7**, 897 (1969).
1739. H. S. Kolesnikov, O. V. Smirnova, V. N. Lamm, A. A. Kudryashov, Polym. Sci. USSR, **11**, 2518 (1969); Vysokomol. Soedin. A, **11**, 2211 (1969).
1740. H. S. Kolesnikov, O. V. Smirnova, A. K. Mikitayev, V. M. Gradyshev, Polym. Sci. USSR, **12**, 1617 (1970); Vysokomol. Soedin. A, **12**, 1424 (1970).
1741. M. Kolínský, V Jisová, D. Lim, IUPAC Helsinki Preprints, Vol. II, 1/88 (1972).
1742. M. Kolínský, J. Janča, J. Polym. Sci., Polym. Chem. Ed., **12**, 1181 (1974).
1743. I. I. Kolodkina, A. M. Yurkevich, Ye. F. Panarin, Polym. Sci. USSR, **18**, 558 (1976); Vysokomol. Soedin. A, **18**, 490 (1976).
1744. Z. Komiya, C. Pugh, R. R. Schrock, Macromolecules, **25**, 6586 (1992).
1745. R. A. Komoroski, R. G. Parker, M. H. Lehr, Macromolecules, **15**, 844 (1982).
1746. G. N. Komratov, R. A. Barzykina, G. V. Korovina, Polym. Sci. USSR, **20**, 686 (1979); Vysokomol. Soedin., A, **20**, 608 (1978).
1747. S. Kondo, T. Nakamatsu, K. Tsuda, J. Macromol. Sci. Chem. A, **11**, 719 (1977).
1748. R. Koningsveld, A. J. Pennings, Rec. Trav. Chim., **83**, 552 (1964).
1749. R. Koningsveld, Fortsch, Hochpolym. Forsch., **7**, 1 (1970).
1750. H. Konishi, Sen-i-Gakkaishi, **17**, 1170 (1961).
1751. Y. Konishi, J. W. Van Nispen, G. Davenport, H. A. Scheraga, Macromolecules, **10**, 1265 (1977).
1752. H. Konishi, Y. Shinagawa, A. Azuma, T. Okano, J. Kiji, Makromol. Chem., **183**, 2941 (1982).
1753. T. Konishi, T. Yoshizaki, J. Shimada, H. Yamakawa, Macromolecules, **22**, 1921 (1989).
1754. J. Kops, H. Spanggaard, Makromol. Chem., **151**, (1972).

1755. J. Kops, H. Spanggaard, Polym. Preprints, **13** (1), 90 (1972).
1756. J. Kops, S. Hvilsted, H. Spanggaard, Macromolecules, **13**, 1058 (1980).
1757. J. Kops, H. Spanggaard, Macromolecules, **15**, 1200 (1982).
1758. J. Kops, H. Spanggaard, Macromolecules, **15**, 1225 (1982).
1759. E. V. Korneeva, I. N. Shtennikova, V. P. Shibaev, S. I. Klenin, G. F. Kolbina, I. V. Ekaeva, J. A. Didenko, Eur. Polym. J., **26**, 781 (1990).
1760. Ye. V. Korneyeva, V. N. Tsvetkov, P. N. Lavrenko, Polym. Sci. USSR, **12**, 1554 (1970); Vysokomol. Sodin. A, **12**, 1369 (1970).
1761. G. V. Korovina, D. Ya, Rossina, D. D. Novikov, S. G. Entelis, Polym. Sci. USSR, **16**, 1471 (1974); Vysokomol. Soedin. A, **16**, 1274 (1974).
1762. V. V. Korshak, S. A. Pavlova, L. V. Boika, T. M. Babchinitser, S. V. Vinogradova, Ya. S. Vygodskii, N. A. Gulebeva, Polym. Sci. USSR, **12**, 63 (1970); Vysokomol. Soedin. A, **12**, 56 (1970).
1763. V. V. Korshak, G. I. Tarasova, S. A. Pavlova, Polym. Sci. USSR, **13**, 1179 (1971); Vysokomol. Soedin. A, **13**, 1047 (1971).
1764. V. V. Korshak, S.-S. A. Pavlova, S. V. Rogozhin, V. A. Davankov, T. N. Sentsova, A. A. Kurganov, G. I. Timofeyeva, Yu. A. Davisovich, Polym. Sci. USSR, **22**, 45 (1980); Vysokomol. Soedin. A, **22**, 41 (1980).
1765. R. A. Korus, R. W. Rosser, Anal. Chem., **50**, 249 (1978).
1766. T. Kotaka, V. Murakami, H. Inagaki, J. Phys. Chem., **72**, 829 (1968).
1767. T. Kotaka, N. Donkai, J. Polym. Sci., A-2, **6**, 1457 (1968).
1768. T. Kotaka, H. Ohnuma, H. Inagaki, Polymer, **10**, 517 (1969).
1769. T. Kotaka, T. Tanaka, H. Ohnuma, Y. Muzakami, H. Inagaki, Polym. J., **1**, 245 (1970).
1770. T. Kotaka, J. L. White, Macromolecules, **7**, 106 (1974).
1771. T. Kotaka, T. Ljda, T. Tanaka, H. Inagaki, Makromol. Chem., **176**, 1273 (1975).
1772. T. Kotaka, Angew. Makromol. Chem., **56**, 77 (1976).
1773. A. Kotera, T. Saito, Y. Watanabe, M. Ohame, Makromol. Chem., **87**, 195 (1965).
1774. J. Kovářová-Lerchová, J. Pilař, G. Samay, J. Pospíšil, Eur. Polym. J., **14**, 601 (1978).
1775. B. M. Kovarskaya, L. I. Golubenkova, M. S. Akutin, I. I. Levantovskaya, Vysokomol. Soedin., **1**, 1042 (1959).
1776. M. Kowalczuk, G. Adamus, Z. Jedlinski, Macromolecules, **27**, 572 (1994).
1777. M. Kozhokaryu, V. S. Skazka, K. G. Berdnikova, Polym. Sci. USSR, **8**, 1167 (1966); Vysokomol. Soedin., **8**, 1063 (1966).
1778. P. V. Kozlov et al., Vysokomol. Soedin., **2**, 1575 (1960).
1779. J. J. Krackeler, H. Naidus, J. Polym. Sci. C, **27**, 207 (1969).
1780. D. Kranz, Kolloid. Z.-Z. Polym., **227**, 41 (1968).
1781. D. Kranz, H. Uwe-Pohl, H. Baumann, Angew. Makromol. Chem., **26**, 59 (1972).
1782. F. Krasovek, J. Stefan, Inst. Ljubljana 203 (1956).
1783. G. Kraus, C. J. Stacy, J. Polym. Sci. A-2, **10**, 657 (1972).
1784. S. Krause, J. Phys. Chem., **65**, 1618 (1961).
1785. S. Krause, E. Cohn-Ginsberg, Polymer, **3**, 566 (1962).
1786. S. Krause, E. Cohn-Ginsberg, J. Polym. Sci. A-1, **2**, 1393 (1964).
1787. S. Krause, D. J. Dunn, A. Seyed-Mozzaffari, A. M. Biswas, Macromolecules, **10**, 786 (1977).
1788. S. Krause, Z. Lu, M. Iskandar, Macromolecules, **15**, 1076 (1982).
1789. J. Krejse, Makromol. Chem., **33**, 244 (1959).
1790. H.-J. Kress, W. Stix, W. Heitz, Makromol. Chem., **185**, 173 (1984).
1791. H. R. Kricheldorf, Q.-Z. Zang, G. Schwarz, Polymer, **23**, 1821 (1982).
1792. H. R. Kricheldorf, G. Bier, Polymer, **25**, 1151 (1984).
1793. H. R. Kricheldorf, M. V. Sumbél, I. Kreiser-Saunders, Macromolecules, **24**, 1944 (1991).
1794. H. R. Kricheldorf, S.-R. Lee, N. Scharnagl, Macromolecules, **27**, 3139 (1994).
1795. H. R. Kricheldorf, B. Weegen-Schulz, J. Polym. Sci. Part A, Polym. Chem., **33**, 2193 (1995).
1796. W. R. Krigbaum, P. J. Flory, J. Am. Chem. Soc., **75**, 1775 (1953).
1797. W. R. Krigbaum, K. J. Krause, J. Polym. Sci., Polym. Chem. Ed., **16**, 3151 (1978).
1798. W. R. Kribaum, T. Tanaka, Macromolecules, **21**, 743 (1988).
1799. W. R. Krigbaum, G. Brelsord, G. Macromolecules, **21**, 2502 (1988).
1800. W. R. Krigbaum, A. M. Kotliar, J. Polym. Sci., **32**, 323 (1958).
1801. E. Kroeze, G. Ten Brinke, G. Hadziioannou, Macromolecules, **28**, 6650 (1995).
1802. E. Kroeze, G. Ten Brinke, G. Hadziioannou, J. Macromol. Sci. A, Pure Appl. Chem., A, **34**, 439 (1997).
1803. H. Krömer, R. Kuhn, H. Pielartzik, W. Siebke, V. Eckhardt, M. Schmid, Macromolecules, **24**, 1950 (1991).
1804. S. Krozer, M. Vainryb, L. Silina, Vysokomol. Soedin., **2**, 1876 (1960).
1805. S. Krozer, P. Carstensen, M. Bodelsen, A. Thirsing, Makromol. Chem., **175**, 1893 (1974).
1806. M. Kubin, P. Špacek, R. Chromecek, Coll. Czech. Chem. Commun., **32**, 3881 (1967).
1807. P. Kubisa, S. Penczek, Makromol. Chem., **179**, 445 (1978).
1808. K. Kubo, K. Ogino, Sci. Pap. Coll. Gen. Educ., **16**, 193 (1966).
1809. H. Kubota, Y. Ogiwara, K. Matsuzaki, J. Polym. Sci., Polym. Chem. Ed., **12**, 2809 (1974).
1810. K. Kubota, B. Chu, Macromolecules, **16**, 105 (1983).
1811. Z. Küçükyavuz, S. Küçükyavuz, Eur. Polym. J., **14**, 867 (1978).
1812. L. G. Kudryavtseva, A. D. Litmanovich, A. V. Topchiyev, V. Ya. Shtern, Neftekhim., **3**, 343 (1963).
1813. J. Kugler, E. W. Fisher, M. Peuscher, C. D. Eisenbach, Makromol. Chem., **184**, 2325 (1983).
1814. S. Kühling, H. Keul, H. Höcker, Macromolecules, **23**, 4192 (1990).
1815. R. Kuhn, H. Kroemer, G. Rossmanith, Angew. Makromol. Chem., **40/41**, 361 (1974).
1816. R. Kuhn, Makromol. Chem., **177**, 1525 (1976).

1817. R. Kuhn, Makromol. Chem., **181**, 725 (1980).
1818. R. Kuhn, Makromol. Chem., **185**, 1003 (1984).
1819. R. Kuhn, J. Appl. Polym. Sci., Appl. Polym. Symp., **52**, 19 (1993).
1820. E. Kukufuta, S. Fujii, Y. Harai, I. Nakamura, Polymer, **23**, 452 (1982).
1821. A. M. Kulakova, I. K. Nekrasov, Polym. Sci. USSR, **18**, 146 (1976); Vysokomol. Soedin. A, **18**, 127 (1976).
1822. W. M. Kulicke, M. Elsabee, C. D. Eisenbach, M. Peuscher, Polym. Bull., **9**, 190 (1983).
1823. L. I. Kulin, N. L. Meijerink, P. S. Sturck, Pure Appl. Chem., **60**, 1404 (1988).
1824. V. N. G. Kumar, K. Hummel, Angew. Makromol. Chem., **102**, 167 (1982).
1825. V. G. Kumar, M. Rama Rao, P. V. Ravindrau, K. V. C. Rao, J. Polym. Sci., Polym. Chem. Ed., **21**, 2133 (1983).
1826. M. Kunioka, Y. Doi, Macromolecules, **23**, 1933 (1990).
1827. T. Kunitake, K. Takarabe, S. Tsugawa, Polym. J., **8**, 363 (1976).
1828. C.-C. Kuo, A. E. Woodward, Macromolecules, **17**, 1034 (1984).
1829. W. Kuran, A. Wieslochowski, Polym. Bull., **2**, 411 (1980).
1830. N. Kuroda, Y. Nishitani, K. Matsuura, M. Miyooshi, Makromol. Chem., **188**, 1897 (1987).
1831. N. Kuroda, Y. Nishitani, K. Matsuura, N. Ikegami, Macromolecules, **25**, 2820 (1992).
1832. J. E. Kurz, J. Polym. Sci. A-1, **3**, 1895 (1965).
1833. M. M. Kusakov, A. Yu. Koshevnik, D. N. Nekrasov, V. F. Chirkova, L. M. Shol'pina, Polym. Sci. USSR, **8**, 1141 (1966); Vysokomol. Soedin., **8**, 1040 (1966).
1834. P. P. Kushch, A. I. Kuzayev, B. A. Komarov, B. A. Rozenberg, Polym. Sci, USSR, **22**, 2205 (1980); Vysokomol. Soedin. A, **22**, 2012 (1980).
1835. N. Kuwahara, M. Kaneko, Makromol. Chem., **82**, 205 (1965).
1836. N. Kuwahara, M. Kaneko, K. Kubo, Makromol. Chem., **110**, 294 (1967).
1837. N. Kuwahara, T. Okazawa, M. Kaneko, J. Polym. Sci. C, **23**, 543 (1968).
1838. N. Kuwahara, S. Higashida, M. Nakata, M. Kaneko, J. Polym. Sci. A-2, **7**, 285 (1969).
1839. N. Kuwahara, S. Saeki, T. Chiba, M. Kaneko, Polymer, **15**, 777 (1974).
1840. A. I. Kuzayev, G. N. Komratov, G. V. Korovina, S. G. Entelis, Polym. Sci. USSR, **12**, 1124 (1970); Vysokomol. Soedin. A, **12**, 995 (1970).
1841. A. I. Kuzayev, G. N. Komratov, G. V. Korovina, S. G. Entelis, Polym. Sci. USSR, **12**, 1167 (1970); Vysokomol. Soedin. A, **12**, 1033 (1970).
1842. A. I. Kuzayev, V. A. Linde, Ya. I. Estrin, N. A. Afanasyev, S. M. Baturin, S. G. Entelis, Polym. Sci. USSR, **18**, 667 (1976); Vysokomol. Soedin. A, **18**, 585 (1976).
1843. A. I. Kuzayev, Polym. Sci. USSR, **22**, 2284 (1980); Vysokomol. Soedin. A, **22**, 2082 (1980).
1844. V. N. Kuznetsov, V. B. Kogan, L. A. Venkstern, T. A. Usatova, V. A. Morozova, Polym. Sci. USSR, **12**, 3143 (1970); Vysokomol. Soedin. A, **12**, 2768 (1970).
1845. Y. Kyogoku, Techn. Rept. Osaka Univ., **7**, 147 (1958), quoted by R. E. Glegg, L. J. Tanghe, R. J. Brewer, in "Fractionation", in "Cellulose and Cellulose Derivatives", Vol. V, Part IV, Editors, N. M. Bikales, L. Segal, Wiley-Interscience, New York, 1971.
1846. Y. Kyoo Han, M. V. Jin, S. K. Choi, J. Polym. Sci., Polym. Chem. Ed., **21**, 73 (1983).
1847. M. Kyotani, H. Kanetsuna, Keukyu Hokoku-Sen'i Kobunshi Zairo Kenkyusho, **172**, 37 (1992).
1848. F. Laborie, J. Polym. Sci., Polym. Chem. Ed., **15**, 1275 (1977).
1849. M. G. Lachtermarcher, A. Rudin, J. Appl. Polym. Sci., **58**, 2077 (1995).
1850. J. H. Lai, Macromolecules, **17**, 1010 (1984).
1851. S.-T. Lai, L. Saugermano, J. Chromatog., **322**, 338 (1985).
1852. D. Lala, J. F. Rabek, B. Rånby, Eur. Polym. J., **16**, 735 (1980).
1853. A. Lambert, Polymer, **10**, 213 (1969).
1854. A. Lambert, Brit. Polym. J., **3**, 13 (1971).
1855. F. Lambert, M. Milas, M. Rinaudo, Polym. Bull., **7**, 185 (1982).
1856. R. F. Landel, J. D. Ferry, J. Phys. Chem., **59**, 658 (1955).
1857. I. Landler, Compt. Rend., **225**, 629 (1947).
1858. H. Lange, W. Scholtan, Angew. Makromol. Chem., **26**, 59 (1972).
1859. H. Lange, H. Baumann, Angew. Makromol. Chem., **43**, 167 (1975).
1860. W. L. Langford, D. J. Vaughan, Nature (London), **184**, 116 (1959).
1861. G. Langhammer, Kolloid. Z., **146**, 44 (1956).
1862. G. Langhammer, Makromol. Chem., **21**, 74 (1956).
1863. G. Langhammer, J. Polym. Sci., **29**, 505 (1958).
1864. G. Langhammer, H. Pfennig, K. Quitzsch, Z. elektrochem., **62**, 458 (1958).
1865. G. Langhammer, H. Foerster, Z. Physik. Chem. N. F. (Frankfurt), **15**, 212 (1958).
1866. G. Langhammer, K. Quitzsch. Makromol. Chem., **43**, 160 (1961).
1867. G. Langhammer, H. Seide, Kolloid. Z.-Z. Polym., **216/217**, 264 (1967).
1868. H. G. Langhammer, Svensk. Kem. Tidskr., **69**, 328 (1957).
1869. H. Langner, Makromol. Chem., **119**, 37 (1968).
1870. J. Lániková, M. Hloušek, Eur. Polym. J., **6**, 25 (1970).
1871. J. Lániková, M. Hloušek, M. Nováková, Pardubice Symposium Preprints, 299 (1970).
1872. H. M. Lanovskaya, N. A. Pravikova, Polym. Sci. USSR, **11**, 2208 (1969); Vysokomol. Soedin. A, **11**, 1938 (1969).
1873. E. Lanza, H. Berghmans, G. Smets, J. Polym. Sci., Polym. Phys. Ed., **11**, 95 (1973).
1874. A. Lapp, G. Beinert, C. Picot, Makromol. Chem., **185**, 453 (1984).
1875. R. Larrain, L. H. Tagle, F. R. Diaz, Polym. Bull., **4**, 487 (1981).
1876. R. D. Law, Polym. Preprints, **8** (2), 1348 (1967).
1877. R. D. Law, J. Polym. Sci. C, **21**, 225 (1968).
1878. R. D. Law, J. Polym. Sci. A-1, **7**, 2097 (1969).
1879. R. D. Law, J. Polym. Sci. A-1, **9**, 589 (1971).

1880. R. D. Law, J. Polym. Sci., Polym. Chem. Ed., **11**, 175 (1973).
1881. J. M. Lawter, R. Sun, W. B. Banks, J. Agric. Food Chem., **43**, 667 (1995).
1882. S. Lazcano, C. Marco, J. G. Fatou, A. Bello, Eur. Polym. J., **24**, 991 (1988).
1883. S. Lazcano, A. Bello, C. Marco, J. G. Fatou, Eur. Polym. J., **24**, 985 (1988).
1884. M. Le Page, R. Beau, A. J. de Vries, Polym. Preprints **8** (2), 1211 (1967).
1885. M. Le Page, R. Beau, A. J. de Vries, J. Polym. Sci. C, **21**, 119 (1968).
1886. V. P. Lebedev. D. Ya. Tsvankin, Yu. V. Glazkovskii, Polym. Sci. USSR, **14**, 1123 (1972); Vysokomol. Soedin. A, **14**, 1010 (1972).
1887. R. G. Lebel, D. A. I. Goring, J. Polym. Sci. C, **2**, 29 (1963).
1888. P. Lebel, C. Job, J. Polym. Sci. C, **4**, 649 (1963).
1889. A. Leborgne, S. L. Malhotra, L.-P. Blanchard, J. Polym. Sci., Polymer Chem. Ed., **14**, 2853 (1976).
1890. J.-J. Lebrun, G. Sauvet, P. Sigwalt, Makromol. Chem., Rapid Commun., **3**, 757 (1982).
1891. G. Lechermeier, A. Revillon, G. Pillot, J. Appl. Polym. Sci., **19**, 1989 (1975).
1892. G. Lechermeier, G. Pillot, J. Gole, A. Revillon, J. Appl. Polym. Sci., **19**, 1979 (1975).
1893. J. Lecourtier, G. Chaureteau, Macromolecules, **17**, 1340 (1984).
1894. M. Ledbetter, J. Cucolo, P. Tucker, J. Polym. Sci., Polym. Chem. Ed., **22**, 1435 (1984).
1895. C. L. Lee, F. A. Emerson, J. Polym. Sci. A-2, **5**, 829 (1967).
1896. C. L. Lee, C. L. Frye, O. K. Johannson, Polym. Preprints, **10** (2), 1361 (1969).
1897. C. L. Lee, O. K. Johannson, J. Polym. Sci., Polym. Chem. Ed., **14**, 729 (1978).
1898. H. Ch. Lee, M. Ree, T. Chang, Polymer, **36**, 2215 (1995).
1899. H. S. J. Lee, W. P. Weber, Macromolecules, **24**, 4749 (1991).
1900. K. S. Lee, V. T. Stannett, R. D. Gilbert, J. Polym. Sci., Polym. Chem. Ed., **20**, 997 (1982).
1901. N.-J. Lee, Y.-A. Kim, S.-H. Kim, W.-M. Choi, J. Macromol. Sci. A: Pure Appl. Chem., **34**, 1 (1997).
1902. S. Lee, J. C. Giddings, Anal. Chem., **60**, 2328 (1988).
1903. S. Lee, M. N. Myers, R. Beckett, J. C. Giddings, Anal. Chem., **60**, 1129 (1988).
1904. S. Lee, A. Molnar, Macromolecules, **28**, 6354 (1995).
1905. P. M. Lefebvre, R. Jérôme, Ph. Teyssié, J. Polym. Sci., Chem. Ed., **21**, 789 (1983).
1906. R. Legras, C. Bailly, M. Daumerie, J. M. Dekoninck, J. P. Mercier, V. Zichy, E. Nield, Polymer, **25**, 835 (1984).
1907. A. Lehtinen, R. Paukkeri, Makromol. Chem. Phys., **195**, 1539 (1994).
1908. G. Y. Lei, J. Polym. Sci., Polym. Chem. Ed., **19**, 389 (1981).
1909. P. J. Lemstra, G. Challa, J. Polym. Sci., Polym. Chem. Ed., **13**, 1809 (1975).
1910. J.-C. Lenain, J.-C. Brosse, Makromol. Chem., Rapid Commun., **3**, 609 (1982).
1911. R. W. Lenz, L. C. Westfelt, J. Polym. Sci., Polym. Chem. Ed., **14**, 2147 (1976).
1912. A. León, L. Gargallo, A. Horta, D. Radić, J. Polym. Sci., Part B: Polym. Phys., **27**, 2337 (1989).
1913. A. León, L. Gargallo, D. Radić, A. Horta, Polymer, **32**, 761 (1991).
1914. A. León, L. Gargallo, D. Radić, J. Bravo, A. Horta, Makromol. Chem. **193**, 393 (1992).
1915. L. M. León, I. Katime, M. Gonzalez, J. Figueruelo, Eur. Polym. J., **14**, 671 (1978).
1916. J. Leonard, H. Daoust, J. Phys. Chem., **69**, 1174 (1965).
1917. J. Leonard, S. L. Malhotra, J. Macromol. Sci., Part A: Chem., **10**, 1279 (1976).
1918. J. Leonard, S. L. Malhotra, J. Macromol. Sci., Part A: Chem., **11**, 2087 (1977).
1919. K.-W. Leong, G. B. Butler, J. Macromol. Sci., Part A: Chem., **14**, 287 (1980).
1920. C. Lepilleur, E. J. Beckman, Macromolecules, **30**, 745 (1997).
1921. J. B. Lesh, K. Schultz, J. D. Porsche, Ind. Eng. Chem., **42**, 1376 (1950).
1922. H. Leth Pedersen, J. Polym. Sci. B, **5**, 239 (1967).
1923. M. Levi, S. Turri, J. Appl. Polym. Sci., **55**, 989 (1995).
1924. G. Levin, J. B. Carmichael, J. Polym. Sci. A-1, **6**, 1 (1968).
1925. R. Lew, D. Suwanda, S. T. Balke, T. H. Mourey, J. Appl. Polym. Sci., Appl. Polym. Symp., **52**, 125 (1993).
1926. D. Li, R. Faust, Macromolecules, **28**, 4893 (1995).
1927. D. Li, R. Faust, Macromolecules, **28**, 1383 (1995).
1928. R. H. Liang, F.-D. Tsay, A. Gupta, Macromolecules, **15**, 974 (1982).
1929. T.-P. Liao, J. P. Kennedy, Polym. Bull., **7**, 233 (1982).
1930. Ch. X. Liao, W. P. Weber, Macromolecules, **25**, 1639 (1992).
1931. F. C. Lin, S. S. Stivala, J. A. Biesenberger, J. Appl. Polym. Sci., **17**, 1073 (1973).
1932. J. J. Lindberg, P. Sterck, B. Hortling, Suomen Kemistilehti. B, **42**, 451 (1969).
1933. C. V. Linden, Polymer, **21**, 171 (1980).
1934. A. D. Litmanovich, V. Ya. Shtern, Polym. Sci. USSR, **7**, 1478 (1965); Vysokomol. Soedin., **7**, 1332 (1965).
1935. A. A. Litmanovich, I. M. Papisov, V. A. Kabanov, Polym. Sci. USSR, **22**, 1297 (1980); Vysokomol. Soedin. A, **22**, 1180 (1980).
1936. J. N. Little, J. L. Waters, K. J. Bombaugh, W. J. Pauplis. Polym. Preprints **10** (1), 326 (1969).
1937. H. Q. Liu, V. T. Stannett, Macromolecules, **23**, 140 (1990).
1938. G. Liu, N. Hu, X. Xu, H. Yao, Macromolecules, **27**, 3892 (1994).
1939. N. Ljungqvist, T. Hjerberg, Macromolecules, **28**, 5993 (1995).
1940. G. H. Llinas, S.-H. Dong, D. T. Mallin, M. D. Rauseh, Y.-G. Lin, H. H. Winter, J. C. W. Chien, Macromolecules, **25**, 1242 (1992).
1941. J. Llopis, A. Albert, P. Usobiaga, Anales Real Soc. Esp. Fis. Quím. B, **62**, 1103 (1966).
1942. J. Llopis, A. Albert, P. Usobiaga, Anales Real Soc. Esp. Fis. Quím. B, **63**, 413 (1967).

1943. J. Llopis, A. Albert, P. Usobiaga, Eur. Polym. J., **3**, 259 (1967).
1944. M. A. Llorente, E. Riande, J. Guzmán, Macromolecules, **17**, 1048 (1984).
1945. P. Longi, F. Greco, U. Rossi, Makromol. Chem., **116**, 113 (1968).
1946. P. Locatelli, M. C. Sacchi, E. Rigamonti, A. Zambelli, Macromolecules, **17**, 123 (1984).
1947. P. Locatelli, M. C. Sacchi, I. Tritto, F. Forlini, Macromolecules, **23**, 2406 (1990).
1948. J. D. Loconti, J. W. Cahill, J. Polym. Sci., **49**, S2 (1961).
1949. J. D. Loconti, J. W. Cahill, J. Polym. Sci. A, **1**, 3163 (1963).
1950. G. Loehr, G. V. Schulz, Makromol. Chem., **117**, 283 (1968).
1951. G. Loehr, A. H. E. Mueller, V. Warzelhan, G. V. Schulz, Makromol. Chem., **175**, 497 (1973).
1952. A. L. Logothetis, J. M. Mc Kenna, J. Polym. Sci., Polym. Chem. Ed., **16**, 2797 (1978).
1953. F. Lombard, Makromol. Chem., **8**, 201 (1952).
1954. J. D. Londono, A. H. Narten, G. D. Wignall, K. G. Honnell, E. T. Hsieh, T. W. Johnson, F. S. Bates, Macromolecules, **27**, 2864 (1994).
1955. V. C. Long, G. C. Berry, L. M. Hobbs, Polymer, **5**, 517 (1964).
1956. P. Longi, A. Roggero. Ann. Chim. (Rome), **51**, 1013 (1961).
1957. P. Longo, A. Grassi, L. Oliva, P. Ammendola, Makromol. Chem., **191**, 237 (1990).
1958. B. E. López Madruga, J. M. Barrales-Rienda, G. M. Guzmán, Anales Quím. (Madrid), **67**, 153 (1971).
1959. B. V. Losikov, N. I. Kaverina, A. A. Fedyantseva, Khim. Tekhnol. Topliva, **3**, 51 (1956).
1960. C. Loucheux, Z. Czlonkowska, J. Polym. Sci. C, **16**, 4001 (1968).
1961. P. A. Lovell, R. H. Still, Makromol. Chem., **188**, 1561 (1987).
1962. E. G. Lovering, W. B. Wright, J. Polym. Sci. A-1, **6**, 2221 (1968).
1963. E. G. Lovering, J. Polym. Sci. C, **30**, 329 (1970).
1964. E. G. Lovering, D. C. Wooden, Polym. Preprints, **11** (2), 935 (1970).
1965. A. J. Lovinger, Polymer, **21**, 1317 (1980).
1966. I. L. Lovric, J. Polym. Sci. A-2, **7**, 1357 (1969).
1967. I. L. Lovric, J. Polym. Sci. A-2, **8**, 807 (1970).
1968. L. Lovrić, Z. Grubišic-Gallot, B. Kunst, Eur. Polym. J., **12**, 189 (1976).
1969. Lj. Lovrić, IUPAC Helsinki Preprints, Vol. III, II-41 (1972).
1970. H. Lu, S. Xiao, Makromol. Chem., **194**, 2095 (1993).
1971. H. Luetje, Z. Physik. Chem. N. F. (Frankfurt), **43**, 11 (1964).
1972. P. L. Luisi, E. Chiellini, P. F. Franchini, M. Orienti, Makromol. Chem., **112**, 197 (1968).
1973. P. L. Luisi, F. Pezzana, Eur. Polym. J., **6**, 259 (1970).
1974. R. Lukáš, J. Janča L. Mrkvicková, S. Pokorný, M. Kolínský, J. Polym. Sci., Polym. Chem. Ed., **16**, 2539 (1978).
1975. S. Luňák, M. Bohdanecký, Coll. Czech, Chem. Commun., **30**, 2756 (1965).
1976. P. Lutz, G. Beinert, P. Rempp, Makromol. Chem., **183**, 2787 (1982).
1977. A. R. Luxton, A. Quig, M.-J. Delvaux, L. J. Fetters, Polymer, **19**, 1320 (1978).
1978. M. M. Lynn, V. T. Stannett, R. G. Gilbert, J. Polym. Sci., Polym. Chem. Ed., **18**, 1967 (1980).
1979. A. R. Lyons, E. Catterall, Makromol. Chem., **146**, 141 (1971).
1980. R. A. Lyons, J. Hutovic, M. C. Piton, D. I. Christie, P. A. Clay, B. G. Manders, S. H. Kable, R. G. Gilbert, Macromolecules, **29**, 1918 (1996).
1981. B. J. Lyons, A. S. Fox, Polym. Preprints, **8** (2), 1241 (1967).
1982. B. J. Lyons, A. S. Fox, J. Polym. Sci. C, **21**, 159 (1968).
1983. S. Ya. Lyubina, V. S. Skazka, I. A. Strelina, G. V. Tarasova, V. M. Yamschikov, Polym. Sci. USSR, **14**, 1539 (1972); Vysokomol. Soedin. A, **14**, 1371 (1972).
1984. D. Mac Callum, Makromol. Chem., **100**, 117 (1967).
1985. S. Machi, J. Silverman, D. J. Metz, J. Phys. Chem., **76**, 930 (1972).
1986. J. Q. G. Maclaine, C. Booth, Polymer, **16**, 191 (1975).
1987. J. Q. G. Maclaine, P. C. Ashman, C. Booth, Polymer, **17**, 109 (1976).
1988. C. Macosko, K. E. Weale, Polym. Preprints, **10** (2), 562 (1969).
1989. T. B. MacRury, M. L. Mc Connell, J. Appl. Polym. Sci., **24**, 651 (1979).
1990. E. Maderek, G. R. Strobl, Colloid Polym. Sci., **261**, 471 (1983).
1991. E. L. Madruga, J. San Román, M. J. Rodriguez, Anales Quim. (Madrid), **79**, 188 (1983).
1992. E. L. Madruga, J. San Román, J. Macromol. Sci., Part A: Chem., **21**, 167 (1984).
1993. W. Maechtle, Makromol. Chem., **183**, 2515 (1982).
1994. Y. Maeda, K. Nakayama, H. Kanetsuna, Polym. J., **14**, 295 (1982).
1995. C. Maes, P. Godard, D. Daoust, R. Legras, C. Strazielle, Makromol. Chem. Phys, **196**, 1523 (1995).
1996. P. L. Magagnini, F. Andruzzi, G. F. Benetti, Macromolecules, **13**, 12 (1980).
1997. S. Ya. Magarik, G. M. Pavlov, G. A. Fomin, Macromolecules, **11**, 294 (1978).
1998. S. Mah, Y. Yamamoto, K. Hayashi, J. Polym. Sci., Polym. Chem. Ed., **20**, 1709 (1982).
1999. H. K. Mahabadi, L. Alexandru, Can. J. Chem., **63**, 221 (1985).
2000. A.-F. de Mahieu, D. Daoust, J. Devaux, M. de Valck, Eur. Polym. J., **28**, 685 (1992).
2001. B. Maitra, A. K. Nandí, Polymer, **34**, 1260 (1993).
2002. J. Majer, Makromol. Chem., **86**, 253 (1965).
2003. S. B. Makarova, Ye. A. Gukasova, A. I. Kuzayev, S. P. Davtyan, Polym. Sci. USSR, **18**, 3141 (1976); Vysokomol. Soedin. A, **18**, 2747 (1976).
2004. A. I. Maklakov, E. I. Nagumanova, Polym. Sci. USSR, **7**, 2302 (1965); Vysokomol. Soedin., **7**, 2101 (1965).
2005. N. V. Makletsova, I. V. Epel'baum, B. A. Rozenberg, Ye. B. Lyudvig, Polym. Sci. USSR, **7**, 73 (1965); Vysokomol. Soedin., **7**, 70 (1965).
2006. N. G. Mal'tseva, I. F. Tibanov, E. M. Aisenshtein, Khim. Volokna., 26 (1971).

2007. S. L. Malhotra, C. Baillet, H. H. Lam-Tran, L. P. Blanchard, J. Macromol. Sci., Part A: Chem., **12**, 103 (1976).

2008. S. L. Malhotra, J. Leonard, E. Ahad, Eur. Polym. J., **14**, 257 (1978).

2009. S. L. Malhotra, A. Leborgne, L.-P. Blanchard, J. Polym. Sci., Polym. Chem. Ed., **16**, 561 (1978).

2010. S. L. Malhotra, P. Lessard, L. Minh, L. P. Blanchard, J. Macromol. Sci., Part A: Chem., **14**, 915 (1980).

2011. S. L. Malhotra, J. M. Gauthier, J. Macromol. Sci., Part A: Chem., **18**, 783 (1982).

2012. S. L. Malhotra, J. Macromol. Sci., Part A: Chem., **18**, 1055 (1982/83).

2013. S. L. Malhotra, M. Breton, J. M. Gauthier, J. Macromol. Sci., Part A: Chem., **18**, 1151 (1982/83).

2014. B. Malm, J. J. Lindberg, Makromol. Chem., **182**, 2747 (1981).

2015. C. P. Malone, H. L. Suchan, W. W. Yau, J. Polym. Sci. B, **7**, 781 (1969).

2016. C. Maltesh, P. Somasundaran, Pradip, R. A. Kulkarni, S. Gundiah, Macromolecules, **24**, 5775 (1991).

2017. P. V. Mangalam, V. Kalpagam, J. Polym. Sci., Phys. Ed., **20**, 773 (1982).

2018. P. V. Manglam, V. Kalpagam, Polymer, **23**, 991 (1982).

2019. D. Mangaraj, S. K. Patra, Makromol. Chem., **107**, 230 (1967).

2020. D. Mangaraj, S. B. Rath, Polym. Preprints, **12** (1), 573 (1971).

2021. P. K. Manna, P. K. Choudhury, J. Polym. Sci., Polym. Chem. Ed., **10**, 77 (1972).

2022. P. K. Manna, P. K. Choudhury, J. Polym. Sci., Polym. Chem. Ed., **10**, 2159 (1972).

2023. P. K. Manna, P. K. Choudhury, J. Macromol. Sci. A: Chem., **8**, 909 (1974).

2024. N. E. Manolova, I. Gitsov, R. S. Velichkova, I. B. Rashkov, Polym. Bull., **13**, 285 (1985).

2025. J. A. Manson, G. J. Arquelte, Makromol. Chem., **37**, 187 (1960).

2026. P. Månsson, J. Polym. Sci., Polym. Chem. Ed., **18**, 1945 (1980).

2027. K. Mara, T. Imoto, Kolloid, Z.-Z. Polym., **237**, 297 (1970).

2028. C. Marco, A. Bello, J. Gómez Fatou, Makromol. Chem., **179**, 1333 (1978).

2029. C. Marco, J. G. Fatou, A. Bello, A. Blanco, Polymer, **20**, 1250 (1979).

2030. C. Marco, J. Gómez Fatou, A. Bello, A. Blanco, Makromol. Chem., **181**, 1357 (1980).

2031. C. Marco, J. M. Pereña, A. Bello, J. G. Fatou, Brit. Polym. J., **14**, 117 (1982).

2032. C. Marco, A. Bello, J. G. Fatou, Eur. Polym. J., **17**, 41 (1983).

2033. C. Marco, J. G. Fatou, A. Bello, A. López, Brit. Polym. J., **16**, 11 (1984).

2034. C. Marco, J. G. Fatou, A. Bello, J. M. Pereña, Makromol. Chem., **185**, 1255 (1984).

2035. C. Marco, A. Bello, J. G. Fatou, J. Garza, Makromol. Chem., **187**, 177 (1986).

2036. C. Marco, J. G. Fatou, M. A. Gómez, H. Tanaka, A. E. Tonelli, Macromolecules, **23**, 2183 (1990).

2037. D. Mardare, K. Matyjaszewski, Macromolecules, **27**, 645 (1994).

2038. S. Margel, A. Rembaum, Macromolecules, **13**, 19 (1980).

2039. P. Marie, J. P. Lingelser, Y. Gallot, Makromol. Chem., **183**, 2961 (1982).

2040. N. G. Marina, Yu. B. Monakov, S. R. Rafikov, V. P. Budtov, N. V. Duvakina, O. A. Ponomarev, Yu. I. Lysikov, E. S. Filatova, Polym. Sci. USSR, **18**, 620 (1976); Vysokomol. Soedin. A, **18**, 542 (1976).

2041. C. Marize, F. Oliveria, A. S. Gomez, Polym. Bull., **22**, 401 (1989).

2042. J. E. Mark, R. A. Wessling, R. E. Hughes, J. Phys. Chem., **70**, 1895 (1966).

2043. J. E. Mark, G. B. Thomas, J. Phys, Chem., **70**, 3588 (1966).

2044. J. E. Mark, G. B. Thomas, Polym. Preprints, **7** (2), 438 (1966).

2045. M. J. Marks, J. K. Sekinger, Macromolecules, **27**, 4106 (1994).

2046. K. Marquardt, P. Mehnert, Angew. Makromol. Chem., **53**, 125 (1976).

2047. W. Marrs, R. H. Peters, R. H. Still, J. Appl. Polym. Sci., **23**, 1063 (1979).

2048. C. A. Marshali, R. A. Mock, J. Polym. Sci., **17**, 591 (1955).

2049. A. Marshall, R. H. Mobbs, C. Booth, Eur. Polym. J., **16**, 881 (1980).

2050. G. Martinez, J. Millán, M. Bert, A. Michel, A. Guyot, J. Macromol. Sci., Part A: Chem., **12**, 489 (1978).

2051. M. Marx-Figini, J. Polym. Sci., **30**, 119 (1958).

2052. M. Marx-Figini, Makromol. Chem., **32**, 233 (1959).

2053. M. Marx-Figini, G. V. Schulz, Makromol. Chem., **62**, 49 (1963).

2054. M. Marx-Figini, E. Penzel, Makromol. Chem., **87**, 307 (1965).

2055. M. Marx-Figini, J. Polym. Sci. C, **28**, 57 (1969).

2056. M. Marx-Figini, B. Pion, Makromol. Chem., **177**, 1013 (1976).

2057. M. Marx-Figini, O. Soubelet, Polym. Bull., **6**, 501 (1982).

2058. M. Marx-Figini, O. Soubelet, Polym. Bull., **11**, 281 (1984).

2059. R. M. Masegosa, I. Hernández-Fuentes, E. A. Ojalvo, E. Saiz, Macromolecules, **12**, 862 (1979).

2060. R. M. Masegosa, M. R. Gómez-Antón, A. Horta, Makromol. Chem., **187**, 163 (1986).

2061. T. Masuda, K. Kitagawa, T. Inoue, S. Onogi, Macromolecules, **3**, 116 (1970).

2062. T. Masuda, K. Kitagawa, S. Onogi, Polym. J., **1**, 418 (1970).

2063. T. Masuda, Y. Nakagawa, Y. Ohta, S. Onogi, Polym. J., **3**, 92 (1972).

2064. T. Masuda, M. Sawamoto, T. Higashimura, Makromol. Chem., **177**, 2981 (1976).

2065. T. Masuda, E. Isobe, T. Higashimura, Macromolecules, **18**, 841 (1985).

2066. T. Masuda, Y. Ohta, S. Onogi, Macromolecules, **19**, 2524 (1986).

2067. T. Masuda, T. Hamano, T. Higashimura, T. Ueda, H. Muramatsu, Macromolecules, **21**, 283 (1988).

2068. T. Masuda, T. Matsumoto, T. Yoshimura, T. Higashimura, Macromolecules, **23**, 4902 (1990).
2069. T. Masuda, T. Hamano, K. Tsuchihara, T. Higashimura, Macromolecules, **23**, 1374 (1990).
2070. T. Masuda, K. Mishima, J.-I. Fujimori, M. Nishida, H. Muramatsu, T. Higashimura, Macromolecules, **25**, 1401 (1992).
2071. T. Masuda, H. Izumikawa, Y. Msiumi, T. Higashimura, Macromolecules, **29**, 1167 (1996).
2072. V. J. Masura, Faserforsch, Textiltech., **13**, 517 (1962).
2073. R. D. Mate, H. S. Lundstrom, Polym. Preprints, **8** (2), 1397 (1967).
2074. R. D. Mate, H. S. Lundstrom, J. Polym. Sci. C, **21**, 317 (1968).
2075. A. R. Mathieson, J. Colloid Sci, **15**, 387 (1960).
2076. V. B. F. Mathot, M. F. J. Pijpers, Polym. Bull., **11**, 297 (1984).
2077. J. D. Matlack, S. N. Chinal, R. A. Guzzi, D. W. Levi, J. Polym. Sci., **49**, 533 (1961).
2078. H. Matsuda K. Yamano, H. Inagaki, J. Polym. Sci. A-2, **7**, 609 (1969).
2079. H. Matsuda, H. Aonuma, S. Kuroiwa, J. Appl. Polym. Sci., **14**, 335 (1970).
2080. H. Matsuda, M. Takeshima, S. Shimizu, S. Kuroiwa, Makromol. Chem., **134**, 309 (1970).
2081. H. Matsuda, I. Yamada, S. Kuroiwa, Polym. J., **8**, 415 (1976).
2082. O. Matsuda, J. Okamoto, N. Suzuki, M. Ito, Y. Tabata, J. Macromol. Sci., Part A: Chem., **8**, 775 (1974).
2083. A. Matsumoto, T. Kitamura, M. Oiwa, G. B. Butler, J. Polym. Sci., Polym. Chem. Ed., **19**, 2531 (1981).
2084. Y. Matsumoto, Shizuoka-ben Kogyo Gijustsu Senta, Kenkyu Hokoku, **33**, 79 (1988).
2085. A. Matsumoto, T. Tarui, T. Otsu, Macromolecules, **23**, 5102 (1990).
2086. T. Matsumoto, N. Nishioka, H. Fujita, J. Polym. Sci. A-2, **10**, 23 (1972).
2087. K. Matsumura, Bull. Chem. Soc. Japan, **42**, 1874 (1969).
2088. K. Matsumura, Makromol. Chem., **124**, 204 (1969).
2089. K. Matsumura, M. Fukaya, K. Mizuno, Bull. Chem. Soc. Japan, **43**, 1881 (1970).
2090. T. Matsuo, H. Inagaki, Makromol. Chem., **55**, 150 (1962).
2091. K. Matsuo, W. H. Stockmayer, S. Mashimo, Macromolecules, **15**, 606 (1982).
2092. Y. Matsushita, H. Furuhashi, H. Choshi, I. Noda, M. Nagasawa, T. Fujimoto, C. C. Han, Polym. J., **14**, 489 (1982).
2093. K. Matsuzaki, C. Ichijo, T. Kawai, Makromol. Chem., **182**, 2919 (1981).
2094. K. Matsuzaki, T. Kanai, C. Ichijo, M. Yuzawa, Makromol. Chem., **185**, 2291 (1984).
2095. W. L. Mattice, J.-T. Lo, Macromolecules, **5**, 734 (1972).
2096. A. Mattiussi, G. B. Gechele, M. Pizzoli, Eur. Polym. J., **2**, 383 (1966).
2097. A. Mattiussi, F. Manescalchi, G. B. Gechele, Eur. Polym. J., **5**, 105 (1969).
2098. M. Matzner, L. M. Robeson, R. J. Greff, J. E. McGrath, Angew. Makromol. Chem., **26**, 137 (1972).
2099. J.-P. Mautekker, S. K. Varsahney, R. Fayt, C. Jacobs, R. Jérôme, Ph. Teyssié, Macromolecules, **23**, 3893 (1990).
2100. J. A. May Jr., W. B. Smith, J. Phys. Chem., **72**, 216, 2993 (1968).
2101. S. Mayeda, M. Nagata, J. Appl. Polym. Sci., Appl. Polym. Symp., **52**, 473 (1993).
2102. R. Mayer, Polymer, **15**, 137 (1974).
2103. G. Mayer, R. Y. M. Huang, Makromol. Chem., **181**, 1727 (1980).
2104. J. F. Mayoral, M. Levy, J. Polym. Sci., Polym. Chem. Ed., **20**, 2755 (1982).
2105. J. W. Mays, N. Hadjichristidis, L. J. Fetters, Macromolecules, **17**, 2723 (1984).
2106. J. W. Mays, W. M. Ferry, N. Hadjichristidis, W. G. Funk, Polymer, **27**, 129 (1986).
2107. J. W. Mays, Macromolecules, **21**, 3179 (1988).
2108. J. W. Mays, S. Man, M. E. Lewis, Macromolecules, **24**, 4857 (1991).
2109. J. W. Mays, S. Man, W. Yunan, J. Li, Macromolecules, **24**, 4469 (1991).
2110. A. M. Mazafarov, M. Golly, M. Moller, Macromolecules, **28**, 8444 (1995).
2111. R. A. D. Mbachu, J. Manley, J. Polym. Sci., Polym. Chem. Ed., **19**, 2065 (1981).
2112. C. L. Mc Cormick, L. S. Park, J. Polym. Sci., Polym. Chem. Ed., **21**, 2229 (1983).
2113. C. L. Mc Cormick, L. S. Park, J. Polym. Sci., Polym. Chem. Ed., **22**, 49 (1984).
2114. C. L. Mc Cormick, G. S. Chen, J. Polym. Sci., Polym. Chem. Ed., **22**, 3649 (1984).
2115. H. W. Mc Cormick, J. Polym. Sci., **36**, 341 (1959).
2116. H. W. Mc Cormick, J. Polym. Sci., **41**, 327 (1959).
2117. H. W. Mc Cormick, J. Polym. Sci. A, **1**, 341 (1959).
2118. R. H. Mc Dowell, Chem. Ind., 1401 (1958).
2119. A. J. Mc Hugh, E. P. Vrahopoulou, B. J. Edwards, J. Polym. Sci., Part B: Polym. Phys., **25**, 953 (1987).
2120. G. B. Mc Kenna, G. Hadziioannou, P. Lutz, G. Hild, C. Strazielle, C. Stranpe, P. Rempp, A. J. Kovas, Macromolecules, **20**, 498 (1987).
2121. E. Meehan, L. L. Lloyd, J. A. Mcconville, F. P. Warner, N. P. Gabbott, J. V. Dawkins J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 3 (1991).
2122. E. Meehan, S. O'Donohue, A. G. Williams, J. V. Dawkins, J. Appl. Polym. Sci., Appl. Polym. Symp., **51**, 151 (1992).
2123. J. W. Meeks, T. F. Banigan, R. W. Planck, India Rubber World, **122**, 301 (1950).
2124. M. Meewes, J. Ricka, M. de Silva, R. Nyffnegger, Th. Binkert, Macromolecules, **24**, 5811 (1991).
2125. A. M. Meffroy-Bigget, Bull. Soc. Chim. (France), **458**, 465 (1954).
2126. A. M. Meffroy-Bigget, Compt. Rend., **240**, 1707 (1955).
2127. M. A. Meilchen, B. M. Hasch, M. A. McHugh, Macromolecules, **24**, 4874 (1991).
2128. A. Mele, R. Rufo, L. Santonato, Polymer, **10**, 233 (1969).
2129. L. G. Melkonyan, R. V. Bagdasaryan, Izv. Akad. Nauk. Arm. SSR Khim. Nauki. **18**. 333 (1965).

2130. A. D. Meltzar, D. A. Tirrell, A. A. Jones, P. T. Jones, P. T. Ingelfield, Macromolecules, **25**, 4541 (1992).
2131. H. W. Melville, B. D. Stead, J. Polym. Sci., **16**, 505 (1956).
2132. Z. Mencík, Chem. Zvesti, **9**, 165 (1955).
2133. H. J. Mencer, Z. Grübisic-Gallot, J. Liquid. Chromatogr., **2**, 649 (1979).
2134. H. J. Mencer, B. Kunst, Makromol. Chem., **180**, 2463 (1979).
2135. H. J. Mencer, Z. Vajnaht, J. Chromatogr., **241**, 205 (1982).
2136. H. J. Mencer, Z. Gomzi, Angew. Makromol. Chem., **162**, 163 (1988).
2137. R. A. Mendelson, J. Polym. Sci. A, **1**, 2361 (1963).
2138. R. A. Mendelson, E. E. Drott, J. Polym. Sci., B, **6**, 795 (1968).
2139. R. A. Mendelson, F. L. Finger, Polym. Preprints, **12** (1), 370 (1971).
2140. J.-P. Merce, A. Sarko, Macromolecules, **5**, 132 (1972).
2141. G. R. Meredith, J. G. Van Dusen, D. J. Williams, Macromolecules, **15**, 1385 (1982).
2142. Y. Merle, L. Merle-Aubry, J. E. Guillet, Macromolecules, **17**, 288 (1984).
2143. R. Mertsch, B. A. Wolf, Macromolecules, **27**, 3289 (1994).
2144. E. H. Merz, J. Polym. Sci., **3**, 790 (1948).
2145. E. H. Merz, R. W. Raetz, J. Polym. Sci., **5**, 587 (1950).
2146. J. C. Meunier, R. Van Leemput, Makromol. Chem., **142**, 1 (1971).
2147. J.-C. Meunier, Z. Gallot, Makromol. Chem., **156**, 117 (1972).
2148. G. Meyerhoff, Makromol. Chem., **12**, 45 (1954).
2149. G. Meyerhoff, J. Romatowski, Makromol. Chem., **74**, 222 (1964).
2150. G. Meyerhoff, Ber. Bunsenges. Phys. Chem., **69**, 866 (1965).
2151. G. Meyerhoff, Makromol. Chem., **89**, 282 (1965).
2152. G. Meyerhoff, N. Suetterlin, Makromol. Chem., **87**, 258 (1965).
2153. G. Meyerhoff, W. Meier, K. Berger, Makromol. Chem., **91**, 290 (1966).
2154. G. Meyerhoff, S. Jovanovic, J. Polym. Sci. B, **5**, 495 (1967).
2155. G. Meyerhoff, S. Shimotsuma, Makromol. Chem., **109**, 263 (1967).
2156. G. Meyerhoff, Polym. Preprints **8** (2), 1295 (1967).
2157. G. Meyerhoff, J. Polym. Sci. C, **21**, 31 (1968).
2158. G. Meyerhoff, Angew. Makromol. Chem., **4/5**, 268 (1968).
2159. G. Meyerhoff, Makromol. Chem., **134**, 129 (1970).
2160. G. Meyerhoff, S. Shimotsuma, Makromol. Chem., **135**, 195 (1970).
2161. G. Meyerhoff, U. Moritz, R. G. Kirste, W. Heitz, Eur. Polym. J., **7**, 933 (1971).
2162. G. Meyerhoff, S. Shimotsuma, IUPAC Boston, Preprints, Vol. 11, 1157 (1971).
2163. R. Meza, L. Gargallo, Eur. Polym. J., **13**, 235 (1977).
2164. F. Michel, A. Bockmann, W. Mechstroth, Makromol. Chem., **48**, 15 (1961).
2165. F. Michel, W. Meckstroth, Makromol. Chem., **51**, 107 (1962).
2166. C. Mijangos, G. Martinez, A. Michel, J. Millán, A. Guyot, Eur. Polym. J., **20**, 1 (1984).
2167. N. V. Mikhailov, S. G. Zelikman, Kolloidn. Zh., **18**, 717 (1956).
2168. J. Millán, E. L. Madruga, G. Martínez, Angew. Makromol. Chem., **45**, 177 (1975).
2169. B. Millaud, C. Strazielle, Polymer, **20**. 563 (1979).
2170. B. Miller, E. Pacsu, J. Polym. Sci., **41**, 97 (1959).
2171. G. W. Miller, S. A. D. Visser, Polym. Preprints, **8** (1), 641 (1967).
2172. L. E. Miller, F. A. Hamm, J. Phys. Chem., **57**, 110 (1953).
2173. M. L. Miller, J. Polym. Sci., **56**, 203 (1962).
2174. N. J. Mills, Eur. Polym. J., **5**, 675 (1969).
2175. T. I. Min, H. Inagaki, Polymer, **21**, 309 (1980).
2176. A. Minakata, N. Imai, Biopolymers, **11**, 329 (1972).
2177. Y. Minoura, M. Mitoh, Makromol. Chem., **99**, 186 (1966).
2178. Y. Minoura, M. Shundo, Y. Enomoto, J. Polym. Sci. A-1, **6**, 979 (1968).
2179. Y. Minoura, M. Mitoh, Makromol. Chem., **124**, 241 (1969).
2180. S. A. Minyailo, N. P. Zakurdaeve, Khim. Volokna, **15**, 44 (1973).
2181. F. M. Mirabella Jr., E. M. Barrall, J. F. Johnson, J. Appl. Polym. Sci., **19**, 2131 (1975).
2182. F. M. Mirabella Jr., E. M. Barrall II, J. F. Johnson, Polymer, **17**, 17 (1976).
2183. F. M. Mirabella Jr., EA Ford, J. Polym. Sci. A-2: Polym. Phys., **25**, 777 (1987).
2184. F. M. Mirabella Jr., J. Appl. Polym. Sci., Appl. Polym. Symp., **51**, 117 (1992).
2185. F. M. Mirabella Jr., Polymer, **34**, 1729 (1993).
2186. F. M. Mirabella Jr., J. Liquid. Chromatogr., **17**, 3201 (1994).
2187. I. M. Mirkamilov, U. Azizov, M. U. Sadykov, Kh. U. Usmanov, Polym. Sci. USSR, **14**, 1904 (1972); Vysokomol. Soedin. A, **14**, 1704 (1972).
2188. V. M. Misin, V. I. Fartunin, A. M. Fomin, M. I. Cherkashin, Polym. Sci. USSR, **22**, 2171 (1980); Vysokomol. Soedin. A, **22**, 1981 (1980).
2189. B. N. Misra, P. S. Chandel, R. Dogra, J. Polym. Sci., Polym. Chem. Ed., **16**, 1801 (1978).
2190. B. N. Misra, R. Dogra, J.Macromol. Sci. A, **14**, 763 (1980).
2191. B. N. Misra, R. Dogra, I. K. Mehta, J. Polym. Sci., Polym. Chem. Ed., **18**, 749 (1980).
2192. B. N. Misra, P. S. Chandel, J. Polym. Sci., Polym. Chem. Ed., **18**, 1171 (1980).
2193. B. N. Misra, I. K. Mehta, J. Polym. Sci., Polym. Chem. Ed., **18**, 1911 (1980).
2194. B. N. Misra, R. Dogra, I. Kaur, D. Sood, J. Polym. Sci., Polym. Chem. Ed., **18**, 341 (1980).
2195. B. N. Misra, R. Dogra, I. K. Mehta, A. S. Singha, Angew. Makromol. Chem., **90**, 83 (1980).
2196. B. N. Misra, I. K. Mehta, R. K. Sharma, Polym. Bull., **3**, 115 (1980).
2197. B. N. Misra, D. S. Sood, I. K. Mehta, J. Macromol. Sci., Part A: Chem., **18**, 209 (1982).
2198. G. S. Misra, V. P. Gupta, Makromol. Chem., **71**, 110 (1964).
2199. G. S. Misra, S. N. Battacharya, Eur. Polym. J., **15**, 125 (1979).

2200. I. Mita, J. Chim. Phys., **59**, 530 (1962).
2201. K. Mitani, T. Suzuki, A. Matsuo, Y. Takegami, J. Polym. Sci., Polym. Chem. Ed., **12**, 771 (1974).
2202. R. L. Mitchell, Ind. Eng. Chem., **45**, 2526 (1953).
2203. J. C. Mitchell, A. E. Woodard, P. Doty, J. Am. Chem. Soc., **79**, 3955 (1957).
2204. H. Mitomo, N. Morishita, Y. Doi, Polymer, **36**, 2573 (1995).
2205. S. P. Mitsengendler et al., Vysokomol. Soedin., **4**, 1366 (1962).
2206. S. P. Mitsengendler, K. I. Sokolova, G. A. Andreyeva, A. A. Korotkov, T. Kadyrov, S. I. Klenin, S. Ya. Magarik, Polym. Sci. USSR, **9**, 1261 (1967); Vysokomol. Soedin., **9**, 1133 (1967).
2207. M. Miura, Y. Kubota, T. Masuzukawa, Bull. Chem. Soc. Japan., **38**, 316 (1965).
2208. Y. Miyaki, Y. Einaga, T. Hirosye, H. Fujita, Macromolecules, **10**, 1356 (1977).
2209. Y. Miyaki, M. Iwata, Y. Fujita, H. Tanisugi, Y. Isono, T. Fujimoto, Macromolecules, **17**, 1907 (1984).
2210. T. Miyamoto, H. Inagaki, Macromolecules, **2**, 554 (1969).
2211. T. Miyamoto, S. Tomoshige, H. Inagaki, Polym. J., **6**, 564 (1974).
2212. M. Miyamoto, M. Sawamoto, T. Higashimura, Macromolecules, **17**, 265 (1984).
2213. M. Miyamoto, K. Hayashizaki, M. Tokumizu, T. Saegusa, Macromolecules, **23**, 4718 (1990).
2214. K. Miyata, K. Muraoka, T. Itaya, T. Tanigaki, K. Inour, Eur. Polym. J., **32**, 1257 (1996).
2215. R. Miyatake, J. Kumanotani, Makromol. Chem., **177**, 2749 (1976).
2216. A. Mizono, T. Abiru, M. Motowoka, M. Kioka, M. Onda, J. Appl. Polym. Sci., Appl. Polym. Symp., **52**, 159 (1993).
2217. K. Mizuhara, K. Hara, T. Imoto, Kolloid. Z.-Z. Polym., **229**, 17 (1969).
2218. A. Mizuno, M. Onda, T. Sogane, Polymer, **32**, 2953 (1991).
2219. J. Moacanin, N. Nelson, E. Back, V. F. Felicetta, J. L. Mc Carthy, J. Am. Chem. Soc., **81**, 2054 (1959).
2220. J. Moacanin, A. Rembaum, R. K. Laudenslager, R. Adler, J. Macromol. Sci., Part A: Chem., **1**, 1497 (1967).
2221. W. E. Mochel, J. B. Nichols, C. J. Mighton, J. Am. Chem. Soc., **70**, 2185 (1948).
2222. W. E. Mochel J. B. Nichols, J. Am. Chem. Soc., **71**, 3435 (1949).
2223. W. E. Mochel, J. B. Nichlos, Ind. Eng. Chem., **43**, 154 (1951).
2224. R. A. Mock, C. A. Marshall, V. Deane Floria, J. W. Sanders, J. Polym. Sci., **11**, 447 (1953).
2225. D. Mohan, G. Radhakrishanan, S. Rajadurai, Makromol. Chem., **183**, 1659 (1982).
2226. R. B. Mohite, S. Gundiah, S. L. Kapur, Makromol. Chem., **108**, 52 (1967).
2227. R. B. Mohite S. Gundiah, S. L. Kapur, Makromol. Chem., **116**, 280 (1968).
2228. R. B. Mohite, G. Meyerhoff, Angew. Makromol. Chem., **25**, 41 (1972).
2229. E. Mokhtari-Nejad, K. C. Berger, R. Hammel, K. Lederer, Makromol. Chem., **179**, 159 (1978).
2230. A. Molenberg, H.-A. Klok, M. Möller, S. Boileau, D. Teyssié, Macromolecules, **30**, 792 (1997).
2231. W. L. H. Moll, Kolloid, Z.-Z. Polym., **223**, 48 (1968).
2232. Ye. D. Molodtsova, S. A. Pavlova, G. I. Timofeyeva, Ya. S. Vygodskii, S. V. Vinogradova, V. V. Korshak, Polym. Sci. USSR, **16**, 2529 (1974); Vysokomol. Soedin. A, **16**, 2183 (1974).
2233. P. Molyneux, Makromol. Chem., **43**, 31 (1961).
2234. P. Molyneux, Kolloid. Z.-Z. Polym., **226**, 15 (1968).
2235. B. Monrabnal, J. Appl. Polym. Sci., **52**, 491 (1994).
2236. B. Monrabal, Makromol. Chem. Macromol. Symposia., **110**, 81 (1994).
2237. B. Monrabal, Analysis of Metallocene Type Polyolefins by CRYSTAF (Crystallization Analysis Fractionation), International GPC Symposium, San Diego, 1996.
2238. V. M. Monroy Soto, J. C. Galin, Polymer, **25**, 254 (1984).
2239. P. G. Montague, F. W. Peaker, P. Bosworth, P. Lemon, Brit. Polym. J., **3**, 93 (1971).
2240. G. Montaudo, P. Maravigna, P. Finocchiaro, C. G. Overberger, Macromolecules, **5**, 203 (1972).
2241. G. Montaudo, D. Garrozo, M. S. Montando, G. Pugliss, F. Samperi, Macromolecules, **28**, 7983 (1995).
2242. S. Montserrat, P. Colomer, Polym. Bull., **12**, 173 (1984).
2243. J. C. Moore, J. Polym. Sci. A, **2**, 835 (1964).
2244. L. D. Moore Jr., G. R. Greear, J. O. Sharp, J. Polym. Sci., **59**, 339 (1962).
2245. W. R. Moore, J. A. Epstein, J. Appl. Chem., **6**, 168 (1956).
2246. W. R. Moore, B. M. Tidswell, J. Appl. Chem., **8**, 232 (1958).
2247. W. R. Moore, R. P. Sheldon, J. Textile Inst., **50**, T 294 (1959).
2248. W. R. Moore, G. P. Pearson, Polymer, **1**, 144 (1960).
2249. W. R. Moore, M. Murphy, J. Polym. Sci., **56**, 519 (1962).
2250. W. R. Moore, R. J. Fort, J. Polym. Sci. A-1, **1**, 929 (1963).
2251. W. R. Moore, R. J. Hutchinsons, Nature (London), **200**, 1095 (1963).
2252. W. R. Moore, G. F. Boden, J. Appl. Polym. Sci., **9**, 2019 (1965).
2253. W. R. Moore, K. Saito, Eur. Polym. J., **3**, 65 (1967).
2254. W. R. Moore, M. A. Uddin, Eur. Polym. J., **3**, 673 (1967).
2255. W. R. Moore, D. Sanderson, Polymer, **9**, 153 (1968).
2256. W. R. Moore, M. Uddin, Eur. Polym. J., **5**, 185 (1969).
2257. W. R. Moore, M. A. Uddin, Eur. Polym. J., **6**, 121 (1970).
2258. M. A. R. Moraes, A. C. F. Moreira, R. V. Barbosa, B. G. Soares, Macromolecules, **29**, 416 (1996).
2259. G. Moraglio, Chim. Ind. (Milan), **44**, 352 (1962).
2260. G. Moraglio, Eur. Polym. J., **1**, 103 (1965).
2261. G. Moraglio, G. Gianotti, F. Danusso, Eur. Polym. J., **3**, 251 (1967).
2262. G. Moraglio, G. Gianotti, Eur. Polym. J., **5**, 781 (1969).
2263. G. Moraglio, G. Gianotti, F. Zoppi, U. Bonicelli, Eur. Polym. J., **7**, 303 (1971).
2264. G. Moraglio, G. Gianotti, U. Bonicelli, Eur. Polym. J., **9**, 623 (1973).
2265. G. Moraglio, U. Bonicelli, Polymer, **19**, 1363 (1978).
2266. B. Morèse-Séguéla, M. St-Jacques, J. M. Renaud, J. Prud'homme, Macromolecules, **13**, 100 (1980).

2267. D. R. Morey, J. W. Tamblyn, J. Appl. Phys., **16**, 419 (1945).
2268. D. R. Morey, J. W. Tamblyn, J. Phys. Chem., **50**, 12 (1946).
2269. D. R. Morey, J. W. Tamblyn, J. Phys. Chem., **51**, 721 (1947).
2270. D. R. Morey, E. W. Taylor, G. P. Waugh, J. Colloid Sci., **6**, 470 (1951).
2271. P. J. Morgan, S. E. Harding, K. Petrak, Macromolecules, **23**, 4461 (1990).
2272. P. W. Morgan, S. L. Kwolek, J. Polym. Sci., **62**, 33 (1962).
2273. H. Mori, A. Hirao, S. Nakahama, K. Senshu, Macromolecules, **27**, 4093 (1994).
2274. H. Mori, A. Hirao, S. Nakahama, Moromolecules, **27**, 35 (1994).
2275. S. Mori, Anal. Chem., **53**, 1813 (1981).
2276. S. Mori, M. Suzuki, Anal. Chem., **56**, 1708 (1984).
2277. S. Mori, Anal. Chem., **60**, 1125 (1988).
2278. S. Mori, M. Mouri, Anal. Chem., **61**, 2171 (1989).
2279. S. Mori, J. Appl. Polym. Sci., Appl. Polym. Symp., **43**, 65 (1989).
2280. S. Mori, Anal. Chem., **62**, 1902 (1990).
2281. S. Mori, J. Appl. Polym. Sci., Appl. Polym. Symp., **45**, 71 (1990).
2282. S. Mori, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 133 (1991).
2283. S. Mori, TRIP, **2**, 208 (1994).
2284. Y. Morishima, T. Hashinoto, Y. Itoh, M. Kamachi, S. Nizakura, L. Polym. Sci., Polym. Chem. Ed., **20**, 299 (1982).
2285. U. Moritz, G. Meyerhoff, Makromol. Chem., **139**, 23 (1970)
2286. G. A. Morneau, P. I. Roth, A. R. Shultz, J. Polym. Sci., **55**, 609 (1961).
2287. A. G. Morozov, E. D. Al'tshuler, A. Y. Pavlov, M. K. Dobrokhotova, Plast Massy, **8**, 72 (1972); Soviet Plast., **8**, 85 (1972).
2288. R. J. Morris, H. E. Persinger, J. Polym. Sci. A, **1**, 1041 (1963).
2289. M. Morton, R. F. Kammereck, L. J. Fetters, Macromolecules, **4**, 11 (1971).
2290. T. M. Moshkina, A. N. Pudovik, Vysokomol. Soedin., **5**. 1106 (1963).
2291. M. Motowoka, T. Norisuye, H. Fujita, Polym. J., **9**, 613 (1977).
2292. M. Motowoka, H. Fujita, T. Norisuye, Polym. J., **10**, 331 (1978).
2293. T. H. Moury, J. Chromatogr., **357**, 101 (1986).
2294. L. Mrkvičková, P. Lopour, S. Pokorný, J. Janča, Angew. Makromol. Chem., **90**, 217 (1980).
2295. L. Mrkvičková, J. Kálal, B. Bednár, J. Janča, Makromol. Chem., **183**, 203 (1982).
2296. H. Mueller, Patent, DE 86-3607946 860311.
2297. O. Mueller, unpublished report to Farbenfabriken Bayer 1962 quoted by H. Giesekus, in: M. J. R. Cantow, (Ed.), "Polymer Fractionation", Academic Press, New York, 1967, p. 191.
2298. R. Muggli, H.-G. Elias, K. Muehlethaler, Makromol. Chem., **121**, 290 (1969).
2299. A. N. Mujumdar, S. G. Young, R. L. Merker, J. H. Magill, Macromolecules, **23**, 14 (1990).
2300. A. K. Mukherjee, J. C. Bertrand, D. A. Allen, Angew. Makromol. Chem., **72**, 213 (1978).
2301. A. K. Mukherjee, H. R. Goel, J. Macromol. Sci. A: Chem., **17**, 545 (1982).
2302. A. Mula L. Chinellato, Eur. Polym. J., **6**, 1 (1970).
2303. F. J. Muller, U. Altenhofen, F. J. Wortmann, H. Zahn, Angew. Makromol. Chem., **83**, 171 (1979).
2304. T. E. Muller, W. J. Alexander, Polym. Preprints, **8** (2), 1371 (1967).
2305. T. E. Muller, W. J. Alexander, J. Polym. Sci. C, **21**, 283 (1968).
2306. W. A. Muller, L. N. Rogers, Ind. Eng. Chem., **45**, 2522 (1953).
2307. Y.-U. Mun, T. Sato, T. Otsu, Makromol. Chem., **185**, 1507 (1984).
2308. P. Munk, Coll. Czech. Chem. Commun., **32**, 787 (1967).
2309. H. Murakami, T. Norisuye, H. Fujita, Macromolecules, **13**, 345 (1980).
2310. K. Murata, J. Polym. Sci., Polym. Chem. Ed., **10**, 3673 (1972).
2311. J. B. Murray, G. J. Perry, M. T. Vertessy, J. Macromol. Sci., Part A: Chem., **17**, 265 (1982).
2312. M. M. Musiani, G. Mengoli, B. Pelli, C. Folonari, Makromol. Chem., **183**, 1869 (1982).
2313. C. Mussa, I. Vercelli, E. Cernia, Ann. Chim. (Rome), **41**, 130 (1951).
2314. C. Mussa, J. Polym. Sci., **28**, 587 (1958).
2315. M. I. Mustafayev, K. V. Aliev, V. A. Kabanov, Polym. Sci USSR, **12**, 968 (1970); Vysokomol. Soedin. A, **12**, 855 (1970).
2316. V. A. Myagchenkov, Ye. V. Kuznetsov, N. I. Dominova, Polym. Sci. USSR, **6**, 1786 (1964); Vysokomol. Soedin, **6**, 1612 (1964).
2317. V. A. Myagchenkov, Ye. V. Kuznetsov, V. Ya. Kitkevich, Polym. Sci. USSR, **6**, 1507 (1964); Vysokomol. Soedin., **6**, 1366 (1964).
2318. V. A. Myagchenkov, Ye. V. Kuznetsov, L. A. Zaltsgendler, Polym. Sci. USSR, **7**, 2275 (1965); Vysokomol. Soedin., **7**, 2077 (1965).
2319. V. A. Myagchenkov, V. F. Kurenkov, E. V. Kuznetsov, S. Ya. Frenkel, Eur. Polym. J., **6**, 63 (1970).
2320. V. A. Myagchenkov, V. F. Kurenkov, L. F. Antonova, A. Ya. Belobokova, Polym. Sci. USSR, **12**, 1980 (1970); Vysokomol. Soedin. A, **12**, 1745 (1970).
2321. V. A. Myagchenkov, A. K. Vagapova, V. F. Kurenkov, S. Ya. Frenkel, Eur. Polym. J., **14**, 169 (1978).
2322. G. E. Myers, J. R. Dagon, J. Polym. Sci. A-1, **2**, 2631 (1964).
2323. C. D. Myrat, J. S. Rowlinson, Polymer **6**, 645 (1965).
2324. T. D. N'Guyen, S. Boileau, Polym. Bull., **1**, 817 (1979).
2325. T. Nagabhushanam, K. T. Joseph, M. Santappa, J. Polym. Sci., Polym. Chem. Ed., **16**, 3287 (1978).
2326. S. Nagarajan, K. S. V. Srinivasan, J. Polym. Sci. Part A: Polym. Chem., **33**, 2925 (1995).
2327. Y. Nagasaki, M. Taniuchi, T. Tsuruta, Makromol. Chem., **189**, 723 (1988).
2328. Y. Nagasaki, N. Yamazaki, K. Takeda, N. Kata, M. Kato, Macromolecules, **27**, 3702 (1994).

2329. T. Nagata, J. Appl. Polym. Sci., **13**, 2601 (1969).
2330. K. Nagayama, S. Okada, Polym. J., **5**, 97 (1973).
2331. J. Naghizadeh, J. Springer, Kolloid. Z.-Z. Polym., **215**, 21 (1967).
2332. S. Nair, B. C. Sekhar, Chem. Soc. (London) Spec. Publ., **23**, 105 (1968).
2333. P. R. M. Nair, R. M. Gohil, K. C. Patel, R. D. Patel, Eur. Polym. J., **13**, 273 (1977).
2334. H. Nakahara, M. Shihanda, J. Soc. Text. Cell. Ind. (Japan), **8**, 438 (1952).
2335. A. Nakai, W. Wang, T. Hashimoto, A. Blumstein, Y. Maeda, Macromolecules, **27**, 6963 (1994).
2336. A. Nakajima, I. Sakurada, Chem. High Polym. (Japan), **11**, 110 (1954).
2337. A. Nakajima, H. Fujiwara, Bull. Chem. Soc. Japan, **37**, 909 (1964).
2338. A. Nakajima, F. Hamada, S. Hayashi, A. Saijoo, Discussion Meeting of Soc. Polym. Sci., Japan, July, **10**, 1965, Kobe, Japan.
2339. A. Nakajima, F. Hamada, Kolloid Z.-Z. Polym., **205**, 55 (1965).
2340. A. Nakajima, F. Hamada, S. Hayashi, J. Polym. Sci. C, **15**, 285 (1966).
2341. A. Nakajima, F. Hamada, T. Shimizu, Makromol. Chem., **90**, 229 (1966).
2342. A. Nakajima, K. Kato, Makromol. Chem., **95**, 52 (1966).
2343. A. Nakajima, S. Hayashi, Kolloid. Z.-Z. Polym., **225**, 116 (1968).
2344. A. Nakajima, S. Hayashi, T. Korenaga, T. Sumida, Kolloid. Z.-Z. Polym., **222**, 124 (1968).
2345. A. Nakajima, T. Hayashi, K. Kugo, K. Shinoda, Macromolecules, **12**, 840 (1979).
2346. N. Nakajima, J. Polym. Sci. A-2, **5**, 101 (1966).
2347. N. Nakajima, Polym. Preprints, **8** (2), 1282 (1967).
2348. N. Nakajima, J. Polym. Sci. C, **21**, 153 (1968).
2349. N. Nakajima, Polym. Preprints, **12** (2), 804 (1971).
2350. M. Nakakata, K. Kawate, Y. Ishitaka, Macromolecules, **27**, 1825 (1994).
2351. Y. Nakamura, T. Norisuye, A. Teramoto, J. Polym. Sci., Part B: Polym. Phys., **29**, 153 (1991).
2352. K. Nakamura, R. Endo, M. Takeda, J. Polym. Sci., Polym. Phys. Ed., **14**, 135 (1976).
2353. Y. Nakamura, K. Akasaka, K. Katayama, T. Norisuye, A. Teramoto, Macromolecules, **25**, 1134 (1992).
2354. S. Nakano, J. Polym. Sci., Polym. Phys. Ed., **12**, 1499 (1974).
2355. S. Nakano, Y. Goto, J. Appl. Polym. Sci., **26**, 4217 (1981).
2356. M. Nakano, T. Masuda, T. Higashimura, Macromolecules, **27**, 1344 (1994).
2357. M. Nakata, Makromol. Chem., **149**, 49 (1971).
2358. N. S. Nametkin, O. B. Semenov, S. G. Durgar'yan, S. Skazka, V. G. Filippova, V. Ya. Nikolayev, Polym. Sci. USSR, **17**, 1115 (1975); Vysokomol. Soedin. A, **17**, 973 (1975).
2359. A. K. Nandi, L. Mandelkern, J. Polym. Sci. Part B, Polym. Phys., **29**, 1287 (1991).
2360. V. Narasimhan, R. Y. M. Huang, C. M. Burns, J. Polym. Sci., Polym. Chem. Ed., **22**, 3895 (1984).
2361. H. Narita, S. Machida, Makromol. Chem., **97**, 209 (1966).
2362. T. Narita, T. Hagiwara, H. Hamana, S. Maesaka, Polym. J., **20**, 519 (1988).
2363. D. W. Nash, D. C. Pepper, Polymer, **16**, 105 (1975).
2364. A. Nasini, C. Mussa, Makromol. Chem., **22**, 59 (1957).
2365. M. Natov, D. Kelchajov, M. Michailova, Angew. Chem., **102**, 175 (1982).
2366. G. Natta, F. Danusso, G. Moraglio, Makromol. Chem., **20**, 37 (1956).
2367. G. Natta, M. Pegoraro, M. Peraldo, Ricerca Sci., **28**, 1473 (1958).
2368. G. Natta et al., Atti. Accad. Nazl, Lincei, **28**, 18 (1960).
2369. G. Natta, A. Zambelli, I. Pasquon, F. Ciampelli, Makromol. Chem., **79**, 161 (1964).
2370. G. Natta, L. Porri, A. Carbonaro, G. Stoppa, Makromol. Chem., **77**, 114 (1964).
2371. G. Natta, L. Porri, A. Carbonaro, Makromol. Chem., **77**, 126 (1964).
2372. G. Natta, I. Pasquon, A. Zambelli, G. Gatti, Makromol. Chem., **70**, 191 (1964).
2373. L. A. Neboinova, V. A. Grechanovskii, I. Ya. Poddubnyi, L. R. Avdeyeva, Polym. Sci. USSR, **16**, 216 (1974); Vysokomol. Soedin. A, **16**, 183 (1974).
2374. I. K. Nekrasov, A. M. Kurakova, S. V. Kuz'minskaya, K. G. Khabarova, Ye. P. Krasnov, Polym. Sci. USSR, **22**, 410 (1980); Vysokomol. Soedin. A, **22**, 370 (1980).
2375. I. K. Nekrasov, Polym. Sci. USSR, **17**, 510 (1975); Vysokomol. Soedin. A, **17**, 439 (1975).
2376. I.K. Nekrasov, K. G. Khabarova, S. V. Kuz'minskaya, Ye. P. Krasnov, S. N. Khar'kov, Polym. Sci. USSR, **22**, 699 (1980); Vysokomol. Soedin. A, **22**, 633 (1980).
2377. T. N. Nekrasova, O. B. Ptitsyn, M. S. Shikanova, Polym. Sci. USSR, **10**, 1771 (1968); Vysokomol. Soedin. A, **10**, 530 (1968).
2378. T. N. Nekrasova, E. Churylo, Polym. Sci. USSR, **11**, 1249 (1969); Vysokomol. Soedin. A, **11**, 1103 (1969).
2379. V. V. Nesterov, V. D. Krasikov, Ye. V. Chubarova, L. D. Turkova, E. S. Gankina, B. G. Belen'kii, Polym. Sci. USSR, **20**, 2608 (1978); Vysokomol. Soedin. A, **20**, 2320 (1978).
2380. M. Netopilík, M. Bohdanecký, C. Wehrstedt, R. Ruhmann, G. Schulz, Makromol. Chem., **194**, 2199 (1993).
2381. C. Neuchl, A. Mersmann, Chemical Engineering Science, **50**, 951 (1995).
2382. P. Neuenschwander, P. Pino, Makromol. Chem., **181**, 737 (1980).
2383. E. W. Neuse, K. Koda, E. Carter, Makromol. Chem., **84**, 213 (1965).
2384. C. J. Neves, E. Monteiro, A. C. Habert, J. Appl. Polym. Sci., **50**, 817 (1993).
2385. S. Newman, J. Polym. Sci., **47**, 111 (1960).
2386. H. A. Nguyen, J. P. Kennedy, Polym. Bull., **10**, 74 (1983).
2387. H. A. Nguyen, E. Maréchal, Polym. Bull., **11**, 99 (1984).
2388. T. Q. Nguyen, G. Yu., H. H. Kausch, Macromolecules, **28**, 4851 (1995).
2389. E. Nichols, Polym. Preprints, **12** (2), 828 (1971).

2390. L. Nicolás, Compt. Rend., **236**, 809 (1953).
2391. L. Nicolás, Compt. Rend., **242**, 2720 (1956).
2392. L. Nicolás, Makromol. Chem., **24**, 173 (1957).
2393. P. Nicolet, J. Macromol. Sci., Part A: Chem., **11**, 577 (1977).
2394. J. Niezette, N. Hadjichristidis, V. Desreux, Makromol. Chem., **177**, 2069 (1976).
2395. J. Niezette, A. Linkens, Polymer, **19**, 939 (1978).
2396. J. Niezette, J. Vauderschueren, G. Yianakopoulas, Eur. Polym. J., **26**, 775 (1990).
2397. A. Ninagawa, I. Ijichi, M. Imoto, Makromol. Chem., **107**, 196 (1967).
2398. N. Nishida, D. G. Salladay, J. L. White, J. Appl. Polym. Sci., **15**, 1181 (1971).
2399. N. Nishida, D. G. Salladay, J. L. White, Polym. Preprints, **12** (2), 522 (1971).
2400. M. Nishide, M. Sera, Kogyo Kagaku Zasshi, **64**, 1145 (1961).
2401. H. Nishide, N. Yoshioka, K. Inagaki, T. Kaku, E. Tsuchida, Macromolecules, **25**, 569 (1992).
2402. S. Nishimoto, K. Yamamoto, T. Kagiya, Macromolecules, **15**, 720 (1982).
2403. Y. Nitadori, T. Tsuruta, Makromol. Chem., **179**, 2069 (1978).
2404. O. V. Noa, V. P. Torchilin, A. D. Litmanovich, N. A. Platé, Polym. Sci. USSR, **16**, 775 (1974); Vysokomol. Soedin. A, **16**, 668 (1974).
2405. I. Noda, S. Saito, T. Fujimoto, M. Nagasawa, J. Phys. Chem., **71**, 4048 (1967).
2406. I. Noda, T. Horikawa, T. Kato, T. Fujimoto, M. Nagasawa, Macromolecules, **3**, 795 (1970).
2407. I. Noda, H. Ishikawa, Y. Miyazaki, K. Kamide, Polym. J., **12**, 87 (1980).
2408. I. Noda, T. Imai, T. Kitano, M. Nagasawa, Macromolecules, **14**, 1303 (1981).
2409. K. Noguchi, Y. Yanagihara, M. Kasai, B. Katayama, J. Appl. Polym. Sci., Appl. Polym. Symp., **51**, 161 (1992).
2410. K. Nollen, W. Funke, K. Hamann, Makromol. Chem., **94**, 248 (1966).
2411. R. Nomura, T. Endo, Macromolecules, **28**, 1754 (1995).
2412. R. Nomura, T. Endo, Macromolecules, **28**, 5372 (1995).
2413. R. Nomura, T. Endo, Polym. Bull., **35**, 683 (1995).
2414. K. Noro, H. Kawazura, T. Moriyama, S. Yoshioka, Makromol. Chem., **83**, 35 (1965).
2415. N. T. Notley, J. Polym. Sci., A, **1**, 227 (1963).
2416. T. E. Nowlin, W. H. Boyd, J. Polym. Sci., Polym. Chem. Ed., **14**, 2341 (1976).
2417. S. Nozakura, Y. Morishima, S. Murahashi, J. Polym. Sci., Polym. Chem. Ed., **10**, 2781 (1972).
2418. S. Nozakura, Y. Morishima, S. Murahashi, J. Polym. Sci., Polym. Chem. Ed., **10**, 2767 (1972).
2419. J. S. U. Nu, I. Piirma, J. Appl. Polym. Sci., **33**, 727 (1987).
2420. M. Nukui, Y. Ohkatsu, T. Tsuruta, Makromol. Chem., **183**, 1457 (1982).
2421. N. Numasawa, K. Kuwamoto, T. Nose, Macromolecules, **19**, 2593 (1986).
2422. O. Nuyken, M. Hofinger, R. Kerber, Polym. Bull., **1**, 679 (1979).
2423. O. Nuyken, R. Rengel, R. Kerber, Makromol. Chem., **181**, 1565 (1980).
2424. K. Nyitrai, F. Cser, G. Csermely, B. D. Ngoc, L. Füzes, G. Samay, G. Hardy, Eur. Polym. J., **14**, 467 (1978).
2425. J. J. O'Malley, P. H. Marchessault, J. Phys. Chem., **70**, 3235 (1966).
2426. J. J. O'Malley, W. J. Stauffer, J. Polym. Sci., Polym. Chem. Ed., **12**, 865 (1974).
2427. T. O'Neill, J. Hoigné, J. Polym. Sci., Polym. Chem. Ed., **10**, 581 (1972).
2428. D. B. O'Connor, G. W. Scott, D. R. Coulter, A. Yavrouian, Macromolecules, **24**, 2355 (1991).
2429. H. Ochiai, Y. Fujino, Y. Tadokoro, I. Murakami, Polymer, **21**, 485 (1980).
2430. H. Ochiai, Y. Kurita, I. Murakami, Makromol. Chem., **189**, 351 (1988).
2431. E. A. Ofstead, Polym. Preprints, **6** (2), 674 (1965).
2432. N. Ogata, K. Sanvi, S. Kitayama, J. Polym. Sci., Polym. Chem. Ed., **22**, 863 (1984).
2433. T. Ogawa, Y. Suzuki, S. Tanaka, S. Hoshino, Kobunshi Kagaku, **27**, 356 (1970).
2434. T. Ogawa, Y. Suzuki, T. Inaba, J. Polym. Sci., Polym. Chem. Ed., **10**, 737 (1972).
2435. T. Ogawa, S. Hoshino, J. Appl. Polym. Sci., **17**, 2235 (1973).
2436. T. Ogawa, T. Inaba, J. Appl. Polym. Sci., **21**, 2979 (1977).
2437. T. Ogawa, M. Sakai, J. Polym. Sci., Polym. Phys. Ed., **19**, 1377 (1981).
2438. T. Ogawa, H.-G. Elias, J. Macromol. Sci., Part A: Chem., **17**, 727 (1982).
2439. T. Ogawa, W. Ishitobi, J. Polym. Sci., Polym. Chem. Ed., **21**, 781 (1983).
2440. E. Ogawa, M. Sato, M. Shima, Eur. Polym. J., **17**, 497 (1981).
2441. N. Oguni, J. Hyoda, Macromolecules, **13**, 1687 (1980).
2442. M. Ohata, S. Ikeda, S. Akatani, Y. Isono, Macromolecules, **25**, 5131 (1992).
2443. H. Ohnuma, T. Kotaka, H. Inagaki, Polymer, **10**, 501 (1969).
2444. H. Ohnuma, K. Inoue, K. Se, T. Kotaka, Macromolecules, **17**, 1285 (1984).
2445. H. Ohta, I. Ando, S. Fujishige, K. Kubota, J. Polym. Sci. Part B: Polym. Phys., **29**, 963 (1991).
2446. T. Ojeda, D. Radić, L. Gargallo, Makromol. Chem., **181**, 2237 (1980).
2447. M. Okada, H. Sumitomo, T. Kakezawa, Makromol. Chem., **162**, 285 (1972).
2448. S. Okajima, T. Kikuchi, Kogyo Kagaku Zasshi, **63**, 1772 (1960).
2449. H. Okamoto, J. Polym. Sci., **37**, 173 (1959).
2450. H. Okamoto, K. Sekikawa, J. Polym. Sci., **55**, 597 (1961).
2451. H. Okamoto, J. Polym. Sci. A-1, **2**, 3451 (1964).
2452. N. Okamoto, F. Mukaida, H. Gu, Y. Nakamura, T. Sato, A. Taremoto, M. M. Green, C. Andreola, N. C. Peterson, S. Lifson, Macromolecules, **29**, 2878 (1996).

Previous Page

2453. Y. Okamoto, K. Suzuki, H. Yuki, J. Polym. Sci., Polym. Chem. Ed., **18**, 3043 (1980).

2454. K. Okita, A. Teramoto, H. Fujita, Biopolymers, **9**, 717 (1970).

2455. N. A. Okladnov, V. I. Zegel'man, V. P. Lebedev, S. V. Svetozarskii, Ye. N. Zil'berman, Polym. Sci. USSR, **12**, 349 (1970); Vysokomol. Soedin. A, **12**, 306 (1970).

2456. N. Okui, J. H. Magill, Polymer, **17**, 1086 (1976).

2457. O. F. Olaj, J. W. Breitenbach, B. Wolf, Monatsh. Chem., **95**, 1646 (1964).

2458. O. F. Olaj, J. W. Breitenbach, I. Hofreiter, Makromol. Chem., **91**, 264 (1966).

2459. J. Y. Olayemi, Makromol. Chem., **183**, 2547 (1982).

2460. J. Y. Olayemi, A. K. Yusuf, J. Appl. Polym. Sci., **38**, 1779 (1989).

2461. T. Olcese, J. Ackermann, P. Radici, U. Bianchi, Makromol. Chem., **174**, 137 (1973).

2462. A. V. Olenin, M. B. Lachinov, V. A. Kasaikin, V. P. Zubov, V. A. Kabanov, Polym. Sci. USSR, **20**, 460 (1978); Vysokomol. Soedin. A, **20**, 407 (1978).

2463. H. Omichi, K. Araki, T. Takahashi, T. Yasukawa, K. Murakami, J. Polym. Sci., Polym. Chem. Ed., **14**, 2365 (1976).

2464. I. Omura, D. C. Lee, S. Itov, A. Teramoto, H. Fujita, Polymer, **17**, 847 (1976).

2465. N. Onda, K. Furusawa, N. Yamaguchi, S. Komuro, J. Appl. Polym. Sci., **23**, 3631 (1979).

2466. M. Ondás, E. Spirk, Angew. Makromol. Chem., **204**, 37 (1993).

2467. Y. Onishi, S. Maruno, S. Kamiya, S. Horkoku, M. Hasegawa, Polymer, **19**, 1325 (1978).

2468. S. Onogi, S. Kimura, T. Kato, T. Masuda, N. Miyanaga, J. Polym. Sci. C, **15**, 381 (1966).

2469. S. Onogi, T. Masuda, K. Kitagawa, Macromolecules, **3**, 109 (1970).

2470. M. Orbay, R. Laible, L. Dulog, Makromol. Chem., **183**, 47 (1982).

2471. J. Orn, T. Hasegawa, T. Kawakita, Y. Kondo, M. Takaki, Macromolecules, **24**, 1253 (1991).

2472. R. J. Orr, H. L. Williams, J. Am. Chem. Soc., **79**, 3137 (1957).

2473. H. Orth, Angew. Makromol. Chem., **23**, 83 (1972).

2474. Z. Osawa, Y. Uchida, J. Polym. Sci., Polym. Chem. Ed., **20**, 2259 (1982).

2475. F. T. Osborne, S. Omi, V. T. Stannett, E. P. Stahel, J. Polym. Sci. A-1, **8**, 1657 (1970).

2476. H. W. Osterhoudt, L. N. Ray, Polym. Preprints, **8** (2), 1220 (1967).

2477. A. Oth, J. Bisschops, Ind. Chim. Belges, **20**, 426 (1955).

2478. J. Oth, V. Desreux, Bull. Soc. Chim. Belges, **63**, 261 (1954).

2479. J. Oth, V. Desreux, Ricerca Sci., **25**, 447 (1955).

2480. E. P. Otocka, M. Y. Hellman, Macromolecules, **3**, 362 (1970).

2481. E. P. Otocka, Macromolecules, **3**, 691 (1970).

2482. E. P. Otocka, R. J. Roe, M. Y. Hellman, P. M. Muglia, Macromolecules, **4**, 507 (1971).

2483. E. P. Otocka, Polym. Preprints, **12** (1), 645 (1971).

2484. E. P. Otocka, M. Y. Hellman, P. M. Muglia, Macromolecules, **5**, 227 (1972).

2485. E. P. Otocka, M. Y. Hellman, J. Polym. Sci., Polym. Lett. Ed., **12**, 331 (1974).

2486. T. Otsu, H. Ohnishi, M. Koyama, J. Macromol. Sci., Part A: Chem., **14**, 853 (1980).

2487. A. C. Ouano, P. L. Mercier, Polym. Preprints **8** (2), 1389 (1967).

2488. A. C. Ouano, P. L. Mercier, J. Polym. Sci. C, **21**, 309 (1968).

2489. A. C. Ouano, J. Polym. Sci. A-2, **9**, 377 (1971).

2490. A. C. Ouano, J. Polym. Sci. B, **9**, 2179 (1971).

2491. A. C. Ouano, A. Broido, E. M. Barrall, II, A. C. Javier-Son, Polym. Preprints, **12** (2), 859 (1971).

2492. A. C. Ouano, E. M. Barrall, II, J. F. Johnson, in: P. E. Siade Jr. (Ed.), "Polymer Molecular Weight", Marcel Dekker, New York, Chapter 6, 1975.

2493. A. C. Ouano, D. E. Johnson, B. Dawson, L. A. Pederson, J. Polym. Sci., Polym. Chem. Ed., **14**, 701 (1976).

2494. A. C. Ouano, Rubber Chem. Technol., **54**, 535 (1981).

2495. G. J. J. Out, A. A. Turestskii, M. Möller, D. Oelfin, Macromolecules, **27**, 3310 (1994).

2496. G. J. J. Out, A. A. Turetskii, M. Möller, D. Oelfin, Macromolecules, **28**, 596 (1995).

2497. C. G. Overberger, A. E. Borchert, A. Katchman, J. Polym. Sci., **44**, 491 (1960).

2498. C. G. Overberger, S. Nozakura, J. Polym. Sci. A, **1**, 1439 (1963).

2499. C. G. Overberger, K. Mizamichi, J. Polym. Sci. A, **1**, 2023 (1963).

2500. C. G. Overberger, P. A. Jarovitzky, H. Mukamal, Polym. Preprints, **8** (1), 401 (1967).

2501. C. G. Overberger, G. W. Halek, J. Polym. Sci. A-1, **8**, 359 (1970).

2502. H. S. Owens, J. C. Miers, W. D. Maclay, J. Colloid, Sci., **3**, 277 (1948).

2503. H. Ozaki, A. Hirao, S. Nakahama, Makromol. Chem. Phys., **196**, 2099 (1995).

2504. A. Pacter, K. A. Sharif, J. Polym. Sci. B, **9**, 435 (1971).

2505. C. S. Paik Sung, C. B. Hu, C. S. Wu, Macromolecules, **13**, 111 (1980).

2506. F. Pailhes, M. Alliot-Lugaz, N. Duveau, J. Kyritsos, J. Polym. Sci. C, **16**, 1177 (1967).

2507. R. Panaris, G. Pallas, J. Polym. Sci. B, **8**, 441 (1970).

2508. R. Panaris, A. Peyrouset, IUPAC Helsinki, Preprints, Vol. III, II-45 (1972).

2509. R. Panaris, A. Peyrouset, IUPAC Helsinki, Preprints, vol. III, II-46 (1972).

2510. L. A. Pankratova, G. I. Timofeyeva, V. V. Korshak, S. A. Pavlova, D. R. Tur, S. V. Vinogradova, Polym. Sci. USSR, **16**, 3105 (1974); Vysokomol. Soedin. A, **16**, 2666 (1974).

2511. J. Pannell, Polymer, **13**, 2 (1972).

2512. J. Pannell, Polymer, **17**, 351 (1976)..

2513. C. J. Panton, P. H. Plesch, P. P. Rutherford, J. Chem. Soc., 2586 (1964).

2514. G. Paradossi, D. A. Brant, Macromolecules, **15**, 874 (1982).

2515. J. P. Parakka, A. P. Chacko, D. E. Nikles, P. Wang, S. Hasegawa, Y. Murayama, R. M. Metzger, M. P. Cava, Macromolecules, **29**, 1928 (1996).

2516. I. H. Park, Q.-W. Wang, B. Chu, Macromolecules, **20**, 1965 (1987).

2517. Y. T. Park, G. Manuel, W. P. Weber, Macromolecules, **23**, 1911 (1990).

2518. E. J. Parks, F. E. Brinckman, C. E. Mullin, D. M. Andersen, V. J. Castelli, J. Appl. Polym. Sci., **26**, 2967 (1981).

2519. P. Parrini, F. Sebastiano, G. Messina, Makromol. Chem., **38**, 27 (1960).

2520. L. Pasti, S. Roccasalvo, F. Dondi, P. Reschigliau, J. Polym. Sci., Part B: Polym. Phys., **33**, 1225 (1995).

2521. E. E. Paschke, B. A. Bidlingmeyer, J. G. Bergmann, J. Polym. Chem. Ed., **15**, 983 (1977).

2522. G. Pastuska, U. Just, Angew. Makromol. Chem., **81**, 11 (1979).

2523. G. Pastuska, U. Just, H. August, Angew. Makromol. Chem., **107**, 173 (1982).

2524. F. Patat, J. Klein, Makromol. Chem., **93**, 230 (1966).

2525. C. K. Patel, R. D. Patel, Makromol. Chem., **128**, 157 (1969).

2526. C. K. Patel, R. D. Patel, Staerke, **21**, 166 (1969).

2527. C. K. Patel, R. D. Patel, Makromol. Chem., **131**, 281 (1970).

2528. G. N. Patel, J. Polym. Sci., Polym. Phys. Ed., **17**, 1591 (1979).

2529. J. R. Patel, C. K. Patel, R. D. Patel, Staerke, **19**, 330 (1967).

2530. J. R. Patel, R. D. Patel, Polymer, **10**, 167 (1969).

2531. J. R. Patel, Makromol. Chem., **134**, 263 (1970).

2532. K. Patel, P. P. Shah, B. Suthar, Eur. Polym. J., **22**, 63 (1986).

2533. V. M. Patel, C. K. Patel, K. C. Patel, R. D. Patel, Angew. Makromol. Chem., **18**, 39 (1971).

2534. S. K. Patra, S. Ghosh, B. K. Patnaik, R. T. Thampy, Chem. Soc. (London) Spec. Pub., **23**, 233 (1968).

2535. E. Patrone, U. Bianchi, Makromol. Chem., **94**, 52 (1966).

2536. R. Paukkeri, A. Lehtinen, Polymer, **35**, 1673 (1994).

2537. S. Paul, B. Rånby, J. Polym. Sci., Polym. Chem. Ed., **14**, 2449 (1976).

2538. P. Paulusma, D. Vermaas, J. Polym. Sci. C, **2**, 488 (1963).

2539. A. Pavan, A. Provasoli, G. Moraglio, A. Zambelli, Makromol. Chem., **178**, 1099 (1977).

2540. W. A. Pavelich, R. A. Livigni, Polym. Preprints, **8** (2), 1343 (1967).

2541. W. A. Pavelick, R. A. Livigni, J. Polym. Sci. C, **21**, 215 (1968).

2542. A. V. Pavlov, S. Ye. Bresler, S. R. Rafikov, Polym. Sci. USSR, **6**, 2293 (1964); Vysokomol. Soedin. A, **6**, 2068 (1964).

2543. A. V. Pavlov, V. G. Aldoshin, S. Ya. Frenkel, Polym. Sci. USSR, **6**, 1773 (1964); Vysokomol. Soedin. A, **6**, 1600 (1964).

2544. G. M. Pavlov, Ye. V. Korneyeva, T. I. Garmonova, D. Hardy, K. Nitrai, Polym. Sci. USSR, **20**, 1844 (1978); Vysokomol. Soedin. A, **20**, 1634 (1978).

2545. G. M. Paulov, E. F. Panarin, E. V. Korneeva, C. V. Kurochkin, V. E. Baikov, V. N. Ushakova, Makromol. Chem., **191**, 2889 (1990).

2546. S. A. Pavlova, T. A. Soboleva, A. P. Suprun, Polym. Sci. USSR, **6**, 144 (1964); Vysokomol. Soedin., **6**, 122 (1964).

2547. S. A. Pavlova, V. I. Pakhomov, I. I. Tverdokhlebova, Polym. Sci. USSR, **6**, 1408 (1964); Vysokomol. Soedin. A, **6**, 1275 (1964).

2548. S. A. Pavlova, T. M. Frunze, L. V. Dubrovina, V. Y. Kurashev, V. I. Zaitsev, V. V. Korshak, Polym. Sci. USSR, **16**, 23 (1974); Vysokomol. Soedin. A, **16**, 22 (1974).

2549. S. A. Pavlova, L. V. Dubrovina, T. M. Orlova, Polym. Sci. USSR, **16**, 1940 (1974); Vysokomol. Soedin. A, **16**, 1673 (1974).

2550. S. A. Pavlova, G. I. Timofeyeva, G. V. Akopyan, Polym. Sci. USSR, **24**, 185 (1982); Vysokomol. Soedin. A, **24**, 161 (1982).

2551. F. W. Peaker, J. C. Robb, Nature (London), **182**, 1591 (1958).

2552. F. W. Peaker, Analyst, **85**, 235 (1960).

2553. F. W. Peaker, J. M. Patel, J. Appl. Polym. Symp., **8**, 125 (1969).

2554. D. S. Pearson, W. W. Graessley, Macromolecules, **13**, 1001 (1980).

2555. D. S. Pearson, L. J. Fetters, W. W. Graessley, G. V. Strate, E. von Meerwall, Macromolecules,, **27**, 711 (1994).

2556. M. Pechočová, V. Pechoč, Makromol. Chem., **163**, 235 (1973).

2557. T. J. Peckman, J. A. Massey, M. Edwards, I. Manners, D. A. Foucher, Macromolecules, **29**, 2396 (1996).

2558. H. G. Peer, Rec. Trav. Chim., **78**, 631 (1959).

2559. A. J. Pennings, A. M. Kiel, Kolloid Z.-Z. Polym., **205**, 160 (1965).

2560. A. J. Pennings, J. Polym. Sci. C, **16**, 1799 (1967).

2561. R. W. Pennisi, L. J. Fetters, Macromolecules, **21**, 1094 (1988).

2562. E. Penzel, G. V. Schulz, Makromol. Chem., **113**, 64 (1968).

2563. D. C. Pepper, P. P. Rutherford, J. Appl. Polym. Sci., **2**, 100 (1959).

2564. V. Percec, B. C. Auman, Makromol. Chem., **185**, 2319 (1984).

2565. V. Percec, B. C. Auman, Makromol. Chem., **185**, 1867 (1984).

2566. V. Percec, R. Yourd, Macromolecules, **21**, 3379 (1988).

2567. V. Percec, S. Okita, J. H. Wang, Macromolecules, **25**, 64 (1992).

2568. V. Percec, S. Okita, R. Weiss, Macromolecules, **25**, 1816 (1992).

2569. V. Percec, H. Oda, Macromolecules, **27**, 4454 (1994).

2570. V. Percec, D. Schlueter, J. C. Ronda, G. Johansson, G. Ungar, J. P. Zhou, Macromolecules, **29**, 1464 (1996).

2571. V. Percec, P. J. Turkaly, A. D. Asandei, Macromolecules, **30**, 943 (1997).

2572. E. B. Perelman, L. Z. Oblasova, G. D. Kharlanspovieh, E. A. Kineva, B. A. Popov, R. F. Valetdinov, A. I. Vovk, V. R. Falikman, Izobreteniya, **7**, 159 (1992).

2573. J. M. Pereña, E. Riande, G. M. Guzmán, Anales Quím. (Madrid), **68**, 441 (1972).

2574. J. M. Pereña, C. Marco, Makromol. Chem., **181**, 1525 (1980).

2575. J. M. Pereña, C. Marco, A. Bello, J. G. Fatou, Macromolecules, **17**, 348 (1984).
2576. C. Pérez, J. Guzmán, E. Riande, J. G. de la Campa, J. de Abajo, Makromol. Chem., **189**, 691 (1988).
2577. E. Pérez, A. Bello, J. G. Fatou, J. A. Rausell-Colom, Makromol. Chem., **190**, 613 (1989).
2578. E. Pérez, A. Bello, J. G. Fatou, Polym. Bull., **18**, 59 (1987).
2579. E. Pérez, A. Bello, J. G. Fatou, Eur. Polym. J., **24**, 431 (1988).
2580. B. Perly, A. Douy, B. Gallot, Makromol. Chem., **177**, 2569 (1976).
2581. R. Perret, A. Skoulios, C. R. Acad. Sci. (Paris) C, **268**, 230 (1969).
2582. R. Perret, A. Skoulios, Makromol. Chem., **156**, 143 (1972).
2583. R. Perret, A. Skoulios, Makromol. Chem., **152**, 291 (1972).
2584. T. D. Perrine, P. F. Goolsby, J. Polym. Sci. A-1, **3**, 3031 (1965).
2585. N. V. Pertsova, K. A. Andrianov, I. I. Tverdokhlebova, S. A. Pavlova, Polym. Sci. USSR, **12**, 1131 (1970); Vysokomol. Soedin. A, **12**, 1001 (1970).
2586. K. A. Peterson, A. D. Stein, M. D. Fayer, Macromolecules, **23**, 111 (1990).
2587. H. M. Petri, A. Stammer, B. A. Wolf, Macromol. Chem. Phys., **196**, 1453 (1995).
2588. V. Petrus, Coll. Czech. Chem. Commun., **33**, 119 (1968).
2589. V. Petrus, I. Danihel, M. Bohdanecký, Eur. Polym. J., **7**, 143 (1971).
2590. J. Peyroche, Y. Girard, R. Laputte, A. Guyot, Makromol. Chem., **129**, 215 (1969).
2591. G. Pezzin, G. Talamini, G. Vidotto, Makromol. Chem., **43**, 12 (1961).
2592. G. Pezzin, G. Sanmartin, F. Zilio-Grandi, J. Appl. Polym. Sci., **11**, 1539 (1967).
2593. G. Pezzin, S. Lora, L. Busulini, Polym. Bull., **5**, 543 (1981).
2594. B. Pfannemueller, H. Meyerhoefer, R. C. Schulz, Makromol. Chem., **121**, 147 (1969).
2595. B. Pfannemueller, W. Burchard, Makromol. Chem., **121**, 1 (1969).
2596. R. J. Phillipp, C. F. Bjork, J. Polym. Sci., **6**, 383, 549 (1951).
2597. R. J. Phillipp, N. Vatansever, Macromolecules, **20**, 2138 (1987).
2598. J. Prud'homme, S. Bywater, Polym. Preprints, **10** (2), 518 (1969).
2599. J. Di Pietro, A. Di Edwardo, Makromol. Chem., **98**, 275 (1966).
2600. I. Piirma, L. P. H. Chou, J. Appl. Polym. Sci., **24**, 2051 (1979).
2601. C. Pinazzi, J.-C. Lenain, J.-C. Brosse, G. Legeay, Makromol. Chem., **177**, 3139 (1976).
2602. P. Pino, F. Ciardelli, G. P. Lorenzi, Makromol. Chem., **70**, 182 (1964).
2603. P. Pino, G. Montagnoli, F. Ciardelli, E. Benedetti, Makromol. Chem., **93**, 158 (1966).
2604. Ye. P. Piskareva, Ye. G. Erenburg, M. A. Yeremina, S. B. Dolgoplosk, I. Ya. Poddubnyi, Polym. Sci. USSR, **14**, 2557 (1972); Vysokomol. Soedin. A, **14**, 2183 (1972).
2605. S. Pispas, N. Hadjichristidis, J. W. Mays, Macromolecules, **27**, 6307 (1994).
2606. S. Pispas, N. Hadjichristidis, Macromolecules, **27**, 1891 (1994).
2607. J. Pitha, Macromolecules, **9**, 771 (1976).
2608. M. Pitsikalis, N. Hadjichristidis, Macromolecules, **28**, 3904 (1995).
2609. C. U. Pittman, J. C. Lai, D. P. Vanderpool, M. Good, R. Prado, Macromolecules, **3**, 746 (1970).
2610. C. U. Pittman, M. Good, J. Lai, D. Vanderpool, in: C. D. Craver (Ed.), "Polymer Characterization", Plenum Press, New York, 97 (1971).
2611. C. U. Pittman, R. L. Voges, J. Elder, Macromolecules, **4**, 302 (1971).
2612. C. U. Pittman, R. L. Voges, W. R. Jones, Macromolecules, **4**, 291 (1971).
2613. C. U. Pittman, Jr., W. J. Patterson, S. P. McManus, J. Polym. Sci., Polym. Chem. Ed., **14**, 1715 (1976).
2614. C. U. Pittman, Jr., A. Hirao, J. Polym. Sci., Polym. Chem. Ed., **15**, 1677 (1977).
2615. C. U. Pittman, Jr., A. Hirao, J. Polym. Sci., Polym. Chem. Ed., **16**, 1197 (1978).
2616. C. U. Pittman, Jr., M. Iqbal, C. Y. Chen, J. N. Helbert, J. Polym. Sci., Polym. Chem. Ed., **16**, 2725 (1978).
2617. H. Pitz, D. Le-Kim, Chromatographia, **12**, 155 (1979).
2618. M. Pizzoii, G. Ceccorulii, Eur. Polym. J., **8**, 769 (1972).
2619. M. Pizzoli, G. Stea, G. Ceccorulli, G. B. Gechele, Eur. Polym. J., **6**, 1219 (1970).
2620. C. Plachetta, N. O. Rau, A. Hauck, R. C. Schulz, Makromol. Chem., Rapid Commun., **3**, 249 (1982).
2621. K. E. B. Platzer, V. S. Ananthanarayanan, R. H. Andreatta, H. A. Scheraga, Macromolecules, **5**, 177 (1972).
2622. I. Y. Poddubnyi, V. A. Grechanovskii, M. I. Mosevitskii, Vysokomol. Soedin., **5**, 1042 (1963).
2623. I. Ya. Poddubnyi, M. A. Rabinerzon, J. Appl. Polym. Sci., **9**, 2527 (1965).
2624. I. Ya. Poddubnyi, V. A. Grechanovskii, E. G. Ehrenburg, Makromol. Chem., **94**, 268 (1966).
2625. I. Ya. Poddubnyi, A. V. Podalinskii, V. A. Grechanovskii, Polym. Sci. USSR, **8**, 1715 (1966); Vysokomol. Soedin., **8**, 1556 (1966).
2626. I. Ya. Poddubnyi, A. V. Podalinskii, Polym. Sci. USSR, **11**, 450 (1969); Vysokomol. Soedin. A, **11**, 400 (1969).
2627. I. Ya. Poddubnyi, A. V. Podalinskii, V. A. Grechanovskii, Polym. Sci. USSR, **14**, 799 (1972); Vysokomol. Soedin. A, **14**, 714 (1972).
2628. J. Podešva, M. Bohdanecký, P. Kratochvil, G. Samay, J. Polym. Sci., Polym. Phys. Ed., **15**, 1521 (1977).
2629. J. Podešva, O. Prochazka, J. Stejskal, P. Spacek, S. Enders. J. Appl. Polym. Sci., **49**, 1265 (1993).
2630. J. Podešva, P. Spacek, A. Sikova, J. Polym. Sci., Polym. Chem. Ed., **22**, 3343 (1984).
2631. J. Podešva, J. Stejskal, O. Procházka, P. Spacek, S. Enders, J. Appl. Polym. Sci., **48**, 1127 (1993).
2632. S. Podzimek, J. Appl. Polym. Sci., **54**, 91 (1994).
2633. N. V. Pogodina, P. N. Lavrenko, K. S. Pozhivilko, A. B. Mel'nikov, T. A. Kolobova, G. N. Marchenko, V. N. Tsevt-

kov, Polym. Sci. USSR, **24**, 358 (1982); Vysokomol. Soedin. A, **24**, 332 (1982).
2634. J. Poláček, Coll. Czech. Chem. Commun., **28**, 1838 (1963).
2635. J. Poláček, Coll. Czech. Chem. Commun., **28**, 3011 (1963).
2636. J. Poláček, I. Koessler, J. Vodehnal, J. Polym. Sci. A-1, **3**, 2511 (1965).
2637. J. Poláček, L. Schulz, I. Koessler, J. Polym. Sci. C, **16**, 1327 (1967).
2638. J. Poláček, Eur. Polym. J., **6**, 81 (1970).
2639. D. J. Pollock, L. J. Elyash, T. W. De Witt, J. Polym. Sci., **15**, 87 (1955).
2640. D. J. Pollock, L. J. Elyash, T. W. De Witt, J. Polym. Sci., **15**, 335 (1955).
2641. S. Polowinski, Eur. Polym. J., **14**, 563 (1978).
2642. V. M. Polyakova, A. Ye. Fainerman, R. T. Voitsekhouskii, Polym. Sci. USSR, **6**, 479 (1964); Vysokomol. Soedin., **6**, 432 (1964).
2643. G. R. Polyakova, N. A. Pravikova, Polym. Sci. USSR, **9**, 1578 (1967); Vysokomol. Soedin., **9**, 1405 (1967).
2644. V. A. Ponomarenko, A. M. Khomutov, A. P. Alimov, Polym. Sci. USSR, **10**, 1203 (1968); Vysokomol. Soedin. A, **10**, 1038 (1968).
2645. V. A. Ponomarenko, A. M. Khomutov, S. I. Il'chenko, G. N. Goseshkova, V. S. Bogdanov, Polym. Sci. USSR, **11**, 206 (1969); Vysokomol. Soedin. A, **11**, 182 (1969).
2646. M. T. Pope, T. J. Weakley, P. J. P. Williams, J. Chem. Soc. 3442 (1959).
2647. I. G. Popovic, L. Katsikas, W. Schnabel, Eur. Polym. J., **25**, 465 (1989).
2648. R. S. Porter, J. F. Johnson, Polymer, **5**, 201 (1964).
2649. R. S. Porter, M. J. R. Cantow, J. F. Johnson, Makromol. Chem., **94**, 143 (1966).
2650. R. S. Porter, M. J. R. Cantow, J. F. Johnson, J. Appl. Polym. Sci., **11**, 335 (1967).
2651. V. A. Postinkov, N. J. Lukin, B. V. Maslov, N. A. Platé Polym. Bull., **3**, 75 (1980).
2652. J. Pouradier, A. M. Venet, J. Chim. Phys., **47**, 11 (1950).
2653. A. Pourdjavadi, P. J. Madec, E. Maréchal, Eur. Polym. J., **20**, 305 (1984).
2654. G. Pouyet, F. Candau, J. Dayantis, Makromol. Chem., **177**, 2973 (1976).
2655. E. Powell, Polymer, **8**, 211 (1967).
2656. P. O. Powers, Ind. Eng. Chem., **42**, 2558 (1950).
2657. T. I. Poznyak, D. M. Lisitsyn, D. D. Novikov, A. A. Berlin, F. S. D'yachkouskii, Yu. A. Prochukham, Yu. A. Sangalov, K. S. Minsker, Polym. Sci. USSR, **22**, 1565 (1980); Vysokomol. Soedin. A, **22**, 1424 (1980).
2658. W. Prakellok, O. Vogl, A. Gupta, J. Polym. Sci., Polym. Chem. Ed., **19**, 3307 (1981).
2659. J. A. Pratt, S-H Lee, M:A. McHugh, J. Appl. Polym. Sci., **49**, 953 (1993).
2660. N. A. Pravikova, Y. U. Ryabova, P. Vyrskyi, Vysokomol. Soedin., **5**, 1165 (1963).
2661. R. Prechner, R. Panaris, H. Benoît, Makromol. Chem., **156**, 39 (1972).
2662. D. C. Prevorsek, B. T. Debona, Y. Kesten, J. Polym. Sci. Polym. Chem. Ed., **18**, 75 (1980).
2663. C. Price, A. G. Watson, M. T. Chow, Polymer, **13**, 333 (1972).
2664. C. Price, A. L. Hudd, C. Booth, B. Wright, Polymer, **23**, 650 (1982).
2665. R. B. Prime, B. Wunderlich, J. Polym. Sci. A-2, **7**, 2061 (1969).
2666. A. Priola, N. Passerini, M. Bruzzone, S. Cesca, Angew. Makromol. Chem., **88**, 21 (1980).
2667. J. Probst, K. Unger, H.-J. Cantow, Angew. Makromol. Chem., **35**, 177 (1974).
2668. J. Probst, G. Kolb, Makromol. Chem., **177**, 2681 (1976).
2669. K. Procházka, G. Glöckner, M. Hoff, Z. Tuzar, Makromol. Chem., **185**, 1187 (1984).
2670. A. Proto, L. Oliva, C. Pellecchia, A. J. Sivak, L. A. Cullo, Macromolecules, **23**, 2904 (1990).
2671. T. Provder, J. H. Clark, E. E. Drott, Polym. Preprints, **12** (2), 819 (1971).
2672. T. Provder, J. C. Woodbrey, J. H. Clark, Sep. Sci., **6**, 101 (1971).
2673. V. S. Pshezhetskii, G. A. Murtazaeva, V. A. Kabanov, Eur. Polym. J., **10**, 584 (1974).
2674. R. Puakkeri, A. Lehtinen, Polymer, **34**, 4075 (1993).
2675. N. G. Puchkova, A. V. Nekrasov, Ye. F. Razvodovskii, B. S. El'tsefon, Polym. Sci. USSR, **22**, 1407 (1980); Vysokomol. Soedin. A, **22**, 1281 (1980).
2676. C. Pugh, R. R. Schrock, Macromolecules, **25**, 6593 (1992)
2677. J. P. Purdon, Jr., R. D. Mate, J. Polym. Sci. A-1, **6**, 243 (1968).
2678. J. Puskás, G. Kaszas, J. P. Kennedy, T. Kelen, F. Tüdös, J, Macromol. Sci., Part A: Chem., **18**, 1263 (1982/83).
2679. J. Puskás, G. Kaszas, J. P. Kennedy, T. Kelen, F. Tüdös, J, Macromol. Sci., Part A: Chem., **18**, 1229 (1982/83).
2680. J. Puskás, G. Kaszas, J. P. Kennedy, T. Kelen, F. Tüdös, J, Macromole. Sci., Part A: Chem., **18**, 1245 (1982/83).
2681. J. Puskás, G. Kaszas, J. P. Kennedy, T. Kelen, F. Tüdös, J, Macromole. Sci., Part A: Chem., **18**, 1315 (1982/83).
2682. A. Qin, M. Tian, C. Ramireddy, S. E. Webber, P. Munk, Z. Tuzar, Macromolecules, **27**, 120 (1994).
2683. O. Quadrat, M. Bohdanecky, Coll. Czech. Chem. Commun., **33**, 2130 (1968).
2684. O. Quadrat, Coll. Czech. Chem. Commun., **35**, 2564 (1970).
2685. O. Quadrat, Coll. Czech. Chem. Commun., **36**, 2042 (1971).
2686. D. V. Quayle, Polymer, **8**, 217 (1967).
2687. M. Y. Qureshi, M. Ochel, Eur. Polym. J., **32**, 691 (1996).
2688. W. Rabel, K. Ueberreiter, Kolloid, Z.-Z. Polym., **198**, 1 (1964).
2689. D. Radić, L. Gargallo, J. Polym. Sci., Polym. Phys. Ed., **16**, 977 (1978).
2690. D. Radić, N. Hamidi, L. Gargallo, Eur. Polym. J., **24**, 799 (1988).
2691. D. Radić, L. Gargallo, A. León, A. Horta, J. Macromol. Sci., Part B: Phys., **31**, 215 (1992).
2692. D. Radić, L. Gargallo, Macromolecules, **30**, 817 (1997).
2693. S. R. Rafikov, V. V. Korshak, G. N. Chelnokova, Izv. Akad. Nauk, SSSR, 642 (1948).
2694. G. Rafler, G. Reinisch, Angew. Makromol. Chem., **20**, 57 (1971).

2695. D. Rahlwes, R. G. Kirste, Makromol. Chem., **178**, 1793 (1977).
2696. A. Rakotomanga, P. Hemery, S. Boileau, B. Lotz, Eur. Polym. J., **14**, 581 (1978).
2697. A. Ram, J. Miltz, J. Appl. Polym. Sci., **15**, 2639 (1971).
2698. A. Ram, J. Miltz, IUPAC Helsinki, Preprints, vol. III, II-47 (1972).
2699. S. Ramakrishnan, E. Berluche, T. C. Chung, Macromolecules, **23**, 378 (1990).
2700. S. Ramakrishnan, T. C. Chung, Macromolecules, **23**, 4519 (1990).
2701. N. K. Raman, J. J. Hermans, J. Polym. Sci., **35**, 71 (1959).
2702. K. R. Ramazanov, N. G. Khiebtsov, V. I. Klenin, Polym. Sci. USSR, **25**, 1279 (1983); Vysokomol. Soedin. A, **25**, 1102 (1983).
2703. C. Ramireddy, Z. Tuzar, K. Procházka, S. E. Webber, P. Munk, Macromolecules, **25**, 2541 (1992).
2704. S. Ramesh, G. Radhakrishnan, Eur. Polym. J., **32**, 993 (1996).
2705. W. G. Rand, A. K. Mukherji, J. Polym. Sci., Polym. Lett. Ed., **20**, 501 (1982).
2706. M. R. Rao, V. Kalpagam, J. Sci. Ind. Res. (India), **20**, 207 (1961).
2707. S. P. Rao, M. Santappa, J. Polym. Sci. A-1, **6**, 95 (1968).
2708. M. R. Rao, S. Packirisamy, P. V. Ravindran, P. K. Nasendranath, Macromolecules, **25**, 5165 (1992).
2709. N. S. Rapp, J. D. Ingham, J. Polym. Sci. A, **2**, 689 (1964).
2710. I. B. Rashkov, G. Merle, Q. T. Pham, V. C. Shishkova, J. P. Pascault, I. M. Panayotv, Eur. Polym. J., **18**, 37 (1982).
2711. I. B. Rashkov, I. Gitsov, J. Polym, Sci., Polym. Chem. Ed., **22**, 905 (1984).
2712. I. B. Rashkov, T. V. Kakuliva, N. G. Vladiminov, I. Gitsoc, Eur. Polym. J., **23**, 407 (1986).
2713. I. B. Raskkov, I. Gitsov, I. M. Panayotov, J. P. Pascault, J. Polym. Sci., Polym. Chem. Ed., **21**, 923 (1983).
2714. G. B. Rathmann, F. A. Bovey, J. Polym. Sci., **15**, 544 (1955).
2715. A. Ravve, J. T. Khamis, J. Polym. Sci., **61**, 185 (1962).
2716. A. Ravve, J. T. Khamis, L. X. Mallavarapu, J. Polym. Sci. A-1, **3**, 1775 (1965).
2717. M. G. Rayner, Polymer, **10**, 827 (1969).
2718. S. F. Redd, Jr., J. Polym. Sci., Polym. Chem. Ed., **10**, 2025 (1972).
2719. C. R. Reddy, V. Kalpagam, J. Polym. Sci., Polym. Phys. Ed., **14**, 749 (1976).
2720. O. Redlich, A. L. Jacobson, W. H. McFadden, J. Polym. Sci. A, **1**, 393 (1963).
2721. G. Reinnisch, G. Rafler, G. I. Timofejeva, Angew. Makromol. Chem., **7**, 110 (1969).
2722. P. Rempp, Polym. Preprints, **7** (1), 141 (1966).
2723. C. N. Renn, R. E. Synovec, Anal. Chem., **60**, 200 (1988).
2724. A. Revillon, L. E. St. Pierre, J. Appl. Polym. Sci., **14**, 373 (1970).
2725. A. Revillon, B. Dumont, A. Guyot, J. Polym. Sci., Polym. Chem. Ed., **14**, 2263 (1976).
2726. A. Revillon, T. Hamaide, Polym. Bull., **6**, 235 (1982).
2727. A. Revillon, P. Boucher, J. Appl. Polym. Sci., Appl. Polym. Symp., **43**, 115 (1989).
2728. A. Revillon, P. Boucher, J.-F. Guilland, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 243 (1991).
2729. R. Rew, D. Suwande, S. T. Balke, T. H. Mousey, J. Appl. Polym. Sci., Appl. Polym. Symp., **52**, 125 (1993).
2730. G. E. J. Reynolds, J. Polym. Sci. C, **16**, 3957 (1968).
2731. E. Riande, M. Alonso, J. M. G. Fatou, Rev. Plast. (Madrid), **186**, 1834 (1971).
2732. E. Riande, J. M. Pereña, J. Macromol. Sci., Part A: Chem., **8**, 1413 (1974).
2733. E. Riande, J. M. Pereña, Makromol. Chem., **175**, 2923 (1974).
2734. E. Riande, J. Polym. Sci., Polym. Phys. Ed., **14**, 2231 (1976).
2735. E. Riande, J. E. Mark, Macromolecules, **11**, 956 (1978).
2736. E. Riande, J. Guzmán, Macromolecules, **12**, 952 (1979).
2737. E. Riande, J. Guzmán, E. Sáiz, Polymer, **22**, 465 (1981).
2738. E. Riande, J. Guzmán, M. A. Llorente, Polym. Bull, **9**, 369 (1983).
2739. E. Riande, J. G. de la Campa, J. Guzmán J. De Abajo, Macromolecules, **17**, 1431 (1984).
2740. E. Riande, J. Guzmán, J. Abajo, Makromol. Chem., **185**, 1943 (1984).
2741. E. Riande, J. Guzmán, J. Polym. Sci., Polym. Phys. Ed., **23**, 1235 (1985).
2742. E. Riande, J. Guzmán, H. Adabbo, Macromolecules, **19**, 2567 (1986).
2743. E. Riande, J. Guzmán, J. G. de la Campa, J. de Abajo, J. Polym. Sci. Part B: Polym. Phys., **25**, 2403 (1987).
2744. E. Riande, J. G. de la Campa, D. D. Schlereth, J. de Abajo, J. Guzmán, Macromolecules, **20**, 1641 (1987).
2745. E. Riande, J. Guzmán J. G. de la Campa, Macromolecules, **21**, 2128 (1988).
2746. D. H. Richards, S. B. Kingston, T. Sovel, Polymer, **19**, 68 (1978).
2747. D. H. Richards, S. B. Kingston, T. Sovel, Polymer, **19**, 806 (1978).
2748. G. N. Richards, J. Appl. Polym. Sci., **5**, 540 (1961).
2749. R. W. Richards, Polymer, **18**, 114 (1977).
2750. M. Ricker, M. Schmidt, Makromol. Chem., **192**, 679 (1991).
2751. B. Rieger, J. C. W. Chien, Polym. Bull., **21**, 159 (1989).
2752. B. Rieger, X. Mu, D. T. Mallin, M. D. Rausch, J. C. W. Chien, Macromolecules, **23**, 3559 (1990).
2753. J. K. Rieke, G. M. Hart, J. Polym. Sci. C, **1**, 117 (1963).
2754. R. Rigamonti, E. Meda, Ricerca Sci., **25**, 457 (1955).
2755. M. Rinaudo, J. P. Merle, Eur. Polym. J., **6**, 41 (1970).
2756. W. Ring, H.-J. Cantow, Makromol. Chem., **89**, 138 (1965)
2757. W. Ring, H.-J. Cantow, W. Holtrup, Eur. Polym. J., **2**, 151 (1966).
2758. W. Ring, W. Holtrup, Makromol. Chem., **103**, 83 (1967).
2759. M. Riou, R. Pibarot, Rev. Gen. Caoutchouc, **27**, 596 (1950).
2760. L. A. Rishina, E. I. Vizen, Eur. Polym. J., **16**, 965 (1980.
2761. L. Robitaille, N. Turcotte, S. Fortin, Macromolecules, **24**, 2413 (1991).
2762. C. Rochas, A. Domard, M. Rinaudo, Eur. Polym. J., **16**, 135 (1980).

2763. W. E. Rochefort, G. W. Heffner, D. S. Pearson, R. D. Miller, P. M. Cotts, Macromolecules, **24**, 4861 (1991).

2764. J. Roda, J. Kralicek, Z. Bouskava, Eur. Polym. J., **13**, 119 (1977).

2765. V. V. Rode, P. N. Gribkova, V. V. Korshak, Polym. Sci. USSR, **11**, 60 (1969); Vysokomol. Soedin. A, **11**, 57 (1969).

2766. D. Roessner, W.-M. Kulicke, J. Chromatogr., **687**, 249 (1994).

2767. R. Roestamsjah, L. A. Wall, R. E. Florin, M. H. Aldridge, L. J. Fetters, J. Polym. Sci., Polym. Phys. Ed., **13**, 1783 (1975).

2768. 'T. W. Rogers, ACS Div. Org. Coating Plastics Chem., **33**, 65 (1973).

2769. C. L. Rohn, J. Polym. Sci. A-2, **5**, 547 (1967).

2770. A. Roig, J. E. Figueruelo, E. Llano, J. Polym. Sci. B, **3**, 171 (1965).

2771. A. Roig, J. E. Figueruelo, E. Llano, J. Polym. Sci. C, **16**, 4141 (1968).

2772. A. Roig, E. Llano, J. E. Figueruelo, I. Katime, Anales Quím. (Madrid), **67**, 343 (1971).

2773. A. Roig, E. Llano, J. E. Figueruelo, Anales Quím. (Madrid), **67**, 699 (1971).

2774. J. M. Rooney, Makromol. Chem., **179**, 2419 (1978).

2775. J. E. L. Roovers, S. Bywater, Polym. Preprints, **12** (1), 290 (1971).

2776. J. E. L. Roovers, S. Bywater, Macromolecules, **5**, 384 (1972).

2777. J. Roovers, N. Hadjichristridis, L. J. Fetters, Macromolecules, **16**, 214 (1983).

2778. J. Roovers, J. D. Cooney, P. M. Toporowski, Macromolecules, **23**, 1611 (1990).

2779. J. L. Rosenberg, C. O. Beckmann, J. Colloid. Sci., **3**, 483 (1948).

2780. A. J. Rosenthal, B. B. White, Ind. Eng. Chem., **44**, 2693 (1952).

2781. D. Rosenvasser, A. S. Casas, R. V. Figini, Makromol. Chem., **183**, 3067 (1982).

2782. W. E. Roseveare, L. Poore, Ind. Eng. Chem., **45**, 2518 (1953).

2783. J. H. Ross, M. E. Casto, J. Polym. Sci. C, **21**, 143 (1968).

2784. A. Rossi, J. Zhang, G. Odian, Macromolecules, **29**, 2231 (1996).

2785. J. L. Roubaty, A. Revillion, Brit. Polym. J., **8**, 95 (1976).

2786. B. A. Rozenberg, N. V. Makletsova, I. V. Epel'baum, Ye. B. Lyudvig, S. S. Medvedev, Polym. Sci., USSR, **7**, 1163 (1965); Vysokomol. Soedin., **7**, 1051 (1965).

2787. D. N. Rubingh, H. Yu, Macromolecules, **9**, 681 (1976).

2788. A. Rudin, G. W. Bennett, J. R. McLaren, J. Appl. Polym. Sci., **13**, 2371 (1969).

2789. A. Rudin, H. L. W. Hoegy, J. Polym. Sci., Polym. Chem. Ed., **10**, 217 (1972).

2790. A. Rudin, C. A. Fyfe, S. Martin Vines, J. Appl. Polym. Sci., **28**, 2611 (1983).

2791. C. A. Rue, M. E. Schimpfuraf, Anal. Chem., **66**, 4054 (1994).

2792. J. Rueter, K.-H. Magosch, R. Feinaur, Angew. Makromol. Chem., **56**, 139 (1976).

2793. J. M. Ruiz, J. P. Busnell, J. Benoit, Pharm. Res., **7**, 928 (1990).

2794. J. P. Ruiz, J. R. Dharia, J. R. Reynolds, L. J. Buckley, Macromolecules, **25**, 849 (1992).

2795. J. R. Runyon, D. E. Barnes, J. D. Rudd, L. H. Tung, J. Appl. Polym. Sci., **13**, 2359 (1969).

2796. A. M. Ruskin, G. Parravano, J. Appl. Polym. Sci., **8**, 565 (1964).

2797. C. A. Russell, J. Appl. Polym. Sci., **4**, 219 (1960).

2798. D. R. Rutherford, J. K. Stille, C. M. Elliot, V. R. Reichart, Macromolecules, **25**, 2294 (1992).

2799. L. G. Ryabova, Z. Ya. Berestneva, N. A. Pravikova, Polym. Sci. USSR, **7**, 1976 (1965); Vysokomol. Soedin., **7**, 1796 (1965).

2800. M. C. Sacchi, Z.-Q. Fan, F. Forlini, I. Tritto, P. Locatelli, Makromol. Chem. Phys., **195**, 2805 (1994).

2801. R. Sack, G. V. Schulz, G. Meyorhoff, Macromolecules, **21**, 3345 (1988).

2802. J. Sadanobu, T. Norisuye, H. Fujita, Polym. J., **13**, 75 (1981).

2803. M. U. Sadykov, U. Azizov, Kh. U. Usmanov, I. M. Mirkamilov, Polym. Sci. USSR, **10**, 376 (1968); Vysokomol. Soedin. A, **10**, 322 (1968).

2804. S. Saeda, J. Yotsuyanagi, K. Yamaguchi, J. Appl. Polym. Sci., **15**, 277 (1971).

2805. T. Saegusa, T. Yatsu, H. Fujii, Polym. J., **8**, 593 (1976).

2806. T. Saegusa, T. Tokuzawa, S. Kobayashi, Polym. Bull., **1**, 341 (1979).

2807. S. Saeki, N. Kuwahara, M. Nakata, M. Kaneko, Polymer, **17**, 685 (1976).

2808. T. Saito, K. Yamaguchi, Polymer, **15**, 219 (1974).

2809. M. Sakai, I. Noda, M. Nagasawa, J. Polym. Sci., Polym. Phys. Ed., **10**, 1047 (1972).

2810. M. Sakai, T. Fujimoto, M. Nagasawa, Macromolecules, **5**, 786 (1972).

2811. W. Sakai, A. Tsuchida, M. Yamamoto, J. Yamauchi, J. Polym. Sci. Part A, Polym. Chem., **33**, 1969 (1995).

2812. M. Sakamoto, Y. Kuroyanagi, J. Polym. Sci., Polym. Chem. Ed., **16**, 1107 (1978).

2813. N. Sakota, H. Nakamura, K. Nishihara, Makromol. Chem., **129**, 56 (1969).

2814. I. Sakurada, S. Matuzawa, Y. Kubota, Makromol. Chem., **68**, 115 (1963).

2815. I. Sakurada, A. Nakajima, K. Shibatani, J. Polym. Sci. A, **2**, 3545 (1964).

2816. K. Sakurai, K. Ochi, T. Norisuye, H. Fujita, Polym. J., **16**, 559 (1984).

2817. P. Salemis, M. Rinaudo, Polym. Bull., **12**, 283 (1984).

2818. P. Salemis, M. Rinaudo, Polym. Bull., **11**, 397 (1984).

2819. C. Salom, J. J. Freire, I. Hernández-Fuentes, Polym. J., **20**, 1109 (1988).

2820. R. Salovey, M. Y. Hellman, J. Polym. Sci. A-2, **5**, 333 (1967).

2821. R. Salovey, M. Y. Hellman, J. Polym. Sci. B, **5**, 647 (1967).

2822. R. Salovey, R. C. Gebauer, J. Polym. Sci., Polym. Chem. Ed., **10**, 1533 (1972).

2823. I. O. Salyer, A. S. Kenyon, M. Ohta, J. Polym. Sci., Polym. Chem. Ed., **10**, 419 (1972).
2824. G. Samay, M. Kubin, J. Podêsva, Angew. Makromol. Chem., **72**, 185 (1978).
2825. S. M. Samoilov, V. N. Monastyrskii, A. F. Zhigach, Polym. Sci. USSR, **18**, 829 (1976); Vysokomol. Soedin. A, **18**, 731 (1976).
2826. G. V. Samsonov, Uspedhi. Khim., **30**, 1410 (1961).
2827. A. Sánchez, C. Marco, J. G. Fatou, A. Bello, Eur. Polym. J., **24**, 355 (1988).
2828. G. Sánchez, G. Weill, R. Knoesel, Makromol, Chem., **179** 131 (1978).
2829. A. Sánchez, C. Marco, J. G. Fatou, A. Bello, Makromol. Chem., **188**, 1205 (1987).
2830. A. Sánchez, A. Bello, C. Marcos, J. G. Fatou, Makromol. Chem., **189**, 399 (1988).
2831. A. Sanguineti, P. A. Guarda, G. Mardionni, G. Ajroldi, Polymer, **36**, 3697 (1995).
2832. K. Sanui, R. W. Lenz, W. J. Mac Knight, J. Polym. Sci., Polym. Chem. Ed., **12**, 1965 (1974).
2833. M. Sanviat, J. P. Cohen-Addad, Polymer, **22**, 461 (1981).
2834. K. Sardelis, H. J. Michels, G. Allen, Polymer, **25**, 1011 (1984).
2835. S. Sarkanen, D. C. Teller, J. Hall, J. L. Mc Carthy, Macromolecules, **14**, 426 (1981).
2836. S. Sarkanen, D. C. Teller, E. Abramowski, J. L. McCarthy, Macromolecules, **15**, 1098 (1982).
2837. S. Sarkanen, D. C. Teller, C. R. Stevens, J. L. McCarthy, Macromolecules, **17**, 2588 (1984).
2838. Y. Sasaki, L. L. Walter, E. L. Hurst, C. U. Pittman Jr., J. Polym. Sci., Chem. Ed., **11**, 1213 (1973).
2839. T. Sasaki, T. Ikeda, K. Ichimura, Macromolecules, **25**, 3807 (1992).
2840. V. R. Sastri, R. Schulman, D. C. Roberts, Macromolecules, **15**, 939 (1982).
2841. K. S. Sastry, R. D. Patel, Eur. Polym. J., **5**, 79 (1969).
2842. H. Sato, Bull. Chem. Soc. Japan, **39**, 2335 (1966).
2843. H. Sato, K. Saito, K. Miyashita, Y. Tanaka, Makromol. Chem., **182**, 2259 (1981).
2844. H. Sato, H. Takeuchji, S. Suzuki, Y. Tanaka, Makromol. Chem., Rapid Commun., **5**, 719 (1984).
2845. H. Sato, H. Takeuchi, Y. Tanaka, Macromolecules, **19**, 2613 (1986).
2846. H. Sato, K. Mitsutami, I. Shimizu, Y. Tanaka, J. Chromatog., **447**, 387 (1988).
2847. K. Sato, F. R. Eirich, J. E. Mark, J. Polym. Sci., Polym. Phys. Ed., **14**, 619 (1976).
2848. T. Sato, M. Yoshioka, T. Otsu, Makromol. Chem., **153**, 47 (1972).
2849. T. Sato, K. Takahashi, H. Tanaka, T. Ota, K. Kato, Macromolecules, **24**, 2330 (1991).
2850. T. Sato, Y. Tsujii, T. Fukuda, T. Miyamoto, Macromolecules, **25**, 3890 (1992).
2851. T. Sato, Y. Jinbo, A. Teramoto, Macromolecules, **30**, 590 (1997).
2852. K. G. Saunders, W. J. Macknight, R. W. Lenz, Macromolecules, **15**, 1 (1982).
2853. M. I. Savitskaya, S. Ya. Frenkel, Zh. Fiz. Khim., **32**, 1063 (1958).
2854. M. Sawamoto, T. Masuda, T. Higashimura, Makromol. Chem., **177**, 2995 (1976).
2855. M. Sawamoto, J. P. Kennedy, J. Macromol. Sci., Part A: Chem., **18**, 1293 (1982/83).
2856. M. Sawamoto, J. P. Kennedy, J. Macromol. Sci., Part A: Chem., **18**, 1275 (1982/83).
2857. M. Sawamoto, J. P. Kennedy, J. Macromol. Sci., Part A: Chem., **18**, 1301 (1982/83).
2858. S. P. Sawan, Y.-G. Tsai, H.-Y. Huang, T.-M. Hsu, Polym. Prepr. Am. Chem. Soc., Div. Polym. Chem., **29** (1), 297 (1988).
2859. E. Scamporrino, D. Vitalini, Macromolecules, **25**, 1625 (1992).
2860. W. Schabel, E. Schamberg, Makromol. Chem., **104**, 9 (1967).
2861. P. C. Scherer, B. P. Rouse, Rayon Synth. Text., **29**, 55, 85 (1948).
2862. P. C. Scherer, R. D. McNeer, Rayon Synth. Text., **30**, 56 (1949).
2863. P. C. Scherer, R. B. Thompson, Rayon Synth. Text., **31**, 51 (1950).
2864. H. G. Schild, D. A. Tirrell, Macromolecules, **25**, 4553 (1992).
2865. C. Schildknecht, S. Gross, H. Davidson, Ind. Eng. Chem., **40**, 2104 (1948).
2866. M. E. Schimpf, P. S. Williams, J. C. Giddings, J. Appl. Polym. Sci., **37**, 2059 (1989).
2867. A. Schindler, Monatsh Chem., **95**, 868 (1964).
2868. A. Schindler, R. B. Strong, Makromol. Chem., **114**, 77 (1968).
2869. A. Schindler, Y. M. Hibionada, C. G. Pitt, J. Polym. Sci., Polym. Chem. Ed., **20**, 319 (1982).
2870. S. Schlick, M. Levy, J. Phys. Chem., **64**, 883 (1960).
2871. P. J. Schmitt, G. V. Schulz, Makromol. Chem., **121**, 184 (1969).
2872. F. P. Schmitz, E. Klesper, Makromol. Chem., Rapid Commun., **2**, 735 (1981).
2873. W. Schnabel, H.Sotobayashi, Polym. J., **8**, 423 (1976).
2874. W. Schnabel, S. Klaumuenzer, H. Sotobayashi, F. Asmussen, Y. Tabata, Macromolecules, **17**, 2108 (1984).
2875. N. S. Schneider, L. G. Holmes, J. Polym. Sci., **38**, 552 (1959).
2876. N. S. Schneider, L. G. Holmes, C. F. Mijal, J. D. Loconti, J. Polym. Sci., **37**, 551 (1959).
2877. N. S. Schneider, J. D. Loconti, L. G. Holmes, J. Appl. Polym. Sci., **3**, 251 (1960).
2878. N. S. Schneider, Anal. Chem., **33**, 1829 (1961).
2879. N. S. Schneider, J. D. Loconti, L. G. Holmes, J. Appl. Polym. Sci., **5**, 354 (1961).
2880. N. S. Schneider, J. Polym. Sci. C, **8**, 179 (1965).
2881. N. S. Schneider, R. J. Traskos, A. S. Hoffman, J. Appl. Polym. Sci., **12**, 1567 (1968).
2882. K. M. Scholsky, L.-W. Morgan, J. Polym. Sci. C, Polym. Letters, **26**, 181 (1988).

2883. K. M. Scholsky, L. W. Morgan, J. Polym. Sci. C, Polym. Lett., **26**, 181 (1988).
2884. K. M. Scholsky, J. Appl. Polym. Sci., **47**, 1633 (1993).
2885. W. Scholtan, Makromol. Chem., **24**, 104 (1957).
2886. W. Scholtan, Makromol. Chem., **24**, 83 (1957).
2887. W. Scholtan, Makromol. Chem., **36**, 162 (1960).
2888. W. Scholtan, H. Marzolph, Makromol. Chem., **55**, 52 (1962).
2889. W. Scholtan, D. Kranz, Makromol. Chem., **110**, 150 (1967).
2890. W. Scholtan, H. Lange, S. Y. Lie, K. Dinges, R. Meyer-Mader, Angew. Makromol. Chem., **14**, 43 (1970).
2891. W. Scholtan, H. Lange, R. Casper, H. Uwe Pohl, D. Wendisch, R. Meyer-Mader, Angew. Makromol. Chem., **27**, 1 (1992).
2892. W. Scholtan, F. J. Kwoll, Makromol. Chem., **151**, 33 (1972).
2893. Th. G. Scholte, J. Polym. Sci. A-2, **6**, 111 (1968).
2894. Th. G. Scholte, Eur. Polym. J., **6**, 51 (1970).
2895. P. Schouterdan, G. Groeninck, B. van der Heijden, F. Jansen, Polymer, **28**, 2099 (1987).
2896. P. Schouterdan, G. Groeninckx, R. Reynaers, H. Riekel, M. H. J., Polym. Bull., **13**, 553 (1985).
2897. G. V. Schuiz, M. Marx-Figini, Makromol. Chem., **14**, 52'(1954).
2898. K. Schultes, B. A. Wolf, W. H. Meyer, G. Wegner, Acta Polym., **46**, 367 (1995).
2899. K. Schultes, B. A. Wolf, W. H. Meyer, G. Wegner, Macromol. Chem. Phys., **196**, 1005 (1995).
2900. D. N. Schulz, A. F. Halasa, J. Polym. Sci., Polym. Chem. Ed., **15**, 2401 (1977).
2901. D. N. Schulz, J. W. Spiewak, J. K. Valaitis, V. D. Mochel, M. L. Barzan, Macromolecules, **13**, 1367 (1980).
2902. G. V. Schulz, in: H. A. Stuart (Ed.), "Die Physik der Hochpolymeren", Springer, Berlin, vol. 2 (1953).
2903. G. V. Schulz, M. Hoffmann, Makromol. Chem., **23**, 220 (1957).
2904. G. V. Schulz, A. Scholz, R. V. Figini, Makromol. Chem., **57**, 220 (1962).
2905. G. V. Schulz, W. Wunderlich, R. Kirste, Makromol. Chem., **75**, 22 (1964).
2906. G. V. Schulz, K. C. Berger, A. G. R. Schloz, Ber. Bunsenges. Phys. Chem., **68**, 856 (1965).
2907. G. V. Schulz, J. Romatowski, Makromol. Chem., **85**, 195 (1965).
2908. G. V. Schulz, E. Penzel, Makromol. Chem., **112**, 260 (1968).
2909. M. F. Schulz, A. K. Khandpur, F. S. Bates, K. Almdal, K. Mortensen, D. A. Hajduk, S. M. Gruner, Macromolecules, **29**, 2857 (1996).
2910. R. C. Schulz, G. Renner, A. Henglein, W. Kern, Makromol. Chem., **12**, 20 (1954).
2911. R. C. Schulz, E. Mueller, W. Kern, Makromol. Chem., **30**, 39 (1959).
2912. R. C. Schulz, N. Vollkommer, Makromol. Chem., **116**, 288 (1968).
2913. R. Schulz, H. Engelhardt, Chromatographia, **29**, 325 (1990).
2914. W. W. Schulz, W. C. Purdy, Anal. Chem., **35**, 2044 (1963).
2915. J. Schurz, Oesterr. Chem. Ztg., **56**, 311 (1955).
2916. J. Schurz, Th. Steiner, H. Streitzig, Makromol. Chem., **23**, 141 (1957).
2917. J. Schurz, M. Rebek, H. Spoerk, Angew. Makromol. Chem., **1**, 42 (1967).
2918. H. Schuster, M. Hoffmann, K. Dinges, Angew. Makromol. Chem., **9**, 35 (1969).
2919. H. J. L. Schuurmans, J. Polym. Sci., **57**, 557 (1962).
2920. T. Schwartz, J. Francois, Makromol. Chem., **182**, 2757 (1981).
2921. J. Schwelzer, J. Springer, Chromatographia, **20**, 347 (1985).
2922. U. Schwenk, Angew. Makromol. Chem., **47**, 43 (1975).
2923. J. Scornaux, R. Van Leemput, Makromol. Chem., **177**, 2721 (1976).
2924. R. M. Screaton, in: B. Ke (Ed.), "Newer Methods in Polymer Characterization", Interscience, New York, 1964.
2925. R. M. Screaton, Polym. Preprints **8** (2), 1379 (1967).
2926. R. M. Screaton, R. W. Seemann, J. Polym. Sci. C, **21**, 297 (1968).
2927. R. M. Screaton, R. W. Seemann, J. Appl. Polym. Symposia, **8**, 81 (1969).
2928. K. Se, H. Ohnuma, T. Kotaka, Polym. J., **14**, 895 (1982).
2929. K. Se, H. Ohnuma, T. Kotaka, Macromolecules, **16**, 1581 (1983).
2930. J. Sebban-Danon, J. Polym. Sci., **48**, 121 (1960).
2931. B. Sébille, J. Néel, J. Chim. Phys., **60**, 475 (1963).
2932. J. Sedlacek, J. Vohlidal, Z. Grubisic-Gallot, Makromol. Chem. Rapid Commun., **14**, 51 (1993).
2933. B. See, T. G. Smith, J. Appl. Polym. Sci., **10**, 1625 (1966).
2934. B. See, T. G. Smith, Eur. Polym. J., **7**, 727 (1971).
2935. C. G. Seefried, Jr., J. V. Koleske, F. E. Critchfield, C. R. Pfaffenberger, J. Polym. Sci., Polym. Phys. Ed., **18**, 817 (1980).
2936. L. Segal, J. Polym. Sci. B, **4**, 1011 (1966).
2937. L. Segal, Polym. Preprints, **8** (2), 1365 (1967).
2938. L. Segal, J. Polym. Sci. C, **21**, 267 (1968).
2939. L. Segal, J. D. Timpa, J. I. Wadsworth, J. Polym. Sci. A-1, **8**, 25 (1970).
2940. L. Segal, J. D. Timpa, J. I. Wadsworth, J. Polym. Sci. A-1, **8**, 3577 (1970).
2941. R. Ségnéla, J. Prud'homme, Macromolecules, **11**, 1007 (1978).
2942. A. L. Segre, M. Delfini. M. Paci, A. M. Raspolli-Galletti, R. Solaro, Macromolecules, **18**, 44 (1985).
2943. N. Segudovic, S. Sertic, M. Kovac-Filipovic, V. Jarm, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 185 (1991).
2944. J. Selb, Y. Gallot, Makromol. Chem., **181**, 809 (1980).
2945. J. Selb, Y. Gallot, Makromol. Chem., **181**, 2605 (1980).
2946. E. Selegny, M. Vert, Eur. Polym. J., **7**, 1307 (1971).
2947. Y. D. Semchikov, L. A. Smirnova, T. Y. Knyazeva, S. A. Bulgakova, V. I. Sherstyankh, Eur. Polym. J., **26**, 883 (1990).
2948. Y. D. Semchikov, L. A. Smirnova, N. A. Kopylova, V. V. Izvolenskii, Eur. Polym. J., **32**, 1213 (1996).
2949. T. K. Sengupta, S. R. Palit, J. Polym. Sci., Polym. Chem. Ed., **16**, 713 (1978).

2950. F. R. Senti, N. N. Hellman, N. H. Ludwig, G. E. Babcock, R. Tobin, C. A. Glass, B. L. Lamberts, J. Polym. Sci., **17**, 527 (1955).
2951. P. K. Seow, Y. Gallot, A. Skoulios, Makromol. Chem., **176**, 3153 (1975).
2952. K. Seow, J. P. Lingelser, Y. Gallot, Makromol. Chem., **178**, 107 (1977).
2953. G. Serpe, N. Chaupart, J. Polym. Sci.: Part B, Polym. Phys., **34**, 2351 (1996).
2954. S. L. N. Seung, R. N. Young, Polym. Bull., **1**, 481 (1979).
2955. V. P. Shaboldin, A. G. Sukhomudrenko, A. I. Krasheninnikov, V. A. Morozov, Polym. Sci. USSR, **14**, 1638 (1972); Vysokomol. Soedin. A, **14**, 1462 (1972).
2956. A. C. Shah, I. Holdaway. I. W. Parsons, R. N. Haward, Polymer, **19**, 1067 (1978).
2957. P. P. Shah, Eur. Polym. J., **20**, 519 (1984).
2958. A. S. Shaikh, S. Dumitriu, E. Comânitâ, C. I. Simionescu, Polym. Bull., **3**, 363 (1980).
2959. L. I. Shakhovskaya, L. V. Krayeva, L. I. Menzentseva, Polym. Sci. USSR, **22**, 2915 (1980); Vysokomol. Soedin. A, **22**, 2660 (1980).
2960. G. V. Shatalov, Ye. N. Pozina, B. I. Mikhant'ev, S. A. Preobrazhenskii, Polym. Sci. USSR, **22**, 995 (1980); Vysokomol. Soedin. A, **22**, 900 (1980).
2961. M. T. Shaw, F. Rodrihuez, Polym. Preprints, **7** (2), 1053 (1966).
2962. K. P. Shen, F. R. Eirich, J. Polym. Sci., **53**, 81 (1961).
2963. P. J. Sheth, J. F. Johnson, R. S. Porter, Polym. Preprints, **12** (2), 513 (1971).
2964. Y. Shibasaki, K. Fukuda, J. Polymer Sci., Polym. Chem. Ed., **19**, 257 (1981).
2965. E. Shiedermaier, J. Klein, Angew. Makromol. Chem., **10**, 169 (1970).
2966. J. S. Shih, J. F. Brandt, M. P. Zussman, D. A. Tirrell, J. Polym. Sci., Polym. Chem. Ed., **20**, 2839 (1982).
2967. M. Shima, E. Ogawa, K. Konishik, Makromol. Chem., **177**, 241 (1976).
2968. E. Shimamura, M. Scandola, Y. Doi, Macromolecules, **27**, 4429 (1994).
2969. T. Shimizu, A. Minakata, T. Tomiyama, Polymer, **21**, 1427 (1980).
2970. J. Shimura, I. Mita, H. Kambe, J. Polym. Sci. B, **2**, 403 (1964).
2971. Y. Shimura, Bull. Chem. Soc. Japan, **40**, 273 (1967).
2972. S. Shiono, J. Polym. Sci., Polym. Chem. Ed., **17**, 4123 (1979).
2973. K. Shirayama, T. Okada, S. Kita, J. Polym. Sci. Part A-2, **3**, 907 (1965).
2974. K. Shiriyama, T. Matsuda, S. I. Kita, Makromol. Chem., **147**, 155 (1971).
2975. Ye. I. Shklyarova, L. B. Itsikson, A. M. Kaplan, V. B. Golubev, V. P. Zubov, V. A. Kabanov, Polym. Sci. USSR, **20**, 517 (1978); Vysokomol. Soedin. A, **20**, 456 (1978).
2976. K. V. Shooter, J. A. V. Butler, J. Polym. Sci., **23**, 705 (1957).
2977. M. F. Shostakovskii, V. Z. Annenkova, A. K. Khaliullin, A. Inyutkin, E. A. Gaitseva, V. N. Salaurov, Polym. Sci. USSR, **11**, 2257 (1969); Vysokomol. Soedin. A, **11**, 1979 (1969).
2978. I. N. Shtennikova, V. N. Tsvetkov, T. V. Peker, Ye. I. Ryumtsev, Yu. P. Getmanchuk, Polym. Sci. USSR, **16**, 1256 (1974); Vysokomol. Soedin. A, **16**, 1086 (1974).
2979. I. N. Shtennikova, E. V. Korneeva, G. F. Kolbina, I. A. Strelina, P. N. Lavrenko, V. P. Shibaev, I. V. Ekaeva, Eur. Polym. J., **28**, 353 (1992).
2980. A. R. Shultz, J. Am. Chem. Soc., **76**, 3422 (1954).
2981. A. R. Shultz, A. L. Bridgman, E. M. Hadsell, C. R. McCullough, J. Polym. Sci. A-2, **10**, 273 (1972).
2982. N. V. Shulyatikova, D. I. Mandel'baum, Zh. Prikl. Khim., **24**, 264 (1951).
2983. S. Shyluk, J. Polym. Sci., **62**, 317 (1962).
2984. W. Siebourg, R. D. Lundberg, R. W. Lenz, Macromolecules, **13**, 1013 (1980).
2985. R. Signer, H. Gross, Helv. Chim. Acta, **17**, 726 (1934).
2986. R. Signer, R. Glanzmann, Makromol. Chem., **5**, 257 (1951).
2987. H. Sihtola. E. Kaila, N. Virkola, Makromol. Chem., **11**, 70 (1953).
2988. H. Sihtola, E. Kaila, L. Laamanen, J. Polym. Sci., **23**, 809 (1957).
2989. D. J. Sikkema, H, Angad-Gaur, Makromol. Chem., **181**, 2259 (1980).
2990. L. Šimek, J. Hrnčivk, M. Mládek, Angew. Markromol. Chem., **92**, (1980).
2991. L. Šimek, Z. Tuzar, M. Bohdanecký, Makromol, Chem., Rapid Commun., **1**, 215 (1980).
2992. L. Šimek, M. Bohdanecký, J. Appl. Polym Sci., **58**, 897 (1995).
2993. B. G. Siminionescu, S. Ioian, C. I. Simionescu, J. Polym. Sci. Part B: Polym. Phys., **25**, 829 (1987).
2994. B. C. Simionescu, S. Ioan, A. Chiriac, C. I. Simionescu, Polym. Bull., **17**, 439 (1987).
2995. C. C. Simionescu, E. Calistru, Faserforsch. Textiltch., **7**, 171 (1956).
2996. C. Simionescu, N. Cobianu, Eur. Polym. J., **15**, 173 (1979).
2997. C. J. Simionescu, S. Ioan, B. C. Simionescu, Eur. Polym. J., **23**, 69 (1987).
2998. C. I. Simionescu, S. Ioan, A. Flondor, B. C. Simionescu, Makromol. Chem., **189**, 2331 (1988).
2999. C. I. Simionescu, A. Chiriac, I. Neamtu, Polym. Bull., **27**, 31 (1991).
3000. A. Simmons, A. Natansohn, Macromolecules, **25**, 3881 (1992).
3001. A. Simmons, A. Natansohn, Macromolecules, **25**, 1272 (1992).
3002. R. P. Simonds, E. J. Goethals, Makromol. Chem., **179**, 1689 (1978).
3003. O. P. Singh, N. K. Scandle, I. K. Varma, Angew, Makromol. Chem., **122**, 193 (1984).
3004. R. Singh, A. S. Hay, Macromolecules, **25**, 1033 (1992).
3005. E. J. Siochi, J. M. DeSimone, A. M. Hellstern, J. E. McGrath, T. C. Ward, Macromolecules, **23**, 4696 (1990).
3006. D. F. Siqueira, S. P. Nunes, B. A. Wolf, Macromolecules, **27**, 1045 (1994).
3007. D. F. Siqueira, S. P. Nunes, B. A. Wolf, Macromolecules, **27**, 234 (1994).
3008. R. M. Sisson, J. C. Giddings, Anal. Chem., **66** 4043 (1994).

3009. G. Sitaramaiah, J. Polym. Sci. A-1, **3**, 2743 (1965).
3010. G. Sitaramaiah, D. Jacobs, Polymer, **11**, 165 (1970).
3011. G. Sitaramaiah, D. Jacobs, Makromol. Chem., **164**, 237 (1973).
3012. K. Sivadasan, S. Gundiah, Polymer., **28**, 1426 (1987).
3013. V. S. Skazka, S. Ya. Magarik, G. M. Pavlov, Kh. Rout, G. V. Tarasova, Polym. Sci. USSR, **20**, 180 (1978): Vysokomol. Soedin. A, **20**, 157 (1978).
3014. E. L. Slagowski, R. G. Gebauer, G. J. Gaesser, J. Appl. Polym. Sci., **21**, 2293 (1977).
3015. D. F. Slonaker, R. L. Combs, J. E. , Guillet, H. W. Coover Jr., J. Polym. Sci. A-2, **4**, 523 (1966).
3016. D. F. Slonaker, R. L. Combs, H. W. Coover Jr., Macromol. Sci., Part A: Chem., **1**, 539 (1967).
3017. G. Smets, M. Claesen, J. Polym. Sci., **8**, 289 (1952).
3018. G. Smets, A. Hertoghe, Makromol. Chem., **17**, 189 (1956).
3019. G. Smets, J. Roovers, W. Van Humbeek, J. Appl. Polym Sci., **5**, 149 (1961).
3020. G. Smets, W. De Winter, G. Delzenne, J. Polym. Sci., **55**, 767 (1961).
3021. G. Smets, A. Poot, G. L. Duncan, J. Polym. Sci., **54**, 65 (1961).
3022. G. Smets, E. Dysseleer, Makromol. Chem., **91** 160 (1966).
3023. W. B. Smith, J. A. May, Ch. W. Kim, J. Polym. Sci. A-2, **4**, 365 (1966).
3024. W. B. Smith, Polym. Preprints, **11** (2), 1019 (1970).
3025. T. W. Smith, J. E. Kuder, D. Wychick, J. Polym. Sci., Polym. Chem. Ed., **14**, 2433 (1976).
3026. J. B. P. Soares, A. E. Hamielec, Polymer, **36**, 1639 (1995).
3027. T. A. Soboleva, A. P. Suprun, S. A. Pavlova, Polym. Sci. USSR, **6**, 104 (1969); Vysokomol. Soedin., **6**, 89 (1964).
3028. A. A. Soeteman, J. P. M. Roels, J. A. P. P. Van Dijk, J. A. M. Smit, J. Polym. Sci., Polym. Phys. Ed., **16**, 2147 (1978).
3029. K. Soga, S. Hosoda, S. Ikeda, J. Polym. Sci., Polym. Chem. Ed., **12**, 729 (1974).
3030. K. Soga, S. Kawakami, H. Shirakawa, S. Ikeda, Makromol. Chem. Rapid Commun., **1**, 523 (1980).
3031. K. Soga, K. Yamamoto, Polym. Bull., **6**, 263 (1982).
3032. K. Soga, M. Ohgizawa, T. Shiono, Makromol. Chem., Rapid Commun., **10**, 503 (1989).
3033. L. B. Sokolov, V. Z. Nikonov, G. N. Shilyakova, Polym. Sci. USSR, **11**, 699 (1969); Vysokomol. Soedin. A, **11**, 616 (1969).
3034. A. A. Solov'yanov, K. S. Kazanskii, Polym. Sci. USSR, **14**, 1186 (1972); Vysokomol. Soedin. A, **14** , 1063 (1972).
3035. L. Šolé, D. Berek, D. Mikulášová, Colloid, Polym Sci., **258**, 702 (1980).
3036. C. X. Song, X. D. Feng, Macromolecules, **17**, 2764 (1984).
3037. A. M. Sookne, M. Harris, Ind. Eng. Chem., **37**, 475, 478 (1945).
3038. B. Soper, R. N. Haward, E. F. T. White, J. Polym. Sci., Polym. Chem. Ed., **10**, 2545 (1972).
3039. M. F. Sorokin, M. M. Babkina, Polym. Sci. USSR, **8**, 122 (1966); Vysokomol. Soedin., **8**, 115 (1966).
3040. H. Sotobayashi, K. Ueberreiter, Z. Elektrochem., **67**, 178 (1963).
3041. T. N. Soundararajan, M. Wayman, IUPAC Toronto, Preprints A, **11**, 3 (1968).
3042. I. Soutar, L. Swanson, F. G. Thorpe, C. Zhu, Macromolecules, **29**, 918 (1996).
3043. G. G. Sowash, S. E. Webber, Macromolecules, **21**, 1608 (1988).
3044. R. W. Sparidens, H. A. Claessens, G. H. J. van Doremaele, A. M. van Herk, J. Chromatogr., **508**, 319 (1990).
3045. A. L. Spatorico, J. Polym. Sci., Polym. Phys. Ed., **12**, 159 (1974).
3046. A. L. Spatorico, G. L. Beyer, J. Appl. Polym. Sci., **19**, 2933 (1975).
3047. J. Speváček, Makromol. Chem., Rapid Commun., **3**, 697 (1982).
3048. P. P. Spiegelman, G. Parravano, J. Polym. Sci. A, **2**, 2245 (1964).
3049. J. Springer, K. Ueberreiter, E. Moeller, Ber. Bunsenges. Phys., Chem., **69**, 494 (1965).
3050. J. Springer, K. Ueberreiter, R. Wenzel, Makromol. Chem., **96**, 122 (1966).
3051. J. Springer, K. Ueberreiter, W. Weinle, Eur. Polym. J., **6**, 87 (1970).
3052. J. Springer, J. Schmelzer, T. Zeplichal, Chromatographia, **13**, 1644 (1980).
3053. J. Springer, F. W. Weigelt, Markomol. Chem., **184**, 1489 (1983).
3054. T. Spychaj, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 199 (1991).
3055. K. S. V. Srinivasan, M. Santappa, J. Polym. Sci., Polym. Phy. Ed., **11**, 331 (1973).
3056. K. S. V. Srinivasan, M. Santappa, Polymer, **14**, 5 (1973).
3057. K. S. V. Srinivasan, D. Sudhakar, Polym. Bull., **2**, 321 (1980).
3058. C. J. Stacy, J. F. Foster, J. Polym. Sci., **25**, 39 (1957).
3059. C. J. Stacy, J. Appl. Polym. Sci., **21**, 2231 (1977).
3060. C. J. Stacy, Polym. Preprints, **26** (1), 180 (1985).
3061. G. D. Staffin, C. C. Price, J. Am. Chem. Soc., **82**, 3632 (1960).
3062. J. W. Stafford, Makromol. Chem., **134**, 87 (1970).
3063. J. W. Stafford, Makromol. Chem., **134**, 73 (1970).
3064. J. W. Stafford, Makromol. Chem., **134**, 57 (1970).
3065. J. W. Stafford, Makromol. Chem., **134**, 113 (1970).
3066. J. W. Stafford, Makromol. Chem., **134**, 99 (1970).
3067. G. Stainsby, Discussion Faraday Soc., **18**, 288 (1954).
3068. D. Statman, B. Chu, Macromolecules, **17**, 1537 (1984).
3069. H. Staudinger, M. Haberle, Makromol. Chem., **9**, 48 (1952).
3070. H. Staudinger, T. Eiche, Makromol. Chem., **10**, 235 (1953).
3071. R. Stefani, M. Chevreton J. Terrier, C. Eyraud, Compt. Rend., **248**, 2006 (1959).
3072. Ya. Steiny, Polym. Sci. USSR, **10**, 2184 (1968).; Vysokomol. Soedin. A, **10**, 1883 (1968).
3073. J. Stejny, J. Macromol. Sci., Part A: Chem., **7**, 1435 (1973).
3074. J. Stejskal, J. Janča, P. Kratochvil, Polym. J., **8**, 549 (1976).
3075. J. Stejskal, P. Kratochvil, Macromolecules, **11**, 1097 (1978).
3076. J. Stejskal, P. Kratochvil, D. Strakova, Macromolecules, **14**, 150 (1981).

3077. J. Stejskal, D. Strakova, P. Kratochvil, S. D. Smith, J. E. McGrath. Macromolecules, **22**, 861 (1989).
3078. T. Stelter, J. Springer, Makromol. Chem., **185**, 1719 (1984).
3079. M. Stickler, G. Meyerhoff, Makromol. Chem., **179**, 2729 (1978).
3080. M. Stickler, F. Eisenbeiss, Eur. Polym. J., **20**, 849 (1984).
3081. M. Stickler, Angew, Makromol. Chem., **123/124**, 85 (1984).
3082. J. K. Stille, M. Kamachi, M. Kurihara, Polym. Preprints, **12** (2)., 223 (1971).
3083. S. S. Stivala, R. J. Valles, D. W. Levi, J. Appl. Polym. Sci., **7**, 97 (1963).
3084. W. H. Stockmayer et al., J. Polym. Sci., **16**, 517 (1955).
3085. G. ver Strate, C. Cozewith, S. Ju, Macromolecules, **21**, 3360 (1988).
3086. C. Strazielle, Makromol. Chem., **119**, 50 (1968).
3087. C. Strazielle, H. Benoît, O. Vogl, Eur. Polym. J., **14**, 331 (1978).
3088. J. Strohalm, J. Kopeček, Angew, Makromol. Chem., **70**, 109 (1978).
3089. K. Su, E. E. Remsen, H. M. Thompson, L. G. Sneddon, Macromolecules, **24**, 3760 (1991).
3090. N. N. Sudareva, O. I. Kurenbin, G. B. Belen'kii, V. M. Mal'tsev, Polym. Sci. USSR, **31**, 1915 (1989); Vysokomol. Soedin. A, **31**, 1739 (1989).
3091. H. Sue, E. Ueno, Y, Nakamoto, S. I. Ishida, Macromolecules, **38**, 1305 (1989).
3092. H. Sue, Y. Nakamoto, S-I, Ishidu, Polym. Bull., **21**, 97 (1989).
3093. N. Suetterlin, G. Meyerhoff, Makromol. Chem., **165**, 223 (1973).
3094. K. Sugamiya, N. Kuwahara, M. Kaneko, Macromolecules, **7**, 66 (1974).
3095. T. Sunga, K. J. Ivin, G. Hofmeister, J. H. Oskan, R. S. Schrock, Macromolecules, **27**, 4043 (1994).
3096. T. Suminoe, N. Yamazoki, S. Kambara, Chem. High Polym. (Japan), **20**, 461 (1963).
3097. H. Sumitomo, K. Kobayashi, J. Polym. Sci. A-1, **4**, 907 (1966).
3098. R. M. Summers, J. Polym. Sci., Polym. Chem. Ed., **16**, 1669 (1978).
3099. S.-J. Sun, T. Chang, J. Polym. Sci., Part A, **31**, 2711 (1993).
3100. C. S. P. Sung, I. R. Gould, N. J. Turro, Macromolecules, **17**, 1447 (1984).
3101. M. Super, E. Berluche, C. Costello, E. Beckman, Macromolecules, **30**, 368 (1997).
3102. B. Suryanayanan, B. W. Peace, K. G. Mayhan, J. Polym. Sci., Polym. Chem. Ed., **12**, 1109 (1974).
3103. B. Suryanayanan, B. W. Peace, K. G. Mayhan, J. Polym. Sci., Polym. Chem. Ed., **12**, 1109 (1974).
3104. W. Sutter, W. Burchard, Makromol. Chem., **179**, 1961 (1978).
3105. H. Suzuki, C. G. Leoni, Brit, Polym. J., **5**, 485 (1973).
3106. T. Suzuki, Y. Tsuji, Y. Takegami, Macromlecules, **11** 639 (1978).
3107. T. Suzuki, V. Murakami, Y. Takegami, Polym. J., **12**, 183 (1980).
3108. T. Suzuki, Y. Murakami, O. Yamada, Y. Takegami, J. Macromol. Sci. Part A: Chem, **18**, 817 (1982).
3109. T. Suzuki, O. Yamada, Y. Murakami, Y. Takegami, Y. Watanabe, Macromolecules, **15**, 223 (1982).
3110. T. Suzuki, Y. Murakami, Y. Takegami, Polym. J., **14**, 431 (1982).
3111. F. Švec, J. Labský, L. Lányová, J. Hradil, S. Pokorný, J. Kálal, Angew, Makromol. Chem., **90**, 47 (1980).
3112. A. X. Swamikannu, M. H. Litt, J. Polym. Sci., Polym. Chem. Ed., **22**, 1623 (1984).
3113. D. L. Swanson, J. W. Williams, J. Appl. Phys., **26**, 810 (1955).
3114. S. A. Swanson, P. M. Cotts, R. Siemens, S. H. Kim, Macromolecules, **24**, 1352 (1991).
3115. H. A. Swenson, Acta Chem. Scand., **9**, 572 (1955).
3116. H. A. Swenson, A. Rosenberg, Acta Chem. Scand., **10**, 1393 (1956).
3117. H. A. Swenson, H. M. Kaustinen, O. A. Kaustinen, N. S. Thompson, J. Polym. Sci. A-2, **6**, 1593 (1968).
3118. H. A. Swenson, H. M. Kaustinen, J. J. Bachhuber, Macromolecules, **2**, 142 (1969).
3119. H. A. Swenson, H. M. Kaustinen, K. E. Almin, J. Polym. Sci. B, **9**, 261 (1971).
3120. H. A. Swenson, J. A. Carlson, H. M. Kaustinen, J. Polym. Sci. C, **36**, 293 (1971).
3121. M. Szesztay, Z. László-Hedvig, F. Tüdós, J. Appl. Polym. Sci., Appl. Polym. Symp., **48**, 227 (1991).
3122. P. Szweczyk, M. Kalfus, J. Mackromol. Sci. Part A: Chem., **7**, 737 (1973).
3123. K. Tada, Y. Yamada, T. Saegusa, J. Furukawa, Makromol. Chem., **99**, 232 (1966).
3124. K. Tada, T. Saegusa, J. Furukawa, Makromol. Chem., **102**, 47 (1967).
3125. N. G. Taganov, A. P. Klinov, G. N. Komratov, R. A. Barzykina, G. V. Korovina, G. V. Isagulyants, S. G. Entelis, Polym. Sci., USSR, **22**, 1762 (1980); Vysokomol. Soedin. A, **22**, 1607 (1980).
3126. A. A. Tager, A. A. Anikeyeva, V. M. Andreyeva, T. Ya. Gomarova, L. A. Chernoskutova, Polym. Sci. USSR, **10**, 1926 (1968).; Vysokomol. Soedin. A, **10**, 1661 (1968).
3127. A. A. Tager, A. A. Anikeyeva, L. V. Adamova, V. M. Andreyeva, T. A. Kuz'mina, M. V. Tsilipotkina, Polym. Sci. USSR, **13**, 751 (1971).; Vysokomol Soedin. A, **13**, 659 (1971).
3128. L. H. Tagle, F. R. Diaz, R. J. Vega, Polym. Bull., **11**, 523 (1984).
3129. A. Takacs, R. Faust, Macromolecules, **28**, 7266 (1995).
3130. A. Takada, T. Fukuda, J. Watanabe, T. Miyamoto, Macromolecules, **28**, 3394 (1995).
3131. M. Takahashi, M. Watanabe, Sen-i-Gakkaishi, **17**, 122 (1961).
3132. M. Takahashi, M. Watanabe, Sen-i-Gakkaishi, **17**, 111 (1961).
3133. A. Takahashi, M. Obara, I. Kagawa Kogyo Kagaku Zasshi, **66**, 960 (1963).
3134. Y. Takahashi, J. E. Mark, J. Am. Chem. Soc., **98**, 3756 (1976).
3135. K. Takamizawa, Bull. Chem. Soc. Japan, **39**, 1186 (1996).

3136. M. Takeda, R. Endo, Y. Matsuura, J. Polym. Sci. C, **23**, 487 (1968).
3137. G. Talamini, G. Vidotto, Makromol. Chem., **110**, 111 (1967).
3138. Y. Tamai, T. Koniski, Y. Einaga, M. Fujii, H. Yamakawa, Macromolecules, **23**, 4067 (1990).
3139. J. W. Tamblyn, D. R. Morey, R. H. Wagner, Ind. Eng. Chem., **37**, 573 (1945).
3140. P. E. Tampellini, E. Occhiello, F. Garbassi, Polym. Bull., **30**, 551 (1993).
3141. E. Tampellini, L. Canova, P. Cinquina, G. Gianotti, Polymer, **35**, 367 (1944).
3142. J. S. Tan, S. P. Gasper, J. Polym. Sci., Polym. Phys. Ed., **12**, 1785 (1974).
3143. J. S. Tan, S. P. Gasper, J. Polym. Sci., Polym. Phys. Ed., **13**, 1705 (1975).
3144. J. S. Tan, A. R. Sochor, Macromolecules, **14**, 1700 (1981).
3145. S. Tanaka, H. Morikawa, J. Polym. Sci. A-1, **3**, 3147 (1965).
3146. S. Tanaka, A. Nakamura,H. Morikawa, Makromol. Chem., **85**, 164 (1965).
3147. S. Tanaka, M. Uno, S. Teramachi, Y. Tsukahara, Polymer, **36**, 2219 (1995).
3148. T. Tanaka, N. Donkai, H. Inagaki, Macromolecules, **13**, 1021 (1980).
3149. L. J. Tangle, W. J. Rebel, R. J. Brewer, J. Polym. Sci. A-1, **8**, 2935 (1970).
3150. S. Tani, F. Hamada, A. Nakajima, Polym. J., **5**, 86 (1973).
3151. K. Tanikawa, T. Enomoto, M. Hatano, K. Moteyi, Z. Okuno, Makromol. Chem., **176**, 3025 (1975).
3152. T. Taninaka, H. Uemura, Y. Minoura, Eur. Polym. J., **14**, 197 (1978).
3153. D. W. Tanner, G. C. Berry, J. Polym. Sci., Polym. Phys. Ed., **12**, 941 (1974).
3154. H. Tanzawa, T. Tanaka, A. Soda, J. Polym. Sci. A-2 , **7**, 929 (1969).
3155. J. Tao, S. Stewart, G. Liu, M. Yaang, Macromolecules, **30**, 2738 (1997).
3156. A. Taquet, R. Jerome, Ph. Teyssie, J. P. Masy, E. Goethals, J. Polym. Sci. Part A: Polym. Chem., **33**, 1169 (1995).
3157. G. I. Tarasova, S. A. Pavlova, V. V. Korkshak, Polym. Sci. USSR, **15**, 1046 (1973); Vysokomol. Soedin. A, **15** 929 (1973).
3158. S. A. Tashmukhamedov, P. P. Larin R. S. Tillayev, Yu. T. Tashpulatov, Kh. U. Usmanov, Polym. Sci. USSR, **10**, 2113 (1968); Vysokomol. Soedin. A, **10**, 1823 (1968).
3159. S. A. Tashmukhamedov, P. P. Larin R. S. Tillayev, Yu. T. Tashpulatov, Kh. U. Usmanov, Polym. Sci. USSR, **11**, 515 (1969); Vysokomol. Soedin. A, **11**, 453 (1968).
3160. S. A. Tashmukhamedov, B. V. Sagdullayev, P. P. Larin, R. S. Tillayev, Kh. U. Usmanov, Polym. Sci. USSR, **16**, 1620 (1974); Vysokomol. Soedin. A, **16**, 1395 (1974).
3161. S. A. Tashmukhamedov, Sh. A. Azizov, A. Sh. Karabayev, E. T. Tsagarayev, R. S. Tillayev, Kh. U. Usmanov, Polym. Sci. USSR, **18**, 2478 (1976); Vysokomol. Soedin. A, **18**, 2171 (1976).
3162. W. C. Taylor, L. H. Tung, Soc. Plastics Eng. Trans., **2**, 1 (1962).
3163. W. C. Taylor, L. H. Tung, J. Polym. Sci., B, **1**, 157 (1963).
3164. D. L. Taylor, J. Polym. Sci. A, **2**, 611 (1964).
3165. G. B. Taylor, J. Am. Chem. Soc., **69**, 638 (1947).
3166. S. Tazuke, H. Nagahara, Makromol. Chem., **181**, 2207 (1980).
3167. S. Tazuke, O. Supakorn, T. Inoue, J. Polym. Sci., Polym. Chem. Ed., **20**, 2239 (1982).
3168. H. Tebbe, F. von Ackern, B. Tieke, Makromol. Chem. Phys., **196**, 1475 (1995).
3169. N. B. Tennikov, P. P. Nefyedov, Polym. Sci. USSR, **22**, 513 (1980); Vysokomol. Soedin. A, **22**, 461 (1980).
3170. M. Tepelekian, P. Quang Tho, A. Guyot, Eur. Polym. J., **5**, 795 (1969).
3171. S. Teramachi, M. Nagasawa, J. Macromol. Sci., Part A: Chem., **2**, 1169 (1968).
3172. S. Teramachi, Y. Kato, J. Macromol. Sci., Part A: Chem., **4**, 1785 (1970).
3173. S. Teramachi, Y. Kato, Macromolecules, **4**, 54 (1971).
3174. S. Teramachi, H. Tomioka, M. Sotokawa, J. Macromol. Sci. Part A: Chem., **6**, 97 (1972).
3175. S. Teramachi, T. Fujikawa, J. Macromol. Sci. PartA Chem., **6**, 1393 (1972).
3176. S. Teramachi, H. Uchiyama, Polym. J., **5**, 243 (1973).
3177. S. Teramachi, T. Fukao, Polym. J., **6**, 532 (1974).
3178. S. Teramachi, H. Eaki, Polym. J., **7**, 593 (1975).
3179. S. Teramachi, R. Obata, K. Yamashita, N. Takemoto, J. Macromol. Sci. Part A: Chem., **11**, 535 (1977).
3180. S. Teramachi, A. Hasegawa, M. Akatsuka, A. Yamashita, N. Takemoto, Macromolecules, **11**, 1206 (1978).
3181. S. Teramachi, A. Haegawa, Y. Shime, M. Akatsuka, M. Nakayama, Macromolecules, **12**, 992 (1979).
3182. S. Teramachi, A. Hasegawa, S. Hasegawa, T. Ishibe, Polym. J., **13**, 319 (1981).
3183. S. Teramachi, A. Hasegawa, S. Yoshida, Polym. J., **14**, 161 (1982).
3184. S. Teramachi, A. Hasegawa, S. Yoshida, Macromolecules, **16**, 542 (1983).
3185. S. Teramachi, A. Hasegawa, Y. Shigekuni, S. Matsunaga, Polym. J., **21**, 803 (1989).
3186. S. Teramachi, A. Hasegawa, Y. Shigekuni, S. Matsunaga, Polym. J., **21**, 803 (1989).
3187. S. Teramachi, A. Hasegawa, Y. Shigekuni, K. Zenta, M. Hahimoto. J. Appl. Polym. Sci., Appl. Polym. Symp., **45**, 87 (1990).
3188. S. Teramachi, A. Hasegawa, K. Motoyama, Polym. J., **22**, 489 (1990).
3189. S. Teramachi, A. Hasegawa, T. Matsumoto, K. Kitahara, Y. Tsukahara, Y. Yamashita, Macromlecules, **25**, 4025 (1992).
3190. S. Teramachi, S. Sato, H. Shimura, S. Watanabe, Y. Tsukahara, Macromolecules, **28**, 6138 (1995).
3191. K. Teramachi, M. Iida, T. Araku, S. Yamashita, H. Tani, Macromolecules, **7**, 421 (1974).
3192. C. C. Teraú, E. M. Macchi, R. V. Figini, J. Appl. Polym. Sci., **34**, 2433 (1987).
3193. M. Terbojevich, A. Casani, M. Camilot, B. Focher, J. Appl. Polym. Sci., **55**, 1663 (1995).
3194. D. R. Terrell, Polymer, **23**, 1045 (1982).

3195. S. L. Terry, Rodriguez, Polym. Preprints, **8** (2), 1270 (1967).
3196. Y. Tezuka, S. Hayashi, Macromolecules, **28**, 3038 (1995).
3197. N. G. Thame, R. D. Lundberg, J. P. Kennedy, J. Polym. Sci., Polym. Ed., **10**, 2507 (1972).
3198. R. C. Thamm, W. H. Buck, J. Polym. Sci., Polym. Chem. Ed., **16**, 539 (1978).
3199. M. Theurig, S. J. Sargeant, G. Manuel, W. P. Weber, Macromolecules, **25**, 3834 (1992).
3200. A. Thierry, A. Skoulios, Makromol. Chem., **177**, 319 (1976).
3201. B. B. Thomas, W. J. Alexander, J. Polym. Sci., **15**, 361 (1955).
3202. L. Thomas, A. Polton, M. Tardi, P. Sigwalt, Macromolecules, **25**, 5886 (1992).
3203. G. H. Thompson, M. N. Myers, J. C. Gidding, Anal. Chem., **41**, 1219 (1969).
3204. J. O. Threlkeld, H. A. Ende, J. Polym. Sci. A-2, **4**, 663 (1966).
3205. C. D. Thurmond, B. H. Zimm, J. Polym. Sci., **8**, 477 (1952).
3206. E. J. Tijsma, L. vad der Does, A. Bantjes, I. Vulic, G. H. W. Bunning, Macromolecules, **27**, 179 (1994).
3207. E. I. Tiktopulo, V. N. Uversky, V. B. Lushchik, S. I. Klenin, V. E. Bychkova, O. B. Ptitsyn, Macromolecules, **28**, 7519 (1995).
3208. T. E. Timell, Ind. Eng. Chem., **47**, 2166 (1955).
3209. G. I. Timofeyeva, S. A. Pavlova, IUPAC Helsinki, Preprints Vol. III, II-64 (1972).
3210. G. I. Timofeyeva, Ye. S. Krongauz, A. M. Berlin, V. V. Kroshak, S.-S. Pavlova, Polym. Sci. USSR, **18**, 984 (1976).; Vysokomol Soedin. A, **18**, 863 (1976).
3211. L. V. Titkova, E. Prokopová, B. Sedlácek, V. Petrus, K. Dušek, M. Bohdanecký, Eur. Polym. J., **14**, 145 (1978).
3212. F. L. Tobiason, C. Chandler, F. E. Schwarz, Macromolecules, **5**, 321 (1972).
3213. F. L. Tobiason, G. H. Cain, J. W. Anderson, J. Polym. Sci., Polym. Chem. Ed., **16**, 275 (1978).
3214. M. Tokita, T. Takahashi, M. Hayashi, K. Inomata, J. Watanabe, Macromolecules, **29**, 1345 (1996).
3215. M. P. Tombs, Biochem. J., **67** (1957).
3216. M. Tomikawa, Chem. High Polym. (Japan), **20**, 11 (1963).
3217. H. Tompa, "Polymer Solutions", Butterworths, London, 1956.
3218. A. V. Topchiev, G. I. Chernyi, V. N. Andronov, Dokl. Akad. Nauk SSR, **143**, 879 (1962).
3219. J. P. Tovborgjensen, J. Kops, J. Polym. Sci., Polym. Chem. Ed., **18**, 2737 (1980).
3220. M. S. Toy, J. M. Newman, J. Polym. Sci. A, **1**, 2333 (1969).
3221. M. S. Toy, Polym. Preprints, **12** (1), 385 (1971).
3222. A. H. Traaen, J. Appl. Polym. Sci., **7**, 581 (1963).
3223. G. Trafara, J. Polym. Sci., Polym. Chem. Ed., **18**, 321 (1980).
3224. R. T. Traskos, N. S. Schneider, A. S. Hoffman, J. Appl. Polym. Sci., **12**, 509 (1968).
3225. B. Trathnigg, H. Junek, Angew, Makromol. Chem., **53**, 81 (1968).
3226. B. Trathnigg, G. Hippmann, Angew. Makromol. Chem., **105**, 9 (1982).
3227. B. Trathnigg, X. Yan, J. Appl. Polym. Sci., Appl. Polym., Symp., **52**, 193 (1993).
3228. W. Trautvetter, Makromol. Chem., **101**, 214 (1967).
3229. V. Trefiletti, G. Conio, F. Pioli, B. Cavazza, A. Perico, E. Patrone, Makromol. Chem., **181**, 1159 (1980).
3230. N. M. Tret'yakova, L. V. Kosmodem'yanskii, R. G. Romanova, E. G. Lazaryants, Polym. Sci. USSR, **12**, 3127 (1970); Vysokomol. Soedin. A, **12**, 2754 (1970).
3231. M. Tricot, V. Desreux, Makromol. Chem., **149**, 185 (1971).
3232. M. Tricot, J. P. Bleus, J. P. Riga, V. Desreux, Makromol. Chem., **175**, 913 (1974).
3233. M. Tricot, Polymer, **24**, 366 (1983).
3234. C. K. Trinh, W. Schnabel, Makromol. Chem., Phys., **198**, 1319 (1997).
3235. P. Trivedi, J. Macromol. Sci., Part A: Chem., **14**, 589 (1980).
3236. P. D. Trivedi, D. N. Shulz, Polym. Bull., **3**, 37 (1980).
3237. P. D. Trivedi, R. J. Ambrose, A. G. Altenau, J. Macromol. Sci. A: Chem., **17**, 1159 (1982).
3238. E. Trommdorff, H. Koehle, P. Lagally, Makromol. Chem., **1**, 169 (1948).
3239. L. D. Trung, J. Mordini, P. Q. Tho, A. Guyot, Eur. Polym. J., **6**, 1187 (1970).
3240. S.-L. Tseng, G. V. Laivins, D. G. Gray, Macromolecules, **15**, 1262 (1982).
3241. S. Tseng, W. Herman, A. E. Woodward, B. A. Newman, Macromolecules, **15**, 338 (1982).
3242. S. Tseng, A. E. Woodward, Macromolecules, **15**, 343 (1982).
3243. M. V. Tsilipotkina, A. A. Tager, E. B. Makovskaya, V. Partina, Polym. Sci., USSR, **12**, 1222 (1970); Vysokomol. Soedin. A, **12**, 1082 (1970).
3244. M. V. Tsilipotkina, A. A. Tager, O. V. Nechayeva, A. V. Vasiliev, L. M. Bykova, S. M. Bulygina, V. A. Vasnsev, S. N. Salazkin, Polym. Sci. USSR, **14**, 2304 (1972); Vysokomol. Soedin. A, **14**, 2053 (1991).
3245. C. Tsitsilianis, P. Lutz, S. Graff, J. P. Lamps, P. Remp. Macromolecules, **24**, 5897 (1991).
3246. C. Tsitsilianis, D. Papanagopoullos, P. Lutz, Polymer, **36**, 3745 (1995).
3247. K. Tsubakiyama, T. Matsuo, T. Sakaki, K. Yoshida, T. Fujimura, K. Azaki, J. Polym. Sci., Polym. Chem., Ed., **17**, 173 (1979).
3248. E. Tsuchida, E. Shouji, K. Yamamoto, Macromolecules, **26**, 7144 (1993).
3249. T. Tsuruta, R. Fuijo, J. Furukawa, Makromol. Chem., **80**, 172 (1964).
3250. T. Tsuruta, S. Inoue, N. Yoshida, Y. Yokota, Makromol. Chem., **81**, 191 (1965).
3251. R. Tsushima, T. Tsuruta, J. Polym. Sci., Polym. Chem. Ed., **12**, 183 (1974).
3252. T. Tsutsui, A. Mizuno, N. Kashiwa, Makromol. Chem., **190**, 1177 (1989).
3253. V. N. Tsvetkov, V. S. Skazka, N. M. Krivoruchko, Vysokomol. Soedin., **2**, 1045 (1960).
3254. V. N. Tsvetkov, N. N. Boitsova, Vysokomol. Soedin., **5**, 1263 (1963).
3255. V. N. Tsvetkov, S. Ya. Lyubina, T. V. Barskaya, Polym. Sci., USSR **6**, 886 (1964); Vysokomol. Soedin., **6**, 806 (1964).

3256. V. N. Tsvetkov, V. P. Budtov, Polym. Sci. USSR, **6**, 1332 (1964); Vysokomol. Soedin., **6**, 1209 (1964).

3257. V. N. Tsvetkov, I. N. Shtennikova, Ye. I. Ryumtsev, G. I. Okhrimenko, Polym. Sci. USSR, **7**, 1223 (1965); Vysokomol. Soedin., **7**, 1104 (1965).

3258. V. N. Tsvetkov, Yu. V. Mitin, I. N. Shtennikova, V. R. Glushenkova, G. V. Tarasova, V. S. Skazka, N. A. Nikitin, Polym. Sci., USSR, **7**, 1216 (1965); Vysokomol. Soedin., **7**, (1965).

3259. V. N. Tsvetkov, I. N. Shtennikova, Ye. I. Ryumtsev, V. S. Skazka, Polym. Sci. USSR. 7, 1231 (1965); Vysokomol. Soedin., **7**, 1111 (1965).

3260. V. N. Tsvetkov, S. Ya. Lubina, V. Ye Bychkova, I. A. Strelina, Polym. Sci. USSR, **8**, 928 (1966); Vysokomol. Soedin., **8**, 846 (1996).

3261. V. N. Tsvetkov, K. A. Andrianov, I. N. Shtennikova, G. I. Okhrimenko, L. N. Andreyeva, G. A. Fomin, V. I. Pakhomov, Polym. Sci. USSR, **10**, 636 (1968); Vysokomol. Soedin. A, **10**, 547 (1968).

3262. V. N. Tsvetkov, G. A. Fomin, P. N. Lavrenko, I. N. Shten nikova, T. V. Sheremeteva, L. I. Godunova, Polym. Sci. USSR, **10**, 1051 (1968); Vyokomol. Soedin. A, **10**, 93 (1968).

3263. V. N. Tsvetkov, V. S. Skazka, G. V. Tarasova, V. M. Yamshchikov, S. Ya. Lyubina, Polym. Sci. USSR, **10**, 81 (1968); Vyokomol, Soedin. A, **10**, 74 (1968).

3264. V. N. Tsvetkov, I. N. Shtennikova, E. I. Riumtsev, Yu. P. Getmanchuk, IUPAC Leiden, Preprints, Vol I, 51 (1970).

3265. V. N. Tsvetkov, K. A. Andrianov, G. I. Okhrimenko, I. N. Shtennikova, G. A. Fomin, M. G. Vitovskaya, V. I. Pakhomov, A. A. Yarosh, D. N. Andreyev, Polym. Sci USSR, **12**, 2146 (1970); Vysokomol. Soedin. A, **12**, 1892 (1970).

3266. V. N. Tsvetkov, N. N. Kupriyanova, G. V. Tarasova, P. N. Lavrnko, I. I. Migunova, Polym. Sci. USSR, **12**, 2238 (1970); Vysokomol. Soedin. A, **12**, (1970).

3267. V. N. Tsvetkov, et al., Polym. Sci. USSR, **13**, 705 (1971); Vysokomol. Soedin. A, **13**, 620 (1971).

3268. V. N. Tsvetkov, E. N. Zakharova, G. A. Fomin, P. N. Lavrenko, Polym. Sci. USSR, **14**, 2192 (1972); Vysokomol. Soedin. A, **14**, 1956 (1972).

3269. V. N. Tsvetkov, L. N. Andreyeva, Ye V. Korneyeva, P. N. Lavrenko, N. A. Platé, V. P. Shibayev, B. S. Petrukhin, Polym. Sci. USSR, **14**, 1944 (1972).

3270. V. N. Tsvetkov, K. A. Andrianov, M. G. Vitovskaya, N. N. Makarova, E. N. Zakharova, S. V. Bushin, P. N. Lavrenko, Polym. Sci. USSR, **14**, 412 (1972); Vysokomol. Soedin. A, **14**, 369 (1972).

3271. V. N. Tsvetkov, Ye. V. Korneyeva, I. N. Shtennikova, P. N. Lavrenko, G. F. Kolbina, D. Khardi, K. Nitrai, Polym. Sci. USSR, **14**, 478 (1972); Vysokomol. Soedin. A, **14**, 427 (1972).

3272. V. N. Tsvetkov, K. A. Andrianov, M. G. Vitovskaya, E. P. Astapenko, Polym. Sci. USSR, **14**, 3032 (1972); Vysokomol Soedin. A, **14**, 2603 (1972).

3273. V. N. Tsvetkov, E. I. Rjumtsev, L. N. Andreeva, N. V. Pogodina, P. N. Lavrenko, L. I. Kutsenko, Eur. Polym. J., **10**, 563 (1974).

3274. V. N. Tsvetkov, I. N. Shtennikova, M. G. Vitovskaya, Ye. I. Ryumtsev, T. V. Pekker, Yu. P. Getmanchuk, P. N. Lavrenko, S. V. Bushin, Polym. Sci. USSR, **16**, 655 (1974); Vysokomol. Sodien. A, **16**, 566 (1974).

3275. V. N. Tsvetkov, I. N. Shtennikova, Ye. I. Ryumtsev, N. V. Pogodina, G. F. Kobina, Ye. V. Korneyeva, P. N. Lavrenko, O. V. Okatova, Yu. B. Amerik, A. A. Baturin, Polym. Sci. USSR, **18**, 2302 (1976); Vysokomol. Soedin. A, **18**, 2016 (1976).

3276. V. N. Tsvetkov, P. N. Lavrenko, G. M. Paulov, S. V. Bushin, E. P. Astapenko, A. A. Boikov, N. A. Shil'dyayeva, S. A. Didenko, B. F. Malichenko, Polym. Sci. USSR, **24**, 2687 (1982); Vysokomol. Soedin. A, **24**, 2343 (1982).

3277. V. N. Tsvetkov, L. N. Andreeva, P. N. Lavrenko, E. V. Believa, O. V. Okatova, A. Y. Bilibin, S. S. Skorokhodov, Eur. Polym. J., **20**, 817 (1984).

3278. V. N. Tsvetkov, P. N. Lavrenko, L. N. Andreeva, A. I. Mashoshin, O. V. Okatova, O. I. Mikriukova, L. I. Kutsenko, Eur. Polym. J., **20**, 823 (1984).

3279. L. H. Tung, J. Polym. Sci., **20**, 495 (1956).

3280. L. H. Tung, J. Polym. Sci., **24**, 333 (1956).

3281. L. H. Tung, S. P. E. Journal, **14**, 25 (1958).

3282. L. H. Tung, J. C. Moore, G. W. Knight, J. Appl. Polym. Sci., **10**, 1261 (1966).

3283. L. H. Tung, J. Polym. Sci., A-2, **7**, 47 (1969).

3284. L. H. Tung, J. Macromol. Sci. Rev. Macromol. Chem. C, **6**, 51 (1971).

3285. L. H. Tung, R. M. Wiley, J. Polym. Sci., Polym. Phys. Ed., **11**, 1413 (1973).

3286. L. H. Tung (Ed.), "Fractionation of Synthetic Polymers Principle and Practices", Marcel Dekker, New York, 1977.

3287. L. H. Tung, J. Appl. Polym. Sci., **24**, 953 (1979).

3288. S. R. Turner, M. Stolka, J. Polym. Sci., Polym. Chem. Ed., **16**, 1167 (1978).

3289. E. Turska, M. Laczkowski, J. Polym. Sci., **23**, 285 (1957).

3290. E. Turska, T. Skwarsi, Zeszyty Nauk. Plitech. Lodz Chem., **5**, 21 (1957).

3291. E. Trsuka, A. Dems, M. Siniarska, Bull. Acad. Polon. Sci. Ser. Sci. Chim, **13**, 189 (1965).

3292. E. Trsuka, T. Siniarska-Kapuścińka, Eur. Polym. J., **5**, (Suppl.)., 431 (1969).

3293. E. Trsuka, A. M. Wróbel, Polymer, **11**, 408 (1970).

3294. Z. Tuzar, P. Kratochvi, M. Bohdanecký, J. Polym. Sci. C, **16**, 633 (1967).

3295. Z. Tuzar, V. Petrus, P. Kratochvil, Makromol. Chem., **175**, 3181 (1974).

3296. Z. Tuzar, B. Bednar, C. Konák, M. Kubin, S. Svobodová, K. Procházka, Markomol. Chem., **183**, 39 (1982).

3297. M. A. Twaik, M. Tahan, A. Zilkha, J. Polym. Sci., A-1, **7**, 2469 (1969).

3298. US Patent, **2**, 457, 238 (1948).

3299. K. Ueberreiter, H. J. Orthmann, G. Sorge, Makromol. Chem., **8**, 21 (1952).

3300. K. Ueberreiter, R. Goetze, Makromol. Chem., **29**, 61 (1959).

3301. K. Ueberreiter, J. Sringer, Z. Phys. Chem. N. F. (Frankfurt). **36**, 299 (1963).

3302. K. Ueberreiter, W. Bruns, Ber Bunsenge Phys. Chem., **68**, 541 (1964).

3303. K. Ueberreiter, J. Melheimer, J. Springer, Kolloid, Z.-Z. Polym., **234**, 989 (1969).
3304. M. Ueda, Polym. J., **3**, 431 (1972).
3305. M. Ueda, T. Abe, H. Awano, Macromolecules, **25**, 5125 (1992).
3306. A. Uede, S. Nagai, J. Polym. Sci. Part A: Polym. Chem., **24**, 405 (1986).
3307. N. Ueyama, T. Araki, H. Tani, Macromolecules, **7**, 153 (1974).
3308. C. V. Uglea, Makromol. Chem., **166**, 275 (1973).
3309. C. V. Uglea, Makromol. Chem., **175**, 1535 (1974).
3310. C. V. Uglea, V. N. Stan, Brit. Polym. J., **14**, 39 (1982).
3311. N. Ugur, K. Alyürük, J. Polym. Sci. Part A: Polym. Chem., **27**, 1749 (1989).
3312. H. I. Ünal, C. Price, P. M. Budd, R. H. Mobbs, Eur. Polym. J., **30**, 1037 (1994).
3313. C. A. Uraneck, J. E. Burleigh, J. Appl. Polym. Sci., **14**, 267 (1970).
3314. C. A. Uraneck, J. N. Short, J. Appl. Polym. Sci., **14**, 1421 (1970).
3315. K. Urushido, A. Matsumoto, M. Oiwa, J. Polym. Sci., Polym. Chem. Ed., **18**, 1771 (1980).
3316. J. R. Urwin, J. M. Stearn. Makromol. Chem., **78**, 194 (1964).
3317. J. R. Urwin, J. Mckenzie, Stearne, D. O. Jordan, R. A. Mills, Makromol. Chem., **72**, 53 (1964).
3318. T. Usami, Y. Gotoh, S. Takayama, Macromolecules, **19**, 2722 (1986).
3319. T. Usami, Y. Gotoh, S. Takayama, H. Ohtami, S. Tsuge, Macromolecules, **20**, 1557 (1987).
3320. T. Usami, Y. Kogawa, S. Takayama, T. Koyama, Eur. Polym. J., **26**, 699 (1990).
3321. T. Usami, Y. Gotoh, H. Umemoto, S. Takayama, J. Appl. Polym. Sci., Appl. Polym. Symp., **52**, 145 (1993).
3322. S. N. Ushakov, S. P. Mitsengendier, V. N. Krasulina, Izv. Akad. Nauk SSSR, **3**, 366 (1957).
3323. Kh. U. Usmanov, A. A. Yul'Chibaev, T. Sirlibaev, Plast. Massy, **2**, 73 (1966).
3324. K. Ute, T. Asada, K. Hatada, Macromolecules, **29**, 1904 (1996).
3325. H. Utiyama, J. Phys. Chem., **69**, 4138 (1965).
3326. I. A. Vakhtina, O. G. Tarakanov, R. I. Khrenova, Polym. Sci. USSR, **16**, 3022 (1974); Vysokomol. Soedin. A, **16**, 2598 (1974).
3327. I. A. Vakhtina, A. G. Okuneva, K. Tekhrits, O. G. Tarakanov, Polym. Sci. USSR, **18**, 541 (1976); Vysokomol. Soedin. A, **18**, 471 (1976).
3328. I. A. Vakhtina, T. Bettger, A. P. Andreyev, O. S. Novozhilova, O. S. Tarakanov, Polym. Sci. USSR, **18**, 2447 (1976); Vysokomol. Soedin. A, **18**, 2138 (1976).
3329. I. K. Vakhtina, Ye. A. Petrakova, H. Knopp, G. A. Gladkovskii, Polym. Sci. USSR, **20**, 2689 (1978); Vysokomol. Soedin. A, **20**, 2388 (1978).
3330. I. K. Vakhtina, Ye. A. Petrakova, U. Pentsel', O. G. Tarakanov, Polym. Sci., USSR, **22**, 1833 (1980); Vysokomol. Soedin. A, **22**, 1671 (1980).
3331. I. A. Vakhtina, G. V. Shirokova, A. G. Okuneva, Polym. Sci. USSR, **24**, 2290 (1982); Vysokomol. Soedin. A, **24**, 2002 (1982).
3332. D. G. Val'kovkii, S. L. Sosin, V. V. Krsohak, S. A. Pavlova, Polym. Sci. USSR, **6**, 2044 (1964); Vysokomol. Soedin., **6**, 1848 (1964).
3333. R. H. Valentine, J. D. Ferry, T. Homma, K. Ninomiya, J. Polym. Sci. A-2, **6**, 479 (1968).
3334. R. J. Vallés, E. C. Schramm, J. Polym. Sci. A-1, **3**, 3853 (1965).
3335. R. J. Vallés, Polym. Preprints, **6** (2), 1041 (1965).
3336. R. J. Vallés, Makromol. Chem., **113**, 147 (1968).
3337. R. J. Vallés, Polym. Preprints, **9** (1), 752 (1968).
3338. I. Valyi, A. G. Janssen, H. Mark. J. Phys. Chem., **49**, 461 (1945).
3339. A. C. van Asten, W. Th. Kok, R. Tijssen, H. Poppe, J. Chromatogr. A, **676**, 361 (1994).
3340. A. C. Van Asten, H. F. M. Boelens, W. T. Kok, H. Poppe, P. S. Williams, J. C. Giddings, Sep. Sci. Technol, **29**, 573 (1994).
3341. J. A. P. P. Van Dijk, W. C. M. Henkens, J. A. M. Smit, J. Polym. Sci., Polym. Phy. Ed., **14**, 1485 (1976).
3342. J. A. P. P. Van Dijk, J. A. M. Smit, F. E. Kohn, J. Feijen, J. Polym. Sci., Polym. Chem. Ed., **21**, 197 (1983).
3343. W. N. E. Van Dijk-Wolthuis,O. Franssen, H. Talsma, M. J. Van Steenbergen, J. J. Kettenes-vanden Bosch, W. E. Hennink, Macromolecules, **28**, 6317 (1995).
3344. J. G. Van Dusen, J. F. O'Malley, J. Polym. Sci., Polym. Lett. Ed., **18**, 493 (1980).
3345. M. E. Van Kreveld, N. Van Den Hoed, J. Chromatogr., **149**, 71 (1978).
3346. D. C. Van Landuyt, C. W. Huskins, J. Polym. Sci. B, **6**, 643 (1968).
3347. D. C. Van Landuyt, J. Polym. Sci., Polym. Lett. Ed., **10**, 125 (1972).
3348. R. Van Leemput, R. Stein, J. Polym. Sci. A, **1**, 985 (1963).
3349. J. G. Van Ommen, H. A. Polman, P. C. J. M. VanBerkel, P. J. Gellings, Eur. Polym. J., **16**, 1051 (1980).
3350. J. Van Schooten, P. W. O. Wijga, Makromol. Chem., **43**, 23 (1961).
3351. J. Van Schooten, H. Van Hoorn, J. Boerma, Polymer, **2**, 161 (1961).
3352. E. J. Vandenberg, J. Polym. Sci., C, **1**, 207 (1963).
3353. I. K. Varma, S. S. Grover, J. Appl. Polym. Sci., **14**, 2965 (1970).
3354. I. K. Varma, K. K. Sharma, Angew, Makromol. Chem., **79**, 147 (1979).
3355. I. K. Varma, S. Patnaik, J. Polym. Sci., Polym. Chem. Ed., **17**, 3279 (1979).
3356. D. S. Varma, R. K. Sarkar, Angew, Makromol. Chem., **35**, 203 (1974).
3357. S. K. Varshney, Ch. Jacobs, J. P. Hautekeer, P. Bayard, R. Jerome, R. Fayt, P. Teyssie, Macromolecules, **24**, 4997 (1991).
3358. S. K. Varshney, Z. Gao, X. F. Zhong, A. Eisenberg, Macromolecules, **27**, 1076 (1994).
3359. N. I. Vasil'ev, V. I. Irzhak, V. I. Kartsovnik, N. S. Yenikolopyan, Polym. Sci. USSR, **12**, 2276 (1970); Vysokomol. Soedin. A, **12**, 2006 (1970).

3360. O. I. Vasilova, V. V. Zaitseva, R. V. Kucher, Polym. Sci. USSR **23**, 657 (1981); Vysokomol. Soedin. A, **23**, 587 (1981).
3361. O. I. Vasilova, R. V. Kucher, Polym. Sci. USSR, **25**, 2276 (1983); Vysokomol. Soedin., **25**, 2276 (1983).
3362. P. Vasudevan, M. Santappa, Makromol. Chem., **137**, 261 (1970).
3363. P. Vasudevan, M. Santappa, J. Polym. Sci. A-2, **9**, 483 (1971).
3364. G. Vaughan, D. E. Eaves, W. Cooper, Polymer, **2**, 235 (1961).
3365. M. F. Vaughan, M. A. Francis, J. Appl. Polym. Sci., **21**, 2409 (1977).
3366. J. M. Vela Estrada, A. E. Hamielec, Polym. Reaction Eng., **1**, 171 (1992–93).
3367. J. Veličković, D. Jovanović J. Vukajlovic, Makromol. Chem., **129**, 203 (1969).
3368. J. Veličković, V. Juraničová, J. Filipovic, Angew. Makromol. Chem., **24**, 77 (1972).
3369. J. Veličković, J. Filipović, IUPAC Helsinki, Preprints, Vol. III, II-67 (1972).
3370. J. Veličković, S. Vasovič, Makromol. Chem., **153**, 207 (1972).
3371. J. Veličković, S. Coseva, R. J. Fort, Eur. Polym. J., **11**, 377 (1975).
3372. J. Veličković, M. Plavšič, Eur. Polym. J., **12**, 151 (1976).
3373. J. Veličković, J. Filipvić, Macromolecules, **17**, 611 (1984).
3374. J. Veličković, J. Filipović, Makromol. Chem., **185**, 569 (1984).
3375. K. Venkatarao, M. Santappa, J. Polym. Sci. A-1, **8**, 1785 (1970).
3376. J. C. Via, C. L. Brave, L. T. Taylor, Anal. Chem., **66**, 603 (1994).
3377. J. L. Vidal, P. Crouzet, MA. Martens, J. Chromatogr. Sci., **20**, 252 (1982).
3378. G. Vidotto, R. Zannetti, L. Cavalli, Makromol. Chem., **146**, 159 (1971).
3379. L. Vilenchik, R. Ayotte, J. Appl. Polym. Sci., Appl. Polym. Symp., **51**, 73 (1992).
3380. M. Villacampa, J. R. Quintana, R. Salazar, I. Katime, Macromolecules, **28**, 1025 (1995).
3381. H. Vink, R. Wikstrom, Svensk. Papperstid., **66**, 55 (1963).
3382. F. Vira, K. Viras, F. Aroni, A. Dondos, Eur. Polym. J., **10**, 891 (1974).
3383. F. Viras, Y.-Z. Luo, K. Viras, R. H. Mobbs, T. A. King. C. Booth, Makromol. Chem., **189**, 459 (1988).
3384. D. Vitalini, E. Scamporrino, P. Maravigna, Macromolecules, **27**, 2291 (1994).
3385. M. G. Vitovskaya, E. P. Astapenko, V. Ya. Nikolayev, S. A. Didenko, V. N. Tsvetkov, Polym. Sci. USSR, **18**, 788 (1976); Vysokomol. Soedin. A, **18**, 691 (1976).
3386. Ye. I. Vizen, F. I. Yakobson, Polum. Sci. USSR, **20**, 1046 (1978); Vysokomol. Soedin. A, **20**, 927 (1976).
3387. J. Vizzini, F. Ciardelli, J. C. W. Chein, Macromolecules, **25**, 108 (1992).
3388. P. Viček, J. Janča, J. Trekoval. Makromol. Chem., Rapid. Commun., **1**, 485 (1980).
3389. P. Viček, J. Otoupalova, A. Sikora, J. Kriz, Macromolecules, **28**, 7262 (1995).
3390. R. E. Vogel, Kunststoffe, **42**, 17 (1952).
3391. R. E. Vogel, Kunststoffe, **44**, 335 (1954).
3392. L. A. Volkova, Ye. A. Andreyeva, V. Ye. Eskin, Polym. Sci. USSR, **20**, 986 (1978); Vysokomol. Soedin. A, **20**, 874 (1978).
3393. B. Vollmert, J.-X. Huang, Makromol. Chem., Rapid Commun, **1**, 333 (1980).
3394. W. C. Von Dohlen, T. P. Wilson, J. Polym. Sci., Polym. Chem., Ed., **17**, 2511 (1979).
3395. P. H. Von Dreele, N. Lotan, V. S. Ananthanarayanan, R. H. Andreatta, D. Poland, H. A. Scheraga, Macromolecules, **4**, 408 (1971).
3396. E. Von Meerwall, D. H. Tomich, N. Hadjichristidis, L. J. Fetters, Macromolecules, **15**, 1157 (1982).
3397. M. J. Voorn, Fortsch. Hochpolym. Forsch., **1**, 192 (1959).
3398. B. Ya. Voronov, G. I. Volkova, Zarodsk. Lab., **30**, 1411 (1964).
3399. N. G. Vyas, S. Shashikant, C. K. Patel, R. D. Patel. J. Polym. Sci., Polym. Phys. Ed., **17**, 2021 (1979).
3400. Yu, P. Vyrskii, O. A. Klapovskaya, N. V. Andrianova, O. F. Alkayeva, Polym. Sci. USSR, **10**, 1959 (1968); Vysokomol Soedin. A, **10**, 1688 (1968).
3401. T. Wada, S. Machi, J. Polym. Sci., Polym. Lett. Ed., **11**, 115 (1973).
3402. T. Wada, M. Takehisa, S. Machi, J. Polym. Sci., Polym. Chem. Ed., **12**, 1629 (1974).
3403. T. Wada, T. Watanabe, M. Takehisa, S. Machi, J. Polym. Sci., Polym. Chem. Ed., **12**, 1585 (1974).
3404. J. I. Wadsworth, L. Segal, J. D. Timpa, Polym. Preprints, **12**, (2)., 854 (1971).
3405. R. H. Wanger, J. Polym. Sci., **2**, 21 (1974).
3406. H. L. Wanger, P. J. Flory, J. Am. Chem. Soc., **74**, 195 (1952).
3407. E. R. Wanger, R. J. Greff, J. Polym. Sci., A-1, **9**, 2193 (1971).
3408. K.-G. Wahlund, H. S. Winegarner, K. D. Caldwell, J. C. Giddings, Anal. Chem., **58**, 573 (1986).
3409. M. Wales, J. W. Williams, J. O. Thompson, R. H. Ewart, J. Phys, Colloid. Chem., **52**, 983 (1948).
3410. M. Wales, F. T. Adler, K. E. Van Holde, J. Phys, Colloid, Chem., **55**, 145 (1951).
3411. M. Wales, D. L. Swanson, J. Phys. Colloid. Chem., **55**, 203 (1951).
3412. M. Wales, S. J. Rehfield, J. Polym. Sci., **62**, 179 (1962).
3413. M. L. Wallach, Polym. Preprints, **10** (2), 1248 (1969).
3414. A. Waly, N. Y. Abou-Zeid, E. A. El-Alfy, A. Hebeish, Angew. Makromol. Chem., **103**, 61 (1982).
3415. R. Walz, W. Heitz, J. Polym. Sci., Polym. Chem. Ed., **16**, 1807 (1978).
3416. P-g. Wand, A. Woodward, Macromolecules, **20**, 1818 (1987).
3417. F. Wang, J. Roovers, P. M. Toporowski, Macromolecules, **26**, 2826 (1993).
3418. J.-S. Wang, J. R. Knox, R. S. Porter, J. Polym. Sci., Polym. Phys. Ed., **16**, 1709 (1978).

3419. J. Wang, P. Bayard, R. Jermoe, P. Teyster, S. V. Varshney, Macromolecules, **26**, 2386 (1993).
3420. L. Wang, Y.-H. Ko. W. P. Weber, Macromolecules, **25**, 2828 (1992).
3421. S. Wang, J. E. Mark. Polym. Bull., **31**, 205 (1993).
3422. T. P. Wang, G. Heidemann, G. Walk, Angew. Makromol. Chem., **25**, 69 (1972).
3423. S. Wanigatunga, K. B. Wagener, Macromolecules, **22**, 4156 (1989).
3424. G. Wanka, H. Hoffmann, W. Ulbricht, Macromolecules, **27**, 4145 (1994).
3425. H. A. Wannow, F. Thormann, Kolloid, Z., **112**, 94 (1949).
3426. V. Warzelhan, H. Hoecker, G. V. Schulz, Makromol. Chem., **179**, 2221 (1978).
3427. H. Watanabe, T. Kotaka, Polym. J., **14**, 739 (1982).
3428. H. Watanabe, H. Yashida, T. Kotaka, Macromolecules, **21**, 2175 (1988).
3429. H. Watanabe, T. Shimura, T. Kotaka, M. Tirrell, Macromolecules, **26**, 6338 (1993).
3430. M. Watanabe, M. Togo, K. Sanui, N. Ogata, T. Kobayashi, E. Ohtaki, Macromolecules, **17**, 2908 (1984).
3431. M. Watanabe, M. Rikukawa, K. Sanui, N. Ogata, N. Kato, T. Kobayashi, Z. Ohtaki, Macromolecules, **17**, 2902 (1984).
3432. S. Watanabe, J. Hayashi, T. Akahori, J. Polym, Sci., Polym. Chem Ed., **12**, 1065 (1974).
3433. T. Watanabe, T. Wada, S. Machi, M. Takeshisa, J. Polym. Sci., Polym. Lett. Ed., **10**, 741 (1972).
3434. T. Watanabe, T. Wada, M. Takehisa, S. Machi, J. Polym. Sci., Polym. Chem. Ed., **12**, 1609 (1974).
3435. T. Watanabe, T. Wada, M. Takeshia, S. Machi, J. Polym. Sci., Polym. Ed., **12**, 1619 (1974).
3436. Y. Watanabe, T. Aida, S. Inoue, Macromolecules, **23**, 2612 (1990).
3437. J. L. Waters, Polym. Preprints, **6** (2), 1061 (1965).
3438. W. F. Watson, J. Polym. Sci., **13**, 595 (1954).
3439. J. W. Ways, S. Nan, W. Yuman, J. Li. N. Hadjichristidis, Macromolecules, **24**, 4469 (1991).
3440. T. J. R. Weakley, R. J. P. Williams, J. D. Wilson, J. Chem. Soc., 3963 (1960).
3441. N. E. Weeks, S. Mori, R. S. Porter, J. Polym. Sci., Polym. Phys. Ed., **13**, 2031 (1975).
3442. G. Weinand, G. Smets, J. Polym. Sci., Polym. Chem. Ed., **16**, 3091 (1978).
3443. K. Weinmann, B. A. Wolf, M. T. Rätzch, L. Tschersich, J. Appl. Polym. Sci., **45**, 1265 (1992).
3444. J. K. Weise, Polym. Preprints, **12** (1)., 512 (1971).
3445. P. Weiss, J. Herz, P. Rempp, Z. Gallot, H. Benoît, Makromol. Chem., **145**, 105 (1971).
3446. G. J. Welch, Polymer, **15**, 429 (1974).
3447. J. D. Welions, A. Schindler, V. Stannett, Polymer, **5**, 499 (1964).
3448. F. Wenger, S. P. S. Yen, Makromol. Chem., **43**, 1 (1961).
3449. F. Wenger, Makromol. Chem., **64**, 151 (1963).
3450. G. Wenz, G. Wegner, Makromol. Chem., Rapid Commun., **3**, 231 (1982).
3451. H. Wesslau, Makromol. Chem., **20**, 111 (1956).
3452. H. Wesslau, Makromol. Chem., **26**, 102 (1958).
3453. H. Wesslau, Makromol. Chem., **26**, 96 (1958).
3454. B. Wesslin, G. Gunneby, G. Hellstroem, P. Svedling, IUPAC Helsinki, Preprints, Vol. I, I/46 (1972).
3455. B. Westall, Polymer, **9**, 423 (1968).
3456. L. Westerman, J. C. Clark, J. Poly Sci., Polym. Phys. Ed., **11**, 559 (1973).
3457. R. L. Whistler, H. J. Roberts, J. Org. Chem., **26**, 2458 (1961).
3458. H. Whitaker, G. Hills, J. Appl. Polym. Sci., **13**, 1921 (1969).
3459. D. M. White, Polym. Preprints, **12** (1), 155 (1971).
3460. D. M. White, Macromolecules, **12**, 1008 (1979).
3461. J. E. White, M. D. Scaia, Polymer, **25**, 850 (1984).
3462. B. A. Whitehouse, Macromolecules, **4**, 463 (1971).
3463. L. R. Whitlock, R. S. Porter, J. Polym. Sci., Polym. Phys. Ed., **10**, 877 (1972).
3464. N. M. Wiederhorn, A. R. Brown, J. Polym. Sci., **8**, 651 (1952).
3465. N. Wiemers, G. Wegner, Makromol. Chem., **175**, 2719 (1974).
3466. D. R. Wiff, M. Gehatia, A. Wereta, J. Polym. Sci., Poly Phys. Ed., **13**, 275 (1975).
3467. P. W. O. Wijga, J. Van Schooten, J. Boerma, Makromol. Chem., **36**, 115 (1960).
3468. J. G. Wijmans, C. A. Smolders, Eur. Polym. J., **19**, 1143 (1983).
3469. L. Wild, R. Guliana, J. Polym. Sci. A-2, **5**, 1087 (1967).
3470. L. Wild, R. Ranganath, T. Ryle, J. Polym. Sci. A-2, **9**, 2137 (1971).
3471. L. Wild, R. Ranganath, T. Ryle, Polym. Preprints, **12** (1)., 266 (1971).
3472. L. Wild, R. Ranganath, A. Barlow, J, Appl. Polym. Sci., **21**, 3331 (1977).
3473. L. Wild, T. R. Ryle, Polym. Preprints, Am. Chem. Sco., Polym. Chem. Div., **18**, 182 (1977).
3474. L. Wild, T. R. Ryle, D. C. Knobeloch, I. R. Peat, J. Polym. Sci., Polym. Phys. Ed., **20**, 441 (1982).
3475. L. Wild, Adv. Polym. Sci, **98**, 1 (1990).
3476. L. Wild, TRIP, **1**, 50 (1993).
3477. C.-E. Wilén, J. H. Näsman, Macromolecules, **27**, 4051 (1994).
3478. D. L. Wilfong, G. W. Knight, J. Polym. Sci., Polym. Phys. Ed., **28**, 861 (1990).
3479. T. Williams, Y. Udagawa, A. Keller, I. M. Ward, J. Polym. Sci., A-2, **8**, 35 (1970).
3480. G. R. Williamson, A. Cervenka, Eur. Polym. J., **8**, 1009 (1972).
3481. F. Wiloth, Makromol. Chem., **8**, 111 (1952).
3482. F. Wiloth, Makromol. Chem., **14**, 156 (1954).
3483. J. E. Wilson, J. Macromol. Sci. Part A: Chem., **8**, 307 (1974).
3484. J. C. Wilson, J. Polym. Sci., Polym. Chem. Ed., **14**, 2927 (1976).
3485. W. W. Wilson, P. Fanz, G. D. McGinnis, J. Appl. Polym. Sci., **24**, 2195 (1979).
3486. E. Wipfelder, H. Heusinger, J. Poly Sci., Polym. Chem. Ed., **16**, 1779 (1978).

3487. M. Wirick, M. Waldman, J. Appl. Polym. Sci., **14**, 579 (1970).
3488. A. Wirsén, Makromol. Chem., **189**, 833 (1988).
3489. P. Wisian-Neilson, M. Bahadur, J. M. Iriarte, R. R. Ford, Ch. E. Wood, Macromolecules, **27**, 4471 (1994).
3490. J. Witte, M. Hoffman, Makromol. Chem., **179**, 641 (1978).
3491. U. Wittenberger, J. Klein, Angew. Makromol. Chem., **8**, 133 (1969).
3492. J. F. Wolfe, J. K. Stille, Macromolecules, **9**, 489 (1976).
3493. E. Wolfram, M. Nagy, Kolloid, Z.-Z. Polym., **227**, 86 (1968).
3494. R. H. Wondraczek, J. P. Kennedy, Polym. Bull., **2**, 675 (1980).
3495. R. H. Wondraczek, H. E. Hotzel. G. Heublein, Polym. Bull., **7**, 377 (1982).
3496. P. V. Wright, J. Polym. Sci., Polym. Phys. Ed., **11**, 51 (1973).
3497. C. Wu, K. F. Woo, X. Luo, D.-Z. Ma, Macromolecules, **27**, 6055 (1994).
3498. C. Wu, M. Siddiq, S. Jiang. Y. Huang, J. Appl. Polym. Sci., **58**, 1779 (1995).
3499. C. Wu, S. Bo, M. Siddiq, G. Yang. T. Chen, Macromolecules, **29**, 2989 (1996).
3500. X. Y. Wu, D. Hunkeler, A. E. Hamielec, R. H. Pelton, D. R. Woods, J. Appl. Polym. Sci., **42**, 2081 (1991).
3501. W. Wunderlich, Makromol. Chem., **177**, 973 (176).
3502. D. P. Wyman, T. Altares, Jr., Marokmol. Chem., **72**, 68 (1964).
3503. D. P. Wyman, L. J. Elyash, W. J. Frazer, J. Polym. Sci. A-1, **3**, 861 (1965).
3504. D. P. Wyman, I. H. Song, Makromol. Chem., **115**, 64 (1968).
3505. M. G. Wyzgoski, J. Appl. Polym. Sci., **26**, 1689 (1981).
3506. X. Xie, T. E. Hogen-Esch, Macromolecules, **29**, 1734 (1996).
3507. J. Xu, A. E. Woodward, Macromolecules, **19**, 1114 (1986).
3508. G. Xu, S. Lin, Macromolecules, **30**, 685 (1997).
3509. G. Xue, Y. Wang, S. Liu, Y. Liao, Macromolecules, **28**, 3476 (1995).
3510. A. I. Yakubchik, B. I. Tikhumirov, L. N. Mikhailova, Polym. Sci. USSR, **7**, 1728 (1965); Vysokomol. Soedin., **7**, 1562 (1965).
3511. A. Yamada, Y. Yamamuro, M. Yanagita, Bull. Chem. Soc. Japan, **35**, 609 (1962).
3512. B. Yamada, T. Otsu, Makromol. Chem., **190**, 915 (1989).
3513. M. Yamada, T. Iguchi, A. Hirao, S. Nakahama, J. Watanabe, Macromolecules, **28**, 50 (1995).
3514. T. Yamada, T. Yoshizaki, H. Yamakawa, Macromolecules, **25**, 1487 (1992).
3515. R. Yamadera, C. Sonode, J. Polym. Sci. B, **3**, 411 (1965).
3516. K. Yamaguchi, S. Saeda, J. Polym. Sci., A-2, **7**, 1303 (1969).
3517. K. Yamaguchi, Makromol. Chem., **128**, 19 (1969).
3518. K. Yamaguchi, Makromol. Chem., **132**, 143 (1970).
3519. K. Yamaguchi, T. Wada, S. Maruyama, M. Takehisa, J. Polym. Sci., Polym. Phys. Ed., **11**, 1573 (1973).
3520. K. Yamaguchi, A. Hirao, K. Suzuki, K. Takenaka, S. Nakahama, N. Yamazaki, J. Polym. Sci., Polym. Lett. Ed., **21**, 395 (1983).
3521. S. Yamaguchi, H. Hayashi, F. Hamada, A. Nakajima, Macromolecules, **17**, 2131 (1984).
3522. K. Yamamoto, H. Fujita, Polymer, **7**, 557 (1966).
3523. K. Yamamoto, H. Fujita, Polymer, **8**, 517 (1967).
3524. K. Yamamoto, N. Bessho, T. Shibashi, E. Maekawa, Polym. J., **13**, 555 (1981).
3525. K. Yamamoto, N. Jikei, K. Miyatake, J. Katoh, H. Nishida, E. Tsuchida, Macromolecules, **27**, 4312 (1994).
3526. T. Yamamoto, D. Oguro, K. Kubota, Macromolecules, **29**, 1833 (1996).
3527. Y. Yamamoto, I. Noda, M. Nagasawa, Polym. J., **1**, 304 (1970).
3528. S. Yamashita, H. Tani, Macromolecules, **7**, 406 (1974).
3529. Y. Yamashita, S. Miura, M. Nakamura, Makromol. Chem., **68**, 31 (1963).
3530. Y. Yamashita, K. Nobutoki, Y. Nakamura, M. Hirota, Macromolecules, **4**, 548 (1971).
3531. Y. Yamashita, M. Hirota, H. Matsui, A. Hirao, K. Nobutoki, Polym. J., **2**, 43 (1971).
3532. Y. Yamashita, K. Ito, K. Chiba, S. Kozawa, Polym. J., **3**, 389 (1972).
3533. Y. Yamashita, Polym. Preprints, **13** (1), 151 (1972).
3534. Y. Yamashita, Y. Tsukahara, H. Ito, Polym. Bull., **7**, 289 (1982).
3535. K. Yamaura, S. Matsuzawa, Y. Go. Kolloid, Z.-Z. Polym. **240**, 820 (1970).
3536. K. Yamaura, S. Kinugasa, S. Matsuzawa, Kolloid. Z.-Z. Polym. **248**, 893 (1971).
3537. K. Yamaura, Y. Höe, S. Matsuzawa, Y. Go, Kolloid, Z.-Z. Polym. **243**, 7 (1971).
3538. M. Yamawaki, A. Matsumoto, M. Oiwa, J. Polym. Sci., Polym. Chem. Ed., **22**, 1541 (1984).
3539. N. Yamazaki et al., Kôgyô Kagaku Zasshi, **64**, 1687 (1961).
3540. Y. C. Yang, J. C. Lim, H. G. Woo, S. K. Choi, J. Macromol. Sci. Part A: Chem., **18**, 677 (1982).
3541. C. B. Yang, C. C. Hsu, J. Appl. Polym. Sci., **58**, 1245 (1995).
3542. C.-W. Yang, Z. Yang, Z.-K. Zhou, D. A. Attwood, C. Booth, Macromolecules, **29**, 670 (1996).
3543. Z. Yang, S. Pickard, N.-J. Deng, R. J. Barlow, D. Attwood, C. Booth, Macromolecules, **27**, 2371 (1994).
3544. J. A. Yanko, J. Polym. Sci., **3**, 576 (1948).
3545. J. A. Yanko, J. Polym. Sci., **22**, 153 (1956).
3546. J. F. Yanus, J. M. Pearson, Macromolecules, **7**, 716 (1974).
3547. H. Yasuda, J. A. Wray, V. Stannett, J. Polym. Sci. C, **2**, 387 (1963).
3548. T. Yasuda, T. Aida, S. Inoue, Makromol. Chem., Rapid Commun., **3**, 585 (1982).
3549. T. Yasuda, T. Aida, S. Inoue, Macromolecules, **16**, 1792 (1983).
3550. T. Yasuda, T. Aida, S. Inoue, Macromolecules, **17**, 2217 (1984).
3551. W. W. Yau, C. P. Halone, J. Polym. Sci. B, **5**, 663 (1967).

3552. W. W. Yau, J. J. Kirkland, D. D. Bly, "Modern Size Exclusion Chromatography", Wiley-Interscience, New York, 1979.
3553. W. W. Yau, J. Appl Polym. Sci., Appl. Polym. Symp., **48**, 85 (1991).
3554. M. Yazdani-Pedram, L. Gargallo, D. Radić, Eur. Polym. J., **21**, 461 (1985).
3555. S. J. Yeh, H. L. Frisch, J. Polym. Sci., **27**, 149 (1958).
3556. S.-P. S. Yen, Makromol. Chem., **81**, 152 (1965).
3557. D. R. Yen, S. Raghavan, E. W. Merrill, Macromolecules, **29**, 8977 (1996).
3558. M. A. Yeremina, Ye. G. Erenburg, I. Ya. Poddubnyi, Polym. Sci. USSR, **20**, 2480 (1978); Vysokomol. Soedin. A, **20**, 2207 (1978).
3559. V. V. Yevreinov, V. I. Gerbich, L. I. Sarynina, S. G. Entelis, Polym. Sci USSR, **12**, 938 (1970); Vysokomol. Soedin. A, **12**, 829 (1970).
3560. R. Yin. T. Hogen-Esch, Macromolecules, **26**, 6952 (1993).
3561. O. Ying, P. Xie, M. Je, Makromol. Chem., Rapid Commun., **6**, 105 (1985).
3562. M. Yokouchi, Y. Kobayashi, J. Appl. Polym. Sci., **26**, 431 (1981).
3563. Y. Yokoyama, M. Okada, H. Sumitomo, Polym. J., **11**, 629 (1979).
3564. K. Yoshida, Y. Yamashita, Makromol. Chem., **100**, 175 (1967).
3565. K. Yoshida, K. Ishigure, H. Garreau, V. Stannett, J. Macromol. Sci. Part A: Chem., **14**, 739 (1980).
3566. H. Yoshida, K. Nakamura, Y. Kobayashi, Polym. J., **14**, 855 (1982).
3567. N. Yoshie, H. Menju, H. Sato, Y. Inoue, Macromolecules, **28**, 6516 (1995).
3568. T. Yoshimura, T. Masuda, T. Higahimura, Macromolecules, **21**, 1899 (1988).
3569. M. Yoshioka, T. Otsu, Macromolecules, **25**, 559 (1992).
3570. J. F. S. Yu, J. L. Zakin, G. K. Patterson, J. Appl. Polym. Sci., **23**, 2493 (1973).
3571. T. Tu, S. Fu, S. Li, C. Ji, W. Cheng, Polymer, **25**, 1363 (1984).
3572. Y. S. Yu, R. Jérôme, R. Fary, Ph. Teyssié, Macromolecules, **27**, 5957 (1994).
3573. A. A. Yulchibayev, Kh. U. Usmanov, A. Kh. Gufarov, A. Matyakubov, IUPAC Helsinki Preprints, Vol. II, I/153 (1972).
3574. M. M. Zafar, R. Mahmood, S. A. Wadood, Makromol. Chem., **160**, 313 (1972).
3575. M. M. Zafar, R. Mahmood, Makromol, Chem., **175**, 903 (1974).
3576. Yu, S. Zaitsev, V. D. Yenal'ev, A. I. Yurzhenko, Polym. Sci. USSR, **12**, 2831 (1970); Vysokomol. Soedin. A, **12**, 2500 (1970).
3577. A. Zambelli, P. Ammendola, A. J. Sivak, Makromol. Chem., **17**, 461 (1984).
3578. L. V. Zamoiskaya, E. S. Gankina, Ye. B. Milovskaya, Polym. Sci. USSR, **18**, 1873 (1976); Vysokomol. Soedin. A, **18**, 1635 (1976).
3579. G. Zanocco, E. Abuin, E. Lissi, L. Gargallo, D. Radić, Eur. Polym. J., **18**, 1037 (1982).
3580. F. Zapf, Makromol. Chem., **3**, 164 (1949).
3581. F. Zapf, Makromol. Chem., **10**, 35 (1953).
3582. F. Zapf, Makromol. Chem., **10**, 61 (1953).
3583. I. N. Zaripov, V. V. Beresnev, P. A. Kirpichnikov, Polym. Sci., USSR, **18**, 2547 (1976); Vysokomol. Soedin. A, **18**, 2228 (1976).
3584. B. Zaschker, J. P. Kennedy, Macromolecules, **28**, 4426 (1995).
3585. V. I. Zegel'man, M. N. Shlykova, S. V. Svetozarskli, Ye, N. Zil'berman, Polym. Sci. USSR, **10**, 133 (1968); Vysokomol Soedin. A, **10**, 114 (1968).
3586. R. P. Zelinski, C. F. Wofford, J. Polym. Sci. A-1, **3**, 93 (1965).
3587. A. C. Zettlemoyer, E. T. Pieski, Ind. Eng. Chem. **45**, 165 (1953).
3588. X. Zhang, Q. Zhou, W. P. Weber, R. F. Horvath, T. H. Chan, G. Manuel, Macromolecules, **21**, 1563 (1988).
3589. X. F. Zhang, E. Eisenberg, Macromolecules, **27**, 4915 (1994).
3590. X.-H. Zhang, R. West, J. Polym. Sci., Polym. Chem. Ed., **22**, 159 (1984).
3591. X.-H. Zhang, R. West, J. Polym. Sci., Polym. Chem. Ed., **22**, 225 (1984).
3592. Y. Zhang, R. A. Gross, R. W. Lenz, Macromolecules, **23**, 3206 (1990).
3593. Z. Zhao, R. Wartkins, S. W. Barton, J. Appl. Polym. Sci., **55**, 773 (1995).
3594. J. Q. Zhao, E. M. Pearce, T.K. Kwei, H. S. Jeon, P. K. Kesani, N. P. Balsara, Macromolecules, **28**, 1972 (1995).
3595. S. Ya Zhelobayeva, A. V. Troitskaya, T. N. Osipova, S. I. Klenin, A. F. Nikolayev, Polym. Sci. USSR, **13**, 951 (1971); Vysokomol. Soedin. A, **13**, 842 (1971).
3596. H. Zhiduan, Y. Meina, Z. Xuanqui, W. Xiaobing, J. Xiaoming. H. Jiaxian, L. Chuynrong. W. Linfu, Eur. Polym. J. **22**, 597 (1986).
3597. S. Zhiquan, O. Jun, W. Fusong, H. Zhenya, V. Fusheng, Q. Baogong, J. Polym. Sci., Polym., Chem. Ed., **18**, 3345 (1980).
3598. X. Zhongde, S. Mingshi, N. Hadjichristidis, L. J. Fetters, Macromolecules, **14**, 1591 (1981).
3599. X. Zhongde, N. Hadjichristidis, L. J. Fetters, Macromolecules, **17**, 2303 (1984).
3600. X. Zhonge, Y. Ping, Z. Jinngguo, J. Erfand, W. Meiyan, J. Appl. Polym. Sci., **37**, 3195 (1989).
3601. S. Q. Zhou, W. P. Weber, Macromolecules, **23**, 1915 (1990).
3602. X. Q. Zhou, J. N. Hay, Eur. Polym. J., **29**, 291 (1993).
3603. X.-Q. Zhou, J. N. Hay, Polymer, **34**, 2283 (1993).
3604. H. Zhuang, J. A. Gardella, Jr, D. M. Hercules, Macromolecules, **300**, 1153 (1997).
3605. M. Zief, G. Brunner, J. Metzendorff, Ind. Eng. Chem., **48**, 119 (1956).
3606. J.-G. Zilliox, P. Rempp, J. Parrod, J. Polym. Sci. C, **22**, 145 (1968).
3607. A. Živný, J.Pouchlý, K. Šolc, Coll. Czech. Chem. Commun. **32**, 2753 (1967).
3608. W. X. Zukas, W. J. Mac Knight, R. W. Lenz, Polym. Bull., **2**, 263 (1980).

Solvents and Non Solvents for Polymers

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A. INTRODUCTION

The tables contain qualitative solubility data for a selected number of polymers. Since no standard definition for solvent–nonsolvent systems has been used in most of the original sources, the recognition of a certain compound as a solvent or nonsolvent is to some extent influenced by personal interpretation. No attempt has been made to edit the original information. Division into only two classes, solvents and nonsolvents, is dictated by the practical point of view. For more quantitative information, the user is referred to the tables of theta solvents and fractionation of polymers in this Handbook.

The arrangement of polymers into classes is based on chemical structure. Since properties change gradually within a series of homologs, polymers are arranged according to increasing complexity, as regards the size of the monomer unit, and the kind and number of substituents. Only when this principle could not be applied was an alphabetical listing chosen. We believe that a typical solution behavior of similar polymers may be recognized more easily by this arrangement. When formulas are given, they refer to the main structures present in the polymers.

Polymers are named in a manner thought to be most recognizable to the reader. IUPAC or common names (and sometimes both) are used.

The table has nearly doubled in size since the last edition. New polymer categories include copolymers, main-chain carbonyl polymers, block copolymers and dendrimers, emphasizing recently developed materials. Copolymers are generally named with the dominant monomer given first, but the reader should check alternate names, since each polymer is listed only once. Solubility of copolymers generally resemble more or less the properties of the dominating monomer, although they often exhibit higher solubilities than the corresponding homopolymers. The solubility of block copolymers may reflect the solubilizing/dispersing power of one of the polymer segments rather than true solubility of all segments. Dendrimers are shown as the repeating monomer unit, with the terminal unit specified when known.

Solubility normally increases with rising temperature. However, negative temperature coefficients in some solvents are observed. This is particularly true for cellulose and poly(ethylene oxide)s.

Increasing molecular weight generally reduces solubility.

Increased branching increases the solubility compared to a linear polymer of the same molecular weight.

Certain combinations of two or more solvents may become nonsolvents. Conversely, mixtures of two or more nonsolvents may sometimes become solvents. These possibilities should be considered if new solvent–nonsolvent combinations are to be examined. It is recommended that the reader consults the solubility parameter section in this Handbook.

The classification of a certain compound as a nonsolvent does not necessarily imply its ability to act as a precipitant since this is also influenced by the nature of the particular solvent of a solvent–nonsolvent pair. However, most nonsolvents combine both properties.

The list of solvents and nonsolvents for each polymer is arranged alphabetically. If class names are used, they refer to the most common compounds. Less common compounds, although falling into a class already mentioned, are additionally cited.

Water is a nonsolvent for most polymers and is, therefore, often not mentioned or only mentioned if similar polymers or derivatives are water soluble.

The data refer to room temperature unless otherwise stated.

Homologs and closely related compounds generally have similar solubility properties. When specific solvents or nonsolvents are cited, it is done so with the understanding that homologs and compounds with similar structures can be expected to exhibit similar properties.

B. ABBREVIATIONS

bzn.	Benzene
DMA	*N*,*N*-Dimethylacetamide
DMF	*N*,*N*-Dimethylformamide
DMSO	Dimethyl sulfoxide
HMPT	Hexamethylphosphoric triamide
TMS	Tetramethylene sulfone
THF	Tetrahydrofuran
W	Water
aqu.	Aqueous
conc.	Concentrated
DS	Degree of substitution
degrad.	Degradation
dil.	Diluted
mol. wt.	Molecular weight
rt	Room temperature
SC	Substituent content
sw	Swelling
temp.	Temperature

C. TABLES OF SOLVENTS AND NONSOLVENTS

TABLE 1. MAIN-CHAIN ACYCLIC CARBON POLYMERS

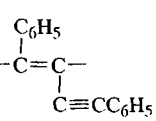
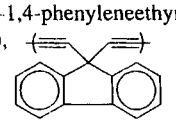
Polymer	Solvents	Nonsolvents	Refs.
1.1. POLY(DIENES), POLY(ACETYLENES) (see also 6.1, 6.2)			
Poly(dienes) unsubstituted			
Poly(allene)	Bzn., halogenated hydrocarbons	Hexane, methanol	1
Poly(1,2-butadiene)	Toluene		2
57% 1,2-	Chloroform		3
Poly(1,3-butadiene)	Higher aliphatic ethers, hydrocarbons, higher ketones, THF	Alcohol, dil. acids, dil. alkalies, hypochlorite solutions, lower ketones and esters, nitromethane, propionitrile, W	4–6
98% *cis*-, 2% 1,2-	Bzn., toluene		7
95% *cis*-, 1% *trans*-	Cyclohexane		7
100% *cis*-	Bzn., heptane/hexane (1/1)		7
97% *trans*-, 3% 1,2-	Cyclohexane, toluene		7
Poly(isoprene),	See Poly(1,3-butadiene)		
cis-, with chlorosulfonylisocyanato groups	Toluene	Diethyl ether, hexane	8
Poly(dienes) substituted			
Poly(2-*tert*-butyl-1,3-butadiene)	Bzn., carbon disulfide, carbon tetrachloride, chloroform, diethyl ether, heptane	Acetone, alcohol	9,10
Poly(dicyclopentadiene)	Toluene	Methanol	11
Poly(5,7-dimethyl-1,6-octadiene)	See Poly(2-*tert*-butyl-1,3-butadiene)		
Poly(1-methoxybutadiene), crystalline		Acetone, bzn., dioxane, heptane, methanol	12
Poly(2-chlorobutadiene), chloroprene	Bzn., carbon tetrachloride, chlorinated hydrocarbons, cyclohexane/toluene, cyclohexanone, dioxane, ethyl acetate, pyridine	Alcohols, aliphatic hydrocarbons, hydrogen fluoride, ketones, mineral oils, nonoxidizing conc. acids incl. hydrogen fluoride, toluene (sw), W	4–7,13
1,4-*cis*-	Bzn., chloroform, ether, hexane, THF	Acetone, ethanol, methanol	14
Poly[2-(chloromethyl)butadiene]	Dichloromethane, THF, toluene	Ethanol, methanol	686
Poly(perfluoro-1,4-pentadiene)	Hexafluorobenzene		
Poly(2,5-diheptyl-1,4-phenylenehexa-1,3,5-triene-1,6-diyl)	Acetone, bzn., chloroform		16
Poly(1,5-pentamethyl-2-enediyl)	Methylene chloride	Methanol	17
Poly(butadiene-*co*-styrene) (7 : 3–95 : 5) Buna-S GR-S SBR rubber	Bzn., cyclohexane, toluene		7
Poly(acetylenes)			
Poly(acetylene)	Aniline, DMF, isopropylamine	Acetone, bzn., carbon tetrachloride, cyclohexane, methanol, methylene chloride, pyridine	18
Poly(phenylacetylene)			
low mol. wt.	Acetone, carbon tetrachloride, methanol		19,20
high mol. wt.	Bzn., cyclohexane (hot), toluene	Methanol	19–21
Poly(diphenyldiacetylene), poly-(1,4-diphenyl-1,3-butadiene), 	Chloroform, cyclohexane, dioxane	Methanol	22
Poly[bis(alkylthio)acetylene]	Aromatic hydrocarbons, halogenated hydrocarbons		23
Poly[3-(*tert*-butoxycarbonylmethyleneiminocarbonyloxy)-8-(5-pyrimidyl)-oct-5-yn-7-ene-1,8-diyl]	Chloroform, hexane, THF		24
Poly[2-(dodecyloxycarbonyl)-1,4-phenyleneethynylene]	THF (hot)	Methanol	25
Poly(9,9-dipropargylfluorene), 	Dichlorobenzene		26
Poly(1-propynyleneoxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy-2-propynylene)	DMF/pyridine	W	27,247
Poly(ethynylene-1,4-phenyleneoxy-1,4-phenyleneethynylene)	Dioxane, pyridine	Methanol	28
Poly(butadiene-*co*-acrylonitrile)	Acetone, bzn., chloroform		7

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TABLE 1. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
1.2. POLY(ALKENES)			
Poly(methylene)	See Polyethylene		
Polyethylene			
high density	Above 80°C: 1,2,4-trichlorobenzene, decalin, di-*n*-amyl ether, halogenated hydrocarbons, higher aliphatic esters and ketones, hydrocarbons, xylene	All common organic solvents at room temperature, more polar organic solvents even at elevated temperatures, inorganic solvents	4–7,13, 29,30
low density	As above, but temperature 20–30°C lower, depending on degree of branching	As above	5,7
chlorinated, 40% Cl	At elevated temperature: chlorobenzene, cyclohexanone, tetrachloroethane, tetrahydronaphthalene, toluene, xylene	Acetone, butanol, chloroform, dioxane, MEK, methanol, methyl acetate, THF	31
chlorinated, 60% Cl	At elevated temperature: acetone/carbon disulfide (1/1), aromatic hydrocarbons, chloroform, cyclohexane, THF	Acetone, aliphatic and cycloaliphatic hydrocarbons, ethanol, methanol, methyl acetate	31
chlorosulfonated, 30% Cl, 1.5% S	Chlorinated hydrocarbons, MEK, THF, toluene		32,33
Poly(ethylene *g*-maleic anhydride) (97 : 3)	Toluene, xylene	W	34
Poly(ethylene-*co*-propylene) (3 : 2)	Bzn., xylene (hot)		32
Poly(ethylene-*co*-acrylic acid) (6 : 1)	Xylene		32
Poly(vinyl chloride-*co*-ethylene) (1 : 9)	Cyclohexanone (hot), nitrobenzene/chloroform (1/1)		127
Poly(ethylene-*co*-propylene-*co*-5-methylene-2-norbornene) (40 : 55 : 5), PolyEPDM	1,2,4-Trichlorobenzene		32
Poly(propylene)			5,7
atactic	Bzn., chlorinated hydrocarbons at rt., cyclohexane, diethyl ether, ethylcyclohexane, hydrocarbons, isoamyl acetate, toluene	More polar organic solvents with small hydrocarbon group even at elevated temperature	
isotactic	See Polyethylene		
Poly(1-butene)			7,32
atactic	Bzn., chloroform, decalin, ethylcyclohexane, toluene		
isotactic	See Polyethylene, ethylcyclohexane		7
Poly(isobutene)	Aliphatic ethers, anisole, carbon disulfide, chlorinated hydrocarbons, di(2-chloroethyl) ether, diethylsulfide, higher alcohols, higher esters, hydrocarbons, THF	Lower alcohols, lower ketones and esters, nitromethane, lower organic acids, propionitrile	5,6,13, 30,35
Poly(4-methyl-1-pentene), isotactic	See Polyethylene		
Poly(2-methyl-3-hydroxy-1,4-butylene)	THF		36
Poly(1-hexene)	Chloroform, THF		37
Poly(cyclopentylethylene)	Chloroform, diethyl ether, toluene	Methanol	38
Poly(cyclohexylethylene)			
atactic	Chlorinated hydrocarbons, hydrocarbons, THF	Alcohols, ethers, dioxane, esters, ketones	39
stereospecific	At elevated temperature: bzn., chlorobenzene, decahydronaphthalene, *o*-dichlorobenzene, methylcyclohexane, tetrahydronaphthalene, xylene	Heptane, MEK, nitrobenzene	40
Poly(cyclohexylalkenes)	Chloroform, diethyl ether (partially), toluene	Methanol	41
Poly(cyclohexenylethylene)			
atactic	Aliphatic hydrocarbons	Acetone, ethanol	42
isotactic	Aromatic hydrocarbons, halogenated hydrocarbons (partially)	Acetone, aliphatic hydrocarbons, diethyl ether, ethanol	42
Poly(pentenamer), Poly(cyclopentene) $-\{CH=CH(CH_2)_3\}-$	Chlorinated hydrocarbons, hydrocarbons	Alcohols, aliphatic ketones, ethers	43
Poly(3,4-dicarboxypent-1-ene-1,5-diyl)	Chloroform, methanol	Diethyl ether	44

TABLE 1. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(3,4-dihydroxypent-1-ene-1,5-diyl)	DMSO, methanol	Ethanol	44
Poly(1,1-diphenyl-2-vinylcyclopropane)	Chloroform	Methanol	45
Poly(*p*-anthrylphenylethylene)	Bzn., chlorobenzene, methylene chloride	Methanol, ethanol, hexane, W	46
Poly(1-methylbicyclo-[2.2.1]-hept-2-ene)	Chlorobenzene, *p*-xylene	Methanol	47
Poly(vinylborazine)	Bzn.	Pentane	48
Poly(methylene-*alt*-5-hydroxycyclohexene-1,3-diyl)	Hexanes		49
Poly(ethylene-*co*-1,4-hexadiene) (97 : 3)	Toluene		50
Poly(ethylene-*co*-1-cyano-1,4-butylene) (1 : 2)	Acetone (hot), chloroform, DMA, DMSO	Methanol	51
Poly(isobutylene-*co*-4-methylstyrene) (95 : 5)	Chloroform		52
Poly[hexene-*co*-*N*,*N*-di(isopropyl)amino-1-pentene] (2 : 3)	Hexane		53

1.3. POLY(ACRYLICS), POLY(METHACRYLICS)

1.3.1. POLY(ACRYLIC ACIDS)

Poly(acrylic acid)			
atactic	Alcohols, dil. alkali solutions, formamide, DMF, W	Dioxane (hot) (sw), esters, hydrocarbons, ketones	4–6, 13,54
isotactic	Dioxane/water (80/20)	Dioxane	55
Poly(methacrylic acid)			
atactic	Alcohols, aqu. hydrogen chloride (0.002 M, >30°C), dil. aqu. sodium hydroxide, W	Carboxylic acids, esters, hydrocarbons, ketones	4,5
isotactic, syndiotactic	W(partially)		56
Poly(acrylic acid-*co*-9-vinylphenanthrene) (3 : 2)	THF		57
Poly(itaconic acid)	DMF, methanol, W	Acetone, aniline, bzn., carbon disulfide, chloroform, ethanol, ethyl acetate, THF	58

1.3.2. POLY(ACRYLATES)

General	Aromatic hydrocarbons, chlorinated hydrocarbons, esters, ketones, THF	Aliphatic hydrocarbons, hydrogenated naphthalenes, diethyl ether	
Poly(methyl acrylate)	See General, acetone, bzn., ethyl acetate, glycol ester ethers, phosphorus trichloride	Alcohols, carbon tetrachloride	5,7, 30,59
Poly(ethyl acrylate)	See General, acetone, butanol, bzn., glycol ether, methanol, THF, *p*-xylene	Aliphatic alcohols C>4, cyclohexanol, tetrahydrofurfuryl alcohol	5,7, 30,59
Poly(2-hydroxyethyl acrylate)	Methanol, W		32
Poly(*n*-butyl acrylate)	See General, butanol, turpentine	Cyclohexyl acetate, ethanol, ethyl acetate, methanol	30
Poly(*tert*-butyl acrylate)	Acetone, methanol		7
Poly(4-hydroxybutyl acrylate)	Isopropanol		32
Poly(isobornyl acrylate)	THF		60
Poly(cyclohexyl acrylate)	Bzn., THF, toluene		32
Poly(2-ethylhexyl acrylate)	Esters, ketones, THF, toluene		32
Poly(benzyl acrylate)	Bzn., THF, toluene		32
Poly(5-cyano-3-thiapentyl acrylate)	Acetone, acetonitrile, dioxane, pyridine	Solvents of low solubility parameter	61
Poly[1-(*N*-ethyl-*N*-(1,4,7,10,13-pentaoxacyclopentadecyl)carbamoyl)ethylene]	Bzn., chloroform	Hexane	62
Poly(pentachlorophenyl acrylate)	Bzn., THF	Hexane, methanol, W	63
Poly(4–10,15,20-triphenylporphorin-21H,23H-5-ylphenyl acrylate)	Bzn., chloroform, THF, trifluoroacetic acid	Acetone, ethanol, hexane, methanol, W	64
Poly(vinylcyclohexane-*co*-methyl acrylate) (9 : 1)	THF	W	65

1.3.3. POLY(METHACRYLATES)

General	Acetic acid, acetone, bzn., chlorobenzene, chloroform, cyclohexanol (hot), cyclohexanone, cyclohexyl acetate, dioxane, 2-ethoxyethanol, ethyl acetate, isobutanol (hot), isobutyric acid, MEK, methylene chloride, xylene	Castor oil, cyclohexane, diethyl ether, ethylene glycol, formamide, gasoline, glycerol, hexane, methanol, nujol	

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TABLE 1. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(methyl methacrylate)			
atactic	See General, ethanol/water, ethanol/carbon tetrachloride, isopropanol/MEK (1/1) above 25°C, formic acid, nitroethane	Butylene glycol, carbon tetrachloride, *m*-cresol, diethyl ether, ethanol (absolute), higher esters, hydrogenated naphthalenes, linseed oil, turpentine	4,5,13, 30,66,67
isotactic	Acetone, acetonitrile, bzn., MEK, THF		7,32
syndiotactic	See isotactic		32
Poly(methyl methacrylate-*co*-2-acrylamido-2-methyl-1-propanesulfonic acid) (95 : 5)	Bzn., toluene		A32
Poly(ethyl methacrylate)	See General, carbon tetrachloride, ethanol (hot), ethyl ether, ethyl acetate, formic acid, isopropanol above 37°C, tetralin, turpentine (hot)	Alcohols, cyclohexane	5,7,67,68
Poly(2-hydroxyethyl methacrylate)	DMF, methanol, methyl Cellosolve®		32
Poly(*n*-propyl methacrylate)	See General, carbon tetrachloride, castor oil (hot), cyclohexane (hot), diethyl ether, ethanol, gasoline (hot), linseed oil (hot), turpentine	Formic acid	67
Poly(*n*-butyl methacrylate) and Poly(isobutyl methacrylate)	See General, carbon tetrachloride, castor oil (hot), cyclohexane, diethyl ether, ethanol (hot), gasoline, hexane, isopropanol above 23.7°C, linseed oil (hot), nujol (hot), turpentine	Ethanol (cold), formic acid	5,13,30, 67–69
Poly(butyl methacrylate-*co*-*i*-butyl methacrylate) (1 : 1)	Acetone, MEK, methylene chloride, THF, toluene		32
Poly(*n*-hexyl methacrylate)	Isopropanol above 33°C, MEK		5
Poly(cyclohexyl methacrylate)	Bzn., dioxane, THF		32
Poly(cyclohexylmethyl methacrylate)	Bzn., THF	Methanol	70
Poly(*n*-octyl methacrylate)	*n*-Butanol, MEK		5
Poly(2-ethylhexyl methacrylate)	Bzn., MEK, THF		32
Poly(*n*-decyl methacrylate)	Bzn., THF		32
Poly(*n*-lauryl methacrylate)	*n*-Pentanol above 29°C, isopropyl acetate, MEK	Ethyl acetate, methanol	5,71
Poly(phenyl methacrylate)	Bzn., MEK, THF, toluene		32
Poly(benzyl methacrylate)	Bzn., chloroform, dioxane, THF		32
Poly(4-*tert*-butylphenyl methacrylate)	Acetone		5
Poly[(4-(*tert*-butyoxycarbonyloxy)phenyl) methacrylate]	Bzn.		696
Poly(bornyl methacrylate)	Bzn.	Methanol	72
Poly(isobornyl methacrylate)	Bzn., THF, toluene		32
Poly[2-(dimethylamino)ethyl methacrylate]	Bzn., toluene		32
Poly[(2-trimethylsiloxy)ethyl methacrylate]	THF		73
Poly[2-(*N*-carbazoyl)ethyl methacrylate], crystalline	Aniline (hot, partially), diphenyl ether (hot), nitrobenzene (hot)	MEK	74
Poly(cyanomethyl methacrylate)	Acetone, acetonitrile, ethyl acetate, THF	Bzn., chloroform, methanol	75
Poly[2-(4-phenylazophenyl)ethyl methacrylate]	THF	Methanol	76
Poly[(methacryloyloxyundecyloxycarbonyldecyl)-2-(trimethylamino)ethyl phosphate]	Ethanol (hot), methanol	Acetone, bzn.	77
Poly[4-(4-methoxyphenyloxycarbonyl) phenoxyhexamethylene methacrylate]	Chloroform, THF, bzn., methylene chloride, *o*-dichlorobenzene,	Hexane, methanol	78
Poly[2-(dimethylamino)ethyl methacrylate-*co*-acrylamide] (4 : 1 to 2 : 3)	W		79
1.3.4. POLY(DISUBSTITUTED ESTERS)			
Poly(dimethyl itaconate)	Acetone, acetonitrile, bzn., DME, furfurol, halogenated hydrocarbons, methyl acetate, nitrobenzene, nitromethane, THF	Aliphatic hydrocarbons, carbon tetrachloride, diisopropyl ether, ethanol, ethyl acetate, methanol, propylene carbonate, toluene, W	80–82
Poly(di-*n*-butyl itaconate)	Acetone, amyl acetate, dioxane, ethanol, halogenated hydrocarbons, hydrocarbons, methyl acetate, THF	Acetonitrile, DMF, furfurol, methanol, nitrobenzene, nitromethane, propylene carbonate	81,82
Poly(diphenyl itaconate)	Bzn., chloroform, dioxane, DMF, ethyl acetate, THF		83

TABLE 1. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(diamyl fumarate)	Chloroform, THF, toluene	Hexane, W	84
Poly(methyl *tert*-butyl fumarate)	Bzn., chloroform, methanol, THF	Hexane, W	85
1.3.5. POLY(ACRYLAMIDES), POLY(METHACRYLAMIDES)			
Polyacrylamide	Ethylene glycol (partially), morpholine, W	Alcohols, diethyl ether, DMF, esters, higher glycols, hydrocarbons, nitrobenzene, propylene glycol, THF	4,5,13
Poly(*N*-isopropylacrylamide)	W (cold)	W (hot)	86,87
Poly(*N*,*N*-dimethylacrylamide)	Methanol, W (40°C)	W (sw)	5
Poly(*N*,*N*-diethylacrylamide)	THF		88
Poly[*N*-(1,1-dimethyl-3-oxobutyl) acrylamide]	Butanol, MEK, toluene	W (sw)	89
Poly(2-acrylamido-2-methyl-1-propanesulfonic acid), PolyAMPS	W		32
Poly(*N*-*tert*-butyl-*N*-allylacrylamide)	Chloroform	Petroleum ether	90
Poly(*N*-dodecylacrylamide-*co*-*N*-vinylcarbazole) (15 : 1 to 2 : 1)	Chloroform	Acrylonitrile	91
Polymethacrylamide	Acetone, ethylene glycol, methanol, W	Diethyl ether, hydrocarbons	4,5,13
Poly(*N*-carbazolylcarbonylethylene), Poly(9-acryloylcarbazole)			
atactic	Bzn. (partially)	Methanol	92
tactic	Chloroform (partially), conc. or moderately conc. sulfuric acid		5
Poly(morpholinocarbonylethylene), Poly(*N*-acryloylmorpholine)	DMF		5
Poly(piperidinocarbonylethylene), Poly(acryloylpiperidide)	DMF		5
1.4. POLY(VINYL ETHERS)			
General	Bzn., *n*-butanol, cyclohexanone, halogenated hydrocarbons, MEK	Heptane	
Unsubstituted			
Poly(methoxyethylene), Poly(methyl vinyl ether)			
amorphous	See General, acetone, ethanol, ethyl acetate, methylene chloride, THF, W (cold)	Diethyl ether, ethylene glycol, hexane, W (hot)	4,5,13,30,93–95
crystalline		Acetone, methanol, W	95,96
Poly(ethoxyethylene), Poly(ethyl vinyl ether),	See Poly(methoxyethylene)		
Poly(propoxyethylene), crystalline		Acetone, heptane	95
Poly(propyl vinyl ether)			
Poly(isopropoxyethylene), crystalline, Poly(isopropyl vinyl ether)	See Poly(propoxyethylene)		
Poly(butoxyethylene), Poly(butyl vinyl ether)	See General, bis(2-ethoxyethyl) ether, cyclohexane, diethyl ether, *n*-heptane	Ethanol, 2-ethoxyethanol	95,97
Poly(isobutoxyethylene), Poly(isobutyl vinyl ether)			
amorphous	See General, aliphatic alcohols C > 2, bis(2-ethoxyethyl) ether, carbon disulfide, cyclohexane, diethyl ether, *n*-heptane, isopropyl acetate, MEK, methylene chloride	Ethanol 2-ethoxyethanol, methanol	4,13,30, 93,98,99
crystalline	Chloroform above 50°C	Heptane, bis(2-ethoxyethyl) ether, bzn. (sw at 20°C), chloroform (sw at 20°C), isopropanol (hot), MEK	95
Poly(*tert*-butoxyethylene), Poly(*tert*-butyl vinyl ether)			
amorphous	Acetone, MEK		95,98
crystalline		Bzn., heptane	95,98
Poly(neopentyloxyethylene), Poly(neopentyl vinyl ether), crystalline		Bzn., heptane	95
Poly(cyclohexyloxyethylene), Poly(cyclohexyl vinyl ether)	Methylene chloride		100
Poly(benzyloxyethylene), Poly(benzyl vinyl ether) atactic	Acetone, bzn., toluene	Ethanol, methanol, W	101

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TABLE 1. *cont'd*

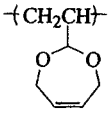
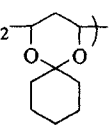
Polymer	Solvents	Nonsolvents	Refs.
Substituted			
Poly(carbomethoxymethoxyethylene)	Chloroform, methylene chloride	Diethyl ether	102
Poly[2-(methoxyethoxy)ethylene], crystalline	W	Diethyl ether	95
Poly[2-(chloroethoxy)ethylene], crystalline		Acetone	95
Poly(2,2,2-trifluoroethoxyethylene), crystalline	Bzn., dioxane, heptane		95
Poly(methyl vinyl ether-*alt*-methyl tricyanoethylenecarboxylate)	DMA, DMF, nitromethane	Diethyl ether	103
Poly(phenyl vinyl ether-*alt*-2-phenyl-1,1-dicyanoethylene)	Acetone, carbon tetrachloride, chloroform, DMF, DMSO, THF	Hexane methanol	104
1.5. POLY(VINYL ALCOHOLS), POLY(ACETALS), POLY(VINYL KETONES)			
Poly(vinyl alcohol)			
atactic	Acetamide, DMF, DMSO (hot), formamide, glycerol (hot), glycols (hot), HMTP, piperazine, triethylenediamine, W	Carboxylic acids, chlorinated hydrocarbons, esters, hydrocarbons, ketones, lower alcohols, conc. aqu. salt solutions, THF	4–6,13, 30,105
syndiotactic	1,3-Propandiol above 160°C, W above 160°C		106
12% acetyl	W (cold)	Carboxylic acids, esters, halogenated hydrocarbons, hydrocarbons, ketones, W (hot)	4
35% acetyl	Alcohols, aqu. sol. of tetraalkyl-ammonium bromide and iodide	W	38
Poly(vinyl alcohol-*co*-ethylene) (1 : 1)	DMF (hot), DMSO (hot)	Methylene chloride, THF	107
Poly(vinyl alcohol-*co*-maleic anhydride) (1 : 1)	Acetone		108
Poly(allyl alcohol)			
lower mol. wt.	Cresol, dioxane, glycerol, methanol, pyridine, THF		13
high mol. wt. (DP > 350)	Mixtures of conc. hydrochloric acid and dioxane, methanol, THF	Most organic solvents, W	13
Poly(vinyl formal)	Acetic acid, benzyl alcohol, bzn./alcohol (70/30), carbon tetrachloride/alcohol (70/30), 2-chloroethanol, chloroform, cyclohexanone, dichloroethylene/diacetone alcohol (50/50), dioxane, DMF, furfurol, THF, toluene	Acetone (sw), aliphatic hydrocarbons, aromatic hydrocarbons (sw), dilute acids, dioxane, esters, ethanol, methanol, pyridine (sw), W	4,6,30
Poly(vinyl acetal)	Acetone, benzyl acetate, bzn./ethanol (1/1), bzn., butanol, carbon tetrachloride, 2-chloroethanol, chloroform, chloroform/methanol (9/1), cyclohexanone, dioxane, ethanol, ethyl acetate, ethylene glycol, THF	Dil. acids, aliphatic hydrocarbons, diethyl ether (sw), pyridine (sw), W	6,30
high degree of acetalization	Acetic acid (glacial), dioxane, ethanol, ethylene dichloride, methanol, nitromethane, pyridine	Acetone, bzn.	4
Poly(vinyl butyral)			
acetalization 70%	Alcohols, cyclohexanone, ethyl glycol acetate, ethyl lactate	Aliphatic ketones, hydrocarbons, methylene chloride, most esters, W	109
acetalization 77%	Acetone, alcohols, cyclohexanone, lower esters, methylene chloride	Higher esters, hydrocarbons, methyl isobutyl ketone	109
acetalization 83%	Alcohols, lower esters, ketones, methylene chloride	Higher esters, hydrocarbons, methanol	109
Poly(vinyl alcohol-*co*-vinyl pentanal)	Bzn., chloroform	Methanol	110
Poly(2-vinyl-4,7-dihydro-1,3-dioxepine), 	Acetone, 2-buten-1,2-diol	W	111
Poly(vinyl cyclohexanone ketal), 	Butanol, chloroform, cyclohexanone, ethanol, ethylene glycol, ethylene chloride, THF	Aliphatic hydrocarbons, carbon tetrachloride (sw), diethyl ether (sw)	30

TABLE 1. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(vinyl methyl ketone)	Acetic acid, acetone, chloroform, dioxane, DMF, ethyl acetate, pyridine, THF	Alcohol, carbon tetrachloride, diethyl ether, petroleum ether, W	4,6,13
Poly(methyl isopropenyl ketone)	Acetone, dioxane, esters	Alcohol, petroleum ether, W	6,13
1.6. POLY(VINYL HALIDES)			
Poly(vinyl chloride)			
high mol. wt.	Acetone/carbon disulfide, chlorobenzene, cyclohexanone, cyclopentanone, DMF, DMSO, MEK, nitrobenzene, THF	Acetic anhydride (sw), acetone (sw), nonoxidizing acids, alcohols, conc. alkalies, aniline (sw), carbon disulfide, carboxylic acids, esters, hydrocarbons, nitroparaffins (sw), vinyl chloride	4–7,13, 30,112
lower mol. wt.	Acetone/carbon disulfide, cyclohexanone, 1,2-dichlorobenzene, diisopropyl ketone, dioxane, DMF, ethylene chloride, HMPT, isophorone, mesityl oxide, nitrobenzene, perchloroethylene/acetone, toluene, tricresyl phosphate		4–6,13,30
chlorinated, 63% Cl	Acetone, aromatic hydrocarbons, butyl acetate, chlorobenzene, chloroform, cyclohexanone, dioxane, DMF, DMSO, nitrobenzene, THF	Aliphatic and cycloaliphatic hydrocarbons, carbon tetrachloride, methyl acetate, nitromethane, organic and inorganic acids	6
Poly(vinylidene chloride)	Benzonitrile, butyl acetate, cyclohexanone, 1,2-dichlorobenzene, dioxane, DMA, DMF, NMP, tetrahydrofurfuryl alcohol, tetralin (hot), THF (hot), trichloroethane	Conc. and moderately conc. acids and alkalies (except ammonia), alcohols, carbon disulfide, chloroform, cyclohexanone (cold, sw), dioxane (cold, sw), ethyl bromide, hydrocarbons, phenols, THF (cold, sw)	4,6,30, 113,114
Poly(vinyl bromide)	Cyclohexanone, THF	DMF, ethanol, HMPT, hydrocarbons, methanol	5
Poly(vinyl fluoride)	Cyclohexanone (hot), dinitrile, DMA (hot), DMF, DMSO (hot)	Aliphatic, cycloaliphatic, and aromatic hydrocarbons	115
chlorinated, 30% Cl	DMF	Aliphatic, cycloaliphatic, and aromatic hydrocarbons	115
chlorinated, 60% Cl	Carbon tetrachloride	Aliphatic, cycloaliphatic, and aromatic hydrocarbons	115
Poly(vinylidene fluoride)	γ-Butyrolactone, cyclohexanone, DMA, DMF, DMSO, ethylene carbonate, NMP	Acetone, alcohols, aliphatic and cycloaliphatic hydrocarbons, chlorinated solvents, methyl isobutyl ketone	116–119
Poly(1,2-difluoroethylene)	Acetone, DMF, methyl isobutyl ketone	Ethanol, hydrocarbons, methanol	118
Poly(1-chloro-1-fluoroethylene)	Toluene		120
Poly(chlorotrifluoroethylene)	2,5-Dinitrotrifluorobenzene (130°C), bzn. (200°C), carbon tetrachloride (114°C), cyclohexane (235°C), 1,2-dichlorotrifluorobenzene, HMPT, mesitylene (140°C), toluene (142°C), 1,1,1-trichloroethane (120°C), 1,2,3-trifluoropentachloropentane	Common organic solvents at RT	4,5,13, 121,674
Poly(tetrafluoroethylene)	Perfluorokerosene (350°C), no other solvent known		4,13,122,123
Poly(3,3,3-trifluoropropylene)	Acetone, hexafluorobenzene	Ethanol, hydrocarbons, methanol	124
Poly(hexafluoropropylene)	Hexafluorobenzene, perfluorodibutylamine, perfluorodibutyl ether		125
Poly(3,3,4,4,5,5,5-heptafluoro-1-pentene)	Perfluorohexane		126
Poly(vinylidene chloride cyanide-*alt*-1,3-cyclohexadiene)	Chloroform	Pentane	128
Poly(chlorotrifluoroethylene-*co*-vinylidene fluoride) (3 : 1)	MEK	Amyl acetate	129
Poly(tetrafluoroethylene-*co*-hexafluoropropylene) (4 : 1)	Carbon dioxide (>200°C)		130
Poly[tetrafluoroethylene-*co*-4,5-difluoro-2,2-bis(trifluoromethyl)-1,3-dioxole] (13 : 87 and 35 : 65)	Hexafluorobenzene, perfluorooctane		131

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TABLE 1. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
1.7. POLY(VINYL NITRILES)			
Polyacrylonitrile	Acetic anhydride, γ-butyrolactone, ε-caprolactam, bis(2-cyanoethyl)ether, bis(4-cyanobutyl)sulfone, chloroacetonitrile/water, chloroacetonitrile, cyanoacetic acid, dimethyl phosphite, dimethyl sulfone, dioxanone, DMA, DMF, DMSO, ethylene carbonate, *N*-formylhexamethyleneimine, 2-hydroxyethyl methyl sulfone, *N*-methyl-β-cyanoethylformamide, methylene dithiocyanate, *N*-methyl-α,α,α,-trifluoroacetamide, 1-methyl-2-pyridone, 3-,4-nitrophenol, nitromethane/W (94 : 6), *N*-nitrosopiperidine, 2-oxazolidone, 1,3,3,5-tetracyanopentane, tetramethylene sulfoxide, 1,1,1-trichloro-3-nitro-2-propane, sulfuric acid, nitric acid, *p*-phenol sulfonic acid, conc. aqu. sodium thiocyanate, molten quat. ammonium salts and their aqu. solutions	Acetonitrile (sw), acrylonitrile, alcohols, aliphatic nitro compounds, bis(2-hydroxyethyl) sulfoxide, chlorinated hydrocarbons, 3,4-dimethyl sulfolane, diethyl ether, diethyl sulfone (sw), diethyl sulfoxide, diethylformamide, dimethylmalonitrile, dimethyloxamide, 1,1-dimethylsuccinonitrile, ethylene urea, formamide, hexamethylene dithiocyanate, 1,6-hexanediamine, hydrocarbons, 1-hydroxypropionitrile, ketones, methanol, methoxyacetamide, methoxyacetonitrile, methyl thiocyanate, 1-nitrophenol, propyl formate	5,6,13,132, 133,675
ultrahigh MW	DMF (100°C)		134
Poly(2-fluoroacrylonitrile)	DMF	THF	135
Poly(acrylonitrile-*co*-vinyl acetate) (98 : 2)	DMF		136
Poly(acrylonitrile-*alt*-1,3-isoprene)	1,2-Dichloroethane	Methanol	137
Poly(vinylidine fluoride-*co*-ethyl vinyl ether) (1 : 1)	THF		135
Poly(acrylonitrile-*co*-butadiene) (2 : 3 to 1 : 4)	MEK, THF (partially), toluene		32
Polymethacrylonitrile	Acetic anhydride, acetone, benzonitrile, cyanoacetic acid, cyclohexanone, dinitriles, DMF (>20°C), DMSO, ethylene carbonate, furfural, HMPT, methylene chloride, nitromethane, pyridine	Aliphatic hydrocarbons, alcohols, esters, methacrylonitrile, toluene	4,5
1.8. POLY(VINYL ESTERS)			
Poly(vinyl acetate)			
atactic	Acetic acid, acetone, acetonitrile, allyl alcohol, benzyl alcohol, bzn., carbon tetrachloride/ethanol, chlorobenzene, chloroform, dichloroethylene/ethanol (20/80), 2,4-dimethyl-3-pentanol, dioxane, DMF, DMSO, lower aliphatic esters, ethanol/W, glycol ether esters, glycol ethers, methanol, nitromethane, tetrahydrofurfuryl alcohol, THF, toluene	Dil. acids, dil. alkalies, carbon tetrachloride (sw), carbon disulfide, cyclohexanol, diethyl ether (anhydrous, alcohol free), higher esters C>5, ethanol (anhydrous, sw), ethylene glycol, saturated hydrocarbons, mesitylene, W (sw)	4,5,13
syndiotactic	Chlorobenzene, chloroform	Acetone, bzn.	106
Poly(vinyl trifluoroacetate)	Acetone, chloroform, ethyl acetate, hexamethyldisiloxane		138
Poly(vinyl propionate)	See Poly(vinyl acetate)		
Poly(vinyl *n*-butyrate)	See Poly(vinyl acetate), amyl alcohol, cyclohexane, hexyl acetate		31
Poly(vinyl isobutyrate)	See Poly(vinyl acetate), amyl alcohol, cyclohexane, hexyl acetate		31
Poly(vinyl pivalate)	Acetone, butanone, bzn., ethyl acetate, toluene	Hexane, methanol, W	139
Poly(vinyl caproate)	Bzn.		5

TABLE 1. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(vinyl caprylate)	Acetone, aliphatic and aromatic hydrocarbons	Lower alcohols	4
Poly(vinyl laurate)	Aliphatic and aromatic hydrocarbons	Lower alcohols, acetone	4
Poly(vinyl benzoate),	See Poly(vinyl acetate)		5,87
Poly(vinyl chloroacetate)	Chlorobenzene, chloroform, cyclohexanone, dioxane, ethyl acetate, pyridine	Acetone (sw), saturated hydrocarbons	4
Poly(vinyl acetylacetate)	Acetic acid, acetone, chloroform, dioxane, DMF, ethyl acetate, pyridine, THF, saturated aqu. sol. of magnesium perchlorate	Alcohols, bzn., diethyl ether, W	140
Poly(vinyl-*N*-octadecylcarbamate)	Toluene (hot)		327
Poly(vinyl acetate-*alt*-*N*-alkyl/arylmaleimide)	Bzn., chloroform	Petroleum ether	142
1.9. POLY(STYRENES)			
Poly(styrene)			
atactic	Bzn., carbon disulfide, lower chlorinated aliphatic hydrocarbons, chloroform, cyclohexane (above 35°C), cyclohexanone, cyclohexane/acetone, decahydronaphthalene/diethyl oxalate, dimethyl phthalate, dioxane, ethyl acetate, ethylbenzene, glycol formal, MEK, methylene chloride, NMP, 1-nitropropane, phenol/acetone, phosphorus trichloride, THF, tributyl phosphate	Acetic acid, acetone, alcohols, diethyl ether, diols, ethylene chlorohydrin, glycol ethers, saturated hydrocarbons, isobutyl phthalate, phenol, 1,2,3, 4-tetrafluorobenzene (lower than 10°C), tri(chloroethyl) phosphate, tricresyl phosphate	4,5,7, 13,30,143
isotactic	Bzn., chloroform, *o*-dichlorobenzene, toluene		7
syndiotactic	*n*-Tetradecane/decahydronaphthalene (2/1)		144
Poly(4-acetoxystyrene)	THF	Methanol	144
Poly(α-methylstyrene),	See Polystyrene		
Poly(2-,3-, or 4-chlorostyrene)	MEK, THF, toluene		32
Poly(chloromethylstyrene)	Bzn., THF, toluene		32
Poly(4-cyanostyrene)	Acetone, acetonitrile, DMF, DMSO, nitromethane	Bzn., carbon tetrachloride, ethyl acetate, diethyl ether, hexane, methanol	146
Poly(2-,3-, or 4-bromostyrene)	Bzn., THF		5,32
Poly(4-iodostyrene)	Bzn., dioxane, THF		5,32
Poly(2,5-dichlorostyrene)	Ethyl acetate/ethanol (15/1) toluene		5
Poly(2,4,6-tribromostyrene)	THF		32
Poly(4-hydroxystyrene), Poly(*p*-hydroxyphenylethylene)	Methanol and higher alcohols, methyl acetate, THF, THF/chloroform (1/1)	Chloroform, toluene	147,148
Poly(4-methoxystyrene)	MEK, toluene		5
Poly(2,5-dimethoxystyrene)	Bzn., toluene, chloroform	Hexane, methanol	149
Poly(3,4-dimethoxystyrene)	Chloroform		150
Poly(*tert*-butyl-4-vinyl perbenzoate)	Acetone, bzn., chlorohexane, DMF, methylene chloride	Hexane	141
Poly(4-formylphenyl)ethylene, Poly(*p*-formylstyrene)	THF	Hexane, methanol, W	151
Poly(styrenesulfonic acid)	Ethanol, lower glycols, W		32
Poly(styrene-*co*-styrenesulfonic acid) (95 : 5) sodium salt	DMF, methylene chloride, THF		152
	Lower glycols, W		153
Poly(4-trimethylsilylethynylenestyrene)	THF		154
Poly(styrene-*co*-allyl alcohol) (94 : 6)	Alcohols, aromatic hydrocarbons, esters, THF		32
Poly(styrene-*co*-acrylonitrile) (3 : 2)	Chloroform, MEK, methylene chloride, THF	Methanol	7,32,155
Poly(styrene-*co*-*tert*-butoxycarbonyloxystyrene) (4 : 5)	THF, pyridine	Methanol, W	156
Poly(styrene-*co*-butyl methacrylate) (1 : 1)	MEK, toluene		32
Poly(styrene-*alt*-2,4-diamino-6-vinyl-1,3,5-triazine)	DMSO, hexafluoroisopropanol		157

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TABLE 1. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(styrene-*co*-maleic anhydride)			
(9 : 1)	Cyclohexane/ethyl acetate (3/2)	Methanol	65
(1 : 1)	Acetone, acetonitrile, MEK		32,158
Poly(styrene-*alt*-maleimide)	THF		157
Poly(styrene-*co*-4-thiomethylstyrene) (95 : 5)	THF	Ethanol, hexane	159
Poly(styrene-*co*-4-vinylpyridine) (1 : 1)	DMF, THF		32
1.10. OTHER COMPOUNDS (alphabetically ordered)			
Polyacrolein, (redox polymerization)	Above 60°C: pyridine/water (55/45 to 90/10), above 130°C: nitrobenzene, sat. stannous chloride solution, γ-butyrolactone (160–170°C), ethylene carbonate (130–135°C), DMF (153°C), DMSO (160–170°C), divinyl sulfone (150–155°C), TMS (160–165°C)	Lower alcohols, hydrocarbons, chlorinated hydrocarbons, diethyl ether, esters, aromatic ketones	160,161
Polyacrolein (ionic polymerization), See 3.1.1			
Poly(anthrylethylene)	Methylene chloride	Methanol	15
Poly(*N*-benztriazolylethylene)	Chlorinated hydrocarbons, cyclohexanol, DMF, glacial acetic acid	Alcohols, esters, hydrocarbons, ketones, W	162
Poly(biphenylyethylene)	Bzn., dimethoxyethane, toluene	Methanol	163
Poly(*N*-carbazolyethylene), Poly(*N*-vinylcarbazole)	Benzyl acetate, bzn., chlorobenzene, chloroform, cyclohexanone, dioxane, conc. nitric acid, conc. sulfuric acid, tetrachloroethane	Acetals, alcohols, dil. alkalies, dil. carboxylic acids, chlorohydrin, 1-chlorotoluene (sw), diethyl ether, dimethyltetrahydrofuran, diols, aliphatic esters, glycol monoethers, aliphatic ketones, saturated hydrocarbons, trichloroethylene (sw), W	4,6,13, 30,164
Poly[6-(*N*-carbazolyl)hexylethylene]	Acetone, methanol (partially), NMP (123°C)	Methanol	165,166
Poly(diallyldimethylsilane)	Bzn.		167
Poly(3,6-dichloro-*N*-carbazolylethylene)	Dichloroethane	Methanol	168
Poly(2,4-dimethyl-6-triazinylethylene)	Alcohols, bzn., esters, ketones, W (cold)	Aliphatic hydrocarbons, W (ppt. below 73°C)	169
Poly(diphenylphosphinylideneethylene), Poly(vinyldiphenylphosphine oxide)	Bzn., ethanol, methanol, toluene	Hexane, W	170,171
Poly(diphenylthiophosphinylideneethylene), Poly (vinyldiphenylphosphine sulfide)	Bzn., chloroform	Methanol	171
Poly(2-methyl-5-pyridylethylene)	Methanol, DMF		5
Poly(3-morpholinylethylene)	W		172
Poly(1-nitropropylene)	DMF (partially)		173
Poly(2-pyridylethylene), Poly(2-vinylpyridine)	Acetone, alcohols/water, bzn., chloroform, dioxane, ethanol, glacial acetic acid, aqu. mineral acids, nitroethane, pyridine, THF	Carbon tetrachloride toluene, W (sw)	4,13,35, 174,175
crystalline	Aromatic hydrocarbons (reflux), chlorinated solvents, methanol	Aliphatic hydrocarbons (reflux), diethyl ether, MEK	
Poly(4-pyridylethylene), Poly(4-vinylpyridine)	Acetone/W (1/1), benzyl alcohol, bzn., cyclohexanol, dioxane/W (1/1), ethanol/water (92/8), isopropanol/MEK (86/14), methanol, aqu. mineral acids, nitromethane, pyridine, THF	Acetone, diethyl ether, dioxane, ethyl acetate, MEK, petroleum ether, W	5,13,175
Poly(*N*-pyrrolidonylethylene), Poly(*N*-vinylpyrrolidone)	(solubility depending on small amounts of W): acetone, dil. acids, aromatic alcohols, chloroacetic acid esters, chloroform, chlorohydrins, ethanol, glacial acetic acid, methanol, nitromethane, pyridine	Acetic esters, acetone, acid esters, carbon tetrachloride, 1-chlorotoluene, diethyl ether, hydrocarbons, methoxybutyrate esters, nitomethane/W	4–6,13,30

TABLE 1. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(*N*-1,2,4-triazolylethylene)	Glacial acetic acid, DMF, DMSO, W	Alcohols, chlorinated hydrocarbons, esters, hydrocarbons ketone	162
Poly(*N*-vinylthiopyrrolidone)	DMF	Butanol, hydrocarbons, MEK, W	176
Poly(5-vinyl-1,3-benzodioxole)	Chloroform		150
Poly(2-vinylthiophene)	Chloroform	Methanol	177
Poly(vinyl sulfate)	Aqu. NaCl (0.5 M), W		5
Poly(vinylsulfofluoride)	Acetone, DMF, ethyl acetate, THF	Diethyl ether	13
Poly(vinylsulfonic acid)	Acetone, dioxane, ethanol, W	Esters, hydrocarbons, ketones	4,13,179
sodium salt	Methanol, W, aqu. NaCl (0.5 M)	Acetone, methanol	13,179
Poly(vinyl sulfoxide)	Chloroform, DMSO, methanol, W	Acetone, bzn.	180,181
Poly(vinyl 4-methoxyphenyl sulfoxide)	THF	Diethyl ether	182
Poly(*N*-vinylpyrrolidone-*co*-acrylic acid) (3 : 1 to 1 : 3)	Acetone/W (1/1), isopropanol/W (1/1), NMP	Methanol, MEK, THF	183
Poly(vinyl amine), Poly(aminoethylene)	Butanol, DMSO, ethylene glycol, formamide, methanol, W		179,184,185
Poly(vinylformamide)	DMSO, ethylene glycol (partially), formamide, W	Acetone, isopropanol	179,185
Poly(*N*-vinylpyrrolidone-*co*-1-hexadecene) (1 : 2)	Saturated hydrocarbons	*n*-Propanol, W	186
Poly(vinylnaphthalene)	Bzn., dioxane, THF		32
Poly(*N*-vinylpyrrolidone-*co*-vinyl acetate) (1 : 4 to 3 : 2)	Ethanol, isopropanol		186
Poly(*N*-vinylpyrrolidone-*co*-vinyl alcohol) (13 : 1)	Ethanol	Diethyl ether	187
Poly(*N*-vinylpyrrolidone-*co*-vinyl acetate) (1 : 1)	Isopropanol, W		32,681
Poly(*N*-vinylformamide-*co*-methyl 3-phenyl-2-cyanopropenoate) (1 : 1)	DMF, DMSO (partially), NMP	Chloroform, diethyl ether, ethyl acetate, methylene chloride	697
Poly(2-vinylpyridine-*co*-styrene) (7 : 3)	Bzn., chloroform, MEK, THF		32
Poly(4-vinylpyridine-*co*-styrene) (3 : 2)	DMF, THF		32

TABLE 2. MAIN-CHAIN CARBOCYCLIC POLYMERS

Polymer	Solvents	Nonsolvents	Refs.
2.1. POLY(PHENYLENES)			
Poly(2-hydroxy-1,4-phenylene)	Methylene chloride		188
Poly(dihydroxyl-1,4-phenylene), Polyhydroquinone	DMF (partially)	Acetone, chloroform, ethanol	181
Poly(2,5-di-*n*-butoxy-1,4-phenylene)	1,2-Dichloroethane, methylene chloride, nitrobenzene	Methanol	189
Poly(2,5-di-*n*-decyl-1,4-phenylene)	Toluene	Acetone	190
Poly(*N*,*N*-dimethylaniline)	Acetic acid, chloroform, DMF, ethanol	Aqu. sodium hydroxide	191
Poly(2,1′ : 4′,1″ : 4″,1‴-quaterphenylene-1,4), Poly [2-(phenyl-1,4-phenylene-1,4-phenylene)-1,4-phenylene]	Toluene	Methanol	192
Poly(2-benzoyl-1,4-phenylene), Poly(2,5-benzophenone)	*m*-Cresol, DMA, methylene chloride, NMP	Methanol	193,194
Poly(nitrophenylene)		All common organic compounds	195
Poly[2-(4-methylbenzoyl)-1,4-phenylene]	DMA		196
Poly(2,5-dihydroxy-1,4-phenylenethenylene)		Common organic solvents, aqu. potassium hydroxide	197
Poly(1,4-benzophenone-*N*-phenyloxime)	DMA, NMP		194
Poly(2,5-dimethoxy-1,4-phenylenethenylene)	Bromoform	Common organic solvents	197,198
Poly(2,5-dimethyl-1,4-phenylenethenylene) low mol. wt.	Chloroform, 1,2- and 1,4-dichlorobenzene		199
Poly(2,5-dimethyl-1,4-phenylenemethylene) low mol. wt.	Chloroform, 1,2- and 1,4-dichlorobenzene (hot)		199
Poly(1,4-phenyleneethenylene), Poly(phenylenevinylene)	Biphenyl, benzyl benzoate, chlorinated biphenyls, diphenyl ether, nitrobenzene		197,200
Poly(1,4-phenylene-1,2-diphenylethenylene)	Chloroform	Methanol	201
Poly[1,4-phenyleneethenylene-1,4-phenyleneethynylene-2,5-(dodecyloxy)-1,4-phenyleneethynylene]	Chloroform, THF, toluene		202
Poly[3-nitro-1,4-phenyleneethenylene-2-(dihexylamino)-1,4-phenyleneethenylene]	Chloroform, 1,2-dichloroethane, THF		203

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TABLE 2. *cont'd*

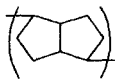
Polymer	Solvents	Nonsolvents	Refs.
Poly(tetramethyl-1,4-phenylenethenylene)		Bzn., 1-chloronaphthalene (sw), ethanol	197
Poly(5-*n*-dodecyloxy-1,4-phenylene)	DMA, DMF		204
Poly(2,5-di-*n*-decyloxy-1,4-phenylene)	THF	Methanol	205
Poly(2,5-di-*n*-decyloxy-4,4′-biphenylylene)	THF	Methanol	205
Poly[2-bis(methoxycarbonyl)-1,4-phenylenethenylene]	Methylene chloride	Methanol	206
Poly[1,4-phenylene-1,2-di(4-phenoxyphenyl)ethenylene]	Chlorobenzene, chloroform, toluene		207
Poly[2-(2-phenylethenyl)-1,4-phenyleneethenylene]	DMF		208
Poly(6-*n*-hexyl-1,4-naphthalenediylethenylene)	Chloroform	Methanol	209
Poly(2,6-naphthalenediylethenylene)	Tetrachloroethylene (partially)		210
2.2. OTHER COMPOUNDS (alphabetically ordered)			
Poly(acenaphthylene)	Bzn., carbon tetrachloride, chloroform, 1,2-dichloroethane (>30°C), THF, toluene	Acetone, acids, alcohols, dimethoxyethane, ethers	4,32,211
Poly[*N*-(3-bromophenyl)norbornenedicarboximide]	1,2-Dichloroethane, methylene chloride	Methanol	212
Poly(1,3-cyclohexadiene)	Bzn., xylene	Methanol	213
Poly(1,3-cyclohexyleneethenylene)	Bzn.	Methanol	214
Poly(1,5-cyclooctadiene)	Aromatic hydrocarbons, chlorinated hydrocarbons, diethyl ether (partially) methanol (oligomers)	Methanol	215
Poly(cyclopentadiene), 	Bzn., chloroform, dioxane, THF	Ether, hexane, methanol, petrol	216
Poly(1,3-cyclophenylenemethylene)	Toluene (hot)		217,218
Poly(*endo*-dicyclopentadiene)	Bzn., carbon disulfide, chlorobenzene	Acetone, hexane, methanol	219
Poly(2,3-dimethoxycarbonyl-2,5-cyclohexadien-1,4-diylethenylene), Poly(dimethyl barrelene-2,3-carboxylate)	Bzn.	Methanol	206
Poly(1,2-dihydronaphthalene)	Bis(phenoxyphenyl) ether (above 290°C)	Amyl acetate, chloronaphthalene, diphenyl ether, tetrachloroethane, tetrahydronaphthalene	220
Poly(dimethylfulvene)	Acetone, bzn., chlorinated hydrocarbons, diethyl ether, hexane	Alcohol	221
Poly(indene)	Aromatic chlorinated hydrocarbons, diethyl ether, dioxane, drying oils, ketones, pyridine	Acids, aliphatic hydrocarbons (sw), alkalies, lower alcohols, W	6
Poly(1-isopropylidene-3a,4,7,7a-tetrahydroindene)	Bzn., carbon disulfide, chloroform, THF	Acetone, cyclohexane, ethanol, hexane	222
Poly[1,4-phenylene-1,2-dicyano-1,2-bis(propoxycarbonyl)ethenylene]	Acetone, acetonitrile, bzn., chloroform, DMF, DMSO, THF	Diethyl ether, hexane, methanol	164

TABLE 3. MAIN-CHAIN CARBONYL POLYMERS

Polymer	Solvents	Nonsolvents	Refs.
Poly(carbonylethylene)	*m*-Cresol		328
Poly(carbonyl-1,2-propylene)	Hexafluoroisopropanol	Methanol	329
Poly(carbonylethylene-*co*-carbonyl-1,2-propylene) (7 : 3)	Cresol, hexafluoropropylene, methylene chloride		330,331
Poly(carbonyl-1-furfuryltrimethylene)	Acetone, dioxane	Chloroform, ethanol, W	332
Poly(carbonylphenylethylene), Poly(carbon monoxide-*co*-styrene)	Chlorobenzene, hexafluoroisopropanol, MEK (hot), methylene chloride	Methanol	333–335
Poly(carbonyl-1-phenyltrimethylene)	Chloroform		336
Poly(carbonyl-1,4-cyclohexan-3-onediyl)	Chloroform	Methanol	337
Poly(carbonyl-1,4-cyclohexan-3-onediylmethylene)	Chloroform		338
Poly[carbonyl-2,3-di(ethoxycarbonyl)bicyclo[2.2.2]hepta-2,5-diene-5,6-diyl]	Acetone, common organic solvents	Alkyl hydrocarbons	339
Poly(carbonyl-4,4′-biphenylylene)		DMA, NMP	194
Poly(carbonyl-1,4-phenyleneoxy-1,4-phenylene)	Chloroform, DMA, 1,1,2,2-tetrachloroethane	MEK, methanol, toluene	340

TABLE 3. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(carbonyl-1,3-phenyleneoxy-1,4-phenylene)	*o*-Chlorophenol		341
Poly[carbonyl-1,4-phenyleneoxy-2-(methoxymethylene)-1,4-phenyleneoxy-1,4-phenylene]	Chloroform, THF		342
Poly(carbonyl-1,3-phenylenecarbonyl-1,4-phenyleneoxy-1,4-phenylene), PEKK	Pentafluorophenol (80°)		343,344
Poly(carbonyl-1,4-phenylenecarbonyl-1,4-phenyleneoxy-1,4-phenylene), PEKK	Pentafluorophenol		344
Poly(carbonyl-1,3-phenylenecarbonyl-1,4-phenyleneoxy-1,3-phenylenemethylene-1,3-phenyleneoxy-1,4-phenylene)	DMA, DMF, NMP		345
Poly(carbonyl-1,4-phenylenecarbonyl-1,4-phenyleneoxy-1,4-phenylene-2-pentylidene-1,4-phenyleneoxy-1,4-phenylene)	DMA	Methanol	346
Poly(carbonyl-1,4-phenyleneoxy-1,4-phenyleneoxy-1,3-pyridinediyloxy-1,4-phenyleneoxy-1,4-phenylene)	DMA		347
Poly(carbonyl-1,4-phenyleneoxy-4,4′-biphenylyleneoxy-1,4-phenylene-*co*-carbonyl-1,4-phenyleneoxy-1,4-phenyleneoxy-1,4-phenylene) (1 : 1 to 1 : 3)	Sulfuric acid, 0.1 wt.%		348
Poly(carbonyl-1,3-phenylenecarbonyl-1,4-phenyleneoxy-1,3-phenylenemethylene-1,3-phenyleneoxy-1,4-phenylene)	DMA, DMSO, NMP		345

TABLE 4. MAIN-CHAIN ACYCLIC HETEROATOM POLYMERS

Polymer	Solvents	Nonsolvents	Refs.
4.1. MAIN-CHAIN -C-O-C- POLYMERS			
4.1.1. POLY(OXIDES)			
Unsubstituted			
Poly(oxymethylene), Polyformaldehyde	At elevated temp.: aniline, benzyl alcohol, benzyl benzoate, bromobenzene, γ-butyrolactone, chlorophenols, diphenyl ether, DMF, ethylene carbonate, formamide, malodinitrile, phenol	Lower alcohols, diethyl ether, lower esters, hydrocarbons	4,224,225
Poly(oxyethylene), Poly(ethylene glycol), Poly(ethylene oxide)	Acetonitrile, alcohols, bzn., chloroform, cyclohexanone, esters, DMF, W (cold), aqu. K_2SO_4 (0.45 M above 35°C), aqu. Mg_2SO_4 (0.39 × M above 45°C)	Aliphatic hydrocarbons, ethers, dioxane (sw), W (hot)	4,5,13
Poly(oxyethylidene), Polyacetaldehyde			
amorphous	Alcohols, aromatic hydrocarbons, chloroform, esters, ketones	Aliphatic hydrocarbons	226
crystalline	Chloroform (partially)	Alcohols, esters, hydrocarbons, ketones	226
Poly(oxypropylene), Poly(1,2-propylene glycol), Poly(propylene oxide), crystalline	Acetone, bzn., chloroform, dioxane, DMF, ethanol, methanol (hot), THF	2-Aminoethanol, diethyl ether (sw), ethyl acetate (sw)	13
Poly(oxypropylidene), crystalline	Chloroform (partially)	Common organic solvents	227
Poly(oxyisopropylidene), Polyacetone	Acetone, chloroform		228
Poly(1,3-dioxolane)	Methylene chloride, W (70°C)		229
Poly(oxytetramethylene), Poly(tetrahydrofuran)	Bzn., chloroform, ethanol, THF	Hexane, methanol, petroleum ether, W	230–233
Poly(oxycyclopentenylene)	Bzn., carbon disulfide, chloroform	Methanol	677
Poly(oxycyclohexenylene)	Aliphatic hydrocarbons, chinese wood oil, methyl acetate, toluene	Acetone, ethanol, 2-ethoxyethanol	13
Poly(oxymethylenecyclopropylidenemethyleneoxyhexamethylene)	Methylene chloride	Hexanes	234
Poly(oxy-1,3-phenylene)	Benzophenone, bzn., diphenyl ether, DMF, DMSO, nitrobenzene, 3-pentanol, pyridine	Methanol	235

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TABLE 4. *cont'd*

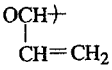
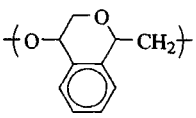
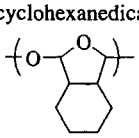
Polymer	Solvents	Nonsolvents	Refs.
Substituted			32,94,236,237
Polyepichlorohydrin, Poly(oxy-3-chloro-1,2-propylene)	Acetone, bzn., chloroform, diethyl ether, MEK, THF, toluene	Methanol	32,94, 236,237
Poly(oxy-2-chloroethylidene)			
amorphous	Chloroform	Methanol	238
crystalline		Chloroform, methanol	238
Poly(oxy-2-methyltetramethylene)	Bzn., Cloroform	Methanol	239
Poly(oxy-2,2,2-trichloroethylidene), crystalline Polychloral		Chloroform, methanol, other common solvents	227
Poly[oxy-2,2-bis(chloromethyl)trimethylene,	Cyclohexanone	Methanol	237,240
Poly(oxy-2,6-dimethyl-1,4-phenylene), Poly(2,6-dimethyl-1,4-phenylene oxide)			241–244
amorphous	α-Pinene (hot)	α-Pinene (cold), methanol, ethanol	241–244
crystalline	Bzn., chlorobenzene, chloroform, toluene	Ethanol, methanol, nitromethane, α-pinene (hot)	241–244
Poly(oxy-2,6-dichloro-1,4-phenylene)	2-Methoxymethanol	Methanol	688
Poly(oxy-1,4-phenylene-1,2-cyclopropylene-1,4-phenyleneoxymethylene)	Chloroform, DMA, NMP	Methanol	245
Poly(oxy-4,4′-biphenylylene-1,2-propylene-1,4-phenyleneoxydodecamethylene)	Chloroform		246
Poly(1-propynylene-1,3-diyloxy-1,4-phenylene-isopropylidene-1,4-phenyleneoxy-2-propynylene 1,3-diyl)	Toluene		247
Poly[oxyethylene-*co*-oxy(chloromethyl)ethylene] (1 : 1)	Chloroform	Methanol	237
Poly[oxy-3,5-bis(4-carboxyphenyl)-1,4-phenylene-*co*-oxy-3,5-bis(phenyl)-1,4-phenylene] (13 : 87 to 62 : 38)	DMF, DMSO	Chloroform, toluene	248
Poly(oxy-1,4-phenyleneisopropylidine-1,4-phenyleneoxy-*alt*-2-hydroxy-1,3-propylene), Poly(Bisphenol A), epoxy resin	Esters, ethers, ketones		249
Polymeric structures, not 100% specific			
Poly(dimethylketene),			
low mol. wt.	Diethyl ether	Methanol	250,251
higher mol. wt.	Bzn., carbon tetrachloride	Chloroform (partially)	250, 251
Polyacrolein, (ionic polymerization) 	Acetone, bzn., dioxane, THF, DMF	Methanol, petroleum ether	252
Polyacrolein, (redox polymerization) see 1.10			
Poly(α-methylacrolein)	Aniline, γ-butyrolactone, DMF, nitrobenzene, pyridine	Alcohols, hydrocarbons	13
Poly[1,2-di(epoxyethyl)benzene], 	Chloroform, THF		253
Poly(phenyl glycidyl ether)	1,2-Dichlorobenzene (hot), DMF (hot), xylene (hot)	Common organic solvents at rt	254
Poly(glutardialdehyde)			
low mol. wt.	Bzn., chloroform, diethyl ether, THF	Petroleum ether, W	133,197, 255,256
high mol. wt.	Methylene chloride, THF, pyridine	Bzn., (partially)	133,197, 255,256
Poly(β-methylglutardialdehyde), low mol. wt.	Chloroform, diethyl ether, dioxane, THF	Petroleum ether, W	255
Poly(β-phenylglutardialdehyde), low mol. wt.	Bzn., chloroform, diethyl ether, THF, dioxane	Petroleum ether, W	255
Poly(*trans*-1,2-cyclohexanedicarboxaldehyde), 	Bzn., chloroform, hexane	Methanol	257
Poly(2-formyl-Δ^5-dihydropyran)	Aromatic hydrocarbons, chloroform, pyridine, THF		258

TABLE 4. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
4.1.2. POLY(CARBONATES)			
Poly(oxycarbonyloxypropylene)	Dioxane, methylene chloride	Methanol, W	678
Poly(oxycarbonyloxyhexamethylene)	Acetone, bzn., chloroform, toluene	Ethanol, ether	13
Poly(oxycarbonyloxy-2,4-dihydroxy-1,4-butylene)	Methylene chloride	Ethanol	260
Poly(oxycarbonyloxy-1,4-butyleneoxycarbonyloxy-1,4-cyclohexylene)	Chloroform, THF		261
Poly(oxycarbonyloxymethylene-1,2-cylcohex-4-enylenemethylene)	Chloroform		262
Poly(oxycarbonyloxy-1,3-phenylene)		Common organic solvents	13
Poly(oxycarbonyloxy-1,4-phenylene)		Common organic solvents	13
Poly(oxycarbonyloxy-1,4-phenylene-isopropylidene-1,4-phenylene), Poly(Bisphenol A carbonate)	Chloroform, *m*-cresol, cyclohexanone, dioxane, DMF, pyridine, THF	Acetone, carbon tetrachloride, lower esters, hydrocarbons, styrene	13,263–265
chlorinated (8% Cl and more)	Bzn., methyl methacrylate, styrene		265
Poly(oxycarbonyloxy-1,4-phenylene-2-pentylidene-1,4-phenylene)	Bzn., chloroform, ethyl acetate		13
Poly(carbon dioxide-*co*-propylene oxide) (1 : 1)	THF	Methanol	266
Poly(propylene carbonate-*co*-propylene oxide) (9 : 1)	Chlorobenzene		267
Poly(oxycarbonyloxy-1,2-cyclohexyleneoxy-1,2-cylclohexylene)	Methylene chloride, THF	Methanol	268
4.1.3. POLY(ESTERS) (alphabetically ordered)			
Poly(ethylene terephthalate), see Poly(oxyterephthaloyloxyethylene)			
Poly(oxyadipoyloxyethylene, Poly(ethylene adipate)	Bzn., THF		32
Poly(oxyadipoyloxy-1,4-phenylene-1,2-propenylene-1,4-phenylene)	Chloroform		269
Poly(oxyadipoyloxytetramethylene), Poly(1,4-butylene adipate)	Chloroform, DMF, THF	Methanol	32,270
Poly(oxyadipoyloxytrimethylene), Poly(1,3-propylene adipate)	Chloroform, MEK, THF		32
Poly(oxyazelaoyloxyethlylene), Poly(ethylene azelate)	Bzn., THF		32
Poly(oxycarbonyl-4,4′-biphenylylenecarbonyloxy-hexamethylene)	Pentafluorophenol/methylene chloride (3/1), THF		271,272
Poly(oxycarbonyl-1,4-cyclohexylene-carbonyloxyoctamethylene), *cis* and *trans*	Chloroform		5
Poly(oxycarbonyldodecamethylenecarbonyloxy-4,4′-biphenylylene)	Phenol/tetrachloroethane (3/2)		273
Poly(oxycarbonylethylene), Poly(β-propiolactone)	Chloroform, formic acid	Hydrocarbons	274,275
Poly(oxycarbonylethyleneoxyethylene)	THF, toluene		278
Poly(oxycarbonylethylidene), Poly(DL-lactic acid)	Acetone, chloroform, toluene	Ethanol	276,277
Poly(oxycarbonylethynylene-1,4-phenylene-ethynylenecarbonyltetramethylene)	DMA	Methanol	279
Poly(oxycarbonyl-2,6-naphthalenediylcarbonyloxy-butylene, Poly(butylene 2,6-naphthalate)	Methylene chloride	Chloroform THF	280,281
Poly(oxycarbonyl-2,6-naphthalenediylcarbonyloxy-ethylene, Poly(ethylene 2,6-napthalenedicarboxylate)	Hexafluoroisopropanol		282
Poly(oxycarbonyl-2,6-naphthalenediyl-*co*-oxycarbonyl-1,4-phenylene) (1 : 3)	Pentafluorophenol/hexafluoro-isopropanol		283
Poly(oxycarbonyloctadecamethylenecarbonyloxyethylene)	Toluene	Methanol	284
Poly(oxycarbonylpentamethylene), Poly(ε-caprolactone), Poly(oxyhexanoyl)	Chloroform, methylene chloride, THF, toluene	Hexane, methanol	285–287
Poly(oxycarbonylpentamethylene-*co*-oxycarbonyl-3-hydroxypentamethylene) (9 : 1 to 1 : 9), Poly(5-hydroxyvaleric acid-*co*-4-hydroxyvaleric acid)	Methylene chloride	Methanol (cold)	288
Poly(oxycarbonyl-1,4-phenylene), Poly(4-hydroxybenzoic acid)	No known solvent		32
Poly(oxycarbonyl-1,4-phenyleneoxyhexamethyleneoxy-1,4-phenylene)	Phenol/*p*-chlorophenol/ trichloroethylene (25/40/35)		289

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TABLE 4. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(oxycarbonyl-1,4-phenyleneoxyterephthaloyloxy-tetramethylene) (7 : 3 to 2 : 3), Poly(butylene terephthalate-*co*-4-hydroxybenzoic acid)	*p*-Chlorophenol/tetrachloroethane, trifluoroacetic acid		291
Poly(oxycarbonyl-2-phenyl-1,3-propylene)	Chloroform	Hexane	290
Poly(oxycarbonylpolyethylenecarbonylpolyethylene)	Cyclohexane		292
Poly(oxycarbonyl-1,2-propylene), Poly(3-hydroxybutyric acid)	Chloroform		7
Poly(oxycarbonyl-1,3-propylene), Poly(γ-butyrolactone), Poly(4-hydroxybutyric acid)	Carbon tetrachloride, chloroform, dichloroethylene, trifluoroethanol		293,294
Poly(oxycarbonyl-1,2-propylene-*co*-oxyethylenecarbonyloxyethylene) (9 : 1 to 1 : 9)	Chloroform	Diethyl ether/hexane (1 : 3)	295
Poly(oxycarbonyltetradecamethylene)	Chloroform		296
Poly(oxycarbonyltrimethyleneoxyethylene-*co*-methyl vinyl ether) (1 : 2 to 4 : 1)	Bzn.	Diethyl ether	297
Poly(oxycarbonyl-1-vinylethylene), Poly(β-vinyl-β-propiolactone)	Butyrolactone, chlorinated hydrocarbons, DMF		298
Poly(oxy-α,α-dibutylsebacoyloxyhexamethylene)	Bzn.		5
Poly(oxydodecanedioyloxy-1,4-phenylene-2-methylethenylene-1,4-phenylene)	Chloroform	Hexane, W	299
Poly(oxyfumaroyloxyhexamethylene)	Chloroform		5
Poly(oxyfumaroyloxytetramethylene)	Chloroform, NMP	Methanol, W	300
Poly(oxyglutaroyloxyhexamethylene)	Bzn., chlorobenzene, chloroform, THF		301
Poly(oxyglutaroyloxypropylene), Poly(1,3-propylene glutarate)	Chloroform, MEK, THF, toluene		32
Poly(oxyisophthyloyloxy-4,4′-biphenylylene)	See poly(oxyisophthaloyloxy-1,4-phenylene)		
Poly(oxyisophthaloyloxy-1,4-phenylene), Poly(phenylene isophthalate)	At elevated temp.: benzophenone, halogenated biphenyls, halogenated diphenyl oxides, halogenated naphthalenes, *m*-terphenyl, 2,4,6-trichlorophenol	Common organic solvents	302
Poly(oxyisophthaloyloxy-1,4-phenylene-fluorenylidene-1,4-phenylene)	Dichloroethane	Methanol, W	303
Poly(oxyisophthaloyloxy-1,1,3-trimethylindan-3,5-diyl-1,4-phenylene)	Chloroform, DMF, *m*-cresol, NMP, nitrobenzene, pyridine, THF	DMSO, hexane, methanol	304
Poly(oxymaleoyloxyhexamethylene), Poly(hexamethylene malonate)	Bzn., chloroform, THF		5
Poly(oxymaleoyloxytetramethylene), Poly(butylene malonate)	Chloroform, NMP, THF	Methanol	300
Poly(oxyoxalyloxyethylene), Poly(ethylene oxalate)		Bzn., Chlorobenzene, chloroform, THF	301
Poly(oxyoxalyloxyhexamethylene), Poly(hexamethylene oxalate)	Chlorobenzene, chloroform, dioxane, nitromethane, toluene	Bzn., methanol, toluene, THF	301,305
Poly(oxy-1-oxo-3-methyltrimethlylene-*co*-oxy-3-ethyltrimethylene) (9 : 1), Poly(3-hydroxybutryic acid-*co*-3-hydroxyvaleric acid)	Chloroform		306
Poly(oxy-1-oxo-3-methyltrimethylene-*co*-oxy-1-oxotrimethylene) (1 : 0 to 3 : 1), Poly(3-hydroxybutyric acid-*co*-3-hydroxypropionic acid)	Chloroform	Hexane, methanol	307
Poly(oxy-1-oxo-1,2-propylene), Poly(lactic acid)	Acetone, chloroform, dioxane, DMF, methylene chloride, pyridine	Ethanol, hexane, methanol	276,277,308,690
Poly(oxy-1-oxo-1,2-propylene-*co*-oxycarbonylmethylene), Poly(lactic acid-*co*-glycolic acid) (1 : 1 to 1 : 3)	Chloroform		309
Poly(oxysuccinoyloxy-2,2-dimethyl-1,3-propylene)	Bzn., chloroform, THF		32
Poly(oxysuccinoyloxydoeicosan-10,12-diyne-1,22-diyl)	Chloroform, THF		310
Poly(oxysebacoyloxyhexamethylene), Poly(hexamethylene sebacate)	Bzn., chlorobenzene, chloroform, THF		5,301
Poly(oxysebacoyloxyhexadecamethylene)	Chloroform		5
Poly(oxysuccinoyloxyhexamethylene), Poly(hexamethylene succinate)	Bzn., chlorobenzene, chloroform, nitromethane, THF	Methanol	5,301,305
Poly(oxysuccinoyloxymethylene-1,4-cyclohexylene-methylene)	Bzn., chloroform, THF (hot)		32
Poly(oxysuccinoyloxynonamethylenebutadyne-1,3-diylnonamethylene)	Chloroform, THF		310

TABLE 4. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(oxysuccinoyloxytetramethylene), Poly(tetramethylene succinate)	Chloroform, toluene	Diethyl ether	311
Poly(oxysuccinoyloxytrimethylene), Poly(1,3-propylene succinate)	Chloroform, MEK, THF, toluene		32
Poly[oxyterephthaloyloxy-1,4-(2-*n*-alkyl)phenylene]	*p*-Chlorophenol, methylene chloride	Acetone, ethanol, hexane	312
Poly(oxyterephthaloyloxytetramethylene), Poly(butylene terephthalate)	Phenol/tetrachloroethane		313
Poly[oxyterephthaloyloxybutylene-*co*-oxycarbony-2,6-naphthanenediylcarbonylpoly(oxyethylene)] (1 : 1)	*m*-Cresol		314
Poly(oxyterephthaloyloxyethylene), crystalline, Poly(ethylene terephthalate), Poly(oxyethyleneoxyterephthaloyl)	Chloral hydrate, chlorophenol, DMSO (hot), halogenated aliphatic carboxylic acids, anhydrous hydrogen fluoride, nitrobenzene, phenol/2,4,6-trichlorophenol (10/7), phenol, phenol/tetrachloroethane (1/1)	Aliphatic alcohols, carboxylic esters, chlorinated hydrocarbons, ethers, hydrocarbons, ketones	4,5,315, 316
Poly(oxyterephthaloyloxyethylene-*co*-oxyhexanoyl) (1 : 1), Poly(ethylene terephthalate-*co*-ε-caprolactone)	Chloroform	Methanol	317
Poly(oxyterephthaloyloxyethylene-*co*-oxycarbonyl-naphthalene-2,6-diylcarbonyloxyethylene) (<4 : 1, >4 : 1)	Phenol/tetrachloroethane (3/2)		318
Poly(oxyterephthaloyloxy-1,4-phenylenecarbonyloxy tetraethylene) (2 : 1 : 1)	*p*-Chlorophenol/tetrachloroethane, trifluoroacetic acid		291
Poly(oxyundecanoyl)	Bzn., chloroform		5
4.1.4. POLY(ANHYDRIDES)			
Poly(oxysebacoyl)	Chloroform		276
Poly(oxyterephthaloyl)	DMF		319,320
4.1.5. POLY(URETHANES) (alphabetically ordered)			
Poly(urethanes) (general)	Phenol, *m*-cresol, formic acid, sulfuric acid	Alcohols, diethyl ether, saturated hydrocarbons	4
Poly(oxy-1,4-butyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl), Poly(butanediol-*alt*-diphenylmethane-1,4-diisocyanate)	DMA		321
Poly(oxy-1,4-cyclohexyleneoxycarbonylimino-1,4-phenylenemethylene-1,4-phenylene-iminocarbonyl)	Dioxane (hot), DMF, THF	1,1,2-Trichloroethane	322
Poly(oxyethyleneoxycarbonyliminoethylene-iminocarbonyl)	*m*-Cresol	Chloroform, formic acid, methylene chloride	323
Poly(oxyethyleneoxycarbonyliminohexa-methyleneiminocarbonyl)	DMF	Diethyl ether	324
Poly(oxyethyleneoxycarbonylimino-1,3-phenyleneiminocarbonyl)	*m*-Cresol (hot), DMSO, sulfuric acid	Chlorobenzene, chloroform, cyclohexanone (sw), DMF (sw), xylene	325
Poly(oxytrimethyleneiminocarbonyl)	DMSO		326

4.2. MAIN-CHAIN -O- HETEROATOM POLYMERS

4.2.1. POLY(SULFONATES)

Poly(aryl sulfonates)

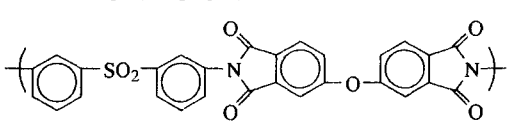
$-(SO_2ArSO_2OAr'O-)$ Ar = $-C_6H_4-$ = $-C_6H_4C_6H_4-$ = $-C_6H_4CH_2C_6H_4-$ = $-C_6H_4SO_2C_6H_4-$	DMF	Methanol	349
	DMA, DMF, NMP	Chloroform, methanol, methyl ethyl ketone, tetrahydrofuran	676

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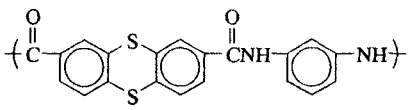
TABLE 4. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
4.2.2. POLY(SILOXANES) (ordered according to increasing complexity)			
Poly(siloxanes) (general)			
fluids and greases	Aromatic and chlorinated hydrocarbons, esters	Moderately conc. acids and alkalies, lower alcohols, W	6
rubbers		Aromatic and chlorinated hydrocarbons (sw), esters (sw)	6
Poly(oxydimethylsilylene), Poly(dimethylsiloxane)	Amyl acetate, chlorobenzene, chloroform, cyclohexyl acetate, dichlorobenzene, 1,2-dimethoxyethane, ethyl bromide, ethyl acetate, *o*-fluorotoluene, hydrocarbons, hydrogenated xylene, isopropyl acetate, MEK (above 20°C), octylamine, phenetol (above 13°C), trichloroethylene	Acetone, acetonitrile, acetophenone, aniline, benzyl alcohol, benzyl acetate, bromobenzene, γ-butyrolactone, Carbitol®, cyclohexanone, cyclohexanol, 1,4-dibromobutane, 1,2-dichloroethane, di(chloroethyl) ether, dichlorobenzene, dimethylnaphthalene, dioxane, diphenyl oxide, ethyl formate, ethyl benzoate, 2-isopropoxyethanol, mesityl oxide, methanol, nitrobenzene, 1-nitropropane	4,5
Poly(oxydiethylsilylene), Poly(diethylsiloxane)	Toluene	W	32
Poly(oxyhydrophenylsilylene)	THF		351
Poly(oxymethylphenylsilylene), Poly(methylphenylsiloxane)	Acetone (hot), butanol, chloroform, diethyl ether, ethyl acetate, toluene	Ethanol, ethylene glycol, methanol	352
Poly(oxymethylhexadecylsilylene)	Chloroform, THF, toluene	W	32
Poly(oxyalkylmethylsilylene)-ω-4-(4-cyanobenzoyl)phenoxy	DMF, toluene	Hexane, methanol	353
Poly(oxydiphenylsilylene), Poly(diphenylsiloxane)	Acetone (hot)	Acetone (cold)	352
Poly[oxybis(3,4-dichlorophenyl)silylene]	Acetone, bzn.		354
Poly(oxydimethylsilyleneethyleneoxydimethylsilylene)	Toluene		355
Poly(oxydimethylsilyleneoxydimethylsilylene-1,4-phenylenedimethylsilylene)	THF		356
Poly[oxymethyl(5-*t*-butoxycarbonyloxy-2-norbornyl)silylene]	Chloroform, toluene	Methanol	357,358
Poly(oxydimethylsilylene-*co*-oxyalkylmethylsilylene), Poly(dimethylsiloxane-*co*-alkylmethylsiloxane)	Aliphatic hydrocarbons, aromatic hydrocarbons, isopropanol	Ethanol, W	359
Poly(oxydimethylsilylene-*co*-oxycarbonyl-1,4-phenyleneisopropylidene-1,4-phenylene) (1 : 3 to 1 : 1)	Bzn., carbon tetrachloride, chloroform, methylene chloride, pyridine, THF		360
4.3. MAIN-CHAIN -C-S-C- AND -C-S-N-POLYMERS			
4.3.1. POLY(SULFIDES) (alphabetically ordered)			
Poly(2,5-dioxopyrrolidine-1,3-diylthio-1,4-phenylene)	Tricresol	Methanol	361
Poly(dithioethyleneoxyethylene)	1,1,2-Trichloroethane (partially), other chlorinated hydrocarbons (partially)	Dil. acids, alcohols, dil. alkalies, bzn. (sw), carbon disulfide, carbon tetrachloride (sw), gasoline (sw), turpentine (sw)	6,178
liquid polymers	Bzn., ethylene dichloride		6,178
Poly(tetrathioethylene)	No solvent known	Dil. acids, alcohols, dil. alkalies, bzn. (sw), carbon tetrachloride, carbon disulfide, gasoline	6,178
Poly(tetrathioethyleneoxyethylene)	1,1,2-Trichloroethane (partially), other chlorinated hydrocarbons (partially)	Dil. acids, alcohols, dil. alkalies, bzn. (sw), carbon tetrachloride, carbon disulfide, gasoline	6,178
liquid polymers	Bzn., ethylene dichloride		6,178
Poly(thio-4,4-biphenylylene)	Biphenyl, dichlorobiphenyl, dimethyl-*p*-terphenyl		362
Poly(thiodifluoromethylene)	Chloroform/methanol	Conc. nitric acid (reflux), aqu. sodium hydroxide (10%, reflux)	363
Poly(thio-2,5-dimethyl-1,4-phenylene)	Hexane, methylene chloride		364
Poly(thiomethylene-1,4-*trans*-cyclohexylene-methylenethiomethylene)	Bzn., chloroform	Methanol, W	365
Poly(thio-1,4-phenylene), Poly(phenylene sulfide)	Biphenyl, *m*-dichlorobenzene, dichlorobiphenyl, dimethyl-*p*-terphenyl, hexachlorobiphenyl	At reflux temp.: 2,4-lutidine, phenyl sulfide, phenyl oxide, pyridine, toluene	362,366, 367

TABLE 4. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(thio-1,4-phenylenecarbonyl-1,4-phenylene-isopropylidene-1,4-phenylenecarbonyl-1,4-phenylene)	Chloroform, NMP	Methanol	368
Poly(thio-1,4-phenyleneiminocarbonyl-1,4-phenyleneoxy carbonylethylene)	Chloroform, NMP	Hexane, methanol	369
Poly(thio-1,4-phenyleneoxymethyleneoxy-1,4-phenylene)	Chloroform, methylene chloride	Methanol	364
Poly(thio-1,4-phenyleneoxy-1,4-phenylene)	Hexane, methylene chloride		364
Poly(thiotetramethyl-1,4-phenylene)	Durene		364
	DMA, DMF, NMP	Methanol	370

4.3.2. POLY(SULFONES)
(ordered according to increasing complexity)

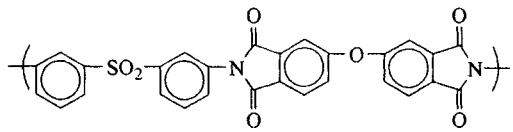
Poly(ethylene-*co*-sulfur dioxide)	No solvent known		13,178
Poly(propylene-*co*-sulfur dioxide)	Conc. nitric acid, conc. sulfuric acid	Common organic solvents	13,178
Poly(1-butene-*co*-sulfur dioxide)	Acetone, cyclohexanone	Paraffins, cycloparaffins, aromatic hydrocarbons	13
Poly(butylethylene-*co*-sulfur dioxide)	Bzn., chloroform, dioxane		7
Poly(1,3-hexadiene-*co*-sulfur dioxide) (1 : < 3)	Acetone, chloroform, DMA, DMF, THF		371
Poly(dimethylbutadiene-*co*-sulfur dioxide)	Conc. nitric acid, conc. sulfuric acid	Common organic solvents	13
Poly(1-pentyne-*co*-sulfur dioxide)	Dioxane		13
Poly(1-hexyne-*co*-sulfur dioxide)	Acetone, dioxane		13
Poly(1-heptyne-*co*-sulfur dioxide)	Acetone, dioxane		13
Poly[4-(*tert*-butoxycarbonyloxy)styrene-*co*-sulfur dioxide] (5 : 1)	Toluene		372
Poly(1,4-phenyleneoxymethyleneoxy-1,4-phenylenesulfonyl)	Chloroform	Methanol	364
Poly(1,4-phenylenesulfonyl)	Aniline, DMA, pyridine		373
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenylene)	Aniline, aromatic amines, DMF, DMSO, methylene chloride, NMP, pyridine	Alcohols, halogenated hydrocarbons, hydrocarbons, silicone oil	32,373–375
Poly(thio-1,4-phenylenesulfonyl-1,4-phenylene)	DMF, DMSO, NMP		376
Poly(carbonyl-1,3-phenylenemethylene-1,4-phenylenesulfonyl-1,4-phenylene)	No known solvent	Acetone, DMF, THF, toluene	377
Poly(thio-1,4-phenylenesulfonyl-1,4-phenylenethio-1,4-phenyleneisopropylidine-1,4-phenylene)	DMA/toluene		378
Poly(3-amino-1,4-phenylenesulfonyl-2-amino-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenylene)	THF	Ethanol/W (1 : 1)	379
Poly(2,5-thiophenediylsulfonyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy)	DMA, methylene chloride, THF	Methanol, W	380
Poly(1,4-phenylenesulfonyl-1,4-phenyleneoxy–4,4′-biphenylyleneoxy)	Aniline, DMA, pyridine		373
Poly(1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenyleneoxy-1,4-phenyleneoxy)	Chloroform, sulfolane	Methanol	381
Poly(1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,3-phenyleneethenylene-1,3-phenyleneoxy)	DMA		382
Poly(1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylenevinylidene-1,4-phenylene)	Chloroform, DMA		181
Poly(1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenyleneoxy-1,3-pyridinediyloxy-1,4-phenyleneoxy)	DMA		347
Poly(1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylenemethylphosphineoxide-1,4-phenyleneoxy)	DMA	Methanol/W	383
	Dichlorobenzene	Isopropanol, methanol	384
Poly(1,3,4,6-tetraoxyperhydrocyclopentene-(1,2-c : 3,4-c′)-dipropyl-2,5-diylethylenesulfonylethylene)	DMF, DMSO, formic acid, NMP, sulfuric acid	Hexane	385

TABLE 4. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
4.3.3. OTHERS			
Poly(trimethylenesulfite)	Methylene chloride		386
4.4. MAIN-CHAIN -C-N-C- POLYMERS			
4.4.1. POLY(AMIDES)			
Poly(amides) 1			
Poly(*N*-ethyliminocarbonyl), Poly(ethyl isocyanate)	Sulfuric acid, trifluoroacetic acid		387
Poly(*N*-butyliminocarbonyl) Poly(butyl isocyanate)	Bzn., carbon tetrachloride, THF		387,388
Poly(*N*-*n*-hexyliminocarbonyl), Poly(*n*-hexyl isocyanate)	Bzn., chloroform, DMF	Methanol	389–391
Poly(*N*-*m*-tolyliminocarbonyl)	DMF		387
Poly(*N*-vinyliminocarbonyl)	DMA, DMF, DMSO, pyridine	Alcohols, carbon disulfide, chloroform, dioxane	392
Poly(amide) 1,5			
Poly(iminoglutaroyliminomethylene)	1,4-Butanediol, dichloroacetic acid		393,394
Poly(amide) 1,6			
Poly(iminoadipoyliminomethylene), Poly(methylene adipamide)	1,4-Butanediol		394
Poly(amide) 1,12			
Poly(iminododecanoyliminomethylene)	2-Methyl-2,4-pentanediol		394
Poly(amides) 2			
Poly(γ-benzyl glutamamide)	*m*-Cresol, DMF		395,396
Poly(lysine-*co*-lactic acid) (1 : 19)	Chloroform	Methanol	397
Poly(amides) 3			
Poly[imino-(1-oxo-2,2-dimethyl-3-phenyltrimethylene)], Poly(3,3-dimethyl-4-phenylazetidine-2-one)	Phenol, sulfuric acid, trifluoroethanol	Chloroform, HMPT, methanol, sol. of $Ca(CNS)_2$ in methanol	399
Poly[imino-(1-oxo-3-methyltrimethylene)], Poly (4-methylazetidine-2-one)			399,691
optical activity < 75%	Formic acid, phenol, sulfuric acid, sol. of $Ca(SCN)_2$ in methanol	Chloroform, HMPT, methanol, trifluoroethanol	
optical activity > 80%	Chlorobenzene/trichloroacetic acid (1/1), dichloroacetic acid, formic acid/trichloroacetic acid (3/7)	Formic acid, sulfuric acid	691
Poly[imino-(1-oxotrimethylene)]	Aqu. chloral hydrate, aqu. sol. of $CaBr_2$ or $FeCl_3$, chloroacetic acid, cyanoacetic acid, formic acid, glycerol (hot), nitric acid, nitrophenols, phenol (hot), sulfuric acid, W>140°C	Most organic solvents, dilute bases and acids, DMF, DMSO, phenol (cold)	13,400,401
Poly(aspartic acid-*co*-lactic acid) (1 : 9)	Methylene chloride		402
Poly(amides) 4			
Poly[imino-(1-oxotetramethylene)], Polypyrrolidone	α-Pyrrolidone (hot), acetic acid (hot), aqu. sol. of $Ca(CNS)_2$, benzyl alcohol (hot), *o*-chlorophenol, *m*-cresol, dichloroacetic acid, formic acid, methanol, sulfuric acid	DMF, formamide, W	403, 404
Poly(iminooxalyliminotetramethylene)	Dichloroacetic acid, sulfuric acid, trifluoroacetamide	Formic acid, phenol, W	405
Poly(imino-1-carboxy-1,3-propylenecarbonyl), Poly (γ-glutamic acid)	W	Ethanol	406
Poly(amide) 4,6			
Poly(iminoadipoyliminotetramethylene), Poly(butylene adipamide)	Hexafluoroisopropanol		407
Poly(amides) 5,5			
Poly(iminoglutaroyliminopentamethylene), Poly (1,5-pentamethylene glutaramide)	Dichloroacetic acid		408
Poly(amide) 5,6			
Poly(iminoadipoyliminopentamethylene), Poly (pentamethylene adipamide)	Dichloroacetic acid, trifluoroethanol	Methanol	409,410
Poly(amides) 6			
Poly[imino-(1-oxohexamethylene)], Poly (ε-caprolactam)	Acetic acid, chlorophenol, *m*-cresol, ethylene carbonate, formic acid, HMPT, phosphoric acid, sulfuric acid, trichloroacetic acid	Alcohols, chloroform, esters, ethers, hydrocarbons, ketones	4,5
chlorinated	*o*-Chlorophenol, DMSO, sulfuric acid	Formic acid	411

TABLE 4. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly[imino-(1-oxohexamethylene)], partially *N*-trifluoroacetic acid substituted	Acetone, chloroform, THF	Hexane, W	412
Poly[imino-(1-oxohexamethylene)-*co*-imino-(1-oxyundecamethylene)] (1 : 1)	2,2,2-Trifluoroethanol/methylene chloride (3/2)		413
Poly[imino-(1-oxohexamethylene)-*co*-iminosuccinoyl], (9 : 1 to 2 : 8)	Mesitylene/sulfolane		414
Poly(amide) 6,6			
Poly(iminoadipoyliminohexamethylene), Poly(hexamethylene adipamide)]	rt: Chloral hydrate, chloroacetic acids, formic acid, HCl, HF, liquid sulfur dioxide, methanol, phenols, phosphoric acid, sulfuric acid, trichloroethanol, trifluoroethanol, saturated solutions of: alcohol-soluble salts, e.g. calcium chloride, magnesium chloride in methanol, at 120–180°C: acetic acid, *N*-acetylmorpholine, benzyl alcohol, 2-butene-1,4-diol, 1,3-chloropropanol, di(ethylene glycol), DMSO, ethylene chlorohydrin, formamide	Aliphatic alcohols, chloroform, diethyl ether, hydrocarbons, aliphatic ketones, aliphatic esters	13
Poly(amide) 6,9			
Poly(iminoazelaoyliminohexamethylene)	*m*-Cresol		32
Poly(amide) 7			
Poly[imino-(1-oxohepamethylene)-*co*-imino-(1-oxodedecamethylene)], (1 : 1)	2,2,2-Trifluoroethanol/methylene chloride (3/2)		413
Poly(amide) 8,6			415
Poly(iminoadipoyliminooctamethylene)	1,4-Butanediol, hot		
Poly(amide) 10,6	Chlorobenzene, diethyl succinate (79°C)		5
Poly(iminoadipoyliminodecamethylene), Poly(decamethylene adipamide)			
Poly(amide) 11			
Poly[imino-(1-oxoundecamethylene)], Poly(11-aminoundecanoic acid)	Higher primary alcohols, DMF, DMSO, hexafluoropropanol		13,692
Poly(amide) 12			
Poly[imino-(1-oxododecamethylene)], Poly(12-aminododecanoic acid)	Cresol		416
Poly(amide) 12/2			
Poly(iminocarbonylmethyleneiminocarbonylundecamethylene)	DMSO	Diethyl ether	398
Poly(amide) 12,6			
Poly(iminoadipoyliminododecamethylene)	Cresol		416
Poly(amide) 18			
Poly[amino-(1-oxooctadecamethylene)], Poly(18-aminostearic acid)	DMF, DMSO, pyridine	Hexane, methanol, W	417
Other aliphatic poly(amides)			
Poly(iminotartronoyliminohexamethylene)	Chloroacetic acid, DMSO	Acetone, hexane, methanol, W	418
Aromatic poly(amides)			
Poly(imino-1,4-phenylenecarbonyl-*co*-oxy-1,4-phenylenecarbonyl) (1 : 1 to 1 : 9), Poly(*p*-aminobenzoic acid-*co*-*p*-hydroxybenzoic acid)	Conc. sulfuric acid (decomp.)	Insoluble in all common solvents	419
Poly(iminocarbonyl-1,4-phenyleneoxyterephthaloyloxy-1,4-phenylenecarbonylimino-1,4-phenylenehexafluoroisopropylidene-1,4-phenylene)	NMP		693
Poly(iminocarbonylpyridine-2,6-diyl-carbonylimino-1,4-phenyleneoxy-1,4-phenylene)	NMP, sulfuric acid	Methanol	420
Poly(iminoterephthaloyliminoethylene), Poly(ethylene terephthalamide)	Sulfuric acid	Acetone, chloroform, *m*-cresol, DMF, ethanol/W (4/1), formamide, trifluoroacetic acid	421
Poly(*N*-methyliminoterephthaloyl-*N*-methyliminoethylene)	Acetone, chloroform, *m*-cresol, DMF, formamide, sulfuric acid, trifluoroacetic acid	Ethanol/W (4/1)	421
Poly(*N*-methyliminoterephthaloyl-	Chloroform, *m*-cresol (hot),	Acetone, DMF	421

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TABLE 4. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
N-methyliminotetramethylene)	ethanol/W (4/1) (hot), formamide, sulfuric acid		
Poly(iminoisophthaloylimino-1,3-phenylene), *N*-methyl substituted	DMSO, sulfuric acid	Hexane, W	422
Poly(iminoisophthaloylimino-1,4-phenyleneoxy-1,4-phenylene)	NMP, sulfuric acid	Methanol	420
Poly(iminoisophthaloylimino-2,4-pentylene)	DMF, DMSO, methanol, trifluoroacetic acid, trifluoroethanol	Hexane, methanol	423
Poly(iminoisophthaloylimino-4,4′-biphenylylene)	Conc. sulfuric acid	*m*-Cresol (hot), DMA (sw), DMF (sw), DMSO (sw), formic acid (85%) (sw), glacial acetic acid (sw), methanol, NMP (sw)	424
Poly(iminoisophthaloylimino-1,4-phenylene)	HMPT		425
Poly(iminoterephthaloyliminoethylene oxyisophthaloyl-oxyethylene)	DMSO, NMP, trifluoroacetic acid	Methanol, DMF	426
Poly(iminoterephthaloyliminotetramethylene) Poly(butylene terephthalamide)	DMF, formamide, sulfuric acid, trifluoroacetic acid	Acetone, chloroform, *m*-cresol	421
Poly(iminoterephthaloylimino-2,4-pentylene)	LiCl, sulfuric acid, trifluoroacetic acid	DMF, DMSO, trifluoroethanol, W	423
Poly(iminoterephthaloylimino-1,4-phenylene), Poly (*p*-phenylene tetrephthalamide)	DMF/LiCl, NMP		427,428
Poly(iminocarbonyl-1-methyl-4,5-imidazolediyl-carbonyliminoethylene)	Chloroform, DMF, methylene chloride, NMP		429
Poly(iminocarbonyl-1,4-phenyleneiminocarbonyl-pentamethylene)	Ethyl acetate		430
Poly(iminoterephthaloylimino-3-methyl-1,4-phenyleneoxyethyleneoxy-2-methyl-1,4-phenylene)	Tetrachloroethane		431
Poly(iminocarbonyl-1,4-phenylenehexafluoroiso-propylidene-1,4-phenylenecarbonylimino-1,4-phenyleneoxy-1,4-phenylene)	DMA	Methanol	432
Poly(iminocarbonyl-1,4-phenyleneoxyterephthaloyloxy-1,4-phenylenecarbonylimino-1,4-phenylenehexafluoro-isopropylidene-1,4-phenylene)	NMP		693
Poly(imino-3-alkylisophthaloylimino-1,4-phenylene)	NMP		433
Poly(iminotetra(octylthio)terephthaloylimino-1,4-phenylene)	Methylene chloride/methanesulfonic acid	Chlorobenzene, chloroform, dioxane, THF, trifluoroacetic acid	434
Poly(pyromellitic dianhydride-*co*-aromatic diamines), Poly(amic acids)	DMF, DMA, DMSO, tetramethylurea		435
Sulfur and phosphorus containing poly(amides)			
Poly(iminoadipoyliminotrimethylene-1,4-phenylene-phosphinidene-1,4-phenylenetrimethylene)	Aqu. hydrochloric acid (5%), cresol, DMF, ethanol, formic acid	Acetone, acetonitrile, chlorobenzene, chloroform, C-7 fluoroalcohol, hexane (sw), W	436
Poly(iminoethyleneiminocarbonyl-1,4-phenylene-sulfonyl-1,4-phenylenecarbonyl)	Formic acid, tetrachloroethane/phenol (2/3), 1,1,2-trichloroethane/formic acid (3/2)	2-Chloroethanol, *m*-cresol, DMF	437
Poly(iminohexamethyleneiminocarbonylethylene-1,4-phenylenephosphinylidene-1,4-phenyleneethylenecarbonyl)	Cresol, DMF, ethanol, C-7 fluoroalcohol, formic acid	Acetone, acetonitrile, aqu. hydrochloric acid (5%, sw), chlorobenzene, chloroform (sw), HX, W (sw)	
Poly(iminohexamethyleneiminocarbonyl-ethylenethioethylenecarbonyl)	Cresol, DMF, formic acid	Acetone, acetonitrile, aqu. hydrochloric acid (5%), chlorobenzene, chloroform, ethanol, C-7 fluoroalcohol (sw), hexane, W	436
Poly(iminoisophthaloylimino-1,3-phenylene-sulfonyl-1,3-phenylene)	DMA, DMF, NMP		438
Poly(iminoterephthaloylimino-1,3-phenylene-sulfonyl-1,3-phenylene)	DMA, DMF	NMP	438
Poly(iminoterephthaloylimino-1,4-phenylene-sulfonyl-1,4-phenylene)	Sol. of LiCl in DMF (5%)	DMF, NMP	438
Poly(iminoterephthaloyliminotrimethylene-1,4-phenylenephosphinidene-1,4-phenylenetrimethylene	Cresol, DMF, formic acid, hydrochloric acid (5% sw)	Acetone, acetonitrile, chlorobenzene, chloroform, ethanol (sw), hexane, W	436
4.4.2. POLY(HYDRAZIDES)			
Poly[diazo-1,4-phenylenecarbonyltri(oxyethylene)-oxycarbonyl-1,4-phenylene]	Chloroform, *o*-dichlorobenzene, nitrobenzene, THF	Acetone, hexane, methanol	439

TABLE 4. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(hydrazoadipoylhydrazoisophthaloyl)	DMSO, HMPT		440
Poly(hydrazoadipoylhydrazosuccinoyl)		DMSO (sw)	440
Poly(hydrazoisophthaloylhydrazoterephthaloyl)	*m*-Cresol, dichloroacetic acid (degrad.), DMA, DMSO (cold), formic acid (degrad.), NMP, sulfuric acid (degrad.), trifluoroacetic acid (degrad.), trifluoropropanol	Chloroform, nitrobenzene, trichloroethane (sw), trifluoropropanol, W	441
Poly(4,4′-benzaldehydeazineoxy-1,4-phenylene-isopropylidene-1,4-phenyleneoxy)	Toluene	Methanol	442
4.4.3. POLY(UREAS)			
Poly(ureas) (general)	*m*-cresol, formic acid, phenol, sulfuric acid	Alcohols, diethyl ether	30
Poly(iminocarbonyliminononamethylene), Poly(ureylene-nonamethylene)	Sulfuric acid (60% and 95%)	*m*-Cresol (sw), glacial acetic acid, aqu. NaOH (45%, sw)	679
Poly(iminocarbonylimino-1,3-phenyleneimino-carbonylimino-2-methyl-5-sulfo-1,3-phenylene), Poly(ureylene-1,3-phenyleneureylene-2-methyl-5-sulfo-1,3-phenylene)	Aqu. NaOH	W	443
4.4.4. POLY(CARBODIIMIDES) (arranged according to increasing complexity)			
Poly(diethylcarbodiimide)	Formic acid, aqu. mineral acids	Heptane	444
Poly(diallylcarbodiimide)	Formic acid	Heptane	444
Poly(di-*n*-butylcarbodiimide)	Formic acid	Heptane	444
Poly(diisobutylcarbodiimide)	Formic acid		444
Poly(methylhexylcarbodiimide)	Toluene		445
Poly(di-*n*-hexylcarbodiimide)	Formic acid, *n*-heptane, toluene		445,446
Poly(diphenylcarbodiimide)	Formic acid		444
−N=C=N−R−			
Poly(hexamethylenecarbodiimide)	No solvent known		447
Poly(4,4′-biphenylylenemethylenecarbodiimide)	*m*-Cresol, trichlorophenol/phenol (7/10)		447
Poly(3-methyl-1,4-phenylenecarbodiimide)	No solvent known		447
Poly(1,3-xylylenecarbodiimide)	No solvent known	Chlorobenzene, chloroform, cyclohexanone, DMF, DMSO, ethylene chlorohydrin, formic acid, trichloroethylene, xylene	447
Poly(2,2′-dimethyl-4,4′-biphenylylenecarbodiimide)	No solvent known		447
Poly(2,2′-dimethoxy-1,4-biphenylylenecarbodiimide)	No solvent known		447
Poly(1,5-naphthylenediylcarbodiimide)	No solvent known		447
4.4.5. OTHER COMPOUNDS			
Poly(*N*-propanoyliminoethylene), Poly(2-ethyl-2-oxazoline)	Ethanol, W		448,449
Poly[*N*-(perfluorooctylethylenecarbonyl)iminoethylene]	Hexafluoroisopropanol	Methanol	450
Poly(*N*-benzoyliminoethylene)	Chloroform	Diethyl ether	451
Poly[*N*-((9-carbazolyl)acetyl)iminoethylene]	Chloroform, THF	Ethanol, diethyl ether	452
Poly[*N*-2-((9-carbazolyl)propanoyl)iminoethylene]	Chloroform, THF	Ethanol, diethyl ether	452
Polysuccinimide	Mesitylene sulfolane		453
Poly(*tert*-butylmaleimide)	Bzn., carbon tetrachloride, DMF, THF	Hexane	454
Poly(phenylmaleimide)	DMF, DMSO	Bzn., hexane	454
Poly(*m*-aniline)	DMF, DMSO, methylene chloride (partially), NMP		455
Polyaniline, camphorsulfonic acid doped	Benzyl alcohol, chloroform, *m*-, *p*-cresol, DMF, NMP		456,457
Poly[*N*-(*p*-*tert*-butylbenzoyl)aniline]	Acetone (partially), chloroform (partially), DMSO, THF		458
Poly(2-pentadecylaniline)	Chloroform, THF		459
Poly(3-trifluoromethylaniline)	Acetone, acrylonitrile, toluene		460
Poly(4-amino-3-sulfoaniline)	Methanol, W		461

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TABLE 4. *cont'd*

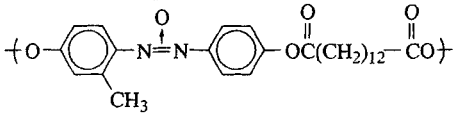
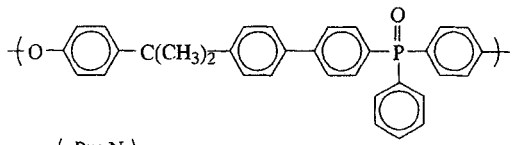
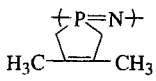
Polymer	Solvents	Nonsolvents	Refs.
Poly[*N*-(*p*-sulfophenyl)imino-3-hydroxymethyl-1,4-phenyleneimino-1,4-phenylene]	DMF (partially), DMSO (partially), NMP (partially), W		462
Poly[*N*-(4-methylphenyl)-6-iminohexamethylenecarbonyl-1,4-phenylene]	Methylene chloride, THF	Diethyl ether, methanol	463
Poly(imino-1,4-phenylenecarbonyloxy-1,4-phenyleneoxycarbonyl-1,4-phenylene)	*m*-Cresol, DMA (partially), DMSO (partially), NMP (partially), sulfuric acid	Chloroform, hexane	464
Poly[*N*-(methyl)imino-3-phenyltriazine-1,5-diyl]	Bzn., cresols, phenols	Hexane	465
Poly(9-nitriloanthraquinone-10-nitrilo-1,4-phenylene)	Chlorobenzene, chloroform	Hexanes	466
Poly(9-nitriloanthraquinone-10-nitrilo-1,4-phenylene-methylene-1,4-phenylene)	Chlorobenzene		467
Poly[carbonyliminocarbonyl-1,4-phenylene-carbonyliminocarbonyloxy-2,2-(dimethyl) tetramethyleneoxy]	DMA	Diethyl ether	468
Poly(oxy-3-methyl-1,4-phenylene-*ONN*-azoxy-2-methyl-1,4-phenyleneoxy-1,12-dodecanoyl), 	Chloroform		469
Poly[*N*-(acetyl)iminoethylene-*co*-*N*-(3-thiopheneylcarbonyl)iminoethylene]		Acetone, acetonitrile, DMF, DMSO, THF	470
4.5. POLY(PHOSPHAZENES) AND RELATED COMPOUNDS			
Poly(dichlorophosphazene)	Acetonitrile, cyclohexane, methylene chloride, THF		682
Poly[bis(trifluoroethoxy)phosphazene]	Acetone, cyclohexanone (hot), dimethyl ether, ethyl acetate, glycol, methanol (hot), THF	Diethyl ether, dioxane, ethanol, hydrocarbons, W	471–473
Poly(dimethoxyphosphazene)	Acetonitrile, chloroform, dimethoxyethane, dioxane, DMF, methanol, pyridine, THF	Acetone, bzn., diethyl ether, ethanol, W	471
Poly(dihexyloxyphosphazene)	Bzn.		7
Poly(methylphenylphosphazene)	THF		474
Poly(diphenoxyphosphazene)	Bzn. (hot), chlorobenzene, DMF, THF	Acetone, DMSO, ethanol, hexane, W	471
Poly(4-nitrophenoxyphenoxyphosphazene)	Dioxane, THF	Hexane, methanol, W	475
Poly[(phenylthiophosphonoyl)trimethyene]	Acetone (partially), acetonitrile (partially), chloroform, DMF, DMSO	Toluene, ethyl acetate, methanol, W	476
	Chloroform	Methanol/W (9/1)	477
	Methylene chloride	Diethyl ether	478
4.6. POLY(SILANES), POLY(SILAZANES)			
Poly(alkyltrisilazane)	Aromatic hydrocarbons, carbon tetrachloride, cyclohexane	Acetone (partially), DMSO (partially)	479
Poly(dimethylsilylene-1,4-phenylene)	Aromatic hydrocarbons	Aliphatic hydrocarbons, ethanol	480
Poly(dimethlylsilylenetrimethylene)	Diethyl ether, THF		481
Poly(*n*-pentyl-*n*-hexylsilylene)	Toluene	Isopropanol/ethyl acetate	482
Poly(di-*n*-hexylsilylene)	Isooctane		483
Poly(methylphenylsilylene)	Chloroform		484
Poly(phenylchlorosilylene)	Carbon tetrachloride		485
Poly(1,1,2,2-tetramethyldisilylenemethylene)	Toluene	Ethanol	486
Poly(diethoxysilylene-1,4-phenylene)	THF	Chloroform/methanol	487
Poly(octamethyltetrasilylenedimethylsilylene)	Toluene	Methanol	488
Poly(octamethyltetrasilylenemethylene)	Toluene	Methanol	488
Poly(dihydrosilylene-1,4-phenylene)	Chloroform, diethyl ether	Ethanol	489
Poly(hydrophenylsilyleneethynylene-1,3-phenyleneethynylene)	Bzn., THF	Hexane, methanol	490,491

TABLE 4. *cont'd*

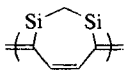
Polymer	Solvents	Nonsolvents	Refs.
Poly(dibutylsilylene-1,4-phenyleneethenylene-1,4-phenyleneethenylene-1,4-phenylene)	Chloroform	THF	492
Poly(hexamethylcyclotrisilazane)	Carbon tetrachloride, hydrocarbons	Acetone, DMSO	493
Poly(dihexylsilyleneferrocenediyl)	THF		680
Poly(dimethylsilylene-*N*-methylimino)	THF		494
Poly(dimethylsilyleneiminoethyleneimino)	Xylene		495
Poly(dimethylsilylene-*co*-1,3-dihydro-1,3-dimethyldisilazane) (1 : 1 to 3 : 1)	Decaline, toluene		496
	Toluene	Methanol	497

TABLE 5. MAIN-CHAIN HETEROCYCLIC POLYMERS

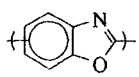
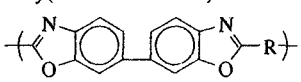
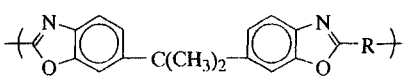
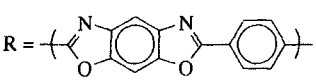
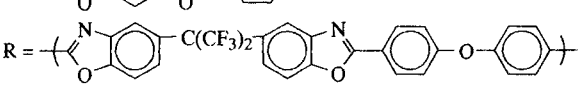
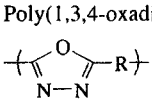
Polymer	Solvents	Nonsolvents	Refs.
5.1. POLY(BENZOXAZOLES), POLY(OXADIAZOLES), POLY(OXADIAZOLIDINES)			
Poly(benzoxazoles) 			
Poly(2,5-benzoxazolediyl)	Sulfuric acid		498
Poly(2,6-benzoxazolediyl)	Sulfuric acid		498
Poly(dibenzothiazoles) 			499
R = 1,3-phenylene	Sulfuric acid	Organic solvents	499
R = 1,4-phenylene	Sulfuric acid	Organic solvents	499,500
R = octamethylene	*m*-Cresol, formic acid, sulfuric acid		500
R = 1,4-cyclohexylene	Sulfuric acid	Acetic acid, DMA, DMF, DMSO, formamide, NMP	501
			
R = tetramethylene	Chloroform, *m*-cresol, DMF (hot), pyridine (hot), sulfuric acid (hot), tetrachloroethane (hot), THF (hot)	Benzyl alcohol, formic acid	502
R = 1,3-phenylene	Benzyl alcohol (hot), chloroform, *m*-cresol (hot), pyridine, sulfuric acid, tetrachloroethane (hot), THF	DMF (partially), formic acid (partially)	502
R = 1,4-phenylene	Chloroform, *m*-cresol, formic acid (hot), sulfuric acid (hot)	Benzyl alcohol, DMF, pyridine, tetrachloroethane, THF	502
R = 	Polyphosphoric acid		503
R = 	DMA, NMP	Methanol	504–506
Poly(1,3,4-oxadiazoles) 			
R = octamethylene	Bzn., chloroform, *m*-cresol, dichloroethylene, DMA (hot), DMF, DMSO (hot), NMP (hot), sulfuric acid		507–509
R = methylene to eicosamethylene	DMA (hot), DMSO (hot), NMP (hot), DMSO (sw), NMP (sw) sulfuric acid (95%)		509

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TABLE 4. *cont'd*

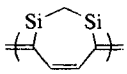
Polymer	Solvents	Nonsolvents	Refs.
Poly(dibutylsilylene-1,4-phenyleneethenylene-1,4-phenyleneethenylene-1,4-phenylene)	Chloroform	THF	492
Poly(hexamethylcyclotrisilazane)	Carbon tetrachloride, hydrocarbons	Acetone, DMSO	493
Poly(dihexylsilyleneferrocenediyl)	THF		680
Poly(dimethylsilylene-*N*-methylimino)	THF		494
Poly(dimethylsilyleneiminoethyleneimino)	Xylene		495
Poly(dimethylsilylene-*co*-1,3-dihydro-1,3-dimethyldisilazane) (1 : 1 to 3 : 1)	Decaline, toluene		496
	Toluene	Methanol	497

TABLE 5. MAIN-CHAIN HETEROCYCLIC POLYMERS

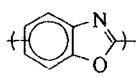
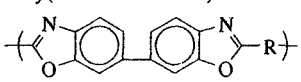
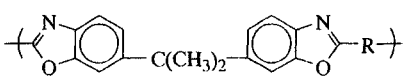
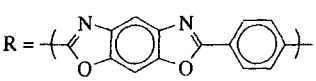
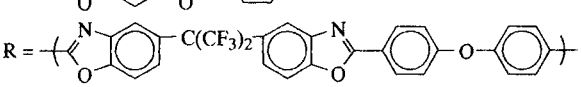
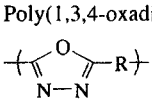
Polymer	Solvents	Nonsolvents	Refs.
5.1. POLY(BENZOXAZOLES), POLY(OXADIAZOLES), POLY(OXADIAZOLIDINES)			
Poly(benzoxazoles) 			
Poly(2,5-benzoxazolediyl)	Sulfuric acid		498
Poly(2,6-benzoxazolediyl)	Sulfuric acid		498
Poly(dibenzothiazoles) 			499
R = 1,3-phenylene	Sulfuric acid	Organic solvents	499
R = 1,4-phenylene	Sulfuric acid	Organic solvents	499,500
R = octamethylene	*m*-Cresol, formic acid, sulfuric acid		500
R = 1,4-cyclohexylene	Sulfuric acid	Acetic acid, DMA, DMF, DMSO, formamide, NMP	501
			
R = tetramethylene	Chloroform, *m*-cresol, DMF (hot), pyridine (hot), sulfuric acid (hot), tetrachloroethane (hot), THF (hot)	Benzyl alcohol, formic acid	502
R = 1,3-phenylene	Benzyl alcohol (hot), chloroform, *m*-cresol (hot), pyridine, sulfuric acid, tetrachloroethane (hot), THF	DMF (partially), formic acid (partially)	502
R = 1,4-phenylene	Chloroform, *m*-cresol, formic acid (hot), sulfuric acid (hot)	Benzyl alcohol, DMF, pyridine, tetrachloroethane, THF	502
R = 	Polyphosphoric acid		503
R = 	DMA, NMP	Methanol	504–506
Poly(1,3,4-oxadiazoles) 			
R = octamethylene	Bzn., chloroform, *m*-cresol, dichloroethylene, DMA (hot), DMF, DMSO (hot), NMP (hot), sulfuric acid		507–509
R = methylene to eicosamethylene	DMA (hot), DMSO (hot), NMP (hot), DMSO (sw), NMP (sw) sulfuric acid (95%)		509

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TABLE 5. *cont'd*

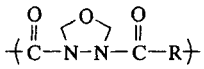
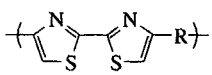
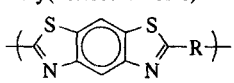
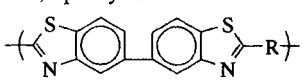
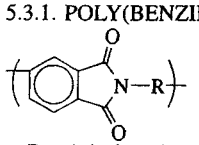
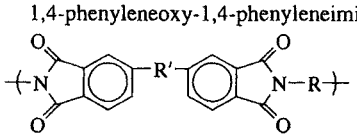
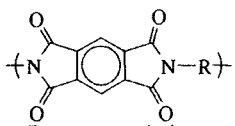
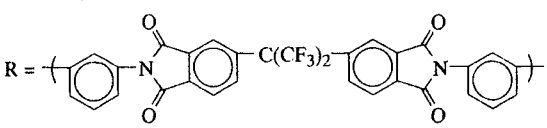
Polymer	Solvents	Nonsolvents	Refs.
R = 1,4-cyclohexylene	Formic acid (80%), sulfuric acid	DMA, DMSO, NMP	510
R = 1,3-cyclohexylene	DMA, DMSO, formic acid (80%), NMP, sulfuric acid		510
R = 1,2-phenylene	Sulfuric acid		511
R = 1,3-phenylene	Sulfuric acid, trifluoroacetic acid	Chloroform, *m*-cresol, DMF, nitrobenzene, tetrafluoropropanol	512–514
Poly(oxadiazolidines)			
			
R = 1,3-phenylene	DMSO	Methanol	515
5.2. POLY(DITHIAZOLES), POLY(BENZOTHIAZOLES)			
Poly(dithiazoles)			
			
R = 1,4-phenylene	Sulfuric acid (cold)	Diethyl ether, diphenyl ether, diphenylmethane, DMA, DMF, DMSO, ethanol, formic acid, quinoline	516
R = 1,4-phenylenehexamethylene-1,4-phenylene	Sulfuric acid (cold)		516
$M_w = 11000$	DMF (cold)		516
$M_w = 12000$	DMF (hot)		516
$M_w = 27000$		DMF (hot)	516
Poly(benzothiazoles)			
			
R = 1,4-phenylene	Methanesulfonic acid		517
R = 5-alkoxy-1,3-phenylene	Methanesulfonic acid		433
R = 1,4-bicyclo[2.2.2]octane	Methanesulfonic acid, sulfuric acid		518
R = 1,4-phenylene and 1,4-phenyleneimino-1,4-phenylene	Methanesulfonic acid, polyphosphoric acid		519
			
R = 1,3-phenylene	Conc. sulfuric acid		520
R = 3,5-pyridinediyl	Conc. sulfuric acid		520
R = 1,4-phenyleneoxy-1,4-phenylene	Conc. sulfuric acid		520
R = 1,4-phenylenecarbonyl-1,4-phenylene	Conc. sulfuric acid		520
R = 1,4-phenylene	Conc. sulfuric acid		500
R = 1,6-anthracenediyl	Sulfuric acid (partially)		521
Poly(2,6-benzothiazolediyl)	Sulfuric acid		498
5.3. POLY(IMIDES)			
5.3.1. POLY(BENZIMIDES)			
			
R = 1,4-phenylenecarbonyloxyethyleneoxy-ethyleneoxycarbonyl	Methylene chloride/trifluoroacetic acid (4/1)	Methanol	522
R = poly(isobutylene)iminocarbonyl	*n*-Butyl acetate, diethyl ether	Ethanol, methanol	313
R = 1,4-phenyleneoxy-1,4-phenyleneiminocarbonyl	NMP		523
R = 4,4′-biphenylyleneiminocarbonyl	NMP/xylene		524
R = 1,4-phenylenemethylene-1,4-phenyleneiminocarbonyl	DMA, DMSO, NMP		525,539
R = 1,4-phenylenecarbonyloxymethyleneoxycarbonyl	*m*-Cresol (150°C), trichlorobenzene (reflux)		526
R = 1,4-phenyleneoxy-1,4-phenyleneiminocarbonyl	DMA	Methanol	527
R = 1,4-phenyleneoxy-1,4-phenylenecarbonylimino-1,4-phenyleneoxy-1,4-phenyleneiminocarbonyl	DMA, NMP	Methanol	528
			

TABLE 5. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
R = 1,4-phenylene R′ = 5-oxy-1,4-phenyleneisopropylidene-1,4-phenylene-5′-oxy	DMA		529
R = hexamethylene R′ = 5,5′-oxy	Conc. sulfuric acid		530
R = 2,2′-di(trifluoromethyl)-4,4′-biphenylylene R′ = 5,5′-hexafluoroisopropylidene	Acetone, chloroform, *m*-cresol		531
R = 1,4-phenyleneoxy-1,4-phenylene R′ = 5,5′-phenylphosphineoxide	NMP		532
R = 2,2′-ditrifluoromethyl-4,4′-biphenylylene R′ = 5,5′-idyl	*m*-Cresol (hot)		533
R = 1,4-phenyleneoxy-1,4-phenylene R′ = 5,5′-carbonyl	DMA		534
R = 1,3-phenylene R′ = 5,5′-oxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy	NMP		535

5.3.2. POLY(PYROMELLITIMIDES)



R = nonamethylene	*m*-Cresol	Common organic solvents	13,540
R = 1,3-phenylene; 1,4-phenylene	Conc. sulfuric acid	Common organic solvents	13,540
R = 4,4′-biphenylylene	Fuming nitric acid	Common organic solvents	13,540
R = 1,4-phenylenemethylene-1,4-phenylene; 1,4-phenyleneisopropylidene-1,4-phenylene	Conc. sulfuric acid	Common organic solvents	13,540
R = 1,4-phenyleneoxy-1,4-phenylene amic acid	DMA, *N*-methylcaprolactone, fuming nitric acid	Common organic solvents	13,540, 541
	DMA, DMF, DMSO		542
R = 1,4-phenylenethio-1,4-phenylene	Fuming nitric acid	Common organic solvents	13,540
R = 1,4-phenylenesulfonyl-1,4-phenylene	Conc. sulfuric acid	Common organic solvents	13,540
R = 1,3-phenylenesulfonyl-1,3-methylene	Conc. sulfuric acid	Common organic solvents	13,540
R = pentaerythrityl	No solvent known		543
R = 4-hydroxy-1,3-phenyleneisopropylidene-4-hydroxy-1,3-phenylene	DMF, DMA, DMSO, nitrobenzene, tetramethylene sulfone		223
R = ethylidenecarbonyloxysebacoyloxy-carbonylethylidene, <40% imide	Chloroform		544
R = hexa-2,4-diyne-1,6-diyl	DMSO	W	545
R = 1,3-phenyleneoxy-1,3-phenyleneoxy-1,3-phenylene	DMA		525
R = 	DMA (partial), DMSO (partial), NMP (partial)		546

5.3.3. OTHERS

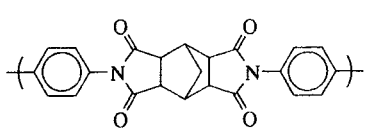
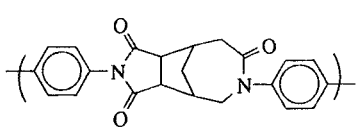
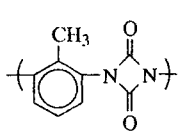
	Cresol, DMA, DMF, DMSO, NMP		536
	4-Butyrolactone		537
	No known solvent		538

TABLE 5. *cont'd*

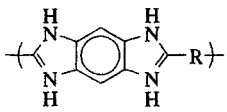
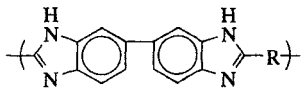
Polymer	Solvents	Nonsolvents	Refs.
5.4. POLY(QUINOXALINES)			
Poly(quinoxaline-2,6-diylquinoxaline-6,2-diyl-1,4-phenylene)	Sulfuric acid		547
Poly(quinoxalines) several different compositions	1,3-Dichloro-1,1,3,3-tetrafluoro-2,2-dihydroxypropane, methanesulfonic acid		548,549
5.5. POLY(BENZIMIDAZOLES)			
Poly(benzimidazoles) 			
R = tetramethylene	DMSO (partially), formic acid	DMF	550
R = 1,4-phenylene		DMF, DMSO, formic acid	550
R = 1,3-phenylene	Formic acid (partially)	DMF, DMSO	550
Poly(dibenzimidazoles) 			
R = tetramethylene to eicosane-1,20-diyl	*m*-Cresol, formic acid (85%), DMA, DMSO, NMP, conc. sulfuric acid	Glacial acetic acid, DMF (sw)	551
R = cyclohexylene	Acetic acid, *m*-cresol, DMF, DMSO, formic acid, NMP, sulfuric acid		501
R = 1,3-phenylene	DMA, DMSO, formic acid, NMP, conc. sulfuric acid	Glacial acetic acid, *m*-cresol, DMF (partially), formic acid (85%, sw)	550,551
R = 1,4-phenylene	Formic acid (2–3% soluble)	DMF, DMSO	550,551
R = 2,2′-biphenylylene	DMF, DMSO, formic acid		550,551
R = 4,4′-biphenylylene	Formic acid (partially)	DMF, DMSO	550,551
R = 3,5-pyridyenediyl	DMF (partially), DMSO, formic acid (partially)		550,551
R = 2,5-furandiyl	DMF, DMSO, formic acid (partially)		550,551
R = 1,4-phenyleneiminocarbonyl-R′-carbonylimino-1,4-phenylene; R′ = octamethylene; 1,4-phenylene; 1,3-phenylene	DMA, DMF, DMSO, NMP, conc. sulfuric acid	Glacial acetic acid (sw), *m*-cresol (sw), formic acid (sw), methanol	550,551
5.6. POLY(PIPERAZINES)			
Poly(2,5-diketo-1,4-piperazinediyl-1,4-phenylene sulfonyl-1,4-phenylene)	Dichloroacetic acid, formic acid, trifluoroacetic acid	DMA, DMF, DMSO	552
Poly(2,5-dimethyl-1,4-piperazinediylcarbonyl-1,4-phenylenesulfonyl-1,4-phenylene)	DMF, *sym*-tetrachloroethane/phenol (2/3)	Methanol, W	13
Poly(1,4-piperazinediylcarbonylethylene-1,4-phenylenephosphinylidene-1,4-phenyleneethylenecarbonyl)	Chloroform, cresol, DMF, ethanol, C-7 fluoroalcohol, formic acid, aqu. hydrochloric acid (5%)	Acetone, acetonitrile (sw), chlorobenzene, hexane	436
Poly(piperazinediylcarbonyloxy-*cis*-cyclohexyleneoxycarbonyl)	Chloroform, *m*-cresol, formic acid		323
Poly(1,4-piperazinediylcarbonyl-2,5-pyridinediylcarbonyl)	*m*-cresol, formic acid, conc. aqu. HCl, phenol, sulfuric acid		553
Poly(1,4-piperazinediyl-2-hydroxytrimethyleneoxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy-1-hydroxyethylene)	DMF		554
Poly(1,4-piperazinediylcarbonyl-2,5-norbornadiene-2,3-diylcarbonyl)	Chloroform, *o*-cresol	Methanol	555

TABLE 5. *cont'd*

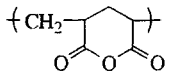
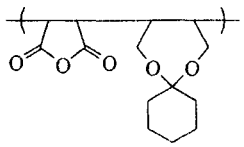
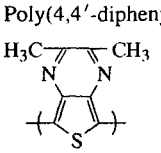
Polymer	Solvents	Nonsolvents	Refs.
5.7. POLY(ANHYDRIDES)			
Poly(maleic anhydride)	Acetic anhydride, acetonitrile, acetophenone, lower aliphatic alcohols, butanone, dioxane, DMF, ethers, ketones, nitroparaffins	Hydrocarbons, most chlorinated hydrocarbons	556
Poly(acrylic anhydride)	DMF, DMSO	Alcohols, bzn., diethyl ether	13
			
Poly(methacrylic anhydride)	DMF, DMSO	Aliphatic hydrocarbons, diethyl ether, methanol, W	557,558
Poly(itaconic acid anhydride)	Acetic acid, acetone, cyclohexanone, THF	Chloroform, diethyl ether	687,700
Poly(maleic anhydride-*alt*-2-cyclohexylene-1,3-dioxepin-5-ene)	THF	Hexenes	559
			
Poly(maleic anhydride-*alt*-isopropenylisocyanate)	Acetone, DMF		560
Poly(maleic anhydride-*alt*-2-propenylisocyanate)	MEK	Bzn.	561
Poly(maleic anhydride-*co*-styrene) (3 : 1)	Acetone, MEK		153
Poly(maleic anhydride-*alt*-methyl vinyl ether)	MEK, THF	Carbon tetrachloride, diethyl ether, hexane	186
Poly(maleic anhydride-*alt*-1-octadecene)	Acetone, bzn., carbon tetrachloride, ethyl acetate	Ethanol	562
Poly(maleic anhydride-*co*-*N*-vinylformamide) (1 : 1)	THF		563
5.8. POLY(THIOPHENES)			
Poly(3-alkyl-2,5-thiophenediyl)	Chloroform, methylene chloride, THF (partially)		564
Poly(3-hexyl-2,5-thiophenediyl)	THF, toluene		7
Poly(3-hexyloxycarbonyl-2,5-thiophenediyl)	Chloroform, DMF	Methanol	565
Poly(3,4-oxyethyleneoxy-2,5-thiophenediyl)	Chloroform	Methanol	566
Poly(2,5-thiophenediyl-*co*-3-undecyl-2,5-thiophenediyl) (1 : 1)	Nitrobenzene, THF	Methanol	567
Poly(3-methyl-2,5-thiophenediyl-*co*-3-oligodimethylsiloxane-2,5-thiophenediyl)	Chloroform		568
Poly(2,5-thiophenediylethylene)	THF		569
Poly(2,5-thiophenediyl-*co*-dimethylsilylene) (3 : 1 to 3 : 2)	Dimethoxyethane	Methanol	570
Poly(2,5-thiophenediyl-*co*-*N*-*n*-dodecyl-2,5-pyrrolediyl) (4 : 1)	THF		571
Poly(2,5-thiophenediyl-1,2-dimethoxyethenylene)	Chloroform	Hexane	572
Poly(2,5-thiophenediylethynylene-2,5-dioctyloxy-1,4-phenylene)	Chloroform, THF, toluene		573
Poly(2,5-thiophenediylethynylene-1,4-tetrafluoro phenyleneethynylene)	Chloroform, THF, toluene		573
Poly(thieno[3.2-b]thiophene-2,5-diyl)	Chloroform	Methanol	574
Poly(2,5-thiophenediylcarbonyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy)	Chloroform, DMA, NMP	Methanol	575–577
Poly(4,4′-diphenyl-2,2′-bisthiazole-5,5′-diyl)	Hexane	Methanol	578
	Chloroform		579
5.9. OTHERS			
Poly(2,3-dihydrofuran-3,4-diyl)	Toluene	Methanol	580
Poly(2,5-furandiylethylene)	THF		569
Poly(3,5-dicarboxytetrahydropyran-3,5-diylmethylene)	DMSO	Acetone	581

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TABLE 5. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Poly(*N*-*n*-butylitaconimide)	Bzn., THF, toluene		7
Poly(*N*-phenylmaleimide)	DMF, DMSO, THF	Methanol	582
Poly(2,5-pyridinediylethenylene)	Formic acid, sulfuric acid		583
Poly(*N*-2-ethylhexyl-3,6-carbazolediylethenylene)	1 M LiCl in NMP	Methanol	584
Poly(*N*-methyleneimidazoleimide-1,5-diyl)	DMA (partially), sulfuric acid (partially)		585
Poly(3-phenyl-3,5-triazolediyl-1,4-phenyleneoxy-1,4-phenyleneisopropylidene-1,4-phenyleneoxy-1,4-phenylene)	Toluene	Methanol	586
Poly[3-(2′-pyridyl)-1,2,4-triazine-5,6-diyl-1,4-phenyleneoxy-4,4′-biphenylyleneoxy-1,4-phenylene]	Chlorobenzene, NMP	Methanol	587
Poly(3,5-pyrazolediyl-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenyleneoxy-1,4-phenylene)	DMA, NMP	Chloroform, methanol	588

TABLE 6. FORMALDEHYDE RESINS

Polymer	Solvents	Nonsolvents	Refs.
Aniline-formaldehyde resins			
low mol. wt.	Methylcyclohexanone	Alcohols, aniline (sw), hydrocarbons	6
high mol. wt.	No solvent known	Aniline, chlorinated hydrocarbons (sw)	6
Melamine-formaldehyde resins			
very low mol. wt.	Alcohol, W		6
intermediate mol. wt.	Dil. and conc. acids, formalin, formic acids, pyridine		6
final resin, high mol. wt.	No solvent known		6
Phenol-formaldehyde resins			
Novolaks and low mol. wt.	Acetone, diethyl ether, esters, hydrocarbons		6
4-*tert*-butylphenol and 4-phenylphenol polymers	Drying oils		6
final resins	Molten phenols (with some decomposition)	Most common organic compounds	6
fully cured resins		Dil. acids (sw), dil. alkalies (sw), alcohol, W (sw)	6
p-Toluenesulfonamide-formaldehyde resins	Alcohols, bzn., esters, ethers, ketones	Aliphatic hydrocarbons, W	6,589
Urea-formaldehyde resins			
very low mol. wt.	Alcohol, W		6
intermediate mol. wt.	Dil. and conc. acids, formalin, formic acid, pyridine		6
final resin, high mol. wt.	No solvent known		6
Phenyl glycidyl ether-formaldehyde resins, low mol. wt.	Acetone, MEK, methyl isobutyl ketone		590

TABLE 7. NATURAL POLYMERS AND MODIFIED NATURAL POLYMERS

Polymer	Solvents	Nonsolvents	Refs.
7.1. NATURAL RUBBER AND DERIVATIVES (see also 1.1)			
Natural rubber	Bzn., chlorinated hydrocarbons, cyclohexane, hydrocarbons, methyl isobutyl ketone (low mol. wt.), *n*-propyl ketone (above 14.5°C), toluene (low mol. wt.)	Acetone, alcohols, carboxylic acids	4,5,7, 591
cyclic	Aromatic and chlorinated hydrocarbons	Acetone, acids, alcohol, alkalies, diethyl ether, aliphatic hydrocarbons, oils (sw), W	6
hydrochlorinated	Chlorinated hydrocarbons	Alcohol, W	6
chlorinated	Aromatic and chlorinated hydrocarbons, hydrogenated naphthas	Conc. acids, oxidizing acids, acetone (sw), alcohol, moderately conc. alkalies, aliphatic hydrocarbons, diethyl ether (sw), W	6
dichlorocarbene modified	Chloroform	Methanol	592

TABLE 7. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
7.2. GUTTA PERCHA			
Gutta percha	Bzn., cloroform, hot petroleum ether	Alcohol, W	6
7.3. CELLULOSE AND DERIVATIVES			
7.3.1. CELLULOSE			
Cellulose	*N*-Ethylpyridinium chloride/DMF (1/1), *N*-ethylpyridinium chloride/pyridine (1/1), trifluoroacetic acid, aqu. solutions of: alkalies (ice-cold), beryllium perchlorate, calcium thiocyanate, cupriammonium hydroxide, cupriethylenediamine, sodium xanthate, tetramethylammonium hydroxide, zinc chloride/hydrochloric acid (cold), zinc chloride (hot), conc. phosphoric acid, conc. sulfuric acid. See also table on cellulose in this Handbook	Dil. aqu. alkalies, hydrocarbons, mineral oils, W (sw)	4–6, 593,594
7.3.2. CELLULOSE ETHERS			
Methyl cellulose			
SC = 3–10%	Aqu. alkalies	W	6,595
= 22–32%	Aqu. alkali, *N*-ethylpyridinium chloride/DMF, methanol/methylene chloride, W (cold)	Diethyl ether, methanol, methylene chloride, W (hot)	6,594,595
= >40%	Acetone, chloroform, cyclohexanone, esters, pyridine	Aliphatic hydrocarbons, diethyl ether, ethanol, W	6,595
Ethyl cellulose			
DS = 0.5–0.7	Aqu. alkali	W (cold)	5,6,30, 595,596
= 1.0–1.5	Acetic acid, cuoxam, formic acid, pyridine, W (cold)	Ethanol	
= 2	Chloroform, chlorohydrins, dichloroethylene, ethanol, methylene chloride, THF	Alcohols, diethyl ether, esters, hydrocarbons, ketones, tetrachloroethylene, W	
= 2.3	Acetic esters, alcohols, alkyl halides, bzn., carbon disulfide, furan derivatives, ketones, nitromethane	Acetone (cold), ethylene glycol	
= 3	Alcohols, bzn., esters, methylene chloride	Carbon tetrachloride, diols, di-*n*-propyl ether, hydrocarbons, tetrahydrofurfuryl alcohol	
Isopropyl cellulose			
DS = 0.5	W (cold)	W (hot)	595
= 2.5	Acetone, butanol, bzn., chloroform, cyclohexanone, dichloroethane, dioxane, glycol formal, methyl acetate, methylcyclohexanone, THF	Ethanol, ethanol/diethylether (1/3), hydrocarbons, methanol, propanol	30
Butyl cellulose			
DS = 0.8	Polar organic solvents	Hydrocarbons, W	595
= 1.9	Aromatic hydrocarbons, polar organic solvents	Aliphatic hydrocarbons, W	595
Benzyl cellulose			
DS = 1.65	Methylene chloride, methyl ether glycol acetate, nitrobenzene	Bzn., chlorobenzene, butyl acetate	210
= 2	Acetates (<C5), acetone, bzn./acetone, bzn./alcohol, chlorobenzene, chlorohydrins, dichloroethylene, ethanol/carbon tetrachloride (1/1), MEK, methylene chloride, pyridine, tetrahydrofurfuryl alcohol, tetralin, trichloroethylene	Alcohols, (>C5), carbon tetrachloride, *o*-chlorotoluene, diethyl ether, dioxane, dipropyl ketone, hydrocarbons	6,30

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TABLE 7. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
DS = 3	Acetone, butyl acetate, chlorobenzene, chloroform, cyclohexanone, ethyl acetate, MEK	Aliphatic hydrocarbons, aromatic hydrocarbons (sw), carbon tetrachloride, diethyl ether, dioxane, ethanol	199
Hydroxyethyl cellulose			
SC = < 30%	Alkali		
= 35–55%	Ethylene chlorohydrin, DMSO, formic acid, W	Cellosolve®, hexane, methanol, toluene	597
= > 55%	Bzn./acetone, bzn./alcohol, chloroform, pyridine, esters	Acetone, bzn., methanol (sw), methyl acetate, THF	
Hydroxypropyl cellulose	Cellosolve®, dioxane, ethanol, methanol, W (< 45°C)	W (> 45°C)	597
Carboxymethyl cellulose			
SC = 5–10%	Alkali		
= 15–30%	W (sodium salt)		
= high	Acetone, bzn./acetone, bzn./alcohol, chloroform, esters, pyridine		
Cyanoethyl cellulose			6
SC = 8–12%	Alkali		
= 24–32%	W		
= > 50%	Acetone, bzn./acetone, bzn./alcohol, chloroform, esters, pyridine		
7.3.3. CELLULOSE ESTERS			
Cellulose triformate	Formic acid, pyridine		30
Cellulose acetate			
DS = 0.6–0.8	W		
= 1.3–1.7	2-Methoxyethanol	Acetone, W	595
= 2–2.3	Acetic acid (glacial), acetone, aniline, benzyl alcohol, chloroform/methanol, cyclohexanone, diethanolamine, dioxane, ethyl acetate/nitrobenzene, ethylene glycol ethers, formic acid, glycol monoethyl ether acetate, methyl acetate, methylene chloride/methanol (4/1), nitromethane, phenols, pyridine	Aliphatic ethers, hydrocarbons, weak mineral acids	4,6
Cellulose triacetate	Acetone/W (4/1), chloroform/alcohol, chloroform, dioxane, ethyl acetate, ethylene glycol ether acetates, ethylene carbonate, methyl acetate, methylene chloride/ethanol (4/1), methylene chloride, THF, trichloroethane	Acetone (partially), aliphatic hydrocarbons, aliphatic ethers, bzn., chlorobenzene, *o*-chlorotoluene, dichloroethane, ethanol (absolute), glycol ethers, methyl *i*-butyl ketone, weak mineral acids	4–6,30, 598,599
Cellulose acetate hydrogen phthalate	Acetone		32
Cellulose tripropionate	Acetone, bzn., chlorobenzene, dichloroethane, ethyl acetate	*o*-Chlorotoluene, ethanol, hydrocarbons	30
Cellulose tributyrate	Acetone, amyl acetate, bzn., chlorobenzene, chloroform, cyclohexanone, dodecane/tetralin (3/1 above 130°C), ethanol (hot), ethyl acetate, nitrobenzene, THF, xylene (hot)	2-Ethylhexanol, cyclohexane, diethyl ether, hexane, methanol	595
Cellulose tricaproate	Acetone, bzn., chlorobenzene, chloroform, cyclohexanone, diethyl ether, dioxane, DMF, ethanol (hot), ethyl acetate, glacial acetic acid (hot), hexane, nitroethane, THF, xylene	2-Hydroxyethyl acetate, formamide, methanol, nitromethane	595
Cellulose tricaprylate	DMF (above 140°C), 3-phenylpropanol (above 48°C), toluene		5
Cellulose stearate	bzn., tetrachloroethane, toluene	Acetone, butyl acetate, dichloroethane, ethyl acetate	595
DS = 3			

TABLE 7. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Cellulose acetate butyrate			
DS (acetate) = 0.5 DS (butyrate) = 2.35	Acetone, amyl acetate, bzn., chloroform, cyclohexanone, dioxane, esters, methanol (hot), nitroethane, tetrachloroethane, toluene (hot)	Aliphatic hydrocarbons, diethyl ether, ethanol, methanol	595
DS (acetate) = 2.1 DS (butyrate) = 0.7	Acetone, chloroform, cyclohexanone, dichloroethane, dioxane, ethyl acetate, nitroethane	Amyl acetate, bzn./methanol (1/4 and 4/1), carbon tetrachloride, diethyl ether, ethanol, hydrocarbons, methanol, W	595
Cellulose tricarbanilate	Acetone, bzn., chloroform, dioxane, DMF, *N*-ethylpyridinium chloride/DMSO (1/1), *N*-ethylpyridinium chloride/pyridine (1/1), pyridine, THF	Methanol, W	7,594, 595
Propyl cellulose benzoate	Acetone, bzn., chloroform, dioxane, DMA, ethyl acetate, pyridine, TMF, toluene, xylene		600
Hydroxypropyl cellulose perfluorooctanoate DS = 3 (ester), DS = 4 (ether)	Acetone	Ethanol	601
7.3.4. CELLULOSE NITRATE AND SULFATE			
Cellulose nitrate			
N = 6.8%	W		602
N = 10.5–12%	Acetic acid (glacial), acetone, alcohol/diethyl ether, alcohol (lower), amyl acetate, ethylene glycol ethers	Higher alcohols, higher carboxylic acids, higher ketones, tricresyl phosphate	4,6,595
N = 12.7%	Acetone, cyclohexanone, ethanol/diethyl ether, ethyl lactate, ethyl butyrate, ethylene carbonate, ethylene glycol ether acetates, furan derivatives, halogenated hydrocarbons, methyl acetate, methyl amyl ketone, nitrobenzene	Dil. carboxylic acids, aliphatic hydrocarbons, aromatic hydrocarbons (sw), diethyl ether, ethylene glycol, higher alcohols (sw), lower alcohols, W	4–6,595
Cellulose sulfate, sodium salt			
DS = 1	W	Ethanol	595
7.4. STARCH AND DERIVATIVES			
Amylose	Ethylenediamine, *N*-ethylpyridinium chloride/pyridine (1/1), DMSO, nitromethane, W (hot), aqu. KOH, aqu. KCl (0.33 M, above 25°C)	*n*-Butanol, diethyl ether	5,7,363,594,595,603
Amylose acetate DS = 2	Chloroform, methyl acetate, nitromethane	W	5
Amylose tricarbanilate	Acetone, dioxane, pyridine		5
Amylopectin	W, W/butanol	Acetone, diethyl ether, ethanol	604
Benzyl amylose DS = 3	Bzn., chloroform, dioxane, DMF, DMSO, THF	Ethanol, methanol, W	605
Starch	Liquid ammonia, aqu. sol. of cupriethylenediamine, fused chloral hydrate, *N*-ethylpyridinium chloride/pyridine (1/1), W (hot, pressure)	Alkali (sw), diethyl ether, W (sw)	6,363, 594,595
Starch, methyl ether			
DS = 0.5–2	W (cold)	W (hot)	595
DS = 3	Chloroform		595
7.5. OTHER POLY(SACCHARIDES)			
Alginic acid	Aqu. alkalies, aqu. sodium carbonate, W (sparingly)	Dil. acids, aqu. calcium salt solutions, hydrophilic organic compounds (sw), hydrophobic organic compounds	6,363
alkali and ammonium salts	Aqu. alkali, W	Organic hydrophobic compounds	6
magnesium and ferrous salts	W	Organic hydrophobic compounds	

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TABLE 7. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
calcium salts	Weak aqu. alkali	Organic hydrophobic compounds, W	
copper, zinc, aluminum salts	Aqu. ammonium hydroxide	Organic hydrophobic compounds, W	
beryllium, chromium, ferric salts		Aqu. alkali, organic hydrophobic compounds	
ι-Carrageenan	W		694
Chitin	Formic acid, conc. mineral acids (degrad.)	Dil. acids, organic solvents	363
Chitosan	Acetic acid, W (pH <6.5)		606
N-[(3′-hydroxy-2′,3′-dicarboxy)ethyl]chitosan	W		607
Glycogen	Dil. trichloroacetic acid	Alcohol	363
Gum arabic	W (warm)	Alcohol	363
Gum tragacanth, Tragacanthin, tragacanthic acid	W	Acetone, alcohol	363
2,3-Di-O-benzyl-(1,5)-α-D-furanan	*o*-Dichlorobenzene (hot)	Acetone, cyclohexanone, ethyl acetate, hexane, methanol	608
Heparin	Dil. alkali solution, alkaline solution of ammonium sulfate, W	Acetic acid, acetone, alcohol	363
Konjac glucomannan	W		609
Pectin	Dil. acids, W	Alcohol	363
Pullulan	W		609
Xanthan gum	W		610
7.6. NATURAL RESINS			
Colophony	Acetone, aromatic hydrocarbons (partially), carbon tetrachloride, chloroform, diethyl ether, ethanol, ethyl acetate, methanol	Aliphatic hydrocarbons	30
Copal	Acetone (25–90%)[a], bzn. (35–60%)[a], carbon tetrachloride (15–40%)[a], chloroform (30–90%)[a], diethyl ether (10–90%)[a], ethanol (20–100%)[a], hydrocarbons (10–55%)[a], methanol (20–70%)[a]		30
Shellac	Acetone (50–80%)*, bzn. (10–20%)[a], carbon tetrachloride (5–15%)[a], chloroform (25–40%)[a], diethyl ether (10–25%)[a], ethanol (85–98%)[a], methanol (100%)	Aliphatic hydrocarbons	30
Zein	Aqu. alcohols, glycols, DMF		32

[a] Percentage soluble, depending on source.

TABLE 8. INORGANIC POLYMERS

Polymer	Solvents	Nonsolvents	Refs.
Poly(phosphorus nitrile chloride)	Bzn., carbon tetrachloride, chloroform, cyclohexane	Aliphatic hydrocarbons	611
Poly(sodium-*m*-phosphate)	Aqu. sol. of LiCl, NaCl or NH_4Cl	Acetone, ethanol, W	612
Poly(tri-*n*-butyl tin fluoride)	Aliphatic hydrocarbons		259
Poly(phosphoryldimethylamide)	Carbon tetrachloride (hot), W	Acetone	613
Poly(2,5-selenienylene)	Chloroform, THF	Methanol, W	614
Poly[η^5-(vinylcyclopentadienyl)tricarbonylmethyltungsten]	Bzn., chloroform, THF	Hexane	615
Poly(*n*-dibutyl stannane)	Chloroform, THF, toluene	Methanol	616

TABLE 9. POLY(ELECTROLYTES)

Polymer	Solvents	Nonsolvents	Refs.
Poly(allyammonium chloride)	W	Methanol	617
Poly(allylammonium phosphate)	HCl, W	Methanol, W	617
Poly(diallyldimethylammonium chloride)	W		618
Poly(benzyltrimethylammonium chloride)	W		32
Poly(diallyldimethylammonium chloride-*co*-*N*-isopropylacrylamide), (2 : 98)	W		619
Poly(sodium trimethyleneoxyethylenesulfonate)	DMSO, W	Bzn., hexane	620
Poly[dimethyldodecyl(2-acrylamidoethyl) ammonium bromide]	W		701
Poly(4-*N*-butylpyridiniumethylene bromide)	Benzyl alcohol, DMSO, nitromethane, W	Bzn., hexane	621
Poly(2-*N*-methylpyridiniumethylene iodine)	Methanol, W	Methylene chloride	687
Poly(*N*-methylpyridinium-2,5-diylethenylene)	DMF, DMSO, NMP, W		583
Poly(4,4′-bipyridiniumdiyl-*N*,*N*′-decamethylene dibromide)	DMF/methanol, W	*tert*-Butyl methyl ether	622
Poly(1,4-pyridiniumdiyl-*N*,*N*′-tetramethylene-4,1-pyridiniumdiylethenylene methylsulfate)	Ethanol	Ethyl acetate	623
Poly(2,6-diphenyl-1,4-pyridiniumdiyl-*N*,*N*′-1,4-phenylene-4,1-pyridiniumdiyl-1,4-phenylene)	DMA, DMF, DMSO, NMP	Diethyl ether	624
Poly[ethyldipropargyl-4-sulfobutyl ammonium betaine]	Formic acid, W	Hexane, methanol, THF, toluene	625

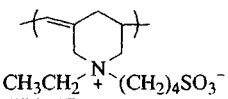

TABLE 10. BLOCK COPOLYMERS
(alphabetically listed)

Polymer	Solvents	Nonsolvents	Refs.
Poly(*t*-butyl methacrylate)-b-polystyrene-b-poly(*t*-butyl methacrylate) (1 : 6 : 1)	Dioxane	Hexane	626
Poly(*t*-butyl methacrylate)-b-poly(oxytetramethylene)-b-poly(butyl methacrylate) (2 : 6 : 2 to 4 : 2 : 4)	Toluene		627
Poly(butyl acrylate)-b-poly[3-(pentamethyldisiloxy)propyl methacrylate] (3 : 1)	THF		628
Poly(butyl vinyl ether)-b-poly(2-ethyloxazoline)	Chlorobenzene, THF	Hexanes	629
Poly(2-cinnamoylethyl methacrylate)-b-poly(acrylic acid) (4 : 5)	Methanol, methylene chloride	Isopropanol/hexane (3/7)	630
Poly(ε-caprolactone)-b-poly(oxytetramethylene)-b-poly(ε-caprolactone) (4 : 2 : 4 to 1 : 3 : 1)	THF, toluene		631
Poly(dimethylsiloxane)-b-poly(6-aminohexanoic acid) (1 : 2)	2,2,2-Trifluoroethanol		632
Poly(2-ethyl-2-oxazoline)-b-poly(lactic acid) (3 : 2 to 2 : 4), Poly(*N*-proponyliminoethylene)-b-poly(oxy-1-oxo-2-methylethylene)	Chlorobenzene	Hexane	633
Poly(3-hydroxybutyric acid)-b-poly(ε-caprolactone) (3 : 5 to 2 : 1)	Chloroform	Methanol	634
Polyisobutylene-b-poly(α-methylstyrene) (9 : 1)	Chloroform	Methanol	635
Poly(methacrylic acid)-b-polystyrene-b-poly(methacrylic acid) (1 : 8 : 1)	Dioxane	Hexane	626
Poly(3-methyl-2,5-thiophenediyl)-b-poly(2-vinylpyrrolidone)	Anisole		636
Poly(oxy-2,2-di(chloromethyl)trimethlylene)-b-poly(oxytetramethylene)-b-polyoxy-2,2-di(chloromethyl)trimethylene (7 : 86 : 7)	Methylene chloride		637
Poly(oxytetramethylene)-b-poly(ε-caprolactone) (1 : 4)	THF, toluene		638
Polystyrene-b-polybutadiene-b-polystyrene) (7 : 86 : 7 to 15 : 70 : 15)	Toluene		639
Polystyrene-b-poly(*tert*-butyl methacrylate) (2 : 3 to 4 : 1)	Dioxane	Hexane	626
Polystyrene-b-poly(dimethylsiloxane) (1 : 1)	THF	Methanol	350
(1 : 9 to 7 : 3)	Bzn./2-methoxyethyl ether, dioxane	Methanol, W	640

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TABLE 10. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
Polystyrene-b-poly(ethylene/butylene)-b-polystyrene (1 : 8 : 1 to 1 : 3 : 1)	Toluene		641,642
Polystyrene-b-poly(ethylene/propylene) (3 : 7)	THF		643
Polystyrene-b-(*N*-ethyl-2-ethynylpyridinium trifluoromethanesulfonate) (8 : 1)	Diethyl ether/petroleum ether (1/1), THF		644
Polystyrene-b-poly(hexyl isocyanate) (1 : 4)	Carbon tetrachloride, chloroform, toluene		645
Polystyrene-b-poly(hydroxyethyl methacrylate) (2 : 3)	DMF, THF		646
Polystyrene-b-polyisoprene (1 : 1)	Dioctyl phthalate		647
Polystyrene-b-polyisoprene-b-polystyrene (1 : 12 : 1 to 2 : 1 : 2)	MEK/isobutyl isobutyrate, methylcylcohexane/methylene chloride (3/2), methyl amyl ketone, methylene chloride, xylene		639, 548–650
Polystyrene-b-poly(methyl methacrylate) (7 : 3)	Toluene		7,698
Polystyrene-b-poly(methylphenylsiloxane) (3 : 2)	Cyclohexane		651
Polystyrene-b-poly(methyl vinyl ether) (2 : 1)	Methylene chloride		652
Polystyrene-b-poly(4-vinylpyridine) (5 : 1)	THF	Hexanes	653
Polystyrene-b-poly(2-vinylpyridine) (1 : 1 to 6 : 1)	Bzn.	Hexane/methanol (8/2)	654
Polystyrene-b-poly(vinylpyridine) (1 : 1)	THF	Hexane	655
Polypropylene-b-poly(methacrylic acid) (2 : 1)	Dioxane, THF, toluene	Methanol	653,626

TABLE 11. DENDRIMERS

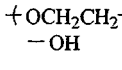
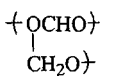
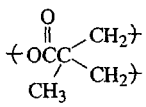
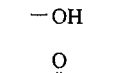
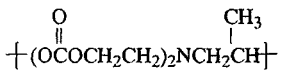
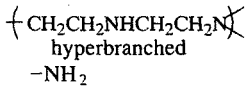
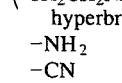
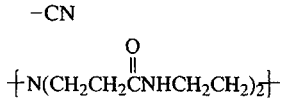
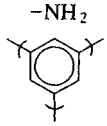
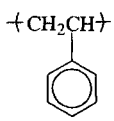
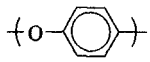
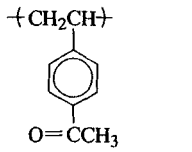
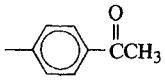
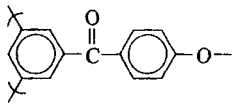
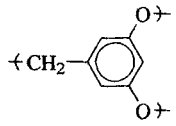
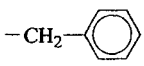
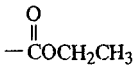
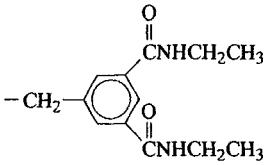
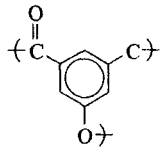
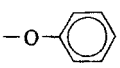
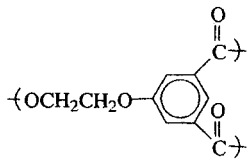
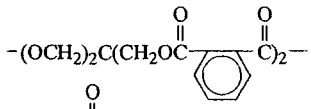
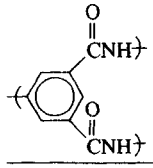
Polymer	Solvents	Nonsolvents	Refs.
	DMSO, methanol, THF (partially), W		656
	THF		657
	Methylene chloride, most organic solvents	Hexane, W	658,684
	Acetone, DMSO		689
	Toluene		659
 hyperbranched	Lower alcohols, ethyl acetate, W	Hydrocarbons	683
	Alcohol, glycols, W		660
	Acetone, chloroform		
	W		661,685
	THF, toluene	Methanol	662
	Bzn., chloroform, cyclohexane, methylene chloride		287

TABLE 11. *cont'd*

Polymer	Solvents	Nonsolvents	Refs.
			
—I	THF	Methanol	663
			
	THF	Methanol	664
			
—OH, —F	DMF, NMP, THF		665
			666,667
	Chloroform	Methanol	668
	THF	Methanol	669
	THF		
	Ethyl acetate, chloroform, DMF, DMSO, methylene chloride, THF, toluene		670
—OH	DMF, DMSO, THF		671
	Methylene chloride		524
			
—OH	THF	Diethyl ether, W	672
	DMF		673
	NMP, THF		695

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D. REFERENCES

1. W. P. Baker, J. Polym. Sci. A, **1**, 655 (1963).
2. A. A. Arest-Yakubovich et al., ACS Polymer Preprints **37** (2), 634 (1996).
3. T. Yasuyuki et al., Macromolecules, **26**, 5253 (1993).
4. H. Dexheimer, O. Fuchs, in: R. R. Nitsche, K. A. Wolf, (Eds.), "Struktur und Physikalisches Verhalten der Kunststoffe", vol. 1, Springer, Heidelberg, 1961.
5. M. Kurata W. H. Stockmayer, in: Adv. Polym. Sci. vol. 3, Springer, Heidelberg, 1963, p. 196.
6. W. J. Roff, in: "Fibers, Plastics, and Rubbers", Academic Press, New York, 1956.
7. Y. Du et al., in: J. E. Mark (Ed.), "Physical Properties of Polymers Handbook", Ch. 17, AIP Press, New York, 1996.
8. L. van der Does, J. F. van Duijd et al., J. Polym. Sci. Chem. Ed., **17**, 53 (1980).
9. W. Marcony, A. Mazzei, S. Cucinella, M. J. Cesari, Polym. Sci. A, **2**, 4261 (1964).
10. Jr. J. M. Wilbur, C. S. Marvel, J. Polym. Sci. A, **2**, 4415 (1964).
11. T. A. Davidson et al., Macromolecules, **29**, 786 (1996).
12. R. F. Heck, D. S. Breslow, J. Polym. Sci., **41**, 521 (1959).
13. Houben-Weyl, in: "Methoden der Organischen Chemie", Band XIV/I, 2, 4, Auflage, ThiemeVerlag, Stuttgart, 1961.
14. Jr. C. A. Aufdermarsh, R. Pariser, J. Polym. Sci. A, **2**, 4727 (1964).
15. R. H. Michel, J. Polym. Sci. A, **2**, 2533 (1964).
16. Y. Sonoda et al., Macromolecules, **29**, 288 (1996).
17. J-i. Sugiyama et al., Macromolecules, **27**, 5543 (1994).
18. Y. Tabata, B. Saito, H. Shibano, H. Sobue, K. Oshima, Makromol. Chem., **76**, 89 (1964).
19. S. Kambara, H. Noguchi, Makromol. Chem., **73**, 24 (1964).
20. I. M. Paushkin, S. A. Nizova, J. Polym. Sci. A, **2**, 2783 (1964).
21. A. Teramoto, ACS Polymer Preprints, **31** (2), 135 (1990).
22. P. Teyssie, A. C. Korn-Girard, Polym. Sci. A, **2**, 2849 (1964).
23. H.-KK. Roth, H. Gruber, Synthetic Metals, **37**, 151 (1990).
24. W. H. Kim et al., Macromolecules, **27**, 1819 (1994).
25. M. Moroni, J. LeMoigne, Macromolecules, **27**, 562 (1994).
26. G. J. Tregre, L. J. Mathias, ACS Polymer Preprints, **36** (2), 310 (1995).
27. S. Dirlikow et al., ACS Polymer Preprints, **33** (1), 483 (1992).
28. E. W. Kwock et al., ACS Polymer Preprints, **34** (1), 900 (1993).
29. K. J. Saunders, "Organic Polymer Chemistry", 2nd ed., Chapman and Hall, New York, 1988, p. 56.
30. K. Thinius, in: "Analytische Chemie der Plaste", Springer, Heidelberg, 1952.
31. O. Fuchs, unpublished results.
32. Scientific Polymer Products, Inc. Catalog 1996/97, Ontario, NY.
33. K. J. Saunders, "Organic Polymer Chemistry", 2nd ed., Chapman and Hall, New York, 1988; p. 61.
34. Trade literature, AlliedSignal Inc., Morrestown, NJ.
35. S. Arichi, H. Matsuura, Y. Tamimoto, H., Murata, Bull. Chem. Soc. (Japan), **39**, 434 (1966).
36. S. Sanganeria, ACS Polymer Preprints, **34** (2), 378 (1993).
37. G. N. Babu et al., Macromolecules, **27**, 3383 (1994).
38. S. Saito, J. Polym. Sci. A-1, **7**, 1789 (1969).
39. H. G. Elias, O. Etter, J. Polym. Sci. A-1, **5**, 947 (1967).
40. W. H. McCarty, G. Parravano, J. Polym. Sci. A-1, **3**, 4029 (1965).
41. Y. Iwakura, K. Uno, K. Ichikawa, J. Polym. Sci. A, **2**, 3387 (1964).
42. W. Marconi, S. Cesca, G. Della Fortuna, J. Polym. Sci. B, **2**, 301 (1964).
43. G. Dall'Asta, G. Natta, G. Mazzanti, Chem. Eng. News, **42** (14), 42 (1964).
44. M. Perrott, B. M. Novak, Macromolecules, **29**, 1817 (1996).
45. I. Cho, J. Y. Lee, Makromol. Chem. Rapid Commun., **5**, 263 (1984).
46. A. Mukoh, H. Morishita, J. Polym. Sci. Letters, **18**, 35 (1980).
47. J. G. Hamilton, K. J. Ivin, J. J. Rooney, Brit. Polym. J., **16**, 21 (1984).
48. K. Su et al., ACS Polymer Preprints, **32** (3), 481 (1991).
49. K. M. Stein et al., ACS Polymer Preprints, **35** (1), 480 (1994).
50. T. C. Chung et al. Macromolecules, **27**, 7533 (1994).
51. C. D. Eisenbach, B. Sperlich, Macromolecules, **29**, 7748 (1996).
52. S. A. Haque et al., ACS Polym. Mater. Sci. Eng., **76**, 87 (1997).
53. A. Mogstad et al., ACS Polymer Preprints, **34** (1), 211 (1993).
54. P. J. Flory, J. E. Osterheld, J. Phys. Chem., **58**, 653 (1954).
55. M. L. Miller, K. O'Donnell, J. Skogman, J. Colloid Sci., **17**, 649 (1962).
56. E. M. Loebl, J. O'Neill, J. Polym. Sci., **45**, 538 (1960).
57. D. J. Diserow et al., ACS Polymer Preprints, **33** (1), 810 (1992).
58. H. Nakamoto, Y. Ogo, T. Imoto, Makromol. Chem., **111**, 104 (1968).
59. G. F. L. Ehlers, J. Polym. Sci. A, **2**, 4989 (1964).
60. S. Coca et al., ACS Polymer Preprints, **38** (1), 689 (1997).
61. J. Prager, R. M. McCurdy, G. B. Rathmann, J. Polym. Sci. A, **2**, 1941 (1964).
62. Y. Nakatsuji, S. Furuyoshi, M. Okahara, Makromol. Chem., **87**, 105 (1986).
63. C. Potin, A. Pleurdeau, C. M. Bruneau, Angew. Makromol. Chem., **138**, 75 (1986).
64. M. Kamachi, H. Akimoto, S. Nozakura, J. Polym. Sci. Letters Ed., **21**, 693 (1983).
65. J.-S. Kim et al., Macromolecules, **27**, 814 (1994).
66. C. Simionescu, D. Feldman, I. Theil, Plaste Kautschuk, **15**, 714 (1968).
67. D. E. Strain, R. G. Kennelly, H. R. Dittmar, Ind. Eng. Chem., **31**, 382 (1939).
68. Trade literature, ICI Acrylics, St. Louis, MO.

69. Trade literature, Rohm Tech, Malden, MA.
70. L. Gargallo, I. Mendez, D. Radic, Makromol. Chem., **184**, 1053 (1983).
71. I. Reynolds et al., Macromolecules, **28**, 7845 (1995).
72. M. Imoto, T. Otsu, K. Tsuda, T. Ito, J. Polym. Sci. A, **2**, 1407 (1964).
73. Y. Nagasaki et al., ACS Polymer Preprints, **38** (1), 514 (1997).
74. G. Natta, P. Longi, E. Pellino, Makromol. Chem., **71**, 212 (1964).
75. M. Ueda, T. Suzuki et al., J. Polym. Sci. Chem. Ed., **21**, 2997 (1983).
76. A. Natansohn et al., Macromolecules, **28**, 4179 (1995).
77. A. Furukawa, T. Nakaya, M. Imoto, Makromol. Chem., **187**, 311 (1986).
78. J. Springer, F. W. Weigelt, Makromol. Chem., **184**, 1489 (1983).
79. S. H. Yuk, ACS Polym. Mater. Sci. Eng., **76**, 231 (1997).
80. R. D Calleja et al., Macromolecules, **28**, 6963 (1995).
81. J. Velickovic, Vukajlovic-Filipovic, J. Angew. Makromol. Chem., **13**, 79 (1970).
82. J. Velickovic, J. Juranicova, J. Filipovic, IUPAC Intern. Symp. Macromolecules; Leiden, vol. I, 289 (1970).
83. J. Velickovic, M. Plavsic, Eur. Polym. J., **19**, 1171 (1983).
84. T. Otsu, K. Shiraishi, A. Matsumoto et al., Polym. Commun., **26**, 367 (1985).
85. N. Toyoda, T. Otsu, J. Macromol. Sci. A, **19**, 1011 (1983).
86. J. S. Scarpa, D. D. Mueller, I. M. Klotz, J. Am. Chem. Soc., **89**, 6024 (1967).
87. C. K. Chiklis, J. M. Grasshoff, J. Polym. Sci. A-2, **8**, 1617 (1970).
88. S. Nakahama et al., ACS Polym. Mater. Sci. Eng., **76**, 11 (1997).
89. L. E. Coleman, J. F. Bork, D. P. Wyman, ACS Polymer Preprints, **5**, 250 (1964).
90. T. Kodaira et al., Macromolecules, **27**, 1320 (1994).
91. R. Miyashita et al., Macromolecules, **27**, 513 (1994).
92. J. Heller, C. B. Kingsley, Makromol. Chem., **78**, 47 (1964).
93. O. Nuyken et al., ACS Polymer Preprints, **37** (1), 343 (1996).
94. A. Alegria et al., Macromolecules, **28**, 8819 (1995).
95. E. J. Vandenberg, R. F. Heck, D. S. Breslow, J. Polym. Sci., **41**, 519 (1959).
96. G. F. L. Ehlers, J. D. Ray, J. Polym. Sci. A, **2**, 4989 (1964).
97. C. E. Schildknecht, S. T. Gross, H. R. Davidson, J. M. Lambert, A. O. Zoss, Ind. Eng. Chem., **40**, 2104 (1948).
98. R. J. Kern, J. J. Hawkins, J. D. Calfee, Makromol. Chem., **66**, 127 (1963).
99. G. Natta, G. Mazzanti, P. Longi, F. Bernardini, J. Polym. Sci., **31**, 181 (1958).
100. H. Cramail, A. Deffieux, Macromolecules, **27**, 1401 (1994).
101. H. Yuki, K. Hatada, S. Murahashi, K. Hibino, Polym. J., **1**, 271 (1970).
102. D. D. Coffman, G. H. Kalb, A. B. Ness, J. Org. Chem., **13**, 223 (1948).
103. T. Yokozawa, E. Tsuruta, Macromolecules, **29**, 8053 (1996).
104. G. B. Dharas et al., ACS Polymer Preprints, **33** (1), 211 (1992).
105. K. Saunders, J. Organic Polymer Chemistry, 2nd ed., Chapman and Hall, New York, 1988, p. 121.
106. M. Sumi, K. Matsumura, R. Obno, S. Nozakura, S. Murahashi, Chem. High Polymers (Tokyo), **24**, 606 (1967).
107. D. Y. Sogah, J. F. Harris, ACS Polymer Preprints, **32** (1), 435 (1991).
108. B. M. Novak, A. K. Cederstav, ACS Polymer Preprints, **36** (1), 548 (1995).
109. Brochure "Mowital" of Hoechst AG, Frankfurt/M., 1971.
110. T. Hayashi et al., Macromolecules, **27**, 2270 (1994).
111. H. Rittner, R. Sperber, Macromolecules, **27**, 5919 (1994).
112. Skillicorn et al., J. of Vinyl Technology, **15** (2), 105 (1993).
113. R. A. Wessling, J. Appl. Polym. Sci., **14**, 1531 (1970).
114. R. A. Wessling, J. Appl. Polym. Sci., **14**, 2263 (1970).
115. G. H. Kalb, D. D. Coffman, T. A. Ford, F. L. Johnston, J. Appl. Polym. Sci., **4**, 55 (1960).
116. R. Advincula et al., ACS Polymer Preprints, **38** (1), 912 (1997).
117. Trade literature, Elf Atochem, Philadelphia, PA.
118. W. S. Durrell, G. Westmoreland, M. G. Moshonas, J. Polym. Sci. A, **3**, 2975 (1965).
119. E. L. Gal'perin et al., Vysokomol Soedin. A, **12**, 1654 (1970).
120. P. L. Rinaldi, ACS Polymer Preprints, **38** (1), 770 (1997).
121. H. T. Hall, J. Am. Chem. Soc., **71**, 68 (1952).
122. R. C. Doban, A. C. Knight, J. J. Peterson, C. A. Sperati, Am. Chem. Soc. Meeting Atlantic City, 1956.
123. N. K. I. Symons, J. Polym. Sci., **51**, S21 (1961).
124. D. McIntyre, D. R. Valentine, ACS Polymer Preprints, **7**, 1133 (1966).
125. R. E. Lowry, D. W. Brown, L. A. Wall, J. Polym. Sci. A-1, **4**, 2229 (1966).
126. D. W. Brown, R. E. Lowry, L. A. Wall, J. Polym. Sci. A-1, **8**, 2441 (1970).
127. D. Lopez et al., Macromolecules, **27**, 7415 (1994).
128. V. Panchalingam et al., ACS Polymer Preprints, **32** (3), 185 (1991).
129. Trade literature, 3M Industrial Chemicals, St. Paul, MN.
130. W. Tuminello et al., Macromolecules, **28**, 1506 (1995).
131. Dupont Specialty Polymers, Wilmington, DE.
132. S. M. Jo et al., ACS Polymer Preprints, **38** (1), 697 (1997).
133. Jr. W. W. Moyer, D. A. Grev, J. Polym. Sci. B, **1**, 29 (1963).
134. A. Yamane et al., Macromolecules, **30**, 4170 (1997).
135. M. Victor et al., ACS Polymer Preprints, **33** (1), 1162 (1992).
136. M. Ulbricht, G. Belfort, ACS Polymer Preprints, **36** (1), 135 (1995).
137. M. G. Mikhael et al., Macromolecules, **27**, 7499 (1994).
138. G. L. Eian, ACS Polymer Preprints, **36** (1), 459 (1995).
139. H. Hopff, J. Dohany, Makromol. Chem., **69**, 131 (1963).
140. H. Staudinger, M. Haeberle, Makromol. Chem., **9**, 52 (1953).
141. K. Schaefers, C. Schneider, J. Polym. Sci. Chem. Ed., **23**, 2879 (1985).

142. T. M. Pyriadi, ACS Polymer Preprints, **36** (1), 245 (1995).
143. R. C. Schulz, R. Wolf, Makromol. Chem., **99**, 76 (1966).
144. M. Tosaka et al., Macromolecules, **30**, 4132 (1997).
145. P. Lu et al., Macromolecules, **29**, 8583 (1996).
146. T. Ishizone et al., Macromolecules, **30**, 761 (1997).
147. P. Xing et al., Macromolecules, **30**, 2726 (1997).
148. Trade literature, Hoechst Celanese, Dallas, TX.
149. H. Kamogawa, H. G. Cassidy, J. Polym. Sci. A, **2**, 2409 (1964).
150. T. Ishizone, Macromolecules, **28**, 3787 (1995).
151. A. Hirao, Y. Ishino, S. Nakahama, Makromol. Chem., **187**, 141 (1986).
152. Y. Fu, R. A. Weiss, ACS Polymer Preprints, **38** (1), 270 (1997).
153. Trade literature, ALCO Chemical, Chattanooga, TN.
154. T. Ishizone et al., ACS Polymer Preprints, **35** (2), 540 (1994).
155. D. Liu et al., Macromolecules, **28**, 622 (1995).
156. B. D. Edgecombe et al., ACS Polym. Mater. Sci. Eng., **75**, 478 (1996).
157. R. F. M. Lange, E. W. Meijer, Macromolecules, **28**, 782 (1995).
158. R. A. Sanayei et al., Macromolecules, **27**, 5577 (1994).
159. J. B. Schlenoff et al., Macromolecules, **28**, 4290 (1995).
160. R. C. Schulz, J. Kovacs, W. Kern, Makromol. Chem., **52**, 236 (1962).
161. R. Hank, Makromol. Chem., **52**, 108 (1962).
162. H. Hopff, M. Lippay, Makromol. Chem., **66**, 157 (1963).
163. J. Moacanin, A. Rembaum, R. K. Laudenslager, ACS Polymer Preprints, **4**, 179 (1963).
164. G. Sitaramaiah, D. Jacobs, Polymer, **11**, 165 (1970).
165. J. Heller, D. J. Lymann, W. A. Hewett, Makromol. Chem., **73**, 48 (1964).
166. Y. L. Gefter, International Series of Monographs on Organic Chemistry, Vol. 6, "Organophosphorus Monomers and Polymers", Pergamon Press: New York, 1962.
167. G. B. Butler, Proc. Conf. "High Temperature Polymer and Fluid Research", Dayton, Ohio, 1962, p. 53.
168. J. Pielichowski, A. Romaniszyn, J. Polym. Sci. Letters Ed., **22**, 395 (1984).
169. A. T. Coscia, R. L. Kugel, J. Pellon, J. Polym. Sci., **55**, 303 (1961).
170. H. R. Allcock, J. Polym. Sci. A, **2**, 4087 (1964).
171. R. Rabinowitz, R. Marcus, J. Pellon, J. Polym. Sci. A, **2**, 1233 (1964).
172. Br. Patent 849038.
173. A. V. Topchiev, V. P. Alaniya, J. Polym. Sci. A, **1**, 559 (1963).
174. G. Natta, G. Mazzanti, G. Dall'Asta, P. Longi, Makromol. Chem., **37**, 160 (1960).
175. P. Grosius, Y. Gallot, A. Skoulios, Makromol. Chem., **136**, 191 (1970).
176. L. E. Coleman, J. F. Bork, J. Polym. Sci. A-1, **8**, 2073 (1970).
177. R. M. O'Malley et al., Macromolecules, **27**, 5107 (1994).
178. N. Gaylord, in: "Polyethers, Polyalkylene Sulfides and Other Polythioethers", High Polymers, vol. XIII, Part III, Interscience, New York-London, 1962.
179. Trade literature, Air Products and Chemicals, Allentown, PA.
180. K. Imai, T. Shiomi et al., J. Macromol. Sci. A, **22**, 1347 (1985).
181. A. Patil, ACS Polym. Mater. Sci. Eng., **75**, 287 (1996).
182. A. Bader, J. R. Wunsch, Macromolecules, **28**, 3794 (1995).
183. J. C. Hornby et al., Soap/Cosmetics/Chemical Specialities June 1993, p. 40.
184. L. A. Belfiore, ACS Polym. Mater. Sci. Eng., **76**, 97 (1997).
185. R. J. Badesso et al., ACS Polym. Mater. Sci. Eng., **69**, 251 (1993).
186. Trade literature, ISP Technologies, Wayne, NJ.
187. F. Shi, R. Subramanian, ACS Polymer Preprints, **37** (1), 637 (1996).
188. Q. Pei et al., ACS Polymer Preprints, **36** (1), 213 (1995).
189. T. Okada et al., Macromolecules, **29**, 7645 (1996).
190. T. F. McCarthy, Macromolecules, **28**, 8350 (1995).
191. R. Zhang et al., Macromolecules, **29**, 7627 (1996).
192. M. Pomerantz, J. Wang, ACS Polymer Preprints, **35** (1), 210 (1994).
193. Trade literature, Maxdem Inc., San Dimas, CA.
194. R. W. Phillips et al., Macromolecules, **27**, 2354 (1994).
195. H. O. Wirth, R. Mueller, W. Kern, Makromol. Chem., **77**, 90 (1964).
196. A. J. Pasquale et al., ACS Polymer Preprints, **38** (1), 170 (1997).
197. C. Aso, Y. Aito, Makromol. Chem., **58**, 195 (1962).
198. L. D. Taylor, H. S. Kolesinski, J. Polym. Sci. B, **1**, 117 (1963).
199. J. A. Moore, ACS Polymer Preprints, **5**, 250 (1964).
200. J. R. Schaefgen, J. Polym. Sci., **41**, 133 (1959).
201. B. R. Hsich, ACS Polym. Mater. Sci. Eng., **67**, 252 (1992).
202. W. Chen et al., ACS Polymer Preprints, **36** (1), 495 (1995).
203. M. Pan et al., ACS Polymer Preprints, **36** (1), 618 (1995).
204. J. L. Reddiger, J. R. Reynolds, ACS Polymer Preprints, **37**(1), 530 (1996).
205. Y. Fu, ACS Polymer Preprints, **38** (1), 410 (1997).
206. M. Wagaman, R. H. Grubbs, Macromolecules, **30**, 3978 (1997).
207. Y. Baek, G. I. Stegeman, ACS Polym. Mater. Sci. Eng., **75**, 221 (1996).
208. J.-I. Jin et al., Macromolecules, **27**, 5239 (1994).
209. L. Pu et. al., Macromolecules, **29**, 1138 (1996).
210. C.-S. Wang, Y.-M. Sun, ACS Polymer Preprints, **36** (1), 197 (1995).
211. R. Kulkarni, ACS Polymer Preprints, **32** (1), 128 (1991).
212. J. Asrar, Macromolecules, **27**, 4036 (1994).
213. D. A. Frey, M. Hasegawa, C. S. Marvel, J. Polym. Sci. A, **1**, 2057 (1963).
214. J. Konzelman, K. B. Wagener, ACS Polymer Preprints, **33** (2), 110 (1992).
215. R. Reichel, C. S. Marvel, R. Z. Greenley, J. Polym. Sci. A, **1**, 2935 (1963).
216. S. P. Yen, ACS Polymer Preprints, **4**, 82 (1963).
217. G. W. Coates, R. W. Waymough, ACS Polym. Mater. Sci. Eng., **67**, 92 (1992).

218. G. W. Coates et al., ACS Polymer Preprints, **33** (1), 1230 (1992).
219. S. Cesca, A. Priola, G. Santi, J. Polym. Sci. B, **8**, 573 (1970).
220. L. A. Wall, L. J. Fetters, S. Straus, J. Polym. Sci. B, **5**, 721 (1967).
221. H. Mains, J. H. Day, J. Polym. Sci. B, **1**, 347 (1963).
222. S. Cesca, A. Roggero, N. Palladino, A. De Chivico, Makromol. Chem., **136**, 23 (1970).
223. V. V. Korshak, S. M. Tseitlin, V. I. Azarov, A. I. Parlov, Izv. Akad. Nauk SSSR, Ser. Chim., **1**, 226 (1968).
224. R. G. Alsup, J. O. Punderson, G. F. Leverett, J. Appl. Polym. Sci., **1**, 185 (1959).
225. N. G. Gaylord, Polyethers; Interscience: New York, Part I, p. 46.
226. J. Furukawa, T. Saegusa, H. Fujii, A. Kawasaki, T. Tatano, Makromol. Chem., **33**, 32 (1959).
227. J. Furukawa, T. Saegusa, H. Fujii, Makromol. Chem., **44/46**, 398 (1961).
228. J. Furukawa, T. Saegusa, in: "Polymerization of Aldehydes and Oxides", Polymer Reviews, Interscience, New York, Vol. 3, 1963.
229. A. Benkhira et al., Macromolecules, **27**, 3963 (1994).
230. K. Weissermel, E. Noelken, Makromol. Chem., **68**, 140 (1963).
231. V. N. Kuznetsov, S. D. Vogman et al., Vysokomol. Soedin. A, **12**, 2635 (1970).
232. R. C. Schulz, R. Wolf, Makromol. Chem., **99**, 76 (1966).
233. T. Saegusa, S. Matsumoto, Y. Hashimoto, Macromol., **3**, 735 (1970).
234. M. E. Wright, G. D. Allred, ACS Polymer Preprints, **33** (1), 1123 (1992).
235. G. B. Brown, A. Goldman, ACS Polymer Preprints, **4** (2), 39 (1963).
236. S. Brochu, G. Ampleman, Macromolecules, **29**, 5539 (1996).
237. J.-C. Lee et al., Macromolecules, **30**, 3766 (1997).
238. T. Iwata, G. Wasai, T. Saegusa, Furukawa, J. Makromol. Chem., **77**, 229 (1964).
239. E. Riande, J. Guzman, L. Garillo, Macromolecules, **17**, 1234 (1984).
240. I. Penczek, S. Penzcek, Makromol. Chem., **67**, 203 (1963).
241. W. A. Butte, C. C. Price, R. E. Hughes, J. Polym. Sci., **61**, 528 (1962).
242. G. F. Endres, A. S. Hay, J. W. Eustance, J. Org. Chem., **28**, 1300 (1963).
243. G. D. Staffin, C. C. Price, J. Am. Chem. Soc., **82**, 3632 (1960).
244. J. M. Barrales-Rienda, D. C. Pepper, Eur. Polym. J., **3**, 535 (1967).
245. R. W. Phillips et al., Macromolecules, **27**, 3451 (1994).
246. R.-M. Ho et al., Macromolecules, **30**, 3349 (1997).
247. Z. Chen et al., ACS Polym. Mater. Sci. Eng., **70**, 40 (1994).
248. M. E. Denker, F. W. Harris, ACS Polymer Preprints, **35** (1), 521 (1994).
249. K. J. Saunders, "Organic Polymer Chemistry", 2nd ed., Chapman and Hall, New York, p. 275.
250. G. F. Pregaglia, M. Binaghi, M. Cambini, Makromol. Chem., **67**, 10 (1963).
251. G. Natta, G. Mazzanti, G. F. Pregaglia, M. Binaghi, M. Cambini, Makromol. Chem., **51**, 148 (1962).
252. R. C. Schulz, W. Passmann, Makromol. Chem., **60**, 139 (1963).
253. C. Aso, Y. Aito, Makromol. Chem., **73**, 141 (1964).
254. A. Takahashi, S. Kambara, Makromol. Chem., **72**, 92 (1964).
255. K. Meyersen, R. C. Schulz, W. Kem, Makromol. Chem., **58**, 204 (1962).
256. C. G. Overberger, S. Ishida, H. Ringsdorf, J. Polym. Sci., **62**, 51 (1962).
257. C. G. Overberger, S. Ishida, ACS Polymer Preprints, **5**, 210 (1964).
258. H. Ohse, H. Cherdron, F. Korte, Makromol. Chem., **76**, 147 (1964).
259. P. Dunn, D. Oldfield, J. Macromol. Sci. A, **4**, 1157 (1970).
260. M. Acemoglu et al., Macromolecules, **28**, 3030 (1995).
261. J. Labadie, ACS Polymer Preprints, **36** (1), 735 (1995).
262. X. Chen et al., Macromolecules, **30**, 3470 (1997).
263. W. R. Moore, M. Uddin, Eur. Polym. J., **5**, 185 (1969).
264. W. R. Moore, M. Uddin, Eur. Polym. J., **6**, 121 (1970).
265. G. S. Kolesnikov et al., Vysokomol. Soedin. B, **12**, 548 (1970).
266. C.-S. Tan, T.-I. Hsu, Macromolecules, **30**, 3147 (1997).
267. J. M. Bronk, J. S. Riffle, ACS Polymer Preprints, **35** (1), 915 (1994).
268. M. Super et al., Macromolecules, **30**, 368 (1997).
269. P. Cebe, J. Carbeck, ACS Polymer Preprints, **33** (1), 331 (1992).
270. J. P. Penning, R. St. John Manley, ACS Polymer Preprints, **36** (1), 330 (1995).
271. M. Tokita et al., Macromolecules, **29**, 1345 (1996).
272. X. Li, F. Brisse, Macromolecules, **27**, 7718 (1994).
273. Y. Maeda, J. Watanabe, Macromolecules, **28**, 1661 (1995).
274. S. Inoue, Y. Tomoi, T. Tsuruta, J. Furukawa, Makromol. Chem., **48**, 229 (1961).
275. H. Cherdron, H. Ohse, F. Korte, Makromol. Chem., **56**, 179, 187 (1962).
276. K. M. Shakesheff et al., ACS Polymer Preprints, **36** (1), 71 (1995).
277. S. M. Li et al., ACS Polymer Preprints, **37** (2), 133 (1996).
278. A. Lofgren et al., Macromolecules, **27**, 5556 (1994).
279. S. Oi et al., Macromolecules, **29**, 2694 (1996).
280. P. Hubbard et al., Macromolecules, **29**, 8304 (1996).
281. P. Wilson, W. J. Brittain, ACS Polymer Preprints, **36** (2), 136 (1995).
282. C. Spies, R. Gehrke, Macromolecules, **30**, 1701 (1997).
283. A. Romo-Uribe, A. H. Windle, Macromolecules, **28**, 7085 (1995).
284. C. G. Bauch et al., ACS Polymer Preprints, **32** (1), 377 (1991).
285. Y. Wei et al., ACS Polymer Preprints, **36** (2), 241 (1995).
286. H.-L. Chen et al., Macromolecules, **30**, 1718 (1997).
287. H. A. Al-Maullen, D. M. Knauss, ACS Polymer Preprints, **38** (1), 68 (1997).

288. D. Tian et al., Macromolecules, **30**, 406 (1997).
289. J.-H. Chang et al., ACS Polymer Preprints, **36** (1), 829 (1995).
290. K. Belfield, G. Zhong, ACS Polymer Preprints, **36** (2), 269 (1995).
291. F. Ignatious et al., ACS Polymer Preprints, **34** (2), 586 (1993).
292. B.-H. Chang, J. Kwiatek, ACS Polymer Preprints, **36** (2), 146 (1995).
293. S. Davies, B. Tighe, ACS Polymer Preprints, **36** (1), 103 (1995).
294. R. H. Marchessault, K. Okamura, C. J. Su, Macromol., **3**, 735 (1970).
295. Y. Hori, A. Yamaguchi, Macromolecules, **28**, 406 (1995).
296. K. S. Bisht et al., Macromolecules, **30**, 2705 (1997).
297. Y. Hiragui, Y. Tokiwa, Macromolecules, **30**, 3691 (1997).
298. H. Ohse, H. Cherdron, Makromol. Chem., **95**, 283 (1966).
299. A. Roviello, A. Sirigu, Makromol. Chem., **181**, 1799 (1980).
300. N. Lacoudre, A. Leborgue et al., Makromol. Chem., **187**, 341 (1986).
301. H. Batzer, H. Lang, Makromol. Chem., **15**, 211 (1955).
302. R. Nitsche, K. A. Wolf, US Patent 3 036 990, 3 036 991, 1962.
303. J. Lo, S. N. Lee, E. M. Pearce, J. Appl. Polym. Sci., **29**, 35 (1984).
304. Y. Imai, S. Tassavori, J. Polym. Sci. Chem. Ed., **22**, 1319 (1984).
305. J. H. Pawlow et al., ACS Polymer Preprints, **37** (2), 198 (1996).
306. J. Hori et al., Macromolecules, **26**, 4388 (1993).
307. E. Shimamura, Macromolecules, **27**, 4429 (1994).
308. H. R. Kricheldorf, J. M. Jonte, Polym. Bull., **9**, 276 (1983).
309. A. G. Shard et al., ACS Polymer Preprints, **36** (1), 74 (1995).
310. R. C. Schulz et al., ACS Polymer Preprints, **31** (2), 402 (1990).
311. Y. K. Sung, D. K. Song, ACS Polymer Preprints, **37** (2), 107 (1996).
312. J. Majnusz, J. Catala, R. W. Lenz, Eur. Polym. J., **19**, 1043 (1983).
313. Trade literature, Amoco Chemical Co., Chicago, IL.
314. J. Jager et al., ACS Polymer Preprints, **37** (1), 236 (1996).
315. K. J. Saunders, "Organic Polymer Chemistry", 2nd ed., Chapman and Hall, New York, 1988, p. 256.
316. H. Prueckner, Erdoel, Kohle, Erdgas, Petrochemie, **16**, 188 (1959).
317. C. Wu et al., Macromolecules, **27**, 6055 (1994).
318. A. S. Jones et al., ACS Polymer Preprints, **37** (1), 229 (1996).
319. Y.-H. So, ACS Polymer Preprints, **35** (1), 819 (1994).
320. F. H. Henglein, H. E. Tarrach, Kunststoffe-Plastics, **6**, 5 (1959).
321. S. Nomula, S. L. Cooper, ACS Polymer Preprints, **38** (1), 78 (1997).
322. D. J. Lyman, J. Polym. Sci., **55**, 507 (1961).
323. E. L. Wittbecker, M. Katz, J. Polym. Sci., **40**, 367 (1959).
324. T. M. Gricenko et al., Khim. Volokna, **5**, 18 (1970).
325. D. J. Lyman, J. Polym. Sci., **45**, 49 (1960).
326. S. Neffgen et al., Macromolecules, **30**, 1289 (1997).
327. Trade literature, Anderson Development Co., Adrian, MI.
328. J. C. Wittmann et al., ACS Polymer Preprints, **36** (1), 257 (1995).
329. S. Bronco, Macromolecules, **27**, 4436 (1994).
330. Z. Jiang et al., ACS Polymer Preprints, **35** (1), 676 (1994).
331. J. G. Bonner, A. K. Powell, ACS Polym. Mater. Sci. Eng., **76**, 108 (1997).
332. A. A. Patel, S. R. Patel, Eur. Polym. J., **19**, 17 (1983).
333. M. Barsacchi et al., ACS Polym. Mater. Sci. Eng., **67**, 63 (1992).
334. M. Brookhart et al., ACS Polymer Preprints, **35** (2), 530 (1994).
335. R. A. Koster, R. H. Birk, ACS Polymer Preprints, **37** (2), 525 (1996).
336. S. Murtuza, A. Sen, ACS Polymer Preprints, **37** (2), 194 (1996).
337. S. L. Borkowsky, R. M. Waymouth, Macromolecules, **29**, 6377 (1996).
338. S. L. Borkowsky, R. M. Waymough, ACS Polymer Preprints, **35** (1), 792 (1994).
339. A. L. Safir, B. M. Novak, ACS Polymer Preprints, **36** (1), 227 (1995).
340. M. F. Teasley, B. S. Hsiao, Macromolecules, **29**, 6432 (1996).
341. M. Teasley et al., ACS Polymer Preprints, **38** (1), 125 (1997).
342. F. Wang, J. Roovers, ACS Polym. Mater. Sci. Eng., **69**, 44 (1993).
343. R.-M. Ho et al., Macromolecules, **27**, 5787 (1994).
344. R.-H. Ho et al., ACS Polymer Preprints, **36** (1), 338 (1995).
345. S. Ahmed et al., ACS Polymer Preprints, **38** (1), 172 (1997).
346. U. Hoffmann et al., Macromolecules, **27**, 3575 (1994).
347. J. V. Facinelli et al., ACS Polymer Preprints, **36** (1), 759 (1995).
348. Z. Wu et al., ACS Polymer Preprints, **36** (2), 215 (1995).
349. D. W. Thomson, G. F. L. Ehlers, J. Polym. Sci. A, **2**, 1051 (1964).
350. A. Mochizuki et al., Macromolecules, **28**, 365 (1995).
351. J. V. Beach, ACS Polym. Mater. Sci. Eng., **72**, 205 (1995).
352. B. A. Kiselov, I. A. Stepina, Z. P. Ablekova, Soviet Plastics, 13 (1970).
353. G. W. Gray, D. Lacey et al., Makromol. Chem. Rapid Commun., **7**, 71 (1986).
354. J. B. Ganci, F. A. Bettelheim, J. Polym. Sci. A, **2**, 4011 (1964).
355. P. R. Dvornic et al., Macromolecules, **27**, 7575 (1994).
356. P. R. Dvornic, R. W. Lenz, Macromolecules, **27**, 5833 (1994).
357. P. R. Dvornic, V. V. Gerov, Macromolecules, **27**, 1068 (1994).
358. R. Puyerbrock et al., ACS Polym. Mater. Sci. Eng., **72**, 110 (1995).
359. Trade literature, GE Silicones, Waterford, NY.
360. H. Zhuang, Jr., J. A. Gardella, Macromolecules, **30**, 3632 (1997).

361. A. Sergeyev, V. I. Ness'kin, S. S. Arustamyan, Polymer Commun., **26**, 365 (1985).
362. Phillips Petroleum US Patent 3 380 951.
363. B. Jirgensons, in: "Natural Organic Polymers", Pergamon Press, New York, 1962.
364. K. Yamamoto et al., Macromolecules, **27**, 4312 (1994).
365. J. L. de la Pena, E. Riande, J. Guzman, Macromolecules, **18**, 2739 (1985).
366. S. S. Wu et al., ACS Polymer Preprints, **36** (1), 267 (1995).
367. H. A. Smith, C. E. Handlovits, Proc. Conf. High Temperature Polymer and Fluid Research, Dayton, Ohio, 1962, p. 123.
368. M. L. Cameron et al., ACS Polymer Preprints, **31** (1), 297 (1990).
369. M. Ueda, T. Okada, Macromolecules, **27**, 3449 (1994).
370. R. A. Johnson, L. J. Mathais, Macromolecules, **28**, 79 (1995).
371. J. D. Davies, W. H. Daley, ACS Polymer Preprints, **37** (1), 745 (1996).
372. M. P. Bohrer, D. A. Mixon, ACS Polymer Preprints, **34** (1), 455 (1993).
373. K. J. Saunders, "Organic Polymer Chemistry", 2nd ed., Chapman and Hall, New York, 1988; p. 283.
374. H. A. Vogel, J. Polym. Sci. A-1, **8**, 2035 (1970).
375. M .E. A. Cudby, R. G. Feasy, B. E. Jennings, M. E. B. Jones, J. B. Rose, Polymer, **6**, 589 (1965).
376. A. Bhatnager, ACS Polymer Preprints, **38** (1), 111 (1997).
377. A. Pandya et al., Macromolecules, **27**, 1367 (1994).
378. S. G. Gaynor, S. Z. Edelman, ACS Polymer Preprints, **38** (1), 703 (1997).
379. M. D. Guiver et al., Macromolecules, **28**, 7612 (1995).
380. R. S. Achibald et al., ACS Polym. Mater. Sci. Eng., **69**, 293 (1993).
381. Y.-K. Han et al., Macromolecules, **28**, 916 (1995).
382. C. Gao, A. S. Hay, ACS Polymer Preprints, **36** (2), 102 (1995).
383. D. J. Riley et al., ACS Polymer Preprints, **37** (1), 489 (1996).
384. T. M. Moy et al., ACS Polymer Preprints, **33** (1), 489 (1992).
385. M. Russo, S. Hara, Chem. High Polymers (Tokyo), **27**, 661 (1970).
386. N. Azuma, Macromolecules, **28**, 7331 (1995).
387. V. E. Shashoua, W. Sweeny, R. F. Tietz, J. Am. Chem. Soc., **82**, 866 (1966).
388. W. Buchard, Makromol. Chem., **67**, 182 (1963).
389. A. Ohshima et al., Macromolecules, **28**, 6095 (1995).
390. Y. Jimbo et al., Macromolecules, **27**, 6080 (1994).
391. C. Khatri et al., ACS Polymer Preprints, **33** (2), 200 (1992).
392. R. C. Schultz, R. Stenner, Makromol. Chem., **12**, 202 (1964).
393. L. Franco et al., ACS Polymer Preprints, **33** (1), 325 (1992).
394. L. Franco et al., Macromolecules, **27**, 4284 (1994).
395. J. A. Muller et al., ACS Polymer Preprints, **36** (1), 166 (1995).
396. J. Helfrich et al., Macromolecules, **27**, 472 (1994).
397. D. Barrera et al., Macromolecules, **28**, 425 (1995).
398. M. Bermudez et al., Macromolecules, **27**, 6325 (1994).
399. R. Graf, G. Lohaus, K. Boerner, E. Schmidt, H. Bestian, Angew. Chem., **74**, 523 (1962).
400. J. Masamoto, Y. Kaneko, K. Sasaguri, H. Kobayashi, J. Soc. Fiber Sci. Technol. Japan, **25**, 525 (1969).
401. J. Masamoto, K. Sasaguri, C. Ohizumi, H. Kobayashi, J. Polym. Sci. A-2, **8**, 1703 (1970).
402. S. Jin, K. E. Gonsalves, ACS Polym. Mater. Sci. Eng., **76**, 15 (1997).
403. C.-W. Ng et al., ACS Polym. Mater. Sci. Eng., **69**, 345 (1993).
404. H. Sekiguchi, Bull. Soc. Chim. France, 1839 (1960).
405. R. J. Gaymans, V. S. Venkatraman, J. Schuijer, J. Polym. Sci. Chem. Ed., **22**, 1373 (1984).
406. S. A. Giannos, ACS Polym. Mater. Sci. Eng., **62**, 236 (1990).
407. K. L. L. Eerisels et al., Macromolecules, **29**, 6744 (1996).
408. E. Navarro et al., Macromolecules, **29**, 5406 (1996).
409. E. Navarro et al., Macromolecules, **28**, 8742 (1995).
410. S. Steadman, L. J. Mathias, ACS Polymer Preprints, **37** (2), 256 (1996).
411. G. Reinisch, K. Dietrich, Eur. Polym. J., **6**, 1269 (1970).
412. K. Weiskopf, G. Meyerhoff, Polymer, **24**, 72 (1983).
413. C. G. Johnson, L. J. Mathias, ACS Polymer Preprints, **34** (2), 348 (1993).
414. T. Kakuchi et al., ACS Polymer Preprints, **37** (1), 561 (1996).
415. M. J. Hill, E. D. T. Atkins, Macromolecules, **28**, 604 (1995).
416. Trade literature, Huls, Marl, Germany.
417. G. Cojazzi, A. M. Drusiani et al., Eur. Polym. J., **17**, 1241 (1981).
418. N. Ogata, K. Sanui et al., J. Polym. Sci. Chem. Ed., **22**, 793 (1984).
419. H. R. Dricheldorf et al., Macromolecules, **27**, 1669 (1994).
420. M. Ueda, A. Mochizuki, Macromolecules, **18**, 2353 (1985).
421. V. E. Shaskoua, W. M. Eareckson, J. Polym. Sci., **40**, 343 (1959).
422. A. C. Karydas, M. Kapuscinska, E. M. Pearce, Eur. Polym. J., **19**, 935 (1983).
423. C. G. Overberger, C. X. Lu, C. C. Chen, J. Polym. Sci. Chem. Ed., **24**, 75 (1986).
424. Y. Iwakura, K. Uno, Y. Imai, M. Fukui, Makromol. Chem., **77**, 41 (1964).
425. K. J. Saunders, "Organic Polymer Chemistry", 2nd ed., Chapman and Hall, New York, 1988; p. 210.
426. C. X. Lu, Q. Li, J. Pan, J. Polym. Sci. Chem. Ed., **23**, 30 (1985).
427. D. J. Schaefer et al., ACS Polymer Preprints, **35** (1), 118 (1994).
428. K. J. Saunders, "Organic Polymer Chemistry", 2nd ed., Chapman and Hall, New York, 1988; p 209.
429. K. J. Bouck, P. G. Rasmussen, ACS Polymer Preprints, **34** (1), 463 (1993).
430. J. Rosch, R. Mulhaupt, ACS Polymer Preprints, **33** (1), 370 (1992).
431. D. Lin et al., ACS Polymer Preprints, **33** (1), 335 (1992).
432. P. E. Cassidy et al., ACS Polymer Preprints, **31** (2), 582 (1990).
433. K. Kimura et al., Macromolecules, **27**, 1303 (1994).

434. H. Kricheldorf, A. Domschke, Macromolecules, **27**, 1509 (1994).
435. C. E. Sroog, S. V. Abramo, C. E. Berr, W. M. Edwards, A. L. Endrey, K. L. Olivier, ACS Polymer Preprints, **5** (1), 132; J. Polym. Sci. A, **3**, 1373 (1965).
436. J. Pellon, W. G. Carpenter, J. Polym. Sci. A, **1**, 863 (1963).
437. B. L. Kwolek, P. W. Morgan, J. Polym. Sci. A, **2**, 269 (1964).
438. S. A. Suncket, W. A. Murphey, S. B. Spenk, J. Polym. Sci., **40**, 389 (1959).
439. K. Iimura, N. Koide et al., Makromol. Chem., **182**, 2563 (1981).
440. A. H. Frazer, F. T. Wallenberger, J. Polym. Sci. A, **2**, 1137 (1964).
441. A. H. Frazer, F. T. Wallenberger, J. Polym. Sci. A, **2**, 1147 (1964).
442. K. R. Carter, J. L. Hedrich, ACS Polymer Preprints, **34** (1), 521 (1993).
443. H. Rinke, E. Istel, in: E. Mueller (Ed.), "Houben-Weyl Methoden der Organischen Chemie", vol. 14 (2), 412 ed., Georg Thieme, Stuttgart, 1963.
444. G. C. Robinson, J. Polym. Sci. A, **2**, 390 (1964).
445. A. Goodwin, B. M. Novak, Macromolecules, **27**, 5520 (1994).
446. K. Shibayama et al., Macromolecules, **30**, 3159 (1997).
447. D. J. Lyman, N. Sadri, Makromol. Chem., **67**, 1 (1963).
448. C. H. Chen et al., Macromolecules, **27**, 6376 (1994).
449. Trade literature, Polymer Chemistry Innovations, Tuscon, AZ.
450. J. M. Rodriguey-Parada et al., Macromolecules, **27**, 1571 (1994).
451. M. A. Mitchell et al., ACS Polym. Mater. Sci. Eng., **68**, 22 (1993).
452. K. S. Arora, C. G. Overberger, J. Polym. Sci. Chem. Ed., **24**, 37 (1986).
453. T. Nakato et al., ACS Polymer Preprints, **37** (1), 555 (1996).
454. O. Takayuki, A. Tatsumi, A. Matsumoto, J. Polym. Sci. Letters Ed., **24**, 113 (1986).
455. A. Ito et al., Macromolecules, **28**, 5618 (1995).
456. Y. Xia, A. G. MacDiarmid, Macromolecules, **27**, 7212 (1994).
457. M. R. Majidi et al., ACS Polymer Preprints, **36** (2), 51 (1995).
458. M. G. Mikheal et al., ACS Polymer Preprints, **37** (1), 523 (1996).
459. T. E. Herold, R. S. Duran, ACS Polymer Preprints, **36** (1), 576 (1995).
460. W. E. Rudzinski et al., ACS Polymer Preprints, **34** (1), 449 (1993).
461. K. Shirdhara et al., Macromolecules, **30**, 4024 (1997).
462. M. T. Nguyen, A. F. Diaz, Macromolecules, **27**, 7003 (1994).
463. T. Tsuda, A. Tobisawa, Macromolecules, **28**, 1360 (1995).
464. J. Gopal, M. Srinivasan, Makromol. Chem., **187**, 1 (1986).
465. G. F. L. Ehlers, J. D. Ray, J. Polym. Sci. A, **2**, 4989 (1964).
466. H. K. Hall, Jr., ACS Polym. Mater. Sci. Eng., **72**, 129 (1995).
467. H. K. Hall et al., Macromolecules, **28**, 1 (1995).
468. T. Endo et al., Macromolecules, **27**, 3694 (1994).
469. A. Nakai et al., Macromolecules, **27**, 6963 (1994).
470. B. M. Culbertson, X.-L. Want, ACS Polymer Preprints, **37** (2), 279 (1996).
471. H. R. Allcock, R. L. Kugel, K. J. Valan, Inorg. Chem., **5**, 1709 (1966).
472. G. Allen, C. J. Lewis, S. M. Todd, Polymer, **11**, 31 (1970).
473. G. Allen, C. J. Lewis, S. M. Todd, Polymer, **11**, 44 (1970).
474. G.-F. Xu, P. Wilsian-Neilson, ACS Polymer Preprints, **36** (1), 219 (1995).
475. H. R. Allcock, P. E. Austin, Macromolecules, **14**, 1616 (1981).
476. S. Kobayashi, M. Suzuki, T. Saegusa, Macromolecules, **19**, 462 (1986).
477. E. Bonaplata et al., ACS Polym. Mater. Sci. Eng., **70**, 405 (1994).
478. J. Gruneich, P. Wisian-Neilsen, ACS Polymer Preprints, **37** (1), 611 (1996).
479. L. W. Breed, R. L. Elliott, A. F. Ferris, J. Polym. Sci. A, **2**, 45, 1964; J. Org. Chem., **27**, 1114 (1962).
480. K. A. Andrianov, V. I. Pachomov, V. M. Gel'perina, Dokl. Akad. Nauk SSSR, **162**, 79 (1965).
481. K. Matsumota, H. Yamaoka, Macromolecules, **28**, 7029 (1995).
482. B. M. Klemann et al., Macromolecules, **29**, 198 (1996).
483. M. M. Despotopoulou et al., Macromolecules, **29**, 5797 (1996).
484. T. Hiraoka et al., ACS Polymer Preprints, **37** (1), 483 (1996).
485. J. P. Banovetz et al., ACS Polymer Preprints, **34** (1), 228 (1993).
486. H. Isaka et al., Macromolecules, **28**, 4733 (1995).
487. J. Ohshita et al., Macromolecules, **30**, 1540 (1997).
488. H. Isaka et al., Macromolecules, **30**, 344 (1997).
489. J. Ohshita et al., Macromolecules, **27**, 5583 (1994).
490. M. Itoh et al., Macromolecules, **30**, 694 (1997).
491. M. Itoh et al., Macromolecules, **27**, 7917 (1994).
492. M.-K. Ryu et al., ACS Polym. Mater. Sci. Eng., **75**, 408 (1996).
493. C. R. Krueger, E. G. Rochow, J. Polym. Sci. A, **2**, 3179 (1964).
494. H. Tang et al., Macromolecules, **30**, 1400 (1997).
495. R. Minne, E. G. Rochow, J. Am. Chem. Soc., **82**, 5625 (1960).
496. J. P. Pillot et al., ACS Polymer Preprints, **34** (1), 562 (1993).
497. Y. Pang, ACS Polymer Preprints, **37** (2), 382 (1996).
498. Y. Imai, K. Uno, Y. Iwakura, Makromol. Chem., **83**, 17 (1965).
499. T. Kubota, R. Nakanishi, J. Polym. Sci. B, **2**, 655 (1964).
500. Y. Imai, I. Kaoka, K. Uno, Y. Iwakura, Makromol. Chem., **83**, 167 (1965).
501. S. Inoue, Y. Imai, K. Uno, Y. Iwakura, Makromol. Chem., **95**, 236 (1966).
502. V. V. Korshak, G. M. Tseitlin, A. I. Pavlow, Vysokomol. Soedin., **8**, 1599 (1966).
503. D. M. Husband et al., ACS Polymer Preprints, **36** (1), 221 (1995).

504. W. D. Joseph et al., ACS Polymer Preprints, **34** (1), 397 (1993).
505. R. J. Parry, B. K. Wilson, Macromolecules, **27**, 40 (1994).
506. W. D. Joseph et al., ACS Polymer Preprints, **35** (1), 551 (1994).
507. A. W. Frazer, W. Sweeny, F. T. Wallenberger, J. Polym. Sci. A, **2**, 1157 (1964).
508. M. Hasegawa, T. Unishi, J. Polym. Sci. B, **2**, 237 (1964).
509. Y. Iwakura, K. Uno, S. Hara, Makromol. Chem., **94**, 103 (1966).
510. Y. Iwakura, K. Uno, S. Hara, Makromol. Chem., **95**, 248 (1966).
511. C. J. Abshire, C. S. Marvel, Makromol. Chem., **44/46**, 388 (1961).
512. A. H. Frazer, W. Sweeny, F. T. Wallenberger, J. Polym. Sci. A, **2**, 1157 (1964).
513. M. Hasegawa, R. Huiski, J. Polym. Sci. B, **2**, 237 (1964).
514. Y. Iwakura, K. Uno, S. Hara, J. Polym. Sci. A, **3**, 45 (1965).
515. A. H. Frazer, F. T. Wallenberger, J. Polym. Sci. A, **2**, 1181 (1964).
516. D. T. Longone, H. H. Un, J. Polym. Sci. A, **3**, 3117 (1965).
517. M. Srinivasarao, ACS Polymer Preprints, **33** (1), 777 (1992).
518. M. Dotrong et al., ACS Polymer Preprints, **32** (3), 201 (1991).
519. K. R. Srinivansan et al., ACS Polymer Preprints, **36** (2), 249 (1995).
520. P. Hergenrother, W. Wrasidlo, H. H. Levine, ACS Polymer Preprints, **5**, 153 (1964).
521. H. Kokelenberg, C. S. Marvel, J. Polym. Sci. A-1, **8**, 3235 (1970).
522. H. R. Kircheldorf et al., Macromolecules, **27**, 2540 (1994).
523. V. N. Sekharipuram et al., ACS Polymer Preprints, **34** (1), 618 (1993).
524. P. Cebe, ACS Polymer Preprints, **37** (1), 140 (1996).
525. D. C. Rich et al., ACS Polymer Preprints, **38** (1), 414 (1997).
526. R. Pardey et al., Macromolecules, **27**, 5794 (1994).
527. R. J. Perry et al., Macromolecules, **28**, 2607 (1995).
528. R. J. Perry et al., Macromolecules, **27**, 4058 (1994).
529. S.-H. Hsiao et al., Macromolecules, **30**, 165 (1997).
530. K. Itoya et al., Macromolecules, **27**, 4101 (1994).
531. A. E. Hoylt, S. J. Haung, ACS Polymer Preprints, **35** (1), 343 (1994).
532. Y.-N. Lin et al., ACS Polymer Preprints, **34** (1), 515 (1993).
533. D. Cheng et al., ACS Polymer Preprints, **34** (2), 718 (1993).
534. R. E. Southwark et al., ACS Polym. Mater. Sci. Eng., **76**, 183 (1997).
535. B. Conger et al., Macromolecules, **30**, 4049 (1997).
536. T. Matsumoto, T. Kurosaki, Macromolecules, **28**, 5684 (1995).
537. N. A. J. M. van Aerle, A. J. W. Tol, Macromolecules, **27**, 6520 (1994).
538. K. Itoya et al., Macromolecules, **27**, 7231 (1994).
539. Trade literature, Amoco Performance Products, Ridgefield, CT.
540. C. E. Sroog, S. E. Abramo, C. E. Berr, W. M. Edwards, A. L. Endrey, K. L. Olivier, ACS Polymer Preprints, **5**, 132 (1964).
541. J. W. Yu et al., Macromolecules, **30**, 1845 (1997).
542. K. J. Saunders, "Organic Polymer Chemistry", 2nd ed., Chapman and Hall, New York, 1988, p. 217.
543. J. Heller, J. H. Hodgkin, F. J. Martinelli, J. Polym. Sci. B, **6**, 153 (1968).
544. K. E. Uhrich et al., Macromolecules, **28**, 2184 (1995).
545. A. N. Thuo, H. W. Beckman, ACS Polymer Preprints, **36** (2), 274 (1995).
546. J. W. Park et al., Macromolecules, **27**, 3459 (1994).
547. J. K. Stille, J. R. Williamson, J. Polym. Sci. A, **2**, 3867 (1964).
548. J. K. Stille, E. L. Mainen, Macromolecules, **1**, 36 (1968).
549. F. De Schryver, C. S. Marvel, J. Polym. Sci. A, **1**, **5**, 545 (1967).
550. H. H. Levine, J. K. Kjeller, C. G. Delano, ACS Polymer Preprints, **5**, 160 (1964).
551. Y. Iwakura, K. Uno, Y. Imai, Makromol. Chem., **77**, 33 (1964).
552. Y. Iwakura, S. I. Izawa, F. Hayano, J. Polym. Sci. A-1, **6**, 1097 (1968).
553. V. P. Sarzevskaja et al., Ukr. Khim. Zh., **35**, 390 (1969).
554. J. E. White et al., ACS Polymer Preprints, **34** (1), 904 (1993).
555. T. Nishikubo et al., Macromolecules, **27**, 1087 (1994).
556. J. L. Lang, W. A. Pavelich, H. D. Clarey, J. Polym. Sci. A, **1**, 1123 (1963).
557. J. C. L. Hwa, W. A. Fleming, L. Miller, J. Polym. Sci. A, **2**, 2395 (1964).
558. F. Xi, O. Vogl, J. Macromol. Sci. A, **20**, 139 (1983).
559. J. L. Ding, R. M. Ottenbrite, ACS Polymer Preprints, **34** (2), 508 (1993).
560. W. Mormann, K. Schmalz, ACS Polymer Preprints, **33** (1), 968 (1992).
561. W. Mormann, K. Schmalz, Macromolecules, **27**, 7115 (1994).
562. Trade literature, Chevron Chemical Co., Houston, TX.
563. Y. Chang, C. L. McCormick, Macromolecules, **26**, 4814 (1993).
564. Trade literature, Rieke Metals, Lincoln, NE.
565. M. Pomerantz et al., ACS Polym. Mater. Sci. Eng., **72**, 271 (1995).
566. A. Kumar, S. R. Reynolds, ACS Polymer Preprints, **37** (2), 337 (1996).
567. J. Kowalik et al., ACS Polym. Mater. Sci. Eng., **72**, 325 (1995).
568. S. Arnold et al., ACS Polym. Mater. Sci. Eng., **72**, 170 (1995).
569. Y.-H. Lu et al., Macromolecules, **29**, 5546 (1996).
570. J. Parakka et al., Macromolecules, **29**, 1928 (1996).
571. J. K. Herrema et al., Macromolecules, **28**, 8102 (1995).
572. L. D. Peeters et al., Macromolecules, **29**, 4216 (1996).
573. Y. Pang et al., ACS Polymer Preprints, **37** (2), 333 (1996).
574. D. R. Rutherford et al., ACS Polymer Preprints, **31** (2), 643 (1990).

575. V. V. Sheares et al., ACS Polymer Preprints, **33** (2), 170 (1992).
576. V. V. Sheares et al., ACS Polym. Mater. Sci. Eng., **69**, 236 (1993).
577. V. V. Sheares, ACS Polym. Mater. Sci. Eng., **73**, 258 (1995).
578. M. D. Curtin et al., ACS Polymer Preprints, **36** (1), 191 (1995).
579. J. Kastner et al., Macromolecules, **28**, 2922 (1995).
580. F. Sanda, M. Matsumoto, Macromolecules, **28**, 6911 (1995).
581. T. Tsuda, L. J. Mathias, ACS Polymer Preprint, **34** (2), 346 (1993).
582. A. Saito, D. A. Tirrell, ACS Polymer Preprints, **34** (1), 152 (1993).
583. M. J. Marsella, T. M. Swager, ACS Polymer Preprints, **33** (1), 1196 (1992).
584. S.-W. Park et al., ACS Polymer Preprints, **36** (2), 259 (1995).
585. Y.-K. Kim et al., ACS Polymer Preprints, **32** (3), 590 (1991).
586. K. R. Carter et al., ACS Polymer Preprints, **34** (1), 481 (1993).
587. M. Strukilj, J. C. Hedrick, Macromolecules, **27**, 7511 (1994).
588. R. G. Bass, K. R. Srinivasan, ACS Polymer Preprints, **32** (1), 619 (1991).
589. Trade literature, AKZO, Dobbs Ferry, NY.
590. Trade literature, Dow Chemical Co., Midland, MI.
591. Trade literature, Hardman Inc., Belleville, NJ.
592. J. H. Bradbury, M. C. S. Perera, Brit. Polym. J., **18**, 127 (1986).
593. H. Valdsar, R. Dunlap, Am. Chem. Soc. Meeting, Atlantic City, N. J., Sept. 1952.
594. E. Husemann, E. Siefert, Makromol. Chem., **128**, 288 (1969).
595. E. Ott, H. M. Spurlin, (Eds.), "Cellulose and Cellulose Derivatives" Interscience, New York, 2nd ed., 1954. This book contains additional references.
596. A. Muenster, H. in: A. Stuart (Ed.), "Physik der Hochpolymeren", vol. II, Springer, Heidelberg, 1953, p. 193.
597. Trade literature, Hercules, Inc., Wilmington, DE.
598. Trade literature, Eastman Chemical Co., Kingsport, TN.
599. P. Howard, R. S. Pankh, Polym. Sci. A-1, **6**, 537 (1968).
600. S. N. Bhadani, D. G. Gray, Makromol. Chem. Rapid Commun., **3**, 449 (1982).
601. F. Guittard et al., Macromolecules, **27**, 6988 (1994).
602. C. E. Schildknecht (Ed.), "Polymer Processes", Interscience: New York, 1956, p. 427.
603. C. T. Greenwood, D. J. Hourston, A. R. Proeter, Eur. Polyrn. J., **6**, 293 (1970).
604. K. R. Meyer, Angew. Chem., **63**, 155 (1951).
605. G. Keilich, N. Frank, E. Husemann, Makromol. Chem., **143**, 275 (1971).
606. Trade literature, Pronova Biopolymers, Portsmouth, NH.
607. J. V. Gruber et al., Macromolecules, **28**, 8865 (1995).
608. A. Hagino, S. Yoshida et al., Macromolecules, **19**, 1 (1986).
609. C. L. Yue et al., ACS Polymer Preprints, **36** (1), 416 (1995).
610. Trade literature, Kelco, Chicago, IL.
611. R. Knoesel, J. Perrod, H. Benoit, Compt. Rend., **251**, 2944 (1960).
612. J. F. McCullough, I. R. Van Walzer, E. J. Griffith, J. Am. Chem. Soc., **78**, 4528 (1956).
613. E. Schwarzmann, I. R. Van Walzer, J. Am. Chem. Soc., **82**, 6009 (1960).
614. H. Guenther, M. D. Bezoavi, P. Kovacic, J. Polym. Sci. Letters Ed., **22**, 65 (1984).
615. C. V. Pittman, R. D. Priester, T. V. Jagaramann, J. Polym. Sci. Chem. Ed., **19**, 3351 (1981).
616. W. K. Zou, N.-L. Yang, ACS Polymer Preprints, **33** (2), 188 (1992).
617. S. Harada, S. Hasegawa, Makromol. Chem. Rapid Commun., **5**, 27 (1984).
618. Trade literature, CPS Chemical Co., Old Bridge, NJ.
619. Y. Deng, R. Pelton, Macromolecules, **28**, 4617 (1995).
620. J. C. Gahin, J. Appl. Polym. Sci., **15**, 213 (1970).
621. M. Tricot, F. Debeauvais, C. Houssier, Eur. Polym. J., **11**, 589 (1975).
622. L. P. Meier et al., Macromolecules, **29**, 718 (1996).
623. P. Cheng et al., ACS Polymer Preprints, **32** (1), 138 (1991).
624. S. A. X. Haung, F. W. Harris, ACS Polymer Preprints, **35** (1), 519 (1994).
625. D.-C. Choi et al., Macromolecules, **30**, 176 (1997).
626. A. Qin et al., Macromolecules, **27**, 120 (1994).
627. R. Nomura et al., Macromolecules, **28**, 86 (1995).
628. A. H. Gabor, C. K. Ober, ACS Polym. Mater. Sci. Eng., **72**, 104 (1995).
629. O. Liu et al., ACS Polymer Preprints, **33** (1), 215 (1992).
630. F. Henselwood, G. Liu, Macromolecules, **30**, 488 (1997).
631. R. Numora, T. Endo, Macromolecules, **28**, 5372 (1995).
632. X. Chen, J. A. Gardella, ACS Polymer Preprints, **33** (2), 312 (1992).
633. C. Kim et al., ACS Polymer Preprints, **37** (2), 159 (1996).
634. M. S. Reeve, ACS Polym. Mater. Sci. Eng., **67**, 182 (1992).
635. D. Li, R. Faust, Macromolecules, **28**, 1383 (1995).
636. J. Li et al., ACS Polymer Preprints, **36** (2), 388 (1995).
637. G.-H. Hsiue et al., ACS Polymer Preprints, **33** (2), 212 (1992).
638. R. Nomura, T. Endo, Macromolecules, **28**, 1754 (1995).
639. Trade literature, Shell Chemical Co., Houston, TX.
640. J. H. Chu et al., ACS Polymer Preprints, **36** (2), 122 (1995).
641. M. Motomatsu et al., ACS Polymer Preprints, **37** (2), 579 (1996).
642. B. S. Pinheiro, K. I. Winey, ACS Polymer Preprints, **36** (1), 174 (1995).
643. D. Hajduk et al., Macromolecules, **27**, 490 (1994).
644. L. Balogh, L. A. Blumstein, ACS Polymer Preprints, **36** (2), 211 (1995).
645. J. T. Chen et al., Macromolecules, **28**, 1688 (1995).
646. S. Nakahama et al., ACS Polymer Preprints, **36** (1), 57 (1995).
647. X. Jin, T. Lodge, ACS Polymer Preprints, **36** (1), 168 (1995).
648. R. F. Story, D. W. Baugh, ACS Polymer Preprints, **36** (1), 414 (1995).
649. Z. Fodor, R. Faust, ACS Polymer Preprints, **36** (2), 176 (1995).
650. X. Chen et al., ACS Polymer Preprints, **38** (1), 715 (1997).

651. S. J. Clarson et al., Macromolecules, **28**, 674 (1995).
652. T. Ohmura et al., Macromolecules, **27**, 3714 (1994).
653. D. Nguyen et al., Macromolecules, **27**, 5173 (1994).
654. Y. Gan, T. E. Hogen-Esch, ACS Polymer Preprints, **36** (1), 406 (1995).
655. Y. Gan et al., ACS Polymer Preprints 1993, **34** (1), 69 (1993).
656. T. K. Bera et al., ACS Polym. Mater. Sci. Eng., **77**, 126 (1997).
657. H. R. Allcock et al., Macromolecules, **30**, 3184 (1997).
658. I. Ihre, A. Hult, ACS Polym. Mater. Sci. Eng., **77**, 71 (1997).
659. S. Rannard, N. Davis, ACS Polym. Mater. Sci. Eng., **77**, 63 (1997).
660. Trade literature, DSM Fine Chemicals, Heerlen, NL.
661. L. J. Hobson et al., ACS Polym. Mater. Sci. Eng., **77**, 220 (1997).
662. Y. H. Kim, ACS Polym. Mater. Sci. Eng., **73**, 175 (1995).
663. C. Kim et al., Macromolecules, **29**, 6353 (1996).
664. P. Lu et al., ACS Polymer Preprints, **37** (2), 342 (1996).
665. C. J. Hawker, F. Chu, Macromolecules, **29**, 4370 (1996).
666. C. J. Hawker, ACS Polym. Mater. Sci. Eng., **76**, 152 (1997).
667. J. P. Kampf, ACS Polymer Preprints, **38** (1), 908 (1997).
668. M. R. Leduc et al., ACS Polym. Mater. Sci. Eng., **77**, 1 (1997).
669. W. Hayes et al., ACS Polym. Mater. Sci. Eng., **77**, 143 (1997).
670. E. W. Kwock et al., ACS Polymer Preprints 1991, **32** (3), 635 (1991).
671. F. Walter et al., ACS Polymer Preprints, **34** (1), 79 (1993).
672. S. R. Turner et al., Macromolecules, **27**, 1611 (1994).
673. C. Wu et al., Macromolecules, **29**, 228 (1996).
674. DBP 926. 163; 928, 793; 929, 931; 929. 932; 1955, Hoechst AG, O. Fuchs et al., Chem. Z. 1169, 1956; 3756, 1958.
675. G. E. Ham, Ind. Eng. Chem., **46**, 390 (1954).
676. S. Petitjean et al., Macromolecules, **27**, 4127 (1994).
677. R. Bacskai, J. Polym. Sci. A, **1**, 2777 (1963).
678. W. Kuran, P. Gorecki, Makromol. Chem., **184**, 907 (1983).
679. T. Ikemura, Chem. High Polymers (Tokyo), **26**, 845 (1969).
680. M. T. Nguyen et al., ACS Polym. Mater. Sci. Eng., **70**, 22 (1994).
681. Y. Xhong, P. Wolf, ACS Polymer Preprints, **36** (2), 370 (1995).
682. H. R. Allcock et al., Macromolecules, **29**, 7740 (1996).
683. Trade literature, BASF, Rensselaer, NY.
684. E. Malstrom, A. Hult, Macromolecules, **29**, 1222 (1996).
685. Trade literature, Dendritech Inc., Midland, MI.
686. J. F. Klebe, J. Polym. Sci. A, **2**, 2673 (1964).
687. K. I. Lee, H. Japson, P. Choleva, Internat. Polym. Sci. Technol., **8**, 143 (1981).
688. M. Kubo et al., Macromolecules, **29**, 4447 (1996).
689. E. Malstrom et al., ACS Polym. Mater. Sci. Eng., **77**, 151 (1997).
690. H. R. Kricheldorf, A. Serra, Polym. Bull., **14**, 497 (1985).
691. E. Schmidt, Angew. Makromol. Chem., **14**, 185 (1970).
692. N. Tsutsumi et al., Macromolecules, **30**, 1637 (1997).
693. M. Bruma et al., ACS Polymer Preprints, **36** (1), 727 (1995).
694. K. Vanneste et al., Macromolecules, **27**, 7496 (1994).
695. S. C. E. Backson et al., ACS Polymer Preprints, **34** (1), 50 (1993).
696. B. A. Howell, B. Pan, ACS Polym. Mater. Sci. Eng., **76**, 401 (1997).
697. M. Eaker et al., ACS Polymer Preprints, **36** (1), 473 (1995).
698. K. Char et al., Macromolecules, **26**, 4165 (1993).
699. S. Hadjikyriacou, R. Foust, Macromolecules, **28**, 7893 (1995).
700. S. Nagai, Bull. Chem. Soc. Japan, **37**, 369 (1964).
701. Y. Chang et al., Macromolecules, **27**, 2145 (1994).

Specific Refractive Index Increments of Polymers in Dilute Solution*

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A. INTRODUCTION

This section presents a compilation of the specific index of refraction increments, $\nu = dn/dc$, of polymer solutions. This table has been updated since the third edition. The trend of using $\lambda_0 = 633$ nm from the HeNe laser continues and thus the 633 nm data has been given its own column in this compilation. As in the previous edition, most copolymer data has not been included.

Perhaps the most significant change since the previous edition is the coupling of size exclusion chromatography (SEC) with multiple detectors, largely eliminating the need for ν for obtaining molecular weights. In addition, the measurement of the specific refractive index increment has become so routine that many authors no longer report the results of these measurements. These two factors combine to reduce the number of new entries to a mere 46 new references and about 160 new entries.

Finally, the excellent introduction written by Huglin for the third edition has been retained in its entirety and follows this brief introduction.

B. INTRODUCTION FROM THE THIRD EDITION

This is the limiting value at $c = 0$ of the quantity $(n - n_0)/c$, where n and n_0 are the refractive indices of the solution and solvent, respectively, and c (g/ml) denotes the concentration of the polymer in solution. Conditions of constant pressure p, constant temperature T, and a fixed wavelength of light in vacuum, λ_0, are implied. Specific refractive index increments are required to determine the weight-average molecular weight, $\overline{M}_w$, of a polymer and the second virial coefficient, A_2, of a polymer in solution, using the light scattering equation.

$$\frac{K^* c}{R_\Theta} = \frac{1}{\overline{M}_w P(\Theta)} + 2A_2 c \qquad \text{(A1)}$$

where

$$K^* = K(n_0 \nu)^2 \quad \text{and} \quad K = 2\pi^2 / N_A \lambda_0^4 \qquad \text{(A2)}$$

Here, c is the concentration of the polymer in solution (g/ml) R_Θ (cm^{-1}) is the reduced intensity at an angle Θ, $P(\Theta)$, is the particle scattering function, A_2 is expressed in mol cm^3/g^2, N_A (mol^{-1}) is the Avogadro number. Because

* Based on a table by M. B. Huglin in the 3rd ed. of the Polymer Handbook.

ν appears as a squared term, an error of $E\%$ in ν will result in an error of $2E\%$ in the derived molecular weight, and consequently an accurate determination of ν is imperative. Examination of reported data discloses that about 20 different values of λ_0 have been used by different workers, this increase over the earlier more restricted number being due to growing use of various laser light sources. The values of the constant K appropriate to nine common values of λ_0 are as follows:

$10^7 \times \lambda_0$ (cm)	$10^6 \times K$ (cm^{-4} mol)
366	18.3
403	12.4
436	9.07
488	5.78
514	4.69
546	3.69
578	2.94
633	2.04
1086	0.236

For a copolymer comprising monomers A and B or a terpolymer comprising units A, B and C, the specific increment is an additive function of the composition of the polymer expressed in terms of weight fractions w. Thus,

$$\nu = w_A\nu_A + w_B\nu_B \quad \text{(A3)}$$

$$= w_A\nu_A + w_B\nu_B + w_C\nu_C \quad \text{(A4)}$$

Equations (A3) and (A4) have been verified with copolymer samples of known composition (5,6). In studies on terpolymerization, Kambe and Honda (7,897) have found that the value of ν, from monomer feeds other than the azeotropic composition, changes with conversion, on account of a composition distribution. On the other hand, a constant value of ν, independent of conversion, is yielded for the azeotropic case in which feed and polymer have the same composition. When ν_A and ν_B differ appreciably in a given solvent, Eq. (A3) affords a convenient means of estimating composition from the measured value of ν. If this is not the situation, it is possible to introduce changes in the specific increment by using mixed solvents. The appropriate specific increment to be utilized in light scattering is then ν_μ, that is, one at constant potential of low molecular weight solvent. ν_μ is measured after dialysis equilibrium between solution and pure mixed solvent. ν and ν_μ allow one to obtain the coefficient of preferential adsorption γ_1 (ml/g) of the liquid (solvent 1) in the binary solvent mixture, which has a greater affinity for the polymer:

$$\gamma_1 = (\nu_\mu - \nu)/(dn_0/d\phi_1) \quad \text{(A5)}$$

In Eq. (A5), $dn_0/d\phi_1$ is the variation of the refractive index of the mixed solvent, with its composition expressed in terms of the volume fraction ϕ_1 of liquid 1.

For a copolymer derived from A and B, light scattering always yields the true value of the molecular weight M if $\nu_A = \nu_B$. However, it is not possible to derive the molecular weights M_A and M_B of the constituent portions or the heterogeneity parameters P and Q. If $\nu_A \neq \nu_B$, an apparent value of M is obtained. Tuzar et al. (8) have summarized the criteria for the most sensitive characterization of a copolymer:

Quantity to be determined	Criterion
M	ν_A and ν_B both large and of same sign; or $\nu_A = \nu_B$
M_A	ν_A large and $\nu_B = 0$
M_B	ν_B large and $\nu_A = 0$
Q	$\nu = 0$

Normally, these conditions cannot be satisfied by using single solvents alone, and recourse must be made to using mixed solvents and employing ν_μ, $(\nu_A)_\mu$, and $(\nu_B)\mu$ of the appropriate composition to satisfy them.

The concentration dependence of ν is expressed by the coefficient a_2, in Eq. (A6).

$$(n - n_0)/c = \nu + a_2c \quad \text{(A6)}$$

Lorimer (1) has derived an expression for coefficient a_2 in terms of n_0, ν, and the concentration dependence of the partial specific volume of the polymer in solution, $\bar{v}_2$. However, within the range of c normally used to measure v, the value of a_2 is usually close to zero. Expressions have also been formulated for v in terms of n_0, v_2, and the refractive index of the polymer in solution, n_2. These afford an indirect means of obtaining $\bar{v}_2$ from an experimental value of v (see Refs. 2–5). Actually $\bar{v}_2$ can now be measured to a very high level of accuracy with the Kratky digital densimeter, provided there is an appreciable difference between the densities of polymer and solvent. The main uncertainty or difficulty lies in the value of n_2, when one wishes to calculate ν via $\bar{v}_2$.

1. Measurement and Calibration

Experimental methods and refinements have been summarized elsewhere (4). The main approaches are given below.

Interferometry In this absolute method, light beams of identical geometrical paths traverse two different optical paths and the number (f) of displaced fringes is measured by an optical compensator. If l is the length of the troughs, $(n - n_0)$ is given by

$$(n - n_0) = f\lambda_0/l \quad \text{(A7)}$$

When $(n - n_0)$ is very small, for example, in extremely dilute solutions, the accuracy can be enhanced by increasing the trough length. A range of troughs of different lengths is normally available with a commercial instrument. The slope of a plot f versus c, when multiplied by (l/λ_0),

yields the value of ν. Baltog et al. (10) have developed an improved cell which prevents the formation of a liquid film at the central surface with the trough, thus enabling accurate measurements to be made with volatile liquids.

Differences in the dispersion of the refractive index of the solvent and the solute can lead to discontinuities in the plot of f versus c. This problem can be overcome by measuring f for a large number of solutions of gradually increasing concentration, i.e., by the dropwise addition of concentrated polymer solution to pure solvent (5,11).

Differential Refractometry This relies on the image displacement ΔX of beams from the light source, which traverse pure solvent and polymer solution in a divided cell. Adequate temperature control is necessary and it is essential to utilize a stoppered cell to prevent solvent "creep". The method is not an absolute one and necessitates calibration with solutions of accurately known $(n - n_0)$, so that the calibration constant k can be determined for the apparatus:

$$k = (n - n_0)/\Delta X \qquad \text{(A8)}$$

In Eq. (A8), n is the refractive index of the solution of calibrant, and n_0 that of the solvent. The value of k decreases slightly with increase in λ_0 and must therefore be re-determined if measurements are to be conducted at a different wavelength. A selection of useful calibration data (all involving aqueous solutions) has been gathered from several sources in the literature and is given in Section C. The Brice–Phoenix differential cell is satisfactory for liquids having refractive indices not in excess of about 1.60. At higher values than this, the slit image disappears from the field of vision because of total reflection in the glass partition dividing the cell. Kratochvil and Babka (9) have described a useful modification whereby the partition is replaced with glass of a higher refractive index. This extends the useful range of the different refractometers so that even such highly refractive liquids as 1-bromonaphthalene can be accomodated.

The calibration data at $\lambda_0 = 436$ nm for aq. KCl in Section C have been fitted by Abdel-Azim and Huglin (676) to the following polynomial in which C is the concentration expressed is g salt/100 g water

$$(n - n_0) \times 10^6 = 1.862 + 1415.8C - 17.699C^2 + 0.428C^3$$

Some light scattering and refractometric data relating to wavelengths different from those in Section C have been summarized elsewhere (833). The most common light source for recent light scattering studies is probably the He–Ne laser affording $\lambda_0 = 633$ nm. At this λ_0, directly measured specific refractive index increments for certain aqueous salt solutions have been reported. On the assumption of direct proportionality between $(n - n_0)$ and concentration, these may be represented as follows for $\lambda_0 = 633$ nm and C in g salt/100 g water (Ref. 942):

$$\text{Aq. } NH_4HO_3 \quad (n - n_0) \times 10^6 = 1239C$$
$$\text{Aq. KCl} \quad (n - n_0) \times 10^6 = 1371C$$
$$\text{Aq. NaCl} \quad (n - n_0) \times 10^6 = 1766C$$

Also, for $\lambda_0 = 1086$ nm and C expressed in g salt/100 ml solution, the following holds at 20°C (Ref. 943):

$$\text{Aq. KCl} \quad (n - n_0) \times 10^6 = 1351C$$

Mandel and coworkers (722) have calibrated the Chromatix differential refractometer for $\lambda_0 = 633$ nm and have also indicated the range over which ΔX is directly proportional to $(n - n_0)$. Values of the latter quantity were interpolated to $\lambda_0 = 633$ nm via a Cauchy dispersion from the values at other wavelengths in Section C. Within the interval $3.6 > C > 0.46$ g salt/100 g water, the following polynomial was found for aq. NaCl at 633 nm

$$(n - n_0) \times 10^6 = 1735C - 27C^2 + C^3$$

Abbreviations/Symbols Used

Arochlor	(approximately) Chlorinated diphenyl
(B)	Branched polymer
Cadoxen	Triethylenediamine cadmium hydroxide
Calgon	Sodium hexametaphosphate
Cuoxam	Cuprammonium hydroxide
Cuoxen	Cupriethylenediamine hydroxide
DCA	Dichloroacetic acid
DMA	*N,N*-Dimethylacetamide
DMSO	Dimethyl sulfoxide
DN	Degree of neutralization (also α)
DS	Degree of substitution
EDTA	Ethylenediamine tetraacetic acid
FeTNa	Iron–tartaric acid–sodium complex solution
Freon 113	1-Fluorodichloro-2-chlorodifluoroethane
GHCl	Guanidine hydrochloride
Halothane	1,1,1-Trifluoro-2-chloro-2-bromoethane
HEMA	2-Hydroxyethyl methacrylate
HMP	Hexamethyl phosphoramide
I	Ionic strength
M	Relative molar mass (M) or molarity (M), according to context
MEK	Methyl ethyl ketone
MMA	Methyl methacrylate
TEAB	Tetraethyl ammonium bromide
TFP	2,2′,3,3′-Tetrafluoropropanol
THF	Tetrahydrofuran
Versene	Sodium salt of EDTA
(μ)	Specific increment measured at dialysis equilibrium

C. REFRACTOMETRIC CALIBRATION DATA

Solute	Concentration (g/100 ml solution)	Concentration (g/100 g water)	Temp. (°C)	$(n - n_0)$ ($\times 10^6$) 436 nm	546 nm	578 nm	589 nm
NaCl	0.0938	0.09412	25	173	166	165	165
	0.1034	0.1037	25	190	184	182	182
	0.3362	0.3375	25	614	592	588	587
	0.5602	0.5627	25	1019	982	976	974
	0.6866	0.6900	25	1246	1202	1194	1191
	0.9090	0.9142	25	1645	1586	1576	1572
	1.1240	1.1311	25	2028	1956	1943	1939
	1.6465	1.6595	25	2954	2848	2830	2824
	2.0327	2.0513	25	3634	3504	3481	3474
	3.7307	3.7854	25	6583	6345	6306	6293
	6.7405	6.9092	25	11673	11247	11174	11151
NH_4NO_3	0.07878	0.07906	25	103.5	98.77	–	97.73
	0.1671	0.1677	25	218.9	208.9	–	206.7
	0.3433	0.3450	25	448.5	428.0	–	423.5
	1.0314	1.0408	25	1336	1275	–	1262
	2.8884	2.9479	25	3702	3531	–	3494
	5.8417	6.0732	25	7410	7066	–	6992
Na_2SO_4	0.09566	0.09595	25	153.1	151.0	–	150.5
	0.2198	0.2205	25	349.0	344.1	–	343.0
	0.2686	0.2694	25	425.3	419.3	–	417.9
	0.3939	0.3952	25	620.1	611.5	–	609.4
	0.7508	0.7536	25	1168	1151	–	1148
	1.2640	1.2693	25	1940	1913	–	1906
	1.5417	1.5488	25	2352	2319	–	2311
	2.5322	2.5471	25	3796	3744	–	3731
	5.7875	5.8500	25	8309	8195	–	8168
$SrCl_2$	0.08474	0.08506	25	159.8	155.1	–	154.1
	0.1749	0.1755	25	328.5	318.9	–	316.7
	0.2430	0.2437	25	455.5	442.2	–	439.1
	0.3580	0.3591	25	669.2	649.6	–	645.1
KCl	0.0696	0.0699	25	100	96	96	96
	0.1067	0.1070	25	153	148	147	146
	0.2799	0.2812	25	399	386	383	382
	0.5964	0.5994	25	845	817	812	810
	1.0794	1.0869	25	1521	1469	1460	1457
	1.4911	1.5037	25	2093	2022	2009	2005
	2.9821	3.0250	25	4135	3994	3969	3962
	3.9969	4.0703	25	5500	5314	5281	5271
	4.4732	4.5647	25	6136	5926	5890	5879
	5.9642	6.1217	25	8105	7828	7781	7766
	6.4680	3.6526	25	8763	8465	8414	8398
Sucrose	0.100	0.100	20	–	–	–	140
	–	–	28	–	–	–	140
	0.200	0.200	20	–	–	–	290
	–	–	28	–	–	–	290
	0.300	0.301	20	–	–	–	430
	–	–	28	–	–	–	430
	0.400	0.402	20	–	–	–	580
	–	–	28	–	–	–	570
	0.500	0.502	20	–	–	–	720
	–	–	28	–	–	–	720
	0.600	0.604	20	–	–	–	860
	–	–	28	–	–	–	860
	0.701	0.705	20	–	–	–	1010
	–	–	28	–	–	–	1000
	0.801	0.806	20	–	–	–	1150
	–	–	28	–	–	–	1140
	0.902	0.9082	20	–	–	–	1300
	–	–	28	–	–	–	1290
	1.002	1.010	20	–	–	–	1440
	–	–	28	–	–	–	1430
	1.506	1.523	20	–	–	–	2160

Solute	Concentration (g/100 ml solution)	Concentration (g/100 g water)	Temp. (°C)	$(n-n_0)$ ($\times 10^6$) 436 nm	546 nm	578 nm	589 nm
	–	–	28	–	–	–	2150
	2.012	2.041	20	–	–	–	2890
	–	–	28	–	–	–	2870
	3.030	3.093	20	–	–	–	4340
	–	–	28	–	–	–	4300
	3.52	–	25	5100	5000	–	–
	4.056	4.167	20	–	–	–	5810
	–	–	28	–	–	–	5760
	5.089	5.263	20	–	–	–	7280
	–	–	28	–	–	–	7220
	6.14	–	25	8900	8800	–	–
	7.181	7.527	20	–	–	–	10270
	–	–	28	–	–	–	10190
	8.240	8.696	20	–	–	–	11780
	–	–	28	–	–	–	11680
	9.306	9.890	20	–	–	–	13300
	–	–	28	–	–	–	13190
	10.6	–	25	15400	15200	–	–
	16.3	–	25	23600	23200	–	–

D. TABLES OF SPECIFIC REFRACTIVE INDEX INCREMENTS OF POLYMER SOLUTIONS

TABLE 1. MAIN-CHAIN ACYCLIC CARBON POLYMERS

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
1.1. POLY(DIENES)							
Poly(1,3-butadiene)	Benzene		0.011				57
		0.0117	0.0086			20	58
		0.022	0.021			25	647
	Bromoform		−0.115				58
	1-Bromonaphthalene		−0.185				58
	Chloroform		0.083				58
	Cyclohexane	0.144					59,60
		0.126	0.118				61
pure 1,4		0.121	0.113			25	62
pure 1,2		0.092	0.087			25	62
	Diethyl ketone	0.156				25	63
	n-Heptane	0.151	0.141				61
	THF		0.132				57
hydroxyl terminated and acetyl terminated	Chloroform	0.092	0.087			25	859
	Chloroform	0.096	0.093			37	859
	Cyclohexane			0.106		20	835
	Cyclohexane	0.127	0.118			25	647
98% *cis* 1,4	Cyclohexane			0.111		30	836
98% *trans* 1,4	Cyclohexane			0.110		30	836
7% vinyl-99% vinyl	Cyclohexane			0.110–0.087		30	836
high 1,4; $M = 0.96 \times 10^3$ to 90.6×10^3	Cyclohexane			0.094–0.111		30	836
	Cyclohexanone		0.081			25	647
	1,2-Dichloroethane	0.092	0.088			25	647
hydroxyl terminated and acetyl terminated	*n*-Heptane	0.157	0.150			25	859
	Hexane	0.177	0.169			35	859
	Isobutyl acetate	0.149	0.143			20.5	859
	MEK	0.165	0.159			26	859
	5-Methyl-2-hexanone	0.125	0.120			12.6	859
	3-Pentanone	0.145	0.139			10.3	859
	THF			0.130		23	830
	Toluene	0.034	0.031			25	647

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Poly(chloroprene)	*n*-Butyl acetate	0.146				25	64
	Carbon tetrachloride	0.0976				25	64
	trans-Decalin	0.0799				25	64
		0.0820				40	64
		0.0777				10	64
	MEK	0.160				25	64
		0.162	0.156			25	65
					0.152		66
	THF				0.138		66
Poly(cyclopentadiene)	Benzene		0.0394			21	67
	Carbon tetrachloride		0.125			21	67
Poly(isoprene)							
97.8% *cis* 1,4	Benzene	0.0143	0.0194			20	58
73–80% *cis* 1,4	Bromobenzene		−0.041			20	73
Poly(isoprene) anionically polymerized	Carbon tetrachloride			0.082		25	964
	Chloroform	0.100	0.095			25	68
73–80% *cis* 1,4	*o*-Chlorotoluene		−0.003			20	73
73–80% *cis* 1,4	Cyclohexane		0.104			20	73
70–75% *cis* 1,4			0.1034			23	34
trans		0.117					59
cyclic			0.131				74,75
		0.1305	0.1238			25	69
			0.124			27	69
			0.117			7	69
	Cyclohexene	0.0988	0.0943			25	68
		0.0992	0.0947			27	68
		0.0953	0.0909			17	68
		0.0914	0.0869			7	68
(non-extracted)			0.145			20	70
(extracted)			0.0946			20	70
	Decalin	0.0669	0.0605			25	69
73–80% *cis* 1,4	*n*-Hexane		0.168			20	73
		0.1886	0.1802			25	68
73–80% *cis* 1,4	Methyl isobutyl ketone		0.139			20	73
cyclic	THF		0.156				73,74
cis 1,4			0.128			20	70
			0.128			20	70
		0.160 (calc.)				19–21	71
70–75% *cis* 1,4	Toluene		0.0308			35	34
73–80% *cis* 1,4			0.028			20	73
		0.0308	0.0339			25	68
chlorinated	MEK	0.131				35	72
synthetic high *cis*	Chloroform	0.104	0.100			25	655
natural hevea		0.106	0.104			25	655
natural guayule		0.108	0.101			25	655
	Cyclohexane	0.117				35	704
				0.106		20	835
trans 1,4					0.117	30	838
star branched				0.104–0.113		25	833
93% 1,4- $M = 0.9 \times 10^3$ – 10.1×10^3				0.093–0.105		25	833
93% 1,4- $M \geq 19.5 \times 10^3$				0.106		25	833
synthetic high *cis*	*n*-Hexane	0.192	0.191			25	655
natural hevea		0.200	0.198			25	655
natural guayule		0.198	0.193			25	655
	THF			0.124		20	835
synthetic, high *cis*		0.153	0.148			25	655
natural hevea		0.160	0.156			25	655
natural guayule		0.157	0.153			25	655
synthetic	Toluene	0.030	0.028			25	655
natural hevea		0.034	0.032			25	655
natural guayule		0.033	0.030			25	655
Poly(trichlorobutadiene)	Benzene				0.06	25	76

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
1.2. POLY(ALKENES)							
Poly(1-butene)	*n*-Nonane	0.092				35	13
		0.108				80	13
isotactic	1-Chloronaphthalene	−0.206	−0.187			135	867
					−0.183 (578)	135	867
				−0.177		135	867
	Cyclohexane		0.074			25	621
	Cyclohexane/*n*-propanol						
	(10/90, v/v)		0.068			35	767
			0.072			35	767
	(20/80, v/v)		0.072			35	767
	(30/70, v/v)		0.075			35	767
	(35/65, v/v)		0.079			35	767
	1,2,4-Trichlorobenzene	−0.120	−0.108		−0.127 (403)	135	867
				−0.103	−0.105 (578)	135	867
Poly(ethylene)	Biphenyl		−0.174			123.2	14
	Bromobenzene	−0.101	−0.089		−0.107 (403)	135	863
				−0.083	−0.085 (578)	135	863
B		−0.095	−0.083		−0.101 (403)	135	863
B				−0.078	−0.080 (578)	135	863
B							
	1-Bromonaphthalene		−0.208			135	863
				−0.205	−0.207 (578)	135	863
B			−0.210			135	863
B					−0.208 (578)	135	863
B				−0.206		135	863
B	*o*-Chlorobenzene			−0.076		135	863
	1-Chloronaphthalene		−0.195			65–80	15
			−0.199			90	16,17
			−0.198			90	18
			−0.197			100	19
			−0.188			105	20,21
			−0.195			110	19
			−0.1957			114	23
			−0.193			120	19
			−0.192			125	24
			−0.195			125	25,18
			−0.191			125	16
			−0.214			125	26
		−0.245				125	27
			−0.1967			127	23
			−0.1955			128	23
			−0.191			130	19
M_n = 12300			−0.1931			135	22
13500			−0.1929			135	22
18300			−0.1932			135	22
25200			−0.1916			135	22
110000			−0.1883			135	22
112000			−0.1879			135	22
			−0.190			135	17,28
			−0.1956			139	23
			−0.1961			140	23
			−0.191			140	25
			−0.189			140	19
			−0.1967			145	23
			−0.187			150	19
			−0.1943			151	23
			−0.193			115	832,917
			−0.192			120	931
			−0.191			125	931
			−0.191			127	610

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
		−0.215	−0.192	−0.183	−0.188 (578)	135	863,867
			−0.192			135	866
			−0.191			135	932
			−0.193			135	832,917,934
			−0.196			135	631
			−0.188 to −0.193			135	815,914
			0.195 to −0.192			135	933
increase with M			−0.193 to −0.188			135	914
B		−0.212	−0.189			135	863,867
B				−0.178	−0.184 (578)	135	863,867
B			−0.190			140	935,936
			−0.196			145	934
				−0.177		145	864
			−0.195				937,938
	n-Decane		0.0881			65	15
			0.0890			70	15
			0.0901			75	15
			0.0917			80	15
			0.0870			111	23
			0.0881			114	23
			0.095			115	24
			0.0915			120	23
			0.0922			125	23
			0.0937			130	23
			0.0955			138	23
			0.0971			139	23
			0.0985			143	23
			0.0979			146	23
			0.0995			149	23
		0.114	0.113		0.114 (404)	106	632
					0.112 (577)	106	632
		0.121	0.117		0.119 (404)	106	632
					0.116 (577)	106	832
		0.126	0.121		0.123 (404)	106	832
					0.120 (577)	106	832
		0.132	0.126		0.130 (404)	106	832
					0.124 (577)	106	832
		0.117	0.114		0.119 (403)	135	863
				0.112	0.113 (578)	135	863
B		0.120	0.118		0.122 (403)	135	863
B				0.116	0.117 (578)	135	863
B			0.098			140	935
	p-Dibromobenzene	−0.179	−0.162		−0.184 (403)	135	863
				−0.155	−0.159 (578)	135	863
B		−0.176	−0.159		−0.182 (403)	135	863
B				−0.153	−0.157 (578)	135	863
B							
	o-Dichlorobenzene		−0.078			120	939
			0.078			120	29
				−0.056		135	864
		−0.095	−0.083		−0.101 (403)	135	863
				−0.078	−0.080 (578)	135	863
B		−0.091	−0.081		−0.099 (403)	135	863
B					−0.078 (578)	135	863
	Diphenyl		−0.174			123	939
		−0.202	−0.176		−0.215 (403)	127	863
				−0.166	−0.172 (578)	127	863
B		−0.199	−0.173		−0.212 (403)	127	863
B				−0.165	−0.169 (578)	127	863
B							
		−0.195	−0.172		−0.209 (403)	135	863
				−0.163	−0.168 (578)	135	863

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
B		−0.192	−0.167		−0.207 (403)	135	863
B				−0.158 (633)	−0.163 (578)	135	863
B							
	Diphenylmethane	−0.146	−0.129		−0.155 (403)	142	863
			−0.126			142	866
				−0.126	−0.127 (578)	142	863
B		−0.135	−0.120		−0.142 (403)	142	863
B				−0.116	−0.118 (578)	142	863
B							
	1-Dodecanol	0.048	0.046		0.049 (403)	137	863
				0.045	0.046 (578)	137	863
B		0.049	0.047		0.050 (403)	137	863
B				0.047	0.047 (578)	137	863
B							
	1-Methylnaphthalene	−0.206	−0.177		−0.220 (403)	135	863
				−0.167	−0.172 (578)	135	863
B		−0.199	−0.170		−0.211 (403)	135	863
B				−0.159	−0.164 (578)	135	863
B							
	Paraffin oil	−0.001	−0.001		−0.001 (403)	135	863
			–	−0.001	−0.001 (578)	135	863
B		0.001	0.001		0.001 (403)	135	863
B				0.001	0.001 (578)	135	863
B							
	Tetralin		−0.075			120	931
			−0.074			125	931
			−0.094			125	940
			−0.094			129	941
		−0.087	−0.077		−0.091 (403)	135	863
				−0.074	−0.075 (578)	135	863
B		−0.085	−0.075		−0.089 (403)	135	863
B				−0.071	−0.073 (578)	135	863
B							
			−0.0834			50	15
			−0.0811			80	15
		0.104				81.5	30,20
		−0.0887	−0.075			81.5	25
			−0.0910			95	23
			−0.0918			95	23
			−0.0903			100	23
			−0.893			104	23
		−0.0805	−0.0691			105	25
			−0.100			105	21*
			−0.078			105	29*
			−0.072			105	29*
			−0.083			105	29*
			−0.088			110	23
			−0.0866			116	23
			−0.0858			119	23
			−0.075			125	15
			−0.0837			128	23
			−0.0813			138	23
			−0.0801			144	23
	1,2,4-Trichlorobenzene		−0.106			135	631
					−0.109 (488)	135	746,900
		−0.125	−0.116		−0.130 (403)	135	863,867
				−0.107	−0.108 (578)	135	863,867
				−0.104		135	864
			−0.111			135	866
B		−0.177	−0.105		−0.122 (403)	135	863,867
B				−0.102	−0.103 (578)	135	863,867

*Values reported erroneously as positive in these references.

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
B							
increase with M			−0.109 to −0.106			135	914
M_n = 2000			−0.1085			135	22
18400			−0.1085			135	22
34000			−0.1082			135	22
77000			−0.1073			135	22
110000			−0.1063			135	22
Poly(hexene-1)	Cyclohexane	−0.063				25	858
	Toluene	−0.042				25	858
Poly(isobutylene)	Benzene	−0.007				25	956
	Cyclohexane		0.105			20	31
					0.099	25	32
	Cyclohexane	0.067				25	872
	Cyclohexane	$0.0285 + 2.08 \times 10^{-4} T$ (K)	$0.0198 + 2.29 \times 10^{-4} T$ (K)				956
	Cyclohexane	0.0991	0.0966			25	957
	1-Chloronaphthalene	−0.146	−0.131			135	867
	Decalin	0.043	0.0374			25	33
	n-Decane		0.1036			35	34
	Dibutyl ether		0.124			22.5	35
			0.167			120	35
	n-Heptane		0.135			25	36
		0.140				25	37
		0.1422				25	38
	n-Hexane	0.1008				25	38
		0.155				25	872
	Isoamyl isovalerate	0.1066				15	38
		0.1091				25	38
		0.1129				40	38
	Isooctane	0.140				25.5	39
					0.135	25	32
			0.139			25	40
				−0.122	−0.128 (578)	135	867
	Toluene	0.001				25	872
	1,2,4-Trichlorobenzene	−0.066	−0.054		−0.070 (403)	135	867
				−0.051	−0.053 (578)	135	867
Poly(1-octene)	Bromobenzene		−0.105			30	41
trans-poly(pentenamer)							
	n-Hexane	0.175	0.171			25	894
Poly(propylene)	Benzene		−0.0497			25	42
	Bromobenzene		−0.1125			40	42
			−0.0599			25	43
	n-Butyl chloride		−0.1082			25	43
	n-Butyl propionate		0.1101			25	43
	Carbon tetrachloride		0.0112			25	43
	1-Chloronaphthalene	−0.228	−0.227			125	44
			−0.195			140	25
		−0.227				145	27
		−0.231				145	45
isotactic			−0.189			125	46
			−0.188			140	28,47
		−0.2275	−0.2157			145	48
		−0.195	−0.184		−0.200 (365)	150	49
	Cyclohexane		0.0570			25	42
	Decalin				0.023	120	50
isotactic	*n*-Heptane	0.1105				30	51
syndiotactic		0.1077				30	52
	n-Hexane		0.1152			25	42
	Methylcyclohexane		0.0652			25	42
	Toluene		−0.0368			25	42
			−0.0027			25	43
	p-Xylene		−0.0380			25	43
atactic	1-Chloronaphthalene	−0.211	−0.188	−0.177	−0.183 (578)	135	867

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
isotactic		−0.205	−0.184	−0.173	−0.180 (578)	135	867
atactic	1,2,4-Trichlorobenzene	−0.123	−0.111		−0.130 (403)	135	867
				−0.105	−0.109 (578)	135	867
isotactic		−0.121	−0.107		−0.128 (403)	135	867
				−0.102	−0.105 (578)	135	867
				−0.092 to −0.094		145	646
Poly(1-phenyl-1-propyne)	Toluene	0.1861	0.1656			25	979
	Cyclohexane	$0.264 + 5.3 \times 10^{-4}$(T°C-30)	$0.234 + 4.7 \times 10^{-4}$(T°C-30)			30–45	979
Poly(trifluorochloroethylene)	Mesitylene		0.026			145	53
Poly(vinylcyclohexane)	Cyclohexane		0.101			25	305
	Cyclohexane/dioxane (50.1/49.9, v/v)		0.103			25	305
	Cyclohexane/MEK (64.6/35.4, v/v)		0.138			25	305
	Decalin/dioxane (50.6/49.4, v/v)		0.078			25	305
	Decalin/MEK (59.7/40.3, v/v)		0.075			25	305
	n-Heptane/dioxane (56.8/43.2, v/v)		0.147			25	305
	n-Heptane/MEK (82.6/17.4, v/v)		0.167			25	305
	n-Hexane		0.170			25	305
	THF		0.131			25	305
1.3. POLY(ACRYLIC ACID) AND DERIVATIVES							
Poly(acrylamide)	Acetic acid	0.199	0.194	0.192		25	627
	DMSO	0.091	0.089	0.089		25	627
	Ethanolamine	0.093	0.089	0.088		25	627
	Ethylene glycol		0.177			25	657
			0.095–0.110			25	657
			0.084			25	620
		0.117	0.113	0.111		25	627
	Formamide	0.109	0.109	0.109		25	627
		0.103				30	898
			0.127			65	928
	Formic acid	0.172	0.169	0.166		25	627
	Aq. 1% LiCl				0.181	25	928
	Aq. 5% LiCl		0.110			65	928
	Aq. 1 M $Mg(ClO_4)_2$		0.165			25	728
	Aq. 2.8 M $Mg(ClO_4)_2$		0.145			25	728
	Aq. 3 M $Mg(ClO_4)_2$		0.138			25	728
	Aq. 1 M NaCl		0.175			RT	674
	Aq. 1 M NaCl		0.175 (μ)			20	714
	Aq. 1 M NaCl			0.156–0.159		25	926
	Aq. 0.2 M NaCl	0.186	0.182			20	714
	Aq. 0.2 M NaCl		0.172 (μ)			20	714
	Aq. 0.1 M NaCl		0.17			20	929
	Aq. 0.1 M NaCl				0.164		927
	Aq. 1 M $NaNO_3$	0.182				30	899
	Aq. 0.02 M Na_2SO_4			0.1869		25	980
	Water	0.185	0.182			20	714
					0.149	20–65	77
			0.163				78
					0.195 (589)	30	79
					0.179		80
			0.173 (μ)			20	714
		0.167	0.160		0.172 (405)	25	639
					0.157 (579)	25	639
			0.187			25	657
			0.183			25	728
			0.170			25	845
			0.169			25	785
			0.162			25	657

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
			0.161			25	804
				0.183		25	926
		0.18				25	888
					0.165		927
		0.193	0.187	0.185		25	627
				0.146		23	924
				0.142		23	925
On aging containing carboxyl groups	Water		0.197–0.326			25	620
	Water			0.149 (488)		53.7	604
Poly[*N*-(1,1-dimethyl-3-imidazolylpropyl)acrylamide]							
	0.01 M Tetrabutyl-ammonium	0.209				25	944
Poly(acrylic acid)							
DN = 0.102, I = 0.10	Aqueous NaCl	0.158				30	85
0.335, 0.10		0.179				30	85
0.994, 0.10		0.261				30	85
	Dioxane				0.090	25	86
					0.088	25	86
	HCl 0.1 N				0.140 (589)		87
	0.2 N				0.146	20–65	77
	Water				0.137		80
			0.156			25	617
	Water/dioxane (90/10, v/v)		0.160			25	617
			0.112 (μ)			25	617
	(80/20, v/v)		0.164			25	617
			0.105 (μ)			25	617
	(60/40, v/v)		0.171			25	617
			0.119 (μ)			25	617
	(50/50, v/v)		0.174			25	617
			0.117 (μ)			25	617
	(40/60, v/v)		0.178			25	617
			0.117 (μ)			25	617
	(35/65, v/v)		0.166			25	617
			0.112 (μ)			25	617
	(30/70, v/v)		0.159			25	617
			0.101 (μ)			25	617
	(25/75, v/v)		0.141			25	617
			0.079 (μ)			25	617
	(10/90, v/v)		0.204			25	617
			0.138 (μ)			25	617
	(5/95, v/v)		0.092			25	617
			0.075 (μ)			25	617
	(1.5/98.5, v/v)		0.078			25	617
			0.084 (μ)			25	617
–, lithium salt	Methanol $\alpha = 0$			0.191		20	743
	0.04			0.198		20	743
	0.09			0.199		20	743
	0.21			0.233		20	743
	0.40			0.235		20	743
	0.60			0.251		20	743
	0.69			0.224		20	743
–, sodium salt	Water		0.231			25	617
	Water/dioxane (90/10, v/v)		0.246			25	617
			0.126 (μ)			25	617
	(80/20, v/v)		0.258			25	617
			0.100 (μ)			25	617
	(60/40, v/v)		0.238			25	617
			0.073 (μ)			25	617
	(50/50, v/v)		0.219			25	617
			0.082 (μ)			25	617
	(40/60, v/v)		0.203			25	617
			0.069 (μ)			25	617

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(30/70, v/v)		0.184			25	617
			0.066 (μ)			25	617
	(20/80, v/v)		0.223			25	617
	Aq. 0.3 M NaCl				0.159 (μ) (488)	25	678
	Aq. 1 M NaCl				0.152 (μ) (488)	25	678
DN = 0.2	Aq. NaBr (M = 0.01, 0.025, 0.1)	0.128 (μ)				25	711
DN = 0.2	(M = 0.5)	0.122 (μ)				25	711
DN = 0.6	(M = 0.01, 0.025, 0.1)	0.156 (μ)				25	711
DN = 0.6	(M = 0.5)	0.138 (μ)				25	711
DN = 1	(M = 0.01, 0.025, 0.1)	0.183 (μ)				25	711
DN = 1	(M = 0.5)	0.166 (μ)				25	711
DN = 1	(M = 1.5)	0.140 (μ)				25	711
Poly(benzyl acrylate)	Benzene	0.062	0.062			20	6
	DMF	0.124	0.119		0.132 (365)	20	6
					0.118 (578)	20	6
					0.117 (589)	20	6
					0.116 (644)	20	6
	MEK	0.1760				20	88
		0.1746	0.1665			20	6
		0.1746	0.1665		0.1653 (578)	20	5
					0.1647 (589)	20	5
					0.1623 (644)	20	5
Poly(*n*-butyl acrylate)	Acetone	0.112				20	89
	Benzene		−0.0292			30	84
	Chlorobenzene		−0.0525			30	84
	Cyclohexanone				0.021 (488)	30	729
	Dioxane				0.043 (488)	30	729
	n-Hexane		0.0885			30	84
	MEK				0.088 (488)	30	729
	THF		0.0651			30	84
			0.062				803
	Toluene		−0.0239			30	84
Poly(*tert*-butyl acrylate)	MEK		0.0818			25	90
Poly(1,1-dihydroperfluoro-butyl acrylate)	Benzotrifluoride				0.30	25	91
					0.0311	25	91
Poly(*N*,*N*-dimethyl-acrylamide)	Methanol				0.194	25	83
	Water				0.150	25	83
Poly(ethyl acrylate)	Acetone		1.107			25	10,92,93
		0.109				20	89
	Benzene		−0.042			25	10
	Chloroform		0.0363			25	10
	Cyclohexanone				0.017 (488)	30	729
	Dichloroethane		0.23			25	10
	Dioxane		0.036			25	10,93
					0.045 (488)	30	729
	DMF		0.032			25	10
	Ethyl acetate		0.0916			25	10
	MEK	0.0870	0.0856		0.0854 (578)	20	5
					0.088 (488)	30	729
					0.0853 (589)	20	5
					0.0852 (644)	20	5
			0.086			25	10,93
			0.1407			30	94
	n-Methylpyrrolidone		−0.027			25	10
	Water				0.131	25	95
Poly(2-ethylhexyl acrylate)	Butanone	0.101					982
	n-Heptane	0.096					982
	Tetrahydrofuran	0.070					982
	Toluene	−0.040					982
	Chlorobenzene	−0.072					982
Poly(isopropyl acrylate) isotactic, syndiotactic and atactic	Bromobenzene	−0.103	−0.0870			60	101

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
isotactic and atactic	TFP		0.124			25	101
Poly(methyl acrylate)	Acetone	0.113				20	89
	Acetonitrile	0.120	0.118			25	96
		0.121	0.119			32	96
		0.122	0.120			40	96
		0.123	0.122			48	96
		0.123				30	96
		0.127	0.125			56	96
	Benzene	−0.019				30	97
	o-Chlorophenol	−0.056				30	916–918
	m-Cresol	−0.035				30	916–918
	1,2-Dibromoethane	−0.0606	−0.560			25	96
		−0.0598	−0.0528			32	96
		−0.0578	−0.0532			40	96
		−0.0579	−0.0523			48	96
		−0.062				30	916–918
	Dioxane	0.0503	0.0497			25	96
		0.052				30	96
		0.055				30	97
		0.0524	0.0515			32	96
		0.0533	0.0526			40	96
		0.0547	0.0537			48	96
		0.0566	0.0562			56	96
	Ethyl acetate	0.0940	0.0920			19.5	96
		0.0989	0.0967			25	96
		0.0967	0.0964			32	96
		0.100				30	96
		0.102				35	98
		0.102	0.101			40	96
		0.105	0.103			48	96
	Isoamyl acetate	0.077				30	97
	MEK	0.090				RT	99
		0.0938	0.0923		0.0921 (578)	20	5
					0.0919 (589)	20	5
					0.0915 (644)	20	5
		0.097				30	97,100
		0.090				35	72
	TFP	0.126				30	916–918
	TFP/*o*-chlorophenol						
	(10/90, v/v)	−0.050				30	916–918
		−0.050 (μ)				30	916–918
	(26/74, v/v)	−0.014				30	916–918
		−0.004 (μ)				30	916–918
	(32/68, v/v)	−0.005				30	916–918
		−0.016 (μ)				30	916–918
	(40/60, v/v)	0.004				30	916–918
		0.029 (μ)				30	916–918
	(50/50, v/v)	0.043				30	916–918
		0.064 (μ)				30	916–918
	(70/30, v/v)	0.075				30	916–918
		0.123 (μ)				30	916–918
	(90/10, v/v)	0.102				30	916–918
		0.135 (μ)				30	916–918
	Toluene	−0.009				30	97
Poly(*n*-octadecyl acrylate)	Benzene		−0.0505			30	84
			−0.0269			30	84
			−0.0363			30	84
	n-Butyl acetate		0.0883			30	84
	Carbon tetrachloride		0.0114			30	84
	Chlorobenzene		−0.0682			30	84
			−0.0489			30	84
			−0.0563			30	84
	Cyclohexane		0.0175			30	84
	n-Hexane		0.1096			30	84
			0.1292			30	84

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	THF		0.0702			30	84
			0.0898			30	84
			0.0723			30	84
	Toluene		−0.0425			30	84
Poly(phenyl acrylate)	MEK	0.183				30	634
Poly(*n*-propyl acrylate)	MEK		0.1102			30	94
Poly(trimethylammoniumethyl chloride acrylate)							
	Aq. 1 M NaCl		0.154			RT	674
1.4. POLY(METHACRYLIC ACID) AND DERIVATIVES							
Poly(allyl methacrylate)	Carbon tetrachloride				0.0508 (589)	25	982
Poly(benzyl methacrylate-*co*-cyclohexylmethyl methacrylate) (1/1, (mol/mol)							
	Di-isobutyl ketone				0.123	12	922
Poly(*N*-*tert*-butylacrylamide)	Methanol	0.249	0.234				81
			0.234			20	82
Poly(*n*-butyl methacrylate)	Acetone	0.1249	0.1236		0.1257 (366)	25	105
		0.122				20	89
	Benzene	−0.023	−0.014			25	106
		−0.024	0.015			25	107
	Bromobenzene	−0.084	−0.073			25	106,107
	Bromoform	−0.129	0.118			25	106
	n-Butyl chloride	0.085				25	911
	Carbon tetrachloride	0.026	0.027			25	106
	m-Cresol/*n*-heptane (30/70, v/v)		0.155			25	789
			0.358 (μ)			25	789
	DMF			0.016		20	769
	Isopropanol	0.102	0.102			23	108
		0.1059				20	109
		0.1066				23.7	109
		0.1068				25	109
		0.1076				31	109
		0.1097				45	109
	MEK	0.103	0.102			25	106,107
		0.1059	0.1046		0.1064 (366)	25	105
			0.104			23	108
				0.097		20	869
			0.100			25	807
	TFP		0.169			25	789
				0.067		20	869
	Toluene			−0.024		20	869
		−0.017				25	911
Poly(*tert*-butyl methacrylate)	*n*-Butyl acetate				0.09		110
Poly(*p*-*tert*-butylphenyl methacrylate)	Acetone				0.183		111
	Chloroform/THF (90/10, v/v)				0.110	25	665
					0.107 (μ)	25	655
	(70/30, v/v)				0.120	25	665
					0.100 (μ)	25	665
	(50/50, v/v)				0.130	25	665
					0.129 (μ)	25	665
	(30/70, v/v)				0.133	25	665
					0.206 (μ)	25	655
	(10/90, v/v)				0.164	25	665
					0.266 (μ)	25	655
	Water/THF (2/98, v/v)				0.142	25	796
					0.124 (μ)	25	796
	(4/96, v/v)				0.208	25	796
					0.143 (μ)	25	796
	(7/93, v/v)				0.177	25	796
					0.194 (μ)	25	796
	(9/91, v/v)				0.192	25	796
					0.205 (μ)	25	796

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) λ_0 = 436 nm	λ_0 = 546 nm	λ_0 = 633 nm	Others	T (°C)	Refs.
Poly(cetyl methacrylate)	Benzene				0.02		112
	n-Butyl acetate	0.095				25	113
	Carbon tetrachloride				−0.015		112
	n-Heptane				0.114	21	112
Poly(2-chloroethyl methacrylate)	THF			0.096		30	749
Poly(4-chlorophenyl methacrylate)	Benzene		0.0758			25	114
Poly(cyclohexyl methacrylate)	*n*-Butyl alcohol		0.098			25	762
	n-Butyl chloride	0.116				25	911
	Cyclohexane		0.0845			25	115
	Toluene	0.019				25	911
Poly(decahydro-β-naphthyl methacrylate)	Cyclohexane		0.104			25	856
Poly(*n*-decyl methacrylate)	Cyclohexane			0.054		30	744
	Ethyl acetate	0.115				25	744
Poly-[2-(2-diethylaminoethoxy)ethyl methacrylate]							
	Methanol			0.163		RT	680
Poly(dimethylaminoethyl methacrylate)	Methanol			0.157		RT	680
Poly(*N*,*N*-dimethylaminoethyl methacrylate *N*-oxide)							
	Aq. 0.1 M NaCl			0.119		25	679
Poly(*n*-docosyl methacrylate)	*o*-Dichlorobenzene	0.102	0.090			25	116
	Tetralin	0.094	0.084			25	116
		0.0935				29	116
		0.093	0.082			35	116
		0.091	0.081			50	116
		0.088	0.077			65	116
		0.084	0.070			75	116
		0.076	0.068			85	116
	THF	0.0745	0.0745			25–48	116
	n-Amyl acetate	0.090				31	744
	Cyclohexane			0.054		30	744
Poly(2-ethylbutyl methacrylate)	Isopropanol	0.109	0.105			25	117
	MEK	0.104	0.102			25	117
Poly(ethyl methacrylate)	*n*-Butyl chloride	0.087				25	911
	Ethyl acetate	0.109					122
	Ethyl acetate/ethanol (1/4.5 – 1/6, v/v)	0.113					122
	MEK		0.104			23	123
			0.101			25	807
	MEK/isopropanol (1/7, v/v)		0.107			23	123
	Methyl cellusolve		0.075			25	807
	Toluene	−0.009				25	911
Poly(*n*-hexyl methacrylate)	MEK	0.105	0.105			23	124
	Isopropanol	0.108	0.106			32.6	124
Poly(2-hydroxyethyl methacrylate)	2-Chloroethanol		0.048			25	616
	DMF		0.076			25	807
			0.076				120
	Dioxane/water (90/10, v/v)		0.083			25	119
			0.066 (μ)			25	119
	(80/20, v/v)		0.086			25	119
			0.071 (μ)			25	119
	(60/40, v/v)		0.095			25	119
			0.100 (μ)			25	119
	(40/60, v/v)		0.108			25	119
			0.120 (μ)			25	119
	(20/80, v/v)		0.123			25	119
			0.130 (μ)			25	119
	Ethanol/water (80/20, v/v)		0.124			25	119
			0.121 (μ)			25	119

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(70/30, v/v)		0.121			25	119
			0.119 (μ)			25	119
	(60/40, v/v)		0.120			25	119
			0.120 (μ)			25	119
	(40/60, v/v)		0.123			25	119
			0.127 (μ)			25	119
	(30/70, v/v)		0.125			25	119
			0.132 (μ)			25	119
Formamide			0.045				120
n-Heptane/*m*-cresol	(50/50, v/v)		0.033			25	698
	(50/50, v/v)		0.361 (μ)			25	698
Methanol			0.162				120
Methanol/HEMA	(25/75, v/v)		0.078			25	726
	(25/75, v/v)		0.070 (μ)			25	726
	(35/65, v/v)		0.090			25	726
	(35/65, v/v)		0.081 (μ)			25	726
	(50/50, v/v)		0.110 (μ)			25	726
	(50/50, v/v)		0.093 (μ)			25	726
	(62.5/37.5, v/v)		0.120			25	726
			0.107 (μ)			25	726
	(75/25, v/v)		0.135			25	726
			0.122 (μ)			25	726
	(87.5/12.5, v/v)		0.151			25	726
			0.140 (μ)			25	726
2-Methoxy ethanol			0.091				769
Methyl cellusolve			0.095			25	807
n-Propanol/dioxane							
(85/15, v/v)			0.105			25	119
			0.108 (μ)			25	119
(70/30, v/v)			0.100			25	119
			0.103 (μ)			25	119
(60/40, v/v)			0.096			25	119
			0.097 (μ)			25	119
(40/60, v/v)			0.091			25	119
			0.090 (μ)			25	119
(30/70, v/v)			0.087			25	119
			0.086 (μ)			25	119
n-Propanol/water	(95/5, v/v)		0.110			25	121
			0.105 (μ)			25	121
	(90/10, v/v)		0.113			25	121
			0.105 (μ)			25	121
	(80/20, v/v)		0.115			25	121
			0.106 (μ)			25	121
	(75/25, v/v)		0.103 (μ)				120
	(70/30, v/v)		0.119			25	121
			0.109 (μ)			25	121
	(60/40, v/v)		0.122			25	121
			0.110 (μ)			25	121
	(50/50, v/v)		0.125			25	121
			0.123 (μ)			25	121
	(40/60, v/v)		0.128			25	121
			0.138 (μ)			25	121
	(30/70, v/v)		0.131			25	121
			0.149 (μ)			25	121
	(20/80, v/v)		0.135			25	121
			0.153 (μ)			25	121
TFP/HEMA	(50/50, v/v)		0.105			25	726
			0.083 (μ)			25	726
Toluene/methanol	(42/58, v/v)		0.101			25	118
			0.063 (μ)			25	118
Toluene/isopropanol			0.087			25	118
(38/62, v/v)			0.064 (μ)			25	118

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Water/2-chloroethanol						
	(80/20, v/v)		0.031			25	616
			0.131 (μ)			25	616
	(72/25, v/v)		0.046			25	616
			0.127 (μ)			25	616
	(70/30, v/v)		0.023			25	616
			0.120 (μ)			25	616
	(60/40, v/v)		0.015			25	616
			0.101 (μ)			25	616
	(50/50, v/v)		0.016			25	616
			0.103 (μ)			25	616
	(40/60, v/v)		0.020			25	616
			0.091 (μ)			25	616
	(20/80, v/v)		0.034			25	616
			0.066 (μ)			25	616
	Water/Hema						
	(20/80, v/v)		0.068			25	726
			0.049 (μ)			25	726
			0.069			30	726
			0.051 (μ)			30	726
			0.071			40	726
			0.053 (μ)			40	726
			0.074			50	726
			0.055 (μ)			50	726
			0.076			60	726
			0.057 (μ)			60	726
			0.079			70	726
			0.059 (μ)			70	726
	(60/40, v/v)		0.103			25	726
			0.143 (μ)			25	726
			0.104			30	726
			0.144 (μ)			30	726
			0.107			40	726
			0.147 (μ)			40	726
			0.109			50	726
			0.149 (μ)			50	726
			0.111			60	726
			0.151 (μ)			60	726
			0.114			70	726
			0.154 (μ)			70	726
Poly [1(2-hydroxyethyl)pyridinium methacrylate]], benzenesulfonate							
	Aq. 0.5 M KBr		0.178			25	818
		0.136 (μ)	0.129 (μ)			25	813
	0.5 M KCl		0.178			25	818
		0.122 (μ)	0.117 (μ)			25	813
	1M KCl		0.174			25	818
		0.118 (μ)	0.114 (μ)			25	813
	2 M KCl		0.167			25	818
		0.113 (μ)	0.108 (μ)			25	813
	3M KCl		0.161			25	818
		0.104 (μ)	0.100 (μ)			25	813
	1.86 M KSCN		0.119 (μ)			25	813
	0.05 M NaF	0.123 (μ)	0.118 (μ)			25	813
	0.05 M NaI		0.183			25	818
			0.155 (μ)			25	818
	4 M NaI		0.121			25	818
			0.119 (μ)			25	818
Poly [1(2-hydroxyethyl)pyridinium methacrylate], toluenesulfonate							
	Benzyl alcohol		0.039			25	813
	Benzyl alcohol/methyl-α-naphthyl ether (45/55, v/v)		0.000			25	813
	DMF		0.130			25	813

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Ethanol		0.182			25	813
	Aq. 0.5 M KBr		0.181			25	813
			0.129 (μ)			25	818
	0.5 M KCl		0.182			25	813
		0.121 (μ)	0.114 (μ)			25	813,818
	1 M KCl		0.178			25	813
		0.117 (μ)	0.111 (μ)			25	813,818
	2 M KCl		0.171			25	813
		0.109 (μ)	0.105 (μ)			25	813,818
	3 M KCl		0.163			25	813
		0.100 (μ)	0.097 (μ)			25	813,818
	Methanol		0.209			25	813
	Methanol + 0.5 M LiCl		0.205			25	813
	0.5 M NaF		0.186			25	813
	4 M NaI		0.121			25	813
			0.116 (μ)			25	818
	7 M NaI		0.072			25	813
	n-Propanol		0.159			25	813
	Tetrahydrofurfuryl alcohol		0.106			25	813
	TFP		0.196			25	813
	Water		0.187			25	813
Poly[1(2-hydroxyethyl)trimethylammonium methacrylate], benzene sulfonate							
	Aq. 0.5 M KCl		0.101 (μ)			25	813
Poly[1(2-hydroxyethyl)trimethylammonium methacrylate], toluene sulfonate							
	Aq. 0.5 M KCl		0.095 (μ)			25	813
Poly(*N*-2-hydroxypropyl methacrylamide)							
	Aq. 0.1 M KCl		0.166			25	755
Poly(isobutyl methacrylate)	Acetone	0.122	0.117			25	103,104
	n-Butyl chloride	0.084				25	911
	Toluene	−0.021				25	911
Poly(isooctyl methacrylate)	*n*-Heptane				0.109		168
Poly(*n*-lauryl methacrylate)	*n*-Amyl alcohol	0.0744	0.0781				125
	n-Butyl acetate	0.092	0.090			23	126
	Isopropyl acetate	0.107	0.104			13	126
Poly(methacrylamide)	Aq. 0.4 M Mg perchlorate		0.180			25	775
			0.180 (μ)			25	775
	Water				0.209	20–65	77
Poly(methacrylic acid)	Water		0.175			25	628
	Water/dioxane (95/5, v/v)		0.170			25	628
			0.179 (μ)			25	628
	(90/10, v/v)		0.165			25	628
			0.180 (μ)			25	628
	(80/20, v/v)		0.163			25	628
			0.200 (μ)			25	628
	(70/30, v/v)		0.150			25	628
			0.191 (μ)			25	628
	(60/40, v/v)		0.150			25	628
			0.196 (μ)			25	628
	(50/50, v/v)		0.139			25	628
			0.169 (μ)			25	628
	(40/60, v/v)		0.134			25	628
			0.155 (μ)			25	628
	(35/65, v/v)		0.148 (μ)			25	628
	(25/75, v/v)		0.125			25	628
			0.122			25	628
	(15/85, v/v)		0.115			25	628
			0.109 (μ)			25	628
	Dimethyl formamide/ 1,4-dioxane (5 : 7, v/v)	0.099				26.9	977
	Alcoholic LiBr	0.154				25	127
	Ethanol	0.154				25	127
	HCl 0.1 N				0.159 (589)		87
	0.5 N		0.137			25	128

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	0.045 N				0.156 (560)	20	129
					0.162 (μ) (560)	20	129
	0.02 N				0.158	20–65	77
	0.001 N		0.140			25	128
	Methanol		0.134			25	128
	NaBr 0.1 M				0.234 (560)	20	129
					0.209 (μ) (560)	20	129
	NaCl 0.1 M				0.213 (μ) (560)	20	129
	NaF 0.1 M				0.229 (560)	20	129
					0.219 (μ)(560)	20	129
	NaI 0.1 M				0.197 (μ) (560)	20	129
	Na_2SO_4 0.1 M				0.210 (μ) (560)	20	129
	$(NH_4)_2Mo_7O_{24}$ 0.01 M				0.227	20	129
					0.169 (μ) (560)	20	129
	Water		0.142			25	128
					0.153		80
Poly(methacrylic acid), sodium salt							
	Water		0.260			25	628
	Water/dioxane (95/5, v/v)		0.256			25	628
			0.230 (μ)			25	628
	(90/10, v/v)		0.240			25	628
			0.189 (μ)			25	628
	(80/20, v/v)		0.250			25	628
			0.154 (μ)			25	628
	(75/25, v/v)		0.252			25	628
			0.121 (μ)			25	628
	(70/30, v/v)		0.238			25	628
			0.108 (μ)			25	628
Poly(methoxybiphenyl methacrylate)							
	THF		0.196				958
50/50 Poly(methoxy-biphenyl methacrylate-*co*-methoxybiphenyl ethylene oxide methacrylate)							
	THF		0.194				958
Poly(methoxybiphenyl ethylene oxide methacrylate)							
	THF		0.212				958
50/50 Poly(methoxy-biphenyl ethylene oxide methacrylate-*co*-methoxy-biphenyl bis(ethylene oxide)methacrylate)							
	THF		0.188				958
Poly(methoxybiphenyl bis(ethylene oxide) methacrylate							
	THF		0.205				958
Poly(methoxybiphenyl methacrylate-*co*-methoxybiphenyl tris(ethylene oxide)methacrylate)							
	THF		0.170				958
Poly(methoxybiphenyl tris(ethylene oxide)methacrylate)							
	THF		0.181				958
50/50 poly(methoxybiphenyl methacrylate-*co*-methoxybiphenyl tetrakis(ethylene oxide) methacrylate							
	THF		0.153				958
Poly(methoxybiphenyl tetrakis(ethylene oxide) methacrylate)							
	THF		0.171				958
Poly(methoxybiphenyl methacrylate)							
	$CHCl_3$		0.163				958
Poly(methoxybiphenyl ethylene oxide methacrylate)							
	$CHCl_3$		0.180				958
50/50 Poly(methoxybiphenyl ethylene oxide methacrylate-*co*-methoxybiphenyl bis(ethylene oxide)methacrylate)							
	$CHCl_3$		0.153				958
Poly(methoxybiphenyl bis(ethylene oxide) methacrylate)							
	$CHCl_3$		0.167				958
Poly(methoxybiphenyl tris(ethylene oxide) methacrylate)							
	$CHCl_3$		0.152				958
Poly(methoxybiphenyl tetrakis(ethylene oxide) methacrylate)							
	$CHCl_3$		0.152				958
Poly(2-methacryloyloxyethyl trimethylammonium chloride)							
	1 M aqueous NaCl			0.157			971
Poly(2-methoxyethyl methacrylate)	Halothane		0.125			25	589
	MEK		0.108			25	589,902

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Nitrobenzene		−0.058			25	589
	THF		0.078			25	589
	Toluene		0.000			25	589
	1,2,4, Trichlorobenzene		−0.067			25	653
Poly(methyl methacrylate)	Acetone	0.1391				27	130
		0.107				27	131
		0.137	0.134			23 ± 2	132
		0.129					133
					0.131		134
		0.1313	0.1293		0.134 (366)	25	135
		0.136	0.134			25	10,136
					0.132		92
		0.132				20	89
		0.1305	0.1285		0.1281 (578)	20	5
					0.1279 (589)	20	5
					0.1276 (644)	20	5
			0.132			20	93
			0.135			20	137,138
		0.1312	0.1292			20	139
			0.1330			25	140,141
	Acetone		0.135			25	759
				0.130			618
				0.129		25	689
$M_w > 5060$	Acetone	0.134				25	947
isotactic	Acetone	0.136	0.134			20	142
	Acetone/benzene (88/12, v/v)		0.1170			25	141
	(76/24, v/v)		0.1005			25	141
	(64/36, v/v)		0.0861			25	141
	(52/48, v/v)		0.0734			25	141
	(40/60, v/v)		0.0590			25	141
	Acetone/chloroform						
	(80/20, v/v)		0.1170			25	140
	(60/40, v/v)		0.1005			25	140
	(52/48, v/v)		0.091			25	140
	(40/60, v/v)		0.0861			25	140
	(20/80, v/v)		0.0734			25	140
	Acetone/water (62.2/37.8, mol)	0.136				20	626
	Acetonitrile		0.133			25	721
		0.140	0.137			20	136
$M_w > 10^4$	Acetonitrile	0.144				44	947
	Acetonitrile/MMA						
	(17/83, v/v)		0.094			25	726
						25	726
			0.078 (μ)			25	726
	(32/68, v/v)		0.100			25	726
			0.085 (μ)			25	726
	(50/50, v/v)		0.110			25	726
			0.097 (μ)			25	726
	(68/32, v/v)		0.119			25	726
			0.113 (μ)			25	726
	Acetophenone/MMA						
	(50/50, v/v)		0.034			25	726
			0.033 (μ)			25	726
	Acetophenone/isopropanol						
	(30/70, v/v)		0.074			25	825
			0.115 (μ)			25	825
	(40/60, v/v)		0.059			25	825
			0.104 (μ)			25	825
	(50/50, v/v)		0.044			25	825
			0.084 (μ)			25	825
	(60/40, v/v)		0.027			25	825
			0.050 (μ)			25	825

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(70/30, v/v)		0.017			25	825
			0.030 (μ)			25	825
	(80/20, v/v)		0.001			25	825
			0.005 (μ)			25	825
	(90/10, v/v)		−0.012			25	825
			−0.012 (μ)			25	825
	Arochlor	−0.124				20	144
	Benzene	0.0398				27	130
			−0.010				145
		0.002				25	146
		0.0039	0.0110			25	139
			0.004				147
					~0		148
			−0.016			25	10
		0.001	0.007			20	6
		0.006				30	677
				0.015		25	689
	Benzene/MMA (30/70, v/v)		0.061			25	726
			0.054 (μ)			25	726
	(50/50, v/v)		0.046			25	726
			0.044 (μ)			25	726
	(70/30, v/v)		0.033			25	726
			0.032 (μ)			25	726
	Benzyl alcohol/MMA		0.026			25	726
	(50/50, v/v)		0.027 (μ)			25	726
	Benzyl alcohol/*n*-butanol						
	(90/10, v/v)		−0.036			25	782
			−0.034 (μ)			25	782
	(80/20, v/v)		0.012			25	782
			0.017 (μ)			25	782
	(70/30, v/v)		0.016			25	782
			0.024 (μ)			25	782
	(60/40, v/v)		0.030			25	782
			0.043 (μ)			25	782
	(50/50, v/v)		0.041			25	782
			0.060 (μ)			25	782
	(40/60, v/v)		0.057			25	782
			0.084 (μ)			25	782
	Benzyl alcohol/*sec*-butanol						
	(90/10, v/v)		−0.037			25	782
			−0.070 (μ)			25	782
	(80/20, v/v)		0.011			25	782
			−0.012 (μ)			25	782
	(70/30, v/v)		0.012			25	782
			0.013 (μ)			25	782
	(60/40, v/v)		0.029			25	782
			0.082 (μ)			25	782
	(50/50, v/v)		0.040			25	782
			0.160 (μ)			25	782
	(40/60, v/v)		0.056			25	782
			0.254 (μ)			25	782
	(30/70, v/v)		0.074			25	782
			0.316 (μ)			25	782
	Bromobenzene		−0.058				145
		−0.050				25	624
	Bromoform/isopropanol		0.056			25	825
	(30/70, v/v)						
			0.196 (μ)			25	825
	(40/60, v/v)		0.036			25	825
			0.166 (μ)			25	825
	(50/50, v/v)		0.019			25	825
			0.136 (μ)			25	825

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(60/40, v/v)		0.002			25	825
			0.092 (μ)			25	825
	(70/30, v/v)		−0.017			25	825
			0.034 (μ)			25	825
	(80/20, v/v)		−0.036			25	825
			−0.031 (μ)			25	825
	(90/10, v/v)		−0.059			25	825
			−0.067 (μ)			25	825
	Bromoform/acetophenone/ isopropanol	Values for numerous solvent ratios given in reference					825
	1-Bromonaphthalene		−0.147				145
			−0.148			25	9
	2-Butanol	0.0802 + 0.0004* T (°C)				55 ≤ T ≤ 76	954
	n-Butyl acetate	0.0987	0.0970		0.101 (366)	25	135
		0.0986	0.0969			25	139
		0.1				25	113
			0.0970			25	149
			0.097			25	670
					0.096	25	689
					0.096	25	802
syndiotactic					0.097	25	742
	n-Butyl acetate/chloroform						
	(20/80, v/v)				0.063	25	802
	(40/60, v/v)				0.070	25	802
	(60/40, v/v)				0.073	25	802
	(80/20, v/v)				0.083	25	802
	(90/10, v/v)				0.094	25	802
	n-Butyl chloride	0.0913	0.0898			20	143
		0.0948	0.0934			30	143
		0.0984	0.0970			40	143
					0.096	40	615
			0.093			25	721
		0.098				40	731
atactic		0.0931	0.0914			25	150
		0.1021	0.1001			50	150
syndiotactic		0.1001	0.0986			50	150
isotactic		0.0902	0.0875			25	150
		0.0965	0.0954			50	150
	n-Butyl chloride/toluene						
	(83/17, v/v)	0.080				25	911
	(56.9/43.1, v/v)	0.058				25	911
	(32.1/67.9, v/v)	0.035				25	911
	(20/80, v/v)	0.026				25	911
	Carbon tetrachloride		0.023				145
		0.033				30	677
	Chlorobenzene		−0.026				145
		−0.004				35	697
					−0.010	25	689
		−0.018				25	146
					−0.026		148
	1-Chlorobutane	0.098				40	955
	Chloroform	0.0631	0.0629		0.0635 (366)	25	135
					0.056		134
		0.0630	0.0628			25	139
					0.050		148
			0.059			25	10,140
		0.056				30	677
					0.062	25	689
					0.059	25	802
	Cyclohexanone				0.046	30	729
	o-Dichlorobenzene				−0.037	25	689

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	1,2-Dichloroethane	0.050	0.051				151
		0.050				30	152
			0.050				147
			0.046			25	10
			0.079			30	826
					0.050	25	689
	Dioxane		0.075				154
		0.0720	0.0707		0.074 (366)	25	135,139
			0.068				145
			0.071			25	93
					0.0703 (589)	25	155
			0.07			25	121,147
					0.070		148
			0.066			25	10
			0.0707			25	156,157
					0.072 (488)	30	729
				0.068			618
				0.069		25	689
			0.070			25	807,822
	Dioxane/water (95/5, v/v)		0.072			25	121
			0.076 (μ)			25	121
	(90/10, v/v)		0.076			25	121
			0.083 (μ)			25	121
	(85/15, v/v)		0.079			25	121
			0.072			25	118
			0.090 (μ)			25	118,121
			0.090			30	158
	(82/18, v/v)		0.082 (μ)			25	121
			0.096 (μ)			25	121
	DMF	0.0600	0.0614		0.0617 (578)	20	5,6
					0.0168 (589)	20	5,6
					0.0619 (644)	20	5,6
					0.055 (365)	20	6
			0.055			25	10
			0.069			30	159
			0.095			90	159
					0.042		148
		0.060	0.058			20	897
			0.086			30	826
			0.058			25	807
					0.063	25	689
Polymer modified with RCH_2Li							
$R = SC_6H_5$					0.079–0.096	25	700
$R = SO_2CH_3$			0.070–0.081			25	700
$R = SO_2N(CH_3)_2$			0.075–0.107			25	700
	Ethyl acetate	0.1200	0.1180		0.123 (366)	25	135,139
					0.1174 (589)	25	155
			0.118			25	10
				0.118			618
isotactic					0.117	25	771
			0.138			30	826
			0.116			25	720
	Furfuryl alcohol		0.010			30	691
	n-Heptane/*m*-cresol		0.040			25	698
	(50/50, v/v)		0.349 (μ)			25	698
	Isoamyl acetate	0.0931	0.0911			20	143
		0.0945	0.0926			30	143
		0.0959	0.0942			40	143
		0.0974	0.0957			50	143
	MEK		0.111				154
			0.112			RT	160
		0.093					131
					0.111		134

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
			0.1173			RT	163
		0.100				RT	99
					0.110		148
		0.1131	0.1112		0.1110 (578)	20	5,6,88
					0.1107 (589)	20	5
					0.1102 (644)	20	5
		0.113	0.111			23 ± 2	132
			0.117			25	162
		0.111				25	146
			0.111			25	132,161
			0.110			25	10,93
			0.108			25	165
		0.01385				27	130
					0.111 (488)	28	134
		0.114	0.113				151
			0.111				145
		0.113				30	152
					0.059	30	164
					0.131		153
			0.113				147
			0.1147			35	34
		0.113				35	72
				0.111		25	689
			0.112			25	807
			0.134			30	826
		0.110	0.108			20	897
	MEK/isopropanol	0.1090	0.1075			10	143
		0.1130	0.1114			20	143
		0.1167	0.1153			30	143
		0.1208	0.1192			40	143
atactic	(55/54, v/v)	0.1169	0.1153			30	150
syndiotactic		0.1165	0.1155			30	150
isotactic		0.1167	0.1152			30	150
	MEK/2-propanol (43.6/56.4, mol/mol)	0.109				20	626
	2-Methoxy ethanol	0.089				40	731
	Methyl cellusolve		0.082			25	807
	MMA/styrene (50/50, v/v)		0.025			25	726
			0.025 (μ)			25	726
	Methyl methacrylate		0.075			30	158
	N-Methylpyrrolidone		0.0176			25	10
	Nitroethane		0.100				154
			0.094			25	161
	2-Propanol/isopropylamine (5.6/94.4, mol/mol)	0.109				20	626
	TFP		0.141			25	121
	TFP/water (92/5, v/v)		0.141			25	121
			0.140 (μ)			25	121
	(90/10, v/v)		0.139			25	121
			0.137 (μ)			25	121
	(80/20, v/v)		0.135			25	121
			0.131 (μ)			25	121
	(70/30, v/v)		0.133			25	121
			0.124 (μ)			25	121
	(65/35, v/v)		0.132			25	121
			0.121 (μ)			25	121
	THF	0.0887	0.0871		0.091 (366)	25	135
		0.0886	0.0870			25	139
			0.0902			36	34
		0.09					166
				0.087			618
				0.098		25	637
				0.086		25	689

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
$M_w > 10^4$				0.110–0.114			984
atactic				0.084		25	811
isotactic				0.086		25	811
varying tacticities				0.084–0.088		25	686
	THF/cyclopentane (65.7/34.3, mol/mol)	0.089				20	626
	Toluene	0.004				30	152
		0.010				25	146
		0.0094	0.0157			25	139
					0.004		153
			0.005				147
			0.003			30	691
		0.004				30	697,699,834
		0.009				50	699
		0.014				70	699
		−0.003				15	834
		0.001				20	834
		0.003				25	834
		0.007				25	911
atactic		0.010				25	146
syndiotactic		0.007				25	146
isotactic		0.014				25	146
	Toluene/methanol		0.094			25	118
	(42/58, v/v)		0.152 (μ)			25	118
			0.096			25	167
			0.154 (μ)			25	167
	Toluene/isopropanol		0.073			25	118
	(38/62, v/v)		0.121 (μ)			25	118
	Toluene/*p*-cymene						
	(73.4/26.6, w/w)	0.006				30	699
	(49.7/503, w/w)	0.008				30	699
Poly(*o*-methylphenylthio methacrylate	Benzene	0.133			0.133	35	666
Poly(β-naphthyl methacrylate)	Benzene		0.142			25	856
Poly(nonyloxy-1,4-phenyleneoxycarbonylphenyl methacrylate)							
	Carbon tetrachloride				0.096	24	890
Poly(*N*-octadecylacrylamide)	Benzene		0.0269			30	84
	Chlorobenzene		0.0489			30	84
	Cyclohexane		0.0651			30	84
	n-Hexane		0.1292			30	84
			0.1168			30	84
	THF		0.0898			30	84
	Toluene		0.0154			30	84
Poly(*n*-octyl methacrylate)	*n*-Butanol	0.083	0.080			16.8	169
	MEK	0.107	0.107			28	169
Poly(4-phenylbutyl-1-methacrylate)	1-Chloroundecane			0.100	0.105 (488)	23	965
Poly(2-phenylethyl-1-methacrylate)	1-Chloroheptane			0.128	0.134 (488)	27.5	965
Poly(phenyl methacrylate)	Acetone		0.234				170
	MEK		0.182			25	115
Poly(phenylthiol methacrylate)	Benzene	0.142				35	666
Poly[2-(3-sodium sulfonato-2-methylpropyl) methacrylamide]							
	Aq. 0.05 M NaCl	0.142 (μ)				RT	701
	Aq. 5 M NaCl	0.097 (μ)				RT	701
Poly(3-sodium sulfonatopropyl methacrylate)							
	Aq. 0.01 M NaCl	0.11				25	824
	Aq. 0.1 M NaCl	0.13				25	824
Poly(*n*-stearyl methacrylate)	Cyclohexane			0.054		20	744
	n-Propyl acetate	0.107				36	744

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Poly(tetrahydro-4H-pyranyl-2-methacrylate)	THF			0.092		20	840
Poly(*n*-tridecyl methacrylate)	Ethyl acetate	0.121				27	744
	THF			0.083		30	744
Poly(3-trimethylammonium ethyl chloride methacrylate)							
	Aq. 1 M NaCl	0.157	0.155	0.154		25	882
		0.158 (μ)	0.155 (μ)	0.154 (μ)		25	882
Poly[2-(triphenylmethoxy)ethyl methacrylate]							
	Dioxane		0.180			25	770
1.5. OTHER α- OR β-SUBSTITUTED POLY(ACRYLICS) AND POLY(METHACRYLICS)							
Poly[bis(phenylethyl) itaconate]	MEK	0.180				20	738
Poly[bis(phenyl-*n*-propyl) itaconate]	MEK	0.173				20	738
Poly(*tert*-butyl crotonate)	*n*-Butyl chloride	0.083				25	718
	Ethyl acetate	0.111				25	718
Poly(di-*n*-amyl itaconate)	*n*-Amyl acetate	0.074				20	102
	Benzene	−0.035				20	102
	Carbon tetrachloride	0.018				20	102
	Chloroform	0.036				20	102
	Ethyl acetate	0.102				20	102
	MEK	0.096				20	102
	Toluene	−0.027				20	102
Poly(dibenzyl itaconate)	MEK	0.183				20	738
Poly(di-*n*-butyl itaconate)	Acetone	0.109				20	102
	n-Amyl acetate	0.074				20	102
	Benzene	−0.030				20	102
	Carbon tetrachloride	0.020				20	102
	Chlorobenzene	−0.050				20	102
	Chloroform	0.036				20	102
	Ethyl acetate	0.101				20	102
	Ethylbenzene	−0.021				20	102
	n-Hexane	0.101				20	102
	MEK	0.096				20	102
	Methyl acetate	0.112				20	102
	Styrene	−0.077				20	102
	Toluene	−0.023				20	102
Poly(di-cyclohexyl itaconate)	*n*-Amyl acetate	0.109				20	763
	n-Butyl acetate	0.126				20	763
	Carbon tetrachloride	0.058				20	763
	Chlorobenzene	−0.005				20	763
	n-Propyl acetate	0.132				20	763
	Toluene	0.017				20	763
Poly(di-*n*-decyl itaconate)	*n*-Amyl acetate	0.079				20	102
	Benzene	−0.041				20	102
	Carbon tetrachloride	0.015				20	102
	Toluene	−0.032				20	102
Poly(diethyl itaconate)	*n*-Amyl acetate	0.074				20	102
	Benzene	−0.018				20	102
	Carbon tetrachloride	0.024				20	102
	Ethyl acetate	0.100				20	102
	MEK	0.096				20	102
	Toluene	−0.014				20	102
Poly(di-*n*-heptyl itaconate)	*n*-Amyl acetate	0.075				20	102
	Benzene	−0.037				20	102
	Benzene	−0.036				20	102
	Carbon tetrachloride	0.017				20	102
	Chloroform	0.035				20	102
	Ethyl acetate	0.102				20	102
	Ethyl acetate	0.102				20	102

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	MEK	0.095				20	102
	MEK	0.095				20	102
	Toluene	−0.029				20	102
	Toluene	−0.028				20	102
Poly(dimethyl itaconate)	Acetone	0.118				20	102
	Benzene	0.000				20	102
	Chlorobenzene	−0.022				20	102
	Chloroform	0.041				20	102
	MEK	0.096				20	102
	Methyl acetate	0.110				20	102
Poly(*N*-1,1-dimethyl-3-oxobutylacrylamide)	MEK	0.137				25	622
Poly(di-*n*-nonyl itaconate)	*n*-Amyl acetate	0.079				20	102
	Benzene	−0.030				20	102
	Toluene	−0.039				20	102
Poly(di-*n*-octyl itaconate)	*n*-Amyl acetate	0.078				20	102
	Benzene	−0.039				20	102
	Carbon tetrachloride	0.016				20	102
	Chloroform	0.033				20	102
	Ethyl acetate	0.105				20	102
	MEK	0.096				20	102
	Toluene	−0.030				20	102
Poly(diphenyl itaconate)	Benzene	0.083				20	768
	Chlorobenzene	0.060				20	768
	Ethyl acetate	0.191				20	768
	MEK	0.189				20	768
	Styrene	0.040				20	768
	Toluene	0.087				20	768
Poly(di-*n*-propyl itaconate)	*n*-Amyl acetate	0.074				20	102
	Benzene	−0.025				20	102
	Carbon tetrachloride	0.021				20	102
	Chloroform	0.036				20	102
	MEK	0.096				20	102
	Toluene	−0.019				20	102
Poly(di-*n*-undecyl itaconate)	*n*-Amyl acetate	0.079				20	102
	Benzene	−0.042				20	102
	Carbon tetrachloride	0.014				20	102
	Chlorobenzene	−0.064				20	102
	Chloroform	0.030				20	102
	Ethyl benzene	−0.030				20	102
	n-Hexane	0.112				20	102
	Styrene	−0.088				20	102
	Toluene	−0.032				20	102
Poly(methyl α-chloroacrylate)	Chloroform	0.072					707
Poly(methyl *n*-butacrylate)	*n*-Butanol	0.0878				13	286
	MEK	0.1192				30	286
Poly(methyl ethacrylate)	Diisobutyl ketone	0.0791				11.4	286
	MEK	0.1357				30	286

1.6. POLY(STYRENES)

Poly(4-acetoxystyrene)	Dioxane	0.139				25	654
Poly(4-acetoxymethyl styrene)	Toluene			0.0643		25	974
Poly(0.05 wt.% 4-acetoxymethyl styrene-*co*-0.95 wt.% *tert*-butyloxy carbonyloxy styrene)	Toluene			0.0344		25	974
Poly(0.10 wt.% 4-acetoxymethyl styrene-*co*-0.90 wt.% *tert*-butyloxy carbonyloxy styrene)	Toluene			0.0368		25	974
Poly(0.20 wt.% 4-acetoxymethyl styrene-*co*-0.80 wt.% *tert*-butyloxy carbonyloxy styrene)	Toluene			0.0391		25	974
Poly(0.35 wt.% 4-acetoxymethyl styrene-*co*-0.65 wt.% *tert*-butyloxy carbonyloxy styrene)	Toluene			0.0443		25	974

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Poly(0.50 wt.% 4-acetoxymethyl styrene-*co*-0.50 wt.% *tert*-butyloxy carbonyloxy styrene)							
	Toluene			0.0486		25	974
Poly(*p*-*tert*-butylstyrene)	Cyclohexane		0.139			35	778
	1-Nitropropane		0.162			35	778
Poly(*o*-chlorostyrene)	MEK		0.191			25	765
Poly(*p*-chlorostyrene)	Benzene	0.095					171
	Chloroform	0.157					171
	Dioxane	0.164					171
	MEK	0.207	0.197			20	172
	Toluene	0.121				35	173
Poly(*p*-iodostyrene)	Dioxane	0.156	0.144		0.166 (365)	20	174
					0.143 (578)	20	174
					0.142 (589)	20	174
					0.140 (644)	20	174
14% iodine	Bromoform				0.032	874	
27% iodine					0.050	874	
Poly(*p*-isopropyl-α-methylstyrene)	Toluene		0.084			25	776
Poly(*p*-isopropylstyrene)	Chloroform	0.134	0.124			25	187
	MEK	0.198	0.186			25	187
	THF	0.177	0.167			25	188
Poly(*p*-isopropylstyrene)	MEK	0.195–0.203	0.184–0.197			25	187
hydroperoxide	THF	0.177	0.167			25	189
Poly(*p*-methoxystyrene)	Benzene/cyclohexanol (61.19/38.81, v/v)		0.112				175
	Benzene/methanol (73.69/26.31, v/v)		0.125				175
	Chloroform/cyclohexane (36.08/63.92, v/v)		0.163				175
	Chloroform/methanol (66.49/33.51, v/v)		0.190				175
	MEK/*n*-heptane (80.01/19.99, v/v)		0.211				175
	MEK/*n*-propanol (80.64/19.36, v/v)		0.215				175
	Toluene	0.105	0.101			25	176
	Toluene/cyclohexanol (62.8/37.2, v/v)		0.113				175
Poly(α-methylstyrene)	Benzene	0.1238				25	177
			0.138			25	178
		0.134				30	179
		0.131				30	180
Poly(α-methylstyrene)	Carbon tetrachloride	0.178	0.168		0.193 (366)	23	181
	Cyclohexane	0.206				32	183
		0.208				36	183
		0.199				37	182
		0.200				39	179
		0.1967				39	177
		0.210				40	183
		0.212				45	183
		0.214				50	183
		0.204	0.192		0.219 (366)	35	181
	1,2-Dichloroethane	0.1764				25	177
	Toluene		0.1165			23	34
		0.131				25	184
		0.1307				25	177
		0.130	0.126		0.135 (366)	23	181
					0.129	25	185
					0.137		186
		0.137				25	613
Poly(*p*-methylstyrene)	Toluene			0.093		20	976
Poly(styrene)	Acetone/cyclohexane (26.6/73.4, v/v)		0.192			25	783

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(38.2/61.8, v/v)		0.202			25	783
	(49.1/50.9, v/v)		0.207			25	783
	(59.1/40.9, v/v)		0.214			25	783
	(68.4/31.6, v/v)		0.220			25	783
	(77.2/22.8, v/v)		0.225			25	783
	Isoamyl acetate	0.205				30	97
	n-Amyl acetate	0.205				20	102,190
	Arochlor		−0.017				144
	Benzene	0.1151					191
		0.111				20	102
		0.1066	0.1034			20	6
		0.113				20	203
		0.1066	0.1034		0.1030 (578)	20	5
					0.1028 (589)	20	5
					0.1021 (644)	20	5
		0.115				25	205
		0.108	0.105			25	106
		0.109	0.105			25	107
					0.109		32
		0.112	0.109		0.107 (589)	25	206
			0.110			25	193
					0.115		194
			0.108				57
			0.1025			25	195
		0.108				25	146,197
		0.1094	0.1062			25	139
					0.106		148
			0.105			25	10
		0.111				25	198
			0.106			25	145,196, 199,201
		0.113				28	192
			0.109			30	204
			0.106				147
		0.101				30	97
		0.115				30	200
			0.1123			30	84
		0.112				35	202
		0.131				70	203
		0.107	0.103			20	709
			0.111			25	664
				0.105		25	593,705
			0.106			25	779,751
			0.103			25	774
		0.109	0.106			25	890,647
					0.109 (488)	25	812
increase with M:			0.093–0.109			25	648
$M = 1.3 \times 10^3 - 1.3 \times 10^5$			0.091–0.106			25	612
$M > 1.05 \times 10^6$			0.108			25	612
star shaped			0.104			25	612
star shaped			0.126			35	683
comb shaped			0.104–0.109			25	612
				0.102		30	671
		0.116				30	677
		0.113				30	821
			0.112			35	683
				0.105			618
	Benzene	$0.0495 + 2.20 \times 10^{-4} T$ (K)					962
	Benzene/bromoform						
	(95/5, v/v)	0.101				25	198
	(90/10, v/v)	0.096				25	198
	(85/15, v/v)	0.099				25	198

TABLE 1. *cont'd*

Polymer	Solvent	d*n*/d*c* (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	*T* (°C)	Refs.
	(75/25, v/v)	0.083				25	198
	(50/50, v/v)	0.074				25	198
	(25/75, v/v)	0.036				25	198
	Benzene/$(CH_2Cl)_2CHOH$						
	(33/67, v/v)		0.112				207
	Benzene/cyclohexane						
	(82.9/18.1, mol/mol)	0.137				20	208
		0.155				70	208
	(65.3/34.7, mol/mol)	0.155				20	208
		0.172				70	208
	(44.8/55.2, mol/mol)	0.170				20	208
		0.186				70	208
	(23.6/76.4, mol/mol)	0.183				20	208
		0.197				70	208
	(10/90, mol/mol)	0.189				20	208
		0.202				70	208
	(4.8/95.2, mol/mol)	0.191				20	208
	(90/10, v/v)	0.116	0.113			20	709
	(80/20, v/v)	0.126	0.120			20	709
		0.138				30	692
		0.142 (μ)				30	692
	(75/25, v/v)		0.124			25	751
	(70/30, v/v)	0.135	0.129			20	709
	(60/40, v/v)	0.144	0.135			20	709
			0.137			25	751
	(50/50, v/v)	0.151	0.142			20	709
			0.145			25	751
		0.150				30	692
		0.159 (μ)				30	692
	(40/60, v/v)	0.158	0.148			20	709
	(35/65, v/v)		0.153			25	751
	(30/70, v/v)	0.165	0.154			20	709
	(20/80, v/v)	0.171	0.158			20	709
			0.162			25	751
		0.166				30	692
		0.175 (μ)				30	692
	(10/90, v/v)	0.175	0.162			20	709
	Benzene/cyclohexanol						
	(61.6/38.4, v/v)		0.108			25	209
	(81.8/18.2, mol/mol)	0.121				20	208
		0.134				70	208
	(90.9/9.1, mol/mol)	0.119				20	208
	(99.628/0.372, mol/mol)	0.124				20	208
		0.134				70	208
	Benzene/*n*-docosane						
	(98.7/1.3, mol/mol)	0.122				20	210
		0.140				70	210
	(95.7/4.3, mol/mol)	0.135				20	210
		0.147				70	210
	(90.3/9.7, v/v)	0.147				20	210
		0.150				70	210
	Benzene/*n*-dodecane						
	(96.2/3.8, mol/mol)	0.123				20	210
		0.139				70	210
	(92.7/7.3, mol/mol)	0.131				20	210
		0.145				70	210
	(86.4/13.6, mol/mol)	0.140				20	210
		0.149				70	210
	Benzene/1-dodecanol						
	(98.1/1.9, mol/mol)	0.116				20	210
	(94.6/5.4, mol/mol)	0.122				20	210
	(89.7/10.3, mol/mol)	0.132				20	210

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Benzene/ethanol						
	(70.4/29.6, v/v)	0.156				19.6	611
	(70.4/29.6, v/v)	0.155				25	611
	(94.3/5.7, v/v)	0.115				20	210
	(89.1/10.9, v/v)	0.119				20	210
		0.131				70	210
	(62.1/37.9, v/v)	0.162				20	210
		0.151				70	210
	Benzene/*n*-heptane						
	(44/56, v/v)	0.130				35	211
	(90/10, v/v)	0.124				35	202
	(50/50, v/v)		0.157			25	751
	(60/40, v/v)		0.152			25	751
	Benzene/*n*-hexane						
	(88/12, mol/mol)	0.141				20	210
	(78.6/21.4, mol/mol)	0.160				20	210
		0.174				70	210
	(63.2/36.8, mol/mol)	0.188				20	210
		0.199				70	210
	(55.1/44.9, mol/mol)	0.197				20	210
		0.205				70	210
	Benzene/1-hexanol						
	(92.7/7.3, mol/mol)	0.119				20	210
	(84.9/15.1, mol/mol)	0.126				20	210
		0.135				70	210
	(76.6/23.4, mol/mol)	0.136				20	210
	(67.8/32.2, mol/mol)	0.148				20	210
		0.144				70	210
	Benzene/MEK						
	(25.1/74.9, mol/mol)	0.210				20	203
		0.224				70	203
	(50.2/49.8, mol/mol)	0.178				20	203
		0.193				70	203
	(75.1/24.9, mol/mol)	0.145				20	203
		0.162				70	203
	Benzene/methanol						
	(70.9/29.1, mol/mol)	0.154				20	208
	(83.9/16.1, mol/mol)	0.124				20	208
	(92.4/7.6, mol/mol)	0.114				20	208
	(22.2/77.8, v/v)		0.145			25	209
	(76.6/23.4, v/v)		0.122			25	193
	(80/20, v/v)		0.137				207
	(74.7/25.3, v/v)		0.159			35	211
	(95/5, v/v)		0.114			25	751
	(80/15, v/v)		0.127			25	751
	(83/17, v/v)		0.132			25	751
	(80/20, v/v)		0.136			25	751
	(78/22, v/v)		0.141			25	751
	(76.8/23.2, v/v)	0.190				16.7	611
	(76.8/23.2, v/v)	0.187				25	611
	(75/25, v/v)		0.144			25	751
	Benzene/1-octadecanol	0.116				20	210
	(99.15/0.85, mol/mol)	0.134				70	210
	Benzene/poly(oxyethylene)						
	420 (95.9/4.1, mol/mol)	0.125				20	210
	2100						
	(99.41/0.59, mol/mol)	0.120				20	210
	Benzene/isopropanol						
	(35.8/64.2, v/v)		0.159			25	212
	(65/35, v/v)		0.154				207
	(61/39, v/v)	0.142				35	211
	(64/36, v/v)	0.150				30	692
		0.216 (μ)				30	692

TABLE 1. *cont'd*

Polymer	Solvent	d*n*/d*c* (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	*T* (°C)	Refs.
	Benzene/*n*-propanol	0.131				21.7	611
	(63.7/36.3, v/v)	0.130				29.7	611
	Bromobenzene	0.0492				18	213
		0.043	0.042			25	106
		0.0495	0.0485				107
					0.0455	25	214
			0.042			25	145,215
		0.047				25	624
			−0.005				57
		0.015	0.012			25	106
		0.014				30	216
	Bromoform		0.012				874
		0.014				30	216,919
	1-Bromonaphthalene		−0.051				145
			−0.091				57
			−0.0517			25	9
	n-Butyl chloride		0.1997			25	43
	n-Butyl propionate		0.2018			25	43
	Carbon tetrachloride	0.156				20	102
		0.161				25	217
			0.144				154
			0.146				145
		0.159	0.147			25	106
		0.160	0.150		0.145 (589)	25	206
			0.1518			30	84
			0.148			30	204
		0.162				35	202
		0.159	0.148			25	647
			0.15			25	651
		0.162				30	677
				0.156		30	671
Poly(styrene) anionically polymerized	Carbon tetrachloride			0.145		25	964
	Carbon tetrachloride/ *n*-butanol (65/35, v/v)	0.176				35	211
	Carbon tetrachloride/ *n*-heptane (53/47, v/v)	0.189				35	211
	Carbon tetrachloride/ methanol (18.3/81.7, v/v)		0.162			25	209
	Chlorobenzene		0.079				145
		0.0833				18	213
		0.082				20	102
		0.079				25	146
			0.079			25	215
			0.0848			30	84
					0.079		148
					0.081		891
		0.083				35	697
	Chlorobenzene/isopropyl ether (68/32, v/v)		0.192			25	209
	1-Chlorodecane	0.165					
	1-Chlorodecane/3-methyl-cyclohexanol					6.2	661
	(78.2/21.8, v/v)	0.165				22.8	661
	(75/25, v/v)	0.168				35	661
	(49.5/50.5, v/v)	0.166				59.6	661
	(10/90, v/v)	0.155				87.8	661
	Chloroform		0.158				57
		0.1554				18	213
		0.165				20	102
			0.169			25	26
					0.152		148
		0.161				25	146
		0.157				30	677

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Cyclohexane	0.1795	0.1682			20	219
		0.1798	0.1685			21	219
		0.1801	0.1687			22	219
		0.1804	0.1690			23	219
		0.1810	0.1695			25	139,219
$\overline{M}_w = 2.18 \times 10^3$		0.168	0.156			25	220
2.92×10^3		0.171	0.160			25	220
4.69×10^3		0.174	0.163			25	220
11.8×10^3		0.177	0.165			25	220
22.2×10^3		0.180	0.167			25	220
167×10^3		0.181	0.167			25	220
		0.1825	0.1709			30	219
		0.181				34.5	221
		0.1840	0.1723			35	219
			0.171			35	209
		0.181				35	202,222
			0.179			40	218
		0.1855	0.1738			40	219
		0.1870	0.1752			45	219
		0.188				50	223
					0.181		224
				0.163			618
		0.179	0.166			20	709
		0.181	0.170			25	647
			0.169			25	751
		0.191				30	677
		0.181	0.171			34.5	712,705
		0.181	0.169			34.5	809
$M = 1.2 \times 10^3 - 1.07 \times 10^5$				0.156–0.168		34.5	732
star shaped				0.157–0.163		34.5	737
after photoisomerization: *trans* → *cis*			0.100			28	736
		0.181				34.8	611
			0.171			34.8	664
					0.175 (488)	35	642
		0.181				35	659,694,704
		0.184				35	661
				0.161		35	671
			0.168			35	683
			0.170			35	683
	Cyclohexane			0.187		30	943
				0.191		40	943
				0.195		50	943
	Cyclohexane/cyclohexanol						
	(75.3/24.7, v/v)	−0.043				36	611
		−0.044				38	611
		−0.045				44	611
		−0.046				52.2	611
		−0.047				58.2	611
	(37.3/62.7, v/v)	−0.048				65.1	611
		−0.049				74	611
		−0.050				82	611
		−0.051				86	611
	Cyclohexane/ethanol	0.065				29.2	611
	(97/3, v/v)	0.064				39.8	611
	Cyclohexane/ethyl acetate						
	(10/90, v/v)	0.229	0.216			20	709
	(20/80, v/v)	0.226	0.213			20	709
	(30/70, v/v)	0.223	0.210			20	709
	(40/60, v/v)	0.219	0.206			20	709
	(50/50, v/v)	0.216	0.202			20	709
	(60/40, v/v)	0.211	0.197			20	709
	(70/30, v/v)	0.206	0.191			20	709

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(80/20, v/v)	0.198	0.184			20	709
	(90/10, v/v)	0.189	0.176			20	709
	Cyclohexane/methanol	0.099				29.2	611
	(95.5/4.5, v/v)	0.098				39.8	611
	Cyclohexane/*n*-propanol	0.041				28.2	611
	(97.6/2.4, v/v)	0.040				37.8	611
	Cyclohexane/styrene						
	(15/85, v/v)		0.075			25	726
			0.086 (μ)			25	726
	(30/70, v/v)		0.092			25	726
			0.112 (μ)			25	726
	(40/60, v/v)		0.104			25	726
			0.129 (μ)			25	726
	(50/50, v/v)		0.120			25	726
			0.152 (μ)			25	726
	(60/40, v/v)		0.129			25	726
			0.160 (μ)			25	726
	(70/30, v/v)		0.138			25	726
			0.166 (μ)			25	726
	(85/15, v/v)		0.155			25	726
			0.173 (μ)			25	726
	Cyclohexanol	0.165				84.2	611
		0.141				87.8	661
		0.169				92.5	611
		0.169				100	611
	Cyclohexanone	0.151				30	216
		0.151				25	647
		0.151				30	216,919
	Cyclohexene	0.165				25	23
	Decalin		0.129			24	217
			0.129			25	215
		0.118				25	23
			0.137			52	23
			0.138			60	23
			0.143			87	23
			0.147			104	23
			0.152			123	23
			0.155			139	23
				0.120		20	615,713
			0.125		0.123 (579)	20	625
		0.119	0.110			20	827
		0.137	0.130			20.4	809
		0.141				22.8	661
		0.120	0.110			25	827
			0.127		0.125 (579)	30	625
		0.120	0.111			30	827
			0.128		0.126 (579)	40	625
		0.120	0.111			40	827
			0.130		0.128 (579)	50	625
			0.132		0.130 (579)	60	625
50 bar pressure			0.119			20	625
			0.120			30	625
			0.121			40	625
50 bar pressure			0.122			50	625
50 bar pressure			0.123			60	625
	cis-Decalin	0.124				12	221
	Decalin, 50% *trans*		0.115			20	885
			0.115			30	885
			0.116			40	885
			0.117			50	885
			0.118			60	885
	1,2-Dibromoethane	0.065				30	216,919

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Dibutyl phthalate		0.108			25	863
			0.120			135	863
	1,2-Dichloroethane	0.168	0.155			25	647
		0.168	0.155				151
			0.161			20	218
		0.168				30	152
			0.168				147
					0.168		152
		0.166				30	200
			0.158			30	826
	Diethyl ether/methylcyclohexane						
	(70/30, v/v)	0.215				25	693
	(65/35, v/v)	0.220				25	693
	(60/40, v/v)	0.215				25	693
	(50/50, v/v)	0.225				25	693
	(40/60, v/v)	0.240				25	693
	(35/65, v/v)	0.245				25	693
	Diethyl oxalate	0.195				59.6	661
	Diethyl fumarate/CH_2ClCH_2OH						
	(80/20, v/v)		0.152				207
	Dioxane		0.171				154
			0.176				225
			0.168				145
			0.172				196
		0.185	0.173		0.204 (365)	20	174
					0.171 (578)	20	174
					0.170 (589)	20	174
					0.168 (644)	20	174
		0.1852	0.1731			25	139
					0.1702 (589)	25	155
		0.193	0.182		0.176 (589)	25	206
					0.170		148
		0.181				35	97
		0.190				35	202
			0.184				147
				0.166			618
			0.170			25	807,822
			0.173			25	751
				0.181		30	671
			0.170			35	683
star shaped			0.195			35	683
deuterated polymer	Dioxane/1-bromonaphthalene		0.164				645
	(96/4, v/v)	0.134 (μ)				30	216,919
	(90/10, v/v)	0.075 (μ)				30	216,919
	(83/17, v/v)	0.000 (μ)				30	216,919
	(81.5/18.5, v/v)	−0.010 (μ)				30	216,919
	(50/50, v/v)	−0.298 (μ)				30	216,919
	Dioxane/*n*-heptane	0.212				35	211
	(41.5/58.5, v/v)						
	Dioxane/methanol						
	(28.6/71.4, v/v)		0.191			25	209
	(66.5/33.5, v/v)	0.218				35	211
	Dioxane/methanol					25	
	(71.4/28.6, v/v)		0.222			35	683
	(71.4/28.6, v/v)		0.197			35	683
	Dioxane/isopropanol	0.212				35	211
	(51.5/48.5, v/v)						
star shaped	Dioxane/styrene						
	(15/85, v/v)		0.076			25	726
	(15/85, v/v)		0.084 (μ)			25	726
	(30/70, v/v)		0.092			25	726
			0.102 (μ)			25	726

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(40/60, v/v)		0.104			25	726
			0.112 (μ)			25	726
	(50/50, v/v)		0.116			25	726
			0.129 (μ)			25	726
	(60/40, v/v)		0.126			25	726
			0.138 (μ)			25	726
	(70/30, v/v)		0.137			25	726
			0.147 (μ)			25	726
	(85/15, v/v)		0.155			25	726
			0.163 (μ)			25	726
	DMF	0.1703				18	213
		0.1739	0.1648		0.1632 (578)	20	5,6
					0.1625 (589)	20	5,6
					0.1604 (644)	20	5,6
					0.145		148
					0.189 (365)	20	6
			0.165			25	215
		0.175				25	146
			0.166				658
					0.171		658
				0.165		20	869
		0.168	0.158			20	897
			0.166			25	638
			0.178			30	826
Microgel			0.174				608
	DMF/tetralin (35.2/64.8, v/v)		0.119 (μ)			25	638
			0.119			80	638
	Ethyl acetate	0.2310				18	213
		0.234				20	102
			0.2231			25	226
		0.231	0.218			20	709
			0.223			25	779
			0.226			30	826
				0.216			618
	Ethyl acetate/ethanol						
	(94.81/5.19, v/v)		0.2196			25	226
	(91.76/8.24, v/v)		0.2210			25	226
	(89.34/10.66, v/v)		0.2238			25	226
	Ethylbenzene	0.111				25	102
			0.093			20	885
			0.093			30	885
			0.096			40	885
			0.102			50	885
			0.114			60	885
	Ethylcyclohexane		0.128			70	806
	Furfuryl alcohol		0.117			30	691
	n-Heptane/styrene						
	(10/90, v/v)		0.075			25	726
			0.089 (μ)			25	726
	(20/80, v/v)		0.091			25	726
			0.109 (μ)			25	726
	(30/70, v/v)		0.106			25	726
			0.136 (μ)			25	726
	(40/60, v/v)		0.122			25	726
			0.162 (μ)			25	726
	(50/50, v/v)		0.137			25	726
			0.191 (μ)			25	726
	(60/40, v/v)		0.154			25	726
			0.195 (μ)			25	726
	MEK	0.2258	0.2138			7.5	219
		0.2275	0.2155			15	219
		0.2290				18	213
		0.2287	0.2167			20	219

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
		0.2296	0.2170			20	6
			0.214			20	227
		0.243				20	203
					0.232		153
		0.229				20	88,231
		0.228				20	102,190
		0.2296	0.2170		0.2150 (578)	20	5
					0.2143 (589)	20	5
					0.2110 (644)	20	5
		0.2298	0.2175			25	139
					0.2146 (589)	25	155
			0.2165			25	165
		0.2298	0.2178			25	219
			0.214			25	195
			0.220				145,154,196
		0.229					146
		0.226				25	217
		0.231	0.220			25	212
		0.231	0.218		0.208 (589)		151,232
					0.220 (589)		232
		0.226				25	228
					0.220		148
			0.219			25	215
			0.230				147
		0.227	0.215			25	106
					0.210*		224
		0.229	0.217				107
					0.222 (437)	25	233
		0.232	0.223		0.216 (589)	25	206
		0.238				30	216
		0.232				30	152
		0.223				30	229
		0.229	0.219				230
					0.170	30	164
		0.2309	0.2189			30	219
		0.227				35	97
		0.2321	0.2201			35	219
		0.2333	0.2213			40	219
		0.2345	0.2225			45	219
			0.230			67	218
		0.254				70	203
				0.212			618
					0.222		891
				0.214		20	869
		0.225	0.207			20	897
			0.217			25	751
			0.220			25	807
		0.230	0.217			25	809
			0.218			25	902
increase with *M*			0.207–0.211			25	648
			0.220			30	826
		0.238				30	216,919
randomly branched		0.235					599
branched with divinyl benzene		0.240				35	598
	MEK/$(CH_2Cl)_2CHOH$ (46/54, v/v)		0.165				207
	MEK/1-dodecanol (95.4/4.6, mol/mol)	0.238				20	203
		0.253				70	203

* Given erroneously as 0.110 in journal.

TABLE 1. *cont'd*

Polymer	Solvent	d*n*/d*c* (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	*T* (°C)	Refs.
	(91.2/8.8, mol/mol)	0.232				20	203
		0.248				70	203
	MEK/1-hexanol						
	(93.3/6.7, mol/mol)	0.241				20	203
	(84.8/15.2, mol/mol)	0.238				20	203
		0.251				70	203
	(73.7/26.3, mol/mol)	0.234				20	203
		0.249				70	203
	MEK/methanol						
	(11.3/88.7, v/v)		0.214			25	209
	MEK/isopropanol		0.211			25	209
	(11.4/88.6, v/v)						
	(85/15, v/v)		0.219				207
	MEK/styrene (50/50, v/v)		0.140			25	726
			0.149 (μ)			25	726
	Methylcyclohexane		0.188				26
	Methylcyclohexane/acetone						
	(17/83, v/v)	0.242				22	234
	(22.2/77.8, v/v)	0.242				22	234
	(28/72, v/v)	0.239				22	234
	(39/61, v/v)	0.231				22	234
	(50/50, v/v)	0.228				22	234
	(61/39, v/v)	0.223				22	234
	(72/28, v/v)	0.222				22	234
	(83/17, v/v)	0.217				22	234
	(88.3/11.7, v/v)	0.210				22	234
	3-Methyl cyclohexanol	0.154				98.4	661
	3-Methyl cyclohexanol/1-chlorodecane						
	(21.8/78.2, v/v)	0.165				22.8	661
	(25/75, v/v)	0.168				35	661
	(50.5/49.5, v/v)	0.166				59.6	661
	(90/10, v/v)	0.155				87.8	661
	3-Methylcyclohexanol/tetralin						
	(23.2/76.8, v/v)	0.096				98.4	903,676
	(43.4/56.6, v/v)	0.115				98.4	903,676
	(49.5/50.5, v/v)	0.112				98.4	903,676
	(61/39, v/v)	0.131				98.4	903,676
	(77/23, v/v)	0.145				98.4	903,676
	Nitropropane	0.208				35	202
	Nitropropane/*n*-heptane						
	(42/58, v/v)	0.216				35	202
		0.218				35	211
	Phenylcyanide	0.0775				25	146
	Styrene	0.054				20	102
	Tetralin	0.072				34.5	903
		0.075				98.4	676,903
	Tetralin/cyclohexane						
	(89.4/10.6, v/v)	0.082 (μ)				34.5	903
	(88/12, v/v)	0.087				34.5	903
	(81/19, v/v)	0.093				34.5	903
	(79.4/20.6, v/v)	0.095 (μ)				34.5	903
	(69.7/30.3, v/v)	0.119 (μ)				34.5	903
	(59.7/40.3, v/v)	0.139 (μ)				34.5	903
	(39.6/60.4, v/v)	0.159 (μ)				34.5	903
	(20/80, v/v)	0.173 (μ)				34.5	903
	THF	0.195				30	216
			0.1926			30	84
					0.190		235
			0.198				57
					0.194		166
$\bar{M}_w =$ 2.4×10^4		0.205	0.193			25	236
5.31×10^4		0.202	0.189			25	236
16.5×10^4		0.207	0.194			25	236

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
54.1×10^4		0.208	0.195			25	236
150×10^4		0.209	0.196			25	236
570×10^4		0.211	0.198			25	236
				0.186			618,798
				0.193		20	869
				0.192		23	830
				0.186		25	637
		0.195				30	216,919
		0.208	0.193	0.188		RT	630
100% mol deuterated		0.186	0.171	0.165		RT	630
62% mol deuterated		0.190	0.177	0.172		RT	630
25% mol deuterated		0.193	0.182	0.178		RT	630
	THF/methanol (28.7/71.3, v/v)		0.179			25	209
	Toluene	0.111	0.109			RT	232
		0.112				12	221
			0.104			20	31,227
		0.112				20	102
			0.109			20	238
		0.1113	0.1079		0.1073 (578)	20	5
					0.1071 (589)	20	5
					0.1060 (644)	20	5
					0.113		153
		0.112	0.108				230
		0.195				20	231
			0.110			20	240
		0.111				23	237
		0.112				25	239
			0.109				225
		0.1188	0.1079			25	241
		0.1129	0.1091			25	242
			0.1130			25	43
			0.110			25	243
			0.111			25	240
		0.111				25	217,228
		0.112				25	146
		0.1150	0.1111			25	139
			0.111			28	29
			0.110				147
		0.113				30	152,216
		0.118				30	229
		0.114				35	97
			0.118			45	243
			0.125			65	243
				0.110			244
			0.118			67	218
				0.106			618
				0.101			842
		0.113				15	834
		0.115				20	834
				0.111		20	869
					0.103 (1086)	20	915
$M = 1.2 \times 10^3 - 1.07 \times 10^5$				0.089–0.105		20	732
star shaped polymer				0.103		20	737
					0.105 (1086)	25	915
			0.109		0.113	25	647
			0.111			25	751
			0.110			25	601
		0.112				25	613
			0.105			25	633
increase with M			0.110–0.112	0.107–0.110		25	635
increase with M:			0.093–0.109			25	648
			0.112			25	664
		0.117				25	834

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
$M > 2 \times 10^4$				0.108		25	833
star branched				0.103–0.106		25	833
				0.107–0.112		23–27	652
				0.108		30	671
			0.110			30	691
		0.113				30	697,699,216
		0.118				30	834
$M = 2 \times 10^3 - 18 \times 10^5$		0.101–0.118				30	837
		0.113				35	659
$M = 2 \times 10^3 - 18 \times 10^5$		0.114				35	704
		0.119				50	699
		0.125				70	699
			0.125			80	638
			0.126			90	609
isotactic		0.111				30	245
$M_w = 0.02 \times 10^5 - 0.5 \times 10^5$	Toluene	0.101–0.107				30	962
$M_w = 1.10 \times 10^5$	Toluene	0.106				30	962
$M_w = 1.60 \times 10^5$	Toluene	0.107				30	962
$M_w = 2.00 \times 10^5 - 8.67 \times 10^5$	Toluene	0.110–0.117				30	962
$M_w = 18 \times 10^5$	Toluene	0.118				30	962
$M_w = 1.25 \times 10^5 - 2.5 \times 10^5$	Toluene	0.117				30	962
dispersion	Water				0.250	25	95
					0.241		246
		0.256				25	247
latex		0.257					248
sulfonated	Neutralization equiv.						
	= 0.9 mEq/g: 0.05 M NaCl		0.212			25	247
	1.8 mEq/g: 0.056 M NaCl		0.207			25	247
	2.0 mEq/g: 0.10 M NaCl		0.195			25	247
	2.7 mEq/g: 0.10 M NaCl		0.205			25	247
	Toluene/*p*-cymene						
	(73.4/26.6 w/w)	0.115				30	699
	(49.7/50.3 w/w)	0.117				30	699
	Trichlorobenzene		0.028			25	653
	Water/THF (1/99, v/v)				0.183	25	796
					0.166 (μ)	25	796
	(3/97, v/v)				0.192	25	796
					0.183 (μ)	25	796
	(4.5/95.5, v/v)				0.180	25	796
					0.177 (μ)	25	796
	(7/93, v/v)				0.180	25	796
					0.188 (μ)	25	796
Poly(styrene-*p*-sulfonic acid)	Aq. KCl 2.5 M		0.0347*			25	249
–, K-salt, DN = 1.0	Aq. KCl; $I = 0.10$	0.197				30	85
–, Na-salt	THF/ethanol (76/24, v/v)		0.124			25	675
			0.128 (μ)			25	675
	Aq. 0.1 M Na_2SO_4		0.152 (μ)			25	879
	Aq. 1 M Na_2SO_4		0.122 (μ)			25	879
Poly(styrene-co-acrylonitrile), (1/1, mol/mol)							
	MEK		0.198			25	753
Poly(styrene-*co*-*n*-butyl methacrylate) (1/1, mol/mol)							
	MEK	0.164				35	814
Poly(styrene-*co*-*p*-chlorostyrene), ($\sim$1/1, mol/mol)							
	Cumene	0.122				25	795
		0.124				40	795
		0.125				55	795
Poly(styrene-*co*-ethylene-*co*-butene) (Kraton G)							
	THF		0.116			25	675
	THF/ethanol (94/6, v/v)		0.118			25	675
			0.120 (μ)			25	675
	(88/12, v/v)		0.120			25	675
			0.124 (μ)			25	675

*Expressed in l/g-segment.

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TABLE 1. *cont'd*

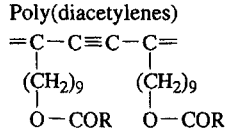
Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
1.7. OTHER COMPOUNDS							
Poly(acrylonitrile)	γ-Butyrolactone		0.079			25	276
			0.078			30	277
	Dimethylacetamide	0.0767	0.0769			25	278
	Chlorobenzene				0.00		891
	DMF		0.080				643
					0.085		891
		0.870				18	213
			0.082			20	238
			0.0874			20	281
		0.0877				20	283
		0.0914				20	283
		0.0924				20	283
			0.196			25	284
			0.083			25	215,276, 278,280
			0.089				279
		0.080				25	282
			0.084			25	277
			0.087			25	10
		0.085	0.084			20	897
			0.078			25	985
	DMSO	0.029	0.031			25	278
			0.043			25	285
			0.092			140	278
			0.0263			25	284
	DMSO/acrylonitrile						
	(65/35, v/v)		0.061			25	726
			0.067 (μ)			25	726
	(73/27, v/v)		0.055			25	726
			0.062 (μ)			25	726
	(82/18, v/v)		0.048			25	726
			0.053 (μ)			25	726
	(91/9, v/v)		0.040			25	726
			0.044 (μ)			25	726
	MEK				0.132		891
Poly(diacetylenes) 							
R = isopropyl	Chloroform				0.174 (580)		644
R = benzyl					0.198 (580)		644
R = 1-methylene naphthalene					0.228 (580)		644
R = 1-methylene naphthalene							
	1,2-Dichloroethane		0.267		0.220 (578)	RT	740
				0.190		RT	740
Poly(*tert*-butyl vinyl ketone)	*sec*-Butyl alcohol		0.097			25	715
Poly(*N*-isobutyl maleimide)	Chlorobenzene				0.01		577
Poly(maleic acid-*co*-ethyl vinyl ether) (1/1, mol/mol)							
	THF			0.091		25	663
Poly(maleic acid-*co*-ethyl vinyl ether) (1/1 mol), sodium salt							
	Aq. 0.18 M NaCl			0.180		25	663
Poly(maleic anhydride-*co*-ethyl vinyl ether (1/1, mol/mol)							
	THF			0.088		25	663
	Acetone	0.133				30	896
Poly(maleic anhydride-*co*-octadecene) (1/1, mol/mol)							
	Ethyl acetate		0.119			25	733
Poly(2-methyl 5-vinylpyridine; *N*-oxide)							
	Methanol			0.262		25	679
Poly(2-methyl-5-vinylpyridine)							
	n-Butyl acetate	0.200	0.188			25	269
	MEK	0.212	0.202			25	269

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Methanol	0.273	0.257			25	270
				0.250		RT	680
Poly(phenyl vinyl ketone)	Benzene		0.102			25	759
Poly(vinyl acetate)	Acetone		0.095			20	238
		0.104					287
					0.116	25	32
			0.104			25	289
		0.104				25	288
			0.105			25	201
			0.1021			35	34
	Acetonitrile				0.104		290
	Benzene				0.030		194
			−0.0225			30	84
	n-Butyl acetate		0.0716			30	84
	Chlorobenzene		−0.0426			30	84
					−0.049	25	32
	Dioxane				0.028		290
	Ethyl acetate				0.087	25	32
	Ethyl formate				0.095		290
	MEK				0.080	15	32
			0.080				291
		0.080					252,287,292
					0.080	25	32,293
		0.083				25	205
			0.080			25	289
					0.080	35	32
				0.075			618
	Methanol	0.107				25	287
		0.130				25	288
		0.132				25	205,294
					0.124	25	32
			0.132				289
		0.1319				34–38	295
				0.119			618
		0.130	0.132		0.128 (366)		881
	Methyl isobutyl ketone				0.068		290
	Methyl isopropyl ketone/ *n*-heptane (3/1–6/1, v/v)	0.075	0.075			25	252
	THF		0.0582			30	84
				0.050			618
Poly(vinyl acetate)							
				0.057		25	637
			0.059				803
				0.054		25	865
	Trichlorobenzene	0.1030				34–38	296
Poly(vinyl acetate) (partially hydrolyzed)							
Deg. of Hydrolysis							
= 0.840	Water		0.151				297
0.883			0.155				297
0.964			0.161				297
0.008	Acetone		0.103			25	289
0.029			0.101			25	289
0.049			0.098			25	289
0.119			0.090			25	289
0.008	MEK		0.083			25	289
0.029			0.087			25	289
0.049			0.092			25	289
0.119			0.110			25	289
0.008	Methanol		0.128			25	289
0.029			0.115			25	289
0.049			0.107			25	289
0.119			0.071			25	289
Poly(9-vinyl adenine)	Aq. 0.1 M NaCl+Na cacodylate		0.326			26	823
			0.331			40	823

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Poly(vinyl alcohol)	Water/*n*-propanol (98/2, w/w)	0.134	0.135		0.128 (366)	25	881
					0.133 (404)	25	881
		0.145	0.147		0.142 (366)	40	881
					0.144 (404)	40	881
		0.158	0.160		0.151 (366)	70	881
					0.153 (404)	70	881
	Water		0.150			20	298
		0.144				20	299
		0.168	0.164		0.164 (578)	30	287
$M_w = 39900$	Water			0.151		25	948
$M_w = 120000$				0.153		25	948
$M_w = 160000$				0.153		25	948
Poly(vinylbiphenyl)	Benzene	0.182	0.165			24	300
	Carbon tetrachloride	0.220	0.207			24	300
	Dioxane	0.257	0.233			24	300
	Ethylene glycol dimethyl ether	0.301	0.277			24	300
	Ethylene glycol dimethyl ether/ethylene glycol monomethyl ether (56/44, v/v)	0.297–0.301	0.271–0.275			30	300
Poly(vinyl bromide)	THF		0.112				251
					0.112 (644)		252
Poly(vinyl *n*-butyl ether)	MEK	0.0792				25	250
Poly(vinyl butyral) – acetylated							
	Acetone		0.119			25	880
	Acetonitrile		0.130			25	880
	n-Butyl acetate		0.084			25	880
	n-Butyl chloride		0.075			25	880
	THF		0.071			25	880
Poly(vinyl *n*-butyrate)	MEK				0.0684	20	301
Poly(vinyl carbanilate)	Dioxane	0.159	0.155			20	302
Poly(*N*-vinylcarbazole)	Benzene	0.210	0.190			25	304
		0.219	0.195			25	303
		0.222	0.198			30	303
		0.225	0.201			35	303
		0.228	0.204			40	303
		0.231	0.207			45	303
	Chloroform	0.268	0.232			25	304
	Tetrachloroethane	0.214	0.185			25	304
	THF	0.282	0.262			25	304
Poly(9-vinylcarbazole-*co*-diethyl fumarate)							
	Benzene		0.128			25	781
Poly(*N*-vinyl-3,6-dibromocarbazole)							
	o-Dichloroenzene		0.113			25	800
	THF		0.174			25	800
Poly(vinyl chloride)	Acetone		0.138			20	238
	iso-Amyl acetate			0.098		20	682
	Aq. NH_4OH; pH 9–10 (latex)		0.200				253
	n-Butyl acetate			0.112		20	682
	Cyclohexanone	0.0718	0.0705		0.0701 (578)	15.5	259
		0.0723	0.0714		0.0708 (578)	20	259
			0.0840			25	255,256
		0.079	0.077		0.076 (589)	25	206
			0.075			25	10
			0.074			25	254,257
					0.078		258
		0.0739	0.0727		0.0716 (578)	25	259
		0.0758	0.0742		0.0740 (578)	25	259
		0.0764	0.0746		0.0745 (578)	30	259
		0.0767	0.0756		0.0753 (578)	35	259

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
			0.0905			50	255
			0.0840			50	256
			0.0900			75	256
			0.0970			75	255
			0.0940			95	256
				0.075		20	682
			0.074			25	851
	Di-*n*-butyl phthalate		0.0380			25	256
	Diethyl adipate			0.085		20	682
	Di-2-ethylhexyl phthalate		0.0435			25	256
			0.0465			50	256
			0.0494			80	256
			0.0518			105	256
	Diethyl malonate			0.094		20	682
	Diethyl oxalate			0.103		20	682
	Diethyl succinate			0.094		20	682
	Di-2-heptylnoyl phthalate		0.0420			25	256
	Di-*n*-hexyl phthalate		0.0250			25	256
			0.0260			50	256
			0.0280			70	256
			0.0300			105	256
	Dimethyl phthalate		0.0260			25	256
	Dioxane		0.0900			25	256
			0.0925			25	260
		0.107	0.104		0.107 (589)	25	206
			0.11			25	261
$M_w = 2.04 \times 10^6$			0.0850			28.5	262
1.78×10^6			0.0865			28.5	262
	Di-*n*-propyl phthalate		0.0395			25	256
	DMF		0.069			25	256
			0.078			50	256
			0.0875			75	256
			0.095			95	256
		0.084					258
		0.0816	0.0810		0.0813 (578)	25	259
	Ethyl heptanoate			0.102		20	682
	Tetrachloroethane		0.0432			20	255
			0.0450			25	255
			0.0540			50	255
			0.0630			75	255
	THF	0.1040	0.1017		0.1010 (578)	16.5	259
		0.115	0.110			RT	265
					0.102		258
			0.106				266
		0.1124				20	263
			0.105			25	261
			0.1123			25	256
		0.119	0.116		0.115 (589)	25	206
			0.1065			25	267
		0.1079	0.1052		0.1068 (578)	25	259
		0.1067	0.1053		0.1046 (578)	25	259
		0.1093	0.1064		0.1059 (578)	30	259
		0.1118	0.1087		0.1093 (578)	35	259
			0.1045			35	34
			0.1065				264
				0.111		20	682
				0.102		25	637
			0.106			25	851
	THF/water (91/9, v/v)	0.112 (μ)	0.1085 (μ)			25	267
–, chlorinated 56.4, wt.% Cl	Tetrachloroethane		0.045			25	255
58.3, wt.% Cl			0.0424			25	255
61.5, wt.% Cl			0.525			25	255
63.4, wt.% Cl			0.0606			25	255
66.6, wt.% Cl			0.073			25	255

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TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Poly(vinyl dimethylamylsilane)	Cyclohexane		0.070			25	884
Poly(vinyl dimethylheptylsilane)	Cyclohexane		0.085			25	884
Poly(vinyl dimethylphenylsilane)	Cyclohexane		0.142			25	884
Poly(vinylidene chloride)	1-Methyl-2-pyrrolidone		0.080			RT	695
	Tetramethylene sulfoxide		0.049			RT	695
Poly(vinyl ethyl ether)	MEK	0.0736				25	250
Poly(vinyl fluoride)	DMF				≈0.02		268
Poly(vinylidene fluoride)	DMA		−0.033	−0.031		25	986
	DMF		−0.023	−0.021		25	986
	DMSO		−0.062	−0.060		25	986
	N,N'-dimethylethyleneurea		−0.055	−0.053		25	986
	Hexamethylphosphoric triamide		−0.046	−0.044		25	986
	N-methyl pyrrolidone		−0.052	−0.50		25	986
	3-methyltetramethylene sulfone		−0.064	−0.063		25	986
	Tetramethylurea		−0.042	−0.040		25	986
Poly(*N*-vinyl-imidazole) incr. with mol. wt.	Ethanol		0.180–0.192			25	887
	Methanol	0.235				25	719
	Tetrabutyl ammonium bromide	0.224				25	719
	Tetrabutyl ammonium bromide	0.224 (μ)				25	719
Poly(vinyl isopropyl ether)	MEK	0.0827				25	250
Poly(vinyl methyl ether)	MEK	0.0944				25	250
	Chloroform			0.044			842
	THF			0.063			842
	Toluene			0.001			842
Poly(α-vinylnaphthalene)	Benzene	0.192	0.175			24	300
	Ethylene dichloride	0.239	0.217			24	300
	Ethylene glycol dimethyl ether	0.268	0.248			24	300
	Toluene	0.162–0.192				24	300
Poly(β-vinylnaphthalene)	Benzene	0.182–0.197	0.167–0.181			24	300
	Carbon tetrachloride	0.215	0.194			24	300
	Ethylene glycol dimethyl ether	0.309	0.284			24	300
	Ethylene glycol dimethyl ether/ethylene glycol monomethyl ether (56/44, v/v)	0.283	0.258			24	300
Poly(2-vinylpyridine)	Aniline	0.023				25	907
	Benzene	0.107				25	907
	n-Butanol	0.209				25	907
	2-Butanol	0.216				25	907
	Chloroform		0.1372			23	34
		0.168				25	907
			0.1215			35	39
	Cyclohexanone	0.144				25	907
	1,2-Dichloroethane	0.148				25	907
	Dioxane	0.175				25	907
	DMF	0.165				25	907
atactic			0.157				271
isotactic			0.163				271
	DME/methacrylic acid (95/5, v/v)				0.187 (589)	20	797
					0.195 (μ) (589)	20	797
	Ethanol	0.245				25	907
atactic			0.221				271
isotactic			0.221				271
isotactic	Ethanol + 0.1 M sodium acetate		0.222				271

TABLE 1. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	MEK	0.215				25	907
	Methanol			0.254		RT	680
		0.270				25	907
$M = 2.1 \times 10^3$–22.9×10^3		0.259–0.263				25	909
Poly(vinyl-2-pyridine)	0.950 Nitromethane/0.050 carbon tetrachloride		0.2020			25	967
	0.900 Nitromethane/0.100 carbon tetrachloride		0.1989			25	967
	0.800 Nitromethane/0.200 carbon tetrachloride		0.1918			25	967
	0.750 Nitromethane/0.250 carbon tetrachloride		0.1889			25	967
	0.700 Nitromethane/0.300 carbon tetrachloride		0.1847			25	967
	0.625 Nitromethane/0.375 carbon tetrachloride		0.1793			25	967
	0.565 Nitromethane/0.435 carbon tetrachloride		0.1740			25	967
	0.500 Nitromethane/0.500 carbon tetrachloride		0.1705			25	967
	0.470 Nitromethane/0.530 carbon tetrachloride		0.1684			25	967
	0.400 Nitromethane/0.600 carbon tetrachloride		0.1560			25	967
	0.250 Nitromethane/0.750 carbon tetrachloride		0.1525			25	967
	0.214 Nitromethane/0.786 carbon tetrachloride		0.1500			25	967
	Nitroethane	0.197				25	907
	n-Propanol	0.215				25	907
	2-Propanol	0.221				25	907
	1,1,2,2-Tetrachloroethane	0.109				25	907
	THF	0.195				25	907
					0.182		272
	1,1,1-Trichloroethane	0.162				25	907
Poly(4-vinylpyridine)	Chloroform	0.150				25	146
	DMF	0.160				25	146
			0.153				658
				0.153		RT	680
	Ethanol	0.231					273
	Ethanol/water (95/5, w/w)	0.231					273
	Isopropanol	0.224				25	274
	MEK/isopropanol (85/15, w/w)	0.221				25	274
	(86/14, w/w)	0.224				25	274
	Methanol	0.267				25	274
	Nitromethane				0.180		275
					0.159		275
					0.152		275
	Phenyl cyanide	0.052				25	146
Poly(2-vinylpyridine 1-oxide)	Aq. 1 M NaCl			0.220		25	679
	2-Propanol	0.209				25	640,987
Poly(4-vinylpyridine *N*-oxide)	Aq. 0.1 M NaCl			0.241		25	679
Poly(vinylpyrrolidone)	Chloroform	0.108	0.108				306
	Methanol		0.183			25	201
			0.176			25	193
	Water		0.135			25	193
		0.185	0.185				306
		0.181	0.176		0.176 (578)	20	792
		0.180	0.176		0.175 (578)	25	792
		0.179	0.175		0.174 (578)	30	792
		0.179	0.174		0.173 (578)	35	792
$M_w = 10100$	Water			0.181		25	948
$M_w = 35100$				0.181		25	948
$M_w = 46400$				0.185		25	948
$M_w = 1540000$				0.179		25	948
Poly(vinylsulfonic acid), potassium salt	KCl 0.35 M		0.01664 (μ)*			23	308
	0.65 M		0.01638 (μ)*			23	308
Poly(vinyltrimethyl silane)	Carbon tetrachloride		0.048				307
	Cyclohexane		0.083				307
	Cyclohexane		0.087			25	884

* Expressed in i/g-segment.

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TABLE 2. MAIN-CHAIN CYCLIC CARBON POLYMERS

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	dn/dc (ml/g) $\lambda_0 = 546$ nm	dn/dc (ml/g) $\lambda_0 = 633$ nm	dn/dc (ml/g) Others	T (°C)	Refs.
Poly(acenaphthylene)	Benzene	0.225	0.119			25	55
	1,2-Dichloroethane	0.264	0.238			35	55
	Toluene		0.190			25	56
Poly(4,7-dimethyl-indene)	Benzene				0.100		750
Poly(6-methyl-indene)	Benzene				0.100		750
Poly(phenylenes)-phenyl substituted							
	Benzene		0.250			25	848
	THF		0.294			25	848

TABLE 3. MAIN-CHAIN HETEROATOM POLYMERS

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	dn/dc (ml/g) $\lambda_0 = 546$ nm	dn/dc (ml/g) $\lambda_0 = 633$ nm	dn/dc (ml/g) Others	T (°C)	Refs.
3.1. C–O–C POLYMERS (POLY(OXIDES), POLY(ESTERS) AND OTHERS)							
3.1.1. POLY(OXIDES)							
Poly(formaldehyde) see Poly(oxymethylene)							
Poly(oxybutylethylene) [Poly(hexamethylene oxide)]							
	Diethyl ketone					0.08825	340
Poly(oxy-*tert*-butylethylene)	*o*-Dichlorobenzene				0.095	80	819
	Isooctane		0.114			75	328
Poly(oxy-2,6-dimethyl-1,4-phenylene) (PPO)							
	Benzene		0.114			25	330
	Chlorobenzene		0.098			25	330
	Chloroform		0.124			25	330
			0.169			25	331
	Toluene		0.1153			25	331
			0.118			25	330
			0.116			25	787
			0.128			35	787
			0.133			40	787
			0.136			60	787
	Trichloroethane		0.139			25	787
			0.145			35	787
			0.152			40	787
			0.160			60	787
	Xylene		0.1136			25	331
Poly(oxy-2,6-diphenyl-1,4-phenylene)							
	Chloroform		0.226			30	921
	Toluene		0.185			25	331
Poly(dichlorophenylene oxide)	Toluene	0.123				30	963
$M_w = 5 \times 10^4$							
$M_w = 0.47 \times 10^5$	Toluene	0.119				30	983
$M_w = 1.14 \times 10^5$	Toluene	0.117				30	983
$M_w = 0.47 \times 10^5$	Toluene	0.119				30	983
$M_w = 0.47 \times 10^5$	Toluene	0.124				30	983
Poly(oxyethylene) glycol							
$M_w =$ 62	Acetonitrile		0.0964			25	332
100			0.106			25	332
161			0.114			25	332
205			0.121			25	332
316			0.123			25	332
407			0.130			25	332
970			0.135			25	332
9400			0.135			25	332
106	Benzene	−0.086					333
194		−0.073					333

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
282		−0.066					333
810		−0.059					333
3510			−0.016			25	332
		−0.018				30	846
		−0.013				54	846
$M = 1.5 \times 10^3 - 5.3 \times 10^5$	Benzene		−0.017 to −0.010			25	612
$M_w = 4000$		−0.022				30	962
	Bromofrom	−0.108	−0.090			23	860
	n-Butanol		0.076				334
	Carbon tetrachloride/ methanol (75/25, v/v)		0.066			25	751
	(50/50, v/v)		0.091			25	751
	(20/80, v/v)		0.128			25	751
	Chlorobenzene	−0.039	−0.030			23	860
	Chloroform	0.054	0.053			23	860
		0.054				30	846
	Chloroform/*n*-hexane (47/53, v/v)		0.091			20	193
	1,2-Dibromoethane	−0.048	−0.044			23	860
$M_w = 10000$	Dioxane		0.045				333
		0.061				45	846
	DMF		<0.05			25	335
$M_w = 9400$			0.044			25	332
$M_w = 810$	MEK		0.092				333
10000			0.094				333
	Methanol		0.150				334
$M_w = 62$			0.118			25	332
100			0.127			25	332
161			0.135			25	332
205			0.139			25	332
316			0.141			25	332
445			0.142			25	332
810			0.143				333
1020			0.144			25	332
3000			0.149				333
6000			0.150				333
9400			0.150			25	332
10000			0.148				333
31000			0.150			25	332
			0.143			25	193
			0.152			45	328
		0.145	0.142			30	846
		0.150				45	846
$M_w = 0.68 \times 10^5$	Methanol			0.142		25	963
$M_w = 6700$	Methyl acetate		0.111			25	332
	Pyridine	−0.026	−0.018			23	860
	1,1,2,2-Tetrachloroethane	0.006	0.007			23	860
	THF		0.068				334
	Water	0.134	0.132			27	337
			0.139				334
		0.138	0.135	0.134	0.135 (578)	20	725
				0.132		25	685
		0.138		0.132		25	688
		0.136	0.134	0.133	0.133 (578)	30	725
		0.134	0.132	0.132	0.132 (578)	40	725
		0.133	0.131	0.129	0.130 (578)	50	725
$M_w = 62$		0.093					333
106		0.108					333
194		0.124					333
300		0.126	0.123				333
600		0.135	0.131				333

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
600					0.128 (589)	25	338
810		0.136	0.128				333
1200		0.139	0.134				333
3000		0.141					333
4000					0.134 (589)	25	338
6000					0.134 (589)	25	338
6000		0.145	0.139				333
9000					0.134 (589)	25	338
9400			0.135			25	332
10000		0.142					333
14400			0.139			25	336
14400			0.115			80	336
31000			0.135			25	332
340000		0.149				25	339
Poly(ethylene glycol) carbamate $M_w = 5500$							
	Benzene	− 0.035				30	962
Poly(oxyethylethylene) [Poly(1-butene oxide)]							
	Benzene		− 0.0403			25	326
	Chlorobenzene		− 0.0784			25	326
	Ethyl acetate		0.1125			25	326
	n-Hexane		0.1008			25	326
	MEK	0.0863				30	327
	Toluene		− 0.0357			25	326
Poly(oxymethylene)	Hexafluoroacetone/water (1/1.7, mol/mol) + 1% (v/v) triethylamine buffer, pH 2.0	0.134	0.128				341
		0.114 (μ)	0.112 (μ)				341
	1H,1H,5H-Octafluoropentanol-1		0.156			110	325
Poly(oxymethyleneoxyethylene) [Poly(dioxolane)]							
MW	Chlorobenzene		0.051				578
	o-Dichlorobenzene		0.075				578
	Dichloroethane		0.020				578
	Dioxane		0.050				578
Poly(oxy-1-octylethylene))	Chloroform				0.0357	25	329
Poly(oxy(1-phenyl)ethylene) [Poly(styrene oxide)]							
	Benzene		0.085			30	328
$M_w =$ 67	Acetone	0.085				25	342
125		0.0915				25	342
450		0.096				25	342
1100		0.099				25	342
1240		0.099				25	342
2100		0.100				25	342
3270		0.100				25	342
3850		0.101				25	342
	Benzene	− 0.0530	− 0.0448			25	343
Poly(propylene glycol)							
$M_w = 4000$	Benzene	− 0.055				30	963
	Chlorobenzene	− 0.0658	− 0.0638			25	343
	n-Hexane	0.0775	0.0775			25	343
		0.0460	0.0460			40	343
$M_w = 9.6 \times 10^5$		0.0887	0.0887			46	343
		0.101	0.101			57	343
2.0×10^5		0.0895	0.0895			46	343
		0.104	0.104			57	343
	Feron 113	0.118	0.115			25	343
	Isooctane	0.0655	0.0655			35	343
	Methanol	0.137	0.135			24 ± 1	300
$M_w = 12.2 \times 10^5$		0.118	0.118			25	343
12.5×10^5		0.115	0.115			25	343
Poly(oxytetramethylene) [Poly(tetrahydrofuran)]							
	Chlorobenzene	0.070					347
	Ethyl acetate	0.110				25	348

TABLE 3. *cont'd*

Polymer	Solvent	d*n*/d*c* (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	*T* (°C)	Refs.
		0.114				30	349
	Ethyl acetate/*n*-hexane (22.7/77.3, w/w)	0.114				31.8	349
	Isopropanol	0.108				46	351
	Isopropyl acetate	0.098				22.5	352
	MEK	0.102				30	348
			0.091			25	350
			0.095				347
	Methyl acetate		0.101			25	350
	3-Methyl-2-heptanone		0.056			25	350
	2-Pentanone		0.084			25	350
	THF	0.0625	0.0625			25	348
			0.064			25	350
Poly(oxytrimethylene)	MEK				0.0946		353
3.1.2. POLY(ESTERS)							
Poly(ethylene terephthalate) see Poly(oxyethyleneoxyterephthaloyl)							
Poly(β-hydroxybutyric acid)	2,2,2-trifluoroethanol	0.153	0.149	0.146	0.148 (578)	25	988
Poly(oxyadipoyloxytetramethylene) [Poly(butylene adipate)]							
	Acetic acid		0.100			50	317
	Diethyl carbonate		0.084			35	317
			0.091			70	317
	Dioxane		0.0518			35	317
	THF		0.0784			35	317
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene) [Poly(carbonate) from Bisphenol A]							
	Bromoform	0.004	0.004			23	860
	Chloroform		0.162				319
		0.164				20	318
			0.156			20	920
			0.153			23	793
		0.157	0.147			23	860
		0.167	0.155			25	588
	Chloroform + 2% (w/w) ethanol		0.154			25	758
	1,2-Dibromomethane	0.054	0.050			23	860
	Dioxane		0.168			25	320
	Dichloroethane	0.166	0.154			25	588
	Methylene chloride	0.1660			0.1547 (553)	7	322
					0.1538 (574)	7	322
		0.177				25	321
		0.1773			0.1654 (533)	27	322
					0.1643 (574)	27	322
			0.177		0.165 (553)	25	322
	Pyridine	0.077	0.073			23	860
	1,1,2,2-Tetrachloroethane	0.106	0.094			23	860
	THF	0.1889			0.1765 (553)	7	322
					0.1753 (574)	7	322
		0.1943			0.1815 (553)	27	322
					0.1805 (574)	27	322
		0.198	0.186			23	860
		0.194	0.182			25	588
Poly(oxyethyleneoxyterephthaloyl) [Poly(ethylene terephthalate)]							
	Dichloroacetic acid		0.1021			35	34
	Trifluoroacetic acid		0.278				323
	1,1,1,3,3,3-Hexafluoropropanol-2			0.243		25	875
		0.308	0.287			25	843
Poly(oxy-1-oxohexamethylene) [Poly(ε-caprolactone)] Prepared with 1 mol/mol $(C_2H_5)_3Al$							
	Acetic acid		0.108			50	317
	o-Dichlorobenzene		−0.0595			35	317

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Diethyl carbonate		0.092			35	317
			0.0990			70	317
	Dioxane		0.0489			25	317
			0.0507			35	317
			0.054			50	317
			0.0581			70	317
Prepared with 1 mol% $(C_2H_5)_2AlCl$							
	Acetic Acid		0.0940			50	317
	o-Dichlorobenzene		−0.057			35	317
	Diethyl carbonate		0.093			35	317
			0.0939			40	317
			0.101			70	317
	Dioxane		0.0489			25	317
			0.0504			35	317
			0.0581			70	317
	THF		0.0795			35	317
Prepared with 0.61 mol% n-C_4H_9OK							
	Diethyl carbonate		0.0917			35	317
			0.1021			70	317
Poly(D-oxy-1-oxo-3-methyl tetramethylene)							
	Trifluoroethanol	0.152	0.149			25	703
Poly(D,L-oxy-1-oxo-3-methyl tetramethylene)							
	Trifluoroethanol	0.153	0.151			25	702
	Tetrahydrofuran	0.153	0.151			25	595
Poly(oxy-1-oxo-2-methylethylene) [poly(lactic acid)]							
	Bromobenzene		−0.06			85	324
Poly(oxypropylenetriol-*co*-diphenylmethane diisocyanate)							
	DMF		0.039			20	636
Poly(oxysebacoyloxyoctamethylene)							
	Acetic acid		0.100			50	317
	o-Dichlorobenzene		−0.071			35	317
	Diethyl carbonate		0.090			45	317
			0.100			70	317
	Dioxane		0.050			35	317
			0.054			50	317
			0.054			70	317
	THF		0.076			35	317
Poly(oxytetramethyleneoxyadipoyl-*co*-butandiol-*co*-diphenylmethane diisocyanate)							
	DMF		0.110			25	877
Poly(oxytetramethyleneoxyterephthaloyl)							
	1,1,1,3,3,3 Hexafluoropropanol-2			0.236		25	875
Poly(oxytriphenylstibyleneoxytetramethylterephthaloyl)							
	HMP				0.56		873
Poly(oxytriphenylstibyleneoxyterephthaloyl)							
	HMP				0.86		873
Poly(oxytriphenylstibyleneoxybromoterephthaloyl)							
	HMP				2.2		873
Poly(oxytriphenylstibyleneoxynitroterephthaloyl)							
	HMP				0.45		873
Poly(oxytriphenylstibyleneoxydichloroterephthaloyl)							
	HMP				0.18		873
Poly(oxytriphenylstibyleneoxyoxalyl)							
	HMP				2.1		873
Poly(oxytriphenylstibyleneoxyfumaroyl)							
	HMP				0.68		873
Poly(oxytriphenylstibyleneoxyferrocene-1,1-dicarboxoyl)							
	HMP				0.88		873
3.1.3. OTHER COMPOUNDS							
Poly(oxy-1,4-phenyleneoxy-1,4-phenylene-carbonyl-1,4-phenylene)							
	97.4% H_2SO_4				0.525 (514)	RT	747
				0.385		RT	747

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
3.2. C–S–C POLYMERS (POLY(SULFONES) AND OTHERS)							
3.2.1. POLY(SULFONES)							
Poly(bicycloheptene-*co*-sulfur dioxide)							
	Chloroform			0.102		25	696
	Cyclohexanone			0.112		25	696
Poly(1-butene-*co*-sulfur dioxide)							
	Chloroform	0.0970			0.0949 (578)	20	355
	MEK	0.133	0.129			26	904
Poly(cyclohexene-*co*-sulfur dioxide)							
	Chloroform	0.1133			0.1104 (578)	20	355
Poly(1-dodecene-*co*-sulfur dioxide)							
Phenolphthalein poly(ether sulfone)							
	$CHCl_3$				0.204 (532)	25	967
	Chloroform	0.0695			0.0581 (578)	20	355
Poly(1-hexene-*co*-sulfur dioxide)							
	Chloroform	0.0790			0.0782 (578)	20	355
	n-Hexyl chloride	0.0796				7	355
		0.0834				20	355
	MEK/isopropanol (37/63, v/v)	0.1242				27	355
		0.1261				37	355
	(41.5/58.5, v/v)	0.1208				7	355
		0.1233				20	355
Poly(1-octene-*co*-sulfur dioxide)							
	Chloroform	0.0732			0.0718 (578)	20	355
Poly(oxy-1,4-phenylenesulfonyl-1,4-phenylene) [Poly(sulfone)]							
	DMF		0.194			25	580
3.2.2. OTHER COMPOUNDS							
Poly(2-acrylamido, 2-methyl propane sulfonic acid)							
	4.5 M aqueous NaCl			0.102		25	966
Poly(2-acrylamido, 2-methyl propane sulfonamide)							
	Formamide			0.082		25	972
	Water			0.157		25	972,966
Poly(2-acrylamido-2-methylpropane sulfonate)							
	1 M aqueous NaCl			0.154			971
Poly(2-ethyl-2-oxazoline)	Water				0.184 (510)	5	970
	Water				0.176 (510)	25	970
	Water				0.174 (510)	45	970
	Ethanol				0.155 (510)	25	970
	i-Propanol				0.155 (510)	25	970
	n-Butanol				0.110 (510)	25	970
Poly(2-methyl-2-oxazoline)	Water				0.182 (510)	5	970
	Water				0.172 (510)	25	970
	Water				0.172 (510)	45	970
Poly(2-methoxycyanurate) of bisphenol F							
	Chloroform	0.173				30	973
Poly(*p*-phenylene-*cis*-benzobisoxazole)							
	Methane sulfonic acid			0.480		25	947
Poly(thio-carbonyl-1-(4-methylphenyl sulfonylamino)-ethylene)							
	DMF		0.126–0.142				853
Poly(thiopropylene)	Benzene		0.100			20	344
			0.0935				345
		0.08095					346
3.3. C–N–C POLYMERS (POLY(AMIDES), POLY(ISOCYANATES), POLY(URETHANES) AND OTHERS)							
3.3.1. POLY(AMIDES)							
Poly(iminoadipoyliminohexamethylene) (Nylon 66)							

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TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Formic acid 90% + 0.5 M sodium formate	0.137				25	870
	Trifluoroethanol	0.228				25	870
Trifluoroacetylated	Acetone	0.076				25	669
	Formic acid 75%				0.144	25	311
	80%				0.145	25	311
	85%	0.141				25	312
	90%				0.145	25	311,313
			0.145			25	309
		0.145				25	312
	95%				0.150		311
	100%				0.157	25	311
		0.1525					310
	Formic acid + KCl						
	85% + 2.0 M KCl				0.124		313
	90% + 0.2 M KCl				0.143		313
	0.5 M KCl				0.140		313
	1.0 M KCl				0.136		313
	1.5 M KCl				0.131		313
	2.0 M KCl				0.126		313
	2.5 M KCl				0.122		313
	95% + 2.0 M KCl				0.129		313
	Formic acid + sodium formate						
	75% + 0.5 M				0.138	25	311
	80% + 0.5 M				0.136	25	311
	90% + 0.02 M				0.147	25	311
	0.05 M				0.146	25	311
	0.10 M				0.142	25	311
	0.2 M				0.142	25	311
	0.5 M				0.136	25	311
	0.75 M				0.130	25	311
	1.0 M				0.124	25	311
	95% + 0.5 M				0.136	25	311
	100% + 0.5 M				0.136	25	311
	Tetrafluoropropanol		0.190			25	314,315
	Tetrafluoropropanol + 0.1 N sodium trifluoroacetate buffer		0.190			25	314
Poly(4-iminobenzoyl)	Dichloroacetic acid		0.151			RT	808
	DMA		0.188			RT	808
	Trifluoroacetic acid		0.226			RT	808
	96% Sulfuric acid		0.360				656
Poly(iminocarbonyl-(4,6-dicarboxy-1,3-phenylene)-carbonylimino-1,4-phenylene) [Poly(amic acid)]							
	DMF	0.269	0.230			25	816
Poly(iminocarbonyl-(3,4-dicarboxy-1,5-phenylene)carbonylimino-1,4-phenyleneisopropylidene-1,4-phenylene) [Poly(amic acid)]							
	LiBr 0.1 N in DMA		0.206				574
			0.210				573
Poly(iminohexamethyleneiminoterephthaloyl)							
	1,1,1,3,3,3-Hexafluoro-isopropanol-2			0.231		25	875
Poly(imino(1-oxododecamethylene)) (Nylon 12)							
	m-Cresol		−0.030			25	766
	m-Cresol		−0.020			25	698
	m-Cresol/*n*-heptane						
	(90/10, v/v)		−0.004			25	698
			0.036 (μ)			25	698
	(80/20, v/v)		0.006			25	698
			0.075 (μ)			25	698
	(70/30, v/v)		0.022			25	698
			0.182 (μ)			25	698
	(60/40, v/v)		0.043			25	698,790
			0.340 (μ)			25	698,790
	(40/60, v/v)		0.083			25	791
			0.254 (μ)			25	791

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	m-Cresol/TFP (40/60, v/v)		0.254 (μ)			25	766
	Hexafluoroisopropanol	0.266	0.255			25	614
Trifluoroacetylated	Acetone	0.088				25	669
Poly(imino-(1-oxohexamethylene)) (Nylon 6)							
	Carbon disulfide 95%		0.0815			25	309
	Chloral hydrate	0.075	0.072		0.071 (578)	60	871
	o-Chlorophenol	0.024				30	916–918
	m-Cresol		−0.016			25	309
		0.003				30	916–918
			0.005			25	698
	m-Cresol/*n*-Heptane						
	(90/10, v/v)		0.020			25	698
			0.047 (μ)			25	698
	(80/20, v/v)		0.031			25	698
			0.032 (μ)			25	698
	(70/30, v/v)		0.246 (μ)			20	698
			0.045			25	698
			0.243 (μ)			25	698
			0.241 (μ)			37	698
			0.240 (μ)			45	698
	(50/40, v/v)		0.053			25	698
			0.471 (μ)			25	698
	(50/50, v/v)		0.853 (μ)			20	698
			0.083			25	698,790
			0.816 (μ)			25	698,790
			0.711 (μ)			37	698
			0.640 (μ)			45	698
	Dichloroacetic acid		0.098			RT	808
			0.098			25	309
			0.099			50	309
			0.104			80	309
	DMA		0.136			RT	808
	Formic acid 90% ± 1.0 M KCl		0.136			25	309
	2.0 M KCl		0.126			25	309
	90% Formic acid + 0.5 M Na formate	0.138	0.135	0.134		25	623
	1,1,1,3,3,3-Hexafluoro-propanol-2			0.253		25	876
	Octafluoropentanol		0.192			25	309
	Tetrafluoropropanol/water						
	(95/5, v/v)		0.183			25	316
	(90/10, v/v)		0.182			25	316
	(80/20, v/v)		0.180			25	316
	(70/30, v/v)		0.179			25	316
	(90/10, v/v)		0.18			25	316
	Tetrafluoropropanol + 1.27×10^{-3} g/ml LiCl (extrapolated to 0% H_2O		0.183			25	316
	2.54×10^{-3} g/ml LiCl (extrapolated to 0% H_2O)		0.183			25	316
	4.24×10^{-3} g/ml LiCl (extrapolated to 0% H_2O)		0.183			25	316
	6.36×10^{-3} g/ml LiCl (extrapolated to 0% H_2O)		0.182			25	316
	Trifluoroacetic acid		0.172			RT	808
	TFP	0.187				30	916–918
		0.038				30	916–918
	TFP/*o*-chlorophenol						
	(10/90, v/v)	0.027 (μ)				30	916–918
	(27/73, v/v)	0.061				30	916–918
	(27/73, v/v)	0.100 (μ)				30	916–918
	(50/50, v/v)	0.092				30	916–918

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TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(50/50, v/v)	0.220 (μ)				30	916–918
	(70/30, v/v)	0.143				30	916–918
	(70/30, v/v)	0.290 (μ)				30	916–918
	(90/10, v/v)	0.174				30	916–918
		0.254 (μ)				30	916, 918
	TFP/water (90/10, v/v)		0.180			25	761
	+ 0.1 M LiCl		0.182			25	791
			0.182 (μ)			25	791
	Trifluoroethanol	0.236				25	870
Trifluoroacetylated	Acetone	0.075				25	669
Poly(imino-(1-oxooctamethylene))							
	TFP/5 vol% water/ 0.05 M LiCl		0.173 (μ)			25	878
Poly[imino-(1-oxotetramethylene)]							
	m-Cresol		0.015			25	698
	m-Cresol/*n*-Heptane						
	(90/10, v/v)		0.028			25	698
			0.042 (μ)			25	698
	(80/20, v/v)		0.035			25	698
			0.121 (μ)			25	698
	(70/30, v/v)		0.055			25	698
			0.234 (μ)			25	698
	(60/40, v/v)		0.079			25	698
			0.415 (μ)			25	698
	(50/50, v/v)		0.089			25	698
			0.710 (μ)			25	698
Poly(imino-1,3-phenylene-imino-isophthaloyl)							
	DMA		0.245			RT	862
	DMA + 3% LiCl		0.219			RT	862
	DMA + 5% LiCl		0.217			RT	862
	DMA + 6.5% LiCl		0.214			RT	862
	DMA + 8% LiCl		0.200			RT	862
Poly(imino-1,4-phenylene-imino-terephthaloyl)							
	Chlorosulfonic acid			0.275		25	841
	Chlorosulfonic acid + 0.01 M $LiClSO_3$			0.287		25	850
	H_2SO_4			0.278		25	841
	96% H_2SO_4		0.309				656
	Methane sulfonic acid			0.254		25	850
Poly(imino-trimethylhexamethylene-imino-terephthaloyl) (Trogamid T Polyamide)							
Trifluoroacetylated	Acetone	0.117				25	669
Trifluoroacetylated	Acetonitrile	0.105				25	669
	Chloroform/ethanol (53/47, v/v)	0.191				25	669
	DMF	0.153				20	784
		0.144				25	669

3.3.2. POLY(ISOCYANATES)

Poly(*N*-butyliminocarbonyl) [Poly(butyl isocyanate)]							
	Chloroform		0.054				356
	THF		0.084			40	706
Poly(*N*-hexyliminocarbonyl)	*n*-Butyl chloride		0.092			25	745
	Chloroform	0.054					358
	n-Hexane	0.134	0.129			25	717
	THF	0.0978					358
		0.100	0.097				360
		0.099					708
$M = 38 \times 10^3$	THF				0.088	25	854
$M = 110 \times 10^3$	THF				0.093	25	854
Poly(*N*-octyliminocarbonyl)	THF		0.098				708

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Poly(*N*-tolyliminocarbonyl) [Poly(tolyl isocyanate)]							
	Bromoform				0.017		359
3.3.3. POLY(URETHANES)							
Poly(oxycarbonylimino-1,4-phenylenemethylene-1,4-phenylene-iminocarbonyloxyhexamethylene)							
	DMF				0.159		357
	DMF/acetone (90/10, v/v)				0.166		357
					0.163 (μ)		357
	(80/20, v/v)				0.173		357
					0.178 (μ)		357
	(70/30, v/v)				0.181		357
					0.188 (μ)		357
	(60/40, v/v)				0.189		357
					0.200 (μ)		357
	(50/50, v/v)				0.195		357
					0.205 (μ)		357
	(40/60, v/v)				0.203		357
	DMF/toluene (90/10, v/v)				0.154		357
					0.159 (μ)		357
	(80/20, v/v)				0.147		357
					0.160 (μ)		357
	(70/30, v/v)				0.143		357
					0.149 (μ)		357
	(60/40, v/v)				0.139		357
					0.135 (μ)		357
	(50/50, v/v)				0.134		357
					0.125 (μ)		357
	(40/60, v/v)				0.130		357
					0.120 (μ)		357
	(30/70, v/v)				0.125		357
	(25/75, v/v)				0.123		357
Poly(oxycarbonylimino-2,4-tolylene-iminocarbonyl-oxy(poly(oxypropylene))							
	Methanol	0.152	0.150			22	361
		0.148	0.145			22	361
3.3.4. OTHER COMPOUNDS							
Poly(aryl ether ether ketone)							
$M_w = 123600$	NMP			0.188		35	952
$M_w = 33800$	NMP			0.185		35	952
$M_w = 23800$	NMP			0.185		35	959
$M_w = 3450$	NMP			0.180		35	952
Phenolphthalein poly(aryl ether ketone)							
	$CHCl_3$				0.231 (532)	25	960
Poly(hydrazocarbonyl-1,4-phenylene-imino-terephthaloyl)							
	DMSO		0.175			25	799
	DMA	0.564				20	868
	DMA + 5% LiCl	0.292	0.254			25	855
	DMSO		0.180			25	831
			0.166			25	916
		0.195	0.166			25	855
	96% H_2SO_4		0.305			25	855
	DMA	0.288	0.248			25	607
	DMSO	0.240	0.205			25	607
Poly(nitrilo-2-methyl-1,4-phenylene-nitrilo-methylidine-1,4-phenylene-methylidine) (Poly(azomethine))							
	96% H_2SO_4			0.62			660
	H_2SO_4		1.05	0.78		22	662
Polyethyleneimine	Water			0.210		35	969
	1 M aqueous NaCl			0.195		35	969
	Methanol			0.225		35	969

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TABLE 3. *cont'd*

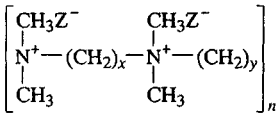
Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Poly(oxydianiline-pyromellitic dianhydride) poly(amic acid)	*N*-methylpyrrolidone			0.192			968
Ionene Polymers							
	(Polymer 3,4 Br with $x=3$, $y=4$ and Z = Br)						
	(Polymer 6,6 Br with $x=6$, $y=6$ and Z = Br)						
	(Polymer 6,6 Br with $x=6$, $y=6$ and Z = Cl)						
	3,4 Br 0.4 M aq. KBr				0.1560–0.1601		572
	0.30 M KBr				0.160		572
	0.20 M				0.160		572
	0.10 M				0.162		572
	0.05 M				0.168		572
	6,6 Br, 0.4 M KBr				0.1567–0.1609		572
	6,6 Cl, 0.4 M KBr				0.1546–0.1599		572
	6,6 Cl, 0.4 M KCl				0.1636–0.1668		572
3.4. POLY(AMINO ACIDS)							
Poly(γ-benzyl-L-glutamate)	Dichloroacetic acid (DCA)		0.085			25	619,629
	DCA/dioxane (20/80, v/v)		0.103			25	619,629
			0.102 (μ)			25	619,629
	(40/60, v/v)		0.097			25	619,629
			0.098 (μ)			25	619,629
	(50/50, v/v)		0.096			25	619,629
			0.100 (μ)			25	619,629
	(60/40, v/v)		0.093			25	619,629
			0.103 (μ)			25	619,629
	(66/34, v/v)		0.081			25	619,629
			0.108 (μ)			25	619,629
	(70/30, v/v)		0.077			25	619,629
			0.115 (μ)			25	619,629
	(80/20, v/v)		0.083			25	619,629
			0.124 (μ)			25	619,629
	(83/14, v/v)		0.084			25	619,629
			0.124 (μ)			25	619,629
	Dioxane		0.114			25	619,629
Bovine Serum Albumin	Native: Aq. buffer, pH 6.2–7.4		0.193			25	906
	Heated 100 min at 60°C; pH 6.2–7.4		0.190			25	906
	Heated 100 min at 70°C; pH 6.2–7.4		0.180			25	906
	Heated 100 min at 90°C; pH 6.2–7.4		0.147			25	906
Poly(γ,*N*-carbobenzoxy-L-α,γ-diaminobutyric acid)							
	DMSO	0.087	0.083			25	600
DNA-Sodium salt							
	Aq. 0.2 M NaCl, $I=0.5$	0.170 (μ)				25	710
	$I=0.2$	0.166 (μ)				25	710
	$I=0.02$	0.172 (μ)				25	710
	$I=0.01$	0.176 (μ)				25	710
	$I=0.005$	0.180 (μ)				25	710
Poly(L-glutamic acid)							
	Aq. 0.5 M NaCl		0.170 (μ)			25	739
	Aq. 0.2 M NaCl		0.183 (μ)			25	739
	Aq. 0.1 M NaCl		0.190 (μ)			25	739
	Aq. 0.05 M NaCl		0.194 (μ)			25	739
	Aq. 0.02 M NaCl		0.195 (μ)			25	739
–, Sodium salt	1-Propanol/water						
	(10/90, v/v)				0.191	25	910
					0.174 (μ)	25	910
	(20/80, v/v)				0.181	25	910
					0.143 (μ)	25	910
	(30/70, v/v)				0.171	25	910
					0.133 (μ)	25	910
	(40/60, v/v)				0.157	25	910
					0.111 (μ)	25	910
	(60/40, v/v)				0.141	25	910
					0.089 (μ)	25	910

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(70/30, v/v)				0.136	25	910
					0.103 (μ)	25	910
	(72/25, v/v)				0.131	25	910
					0.107 (μ)	25	910
	(85/15, v/v)				0.128	25	910
					0.119 (μ)	25	910
	2-Propanol/water						
	(10/90, v/v)				0.185	25	910
					0.167 (μ)	25	910
	(20/80, v/v)				0.174	25	910
					0.140 (μ)	25	910
	(30/70, v/v)				0.166	25	910
					0.130 (μ)	25	910
	(40/60, v/v)				0.151	25	910
					0.109 (μ)	25	910
	(50/50, v/v)				0.143	25	910
					0.103 (μ)	25	910
	(60/40, v/v)				0.135	25	910
					0.097 (μ)	25	910
	(70/30, v/v)				0.129	25	910
					0.117 (μ)	25	910
	(72/25, v/v)				0.127	25	910
					0.119 (μ)	25	910
	tert-Butanol/water						
	(10/90, v/v)				0.191	25	910
					0.171 (μ)	25	910
	(20/80, v/v)				0.176	25	910
					0.142 (μ)	25	910
	(30/70, v/v)				0.168	25	910
					0.127 (μ)	25	910
	(50/50, v/v)				0.138	25	910
					0.084 (μ)	25	910
	(60/40, v/v)				0.132	25	910
					0.082 (μ)	25	910
	(70/30, v/v)				0.128	25	910
					0.090 (μ)	25	910
	(75/25, v/v)				0.124	25	910
					0.100 (μ)	25	910
	(80/20, v/v)				0.123	25	910
					0.103 (μ)	25	910
	Glycol/water						
	(10/90, v/v)				0.172	25	910
					0.182 (μ)	25	910
	(20/80, v/v)				0.161	25	910
					0.176 (μ)	25	910
	(30/70, v/v)				0.152	25	910
					0.166 (μ)	25	910
	(40/60, v/v)				0.142	25	910
					0.150 (μ)	25	910
	(50/50, v/v)				0.137	25	910
					1.138 (μ)	25	910
	(60/40, v/v)				0.132	25	910
					0.128 (μ)	25	910
	(70/30, v/v)				0.130	25	910
					0.120 (μ)	25	910
	(80/20, v/v)				0.126	25	910
					0.112 (μ)	25	910
	(90/10, v/v)				0.121	25	910
					0.100 (μ)	25	910
	DMSO/water						
	(10/90, v/v)				0.180	25	910
					0.157 (μ)	25	910

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TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	(20/80, v/v)				0.170	25	910
					0.131 (μ)	25	910
	(30/70, v/v)				0.161	25	910
					0.120 (μ)	25	910
	(40/60, v/v)				0.151	25	910
					0.092 (μ)	25	910
	(50/50, v/v)				0.141	25	910
					0.073 (μ)	25	910
	(60/40, v/v)				0.130	25	910
					0.045 (μ)	25	910
Glycogen	Aq. 1% cetyl trimethyl ammonium bromide	0.154	0.150				650
	Aq. 8 M GHCl	0.131	0.127				650
	Aq. 15% $MgCl_2$	0.102	0.097				650
	Aq. 0.1 M NaCl	0.158	0.156				650
	Aq. 8 M Urea	0.102	0.095				650
	Water	0.153	0.151				650
Poly(N^5-(2-hydroxyethyl)L-glutamine)							
	Water	0.186				25	590
	Water/2-propanol						
	(75/25, v/v)	0.161				25	590
	(60/40, v/v)	0.158				25	590
	(40/60, v/v)	0.151				25	590
Poly(4-nitrol-L-phenylalanine)	*N,N*-Dimethylacetamide			0.1398			961
Poly(L-proline)	Form I: propionic acid	0.158				24	857
	Form II: propionic acid	0.167				24	857
	Collapsed form: propionic acid + 3 M LiBr	0.122				24	857
Phycocyanin	Aq. pH 6.8, $I = 0.1$		0.180			25	908
Poly(sulfopropyl betaines)	Aq. 0.1 M NaCl			0.197		25	679
Tobacco mosaic virus	Water		0.184				587
α-Tropomyosin	Aq. NaCl	0.187				60	734
3.5. POLY(SILANES) AND POLY(SILAZANES)							
Ludox	Diluted with water	0.0620	0.0608		0.0606 (578)	23 ± 2	362
	Diluted with NaCl (0.05 M)	0.0592	0.0587		0.0585 (578)	23 ± 2	362
Poly(di-*n*-hexylsilane)	Hexane			0.177		25	947
	THF			0.138		25	947
	Cyclohexane			0.123		25	947
	Toluene			0.044		25	947
Poly(dimethyl siloxane)	Bromocyclohexane		−0.825			28	978
	Toluene		−0.0938			25	978
Poly(dimethylsilmethylene)	*n*-Heptane	0.098				25	37
	n-Heptane/*n*-propanol						
	(75/25, v/v)	0.115				25	37
	(71/29, v/v)	0.092				25	37
	(69.5/30.5, v/v)	0.103				25	37
Poly(dimethylsiltrimethylene)	*n*-Heptane	0.123				25	37
Poly(diphenylsiltrimethylene)	Toluene	0.112				25	37
	Toluene/cyclohexanol						
	(79.5/20.5, v/v)	0.133				25	37
	(75/25, v/v)	0.135				25	37
	(70/30, v/v)	0.145				25	37
	(40/60, v/v)	0.172				25	37
	(37.6/62.4, v/v)	0.175				25	37
Poly(oxydimethylsilylene)	Benzene		0.098			25	201
			−0.096			25	779
$M = 1100$–8700			−0.108 to −0.099			25	777
	Benzene/triethyl borate (94.44/5.56, v/v)		0.115			25	138

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
$M_w = 1.06 \times 10^6$	Bromocyclohexane		0.078			25	363
0.078×10^6			0.079			25	363
0.545×10^6			−0.083			29	364
0.34×10^6			0.081			25	363
0.085×10^6			0.090			25	363
0.033×10^6			0.090			25	363
1892			−0.086			29	364
978			−0.092			29	364
682			−0.098			29	364
150			−0.111			29	364
$M = 930–32600$		−0.105 to 0.082				28	667
	Ethyl acetate		0.036			25	779
	MEK			0.029		20	673
	MEK/*n*-undecane						
	(80/20, v/v)			0.023		20	673
	(70/30, v/v)			0.020		20	673
	(60/40, v/v)			0.016		20	673
	(50/50, v/v)			0.011		20	673
	(40/60, v/v)			0.009		20	673
	(10/90, v/v)			−0.010		20	673
	(5/95, v/v)			−0.013		20	673
$M = 481–1132$	*n*-Octadecane	−0.050 to −0.030			70–100	748	
	Toluene	0.103	0.094				365
		−0.1042	−0.0947			14.6	366
		−0.1037	−0.0941			19.9	366
		−0.1029	−0.0933			25.4	366
		−0.1027	−0.0933			29.9	366
		−0.1023	−0.0927			34.8	366
$M_w = 0.545 \times 10^6$			−0.084			25	364
0.21×10^6			0.092			25	367
1892			−0.085			25	364
978			−0.094			25	364
682			−0.099			25	364
150			−0.134			25	364
			−0.093	−0.090		25	641
$M = 1430$		−0.105 to −0.107				16.5–35	801
$M = 850–301000$		−0.119 to 0.084				25	667,672, 748,801
$M = 481–677$		−0.130 to −0.122				25–70	748
			−0.094			25	752
Cyclic polymer, $M = 1980–21800$		−0.087 to −0.084				25	667
	n-Undecane			−0.016		20	673
Poly(oxymethylphenylsilylene)							
	Acetone		0.179			25	690
	Benzene		0.053			25	690
	n-Butyl chloride		0.142			25	690
	Carbon tetrachloride		0.089			25	690
	Chlorobenzene		0.030			25	690
	Chloroform		0.102			25	690
	Cyclohexane		0.111			25	690
	MEK		0.163			25	690
Poly(oxymethyl-3,3,3-trifluoropropylsilylene)							
	Cyclohexyl acetate	0.052				35	368
Poly(oxyphenylsilylene)	Benzene				0.08		369
Poly(phenylsilsesquioxane)							
cis-syndio	Chloroform	0.119	0.111			20	847
cis-syndio	THF			0.133		20	847
Poly(silicic acid)	Water	0.0592					362
					0.65		370
Poly(trimethylsiloxanotitanoxanes)							
	Cyclohexane		0.156			25	371

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
3.6. POLY(HETEROCYCLICS) AND OTHERS							
3.6.1. POLY(HETROCYCYCLICS)							
Poly(benzimidazolediyl-benzimidazolediyl-1,4-phenyleneoxy-1,4-phenylene)							
	DMA		0.326			30	828
Poly(benzo[1,2-d: 5,4-d′] Bisoxazol-2,6-diyl-1,4-phenylene)							
	Chlorosulfonic acid			0.502		25	850
	+ 0.01 M $LiClSO_3$						
	Methane sulfonic acid			0.0474		25	850
Poly(benzo[1,2-d: 5,4-d′]-bisthiazol-2,6-diyl-1,4-phenylene)							
	Chlorosulfonic acid			0.540		25	841
	Methane sulfonic acid			0.564		25	841
Poly(benzoxazolediyl-iminoterephthaloylimino-1,3-(4-hydroxyphenylene))							
	96% H_2SO_4		0.30				895
Poly(3,9-bis(ethylidene-2,4,8,10-tetraoxaspiro[5,5]undecane-per-cyclohexanedimethanol) (TCHD)							
Poly(3,9-bis(ethylidene-2,4,8,10-tetraoxaspiro[5,5]undecane-per-hexanediol) (HD)							
TCHD/HD = 100/0, mol/mol	THF			0.085			839
= 70/30, mol/mol				0.083			839
= 60/40, mol/mol				0.083			839
= 50/50, mol/mol				0.082			839
= 29/71, mol/mol				0.081			839
= 0/100, mol/mol				0.079			839
Poly(*N*,*N*-dimethylpiperidine-3,5-diyl-methylene), hydrochloride							
	Methanol + 0.0093 M TEAB	0.211				21	780
		0.258 (μ)				21	780
	+ 0.0186 M TEAB	0.215				21	780
		0.262 (μ)				21	780
	+ 0.030 M TEAB	0.220				21	780
		0.270 (μ)				21	780
	+ 0.0352 TEAB	0.230				21	780
		0.300 (μ)				21	780
	+ 0.046 M TEAB	0.210				21	780
		0.305 (μ)				21	780
	Aq. 0.1 M NaCl	0.183				21	772
		0.182 (μ)				21	772
	Aq. 0.2 M NaCl	0.200				21	772
		0.186 (μ)				21	772
	Aq. 0.3 M NaCl	0.173				21	772
		0.170 (μ)				21	772
	Aq. 0.4 M NaCl	0.171				21	772
		0.170 (μ)				21	772
	Aq. 0.5 M NaCl	0.181				21	772
		0.167 (μ)				21	772
Poly(2,5-dimethyl-1,4-piperazinediyl-isophthaloyl)							
	N-Methyl-2-pyrrolidone	0.131	0.123			25	603
	Trifluoroethanol	0.279	0.267			25	603
Poly(isophthaloyl-*trans*-2,5-dimethylpiperazine)							
	NMP	0.131	0.123			25	951
Poly(2,5-dimethyl-1,4-piperazinediyl-phthaloyl)							
	Acetic acid (glacial)	0.212	0.200			25	591
	Chloroform	0.160	0.150			25	591
	m-Cresol	0.066	0.064			25	591
	DMA	0.152	0.146			25	591
	Hexafluoro-propanol-2	0.290	0.278			25	591
	N-Methyl-2-pyrrolidone	0.130	0.121			25	591
Poly(2,5-dimethyl-1,4-piperazinediyl-terephthaloyl)							
	Trifluoroethanol	0.278	0.266			25	592
Poly(iminobenzimidazole-diyl-1,4-phenylene-iminoterephthaloyl)							
	DMF		0.29			25	889
	Sulfuric acid		0.29			25	889
Poly(pyrazole-3,5-diyl-methylenephenylene)							
	DMF		0.223			25	687

TABLE 3. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Poly[(1,3,5,7-tetraoxo-2,3,6,7-tetrahydro-1H,5H-benzo[1,2,-c: 4,5-c']dipyrrol-2,6-diyl)-1,4-phenyleneisopropylidene-1,4-phenylene] (Polyimide from pyromellitic anhydride and 4,4'-diaminodiphenylpropane)							
	LiBr 0.1 N in DMA		0.196				574
3.6.2. OTHER COMPOUNDS							
Poly(2,6-dichloroquinoline)	Formic acid				0.57		951
Poly [*trans*-bis(tributylphosphine)-platinum-1,4-butadiynediyl]							
	n-Heptane		0.199			25	597
Poly[bis(*m*-chlorophenoxy)phosphazene]							
	Chloroform		0.114			25	861
Poly(fluoroalkoxyphosphazenes)	Frezon E2/acetone (10/1, v/v)	0.048				30	579
Sodium polyphosphate	Aq. NaBr 0.1 N				0.109		581
	0.25 N				0.105		581
	0.35 N				0.102		581
	0.40 N				0.101		581

TABLE 4. CELLULOSE AND DERIVATIVES

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Acetoxypropyl cellulose							
	Acetone		0.100			25	730
	THF		0.056			25	730
Carboxymethyl cellulose	NaCl 0.1 M		0.147			25	373
Carboxymethyl cellulose, sodium salt							
DS = 0.96	Cadoxen 0.237 M Cd.	0.1398	0.1896			25	377
		0.1373 (μ)	0.1861 (μ)			25	377
0.94		0.147	0.145			25	404
0.44		0.164	0.162			25	404
0.21		0.175	0.171			25	404
	Aq. NaCl	0.136					128
		0.154					405
		0.158					405
Cellulose	Acetone	0.111				25	374
	Cadoxen	0.186	0.183			25	375,376
	Cadoxen (5% Cd)/water (1/1, v/v)	0.190 (μ)	0.189 (μ)			25	912
	0.237 M Cd	0.1317	0.1927			25	377
		0.1925 (μ)	0.1890			25	377
	Cuoxam 0.205 M Cu	0.117	0.233			25	378
	Cuoxen 0.0518 M Cu	0.1352	0.2574			25	377
	0.0776 M Cu		0.2653			25	377
	FeTNa	0.110 (μ)	0.245 (μ)			25	379
	DMA + 5%w/w LiCl		0.08			25	735
	DMA + 9%w/w LiCl (incr. with M_w)			0.040–0.027		30	741
Cellulose acetate							
DS = 0.49	DMA	0.068					602
DS = 0.49	Formamide	0.069					602
DS = 0.49	Water	0.131					602
DS = 1.75	DMA	0.046				25	605
DS = 2.45	THF	0.071				25	817
DS = 2.45	Trifluoroethanol	0.157				25	817
Cellulose carbanilate							
DS 1.6–2.3	Dioxane				0.141		757
DS 1.6–2.3	Ethyl acetate				0.180		757

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TABLE 4. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Cellulose diacetate	Acetone	0.112				25	786
	Acetone		0.109			25	913
	DMA	0.042				25	594
	MEK	0.104				50	794
Cellulose nitrate							
10.98% N	Acetone	0.1022	0.0998			25	380
11.89% N		0.1010	0.0985			25	380
12.55% N		0.0968	0.0950			25	380
12.9% N		0.1151				25	382
11.3% N	Ethyl acetate		0.103			20	381
13.49% N		0.102				25	383
12–12.3% N	Acetone	0.107				20	756
	DMA + 6% LiCl				0.055	25	892
	Ethyl acetate				0.092	25	892
Cellulose triacetate	Chloroform				0.0406	25	384
					0.0496	55	384
	DMA	0.040				25	594
Cellulose nitrate	Methylene chloride/ methanol (50/50, v/v)	0.130 (μ)					773
	Trifluoroacetic acid				0.185 (589)		668
Cellulose tricaproate	1-Chloronaphthalene		0.147			24	385
Cellulose tributyrate	Bromoform		−0.11			21	886
	MEK		0.078			21	886
	Dioxane/water (100/7, v/v)		0.104			63	385
	DMF	0.0442	0.478			41	385
Cellulose tricarbanilate	Acetone	0.2069	0.1966			7	385
		0.2155	0.2047			25	386
		0.2176	0.2033			27	385
		0.215				30	216
			0.205			25	844
	Benzophenone				0.0148 (589)	55	387
	Cyclohexanone	0.1556	0.1666			25	386
		0.137				30	216,919
		0.137				30	216
	Dibutyl ketone		0.210			25	388
	Diethyl ketone	0.1911	0.1797			25	386
	Dioxane	0.1651	0.1558			25	386
					0.156 (488)	20	681
		0.168				30	764
		0.168				30	216,919
	Dioxane/1-bromonaphthalene						
	(97/3, v/v)	0.162 (μ)				30	216,919
	(96/4, v/v)	0.140 (μ)				30	216,919
	(90/10, v/v)	0.000 (μ)				30	216,919
	(83/17, v/v)	−0.150 (μ)				30	216,919
		0.168	0.156		0.181 361	20	389
	Furfural	0.070				30	216
		0.070				30	216,919
	MEK	0.188				30	216
		0.188				30	216,919
	Pyridine	0.089				30	216
		0.089				30	216,919
	THF	0.182				30	216
			0.169			25	844
		0.182				30	216,919
	Nitrobenzene	0.050				30	216,919
Cellulose trinitrate (fully substituted)	Acetone	0.105					253
		0.102	0.101			RT	390
		0.1116				25	382
		0.0930	0.0903			25	380
					0.098 (1086)	25	915

TABLE 4. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Ethyl acetate		0.102			30	383
		0.107				25	391
			0.103			20	381
		0.105	0.105				852
Cellulose xanthate	Aq. NaOH		0.212				392
			0.230				393
			0.190				394
					0.20		395
	1.0 M		0.231				378
	2.5 M		0.185				396
Cellulose xanthate, sodium salt							
DS = 0.25	Aq. NaOH 1.0 N	0.205					406
0.48		0.275					406
0.77		0.300					406
0.91		0.425					406
Diethylacetamide cellulose xanthate							
DS = 0.40	Aq. DMSO 90%				0.144		397
0.49					0.098		397
0.60					0.144		397
0.80					0.144		397
0.92					0.098		397
1.00					0.096		397
1.22					0.079		397
Ethyl cellulose	Methanol	0.130				25	398
Ethyl hydroxyethyl cellulose	Methanol	0.147					399
Hemicellulose	KOH 0.36 M		0.131				400
	NaCl 0.25 M/0.5% Calgon	0.127					400
	0.5 M		0.145				400
	0.5 M/0.5% Versene	0.136					400
Hydroxypropyl cellulose	Anhydrous ethanol		0.120			25	401
	Water	0.146	0.143		0.143 (578)	25	684
Lignin	Aq. $NaHCO_3$ pH 9.65		0.218			25	372
Ligninsulfonic acid, sodium salt	Aq. NaCl 0.1 M				0.197		407
					0.200		407
	DMSO				0.117		407
Methyl cellulose	Water	0.154					402
$M_w = 160000$	Water			0.134		25	948
$M_w = 210000$	Water			0.135		25	948
$M_w = 234000$	Water			0.132		25	948
Sodium carboxymethyl cellulose	Water	0.153	0.150		0.150 (578)	25	684
Sodium cellulose disulfate	Aq. 0.5 M NaCl	0.081 (μ)				25	788

TABLE 5. POLY(SACCHARIDES)

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
Amylopectin	DMSO		0.066				408,409
	DMSO/water (90/10, v/v)		0.074				408,409
	(80/20, v/v)		0.083				408,409
	(60/40, v/v)		0.101				408,409
	(50/50, v/v)		0.110				408,409
	Ethylenediamine	0.098	0.098				410
	Ethylenediamine hydrate	0.092	0.092				410
	Formamide	0.069	0.069				410
	KOH 1.0 N	0.142	0.142				410
	NaOH 0.2 N	0.142	0.142				412

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TABLE 5. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Water	0.156	0.154				412
		0.155	0.0151				413
			0.152				408,409,411
	Aq. 1% cetyl trimethyl ammonium bromide	0.146	0.143				650
	Aq. 0.33 M KCl	0.153					650
	Aq. 0.1 M KCl	0.156					650
	Aq. 15% $MgCl_2$	0.112	0.111				650
	Aq. 1% Na lauryl sulfate	0.158	0.153				650
	Aq. 8 M Urea	0.106	0.100				650
	Water	0.160	0.157				650
Amylopectin acetate	Acetone	0.118					412
	Acetonitrile	0.128					412
	Chloroform	0.051					412
	Dioxane	0.057					412
	Nitromethane	0.088					412
Amylose	DMSO/water (30/70, v/v)	0.127 (μ)				25	716
	(35/65, v/v)	0.128 (μ)				25	716
	(45/55, v/v)	0.119 (μ)				15	716
		0.108 (μ)				25	716
		0.118 (μ)				35	716
		0.147 (μ)				45	716
	(50/50, v/v)	0.112 (μ)				25	716
	(60/40, v/v)	0.093 (μ)				25	716
	(70/30, v/v)	0.080 (μ)				25	716
	(80/20, v/v)	0.076 (μ)				25	716
	(90/10, v/v)	0.062 (μ)				25	716
	DMSO	0.0676	0.0659				414
	GHCl 4.2 M	0.118	0.116			25	415
	Aq. KCl				0.146		416
	KOH 1.0 M				0.146		416
	Aq. $NaIO_3$	0.153	0.152				417
Amylose acetate	Nitromethane	0.0875				30	418
		0.0857				RT	419
			0.0835			20	420
			0.0835			20	421
		0.088	0.083				650
DS = 2.9		0.087				30	760
Amylose tributyrate	Ethyl acetate	0.098				25	754
	MEK	0.091				25	754
Amylose tricarbonilate	Acetone	0.2279	0.2164			27	422
		0.2218	0.2094			7	422
	Dioxane	0.163	0.151		0.178 (361)	20	389
	Pyridine		0.0973			25	423
Amylose tripropionate	Ethyl acetate	0.092				25	820
	MEK	0.086				25	820
Araban	Aq. DMF 1%				0.142		424
	NaCl 2 M				0.130		424
Araban acetate	DMF		0.126				425
	MEK		0.0886				425
Chitosan	Water				0.166		426
Curdlan (bacterial β-1,3 glucan)							
	Cadoxen/water (1/1, v/v)	0.107				25	596
		0.163 (μ)				25	596
Dextran	Water				0.148		649
	$Ba(OH)_2$ 0.0503 M	0.1465	0.1441			25	377
		0.1642 (μ)	0.1618 (μ)			25	377
	0.108 M	0.1415	0.1394			25	377
		0.1738 (μ)	0.1708 (μ)			25	377
	Cadoxen 0.237 M Cd	0.1235	0.1219			25	377
		0.1842 (μ)	0.1809 (μ)			25	377
		0.1907 (μ)	0.1875 (μ)			25	377

TABLE 5. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Cuoxen 0.0518 M Cu	0.1323				25	377
		0.2525 (μ)				25	377
	0.0776 M Cu	0.2625 (μ)				25	377
	LiSCN 2.5 M		0.129				409
	NaCl 0.1 M			0.151			427
	0.5 M	0.1476	0.1454			25	377
		0.1470 (μ)	0.1445 (μ)			25	377
	1.0 M	0.1447	0.1424			25	377
		0.1435 (μ)	0.1413 (μ)			25	377
	2.0 M	0.1361	0.1342			25	377
		0.1374 (μ)	0.1351 (μ)			25	377
	NaOH 0.25 M	0.1456	0.1433			25	377
		0.1584 (μ)	0.1558 (μ)			25	377
	0.75 M	0.1437	0.1415			25	377
		0.1646 (μ)	0.1621 (μ)			25	377
	$ZnCl_2$ 4 M	0.151	0.147				428
	Water	0.154	0.151				429
			0.151				430
			0.15			25	431
		0.1518	0.1481		0.1476 (578)	25	432
		0.154	0.151				428
		0.161				23	237
		0.150				20	433
			0.152				409
		0.1504	0.1481				377
		0.1494 (μ)	0.1472 (μ)				377
Dextran							
$M_w = 10100$	Water			0.152		25	948
$M_w = 43400$	Water			0.151		25	948
$M_w = 68100$	Water			0.147		25	948
$M_w = 508900$	Water			0.147		25	948
Dextran tricarbanilate	DMF	0.161	0.151				428
Dialdehyde starch	Water	0.153	0.152				434
Glucomannan triacetate	Nitroethane		0.076			25	435
D-Gluose (products from irradiated aq. glucose solutions)							
$M = 9.8 \times 10^3$	Water		0.166				436
14.4×10^4			0.200				436
50×10^5			0.215				436
25×10^5	DMSO/water (98/2, w/w)		−0.135				436
Guaran triacetate	Acetonitrile		0.1200			22.5	437
Gum arabic	Aq. HCl	0.152	0.150			25	438
	NaCl 0.35 M; NaOH 0.02 N; 0.25% Na salt of EDTA	0.149					439
	Water	0.145					439
Hyaluronic Acid	Water		0.166–0.170			25	905
Hydroxyethyl starch	Aq. NaCl 0.9%	0.136				25	440
DS = 0.8		0.160	0.161			25	441
0.06–1.2 (no change, av. value)	Aq. NaCl 0.9%	0.151				20	442
	Water	0.151				20	442
Related compounds:							
Acetyl (DS = 0.1)	Water	0.150				20	442
(DS = 0.7)		0.147				20	442
(DS = 1.0)		0.141				20	442
Hydroxy butyl starch (DS = 0.2)		0.151				20	442
Hydroxy propyl starch							
(DS = 0.5)		0.151				20	442
(DS = 1.5)		0.152				20	442
Hydrolyzed waxy maize starch		0.152				20	442
α-Keratose	Aq. Phosphate buffer; $I = 0.2$; pH 6.7		0.182				443

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TABLE 5. *cont'd*

Polymer	Solvent	dn/dc (ml/g) $\lambda_0 = 436$ nm	$\lambda_0 = 546$ nm	$\lambda_0 = 633$ nm	Others	T (°C)	Refs.
	Dichloroacetic acid		0.088				444
	Dichloroacetic acid 0.1 M KCl		0.088				444
	Formic acid		0.156				444
	0.1 M KCl		0.155				444
	0.5 M KCl		0.152				444
Lipopolysaccharide	Water	0.151					445
Mannan	Acetone	0.102	0.102			RT	446
	Ethyl acetate	0.101	0.101			RT	446
Magnesium alginate	Aq. $MgCl_2$	0.158	0.157			20–25	447
	Aq. 0.033 M $MgCl_2$	0.158 (μ)				RT	805
Mucopolysaccharides	Phosphate buffer 0.15 M				0.166		448
	+ NaCl 0.2 M				0.166		448
	Water				0.181		448
Potassium alginate	Aq.; in presence of KCl	0.160	0.159			20–25	447
Schizophyllan	DMSO		0.057			25	727
(Polysaccharide)		0.062	0.061			25	829
	DMSO/Water						
	(86.5/13.5, w/w)		0.066			25	727
	(86.5/13.5, w/w)		0.071 (μ)			25	727
	Water	0.145	0.142			25	829
			0.142			25	727
Sodium Alginate	Aq. 0.1 M NaCl	0.150 (μ)				RT	805
	Aq. NaCl (+ Na EDTA)				0.15	RT	724
	Aq.; in presence of NaCl	0.165	0.163			20–25	447
Sodium amylose xanthate							
(DS = 0.4–0.5)	Aq. NaCl 0.1 M		−0.1460			25	449
	0.25 M		−0.1480			25	449
	0.50 M		−0.1555			25	449
	1.0 M		−0.1741			25	449
(Ripening time = 0 hr) (in air)	Aq. NaOH 4%		0.1700				450
22			0.1547				450
46			0.1471				450
94			0.1522				450
142			0.1938				450
190			0.2564				450
0 (under N_2)			0.1700				450
45			0.1785				450
93			0.1899				450
140			0.2181				450
188			0.2963				450
Sodium carboxymethyl amylose	Aq. NaCl 0.0065 M	0.138				30	450
	0.01 M	0.138				30	450
	0.03 M	0.138				30	450
	0.035 M	0.134				30	450
	1.0 M	0.124				30	450
	2.5 M	0.103				35	450
	0.35 M	0.1325				30	451
Xanthan	Aq. 0.1 M NaCl	0.144 (μ)				27	723
	Na salt: Aq. 0.1 M NaCl	0.144 (μ)	0.141 (μ)			25	606
	Na salt: cadoxen	0.167 (μ)	0.164 (μ)			25	606
Xylan	Cadoxen				0.178	25	452
	DMSO				0.064		453
					0.062		454
	Ethyl acetate	0.097	0.97				455

E. REFERENCES

1. J. W. Lorimer, Polymer, **13**, 274 (1972).
2. M. B. Huglin, in: M. B. Huglin, (Ed.), "Light Scattering from Polymer Solutions" Academic Press, London/New York, 1972.
3. W. Heller, J. Polym. Sci. A-2, **4**, 209 (1966).
4. M. B. Huglin, J. Appl. Polym. Sci., **9**, 4003 (1965).
5. W. Mächtle, H. Fischer, Angew. Makromol. Chem., **7**, 147 (1969).
6. W. Mächtle, Angew. Makromol. Chem., **15**, 17 (1971).
7. Y. Kambe, C. Honda, Angew. Makromol. Chem., **25**, 163 (1972).
8. Tuzar, P. Kratochvíl, D. Strakova, Eur. Polym. J., **6**, 1113 (1970).
9. P. Kratochvíl, J. Babka, J. Appl. Polym. Sci., **16**, 1053 (1972).
10. I. Baltog, C. Ghita, L. Ghita, Eur. Polym. J., **6**, 1299 (1970).
11. P. Kratochvíl, B. Sedláček, D. Straková, Z. Tuzar, Makromol. Chem., **148**, 271 (1971).
12. M. B. Huglin, J. Appl. Polym. Sci., **9**, 3963 (1965).
13. W. R. Krigbaum, J. E. Kurz, P. Smith, J. Phys. Chem., **65**, 1984 (1961).
14. Th. G. Scholte, J. Polym. Sci. A-2, **6**, 91 (1968).
15. R. Chiang, J. Polym. Sci. C, **8**, 295 (1965).
16. F. W. Billmeyer, Jr., J. Am. Chem. Soc., **75**, 6118 (1953).
17. R. Chiang, J. Polym. Sci., **36**, 91 (1959).
18. L. H. Tung, J. Polym. Sci., **36**, 287 (1959).
19. H. P. Schreiber, M. H. Waldman, J. Polym. Sci., A-2, 1655 (1964).
20. Q. A. Trementozzi, J. Polym. Sci., **23**, 887 (1957).
21. Q. A. Trementozzi, J. Polym. Sci., **36**, 113 (1959).
22. H. L. Wagner, C. A. J. Hoeve, J. Polym. Sci. A-2, **9**, 1763 (1971).
23. J. Ehl, C. Loucheux, C. Reiss, H. Benoit, Makromol. Chem., **75**, 35 (1964).
24. V. Kokle, F. W. Billmeyer, Jr., L. T. Muus, E. J. Newitt, J. Polym. Sci., **62**, 251 (1962).
25. E. E. Drott, R. A. Mendelson, J. Polym. Sci. B-2, 187 (1964).
26. A. Kotera, N. Iso, A. Senuma, T. Hamada, J. Polym. Sci. A-2, **5**, 277 (1967).
27. A. Kotera, T. Saito, H. Matsuda, A. Wada, Rept. Progr. Polym. Phys. Japan, **8**, 5 (1965).
28. R. Chiang, J. Polym. Sci., **28**, 235 (1958).
29. J. E. Kurz, J. Polym. Sci. A, **3**, 1895 (1965).
30. R. A. Mendelson, J. Polym. Sci., **46**, 493 (1960).
31. D. A. Albright, J. W. Williams, J. Phys. Chem., **71**, 2780 (1967).
32. A. A. Tager, A. A. Anikeyeva, V. M. Andreyeva, T. Ya. Gumarova, L. A. Chernoskutova, Polym. Sci. USSR, **10**, 1962 (1968).
33. F. R. Cottrell, E. W. Merrill, K. A. Smith, J. Polym. Sci. A-2, **7**, 1415 (1969).
34. D. Coupe, P. A. Curnuck, J. V. Dawkins, ICI Limited, Petrochemicals + Polymer Lab., Runcorn, England–private communication (1969).
35. G. Delmas, D. Patterson, Polymer, **7**, 513 (1966).
36. A. Červenka, M. Marek, K. Šolc, P. Kratochvíl, Coll. Czech, Chem. Commun., **33**, 4248 (1968).
37. A. Yu. Koshevnik, M. M. Kusakov, E. A. Razumovskaya, Polym. Sci. USSR, **10**, 3242 (1968).
38. T. Matsumoto, N. Nishioka, H. Fujita, J. Polym. Sci. A-2, **10**, 23 (1972).
39. A. Sakanishi, J. Chem. Phys. **48**, 3850 (1968).
40. K. E. van Holde, J. W. Williams, J. Polym. Sci., **11**, 243 (1953).
41. J. B. Kinsinger, L. E. Ballard, J. Polym. Sci. A, **3**, 3963 (1965).
42. H.-G. Elias, H. Dietschy, Makromol. Chem., **105**, 102 (1967).
43. S. Florián, D. Lath, Z. Maňásek, Angew. Makromol. Chem., **13**, 43 (1970).
44. J. B. Kinsinger, R. E. Hughes, J. Phys. Chem., **63**, 2002 (1959).
45. K. Yamaguchi, Makromol. Chem., **128**, 19 (1969).
46. W. E. Weston, F. W. Billmeyer, Jr., J. Phys. Chem., **65**, 576 (1961).
47. L. Westerman, J. Polym. Sci. A, **1**, 411 (1963).
48. P. P. Parrini, F. Sebastiano, G. Messina, Makromol. Chem., **38**, 27 (1960).
49. D. K. Carpenter, J. Polym. Sci. A-2, **4**, 923 (1966).
50. H. Janeschitz–Kriegl, U. Daum, Kolloid Z.-Z. Polym., **210**, 112 (1966).
51. T. Miyamoto, H. Inagaki, J. Polym. Sci. A-2, **7**, 963 (1969).
52. H. Inagaki, T. Miyamoto, S. Ohta, J. Phys. Chem., **70**, 3421 (1966).
53. H. T. Hall, J. Polym. Sci., **7**, 443 (1951).
54. L. Saunders, L. Spirer, Polymer, **6**, 635 (1965).
55. J. Moacanin, A. Rembaum, R. K. Laudenslager, R. Adler, J. Macromol. Sci. A, **1**, 1497 (1967).
56. J. M. Barrales–Rienda, D. C. Pepper, Polymer, **8**, 337 (1967).
57. R. J. Angelo, R. M. Ikeda, M. L. Wallach, Polymer, **6**, 141 (1965).
58. R. Kuhn, S. B. Liang, H.-J. Cantow, Angew. Makromol. Chem., **18**, 101 (1971).
59. W. Cooper, D. E. Eaves, G. Vaughan, J. Polym. Sci., **59**, 241 (1962).
60. W. Cooper, G. Vaughan, D. E. Eaves, R. W. Madden, J. Polym. Sci., **50**, 159 (1961).
61. Ph. Ribeyrolles, A. Guyot, H. Benoit, J. Chim. Phys., **56**, 377 (1959).
62. R. Kratochvíl, D. Straková, P. Schmidt, Angew. Makromol. Chem., **23**, 169 (1972).
63. S. Abe, Y. Murakami, H. Fujita, J. Appl. Polym. Sci., **9**, 2549 (1965).
64. T. Norisaye, K. Kawahara, A. Teramoto, H. Fujita, J. Chem. Phys., **49**, 4330 (1968).
65. K. Hanafusa, A. Teramoto, H. Fujita, J. Phys. Chem., **70**, 4004 (1966).
66. J. Curchod, T. Ve, J. Appl. Polym. Sci., **9**, 3541 (1965).

67. P. V. French, A. Wassermann, J. Chem. Soc. 1044 (1963).
68. G. V. Schulz, K. Altgelt, H.-J. Cantow, Makromol. Chem., **21**, 13 (1956).
69. K. Altgelt, G. V. Schulz, Makromol. Chem., **36**, 209 (1959).
70. J. Vavra, J. Polym. Sci. C, **16**, 1103 (1967).
71. G. M. Bristow, B. Westall, Polymer, **8**, 609 (1967).
72. S. P. Rao, M. Santappa, J. Polym. Sci. A-1, **6**, 95 (1968).
73. P. J. Reed, J. R. Urwin, Austral, J. Chem., **23**, 1743 (1970).
74. V. Boháčkova, J. Polacek, H. Benoit, J. Chim. Phys., **66**, 197 (1969).
75. V. Boháčkova, J. Polacek, Grubisic, H. Benoit, J. Chim. Phys., **66**, 207 (1969).
76. S. A. Pavlova, T. A. Soboleva, A. P. Suprun, Vysokomol, Soedin., **6**, 122 (1964).
77. A. Silberberg, J. Eliassaf, A. Katchalsky, J. Polym. Sci., **23**, 259 (1957).
78. H.-J. Cantow, Z. Naturforsch., **7B**, 485 (1952).
79. T. A. Fadner, H. Morawetz, J. Polym. Sci., **45**, 475 (1960).
80. V. A. Myagchenkov, A. K. Vagapova, E. V. Kuznetsov, Vysokomol–Soedin., **11**, 673 (1969).
81. E. A. S. Cavell, I. T. Gilson, B. R. Jennings, H. G. Jerrard, J. Polym. Sci. A-2, 3615 (1964).
82. B. R. Jennings, Thesis–Univ. Southampton, 1964.
83. L. Trossarelli, M. Meirone, J. Polym. Sci., **57**, 445 (1962).
84. E. F. Jordan, Jr., J. Polym. Sci. A-1, **6**, 2209 (1968).
85. T. A. Orofino, P. J. Flory, J. Phys. Chem., **63**, 283 (1959).
86. S. Newman, W. R. Krigbaum, C. Langier, P. J. Flory, J. Polym. Sci., **14**, 451 (1954).
87. Z. Alexandrowicz, J. Polym. Sci., **40**, 91 (1959).
88. D. Braun, G. Mott, Angew. Makromol. Chem., **18**, 183 (1971).
89. W. Wunderlich, Angew. Makromol. Chem., **11**, 189 (1970).
90. R. Jerome, V. Desreux, Eur. Polym. J., **6**, 411 (1970).
91. G. B. Rathman, F. A. Bovey, J. Polym. Sci., **15**, 544 (1955).
92. M. Giurgea, C. Ghita, I. Baltog, A. Lupu, J. Polym. Sci., **4**, 529 (1966).
93. M. Giurgea, L. Pop, I. Baltog, A. Belea, Rev. Roumaine Chim., **11**, 467 (1966).
94. D. Mangaraj, S. K. Patra, Makromol. Chem., **107**, 230 (1967).
95. H. Wesslau, Makromol. Chem., **69**, 213 (1963).
96. K. Harvey, B. Sc. Thesis University of Salford, 1969.
97. H. Matsuda, K. Yamano, H. Inagaki, J. Polym. Sci. A-2, **7**, 609 (1969).
98. K. Karunakaran, M. Santappa, J. Polym. Sci. A-2, **6**, 713 (1968).
99. A. Kotera, T. Saito, Y. Watanabe, M. Ohama, Makromol. Chem., **87**, 195 (1965).
100. H. Matsuda, H. Inagaki, Bull. Inst. Chem. Res. Kyoto Univ., **46**, 48 (1968).
101. R. A. Wessling, J. E. Mark, E. Hamori, R. E. Hughes, J. Phys. Chem., **70**, 1903 (1966).
102. J. Velicković, J. Vukajlović–Filipović, Angew, Makromol. Chem., **13**, 79 (1970).
103. R. J. Valles, J. Polym. Sci. A, **3**, 3853 (1965).
104. R. J. Valles, Makromol. Chem., **100**, 167 (1967).
105. R. van Leemput, R. Stein, J. Polym. Sci. A, **1**, 985 (1963).
106. V. E. Eskin, A. L. Izyumnikov, E. D. Rogozhkina, Yu. P. Vyrskii, Polym. Sci. USSR, **7**, 1310 (1965).
107. V. E. Eskin, A. L. Izyumnikov, E. D. Rogozhkina, Yu. P. Vryskii, Polym. Sci. USSR, **9**, 591 (1967).
108. S. N. Chinai, R. A. Guzzi, J. Polym. Sci., **21**, 417 (1956).
109. R. van Leemput, R. Stein., J. Polym. Sci. A, **2**, 4039 (1964).
110. M. Kozhokaryu, V. S. Skazka, K. G. Berdnikova, Polym. Sci., USSR, **8**, 1167 (1966).
111. V. N. Tsvetkov, O. V. Kallistov, Zh. Fiz. Khim, **33**, 710 (1959).
112. V. N. Tsevtkov, D. Khardi, I. N. Shtennikova, Ye. V. Korneyeva, G. F. Pirogova, K. Nitrai, Polym. Sci. USSR, **11**, 392 (1969).
113. S. V. Bereza, S. R. Rafikov, Ye. A. Bekturov, R. Ye. Legkunets, Polym. Sci. USSR, **10**, 2945 (1968).
114. M. Tricot, V. Desreux, Makromol. Chem., **149**, 185 (1971).
115. N. Hadjichristidis, M. Devaleriola, V. Desreux, Eur. Polym. J., **8**, 1193 (1972).
116. D. W. Levi, R. J. Valles, E. C. Schramm, J. Polym. Sci., **56**, 305 (1962).
117. F. E. Didcot, S. N. Chinai, D. W. Levi, J. Polym. Sci., **43**, 557 (1960).
118. Z. Tuzar, P. Kratochvíl, Polym. Letters, **7**, 825 (1969).
119. Z. Tuzar, M. Bohdanecký, Coll. Czech. Chem. Commun., **34**, 289 (1969).
120. M. Bohdanecký, Z. Tuzar, M. Stoll, R. Chromocek, Coll. Czech. Chem. Commun., **33**, 4140 (1968).
121. Z. Tuzar, P. Kratochvíl, Coll. Czech. Chem. Commun., **32**, 3358 (1967).
122. K. Karunakaran, M. Santappa, Makromol. Chem., **111**, 20 (1968).
123. S. N. Chinai, R. J. Samuels, J. Polym. Sci., **19**, 463 (1956).
124. S. N. Chinai, J. Polym. Sci., **25**, 413 (1957).
125. H. T. Lee, D. W. Levi, J. Polym. Sci., **47**, 449 (1960).
126. S. N. Chinai, R. A. Guzzi, J. Polym. Sci., **41**, 475 (1959).
127. Q. A. Trementozzi, J. Am. Chem. Soc, **76**, 5273 (1954).
128. H. J. L. Trapp, J. J. Hermans, J. Phys. Chem., **58**, 757 (1954).
129. A. Vrij, J. Th. G. Overbeek, J. Colloid Sci., **17**, 570 (1962).
130. H. L. Bhatnagar, A. R. Biswas, J. Polym. Sci., **13**, 461 (1954).
131. R. Tremblay, Y. Sicotte, M. Rinfret, J. Polym. Sci., **14**, 310 (1954).
132. E. S. Cohn, E. M. Schuele, J. Polym. Sci., **14**, 309 (1954).
133. J.-Y. Chien, L.-H. Shih, S.-C. Yu, J. Polym. Sci., **29**, 117 (1958).
134. J. Bischoff, V. Desreux, Bull. Soc. Chem. Belges, **61**, 10 (1952).
135. H.-J. Cantow, O. Bodmann, Z. Physik Chem. (Frankfurt), **3**, 65 (1955).
136. E. Cohn–Ginsberg, T. G. Fox, H. F. Mason, Polymer, **3**, 97 (1962).
137. H.-G. Elias, O. Etter, Makromol. Chem., **89**, 228 (1965).
138. W. Schnabel, Makromol. Chem., **86**, 9 (1965).
139. O. Bodmann, Makromol. Chem., **122**, 196 (1969).

140. S. Fujishige, H.-G. Elias, Makromol. Chem., **155**, 127 (1972).
141. S. Fujishige, H.-G. Elias, Makromol. Chem., **155**, 137 (1972).
142. S. Krause, E. Cohn–Ginsberg, Polymer, **3**, 565 (1962).
143. R. Kirste, G. V. Schulz, Z. Physik Chem. (Frankfurt), **27**, 301 (1961).
144. T. Matsuo, A. Pavan, A. Peterlin, D. T. Turner, J. Colloid Interfac. Sci., **24**, 241 (1967).
145. W. Bushuk, H. Benoit, Canad. J. Chem., **36**, 1616 (1958).
146. A. Bello., G. M. Guzman, Eur. Polym. J., **2**, 85 (1966).
147. T. Kotaka, N. Donkai, H. Ohnuma, H. Inagaki, J. Polym. Sci. A-2, **6**, 1803 (1968).
148. J. Lamprecht, C. Strazielle, J. Dayantis, H. Benoit, Makromol. Chem., **148**, 285 (1971).
149. L. Mrkvicková–Vaculová, P. Kratochvíl, Coll. Czech. Commun., **37**, 2015 (1972).
150. G. V. Schulz, W. Wunderlich, R. Kirste, Makromol. Chem., **75**, 22 (1964).
151. S. Krause, J. Phys. Chem., **65**, 1618 (1961).
152. H. Ohnuma, T. Kotaka, H. Inagaki, Polymer, **10**, 501 (1969).
153. H. Ohnuma, T. Kotaka, H. Inagaki, Rept. Progr. Polym. Phys. Japan, **11**, 29 (1968).
154. W. H. Stockmayer, L. D. Moore, Jr., M. Fixman, B. N. Epstein, J. Polym. Sci., **16**, 517 (1955).
155. O. Bodmann, Makromol. Chem., **122**, 210 (1969).
156. L. Mrkvicková–Vaculová, P. Kratochvíl, Coll. Czech. Chem. Commun., **37**, 2029 (1972).
157. J. Trekoval, P. Kratochvíl J. Polym. Sci. A-1, **10**, 1391 (1972).
158. A. Červenka, P. Kratochvíl, Coll. Czech. Chem. Commun., **34**, 2568 (1969).
159. R. Chiang, J. J. Burke, J. O. Threlkeld, T. A. Orofino, J. Phys. Chem., **70**, 3591 (1966).
160. F. W. Billmeyer, Jr., C. B. de Than, J. Am. Chem. Soc., **77**, 4763 (1955).
161. E. F. Casassa, W. H. Stockmayer, Polymer, **3**, 53 (1962).
162. S. N. Chinai, J. D. Matlock, A. L. Resnick, R. J. Samuels, J. Polym. Sci., **17**, 391 (1955).
163. S. N. Chinai, J. Polym. Sci., **14**, 408 (1954).
164. Y. Shimura, H. Kambe, Rept. Progr. Polym. Phys. Japan, **9**, 103 (1966).
165. U. Gruber, H.-G. Elias, Makromol. Chem., **84**, 168 (1965).
166. A. Dondos, Makromol. Chem., **147**, 123 (1971).
167. J. T. Guthrie, M. B. Huglin, G. O. Phillips, Makromol. Chem., **149**, 309 (1971).
168. M. Kalfus, J. Mitus, J. Polym. Sci. A-1, **4**, 953 (1966).
169. S. N. Chinai, A. L. Resnick, H. T. Lee, J. Polym. Sci., **33**, 471 (1958).
170. V. N. Tsvetkov, V. G. Aldoshin, Russ. J. Phys. Chem., **33**, 619 (1959).
171. R. B. Mohite, S. Gundiah, S. L. Kapur, Makromol. Chem., **116**, 280 (1968).
172. G. Greber, J. Tolle, W. Burchard, Makromol. Chem., **71**, 47 (1964).
173. K. Matsumura, Makromol. Chem., **124**, 204 (1969).
174. D. Braun, W. Chaudhari, W. Mächtle, Angew. Makromol. Chem., **15**, 83 (1971).
175. A. Matiussi, E. Conti, G. B. Gechele, Eur. Polym. J., **8**, 429 (1972).
176. M. Pizzoli, G. Stea, G. Ceccorulli, G. B. Gechele, Eur. Poly. J., **6**, 1219 (1970).
177. K. Okita, A. Teramoto, K. Kawahara, H. Fujita, J. Phys. Chem., **72**, 278 (1968).
178. A. M. Kotliar, J. Polym. Sci., **55**, 71 (1961).
179. M. Abe, K. Sakato, T. Kageyama, M. Fukatsu, M. Kurata, Bull. Chem. Soc. Japan, **41**, 2330 (1968).
180. K. Sakato–quoted in Ref. 584.
181. P. F. Mijnlieff, D. J. Coumou, J. Colloid Interf. Sci., **27**, 553 (1968).
182. J. M. G. Cowie, S. Bywater, D. J. Worsfold, Polymer, **8**, 105 (1967).
183. T. Kato, K. Miyasu, M. Nagasaawa, J. Phys. Chem., **72**, 2161 (1968).
184. T. Fujimoto, N. Ozaki, M. Nagasawa, J. Polym. Sci. A, **3**, 2259 (1965).
185. D. E. Burge, D. B. Bruss, J. Polym. Sci. A, **1**, 1927 (1963).
186. A. F. Sirianni, D. G. Worsfold, S. Bywater, Trans. Faraday Soc., **55**, 2124 (1959).
187. J. D. Matlock, S. N. Chinai, R. A. Guzzi, D. W. Levi, J. Polym. Sci., **49**, 533 (1961).
188. F. S. Holahan, S. S. Stivala, D. W. Levi, J. Polym. Sci. A, **3**, 3987 (1965).
189. F. S. Holahan, S. S. Stivala, D. W. Levi, J. Polym. Sci. A, **3**, 3993 (1965).
190. J. Velickovíc, D. Jovanovíc, J. Vukajlovíc, J. Makromol. Chem., **129**, 203 (1969).
191. M. Morton, T. E. Helminiak, S. D. Gadkary, F. Bueche, J. Polym. Sci., **57**, 471 (1962).
192. J. C. Spitzbergen, H. C. Beachell, J. Polym. Sci. A, **2**, 1205 (1964).
193. H.-G. Elias, Makromol. Chem., **50**, 1 (1961).
194. V. V. Varadaiah, J. Polym. Sci., **19**, 477 (1956).
195. A. J. Hyde, J. H. Ryan, F. T. Wall, T. F. Schatzki, J. Polym. Sci., **33**, 129 (1958).
196. B. R. Jennings, H. Plummer, Brit. J. Appl. Phys., **1**, 1201 (1968).
197. L. M. Alberino, W. W. Graessley, J. Phys. Chem., **72**, 4229 (1968).
198. J. M. G. Cowie, R. Dey, J. T. McCrindle, Polym. J., **2**, 88 (1971).
199. R. S. Roche, A. G. Tanner, Angew. Makromol. Chem., **13**, 183 (1970).
200. A. Yamamoto, M. Fujii, G. Tanaka, H. Yamakawa, Polym. J., **2**, 799 (1971).
201. Y. Ikada, W. Schnabel, Makromol. Chem., **86**, 20 (1965).
202. J. M. G. Cowie, S. Byater, J. Macromol. Chem., **1**, 581 (1966).
203. H. Lange, Makromol. Chem., **86**, 192 (1965).
204. T. E. Smith, D. K. Carpenter, Macromolecules, **1**, 204 (1968).
205. W. W. Graessley, R. D. Hartung, W. C. Uy, J. Polym. Sci. A-2, **7**, 1919 (1969).

206. G. Wigand, J.-J. Veith, Plaste Kautschuk, **16**, 671 (1969).
207. R. H. Ewart, C. P. Roe, P. Debye, J. R. McCartney, J. Chem. Phys., **14**, 687 (1946).
208. H. Lange, Kolloid Z.-Z. Polym., **201**, 123 (1965).
209. H.-G. Elias, O. Etter, Makromol. Chem., **66**, 56 (1963).
210. H. Lange, Kolloid Z.-Z. Polym., **199**, 128 (1964).
211. J. M. G. Cowie, J. Polym. Sci. C, **23**, 267 (1968).
212. B. A. Brice, M. Halwer, R. Speiser, J. Opt Soc. Am., **40**, 768 (1950).
213. H. Kambe, I. Mita, Y. Shimura, Rept. No. 381 (Vol. 29, No. 3). Aeronaut. Res. Inst., Univ. of Tokyo (1964).
214. H. Janeschitz-Kriegl, Kolloid-Z., **203**, 119 (1965).
215. H. Baumann, H. Lange, Angew. Makromol. Chem., **9**, 16 (1969).
216. J. T. Guthrie. Ph.D. Thesis, Univ. Salford, 1971.
217. J. W. Breitenbach, H. Gabler, Makromol. Chem., **37**, 53 (1960).
218. P. Outer, C. J. Carr, B. H. Zimm, J. Chem. Phys., **18**, 830 (1950).
219. H.-J. Cantow, Z. Phys. Chem. (Frankfurt), **7**, 58 (1956).
220. J. P. Kratohvíl in "Characterization of Macromolecular Structure" -Proc. Conf. at Warrenton, Virginia, USA, Publication No. 1573, Nat. Acad. Sci., Washington DC, USA, 1968.
221. G. C. Berry, J. Chem. Phys., **44**, 4550 (1966).
222. J. M. G. Cowie, E. L. Cussler, J. Chem. Phys., **46**, 4886 (1967).
223. L. A. Papazian, Polymer, **10**, 399 (1969).
224. T. Alteres, Jr., D. P. Wyman, V. R. Allen, K. Meyersen, J. Polym. Sci. A, **3**, 4131 (1965).
225. Q. A. Trementozzi, R. F. Steiner, P. Doty, J. Am. Chem. Soc., **74**, 2070 (1952).
226. R. M. Johnsen, Chem. Scripta, **1**, 81 (1971).
227. P. M. Doty, B. H. Zimm, H. F. Mark, J. Chem. Phys., **12**, 144 (1944).
228. J. W. Breitenbach, H. Gabler, O. F. Olaj, Makromol. Chem., **81**, 32 (1965).
229. D. K. Carpenter, W. R. Krigbaum, J. Chem. Phys., **24**, 1041 (1956).
230. S. M. Maron, R. L. H. Lou, J. Polym. Sci., **14**, 229 (1955).
231. J. Oth, V. Desreux, Bull. Soc. Chim. Belges, **63**, 285 (1954).
232. H. P. Frank, H. F. Mark, J. Polym. Sci., **17**, 1 (1955).
233. P. Doty, R. F. Steiner, J. Chem. Phys., **18**, 1211 (1950).
234. J. M. G. Cowie, J. T. McCrindle, Eur. Polym. J., **8**, 1185 (1972).
235. P. Grosius, Y. Gallot, A. Skoulios, Makromol. Chem., **127**, 94 (1969).
236. G. V. Schulz, H. Baumann, Makromol. Chem., **114**, 122 (1968).
237. Y. Tomimatsu, L. Vitello, K. Fong, J. Colloid Interf. Sci., **27**, 573 (1968).
238. E. Elbing, A. G. Parts, Makromol. Chem., **82**, 270 (1965).
239. J. M. G. Cowie, D. J. Worsfold, S. Bywater, Trans. Faraday Soc., **57**, 705 (1961).
240. O. Bodmann, Thesis, Mainz (1955).
241. J. H. O'Mara, D. McIntyre, J. Phys. Chem., **63**, 1435 (1959).
242. P. N. Norberg, L.-O. Sundelöf, Makromol. Chem., **77**, 77 (1964).
243. T. G. Scholte, J. Polym. Sci. A-2, **10**, 519 (1972).
244. B. Chincoli, A. J. Havlik, R. D. Vold-unpublished results quoted in Ref. 586.
245. L. Trossarelli, E. Campi, G. Saini, J. Polym. Sci., **35**, 205 (1959).
246. T. Alfrey, Jr., E. B. Bradford, J. W. Vanderhoff, J. Opt. Soc. Am., **44**, 603 (1954).
247. F. S. Chan, D. A. I. Goring, Can. J. Chem., **44**, 725 (1966).
248. W. B. Dandliker, J. Am. Chem. Soc. **72**, 5110 (1950).
249. W. R. Carroll, H. Eisenberg, J. Polym. Sci. A-2, **4**, 599 (1966).
250. J. A. Manson, G. J. Arquette, Makromol. Chem., **37**, 187 (1960).
251. A. Ciferri, M. Kryszewski, G. Weill, J. Polym. Sci., **27**, 167 (1958).
252. A. R. Shultz, J. Am. Chem. Soc., **76**, 3422 (1954).
253. H. Benoit, A. M. Holtzer, P. Doty, J. Phys. Chem., **58**, 624 (1954).
254. P. Kratochvíl, Coll. Czech. Chem. Commun., **29**, 2767 (1964).
255. G. Kalz, Plaste Kautschuk, **17**, 331 (1970).
256. E. Schroeder, Plaste Kautschuk, **15**, 2 (1968).
257. M. Hlousek, J. Lániková, Chem. Prumysel, **16**, 160 (1966).
258. C. Wippler-through SOFICA Manual (1962).
259. L. Lapčik, L.-O. Sundelöf, Chem. Scripta, **2**, 41 (1972).
260. M. Bohdanecký, V. Petrus, P. Kratochvíl, Coll. Czech. Chem. Commun., **33**, 4089 (1968).
261. G. Kalz-unpublished results quoted in Ref. 255.
262. P. Doty, H. Wagner, S. Singer, J. Phys. Chem., **51**, 32 (1947).
263. M. Freeman, P. P. Manning, J. Polym. Sci., A, **2**, 2017 (1964).
264. D. Laker, J. Polym. Sci., **25**, 122 (1957).
265. T. Kobayashi, Bull. Chem. Soc. Japan, **35**, 636 (1962).
266. G. Palma, G. Pezzin, S. Zaramella, J. Polym. Sci. C, **33**, 23 (1971).
267. P. Kratochvíl, D. Strakova, Makromol. Chem., **154**, 325 (1972).
268. M. L. Wallach, M. A. Kabayama, J. Polym. Sci. A-1, **4**, 2667 (1966).
269. C. Garbuglio, L. Crescentini, A. Mula, G. B. Gechele, Makromol. Chem., **97**, 97 (1966).
270. M. Miura, Y. Kubota, T. Masuzukawa, Bull. Chem. Soc. Japan, **38**, 316 (1965).
271. C. Loucheux, Z. Czlonkowska, J. Polym. Sci. C, **16**, 4001 (1968).
272. G. Muller-quoted in Ref. 235
273. J. B. Berkowitz, M. Yamin, R. M. Fuoss, J. Polym. Sci., **28**, 69 (1958).
274. A. G. Boyes, U. P. Strauss, J. Polym. Sci., **22**, 463 (1956).
275. G. R. Seely, Macromolecules, **2**, 302 (1969).
276. R. L. Cleland, W. H. Stockmayer, J. Polym. Sci., **17**, 473 (1955).

277. T. Shibukawa, M. Sone, A. Uchida, K. Iwahori, J. Polym. Sci. A-1, **6**, 147 (1968).
278. R. Chiang, J. C. Stauffer, J. Polym. Sci. A-2, **5**, 101 (1967).
279. W. R. Krigbaum, A. M. Kotliar, J. Polym. Sci., **32**, 323 (1958).
280. P. F. Onyon, J. Polym. Sci., **37**, 315 (1959).
281. N. Fujisaki, Kobunshi Kagaku, **18**, 581 (1961).
282. L. H. Peebles, J. Polym. Sci. A, **3**, 361 (1965).
283. K. Kamide, H. Kobayashi, Y. Miyazaki, C. Nakayama, Chem. High Polymer (Tokyo), **24**, 679 (1967).
284. M. P. Zverev, A. N. Barash, L. P. Nikonorova, L. V. Ivanova, P. I. Zubov, Polym. Sci. USSR, **9**, 1038 (1967).
285. H. Lange, H. Baumann, Angew. Makromol. Chem., **14**, 25 (1970).
286. M. Iwama, H. Utiyama, M. Kurata, J. Macromol. Chem., **1**, 701 (1966).
287. M. Matsumoto, Y. Ohyanagi, J. Polym. Sci., **46**, 441 (1960).
288. M. Ueda, K. Kajitani, Makromol. Chem., **108**, 138 (1967).
289. H.-G. Elias, P. Vogt, Makromol. Chem., **157**, 263 (1972).
290. V. Kalpagam, M. Ramakishna, V. S. R. Rao, J. Polym. Sci. A, **1**, 233 (1963).
291. H.-G. Elias, F. Patat, Makromol. Chem., **25**, 13 (1958).
292. L. M. Hobbs, V. C. Long, Polymer, **4**, 479 (1963).
293. R. U. Howard, Thesis, M. I. T., Cambridge, Mass. USA (1952).
294. W. W. Graessley, H. M. Mittelhauser, J. Polym. Sci. A-2, **5**, 431 (1967).
295. F. W. Billmeyer, Jr.- quoted in Ref. 582.
296. W. W. Graesssley–quoted in Ref. 582.
297. A. Beresniewicz, J. Polym. Sci., **39**, 63 (1959).
298. H. Inagaki, T. Oyama, Kobunshi Kagaku, **1**, 20 (1952).
299. H. A. Dieu, J. Polym. Sci., **12**, 417 (1954).
300. W. H. Beattie, R. K. Laudenslager, J. Moacanin, N. A. S. A. Tech. Memo No. 33-242, Jet Prop. Lab, California, 1966.
301. S. Matsuzawa, H. Tohara, A. Harada, Rept. Progr. Polym. Phys. Japan, **12**, 29 (1969).
302. W. Burchard, M. Nosseir, Makromol. Chem., **82**, 109 (1965).
303. K. Ueberreiter, J. Springer, Z. Physik, Chem. (N. F.), **36**, 299 (1963).
304. G. Sitaramaiah, D. Jacobs, Polymer, **11**, 165 (1970).
305. H.-G. Elias, O. Etter, J. Macromol. Chem., **1**, 431 (1966).
306. G. B. Levy, J. Polym. Sci., **17**, 247 (1955).
307. A. B. Duchkova, N. S. Nametkin, V. S. Khotimskii, L. E. Gusel'nikov, S. G. Durgar'yan, Polym. Sci. USSR, **8**, 2002 (1966).
308. H. Eisenberg, J. Chem. Phys., **44**, 137 (1966).
309. H.-G. Elias, R. Schumacher, Makromol. Chem., **76**, 23 (1964).
310. A. G. Nasini, C. Ambrosino, L. Trossarelli, Ricerca Sci., **25**, 625 (Int. Symp. Macromol. Chem., Milan–Turin, 1954) (1955).
311. P. R. Saunders, J. Polym. Sci. A-2, 3755 (1964).
312. H. G. Fendler, H. A. Stuart, Makromol. Chem., **25**, 159 (1958).
313. P. R. Saunders, J. Polym. Sci., **57**, 131 (1962).
314. H. C. Beachell, D. W. Carlson, J. Polym. Sci., **40**, 543 (1959).
315. D. W. Carlson, Thesis, Univ. Delaware, USA, 1959.
316. Z. Tuzar, P. Kratochvíl M. Bohdanecký, J. Polym. Sci. C, **16**, 633 (1967).
317. M. K. Knecht, H.-G. Elias, Makromol. Chem., **157**, 1 (1972).
318. A. De Chirico, Chim. Ind. (Milan), **42**, 248 (1960).
319. G. Sitaramaiah, J. Polym. Sci. A, **3**, 2743 (1965).
320. A. R. Shultz, P. L. Wineman, M. Kramer, Macromolecules, **1**, 488 (1968).
321. G. C. Berry, H. Nomura, K. G. Mayhan, J. Polym. Sci. A-2, **5**, 1 (1967).
322. G. V. Schulz, A. Horbach, Makromol. Chem., **29**, 93 (1959).
323. M. L. Wallach, Makromol. Chem., **103**, 19 (1967).
324. A. E. Tonelli, P. J. Flory, Macromolecules, **2**, 225 (1969).
325. H. L. Wagner, K. F. Wissbrun, Makromol. Chem., **81**, 14 (1965).
326. C. Booth, R. Orme, Polymer, **11**, 626 (1970).
327. M. Matsushima, M. Fakatsu, M. Kurata, Bull. Chem. Soc. Japan, **41**, 2570 (1968).
328. G. Allen, C. Booth, S. J. Hurst, M. N. Jones, C. Price, Polymer, **8**, 391 (1967).
329. K. Yamamoto, H. Fujita, Polymer, **8**, 517 (1967).
330. J. M. Barrales–Riendo, D. C. Pepper, Eur. Polym. J., **3**, 535 (1967).
331. P. J. Akers, G. Allen, M. J. Bethell, Polymer, **9**, 575 (1968).
332. H.-G. Elias, H. Lys, Makromol. Chem., **92**, 1 (1966).
333. P. Rempp, J. Chim. Phys., **54**, 421 (1957).
334. C. Strazielle Makromol. Chem., **119**, 50 (1968).
335. T. A. Ritscher, H.-G. Elias, Makromol. Chem., **30**, 48 (1959).
336. W. Schnabel, U. Borgwardt, Makromol. Chem., **123**, 73 (1969).
337. W. Ring, H.-J. Cantow, W. Holtrup, Eur. Polym. J., **2**, 151 (1966).
338. W. Heller, T. L. Pugh, J. Polym. Sci., **47**, 203 (1960).
339. A. Teramoto, H. Fujita, Makromol. Chem., **85**, 261 (1965).
340. K. Yamamoto, H. Fujita, Polymer, **7**, 557 (1966).
341. W. H. Stockmayer, L.-L. Chan, J. Polym. Sci. A-2, **4**, 437 (1966).
342. G. Meyerhoff, U. Moritz, Makromol, Chem., **109**, 143 (1967).
343. G. Allen, C. Booth, M. N. Jones, Polymer, **5**, 195 (1964).
344. V. E. Eskin, A. E. Nesterov, Polym. Sci. USSR, **8**, 152 (1966).
345. A. Von Raven, H. Heusinger, Paper No. P-53, IUPAC Internat. Conf. on "Chemical Transformation of Polymers", Bratislava, June 22-24 (1971).
346. A. Stokes, Eur. Polym. J., **6**, 719 (1970).
347. N. V. Makletsova, I. V. Epel'baum, B. A. Rozenberg, E. B. Lyudvig, Polym. Sci. USSR, **7**, 73 (1965).
348. D. Sims, Ministry of Defence, E. R. D. E. Waltham Abbey, Essex, England–Unpublished results (1966).
349. M. Kurata, H. Utiyama, K. Kamada, Makromol, Chem., **88**, 281 (1965).

350. H.-G. Elias, G. Adank, Makromol, Chem., **103**, 230 (1967).
351. J. M. Evans, M. B. Huglin, Makromol. Chem., **127**, 141 (1969).
352. J. M. Evans, M. B. Huglin, Eur. Polym. J., **6**, 1161 (1970).
353. K. Yamamoto, A. Teramoto, H. Fujita, Polymer, **7**, 267 (1966).
355. K. V. Ivin, H. A. Ende, G. Meyerhoff, Polymer, **3**, 129 (1962).
356. A. J. Bur, D. E. Roberts, J. Chem. Phys., **51**, 406 (1969).
357. Z. Tuzar, H. C. Beachell, Polym. Letters, **9**, 37 (1971).
358. N. S. Schneider, S. Furusaki, R. W. Lenz, J. Polym. Sci. A-1, **3**, 933 (1965).
359. V. N. Tsvetkov, I. N. Shtennikova, E. I. Ryumtsev, L. N. Andreyeva, Yu. P. Getmanchuk, Yu. L. Spirin, R. I. Dryagileva, Polym. Sci. USSR, **10**, 2482 (1968).
360. H. Plummer, B. R. Jennings, Eur. Polym. J., **6**, 171 (1970).
361. J. Moacanin, J. Appl. Polym. Sci., **1**, 272 (1959).
362. Gj. Deželić, J. P. Kratohvíl, Kolloid-Z., **173**, 38 (1960).
363. G. V. Schulz, A. Haug, Z. Phys. Chem. (Frankfurt), **34**, 328 (1962).
364. G. Adank, H.-G. Elias, Makromol. Chem., **102**, 151 (1967).
365. H. H. Tokimoto, C. T. Forbes, R. K. Laudenslager, J. Appl. Polym. Sci., **5**, 153 (1961).
366. R. Nilson, L.-O. Sundelöf, Makromol. Chem., **66**, 11 (1963).
367. W. Schnabel, Makromol. Chem., **103**, 103 (1967).
368. R. R. Buch, H. M. Klimisch, O. K. Johannson, J. Polym. Sci. A-2,, **7**, 563 (1969).
369. V. N. Tsvetkov, K. A. Andrianov, I. N. Shtennikova, G. I. Okhrimenko, L. N. Andreyeva, G. A. Fomin, V. I. Pakhomov, Polym. Sci. USSR, **10**, 636 (1968).
370. A. Audsley, J. Aveston, J. Chem. Soc., 2320 (1962).
371. D. E. G. Jones, J. W. Lorimer, Polymer, **13**, 265 (1972).
372. P. R. Gupta, D. A. I. Goring, Canad. J. Chem., **38**, 270 (1960).
373. G. Sitaramaiah, D. A. I. Goring, J. Polym. Sci., **58**, 1107 (1962).
374. M. Marx-Figini, E. Penzel, Makromol. Chem., **87**, 307 (1965).
375. W. Brown, R. Wikstroem, Eur. Polym. J., **1**, 1 (1965).
376. D. Henley, Arkiv Kemi, **18**, 327 (1961).
377. H. Vink, G. Dahlström, Makromol. Chem., **109**, 249 (1967).
378. N. Grálen, Thesis, Uppsala (1944), quoted in Ref. 583.
379. L. Valtasaari, Tappi, **48**, 627 (1965).
380. R. M. Badger, R. H. Blaker, J. Phys. Colloid Chem., **53**, 1056 (1949).
381. C. Wippler, J. Chim. Phys., **53**, 346 (1956).
382. G. V. Schulz, E. Penzel, Makromol. Chem., **112**, 260 (1968).
383. M. L. Hunt, S. Newman, H. A. Scheraga, P. J. Flory, J. Phys. Chem., **60**, 1278 (1956).
384. A. Sharples, F. L. Swinton, J. Polym. Sci., **50**, 53 (1961).
385. W. R. Krigbaum, L. H. Sperling, J. Phys. Chem., **64**, 99 (1960).
386. J. Oehman, Arkiv Kemi., **31**, 125 (1969).
387. H. Janeschitz-Kriegl, W. Burchard, J. Polym. Sci. A-2, **6**, 1953 (1968).
388. J. Oehman, V. P. Shanbhag, Arkiv Kemi., **31**, 137 (1969).
389. W. Burchard, Makromol. Chem., **88**, 11 (1965).
390. S. B. Sellen, M. P. Levi, Polymer, **8**, 633 (1967).
391. M. M. Huque, D. A. I. Goring, S. G. Mason, Canad. J. Chem., **36**, 952 (1958).
392. C. W. Tait, R. J. Vetter, J. M. Swanson, P. Debye, J. Polym. Sci., **7**, 261 (1951).
393. S. Claesson, H. H. Bruun, Svensk. Papperstid., **60**, 336 (1957).
394. K. K. Ghosh, P. K. Choudhury, Makromol. Chem., **102**, 217 (1967).
395. P. F. Onyon, J. Polym. Sci., **37**, 295 (1959).
396. K. K. Ghosh, J. Appl. Polym. Sci., **15**, 537 (1971).
397. R.H. Cornell, H. A. Swenson, J. Appl. Polym. Sci. **5**, 641 (1961).
398. P. C. Scherer, A. Tannenbaum, D. W. Levi, J. Polym. Sci., **43**, 531 (1960).
399. R. St. J. Manley, Arkiv Kemi **9**, 519 (1956).
400. H. A. Swenson, A. J. Morak, S. Kurath, J. Polym. Sci., **51**, 231 (1961).
401. M. G. Wirick, M. H. Waldman, J. Appl. Polym. Sci., **14**, 579 (1970).
402. W. B. Neely, J. Polym. Sci. A, **1**, 311 (1963).
403. W. J. Hillend, H. A. Swenson, J. Polym Sci. A, **2**, 4921 (1964).
404. W. Brown, D. Henley, J. Oehman, Makromol. Chem., **62**, 164 (1963).
405. N. S. Schneider, P. Doty, J. Phys. Chem., **58**, 762 (1954).
406. B. Das, P. K. Choudhury, J. Polym. Sci. A-1, **5**, 769 (1967).
407. W. G. Yean, D. A. I. Goring, J. Appl. Polym. Sci., **14**, 115 (1970).
408. S. R. Erlander, R. Tobin, Makromol. Chem., **111**, 194 (1968).
409. S. R. Erlander, R. Tobin, Makromol. Chem., **111**, 212 (1968).
410. C. J. Stacy, J. F. Foster, J. Polym. Sci., **20**, 57 (1956).
411. S. R. Erlander, H. L. Griffin, Stärke, **19**, 134 (1967).
412. L. P. Witnauer, F. R. Senti, M. D. Stern, J. Polym. Sci., **16**, 1 (1955).
413. P. Debye, J. Phys. Colloid Chem., **51**, 18 (1947).
414. W. B. Everett, J. F. Foster, J. Am. Chem. Soc., **81**, 3459 (1959).
415. H. L. Griffin, S. R. Erlander, F. R. Senti, Stärke, **19**, 8 (1967).
416. E. F. Paschall, J. F. Foster, J. Polym. Sci., **9**, 85 (1952).
417. S. R. Erlander, H. L. Griffin, Stärke, **19**, 139 (1967).
418. J. M. G. Cowie, J. Polym. Sci., **49**, 455 (1961).
419. R. S. Patel, R. D. Patel, Makromol. Chem., **90**, 262 (1966).
420. W. Banks, C. T. Greenwood, Stärke, **19**, 394 (1967).
421. W. Banks, C. T. Greenwood, D. J. Hourston, Trans. Faraday Soc., **64**, 363 (1968).
422. W. Burchard, E. Husemann, Makromol. Chem., **44**, 358 (1961).

423. W. Banks, C. T. Greenwood, J. Sloss, Eur. Polym. J., **7**, 263 (1971).
424. Y. Tomimatsu, K. J. Palmer, A. E. Goodban, W. H. Ward, J. Polym. Sci., **36**, 129 (1959).
425. Y. Tomimatsu, K. J. Palmer, J. Polym. Sci. A, **1**, 1005 (1963).
426. P. J. Van Duin, J. J. Hermans, J. Polym. Sci., **36**, 295 (1959).
427. B. Sedláček, I. Fric, Coll. Czech. Chem. Commun., **24**, 2834 (1959).
428. W. Burchard, B. Pfannemueller, Makromol. Chem., **121**, 18 (1969).
429. F. R. Senti, N. N. Hellman, N. H. Ludwig, G. E. Babcock, R. Tobin, C. A. Glass, B. L. Lamberts, J. Polym. Sci., **17**, 527 (1955).
430. L. H. Arond, H. P. Frank, J. Phys. Chem., **58**, 953 (1954).
431. K. Frombling, F. Patat, Makromol. Chem., **25**, 41 (1958).
432. M. Zebec, Gj. Deželić, J. Kratohvíl, K. F. Schulz, Croat. Chem. Acta, **30**, 251 (1958).
433. E. Antonini, L. Bellelli, M. R. Bruzzesi. A. Caputo, E. Chiancone, A. Rossi-Fanelli, Biopolymers, **2**, 27 (1964).
434. S. Levine, H. L. Griffin, F. R. Senti, J. Polym. Sci., **35**, 31 (1959).
435. D. H. Juers, H. A. Swenson, S. F. Kurath, J. Polym. Sci. A-2, **5**, 361 (1967).
436. J. B. Snell, J. Polym. Sci. A, **3**, 2591 (1965).
437. G. V. Koleske, S. F. Kurath, J. Polym. Sci. A, **2**, 4123 (1964).
438. A. Veis, D. N. Eggenberger, J. Am. Chem. Soc., **76**, 1560 (1954).
439. H. A. Swenson, H. M. Kaustinen, O. A. Kaustinen, N. S. Thompson, J. Polym. Sci. A-2, **6**, 1593 (1968).
440. L. C. Cerny, R. C. Graham, H. James, Jr., J. Appl. Polym. Sci., **11**, 1941 (1967).
441. C. T. Greenwood, D. J. Hourston, Stärke, **19**, 243 (1967).
442. K. A. Granath, R. Stroemberg, A. N. de Belder, Stärke, **21**, 251 (1969).
443. B. S. Harrap, E. F. Woods, Austral. J. Chem., **11**, 581 (1958).
444. B. S. Harrap, E. F. Woods, Austral. J. Chem., **11**, 592 (1958).
445. F. C. McIntire, H. W. Sievert, G. H. Barlow, R. A. Finley, A. Y. Lee, Biochem., **6**, 2363 (1967).
446. W. Mackie, S. B. Sellen, Polymer, **10**, 621 (1969).
447. P. Buchner, R. E. Cooper, A. Wasserman, J. Chem. Soc., 3974 (1961).
448. M. B. Mathews. Biochem. Biphys. Acta, **35**, 9 (1959).
449. A. Pramanik, P. K. Choudhury, J. Polym. Sci. A-1, **6**, 1121 (1968).
450. A. Pramanik, P. K. Choudhury, J. Polym. Sci. A-1, **7**, 1055 (1969).
451. J. R. Patel, C. K. Patel, R. D. Patel, Stärke, **19**, 330 (1967).
452. R. Wikström, Svensk Papperstid, **71**, 399 (1968).
453. D. A. I. Goring, T. E. Timell, J. Phys. Chem., **64**, 1426 (1960).
454. R. G. Le Bel, D. A. I. Goring, J. Polym. Sci. C, **2**, 29 (1963).
455. W. Mackie, D. B. Sellen, Biopolymers, **10**, 1 (1971).
456. G. Cohen, H. Eisenberg, Biopolymers, **4**, 429 (1966).
571. D. Casson, A. Rembaum, Macromolecules, **5**, 75 (1972).
572. M. L. Wallach, J. Polym. Sci. A-2, **5**, 653 (1967).
573. M. L. Wallach, J. Polym. Sci. A-2, **7**, 1995 (1969).
574. L. V. Dubrovina, S. A. Pavlova, V. V. Korshak, Polym. Sci. USSR, **8**, 827 (1966).
575. S. A. Pavlova, L. V. Dubrovina, G. I. Timoveeva, J. Polym. Sci. C, **16**, 2649 (1967).
576. V. N. Tsvetkov, G. A. Fomin, P. N. Lavrenko, I. N. Shtennikova, T. V. Sheremeteva, L. I. Godunova, Polym. Sci. USSR, **10**, 1051 (1968).
577. N. A. Pravikova, Ye. B. Berman, Ye. B. Lyudvig, A. G. Davtyan, Polym. Sci. USSR, **12**, 653 (1970).
578. G. L. Hagnauer, N. S. Schneider, J. Polym. Sci. A-2, **10**, 699 (1972).
579. G. Allen, J. McAinsh, Eur. Polym. J., **6**, 1635 (1970).
580. U. P. Strauss, P. L. Wineman, J. Am. Chem. Soc., **80**, 2366 (1958).
581. G. C. Berry, L. M. Hobbs, V. C. Long, Polymer, **5**, 31 (1964).
582. R. St. John Manley, A. Bengtsson, Svensk, Papperstid., **61**, 471 (1958).
583. H. Utiyama, N. Sugi, M. Kurata, M. Tamura, Bull. Inst. Chem. Res., Kyoto Univ., **46**, 198 (1968).
584. J. P. Kratohvíl, Kolloid-Z, -Z Polym., **238**, 455 (1970).
585. W. Kaye, A. J. Havlik, J. B. McDaniel, Polym. Let., **9**, 695 (1971).
586. J. C. Ravey, P. Mazeron, Polym. J., **6**, 279 (1974).
587. T. Tsui, T. Norisuye, H. Fujita, Polym. J., **7**, 558 (1975).
588. J. Stejskal, J. Janča, P. Kratochvíl, Polym. J., **8**, 549 (1976).
589. T. Ohta, T. Norisuye, A. Teramoto, H. Fujita, Polym. J., **8**, 281 (1976).
590. M. Motowoka, T. Norisuye, H. Fujita, Polym. J., **9**, 613 (1977).
591. M. Motowoka, H. Fujita, T. Norisuye, Polym. J., **10**, 331 (1978).
592. Y. Einaga, T. Mitani, J. Hashizume, H. Fujita, Polym. J., **11**, 565 (1979).
593. K. Kamide, Y. Mijazaki, T. Abe, Polym. J., **11**, 523 (1979).
594. T. Hirosye, Y. Einaga, H. Fujita, Polym. J., **11**, 819 (1979).
595. I. Hirano, Y. Einaga, H. Fujita, Polym. J., **11**, 901 (1979).
596. M. Motowoka, T. Norisuye, A. Teramoto, H. Fujita, Polym. J., **11**, 665 (1979).
597. H. Matsuda, I. Yamada, M. Okabe, S. Kuroiwa, Polym. J., **9**, 527 (1977).
598. M. Kurata, M. Abe, M. Iwama, M. Matshushima, Polym. J., **3**, 729 (1972).
599. S. Saruta, Y. Einaga, A. Teramoto, Polym. J., **12**, 161 (1980).
600. M. Hara, A. Nakajima, Polym. J., **12**, 701 (1980).
601. K. Kamide, M. Saito, T. Abe. Polym. J., **13**, 421 (1981).
602. J. Sadanobu, T. Norisuye, H. Fujita, Polym. J., **13**, 75 (1981).
603. L. Van Minh, T. Nose, Polym. J., **15**, 159 (1983).
604. M. Saito, Polym. J., **15**, 249 (1983).
605. T. Sato, T. Norisuye, H. Fujita, Polym. J., **16**, 341 (1984).
606. K. Sakarai, K. Ochi, T. Norisuye, H. Fujita, Polym. J., **16**, 559 (1984).
607. M. Hoffmann, Makromol. Chem., **175**, 613 (1974).

608. R. Kuhn, H. Alberts, H. Bartl, Makromol. Chem., **175**, 1471 (1974).
609. R. Casper, U. Biskup, H. Lange, U. Pohl, Makromol. Chem., **177**, 1111 (1976).
610. J. Scornaux, R. van Leemput, Makromol. Chem., **177**, 2721 (1976).
611. F. Candau, C. Dufour, J. Francois, Makromol. Chem., **177**, 3359 (1976).
612. D. Rahlwes, R. G. Kirste, Makromol. Chem., **178**, 1793 (1977).
613. R. Hammel, C. Gerth, Makromol. Chem., **178**, 2697 (1977).
614. H. Hack, G. Meyerhoff, Makromol. Chem., **179**, 2475 (1978).
615. B. Carton, V. Bottiglione, M. Morcellet, C. Loucheux, Makromol. Chem., **179**, 2931 (1978).
616. M. Morcellet, C. Loucheux, Makromol. Chem., **179**, 2439 (1979).
617. B. Millaud, C. Strazielle, Makromol. Chem., **180**, 441 (1979).
618. M.-K. Lai, M. Sc. Thesis, Univ. of Salford (1977).
619. W.-M. Kulicke, R. Kniewska, Makromol. Chem., **181**, 823 (1980).
620. R. Chandra, R. P. Singh, Makromol. Chem. **181**, 1637 (1980).
621. O. Chiantore, L. Trossarelli, M. Guaita, Makromol. Chem., **181**, 1115 (1980).
622. H. Müller, D. Neuray, A. Horbach, Makromol. Chem., **182**, 177 (1981).
623. T. Tanaka, M. Omoto, H. Inagaki, Makromol. Chem., **182**, 2889 (1981).
624. G. L. Hammel, G. V. Schulz, Makromol. Chem., **182**, 1835 (1981).
625. W. Wunderlich, Makromol. Chem., **182**, 2465 (1981).
626. J. Stejskal, J. Horská, Makromol. Chem., **183**, 2527 (1982).
627. G. Vorreux, M. Morcellet, C. Loucheux, Makromol. Chem., **183**, 711 (1982).
628. R. W. Holmes, M. B. Huglin, M.-K. Lai, Int. J. Biol. Macromols. **1**, 23 (1979).
629. A. Lapp, C. Strazielle, Makromol. Chem., **184**, 1877 (1983).
630. M. Hert, C. Strazielle, Makromol. Chem., **184**, 135 (1983).
631. L. L. Böhm, U. Lanver, M. D. Lechner, Makromol. Chem., **184**, 585 (1983).
632. O. Procházka, Z. Tuzar, P. Kratochvíl, Makromol. Chem., **184**, 2097 (1983).
633. F. Hooshmand-Mozaffar, M. K. Hosseinalizadeh Khorasani, M. B. Huglin, Polymer, **21**, 413 (1980).
634. R. Kniewske, W.-M. Kulicke, Makromol. Chem., **184**, 2173 (1983).
635. K. Kajiwara, W. Burchard, M. Kowalski, D. Nerger, K. Dušek, L. Matějka, Z. Tuzar, Makromol. Chem., **185**, 2543 (1984).
636. L. Beltzung, C. Strazielle, Makromol. Chem., **185**, 1155 (1984).
637. R. Kuhn. Makromol. Chem., **185**, 1003 (1984).
638. M. D. Lechner, D. G. Steinmeir, Makromol. Chem. Rapid Commun., **2**, 421 (1981).
639. N. Sakamoto, H. Tanaka, K. Yamaguchi, M. Yoshida, S. Arichi, Makromol. Chem., **186**, 1881 (1985).
640. A. Lapp. J. Herz. C. Strazielle, Makromol. Chem., **186**, 1919 (1985).
641. G. Bült. G. Meyerhoff, Makromol. Chem., Suppl., **1**, 359 (1975).
642. E. Penzel, H. Pohleman, J. Swoboda, Makromol. Chem. Suppl., **1**, 367 (1975).
643. C. Plachetta, R. C. Schulz, Makromol. Chem. Rapid Commun., **3**, 815 (1982).
644. S. F. Sun, Makromol. Chem. Rapid Commun., **4**, 203 (1983).
645. V. Grinshpun, A. Rudin, Makromol. Chem. Rapid Commun., **6**, 219 (1985).
646. T. C. Chau, A. Rudin, Polymer, **15**, 593 (1974).
647. F. Candau, J. Francois, H. Benoit, Polymer, **15**, 626 (1974).
648. D. B. Sellen, Polymer, **16**, 561 (1975).
649. C. T. Greenwood, D. J. Hourston, Polymer, **16**, 474 (1975).
650. J. V. Champion, G. H. Meeten, G. W. Southwell, Polymer, **17**, 651 (1976).
651. T. Dumelow, S. R. Holding, L. J. Maisey, Polymer, Commun., **24**, 307 (1983).
652. P. Kratochvíl, D. Straková, J. Stejskal, Polymer Commun., **26**, 202 (1985).
653. S. Arichi, N. Sakamoto, M. Miki, M. Yoshida, Polymer, **26**, 1175 (1985).
654. J. L. Angulo-Sánchez, A. Gallegos, M. A. Ponce-Vélez, E. Campos-López, Polymer, **18**, 922 (1977).
655. M. Arpin, C. Strazielle, Polymer, **18**, 591 (1977).
656. J. Klein, K.-D. Conrad, Makromol. Chem., **179**, 1635 (1978).
657. J. Selb. Y. Gallot, Polymer, **20**, 1273 (1979).
658. J. Roovers, Polymer, **20**, 843 (1979).
659. B. Millaud, C. Strazielle, Polymer, **20**, 563 (1979).
660. A. Bazuaye, M. B. Huglin, Polymer, **20**, 44 (1979).
661. B. Millaud, C. Strazielle, G. Weill, Polymer **21**, 639 (1980).
662. T. Shimizu, A. Minakata, T. Tomiyama, Polymer, **21**, 1427 (1980).
663. V. Chrástová, D. Mikulásová, J. Lacok, C. Citovicky, Polymer, **22**, 1055 (1981).
664. D. Radic, L. Gargallo, Polymer, **22**, 1045 (1981).
665. D. Kokkiaris, C. Touloupis, N. Hadjichristidis, Polymer **22**, 63 (1981).
666. C. J. C. Edwards, R. F. T. Stepto, J. A. Semlyen, Polymer **23**, 869 (1982).
667. G. H. Meeten, P. Navard, Polymer, **23**, 483 (1982).
668. K. Weisskopf, G. Meyerhoff, Polymer, **24**, 72 (1983).
669. L. Mrkvičková, J. Stejskal. J. Spěváček, J. Horská, O. Quadrat, Polymer, **24**, 700 (1983).
670. E. Ozdemir, R. W. Richards, Polymer, **24**, 1097 (1983).
671. M. B. Huglin, I. Mohd. Idris, Polymer, **24**, 1434 (1983).
672. A. Campos, B. Celda, J. Mora, J. E. Figueruelo, Polymer, **25**, 1479 (1984).

673. F. Mabire, R. Audebert, C. Quivoron, Polymer, **25**, 1317 (1984).
674. B. Bednář, J. Devátý, B. Koupalová, J. Králiček, Z. Tuzar, Polymer, **25**, 1178 (1984).
675. A. A. Abdel-Azim, M. B. Huglin, Polymer, **25**, 803 (1984).
676. D. Kisakurek, B. M. Baysal, Polymer, **25**, 693 (1984).
677. T. Kato, T. Kotuya, T. Nozaki, A. Takahashi, Polymer, **25**, 218 (1984).
678. V. M. Monroy Soto, J. C. Galin, Polymer, **25**, 254 (1984).
679. V. M. Monroy Soto, J. C. Galin, Polymer, **25**, 121 (1984).
680. M. Wenzel, W. Burchard, K. Schätzel, Polymer, **27**, 195 (1986).
681. P. H. Mutin, J. M. Guenet, Polymer, **27**, 1098 (1986).
682. S. F. Sun, D. Chao, S. Lin, Polym. Commun., **24**, 84 (1983).
683. R. Rymdén, J. Carlfors, Polym. Commun., **24**, 114 (1983).
684. Y. Kambe, C. Honda, Polym. Commun., **24**, 208 (1983).
685. R. Jenkins, R. S. Porter, Polymer, **23**, 105 (1982).
686. M. Tricot, Polym. Commun., **24**, 366 (1983).
687. Y. Kambe, C. Honda, Polym. Commun., **25**, 154 (1984).
688. H. Ochiai, K. Kamata, I. Murakami, Polym. Commun., **25**, 158 (1984).
689. M. R. Gómez-Antón, A. Horta, I. Hernández-Fuentes Polym. Commun., **27**, 5 (1986).
690. H. Utiyama, K. Takenaka, M. Mizumori, M. Fukuda, Y. Tsunashima, M. Kurata, Macromolecules, **7**, 515 (1974).
691. K. Takashima, K. Nakae, M. Shibata, H. Yamakawa, Macromolecules, **7**, 641 (1974).
692. J. M. G. Cowie, I. J. McEwen, Macromolecules, **7**, 291 (1974).
693. J. E. L. Roovers, S. Bywater, Macromolecules, **7**, 443 (1974).
694. K. Matsuo, W. H. Stockmayer, Macromolecules, **8**, 660 (1975).
695. A. C. Ouano, E. Gipstein, W. Kaye, B. Dawson, Macromolecules, **8**, 558 (1975).
696. T. Tanaka, T. Kotaka, H. Inagaki, Macromolecules, **9**, 561 (1976).
697. Z. Tuzar, P. Kratochvíl, Macromolecules, **10**, 1108 (1977).
698. T. Tanaka, T. Kotaka, K. Ban, M. Hattori, H. Inagaki, Macromolecules, **10**, 960 (1977).
699. J. J. Bourguignon, J. C. Galin, Macromolecules, **10**, 804 (1977).
700. L. W. Fisher, A. R. Sochor, J. S. Tan, Macromolecules, **10**, 949 (1977).
701. Y. Miyaki, Y. Einaga, T. Hirosye, H. Fujita Macromolecules, **10**, 1356 (1977).
702. S. Akita, Y. Einaga, Y. Miyaka, H. Fujita, Macromolecules, **9**, 774 (1976).
703. P. M. Toporowski, J. Roovers, Macromolecules, **11**, 365 (1978).
704. Y. Miyaki, Y. Einaga, H. Fujita, Macromolecules, **11**, 1180 (1978).
705. M. R. Ambler, D. McIntyre, L. J. Fetters, Macromolecules, **11**, 300 (1978).
706. J. N. Helbert, C.-Y. Chen, C. U. Pittman Jr., G. L. Hagnauer, Macromolecules, **11**, 1104 (1978).
707. L. J. Fetters, H. Yu, Macromolecules, **4**, 385 (1971).
708. T. M. Aminabhavi, P. Munk, Macromolecules, **12**, 1186 (1979).
709. M. Mandel, J. Schouten, Macromolecules, **13**, 1247 (1980).
710. T. Kitano, A. Taguchi, I. Noda, M. Nagasawa Macromolecules, **13**, 57 (1980).
711. Y. Miyaki, Y. Einaga, H. Fujita, M. Fukuda Macromolecules, **13**, 588 (1980).
712. B. Appelt, G. Meyerhoff, Macromolecules, **13**, 657 (1980).
713. P. Munk, T. M. Aminabhavi, P. Williams, D. E. Hoffman, M. Chmelir, Macromolecules, **13**, 871 (1980).
714. J. A. Guest, K, Matsuo, W. H. Stockmayer, U. W. Suter, Macromolecules, **13**, 560 (1980).
715. R. C. Jordan, D. A. Brant, Macromolecules, **13**, 491 (1980).
716. H. Murakawi, T. Norisuye, H. Fujita, Macromolecules, **13**, 345 (1980).
717. I. Noda, T. Imai, T. Kitano, M. Nagasawa, Macromolecules, **14**, 1303 (1981).
718. J. S. Tan, A. R. Sochor, Macromolecules, **14**, 1700 (1981).
719. M. G. Prolongo, R. M. Masegosa, I. Hernandez-Fuentes, A. Horta, Macromolecules, **14**, 1526 (1981).
720. A. Horta, I. Fernández-Piérola, Macromolecules, **14**, 1519 (1981).
721. M. Mandel, F. A. Varkevisser, C. J. Bloys Van Treslong, Macromolecules, **15**, 675 (1982).
722. G. Paradossi, D. A. Brant, Macromolecules, **15**, 874 (1982).
723. K. N. Strand, A. Boe, P. S. Dalberg, T. Sikkeland, O. Smidsrd, Macromolecules, **15**, 570 (1982).
724. W. F. Polik, W. Burchard, Macromolecules, **16**, 978 (1983).
725. P. Kratochvil, D. Straková, J. Stejskal, Z. Tuzar, Macromolecules, **16**, 1136 (1983).
726. T. Sato, T. Norisuye, H. Fujita, Macromolecules, **16**, 185 (1983).
727. B. Sedláček, V. Petrus, M Bohdanecký, IUPAC Int. Symp. Macromolecules Tashkent (1978). Abstract 5–198.
728. B. Chu, Q. Ying, D.-G. Lee, D.-Q. Wu, Macromolecules, **18**, 1962 (1985).
729. G. V. Laivins, D. G. Grug, Macromolecules, **18**, 1746 (1985).
730. M. Nakata, N. Numasawa, Macromolecules, **18**, 1736 (1985).
731. K. Huber, S. Bantle, P. Lutz, W. Burchard, Macromolecules, **18**, 1461 (1985).
732. K. Matsuo, W. H. Stockmayer, F. Bangerter, Macromolecules, **18**, 1346 (1985).
733. S. Yukioka, I. Noda, M. Nagasawa, M. E. Holtzer, A. Holtzer, Macromolecules, **18**, 1083 (1985).
734. M. Terbojevich, A. Cosani, G. Conio, A. Ciferri, E. Bianchi, Macromolecules, **18**, 640 (1985).
735. M. Irie, W. Schnabel, Macromolecules, **18**, 394 (1985).
736. K. Huber, W. Burchard, L. J. Fetters, Macromolecules, **17**, 541 (1984).
737. J. Velicković, J. Filipović, Macromolecules, **17**, 611 (1984).
738. A. Kidera, A. Nakajima, Macromolecules, **17**, 659 (1984).
739. G. Wenz, M. A. Müller, M. Schmidt, G. Wegner, Macromolecules, **17**, 837 (1984).
740. C. L. McCormick, P. A. Callais, B. H. Hutchinson Jr., Macromolecules, **18**, 2394 (1985).

741. B. Sedláček, J. Spěváček, L. Mrkvičková, J. Stejskal, J. Horská, J. Baldrian, O. Quadrat, Macromolecules, **17**, 825 (1984).
742. N. T. M. Klooster, F. van der Touw, M. Mandel, Macromolecules, **17**, 2078 (1984).
743. X. Zhongde, N. Hadjichristidis, L. J. Fetters, Macromolecules, **17**, 2303 (1984).
744. M. Kuwata, H. Murakami, T. Norisuye, H. Fujita, Macromolecules, **17**, 2731 (1984).
745. J. W. Pope, B. Chu, Macromolecules, **17**, 2633 (1984).
746. M. T. Bishop, F. E. Karasz, P. S. Russo, K. H. Langley, Macromolecules, **18**, 86 (1985).
747. M. B. Huglin, M. B. Sokro, Polymer, **21**, 651 (1980).
748. J. W. Mays, W. Ferry, N. Hadjichristidis, L. J. Fetters, Macromolecules, **18**, 2330 (1985).
749. H. Garreau, E. Marechal, Eur. Polym. J., **9**, 263 (1973).
750. M. Hert, C. Strazielle, Eur. Polym. J., **9**, 543 (1973).
751. J. Brzezinski, Z. Czlonkowska-Kohutnicka, B. Czarnecka, A. Kornas-Calka, Eur. Polym. J., **9**, 733 (1973).
752. C. Cincu, Eur. Polym. J., **10**, 29 (1974).
753. J. M. G. Cowie, A. Maconnachie, Eur. Polym. J., **10**, 367 (1974).
754. M. Bohdanecký, H. Bažilová, J. Kopeček, Eur. Polym. J., **10**, 405 (1974).
755. B. R. Jennings, J. F. Schweitzer, Eur. Polym. J., **10**, 459 (1974).
756. V. N. Tsvetkov, E. I. Rjumtsev, L. N. Andreeva, N. V. Pogodina, P. N. Lavrenko, L. I. Kutsenko, Eur. Polym. J., **10**, 563 (1974).
757. R. L. Bartosiewicz, C. Booth, A. Marshall, Eur. Polym. J., **10**, 783 (1974).
758. G. Beck, J. Kiwi, D. Lindenau, W. Schnabel, Eur. Polym. J., **10**, 1069 (1974).
759. J. M. G. Cowie, A. Maconnachie, Eur. Polym. J., **11**, 79 (1975).
760. P. Biernacki, M. Wlodarczyk, Eur. Polym. J., **11**, 107 (1975).
761. I. A. Katime Amashta, G. Sanchez, Eur. Polym. J., **11**, 223 (1975).
762. J. Veličković, S. Coseva, R. J. Fort, Eur. Polym. J., **11**, 377 (1975).
763. J. T. Guthrie, M. B. Huglin, R. W. Richards, V. I. Shah, A. H. Simpson, Eur. Polym. J., **11**, 527 (1975).
764. I. Katime, A. Roig, Eur. Polym. J., **11**, 603 (1975).
765. Z. Tuzar, M. Bohdanecký, R. Puffr, J. Šebenda, Eur. Polym. J., **11**, 851 (1975).
766. I. Katime, P. Garro, J. M. Teijon, Eur. Polym. J., **11**, 881 (1975).
767. J. Veličkovič, M. Plavšić, Eur. Polym. J., **12**, 151 (1976).
768. R. J. Fort, T. M. Polyziodis, Eur. Polym. J., **12**, 685 (1976).
769. M. Doležalova, V. Petrus, Z. Tuzar, M. Bohdanecký, Eur. Polym. J., **12**, 701 (1976).
770. I. Katime, A. Roig, L. M. Leon, S. Montero, Eur. Polym. J., **13**, 59 (1977).
771. S. Ioan, E. Dumitriu, S. Maxim, A. Carpov, Eur. Polym. J., **13**, 109 (1977).
772. P. R. M. Nair, R. M. Gohil, K. C. Patel, R. D. Patel, Eur. Polym. J., **13**, 273 (1977).
773. I. A. Schneider, A. Onu, N. Aelene, Eur. Polym. J., **13**, 725 (1977).
774. L. V. Titkova, E. Prokopová, B. Sedláček, V. Petrus, K. Dušek, M. Bohdanecký, Eur. Polym. J., **14**, 145 (1978).
775. S. L. Malhotra, J. Leonard, E. Ahad, Eur. Polym. J., **14**, 257 (1978).
776. A. Campos Munoz, C. Strazeille, Eur. Polym. J., **14**, 517 (1978).
777. Z. Kücükyavuz, S. Kücükyavuz, Eur. Polym. J., **14**, 867 (1978).
778. C. Strazielle, Eur. Polym. J., **15**, 55 (1979).
779. S. Ioan, S. Maxim, Eur. Polym. J., **15**, 161 (1979).
780. C. Simionescu, N. Cobianu, Eur. Polym. J., **15**, 173 (1979).
781. I. Katime, R. Valenciano, J. M. Teijon, Eur. Polym. J., **15**, 261 (1979).
782. H. Maillols, L. Bardet, S. Gromb, Eur. Polym. J., **15**, 307 (1979).
783. J. Herold, G. Meyerhoff, Eur. Polym. J., **15**, 525 (1979).
784. A. Blau, Dissertation, Techn. Hochschule Karlsruhe (1977).
785. H. Suzuki, Y. Miyazaki, K. Kamide, Eur. Polym. J., **16**, 703 (1980).
786. J. W. A. Van Den Berg, G. van de Ridder, C. A. Smolders, Eur. Polym. J., **17**, 935 (1981).
787. K. Kishino, T. Kawai, T. Nose, M. Saitoh, K. Kamide, Eur. Polym. J., **17**, 623 (1981).
788. Z. Tuzar, O. Procházka, P. Kratochvíl, Eur. Polym. J., **17**, 87 (1981).
789. Z. Tuzar, P. Kratochvíl, Macromolecules, **10**, 1108 (1977).
790. Z. Tuzar, P. Kratochvíl, M. Bohdanecký, Adv. Polym. Sci., **30**, 117 (1979).
791. J. Carlfors, R. Rymdén, Eur. Polym. J., **18**, 933 (1982).
792. Z. Czlonkowska-Kohutnicka, Z. Dubkowski, Eur. Polym. J., **18**, 911 (1982).
793. H. Suzuki, Y. Muraoka, M. Saito, K. Kamide, Eur. Polym. J., **18**, 831 (1982).
794. E. Ogawa, M. Sato, N. Yamaguchi, M. Shima, Eur. Polym. J., **18**, 257 (1982).
795. D. Radic, L. Gargallo, Eur. Polym. J., **18**, 151 (1982).
796. J. Smid, Y. Y. Tan, G. Challa, Eur. Polym. J., **19**, 853 (1983).
797. G. Hild, C. Strazielle, P. Rempp, Eur. Polym. J., **19**, 721 (1983).
798. V. N. Tsvetkov, S. O. Tsepelevich, Eur. Polym. J., **19**, 267 (1983).
799. J. M. Barrales-Rienda, P. A. Galera Gómez, Eur. Polym. J., **20**, 1213 (1984).
800. I. Mohd. Idris, Doctoral Thesis, University of Salford (1983).
801. I. Katime, R. B. Valenciano, M. S. Anasagasti, D. Yoldi, Eur. Polym. J., **21**, 485 (1985).
802. K. S. Minsker, R. B. Pancheshnikova, Ya. B. Monakov, G. E. Zaikov, Eur. Polym. J., **21**, 981 (1985).
803. T. Ishige, A. E. Hamielec, J. Appl. Polym. Sci., **17**, 1479 (1973).
804. E. Dingsøyr, O. Smidsrød, Brit, Polym. J., **9**, 56 (1977).
805. J. Kratochvíl, B. Sedlácek, Brit. Polym. J., **9**, 206 (1977).
806. P. Kratochvíl, D. Straková, Z. Tuzar, Brit, Polym. J., **9**, 217 (1977).

807. S. Ali, Brit. Polym. J., **10**, 108 (1978).
808. H. Suzuki, Brit. Polym. J., **11**, 41 (1979).
809. R. Muqbill, G. Muller, J. C. Fenyo, E. Selegny, J. Polym. Sci., Polym. Lett., **17**, 369 (1979).
810. R. Jenkins, R. S. Porter, J. Polym. Sci., Polym. Letters, **18**, 742 (1980).
811. N. Numasawa, T. Hamada, T. Nose, J. Polym. Sci., Polym. Lett., **23**, 1 (1985).
812. J. Stejskal, M. J. Beneš, P. Kratochvíl, J. Peška, J. Polym. Sci. Polym. Phys. Ed., **11**, 1803 (1973); and Idem, *Ibid.*, **12**, 1941 (1974).
813. K. S. V. Srinivasan, M. Santappa, J. Polym. Sci. Polym. Phys. Ed., **11**, 331 (1973).
814. H. L. Wagner, C. A. J. Hoeve, J. Polym. Sci. Polym. Phys. Ed., **11**, 1189 (1973).
815. C. W. Tsimpris, K. G. Mayhan, J. Polym. Sci. Polym. Phys. Ed., **11**, 1151 (1973).
816. D. W. Tanner, G. C. Berry, J. Polym. Sci. Polym. Phys. Ed., **12**, 941 (1974).
817. J. Stejskal, P. Kratochvíl, J. Pouchly, J. Polym. Sci. Polym. Phys. Ed., **12**, 891 (1974).
818. N. Doddi, W. C. Forsman, C. C. Price, J. Polym. Sci. Polym. Phys. Ed., **12**, 1395 (1974).
819. J. M. G. Cowie, A. Maconnachie, J. Polym. Sci. Polym. Phys. Ed., **12**, 407 (1974).
820. M. Fukuda, M. Fukutomi, Y. Kato, T. Hashimoto, J. Polym. Sci. Polym. Phys. Ed., **12**, 871 (1974).
821. P. Kratochvíl, J. Vorlíček, D. Straková, Z. Tuzar, J. Polym. Sci. Polym. Phys. Ed., **13**, 2321 (1975).
822. H. Kaye, H. J. Chou, J. Polym. Sci. Polym. Phys. Ed., **13**, 477 (1975).
823. J. S. Tan, S. P. Gasper, J. Polym. Sci. Polym. Phys. Ed., **13**, 1705 (1975).
824. J. Stejskal, P. Kratochvíl, J. Polym. Sci. Polym. Phys. Ed., **13**, 715 (1975).
825. C. R. Reddy, V. Kalpagam, J. Polym. Sci. Polym. Phys. Ed., **14**, 749 (1976).
826. J. Roots, B. Nyström, J. Polym. Sci. Polym. Phys. Ed., **16**, 695 (1978).
827. T. Kojima, R. Yokota, M. Kochi, H. Kambe, J. Polym. Sci. Polym. Phys. Ed., **18**, 1673 (1980).
828. T. Norisuve, T. Yanaki, H. Fujita, J. Polym. Sci. Polym. Phys. Ed., **18**, 547 (1980).
829. C. J. Stacy, G. Kraus, J. Polym. Sci. Polym. Phys. Ed., **17**, 2007 (1979).
830. E. Bianchi, A. Ciferri, A. Tealdi, W. R. Krigbaum, J. Polym. Sci. Polym. Phys. Ed., **17**, 2091 (1979).
831. H. Suzuki, Y. Murnoka, H. Inagaki, J. Polym. Sci. Polym. Phys. Ed., **19**, 189 (1981).
832. N. Hadjichristidis, L. J. Fetters, J. Polym. Sci. Polym. Phys. Ed., **20**, 2163 (1982).
833. C.-Y. Lin, S. L. Rosen, J. Polym. Sci. Polym. Phys. Ed., **20**, 1497 (1982).
834. N. Hadjichristidis, X. Zhongde, L. J. Fetters, J. Polym. Sci. Polym. Phys. Ed., **20**, 743 (1982).
835. X. Chen. Z. Chu, N. Hadjichristidis, L. J. Fetters, J. Carella, W. W. Graessley, J. Polym. Sci. Polym. Phys. Ed., **22**, 777 (1984).
836. N. Binboga, D. Kisakürek, B. M. Baysal. J. Polym. Sci. Polym. Phys. Ed., **23**, 925 (1985).
837. P. N. Chaturvedi, C. K. Patel, J. Polym. Sci. Polym. Phys. Ed., **23**, 1255 (1985).
838. D. B. Cotts, J. Polym. Sci. Polym. Phys. Ed., **23**, 2217 (1985).
839. N. Hadjichristidis, J. Mays, R. D. Vargo, L. J. Fetters, J. Polym. Sci. Polym. Phys. Ed., **21**, 189 (1983).
840. P. M. Cotts, G. C. Berry, J. Polym. Sci. Polym. Phys. Ed., **21**, 1255 (1983).
841. D. B. Cotts, J. Polym. Sci. Polym. Phys. Ed., **21**, 1381 (1983).
842. N. Naoki, I.-H. Park, S. I. Wunder, B. Chu, J. Polym. Sci. Polym. Phys. Ed., **23**, 2567 (1985).
843. J. Danhelka, I. Kössler, V. Boháčková, J. Polym. Sci. Chem. A-1, **14**, 287 (1976).
844. G. Muller, J. P. Laine, J. C. Fenyo, J. Polym. Sci. Chem. A-1, **17**, 659 (1979).
845. D. K. Carpenter, G. Santiago, A. H. Hunt, J. Polym. Sci. Polym. Symp., **44**, 75 (1974).
846. T. E. Helminiak, G. C. Berry, J. Polym. Sci. Polym. Symp., **65**, 107 (1978).
847. J. L. Work, G. C. Berry, E. F. Casassa, J. K. Stille, J. Polym. Sci. Polym. Symp., **65**, 125 (1978).
848. G. C. Berry, J. Polym. Sci. Polym. Symp., **65**, 143 (1978).
849. C.-P. Wong, H. Ohnuma, G. C. Berry, J. Polym. Sci. Polym. Symp., **65**, 173 (1978).
850. M. Carenza, G. Palma, M. Tavan, J. Polym. Sci. Polym. Sym., **42**, 1031 (1973).
851. C. Holt, W. Mackie, D. B. Sellen, J. Polym. Sci. Polym. Symp., **42**, 1505 (1973).
852. N. Šegudović, Gj. Deželić, D. Flēs, J. Polym. Sci. Polym. Symp., **42**, 1209 (1973).
853. M. N. Berger, B. M. Tidswell, J. Polym. Sci. Polym. Symp., **42**, 1063 (1973).
854. J. J. Burke, J. Macromol. Sci. Chem. A, **7**, 187 (1973).
855. N. Hadjichristidis, V. Desreux, J. Macromol. Sci. Chem. A., **6**, 1227 (1972).
856. S. H. Carr, E. Baer, A. G. Walton, J. Macromol, Sci. Phys. B, **6**, 15 (1972).
857. F. C. Lin, S. S. Stivala, J. A. Biesenberger, J. Appl. Polym. Sci., **17**, 1073 (1973).
858. E. Ahad, J. Appl. Polym. Sci., **17**, 365 (1973).
859. A. L. Spatorico, J. Appl. Polym. Sci., **18**, 1793 (1974).
860. G. L. Hagnauer, B. R. Laliberte, J. Appl. Polym. Sci., **20**, 3073 (1976).
861. D. Harwood, H. Aoki, Y.-D. Lee, J. F. Fellers, J. L. White, J. Appl. Polym. Sci., **23**, 2155 (1979).
862. J. Horská, J. Stejskal, P. Kratochvíl, J. Appl. Polym. Sci., **24**, 1845 (1979).
863. T. B. MacRury, M. L. McConnell, J. Appl. Polym. Sci., **24**, 651 (1979).
864. S. H. Agarwal, R. F. Jenkins, R. S. Porter, J. Appl. Polym. Sci., **27**, 113 (1982).
865. J. Stejskal, J. Horská, P. Kratochvíl, J. Appl. Polym. Sci., **27**, 3929 (1983).
866. J. Horska, J. Stejskal, P. Kratochvíl, J. Appl. Polym. Sci., **28**, 3873 (1983).

867. P. R. Dvornic, J. Appl. Polym. Sci., **28**, 2729 (1983).
868. F. B. Malihi, C.-Y. Kou, T. Provder, J. Appl. Polym. Sci., **29**, 925 (1984).
869. W. Dietrich, A. Basch, Angew, Makromol. Chem., **40/41**, 159 (1974).
870. M. B. Huglin, R. W. Richards, Angew Makromol. Chem., **38**, 207 (1974).
871. J. Schurz, H. Hochberger, K. H. Schmidt, Angew Makromol. Chem., **52**, 165 (1976).
872. C. E. Carraher Jr., H. S. Blaxall, Angew Makromol. Chem., **83**, 37 (1979).
873. S. M. Aharoni, E. K. Walsh, Angew. Makromol. Chem., **81**, 75 (1979).
874. H. Horbach, R. Binsack, H. Müller, Angew. Makromol. Chem., **98**, 35 (1981).
875. G. Pastuska, U. Just, H. August, Angew. Makromol. Chem., **107**, 173 (1982).
876. L. Simek, J. Hrncirík, M. Mladek, Z. Tuzar, Angew. Makromol. Chem., **107**, 185 (1982).
877. I. Prokopová, Z. Tuzar. Kondelíková, D. Schmiedová, J. Králíček. Angew Makromol. Chem., **102**, 39 (1982).
878. W.-M. Kulicke, H. H. Hörl, Angew. Makromol. Chem., **116**, 149 (1983).
879. J. Daňhelka, L. Mrkvičková, V. Štěpán, I. Kössler, Angew. Makromol Chem., **122**, 77 (1984).
880. M. D. Lechner, R. Mattern, Angew. Makromol. Chem., **123/124**, 45 (1984).
881. M. Stickler, Angew. Makromol. Chem., **123/124**, 85, (1984).
882. M. B. Huglin, Topics Curr. Chem., **77**, 141 (1978).
883. F. F. Khodzhevanov, N. S. Nametkin, S. G. Durgaryan, I. Ye. Chernyakov, Polym. Sci. USSR, **14**, 1022 (1972).
884. V. M. Andreyeva, A. A. Tager, I. S. Fominykh, O. L. Zamarayeva, Polym. Sci. USSR, **18**, 328 (1976).
885. V. S. Skazka, S. Ya. Lyubina, G. V. Tarasova, S. I. Klenin, V. N. Tsvetkov, Polym. Sci. USSR, **18**, 1583 (1976).
886. V. Ye. Eskin, S. Ya. Magarik, U. B. Zhurayev, G. D. Rudkovskaya, Polym. Sci. USSR, **20**, 2494 (1979).
887. O. V. Klenina, M. Yu. Prozorova, V. I. Klenin. G. Z. Aksel'rod, Polym. Sci. USSR, **22**, 325 (1980).
888. M. G. Vitovskaya, P. N. Lavrenko, O. V. Okatova, E. P. Astapenko, V. B. Novakovskii, S. V. Bushin, S. A. Didenko, L. V. Avrorova, A. V. Tokarov, G. I. Kudryavtsev, V. N. Tsvetkov, Polym. Sci. USSR, **23**, 2136 (1982).
889. S. V. Bushin, Ye. V. Korneyeva, I. I. Konstantinov, Yu. B. Amerik, S. A. Didenko, I. N. Shtennikova, V. N. Tsvetkov, Polym. Sci. USSR, **24**, 1671 (1982).
890. S. I. Ganicheva, V. M. Belyayev, V. N. Krivobokov, O. V. Kallistov, S. Ya. Frenkel, Polym. Sci. USSR, **24**, 1734 (1982).
891. S. V. Bushin, Ye. B. Lysenko, V. A. Cherkasov, K. P. Smirnov, S. A. Didenko, G. N. Marchenko, V. N. Tsvetkov, Polym. Sci. USSR, **25**, 2212 (1983).
892. V. N. Tsvetkov, S. O. Tsepelevich, Polym. Sci. USSR, **25**, 2221 (1983).
893. A. L. Izyumnikov, G. R. Polyakova, A. R. Gantmakher, Polym. Sci. USSR, **25**, 2721 (1983).
894. P. N. Lavrenko, V. B. Novakovskii, O. I. Mikryukova, V. N. Kolot, G. I. Kudryavtsev, V. N. Tsvetkov, Polym. Sci. USSR, **26**, 875 (1984).
895. H. Matsuda, S. Otora, I. Yamada, S. Kuroiwa, Reports Progress Polym. Phys. Japan, **15**, 55 (1972).
896. Y. Kambe, C. Honda, Reports Progr. Polym. Phys. Japan, **15**, 425 (1972).
897. A. Kotera, K. Furasawa, N. Onda, Y. Koya, Rep. Progr. Polym. Phys. Japan, **17**, 43 (1974).
898. C. Okada, K. Kamada, T. Yoshihara, Reports Progr. Polym. Phys. Japan, **20**, 5 (1977).
899. B. Chu, M. Onclin, J. R. Ford, J. Phys. Chem., **88**, 6566 (1984).
900. M. B. Huglin in: J. Brandrup, E. H. Immergut (Eds.), "Polymer Handbook", 2nd ed., Wiley, New York, 1974, p. IV-267.
901. J. Stejskal, D. Straková, O. Procházka, P. Kratochvíl, Coll. Czech. Chem. Commun., **48**, 2656 (1983).
902. A. A. Abdel-Azim, Doctoral Thesis, University of Salford, 1983.
903. J. R. Brown, J. H. O'Donnell, J. Macromol. Sci. A: Chem., **6**, 1411 (1972).
904. G. Ribitsch, J. Schurz, V. Ribitsch, Colloid Polym. Sci., **258**, 1322 (1980).
905. M. Nakagaki, Y. Sano, Bull. Chem. Soc. Japan, **46**, 791 (1973).
906. S. Arichi, M. Yoshida, Y. Ogawa, Bull. Chem. Soc. Japan, **48**, 1417 (1975).
907. N. Iso, H. Mizuno, T. Saito, N. Nitta, K. Yoshikazi, Bull. Chem. Soc. Japan, **50**, 2892 (1977).
908. M. Yoshida, S. Yamaguchi, N. Sakamoto, S. Arichi, Bull. Chem. Soc. Japan, **58**, 1023 (1985).
909. T. Mori, J. Komiyama, T. Iijima, J. Chem. Soc. Faraday Trans I, **74**, 2583 (1978).
910. P. M. Sasia, University of Bilbao and University of Salford– to be published.
911. M. B. Huglin, S. J. O'Donohue, P. M. Sasia, J. Polym. Sci., Polym. Phys. Ed., **26**, 1067 (1988).
912. K. Kamide, T. Terakawa, Y. Miyazaki, Polym. J., **11**, 285 (1979).
913. H. L. Wagner, J. Res. Natl. Bur. Stand. A, **76**, 151 (1972).
914. A. Burmeister, G. Meyerhoff, Ber. Bunsen. Ges. Phys. Chem., **78**, 1366 (1974).
915. R. W. Richards, Doctoral thesis, Univ, of Salford, 1974.
916. M. B. Huglin, R. W. Richards, J. Appl. Polym. Sci., **25**, 2513 (1980).
917. M. B. Huglin, R. W. Richards, Polymer, **17**, 587 (1976).
918. J. T. Guthrie, M. B. Huglin, G. O. Phillips, Polymer, **18**, 521 (1977).
919. W. R. Moore, M. Uddin, Eur. Polym. J., **5**, 185 (1969).
920. A. R. Schultz, A. L. Bridgman, E. M. Hadsell, C. R. McCullough, J. Polym. Sci. A-2, **10**, 273 (1972).
921. L. Gargallo, C. L. Guemos, D. Radic, Eur. Polym. J., **20**, 483 (1984).
922. M. B. Sokro, Doctoral thesis, Univ. of Salford (1980).
923. M. L. McConnell, Am. Lab., **10**, 63 (1978).
924. Application Notes LS-7, Chromatix, Mountain View, Calif, USA.

925. *Ibid*–Appendix 1.
926. J. Francois, D. Sarazin, T. Schwartz, G. Weil, Polymer, **20**, 969 (1979).
927. R. Boyadjian, G. Seytre, P. Berdicat, G. Vallet, Eur. Polym. J., **12**, 401 (1976).
928. O. Z. Korotkina, V. V. Magbinec, M. N. Savickaya, V. E. Eskin, Zh. Prikl. Khim (Leningrad), **38**, 2533 (1965).
929. K.-D. Conrad, Dissertation, Technisch. Univ. Braunschweig, (1978).
930. J. Brandrup, E. H. Immergut, (Eds.) "Polymer Handbook" Interscience, New York, p. IV-282, 1967.
931. R. Chiang, J. Phys. Chem., **69**, 1645 (1965).
932. C. C. Han, P. H. Verdier, H. L. Wagner, J. Res. Natl. Bur. Stand., **83**, 185 (1978).
933. A. Peyrouset, R. Prechner, R. Panaris, H. Benoit, J. Appl. Polym. Sci., **19**, 1363 (1975).
934. L. F. Shalaeva, V. V. Arafeva, Vysokomol. Soedin Ser. B., **16**, 802 (1974).
935. L. F. Shalaeva, A. I. Barkova, K. E. Kosmatych, E. E. Veselovskaya, A. V. Shagilova, Plast. Massy, **12**, 59 (1973).
936. T. Koboyashi, A. Chitale, H. P. Frank, J. Polym. Sci., **24**, 156 (1957).
937. E. Brauer, H. Wiegleb, M. Helmstedt, W. Eifert, Plaste Kautsch. **24**, 463 (1977).
938. J. E. Kurz, J. Polym. Sci. A, **3**, 1895 (1965).
939. L. Nicolas, C. R. Acad. Sci., **244**, 80 (1957).
940. L. Nicolas, J. Polym. Sci., **29**, 191 (1958).
941. Application Notes LS-1, Chromatix, Mountain View, Calif., USA.
942. F. J. Schoenes, Z. Angew. Phys., **28**, 363 (1970).
943. P. Metzger Cotts, Macromolecules, **19**, 488 (1986).
944. T. M. Handel, I. S. Ponticello, J. S. Tan, Macromolecules, **20**, 264 (1987).
945. P. M. Cotts, S. Ferline, G. Dagli, D. S. Pearson, Macromolecules, **24**, 6730 (1991).
946. H. Hasse, H.-P. Kany, R. Tintinger, G. Maurer, Macromolecules, **28**, 3540 (1995).
947. Y. Tamai, T. Konishi, Y. Einaga, M. Fujii, H. Yamakawa, Macromolecules, **23**, 4067 (1990).
948. S. J. Rathbone, C. A. Haynes, H. W. Blanch, J. M. Prausnitz, Macromolecules, **23**, 3944 (1990).
949. D. B. Roitman, R. A. Wessling, J. McAlister, Macromolecules, **26**, 5174 (1993).
950. N. Saito, T. Kanbara, Y. Nakamura, T. Yamamoto, K. Kubota, Macromolecules, **27**, 756 (1994).
951. T. Kitagawa, J. Sadanobu, T. Norisuye, Macromolecules, **23**, 602 (1990).
952. F. Wang, J. Roovers, P. M. Toporowski, Macromolecules, **26**, 3826 (1993).
953. S. Vanhee, R. Rulkens, U. Lehmann, C. Rosenauer, M. Schulze, W. Koehler, G. Wegner, Macromolecules, **29**, 5136 (1996).
954. M. Nakata, A. Kaji, Macromolecules, **21**, 2514 (1988).
955. M. Nakata, Y. Nakano, K. Kawate, Macromolecules, **21**, 2509 (1988).
956. Z. Tong, Y. Einaga, T. Kitagawa, H. Fujita, Macromolecules, **22**, 450 (1989).
957. Y. Nakamura, K. Akasaka, K. Katayama, T. Norisuye, A. Teramoto, Macromolecules, **25**, 1134 (1992).
958. R. Duran, C. Strazielle, Macromolecules, **20**, 2853 (1987).
959. C. Wu, M. Siddiq, S. Bo, T. Chen, Macromolecules, **29**, 3157 (1996).
960. C. Wu, S. Bo, M. Siddiq, G. Wang, T. Chen, Macromolecules, **29**, 2989 (1996).
961. J. N. Zeng, D. H. Magiera, I. S. Krull, J. Polym. Sci. A: Polym. Chem. Ed., **30**, 1809 (1992).
962. N. Binboga, D. Kisakürek, B. M. Baysal, J. Polym. Sci. Polym. Phys. Ed., **23**, 925 (1985).
963. S. Kiugasa, H. Nakahara, J.-I. Kawahara, Y. Koga, H. Takaya, J. Polym. Sci. Part B: Polym. Phys. Ed., **34**, 583 (1996).
964. S. Alioria, S. Pispas, E. Siakali-Kioulafa, N. Hadjichristidis, J. Polym. Sci. B: Polym. Phys. Ed., **33**, 2229 (1995).
965. Y. Ju Chen, J. W. Mays, N. Hadjichristidis, J. Polym. Sci. B: Polym. Phys. Ed., **32**, 715 (1994).
966. S. Gooda, M. B. Huglin, J. Polym. Sci. A: Polym. Chem. Ed., **30**, 1549 (1992).
967. F. Viras, K. Viras, J. Polym. Sci. B: Polym. Phys. Ed., **26**, 2525 (1988).
968. W. Volksen, P. Cotts, D. Y. Yoon, J. Polym. Sci. B: Polym. Phys. Ed., **25**, 2487 (1987).
969. L. H. Park, E.-J. Choi, Polymer, **37**, 313 (1996).
970. C. H. Chen, J. Wilson, W. Chen, R. M. Davis, J. S. Riffle, Polymer, **35**, 3587 (1994).
971. J. M. Corpart, J. Selb, F. Candau, Polymer, **34**, 3873 (1993).
972. S. R. Gooda, M. B. Huglin, Polymer, **34**, 1913 (1993).
973. S. Kansara, N. K. Patel, C. K. Patel, Polymer, **34**, 1303 (1993).
974. P. M. Cotts, R. Siemens, Polymer, **32**, 3052 (1991).
975. H. Mutschler, U. Schoeder, E. Fahner, K. H. Ebert, A. E. Harnielec, Polymer, **26**, 935 (1985).
976. D. Kisakürek, N. Binboga, J. F. Harrod, Polymer, **28**, 1767 (1987).
977. K. Sivadasan, S. Gundiah, Polymer, **28**, 1426 (1987).
978. J. Horska, V. Petrus, M. Bohdanecký, Polymer, **28**, 1207 (1987).
979. T. Hiraro, A. Teramoto, T. Sato, T. Norisuye, T. Masuda, T. Higashimura, Polymer Journal (Tokyo), **23**, 925 (1991).
980. D. Hunkele, A. E. Hamielec, J. Appl. Polym. Sci., **35**, 1603 (1988).
981.
982. E. Lathova, D. Lath, J. Pavlinec, Polymer Bulletin (Berlin), **30**, 713 (1993).
983. C. Erbil, H. Çatalgil-Giz, N. Uyanik, Polymer International, **33**, 377 (1994).
984. Y.-J. Chen, J. Li, N. Hadjichristidis, J. W. Mays, Polymer Bulletin (Berlin), **30**, 575 (1993).
985. K. Kamide, Y. Miyazaki, H. Kobayashi, Polymer Journal (Tokyo), **17**, 607 (1985).
986. J. C. Galin, G. Lutringer, M. Galin, J. Appl Polym. Sci., **37**, 487 (1989).
987. N. Sakamoto, H. Tanaka, K. Yamaguchi, M. Yoshida, S. Arichi, Makromol. Chem., **186**, 1881 (1985).
988. M. B. Huglin, M. A. Radwan, Polymer International, **24**, 119 (1991).

Particle Scattering Factors in Rayleigh Scattering

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A. INTRODUCTION: GENERAL RELATIONS FOR A HOMOGENEOUS SOLUTE (1,2)

When electromagnetic radiation interacts with any fluid medium, inhomogeneities in refractive index (dielectric constant) arising from microscopic thermal fluctuations in density and composition cause scattering of a small fraction of the incident radiation. If scattering elements are spatially correlated over distances of the order of magnitude of the wavelength of the exciting radiation, the scattering is subject to optical interferences that affect the angular intensity distribution. This distribution can provide information on the size and shape of macromolecular solutes or suspended colloidal particles: in principle, the molecular radius of gyration is determinable unambiguously, and calculated scattering distributions for idealized models can be compared with experimental data to infer an acceptable molecular conformation. However, an experimental scattering distribution does not permit a unique model assignment in the absence of additional information or supposition.

The limiting dilute-solution behavior of the reduced intensity $R(q, c)$ (Rayleigh's ratio) of light scattered through an angle θ is given by (3)

$$Kc/R(q,c) = 1/MP(q) + 2A_2Q(q,c)c + O(c^2) \quad \text{(A1)}$$

where M is the molecular weight of the solute, c is the concentration in units of mass/volume, A_2 is the second virial coefficient, and q is an angle variable

$$q = (4\pi\tilde{n}/\lambda)\sin(\theta/2) \quad \text{(A2)}$$

$\tilde{n}$ denoting the refractive index of the solution, and λ the vacuum wavelength of the incident radiation. In the optical constant (assuming incident radiation polarized perpendicular to the plane containing angle θ)

$$K = \frac{2\pi^2\tilde{n}^2(\partial\tilde{n}/\partial c^2)}{N_A\lambda^4} \quad \text{(A3)}$$

$\partial\tilde{n}/\partial c$ is the specific refractive index increment at constant pressure and temperature, and N_A is Avogadro's number. The intensity $R(q, c)$ is that due to the solute; the (relatively weak) background solvent scattering is assumed to be that measured for the pure solvent.

The first term on the right-hand side of Eq. (A1) contains the *intramolecular* interference function, or particle scattering factor, defined by (4,5)

$$P(q) = [R(q,c)/R(0,c)]^0 \quad \text{(A4)}$$

the superscript zero denoting the limit of infinite dilution. Thus the intensity distribution $P(q)$ is normalized to unity at zero scattering angle. Unless particle dimensions are infinitesimally small compared with λ, $P(q)$ is less than unity for $q < 0$; and for particles of less than colloidal dimensions, it is almost always a monotonously decreasing function of q throughout the accessible range of q with visible light.

The second term of Eq. (A1) contains the effects of pairwise interactions between solute molecules (and, implicitly, of interactions involving solvent). The dimensionless angle factor, $Q(q)$, is a function generally unknown, dependent on both *intermolecular* and *intramolecular* correlations (3,6). Like $P(q)$ it is normalized to unity at $\theta = 0$, but usually the approximation is made that $Q(q)$ is unity at all angles so that

$$Kc/R(q,c) = 1/MP(q) + 2A_2c + \cdots \quad \text{(A5)}$$

replaces Eq. (A1). In all that follows, Eq. (A5), which is Zimm's "single contact" approximation (4), is assumed to hold for any macromolecular species in solution. At $\theta = 0$, Eqs. (A1) and (A5) revert to a thermodynamic equation of a state in virial form.

A scattering entity (polymer molecule, particle) may be represented by n optically isotropic point scattering centers whose spatial arrangement in any given conformation is specified by $n(n-1)/2 \approx n^2/2$ vectors $\mathbf{r}_{ij}$, separating pairs of scattering centers i, j. The scattering function $P(q)$ is then

$$P(q) = n^{-2} \sum_i \sum_j \int F(\mathbf{r}_{ij}) \exp(i\mathbf{q} \times \mathbf{r}_{ij}) d\mathbf{r}_{ij} \qquad \text{(A6)}$$

The vector $\mathbf{q}$ is the difference between vectorial wave numbers of incident and scattered rays such that $|\mathbf{q}| = q$, as defined above. The nonsubscript i has the usual significance of $(-1)^{1/2}$, and $F(\mathbf{r}_{ij})$ is the probability distribution function (probability density) for the vector $\mathbf{r}_{ij}$. Since the molecules in solution exhibit no preferential orientation in space, $F(\mathbf{r}_{ij})$ is spherically symmetric, and $P(q)$ assumes the form (7)

$$P(q) = \sum_i \sum_j \left\langle \frac{\sin(qr_{ij})}{qr_{ij}} \right\rangle \qquad \text{(A7)}$$

where the pointed brackets indicate averaging over molecular conformations. It is evident from Eq. (A7) that $P(q)$ can be expanded in the even moments of the intramolecular distances $\mathbf{r}_{ij}$. It follows that (7)

$$P(q) = 1 - \frac{1}{3}\left(\frac{4\pi\tilde{n}}{\lambda} \sin\frac{\theta}{2}\right)^2 R_g^2 + \cdots \qquad \text{(A8)}$$

where R_g denotes the root-mean-square (rms) radius of gyration of the molecule about its center of mass

$$R_g^2 \equiv \frac{1}{2n^2} \sum_i \sum_j \langle r_{ij}^2 \rangle^2 \qquad \text{(A9)}$$

B. SCATTERING FACTORS FOR VARIOUS MOLECULAR MODELS (1,2)

The scattering factor $P(q)$ has been calculated for a number of simple molecular geometries. For monodisperse random-flight (Gaussian) coils it is given by (7)

$$P(q)_{\text{coil}} = 2u^{-2}(e^{-u} + u - 1) = 1 - u/3 + \cdots \qquad \text{(B1)}$$

$$u \equiv (nb^2/6)q^2 = R_g^2 q^2 \qquad \text{(B2)}$$

where n is the number of equivalent statistical segments of root-mean-square length b in the chain.

For a flexible-ring model (a long chain of Gaussian segments constrained to have ends joined), $P(q)$ is given by (8)

$$P(q)_{\text{ring}} = 2u^{-1/2} I_D(u^{1/2}/2) \qquad \text{(B3)}$$

where

$$I_D(x) = \exp(-x^2) \int_0^x \exp(t^2) dt \qquad \text{(B4)}$$

is Dawson's integral, for which tabulations are available (9). The variable u is defined as in the first equality in Eq. (B2), so that in this case $u = 2q^2 R_{g,\text{ring}}^2$ since the mean-square radius $R_{g,\text{ring}}^2$ is $nb^2/12$.

For flexible regular-star branched molecules with f identical arms (f random-flight chains starting from a common node), $P(q)$ is given by (10,11)

$$P(q)_{\text{star}} = \frac{2}{u^2}\left[\frac{f(f-1)}{2} e^{-2u/f} - f(f-2)e^{-u/f} + u + \frac{f(f-3)}{2}\right] \qquad \text{(B5)}$$

The mean-square radius of the star model is $[(3f-2)/f^2][nb^2/6]$. Scattering functions have also been calculated for certain random-flight comb-shaped models of flexible backbones with uniform side chains (11). Provided $q^2 R_g^2 \leq 2$, a good approximation to $P(q)$ for a variety of branched chains and for the ring model is afforded by the form of Eq. (B1) for a linear chain with u replaced by gu, g denoting the ratio of the mean-square radius of the nonlinear structure to that of the linear chain of the same mass (2,10).

Strictly, Eqs. (B1–B5) apply only to flexible chain molecules in a theta solvent. To account approximately for the chain expansion in a good solvent, the most obvious expedient is to increase the statistical mean-square segment length b^2 by the same factor as that for the increase of the overall mean-square radius over the unperturbed value and assume that the foregoing equations still apply. Several derivations (12–15) however, recognize the non-Gaussian character of the segment distribution perturbed by the excluded volume effect. Scattering factors have also been obtained for another type of modification of the random flight, the "wormlike" chain of Kratky and Porod, which introduces a persistence length to account for chain "stiffness" (1,16).

For thin cylindrical rods of length L, $P(q)$ is (7,17,18)

$$P(q)_{\text{rod}} = \frac{1}{y}\text{Si}(2y) - \left(\frac{\sin y}{y}\right)^2 = 1 - \frac{1}{9}y^2 + \frac{2}{225}y^4 - \frac{1}{2205}y^6 + \cdots \qquad \text{(B6)}$$

where $y = qL/2$, and

$$\text{Si}(x) = \int_0^x \frac{\sin w}{w} dw \qquad \text{(B7)}$$

The "sine integral" function $\text{Si}(x)$ is given in standard tabulations (9). For large y this $P(q)$ function asymptotically approaches $(\pi/2y) - (1/2y^2)$; and, in fact, the asymptotic relation is less than 1% in error for $y > 4$. The four terms of the series form given in Eq. (B6) are within 1% of the correct value for $y < 2$.

For spheres of diameter D, $P(q)$ is (7,18)

$$P(q)_{\text{sphere}} = [(3/v^3)(\sin v - v\cos v)]^2 = 1 - \frac{1}{5}v^2 + \cdots \tag{B8}$$

where $v = qD/2$; and for thin disks of diameter D, the function is (18)

$$P(q)_{\text{disk}} = \frac{2}{w^2}\left[1 - \frac{1}{w}J_1(2w)\right] = 1 - \frac{1}{6}w^2 + \cdots \tag{B9}$$

with $w = qD/2$, and $J_1(x)$, the Bessel function of the first kind, of order one (9).

Scattering factors have been calculated for a variety of more complicated particles; e.g., oblate and prolate ellipsoids, cylinders of finite thickness, shells of different forms, particles of nonuniform density, and assemblages of simple geometrical shapes (see Ref. 19 and citations therein).

C. EFFECTS OF DISPERSION IN MOLECULAR WEIGHT (2)

For a mixture of solutes all having the same specific refractive increment, e.g., a polymer heterogeneous in molecular weight, the reciprocal scattering function in Eq. (A5), for the single-contact approximation, is replaced by (2, 20–22)

$$\frac{Kc}{R(q,c)} = \frac{1}{M_w P_X(q)} + 2A_{2,\text{app}}c + \cdots \tag{C1}$$

Here, M_w denotes the weight-average molecular weight $\sum_i M_i w_i$, M_i being the molecular weight of the ith solute and w_i its weight fraction in the solute mixture; c is the total solute concentration. The measured particle scattering factor is the average

$$P_X(q) = M_w^{-1}\sum_i M_i P_i(q) w_i \tag{C2}$$

where $P_i(q)$ is the scattering factor for the ith polymer. The quantity $A_{2,\text{app}}$ is an apparent, angle-dependent second virial coefficient

$$A_{2,\text{app}} = \sum_i \sum_j B_{ij} M_i w_i P_i(q) M_j w_j P_j(q)/[M_w P_X(q)]^2 \tag{C3}$$

where B_{ij} is the virial coefficient for interaction between components i and j (B_{ii} is just A_2 for a solution of component i alone). At the limit $\theta \to 0$, $A_{2,\text{app}}$ reduces to a true thermodynamic quantity:

$$\lim_{\theta\to 0} A_{2,\text{app}} = A_2^{(\text{LS})} = M_w^{-2}\sum_i \sum_j B_{ij} M_i w_i M_j w_j \tag{C4}$$

which might be described as a "light-scattering-average" second virial coefficient (to distinguish it from the analogous average obtained from an osmotic pressure experiment).

Series expansion of $P_X(q)$ gives an average over the mean-square radii of gyration R_{gi}^2 of the solute species

$$P_X(q) = 1 - (q^2/3)R_X^2 + \cdots \tag{C5}$$

$$R_X^2 = M_w^{-1}\sum_i R_{gi}^2 M_i w_i \tag{C6}$$

The average R_X^2 is sometimes called the Z-average mean-square radius (for a distribution of random-flight chain species R_X^2 is proportional to the so-called Z-average molecular weight

$$M_z = \sum_i (M_i^2 w_i/M_w) \tag{C7}$$

Angular scattering functions $P_X(q)$ have been calculated for various molecular size distributions. In particular, distributions of linear random-flight chains have been studied for the exponential weight-fraction distribution of Schulz and Zimm (20)

$$w(M) = \frac{b^{h+1}}{\Gamma(h+1)}M^h e^{-bM} \tag{C8}$$

with $b = (h+1)/M_w = h/M_n$, $\Gamma(x)$ the gamma function, and the dispersion parameter $h = 1/[M_w/M_n) - 1]$. For a system of random flight chains with this distribution, the averaged scattering factor is

$$P_X(q) = \frac{2}{h(h+1)}\left(\frac{h+2}{u_X}\right)^2\left[\left(\frac{u_X}{h+2}+1\right)^{-h} + \frac{h}{h+2}u_X - 1\right] \tag{C9}$$

where $u_X = q^2R_X^2$. For a monodisperse polymer, h is infinite; and for increasing dispersion in molecular weight it decreases towards zero. It is unity for the "most probable" distribution (characteristic of the products of step-growth polymerization at high conversion and of free-radical polymerization at low conversion with chain transfer). In this special case, $P_X(q)$ for random-flight linear chains is given by the very simple relation (20)

$$[P_X(q)]^{-1} = 1 + \frac{1}{3}u_X \tag{C10}$$

Plots of $[P_X(q)]^{-1}$ versus u_X for $h < 1$ curve upward, and for $h > 1$ curve downward.

The asymototic dependence of $[P_X(q)]^{-1}$ for heterogeneous chains at large q is (23)

$$\frac{1}{P_X(q)} \sim \frac{M_w}{2M_n} + \frac{M_w}{2M_z}u_X \tag{C11}$$

so that, in principle, this dependence could provide information on M_n. In practice, the experimental limit on q imposed by the upper bound on θ of 180° makes it difficult to ascertain whether the asymptotic behavior is accessible, even for polymers of high molecular weight.

Calculations for branched monodisperse Gaussian chain models and for systems with certain distributions of

References page VII-635

branching and molecular weight suggest as a reasonable approximation for such heterogeneous systems in general, the $P_X(q)$ obtained by replacement of the variable u in Eq. (B1) by $(u_X)_1 R_X^2/(R_X^2)_1$, where $(u_X)_1$ and $(R_X^2)_1$ denote u_X and R_X^2, as defined above, for the system of linear chains with the same weight distribution as the structurally heterogeneous system. Derivations of $P(q)$ for randomly branched systems have been based on cascade models and moment generating functions (24–26). Branching and molecular-weight dispersion have opposing effects on the $P(q)$ function and in randomly branched systems are largely compensatory (2,26). Interestingly, however, the asymptotic behavior of $P(q)$ for Gaussian chains is unaffected by chain branching; it depends only on molecular weight and weight distribution (10).

$P_X(q)$ has been obtained in closed analytical form for systems of thin rodlike (cylindrical) particles distributed in length (and therefore in mass) according to Eq. (C8) for integral values of the parameter h (27). For $h = 1$, the result is

$$P_X(q) = y_w^{-1} \arctan y_w \qquad \text{(C12)}$$

where $y_w = (q/2)\sum_i L_i w_i$, L_i being the length of the ith rod species. For large qL, the asymptotic dependence of $[Kc/R(\theta)]^\circ$ on q (not q^2) for rods of uniform diameter and density is linear, and the slope of the plot is $(L_i/M_i)/\pi$: i.e., the slope yields the linear density or mass per unit rod length, independent of any dispersion in L (28). If the rods are heterogeneous in diameter, the limiting slope is $[\pi \sum_i (M_i w_i/L_i)]^{-1}$. Unlike the asymptotic behavior for Gaussian coils, this asymptotic dependence for rods is manifested in practical situations with both synthetic polymers and biopolymers. Some care in interpretation is needed, however. For thin oblate ellipsoids ("needles"), a model seemingly scarcely different from thin cylinders, the asymptotic dependence of $[(Kc/R(\theta)]^\circ$ on q is also linear but with slope 5/6 that for the cylinder system (28).

D. DETERMINATION OF MOLECULAR WEIGHT AND RADIUS OF GYRATION

In general, information from a light scattering experiment on a polymer/solvent system consists of an array of intensity data determined at a number of concentrations and a number of angles. These are usually represented as values of the reciprocal intensity function $Kc/R(\theta, c)$. The form of Eq. (A5) suggests use of the popular Zimm plot: i.e., the values of $Kc/R(\theta, c)$ are plotted against $C_1 c + C_2 \sin^2(\theta/2)$, with C_1 and C_2 being arbitrary constants chosen merely to separate data points conveniently, to form a gridlike pattern (4,20). In conformity with Eq. (A5), a homogeneous polymer gives contours at constant angles (θ) that form a family of parallel straight (or nearly parallel and straight) lines, and contours at constant concentrations that form another family of curves of similar shape, displaced from one another by translations. This representation, shown schematically in Figure 1, facilitates simultaneous extrapolations of angular data to zero concentration and concentration data to zero angle, and extrapolation of both limiting dependences to a common intercept $[Kc/R(0,c)]^\circ$. In general, the ratio of the initial slope to the intercept of the angular plot extrapolated to $c = 0$ is $16\pi^2 \tilde{n}^2 R_X^2/3\lambda^2$; the slope of the concentration dependence extrapolated to $\theta = 0$ is twice the averaged second virial coefficient $A_2^{(LS)}$; and the double limit $[Kc/R(0,c)]^\circ$ is M_w^{-1}. The dominant cause of lack of parallelism of the contours in experimental Zimm plots is dispersion in molecular weight (cf. Eqs. C19, C21).

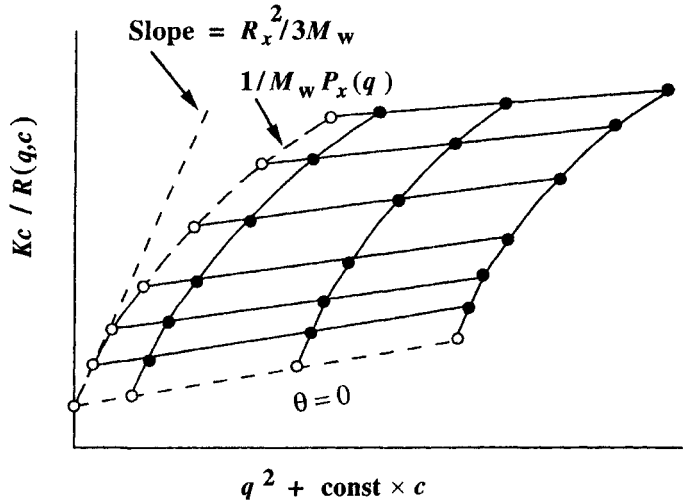

Figure 1. Schematic Zimm plot for a solute polydisperse in molecular weight, illustrating empirical determination of averaged molecular weight M_w, scattering factor $P_x(q)$, and radius of gyration R_x according to Eq. 19. The filled circles simulate experimental data; the open circles represent extrapolations to zero scattering angle and zero concentration.

If both the molecular architecture of a polymer in solution and its molecular weight dispersion are known, the scattering function $P_X(q)$ is known (in principle) and can be combined with limited angular scattering data to deduce M_w and R_X^2. Thus, in the "dissymmetry" method (7) scattered intensities at 90° and the ratio of intensities at 45° and 135° (or another pair of complementary angles) are measured for each concentration. Then the extrapolated transverse scattering function $[Kc/R(q_{90})]^\circ$ and the extrapolated dissymmetry ratio

$$Z_{45} = [R(q_{45}, c)]/R(q_{135}, c)]^\circ \qquad \text{(D1)}$$

determine the molecular weight and radius of gyration, given the $P_X(q)$ function. Tables correlating Z_{45} with $P^{-1}(q_{90})$ and the molecular dimensions have been compiled for monodisperse coils (5,29), rods (5,29), spheres (5,29), and disks (5,29,30), and for polydisperse coils (5,31).

Except for routine characterization of systems of well known character exhibiting low dissymmetry, the dissymmetry method cannot be recommended if a photometer is available that is capable of a full complement of angular readings. Apart from the desirability of determining M_w and R_X^2 without assumptions, it is

helpful to have the maximum amount of information as an aid in detecting any spurious angular dependence of scattering due to instrumental artifacts or to contamination by "dust".

In current practice, photometers that do not provide scattering measurements over a continuous angular range, are typically dedicated to the limited task of monitoring the output stream in size exclusion chromatography, perhaps on line with other detection modes. In this application, passage of the polymer through the chromatographic column effectively eliminates the problem of contamination by foreign particulate matter. One popular instrument of this kind reads scattering only at right angles and at 15°.

E. CALCULATIONS OF SCATTERING FACTORS

Figure 2 and Tables 1 and 2 illustrate some quantitative comparisons among scattering factors for several molecular models and the effects of dispersion in molecular weight for Gaussian coils. Tables of $P(q)$ and Z_{45} in the older literature are much more extensive and reflect the needs of a time without widely available computer technology. Today, for careful work to compare experimental $R(q, 0)$ data with theoretical scattering functions, investigators will find it preferable to return to the mathematical functions and write computer routines to do calculations for data analysis rather than to routinely interpolate in tables. For those who may wish to use a traditional programming language such as FORTRAN, the following rational approximation formulas will facilitate accurate calculation of the special functions that appear in the various expressions for $P(q)$.

The sine integral $\mathrm{Si}(x)$ for $x \leq 1$ (Refs. 32,33):

$$\mathrm{Si}(x) = \frac{\pi}{2} - f(x)\cos x - g(x)\sin x$$

$$f(x) = \frac{1}{x}\left(\frac{a_0 + a_2x^2 + a_4x^4 + x^6}{b_0 + b_2x^2 + b_4x^4 + x^6}\right) + \epsilon$$

$$a_0 = 8.493336 \quad a_2 = 47.411538 \quad a_4 = 19.394119$$
$$b_0 = 30.038277 \quad b_2 = 70.376496 \quad b_4 = 21.361055$$
$$10^6|\epsilon| < 8$$

$$g(x) = \frac{1}{x^2}\left(\frac{a_0 + a_2x^2 + a_4x^4 + x^6}{b_0 + b_2x^2 + b_4x^4 + x^6}\right) + \epsilon$$

$$a_0 = 5.08904 \quad a_2 = 49.719775 \quad a_4 = 21.383724$$
$$b_0 = 76.707878 \quad b_2 = 119.918932 \quad b_4 = 27.177958$$
$$10^6|\epsilon| < 6$$

The Bessel function $J_1(x)$ for $0 \leq x \leq 3$ (Ref. 34):

$$x^{-1}J_1(x) = \frac{1}{2} - 0.56249985(x/3)^2 + 0.21093573(x/3)^4$$
$$- 0.03954289(x/3)^6$$
$$+ 4.43319 \times 10^{-3}(x/3)^8 - 3.1761 \times 10^{-4}(x/3)^{10}$$
$$+ 1.109 \times 10^{-5}(x/3)^{12} + \epsilon$$
$$10^8|\epsilon| < 1.3$$

The Bessel function $J_1(x)$ for $x \geq 3$ (Ref. 34):

$$x^{1/2}J_1(x) = f_1\cos\theta_1$$
$$f = 0.79788456 + 1.56 \times 10^{-6}(3/x)$$
$$+ 0.01659667(3/x)^2$$
$$+ 1.7105 \times 10^{-4}(3/x)^3 - 2.49511 \times 10^{-3}(3/x)^4$$
$$+ 1.13653 \times 10^{-3}(3/x)^5$$
$$- 2.0033 \times 10^{-4}(3/x)^6 + \epsilon$$
$$10^8|\epsilon| < 4$$
$$\theta_1 = x - 2.35619449 + 0.12499612(3/x)$$
$$+ 5.650 \times 10^{-5}(3/x)^2$$
$$- 6.37879 \times 10^{-3}(3/x)^3$$
$$+ 7.4348 \times 10^{-4}(3/x)^4$$
$$+ 7.9824 \times 10^{-4}(3/x)^5$$
$$- 2.9166 \times 10^{-4}(3/x)^6 + \epsilon$$
$$10^8|\epsilon| < 9$$

Modern mathematics programs such as Mathematica and Maple constitute higher level languages that make possible direct operations with special functions on a personal computer (35,36).

F. PARTICLE SCATTERING FACTORS AND DISSYMMETRIES

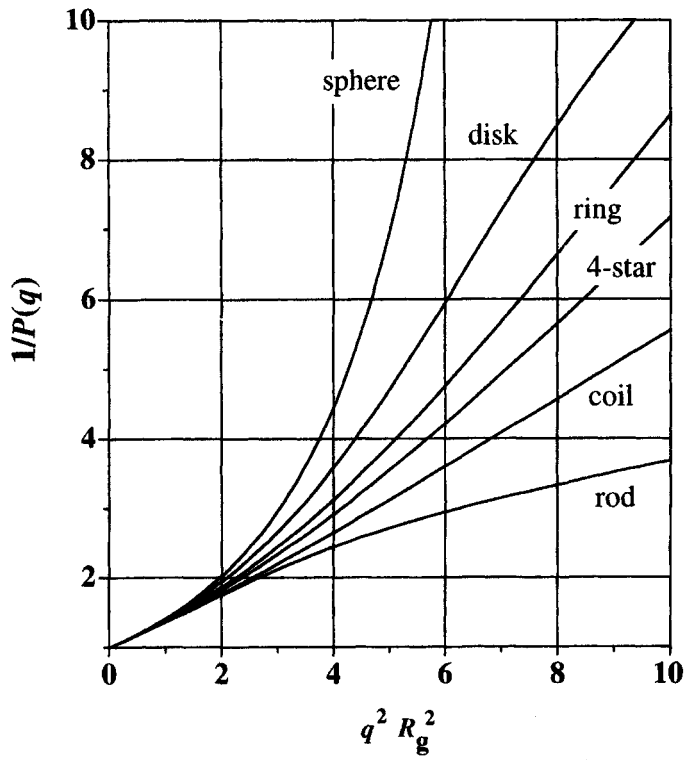

Figure 2. Reciprocal scattering factors $1/P(q)$ versus $q^2R_g^2$ for monodisperse systems of the models indicated. R_g^2 is: $nb^2/6$ (linear Gaussian coil); $5nb^2/48$ (4-branch regular star); $nb^2/12$ (flexible ring); $L^2/12$ (thin rod); $D^2/8$ (disk); $3D^2/20$ (sphere).

TABLE 1. DISSYMMETRIES FOR MONODISPERSE SYSTEMS

	Z_{45} and $P^{-1}(q_{90})$				
$R_g\tilde{n}/\lambda$	Coil	4-Star[a]	Rod	Disk	Sphere
0.00	1.000 (1.000)[b]	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
0.01	1.004 (1.003)	1.004 (1.003)	1.004 (1.003)	1.004 (1.003)	1.004 (1.003)
0.02	1.015 (1.011)	1.015 (1.011)	1.015 (1.011)	1.015 (1.011)	1.015 (1.011)
0.03	1.034 (1.024)	1.034 (1.024)	1.034 (1.024)	1.034 (1.024)	1.034 (1.024)
0.04	1.060 (1.043)	1.061 (1.043)	1.060 (1.043)	1.061 (1.043)	1.062 (1.043)
0.05	1.094 (1.067)	1.095 (1.067)	1.093 (1.067)	1.097 (1.068)	1.098 (1.068)
0.06	1.136 (1.097)	1.139 (1.098)	1.137 (1.097)	1.143 (1.099)	1.146 (1.100)
0.07	1.186 (1.133)	1.191 (1.135)	1.187 (1.133)	1.200 (1.138)	1.204 (1.139)
0.08	1.245 (1.175)	1.253 (1.178)	1.245 (1.176)	1.268 (1.183)	1.277 (1.186)
0.09	1.311 (1.224)	1.325 (1.229)	1.311 (1.225)	1.351 (1.237)	1.365 (1.242)
0.10	1.386 (1.280)	1.407 (1.287)	1.384 (1.280)	1.448 (1.301)	1.473 (1.308)
0.11	1.469 (1.342)	1.500 (1.353)	1.463 (1.343)	1.564 (1.374)	1.604 (1.386)
0.12	1.559 (1.412)	1.604 (1.428)	1.546 (1.412)	1.700 (1.459)	1.765 (1.477)
0.13	1.657 (1.490)	1.720 (1.512)	1.632 (1.488)	1.859 (1.558)	1.963 (1.584)
0.14	1.761 (1.575)	1.848 (1.607)	1.717 (1.570)	2.044 (1.671)	2.208 (1.710)
0.15	1.872 (1.669)	1.987 (1.712)	1.801 (1.659)	2.258 (1.800)	2.514 (1.858)
0.16	1.987 (1.771)	2.138 (1.828)	1.879 (1.753)	2.504 (1.949)	2.901 (2.034)
0.17	2.108 (1.881)	2.299 (1.957)	1.951 (1.853)	2.783 (2.120)	3.397 (2.242)
0.18	2.231 (2.001)	2.470 (2.098)	2.013 (1.956)	3.096 (2.315)	4.045 (2.490)
0.19	2.357 (2.129)	2.650 (2.253)	2.067 (2.063)	3.439 (2.538)	4.906 (2.789)
0.20	2.485 (2.266)	2.837 (2.422)	2.110 (2.171)	3.807 (2.791)	6.080 (3.149)
0.21	2.613 (2.413)	3.030 (2.606)	2.145 (2.279)	4.190 (3.079)	7.724 (3.589)
0.22	2.741 (2.568)	3.277 (2.806)	2.173 (2.387)	4.574 (3.403)	10.112 (4.130)
0.23	2.868 (2.733)	3.427 (3.022)	2.195 (2.493)	4.942 (3.768)	
0.24	2.993 (2.906)	3.627 (3.254)	2.212 (2.597)	5.278 (4.174)	
0.25	3.116 (3.089)	3.826 (3.502)	2.226 (2.698)	5.566 (4.622)	
0.26	3.236 (3.280)	4.022 (3.768)	2.238 (2.796)	5.797 (5.112)	
0.27	3.353 (3.481)	4.214 (4.051)	2.249 (2.892)	5.970 (5.640)	
0.28	3.466 (3.690)	4.400 (4.350)	2.259 (2.985)	6.089 (6.202)	
0.29	3.576 (3.908)	4.579 (4.667)	2.268 (3.076)	6.163 (6.788)	

[a] Regular star with four arms.
[b] Entries in parentheses indicate $P^{-1}(q_{90})$ values.

TABLE 2. DISSYMMETRIES FOR MONODISPERSE AND POLYDISPERSE COILS[a]

	$P^{-1}(q_{90})$ and $\left(\frac{1000\tilde{n}}{\lambda}(R_g^2)_w^{1/2}\right)$				
Z_{45}	$\rho = 1$	$\rho = 1.2$	$\rho = 1.5$	$\rho = 2.0$	$\rho = 5.0$
1.00	1.000 (0)[b]	1.000 (0)	1.000 (0)	1.000 (0)	1.000 (0)
1.02	1.014 (23)	1.014 (21)	1.014 (20)	1.014 (19)	1.014 (17)
1.04	1.028 (33)	1.028 (30)	1.028 (29)	1.028 (27)	1.028 (24)
1.06	1.042 (40)	1.042 (37)	1.042 (35)	1.042 (33)	1.042 (30)
1.08	1.056 (46)	1.056 (43)	1.056 (40)	1.056 (38)	1.056 (35)
1.10	1.070 (51)	1.070 (48)	1.071 (45)	1.071 (43)	1.071 (39)
1.12	1.085 (56)	1.085 (53)	1.086 (49)	1.086 (47)	1.087 (43)
1.14	1.099 (61)	1.099 (57)	1.100 (53)	1.101 (51)	1.102 (47)
1.16	1.113 (65)	1.114 (61)	1.115 (57)	1.116 (54)	1.116 (50)
1.18	1.128 (69)	1.129 (65)	1.130 (61)	1.131 (58)	1.133 (53)
1.20	1.142 (72)	1.143 (68)	1.145 (64)	1.146 (61)	1.148 (56)
1.22	1.157 (76)	1.159 (71)	1.161 (67)	1.162 (64)	1.164 (59)
1.24	1.171 (79)	1.173 (75)	1.176 (71)	1.177 (67)	1.180 (62)
1.26	1.186 (82)	1.189 (78)	1.191 (73)	1.193 (70)	1.196 (65)
1.28	1.201 (85)	1.204 (80)	1.207 (76)	1.210 (73)	1.213 (68)

TABLE 2. *cont'd*

	$P^{-1}(q_{90})$ and $\left(\frac{1000\tilde{n}}{\lambda}(R_g^2)_w^{1/2}\right)$				
Z_{45}	$\rho=1$	$\rho=1.2$	$\rho=1.5$	$\rho=2.0$	$\rho=5.0$
1.30	1.215 (88)	1.219 (83)	1.222 (79)	1.225 (76)	1.229 (70)
1.40	1.290 (102)	1.297 (97)	1.303 (92)	1.308 (88)	1.314 (82)
1.50	1.366 (113)	1.377 (109)	1.387 (104)	1.394 (100)	1.404 (93)
1.60	1.444 (124)	1.460 (119)	1.473 (115)	1.484 (111)	1.498 (104)
1.70	1.524 (134)	1.546 (129)	1.564 (125)	1.578 (121)	1.597 (114)
1.80	1.607 (144)	1.636 (139)	1.659 (135)	1.677 (131)	1.703 (125)
1.90	1.693 (152)	1.729 (149)	1.758 (145)	1.782 (141)	1.815 (134)
2.00	1.781 (161)	1.826 (158)	1.861 (154)	1.890 (150)	1.932 (144)
2.10	1.874 (169)	1.927 (167)	1.971 (163)	2.007 (160)	2.058 (154)
2.20	1.969 (178)	2.032 (175)	2.084 (172)	2.128 (169)	2.190 (163)
2.30	2.069 (185)	2.143 (184)	2.205 (181)	2.257 (178)	2.332 (173)
2.40	2.174 (194)	2.260 (192)	2.332 (190)	2.393 (188)	2.484 (183)
2.50	2.283 (201)	2.382 (201)	2.466 (199)	2.538 (198)	2.645 (193)
2.60	2.397 (209)	2.510 (209)	2.607 (209)	2.691 (207)	2.817 (203)
2.70	2.517 (217)	2.645 (218)	2.757 (218)	2.855 (217)	3.003 (214)
2.80	2.643 (225)	2.788 (227)	2.916 (227)	3.029 (227)	3.202 (225)
2.90	2.776 (233)	2.939 (235)	3.085 (236)	3.215 (237)	3.416 (236)
3.00	2.916 (240)	3.099 (244)	3.264 (246)	3.414 (247)	3.648 (247)
3.50	3.755 (283)	4.070 (292)	4.375 (299)	4.665 (305)	5.145 (312)
4.00	4.948 (333)	5.488 (349)	6.044 (364)	6.601 (377)	7.588 (397)

[a] Polydisperse systems characterized by the Schulz–Zimm distribution: $\rho = M_w/M_n$.
[b] Entries in parentheses denote weight-average value of R_g^2.

G. REFERENCES

1. H. Yamakawa, "Modern Theory of Polymer Solutions", Harper and Row, New York, 1971, Ch. 5.
2. E. F. Casassa, G. C. Berry, in: P. E. Slade Jr., (Ed.), "Polymer Molecular Weights", Marcel Dekker, New York, 1975, C. 5.
3. A. C. Albrecht, J. Chem. Phys., **27**, 1014 (1957).
4. B. H. Zimm, J. Chem. Phys., **16**, 1093 (1948).
5. P. Doty, R. F. Steiner, J. Chem. Phys., **18** 1211 (1950).
6. E. F. Casassa, J. Polym. Sci., Polym. Phys. Ed., **17**, 2077 (1979).
7. P. Debye, J. Phys. Colloid Chem., **51**, 18 (1947).
8. E. F. Casassa, J. Polym. Sci. A, **3**, 605 (1965).
9. M. Abramovitz, I. A. Stegun (Eds.), "Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables", National Bureau of Standards Applied Mathematics Series 55, US Government Printing Office, Washington DC, 10th printing, 1972.
10. H. Benoit, J. Polym. Sci., **11**, 507 (1953).
11. E. F. Casassa, G. C. Berry, J. Polym. Sci. A-2, **4**, 881 (1966).
12. H. Benoit, Compt. Rend., **245**, 2244 (1957).
13. O. B. Ptitsyn, Zh. Fiz. Khim., **31**, 1091 (1957).
14. A. J. Hyde, J. H. Ryan, F. T. Wall, J. Polym. Sci., **33**, 129 (1958).
15. P. J. Flory, R. L. Jernigan, J. Am. Chem. Soc., **90**, 3128 (1968).
16. A. Peterlin, in: M. Kerker, (Ed.), "Electromagnetic Scattering", Pergamon Press, New York, 1963, p. 357.
17. T. Neugebauer, Ann. Phys., **42**, 509 (1943).
18. O. Kratky, J. Porod, J. Colloid Sci., **4**, 35 (1949).
19. M. Kerker, "The scattering of Light and other Electromagnetic Radiation", Academic Press, New York, 1969, Ch. 8.
20. B. H. Zimm, J. Chem. Phys., **16**, 1099 (1948).
21. J. G. Kirkwood, R. J. Goldberg, J. Chem. Phys., **18**, 54 (1950).
22. W. H. Stockmeyer, J. Chem. Phys., **18**, 58 (1950).
23. H. Benoit, J. Polym. Sci., **11**, 507 (1953).
24. A. H. Dautzenberg, C. Ruscher, J. Polym. Sci. C, **16**, 2913 (1967).
25. K. Kajiwara, W. Burchard, M. Gordon, Brit. Polym. J., **2**, 110 (1970).
26. W. Burchard, Macromolecules, **10**, 919 (1977).
27. M. Goldstein, J. Chem. Phys., **21**, 1255 (1953).
28. E. F. Casassa, J. Am. Chem. Soc., **78**, 3980 (1956).
29. W. H. Beattie, C. Booth, J. Phys. Chem., **64**, 696 (1960).
30. P. Becher, J. Phys. Chem., **63**, 1213 (1959).
31. W. H. Beattie, C. Booth, J. Polym. Sci., **44**, 81 (1960).
32. M. Abramovitz, I. A. Stegun (Eds.), "Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables", National Bureau of Standards Applied Mathematics Series 55, US Government Printing Office, Washington DC, p. 232.
33. C. Hastings, Jr., "Approximations for Digital Computers", Princeton University Press, Princeton, 1955, pp. 197, 199.

34. M. Abramovitz, I. A. Stegun (Eds.), "Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables", National Bureau of Standards Applied Mathematics Series 55, US Government Printing Office, Washington DC, p. 370.
35. S. Wolfram, The Mathematica Book, Wolfram Media, Champaign, Illinois, 1996.
36. B. W. Char, K. O. Geddes, K. H. Gonnet, B. L. Leong, M. B. Monagan, S. M, Watt, "First Leaves: A Tutorial Introduction to Maple V", Springer, New York, 1992.

Dipole Moments of Polymers in Solution

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A. INTRODUCTION

Each conformation of a polymer chain has a dipole moment, μ, in Debye units D ($=10^{-18} \times$ u.e.e. $\times$ 1 cm or 3.338×10^{-30} C $\times$ m), equal to the vectorial sum of the moments of N dipolar groups along the chain, that is, $\mu(\zeta) = \sum_i m_i$, where m_i is the dipole moment associated with the skeletal bond i in the conformation ζ. The mean-square dipole moment $\langle \mu^2 \rangle$ can be written as (22,91)

$$\langle \mu^2 \rangle = \left\langle \sum_i \sum_j m_i m_j \right\rangle = \sum_i m_i^2 + \left\langle \sum_i \sum_{i \neq j} m_i m_j \right\rangle \tag{A1}$$

where m_i and m_j are the dipole moments associated with the skeletal bonds i and j, respectively, and the brackets mean averages. For a freely jointed chain, the last term in Eq. (A1) averages to zero, and the corresponding mean-square dipole moment is given by

$$\langle \mu^2 \rangle = \sum_i^N m_i^2 \tag{A2}$$

The values of the mean-square dipole moment are currently expressed in terms of the dipole moment ratio, $D_n = \langle \mu^2 \rangle / \sum m_i^2$, or the dipolar correlation coefficient, g, by means of the following expression:

$$g = D_n = 1 + \frac{\left\langle \sum_i \sum_{i \neq j} m_i m_j \right\rangle}{\sum_i m_i^2} \tag{A3}$$

Debye (13) pioneered the theories which allow calculation of the dipole moments of molecules in the vapor or gas phase. Other theories (23,32,53) developed to determine the dipole moments of molecules in the liquid state were not so successful because of the difficulties involved in defining the internal field in a polar, condensed medium. This problem was circumvented by calculating the dipole moment of polymers and polar liquids in very dilute solutions using nonpolar solvents. In this way, the solute molecules are separated from one another by nonpolar molecules, and the dielectric behavior of the system resembles that of a gaseous condition. The method developed by Guggenheim (27) and Smith (106), based on this approach, permits calculation of the mean-square dipole moments of polymers by means of the following equation:

$$\langle \mu^2 \rangle = \frac{27 k_B M T}{4\pi \rho N_A (\varepsilon_1 + 2)^2} \left[\frac{d\varepsilon}{dw} - \frac{dn^2}{dw} \right] \tag{A4}$$

where k_B and N_A are, respectively, the Boltzmann constant and Avogadro's number, M is the molecular weight of the solute, T the absolute temperature, ρ and ε_1 the density and the dielectric permitivity of the solvent, respectively. The term $d\varepsilon/dw$ (proportional to the total polarization of the solute) can be obtained from plots of the dielectric permitivity of the solution, ε, against the weight fraction of solute w, in the limit $w \to 0$. The values of dn^2/dw (proportional to the electronic polarization) can be obtained from the increments of the indices of refraction of the solution n with respect to that of the solvent n_1, also in the limit $w \to 0$. A shortcoming of this equation is that a ficticious atomic polarization (instead of the true one) is assumed. However, with the exception of the silicones, the atomic polarization for most systems is less than 10% of the electronic polarization and, consequently, the error involved in the determination of $\langle \mu^2 \rangle$ with Eq. (A4) is negligible. The mean-square dipole moment presents some advantages over other more traditional conformation-dependent properties such as the mean-square end-to-end distance $\langle r^2 \rangle_0$. For example, since the skeletal bonds change much more in polarity than they do in length, $\langle \mu^2 \rangle$ is usually more sensitive to structure than $\langle r^2 \rangle_0$. Moreover, with the exception of a relatively few number of polymers that have dipole moments in their structure with a component parallel and the other perpendicular to the chain contour, excluded volume effects are negligible.

In the tables, the dipole moments are expressed in terms of either the dipolar correlation coefficient g, or the mean-square dipole moment per repeating unit, $\langle \mu^2 \rangle / x$. For relatively long chains in which dipole contributions of the end groups are not important, $\langle \mu^2 \rangle / x$ and g are related by

$$\frac{\langle \mu^2 \rangle}{x} = g\mu_0^2 \qquad \text{(A5)}$$

where x is the degree of polymerization, and μ_0^2 is the sum of the squares of the dipole moments associated with the repeating unit. Below the generic name of each family of polymers, the dipole moments associated with the backbone are indicated.

B. DIPOLE MOMENTS OF POLYMERS IN SOLUTION

TABLE 1. POLYOXIDES (POLYETHERS)*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle \mu^2 \rangle / x$, D^2	Remarks	Refs.
Poly(oxymethylene)						
Polyformaldehyde						
With $x = 1$	Benzene	25	0.1			109
With $x = 2$	Benzene	25	0.29			109
With $x =$ infinite	Melt	202	0.3			56
Poly(oxyethylene)						
Poly(ethylene oxide)						
$x = 2$–6	Benzene	20		1.32–1.23	End groups: $-OC_2H_5$	47
$M_n = 4000$	Benzene	20–60	0.603–0.644		$m(CH_2\text{–}O) = 0.99D$	59
$x = 1$–6	Benzene	25		1.30–1.14	End groups: $-OC_2H_5$	34
$x = 4.0$–176.2	Benzene	25		2.82–1.28		46
$x = 4.1$–153.0	Benzene	25		2.59–1.28		2
$x = 1$–33.6	Benzene	20		1.99–1.19		45
$M_n = 4000$	Benzene	20–60	0.621–0.693		$m(CH_2\text{–}O) = 0.99D$	4
$x = 1$–7	Dioxane	25		2.82–1.66		110
$x = 2$–227	Benzene	20–30		2.13–1.14		47
Poly(oxypropylene)						
Poly(propylene oxide)						
$x = 6.6$–69.0	Benzene	25		1.96–1.04		45
	Benzene	25	0.54		Isotactic	1
	Dioxane	25	0.49		Isotactic	1
Poly(oxytrimethylene)						
Poly(trimethylene oxide)	Benzene	25	0.41			50
Poly[oxy(2,2-dimethyltrimethylene)]						
Poly(3,3-dimethyloxetane)						
$M_n = 8500$	Benzene	20–60	0.25–0.30			94
$M_n = 19000$	Benzene	30–60	0.20–0.22			25
$M_n = 35000$	Benzene	30–60	0.21–0.23			25
Poly[oxy(1-methyltrimethylene)]						
Poly(2-methyloxetane)						
$M_n = 13000$	Cyclohexane	30–60	0.36–0.41			78
Poly[oxy(2-methyltrimethylene)]						
Poly(3-methyloxetane)						
$M_n = 125000$	Cyclohexane	30–60	0.35–0.38			77

*If not indicated otherwise, the dipole moments associated with the skeletal bonds used in the calculation of g were $m(CH_2\text{–}O) = 1.07D$, $m(CH_2\text{–}CH_2) = 0D$, and $m(O\text{–}H) = 1.7D$.

TABLE 1. *cont'd*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
Poly(oxytetramethylene)					$m(CH_2-O) = 0.99D$	
Poly(tetramethylene oxide)						
$M_n = 2500$	Benzene	20–60	0.60–0.65			61
$M_n = 408000$	Benzene	20–60	0.59–0.65			4
Poly[oxy(2-methyltetramethylene)]						
Poly(3-methyltetrahydrofuran)						
$M_n = 6700$	Benzene	30–60	0.52–0.55			81
Poly(oxyhexamethylene)					$m(CH_2-O) = 0.99D$	
Poly(hexamethylene oxide)						
$M_n = 5500$	Benzene	35–60	0.65–0.68			59
$M_n = 2460$	Benzene	35–60	0.64–0.67			80
Poly(oxydecamethylene)					$m(CH_2-O) = 0.99D$	
Poly(decamethylene oxide)						
$M_n = 2000$	Benzene	35–60	0.64–0.68			61
Poly(oxyethyleneoxymethylene)					Polyformals: alternating copolymer of methylene oxide and alkylene oxide	
Poly(1,3-dioxolane)						
$M_n = 12000$	Benzene	20–60	0.160–0.204			62
$M_n = 16000$	Benzene	20–60	0.164–0.213			62
$M_n = 42000$	Benzene	20–60	0.177–0.227			62
Poly(oxytetramethyleneoxymethylene)						
Poly(1,3-dioxepane)						
$M_n = 17000$	Benzene	25–60	0.160–0.190			63
Poly(oxypentamethyleneoxymethylene)						
Poly(1,3-dioxocane)						
$M_n = 1850$	Benzene	20–60	0.190–0.232			64
$M_n = 2560$	Benzene	20–60	0.183–0.207			64
$M_n = 3700$	Benzene	20–60	0.168–0.207			64
Poly(oxyhexamethyleneoxymethylene)						
Poly(1,3-dioxonane)						
$M_n = 4020$	Benzene	20–60	0.190–0.230			64
$M_n = 2600$	Benzene	20–60	0.180–0.217			64
Poly(*trans*-oxymethylene-1,4-cyclohexyleneoxymethylene)						
Poly(trans-1,4-cyclohexane dimethanol-alt-formaldehyde)						
$M_n = 6500$	Benzene	20–60	0.17–0.21			71
Poly(oxypropyleneoxymethylene)						
Poly[(4-methyl)-1,3-dioxolane]						
$M_n = 40000$	Benzene	20–60	0.237–0.281			73
$M_n = 48000$	Benzene	20–60	0.216–0.256			73
Poly(*trans*-7-oxabicyclo-[4.3.0]nonane)						
Poly(trans-7-oxabicyclo-[4.3.0]nonane)	Benzene	30–60	0.63–0.66			111

TABLE 2. POLYSULFIDES (POLYTHIOETHERS)*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
Poly(thiotrimethylene)						
Poly(trimethylene sulfide)						
$M_n = 14000$	Benzene	20–60	0.60–0.63			75
Poly(thiopentamethylene)						
Poly(pentamethylene sulfide)						
$M_n = 3200$	Benzene	20–60	0.75–0.78		End bonds: CH_2-Br; $m(CH_2-Br) = 1.96D$	74
Poly(thiopropylene)						
Poly(propylene sulfide)						
	Benzene	20–60	0.434–0.449			67
$M_n = 5000000$	Carbon tetrachloride	20–60	0.362–0.407			68
$M_n = 5000000$	Benzene	20–60	0.429–0.458			68
$M_w = 1.6 \times 10^6$	Carbon tetrachloride	20–60	0.325–0.347	Isotactic		68
$M_n = 5000$	Benzene	20–60	0.448–0.460			67
$M_n = 6000$	Carbon tetrachloride	20–60	0.357–0.389			67
$M_n = 1.6 \times 10^6$	Benzene	20–60	0.386–0.420	Isotactic		68

* $m(CH_2-CH_2) = 0D$, $m(CH_2-S) = 1.21D$, $m(O-H) = 1.7D$.

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TABLE 2. *cont'd*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
Poly[thio(2,2-dimethyltrimethylene)]						
Poly(3,3-dimethylthietane)						
$M_n = 8200$	Benzene	30–60	0.62–0.63			72
Poly(thiopentamethylenethioethylene)						
Poly(ethylene sulfide-alt-pentamethylene sulfide)						
$M_n = 3800$	Benzene	30–60	0.64–0.67		End bonds: CH_2–Br, $m(CH_2\text{–Br}) = 1.96D$	70
Poly(thiopentamethylenethiomethylene)					Alternating copolymer of pentamethylene sulfide and methylene sulfide	
Poly(1,3-dithiocane)						
$M_n = 14000$	Benzene	20–60	0.26–0.31			112
Poly(thiomethylene-1,4-cyclohexylenemethylenethiomethylene) *trans*						
$M_n = 8000$	Benzene	30–60	0.176–0.207			54

TABLE 3. HYDROXYL TERMINATED OXIDE/SULFIDE COPOLYMERS[a]

Polymer	Solvent	Temp. (°C)	g or D_n	Remarks	Refs.
Poly(thioethyleneoxyethylene)				Alternating copolymer of ethylene sulfide and methylene sulfide	
Poly(thiodiethylene glycol)					
$M_n = 13000$	Benzene	20–60	0.607–0.648		66
$M_n = 8000$	Benzene	20–60	0.627–0.659		66
Poly(thiomethyleneoxymethylenethiopentamethylene)					
Poly(3-oxa-1,5-dithiadeca-methylene)					
$M_n = 5200$	Benzene	30–60	0.323–0.358		84
Poly(oxymethyleneoxyethylenethioethylene)					
Poly(1,3-dioxa-6-thiocane)					
$M_n = 2500$	Benzene	20–60	0.419–0.432		65
$M_n = 3000$	Benzene	20–60	0.419–0.432		65
Poly(oxymethyleneoxyethylenethiomethylenethioethylene)					
Alternating copolymer of *1,3-dioxolane and 1,3-dithiolane*					
	Benzene	30	0.26		66

[a] $m(CH_2\text{–}CH_2) = 0D$, $m(CH_2\text{–O}) = 1.07D$, $m(CH_2\text{–S}) = 1.21D$.

TABLE 4. POLYESTERS DERIVED FROM AROMATIC DIACIDS*

Polymer	Solvent	Temp. (°C)	g or D_n	Remarks	Refs.
Poly[di(oxyethylene) oxyterephthaloyl]					
Poly(diethyleneglycol terephthalate)					
$M_n = 2500$	Benzene	35–60	0.688–0.692		60
Poly[di(oxyethylene) oxyisophthaloyl]					
Poly(diethyleneglycol isophthalate)					
$M_n = 5900$	Benzene	30–60	0.691–0.692		79
Poly[di(oxyethylene) oxyphthaloyl)					
Poly(diethyleneglycol phthalate)					
$M_n = 10300$	Benzene	30–60	0.637–0.679		86
Poly(oxypropyleneoxyterephthaloyl)					
Poly(propyleneglycol terephthalate)					
$M_n = 11000$	Benzene	30–60	0.582–0.609		87
Poly[di(oxypropylene) oxyterephthaloyl)					
Poly(dipropyleneglycol terephthalate)					
$M_n = 10000$	Benzene	30–60	0.715–0.750		14
Poly(tri(oxytrimethylene) oxyterephthaloyl)					
Poly(triethyleneglycol terephthalate)					
$M_n = 7800$	Benzene	30–60	0.677–0.695		82
Poly[di(oxytrimethylene) oxyterephthaloyl]					
Poly(ditrimethyleneglycol terephthalate)					
$M_n = 15500$	Benzene	30–60	0.805–0.798		26
(Poly(thioethyleneoxyethyleneoxyterephthaloyl)					
Poly(thiodiethyleneglycol terephthalate)					
$M_n = 6400$	Dioxane	40–70	0.540–0.552	$m(CH_2\text{–S}) = 1.21D$	69

*Hydroxyl terminated polyesters $m(C_6H_5COOCH_2) = 1.98D$; $m(CH_2\text{–O})$ (ether) $= 1.07D$; $m(CH_2\text{–}CH_2) = 0D$; $m(O\text{–H}) = 1.7D$.

TABLE 5. ALIPHATIC AND CYCLOALIPHATIC POLYESTERS*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x$	Refs.
Poly[oxy (2,2-dimethyltrimethylene) oxysuccinoyl]					
Poly(neopentylglycol succinate)					
$M_n = 6900$	Benzene	30–60	0.548–0.586		88
Poly(oxysuccinoyloxymethylene-1,4-cyclohexylenemethylene) *trans*					
Poly(trans-1,4-cyclohexane dimethanol succinate)					
$M_n = 5800$	Dioxane	70	0.454		90
Poly[oxy(2,2-dimethyltrimethylene) oxyadipoyl]					
Poly(neopentylglycol adipate)					
$M_n = 5500$	Benzene	30–60	0.653–0.688		85
Poly(oxyadipoyloxymethylene-1,4-cyclohexylenemethylene) *cis*					
Poly(cis-1,4-cyclohexane dimethanol adipate)					
$M_n = 11000$	Benzene	30–60	0.950–0.972		90
$M_n = 11000$	Dioxane	30–60	0.914–0.959		90
Poly(oxyadipoyloxymethylene-1,4-cyclohexylenemethylene) *trans*					
Poly(trans-1,4-cyclohexane dimethanol adipate)					
$M_n = 10000$	Dioxane	50–70	0.596–0.65		90
Poly(oxysebacoyloxymethylene-1,4-cyclohexylenemethylene) *trans*					
Poly(trans-1,4-cyclohexane dimethanol sebacate)					
$M_n = 4600$	Benzene	30–60	0.631–0.709		83
Poly[oxy(2,2-dimethyltrimethylene) oxyhexafluoroglutaroyl]					
Poly(neopentylglycol hexafluoroglutarate)					
$M_n = 3640$	Benzene	30–60		9.16–9.78	58
Poly[oxy(2,2-dimethyltrimethylene) oxyglutaroyl]					
Poly(neopentylglycol glutarate)					
$M_n = 4500$	Benzene	30–60		4.26–4.52	58
Poly(oxysebacoyloxymethylene-1,4-cyclohexylenemethylene) *cis*					
Poly(cis-1,4-cyclohexane dimethanol sebacate)					
$M_n = 5200$	Benzene	300–60	0.983–1.04		83
Poly[di(oxyethylene) oxycarbonyl-1,4-cyclohexylenecarbonyl] *trans*					
Poly(diethyleneglycol-1,4-trans-cyclohexane dicarboxylate)					
$M_n = 5300$	Benzene	30–60	0.585–0.608		89

* Hydroxyl terminated polyesters $m(CH_2COOCH_2) = 1.89D$.

TABLE 6. POLYCARBONATES

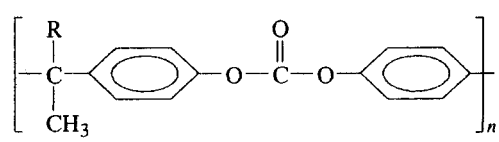
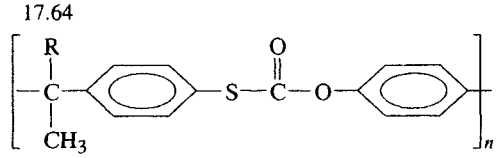
Polymer	Solvent	Temp. (°C)	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
Poly(oxycarbonyloxy-1,4-phenylene-1-(R)-1-methylmethylene-1,4-phenylene)					
Polycarbonate					
$x = 97$	Benzene	25	9.36	R = 3-chlorophenyl	28
$x = 88$	Benzene	25	11.70	R = 4-chlorophenyl	28
$x = 73$	Benzene	25	17.64		28
Poly(thiocarbonate)					
Poly(thiocarbonate)					
$x = 128$	Benzene	25	1.69	R = isobutyl	21
$x = 75$	1,4-Dioxane	25	1.88	R = H	21
$x = 280$	1,4-Dioxane	25	2.07	R = phenyl	21
$x = 75$	Benzene	25	2.76	R = H	21
$x = 49$	Benzene	25	2.13	R = ethyl	21
Poly(thiocarbonate of 2,2/-bis(4-hydroxylphenyl)propane)					
Poly(thiocarbonate of 2,2/-bis(4-hydroxylphenyl)propane)					
$M_n = 9400$	1,4-Dioxane	25	0.92		97
Poly(thiocarbonate of 1,1/-bis(4-hydroxyphenyl)-1-(3-chlorophenyl)ethane)					
Poly(thiocarbonate of 1,1/-bis(4-hydroxyphenyl)-1-(3-chlorophenyl)ethane)					
With $x = 97$	Benzene	25	7.40		99,28
Poly(thiocarbonate of 1,1/-bis(4-hydroxyphenyl)-1-(4-chlorophenyl)ethane)					
Poly(thiocarbonate of 1,1/-bis(4-hydroxyphenyl)-1-(4-chlorophenyl)ethane)					
With $x = 88$	Benzene	25	6.05		99,28

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TABLE 6. *cont'd*

Polymer	Solvent	Temp. (°C)	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
Poly(thiocarbonate of 1,1/-bis(4-hydroxyphenyl)-1-(3,4-dichlorophenyl)ethane)					
Poly(thiocarbonate of 1,1/-bis(4-hydroxyphenyl)-1-(3,4-dichlorophenyl)ethane)					
With $x = 73$	Benzene	25	12.25		99,28

TABLE 7. VINYL POLYMERS*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
Poly(1-phenylethylene)						
Poly(styrene)						
Atactic	Toluene	38.4	0.36		$\mu_0 = 0.60D$	35
Atactic	Carbon tetrachloride	25	0.56		$\mu_0 = 0.35D$	37
Isotactic	Toluene	38.4	0.53		$\mu_0 = 0.60D$	35
Poly(2-bromophenylethylene)						
Poly(o-bromostyrene)			0.50			57
			1.21			57
Poly(4-bromophenylethylene)						
Poly(p-bromostyrene)						
PPBSI	*p*-Dioxane	20–50		2.20–2.49		108
PPBSI	Carbon tetrachloride	20–50		2.12–2.54		108
PPBSII	*p*-Dioxane	20–50		2.44–2.72		108
Poly(1-phenylethylene/(4-pyridyl) ethylene)						
Poly(styrene/4-vinylpyridine)	Benzene	25	0.73		$\mu_0 = 2.65D$	5
Poly[1-methyl-1-(4-fluorophenyl) ethylene]						
Poly(p-fluoromethylstyrene)	Benzene	25	0.62		$\mu_0 = 1.82D$	6
	Benzene	25	0.27	0.89	$\mu_0 = 1.82D$	5
Poly[(4-methoxyphenyl) ethylene]						
Poly(p-methoxystyrene)	Toluene	25	0.54	0.78	$\mu_0 = 1.21D$	31
	Benzene	25	0.72	1.06	$\mu_0 = 1.21D$	5
	Benzene	25	0.64	0.94	$\mu_0 = 1.21D$	6
Poly[(4-fluorophenyl) ethylene]						
Poly(p-fluorostyrene)	Benzene	25	0.62	2.05	$\mu_0 = 1.82D$	6
	Benzene	25	0.51	1.69	$\mu_0 = 1.82D$	5
Poly[(2-chlorophenyl) ethylene]						
Poly(o-chlorostyrene)						
				2.85		57
Poly[(3-chlorophenyl) ethylene]						
(Poly(m-chlorostyrene)	Toluene	25	0.63	1.61	$\mu_0 = 1.60D$	93
Poly[(4-chlorophenyl) ethylene]						
Poly(p-chlorostyrene)	Xylene	30	0.71	2.00	$\mu_0 = 1.68$	11
		30–50	0.42–0.56			57
	Benzene	30	0.56	2.10	$\mu_0 = 1.93$	12
	Carbon tetrachloride	50		1.89		108
	p-Dioxane	20–50		2.11–2.46		108
	Toluene	25		1.82		113
	Cumene	25–65		1.87–1.73		113
	Toluene	30	0.77	2.17	$\mu_0 = 1.68$	11
	Isopropylbenzene	30–90	0.73–0.70	2.06–1.98	$\mu_0 = 1.68$	9
	Toluene	30–60	0.74–0.75	2.09–2.12	$\mu_0 = 1.68$	9
	Benzene	25	0.64		$\mu_0 = 2.0$	5
	Toluene	25		1.65		31
	Isopropylbenzene	30	0.53	1.50	$\mu_0 = 1.68$	11
	n-Propylbenzene	20–90	0.436–0.499	1.641–1.878		114
	Carbon tetrachloride	25	0.57		$\mu_0 = 1.88D$	6
	Benzene	30		1.49		55
	Benzene	30		1.49	Isotactic	55
	Toluene	25	0.60	1.59	$\mu_0 = 1.63$	93
	Isopropylbenzene	30–90	0.493–0.520	11.854–1.956		114
	Benzene	15–70	0.599–0.562	2.373–2.224		114

*In vinyl polymers the dipolar correlation coefficient is expressed by $g = \langle\mu^2\rangle/x\mu_0^2$, where x is the degree of polymerization and μ_0 is the dipole moment of a model compound of the repeating unit.

TABLE 7. *cont'd*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
Poly(1-acetoxyethylene)						
Poly(vinyl acetate)	Benzene and CCl_4	25	0.89 and 0.94	2.59–2.89	$\mu_0 = 1.80D$	38
	Benzene	20	0.84	2.89	$\mu_0 = 1.86$	101
			0.75–0.80			107
Poly(1-bromoethylene)						
Poly(vinyl bromide)						
Several fractions	*p*-Dioxane	25		1.97–3.20	$\mu_0 = 1.93D$	40
	p-Dioxane	30–60	0.53–0.86	2.29–2.38	$\mu_0 = 2.08D$	95
	1-Methylnaphthalene	30–60	0.45–0.46		$\mu_0 = 2.08D$	95
Poly(1-chloroethylene)						
Poly(vinyl chloride)						
Atactic fraction	*p*-Dioxane	20 and 40	0.59	2.65	$\mu_0 = 2.12D$	33
	p-Dioxane	20–65	0.70–0.75	2.79–3.06	$\mu_0 = 2.00D$	29
Atactic fraction	Tetrahydrofuran	40		1.716	$\mu_0 = 1.64D$	33
Atactic fraction	Tetrahydrofuran	20 and 40	0.64–0.62	1.72–1.67	$\mu_0 = 1.64D$	33
Eight fractions (atactic)	*p*-Dioxane	25	0.66–0.72	2.59–2.82	$\mu_0 = 2.00D$	39
Syndiotactic fraction	Tetrahydrofuran	20	0.71	1.93	$\mu_0 = 1.64D$	33
$M_n = 26000$ (atactic)	*p*-Dioxane	25–55	0.68–0.65	2.72–2.60	$\mu_0 = 2.00D$	8
			0.75			12
$M_n = 47000$ (atactic)	*p*-Dioxane	25–55	0.70–0.68	2.80–2.72	$\mu_0 = 2.00D$	8
Poly(isobutoxyethylene)						
Poly(vinyl isobutylether)						
Isotactic fraction	Benzene	25 and 50	0.90 and 0.98	1.345 and 1.464	$\mu_0 = 1.22D$	107
Atactic fraction	Benzene	25	0.77 and 0.83		$\mu_0 = 1.22D$	107
Isotactic fraction	Benzene	30		0.98		55
Atactic fraction	Benzene	30	0.98	0.96		55
Poly(carbazolylethylene)						
Poly(vinyl carbazole)	Toluene	25	0.36	3.71	$\mu_0 = 3.21D$	30
	p-Dioxane	30–60	0.41–0.44	3.52–3.78	$\mu_0 = 2.93D$	102
Poly(ethylenethiophenylene)						
Poly(vinylphenyl sulfide)	Benzene	20–50	1.10–1.06	2.34–2.26	$\mu_0 = 1.46D$	100
Poly[1-methyl-1-(4-bromophenyl) ethylene]						
Poly(p-bromo-α-methylstyrene)	Benzene	25	0.28	1.21	$\mu_0 = 2.08D$	6

TABLE 8. ACRYLIC POLYMERS*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
Poly[1-methoxycarbonyl) ethylene]						
Poly(methyl acrylate)	Benzene	20	0.717	2.22	$\mu_0 = 1.76D$	51
	Benzene	25	0.64–0.67	1.99–2.07	$\mu_0 = 1.76D$	43
Poly[2-(biphenyloxycarbonyl) ethylene]						
Poly(2-biphenyl acrylate)	Benzene	30–60	0.69–0.74	2.05–2.22	$\mu_0 = 1.723–1.732D$	16
Poly[1-(cyclohexyloxycarbonyl) ethylene]						
Poly(cyclohexyl acrylate)	Benzene	30–60	0.75–0.81	2.79–3.01	$\mu_0 = 1.93D$	15
Poly[1-(ethoxycarbonyl) ethylene]						
Poly(ethyl acrylate)						
Atactic fraction	Benzene	30	1.04			55
Syndiotactic fraction	Benzene	30	1.04			55
Poly[1-(phenoxycarbonyl) ethylene]						
Poly(phenyl acrylate)	Benzene	30–60	0.62–0.66	1.73–1.91	$\mu_0 = 1.669–1.700D$	98
Poly[1-((3-chloro) phenoxycarbonyl) ethylene]						
Poly(m-chlorophenyl acrylate)	Benzene	30–60	0.57–0.61	1.85–2.02	$\mu_0 = 1.800–1.824D$	98
Poly[1-((2-chloro) phenoxycarbonyl) ethylene]						
Poly(o-chlorophenyl acrylate)	Benzene	30–60	1.09–1.09	3.30–3.97	$\mu_0 = 1.74–1.91D$	98
Poly[1-((4-chloro) phenoxycarbonyl) ethylene]						
Poly(p-chlorophenyl acrylate)	Benzene	30–60	0.524–0.501		$\mu_0 = 2.590–2.694D$	98
Poly[1-((4-benzoyl) phenoxycarbonyl) ethylene]						
Poly(4-benzoylphenyl acrylate)	1,4-Dioxane	30–60	0.81–0.84	10.27–10.41	$\mu_0 = 3.561–3.521D$	17
Poly[1-(4-phenoxy) phenyloxycarbonyl) ethylene]						
Poly(4-phenoxyphenyl acrylate)	Benzene	30–60	0.82–0.84	3.61–3.74	$\mu_0 = 2.10–2.11D$	18
Poly[1-(hydroxyethylenedi(oxyethylene) oxycarbonyl) ethylene]						
Poly(triethyleneglycol acrylate)	Benzene	30–60		7.97–8.16		92

* μ_0 is the dipole moment of a model compound of the side group.

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TABLE 9. METHACRYLIC POLYMERS*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
Poly[1-methoxycarbonyl)-1-methylethylene]						
Poly(methyl methacrylate)						
Atactic fraction	Five solvents	25	0.62		$\mu_0 = 1.73D$	48
Atactic fraction	Benzene	25	0.53–0.66			57,107
Atactic fraction	Benzene	25	0.55–0.72	1.77–2.31	$\mu_0 = 1.80D$	42
Syndiotactic fraction	Toluene	30–90	0.67–0.79	1.72–2.01	$\mu_0 = 1.60D$	105
Isotactic fraction	Toluene	25–65	0.65–0.75	1.66–1.93	$\mu_0 = 1.60D$	105
Syndiotactic fraction	Benzene	25–65	0.70–0.78	1.79–1.99	$\mu_0 = 1.60D$	105
Atactic fraction	Benzene	25–65	0.69–0.78	1.77–1.99	$\mu_0 = 1.60D$	105
Isotactic fraction	Benzene	25–65	0.77–0.81	1.96–2.07	$\mu_0 = 1.60D$	105
Isotactic fraction	Benzene	30	0.67	2.00	$\mu_0 = 0.73D$	3
Atactic fraction	Benzene	30	0.56–0.61	1.66–1.82	$\mu_0 = 1.73D$	3
Atactic fraction	Toluene	25–65	0.66–0.78	1.69–1.99	$\mu_0 = 1.60D$	105
Syndiotactic fraction	Benzene	30	0.53	1.61	$\mu_0 = 1.73D$	3
Atactic fraction	*p*-Dioxane	23		2.25		10
Poly[1-(ethoxycarbonyl)-1-methylethylene]						
Poly(ethyl methacrylate)	1,4-Dioxane	25	0.58	2.00	$\mu_0 = 1.86D$	36
	Carbon tetrachloride	25	0.54	1.87	$\mu_0 = 1.86D$	36
	Toluene	25	0.62	2.14	$\mu_0 = 1.86D$	36
			0.59–0.62			57
	Benzene	25	0.58	2.00	$\mu_0 = 1.86D$	
Poly[1-(propoxycarbonyl)-1-methylethylene]						
Poly(propyl methacrylate)			0.56–0.59			57
Poly[1-(isopropoxycarbonyl)-1-methylethylene]						
Poly(isopropyl methacrylate)			0.61–0.66			57
Poly(butyl methacrylate)						
Poly(butyl methacrylate)				1.93–2.72		10
	Four solvents	23	0.55–0.59			57
Poly(phenylmethacrylate)						
Poly(phenylmethacrylate)			0.55			57
Poly(*p*-chlorophenylmethacrylate)						
Poly(p-chlorophenylmethacrylate)			0.35			57
Poly(dichlorophenylmethacrylate)						
Poly(dichlorophenylmethacrylate)			0.38			52

*μ_0 is the dipole moment of a model compound of the side group.

TABLE 10. ITACONATE POLYMERS

Polymer	Solvent	Temp. (°C)	$\langle\mu^2\rangle/x$, D^2	Refs.
Poly[1-(cyclohexylmethylene oxycarbonylmethylene)-1-(carboxy)ethylene]				
Poly(monocyclohexylmethylene itaconate)	1,4-Dioxane	20–50	5.75–5.70	20
Poly[1-(cyclohexylmethyleneoxycarbonylmethylene)-1-(cyclohexylmethyleneoxycarbonylmethylene) ethylene]				
Poly(dicyclohexylmethylene itaconate)	Benzene	30–60	6.89–7.00	19
Poly[1-benzyloxycarbonylmethylene)-1-(carboxy) ethylene]				
Poly(monobenzyl itaconate)	1,4-Dioxane	25	3.84	96
Poly[1-(benzyloxycarbonylmethylene)-1-(benzyloxycarbonylmethylene)ethylene]				
Poly(dibenzyl itaconate)	1,4-Dioxane	25–33	2.31–2.79	24

TABLE 11. POLYDIENES

Polymer	Solvent	Temp. (°C)	g or D_n	Remarks	Refs.
Poly(1-methyl-1-butenylene)					
Poly(isoprene)					
cis 1,4	Benzene	25	0.70	$\mu_0 = 0.34D$	41
trans 1,4	Benzene	25	0.82	$\mu_0 = 0.34D$	41
Poly(1-chloro-1-butenylene)					
Poly(chloroprene)	Benzene	20	2.70		101

TABLE 12. SILICON BASED POLYMERS[a]

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x, D^2$	Remarks	Refs.
Linear Poly(dimethylsiloxane)						
Linear Poly(dimethylsiloxane)						
$x+1=188$	Undiluted	10–60	0.306–0.297			44
$x=9.85$	Cyclohexane, benzene	30	0.18–0.21		Cyclic PDMS	76
$x=1$, 102	Undiluted	25			Linear PDMS; $\langle\mu^2\rangle^{1/2}=2.07D$, 14.4$D$	7
$x=4$, 95.0	Undiluted	25			Cyclic PDMS; $\langle\mu^2\rangle^{1/2}=1.50D$, 14.3$D$	7
$x=69.6$	Cyclohexane, benzene	30	0.20,0.33			76
$x=14.8$	Cyclohexane, benzene	30	0.21,0.28		Cyclic PDMS	76
$x=70.55$	Cyclohexane, benzene	30	0.21,0.30			76
$x=15.8$	Cyclohexane, benzene	30	0.19,0.26			76
$x=10.5$	Cyclohexane, benzene	30	0.18,0.27			76
$x+1=239$	Undiluted	10–60	0.305–0.297			44
$x+1=97$	Undiluted	10–60	0.314–0.306			44
$x+1=1038$	Cyclohexane	10–60	0.398–0.444			44
$x+1=239$	Cyclohexane	10–60	0.377–0.412			44
$x+1=188$	Cyclohexane	10–60	0.377–0.369			44
$x+1=97$	Cyclohexane	10–60	0.387–0.407			44
$x+1=1038$	Undiluted	10–60	0.308–0.301			44
Poly(1,1-dimethylsilazane)						
Poly(1,1-dimethylsilazane)	Cyclohexane	20–50		0.370–0.326		104
	Hexamethyldisilazane	25		0.246		104
$M_n=750$	Hexane	20		0.330		104
	Carbon tetrachloride	20–50		0.310–0.310		104
	Benzene	20–50		0.342–0.359		104
Poly(methylphenyl siloxane)						
Poly(methylphenyl siloxane)						
$M_n=45000$	Cyclohexane	25	0.31			103
Poly(dimethyl siloxane-*co*-methylphenyl siloxane) (35.7/54.3)						
Poly(dimethyl siloxane-co-methylphenyl siloxane) (35.7/54.3)						
$M_n=4200$	Cyclohexane	25	0.22			103
Poly(dimethylsiloxane-*co*-methylphenylsiloxane) (90.5/9.5)						
Poly(dimethylsiloxane-co-methylphenylsiloxane) (90.5/9.5)						
$M_n=6400$	Cyclohexane	25	0.27			103
Poly(methylsilmethylene)						
Poly(methylsilmethylene)	Cyclohexane	20	0.391		m(Si–C) = 0.50D	49

[a] m(Si–O) = 0.60D.

TABLE 13. STYRENE COPOLYMERS*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x, D^2$	Remarks	Refs.
Poly[1-phenylethylene-*co*-(4-methoxyphenyl) ethylene]						
Poly(styrene-co-p-methoxystyrene)						
$x_p=0.157$	Toluene	25	1.07	1.56	$\mu_0=1.21D$; $\langle\mu^2\rangle/x$ is referred to the polar unit	31 30
$x_p=0.328$	Toluene	25	0.84	1.23		31 30
$x_p=0.480$	Toluene	25	0.74	1.08		31 30
$x_p=0.721$	Toluene	25	0.69	1.01		31 30
$x_p=1$	Toluene	25	0.54	0.78		31 30
Poly(1-phenylethylene-*co*-carbazolylethylene)						
Poly(styrene-co-vinylcarbazole)						
$x_p=0.074$	Toluene	25	0.53		$\mu_0=3.21D$	30
$x_p=1.00$	Toluene	25	0.36			30
$x_p=0.368$	Toluene	25	0.53			30
$x_p=0.665$	Toluene	25	0.39			30
Poly[1-phenylethylene-*co*-(4-pyvidyl) ethylene]						
Poly(styrene-co-4-vinylpyridine)						
$x_p=0.028$	Toluene	25	0.31		$\mu_0=2.655D$	30
$x_p=0.183$	Toluene	25	0.50			30
$x_p=0.154$	Toluene	25	0.34			30

* x_p is the molar fraction of polar comonomer and μ_0 is the dipole moment of the polar repeating unit.

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TABLE 13. *cont'd*

Polymer	Solvent	Temp. (°C)	g or D_n	$\langle\mu^2\rangle/x$, D^2	Remarks	Refs.
$x_p = 0.118$	Toluene	25	0.29			30
$x_p = 0.051$	Toluene	25	0.29			30
$x_p = 0.082$	Toluene	25	0.23			30
Poly[1-phenylethylene-*co*-(4-chlorophenyl) ethylene]						
Poly(styrene-co-p-chlorostyrene)						
$x_p = 0.247$	Toluene	25	0.67	2.67	$\mu_0 = 2.0D$	31,30
$x_p = 0.802$	Toluene	25	0.44	1.76		31,30
$x_p = 1.00$	Toluene	25	0.41	1.4		31,30

C. REFERENCES

1. A. Abe, T. Hirano, T. Tsuruta, Macromolecules, **12**, 1092 (1979).
2. M. Aroney, R. J. W. Le Fèvre, G. M. Parkins, J. Chem. Soc. 2890 (1960).
3. R. Bacskai, H. A. Pohl, J. Polym. Sci., **42**, 151 (1960).
4. K. Bak, G. Elefante, J. E. Mark, J. Phys. Chem., **71**, 4007 (1967).
5. B. M. Baysal, H. Yu, W. H. Stockmayer, "Dielectric Dispersion in Dilute Solutions of Several Parasubstituted Polystyrenes", in: F. E. Karasz (Ed.), "Dielectric Properties of Polymers", 1972: p. 329. Plenum Press, New York, 1971.
6. B. M. Baysal, L. Aras, Macromolecules, **18**, 1693 (1985).
7. M. S. Beevers, S. J. Mumby, S. J. Carlson, A. J. Semlyen, Polymer, **24**, 1565 (1983).
8. F. Blasco-Cantera, E. Riande, J. P. Almendro, E. Saiz, Macromolecules, **14**, 138 (1981).
9. F. Blasco, E. Riande, J. Polym. Sci., Polym. Phys. (Ed.), **21**, 835 (1983).
10. L. de Brouckère, M. Mandel, "Dielectric Properties of Dilute Polymer Solutions", in: I. Prigogine I(Ed.), "Advances in Chemical Physics", Interscience, New York, 1958, p. 77.
11. L. L. Burshtein, T. P. Stepenova, Polym. Sci. USSR, **11**, 2885 (1969).
12. P. Debye, F. Bueche, J. Chem. Phys., **19**, 589 (1951).
13. P. Debye, Phys. Z., **13**, 97 (1912); "Collected papers, Polar Molecules", Wiley Interscience, New York, 1954, p. 173.
14. R. Díaz-Calleja, E. Riande, J. Guzmán, Macromolecules, **22**, 3654 (1989).
15. R. Díaz-Calleja, E. Riande, J. San Román, Macromolecules, **25**, 2875 (1992).
16. R. Díaz-Calleja, E. Riande, J. San Román, J. Phys. Chem., **96**, 6843 (1992).
17. R. Díaz-Calleja, E. Riande, J. San Román, J. Phys. Chem., **97**, 4848 (1993).
18. R. Díaz-Calleja, E. Riande, J. San Román, Polymer, **34**, 3456 (1993).
19. R. Díaz-Calleja, E. Riande, L. Gargallo, D. Radic, Macromolecules, **26**, 3795 (1993).
20. R. Díaz-Calleja, E. Riande, L. Gargallo, D. Radic, J. Polym. Sci., Part B: Polym. Phys., **32**, 1069 (1994).
21. M. J. Fabre, I. Hernandez-Fuentes, L. Gargallo, D. Radic, Polymer, **33**, 194 (1992).
22. P. J. Flory, "Statistical Mechanics of Chain Molecules", Wiley-Interscience, New York, 1969.
23. H. Fröhlich, Trans. Faraday Soc., **44**, 238 (1948); "Theory of Dielectrics", Oxford University Press, London, 1958.
24. L. Gargallo, M. Yazdani-Pedram, D. Radic, I. Hernandez-Fuentes, Makromol. Chem., **189**, 145 (1988).
25. L. Garrido, E. Riande, J. Guzmán, J. Polym. Sci., Polym. Phys. Ed., **20**, 1805 (1982).
26. C. González, E. Riande, A. Bello, J. M. Pereña, Macromolecules, **21**, 3230 (1988).
27. E. A. Guggenheim, Trans. Faraday Soc., **45**, 714 (1949); *ibid.* **47**, 573 (1951).
28. I. Hernandez-Fuentes, F. Rey-Stolle, L. Hernán Tagle, E. Saiz, Macromolecules, **25**, 3291 (1992).
29. Y. Imamura, J. Chem. Soc. Japan, **76**, 217 (1955).
30. Z. Kücükyavuz, B. M. Baysal, Macromolecules, **21**, 2287 (1988).
31. K. Kücükyavuz, B. M. Baysal, Polymer, **32**, 3418 (1991).
32. J. G. Kirkwood, J. Chem. Phys., **7**, 911 (1939).
33. A. Kotera, M. Shima, N. Fujisaki, T. Kobayashi, Bull. Chem. Soc. Japan, **35**, 1117 (1962).
34. A. Kotera, K. Suzuki, K. Matsumura, T. Nakano, Y. Oyama, U. Kabayashi, Bull. Chem. Soc. Japan, **35**, 797 (1962).
35. W. R. Krigbaum, A. Roig, J. Chem. Phys., **31**, 544 (1959).
36. A. Kuntman, I. Bahar, B. M. Baysal, Macromolecules, **23**, 4959 (1990).
37. C. G. Le Fèvre, R. J. W. Le Fèvre, G. M. Parkins, J. Chem. Soc., 1468 (1958).
38. C. G. Le Fèvre, R. J. W. Le Fèvre, G. M. Parkins, J. Chem. Soc., 1814 (1960).
39. R. J. W. Le Fèvre, K. M. S. Sundaram, J. Chem. Soc., 1494 (1962).
40. R. J. W. Le Fèvre, K. M. S. Sundaram, J. Chem. Soc., 4003 (1962).
41. R. J. Le Fèvre, K. M. S. Sundaram, J. Chem. Soc., 3547 (1963).
42. R. J. W. Le Fèvre, K. M. S. Sundaram, J. Chem. Soc., 1880 (1963).
43. R. J. W. Le Fèvre, K. M. S. Sundaram, J. Chem. Soc., 3188 (1963).
44. S. C. Liao, J. E. Mark, J. Chem. Phys., **59**, 3825 (1973).
45. G. D. Loveluck, J. Chem. Soc., 4729 (1961).
46. V. Magnasco, G. Dellepiane, C. Rossi, Makromol. Chem., **65**, 16 (1963).

47. J. Marchal, H. Benoit, J. Polym. Sci., **23**, 223 (1957).
48. J. Marchal, C. Lapp, J. Polym. Sci., **27**, 571 (1958).
49. J. E. Mark, J. H. Ko Macromolecules, **8**, 869 (1975).
50. J. E. Mark, D. S. Chiu, J. Chem. Phys., **66**, 1901 (1977).
51. R. M. Masegosa, I. Hernandez-Fuentes, E. A. Ojalvo, E. Saiz, Macromolecules, **12**, 862 (1979).
52. G. P. Mikhailov, J. Polym. Sci., **30**, 605 (1958).
53. L. Onsager, J. Am. Chem. Soc., **7**, 911 (1936).
54. J. L. de la Peña, E. Riande, J. Guzmán, Macromolecules, **18**, 2739 (1985).
55. H. A. Pohl, H. H. Zabusky, J. Phys. Chem., **66**, 1390 (1962).
56. C. H. Porter, J. H. Lawler, R. H. Boyd, Macromolecules, **3**, 308 (1970).
57. O. B. Ptitsyn, Soviet Phys.-Usp. (English transl.), **2**, 797 (1960).
58. R. C. Reis-Nunes, E. Riande, N. Chavez, J. Guzmán, Macromolecules, **29**, 7989 (1996).
59. E. Riande, J. Polym. Sci., Polym. Phys. Ed., **14**, 2231 (1976).
60. E. Riande, J. Polym. Sci., Polym. Phys. Ed., **15**, 1397 (1977).
61. E. Riande, Makromol. Chem., **178**, 2001 (1977).
62. E. Riande, J. E. Mark, Macromolecules, **11**, 956 (1978).
63. E. Riande, J. E. Mark, J. Polym. Sci., Polym. Phys. Ed., **17**, 2013 (1979).
64. E. Riande, J. E. Mark, Polymer, **20**, 1188 (1979).
65. E. Riande, J. Guzmán, Macromolecules, **12**, 1117 (1979).
66. E. Riande, J. Guzmán, Macromolecules, **12**, 952 (1979).
67. E. Riande, S. Boileau, P. Hemery, J. E. Mark, J. Chem. Phys., **71**, 4206 (1979).
68. E. Riande, S. Boileau, P. Hemery, J. E. Mark, Macromolecules, **12**, 702 (1979).
69. E. Riande, J. Guzmán, J. San Román, J. Chem. Phys., **72**, 5263 (1980).
70. E. Riande, J. Guzmán, Macromolecules, **14**, 1234 (1981).
71. E. Riande, J. Guzmán, E. Saiz, Polymer, **22**, 465 (1981).
72. E. Riande, J. Guzmán, E. Saiz, J. de Abajo, Macromolecules, **14**, 608 (1981).
73. E. Riande, M. García, J. Mark, J. Polym. Sci., Polym. Phys. Ed., **19**, 1739 (1981).
74. E. Riande, J. Guzmán, W. J. Welsh, J. E. Mark, Makromol. Chem., **183**, 2555 (1982).
75. J. Guzmán, E. Riande, W. J. Welsh, J. E. Mark, Makromol. Chem., **183**, 2573 (1982).
76. E. Riande, J. E. Mark, Eur. Polym. J., **20**, 517 (1984).
77. E. Riande, J. G. de la Campa, J. Guzmán, J. de Abajo, Macromolecules, **17**, 1431 (1984).
78. E. Riande, J. G. de la Campa, J. Guzmán, J. de Abajo, Macromolecules, **17**, 1891 (1984).
79. E. Riande, J. Guzmán, J. de Abajo, Makromol. Chem., **185**, 1943 (1984).
80. E. Riande, J. Guzmán, L. Garrido, Macromolecules, **17**, 1234 (1984).
81. E. Riande, J. Guzmán, L. Garrido, Macromolecules, **17**, 134 (1984).
82. E. Riande, J. Guzmán, J. Polym. Sci., Polym. Phys. Ed., **23**, 1235 (1985).
83. E. Riande, J. Guzmán, J. G. de la Campa, J. de Abajo, Macromolecules, **18**, 1583 (1985).
84. E. Riande, J. Guzmán, H. Addabo, Macromolecules, **19**, 2567 (1986).
85. E. Riande, J. E. Mark, Macromolecules, **19**, 2956 (1986).
86. E. Riande, J. Guzmán, J. G. de la Campa, J. de Abajo, J. Polym. Sci., Polym. Phys. Ed., **25**, 2403 (1987).
87. E. Riande, J. G. de la Campa, D. D. Schlereth, J. de Abajo, J. Guzmán, Macromolecules, **20**, 1641 (1987).
88. E. Riande, J. Guzmán, J. Chem. Soc. Perkin Trans., **2**, 299 (1988).
89. E. Riande, J. Guzmán, J. G. de la Campa, Macromolecules, **21**, 2128 (1988).
90. E. Riande, J. Guzmán, J. de Abajo, Macromolecules, **22**, 4026 (1989).
91. E. Riande, E. Saiz, "Dipole Moments and Birefringence of Polymers", Prentice Hall, Englwood Cliff, NJ, 1992.
92. E. Riande, J. Guzmán, Macromolecules, **29**, 1728 (1996).
93. A. Roig, I. Hernandez-Fuentes, An. Quím., **70**, 668 (1974).
94. E. Saiz, E. Riande, J. Guzmán, J. de Abajo, Chem. Phys., **73**, 958 (1980).
95. E. Saiz, E. Riande, M. P. Delgado, J. M. Barrales-Rienda, Macromolecules, **15**, 1152 (1982).
96. E. Saiz, A. Horta, L. Gargallo, I. Hernandez-Fuentes, D. Radic, Macromolecules, **21**, 1736 (1988).
97. E. Saiz, M. J. Fabre, L. Gargallo, D. Radic, I. Hernandez-Fuentes, Macromolecules, **22**, 3660 (1989).
98. E. Saiz, E. Riande, J. San Román, E. L. Madruga, Macromolecules, **23**, 785 (1990).
99. E. Saiz, C. Abradelo, J. Mogin, L. Hernán Tagle, I. Hernandez-Fuentes, Macromolecules, **24**, 5594 (1991).
100. E. Saiz, C. Mijangos, E. Riande, D. López, Macromolecules, **27**, 5716 (1994).
101. I. Sakurada, S. Lee, Z. Phys. Chem. B, **43**, 245 (1939).
102. M. Salmerón, J. M. Barrales-Rienda, E. Riande, E. Saiz, Macromolecules, **17**, 2728 (1984).
103. C. Salom, J. J. Freire, I. Hernandez-Fuentes, Polymer ., **20**, 1109 (1988).
104. C. Salom, E. Riande, I. Hernández-Fuente, R. Díaz-Calleja, J. Polym. Sci., Part B: Polym. Phys., **31**, 1591 (1993).
105. R. Salovey, J. Polym. Sci., **50**, S7 (1961).
106. E. A. Smith, Trans. Faraday Soc., **46**, 394 (1950).
107. M. Takeda, Y. Imamura, S. Okamura, T. Higashimura, J. Chem. Phys., **33**, 631 (1960).
108. A. E. Tonelli, L. A. Belfiore, Macromolecules, **16**, 1740 (1983).
109. T. Uchida, Y. Kurita, M. Kubo, J. Polym. Sci., **19**, 365 (1956).
110. T. Uchida, Y. Kurita, N. Koizumi, M. Kubo, J. Polym. Sci., **21**, 313 (1956).
111. S. Vega, E. Riande, J. Guzmán, Macromolecules, **23**, 3573 (1990).
112. W. J. Welsh, J. E. Mark, J. Guzmán, E. Riande, Makromol. Chem., **183**, 2562 (1982).
113. N. Yamaguchi, M. Sato, E. Ogawa, M. Shima, Polymer, **22**, 1464 (1981).
114. F. Yilmaz, B. M. Baysal, J. Polym. Sci., Part B: Polym. Phys., **30**, 197 (1992).

Heat, Entropy, and Volume Changes for Polymer–Liquid Mixtures

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A. INTRODUCTION

Tabulated below are references to the studies of the enthalpy, entropy, and volume changes that occur when polymer and solvent are mixed to form a homogeneous liquid solution. This bibliography follows closely the form employed by C. Booth (Manchester) in his contribution under the same title in the second edition of this Handbook. With several exceptions, references to work reported before 1972 were compiled by Professor Booth for the second edition.

1. Scope of the Table

This table includes literature citations to studies from which the thermodynamic properties named in the title can be deduced. The experimental methods utilized are also identified.

This tabulation is limited to reports of measurements on solutions that contain only one nonelectrolyte homopolymer; solutions with a copolymer, polyelectrolyte, or polymer mixture have been excluded. Solvents for the solutions in this compilation are liquids comprising single molecular entities; i.e., references to studies in which the solvent is a binary or ternary mixture or a mixture of oligomers are not included.

Most studies whose primary goal is to characterize the individual polymer molecule and its interaction in an infinitely dilute solution will not be found in this table. However, references to studies of dilute solutions are presented in some of the other tables of Part VII ("Solution Properties"). Most of the reports of changes in free energy on mixing or on dilution cited here have been limited to those that also provide sufficient data to determine the change in the enthalpy (or entropy). References to measurements of solvent activities which can yield information about changes in free energy can be found in "Polymer–Solvent Interaction Parameters".

These references were collected without regard for the quality of data.

2. Experimental Methods

Heat Several different kinds of enthalpy changes are recognized in the formation of solutions. The *integral heat of mixing or integral heat of solution,*

$$\Delta H_{\text{mix}} = H_{\text{soln}} - n_1 H_1^0 - n_2 H_2^0 \qquad \text{(A1)}$$

is the heat absorbed when pure solvent and pure polymer are combined to form a solution at constant temperature and pressure. Here H_{soln}, H_1^0, and H_2^0 are the molar enthalpies of solution, pure solvent, and pure solute respectively; and n_1 and n_2 the number of moles of each component. The value of ΔH_{mix} is dependent on the degree of crystallinity for semicrystalline polymers and, usually to a lesser extent, on the thermal history of glassy polymers. However, if a

polymer solution already exists and more solvent is added, the accompanying enthalpy change, which is called the *integral heat of dilution,* is independent of any crystalline or glassy aspect of the pure polymeric solute. The integral heats, both of mixing and of dilution, are usually determined calorimetrically. The latter is often preferable for experimental reasons, because its measurement is not complicated by the long dissolution times usually required in the direct measurements of integral heats of solution.

The *partial* or *differential heat of solution,* $\overline{\Delta H_2}$, is the change in enthalpy when a very small amount of pure solute is added to a large amount of either solution or pure solvent. In the latter instance, the resultant quantity which is properly identified as the *partial heat of solution at infinite dilution* is sometimes referred to more simply as the heat of solution. For polymer solutions, $\overline{\Delta H_2}$ is often expressed as the heat absorbed per unit mass of solute added and can be found as the derivative of the integral heat of mixing:

$$\overline{\Delta H_2} = (\partial \Delta H_{\text{mix}}/\partial m_2)_{T,p,n_1} \qquad \text{(A2)}$$

Here m_2 is the mass of polymeric solute. Values of $\overline{\Delta H_2}$ at infinite dilution for some polymer–solvent systems are listed in "Heats of Solution of Some Common Polymers".

The *partial or differential heat of dilution* $\overline{\Delta H_1}$ differs from $\overline{\Delta H_2}$ in that it is a consequence of adding solvent, rather than solute, to a solution and is almost always expressed as the heat per mol of added solvent. Accordingly,

$$\overline{\Delta H_1} = (\partial \Delta H_{\text{mix}}/\partial n_1)_{T,p,m_2} = (\partial H_{\text{soln}}/\partial n_1)_{T,p,m_2} - H_1^0 \qquad \text{(A3)}$$

It can also be obtained from the variation with concentration of the integral heat of dilution. Or it may be determined from the temperature dependence of the activity a_1 of solvent according to the Gibbs–Helmholtz equation

$$\Delta H_1 = \mathrm{R}[\partial \ln a_1/\partial(1/T)]_{p,n_1,m_2} \qquad \text{(A4)}$$

I turn, the activity a_1 of solvent in a polymer solution has a variety of sources including the osmotic pressure of the solution, the vapor pressure of solvent above the solution relative to that of the pure solvent, the gas-chromatographic retention volumes of volatile solvent on columns with polymeric solute as the stationary phase, swelling equilibria for polymer networks, the concentration dependence of Rayleigh scattering, and sedimentation equilibria. The first three methods are the ones used most frequently.

Entropy The entropy of mixing pure polymer with pure solvent can be obtained from the difference between the integral heat of mixing and the free energy of mixing

$$\Delta S_{\text{mix}} = (\Delta H_{\text{mix}} - \Delta G_{\text{mix}})/T \qquad \text{(A5)}$$

Free energies of mixing can be determined by integrating concentration-dependent functions of the chemical potential of the solvent.

More often, the partial molar entropy of solvent

$$(\partial \Delta S_{\text{mix}}/\partial n_1)_{T,p,m_2} = \overline{\Delta S_1} \qquad \text{(A6)}$$

is reported, using the temperature dependence of the activity of solvent

$$\overline{\Delta S_1} = -R[\ln a_1 + (\partial \ln a_1/\partial T)_{p,n_1,m_2}] \qquad \text{(A7)}$$

Volume Volume changes reported in the literature cited below are measured directly (dilatometry) or from density determinations.

B. KEY FOR THE TABLES

The following abbreviations are used to identify the experimental methods utilized for the studies cited:

CD	Calorimetry: integral heat of dilution
CM	Calorimetry: integral heat of mixing
D	Density
DL	Dilatometry
GC	Gas chromatography with polymeric stationary phase
LS	Rayleigh light scattering
OS	Osmotic pressure
SD	Sedimentation equilibrium
SW	Swelling of polymer network
VP	Vapor pressure including manometry, isopiestic distillation, vapor phase osmometry, etc.
XS	X-ray scattering

A Chemical Abstracts Service registry number accompanies most of the polymers named in the table. Although some polymers have more than one registry number owing to particular stereochemistries, different terminal groups, etc., only the most common number appears in the table for each polymer.

C. TABLES OF LITERATURE REFERENCES FOR HEAT, ENTROPY, AND VOLUME CHANGES FOR POLYMER–SOLVENT MIXTURES

TABLE 1. MAIN-CHAIN CARBON POLYMERS

Polymer–Solvent	Method	Refs.
1.1. POLY(DIENES)		
Poly(1,3-butadiene) [9003-17-2]		
Benzene	CD	140
	CM	19,84
	GC	308,327
	VP	73
cis-Bicyclo[4.4.0]decane	CM	293
trans-Bicyclo[4.4.0]decane	CM	293
1,3-Butadiene	DL	374
	GC	327
Butane	DL	374
	GC	327

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Poly(1,3-butadiene) *cont'd*		
1-Butene	DL	374
cis-2-Butene	DL	374
trans-2-Butene	DL	374
Carbon dioxide	SW	350
Carbon tetrachloride	GC	308
1-Chlorobutane	GC	308
Chloroform	GC	308
	VP	119
Cyclohexane	CM	293
	GC	327
Cyclooctane	CM	293
Cyclopentane	CM	293
	GC	327
Decane	GC	327
3,3-Diethylpentane	CM	293
2,2-Dimethylbutane	GC	327
2,2-Dimethylpentane	CM	293
2,3-Dimethylpentane	CM	293
2,4-Dimethylpentane	CM	293
3,3-Dimethylpentane	CM	293
1,4-Dioxane	VP	165
Dodecane	CM	294
	GC	327
Ethane	SW	350
Ethylbenzene	GC	327
Ethylene	SW	350
3-Ethylpentane	CM	293
2,2,4,4,6,8,8-Heptamethylnonane	CM	294
Heptane	CM	294
	GC	327
Hexadecane	CM	294
Hexane	GC	327
2-Methylbutane	GC	327
3-Methylhexane	CM	293
2-Methylpropane	DL	374
	GC	327
2-Methylpropene	DL	374
Nonane	CM	294
	GC	327
Octane	CM	294
	GC	308,327
1-Octene	GC	308
2,2,4,6,6-Pentamethylheptane	CM	294
Pentane	GC	308,327
2,2,4,4-Tetramethylpentane	CM	293,294
2,3,3,4-Tetramethylpentane	CM	293
Toluene	CD	140
	GC	308,327
1,3,5-Trimethylbenzene	GC	327
2,2,4-Trimethylpentane	CM	294
m-Xylene	GC	327
Water	SW	388
Poly(chloroprene) [9010-98-4]		
Benzene	CM	19
	GC	308
Butyl alcohol	GC	308
Carbon tetrachloride	GC	308
2-Chloro-1,3-butadiene	VP	126
1-Chlorobutane	GC	308
Chloroform	GC	308
Decane	GC	308
Octane	GC	308
1-Octene	GC	308
2-Pentanone	GC	308

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Poly(*cis*-1,3-isoprene) [9003-31-0] and [104389-31-3]		
Acetone	SW	9,120
	VP	120
Benzene	CM	6,8,55
	DL	5,166
	OS	5,166
	SW	9,121,130
	VP	5,8,74,130,166
Butane	VP	46
2-Butanone	DL	120
	SW	9,120
	VP	120
Butyl acetate	SW	9
Butyl butyrate	SW	9
Carbon disulfide	SW	9
Carbon tetrachloride	SW	9
Chloroform	SW	9
Cyclohexane	SW	9
2,4-Dimethyl-3-pentanone	SW	9
2,2-Dimethylpropane	VP	46
Ethyl acetate	DL	120
	SW	9,120
	VP	63,120
Heptane	CM	6
	SW	9
Hexane	VP	59
Isoprene	VP	126
Methyl acetate	SW	9,120
Methyl alcohol	VP	6
2-Methylbutane	VP	46
2-Methylpropane	VP	46
Octane	VP	177
Pentane	SW	9
	VP	46
2-Pentanone	SW	120
Propyl acetate	SW	9
Toluene	OS	1
	SW	9
	VP	1
2,2,4-Trimethylpentane	VP	177
Poly(*trans*-1,3-isoprene) [9003-31-0] and [104389-32-4]		
Octane	VP	177
Toluene	OS	2
2,2,4-Trimethylpentane	VP	167,177

1.2. POLY(ALKENES)

Poly(1-butene) [9003-28-5]		
cis-Bicyclo[4.4.0]decane	CM	293
trans-Bicyclo[4.4.0]decane	CM	293
Cyclohexane	CM	293
Cyclooctane	CM	293
Cyclopentane	CM	293
Decane	CM	248
3,3-Diethylpentane	CM	293
2,2-Dimethylpentane	CM	293
2,3-Dimethylpentane	CM	293
2,4-Dimethylpentane	CM	293
3,3-Dimethylpentane	CM	293
Dodecane	CM	248
	VP	107
3-Ethylpentane	CM	293
2,2,4,4,6,8,8-Heptamethylnonane	CM	294
Heptane	CM	248
Hexadecane	CM	248

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TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Hexane	CM	248
3-Methylhexane	CM	293
Nonane	CM	248
Octane	CM	248
	VP	102
2,2,4,6,6-Pentamethylheptane	CM	294
Pentane	CM	248
Tetradecane	CM	248
	GC	281
2,2,4,4-Tetramethylpentane	CM	293,294
2,3,3,4-Tetramethylpentane	CM	293
2,2,4-Trimethylpentane	CM	294
Poly(1-decene) [25189-70-2]		
Toluene	VP	195
Poly(1-dodecene) [25067-08-7]		
Toluene	VP	195
Poly(ethylene) [9002-88-4]		
Benzene	GC	299
Bromomethane	VP	87
2-Butanone	GC	351
Butyl acetate	GC	351
Butyl alcohol	GC	299
Carbon tetrachloride	GC	299
Chlorobenzene	GC	299,351
1-Chlorobutane	GC	351,299
Chloroform	GC	299,351
1-Chlorohexane	GC	351
1-Chloronaphthalene	CM	151,367
Cyclohexane	CM	367
	GC	299,351
Cyclohexanol	GC	299
Cyclohexanone	GC	299,351
Cyclohexyl chloride	GC	299
Cyclopentane	CM	367
Decahydronaphthalene	CD	246
	CM	367,369
Decane	DL	252
	GC	299,351
2,6-Di-*tert*-butyl-4-methylphenol	VP	361
1,5-Dichloropentane	GC	351
N,N-Dimethylacetamide	GC	351
N,N-Dimethylformamide	GC	351
1,2-Dimethylnaphthalene	GC	234
1,3-Dimethylnaphthalene	GC	234
1,4-Dimethylnaphthalene	GC	234
1,5-Dimethylnaphthalene	GC	234
1,6-Dimethylnaphthalene	GC	234
1,7-Dimethylnaphthalene	GC	234
1,8-Dimethylnaphthalene	GC	234
2,3-Dimethylnaphthalene	GC	234
2,6-Dimethylnaphthalene	GC	234
2,7-Dimethylnaphthalene	GC	234
2,4-Dimethylpentane	CM	367
Dodecane	VP	100,107,351
1-Ethylnaphthalene	GC	234
2-Ethylnaphthalene	GC	234
Ethyl propionate	GC	351
2,2,4,4,6,8,8-Heptamethylnonane	CM	367
Heptane	GC	351
	VP	38
Hexadecane	CM	367
Hexane	VP	87
2-Methylbutane	CM	367
1-Methylnaphthalene	GC	234
2-Methylnaphthalene	GC	234

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Methyl sulfoxide	GC	351
Nonane	GC	299
Octacosane	SW	93
Octane	GC	281,351
1-Octene	GC	299
2-Pentanone	GC	299
Phenol	GC	299
Phenyl ether	VP	180
Propyl ether	GC	351
Propyl sulfide	GC	351
1,1,2,2-Tetrachloroethane	GC	351
Tetradecane	GC	281
1,2,3,4-Tetrahydronaphthalene	CM	151
Toluene	GC	351
1,2,4-Trichlorobenzene	CM	367
1,1,2-Trichloroethane	GC	351
p-Xylene	VP	131
Poly(1-heptene) [25511-64-2]		
Toluene	VP	195
Poly(isobutene) [9003-27-4], polyisobutylene		
Benzaldehyde	GC	291
Benzene	CD	72,98
	CM	19,88,94,116, 141,157,184,278
	D	73,82,128,211,254
	DL	69,137,159,184
	GC	263,287,291,378
	OS	61,159
	VP	56,73,159,259,318
cis-Bicyclo[4.4.0]decane	CM	293
trans-Bicyclo[4.4.0]decane	CM	293
Butane	VP	35
2-Butanone	GC	291
Butyl ether	CM	95
Butyl propionate	VP	105
Carbon tetrachloride	CD	104
	CM	104,116,278
	D	254
	VP	105
Chlorobenzene	CD	98
	CM	88,94,184
	D	82,254
	DL	137,184
	OS	96
Chloroform	D	254
	GC	291
Cyclohexane	CD	255
	CM	88,94,104,116, 184,278,293
	D	128
	DL	137,155,184
	GC	263,378
	LS	158
	OS	61
	VP	26,56,105,155
Cyclooctane	CM	293
Cyclopentane	CM	293
Decahydronaphthalene	D	254
Decane	CM	94
	D	254
	DL	137,160
Decyl alcohol	GC	291
o-Dichlorobenzene	D	254
p-Dichlorobenzene	D	254
3,3-Diethylpentane	CM	293

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
2,2-Dimethylbutane	GC	378
2,3-Dimethylbutane	GC	378
2,2-Dimethylhexane	GC	378
2,5-Dimethylhexane	GC	378
2,2-Dimethylpentane	CM	293
	GC	378
2,3-Dimethylpentane	CM	293
	GC	378
2,4-Dimethylpentane	CM	293
	GC	378
3,3-Dimethylpentane	CM	293
2,2-Dimethylpropane	VP	35
Dodecane	CM	94,278,294
	D	254
Ethyl acetate	GC	291
Ethylbenzene	CM	88,297,378
Ethyl decanoate	CM	95
Ethyl ether	CM	94,95
Ethyl heptanoate	CM	95
Ethyl hexadecanoate	CM	95
Ethyl hexanoate	CM	95
Ethyl nonanoate	CM	95
Ethyl octanoate	CM	95
3-Ethylpentane	CM	293
Ethyl tetradecanoate	CM	95
Heptadecane	D	254
2,2,4,4,6,8,8-Heptamethylnonane	CM	294
Heptane	CM	19,88,94,116,157, 184,278,294
	DL	69,137,160,184
	GC	263,378
	OS	221
Hexadecane	CM	94,278,294
	DL	160
Hexane	CM	94,184
	D	254
	DL	137,160,184
	GC	263,378
	OS	221
Hexyl ether	CM	95
Methyl alcohol	VP	105
2-Methylbutane	CM	94
	VP	35
Methylcyclohexane	CM	94,116,184
	DL	137,184
2-Methylheptane	GC	378
4-Methylheptane	GC	378
2-Methylhexane	GC	378
3-Methylhexane	CM	293
	GC	378
2-Methylpentane	GC	378
3-Methylpentane	CM	94
	GC	378
2-Methylpropane	VP	35
Naphthalene	GC	291
Nitrobenzene	GC	291
Nonane	CM	94,294
	GC	291,378
Octane	CM	94,294
	D	254
	DL	160
	GC	263,378
	OS	160
2,2,4,6,6-Pentamethylheptane	CM	294
Pentanal	GC	291

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Pentane	CD	255
	CM	94,278
	D	254,338
	DL	97,160
	GC	263,378
	LS	238
	OS	221
	VP	35,97,156
Pentyl ether	CM	95
Propyl alcohol	GC	291
Propyl ether	CM	95
Pyridine	GC	291
Tetradecane	CM	94
1,2,3,4-Tetrahydronaphthalene	D	254
2,2,4,4-Tetramethylpentane	CM	293,294
2,3,3,4-Tetramethylpentane	CM	293
Toluene	CD	98,104
	CM	19,104,184
	D	254
	DL	69,137,184
	GC	291,378
	VP	105
Tridecane	CM	94
2,2,4-Trimethylpentane	CM	19,94,104,157,294
	D	211
	DL	69
	GC	287,378
	LS	145,158,238
	VP	14,237,259
2,3,4-Trimethylpentane	GC	378
Undecane	CM	94
Poly(1-octadecene) [25511-67-5]		
Toluene	VP	195
Poly(propylene) [9003-07-0]		
Benzene	CM	315
	D	315
	VP	101
cis-Bicyclo[4.4.0]decane	CM	293
trans-Bicyclo[4.4.0]decane	CD	293
Carbon tetrachloride	CD	280
	CM	315
	D	315
	VP	216
Chloroform	CD	280
	CM	315
	D	315
1-Chloronaphthalene	CM	151
Cyclohexane	CM	293,315
	D	315
Cyclooctane	CM	293
Cyclopentane	CM	293
Decane	CM	315
	GC	281
3,3-Diethylpentane	CM	293
2,2-Dimethylpentane	CM	293
2,3-Dimethylpentane	CM	293
2,4-Dimethylpentane	CM	293
3,3-Dimethylpentane	CM	293
2,4-Dimethyl-3-pentanone	VP	123
Dodecane	CM	294
	GC	281
	VP	125,171
Ethylbenzene	CM	315
3-Ethylpentane	CM	293
2,2,4,4,6,8,8-Heptamethylnonane	CM	294

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TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Heptane	CM	294,315
	D	315
	VP	101,217
Hexadecane	CM	294
Hexane	CM	315
	D	315
	VP	101
3-Methylhexane	CM	293
Nonane	CM	294,315
Nonyl alcohol	VP	134
Octane	CM	294,315
2,2,4,6,6-Pentamethylheptane	CM	294
Pentane	CM	315
	D	315
	VP	217
3-Pentanone	VP	117,123
Tetradecane	GC	281
1,2,3,4-Tetrahydronaphthalene	CM	151
2,2,4,4-Tetramethylpentane	CM	293,294
2,3,3,4-Tetramethylpentane	CM	293
Toluene	CM	315
2,2,4-Trimethylpentane	CM	294
	VP	101
m-Xylene	CM	315
o-Xylene	CM	315
p-Xylene	CM	315

1.3. POLY(ACRYLICS)

Poly(acrylamide) [9003-05-8]		
Water	CD	313
	D	196,313
Poly(acrylic acid) [9003-01-4]		
Butyl alcohol	CM	381
N,N-Dimethylformamide	CM	381
Ethyl alcohol	CD	343
	D	343
Ethylene glycol	CD	292
	CM	292
Formamide	CD	292
	CM	292
Water	CD	233,292,343
	CM	292,381
	D	196,343
	VP	366
Poly(butyl methacrylate) [9003-63-8]		
Benzene	GC	307
Butyl acetate	CM	55
	GC	307
Butyl alcohol	GC	307
Butylbenzene	GC	307
tert-Butylbenzene	GC	307
Butylcyclohexane	GC	307
Carbon tetrachloride	GC	307
Chlorobenzene	GC	307
1-Chlorobutane	GC	307
Chloroform	CM	260
	GC	307
Cyclohexane	GC	307
Cyclohexanol	GC	307
Decane	GC	307
Dichloromethane	GC	307
Octane	GC	307
2-Pentanone	GC	307
3,4,5-Trimethylheptane	GC	307
2,2,4 ~ -Trimethylpentane	GC	307

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Poly(*N,N*-dimethylacrylamide) [26793-31-0]		
Water	CD	313
	D	313
Poly(ethyl methacrylate) [9003-42-3]		
Acetone	CD	284
Acetonitrile	CD	284
Acrylonitrile	CD	284
Benzene	CD	284
Bromobenzene	CD	284
1-Bromobutane	CD	284
2-Butanone	CD	284
Butyl acetate	CD	284
Butyl alcohol	CD	284
Butylamine	CD	284
Butyraldehyde	CD	284
Carbon tetrachloride	CD	284
Chlorobenzene	CD	284
Chloroform	CD	284
Cyclohexane	CD	284
Decane	CD	284
Ethane	D	360
Ethylbenzene	CD	284
Ethylcyclohexane	CD	284
Ethylene	D	360
Heptane	CD	284
Methylcyclohexane	CD	284
Nitroethane	CD	284
Nitromethane	CD	284
1-Nitropropane	CD	284
Nonane	CD	284
Octane	CD	284
1-Octene	CD	284
Pentyl alcohol	CD	284
Propane	D	360
	VP	341
Propionic acid	CD	284
Propionitrile	CD	284
Propyl acetate	CD	284
Propyl alcohol	CD	284
Propyl ether	CD	284
Toluene	CD	284
Poly(*N*-isopropylacrylamide) [25189-55-3]		
Acetic acid	GC	198
Butyl alcohol	GC	198,281
1-Chloronaphthalene	GC	198
Hexadecane	GC	198
Naphthalene	GC	198
Water	CM	353
	OS	368
Poly(methacrylic acid) [54193-36-1]		
Water	CM	381
Poly(*N*-methylacrylamide) [25722-14-9]		
Water	CD	313
	D	313
Poly(methyl acrylate) [9003-21-8]		
Benzene	GC	282
Butylbenzene	GC	282
tert-Butylbenzene	GC	282
Butylcyclohexane	GC	282
Cyclohexane	GC	282
cis-Decahydronaphthalene	GC	282
trans-Decahydronaphthalene	GC	282
Decane	GC	282
Dodecane	GC	282
Ethylbenzene	GC	282
Naphthalene	GC	282

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Octane	GC	282
Tetradecane	GC	282
1,2,3,4-Tetrahydronaphthalene	GC	282
3,3,4,4-Tetramethylhexane	GC	282
Toluene	GC	282
3,4,5-Trimethylheptane	GC	282
2,2,5-Trimethylhexane	GC	282
2,2,4-Trimethylpentane	GC	282
Poly(methyl methacrylate) [9011-14-7]		
Acetone	CD	148,161
	D	60,178
	OS	31,178,261
	VP	220
Benzene	CD	162,186
	CM	224
	D	60,178
	OS	31,178,261
	VP	220
2-Butanone	D	60
	OS	261
Butyl acetate	D	60
	OS	36
Carbon dioxide	DL	387
	VP	387
Carbon tetrachloride	D	60
Chlorobenzene	CD	162
	D	60
Chloroform	CD	212
	CM	193
	D	60
	DL	210
	OS	31
Cyclohexane	OS	261
o-Dichlorobenzene	CD	162
	CM	266
	D	60
Dichloroethane	CM	83
	VP	83
1,2-Dichloroethylene	OS	149
1,4-Dioxane	CD	148
	D	60
	OS	31,149
Ethyl acetate	D	60
	OS	36,261
Methyl isobutyrate	CM	83
	VP	83
3-Pentanone	D	60
	OS	31
Tetrahydrofuran	D	60
	OS	31
Toluene	CD	148,212
	D	60
	DL	210
	OS	31,261
m-Xylene	D	60
	OS	31
1.4. VINYL POLYMERS		
Poly(acrylonitrile) [25014-41-9]		
Acrylonitrile	VP	126
Benzene	CM	47
	VP	47
N,N-Dimethylformamide	CM	47
Poly(vinyl acetate) [9003-20-7]		
Acetone	CM	75,184,345
	DL	184

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
	GC	354
	LS	158
	OS	261
	VP	37,75
Acetonitrile	GC	354
Benzene	CD	170
	CM	41,224
	GC	288,295,299,354
	OS	261
	VP	37,76,85
2-Butanone	CD	170
	CM	184
	DL	184
	GC	354
	LS	158
	OS	15,261
Butyl acetate	CM	55,184
	DL	184
Butyl alcohol	GC	299
Butylbenzene	GC	288
Butylcyclohexane	GC	288
Carbon tetrachloride	GC	299,354
Chlorobenzene	CM	41
	D	49,82
	GC	299,354
	LS	158
1-Chlorobutane	GC	299
Chloroform	CM	41,345
	GC	299,354
1-Chlorohexane	GC	354
1-Chloropropane	VP	37
3-Chloropropene	VP	37
Cyclohexane	GC	288,295,299,354
	OS	261
Cyclohexanol	GC	299
Cyclohexanone	GC	299
Cyclohexyl chloride	GC	299
cis-Decahydronaphthalene	GC	288
Decane	GC	288,299
Dichloroethane	CD	232
1,2-Dichloroethane	CM	41
	GC	295,354
1,5-Dichloropentane	GC	354
Diethyl adipate	CD	27
	D	27
Diethyl malonate	CD	27
	D	27
Diethyl oxalate	CD	27
	D	27
Diethyl sebacate	CD	27
	D	27
Diethyl succinate	CD	27
	D	27
N,N-Dimethylacetamide	GC	354
N,N-Dimethylformamide	GC	354
1,4-Dioxane	CM	342
	GC	295,354
Dodecane	GC	288
Ethyl acetate	CM	30,184
	D	211
	DL	184
	GC	354
	LS	158
	OS	261
	VP	237

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TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Heptane	GC	295,354
3-Heptanone	CM	141,184
	DL	184
Hexane	GC	354
Isopropyl alcohol	GC	295
Isopropylamine	VP	37
Methyl acetate	CM	184
	DL	184
Methyl alcohol	CD	271
	CM	41,48,345
	D	211,271
	LS	158
Methyl sulfoxide	GC	354
Nonane	GC	288,299
1-Octene	GC	299
2-Pentanone	CM	184
	DL	184
	GC	299
Propyl acetate	CM	184
	DL	184
Propyl alcohol	VP	37
Propylamine	VP	37
Propyl ether	GC	354
Propyl sulfide	GC	354
1,1,2,2-Tetrachloroethane	CM	41,48
Tetrahydrofuran	GC	354
1,2,3,4-Tetrahydronaphthalene	GC	288
3,3,4,4-Tetramethylhexane	GC	288
Toluene	CD	64,232
	CM	41
	GC	295,354
	OS	261
1,2,3-Trichloropropane	OS	15
Undecane	GC	288
Vinyl acetate	VP	85
Water	CM	342
Poly(vinyl alcohol) [9002-89-5]		
Carbon tetrachloride	CM	359
Chloroacetic acid	CM	112
1,4-Dioxane	CD	342
Ethyl alcohol	CM	30
Water	CD	57,58
	CM	30,55,342
	LS	219
	VP	86,219
Poly(*N*-vinylcarbazole) [25067-59-8]		
Benzene	CD	161
	VP	124
Poly(vinyl chloride) [9002-86-2]		
Benzene	GC	295
Cyclohexane	GC	295
Cyclohexanone	CD	231
	CM	106,231
	D	273
	DL	106
Cyclopentanone	CD	243
	CM	243
Dibutyl phthalate	VP	250
Dichloroethane	CM	55
1,2-Dichloroethane	GC	295
Diethyl phthalate	VP	250
Dioctyl phthalate	VP	250
1,4-Dioxane	GC	295
Dodecane	GC	281
Heptane	GC	295

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Isopropyl alcohol	GC	295
Tetrahydrofuran	CD	244
	CM	244
Toluene	GC	295
Poly(vinylidene fluoride) [24937-79-9]		
Acetophenone	GC	354
2-Butanone	GC	354
Butyl acetate	GC	354
Butylbenzene	GC	354
γ-Butyrolactone	GC	354
Chlorobenzene	GC	354
1-Chlorodecane	GC	354
2-Chloroethyl ether	GC	354
1-Chlorohexane	GC	354
1-Chlorooctane	GC	354
Cyclohexanone	GC	354
cis-Decahydronaphthalene	GC	354
1,2-Dichlorobenzene	GC	354
1,2-Dichloroethane	GC	354
1,5-Dichloropentane	GC	354
N,N-Dimethylacetamide	GC	354
N,N-Dimethylethyleneurea	GC	354
N,N-Dimethylformamide	GC	354
N,N-Dimethylpropyleneurea	GC	354
Dodecane	GC	354
Ethyl acetate	GC	354
Ethylbenzene	GC	354
Ethyl propionate	GC	354
Hexadecane	GC	354
Hexamethylphosphoramide	GC	354
1-Methyl-2-pyrrolidinone	GC	354
Methyl sulfoxide	GC	354
Pentachloroethane	GC	354
Propylene carbonate	GC	354
Propyl sulfide	GC	354
Pyridine	GC	354
1,1,2,2-Tetrachloroethane	GC	354
Tetradecane	GC	354
Tetrahydronaphthalene	GC	354
Tetramethylene sulfone	GC	354
1,1,3,3-Tetramethylurea	GC	354
Toluene	GC	354
1,1,2-Trichloroethane	GC	354
m-Xylene	GC	354
Poly(vinyl methyl ether) [9003-09-2]		
Cyclohexane	VP	372
Ethylbenzene	CM	332
	D	332
	OS	332
Toluene	CM	332
	D	332
	OS	332
Water	CM	353
Poly(2-vinylpyridine) [25014-15-7]		
Benzene	D	267
2-Butanone	D	267
Butyl alcohol	D	267
Cyclohexanone	D	267
1,4-Dioxane	D	267
Ethyl alcohol	D	267
	LS	111
Methyl alcohol	D	267
Propyl alcohol	D	267
Poly(*N*-vinylpyrrolidone) [9003-39-8]		
Butyl alcohol	CD	309

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Chloroform	CD	330
	CM	174
	D	174
Ethyl alcohol	CD	309
Methyl alcohol	CD	269
Propyl alcohol	CD	309
Water	CD	269
	CM	174
	D	174
	OS	214

1.5. POLY(STYRENES)

Poly(*p*-bromostyrene) [24936-50-3]		
Toluene	VP	103
Poly(*p*-chlorostyrene) [24991-47-7]		
Benzene	GC	363
Ethyl acetate	VP	199
Heptane	GC	363
Hexane	GC	363
Isopropylbenzene	GC	363
Pentane	GC	363
Propylbenzene	GC	363
Toluene	GC	363
	VP	103
Poly(α-methylstyrene) [25014-31-7]		
Benzene	CM	223
Isopropylbenzene	VP	136
α-Methylstyrene	VP	136,355
Toluene	VP	355
Poly(*p*-methylstyrene) [24936-41-2]		
Toluene	VP	103
Poly(styrene) [9003-53-6]		
Acetic acid	GC	306
Acetone	CM	29,34,43
	D	43,60
	GC	306
	OS	261
	VP	51,220
Acetonitrile	GC	306,354
Acetophenone	CD	279
Aniline	GC	306
Anisole	CD	135,279,286
	GC	336
Benzaldehyde	GC	306
Benzene	CD	28,70,81,147, 191,192,225
	CM	19,29,32,44,55, 90,184,192 222, 333
	D	24,39,60,128,247, 283,296,329
	DL	12,69,184
	GC	235,282,305,306, 307,325,336,354
	LS	23,114,143, 145,158
	OS	12,33,147,261
	SW	147
	VP	13,28,32,147, 192,220
	XS	200
Benzyl alcohol	GC	306
Benzyl ether	CD	279
Bromobenzene	D	25
Butane	VP	337
2-Butanone	CD	135,249,255
	CM	29,68,146
	D	24,39,45,60
	DL	12,207,256
	GC	235,306,354
	OS	12,18,33, 207,256,261
	VP	17
Butyl acetate	CM	29
	D	60
	GC	306
	OS	33
	VP	13
Butyl alcohol	GC	306
sec-Butyl alcohol	SW	115
Butylbenzene	CM	184
	DL	184
	GC	282,325
tert-Butylbenzene	GC	307
Butylcyclohexane	GC	282
Carbon dioxide	DL	339
Carbon tetrachloride	CD	104,240
	CM	104,240
	D	60
	GC	235,306,307
	LS	114
	OS	33
	VP	13,220
Chlorobenzene	CD	53
	CM	53,146
	D	25,60,82
	GC	306,354
	LS	114
	OS	127
	SW	115
1-Chlorobutane	GC	307
Chloroform	CD	54,191,225
	CM	29
	D	45,60,82
	GC	306,307,354
	OS	33
	SW	115
	VP	51
1-Chlorohexane	GC	354
Cyclohexane	CD	53,70,81,192,285
	CM	29,34,43,53, 184,192,359
	D	43,45,60,189, 211,296,329
	DL	12,69,184, 189,209
	GC	282,306, 307,325,354
	LS	114,143,190,206
	OS	18,33,71,80,261
	SD	189,229
	SW	115
	VP	13,52,80, 192,237,259
Cyclohexanone	CD	289
	GC	306,354
Cyclohexene	CD	192
	CM	29,192
	VP	201

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TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Decahydronaphthalene	CD	192
	CM	184,192,359
	D	24
	DL	184
	LS	114,145
	VP	201
cis-Decahydronaphthalene	GC	282
trans-Decahydronaphthalene	GC	282
	LS	238
	OS	290
Decane	GC	281,282,305,307
o-Dichlorobenzene	D	24,60
	GC	235
1,2-Dichloroethane	GC	306,354
Dichloromethane	GC	306,307
1,5-Dichloropentane	GC	354
N,N-Dimethylformamide	GC	354
2,2-Dimethylpropane	VP	245
1,4-Dioxane	CD	135
	CM	55,184
	D	24,60
	DL	184
	GC	235,306,354
	OS	127
	VP	13
1,3-Diphenylbutane	CD	113,147
	SW	113,147
Dodecane	GC	281,282
Ethane	VP	337
2-Ethoxyethyl acetate	CM	29
Ethyl acetate	CD	135,257
	CM	29,68
	D	24,60,296
	DL	257
	GC	306
	OS	18,33,261
	VP	52
Ethyl alcohol	GC	306
Ethylbenzene	CD	50,53,54,81,113, 147,192,228,255
	CM	30,44,53,78,90, 104,184,192, 228,264
	D	24,39,113,147,211
	DL	69,184,208
	GC	235,325,336,354
	LS	114
	OS	113,147,208
	SW	113,115,147
	VP	50,90.113,147, 237,259
Ethylene glycol	GC	306
Ethyl ether	GC	306
Formamide	GC	306
Heptane	GC	306,325,335,354
	SW	115
Hexadecane	GC	281,282
Hexane	GC	306,325,335,354
	SW	115
Hexyl alcohol	SW	115
Hexylbenzene	GC	325
Isobutyl alcohol	GC	306
Isopentyl alcohol	SW	115
Isopropyl alcohol	GC	306

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Isopropyl ether	GC	306
Methyl alcohol	GC	306
2-Methylbutane	VP	245
2-Methylheptane	GC	325
2-Methylpentane	GC	325
3-Methylpentane	GC	325
Naphthalene	GC	282
Nitrobenzene	GC	306
Nitromethane	VP	13
Nonane	GC	305,325
Octane	GC	305,306,307,325
	SW	115
	VP	13
1-Octene	GC	325
cis-2-Octene	GC	325
Octyl alcohol	GC	306
Pentane	GC	306
	SW	115
	VP	245
3-Pentanone	CD	319
	D	60
	VP	13
Pentyl alcohol	GC	306
Pentylbenzene	GC	325
Propane	VP	337
Propyl acetate	VP	51
Propyl alcohol	GC	306
	SW	115
Propylbenzene	CM	184
	DL	184
	GC	325
Propyl ether	GC	354
	VP	13
Propyl sulfide	GC	354
Pyridine	GC	306,336,354
Styrene	CM	10,29
	GC	235
	VP	126
Tetradecane	GC	281,282
Tetrahydrofuran	D	60
	GC	306
1,2,3,4-Tetrahydronaphthalene	CM	192
	GC	282
3,3,4,4-Tetramethylhexane	GC	282
Toluene	CD	28,53,54,81,135, 183,188,191,192, 225,228,240, 255,286
	CM	29,34,43,53,55,90, 146,154,184,192, 226,228,240, 264,332,333
	D	11,24,39,43, 45,60,189,332
	DL	12,69,184, 189,210
	GC	235,305,306, 325,336,354
	LS	114,190
	OS	12,18,33,71,144, 203,261,332
	SD	189
	SW	115
	VP	13,17,28,52,103

TABLE 1. *cont'd*

Polymer–Solvent	Method	Refs.
Trichloroethylene	GC	306
1,3,5-Trimethylbenzene	CM	29,184
	D	60
	DL	184
	GC	305
3,4,5-Trimethylheptane	GC	307
2,2,4-Trimethylpentane	GC	306,307,325
Water	GC	306
m-Xylene	CM	29,184
	D	60
	DL	184
	GC	305
	OS	33
	VP	13
o-Xylene	CM	29
	GC	235,306
	OS	33
1.6. OTHER COMPOUNDS		
Poly(nitro-1,4-phenylene)	VP	187
1,4-Dioxane		

TABLE 2. MAIN-CHAIN CARBON HETEROATOM POLYMERS

Polymer–Solvent	Method	Refs.
2.1. POLY(OXIDES) AND POLY(ESTERS)		
Epoxy from bis(4-glycidylphenyl) propane and 1,6 hexanediamine		
Chloroform	SW	110
	VP	110
1,4-Dioxane	SW	110
	VP	110
Poly(oxy-1,4-butanediyl) [25190-06-1], polytetrahydrofuran		
Acetone	GC	376
Acetonitrile	GC	376
Benzene	GC	376
2-Butanone	GC	376
Butyl acetate	GC	376
Carbon tetrachloride	GC	376
Chlorobenzene	GC	376
Chloroform	GC	376
1-Chlorohexane	GC	376
Cyclohexane	GC	376
Decane	GC	376
1,2-Dichloroethane	GC	376
Dichloromethane	GC	376
1,5-Dichloropentane	GC	376
1,4-Dioxane	GC	376
Ethyl acetate	GC	376
Ethylbenzene	GC	376
Heptane	GC	376
Propyl ether	GC	376
Propyl sulfide	GC	376
Pyridine	GC	376
Tetrahydrofuran	GC	376
Toluene	GC	376
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene) [24936-68-3] and [25037-45-0], poly(bisphenol A carbonate), poly(carbonate)		
Acetone	DL	386
Benzene	GC	362
Butyl acetate	GC	362
Butylbenzene	GC	362
tert-Butylbenzene	GC	362

TABLE 2. *cont'd*

Polymer–Solvent	Method	Refs.
Butylcyclohexane	GC	362
Carbon dioxide	CM	377,385
	DL	338,386
Chlorobenzene	GC	310,362
Chloroform	GC	362
Cyclohexanol	GC	362
Cyclohexanone	GC	362
Decane	GC	310,362
o-Dichlorobenzene	GC	310
Dichloromethane	GC	362
1,4-Dioxane	CD	161
Dodecane	GC	362
Ethylbenzene	GC	362
Methane	DL	386
Methylcyclohexane	GC	362
2-Pentanone	GC	362
Propyl alcohol	GC	362
1,1,2,2-Tetrachloroethane	GC	362
Tetradecane	GC	362
Toluene	GC	362
Water	DL	386
Poly(oxyethylene) [25322-68-3] and [9004-74-4], poly(ethylene glycol), poly(ethylene oxide)		
Acetone	GC	268,302,317,376
Acetonitrile	GC	268,320,376
Acetophenone	GC	316
Aniline	GC	239
Benzaldehyde	GC	239
Benzene	CD	262,312
	CM	138,277,312
	D	213,382
	GC	239,268,274,276, 302,317,320,376
	VP	175,213
Benzyl alcohol	GC	239
p-Bromoacetophenone	GC	316
2-Butanone	GC	268,302,320,376
Butyl acetate	GC	268,376
Butyl alcohol	CD	218
	GC	258,268,302
sec-Butyl alcohol	GC	258,302
Carbon tetrachloride	CM	129,185,277,357
	DL	185
	GC	268,302,376
	VP	185,357
p-Chloroacetophenone	GC	316
Chlorobenzene	GC	239,317
Chloroform	CD	230
	CM	185,230
	DL	185
	GC	268,302,320,376
	VP	117,185
1-Chlorohexane	GC	376
Chloromethane	CD	230
	CM	230
Cyclohexane	GC	302,317,320
Decane	GC	239,302,320,376
1,2-Dichloroethane	GC	268,376
Dichloromethane	GC	276,376
1,5-Dichloropentane	GC	376
1,2-Dimethoxyethane	GC	320
1,2-Dimethylnaphthalene	GC	234
1,3-Dimethylnaphthalene	GC	234
1,4-Dimethylnaphthalene	GC	234
1,5-Dimethylnaphthalene	GC	234

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TABLE 2. *cont'd*

Polymer–Solvent	Method	Refs.
1,6-Dimethylnaphthalene	GC	234
1,7-Dimethylnaphthalene	GC	234
1,8-Dimethylnaphthalene	GC	234
2,3-Dimethylnaphthalene	GC	234
2,6-Dimethylnaphthalene	GC	234
2,7-Dimethylnaphthalene	GC	234
1,4-Dioxane	CM	129
	GC	302,320,376
Dodecane	GC	239,302,320
2-Ethoxyethyl ether	GC	317
Ethyl acetate	GC	276,317,320,376
Ethyl alcohol	CD	218
	CM	138,277
	GC	258,276,302,317
Ethylbenzene	GC	320,376
Ethyl ether	GC	302
1-Ethylnaphthalene	GC	234
2-Ethylnaphthalene	GC	234
o-Fluoroacetophenone	GC	316
p-Fluoroacetophenone	GC	316
Heptane	GC	239,302,320,376
3-Heptanone	GC	302
1-Heptene	GC	302
Heptyl alcohol	GC	258
Hexane	GC	239,276,302,317
2-Hexanol	GC	258
3-Hexanol	GC	258
1-Hexene	GC	302
Hexyl alcohol	GC	258
o-Hydroxyacetophenone	GC	316
p-Hydroxyacetophenone	GC	316
Isopropyl alcohol	GC	258,320
o-Methoxyacetophenone	GC	316
p-Methoxyacetophenone	GC	316
2-Methoxyethyl ether	GC	320
Methyl acetate	GC	302
o-Methylacetophenone	GC	316
p-Methylacetophenone	GC	316
Methyl alcohol	CD	218
	CM	138,277
	GC	258,268,274,317
	LS	242
1-Methylnaphthalene	GC	234
2-Methylnaphthalene	GC	234
o-Nitroacetophenone	GC	316
p-Nitroacetophenone	GC	316
Nitroglycerin	GC	346
Nitromethane	GC	302
Nonane	GC	239,302,344
Octane	GC	239,302,317
Octyl alcohol	GC	258
Pentane	GC	302
2-Pentanol	GC	258
Pentyl alcohol	GC	258
Phenol	GC	239
Phenyl acetate	GC	239
Propyl alcohol	GC	258,317,320,344
Propyl ether	GC	376
Propyl sulfide	GC	376
Pyridine	GC	302,320,376
Tetrahydrofuran	GC	268,376
Toluene	GC	268,302,317, 320,376
	LS	314
1,1,2-Trichloroethane	GC	268

TABLE 2. *cont'd*

Polymer–Solvent	Method	Refs.
2,2,2-Trifluoroethanol	GC	320
Undecane	GC	239,302
Water	CD	150,230,303,312, 324,352,365
	CM	62,91,139,194, 230,311,312,324, 348,356
	D	62,176,300, 365,382
	GC	268
	OS	214,368
	VP	62,176,375
m-Xylene	GC	302
o-Xylene	GC	268
Poly(oxyethyleneoxyadipoyl) [24937-05-1]		
1,2-Dimethylnaphthalene	GC	234
1,3-Dimethylnaphthalene	GC	234
1,4-Dimethylnaphthalene	GC	234
1,5-Dimethylnaphthalene	GC	231
1,6-Dimethylnaphthalene	GC	234
1,7-Dimethylnaphthalene	GC	234
1,8-Dimethylnaphthalene	GC	234
2,3-Dimethylnaphthalene	GC	231
2,6-Dimethylnaphthalene	GC	234
2,7-Dimethylnaphthalene	GC	234
1,4-Dioxane	CD	265
	CM	132
	D	265
	VP	132,165
1-Ethylnaphthalene	GC	234
2-Ethylnaphthalene	GC	234
1-Methylnaphthalene	GC	234
2-Methylnaphthalene	GC	234
Poly(oxyethyleneoxysebacoyl) [26762-06-11]		
Butyl butyrate	VP	197
1,4-Dioxane	CM	132
	VP	132
Poly(oxyethyleneoxysuccinyl) [25667-11-2]		
1,4-Dioxane	CM	132
	VP	132
Poly(oxyhexamethyleneoxyadipoyl) [24937-10-8]		
1,4-Dioxane	CD	265
	D	265
Poly(oxy-1-oxopentamethylene) [26354-94-9]		
1,4-Dioxane	CD	265
	D	265
Poly(oxypropylene) [25322-69-4], poly(propylene glycol), poly(propylene oxide)		
Benzene	CM	347,349
	GC	331
	VP	215
Butyl alcohol	CM	118
	DL	118
	SW	118
sec-Butyl alcohol	CM	118
	SW	118
tert-Butyl alcohol	CM	118
	SW	118
Carbon tetrachloride	CM	118,163,371
	DL	163
	SW	118
	VP	163,371
Chloroform	CM	163
	DL	163
	VP	163

TABLE 2. *cont'd*

Polymer–Solvent	Method	Refs.
Cyclohexane	CM	118
	GC	331
	SW	118
Cyclohexene	GC	331
Ethyl alcohol	CM	118,347,348
	DL	118
	SW	118
Heptane	GC	331
1-Heptene	GC	331
2-Heptene	GC	331
Hexane	CD	241
	CM	118,241
	D	241
	GC	331
	SW	118
1-Hexene	GC	331
cis-2-Hexene	GC	331
trans-2-Hexene	GC	331
Isobutyl alcohol	CM	118
	SW	118
Isopropyl alcohol	CM	118
	SW	118
Methyl alcohol	CM	89
	D	89
	VP	89
2-Methylhexane	GC	331
2-Methylpentane	GC	331
Nonane	GC	344
2-Pentanol	CM	118
	SW	118
Pentyl alcohol	CM	118
	SW	118
Propyl alcohol	CM	118
	GC	344
	SW	118
Pyridine	CM	347
Water	CD	241
	CM	62,91,164,241,272, 311,322,348,353,373
	D	62,241
	VP	62,323
Poly(oxytrimethyleneoxyadipoyl) [36221-42-8]		
1,4-Dioxane	CM	132
	VP	132

2.2. POLY(AMIDES)

Poly[L-iminocarbonyl(benzyloxycarbonylethyl) methylene] [25038-53-3], poly(γ-benzyl-L-glutamate)		
Chloroform	CD	179
Dichloroacetic acid	CD	179
	CM	142
1,2-Dichloroethane	CM	142
	D	133
	VP	133
N,N-Dimethylformamide	VP	227
Pyridine	D	133
	VP	133
Poly[L-iminocarbonyl(benzyloxycarbonylmethyl) methylene] [25736-41-8], poly(γ-benzyl-L-aspartate)		
Chloroform	D	133
	VP	133
Poly[imino(1-oxohexamethylene)] [25038-54-4], nylon 6		
Decyl alcohol	VP	197
Water	VP	370

TABLE 2. *cont'd*

Polymer–Solvent	Method	Refs.

2.3. POLY(SILOXANES)

Poly(oxydimethylsilylene) [9016-00-6], poly(dimethylsiloxane)		
Acetone	SW	182
Benzene	CD	253
	CM	20,116,169
	D	169
	SW	182
	VP	20,204
cis-Bicyclo[4.4.0]decane	CM	293
trans-Bicyclo[4.4.0]decane	CM	293
Bromocyclohexane	CM	184
	DL	184
Butane	VP	99
2-Butanone	CM	141,184,205,380
	DL	184
	GC	383
	LS	380
	SW	182
Butyl acetate	CM	122
Butyl ether	CM	122
Butyl propionate	CM	122
Carbon dioxide	CM	377
Carbon tetrachloride	CM	116
Chlorobenzene	CM	116
	GC	310
	OS	168
Cyclohexane	CM	116,169,293
	D	169
	OS	168
Cyclopentane	CM	293
Cyclooctane	CM	293
Decane	CD	253
	CM	95
	GC	270,310
Decyl acetate	CM	122
o-Dichlorobenzene	GC	310
Diethoxymethane	CM	122
3,3-Diethylpentane	CM	293
1,2-Dimethoxyethane	CM	122
2,3-Dimethylbutane	GC	270
2,5-Dimethylhexane	GC	270
2,2-Dimethylpentane	CM	293
	GC	270
2,3-Dimethylpentane	CM	293
2,4-Dimethylpentane	CM	293
3,3-Dimethylpentane	CM	293
2,2-Dimethylpropane	VP	99
Dodecane	CM	95,294
Ethyl acetate	CM	122,184,205
	DL	184
Ethylbenzene	CD	253
	CM	169
	D	169
	VP	204
Ethyl butyrate	CM	122
Ethyl decanoate	CM	122
Ethyl dodecanoate	CM	122
Ethyl ether	CM	122,205
Ethyl heptanoate	CM	122
Ethyl hexanoate	CM	122
Ethyl nonanoate	CM	122
Ethyl octanoate	CM	122
3-Ethylpentane	CM	293
Ethyl pentanoate	CM	122

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TABLE 2. *cont'd*

Polymer–Solvent	Method	Refs.
Ethyl propionate	CM	122
Ethyl undecanoate	CM	122
2,2,4,4,6,8,8-Heptamethylnonane	CM	294
Heptane	CD	253
	CM	95,116,169,184
	D	169
	DL	184
	GC	270
Heptyl acetate	CM	122
Hexadecane	CM	95,294
Hexamethyldisiloxane	CD	253
	CM	181,184
	DL	184
Hexane	CD	253
	CM	95,184
	DL	184
	GC	270
	VP	204
Hexyl acetate	CM	122
Hexyl alcohol	SW	182
Hexyl ether	CM	122
Isopropylbenzene	CM	169
	D	169
Methyl alcohol	SW	182
Methyl butyrate	CM	122
Methylcyclohexane	CM	169
	D	169
Methyl decanoate	CM	122
Methyl heptanoate	CM	122
3-Methylhexane	CM	293
Methyl heptanoate	CM	122
Methyl nonanoate	CM	122
Methyl octanoate	CM	122
Methyl pentanoate	CM	122
2-Methylpropane	VP	99
Methyl propionate	CM	122
Nonane	CM	95
	GC	270
Nonyl acetate	CM	122
Octamethylcyclotetrasiloxane	CM	40,184
	DL	184
	VP	40
Octamethyltrisiloxane	CD	253
Octane	CD	253
	CM	95,294
	GC	270
	VP	204
Octyl acetate	CM	122
2,2,4,6,6-Pentamethylheptane	CM	294
Pentane	CD	253
	CM	95
	GC	270
	VP	99
2-Pentanone	SW	182
Pentyl acetate	CM	122
Pentyl ether	CM	122
Pentyl propionate	CM	122
Propyl acetate	CM	122,184
	DL	184
Propyl alcohol	SW	182
Propyl ether	CM	122
Propyl propionate	CM	122
Tetradecane	CD	253
	CM	95
2,2,4,4-Tetramethylpentane	CM	293,294

TABLE 2. *cont'd*

Polymer–Solvent	Method	Refs.
2,3,3,4-Tetramethylpentane	CM	293
Toluene	CD	253
	CM	169
	D	169
	OS	168
	SW	182
	VP	204,372
Tridecane	CM	95,205
1,3,5-Trimethylbenzene	CM	169
	D	169
2,2,4-Trimethylpentane	CM	294
	GC	270
Undecane	CM	95
m-Xylene	CM	169
	D	169
	SW	182
o-Xylene	CM	169
	D	169
	SW	182
p-Xylene	CD	253
	CM	169
	D	169
	SW	182

2.4. DERIVATIVES OF CELLULOSE

Cellulose acetate [9004-35-7]		
Acetone	D	65,152
	LS	304
	OS	65
	VP	65,108
Acetonitrile	D	152
Aniline	D	152
Chloroform	D	152
	VP	109
Chloromethane	D	152
N,N-Dimethylacetamide	CM	251
1,4-Dioxane	D	152
	VP	108
Ethyl alcohol	DL	335
Methyl acetate	D	152
	VP	108
Methyl sulfoxide	CM	251
	VP	109
Phenol	D	65
	OS	65
	VP	65
Pyridine	D	152
	VP	108
Tetrachloroethane	D	3
	OS	3
Cellulose nitrate [9004-70-0], nitrocellulose		
Acetone	CD	4,7
	CM	21,55
	D	4,7,128
	OS	202
	VP	4,7,16,42,67,108
Acetonitrile	VP	42
2-Butanone	D	128
Butyl acetate	CM	21,55
	D	128
Cyclopentanone	VP	42
Dibutyl phthalate	CM	21,328,379
2,4-Dimethyl-3-pentanone	VP	42
1,4-Dioxane	VP	42

TABLE 2. *cont'd*

Polymer–Solvent	Method	Refs.
Ethyl acetate	CM	21,22
	D	128
Ethyl propyl ether	VP	42
2-Heptanone	D	128
Methyl acetate	D	128
	VP	108
Methyl alcohol	CM	22,55
3-Methyl-2-butanone	VP	42
Nitroglycerin	VP	384
Nitromethane	VP	42
2-Pentanone	D	128
	VP	42
Pinacolone	VP	42
Propyl acetate	D	128
Triacetin	CM	326,334
Tritolyl phosphate	CM	21
Ethyl cellulose [9004-57-3]		
Acetone	D	172
	VP	66
Benzene	D	172
	VP	66
Butane	VP	77
2-Butanone	D	172
Butyl acetate	D	172
Carbon tetrachloride	D	172
Chloroform	D	172
	GC	364
Dichloromethane	GC	364
2,2-Dimethylpropane	VP	77
Ethyl acetate	D	172
Methyl alcohol	VP	66
2-Methylbutane	VP	77
Pentane	VP	77
1,1,1-Trichloroethane	GC	364
Water	VP	79
Hydroxyethyl cellulose [9004-62-0]		
Water	OS	214
Hydroxypropyl cellulose [9004-64-2]		
Water	CM	353
Methyl cellulose [9004-67-5]		
Water	CM	321,340
Cellulose tricarbanilate [9047-07-8]		
Cyclohexanol	CD	271
	D	271
5-Nonanone	CD	271
	D	271
2.5. OTHER COMPOUNDS		
Dextran [9004-54-0]		
Ethanolamine	CM	301
Ethylene glycol	CM	298,301
Formamide	CM	298,301
Glycerol	CM	301
Methyl sulfoxide	CD	298,301
	CM	298,301
Water	CD	298,301
	CM	275,298,301
	OS	214,358
	VP	236,375
Lignin [9005-53-2]		
N,N-Dimethylformamide	VP	153
1,4-Dioxane	VP	153
Methyl sulfoxide	VP	153
Starch [9005-25-8]		
Water	VP	173

D. REFERENCES

1. K. H. Meyer, E. Wolff, C. G. Boissonas, Helv. Chim. Acta, **23**, 430 (1940).
2. E. Wolff, Helv. Chim. Acta, **23**, 439 (1940).
3. O. Hagger, A. J. A. van der Wyk, Helv. Chim. Acta, **23**, 488 (1940).
4. E. Calvet, Compt. Rend., **213**, 126 (1941); **214**, 767 (1942).
5. G. Gee, L. R. G. Treloar, Trans. Faraday Soc., **38**, 147 (1942).
6. J. Ferry, G. Gee, L. R. G. Treloar, Trans. Faraday Soc., **41**, 340 (1945).
7. E. Calvet, Bull. Soc. Chim., **12**, 553 (1945).
8. G. Gee, W. J. C. Orr, Trans. Faraday Soc., **42**, 507 (1946).
9. G. Gee, Trans. Faraday Soc., **42**, 585 (1946).
10. D. E. Roberts, W. W. Walton, R. S. Jessup, J. Polym. Sci., **2**, 420 (1947).
11. R. F. Boyer, R. S. Spencer, J. Polym. Sci., **3**, 97 (1948).
12. J. W. Breitenbach, H. P. Frank, Monatsh., **79**, 531 (1948).
13. E. C. Baughan, Trans. Faraday Soc., **44**, 495 (1948).
14. A. A. Tager, V. A. Kargin, Kolloidn. Zh., **10**, 455 (1948).
15. G. V. Browning, J. D. Ferry, J. Chem. Phys., **17**, 1107 (1949).
16. H. Campbell, P. Johnson, J. Polym. Sci., **4**, 247 (1949).
17. C. E. H. Bawn, R. F. J. Freeman, A. R. Kamaliddin, Trans. Faraday Soc., **46**, 677 (1950).
18. M. J. Schick, P. Doty, B. H. Zimm, J. Am. Chem. Soc., **72**, 530 (1950).
19. A. A. Tager, V. Sanatina, Kolloidn. Zh., **12**, 474 (1950); Rubber Chem. Technol., **24**, 773 (1951).
20. M. J. Newing, Trans. Faraday Soc., **46**, 613 (1950).
21. S. M. Lipatov, S. I. Meerson, Kolloidn. Zh., **12**, 122 (1950).
22. S. M. Lipatov, S. I. Meerson, Kolloidn. Zh. **12**, 427 (1950).
23. P. Debye, A. M. Bueche, J. Chem. Phys., **18**, 1423 (1950).
24. D. J. Streeter, R. F. Boyer, Ind. Eng. Chem., **43**, 1790 (1951).
25. W. Heller, A. C. Thompson, J. Colloid Sci., **6**, 57 (1951).
26. L. der Minassian, M. Magat, J. Chim. Phys., **48**, 574 (1951).
27. P. Meares, Trans. Faraday Soc., **47**, 699 (1951).
28. H. Tompa, J. Polym. Sci., **8**, 51 (1952).
29. H. Hellfritz, Makromol. Chem., **7**, 191 (1952).
30. A. A. Tager, V. A. Kargin, Kolloidn. Zh., **14**, 367 (1952).
31. G. V. Schulz, H. Doll, Z. Elektrochem., **56**, 248 (1952).
32. A. A. Tager, Zh. H Dombek, Kolloidn. Zh., **15**, 69 (1953).
33. G. V. Schulz, H. Hellfritz, Z. Elektrochem., **57**, 835 (1953).
34. K. von Guenner, G. V. Schulz, Naturwissensch., **40**, 164 (1953).
35. S. Prager, E. Bagley, F. A. Long, J. Am. Chem. Soc., **75**, 2742 (1953).
36. G. V. Schulz, H. Doll, Z. Elektrochem., **57**, 841 (1953).
37. R. J. Kokes, A. R. DiPietro, F. A. Long, J. Am. Chem. Soc., **75**, 6319 (1953).
38. J. H. van der Waals, J. J. Hermans, Rec. Trav. Chim. Pays-Bas, **69**, 971 (1953).
39. M. Griffel, R. S. Jessup, J. A. Cogliano, R. P. Park, J. Res. Nat. Bur. Std., **52**, 217 (1954).
40. R. C. Osthoff, W. T. Grubb, J. Am. Chem. Soc., **76**, 399 (1954).

41. H. Daoust, M. Rinfret, Can. J. Chem., **32**, 492 (1954).
42. E. C. Baughan, A. L. Jones, K. Stewart, Proc. Roy. Soc., A, **225**, 478 (1954).
43. G. V. Schulz, K. von Guenner, H. Gerrens, Z. Phys. Chem., NF, **4**, 192 (1955).
44. A. A. Tager, R. V. Krivokorytova, P. M. Khodorov, Dokl. Akad. Nauk, SSSR, **100**, 741 (1955).
45. B. Rosen, J. Polym. Sci., **17**, 559 (1955).
46. A. Aitken, R. M. Barrer, Trans. Faraday Soc., **51**, 116 (1955).
47. A. A. Tager, L. K. Kosova, D. Y. Karlinskaya, I. A. Yurina, Kolloidn. Zh., **17**, 315 (1955).
48. M. Parent, M. Rinfret, Can. J. Chem., **33**, 971 (1955).
49. A. Horth, M. Rinfret, J. Am. Chem. Soc., **77**, 503 (1955).
50. T. V. Gatovskaya, V. A. Kargin, A. A. Tager, Zh. Fiz. Khim., **29**, 883 (1955).
51. C. E. H. Bawn, M. A. Wajid, Trans. Faraday Soc., **52**, 1658 (1956).
52. K. Schmoll, E. Jenckel, Z. Elektrochem., **60**, 756 (1956).
53. E. Jenckel, K. Gorke, Z. Elektrochem., **60**, 579 (1956).
54. K. Amaya, R. Fujishiro, Bull. Chem. Soc. Japan, **29**, 270 (1956).
55. S. I. Meerson, S. M. Lipatov, Kolloidn. Zh., **18**, 447 (1956).
56. C. E. H. Bawn, R. D. Patel, Trans. Faraday Soc., **52**, 1664 (1956).
57. K. Amaya, R. Fujishiro, Bull. Chem. Soc. Japan, **29**, 361 (1956).
58. K. Amaya, R. Fujishiro, Bull. Chem. Soc. Japan, **29**, 830 (1956).
59. V. A. Kargin, T. V. Gatovskaya, Zh. Fiz. Khim., **30**, 1852 (1956).
60. G. V. Schulz, M. Hoffmann, Makromol. Chem., **23**, 220 (1957).
61. P. J. Flory, H. Daoust, J. Polym. Sci., **25**, 429 (1957).
62. G. N. Malcolm, J. S. Rowlinson, Trans. Faraday Soc., **53**, 921 (1957).
63. C. Booth, G. Gee, G. R. Williamson, J. Polym. Sci., **23**, 3 (1957).
64. H. J. L. Schuurmans, J. J. Hermans, J. Phys. Chem., **61**, 1496 (1957).
65. R. Jeffries, Trans. Farday Soc., **53**, 1592 (1957).
66. R. M. Barrer, J. A. Barrie, J. Polym. Sci., **23**, 331 (1957).
67. H. Takenaka, J. Polym. Sci., **24**, 321 (1957).
68. A. A. Tager, L. A. Galkina, Nauch. Dokl. Vysshei Skoly, Khim. Khim. Tekhno., **1**, 357 (1958).
69. A. A. Tager, A. Smirnova, N. Sysueva, Nauch. Dokl. Vysshei Skoly, Khim. Khim. Tekhnol., **1**, 135 (1958).
70. K. Amaya, R. Fujishiro, Bull. Chem. Soc. Japan, **31**, 19 (1958).
71. G. Rehage, H. Meys, J. Polym. Sci., **30**, 271 (1958).
72. M. A. Kabayama, H. Daoust, J. Phys. Chem., **62**, 1127 (1958).
73. R. S. Jessup, J. Res. Nat. Bur. Std., **60**, 47 (1958).
74. R. M. Barrer, R. R. Ferguson, Trans. Faraday Soc., **54**, 989 (1958).
75. A. A. Tager, M. Iovleva, Zh. Fiz. Khim., **32**, 1774 (1958).
76. A. F. Sirianni, R. Tremblay, I. E. Puddington, Can. J. Chem., **36**, 543 (1958).
77. R. M. Barrer, J. A. Barrie, J. Slater, J. Polym. Sci., **27**, 177 (1958).
78. A. A. Tager, M. M. Gur'yanova, Zh. Fiz. Khim., **32**, 1958 (1958).
79. R. M. Barrer, J. A. Barrie, J. Polym. Sci., **28**, 377 (1958).
80. W. R. Krigbaum, D. O. Geymer, J. Am. Chem. Soc., **81**, 1859 (1959).
81. G. V. Schulz, A. Horbach, Z. Phys. Chem., NF, **22**, 377 (1959).
82. A. Horth, D. Patterson, M. Rinfret, J. Polym. Sci., **39**, 189 (1959).
83. A. A. Tager, M. V. Tsilipotkina, V. K. Doronina, Zh. Fiz. Khim., **33**, 335 (1959).
84. R. S. Jessup, J. Res. Nat. Bur. Std., **62**, 1 (1959).
85. A. Nakajima, H. Yamakawa, I. Sakurada, J. Polym. Sci., **35**, 489 (1959).
86. I. Sakurada, A. Nakajima, H. Fujiwara, J. Polym. Sci., **35**, 497 (1959).
87. C. E. Rogers, V. Stanett, M. Szwarc, J. Phys. Chem., **63**, 1406 (1959).
88. C. Watters, H. Daoust, M. Rinfret, Can. J. Chem., **38**, 1087 (1960).
89. M. L. Lakhanpal, B. E. Conway, J. Polym. Sci., **46**, 75 (1960).
90. S. I. Meerson, S. M. Lipatov, Kolloidn. Zh., **21**, 531 (1959).
91. R. G. Cunninghame, G. N. Malcolm, J. Phys. Chem., **65**, 1454 (1961).
92. G. Allen, G. Gee, J. P. Nicholson, Polymer, **2**, 8 (1961).
93. P. J. Flory, A. Ciferri, R. Chiang, J. Am. Chem. Soc., **83**, 1023 (1961).
94. G. Delmas, D., Patterson, T. Somcynsky, J. Polym. Sci., **57**, 79 (1962).
95. G. Delmas, D. Patterson, D. Bohme, Trans. Faraday Soc., **58**, 2116 (1962).
96. J. Leonard, H. Daoust, J. Polym. Sci., **57**, 53 (1962).
97. C. H. Baker, W. B. Brown, G. Gee, J. S. Rowlinson, D. Stubley, R. E. Yeadon, Polymer, **3**, 215 (1962).
98. M. Senez, H. Daoust, Can. J. Chem., **40**, 734 (1962).
99. R. M. Barrer, J. A. Barrie, N. K. Raman, Polymer, **3**, 595 (1962).
100. T. V. Gatovskaya, G. M. Pavlyuchenko, V. A. Berestnev, V. A. Kargin, Dokl. Akad. Nauk, SSSR, **143**, 590 (1962); Proc. Acad. Sci. SSSR, **143**, 209 (1962).
101. K. P. Kwei, T. K. Kwei, J. Phys. Chem., **66**, 2146 (1962).
102. G. M. Pavlyuchenko, T. V. Gatovskaya, V. A. Kargin, Dokl. Akad. Nauk, SSSR, **147**, 150 (1962); Proc. Acad. Sci. USSR, **147**, 790 (1962).
103. R. Corneliussen, S. A. Rice, H. Yamakawa, J. Chem. Phys., **38**, 1768 (1963).
104. A. A. Tager, A. I. Podlesnyak, Vysokomol, Soedin. A, **5**, 87 (1963); Polym. Sci. USSR, **4**, 698 (1963).
105. A. A. Tager, M. V. Tsilipotkina V. Ye. Dreval, O. V. Nechayeva, Vysokomol. Soedin. A, **5**, 94 (1963); Polym. Sci. USSR, **4**, 706 (1963).

106. S. I. Meerson, I. M. Zagrayevskaya, Kolloidn. Zh., **25**, 202 (1963).
107. T. V. Gatovskaya, G. M. Pavlyuchenko, V. A. Berestnev, V. A. Kargin, Vysokomol. Soedin. A, **5**, 960 (1963); Polym. Sci. USSR, **5**, 9 (1963).
108. W. R. Moore, R. Shuttleworth, J. Polym. Sci. A, **1**, 733 (1963).
109. W. R. Moore, R. Shuttleworth, J. Polym. Sci. A, **1**, 1985 (1963).
110. T. K. Kwei, J. Polym. Sci. A, **1**, 2977 (1963).
111. A. J. Hyde, R. B. Taylor, Makromol. Chem., **62**, 204 (1963).
112. S. I. Meerson, I. M. Zagrayevskaya, Kolloidn. Zh., **25**, 197 (1963).
113. J. Biros, K. Šolc, J. Pouchly, Faserforsch. Textil., **15**, 608 (1964).
114. A. A. Tager, V. M. Andreyeva, E. M. Evsina, Vysokomol. Soedin. A, **6**, 1901 (1964); Polym. Sci. USSR, **6**, 2107 (1964).
115. G. Rehage, Kolloid Z., **196**, 97 (1964); **199**, 1 (1964).
116. G. Delmas, D. Patterson, S. N. Bhattacharyya, J. Phys. Chem., **68**, 1468 (1964).
117. G. Allen, C. Booth, G. Gee, M. N. Jones, Polymer, **5**, 367 (1964).
118. B. E. Conway, J. P. Nicholson, Polymer, **5**, 387 (1964).
119. C. Booth, G. Gee, M. N. Jones, W. D. Taylor, Polymer, **5**, 353 (1964).
120. C. Booth, G. Gee, G. Holden, G. R. Williamson, Polymer, **5**, 343 (1964).
121. G. Butenuth, Rubber, Chem. Technol., **37**, 326 (1964).
122. D. Patterson, J. Polym. Sci. A, **2**, 5177 (1964).
123. W. B. Brown, G. Gee, W. D. Taylor, Polymer, **5**, 362 (1964).
124. K. Ueberreiter, W. Bruns, Ber. Bunsenges. Phys. Chem., **68**, 541 (1964).
125. G. M. Pavlyuchenko, T. V. Gatovskaya, V. A. Kargin, Vysokomol. Soedin. A, **6**, 1190 (1964); Polym. Sci. USSR, **6**, 1309 (1964).
126. A. N. Gudkov, N. A. Fermor, N. I. Smirnov, Zh. Prikl. Khim., **37**, 2204 (1964); J. Appl. Chem. USSR, **37**, 2179 (1964).
127. J. Leonard, H. Daoust, J. Phys. Chem., **69**, 1174 (1965).
128. W. R. Moore, B. M. Tidswell, Makromol. Chem., **81**, 1 (1965).
129. M. L. Lakhanpal, M. Lal, R. K. Sharma, Indian J. Chem., **3**, 547 (1965).
130. G. Gee, J. B. M. Herbert, R. C. Roberts, Polymer, **6**, 541 (1965).
131. A. S. Michaels, R. W. Hausslein, J. Polym. Sci. C, **10**, 61 (1965).
132. A. A. Tager, L. Ya. Kasas', Dokl. Akad. Nauk, SSSR, **165**, 1122 (1965); Proc. Acad. Sci. USSR, **165**, 892 (1965).
133. P. J. Flory, W. J. Leonard, J. Am. Chem. Soc., **87**, 2102 (1965).
134. G. M. Pavlyuchenko, T. V. Gatovskaya, V. A. Kargin, Vysokomol. Soedin. A, **7**, 647 (1965); Polym. Sci. USSR, **7**, 714 (1965).
135. A. Kagemoto, S. Murakami, R. Fujishiro, Bull. Chem. Soc. Japan, **39**, 15 (1966).
136. S. G. Canagaratna, D. Margerison, J. P. Newport, Trans. Faraday Soc., **62**, 3058 (1966).
137. C. Cuniberti, U. Bianchi, Polymer, **7**, 151 (1966).
138. M. L. Lakhanpal, H. L. Taneja, R. K. Sharma, Indian J. Chem., **4**, 12 (1966).
139. M. L. Lakhanpal, V. Kapoor, R. K. Sharma, S. C. Sharma, Indian J. Chem., **4**, 59 (1966).
140. A. Kagemoto, S. Murakami, R. Fujishiro, Bull. Chem. Soc. Japan, **39**, 1814 (1966).
141. U. Bianchi, E. Pedemonte, C. Rossi, Makromol. Chem., **92**, 114 (1966).
142. G. Giacometti, A. Turolla, Z. Phys. Chem., NF, **51**, 108 (1966).
143. H. Benoit, C. Picot, Pure Appl. Chem., **12**, 545 (1966).
144. N. Kuwahara, T. Okazawa, M. Kaneko, J. Chem. Phys., **47**, 3357 (1967).
145. A. A. Tager, V. M. Andreyeva, J. Polym. Sci. C, **16**, 1145 (1967)
146. U. Bianchi, C. Cuniberti, E. Pedemonte, C. Rossi, J. Polym. Sci. A-2, **5**, 743 (1967).
147. J. Pouchly, J. Biros, K. Šolc, J. Vondrejsova, J. Polym. Sci. C, **16**, 679 (1967).
148. A. Kagemoto, S. Murakami, R. Fujishiro, Bull. Chem. Soc. Japan, **40**, 11 (1967).
149. J. Leonard, H. Daoust, Can. J. Chem., **45**, 409 (1967).
150. A. Kagemoto, S. Murakami, R. Fujishiro, Makromol. Chem., **105**, 154 (1967).
151. H. P. Schreiber, M. H. Waldman, J. Polym. Sci. A-2, **5**, 555 (1967).
152. W. R. Moore, J. Polym. Sci. C, **16**, 571 (1967).
153. W. Brown, J. Appl. Polym. Sci., **11**, 2381 (1967).
154. G. Allen, R. C. Ayerst, J. R. Cleveland, G. Gee, C. Price, J. Polym. Sci. C, **23**, 127 (1968).
155. B. E. Eichinger, P. J. Flory, Trans. Faraday Soc., **64**, 2061 (1968).
156. B. E. Eichinger, P. J. Flory, Trans. Faraday Soc., **64**, 2066 (1968).
157. A. A. Tager, A. I. Podlesnyak, L. V. Demidova, Vysokomol. Soedin. B, **10**, 601 (1968).
158. A. A. Tager, A. A. Anikeyeva, V. M. Andreyeva, T. Ya. Gumarova, L. A. Chemoskutova, Vysokomol. Soedin. A, **10**, 1661 (1968); Polym. Sci. USSR, **10**, 1926 (1968).
159. B. E. Eichinger, P. J. Flory, Trans. Faraday Soc., **64**, 2053 (1968).
160. P. J. Flory, J. L. Ellenson, B. E. Eichinger, Macromolecules, **1**, 279 (1968).
161. W. Bruns, F. Mehdom, K. Mirus, K. Ueberreiter, Kolloid Z., **224**, 17 (1968).
162. H. Daoust, A. Hade, Polymer, **9**, 47 (1968).
163. R. W. Kershaw, G. N. Malcolm, Trans. Faraday Soc., **64**, 323 (1968).
164. M. L. Lakhanpal, H. G. Singh, H. Singh, S. C. Sharma, Indian J. Chem., **6**, 95 (1968).
165. Yu. S. Lipatov, L. M. Sergeyeva, G. F. Kovalenko, Vysokomol. Soedin. B, **10**, 205 (1968).
166. B. E. Eichinger, P. J. Flory, Trans. Faraday Soc., **64**, 2035 (1968).

167. T. A. Bogayevskaya, T. V. Gatovskaya, V. A. Kargin, Vysokomol. Soedin. B, **10**, 376 (1968).
168. N. Kuwahara, T. Okazawa, M. Kaneko, J. Polym. Sci. C, **23**, 543 (1968).
169. S. Morimoto, J. Polym. Sci. A-1, **6**, 1547 (1968).
170. A. Kagemoto, R. Fujishiro, Bull. Chem. Soc. Japan, **41**, 2201 (1968).
171. T. A. Bogayevskaya, T. V. Gatovskaya, V. A. Kargin, Vysokomol. Soedin. A, **10**, 1357 (1968); Polym. Sci. USSR, **10**, 1574 (1968).
172. W. R. Moore, Contribution to Chem. Soc. Symposium, "Solution Properties of Natural Polymers", Edinburgh 1968; Chem. Soc. Special Publication No. 23, 185 (1968).
173. M. Masuzawa, C. Stirling, J. Appl. Polym. Sci., **12**, 2023 (1968).
174. J. Goldfarb, S. Rodriguez, Makromol. Chem., **116**, 96 (1968).
175. M. L. Lakhanpal, H. G. Singh, S. C. Sharma, Indian J. Chem., **6**, 436 (1968).
176. M. L. Lakhanpal, K. S. Chhina, S. C. Sharma, Indian J. Chem., **6**, 505 (1968).
177. T. A. Bogayevskaya, T. V. Gatovskaya, Vysokomol. Soedin. B, **10**, 555 (1968).
178. N. Kuwahara, T. Oikawa, M. Kaneko, J. Chem. Phys., **49**, 4972 (1968).
179. A. Kagemoto, R. Fujishiro, Makromol. Chem., **114**, 139 (1968).
180. R. Koningsveld, A. J. Staverman, J. Polym. Sci. A-2, **6**, 325 (1968).
181. D. Patterson, S. N. Bhattacharyya, P. Picker, Trans. Faraday Soc., **64**, 648 (1968).
182. R. D. Seeley, Rubber Chem. Technol., **41**, 608 (1968).
183. G. Lewis, A. F. Johnson, J. Chem. Soc. A, 1816 (1969).
184. U. Bianchi, C. Cuniberti, E. Pedemonte, C. Rossi, J. Polym. Sci., A-2, **7**, 855 (1969).
185. G. N. Malcolm, C. E. Baird, G. R. Bruce, K. G. Cheyne, R. W. Kershaw, M. C. Pratt, J. Polym. Sci. A-2, **7**, 1495 (1969).
186. S. Takagi, R. Fujishiro, Rep. Prog. Polym. Phys. (Japan), **12**, 39 (1969).
187. M. Hyodo, K. Kubo, K. Ogino, Rep. Prog. Polym. Phys. (Japan), **12**, 21 (1969).
188. G. Lewis, A. F. Johnson, Polymer, **11**, 336 (1970).
189. Th. G. Scholte, J. Polym. Sci. A-2, **8**, 841 (1970).
190. Th. G. Scholte, Eur. Polym. J., **6**, 1063 (1970).
191. S. Morimoto, Nippon Kagaku Zasshi, **91**, 31, 117 (1970).
192. A. A. Tager, A. I. Podlesnayak, M. V. Tsilipotkina, L. V. Adamova, A. A. Bakhareva, L. V. Demidova, Vysokomol. Soedin. A, **12**, 1320 (1970); Polym. Sci. USSR, **12**, 1497 (1970).
193. C. Gerth, F. H. Mueller, Kolloid. Z., **241**, 1071 (1970).
194. H. Nakayama, Bull. Chem. Soc. Japan., **43**, 1683 (1970).
195. P. J. T. Tait, P. J. Livesey, Polymer, **11**, 359 (1970).
196. P. Roy-Chowdhury, K. M. Kale, J. Appl. Polym. Sci., **14**, 2397 (1970).
197. T. A. Bogayavskaya, T. V. Gatovskaya, V. A. Kargin, Vysokomol. Soedin. A, **12**, 243 (1970); Polym. Sci. USSR, **12**, 279 (1970).
198. O. Smidsrød, J. E. Guillet, Macromolecules, **2**, 272 (1969).
199. K. Kubo K. Yamatsuta, K. Ogino, Rep. Prog. Polym. Phys. (Japan), **13**, 19 (1970).
200. Y. Taru, K. Yosizaki, E. Wada, K. Kanamaru, Rep. Prog. Polym. Phys. (Japan), **13**, 21 (1970).
201. S. Higashida, N. Kuwahara, M. Kaneko, Rep. Prog. Polym. Phys. (Japan), **13**, 5 (1970).
202. T. Oikawa, N. Kuwahara, M. Kaneko, Rep. Prog. Polym. Phys. (Japan), **13**, 9 (1970).
203. K. Kamada, H. Sato, Rep. Prog. Polym. Phys. (Japan), **13**, 47 (1970).
204. A. Muramoto, Polym. J., **1**, 450 (1970).
205. S. Morimoto, Makromol. Chem., **133**, 197 (1970) Rep. Prog. Polym. Phys. (Japan), **13**, 29 (1970).
206. Th. G. Scholte, J. Polym. Sci. A-2, **9**, 1553 (1971).
207. P. J. Flory, H. Hoecker, Trans. Faraday Soc., **67**, 2258 (1971).
208. H. Hoecker, P. J. Flory, Trans. Faraday Sco., **67**, 2270 (1971).
209. H. Hoecker, H. Shih, P. J. Flory, Trans. Faraday Soc., **67**, 2275 (1971).
210. G. Lewis, A. F. Johnson, J. Chem. Soc. A, 3528 (1971).
211. A. A. Tager, L. V. Adamova, M. V. Tsilipotkina, G. I. Florova, Vysokomol. Soedin. A, **13**, 654 (1971); Polym. Sci. USSR, **13**, 745 (1971).
212. G. Lewis, A. F. Johnson, J. Chem. Soc. A, 3524 (1971).
213. C. Booth, C. J. Devoy, Polymer, **12**, 309 (1971).
214. H. Vink, Eur. Polym. J., **7**, 1411 (1971).
215. C. Booth, C. J. Devoy, Polymer, **12**, 320 (1971).
216. H. Ochiai, K. Gekko, H. Yamamura, J. Polym. Sci., A-2, **9**, 1629 (1971).
217. A. Takizawa, T. Negishi, K. Ishikawa, Sen-i Gakkaishi, **26**, 567 (1971).
218. A. Kagemoto, Y. Itoi, Y. Baba, R. Fujishiro, Makromol. Chem., **150**, 255 (1971).
219. A. A. Tager, A. A. Anikeyeva, L. V. Adamova, V. M. Andreyeva, T. A. Kuz'mina, M. V. Tsilipotkina, Vysokomol. Soedin. A, **13**, 659 (1971); Polym. Sci. USSR, **13**, 751 (1971).
220. K. Yamatsuta, K. Kubo, K. Ogino, Rep. Prog. Polym. Phys. (Japan), **14**, 29 (1971).
221. T. Okazawa, M. Kaneko, Rep. Prog. Polym. Phys. (Japan), **14**, 7 (1971); Polym. J., **2**, 747 (1971).
222. S. Ichihara, A. Komatsu, T. Hata, Polym. J., **2**, 644 (1971).
223. S. Ichihara, A. Komatsu, T. Hata, Polym. J., **2**, 650 (1971).
224. S. Ichihara, A. Komatsu, T. Hata, Polym. J., **2**, 640 (1971).
225. S. Morimoto, Bull. Chem. Soc. Japan, **44**, 879 (1971).
226. C. Price, R. C. Williams, R. C. Ayerst, "Amorphous Polymers", Wiley, London, 1972, p. 117.
227. J. H. Rai, W. G. Miller, Macromolecules, **5**, 45 (1972).
228. S. H. Maron, F. E. Filisko, J. Macromol. Sci., B, **6**, 57 (1972).
229. B. J. Reitveld, Th. G. Scholte, J. P. L. Pijpres, Brit. Polym. J., **4**, 109 (1972).
230. S. H. Maron, F. E. Filisko, J. Macromol. Sci., B, **6**, 79 (1972).

231. S. H. Maron, F. E. Filisko, J. Macromol. Sci. B, **6**, 413 (1972); S. H. Maron, M.-S. Lee, J. Macromol. Sci., Phys., **7**, 29 (1973).
232. A. A. Tager, A. I. Suvorova, Yu. S. Bessonov, A. I. Podlesnyak, I. A. Koroleva, L. V. Adamova, M. V. Tsilipotkina, Vysokomol. Soedin. A, **13**, 2454 (1971); Polym. Sci. USSR, **13**, 2755 (1971).
233. J. P. Cartier, H. Daoust, Can. J. Chem., **49**, 3935 (1971).
234. S. L. Mel'nikova, A. N. Korol, Zh. Fiz. Khim., **46**, 30 (1972); Russian J. Phys. Chem., **46**, 17 (1972).
235. F. H. Covitz, J. W. King, J. Polym. Sci. A-1, **10** 689 (1972).
236. K. Gekko, J. Sci. Hiroshima Univ., Ser. A-2, **35**, 111 (1971).
237. A. A. Tager, L. V. Adamova, M. V. Tsilipotkina, R. K. Kuleshova, S. V. Fedotova, Vysokomol. Soedin. B, **14**, 235 (1972).
238. M. D. Lechner, G. V. Schulz, B. A. Wolf, J. Colloid Interface Sci., **39**, 462 (1972).
239. M. Mitooka, Buneski Kagaku, **21**, 717 (1972).
240. P. Davalloo, J. L. Gainer, K. R. Hall, J. Chem. Thermodyn., **4**, 691 (1972).
241. A. A. Tager, L. V. Adamova, Yu. S. Bessonov, V. N. Kuznetsov, T. A. Plyusnina, V. V. Soldatov, M. V. Tsilipotkina, Vysokomol. Soedin. A, **14**, 1991 (1972); Polym. Sci. USSR, **14**, 2233 (1972).
242. B. M. Fechner, C. Strazielle, Makromol. Chem., **160**, 195 (1972).
243. S. H. Maron, M.-S. Lee, J. Macromol. Sci. Phys., **7**, 47 (1973).
244. S. H. Maron, F. E. Filisko, J. Macromol. Sci. Phys., **6**, 413 (1972); S. H. Maron, M.-S. Lee, J. Macromol. Sci. Phys., **7**, 61 (1973).
245. H. Horacek, Kolloid-Z. Z. Polym., **250**, 863 (1972).
246. D. A. Blackadder, T. L. Roberts, Angew. Makromol. Chem., **27**, 165 (1972).
247. F. Candau, J. Francois, C. R. Acad. Sci., Ser. C, **276**, 571 (1973).
248. G. Delmas, P. Tancrede, Eur. Polym. J., **9**, 199 (1973).
249. K. Tamura, S. Murakami, R. Fujishiro, Polymer, **14**, 237 (1973).
250. A. Franck, Angew. Makromol. Chem., **29/30**, 179 (1973).
251. E. M. Khalik, L. S. Gal'braikh, N. I. Ilieva, S. I. Meerson, Z. A. Rogovin, Vysokomol. Soedin. B, **15**, 452 (1973).
252. R. A. Orwoll, J. A. Small, Macromolecules, **6**, 755 (1973).
253. R. S. Chahal, W.-P. Kao, D. Patterson. J. Chem. Soc., Faraday Trans. I, **69**, 1834 (1973).
254. T. N. Nekrasova, V. E. Eskin, Vysokomol. Soedin. A, **15**, 2429 (1973); Polym. Sci. USSR, **15**, 2749 (1973).
255. D. Gaeckle, W.-P. Kao, Patterson, M. Rinfret, J. Chem. Soc., Faraday Trans., I, 1849 (1943).
256. A. Nakajima, F. Hamada, K. Yasue, K. Fujisawa, T. Shiomi, Makromol. Chem., **175**, 197 (1974).
257. Y. Baba, H. Katayama, A. Kagemoto. Makromol. Chem., **175**, 209 (1974).
258. G. Castello, G. D'Amato, J. Chromatogr., **90**, 291 (1974).
259. A. A. Tager, L. V. Adamova, V. V. Serpinskii, M. V. Tsilipotkina, Vysokomol. Soedin. A, **16**, 203 (1974); Polym. Sci. USSR, **16**, 240 (1974).
260. Z. Masa, A. Zivny, J. Biros, Collect. Czech. Chem. Commun., **38**, 3123 (1973).
261. H. Vink, Eur. Polym. J., **10**, 149 (1974).
262. Y. Baba, H. Katayama, A. Kagemoto, Polym. J., **6**, 230 (1974).
263. Y.-K. Leung, B. E. Eichinger, Macromolecules, **7**, 685 (1974).
264. F. E. Filisko, R. S. Raghava, G. S. Y. Yeh, J. Macromol. Sci. B: Phys., **10**, 371 (1974).
265. G. Manzini, V. Crescenzi, R. Furlanetto, Macromolecules, **8**, 198 (1975).
266. R. S. Raghava, F. E. Filisko, J. Appl. Phys., **45**, 4155 (1974).
267. S. Arichi, M. Yoshida, Y. Ogawa, Bull. Chem. Soc. Japan, **48**, 1417 (1975).
268. Y. H. Chang, D. C. Bonner, J. Appl. Polym. Sci., **19**, 2439 (1975).
269. S. Murakami, F. Kimura, R. Fujishiro, Makromol. Chem., **176**, 3425 (1975).
270. W. E. Hammers, B. C. Bos, L. H. Vaas, Y. J. W. A. Loomans, C. L. De Ligny, J. Polym. Sci., Polym. Phys. Ed., **13**, 401 (1975).
271. A. A. Tager, Yu. S. Bessonov, Vysokomol. Soedin. A, **17**, 2377 (1975); Polym. Sci. USSR, **17**, 2734 (1975).
272. A. A. Tager, Yu. S. Bessonov, Vysokomol. Soedin. A, **17**, 2383 (1975); Polym. Sci. USSR, **17**, 2741 (1975).
273. E. Schroeder, U. Duesedau, J. T. Haponiuk, Faserforsch. Textiltech., **27**, 67 (1976).
274. M. Kawakami, S. Kagawa, M. Egashira, Nagasaki Daigaku Kogakubu Kenkyu Hokoku, **6**, 135 (1975).
275. V. P. Kiselev, E. M. Shakhova, E. Z. Fainberg, A. D. Virnik, Z. A. Rogovin, Vysokomol. Soedin. B, **18**, 847 (1976).
276. M. Kawakami, M. Egashira, S. Kagawa, Bull. Chem. Soc. Jpn., **49**, 3449 (1976).
277. M. L. Lakhanpal, S. C. Sharma, B. Krishan, R. N. Parashar, Indian J. Chem. A, **14**, 642 (1976).
278. D. D. Deshpande, C. S. Prabhu, Macromolecules, **10**, 433 (1977).
279. Y. Baba, K. Fujimoto, A. Kagemoto, R. Fujishiro, Makromol. Chem., **178**, 1439 (1977).
280. H. Ochiai, Y. Nishihara, S. Yamaguchi, I. Murackami, J. Sci. Hiroshima Univ., Ser. A: Phys. Chem., **41**, 157 (1977).
281. G. DiPaola-Baranyi, J. M. Braun, J. E. Guillet, Macromolecules, **11**, 224 (1978).
282. G. DiPaola-Baranyi, J. E. Guillet, Macromolecules, **11**, 228 (1978).
283. J. Francois, F. Candau, D. Sarazin, C. R. Hebd. Seances Acad. Sci., Ser. C, **286**, 325 (1978).
284. K. A. Karim, D. C. Bonner, J. Appl. Polym. Sci., **22**, 1277 (1978).
285. M. Okubo, K. Ueberreiter, Colloid Polym. Sci., **256**, 941 (1978).
286. Y. Baba, A. Kagemoto, Netsu Sokutei, **4**, 57 (1977).
287. A. A. Tager, T. I. Kirillova, T. V. Ikanina, Vysokomol. Soedin. A, **20**, 2543 (1978).
288. G. DiPaola-Baranyi, J. E. Guillet, J. Klein, H.-E. Jeberien, J. Chromatogr., **166**, 349 (1978).

289. I. Fujihara, K. Tamura, S. Murakami, R. Fujishiro, Polym. J., **11**, 153 (1979).
290. J. Roots, B. Nystrom, Eur. Polym. J., **14**, 773 (1978).
291. A. N. Korol, J. Chromatogr., **172**, 77 (1979).
292. J. Klein, W. Scholz, Makromol. Chem., **180**, 1477 (1979).
293. H. Phuong-Nguyen, G. Delmas, Macromolecules, **12**, 746 (1979).
294. H. Phuong-Nguyen, G. Delmas. Macromolecules, **12**, 740 (1979).
295. A. Heintz, R. N. Lichtenthaler, J. M. Prausnitz, Ber. Bunsenges. Phys. Chem., **83**, 926 (1979).
296. T. M. Aminabhavi, P. Munk, Macromolecules, **12**, 1186 (1979).
297. J.-G. Lee, M. Ono, F. Hamada, A. Nakajima, Polym. Bull. (Berlin), **1**, 763 (1979).
298. A. M. Basedow, K. H. Ebert, J. Polym. Sci., Polym. Symp., **66**, 101 (1979).
299. G. DiPaola-Baranyi, J. F. Guillet, H.-E. Jeberien, J. Klein, Makromol. Chem., **181**, 215 (1980).
300. K. Ueberreiter, Makromol. Chem., Rapid Commun., **1**, 143 (1980).
301. A. M. Basedow, K. H. Ebert, W. Feigenbutz, Makromol. Chem., **181**, 1071 (1980).
302. J. Klein, H. E. Jeberrien, Makromol. Chem., **181**, 1237 (1980).
303. H. Schoenert, F. Monshausen, Colloid Polym. Sci., **258**, 578 (1980).
304. H. Suzuki, K. Kamide, Y. Miyazaki, Netsu Sokutei, **7**, 37 (1980).
305. Y. Iwai, M. Nagafuji, Y. Arai, Sekiyu Gakkaishi, **23**, 215 (1980).
306. S. Gunduz, S. Dincer, Polymer, **21**, 1041 (1980).
307. G. DiPaola-Baranyi, Macromolecules, **14**, 683 (1981).
308. J. E. G. Lipson, J. E. Guillet, J. Polym Sci., Polym. Phys. Ed., **19**, 1199 (1981).
309. H. Ogawa, Y. Baba, A. Kagemoto, Makromol. Chem., **182**, 2495 (1981).
310. T. C. Ward, D. P. Sheehy, J. E. McGrath, J. S. Riffle, Macromolecules, **14**, 1791 (1981).
311. Z. N. Medved, N. A. Starikova, Zh. Fiz. Khim., **55**, 2438 (1981); Russian J. Phys. Chem., **55**, 1380 (1981).
312. J. Koller, E. Killmann, Makromol. Chem., **182**, 3579 (1981).
313. J. C. Day, I. D. Robb, Polymer, **22**, 1530 (1981).
314. P. Volatie, D. Sandulescu, Rev. Roum. Phys., **27**, 81 (1982).
315. H. Ochiai, T. Ohashi, Y. Tadokoro, I. Murakami, Polym. J. (Tokyo), **14**, 457 (1982).
316. A. E. Habboush, N. S. Nassory, Iraqi J. Sci., **22**, 362 (1981).
317. M. J. Fernandez-Berridi, T. F. Otero, G. M. Guzman, J. M. Elorza, Polymer, **23**, 1361 (1982).
318. M. J. Fernandez-Berridi, J. I. Eguiazabal, J. M. Elorza, J. J. Iruin, J. Polym. Sci., Polym. Phys. Ed., **21**, 859 (1983).
319. I. Fujihara, T. Sakuta, A. Kagemoto, Makromol. Chem., **184**, 1231 (1983).
320. M. Galin, Polymer, **24**, 865 (1983); Polymer, **25**, 1784 (1984).
321. T. Nagao, Hakodate Kogyo Senmon Gakko Kiyo, **17**, 103 (1983).
322. Z. N. Medved, N. A. Starikova, Zh. Fiz. Khim., **57**, 2590 (1983); Russian J. Phys. Chem., **57**, 1563 (1983).
323. Z. N. Meved, N. I. Petrova, O. G. Tarakanov, Vysokomol. Soedin. B, **25**, 764 (1983).
324. H. Daoust, D. St-Cyr, Macromolecules, **17**, 596 (1984).
325. R. H. Schuster, H. Graeter, H.-J. Cantow, Macromolecules, **17**, 596 (1984); Polym. Bull. (Berlin), 14, 379 (1985).
326. K. V. Kir'yanov, Termodin. Org. Soedin, 58 (1983).
327. S. Ituno, M. Ohzono, Y. Iwai, Y. Arai, Kobunshi Ronbunshu, **42**, 73 (1985).
328. I. B. Rabinovich, T. B. Khlyustova, A. N. Mochalov, Vysokomol. Soedin. A, **27**, 525 (1985).
329. D. Sarazin, J. Francois, Macromolecules, **18**, 1190 (1985).
330. E. Killmann, M. Bergmann, Colloid Polym. Sci., **263**, 381 (1985).
331. K. Naito, N. Ohwada, S. Moriguchi, S. Takei, J. Chromatogr., **330**, 193 (1985).
332. T. Shiomi, K. Kohno, K. Yoneda, T. Tomita, M. Miya, K. Imai, Macromolecules, **18**, 414 (1985).
333. N. Aelenei, Mater. Plast. (Bucharest), **22**, 92 (1985).
334. I. B. Rabinovich, T. B. Khlyustova, A. N. Mochalov, Vysokomol. Soedin. A, **27**, 1724 (1985).
335. B. A. Bolton, S. Kint, G. F. Bailey, J. R. Scherer, J. Phys. Chem., **90**, 1207 (1986).
336. K. Inoue, R. Fujii, Y. Baba, A. Kagemoto, C. L. Beatty, Makromol. Chem., **187**, 923 (1986).
337. E. Casur, T. G. Smith, J. Appl. Polym. Sci., **31**, 1425 (1986).
338. M. D. Sefcik, J. Polym. Sci., Polym. Phys. Ed., **24**, 935 (1986).
339. M. D. Sefcik, J. Polym. Sci., Polym. Phys. Ed., **24**, 957 (1986).
340. T. Nagao, Hakodate Kogyo Koto Senmon Gakko Kiyo, **19**, 161 (1985).
341. U. B. Goradia, H. G. Spencer, J. Appl. Polym. Sci., **33**, 1525 (1987).
342. P. Schneider, A. Heintz, Thermochim. Acta, **119**, 47 (1987).
343. H. Azuma, K. Hanada, Y. Yoshikawa, Y. Baba, A. Kagemoto, Thermochim. Acta, **123**, 271 (1988).
344. H. Becker, R. Gnauck, J. Chromatogr., **410**, 267 (1987).
345. M. Bender, A. Heintz, J. Solution Chem., **18**, 1 (1989).
346. A. Edelman, A. Fradet, Polymer, **30**, 324 (1989).
347. R. Parashar, H. G. Singh, S. C. Sharma, Indian J. Chem. A, **28**, 317 (1989).
348. A. Zivny, J. Biros, J. Pouchly, Makromol. Chem., **190**, 1345 (1989).
349. R. Parashar, S. C. Sharma, Indian J. Chem. A, **28**, 1042 (1988).
350. Y. Kamiya, Y. Naito, K. Mizoguchi, J. Polym. Sci., Polym. Phys. Ed., **27**, 2243 (1989).
351. M. Galin, Polymer, **30**, 2074 (1989).
352. B. Andersson, G. Olofsson, J. Solution Chem., **18**, 1019 (1989).
353. H. G. Schild, D. A. Tirrell, J. Phys. Chem., **94**, 4352 (1990).
354. M. Galin, Macromolecules, **23**, 3006 (1990).
355. A. Hamdouni, J. Leonard, V. T. Bui, Polym. Commun., **31**, 258 (1990).
356. M. Labadie, J. C. Bollinger, Thermochim. Acta, **162**, 445 (1990).

357. E. Killmann, F. Cordt, F. Moeller, Makromol. Chem., **191**, 2929 (1990).
358. M. M. Selin, Khim.-Farm. Zh., **24**, 72 (1990).
359. A. P. Safronov, A. A. Tager, Vysokomol. Soedin. A, **33**, 2198 (1991).
360. Y. Kamiya, D. Bourbon, K. Mizoguchi, Y. Naito, Polym. J. (Tokyo), **24**, 443 (1992).
361. A. P. Mar'in, Yu. A. Shlyapnikov, A. Zh. Makhkamov, A. T. Dzhalilov, Int. J. Polym. Mater., **16**, 37 (1992); Oxid. Commun., **20**, 57 (1997).
362. G. DiPaola-Baranyi, C. K. Hsiao, M. Spes, P. G. Odell, R. A. Burt, J. Appl. Polym. Sci., Appl. Polym. Symp. **51**, 195 (1992).
363. F. Yilmaz, O. G. Cankurtaran, B. M. Baysal, Polymer, **33**, 4563 (1992).
364. J. Y. Wang, G. Charlet, Macromolecules, **26**, 2413 (1993).
365. D. Eagland, N. J. Crowther, C. J. Butler, Polymer, **34**, 2804 (1993); **35**, 4704 (1994).
366. A. P. Safronov, A. A. Tager, E. S. Klyuzhin, L. V. Adamova, Vysokomol. Soedin. A, **35**, 700 (1993).
367. H. Phuong-Nguyen, G. Delmas, J. Solution Chem., **23**, 249 (1994).
368. K. Nagahama, H. Inomata, S. Saito, Fluid Phase Equilib., **96**, 203 (1994).
369. H. Phuong-Nguyen, G. Delmas, Thermochim. Acta, **238**, 257 (1994).
370. R. J. Hernandez, R. Gavara, J. Polym. Sci., Polym. Phys. Ed., **32**, 2367 (1994).
371. E. Killmann, F. Cordt, F. Moeller, H. Zellner, Macromol. Chem. Phys., **196**, 47 (1995).
372. H.-M. Petri, N. Schuld, B. A. Wolf, Macromolecules, **28**, 4975 (1995).
373. M. Carlsson, D. Hallen, P. Linse, J. Chem. Soc., Faraday Trans., **91**, 2081 (1995).
374. Y. Kamiya, K. Terada, Y. Naito, J.-S. Wang, J. Polym. Sci., Polym. Phys. Ed., **33**, 1663 (1995).
375. C. Grossmann, R. Tintinger, J. Zhu, G. Maurer, Fluid Phase Equilib., **106**, 111 (1995).
376. M. Galin, Polymer, **36**, 3533 (1995).
377. T. Banerjee, M. Chhajer, G. G. Lipscomb, Macromolecules, **28**, 8563 (1995).
378. R. C. Castells, L. M. Romero, A. M. Nardillo, Macromolecules, **29**, 4278 (1996).
379. Y. N. Matyushin, T. S. Konkova, Int. Annu. Conf. ICT, 27th, 122.1 (1996).
380. S. A. Vshivkov, E. V. Rusinova, G. B. Zarubin, V. N. Dubchak, Vysokomol. Soedin., **38AB**, 868 (1996).
381. A. P. Safronov, A. A. Tager, E. V. Koroleva, Vysokomol. Soedin., **38AB**, 900 (1996).
382. E. Sabadini, E. M. Assano, T. D. Z. Atvars, J. Appl. Polym. Sci., **63**, 301 (1997).
383. F. Yilmaz, O. Cankurtaran, Polymer, **38**, 3539 (1997).
384. Y. M. Lotmentsev, D. V. Pleshakov, Propellants, Explos., Pyrotech., **22**, 203 (1997).
385. T. Banerjee, G. G. Lipscomb, Polymer, **38**, 5807 (1997).
386. P. Gotthardt, A. Grueger, H. G. Brion, R. Plaetschke, R. Kirchleim, Macromolecules, **30**, 8058 (1997).
387. Y. Kamiya, K. Mizoguchi, K. Terada, Y. Fujiwara, J.-S. Wang, Macromolecules, **31**, 472 (1998).
388. S. Erdal, I. Bahar, B. Erman, Polymer, **39**, 2035 (1998).

Heats of Solution of Some Common Polymers

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A. INTRODUCTION

Heats of solution (at infinite dilution) are tabulated below for some common polymers dissolved in some common solvents. The values were determined calorimetrically and are reported here as the heat (in Joules) absorbed per gram of polymer as pure polymer and a large excess of pure solvent mixed isothermally to form a homogeneous solution.

The state of the polymer before dissolution is indicated in the fourth column. It can significantly affect the heat of solution. An amorphous polymer below its glass transition temperature (G) often dissolves with release of heat. The heat of solution for a glass is usually dependent on temperature and, to some extent, on the thermal history of the glass. The dissolving of a semicrystalline polymer (C) requires an additional amount of heat associated with the disordering of crystalline regions. Consequently its heat of solution is usually positive and depends on the extent of crystallinity of the sample. An amorphous polymer above its glass transition temperature (L) usually dissolves with the absorption of a small amount of heat but may also dissolve with the release of heat.

Most of the references to measurements made before 1972 were collected by D. R. Cooper (Manchester) for the second edition of this Handbook. A more complete listing of references to heats of solution can be found in the chapter "Heat, Entropy, and Volume Changes for Polymer–Liquid Systems."

B. TABLE OF HEATS OF SOLUTION

Solvent	ΔH (J/g)	Temp. (°C)	State	MW × 10^{-4} (g/mol)	Refs.
Poly(butadiene)					
Benzene	13	27	L	8	5
Cyclohexane	5.4	25	L	–	23
n-Hexadecane	4.9	25	L	–	24
Poly(dimethyl siloxane)					
Benzene	14	25	L	10	11
Carbon tetrachloride	2.4	25	L	10	11
Cyclohexane	5.1	25	L	10	11
Cyclohexane	5.2	25	L	2.0	23
n-Decane	3.9	25	L	10	7
Ethyl ether	−1.3	25	L	8	10
n-Hexadecane	5.5	25	L	10	7
n-Hexadecane	5.5	25	L	2	24
n-Pentane	−0.9	25	L	10	7
Poly(ethylene)					
Cyclohexane	205	106	C	90	32
1,2,3,4-Tetrahydronaphthalene[a]	39	130	L	67	12
1,2,3,4-Tetrahydronaphthalene[a]	560	80	C	67	12
1,2,3,4-Tetrahydronaphthalene[b]	88	150	L	14	12
1,2,3,4-Tetrahydronaphthalene[b]	1200	110	C	14	12
Toluene	262	126	C	90	32
Poly(isobutylene)					
Benzene	19	25	L	3	8
Benzene	19	30	L	4.8	25

Solvent	ΔH (J/g)	Temp. (°C)	State	MW × 10^{-4} (g/mol)	Refs.
Benzene	23	25	L	0.13	21
Carbon tetrachloride	4.1	–	L	200	9
Carbon tetrachloride	5.0	25	L	0.13	21
Cyclohexane	−0.7	25	L	3	8
Cyclohexane	−0.6	25	L	4.5	24
Cyclohexane	1.1	25	L	0.13	21
n-Decane	−0.5	25	L	3	8
Ethylbenzene	3.5	69	L	4.8	25
Ethylbenzene	9.3	18	L	4.8	25
Ethyl ether	2.8	25	L	3	8
n-Hexadecane	0.04	25	L	3	1
n-Hexadecane	0.9	25	L	0.45	24
n-Hexadecane	1.0	25	L	0.13	21
n-Pentane	−3.6	25	L	3	8
n-Pentane	−3.2	25	L	0.13	21
Poly(isoprene)					
Benzene	12	16	L	0.4	2
Poly(methyl methacrylate)					
Chloroform	−84	25	G	10	14
Chloroform	−37	25	C	–	14
Poly(oxyethylene)					
Benzene	170	30	C	4.3	26
Carbon tetrachloride[c]	16	30	L	0.06	31
Carbon tetrachloride[d]	−12	30	L	0.04	31
Chloroform	52	30	C	0.6	18
Water	−120	80	L	0.5	4
Water[c]	−160	25	L	0.02	30
Water[d]	−200	25	L	0.02	30
Water	10	25	C	2	28
Water	24	30	C	0.6	18
Water	40	30	C	4.3	26
Poly(oxypropylene)					
Carbon tetrachloride	−20	6	L	0.2	13
Carbon tetrachloride[c]	−8	30	L	0.2	33
Carbon tetrachloride[d]	−14	30	L	0.16	33
Chloroform	−100	6	L	0.2	13
Methyl alcohol	−7	27	L	0.1	6
Poly(propylene)					
Benzene	31	25	L	1.8	27
Carbon tetrachloride	6.4	25	L	1.8	22
Carbon tetrachloride	6.6	25	L	1.8	27
Chloroform	17	25	L	1.8	22
Chloroform	17	25	L	1.8	27
Cyclohexane	2.3	25	L	0.6	23
Cyclohexane	3.9	25	L	1.8	27
n-Decane	3.1	25	L	1.8	27
Ethylbenzene	14	25	L	1.8	27
n-Hexadecane	2.3	25	L	0.6	24
n-Pentane	−4.7	25	L	1.8	27
1,2,3,4-Tetrahydronaphthalene	320	120	C	–	12
Toluene	16.5	25	L	1.8	27
Poly(styrene)					
Carbon tetrachloride	−20	25	G	50	19
Chloroform	−24	45	G	10	16
Cyclohexane	−8.2	26	G	70	15
Ethylbenzene	−6	110	L	11	20
Ethylbenzene	−24	35	G	11	20
Ethylbenzene	−22	30	G	6	17
Ethylbenzene	−5	78	G	11	20
Toluene	−3	100	L	11	20
Toluene	−26	35	G	11	20
Toluene	−22	26	G	70	15
Toluene	−21	30	G	6	17
Toluene	−7	78	G	11	20
Toluene	−2.5	25	G	–	29
Poly(vinyl acetate)					

Solvent	ΔH (J/g)	Temp. (°C)	State	MW $\times 10^{-4}$ (g/mol)	Refs.
Benzene	2.3	25	G	14	3
Chloroform	−45	25	G	14	3
Methyl alcohol	28	25	G	14	3

[a] High-pressure poly(ethylene).
[b] Low-Pressure poly(ethylene).
[c] Hydroxy terminated.
[d] Methoxy terminated.

C. REFERENCES

1. E. Calvet, Bull. Soc. Chim., **12**, 553 (1945).
2. G. Gee, W. J. C. Orr, Trans. Faraday Soc., **42**, 507 (1946).
3. H. Daoust, M. Rinfret, Can. J. Chem., **32**, 492 (1954).
4. G. N. Malcolm, J. S. Rowlinson, Trans. Faraday Soc., **53**, 921 (1957).
5. R. S. Jessup, J. Res. Natl. Bur. Std., **62**, 1 (1959).
6. M. L. Lakhanpal, B. E. Conway, J. Polym. Sci., **46**, 75 (1960).
7. G. Delmas, D. Patterson, D. Bohme, Trans. Faraday Soc., **58**, 2116 (1962).
8. G. Delmas, D. Patterson, T. Somcynsky, J. Polym. Sci., **57**, 79 (1962).
9. A. A. Tager, A. I. Podlesnyak, Polym. Sci. USSR, **4**, 698 (1963).
10. D. Patterson. J. Polym. Sci. A, **2**, 5177 (1964).
11. G. Delmas, D. Patterson. S. N. Bhattacharyya, J. Phys. Chem., **68**, 1468 (1964).
12. H. P. Schreiber, M. H. Waldman, J. Polym. Sci. A-2, **5**, 555 (1967).
13. R. W. Kershaw, G. N. Malcolm, Trans. Faraday Soc., **64**, 323 (1968).
14. C. Gerth, F. H. Mueller, Kolloid-Z., **241**, 1071 (1970).
15. A. A. Tager, A. I. Podlesnayak, M. V. Tsilipotkina, L. V. Adamova, A. A. Bakhareva, L. V. Demidova, Polym. Sci. USSR, **12**, 1497 (1970).
16. S. Morimoto, Bull. Chem. Soc. Japan, **44**, 879 (1971).
17. S. H. Maron, F. E. Filisko, J. Macromol. Sci. B, **6**, 57 (1972).
18. S. H. Maron, F. E. Filisko, J. Macromol. Sci. B, **6**, 79 (1972).
19. P. Davalloo, J. L. Gainer, K. R. Hall, J. Chem. Thermodyn., **4**, 691 (1972).
20. F. E. Filisko, R. S. Raghava, G. S. Y. Yeh, J. Macromol. Sci. B: Phys., **10**, 371 (1974).
21. D. D. Deshpande, C. S. Prabhu, Macromolecules, **10**, 433 (1977).
22. H. Ochiai, Y. Nishihara, S. Yamaguchi, I. Murakami, J. Sci. Hiroshima Univ. Ser. A: Phys. Chem., **41**, 157 (1977).
23. H. Phuong-Nguyen, G. Delmas, Macromolecules, **12**, 746 (1979).
24. H. Phuong-Nguyen, G. Delmas, Macromolecules, **12**, 740 (1979).
25. J.-G. Lee, M. Ono, F. Hamad, A. Nakajima, Polym. Bull. (Berlin), **1**, 763 (1979).
26. J. Koller, E. Killmann, Makromol. Chem., **182**, 3579 (1981).
27. H. Ochiai, T. Ohashi, Y. Tadokoro, I. Murakami, Polym. J. (Tokyo), **14**, 457 (1982).
28. H. Daoust, D. St-Cyr, Macromolecules, **17**, 596 (1984)
29. S. Morimoto, Thesis, Osaka City Univ, 1970, as cited in T. Shiomi, K. Kohno, K. Yoneda, T. Tomita, M. Miya, K. Imai, Macromolecules, **18**, 414 (1985).
30. J. Biros, J. Pouchly, A. Zivny, Makromol. Chem., **188**, 379 (1987).
31. E. Killmannn, F. Cordt, F. Möller, Makromol. Chem., **191**, 2929 (1990).
32. H. Phuong-Nguyen, G. Delmas, J. Soln. Chem., **23**, 249 (1994).
33. E. Killmannn, F. Cordt, F. Möller, H. Zellner, Makromol. Chem. Phys. **196**, 47 (1995).

Solubility Parameter Values

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A. INTRODUCTION

Applications of solubility parameters include selecting compatible solvents for coating resins, predicting the swelling of cured elastomers by solvents, estimating solvent vapor pressure in polymer solutions for devolatilization and reaction systems (16), and predicting phase equilibria for polymer–polymer (107), polymer–binary (93), random copolymer (102), and multicomponent solvents (38, 98,108,109).

Cohesive energy density and solubility parameters are defined in the section on miscibility of solvents and polymers (Section B). In addition, the applicability of solubility parameters to thermodynamic calculations and their limitations are discussed. Section C contains methods for measuring, calculating and correlating solubility parameters of solvents and polymers. Section D contains

alphabetical listings of solubility parameters (Table 7), a list of solubility parameters in rank order (Table 8), a list of three-component solubility parameters of solvents (Table 9), and a list of solubility parameters of polymers (Table 10). With the exception of Table 7, solubility parameter values are reported in $MPa^{1/2}$ units. The table showing solubility parameter value ranges for polymers (Table 3.4 in the third edition) has not been reproduced here.

B. MISCIBILITY OF SOLVENTS AND POLYMERS

1. Cohesive Energy Density and the Solubility (Hildebrand) Parameter

Dissolution of an amorphous polymer in a solvent is governed by the free energy of mixing

$$\Delta G_m = \Delta H_m - T\Delta S_m \tag{B1}$$

where ΔG_m is the Gibbs free energy change on mixing, ΔH_m is the enthalpy change on mixing, T the absolute temperature, and ΔS_m is the entropy change on mixing. A negative value of the free energy change on mixing means that the mixing process will occur spontaneously. Otherwise, two or more phases result from the mixing process. Since the dissolution of a high molecular weight polymer is always connected with a small or modest increase in entropy, the enthalpy term (the sign and magnitude of ΔH_m) is the deciding factor in determining the sign of the Gibbs free energy change. Solubility parameters were developed to describe the enthalpy of mixing of simple liquids (nonpolar, nonassociating solvents), but have been extended to polar solvents and polymers.

Hildebrand and Scott (59) and Scatchard (101) proposed that

$$\Delta H_m = V\left(\left(\frac{\Delta E_1^v}{V_1}\right)^{1/2} - \left(\frac{\Delta E_2^v}{V_2}\right)^{1/2}\right)^2 \phi_1\phi_2 \tag{B2}$$

where V is the volume of the mixture, ΔE_i^v the energy of vaporization of species i, V_i the molar volume of species i, and ϕ_i the volume fraction of i in the mixture. ΔE_i^v is the energy change upon isothermal vaporization of the saturated liquid to the ideal gas state at infinite volume (94). The cohesive energy density (CED), ΔE_i^v, is the energy of vaporization per cm^3. The solubility parameter has been defined as the square root of the cohesive energy density and describes the attractive strength between molecules of the material.

$$\delta_i = \left(\frac{\Delta E_i^v}{V_i}\right)^{1/2} \tag{B3}$$

The dimensions of δ_i are

$$(cal/cm^3)^{1/2} = (4.187\,J/10^{-6}\,m^3)^{1/2} = 2.046 \times 10^3 (J/m^3)^{1/2} = 2.046\,MPa^{1/2} \tag{B4}$$

The solubility parameter can be interpreted as the "internal pressure" of the solvent (9–11). δ_i is called the Hildebrand parameter by some authors. Other researchers (13) prefer the term, "cohesion parameter", since it correlates with a large number of physical and chemical properties, and not just the miscibility of the components. The solubility parameter of a mixture is often taken as the sum of the products of the component solubility parameters with their volume fractions:

$$\delta_{mixture} = \sum_i \delta_i\phi_i \tag{B5}$$

Relation between δ_1 and ΔH_m Equation (B2) can be rewritten to give the heat of mixing per unit volume for a binary mixture:

$$\frac{\Delta H_m}{V} = (\delta_1 - \delta_2)^2\phi_1\phi_2 \tag{B6}$$

Equation (B6) gives the heat of mixing of regular solutions in which the components mix with: (a) no volume change on mixing at constant pressure, (b) no reaction between the components, and (c) no complex formation or special associations (114). The heat of mixing must be smaller than the entropic term in Eq. (B1) for polymer–solvent miscibility ($\Delta G_m \leq 0$). When $\delta_1 = \delta_2$, the free energy of mixing will always be less than zero for regular solutions and the components will be miscible in all proportions. In general, the solubility parameter difference, $(\delta_1 - \delta_2)$ must be small for miscibility over the entire volume fraction range.

Relation between δ_1 and ΔH_1^v The energy change on isothermal vaporization can be related to the enthalpy of vaporization:

$$\Delta E_i^v = \Delta H_i^v + \Delta H_i^\infty - RT + p_i^s V_i \tag{B7}$$

where ΔH_i^v is the enthalpy of vaporization at standard conditions, ΔH_i^∞ the molar increase in enthalpy on isothermally expanding the saturated vapor to zero pressure, R the ideal gas constant, and p^s the saturation vapor pressure at temperature, T. At pressures below 1 atm, the ΔH_i^∞ and $p^s V_i$ terms are usually much less than the ΔH_i^v and RT terms, and Eq. (B7) reduces to Eq. (B8):

$$\Delta E_i^v = \Delta H_i^v - RT \tag{B8}$$

The solubility parameter of volatile materials (solvents for example) can be determined by measuring their enthalpy of vaporization or using a correlation for this quantity, and using Eq. (B9):

$$\delta_i = \left(\frac{\Delta H_i^v - RT}{V_i}\right)^{1/2} \tag{B9}$$

Equation (B9) should be used at pressures near 1 atm. Near the critical point, $\Delta H_i^v = 0$, and Eq. (B8) incorrectly

predicts a negative value for the cohesive energy density while Eq. (B7) yields a small positive value.

Dissolution of crystalline polymers The free energy of mixing for crystalline polymers contains terms for the free energy of fusion for the crystalline volume fraction of the material. The free energy of fusion may contain terms that account for the loss of crystallite surface area and the mixing of amorphous material with oriented material. These terms would be added to Eq. (B1). The enthalpy of mixing could still be modeled using Eqs. (B2) or (B6). Some crystalline polymers obey the solubility parameter model at temperatures near their melting point, $T > 0.9T_m$ (123). Solvent swelling experiments with crystalline polymers may fit Eq. (B1), particularly if the solvent is a poor one for the polymer and does not significantly dissolve crystalline regions.

2. Cohesive Energy Parameters for Polar Systems

The solubility parameter describes well the enthalpy change on mixing of nonpolar solvents but does not always give reliable results when extended to polar systems. The free energy change of mixing for polar systems is dominated by hydrogen-bonding forces between various groups in the solvent and polymer. Hydrogen-bonding forces are much stronger than van der Waals or dipole forces and often dominate the free energy of mixing. Complete miscibility is expected to occur if the solubility parameters are similar and the degree of hydrogen bonding (p: poor, m: moderate and s: strong) is similar between the components. Hydrocarbons, chlorinated hydrocarbons and nitrohydrocarbons are considered to be poor hydrogen-bonding solvents. Ketones, esters, and glycol monoethers give moderate hydrogen bonding. Alcohols, amines, acids, amides and aldehydes are strong hydrogen-bonding solvents. Table 7 classifies materials using these categories (21–24). Alternative classifications have been given by Lieberman (69), Gardon (41,85,86) and Dyck and Hoyer (32).

Other investigators have decomposed the Hildebrand parameter into several terms representing different contributions to the free energy of mixing. Hildebrand (59) used dispersive and polar solubility terms for solvents, with the complete parameter being given by

$$\delta^2 = \delta_d^2 + \delta_p^2 \tag{B10}$$

where δ_d is the dispersive term and δ_p the polar term. The additional term improved agreement between δ and experimental data. Prausnitz and coworkers accounted for polar bonding by including parameters for permanent dipole interactions and dispersion type interactions. This approach has been applied to polymer solutions (15) and complex formation (57). Crowley et al. (26,27) proposed a three-parameter system.

Hansen (49–53,56) and Hansen and Skaarup (54) assume that the cohesive energy arises from dispersive, permanent dipole–dipole interactions and hydrogen bonding forces:

$$\delta^2 = \delta_d^2 + \delta_p^2 + \delta_h^2 \tag{B11}$$

where δ_h accounts for a variety of association bonds, including hydrogen bonds and permanent dipole-induced dipole (47). The values of these components for solvents were calculated from a large number of solubility data sets. Polymer solubility parameters can also be decomposed to a three-term set. The Hansen parameters give improved agreement with data but are still not completely accurate in predicting solutions thermodynamics for every system. Peiffer (172) has related dispersion, polar and induced solubility parameters to intermolecular forces and molecular size. The dispersive component increases with molecular size, while the polar component decreases with molecular size.

Temperature effects The solubility parameter decreases with temperature. The individual terms have varied dependence on temperature, namely (Ref. 55)

$$\begin{aligned} \frac{d\delta_d}{dT} &= -1.25 \times \alpha \times \delta_d \\ \frac{d\delta_p}{dT} &= -\frac{\delta_p \times \alpha}{2} \\ \frac{d\delta_p}{dT} &= -\delta_h\left(1.22 \times 10^{-3} + \frac{\alpha}{2}\right) \end{aligned} \tag{B12}$$

where $\alpha(K^{-1})$ is the volumetric thermal expansion coefficient.

3. Relationship between Solubility Parameters and other Thermodynamic Parameters

Hildebrand and Hansen parameters can be calculated using other thermodynamic quantities. This section contains some of the relationships for binary systems. Extensions to multicomponent systems are described by Flory (37) and Olabisi et al. (89).

Activity coefficients Excess free energy calculations for regular solutions (93) can be used to relate solubility parameters and solvent activity coefficients:

$$RT \ln\gamma_1 = V_1\phi_2^2(\delta_1 - \delta_2)^2 \tag{B13}$$

$$RT \ln\gamma_2 = V_2\phi_1^2(\delta_1 - \delta_2)^2 \tag{B14}$$

where γ_i is the activity coefficient of component i. Equations (B13) and (B14) depend on the use of the geometric mean, $\delta_{12} = (\delta_1 \times \delta_2)^{1/2}$. Funk and Prausnitz (40) show that there are deviations from this rule for aromatic hydrocarbons.

Molar excess free energy of mixing The molar excess free energy of mixing for binary solutions is

$$\Delta G^E = \sum_i (\delta_{1-i} - \delta_{2-i})^2 \tag{B15}$$

where δ_{j-i} is the solubility term of group i for species j.

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Polymer–solvent interaction parameter The polymer–solvent interaction parameter, χ, is modeled as the sum of entropic and enthalpic components:

$$\chi = \chi_H + \chi_S \tag{B16}$$

where χ_H is the enthalpic component and χ_S is the entropic component. χ_S is usually taken to be a constant between 0.3 and 0.4 for nonpolar systems: $\chi_S = 0.34$ is often used (15,104). The enthalpic component can be related to the Hildebrand parameters:

$$\chi_H = \frac{V_1}{RT}(\delta_1 - \delta_2)^2 \tag{B17}$$

Substituting Eq. (B17) into Eq. (B16)

$$\chi = \frac{V_1}{RT}(\delta_1 - \delta_2)^2 + 0.34 \tag{B18}$$

Equation (B18) permits only positive values of the interaction parameter. Since the Flory–Huggins criterion for complete solvent–polymer miscibility is $\chi < 0.5$, the enthalpic contribution must be small and the solubility parameters of the solvent and polymer must be similar (21). The molar volume of the solvent also affects miscibility and phase equilibria: a Hildebrand parameter needed for phase separation or miscibility cannot be specified without specifying V_1.

Interchange energy density Equation (B18) works well for nonpolar systems for which Eq. (B17) is a good description of the enthalpic component of the interaction parameter. The geometric mean assumption of regular solution theory is not appropriate for polar systems, and better models include an extra term describing the interchange energy density for the solvent–polymer pair. For example

$$\chi = \chi_S + \frac{V_1 A_{12}}{RT} \tag{B19}$$

with

$$A_{12} = \delta_1^2 + \delta_2^2 - 2 \times l_{12}\delta_1\delta_2 \tag{B20}$$

where l_{12} characterizes the intermolecular forces between molecules (rather than using the geometric mean assumption). Equation (B20) allows the modeling of specific interactions between components 1 and 2. Mixed solvents can be treated as a single solvent by determining the solubility parameter of the solvent mixture, and then using this value in Eq. (B18). If both the solvents and the polymers interact, the description is more complicated. An alternative definition of A_{12} uses the two-component solubility parameter (Ref. 25)

$$A_{12} = (\delta_{1d} - \delta_{2d})^2 + (\delta_{1p} - \delta_{2p})^2 \tag{B21}$$

The Hansen parameters may also be used to model A_{12}.

Several recent theories for polymer solution thermodynamics include entropic, enthalpic and free-volume contributions to the free energy of mixing. The free-volume contributions modify the entropic components, and have the opposite temperature dependence from the combinatorial term, helping to explain the lower critical solution temperature. Since Eq. (B18) assumes a constant value for the entropic component, it may not be valid over large temperature ranges. However, Eq. (B18) will predict the upper critical solution temperature, and, from this standpoint, is adequate for a number of phase equilibria applications near this condition.

The Lattice Fluid model (170) can be used to predict the solubilities of hydrocarbon and chlorinated hydrocarbons in nonpolar polymers. Three-dimensional solubility parameters can be used to provide an empirical correction to the geometric mean approximation (171). This correction predicts the solubility of polar and nonpolar solvents in polymers using only the pure component equation-of-state and solubility parameters.

Guide to polymer–solvent miscibility A region of solubility has been characterized by the distance between solvent and solute coordinates (Ref. 14)

$$R_{ij} = \left(4(\delta_{1d} - \delta_{2d})^2 + (\delta_{1p} - \delta_{2p})^2 + (\delta_{1h} - \delta_{2h})^2\right)^{1/2} \tag{B22}$$

A more general definition of the region of solubility is

$$R_{ij} = \delta_1 - \delta_2 = (a(\delta_{1d} - \delta_{2d})^2 + b(\delta_{1p} - \delta_{2p})^2 + b(\delta_{1h} - \delta_{2h})^2)^{1/2} \tag{B23}$$

where a and b are empirical weighing factors. If the distance between the solvent and the solute Hansen coordinate position exceeds R_{ij}, the two components are not soluble, or swelling is less than expected. R_{ij} reduces the need for three-dimensional plots. Some investigators have used two-dimensional plots for polar and hydrogen bonding terms, but the technique may be misleading for materials with large dispersion contributions. Barton (12) gives a number of models and contour map examples. As mentioned previously, Eqs. (B17)–(B19) suggest that the solvent molar volume can have a significant effect on miscibility.

Zeller (131) reviewed graphical three-dimensional solubility parameter estimation methods (12,14,49,55,172,173) as applied to solvent swelling of crosslinked elastomers. In general, the graphical method (Eqs. B22 and B23) does not account for the known influence of molar volume and crosslink density of solubility, and incorrectly assumes a linear relationship between the solubility parameter difference and solubility. An improved method used the Flory–Rehner equation to modify the interaction parameter for the effects of crosslink density (132).

Guide to polymer–polymer miscibility In many polymer–polymer systems, miscibility can be predicted by comparing the solubility parameters (139). The following table proposes misciblity guidelines. Polymer–polymer miscibility is enhanced when hydrogen bonding is present.

TABLE OF MISCIBILITY GUIDELINES[a]

Intermolecular interaction	Polymer blend example	Critical value of the interaction parameter, χ_{crit}	Upper limit of the non-hydrogen bonded solubility parameter difference, $\Delta\delta$ (MPa$^{1/2}$)
Dispersive forces only	Polyisoprene–Poly-(vinylethylene)	<0.002	<0.02
Dipole–dipole interactions	Poly(methyl methyacrylate)–poly(ethylene oxide)	0.002–0.001	0.2–1.0
Weak hydrogen bonds	Poly(vinyl chloride)–polycaprolactone	0.02–0.2	1.0–2.0
Moderate hydrogen bonds	Poly(vinyl phenol)–poly(vinyl acetate)	0.2–1.0	2.0–5.0
Strong hydrogen bonds	Poly(vinyl phenol)–poly(vinyl methyl ether	1.0–1.5	5.0–6.0

[a] Ref. 139.

C. SOLUBILITY PARAMETER MEASUREMENTS, CALCULATIONS, AND CORRELATIONS

Solubility parameters can be determined by direct measurements, indirect calculations, or correlations with other physical parameters. The solubility parameters of solvents usually can be determined directly by measuring the energy of vaporization. The solubility parameters of polymers can only be determined indirectly and may be affected by variations in their chemical constitutions, i.e., the number of crosslinks and the distribution of chain branches or substitutive groups along the polymer backbone. The methods presented in this section can be used to develop correlations of solubility parameters with other physical properties for specific commercial polymer products or to estimate the solubility parameters of new polymers.

1. Solvents

Calculation of the solubility term (δ) by relating the enthalpy of vaporization to the energy of vaporization Equation (B8) can be used to calculate ΔE_i^v when ΔH_i^v is available. The molar volume of the solute is needed to complete the calculation of δ, using Eq. (B3). This is the most direct and accurate method of determining the solubility parameter. However, δ's determined in this way may not give the best prediction of solution behavior. The solubility parameter values in Tables 7–9 reflect the solvent's behavior in a variety of systems.

Correlations Both the enthalpy of vaporization and the molar volume of a solvent can be estimated from correlations. When ΔH_i^v is known at the normal boiling point, it can be converted to the appropriate value at a second temperature using (126)

$$\frac{\Delta H_i^v(T_2)}{\Delta H_i^v(T_1)} = \left(\frac{T_c - T_2}{T_c - I_1}\right)^{0.38} \qquad \text{(C1)}$$

This corresponding state-type procedure gives estimates within 2% of experimental values over a wide range of temperatures (13). Hildebrand and Scott (59) proposed an empirical correlation between ΔH_i^v at 25°C and the normal boiling point, T_b:

$$\Delta H_i^v(298K) = 0.2950 + 23.6\,T_b + 0.020\,T_b^2 \qquad \text{(C2)}$$

with enthalpy units of kcal/mol. Lawson (67) suggests different coefficients to include fluorocarbon liquids. The Clausius–Clapeyron equation can be applied if vapor pressure data are available. Eq. (C2) is reasonably accurate only for liquids that are not hydrogen bonded. The Hildebrand parameter calculated by Eq. (C2) should be adjusted as follows for different solvents (24): add 1.4 (cal/cm^3)$^{1/2}$ for alcohols, add 0.6 (cal/cm^3)$^{1/2}$ for esters, and add 0.5 (cal/cm^3)$^{1/2}$ for ketones if the boiling point is less than 100°C. Jayasri and Yaseen (64) suggest adding 1.7 (cal/cm^3)$^{1/2}$ for alcohols.

Solvent molar volumes are available or can be calculated by group molar volume methods at 25°C (35,163–165). The molar volume for solids at 25°C can be extrapolated from liquid state values (the liquid is assumed to be subcooled).

Thermodynamic coefficients The internal pressure, π, is defined as (Ref. 88)

$$\pi = \frac{\partial E}{\partial V} = T\beta - P \qquad \text{(C3)}$$

where β is the constant volume thermal pressure coefficient (the ratio of the coefficient of thermal expansion, α, to the isothermal compressibility, κ). Since the external pressure is usually small with respect to $T\beta$, the internal pressure is approximated by

$$\pi = T\beta = T\frac{\alpha}{\kappa} = T\left(\frac{\partial P}{\partial T}\right) \qquad \text{(C4)}$$

The Hildebrand parameter is the square root of the internal pressure. Eq. (C4) provides a method for scaling the Hildebrand parameter with temperature. This equation also provides a method for the direct estimation for δ of polymers; α and κ are measurable when ΔH_i^v is not meaningful (B7). The thermal pressure coefficient (C4) can be evaluated from vapor-pressure data and is easier to apply than the Clausius–Clapeyron equation. For high pressure applications, such as phase equilibria in reverse osmosis membranes, the external pressure term in Eq. (C3) may not be negligible. Solubility parameters generally decrease with increasing pressure.

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van der Waals gas constant Tables are available in many Handbooks for the van der Waals correction constants to the ideal gas law, a and b, where a has units of l^2/atm. For some liquids, these values may be at hand when other data are not available. They can be used to check Hildebrand parameter values obtained from other sources.

$$\delta = 1.2\frac{\sqrt{a}}{V} \tag{C5}$$

The form of Eq. (C5) can be obtained by substituting the van der Waals equation of state into Eq. (C3).

Critical pressure The solubility parameter is related to the critical pressure, P_c of a substance through the empirical equation

$$\delta = 1.25\,P_c^{1/2} \tag{C6}$$

where the critical pressure is expressed in atmospheres. Equation (C6) is not very accurate, but is simple to apply when critical pressure data are available.

Surface tension Michaels (83) has shown that the surface tension can be related to the cohesive energy density:

$$\frac{\Delta E_i^v}{V_i} = A\left(\frac{1}{V_i}\right)^{1/3}\gamma_L \tag{C7}$$

where γ_L is the surface tension and A is a constant. Koenhen and Smolders (65) correlated surface tension and two Hansen parameters:

$$\delta_d^2 + \delta_p^2 = 13.8\left(\frac{1}{V_i}\right)^{1/3}\gamma_L \tag{C8}$$

δ_h is probably not related to the liquid–vapor interfacial energy as these interactions do not involve breaking hydrogen bonds. Eq. (C8) does not describe cyclic compound, acetonitrile, carboxylic acids, polyfunctional alcohols, and other polar compounds well. Hildebrand and Scott (59) have proposed a different equation, which has been discussed by Lee (68).

Index of refraction The dispersive Hansen parameter can be related to the index of refraction, n_D (65,105).

$$\delta_d = 9.55\,n_D - 5.55 \tag{C9}$$

The interaction energy between nonpolar molecules should depend on polarizability (London dispersion forces) and, therefore, on the index of refraction.

Dipole moment Hansen and Skaarup (54) related the polar Hansen parameter to the dielectric constant, ε, and the dipole moment, μ:

$$\delta_p^2 = \frac{12108}{V^2}\frac{\varepsilon - 1}{2\varepsilon + n_D^2}(n_D^2 + 2)\mu^2 \tag{C10}$$

Beerbower and Dicky (14) proposed an empirical relationship:

$$\delta_p^2 = 9.5\frac{\mu}{V_i^{1/2}} \tag{C11}$$

Kauri butanol values Proprietary hydrocarbon solvents are usually mixtures with boiling ranges and unknown molecular weights. Solubility parameter values may be estimated from Kauri butanol values (ASTM Method D1133-54T) using the equation

$$\delta = 6.3 + 0.03\,\mathrm{KB} \tag{C12}$$

2. Polymers

Hildebrand parameters cannot be calculated for polymers from heat of vaporization data because of their nonvolatility. Indirect methods are described below. The solubility parameter values in Table 10 may only be representative of a given polymer since variations in compositions can lead to changes in δ.

2.1. Indirect Measurements

Solvency testing (screening procedure) The classic method for determining solubility parameter of a commercial polymer is solvency testing, or the solvency screening procedure. This technique compares polymer solubility in solvent groups with different hydrogen bonding characteristics (poor, moderate and strong). The midpoints of the polymer's solubility range may be used as single-valued quantities for some purposes, but may not agree with values determined by other methods (43). A gram or two of solid polymer is placed in a test tube and an approximate amount of a selected solvent is added such that the final solution would have about the correct solid content for the expected commercial use, e.g., 50% for alkyds, 20% for vinyls, etc. The exact amount is often unimportant, except for poor solvents. A typical phase diagram for binary polymer–solvent systems may show upper and lower critical solution temperatures. The one-phase region expands away from the UCST and LCST points, which often occur in the range of 20–80 wt.% polymer. The usual purpose of these experiments is to find good solvents for commercial product formulation, and this screening procedure will help identify such solvents. The mixture may be warmed and stirred to speed up solution, but it should be cooled and observed at room temperature. The resulting mixture should be a single phase, clear and free of gel particles or cloudiness, otherwise the polymer is judged insoluble. The solvents to be used are selected from Table 1.

This grouping of solvents has been selected so that the Hildebrand parameter values increase by reasonably constant steps within each hydrogen bonding class. The object of using these solvent spectra is to establish a solubility parameter range for a polymer rather than a single-valued number. This range has the advantage of showing the difference, $(\delta_1 - \delta_2)^2$ (see Eqs. B13, B14 and

B18), which can be tolerated between the solubility values of the polymer and solvent for miscibility. Van Dyk et al. (120) have shown that the molar volume improves the correlation between solvency and the solubility parameters. A better measure of solubility might be the group, $V_1(\delta_1 - \delta_2)^2$.

In carrying out the procedure, it is convenient to select the first trials about 1/2 and 2/3 of the way down the column; for example, in the poorly hydrogen bonded group, toluene and nitroethane would be chosen. If the polymer is soluble in both, there is no need to try intermediate solvents because experience has shown that the polymer will be soluble in every case. The solvents at the end of the spectrum should be tried next. If the polymer is soluble in one but not both of the initial trials, the third trial should be half-way between the two. By successive choices, sets of two adjacent solvents will be found, one of which dissolves the polymer and one that does not. The parameter values of the solvents which do dissolve the polymer mark the ends of the range. The procedure is repeated for the other two hydrogen bonding classes.

Osmotic pressure Fedors (35) used the osmotic pressure of polymer solutions to determine solubility parameters.

Swelling values Another method for measuring the solubility parameter of polymers is to prepare a sparsely crosslinked sample and immerse it in a series of liquids of varying δ (166,167). The crosslinked material will swell to varying degrees, with the maximum amount of swelling occurring when the solvent has the same Hildebrand parameter as the polymer. The amount of swelling can be measured by length, weight, or volume changes. By inference, the soluble, uncrosslinked materials has the same value. An example of a crosslinked sample is styrene polymerized with 1% divinyl benzene.

Turbidity Polymer can be precipitated from dilute solution by adding a non-solvent. The cloud point defines the onset of two phases. The enthalpic component of the interaction parameter, χ_H, can be related to the solubility parameter difference between the solvent and the polymer (Eq. B17). Two different non-solvents are used: one having a solubility parameter above that of the solvent, and another having a solubility parameter below that of the solvent. The enthalpic components of the interaction parameter for each phase are equal at the cloud point, which can be used to measure the solubility parameter of the polymer (116). The two expressions for χ_H are

$$\frac{V_{ml}(\delta_2 - \delta_{ml})^2}{RT} = \frac{V_{mh}(\delta_2 - \delta_{mh})^2}{RT} \qquad \text{(C13)}$$

where V_{ml} and V_{mh} represent the molar volumes at the cloud points of the solvent-lower solubility parameter nonsolvent and solvent-higher solubility parameter nonsolvent, respectively. δ_{ml} and δ_{mh} are the solubility parameters of the mixed solvents. The mixture properties are calculated using

$$\delta_m = \phi_1\delta_1 + \phi_3\delta_3 \qquad \text{(C14)}$$

$$V_m = \frac{V_1V_3}{\phi_1V_3 + \phi_3V_1} \qquad \text{(C15)}$$

where the subscripts, 1 and 3, refer to the solvent and the nonsolvent. Eq. (C13) is simplified to determine the solubility parameter of the polymer:

$$\delta_2 = \frac{(\sqrt{V_{ml}}\delta_{ml} + \sqrt{V_{mh}}\delta_{mh})}{(\sqrt{V_{ml}} + \sqrt{V_{mh}})} \qquad \text{(C16)}$$

The volume fraction of the nonsolvent at the cloud point is used to compute the solvent mixture solubility parameter and molar volume. Cloud point experiments are done for a series of solvents with the two nonsolvents, generating a set of points, (δ_1, δ_2). The intersection of the line defined by this set of points with the line, $\delta_1 = \delta_2$, determines the polymer's solubility parameter.

Specific volume The specific volume of polymers varies with the solvent. Good solvents give higher values of the specific volume while poor solvents give lower values. A plot of the polymer specific volume as a function of solubility parameter gives a maximum value, which is taken to be the solubility parameter of the polymer. High precision density measurements at carefully controlled temperatures are needed for this method (168). The partial specific volume is defined as

$$\bar{V}_i = \left.\frac{\partial V}{\partial c_i}\right|_{T,p,i\neq j} \qquad \text{(C17)}$$

The specific volume of a nonideal two-component system is

$$V_{spec} = \phi_1\bar{V}_1 + \phi_2\bar{V}_2 \qquad \text{(C18)}$$

where $\bar{V}_2$ is the rate of change in volume of the solution when a very small amount of polymer is added. As the solvent volume fraction goes to one, the partial molar volume of solvent is constant and Eq. (C18) becomes

$$V_{spec} = \bar{V}_{1(c_2=0)} + \phi_2\bar{V}_2 \qquad \text{(C19)}$$

The maximum of the polymer partial specific volume is estimated by fitting a quadratic to the data (Ref. 142):

$$\bar{V}_2 = \bar{V}_{2,max} - A(\delta_1 - \delta_2)^2 \qquad \text{(C20)}$$

It is possible to estimate the polymer solubility parameter with only two reference solvents (168), but more reliable values are found by using a range of solvents.

Intrinsic viscosity A number of researchers have used intrinsic viscosity to estimate Hildebrand parameters (74).

Flory (37) related intrinsic viscosity to polymer molecular weight and the chain-expansion factor. The chain-expansion factor can, in turn, be related to the polymer–solvent interaction parameter using the Flory–Huggins theory. Maximum coil extension should occur in solvents near the polymer's solubility parameter. A variety of models can be used to relate the interaction parameter to Hildebrand parameters (19,80,115,167–169). These equations are quadratic (Eq. C21) or take the form of a gaussian curve (Eq. C22):

$$[\eta] = [\eta]_{max} - A(\delta_1 - \delta_2)^2 \quad \text{(C21)}$$

$$[\eta] = [\eta]_{max}\exp(-A(\delta_1 - \delta_2)^2) \quad \text{(C22)}$$

where $[\eta]$ is the intrinsic viscosity, $[\eta]_{max}$ the maximum intrinsic viscosity, and A the constant. These researchers have shown that Eq. (C14) gives a good correlation between solvency and Hansen parameters for methacrylate polymers.

Inverse phase gas chromatography This method has been used by a number of investigators to measure infinite dilution weight fraction activity coefficients (6,29–31,63,71,72,84,90,91,113). These coefficients can be related to Hildebrand parameters by using a thermodynamic theory for polymer solutions, such as Flory–Huggins theory. The polymer is the stationary phase in a gas chromatography column. Both binary and multicomponent equilibria (46,99,100) can be studied using this method. Chromatographic techniques have the advantage of rapid measurement of thermodynamic values once the columns have been made. The solubility parameter can be related to the specific retention volume of the solvent on the column. Both V_i and ΔH_i^v must be known at the temperature of the column. Molar volumes for the solvents can be determined by using literature density equations or generating equations from density data. Values for the enthalpy of vaporization can be determined at the experimental temperature (71).

The inverse phase gas chromatography method has the advantage of providing values for the infinite dilution solubility parameter, δ_i^∞, over a range of temperatures. This is particularly valuable for the prediction of phase equilibria at elevated temperatures. The value of δ_i^∞ at 25°C can be estimated by using the expected temperature dependence for χ (Eq. B18) of

$$\chi = \alpha + \frac{\beta}{T} \quad \text{(C23)}$$

One potential problem with this technique is that χ is known to be a function of concentration and the polymer Hildebrand parameter is determined at infinite dilution of solvent. For a number of binary systems, the change in χ with solvent weight fraction is the largest as $\omega \to 0$. The concentration dependence of the interaction parameter can be modeled using methods given in the chapter on interaction parameters in this Handbook.

2.2. Correlation Methods

Refractive index Koenhen and Smolders (65) have predicted dispersive Hansen parameters from refractive index measurements of polymers. Wu (128) has suggested an effective cross-sectional area to relate the cohesive energy density and surface tension

$$\delta_d^2 = A\left(\frac{n_s}{V_{i,s}}\right)^{1/3}\gamma_s^d \quad \text{(C24)}$$

where n_s is the number of atoms in a segment, $V_{i,s}$ the molar volume of a segment, and γ_s^d the dispersion contribution to the free surface energy of the polymer.

Dipole moment Equation (C10) has been applied to polymers by Koenhen and Smolders (65). The dipole moments of polymers are between 70% and 90% of those of the corresponding monomer units.

Hydrogen-bonding parameter Hansen and Beerbower (55) compiled enthalpy data for hydrogen bonding groups. The following values are suggested:

Group	Cohesive energy, E_h, (kJ/mol)	Refs.
Alcohol	20.9	–
Amide	16.3	83
Ester	5.2	54
Nitrile	2.1	54
Ether	2.3	54
Monochloro substituent	0.4	54
Phenylene ring	0.4	54

The hydrogen-bonding parameter is given by

$$\delta_h^2 = \frac{E_h}{V_i} \quad \text{(C25)}$$

Group contribution methods These methods have been used to estimate the solubility parameter (17,20,28,33,35,58,60,61,96,112,121,122). van Krevelen (123), Fedors (35), and Barton (12) have reviewed these techniques and given tables of group values. The molar volume of solvents and polymers can also be estimated by group contribution techniques (108).

The sets of group constants of Small (112), Hoy (61) van Krevelen (121) and van Krevelen and Hoftyzer (122) seem to be most comprehensive. Table 2 gives the group molar attraction constants at 25°C. Small's values were derived from measurements of the heat of vaporization. Hoy's values were derived from vapor pressure measurements. The group contribution values of van Krevelen and Hoftyzer are based on cohesive energy data of polymers.

The group contribution techniques are based on the assumption that the contributions of different functional groups to the thermodynamic property are additive. The energy of vaporization of a solvent or polymer is

$$\Delta E_i^v = \sum_j n_j \Delta e_j \quad \text{(C26)}$$

where Δe_j is the energy of vaporization contribution of group j, and n_j the number of groups of type j in the molecule. The solubility parameter is obtained by combining Eqs. (B3) and (C26):

$$\delta_i = \left(\frac{\Delta E_i^v}{V_i}\right)^{1/2} = \left(\frac{\sum_j n_j \Delta e_j}{V_i}\right)^{1/2} \quad \text{(C27)}$$

Small (112) defined the molar attraction constant as

$$F_j = (\Delta E_{i,j}^v V_{i,j})^{1/2} \quad \text{(C28)}$$

that can be used to calculate the solubility parameter by

$$\delta_i = \left(\frac{\Delta E_i^v V_i}{V_i^2}\right)^{1/2} = \frac{\sum_j F_j}{V_i} = \frac{\rho_i \sum_j F_j}{M_i} \quad \text{(C29)}$$

where ρ_i is the polymer density and M_i is the polymer molecular weight. δ_i can be evaluated for polymer repeating group by using group contribution calculations for the molar volume and the cohesive energy density or molar attraction constant. Table 2 gives sets of molar attraction constants and the energy of vaporization provided by several different authors. It is preferrable to use parameters from only one set when determining the solubility parameter of a repeating unit. Table 3 gives the cohesive energy densities and group molar volumes determined by Fedors. This set may give less accurate estimates than those in Table 2, but it has value because it is more comprehensive and can be applied to more systems.

Hansen parameters The terms in Eq. (B11) can be estimated by group contribution methods. In general, the resulting factors are known less accurately than the group molar attraction constants or energy of vaporizations. The interaction of structural groups within molecules may not follow simple additive rules. However, the estimate of Hansen parameters can be very useful. Under the method of Hoftyzer and van Krevelen (Table 4), the terms are estimated as

$$\delta_d = \frac{\sum F_{di}}{V}$$
$$\delta_p = \frac{\sqrt{\sum F_{pi}^2}}{V} \quad \text{(C30)}$$
$$\delta_h = \sqrt{\frac{\sum E_{hi}}{V}}$$

The dispersive Hansen parameter treats the molar attraction constants as additive. The polar Hansen parameters also are additive, unless more than one polar group is present. Methods for treating additional polar groups are given in the last three rows of Table 4: additional polar groups do not add linearly to the polar Hansen term. The molar attraction constant is not applied to the hydrogen-bonding Hansen parameter. Rather, a hydrogen bonding energy, E_{hi}, is used.

Hoy (178–179) has an alternative group contribution method (Table 5). It includes a molar attraction function, $F_{t,i}$ a polar component of this function, $F_{p,i}$, the molar volume of the solvent or polymer structural unit, the Lyderson correction for solvent non-ideality, $\Delta_{T,i}$, and a similar correction for polymer non-ideality, $\Delta_{T,i}^{(P)}$.

Bicerano (164) and Porter (165) have developed new group contribution techniques for a wide variety of polymer properties. These approaches consider how the different functional groups are connected in the molecule or in the polymeric repeating unit. Bicerano's method uses Fedor's (35) and van Krevelen's (123,163) group contribution values. Both references provide solubility parameter predictions of a number of polymers.

TABLE 1. SELECTED SOLVENTS FOR USE IN POLYMER SOLVENCY TESTING

Solvent	δ (MPa$^{1/2}$)
POORLY HYDROGEN BONDED	
n-Pentane	14.3
n-Heptane	15.1
Methylcyclohexane	16.0
Solvesso 150	17.4
Toluene	18.2
Tetrahydronaphthalene	19.4
o-Dichlorobenzene	20.5
1-Bromonaphthalene	21.7
Nitroethane	22.7
Acetonitrile	24.1
Nitromethane	26.0
MODERATELY HYDROGEN BONDED	
Diethyl ether	15.2
Diisobutyl ketone	16.0
n-Butyl acetate	17.4
Methyl propionate	18.2
Dibutyl phthalate	19.0
Dioxane	20.3
Dimethyl phthalate	21.9
2,3-Butylene carbonate	24.8
Propylene carbonate	27.2
Ethylene carbonate	30.1
STRONGLY HYDROGEN BONDED	
2-Ethyl hexanol	19.4
Methyl isobutyl carbinol	20.5
2-Ethyl butanol	21.5
n-Pentanol	22.3
n-Butanol	23.3
n-Propanol	24.3
Ethanol	26.0
Methanol	29.7

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TABLE 2. GROUP CONTRIBUTIONS TO COHESIVE ENERGY DENSITY

Group	F [MPa$^{1/2}$ cm^3/mol]		ΔE_i^v(J/mol)	
	Small (112)	van Krevelen (121)	Hoy (61)	van Krevelen and Hoftyzer (163)
2.1. CARBON-CONTAINING GROUPS				
$-CH_3$	437	420	303	9630
$\rangle CH_2$	272	280	269	4190
$\rangle CH-$	57	140	176	420
$\rangle C\langle$	−190	0	65	−5580
$=CH_2$	388	–	259	–
$=CH-$	277	222	249	–
$=C\langle$	39	82	173	–
−CH=(aromatic)	–	–	240	–
−C=(aromatic)	–	–	201	–
$-CH(CH_3)-$	495	560	(479)	(10060)
$-C(CH_3)_2-$	685	841	(672)	(13700)
−CH=CH−	454	444	497	10200
−C=CH−	265	304	421	4860
CH_3 H, $C=C$	(704)	724	(725)	(14500)
H−C≡C−	583	–	–	–
−C≡C−	454	–	–	–
Cyclopentyl	–	1380	1300	–
Cyclohexyl	–	1660	1470	–
Phenyl	1500	1520	(1400)	31000
p-Phenylene	1350	1380	(1440)	25140
Naphthyl	2340	–	–	–
2.2. OXYGEN-CONTAINING GROUPS				
−O−, ether	143	255	235	6290
−O−, epoxide	–	–	360	–
−OH	–	754	462	–
−OH, aromatic	–	–	350	–
−(C=O)−	562	685	538	–
−(C=O)−O−	634	511	688	13410
−(C=O)−OH	–	651	(998)	–
−O−(C=O)−O−	–	767	(904)	–
−(C=O)−O−(C=O)−	–	767	1160	–
2.3. NITROGEN-CONTAINING GROUPS				
$-NH_2$	–	–	464	–
−NH−	–	–	368	–
$-N\langle$	–	–	125	–
−CH−CN	(896)	1120	(901)	25420
−CN	839	982	725	25000
−(C=O)−NH−	–	1290	(906)	60760
−O−(C=O)−NH	–	1480	(1040)	–
−N=C=O	–	–	734	–
2.4. OTHER GROUPS				
−H	164–205	140	−103 (acidic dimer)	–
−S−	460	460	428	8800
−SH	644	–	–	–
−F	(250)	164	84	4470
−Cl (primary)	552	471	420	12990
−Br (primary)	695	614	528	15500
−I	870	–	–	–
$-CF_2-$	307	–	–	–
$-CF_3$	561	–	–	–
−O−N=O	900	–	–	–
$-NO_2$	900	–	–	–
$-PO_4$	1020	–	–	–
−Si−	−77	–	–	–

TABLE 2. *cont'd*

Group	F [MPa$^{1/2}$ cm^3/mol] Small (112)	F [MPa$^{1/2}$ cm^3/mol] van Krevelen (121)	ΔE_i^v(J/mol) Hoy (61)	ΔE_i^v(J/mol) van Krevelen and Hoftyzer (163)
2.5. STRUCTURAL FEATURES				
Conjugation	41–61	–	48	–
cis	–	–	−15	–
trans	–	–	−28	–
Ring, 4 member	–	–	159	–
5 member	215–235	–	43	–
6 member	194–215	–	−48	–
Subscription, ortho	–	–	20	–
meta	–		14	–

TABLE 3. CONTRIBUTIONS TO E_{coh} AND V[a]

Group	E_{coh} (J/mol)	V (cm^3/mol)
$-CH_3$	4710	33.5
$\diagdown CH_2$	4940	16.1
$\diagdown CH-$	3430	−1.0
$\diagdown C\diagup$	1470	−19.2
$=CH_2$	4310	28.5
$=CH-$	4310	13.5
$=C\diagup$	4310	−5.5
$H-C\equiv$	3850	27.4
$-C\equiv$	7070	6.5
Phenyl	31940	71.4
Phenylene: *o*, *m*, *p*	31940	52.4
Phenyl (trisubstituted)	31940	33.4
(tetrasubstituted)	31940	14.4
(pentasubstituted)	31940	−4.6
(hexasubstituted)	31940	−23.6
Ring closure, 5+ atoms	1050	16.0
3 or 4 atoms	3140	18.0
Conjugation in ring for each double bond	1670	−2.2
Halogen attached to carbon atom with double bond	−20% of E_{coh} of halogen	4.0
$-F$	4190	18.0
−F, disubstituted	3560	20.0
trisubstituted	2300	22.0
$-CF_2-$, perfluoro compounds	4270	23.0
$-CF_3$, perfluoro compounds	4270	57.5
$-Cl$	11550	24.0
−Cl, disubstituted	9630	26.0
trisubstituted	7530	27.3
$-Br$	15490	30.0
−Br, disubstituted	12350	31.0
trisubstituted	10670	32.4
$-I$	19050	31.5
−I, disubstituted	16740	33.5
trisubstituted	16330	37.0
$-CN$	25530	24.0
$-OH$	29800	10.0
−OH, disubstituted or adjacent C atoms	21850	13.0
$-O-$	3350	3.8
−CHO, aldehyde	21350	22.3
$-CO-$	17370	10.8
$-COOH$	27630	28.5
$-COO-$	18000	18.0
$-CO_3-$, carbonate	17580	22.0
$-C_2O_3-$, anhydride	30560	30.0

TABLE 3. *cont'd*

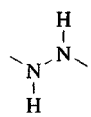
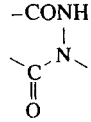
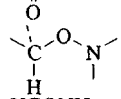
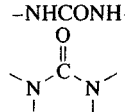
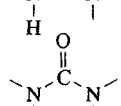
Group	E_{coh} (J/mol)	V (cm^3/mol)
HCOO−, formate	18000	32.5
$-CO_2CO_2-$, oxalate	26790	37.3
$-HCO_3$	12560	18.0
$-COF$	13400	29.0
$-COCl$	17580	38.1
$-COBr$	24150	41.6
$-COI$	29300	48.7
$-NH_2$	12560	19.2
$-NH-$	8370	4.5
$-N\diagup$	4190	−9.0
$-N=$	11720	5.0
$-NHNH_2$	21980	–
$-N(NH_2)-$	16740	16.0
	16740	16.0
$-N_2$, diazo	8370	23
$-N=N-$	4190	–
$\diagdown C-N=N-C\diagup$	20090	–
$-N=C=N-$	11470	–
$-N\equiv C$	18840	23.1
$-NF_2$	7660	33.1
$-NF-$	5070	24.5
$-CONH_2$	41860	17.5
$-CONH-$	33490	9.5
	29510	−7.7
	27630	11.3
HCONH−	43950	27.0
$-NHCOO-$	26370	18.5
$-NHCONH-$	50230	–
	41860	–
	20930	−14.5
NH_2COO-	37000	–
$-NCO$	28460	35.0
$-ONH_2$	19050	20.0

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TABLE 3. *cont'd*

Group	E_{coh} (J/mol)	V (cm^3/mol)
	25120	11.3
–CH=NOH	25120	24.0
$-NO_2$, aliphatic	29300	24.0
$-NO_2$, aromatic	15360	32.0
$-NO_3$	20930	33.5
$-NO_2$, nitrate	11720	33.5
$-NHNO_2$	39770	28.7
–NNO–	27210	10.0
–SH	14440	28.0
–S–	14150	12.0
$-S_2-$	23860	23.0
$-S_3-$	13400	47.2
	39140	–
SO_3	18840	27.6
SO_4	28460	31.6
$-SO_2Cl$	37070	43.5
–SCN	20090	37.0
–NCS	25120	40.0
P	9420	– 1.0
PO_3	14230	22.7
PO_4	20930	28.0
PO_3–OH	31810	32.2
Si	3390	0
SiO_4	21770	20.0
B	13810	– 2.0
BO_3	0	20.4
Al	13810	– 2.0
Ga	13810	– 2.0
In	13810	– 2.0
Tl	13810	– 2.0
Ge	8080	– 1.5
Sn	11300	1.5
Pb	17160	2.5
As	12980	7.0
Sb	16330	8.9
Bi	21350	9.5
Se	17160	16.0
Te	20090	17.4
Zn	14480	2.5
Cd	17790	6.5
Hg	22810	7.5

[a] Ref. 35.

TABLE 4. SOLUBILITY PARAMETER I: COMPONENT GROUP CONTRIBUTIONS[a]

Structural Group	F_{di} (J$^{1/2}$ cm$^{3/2}$/mol)	F_{pi} (J$^{1/2}$ cm$^{3/2}$/mol)	E_{hi} (J/mol)
$-CH_3$	420	0	0
	270	0	0
	80	0	0
	– 70	0	0
$=CH_2$	400	0	0
=CH–	200	0	0
	70	0	0
	1620	0	0
	1430	110	0
(*o*, *m*, *p*)	1270	110	0
–F	(220)	-	-
–Cl	450	550	400
–Br	(550)	-	-
–CN	430	1100	2500
–OH	210	500	20000
–O–	100	400	3000
–COH	470	800	4500
–CO–	290	770	2000
–COOH	530	420	10000
–COO–	390	490	7000
HCOO–	530	–	–
$-NH_2$	280	–	8400
–NH–	160	210	3100
	20	800	5000
$-NO_2$	500	1070	1500
–S–	440	–	–
$=PO_4-$	740	1890	13000
Ring	190	–	–
One plane of symmetry	–	0.50 ×	
Two planes of symmetry	–	0.25 ×	–
More planes of symmetry	–	0 ×	0 ×

[a] Ref. 163.

TABLE 5. SOLUBILITY PARAMETER II: COMPONENT GROUP CONTRIBUTIONS[a]

Group	F_{ti} ((J cm^3)$^{1/2}$/mol)	F_{pi} ((J cm^3)$^{1/2}$/mol)	V_i (cm^3/mol)	$\Delta_{Ti}{}^*$	$\Delta_{Ti}^{(P)}$
$-CH_3$	303.5	0	21.55	0.023	0.022
	269.0	0	15.55	0.020	0.020
	176.0	0	9.56	0.012	0.013
	65.5	0	3.56	0	0.04
$=CH_2$	259	67	19.17	0.018	0.019
=CH–	249	59.5	13.18	0.018	0.0185
	173	63	7.18	0	0.013
$CH_{aromatic}$	241	62.5	13.42	0.011	0.018
$C_{aromatic}$	201	65	7.42	0.011	0.015
–HC=O	600	532	23.3	0.048	0.045

TABLE 5. *cont'd*

Group	$F_{t,i}$ ((J cm^3)$^{1/2}$/mol)	$F_{p,i}$ ((J cm^3)$^{1/2}$/mol)	V_i (cm^3/mol)	$\Delta_{T,i}{}^*$	$\Delta_{T,i}^{(P)}$
–CO–	538	525	17.3	0.040	0.040
–COOH	565	415	26.1	0.039	0.039
–COO–	640	528	23.7	0.047	0.050
–CO–O–CO–	1160	1160	41.0	0.086	0.086
–CN	725	725	23.1	0.060	0.054
–N=C=O	736	8.2	25.9	0.054	0.054
–N(HCO)–	1020	725	35.8	0.062	0.055
$-CONH_2$	1200	900	34.3	0.071	0.084
–CONH–	1131	895	28.3	0.054	0.073
–OCONH–	1265	890	34.8	0.078	0.094
–OH→H-bonded	485	485	10.65	0.082	0.034
–OH, primary	675	675	12.45	0.082	0.049
secondary	591	591	12.45	0.082	0.049
tertiary	(500)	(500)	12.45	0.082	0.049
phenolic	350	350	12.45	0.031	0.006
–O–, ether	235	216	6.45	0.021	0.018
acetal	236	102	6.45	0.018	0.018
epoxide	361	156	6.45	0.027	0.027
$-NH_2$	464	464	17.0	0.031	0.035
–NH–	368	368	11.0	0.031	0.275
$-N\diagup\diagdown$	125	125	12.6	0.014	0.009
–S–	428	428	18.0	0.015	0.032
–F	845	73.5	11.2	0.018	0.006
–Cl, primary	419.5	307	19.5	0.017	0.031
secondary	426	315	19.5	0.017	0.032
aromatic	330	81.5	19.5	0.017	0.025
$\diagup$Cl $\diagdown$Cl	705	572	39.0	0.034	0.052
–Br, aliphatic	528	123	25.3	0.010	0.039
aromatic	422	100	25.3	0.010	0.031
Base value, B	277				
Ring, non-aromatic 4-member	159	203	–	0	0.012
5-member	43	85	–	0	0.003
6-member	–48	61	–	0	–0.0035
7-member	92	0	–	0	0.007
Conjugation, isomerism	47.5	–19.8	–	0	0.0035
cis	–14.6	–14.6	–	0	–0.001
trans	–27.6	–27.6	–	0	–0.002
Aromatic substitution, ortho	20.2	–13.3	–	0	0.0015
meta	13.5	–24.3	–	0	0.001
para	83	–34.0	–	0	0.006

[a] Ref. (178–179).

TABLE 6. EQUATIONS TO BE USED FOR HOY'S SYSTEM[a]

Formulae	Solvents	Amorphous polymers
Additive molar functions	$F_t = \sum n_i F_{t,i}$	$F_t = \sum n_i F_{t,i}$
	$F_p = \sum n_i F_{p,i}$	$F_p = \sum n_i F_{p,i}$
	$V = \sum n_i V_i$	$V = \sum n_i V_i$
	$\Delta_T = \sum n_i \Delta_{T,i}$	$\Delta_T^{(P)} = \sum n_i \Delta_{T,i}^{(P)}$
Auxiliary equations	$\log \alpha = 3.39 \frac{T_b}{T_{cr}} - 0.1585 - \log V$,	$\frac{\alpha^{(P)} = 777 \Delta_T^{(P)}}{V}$
	T_b = boiling point, T_{cr} = critical temp.	
	$\left(\frac{T_b}{T_{cr}}\right) = 0.567 + \Delta_T - (\Delta_T)^2$,	$\bar{n} = \frac{0.5}{\Delta_T^{(P)}}$
	Lyderson equation	
Expressions for δ and δ-components	$\delta_t = \frac{F_t + B}{V}$; $B = 277$	$\delta_t = \frac{F_t + \frac{B}{\bar{n}}}{V}$; $B = 277$

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TABLE 6. *cont'd*

Formulae	Solvents	Amorphous polymers
	$\delta_p = \delta_t\left(\frac{1}{\alpha}\frac{F_p}{F_t + B}\right)^{1/2}$	$\delta_p = \delta_t\left(\frac{1}{\alpha^{(P)}}\frac{F_p}{F_t + B/\bar{n}}\right)^{1/2}$
	$\delta_h = \delta_t\left(\frac{\alpha - 1}{\alpha}\right)^{1/2}$	$\delta_h = \delta_t\left(\frac{\alpha^{(P)} - 1}{\alpha^{(P)}}\right)^{1/2}$
	$\delta_d = (\delta_t^2 - \delta_p^2 - \delta_h^2)^{1/2}$	$\delta_d = (\delta_t^2 - \delta_p^2 - \delta_h^2)^{1/2}$

[a] In the equations given here, F_t is always combined with a Base value, α is the molecular aggregation number, and $\bar{n}$ the number of repeating units per polymer chain segment.

D. SOLUBILITY PARAMETER TABLES

TABLE 7. SOLUBILITY PARAMETERS OF SOLVENTS IN ALPHABETICAL ORDER

Solvent	CAS number	Solubility parameter δ		H-bonding group
		$(MPa^{1/2})$	$(cal/cm)^{1/2}$	
Acetaldehyde	75-07-0	21.1	10.3	m
Acetic acid	64-19-7	20.7	10.2	s
Acetic anhydride	108-24-7	21.1	10.3	s
Acetone	67-64-1	20.3	9.9	m
Acetonitrile	75-07-8	24.3	11.9	p
Acetophenone	98-86-2	21.7	10.6	m
Acetyl chloride	75-36-5	19.4	9.5	m
Acetylmorpholine (*N*)	1696-20-4	23.7	11.6	m
Acetylpiperidine (*N*)	618-42-8	22.9	11.2	s
Acetylpyrrolidine (*N*)	1072-83-9	23.3	11.4	s
Acrolein	107-02-8	20.1	9.8	s
Acrylic acid	79-10-7	24.6	12.0	s
Acrylonitrile	107-13-1	21.5	10.5	p
Allyl acetate	591-87-7	18.8	9.2	m
Allyl alcohol	107-18-6	24.1	11.8	s
Allyl chloride	107-05-1	18.0	8.8	m
Ammonia	7664-41-7	33.4	16.3	s
Amyl acetate (iso)	625-16-1	16.0	7.8	m
(normal)	628-63-7	17.4	8.5	m
(secondary)	626-38-0	17.0	8.3	m
Amyl alcohol	75-85-4	20.5	10.0	s
Amyl alcohol (normal)	71-41-0	22.3	10.9	s
Amylamine (normal)	110-58-7	17.8	8.7	s
Amyl bromide (1-bromopentane)	110-53-2	15.6	7.6	m
Amyl chloride (1-chloropentane)	543-59-9	17.0	8.3	m
Amylene (2-methyl-2-butene)	513-35-9	14.1	6.9	p
Amyl ether (pentyl ether)	693-65-2	14.9	7.3	m
Amyl formate (iso)	110-45-2	16.4	8.0	m
(normal)	638-49-3	17.2	8.5	m
Amyl iodide (1-iodopentane)	628-17-1	17.2	8.4	m
Anethole (*para*)	4180-23-8	17.2	8.4	m
Aniline	65-53-3	21.1	10.3	s
Anthracene	120-12-7	20.3	9.9	p
Apco #18 solvent		15.3	7.5	p
Apco #140 solvent		14.9	7.3	p
Apco thinner		16.0	7.8	p
Aroclor 1248	12672-29-6	18.0	8.8	p
Benzaldehyde	100-52-7	19.2	9.4	m
Benzene	71-43-2	18.8	9.2	p
Benzonitrile	100-47-0	17.2	8.4	p
Benzyl alcohol	100-51-6	24.8	12.1	s
Bicyclohexyl	92-51-3	17.4	08.5	p
Bromobenzene	108-86-1	20.3	09.9	p
1-Bromonaphthalene	90-11-9	21.7	10.6	p
Bromostyrene (*ortho*)	2039-88-5	20.1	9.8	p
1,3-Butadiene	106-99-0	14.5	7.1	p

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TABLE 6. *cont'd*

Formulae	Solvents	Amorphous polymers
	$\delta_p = \delta_t\left(\frac{1}{\alpha}\frac{F_p}{F_t + B}\right)^{1/2}$	$\delta_p = \delta_t\left(\frac{1}{\alpha^{(P)}}\frac{F_p}{F_t + B/\bar{n}}\right)^{1/2}$
	$\delta_h = \delta_t\left(\frac{\alpha - 1}{\alpha}\right)^{1/2}$	$\delta_h = \delta_t\left(\frac{\alpha^{(P)} - 1}{\alpha^{(P)}}\right)^{1/2}$
	$\delta_d = (\delta_t^2 - \delta_p^2 - \delta_h^2)^{1/2}$	$\delta_d = (\delta_t^2 - \delta_p^2 - \delta_h^2)^{1/2}$

[a] In the equations given here, F_t is always combined with a Base value, α is the molecular aggregation number, and $\bar{n}$ the number of repeating units per polymer chain segment.

D. SOLUBILITY PARAMETER TABLES

TABLE 7. SOLUBILITY PARAMETERS OF SOLVENTS IN ALPHABETICAL ORDER

Solvent	CAS number	Solubility parameter δ		H-bonding group
		$(MPa^{1/2})$	$(cal/cm)^{1/2}$	
Acetaldehyde	75-07-0	21.1	10.3	m
Acetic acid	64-19-7	20.7	10.2	s
Acetic anhydride	108-24-7	21.1	10.3	s
Acetone	67-64-1	20.3	9.9	m
Acetonitrile	75-07-8	24.3	11.9	p
Acetophenone	98-86-2	21.7	10.6	m
Acetyl chloride	75-36-5	19.4	9.5	m
Acetylmorpholine (*N*)	1696-20-4	23.7	11.6	m
Acetylpiperidine (*N*)	618-42-8	22.9	11.2	s
Acetylpyrrolidine (*N*)	1072-83-9	23.3	11.4	s
Acrolein	107-02-8	20.1	9.8	s
Acrylic acid	79-10-7	24.6	12.0	s
Acrylonitrile	107-13-1	21.5	10.5	p
Allyl acetate	591-87-7	18.8	9.2	m
Allyl alcohol	107-18-6	24.1	11.8	s
Allyl chloride	107-05-1	18.0	8.8	m
Ammonia	7664-41-7	33.4	16.3	s
Amyl acetate (iso)	625-16-1	16.0	7.8	m
(normal)	628-63-7	17.4	8.5	m
(secondary)	626-38-0	17.0	8.3	m
Amyl alcohol	75-85-4	20.5	10.0	s
Amyl alcohol (normal)	71-41-0	22.3	10.9	s
Amylamine (normal)	110-58-7	17.8	8.7	s
Amyl bromide (1-bromopentane)	110-53-2	15.6	7.6	m
Amyl chloride (1-chloropentane)	543-59-9	17.0	8.3	m
Amylene (2-methyl-2-butene)	513-35-9	14.1	6.9	p
Amyl ether (pentyl ether)	693-65-2	14.9	7.3	m
Amyl formate (iso)	110-45-2	16.4	8.0	m
(normal)	638-49-3	17.2	8.5	m
Amyl iodide (1-iodopentane)	628-17-1	17.2	8.4	m
Anethole (*para*)	4180-23-8	17.2	8.4	m
Aniline	65-53-3	21.1	10.3	s
Anthracene	120-12-7	20.3	9.9	p
Apco #18 solvent		15.3	7.5	p
Apco #140 solvent		14.9	7.3	p
Apco thinner		16.0	7.8	p
Aroclor 1248	12672-29-6	18.0	8.8	p
Benzaldehyde	100-52-7	19.2	9.4	m
Benzene	71-43-2	18.8	9.2	p
Benzonitrile	100-47-0	17.2	8.4	p
Benzyl alcohol	100-51-6	24.8	12.1	s
Bicyclohexyl	92-51-3	17.4	08.5	p
Bromobenzene	108-86-1	20.3	09.9	p
1-Bromonaphthalene	90-11-9	21.7	10.6	p
Bromostyrene (*ortho*)	2039-88-5	20.1	9.8	p
1,3-Butadiene	106-99-0	14.5	7.1	p

TABLE 7. *cont'd*

Solvent	CAS number	Solubility parameter δ		H-bonding group
		$(MPa^{1/2})$	$(cal/cm)^{1/2}$	
n-Butane	106-97-8	13.9	6.8	p
1,3-Butanediol	107-88-0	23.7	11.6	s
1,4-Butanediol	110-63-4	24.8	12.1	s
2,3-Butanediol	513-85-9	22.7	11.1	s
Butyl acetate (iso)	540-88-5	17.0	8.3	m
n-Butyl acetate	123-86-4	17.4	8.5	m
sec-Butyl acetate	105-46-4	16.8	8.2	m
Butyl acrylate (iso)	1663-39-4	17.4	8.5	m
n-Butyl acrylate	141-32-2	18.0	8.8	m
Butyl alcohol (iso) (2-methyl-1-propanol)	78-83-1	21.5	10.5	s
n-Butyl alcohol (1-butanol)	71-36-3	23.3	11.4	s
sec-Butyl alcohol (2-butanol)	15892-23-6	22.1	10.8	s
tert-Butyl alcohol (2-methyl-2-propanol)	75-65-0	21.7	10.6	s
n-Butylamine	109-73-9	17.8	8.7	s
n-Butyl bromide (1-bromobutane)	109-65-9	17.8	8.7	m
sec-Butyl bromide (2-bromobutane)	78-76-2	17.2	8.4	m
Butyl (iso) butyrate (normal)	539-90-2	16.0	7.8	m
Butyl (normal) butyrate (normal)	109-21-7	16.6	8.1	m
Butyl chloride (iso) (2-chloro-2-methylpropane)	507-20-0	16.6	8.1	m
Butylene-2,3 carbonate	4437-85-9	24.8	12.1	m
Butylene (iso) (2-methylpropene)	115-11-7	13.7	6.7	p
Butyl ether	142-96-1	16.0	7.8	m
Butyl formate (iso)	762-75-4	16.8	8.2	m
(normal)	592-84-7	18.2	8.9	m
Butyl iodide (normal) (1-iodobutane)	542-69-8	17.6	8.6	m
Butyl lactate (normal)	138-22-7	19.2	9.4	m
Butyl methacrylate	97-88-1	16.8	8.2	m
Butyl propionate	590-01-2	18.0	8.8	m
Butyl stearate	123-95-5	15.3	7.5	m
Butyraldehyde	123-72-8	18.4	9.0	m
Butyric acid (iso) (2-methylpropionic acid)	79-31-2	21.1	10.3	s
(normal)	107-92-6	21.5	10.5	s
Butyrolactone, γ	96-48-0	25.8	12.6	m
Butyronitrile (iso)	78-82-0	20.1	9.8	p
(normal)	109-74-0	21.5	10.5	p
Caprolactam, ε	105-60-2	26.0	12.7	m
Caprolactone, ε	502-44-3	20.7	10.1	m
Capronitrile (heptyl cyanide)	124-12-9	19.2	9.4	p
Carbon disulfide	75-15-0	20.5	10.0	p
Carbon tetrachloride	56-23-5	17.6	8.6	p
Celanese solvent 601		18.8	9.2	m
Chloroacetonitrile	920-37-6	25.8	12.6	p
Chlorobenzene	108-90-7	19.4	9.5	p
Chloroethyl acetate (β)	542-58-5	19.8	9.7	m
Chloroform	67-66-3	19.0	9.3	p
Chlorostyrene (*ortho* or *para*)	2039-87-4	19.4	9.5	p
Chlorotoluene (*para*)	106-43-4	18.0	8.8	p
Cresol (meta)	108-39-4	20.9	10.2	s
Cyclobutanedione	14406-53-3	22.5	11.0	m
Cyclohexane	110-82-7	16.8	8.2	p
Cyclohexanol	108-93-0	23.3	11.4	s
Cyclohexanone	108-94-1	20.3	9.9	m
Cyclopentane	287-92-3	17.8	8.7	p
Cyclopentanone	120-92-3	21.3	10.4	m
Cymene (*para*) (2-isopropyl toluene)	99-87-6	16.8	8.2	p
Decahydronaphthalene (*cis* and *trans*) (Decalin)	91-17-8	18.0	8.8	p
Decane (normal)	124-18-5	13.5	6.6	p
Decyl acrylate (iso)	2156-96-9	16.8	8.2	m
Diacetone alcohol (4-hydroxy-4-methyl-2-pentanone)	123-42-2	18.8	9.2	m
Diacetone alcohol methyl ether (Pentoxone)	107-70-0	16.8	8.2	m
Diacetylpiperazine (*N,N*)	18940-57-3	28.0	13.7	m
Diamyl phthalate	131-18-0	18.6	9.1	m
Dibenzyl ether	103-50-4	19.2	9.4	m

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TABLE 7. *cont'd*

Solvent	CAS number	Solubility parameter δ		H-bonding group
		$(MPa^{1/2})$	$(cal/cm)^{1/2}$	
Dibromoethane-1,2	106-93-4	21.3	10.4	p
Dibromoethylene-1,2 (*cis* and *trans*)	540-49-8	20.7	10.1	p
Dibutoxyethyl phthalate (Kronisol)	117-83-9	16.4	8.0	m
Dibutylamine	111-92-2	16.6	8.1	s
Dibutyl fumarate	105-75-9	18.4	9.0	m
Dibutyl maleate	105-76-0	18.4	9.0	m
Dibutyl phenyl phosphate	2528-36-1	17.8	8.7	m
Dibutyl phthalate	84-74-2	19.0	9.3	m
Dibutyl sebacate	109-43-3	18.0	9.2	m
Dichloroacetic acid	79-43-6	22.5	11.0	s
Dichlorobenzene (*ortho*)	95-50-1	20.5	10.0	p
Dichlorodifluoromethane (Freon 12)	75-71-8	11.3	5.5	p
Dichloroethyl ether	623-46-1	20.1	9.8	m
Dichloroethylene, *cis*-1,2	156-59-2	18.6	9.1	p
trans-1,2	156-60-5	18.4	9.0	p
Dichlorofluoromethane (Freon 21)	75-43-4	17.0	8.3	p
Dichloropropane-1,2	78-87-5	18.4	9.0	p
Dichloropropane-2,2	594-20-7	16.8	8.2	p
Diethylacetamide (*N,N*)	685-91-6	20.3	9.9	m
Diethylamine	109-89-7	16.4	8.0	s
Diethyl carbonate	105-58-8	18.0	8.8	m
Diethylene glycol	111-46-6	24.8	12.1	s
Diethylene glycol monobutyl ether (normal)	112-34-5	19.4	9.5	m
Diethylene glycol monobutyl ether	112-34-5	20.9	10.2	m
Diethylene glycol monobutyl ether acetate	124-17-4	17.4	8.5	m
Diethylene glycol monolaurate	141-20-8	17.8	8.7	m
Diethyl ether	60-29-7	15.1	7.4	m
Diethylformamide (*N,N*)	617-84-5	21.7	10.6	m
Diethyl ketone	96-22-0	18.0	8.8	m
Diethyl maleate	141-05-9	20.3	9.9	m
Diethyl oxalate	95-92-1	17.6	8.6	m
Diethyl phthalate	84-66-2	20.5	10.0	m
Diethyl-2,2-propanediol-1,2 (heptylene glycol)	115-76-4	20.3	9.9	s
Diethyl sulfone	597-35-3	25.4	12.4	m
Difluorotetrachloroethane (Freon 112)	76-11-9	16.0	7.8	p
Diformylpiperazine (*N,N*)	4164-39-0	31.5	15.4	m
Dihexyl ether	112-58-3	16.4	8.0	m
Di-*n*-hexyl phthalate	108-83-8	18.2	8.9	m
Diisobutylene	107-39-1	15.8	7.7	p
Diisobutyl ketone	108-83-8	16.0	7.8	m
Diisodecyl phthalate	26761-40-0	14.7	7.2	m
Diisopropyl ether	108-20-3	14.1	6.9	m
Diisopropyl ketone	565-80-0	16.4	8.0	m
Dimethylacetamide (*N,N*)	127-19-5	22.1	10.8	m
Dimethylaniline	121-69-7	19.8	9.7	m
Dimethyl-2,2-butanediol-1,2 (isobutylene glycol)	558-43-0	22.9	11.2	s
Dimethyl-2,2-butanediol-1,3	76-09-3	20.5	10.0	s
Dimethyl carbonate	616-38-6	20.3	9.9	m
Dimethyl ether	115-10-6	18.0	8.8	m
Dimethylformamide (*N,N*)	68-12-2	24.8	12.1	m
Dimethyl malonate	108-59-8	22.5	11.0	m
Dimethylnitroamine (*N,N*)	62-75-9	26.8	13.1	s
Dimethyl oxalate	553-90-2	22.5	11.0	m
Dimethyl phosphite	868-85-9	25.6	12.5	m
Dimethyl phthalate	131-11-3	21.9	10.7	m
Dimethyl siloxane	9016-00-6	10.0–12.1	4.9–5.9	p
Dimethyl sulfide	75-18-3	19.2	9.4	p
Dimethyl sulfoxide	67-68-5	29.7	14.5	m
Dimethyltetramethylene sulfone	1003-78-7	24.6	12.0	m
Dioctyl adipate	103-23-1	17.8	8.7	m
Dioctyl phthalate	117-81-7	24.8	12.1	m
Dioctyl sebacate	1222-62-3	17.8	8.7	m
Dioxane-1,4	123-91-1	20.5	10.0	m

TABLE 7. *cont'd*

Solvent	CAS number	Solubility parameter δ		H-bonding group
		$(MPa^{1/2})$	$(cal/cm)^{1/2}$	
Dioxolane-1,3	646-06-0	17.6	8.6	m
Dipentene	138-86-3	20.5	10.0	m
Diphenyl ether	101-84-8	20.9	10.2	m
Diphenyl 2-ethylhexyl phosphate	1241-94-7	17.4	8.5	p
Dipropylene glycol	110-98-5	20.7	10.1	m
Dipropylene glycol monomethyl ether	34590-94-8	17.6	8.6	m
Dipropyl phthalate	131-16-8	20.5	10.0	s
Dipropyl sulfone	598-03-8	23.1	11.3	m
Dodecane	112-40-3	16.2	7.9	p
Dodecanol-1	112-53-8	20–21	9.8	s
Epichlorohydrin	106-89-8	22.5	11.0	s
Ethane	74-84-0	12.3	6.0	p
Ethylacetamide (*N*)	625-50-3	25.2	12.3	s
Ethyl acetate	141-78-6	18.6	9.1	m
Ethyl acrylate	140-88-5	17.6	8.6	m
Ethyl alcohol	64-17-5	26.0	12.7	s
Ethylamine	75-04-7	20.5	10.0	s
Ethyl amyl ketone (3-octanone)	106-68-3	16.8	8.2	m
Ethylbenzene	100-41-4	18.0	8.8	p
Ethyl benzoate	93-89-0	16.8	8.2	m
Ethyl bromide (bromoethane)	74-96-4	19.6	9.6	m
Ethyl-2-butanol-1	97-95-0	21.5	10.5	s
Ethyl *n*-butyrate	105-54-4	17.4	8.5	m
Ethyl caprylate	106-32-1	14.9	7.3	m
Ethyl chloride	75-00-3	18.8	9.2	m
Ethyl cyanoacetate	105-56-6	22.5	11.0	m
Ethylene bromide	593-60-2	19.8	9.7	p
Ethylene carbonate	96-49-1	30.1	14.7	m
Ethylene chlorohydrin (2-chloroethanol)	107-07-3	25.0	12.2	s
Ethylene cyanohydrin (3-hydroxypropionitrile)	109-78-4	31.1	15.2	s
Ethylenediamine	107-15-3	25.2	12.3	s
Ethylene dichloride (1,2-dichloroethane)	107-06-2	20.1	9.8	p
Ethylene glycol	107-21-1	29.9	14.6	s
Ethylene glycol diacetate	111-55-7	20.5	10.0	m
Ethylene glycol diethyl ether	629-14-1	17.0	8.3	m
Ethylene glycol dimethyl ether	110-71-4	17.6	8.6	m
Ethylene glycol methyl ether acetate	110-49-6	18.8	9.2	m
Ethylene glycol monobenzyl ether	622-08-2	22.3	10.9	m
Ethylene glycol monobutyl ether	111-76-2	19.4	9.5	m
Ethylene glycol monoethyl ether (2-ethoxyethanol; Cellusolve)	110-80-5	21.5	10.5	m
Ethylene glycol monomethyl ether (2-methoxyethanol)	109-86-4	23.3	11.4	m
Ethylene glycol monophenyl ether	122-99-6	23.5	11.5	m
Ethylene oxide	75-21-8	22.7	11.1	m
Ethylene formamide (*N*)		28.4	13.9	s
Ethyl formate	109-94-4	19.2	9.4	m
Ethyl-2-hexanediol-1,3 (octylene glycol)	94-96-2	19.2	9.4	s
Ethylhexanol	104-76-7	19.4	9.5	s
Ethyl hexyl acrylate	103-11-7	16.0	7.8	m
Ethylidene chloride	75-34-3	18.2	8.9	p
Ethyl iodide (iodoethane)	75-03-6	19.2	9.4	m
Ethyl isobutyl ether	627-02-1	15.3	7.5	m
Ethyl isobutyrate	97-62-1	16.2	7.9	m
Ethyl lactate	687-47-8	20.5	10.0	m
Ethyl mercaptan (ethane thiol)	75-08-1	18.8	9.2	p
Ethyl methacrylate	97-63-2	17.0	8.3	m
Ethyl morpholine (*N*)	100-74-3	18.2	8.9	s
Ethyl orthoformate (triethyl orthoformate)	122-51-0	17.0	8.3	m
Ethyl propionate	105-37-3	17.2	8.4	m
Fluorocarbons, aliphatic		11.3–12.7	5.5–6.2	p
aromatic		15.3–16.8	7.5–8.2	p
Formamide	75-12-7	39.3	19.2	s
Formic acid	64-18-6	24.8	12.1	s

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TABLE 7. *cont'd*

Solvent	CAS number	Solubility parameter δ		H-Bonding Group
		$MPa^{1/2}$	$(cal/cm)^{1/2}$	
Formylmorpholine (*N*)	4394-85-8	26.6	13.0	m
Formylpiperidine (*N*)	2591-86-8	23.5	11.5	m
Furan	110-00-9	19.2	9.4	m
Furfural (2-furaldehyde)	98-01-1	22.9	11.2	m
Furfuryl alcohol	98-00-0	25.6	12.5	s
Glycerol	56-81-5	33.8	16.5	s
Heptane (normal)	142-82-5	15.1	7.4	p
Heptyl alcohol (normal)	111-70-6	21.7	10.6	s
Hexamethylphosphoramide	680-31-9	21.5	10.5	s
Hexane (normal)	110-54-3	14.9	7.3	p
Hexanediol-2,5	2935-44-6	21.2	10.3	s
Hexene-1	592-41-6	15.1	7.4	p
Hexyl alcohol (normal)	111-27-3	21.9	10.7	s
Hydrazine	302-10-2	37.3	18.1	s
Hydrogen	1333-74-0	6.9	3.0	p
Hydrogenated terphenyl (Monsanto HB-40)		18.4	9.0	p
Hydrogen cyanide	74-90-8	24.8	12.1	s
Iodobenzene	591-50-4	20.7	10.1	p
Isophorone	78-59-1	18.6	9.1	m
Isoprene	78-79-5	15.1	7.4	p
Lauryl alcohol	112-53-8	16.6	8.1	s
Low odor mineral spirits		14.1	6.9	p
Maleic anhydride	108-31-6	27.8	13.6	s
Malononitrile	109-77-3	30.9	15.1	p
Mesitylene	108-67-8	18.0	8.8	p
Mesityl oxide	141-79-7	18.4	9.0	m
Methacrylic acid	79-41-4	22.9	11.2	s
Methane	74-82-8	11.0	5.4	p
Methanol	67-56-1	29.7	14.5	s
Methoxy-4-methyl-4-pentanol-2		17.4	8.5	s
Methoxy-4-methyl-4-pentanone-2	107-70-0	17.0	8.3	m
Methylacetamide	79-16-3	29.9	14.6	s
Methyl acetate	79-20-9	19.6	9.6	m
Methyl acrylate	96-33-3	18.2	8.9	m
Methylamine	74-89-5	22.9	11.2	s
Methyl amyl acetate	108-84-9	16.4	8.0	m
Methyl amyl ketone	110-43-0	17.4	8.5	m
Methyl benzoate	95-58-3	21.5	10.5	m
Methyl bromide	74-83-9	19.6	9.6	m
Methyl *n*-butyl ketone	591-78-6	17.0	8.3	m
Methyl *n*-butyrate	623-42-7	18.2	8.9	m
Methyl caprolactone		18.2	8.9	m
Methyl chloride	74-87-3	19.8	9.7	m
Methylcyclohexane	10120-28-2	16.0	7.8	p
Methylcyclohexanone	589-92-4	19.0	9.3	m
Methylene chloride	75-09-2	19.8	9.7	p
Methylene glycolate	61192-32-3	25.4	12.4	m
Methylene iodide	75-11-6	24.1	11.8	p
Methyl ethyl ketone	78-93-3	19.0	9.3	m
Methyl ethyl sulfone	594-43-4	27.4	13.4	m
Methylformamide (*N*)	123-39-7	32.9	16.1	s
Methyl formate	107-31-3	20.9	10.2	m
Methyl *n*-hexyl ketone	111-13-7	17.0	8.3	m
Methyl iodide	74-88-4	20.9	10.2	m
Methyl isoamyl ketone	110-12-3	17.2	8.4	m
Methyl isobutyl carbinol	108-11-2	20.5	10.0	s
Methyl isobutyl ketone	108-10-1	17.2	8.4	m
Methyl isobutyrate	547-63-7	17.0	8.3	m
Methyl isopropyl ketone	563-80-4	17.4	8.5	m
Methyl isovalerate	624-24-8	16.2	7.9	m
Methyl methacrylate	80-62-6	18.0	8.8	m
Methyl nonyl ketone	112-12-9	16.0	7.8	m
Methyl-2-pentanediol-1,3	149-31-5	21.1	10.3	s

TABLE 7. *cont'd*

Solvent	CAS number	Solubility parameter δ $(MPa^{1/2})$	Solubility parameter δ $(cal/cm)^{1/2}$	H-bonding group
Methyl-2-pentanediol-2,4	107-41-5	19.8	9.7	s
Methyl-2-pentanediol monoethyl ether (Pentoxol)		17.4	8.5	m
Methyl propionate	554-12-1	18.2	8.9	m
Methyl propyl ketone	107-87-9	17.8	8.7	m
Methyl propyl sulfone	1977-37-3	25.6	12.5	m
Methyl pyrrolidone-2 (1-methyl-2-pyrrolidinone)	872-50-4	23.1	11.3	m
Methyl salicylate	119-36-8	21.7	10.6	m
Methyl styrene (α)	98-83-9	17.4	8.5	p
Methyltetramethylene sulfone		26.4	12.9	m
Methyl *n*-valerate	624-24-8	16.2	7.9	m
Morpholine	110-91-8	22.1	10.8	s
Naphthalene	91-20-3	20.3	9.9	p
Neopentane	463-82-1	12.9	6.3	p
Neopentyl glycol	126-30-7	22.5	11.0	s
Nitrobenzene	98-95-3	20.5	10.0	p
Nitroethane	79-24-3	22.7	11.1	p
Nitromethane	75-52-5	26.0	12.7	p
Nitro-*n*-octane		14.3	7.0	p
Nitro-1-propane	108-03-2	21.1	10.3	p
Nitro-2-propane	79-46-9	20.3	9.9	p
Nonyl phenol	25154-52-3	19.2	9.4	s
Octane (normal)	111-65-9	15.6	7.6	p
Octyl alcohol (normal)	111-87-5	21.1	10.3	s
Pentachloroethane	76-01-7	19.2	9.4	p
Pentane (normal)	109-66-0	14.3	7.0	p
Pentanediol-1,5	111-29-5	23.5	11.5	s
Pentanediol-2,4	625-69-4	22.1	10.8	s
Perchloroethylene	127-18-4	19.0	9.3	p
Perfluoroheptane	335-57-9	11.9	5.8	p
Perfluoromethylcyclohexane	355-02-2	12.3	6.0	p
Phenanthrene	85-01-8	20.1	9.8	p
Phenylhydrazine	100-63-0	25.6	12.5	s
Pine oil		17.6	8.6	p
Piperidine	110-89-4	17.8	8.7	p
Piperidone	675-20-7	27.8	13.6	s
Propane	74-98-6	13.1	6.4	p
Propiolactone	57-57-8	27.2	13.3	m
Propionic acid	79-09-4	20.3	9.9	s
Propionic anhydride	123-62-6	20.5	10.0	s
Propionitrile	107-12-0	22.1	10.8	p
Propyl acetate (iso)	108-21-4	17.2	8.4	m
Propyl acetate (normal)	109-60-4	18.0	8.8	m
Propyl alcohol (2-propanol)	67-63-0	23.5	11.5	s
Propyl alcohol (1-propanol)	71-23-8	24.3	11.9	s
Propylbenzene (normal)	103-65-1	17.6	8.6	p
Propyl bromide (1-bromopropane)	106-94-5	18.2	8.9	m
Propyl butyrate	105-66-8	17.2	8.4	m
Propyl butyrate (iso)	638-11-9	16.2	7.9	m
Propyl chloride (2-chloropropane)	75-29-6	16.6	8.1	m
Propyl chloride (1-chloropropane)	540-54-5	17.4	8.5	m
Propylene-1,2-carbonate	108-32-7	27.2	13.3	m
Propylene glycol (1,2-propanediol)	57-55-6	25.8	12.6	s
Propylene glycol methyl ether (1,2-dimethoxypropane)	7778-85-0	20.7	10.1	m
Propylene oxide	75-56-9	18.8	9.2	m
Propyl ether (di-, normal)	111-43-3	16.0	7.8	m
Propyl ether (iso)	108-20-3	14.5	7.1	m
Propyl formate	110-74-7	18.8	9.2	m
Propyl propionate	106-36-5	17.4	8.5	m
Pyridine	110-86-1	21.9	10.7	s
Pyrone (γ)	504-31-4	27.4	13.4	m
Pyrrolidine (α)	123-75-1	30.1	14.7	s
Quinoline	00-02-2	22.1	10.8	s
Santicizer 8		24.3	11.9	m

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TABLE 7. *cont'd*

Solvent	CAS number	Solubility parameter δ		H-bonding group
		$(MPa^{1/2})$	$(cal/cm)^{1/2}$	
Shell Sol 72		14.7	7.2	p
Shell TS28 solvent		15.1	7.4	p
Silicon tetrachloride		15.1	7.4	p
Socal solvent No. 1		16.6	8.1	p
Socal solvent No. 2		16.2	7.9	p
Socal solvent No. 3		15.8	7.7	p
Solvesso 100		17.6	8.6	p
Solvesso 150		17.4	8.5	p
Styrene	100-42-5	19.0	9.3	p
Styrene oxide	96-09-3	21.5	10.5	m
Succinic anhydride	108-30-5	31.5	15.4	s
Terpene B		17.2	8.4	p
Tetrachloroethane-1,1,2,2	79-34-5	19.8	9.7	p
Tetrachloroethylene (perchloroethylene)	127-18-4	19.0	9.3	p
Tetraethylene glycol	112-60-7	20.3	9.9	s
Tetrahydrofuran	109-99-9	18.6	9.1	m
Tetrahydronaphthalene (tetralin)	119-64-2	19.4	9.5	p
Tetramethylene sulfone (sulfolane, tetrahydrothiophene 1,1-dioxide)	126-33-0	27.4	13.4	m
Tetramethyloxamide		23.3	11.4	m
Thiophene	110-02-1	20.1	9.8	m
Toluene	108-88-3	18.2	8.9	p
Tolylene diisocyanate (4-methyl-1,3-phenylene diisocyanate)	584-84-9	23.7	11.6	s
Tributylamine	102-82-9	15.8	7.7	s
Trichloroethane-1,1,2	79-00-5	19.6	9.6	p
Trichloroethylene	79-01-6	18.8	9.2	p
Trichlorofluoromethane	75-69-4	15.5	7.6	p
Trichlorotrifluoroethane (1,1,2-trichlorotrifluoroethane)	76-13-1	14.9	7.3	p
Tricresyl phosphate (tritolyl phosphate)	1330-78-5	17.2	8.4	m
Triethylamine	121-44-8	15.1	7.4	s
Triethylene glycol	112-27-6	21.9	10.7	s
Triethylenetetramine	112-24-3	22.7	11.1	s
Trimethyl-3,5,5-hexanol (nonyl alcohol)	3452-97-9	17.2	8.4	s
Triphenyl phosphate	115-86-6	17.6	8.6	m
Triphenyl phosphite	101-02-0	19.0	9.3	m
Tripropylene glycol	24800-44-0	18.8	9.2	s
Tripropylene glycol methyl ether	20324-33-8	17.8	8.7	m
Turpentine	8006-64-2	16.6	8.1	p
Valeric acid (normal)	109-52-4	20.1	9.8	s
Valeronitrile (normal)	110-59-8	19.6	9.6	p
Varnolene (Varsol #2)		15.6	7.6	p
Vinyl acetate	108-05-4	18.4	9.0	m
Vinyl chloride	75-01-4	16.0	7.8	m
Vinyl toluene	622-97-9	18.6	9.1	p
V M & P naphtha		15.6	7.6	p
Water	7732-18-5	47.9	23.4	s
Xylene (*p*-xylene)	106-42-3	18.0	8.8	p

TABLE 8. SOLUBILITY PARAMETERS OF SOLVENTS IN INCREASING ORDER OF δ

Solvent	δ $(MPa^{1/2})$
Hydrogen	6.9
Dimethyl siloxane	10.0–12.1
Methane	11.0
Dichlorodifluoromethane (Freon 12)	11.3
Fluorocarbons, aliphatic	11.3–12.7
Perfluoroheptane	11.9
Ethane	12.3
Perfluoromethylcyclohexane	12.3
Neopentane	12.9

TABLE 8. *cont'd*

Solvent	δ $(MPa^{1/2})$
Propane	13.1
Decane (normal)	13.5
Butylene (iso) (2-methylpropene)	13.7
n-Butane	13.9
Amylene (2-methyl-2-butene)	14.1
Diisopropyl ether	14.1
Low odor mineral spirits	14.1
Nitro-*n*-octane	14.3
Pentane (normal)	14.3
1,3-Butadiene	14.5

TABLE 8. *cont'd*

Solvent	δ (MPa$^{1/2}$)
Propyl ether (iso)	14.5
Diisodecyl phthalate	14.7
Shell Sol 72	14.7
Amyl ether (pentyl ether)	14.9
Apco #140 solvent	14.9
Ethyl caprylate	14.9
Hexane (normal)	14.9
Trichlorotrifluoroethane (1,1,2-trichlorotrifluoroethane)	14.9
Diethyl ether	15.1
Heptane (normal)	15.1
Hexene-1	15.1
Isoprene	15.1
Shell TS28 solvent	15.1
Silicon tetrachloride	15.1
Triethylamine	15.1
Apco #18 solvent	15.3
Butyl stearate	15.3
Ethyl isobutyl ether	15.3
Fluorocarbons (aromatic)	15.3–16.8
Trichlorofluoromethane	15.5
Amyl bromide (1-bromopentane)	15.6
Octane (normal)	15.6
Varnolene (varsol #2)	15.6
V M & P naphtha	15.6
Diisobutylene	15.8
Socal solvent No. 3	15.8
Tributylamine	15.8
Amyl acetate (iso)	16.0
Apco thinner	16.0
Butyl (iso) butyrate (normal)	16.0
Butyl ether	16.0
Difluorotetrachloroethane (Freon 112)	16.0
Diisobutyl ketone	16.0
Ethyl hexyl acrylate	16.0
Methylcyclohexane	16.0
Methyl nonyl ketone	16.0
Propyl ether (di-, normal)	16.0
Vinyl chloride	16.0
Dodecane	16.2
Ethyl isobutyrate	16.2
Methyl isovalerate	16.2
Methyl *n*-valerate	16.2
Propyl butyrate (iso)	16.2
Socal solvent No. 2	16.2
Amyl formate (iso)	16.4
Dibutoxyethyl phthalate (Kronisol)	16.4
Diethylamine	16.4
Dihexyl ether	16.4
Diisopropyl ketone	16.4
Methyl amyl acetate	16.4
Butyl (normal) butyrate (normal)	16.6
Butyl chloride (iso) (2-chloro-2-methylpropane)	16.6
Dibutylamine	16.6
Lauryl alcohol	16.6
Propyl chloride (2-chloropropane)	16.6
Socal solvent No. 1	16.6
Turpentine	16.6
sec-Butyl acetate	16.8
Butyl formate (iso)	16.8
Butyl methacrylate	16.8
Cyclohexane	16.8
Cymene (*para*) (2-isopropyl toluene)	16.8
Decyl acrylate (iso)	16.8
Diacetone alcohol methyl ether (Pentoxone)	16.8
Dichloropropane-2,2	16.8

TABLE 8. *cont'd*

Solvent	δ (MPa$^{1/2}$)
Ethyl amyl ketone (3-octanone)	16.8
Ethyl benzoate	16.8
Ethyl benzoate (secondary)	17.0
Amyl chloride (1-chloropentane)	17.0
Butyl acetate (iso)	17.0
Dichlorofluoromethane (Freon 21)	17.0
Ethylene glycol diethyl ether	17.0
Ethyl methacrylate	17.0
Ethyl orthoformate (triethyl orthoformate)	17.0
Methoxy-4-methyl-4-pentanone-2	17.0
Methyl *n*-butyl ketone	17.0
Methyl *n*-hexyl ketone	17.0
Methyl isobutyrate	17.0
Amylformates (normal)	17.2
Methyl isobutyrate (normal)	17.2
Amyl iodide (1-iodopentane)	17.2
Anethole (*para*)	17.2
Benzonitrile	17.2
sec-Butyl bromide (2-bromobutane)	17.2
Ethyl propionate	17.2
Methyl isoamyl ketone	17.2
Methyl isobutyl ketone	17.2
Propyl acetate (iso)	17.2
Propyl butyrate	17.2
Terpene B	17.2
Tricresyl phosphate (tritolyl phosphate)	17.2
Trimethyl-3,5,5-hexanol (nonyl alcohol)	17.2
Trimethyl-3,5,5-hexanol (normal)	17.4
Bicyclohexyl	17.4
n-Butyl acetate	17.4
Butyl acrylate (iso)	17.4
Diethylene glycol monobutyl ether acetate	17.4
Diphenyl 2-ethylhexyl phosphate	17.4
Ethyl *n*-butyrate	17.4
Methoxy-4-methyl-4-pentanol-2	17.4
Methyl amyl ketone	17.4
Methyl isopropyl ketone	17.4
Methyl-2-pentanediol monoethyl ether (Pentoxol)	17.4
Methyl styrene (α)	17.4
Propyl chloride (1-chloropropane)	17.4
Propyl propionate	17.4
Solvesso 150	17.4
Butyl iodide (normal) (1-iodobutane)	17.6
Carbon tetrachloride	17.6
Diethyl oxalate	17.6
Dioxolane-1,3	17.6
Dipropylene glycol monomethyl ether	17.6
Ethyl acrylate	17.6
Ethylene glycol dimethyl ether	17.6
Pine oil	17.6
Propylbenzene (normal)	17.6
Solvesso 100	17.6
Triphenyl phosphate	17.6
Amylamine (normal)	17.8
n-Butylamine	17.8
n-Butyl bromide (1-bromobutane)	17.8
Cyclopentane	17.8
Dibutyl phenyl phosphate	17.8
Diethylene glycol monolaurate	17.8
Dioctyl adipate	17.8
Dioctyl sebacate	17.8
Methyl propyl ketone	17.8
Piperidine	17.8
Tripropylene glycol methyl ether	17.8
Aroclor 1248	18.0

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TABLE 8. *cont'd*

Solvent	δ (MPa$^{1/2}$)
Allyl chloride	18.0
n-Butyl acrylate	18.0
Butyl propionate	18.0
Chlorotoluene (*para*)	18.0
Decahydronaphthalene (*cis* and *trans*) (decalin)	18.0
Dibutyl sebacate	18.0
Diethyl carbonate	18.0
Diethyl ketone	18.0
Dimethyl ether	18.0
Ethylbenzene	18.0
Mesitylene	18.0
Methyl methacrylate	18.0
Propyl acetate (normal)	18.0
Xylene (*p*-xylene)	18.0
Butyl formate(normal)	18.2
Di-*n*-hexyl phthalate	18.2
Ethylidene chloride	18.2
Ethyl morpholine (normal)	18.2
Methyl acrylate	18.2
Methyl *n*-butyrate	18.2
Methyl caprolactone	18.2
Methyl propionate	18.2
Propyl bromide (1-bromopropane)	18.2
Toluene	18.2
Butyraldehyde	18.4
Dibutyl fumarate	18.4
Dibutyl maleate	18.4
Dichloroethylene *trans*-1,2	18.4
Dichloropropane-1,2	18.4
Hydrogenated terphenyl (Monsanto HB-40)	18.4
Mesityl oxide	18.4
Vinyl acetate	18.4
Diamyl phthalate	18.6
Dichloroethylene, *cis*-1,2	18.6
Ethyl acetate	18.6
Isophorone	18.6
Tetrahydrofuran	18.6
Vinyl toluene	18.6
Allyl acetate	18.8
Benzene	18.8
Celanese solvent 601	18.8
Diacetone alcohol (4-hydroxy-4-methyl-2-pentanone)	18.8
Ethyl chloride	18.8
Ethylene glycol methyl ether acetate	18.8
Ethyl mercaptan (ethane thiol)	18.8
Propylene oxide	18.8
Propyl formate	18.8
Trichloroethylene	18.8
Tripropylene glycol	18.8
Chloroform	19.0
Dibutyl phthalate	19.0
Methylcyclohexanone	19.0
Methyl ethyl ketone	19.0
Perchloroethylene	19.0
Styrene	19.0
Tetrachloroethylene (perchloroethylene)	19.0
Triphenyl phosphite	19.0
Benzaldehyde	19.2
Butyl lactate (normal)	19.2
Capronitrile (heptyl cyanide)	19.2
Dibenzyl ether	19.2
Dimethyl sulfide	19.2
Ethyl formate	19.2
Ethyl-2-hexanediol-1,3 (octylene glycol)	19.2
Ethyl iodide (iodoethane)	19.2

TABLE 8. *cont'd*

Solvent	δ (MPa$^{1/2}$)
Furan	19.2
Nonyl phenol	19.2
Pentachloroethane	19.2
Acetyl chloride	19.4
Chlorobenzene	19.4
Chlorostyrene (*ortho* or *para*)	19.4
Diethylene glycol monobutyl ether (normal)	19.4
Ethylene glycol monobutyl ether	19.4
Ethylhexanol	19.4
Tetrahydronaphthalene (tetralin)	19.4
Ethyl bromide (bromoethane)	19.6
Methyl acetate	19.6
Methyl bromide	19.6
Trichloroethane-1,1,2	19.6
Valeronitrile (normal)	19.6
Chloroethyl acetate (β)	19.8
Dimethylaniline	19.8
Ethylene bromide	19.8
Methyl chloride	19.8
Methylene chloride	19.8
Methyl-2-pentanediol-2,4	19.8
Tetrachloroethane-1,1,2,2	19.8
Dodecanol-1	20–21
Acrolein	20.1
Bromostyrene (*ortho*)	20.1
Butyronitrile (iso)	20.1
Dichloroethyl ether	20.1
Ethylene dichloride (1,2-dichloroethane)	20.1
Phenanthrene	20.1
Thiophene	20.1
Valeric acid (normal)	20.1
Acetone	20.3
Anthracene	20.3
Bromobenzene	20.3
Cyclohexanone	20.3
Diethylacetamide (*N,N*)	20.3
Diethyl maleate	20.3
Diethyl-2,2-propanediol-1,2 (heptylene glycol)	20.3
Dimethyl carbonate	20.3
Naphthalene	20.3
Nitro-2-propane	20.3
Propionic acid	20.3
Tetraethylene glycol	20.3
Amyl alcohol	20.5
Carbon disulfide	20.5
Dichlorobenzene (*ortho*)	20.5
Diethyl phthalate	20.5
Dimethyl-2,2-butanediol-1,3	20.5
Dioxane-1,4	20.5
Dipentene	20.5
Dipropyl phthalate	20.5
Ethylamine	20.5
Ethylene glycol diacetate	20.5
Ethyl lactate	20.5
Methyl isobutyl carbinol	20.5
Nitrobenzene	20.5
Propionic anhydride	20.5
Acetic acid	20.7
Caprolactone (ε)	20.7
Dibromoethylene-1,2 (*cis* and *trans*)	20.7
Dipropylene glycol	20.7
Iodobenzene	20.7
Propylene glycol methyl ether (1,2-dimethoxypropane)	20.7
Cresol (*meta*)	20.9
Diethylene glycol monobutyl ether	20.9

TABLE 8. *cont'd*

Solvent	δ (MPa$^{1/2}$)
Diphenyl ether	20.9
Methyl formate	20.9
Methyl iodide	20.9
Acetaldehyde	21.1
Acetic anhydride	21.1
Aniline	21.1
Butyric acid (iso) (2-methylpropionic acid)	21.1
Methyl-2-pentanediol-1,3	21.1
Nitro-1-propane	21.1
Octyl alcohol (normal)	21.1
Hexanediol-2,5	21.2
Cyclopentanone	21.3
Dibromoethane-1,2	21.3
Acrylonitrile	21.5
Butyl alcohol (iso) (2-methyl-1-propanol)	21.5
Butyric acid (normal)	21.5
Butyronitrile (normal)	21.5
Ethyl-2-butanol-1	21.5
Ethylene glycol monoethyl ether (2-ethoxyethanol; Cellusolve)	21.5
Hexamethylphosphoramide	21.5
Methyl benzoate	21.5
Styrene oxide	21.5
Acetophenone	21.7
1-Bromonaphthalene	21.7
tert-Butyl alcohol (2-methyl-2-propanol)	21.7
Diethylformamide (*N,N*)	21.7
Heptyl alcohol (normal)	21.7
Methyl salicylate	21.7
Dimethyl phthalate	21.9
Hexyl alcohol (normal)	21.9
Pyridine	21.9
Triethylene glycol	21.9
sec-Butyl alcohol (2-butanol)	22.1
Dimethylacetamide (*N,N*)	22.1
Morpholine	22.1
Pentanediol-2,4	22.1
Propionitrile	22.1
Quinoline	22.1
Quinoline (normal)	22.3
Ethylene glycol monobenzyl ether	22.3
Cyclobutanedione	22.5
Dichloroacetic acid	22.5
Dimethyl malonate	22.5
Dimethyl oxalate	22.5
Epichlorohydrin	22.5
Ethyl cyanoacetate	22.5
Neopentyl glycol	22.5
2,3-Butanediol	22.7
Ethylene oxide	22.7
Nitroethane	22.7
Triethylenetetramine	22.7
Acetylpiperidine (*N*)	22.9
Dimethyl-2,2-butanediol-1,2 (isobutylene glycol)	22.9
Furfural (2-furaldehyde)	22.9
Methacrylic acid	22.9
Methylamine	22.9
Dipropyl sulfone	23.1
Methyl pyrrolidone-2 (1-methyl-2-pyrrolidinone)	23.1
Acetylpyrrolidine (*N*)	23.3
n-Butyl alcohol (1-butanol)	23.3
Cyclohexanol	23.3
Ethylene glycol monomethyl ether (2-methoxyethanol)	23.3
Tetramethyloxamide	23.3
Ethylene glycol monophenyl ether	23.5
Formylpiperidine (normal)	23.5

TABLE 8. *cont'd*

Solvent	δ (MPa$^{1/2}$)
Pentanediol-1,5	23.5
Propyl alcohol (2-propanol)	23.5
Acetylmorpholine (*N*)	23.7
1,3-Butanediol	23.7
Tolylene diisocyanate (4-methyl-1,3-phenylene diisocyanate)	23.7
Allyl alcohol	24.1
Methylene iodide	24.1
Acetonitrile	24.3
Propyl alcohol (1-propanol)	24.3
Santicizer 8	24.3
Acrylic acid	24.6
Dimethyltetramethylene sulfone	24.6
Benzyl alcohol	24.8
1,4-Butanediol	24.8
Butylene-2,3 carbonate	24.8
Diethylene glycol	24.8
Dimethylformamide (*N,N*)	24.8
Dioctyl phthalate	24.8
Formic acid	24.8
Hydrogen cyanide	24.8
Ethylene chlorohydrin (2-chloroethanol)	25.0
Ethylacetamide (normal)	25.2
Ethylenediamine	25.2
Diethyl sulfone	25.4
Methylene glycolate	25.4
Dimethyl phosphite	25.6
Furfuryl alcohol	25.6
Methyl propyl sulfone	25.6
Phenylhydrazine	25.6
Butyrolactone (γ)	25.8
Chloroacetonitrile	25.8
Propylene glycol (1,2-propanediol)	25.8
Caprolactam (ε)	26.0
Ethyl alcohol	26.0
Nitromethane	26.0
Methyltetramethylene sulfone	26.4
Formylmorpholine (*N*)	26.6
Dimethylnitroamine (*N,N*)	26.8
Propiolactone	27.2
Propylene-1,2-carbonate	27.2
Methyl ethyl sulfone	27.4
Pyrone (γ)	27.4
Tetramethylene sulfone (sulfolane, tetrahydrothiophene 1,1-dioxide)	27.4
Maleic anhydride	27.8
Piperidone	27.8
Diacetylpiperazine (*N,N*)	28.0
Ethylene formamide (*N*)	28.4
Dimethyl sulfoxide	29.7
Methanol	29.7
Ethylene glycol	29.9
Methylacetamide	29.9
Ethylene carbonate	30.1
Pyrrolidine (α)	30.1
Malononitrile	30.9
Ethylene cyanohydrin (3-hydroxypropionitrile)	31.1
Diformylpiperazine (*N,N*)	31.5
Succinic anhydride	31.5
Methylformamide (*N*)	32.9
Ammonia	33.4
Glycerol	33.8
Hydrazine	37.3
Formamide	39.3
Water	47.9

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TABLE 9. HANSEN SOLUBILITY PARAMETERS OF LIQUIDS AT 25°C

Solvent	CAS Number	Molar volume (cm^3/mol)	Solubility parameters ($MPa^{1/2}$)			
			δ_d	δ_p	δ_h	δ
9.1. PARAFFINIC HYDROCARBONS						
n-Butane	106-97-8	101.4	14.1	0	0	14.1
n-Pentane	109-66-0	116.2	14.5	0	0	14.5
Pentane (iso)	78-78-4	117.4	13.7	0	0	13.7
n-Hexane	110-54-3	147.4	14.9	0	0	14.9
n-Heptane	142-82-5	147.4	15.3	0	0	15.3
n-Octane	111-65-9	163.5	15.6	0	0	15.6
2,2,4-Trimethylpentane	540-84-1	166.1	14.3	0	0	14.3
n-Nonane	111-84-2	179.7	15.8	0	0	15.8
n-Decane	124-18-5	195.9	15.8	0	0	15.8
n-Dodecane	112-40-3	228.6	16	0	0	16
n-Hexadecane	544-76-3	294.1	16.4	0	0	16.4
n-Eicosane	112-95-8	359.8	16.6	0	0	16.6
Cyclohexane	110-82-7	108.7	16.8	0	0.2	16.8
Methylcyclohexane	108-87-2	128.3	16	0	1	16
cis-Decahydronaphthalene	493-01-6	156.9	18.8	0	0	18.8
trans-Decahydronaphthalene	493-02-7	159.9	18.8	0	0	18.8
9.2. AROMATIC HYDROCARBONS						
Benzene	71-43-2	89.4	18.4	0 [a]	2	18.6
Toluene	108-88-3	106.8	18	1.4	2	18.2
Naphthalene [b]	91-20-3	111.5	19	2	5.9	20
Styrene	1-42-5	115.6	18.6	1	4.1	19
o-Xylene	95-47-6	121.2	17.8	1	3.1	18
Ethylbenzene	1-41-4	123.1	17.8	0.6	1.4	17.8
1-Methylnaphthalene	90-12-0	138.8	20.6	0.8	4.7	21.2
Mesitylene	108-67-8	139.8	18	0	0.6	18
Tetrahydronaphthalene	119-64-2	136.0 [a]	19.6 [a]	0	2.9	20
Biphenyl	92-52-4	154.1	21.5	1	2	21.7
p-Diethylbenzene	105-05-5	156.9	18	0	0.6	18
9.3. HALOHYDROCARBONS						
Methyl chloride	74-87-3	55.4	15.3	6.1	3.9	17
Methylene dichloride	75-09-2	63.9	18.2	6.3	6.1	20.3
Bromochloromethane	74-97-5	65	17.4	5.7	3.5	18.6
Chlorodifluoromethane	75-45-6	72.9	12.3	6.3	5.7	14.9
Dichlorofluoromethane	75-43-4	75.4	15.8	3	5.7	17
Ethyl bromide	74-96-4	76.9	16.6	8	5.1	19
1,1-Dichloroethylene	75-35-4	79	17	6.8	4.5	18.8
Ethylene dichloride	107-06-2	79.4	19.0 [a]	7.4	4.1	20.9
Methylene diiodide [c]	75-11-6	80.5	17.8	3.9	5.5	19
Chloroform	67-66-3	80.7	17.8	3.1	5.7	19
1,1-Dichloroethane	75-34-3	84.8	16.6	8.2	0.4	18.4
Ethylene dibromide	106-93-4	87	19.6	6.8	12.1	23.9
Bromoform	75-25-2	87.5	21.5 [a]	4.1	6.1 [a]	22.7
n-Propyl chloride	540-54-5	88.1	16	7.8	2	17.8
Trichloroethylene	79-01-6	90.2	18	3.1	5.3	19
Dichlorodifluoromethane	75-71-8	92.3	12.3	2	0	12.5
Trichlorofluoromethane	75-69-4	92.8	15.3	2	0	15.5
Bromotrifluoromethane	75-63-8	97	9.6	2.5	0	10
Carbon tetrachloride	56-23-5	97.1	17.8	0	0.6	17.8
1,1,1-Trichloroethane	71-55-6	1.4	17	4.3	2	17.6
Tetrachloroethylene	127-18-4	101.1	19	6.5 [a]	2.9 [a]	20.3
Chlorobenzene	108-90-7	102.1	19	4.3	2	19.6
n-Butylchloride	109-69-3	104.9	16.4	5.5	2	17.4
1,1,2,2-Tetrachloroethane	79-34-5	105.2 [a]	18.8	5.1	9.4	21.7
Bromobenzene	222-22-2	105.3	20.5	5.5	4.1	21.7
o-Dichlorobenzene	95-50-1	112.8	19.2	6.3	3.3	20.5
Benzyl chloride	100-44-7	115	18.8 [a]	7.2	2.7	20.3
1,1,2,2-Tetrabromoethane [c]	79-27-6	116.8	22.7	5.1	8.2	24.8
1,2-Dichlorotetrafluoroethane [c]	76-14-2	117	12.7	1.8	0	12.9
1,1,2-Trichlorotrifluoroethane	76-13-1	119.2	14.7	1.6	0	14.7
Cyclohexyl chloride	542-18-7	121.3	17.4	5.5	2	18.4

TABLE 9. *cont'd*

Solvent	CAS Number	Molar volume (cm^3/mol)	Solubility parameters ($MPa^{1/2}$)			
			δ_d	δ_p	δ_h	δ
1-Bromonaphthalene	90-11-9	140	20.3	3.1	4.1	20.9
Trichlorobiphenyl[d]	7012-37-5	187	19.2	5.3	4.1	20.5
Perfluoromethylcyclohexane	355-02-2	196	12.5	0	0	12.5
Perfluorodimethylcyclohexane	335-27-3	217.4	12.5	0	0	12.5
Perfluoro-*n*-heptane	335-57-9	227.3	12.1	0	0	12.5
9.4. ETHERS						
Furan	110-00-9	72.5	17.8	1.8	5.3	18.6
Epichlorohydrin	106-89-8	79.9	19	10.2	3.7	21.9
Tetrahydrofuran	109-99-9	81.7	16.8	5.7	8	19.4
1,4-Dioxane	123-91-1	85.7	19	1.8	7.4	20.5
Methylal (dimethoxymethane)	109-87-5	88.8	15.1	1.8	8.6	17.6
Diethyl ether	60-29-7	104.8	14.5	2.9	5.1	15.8
Bis(2-chloroethyl) ether	107-30-2	117.6	18.8	9	5.7	21.7
Anisolec	100-66-3	119.1	17.8	4.1	6.8	19.4
Bis-(2-methoxyethyl) ether	111-96-6	142	15.8	6.1	9.2	19.2
Dibenzyl ether[c]	103-50-4	192.7	17.4	3.7	7.4	19.2
Di-(chloro-iso-propyl) ether	108-60-1	146	19	8.2	5.1	21.3
Bis-(*m*-phenoxyphenyl) ether	748-30-1	373	19.6	3.1	5.1	20.5
9.5. KETONES						
Acetone	67-64-1	74	15.5	10.4	7	20.1
Methyl ethyl ketone (2-butanone)	78-93-3	90.1	16	9	5.1	19
Cyclohexanone	108-94-1	104	17.8	6.3	5.1	19.6
Diethyl ketone (3-pentanone)	96-22-0	106.4	15.8	7.6	4.7	18.2
Mestiyl oxide (4-methyl-3-pentene-2-one)	141-79-7	115.6	16.4	7.2	6.1	18.8
Acetophenone	98-86-2	117.4	19.6[a]	8.6	3.7	21.7
Methyl iso-butyl ketone (2-hexanone)	591-78-6	125.8	15.3	6.1	4.1	17
Methyl iso-amyl ketone		142.8	16	5.7	4.1	17.4
Isophorone	78-59-1	150.5	16.6	8.2	7.4	19.8
Di-iso-butyl ketone (2,6-dimethyl-4-heptanone)	108-83-8	177.1	16	3.7	4.1	16.8
9.6. ALDEHYDES						
Acetaldehyde	75-07-0	57.1	14.7	8	11.3	20.3
2-Furfuraldehyde (furfural)	98-01-1	83.2	18.6	14.9	5.1	24.3
Butyraldehyde	123-72-8	88.5	14.7	5.3	7	17.2
Benzaldehyde	100-52-7	101.5	19.4	7.4	5.3	21.5
9.7. ESTERS						
Ethylene carbonate (1,3-dioxolan-2-one)	96-41-9	66	19.4	21.7	5.1	29.5
γ-Butyrolactone (4-hydroxybutryic acid γ-lactone)	96-48-0	76.8	19	16.6	7.4	26.2
Methyl acetate	79-20-9	79.7	15.5	7.2	7.6	18.8
Ethyl formate	109-94-4	80.2	15.5	8.4	8.4	19.6
Propylene carbonate (1,2-propanediol cyclic carbonate)	108-32-7	85	20.1	18	4.1	27.2
Ethyl chloroformate	541-41-3	95.6	15.5	10	6.8	19.6
Ethyl acetate	141-78-6	98.5	15.8[a]	5.3	7.2[a]	18.2
Trimethyl phosphate	512-56-1	99.9	16.8	16	10.2	25.4
Diethyl carbonate	105-58-8	121	16.6	3.1	6.1	18
Diethyl sulfate	64-67-5	131.5	15.8	14.7	7.2	22.7
n-Butyl acetate	123-86-4	132.5	15.8	3.7	6.3	17.4
Iso-butyl acetate	540-88-5	133.5	15.1	3.7	6.3	16.8
2-Ethoxyethyl acetate (Cellusolve acetate)	111-15-9	136.2	16	4.7	10.6	19.6
Iso-amyl acetate	628-63-7	148.8	15.3	3.1	7	17.2
Iso-butyl iso-butyrate	109-21-7	163	15.1	2.9	5.9	16.6
Dimethyl phthalate	131-11-3	163	18.6[a]	10.8[a]	4.9	22.1
Ethyl *trans*-cinnamate	103-36-6	166.8	18.4	8.2	4.1	20.5
Triethyl phosphate	78-40-0	171	16.8	11.5	9.2	22.3
Diethyl phthalate	84-66-2	198	17.6	9.6	4.5	20.5
Di-*n*-butyl phthalate	84-74-2	266	17.8[a]	8.6	4.1	20.3
n-Butyl benzyl phthalate	85-68-7	306	19	11.3	3.1	22.3
Tricresyl phosphate (tritolyl phosphate)	1330-78-5	613	19	12.3	4.5	23.1
Tri-*n*-butyl phosphate	126-73-8	345	16.4	6.3	4.3	18

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TABLE 9. *cont'd*

Solvent	CAS Number	Molar volume (cm^3/mol)	Solubility parameters ($MPa^{1/2}$)			
			δ_d	δ_p	δ_h	δ
Iso-propyl palmitate[c]	2239-78-3	330	14.3	3.9	3.7	15.3
Di-*n*-butyl sebacate	109-43-3	339	14.5	3.9	3.7	15.5
Methyl oleate[d]	112-62-9	340	14.5	3.9	3.7	15.5
Dioctyl phthalate (bis(2-ethylhexyl) phthalate)	117-81-7	377	16.6	7	3.1	18.2
Di-butyl stearate[c]		382	14.5	3.7	3.5	15.3
9.8. NITROGEN-CONTAINING COMPOUNDS						
Acetonitrile	75-05-8	52.6	15.3	18	6.1	24.6
Acrylonitrile	107-13-1	67.1	16.5	17.4	6.8	24.8
Propionitrile	107-12-0	70.9	15.3	14.3	5.5	21.7
Butyronitrile	109-74-0	87	15.3	12.5	5.1	20.5
Benzonitrile	100-47-0	102.6	17.4	9	3.3	19.8
Nitromethane	75-52-5	54.3	15.8	18.8	5.1	25
Nitroethane	79-24-3	71.5	16	15.5	4.5	22.7
2-Nitropropane	79-46-9	86.9	16.2	12.1	4.1	20.7
Nitrobenzene	98-95-3	102.7	20.1	8.6	4.1	22.1
Ethanolamine (2-aminoethanol)	141-43-5	60.2	17.2	15.5	21.3	31.5
Ethylene diamine	107-15-3	67.3	16.6	8.8	17	25.4
1,1-Dimethylhydrazine[c]	57-14-7	76	15.3	5.9	11	19.8
2-Pyrrolidone (2-pyrrolidinone)	616-45-5	76.4	19.4	17.4	11.3	28.4
Pyridine	110-86-1	80.9	19	8.8	5.9	21.7
n-Propylamine	107-10-8	83	17	4.9	8.6	19.6
Morpholine	110-91-8	87.1	18.8	4.9	9.2	21.5
Aniline	62-53-3	91.5	19.4	5.1	10.2	22.5
N-Methyl-2-pyrrolidone (1-methyl-2-pyrrolidinone)	872-50-4	96.5	18	12.3	7.2	22.9
n-Butylamine	109-73-9	99	16.2[a]	4.5[a]	8	18.6
Diethylamine	109-87-7	103.2	14.9	2.3	6.1	16.4
Diethylenetriamine	111-40-0	108	16.8	13.3	14.3	25.8
Cyclohexylamine	108-91-8	115.2	17.4	3.1	6.5	18.8
Quinoline	91-22-5	118	19.4	7	7.6	22.1
Di-*n*-propylamine	142-84-7	136.9	15.3	1.4	4.1	16
Formamide	75-12-7	39.8	17.2	26.2	19	36.6
Dimethylformamide	68-12-2	77	17.4	13.7	11.3	24.8
N,N-Dimethylacetamide	127-19-5	92.5	16.8	11.5	10.2	22.7
1,1,3,3-Tetramethylurea	632-22-4	120.4	16.8	8.2	11	21.7
Hexamethyl phosporamide[c] (hexamethyl phosphoric triamide)	680-31-9	175.7	18.4	8.6	11.3	23.3
9.9. SULFUR-CONTAINING COMPOUNDS						
Carbon disulfide	75-15-0	60	20.5	0	0.6	20.5
Dimethyl sulfoxide (methyl sulfoxide)	67-68-5	71.3	18.4	16.4	10.2	26.6
Ethanethiol[c] (ethyl mercaptan)	75-08-1	74.3	15.8	6.5	7.2	18.4
Dimethyl sulfone[b] (methyl sulfone)	67-71-0	75	19	19.4	12.3	29.9
Diethyl sulfide (ethyl sulfide)	352-93-2	108.2	17	3.1	2	17.4
9.10. ACID HALIDES AND ANHYDRIDES						
Acetyl chloride	75-36-5	71	15.8	10.6	3.9	19.4
Succinic anhydride[b]	108-30-5	66.8	18.6	19.2	16.6	31.5
Acetic anhydride	108-24-7	94.5	16.0[a]	11.7[a]	10.2[a]	22.3
9.11. ALCOHOLS						
Methanol	67-56-1	40.7	15.1	12.3	22.3	29.7
Ethanol	64-17-5	58.5	15.8	8.8	19.4	26.6
Ethylene cyanohydrin (hydracrylonitrile; 3-hydroxypropionitrile)	109-78-4	68.3	17.2	18.8	17.6	30.9
Allyl alcohol (2-propen-1-ol)	107-18-6	68.4	16.2	10.8	16.8	25.8
1-Propanol	71-23-8	75.2	16	6.8	17.4	24.6
2-Propanol	67-63-0	76.8	15.8	6.1	16.4	23.5
3-Chloropropanol (trimethylene chlorohydrin)	627-30-5	84.2	17.6	5.7	14.7	23.7
Furfuryl alcohol	98-00-0	86.5	17.4	7.6	15.1	24.3
1-Butanol (butyl alcohol)	71-36-3	91.5	16	5.7	15.8	23.1
2-Butanol	4221-99-2	92	15.8	5.7	14.5	22.1

TABLE 9. *cont'd*

Solvent	CAS Number	Molar volume (cm^3/mol)	Solubility parameters ($MPa^{1/2}$)			
			δ_d	δ_p	δ_h	δ
2-Methyl-1-propanol (isobutyl alcohol)	78-83-1	92.8	15.1	5.7	16	22.7
Benzyl alcohol	100-51-6	103.6	18.4	6.3	13.7	23.7
Cyclohexanol	108-93-0	106	17.4	4.1	13.5	22.5
1-Pentanol	71-41-0	109	16	4.5	13.9	21.7
2-Ethyl-1-butanol	97-95-0	123.2	15.8	4.3	13.5	21.3
Diacetone alcohol (4-hydroxy-4-methyl-2-pentanone)	123-42-2	124.2	15.8	8.2	10.8	20.9
1,3-Dimethyl-1-butanol	105-30-6	127.2	15.3	3.3	12.3	19.8
Ethyl lactate	687-47-8	115	16	7.6	12.5	21.7
n-Butyl lactate	138-22-7	149	15.8	6.5	10.2	19.8
Ethylene glycol monomethyl ether (2-methoxyethanol)	109-86-4	79.1	16.2	9.2	16.4	24.8
Ethylene glycol monoethyl ether (2-ethoxyethanol) Cellusolve	111-15-9	97.8	16.2	9.2	14.3	23.5
Diethylene glycol monomethyl ether (2-(2-methoxyethoxy)ethanol)	111-77-3	118	16.2	9.2	12.3	22.3
Diethylene glycol monoethyl ether (2-(2-ethyoxyethyoxy)ethanol)	111-90-0	130.9	16.2	9.2	12.3	21.9
Ethylene glycol mono-*n*-butyl ether (2-butoxyethanol) Butyl Cellusolve	111-76-2	131.6	16	5.1	12.3	22.3
2-Ethyl-1-hexanol	104-76-7	157	16	3.3	11.9	20.1
1-Octanol (capryl alcohol)	111-87-5	157.7	17	3.3	11.9	20.9
2-Octanol	123-96-6	159.1	16.2	4.9	11	20.3
Diethylene glycol mono-*n*-butyl ether (2-(2-butoxyethoxy)ethanol)	112-34-5	170.6	16	7	10.6	20.5
1-Decanol	112-30-1	191.8	17.6	2.7	10	20.5
Tridecyl alcohol[d]	112-70-9	242	14.3	3.1	9	17.2
Nonyl phenoxy ethanol[d]	27986-36-3	275	16.8	10.2	8.4	21.3
Oleyl alcohol[d]	143-28-2	316	14.3	2.7	8	16.6
Triethylene glycol mono-oleyl ether		418.5	13.3	3.1	8.4	16
9.12. ACIDS						
Formic acid	64-18-6	37.8	14.3	11.9	16.6	25
Acetic acid	64-19-7	57.1	14.5	8	13.5	21.3
Benzoic acid[b]	65-85-0	100	18.2	7	9.8	21.9
n-Butyric acid[c]	107-92-6	110	14.9	4.1	10.6	18.8
n-Octanoic acid[c]	124-07-2	159	15.1	3.3	8.2	17.6
Oleic acid	112-80-1	320	14.3	3.1	5.5	15.8
Stearic acid[b]	57-11-4	326	16.4	3.3	5.5	17.6
9.13. PHENOLS						
Phenol	108-95-2	87.5	18	5.9	14.9	24.1
1,3-Benzenediol[b] (resorcinol)	108-46-3	87.5	18	8.4	21.1	28.8
m-Cresol	108-39-4	104.7	18	5.1	12.9	22.7
o-Methoxyphenol (guaiacol)	90-05-1	109.5	18	8.2	13.3	23.7
Methyl salicylate	119-36-8	129	16	8	12.3	21.7
Nonyl phenol[d]	25154-52-3	231	16.6	4.1	9.2	19.4
9.14. WATER						
Water[c]	7732-18-5	18	15.5[a]	16.0[a]	42.4[a]	47.9
9.15. POLYHYDRIC ALCOHOLS						
Ethylene glycol	107-21-1	55.8	17	11	26	32.9
Glycerol	56-81-5	73.8	17.4	12.1	29.3	36.2
Propylene glycol (1,2-propanediol)	57-55-6	73.6	16.8	9.4	23.3	30.3
1,3-Butanediol	107-88-0	89.9	16.6	10	21.5	28.8
Diethylene glycol	111-46-6	95.3	16.2	14.7	20.5	29.9
Triethylene glycol	112-27-6	114	16	12.5	18.6	27.4
Hexylene glycol (2-methyl-2,4-pentanediol)	107-41-5	123	15.8	8.4	17.8	25.2
Dipropylene glycol[d]	110-98-5	131.3	16	20.3	18.4	31.7

[a] Altered from previously published value.
[b] Solid, treated as supercooled liquid.
[c] Values uncertain.
[d] Impure commercial product of this nominal formula.

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TABLE 10. SOLUBILITY PARAMETERS OF POLYMERS

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
10.1. MAIN CHAIN CARBON POLYMERS								
10.1.1. POLY(DIENES)								
Poly(butadiene)					14.65	Calc.		28
					17.19			66
					17.09			110
					17.15	Calc.		112
					17.2–17.6	Obs.		112
					16.6			118
					17.6			72
					16.2 ± 0.2	IPGC	75	70
Emulsion					17.19			79
Sodium					17.6			79
					16.6	Calc.		62
					17.09			104
Hydrogenated					16.6	Swelling		75
					16.47			75
					16.6	Av.		75
		16.98	–[a]	1.02	17.02			98
					16.1	^{129}Xe shift		152
Poly(1,2-butadiene)	58.85				17.06	Calc.[HK b]		3
Poly(3-methyl 1,2-butadiene)	73.88				17.39	Calc.[HK]		3
Poly(1,3-butadiene)	58.85				17.06	Calc.[HK]		3
Poly(2,3-dimethyl 1,3 butadiene)	88.91				17.60	Calc.[HK]		3
Poly(*cis*-butadiene) elastomer (Bunahyls CB10, Chemische Werke Huels)		17.53	2.25	3.42	18			56
Poly(butadiene-*co*-acrylonitrile)								
BUNA N (82/18)					17.90–17.72			19
(82/20)					18.4	Calc.		66
					19.4	Obs.		66
BUNA N (75/25)					18.93	Calc.	25	112
					19.19	Obs.		42
					19.4	Obs.		104
					18.2			118
					19.4			79
(70/30)					20.11–20.26			19
					19.19			43
(61/39)					21.1			19
					21.28–21.38			19
					20.5 ± 0.6	IPGC	75	70
Hycar (BFGoodrich)		18.6	8.8	4.2	21			56
Poly(butadiene-*co*-styrene)								
BUNA S (94/6)					16.45–16.64			19
(90/10)					17.13			110
(87.5/12.5)					16.39–16.57			19
					16.55			43
					17.6			104
(85/15)					17.31			112
					17.19			110
					17.35	Calc.		112
					17.4	Obs.		112
					17.39			79
					17.41	Calc.		66
					17.39	Obs.		66
					17.5			104
(75/25)					17.29			110
					17.47	Calc.		112
					16.55	Obs.		112
					16.49	Obs.		112
					16.6			118
					17.5			79
					17.56	Calc.		66
					17.5	Obs.		66
					17.6			104

TABLE 10. *cont'd*

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
(71.5/28.5)					16.55–16.72			19
					17.51			112
(70/30)					17.35			110
(60/40)					17.5			110
					17.7	Calc.		112
					17.74	Obs.		112
					17.74			79
					17.76	Calc.		66
					17.74	Obs.		66
					17.8			104
Poly(butadiene-*co*-styrene) (Polysar 5630, Polymer Corp.)		17.55	3.36	2.7	18.07			56
Poly(butadiene-*co*-vinylpyridine)					19.13			79
Poly(chloroprene)					18.42		25	110
					16.59	Calc.		28
					19.19	Calc.		112
					16.74	Obs.		42
					18.93	Obs.		104
					17.6			118
					16.8			44
					18.8			72
					16.6			79
					17.6	Swelling		43
					17.54–17.74			19
					15.18			19
					18.0 ± 0.4	IPGC	75	70
					17.6	^{129}Xe shift		152
Poly(1,4-*cis*-isoprene)					15.18	Calc.	25	28
					16.64			42
					16.68		25	74
					16.57		35	74
					20.46	Swelling	35	74
					16.47	Av.	35	74
					16.47	Swelling	35	74
					16.57	Swelling	35	74
					16.68	Calc.	35	74
					16.6	Swelling		75
					16.4			76
					16.47	Av.		76
					16.82		25	110
					16.68	Calc.	25	112
					16.2	Obs.		112
					16.33	Obs.		112
					17.09	Obs.		112
					16.2			72
Poly(isoprene) elastomer (Cariflex IR 305, Shell)		16.57	1.41	−0.82	16.65			56
Natural Rubber					16.2			43
					16.6			19
					16.68			112
					17.09			104
					17			124
					16.6			118
					16.4			19,10
					17.09			79
					16.33			43
					16.6			19
					16.42–16.49			19
					16.2	^{129}Xe shift		152
Gutta percha					16.6	Calc.		73
Chlorinated					19.2			21–24
10.1.2. POLY(ALKENES)								
Poly(2,3-dimethyl 1-butene)	93.15				18.05	Calc.^{HK}		3
Poly(3,3-dimethyl 1-butene)	93.15				18.05	Calc.^{HK}		3

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TABLE 10. *cont'd*

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
Poly(3-methyl 1-butene)	78.12				17.76	Calc.[HK]		3
Poly(2-methyl 1-butene)	78.12				17.92	Calc.[HK]		3
Poly(2-methyl 2-butene)	78.12				17.68	Calc.[HK]		3
Polyethylene					15.76			110
					16.6	Calc.		112
					16			58
					16.2			97
					17.09			8
					16.4	Calc.		124
					16.2			118
					16.2			72
					16.8	Calc.		97
					16.2	Obs.		73
					18.4	^{129}Xe shift		152
					17.99	Calc.		119
Poly(ethylene)	33.03				16.96	Calc.[HK]		3
Poly(1,1-diphenyl ethylene)	166.29				19.93	Calc.[HK]		3
Poly(methylene)					14.3	Extrapol.	20	45
Poly(ethylene-*co*-vinyl acetate)					18.6 ± 0.9	IPGC	25	71
					17.0 ± 0.4	IPGC	75	70
Poly(isobutene)					14.5	Calc.		28
					16.06	Av.	35	74
					16	Swelling		74
					16.47	Swelling		74
					16.25			19,10
					16.06		25	110
					15.76	Calc.		112
					16.47	Obs.		112
					16.4			58
					16.47			104
					16.6			21–24
					16			19
					17	Calc.		124
					16.47			72
					16.47			79
Poly(isobutylene) (Lutonal IC/123, BASF)		14.53	2.52	4.66	15.47			56
					15	^{129}Xe shift		152
Poly(isobutene-*co*-isoprene) butyl rubber					15.9–16.06			19
					16.47			104
					15.76			112
Poly(propylene)					18.8		25	58
					19.2	Calc.		124
Poly(propylene)	48.06				17.49	Calc.[HK]		3
Poly(propylene), isotactic (Profax 6701, Hercules)		17.19	–	–	17.19			81
Poly(propylene), chlorinated (Parlon P10, Hercules)		20.26	6.32	5.4	21.89			56
Poly(2-methyl propene)	63.09				17.72	Calc.[HK]		3
10.1.3. POLY(ACRYLICS) AND POLY(METHACRYLICS)								
Poly(acrylic acid), butyl ester					18		35	75
					18.01	Av.		75
					18.52	Swelling		75
					17.4	Calc.		75
					18.6	Swelling		77
					18.52			77
					19.77	Calc.		119
Poly(acrylic acid), ethyl ester					19.13	Av.		75
					19.2	Swelling		75
					19.8	Calc.		75
					19.2	Swelling		77
					19.13			77
					19.19			45
					18.8	Calc.		62
					20.4	Calc.		119

TABLE 10. *cont'd*

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
Poly(acrylic acid), isobornyl ester					16.8	Calc.		62
Poly(acrylic acid), methyl ester					20.7	Av.		75
					20.77	Swelling		75
					20.1	Calc.		75
					20.77	Swelling		77
					21.3			77
					20.7	Swelling		77
Poly(acrylic acid), propyl ester					18.52			75
					18.42	Av.		75
					18.4	Calc.		75
Poly(α-chloroacrylic acid), methyl ester					20.7	Calc.		112
Poly(methacrylic acid), butyl ester					17.9	Swelling		77
					17.8			77
					18.01	Swelling		77
					17.9			77
					17	Calc.		73
					14.7	IPGC	140	31
Poly(methacrylic acid), isobutyl ester					14.7	IPGC	140	31
Poly(methacrylic acid), *sec*-butyl ester					14.7	IPGC	140	31
Poly(methacrylic acid), ethoxyethyl ester					18.4	Swelling		77
					20.3			77
Poly(methacrylic acid), ethyl ester					18.31	Swelling		77
					18.2			77
					18.6	Calc.		73
Poly(ethyl methacrylate) (Lucite 2042, DuPont)		18.64	10.52	7.51	22.69			56
					20.4	^{129}Xe shift		152
Poly(methacrylic acid), *n*-hexyl ester					17.6	Calc.		73
Poly(methacrylic acid), isobornyl ester					16.6			62
Poly(methacrylic acid), lauryl ester					16.8	Calc.		73
Poly(methacrylic acid), methyl ester					18.58		25	19
					18.52–18.66			19
					19.4	Swelling		77
					19.34			77
					26.27			110
					18.93	Calc.		112
					18.4–19.4			7
					18.58			118
					19.5	Calc.		119
Poly(methyl methacrylate) (Rohm and Haas)		18.64	10.52	7.51	22.69			56
Poly(methacrylic acid), octyl ester					17.2	Calc.		73
Poly(methacrylic acid), propyl ester					18	Calc.		73
Poly(methacrylic acid), stearyl ester					16	Calc.		73
Poly(methacrylonitrile)					21			118
					21.9	Calc.		112
10.1.4. POLY(VINYL ALCOHOLS)								
Poly(vinyl alcohol)					25.78			110
					21.7			139
Poly(4-vinyl phenol)					22.5			139
10.1.5. POLY(VINYL ESTERS)								
Poly(vinyl acetate)					19.62	Calc.	25	28
					19.13		35	78
					20.93	Calc.		119
					19.2	Calc.		78
					18			78
					22.61		25	110
					19.2	Calc.		112
					19.2			118
Poly(vinyl acetate) (Mowilith 50, Hoechst)		20.93	11.27	9.66	25.66			56
Poly(vinyl acetate-*co*-vinyl alcohol)					21.94	Calc.		119
Poly(vinyl butyral) (Butvar B76, Shawinigan)		18.6	4.36	13.03	23.12			56
Poly(vinyl propionate)					18.01		35	78
					18.52	Calc.		78

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TABLE 10. *cont'd*

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
10.1.6. POLY(ALLYL ETHERS) AND POLY(VINYL ETHERS)								
Poly(allyl methyl ether)	70.79				19.44	Calc.[HK]		3
Poly(allyl ethyl ether)	85.82				19.29	Calc.[HK]		3
Poly(allyl propyl ether)	100.85				19.21	Calc.[HK]		3
Poly(allyl isopropyl ether)	100.85				19.21	Calc.[HK]		3
Poly(allyl phenyl ether)	122.39				20.19	Calc.[HK]		3
Poly(allyl 2, tolyl ether)	137.41				20.03	Calc.[HK]		3
Poly(allyl 3, tolyl ether)	137.41				20.03	Calc.[HK]		3
Poly(allyl 4, tolyl ether)	137.41				20.03	Calc.[HK]		3
Poly(diallyl ether)	96.61				18.84	Calc.[HK]		3
Poly(vinyl methyl ether)	55.76				19.66	Calc.[HK]		3
Poly(vinyl ethyl ether)	70.79				19.44	Calc.[HK]		3
Poly(vinyl propyl ether)	85.82				19.29	Calc.[HK]		3
Poly(vinyl butyl ether)	100.85				19.21	Calc.[HK]		3
Poly(vinyl isopropyl ether)	85.82				19.29	Calc.[HK]		3
Poly(vinyl isobutyl ether)	100.85				19.21	Calc.[HK]		3
Poly(vinyl isoamyl ether)	115.88				19.13	Calc.[HK]		3
Poly(vinyl-1-amyl methyl ether)	119.91				20.83	Calc.[HK]		3
Poly(vinyl-2-ethyl hexyl ether)	160.96				18.99	Calc.[HK]		3
Poly(vinyl-2-methoxyethyl ether)	93.52				20.44	Calc.[HK]		3
Poly(vinyl phenyl ether)	107.36				20.19	Calc.[HK]		3
Poly(vinyl-1-phenyl methyl ether)	122.39				20.19	Calc.[HK]		3
Poly(vinyl-1-methyl phenyl ether)	122.39				20.19	Calc.[HK]		3
Poly(vinyl-1-phenyl phenyl ether)	174.01				20.50	Calc.[HK]		3
Poly(1-methyl vinyl ethyl ether)	85.82				19.29	Calc.[HK]		3
Poly(1-ethyl vinyl ethyl ether)	100.85				19.21	Calc.[HK]		3
Poly(1-phenyl vinyl ethyl ether)	137.41				20.03	Calc.[HK]		3
Poly(divinyl ether)	66.55				18.93	Calc.[HK]		3
10.1.7. POLY(VINYL HALIDES) AND POLY(VINYL NITRILES)								
Poly(acrylonitrile)					25.27	Calc.		119
					25.6			66
					26.09	Calc.	25	112
					31.5			118
		18.21	16.16	6.75	25.27			98
Poly(allyl acetonitrile)	81.13				24.18	Calc.[HK]		3
Poly(1-methyl acrylonitrile)	66.09				25.45	Calc.[HK]		3
Poly(2-methyl acrylonitrile)	66.09				25.45	Calc.[HK]		3
Poly(acrylonitrile-*co*-isopropyl methacrylate)					19.84			141
Poly(tetrafluoroethylene)					12.7	Calc.		112
					12.7			118
Poly(vinyl bromide)					19.42			34
					19.6			112
Poly(vinyl chloride)					19.19–19.34			19
					19.28	Calc.		28
					20.67			110
					19.54	Calc.		112
					19.8	Obs.		72
					19.2			19
					19.8			21–24
					22.1			58
					20.1			124
					19.5			118
					19.8			72
					20.32	Calc.		119
Poly(vinyl chloride) (Vipla KR, K = 50, Montecatini)		18.23	7.53	8.35	21.42			56
Poly(vinyl chloride)		18.72	10.03	3.07	21.46			33
		18.82	10.03	3.07	21.54			98
Poly(vinyl chloride), high molecular weight					20.13	Turbidity		162
					20.25	Swelling		162
					20.3	Visc.		162
Poly(vinyl chloride), chlorinated					19	Visc.	25	117

TABLE 10. *cont'd*

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
Poly(vinyl chloride-*co*-vinyl acetate) (87/13)					21.7	Calc.		28
					21.3			21–24
VYHH					20.42	Calc.		119
Poly(vinyl chloride-*co*-vinyl acetate-*co*-maleic acid)					20.44	Calc.		119
Poly(vinyl chloride-*co*-vinyl acetate-*co*-vinyl alcohol)					20.77	Calc.		119
Poly(allyl cyanide)	66.09				25.45	Calc.[HK]		3
Poly(allyl isocyanide)	66.09				25.45	Calc.[HK]		3
Poly(vinylidene chloride)					25			21–24
Poly(vinylidene cyanide-*co*-vinyl acetate)					22.67	Calc.		128
Poly(vinylidene fluoride)		–	–	–	23.2	Visc.		147
		17.2	12.5	9.2	23.2	Obs.		147
		16.6, 17.9				Calc.		147
		16.6				Refract. index		147
		13.7	10.6	8.2	19.2	Contact ang.		149
					24.6, 25.2	Swelling		148
10.1.8. POLY(STYRENES)								
Poly(styrene)					17.52			19
					17.45–17.58			19
					20.16	Calc.		28
					17.86–17.92			48
					17.84–18.56	Visc.		116
					18.6			66
					18.72		35	74
					18.62	Av.		74
					18.62	Swelling		74
					18.66	Calc.		74
					19.09		25	110
					18.66	Calc.		112
					18.6	Obs.		112
					17.6–19.8	Obs.		112
					17.4			18
					17.6			19
					18.4			58
					19.28	Calc.		119
					21.1	Calc.		124
					17.52			118
					17.84			19,97
					18.6			104
					17.6–17.8			72
					18.6			79
					15.6	IPGC	140	31
Poly(styrene) (Polystyrene LG, BASF)		21.28	5.75	4.3	22.47			56
Poly(*o*-methyl styrene)	114.69				19.33	Calc.[HK]		3
Poly(*m*-methyl styrene)	114.69				19.33	Calc.[HK]		3
Poly(*p*-methyl styrene)	114.69				19.33	Calc.[HK]		3
Poly(1-methyl styrene)	114.69				19.33	Calc.[HK]		3
Poly(methoxy styrene)	122.39				20.19	Calc.[HK]		3
Poly(cyano styrene)	124.28				22.36	Calc.[HK]		3
Poly(nitro styrene)	113.29				22.71	Calc.[S']		3
Poly(2-nitro styrene)	113.29				22.20	Calc.[S]		3
Poly(styrene-*co*-divinylbenzene)					18.6	Obs.		112
					17.39			18
5% crosslinking					14.8			137
10% crosslinking					15.7			137
20% crosslinking					17.8			137
Poly(styrene-*co*-*n*-butyl methacrylate)					15.1	IPGC	140	31
Poly(styrene-*co*-isobutyl methacrylate)					15.1	IPGC	140	31
Poly(4-chlorostyrene)		17.6	6.1	4.1	19.02			151
Poly(α-methylstyrene-*co*-acrylonitrile)					16.4	IPGC	140	111
Poly(4-acetoxystyrene)		17.8	9	8.4	21.7			138
Poly(4-hydroxystyrene)		17.6	10	13.7	24.55			138

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TABLE 10. *cont'd*

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
Poly(4-α-acetoxystyrene), $-CH_2-C-(OCOCH_3)-(C_6H_5-OCOR)-$, $R=CH_3$, C_2H_5, n-C_3H_7, OCH_3		15.7	10.2	7.4	21.9			161
Poly(vinylidene cyanide-*co*-4-α-acetoxystyrene), $-CH_2-C(CN)_2-CH_2-C-(OCOCH_3)-(C_6H_5-OCOR)-$, $R=CH_3$		21.5	11.3	7.2	25.3			161
Poly(vinylidene cyanide-*co*-4-α-acetoxystyrene), $R=OCOCH_3$		17	13.5	6.5	23.5			161
Poly(vinylidene cyanide-*co*-4-α-acetoxystyrene), $R=OCOCH_3$		17	12.1	8.2	22.4			161
Poly(vinylidene cyanide-*co*-4-α-acetoxystyrene), $R=OCOCH_3$		17	11.9	7.2	21.9			161
10.2. MAIN CHAIN C–O POLYMERS								
10.2.1. POLY(VINYL ESTERS)								
Poly(vinyl acetate)	74.25				18.21	Calc.[HK]		3
Poly(allyl acetate)	89.28				18.27	Calc.[HK]		3
Poly(diethylaminoethyl acrylate)-palladium ion					20.50			136
Poly(methyl acrylate)	74.25				18.21	Calc.[HK]		3
Poly(ethyl acrylate)	89.28				18.27	Calc.[HK]		3
Poly(allyl acrylate)	100.07				17.94	Calc.[HK]		3
Poly(butyl acrylate)	119.34				20.42	Calc.[HK]		3
Poly(isobutyl acrylate)	119.34				18.37	Calc.[HK]		3
Poly(2-ethyl hexyl acrylate)	179.46				18.45	Calc.[HK]		3
Poly(2-cyanoethyl acrylate)[a]	89.36				31.77	Calc.[HK]		3
Poly(benzyl acrylate)	140.88				19.38	Calc.[HK]		3
Poly(vinyl butyrate)	104.31				18.33	Calc.[HK]		3
Poly(dimethyl citraconate)	130.5				18.58	Calc.[HK]		3
Poly(diethyl citraconate)	160.58				18.58	Calc.[HK]		3
Poly(vinyl crotonate)	100.07				17.94	Calc.[HK]		3
Poly(methyl dimethyl fumarate)	130.5				18.58	Calc.[HK]		3
Poly(methyl diethyl fumarate)	160.56				18.58	Calc.[HK]		3
Poly(dimethyl fumarate)	115.47				18.56	Calc.[HK]		3
Poly(diethyl fumarate)	145.53				18.58	Calc.[HK]		3
Poly(dipropyl fumarate)	165.58				19.72	Calc.[HK]		3
Poly(di-*n*-butyl fumarate)	205.65				18.60	Calc.[HK]		3
Poly(di-*n*-amyl fumarate)	235.71				18.60	Calc.[HK]		3
Poly(di-isopropyl fumarate)	165.58				19.72	Calc.[HK]		3
Poly(di-isobutyl fumarate)	205.65				18.60	Calc.[HK]		3
Poly(di-isoamyl fumarate)	235.71				18.60	Calc.[HK]		3
Poly(dinitrile fumarate)	46.32				48.45	Calc.[HK]		3
Poly(diphenyl fumarate)	217.67				19.91	Calc.[HK]		3
Poly(vinyl-2-ethyl hexoate)	157.83				19.21	Calc.[HK]		3
Poly(dimethyl maleate)	115.47				18.56	Calc.[HK]		3
Poly(diethyl maleate)	145.53				18.58	Calc.[HK]		3
Poly(di-*n*-propyl maleate)	165.58				19.70	Calc.[HK]		3
Poly(di-*n*-butyl maleate)	205.65				18.60	Calc.[HK]		3
Poly(di-*n*-amyl maleate)	235.71				18.60	Calc.[HK]		3
Poly(di-isobutyl maleate)	205.65				18.60	Calc.[HK]		3
Poly(di-isoamyl maleate)	235.71				18.60	Calc.[HK]		3
Poly(diphenyl maleate)	217.67				19.91	Calc.[HK]		3
Poly(3-chloropropyl methacrylate)					19.60			160
Poly(methyl methacrylate)	89.28				18.27	Calc.[HK]		3
Poly(ethyl methacrylate)	110.9				17.25	Calc.[HK]		3
Poly(propyl methacrylate)	119.34				18.37	Calc.[HK]		3
Poly(butyl methacrylate)	134.37				18.41	Calc.[HK]		3
Poly(hexyl methacrylate)	191.91				15.80	Calc.[HK]		3
Poly(allyl methacrylate)	115.1				17.51	Calc.[S]		3
Poly(isopropyl methacrylate)	119.34				18.39	Calc.[HK]		3
Poly(isobutyl methacrylate)	134.37				18.39	Calc.[HK]		3
Poly(vinyl propionate)	89.32				18.27	Calc.[HK]		3

TABLE 10. *cont'd*

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
10.3. MAIN CHAIN C–N POLYMERS								
10.3.1. POLY(AMIDES)								
Poly(lactams)								
Nylon 3					26.2	Calc.		139
Nylon 4					23.9	Calc.		139
		19.44	–[a]	14.12	24.02			81
Nylon 5					22.5	Calc.		139
Nylon 6					21.5	Calc.		139
Nylon 7					20.7	Calc.		139
Nylon 8					20.3	Calc.		139
Nylon 9					19.8	Calc.		139
Nylon 10					19.4	Calc.		139
Nylon 11					19.2	Calc.		139
Nylon 12					19	Calc.		139
Nylon 2-I					27.2	Calc.		139
Nylon 3-I					26	Calc.		139
Nylon 4-I					25	Calc.		139
Nylon 5-I					24.1	Calc.		139
Nylon 6-I					23.5	Calc.		139
Nylon 6I, poly(hexamethylene isophthalamide)					26.3	Calc.[HK]		150
					23.2	Calc.[H']		150
Nylon 7-I					22.9	Calc.		139
Nylon 8-I					22.5	Calc.		139
Nylon 9-I					22.1	Calc.		139
Nylon 10-I					21.7	Calc.		139
Nylon 12-I					21.1	Calc.		139
Nylon 66		18.62	5.11	12.28	22.87			98
Nylon 66 (Zytel, DuPont)		18.62	0	14.12	23.37			56
Poly(iminohexamethylene iminoadipoyl)					27.8			118
Poly(amide), thermoplastics (Versamid 930, General Mills)		17.43	– 1.92	14.89	23.02			56
Poly(*p*-benzamide)		18	11.9	7.9	23	Swelling		143
Poly(*p*-phenylene terephthalamide)		18	11.9	7.9	23	Swelling		143
Poly(*p*-benzanilide terephthalamide)		18	11.9	7.9	23	Swelling		143
Poly(*n*-isopropyl acrylamide)					22.8	Calc.		154
Gel					23.5	Swelling		153
10.3.2. CELLULOSE AND DERIVATIVES								
Benzyl cellulose					25.23			110
Cellulose					32.02			110
		12.69	–[a]	22.3	25.66			81
Cellulose		8.27	8	13.88	18.03			140
Cellulose–alkylketene dimer treatment		8.1	12.35	7.96	16.78			140
Cellulose acetate (56% acetate groups)					27.83			110
(48% acetate groups)					27.19			110
Cellulose acetate (Cellidore A. Bayer)		18.6	12.73	11.01	25.08			56
Cellulose acetate		15.55	–[a]	11.87	19.56			81
Cellulose acetate butyrate		15.75	–[a]	8.59	17.94			81
Cellulose acetate propionate		15.75	–[a]	10.23	18.76			18
Cellulose diacetate					23.22	Calc.		112
					22.3			118
					22.3			72
Cellulose nitrate (11.83% N)					30.39			110
					21.44			112
					21.7			72
					23.5			21–24
(11.4% N)					21.93			72
Cellulose nitrate (1/2 sec; H-23 Hagedon)		15.41	14.73	8.84	23.08			56
Cellulose triacetate		15.55	–[a]	10.64	18.84			81
Cellulose tridecanoate		19.85	–[a]	6.14	20.77			81
Ethyl cellulose					21.1			21–24

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TABLE 10. *cont'd*

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
10.4. OTHER POLYMERS								
Alcohol soluble resin (Pentalyn 255, Hercules)		17.5	9.3	14.3	24.4			56
Alcohol soluble rosin resin (Pentalyn 830, Hercules)		20	5.8	10.9	23.5			56
Alkyd, short oil (coconut oil 34% phthalic anhydride, Plexal C34)		18.5	9.21	4.91	21.24			56
Alkyd resin, medium oil length					19.2			21–24
Alkyd, long oil (66% oil length, Plexal P65, Polyplex)		20.42	3.44	4.56	12.2			56
Coumarone-indene resin (Piccoumarone 450L, Penn. Ind. Chem.)		19.42	5.48	5.77	20.99			56
Epoxy resin					22.3			118
Epoxy (Epikote 1001, Shell)		20.36	12.03	11.48	26.29			56
Epoxy DBEBA–DDMe*	32				22		35	135
DGEBA–DMM	22				22.2		35	135
TGAP–DDMe	22.4				22.5		35	135
TGAP–ANI					23.9		35	135
TGAP–DDM					24.8		35	135
TGAP–DDS					27.1		35	135
Ester gum (ester gum BL, Hercules)		19.64	4.73	7.77	21.65			56
Furfuryl alcohol resin (Durez 14383, Hooker Chemical)		21.16	13.56	12.81	28.21			56
Hexamethoxymethyl melamine (Cymel 300, American Cyanimid)		20.36	8.53	10.64	24.51			56
Pentaerythritol ester of rosin, modified (Cellolyn 102, Hercules)		21.73	0.94	8.53	23.37			56
Petroleum hydrocarbon resin (Piccopale 110, Penn. Ind. Chem.)		17.55	1.19	3.6	17.96			56
Phenolic resin (resole, Phenodur 373U Chemische Werke Albert)		19.74	11.62	14.59	27.15			56
Phenolic resin (Super Beckacite 1001, Reichhold)		23.26	6.56	8.35	25.57			56
Poly(2-acrylamide-2-methyl propane sulphonamide) (tested in formamide/water mixtures)					43.6	Visc.	25	159
Poly(3,3-dimethyl oxetane)		15.35	2.46	4.09	16.16	Visc.		92
Poly(3,3-diethyl oxetane)		15.55	2.05	3.48	16.16	Visc.		92
Poly(1,3-dioxolane)					20.66	Visc.	25	175
Poly(ether urethane)					$17.99 \pm .02$	IPGC	25	158
					17.38 ± 0.16	IPGC	100	158
					17.20 ± 0.15	IPGC	120	158
					17.14 ± 0.14	IPGC	130	158
					17.06 ± 0.15	IPGC	140	158
					16.96 ± 0.15	IPGC	150	158
Poly(DL-lactic acid)					21.0 ± 0.3	Density	25	142
					20.5 ± 0.4	Visc.	25	142
					19.8	Calc.^{S}	25	142
					20.3	Calc.^{H}	25	142
					19.2	Calc.^{HK}	25	142
Poly(oxetane)		17.39	5.12	5.12	19.23			92
Poly(oxydimethylsilylene)					15.04			110
					14.9	Obs.		21–24
					15.45	Swelling		76
					15.6	Av.		76
					14.9			118
					15.59			45
					15.4		25	125
					15	Vapor sorp.		145
					15.1	IPGC		146

* DGEBA – diglycidyl ether of bisphenol A; DDMe – tetraethyl derivative of DDM; DDM – diamino diphenyl methane; TGAP – triglycidyl derivative of amino phenol; ANI – aniline; DDS – diamino diphenylsulfone.

TABLE 10. *cont'd*

Polymer	Molar volume (cm^3/gmol)	Solubility parameters $(MPa)^{1/2}$ δ_d	δ_p	δ_h	δ	Method	T(°C)	Refs.
$\delta_2 = (15.7 - 0.026)t$, t in °C					15.1	IPGC		144
					17.5	^{129}Xe shift		152
Poly(ethylene adipate)		16.3 ± 2	6.1 ± 0.5	9.4 ± 0.1	19.8 ± 2			157
Poly(ethylene oxide)					20.2 ± 2	IPGC	25	31
		17.3 ± 2	3.0 ± 1	9.4 ± 0.5	19.9 ± 2.2			157
					17.8	Calc.		157
					19.64	IPGC	70	176
Poly(oxyethyleneoxyterephthaloyl)					21.9			118
					21.9			112
Poly(oxytrimethylene) polyoxetane					19.2	Visc.	25	31
-3,3 dimethyl					16.2	Visc.	25	92
-3,3 diethyl					16.2	Visc.	25	92
Poly(propylene oxide)					16.3		81	156
					16.2		86	156
					16.1		91	156
		16.3 ± 1	4.7 ± 0.5	7.4 ± 0.5	18.5 ± 1.2			157
					15.9	Calc.		157
Poly(sulfone), Bisphenol A (Udel)		19.03	0	6.96	20.26			81
Poly(thioethylene)					19.23			110
					19.19	Swelling		43
					18.4	Swelling		43
Poly(thiophenylethylene)					19.0 ± 1.0			87
Poly(urethane) (unknown composition)					20.5	Swelling		76
					20.5	Av.		76
Poly(urethane), amorphous					22.9			155
Poly(vinyl ethylene)					16.5	^{129}Xe shift		152
Poly(vinyl pyrrolidone)					25.6	Visc.		177
Poly(vinyl methyl ketone)	66.59				22.92	Calc.^{HK}		3
Poly(vinyl ethyl ketone)	81.58				22.14	Calc.^{HK}		3
Poly(vinyl methyl sulfide)	66.63				19.52	Calc.^{HK}		3
Poly(vinyl phenyl sulfide)	118.35				20.28	Calc.^{HK}		3
Terpene resin (Piccolyte S-1000, Penn. Ind. Chem.)		16.47	0.37	2.84	16.72			56
Urea-formaldehyde resin (Plastopal H, BASF)		20.81	8.29	12.71	25.74			56
Viton		17	10.3	6.1	20.8			130

[a] Altered from previously published value.
[b] Method of van Krevelen and Hoftyzee.
[c] Method of Small.
[d] Method of Hoy.

E. REFERENCES

1. H. Ahmad, M. Yaseen, J. Oil Chem. Assoc., **60**, 488, (1977).
2. H. Ahmad, M. Yaseen, Farbe Lack, **85**, 356, (1979).
3. H. Ahmad, M. Yaseen, Polym. Eng. Sci., **19** (2), 858 (1979).
4. H. Ahmad, J. Oil Col. Chem. Assoc., **63**, 263 (1980).
5. H. Ahmad, J. Colour Soc., **20**, 108 (1981).
6. P. Alessi, I. Kikic, G. Torriano, A. Papo, J. Coatings Technol., **51** (560), 62 (1979).
7. J. Alfrey, A. O. Goldberg, J. A. Price, J. Colloid Sci., **5**, 251 (1950).
8. G. Allen, G. Geen, D. Mangaraj, D. Dims, G. J. Wilson, Polymer, **1**, 467 (1960).
9. E. Bagley et al., J. Paint Tech., **41**, 495 (1969).
10. E. B. Bagley, T. P. Nelson, J. W. Barlow, S. A. Chen, Ind. Eng. Chem. Fund., **9**, 93 (1970).
11. E. Bagley et al., J. Paint Tech., **43**, 35 (1971).
12. A. F. M. Barton, "CRC Handbook of Solubility Parameters and Other Cohesion Parameters", CRC Press, Boca Raton, FL, 1983, Ch. 15.
13. A. F. M. Barton, Pure Appl. Chem., **57** (7), 905 (1985).
14. A. Beerbower, J. R. Dickey, Am. Soc. Lubr. Eng., Trans., **12**, 1 (1969).
15. R. G. Blanks, J. M. Prausnitz, Ind. Eng. Chem. Fund., **3** (1), 1 (1964).
16. R. F. Blanks, Polym.-Plast. Technol. Eng., **8** (1), 13 (1977).
17. S. T. Bowden, W. J. Jones, Phil. Mag., **39**, 155 (1948).
18. R. F. Boyer, R. S. Spencer, "High Polymer Physics", Paper 5, Part III, Remoen Press, New York, 1948.
19. G. M. Bristow, W. F. Watson, Trans. Faraday Soc., **54**, 1731 (1958).
20. C. W. Bunn, J. Polym. Sci., **16**, 323 (1955).
21. H. Burrell, Interchem. Rev., **144**, 3 and 31 (1955).

22. H. Burrell, Off. Dig., Fed. Paint Varn. Prod. Clubs, **27**, 726 (1955).
23. H. Burrell, Off. Dig., Fed. Paint Varn. Prod. Clubs, **29**, p. 1069, p. 1159 (1955).
24. H. Burrell, J. Paint Technol., **40**, 197 (1968).
25. S. A. Chen, J. Appl. Polym. Sci., **15**, 1247 (1971).
26. J. D. Crowley, G. S. Teague, J. W. Lowe, J. Paint Technol., **38** (496), 296 (1967).
27. J. D. Crowley, G. S. Teague, J. W. Lowe, J. Paint Technol., **39** (504), 19 (1967).
28. A. T. DiBenedetto, J. Polym. Sci. A, **1**, 3459 (1963).
29. G. DiPaola-Baranayi, J. E. Guillet, Macromolecules, **11**, 228 (1978).
30. G. DiPaola-Baranayi, J. E. Guillet, J. Klein, H.-E. Jeberien, J. Chromatogr., **166**, 349 (1978).
31. G. DiPaola-Baranayi, Macromolecules, **15**, 622 (1982).
32. M. Dyck, P. Hower, Farbe Lack, **70**, 522 (1964).
33. M. Dunkel, J. Phys. Chem. A, **138**, 42 (1928).
34. D. Edelson, R. M. Fuoss, J. Am. Chem. Soc., **71**, 3548 (1949).
35. R. F. Fedors, Polym. Eng. Sci., **14** (2), p. 147, p. 472 (1974).
36. M. J. Fernandez-Berridi, T. F. Otero, G. M. Guzman, J. M. Elorza, Polymer, **23**, 1361 (1982).
37. P. J. Flory, "Principles of Polymer Chemistry", Cornell University Press, Ithaca, NY, 1953.
38. P. E. Froehling, L. T. Hillegers, Polymer, **22**, 261 (1981).
39. P. E. Froehling, D. M. Koenhen, A. Bantjes, C. A. Smolders, Polymer, **17**, 835 (1976).
40. E. W. Funk, J. M. Prausnitz, Ind. Eng. Chem., **62** (9), 8 (1970).
41. J. L. Gardon, J. Paint Technol., **38**, 43 (1966).
42. G. Gee, Trans. Faraday Soc., **38**, 418 (1942).
43. G. Gee, Trans. Inst. Rubber Ind., **18**, 266 (1943).
44. G. Gee, Trans. Faraday Soc., **40**, 468 (1944).
45. G. Gee, G. Allen, G. Wilson, Polymer (London), **1**, 456 (1960).
46. G. J. Glover, W. R. Lau, AIChE Jr., **29** (1), 73 (1983).
47. R. J. Good, in: R. L. Patrick (Ed.), "Treatise on Adhesion and Adhesives", Marcel Dekker, NY, 1967.
48. J. H. S. Green, Nature, **183**, 818 (1959).
49. C. M. Hansen, Ind. Eng. Chem. Prod. Res. Dev., **8**, 2 (1969).
50. C. M Hansen, Doctoral Dissertation, Technical University of Denmark, Danish Technical Press, Copenhagen, 1967.
51. C. M. Hansen, Farg Lack, **14** (1), 18;(2),23 (1968).
52. C. M.Hansen, J. Paint Technol., **39** (505), 104 (1967).
53. C. M. Hansen, J. Paint Technol., **39** (511), 505 (1967).
54. C. M. Hansen, K. Skaarup, J. Paint Technol., **39** (511), 511 (1967).
55. C. Hansen, A. Beerbower, Solubility Parameters, in: Kirk-Othmer (Ed.), "Encyclopedia of Chemical Technology", Supp. vol. (A. Standen (Ed.)),, 2nd ed., Interscience, New York, 1971, p. 889.
56. C. M. Hansen, Skaud. Tidskr. Faerg. Lack, **17**, 69 (1971).
57. H. G. Harris, J. M. Prausnitz, Ind. Eng. Chem. Fundamentals, **8**, 180 (1969).
58. R. A. Hayes, J. Appl. Polym. Sci., **5**, 318 (1961).
59. J. H. Hildebrand, R. L. Scott, "The Solubility of Nonelectrolytes", (3rd ed.) Reinhold, NY 1959.
60. P. J. Hoftyzer, D. W. van Krevelen, Paper No. IIIs-15, presented at International Symposium on Macromolecules of IUPAC, Lyden, 1970.
61. K. L. Hoy, J. Paint Technol., **42**, 76 (1970).
62. I. J. Hughes, private communication to H. Burrell, "Polymer Handbook", 1st ed., J. Brandrup, E. H. Immergut (Eds.), Wiley, New York, 1966.
63. K. Ito, J. E. Guillet, Macromolecules, **12**, 1163 (1979).
64. A. Jayasri, M. Yaseen, J. Oil Col. Chem. Assoc., **63**, 61 (1980).
65. D. M. Koenhen, C. A. Smolders, J. Appl. Polym. Sci., **19**, 1163 (1975).
66. M. Lautout, M. Magat, Z. Physik. Chem. (Frankfurt), **16**, 292 (1958).
67. D. D. Lawson, Appl. Energy, **6**, 241 (1980).
68. L.-H. Lee, J. Paint Technol., **42**, 365 (1970).
69. E. P. Lieberman, Off. Dig., Fed Soc. Paint Technol., **349** (444), 30 (1962).
70. J. E. G. Lipson, J. E. Guillet, J. Polym. Sci. Polym. Phys. Ed., **19**, 1199 (1981).
71. J. E. G. Lipson, J. E. Guillet, J. Coatings Technol., **54** (684), 90 (1982).
72. M. Magat, J. Chem. Phys., **46**, 344 (1949).
73. L. Mandelkern, private communcation to H. Burrell, "Polymer Handbook", 1st ed., J. Brandrup, E. H. Immergut (Eds.), Wiley, New York, 1966.
74. D. Mangaraj, S. K. Bhatnagar, S. B. Rath, Makromol. Chem., **67**, 75 (1963).
75. D. Mangaraj, S. Patra, S. B. Rath, Makromol. Chem., **67**, 84 (1963).
76. D. Mangaraj, Makromol. Chem., **65**. 29 (1963).
77. D. Mangaraj, S. Patra, S. Rashid, Makromol. Chem., **65**, 39 (1963).
78. D. Mangaraj, S. Patra, P. C. Roy, S. K. Bhatnagar, Makromol. Chem., **84**, 225 (1965).
79. H. Mark, A. V. Tobolsky, "Physical Chemistry of High Polymers", Interscience, New York, 1950, p. 263.
80. T. Matsuo, Preprints ACE/JSC, **20**, 895 (1979).
81. T. Matsuura, P. Blais, S. Sourirajan, J. Appl. Polym. Sci., **20**, 1515 (1976).
82. A. McLellan, G. Pimentel, The Hydrogen Bond, Freeman, San Francisco, 1960.
83. A. S. Michaels, ASTM Technical Publications, 340, 1963, p. 3.
84. W. Merk, R. N. Lichtenthaler, J. M Prausnitz, J. Phys Chem., **84**, 1694 (1980).
85. R. C. Nelson, R. W. Hemwall, G. D. Edwards, J. Paint Technol., **42** (550), 636 (1970).
86. R. C. Nelson, V. F. Figurelli, J. G. Walsham, G. D. Edwards, J. Paint Technol., **42** (550), 644 (1970).
87. A. Noshay, C. C. Price, J. Polym. Sci., **54**, 533 (1961).
88. O. Olabisi, R. Simha, J. Appl. Polym. Sci., **21**, 149 (1977).
89. O. Olabisi, L. M. Robeson, M. T. Shaw, "Polymer–Polymer Miscibility", Academic Press, NY, 1979.
90. D. Patterson, Macromolecules, **4**, 30 (1971).
91. D. Patterson, Y. B. Tewari, H. P. Schreiber, J. E. Guillet, Macromolecules, **4**, 356 (1971).

92. E. Perez, M. A. Gomez, A. Bello, J.G. Fatou, J. Appl. Polym. Sci., **27**, 3721 (1982).
93. S. Piccarolo, G. Titomanlio, Ind. Eng. Chem. Proc. Des. Dev., **22**, 146 (1982).
94. J. M. Prausnitz, "Molecular Thermodynamics of Fluid Phase Equilibria", Prentice-Hall, Englewood Clifffs, NJ, 1969, p. 269.
95. J. Polak, Coll. Czech. Chem. Commun., **31**, 1483 (1966).
96. A. E. Theineck, K. F. Lin, J. Paint Technol., **40**, 611, (1968).
97. R. B. Richards, Trans. Faraday Soc., **42**, 10 (1946).
98. Z. Ribgi, Polymer, **19**, 1229 (1978).
99. W. A. Ruff, C. J. Glover, A. T. Watson, AIChE Jr., **32** (12), 1948 (1986).
100. W. A. Ruff, C. J. Glover, A. T. Watson, W. R. Lau, J. C. Holste, AIChE Jr., **32** (12), 1954 (1986).
101. G. Scatchard, Chem. Rev., **44**, 7 (1949).
102. B. Schneier, Polym. Lett., **10**, 245 (1972).
103. R. L. Scott, M. Magat, J. Chem. Phys., **13**, p. 172, p. 178 (1945).
104. R. L. Scott, M. Magat, J. Polym Sci., **4**, 555 (1949).
105. J. H. Sewell, R. A. E. Technical Report No. 66185, June, 1966.
106. J. H. Sewell. J. Appl. Polymer. Sci., **17**, 1741 (1973).
107. M. T. Shaw, J. Appl. Polym. Sci., **18**, 449 (1974).
108. Shell Chemicals, Solubility Parameters, 2nd Ed., Tech. Bull., ICS(X)/78/1, 1978.
109. Shell Chemicals, Solvent Systems Design, Tech. Bull. ICS (X)/78/2, 1987.
110. A. G. Shvarts, Kolloid Zh., **18**, 755 (1956).
111. K. S. Siow, S. H. Goh, K. S. Yap, Polym. Materials Sci. Eng., **51**, 523 (1984).
112. P. A. Small, J. Appl. Chem., **3**, 71 (1953).
113. C. Smidsrod, J. E. Guillet, Macromolecules, **2**, 272 (1968).
114. H. M. Spurlin, J. Polym. Sci., **3**, 714 (1948).
115. W. H. Stockmayer, M. Fixman, J. Polym. Sci. C, **1**, 137 (1963).
116. K. W. Suh, D. H. Clarke, J. Polym. Sci. A-1, **5**, 1671 (1967).
117. R. S. Tillaev, M. Khasaukhanova, S. A. Tashmukhamedov, Kk. U. Usmanov, J. Polym. Sci. C, **39**, 107 (1972).
118. A. V. Tobolsky, "Properties and Structure of Polymers", Wiley, NY, 1960, p. 64, p. 66.
119. Union Carbide Corp., private communication to H. Burrell, "Polymer Handbook", 1st ed., J. Brandrup, E. H. Immergut (Eds.), Wiley, New York 1966.
120. J. W. van Dyk, H. L. Frisch, D. T. Wu, Ind. Eng. Chem. Prod. Res. Dev., **24** (3), 473 (1985).
121. D. W. van Krevelen, Fuel, **44**, 229 (1965).
122. D. W. van Krevelen, P. J. Hoftyzer, J. Appl. Polym. Sci., **11**, 2189 (1967).
123. D. W. van Krevelen, P. H. Hoftyzer, Properties of Polymers. Correlations with Chemical Structure, Elsevier, NY, 1972, Chs. 6 and 8.
124. F. Vocks, J. Polym. Sci. A, **2**, 5319 (1964).
125. H. Watanabe, T. Miyauchi, J. Chem. Eng. Japan, **6** (2), 109 (1973).
126. K. M. Watson, Ind. Eng. Chem., **23**, 360 (1931); **35**, 398 (1948).
127. S. Wu, J. Phys. Chem., **27**, 3332 (1968).
128. J. A. Yanko, J. Polym. Sci., **22**, 153 (1956).
129. T. Yamaguchi, S-I. Nakao, S. Kimura, Ind Eng. Chem. Res., **32**, 848 (1993).
130. E. T. Zellers, J. Appl. Polym. Sci., **50**, 513 (1993).
131. E. T. Zellers, G-Z. Zhang, J. Appl. Polym. Sci., **50**, 531 (1993).
132. E. T. Zellers, D. H. Anna, R. Sulewski, X. Wei, J. Appl. Polym. Sci., **62** (12), 2069 (1996).
133. E. T. Zellers, D. H. Anna, R. Sulewski, X. Wei, J. Appl. Polym. Sci., **62** (12), 2081 (1996).
134. D. G. Peiffer, J. Appl. Polym. Sci., **25** (3), 369 (1980).
135. V. Bellenger, E. Morel, J. Verdu, J. Appl. Polym. Sci., **37** (9), 2563 (1989).
136. H. Jiadong, W. Ruofend, L. Lu, F. Peirong, J. Appl. Polym. Sci., **38** (7), 1211 (1989).
137. L. A. Errede, J. Phys. Chem. **94** (1), 466 (1990).
138. S. Arichi, S. Himuro, Polymer, **30** (4), 686 (1989).
139. M. M. Coleman, C. J. Serman, D. E. Bhagwagar, P. C. Painter, Polymer, **31** (7), 1187 (1990).
140. H. L. Lee, P. Luner, J. Wood Chem. Technol., **11** (2), 247 (1991).
141. B. Bixamaiah, R. P. Raghunath, E. V. Sundaram, J. Appl. Polym. Sci., **44** (8), 1415 (1992).
142. U. Siemann, Europ. Polymer J., **28** (3), 293 (1992).
143. S. M. Aharoni, J. Appl. Polym. Sci., **45** (5), 813 (1992).
144. M. Roth, J. Polym. Sci., Phys. Ed., **28**(13), 2715 (1990).
145. A. J. Ashworth, G. J. Price, Macromolecules, **19**, 362 (1986).
146. G. J. Price, J. E. Guillet, J. Solution Chem., **16**, 605 (1987).
147. A. Bottino, G. Capannelli, S. Munari, A. Turturro, J. Polym. Sci., Polym. Phys., **26** (4), 785 (1988).
148. A. Chapiro, Z. Mankowski, N. Schmitt, J. Polym. Sci., Polym. Chem. Ed., **20**, 1791 (1982).
149. C. M. Hansen, J. Paint Technol., **42**, 660 (1970).
150. W. Liu, B. Breault, J. Brisson, J. Polym. Sci., Part B: Polym. Physics, **33** (4), 619 (1995).
151. Y. Izumi, Y. Miyake, Polym. J., **3**, 647 (1972).
152. J. B. Miller, J. H. Walton, C. M. Roland, Macromolecules, **26** (21), 5602 (1993).
153. Y. Yagi, H. Inomata, S. Saito, Macromolecules, **25** (11), 2997 (1992).
154. H. Ahmad, J. Macromol. Sci. A: Chem., **17**, 585 (1982).
155. M. M. Coleman, J. Hu, Y. Park, P. C. Painter, Polymer, **29**, 1659 (1988).
156. E. Morales, J. L. Acosta, Polymer J., **28** (2), 127 (1996).
157. R. Mieczkowski, Eur. Polym. J., **27** (4/5), 377 (1991).
158. A. M. Farooque, D. D. Deshpande, Eur. Polym. J., **28** (12), 1597 (1992).
159. S. R. Gooda, M. B. Huglin, Eur. Polym. J., **29** (2/3), 365 (1993).
160. S. M. Low, S. Y. Lee, S. H. Goh, Eur. Polym. J., **29** (8), 1075 (1993).

161. A. Belfkira, J.-P. Montheard, J. Appl. Polym. Sci., **51** (11), 1849 (1994).
162. J. Zhang, C. Zheng, Y. Yang, Polymeric Mat. Sci. Eng. (China) [Gaofenzi Cailiao Kexue Yu Gongcheng], **10** (5), 50 (1994).
163. D. W. van Krevelen, "Properties of Polymers", 3rd ed., Elsevier, New York, 1990, Ch. 7.
164. J. Bicerano, "Prediction of Polymer Properties", 2nd ed., Marcel Dekker, New York, 1996, Ch. 5.
165. D. Porter, "Group Interaction Modeling of Polymer Properties", Marcel Dekker, New York, 1995.
166. M. B. Huglin, D. J. Pass, J. Appl. Polym. Sci., **12**, 473 (1968).
167. L. A. Errede, Macromolecules, **19**, 1522 (1986).
168. Z. Liron, S. Cohen, J. Pharm. Sci., **72**, 499 (1983).
169. C. Marco, A. Bello, J. G. Faton, J. Garza, Makromolek. Chem., **187**, 177 (1986).
170. I. C. Sanchez, P. A. Rodgers, Pure Appl. Chem., **62**, 2107 (1990).
171. P. A. Rodgers, I. C. Sanchez, J. Polym. Sci., Part B: Polym. Phys., **31** (3), 273 (1993).
172. D. G. Peiffer, J. Appl. Polym. Sci., **25**, 369 (1980).
173. J. L. Perkins, A. D. Tippit, Am. Ind. Hyg. Assoc. J., **46**, 455 (1985).
174. J. L. Perkins, M. C. Ridge, A. B. Holcombe, M. K. Wang, W. E. Nonidez, Am. Ind. Hyg. Assoc. J., **47**, 803 (1986).
175. R. Alamo, J. G. Fatou, A. Bello, Polym. J., **15** (7), 491 (1983).
176. M. Galin, Polymer, **24** (7), 865 (1983).
177. L. S. Huppenthal, U. Danuta, J. Nowacki, Polimery, **27**(11), 427 (1982).
178. K. L. Hoy, "Tables of Solubility Parameters", Solvent and Coatings Materials Research and Development Department, Union Carbide Corporation, 1985.
179. K. L. Hoy, J. Coated Fabrics, **19**, 53 (1989).

Optically Active Polymers

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By definition the optical activity, $[M]_\lambda$, of polymers and their low molecular weight analogs is given by

$$[M]_\lambda = [\alpha]_\lambda \times (\text{mean residue weight})/100$$

A. ABBREVIATIONS USED

AcAc	Acetylacetonate
AIBN	Azobisisobutyronitrile
BPO	Benzoyl peroxide
DBP	Dibenzoyl peroxide
DCA	Dichloroacetic acid
DCM	Dichloromethane
DDB	2,3-Dimethoxy-1,4-bis(dimethylamino)butane
DMAc	Dimethylacetamide
DMF	Dimethylformamide
DMSO	Dimethylsulfoxide
DPEDA-Li	*N,N'*-diphenylethylenediamine monolithium amide
EDC	Ethylene dichloride
M	Monomer
MC	Model compound
NCA	*N*-Carboxylic anhydride
P	Polymer
RT	Room temperature
TMEDA	*N,N,N',N'*-tetramethylethylenediamine
TCA	Trichloroacetic acid
TFA	Trifluoroacetic acid
TFEL	Trifluoroethanol
THF	Tetrahydrofuran

B. OPTICALLY ACTIVE POLYMERS FROM CHIRAL MONOMERS

TABLE 1. MAIN-CHAIN ACYCLIC CARBON POLYMERS

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	T (°C)	Refs.
1.1. POLY(ALKENES)									
Poly[(*R*)(−)3,7-dimethyl-1-octene]									
$Al(i\text{-}C_4H_9)_3/TiCl_4$				−20 to −120		14.4 (MC)	Toluene	25	4,7
$Al(i\text{-}C_4H_9)_3/TiCl_3$				−35.3 to −144		14.4 (MC)	Toluene	25	
$LiAlH_4/TiCl_4$/monomer	Benzene	30–40	−78.1	−111	6.25	8.88 (MC)	Carbon tetrachloride	25	
Poly[(*S*)(+)5-methyl-1-heptene]									
$Al(i\text{-}C_4H_9)_3/TiCl_3$	Isooctane	RT		27.4–68.1		11.7 (MC)	Benzene		2–6
Poly[(*S*)(+)4-methyl-1-hexene]									
$Al(i\text{-}C_4H_9)_3/TiCl_4$	Isooctane	RT		205–288		21.3 (MC)	Benzene (MC); toluene (P)	25	2–6,9
$Al(i\text{-}C_4H_9)_3/TiCl_3$	Isooctane	RT		149–279		21.3(MC)	Benzene (P)	25	
Poly[(*S*)(+)6-methyl-1-octene]									
$Al(i\text{-}C_4H_9)_3/TiCl_3$	Isooctane	RT		16.0		13.3 (MC)	Benzene	25	4
				20.4		13.3 (MC)			5
Poly[(*S*)(+)3-methyl-1-pentene]									
$Al(i\text{-}C_4H_9)_3/TiCl_4$	–	20–25		29.4–158		−11.4 (MC)	Tetralin (P)	25 (P); 20 (MC)	2–6, 8,9
$Al(i\text{-}C_4H_9)_3/TiCl_3$	–	20–25		75.8–157		−11.4 (MC)	Tetralin (P)	25 (P); 20 (MC)	
				161					5
Poly[(*S*)5-phenyl-1-heptene]									
$Al(i\text{-}C_4H_9)_3/TiCl_3$	2,2,4-Trimethylpentane	25–60	26.6		4.80 (M)		Cyclohexane (P); neat (M)	25	118
Poly[(*R*)4-phenyl-1-hexene]									
$Al(i\text{-}C_4H_9)_3/TiCl_3$	2,2,4-Trimethylpentane	60	−61.6		−8.99 (M)		Chloroform (P); neat (M)	25	118
1.2. POLY(ACRYLICS) AND RELATED POLYMERS									
Poly(acryloyl-L-glutamic acid)									
AIBN	Dioxane	60	−21.0		−21.0 (MC)		Dioxane	25	12
Poly(L-bornyl acrylate)									
AIBN	–	65		−46.1		−56.4 (M)	Toluene	25	10
AIBN	Benzene	60		−47.2		−56.4 (M)	Toluene	25	
AIBN	Toluene	35		−47.4		−56.4 (M)	Toluene	25	
C_6H_5MgBr	Toluene	−70		−43.0		−56.4 (M)	Toluene	25	
UV benzoin	Toluene	−78		−47.2		−56.4 (M)	Toluene	25	
Poly(*d-sec*-butyl α-chloroacrylate)									
BPO	–	50–100		23.4		55.5 (M)	Chloroform (P); neat (M)	21 (P); 22 (M)	13
BPO	Dioxane	RT		17.9		42.2 (M)	Dioxane	25	14
Poly(*d-sec*-butyl α-bromoacrylate)									
BPO	Dioxane	RT		15.3		37.4 (M)	Dioxane	30	13
Poly(L-menthyl acrylate)									
AIBN	–	50	−79.0		−99.3 (M)		Benzene	22	11
AIBN	Benzene	60	−73.9		−88.3 (M)		Chloroform (P); neat (M)	25	215

Poly[(+)*N*-methyl-*N*-α-methylbenzyl-acrylamide]									
AIBN	–	60		278		504 (M)	Benzene	25	29
n-BuLi	Toluene	22		397		504 (M)	Benzene	25	
n-BuLi	Toluene	−50		366		504 (M)	Benzene	25	
C_6H_5MgBr	Toluene	−50		337		504 (M)	Benzene	25	
Poly[(−)*N*-propyl-*N*-α-methylbenzyl-acrylamide]									
AIBN	Toluene	70		−167		−444 (M)	Benzene	25	29
n-BuLi	Toluene	22		−318		−444 (M)	Benzene	25	

1.3. POLY(METHACRYLICS) AND RELATED POLYMERS

Poly(L-bornyl methacrylate)									
AIBN	Benzene	55		−70.8		−50.1 (M)	Toluene	20	10
Poly(1,2;5,6-diisopropylidene-D-glucofuranosyl-3-methacrylate)									
BPO	Benzene	80	−49.0		−32.2 (M)		EDC (P); ethanol (M)	25	23,24
AIBN	Benzene	40–50	−48.4		−28.8 (M)		Benzene	20	25–27
Poly(2,3;4,6-diisopropylidene-L-sorbofuranosyl-1-methacrylate)									
AIBN	Carbon tetrachloride	60	−84.5		−8.6 (M)		Benzene	19	26
Poly[(+)1,3-dimethylbutyl methacrylate]									
BPO	–	82–110	18.2	30.9	23.4 (MC)	43.5 (MC)	DCM	20	17,18
Poly[(+)(*S*)-diphenyl(1-methyl-pyrrolidin-2-yl)methyl methacrylate]									
DPEDA-Li/TMEDA	Toluene	−78	780 ($[\alpha]_{365}$)		−82.2 (M) ($[\alpha]_{365}$)		Benzene (M); methanol −0.01% CH_3SO_3H(P)	25	225
			−410 ($[\alpha]_{365}$)				Methanol−0.01%CH_3SO_3H after 1 h	25	
			779 ($[\alpha]_{365}$)				Methanol−13%CH_3SO_3H	25	
Poly(1-menthyl methacrylate)									
AIBN	Benzene	55		−193		−202 (M)	Benzene	25	16,11
BPO	–	40–85		−256.8		−202 (M)	Chloroform	20	
C_6H_5MgBr	Toluene	RT		−158.9		−202 (M)	Chloroform	20	
C_6H_5MgBr	Toluene	−75		−177.0		−202 (M)	Benzene	25	
γ-Ray	–	−75		−194.0		−202 (M)	Benzene	25	
BuLi	THF	−78	−96.9				Chloroform	25	205
Poly(methacryloyl-D-alanine)									
AIBN	Dioxane	60	42.0		48.9 (MC)		Dioxane	25	28
Poly(*N*-methacryloyl-L-alanine)									
AIBN	Dioxane		−42 ($[\alpha]_{546}$)		−41 (M) ($[\alpha]_{546}$)		Water	20	119
Poly(*N*-methacryloyl-L-asparagine)									
AIBN	Dioxane	60	−24.4 ($[\alpha]_{546}$)		−36 (MC) ($[\alpha]_{546}$)		Water	20	215
Poly(methacryloyl-L-glutamic acid)									
AIBN	Dioxane	60	−23.0		−26.0 (MC)		Dioxane	25	28
Poly(ε-*N*-methacryloyl-L-lysine]									
Potassium persulfate	Water	65	12.1		7 (M)		Water	20	222
Poly(1-α-methylbenzyl methacrylate)									
UV AIBN	–	35	−79.5	−151	−54.4 (M)	−103 (M)	Dioxane	25	19–21
UV benzoin	–	−65	−72.5	−138	−54.4 (M)	−103 (M)	Dioxane	25	
n-BuLi	Toluene	−60	−100	−190	−54.4 (M)	−103 (M)	Dioxane	25	
AIBN	Dioxane	35		−147		−78.8 (M)	Dioxane (P); neat (M)	25	
Poly(2-methylbutyl methacrylate)									
BPO	–	100		5.93		6.99 (M)	Chloroform (P); isooctane (M)	20	16,22
C_6H_5MgBr	Toluene	RT		6.84		6.99 (M)	Chloroform (P); isooctane (M)	20	
Poly[(*S*)4-methyl-2-*N*,*N*-dimethylaminopentyl methacrylate]									
AIBN	Benzene	60	2.2		−2.1 (M)		Benzene		120

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TABLE 1. *cont'd*

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	T (°C)	Refs.
Poly[(+)phenyl-2-pyridyl-*o*-tolylmethyl methacrylate]									
n-BuLi	THF	−78	395 −321		88.6 (M)		Benzene (M); THF (P) THF	−78 (P); 25 (M) 25 (P)	223
Diisopropyl peroxydicarbonate	Toluene	40	−617 ($[\alpha]_{365}$)		190.1 (M) ($[\alpha]_{365}$)		Chloroform(M); 90% chloroform-10% TFEL (P)	25	185
Poly[(+)phenyl-2-pyridyl-*m*-tolylmethyl methacrylate]									
DPEDA-Li/(−)DDB	Toluene	−78	−845 ($[\alpha]_{365}$) 1481 ($[\alpha]_{365}$)		14.0 (M) ($[\alpha]_{365}$)		Chloroform (P); benzene (M) Chloroform (after heating at 60°C)	25 25	224
DPEDA-Li/TMEDA	Toluene	−78	262 ($[\alpha]_{365}$) 1370 ($[\alpha]_{365}$)		14.0 (M) ($[\alpha]_{365}$)		Chloroform (P); benzene (M) Chloroform (after heating at 60°C)	25 25	224
1.4. POLY(VINYL ETHERS), POLY(VINYL KETONES), POLY(VINYL ESTERS)									
1.4.1. POLY(VINYL ETHERS)									
Poly[(−)(*S*)2,2′-bis(2-vinyloxyethoxy)-1,1′-binaphthyl]									
$SnCl_4$	DCM	0	−131.6		−47.6 (M)		Chloroform	25	121
$BF_3{\cdot}OEt_2$	Nitromethane	0	−131.9		−47.6 (M)		Chloroform	25	
Poly[(−)(*S*)2,2′-bis[2-(2-vinyloxyethoxy)ethoxy]-1,1′-binaphthyl]									
$SnCl_4$	DCM	0	−93.9		−37.7 (M)		Chloroform	25	121
$BF_3{\cdot}OEt_2$	DCM	0	−95.8		−37.7 (M)		Chloroform	25	
Poly[(−)(*S*)2,2′-bis[2-(2-vinyloxyethoxy)ethoxy]-3,3′-dimethyl-1,1′-binaphthyl]									
$SnCl_4$	DCM	0	44.5		−99.7 (M)		Chloroform	25 (P); 29 (M)	245
Poly(1-bornyloxyethylene)									
$BF_3{\cdot}OEt_2$				−142		−122.4 (M)	Benzene	25	31
Poly(1-cholesteryloxyethylene)				−93		−131.4 (M)			31
Poly(1,2 : 5,6-diisopropylidene-α-D-glucofuranosyl-3-oxyethylene)									
$BF_3{\cdot}OEt_2$	*n*-Hexane + DCM	−78		5.7		−77.2	EDC (P); ethanol (M)	25	36,24
Poly(1-menthyloxyethylene)									
C_4H_9MgBr	Toluene	80		−358		−121.9 (M)	Benzene (P); chloroform (M)	25	31
$BF_3{\cdot}OEt_2$	*n*-Hexane	−78		−396		−121.9 (M)	Benzene (P)	25	
$SnCl_4$	Petroleum ether	25		−373		−121.9 (M)	Benzene (P)	25	
$Mn{-}MoO_3{-}H_2SO_4$	*n*-Hexane	−10		−357		−118.8 (M)	Chloroform (M); benzene (P)	25	33
$Mn{-}MoO_3{-}H_2SO_4$	*n*-Hexane	−30		−353		−118.8 (M)	Benzene (P)	25	
$Al(i\text{-}OC_3H_7)_3/H_2SO_4$	*n*-Pentane	−15 to RT	−198		−72.2 (M)		Neat (M); benzene (P)	25	123
$BF_3{\cdot}OEt_2$	Toluene	−78	−220		−61.5 (M)		Toluene (M); benzene (P)	25 (P)	124
Poly(1-α-methylbenzyloxyethylene)									
$BF_3{\cdot}OEt_2$	Propane	−78		68.6		−71.9 (M)	Benzene	25	35
Poly[(*S*)2-methylbutyloxyethylene]									
$Al(i\text{-}C_4H_9)Cl_2$	Propylene + Toluene	−78		4.9		1.1 (MC) 7.6 (M)	Toluene (P); neat (M,MC)	25	34
$Al(i\text{-}C_4H_9)_3/H_2SO_4$	Ether	15–20		−5.9		1.1 (MC) 7.6 (M)	Toluene (P)	25	
$Al(i\text{-}OC_3H_7)_3/H_2SO_4$	Ethyl acetate	0–20		6.4		1.1 (MC) 7.6 (M)	Toluene (P)	25	

Poly[(*S*)1-methylpropyloxyethylene]									
$Al(i\text{-}C_4H_9)Cl_2$	Propylene + Toluene	−78		246		13.24 (M)	Toluene (P); neat (M)	25	32
$Al(i\text{-}C_4H_9)_3/H_2SO_4$	Diethyl ether	15–20		206		13.67 (M)	Toluene (P)	25	
$Al(i\text{-}OC_3H_7)_3/H_2SO_4$	*n*-Pentane	−15 to RT	285		13.8 (M)		Benzene (P); neat (M)	25	123
Poly[(*S*)2-phenylvinyl 2-methylbutyl ether]									
$BF_3{\cdot}OEt_2$	Toluene	−78	11.4		3 (MC)		DCM	25 (MC)	125
$SnCl_4$	Toluene	−78	12.0		3 (MC)		DCM	25 (MC)	
Poly[(*S*)2-phenylvinyl 3-methylpentyl ether]									
$BF_3{\cdot}OEt_2$	Toluene	−78	7.8		7.12 (MC)		DCM	25 (MC)	125
$SnCl_4$	Toluene	−78	7.8		7.12 (MC)		DCM	25 (MC)	
1.4.2. POLY(VINYL KETONES)									
Poly[(*S*)2-methylbutyl vinyl ketone]									
Spontaneous	–	20–25		−10.0		11.5 (MC)	Chloroform (P); neat (MC)	25	30
$LiAlH_4$	Toluene	−16 to 50		−43.0		11.5 (MC)	Chloroform (P)	25	
Poly[(*S*)3-methylpentyl vinyl ketone]									
Spontaneous	–	20–25		15.6		15.2 (MC)	Chloroform (P); neat (MC)	25	30
$LiAlH_4$	Toluene	0		11.7		15.2 (MC)	Chloroform (P)	25	
Poly[(*S*)1-methylpropyl vinyl ketone]									
$LiAlH_4$	Toluene	0		−118		33.4 (MC)	Chloroform (P); neat (MC)	25	30
Spontaneous	–	RT		−42.5		33.4 (MC)	Chloroform (P)	25	
AIBN	Toluene	60		−39.1	44.08 (M)		EDC	25	126
$LiAlH_4$	Toluene	−15		−139.0	44.08 (M)		EDC	25	
1.4.3. POLY(VINYL ESTERS)									
Poly(*d-sec*-butyl *p*-vinyl-benzoate)									
BPO	–	40	22.9		24.0 (M)		Benzene	50	40
Poly(vinyl l-2-phenylbutyrate)									
BPO	Dioxane	100	−29.1		−20.4 (M)		Dioxane	25	13

1.5. POLY(STYRENES), POLY(CARBAZOLES)

Poly(4-*sec*-butoxymethylstyrene)									
BPO	Dioxane	55	10.15		12.07 (M)		Dioxane	25	41
Poly[(*S*)3-*sec*-butyl-9-vinylcarbazole]									
AIBN	Benzene	60	−2.5		3.5 (M)		DCM (P); acetone (M)	25	127
$EtAlCl_2$	Toluene	−78	−6.2		3.5 (M)		DCM (P)	25	
$C_7H_7SbCl_6$	DCM	0	−3.8		3.5 (M)		DCM (P)	25	
Poly[(*S*)9-(2-methylbutyl)-2-vinylcarbazole]									
$EtAlCl_2$	DCM	−78	−13.4		−11.6 (M)		DCM (P); heptane (M)	25	127
n-BuLi	THF	25	−39.4		−11.6 (M)		DCM (P)	25	
AIBN	–	60	−21.4		−11.6 (M)		DCM (P)	25	
Poly[(*S*)9-(2-methylbutyl)-3-vinylcarbazole]									
AIBN	Benzene	60	−14.5		−7.5 (M)		DCM (P); heptane (M)	25	127
$C_7H_7SbCl_6$	DCM	0	−19.6		−7.5 (M)		DCM (P)	25	
–	–	200	−12.2		−7.5 (M)		DCM (P)	25	
Poly[2-(2-thio-3-methylpentyl)styrene]									
AIBN	–	76	9.57		13.97 (M)		Benzene (P); neat (M)	24.7	42
Poly[*p*-(*p*-tolylsulfinyl)styrene]									
AIBN	Benzene	70	−8.5 to −11.0		2.70 (M)		Chloroform	20	128

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TABLE 1. *cont'd*

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		Refs.
Catalyst or initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	T (°C)	
1.6. POLY(DIENES)									
Poly[1-(*d*-bornylammonium carboxylate)-1,3-butadiene]									
AIBN	Methanol	50	14.4		6.8 (M)		Methanol	25	177
Poly[1-((−)1-cyclohexylethylammonium carboxylate)-1,3-butadiene]									
AIBN	Methanol	50	−3.6		−3.7 (M)		Methanol	25	177
Poly[(3*E*)(5*S*)3,5-dimethyl-1,3-heptadiene]									
$K_2S_2O_8$	Aq. emulsion	55		44.0	42.8 (M)		Benzene	20 (P); 25 (M)	129
$Al(i\text{-}C_4H_9)_3/TiCl_4$	Benzene	25		33.2	42.8 (M)		Benzene	20 (P)	
Poly[(3*Z*)(5*S*)3,5-dimethyl-1,3-heptadiene]									
$K_2S_2O_8$	Aq. emulsion	55		41.4	41.6 (M)		Benzene	20 (P); 25 (M)	129
$Al(i\text{-}C_4H_9)_3/TiCl_4$	Benzene	25		31.7	41.6 (M)		Benzene	20 (P)	
Poly[1-((−)menthyloxycarbonyl)-1,3-pentadiene]									
AIBN	Benzene	50	−71.3		−84.5 (M)		THF (P); benzene (M)	25	190
Poly[1-((*S*)(−)α-phenylethylammonium carboxylate)-1,3-butadiene]									
AIBN	Methanol	50	−11.4		−8.0 (M)		Methanol	20	178
Poly[*trans*-(*R*)6-phenyl-1,3-heptadiene]									
$Al(i\text{-}C_4H_9)_3/TiCl_3$	*n*-Heptane	25		−46.87	−16.6 (M)		Benzene	25 (P); 15 (M)	130
$AlEt_3/VCl_3$	*n*-Heptane	25		−45.82	−16.6 (M)		Benzene	25 (P)	
n-BuLi	*n*-Heptane	25		−40.30	−16.6 (M)		Benzene	25 (P)	
Poly[*trans*-(*R*)5-phenyl-1,3-hexadiene]									
$Al(i\text{-}C_4H_9)_3/TiCl_3$	*n*-Heptane	25		−39.84	−18.74 (M)		Benzene	25 (P); 15 (M)	130
$AlEt_3/VCl_3$	*n*-Heptane	25		−42.86	−18.74 (M)		Benzene	25 (P)	
n-BuLi	*n*-Heptane	25		−37.54	−18.74 (M)		Benzene	25 (P)	
1.7. POLY(ISONITRILES)									
Poly[((*S*)(1-(acetoxymethyl)ethyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	−150 ($[\alpha]_{578}$)	−190 ($[M]_{578}$)	58.3 ($[\alpha]_{578}$)	74.0 (M) ($[M]_{578}$)	Chloroform	20–22	132
Poly[((*S*)(1-(acetoxymethyl)-2-methylpropyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	−111 ($[\alpha]_{578}$)	−172 ($[M]_{578}$)	46.8 ($[\alpha]_{578}$)	72.5 (M) ($[M]_{578}$)	Chloroform	20–22	132
Poly[((*R*)(2-acetoxy-1-phenylethyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	−82 ($[\alpha]_{578}$)	−156 ($[M]_{578}$)	−68.5 ($[\alpha]_{578}$)	−129 (M) ($[M]_{578}$)	Chloroform	20–22	132
Poly[(L-alanyl-(*O*-acetyl)L-serine methylester)methylene]									
$NiCl_2 \cdot 6H_2O$	Chloroform		205		−58.1 (M)		Chloroform–methanol (5 : 2) (P); chloroform (M)	20	231
Poly[(L-alanyl-(*O*-acetyl)D-serine methylester)methylene]									
$NiCl_2 \cdot 6H_2O$	Chloroform		−33		−12.0 (M)		Chloroform–methanol	20	231
Poly[(L-alanyl-(*N*(Im)-tosyl)L-histidine methylester)methylene]									
$NiCl_2 \cdot 6H_2O$	Chloroform/Methanol	RT	144 ($[\alpha]_{578}$)		40.6 ($[\alpha]_{578}$)		Chloroform	22	249
Poly[(D-alanyl-(*N*(Im)-tosyl)L-histidine methylester)methylene]									
$NiCl_2 \cdot 6H_2O$	Chloroform/Methanol	RT	−170 ($[\alpha]_{578}$)		18.6 ($[\alpha]_{578}$)		Chloroform	22	249
Poly[(L-alanyl-(*N*(Im)-tosyl) L-histidyl-(*O*-acetyl)L-serine methylester)methylene]									
$NiCl_2 \cdot 6H_2O$	Chloroform		−58.9		−13.5 (M)		Chloroform–methanol (5 : 2) (P); chloroform (M)	20	230

Poly[(L-alanyl-(*N*(Im)-tosyl)L-histidyl-(*O*-acetyl)D-serine methylester)methylene]									
$NiCl_2 \cdot 6H_2O$	Chloroform		−29.3		−2.1 (M)		Chloroform–methanol (5:2) (P); chloroform (M)	20	230
Poly[(D-alanyl-(*N*(Im)-tosyl)L-histidyl-(*O*-acetyl)L-serine methylester)methylene]									
$NiCl_2 \cdot 6H_2O$	Chloroform		−8.0		9.2 (M)		Chloroform–methanol (5:2) (P); chloroform (M)	20	230
Poly[((*S*)(1-benzylethyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	15 ($[\alpha]_{578}$)	22 ($[M]_{578}$)	48.7 ($[\alpha]_{578}$)	70.6 (M) ($[M]_{578}$)	Chloroform	20–22	132
Poly[(((*S*)1-(*tert*-butoxycarbonyl)-2-methylpropyl)imino)methylene]									
$NiCl_2$	–	45	32.5		29.5 (M)		Chloroform	20	236
Poly[((*S*)(1-carbethoxyethyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	–	RT	−280 ($[\alpha]_{578}$)	−355 ($[M]_{578}$)	16.7 ($[\alpha]_{578}$)	21.2 (M) ($[M]_{578}$)	Chloroform	20–22	132
Poly[(((1*S*,2*S*)1-carbomethoxy-2-methylbutyl)imino)methylene]									
$NiCl_2$	–	45	−32.2		35.0 (M)		Chloroform	20	236
Poly[((*S*)(−)α-cyclohexylethylimino)methylene]									
$NiCl_2 \cdot 6H_2O$	Neat	65	−295		27.4 (MC)		THF (P); hexane (MC)	24 (P); 20 (MC)	209
Poly[(((*S*)2,2-dimethyl-1,3-dioxolane-4-yl)methylimino)methylene]									
$NiCl_2 \cdot 6H_2O$	–	RT	68.5 ($[\alpha]_{546}$)		38.5 (M) ($[\alpha]_{546}$)		Dioxane	25	227
Poly[(((*R*)1,2-dimethylpropyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	16.7 ($[\alpha]_{578}$)		−24.2 (M) ($[\alpha]_{578}$)		Chloroform	22	236
Poly[((*R*)(2-(diphenyl phosphinyl)propyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	−338 ($[\alpha]_{578}$)	−910 ($[M]_{578}$)	13.1 ($[\alpha]_{578}$)	35.3 (M) ($[M]_{578}$)	Chloroform	20–22	132
Poly[((*S*)(1-ethyl-2-propenyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	29 ($[\alpha]_{578}$)	27.6 ($[M]_{578}$)	95 ($[\alpha]_{578}$)	90.2 (M) ($[M]_{578}$)	Chloroform (P); neat (M)	20	132
Poly[((*S*)(1-ethyl-2-propynyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	−110 ($[\alpha]_{578}$)	−102 ($[M]_{578}$)	−12.8 ($[\alpha]_{578}$)	−11.9 (M) ($[M]_{578}$)	Chloroform (P); neat (M)	20–22	132
Poly[(((*S*)1-(isopropyloxycarbonyl)-2-methylpropyl)imino)methylene]									
$NiCl_2$	–	5	−24		32.7 (M)		Chloroform	20	236
Poly[((−)menthylimino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	65	−207		−106.4 (MC)		Hexane	25 (P); 20 (MC)	209
Poly[(((*S*)1-(methoxycarbonyl)-2-methylpropyl)imino)methylene]									
$NiCl_2$	–	RT	−110		5.5 (M)		Chloroform	20	236
Poly[((*R*)3-methyl-2-butyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	−16.7 ($[\alpha]_{578}$)	−16.2 ($[M]_{578}$)	−24.2 ($[\alpha]_{578}$)	−23.5 (M) ($[M]_{578}$)	Chloroform	22	131
Poly[((*R*)4-methyl-2-pentyl)imino)methylene]									
$NiCl_2 \cdot 6H_2O$		RT	62 ($[\alpha]_{578}$)	68.9 ($[M]_{578}$)	−61.9 ($[\alpha]_{578}$)	−68.8 (M) ($[M]_{578}$)	Chloroform	22	131
Poly[((*R*)2-octylimino)methylene]									
$NiCl_2 \cdot 6H_2O$	Methanol	RT	34 ($[\alpha]_{578}$)	47.3 ($[M]_{578}$)	−47.6 ($[\alpha]_{578}$)	−66.3 (M) ($[M]_{578}$)	Chloroform	22	131
Poly[(*d*-α-phenylethylimino)methylene]									
DBP-O_2	Heptane	60	365		37.2 (M)		Toluene	27	39
Poly[(l-α-phenylethylimino)methylene]									
H_2SO_4-O_2	Heptane	50	−382		−35.8 (M)	−179.7 (M)	Toluene	27	39
$NiCl_2 \cdot 6H_2O$	–	0–5	−350 ($[\alpha]_{578}$)	−458 ($[M]_{578}$)	−40.9	−55.0 (M) ($[M]_{578}$)	Chloroform (P); methanol (M)	20	132

1.8. POLY(ALKYNES)

Poly[(*S*)3,4-dimethyl-1-pentyne]									
$Al(i\text{-}C_4H_9)_3/Fe(AcAc)_3$	*n*-Heptane	20–35		470		42.2 (MC)	*n*-Heptane (P); neat (MC)	25	133
Poly[*p*-(L-(−)-menthoxycarbonyl)phenylacetylene]									
[Rh(norbornadiene)Cl]$_2$	Triethylamine	RT	−605		−68.8 (M)		Chloroform	20	197
Poly[*p*-((+)-2-methylbutoxy)phenylacetylene]									
[Rh(norbornadiene)Cl]$_2$	Triethylamine	RT	19.2		5.90 (M)		Chloroform	20	197

TABLE 1. *cont'd*

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	T (°C)	Refs.
Poly[(*S*)4-methyl-1-hexyne]									
Al(*i*-C_4H_9)$_3$/Fe(AcAc)$_3$	Isooctane	20–30	−22.9		−2.97 (MC)		Isooctane	25	1
Al(*i*-C_4H_9)$_3$/Fe(AcAc)$_3$	*n*-Heptane	20–35		−10.6		−3.0 (MC)	*n*-Heptane (P); neat (MC)	25	133
Poly[(*S*)5-methyl-1-heptyne]									
Al(*i*-C_4H_9)$_3$/Fe(AcAc)$_3$	*n*-Heptane	20–35		−146		12.2 (MC)	*n*-Heptane (P); neat (MC)	25	133
Poly[(*S*)6-methyl-1-octyne]									
Al(*i*-C_4H_9)$_3$/Fe(AcAc)$_3$	*n*-Heptane	20–35		−34		13.7 (MC)	*n*-Heptane (P); neat (MC)	25	133
Poly[(*S*)3-methyl-1-pentyne]									
Al(*i*-C_4H_9)$_3$/Fe(AcAc)$_3$	*n*-Heptane	20–35		498		32.1 (MC)	*n*-Heptane (P); neat (MC)	25	133
1.9. POLY(MALEIMIDES) AND RELATED POLYMERS									
Poly(*N*-bornyl maleimide)									
BPO	THF	50	8.4		−9.6 (M)		THF (P); ethanol (M)	25 (P)	43
AIBN	THF	50	10.1		−9.6 (M)		THF (P)	25 (P)	
n-BuLi	THF	50	−9.1		−9.6 (M)		THF (P)	25 (P)	
Poly[*N*-(cholesteroxycarbonylmethyl)maleimide]									
AIBN	THF	70	−25.9		−26.8 (MC)		THF	RT	203
n-BuLi	THF	0	−37.1		−26.8 (MC)		THF	RT	
Poly(*N*-maleoyl-L-alanine)									
AIBN	THF	60	−42.1		−49.9 (MC)		THF	25	201
Poly(*N*-maleoyl-L-phenylalanine)									
AIBN	THF	60	−144.1		−138.3 (MC)		THF	25	201
Poly[*N*-maleoyl-L-phenylalanine ethyl ester]									
AIBN	Dioxane	60	−120.0		−136.3 (MC)		THF	25	252
BuLi	THF	0	−222.0		−136.3 (MC)		THF	25	
Poly[*N*-((*N'*-((*R*)-α-methylbenzyl)aminocarbonyl)methyl)maleimide]									
AIBN	DCM	70	59.9		115.5 (MC)		THF	25	200
Poly[*N*-(*N'*-((*R*)-α-methylbenzyl)aminocarbonyl-*n*-pentyl) maleimide]									
AIBN	Dioxane	60	50.1		56.6 (MC)		THF	25	183
Poly[*N*-(4-*N'*-((*R*)-α-methylbenzyl)aminocarbonylphenyl)maleimide]									
AIBN	THF	50	−141.4		−26.9 (MC)		THF	25	192
n-BuLi	THF	0	−78.0		−26.9 (MC)		THF	25	
Poly[*N*-(4-*N'*-((*S*)-α-methylbenzyl)aminocarbonylphenyl)maleimide]									
AIBN	THF	50	163.4		28.9 (MC)		THF	25	206
Poly[*N*-(*N'*-((*R*)-α-methylbenzyl)aminocarbonylundecyl)maleimide]									
AIBN	Dioxane	60	44.6		44.8 (MC)		THF	25	183
Poly(*N*-α-methylbenzyl citraconimide)									
AIBN	Benzene	50	56.7		55.3 (MC)		THF	25	233
Poly(*N*-α-methylbenzyl maleimide)									
AIBN	THF	50	11.1–13.0		57.9 (M)		THF	25	137
Poly[*N*-(2-phenyl-1-phenyloxycarbonyl)ethyl maleimide]									
AIBN	Benzene	60	−69.7		−106.8 (MC)		THF	25	187
n-BuLi	Toluene	0	−65.2		−106.8 (MC)		THF	25	

1.10. OTHER COMPOUNDS

Poly[(2,2′-bis(hexyloxy)-1,1′-binaphthyl-6,6′-ylene)-1,4-phenylenevinylene-1,4-phenylenevinylene-1,4-phenylene]									
$Pd(PPh_3)_4$	THF/K_2CO_3(aq.)		−351		35.6 (M)		THF (P); DMSO (M)		166
Poly(bis-2-methylbutyl itaconate)									
–	–	RT	4.92		4.97 (M)				15
BPO	–	40–85	4.75		5.57 (M)		Heptane	20	16
Poly[N^3-(1,2-diethoxycarbonylethyl)ureidoethylene]									
AIBN	Benzene	60	24.3	63.1	43.5 (M)	112.5 (M)	Dioxane	25	38
			31.7	82.0	43.5 (M)	112.5 (M)	Benzene	25	
Poly[N^3-(1,3-diethoxycarbonylpropyl)ureidoethylene]									
AIBN	Benzene	60	7.83	21.3	21.6 (M)	58.9 (M)	Benzene	25	38
Poly[(*R*)3,7-dimethyloctanal divinylacetal]									
AIBN	–	80	3.18	7.19	2.48 (MC)		Chloroform	25	72
$BF_3{\cdot}OEt_2$	–	−78	−6.78	−14.32	2.48 (MC)		Chloroform	25	
Poly[N^3-(1-ethoxycarbonyl-3-methylbutyl)ureidoethylene]									
AIBN	Benzene	60	0.5	1.1	20.3 (M)	46.3 (M)	Benzene	25	38
Poly(ethylidene-1-propenylidene) (from (*R*)2,3-pentadiene)									
π-allyl-Ni-iodide		−20	−232.10		−47.39 (M)		Carbon tetrachloride (P); neat (M)	25	243
Poly(L-menthyloxycarbonylaminoethylene)									
AIBN	–	95	−60.4	−136.0	−79.9 (M)	−179.7 (M)	Benzene	25	38
AIBN	Benzene	60	−60.4	−136.0	−79.9 (M)	−179.7 (M)	Benzene	25	
γ-ray	Benzene	22	−58.5	−131.8	−79.9 (M)	−179.7 (M)	Benzene	25	
tert-BuOOH	SO_2	−25	−66.2		−79.9 (M)	−179.7 (M)	Benzene	25	
Poly[*N*-(4-*N*′-((*R*)-α-methylbenzyl)aminocarbonylphenyl)itaconimide]									
AIBN	THF	60	52.1		−23.8 (MC)		THF	25	191
Poly[(−)3-(β-styryloxy)menthane]									
$BF_3{\cdot}OEt_2$	DCM	−72	−114		−15.9 (M)		THF	25 (P)	122
Poly[*N*-(*p*-(*p*-tolylsulfinyl)phenyl)-2,5-dioxo-pyrrolidin-3,4-diyl]									
BPO	THF	60	−22.0		−25.9 (M)		Chloroform	20	135
Poly(*m*-tolyl vinyl sulfoxide)									
n-BuLi/(−)-sparteine	Toluene	−78	311		486 (M)				246
Poly(*p*-tolyl vinyl sulfoxide)									
n-BuMgBr	THF	30	194		443 (M)		Acetone	23	134
Poly[(*S*)*N*-vinyl-3-methyl-2-pyrrolidone]									
AIBN	Water	95	28		69.3 (M)		Methanol (P); benzene (M)	25	136

TABLE 2. MAIN-CHAIN ACYCLIC HETROATOM POLYMERS

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or Initiator	Solvent	*T* (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	*T* (°C)	Refs.
2.1. POLY(OXIDES)									
Poly[(*R*)(+)citronellal]									
$Al(i\text{-}C_4H_9)_3$	Ether	−78	−82.7 to −90.1	−127 to −139		11.2 (MC)	Chloroform	25	44
n-BuLi	*n*-Hexane	−78	−91.1	−140		11.2 (MC)	Chloroform	25	
$BF_3{\cdot}OEt_2$	Ether	−78	−89.5	−138		11.2 (MC)	Chloroform	25	

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TABLE 2. *cont'd*

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or Initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	T (°C)	Refs.
Poly(D-oxy-2,3-dimethylethylene)									
$Al(i\text{-}C_4H_9)_3/H_2O$	*n*-Heptane	−78	0		58.8 (M)		Benzene or chloroform (P); neat (M)	25	47
Poly[(*R*)(+)oxy-5-methoxy-3-methyl-hexylidene]									
$ZnEt_2$	*n*-Hexane	−75	36.7	52.8	2.47 (M)	8.24 (MC)	Chloroform	25	45
Poly[(*R*)oxy-2-methylbutylidene]									
$ZnEt_2$	*n*-Hexane	−75	36.7		30.7 (M)		Chloroform	25	45
Poly[(+)oxy-1-methyl-2-carbomethoxyethylene]									
$BF_3{\cdot}OEt_3/AlEt_3$	–	25	−36.9		24.6 (M)		DCM (P); neat (M)	25	48
Poly[oxymethylene-(*R*,*R*)-dibenzo-19-crown-6]									
KOH	DMSO	RT	5.0 ($[\alpha]_{435}$)		1.22 (M) ($[\alpha]_{435}$)		Chloroform	25 (P); 24 (M)	208
Poly[oxymethylene-(*S*,*S*)-dibenzo-19-crown-6]									
KOH	DMSO	RT	−6.0 ($[\alpha]_{435}$)		−1.07 (M) ($[\alpha]_{435}$)		Chloroform	25 (P); 28 (M)	208
Poly(L-oxypropylene)									
KOH	–	RT	−16 ± 5		15 (M)		Benzene (P); ether (M)	20 (P); 21 (M)	46
			25 ± 5		15 (M)		Chloroform (P)	20 (P)	
$FeCl_3$-monomer	Ether	RT	−20 ± 5		15 (M)		Benzene (P)	20 (P)	
			25 ± 5		15 (M)		Chloroform (P)	20 (P)	
$ZnEt_2/CH_3OH$	Benzene	33	24.9		11.55 (M)		Cyclohexane (P); neat (M)	25	138
			−32.0		11.55 (M)		Benzene (P)	25	
Ferric acetate hydroxide	Heptane	70		−9.50	15.0 (M)		Benzene (P); ethyl ether (M)	25 (P); 27 (M)	139
				7.36	15.0 (M)		Dibutyl ether (P)	25 (P)	
2.2. POLY(ESTERS)									
Poly[L(−)oxycarbonylethylidene]									
$ZnCl_2$	Heptane	175–195	−153.3 ($[\alpha]_{578}$)		−275.8 (M) ($[\alpha]_{578}$)		Chloroform	25	51
Pyridine	Pyridine	20	−140.0		−19.2 (M)		Chloroform	20	175
Poly(L-oxycarbonylisobutylidene)									
ZnO	–	180–190	−20		−50 (MC)		Cyclohexane		52
ZnO	–	180–190	−67		−256 (M)		Chloroform	20 (P); 25 (M)	140
Poly(L-α-oxycarbonylisopentylidene)									
ZnO	–	175	−50		−226 (M)		DCM (P); chloroform (M)	20 (P); 25 (M)	140
Poly[(*R*)(+)oxycarbonyl-3-methylhexamethylene]									
$Sb_2O_3/Zn(CH_3COO)_2$	–	215	8.55		0.94 (M)		TFEL	25	49
Poly[(*R*)(−)oxycarbonyl-3-methylhexamethylene]									
$Sb_2O_3/Zn(CH_3COO)_2$	–	215	3.36		−0.96 (M)		Benzene	25	49
Poly[(*R*)(−)oxycarbonyl-2-methylpentamethylene] (Poly[(*R*)(−)-β-methylcaprolactone])									
$Al(i\text{-}C_4H_9)_3/H_2O$	Benzene	50	32.57 ($[\alpha]_{450}$)		29.78 (MC) ($[\alpha]_{350}$)		TFEL	25	53
Poly[(*R*)oxycarbonyl-3-methylpentamethylene]									
$Al(i\text{-}C_4H_9)_3/H_2O$	Benzene	50	26.95 ($[\alpha]_{450}$)		51.77 (M)		TFEL (P); chloroform (M)	25	53
Poly[(*R*)oxycarbonyl-4-methylpentamethylene]									
$Al(i\text{-}C_4H_9)_3/H_2O$	Benzene	50	8.78 ($[\alpha]_{450}$)		−15.91 (MC) ($[\alpha]_{350}$)		TFEL (P,MC); chloroform (M)	25	53
					−36.11 (M)				
Poly[oxycarbonyl-1,4-phenylene-oxyterephthaloyloxy-1,4-phenylene-carbonyloxy-(*S*)1-methylethylene]									
–	EDC	Reflux		42		22.3 (M)	Fuming sulfuric acid	25	237

Poly[oxycarbonyl-1,4-phenylene-oxyterephthaloyloxy-1,4-phenylene-carbonyloxy-1,2-bis(2,2-dimethyl-1,3-dioxolane-4-yl)ethylene]								
–	DCM	85		228	4.0 (M)	Dioxane (P); methanol (M)	25	226
Poly(D-oxycarbonylpropylene)								
$AlEt_3$	Ether	RT	19 ($[\alpha]_{300}$)			Chloroform		50
Natural polymer			44 ($[\alpha]_{300}$)			Chloroform		
Poly(oxycarbonyl-1,2,3-trimethoxytetramethylene) (Poly(2,3,4-trimethyl-1-arabonolactone))								
CH_3COCl/HCl	–	RT	39 ($[\alpha]_{546}$)		203.5 (M) ($[\alpha]_{546}$)	Benzene	21	54
Poly[oxyisophthaloyloxy-(2-acetamide-1-phenyl)trimethylene]								
–	Chloroform	60	– 30.8		24.9 (M)	DMF (P); methanol (M)	25	167
Poly[oxyterephthaloyloxy-(1*R*,3*R*)-1,3-dimethyltrimethylene]								
–	EDC	82	– 295		– 136 (MC)	Chloroform	25	202
Poly[oxyterephthaloyloxy-(1-methyl-3-oxo)trimethylene-oxyethyleneoxy-(3-methyl-1-oxo)trimethylene-oxycarbonyl]								
–	DCM	Reflux	– 62		– 41.8 (M)	Chloroform	25	189

2.3. POLY(SULFIDES), POLY(THIOESTERS)

Poly(*tert*-butyl thiirane)								
tert-BuOK	DMSO	RT	268		– 39.0 (M)	Benzene (P); neat (M)	25	142
$ZnEt_2$	Toluene	RT	231–240		– 39.0 (M)	Benzene (P); neat (M)	25	
Poly[thio-1-(*N*-(*S*)*sec*-butyl-*N*-methylaminomethyl)ethylene]								
$ZnEt_2$/methanol	–	RT	26.1			Benzene	25	232
Poly[(*S*)thio(carbonyl)-1-(*p*-chlorobenzenesulfonamide)ethylene]								
$PhCH_2SH$	DMF	25	– 85.52		– 11.08 (M)	DMF	24 (M)	143
Poly[(*R*)thio(carbonyl)-1-methyl-1-ethylethylene]								
Tetrabutylammonium versatate	–	100	151.7		56.9 (M)	Benzene	23 (P); 26 (M)	141
Poly[(*S*)thio(carbonyl)-1-(*p*-methyloxybenzenesulfonamide)ethylene]								
$PhCH_2SH$	DMF	25	– 92.44		25.25 (M)	DMF	20 (M)	143
Poly[(*R*)(–)thiocarbonyl-3-methylpentamethylene]								
n-BuLi	–	RT	– 5.39		– 3.21 (M)	Dioxane	25	57
Poly[(*R*)(+)thiocarbonyl-2-methylpentamethylene]								
n-BuLi	–	190	15.66		9.03 (M)	Dioxane	25	57
			18.62			THF	25	
Poly[(*S*)thio(carbonyl)-1-(*p*-nitrobenzenesulfonamide)ethylene]								
$PhCH_2SH$	DMF	25	– 100.25		– 46.01 (M)	DMF	28 (M)	143
Poly[thio(carbonyl)-1-tosylaminoethylene]								
$PhCH_2SH$	DMF	25	– 90.6		19.87 (M)	DMF		143
–	–	112	– 80.6		19.87 (M)	DMF		
Poly[(–)thiopropylene]								
$BF_3 \cdot OEt_2$	DCM		– 62.8		– 32.6 (M)	DCM	22	55
			– 126.9		32.6 (M)	Chloroform	22	
KOH	–	15	– 189		– 35 (MC)	Chloroform	18 (P); 20 (MC)	56
			– 159		– 35 (MC)	Benzene (P)	18 (P)	

2.4. POLY(URETHANES), POLY(UREAS)

2.4.1. POLY(URETHANES)

Poly[oxycarbonylimino-(1-ethoxycarbonyl)pentamethyleneiminocarbonyloxyethylene]								
–	Neat	140–160	– 12			DMSO		174
Poly[oxycarbonylimino-(L)2-isobutylethylene-ureylene-(L)1-isobutylethylene)]								
Na_2CO_3	H_2O/THF	< 0	– 48.2			Methanol	20	63
Poly[oxycarbonylimino-(L)2-isobutylethylene-ureylene-(L)1-isopropylethylene)]								
Na_2CO_3	H_2O/THF	< 0	– 27.2			Methanol	20	63
Poly[oxycarbonylimino-(L)2-isobutylethylene-ureylene-(L)1-methylethylene)]								
Na_2CO_3	H_2O/THF	< 0	– 19.2			Methanol	20	63

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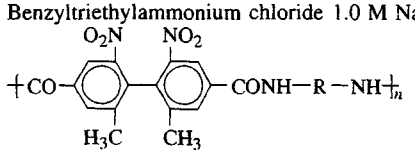
TABLE 2. *cont'd*

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or Initiator	**Solvent**	***T* (°C)**	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	**Solvent**	***T* (°C)**	**Refs.**
Poly[oxycarbonyliminomethylene-1,3-phenylenemethylene-ureylene-(1-(methoxymethyl)-2-phenyl)ethylene]									
Dibutyltin dilaurate	DMSO	80	20.48		15.20 (MC)		DMSO	25	168
Poly[oxycarbonyliminomethylene-1,3-(1,5,5-trimethyl)cyclohexylene-iminocarbonyloxy-2,3-bornanediyl]									
–	Methyl ethyl ketone	80	−7.4		−11.67 (MC)		DMF	30	216
Poly[oxycarbonylimino-(D)2-methylethylene-ureylene-(L)1-isobutylethylene)]									
Na_2CO_3	H_2O/THF	<0	−54.7				Methanol	20	63
Poly[oxycarbonylimino-(L)2-methylethylene-ureylene-(L)1-isobutylethylene)]									
Na_2CO_3	H_2O/THF	<0	−47.9				Methanol	20	63
Poly[oxycarbonylimino-(D)2-methylethylene-ureylene-(L)1-isopropylethylene)]									
Na_2CO_3	H_2O/THF	<0	−53.2				Methanol	20	63
Poly[oxycarbonylimino-(L)2-methylethylene-ureylene-(L)1-isopropylethylene)]									
Na_2CO_3	H_2O/THF	<0	−26.0				Methanol	20	63
Poly[oxycarbonylimino-(D)2-methylethylene-ureylene-(L)1-methylethylene)]									
Na_2CO_3	H_2O/THF	<0	−16.0				*m*-Cresol	20	63
Poly[oxycarbonylimino-(L)2-methylethylene-ureylene-(L)1-methylethylene)]									
Na_2CO_3	H_2O/THF	<0	−50.0				*m*-Cresol	20	63
Poly[oxycarbonylimino-1,4-phenylenemethylene-1,4-phenylene-ureylene-(1-(methoxymethyl)-2-phenyl)ethylene]									
Dibutyltin dilaurate	DMSO	80	−14.40		−8.72 (MC)		DMSO	25	168
Poly[oxycarbonylimino-2,4-toluyleneiminocarbonyloxy-(2-acetamide-1-phenyl)trimethylene]									
–	Chloroform	75	−20.6		−38.6 (MC)		DMF	25	167
Poly[oxycarbonylimino-2,4-toluyleneiminocarbonyloxy-2,3-bornanediyl]									
–	Methyl ethyl ketone	80	−27.6		−14.8 (MC)		DMF	30	216
2.4.2. POLY(UREAS)									
Poly[((*R*)5-ethyl-1,3-diazolidin-2-one-1,3-diyl)ethyleneoxyethylene]									
Methyl iodide	Benzonitrile	150	−19.4		39.4 (M)		Chloroform	24	250
Poly[((*R*)5-ethyl-1,3-diazolidin-2-one-1,3-diyl)methylene-1,2-phenylenemethylene]									
Methyl iodide	Benzonitrile	150	−28.0		32.1 (M)		Chloroform	24	250
Poly[((*R*)5-ethyl-1,3-diazolidin-2-one-1,3-diyl)tetramethylene]									
Methyl iodide	Benzonitrile	150	−54.0		38.8 (M)		Chloroform	24	250
Poly[((*S*)5-isopropyl-1,3-diazolidin-2-one-1,3-diyl)tetramethylene]									
Methyl iodide	Benzonitrile	150	−40.5		−34.4 (M)		Chloroform	24	250
Poly[((*S*)5-phenylmethyl-1,3-diazolidin-2-one-1,3-diyl)tetramethylene]									
Methyl iodide	Benzonitrile	150	−71.4		32.0 (M)		Chloroform	24	250
2.5. POLY(AMIDES)									
Poly[iminocarbonyl-1,2-(3,4-bis(2-hydroxyphenyl))cyclobutylenecarbonyliminohexamethylene]									
–	DMAc	80		−241		−295 (MC)	DMAc		235
Poly[iminocarbonyl-1,2-(3,4-bis(2-hydroxyphenyl))cyclobutylenecarbonylimino-1,4-phenylene]									
–	DMAc	80		−358		−456 (MC)	DMAc		235
Poly[iminocarbonyl-1,2-(3,4-bis(2-methoxyphenyl))cyclobutylenecarbonyliminohexamethylene]									
Benzyltriethylammonium chloride	1.0M NaOH/chloroform	RT	−70.3		−81.0 (MC)		*m*-Cresol	20	218
Poly[iminocarbonyl-1,2-(3,4-bis(2-methoxyphenyl))cyclobutylenecarbonylimino-1,4-phenylene]									
Benzyltriethylammonium chloride	1.0M NaOH/chloroform	RT	5.7		−32.7 (MC)		DMAc	20	218
Poly(*trans*-iminocarbonyl-1,2-cyclohexylene)									
Triethylamine	DMF	60	75.2–84.8		−60.0 (MC)		DCA (P); ethanol (M)	20	144

Poly[iminocarbonyl-(1*S*,2*S*)1,2-*trans*-cyclohexylene-carbonyliminoethylene]								
–	DMAc		56.5	19.4 (M)		DCA (P); acetone (M)	25 (P); 20 (M)	221
Poly[iminocarbonyl-(1*S*,2*S*)1,2-*trans*-cyclohexylene-carbonylimino-1,4-phenylene]								
–	DMAc		239	19.4 (M)		DCA (P); acetone (M)	25 (P); 20 (M)	221
Poly[imino-((2*S*,3*S*)-2,3-dimethoxytetramethylene)-imino-(1,4-dioxotetramethylene)]								
–	Chloroform	50	– 14.1	– 8.2 (M)		Formic acid (P); chloroform (M)	23	184
Poly[imino-(1,5-dioxohexamethylene)imino-(1*R*,2*R*)1,2-*trans*-cyclohexylene]								
–	DMAc		65.4	– 33.4 (M)		DCA (P); water (M)	25 (P); 20 (M)	221
Poly[iminoethylideneimino-(2,3-dimethoxy-1,4-dioxo)tetramethylene]								
–	Chloroform	RT	175.2	– 38.25 (M)		Formic acid (P); chloroform (M)	23	199
Poly[threo-imino-(1-oxo-2,3-dimethyltrimethylene)]								
K-Pyrrolidone	DMSO	25	– 25.0	– 50.6 (M)		TFA (P); neat (M)	25	62
Poly[(*S*)imino-(1-oxo-3-ethyltrimethylene)] (Poly[(*S*)β-aminovalerate])								
K-Pyrrolidone	DMSO	25	– 32.7	– 2.4 (M)		DCA (P); neat (M)	25	62
Poly[(–)imino-(1-oxo-3-methylhexamethylene)]								
H_2O	–	200	39.7	– 36.15 (M)		Cresol (P); water (M)	25	59,60
NaH	–	160	43.07	– 36.15 (M)		Cresol (P); water (M)	25	61
Poly[(+)imino-(1-oxo-4-methylhexamethylene)]								
NaH	–	150	19.24			TFEL	25	61
Poly[(+)imino-(1-oxo-5-methylhexamethylene)]								
NaH	–	194	25.15			TFEL	25	61
Poly[(+)imino-(1-oxo-3-methyl-6-isopropylhexamethylene)]								
K-C_6H_5COCl	–	150	9.5	– 53.6 (M)		Formic acid	19	58
Poly[(*S*)imino-(1-oxo-3-methyltrimethylene)] (Poly[(*S*)-β-aminobutyrate])								
K-Pyrrolidone	DMSO	25	– 9.3	– 8.3 (M)		TCA (P); neat (M)	25	62
			– 26.1	– 8.3 (M)		DCA (P)	25	
			– 33.5	– 8.3 (M)		TCA–chlorobenzene (P)	25	
Poly[(*R*)imino-(1-oxo-3-vinyltrimethylene)]								
K-Pyrrolidone	DMSO	25	– 19.5	25 (M)		DCA (P); neat (M)	25	62
Poly[iminoterephthaloylimino-(1*R*,2*R*)1,2-*trans*-cyclohexylene]								
	DMAc		– 211	– 33.4 (M)		DCA (P); water (M)	25 (P); 20 (M)	221
Poly[(+)iminoterephthaloylimino-2,8-naphthylene-8,2-naphthylene]								
			– 15 ($[\alpha]_{578}$)	66.6 (MC) ($[\alpha]_{578}$)		THF	21	71
Poly[methyliminocarbonyl-1,2-(3,4-bis(2-hydroxyphenyl))cyclobutylenecarbonyl-methylimino-hexamethylene]								
–	DMAc	80		– 437	– 474 (MC)	DMAc		235
Poly[*N*-(phenylethylidene)iminoethylene-*N*-(phenylethylidene)iminoterephthaloyl]								
			– 74 ($[\alpha]_{289}$)	– 294 ($[\alpha]_{289}$)			20	70
Poly[*N*-(phenylethylidene)iminohexamethylene-*N*-(phenylethylidene)iminodipicolinoyl]								
			– 139 ($[\alpha]_{289}$)	– 610 ($[\alpha]_{289}$)			20	70
Poly[1,4-piperazinediylcarbonyl-1,2-(3,4-bis(2-hydroxyphenyl))cyclobutylenecarbonyl]								
–	DMAc	80		116	– 391 (MC)	DMAc		235
Poly[1,4-piperazinediylcarbonyl-1,2-(3,4-bis(2-methoxyphenyl))cyclobutylenecarbonyl]								
Benzyltriethylammonium chloride 1.0 M NaOH/chloroform		RT	– 40.7	– 83.0 (MC)		DMAc	20	218



(–) R = 1,2-phenylene			– 4.94			5% LiCl–DMF	20	73
(+) R = 1,3-phenylene			– 0.50			5% LiCl–DMF	20	73
(+) R = 1,4-phenylene			6.53			5% LiCl–DMF	20	73

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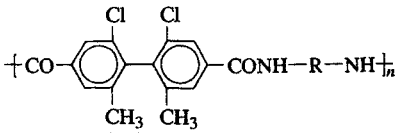
TABLE 2. *cont'd*

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or Initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	T (°C)	Refs.
(−) R = 1,2-pyrazolidene			5.4				TFEL	20	73
(−) R = 2,5-dimethyl-1,4-piperazinediyl			28.0				TFEL	20	73
									
(−) R = 1,2-phenylene			−19.6				5% LiCl–DMF	20	73
(−) R = 1,3-phenylene			−19.4				5% LiCl–DMF	20	73
( ) R = 1,4-phenylene			−18.2				5% LiCl–DMF	20	73
(−) R = 1,2-piperazinediyl			−6.7				TFEL	20	73
(−) R = 1,4-piperazinediyl			−10.0				TFEL	20	73
(−) R = 2,5-dimethyl-1,4-piperazinediyl			−6.7				TFEL	20	73
2.6. POLY(ISOCYANATES)									
Poly[((*R*)1-deuterio-1-hexyl)iminocarbonyl]									
NaCN	DMF	−58	−367		0.65 (M)		Chloroform (P); neat (M)	25	229
Poly[(((*S*)2,2-dimethyl-1,3-dioxolane-4-yl)methyl)iminocarbonyl]									
NaCN	DMF	−65	−311 ($[\alpha]_{546}$)		−21.1 (M) ($[\alpha]_{546}$)		Chloroform(P); ethylether (M)	25	228
Poly[(*S*)(+)*N*-(2-methylbutyl)iminocarbonyl]									
NaCN	DMF	−75	160	180.8	3.08 (M) −6 (MC)	−10.3 (MC)	Chloroform	25	64
Poly[*N*-(3-methyl-1-carbomethoxybutyl)iminocarbonyl]									
Methyllithium	DMF	−50	62				TFA	25	182
Poly[*d*-*N*-(phenylpropyl)iminocarbonyl]									
NaCN	DMF	RT	−468.0	−754.8	35 (M) 114.4 (MC)	44.8 (M)	Chloroform	25	65
2.7. POLY(IMINES)									
Poly(D-conidine)									
$BF_3{\cdot}OEt_2$	Ether	RT	−140.8		71.7 (M)		Chloroform		69
Poly(L-conidine)									
$BF_3{\cdot}OEt_2$	Ether	RT	140.8		−71.0 (M)		Chloroform		69
Poly[(*S*)decyliminoisobutylethylene]									
$BF_3{\cdot}OEt_2$	Neat	−78	−68.6		17.9 (M)		Benzene (P); ethanol (M)	19–21	173
Poly[(*R*)ethyliminoethylethylene]									
$BF_3{\cdot}OEt_2$	Neat	−78	155.13		−41.80 (M)		Dioxane (P); methanol (M)	25	234
Poly[imino-1-butyl-(*R*)2-ethylethylene]									
$BF_3{\cdot}OEt_2$	–	98–100	137.66		2.73 (M)		Benzene (P); ethanol (M)		145
Poly[imino-1-(*R*)2-diethylethylene]									
$BF_3{\cdot}OEt_2$	–	98–100	155.13		−20.33 (M)		Dioxane (P); ethanol (M)	25	145
Poly[(*R*)imino-2-ethylethylene]									
$BF_3{\cdot}OEt_2$	–	100	−127.4		17.6 (M)		Benzene (P); neat (M)	23 (P); 18 (M)	67

$AlCl_3$	–	100	– 108.9	17.6 (M)	Benzene (P)	16 (P)	
From (*R*)-2-amino-1-bromobutane			– 109.2		Benzene	16	
Poly[imino-(*R*)2-ethyl-1-pentylethylene]							
$BF_3 \cdot OEt_2$	–	98–100	132.39	4.16 (M)	Benzene (P); ethanol (M)		145
Poly(imino-*N*-formyl-isopropylethylene)							
Methyl triflate	Acetonitrile	80	199.3	– 130.5 (M)	Chloroform (P); ethanol (M)	25	146
$BF_3 \cdot OEt_2$	Acetonitrile	80	86.8	– 130.5 (M)	Chloroform (P)	25	
Poly(imino-*N*-formyl-propylene)							
Ethyl trifluoromethane-sulfonate	Acetonitrile	80	130	– 148 (M)	Methanol (P); ethanol (M)	25 (P); 30 (M)	148
Dimethyl sulfate	–	120	107	14.4 (M)	H_2O (P); neat (M)	20	149
Poly[(*S*)iminoisobutylethylene]							
$BF_3 \cdot OEt_2$	–	80–100	66.7	– 24.6 (M)	Benzene	20 (P); 32 (M)	68
Poly(iminoisopropylethylene)							
$BF_3 \cdot OEt_2$	–	80	82.8	– 27.1 (M)	Ethanol	24	146
Poly[imino-(*S*)*N*-1-methylbenzylethylene]							
$BF_3 \cdot OEt_2$	–	0–60	7.09	– 81.8 (M)	Benzene (P); neat (M)	20	147
			– 8.15	– 81.8 (M)	Chloroform (P)	20	
Poly(D-iminopropylene)							
$BF_3 \cdot OEt_2$	–	80	– 101.5	12.4 (M)		22	66
Poly(L-iminopropylene)							
$BF_3 \cdot OEt_2$	–	80	109.3	– 12.8 (M)		22	66
Poly[(*S*)methyliminoisobutylethylene]							
$BF_3 \cdot OEt_2$	Neat	– 78	– 163.5	33.7 (M)	Benzene (P); ethanol (M)	19–21	173

2.8. POLY(AMINO ACIDS)

Poly(*O*-acetyl-L-hydroxyproline)							
NCA	Pyridine		25 (Form I)		90% HCOOH	25	110
			– 175 (Form II)		90% HCOOH after 6 h	25	
Poly(*O*-acetyl-L-serine)							
NCA, triethylamine	Nitrobenzene		31 ($[\alpha]_{546}$)		DCA	25	114
			159 ($[\alpha]_{546}$)		10% DCA–90% EDC	25	
Poly(L-alanine)							
NCA, tributylamine	Benzene–dioxane			21 ($[m']_D$)	99% $CHCl_3$–1% DCA		81,84
				– 90 ($[m']_D$)	TFA		
Poly(D-α-amino-*n*-butyric acid)							
NCA, benzylamine	Nitrobenzene			– 19 ($[m']_D$)	90% $CHCl_3$–10% DCA		81,84
				107 ($[m']_D$)	TFA		
Poly(L-arginine·HCl)							
From poly(L-nitroarginine)			– 21.3		H_2O	20	85
Poly(α-L-aspartic acid)							
From poly(β-benzyl-L-aspartate)			6.2		*c* =31 N NaOH		87
(NCA, 120°C in vac.)							
			– 11.2		*c* =51 N NaOH		88
Poly(β-L-aspartic acid)							
Active ester method		140	– 14 ($[\alpha]_{546}$)		0.2 M NaCl, pH 2	25	89
			– 23 ($[\alpha]_{546}$)		0.1 M NaCl, pH 4	25	
Poly(β-benzyl-D-aspartate)							
NCA, NaOMe	Chloroform		174 ($[\alpha]_{546}$)		Chloroform	22	86
			19 ($[\alpha]_{546}$)		DCA	22	
Poly(β-benzyl-L-aspartate)							
NCA, NaOMe	Chloroform		– 168 ($[\alpha]_{546}$)		Chloroform	22	86
			– 18 ($[\alpha]_{546}$)		DCA	22	

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TABLE 2. *cont'd*

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or Initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	T (°C)	Refs.
Poly(γ-benzyl-D-glutamate)									
NCA, NaOMe	Chloroform		−18 ($[\alpha]_{546}$)				Chloroform	22	86
	Chloroform		17 ($[\alpha]_{546}$)				DCA	22	
Poly(γ-benzyl-L-glutamate)									
NCA			−30				Hydrazine	20	96
			−46				TFA	20	
NCA, NaOMe	Chloroform		14 ($[\alpha]_{546}$)				Chloroform	22	86,98
	Chloroform		−15 ($[\alpha]_{546}$)				DCA	22	86
NCA, NaOMe	Chloroform		15				Chloroform	25	97
	Chloroform		−14				DCA	25	
NCA, Ni-dl-2-methyl-butyrate-n-Bu_3P	Dioxane	30	−17.0		−17.1 (NCA)		DCA (P); ethylacetate (NCA)	20	151
			15.3		−17.1 (NCA)		Chloroform (P)	20	
NCA, benzylamine	Dioxane	30	−17.0		−17.1 (NCA)		DCA (P); ethylacetate (NCA)	20	
			16.0		−17.1 (NCA)		Chloroform (P)	20	
NCA, benzylamine	THF	30	−15.3		−17.6 (NCA)		DCA (P); ethylacetate (NCA)	25	152
Poly(1-benzyl-L-histidine)									
NCA, triethylamine	Dioxane		11 ($[\alpha]_{546}$)				Chloroform	25	101
Triethylamine			−208 ($[\alpha]_{546}$)				DCA–$CHCl_3$	25	
NCA, triethylamine	Dioxane		−21.1				Acetic acid	20	102
Poly(ε-*N*-carbobenzoxy-L-lysine)									
NCA, NaOMe	Dioxane		5 ($[\alpha]_{546}$)				Chloroform	25	84,103
			−25 ($[\alpha]_{546}$)				50% $CHCl_3$–50% DCA	25	
Poly(*S*-carbobenzoxymethyl-L-cysteine)									
NCA, NaOMe	Nitrobenzene		−128 ($[\alpha]_{546}$)				0.2 M NaCl, pH 7	25	90
			−125 ($[\alpha]_{546}$)				0.2 M NaCl, pH 5	25	
			−33 ($[\alpha]_{546}$)				0.2 M NaCl, pH 3	25	
Poly(3,4-dehydro-L-proline)									
NCA	Pyridine		−500				99% formic acid	25	109
			−1250				TFA	25	
Poly(L-glutamic acid)									
From poly(γ-benzyl-L-glutamate) (NCA, aliphatic amine, DMF)			3				Dioxane–H_2O, 0.2 M NaCl, pH 4.7	22	84,92
			−77				Dioxane–H_2O, 0.2 M NaCl, pH 6.5	22	
			7				2-Chloroethanol–H_2O, 0.1 M RbCl, pH 5		84,93
			−75				2-Chloroethanol–H_2O, 0.1 M RbCl, pH 7.5		
From poly(γ-benzyl-L-glutamate) (NCA, NaOMe)			−6				0.2 M NaCl, pH 4.5	20	84,94
			−102				0.2 M NaCl, pH 8.0	20	
Poly(γ-D-glutamic acid)									
Active ester method			23.6				H_2O	23	95
Poly(γ-L-glutamic acid)									
Active ester method			−23.6				H_2O	23	95

Poly(L-histidine)						
From poly(1-benzyl-L-histidine)						
(NCA, triethylamine, dioxane)		−65 ($[\alpha]_{546}$)		0.2 M NaCl, pH 2.4	25	101
		−210 ($[\alpha]_{546}$)		0.2 M NaCl, pH 5.85	25	
		−67 ($[\alpha]_{546}$)		0.2 M NaCl, pH 6.00	25	
		28 ($[\alpha]_{546}$)		DCA	25	
		−20		Acetic acid	22	102
Poly(δ-hydroxy-*O*-acetyl-L-α-aminovaleric acid)						
NCA, *triethylamine*	*Nitrobenzene*	10.1 ($[\alpha]_{546}$)		DMF	25	83
		−26.5 ($[\alpha]_{546}$)		TFEL	25	
		−31.4 ($[\alpha]_{546}$)		DCA	25	
Poly(L-hydroxyproline)						
From poly(*O*-acetyl hydroxy-L-proline)		−400		H_2O	25	110
Poly(L-leucine)						
NCA, tributylamine	Benzene		−12 ($[m']_D$)	Benzene		81,84
			−111 ($[m']_D$)	TFA		81
Poly(L-lysine)						
From poly(ε-*N*-carbobenzoxy-L-lysine)		−144		H_2O, pH 7		105
		−35		H_2O, pH 12		
		−51		H_2O, pH 10.10	21.4	
		−77		H_2O, pH 9.98	23.2	
		−108		H_2O, pH 9.72	25.6	
Poly(L-lysine hydrochloride)						
From poly(ε-*N*-carbobenzoxy-L-lysine)		−78.8		6 N HCl	25	106
Poly(L-methionine)						
NCA, NaOMe	Nitrobenzene	24.4 ($[\alpha]_{546}$)		90% $CHCl_3$–10% DCA	25	107
		16.8 ($[\alpha]_{546}$)		DCA	25	
		−115.3 ($[\alpha]_{546}$)		TFA	25	
		22 ($[\alpha]_{546}$)		Chloroform		108
		−107 ($[\alpha]_{546}$)		80% TFA – 20% DCA		
Poly(L-methionine-*S*-carboxymethylthetin)						
From poly(L-methionine)		−44 ($[\alpha]_{546}$)		HCl, H_2O (pH 1.1–2.0)	25	107
Poly(L-methionine-*S*-methylsulfonium bromide)						
From poly(L-methionine)		−74.3 ($[\alpha]_{546}$)		H_2O	24	107
		−38.6($[\alpha]_{546}$)		10M LiBr	24	
Poly(*N*-methyl-L-alanine)						
NCA, benzylamine	Dioxane	−3.04		TFEL	24	82
Poly(γ-methyl-L-glutamate)						
NCA		−31.3		DCA	25	99
		−76.1		TFA	25	
		10		DMF	25	100
Poly(L-nitroarginine)						
NCA	DMF	−16.6		DMF	30	85
Poly(γ-phthalimidomethyl-L-glutamate)						
NCA, triethylamine		12.5		DMF	25	91
Poly(L-pipecolic acid)						
NCA, diethylamine	Benzene or dioxane	−325 (Form I)		Acetic acid	25	109
		−50 (Form II)		Acetic acid after 2 h	25	

TABLE 2. *cont'd*

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or Initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	T (°C)	Refs.
Poly(L-proline)									
			50 (Form I)				Acetic acid		111
			−540 (Form II)				Acetic acid		
NCA, pyridine			48 (Form I)				10% Pyridine 90% H_2O	25	112
			−420 (Form II)				10% Pyridine after 50 h	25	
			33 (Form I)				H_2O		113
			−540 (Form II)				0.1 M KCl		
Poly(β-proline)									
Triethylamine	DMF	30	−10 to −11		7.6 (MC)		Chloroform	25	150
Poly(L-serine)									
From poly(*O*-acetyl-L-serine)			−22 $[\alpha]_{546}$				H_2O	25	114
			−7.3 $[\alpha]_{546}$				10 M LiBr	25	
			−25.2 $[\alpha]_{546}$				8 M urea	25	
			−9.3 $[\alpha]_{546}$				Hydrazine	25	
			−9.7 $[\alpha]_{546}$				DCA	25	
Poly(*O*-*p*-tolylsulfonyloxy-L-proline)									
NCA	Pyridine		0 (Form I)				Acetic acid	25	110
			−120 (Form II)				Acetic acid after 3 h		
Poly(ε-*N*-trifluoroacetyl-L-lysine)									
NCA, diethylamine	DMF		−6.8 ($[\alpha]_{546}$)				Methanol	20	104
			−36.8 ($[\alpha]_{546}$)				DCA	20	
Poly(L-tryptophan)									
NCA, diethylamine	Dioxane		218 ($[\alpha]_{546}$)				DMF	20	115
			343 ($[\alpha]_{546}$)				40% DMF − 60% DCA	20	
Poly(L-tyrosine)									
From poly(*O*-carbobenzoxy-L-tyrosine) (NCA, 110°C in vac.)									
			184				DMF		116
			73				20% DMF–80% DCA		
			150				0.15 M NaCl, pH 10.85		
			28				0.15 M NaCl, pH 12.27		
			8.7				1 N NaOH	20	117

TABLE 3. POLY(SACCHARIDES)

Polymer	Polymerization system		Polymer (P) values		Monomer (M) or model compound (MC) values		Optical activity measured		
Catalyst or initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	$[\alpha]_D$	$[M]_D$	Solvent	T (°C)	Refs.
Poly[(1→6)-2,5-anhydro-3,4-di-*O*-decyl-D-glucitol]									
$BF_3{\cdot}OEt_2$	DCM	0	16.0		6.8 (M)		Chloroform	22	211
tert-BuOK	THF	60	23.7		6.8 (M)		Chloroform	22	
Poly[(1→6)-2,5-anhydro-3,4-di-*O*-ethyl-D-glucitol]									
$BF_3{\cdot}OEt_2$	DCM	−30	32.7		54.1 (MC); −5.2 (M)		Chloroform	22	210

Poly[(1→6)-2,5-anhydro-3,4-di-*O*-methyl-D-glucitol]							
tert-BuOK	Toluene	60	84.5 ($[\alpha]_{546}$)		Chloroform	23	212
KOH	Toluene	60	75.5 ($[\alpha]_{546}$)		Chloroform	23	
$BF_3 \cdot OEt_2$ (from D-mannitol)	DCM	0	41.3 ($[\alpha]_{546}$)	−12.3 (M) ($[\alpha]_{546}$)	Chloroform	22	213
$BF_3 \cdot OEt_2$ (from L-iditol)	DCM	0	43.1 ($[\alpha]_{546}$)	−9.8 (M) ($[\alpha]_{546}$)	Chloroform	22	
Poly(2,5-anhydro-3,4-di-*O*-methyl-D-mannitol)							
$BF_3 \cdot OEt_2$	DCM	0	50.2 ($[\alpha]_{546}$)	−9.6 (M) ($[\alpha]_{546}$); 44.1 (MC)	Chloroform	22 (P, M); 23 (MC)	214
Poly[(1→6)-2,5-anhydro-3,4-di-*O*-pentyl-D-glucitol]							
$BF_3 \cdot OEt_2$	DCM	0	25.8	9.3 (M)	Chloroform	22	211
tert-BuOK	THF	60	32.8	9.3 (M)	Chloroform	22	
Poly[α-(1→6′)-anhydro-D-galactopyranose]							
From 2,3,4-tri-*O*-benzyl derivative			207.1		10% LiOH − 0.5% borate	25	75
Poly(1,6-anhydro-β-D-glucopyranose)							
Chloroacetic acid	–	21–127	91		H_2O	22	77
Poly(α-1,6-anhydro-D-mannopyranose)							
From 2,3,4-tri-*O*-benzyl derivative			122.8		1% DMSO	30	80
Poly(α-1,6-anhydro-2,3,4-tri-*O*-acetyl-β-D-mannopyranose)							
PF_5	DCM	0	96		DMSO	30	80
Poly[α(1→6′)-anhydro-2,3,4-tri-*O*-benzyl-β-D-galactopyranose]							
PF_5	DCM	−78	105.1	−46.1 (M)	Chloroform	25	75
Poly[α-1,6-anhydro-2,3,4-tri-*O*-benzyl-β-D-mannopyranose]							
PF_5	DCM	−78	57.5	−32 (M)	Chloroform	30 (P); 25 (M)	80
Poly[1,6-anhydro-2,3,4-tri-*O*-methyl-β-D-glycopyranose]							
$BF_3 \cdot OEt_2$	Toluene	RT	199	−63.7 (M)	Chloroform	25 (P); 20 (M)	78
$BF_3 \cdot OEt_2$	Toluene	RT	204	−63.7 (M)	Chloroform	25 (P); 20 (M)	
Poly(D-galactose)							
HCl	DMSO	40	79.1		H_2O	20	74
Poly(D-glucose)							
HCl	DMSO	40	90.3		H_2O	20	74
Polyphosphate formamide		50–60	16				76
Poly(D-mannose)							
HCl	DMSO	40	102.9		H_2O	20	74
Poly(2,3,6-tri-*O*-methyl-D-glucose)							
$HBr–HCl–P_2O_5$	DMSO	RT	136.6		Chloroform	20	79
Poly(D-xylose)							
HCl	DMSO	40	96.1		H_2O	20	74

C. OPTICALLY ACTIVE POLYMERS FROM ACHIRAL MONOMERS

TABLE 4. MAIN-CHAIN ACYCLIC CARBON POLYMERS

Polymer	Polymerization system		Polymer (P) Values			Optical activity measured		
Catalyst or initiator	Solvent	*T* (°C)	$[\alpha]_D$	$[M]_D$	Origin of optical activity	Solvent	*T* (°C)	Refs.
4.1. POLY(ACRYLICS) AND RELATED POLYMERS								
Poly(*N*-acryloylcarbazole)								
Fluorenyllithium/(−)-sparteine	Toluene	−97	57.4		Helix-sense asymmetry	Chloroform	25	188
Poly(*N*-acryloyliminodibenzyl)								
Fluorenyllithium/(−)-sparteine	Toluene	−97	−0.5		Helix-sense asymmetry	Chloroform	25	188

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TABLE 4. *cont'd*

Polymer	Polymerization system		Polymer (P) Values			Optical activity measured		
Catalyst or initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	Origin of optical activity	Solvent	T (°C)	Refs.
Poly(*N*-acryloylphenothiazine)								
Fluorenyllithium/(−)-sparteine	Toluene	−97	8.2		Helix-sense asymmetry	Chloroform	25	188
Poly(*N*,*N*-diphenylacrylamide)								
BuLi/(−)-sparteine	Toluene	−96	−44.2 to −50.9		Helix-sense asymmetry	THF	25	153
Fluorenyllithium/(−)-sparteine	Toluene	−97	−101			Chloroform	25	188
Poly[*N*,*N*-di(2-pyridyl)-acrylamide]								
Fluorenyllithium/(−)-sparteine	Toluene	−97	−23.7		Helix-sense asymmetry	Chloroform/DCA (22/1)	25	188
Poly[*N*-methyl-*N*-triphenylmethyl-acrylamide]								
n-BuLi/(−)-sparteine	Toluene	−78	−2.4		Helix-sense asymmetry	THF	25	188
Poly(1-phenyldibenzosuberyl acrylate)								
DPEDA-Li/(+)-DBB	Toluene	−40	−129		Helix-sense asymmetry	Chloroform	25	219
Poly(*N*-phenyl-*N*-1-naphthylacrylamide)								
BuLi/(−)-sparteine	Toluene	−78	8.7		Helix-sense asymmetry	Toluene	25	153,188
Poly(*N*-phenyl-*N*-2-naphthylacrylamide)								
BuLi/(−)-sparteine	Toluene	−78	81.7		Helix-sense asymmetry	Toluene	25	153,188
Poly[*N*-phenyl-*N*-(1-phenylethyl)-acrylamide]								
Fluorenyllithium/(−)-sparteine	Toluene	−97	−0.3		Helix-sense asymmetry	THF	25	188
Poly(triphenylmethyl acrylate)								
DPEDA-Li/(+)1-(2-pyrrolidinylmethyl)pyrrolidine	Toluene	−78	102 ($[\alpha]_{365}$)		Helix-sense asymmetry	Chloroform	25	186
4.2. POLY(METHACRYLICS) AND RELATED POLYMERS								
Poly(benzyl methacrylate)								
(*S*)2,3 : 4,5-bis[1,2-(3-methylnaphtho)]-1,6,9,12,15,18-hexaoxacycloeicosa-2,4-diene-*KO*-*tert*-Bu	95% Toluene–5% THF	−78	350 ($[\alpha]_{578}$)		Miscellaneous	THF	25	164
Poly(*tert*-butyl methacrylate)								
(*S*)2,3 : 4,5-bis[1,2-(3-methylnaphtho)]-1,6,9,12,15,18-hexaoxacycloeicosa-2,4-diene-*KO*-*tert*-Bu	95% Toluene–5% THF	−78	117 ($[\alpha]_{578}$)		Miscellaneous	THF	25	164
Poly(*m*-chlorophenyldiphenylmethyl methacrylate)								
DPEDA-Li/(+)-DDB	Toluene	−78	314		Helix-sense asymmetry	THF	25	220
Poly(cyclohexyldiphenylmethyl methacrylate)								
DPEDA-Li/(+)1-(2-pyrrolidinylmethyl)pyrrolidine	Toluene	−78	758 ($[\alpha]_{365}$)		Helix-sense asymmetry	Chloroform	25	240
			770 ($[\alpha]_{365}$)			Chloroform	60	
			−480 ($[\alpha]_{365}$)			Chloroform after 1 h	60	
Poly(cyclopropyldiphenylmethyl methacrylate)								
DPEDA-Li/(+)-DDB	Toluene	−78	−44 ($[\alpha]_{365}$)		Helix-sense asymmetry	Chloroform	25	240
Poly(diphenyl-*m*-fluorophenylmethyl methacrylate)								
DPEDA-Li/(+)-DDB	Toluene	−78	297		Helix-sense asymmetry	THF	25	220
Poly(diphenyl-2-pyridylmethyl methacrylate)								
DPEDA-Li/(−)-DDB	Toluene	−78	−1403 ($[\alpha]_{365}$)		Helix-sense asymmetry	90% Chloroform – 10% TFEL	25	155
Poly(diphenyl-4-pyridylmethyl methacrylate)								
BuLi/(−)-sparteine	Toluene	−78	−40 ($[\alpha]_{546}$)		Helix-sense asymmetry	90% Chloroform – 10% TFEL	25	154
Poly(methyl methacrylate)								
From poly(methacrylic acid) (potassium persulfate/dextrin, water, 0°C)			6.4 ($[\alpha]_{350}$)		Induced asymmetry	Benzene	25	162

From poly(methacrylic acid) (potassium persulfate/chitosan, water, 30°C)							
			16.6	Induced asymmetry	Chloroform	21	163
(*S*)2,3 : 4,5-bis[1,2-(3-methylnaphtho)]-1,6,9,12,15,18-hexaoxacycloeicosa-2,4-diene-*KO-tert*-Bu							
	95% Toluene–5% THF	−78	250 ([α]$_{578}$)	Miscellaneous	THF	25	164
From poly(2,3-*O*-isopropylidene-1,4-di-*O*-methacryloyl-L-threitol) (AIBN, toluene, 60°C)							
			−4.3 ([α]$_{405}$)	Induced asymmetry	Chloroform	20	244
Poly(1-phenyldibenzosuberyl methacrylate)							
Diisopropyl peroxydicarbonate	(+)Menthol/toluene	50	−200 ([α]$_{365}$)	Helix-sense asymmetry	THF	25	196
Diisopropyl peroxydicarbonate	(−)Menthol/toluene	50	180 ([α]$_{365}$)		THF	25	196
Diisopropyl peroxydicarbonate/ (+)neomenthanethiol	Toluene	40	−140 ([α]$_{365}$)		THF	25	253
(−)Dimethyl peroxydicarbonate	Toluene	50	40 ([α]$_{365}$)		THF	25	253
DPEDA-Li/(−)-sparteine	Toluene	−78	1670 ([α]$_{365}$)		Chloroform	25	196
Poly[1-(2-pyridyl)dibenzosuberyl methacrylate]							
DPEDA-Li/(−)-sparteine	Toluene	−78	1290 ([α]$_{365}$)	Helix-sense asymmetry	Chloroform	25	251
Poly(triphenylmethyl methacrylate)							
Li(R)-*N*-(1-phenylethyl)anilide	THF	−78	−104	Helix-sense asymmetry	Toluene	20	156
BuLi/(−)-sparteine	Toluene	−78	363		Toluene	20	
(*S*)-6-(2-dimethylaminoethyl)-1,11-dimethyl-6,7-dihydro-5H-dibenz[c,e]azepine							
	Toluene	−78	373		THF	25	157
Poly[tris(*m*-chlorophenyl)methyl methacrylate]							
DPEDA-Li/(−)-DDB	Toluene	−78	−251	Helix-sense asymmetry	THF	25	220
Poly[tris(*m*-tolyl)methyl methacrylate]							
DPEDA-Li/(−)-DDB	Toluene	−78	−234 ([α]$_{546}$)	Helix-sense asymmetry	Chloroform	25	220

4.3. POLY(VINYL ETHERS)

Poly(benzofuran)							
$AlCl_3$/phenylalanine	Toluene	−75	24–56.7	Induced asymmetry	Benzene		169
$AlCl_3$/Et_3SnCl/(−)menthoxy-triethyltin	Toluene	−75	73.8		Benzene	18	242
Poly(α-naphthofuran)							
2$AlCl_3$/(−)-β-phenylalanine	Toluene	−75	41	Induced asymmetry	Chloroform	20	170
Poly(β-naphthofuran)							
2$AlCl_3$/(−)-β-phenylalanine	Toluene	−75	−145	Induced asymmetry	Chloroform	20	170
Poly(2-phenyl-1,3-dioxane-4,6-diyl) (Poly(benzaldehyde divinyl acetal))							
$ZnCl_2$·OEt_2/(+)-10-camphor-sulfonic acid	Toluene	0	−17.1 ([α]$_{435}$)	Induced asymmetry	Chloroform	20	198
$ZnCl_2$·OEt_2/(−)-10-camphor-sulfonic acid	Toluene	0	13.7 ([α]$_{435}$)		Chloroform	20	198

4.4. POLY(STYRENES)

Poly(4-iodostyrene)							
From copolymer of 3,4-*O*-cyclohexylidene-D-mannitol-1,2 : 5,6-bis-*O*-[(4-vinylphenyl)boronate] and styrene (AIBN, toluene, 60–70°C)							
			0 to −4.5 ([α]$_{436}$)	Induced asymmetry	THF	30	179,180
Polystyrene							
From copolymer of 3,4-*O*-cyclohexylidene-D-mannitol-1,2 : 5,6-bis-*O*-[(4-vinylphenyl)boronate] and styrene (AIBN, toluene, 60–70°C)							
			−0.5 to −3.5 ([α]$_{365}$)	Induced asymmetry	THF	30	179,180

4.5. POLY(DIENES)

Poly(3-carboxy-1-butenylene)							
From poly[1-(*d*-bornylammonium carboxylate)-1,3-butadiene] (AIBN, methanol, 50°C. See Section B. Table 1, 1.6)							
			0.2	Induced asymmetry	Methanol	25	177

TABLE 4. *cont'd*

Polymer	Polymerization system		Polymer (P) Values			Optical activity measured		
Catalyst or initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	Origin of optical activity	Solvent	T (°C)	Refs.
From poly[1-(*S*(−)α-phenylethylammonium carboxylate)-1,3-butadiene] (AIBN, methanol, 50°C. See Section B. Table 1, 1.6)								
			−4.8	−4.7		Methanol	20	178
Poly(3-carboxy-4-methyl-1-butenylene) (Poly(sorbic acid))								
Potassium persulfate/chitosan	Water	50	2.32		Induced asymmetry	Methanol	20	161
From poly[1-((−)menthyloxycarbonyl)-1,3-pentadiene] (AIBN, benzene, 50°C. See Section B. Table 1, 1.6)								
			−58.3			THF	25	190
Poly[3-(ethoxycarbonyl)-1-butenylene]								
n-BuLi/sodium 1-menthoxide	Toluene	−78	−14.0		Induced asymmetry	Benzene	RT	176
n-BuLi/sodium *d*-bornoxide	Toluene	−78	6.0			Benzene	RT	
Poly[3-(methoxycarbonyl)-1-butenylene]								
n-BuLi/sodium 1-menthoxide	Toluene	−78	−2.4		Induced asymmetry	Benzene	RT	176
n-BuLi/sodium *d*-bornoxide	Toluene	−78	2.7			Benzene	RT	
Poly(*trans*-2-methyl-1,3-pentadiene)								
γ-ray/apocholic acid	–	50	−106		Induced asymmetry	Carbon tetrachloride	20	159
γ-ray/deoxycholic acid	–	50	90			Carbon tetrachloride	20	160
Poly(*cis*-2-methyl-1,3-pentadiene)								
γ-ray/apocholic acid	–	50	−66		Induced asymmetry	Carbon tetrachloride	20	159
γ-ray/deoxycholic acid	–	50	320			Carbon tetrachloride	20	160
Poly(1,3-pentadiene)								
$Al(C_2H_5)$/(−)titanium tetramenthoxide	Benzene	0	−22.8		Induced asymmetry	Hexane	18	171
Poly(*trans*-1,3-pentadiene)								
γ-ray/apocholic acid	–	50	−6.5		Induced asymmetry	Carbon tetrachloride	20	159
γ-ray/deoxycholic acid	–	RT	0.3			Chloroform	26	158
γ-ray/*trans*,anti,*trans*,anti,*trans*-perhydrotriphenylene								
			2.5			Chloroform	25	172
Poly(*cis*-1,3-pentadiene)	–							
γ-ray/apocholic acid	–	50	−3.7		Induced asymmetry	Carbon tetrachloride	20	159
γ-ray/deoxycholic acid	–	RT	−21			Chloroform	26	158
4.6. POLY(ISONITRILES)								
Poly[(*tert*-butylimino)methylene]								
tris(*tert*-butyl isocyanide)[(*S*)(−)-(1-phenylethyl)amino(*tert*-butylamino)carbene]-nickel(II) perchlorate								
	Neat	RT	−28.7		Helix-sense asymmetry	Chloroform	20	207
Poly[5,8-dimethyl-6,7-bis(propoxymethyl)-quinoxaline-2,3-diyl] (Poly[1,2-diisocyano-3,6-dimethyl-4,5-bis(propoxymethyl)benzene])								
(+)oligo(5,8-di-*p*-tolylquinoxaline-2,3-diyl)/Pd(II)								
	THF	RT	−171		Helix-sense asymmetry			204
4.7. POLY(MALEIMIDES)								
Poly(*N*-benzyl maleimide)								
n-BuLi/(−)-sparteine	Toluene	0	11.3		Induced asymmetry	Chloroform or THF		181
Poly(*N*-*n*-butyl maleimide)								
n-BuLi/(−)-sparteine	Toluene	0	−7.1		Induced asymmetry	Chloroform or THF		181
Poly(*N*-sec-butyl maleimide)								
n-BuLi/(−)-sparteine	Toluene	6	−21.0		Induced asymmetry	Chloroform or THF		181

Poly(*N*-*tert*-butyl maleimide)							
n-BuLi/(−)-sparteine	Toluene	0	2.8	Induced asymmetry	Chloroform or THF		181
Poly(*N*-cyclohexyl maleimide)							
n-BuLi/(−)-sparteine	Toluene	0	−39.5	Induced asymmetry	Chloroform or THF		181
Poly(*N*-2-fluorenyl maleimide)							
n-BuLi/(−)-sparteine	Toluene	0	−0.4	Induced asymmetry	Chloroform or THF		181
Poly(*N*-isobutyl maleimide)							
n-BuLi/(−)-sparteine	Toluene	0	−13.4	Induced asymmetry	Chloroform or THF		181
Poly(*N*-isopropyl maleimide)							
n-BuLi/(−)-sparteine	Toluene	0	−18.7	Induced asymmetry	Chloroform or THF		181
Poly(*N*-1-naphthyl maleimide)							
n-BuLi/(−)-sparteine	Toluene	0	0.9	Induced asymmetry	Chloroform or THF		181
Poly(*N*-phenyl maleimide)							
n-BuLi/(−)-sparteine	Toluene	0	−15.4	Induced asymmetry	Chloroform or THF		181
n-BuLi/(−)-sparteine/CuI	Toluene	0	−23.7 ($[\alpha]_{435}$)		Chloroform	25	241
Poly(*N*-*n*-propyl maleimide)							
n-BuLi/(−)-sparteine	Toluene	0	−4.9	Induced asymmetry	Chloroform or THF		181

4.8. OTHER COMPOUNDS

Poly(methylene-1,3-cyclopentylene) (from 1,5-hexadiene)							
(+)(*S*,*S*)ethylene-1,2-bis(η^5-4,5,6,7-tetrahydro-1-indenyl)zirconium-1,1′-bi-2-naphtholate/methylaluminoxane							
	Toluene	RT	51.0 ($[M]_{405}$)	Induced asymmetry	Chloroform	28	239
Poly[1-(*m*-tolylsulfonyl)ethylene]							
From poly(*m*-tolyl vinyl sulfoxide) (*n*-BuLi/(−)-sparteine, toluene, −78°C. See Section B, Table 1, 1.10)							
			42	Miscellaneous			246

TABLE 5. MAIN-CHAIN ACYCLIC HETROATOM POLYMERS

Polymer	Polymerization system		Polymer (P) values			Optical activity measured		
Catalyst or initiator	Solvent	*T* (°C)	$[\alpha]_D$	$[M]_D$	Origin of optical activity	Solvent	*T* (°C)	Refs.
5.1. POLY(OXIDES) AND POLY(SULFIDES)								
Poly(chloral)								
Tetramethylammonium (−) acetyl mandelate								
	–	0–75	242–668		Helix-sense asymmetry	Film (soaked in diphenyl ether)	25	165
Lithium coprostan-3β-oxide	Hexane	<58	2970			Film (soaked in diphenyl ether)	RT	247
Poly(oxy-1,2-cyclohexylidene)								
$ZnEt_2$/(1*S*,2*R*)-ephedrine	–	80	−9.1 to 12.3		Induced asymmetry	Benzene	25	194
Poly(oxy-1,2-dimethylethylene)								
$ZnEt_2$/(1*S*,2*R*)-ephedrine	–	80	15		Induced asymmetry	Benzene	25	238
Poly(thio-1,2-cyclohexylidene)								
$ZnEt_2$/(*R*,*R*)-1,2-diphenyl-1,2-ethandiol								
	Toluene	RT	59		Induced asymmetry	Trichlorobenzene	25	238
Poly(thio-1,2-dimethylethylene)								
$ZnEt_2$/(*R*,*R*)-1,2-diphenyl-1,2-ethandiol								
	–	RT	154		Induced asymmetry	Chloroform	25	238

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TABLE 5. *cont'd*

Polymer	Polymerization system		Polymer (P) values			Optical activity measured		
Catalyst or initiator	Solvent	T (°C)	$[\alpha]_D$	$[M]_D$	Origin of optical activity	Solvent	T (°C)	Refs.
5.2. POLY(ISOCYANATES)								
Poly[(butyl)iminocarbonyl]								
(*R*)*N*-benzyl-1-phenylethylamine lithium amide	Toluene	−78	−40 ($[\alpha]_{365}$)		Helix-sense asymmetry	Chloroform	25	248
Poly[*N*-(*p*-chlorophenyl)iminocarbonyl]								
tert-BuLi/(*S*)(-)-(2-methyoxymethyl)pyrrolidine	THF	−98	377 ($[\alpha]_{365}$)		Helix-sense asymmetry	THF	25	193
Poly[*N*-(3,4-dimethylphenyl)iminocarbonyl]								
tert-BuLi/(*S*)(−)-(2-methyoxymethyl)pyrrolidine	THF	−98	569 ($[\alpha]_{365}$)		Helix-sense asymmetry	DCM	25	193
Poly[*N*-(*p*-methoxyphenyl)iminocarbonyl]								
tert-BuLi/(*S*)(−)-(2-methyoxymethyl)pyrrolidine	THF	−98	588 ($[\alpha]_{365}$)		Helix-sense asymmetry	Chloroform	25	193
Poly[*N*-(*m*-methylphenyl)iminocarbonyl]								
tert-BuLi/(*S*)(−)-(2-methyoxymethyl)pyrrolidine	THF	−98	819 ($[\alpha]_{365}$)		Helix-sense asymmetry	THF	25	193
Poly[*N*-(*p*-methylphenyl)iminocarbonyl]								
tert-BuLi/(*S*)(−)-(2-methyoxymethyl)pyrrolidine	THF	−98	748 ($[\alpha]_{365}$)		Helix-sense asymmetry	Chloroform	25	193
Poly[*N*-(1-naphthyl)iminocarbonyl]								
tert-BuLi/(*S*)(−)-(2-methyoxymethyl)pyrrolidine	THF	−98	529 ($[\alpha]_{365}$)		Helix-sense asymmetry	DCM	25	193
Poly[(4-phenylbutyl)iminocarbonyl]								
(*R*)*N*-benzyl-1-phenylethylamine lithium amide	Toluene	−78	−17 ($[\alpha]_{365}$)		Helix-sense asymmetry	Chloroform	25	248

D. REFERENCES

1. F. Ciardelli, E. Benedetti, O. Pieroni, Makromol. Chem., **103**, 1 (1967).
2. P. Pino, G. P. Lorenzi, L. Lardicci, Chim. Ind. (Milan), **42**, 712 (1960).
3. P. Pino, G. P. Lorenzi, J. Am. Chem. Soc., **82**, 4745 (1960).
4. P. Pino, F. Ciardelli, G. P. Lorenzi, G. Montagnoli, Makromol. Chem., **61**, 207 (1963).
5. P. Pino, P. Salvadori, E. Chiellini, P. L. Luisi, Pure Appl. Chem., **16**, 469 (1968).
6. P. Pino, G. P. Lorenzi, L. Lardicci, F. Ciardelli, Vysokomol. Soedin., **3**, 1567 (1961).
7. M. Goodman, K. J. Clark, M. A. Stake, A. Abe, Makromol. Chem., **72**, 131 (1964).
8. W. J. Bailey, E. T. Yates, J. Org. Chem., **25**, 1800 (1960).
9. S. Nozakura, S. Takeuchi, H. Yuki, S. Murahashi, Bull. Chem. Soc. (Japan), **34**, 1673 (1961).
10. R. C. Schulz, H. Hilpert, Makromol. Chem., **55**, 132 (1962).
11. R. C. Schulz, Z. Naturforsch **19b**, 387 (1964).
12. R. K. Kulkarni, H. Morawetz, J. Polym. Sci., **54**, 491 (1961).
13. C. S. Marvel, J. Dec, H. G. Cooke, Jr., J. Am. Chem. Soc., **62**, 3499 (1940).
14. J. W. C. Crawford, D. Plant, J. Chem. Soc., 4492 (1952).
15. P. Walden, Z. Phys. Chem., **20**, 383 (1896).
16. E. I. Klabunovskii, I. I. Petrov, M. I. Schvartmen, Vysokomol. Soedin., **6**, 1487 (1964).
17. C. L. Arcus, D. W. West, Chem. Ind. (London), 230 (1958).
18. C. L. Arcus, D. W. West, J. Chem. Soc., 2699 (1959).
19. K. J. Liu, J. S. Lignowski, R. Ullman, Makromol. Chem., **105**, 8 (1967).
20. N. Beredjick, C. Schuerch, J. Am. Chem. Soc., **78**, 2646 (1956).
21. N. Beredjick, C. Schuerch, J. Am. Chem. Soc., **80**, 1933 (1958).
22. E. I. Klabunovskii, M. I. Schvartmen, I. I. Petrov, Vysokomol. Soedin., **6**, 1579 (1964).
23. T. P. Bird, W. A. P. Black, E. T. Dewar, D. Rutherford, Chem. Ind. (London), 1331 (1960).
24. W. A. P. Black, E. T. Dewar, D. Rutherford, J. Chem. Soc., 4433 (1963).
25. S. Kimura, M. Imoto, Makromol. Chem., **50**, 155 (1961).
26. S. Kimura, K. Hirano, M. Imoto, Kogyo Kagaku Zasshi (Japan), **65**, 688 (1962).
27. M. Imoto, S. Kimura, Makromol. Chem., **53**, 210 (1962).
28. R. K. Kulkarni, H. Morawetz, J. Polym. Sci., **54**, 491 (1961).
29. V. E. Kaiser, R. C. Schulz, Makromol. Chem., **81**, 273 (1965).
30. O. Pieroni, F. Ciardelli, C. Botteghi, L. Lardidi, P. Salvadori, P. Pino, J. Polym. Sci. C, **22**, 993 (1967).
31. G. Anzuino, V. Crescenzi, M. D'Alagni, A. M. Liquori, F. Quadrifoglio, F. Ascoli, Communication at the IX, Natl. Meeting of Italian Chem. Soc., Naples, 1962.
32. G. P. Lorenzi, E. Benedetti, E. Chiellini, Chim. Ind. (Milan), **46**, 1474 (1964).
33. D. Basagni, A. M. Liquori, B. Pispesa, J. Polym. Sci. B, **2**, 241 (1964).
34. P. Pino, G. P. Lorenzi, Makromol. Chem., **47**, 242 (1961).
35. G. J. Schmitt, C. Schuerch, J. Polym. Sci., **45**, 313 (1960).
36. W. A. P. Black, E. T. Dewar, D. Rutherford, Chem. Ind. (London) 1624 (1962).
37. R. C. Schulz, R. H. Jung, Makromol. Chem., **96**, 295 (1966).
38. R. C, Schulz, H. Hartman, Makromol. Chem., **65**, 106 (1963).
39. F. Millich, G. K. Baker, Macromolecules **2**, 122 (1969).
40. C. S. Marvel, C. G. Overberger, J. Am. Chem. Soc., **68**, 2106 (1946).
41. C. S. Marvel, C. G. Overberger, J. Am. Chem. Soc., **66**, 475 (1944).
42. C. G. Overberger, L. C. Palmer, J. Am. Chem. Soc., **78**, 666 (1956).
43. H. Yamaguchi, Y. Minoura, J. Polym. Sci. A-1, **8**, 929 (1970).
44. A. Abe, M. Goodman, J. Polym. Sci. A, **1**, 2193 (1963).
45. A. Abe, M. Goodman, J. Polym. Sci., **59**, S37 (1962).
46. C. C. Price, M. Osgan, R. E. Hughes, C. Shambelan, J. Am. Chem. Soc., **78**, 690 (1956); 4787 (1956).
47. E. J. Vandenberg, J. Am. Chem. Soc., **83**, 3538 (1961).
48. H. Shimazaki, J. Chem. Soc. (Japan), **87**, 462 (1966).
49. C. G. Overberger, S. Ozaki, D. M. Braunstein, Makromol. Chem., **93**, 13 (1966).
50. J. R. Shelton, J. B. Lando, D. E. Agostini, Polym. Lett. **9**, 173 (1971).
51. R. C. Schulz, J. Schwaab, Makromol. Chem., **87**, 90 (1965).
52. Y. Iwakura, K. Iwata, S. Matsuo, A. Tohara, Makromol. Chem., **122**, 275 (1969).
53. C. G. Overberger, H. Kaye, J. Am. Chem. Soc., **89**, 5649 (1967).
54. H. D. K. Drew, W. N. Haworth, J. Chem. Soc., 775 (1927).
55. N. Spassky, P. Sigwalt, Bull. Chim. Soc. France, 4617 (1967).
56. T. Tsunetsugu, J. Furukawa, T. Fueno, J. Polym. Sci. A-1, **9**, 3541 (1971).
57. C. G. Overberger, J. K. Weise, J. Am. Chem. Soc., **90**, 3538 (1968).
58. M. Imoto, H. Sakurai, T. Kono, J. Polym. Sci., **50**, 467 (1961).
59. C. G. Overberger. H. Jabloner, J. Polym. Sci., **55**, 32 (1961).
60. C. G. Overberger, H. Jabloner, J. Am. Chem. Soc., **85**, 3431 (1963).
61. C. G. Overberger, G. M. Parker, J. Polym. Sci. C, **22**, 387 (1968).
62. E. Schmidt, Angew. Makromol. Chem., **14**, 185 (1970).
63. Y. Iwakura, K. Hayashi, K. Inagaki, Makromol. Chem., **110**, 84 (1967).
64. M. Goodman, S. Chen, Macromolecules, **4**, 625 (1971).
65. M. Goodman, S. Chen, Macromolecules, **3**, 398 (1970).
66. Y. Minoura, S. Takebayashi, C. C. Price, J. Am. Chem. Soc., **81**, 4689 (1959).
67. K. Tsuboyama, S. Tsuboyama, M. Yanagita, Bull. Chem. Soc. (Japan), **40**, 2954 (1967).

68. S. Tsuboyama, Bull. Chem. Soc. (Japan), **35**, 1004 (1962).
69. M. S. Toy, C. C. Price, J. Am. Chem. Soc., **82**, 2613 (1960).
70. A. P. Terentev, V. V. Dunina, E. G. Rukhadze, Vysokomol. Soedin. A, **9**, 599 (1967).
71. R. C. Schulz, R. H. Jung, Makromol. Chem., **116**, 190 (1968).
72. A. Abe, M. Goodman, J. Polym. Sci. A, **2**, 3491 (1964).
73. C. G. Overberger, T. Yoshimura, A. Ohnishi, A. S. Gomes, J. Polym. Sci. A-1, **8**, 2275 (1970).
74. F. Micheel, A. Bockmann, W. Meckstroth, Makromol. Chem., **48**, 1 (1961).
75. T. Uryu, H. Libert, J. Zachoval, C. Schuerch, Macromolecules, **3**, 345 (1970).
76. G. Schramm, H. Grotsch, W. Pollman, Angew. Chem., **74**, 53 (1962).
77. J. Das. Carvalho, W. Trins, C. Schuerch, J. Am. Chem. Soc., **81**, 4054 (1959).
78. C. C. Tu. C. Schuerch, J. Polym. Sci. B, **1**, 163 (1963).
79. F. Micheel, A. Bockmann, Makromol. Chem., **51**, 97, (1962).
80. F. Frechet, C. Schuerch, J. Am. Chem. Soc., **91**, 1161 (1969).
81. A. R. Downie, A. Elliot, W. E. Hanby, B. R. Malcolm, Proc. Roy. Soc. A, **242**, 325 (1957).
82. M. Goodman, Fu Chen, F. R. Prince, Biopolymers, **12**, 2549 (1973).
83. M. Goodman, A. M. Felix, Biochemistry, **3**, 1529 (1964).
84. P. Urnes, P. Doty, Advan. Protein Chem., **16**, 401 (1961).
85. T. Hayakawa, Y. Fujiwara, J. Noguchi, Bull. Chem. Soc. (Japan), **40**, 1205 (1967).
86. E. R. Blout, R. H. Karlson, J. Am. Chem. Soc., **80**, 1259 (1958).
87. M. Frankel, A. Berger, J. Org. Chem., **16**, 1513 (1951).
88. A. Berger, E. Katchalski, J. Am. Chem. Soc., **73**, 4084 (1951).
89. J. Kovacs, R. Ballina, R. L. Rodin, D. Balasurkamanium, H. Applequist, J. Am. Chem. Soc., **87**, 119 (1965).
90. S. Ikeda, Biopolymers, **5**, 359 (1967).
91. M. Wilchek, A. Fransdorff, M. Sela, Arch. Biochem. Biophys., **113**, 742 (1966).
92. W. Moffitt, J. T. Young, Pro. Nat. Acad. Sci. USA, **42**, 596 (1956).
93. L. Goldstein, E. Katchalski, Bull. Research Council Israel, **A9**, 138 (1960).
94. A. Wada, Mol. Phys., **3**, 409 (1960).
95. J. Kovacs, G. H. Schmit, E. J. Jonson, Can. J. Chem., **47**, 3690 (1960).
96. J. T. Yang, P. Doty, J. Am. Chem. Soc., **79**, 761 (1957).
97. E. R. Blout, R. H. Karlson, J. Am. Chem. Soc., **78**, 941 (1958).
98. W. E. Hanby, S. G. Ealey, J. Eatson, J. Chem. Soc., 3239, (1950).
99. M. Goodman, E. Schmitt, I. Listowoky, F. Boardman, I. G. Rosen, M. Stake, in M. A. Stahmann (Ed.), "Polyamino Acids, Polypeptides and Proteins", Univ. Wisconsin Press, Madison, 1962.
100. M. Goodman, E. E. Schmitt, J. Am. Chem. Soc., **81**, 5507 (1959).
101. K. S. Norland, G. D. Fasman, E. Katchalski, E. R. Blout, Biopolymers, **1**, 277 (1961).
102. A. Patchornik, A. Berger, E. Katchalski, J. Am. Chem. Soc., **79**, 5227 (1957).
103. G. D. Fasman, M. Idelson, E. R. Blout, J. Am. Chem. Soc., **83**, 709 (1961).
104. M. Sela, R. Arnon, I. Jacobson, Biopolymers, **1**, 517 (1963).
105. J. Applequist, P. Doty, in: M. A. Stahmann (Ed.), "Polyamino Acids, Polypeptides and Proteins", Univ. Wisconsin Press, Madison, 1962.
106. H. Devoe, I. Tinoco, J. Mol. Biol., **4**, 518 (1962).
107. G. E. Perlmann, E. Katchalski, J. Am. Chem. Soc., **84**, 452 (1962).
108. G. D. Fasman, in: M. A. Stahmann (Ed.) "Polyamino Acids, Polypeptides and Proteins", Univ. Wisconsin Press, Madison, 1962.
109. E. Katchalski, A. Berger, J. Kurtz, in: G. N. Ramachandran (Ed.) ,"Aspects of Protein Structure", Academic Press, New York, 1963, p.205.
110. J. Kurtz, G. D. Fasman, A. Berger, E. Katchalski, J. Am. Chem. Soc., **80**, 393 (1958).
111. E. Katchalski, I. Z. Steinberg, Ann. Rev. Phys. Chem., **12**, 433 (1961).
112. J. Kurtz, A. Berger, E. Katchalski, Nature, **178**, 1066 (1956).
113. W. F. Harrington, M. Sela, Biochem. Biophys. Acta, **27**, 24 (1958).
114. G. D. Fasman, E. R. Blout, J. Am. Chem. Soc., **82**, 2262 (1960).
115. M. Sela, I. Z. Steinberg, E. Daniel, Biochem. Biophys. Acta, **46**, 433 (1961).
116. J. D. Coombes, E. Katchalski, P. Doty, Nature, **185**, 534 (1960).
117. E. Katchalski, M. Sela, J. Am. Chem. Soc., **75**, 5284 (1953).
118. C. Carlini, F. Ciardelli, D. Pino, Makromol. Chem., **174**, 15 (1974).
119. J. M. Sauvage, M. Morcellet, C. Loucheux, Macromolecules **16**, 1564 (1983).
120. T. Sato, T. Kadowaki, H. Takeda, T. Otsu, J. Macromol. Sci., Chem. A, **12**, 1225 (1978).
121. T. Kakuchi, K. Yokota, Makromol. Chem., Rapid Commun., **6**, 551 (1985).
122. H. Yamaguchi, K. Sekioka, M. Kurokawa, T. Doiuchi, Makromol. Chem., **184**, 455 (1983).
123. E. Chiellini, Macromolecules **3**, 527 (1970).
124. M. Kurokawa, H. Yamaguchi, Y. Minoura, J. Polym. Sci., Polym. Chem. Ed., **17**, 485 (1979).
125. R. Vukovic, D. Fles, J. Polym. Sci., Polym. Chem. Ed., **13**, 49 (1975).
126. N. Thien, U. W. Suter, P. Pino, Makromol. Chem., **184**, 2335 (1983).
127. E. Chiellini, R. Solaro, A. Ledwith, Makromol. Chem., **179**, 1929 (1978).
128. N. Kunieda, J. Shibatani, Y. Fijiwara, M. Kinoshita, Makromol. Chem., **175**, 2509 (1974).
129. Z. Janovic, D. Fles, J. Polym. Sci. A-1, **9**, 1103 (1971).
130. R. R. Kiprijanova, D. Fles, J. Polym. Sci., Polym. Chem. Ed., **13**, 1141 (1975).

131. A. J. M. Beijnen, R. J. M. Nolte, W. Drenth, A. M. F. Hezemans, P. J. F. M. Coolwijk, Macromolecules, **13**, 1386 (1980).
132. A. J. M. Beijnen, R. J. M. Nolte, A. J. Naaktgeboren, J. W. Zwikker, W. Drenth, A. M. F. Hezemans, Macromolecules, **16**, 1679 (1983).
133. F. Ciardelli, S. Lanzillo, O. Pieroni, Macromolecules **7**, 174 (1974).
134. N. Kunieda, M. Kinoshita, M. Imoto, J. Polym. Sci. B, **9**, 241 (1971).
135. N. Kunieda, H. Wada, J. Shibatani, M. Kinoshita, Makromol. Chem., **172**, 237 (1973).
136. J. Huguet, M. Vert, Polym. Bull., **5**, 505 (1981).
137. T. Oishi, M. Fujimoto, J. Polym. Sci., Polym. Chem. Ed., **22**, 2789 (1984).
138. T. Hirano, A. Sato, T. Tsuruta, W. C. Johnson, J. Polym. Sci., Polym. Phys. Ed., **17**, 1601 (1979).
139. E. Chiellini, P. Salvadori, M. Osgan, P. Pino, J. Polym. Sci. A-1, **8**, 1589 (1970).
140. Y. Iwakura, K. Iwata, S. Matsuo, A. Tohara, Makromol. Chem., **146**, 21 (1971).
141. B. Jerman, D. Fles, J. Polym. Sci., Polym. Chem. Ed., **14**, 1117 (1976).
142. P. Dumas, N. Spassky, P. Sigwalt, Makromol. Chem., **156**, 55 (1972).
143. D. Fles, V. Tomasic, M. Samsa, D. Ahmetovic, B. Jerman, M. Fles, J. Polym. Sci., Symposium Ed., **42**, 321 (1973).
144. K. Ehara, T. Miura, W. Urahashi, H. Kitahara, H. Nohira, Kobunshi Ronbunshu (Japan), **37**, 463 (1980).
145. N. Yahiro, K. Asakawa, Kobunshi Ronbunshu (Japan), **39**, 507 (1982).
146. Y. Kawakami, T. Sugiura, Y. Mizutani, Y. Yamashita, J. Polym. Sci., Polym. Chem. Ed., **18**, 3009 (1980).
147. A. D. Aliev, E. P. Tiurina, A. Y. Koschevnik, S. L. Aliev, B. A. Krentsel, Eur. Polym. J., **16**, 679 (1980).
148. T. Saegusa, S. Kobayashi, M. Ishiguro, Macromolecules **7**, 958 (1974).
149. J. G. Hamilton, K. J. Ivin, L. C. K. Essig, P. Watt, Macromolecules, **9**, 67 (1976).
150. H. Yuki, Y. Okamoto, Y. Kobayashi, J. Polym. Sci., Polym. Chem. Ed., **17**, 3867 (1979).
151. S. Yamashita, K. Waki, N. Yamawaki, H. Tani, Macromolecules, **7**, 410 (1974).
152. N. Oguni, H. Kuboyama, A. Nakamura, J. Polym. Sci., Polym. Chem. Ed., **21**, 1559 (1983).
153. Y. Okamoto, M. Adachi, H. Shohi, H. Yuki, Polym. J., **13**, 175 (1981).
154. Y. Okamoto, M. Ishikura, K. Hatada, H. Yuki, Polym. J., **15**, 851 (1983).
155. Y. Okamoto, H. Mohri, M. Ishikura, K. Hatada, H. Yuki, J. Polym. Sci., Polym. Symposium Ed., **74**, 125 (1986).
156. Y. Okamoto, K. Suzuki, H. Yuki, J. Polym. Sci., Polym. Chem. Ed., **18**, 3043 (1980).
157. S. Kanoh, H. Suda, N. Kawaguchi, M. Motoi, Makromol. Chem., **187**, 53 (1986).
158. G. Audisio, A. Silvani, J. Chem. Soc., Chem. Commun., 481, (1976).
159. M. Miyata, Y. Kitahara, K. Takemoto, Polym. Bull., **2**, 671 (1980).
160. M. Miyata, Y. Kitahara, K. Takemoto, Polym. J., **13**, 111 (1981).
161. S. Kataoka, T. Ando, Kobunshi Ronbunshu (Japan), **37**, 375 (1980).
162. K. Nozaki, Y. Matsubara, M. Yoshihara, T. Maeshima, Makromol. Chem., Rapid Commun., **5**, 723 (1984).
163. S. Kataoka, T. Ando, Kobunshi Ronbunshu (Japan), **37**, 185 (1980).
164. D. J. Cram, D. Y. Sogah, J. Am. Chem. Soc., **107**, 8301 (1985).
165. W. J. Harris, O. Vogl, Polym. Prepr., Am. Chem. Soc., Div. Polym. Chem., **22** (2), 309 (1981).
166. Q.-S. Hu, D. Vitharana, G.-Y. Liu, V. Jain, M. W. Waganam, L. Zhang, T. R. Lee, L. Pu, Macromolecules, **29**, 1082 (1996).
167. Y. Chen, J.-J. Tsay, Polym. J., **24**, 263 (1992).
168. Y. Chen, H.-H. Tseng, J. Polym. Sci., Polym. Chem. Ed., **31**, 1719 (1993).
169. M. Farina, G. Bressan, Makromol. Chem., **61**, 79 (1963).
170. G. Bressan, M. Farina, G. Natta, Makromol. Chem., **93**, 283 (1966).
171. G. Natta, L. Porri, S. Valenti, Makromol. Chem., **67**, 225 (1963).
172. M. Farina, G. Audisio, G. Natta, J. Am. Chem. Soc., **89**, 5071 (1967).
173. N. Yahiro, S. Ito, Kobunshi Ronbunshu, **42**, 427 (1985).
174. R. D. Katsarava, T. M. Kartvelishvili, M. M. Zaalishvili, Dokl. Chem., **281**, 99 (1985).
175. 175 H. R. Kricheldorf, J. M. Jonte, Polym. Bull., **9**, 276 (1983).
176. T. Tsunetsugu, T. Fueno, J. Furukawa, Makromol. Chem., **112**, 220 (1968).
177. H. Yamaguchi, T. Iwama, T. Hayashi, Makromol. Chem., **191**, 1243 (1990).
178. Y. Bando, H. Yamaguchi, Y. Minoura, Eur. Polym. J., **15**, 497 (1979).
179. G. Wulff, P. K. Dhal, Angew. Chem. Int. Ed. Engl., **28**, 196 (1989).
180. G. Wulff, P. Kemmerer, B. Vogt, J. Am. Chem. Soc., **109**, 7449 (1987).
181. T. Oishi, H. Yamasaki, M. Fujimoto, Polym. J., **23**, 795 (1991).
182. F. Sanda, T. Tanaka, T. Endo, J. Polym. Sci., Polym. Chem. Ed., **33**, 2353 (1995).
183. T. Oishi, K. Kagawa, M. Fujimoto, J. Polym. Sci., Polym. Chem. Ed., **33**, 1341 (1995).
184. J. J. Bou, I. Iribarren, S. M.-Guerra, Macromolecules, **27**, 5263 (1994).
185. Y. Okamoto, M. Nishikawa, T. Nakano, E. Yashima, K. Hatada, Macromolecules, **28**, 5135 (1995).
186. S. Habaue, T. Tanaka, Y. Okamoto, Macromolecules, **28**, 5973 (1995).
187. K. Kagawa, T. Oishi, Polym. J., **27**, 579 (1995).
188. Y. Okamoto, H. Hayashida, K. Hatada, Polym. J., **21**, 543 (1989).

189. T. Kobayashi, M. Kakimoto, Y. Imai, Polym. J., **26**, 758 (1994).
190. T. Doiuchi, T. Dodoh, H. Yamaguchi, Makromol. Chem., **191**, 1253 (1990).
191. T. Oishi, T. Kawamoto, Polym. J., **26**, 920 (1994).
192. T. Oishi, M. Fujimoto, J. Polym. Sci., Polym. Chem. Ed., **30**, 1821 (1992).
193. Y. Okamoto, M. Matsuda, T. Nakano, E. Yashima, J. Polym. Sci., Polym. Chem. Ed., **32**, 309 (1994).
194. M. Sepulchre, A. Kassamaly, N. Spassky, Makromol. Chem., Macromol. Symp., **42/43**, 489 (1991).
195. L. Angiolini, C. Carlini, M. Tramontini, Macromol. Chem., **191**, 2083 (1990).
196. Y. Okamoto, T. Nakano, Y. Shikisai, M. Mori, Macromol. Symp., **89**, 479 (1995).
197. T. Aoki, M. Kokai, K. Shinohara, E. Oikawa, Chem. Lett., 2009, (1993).
198. O. Haba, T. Kakuchi, K. Yokota, Macromolecules, **26**, 1782 (1993).
199. J. J. Bou, A. Rodriquez-Galan, S. Munoz-Guerra, Macromolecules, **26**, 5664 (1993).
200. T. Oishi, K. Kagawa, M. Fujimoto, Macromolecules, **26**, 24 (1993).
201. T. Oishi, K. Matsusaki, M. Fujimoto, Polym. J., **24**, 1281 (1992).
202. T. Kobayashi, M. Kakimoto, Y. Imai, Polym. J., **25**, 741 (1993).
203. T. Oishi, Y. Otsubo, M. Fujimoto, Polym. J., **24**, 527 (1992).
204. Y. Ito, E. Ihara, M. Murakami, Angew. Chem. Int. Ed. Engl., **31**, 1509 (1992).
205. A. Altomare, F. Ciardelli, R. Lima, R. Solaro, Chirality **3**, 292 (1991).
206. T. Oishi, M. Fujimoto, J. Macromol. Chem., Pure Appl. Chem. A, **29**, 1187 (1992).
207. P. C. Kamer, R. J. M. Nolte, W. Drenth, J. Am. Chem. Soc., **110**, 6818 (1988).
208. H. Hashimoto, T. Kakuchi, O. Haba, K. Yokota, Macromolecules, **25**, 1828 (1992).
209. D. Pini, A. Iuliano, P. Salvadori, Macromolecules, **25**, 6059 (1992).
210. H. Hashimoto, T. Kakuchi, K. Yokota, J. Org. Chem., **56**, 6470 (1991).
211. T. Kakuchi, T. Satoh, J. Muta, S. Umeda, H. Hashimoto, K. Yokota, J. Macromol. Sci., Pure Appl. Chem. A, **33**, 352 (1996).
212. T. Satoh, T. Hatakeyama, S. Umeda, H. Hashimoto, K. Yokota, T. Kakuchi, Macromolecules, **29**, 3447 (1996).
213. T. Kakuchi, T. Satoh, S. Umeda, H. Hashimoto, K. Yokota, Macromolecules, **28**, 5643 (1995).
214. T. Satoh, T. Hatakeyama, S. Umeda, K. Yokota, T. Kakuchi, Polym. J., **28**, 520 (1996).
215. A. Lekchiri, J. Morcellet, M. Morcellet, Macromolecules **20**, 49 (1987).
216. J.-H. Liu, F.-R. Tsai, Y.-C. Lai, Angew. Makromol. Chem., **231**, 35 (1995).
217. A. Lekchiri, J. Morcellet, M. Morcellet, Macromolecules **20**, 49 (1987).
218. Y. Chen, K. Saigo, N. Yonezawa, M. Hasegawa, Bull. Chem. Soc. Japan, **60**, 1895 (1987).
219. T. Tanaka, S. Habaue, Y. Okamoto, Polym. J., **27**, 1202 (1995).
220. Y. Okamoto, E. Yashima, M. Ishikura, K. Hatada, Polym. J., **19**, 1183 (1987).
221. Y. Okamoto, N. Nagamura, T. Fukumoto, K. Hatada, Polym. J., **23**, 1197 (1991).
222. A. Castellano, A. Lekchiri, J. Morcellet, M. Morcellet, J. Polym. Sci., Polym. Chem., **25**, 1419 (1987).
223. Y. Okamoto, E. Yashima, K. Hatada, J. Polym. Sci., Polym. Lett., **25**, 297 (1987).
224. Y. Okamoto, T. Nakano, T. Asakura, H. Mohri, K. Hatada, J. Polym. Sci., Polym. Chem., **29**, 287 (1991).
225. Y. Okamoto, T. Nakano, E. Ono, K. Hatada, Chem. Lett., 525, (1991).
226. E. Chiellini, G. Galli, R. Po, Polym. Bull., **23**, 397 (1990).
227. M. M. Green, R. A. Gross, F. C. Schilling, K. Zero, C. Crosby III, Macromolecules, **21**, 1839 (1988).
228. M. M. Green, R. A. Gross, C. Crosby III, F. C. Schilling, Macromolecules, **20**, 992 (1987).
229. S. Lifson, C. Andreola, N. C. Peterson, M. M. Green, J. Am. Chem. Soc., **111**, 8850 (1989).
230. H. G. J. Visser, R. J. M. Nolte, J. W. Zwikker, W. Drenth, J. Org. Chem., **50**, 3133 (1985).
231. H. G. J. Visser, R. J. M. Nolte, J. W. Zwikker, W. Drenth, J. Org. Chem., **50**, 3138 (1985).
232. J. Huguet, D. Vallin, M. Vert, Polym. J., **14**, 335 (1982).
233. T. Oishi, M. Fujimoto, J. Macromol. Sci., Chem. A, **23**, 619 (1986).
234. N. Yahiro, Kobunshi Ronbunshu, **40**, 371 (1983).
235. K. Saigo, Y. Chen, N. Yonezawa, T. Kanoe, K. Tachibana, M. Hasegawa, Macromolecules, **19**, 1552 (1986).
236. P. C. Kamer, M. C. Cleij, R. J. M. Nolte, T. Harada, A. M. F. Hezemans, W. Drenth, J. Am. Chem. Soc., **110**, 1581 (1988).
237. E. Chiellini, R. Po, S. Carrozzino, G. Galli, and B. Gallot, Mol. Cryst. Liq. Cryst., **179**, 405 (1990).
238. N. Spassky, A. Momtaz, A. Kassamaly, M. Sepulchre, Chirality, **4**, 295 (1992).
239. G. W. Coates, R. M. Waymouth, J. Am. Chem. Soc., **115**, 90 (1993).
240. Y. Okamoto, T. Nakano, T. Fukuoka, K. Hatada, Polym. Bull., **26**, 259 (1991).
241. Y. Okamoto, T. Nakano, H. Kobayashi, K. Hatada, Polym. Bull., **25**, 5 (1991).
242. Y. Takeda, Y. Hayakawa, T. Fueno, J. Furukawa, Makromol. Chem., **83**, 234 (1965).
243. L. Porri, R. Rossi, G. Ingrosso, Tetrahedron Lett., 1083, (1971).
244. T. Kakuchi, H. Kawai, S. Katoh, O. Haba, K. Yokota, Macromolecules, **25**, 5545 (1992).
245. K. Yokota, T. Kakuchi, H. Sasaki, H. Ohmori, Makromol. Chem., **190**, 1269 (1989).
246. F. Toda, K. Mori, J. Chem. Soc., Chem. Commun., 1059, (1986).
247. G. D. Jaycox, O. Vogl, Makromol. Chem., Rapid Commun., **11**, 61 (1990).

248. Y. Okamoto, M. Matsuda, T. Nakano, E. Yashima, Polym. J., **25**, 391 (1993).
249. J. M. van der Eijk, R. J. M. Nolte, W. Drenth, A. M. F. Hezemans, Macromolecules, **13**, 1391 (1980).
250. M. Miyamoto, T. Umezawa, Polym. J., **28**, 432 (1996).
251. T. Nakano, A. Matsuda, M. Mori, Y. Okamoto, Polym. J., **28**, 330 (1996).
252. K. Kagawa and T. Oishi, Polym. J., **28**, 1 (1996).
253. T. Nakano, Y. Shikisai, Y. Okamoto, Polym. J., **28**, 51 (1996).

Anisotropy of Segments and Monomer Units of Polymer Molecules

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A. INTRODUCTION

Segmental anisotropy $(\alpha_1 - \alpha_2)^*$ of a chain molecule may be determined experimentally from measurements of flow birefringence in the solution of a polymer with a sufficiently high molecular weight, so that the conformation of its molecules would correspond to that of a Gaussian coil.

For calculating $(\alpha_1 - \alpha_2)^*$, Kuhn's equation was used (Ref. 1):

$$\frac{\Delta n}{\Delta \tau} = \left[\frac{\Delta n}{g(\eta - \eta_0)}\right]_{\substack{g\to 0\\ c\to 0}} \equiv \frac{[n]}{[\eta]} = \frac{4\pi}{45kT} \times \frac{(n^2+2)^2}{n}(\alpha_1 - \alpha_2)^* \qquad \text{(A1)}$$

where $\Delta\tau$ is the tangential flow stress (shearing stress), Δn is the observed flow birefringence of the solution, g is the velocity gradient, c and η are the concentration and the viscosity of the solution, respectively, $[\eta]$ and $[n]$ are intrinsic values of viscosity and flow birefringence of the solution, and η_0 and n are the viscosity and the refractive index of the solvent. The value $\Delta n/\Delta\tau = [n]/[\eta]$ may be called the shear optical coefficient.

Another method of determining segmental anisotropy (which is seldom used) is through the measurement of stress birefringence in swollen polymers (74). The stress optical coefficient $\varepsilon = \Delta n/\Delta p$ (where Δp is the normal stress in the sample) determined experimentally is equal to $\Delta n/2\Delta\tau$ (2), where $\Delta n/\Delta\tau$ is also related to $(\alpha_1 - \alpha_2)^*$ according to Eq. (A1). Data obtained by this method are marked in the table with the symbol sw. p. (swollen polymer).

If the refractive indices of the polymer and the solvent are not equal, the experimental value of $(\alpha_1 - \alpha_2)^*$ includes not only the intrinsic segmental $(\alpha_1 - \alpha_2)$, but also a part produced by the form effect (3,4). In this case the value $\alpha_1 - \alpha_2$ may be calculated using $(\alpha_1 - \alpha_2)^*$ and the theoretical equation for macro- and micro-form effects (3,4).

For determining $\alpha_1 - \alpha_2$, it is preferable to use the "matching" solvent in which the increment of index of the polymer, dn/dc, is equal to zero

The tables give the intrinsic segmental anisotropy, $\alpha_1 - \alpha_2$, obtained directly from measurements in the "matching" solvent, as well as the $(\alpha_1 - \alpha_2)^*$ values including the form effect. The corresponding figures are marked with the symbol *.

In some of the reviewed works in which measurements were carried out in "nonmatching" solvents, the segmental anisotropy was calculated on the basis of the theory of the form effect (3,4). In these cases the figures are marked with the symbol **.

Intrinsic segmental anisotropy of the polymer chain in various solvents, even in the absence of the form effect, may differ owing to a "specific" effect of the solvent (see, e.g., poly(vinyl acetate)) (5–11). "Specific" effect may probably includes phenomena such as a change in the type of rotation of the side groups (5,6,8,12), a change in the polarizability of its bonds, and the short-range ordering of the solvent molecules (11).

Principal values which effect the intrinsic segmental anisotropy of the chain are the anisotropy of the monomer unit and the equilibrium rigidity of the chain.

The Kuhn segment with the length A is the measure of the equilibrium rigidity of the chain. If the chain adopts the conformation of the Gaussian coil, then we have $A = \langle r^2 \rangle / L$, where L is the complete length of the extended chain and $\langle r^2 \rangle$ is the mean square end-to-end distance.

Also, $a_\parallel - a_\perp$ is the difference between the polarizabilities of the monomer unit in the parallel and perpendicular directions of the chain. Values of $a_\parallel - a_\perp$ presented in the table were calculated from

$$a_\parallel - a_\perp = (\alpha_1 - \alpha_2)\lambda / A \qquad \text{(A2)}$$

where λ is the length of the monomer unit in the chain direction. For cellulose esters, $a_\parallel - a_\perp$ and correspondingly, $\alpha_1 - \alpha_2$ depend on the degrees of substitution (DS), which are also given in the table.

For heterocyclic polycondensation polymers, the term "monomer unit" means the repeat unit of the polymer chain.

Many rigid-chain polymers have been investigated in the molecular-weight range in which their molecules exhibit a conformation intermediate between the rodlike conformation and the Gaussian coil. The best model for these polymers is the wormlike chain. Under these conditions the shear optical coefficient $\Delta n/\Delta\tau$ varies with the molecular weight M of the polymer and is given by the equation (100)

$$\Delta n/\Delta\tau = B\langle \gamma_1 - \gamma_2 \rangle \langle h^4 \rangle / \langle h^2 \rangle^2 \qquad \text{(A3)}$$

where $B = (4\pi/45kT)(n^2 + 2)^2/n$, $\langle \gamma_1 - \gamma_2 \rangle$ is the mean value of the wormlike chain optical anisotropy, $\langle h^2 \rangle$ and $\langle h^4 \rangle$ are the mean square and fourth degree end-to-end distance of the wormlike chain.

For the wormlike chain, the value of $\langle \gamma_1 - \gamma_2 \rangle$ as a function of the reduced chain length $x = 2L/A$ is given by the equations

$$\langle \gamma_1 - \gamma_2 \rangle = \frac{3}{5}(\alpha_1 - \alpha_2)\frac{f_1(x)}{1 - 0.8 f_1(x)/x} \qquad \text{(A4)}$$

$$f_1(x) = 1 - (1 - e^{-x})/x$$

According to Eqs. (A3) and (A4) the limit values of $\Delta n/\Delta\tau$ for wormlike chains are as follows: at $x \to 0$, $(\Delta n/\Delta\tau)_0 = B\langle \gamma_1 - \gamma_1 \rangle x \to 0$ as for a thin rod, and at $x \to \infty$, $(\Delta n/\Delta\tau)_\infty = B(\alpha_1 - \alpha_2)$ as for a kinetically flexible Gaussian coil.

The dependence of $\Delta n/\Delta\tau$ on M for the wormlike chains given by the Eqs. (A3) and (A4) may be approximated (100,101) by a linear relationship (A5) between the inverse values $(\Delta n/\Delta\tau)^{-1}$ and M^{-1}:

$$\frac{4\pi}{45kT} \times \frac{(n^2 + 2)^2}{n}(\Delta n/\Delta\tau)^{-1} = (\alpha_1 - \alpha_2)^{-1} + (M_0/\Delta a) \times M^{-1} \qquad \text{(A5)}$$

where M_0 is the molecular weight of the repeat unit of the chain. If the experimental dependence of $(\Delta n/\Delta\tau)^{-1}$ on $1/M$ is plotted, the slope of the straight line obtained makes it possible to calculate $\Delta a = a_\parallel - a_\perp$, and its intercept with the ordinate gives $\alpha_1 - \alpha_2$.

B. TABLES OF ANISOTROPY OF SEGMENTS AND MONOMER UNITS OF POLYMER MOLECULES

TABLE 1. MAIN-CHAIN ACYCLIC CARBON POLYMERS

Polymer	Solvent	$(\alpha_\parallel - \alpha_\perp) \times 10^{25}$ cm^3	$(a_\parallel - a_\perp) \times 10^{25}$ cm^3	Refs.
1.1. POLY(DIENES)				
Poly(butadiene) 1,4-*cis*	Benzene, sw. p.	61.3–63	30.8	52–54
	Carbon tetrachloride, sw. p.	53.5*	31.7*	53
	sw. p.	55.2*		54
	Cyclohexane, sw. p.	57.3	33.9	53
	Toluene, sw. p.	72	42.6	53
	p-Xylene, sw. p.	86.9	51.4	53

TABLE 1. *cont'd*

Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp})$ $\times 10^{25}$ cm^3	Refs.
Poly(butadiene) 1,4-*trans*	Benzene	71	37.4	52
	sw. p.	70.4		53
	Carbon tetrachloride, sw. p.	58.1*		55
	sw. p.	61.1*	36.3*	53
	Cyclohexane, sw. p.	57.3	33.1*	53
	Toluene, sw. p.	81.6	48.6	53
	p-Xylene, sw. p.	101	60.2	53
Poly(chloroprene)	α-Bromonaphthalene	110*		11
	Carbon tetrachloride	33		11
	Chlorobenzene	64		11
	Dichloroethane	39		11
	α-Methylnaphthalene	99*		11
	Tetrachloroethylene	46		11
	Toluene	67		11
	p-Xylene	88		11
Poly(isoprene) *cis*	Benzene	48	30.5	52
	Squalene, sw. p.	53.1		56
trans	Benzene	49	31	52
1.2. POLY(ALKENES)				
Poly(1-butene) atactic	Toluene	33.4		13
isotactic	Toluene	25.2		13
Poly(1-decene) isotactic	Toluene	−82.5		13
Poly(1-dodecene)	Toluene	−120		13
Poly(ethylene)	Tetralin, xylene	60		14
	Decalin	30		14
Poly(1-heptene) isotactic	Toluene	−24.5		13
Poly(1-hexadecene) isotactic	Toluene	−205 to −213		13
Poly(1-hexene) atactic	Toluene	12.1		13
isotactic	Toluene	−6.5		13
Poly(isobutene)	Benzene, chlorobenzene, tetrachloroethylene, *m*-xylene	45–59		11,14,15
	Carbon tetrachloride	35		11
	Decalin	30		14
	p-Xylene	69		11
Poly(1-octadecene) isotactic	Toluene	−257		13
Poly(1-octene) isotactic	Toluene	−39		13
Poly(1-pentene) isotactic	Toluene	8.0		13
		9.3		13
Poly(diphenylpropylene)	Bromoform	80		16
Poly(propylene) atactic	Benzene, xylene	45		14
	Carbon tetrachloride	30	3.5	17
	Decalin	30		14
	Toluene	55		13
isotactic	Carbon tetrachloride	30	3.5	17
Poly(1-tetradecene) isotactic	Toluene	−176		13
		−171		13
1.3. POLY(ACRYLIC ACID) AND DERIVATIVES				
Poly(acrylic acid)	Dioxane	~−0.5**	~−0.1**	18
−, sodium salt	0.0012 M NaCl, pH 7	20**		
	Water, pH 6.1	−4** to +10**		19
Poly(*n*-butyl acrylate)	Benzene	−11	−1.5	20
	Decalin	−17.8	−1.9	21
	Toluene	−10.1	−1.1	21
		−6.5	−0.9	20
Poly(cetyl acrylate)	Decalin	−141	−6.5	22
	Toluene	−164	−7.5	21
Poly(cholesteryl acrylate)	Benzene	−360	−16	23

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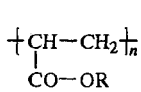
TABLE 1. *cont'd*

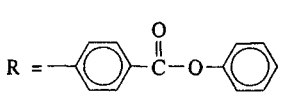
Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp}) \times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp}) \times 10^{25}$ cm^3	Refs.
Poly(decyl acrylate)	Decalin	− 74	− 3.7	21
	Toluene	− 95	− 4.7	21
Poly(ethyl acrylate)	Benzene, sw. p.	3.0	0.36	24
	Bromobenzene, sw. p.	10 **	1.2**	24
	Bromoform, sw. p.	− 37 **	4.5**	24
	Dibromoethane, sw. p.	− 14 **	− 1.7**	24
	Dimethylformamide, sw. p.	− 11 **	− 1.7**	24
Poly(methyl acrylate)	Benzene	17	2.5	20
	Toluene	16	1.9	21
		26	3.6	20
Poly(octadecyl acrylate)	Decalin	− 190	− 6.6	21
	Toluene	− 232	− 8.0	21
Poly(octyl acrylate)	Decalin	− 57.4	− 4.3	21
	Toluene	− 47.9	− 3.6	21

1.4. POLY(METHACRYLIC ACID) AND DERIVATIVES

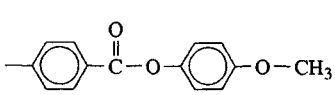
Poly(*n*-butyl methacrylate) atactic	Benzene	− 14	− 2.1	25
isotactic	Benzene	− 2.0	− 0.3	26
Poly(*tert*-butyl methacrylate) atactic	Benzene	2.1	0.3	20
isotactic	Benzene	19.3	3.0	20
Poly(*p*-*tert*-butylphenyl methacrylate)	Bromobenzene	− 90	− 7.5	22,28
Poly(*p*-carbethoxy-*N*-phenylmethacrylamide)	*o*-Toluidine	− 230	− 23	40
Poly(*p*-carboxyphenyl methacrylate)	Dioxane	180 *	8.0*	35
	0.1 M NaCl	370 *		35
Poly(cetyl methacrylate)	Benzene	− 160	− 8.9	29
Poly(*p*-chloro-*N*-phenylmethacrylamide)	*o*-Toluidine	− 160	− 20	40
Poly(glycol methacrylate)	Dimethylformamide	1.0	0.18	30
	Ethyl alcohol	− 6.0**	− 1.1**	30
	Water	− 6.0**	− 1.1**	30
Poly(hexyl methacrylate)	Benzene	− 40	− 4.6	31
	Carbon tetrachloride	− 9.7	− 1.1	31
Poly(methacrylic acid)	Methanol	50 *		32
	0.002 M HCl	150 *		32
–, sodium salt	0.012 M NaCl, pH 7	150 *		33
	0.0012 M NaCl, pH 7	400		33
	Water	56 *–300 *		34
Poly(*p*-methylcarboxy-phenyl methacrylate)	Dibromoethane	− 77	− 7	35
Poly(methyl methacrylate) atactic	Benzene	2.0	0.3	38
isotactic	Benzene	25	3.5	38
Poly(β-naphthyl methacrylate)	Tetrabromoethane	− 60	− 8.5	4
Poly(octyl methacrylate)	Benzene	− 47	− 5.9	31
	Carbon tetrachloride	− 12.5	− 1.6	31
Poly(*N*-phenylmethacrylamide)	*o*-Toluidine	− 103	− 13	40
Poly(phenyl methacrylate)	Bromobenzene	− 10.5	− 1.5	4

1.5. POLY(ACRYLIC ACID) DERIVATIVES WITH MESOGENIC SIDE GROUPS



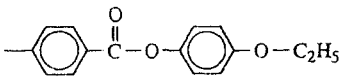
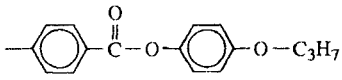
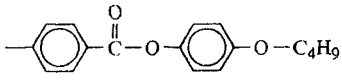
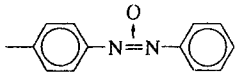
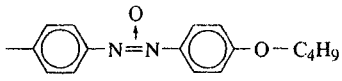
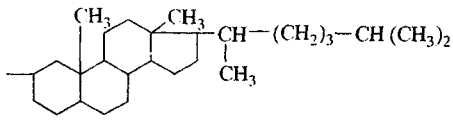


Tetrachloroethane	− 520*		100

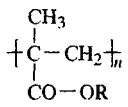


Tetrachloroethane	− 520*		100

TABLE 1. *cont'd*

Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp})$ $\times 10^{25}$ cm^3	Refs.
	Tetrachloroethane	−630*		100
	Tetrachloroethane	−810*		100
	Tetrachloroethane	−800*		100
	Tetrachloroethane	−450*	−19*	100,102
	Tetrachloroethane	−510*	−21	100,102
	Benzene	−300*	−15*	100,103

1.6. POLY(METHACRYLIC ACID) DERIVATIVES WITH MESOGENIC SIDE GROUPS



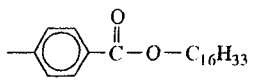
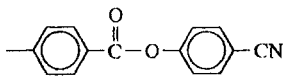
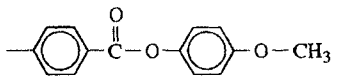
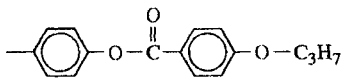
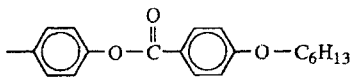
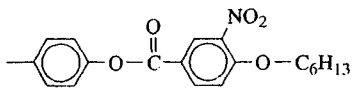
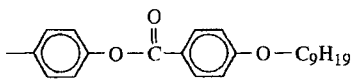
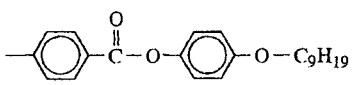
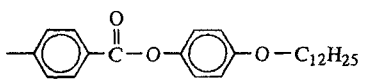
R =				
	Carbon tetrachloride	−445*	−18*	100,105
	Dimethylformamide	−240*		100,106
	Tetrachloroethane	−500*		100
	Dimethylformamide/ toluene (1/1)	−320*		36,100
	Benzene	−370*		100,107
	Dioxane	−1200*		100,106
	Carbon tetrachloride	−600*	−40*	100,108
	Carbon tetrachloride	−2700*	−110*	100,109
	Carbon tetrachloride	−2350*		100

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TABLE 1. *cont'd*

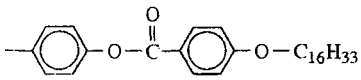
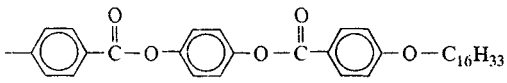
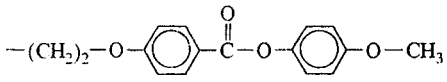
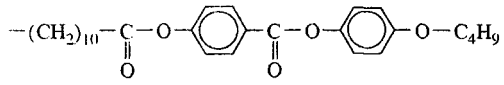
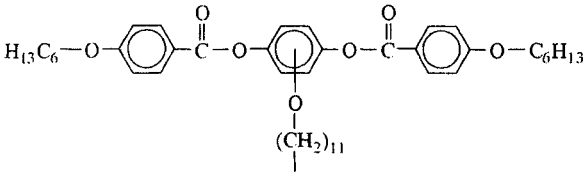
Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp})$ 10^{25} cm^3	Refs.
	Benzene	−1600*		36
	Bromoform	−1000*	−40*	36
	Chloroform	−1400*		36
	Carbon tetrachloride	−2700*	−110*	37,100
	Tetrahydrofuran	−890*		36
	Benzene/heptene (52/48), (66/34)	−4200*		36
Copolymer with cetylmethacrylate				
mol% 70/30	Carbon tetrachloride	−1050*	−44*	100,93
50/50	Carbon tetrachloride	−680*	−28*	100,93
	Benzene	−3000*	−150*	100,111
	Chloroform	−4900*	−100*	100,111
	Carbon tetrachloride	−117*		100,109
	Dioxane	−90*		100,109
	Tetrachloromethane	−180*		153
		−250*		153
1.7. VINYL POLYMERS				
Poly(acrylonitrile)	Dimethylformamide	−23	−1.8	4
Poly(*p*-chlorostyrene)	Bromoform	−230	−35	41
Poly(2,5-dichlorostyrene)	Bromoform	−265	−30	41
Poly(3,4-dichlorostyrene)	Tetrabromoethane	−300	−25	15
Poly(α-methylstyrene)	Tetrabromoethane	−133	−17	128
Poly(*p*-methylstyrene) atactic	Bromoform	−147	−20	42
isotactic	Bromoform	−140	−19	42
Poly(2,5-dimethylstyrene)	Bromoform	−180	−25	41
Poly(2-methyl-5-vinyl-*N*-butylpyridinium bromide)	0.01 M NaCl	−900*		43
	0.1 M NaCl	−270*		43
Poly(2-methyl-5-vinyl-pyridine)	Bromoform	−260	−29	44
Poly(2-methyl-5-vinyl-pyridinium chloride)	0.1 M HCl	−300*		45
Poly(styrene) atactic	Bromoform	−145	−18	46,41
isotactic	Bromoform	−224	−23	46
Poly(vinyl acetate)	Acetone	−20**	−5.3**	7
	sw. p.	−37**	−5.3**	10
	Benzene	4.0–5.9	0.5 to 0.8	6,8,9,11
	sw. p.	5.1**	0.75**	10
	Bromobenzene, sw. p.	9.4**	1.3**	10
	Bromoform	−20**		7
	sw. p.	−18.7**	−2.7**	10
	Carbon tetrachloride	−16	−2.0	9,11
		−26		6
	sw. p.	−25.2	−3.6	10
	Chlorobenzene	14**	1.75**	11
	Chloroform	−34.9**	−2.6**	6
		−24	−3.0	9,11
	sw. p.	−17.9		10
	Cyclohexanone	−23	−2.9	9,11
	Dichloroethane	−36	−4.5	11
		−39	−4.9	11

TABLE 1. *cont'd*

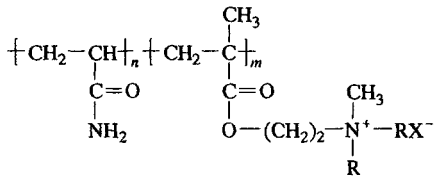
Polymer	Solvent	$(\alpha_\parallel - \alpha_\perp) \times 10^{25}$ cm^3	$(a_\parallel - a_\perp) \times 10^{25}$ cm^3	Refs.
	Tetrabromoethane	−25 **	−3.1 **	11
		−33 **		7
	sw. p.	−46.4 **	−6.6 **	10
	Toluene	10	1.25	9,11
		13.5	2.0	8
		19		6
	sw. p.	9.4 **	1.3 **	10
	o-Xylene	2.0	0.25	9,11
	sw. p.	9.8 **	1.4 **	10
Poly(vinyl butyral)	Chloroform	81		11,47
	Toluene/phenol (79/21)	173		11,47
Poly(vinyl butyrate)	Benzene	−8.0		11,47
	Carbon tetrachloride	−36		11,47
	Chloroform	−48		11,47
Poly(vinyl chloride)	Tetrahydrofuran	40		48
Poly(vinyl cinnamate)	Bromoform	−420	−35	49
Poly(β-vinylnaphthalene)	Tetrabromoethane	−430	−30	4
	Bromoform	−440	−20 to −30	39
Poly(vinyl propionate)	Benzene	−4.4		11,47
	Carbon tetrachloride	−31		11,47
	Chloroform	−40		11,47
	Toluene	1.3		11,47
Poly(4-vinylpyridine)	Bromoform	−240	−22	50
Poly(4-vinylpyridinium chloride)	0.1 M HCl	−260 *		50
	0.05 M HCl	−440 *		50
Poly(vinylpyrrolidone)	Benzyl alcohol	−75	−10	4
Poly(vinyl stearate)	Carbon tetrachloride	−130	−4.7	51

1.8. POLY(VINYL) DERIVATIVES WITH MESOGENIC SIDE GROUPS

Poly[4-(4-nonyloxy-benzamido)styrene]	Benzene	−2500 *	−100 *	100,104

1.9. COPOLYMERS, GRAFT AND BLOCK COPOLYMERS

Poly(acryl amide-*co*-methacrylic acid ether)



$R = -C_2H_5$; $X^- = -Cl^-$

n mol% = 0	0.001 M NaCl	2.5 *		157
6		15		
30		35 *		
69		90 *		
80		80 *		
100	Water	40 *		
Poly(*p*-(4-cetoxybenzoxy)-phenyl methacrylate-*co*-cetyl methacrylate)	Carbon tetrachloride			93
mol% 81/19		−2000 *		
60/40		−920 *		
59/41		540 *		
22/78		−400 *		
15/85		−277 *		
8/92		−180 *		
4/96		−16 *		

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TABLE 1. *cont'd*

Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp})$ $\times 10^{25}$ cm^3	Refs.
Poly(methyl methacrylate-*co*-*p*-*tert*-butylphenyl methacrylate) mol% 91/9	Chlorobenzene	1.5		94
80/29		− 7.4		
50/50		− 30.4		
25/75		− 44		
Poly(phenylbutyl isocyanate-*co*-chloral) mol% 50/50	Carbon tetrachloride	12 *		60
Poly(styrene-*co*-chlorostyrene) mol% 89.7/10.3	Bromoform	− 165		95
83.2/16.8		− 172		
68.7/31.3		− 198		
55/45		− 202		
39.5/60.5		− 226		
Poly(styrene-*co*-*N*-methylcitraconimide) mol% 54/46	Bromoform	− 26		63
48/52		− 34		
Poly(styrene-*co*-methyl methacrylate) mol% 70/30	Bromobenzene	− 88		96
50/50		− 68		
30/70	Chlorobenzene	− 34		
Poly(styrene-*block*-propylene), atactic mol% 64/36	Chlorobenzene	− 8		97
Poly(*n*-butyl methacrylate-*graft*-styrene) mol% 8/92	Bromoform	1190		98
Poly(*tert*-butyl methacrylate-*graft*-styrene) mol% 8/92	Bromoform	540		98
Poly(methyl methacrylate-graft-styrene) mol% 10/90	Bromoform	700–7000		98,99
87/13		30		98
(70–90)/(30–10)		100–1100		16
Poly(propylene-*graft*-atactic styrene) mol% 30/70	Chlorobenzene	22		93
Poly(vinyl chloride-*graft*-styrene) wt.% 5.2/94.8	Benzene	330		48
5.6/94.4		180		
12.1/87.9		155		
30.7/69.3		110		

TABLE 2. MAIN-CHAIN CARBOCYCLIC POLYMERS

Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp})$ $\times 10^{25}$ cm^3	Refs.
Poly(acenaphthylene) helix 4_1	Bromoform	− 300 **	− 13.6**	39
trans	Bromoform	− 300 **	− 20 **	39
Poly(1,2,3-trimethyl-2,3-dihydro-1,6-indendiyl)	Bromoform	78		92
Poly(1,2,3-trimethyl-2,3-dihydro-1,6-indendiyl-1,4-phenylene-ethylene)	Bromoform	142		92

TABLE 3. MAIN-CHAIN HETEROATOM POLYMERS

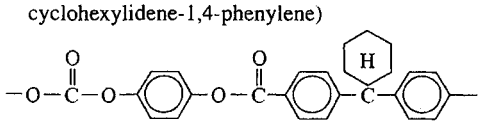
Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp})$ $\times 10^{25}$ cm^3	Refs.
3.1. POLY(OXIDES)				
Poly(oxypropylene)	Cyclohexanone	18	6	57
3.2. POLY(ESTERS)				
Poly(oxycarbonyloxy-1,4-phenylene-cyclohexylidene-1,4-phenylene) 	Bromoform	− 114	− 76	58
Poly(oxyethyleneoxyterephthaloyl)	Dichloroethane/phenol (1/1)	70		59
	Dichloroacetic acid	48.7*	33.7*	89

TABLE 3. *cont'd*

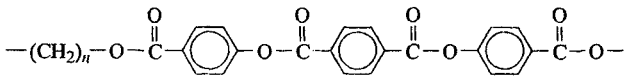
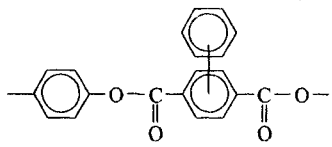
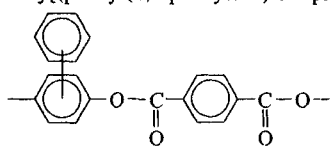
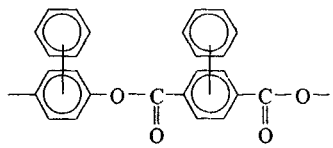
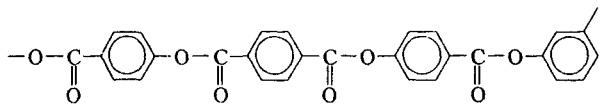
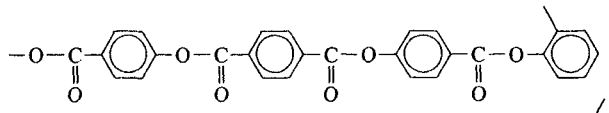
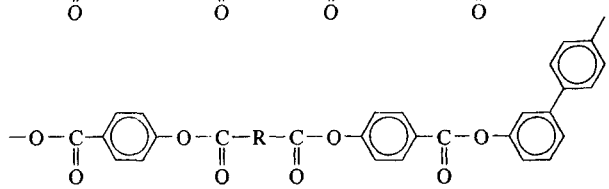
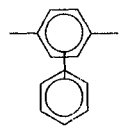
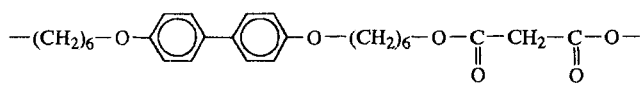
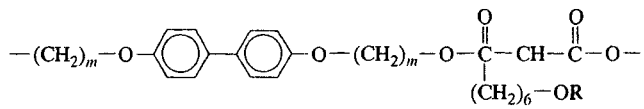
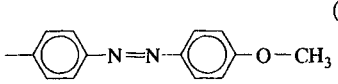
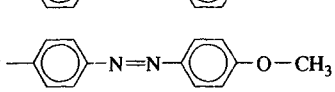
Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp})$ $\times 10^{25}$ cm^3	Refs.
Poly(oxy-*n*-methyleneoxycarbonyl-1,4-phenylene-oxyterephthaloyloxy-1,4-phenylene-carbonyl) 				
$n = 4$	Dichloroacetic acid	280 *		89
$n = 5$	Dichloroacetic acid	250 *		89
$n = 10$	Dichloroacetic acid	200 *	132 * – 165 *	89
Poly[1,4-phenylene (phenylterephthalate)] 	Dichloroacetic acid	1800 *		158
	Tetrachloroethane	1800 *		158
Poly[(phenyl(1,4-phenylene)-terephthalate] 	Dichloroacetic acid	3000 *		159
	Dichloroacetic acid	1600 *		160
	Dioxane	1600 *		160
	Dichloroacetic acid	600 *		161
	Dichloroacetic acid	18.2 *		162
				
R = 	Dioxane	375 *		163
—$(CH_2)_6$—	Dioxane	185 *		163
	Tetrachloroethane	153 *	315 *	164
				
$m = 2$, R = 	Tetrachloroethane	150 *	136 *	165
$m = 6$, R = 	Tetrachloroethane	200 *	365	166

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TABLE 3. *cont'd*

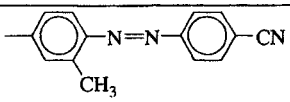
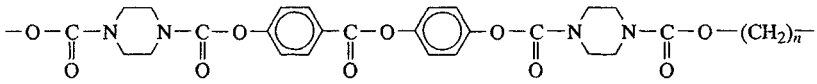
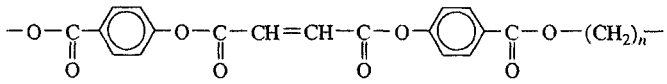
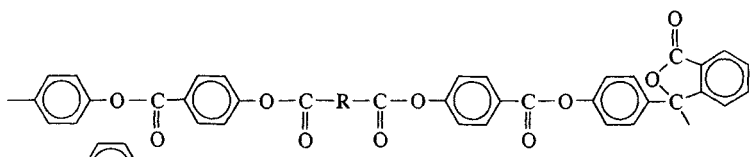
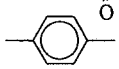
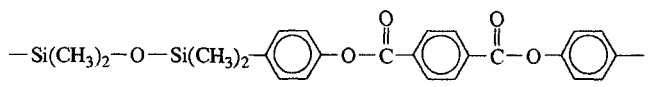
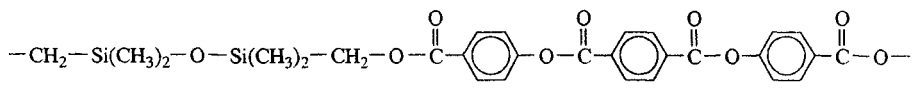
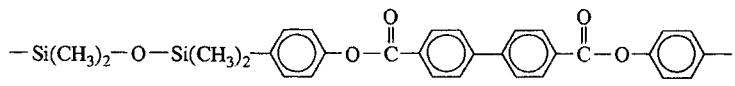
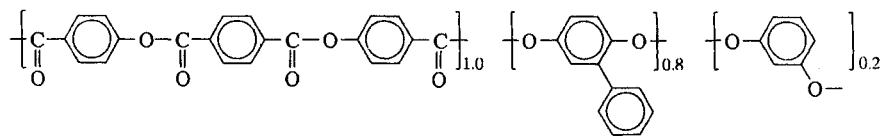
Polymer	Solvent	$(\alpha_{\parallel}-\alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel}-a_{\perp})$ $\times 10^{25}$ cm^3	Refs.
$m = 6$, R = 	Dioxane	260*	513*	167
				
$n = 2$	Tetrachloroethane	186*		168
$n = 9$	Chloroform	270*		168
	Dioxane	254*		168
	Tetrachloroethane	256*		168
				
$n = 3$	Dichloroacetic acid	268*		169
4	Trifluoroacetic acid	594*		
5	Tetrachloroethane	221*		
	Trifluoroacetic acid	396*		
6	Tetrachloroethane	249*		
7	Tetrachloroethane	187*		
8	Tetrachloroethane	179*		
9	Tetrachloroethane	188*		
10	Tetrachloroethane	196*		
				
R = 	Tetrachloroethane	380*		170
	Tetrachloroethane	228*		170
	Tetrachloroethane	57.5*		170
	Tetrachloroethane	259*	144*	171
	Tetrachloroethane	139*	146*	171
	Tetrachloroethane	353*	196*	171
	Dichloroacetic acid	2120*		172

TABLE 3. *cont'd*

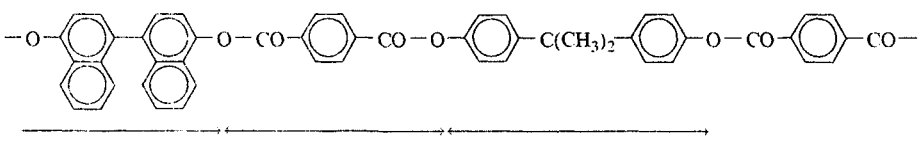
Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp})$ $\times 10^{25}$ cm^3	Refs.
				
N T D				
N : T : D (mol%) = 0 : 1 : 1	Tetrachloroethane	229*		91
0.1 : 1 : 0.9		250*		
0.2 : 1 : 0.8		335*		
0.3 : 1 : 0.7		410*		
0.4 : 1 : 0.6		510*		
0.5 : 1 : 0.5		630*		
0.6 : 1 : 0.4		740*		
0.7 : 1 : 0.3		1100*	280*	91
3.3. POLY(AMIDES)				
Poly(*n*-butyliminocarbonyl) (Poly(butylisocyanate))	Carbon tetrachloride	4100	11	60,100
	Carbon tetrachloride/ pentafluorophenol (0.9/0.1)	800	11	100
Poly(chlorohexyliminocarbonyl)	Carbon tetrachloride	3000		60
Poly(hydrazocarbonyl-1,4-phenylene iminoterephthaloyl)	Dimethylsulfoxide	3630*	85*	100,123
Poly(iminocarbonyl-cyclohexylene)	Sulfuric acid	390*		100,115
Poly(iminocarbonyl-1,4-phenylene)	Sulfuric acid	10500*	103*	100,117,118
Poly[imino-(4,6-dicarboxyisophthaloyl -imino) biphenyl-4,4′-diyl]	Dimethylacetamide	75*		100
Poly[imino (1-oxohexamethylene)] (Poly(ε-caprolactam))	Sulfuric acid	63*		100,115
Poly[imino-1,4-phenyleneimino-(4,6-dicarboxyisophthaloylimino)-1,4-phenyleneimino-terephthaloyl]	Dimethylacetamide	88*		100
Poly(imino-1,3-phenyleneiminoisophthaloyl)	Sulfuric acid	360*	100*	100,121,122
Poly(imino-1,4-phenyleneiminoterephthaloyl)	Sulfuric acid	5250*	206*	100,119,120
Poly(imino-1,4-phenyleneimino-terephthaloyl), copolymer with poly(iminocarbonyl-1,4-phenylene) (ratio 1/9)	Sulfuric acid	4380*		110,119
Poly(iminio-terephthaloylimino-1,4-phenylenediphenylmethyl-1,4-phenylene)	Sulfuric acid	130*		100,127
Poly(*n*-tolyliminocarbonyl)	Bromoform	− 39	− 5	60,100
3.4. POLY(PEPTIDES) AND NUCLEIC ACIDS				
Poly(γ-benzyl-L-glutamate) helix	Dichloroethane	25000*	25*	86,87,100
coil	Dichloroacetic acid	230*	25*	88,100
Poly(*S*-carbobenzoxymethyl-L-cystein)	Dichloroacetic acid	22*		86
Poly(ε,*N*-carbobenzoxy-L-lysine) helix	Dimethylformamide	3600*	12*	86
coil	Dichloroacetic acid	< +		86
Poly(L-glutamic acid)	Phosphate buffer			
helix	0.1 M, pH 4.2	1900*		86
coil	0.1 M, pH 12.5	136*		86
DNA	aqueous 0.2 M NaCl	− 30000*	− 67*	100
	< 2 M NaCl	− 33000*		154,157
	> 3 M NaCl	− 41000*		154,157
3.5. POLY(PHOSPHAZENES)				
Poly(fluoroalkoxy phosphazene)				
$-PR_2=N-$				
R = $-OCH_2-CF_3$	Ethyl acetate	160		155
$-OCH_2-(CF_2)_2-H$	Ethyl acetate	145		156
$-OCH_2-(CF_2)_4-H$	Ethyl acetate	106		156

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TABLE 3. *cont'd*

Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp}) \times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp}) \times 10^{25}$ cm^3	Refs.
3.6. POLY(SILOXANES)				
Poly(oxydimethylsilylene)	Petroleum ether	4.7	0.96	77
Poly(oxyethylphenylsilylene) 14 : 86	Benzene	− 230 *	− 29.5	173
Poly(oxymethylphenylsilylene) 90 : 10	Petroleum ether	− 2.3	− 0.47	77
87.5 : 12.5	Decalin	− 10	− 2.0	78
75 : 25	Benzene	− 13.6	− 2.7	79
	Decalin	− 21	− 4.0	78
62.5 : 37.5	Decalin	− 36 **	− 7.2**	78
50 : 50 atactic	Benzene	− 25.5	− 5.1	79
	Decalin, tetralin	− 52 **	− 10 **	78
isotactic	Decalin, tetralin	− 82 **	− 16 **	78
Poly(oxyphenylsilylene)	Benzene	− 85	− 17	79
LADDER POLYMERS				
Poly(isobutylphenyl-silsesquioxane) (1 : 1)	Benzene	− 840	− 22	80,100
Poly(isohexylphenyl-silsesquioxane) (1 : 1)	Benzene	− 980	− 19.5	80,100
Poly(*m*-chlorophenyl-silsesquioxane)	Carbon tetrachloride	− 4700 *	− 40 *	100,80
Poly(dichlorophenyl-silsesquioxane)	Bromoform	− 4450	− 50	100,112
	Tetrabromoethane	− 4700	− 53	100,112
Poly(3-methyl-1-butene-silsesquioxane)	Benzene	− 570 *	− 6.5 *	81,100
	Butylacetate	− 400 *	− 4.2 *	81,82,100
Poly(phenyl-silsesquioxane)	Bromoform	− 1060 to − 1800	− 23 to − 31	83–85, 100

TABLE 4. MAIN-CHAIN HETEROCYCLIC POLYMERS

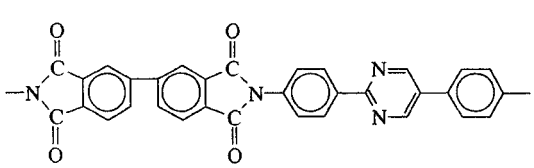
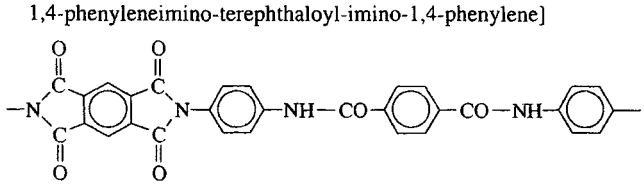
Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp}) \times 10^{25}$ cm^3	$a_{\parallel} - a_{\perp} \times 10^{25}$ cm^3	Refs.
4.1. POLY(IMIDES)				
Poly(*N*-2,4-dimethylphenylmaleimide)	Bromoform	− 200	− 12.5	61,100
Poly(*N*-isobutylmaleimide)	Chlorobenzene	160	10	62,100
Poly(*N*-methylcitraconimide)	Bromoform	150	16.7	63
Poly(*N*,*N*-piperazindiyl-2,5-diketo-1,3-pyrrolidindiyl-hexamethylene-2,5-diketo-1,3-pyrrolidindiyl)	Benzyl alcohol	56	31	64
	Bromoform/cyclohexanol (60 : 40)	30		64
Poly(*p*-tolylmaleimide)	Bromoform	− 160	− 10	65
	H_2SO_4, 101%	500 *	50 *	110
4.2. POLY(PYRROMELLITIMIDES)				
Poly[1,3,5,7-tetraoxo-2,3,6,7-tetrahydro-1H,5H-benzo-[1,2-c,4,5-c′ dipyrrol-2,6-diyl)-1,4-phenylene]	H_2SO_4	2500 * – 4400 *		100
Poly[(1,3,5,7-tetraoxo-2,3,6,7-tetrahydro-1H,5H-benzo [1,2-c,4,5-c′ dipyrrol-2,6-diyl)-1,4-phenyleneimino-terephthaloyl-imino-1,4-phenylene]				
	H_2SO_4	3400 *		100

TABLE 4. *cont'd*

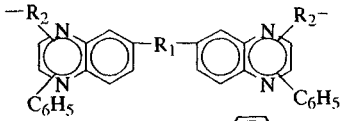
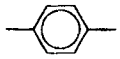
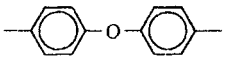
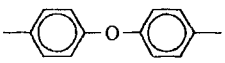
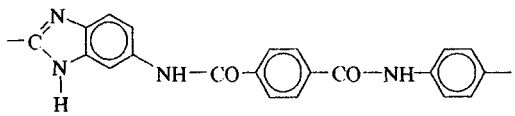
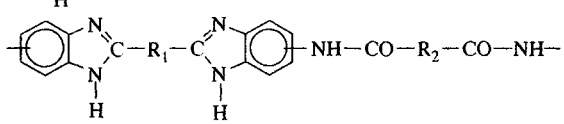
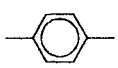
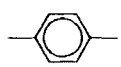
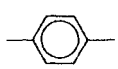
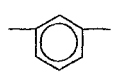
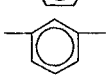
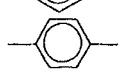
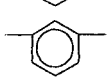
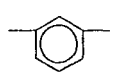
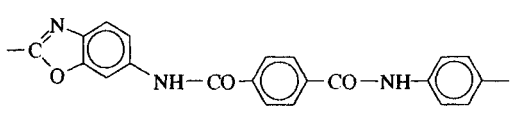
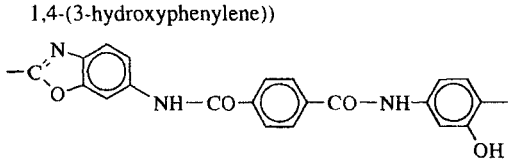
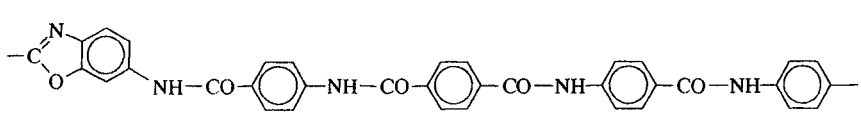
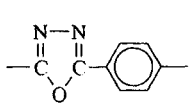
Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp})$ $\times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp})$ $\times 10^{25}$ cm^3	Refs.
4.3. POLY(QUINOXALINES)				
				
R_1 = nought, R_2 = 	Tetrachloroethane, chloroform	1750*		100,116
R_1 = nought, R_2 = 	Tetrachloroethane, chloroform	1060*		100,116
R_1 = —O—, R_2 = 	Tetrachloroethane, chloroform	650*		100,116
4.4. POLY(BENZIMIDAZOLES), POLY(BENZOXAZOLES)				
Poly(benzimidazole-2,6-diyl-iminoterephthaloylimino-1,4-phenylene)				
	H_2SO_4	5080*	360*	100,124
				
R_1 =  , R_2 = 	H_2SO_4 H_2SO_4	4900* 3625*		100 150
R_1 =  , R_2 = 	H_2SO_4	1250*		150
R_1 =  , R_2 = 	H_2SO_4	1250*		150
R_1 =  , R_2 = 	H_2SO_4	750*		150
Poly(benzoxazole-2,6-diyl-iminoterephthaloyliminio-1,4-phenylene)				
	H_2SO_4	5940*	360*	100,125
Poly(benzoxazole-2,6-diyl-iminoterephthaloylimino-1,4-(3-hydroxyphenylene))				
	H_2SO_4	5940*	320*	100,126
Poly(benzoxazole-2,4-diyl-iminoterephtaloylimino-1,3-(4-hydroxyphenylene))	H_2SO_4	2340*	300*	100,126
Poly(benzoxazole-2,6-diyl-iminocarboxy-1,4-phenylene-iminoterephthaloylimino-1,4-phenylene-carboxyimino-1,4-phenylene)				
	H_2SO_4	4100*		152
Poly(oxadiazole-2,5-diyl-1,4-phenylene)				
	H_2SO_4	1750*	80*	100,114

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TABLE 4. *cont'd*

Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp}) \times 10^{25}$ cm^3	$(a_{\parallel} - a_{\perp}) \times 10^{25}$ cm^3	Refs.
4.5. POLY(SACCHARIDES)				
CELLULOSE ESTERS				
Benzyl cellulose, DS = 2.5	Dioxane	294*	5*	67,100
Carboxymethyl cellulose, DS = 0.9	Cadoxene	310*		147
	2 M NaOH	450*		147
	0.5 M NaOH	560*		147
Cellulose acetate, DS = 2.4	Aniline	260*	5.3*	136
	Dioxane	39*	0.78*	136
	Pyridine	0	0	100
	Trichloromethane	−8.4*	−0.21*	136
	Tetrachloroethane	−57*	−1.6*	136
Cellulose acetate benzoate, DS = 55/245	Dioxane	−1330*	−26.6*	137
Cellulose acetate benzhydroxamate, DS = 15/285	Dioxane	−870*	−17.4*	137
60/240	Dioxane	−575*	−11.5*	137
90/170	Dioxane	−350*	−7*	137
Cellulose acetate cinnamate, DS = 60/200	Dioxane	−1900*	−38*	137,138
160/140	Dioxane	−1300*	−26*	137,138
250/50	Dioxane	−450*	−9*	137,138
Cellulose acetate diphenylacetate, DS = 10/290	Dioxane	1280*	25.6*	137,138
40/150	Dioxane	1200*	24.0*	137,138
20/130	Dioxane	1345*	26.9*	137,138
130/115	Dioxane	1020*	20.4*	137,138
270/30	Dioxane	405*	8.1*	137,138
Cellulose acetate myristate, DS = 10/290	Benzene	−110*	−2.2*	137
	o-Dichlorobenzene	−164*	−3.1*	136
	Dichloromethane	−190*	−3.8*	136
	Trichloromethane	−210*	−4.2*	136
	Tetrachloromethane	−223*	−4.5*	136
	Pentachloroethane	−194*	−4.5*	136
	para-Chlorotoluene	−171*	−3.6*	136
	α-Bromonapthaline	142*	3.6*	136
DS = 25/275	Tetrachloromethane	−228*	−4.6*	136
50/250	Tetrachloromethane	−217*	−4.3*	136
100/200	Tetrachloromethane	−240*	−4.8*	136
150/150	Dioxane	−72*	−1.4*	136
	Tetrachloroethane	−150*	−4.3*	136
200/100	Tetrachloroethane	−100*	−3.0*	136
245/55	Tetrachloroethane	−66*	−1.8*	136
290/10	Tetrachloroethane	−25*	−0.6*	136
Cellulose acetate monophenylacetate, DS = 40/260	Dioxane	365*	7.3*	137
100/150	Dioxane	250*	5.0*	137
200/70	Dioxane	90*	1.8*	137
246/32	Dioxane	50*	1.0*	137
285/5	Dioxane	65*	1.3*	137
Cellulose acetate α-naphthylacetate, DS = 250/50	Dioxane	70*	1.4*	137
275/25	Dioxane	70*	1.4*	137
290/10	Dioxane	55*	1.1*	137
Cellulose acetate nitrobenzoate, DS = 275/25	Dioxane	−115*	−2.3*	137
Cellulose acetate triphenylmethylate, DS = 110/145	Dioxane	1100*	22*	137
Cellulose benzoate, DS = 3.0	Dimethylformamide	−617*	−10.4*	66,100
	Chloroform	−763*	−16.4*	66,100
	Bromobenzene	−914*	−19.6*	66,100
	Dimethylphthalate	−830*	−17.8*	66,100
	Dioxane	−447*	67	
2.16	Dioxane	−1350*	−27*	137
Cellulose butyrate, DS = 3.0	Tetrachloroethane	−34	−2.6	72,100
	Bromoform	35*	2.7*	72,100
	Dioxane	61*	1.0*	67
	Methyl ethyl ketone	144*	5.6*	72,100
2.75	Dioxane	90*	1.8*	136
	Trichloromethane	$<\|1\|$*	$<\|0.03\|$	136
	Tetrachloroethane	−47.3*	−1.6*	136

TABLE 4. *cont'd*

Polymer	Solvent	$(\alpha_\parallel - \alpha_\perp)$ $\times 10^{25}$ cm^3	$(a_\parallel - a_\perp)$ $\times 10^{25}$ cm^3	Refs.
Cellulose dimethylphosphonocarbamate	0.01 M NaCl	710 *	16 *	68
DS = 2.0	0.2 M NaCl	640 *	16 *	68
Cellulose diphenylacetate	Acetophenone	1360	22	69
	Dioxane	1030 *	22 *	67
DS = 2.9	Dioxane	1760 *	35.2*	137,138
Cellulose diphenylphosphonocarbamate DS = 2.2	Dioxane	626 *–640	10*– 13.7*	100,129
Cellulose hydroxypropylacetate, DS = 3.3, 2.3	Tetrachloroethane	400	12	144,145
Cellulose hydroxypropylbutyrate, DS = 3.3, 2.2	Tetrachloroethane	270	8.3	145,146
DS = 3.3, 0.6	Tetrachloroethane	490	12	145,146
Cellulose hydroxypropylcaprinate, DS = 3.3, 2.0	Tetrachloroethane	230		145
Cellulose hydroxypropylstearate, DS = 3.3, 0.9	Tetrachloroethane	230		145
Cellulose monophenylacetate	Bromobenzene	600	13	100,129
DS = 2.8	Bromoform	478	10.3	100,129
Cellulose nitrate, DS = 2.8	Cyclohexanone	– 820	– 14	70,100
2.7	Cyclohexanone	– 540	– 9.0	100,130
	Amyl acetate	– 320 *	– 5.4*	100,130
	Acetone	– 115 *	– 1.9*	100,130
	Butyl acetate	– 300 *	– 5.0*	100,130
	Ethyl acetate	– 140 *	– 2.3*	100,130
2.3	Cyclohexanone	– 330	– 8.4	100,131
1.9	Cyclohexanone	– 62	– 2.0	100,132
	Dioxane	149	4.8	100,132
Cellulose phenylcarbamate	Acetophenone	– 1100	– 18	69
DS = 3.0	Benzophenone 55°	– 742		71
	80°	– 572		71
	Dioxane 21°	– 1830 *	– 45.8*	100,134
	65°	– 872 *		100,134
	Ethyl acetate	– 560 *	– 15.2*	100,134
2.2	Dioxane	– 1880 *	– 47.0*	100,133
2.3	Dioxane	– 1700 *	– 34 *	140
	Ethyl acetate/dioxane (0.017/1)	– 1440 *	– 45 *	140
	(0.034/1)	– 1664 *	– 52 *	140
	(0.063/1)	– 1440 *	– 45 *	140
	(0.091/1)	– 1360 *	– 40 *	140
	(0.224/1)	– 1260 *	– 35 *	140
	(0.469/1)	– 962 *	– 26 *	140
2.7	Dioxane	– 2030 *	– 40 *	141
	Ethyl acetate	– 1216 *	– 32 *	141
	Cyclohexanone	– 1254 *	– 38 *	141
	Dimethylacetamide	– 1280 *	– 40 *	141
Cellulose stearate DS = 2.0	Tetrachloroethane	– 500		72
Cyanoethyl cellulose DS = 2.6	Cyclohexanone	900	17.8	100,135
2.66	Cyclohexanone	970 *		139
2.37	Cyclohexanone	715 *		139
Cyanoethylacetyl cellulose	Cyclohexanone, sw. p.	+ 15	0.4	73,74
	Acetone	390 *	17.0*	73,100
	Dimethylformamide	90	3.9	73,100
Cyanoethyl hydroxyethyl cellulose DS = 1.8	Cyclohexanone	900		148
Cyanoethyl hydroxypropyl cellulose DS = 2.4	Cyclohexanone	660		148
Cyanoethyltrityl cellulose	Cyclohexanone, sw. p.	220		74
Cyanoethyl nitrate cellulose DS = 248/26	Cyclohexanone	855 *		139,151
233/52	Cyclohexanone	890 *		139,151
181/79	Cyclohexanone	470 *		139,151
100/117	Cyclohexanone	165 *		139,151
62/192	Cyclohexanone	35 *		139,151
36/213	Cyclohexanone	– 190 *		139,151
34/226	Cyclohexanone	– 115 *		139,151
19/242	Cyclohexanone	– 190 *		139,151
Ethyl cellulose	Carbon tetrachloride	430	11	75
DS = 2.5	Dioxane	512	4.6	100,129
Hydroxypropyl cellulose, DS = 2.5–3.8	Dimethylacetamide	450 *	9.9*	142,143
	Tetrachloroethane	505	11.3	142,143
	Water		12.4*	143

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TABLE 4. *cont'd*

Polymer	Solvent	$(\alpha_{\parallel} - \alpha_{\perp}) \times 10^{25}$ cm^3	$(a_{\parallel} - a_{-}) \times 10^{25}$ cm^3	Refs.
Polymaltotriose	Dimethylsulfoxide	33 to 39	7	149
	Water	109*		149
Sulfate cellulose, sodium salt	Aqueous NaCl 0.2 M	634*	16*	76, 100
DS = 0.4	0.15 M	645*	17*	76,100
	0.10 M	680*	16*	76,100
	0.01 M	980*	16*	76,100
	0.005 M	1300*	16*	76,100
	0.001 M	1750*	16*	76,100
CHITINS				
Chitozane	Acetic acid + NaCl, IS = 0.1 to 0.5	670*		174

C. REFERENCES

1. W. Kuhn, H. Kuhn, Helv. Chim. Acta, **20**, 1394 (1943).
2. A. Lodge, Trans. Faraday Soc., **42**, 273 (1946).
3. V. N. Tsvetkov, Chapter XIV in Bacon Ke (Ed.) "Newer Methods of Polymer Characterization", Interscience, New York, 1964.
4. V. N. Tsvetkov, V. E. Eskin, S. Ya. Frenkel, "Structure of Macromolecules in Solutions", National Lending Library for Science and Technology, Boston Spa, Yorkshire, England, 1971.
5. V. N. Tsvetkov, J. Polym. Sci., **57**, 727 (1962).
6. E. V. Frisman, V. A. Andreichenko, Vysokomol. Soedin., **4**, 1559 (1962).
7. E. V. Frisman, A. B. Chzu, Vysokomol. Soedin., **4**, 1564 (1962).
8. V. N. Tsvetkov, N. N. Boitsova, M. G. Vitovskaja, Vysokomol. Soedin., **6**, 267 (1964).
9. E. V. Frisman, G. A. Dyuzev, A. K. Dadivanyan, Vysokomol. Soedin., **6**, 341 (1964).
10. V. N. Tsvetkov, A. E. Grishchenko, L. E. De-Millo, E. N. Rostovskii, Vysokomol. Soedin., **6**, 384 (1964).
11. E. V. Frisman, A. K. Dadivanyan, J. Polym. Sci. C, **16**, 1001 (1967).
12. V. N. Tsvetkov, Leningrad. Univ. Vestnik, Ser. Fiz. Khim, **N22**, 39 (1961).
13. W. Philippoff, E. G. M. Tornqist, J. Polym. Sci. C, **N23**, 881 (1968).
14. T. I. Garmonova, Leningrad. Univ. Vestnik, Ser. Fiz. Khim, **N22**, 72 (1961).
15. V. N. Tsvetkov, V. E. Bychkova, S. M. Savvon, I. K. Nekrasov, Vysokomol. Soedin., **1**, 1407 (1959).
16. P. Gramain, J. Leray, H. Benoit, J. Polym. Sci. C, **16**, 3983 (1968).
17. V. N. Tsvetkov, O. V. Kallistov, E. V. Korneeva, I. K. Nekrasov, Vysokomol. Soedin., **5**, 1538 (1963).
18. V. N. Tsvetkov, S. Ya. Ljubina, T. V. Barskaya, Vysokomol. Soedin., **6**, 1806 (1964).
19. W. Kuhn, H. Oswald, H. Kuhn, Helv. Chim. Acta, **36**, 1209 (1953).
20. V. N. Tsvetkov, N. N. Boitsova, M. G. Vitovskaja, Vysokomol. Soedin., **6**, 297 (1964).
21. V. N. Tsvetkov, L. N. Andreeva, E. V. Korneeva, P. N. Lavrenko, Dokl. Akad. Nauk SSSR, **205**, 895 (1972).
22. V. N. Tsvetkov, L. N. Andreeva, E. V. Korneeva, P. N. Lavrenko, N. A. Plate, V. P. Shibaev, B. S. Petrukhin, Vysokomol. Soedin., **13A**, 2226 (1971).
23. V. N. Tsvetkov, E. V. Korneeva, I. N. Shtennikova, P. N. Lavrenko, G. F. Kolbina, D. Hardi, K. Nitrai, Vysokomol. Soedin., **14A**, 427 (1972).
24. A. E. Grishchenko, E. P. Vorobjeva, Vysokomol. Soedin., **15A**, 895 (1973).
25. V. N. Tsvetkov, S. Ya. Ljubina, Vysokomol. Soedin., **1**, 577 (1959).
26. V. N. Tsvetkov, M. G. Vitovskaja, S. Ya. Ljubina, Vysokomol. Soedin., **4**, 577 (1962).
27. V. N. Tsvetkov, S. Ya. Magarik, Dokl. Akad. Nauk SSSR, **115**, 911 (1957).
28. V. N. Tsvetkov, I. N. Shtennikova, Zh. Tekh. Fiz., **29**, 885 (1959).
29. V. N. Tsvetkov, D. Hardi, I. N. Shtennikova, E. V. Korneeva, G. F. Pirogova, K. Nitrai, Vysokomol. Soedin., **11A**, 349 (1969).
30. A. E. Grishchenko, R. I. Esrielev, Vysokomol. Soedin., **14A**, 521 (1972).
31. A. E. Grishchenko, M. G. Vitovskaja, V. N. Tsvetkov, L. N. Andreeva, Vysokomol. Soedin., **8A**, 800 (1972).
32. V. N. Tsvetkov, S. Ya. Ljubina, K. L. Bolevskii, Vysokomol. Soedin. Sb. Karbotsepnie Soedin., **4**, 26 (1959).
33. V. N. Tsvetkov, S. Ya. Ljubina, K. L. Bolevskii, Vysokomol. Soedin. Sb. Karbotsepnie Soedin., **4**, 33 (1959).
34. W. Kuhn, O. Kuenzle, A. Katchalsky, Helv. Chim. Acta, **31**, 1994 (1948).
35. E. N. Zakharova, P. N. Lavrenko, G. A. Fomin, I. I. Konstantinov, Vysokomol. Soedin., **13A**, 1870 (1971).
36. V. N. Tsvetkov, I. N. Shtennikova, E. I. Rjumtsev, G. F. Kolbina, I. I. Konstantinov, Ju. B. Amerik, B. A. Krentsel, Vysokomol. Soedin., **11A**, 2528 (1969).

37. V. N. Tsvetkov, I. N. Shtennikova, E. I. Rjumtsev, G. F. Kolbina, I. I. Konstantinov, Ju. B. Amerik, B. A. Krentsel, Vysokomol. Soedin., **15A**, 2158 (1973).
38. V. N. Tsvetkov, N. N. Boitsova, Vysokomol. Soedin., **2**, 1176 (1960).
39. V. N. Tsvetkov, M. G. Vitovskaja, P. N. Lavrenko, E. N. Zakharova, Vysokomol. Soedin., **13A**, 2532 (1971).
40. V. N. Tsvetkov, V. E. Bychkova Vysokomol. Soedin., **6**, 600 (1964).
41. E. V. Frisman, A. M. Maztsinivsky, N. A. Domnicheva, Vysokomol. Soedin., **2**, 1148 (1960).
42. V. N. Tsvetkov, N. N. Boitsova, Vysokomol. Soedin., **5**, 1263 (1963).
43. V. N. Tsvetkov, S. Ya. Ljubina, V. E. Bychkova, I. A. Strelina, Vysokomol. Soedin., **8**, 846 (1966).
44. I. N. Shtennikova, E. V. Korneeva, V. E. Bychkova, G. M. Pavlov, G. S. Sogomonyants, Vysokomol. Soedin., **14B**, 118 (1972).
45. S. Ya. Ljubina, I. A. Strelina, G. S. Sogomonyants, S. I. Dmitrieva, O. Z. Korotkina, G. V. Tarasova, V. S. Skazka, V. M. Jamschchikov, Vysokomol. Soedin., **12A**, 1560 (1970).
46. V. N. Tsvetkov, S. Ja. Magarik, Dokl. Akad. Nauk SSSR, **127**, 840 (1959).
47. E. V. Frisman, A. K. Dadivanyan, G. A. Dyuzhev, Dokl. Akad. Nauk SSSR, **153**, 1062 (1963).
48. S. P. Micengendler, K. I. Sokolova, G. A. Andreeva, A. A. Korotkov, T. Kadirov, S. I. Klenin, S. Ja. Magarik, Vysokomol. Soedin., **9A**, 1133 (1967).
49. V. N. Tsvetkov, E. N. Zakharova, G. A. Fomin, P. N. Lavrenko, Vysokomol. Soedin., **14A**, 1956 (1972).
50. S. Ya. Ljubina, I. A. Strelina, G. V. Tarasova, V. S. Skazka, V. M. Jamschchikov, Vysokomol. Soedin., **14A**, 1371 (1972).
51. V. S. Skazka, G. A. Fomin, G. V. Tarasova, I. G. Kirillova, V. M. Jamschchikov, A. E. Grishchenko, I. A. Shefer, I. S. Alekseeva, Vysokomol. Soedin., **15A**, 2561 (1973).
52. I. Ja. Poddubnyi, E. G. Erenburg, M. A. Eruomina, Vysokomol. Soedin., **10A**, 1381 (1968).
53. M. Fukuda, G. L. Wilkes, R. S. Stein, J. Polym. Sci. A-2, 1417 (1971).
54. T. Ishikava, K. Nagai, J. Polym. Sci. A-2, 1123 (1969).
55. T. Ishikava, K. Nagai, Polym. J. (Japan), **1**, 116 (1970).
56. L. R. G. Treloar, Trans. Faraday Soc., **43**, 234 (1947).
57. V. N. Tsvetkov, T. I. Garmonova, R. P. Stankevitch, Vysokomol. Soedin., **8**, 980 (1966).
58. T. I. Garmonova, M. G. Vitovskaja, P. N. Lavrenko, V. N. Tsvetkov, E. V. Korovina, Vysokomol. Soedin., **13**, 884 (1971).
59. S. M. Savvon, K. K. Turoverov, Vysokomol. Soedin., **6**, 205 (1964).
60. V. N. Tsvetkov, I. N. Shtennikova, E. I. Rjumtsev, Ju. P. Getmanchuk, Eur. Polym. J., **1**, 767 (1971).
61. V. N. Tsvetkov, G. V. Tarasova, E. L. Vinogradov, N. N. Kuprijanova, V. M. Jamshchikov, V. S. Skazka, V. S. Ivanov, V. K. Smirnova, I. I. Migunova, Vysokomol. Soedin., **10A**, 620 (1968).
62. V. N. Tsvetkov, G. A. Fomin, P. N. Lavrenko, I. N. Shtennikova, T. V. Sheremeteva, Vysokomol. Soedin., **10A**, 903 (1968).
63. M. G. Vitovskaja, V. N. Tsvetkov, L. I. Godunova, T. V. Sheremeteva, Vysokomol. Soedin., **9A**, 1682 (1967).
64. T. I. Garmonova, M. G. Vitovskaja, S. V. Bushin, T. V. Sheremeteva, Vysokomol. Soedin., **16A**, 265 (1974).
65. V. N. Tsvetkov, N. N. Kuprijanova, G. V. Tarasova, P. N. Lavrenko, I. I. Migunova, Vysokomol. Soedin., **12A**, 1974 (1970).
66. V. N. Tsvetkov, I. N. Shtennikova, Vysokomol. Soedin., **6**, 1041 (1964).
67. V. N. Tsvetkov, E. I. Rjumtsev, I. N. Shtennikova, T. V. Peker, N. V. Pogodina, Dokl. Akad. Nauk SSSR, **207**, 1173 (1972).
68. E. N. Zakharova, L. I. Kutsenko, V. N. Tsvetkov, V. S. Skazka, G. V. Tarasova, V. M. Jamschikov, Leningrad. Univ. Vestnik, Ser. Fiz. Khim., **N16**, 55 (1970).
69. G. I. Okhrimenko, Thesis, Inst. Macromolecular Compounds Ac. Sci. SSSR, Leningrad, 1969.
70. V. N. Tsvetkov, I. N. Shtennikova, N. A. Megeritskaya, L. S. Bolotnikova, Vysokomol. Soedin., **5**, 74 (1963).
71. H. Janeschitz-Kriegl, W. Burhard, Adv. Polym. Sci., **6**, 170 (1969).
72. V. N. Tsvetkov, S. Ya. Ljubina, I. A. Strelina, S. I. Klenin, V. I. Kurljankina, Vysokomol. Soedin., **15A**, 691 (1973).
73. V. N. Tsvetkov, A. E. Grischchenko, P. A. Slavetskaja, Vysokomol. Soedin., **6**, 856 (1964).
74. V. N. Tsvetkov, A. E. Grischchenko, J. Polym. Sci. C, **16**, 3195 (1968).
75. V. N. Tsvetkov, I. N. Shtennikova, Vysokomol. Soedin., **2**, 808 (1960).
76. V. N. Tsvetkov, E. N. Zakharova, M. M. Krunchak, Vysokomol. Soedin., **10A**, 685 (1960).
77. V. N. Tsvetkov, E. V. Frisman, N. N. Boitsova, Vysokomol. Soedin., **2**, 1001 (1960).
78. V. N. Tsvetkov, K. A. Andrianov, E. L. Vinogradov, S. E. Yakushkina, Ts. V. Vardasanidze, Vysokomol. Soedin., **9B**, 893 (967).
79. V. N. Tsvetkov, K. A. Andrianov, E. L. Vinogradov, V. I. Pakhomov. S. E. Yakushkina, Vysokomol. Soedin., **9A**, 3 (1967).
80. V. N. Tsvetkov, Makromol. Chem., **160**, 1 (1972).
81. V. N. Tsvetkov, K. A. Andrianov, M. G. Vitovskaja, N. N. Makarova, E. N. Zakharova, S. V. Bushin, P. N. Lavrenko, Vysokomol. Soedin., **14A**, 369 (1972).
82. V. N. Tsvetkov, K. A. Andrianov, M. G. Vitovskaja, N. N. Makarova, S. V. Bushin, E. N. Zakharova, P. N. Lavrenko, A. A. Gorbunov, Vysokomol. Soedin., **15A**, 872 (1973).
83. V. N. Tsvetkov, K. A. Andrianov, E. L. Vinogradov, I. N. Shtennikova, S. E. Yakushkina, V. I. Pakhomov, J. Polym. Sci. C, **23**, 385 (1968).
84. V. N. Tsvetkov, K. A. Andrianov, I. N. Shtennikova, G. I. Okhrimenko, L. N. Andreeva, G. A. Fomin, V. I. Pakhomov, Vysokomol. Soedin., **10A**, 547 (1968).
85. V. N. Tsvetkov, K. A. Andrianov, G. I. Okhrimenko, I. N. Shtennikova, G. A. Fomin, M. G. Vitovskaja, V. I. Pakhomov, A. A. Jarosh, D. N. Andreev, Vysokomol. Soedin., **12A**, 1892 (1970).
86. V. N. Tsvetkov, I. N. Shtennikova, V. S. Skazka, E. I. Rjumtsev, J. Polym. Sci. C, **16**, 3205 (1968).

87. V. N. Tsvetkov, E. I. Rjumtsev, I. N. Shtennikova, G. I. Okhrimenko, Vysokomol. Soedin., **8**, 1466 (1966).
88. V. N. Tsvetkov, I. N. Shtennikova, E. I. Rjumtsev, G. F. Pirogova, Vysokomol. Soedin., **9A**, 1575 (1967).
89. V. N. Tsvetkov, L. N. Andreeva, P. N. Lavrenko, E. V. Beliaeva, O. V. Okatova, A. Yu. Bilibin, S. S. Skorokhodov, Eur. Polym. J., **20**, 817 (1984).
90. V. N. Tsvetkov, L. N. Andreeva, P. N. Lavrenko, O. V. Okatova, E. V. Beliaeva, A. Yu. Bilibin, S. S. Skorokhodov, Eur. Polym. J., **21**, 933 (1985).
91. V. N. Tsvetkov, L. N. Andreeva, S. V. Bushin, A. I. Mashoshin, V. A. Cherkasov, Z. Edlinski, D. Sek, Eur. Polym. J., **20**, 371 (1984).
92. V. N. Tsvetkov, S. Ya. Magarik, Dokl. Acad. Nauk SSSR, **115**, 911 (1957).
93. V. N. Tsvetkov, E. I. Rjumtsev, I. N. Shtennikova, E. V. Korneeva, G. I. Okhrimenko, N. A. Mikhailova, A. A. Baturin, Ju. A. Amerik, B. A. Krentsel, Vysokomol. Soedin., **15A**, 2570 (1973).
94. S. Ja. Magarik, Zh. Fiz. Khim., **33**, 835 (1959).
95. T. M. Birshtein, V. P. Budtov, E. V. Frisman, N. K. Janovskaja, Vysokomol. Soedin., **4**, 455 (1962).
96. E. V. Frisman, N. N. Boitsova, Leningrad. Univ. Vestnik, **N4**, 26 (1959).
97. A. Romanov, S. Ja. Magarik, M. Lazar, Vysokomol. Soedin., **9B**, 292 (1967).
98. V. N. Tsvetkov, G. A. Andreeva, I. A. Baranovskaja, V. E. Eskin, S. I. Klenin, S. Ja. Magarik, J. Polym. Sci. C, **16**, 239 (1967).
99. V. N. Tsvetkov, S. Ja. Magarik, T. Kadirov, G. A. Andreeva, Vysokomol. Soedin., **10A**, 943 (1968).
100. V. N. Tsvetkov, "Rigid-Chain Polymers", Consultants Bureau, Plenum, New York, 1989.
101. V. N. Tsvetkov, Vysokomol. Soedin., **25A**, 1571 (1983).
102. G. M. Pavlov, E. V. Korneeva, T. I. Garmonova, D. Hardi, K. Nitrai, Vysokomol. Soedin., **20A**, 1634 (1978).
103. V. N. Tsvetkov, E. V. Korneeva, I. N. Shtennikova, P. N. Lavrenko, G. F. Kolbina, D. Hardi, K. Nitrai, Vysokomol. Soedin., **14A**, 427 (1972).
104. E. I. Rjumtsev, I. N. Shtennikova, N. V. Pogodina, G. F. Kolbina, I. I. Konstantinov, Ju. B. Amerik, Vysokomol. Soedin., **18A**, 439 (1976).
105. V. N. Tsvetkov, I. N. Shtennikova, E. I. Rjumtsev, N. V. Pogodina, G. F. Kolbina, E. V. Korneeva, P. N. Lavrenko, O. V. Okatova, Ju. B. Amerik, A. A. Baturin, Vysokomol. Soedin., **18A**, 2016 (1976).
106. G. F. Kolbina, Theses, Inst. Macromolecular Compounds Ac. Sci. SSSR, Leningrad, 1981.
107. V. N. Tsvetkov, E. I. Rjumtsev, I. I. Konstantinov, Ju. B. Amerik, B. A. Krentsel, Vysokomol. Soedin., **14A**, 67 (1972).
108. V. N. Tsvetkov, I. N. Shtennikova, E. I. Rjumtsev, G. F. Kolbina, E. V. Korneeva, B. A. Krentsel, Ju. B. Amerik, I. I. Konstantinov, Vysokomol. Soedin., **15A**, 2158 (1973).
109. I. N. Shtennikova, T. V. Peker, G. F. Kolbina, V. R. Petrov, V. S. Grebneva, I. I. Konstantinov, Ju. B. Amerik, Vysokomol. Soedin., **24A**, 2047 (1982).
110. T. I. Garmonova, V. N. Artem'eva, Ye. M. Nekrasova, Vysokomol. Soedin., **32A**, 2062 (1990).
111. V. N. Tsvetkov, I. N. Shtennikova, G. F. Kolbina, A. I. Mashoshin, P. N. Lavrenko, S. V. Bushin, A. A. Baturin, Ju. A. Amerik, Vysokomol. Soedin., **27A**, 319 (1985).
112. L. N. Andreeva, E. V. Beliaeva, A. A. Boikov, A. M. Muzafarov, V. N. Emelianov, K. A. Andrianov, V. N. Tsvetkov, Vysokomol. Soedin., **21A**, 362 (1979).
113. V. N. Tsvetkov, K. A. Andrianov, N. N. Makarova, M. G. Vitovskaja, Vysokomol. Soedin., **15A**, 400 (1973).
114. V. N. Tsvetkov, N. A. Mikhailova, V. B. Novakovsky, A. V. Volokhina, A. B. Raskina, Vysokomol. Soedin., **22A**, 1028 (1980).
115. N. V. Pogodina, L. V. Starchenko, V. N. Tsvetkov, Eur. Polym. J., **16**, 387 (1980).
116. V. N. Tsvetkov, V. V. Korshak, I. N. Shtennikova, Kh. Raubakh, E. S. Krongauz, G. M. Pavlov, G. F. Kolbina, S. O. Tsepelevich, Vysokomol. Soedin., **21A**, 83 (1979); Macromolecules, **12**, 645 (1979).
117. V. N. Tsvetkov, G. I. Kudryavtsev, I. N. Shtennikova, T. V. Peker, E. N. Zakharova, V. D. Kalmykova, A. V. Volokhina, Eur. Polym. J., **12**, 517 (1976).
118. V. N. Tsvetkov, N. V. Pogodina, I. N. Bogatova, L. V. Starchenko, Vysokomol. Soedin., **26A**, 122 (1984).
119. N. V. Pogodina, L. V. Starchenko, K. S. Pozhivilko, V. D. Kalmykova, T. A. Kulichikhina, A. V. Volokhina, G. I. Kudryavtsev, V. N. Tsvetkov, Vysokomol. Soedin., **23A**, 2185 (1981).
120. V. N. Tsvetkov, I. N. Shtennikova, T. V. Peker, Eur. Polym. J., **13**, 455 (1977).
121. V. N. Tsvetkov, E. N. Zakharova, N. A. Mikhailova, Dokl. Akad. Nauk SSSR, **224**, 1365 (1975).
122. N. A. Mikhailova, Theses, University of Leningrad, 1982.
123. V. N. Tsvetkov, N. V. Pogodina, L. V. Starchenko, Eur Polym. J., **19**, 837 (1983).
124. I. N. Shtennikova, T. V. Peker, T. I. Garmonova, G. F. Kolbina, L. V. Avrorova, A. B. Tokareva, G. I. Kudryavtsev, V. N. Tsvetkov, Vysokomol. Soedin., **23A**, 2510 (1981).
125. N. A. Mikhailova, V. N. Tsvetkov, V. B. Novakovsky, Vysokomol. Soedin., **24B**, 770 (1982).
126. N. V. Pogodina, L. V. Starchenko, V. N. Tsvetkov, Eur. Polym. J., **19**, 841 (1983).
127. P. D. Sybert, W. H. Beever, J. K. Stille, Macromolecules, **14**, 493 (1981).
128. L. N. Andreeva, E. V. Beliaeva, A. A. Boikov, P. N. Lavrenko, V. N. Tsvetkov, Vysokomol. Soedin., **25A**, 1631 (1983).
129. L. N. Andreeva, T. I. Garmonova, I. N. Shtennikova, V. N. Tsvetkov, The Fifth Meeting on Cellulose, Vladimir 1976.
130. N. V. Pogodina, K. S. Pozhivilko, A. B. Melnikov, S. A. Didenko, G. N. Marchenko, V. N. Tsvetkov, Vysokomol. Soedin., **23A**, 2454 (1981).
131. N. V. Pogodina, P. N. Lavrenko, K. S. Pozhivilko, A. B. Melnikov, T. A. Kolobova, G. N. Marchenko, V. N. Tsvetkov, Vysokomol. Soedin., **24A**, 332 (1982).
132. N. V. Pogodina, Y. B. Tarabukina, L. V. Starchenko, G. N. Marchenko, V. N. Tsvetkov, Vysokomol. Soedin., **22A**, 2219 (1980).
133. L. N. Andreeva, V. Yu. Elohovski, Vysokomol. Soedin., **19B**, 111 (1977).

134. P. N. Lavrenko, E. U. Urinov, L. N. Andreeva, K. J. Linow, H. Dautzenberg, B. Philipp, Vysokomol. Soedin., **18A**, 2579 (1976).
135. V. N. Tsvetkov, P. N. Lavrenko, L. N. Andreeva, A. I. Mashoshin, V. Okatova, O. I. Mikriukova, L. I. Kutsenko, Eur. Polym. J., **20**, 823 (1984).
136. N. V. Tsvetkov, A. K. Khripunov, E. P. Astapenko, S. A. Didenko, Vysokomol. Soedin. **37A**, 1306 (1995).
137. V. N. Tsvetkov, N. V. Tsvetkov, Russian Chem. Rev., **62**, 851 (1994).
138. N. V. Tsvetkov, G. N. Marchenko, M. A. Bezrukova, S. A. Didenko, V. M. Denisov, A. K. Khripunov, A. V. Lezov, V. N. Tsvetkov, Polym. Sci. (Russ.), **35**, 345 (1993).
139. N. V. Tsvetkov, L. I. Kutsenko, S. A. Didenko, Vysokomol. Soedin., **37A**, 1300 (1995).
140. N. V. Tsvetkov, S. A. Didenko, Vysokomol. Soedin., **35A**, 1640 (1993).
141. V. N. Tsvetkov, A. V. Lezov, N. V. Tsvetkov, L. N. Andreeva, Eur. Polym. J., **26**, 1103 (1990).
142. I. N. Stennikova, G. F. Kolbina, V. P. Shibaev, I. V. Ekaeva, Eur. Polym. J., **26**, 787 (1990).
143. I. N. Stennikova, G. F. Kolbina, Vysokomol. Soedin., **37A**, 1314 (1995).
144. I. N. Stennikova. E. V. Korneeva, G. F. Kolbina, I. A. Strelina, P. N. Lavrenko V. P. Shibaev, I. V. Ekaeva, Eur. Polym. J., **28**, 353 (1992).
145. G. F. Kolbina. E. V. Korneeva, I. N. Stennikova, Mol. Mat., **5**, 79 (1995).
146. I. N. Stennikova, G. F. Kolbina, E. V. Korneeva, I. A. Strelina, V. P. Shibaev, I. V. Ekaeva, Polym. Sci., **34**, 1022 (1992).
147. I. N. Stennikova, T. V. Filippova, H. Dautzenberg, Vysokomol. Soedin., **35**, 1459 (1993).
148. P. N. Lavrenko, I. A. Strelina, S. Ya. Magarik, L. I. Kutsenko, Vysokomol. Soedin., **37**, 2007 (1995).
149. G. M. Pavlov, N. P. Evlampieva, Biophisica, **40** (6), 1220 (1995).
150. G. M. Pavlov, A. N. Kozlov, S. M. Yakopson, S. V. Usova, L. S. Efros. Vysokomol. Soedin., **27A**, 30 (1985).
151. N. V. Tsvetkov, S. A. Didenko, V. N. Tsvetko, Dokl. A Nauk. SSSR, 337, 483 (1994).
152. P. N. Lavrenko. T. I. Garmonova, M. M. Gel'mont, L. S. Efros, I. M. Kralina, Vysokomol. Soedin., **34B**, 45 (1992).
153. I. P. Kolomiets, P. N. Lavrenko, A. V. Lezov, A. M. Ovsipyan, H. Finkelmann, Vysokomol. Soedin., **35A**, 1620 (1993).
154. E. V. Frisman, N. A. Kas'yanenko, Mol. Biol., **24** (2), 318 (1990).
155. E. I. Rumtsev, I. N. Shtennikova, D. R. Tur, G. F. Kolbina, E. V. Korneeva, V. G. Kulichikhin, Vysokomol. Soed., **31B**, 648 (1990).
156. E. I. Rumtsev, I. P. Kolomiets, I. N. Shtennikova, G. F. Kolbina, D. R. Tur, V. G. Kulichikhin, Polym. Sci., **34**, 541 (1992).
157. N. A. Kas'yanenko, E. V. Frisman, S. I. Klenin, Ye. N. Bykova, I. S. Kochetkova, Vysokomol. Soedin., **29A**, 1129 (1987).
158. V. N. Tsvetkov, S. V. Bushin, L. N. Andreeva, K. P. Smirnov, E. V. Belyaeva, A. Yu Bilibin, and A. R. Stepanova, Polym. Sci., **34**, 229, (1992).
159. V. N. Tsvetkov, L. N. Andreeva, S. V. Bushin, E. V. Belyaeva, V. A. Cherkasov, A. I. Mashoshin, A. Yu. Bilibin, S. S. Skorokhodov, Vysokomol. Soedin., **30A**, 713 (1988).
160. V. N. Tsvetkov, L. N. Andreeva, A. P. Filippov, E. V. Belyaeva, A. Yu. Bilibin, and A. R. Stepanova, Eur. Polym. J., **27**, 319 (1991).
161. V. N. Tsvetkov, L. N. Andreeva, P. N. Iavrenko, O. V. Okatova, E. V. Beliaeva, A. Yu. Bilibin, and S. S. Skorokhodov, Eur. Polym. J., **21**, 933 (1985).
162. V. N. Tsvetkov, L. N. Andreeva, P. N. Lavrenko, O. V. Okatova, E. V. Beliaeva, A. Yu. Bilibin, and S. S. Skorokhodov, Vysokomol. Soedin. A, **XXX**, 1263 (1933).
163. V. N. Tsvetkov, S. V. Bushin, L. N. Andreeva, K. P. Smirnov, E. V. Belyaeva, E. P. Astapenko, A. Yu. Bilibin, A. V. Ten'kovtsev, A. R. Stepanova, Polym. Sci., **34**, 536 (1992).
164. S. V. Bushin, N. V. Tsvetkov, E. P. Astapenko, V. N. Tsvetkov, E. V. Belyaeva, V. V. Zuev, S. S. Skorokhodov, Vysokomol. Soedin., **38A**, 1308 (1996).
165. N. V. Tsvetkov, V. N. Tsvetkov, S. S. Skorokhodov, Vysokomol. Soedin., **38A**, 1032 (1996).
166. V. N. Tsvetkov, S. V. Bushin, E. P. Astapenko, N. V. Tsvetkov, S. S. Skorokhodov, V. V. Zuev, R. Zentel, H. Potsch, Polym. Sci., **36A**, 813 (1994).
167. N. V. Tsvetkov, V. N. Tevetkov, V. V. Zuev, S. S. Skorokhodov, R. Zentel, Vysokomol. Soedin., **38A**, 1831 (1996).
168. N. V. Tsvetkov, N. V. Pogodina, V. V. Zuev, Vysokomol. Soedin., **38A**, 1133 (1996).
169. N. V. Tsvetkov, V. V. Zuev, I. V. Ksenofontov, S. A. Didenko, Vysokomol. Soedin. **38A**, 1840 (1996).
170. V. N. Tsvetkov, L. N. Andreeva, S. V. Bushin, A. I. Mashoshin, A. Yu. Bilibin, S. S. Skorokhodov, Vysokomol. Soedin., **30A**, 2128 (1988).
171. S. V. Bushin, V. N. Tsvetkov, L. N. Andreeva, E. P. Astapenko, E. V. Belyaeva, D. N. Asinovskaya, S. S. Skorokhodov, Polym. Sci., **39**, 943 (1997).
172. L. N. Andreeva, S. V. Bushin, A. I. Mashoshin, M. A. Bezrukova, V. N. Tsvetkov, A. Yu. Bilibin, S. S. Skorokhodov, Vysokomol. Soedin., **32A**, 1754 (1990).
173. I. N. Stennikova, G. F. Kolbina, S. V. Bozhko, N. N. Makarova, Vysokomol. Soedin., **37B**, 1585 (1995).
174. S. Ya. Lyubina, I. A. Strelina, L. A. Nud'ga, Ye. A. Plisko, I. N. Bogatova, Vysokomol. Soedin., **25A**, 1467 (1983).

Gelation Properties of Polymer Solutions

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Some polymer solutions show the phenomenon of gelation. The polymeric chains form association complexes at widely separated points at a certain reduced solvent power of the medium. This leads to the formation of a continuous physical network structure extending throughout the volume of the system. The association producing quasi-crosslinkages is a reversible process, so that the gel may be liquified and reset many times. The nature of the linkages is rather imperfectly understood, but the phenomenon is usually encountered with more or less crystalline polymers. Sometimes, sharp X-ray diffraction patterns are observed which disappear at the melting point of the gel.

The table contains data for a selected number of synthetic polymers which are capable of forming physical, thermoreversible gels with various solvents. Since no standard definition or standard measurement methods have been established for physical polymer gels, the gelation properties are taken from the references cited.

EXPLANATION OF ABBREVIATIONS USED

M_w The molecular weight of polymer refers to the weight average molecular weight unless otherwise stated.

λ The branch density of branched polymer.

S The syndiotacticity diad content in %.

T_{gel} The temperature (in degrees centigrade) at which the gel forms when the solution is cooled; the sol–gel transition temperature.

T_m The melting temperature (in degrees centigrade); the gel–sol transition temperature.

ΔH_m When DSC was used, this is the enthalpy associated with the endotherm. When other methods are used, ΔH is obtained from the van't Hoff plot.

Methods Used for the Measurement of Melting Temperature of the Gel

1. *Flow:* A U-tube in which the two limbs are joined by a capillary section is used. The gel is allowed to set with different heights in the two links and the melting point is the temperature at which flow starts from one limb to the other.
2. *Ball drop:* A steel ball with a known diameter or weight is placed on the gel and the gel warmed at a certain heating rate. The ball drops at the gel melting temperature.
3. *DSC:* Differential scanning calorimetry.
4. *Tilting:* The gel is warmed in the test tube and the melting point is observed when the gel begins to flow under its own weight when the upright test tube is tilted.

Polymer	Solvent	M_w	Conc. (g/l)	T_{gel} (°C)	T_m (°C)	ΔH_m (kJ/mol)	Method	Refs.
Poly(ethylene), low density								
$\lambda = 2.9$	Decalin	74000	50–140		54–72		Ball drop	6
$\lambda = 6.8$	Decalin	161000	40–140		58–72			
$\lambda = 5.6$	Decalin	137000	50–140		60–74			
$\lambda = 8.9$	Decalin	175000	50–140		50–67			
$\lambda = 6.8$	Toluene	161000	60–150		68–76			
$\lambda = 6.8$	*o*-Xylene	161000	50–140		67–76			
$\lambda = 6.8$		161000	50–150		68–75			
$\lambda = 6.8$	Tetralin	161000	60–140		72–78			
$\lambda = 3.1$	Cyclohexane-CS	30000	40–75		30–41			
$\lambda = 4.7$	(1/1, v/v)	65000	40–75		36–43			
$\lambda = 12.2$		222000	30–75		34–44			
$\lambda = 7.18$	Cyclohexane-CS_2	30000	65–95		40–55		Ball drop	8
$\lambda = 0$	(1/1, v/v)	10100	30–65		73–75			
$\lambda = 4.07$		15100	50–95		55–65			

Polymer	Solvent	M_w	Conc. (g/l)	T_{gel} (°C)	T_m (°C)	ΔH_m (kJ/mol)	Method	Refs.
Poly(ethylene), high density								26
	o-Dichlorobenzene	220000	120	89		129	Tilting	23
Poly(ethylene-*co*-butene-1)	Decalin	113000	50–140		86–91			
		104000	60–140		87–90.5			
		71000	50–140		83–88			
		55000	60–140		82–89			
Poly(ethylene-*co*-		80000	60–140		88.5–92.5			
4-methylpentene-1)		84000	50–120		89.5–93			
		87000	60–120		89–93			
Poly(ethylene-*co*-octene-1)		83000	50–120		91–94.5			
		119000	50–100		92–95			
		80000	60–120		91–94			
		90000	50–115		90.5–94			
Poly(propylene)	Decahydronaphthalene		50	10	97.7			10
Poly(4-methylpentene-1)	Cyclopentane		100		80		Visual observation	22
	Cyclohexane		5–100		45			
Poly(acrylonitrile)	Dimethyl sulfoxide		72		35			15
Poly(acrylonitrile-*co*-	Dimethylacetamide	200000	240–280		42–75		Flow	18
vinyl acetate)	Dimethyl sulfoxide	200000	270–320		34–46			
	Dimethylformamide	200000	290–310		56–66			
	Ethylene carbonate	200000	230–300		41–72			
	γ-Butyrolactone	200000	250–270		66–73			
Poly(cetyl acrylate)	*n*-Aliphatic alcohols							
	C = 7				28		DSC	24
	C = 8				28			
	C = 9				27			
	C = 10				26			
	C = 12				25			
	Hydrocarbons							
	C = 10				12			
	C = 12				15			
	C = 14				24			
	C = 16				26			
	n-Decane		100		17		DSC	24
			200		18			
			400		18			
			500		20			
			700		25			
			800		26			
			900		30			
	Cetane		300		30	35 (J/g)		
			500		31			
			700		32	79		
			900		34	96		
	Alcohols		100		29	8		
			200		29	12		
			300		28	21		
			400		27	29		
			500		27	38		
			600		27	46		
			700		27	50		
			800		28	59		
			900		29			
Poly(docosyl acrylate)	*n*-Aliphatic alcohols							
	C = 7				57		DSC	24
	C = 8				57			
	C = 10				56			
	C = 12				56			
Poly(octadecyl acrylate)	*n*-Aliphatic alcohols							
	C = 7				38		DSC	24
	C = 8				38			
	C = 10				37			
	C = 12				37			
	Hydrocarbons							
	C = 6				18			
	C = 8				20			
	C = 12				28			

Polymer	Solvent	M_w	Conc. (g/l)	T_{gel} (°C)	T_m (°C)	ΔH_m (kJ/mol)	Method	Refs.
Poly(methyl methacrylate)	Dimethylformamide		A very special case of gelation is observed in cosolutions of isotactic and syndiotactic PMMA.					14
Poly(vinyl acetate)	Glycerin/water (40/60)	60000	50		77	55		19
			100		89			
			200		101			
	(60/40)	60000	20		82	54		
			50		94			
			100		99			
			200		106			
	(80/20)	60000	20		96	71		
			50		101			
			100		112			
			200		123			
	Methanol/water (40/60)	60000	100		35	15		
			151		58			
			200		73			
			250		78			
	N-Methylpyrrolidone	60000	50		87.5	54		
			100		101			
			200		120			
	Formamide	60000	20		62	48		
			50		68			
			100		98			
			200		106			
Poly(vinyl alcohol)	Water		90	7			Tilting	27
			140	20				
			200	47				
			300	58				
highly syndiotactic	Phenol/water		0.6		75	58	DSC	20
$S = 65\%$			0.8		82			
			1		86			
			2		102			
			3		105			
			5		118			
57%			0.4		42	51		
			0.8		59			
			1		60			
			2		72			
			3		79			
			4		88			
			5		93			
45%		4		19	28			
			5		20			
			6.5		29			
			10		37			
			12.5		42			
Poly(vinyl chloride)	Bromobenzene	34000	100		27.3		Ball drop	2
		59000	100		48.0			
		145000	100		88.3			
		235000	100		116			
	Dichloroethane	98000	100		44			2
		145000	100		73			
		154000	80–120		42–72	13.0		
	Dimethyl phthalate	34000	100		85.5			2
		62000	100		99.0			
		145000	100		116			
		135000	100		133			
		154000	10–100		52.0–104.0	50.6		
	Dioxane	200000	100 (w/w)		50–67[a]	–		3,4
		154000	50–120		42.0–84.0	50.6		2
		56000	50–140		45–70	36		1
		103000	40–100		50–75	35		
		114200	30–100		50–80	37		
		133000	20–80		60–85	45		
		62000	100		43.5		Ball drop	2
		145000	100		79.8			
		235000	100		96.9			

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Polymer	Solvent	M_w	Conc. (g/l)	T_{gel} (°C)	T_m (°C)	ΔH_m (kJ/mol)	Method	Refs.
	Bromobenzene	154000	40–120		38.0–82.5	21.0		2
		200000	100 (w/w)		40–63[a]	90–110[b]	DSC	3,4
		56000	92		76.2	34.0		1
	1,2-Dichlorobenzene	200000	100 (w/w)		44–68[a]	87–103[b]		3,4
			100 (w/w)		55–90[a]	90–137[b]		3,4
	Nitrobenzene	200000	100 (w/w)		40–55[a]	57–75[b]		3,4
		56000	92		32.7	23.1		1
	Butanone	200000	100 (w/w)		56–76[a]	98–102[b]		3,4
		56000	92		51.0	17.2		1
	Tritolyl phosphate	200000	100 (w/w)		58–87[a]	96–142[b]		3,4
	Di-isopropyl ketone	200000	100 (w/w)		54–89[a]	89–120[b]		3,4
	Dimethyl phthalate	200000	100 (w/w)		54–74[a]	100–150[b]		3,4
	Nitrochlorobenzene		100 (w/w)		43–75[a]	116–139[b]		3,4
	Dibutyl phthalate	148000	40–70		90–105		Flow	25
		56000	92		93.7	22.3	Flow	1
	Dipentyl phthalate	148000	40–70		97–112			25
	Dihexyl phthalate	148000	40–70		101–119			
	Diheptyl phthalate	148000	40–70		107–123			
	Dinonyl phthalate	148000	50		126			
	Pentan-3-one	56000	92		46.7	10.9		1
	Dimethylformamide	56000	92		20.0			1
	Tetrachloroethane	56000	92		69.1	35.7		1
	Chlorobenzene	56000	92		78.4	61.7		1
	Methoxybenzene	56000	92		85.1	43.1		1
Poly(styrene), atactic	Carbon disulfide	2000000	5–135		−53–7	31	Tilting	13
		670000	10–140		−63–0	27		
		410000	17–170		−56 to −10	26		
		200000	20–150		−70 to −20	24		
		30000	70–190		−76 to −12	10		
		20000	90–170		−72 to −22	6		
		4000	120–180		−55 to −23	5		
	n-Amyl acetate	670000	166		−52	11.7		
	Isoamyl acetate	670000	166		−56	11.4		
	n-Butyl acetate	670000	166		−65	10.6		
	n-Propyl acetate	670000	166		−68	10.6		
	1-Chloropentane	670000	166		−76	12.2		
	1-Chlorobutane	670000	166		−79	8.0		
	1-Chloropropane	670000	166		−86	5.8		
Poly(styrene), isotactic	Decalin	190000	50		670	62	DSC	11
		670000	10	12	63		Tilting	12
	1-Chlorodecane	670000	10	25	88			
	Nitrobenzene	670000	10	−2	52			
	Aniline	670000	10	−20	5			
	Benzaldehyde	670000	10	−35	20			
Poly(oxy-2,6-dimethyl-phenylene)	1-Bromonaphthalene	25000 (M_v)		25	110		Tilting	12
	1-Chloronaphthalene	25000 (M_v)		25	86			
	Carbon disulfide	25000 (M_v)			0			
Poly(γ-benzyl L-glutamate)	Benzyl alcohol	50000 (M_v)	30		50		Ball drop	21
		80000	30		55			
		150000	30		58			
		300000	30		58			
		150000	50		55			
		150000	100		56			
		150000	200		58			
		150000	300		58.5			
		150000	400		59			
		150000	500		60			

[a] T_1 endotherm.
[b] T_2 endotherm.

REFERENCES

1. M. A. Harrison, P. H. Morgan, G. S. Park, Eur. Polym. J., **8**, 1361 (1972).
2. Y. C. Yang, P. H. Geil, J. Macromol. Sci. B: Phys., **22**, 463 (1983).
3. S. J. Guerrero, A. Keller, J. Macromol. Sci. B: Phys., **20** (2), 167 (1981).
4. S. J. Guerrero, A. Keller, P. L. Soni, P. H. Geil, J. Polym. Sci., Polym. Phys. Ed., **18**, 1533 (1980).

5. S. J. Guerrero, A. Keller, P. L. Soni, P. H. Geil, J. Macromol. Sci. B: Phys., **20** (2), 161 (1981).
6. H. Matsuda, M. Imaizumi, H. Fujimatsu, S. Kuroiwa, M. Okabe, Polym. J., **16** (2), 151 (1984).
7. H. Matsuda, K. Araki, H. Fujimatsu, S. Kuroiwa, Polym. J., **13** (6), 587 (1981).
8. A. Takahashi, M. Sakai, T. Kato, Polym. J., **12** (5), 335 (1980).
9. P. J. Barham, Polymer, **23** (8), 1112 (1982).
10. JP 57,180,635 [82,180,635] (1982)).
11. M. Girolamo, A. Keller, K. Miyasaka, N. Overberg, J. Polym. Sci., Polym. Phys. Ed., **14**, 39 (1976).
12. S. Wellinghoff, J. Shaw, E. Baer, Macromol., **12** (5), 932 (1979).
13. H. -M. Tan, A. Moet, A. Hiltner, E. Baer, Macromol., **16** (1), 28 (1983).
14. C. F. Ryan, P. C. Fleischer, Jr., J. Phys. Chem., **69**, 3384 (1965).
15. S. Vladimir, Czech. Patent 178, 638 (1979).
16. J. Bisschops, J. Polym. Sci., **17**, 89 (1955).
17. J. Bisschops, J. Polym. Sci., **12**, 583 (1954).
18. D. R. Paul, J. Appl. Polym. Sci., **11**, 439 (1967).
19. L. Z. Rogovina, G. L. Slonimskii, L. S. Gembitskii, Ye. A. Serova, V. A. Grigor'eva, Ye. N. Gubenkova, Vysokomol Soyed. A, **15** (6), 1256 (1973).
20. S. Matsuzawa, K. Yamaura, H. Kobayashi, Colloid Polym. Sci., **259**, 1147 (1981).
21. S. Sasaki, M. Hikata, C. Shiraki, I. Uematsu, Polym. J., **14**, 205 (1982).
22. G. Charlet, H. Phuong-Nguyen, G. Delmas, Macromol., **17**, 1200 (1984).
23. H.-M. Tan, B. H. Chang, E. Baer, A. Hiltner, Eur. Polym. J., **19**, 1021 (1983).
24. R. V. Tal'roze, V. P. Shibaev, N. A. Platé, J. Polym. Sci. Symp., **44**, 35 (1974).
25. S. A. Leharne, G. S. Park, Eur. Polym. J., **21**, 383 (1985).
26. M. Okabe, M. Isayama, H. Matsuda, Polym. J., **17**, 369 (1985).
27. M. Komatsu, T. Inoue, K. Miyasaka, J. Polym. Sci., Polym. Phys. Ed., **24**, 303 (1986).

SECTION VIII

ABBREVIATIONS OF POLYMER NAMES

Abbreviations for Thermoplastics, Thermosets, Fibers, Elastomers, and Additives

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A. INTRODUCTION

Abbreviations and acronyms for the names of thermoplastics, thermosets, fibers, elastomers, and additives have been proposed by many national and international organizations. Some of these abbreviations were introduced by law, others are trademarks in certain countries. As a result, the same abbreviation is often used for different polymers. The same polymer may also have different abbreviations or acronyms. Furthermore, some abbreviations refer to chemical compounds (IUPAC) whereas abbreviations recommended by ASTM, BS, DIN, and ISO usually refer to plastics, fibers, elastomers, etc., that are *based* on the corresponding polymers, i.e., polymers plus additives, modifiers, fillers, etc.

The following tables report abbreviations either found in the scientific or technical literature or recommended or proposed by the organizations listed below. Note that * and ** are part of the recommended ASTM abbreviations and acronyms and *not* symbols for footnotes.

ANSI/ASTM	American National Standards Institute (ASTM D 1600–75; D 1600–86a; D 1418–77)
ASTM	American Society for Testing of Materials (ASTM D 1600–64; D 1600–86a; D 1418–67; D 4020–81)
BS	British Standards (Standard 3502–1862)
DIN	German Industrial Standard (DIN 7723, 7728, 16913, 55950, 60001)
EDV	Data processing key of the European Textile Characterization Law
EC	European Community
ISO	International Organization for Standardization (ISO 1043–1980, 1629–1980)
IUPAC	International Union for Pure and Applied Chemistry [*Pure Appl. Chem.*, **18**, 583 (1969); **40**, 473 (1974); **59**, 691 (1987)]
WTR	Working Group Toxicology of Rubber Auxiliaries

Section B lists, in alphabetical order, the abbreviations and acronyms for thermoplastics, thermosets, fibers, elastomers, and additives. Sections C and D list the abbreviations for polymers, based on their monomeric units and characteristic groups, respectively. Section E contains abbreviations and acronyms for polymers obtained by chemical transformation of base polymers. Section F contains special abbreviations for blends, reinforced polymers, etc., and Section G the ISO and DIN codes for plastics. Section H lists the special abbreviations connected with the recycling of plastics. Recommended abbreviations and acronyms for elastomers and fibers are given in Sections I and J, respectively.

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B. ABBREVIATIONS IN ALPHABETICAL ORDER

AB	Manila hemp fibers (Abaca) (EDV); see also Ma
ABR	Acrylate–butadiene rubber; elastomeric copolymer from an acrylic ester and butadiene (ANSI/ASTM, IUPAC); see also AR
ABS	Acrylonitrile–butadiene–styrene; a thermoplastic from acrylonitrile, butadiene and styrene (ASTM, DIN, ISO, IUPAC)
AC	Fiber from deacetylated cellulose acetate (EDV)
ACM	Elastic copolymer from an acrylic ester and a small proportion of a vulcanizable monomer, e.g., 2-chlorovinyl ether (ANSI, ASTM)
ACS	Thermoplastic blend of a copolymer from acrylonitrile and styrene with chlorinated poly(ethylene)
ACSP	Acetyl cyclohexylsulfonyl peroxide
ADA	Adipic acid
ADC	Azodicarbonamide (WTR), see also ADEC
ADCK	Activated azodicarbonamide (WTR)
ADEC	Ammonium diethyl dithiocarbamate (WTR); see also ADC
ADPA	Condensation product from acetone and diphenylamine (WTR)
AE	Aminoethyl (with cellulosics)
AES	Thermoplastic quaterpolymer from acrylonitrile, ethylene, propylene, and styrene
AF	Other fibers (i.e., those not specified by the EDV system)
AFK	Asbestos fiber reinforced plastic
AFMU	Elastomeric terpolymer from tetrafluoroethylene, trifluoronitrosomethane, and a small amount of nitrosoperfluorobutyric acid; nitroso rubber (ANSI/ASTM); see also CNR
AG	Alginate fiber (EDV)
Ag	1. Silver (IUPAC) 2. Esparto grass fiber (DIN)
AI	Amide-imide polymers
AIBN	*N,N*-Azobisisobutyronitrile
Ak	Angora wool (DIN)
AL	1. Alfalfa fibers (EDV) 2. Alginate fibers (DIN); see also AG
AMMA	Thermoplastic copolymer from acrylonitrile and methyl methacrylate (ASTM); see also A/MMA
A/MMA	See AMMA
An	Calatropis fiber (DIN)
AN	Acrylonitrile
An–AE	Elastomeric copolymer from acrylonitrile and acrylic ester; see also ANM
ANM	See AN–AE
AO	Antioxidant
Ap	Alpaca wool (DIN)
AP	Elastic copolymer from ethylene and propylene; see also APR, E/P, EPM, and EPR (older German literature)
APH	Alkylated phenol
APR	See AP
APT	Elastomeric copolymer from ethylene, propylene, and an unsaturated termonomer; see also APTK, EPD, EPDM, EPT, EPTR
APTK	Elastomeric copolymer from ethylene, propylene, and an unsaturated termonomer; see also APT, EPD, EPDM, EPT, EPTR
Aq	Anthraquinone
AR	1. Elastomeric copolymer from acrylic esters and olefins; see also ABR 2. Ardein fibers (DIN)
As	1. Arsenic (IUPAC). 2. Asbestos (DIN); see also AS
AS	Asbestos fiber (EDV); see also As
ASA	Thermoplastic copolymer from acrylonitrile, styrene, and acrylic esters (ASTM); see also A/S/A
A/S/A	Thermoplastic copolymer from acrylonitrile, styrene, and acrylic esters (DIN, ISO); see also ASA
ASE	Alkylsulfonic acid ester (ISO)
ASR	Alkylene sulfide rubber
ATE	Aluminium triethyl
ATH	Alumina trihydrate
ATO	Antimony trioxide
AU	1. Elastomeric polyurethane with polyester segments (ANSI/ASTM, ISO) 2. Polyester rubbers with carbon, oxygen, and nitrogen in the chain. Both classifications probably refer to the same type of elastomer
AU-I	AU, cross-linkable with isocyanates
AU-P	AU, cross-linkable with peroxides
BA	Condensation product of butyraldehyde and aniline (WTR)
BAA	Condensation product of butyraldehyde and aniline
BBP	Benzylbutyl phthalate (DIN, ISO)
BD	1. 1,4-Butanediol 2. Derivative of benzofuran
BDMA	Butanediol 1,4-dimethacrylate (WTR)
BDP	Butyldecyl phthalate (WTR)
BDSH	Benzene-1,3-disulfonyl hydrazide (WTR)
BFE	Bromotrifluoroethylene polymers

BFK	Boron fiber reinforced plastic (German technical literature)
BHA	Butylated hydroxyanisole; *t*-butyl-4-methoxyphenol
BHET	Bis(2-hydroxyethyl) terephthalate
BHT	Butylated hydroxytoluene; 2,6-di(*t*-butyl)-*p*-cresol
BiDMC	Bismuth dimethyl dithiocarbamate (WTR); see also BMDC
BIIR	Brominated elastomer from isobutene and isoprene; bromobutyl rubber (ANSI/ASTM)
BMC	Bulk molding compound
BMDC	Bismuth dimethyl dithiocarbamate (WTR)
BOA	Benzyl octyl adipate (= benzyl 2-ethylhexyl adipate) (ISO)
BOP	Benzyl octyl phthalate (= benzyl 2-ethylhexyl phthalate)
BOPP	Biaxially oriented poly(propylene) film; balanced oriented poly(propylene) film
BP	Poly(butadiene) rubber; see also BR and CBR
BPH	2,2′-Methylene-bis(4-methyl-6-*t*-butylphenol) (WTR)
BPO	Dibenzoyl peroxide
BR	Elastomeric poly(butadiene); butadiene rubber (ANSI/ASTM, IUPAC); see also BP and CBR
BSH	Benzene sulfonyl hydrazide (WTR)
BT	Thermoplastic poly(1-butene); see also Pb, PBT
BTDA	3,3′,4,4′-Benzophenonetetracarboxylic dianhydride
BTX	Benzene, toluene, and xylenes
Buna	Butadiene elastomers (old German literature). The first types were synthesized by sodium ("*Na*trium") initiated *bu*tadiene polymerizations, hence the name "Buna"; the various grades were differentiated by numbers (hence "number Buna"). Later Buna types were copolymers from the free-radical copolymerization of butadiene with various comonomers. These types were called "letter Buna" (Buna S = copolymer of butadiene and styrene, Buna N = copolymer of butadiene and acrylonitrile, etc.).
Butyl	Elastomeric copolymer from isobutene and isoprene; butyl rubber (BS); see also GR-I, IIR, PIBI
BVE	Butyl vinyl ether (WTR)
Bw	Cotton ("Baumwolle") (DIN); see also CO
C	1. Carbon (IUPAC) 2. Channel black 3. Cellophane (ASTM); "Cellophan®" is a registered trademark in Germany
C*	Saran coated cellophane (ASTM)
CA	1. Cellulose acetate (ASTM, DIN, ISO, IUPAC); see also AC 2. Hemp fibers (EDV)
CAB	Cellulose acetate-butyrate (ASTM, DIN, ISO, IUPAC)
CAP	Cellulose acetate-propionate (ASTM, DIN, ISO, IUPAC)
CAR	Carbon fiber
CBA	Chemical blowing agent
CBR	Butadiene rubber; 1,4-*cis*-poly(butadiene); see also BP and BR
CBS	*N*-Cyclohexyl-2-benzothiazolesulfene amide
CC	1. Cocos fiber (EDV) 2. Cupro (regenerated cellulose fibers *via* the copper/ammonia process (DIN); see also CU
CDB	Conjugated diene butyl elastomer
CdDEC	Cadmium diethyl dithiocarbamate (WTR); also Cd-DEDC
CDP	Cresyl diphenyl phosphate
CdPDC	Cadmium pentamethylene dithiocarbamate (WTR)
CE	Cellulose plastics in general (ASTM)
CEM	Poly(chlorotrifluoroethylene) (ASTM); see also CFM, CTFE, PCTFE
CF	Thermoset from cresol and formaldehyde; cresol-formaldehyde resin (ASTM, DIN, ISO, IUPAC)
CFK	1. Man-made fiber reinforced plastic (German technical literature: "*Chemie*faserverstärkter *K*unststoff") 2. Carbon fiber reinforced plastic (German technical literature); see also CRP
CFM	1. Poly(chlorotrifluoroethylene) (ANSI/ASTM); see also CEM, CTFE, PCTFE 2. Copolymer from chlorotrifluoroethylene and vinylidene fluoride
CFRP	Carbon fiber reinforced plastics
CHC	Elastomeric copolymer from epichlorohydrin and ethylene oxide; see also ECO
CHDM	1,4-Cyclohexanedimethylol
CHR	Elastomeric poly(epichlorohydrin) (ISO); see also CO
CIIR	Postchlorinated elastomeric copolymer from isobutene and isoprene (ANSI/ASTM)
CL	Poly(vinyl chloride) fiber (EWG; EDV); see also PVC
CM	Postchlorinated poly(ethylene) (ANSI/ASTM); see also CPE
CMBT	Copper mercaptobenzothiazole, see also CuMBT
CMC	1. Carboxymethylcellulose (ASTM, DIN, ISO, IUPAC) 2. Critical micelle concentration

CMHEC	(Carboxymethyl/hydroxyethyl) cellulose
CMF	Classified mineral filler
CN	Cellulose nitrate (ASTM, DIN, ISO, IUPAC); see also NC, C/N
C/N	Cellulose nitrate, see also CN, NC
CNR	Elastomeric terpolymer from tetrafluoroethylene, trifluoronitrosomethane, and a small amount of an unsaturated termonomer, e.g., nitrosoperfluorobutyric acid; nitroso rubber, carboxynitroso rubber; see also AFMU
CO	1. Elastomeric poly(chloromethyl)oxirane; poly(epichlorohydrin) (ANSI/ASTM; ISO); see also CHR 2. Cotton (EDV); see also Bw
COX	Carboxylic rubber
CP	Cellulose propionate (ASTM, DIN, ISO, IUPAC)
C.P.	Chemically pure
CP2	Alternating copolymer from vinyl ether and maleic acid
CP4	Copolymer from acrylic acid and maleic acid
CPE	Chlorinated poly(ethylene) (ASTM); see also CM
CPET	Crystallizable poly(ethylene terephthalate) (i.e., a PET with fast-acting nucleating agents) (trade journals)
CPI	*cis*-Poly(isoprene); see also IR, PIP
CPP	Spinning paper (small cut strips from sodium cellulose or sulfite cellulose paper) or cellulon (wet strips from cellulose pulp) (DIN)
CPVC	1. Postchlorinated poly(vinyl chloride) (ASTM); see also PC, PeCe, PVCC 2. Critical pigment concentration
CR	Poly(chloroprene) (ANSI/ASTM, BS, IUPAC)
CRP	Carbon fiber reinforced plastics; see also CFK
CS	Casein (ASTM, DIN, ISO, IUPAC)
CSM	Chlorosulfonated poly(ethylene) (ANSI/ASTM); see also CSPR, CSR
CSPR	Chlorosulfonated poly(ethylene) (BS); see also CSM, CSR
CSR	Chlorosulfonated poly(ethylene); see also CM, CSPR
CT	1. Cellulose triacetate (DIN); see also CTA, TA 2. Continuous thread
CTA	Cellulose triacetate (ASTM); see also CT, TA
CTFE	Poly(chlorotrifluoroethylene) (ASTM); see also CEM, CFM, PCTFE
CTP	*N*-Cyclohexylthiophthalimide (WTR)
CTS	Compact tension specimen
CU	Copper silk ("cupro silk" = regenerated cellulose fibers) (EDV); see also CC
CuDEC	Copper diethyl dithiocarbamate (WTR); also CDC, CuDC
CuDMC	Copper dimethyl dithiocarbamate (WTR); see also CuDMDC
CuMBT	Copper mercaptobenzothiazole; see also CMBT
Culhp	Copper laminated hard paper (German technical literature)
CV	Viscose (rayon) (DIN); see also VI
DABCO®	Triethylenediamine = Diazabicyclo [2.2.2] octane
DAC	Diallyl chlorendate; diallyl 1,4,5,6,7,7-hexachlorobicyclo[2.2.2]hept-5-ene-2,3-dicarboxylate (ASTM)
DADI	Dianisidine diisocyanate (WTR)
DAF	Diallyl fumarate (ASTM)
DAIP	Diallyl isophthalate resin (ASTM)
DAM	Diallyl isophthalate resin (ASTM)
DAP	Diallyl phthalate resin (ASTM, DIN); see also FDAP
DATBP	Diallyl tetrabromophthalate
DBDPO	Decabromodiphenyl oxide
DBEEA	Dibutoxyethoxyethyl adipate (WTR)
DBEP	Bis(2-butoxyethyl) phthalate
DBF	Dibutyl fumarate
DBNPG	Dibromoneopentyl glycol
DBP	Dibutyl phthalate (ASTM, DIN, ISO, IUPAC)
DBPPD	*N*,*N*′-Di(*s*-butyl)-*p*-phenylenediamine (WTR)
DBS	Dibutyl sebacate
DBTDL	Dibutyltin dilaurate
DBTU	Dibutyl urea (WTR)
DCBS	Benzothiazyl-2-dicyclohexylsulfene amide (WTR)
DCHP	Dicyclohexyl phthalate
DCP	Dicapryl phthalate (ASTM, DIN, ISO, IUPAC)
DDP	Didecyl phthalate
DE	Diatomaceous earth
DEA	*N*,*N*-Diethylaniline
DEAE	(Diethylamino)ethyl- (with cellulose)
DEG	Diethylene glycol
DEHP	Bis(2-ethylhexyl) phthalate; see also DOP
DEP	Diethyl phthalate (ISO)
DETA	Diethylenetriamine (WTR)
DETU	Diethyl thiourea (WTR)
DGEBA	Diglycidyl ethers of bisphenol A
DHA	Dehydroacetic acid, acetic anhydride
DHBP	2,4-Dihydroxy benzophenone (WTR)
DHP	Diheptyl phthalate (ISO)
DHS	Dihexyl sebacate (WTR)
DHXP	Dihexyl phthalate (ISO)
DIBA	Diisobutyl adipate
DIBP	Diisobutyl phthalate (DIN, ISO)
DIDA	Diisodecyl adipate (ASTM, DIN, ISO, IUPAC)

DIDG	Diisodecyl glutarate
DIDP	Diisodecyl phthalate (ASTM, DIN, ISO, IUPAC)
DINA	Diisononyl adipate (ISO)
DINP	Diisononyl phthalate (DIN, ISO)
DIOA	Diisooctyl adipate (ASTM, DIN, ISO, IUPAC)
DIOF	Diisooctyl fumarate
DIOM	Diisooctyl maleate (WTR)
DIOP	Diisooctyl phthalate (ASTM, DIN, ISO, IUPAC)
DIOS	Diisooctyl sebacate
DIOZ	Diisooctyl azelate
DIPP	Diisopentyl phthalate
DITDP	Diisotridecyl phthalate (DIN, ISO); see also DITP
DITP	Diisotridecyl phthalate (DIN); see also DITDP
DLTDP	Dilauryl 3,3′-thiodipropionate
DMA	*N,N*-Dimethylacetamide
DMAA	*N,N*-Dimethylacetoacetamide
DMAC	*N,N*-Dimethylacetamide
DMBPPD	*N*-(1,3-Dimethylbutyl)-*N′*-phenyl-*p*-phenylenediamine (WTR)
DMBS	Dimethylbenzothiazylsulfeneamide (WTR)
DMC	Dough molding compound (UP resin) (DIN)
DME	1. Dimethylethanolamine, 2-(dimethylamino)ethanol
DMEP	Bis(2-methoxyethyl) phthalate
DMF	1. *N,N*-Dimethyl phthalate (ISO); 2. *N,N*-dimethyl formamide
DMG	Dimethyl glutarate
DMHPPD	*N,N′*-Di(3-methylheptyl)-*p*-phenylenediamine (WTR)
DMP	Dimethyl phthalate (ISO)
DMPTD	Dimethylphenyl thiuram disulfide (WTR)
DMSO	Dimethyl sulfoxide
DMT	Dimethyl terephthalate
DMTD	2,5-Dimercapto-1,3,4-thiadiazole (WTR)
DMTDP	Dimyristyl-3,3′-thiodipropionate
DMTDS	2,5-Dimercapto-1,3,4-thiadiazole disulfide (WTR)
DNA	1. Deoxyribonucleic acid 2. Dinonyl adipate
DNHZ	Di(*n*-hexyl) azelate
DNODA	Di(*n*-octadecyl) adipate
DNPDP	Di(*n*-octadecyl) phthalate (ASTM)
DNP	Dinonyl phthalate (ASTM, ISO, IUPAC)
DNPT	*N,N*-Dinitrosopentamethylene tetramine
DNS	1. Di(3,5,5-trimethylhexyl) sebacate (WTR) 2. Deoxyribonucleic acid (German: Desoxyribonukleinsäure)
DOA	Dioctyl adipate, bis(2-ethylhexyl)adipate (ASTM, DIN, ISO, IUPAC)
DODP	Di(octadecyl) phthalate (ISO)
DOF	Dioctyl fumarate
DOIP	Dioctyl isophthalate, bis(2-ethylhexyl) isophthalate (DIN, ISO)
DOM	Dioctyl maleate, bis(2-ethylhexyl) maleate (WTR)
DOP	Dioctyl phthalate, bis(2-ethylhexyl) phthalate (ASTM, DIN, ISO, IUPAC)
DOS	Dioctyl sebacate, bis(2-ethylhexyl) sebacate (ASTM, DIN, ISO, IUPAC)
DOTG	*N,N′*-Di(*o*-tolyl)guanidine (WTR)
DOTP	Dioctyl terephthalate, bis(2-ethylhexyl) terephthalate (DIN, ISO)
DOZ	Dioctyl azelate, bis(2-ethylhexyl) azelate (ASTM, DIN, ISO, IUPAC)
DPA	Diphenylamine
DPCF	Diphenyl cresyl phosphate (ASTM, ISO); see also DPCP
DPCP	Diphenyl cresyl phosphate (ASTM, ISO); see also DPCF
DPG	*N,N′*-Diphenylguanidine (WTR)
DPOF	Diphenyl 2-ethylhexyl phosphate (ASTM, ISO); see also EHDPP
DPOP	Diphenyl octyl phosphate
DPP	1. Diphenyl phthalate 2. Dipropyl phthalate
DPPD	*N,N′*-Diphenyl-*p*-phenylene diamine (WTR)
DPQD	Dibenzoyl-*p*-quinone dioxime
DPTD	Di(pentamethylene)thiuram hexasulfide, see also DPTH
DPTH	Di(pentamethylene)thiuram hexasulfide (WTR), see also DPTD
DPDM	Di(pentamethylene)thiuram monosulfide (WTR)
DPTT	Di(pentamethylene)thiuram tetrasulfide (WTR)
DPTU	Diphenyl thiourea (WTR)
DR	Degree of reaction
DRC	Dry rubber content
DS	Degree of substitution
DSTDP	Distearyl 3,3′-dithiopropionate
DTBP	Di(*t*-butyl) peroxide (WTR)
DTDC	Dithiodicaprolactam (WTR)
DTDM	4,4-Dithiodimorpholine (WTR)
DTDP	Di(tridecyl) phthalate
DTPD	1. Di(tridecyl) phthalate 2. *N,N′*-Ditolyl-*p*-phenylenediamine (WTR)
DUP	Di(undecyl) phthalate
DVB	Divinyl benzene
EA	Segmented polyurethane fiber (Elasthan fiber) (EDV); see also PUE
EAA	Ethylene acrylic acid copolymers
EAM	Elastomeric copolymer of ethylene and vinyl acetate (ANSI/ASTM)
EBM	Extrusion blow molding

EC Ethyl cellulose (ASTM, DIN, ISO, IUPAC)

ECB Blends from ethylene copolymers with bitumen

ECO Elastomeric copolymer from ethylene oxide and epichlorohydrin, chloromethyloxirane (ANSI/ASTM); see also CHC

ECTFE Poly(ethylene-*co*-chlorotrifluoroethylene); see also E-CTFE

E-CTFE Poly(ethylene-*co*-chlorotrifluoroethylene); see also ECTFE

EDA Ethylene diamine (WTR)

EDC Ethylene dichloride, 1,2-dichloroethane

EDMA Ethylene glycol dimethacrylate

EDTA Ethylenediaminetetraacetic acid

EEA Elastomeric copolymer from ethylene and ethyl acrylate (ASTM); see also E/EA

E/EA Elastomeric copolymer from ethylene and ethyl acrylate (ISO); see also EEA

EG Ethylene glycol

EHDPP 2-Ethylhexyldiphenyl phosphate; see also DPOF

EHEC Hydroxyethylcellulose

EL Elastodiene fiber (EDV); see also EU, LA

ELO Epoxidized linseed oil (DIN, ISO)

EMA 1. Copolymer from ethylene and methacrylic acid (ASTM)
2. Alternating copolymer from ethylene and maleic anhydride

EOT Ethylene oxide polysulfide

EP Epoxy resin (ASTM, DIN, ISO, IUPAC)

E/P Copolymer from ethylene and propylene (ISO); see also AP, APR, EPM, EPR

EPC Easy processing channel black

EPD Elastomeric terpolymer from ethylene, propylene, and a nonconjugated diene (ASTM); see also APT, EPDM, EPT, EPTR

EPDM Elastomeric terpolymer from ethylene, propylene, and a nonconjugated diene (ASTM); see also, APT, APTK, EPD, EPT, EPTR

EPE Ester of an epoxy resin (DIN)

EP-G-G Prepreg from an epoxy resin and a textile glass fabric (German technical literature)

EP-K-L Prepreg from an epoxy resin and a carbon fiber fabric (German technical literature)

EPM Elastomeric copolymer of ethylene and propylene (ANSI/ASTM, ISO); see also AP, APR, E/P, EPM, EPR

EPR Elastomeric copolymer of ethylene and propylene (BS); see also AP, APR, E/P, EPM

EPS Expanded poly(styrene) [poly(styrene) foam]

EPT Elastomeric terpolymer from ethylene, propylene and a diene; see also APT, APTK, EPD, EPDM, EPTR

EPTD Diethyl diphenyl thiuram disulfide (WTR)

EPTM Diethyl diphenyl thiuram monosulfide (WTR)

EPTR Elastomeric terpolymer from ethylene, propylene and a diene (BS); see also APT, APTK, EPD, EPDM, EPT

E-PVC Emulsion-polymerized poly(vinyl chloride)

ERM Elastin reservoir molding

E-SBR Copolymer from styrene and butadiene, polymerized in emulsion

ESC Environmental stress cracking

ESCR Environmental stress cracking resistance

ESO Epoxidized soy bean oil (DIN, ISO)

ET Ethylene polysulfide

ETCH Ethylene chlorohydrine; 2-chloroethanol

ETFE Copolymer of ethylene and tetrafluoroethylene (ASTM)

E/TFE Copolymer of ethylene and tetrafluoroethylene

ETG Ethylene triglycol

ETMQ 2,2,4-Trimethyl-1,2-dihydro-6-ethoxy quinoline (WTR)

EU 1. Elastomeric polyurethane with polyether segments (ANSI/ASTM)
2. Polyether rubber with carbon, oxygen, and nitrogen in the polymer chain (ISO)
3. Elastodiene fibers (ASTM); see also EL, LA
Classifications 1 and 2 probably refer to the same polymer

EVA Copolymer from ethylene and vinyl acetate (ASTM, DIN, ISO); see also E/VAC, EVM, VAE

E/VAC Copolymer from ethylene and vinyl acetate (ASTM, DIN); see also E/VAC, EVM, VAE

EVAL® Copolymer of ethylene and vinyl alcohol

E/VAL Copolymer of ethylene and vinyl alcohol

EVE Ethylene vinyl ether

EVOH® Copolymer with ethylene and vinyl alcohol units

EVM Elastomer from ethylene and vinyl acetate; see also EVA, E/VAC, VAE

F 1. Fluorine (IUPAC)
2. Furnace black

FC Fluorocarbon

FDAP Diallyl phthalate or resins therefrom, see also DAP

FE Fluorine containing elastomer

FEF Fast extruding furnace black

FEP Copolymer from tetrafluoroethylene and hexafluoropropylene (ASTM, DIN, ISO, IUPAC); see also PFEP

FF 1. Resin from furan and formaldehyde (ASTM)
2. Fine furnace black

Fi	Fique (DIN)
FK	Fiber reinforced plastic ("faserverstärkter Kunststoff"; German technical literature)
FKM	Elastomer with saturated main chain and fluoro, perfluoroalkyl, or perfluoroalkoxy substituents (ANSI/ASTM); see also FPM
FL	Fluorofiber (EDV); probably fibers from poly(tetrafluoroethylene)
Fl	Flax (DIN)
FMQ	Methyl fluorosilicone rubber; see also MFQ
FPM	1. Elastomeric copolymer from vinylidene fluoride and hexafluoropropylene (ASTM) 2. Fluoroelastomers in general 3. Rubbers having fluoro and fluoroalkyl or fluoroalkoxy substituent groups at the polymer chain (ISO)
FPVC	Flexible PVC film
FQ	Elastomeric silicone with fluorine containing substituents (ASTM); see also FSI
FR	Flame retardant
FRP	Glass fiber reinforced polyester
FSI	Fluorinated silicone (ASTM); see also FQ
FT	Fine thermal black
FVMQ	Silicone rubber with fluorine, vinyl, and methyl substituents (ANSI/ASTM)
FVSI	Fluorosilicone
GC	Glass coupled
GEP	Glass fiber reinforced epoxy resin
GF	Glass fiber reinforced plastic; see also GFK
GF-EP	Glass fiber reinforced epoxy resin
GFK	Glass fiber reinforced plastic (German technical literature)
GF-PF	Glass fiber reinforced phenolic resin
GF-UP	Glass fiber reinforced unsaturated polyester (German technical literature)
GI	Broom (fibers from) (EDV)
GL	Glass fiber (DIN, EDV)
GP	1. Gutta percha 2. General purpose (ASTM)
GPF	General purpose furnace black
GPO	Elastomeric copolymer from propylene oxide and allyl glycidyl ether (ANSI/ASTM)
GR	1. Glass fiber reinforced 2. Former US symbol for "government rubber"
GR-A	Former US symbol for nitrile rubber; see also GR-N
GRAS	Generally recognized as safe
GR-I	Former US symbol for butyl rubber
GR-N	Former US symbol for nitrile rubber; see also GR-A
GRP	Glass fiber reinforced plastic
GR-P	Glass fiber reinforced plastic
GR-S	Former US symbol for styrene–butadiene rubber
Gu	Guanaco wool (DIN)
GU	Fibers from diene rubbers (old German technical literature)
GUP	Glass fiber reinforced unsaturated polyester resins (German technical literature)
GUR	Ultrahigh molecular weight poly(ethylene); see also UHMPE
GV	Glass fiber reinforced thermoplastics
HA	Human hair (EDV)
HAF	High abrasion furnace black
HBPA	2,2-bis(4-hydroxycyclohexyl)propane
HDI	Hexamethylene-1,6-diisocyanate (WTR)
HDPE	High-density poly(ethylene) (ASTM); in the older German technical literature, HDPE was sometimes used for poly(ethylene) from high pressure polymerizations ("Hochdruck"), i.e., for poly(ethylene)s with mostly *low* densities; see also PEH
He	1. Helium 2. Henequen (DIN); see also HE
HE	Henequen (EDV); see also He
HEC	Hydroxyethylcellulose
HEMA	2-Hydroxyethyl methacrylate
HEXA	Hexamethylene tetramine (WTR); see also HMT
HF	Half linen (EDV)
HFIP	Hexafluoroisopropanol; 1,1,1,3,3,3-hexafluoro-2-propanol; see also HFP
HFP	1. Hexafluoroisopropanol; see also HFIP 2. Hexafluoropropene
Hgw	Hard fabric ("Hartgewebe", German technical literature)
HIPS	High impact poly(styrene)
Hm	Hard mat (German technical literature)
HM	Hot melt adhesive
HMC	Sheet molding compound with high glass fiber content (UP resin) (DIN)
HMDAC	Hexamethylene diamine carbamate (WTR)
HMDI	Hydrogenated diphenylmethane diisocyanate
HMF	High-modulus furnace black
HMPT	Hexamethylphosphoric acid triamide; see also HPT
HMT	Hexamethylene tetramine
HMWPE	Poly(ethylene) with very high molar mass (molecular weight)
H-NBR	Hydrogenated nitrile rubber
HP	Hard paper (German technical literature)
HPA	Hydroxypropyl acrylate
HPC	1. Hydroxypropyl cellulose 2. Hard processing channel black
HPMC	(Hydroxypropyl)methyl cellulose
HPT	Hexamethyl phosphorotriamide; see also HMPT

HR	High resiliency foam
Hs	Hare hair (DIN)
HS	1. Heat stabilized 2. High percent solids
HTBA	High temperature blowing agent
HTE	Hydroxyl terminated polyether
HTST	High-temperature, short-term processing
Hz	Goat hair ("Haar von der Ziege"; DIN)
ICR	Initial concentration rubber (from undiluted natural rubber latex)
IEN	Interpenetrating elastomeric network
IFR	Intumescent flame retardant
IHPN	Interpenetrating homopolymer network
IIR	Elastomeric copolymer from isobutene and isoprene (ANSI/ASTM, IUPAC); see also Butyl, GR-I, PIBI
IM	Poly(isobutylene); see also PIB
IMW	Intermediate molecular weight
IPA	Isophthalic acid
IPDI	3-Isophorone diisocyanate (WTR)
IPN	Interpenetrating polymer network
IPO	Isopropyl oleate
IPP	Isopropyl percarbonate; diisopropyl peroxydicarbonate
IPPD	*N*-Isopropyl-*N'*-phenyl-*p*-phenylenediamine (WTR)
IR	Synthetic *cis*-1,4-poly(isoprene) (ANSI/ASTM, BS, IUPAC); see also CPI, PIP
IRT	Insect resistant treated
IS	Insoluble sulfur (WTR)
IT	Interrupted or split thread
IVE	Isobutyl vinyl ether
Ju	Jute (DIN); see also JU
JU	Jute (EDV); see also Ju
KA	Casein fibers (DIN)
KE	Kenaf (EDV); see also Kf
KEK	Carbon fiber reinforced plastic (DIN)
Kf	Kenaf (DIN); see also KE
KFA	Potassium fatty acid soap
Km	Camel hair (DIN)
Kn	Rabbit hair (DIN)
Ko	Coir (DIN)
Kp	Kapok (DIN); see also KP
KP	Kapok (EDV); see also Kp
Kz	Cashmere goat wool (DIN)
LA	Fibers from diene rubbers (DIN); see also EL, EU
LDPE	Low-density poly(ethylene) (ASTM); in the older German technical literature, LDPE was sometimes used for poly(ethylene) from low pressure polymerizations, i.e., for poly(ethylene)s with mostly *high* densities; see also NDPE
LI	Linen, flax (EDV)
LIM	Liquid impingement molding (now RIM)
LIPN	Latex interpenetrating polymer network
LLDPE	Linear low-density poly(ethylene) (ASTM)
Lm	Llama wool (DIN)
LRM	Liquid reaction molding (now RIM)
LRMR	Reinforced liquid reaction molding
L-SBR	Solution polymerized SBR
M	Medium thermal carbon black
Ma	Manila hemp fibers (Abaca) (DIN)
MA	1. Modacrylic fiber (EDV); see also PAM 2. Maleic anhydride
MABS	Copolymer from methyl methacrylate, acrylonitrile, butadiene, and styrene
MAN	Methacrylonitrile
MBI	Mercaptobenzimidazole; 2-benzimidazolethiol
MBS	1. Copolymer from methyl methacrylate, butadiene, and styrene 2. Benzthiazyl-2-sulfene morpholide (WTR)
MBSS	2-(4-Morpholinyldithio)benzothiazole (WTR)
MBT	Mercaptobenzothiazole; 2-benzthiazolethiol
MBTS	2-Benzthiazolyl disulfide
MC	Methyl cellulose
MCB	Monochlorobenzene
MD	Modal fiber (EDV)
MDA	1. Methylenediamine 2. 4,4'-Diaminodiphenylmethane
MDAC	4-Methyl-7-(diethylamino)coumarine
MDI	4,4'-Diphenylmethane diisocyanate; methylene di-*p*-phenylene diisocyanate
MDPE	Poly(ethylene) with medium density ($\approx 0.93-0.94\ g/cm^3$) (ASTM)
ME	Metal fiber (EDV)
MEK	Methyl ethyl ketone; 2-butanone
MEKP	Methyl ethyl ketone peroxide
MF	Melamine–formaldehyde resin (ASTM, DIN, ISO, IUPAC)
MFK	Metal fiber reinforced plastic (German technical literature)
MFQ	Silicone rubber having both methyl and fluorine substituent groups on the polymer chain (ISO); see also FMQ
MG	Maguey (EDV)
MMA	Methyl methacrylate (ASTM)
Mo	1. Molybdenum (IUPAC) 2. Mohair wool (DIN)
MOCA	4,4'-Methylenebis(2-chloroaniline); 3,3'-dichloro-4,4'-diaminodiphenylmethane
MOD	Modacryl fiber (EWG)
MP	Melamine–phenol–formaldehyde resin; see also MPF
MPE	Metallized polyester film

MPF	Melamine–phenol–formaldehyde resin (ISO); see also MP
MPPPD	*N*-Methyl-2-pentyl-*N'*-phenyl-*p*-phenylene diamine (WTR)
MPQ	Silicone rubbers having both methyl and phenyl substituent groups on the polymer chain (ISO); see also PMQ
MPTD	Dimethyl diphenyl thiuram disulfide (WTR)
M-PVC	PVC, polymerized in bulk (mass) (German technical literature)
MPVQ	Silicone rubbers having methyl, phenyl, and vinyl substituent groups on the polymer chain (ISO); see also PVMQ
MQ	Elastomeric silicones with methyl substituents (ANSI/ASTM)
MR	Mold resistant
Ms	Silk from the silkworm (mulberry spinner; *Bombyx mori L.*) (DIN)
MSA	Maleic anhydride (German technical literature)
MT	Metal fibers (DIN); see also ME
MTI	Methylene tri(*p*-phenylene isocyanate)
MTM	Mixed tertiary mercaptans
MVQ	Silicone rubbers having both methyl and vinyl substituent groups on the polymer chain (ISO); see also VMQ
MWR	Molding with rotation
NBR	Elastomeric copolymer from butadiene and acrylonitrile; nitrile rubber (ANSI/ASTM, BS, IUPAC); see also GR-N, PBAN
NC	Cellulose nitrate; see also SN
NCR	Elastomeric copolymer from acrylonitrile and chloroprene (ANSI/ASTM, IUPAC)
NDBC	Nickel dibutyl dithiocarbamate (WTR)
NDGA	Nordihydroguaiaretic acid
NDI	1,5-Diisocyanatonaphthalene (WTR)
NDOP	*n*-Decyl-*n*-octyl phthalate; see also ODP
NDPA	*N*-Nitrosodiphenylamine (WTR)
NDPE	Low-density poly(ethylene); see also LDPE
NEM	*N*-Ethylmorpholine
Nf	Phormium (DIN)
NHDP	*n*-Hexyl-*n*-decyl phthalate
NIR	Elastomeric copolymer from acrylonitrile and isoprene (ANSI/ASTM)
NK	Natural rubber; see also NR
NMP	*N*-Methylpyrrolidone
NODA	*n*-Octyl-*n*-decyl adipate
NODP	*n*-Octyl-*n*-decyl phthalate; see also ODP
NPG	Neopentyl glycol
NPP	*n*-Propyl percarbonate; dipropyl peroxydicarbonate
NR	Natural rubber (ANSI/ASTM, IUPAC); see also NK
OBSH	*p,p'*-Oxybis(benzenesulfonyl hydrazide)
ODP	Octadecyl phthalate (ISO); see also NODP
ODPA	Octylated diphenylamine (WTR)
ODPP	Octyldiphenyl phthalate
OER	Oil-extended rubber
OPET	Oriented poly(ethylene terephthalate)
OPP	Oriented poly(propylene) film or bottles; see also PP
OPR	Elastomeric polymer from propylene oxide
OPS	Oriented poly(styrene) film
OPVC	Oriented poly(vinyl chloride)
OT	Polysulfide rubber
OTBG	*o*-Tolyl biguanide (WTR)
OTOS	*N*-Oxydiethylenedithiocarbamyl-*N'*-oxydiethylenesulfamide (WTR)
P3FE	Poly(trifluoroethylene) (IUPAC)
PA	1. Polyamide (ASTM, DIN, ISO, IUPAC); the acronym PA is normally followed by a number, a combination of numbers, a letter, or a combination of letters and numbers. A single number *i* refers to a polyamide from an α,ω-amino acid or its lactam with *i* carbon atoms in the aliphatic chain of the monomeric unit. A combination of two numbers indicates polyamides from the polycondensation of a diamine and a dicarboxylic acid; the first number gives the number of methylene groups of aliphatic diamines, the second number the number of carbon atoms of straight-chain aliphatic dicarboxylic acids. Both numbers are sometimes separated by a period. "I" stands for isphthalic acid, "T" for terephthalic acid. 2. Phthalic anhydride
PA*	Saran coated polyamide (ASTM)
PA**	Metal coated polyamide (ASTM)
PAA	Poly(acrylic acid) (ASTM)
PAB	*p*-Aminobiphenyl
PABM	Polyaminobismaleimide
PAC	Poly(acrylonitrile) fiber (DIN, IUPAC); see also PAN, PC
PAE	Phthalate ester plasticizer
PAI	Polyamide-imide (ASTM)
PAM	Modacryl fiber (DIN); see also MA
PAMS	Poly(α-methyl styrene)
PAN	1. Poly(acrylonitrile) fiber (ASTM, IUPAC, EDV); see also PAC, PC. PAN® is also a German trademark. 2. Phenyl-α-naphthylamine (WTR)
PAPA	Polyazelaic polyanhydride
PAPDC	Pentamethylene ammonium *N*-pentamethylene dithiocarbamate (WTR)
PAPI	Polymethylenepolyphenylene isocyanate; see also PMPPI

PARA	Polyarylamide (ASTM)
PAS	1. Polyaryl sulfone 2. Poly(acrylic acid) (German technical literature)
PAT	Polyaminotriazole
PB	1. Poly(1-butene) (ASTM, DIN, ISO); see also BT, PBT 2. Polyurea fiber (EDV); see also PUA 3. Elastic polydiene fiber (German technical literature)
PBA	Physical blowing agent
PBAN	Copolymer from butadiene and acrylonitrile (ASTM); see also GR-N, NBR
PBB	Polybrominated biphenyl
PBI	Polybenzimidazole
PBMA	Poly(*n*-butyl methacrylate)
PBNA	Phenyl-β-naphthylamine; *N*-phenyl-2-naphthylamine
PBR	Copolymer from butadiene and vinylpyridine (ANSI/ASTM, IUPAC)
PBS	Copolymer from butadiene and styrene (ASTM); see also SB, GR-S, SBR
PBT	1. Poly(butylene terephthalate) (ASTM); see also PBTP, PTMT 2. Poly(1-butene), see also BT, PB 3. Poly(*p*-phenylenebenzobisthiazole)
PBTP	Poly(butylene terephthalate) (DIN, ISO); see also PTMT
PC	1. Polycarbonate (ASTM, DIN, ISO, IUPAC); see also PCO 2. Poly(acrylonitrile) fiber (EWG, EDV); see also PAC, PAN 3. Formerly: chlorinated PVC; see also CPVC, PeCe, PVCC
PCB	Polychlorinated biphenyl
PCD	Polycarbodiimide
PCDT	Poly(1,4-cyclohexylenedimethylene terephthalate)
PCF	Poly(chlorotrifluoroethylene) fiber (DIN)
PCO	Polycarbonate; see also PC
PCTFE	Poly(chlorotrifluoroethylene) (ASTM, DIN, ISO, IUPAC); see also CEM, CFM, CTFE
PCU	Unchlorinated poly(vinyl chloride) (old German technical literature)
PDAP	Poly(diallyl phthalate) (ASTM, DIN, ISO, IUPAC); see also DAP, FDAP
PDMS	Poly(dimethyl siloxane)
PE	1. Poly(ethylene) (ASTM, DIN, ISO, IUPAC); see also PL 2. Polyester fiber (EWG) 3. Pentaerythritol
PEA	Poly(ethyl acrylate)
PEC	Chlorinated poly(ethylene) (DIN, ISO); see also CPE
PeCe	Chlorinated PVC; see also CPVC, PC, PVCC
PEE	Polyester ether fibers (containing diol and *p*-hydroxybenzoate units) (DIN)
PEEK	Polyether ether ketone = Poly(oxy-1,4-phenylene-oxy-1,4-phenylene-carbonyl-1,4-phenylene)
PEEKK	Polyether ether ketone ketone
PEG	Poly(ethylene glycol)
PEH	Poly(ethylene) with high density; see also HDPE
PEHA	Pentaethylenehexamine (WTR)
PE-HD	Poly(ethylene) with high density (DIN, ISO); see also HDPE, PEH
PEK	Polyether ketone
PEKEKK	Polyether ketone ether ketone ketone
PEKK	Polyether ketone ketone
PEL	Poly(ethylene) with low density; see also LDPE, PE-LD
PEM	Poly(ethylene) with medium density; see also MDPE
PEO	Poly(ethylene oxide) (ASTM, IUPAC); see also PEOX
PEOX	Poly(ethylene oxide) (DIN, ISO); see also PEO
PEP	Thermoplastic copolymer from ethylene and propylene
PES	1. Polyester fiber (DIN); see also PET, PL 2. Polyether sulfone (ASTM); see also PSU
PESU	Polyether sulfone (ASTM); see also PES
PET	Poly(ethylene terephthalate) (ASTM); see also PETP, PE, PES
PET*	Saran coated PET (ASTM)
PETP	Poly(ethylene terephthalate) (ASTM, DIN, ISO, IUPAC); see also PET, PE, PES
PETS	Pentaerythryl tetrastearate (WTR)
PF	Phenol–formaldehyde resin (ASTM, DIN, ISO, IUPAC)
PFA	Perfluoroalkoxy resins, copolymers of tetrafluoroethylene and perfluorinated alkyl vinyl ethers
PFEP	Copolymer from tetrafluoroethylene and hexafluoropropylene; see also FEP
PF-P-B	Phenolic resin–paper prepreg
PG	Propylene glycol
PHAS	Poly(α-hydroxy acrylic acid)
PHA	Poly(hydroxyalkanoate)
PHB	1. Poly(*p*-hydroxybenzoate); see also POB 2. Poly(α-hydroxybutyrate)
PHEMA	Poly(2-hydroxyethyl methacrylate)
PHP	Physiological hydrophilic polymers
PI	1. *trans*-1,4-Poly(isoprene), gutta percha (BS) 2. Polyimide (ASTM) 3. Paper (EDV)
PIB	Poly(isobutene) (ASTM, BS, DIN, ISO, IUPAC)

PIBI	Elastomeric copolymer from isobutene and isoprene (butyl rubber); see also Butyl, GR-I, IIR
PIBO	Poly(isobutylene oxide)
PIP	Synthetic *cis*-1,4-poly(isoprene); see also CPI, IR
PIR	Poly(isocyanurate) (DIN)
PIS	Poly(isobutylene)
PL	1. Poly(ethylene) (EWG); see also PE 2. Polyester fiber (EDV); see also PES, PET
PMA	1. Pyromellitic acid 2. Pyromellitic dianhydride 3. Poly(methyl acrylate)
PMAC	Polymethoxy acetal
PMCA	Poly(methyl α-chloromethacrylate) (ASTM)
PMF	Processed mineral fiber
PMI	Poly(methacrylimide)
PMMA	(Poly(methyl methacrylate) (ASTM, DIN, ISO, IUPAC)
PMMI	Poly(pyromellitimide)
PMP	Poly(4-methyl-1-pentene) (ASTM, DIN, ISO); see also TPX
PMPPI	Polymethylenepolyphenylene isocyanate; see also PAPI
PMQ	Silicon rubbers with methyl and phenyl substituents (ANSI/ASTM)
PMTT	3-Phenyl-5-mercapto-1,3,4-thiadiazole-2-thione (WTR)
PNA	Polynuclear aromatics
PNAH	Polynuclear aromatic hydrocarbons
PNF	Phosphorus nitrile fluoroelastomer
PNR	Poly(norbornene) rubber
PO	1. Elastomeric poly(propylene oxide) (ASTM) 2. Poly(olefin) 3. Phenoxy resin
POB	Poly(*p*-hydroxybenzoate); see also PHB
POM	Poly(oxymethylene), poly(formaldehyde) (ASTM, DIN, ISO, IUPAC)
POP	Poly(1,4-phenylene oxide) (ASTM); see also PPO
POR	Elastomeric copolymer from propylene oxide and allyl glycidyl ether
PP	Poly(propylene) (ASTM, DIN, EDV, ISO, IUPAC)
PP*	Metallized poly(propylene) (ASTM)
PPA	1. Poly(parabanic acid) 2. Poly(propylene adipate)
PPC	Chlorinated poly(propylene) (DIN)
PPI	Polymeric polyisocyanate
PPO[R]	Poly(oxy-1,4-phenylene) = "poly(phenylene oxide)"; see also POP
PPOX	Poly(propylene oxide) (ASTM, DIN, ISO)
PPP	Poly(*p*-phenylene)
PPS	Poly(*p*-phenylene sulfide) (ASTM)
PPSU	Poly(phenylene sulfone) (ASTM, DIN, ISO); see also PSU
PPT	Poly(propylene terephthalate)
PQ	Elastomeric silicone with phenyl substituents (ASTM)
PQD	*p*-Quinone dioxime (WTR)
PR	Regenerated protein fiber (EDV)
PS	Poly(styrene) (ASTM, DIN, ISO, IUPAC)
P-S	Pressure-sensitive adhesive
PSA	Pressure-sensitive adhesive
PSAB	Thermoplastic copolymer from styrene, acrylic ester and butadiene (DIN); see also SBV, S/B
PSAN	Thermoplastic copolymer from styrene and acrylonitrile (DIN), see also SAN
PSBR	Elastomeric terpolymer from vinylpyridine, styrene, and butadiene (ANSI/ASTM)
PSF	Polysulfone; see also PSO
PSI	Poly(methyl phenyl siloxane) (ASTM)
PSO	Polysulfone; see also PSF
PST	Poly(styrene) fiber with at least 85% styrene units (DIN)
PS-TSG	Poly(styrene) foam, processed by thermoplastic injection (German technical literature)
PSU	Poly(phenylene sulfone); see also PPSU
PSUL	Polysulfone (ASTM)
PTA	1. Purified terephthalic acid 2. Phthalic anhydride
PTF	Pole(tetrafluoroethylene) fiber
PTFE	Poly(tetrafluoroethylene) (ASTM, DIN, ISO, IUPAC)
PTMA	Poly(tetramethylene adipate)
PTMEG	Poly(tetramethylene ether glycol)
PTMG	Poly(oxytetramethylene glycol)
PTMT	Poly(tetramethylene terephthalate) = poly(butylene terephthalate) see also PBTP
PU	1. Polyurethane elastomer (BS) 2. Polyurethane fiber (EDV)
PUA	Polyurea fiber (DIN); see also PB
PUE	Segmented polyurethane fiber (DIN); see also EA
PUR	Polyurethane (ASTM, DIN, ISO, IUPAC); see also PU
PVA	1. Poly(vinyl acetate); see also PVAC 2. Poly(vinyl alcohol) (ASTM); see also PVAL, PVOH 3. Poly(vinyl alcohol) fiber with at least 85% vinyl alcohol units (DIN); see also VY 4. Poly(vinyl ether) (old German technical literature); see also PVOM, PVME
PVAC	Poly(vinyl acetate) (ASTM, DIN, ISO, IUPAC); see also PVA
PVAL	Poly(vinyl alcohol) (ASTM, DIN, ISO, IUPAC)

PVC	1. Poly(vinyl chloride) (ASTM, DIN, ISO, IUPAC); see also CL 2. Poly(vinyl chloride) fiber with at least 50 wt.% vinyl chloride units (DIN) 3. Pigment volume concentration
PVCA	Copolymer from vinyl chloride and vinyl acetate (ASTM, DIN, IUPAC); see also PVCAC
PVCAC	Copolymer from vinyl chloride and vinyl acetate (ASTM); see also PVCA
PVCC	Chlorinated PVC (DIN, ISO); see also CPVC, PC, PeCe
PVD	Poly(vinylidene chloride) fiber with at least 50 wt.% vinylidene chloride units (DIN); see also PVDC
PVDC	Poly(vinylidene chloride) (ASTM, DIN, ISO, IUPAC); see also PVD
PVDF	Poly(vinylidene fluoride) (ASTM, DIN, ISO, IUPAC); see also PVF
PVF	1. Poly(vinyl fluoride) (ASTM, DIN, ISO, IUPAC) 2. Poly(vinylidene fluoride); see also PVDF
PVFM	Poly(vinyl formal) (ASTM, DIN, ISO, IUPAC); see also PVFO
PVFO	Poly(vinyl formal) (DIN); see also PVFM
PVI	Poly(vinyl isobutyl ether)
PVID	Poly(vinylidene cyanide)
PVK	Poly(*N*-vinylcarbazole) (ASTM, DIN, ISO)
PVM	1. Copolymer from vinyl chloride and vinyl methyl ether 2. Fiber from the copolymer of vinyl and/or vinylidene compounds with a main component of at least 85 wt.% (DIN)
PVME	Poly(vinyl methyl ether)
PVMQ	Silicone rubber with methyl, phenyl, and vinyl substituents (ANSI/ASTM)
PVOH	Poly(vinyl alcohol); see also PVA, PVAL
PVP	Poly(*N*-vinylpyrrolidone) (ASTM, DIN, ISO)
PVSI	Poly(dimethyl siloxane) with phenyl and vinyl substituents (ASTM)
PY	Unsaturated polyester resins (BS)
QMC	Quick molding change
Ra	Ramie (DIN)
RA	Ramie (EDV)
RAM	Restricted area molding
RAN	Residual acrylonitrile (monomer in polymer)
RF	Resorcine–formaldehyde resin
Rh	1. Rhenium (IUPAC) 2. Horse hair ("Rosshaar") (DIN)
RHB	Reheat blow molding
RHC	Rubber hydrocarbon content
Ri	Cow hair ("Rind") (DIN)
RIM	Reaction injection molding
Ro	Rosella (DIN)
RP	Reinforced plastics
RP/C	Reinforced plastics (composite)
RP/M	Reinforced polyester (mortar)
RPVC	Rigid PVC film
RRIM	Reinforced reaction injection molding
RTP	Reinforced thermoplastic
RTV	Room temperature vulcanization
RVCM	Residual vinyl chloride monomer
RUC	Chlorinated isoprene rubber (DIN)
S	Styrene
SAF	Super abrasion furnace black
SAF-HS	High structure SAF
SAF-LS	Low-structure SAF
SAIB	Sucrose acetate isobutyrate
SAL	Salicylic acid
SAN	Thermoplastic copolymer from styrene and acrylonitrile (ASTM, DIN, ISO, IUPAC); see also PSAN
SB	Thermoplastic copolymer from styrene and butadiene (ASTM); see also S/B
S/B	Thermoplastic copolymer from styrene and butadiene (ANSI/ASTM, ISO, IUPAC); see also SB
SBP	1. *s*-Butyl percarbonate; di-*s*-butyl peroxydicarbonate 2. Styrene–butadiene resin
SBR	Elastomeric copolymer from styrene and butadiene (ASTM, BS, IUPAC); see also GR-S
SBS	1. Solid bleached sulfate 2. Styrene–butadiene–styrene triblock copolymer
SCR	Elastomeric copolymer from styrene and chloroprene (ANSI/ASTM, IUPAC)
SDD	Sodium dimethyldithiocarbamate
SDPA	Aralkylated diphenylamines (WTR)
SE	1. Sulfoethyl- (with cellulose) 2. Silk ("Seide") (EDV)
SEBS	Styrene–ethylene/butylene–styrene triblock copolymer (= hydrogenated styrene–isoprene–styrene triblock copolymer)
SEDC	Selenium diethyl dithiocarbamate (WTR)
S-EPDM	Sulfonated ethylene–propylene–diene terpolymer
SF	Structural foam
SFK	Synthetic fiber reinforced plastic (German technical literature)
SFP	Scrapless forming process
SHIPS	Super-high impact poly(styrene)
Si	1. Silicon (IUPAC) 2. Sisal (DIN); see also SI
SI	1. Silicones in general (ASTM, DIN, ISO) 2. Poly(dimethyl siloxane) (ASTM) 3. Thermoplastic silicone (IUPAC) 4. Sisal (EDV); see also SI

SIN	1. Simultaneous interpenetrating network 2. Semi-interpenetrating network
SIPN	Sequential interpenetrating polymer network
SIR	1. Elastomeric copolymer from styrene and isoprene (ANSI/ASTM, IUPAC) 2. Standardized Indonesian rubber
SIS	Styrene–isoprene–styrene triblock copolymer
SL	Slag wool (DIN)
SM	Styrene monomer
SMA	Copolymer from styrene and maleic anhydride (ASTM)
SMC	Sheet molding compound (UP resin)
SMR	Standardized Malaysian rubber
SMS	Copolymer from styrene and α-methylstyrene (ASTM, DIN); see also S/MS
S/MS	Copolymer from styrene and α-methylstyrene (DIN, ISO); see also SMS
Sn	1. Tin (IUPAC) 2. Sunn hemp fibers (DIN); see also SN
SN	Sunn hemp fibers (EDV)
SP	Superior processing rubber (a grade of natural rubber)
SPF	Superior processing furnace black
SPH	Styrenated phenol (WTR)
SPPF	Solid-phase pressure forming
SPSF	Solid-phase stretch forming
S-PVC	Poly(vinyl chloride), polymerized in suspension
SR	Synthetic rubber
SRF	Semi-reinforcing furnace black
SRF-HM	High-modulus SRF
SRF-HM-NS	Nonstaining high-modulus SRF
SRF-LM	Low-modulus SRF
SRF-LM-NS	Nonstaining low-modulus SRF
SRP	Rubber reinforced poly(styrene) (ASTM)
SS	1. Single stage (ASTM) 2. Steam-stripped (to remove residual monomer)
ST	Stone wool (DIN)
SWP	1. Solvent-welded plastics pipe 2. Synthetic wood pulp
T	1. Polysulfide rubber; see also OT, TR 2. Thermal black
TA	Cellulose triacetate (EDV); see also CT, CTA
TAC	Triallyl cyanurate (ASTM)
TATM	Triallyl trimellitate (WTR)
TBBS	Benzothiazyl-2-*t*-butyl sulfenamide (WTR)
TBC	Total Broadhurst Lee process
TBP	*t*-Butyl perbenzoate; see also TBPB
TBPA	Tetrabromophthalic anhydride
TBPB	*t*-Butyl perbenzoate; see also TBP
TBS	*t*-Butylstyrene
TBT	Tetrabutyl titanate
TBTD	Tetrabutyl thiuram disulfide (WTR); see also TBUT
TBTU	Tributyl thiourea (WTR)
TBUT	Tetrabutyl thiuram disulfide (WTR)
TC	Technically classified natural rubber
TCE	Trichloroethylene
TCEF	Trichloroethyl phosphate (ASTM, ISO); see also TCEP
TCF	Tricresyl phosphate, tritolyl phosphate (ASTM, DIN, ISO); see also TCP, TKP, TTP
TCM	Trichloromelamine (WTR)
TCP	Tricresyl phosphate (IUPAC); TCP® is a registered trademark in the United Kingdom; see also TCF, TKP, TTP
TDEC	Tellurium diethyl dithiocarbamate (WTR)
TDI	Tolylene diisocyanate; tolylene isocyanate; methylphenylene diisocyanate; usually an 80:20 mixture of the 2,4- and 2,6-isomers
TDID	Dimeric tolylene diisocyanate
TDM	*t*-Dodecyl mercaptan (WTR)
TEAE	Triethylaminoethyl- (with cellulose)
TEDMA	Triethylene glycol dimethacrylate (WTR)
TEGDA	Triethylene glycol diacetate (WTR)
TEP	Triethyl phosphate
TEPA	Tetraethylene pentamine (WTR)
TETA	Triethylene tetramine
TETD	Tetraethyl thiuram disulfide (WTR)
TFE	Tetrafluoroethylene (ASTM)
THBP	2,4,5-Trihydroxybutyrophenone
THF	Tetrahydrofuran
THT	Tetrahydrothiophene (WTR)
TIOTM	Triisooctyl trimellitate (DIN, ISO)
TKP	Tricresyl phosphate; see also TCE, TCP, TTP
Tl	1. Thallium (IUPAC) 2. American moss (DIN)
TM	Thioplastics
TMA	Trimellitic anhydride
TMC	Thick molding compound
TMETD	Tetramethyl tetraethyl thiuram disulfide (WTR)
TMP	1. Thermal mechanical pulp 2. Trimethylol propane (WTR)
TMPT	Trimethylol propane trimethacrylate (WTR)
TMQ	2,2,4-Trimethyl-1,2-dihydroquinone (WTR)
TMS	Tetramethylsilane
TMTD	Tetramethyl thiuram disulfide (WTR)
TMTS	Tetramethyl thiuram monosulfide (WTR)
TMTT	Tetramethyl thiuram tetrasulfide (WTR)
TMTU	Tetramethyl urea (WTR); see also TMU
TMU	Tetramethyl urea; see also TMTU
TNPP	Trinonyl phenyl phosphate (WTR)
TOF	Trioctyl phosphate = tris(2-ethylhexyl) phosphate (ASTM, DIN, ISO); see also TOP

TOP	Trioctyl phosphate = tris(2-ethylhexyl) phosphate (ASTM, DIN, ISO); see also TOF
TOPM	Tetraoctyl pyromellitate, i.e., tetrakis(2-ethylhexyl) pyromellitate (DIN, ISO)
TOR	*trans*-Poly(octenamer)
TOTM	Trioctyl mellitate = tris(2-ethylhexyl) pyromellitate (DIN, ISO)
TPA	1. 1,4-*trans*-Poly(pentenamer); see also TPR 2. Terephthalic acid
TPE	Thermoplastic elastomer
TPEL	Thermoplastic rubber or elastomer (ASTM)
TPES	Thermoplastic polyester in general (ASTM)
TPF	Triphenyl phosphate (DIN, ISO); see also TPP
TPI	Thermoplastoid (WTR)
TPO	Thermoplastic olefin elastomer
TPP	Triphenyl phosphate (ASTM, IUPAC); see also TPF
TPR	1. 1,5-*trans*-Poly(pentenamer); see also TPA 2. Thermoplastic elastomer; see also TR, TPE
TPS	Toughened poly(styrene) (in the UK for HIPS)
TPTD	Tetraisopropyl thiuram disulfide (WTR)
TPU	Thermoplastic polyurethanes
TPUR	Thermoplastic urethane (ASTM)
TPX	Poly(4-methyl-1-pentene); see also PMP
TR	1. Thermoplastic elastomer 2. Thio rubber (in the UK for polysulfide) 3. Textile residues or fiber or fabric of unknown composition (EDV)
TRIM	Trimethylolpropane trimethacrylate
Ts	Tussah silk (DIN)
TSH	Toluene sulfonyl hydrazide (WTR)
TSUR	Thermoset polyurethane (ASTM)
TTP	Tricresyl phosphate, tritolyl phosphate (ISO); see also TCE, TCP, TKP
TTT	Triethyltrimethylenetriamine (WTR)
TU	Thiourea
TV	Trivinyl fiber (EDV)
Tz	Tibet goat hair (goat = "Ziege") (DIN)
UE	Polyurethane elastomer (ASTM); see also UR
UF	Urea–formaldehyde resin (ASTM, DIN, ISO, IUPAC)
UFS	Urea–formaldehyde foam
UHMPE	Poly(ethylene) with ultrahigh molar mass
UHMW-PE	Poly(ethylene) with ultrahigh molar mass (molar mass over 3 100 000 g/mol) (ASTM)
UP	Unsaturated polyester (ASTM, DIN, ISO, IUPAC)
UP-G-G	Prepreg from unsaturated polyesters and textile glass fibers
UP-G-M	Prepreg from unsaturated polyesters and textile glass mats
UP-G-R	Prepreg from unsaturated polyesters and textile glass rovings
Ur	Urena (DIN)
UR	Polyurethane elastomers (BS); see also UP
VA	Vinyl acetate
VAC	Vinyl acetate
VAE	Vinyl acetate–ethylene copolymers; see also EVA, E/VAC
VC	Vinyl chloride
VC/E	Copolymer from ethylene and vinyl chloride (ISO)
VC/E/MA	Copolymer from ethylene, vinyl chloride, and maleic anhydride (ISO)
VC/E/VAC	Copolymer from ethylene, vinyl chloride, and vinyl acetate (ISO)
VCI	Volatile corrosion inhibitor
VCM	Vinyl chloride (monomer); see also VC
VC/MA	Copolymer from vinyl chloride and methyl acrylate (ISO)
VC/MMA	Copolymer from vinyl chloride and methyl methacrylate (ISO)
VC/OA	Copolymer from vinyl chloride and octyl acrylate (ISO)
VC/VAC	Copolymer from vinyl chloride and vinyl acetate (ISO)
VC/VDC	Copolymer from vinyl chloride and vinylidene chloride (ISO)
VDC	Vinylidene chloride
Vf	1. Vulcan fiber 2. Vinyl fluoride
VF/HFP	Copolymer from vinylidene fluoride and hexafluoropropylene
Vi	Vicuna wool (DIN)
VI	Viscose (rayon) (EDV); see also CV
VMQ	Silicone rubber with methyl and phenyl substituents (ANSI/ASTM)
VPE	Cross-linked poly(ethylene) (DIN)
VQ	Elastomeric silicone with vinyl substituents (ASTM)
VRI	Volatile rust inhibitor
VSI	Poly(dimethyl siloxane) with vinyl groups (ASTM)
VT	Vinyl toluene
VVO	Vulcanized vegetable oil = factice (WTR)
VY	Vinal (= poly(vinyl alcohol) fiber) (ASTM, EDV)
WA	Angora (rabbit) wool (EDV)
WB	Beaver wool (EDV)
WFK	Whisker reinforced plastic (German technical literature)
WG	Vicuna wool (EDV)

WK	Camel wool (EDV)
WL	Llama wool (EDV)
WM	1. Plasticizer (German technical literature: "Weichmacher") 2. Mohair wool (EDV)
Wo	1. Tungsten (IUPAC) 2. Sheep's wool (DIN); see also WO
WO	Sheep's wool (EDV); see also Wo
WP	Alpaca wool (EDV)
WS	Cashmere goat wool (EDV)
WT	Otter wool (EDV)
WU	Guanaco wool (EDV)
WV	Virgin sheep wool (EDV)
WY	Yak wool (EDV)
XABS	Elastomeric copolymer from acrylonitrile, butadiene, styrene, and a carboxylic group containing monomer (ASTM)
XLPE	Crosslinked poly(ethylene)
XNBR	Elastomeric copolymer from acrylonitrile, butadiene, and a carboxylic group containing monomer (ANSI/ASTM)
XPS	Expandable (or expanded) poly(styrene)
XSBR	Elastomeric copolymer from butadiene, styrene, and a carboxylic group containing monomer (ANSI/ASTM)
YSBR	Thermoplastic triblock copolymer from styrene and butadiene (ANSI/ASTM); see also SBS
YXSBR	Thermoplastic block copolymer from styrene and butadiene containing carboxylic groups (ANSI/ASTM)
Z5MC	Zinc pentamethylene dithiocarbamate (WTR)
ZBDC	Zinc dibenzyl dithiocarbamate (WTR)
ZBPD	Zinc dibutyl dithiophosphate (WTR)
ZBX	Zinc dibutyl xanthate (WTR)
ZDBC	Zinc dibutyl dithiocarbamate (WTR)
ZDEC	Zinc diethyl dithiocarbamate (WTR)
ZDMC	Zinc dimethyl dithiocarbamate (WTR)
ZE	Zein fibers (DIN)
ZEH	Zinc ethylhexanoate (WTR)
ZEPC	Zinc ethylphenyl dithiocarbamate (WTR)
ZIX	Zinc isopropyl xanthate (WTR)
ZMBI	Zinc 2-mercaptobenzimidazole
ZMBT	Zinc-2-mercaptobenzthiazole

C. ABBREVIATIONS AND ACRONYMS BASED ON POLY(MONOMER) NAMES

The following table contains the abbreviations and acronyms of names of polymeric materials whose base polymers were obtained by chain polymerization (addition polymerization), copolymerization, polycondensation (condensation polymerization), and polyaddition. Note that these abbreviations and acronyms do not apply in industry to polymers *per se* but to polymeric materials, i.e. polymers with or without additives, fillers, plasticizers, etc.

Abbreviations and acronyms are in part based on the poly(monomer) nomenclature, i.e., on the names of the monomers used in the manufacture of polymers, sometimes however, without a prefix "P" for "poly". The names of monomers for copolymers are given in alphabetical order without regard to their prevalence.

Registered trademarks: a) SAN in Japan and the USA; b) PAN in Europe.

Polymers of	ASTM	DIN	ISO	IUPAC	Other
Acrylic ester + acrylonitrile + butadiene	ABA	A/B/A	A/B/A		
Acrylic ester + acrylonitrile + styrene	ASA	A/S/A	A/S/A		
Acrylic ester + ethylene	EEA		E/EA		
Acrylic acid	PAA				PAS
Acrylonitrile	PAN	PAC		PAN	
Acrylonitrile + butadiene	PBAN				
Acrylonitrile + butadiene + methyl methacrylate + styrene					MABS
Acrylonitrile + butadiene + styrene	ABS	ABS	ABS	ABS	
Acrylonitrile + chlorinated poly(ethylene) + styrene		A/PE-C/S	A/CPE/S		ACS
Acrylonitrile + ethylene–propylene–diene + styrene		A/EPDM/S	A/EPDM/S		
Acrylonitrile + methyl methacrylate	AMMA	A/MMA	A/MMA		
Acrylonitrile + styrene	SAN	SAN	SAN	SAN	AS
Adipic acid + hexamethylenediamine	PA 6.6	PA 66	PA 66	PA 6.6	
Adipic acid + hexamethylenediamine + sebacic acid			PA 66/610		
Adipic acid + tetramethylene glycol					PTMA
Allyl diglycol carbonate	ADC				
p-Aminobenzoic acid					PAB

Polymers of	ASTM	DIN	ISO	IUPAC	Other
Aminotriazol					PAT
Aminoundecanoic acid, 11-		PA 11	PA 11		
Azelaic acid anhydride					PAPA
Bisphenol A + phosgene	PC	PC	PC	PC	
Bitumen + ethylene				ECB	
Butadiene + methyl methacrylate + styrene		MBS	MBS		
Butadiene + styrene (thermoplastic)	PBS	S/B	S/B	S/B	SB
Butene-1	PB	PB	PB		BT
Butyl acrylate		PBA	PBA	PBA	
Butylene glycol + terephthalic acid	PBT	PBTP	PBTP		PTMT
Butyl methacrylate					PBMA
ε-Caprolactam	PA 6	PA 6	PA 6	PA 6	
ε-Caprolactam + ω-dodecanolactam			PA 6/12		
Chlorotrifluoroethylene	PCTFE	PCTFE	PCTFE	PCTFE	
Chlorotrifluoroethylene + ethylene					ECTFE
Cresol + formaldehyde	CF	CF	CF	CF	
1,4-Cyclohexanedimethylol + terephthalic acid					PCDT
Diallyl chloroendate (= diallyl ester of 4,5,6,7,7-hexachlorobicyclo-[2,2,1]-5-heptane-2,3-dicarbonic acid)	PDAC				
Diallyl fumarate	PDAF				
Diallyl isophthalate	PDAIP				
Diallyl maleate	PDAM				
Diallyl phthalate	PDAP	PDAP	PDAP	PDAP	DAP
1,4-Dichlorobenzene + disodium sulfide	PPS	PPS			
2,6-Dimethylphenol + oxygen	POP				PPO™
ω-Dodecanolactam (laurolactam)	PA 12	PA 12	PA 12	PA 12	
Ethyl acrylate					PEA
Ethyl acrylate + ethylene	EEA	E/EA	E/EA		
Ethylene	PE	PE	PE	PE	PL
–, chlorinated polymer	CPE	PE-C	PE-C		
–, cross-linked polymer			PE-X		VPE
–, high-density polymer	HDPE	PE-HD	PE-HD		
–, high impact polymer			PE-HI		
–, linear low-density polymer	LLDPE		PE-LLD		
–, low density polymer	LDPE	PE-LD	PE-LD		
–, medium density polymer	MDPE		PE-MD		
–, very low density linear polymer					VLDPE
–, ultra-high molar mass polymer	UHMW-PE		PE-UHMW		
Ethylene + methacrylic acid	EMA				
Ethylene + maleic anhydride + vinyl chloride		VC/E/MA	VC/E/MA		
Ethylene + methyl methacrylate		E/MMA	E/MMA		
Ethylene + propylene		E/P	E/P		PEP
Ethylene + propylene (+ diene)	EPDM	EPDM	EPDM		
Ethylene + tetrafluoroethylene	ETFE	E/TFE	E/TFE		
Ethylene + vinyl acetate	EVA	E/VA	E/VAC		EVAC
Ethylene + vinyl acetate + vinyl chloride		VC/E/VAC		VC/E/VAC	
Ethylene + vinyl chloride		VC/E	VC/E		
Ethylene glycol					PEG
Ethylene glycol + maleic anhydride	UP	UP	UP	UP	
Ethylene glycol + terephthalic acid(ester)	PET	PET	PET	PETP	PETE
–, with glycol comonomer	PETG				
–, fast crystallizing polymer					CPET
–, oriented polymer					OPET
Ethylene oxide	PEO	PEOX	PEOX	PEO	
Formaldehyde (or trioxane)	POM	POM	POM	POM	
Formaldehyde + furan	FF		FF		
Formaldehyde + melamine	MF	MF	MF	MF	
Formaldehyde + melamine + phenol		MPF			
Formaldehyde + phenol	PF	PF	PF	PF	
Formaldehyde + urea	UF	UF	UF	UF	
Furfural + phenol	PFF				
Hexafluoropropylene + tetrafluoroethylene	FEP	FEP	FEP		
Hexamethylenediamine + sebacic acid	PA 6.10	PA 610	PA 610		PA 6.10

Polymers of	ASTM	DIN	ISO	IUPAC	Other
p-Hydroxybenzoic acid	POB				PHB
3-Hydroxybutyric acid					PHB
2-Hydroxyethylmethacrylate					PHEMA
Isobutene (isobutylene)	PIB	PIB	PIB	PIB	IM
Isocyanurates		PIR	PIR		
Laurolactam (dodecanolactam)	PA 12	PA 12	PA 12	PA 12	
Linseed oil, epoxidized		ELO	ELO		
Maleic anhydride + styrene	SMA	S/MA	SMA		
Methacrylimide		PMI	PMI		
Methyl acrylate					PMA
Methyl acrylate + vinyl chloride		VC/MA	VC/MA		
Methyl α-chloromethacrylate	PMCA				
Methyl methacrylate	PMMA	PMMA	PMMA	PMMA	
Methyl methacrylate + vinyl chloride			VC/MMA		
4-Methylpentene-1	PMP	PMP	PMP		TPX
α-Methylstyrene		PMS	PMS		PAMS
α-Methylstyrene + styrene	SMS	S/MS	S/MS		
Octyl acrylate + vinyl chloride		VC/OA	VC/OA		
Perfluoro alkoxy alkane	PFA	PFA	PFA		
Phosphoric acid (polymer)					PPA
Propylene	PP	PP	PP	PP	
–, oriented polymer					OPP
–, biaxially oriented polymer					BOPP
Propylene + tetrafluoroethylene (*alt.*)					TFE-P
Propylene oxide	PPOX	PPOX	PPOX		
Soybean oil, epoxidized		ESO	ESO		
Styrene	PS	PS	PS	PS	
–, expanded (foamed) polymer					EPS
–, high-impact polymer			PS-HI		HIPS
–, impact resistant polymer	SRP	IPS			
–, oriented polymer					OPS
–, biaxially oriented polymer					BPS
Tetrafluoroethylene	PTFE	PTFE	PTFE	PTFE	
Tetrahydrofuran					PTHF
–, polymer with hydroxy endgroups					PTMEG
Triallyl cyanurate	PTAC				
Trifluoroethylene				P3FE	
Trioxane (+ co-monomers)	POM	POM	POM	POM	
Vinyl acetate	PVAC	PVAC	PVAC	PVAC	PVA
Vinyl acetate + vinyl chloride	PVCA	VC/VAC	VC/VAC		
N-Vinylcarbazole	PVK	PVK	PVK		
Vinyl chloride	PVC	PVC	PVC	PVC	PCU;V
–, polymerized in bulk					M-PVC
–, polymerized in emulsion					E-PVC
–, polymerized in suspension					S-PVC
–, polymer as flexible film					FPVC
–, polymer as oriented film					OPVC
–, polymer as rigid film					RPVC
Vinyl chloride + vinylidene chloride		VC/VDC	VC/VDC		
Vinyl fluoride	PVF	PVF	PVF	PVF	
Vinylidene chloride	PVDC	PVDC	PVDC	PVDC	
Vinylidene fluoride	PVDF	PVDF	PVDF	PVDF	
Vinyl methyl ether					PVME
N-Vinylpyrrolidone	PVP	PVP	PVP		

D. ABBREVIATIONS FOR POLYMERS NAMED AFTER A CHARACTERISTIC POLYMER GROUP

Characteristic polymer group	ASTM	DIN	ISO	IUPAC	Other
Amide	PA	PA	PA		
Amide, aromatic (aryl amide)	PARA				PAR
–, metal coated films	PA**				
–, saran coated films	PA*				
Amide-imide	PAI	PAI	PAI		

Characteristic polymer group	ASTM	DIN	ISO	IUPAC	Other
Arylene sulfone					PAS
Benzimidazole					PBI
Bisphenol A-sulfone					PSF
Carbodiimide					PCD
Carbonate, aromatic	PC	PC	PC	PC	
Epoxide (epoxy)	EP	EP	EP	EP	
–, glass fiber reinforced					GEP
Ester, saturated (polyester)		SP	SP		
–, thermoplastic	TPES				
–,–, metallized polymer film					MPE
Ester, unsaturated (polyester)	UP	UP	UP	UP	
–,–, glass fiber reinforced					FRP
Ester–alkyd	PAK				
Ester–ether (thermoplastic elastomer)	TEEE				
Ester–imide	PEI		PEI		
Ester–urethane			PEUR		
Ether–*block*–amide	PEBA	PEBA	PEBA		
Ether ether ketone	PEEK		PEEK		
Ether–imide		PEI			
Ether–sulfone	PES	PES	PES		PES
	PESU				
Imide	PI	PI	PI		
Parabanic acid					PPA
Phenylene ether	PPE	PPE	PPE		PPO™
Phenylene sulfide	PPS		PPS		
Phenylene sulfone	PPSU	PPSU	PPSU		PSU
Silicone (plastics)	SI	SI	SI		
Sulfone	PSU	PSU	PSU		PSU
Urethane	PUR	PUR	PUR	PUR	
–, thermoplastic	TPUR				TPU
–, thermoset	TSUR				

E. ABBREVIATIONS FOR POLYMERS PRODUCED BY CHEMICAL TRANSFORMATION OF OTHER POLYMERS

Resulting polymer	ASTM	DIN	ISO	IUPAC	Other
Carboxymethyl cellulose	CMC	CMC	CMC	CMC	
Carboxymethyl hydroxyethyl cellulose					CMHEC
Casein, crosslinked with formaldehyde	CS	CS	CS	CS	
Cellulose, as Cellophan[tt]	C				
–, coated with Saran	C*				
Cellulose acetate	CA	CA	CA	CA	
Cellulose acetobutyrate	CAB	CAB	CAB	CAB	
Cellulose acetopropionate	CAP	CAP	CAP	CAP	
Cellulose nitrate	CN	CN	CN	CN	NC
Cellulose plastics, unspecified	CE				
Cellulose propionate	CP	CP	CP	CP	
Cellulose triacetate	CTA	CTA	CTA		TA
Ethyl cellulose	EC	EC	EC	EC	
Hydroxyethyl cellulose					HEC
Hydroxypropyl cellulose					HPC
Hydroxypropyl methyl cellulose					HPMC
Methyl cellulose		MC	MC		
Poly(ethylene), chlorinated	CPE	PE-C	PE-C		
–, chlorosulfonated	CSM				CSR
Poly(ethylene-*co*-vinyl alcohol)		E/VAL	E/VAL		
	EVAL™				
					EVOH
Poly(propylene), chlorinated		PPC			
Poly(vinyl alcohol)	PVAL	PVAL	PVAL	PVAL	PVOH
Poly(vinyl butyral)	PVB	PVB	PVB		
Poly(vinyl chloride), chlorinated	CPVC	PVC-C	PVC-C		PC, PeCe
Poly(vinyl formal)	PVFM	PVFM	PVFM	PVFM	

F. ABBREVIATIONS FOR BLENDS, REINFORCED POLYMERS, ETC.

For mixtures of polymers, ISO recommends that the symbols for the parent polymers be put in parentheses, separated by a plus sign; for example, (PMMA + ABS) for a physical blend of poly(methyl methacrylate) and ABS.

ASTM recommends GP for "general purpose" and SS for "single stage".

Resulting polymer	ASTM	DIN	ISO	IUPAC	Other
Bulk molding compound					BMC
Elastomers, thermoplastic					
–,–, containing ester and ether groups	TEEE				
–,–, olefin-based	TEO				
–,–, styrene-based	TES				
Epoxide, glass fiber reinforced			EP-G		GEP
Plastic, carbon fiber reinforced		KEK			CFP
–, glass fiber reinforced					GFK
–, man-made fiber reinforced					CFK
–, metal fiber reinforced					MFK
Poly(acrylonitrile-co-styrene) + chlorinated poly(ethylene)					ACS[a]
Poly(styrene), rubber-modified	SRP	IPS			HIPS
Sheet molding compound					SMC
–, with high glass fiber content		HMC			

G. ISO CODES FOR THERMOPLASTIC MATERIALS

The standard designation of a thermoplastic material consists of

- a *description block* (giving the type of material),
- a *standard number block* (consisting of the number of the ISO standard or a national standard such as DIN), followed by a hyphen, and
- an *individual item block* (with five data blocks).

An example is

Molding material DIN 7744-PA 12, XF, 22-030, GF 30

The code letters are arranged after the symbol for the base polymer according to ISO and DIN; in the technical literature, however, they are commonly placed in front of the base symbol, following ASTM, which uses similar code letters and asterisks.

A designation system similar to the above ISO/DIN system has been recommended by ASTM. A typical ASTM designation, for example, would read

Molding material ASTM D-4000 PI000G42360

This molding material is a polyimide (PI), whose properties have been specified via cell table G of the ASTM standard D-4000. This particular cell table identifies five different properties by designating cell limits to the five digits following the letter G. According to cell table G, the "4" indicates a tensile strength of at least 85 MPa. The measured flexural modulus is characterized by "2", the Izod impact strength of 50 J/m by "3", and the heat deflection temperature of 300°C by "6". The fifth digit is undetermined; the "0" indicates an unspecified property.

1. Codes for ISO Data Blocks 1–5

1.1. ISO and DIN Codes for Data Block 1 *Data block* 1 of the individual item block contains the abbreviation of the chemical name of the material, e.g., PA 12 (see example). This abbreviation may be followed by analytical data with respect to the composition of the material; an example is the content of vinyl acetate units in ethylene–vinyl acetate copolymers. Such chemical structure data may be composition, such as content of vinyl acetate (VAC) or acrylonitrile units (AN), configuration as measured by isotacticity ("isotaxy") (IT), branching as revealed by density (D).

These analytical data are, however, not the exact analytical data but rather code numbers for the range (called "cell") which will permit this material to substitute for a similar one. Separated by a hyphen, other supplementary may be given too, e.g., H = homopolymer, P = plasticized, E = polymerized in emulsion, Q = mixture of different polymers, etc. (see ISO and DIN codes for data block 1).

Furthermore, ASTM uses * for saran coated polymer films and ** for metallized polymer films. The technical literature indicates biaxially oriented films by BO and oriented films by O (see Abbreviations in alphabetical order).

Code	ISO data block 1	DIN data block 1
A	–	–
B	Block copolymer	–
C	Chlorinated	Chlorinated
D	–	Density
E	Polymerization in emulsion	Expanded or expandable
F	–	Flexible or fluid (liquid state)
G	–	–
H	Homopolymer	High

Code	ISO data block 1	DIN data block 1
I	–	Impact resistant
J	–	–
K	–	–
L	–	Linear or low
M	Bulk polymerization	Medium or molecular
N	–	Normal or novolac
O	–	–
P	Plasticized	Plasticized
Q	Mixtures of different polymers	–
R	Random copolymer	Resol; formerly also: raised (enhanced)
S	Polymerization in suspension	–
T	–	–
U	Unplasticized	Ultra or unplasticized
V	–	Very
W	–	Weight (mass)
X	No indication	Crosslinked or crosslinkable
Y	–	–
Z	–	–

1.2. ISO Codes for Items in Data Block 2

Data block 2 may comprise up to four letters which give qualitative information. The first letter denotes the intended application, e.g., B = blow molding, G = general purpose, X = no indication (see example). The following letters 2, 3, and 4 can code up to three essential additives or supplementary information, for example, D = powder (dry blend), F = special burning characteristics (see example), L = light and weather stabilizer, etc.

Code	Position 1	Code	Positions 2–4
A	Adhesives	A	Antioxidant for processing
B	Blow molding	B	Antiblocking agent
C	Calendering	C	Colored
D	Disc manufacture	D	Powder (dry blend)
E	Extrusion (pipes, profiles, sheets)	E	Additive for expansion
F	Extrusion (films, foils)	F	Special burning characteristics
G	General purpose	G	Granules (pellets)
H	Coating	H	Heat aging stabilizer
K	Cable and wire insulation	K	–
L	Extrusion (monofilament)	L	Light and weather stabilizer
M	Injection molding	M	–
N	–	N	Natural (not colored)
P	Paste resin	P	Polymer modifier
Q	Compression molding	Q	–
R	Rotational molding	R	Mold release agent
S	Powder spray sintering	S	Slip agent, lubricant
T	Tape manufacture	T	Improved transparency
W	–	W	Stabilized against hydrolysis
X	No indication	X	–
Y	Textile yarn	Y	–
Z	–	Z	Antistatic agent

1.3. ISO Codes for Items in Data Block 3

Data block 3 contains quantitative information about the designated properties. Such properties include

Molar mass indicators	Intrinsic viscosity ("viscosity number")	IV
	Fikentscher's K value (viscosity-based quantity for PVC)	FK
Macrostructure data	Bulk density	BD
Rheological data	Melt flow rate (melt flow index)	MFR
Thermal data	Vicat temperature	VST
	Torsional stiffness temperature	TST
Mechanical properties	Modulus of elasticity	E
	Tensile stress at 100% strain	TS
	Shore hardness	SH
	Impact strength (notched)	ISN

Each material is characterized by one or more than one property. The following leading criteria are used for the various polymers:

Poly(ethylene)	D, MFR
Poly(propylene)	I, MFR
Poly(styrene) and acrylonitrile–styrene copolymer	VST, MFR
Styrene–acrylonitrile copolymer	VST, MFR, AN
Styrene–butadiene copolymer	VST, MFR, ISN
Acrylonitrile–butadiene–styrene, and acrylonitrile–styrene–acrylic acid copolymers	VST, MFR, ISN, AN
Ethylene–vinyl acetate copolymer	VAC, MFR
Poly(vinyl chloride)	IV, (FK), BD
Poly(vinyl chloride), unplasticized	VST, E, ISN
Poly(vinyl chloride), plasticized	TS, SH, TST
Polyamide	IV, E
Polycarbonate	IV, MFR, E
Poly(methyl methacrylate)	IV, VST
Poly(ethylene terephthalate)	IV

In the example for a polyamide 12, the code "22-030" of data block 3 indicates a polymer with an intrinsic viscosity of 210 ml/g ("cell 22"), and a modulus of elasticity of 280 MPa ("cell 030"). On the other hand, "20-D050" in data block 3 for a poly(ethylene) tested according to DIN 16776 means a material with a density within cell "020" (e.g., 0.918 g/cm^3) and a melt flow rate within the cell "050" (e.g., 4.2 g/min), measured under condition D (a temperature of 190°C under a load of 2.16 kg). The definition of cells for each material and testing procedure can be found in special tables.

1.4. ISO Codes for Extending and Reinforcing Fillers (Data Block 4)

Data block 4 gives information about the type and content of fillers or reinforcing materials. The first

letter gives the type of filler: C = carbon, G = glass, etc. The second letter indicates the shape of the filler: F = fiber, S = spheres, etc. The third position is a code for the weight content of the filler, e.g., "30" for the "cell" (range) 27.5–32.5 wt.%.

Code	Position 1 (material)	Code	Position 2 (form or structure)
B	Boron	B	Balls, beads, spheres
C	Carbon	C	Chips, cuttings
D	–	D	Powder (dry blend)
E	Clay	E	–
F	–	F	Fiber
G	Glass	G	Ground, grinding stock
H	–	H	Whisker
K	Calcium carbonate	K	Knitted fabric
L	Cellulose	L	Layer
M	Mineral, metal	M	Mat (thick)
N	–	N	Non-woven (fabric, thin)
P	Mica	P	Paper
Q	Silicon	Q	–
R	Aramid	R	Roving
S	Synthetic, organic	S	Scale, flake (spheres: now B)
T	Talcum	T	Cord
V	–	V	Veneer
W	Wood	W	Woven fabric
X	Not specified	X	Not specified
Z	Others	Z	Others

1.5. ISO Codes for Data Block 5 *Data block* 5 is reserved for specifications based on individual agreements between supplier and customer. It may code additional requirements, restrictions, or supplemental information.

H. SPI CODES FOR THE RECYCLING OF PLASTICS

Number code	Letter code	Polymer
1	PETE	Poly(ethylene terephthalate)
2	HDPE	High-density poly(ethylene)
3	V	Vinyl chloride polymers
4	LDPE	Low-density poly(ethylene)
5	PP	Poly(propylene)
6	PS	Poly(styrene)
7	–	All other plastics, including multilayered materials

I. RECOMMENDED ABBREVIATIONS AND ACRONYMS FOR NAMES OF ELASTOMERS

Names of elastomers are listed in alphabetical order of the names of their monomers, regardless of their proportions. A "PM" indicates an after-treatment of the polymer; "Other" refers to old or often used "unofficial" abbreviations; # indicates that the abbreviation is recommended by IUPAC.

Monomers (incl. after-treatment of PM)	ASTM	BS	ISO	Other
Acrylic ester + acrylonitrile	ANM			
Acrylic ester + butadiene	ABR#			AR
Acrylester (+ saturated vulcanizable comonomer)	ACM			
Acrylonitrile + butadiene	NBR#			GR-N
–, PM contains carboxylic groups	XNBR			
–, PM is hydrogenated				H-NBR
Acrylonitrile + butadiene + styrene, PM contains COOH groups	XABS			
Butadiene	BR#			BP, CBR
Butadiene + styrene	SBR#	SBR		GR-S
Butadiene + styrene, PM contains COOH groups	XSBR			
Butadiene + styrene, elastomeric triblock copolymers	YSBR			SBS
Butadiene + styrene, elastomeric block copolymers with COOH groups	YXSBR			
Butadiene + 2-vinylpyridine	PBR#			
Butadiene + 2-vinylpyridine + styrene	PSBR			
Chloroprene	CR#	CR		
Chloroprene + acrylonitrile	NCR#			
Chloroprene + styrene	SCR#			
Cyclooctene (*trans* polymer)				TOR
Cyclopentene (*trans* polymer)				TPA, TPR
Epichlorohydrine	CO#		CO	CHR
Epichlorohydrine + ethylene oxide	ECO			CHC
Ethylene (PM postchlorinated)	CM			
Ethylene (PM perchlorosulfonated)	CSM	CSPR		CSR
Ethylene + acrylic ester	EAM			
Ethylene + ethyl acrylate	EEA		E/EA	
Ethylene + propylene	EPM	EPR	EPM	AP
Ethylene + propylene + non-conjugated diene	EPDM	EPTR		APT, EPT
Ethylene + vinyl acetate	EVM			
Ethylene + tetrafluoroethylene				ETFE
Fluorinated carbon chains (substituents: fluoroperfluoroalkyl-, perfluoroalkoxy-)	FKM		FPM	
Hexafluoropropylene + tetrafluoroethylene				FEP
Hexafluoropropylene + vinylidenefluoride	FPM			
Isobutylene	PIB#	PIB	PIB	PIS

Monomers (incl. after-treatment of PM)	ASTM	BS	ISO	Other
Isobutylene + isoprene	IIR#	Butyl		GR-I, PIBI
Isobutylene + isoprene (postbrominated PM)	BIIR			
Isobutylene + isoprene (postchlorinated PM)	CIIR			
Isoprene (natural rubber)	NR#			NK
Isoprene (natural rubber, easy-processing)				SP
Isoprene (natural rubber, standardized Indonesian)				SIR
Isoprene (natural rubber, standardized Malaysian)				SMR
Isoprene (natural rubber, technical classification)				TC
Isoprene (synthetic rubbers)	IR#	IR		CPI, PIP
Isoprene (chlorinated PM)				RUC
Isoprene + acrylonitrile	NIR			
Isoprene + styrene	SIR#			
Isoprene + styrene (triblock copolymer)				SIS
Isoprene + styrene (hydrogenated triblock copolymer)				SEBS
Norbornene				PNR
Phosphonitrile dichloride, with fluorine-containing substituents				PNF
Propylene oxide	PO			OPR
Propylene oxide + allylglycidyl ether	GPO			POR
Silicone with fluoro substituents	FQ			
Silicone with fluoro + methyl substituents	FMQ		MFQ	
Silicone with fluoro + methyl + vinyl substituents	FVMQ			
Silicone with methyl substituents	MQ			
Silicone with phenyl substituents	PQ			
Silicone with phenyl + methyl substituents	PMQ		MPQ	
Silicone with phenyl + vinyl + methyl substituents	PVMQ		MPVQ	
Silicone with vinyl substituents	VQ			
Silicone with vinyl + methyl substituents	VMQ		MVQ	
Sulfides (thioplastics)				T, TR, OT
Tetrafluoroethylene + trifluoronitrosomethane + nitrosoperfluorobutyric acid (nitroso rubber)	AFMU			CNR
Urethane elastomers	UE	UR		
Urethane elastomers with polyester segments	AU		AU	
Urethane elastomers with polyether segments	EU		EU	
Urethane–polyester elastomers, crosslinkable by isocyanates	AU-I			
Urethane–polyester elastomers, crosslinkable by peroxides	AU-P			
Vinylidenefluoride + hexafluoropropylene	FPM		FPM	
Tetrafluoroethylene + perfluoralkylvinyl ether				PFA
Thermoplastic elastomers, in general	TPEL			TPE, TPR
Thermoplastic olefin elastomers				TPE-O, TPO
Thermoplastic elastomers on basis styrene–diene				TPE-S
Thermoplastic elastomers on basis ester				TPE-E
Thermoplastic elastomers on basis ether–amide				TPE-A

The following abbreviations are also used. ICR: rubber from undiluted latex of natural rubber (initial concentration rubber); OER: oil-extended rubber; RTV: room temperature vulcanization.

J. ABBREVIATIONS FOR TEXTILE FIBERS*

Name	Short Name		Abbreviation		
	ISO	FTC	DIN	ASTM	ETG
NATURAL PROTEIN FIBERS (SILK)					
Mulberry spinner (*Bombyx mori L.*)			Ms		SE
Tussah					Ts
NATURAL PROTEIN FIBERS (WOOL)					
Alpaca (*Llama pacos*)			Ap		WP
Angora rabbit ("Angorakaninchen")			Ak		WA
Beaver					"B

* ISO: International Standardization Organization; FTC: US Federal Trade Commission; DIN: German Standards (Deutsches Institut für Normung), DIN 60001; "..." indicates the German word from which the abbreviation or acronym was derived; ASTM: American Society for Testing and Materials; ETG: European Textile Characterization Law.

Name	Short Name ISO	Short Name FTC	Abbreviation DIN	Abbreviation ASTM	Abbreviation ETG
Cashmere goat ("Kaschmir-Ziege")			Kz		WS
Guanaco					WU
Llama			Lm		WL
Mohair			Mo		WM
Otter					WT
Sheep			Wo		WO
–, virgin wool					WV
Vicuna			Vi		WG
Yak					WY
NATURAL PROTEIN FIBERS (HAIRS)					
Camel ("Kamel")			Km		WK
Cow ("Rind")			Ri		
Goat ("Haar der Ziege")					Hz
Hare ("Hase")					Hs
Horse ("Rosshaar")			Rh		
Human ("Haar")					HA
Rabbit ("Kaninchen")			Kn		
Tibetan goat ("Tibetziege")					Tz
NATURAL CELLULOSE FIBERS (TEXTILE FIBERS)					
Cotton ("Baumwolle")			Bw		CO
Flax			Fl		LI
–, linen					LI
–, half-linen					HF
Ramie			Ra		RA
NATURAL CELLULOSE FIBERS (HARD FIBERS AND OTHERS)					
Alfalfa					AL
Broom ("Ginster")					GI
Calatropis			An		
Coir ("Kokos")			Ko		CC
Esparto grass (*alfa*)			Ag		
Hemp (*Cannabis*)					CA
–, Bombay (Sunn hemp)			Sn		SN
–, Manila (Abaca)			Ma		AB
Henequen (Agave)			He		
Jute			Ju		JU
Kapok			Kp		KP
Kenaf			Kf		KE
Maguey					MG
Mauritius fiber (Fique)			Fi		
Moss, American			Tl		
Phormium (New Zealand flax)			Nf		
Rosella			Ro		
Sisal			Si		SI
Urena			Ur		
REGENERATED POLYSACCHARIDE FIBERS					
Alginate			AL		AG
Cellulose	Cupro	Cupra (rayon)	CC		CU
	Viscose	Rayon	CV		VI
	Modal	Rayon			MD
	Deacetylated Acetate	Rayon			
Cellulose 2 1/2 acetate	Acetate	Acetate	CA	CA	AC
Cellulose triacetate	triacetate	triacetate	CT	CTA	TA
REGENERATED PROTEIN FIBERS					
Ardein fibers			AR		
Casein fibers			KR		CS
Regenerated protein fibers, in general	Protein	Azlon			PR
Zein fibers			ZE		

Name	Short Name		Abbreviation		
	ISO	FTC	DIN	ASTM	ETG
SYNTHETIC FIBERS					
Elastodiene fibers			LA	EU	EL
Fluoro fiber					FL
Poly(acrylonitrile)	Acrylic	Acrylic	PAC	PAN	PAN
Poly(acrylonitrile), copolymeric	Modacrylic	Modacrylic	PAM		MA
Polyamide, aliphatic	Nylon, Polyamide	Nylon, Polyamide	PA	PA	PA
Polyamide, aromatic	Aramide	Aramide			
Polyester [poly(ethylene terephthalate]	Polyester	Polyester	PES	PET	PL
–, [from ethylene glycol, terephthalic acid and *p*-hydroxybenzoic acid]			PEE		
Poly(ethylene)	Polyethylene	Olefin	PE	PE	PE
Poly(propylene)	Polypropylene	Olefin	PP	PP	PP
Poly(styrene) (>88% sty)			PST		
Polyurea			PUA		PB
Polyurethane			PUR	PUR	PU
–, segmented	Elastan	Spandex	PUE		EA
Poly(vinyl alcohol)	Vinylal	Vinal	PVA		VY
Poly(vinyl chloride) (>85% VC)	Chlorofiber		PVC		
Poly(vinyl chloride), copolymeric	Chlorofiber	Vinyon	PVM		
Poly(vinylidene chloride), copolymeric	Chlorofiber	Saran	PVD		
Trivinyl fiber					TV
INORGANIC FIBERS					
Asbestos			As		AS
Glass	Glass	Glass	GL		GL
Metal			MT		ME
Mineral (rock wool)			ST		
Slag wool			SL		
OTHER FIBERS					
Unspecified fibers					AF
Unknown fibers					TR

Chemical Abstract Registry Numbers and Online Database Searching for Polymer Literature

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The advent of computerized database searching for polymer literature has made locating technical information both easier and harder. While computers greatly increase the speed and completeness of a search, the algorithms and conventions used are often quite different from those used for hardcopy (paper) indexes. The following discussion and tables give basic information needed to start identifying and searching the most common polymers in the most used databases and paper indexes.

A. NOMENCLATURE

The *Common Name* is the name used by many scientists and engineers, and is used for listing polymers in catalogs such as Aldrich, and engineering indexes such as Engineering Index. Various other common *Synonyms* or *Trade Names* are also listed in the table and used in the literature. The *Chemical Abstracts (CA) Index Name* is the most recent name used by Chemical Abstracts Service (CAS). In general, Chemical Abstracts follows IUPAC guidelines when naming polymers. The *CAS Registry Number* (CAS Reg. No.) is a unique identifying number given to each of the 18 million chemical compounds CAS has indexed. The *Molecular Formula* is the formula for the smallest repeating unit of the polymer. In some cases, such as cellulose, the exact molecular formula is not known.

B. CAS REGISTRY NUMBERS

CAS Registry Numbers provide a consistent method of identifying and searching for polymers in literature databases, and have been used to develop the table in this section. CAS indexes the chemical and chemical engineering literature from about 80000 worldwide journals, conference papers, books, and patent literature.

The way that CAS (or IUPAC) names a compound is often very different from that used in scientific journals or the popular press. Polyethylene provides an excellent example of this problem. The name in CA is currently Ethene, homopolymer. CAS used to index it under Ethylene, polymers. Over 2000 synonyms or trade names are listed for polyethylene compounds, ranging from Alathon (registered trademark of the Lyondell Petrochemical Company) to Suzulon (a tradename used by Aicello Chemical Corporation, but not trademarked). As CAS understates in its publication, "Polymer Searching", "Unfortunately, naming conventions for polymers have not achieved widespread understanding, and molecular formulas are not unique search terms." CAS Registry Numbers are used not only by CAS in their database indexing but by the US Federal Government in their Toxic Substances Inventory, by the US Department of Transportation for regulating transport of chemicals, and by other indexes such as Merck and Aldrich. They are the most precise and unique compound identification.

Authors of polymer literature may not always be able to report the precise structure of their materials. However, they usually do report the monomers used in the polymerization. Thus, while polymers are represented in two ways, as monomers and by final structure (Structural Repeating Unit or SRU), the primary registration of a polymer is by its component monomers. The Registry Number for most

polymers in the table can be used to find the component Registry Numbers for their corresponding monomers. Similarly, the Registry Number for copolymers will lead to the monomer registry numbers. Searching just by the SRU, the homopolymer, or the copolymer number will not retrieve a complete citation list.

C. INDEXING AND CAS REGISTRY NUMBER ASSIGNMENT

Chemical Abstracts assigns Registry Numbers to polymers based on both their monomers and their final structure. The table here usually lists the Registry Number for the homopolymer or copolymer. However, searches that use only one Registry Number for a polymer are often incomplete. Polymers based on more than one monomer should be searched using all the component monomers. All the copolymer references of a specific monomer can be found by searching the monomer's Registry Number. The polymers having the same final structure can be found by searching the final structure's Registry Number.

Some compounds are so well characterized that they do not follow the rules and are indexed only under their structural repeating unit (SRU). These are marked (R) in the table and include common polymers such as Nylon 6 or PET. Some polymers are not indexed by registry numbers at all but only by subject. These include tung oil, cellulose, and linseed oil, which are marked (S) in the table. These materials are exceptions and, in most searching, registry numbers must be used in the CAS online files. Some registry numbers in the Registry File are marked with an asterisk (*). STN gives the message on-line that these "represent substances of unknown or variable composition, complex reaction products, or biological substances that do not meet the criteria for Chemical Abstracts indexing". These substances generally have been registered for regulatory agencies and receive a CA Index Name. Searching these substances in STN files can be done by using the Registry Number(s) from the record, terms derived from the name, terms that are more or less specific than those in the name (which may include other CAS Registry Numbers), or associated trade names. Every likely term should be tried, and additional help should be sought if other references seem likely.

Section E gives the common names and CA Registry Numbers for a number of important polymers. A list of Registry Numbers in numerical order with their corresponding names is available from the authors. Additional synonyms and trade names are provided, as are the CA Index names. Molecular formulae are provided, and can be helpful in searching with structural-related databases. The Search Guide indices have the following meanings:

S search by CA Index names as a subject in the citation file
* search using every likely term – Registry Numbers, name, synonyms, etc.
R search by Registry Numbers for components, copolymers, SRU's
U search using the single Registry Number provided.

The CAS database system is available commercially, most commonly through STN (1-800-848-6538) and Dialog (1-800-334-2564). Most academic and industrial libraries have ready access to these systems and may be consulted for assistance. For illustrative purposes, searches from each system are demonstrated here. The logic of searching each system is the same, but commands and features can vary. For more information about online searching, consult a librarian or the database vendors. The Chemical Abstracts Service book, "Searching for Polymer Information in CAS Online", American Chemical Society, Chemical Abstracts Service, 1990, p. 173, is the best starting source for polymer information within Chemical Abstracts. It may be ordered by calling the STN toll-free number or using the web page.

D. EXAMPLE SEARCHES

1. Example of a Search Using the Polymer CAS Registry Number

The first step in a CA search is to identify the registry numbers. Section E lists registry numbers for many common polymers. The Registry Number for poly-(acrylamide) is 9003-05-8. Leading zeros should be disregarded when searching online, for example, search 009003-05-8 as 9003-05-8. Using the CA Registry Number to search the chemical compound registry file would return the following information (Fig. 1) from the Dialog database. The Dialog File number at the beginning of each output refers to a specific work session for the user. This particular database provides the molecular formula, related CAS Registry Numbers, the CA name, a list of synonyms including tradenames, the CA Registry Number for the monomer, and the number of literature references over the specified time interval.

2. Example of a Search Using the Monomer CAS Registry Number

The CAS Registry Number for the monomer from the previous example, acrylamide, is 79-06-1. Searching by the component Registry Number in the Registry File gives the following record (Fig. 2).

3. Example of a Combined Search

The results of both searches (polymer and monomer CAS Registry Numbers) would be combined to generate a more complete listing of the literature. The following answers (Fig. 3) are found by searching the abstracts file for CA (399 on Dialog) with the monomer and polymer Registry Numbers.

Searching by the homopolymer Registry Number (RN), 9003-05-8, produced most citations. However, over 6000 additional articles containing information about the monomer would have been missed if the monomer Registry Number, 79-06-1, had not been used. Some of these may contain needed information about the polymer. Only 539 articles were indexed using both Registry Numbers.

(Dialog File 398)
CAS REGISTRY NUMBER: 9003-05-8 (POLYMER)
MOLECULAR FORMULA: (C3H5NO)x
REPLACED CAS REGISTRY NUMBER(S): 9082-06-8 12624-24-7 25038-45-3 27754-57-0 33338-03-3 39355-07-2 39387-77-4 51312-40-4 57679-11-5 68247-81-4 72270-86-1 79079-15-5 104981-89-7 114265-35-9 143180-09-0 143180-13-6 143180-22-7
CA NAME(S):
HP = 2-Propenamide (9CI)
NM = homopolymer
HP = Acrylamide (8CI)
NM = polymers
SYNONYMS: Acrylamide homopolymer; Acrylamide polymer; Alcoflood 1175; Alcoflood 935; American Cyanamid KPAM; American Cyanamid P 250; AMF; Aminogen PA; AP 273; Aron F 40; BanDrift; Boze Floc N 46BT; Calgon 470; Calgon 800; Cogum 20P; Cogum 25H; Cyanamer N 300LMW; Cyanamer P 250; Cyanamer P 35; Cytame 5; Diaclear MA 3000H; Diaclear MN 3000H; DK Dry Capsule ESP; Dow ET 597; Dow J 100; Dow 164; DP 1916; DP 9-6193; Drew Floc 41; E 936; ET 597; Flygtol GB; Formula 358; Gelamide 250; Get-Down ; GPA-u; Himoloc OK 507; Himoloc SS 200; Hiset P 700; Hyset P 700SN; Instar NS; J 100; J 100(polymer); K 4; K 4(acrylic polymer); K-PAM; KAL 13; KW 677; Magnifloc 900N; Maquat 100; Migunon NS; Mirbane 301; Nalco Lp 3033; Nalco 7871; Nalco 8871; NL; P 250; P 250(polymer); P 300; P 300(polyacrylamide); PAA 1; PAA 70L; PAM; PAM 50; PAM(homopolymer); Pamid; PC 305; Percol 333; Plex 4847D; Polias 320; Polyacron KR 143; Polyacrylamide; Polyacrylamide resin; Polyhall 2J; Polyhall 5J; Polymerset C 305; Polysic; Polystron 117; Polystron 145; Polystron 191; Polystron 194-7; Pomosist 117; Praestol 2800; Praestol 2810/73; Praestol 6000; Sanpoly A 520; Santac SP 66; Santac SP 67; Seabetter AD; Sedipur TF; Sedipur TF 514; Sepaflux CE 5174; Solidokoll K; Solvitose 433; Stargum AD-S; Stokopol D 2624; Sumifloc FN 10H; Sumirez A 17; Sumirez A 27; Sursolan P 5; Texapret AM; Tulsepar PNS 1; TY 007; Versicol W 11; Versicol W 17; Versicol W 25; Versicol WN 15; Viterra II ; X-Coat P 130C; X-Coat P 180S; Zonen A; 2J; 3330s; 38F
COMPONENT CAS REGISTRY NUMBER(S):
(79-06-1)

Component RN	Component Molecular Formula
1) 79-06-1	C3H5NO

SUBFILE: CHEMNAME 13873 LITERATURE REFERENCE(S) IN FILE 399.
LAST UPDATE: 199611

Figure 1. Dialog database output from searching CAS Registry No. 9003-05-8.

CAS REGISTRY NUMBER: 79-06-1
MOLECULAR FORMULA: C3H5NO
CA NAME(S):
HP = 2-Propenamide (9CI)
HP = Acrylamide (8CI)
SYNONYMS: Acrylic amide; Ethylenecarboxamide; Propenamide; Vinyl amide
SUBFILE: CHEMNAME 6574 LITERATURE REFERENCE(S) IN FILE 399.
LAST UPDATE: 199608

Figure 2. Dialog database output from searching CAS Registry No. 79-06-1.

```
   Set Items Description
   .... ..... .....
?s rn = 9003-05-8
   S1   14051 RN = 9003-05-8
?s rn = 79-06-1
   S2   6622 RN = 79.06-1
?s s1 or s2
      14051 S1
      6622 S2
   S3   20134 S1 OR S2
?s s1 and s2
      14051 S1
      6622 S2
   S4   539 S1 AND S2
```

Figure 3. Dialog database output from combining the polymer and monomer searches.

4. Example of a Copolymer Search

Poly(acrylonitrile-*co*-butadiene) is a common commercial product. The STN database was searched for this copolymer, using its CAS Registry Number along with those of the monomers. The search method is similar to that for homopolymers; the copolymer and the two monomers are searched separately, and the results are combined. Some records have been edited for clarity and conciseness. In Fig. 4, component 1 is acrylonitrile and component 2 is butadiene.

```
(STN Registry File)

L1 ANSWER 1 OF 1 REGISTRY COPYRIGHT 1997 ACS
RN ***9003-18-3*** REGISTRY
CN 2-Propenenitrile, polymer with 1,3-butadiene (9CI) (CA INDEX NAME)
OTHER CA INDEX NAMES:
CN 1,3-Butadiene, polymer with 2-propenenitrile (9CI)
CN 1,3-Butadiene, polymer with acrylonitrile (8CI)
CN Acrylonitrile, polymer with 1,3-butadiene (8CI)
OTHER NAMES:
CN 1,3-Butadiene-acrylonitrile copolymer
CN 1,3-Butadiene-acrylonitrile polymer
CN Acrylonitrile-1,3-butadiene copolymer
(30 other chemical names or synonyms or trade names are listed)
MF (C4 H6 . C3 H3 N)x

  CM 1 (Component One)

  CRN 107-13-1 (Component Registry Number)
  CMF C3 H3 N (Component Molecular Formula)

  CM 2 (Component Two)

  CRN 106-99-0 (Component Registry Number)
  CMF C4 H6 (Component Molecular Formula)
```

Figure 4. STN database output from a copolymer search.

Searching by the two component Registry Numbers given above, (107-13-1, acrylonitrile; 106-99-0, butadiene), gives the following records in the Registry File (Fig. 5).

Figure 6 shows the output when all the separate searches (Figs. 4 and 5) are combined. Search strings for each command are shown.

In this case, there are fewer articles for the copolymer (9003-18-3) than for either of the component monomers individually. On the other hand, 550 citations have the two components together, with very little overlap between that set and the copolymer. This once again illustrates the importance of including the monomers to get a complete search in CA files online.

A similar search in the paper version of CA could be done by finding the latest CA name for the homopolymer or copolymer and all related monomers in the Chemical Substance Index, and then referring to the abstracts. The Chemical Substance Index also lists the Registry Numbers for all compounds indexed, and should be checked prior to online work.

Additional search methods include using the trade name, chemical name, and molecular formula in the Registry File to start the search for a polymer Registry Number. The table gives a starting point for common polymers, but with over 700000 polymeric entries, it is impossible to list them all.

```
L2 ANSWER 1 OF 2 REGISTRY COPYRIGHT 1997 ACS
RN ***107-13-1*** REGISTRY
CN 2-Propenenitrile (9CI) (CA INDEX NAME)
OTHER CA INDEX NAMES:
CN Acrylonitrile (8CI)
OTHER NAMES:
CN Acrylon
CN Carbacryl
CN Cyanoethene
CN Cyanoethylene
CN Fumigrain
CN Propenenitrile
CN VCN
CN Ventox
CN Vinyl cyanide

L2 ANSWER 2 OF 2 REGISTRY COPYRIGHT 1997 ACS
RN ***106-99-0*** REGISTRY
CN 1,3-Butadiene (8CI, 9CI) (CA INDEX NAME)
OTHER NAMES:
CN .alpha.,.gamma.-Butadiene
CN Biethylene
CN Bivinyl
CN Butadiene
CN Butadiene-1,3
CN Divinyl
CN Erythrene
CN Vinylethylene
MF C4 H6
```

Figure 5. STN database output of comonomer searches.

```
(File CA on STN)

=> s 9003-18-3
L3   7648 9003-18-3
=> s 107-13-1
L4   16353 107-13-1
=> s 106-99-0
L5   12917 106-99-0
=> s l4 and l5
L6   550 L4 AND L5
=> s l3 and l6
L7   19 L3 AND L6
```

Figure 6. STN database output from a combined search for a copolymer and its monomers.

5. Example of an Engineering Database Search

Searching the engineering databases is a less formidable task. In most cases, the searches for polymers are started using the common names of the compound. The indexing of the answer set is then evaluated for other terms to use. The following search is from the Engineering Index (file 8) on Dialog (Fig. 7).

```
  Set Items Description
  ... ..... .....
?s polyoxymethylene (compound entered as one word)
  S1   396 POLYOXYMETHYLENE
?s poly()oxymethylene (compound entered as two words)
     39391 POLY
       155 OXYMETHYLENE
  S2   87 POLY()OXYMETHYLENE
?s s1 or s2
     396 S1
      87 S2
  S3   459 S1 OR S2
?t 1/8/1

1/8/1
DIALOG(R)File 8:(c) 1997 Engineering Info. Inc. All rts. reserv.

04628455
Title: Hybrid effects on mechanical properties of polyoxymethylene
Descriptors: *Polymers; Fracture toughness; Strain rate; Tensile strength;
Bending strength; Impact resistance; Elastic moduli
Identifiers: Polyoxymethylene; Hybrid strength; Impact strength;
Flexural modulus

2/8/2
DIALOG(R)File 8:(c) 1997 Engineering Info. Inc. All rts. reserv.

04572617
Title: Spatially resolved in situ analysis of polymer additives by
two-step laser mass spectrometry
  Descriptors: *Additives; Polyvinyl chlorides; Polypropylenes;
Polyethylene terephthalates; Composition effects; Desorption; Stabilizers
(agents); Carbon dioxide lasers; Laser ablation; Antioxidants
  Identifiers: Polyoxymethylene; Hydroxyphenylbenzotriazole; Polymer
additives
```

Figure 7. Engineering Index output for poly(oxymethylene).

In this case, poly(oxymethylene) is indexed as one word so the search term was appropriate to this database. The two-word search (S2 and S3) was taken from the authors' abstracts, which are searched automatically in this database (unlike in CAS files).

6. Example of a Search with Punctuation in the Polymer Name

Poly(L-proline) is a more difficult search because most databases do not index punctuation. In this case, the compound is entered as if it were three words. In the first search, each word is specified to be in that specific order (adjacent to each other). This search yielded 26 hits (Fig. 8), but the specific records show that the term was found just in the abstract, not in the indexing. Searching by the broader term "polypeptides" would get a larger, less specific, answer set since any polypeptide could be included. The database also shows that "polypeptides" has been an index term since 1977.

Searching poly(L-proline) as three separate words would pull the terms from anywhere in the database record. This method gives more answers than the first search strategy (Fig. 9) but less than the broad polypeptide search.

7. Example of an Engineering Materials Abstracts Search

Similar strategies of searching chemical names and checking indexing should be used for other engineering databases such as Engineering Materials Abstracts. A search of the word, nylon, retrieves a larger answer set than the search of the sequence, nylon 6 (Fig. 10). A check of the indexing determines that nylon 6 is a valid term and would be a more specific, focused search on the topic.

Chemical Abstracts, Engineered Materials, Engineering Index, and other indexes such as Current Contents or the National Technical Information Service are available online through many vendors, on CDROM, at libraries, and over the Internet, via accounts with security and passwords. They are still available in paper copy at many libraries. Before doing any kind of literature search, the researcher must first decide how much information is needed, for what time period, what limits such as language or type of material are to be applied, and how much time is to be devoted to the search. Then each resource can be evaluated for what is covered and how it is indexed so that the proper tools will be used. Finding information is a lot like doing an experiment. Many paths can be explored and different techniques are tried until the researcher is satisfied.

8. Example of a Polymer Trademark Search

A trademark is either a word, phrase, symbol or design, or combination of words, phrases, symbols or designs, which identifies and distinguishes the source of the goods or services of one party from those of others. US trademark rights arise from either (a) actual use of the mark, or (b) the filing of a proper application to register a mark in the Patent

```
?s poly()l()proline
     39391 POLY
     71166 L
       259 PROLINE
  S7   26 POLY()L()PROLINE
?t 7/5

7/5/1
DIALOG(R)File 8:Ei Compendex(R)
(c) 1997 Engineering Info. Inc. All rts. reserv.

03979891 E.I. No: EIP94112411957
Title: Optical activity measurements in solids 7. Polylactides and poly (
beta -hydroxybutyrates)
   Author: Bartus, Jan; Weng, Dexi; Vogl, Otto
   Corporate Source: Polytechnic Univ, Brooklyn, NY, USA
   Source: Polymer International v 34 n 4 Aug 1994. p 433-442
   Publication Year: 1994
   CODEN: PLYIEI ISSN: 0959-8103
   Language: English
   Document Type: JA; (Journal Article) Treatment: X; (Experimental)
   Journal Announcement: 9412W4
   Abstract: The optical activities of poly-(R)-lactide, poly-(S)-lactide,
poly(beta -hydroxybutyrate) and two beta -hydroxyvalerate copolymers were
measured in solution, as solid powders in suspension, and where possible,
as films. Poly-(plus)-3-methyl-1-pentene was also reinvestigated. In some
cases the specific rotation values of powder samples showed significant
differences from the values of the solution measurements. The discrepancies
of the data observed seem to reflect the local environment of the polymer
chains in supermolecular assemblies and consequently the solid state
structure (morphology) of the polymers. The circular dichroism (CD) spectra
of the polymers were also measured in solution and in the form of their
films. For comparison, the CD spectra of the naturally occurring protein
casein and of the synthetic polypeptide poly-(L)-proline were also
measured. (Author abstract) 21 Refs.
   Descriptors: *Polymers; Optical variables measurement; Powders; Plastic
films; Molecular structure; Morphology; Proteins; Casein; Polypeptides;
Sodium compounds
   Identifiers: Polylactides; Polyhydroxybutyrates; Circular dichroism

?s polypeptides
  S1   1783 POLYPEPTIDES (January 1977)
```

Figure 8. Engineering Index output for an adjacent word search.

```
?s poly and l and proline
     39391 POLY
     71166 L
       259 PROLINE
  S8   39 POLY AND L AND PROLINE

Title: Synthesis of crosslinked poly(vinyl alcohol) with L- proline
pendant as the chiral stationary phase for resolution of amino acid
enantiomers
   Abstract: The porous crosslinked poly (vinyl alcohol) beads with the L
-proline pendant was synthesized as the chiral stationary phase (CSP) for
ligand-exchange chromatography of amino...

...and triallyl isocyanurate as a crosslinker, methanolysis of the
copolymer, glycidylation of the formed crosslinked poly (vinyl alcohol),
and final functionalization with L-proline. After the polymer with the
chiral ligand was complexed with copper(II) cations, it was...
```

Figure 9. Engineering Index output for a general word search.

```
?s nylon
  S7   3241 NYLON
?s nylon()6
     3241 NYLON
     18621 6
  S8   1615 NYLON()6

t 8/8/1
   Engineering composites with nylon 6 matrix.
   Descriptors: Conference Paper; Nylon 6- Composite materials; Styrene
butadiene resins- Composite materials; Thermoplastic elastomers-
   Mechanical properties; Tensile strength; Impact strength
   Section Headings: D2 Materials Development
   Subfile: P Polymers
```

Figure 10. Engineering Materials Abstracts output.

ALATHON Stylized Letters
INTL CLASS: 1 (Chemicals)
U.S. CLASS: 6 (Chemicals and Chemical Compositions)
STATUS: Renewed
SERIAL NO.: 71-569,356
REG. NO.: 543,580
REGISTERED: June 12, 1951
PUBLISHED: March 13, 1951
ORIGINAL REGISTRANT: E.I. DU PONT DE NEMOURS AND COMPANY (Delaware Corporation), 1007 MARKET STREET, WILMINGTON, DE (Delaware), USA (United States of America)
3RD NEW OWNER ENTERED AFTER REGISTRATION: OCCIDENTAL CHEMICAL 2400, HOUSTON, TX (Texas), 77046, USA (United States of America)
RENEWAL OWNER: (New Samek Corporation), 5 GREENWAY PLAZA, SUITE
Renewed: June 12, 1971^RENEWAL OWNER: OCCIDENTAL CHEMICAL CORPORATION (New York Corporation), 5 GREENWAY PLAZA, SUITE 2400, HOUSTON, TX (Texas), 77046, USA (United States of America)
Renewed: June 12, 1991
ASSIGNEE(S): CHASE MANHATTAN BANK, THE (NATIONAL ASSOCIATION)
AS
AGENT FOR THE BANKS
Assignor(s): CAIN CHEMICAL INC. (Delaware Corporation)
Reel/Frame: 0564/0786
Recorded: June 8, 1987
Brief: SECURITY INTEREST
ASSIGNEE(S): CAIN CHEMICAL INC. (Delaware Corporation), ELEVEN GREENWAY PLAZA, SUITE 2700, HOUSTON, TX (Texas), USA (United States of America)
Assignor(s): E. I. DU PONT DE MEMOURS AND COMPANY (Delaware Corporation), 1007 MARKET ST., WILMINGTON, DE (Delaware).
USA
(United States of America)
Reel/Frame: 0574/0432
Acknowledged: June 9, 1987
Recorded: August 24, 1987
Brief: ASSIGNS THE ENTIRE INTEREST AND GOOD WILL
ASSIGNEE(S): CAIN CHEMICAL INC. (Delaware Corporation)
Assignor(s): CHASE MANHATTAN BANK, THE (NATIONAL
ASSOCIATION),
AS AGENT
Reel/Frame: 0648/0279
Recorded: December 19, 1988
Brief: RELEASE BY SECURED PARTY OF THE SECURITY AGREEMENT RECORDED ON JUNE 8, 1987, AT REEL 0564, FRAME 786.
ASSIGNEE(S): OXY PETROCHEMICALS INC.
Assignor(s): CAIN CHEMICAL INC. (Delaware Corporation)
Reel/Frame: 0738/0721
Recorded: June 14, 1990
Brief: CHANGE OF NAME EFFECTIVE SEP. 21, 1989
ASSIGNEE(S): OCCIDENTAL CHEMICAL CORPORATION (New York Corporation), FIVE GREENWAY PLAZA, SUITE 2400, HOUSTON, TX (Texas), USA (United States of America)
Assignor(s): OXY PETROCHEMICALS INC. (Delaware Corporation), FIVE GREENWAY PLAZA, SUITE 2500, HOUSTON, TX (Texas), USA (United States of America)
Reel/Frame: 0738/0728
Recorded: June 14, 1990
Brief: ASSIGNS THE ENTIRE INTEREST AND GOOD WILL
ASSIGNEE(S): LYONDELL PETROCHEMICAL COMPANY (New York Corporation), 1221 MCKINNEY ST., SUITE 1600, HOUSTON, TX (Texas), 77010, USA (United States of America)
Assignor(s): OCCIDENTAL CHEMICAL CORPORATION (New York Corporation)
Reel/Frame: 1434/0830
Recorded: February 13, 1996
Brief: ASSIGNS THE ENTIRE INTEREST AND GOOD WILL

Figure 11. Trademark database search for Alathon.

and Trademark Office (PTO) stating that the applicant has a bonafide intention to use the mark in commerce regulated by the US Congress. Federal registration is not required to establish rights in a mark, nor is it required to begin use of a mark. However, federal registration can secure benefits beyond the rights acquired by merely using a mark. For example, the owner of a federal registration is presumed to be the owner of the mark for the goods and services specified in the registration, and to be entitled to use the mark nationwide. There are two related but distinct types of rights in a mark: the right to register and the right to use. Generally, the first party who either uses a mark in commerce or files an application in the PTO has the ultimate right to register that mark. The PTO's authority is limited to determining the right to register. The right to use a mark can be more complicated to determine.

Information on polymer trademarks can be found by searching in trademark databases on Dialog. The search is complicated because a name may be trademarked in one country, and yet not in another. Trademarks may be sold, or they may be assigned by one business to another.

A search for Alathon, a registered trademark for polyethylene, is given as an example (Fig. 11). Dialog's trademark scan software provides a description of the goods or services the trademark describes, gives the trademark status, the application number, the registration number and type, and the dates of the action. Previous owners or assignees of the trademark can be identified.

Usually, the trademark database of each country of interest must be checked to determine whether the name is registered.

E. LIST OF CAS REGISTRY NUMBERS OF COMMON POLYMERS

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Alginic acid		9005-32-7	Alginic acid	R	Unk*
Alginic acid, sodium salt		9005-38-3	Alginic acid, sodium salt	R	Unk
Bayberry wax	Myrtle wax	8038-77-5	Fats and Glyceridic oils, bayberry	S*	Unk
Beeswax, refined, yellow		8012-89-3	Beeswax	S*	Unk
Boron nitride		10043-11-5	Boron nitride	R	BN
Candelilla wax, natural		8006-44-8	Beeswax	S*	Unk
Carboxymethyl cellulose	Carboxymethylated cellulose sodium salt; cellulose glycolate	9004-32-4	Cellulose, carboxymethyl ether, sodium salt	R	$(C_2H_4O_3)_x$ Na_x; Unk
Carboxymethyl cellulose, sodium salt	Ambergum, carboxymethylcellulose; sodium carboxylmethyl cellulose	9004-32-4	Cellulose, carboxymethyl ether, sodium salt	R	$(C_2H_4O_3)_x$ Na_x; Unk
Carnauba wax, refined, No. 1, yellow		8015-86-9		S*	Unk
Carrageenan, type I (κ-carrageenan)		9000-07-1	Carrageenan	R	Unk
Carrageenan, type II (ι-carrageenan)		9062-07-1	*i*-Carrageenan	R	Unk
Cellulose		9004-34-6	Cellulose	R	Unk
Cellulose acetate	Acetic acid, cellulose ester	9004-35-7	Cellulose, acetate	R	$(C_2H_4O_2)_x$; Unk
Cellulose acetate butyrate	Acetobutyrate cellulose; cellulose acetobutyrate	9004-36-8	Cellulose, acetate butanoate	R	$(C_4H_8O_2)_x$ $(C_2H_4O_2)_x$; Unk
Cellulose acetate butyrate, acrylamidomethyl		91313-01-8	Cellulose, acetate butanoate, ((1-oxo-2-propenyl)amino)-methyl ether	R	$(C_4H_8O_2)_x$ $(C_4H_7NO_2)_x$ $(C_2H_4O_2)_x$; Unk
Cellulose acetate hydrogen phthalate	Acetyl phthalyl cellulose; cellulose acetate phthalate	9004-38-0	Cellulose, acetate hydrogen 1,2-benzene dicarboxylate	R	$(C_8H_6O_4)_x$ $(C_2H_4O_2)_x$; Unk
Cellulose acetate propionate	Acetylpropionyl cellulose	9004-39-1	Cellulose, acetate propanoate	R	$(C_3H_6O_2)_x$ $(C_2H_4O_2)_x$; Unk
Cellulose acetate propionate, acrylamidomethyl		97635-64-8	Cellulose, acetate propanoate, ((1-oxo-2-propenyl)amino)-methyl ether	R	$(C_4H_7NO_2)_x$ $(C_3H_6O_2)_x$ $(C_2H_4O_2)_x$; Unk
Cellulose acetate trimellitate		52907-01-4	Cellulose, acetate 1,2,4-benzenetricarboxylate	R	$(C_9H_6O_6)_x$ $(C_2H_4O_2)_x$; Unk
Cellulose diacetate	Diacetyl cellulose	9035-69-2	Cellulose, diacetate	R	$(C_2H_4O_2)_{1/2}$; Unk

* Indicates that the precise molecular formula is unknown.

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Cellulose nitrate	Collodion; soluble gun cotton	9004-70-0	Cellulose, nitrate	R	$(HNO_3)_x$; Unk
Cellulose propionate		9004-48-2	Cellulose, propanoate	R	$(C_3H_6O_2)_x$; Unk
Cellulose sulfate	Sodium cellulose sulfate	9005-22-5	Cellulose, hydrogen sulfate, sodium salt	R	$(H_2O_4S)_x$ Na_x; Unk
Cellulose triacetate	Poroplastic; triacetylcellulose	9012-09-3	Cellulose, triacetate	R	$(C_2H_4O_2)_{1/3}$; Unk
Cellulose tributyrate		39320-16-6	Cellulose, tributanoate	R	$(C_4H_8O_2)_{1/3}$; Unk
Cellulose tricarbanilate		9047-07-8	Cellulose, tris (phenylcarbamate)	R	$(C_7H_7NO_2)_{1/3}$; Unk
Cellulose xanthate	Cellulose carbonodithioate; cellulose hydrogen dithiocarbonate	9032-37-5	Cellulose, hydrogen carbonodithioate	R	$(CH_2OS_2)_x$; Unk
Cellulose, cyanoethylated	Cellulose cyanoethyl ether; cellulose ethylcyanide	9004-41-5	Cellulose, 2-cyanoethyl ether	R	$(C_3H_5NO)_x$; Unk
Cellulose, microcrystalline, colloidal		51395-75-6	Cellulose, mixt. with cellulose carboxymethyl ether sodium salt	R	$(C_2H_4O_3)_x$ Na_x; Unk
Chitin, crab shells		1398-61-4	Chitin	S*	Unk
Chitosan		9012-76-4	Chitosan	R	Unk
Chlorinated poly(vinyl chloride)		68648-82-8	Ethene, chloro-, homopolymer, chlorinated	*	Unk
Chlorinated polyethylene		63231-66-3	Rubber, synthetic, chlorinated polyethylene	S	Unk
Coconut oil	Fats and glyceridic oils	8001-31-8	Coconut oil	S*	Unk
Ethyl cellulose	Cellulose ethyl ether; cellulose ethylate; ethocel, ethyl cellulose ether	9004-57-3	Cellulose, ethyl ether	R	$(C_2H_6O)_x$; Unk
Graphite, fluorinated, polymer		51311-17-2	Carbon fluoride	R	Unk
Guar gum carboxymethyl 2-hydroxypropyl ether, sodium salt	Sodium carboxymethyl hydroxypropyl guar	68130-15-4	Guar gum, carboxymethyl 2-hydroxypropyl ether, sodium salt	R	Unk
Hydroxybutyl methyl cellulose		9041-56-9	Cellulose, hydroxybutyl methyl ether	R	$(C_4H_{10}O_2)_x$ $(CH_4O)_x$; Unk
Hydroxyethyl cellulose (2-hydroxyethyl cellulose)	Cellosize; cellulose hydroxyethyl ether; cellulose hydroxyethylate; glycol cellulose	9004-62-0	Cellulose, 2-hydroxyethyl ether	R	$(C_2H_6O_2)_x$; Unk
Hydroxypropyl cellulose	Cellulose hydroxypropyl ether; hydropropyl cellulose; hydroxypropyl cellulose	9004-64-2	Cellulose, 2-hydroxypropyl ether	R	$(C_3H_8O_2)_x$; Unk
Hydroxypropyl methyl cellulose	Cellulose hydroxypropyl methyl ether	9004-65-3	Cellulose, 2-hydroxypropyl methyl ether	R	$(C_3H_8O_2)_x$ $(CH_4O)_x$; Unk
Hydroxypropyl methyl cellulose phthalate	Cellulose phthalate hydroxypropyl methyl ether	9050-31-1	Cellulose, hydrogen 1,2-benzenedicarboxylate, 2-hydroxypropyl methyl ether	R	$(C_8H_6O_4)_x$ $(C_3H_8O_2)_x$ $(CH_4O)_x$; Unk
Lignin, alkali	Indulin A	8068-05-1	Lignin, alkali	R	Unk
Lignin, alkali, 2-hydroxypropyl ether		88402-77-1	Lignin, alkali, 2-hydroxypropyl ether	R	$(C_3H_8O_2)_x$; Unk
Lignin, alkali, carboxymethyl ether	Kraft lignin carboxymethyl ether	102962-28-7	Lignin, alkali, carboxymethyl ether	R	$(C_2H_4O_3)_x$; Unk
Lignin, hydrolytic		8072-93-3	Lignin, hydrolytic	R	Unk
Lignin, organosolv	Dioxane lignin	8068-03-9	Lignin, organosolv	R	Unk
Lignin, organosolv, 2-hydroxyethyl ether		90881-34-8	Lignin, organosolv, 2-hydroxyethyl ether	R	$(C_2H_6O_2)_x$; Unk
Lignosulfonic acid, sodium salt		8061-51-6	Lignosulfonic acid, sodium salt	R	Unk
Linseed oil	Fats and glyceridic oils	8001-26-1	Linseed oil	S*	Unk
Maltodextrin		9050-36-6	Maltodextrin	R	Unk
Merrifield's peptide resin, cross-linked	Chloromethylstyrene-*p*-divinylbenzene–styrene copolymer; A221481; chloromethylstyrene–styrene–divinylbenzene copolymer	55844-94-5	Benzene, (chloromethyl)ethenyl-, polymer with diethenylbenzene and ethenylbenzene	R	$(C_{10}H_{10} \cdot C_9H_9Cl \cdot C_8H_8)_x$
Methyl 2-hydroxyethyl cellulose	Hydroxyethyl methyl cellulose	9032-42-2	Cellulose, 2-hydroxyethyl methyl ether	R	$(C_2H_6O_2)_x$ $(CH_4O)_x$; Unk
Methyl cellulose	Methyl ether cellulose	9004-67-5	Cellulose, methyl ether	R	$(CH_4O)_x$; Unk

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Net-Poly(acrylic acid)-inter-net-poly(siloxane)		143106-82-5	2-Propenoic acid, 2-methyl-, 3-(dimethoxymethylsilyl)propyl ester, polymer with butyl 2-propenoate, decamethylcyclopentasiloxane, etc.	R	$(C_{10}H_{30}O_5Si_5 \cdot C_{10}H_{20}O_4Si \cdot C_8H_{24}O_4Si_4 \cdot C_7H_{12}O_2 \cdot C_7H_{10}O_2 \cdot C_6H_{18}O_3Si_3 \cdot C_5H_8O_2)_x$
Nylon 6	Polyamide 6	25038-54-4	Poly(imino(1-oxo-1,6-hexanediyl))	U	$(C_6H_{11}NO)_n$
Nylon 9	Polynonanamide; poly(imino-1-oxononamethylene)	25748-72-5	Nonanoic acid, 9-amino-, homopolymer	R	$(C_9H_{17}NO)_x$
Nylon 11	Poly(imino-1-oxoundecamethylene)	25035-04-5	Poly(imino(1-oxo-1,11-undecanediyl))	R	$(C_{11}H_{21}NO)$
Nylon 12	Polydodecanamide; poly(imino-carbonylundecamethylene); poly(imino-1-oxododeca-methylene)	24937-16-4	Poly(imino(1-oxo-1,12-dodecanediyl))	R	$(C_{12}H_{23}NO)_n$
Nylon 46	Adipic acid-1,4-butanediamine copolymer, SRU; poly(tetra-methylene adipamide)	50327-22-5	Poly(imino-1,4-butanediylimino (1,6-dioxo-1,6-hexanediyl))	R	$(C_{10}H_{18}N_2O_2)_n$
Nylon 46, monomer-based		50327-77-0	Hexanedioic acid, polymer with 1,4-butanediamine	R	$(C_6H_{10}O_4 \cdot C_4H_{12}N_2)_x$
Nylon 6, 6	Adipic acid-1, 6-hexanediamine copolymer; adipoyl dichloride-hexamethylenediamine polymer, SRU; polyamides, nylon 66; poly(hexamethylene adipamide); poly(iminoadipoyl iminohexamethylene)	32131-17-2	Poly(imino(1,6-dioxo-1,6-hexanediyl)imino-1,6-hexanediyl)	U	$(C_{12}H_{22}N_2O_2)_n$
Nylon 6, 9	Azelaic acid-1,6-hexanediamine salt polymer, SRU	28757-63-3	Poly(imino-1,6-hexanediylimino (1,9-dioxo-1,9-nonanediyl))	R	$(C_{15}H_{28}N_2O_2)_n$
Nylon 6/66	Adipic acid-ε-caprolactam-1,6-hexanediamine copolymer; nylon 6-nylon 66 copolyamide; nylon 6-nylon 66 copolymer; poly(hexamethyleneadipamide)	24993-04-2	Hexanedioic acid, polymer with hexahydro-2H-azepin-2-one and 1,6-hexanediamine	R	$(C_6H_{16}N_2 \cdot C_6H_{11}NO \cdot C_6H_{10}O_4)_x$
Nylon 610	Poly(iminohexamethylene iminosebacoyl); poly(hexa-methylenesebacamide)	9008-66-6	Poly(imino-1,6-hexanediylimino-1,10-dioxo-1,10-decanediyl))	R	$(C_{16}H_{30}N_2O_2)_n$
Nylon 612		24936-74-1	Poly(imino-1,6-hexanediylimino (1,12-dioxo-1,12-dodecanediyl))	R	$(C_{18}H_{34}N_2O_2)_n$
Nylon 6/T, poly(hexamethylene terephthalamide)	Hexamethylenediamine–terephthalic acid polymer	24938-03-2	1,4-Benzenedicarboxylic acid, polymer with 1,6-hexanediamine	R	$(C_8H_6O_4 \cdot C_6H_{16}N_2)_x$
Paraffin wax, chunks		8002-74-2	Paraffin waxes and hydrocarbon waxes	S*	Unk
Paraformaldehyde	Paraform	30525-89-4	Paraformaldehyde	R	$(CH_2O)_x$
Phenol–formaldehyde resin	Formaldehyde–phenol polymer	9003-35-4	Phenol, polymer with formaldehyde	R	$(C_6H_6O \cdot CH_2O)_x$
Poly(D-alanine)	Poly-D-alanine	26701-36-0	D-Alanine, homopolymer	R	$(C_3H_7NO_2)_x$
Poly(L-alanine)	Polyalanine	25191-17-7	L-Alanine, homopolymer	R	$(C_3H_7NO_2)_x$
Poly(D,L-alanine)		25281-63-4	DL-Alanine, homopolymer	R	$(C_3H_7NO_2)_x$
Poly(γ-aminobutyric acid)	Poly(4-aminobutyric acid)	53504-43-1	Butanoic acid, 4-amino-, homopolymer	R	$(C_4H_9NO_2)_x$
Poly(L-aspartic acid), sodium salt	Poly(aspartic acid) sodium salt	34345-47-6	L-Aspartic acid, homopolymer, sodium salt	R	$(C_4H_7NO_4)_x \cdot xNa$
Poly(acenaphthylene)	Oligoacenaphthylene	25036-01-5	Acenaphthylene, homopolymer	R	$(C_{12}H_8)_x$
Poly(acrylamide)	Acrylamide polymer	9003-05-8	2-Propenamide, homopolymer	R	$(C_3H_5NO)_x$
Poly(acrylamide-*co*-acrylic acid)	Acrylamide–acrylate polymer; acrylamide–acrylic acid polymer; poly(acrylic acid-acrylamide)	9003-06-9	2-Propenoic acid, polymer with 2-propenamide	R	$(C_3H_5NO \cdot C_3H_4O_2)_x$
Poly(acrylamide-*co*-diallyl dimethyl ammonium chloride)	Acrylamide–diallyldimethyl-ammonium chloride polymer	26590-05-6	2-Propen-1-aminium, *N*,*N*-dimethyl-*N*-2-propenyl-, chloride, polymer with 2-propenamide	R	$(C_8H_{16}N \cdot C_3H_5NO \cdot Cl)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(2-acrylamido, 2-methyl, 1-propane sulfonic acid-*co*-acrylonitrile)		54640-82-3	1-Propanesulfonic acid, 2-methyl-2-((1-oxo-2-propenyl)amino)-, polymer with 2-propenenitrile	R	$(C_7H_{13}NO_4S \cdot C_3H_3N)_x$
Poly(2-acrylamido-2-methyl-1-propane sulfonic acid)		27119-07-9	1-Propanesulfonic acid, 2-methyl-2-((1-oxo-2-propenyl)amino)-, homopolymer	R	$(C_7H_{13}NO_4S)_x$
Poly(2-acrylamido-2-methyl-1-propane sulfone acid-*co*-styrene)		51121-85-8	1-Propanesulfonic acid, 2-methyl-2-((1-oxo-2-propenyl)amino)-, polymer with ethenylbenzene	R	$(C_8H_8 \cdot C_7H_{13}NO_4S)_x$
Poly(acrylic acid)	Acrylic acid polymer	9003-01-4	2-Propenoic acid, homopolymer	R	$(C_3H_4O_2)_x$
Poly(acrylic acid), ammonium salt	Ammonium polyacrylate; ammonium polyacrylic acid	9003-03-6	2-Propenoic acid, homopolymer, ammonium salt	R	$(C_3H_4O_2)_x \cdot xH_3N$
Poly(acrylic acid), potassium salt, crosslinked	Potassium polyacrylate	25608-12-2	2-Propenoic acid, homopolymer, potassium salt	R	$(C_3H_4O_2)_x \cdot xK$
Poly(acrylic acid), sodium salt	Acrylic acid polymer, sodium salt	9003-04-7	2-Propenoic acid, homopolymer, sodium salt	R	$(C_3H_4O_2)_x \cdot xNa$
Poly(acrylic acid-*co*-trimethylolpropane), sodium salt	Acrylic acid–sodium acrylate–trimethylolpropane triacrylate copolymer	76774-25-9	2-Propenoic acid, polymer with 2-ethyl-2-(((1-oxo-2-propenyl)-oxy)methyl)-1,3-propanediyl di-2-propenoate and sodium 2-propenoate	R	$(C_{15}H_{20}O_6 \cdot C_3H_4O_2 \cdot C_3H_4O_2 \cdot Na)_x$
Poly(acrylic acid, sodium salt-*graft*-poly(ethylene oxide)), crosslinked	Acrylic acid–vinyl alcohol polymer, sodium salt	27599-56-0	2-Propenoic acid, polymer with ethenol, sodium salt	R	$(C_3H_4O_2 \cdot C_2H_4O)_x \cdot xNa$
Poly(acrylic acid-*co*-acrylamide), potassium salt, crosslinked	Acrylamide–potassium acrylate copolymer	31212-13-2	2-Propenoic acid, potassium salt, polymer with 2-propenamide	R	$(C_3H_5NO \cdot C_3H_4O_2 \cdot K)_x$
Poly(acrylic acid-*co*-maleic acid)		29132-58-9	2-Butenedioic acid (Z)-, polymer with 2-propenoic acid	R	$(C_4H_4O_4 \cdot C_3H_4O_2)_x$
Poly(acrylic acid-*co*-maleic acid), sodium salt	Acrylic acid–maleic anhydride copolymer, sodium salt	52255-49-9	2-Propenoic acid, polymer with 2,5-furandione, sodium salt	R	$(C_4H_2O_3 \cdot C_3H_4O_2)_x \cdot xNa$
Poly(acrylonitrile)	Acrylonitrile polymer; poly(1-cyanoethylene)	25014-41-9	2-Propenenitrile, homopolymer	R	$(C_3H_3N)_x$
Poly(acrylonitrile-*co*-butadiene)	Acrylonitrile–butadiene polymer	9003-18-3	2-Propenenitrile, polymer with 1,3-butadiene	R	$(C_4H_6 \cdot C_3H_3N)_x$
Poly(acrylonitrile-*co*-butadiene), amine terminated		68683-29-4	2-Propenenitrile, polymer with 1,3-butadiene, 1-cyano-1-methyl-4-oxo-4-((2-(1-piperazinyl)ethyl)-amino)butyl-terminated	R	$(C_4H_6 \cdot C_3H_3N)_x$
Poly(acrylonitrile-*co*-butadiene), hydrogenated		88254-10-8	2-Propenenitrile, polymer with 1,3-butadiene, hydrogenated	*	$(C_4H_6 \cdot C_3H_3N)_x$
Poly(acrylonitrile-*co*-butadiene-*co*-acrylic acid), dicarboxy terminated		68891-50-9	2-Propenoic acid, polymer with 1,3-butadiene and 2-propenenitrile, 3-carboxy-1-cyano-1-methylpropyl-terminated	*	$(C_4H_6 \cdot C_3H_4O_2 \cdot C_3H_3N)_x$
Poly(acrylonitrile-*co*-butadiene-*co*-styrene)	Acrylonitrile–butadiene–styrene polymer; cycolac; styrene–acrylonitrile–butadiene polymer	9003-56-9	2-Propenenitrile, polymer with 1,3-butadiene and ethenylbenzene	R	$(C_8H_8 \cdot C_4H_6 \cdot C_3H_3N)_x$
Poly(acrylonitrile-*co*-vinylidene chloride-*co*-methyl methacrylate)	Acrylonitrile–methyl methacrylate–vinylidene chloride polymer	25214-39-5	2-Propenoic acid, 2-methyl-, methyl ester, polymer with 1,1-dichloroethene and 2-propenenitrile	R	$(C_5H_8O_2 \cdot C_3H_3N \cdot C_2H_2C_2)_x$
Poly(allyl amine hydrochloride)	Allylamine hydrochloride	71550-12-4	2-Propen-1-amine, hydrochloride, homopolymer	R	$(C_3H_7N \cdot ClH)_x$
Poly(*p*-aminobenzaldehyde)	Poly(4-aminobenzaldehyde)	28107-09-7	Benzaldehyde, 4-amino-, homopolymer	R	$(C_7H_7NO)_x$
Poly(anethole sulfonic acid), sodium salt	Polyanethole sulfonate; sodium polyanetholesulfonate	52993-95-0	Benzene, 1-methoxy-4-(1-propenyl)-, (E)-, homopolymer, sulfonated, sodium salt	R	Unk
Poly((α,β)-DL-aspartic acid), sodium salt		94525-01-6	DL-Aspartic acid, homopolymer, sodium salt	R	$(C_4H_7NO_4)_x \cdot xNa$
Poly(azelaic anhydride)	Azelaic acid polymer	26776-28-3	Nonanedioic acid, homopolymer	R	$(C_9H_{16}O_4)_x$
Poly(1,4-benzene dicarboxy-*alt*-bis(4-phenoxyphenyl) methanone)		55088-54-5	1,4-Benzenedicarbonyl dichloride, polymer with bis(4-phenoxyphenyl)methanone	R	$(C_{25}H_{18}O_3 \cdot C_8H_4Cl_2O_2)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(benzimidazole)	Diphenyl isophthalate-3,3′,4,4′-tetraaminodiphenyl polymer	25928-81-8	1,3-Benzenedicarboxylic acid, diphenyl ester, polymer with (1,1′-biphenyl)-3,3′,4,4′-tetramine	R	$(C_{20}H_{14}O_4 \cdot C_{12}H_{14}N_4)_x$
Poly(benzoguanamine-*co*-formaldehyde), methylated/ethylated		68037-08-1	Formaldehyde, polymer with 6-phenyl-1,3,5-triazine-2,4-diamine, ethylated/methylated	*	Unk
Poly(benzophenone tetracarboxylic dianhydride-*co*-phenylene diamine)	Benzophenonetetracarboxylic dianhydride-4,4′-oxydianiline-*m*-phenylenediamine copolymer	31942-21-9	1,3-Isobenzofurandione, 5,5′-carbonylbis-, polymer with 1,3-benzenediamine and 4,4′-oxybis(benzenamine)	R	$(C_{17}H_6O_7 \cdot C_{12}H_{12}N_2O \cdot C_6H_8N_2)_x$
Poly(2-(4-benzoyl-3-hydroxyphenoxy) ethyl acrylate)	2-Hydroxy-4-acryloxyethoxybenzophenone polymer	29963-76-6	2-Propenoic acid, 2-(4-benzoyl-3-hydroxyphenoxy)ethyl ester, homopolymer	R	$(C_{18}H_{16}O_5)_x$
Poly(benzyl methacrylate)	Benzyl methacrylate polymer	25085-83-0	2-Propenoic acid, 2-methyl-, phenylmethyl ester, homopolymer	R	$(C_{11}H_{12}O_2)_x$
Poly(β-cyclodextrin-*co*-epichlorohydrin)		25655-42-9	β-Cyclodextrin, polymer with (chloromethyl)oxirane	R	$(C_{42}H_{70}O_{35} \cdot C_3H_5ClO)_x$
Poly(bis(2-chloroethyl)ether-*co*-1,3-bis(3-(dimethylamino)-propyl)urea), quaternary salt		68555-36-2	Urea, *N*,*N*′-bis(3-(dimethylamino)-propyl)-, polymer with 1,1′-oxybis(2-chloroethane)	R	$(C_{11}H_{26}N_4O \cdot C_4H_8Cl_2O)_x$
Poly(bis(benzyl thio)acetylene)		93975-09-8	Benzene, 1,1′-(1,2-ethynediylbis-(thiomethylene))bis-, homopolymer	R	$(C_{16}H_{14}S_2)_x$
Poly(bis(ethyl thio)acetylene)		93975-08-7	Ethyne, bis(ethylthio)-, homopolymer	R	$(C_6H_{10}S_2)_x$
Poly(bis(methyl thio)acetylene)		93975-07-6	Ethyne, bis(methylthio)-, homopolymer	R	$(C_4H_6S_2)_x$
Poly(bisphenol A-*co*-4-nitrophthalic anhydride-*co*-1,3-phenylene diamine)	2,2-Bis(4-(3,4-dicarboxyphenoxy)-phenyl)propane dianhydride-*m*-phenylenediamine copolymer	61128-46-9	1,3-Isobenzofurandione, 5,5′-((1-methylethylidene) bis(4,1-phenyleneoxy)) bis- polymer with 1,3-benzenediamine	R	$(C_{31}H_{20}O_8 \cdot C_6H_8N_2)_x$
Poly(bisphenol A-*co*-epichlorohydrin)		25036-25-3	Phenol, 4,4′-(1-methylethylidene)-bis-, polymer with 2,2′-((1-ethylethylidene) bis(4,1-phenyleneoxymethylene))-bis(oxirane)	R	$(C_{21}H_{24}O_4 \cdot C_{15}H_{16}O_2)_x$
Poly(bisphenol A-*co*-epichlorohydrin)	Bisphenol A epoxy resin; phenoxy resin	25068-38-6	Phenol, 4,4′-(1-methylethylidene)-bis-, polymer with (chloromethyl)oxirane	R	
Poly(bisphenyl tetracarboxylic dianhydride-*co*-phenylene diamine)	Biphenyl dianhydride-*p*-phenylenediamine copolymer	29319-22-0	(5,5′-Biisobenzofuran)-1,1′,3,3′-tetrone, polymer with 1,4-benzenediamine	R	$(C_{16}H_6O_6 \cdot C_6H_8N_2)_x$
Poly(4-bromostyrene)	*p*-Bromostyrene polymer	24936-50-3	Benzene, 1-bromo-4-ethenyl-, homopolymer	R	$(C_8H_7Br)_x$
Poly(butadiene)	Poly(1-butenylene); butadiene homopolymer	9003-17-2	1,3-Butadiene, homopolymer	R	$(C_4H_6)_x$
Poly(butadiene); *cis* and *trans*	Poly(1-butenylene)	25038-44-2	Poly(1-butene-1,4-diyl)	R	$(C_4H_6)_n$
Poly(butadiene), *cis*		40022-03-5	Poly(1-butene-1,4-diyl),	R	$(C_4H_6)_n$
Poly(butadiene), dicarboxy terminated		68891-79-2	1,3-Butadiene, homopolymer, 3-carboxy-1-cyano-1-methylpropyl-terminated	*	Unk
Poly(butadiene), hydroxyl functionalized		69102-90-5	1,3-Butadiene, homopolymer, hydroxy-terminated	*	Unk
Poly(1,4-butanediol bis(4-aminobenzoate))		54667-43-5	Poly(oxy-1,4-butanediyl), α-(4-aminobenzoyl)-ω-((4-aminobenzoyl)oxy)-	R	$(C_4H_8O)_n$ $C_{14}H_{12}N_2O_3$
Poly(1,4-butanediol/neopentyl glycol-*alt*-adipic acid)		36609-31-1	Hexanedioic acid, polymer with 1,4-butanediol and 2,2-dimethyl-1,3-propanediol	R	$(C_6H_{10}O_4 \cdot C_5H_{12}O_2 \cdot C_4H_{10}O_2)_x$
Poly(butene)	Polybutene; polybutylene	9003-29-6	Butene, homopolymer	R	$(C_4H_8)_x$
Poly(1-butene)	Polybutene	9003-28-5	1-Butene, homopolymer	R	$(C_4H_8)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(butenes), hydrogenated		68937-10-0	Butene, homopolymer, hydrogenated	*	Unk
Poly(butenes), monoepoxide		119275-53-5	1-Butene, polymer with 2-butene and 2-methyl-1-propene, epoxidized	*	$(C_4H_8 \cdot C_4H_8 \cdot C_4H_8)_x$
Poly(butyl acrylate)	Butyl acrylate polymer	9003-49-0	2-Propenoic acid, butyl ester, homopolymer	R	$(C_7H_{12}O_2)_x$
Poly(4-*tert*-butylcyclohexyl methacrylate)		34903-89-4	2-Propenoic acid, 2-methyl-, 4-(1,1-dimethylethyl)cyclohexyl ester, homopolymer	R	$(C_{14}H_{24}O_2)_x$
Poly(butyl methacrylate)	Butyl methacrylate polymer	9003-63-8	2-Propenoic acid, 2-methyl, butyl ester, homopolymer	R	$(C_8H_{14}O_2)_x$
Poly(butyl methacrylate-*co*-isobutyl methacrylate)	Butyl methacrylate – isobutyl methacrylate polymer	9011-53-4	2-Propenoic acid, 2-methyl, butyl ester, polymer with 2-methylpropyl 2-methyl-2-propenoate	R	$(C_8H_{14}O_2 \cdot C_8H_{14}O_2)_x$
Poly(butyl methacrylate-*co*-methyl methacrylate)	Methyl methacrylate – butyl methacrylate polymer	25608-33-7	2-Propenoic acid, 2-methyl-, butyl ester, polymer with methyl 2-methyl-2-propenoate	R	$(C_8H_{14}O_2 \cdot C_5H_8O_2)_x$
Poly(4-*tert*-butylphenyl methacrylate)	*p*-*tert*-Butylphenyl methacrylate polymer	29696-27-3	2-Propenoic acid, 2-methyl-, 4-(1,1-dimethylethyl)phenyl ester, homopolymer	R	$(C_{14}H_{18}O_2)_x$
Poly(4-*tert*-butylstyrene)	*p*-*tert*-Butylstyrene polymer	26009-55-2	Benzene, 1-(1,1-dimethylethyl)-4-ethenyl-, homopolymer	R	$(C_{12}H_{16})_x$
Poly(1,4-butylene adipate)	Adipic acid–butylene glycol copolymer; adipic acid-butylene glycol polymer; PBAG; poly(butylene adipate); poly-(tetramethylene adipate)	25103-87-1	Hexanedioic acid, polymer with 1,4-butanediol	R	$(C_6H_{10}O_4 \cdot C_4H_{10}O_2)_x$
Poly(1,4-butylene adipate-*co*-1,4-butylene succinate)	Adipic acid-1,4-butanediol-HMDI-succinic acid copolymer	119553-67-2	Hexanedioic acid, polymer with butanedioic acid, 1,4-butanediol and 1,6-diisocyanatohexane	R	$(C_8H_{12}N_2O_2 \cdot C_6H_{10}O_4 \cdot C_4H_{10}O_2 \cdot C_4H_6O_4)$
Poly(1,2-butylene glycol)	1,2-Epoxybutane–propylene glycol polymer	31923-86-1	1,2-Propanediol, polymer with ethyloxirane	R	$(C_4H_8O \cdot C_3H_8O_2)_x$
Poly(1,2-butylene glycol), monobutyl ether		144437-84-3	Oxirane, ethyl-, homopolymer, mono(2-butoxy-1-methylethyl) ether	R	$(C_7H_{16}O_2 \cdot C_4H_8O)_x$
Poly(1,4-butylene glutarate), hydroxy terminated	1,4-Butanediol–glutaric acid copolymer	52256-54-9	Pentanedioic acid, polymer with 1,4-butanediol	R	$(C_5H_8O_4 \cdot C_4H_{10}O_2)_x$
Poly(1,4-butylene succinate)	Butylene glycol–succinic acid copolymer; poly(tetramethylene succinate)	25777-14-4	Butanedioic acid, polymer with 1,4-butanediol	R	$(C_4H_{10}O_2 \cdot C_4H_6O_4)_x$
Poly(1,4-butylene succinate-*co* HMDI)	1,4-Butanediol-HMDI-succinic acid copolymer	143606-53-5	Butanedioic acid, polymer with 1,4-butanediol and 1,6-diisocyanatohexane	R	$(C_8H_{12}N_2O_2 \cdot C_4H_{10}O_2 \cdot C_4H_6O_4)_x$
Poly(1,4-butylene terephthalate)	Butanediol–terephthalic acid copolymer; poly(oxytetra-methyleneoxy-terephthalic acid–terephthaloyl); tetramethylene glycol polymer, SRU	24968-12-5	Poly(oxy-1,4-butanediyloxycar-bonyl-1,4-phenylenecarbonyl)	R	$(C_{12}H_{12}O_4)_n$
Poly(caprolactone diol)	Caprolactone – diethylene glycol polymer	36890-68-3	2-Oxepanone, polymer with 2,2′-oxybis(ethanol)	R	$(C_6H_{10}O_2 \cdot C_4H_{10}O_3)_x$
Poly(caprolactone triol)	ε-Caprolactone – trimethylolpropane polymer	37625-56-2	2-Oxepanone, polymer with 2-ethyl-2-(hydroxymethyl)-1,3-propanediol	R	$(C_6H_{14}O_3 \cdot C_6H_{10}O_2)_x$
Poly(caprolactone triol)		54735-63-6	Poly(oxy(1-oxo-1,6-hexanediyl)), α-hydro-ω-hydroxy-, ester with 2-ethyl-2-(hydroxymethyl)-1,3-propanediol (3 : 1)	R	$(C_6H_{10}O_2)_n (C_6H_{10}O_2)_n (C_6H_{10}O_2)_n C_6H_{14}O_3$
Poly(caprolactone)		24980-41-4	2-Oxepanone, homopolymer	R	$(C_6H_{10}O_2)_x$
Poly(chloroprene)	Chlorobutadiene polymer	9010-98-4	1,3-Butadiene, 2-chloro-, homopolymer	R	$(C_4H_5Cl)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(chlorostyrene)	Chlorostyrene polymer	9022-52-0	Benzene, chloroethenyl-, homopolymer	R	$(C_8H_7Cl)_x$
Poly(4-chlorostyrene)	*p*-Chlorostyrene polymer 4-Chlorostyrene polymer	24991-47-7	Benzene, 1-chloro-4-ethenyl-, homopolymer	R	$(C_8H_7Cl)_x$
Poly(chlorotrifluoroethylene)	Halocarbon oil	9002-83-9	Ethene, chlorotrifluoro-, homopolymer	R	$(C_2ClF_3)_x$
Poly(chlorotrifluoroethylene-*co*-vinylidene fluoride)	Chlorotrifluoroethylene–vinylidene fluoride polymer; trifluorochloroethylene–vinylidene fluoride polymer	9010-75-7	Ethene, chlorotrifluoro-, polymer with 1,1-difluoroethene	R	$(C_2H_2F_2 \cdot C_2ClF_3)_x$
Poly(coumarone-*co*-indene)		35343-70-5	Benzofuran, polymer with 1H-indene	R	$(C_9H_8 \cdot C_8H_6O)_x$
Poly(*o*-cresol glycidyl ether-*co*-formaldehyde)		29690-82-2	Formaldehyde, polymer with (chloromethyl)oxirane and 2-methylphenol	R	$(C_7H_8O \cdot C_3H_5ClO \cdot CH_2O)_x$
Poly(*o*-cresyl glycidyl ether-*co*-formaldehyde)	Cresol–epichlorohydrin–formaldehyde polymer	37382-79-9	Formaldehyde, polymer with (chloromethyl)oxirane and methylphenol	R	$(C_7H_8O \cdot C_3H_5ClO \cdot CH_2O)_x$
Poly(2-cyanobutyl acrylate)	Butyl 2-cyanoacrylate polymer; poly(butyl cyanoacrylate)	25154-80-7	2-Propenoic acid, 2-cyano-, butyl ester, homopolymer	R	$(C_8H_{11}NO_2)_x$
Poly(2-cyanoethyl acrylate)	Poly(ethyl cyanoacrylate);	25067-30-5	2-Propenoic acid, 2-cyano-, ethyl ester, homopolymer	R	$(C_6H_7NO_2)_x$
Poly(2-cyanoheptyl acrylate)		26936-29-8	2-Propenoic acid, 2-cyano-, heptyl ester, homopolymer	R	$(C_{11}H_{17}NO_2)_x$
Poly(2-cyanohexyl acrylate)	Poly(hexyl cyanoacrylate)	26877-39-4	2-Propenoic acid, 2-cyano-, hexyl ester, homopolymer	R	$(C_{10}H_{15}NO_2)_x$
Poly(2-cyanoisobutyl acrylate)	Isobutyl 2-cyanoacrylate polymer	26809-38-1	2-Propenoic acid, 2-cyano-	R	$(C_8H_{11}NO_2)_x$
Poly(2-cyanoisopropyl acrylate)	Poly(isopropyl cyanoacrylate)	25931-02-6	2-Propenoic acid, 2-cyano-, 1-methylethyl ester, homopolymer	R	$(C_7H_9NO_2)_x$
Poly(cyclododecyl acrylate)		56710-66-8	2-Propenoic acid, cyclododecyl ester, homopolymer	R	$(C_{15}H_{26}O_2)_x$
Poly(1,4-cyclohexane dimethylene terephthalate-*co*-ethylene terephthalate)	Dimethyl terephthalate-ethylene glycol-1,4-bis(hydroxymethyl)-cyclohexane copolymer	25640-14-6	1,4-Benzenedicarboxylic acid, dimethyl ester, polymer with 1,4-cyclohexanedimethanol and 2-ethanediol	R	$(C_{10}H_{10}O_4 \cdot C_8H_{16}O_2 \cdot C_2H_6O_2)$
Poly(cyclohexyl methacrylate)		25768-50-7	2-Propenoic acid, 2-methyl-, cyclohexyl ester, homopolymer	R	$(C_{10}H_{16}O_2)_x$
Poly(di(ethylene glycol) glycerol-*alt*-adipic acid)	Diethylene glycol–glycerol–adipic acid copolymer	26760-54-3	Hexanedioic acid, polymer with 2,2′-oxybis(ethanol) and 1,2,3-propanetriol	R	$(C_6H_{10}O_4 \cdot C_4H_{10}O_3 \cdot C_3H_8O_3)$
Poly(diallyl dimethyl ammonium chloride)	Diallyldimethylammonium chloride polymer	26062-79-3	2-Propen-1-aminium, *N,N*-dimethyl-*N*-2-propenyl-, chloride, homopolymer	R	$(C_8H_{16}N \cdot Cl)_x$
Poly(diallyl isophthalate)	Diallyl isophthalate polymer	25035-78-3	1,3-Benzenedicarboxylic acid, di-2-propenyl ester, homopolymer	R	$(C_{14}H_{14}O_4)_x$
Poly(diallyl phthalate)	Diallyl phthalate polymer	25053-15-0	1,2-Benzenedicarboxylic acid, di-2-propenyl ester, homopolymer	R	$(C_{14}H_{14}O_4)_x$
Poly(diaminobenzophenone-*co*-benzophenone tetracarboxylic dianhydride)	Benzophenone tetracarboxylic acid dianhydride-*m,m′*-diaminobenzophenone polymer, SRU	51518-44-6	Poly((1,3-dihydro-1,3-dioxo-2H-isoindole-2,5-diyl)carbonyl-(1,3-dihydro-1,3-dioxo-2H-isoindole-5,2-diyl)-1,3-phenylene-carbonyl-1,3-phenylene)	R	$(C_{30}H_{14}N_2O_6)_n$
Poly((dibenzo-18-crown-6)-*co*-formaldehyde)		53660-42-7	Formaldehyde, polymer with 6,7,9,10,17,18,20,21-octahydrodibenzo(b,k)(1,4,7,10,13,16) hexaoxacyclooctadecin	R	$(C_{20}H_{24}O_6 \cdot CH_2O)_x$
Poly(dibromostyrene)		62354-98-7	Benzene, dibromoethenyl-, homopolymer	R	$(C_8H_6Br_2)_x$
Poly(dichlorophosphazene)	Dichlorophosphazine polymer; Poly(nitrile(dichlorophosphoranylidyne))	26085-02-9	Poly(nitrilo(dichlorophosphoranylidyne))	R	$(Cl_2NP)_n$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(diethyl fumarate)		2698-06-6	2-Butenedioic acid (E)-, diethyl ester, homopolymer	R	$(C_8H_{12}O_4)_x$
Poly(diethylene glycol adipate)	Adipic acid–diethylene glycol polymer	9010-89-3	Hexanedioic acid, polymer with 2,2′-oxybis(ethanol)	R	$(C_6H_{10}O_4 \cdot C_4H_{10}O_3)_x$
Poly(diethylene glycol/trimethylol propane-*alt*-adipic acid), polyol	Adipic acid–diethylene glycol–trimethylolpropane polymer	28183-09-7	Hexanedioic acid, polymer with 2-ethyl-2-(hydroxymethyl)-1,3-propanediol and 2,2′-oxybis(ethanol)	R	$(C_6H_{14}O_3 \cdot C_6H_{10}O_4 \cdot C_4H_{10}O_3)_x$
Poly(1,2-dihydro-2,2,4-trimethyl quinoline)	Antioxidant HS; trimethyldihydroquinoline polymer	26780-96-1	Quinoline, 1,2-dihydro-2,2,4-trimethyl-, homopolymer	R	$(C_{12}H_{15}N)_x$
Poly(dimer acid-*co*-ethylene glycol), hydrogenated		68855-78-7	Fatty acids, C18-unsatd., dimers, hydrogenated, polymers with ethylene glycol	S	Unk
Poly(*N*,*N*-dimethylacrylamide)	*N*,*N*-Dimethylacrylamide polymer	26793-34-0	2-Propenamide, *N*,*N*-dimethyl-, homopolymer	R	$(C_5H_9NO)_x$
Poly(1,1-dimethyl 3,5-dimethylene piperidinium chloride)		32168-43-7	Poly((1,1-dimethylpiperidinium-3,5-diyl)methylene chloride)	R	$(C_8H_{16}N)_n \cdot Cl$
Poly(2,6-dimethyl-1,4-phenylene oxide)	Poly(2,6-xylenol)	25134-01-4	Phenol, 2,6-dimethyl-, homopolymer	R	$(C_8H_{10}O)_x$
Poly(2,2-dimethyl-1,3-propylene)	Polyneopentyl glycol succinate	28776-65-0	Poly(oxy(2,2-dimethyl-1,3-propanediyl)oxy(1,4-dioxo-1,4-butanediyl))	R	$(C_9H_{14}O_4)_n$
Poly(dimethyl siloxane), distearate terminated		130169-63-0	Siloxanes and silicones, di-Me, ((1-oxooctadecyl)oxy)-terminated	S	Unk
Poly(dimethyl siloxane), hydride terminated	Hydrogen-terminated di-Me silicones	70900-21-9	Siloxanes and silicones, di-Me, hydrogen-terminated	S	Unk
Poly(dimethyl siloxane), methoxy terminated		68951-97-3	Siloxanes and silicones, di-Me, methoxy-terminated	S	Unk
Poly(dimethyl siloxane), vinyl terminated		68951-99-5	Siloxanes and silicones, di-Me, Me vinyl, mono(vinyl group)-terminated	S	Unk
Poly(dimethyl siloxane-*co*-(3-hydroxypropyl)-methyl siloxane), ethoxylated/propoxylated		68937-55-3	Siloxanes and silicones, di-Me, 3-hydroxypropyl Me, ethoxylated propoxylated	S	Unk
Poly(dimethyl siloxane-*co*-methyl(3-OH-propyl) siloxane)-γ-PEG 3-aminopropyl ether)		133779-15-4	Siloxanes and silicones, di-Me, 3-hydroxypropyl Me, ethers with polyethylene glycol mono(3-aminopropyl) ether	S	Unk
Poly(dimethyl siloxane), hydroxy terminated	Silanol-terminated di-Me siloxanes	70131-67-8	Siloxanes and silicones, di-Me, hydroxy-terminated	S	Unk
Poly(dimethylamine-*co*-epichlorohydrin)	Epichlorohydrin–dimethylamine copolymer	25988-97-0	Methanamine, *N*-methyl-, polymer with (chloromethyl)oxirane	R	$(C_3H_5ClO \cdot C_2H_7N)_x$
Poly(dimethylamine-*co*-epichlorohydrin-*co*-ethylene diamine)		42751-79-1	1,2-Ethanediamine, polymer with (chloromethyl)oxirane and *N*-methylmethanamine	R	$(C_3H_5ClO \cdot C_2H_8N_2 \cdot C_2H_7N)_x$
Poly(dimethylsiloxane)	Dimethyl silicone; dimethylsilanediol polymer	9016-00-6	Poly(oxy(dimethylsilylene))	R	$(C_2H_6OSi)_n$
Poly(dimethylsiloxane), vinyl terminated	Dimethylsiloxanes, dimethylvinylsilyl-terminated	68083-19-2	Siloxanes and Silicones, di-Me, vinyl group-terminated	*	Unk
Poly(dimethylsiloxane-*co*-methylphenyl siloxane)	Methylphenylsiloxane polymer	9005-12-3	Poly(oxy(methylphenylsilylene))	R	$(C_7H_8OSi)_n$
Poly(dipropylene glycol), phenyl phosphite		116265-68-0	Phosphorous acid, triphenyl ester, reaction products with dipropylene glycol	R	$C_{18}H_{15}O_3P \cdot C_6H_{14}O_3$
Poly(3-dodecyl thiophene), regioregular	3-Dodecylthiophene homopolymer	104934-53-4	Thiophene, 3-dodecyl-, homopolymer	R	$(C_{16}H_{28}S)_x$
Poly(epichlorohydrin)	Poly(chloromethyl)ethylene oxide	24969-06-0	Oxirane, (chloromethyl)-, homopolymer	R	$(C_3H_5ClO)_x$
Poly(ethyl acrylate)	Ethyl acrylate polymer	9003-32-1	2-Propenoic acid, ethyl ester, homopolymer	R	$(C_5H_8O_2)_x$
Poly(ethyl methacrylate)	Ethyl methacrylate polymer	9003-42-3	2-Propenoic acid, 2-methyl-, ethyl ester, homopolymer	R	$(C_6H_{10}O_2)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(ethyl methacrylate-*co*-methyl acrylate)	Acrylic acid methyl ester, polymer with ethyl methacrylate	26572-20-3	2-Propenoic acid, 2-methyl-	R	$(C_6H_{10}O_2 \cdot C_4H_6O_2)_x$
Poly(ethylene adipate)	Adipic acid-ethylene glycol polymer; PEGA; poly(ethylene glycol adipate)	24938-37-2	Hexanedioic acid, polymer with 1,2-ethanediol	R	$(C_6H_{10}O_4 \cdot C_2H_6O_2)_x$
Poly(ethylene adipate), tolylene 2,4-diisocyanate terminated	Adipic acid–ethylene glycol–toluene diisocyanate copolymer	9019-92-5	Hexanedioic acid, polymer with 1,3-diisocyanatomethylbenzene and 1,2-ethanediol	R	$(C_9H_6N_2O_2 \cdot C_6H_{10}O_4 \cdot C_2H_6O_2)_x$
Poly(ethylene azelate)	Azelaic acid–ethylene glycol polymer	26760-99-6	Nonanedioic acid, polymer with 1,2-ethanediol	R	$(C_9H_{16}O_4 \cdot C_2H_6O_2)_x$
Poly(ethylene glycol methyl ether methacrylate)	Methoxypolyethylene glycol monomethacrylate	26915-72-0	HP = Poly(oxy-1,2-ethanediyl), α-(2-methyl-1-oxo-2-propenyl)-ω-methoxy-	R	$(C_2H_4O)_n$ $C_5H_8O_2$
Poly(ethylene glycol)	Carbowax; oxirane polymer; PEG, poly(ethylene oxide), poly(oxyethylene); poly(vinyl oxide)	25322-68-3	Poly(oxy-1,2-ethanediyl), α-hydro-ω-hydroxy-	U	$(C_2H_4O)_nH_2O$
Poly(ethylene glycol), allyl methyl ether		27252-80-8	Poly(oxy-1,2-ethanediyl), α-methyl-ω-(2-propenyloxy)-	R	$(C_2H_4O)_n$ C_4H_8O
Poly(ethylene glycol), bis-2-ethyl-hexanoate		9004-93-7	Poly(oxy-1,2-ethanediyl), α-(2-ethyl-1-oxohexyl)-ω-((2-ethyl-1-oxohexyl)oxy)-	R	$(C_2H_4O)_n$ $C_{16}H_{30}O_3$
Poly(ethylene glycol), butyl ether	Polyoxyethylene butyl alcohol ether	9004-77-7	Poly(oxy-1,2-ethanediyl), α-butyl-ω-hydroxy-	R	$(C_2H_4O)_n$ $C_4H_{10}O$
Poly(ethylene glycol), diacrylate		26570-48-9	Poly(oxy-1,2-ethanediyl), α-(1-oxo-2-propenyl)-ω-((1-oxo-2-propenyl)oxy)-	R	$(C_2H_4O)n$ $C_6H_6O_3$
Poly(ethylene glycol), dibenzoate	Polyoxyethylene dibenzoate	9004-86-8	Poly(oxy-1,2-ethanediyl), α-benzoyl-ω-(benzoyloxy)-	R	$(C_2H_4O)_n$ $C_{14}H_{10}O_3$
Poly(ethylene glycol), dimethacrylate	Polyethylene glycol methacrylate	25852-47-5	Poly(oxy-1,2-ethanediyl), α-(2-methyl-1-oxo-2-propenyl)-ω-((2-methyl-1-oxo-2-propenyl)oxy)-	R	$(C_2H_4O)_n$ $C_8H_{10}O_3$
Poly(ethylene glycol), dimethyl ether	Polyoxyethylene dimethyl ether	24991-55-7	Poly(oxy-1,2-ethanediyl), α-methyl-ω-methoxy-	R	$(C_2H_4O)_n$ C_3H_6O
Poly(ethylene glycol), distearate	Polyethylene oxide distearate	9005-08-7	Poly(oxy-1,2-ethanediyl), α-(1-oxooctadecyl)-ω-((1-oxooctadecyl)oxy)-	R	$(C_2H_4O)_n$ $C_{36}H_{70}O_3$
Poly(ethylene glycol), methyl ether	Carbowax; methoxypoly(ethylene glycol); methyl polyglycol; monomethoxy poly(ethylene oxide)	9004-74-4	Poly(oxy-1,2-ethanediyl), α-methyl-ω-hydroxy-	R	$(C_2H_4O)_nCH_4O$
Poly(ethylene glycol), methyl ether acrylate	Acrylic acid-methoxy(polyethylene glycol) ester	32171-39-4	Poly(oxy-1,2-ethanediyl), α-(1-oxo-2-propenyl)-ω-methoxy-	R	$(C_2H_4O)_n$ $C_4H_6O_2$
Poly(ethylene glycol), 4-nonylphenyl ether acrylate	Tetraethylene glycol nonylphenyl ether acrylate	87079-63-8	2-Propenoic acid, 2-(2-(2-(2-(nonylphenoxy)ethoxy)ethoxy)-ethoxy)ethyl ester	R	$C_{26}H_{42}O_6$
Poly(ethylene glycol), bis carboxymethyl ether	Carboxymethylpolyethylene glycol	39927-08-7	Poly(oxy-1,2-ethanediyl), α-(carboxymethyl)-ω-(carboxymethoxy)-	R	$(C_2H_4O)_n$ $C_4H_6O_5$
Poly(ethylene glycol), bis(3-aminopropyl) terminated		34901-14-9	Poly(oxy-1,2-ethanediyl), α-(3-aminopropyl)-ω-(3-aminopropoxy)-	R	$(C_2H_4O)_n$ $C_6H_{16}N_2O$
Poly(ethylene glycol), divinyl ether		50856-26-3	Poly(oxy-1,2-ethanediyl), α-ethenyl-ω-(ethenyloxy)-	R	$(C_2H_4O)_n$ C_4H_6O
Poly(ethylene glycol), ethyl ether methacrylate		35625-93-5	Poly(oxy-1,2-ethanediyl), α-(2-methyl-1-oxo-2-propenyl)-ω-ethoxy-	R	$(C_2H_4O)_n$ $C_6H_{10}O_2$
Poly(ethylene glycol), methyl glutarate		79934-70-6	Poly(oxy-1,2-ethanediyl), α-(4-carboxy-1-oxobutyl)-ω-methoxy-	R	$(C_2H_4O)_n$ $C_6H_{10}O_4$
Poly(ethylene glycol), phenyl ether acrylate	Phenol ethoxylate acrylate, polyethylene glycol monophenyl ether acrylate	56641-05-5	Poly(oxy-1,2-ethanediyl), α-(1-oxo-2-propenyl)-ω-phenoxy-	R	$(C_2H_4O)_n$ $C_9H_8O_2$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(ethylene glycol), reacted with bisphenol A diglycidyl ether	Bisphenol A diglycidyl ether–polyethylene glycol polymer	37225-26-6	Oxirane, 2,2′-((1-methylethylidene) bis(4,1-phenyleneoxymethylene))bis-polymer with α-hydro-ω-hydroxypoly(oxy-1,2-ethanediyl)	R	$(C_{21}H_{24}O_4 \cdot (C_2H_4O)_n H_2O)_x$
Poly(ethylene glycol), tetrahydrofurfuryl ether	Glycofurol; tetrahydrofurfuryl alcohol polyethylene glycol	31692-85-0	Poly(oxy-1,2-ethanediyl), α-((tetrahydro-2-furanyl)methyl)-ω-hydroxy-	R	$(C_2H_4O)_n C_5H_{10}O_2$
Poly(ethylene glycol), methacrylate		25736-86-1	Poly(oxy-1,2-ethanediyl), α-(2-methyl-1-oxo-2-propenyl)-ω-hydroxy-	R	$(C_2H_4O)_n C_4H_6O_2$
Poly(ethylene glycol-*co*-propylene glycol)	Polyethylene-polypropylene glycol monobutyl ether	9038-95-3	Oxirane, methyl-, polymer with oxirane, monobutyl ether	R	$C_4H_{10}O \cdot (C_3H_6O \cdot C_2H_4O)_x$
Poly(ethylene imine)	Polyaziridine, polyethenimide	9002-98-6	Aziridine, homopolymer	R	$(C_2H_5N)_x$
Poly(ethylene imine)	Ethylenediamine-ethylenimine polymer	25987-06-8	1,2-Ethanediamine, polymer with aziridine	R	$(C_2H_8N_2 \cdot C_2H_5N)_x$
Poly(ethylene imine), ethoxylated	Aziridine–ethylene oxide copolymer	26658-46-8	Aziridine, polymer with oxirane	R	$(C_2H_5N \cdot C_2H_4O)_x$
Poly(ethylene monoalcohol)		71750-71-5	Alcohols, c > 14	S	Unk
Poly(ethylene succinate)	Ethylene glycol–succinic acid polymer	25569-53-3	Butanedioic acid, polymer with 1,2-ethanediol	R	$(C_4H_6O_4 \cdot C_2H_6O_2)_x$
Poly(ethylene terephthalate)	Dimethyl terephthalate–ethylene glycol polymer; PET; poly(oxyethylene oxyterephthaloyl)	25038-59-9	Poly(oxy-1,2-ethanediyloxycarbonyl-1,4-phenylenecarbonyl)	U	$(C_{10}H_8O_4)_n$
Poly(ethylene)	Poly(methylene)	9002-88-4	Ethene, homopolymer	R	$(C_2H_4)_x$
Poly(ethylene), chlorinated		63231-66-3	Rubber, synthetic, chlorinated polyethylene	S	Unk
Poly(ethylene), chlorosulfonated		68037-39-8	Ethene, homopolymer, chlorinated, chlorosulfonated	*	Unk
Poly(ethylene), oxidized		68441-17-8	Ethene, homopolymer, oxidized	*	Unk
Poly(ethylene-*alt*-chlorotrifluoroethylene)	Chlorotrifluoroethene–ethene copolymer; ethylene–trifluorochloroethylene copolymer	25101-45-5	Ethene, chlorotrifluoro-, polymer with ethene	R	$(C_2H_4 \cdot C_2ClF_3)_x$
Poly(ethylene-*alt*-maleic anhydride)	Maleic anhydride–ethylene polymer	9006-26-2	2,5-Furandione, polymer with ethene	R	$(C_4H_2O_3 \cdot C_2H_4)_x$
Poly(ethylene-*block*-poly(ethylene glycol))		97953-22-5	Alcohols, C > 30, ethoxylated	S	Unk
Poly(ethylene-*co*-1-butene)	Butene–ethylene polymer; ethylene–1-butene copolymer; 1-butene-ethene polymer	25087-34-7	1-Butene, polymer with ethene	R	$(C_4H_8 \cdot C_2H_4)_x$
Poly(ethylene-*co*-1-butene-*co*-1-hexene)	Butene–ethene–hexene-1 copolymer	60785-11-7	1-Hexene, polymer with 1-butene and ethene	R	$(C_6H_{12} \cdot C_4H_8 \cdot C_2H_4)_x$
Poly(ethylene-*co*-1-octene)	Ethylene–1-octene polymer	26221-73-8	1-Octene, polymer with ethene	R	$(C_8H_{16} \cdot C_2H_4)_x$
Poly(ethylene-*co*-acrylic acid)	Acrylic acid–ethylene polymer; ethylene–acrylic acid polymers; primacor	9010-77-9	2-Propenoic acid, polymer with ethene	R	$(C_3H_4O_2 \cdot C_2H_4)_x$
Poly(ethylene-*co*-acrylic acid), sodium salt	Ethylene–acrylic acid polymer, sodium salt	25750-82-7	2-Propenoic acid, polymer with ethene, sodium salt	R	$(C_3H_4O_2 \cdot C_2H_4)_x \cdot x$Na
Poly(ethylene-*co*-acrylic acid), zinc salt		28208-80-2	2-Propenoic acid, polymer with ethene, zinc salt	R	$(C_3H_4O_2 \cdot C_2H_4)_x \cdot x$Zn
Poly(ethylene-*co*-butyl acrylate)	Butyl acrylate–ethylene copolymer	25750-84-9	2-Propenoic acid, butyl ester, polymer with ethene	R	$(C_7H_{12}O_2 \cdot C_2H_4)_x$
Poly(ethylene-*co*-butyl acrylate-*co*-maleic anhydride)		64652-60-4	2-Propenoic acid, butyl ester, polymer with ethene and 2,5-furandione	R	$(C_7H_{12}O_2 \cdot C_4H_2O_3 \cdot C_2H_4)_x$
Poly(ethylene-*co*-carbon monoxide)	Carbon monoxide–ethylene copolymer	25052-62-4	Ethene, polymer with carbon monoxide	R	$(C_2H_4 \cdot CO)_x$
Poly(ethylene-*co*-ethyl acrylate)	Ethyl acrylate–ethylene polymer	9010-86-0	2-Propenoic acid, ethyl ester, polymer with ethene	R	$(C_5H_8O_2 \cdot C_2H_4)_x$
Poly(ethylene-*co*-ethyl acrylate-*co*-maleic anhydride)		41171-14-6	2-Propenoic acid, ethyl ester, polymer with ethene and 2,5-furandione	R	$(C_5H_8O_2 \cdot C_4H_2O_3 \cdot C_2H_4)_x$
Poly(ethylene-*co*-glycidyl methacrylate)	2,3-Epoxypropyl methacrylate–ethylene copolymer	26061-90-5	2-Propenoic acid, 2-methyl-, oxiranylmethyl ester, polymer with ethene	R	$(C_7H_{10}O_3 \cdot C_2H_4)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(ethylene-*co*-methacrylic acid)	Ethylene–methacrylic acid copolymers; Poly(ethylene-methacrylic acid)	25053-53-6	2-Propenoic acid, 2-methyl-, polymer with ethene	R	$(C_4H_6O_2 \cdot C_2H_4)_x$
Poly(ethylene-*co*-methacrylic acid), sodium salt	Ethylene–methacrylic acid polymer, sodium salt	25608-26-8	2-Propenoic acid, 2-methyl, polymer with ethene, sodium salt	R	$(C_4H_6O_2 \cdot C_2H_4)_x \cdot xNa$
Poly(ethylene-*co*-methacrylic acid), zinc salt		28516-43-0	2-Propenoic acid, 2-methyl-, polymer with ethene, zinc salt	R	$(C_4H_6O_2 \cdot C_2H_4)_x \cdot xZn$
Poly(ethylene-*co*-methyl acrylate)	Ethylene–methyl acrylate polymer	25103-74-6	2-Propenoic acid, methyl ester, polymer with ethene	R	$(C_4H_6O_2 \cdot C_2H_4)_x$
Poly(ethylene-*co*-methyl acrylate-*co*-acrylic acid)		41525-41-1	2-Propenoic acid, polymer with ethene and methyl 2-propenoate	R	$(C_4H_6O_2 \cdot C_3H_4O_2 \cdot C_2H_4)_x$
Poly(ethylene-*co*-methyl acrylate-*co*-glycidyl methacrylate)		51541-08-3	2-Propenoic acid, 2-methyl-, oxiranylmethyl ester, polymer with ethene and methyl 2-propenoate	R	$(C_7H_{10}O_3 \cdot C_4H_6O_2 \cdot C_2H_4)_x$
Poly(ethylene-*co*-propylene)	Ethylene–propene copolymer; poly(ethylene-propylene); polypro, polypropylene–polyethylene copolymer	9010-79-1	1-Propene, polymer with ethene	R	$(C_3H_6 \cdot C_2H_4)_x$
Poly(ethylene-*co*-propylene-*co*-5-methylene-2-norbornene)	Ethylene–ethylidenebicycloheptene–propene copolymer; ethylene–ethylidenenorbornene–propylene polymer	25038-36-2	Bicyclo(2.2.1)hept-2-ene, 5-ethylidene-, polymer with ethene and 1-propene	R	$(C_9H_{12} \cdot C_3H_6 \cdot C_2H_4)_x$
Poly(ethylene-*co*-tetrafluoroethylene)	Ethylene copolymer with tetrafluoroethylene	25038-71-5	Ethene, tetrafluoro-, polymer with ethene	R	$(C_2H_4 \cdot C_2F_4)_x$
Poly(ethylene-*co*-vinyl acetate)	Vinyl acetate–ethylene polymer	24937-78-8	Acetic acid ethenyl ester, polymer with ethene	R	$(C_4H_6O_2 \cdot C_2H_4)_x$
Poly(ethylene-*co*-vinyl acetate-*co*-carbon monoxide)	Carbon monoxide–ethylene–vinyl acetate polymer	26337-35-9	Acetic acid ethenyl ester, polymer with carbon monoxide and ethene	R	$(C_4H_6O_2 \cdot C_2H_4 \cdot CO)_x$
Poly(ethylene-*graft*-maleic anhydride)		106343-08-2	2,5-Furandione, polymer with ethene, graft	R	$(C_4H_2O_3 \cdot C_2H_4)_x$
Poly(ethylene-*ran*-butylene), mono-ol		68954-09-6	1,3-Butadiene, homopolymer, hydrogenated	*	Unk
Poly(2-ethylhexyl acrylate)	2-Ethylhexyl acrylate polymer	9003-77-4	2-Propenoic acid, 2-ethylhexyl ester, homopolymer	R	$(C_{11}H_{20}O_2)_x$
Poly(2-ethylhexyl methacrylate)		25719-51-1	2-Propenoic acid, 2-methyl-, 2-ethylhexyl ester, homopolymer	R	$(C_{12}H_{22}O_2)_x$
Poly(2-ethyl-2-oxazoline)	2-Ethyl-2-oxazoline polymer; 2-ethyloxazoline homopolymer	25805-17-8	Oxazole, 2-ethyl-4,5-dihydro-, homopolymer	R	$(C_5H_9NO)_x$
Poly(4-ethylstyrene-*co*-divinylbenzene)	Divinylbenzene–ethylstyrene polymer	9043-77-0	Benzene, diethenyl-polymer with ethenylethylbenzene	R	$(C_{10}H_{12} \cdot C_{10}H_{10})_x$
Poly(furfuryl alcohol)		25212-86-6	2-Furanmethanol, homopolymer	R	$(C_5H_6O_2)_x$
Poly(D-glutamic acid), sodium salt		30811-79-1	D-Glutamic acid, homopolymer, sodium salt	R	$(C_5H_9NO_4)_x \cdot xNa$
Poly(L-glutamic acid), sodium salt	Poly(glutamic acid) sodium salt	26247-79-0	L-Glutamic acid, homopolymer, sodium salt	R	$(C_5H_9NO_4)_x \cdot xNa$
Poly(glycine)	Nylon 2, poly(imino(1-oxoethylene))	25718-94-9	Glycine, homopolymer	R	$(C_2H_5NO_2)_x$
Poly(glycolide)	Hydroxyacetic acid polymer; poly(glycolic acid)	26124-68-5	Acetic acid, hydroxy-, homopolymer	R	$(C_2H_4O_3)_x$
Poly(hexadecyl methacrylate)	Poly(cetyl methacrylate)	25986-80-5	2-Propenoic acid, 2-methyl-, hexadecyl ester, homopolymer	R	$(C_{20}H_{38}O_2)_x$
Poly(hexafluoroisopropylidene diphthalic anhydride-*co*-oxydianiline)	Hexafluoroisopropylidenebis-(phthalic anhydride)-oxydianiline copolymer	032240-73-6	1,3-Isobenzofurandione, 5,5′-(2,2,2-trifluoro-1-(trifluoromethyl)-ethylidene)bis-, polymer with 4,4′-oxybis(benzenamine)	R	$(C_{19}H_6F_6O_6 \cdot C_{12}H_{12}N_2O)_x$
Poly(1,6-hexamethylene adipate)	Adipic acid–hexanediol polymer	25212-06-0	Hexanedioic acid, polymer with 1,6-hexanediol	R	$(C_6H_{14}O_2 \cdot C_6H_{10}O_4)_x$
Poly(1,6-hexanediol/neopentyl glycol/di(ethylene glycol)-*alt*-adipic acid) diol		68492-71-7	Hexanedioic acid, polymer with 2,2-dimethyl-1,3-propanediol, 2-ethyl-2-(hydroxymethyl)-1,3-propanediol, 1,6-hexanediol and 2,2′-oxybis(ethanol)	R	$(C_6H_{14}O_3 \cdot C_6H_{14}O_2 \cdot C_6H_{10}O_4 \cdot C_5H_{12}O_2 \cdot C_4H_{10}O_3)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(1,6-hexanediol/neopentyl glycol-*alt*-adipic acid)	Adipic acid–hexamethylene glycol–neopentyl glycol polymer	25214-14-6	Hexanedioic acid, polymer with 2,2-dimethyl-1,3-propanediol and 1,6-hexanediol	R	$(C_6H_{14}O_2 \cdot C_6H_{10}O_4 \cdot C_5H_{12}O_2)_x$
Poly(hexamethylene diisocyanate)	HMDI homopolymer	28182-81-2	Hexane, 1,6-diisocyanato-, homopolymer	R	$(C_8H_{12}N_2O_2)_x$
Poly(hexyl methacrylate)	Hexyl methacrylate polymer	25087-17-6	2-Propenoic acid, 2-methyl-, hexyl ester, homopolymer	R	$(C_{10}H_{18}O_2)_x$
Poly(1-hexene)		25067-06-5	1-Hexene, homopolymer	R	$(C_6H_{12})_x$
Poly(3-hexyl thiophene), regioregular		140934-50-1	Thiophene, 3-hexyl-, homopolymer	R	$(C_{10}H_{16}S)_x$
Poly(L-histidine)	Polyhistidine	26062-48-6	L-Histidine, homopolymer	R	$(C_6H_7N_3O)_x$
Poly(L-histidine hydrochloride)	Polyhirtidine hydrochloride	61857-39-4	L-Histidine, homopolymer, hydrochloride	R	$(C_6H_9N_3O_2)_x \cdot x$ClH
Poly(4-hydroxybenzoate)	Poly(*p*-acetoxybenzoic acid)	26099-71-8	Poly(oxy-1,4-phenylenecarbonyl)	R	$(C_7H_4O_2)_n$
Poly(4-hydroxy benzoic acid-*co*-6-hydroxy-2-naphthoic acid)	*p*-Acetoxybenzoic acid-6-acetoxy-2-naphthoic acid polymer	70679-92-4	2-Naphthalenecarboxylic acid, 6-(acetyloxy)-, polymer with 4-(acetyloxy)benzoic acid	R	$(C_{13}H_{10}O_4 \cdot C_9H_8O_4)_x$
Poly(3-hydroxybutyric acid)	Poly(β-hydroxybutyric acid)	26063-00-3	Butanoic acid, 3-hydroxy-, homopolymer	R	$(C_4H_8O_3)_x$
Poly(3-hydroxybutyric acid-*co*-3-hydroxy valeric acid)	Bipol PHBV; 3-hydroxy butanoic acid-3-hydroxy valeric acid copolymer	80181-31-3	Pentanoic acid, 3-hydroxy-, polymer with 3-hydroxybutanoic acid	R	$(C_5H_{10}O_3 \cdot C_4H_8O_3)_x$
Poly(2-hydroxyethyl methacrylate)	Ethylene glycol monomethacrylate polymer	25249-16-5	2-Propenoic acid, 2-methyl-, 2-hydroxyethyl ester, homopolymer	R	$(C_6H_{10}O_3)_x$
Poly(2-hydroxypropyl methacrylate)		25703-79-1	2-Propenoic acid, 2-methyl, 2-hydroxypropyl ester, homopolymer	R	$(C_7H_{12}O_3)_x$
Poly(imino(tetrachloroterephthaloyl)-iminohexamethylene)	Hexamethylenediamine-perchloroterephthaloyl dichloride copolymer, SRU	071164-49-3	Poly(iminocarbonyl(2,3,5,6-tetrachloro-1,4-phenylene)-carbonylimino-1,6-hexanediyl)	R	$(C_{14}H_{14}Cl_4N_2O_2)_n$
Poly(isobornyl methacrylate)	Methacrylic acid, isobornyl ester, polymers	64114-51-8	2-Propenoic acid, 2-methyl-, 1,7,7-trimethylbicyclo(2.2.1)hept-2-yl ester, exo-, homopolymer	R	$(C_{14}H_{22}O_2)_x$
Poly(isobutyl methacrylate)	Isobutyl methacrylate polymer	9011-15-8	2-Propenoic acid, 2-methyl-, 2-methylpropylester, homopolymer	R	$(C_8H_{14}O_2)_x$
Poly(isobutylene)	Poly(1,1-dimethylethylene); isobutene polymer; isobutylene polymer; polyisobutene	9003-27-4	1-Propene, 2-methyl, homopolymer	R	$(C_4H_8)_x$
Poly(isobutylene-*alt*-maleic anhydride)		26426-80-2	2,5-Furandione, polymer with 2-methyl-1-propene	R	$(C_4H_8 \cdot C_4H_2O_3)_x$
Poly(isobutylene-*co*-isoprene)	Isobutylene–isoprene polymer	9010-85-9	1,3-Butadiene, 2-methyl, polymer with 2-methyl-1-propene	R	$(C_5H_8 \cdot C_4H_8)_x$
Poly(isobutylene-*co*-isoprene)		68081-82-3	1,3-Butadiene, 2-methyl-, polymer with 2-methyl-1-propene, chlorinated	*	$(C_5H_8 \cdot C_4H_8)_x$
Poly(isobutylene-*co*-isoprene), brominated		68441-14-5	1,3-Butadiene, 2-methyl-, polymer with 2-methyl-1-propene, brominated	*	$(C_5H_8 \cdot C_4H_8)_x$
Poly(isodecyl methacrylate)		27300-12-7	2-Propenoic acid, 2-methyl-, isodecyl ester, homopolymer	R	$(C_{14}H_{26}O_2)_x$
Poly(isophorone diisocyanate)		53880-05-0	Cyclohexane, 5-isocyanato-1-(isocyanatomethyl)-1,3,3-trimethyl-, homopolymer	R	$(C_{12}H_{18}N_2O_2)_x$
Poly(isophorone diisocyanate-*co*-poly(propylene glycol))	Isophorone diisocyanate–polypropylene ether glycol polymer; polypropylene glycol–isophorone diisocyanate copolymer	039323-37-0	Poly(oxy(methyl-1,2-ethanediyl)), α-hydro-ω-hydroxy-, polymer with	R	$(C_{12}H_{18}N_2O_2 \cdot (C_3H_6O)_n H_2O)_x$
Poly(isoprene), *cis*	Poly(1-methyl-1-butenylene)	9003-31-0	1,3-Butadiene, 2-methyl-, homopolymer	R	$(C_5H_8)_x$
Poly(isoprene), *trans*		104389-32-4	Poly(1-methyl-1-butene-1,4-diyl), (E)-	R	$(C_5H_8)n$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(isoprene)-*block*-poly(butadiene)	Butadiene–isoprene polymer; 1,3-Butadiene–isoprene polymer	25102-52-7	1,3-Butadiene, 2-methyl-, polymer with 1,3-butadiene	R	$(C_5H_8 \cdot C_4H_6)_x$
Poly(isoprene-*graft*-maleic acid monomethyl ester)		128000-08-8	1,3-Butadiene, 2-methyl-, homopolymer, maleated, mono-methyl esters	*	Unk
Poly(2-isopropenyl-2-oxazoline-*co*-methyl methacrylate)		27341-60-2	2-Propenoic acid, 2-methyl-, methyl ester, polymer	R	$(C_6H_9NO \cdot C_5H_8O_2)_x$
Poly(isopropyl methacrylate)		26655-94-7	2-Propenoic acid, 2-methyl-, 1-methylethyl ester, homopolymer	R	$(C_7H_{12}O_2)_x$
Poly(isopropyl vinyl ether)	Isopropyl vinyl ether polymer	25585-49-3	Propane, 2-(ethenyloxy)-, homopolymer	R	$(C_5H_{10}O)_x$
Poly(itaconic acid)	Poly(2-methylenesuccinic acid)	25119-64-6	Butanedioic acid, methylene, homopolymer	R	$(C_5H_6O_4)_x$
Poly(lactide)	Lactide polymer; poly(DL-lactide)	26680-10-4	1,4-Dioxane-2,5-dione, 3,6-dimethyl-homopolymer	R	$(C_6H_8O_4)_x$
Poly(D,L-lactide-*co*-glycolide)	Glycolide-DL-lactide polymer; polyglactin	26780-50-7	1,4-Dioxane-2,5-dione, 3,6-dimethylpolymer with 1,4-dioxane-2,5-dione	R	$(C_6H_8O_4 \cdot C_4H_4O_4)_x$
Poly(lauryl lactam)-*block*-poly(tetrahydrofuran)	Maleic anhydride-1-tetradecene polymer	31473-53-7	2,5-Furandione, polymer with 1-tetradecene	R	$(C_{14}H_{28} \cdot C_4H_2O_3)_x$
Poly(lauryl methacrylate)	Poly(dodecyl methacrylate)	25719-52-2	2-Propenoic acid, 2-methyl-, dodecyl ester, homopolymer	R	$(C_{16}H_{30}O_2)_x$
Poly(lauryl methacrylate-*co*-ethylene glycol dimethacrylate)	Ethylene glycol dimethacrylate–lauryl methacrylate copolymer	61181-29-1	2-Propenoic acid, 2-methyl-, 1,2-ethanediyl ester, polymer with dodecyl 2-methyl-2-propenoate	R	$(C_{16}H_{30}O_2 \cdot C_{10}H_{14}O_4)_x$
Poly(D,L-lysine hydrobromide)		73565-55-6	DL-Lysine, homopolymer, hydrobromide	R	$(C_6H_{14}N_2O_2)_x \cdot x$BrH
Poly(L-leucine)	Polyleucine	25248-98-0	L-Leucine, homopolymer	R	$(C_6H_{13}NO_2)_x$
Poly(L-leucine hydrobromide)	Poly-L-lysine hydrogen bromide; polylysine hydrobromide	25988-63-0	L-Lysine, homopolymer, hydrobromide	R	$(C_6H_{14}N_2O_2)_x \cdot x$BrH
Poly(D-lysine hydrobromide)		27964-99-4	D-Lysine, homopolymer, hydrobromide	R	$(C_6H_{14}N_2O_2)_x \cdot x$BrH
Poly(L-lysine hydrochloride)	Lysine homopolymer hydrochloride	28826-16-6	L-Lysine, homopolymer, hydrochloride	R	$(C_6H_{14}N_2O_2)_x \cdot x$ClH
Poly(L-lysine)	Lysine homopolymer	25104-18-1	L-Lysine, homopolymer	R	$(C_6H_{14}N_2O_2)_x$
Poly(maleic anhydride-*alt*-1-octadecene)	1-Octadecene–maleic anhydride polymer	25266-02-8	2,5-Furandione, polymer with 1-octadecene	R	$(C_{18}H_{36} \cdot C_4H_2O_3)_x$
Poly(maleic anhydride-*alt*-α-olefin)		68036-97-5	1,3,5-Triazine-2,4,6-triamine, polymer with formaldehyde, butylated methylated	*	$(C_3H_6N_6 \cdot CH_2O)_x$
Poly(melamine-*co*-formaldehyde), butylated		68002-25-5	1,3,5-Triazine-2,4,6-triamine, polymer with formaldehyde, butylated	*	$(C_3H_6N_6 \cdot CH_2O)_x$
Poly(melamine-*co*-formaldehyde), butylated		68459-79-0	Alkenes, C24-28 α-, polymers with maleic anhydride	S	Unk
Poly(melamine-*co*-formaldehyde), butylated/methylated		70750-53-7	Terpenes and terpenoids, limonene fraction, polymd.	S	Unk
Poly(melamine-*co*-formaldehyde), methylated	Laurolactam–polytetramethylene glycol copolymer	84732-38-7	Azacyclotridecan-2-one, polymer with α-hydro-ω-hydroxypoly-(oxy-1,4-butanediyl)	R	$(C_{12}H_{23}NO \cdot (C_4H_8O)_n H_2O)_x$
Poly(melamine-*co*-formaldehyde), isobutylated		68002-21-1	1,3,5-Triazine-2,4,6-triamine, polymer with formaldehyde, isobutylated	*	$(C_3H_6N_6 \cdot CH_2O)_x$
Poly(methacrylic acid), sodium salt		54193-36-1	2-Propenoic acid, 2-methyl-, homopolymer, sodium salt	R	$(C_4H_6O_2)_x \cdot x$Na
Poly(L-methionine)	Methionine homopolymer	26242-79-0	L-Methionine, homopolymer	R	$(C_5H_{11}NO_2S)_x$
Poly(4-methoxystyrene)	Poly(*p*-methoxystyrene); poly(4-vinylanisole)	24936-44-5	Benzene, 1-ethenyl-4-methoxy-, homopolymer	R	$(C_9H_{10}O)_x$
Polymethylene polyphenylene isocyanate	Poly(phenyl isocyanate-*co*-formaldehyde), poly(methylene poly(phenyl isocyanate))	9016-87-9	Isocyanic acid, polymethylene-polyphenylene ester	R	Unk

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(methyl acrylate)	Methyl acrylate polymer	9003-21-8	2-Propenoic acid, methyl ester, homopolymer	R	$(C_4H_6O_2)_x$
Poly(4,4′-methylene bis(phenyl isocyanate)-*alt*-butanediol/di-PG/polycaprolactone))		68084-39-9	2-Oxepanone, polymer with 1,4-butanediol, 1,1′-methylenebis(isocyanatobenzene) and oxybis(propanol)	R	$(C_{15}H_{10}N_2O_2 \cdot C_6H_{14}O_3 \cdot C_6H_{10}O_2 \cdot C_4H_{10}O_2)_x$
Poly(4,4′-methylene bisphenyl isocyanate)	1,4-Butanediol-methylenedi-*p*-phenylene isocyanate-polytetramethylene–ether glycol copolymer	9018-04-6	1,4-Butanediol, polymer with α-hydro-ω-hydroxypoly(oxy-1,4-butanediyl) and 1,1′-methylenebis(4-isocyanatobenzene)	R	$(C_{15}H_{10}N_2O_2 \cdot C_4H_{10}O_2 \cdot (C_4H_8O)_n H_2O)_x$
Poly(methyl methacrylate)	Methacrylic acid methyl ester; methyl methacrylate polymer	9011-14-7	2-Propenoic acid, 2-methyl, methyl ester, homopolymer	R	$(C_5H_8O_2)_x$
Poly(methyl methacrylate-*co*-butadiene-*co*-styrene)	Butadiene–methyl methacrylate–styrene polymer	25053-09-2	2-Propenoic acid,2-methyl-, methyl ester, polymer with 1,3-butadiene and ethenylbenzene	R	$(C_8H_8 \cdot C_5H_8O_2 \cdot C_4H_6)_x$
Poly(methyl methacrylate-*co*-ethyl acrylate)	Methyl methacrylate–ethyl acrylate polymer	9010-88-2	2-Propenoic acid, 2-methyl-, methyl ester, polymer with ethyl 2-propenoate	R	$(C_5H_8O_2 \cdot C_5H_8O_2)_x$
Poly(3-methyl 1,5-pentanediol adipate)	Adipic acid-3-methyl-1,5-pentanediol copolymer	39751-34-3	Hexanedioic acid, polymer with 3-methyl-1,5-pentanediol	R	$(C_6H_{14}O_2 \cdot C_6H_{10}O_4)_x$
Poly(4-methyl-1-pentene)	Methylpentene polymer; 4-methylpentene homopolymer	25068-26-2	1-Pentene, 4-methyl-, homopolymer	R	$(C_6H_{12})x$
Poly(methyl sorbate)		30813-48-0	2,4-Hexadienoic acid, methyl ester, (E,E)-, homopolymer	R	$(C_7H_{10}O_2)_x$
Poly(4-methyl styrene)	*p*-Methylstyrene polymer; 4-methylstyrene homopolymer	24936-41-2	Benzene, 1-ethenyl-4-methyl-, homopolymer	R	$(C_9H_{10})_x$
Poly(α-methyl styrene)	α-Methylstyrene polymer	25014-31-7	Benzene, (1-methylethenyl)-, homopolymer	R	$(C_9H_{10})_x$
Poly(methyl styrene-*co*-indene), hydrogenated		69430-35-9	Hydrocarbons, C6–C20, polymers, hydrogenated	S	Unk
Poly(methyl vinyl ether-*alt*-maleic acid monoethyl ester)	Ethyl maleate–methyl vinyl ether polymer; maleic acid monoethyl ester–methyl vinyl ether polymer; monoethyl maleate–methyl vinyl ether polymer	25087-06-3	2-Butenedioic acid (Z)-, monoethyl ester, polymer with methoxyethene	R	$(C_6H_8O_4 \cdot C_3H_6O)_x$
Poly(methyl vinyl ether-*alt*-maleic acid monoisopropyl ester)		31307-95-6	2-Butenedioic acid (Z)-, mono(1-methylethyl) ester, polymer with methoxyethene	R	$(C_7H_{10}O_4 \cdot C_3H_6O)_x$
Poly(methyl vinyl ether-*alt*-maleic acid)	Maleic acid–methyl vinyl ether copolymer; maleic acid–methyl vinyl ether polymer	25153-40-6	2-Butenedioic acid (Z)-, polymer with methoxyethene	R	$(C_4H_4O_4 \cdot C_3H_6O)_x$
Poly(methyl vinyl ether-*alt*-maleic anhydride)	Maleic anhydride–methyl vinyl ether polymer	9011-16-9	2,5-Furandione, polymer with methoxyethene	R	$(C_4H_2O_3 \cdot C_3H_6O)_x$
Poly(methyl vinyl ether-*alt*-maleic anhydride)	1,9-Decadiene–maleic anhydride–methyl vinyl ether copolymer	136392-67-1	2,5-Furandione, polymer with 1,9-decadiene and methoxyethene	R	$(C_{10}H_{18} \cdot C_4H_2O_3 \cdot C_3H_6O)_x$
Poly(methylhydrosiloxane)	Methyl hydrogen siloxane	9004-73-3	Poly(oxy(methylsilylene))	R	$(CH_4OSi)_n$
Poly(neopentyl glycol adipate)	Adipic acid–neopentyl glycol copolymer	27925-07-1	Hexanedioic acid, polymer with 2,2-dimethyl-1,3-propanediol	R	$(C_6H_{10}O_4 \cdot C_5H_{12}O_2)_x$
Poly(neopentyl glycol sebacate)		28759-54-8	Poly(oxy(2,2-dimethyl-1,3-propanediyl)oxy(1,10-dioxo-1,10-decanediyl))	R	$(C_{15}H_{26}O_4)_n$
Poly(neopentyl methacrylate)		34903-87-2	2-Propenoic acid, 2-methyl-, 2,2-dimethylpropyl ester, homopolymer	R	$(C_9H_{16}O_2)_x$
Poly(norbornene)	Poly(2-norbornene); polynorbornenylene	25038-76-0	Bicyclo(2.2.1)hept-2-ene, homopolymer	R	$(C_7H_{10})_x$
Poly(octadecyl methacrylate)	Stearyl methacrylate homopolymer	25639-21-8	2-Propenoic acid, 2-methyl-, octadecyl ester, homopolymer	R	$(C_{22}H_{42}O_2)_x$
Poly(octadecyl vinyl ether-*co*-maleic anhydride)	Maleic anhydride–octadecyl vinyl ether polymer	28214-64-4	2,5-Furandione, polymer with 1-(ethenyloxy)octadecane	R	$(C_{20}H_{40}O \cdot C_4H_2O_3)_x$
Poly(octyl acrylate)	Octyl acrylate polymer	25266-13-1	2-Propenoic acid, octyl ester, homopolymer	R	$(C_{11}H_{20}O_2)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(3-octyl thiophene), regioregular		104934-51-2	Thiophene, 3-octyl-, homopolymer	R	$(C_{12}H_{20}S)_x$
Poly(L-ornithine)	Polyornithine	25104-12-5	L-Ornithine, homopolymer	R	$(C_5H_{12}N_2O_2)_x$
Poly(L-ornithine hydrobromide)	Polyorthine hydrobromide	27378-49-0	L-Ornithine, homopolymer, hydrobromide	R	$(C_5H_{12}N_2O_2)_x \cdot xBrH$
Poly(D,L-ornithine hydrobromide)		82682-33-5	DL-Ornithine, homopolymer, hydrobromide	R	$(C_5H_{12}N_2O_2)_x \cdot xBrH$
Poly(L-ornithine hydrochloride)	Poly-L-ornithine-hydrochloride	26982-21-8	L-Ornithine, homopolymer, hydrochloride	R	$(C_5H_{12}N_2O_2)_x \cdot xClH$
Poly[oxy(2,6-dimethyl-1,4-phenylene)]	Poly(2,6-dimethyl-*p*-phenylene ether); poly(2,6-dimethyl-1,4-phenylene oxide). SRU	24938-67-8	Poly(oxy(2,6-dimethyl-1,4-phenylene))	R	
Poly(oxy-1,4-phenylene oxy-1,4-phenylene carbonyl 1,4-phenylene)	4,4′-Difluorobenzophenone–hydroquinone copolymer, SRU; PEEK	31694-16-3	Poly(oxy-1,4-phenyleneoxy-1,4-phenylenecarbonyl-1,4-phenylene)	R	$(C_{19}H_{12}O_3)_n$
Poly(oxycarbonyloxy-1,4-phenyleneisopropylidene-1,4-phenylene)	Bisphenol A polycarbonate; poly (4,4′-isopropylidenediphenyl carbonate)	24936-68-3	Poly(oxycarbonyloxy-1,4-phenylene(1-methylethylidene)-1,4-phenylene)	R	$(C_{16}H_{14}O_3)_n$
Poly(oxymethylene)	Poly(formaldehyde); formaldehyde polymer	9002-81-7	Poly(oxymethylene)	R	$(CH_2O)_n$
Poly(oxymethylene), acetate end-capped	Poly(oxymethylene) diacetate	25231-38-3	Poly(oxymethylene), α-acetyl-ω-(acetyloxy)-	R	$(CH_2O)_n C_4H_6O_3$
Poly(oxymethylene-*co*-1,3-dioxepane)	Butanediol formal–trioxane copolymer; formaldehyde tetramethylene acetal–trioxane copolymer; 1,3-dioxepane-trioxane polymer	25214-85-1	1,3-Dioxepane, polymer with 1,3,5-trioxane	R	$(C_5H_{10}O_2 \cdot C_3H_6O_3)_x$
Poly(PAMAM), starburst dendrimer	Starburst 2nd generation	93376-66-0	4,7,11,14,18,21,25,28,32,35-Decaazaoctatriacontane-diamide, etc	R	$C_{142}H_{288}N_{58}O_{28}$
Poly(perfluoropropylene oxide-*co*-perfluoro formaldehyde)		69991-67-9	1-Propene, 1,1,2,3,3,3-hexafluoro-, oxidized, polymd.	*	Unk
Poly(*n*,*n*′-1,4-phenylene-3,3′,4,4′-benzophenone tetracarboxylic imid/amic acid	Benzophenonetetracarboxylic dianhydride-*p*-phenylenediamine polymer, SRU	26023-21-2	Poly((1,3-dihydro-1,3-dioxo-2H-isoindole-2,5-diyl)carbonyl-(1,3-dihydro-1,3-dioxo-2H-isoindole-5,2-diyl)-1,4-phenylene)	U	$(C_{23}H_{10}N_2O_5)_n$
Poly(*n*,*n*′-1,3-phenylene isophthalamide)	Isophthaloyl chloride-1,3-phenylenediamine polymer	25765-47-3	1,3-Benzenedicarbonyl dichloride, polymer with 1,3- benzenediamine	R	$(C_8H_4Cl_2O_2 \cdot C_6H_8N_2)_x$
Poly(*n*,*n*-1,4-phenylene terephthalamide)	*p*-Phenylenediamine–terephthaloyl chloride copolymer	26125-61-1	1,4-Benzenedicarbonyl dichloride, polymer with 1,4-benzenediamine	R	$(C_8H_4Cl_2O_2 \cdot C_6H_8N_2)_x$
Poly((phenyl glycidyl ether)-*co*-dicyclopentadiene)		119345-05-0	Phenol, polymer with 3a,4,7,7a-tetrahydro-4,7-methano-1H-indene, glycidyl ether	*	$(C_{10}H_{12} \cdot C_6H_6O)_x$
Poly(phenyl glycidyl ether-*co*-formaldehyde)		28064-14-4	Phenol, polymer with formaldehyde, glycidyl ether	R	$(C_6H_6O \cdot CH_2O)_x$
Poly(phenyl vinyl ketone)	Poly(acrylophenone)	26742-84-7	2-Propen-1-one, 1-phenyl-, homopolymer	R	$(C_9H_8O)_x$
Poly(1,4-phenylene diisocyanate-*co*-poly(1,4-butanediol))		89339-41-3	Poly(oxy-1,4-butanediyl),α-hydro-ω-hydroxy-, polymer with 1,4-diisocyanatobenzene	R	$(C_8H_4N_2O_2 \cdot (C_4H_8O)_n H_2O)_x$
Poly(1,4-phenylene ether ether sulfone)		28212-68-2	Poly(oxy-1,4-phenyleneoxy-1,4-phenylenesulfonyl-1,4-phenylene)	R	$(C_{18}H_{12}O_4S)_n$
Poly(1,4-phenylene ether sulfone)		25667-42-9	Poly(oxy-1,4-phenylenesulfonyl-1,4-phenylene)	U	$(C_{12}H_8O_3S)_n$
Poly(phenylene oxide)	Poly(phenylene ether)	9041-80-9	Poly(oxyphenylene)	R	$(C_6H_4O)_n$
Poly(1,4-phenylene sulfide)	*p*-Dichlorobenzene-sodium sulfide polymer; poly(*p*-phenylene sulfide)	25212-74-2	Poly(thio-1,4-phenylene)	U	$(C_6H_4S)_n$
Poly(*p*-phenylene terephthalamide)	PPTA	24938-64-5	Poly(imino-1,4-phenyleneimino-carbonyl-1,4-phenylene-carbonyl)	U	$(C_{14}H_{10}N_2O_2)_n$

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Poly(α-pinene)		70750-57-1	Terpenes and terpenoids, turpentine-oil, α-pinene fraction, polymd.	S	Unk
Poly(poly(tetrahydrofuran) carbonate), diol		92538-66-4	Carbonic acid, diethyl ester, polymer with α-hydro-ω-hydroxypoly(oxy-1,4-butanediyl)	R	$(C_5H_{10}O_3 \cdot (C_4H_8O)_n H_2O)_x$
Poly(L-proline)	Polyproline; proline homopolymer	25191-13-3	L-Proline, homopolymer	R	$(C_5H_9NO_2)_x$
Poly(1,3-propylene adipate)	Ethyl acrylate–ethylene–glycidyl methacrylate polymer	35830-43-4	2-Propenoic acid, 2-methyl-, oxiranylmethyl ester, polymer with ethene and ethyl 2-propenoate	R	$(C_7H_{10}O_3 \cdot C_5H_8O_2 \cdot C_2H_4)_x$
Poly(propylene carbonate)	Carbon dioxide–propylene oxide polymer	25511-85-7	Oxirane, methyl-, polymer with carbon dioxide	R	$(C_3H_6O \cdot CO_2)_x$
Poly(1,3-propylene glutarate)	Poly(trimethylene glutarate)	52256-48-1	Pentanedioic acid, polymer with 1,3-propanediol	R	$(C_5H_8O_4 \cdot C_3H_8O_2)_x$
Poly(propylene glycol bis(2-aminopropyl ether))	Poly(oxypropylene)diamine	9046-10-0	Poly(oxy(methyl-1,2-ethanediyl)), α-(2-aminomethylethyl)-ω-(2-aminomethylethoxy)-	R	$(C_3H_6O)_n C_6H_{16}N_2O$
Poly(propylene glycol)	Oxirane, methyl-, homopolymer; poly(propylene oxide); PPG: propylene glycol–propylene oxide polymer	25322-69-4	Poly(oxy(methyl-1,2-ethanediyl)), α-hydro-ω-hydroxy-	U	$(C_3H_6O)_n H_2O$
Poly(propylene glycol), 4-nonyl phenol ether acrylate		71926-19-7	Poly(oxy(methyl-1,2-ethanediyl)), α-(1-oxo-2-propenyl)-ω-(nonylphenoxy)-	R	$(C_3H_6O)_n C_{18}H_{26}O_2$
Poly(propylene glycol), diacrylate		52496-08-9	Poly(oxy(methyl-1,2-ethanediyl)), α-(1-oxo-2-propenyl)-ω-((1-oxo-2-propenyl)oxy)-	R	$(C_3H_6O)_n C_6H_6O_3$
Poly(propylene glycol), dimethacrylate		25852-49-7	Poly(oxy(methyl-1,2-ethanediyl)), α-(2-methyl-1-oxo-2-propenyl)-ω-((2-methyl-1-oxo-2-propenyl)oxy)-	R	$(C_3H_6O)_n C_8H_{10}O_3$
Poly(propylene glycol), methacrylate		39420-45-6	Poly(oxy(methyl-1,2-ethanediyl)), α-(2-methyl-1-oxo-2-propenyl)-ω-hydroxy-	R	$(C_3H_6O)_n C_4H_6O_2$
Poly(propylene glycol), dibenzoate		72245-46-6	Poly(oxy(methyl-1,2-ethanediyl)), α-benzoyl-ω-(benzoyloxy)-	R	$(C_3H_6O)_n C_{14}H_{10}O_3$
Poly(propylene glycol), diglycidyl ether		26142-30-3	Poly(oxy(methyl-1,2-ethanediyl)), α-(oxiranylmethyl)-ω-(oxiranylmethoxy)-	R	$(C_3H_6O)_n C_6H_{10}O_3$
Poly(propylene glycol), monobutyl ether	Butoxypolypropylene glycol; polyoxypropylene monobutyl ether	9003-13-8	Poly(oxy(methyl-1,2-ethanediyl)), α-butyl-ω-hydroxy-	R	$(C_3H_6O)_n C_4H_{10}O$
Poly(propylene glycol)-*co*-poly(ethylene glycol)	Ethylene glycol–propylene glycol copolymer; ethylene glycol–propylene oxide copolymer; ethylene oxide–propylene oxide copolymer ethylene glycol ether; methyloxirane–oxirane copolymer	9003-11-6	Oxirane methyl, polymer with oxirane	R	$(C_3H_6O \cdot C_2H_4O)_x$
Poly(propylene glycol-*block*-PEG-*block*-PPG bis (2-aminopropyl ether))	Ethylene oxide–propylene oxide copolymer bis(2-aminopropyl)-ether	65605-36-9	Oxirane, methyl-, polymer with oxirane, bis(2-aminopropyl) ether	R	$(C_3H_9NO)_{1/2} (C_3H_6O \cdot C_2H_4O)_x$
Poly(1,3-propylene succinate)	Poly(trimethylene succinate)	28158-21-6	Butanedioic acid, polymer with 1,3-propanediol	R	$(C_4H_6O_4 \cdot C_3H_8O_2)_x$
Poly(propylene)	Poly(propene); propene polymer	9003-07-0	1-Propene, homopolymer	R	$(C_3H_6)_x$
Poly(propylene), chlorinated		68442-33-1	1-Propene, homopolymer, chlorinated	*	Unk
Poly(propylene), isotactic	Isotactic propylene polymer; polypro	25085-53-4	1-Propene, homopolymer isotactic	R	$(C_3H_6)_x$
Poly(propylene-*alt*-ethylene, multiarm		127883-08-3	Benzene, diethenyl-, polymer with 2-methyl-1,3-butadiene, hydrogenated	*	$(C_{10}H_{10} \cdot C_5H_8)_x$
Poly(propylene-*co*-1-butene)		29160-13-2	1-Butene, polymer with 1-propene	R	$(C_4H_8 \cdot C_3H_6)_x$
Poly(propylene-*co*-tetrafluoroethylene)	Propene–tetrafluoroethene polymer	27029-05-6	1-Propene, polymer with tetrafluoroethene	R	$(C_3H_6 \cdot C_2F_4)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(propylene-*graft*-maleic anhydride)	Maleic anhydride–PN 240 graft copolymer	107001-49-0	2,5-Furandione, polymer with 1-propene, graft	R	$(C_4H_2O_3 \cdot C_3H_6)_x$
Poly(pyromellitic dianhydride-*co*-4,4′-oxydianiline)	Diaminodiphenyl ether–pyromellitic dianhydride polymer; oxydianiline–pyromellitic dianhydride polymer	25038-81-7	1H, 3H-Benzo(1,2-c:4,5-c′)-difuran-1,3,5,7-tetrone, polymer with 4,4′-oxybis(benzenamine)	R	$(C_{12}H_{12}N_2O \cdot C_{10}H_2O_6)_x$
Poly(sebacic anhydride)	Sebacic acid polyanhydride	26776-29-4	Decanedioic acid, homopolymer	R	$(C_{10}H_{18}O_4)_x$
Poly(L-serine)	Polyserine	25821-52-7	L-Serine, homopolymer	R	$(C_3H_7NO_3)_x$
Poly(sodium 4-styrene sulfonate)	Poly(sodium *p*-styrenesulfonate); poly(sodium *p*-vinylbenzenesulfonate)	25704-18-1	Benzenesulfonic acid, 4-ethenyl-, sodium salt, homopolymer	R	$(C_8H_8O_3S \cdot Na)_x$
Poly(styrene)	Poly(1-phenylethylene); poly(vinylbenzene)	9003-53-6	Benzene, ethenyl-, homopolymer	R	$(C_8H_8)_x$
Poly(styrene-*alt*-maleic acid), sodium salt	Maleic acid anhydride–styrene polymer, sodium salt; 2,5-furandione–styrene copolymer, sodium salt	25736-61-2	2,5-Furandione, polymer with ethenylbenzene, sodium salt	R	$(C_8H_8 \cdot C_4H_2O_3)_x \cdot x$Na
Poly(styrene-*b*-butadiene)	Butadiene–styrene polymer, polybutadiene–polystyrene copolymer, styrene–butadiene polymer	9003-55-8	Benzene, ethenyl-, polymer with 1,3-butadiene	R	$(C_8H_8 \cdot C_4H_6)x$
Poly(styrene-*b*-poly(ethylene-*ran*-butylene)-*b*-poly(styrene))		66070-58-4	Benzene, ethenyl-, polymer with 1,3-butadiene, hydrogenated	*	Unk
Poly(styrene-*block*-poly(oxyethylene))	Ethylene oxide–styrene polymer	25267-79-2	Oxirane, polymer with ethenylbenzene	R	$(C_8H_8 \cdot C_2H_4O)_x$
Poly(styrene-*block*-polyisoprene-*block*-polystyrene)	Isoprene–styrene polymer	25038-32-8	Benzene, ethenyl-, polymer with 2-methyl-1,3-butadiene	R	$(C_8H_8 \cdot C_5H_8)_x$
Poly(styrene-*co*-acrylonitrile)	Acrylonitrile–styrene polymer; polystyrene–acrylonitrile; styrene–acrylonitrile polymer	9003-54-7	2-Propenenitrile, polymer with ethenylbenzene	R	$(C_8H_8 \cdot C_3H_3N)_x$
Poly(styrene-*co*-allyl alcohol)	Allyl alcohol–styrene polymer	25119-62-4	2-Propen-1-ol, polymer with ethenylbenzene	R	$(C_8H_8 \cdot C_3H_6O)_x$
Poly(styrene-*co*-α-methylstyrene)	Styrene-α–methylstyrene polymer	9011-11-4	Benzene, ethenyl-, polymer with (1-methylethenyl)benzene	R	$(C_9H_{10} \cdot C_8H_8)_x$
Poly(styrene-*co*-divinyl benzene)		69011-14-9	Benzene, diethenyl-, polymer with ethenylbenzene and ethenylethylbenzene, chloromethylated	*	$(C_{10}H_{12} \cdot C_{10}H_{10} \cdot C_8H_8)_x$
Poly(styrene-*co*-divinyl benzene), amine functionalized		69011-17-2	Methanamine, *N*-methyl-, reaction products with chloromethylated divinylbenzene–ethenylethylbenzene-styrene polymer	*	$(C_{10}H_{12} \cdot C_{10}H_{10} \cdot C_8H_8 \cdot C_2H_7N)_x$
Poly(styrene-*co*-maleic acid), isobutyl/methyl mixed ester		63528-92-7	2-Butenedioic acid (Z)-, monomethyl ester, polymer with ethenylbenzene, 2,5-furandione and (Z)-2-methylpropyl hydrogen 2-butenedioate	R	$(C_8H_{12}O_4 \cdot C_8H_8 \cdot C_5H_6O_4 \cdot C_4H_2O_3)_x$
Poly(styrene-*co*-maleic acid), partial 2-butoxyethyl ester		68890-84-6	2,5-Furandione, polymer with ethenylbenzene, 2-butoxyethyl ester	R	Unk
Poly(styrene-*co*-maleic acid), partial 2-butoxyethyl ester, cumene terminated		160611-50-7	2,5-Furandione, telomer with ethenylbenzene and (1-methylethyl)benzene, 2-butoxyethyl ester	R	$C_9H_{12} \cdot (C_8H_8 \cdot C_4H_2O_3)_x \cdot xC_6H_{14}O_2$
Poly(styrene-*co*-maleic acid), partial cyclohexyl/1-PR ester, cumene terminated		160611-51-8	2,5-Furandione, telomer with ethenylbenzene and (1-methylethyl)benzene, cyclohexyl 1-methylethyl ester	R	$C_9H_{12} \cdot (C_8H_8 \cdot C_4H_2O_3)_x \cdot xC_6H_{12}O \cdot xC_3H_8O$
Poly(styrene-*co*-maleic acid), partial cyclohexyl/isopropyl ester		68890-82-4	2,5-Furandione, polymer with ethenylbenzene, cyclohexyl 1-methylethyl ester	R	Unk
Poly(styrene-*co*-maleic acid), partial isobutyl ester	Hydrogen isobutyl maleate–styrene copolymer	28571-95-1	2-Butenedioic acid (Z)-, mono (2-methylpropyl) ester, polymer with ethenylbenzene	R	$(C_8H_{12}O_4 \cdot C_8H_8)_x$
Poly(styrene-*co*-maleic acid), partial isooctyl ester		68441-87-2	2,5-Furandione, polymer with ethenylbenzene, isooctyl ester	R	Unk

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(styrene-*co*-maleic acid), partial propyl ester	Maleic anhydride–styrene polymer, propyl ester	68890-81-3	2,5-Furandione, polymer with ethenylbenzene, propyl ester	R	Unk
Poly(styrene-*co*-maleic acid), partial propyl ester, cumene terminated		160611-48-3	2,5-Furandione, telomer with ethenylbenzene and (1-methyl ethyl)benzene, propyl ester	R	$C_9H_{12} \cdot (C_8H_8 \cdot C_4H_2O_3)_x \cdot xC_3H_8O$
Poly(styrene-*co*-maleic acid), *sec*-butyl/methyl mixed ester		65652-36-0	2-Butenedioic acid (Z)-, monomethyl ester, polymer with ethenylbenzene, 2,5-furandione and (Z)-1-methylpropyl hydrogen 2-butenedioate	R	$(C_8H_{12}O_4 \cdot C_8H_8 \cdot C_5H_6O_4 \cdot C_4H_2O_3)_x$
Poly(styrene-*co*-maleic anhydride)	Maleic anhydride–styrene polymers	9011-13-6	2,5-Furandione, polymer with ethenylbenzene	R	$(C_8H_8 \cdot C_4H_2O_3)_x$
Poly(styrene-*co*-maleic anhydride), cumene terminated		26762-29-8	2,5-Furandione, telomer with ethenylbenzene and (1-methylethyl)benzene	R	$C_9H_{12} \cdot (C_8H_8 \cdot C_4H_2O_3)$
Poly(styrene sulfonic acid-*co*-maleic acid), sodium salt		68037-40-1	2,5-Furandione, polymer with ethenylbenzene, sulfonated, sodium salt	*	$(C_8H_8 \cdot C_4H_2O_3 \cdot Na)_x$
Poly(sulfone)	Bisphenol A–bis(chlorophenyl) sulfone copolymer, SRU	25135-51-7	Poly(oxy-1,4-phenylenesulfonyl-1,4-phenyleneoxy-1,4-phenylene(1-methylethylidene)-1,4-phenylene)	U	$(C_{27}H_{22}O_4S)_n$
Poly(*tert*-butyl methacrylate)		25189-00-8	2-Propenoic acid, 2-methyl-, 1,1-dimethylethyl ester, homopolymer	R	$(C_8H_{14}O_2)_x$
Poly(tetrafluoroethylene)	Poly(difluoromethylene); fluoropolymers; perfluoroethylene polymer	9002-84-0	Ethene, tetrafluoro-homopolymer	R	$(C_2F_4)_x$
Poly(tetrahydrofuran)	Poly(butylene oxide); tetrahydrofuran homopolymer, SRU; THF polymer, SRU	25190-06-1	Poly(oxy-1,4-butanediyl), α-hydro-ω-hydroxy-	R	$(C_4H_8O)_nH_2O$
Poly(tetrahydrofuran), bis(3-aminopropyl) terminated	Polytetramethylene glycol bis(3-aminopropyl) ether	72088-96-1	Poly(oxy-1,4-butanediyl), α-(3-aminopropyl)-ω-(3-aminopropoxy)-	R	$(C_4H_8O)_n C_6H_{16}N_2O$
Poly(tetrahydrofuran), linear	Oxirane–THF copolymer; polyethylene-poly(tetramethylene) glycol	27637-03-2	Furan, tetrahydro-, polymer with oxirane	R	$(C_4H_8O \cdot C_2H_4O)_x$
Poly(thioisobutylene)	Isobutylene sulfide polymer	26373-01-3	Thirane, 2,2-dimethyl-, homopolymer	R	$(C_4H_8S)_x$
Poly(thiophenylene)	Poly(phenylene sulfide)	9016-75-5	Poly(thiophenylene)	U	$(C_6H_4S)_x$
Poly(*p*-toluene sulfonamide-*co*-formaldehyde)	Formaldehyde-*p*-toluenesulfonamide polymer	25035-71-6	Benzenesulfonamide, 4-methyl-, polymer with formaldehyde	R	$(C_7H_9NO_2S \cdot CH_2O)_x$
Poly(tolylene 2,4-diisocyanate)	2,4-Toluene diisocyanate polymer	26006-20-2	Benzene, 2,4-diisocyanato-1-methyl- homopolymer	R	$(C_9H_6N_2O_2)_x$
Poly(tolylene 2,4-diisocyanate-*co*-poly(1,4-butanediol))	Poly(tetramethylene ether glycol)–TDI copolymer; polytetramethylene glycol–tolylene isocyanate polymer	9069-50-5	Poly(oxy-1,4-butanediyl), α-hydro-ω-hydroxy-, polymer with 1,3-diisocyanatomethylbenzene	R	$(C_9H_6N_2O_2 \cdot (C_4H_8O)_n H_2O)_x$
Poly(tolylene 2,4-diisocyanate-*co*-poly(propylene glycol))	Polyoxypropylene glycol–tolylene diisocyanate polymer; polypropylene glycol–TDI polymer	9057-91-4	Poly(oxy(methyl-1,2-ethanediyl)), α-hydro-ω-hydroxy-, polymer with 1,3-diisocyanatomethylbenzene	R	$(C_9H_6N_2O_2 \cdot (C_3H_6O)_n H_2O)_x$
Poly(trimethylolpropane/di(propylene glycol)-*alt*-adipic acid/phthalic anhydride))		68133-07-3	Hexanedioic acid, polymer with 2-ethyl-2-(hydroxymethyl)-1,3-propanediol, 1,3-isobenzofurandione and 1,1′-oxybis-(2-propanol)	R	$(C_8H_4O_3 \cdot C_6H_{14}O_3 \cdot C_6H_{14}O_3 \cdot C_6H_{10}O_4)_x$
Poly(L-tyrosine)	Polytyrosine	25619-78-7	L-Tyrosine, homopolymer	R	$(C_9H_{11}NO_3)_x$
Poly(L-tyrosine hydrobromide)		27378-49-0	L-Tyrosine, homopolymer, hydrobromide	R	$(C_5H_{12}N_2O_2)_x \cdot xBrH$
Poly(urea-*co*-formaldehyde), butylated		68002-19-7	Urea, polymer with formaldehyde, butylated	*	Unk
Poly(urea-*co*-formaldehyde), methylated		68071-45-4	Urea, polymer with formaldehyde, methylated	*	$(CH_4N_2O \cdot CH_2O)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(vinyl acetate)	Poly(1-acetoxyethylene); PVAC	9003-20-7	Acetic acid ethenyl ester, homopolymer	R	$(C_4H_6O_2)_x$
Poly(vinyl acetate-*co*-butyl maleate-*co*-isobornyl acrylate)		136392-68-2	2-Butenedioic acid (Z)-, dibutyl ester, polymer with ethenyl acetate and exo-1,7,7-trimethylbicyclo(2.2.1)hept-2-yl 2-propenoate	R	$(C_{13}H_{20}O_2 \cdot C_{12}H_{20}O_4 \cdot C_4H_6O_2)_x$
Poly(vinyl acetate-*co*-crotonic acid)	Crotonic acid–vinyl acetate polymer	25609-89-6	2-Butenoic acid, polymer with ethenyl acetate	R	$(C_4H_6O_2 \cdot C_4H_6O_2)_x$
Poly(vinyl alcohol)	Poly(1-hydroxyethylene); gohsenal; lemol; polyvinol; polyviol	9002-89-5	Ethenol, homopolymer	R	$(C_2H_4O)_x$
Poly(vinyl alcohol-*co*-ethylene)	Ethylene–vinyl alcohol polymer	25067-34-9	Ethenol, polymer with ethene	R	$(C_2H_4O \cdot C_2H_4)_x$
Poly(vinyl alcohol-*co*-vinyl acetate)	Vinyl alcohol–vinyl acetate polymer	25213-24-5	Acetic acid ethenyl ester, polymer with ethenol	R	$(C_4H_6O_2 \cdot C_2H_4O)_x$
Poly(4-vinyl biphenyl)		25232-08-0	1,1′-Biphenyl, 4-ethenyl-homopolymer	R	$(C_{14}H_{12})_x$
Poly(vinyl bromide)	Bromoethylene polymer	25951-54-6	Ethene, bromo-, homopolymer	R	$(C_2H_3Br)_x$
Poly(vinyl butyral)		63148-65-2	Vinyl acetal polymers, butyrals	S	Unk
Poly(vinyl butyral-*co*-vinyl alcohol-*co*-vinyl acetate)		27360-07-2	Acetic acid ethenyl ester, polymer with 1,1-bis(ethenyloxy)butane and ethenol	R	$(C_8H_{14}O_2 \cdot C_4H_6O_2 \cdot C_2H_4O)_x$
Poly(vinyl butyrate)		24991-31-9	Butanoic acid, ethenyl ester, homopolymer	R	$(C_6H_{10}O_2)_x$
Poly(vinyl chloride)	Poly(1-chloroethylene)	9002-86-2	Ethane, chloro-homopolymer	R	$(C_2H_3Cl)_x$
Poly(vinyl chloride-*co*-vinyl acetate)	Vinyl chloride–vinyl acetate polymer	9003-22-9	Acetic acid ethenyl ester, polymer with chloroethene	R	$(C_4H_6O_2 \cdot C_2H_3Cl)_x$
Poly(vinyl chloride-*co*-vinyl acetate-*co*-2-hydroxypropyl acrylate)		41618-91-1	2-Propenoic acid, 2-hydroxypropyl ester, polymer with chloroethene and ethenyl acetate	R	$(C_6H_{10}O_3 \cdot C_4H_6O_2 \cdot C_2H_3Cl)_x$
Poly(vinyl chloride-*co*-vinyl acetate-*co*-maleic acid)	Maleic acid–vinyl acetate–vinyl; vinyl acetate–vinyl chloride–maleic acid polymer	9005-09-8	2-Butenedioic acid (Z)-, 2-Butenedioic acid (Z)-	R	$(C_4H_6O_2 \cdot C_4H_4O_4 \cdot C_2H_3Cl)_x$
Poly(vinyl chloride-*co*-vinyl acetate-vinyl alcohol)	Vinyl acetate–vinyl alcohol–vinyl chloride polymer; vinyl chloride–vinyl acetate–vinyl alcohol copolymer	25086-48-0	Acetic acid ethenyl ester, polymer with chloroethene and ethenol	R	$(C_4H_6O_2 \cdot C_2H_4O \cdot C_2H_3Cl)_x$
Poly(vinyl cinnamate)		24968-99-8	2-Propenoic acid,3-phenyl-, ethenyl ester, homopolymer	R	$(C_{11}H_{10}O_2)_x$
Poly(vinyl ethyl ether)	Poly(ethyl vinyl ether)	25104-37-4	Ethene, ethoxy-, homopolymer	R	$(C_4H_8O)_x$
Poly(vinyl fluoride)	Monofluoroethylene polymer	24981-14-4	Ethene, fluoro-, homopolymer	R	$(C_2H_3F)_x$
Poly(vinyl formal)	Poly(divinyl formal)	9003-33-2	1,1′-(methylenebis(oxy))bis-, homopolymer	R	$(C_5H_8O_2)_x$
Poly(*N*-vinyl-methylacetamide)	Poly(*N*-methyl-*N*-vinylacetamide)	26616-03-5	Acetamide, *N*-ethenyl-*N*-methyl-, homopolymer	R	$(C_5H_9NO)_x$
Poly(vinyl methyl ether)	Methyl vinyl ether homopolymer	9003-09-2	Ethene methoxy, homopolymer	R	$(C_3H_6O)_x$
Poly(vinyl methyl ketone)	Methyl vinyl ketone polymer	25038-87-3	3-Buten-2-one, homopolymer	R	$(C_4H_6O)_x$
Poly(2-vinyl naphthalene)		28406-56-6	Naphthalene, 2-ethenyl-, homopolymer	R	$(C_{12}H_{10})_x$
Poly(vinyl *n*-octodecylcarbamate)		36671-85-9	Carbamic acid, octadecyl-, ethenyl ester, homopolymer	R	$(C_{21}H_{41}NO_2)_x$
Poly(4-vinyl phenol)	*p*-Vinylphenol polymer; *p*-vinylphenol polymer	24979-70-2	Phenol, 4-ethenyl-, homopolymer	R	$(C_8H_8O)_x$
Poly(vinyl propionate)	Vinyl propionate polymer	25035-84-1	Propanoic acid, ethenyl ester, homopolymer	R	$(C_5H_8O_2)_x$
Poly(2-vinyl pyridine)	2-Vinylpyridine polymer	25014-15-7	Pyridine, 2-ethenyl-, homopolymer	R	$(C_7H_7N)_x$
Poly(2-vinyl pyridine-*co*-styrene)	2-Vinylpyridine-styrene polymer	24980-54-9	Pyridine, 2-ethenyl-, polymer with styrene	R	$(C_8H_8 \cdot C_7H_7N)_x$
Poly(vinyl pyrrolidone)	Poly(*n*-pyrrolidinylethylene); poly(vinylpyrrolidone); vinylpyrrolidone polymer	9003-39-8	2-Pyrrolidinone, 1-ethenyl-, homopolymer	R	$(C_6H_9NO)_x$
Poly(vinyl pyrrolidone), iodine complex		25655-41-8	2-Pyrrolidinone, 1-ethenyl-, homopolymer, compd. with iodine	R	$(C_6H_9NO)_x \cdot xI_2$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(1-vinyl pyrrolidone-*co*-acrylic acid)	Acrylic acid–vinylpyrrolidone polymer	28062-44-4	2-Propenoic acid, polymer with 1-ethenyl-2-pyrrolidinone	R	$(C_6H_9NO \cdot C_3H_4O_2)_x$
Poly(1-vinyl pyrrolidone-*co*-2-dimethylamino-ethyl methacrylate)		30581-59-0	2-Propenoic acid, 2-methyl-, 2-(dimethylamino)ethyl ester, polymer with 1-ethenyl-2-pyrrolidinone	R	$(C_8H_{15}NO_2 \cdot C_6H_9NO)_x$
Poly(1-vinyl pyrrolidone-*co*-styrene)	Styrene-*N*-vinylpyrrolidone polymer; vinylpyrrolidinone–styrene polymer	25086-29-7	2-Pyrrolidinone, 1-ethenyl-, polymer with ethenylbenzene	R	$(C_8H_8 \cdot C_6H_9NO)_x$
Poly(1-vinyl pyrrolidone-*co*-vinyl acetate)	*N*-Vinyl-2-pyrrolidone–vinyl acetate copolymer; poly(vinyl acetate–vinylpyrrolidinone); vinyl acetate–vinylpyrrolidone copolymer	25086-89-9	Acetic acid ethenyl ester, polymer with 1-ethenyl-2-pyrrolidinone	R	$(C_6H_9NO \cdot C_4H_6O_2)_x$
Poly(1-vinyl pyrrolidone-γ-(1-triacontene))		136445-69-7	Alkenes, C > 10 α-, polymers with vinylpyrrolidone	S	Unk
Poly(1-vinyl pyrrolidone-γ-1-hexadecene)		63231-81-2	2-Pyrrolidinone, 1-ethenyl-hexadecyl-, homopolymer	R	$(C_{22}H_{41}NO)_x$
Poly(vinyl stearate)	Vinyl stearate polymer	9003-95-6	Octadecanoic acid, ethenyl ester, homopolymer	R	$(C_{20}H_{38}O_2)_x$
Poly(vinyl sulfate), potassium salt		26182-60-5	Sulfuric acid, monoethenyl ester, potassium salt, homopolymer	R	$(C_2H_4O_4S \cdot K)_x$
Poly(vinyl sulfonic acid), sodium salt	Ethenesulfonic acid polymer sodium salt; ethenesulfonic acid polymer sodium salt	25053-27-4	Ethenesulfonic acid, homopolymer, sodium salt	R	$(C_2H_4O_3S)_x \cdot x$Na
Poly(vinyl toluene)	Poly(methylstyrene); vinyltoluene polymer	9017-21-4	Benzene, ethenylmethyl-homopolymer	R	$(C_9H_{10})x$
Poly(vinyl toluene-*co*-alpha-methylstyrene)	α-Methylstyrene–vinyltoluene polymer	9017-27-0	Benzene, ethenylmethyl-, polymer with (1-methylethenyl)benzene	R	$(C_9H_{10} \cdot C_9H_{10})_x$
Poly(vinylcarbazole); Poly(9-vinyl carbazole)	Poly(*n*-carbazolylethylene); poly(*N*-vinylcarbazole)	25067-59-8	9H-Carbazole, 9-ethenyl-, homopolymer	R	$(C_{14}H_{11}N)_x$
Poly(vinylcyclohexane)		25498-06-0	Cyclohexane, ethenyl-, homopolymer	R	$(C_8H_{14})_x$
Poly(vinylferrocene)		34801-99-5	Ferrocene, ethenyl-, homopolymer	R	$(C_{12}H_{12}Fe)_x$
Poly(vinylidene chloride)	Poly(1,1-dichloroethylene)	9002-85-1	Ethene,1,1-dichloro-, homopolymer	R	$(C_2H_2Cl_2)_x$
Poly(vinylidene chloride-*co*-acrylonitrile)	Acrylonitrile–vinylidene chloride polymer	9010-76-8	2-Propenenitrile, polymer with 1,1-dichloroethene	R	$(C_3H_3N \cdot C_2H_2Cl_2)_x$
Poly(vinylidene chloride-*co*-methyl acrylate)	Methyl acrylate–vinylidene chloride polymer	25038-72-6	2-Propenoic acid, methyl ester, polymer with 1,1-dichloro ethene	R	$(C_4H_6O_2 \cdot C_2H_2Cl_2)_x$
Poly(vinylidene chloride-*co*-vinyl chloride)	Vinyl chloride–vinylidene chloride polymer	9011-06-7	Ethene, 1,1-dichloro-, polymer with chloroethylene	R	$(C_2H_3Cl \cdot C_2H_2Cl_2)_x$
Poly(vinylidene fluoride)	Poly(1,1-difluoroethylene); PVDF	24937-79-9	Ethene, 1,1-difluoro-, homopolymer	R	$(C_2H_2F_2)_x$
Poly(vinylidene fluoride-*co*-hexafluoropropylene)	Hexafluoropropene–vinylidene fluoride polymer	9011-17-0	1-Propene, 1,1,2,3,3,3-hexafluoro-, polymer with 1,1-difluoroethene	R	$(C_3F_6 \cdot C_2H_2F_2)_x$
Poly(vinylidene fluoride-*co*-hexafluoropropylene-*co*-tetrafluoroethylene)	Hexafluoropropene–tetrafluoroethylene–vinylidene fluoride polymer; vinylidene fluoride–tetrafluoroethylene–perfluoropropylene copolymer	25190-89-0	1-Propene, 1,1,2,3,3,3-hexafluoro-, polymer with 1,1-difluoroethene and tetrafluoroethene	R	$(C_3F_6 \cdot C_2H_2F_2 \cdot C_2F_4)_x$
Poly(vinylidene fluoride-*co*-tetrafluoroethylene-*co*-propylene)	Propene–tetrafluoroethylene–vinylidene fluoride copolymer	54675-89-7	1-Propene, polymer with 1,1-difluoroethene and tetrafluoroethene	R	$(C_3H_6 \cdot C_2H_2F_2 \cdot C_2F_4)_x$
Poly(vinylisobutyl ether)	Isobutyl vinyl ether polymer	9003-44-5	Propane, 1-(ethenyloxy)-2-methyl-, homopolymer	R	$(C_6H_{12}O)_x$
Poly(4-vinyl pyridine)	4-Vinylpyridine homopolymer	25232-41-1	Pyridine, 4-ethenyl-, homopolymer	R	$(C_7H_7N)_x$
Poly(4-vinyl pyridine-*co*-butyl methacrylate)	Butyl methacrylate-4-vinylpyridine polymer	53761-76-5	2-Propenoic acid, 2-methyl-, butyl ester, polymer with 4-ethenylpyridine	R	$(C_8H_{14}O_2 \cdot C_7H_7N)_x$
Poly(4-vinyl pyridene-*co*-styrene)		26222-40-2	Pyridine, 4-ethenyl-, polymer with ethenylbenzene	R	$(C_8H_8 \cdot C_7H_7N)_x$
Poly(4-vinyl pyridine hydrochloride), crosslinked		29323-87-3	Pyridine, 4-ethenyl-, hydrochloride, homopolymer	R	$(C_7H_7N \cdot ClH)_x$
Poly(4-vinyl pyridine), crosslinked	Divinylbenzene-4-vinylpyridine polymer	9017-40-7	Pyridine, 4-ethenyl-, polymer with diethenylbenzene	R	$(C_{10}H_{10} \cdot C_7H_7N)_x$

Common name	Synonyms	CAS Registry No.	CA name	Search method	Molecular formula
Poly(4-vinyl pyridinium dichromate)		82246-45-5	Chromic acid ($H_2Cr_2O_7$), salt with diethenylbenzene polymer with 4-ethenylpyridine	R	$(C_{10}H_{10} \cdot C_7H_7N)_x \cdot xCr_2H_2O_7$
Poly(4-vinyl pyridinium tribromide)		91650-35-0	Hydrogen tribromide, compd. with 4-ethenylpyridine (1 : 1), homopolymer	R	$(C_7H_7N \cdot Br_3H)_x$
Poly(*p*-xylylene)		25951-90-0	Benzene, 1,4-dimethyl-, homopolymer	R	$(C_8H_{10})_x$
Rosin, ester with glycerol		8050-31-5	Resin acids and rosin acids, esters with glycerol	S*	Unk
Rosin, ester with pentaerythritol		8050-26-8	Resin acids and rosin acids, esters with pentaerythritol	S*	Unk
Rubber, chlorinated		9006-03-5	Rubber, chlorinated	R	Unk
Safflower seed oil		8001-23-8	Safflower oil	S*	Unk
Silicone oil	Methyl phenyl silicones	63148-58-3	Siloxanes and Silicones, Me Ph	S	Unk
Soybean oil	Fats and glyceridic oils	8001-22-7	Soybean oil	S*	Unk
Starch	Arrowroot starch; potato starch; rice starch; tapioca starch	9005-25-8	Starch	R	Unk
Starch, soluble		9005-84-9	Amylodextrin	R	Unk
Starch-*graft*-poly(acrylic acid), sodium salt	Acrylic acid–starch polymer sodium salt	60323-79-7	Starch, polymer with 2-propenoic acid, sodium salt	R	$(C_3H_4O_2 \cdot Unk)_x \cdot xNa$
Tri-*o*-benzyl cellulose		91104-69-7	Cellulose, tris(phenylmethyl) ether	R	$(C_7H_8O)_{1/3}$; Unk
Tung oil	Fats and glyceridic oils	8001-20-5	Tung oil	S*	Unk
Zein		9010-66-6	Zeins	S*	Unk

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This index lists only physical constants of polymer classes and a few selected specific polymers. Information on specific polymers can be found under the polymer class name. Specific polymers in the index, mostly from Section V, do not necessarily represent the only place of reference of that polymer. Additional references may be found under the respective polymer class name.

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